

The Concepts section helps you learn about the parts of the Kubernetes system and the abstractions Kubernetes uses to represent your [cluster](#), and helps you obtain a deeper understanding of how Kubernetes works.

[Overview](#)

Kubernetes is a portable, extensible, open source platform for managing containerized workloads and services, that facilitates both declarative configuration and automation. It has a large, rapidly growing ecosystem. Kubernetes services, support, and tools are widely available.

[Cluster Architecture](#)

The architectural concepts behind Kubernetes.

[Containers](#)

Technology for packaging an application along with its runtime dependencies.

[Workloads](#)

Understand Pods, the smallest deployable compute object in Kubernetes, and the higher-level abstractions that help you to run them.

[Services, Load Balancing, and Networking](#)

Concepts and resources behind networking in Kubernetes.

[Storage](#)

Ways to provide both long-term and temporary storage to Pods in your cluster.

[Configuration](#)

Resources that Kubernetes provides for configuring Pods.

[Security](#)

Concepts for keeping your cloud-native workload secure.

[Policies](#)

Manage security and best-practices with policies.

[Scheduling, Preemption and Eviction](#)

In Kubernetes, scheduling refers to making sure that Pods are matched to Nodes so that the kubelet can run them. Preemption is the process of terminating Pods with lower Priority so that

Pods with higher Priority can schedule on Nodes. Eviction is the process of proactively terminating one or more Pods on resource-starved Nodes.

[Cluster Administration](#)

Lower-level detail relevant to creating or administering a Kubernetes cluster.

[Windows in Kubernetes](#)

Kubernetes supports nodes that run Microsoft Windows.

[Extending Kubernetes](#)

Different ways to change the behavior of your Kubernetes cluster.

Overview

Kubernetes is a portable, extensible, open source platform for managing containerized workloads and services, that facilitates both declarative configuration and automation. It has a large, rapidly growing ecosystem. Kubernetes services, support, and tools are widely available.

This page is an overview of Kubernetes.

Kubernetes is a portable, extensible, open source platform for managing containerized workloads and services, that facilitates both declarative configuration and automation. It has a large, rapidly growing ecosystem. Kubernetes services, support, and tools are widely available.

The name Kubernetes originates from Greek, meaning helmsman or pilot. K8s as an abbreviation results from counting the eight letters between the "K" and the "s". Google open-sourced the Kubernetes project in 2014. Kubernetes combines [over 15 years of Google's experience](#) running production workloads at scale with best-of-breed ideas and practices from the community.

Going back in time

Let's take a look at why Kubernetes is so useful by going back in time.

Deployment evolution

Traditional deployment era: Early on, organizations ran applications on physical servers. There was no way to define resource boundaries for applications in a physical server, and this caused resource allocation issues. For example, if multiple applications run on a physical server, there can be instances where one application would take up most of the resources, and as a result, the other applications would underperform. A solution for this would be to run each application on a different physical server. But this did not scale as resources were underutilized, and it was expensive for organizations to maintain many physical servers.

Virtualized deployment era: As a solution, virtualization was introduced. It allows you to run multiple Virtual Machines (VMs) on a single physical server's CPU. Virtualization allows applications to be isolated between VMs and provides a level of security as the information of one application cannot be freely accessed by another application.

Virtualization allows better utilization of resources in a physical server and allows better scalability because an application can be added or updated easily, reduces hardware costs, and much more. With virtualization you can present a set of physical resources as a cluster of disposable virtual machines.

Each VM is a full machine running all the components, including its own operating system, on top of the virtualized hardware.

Container deployment era: Containers are similar to VMs, but they have relaxed isolation properties to share the Operating System (OS) among the applications. Therefore, containers are considered lightweight. Similar to a VM, a container has its own filesystem, share of CPU, memory, process space, and more. As they are decoupled from the underlying infrastructure, they are portable across clouds and OS distributions.

Containers have become popular because they provide extra benefits, such as:

- Agile application creation and deployment: increased ease and efficiency of container image creation compared to VM image use.
- Continuous development, integration, and deployment: provides for reliable and frequent container image build and deployment with quick and efficient rollbacks (due to image immutability).
- Dev and Ops separation of concerns: create application container images at build/release time rather than deployment time, thereby decoupling applications from infrastructure.
- Observability: not only surfaces OS-level information and metrics, but also application health and other signals.
- Environmental consistency across development, testing, and production: runs the same on a laptop as it does in the cloud.
- Cloud and OS distribution portability: runs on Ubuntu, RHEL, CoreOS, on-premises, on major public clouds, and anywhere else.
- Application-centric management: raises the level of abstraction from running an OS on virtual hardware to running an application on an OS using logical resources.
- Loosely coupled, distributed, elastic, liberated micro-services: applications are broken into smaller, independent pieces and can be deployed and managed dynamically – not a monolithic stack running on one big single-purpose machine.
- Resource isolation: predictable application performance.
- Resource utilization: high efficiency and density.

Why you need Kubernetes and what it can do

Containers are a good way to bundle and run your applications. In a production environment, you need to manage the containers that run the applications and ensure that there is no downtime. For example, if a container goes down, another container needs to start. Wouldn't it be easier if this behavior was handled by a system?

That's how Kubernetes comes to the rescue! Kubernetes provides you with a framework to run distributed systems resiliently. It takes care of scaling and failover for your application, provides deployment patterns, and more. For example: Kubernetes can easily manage a canary deployment for your system.

Kubernetes provides you with:

- **Service discovery and load balancing** Kubernetes can expose a container using the DNS name or using their own IP address. If traffic to a container is high, Kubernetes is able to load balance and distribute the network traffic so that the deployment is stable.
- **Storage orchestration** Kubernetes allows you to automatically mount a storage system of your choice, such as local storages, public cloud providers, and more.
- **Automated rollouts and rollbacks** You can describe the desired state for your deployed containers using Kubernetes, and it can change the actual state to the desired state at a controlled rate. For example, you can automate Kubernetes to create new containers for your deployment, remove existing containers and adopt all their resources to the new container.
- **Automatic bin packing** You provide Kubernetes with a cluster of nodes that it can use to run containerized tasks. You tell Kubernetes how much CPU and memory (RAM) each container needs. Kubernetes can fit containers onto your nodes to make the best use of your resources.
- **Self-healing** Kubernetes restarts containers that fail, replaces containers, kills containers that don't respond to your user-defined health check, and doesn't advertise them to clients until they are ready to serve.
- **Secret and configuration management** Kubernetes lets you store and manage sensitive information, such as passwords, OAuth tokens, and SSH keys. You can deploy and update secrets and application configuration without rebuilding your container images, and without exposing secrets in your stack configuration.
- **Batch execution** In addition to services, Kubernetes can manage your batch and CI workloads, replacing containers that fail, if desired.
- **Horizontal scaling** Scale your application up and down with a simple command, with a UI, or automatically based on CPU usage.
- **IPv4/IPv6 dual-stack** Allocation of IPv4 and IPv6 addresses to Pods and Services
- **Designed for extensibility** Add features to your Kubernetes cluster without changing upstream source code.

What Kubernetes is not

Kubernetes is not a traditional, all-inclusive PaaS (Platform as a Service) system. Since Kubernetes operates at the container level rather than at the hardware level, it provides some generally applicable features common to PaaS offerings, such as deployment, scaling, load balancing, and lets users integrate their logging, monitoring, and alerting solutions. However, Kubernetes is not monolithic, and these default solutions are optional and pluggable. Kubernetes provides the building blocks for building developer platforms, but preserves user choice and flexibility where it is important.

Kubernetes:

- Does not limit the types of applications supported. Kubernetes aims to support an extremely diverse variety of workloads, including stateless, stateful, and data-processing workloads. If an application can run in a container, it should run great on Kubernetes.
- Does not deploy source code and does not build your application. Continuous Integration, Delivery, and Deployment (CI/CD) workflows are determined by organization cultures and preferences as well as technical requirements.
- Does not provide application-level services, such as middleware (for example, message buses), data-processing frameworks (for example, Spark), databases (for example, MySQL), caches, nor cluster storage systems (for example, Ceph) as built-in services.

Such components can run on Kubernetes, and/or can be accessed by applications running on Kubernetes through portable mechanisms, such as the [Open Service Broker](#).

- Does not dictate logging, monitoring, or alerting solutions. It provides some integrations as proof of concept, and mechanisms to collect and export metrics.
- Does not provide nor mandate a configuration language/system (for example, Jsonnet). It provides a declarative API that may be targeted by arbitrary forms of declarative specifications.
- Does not provide nor adopt any comprehensive machine configuration, maintenance, management, or self-healing systems.
- Additionally, Kubernetes is not a mere orchestration system. In fact, it eliminates the need for orchestration. The technical definition of orchestration is execution of a defined workflow: first do A, then B, then C. In contrast, Kubernetes comprises a set of independent, composable control processes that continuously drive the current state towards the provided desired state. It shouldn't matter how you get from A to C. Centralized control is also not required. This results in a system that is easier to use and more powerful, robust, resilient, and extensible.

What's next

- Take a look at the [Kubernetes Components](#)
- Take a look at the [The Kubernetes API](#)
- Take a look at the [Cluster Architecture](#)
- Ready to [Get Started](#)?

Objects In Kubernetes

Kubernetes objects are persistent entities in the Kubernetes system. Kubernetes uses these entities to represent the state of your cluster. Learn about the Kubernetes object model and how to work with these objects.

This page explains how Kubernetes objects are represented in the Kubernetes API, and how you can express them in .yaml format.

Understanding Kubernetes objects

Kubernetes objects are persistent entities in the Kubernetes system. Kubernetes uses these entities to represent the state of your cluster. Specifically, they can describe:

- What containerized applications are running (and on which nodes)
- The resources available to those applications
- The policies around how those applications behave, such as restart policies, upgrades, and fault-tolerance

A Kubernetes object is a "record of intent"—once you create the object, the Kubernetes system will constantly work to ensure that object exists. By creating an object, you're effectively telling the Kubernetes system what you want your cluster's workload to look like; this is your cluster's *desired state*.

To work with Kubernetes objects—whether to create, modify, or delete them—you'll need to use the [Kubernetes API](#). When you use the kubectl command-line interface, for example, the CLI

makes the necessary Kubernetes API calls for you. You can also use the Kubernetes API directly in your own programs using one of the [Client Libraries](#).

Object spec and status

Almost every Kubernetes object includes two nested object fields that govern the object's configuration: the object *spec* and the object *status*. For objects that have a spec, you have to set this when you create the object, providing a description of the characteristics you want the resource to have: its *desired state*.

The status describes the *current state* of the object, supplied and updated by the Kubernetes system and its components. The Kubernetes [control plane](#) continually and actively manages every object's actual state to match the desired state you supplied.

For example: in Kubernetes, a Deployment is an object that can represent an application running on your cluster. When you create the Deployment, you might set the Deployment spec to specify that you want three replicas of the application to be running. The Kubernetes system reads the Deployment spec and starts three instances of your desired application--updating the status to match your spec. If any of those instances should fail (a status change), the Kubernetes system responds to the difference between spec and status by making a correction--in this case, starting a replacement instance.

For more information on the object spec, status, and metadata, see the [Kubernetes API Conventions](#).

Describing a Kubernetes object

When you create an object in Kubernetes, you must provide the object spec that describes its desired state, as well as some basic information about the object (such as a name). When you use the Kubernetes API to create the object (either directly or via kubectl), that API request must include that information as JSON in the request body. Most often, you provide the information to kubectl in file known as a *manifest*. By convention, manifests are YAML (you could also use JSON format). Tools such as kubectl convert the information from a manifest into JSON or another supported serialization format when making the API request over HTTP.

Here's an example manifest that shows the required fields and object spec for a Kubernetes Deployment:

[application/deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  selector:
    matchLabels:
      app: nginx
  replicas: 2 # tells deployment to run 2 pods matching the template
  template:
    metadata:
      labels:
        app: nginx
```

```
spec:
  containers:
  - name: nginx
    image: nginx:1.14.2
    ports:
    - containerPort: 80
```

One way to create a Deployment using a manifest file like the one above is to use the [kubectl apply](#) command in the kubectl command-line interface, passing the .yaml file as an argument. Here's an example:

```
kubectl apply -f https://k8s.io/examples/application/deployment.yaml
```

The output is similar to this:

```
deployment.apps/nginx-deployment created
```

Required fields

In the manifest (YAML or JSON file) for the Kubernetes object you want to create, you'll need to set values for the following fields:

- `apiVersion` - Which version of the Kubernetes API you're using to create this object
- `kind` - What kind of object you want to create
- `metadata` - Data that helps uniquely identify the object, including a name string, UID, and optional namespace
- `spec` - What state you desire for the object

The precise format of the object spec is different for every Kubernetes object, and contains nested fields specific to that object. The [Kubernetes API Reference](#) can help you find the spec format for all of the objects you can create using Kubernetes.

For example, see the [spec field](#) for the Pod API reference. For each Pod, the `.spec` field specifies the pod and its desired state (such as the container image name for each container within that pod). Another example of an object specification is the [spec field](#) for the StatefulSet API. For StatefulSet, the `.spec` field specifies the StatefulSet and its desired state. Within the `.spec` of a StatefulSet is a [template](#) for Pod objects. That template describes Pods that the StatefulSet controller will create in order to satisfy the StatefulSet specification. Different kinds of object can also have different `.status`; again, the API reference pages detail the structure of that `.status` field, and its content for each different type of object.

Note:

See [Configuration Best Practices](#) for additional information on writing YAML configuration files.

Server side field validation

Starting with Kubernetes v1.25, the API server offers server side [field validation](#) that detects unrecognized or duplicate fields in an object. It provides all the functionality of `kubectl --validate` on the server side.

The `kubectl` tool uses the `--validate` flag to set the level of field validation. It accepts the values `ignore`, `warn`, and `strict` while also accepting the values `true` (equivalent to `strict`) and `false` (equivalent to `ignore`). The default validation setting for `kubectl` is `--validate=true`.

Strict

Strict field validation, errors on validation failure

Warn

Field validation is performed, but errors are exposed as warnings rather than failing the request

Ignore

No server side field validation is performed

When `kubectl` cannot connect to an API server that supports field validation it will fall back to using client-side validation. Kubernetes 1.27 and later versions always offer field validation; older Kubernetes releases might not. If your cluster is older than v1.27, check the documentation for your version of Kubernetes.

What's next

If you're new to Kubernetes, read more about the following:

- [Pods](#) which are the most important basic Kubernetes objects.
- [Deployment](#) objects.
- [Controllers](#) in Kubernetes.
- [kubectl](#) and [kubectl commands](#).

[Kubernetes Object Management](#) explains how to use `kubectl` to manage objects. You might need to [install kubectl](#) if you don't already have it available.

To learn about the Kubernetes API in general, visit:

- [Kubernetes API overview](#)

To learn about objects in Kubernetes in more depth, read other pages in this section:

- [Kubernetes Object Management](#)
- [Object Names and IDs](#)
- [Labels and Selectors](#)
- [Namespaces](#)
- [Annotations](#)
- [Field Selectors](#)
- [Finalizers](#)
- [Owners and Dependents](#)
- [Recommended Labels](#)

Kubernetes Object Management

The `kubectl` command-line tool supports several different ways to create and manage Kubernetes [objects](#). This document provides an overview of the different approaches. Read the [Kubectl book](#) for details of managing objects by `Kubectl`.

Management techniques

Warning:

A Kubernetes object should be managed using only one technique. Mixing and matching techniques for the same object results in undefined behavior.

Management technique	Operates on	Recommended environment	Supported writers	Learning curve
Imperative commands	Live objects	Development projects	1+	Lowest
Imperative object configuration	Individual files	Production projects	1	Moderate
Declarative object configuration	Directories of files	Production projects	1+	Highest

Imperative commands

When using imperative commands, a user operates directly on live objects in a cluster. The user provides operations to the `kubectl` command as arguments or flags.

This is the recommended way to get started or to run a one-off task in a cluster. Because this technique operates directly on live objects, it provides no history of previous configurations.

Examples

Run an instance of the `nginx` container by creating a Deployment object:

```
kubectl create deployment nginx --image nginx
```

Trade-offs

Advantages compared to object configuration:

- Commands are expressed as a single action word.
- Commands require only a single step to make changes to the cluster.

Disadvantages compared to object configuration:

- Commands do not integrate with change review processes.
- Commands do not provide an audit trail associated with changes.
- Commands do not provide a source of records except for what is live.
- Commands do not provide a template for creating new objects.

Imperative object configuration

In imperative object configuration, the `kubectl` command specifies the operation (create, replace, etc.), optional flags and at least one file name. The file specified must contain a full definition of the object in YAML or JSON format.

See the [API reference](#) for more details on object definitions.

Warning:

The imperative `replace` command replaces the existing spec with the newly provided one, dropping all changes to the object missing from the configuration file. This approach should not be used with resource types whose specs are updated independently of the configuration file. Services of type `LoadBalancer`, for example, have their `externalIPs` field updated independently from the configuration by the cluster.

Examples

Create the objects defined in a configuration file:

```
kubectl create -f nginx.yaml
```

Delete the objects defined in two configuration files:

```
kubectl delete -f nginx.yaml -f redis.yaml
```

Update the objects defined in a configuration file by overwriting the live configuration:

```
kubectl replace -f nginx.yaml
```

Trade-offs

Advantages compared to imperative commands:

- Object configuration can be stored in a source control system such as Git.
- Object configuration can integrate with processes such as reviewing changes before push and audit trails.
- Object configuration provides a template for creating new objects.

Disadvantages compared to imperative commands:

- Object configuration requires basic understanding of the object schema.
- Object configuration requires the additional step of writing a YAML file.

Advantages compared to declarative object configuration:

- Imperative object configuration behavior is simpler and easier to understand.
- As of Kubernetes version 1.5, imperative object configuration is more mature.

Disadvantages compared to declarative object configuration:

- Imperative object configuration works best on files, not directories.
- Updates to live objects must be reflected in configuration files, or they will be lost during the next replacement.

Declarative object configuration

When using declarative object configuration, a user operates on object configuration files stored locally, however the user does not define the operations to be taken on the files. Create, update, and delete operations are automatically detected per-object by `kubectl`. This enables working on directories, where different operations might be needed for different objects.

Note:

Declarative object configuration retains changes made by other writers, even if the changes are not merged back to the object configuration file. This is possible by using the patch API operation to write only observed differences, instead of using the replace API operation to replace the entire object configuration.

Examples

Process all object configuration files in the configs directory, and create or patch the live objects. You can first diff to see what changes are going to be made, and then apply:

```
kubectldiff -f configs/  
kubectlapply -f configs/
```

Recursively process directories:

```
kubectldiff -R -f configs/  
kubectlapply -R -f configs/
```

Trade-offs

Advantages compared to imperative object configuration:

- Changes made directly to live objects are retained, even if they are not merged back into the configuration files.
- Declarative object configuration has better support for operating on directories and automatically detecting operation types (create, patch, delete) per-object.

Disadvantages compared to imperative object configuration:

- Declarative object configuration is harder to debug and understand results when they are unexpected.
- Partial updates using diffs create complex merge and patch operations.

What's next

- [Managing Kubernetes Objects Using Imperative Commands](#)
- [Imperative Management of Kubernetes Objects Using Configuration Files](#)
- [Declarative Management of Kubernetes Objects Using Configuration Files](#)
- [Declarative Management of Kubernetes Objects Using Kustomize](#)
- [Kubectl Command Reference](#)
- [Kubectl Book](#)
- [Kubernetes API Reference](#)

Object Names and IDs

Each [object](#) in your cluster has a [Name](#) that is unique for that type of resource. Every Kubernetes object also has a [UID](#) that is unique across your whole cluster.

For example, you can only have one Pod named myapp-1234 within the same [namespace](#), but you can have one Pod and one Deployment that are each named myapp-1234.

For non-unique user-provided attributes, Kubernetes provides [labels](#) and [annotations](#).

Names

A client-provided string that refers to an object in a resource URL, such as `/api/v1/pods/some-name`.

Only one object of a given kind can have a given name at a time. However, if you delete the object, you can make a new object with the same name.

Names must be unique across all [API versions](#) of the same resource. API resources are distinguished by their API group, resource type, namespace (for namespaced resources), and name. In other words, API version is irrelevant in this context.

Note:

In cases when objects represent a physical entity, like a Node representing a physical host, when the host is re-created under the same name without deleting and re-creating the Node, Kubernetes treats the new host as the old one, which may lead to inconsistencies.

Below are four types of commonly used name constraints for resources.

DNS Subdomain Names

Most resource types require a name that can be used as a DNS subdomain name as defined in [RFC 1123](#). This means the name must:

- contain no more than 253 characters
- contain only lowercase alphanumeric characters, '-' or '.'
- start with an alphanumeric character
- end with an alphanumeric character

RFC 1123 Label Names

Some resource types require their names to follow the DNS label standard as defined in [RFC 1123](#). This means the name must:

- contain at most 63 characters
- contain only lowercase alphanumeric characters or '-'
- start with an alphanumeric character
- end with an alphanumeric character

RFC 1035 Label Names

Some resource types require their names to follow the DNS label standard as defined in [RFC 1035](#). This means the name must:

- contain at most 63 characters
- contain only lowercase alphanumeric characters or '-'

- start with an alphabetic character
- end with an alphanumeric character

Note:

The only difference between the RFC 1035 and RFC 1123 label standards is that RFC 1123 labels are allowed to start with a digit, whereas RFC 1035 labels can start with a lowercase alphabetic character only.

Path Segment Names

Some resource types require their names to be able to be safely encoded as a path segment. In other words, the name may not be "." or ".." and the name may not contain "/" or "%".

Here's an example manifest for a Pod named `nginx-demo`.

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-demo
spec:
  containers:
  - name: nginx
    image: nginx:1.14.2
    ports:
    - containerPort: 80
```

Note:

Some resource types have additional restrictions on their names.

UIDs

A Kubernetes systems-generated string to uniquely identify objects.

Every object created over the whole lifetime of a Kubernetes cluster has a distinct UID. It is intended to distinguish between historical occurrences of similar entities.

Kubernetes UIDs are universally unique identifiers (also known as UUIDs). UUIDs are standardized as ISO/IEC 9834-8 and as ITU-T X.667.

What's next

- Read about [labels](#) and [annotations](#) in Kubernetes.
- See the [Identifiers and Names in Kubernetes](#) design document.

Labels and Selectors

Labels are key/value pairs that are attached to [objects](#) such as Pods. Labels are intended to be used to specify identifying attributes of objects that are meaningful and relevant to users, but

do not directly imply semantics to the core system. Labels can be used to organize and to select subsets of objects. Labels can be attached to objects at creation time and subsequently added and modified at any time. Each object can have a set of key/value labels defined. Each Key must be unique for a given object.

```
"metadata": {  
  "labels": {  
    "key1" : "value1",  
    "key2" : "value2"  
  }  
}
```

Labels allow for efficient queries and watches and are ideal for use in UIs and CLIs. Non-identifying information should be recorded using [annotations](#).

Motivation

Labels enable users to map their own organizational structures onto system objects in a loosely coupled fashion, without requiring clients to store these mappings.

Service deployments and batch processing pipelines are often multi-dimensional entities (e.g., multiple partitions or deployments, multiple release tracks, multiple tiers, multiple micro-services per tier). Management often requires cross-cutting operations, which breaks encapsulation of strictly hierarchical representations, especially rigid hierarchies determined by the infrastructure rather than by users.

Example labels:

- "release" : "stable", "release" : "canary"
- "environment" : "dev", "environment" : "qa", "environment" : "production"
- "tier" : "frontend", "tier" : "backend", "tier" : "cache"
- "partition" : "customerA", "partition" : "customerB"
- "track" : "daily", "track" : "weekly"

These are examples of [commonly used labels](#); you are free to develop your own conventions. Keep in mind that label Key must be unique for a given object.

Syntax and character set

Labels are key/value pairs. Valid label keys have two segments: an optional prefix and name, separated by a slash (/). The name segment is required and must be 63 characters or less, beginning and ending with an alphanumeric character ([a-z0-9A-Z]) with dashes (-), underscores (_), dots (.), and alphanumerics between. The prefix is optional. If specified, the prefix must be a DNS subdomain: a series of DNS labels separated by dots (.), not longer than 253 characters in total, followed by a slash (/).

If the prefix is omitted, the label Key is presumed to be private to the user. Automated system components (e.g. kube-scheduler, kube-controller-manager, kube-apiserver, kubectl, or other third-party automation) which add labels to end-user objects must specify a prefix.

The kubernetes.io/ and k8s.io/ prefixes are [reserved](#) for Kubernetes core components.

Valid label value:

- must be 63 characters or less (can be empty),
- unless empty, must begin and end with an alphanumeric character ([a-z0-9A-Z]),
- could contain dashes (-), underscores (_), dots (.), and alphanumerics between.

For example, here's a manifest for a Pod that has two labels `environment: production` and `app: nginx`:

```
apiVersion: v1
kind: Pod
metadata:
  name: label-demo
  labels:
    environment: production
    app: nginx
spec:
  containers:
  - name: nginx
    image: nginx:1.14.2
    ports:
    - containerPort: 80
```

Label selectors

Unlike [names and UIDs](#), labels do not provide uniqueness. In general, we expect many objects to carry the same label(s).

Via a *label selector*, the client/user can identify a set of objects. The label selector is the core grouping primitive in Kubernetes.

The API currently supports two types of selectors: *equality-based* and *set-based*. A label selector can be made of multiple *requirements* which are comma-separated. In the case of multiple requirements, all must be satisfied so the comma separator acts as a logical *AND* (&&) operator.

The semantics of empty or non-specified selectors are dependent on the context, and API types that use selectors should document the validity and meaning of them.

Note:

For some API types, such as ReplicaSets, the label selectors of two instances must not overlap within a namespace, or the controller can see that as conflicting instructions and fail to determine how many replicas should be present.

Caution:

For both equality-based and set-based conditions there is no logical *OR* (||) operator. Ensure your filter statements are structured accordingly.

Equality-based requirement

Equality- or *inequality-based* requirements allow filtering by label keys and values. Matching objects must satisfy all of the specified label constraints, though they may have additional labels as well. Three kinds of operators are admitted `=`, `==`, `!=`. The first two represent *equality* (and are synonyms), while the latter represents *inequality*. For example:

```
environment = production
tier != frontend
```

The former selects all resources with key equal to `environment` and value equal to `production`. The latter selects all resources with key equal to `tier` and value distinct from `frontend`, and all resources with no labels with the `tier` key. One could filter for resources in `production` excluding `frontend` using the comma operator: `environment=production,tier!=frontend`

One usage scenario for equality-based label requirement is for Pods to specify node selection criteria. For example, the sample Pod below selects nodes with the label `"accelerator=nvidia-tesla-p100"`.

```
apiVersion: v1
kind: Pod
metadata:
  name: cuda-test
spec:
  containers:
  - name: cuda-test
    image: "registry.k8s.io/cuda-vector-add:v0.1"
  resources:
    limits:
      nvidia.com/gpu: 1
  nodeSelector:
    accelerator: nvidia-tesla-p100
```

Set-based requirement

Set-based label requirements allow filtering keys according to a set of values. Three kinds of operators are supported: `in`, `notin` and `exists` (only the key identifier). For example:

```
environment in (production, qa)
tier notin (frontend, backend)
partition
!partition
```

- The first example selects all resources with key equal to `environment` and value equal to `production` or `qa`.
- The second example selects all resources with key equal to `tier` and values other than `frontend` and `backend`, and all resources with no labels with the `tier` key.
- The third example selects all resources including a label with key `partition`; no values are checked.
- The fourth example selects all resources without a label with key `partition`; no values are checked.

Similarly the comma separator acts as an *AND* operator. So filtering resources with a partition key (no matter the value) and with environment different than qa can be achieved using `partition,environment notin (qa)`. The *set-based* label selector is a general form of equality since `environment=production` is equivalent to `environment in (production)`; similarly for `!=` and `notin`.

Set-based requirements can be mixed with *equality-based* requirements. For example: `partition in (customerA, customerB),environment!=qa`.

API

LIST and WATCH filtering

LIST and WATCH operations may specify label selectors to filter the sets of objects returned using a query parameter. Both requirements are permitted (presented here as they would appear in a URL query string):

- *equality-based* requirements: `?labelSelector=environment%3Dproduction,tier%3Dfrontend`
- *set-based* requirements: `?labelSelector=environment+in+%28production%2Cqa%29%2Ctier+in+%28frontend%29`

Both label selector styles can be used to list or watch resources via a REST client. For example, targeting apiserver with kubectl and using *equality-based* one may write:

```
kubectl get pods -l environment=production,tier=frontend
```

or using *set-based* requirements:

```
kubectl get pods -l 'environment in (production),tier in (frontend)'
```

As already mentioned *set-based* requirements are more expressive. For instance, they can implement the *OR* operator on values:

```
kubectl get pods -l 'environment in (production, qa)'
```

or restricting negative matching via *notin* operator:

```
kubectl get pods -l 'environment,environment notin (frontend)'
```

Set references in API objects

Some Kubernetes objects, such as [services](#) and [replicationcontrollers](#), also use label selectors to specify sets of other resources, such as [pods](#).

Service and ReplicationController

The set of pods that a service targets is defined with a label selector. Similarly, the population of pods that a replicationcontroller should manage is also defined with a label selector.

Label selectors for both objects are defined in json or yaml files using maps, and only *equality-based* requirement selectors are supported:

```
"selector": {  
  "component" : "redis",  
}
```

or

```
selector:  
  component: redis
```

This selector (respectively in json or yaml format) is equivalent to `component=redis` or `component in (redis)`.

Resources that support set-based requirements

Newer resources, such as [Job](#), [Deployment](#), [ReplicaSet](#), and [DaemonSet](#), support *set-based* requirements as well.

```
selector:  
  matchLabels:  
    component: redis  
  matchExpressions:  
    - { key: tier, operator: In, values: [cache] }  
    - { key: environment, operator: NotIn, values: [dev] }
```

`matchLabels` is a map of {key,value} pairs. A single {key,value} in the `matchLabels` map is equivalent to an element of `matchExpressions`, whose key field is "key", the operator is "In", and the values array contains only "value". `matchExpressions` is a list of pod selector requirements. Valid operators include In, NotIn, Exists, and DoesNotExist. The values set must be non-empty in the case of In and NotIn. All of the requirements, from both `matchLabels` and `matchExpressions` are ANDed together -- they must all be satisfied in order to match.

Selecting sets of nodes

One use case for selecting over labels is to constrain the set of nodes onto which a pod can schedule. See the documentation on [node selection](#) for more information.

Using labels effectively

You can apply a single label to any resources, but this is not always the best practice. There are many scenarios where multiple labels should be used to distinguish resource sets from one another.

For instance, different applications would use different values for the `app` label, but a multi-tier application, such as the [guestbook example](#), would additionally need to distinguish each tier. The frontend could carry the following labels:

```
labels:  
  app: guestbook  
  tier: frontend
```

while the Redis master and replica would have different tier labels, and perhaps even an additional role label:

```
labels:
  app: guestbook
  tier: backend
  role: master
```

and

```
labels:
  app: guestbook
  tier: backend
  role: replica
```

The labels allow for slicing and dicing the resources along any dimension specified by a label:

```
kubectl apply -f examples/guestbook/all-in-one/guestbook-all-in-one.yaml
kubectl get pods -Lapp -Ltier -Lrole
```

NAME	READY	STATUS	RESTARTS	AGE	APP	TIER	ROLE
guestbook-fe-4nlpb	1/1	Running	0	1m	guestbook	frontend	<none>
guestbook-fe-ght6d	1/1	Running	0	1m	guestbook	frontend	<none>
guestbook-fe-jpy62	1/1	Running	0	1m	guestbook	frontend	<none>
guestbook-redis-master-5pg3b	1/1	Running	0	1m	guestbook	backend	master
guestbook-redis-replica-2q2yf	1/1	Running	0	1m	guestbook	backend	replica
guestbook-redis-replica-qgazl	1/1	Running	0	1m	guestbook	backend	replica
my-nginx-divi2	1/1	Running	0	29m	nginx	<none>	<none>
my-nginx-o0ef1	1/1	Running	0	29m	nginx	<none>	<none>

```
kubectl get pods -lapp=guestbook,role=replica
```

NAME	READY	STATUS	RESTARTS	AGE
guestbook-redis-replica-2q2yf	1/1	Running	0	3m
guestbook-redis-replica-qgazl	1/1	Running	0	3m

Updating labels

Sometimes you may want to relabel existing pods and other resources before creating new resources. This can be done with `kubectl label`. For example, if you want to label all your NGINX Pods as frontend tier, run:

```
kubectl label pods -l app=nginx tier=fe
```

```
pod/my-nginx-2035384211-j5fhi labeled
pod/my-nginx-2035384211-u2c7e labeled
pod/my-nginx-2035384211-u3t6x labeled
```

This first filters all pods with the label "app=nginx", and then labels them with the "tier=fe". To see the pods you labeled, run:

```
kubectl get pods -l app=nginx -L tier
```

NAME	READY	STATUS	RESTARTS	AGE	TIER
my-nginx-2035384211-j5fhi	1/1	Running	0	23m	fe
my-nginx-2035384211-u2c7e	1/1	Running	0	23m	fe
my-nginx-2035384211-u3t6x	1/1	Running	0	23m	fe

This outputs all "app=nginx" pods, with an additional label column of pods' tier (specified with -L or --label-columns).

For more information, please see [kubecttl label](#).

What's next

- Learn how to [add a label to a node](#)
- Find [Well-known labels, Annotations and Taints](#)
- See [Recommended labels](#)
- [Enforce Pod Security Standards with Namespace Labels](#)
- Read a blog on [Writing a Controller for Pod Labels](#)

Namespaces

In Kubernetes, *namespaces* provide a mechanism for isolating groups of resources within a single cluster. Names of resources need to be unique within a namespace, but not across namespaces. Namespace-based scoping is applicable only for namespaced [objects](#) (e.g. *Deployments, Services, etc.*) and not for cluster-wide objects (e.g. *StorageClass, Nodes, PersistentVolumes, etc.*).

When to Use Multiple Namespaces

Namespaces are intended for use in environments with many users spread across multiple teams, or projects. For clusters with a few to tens of users, you should not need to create or think about namespaces at all. Start using namespaces when you need the features they provide.

Namespaces provide a scope for names. Names of resources need to be unique within a namespace, but not across namespaces. Namespaces cannot be nested inside one another and each Kubernetes resource can only be in one namespace.

Namespaces are a way to divide cluster resources between multiple users (via [resource quota](#)).

It is not necessary to use multiple namespaces to separate slightly different resources, such as different versions of the same software: use [labels](#) to distinguish resources within the same namespace.

Note:

For a production cluster, consider *not* using the default namespace. Instead, make other namespaces and use those.

Initial namespaces

Kubernetes starts with four initial namespaces:

default

Kubernetes includes this namespace so that you can start using your new cluster without first creating a namespace.

kube-node-lease

This namespace holds [Lease](#) objects associated with each node. Node leases allow the kubelet to send [heartbeats](#) so that the control plane can detect node failure.

kube-public

This namespace is readable by *all* clients (including those not authenticated). This namespace is mostly reserved for cluster usage, in case that some resources should be visible and readable publicly throughout the whole cluster. The public aspect of this namespace is only a convention, not a requirement.

kube-system

The namespace for objects created by the Kubernetes system.

Working with Namespaces

Creation and deletion of namespaces are described in the [Admin Guide documentation for namespaces](#).

Note:

Avoid creating namespaces with the prefix `kube-`, since it is reserved for Kubernetes system namespaces.

Viewing namespaces

You can list the current namespaces in a cluster using:

```
kubectl get namespace
```

NAME	STATUS	AGE
default	Active	1d
kube-node-lease	Active	1d
kube-public	Active	1d
kube-system	Active	1d

Setting the namespace for a request

To set the namespace for a current request, use the `--namespace` flag.

For example:

```
kubectl run nginx --image=nginx --namespace=<insert-namespace-name-here>  
kubectl get pods --namespace=<insert-namespace-name-here>
```

Setting the namespace preference

You can permanently save the namespace for all subsequent `kubectl` commands in that context.

```
kubectl config set-context --current --namespace=<insert-namespace-name-here>  
# Validate it  
kubectl config view --minify | grep namespace:
```

Namespaces and DNS

When you create a [Service](#), it creates a corresponding [DNS entry](#). This entry is of the form `<service-name>.<namespace-name>.svc.cluster.local`, which means that if a container only uses `<service-name>`, it will resolve to the service which is local to a namespace. This is useful for using the same configuration across multiple namespaces such as Development, Staging and Production. If you want to reach across namespaces, you need to use the fully qualified domain name (FQDN).

As a result, all namespace names must be valid [RFC 1123 DNS labels](#).

Warning:

By creating namespaces with the same name as [public top-level domains](#), Services in these namespaces can have short DNS names that overlap with public DNS records. Workloads from any namespace performing a DNS lookup without a [trailing dot](#) will be redirected to those services, taking precedence over public DNS.

To mitigate this, limit privileges for creating namespaces to trusted users. If required, you could additionally configure third-party security controls, such as [admission webhooks](#), to block creating any namespace with the name of [public TLDs](#).

Not all objects are in a namespace

Most Kubernetes resources (e.g. pods, services, replication controllers, and others) are in some namespaces. However namespace resources are not themselves in a namespace. And low-level resources, such as [nodes](#) and [persistentVolumes](#), are not in any namespace.

To see which Kubernetes resources are and aren't in a namespace:

```
# In a namespace
kubectl api-resources --namespaced=true

# Not in a namespace
kubectl api-resources --namespaced=false
```

Automatic labelling

FEATURE STATE: Kubernetes 1.22 [stable]

The Kubernetes control plane sets an immutable [label](#) `kubernetes.io/metadata.name` on all namespaces. The value of the label is the namespace name.

What's next

- Learn more about [creating a new namespace](#).
- Learn more about [deleting a namespace](#).

Annotations

You can use Kubernetes annotations to attach arbitrary non-identifying metadata to [objects](#). Clients such as tools and libraries can retrieve this metadata.

Attaching metadata to objects

You can use either labels or annotations to attach metadata to Kubernetes objects. Labels can be used to select objects and to find collections of objects that satisfy certain conditions. In contrast, annotations are not used to identify and select objects. The metadata in an annotation can be small or large, structured or unstructured, and can include characters not permitted by labels. It is possible to use labels as well as annotations in the metadata of the same object.

Annotations, like labels, are key/value maps:

```
"metadata": {  
  "annotations": {  
    "key1" : "value1",  
    "key2" : "value2"  
  }  
}
```

Note:

The keys and the values in the map must be strings. In other words, you cannot use numeric, boolean, list or other types for either the keys or the values.

Here are some examples of information that could be recorded in annotations:

- Fields managed by a declarative configuration layer. Attaching these fields as annotations distinguishes them from default values set by clients or servers, and from auto-generated fields and fields set by auto-sizing or auto-scaling systems.
- Build, release, or image information like timestamps, release IDs, git branch, PR numbers, image hashes, and registry address.
- Pointers to logging, monitoring, analytics, or audit repositories.
- Client library or tool information that can be used for debugging purposes: for example, name, version, and build information.
- User or tool/system provenance information, such as URLs of related objects from other ecosystem components.
- Lightweight rollout tool metadata: for example, config or checkpoints.
- Phone or pager numbers of persons responsible, or directory entries that specify where that information can be found, such as a team web site.
- Directives from the end-user to the implementations to modify behavior or engage non-standard features.

Instead of using annotations, you could store this type of information in an external database or directory, but that would make it much harder to produce shared client libraries and tools for deployment, management, introspection, and the like.

Syntax and character set

Annotations are key/value pairs. Valid annotation keys have two segments: an optional prefix and name, separated by a slash (/). The name segment is required and must be 63 characters or less, beginning and ending with an alphanumeric character ([a-z0-9A-Z]) with dashes (-), underscores (_), dots (.), and alphanumerics between. The prefix is optional. If specified, the prefix must be a DNS subdomain: a series of DNS labels separated by dots (.), not longer than 253 characters in total, followed by a slash (/).

If the prefix is omitted, the annotation Key is presumed to be private to the user. Automated system components (e.g. kube-scheduler, kube-controller-manager, kube-apiserver, kubectl, or other third-party automation) which add annotations to end-user objects must specify a prefix.

The kubernetes.io/ and k8s.io/ prefixes are reserved for Kubernetes core components.

For example, here's a manifest for a Pod that has the annotation imageregistry: https://hub.docker.com/ :

```
apiVersion: v1
kind: Pod
metadata:
  name: annotations-demo
  annotations:
    imageregistry: "https://hub.docker.com/"
spec:
  containers:
  - name: nginx
    image: nginx:1.14.2
    ports:
    - containerPort: 80
```

What's next

- Learn more about [Labels and Selectors](#).
- Find [Well-known labels, Annotations and Taints](#)

Field Selectors

Field selectors let you select Kubernetes [objects](#) based on the value of one or more resource fields. Here are some examples of field selector queries:

- metadata.name=my-service
- metadata.namespace!=default
- status.phase=Pending

This kubectl command selects all Pods for which the value of the [status.phase](#) field is Running:

```
kubectl get pods --field-selector status.phase=Running
```

Note:

Field selectors are essentially resource *filters*. By default, no selectors/filters are applied, meaning that all resources of the specified type are selected. This makes the `kubectl get pods` and `kubectl get pods --field-selector ""` equivalent.

Supported fields

Supported field selectors vary by Kubernetes resource type. All resource types support the `metadata.name` and `metadata.namespace` fields. Using unsupported field selectors produces an error. For example:

```
kubectl get ingress --field-selector foo.bar=baz
```

Error from server (BadRequest): Unable to find "ingresses" that match label selector "", field selector "foo.bar=baz": "foo.bar" is not a known field selector: only "metadata.name", "metadata.namespace"

List of supported fields

Kind	Fields
Pod	spec.nodeName spec.restartPolicy spec.schedulerName spec.serviceAccountName spec.hostNetwork status.phase status.podIP status.nominatedNodeName
Event	involvedObject.kind involvedObject.namespace involvedObject.name involvedObject.uid involvedObject.apiVersion involvedObject.resourceVersion involvedObject.fieldPath reason reportingComponent source type
Secret	type
Namespace	status.phase
ReplicaSet	status.replicas
ReplicationController	status.replicas
Job	status.successful
Node	spec.unschedulable
CertificateSigningRequest	spec.signerName

Supported operators

You can use the `=`, `==`, and `!=` operators with field selectors (`=` and `==` mean the same thing). This `kubectl` command, for example, selects all Kubernetes Services that aren't in the default namespace:

```
kubectl get services --all-namespaces --field-selector metadata.namespace!=default
```

Note:

[Set-based operators](#) (`in`, `notin`, `exists`) are not supported for field selectors.

Chained selectors

As with [label](#) and other selectors, field selectors can be chained together as a comma-separated list. This `kubectl` command selects all Pods for which the `status.phase` does not equal `Running` and the `spec.restartPolicy` field equals `Always`:

```
kubectl get pods --field-selector=status.phase!=Running,spec.restartPolicy=Always
```

Multiple resource types

You can use field selectors across multiple resource types. This `kubectl` command selects all Statefulsets and Services that are not in the default namespace:

```
kubectl get statefulsets,services --all-namespaces --field-selector metadata.namespace!=default
```

Finalizers

Finalizers are namespaced keys that tell Kubernetes to wait until specific conditions are met before it fully deletes resources marked for deletion. Finalizers alert [controllers](#) to clean up resources the deleted object owned.

When you tell Kubernetes to delete an object that has finalizers specified for it, the Kubernetes API marks the object for deletion by populating `.metadata.deletionTimestamp`, and returns a 202 status code (HTTP "Accepted"). The target object remains in a terminating state while the control plane, or other components, take the actions defined by the finalizers. After these actions are complete, the controller removes the relevant finalizers from the target object. When the `metadata.finalizers` field is empty, Kubernetes considers the deletion complete and deletes the object.

You can use finalizers to control [garbage collection](#) of resources. For example, you can define a finalizer to clean up related resources or infrastructure before the controller deletes the target resource.

You can use finalizers to control [garbage collection](#) of [objects](#) by alerting [controllers](#) to perform specific cleanup tasks before deleting the target resource.

Finalizers don't usually specify the code to execute. Instead, they are typically lists of keys on a specific resource similar to annotations. Kubernetes specifies some finalizers automatically, but you can also specify your own.

How finalizers work

When you create a resource using a manifest file, you can specify finalizers in the `metadata.finalizers` field. When you attempt to delete the resource, the API server handling the delete request notices the values in the `finalizers` field and does the following:

- Modifies the object to add a `metadata.deletionTimestamp` field with the time you started the deletion.
- Prevents the object from being removed until all items are removed from its `metadata.finalizers` field
- Returns a 202 status code (HTTP "Accepted")

The controller managing that finalizer notices the update to the object setting the `metadata.deletionTimestamp`, indicating deletion of the object has been requested. The controller then attempts to satisfy the requirements of the finalizers specified for that resource. Each time a finalizer condition is satisfied, the controller removes that key from the resource's `finalizers` field. When the `finalizers` field is emptied, an object with a `deletionTimestamp` field set is automatically deleted. You can also use finalizers to prevent deletion of unmanaged resources.

A common example of a finalizer is `kubernetes.io/pv-protection`, which prevents accidental deletion of `PersistentVolume` objects. When a `PersistentVolume` object is in use by a Pod, Kubernetes adds the `pv-protection` finalizer. If you try to delete the `PersistentVolume`, it enters a `Terminating` status, but the controller can't delete it because the finalizer exists. When the Pod stops using the `PersistentVolume`, Kubernetes clears the `pv-protection` finalizer, and the controller deletes the volume.

Note:

- When you `DELETE` an object, Kubernetes adds the deletion timestamp for that object and then immediately starts to restrict changes to the `metadata.finalizers` field for the object that is now pending deletion. You can remove existing finalizers (deleting an entry from the `finalizers` list) but you cannot add a new finalizer. You also cannot modify the `deletionTimestamp` for an object once it is set.
- After the deletion is requested, you can not resurrect this object. The only way is to delete it and make a new similar object.

Owner references, labels, and finalizers

Like [labels](#), [owner references](#) describe the relationships between objects in Kubernetes, but are used for a different purpose. When a [controller](#) manages objects like Pods, it uses labels to track changes to groups of related objects. For example, when a [Job](#) creates one or more Pods, the Job controller applies labels to those pods and tracks changes to any Pods in the cluster with the same label.

The Job controller also adds *owner references* to those Pods, pointing at the Job that created the Pods. If you delete the Job while these Pods are running, Kubernetes uses the owner references (not labels) to determine which Pods in the cluster need cleanup.

Kubernetes also processes finalizers when it identifies owner references on a resource targeted for deletion.

In some situations, finalizers can block the deletion of dependent objects, which can cause the targeted owner object to remain for longer than expected without being fully deleted. In these situations, you should check finalizers and owner references on the target owner and dependent objects to troubleshoot the cause.

Note:

In cases where objects are stuck in a deleting state, avoid manually removing finalizers to allow deletion to continue. Finalizers are usually added to resources for a reason, so forcefully removing them can lead to issues in your cluster. This should only be done when the purpose of the finalizer is understood and is accomplished in another way (for example, manually cleaning up some dependent object).

What's next

- Read [Using Finalizers to Control Deletion](#) on the Kubernetes blog.

Owners and Dependents

In Kubernetes, some [objects](#) are *owners* of other objects. For example, a [ReplicaSet](#) is the owner of a set of Pods. These owned objects are *dependents* of their owner.

Ownership is different from the [labels and selectors](#) mechanism that some resources also use. For example, consider a Service that creates EndpointSlice objects. The Service uses [labels](#) to allow the control plane to determine which EndpointSlice objects are used for that Service. In addition to the labels, each EndpointSlice that is managed on behalf of a Service has an owner reference. Owner references help different parts of Kubernetes avoid interfering with objects they don't control.

Owner references in object specifications

Dependent objects have a `metadata.ownerReferences` field that references their owner object. A valid owner reference consists of the object name and a [UID](#) within the same [namespace](#) as the dependent object. Kubernetes sets the value of this field automatically for objects that are dependents of other objects like ReplicaSets, DaemonSets, Deployments, Jobs and CronJobs, and ReplicationControllers. You can also configure these relationships manually by changing the value of this field. However, you usually don't need to and can allow Kubernetes to automatically manage the relationships.

Dependent objects also have an `ownerReferences.blockOwnerDeletion` field that takes a boolean value and controls whether specific dependents can block garbage collection from deleting their owner object. Kubernetes automatically sets this field to true if a [controller](#) (for example, the Deployment controller) sets the value of the `metadata.ownerReferences` field. You

can also set the value of the `blockOwnerDeletion` field manually to control which dependents block garbage collection.

A Kubernetes admission controller controls user access to change this field for dependent resources, based on the delete permissions of the owner. This control prevents unauthorized users from delaying owner object deletion.

Note:

Cross-namespace owner references are disallowed by design. Namespaced dependents can specify cluster-scoped or namespaced owners. A namespaced owner **must** exist in the same namespace as the dependent. If it does not, the owner reference is treated as absent, and the dependent is subject to deletion once all owners are verified absent.

Cluster-scoped dependents can only specify cluster-scoped owners. In v1.20+, if a cluster-scoped dependent specifies a namespaced kind as an owner, it is treated as having an unresolvable owner reference, and is not able to be garbage collected.

In v1.20+, if the garbage collector detects an invalid cross-namespace ownerReference, or a cluster-scoped dependent with an ownerReference referencing a namespaced kind, a warning Event with a reason of `OwnerRefInvalidNamespace` and an involvedObject of the invalid dependent is reported. You can check for that kind of Event by running `kubectl get events -A --field-selector=reason=OwnerRefInvalidNamespace`.

Ownership and finalizers

When you tell Kubernetes to delete a resource, the API server allows the managing controller to process any [finalizer rules](#) for the resource. [Finalizers](#) prevent accidental deletion of resources your cluster may still need to function correctly. For example, if you try to delete a [PersistentVolume](#) that is still in use by a Pod, the deletion does not happen immediately because the PersistentVolume has the `kubernetes.io/pv-protection` finalizer on it. Instead, the [volume](#) remains in the Terminating status until Kubernetes clears the finalizer, which only happens after the PersistentVolume is no longer bound to a Pod.

Kubernetes also adds finalizers to an owner resource when you use either [foreground or orphan cascading deletion](#). In foreground deletion, it adds the foreground finalizer so that the controller must delete dependent resources that also have `ownerReferences.blockOwnerDeletion=true` before it deletes the owner. If you specify an orphan deletion policy, Kubernetes adds the orphan finalizer so that the controller ignores dependent resources after it deletes the owner object.

What's next

- Learn more about [Kubernetes finalizers](#).
- Learn about [garbage collection](#).
- Read the API reference for [object metadata](#).

Recommended Labels

You can visualize and manage Kubernetes objects with more tools than `kubectl` and the dashboard. A common set of labels allows tools to work interoperably, describing objects in a common manner that all tools can understand.

In addition to supporting tooling, the recommended labels describe applications in a way that can be queried.

The metadata is organized around the concept of an *application*. Kubernetes is not a platform as a service (PaaS) and doesn't have or enforce a formal notion of an application. Instead, applications are informal and described with metadata. The definition of what an application contains is loose.

Note:

These are recommended labels. They make it easier to manage applications but aren't required for any core tooling.

Shared labels and annotations share a common prefix: `app.kubernetes.io`. Labels without a prefix are private to users. The shared prefix ensures that shared labels do not interfere with custom user labels.

Labels

In order to take full advantage of using these labels, they should be applied on every resource object.

Key	Description	Example	Type
<code>app.kubernetes.io/name</code>	The name of the application	<code>mysql</code>	string
<code>app.kubernetes.io/instance</code>	A unique name identifying the instance of an application	<code>mysql-abcxyz</code>	string
<code>app.kubernetes.io/version</code>	The current version of the application (e.g., a SemVer 1.0 , revision hash, etc.)	<code>5.7.21</code>	string
<code>app.kubernetes.io/component</code>	The component within the architecture	<code>database</code>	string
<code>app.kubernetes.io/part-of</code>	The name of a higher level application this one is part of	<code>wordpress</code>	string
<code>app.kubernetes.io/managed-by</code>	The tool being used to manage the operation of an application	<code>Helm</code>	string

To illustrate these labels in action, consider the following [StatefulSet](#) object:

```
# This is an excerpt
apiVersion: apps/v1
kind: StatefulSet
metadata:
  labels:
    app.kubernetes.io/name: mysql
    app.kubernetes.io/instance: mysql-abcxyz
```

```
app.kubernetes.io/version: "5.7.21"  
app.kubernetes.io/component: database  
app.kubernetes.io/part-of: wordpress  
app.kubernetes.io/managed-by: Helm
```

Applications And Instances Of Applications

An application can be installed one or more times into a Kubernetes cluster and, in some cases, the same namespace. For example, WordPress can be installed more than once where different websites are different installations of WordPress.

The name of an application and the instance name are recorded separately. For example, WordPress has a `app.kubernetes.io/name` of `wordpress` while it has an instance name, represented as `app.kubernetes.io/instance` with a value of `wordpress-abcxyz`. This enables the application and instance of the application to be identifiable. Every instance of an application must have a unique name.

Examples

To illustrate different ways to use these labels the following examples have varying complexity.

A Simple Stateless Service

Consider the case for a simple stateless service deployed using Deployment and Service objects. The following two snippets represent how the labels could be used in their simplest form.

The Deployment is used to oversee the pods running the application itself.

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  labels:  
    app.kubernetes.io/name: myservice  
    app.kubernetes.io/instance: myservice-abcxyz  
...
```

The Service is used to expose the application.

```
apiVersion: v1  
kind: Service  
metadata:  
  labels:  
    app.kubernetes.io/name: myservice  
    app.kubernetes.io/instance: myservice-abcxyz  
...
```

Web Application With A Database

Consider a slightly more complicated application: a web application (WordPress) using a database (MySQL), installed using Helm. The following snippets illustrate the start of objects used to deploy this application.

The start to the following Deployment is used for WordPress:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app.kubernetes.io/name: wordpress
    app.kubernetes.io/instance: wordpress-abcxyz
    app.kubernetes.io/version: "4.9.4"
    app.kubernetes.io/managed-by: Helm
    app.kubernetes.io/component: server
    app.kubernetes.io/part-of: wordpress
...
```

The Service is used to expose WordPress:

```
apiVersion: v1
kind: Service
metadata:
  labels:
    app.kubernetes.io/name: wordpress
    app.kubernetes.io/instance: wordpress-abcxyz
    app.kubernetes.io/version: "4.9.4"
    app.kubernetes.io/managed-by: Helm
    app.kubernetes.io/component: server
    app.kubernetes.io/part-of: wordpress
...
```

MySQL is exposed as a StatefulSet with metadata for both it and the larger application it belongs to:

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  labels:
    app.kubernetes.io/name: mysql
    app.kubernetes.io/instance: mysql-abcxyz
    app.kubernetes.io/version: "5.7.21"
    app.kubernetes.io/managed-by: Helm
    app.kubernetes.io/component: database
    app.kubernetes.io/part-of: wordpress
...
```

The Service is used to expose MySQL as part of WordPress:

```
apiVersion: v1
kind: Service
metadata:
  labels:
    app.kubernetes.io/name: mysql
    app.kubernetes.io/instance: mysql-abcxyz
    app.kubernetes.io/version: "5.7.21"
    app.kubernetes.io/managed-by: Helm
    app.kubernetes.io/component: database
```

app.kubernetes.io/part-of: wordpress

With the MySQL StatefulSet and Service you'll notice information about both MySQL and WordPress, the broader application, are included.

Kubernetes Components

A Kubernetes cluster consists of the components that are a part of the control plane and a set of machines called nodes.

When you deploy Kubernetes, you get a cluster.

A Kubernetes cluster consists of a set of worker machines, called [nodes](#), that run containerized applications. Every cluster has at least one worker node.

The worker node(s) host the [Pods](#) that are the components of the application workload. The [control plane](#) manages the worker nodes and the Pods in the cluster. In production environments, the control plane usually runs across multiple computers and a cluster usually runs multiple nodes, providing fault-tolerance and high availability.

This document outlines the various components you need to have for a complete and working Kubernetes cluster.

Components of Kubernetes

The components of a Kubernetes cluster

Control Plane Components

The control plane's components make global decisions about the cluster (for example, scheduling), as well as detecting and responding to cluster events (for example, starting up a new [pod](#) when a Deployment's [replicas](#) field is unsatisfied).

Control plane components can be run on any machine in the cluster. However, for simplicity, setup scripts typically start all control plane components on the same machine, and do not run user containers on this machine. See [Creating Highly Available clusters with kubeadm](#) for an example control plane setup that runs across multiple machines.

kube-apiserver

The API server is a component of the Kubernetes [control plane](#) that exposes the Kubernetes API. The API server is the front end for the Kubernetes control plane.

The main implementation of a Kubernetes API server is [kube-apiserver](#). kube-apiserver is designed to scale horizontally—that is, it scales by deploying more instances. You can run several instances of kube-apiserver and balance traffic between those instances.

etcd

Consistent and highly-available key value store used as Kubernetes' backing store for all cluster data.

If your Kubernetes cluster uses etcd as its backing store, make sure you have a [back up](#) plan for the data.

You can find in-depth information about etcd in the official [documentation](#).

kube-scheduler

Control plane component that watches for newly created [Pods](#) with no assigned [node](#), and selects a node for them to run on.

Factors taken into account for scheduling decisions include: individual and collective resource requirements, hardware/software/policy constraints, affinity and anti-affinity specifications, data locality, inter-workload interference, and deadlines.

kube-controller-manager

Control plane component that runs [controller](#) processes.

Logically, each [controller](#) is a separate process, but to reduce complexity, they are all compiled into a single binary and run in a single process.

There are many different types of controllers. Some examples of them are:

- Node controller: Responsible for noticing and responding when nodes go down.
- Job controller: Watches for Job objects that represent one-off tasks, then creates Pods to run those tasks to completion.
- EndpointSlice controller: Populates EndpointSlice objects (to provide a link between Services and Pods).
- ServiceAccount controller: Create default ServiceAccounts for new namespaces.

The above is not an exhaustive list.

cloud-controller-manager

A Kubernetes [control plane](#) component that embeds cloud-specific control logic. The [cloud controller manager](#) lets you link your cluster into your cloud provider's API, and separates out the components that interact with that cloud platform from components that only interact with your cluster.

The cloud-controller-manager only runs controllers that are specific to your cloud provider. If you are running Kubernetes on your own premises, or in a learning environment inside your own PC, the cluster does not have a cloud controller manager.

As with the kube-controller-manager, the cloud-controller-manager combines several logically independent control loops into a single binary that you run as a single process. You can scale horizontally (run more than one copy) to improve performance or to help tolerate failures.

The following controllers can have cloud provider dependencies:

- Node controller: For checking the cloud provider to determine if a node has been deleted in the cloud after it stops responding
- Route controller: For setting up routes in the underlying cloud infrastructure
- Service controller: For creating, updating and deleting cloud provider load balancers

Node Components

Node components run on every node, maintaining running pods and providing the Kubernetes runtime environment.

kubelet

An agent that runs on each [node](#) in the cluster. It makes sure that [containers](#) are running in a [Pod](#).

The [kubelet](#) takes a set of PodSpecs that are provided through various mechanisms and ensures that the containers described in those PodSpecs are running and healthy. The kubelet doesn't manage containers which were not created by Kubernetes.

kube-proxy

kube-proxy is a network proxy that runs on each [node](#) in your cluster, implementing part of the Kubernetes [Service](#) concept.

[kube-proxy](#) maintains network rules on nodes. These network rules allow network communication to your Pods from network sessions inside or outside of your cluster.

kube-proxy uses the operating system packet filtering layer if there is one and it's available. Otherwise, kube-proxy forwards the traffic itself.

Container runtime

A fundamental component that empowers Kubernetes to run containers effectively. It is responsible for managing the execution and lifecycle of containers within the Kubernetes environment.

Kubernetes supports container runtimes such as [containerd](#), [CRI-O](#), and any other implementation of the [Kubernetes CRI \(Container Runtime Interface\)](#).

Addons

Addons use Kubernetes resources ([DaemonSet](#), [Deployment](#), etc) to implement cluster features. Because these are providing cluster-level features, namespaced resources for addons belong within the kube-system namespace.

Selected addons are described below; for an extended list of available addons, please see [Addons](#).

DNS

While the other addons are not strictly required, all Kubernetes clusters should have [cluster DNS](#), as many examples rely on it.

Cluster DNS is a DNS server, in addition to the other DNS server(s) in your environment, which serves DNS records for Kubernetes services.

Containers started by Kubernetes automatically include this DNS server in their DNS searches.

Web UI (Dashboard)

[Dashboard](#) is a general purpose, web-based UI for Kubernetes clusters. It allows users to manage and troubleshoot applications running in the cluster, as well as the cluster itself.

Container Resource Monitoring

[Container Resource Monitoring](#) records generic time-series metrics about containers in a central database, and provides a UI for browsing that data.

Cluster-level Logging

A [cluster-level logging](#) mechanism is responsible for saving container logs to a central log store with search/browsing interface.

Network Plugins

[Network plugins](#) are software components that implement the container network interface (CNI) specification. They are responsible for allocating IP addresses to pods and enabling them to communicate with each other within the cluster.

What's next

Learn more about the following:

- [Nodes](#) and [their communication](#) with the control plane.
- Kubernetes [controllers](#).
- [kube-scheduler](#) which is the default scheduler for Kubernetes.
- Etcd's official [documentation](#).
- Several [container runtimes](#) in Kubernetes.
- Integrating with cloud providers using [cloud-controller-manager](#).
- [kubectl](#) commands.

The Kubernetes API

The Kubernetes API lets you query and manipulate the state of objects in Kubernetes. The core of Kubernetes' control plane is the API server and the HTTP API that it exposes. Users, the different parts of your cluster, and external components all communicate with one another through the API server.

The core of Kubernetes' [control plane](#) is the [API server](#). The API server exposes an HTTP API that lets end users, different parts of your cluster, and external components communicate with one another.

The Kubernetes API lets you query and manipulate the state of API objects in Kubernetes (for example: Pods, Namespaces, ConfigMaps, and Events).

Most operations can be performed through the [kubectl](#) command-line interface or other command-line tools, such as [kubeadm](#), which in turn use the API. However, you can also access the API directly using REST calls. Kubernetes provides a set of [client libraries](#) for those looking to write applications using the Kubernetes API.

Each Kubernetes cluster publishes the specification of the APIs that the cluster serves. There are two mechanisms that Kubernetes uses to publish these API specifications; both are useful to enable automatic interoperability. For example, the kubectl tool fetches and caches the API specification for enabling command-line completion and other features. The two supported mechanisms are as follows:

- [The Discovery API](#) provides information about the Kubernetes APIs: API names, resources, versions, and supported operations. This is a Kubernetes specific term as it is a separate API from the Kubernetes OpenAPI. It is intended to be a brief summary of the available resources and it does not detail specific schema for the resources. For reference about resource schemas, please refer to the OpenAPI document.
- The [Kubernetes OpenAPI Document](#) provides (full) [OpenAPI v2.0 and 3.0 schemas](#) for all Kubernetes API endpoints. The OpenAPI v3 is the preferred method for accessing OpenAPI as it provides a more comprehensive and accurate view of the API. It includes all the available API paths, as well as all resources consumed and produced for every operations on every endpoints. It also includes any extensibility components that a cluster supports. The data is a complete specification and is significantly larger than that from the Discovery API.

Discovery API

Kubernetes publishes a list of all group versions and resources supported via the Discovery API. This includes the following for each resource:

- Name
- Cluster or namespaced scope
- Endpoint URL and supported verbs
- Alternative names
- Group, version, kind

The API is available both aggregated and unaggregated form. The aggregated discovery serves two endpoints while the unaggregated discovery serves a separate endpoint for each group version.

Aggregated discovery

FEATURE STATE: Kubernetes v1.30 [stable]

Kubernetes offers stable support for *aggregated discovery*, publishing all resources supported by a cluster through two endpoints (/api and /apis). Requesting this endpoint drastically reduces

the number of requests sent to fetch the discovery data from the cluster. You can access the data by requesting the respective endpoints with an Accept header indicating the aggregated discovery resource: `Accept: application/json;v=v2;g=apidiscovery.k8s.io;as=APIGroupDiscoveryList`.

Without indicating the resource type using the Accept header, the default response for the `/api` and `/apis` endpoint is an unaggregated discovery document.

The [discovery document](#) for the built-in resources can be found in the Kubernetes GitHub repository. This Github document can be used as a reference of the base set of the available resources if a Kubernetes cluster is not available to query.

The endpoint also supports ETag and protobuf encoding.

Unaggregated discovery

Without discovery aggregation, discovery is published in levels, with the root endpoints publishing discovery information for downstream documents.

A list of all group versions supported by a cluster is published at the `/api` and `/apis` endpoints. Example:

```
{
  "kind": "APIGroupList",
  "apiVersion": "v1",
  "groups": [
    {
      "name": "apiregistration.k8s.io",
      "versions": [
        {
          "groupVersion": "apiregistration.k8s.io/v1",
          "version": "v1"
        }
      ],
      "preferredVersion": {
        "groupVersion": "apiregistration.k8s.io/v1",
        "version": "v1"
      }
    },
    {
      "name": "apps",
      "versions": [
        {
          "groupVersion": "apps/v1",
          "version": "v1"
        }
      ],
      "preferredVersion": {
        "groupVersion": "apps/v1",
        "version": "v1"
      }
    }
  ],
}
```

```
} ...  
}
```

Additional requests are needed to obtain the discovery document for each group version at `/apis/<group>/<version>` (for example: `/apis/rbac.authorization.k8s.io/v1alpha1`), which advertises the list of resources served under a particular group version. These endpoints are used by `kubectl` to fetch the list of resources supported by a cluster.

OpenAPI interface definition

For details about the OpenAPI specifications, see the [OpenAPI documentation](#).

Kubernetes serves both OpenAPI v2.0 and OpenAPI v3.0. OpenAPI v3 is the preferred method of accessing the OpenAPI because it offers a more comprehensive (lossless) representation of Kubernetes resources. Due to limitations of OpenAPI version 2, certain fields are dropped from the published OpenAPI including but not limited to `default`, `nullable`, `oneOf`.

OpenAPI V2

The Kubernetes API server serves an aggregated OpenAPI v2 spec via the `/openapi/v2` endpoint. You can request the response format using request headers as follows:

Valid request header values for OpenAPI v2 queries

Header	Possible values	Notes
Accept-Encoding	gzip	<i>not supplying this header is also acceptable</i>
Accept	application/com.github.proto-openapi.spec.v2@v1.0+protobuf	<i>mainly for intra-cluster use</i>
	application/json	<i>default</i>
	*	<i>serves application/json</i>

OpenAPI V3

FEATURE STATE: Kubernetes v1.27 [stable]

Kubernetes supports publishing a description of its APIs as OpenAPI v3.

A discovery endpoint `/openapi/v3` is provided to see a list of all group/versions available. This endpoint only returns JSON. These group/versions are provided in the following format:

```
{  
  "paths": {  
    ...,  
    "api/v1": {  
      "serverRelativeURL": "/openapi/v3/api/v1?  
hash=CC0E9BFD992D8C59AEC98A1E2336F899E8318D3CF4C68944C3DEC640AF5AB52D864A  
C50DAA8D145B3494F75FA3CFF939FCBDDA431DAD3CA79738B297795818CF"  
    },  
    "apis/admissionregistration.k8s.io/v1": {  
      "serverRelativeURL": "/openapi/v3/apis/admissionregistration.k8s.io/v1?"
```

```

hash=E19CC93A116982CE5422FC42B590A8AFAD92CDE9AE4D59B5CAAD568F083AD07946E6
CB5817531680BCE6E215C16973CD39003B0425F3477CFD854E89A9DB6597"
    },
    ....
  }
}

```

The relative URLs are pointing to immutable OpenAPI descriptions, in order to improve client-side caching. The proper HTTP caching headers are also set by the API server for that purpose (Expires to 1 year in the future, and Cache-Control to immutable). When an obsolete URL is used, the API server returns a redirect to the newest URL.

The Kubernetes API server publishes an OpenAPI v3 spec per Kubernetes group version at the `/openapi/v3/apis/<group>/<version>?hash=<hash>` endpoint.

Refer to the table below for accepted request headers.

Valid request header values for OpenAPI v3 queries

Header	Possible values	Notes
Accept-Encoding	gzip	<i>not supplying this header is also acceptable</i>
Accept	application/com.github.proto-openapi.spec.v3@v1.0+protobuf	<i>mainly for intra-cluster use</i>
	application/json	<i>default</i>
	*	<i>serves application/json</i>

A Golang implementation to fetch the OpenAPI V3 is provided in the package k8s.io/client-go/openapi3.

Kubernetes 1.30 publishes OpenAPI v2.0 and v3.0; there are no plans to support 3.1 in the near future.

Protobuf serialization

Kubernetes implements an alternative Protobuf based serialization format that is primarily intended for intra-cluster communication. For more information about this format, see the [Kubernetes Protobuf serialization](#) design proposal and the Interface Definition Language (IDL) files for each schema located in the Go packages that define the API objects.

Persistence

Kubernetes stores the serialized state of objects by writing them into [etcd](#).

API groups and versioning

To make it easier to eliminate fields or restructure resource representations, Kubernetes supports multiple API versions, each at a different API path, such as `/api/v1` or `/apis/rbac.authorization.k8s.io/v1alpha1`.

Versioning is done at the API level rather than at the resource or field level to ensure that the API presents a clear, consistent view of system resources and behavior, and to enable controlling access to end-of-life and/or experimental APIs.

To make it easier to evolve and to extend its API, Kubernetes implements [API groups](#) that can be [enabled or disabled](#).

API resources are distinguished by their API group, resource type, namespace (for namespaced resources), and name. The API server handles the conversion between API versions transparently: all the different versions are actually representations of the same persisted data. The API server may serve the same underlying data through multiple API versions.

For example, suppose there are two API versions, `v1` and `v1beta1`, for the same resource. If you originally created an object using the `v1beta1` version of its API, you can later read, update, or delete that object using either the `v1beta1` or the `v1` API version, until the `v1beta1` version is deprecated and removed. At that point you can continue accessing and modifying the object using the `v1` API.

API changes

Any system that is successful needs to grow and change as new use cases emerge or existing ones change. Therefore, Kubernetes has designed the Kubernetes API to continuously change and grow. The Kubernetes project aims to *not* break compatibility with existing clients, and to maintain that compatibility for a length of time so that other projects have an opportunity to adapt.

In general, new API resources and new resource fields can be added often and frequently. Elimination of resources or fields requires following the [API deprecation policy](#).

Kubernetes makes a strong commitment to maintain compatibility for official Kubernetes APIs once they reach general availability (GA), typically at API version `v1`. Additionally, Kubernetes maintains compatibility with data persisted via *beta* API versions of official Kubernetes APIs, and ensures that data can be converted and accessed via GA API versions when the feature goes stable.

If you adopt a beta API version, you will need to transition to a subsequent beta or stable API version once the API graduates. The best time to do this is while the beta API is in its deprecation period, since objects are simultaneously accessible via both API versions. Once the beta API completes its deprecation period and is no longer served, the replacement API version must be used.

Note:

Although Kubernetes also aims to maintain compatibility for *alpha* APIs versions, in some circumstances this is not possible. If you use any alpha API versions, check the release notes for Kubernetes when upgrading your cluster, in case the API did change in incompatible ways that require deleting all existing alpha objects prior to upgrade.

Refer to [API versions reference](#) for more details on the API version level definitions.

API Extension

The Kubernetes API can be extended in one of two ways:

1. [Custom resources](#) let you declaratively define how the API server should provide your chosen resource API.
2. You can also extend the Kubernetes API by implementing an [aggregation layer](#).

What's next

- Learn how to extend the Kubernetes API by adding your own [CustomResourceDefinition](#).
- [Controlling Access To The Kubernetes API](#) describes how the cluster manages authentication and authorization for API access.
- Learn about API endpoints, resource types and samples by reading [API Reference](#).
- Learn about what constitutes a compatible change, and how to change the API, from [API changes](#).

Cluster Architecture

The architectural concepts behind Kubernetes.

Components of Kubernetes

Kubernetes cluster architecture

[Nodes](#)

[Communication between Nodes and the Control Plane](#)

[Controllers](#)

[Leases](#)

[Cloud Controller Manager](#)

[About cgroup v2](#)

[Container Runtime Interface \(CRI\)](#)

[Garbage Collection](#)

[Mixed Version Proxy](#)

Nodes

Kubernetes runs your [workload](#) by placing containers into Pods to run on *Nodes*. A node may be a virtual or physical machine, depending on the cluster. Each node is managed by the [control plane](#) and contains the services necessary to run [Pods](#).

Typically you have several nodes in a cluster; in a learning or resource-limited environment, you might have only one node.

The [components](#) on a node include the [kubelet](#), a [container runtime](#), and the [kube-proxy](#).

Management

There are two main ways to have Nodes added to the [API server](#):

1. The kubelet on a node self-registers to the control plane
2. You (or another human user) manually add a Node object

After you create a Node [object](#), or the kubelet on a node self-registers, the control plane checks whether the new Node object is valid. For example, if you try to create a Node from the following JSON manifest:

```
{
  "kind": "Node",
  "apiVersion": "v1",
  "metadata": {
    "name": "10.240.79.157",
    "labels": {
      "name": "my-first-k8s-node"
    }
  }
}
```

Kubernetes creates a Node object internally (the representation). Kubernetes checks that a kubelet has registered to the API server that matches the `metadata.name` field of the Node. If the node is healthy (i.e. all necessary services are running), then it is eligible to run a Pod. Otherwise, that node is ignored for any cluster activity until it becomes healthy.

Note:

Kubernetes keeps the object for the invalid Node and continues checking to see whether it becomes healthy.

You, or a [controller](#), must explicitly delete the Node object to stop that health checking.

The name of a Node object must be a valid [DNS subdomain name](#).

Node name uniqueness

The [name](#) identifies a Node. Two Nodes cannot have the same name at the same time. Kubernetes also assumes that a resource with the same name is the same object. In case of a Node, it is implicitly assumed that an instance using the same name will have the same state (e.g. network settings, root disk contents) and attributes like node labels. This may lead to inconsistencies if an instance was modified without changing its name. If the Node needs to be replaced or updated significantly, the existing Node object needs to be removed from API server first and re-added after the update.

Self-registration of Nodes

When the kubelet flag `--register-node` is true (the default), the kubelet will attempt to register itself with the API server. This is the preferred pattern, used by most distros.

For self-registration, the kubelet is started with the following options:

- `--kubeconfig` - Path to credentials to authenticate itself to the API server.
- `--cloud-provider` - How to talk to a [cloud provider](#) to read metadata about itself.
- `--register-node` - Automatically register with the API server.
- `--register-with-taints` - Register the node with the given list of [taints](#) (comma separated `<key>=<value>:<effect>`).

No-op if `register-node` is false.

- `--node-ip` - Optional comma-separated list of the IP addresses for the node. You can only specify a single address for each address family. For example, in a single-stack IPv4 cluster, you set this value to be the IPv4 address that the kubelet should use for the node. See [configure IPv4/IPv6 dual stack](#) for details of running a dual-stack cluster.

If you don't provide this argument, the kubelet uses the node's default IPv4 address, if any; if the node has no IPv4 addresses then the kubelet uses the node's default IPv6 address.

- `--node-labels` - [Labels](#) to add when registering the node in the cluster (see label restrictions enforced by the [NodeRestriction admission plugin](#)).
- `--node-status-update-frequency` - Specifies how often kubelet posts its node status to the API server.

When the [Node authorization mode](#) and [NodeRestriction admission plugin](#) are enabled, kubelets are only authorized to create/modify their own Node resource.

Note:

As mentioned in the [Node name uniqueness](#) section, when Node configuration needs to be updated, it is a good practice to re-register the node with the API server. For example, if the kubelet is being restarted with a new set of `--node-labels`, but the same Node name is used, the change will not take effect, as labels are only set (or modified) upon Node registration with the API server.

Pods already scheduled on the Node may misbehave or cause issues if the Node configuration will be changed on kubelet restart. For example, already running Pod may be tainted against the new labels assigned to the Node, while other Pods, that are incompatible with that Pod will be scheduled based on this new label. Node re-registration ensures all Pods will be drained and properly re-scheduled.

Manual Node administration

You can create and modify Node objects using [kubectl](#).

When you want to create Node objects manually, set the kubelet flag `--register-node=false`.

You can modify Node objects regardless of the setting of `--register-node`. For example, you can set labels on an existing Node or mark it unschedulable.

You can use labels on Nodes in conjunction with node selectors on Pods to control scheduling. For example, you can constrain a Pod to only be eligible to run on a subset of the available nodes.

Marking a node as unschedulable prevents the scheduler from placing new pods onto that Node but does not affect existing Pods on the Node. This is useful as a preparatory step before a node reboot or other maintenance.

To mark a Node unschedulable, run:

```
kubectcl cordon $NODENAME
```

See [Safely Drain a Node](#) for more details.

Note:

Pods that are part of a [DaemonSet](#) tolerate being run on an unschedulable Node. DaemonSets typically provide node-local services that should run on the Node even if it is being drained of workload applications.

Node status

A Node's status contains the following information:

- [Addresses](#)
- [Conditions](#)
- [Capacity and Allocatable](#)
- [Info](#)

You can use `kubectcl` to view a Node's status and other details:

```
kubectcl describe node <insert-node-name-here>
```

See [Node Status](#) for more details.

Node heartbeats

Heartbeats, sent by Kubernetes nodes, help your cluster determine the availability of each node, and to take action when failures are detected.

For nodes there are two forms of heartbeats:

- Updates to the `.status` of a Node.
- [Lease](#) objects within the kube-node-lease [namespace](#). Each Node has an associated Lease object.

Node controller

The node [controller](#) is a Kubernetes control plane component that manages various aspects of nodes.

The node controller has multiple roles in a node's life. The first is assigning a CIDR block to the node when it is registered (if CIDR assignment is turned on).

The second is keeping the node controller's internal list of nodes up to date with the cloud provider's list of available machines. When running in a cloud environment and whenever a node is unhealthy, the node controller asks the cloud provider if the VM for that node is still available. If not, the node controller deletes the node from its list of nodes.

The third is monitoring the nodes' health. The node controller is responsible for:

- In the case that a node becomes unreachable, updating the Ready condition in the Node's .status field. In this case the node controller sets the Ready condition to Unknown.
- If a node remains unreachable: triggering [API-initiated eviction](#) for all of the Pods on the unreachable node. By default, the node controller waits 5 minutes between marking the node as Unknown and submitting the first eviction request.

By default, the node controller checks the state of each node every 5 seconds. This period can be configured using the `--node-monitor-period` flag on the kube-controller-manager component.

Rate limits on eviction

In most cases, the node controller limits the eviction rate to `--node-eviction-rate` (default 0.1) per second, meaning it won't evict pods from more than 1 node per 10 seconds.

The node eviction behavior changes when a node in a given availability zone becomes unhealthy. The node controller checks what percentage of nodes in the zone are unhealthy (the Ready condition is Unknown or False) at the same time:

- If the fraction of unhealthy nodes is at least `--unhealthy-zone-threshold` (default 0.55), then the eviction rate is reduced.
- If the cluster is small (i.e. has less than or equal to `--large-cluster-size-threshold` nodes - default 50), then evictions are stopped.
- Otherwise, the eviction rate is reduced to `--secondary-node-eviction-rate` (default 0.01) per second.

The reason these policies are implemented per availability zone is because one availability zone might become partitioned from the control plane while the others remain connected. If your cluster does not span multiple cloud provider availability zones, then the eviction mechanism does not take per-zone unavailability into account.

A key reason for spreading your nodes across availability zones is so that the workload can be shifted to healthy zones when one entire zone goes down. Therefore, if all nodes in a zone are unhealthy, then the node controller evicts at the normal rate of `--node-eviction-rate`. The corner case is when all zones are completely unhealthy (none of the nodes in the cluster are healthy). In such a case, the node controller assumes that there is some problem with connectivity between the control plane and the nodes, and doesn't perform any evictions. (If there has been

an outage and some nodes reappear, the node controller does evict pods from the remaining nodes that are unhealthy or unreachable).

The node controller is also responsible for evicting pods running on nodes with NoExecute taints, unless those pods tolerate that taint. The node controller also adds [taints](#) corresponding to node problems like node unreachable or not ready. This means that the scheduler won't place Pods onto unhealthy nodes.

Resource capacity tracking

Node objects track information about the Node's resource capacity: for example, the amount of memory available and the number of CPUs. Nodes that [self register](#) report their capacity during registration. If you [manually](#) add a Node, then you need to set the node's capacity information when you add it.

The Kubernetes [scheduler](#) ensures that there are enough resources for all the Pods on a Node. The scheduler checks that the sum of the requests of containers on the node is no greater than the node's capacity. That sum of requests includes all containers managed by the kubelet, but excludes any containers started directly by the container runtime, and also excludes any processes running outside of the kubelet's control.

Note:

If you want to explicitly reserve resources for non-Pod processes, see [reserve resources for system daemons](#).

Node topology

FEATURE STATE: Kubernetes v1.27 [stable]

If you have enabled the TopologyManager [feature gate](#), then the kubelet can use topology hints when making resource assignment decisions. See [Control Topology Management Policies on a Node](#) for more information.

Swap memory management

FEATURE STATE: Kubernetes v1.28 [beta]

To enable swap on a node, the NodeSwap feature gate must be enabled on the kubelet (default is true), and the `--fail-swap-on` command line flag or `failSwapOn` [configuration setting](#) must be set to false. To allow Pods to utilize swap, `swapBehavior` should not be set to `NoSwap` (which is the default behavior) in the kubelet config.

Warning:

When the memory swap feature is turned on, Kubernetes data such as the content of Secret objects that were written to tmpfs now could be swapped to disk.

A user can also optionally configure `memorySwap.swapBehavior` in order to specify how a node will use swap memory. For example,

memorySwap:
swapBehavior: LimitedSwap

- NoSwap (default): Kubernetes workloads will not use swap.
- LimitedSwap: The utilization of swap memory by Kubernetes workloads is subject to limitations. Only Pods of Burstable QoS are permitted to employ swap.

If configuration for memorySwap is not specified and the feature gate is enabled, by default the kubelet will apply the same behaviour as the NoSwap setting.

With LimitedSwap, Pods that do not fall under the Burstable QoS classification (i.e. BestEffort/ Guaranteed QoS Pods) are prohibited from utilizing swap memory. To maintain the aforementioned security and node health guarantees, these Pods are not permitted to use swap memory when LimitedSwap is in effect.

Prior to detailing the calculation of the swap limit, it is necessary to define the following terms:

- nodeTotalMemory: The total amount of physical memory available on the node.
- totalPodsSwapAvailable: The total amount of swap memory on the node that is available for use by Pods (some swap memory may be reserved for system use).
- containerMemoryRequest: The container's memory request.

Swap limitation is configured as: $(\text{containerMemoryRequest} / \text{nodeTotalMemory}) * \text{totalPodsSwapAvailable}$.

It is important to note that, for containers within Burstable QoS Pods, it is possible to opt-out of swap usage by specifying memory requests that are equal to memory limits. Containers configured in this manner will not have access to swap memory.

Swap is supported only with **cgroup v2**, cgroup v1 is not supported.

For more information, and to assist with testing and provide feedback, please see the blog-post about [Kubernetes 1.28: NodeSwap graduates to Beta1](#), [KEP-2400](#) and its [design proposal](#).

What's next

Learn more about the following:

- [Components](#) that make up a node.
- [API definition for Node](#).
- [Node](#) section of the architecture design document.
- [Graceful/non-graceful node shutdown](#).
- [Cluster autoscaling](#) to manage the number and size of nodes in your cluster.
- [Taints and Tolerations](#).
- [Node Resource Managers](#).
- [Resource Management for Windows nodes](#).

Communication between Nodes and the Control Plane

This document catalogs the communication paths between the [API server](#) and the Kubernetes [cluster](#). The intent is to allow users to customize their installation to harden the network configuration such that the cluster can be run on an untrusted network (or on fully public IPs on a cloud provider).

Node to Control Plane

Kubernetes has a "hub-and-spoke" API pattern. All API usage from nodes (or the pods they run) terminates at the API server. None of the other control plane components are designed to expose remote services. The API server is configured to listen for remote connections on a secure HTTPS port (typically 443) with one or more forms of client [authentication](#) enabled. One or more forms of [authorization](#) should be enabled, especially if [anonymous requests](#) or [service account tokens](#) are allowed.

Nodes should be provisioned with the public root [certificate](#) for the cluster such that they can connect securely to the API server along with valid client credentials. A good approach is that the client credentials provided to the kubelet are in the form of a client certificate. See [kubelet TLS bootstrapping](#) for automated provisioning of kubelet client certificates.

[Pods](#) that wish to connect to the API server can do so securely by leveraging a service account so that Kubernetes will automatically inject the public root certificate and a valid bearer token into the pod when it is instantiated. The kubernetes service (in default namespace) is configured with a virtual IP address that is redirected (via [kube-proxy](#)) to the HTTPS endpoint on the API server.

The control plane components also communicate with the API server over the secure port.

As a result, the default operating mode for connections from the nodes and pod running on the nodes to the control plane is secured by default and can run over untrusted and/or public networks.

Control plane to node

There are two primary communication paths from the control plane (the API server) to the nodes. The first is from the API server to the [kubelet](#) process which runs on each node in the cluster. The second is from the API server to any node, pod, or service through the API server's *proxy* functionality.

API server to kubelet

The connections from the API server to the kubelet are used for:

- Fetching logs for pods.
- Attaching (usually through kubectl) to running pods.
- Providing the kubelet's port-forwarding functionality.

These connections terminate at the kubelet's HTTPS endpoint. By default, the API server does not verify the kubelet's serving certificate, which makes the connection subject to man-in-the-middle attacks and **unsafe** to run over untrusted and/or public networks.

To verify this connection, use the `--kubelet-certificate-authority` flag to provide the API server with a root certificate bundle to use to verify the kubelet's serving certificate.

If that is not possible, use [SSH tunneling](#) between the API server and kubelet if required to avoid connecting over an untrusted or public network.

Finally, [Kubelet authentication and/or authorization](#) should be enabled to secure the kubelet API.

API server to nodes, pods, and services

The connections from the API server to a node, pod, or service default to plain HTTP connections and are therefore neither authenticated nor encrypted. They can be run over a secure HTTPS connection by prefixing `https:` to the node, pod, or service name in the API URL, but they will not validate the certificate provided by the HTTPS endpoint nor provide client credentials. So while the connection will be encrypted, it will not provide any guarantees of integrity. These connections **are not currently safe** to run over untrusted or public networks.

SSH tunnels

Kubernetes supports [SSH tunnels](#) to protect the control plane to nodes communication paths. In this configuration, the API server initiates an SSH tunnel to each node in the cluster (connecting to the SSH server listening on port 22) and passes all traffic destined for a kubelet, node, pod, or service through the tunnel. This tunnel ensures that the traffic is not exposed outside of the network in which the nodes are running.

Note:

SSH tunnels are currently deprecated, so you shouldn't opt to use them unless you know what you are doing. The [Konnectivity service](#) is a replacement for this communication channel.

Konnectivity service

FEATURE STATE: Kubernetes v1.18 [beta]

As a replacement to the SSH tunnels, the Konnectivity service provides TCP level proxy for the control plane to cluster communication. The Konnectivity service consists of two parts: the Konnectivity server in the control plane network and the Konnectivity agents in the nodes network. The Konnectivity agents initiate connections to the Konnectivity server and maintain the network connections. After enabling the Konnectivity service, all control plane to nodes traffic goes through these connections.

Follow the [Konnectivity service task](#) to set up the Konnectivity service in your cluster.

What's next

- Read about the [Kubernetes control plane components](#)
- Learn more about [Hubs and Spoke model](#)

- Learn how to [Secure a Cluster](#)
- Learn more about the [Kubernetes API](#)
- [Set up Konnectivity service](#)
- [Use Port Forwarding to Access Applications in a Cluster](#)
- Learn how to [Fetch logs for Pods](#), [use kubectl port-forward](#)

Controllers

In robotics and automation, a *control loop* is a non-terminating loop that regulates the state of a system.

Here is one example of a control loop: a thermostat in a room.

When you set the temperature, that's telling the thermostat about your *desired state*. The actual room temperature is the *current state*. The thermostat acts to bring the current state closer to the desired state, by turning equipment on or off.

In Kubernetes, controllers are control loops that watch the state of your [cluster](#), then make or request changes where needed. Each controller tries to move the current cluster state closer to the desired state.

Controller pattern

A controller tracks at least one Kubernetes resource type. These [objects](#) have a spec field that represents the desired state. The controller(s) for that resource are responsible for making the current state come closer to that desired state.

The controller might carry the action out itself; more commonly, in Kubernetes, a controller will send messages to the [API server](#) that have useful side effects. You'll see examples of this below.

Control via API server

The [Job](#) controller is an example of a Kubernetes built-in controller. Built-in controllers manage state by interacting with the cluster API server.

Job is a Kubernetes resource that runs a [Pod](#), or perhaps several Pods, to carry out a task and then stop.

(Once [scheduled](#), Pod objects become part of the desired state for a kubelet).

When the Job controller sees a new task it makes sure that, somewhere in your cluster, the kubelets on a set of Nodes are running the right number of Pods to get the work done. The Job controller does not run any Pods or containers itself. Instead, the Job controller tells the API server to create or remove Pods. Other components in the [control plane](#) act on the new information (there are new Pods to schedule and run), and eventually the work is done.

After you create a new Job, the desired state is for that Job to be completed. The Job controller makes the current state for that Job be nearer to your desired state: creating Pods that do the work you wanted for that Job, so that the Job is closer to completion.

Controllers also update the objects that configure them. For example: once the work is done for a Job, the Job controller updates that Job object to mark it Finished.

(This is a bit like how some thermostats turn a light off to indicate that your room is now at the temperature you set).

Direct control

In contrast with Job, some controllers need to make changes to things outside of your cluster.

For example, if you use a control loop to make sure there are enough [Nodes](#) in your cluster, then that controller needs something outside the current cluster to set up new Nodes when needed.

Controllers that interact with external state find their desired state from the API server, then communicate directly with an external system to bring the current state closer in line.

(There actually is a [controller](#) that horizontally scales the nodes in your cluster.)

The important point here is that the controller makes some changes to bring about your desired state, and then reports the current state back to your cluster's API server. Other control loops can observe that reported data and take their own actions.

In the thermostat example, if the room is very cold then a different controller might also turn on a frost protection heater. With Kubernetes clusters, the control plane indirectly works with IP address management tools, storage services, cloud provider APIs, and other services by [extending Kubernetes](#) to implement that.

Desired versus current state

Kubernetes takes a cloud-native view of systems, and is able to handle constant change.

Your cluster could be changing at any point as work happens and control loops automatically fix failures. This means that, potentially, your cluster never reaches a stable state.

As long as the controllers for your cluster are running and able to make useful changes, it doesn't matter if the overall state is stable or not.

Design

As a tenet of its design, Kubernetes uses lots of controllers that each manage a particular aspect of cluster state. Most commonly, a particular control loop (controller) uses one kind of resource as its desired state, and has a different kind of resource that it manages to make that desired state happen. For example, a controller for Jobs tracks Job objects (to discover new work) and Pod objects (to run the Jobs, and then to see when the work is finished). In this case something else creates the Jobs, whereas the Job controller creates Pods.

It's useful to have simple controllers rather than one, monolithic set of control loops that are interlinked. Controllers can fail, so Kubernetes is designed to allow for that.

Note:

There can be several controllers that create or update the same kind of object. Behind the scenes, Kubernetes controllers make sure that they only pay attention to the resources linked to their controlling resource.

For example, you can have Deployments and Jobs; these both create Pods. The Job controller does not delete the Pods that your Deployment created, because there is information ([labels](#)) the controllers can use to tell those Pods apart.

Ways of running controllers

Kubernetes comes with a set of built-in controllers that run inside the [kube-controller-manager](#). These built-in controllers provide important core behaviors.

The Deployment controller and Job controller are examples of controllers that come as part of Kubernetes itself ("built-in" controllers). Kubernetes lets you run a resilient control plane, so that if any of the built-in controllers were to fail, another part of the control plane will take over the work.

You can find controllers that run outside the control plane, to extend Kubernetes. Or, if you want, you can write a new controller yourself. You can run your own controller as a set of Pods, or externally to Kubernetes. What fits best will depend on what that particular controller does.

What's next

- Read about the [Kubernetes control plane](#)
- Discover some of the basic [Kubernetes objects](#)
- Learn more about the [Kubernetes API](#)
- If you want to write your own controller, see [Kubernetes extension patterns](#) and the [sample-controller](#) repository.

Leases

Distributed systems often have a need for *leases*, which provide a mechanism to lock shared resources and coordinate activity between members of a set. In Kubernetes, the lease concept is represented by [Lease](#) objects in the coordination.k8s.io [API Group](#), which are used for system-critical capabilities such as node heartbeats and component-level leader election.

Node heartbeats

Kubernetes uses the Lease API to communicate kubelet node heartbeats to the Kubernetes API server. For every Node, there is a Lease object with a matching name in the kube-node-lease namespace. Under the hood, every kubelet heartbeat is an **update** request to this Lease object, updating the spec.renewTime field for the Lease. The Kubernetes control plane uses the time stamp of this field to determine the availability of this Node.

See [Node Lease objects](#) for more details.

Leader election

Kubernetes also uses Leases to ensure only one instance of a component is running at any given time. This is used by control plane components like kube-controller-manager and kube-scheduler in HA configurations, where only one instance of the component should be actively running while the other instances are on stand-by.

API server identity

FEATURE STATE: Kubernetes v1.26 [beta]

Starting in Kubernetes v1.26, each kube-apiserver uses the Lease API to publish its identity to the rest of the system. While not particularly useful on its own, this provides a mechanism for clients to discover how many instances of kube-apiserver are operating the Kubernetes control plane. Existence of kube-apiserver leases enables future capabilities that may require coordination between each kube-apiserver.

You can inspect Leases owned by each kube-apiserver by checking for lease objects in the kube-system namespace with the name kube-apiserver-`<sha256-hash>`. Alternatively you can use the label selector `apiserver.kubernetes.io/identity=kube-apiserver`:

```
kubectl -n kube-system get lease -l apiserver.kubernetes.io/identity=kube-apiserver
```

NAME	HOLDER	AGE
apiserver-07a5ea9b9b072c4a5f3d1c3702		
apiserver-07a5ea9b9b072c4a5f3d1c3702_0c8914f7-0f35-440e-8676-7844977d3a05		5m33s
apiserver-7be9e061c59d368b3ddaf1376e		
apiserver-7be9e061c59d368b3ddaf1376e_84f2a85d-37c1-4b14-b6b9-603e62e4896f		4m23s
apiserver-1dfef752bcb36637d2763d1868		
apiserver-1dfef752bcb36637d2763d1868_c5ffa286-8a9a-45d4-91e7-61118ed58d2e		4m43s

The SHA256 hash used in the lease name is based on the OS hostname as seen by that API server. Each kube-apiserver should be configured to use a hostname that is unique within the cluster. New instances of kube-apiserver that use the same hostname will take over existing Leases using a new holder identity, as opposed to instantiating new Lease objects. You can check the hostname used by kube-apiserver by checking the value of the `kubernetes.io/hostname` label:

```
kubectl -n kube-system get lease apiserver-07a5ea9b9b072c4a5f3d1c3702 -o yaml
```

```
apiVersion: coordination.k8s.io/v1
kind: Lease
metadata:
  creationTimestamp: "2023-07-02T13:16:48Z"
  labels:
    apiserver.kubernetes.io/identity: kube-apiserver
    kubernetes.io/hostname: master-1
  name: apiserver-07a5ea9b9b072c4a5f3d1c3702
  namespace: kube-system
  resourceVersion: "334899"
  uid: 90870ab5-1ba9-4523-b215-e4d4e662acb1
spec:
```

```
holderIdentity: apiserver-07a5ea9b9b072c4a5f3d1c3702_0c8914f7-0f35-440e-8676-7844977d3a05
leaseDurationSeconds: 3600
renewTime: "2023-07-04T21:58:48.065888Z"
```

Expired leases from kube-apiservers that no longer exist are garbage collected by new kube-apiservers after 1 hour.

You can disable API server identity leases by disabling the `APIServerIdentity` [feature gate](#).

Workloads

Your own workload can define its own use of Leases. For example, you might run a custom [controller](#) where a primary or leader member performs operations that its peers do not. You define a Lease so that the controller replicas can select or elect a leader, using the Kubernetes API for coordination. If you do use a Lease, it's a good practice to define a name for the Lease that is obviously linked to the product or component. For example, if you have a component named Example Foo, use a Lease named example-foo.

If a cluster operator or another end user could deploy multiple instances of a component, select a name prefix and pick a mechanism (such as hash of the name of the Deployment) to avoid name collisions for the Leases.

You can use another approach so long as it achieves the same outcome: different software products do not conflict with one another.

Cloud Controller Manager

FEATURE STATE: Kubernetes v1.11 [beta]

Cloud infrastructure technologies let you run Kubernetes on public, private, and hybrid clouds. Kubernetes believes in automated, API-driven infrastructure without tight coupling between components.

The cloud-controller-manager is a Kubernetes [control plane](#) component that embeds cloud-specific control logic. The [cloud controller manager](#) lets you link your cluster into your cloud provider's API, and separates out the components that interact with that cloud platform from components that only interact with your cluster.

By decoupling the interoperability logic between Kubernetes and the underlying cloud infrastructure, the cloud-controller-manager component enables cloud providers to release features at a different pace compared to the main Kubernetes project.

The cloud-controller-manager is structured using a plugin mechanism that allows different cloud providers to integrate their platforms with Kubernetes.

Design

Kubernetes components

The cloud controller manager runs in the control plane as a replicated set of processes (usually, these are containers in Pods). Each cloud-controller-manager implements multiple [controllers](#) in a single process.

Note:

You can also run the cloud controller manager as a Kubernetes [addon](#) rather than as part of the control plane.

Cloud controller manager functions

The controllers inside the cloud controller manager include:

Node controller

The node controller is responsible for updating [Node](#) objects when new servers are created in your cloud infrastructure. The node controller obtains information about the hosts running inside your tenancy with the cloud provider. The node controller performs the following functions:

1. Update a Node object with the corresponding server's unique identifier obtained from the cloud provider API.
2. Annotating and labelling the Node object with cloud-specific information, such as the region the node is deployed into and the resources (CPU, memory, etc) that it has available.
3. Obtain the node's hostname and network addresses.
4. Verifying the node's health. In case a node becomes unresponsive, this controller checks with your cloud provider's API to see if the server has been deactivated / deleted / terminated. If the node has been deleted from the cloud, the controller deletes the Node object from your Kubernetes cluster.

Some cloud provider implementations split this into a node controller and a separate node lifecycle controller.

Route controller

The route controller is responsible for configuring routes in the cloud appropriately so that containers on different nodes in your Kubernetes cluster can communicate with each other.

Depending on the cloud provider, the route controller might also allocate blocks of IP addresses for the Pod network.

Service controller

[Services](#) integrate with cloud infrastructure components such as managed load balancers, IP addresses, network packet filtering, and target health checking. The service controller interacts with your cloud provider's APIs to set up load balancers and other infrastructure components when you declare a Service resource that requires them.

Authorization

This section breaks down the access that the cloud controller manager requires on various API objects, in order to perform its operations.

Node controller

The Node controller only works with Node objects. It requires full access to read and modify Node objects.

v1/Node:

- get
- list
- create
- update
- patch
- watch
- delete

Route controller

The route controller listens to Node object creation and configures routes appropriately. It requires Get access to Node objects.

v1/Node:

- get

Service controller

The service controller watches for Service object **create**, **update** and **delete** events and then configures Endpoints for those Services appropriately (for EndpointSlices, the kube-controller-manager manages these on demand).

To access Services, it requires **list**, and **watch** access. To update Services, it requires **patch** and **update** access.

To set up Endpoints resources for the Services, it requires access to **create**, **list**, **get**, **watch**, and **update**.

v1/Service:

- list
- get
- watch
- patch
- update

Others

The implementation of the core of the cloud controller manager requires access to create Event objects, and to ensure secure operation, it requires access to create ServiceAccounts.

v1/Event:

- create
- patch
- update

v1/ServiceAccount:

- create

The [RBAC](#) ClusterRole for the cloud controller manager looks like:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: cloud-controller-manager
rules:
- apiGroups:
  - ""
  resources:
  - events
  verbs:
  - create
  - patch
  - update
- apiGroups:
  - ""
  resources:
  - nodes
  verbs:
  - '*'
- apiGroups:
  - ""
  resources:
  - nodes/status
  verbs:
  - patch
- apiGroups:
  - ""
  resources:
  - services
  verbs:
  - list
  - patch
  - update
  - watch
- apiGroups:
  - ""
```

```
resources:
- serviceaccounts
verbs:
- create
- apiGroups:
- ""
resources:
- persistentvolumes
verbs:
- get
- list
- update
- watch
- apiGroups:
- ""
resources:
- endpoints
verbs:
- create
- get
- list
- watch
- update
```

What's next

- [Cloud Controller Manager Administration](#) has instructions on running and managing the cloud controller manager.
- To upgrade a HA control plane to use the cloud controller manager, see [Migrate Replicated Control Plane To Use Cloud Controller Manager](#).
- Want to know how to implement your own cloud controller manager, or extend an existing project?
 - The cloud controller manager uses Go interfaces, specifically, CloudProvider interface defined in [cloud.go](#) from [kubernetes/cloud-provider](#) to allow implementations from any cloud to be plugged in.
 - The implementation of the shared controllers highlighted in this document (Node, Route, and Service), and some scaffolding along with the shared cloudprovider interface, is part of the Kubernetes core. Implementations specific to cloud providers are outside the core of Kubernetes and implement the CloudProvider interface.
 - For more information about developing plugins, see [Developing Cloud Controller Manager](#).

About cgroup v2

On Linux, [control groups](#) constrain resources that are allocated to processes.

The [kubelet](#) and the underlying container runtime need to interface with cgroups to enforce [resource management for pods and containers](#) which includes cpu/memory requests and limits for containerized workloads.

There are two versions of cgroups in Linux: cgroup v1 and cgroup v2. cgroup v2 is the new generation of the cgroup API.

What is cgroup v2?

FEATURE STATE: Kubernetes v1.25 [stable]

cgroup v2 is the next version of the Linux cgroup API. cgroup v2 provides a unified control system with enhanced resource management capabilities.

cgroup v2 offers several improvements over cgroup v1, such as the following:

- Single unified hierarchy design in API
- Safer sub-tree delegation to containers
- Newer features like [Pressure Stall Information](#)
- Enhanced resource allocation management and isolation across multiple resources
 - Unified accounting for different types of memory allocations (network memory, kernel memory, etc)
 - Accounting for non-immediate resource changes such as page cache write backs

Some Kubernetes features exclusively use cgroup v2 for enhanced resource management and isolation. For example, the [MemoryQoS](#) feature improves memory QoS and relies on cgroup v2 primitives.

Using cgroup v2

The recommended way to use cgroup v2 is to use a Linux distribution that enables and uses cgroup v2 by default.

To check if your distribution uses cgroup v2, refer to [Identify cgroup version on Linux nodes](#).

Requirements

cgroup v2 has the following requirements:

- OS distribution enables cgroup v2
- Linux Kernel version is 5.8 or later
- Container runtime supports cgroup v2. For example:
 - [containerd](#) v1.4 and later
 - [cri-o](#) v1.20 and later
- The kubelet and the container runtime are configured to use the [systemd cgroup driver](#)

Linux Distribution cgroup v2 support

For a list of Linux distributions that use cgroup v2, refer to the [cgroup v2 documentation](#)

- Container Optimized OS (since M97)
- Ubuntu (since 21.10, 22.04+ recommended)

- Debian GNU/Linux (since Debian 11 bullseye)
- Fedora (since 31)
- Arch Linux (since April 2021)
- RHEL and RHEL-like distributions (since 9)

To check if your distribution is using cgroup v2, refer to your distribution's documentation or follow the instructions in [Identify the cgroup version on Linux nodes](#).

You can also enable cgroup v2 manually on your Linux distribution by modifying the kernel cmdline boot arguments. If your distribution uses GRUB, `systemd.unified_cgroup_hierarchy=1` should be added in `GRUB_CMDLINE_LINUX` under `/etc/default/grub`, followed by `sudo update-grub`. However, the recommended approach is to use a distribution that already enables cgroup v2 by default.

Migrating to cgroup v2

To migrate to cgroup v2, ensure that you meet the [requirements](#), then upgrade to a kernel version that enables cgroup v2 by default.

The kubelet automatically detects that the OS is running on cgroup v2 and performs accordingly with no additional configuration required.

There should not be any noticeable difference in the user experience when switching to cgroup v2, unless users are accessing the cgroup file system directly, either on the node or from within the containers.

cgroup v2 uses a different API than cgroup v1, so if there are any applications that directly access the cgroup file system, they need to be updated to newer versions that support cgroup v2. For example:

- Some third-party monitoring and security agents may depend on the cgroup filesystem. Update these agents to versions that support cgroup v2.
- If you run [cAdvisor](#) as a stand-alone DaemonSet for monitoring pods and containers, update it to v0.43.0 or later.
- If you deploy Java applications, prefer to use versions which fully support cgroup v2:
 - [OpenJDK / HotSpot](#): jdk8u372, 11.0.16, 15 and later
 - [IBM Semeru Runtimes](#): 8.0.382.0, 11.0.20.0, 17.0.8.0, and later
 - [IBM Java](#): 8.0.8.6 and later
- If you are using the [uber-go/automaxprocs](#) package, make sure the version you use is v1.5.1 or higher.

Identify the cgroup version on Linux Nodes

The cgroup version depends on the Linux distribution being used and the default cgroup version configured on the OS. To check which cgroup version your distribution uses, run the `stat -fc %T /sys/fs/cgroup/` command on the node:

```
stat -fc %T /sys/fs/cgroup/
```

For cgroup v2, the output is `cgroup2fs`.

For cgroup v1, the output is `tmpfs`.

What's next

- Learn more about [cgroups](#)
- Learn more about [container runtime](#)
- Learn more about [cgroup drivers](#)

Container Runtime Interface (CRI)

The CRI is a plugin interface which enables the kubelet to use a wide variety of container runtimes, without having a need to recompile the cluster components.

You need a working [container runtime](#) on each Node in your cluster, so that the [kubelet](#) can launch [Pods](#) and their containers.

The Container Runtime Interface (CRI) is the main protocol for the communication between the [kubelet](#) and Container Runtime.

The Kubernetes Container Runtime Interface (CRI) defines the main [gRPC](#) protocol for the communication between the [node components kubelet](#) and [container runtime](#).

The API

FEATURE STATE: Kubernetes v1.23 [stable]

The kubelet acts as a client when connecting to the container runtime via gRPC. The runtime and image service endpoints have to be available in the container runtime, which can be configured separately within the kubelet by using the `--image-service-endpoint` [command line flags](#).

For Kubernetes v1.30, the kubelet prefers to use CRI v1. If a container runtime does not support v1 of the CRI, then the kubelet tries to negotiate any older supported version. The v1.30 kubelet can also negotiate CRI v1alpha2, but this version is considered as deprecated. If the kubelet cannot negotiate a supported CRI version, the kubelet gives up and doesn't register as a node.

Upgrading

When upgrading Kubernetes, the kubelet tries to automatically select the latest CRI version on restart of the component. If that fails, then the fallback will take place as mentioned above. If a gRPC re-dial was required because the container runtime has been upgraded, then the container runtime must also support the initially selected version or the redial is expected to fail. This requires a restart of the kubelet.

What's next

- Learn more about the CRI [protocol definition](#)

Garbage Collection

Garbage collection is a collective term for the various mechanisms Kubernetes uses to clean up cluster resources. This allows the clean up of resources like the following:

- [Terminated pods](#)
- [Completed Jobs](#)
- [Objects without owner references](#)
- [Unused containers and container images](#)
- [Dynamically provisioned PersistentVolumes with a StorageClass reclaim policy of Delete](#)
- [Stale or expired CertificateSigningRequests \(CSRs\)](#)
- [Nodes](#) deleted in the following scenarios:
 - On a cloud when the cluster uses a [cloud controller manager](#)
 - On-premises when the cluster uses an addon similar to a cloud controller manager
- [Node Lease objects](#)

Owners and dependents

Many objects in Kubernetes link to each other through [owner references](#). Owner references tell the control plane which objects are dependent on others. Kubernetes uses owner references to give the control plane, and other API clients, the opportunity to clean up related resources before deleting an object. In most cases, Kubernetes manages owner references automatically.

Ownership is different from the [labels and selectors](#) mechanism that some resources also use. For example, consider a [Service](#) that creates EndpointSlice objects. The Service uses *labels* to allow the control plane to determine which EndpointSlice objects are used for that Service. In addition to the labels, each EndpointSlice that is managed on behalf of a Service has an owner reference. Owner references help different parts of Kubernetes avoid interfering with objects they don't control.

Note:

Cross-namespace owner references are disallowed by design. Namespaced dependents can specify cluster-scoped or namespaced owners. A namespaced owner **must** exist in the same namespace as the dependent. If it does not, the owner reference is treated as absent, and the dependent is subject to deletion once all owners are verified absent.

Cluster-scoped dependents can only specify cluster-scoped owners. In v1.20+, if a cluster-scoped dependent specifies a namespaced kind as an owner, it is treated as having an unresolvable owner reference, and is not able to be garbage collected.

In v1.20+, if the garbage collector detects an invalid cross-namespace `ownerReference`, or a cluster-scoped dependent with an `ownerReference` referencing a namespaced kind, a warning Event with a reason of `OwnerRefInvalidNamespace` and an `involvedObject` of the invalid dependent is reported. You can check for that kind of Event by running `kubectl get events -A --field-selector=reason=OwnerRefInvalidNamespace`.

Cascading deletion

Kubernetes checks for and deletes objects that no longer have owner references, like the pods left behind when you delete a ReplicaSet. When you delete an object, you can control whether Kubernetes deletes the object's dependents automatically, in a process called *cascading deletion*. There are two types of cascading deletion, as follows:

- Foreground cascading deletion
- Background cascading deletion

You can also control how and when garbage collection deletes resources that have owner references using Kubernetes [finalizers](#).

Foreground cascading deletion

In foreground cascading deletion, the owner object you're deleting first enters a *deletion in progress* state. In this state, the following happens to the owner object:

- The Kubernetes API server sets the object's `metadata.deletionTimestamp` field to the time the object was marked for deletion.
- The Kubernetes API server also sets the `metadata.finalizers` field to `foregroundDeletion`.
- The object remains visible through the Kubernetes API until the deletion process is complete.

After the owner object enters the deletion in progress state, the controller deletes the dependents. After deleting all the dependent objects, the controller deletes the owner object. At this point, the object is no longer visible in the Kubernetes API.

During foreground cascading deletion, the only dependents that block owner deletion are those that have the `ownerReference.blockOwnerDeletion=true` field. See [Use foreground cascading deletion](#) to learn more.

Background cascading deletion

In background cascading deletion, the Kubernetes API server deletes the owner object immediately and the controller cleans up the dependent objects in the background. By default, Kubernetes uses background cascading deletion unless you manually use foreground deletion or choose to orphan the dependent objects.

See [Use background cascading deletion](#) to learn more.

Orphaned dependents

When Kubernetes deletes an owner object, the dependents left behind are called *orphan* objects. By default, Kubernetes deletes dependent objects. To learn how to override this behaviour, see [Delete owner objects and orphan dependents](#).

Garbage collection of unused containers and images

The [kubelet](#) performs garbage collection on unused images every two minutes and on unused containers every minute. You should avoid using external garbage collection tools, as these can break the kubelet behavior and remove containers that should exist.

To configure options for unused container and image garbage collection, tune the kubelet using a [configuration file](#) and change the parameters related to garbage collection using the [KubeletConfiguration](#) resource type.

Container image lifecycle

Kubernetes manages the lifecycle of all images through its *image manager*, which is part of the kubelet, with the cooperation of [cadvisor](#). The kubelet considers the following disk usage limits when making garbage collection decisions:

- HighThresholdPercent
- LowThresholdPercent

Disk usage above the configured HighThresholdPercent value triggers garbage collection, which deletes images in order based on the last time they were used, starting with the oldest first. The kubelet deletes images until disk usage reaches the LowThresholdPercent value.

Garbage collection for unused container images

FEATURE STATE: Kubernetes v1.30 [beta]

As a beta feature, you can specify the maximum time a local image can be unused for, regardless of disk usage. This is a kubelet setting that you configure for each node.

To configure the setting, enable the ImageMaximumGCAge [feature gate](#) for the kubelet, and also set a value for the ImageMaximumGCAge field in the kubelet configuration file.

The value is specified as a Kubernetes *duration*; for example, you can set the configuration field to 3d12h, which means 3 days and 12 hours.

Note:

This feature does not track image usage across kubelet restarts. If the kubelet is restarted, the tracked image age is reset, causing the kubelet to wait the full ImageMaximumGCAge duration before qualifying images for garbage collection based on image age.

Container garbage collection

The kubelet garbage collects unused containers based on the following variables, which you can define:

- MinAge: the minimum age at which the kubelet can garbage collect a container. Disable by setting to 0.
- MaxPerPodContainer: the maximum number of dead containers each Pod can have. Disable by setting to less than 0.
- MaxContainers: the maximum number of dead containers the cluster can have. Disable by setting to less than 0.

In addition to these variables, the kubelet garbage collects unidentified and deleted containers, typically starting with the oldest first.

MaxPerPodContainer and MaxContainers may potentially conflict with each other in situations where retaining the maximum number of containers per Pod (MaxPerPodContainer) would go

outside the allowable total of global dead containers (MaxContainers). In this situation, the kubelet adjusts MaxPerPodContainer to address the conflict. A worst-case scenario would be to downgrade MaxPerPodContainer to 1 and evict the oldest containers. Additionally, containers owned by pods that have been deleted are removed once they are older than MinAge.

Note:

The kubelet only garbage collects the containers it manages.

Configuring garbage collection

You can tune garbage collection of resources by configuring options specific to the controllers managing those resources. The following pages show you how to configure garbage collection:

- [Configuring cascading deletion of Kubernetes objects](#)
- [Configuring cleanup of finished Jobs](#)

What's next

- Learn more about [ownership of Kubernetes objects](#).
- Learn more about Kubernetes [finalizers](#).
- Learn about the [TTL controller](#) that cleans up finished Jobs.

Mixed Version Proxy

FEATURE STATE: Kubernetes v1.28 [alpha]

Kubernetes 1.30 includes an alpha feature that lets an [API Server](#) proxy a resource requests to other *peer* API servers. This is useful when there are multiple API servers running different versions of Kubernetes in one cluster (for example, during a long-lived rollout to a new release of Kubernetes).

This enables cluster administrators to configure highly available clusters that can be upgraded more safely, by directing resource requests (made during the upgrade) to the correct kube-apiserver. That proxying prevents users from seeing unexpected 404 Not Found errors that stem from the upgrade process.

This mechanism is called the *Mixed Version Proxy*.

Enabling the Mixed Version Proxy

Ensure that UnknownVersionInteroperabilityProxy [feature gate](#) is enabled when you start the [API Server](#):

```
kube-apiserver \
--feature-gates=UnknownVersionInteroperabilityProxy=true \
# required command line arguments for this feature
--peer-ca-file=<path to kube-apiserver CA cert>
--proxy-client-cert-file=<path to aggregator proxy cert>,
--proxy-client-key-file=<path to aggregator proxy key>,
```

```
--requestheader-client-ca-file=<path to aggregator CA cert>,  
# requestheader-allowed-names can be set to blank to allow any Common Name  
--requestheader-allowed-names=<valid Common Names to verify proxy client cert against>,  
  
# optional flags for this feature  
--peer-advertise-ip=`IP of this kube-apiserver that should be used by peers to proxy requests`  
--peer-advertise-port=`port of this kube-apiserver that should be used by peers to proxy  
requests`  
  
# ...and other flags as usual
```

Proxy transport and authentication between API servers

- The source kube-apiserver reuses the [existing APIserver client authentication flags](#) `--proxy-client-cert-file` and `--proxy-client-key-file` to present its identity that will be verified by its peer (the destination kube-apiserver). The destination API server verifies that peer connection based on the configuration you specify using the `--requestheader-client-ca-file` command line argument.
- To authenticate the destination server's serving certs, you must configure a certificate authority bundle by specifying the `--peer-ca-file` command line argument to the **source** API server.

Configuration for peer API server connectivity

To set the network location of a kube-apiserver that peers will use to proxy requests, use the `--peer-advertise-ip` and `--peer-advertise-port` command line arguments to kube-apiserver or specify these fields in the API server configuration file. If these flags are unspecified, peers will use the value from either `--advertise-address` or `--bind-address` command line argument to the kube-apiserver. If those too, are unset, the host's default interface is used.

Mixed version proxying

When you enable mixed version proxying, the [aggregation layer](#) loads a special filter that does the following:

- When a resource request reaches an API server that cannot serve that API (either because it is at a version pre-dating the introduction of the API or the API is turned off on the API server) the API server attempts to send the request to a peer API server that can serve the requested API. It does so by identifying API groups / versions / resources that the local server doesn't recognise, and tries to proxy those requests to a peer API server that is capable of handling the request.
- If the peer API server fails to respond, the *source* API server responds with 503 ("Service Unavailable") error.

How it works under the hood

When an API Server receives a resource request, it first checks which API servers can serve the requested resource. This check happens using the internal [StorageVersion API](#).

- If the resource is known to the API server that received the request (for example, GET /api/v1/pods/some-pod), the request is handled locally.

- If there is no internal `StorageVersion` object found for the requested resource (for example, GET `/my-api/v1/my-resource`) and the configured `APIService` specifies proxying to an extension API server, that proxying happens following the usual [flow](#) for extension APIs.
- If a valid internal `StorageVersion` object is found for the requested resource (for example, GET `/batch/v1/jobs`) and the API server trying to handle the request (the *handling API server*) has the batch API disabled, then the *handling API server* fetches the peer API servers that do serve the relevant API group / version / resource (`api/v1/batch` in this case) using the information in the fetched `StorageVersion` object. The *handling API server* then proxies the request to one of the matching peer kube-apiservers that are aware of the requested resource.
 - If there is no peer known for that API group / version / resource, the handling API server passes the request to its own handler chain which should eventually return a 404 ("Not Found") response.
 - If the handling API server has identified and selected a peer API server, but that peer fails to respond (for reasons such as network connectivity issues, or a data race between the request being received and a controller registering the peer's info into the control plane), then the handling API server responds with a 503 ("Service Unavailable") error.

Containers

Technology for packaging an application along with its runtime dependencies.

Each container that you run is repeatable; the standardization from having dependencies included means that you get the same behavior wherever you run it.

Containers decouple applications from the underlying host infrastructure. This makes deployment easier in different cloud or OS environments.

Each [node](#) in a Kubernetes cluster runs the containers that form the [Pods](#) assigned to that node. Containers in a Pod are co-located and co-scheduled to run on the same node.

Container images

A [container image](#) is a ready-to-run software package containing everything needed to run an application: the code and any runtime it requires, application and system libraries, and default values for any essential settings.

Containers are intended to be stateless and [immutable](#): you should not change the code of a container that is already running. If you have a containerized application and want to make changes, the correct process is to build a new image that includes the change, then recreate the container to start from the updated image.

Container runtimes

A fundamental component that empowers Kubernetes to run containers effectively. It is responsible for managing the execution and lifecycle of containers within the Kubernetes environment.

Kubernetes supports container runtimes such as [containerd](#), [CRI-O](#), and any other implementation of the [Kubernetes CRI \(Container Runtime Interface\)](#).

Usually, you can allow your cluster to pick the default container runtime for a Pod. If you need to use more than one container runtime in your cluster, you can specify the [RuntimeClass](#) for a Pod to make sure that Kubernetes runs those containers using a particular container runtime.

You can also use RuntimeClass to run different Pods with the same container runtime but with different settings.

[Container Environment](#)

[Container Lifecycle Hooks](#)

Images

A container image represents binary data that encapsulates an application and all its software dependencies. Container images are executable software bundles that can run standalone and that make very well defined assumptions about their runtime environment.

You typically create a container image of your application and push it to a registry before referring to it in a [Pod](#).

This page provides an outline of the container image concept.

Note:

If you are looking for the container images for a Kubernetes release (such as v1.30, the latest minor release), visit [Download Kubernetes](#).

Image names

Container images are usually given a name such as `pause`, `example/mycontainer`, or `kube-apiserver`. Images can also include a registry hostname; for example: `fictional.registry.example/imagename`, and possibly a port number as well; for example: `fictional.registry.example:10443/imagename`.

If you don't specify a registry hostname, Kubernetes assumes that you mean the [Docker public registry](#). You can change this behaviour by setting default image registry in [container runtime](#) configuration.

After the image name part you can add a *tag* or *digest* (in the same way you would when using with commands like `docker` or `podman`). Tags let you identify different versions of the same series of images. Digests are a unique identifier for a specific version of an image. Digests are

hashes of the image's content, and are immutable. Tags can be moved to point to different images, but digests are fixed.

Image tags consist of lowercase and uppercase letters, digits, underscores (`_`), periods (`.`), and dashes (`-`). It can be up to 128 characters long. And must follow the next regex pattern: `[a-zA-Z0-9_-]{0,127}` You can read more about and find validation regex in the [OCI Distribution Specification](#). If you don't specify a tag, Kubernetes assumes you mean the tag `latest`.

Image digests consists of a hash algorithm (such as `sha256`) and a hash value. For example: `sha256:1ff6c18fbef2045af6b9c16bf034cc421a29027b800e4f9b68ae9b1cb3e9ae07` You can find more information about digests format in the [OCI Image Specification](#).

Some image name examples that Kubernetes can use are:

- `busybox` - Image name only, no tag or digest. Kubernetes will use Docker public registry and latest tag. (Same as `docker.io/library/busybox:latest`)
- `busybox:1.32.0` - Image name with tag. Kubernetes will use Docker public registry. (Same as `docker.io/library/busybox:1.32.0`)
- `registry.k8s.io/pause:latest` - Image name with a custom registry and latest tag.
- `registry.k8s.io/pause:3.5` - Image name with a custom registry and non-latest tag.
- `registry.k8s.io/pause@sha256:1ff6c18fbef2045af6b9c16bf034cc421a29027b800e4f9b68ae9b1cb3e9ae07` - Image name with digest.
- `registry.k8s.io/pause:3.5@sha256:1ff6c18fbef2045af6b9c16bf034cc421a29027b800e4f9b68ae9b1cb3e9ae07` - Image name with tag and digest. Only digest will be used for pulling.

Updating images

When you first create a [Deployment](#), [StatefulSet](#), Pod, or other object that includes a Pod template, then by default the pull policy of all containers in that pod will be set to `IfNotPresent` if it is not explicitly specified. This policy causes the [kubelet](#) to skip pulling an image if it already exists.

Image pull policy

The `imagePullPolicy` for a container and the tag of the image affect when the [kubelet](#) attempts to pull (download) the specified image.

Here's a list of the values you can set for `imagePullPolicy` and the effects these values have:

`IfNotPresent`

the image is pulled only if it is not already present locally.

`Always`

every time the kubelet launches a container, the kubelet queries the container image registry to resolve the name to an image [digest](#). If the kubelet has a container image with that exact digest cached locally, the kubelet uses its cached image; otherwise, the kubelet pulls the image with the resolved digest, and uses that image to launch the container.

`Never`

the kubelet does not try fetching the image. If the image is somehow already present locally, the kubelet attempts to start the container; otherwise, startup fails. See [pre-pulled images](#) for more details.

The caching semantics of the underlying image provider make even `imagePullPolicy: Always` efficient, as long as the registry is reliably accessible. Your container runtime can notice that the image layers already exist on the node so that they don't need to be downloaded again.

Note:

You should avoid using the `:latest` tag when deploying containers in production as it is harder to track which version of the image is running and more difficult to roll back properly.

Instead, specify a meaningful tag such as `v1.42.0` and/or a digest.

To make sure the Pod always uses the same version of a container image, you can specify the image's digest; replace `<image-name>:<tag>` with `<image-name>@<digest>` (for example, `image@sha256:45b23dee08af5e43a7fea6c4cf9c25ccf269ee113168c19722f87876677c5cb2`).

When using image tags, if the image registry were to change the code that the tag on that image represents, you might end up with a mix of Pods running the old and new code. An image digest uniquely identifies a specific version of the image, so Kubernetes runs the same code every time it starts a container with that image name and digest specified. Specifying an image by digest fixes the code that you run so that a change at the registry cannot lead to that mix of versions.

There are third-party [admission controllers](#) that mutate Pods (and pod templates) when they are created, so that the running workload is defined based on an image digest rather than a tag. That might be useful if you want to make sure that all your workload is running the same code no matter what tag changes happen at the registry.

Default image pull policy

When you (or a controller) submit a new Pod to the API server, your cluster sets the `imagePullPolicy` field when specific conditions are met:

- if you omit the `imagePullPolicy` field, and you specify the digest for the container image, the `imagePullPolicy` is automatically set to `IfNotPresent`.
- if you omit the `imagePullPolicy` field, and the tag for the container image is `:latest`, `imagePullPolicy` is automatically set to `Always`;
- if you omit the `imagePullPolicy` field, and you don't specify the tag for the container image, `imagePullPolicy` is automatically set to `Always`;
- if you omit the `imagePullPolicy` field, and you specify the tag for the container image that isn't `:latest`, the `imagePullPolicy` is automatically set to `IfNotPresent`.

Note:

The value of `imagePullPolicy` of the container is always set when the object is first *created*, and is not updated if the image's tag or digest later changes.

For example, if you create a Deployment with an image whose tag is *not* `:latest`, and later update that Deployment's image to a `:latest` tag, the `imagePullPolicy` field will *not* change to `Always`. You must manually change the pull policy of any object after its initial creation.

Required image pull

If you would like to always force a pull, you can do one of the following:

- Set the `imagePullPolicy` of the container to `Always`.
- Omit the `imagePullPolicy` and use `:latest` as the tag for the image to use; Kubernetes will set the policy to `Always` when you submit the Pod.
- Omit the `imagePullPolicy` and the tag for the image to use; Kubernetes will set the policy to `Always` when you submit the Pod.
- Enable the [AlwaysPullImages](#) admission controller.

ImagePullBackOff

When a kubelet starts creating containers for a Pod using a container runtime, it might be possible the container is in [Waiting](#) state because of `ImagePullBackOff`.

The status `ImagePullBackOff` means that a container could not start because Kubernetes could not pull a container image (for reasons such as invalid image name, or pulling from a private registry without `imagePullSecret`). The `BackOff` part indicates that Kubernetes will keep trying to pull the image, with an increasing back-off delay.

Kubernetes raises the delay between each attempt until it reaches a compiled-in limit, which is 300 seconds (5 minutes).

Image pull per runtime class

FEATURE STATE: Kubernetes v1.29 [alpha]

Kubernetes includes alpha support for performing image pulls based on the `RuntimeClass` of a Pod.

If you enable the `RuntimeClassInImageCriApi` [feature gate](#), the kubelet references container images by a tuple of (image name, runtime handler) rather than just the image name or digest. Your [container runtime](#) may adapt its behavior based on the selected runtime handler. Pulling images based on runtime class will be helpful for VM based containers like windows hyperV containers.

Serial and parallel image pulls

By default, kubelet pulls images serially. In other words, kubelet sends only one image pull request to the image service at a time. Other image pull requests have to wait until the one being processed is complete.

Nodes make image pull decisions in isolation. Even when you use serialized image pulls, two different nodes can pull the same image in parallel.

If you would like to enable parallel image pulls, you can set the field `serializeImagePulls` to `false` in the [kubelet configuration](#). With `serializeImagePulls` set to `false`, image pull requests will be sent to the image service immediately, and multiple images will be pulled at the same time.

When enabling parallel image pulls, please make sure the image service of your container runtime can handle parallel image pulls.

The kubelet never pulls multiple images in parallel on behalf of one Pod. For example, if you have a Pod that has an init container and an application container, the image pulls for the two containers will not be parallelized. However, if you have two Pods that use different images, the kubelet pulls the images in parallel on behalf of the two different Pods, when parallel image pulls is enabled.

Maximum parallel image pulls

FEATURE STATE: Kubernetes v1.27 [alpha]

When `serializeImagePulls` is set to false, the kubelet defaults to no limit on the maximum number of images being pulled at the same time. If you would like to limit the number of parallel image pulls, you can set the field `maxParallelImagePulls` in kubelet configuration. With `maxParallelImagePulls` set to n , only n images can be pulled at the same time, and any image pull beyond n will have to wait until at least one ongoing image pull is complete.

Limiting the number parallel image pulls would prevent image pulling from consuming too much network bandwidth or disk I/O, when parallel image pulling is enabled.

You can set `maxParallelImagePulls` to a positive number that is greater than or equal to 1. If you set `maxParallelImagePulls` to be greater than or equal to 2, you must set the `serializeImagePulls` to false. The kubelet will fail to start with invalid `maxParallelImagePulls` settings.

Multi-architecture images with image indexes

As well as providing binary images, a container registry can also serve a [container image index](#). An image index can point to multiple [image manifests](#) for architecture-specific versions of a container. The idea is that you can have a name for an image (for example: `pause`, `example/mycontainer`, `kube-apiserver`) and allow different systems to fetch the right binary image for the machine architecture they are using.

Kubernetes itself typically names container images with a suffix `-${ARCH}`. For backward compatibility, please generate the older images with suffixes. The idea is to generate say `pause` image which has the manifest for all the arch(es) and say `pause-amd64` which is backwards compatible for older configurations or YAML files which may have hard coded the images with suffixes.

Using a private registry

Private registries may require keys to read images from them. Credentials can be provided in several ways:

- Configuring Nodes to Authenticate to a Private Registry
 - all pods can read any configured private registries
 - requires node configuration by cluster administrator
- Kubelet Credential Provider to dynamically fetch credentials for private registries
 - kubelet can be configured to use credential provider exec plugin for the respective private registry.
- Pre-pulled Images
 - all pods can use any images cached on a node
 - requires root access to all nodes to set up

- Specifying ImagePullSecrets on a Pod
 - only pods which provide own keys can access the private registry
- Vendor-specific or local extensions
 - if you're using a custom node configuration, you (or your cloud provider) can implement your mechanism for authenticating the node to the container registry.

These options are explained in more detail below.

Configuring nodes to authenticate to a private registry

Specific instructions for setting credentials depends on the container runtime and registry you chose to use. You should refer to your solution's documentation for the most accurate information.

For an example of configuring a private container image registry, see the [Pull an Image from a Private Registry](#) task. That example uses a private registry in Docker Hub.

Kubelet credential provider for authenticated image pulls

Note:

This approach is especially suitable when kubelet needs to fetch registry credentials dynamically. Most commonly used for registries provided by cloud providers where auth tokens are short-lived.

You can configure the kubelet to invoke a plugin binary to dynamically fetch registry credentials for a container image. This is the most robust and versatile way to fetch credentials for private registries, but also requires kubelet-level configuration to enable.

See [Configure a kubelet image credential provider](#) for more details.

Interpretation of config.json

The interpretation of config.json varies between the original Docker implementation and the Kubernetes interpretation. In Docker, the auths keys can only specify root URLs, whereas Kubernetes allows glob URLs as well as prefix-matched paths. The only limitation is that glob patterns (*) have to include the dot (.) for each subdomain. The amount of matched subdomains has to be equal to the amount of glob patterns (*), for example:

- *.kubernetes.io will *not* match kubernetes.io, but abc.kubernetes.io
- *.*.kubernetes.io will *not* match abc.kubernetes.io, but abc.def.kubernetes.io
- prefix.*.io will match prefix.kubernetes.io
- *-good.kubernetes.io will match prefix-good.kubernetes.io

This means that a config.json like this is valid:

```
{
  "auths": {
    "my-registry.io/images": { "auth": "..."},
    "*.my-registry.io/images": { "auth": "..."}
  }
}
```

Image pull operations would now pass the credentials to the CRI container runtime for every valid pattern. For example the following container image names would match successfully:

- my-registry.io/images
- my-registry.io/images/my-image
- my-registry.io/images/another-image
- sub.my-registry.io/images/my-image

But not:

- a.sub.my-registry.io/images/my-image
- a.b.sub.my-registry.io/images/my-image

The kubelet performs image pulls sequentially for every found credential. This means, that multiple entries in config.json for different paths are possible, too:

```
{
  "auths": {
    "my-registry.io/images": {
      "auth": "..."
    },
    "my-registry.io/images/subpath": {
      "auth": "..."
    }
  }
}
```

If now a container specifies an image my-registry.io/images/subpath/my-image to be pulled, then the kubelet will try to download them from both authentication sources if one of them fails.

Pre-pulled images

Note:

This approach is suitable if you can control node configuration. It will not work reliably if your cloud provider manages nodes and replaces them automatically.

By default, the kubelet tries to pull each image from the specified registry. However, if the `imagePullPolicy` property of the container is set to `IfNotPresent` or `Never`, then a local image is used (preferentially or exclusively, respectively).

If you want to rely on pre-pulled images as a substitute for registry authentication, you must ensure all nodes in the cluster have the same pre-pulled images.

This can be used to preload certain images for speed or as an alternative to authenticating to a private registry.

All pods will have read access to any pre-pulled images.

Specifying imagePullSecrets on a Pod

Note:

This is the recommended approach to run containers based on images in private registries.

Kubernetes supports specifying container image registry keys on a Pod. imagePullSecrets must all be in the same namespace as the Pod. The referenced Secrets must be of type `kubernetes.io/dockerconfig` or `kubernetes.io/dockerconfigjson`.

Creating a Secret with a Docker config

You need to know the username, registry password and client email address for authenticating to the registry, as well as its hostname. Run the following command, substituting the appropriate uppercase values:

```
kubectrl create secret docker-registry <name> \
--docker-server=DOCKER_REGISTRY_SERVER \
--docker-username=DOCKER_USER \
--docker-password=DOCKER_PASSWORD \
--docker-email=DOCKER_EMAIL
```

If you already have a Docker credentials file then, rather than using the above command, you can import the credentials file as a Kubernetes [Secrets](#).

[Create a Secret based on existing Docker credentials](#) explains how to set this up.

This is particularly useful if you are using multiple private container registries, as `kubectrl create secret docker-registry` creates a Secret that only works with a single private registry.

Note:

Pods can only reference image pull secrets in their own namespace, so this process needs to be done one time per namespace.

Referring to an imagePullSecrets on a Pod

Now, you can create pods which reference that secret by adding an imagePullSecrets section to a Pod definition. Each item in the imagePullSecrets array can only reference a Secret in the same namespace.

For example:

```
cat <<EOF > pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: foo
  namespace: awesomeapps
spec:
  containers:
  - name: foo
```



```
image: janedoe/awesomeapp:v1
imagePullSecrets:
  - name: myregistrykey
EOF
```

```
cat <<EOF >> ./kustomization.yaml
resources:
- pod.yaml
EOF
```

This needs to be done for each pod that is using a private registry.

However, setting of this field can be automated by setting the imagePullSecrets in a [ServiceAccount](#) resource.

Check [Add ImagePullSecrets to a Service Account](#) for detailed instructions.

You can use this in conjunction with a per-node .docker/config.json. The credentials will be merged.

Use cases

There are a number of solutions for configuring private registries. Here are some common use cases and suggested solutions.

1. Cluster running only non-proprietary (e.g. open-source) images. No need to hide images.
 - Use public images from a public registry
 - No configuration required.
 - Some cloud providers automatically cache or mirror public images, which improves availability and reduces the time to pull images.
2. Cluster running some proprietary images which should be hidden to those outside the company, but visible to all cluster users.
 - Use a hosted private registry
 - Manual configuration may be required on the nodes that need to access to private registry
 - Or, run an internal private registry behind your firewall with open read access.
 - No Kubernetes configuration is required.
 - Use a hosted container image registry service that controls image access
 - It will work better with cluster autoscaling than manual node configuration.
 - Or, on a cluster where changing the node configuration is inconvenient, use imagePullSecrets.
3. Cluster with proprietary images, a few of which require stricter access control.
 - Ensure [AlwaysPullImages admission controller](#) is active. Otherwise, all Pods potentially have access to all images.
 - Move sensitive data into a "Secret" resource, instead of packaging it in an image.
4. A multi-tenant cluster where each tenant needs own private registry.
 - Ensure [AlwaysPullImages admission controller](#) is active. Otherwise, all Pods of all tenants potentially have access to all images.
 - Run a private registry with authorization required.
 - Generate registry credential for each tenant, put into secret, and populate secret to each tenant namespace.
 - The tenant adds that secret to imagePullSecrets of each namespace.

If you need access to multiple registries, you can create one secret for each registry.

Legacy built-in kubelet credential provider

In older versions of Kubernetes, the kubelet had a direct integration with cloud provider credentials. This gave it the ability to dynamically fetch credentials for image registries.

There were three built-in implementations of the kubelet credential provider integration: ACR (Azure Container Registry), ECR (Elastic Container Registry), and GCR (Google Container Registry).

For more information on the legacy mechanism, read the documentation for the version of Kubernetes that you are using. Kubernetes v1.26 through to v1.30 do not include the legacy mechanism, so you would need to either:

- configure a kubelet image credential provider on each node
- specify image pull credentials using `imagePullSecrets` and at least one Secret

What's next

- Read the [OCI Image Manifest Specification](#).
- Learn about [container image garbage collection](#).
- Learn more about [pulling an Image from a Private Registry](#).

Container Environment

This page describes the resources available to Containers in the Container environment.

Container environment

The Kubernetes Container environment provides several important resources to Containers:

- A filesystem, which is a combination of an [image](#) and one or more [volumes](#).
- Information about the Container itself.
- Information about other objects in the cluster.

Container information

The *hostname* of a Container is the name of the Pod in which the Container is running. It is available through the `hostname` command or the [gethostname](#) function call in `libc`.

The Pod name and namespace are available as environment variables through the [downward API](#).

User defined environment variables from the Pod definition are also available to the Container, as are any environment variables specified statically in the container image.

Cluster information

A list of all services that were running when a Container was created is available to that Container as environment variables. This list is limited to services within the same namespace as the new Container's Pod and Kubernetes control plane services.

For a service named *foo* that maps to a Container named *bar*, the following variables are defined:

```
FOO_SERVICE_HOST=<the host the service is running on>
FOO_SERVICE_PORT=<the port the service is running on>
```

Services have dedicated IP addresses and are available to the Container via DNS, if [DNS add-on](#) is enabled.

What's next

- Learn more about [Container lifecycle hooks](#).
- Get hands-on experience [attaching handlers to Container lifecycle events](#).

Runtime Class

FEATURE STATE: Kubernetes v1.20 [stable]

This page describes the RuntimeClass resource and runtime selection mechanism.

RuntimeClass is a feature for selecting the container runtime configuration. The container runtime configuration is used to run a Pod's containers.

Motivation

You can set a different RuntimeClass between different Pods to provide a balance of performance versus security. For example, if part of your workload deserves a high level of information security assurance, you might choose to schedule those Pods so that they run in a container runtime that uses hardware virtualization. You'd then benefit from the extra isolation of the alternative runtime, at the expense of some additional overhead.

You can also use RuntimeClass to run different Pods with the same container runtime but with different settings.

Setup

1. Configure the CRI implementation on nodes (runtime dependent)
2. Create the corresponding RuntimeClass resources

1. Configure the CRI implementation on nodes

The configurations available through RuntimeClass are Container Runtime Interface (CRI) implementation dependent. See the corresponding documentation ([below](#)) for your CRI implementation for how to configure.

Note:

RuntimeClass assumes a homogeneous node configuration across the cluster by default (which means that all nodes are configured the same way with respect to container runtimes). To support heterogeneous node configurations, see [Scheduling](#) below.

The configurations have a corresponding handler name, referenced by the RuntimeClass. The handler must be a valid [DNS label name](#).

2. Create the corresponding RuntimeClass resources

The configurations setup in step 1 should each have an associated handler name, which identifies the configuration. For each handler, create a corresponding RuntimeClass object.

The RuntimeClass resource currently only has 2 significant fields: the RuntimeClass name (metadata.name) and the handler (handler). The object definition looks like this:

```
# RuntimeClass is defined in the node.k8s.io API group
apiVersion: node.k8s.io/v1
kind: RuntimeClass
metadata:
  # The name the RuntimeClass will be referenced by.
  # RuntimeClass is a non-namespaced resource.
  name: myclass
# The name of the corresponding CRI configuration
handler: myconfiguration
```

The name of a RuntimeClass object must be a valid [DNS subdomain name](#).

Note:

It is recommended that RuntimeClass write operations (create/update/patch/delete) be restricted to the cluster administrator. This is typically the default. See [Authorization Overview](#) for more details.

Usage

Once RuntimeClasses are configured for the cluster, you can specify a runtimeClassName in the Pod spec to use it. For example:

```
apiVersion: v1
kind: Pod
metadata:
  name: mypod
spec:
  runtimeClassName: myclass
# ...
```

This will instruct the kubelet to use the named RuntimeClass to run this pod. If the named RuntimeClass does not exist, or the CRI cannot run the corresponding handler, the pod will enter the Failed terminal [phase](#). Look for a corresponding [event](#) for an error message.

If no `runtimeClassName` is specified, the default `RuntimeHandler` will be used, which is equivalent to the behavior when the `RuntimeClass` feature is disabled.

CRI Configuration

For more details on setting up CRI runtimes, see [CRI installation](#).

[containerd](#)

Runtime handlers are configured through containerd's configuration at `/etc/containerd/config.toml`. Valid handlers are configured under the `runtimes` section:

```
[plugins."io.containerd.grpc.v1.cri".containerd.runtimes.${HANDLER_NAME}]
```

See containerd's [config documentation](#) for more details:

[CRI-O](#)

Runtime handlers are configured through CRI-O's configuration at `/etc/crio/crio.conf`. Valid handlers are configured under the [crio.runtime table](#):

```
[crio.runtime.runtimes.${HANDLER_NAME}]
runtime_path = "${PATH_TO_BINARY}"
```

See CRI-O's [config documentation](#) for more details.

Scheduling

FEATURE STATE: Kubernetes v1.16 [beta]

By specifying the `scheduling` field for a `RuntimeClass`, you can set constraints to ensure that Pods running with this `RuntimeClass` are scheduled to nodes that support it. If scheduling is not set, this `RuntimeClass` is assumed to be supported by all nodes.

To ensure pods land on nodes supporting a specific `RuntimeClass`, that set of nodes should have a common label which is then selected by the `runtimeclass.scheduling.nodeSelector` field. The `RuntimeClass`'s `nodeSelector` is merged with the pod's `nodeSelector` in admission, effectively taking the intersection of the set of nodes selected by each. If there is a conflict, the pod will be rejected.

If the supported nodes are tainted to prevent other `RuntimeClass` pods from running on the node, you can add tolerations to the `RuntimeClass`. As with the `nodeSelector`, the tolerations are merged with the pod's tolerations in admission, effectively taking the union of the set of nodes tolerated by each.

To learn more about configuring the node selector and tolerations, see [Assigning Pods to Nodes](#).

Pod Overhead

FEATURE STATE: Kubernetes v1.24 [stable]

You can specify *overhead* resources that are associated with running a Pod. Declaring overhead allows the cluster (including the scheduler) to account for it when making decisions about Pods and resources.

Pod overhead is defined in `RuntimeClass` through the `overhead` field. Through the use of this field, you can specify the overhead of running pods utilizing this `RuntimeClass` and ensure these overheads are accounted for in Kubernetes.

What's next

- [RuntimeClass Design](#)
- [RuntimeClass Scheduling Design](#)
- Read about the [Pod Overhead](#) concept
- [PodOverhead Feature Design](#)

Container Lifecycle Hooks

This page describes how kubelet managed Containers can use the Container lifecycle hook framework to run code triggered by events during their management lifecycle.

Overview

Analogous to many programming language frameworks that have component lifecycle hooks, such as Angular, Kubernetes provides Containers with lifecycle hooks. The hooks enable Containers to be aware of events in their management lifecycle and run code implemented in a handler when the corresponding lifecycle hook is executed.

Container hooks

There are two hooks that are exposed to Containers:

PostStart

This hook is executed immediately after a container is created. However, there is no guarantee that the hook will execute before the container `ENTRYPOINT`. No parameters are passed to the handler.

PreStop

This hook is called immediately before a container is terminated due to an API request or management event such as a liveness/startup probe failure, preemption, resource contention and others. A call to the `PreStop` hook fails if the container is already in a terminated or completed state and the hook must complete before the `TERM` signal to stop the container can be sent. The Pod's termination grace period countdown begins before the `PreStop` hook is executed, so regardless of the outcome of the handler, the container will eventually terminate within the Pod's termination grace period. No parameters are passed to the handler.

A more detailed description of the termination behavior can be found in [Termination of Pods](#).

Hook handler implementations

Containers can access a hook by implementing and registering a handler for that hook. There are three types of hook handlers that can be implemented for Containers:

- Exec - Executes a specific command, such as `pre-stop.sh`, inside the cgroups and namespaces of the Container. Resources consumed by the command are counted against the Container.
- HTTP - Executes an HTTP request against a specific endpoint on the Container.
- Sleep - Pauses the container for a specified duration. This is a beta-level feature default enabled by the `PodLifecycleSleepAction` [feature gate](#).

Hook handler execution

When a Container lifecycle management hook is called, the Kubernetes management system executes the handler according to the hook action, `httpGet`, `tcpSocket` and `sleep` are executed by the kubelet process, and `exec` is executed in the container.

Hook handler calls are synchronous within the context of the Pod containing the Container. This means that for a `PostStart` hook, the Container `ENTRYPOINT` and hook fire asynchronously. However, if the hook takes too long to run or hangs, the Container cannot reach a running state.

`PreStop` hooks are not executed asynchronously from the signal to stop the Container; the hook must complete its execution before the `TERM` signal can be sent. If a `PreStop` hook hangs during execution, the Pod's phase will be `Terminating` and remain there until the Pod is killed after its `terminationGracePeriodSeconds` expires. This grace period applies to the total time it takes for both the `PreStop` hook to execute and for the Container to stop normally. If, for example, `terminationGracePeriodSeconds` is 60, and the hook takes 55 seconds to complete, and the Container takes 10 seconds to stop normally after receiving the signal, then the Container will be killed before it can stop normally, since `terminationGracePeriodSeconds` is less than the total time (55+10) it takes for these two things to happen.

If either a `PostStart` or `PreStop` hook fails, it kills the Container.

Users should make their hook handlers as lightweight as possible. There are cases, however, when long running commands make sense, such as when saving state prior to stopping a Container.

Hook delivery guarantees

Hook delivery is intended to be *at least once*, which means that a hook may be called multiple times for any given event, such as for `PostStart` or `PreStop`. It is up to the hook implementation to handle this correctly.

Generally, only single deliveries are made. If, for example, an HTTP hook receiver is down and is unable to take traffic, there is no attempt to resend. In some rare cases, however, double delivery may occur. For instance, if a kubelet restarts in the middle of sending a hook, the hook might be resent after the kubelet comes back up.

Debugging Hook handlers

The logs for a Hook handler are not exposed in Pod events. If a handler fails for some reason, it broadcasts an event. For PostStart, this is the FailedPostStartHook event, and for PreStop, this is the FailedPreStopHook event. To generate a failed FailedPostStartHook event yourself, modify the [lifecycle-events.yaml](#) file to change the postStart command to "badcommand" and apply it. Here is some example output of the resulting events you see from running `kubectl describe pod lifecycle-demo`:

Events:

Type	Reason	Age	From	Message
Normal	Scheduled	7s	default-scheduler	Successfully assigned default/lifecycle-demo to ip-XXX-XXX-XX-XX.us-east-2...
Normal	Pulled	6s	kubelet	Successfully pulled image "nginx" in 229.604315ms
Normal	Pulling	4s (x2 over 6s)	kubelet	Pulling image "nginx"
Normal	Created	4s (x2 over 5s)	kubelet	Created container lifecycle-demo-container
Normal	Started	4s (x2 over 5s)	kubelet	Started container lifecycle-demo-container
Warning	FailedPostStartHook	4s (x2 over 5s)	kubelet	Exec lifecycle hook ([badcommand]) for Container "lifecycle-demo-container" in Pod "lifecycle-demo_default(30229739-9651-4e5a-9a32-a8f1688862db)" failed - error: command 'badcommand' exited with 126: , message: "OCI runtime exec failed: exec failed: container_linux.go:380: starting container process caused: exec: \"badcommand\": executable file not found in \$PATH: unknown\r\n"
Normal	Killing	4s (x2 over 5s)	kubelet	FailedPostStartHook
Normal	Pulled	4s	kubelet	Successfully pulled image "nginx" in 215.66395ms
Warning	BackOff	2s (x2 over 3s)	kubelet	Back-off restarting failed container

What's next

- Learn more about the [Container environment](#).
- Get hands-on experience [attaching handlers to Container lifecycle events](#).

Workloads

Understand Pods, the smallest deployable compute object in Kubernetes, and the higher-level abstractions that help you to run them.

A workload is an application running on Kubernetes. Whether your workload is a single component or several that work together, on Kubernetes you run it inside a set of [pods](#). In Kubernetes, a Pod represents a set of running [containers](#) on your cluster.

Kubernetes pods have a [defined lifecycle](#). For example, once a pod is running in your cluster then a critical fault on the [node](#) where that pod is running means that all the pods on that node fail. Kubernetes treats that level of failure as final: you would need to create a new Pod to recover, even if the node later becomes healthy.

However, to make life considerably easier, you don't need to manage each Pod directly. Instead, you can use *workload resources* that manage a set of pods on your behalf. These resources configure [controllers](#) that make sure the right number of the right kind of pod are running, to match the state you specified.

Kubernetes provides several built-in workload resources:

- [Deployment](#) and [ReplicaSet](#) (replacing the legacy resource [ReplicationController](#)). Deployment is a good fit for managing a stateless application workload on your cluster, where any Pod in the Deployment is interchangeable and can be replaced if needed.
- [StatefulSet](#) lets you run one or more related Pods that do track state somehow. For example, if your workload records data persistently, you can run a StatefulSet that matches each Pod with a [PersistentVolume](#). Your code, running in the Pods for that StatefulSet, can replicate data to other Pods in the same StatefulSet to improve overall resilience.
- [DaemonSet](#) defines Pods that provide facilities that are local to nodes. Every time you add a node to your cluster that matches the specification in a DaemonSet, the control plane schedules a Pod for that DaemonSet onto the new node. Each pod in a DaemonSet performs a job similar to a system daemon on a classic Unix / POSIX server. A DaemonSet might be fundamental to the operation of your cluster, such as a plugin to run [cluster networking](#), it might help you to manage the node, or it could provide optional behavior that enhances the container platform you are running.
- [Job](#) and [CronJob](#) provide different ways to define tasks that run to completion and then stop. You can use a [Job](#) to define a task that runs to completion, just once. You can use a [CronJob](#) to run the same Job multiple times according a schedule.

In the wider Kubernetes ecosystem, you can find third-party workload resources that provide additional behaviors. Using a [custom resource definition](#), you can add in a third-party workload resource if you want a specific behavior that's not part of Kubernetes' core. For example, if you wanted to run a group of Pods for your application but stop work unless *all* the Pods are available (perhaps for some high-throughput distributed task), then you can implement or install an extension that does provide that feature.

What's next

As well as reading about each API kind for workload management, you can read how to do specific tasks:

- [Run a stateless application using a Deployment](#)
- Run a stateful application either as a [single instance](#) or as a [replicated set](#)
- [Run automated tasks with a CronJob](#)

To learn about Kubernetes' mechanisms for separating code from configuration, visit [Configuration](#).

There are two supporting concepts that provide backgrounds about how Kubernetes manages pods for applications:

- [Garbage collection](#) tidies up objects from your cluster after their *owning resource* has been removed.
- The [time-to-live after finished controller](#) removes Jobs once a defined time has passed since they completed.

Once your application is running, you might want to make it available on the internet as a [Service](#) or, for web application only, using an [Ingress](#).

Pods

Pods are the smallest deployable units of computing that you can create and manage in Kubernetes.

A *Pod* (as in a pod of whales or pea pod) is a group of one or more [containers](#), with shared storage and network resources, and a specification for how to run the containers. A Pod's contents are always co-located and co-scheduled, and run in a shared context. A Pod models an application-specific "logical host": it contains one or more application containers which are relatively tightly coupled. In non-cloud contexts, applications executed on the same physical or virtual machine are analogous to cloud applications executed on the same logical host.

As well as application containers, a Pod can contain [init containers](#) that run during Pod startup. You can also inject [ephemeral containers](#) for debugging a running Pod.

What is a Pod?

Note:

You need to install a [container runtime](#) into each node in the cluster so that Pods can run there.

The shared context of a Pod is a set of Linux namespaces, cgroups, and potentially other facets of isolation - the same things that isolate a [container](#). Within a Pod's context, the individual applications may have further sub-isolations applied.

A Pod is similar to a set of containers with shared namespaces and shared filesystem volumes.

Pods in a Kubernetes cluster are used in two main ways:

- **Pods that run a single container.** The "one-container-per-Pod" model is the most common Kubernetes use case; in this case, you can think of a Pod as a wrapper around a single container; Kubernetes manages Pods rather than managing the containers directly.
- **Pods that run multiple containers that need to work together.** A Pod can encapsulate an application composed of [multiple co-located containers](#) that are tightly coupled and need to share resources. These co-located containers form a single cohesive unit.

Grouping multiple co-located and co-managed containers in a single Pod is a relatively advanced use case. You should use this pattern only in specific instances in which your containers are tightly coupled.

You don't need to run multiple containers to provide replication (for resilience or capacity); if you need multiple replicas, see [Workload management](#).

Using Pods

The following is an example of a Pod which consists of a container running the image `nginx:1.14.2`.

[pods/simple-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
  - name: nginx
    image: nginx:1.14.2
    ports:
    - containerPort: 80
```

To create the Pod shown above, run the following command:

```
kubectl apply -f https://k8s.io/examples/pods/simple-pod.yaml
```

Pods are generally not created directly and are created using workload resources. See [Working with Pods](#) for more information on how Pods are used with workload resources.

Workload resources for managing pods

Usually you don't need to create Pods directly, even singleton Pods. Instead, create them using workload resources such as [Deployment](#) or [Job](#). If your Pods need to track state, consider the [StatefulSet](#) resource.

Each Pod is meant to run a single instance of a given application. If you want to scale your application horizontally (to provide more overall resources by running more instances), you should use multiple Pods, one for each instance. In Kubernetes, this is typically referred to as *replication*. Replicated Pods are usually created and managed as a group by a workload resource and its [controller](#).

See [Pods and controllers](#) for more information on how Kubernetes uses workload resources, and their controllers, to implement application scaling and auto-healing.

Pods natively provide two kinds of shared resources for their constituent containers: [networking](#) and [storage](#).

Working with Pods

You'll rarely create individual Pods directly in Kubernetes—even singleton Pods. This is because Pods are designed as relatively ephemeral, disposable entities. When a Pod gets created (directly by you, or indirectly by a [controller](#)), the new Pod is scheduled to run on a [Node](#) in your cluster. The Pod remains on that node until the Pod finishes execution, the Pod object is deleted, the Pod is *evicted* for lack of resources, or the node fails.

Note:

Restarting a container in a Pod should not be confused with restarting a Pod. A Pod is not a process, but an environment for running container(s). A Pod persists until it is deleted.

The name of a Pod must be a valid [DNS subdomain](#) value, but this can produce unexpected results for the Pod hostname. For best compatibility, the name should follow the more restrictive rules for a [DNS label](#).

Pod OS

FEATURE STATE: Kubernetes v1.25 [stable]

You should set the `.spec.os.name` field to either `windows` or `linux` to indicate the OS on which you want the pod to run. These two are the only operating systems supported for now by Kubernetes. In the future, this list may be expanded.

In Kubernetes v1.30, the value of `.spec.os.name` does not affect how the [kube-scheduler](#) picks a Pod to run a node. In any cluster where there is more than one operating system for running nodes, you should set the [kubernetes.io/os](#) label correctly on each node, and define pods with a `nodeSelector` based on the operating system label, the kube-scheduler assigns your pod to a node based on other criteria and may or may not succeed in picking a suitable node placement where the node OS is right for the containers in that Pod. The [Pod security standards](#) also use this field to avoid enforcing policies that aren't relevant to the operating system.

Pods and controllers

You can use workload resources to create and manage multiple Pods for you. A controller for the resource handles replication and rollout and automatic healing in case of Pod failure. For example, if a Node fails, a controller notices that Pods on that Node have stopped working and creates a replacement Pod. The scheduler places the replacement Pod onto a healthy Node.

Here are some examples of workload resources that manage one or more Pods:

- [Deployment](#)
- [StatefulSet](#)
- [DaemonSet](#)

Pod templates

Controllers for [workload](#) resources create Pods from a *pod template* and manage those Pods on your behalf.

PodTemplates are specifications for creating Pods, and are included in workload resources such as [Deployments](#), [Jobs](#), and [DaemonSets](#).

Each controller for a workload resource uses the PodTemplate inside the workload object to make actual Pods. The PodTemplate is part of the desired state of whatever workload resource you used to run your app.

When you create a Pod, you can include [environment variables](#) in the Pod template for the containers that run in the Pod.

The sample below is a manifest for a simple Job with a template that starts one container. The container in that Pod prints a message then pauses.

```
apiVersion: batch/v1
kind: Job
metadata:
  name: hello
spec:
  template:
    # This is the pod template
    spec:
      containers:
        - name: hello
          image: busybox:1.28
          command: ['sh', '-c', 'echo "Hello, Kubernetes!" && sleep 3600']
          restartPolicy: OnFailure
    # The pod template ends here
```

Modifying the pod template or switching to a new pod template has no direct effect on the Pods that already exist. If you change the pod template for a workload resource, that resource needs to create replacement Pods that use the updated template.

For example, the StatefulSet controller ensures that the running Pods match the current pod template for each StatefulSet object. If you edit the StatefulSet to change its pod template, the StatefulSet starts to create new Pods based on the updated template. Eventually, all of the old Pods are replaced with new Pods, and the update is complete.

Each workload resource implements its own rules for handling changes to the Pod template. If you want to read more about StatefulSet specifically, read [Update strategy](#) in the StatefulSet Basics tutorial.

On Nodes, the [kubelet](#) does not directly observe or manage any of the details around pod templates and updates; those details are abstracted away. That abstraction and separation of concerns simplifies system semantics, and makes it feasible to extend the cluster's behavior without changing existing code.

Pod update and replacement

As mentioned in the previous section, when the Pod template for a workload resource is changed, the controller creates new Pods based on the updated template instead of updating or patching the existing Pods.

Kubernetes doesn't prevent you from managing Pods directly. It is possible to update some fields of a running Pod, in place. However, Pod update operations like [patch](#), and [replace](#) have some limitations:

- Most of the metadata about a Pod is immutable. For example, you cannot change the namespace, name, uid, or creationTimestamp fields; the generation field is unique. It only accepts updates that increment the field's current value.
- If the `metadata.deletionTimestamp` is set, no new entry can be added to the `metadata.finalizers` list.

Pod updates may not change fields other than `spec.containers[*].image`,

- `spec.initContainers[*].image`, `spec.activeDeadlineSeconds` or `spec.tolerations`. For `spec.tolerations`, you can only add new entries.

- When updating the `spec.activeDeadlineSeconds` field, two types of updates are allowed:

1. setting the unassigned field to a positive number;
2. updating the field from a positive number to a smaller, non-negative number.

Resource sharing and communication

Pods enable data sharing and communication among their constituent containers.

Storage in Pods

A Pod can specify a set of shared storage [volumes](#). All containers in the Pod can access the shared volumes, allowing those containers to share data. Volumes also allow persistent data in a Pod to survive in case one of the containers within needs to be restarted. See [Storage](#) for more information on how Kubernetes implements shared storage and makes it available to Pods.

Pod networking

Each Pod is assigned a unique IP address for each address family. Every container in a Pod shares the network namespace, including the IP address and network ports. Inside a Pod (and **only** then), the containers that belong to the Pod can communicate with one another using localhost. When containers in a Pod communicate with entities *outside the Pod*, they must coordinate how they use the shared network resources (such as ports). Within a Pod, containers share an IP address and port space, and can find each other via localhost. The containers in a Pod can also communicate with each other using standard inter-process communications like SystemV semaphores or POSIX shared memory. Containers in different Pods have distinct IP addresses and can not communicate by OS-level IPC without special configuration. Containers that want to interact with a container running in a different Pod can use IP networking to communicate.

Containers within the Pod see the system hostname as being the same as the configured name for the Pod. There's more about this in the [networking](#) section.

Pod security settings

To set security constraints on Pods and containers, you use the `securityContext` field in the Pod specification. This field gives you granular control over what a Pod or individual containers can do. For example:

- Drop specific Linux capabilities to avoid the impact of a CVE.
- Force all processes in the Pod to run as a non-root user or as a specific user or group ID.
- Set a specific seccomp profile.
- Set Windows security options, such as whether containers run as HostProcess.

Caution:

You can also use the Pod securityContext to enable [privileged mode](#) in Linux containers. Privileged mode overrides many of the other security settings in the securityContext. Avoid using this setting unless you can't grant the equivalent permissions by using other fields in the securityContext. In Kubernetes 1.26 and later, you can run Windows containers in a similarly privileged mode by setting the windowsOptions.hostProcess flag on the security context of the Pod spec. For details and instructions, see [Create a Windows HostProcess Pod](#).

- To learn about kernel-level security constraints that you can use, see [Linux kernel security constraints for Pods and containers](#).
- To learn more about the Pod security context, see [Configure a Security Context for a Pod or Container](#).

Static Pods

Static Pods are managed directly by the kubelet daemon on a specific node, without the [API server](#) observing them. Whereas most Pods are managed by the control plane (for example, a [Deployment](#)), for static Pods, the kubelet directly supervises each static Pod (and restarts it if it fails).

Static Pods are always bound to one [Kubelet](#) on a specific node. The main use for static Pods is to run a self-hosted control plane: in other words, using the kubelet to supervise the individual [control plane components](#).

The kubelet automatically tries to create a [mirror Pod](#) on the Kubernetes API server for each static Pod. This means that the Pods running on a node are visible on the API server, but cannot be controlled from there. See the guide [Create static Pods](#) for more information.

Note:

The spec of a static Pod cannot refer to other API objects (e.g., [ServiceAccount](#), [ConfigMap](#), [Secret](#), etc).

Pods with multiple containers

Pods are designed to support multiple cooperating processes (as containers) that form a cohesive unit of service. The containers in a Pod are automatically co-located and co-scheduled on the same physical or virtual machine in the cluster. The containers can share resources and dependencies, communicate with one another, and coordinate when and how they are terminated.

Pods in a Kubernetes cluster are used in two main ways:

- **Pods that run a single container.** The "one-container-per-Pod" model is the most common Kubernetes use case; in this case, you can think of a Pod as a wrapper around a single container; Kubernetes manages Pods rather than managing the containers directly.
- **Pods that run multiple containers that need to work together.** A Pod can encapsulate an application composed of multiple co-located containers that are tightly coupled and need to share resources. These co-located containers form a single cohesive unit of service—for example, one container serving data stored in a shared volume to the public, while a separate [sidecar container](#) refreshes or updates those files. The Pod wraps

these containers, storage resources, and an ephemeral network identity together as a single unit.

For example, you might have a container that acts as a web server for files in a shared volume, and a separate [sidecar container](#) that updates those files from a remote source, as in the following diagram:

Pod creation diagram

Some Pods have [init containers](#) as well as [app containers](#). By default, init containers run and complete before the app containers are started.

You can also have [sidecar containers](#) that provide auxiliary services to the main application Pod (for example: a service mesh).

FEATURE STATE: Kubernetes v1.29 [beta]

Enabled by default, the SidecarContainers [feature gate](#) allows you to specify `restartPolicy`: Always for init containers. Setting the Always restart policy ensures that the containers where you set it are treated as *sidecars* that are kept running during the entire lifetime of the Pod. Containers that you explicitly define as sidecar containers start up before the main application Pod and remain running until the Pod is shut down.

Container probes

A *probe* is a diagnostic performed periodically by the kubelet on a container. To perform a diagnostic, the kubelet can invoke different actions:

- `ExecAction` (performed with the help of the container runtime)
- `TCPSocketAction` (checked directly by the kubelet)
- `HTTPGetAction` (checked directly by the kubelet)

You can read more about [probes](#) in the Pod Lifecycle documentation.

What's next

- Learn about the [lifecycle of a Pod](#).
- Learn about [RuntimeClass](#) and how you can use it to configure different Pods with different container runtime configurations.
- Read about [PodDisruptionBudget](#) and how you can use it to manage application availability during disruptions.
- Pod is a top-level resource in the Kubernetes REST API. The [Pod](#) object definition describes the object in detail.
- [The Distributed System Toolkit: Patterns for Composite Containers](#) explains common layouts for Pods with more than one container.
- Read about [Pod topology spread constraints](#)

To understand the context for why Kubernetes wraps a common Pod API in other resources (such as [StatefulSets](#) or [Deployments](#)), you can read about the prior art, including:

- [Aurora](#)
- [Borg](#)
- [Marathon](#)

- [Omega](#)
- [Tupperware](#).

Pod Lifecycle

This page describes the lifecycle of a Pod. Pods follow a defined lifecycle, starting in the Pending [phase](#), moving through Running if at least one of its primary containers starts OK, and then through either the Succeeded or Failed phases depending on whether any container in the Pod terminated in failure.

Whilst a Pod is running, the kubelet is able to restart containers to handle some kind of faults. Within a Pod, Kubernetes tracks different container [states](#) and determines what action to take to make the Pod healthy again.

In the Kubernetes API, Pods have both a specification and an actual status. The status for a Pod object consists of a set of [Pod conditions](#). You can also inject [custom readiness information](#) into the condition data for a Pod, if that is useful to your application.

Pods are only [scheduled](#) once in their lifetime. Once a Pod is scheduled (assigned) to a Node, the Pod runs on that Node until it stops or is [terminated](#).

Pod lifetime

Like individual application containers, Pods are considered to be relatively ephemeral (rather than durable) entities. Pods are created, assigned a unique ID ([UID](#)), and scheduled to nodes where they remain until termination (according to restart policy) or deletion. If a [Node](#) dies, the Pods scheduled to that node are [scheduled for deletion](#) after a timeout period.

Pods do not, by themselves, self-heal. If a Pod is scheduled to a [node](#) that then fails, the Pod is deleted; likewise, a Pod won't survive an eviction due to a lack of resources or Node maintenance. Kubernetes uses a higher-level abstraction, called a [controller](#), that handles the work of managing the relatively disposable Pod instances.

A given Pod (as defined by a UID) is never "rescheduled" to a different node; instead, that Pod can be replaced by a new, near-identical Pod, with even the same name if desired, but with a different UID.

When something is said to have the same lifetime as a Pod, such as a [volume](#), that means that the thing exists as long as that specific Pod (with that exact UID) exists. If that Pod is deleted for any reason, and even if an identical replacement is created, the related thing (a volume, in this example) is also destroyed and created anew.

Pod diagram

A multi-container Pod that contains a file puller and a web server that uses a persistent volume for shared storage between the containers.

Pod phase

A Pod's status field is a [PodStatus](#) object, which has a phase field.

The phase of a Pod is a simple, high-level summary of where the Pod is in its lifecycle. The phase is not intended to be a comprehensive rollout of observations of container or Pod state, nor is it intended to be a comprehensive state machine.

The number and meanings of Pod phase values are tightly guarded. Other than what is documented here, nothing should be assumed about Pods that have a given phase value.

Here are the possible values for phase:

Value	Description
Pending	The Pod has been accepted by the Kubernetes cluster, but one or more of the containers has not been set up and made ready to run. This includes time a Pod spends waiting to be scheduled as well as the time spent downloading container images over the network.
Running	The Pod has been bound to a node, and all of the containers have been created. At least one container is still running, or is in the process of starting or restarting.
Succeeded	All containers in the Pod have terminated in success, and will not be restarted.
Failed	All containers in the Pod have terminated, and at least one container has terminated in failure. That is, the container either exited with non-zero status or was terminated by the system.
Unknown	For some reason the state of the Pod could not be obtained. This phase typically occurs due to an error in communicating with the node where the Pod should be running.

Note:

When a Pod is being deleted, it is shown as Terminating by some kubectl commands. This Terminating status is not one of the Pod phases. A Pod is granted a term to terminate gracefully, which defaults to 30 seconds. You can use the flag `--force` to [terminate a Pod by force](#).

Since Kubernetes 1.27, the kubelet transitions deleted Pods, except for [static Pods](#) and [force-deleted Pods](#) without a finalizer, to a terminal phase (Failed or Succeeded depending on the exit statuses of the pod containers) before their deletion from the API server.

If a node dies or is disconnected from the rest of the cluster, Kubernetes applies a policy for setting the phase of all Pods on the lost node to Failed.

Container states

As well as the [phase](#) of the Pod overall, Kubernetes tracks the state of each container inside a Pod. You can use [container lifecycle hooks](#) to trigger events to run at certain points in a container's lifecycle.

Once the [scheduler](#) assigns a Pod to a Node, the kubelet starts creating containers for that Pod using a [container runtime](#). There are three possible container states: Waiting, Running, and Terminated.

To check the state of a Pod's containers, you can use `kubectl describe pod <name-of-pod>`. The output shows the state for each container within that Pod.

Each state has a specific meaning:

Waiting

If a container is not in either the `Running` or `Terminated` state, it is `Waiting`. A container in the `Waiting` state is still running the operations it requires in order to complete start up: for example, pulling the container image from a container image registry, or applying [Secret](#) data. When you use `kubectl` to query a Pod with a container that is `Waiting`, you also see a `Reason` field to summarize why the container is in that state.

Running

The `Running` status indicates that a container is executing without issues. If there was a `postStart` hook configured, it has already executed and finished. When you use `kubectl` to query a Pod with a container that is `Running`, you also see information about when the container entered the `Running` state.

Terminated

A container in the `Terminated` state began execution and then either ran to completion or failed for some reason. When you use `kubectl` to query a Pod with a container that is `Terminated`, you see a reason, an exit code, and the start and finish time for that container's period of execution.

If a container has a `preStop` hook configured, this hook runs before the container enters the `Terminated` state.

How Pods handle problems with containers

Kubernetes manages container failures within Pods using a [restartPolicy](#) defined in the Pod spec. This policy determines how Kubernetes reacts to containers exiting due to errors or other reasons, which falls in the following sequence:

1. **Initial crash:** Kubernetes attempts an immediate restart based on the Pod `restartPolicy`.
2. **Repeated crashes:** After the initial crash Kubernetes applies an exponential backoff delay for subsequent restarts, described in [restartPolicy](#). This prevents rapid, repeated restart attempts from overloading the system.
3. **CrashLoopBackOff state:** This indicates that the backoff delay mechanism is currently in effect for a given container that is in a crash loop, failing and restarting repeatedly.
4. **Backoff reset:** If a container runs successfully for a certain duration (e.g., 10 minutes), Kubernetes resets the backoff delay, treating any new crash as the first one.

In practice, a `CrashLoopBackOff` is a condition or event that might be seen as output from the `kubectl` command, while describing or listing Pods, when a container in the Pod fails to start properly and then continually tries and fails in a loop.

In other words, when a container enters the crash loop, Kubernetes applies the exponential backoff delay mentioned in the [Container restart policy](#). This mechanism prevents a faulty container from overwhelming the system with continuous failed start attempts.

The `CrashLoopBackOff` can be caused by issues like the following:

- Application errors that cause the container to exit.
- Configuration errors, such as incorrect environment variables or missing configuration files.

- Resource constraints, where the container might not have enough memory or CPU to start properly.
- Health checks failing if the application doesn't start serving within the expected time.
- Container liveness probes or startup probes returning a Failure result as mentioned in the [probes section](#).

To investigate the root cause of a CrashLoopBackOff issue, a user can:

1. **Check logs:** Use `kubectl logs <name-of-pod>` to check the logs of the container. This is often the most direct way to diagnose the issue causing the crashes.
2. **Inspect events:** Use `kubectl describe pod <name-of-pod>` to see events for the Pod, which can provide hints about configuration or resource issues.
3. **Review configuration:** Ensure that the Pod configuration, including environment variables and mounted volumes, is correct and that all required external resources are available.
4. **Check resource limits:** Make sure that the container has enough CPU and memory allocated. Sometimes, increasing the resources in the Pod definition can resolve the issue.
5. **Debug application:** There might exist bugs or misconfigurations in the application code. Running this container image locally or in a development environment can help diagnose application specific issues.

Container restart policy

The spec of a Pod has a `restartPolicy` field with possible values `Always`, `OnFailure`, and `Never`. The default value is `Always`.

The `restartPolicy` for a Pod applies to [app containers](#) in the Pod and to regular [init containers](#). [Sidecar containers](#) ignore the Pod-level `restartPolicy` field: in Kubernetes, a sidecar is defined as an entry inside `initContainers` that has its container-level `restartPolicy` set to `Always`. For init containers that exit with an error, the kubelet restarts the init container if the Pod level `restartPolicy` is either `OnFailure` or `Always`:

- `Always`: Automatically restarts the container after any termination.
- `OnFailure`: Only restarts the container if it exits with an error (non-zero exit status).
- `Never`: Does not automatically restart the terminated container.

When the kubelet is handling container restarts according to the configured restart policy, that only applies to restarts that make replacement containers inside the same Pod and running on the same node. After containers in a Pod exit, the kubelet restarts them with an exponential backoff delay (10s, 20s, 40s, ...), that is capped at 300 seconds (5 minutes). Once a container has executed for 10 minutes without any problems, the kubelet resets the restart backoff timer for that container. [Sidecar containers and Pod lifecycle](#) explains the behaviour of init containers when specify `restartPolicy` field on it.

Pod conditions

A Pod has a `PodStatus`, which has an array of [PodConditions](#) through which the Pod has or has not passed. Kubelet manages the following `PodConditions`:

- `PodScheduled`: the Pod has been scheduled to a node.
- `PodReadyToStartContainers`: (beta feature; enabled by [default](#)) the Pod sandbox has been successfully created and networking configured.

- ContainersReady: all containers in the Pod are ready.
- Initialized: all [init containers](#) have completed successfully.
- Ready: the Pod is able to serve requests and should be added to the load balancing pools of all matching Services.

Field name	Description
type	Name of this Pod condition.
status	Indicates whether that condition is applicable, with possible values "True", "False", or "Unknown".
lastProbeTime	Timestamp of when the Pod condition was last probed.
lastTransitionTime	Timestamp for when the Pod last transitioned from one status to another.
reason	Machine-readable, UpperCamelCase text indicating the reason for the condition's last transition.
message	Human-readable message indicating details about the last status transition.

Pod readiness

FEATURE STATE: Kubernetes v1.14 [stable]

Your application can inject extra feedback or signals into PodStatus: *Pod readiness*. To use this, set readinessGates in the Pod's spec to specify a list of additional conditions that the kubelet evaluates for Pod readiness.

Readiness gates are determined by the current state of status.condition fields for the Pod. If Kubernetes cannot find such a condition in the status.conditions field of a Pod, the status of the condition is defaulted to "False".

Here is an example:

```
kind: Pod
...
spec:
  readinessGates:
    - conditionType: "www.example.com/feature-1"
status:
  conditions:
    - type: Ready                # a built in PodCondition
      status: "False"
      lastProbeTime: null
      lastTransitionTime: 2018-01-01T00:00:00Z
    - type: "www.example.com/feature-1"    # an extra PodCondition
      status: "False"
      lastProbeTime: null
      lastTransitionTime: 2018-01-01T00:00:00Z
  containerStatuses:
    - containerID: docker://abcd...
      ready: true
...
```

The Pod conditions you add must have names that meet the Kubernetes [label key format](#).

Status for Pod readiness

The `kubectl patch` command does not support patching object status. To set these status.conditions for the Pod, applications and [operators](#) should use the PATCH action. You can use a [Kubernetes client library](#) to write code that sets custom Pod conditions for Pod readiness.

For a Pod that uses custom conditions, that Pod is evaluated to be ready **only** when both the following statements apply:

- All containers in the Pod are ready.
- All conditions specified in `readinessGates` are True.

When a Pod's containers are Ready but at least one custom condition is missing or False, the kubelet sets the Pod's [condition](#) to `ContainersReady`.

Pod network readiness

FEATURE STATE: Kubernetes v1.29 [beta]

Note:

During its early development, this condition was named `PodHasNetwork`.

After a Pod gets scheduled on a node, it needs to be admitted by the kubelet and to have any required storage volumes mounted. Once these phases are complete, the kubelet works with a container runtime (using [Container runtime interface \(CRI\)](#)) to set up a runtime sandbox and configure networking for the Pod. If the `PodReadyToStartContainersCondition` [feature gate](#) is enabled (it is enabled by default for Kubernetes 1.30), the `PodReadyToStartContainers` condition will be added to the `status.conditions` field of a Pod.

The `PodReadyToStartContainers` condition is set to False by the Kubelet when it detects a Pod does not have a runtime sandbox with networking configured. This occurs in the following scenarios:

- Early in the lifecycle of the Pod, when the kubelet has not yet begun to set up a sandbox for the Pod using the container runtime.
- Later in the lifecycle of the Pod, when the Pod sandbox has been destroyed due to either:
 - the node rebooting, without the Pod getting evicted
 - for container runtimes that use virtual machines for isolation, the Pod sandbox virtual machine rebooting, which then requires creating a new sandbox and fresh container network configuration.

The `PodReadyToStartContainers` condition is set to True by the kubelet after the successful completion of sandbox creation and network configuration for the Pod by the runtime plugin. The kubelet can start pulling container images and create containers after `PodReadyToStartContainers` condition has been set to True.

For a Pod with init containers, the kubelet sets the `Initialized` condition to True after the init containers have successfully completed (which happens after successful sandbox creation and network configuration by the runtime plugin). For a Pod without init containers, the kubelet sets the `Initialized` condition to True before sandbox creation and network configuration starts.

Pod scheduling readiness

FEATURE STATE: Kubernetes v1.26 [alpha]

See [Pod Scheduling Readiness](#) for more information.

Container probes

A *probe* is a diagnostic performed periodically by the [kubelet](#) on a container. To perform a diagnostic, the kubelet either executes code within the container, or makes a network request.

Check mechanisms

There are four different ways to check a container using a probe. Each probe must define exactly one of these four mechanisms:

exec

Executes a specified command inside the container. The diagnostic is considered successful if the command exits with a status code of 0.

grpc

Performs a remote procedure call using [gRPC](#). The target should implement [gRPC health checks](#). The diagnostic is considered successful if the status of the response is SERVING.

httpGet

Performs an HTTP GET request against the Pod's IP address on a specified port and path. The diagnostic is considered successful if the response has a status code greater than or equal to 200 and less than 400.

tcpSocket

Performs a TCP check against the Pod's IP address on a specified port. The diagnostic is considered successful if the port is open. If the remote system (the container) closes the connection immediately after it opens, this counts as healthy.

Caution:

Unlike the other mechanisms, exec probe's implementation involves the creation/forking of multiple processes each time when executed. As a result, in case of the clusters having higher pod densities, lower intervals of `initialDelaySeconds`, `periodSeconds`, configuring any probe with exec mechanism might introduce an overhead on the cpu usage of the node. In such scenarios, consider using the alternative probe mechanisms to avoid the overhead.

Probe outcome

Each probe has one of three results:

Success

The container passed the diagnostic.

Failure

The container failed the diagnostic.

Unknown

The diagnostic failed (no action should be taken, and the kubelet will make further checks).

Types of probe

The kubelet can optionally perform and react to three kinds of probes on running containers:

livenessProbe

Indicates whether the container is running. If the liveness probe fails, the kubelet kills the container, and the container is subjected to its [restart policy](#). If a container does not provide a liveness probe, the default state is Success.

readinessProbe

Indicates whether the container is ready to respond to requests. If the readiness probe fails, the endpoints controller removes the Pod's IP address from the endpoints of all Services that match the Pod. The default state of readiness before the initial delay is Failure. If a container does not provide a readiness probe, the default state is Success.

startupProbe

Indicates whether the application within the container is started. All other probes are disabled if a startup probe is provided, until it succeeds. If the startup probe fails, the kubelet kills the container, and the container is subjected to its [restart policy](#). If a container does not provide a startup probe, the default state is Success.

For more information about how to set up a liveness, readiness, or startup probe, see [Configure Liveness, Readiness and Startup Probes](#).

When should you use a liveness probe?

If the process in your container is able to crash on its own whenever it encounters an issue or becomes unhealthy, you do not necessarily need a liveness probe; the kubelet will automatically perform the correct action in accordance with the Pod's restartPolicy.

If you'd like your container to be killed and restarted if a probe fails, then specify a liveness probe, and specify a restartPolicy of Always or OnFailure.

When should you use a readiness probe?

If you'd like to start sending traffic to a Pod only when a probe succeeds, specify a readiness probe. In this case, the readiness probe might be the same as the liveness probe, but the existence of the readiness probe in the spec means that the Pod will start without receiving any traffic and only start receiving traffic after the probe starts succeeding.

If you want your container to be able to take itself down for maintenance, you can specify a readiness probe that checks an endpoint specific to readiness that is different from the liveness probe.

If your app has a strict dependency on back-end services, you can implement both a liveness and a readiness probe. The liveness probe passes when the app itself is healthy, but the readiness probe additionally checks that each required back-end service is available. This helps you avoid directing traffic to Pods that can only respond with error messages.

If your container needs to work on loading large data, configuration files, or migrations during startup, you can use a [startup probe](#). However, if you want to detect the difference between an app that has failed and an app that is still processing its startup data, you might prefer a readiness probe.

Note:

If you want to be able to drain requests when the Pod is deleted, you do not necessarily need a readiness probe; on deletion, the Pod automatically puts itself into an unready state regardless of whether the readiness probe exists. The Pod remains in the unready state while it waits for the containers in the Pod to stop.

When should you use a startup probe?

Startup probes are useful for Pods that have containers that take a long time to come into service. Rather than set a long liveness interval, you can configure a separate configuration for probing the container as it starts up, allowing a time longer than the liveness interval would allow.

If your container usually starts in more than $\text{initialDelaySeconds} + \text{failureThreshold} \times \text{periodSeconds}$, you should specify a startup probe that checks the same endpoint as the liveness probe. The default for `periodSeconds` is 10s. You should then set its `failureThreshold` high enough to allow the container to start, without changing the default values of the liveness probe. This helps to protect against deadlocks.

Termination of Pods

Because Pods represent processes running on nodes in the cluster, it is important to allow those processes to gracefully terminate when they are no longer needed (rather than being abruptly stopped with a KILL signal and having no chance to clean up).

The design aim is for you to be able to request deletion and know when processes terminate, but also be able to ensure that deletes eventually complete. When you request deletion of a Pod, the cluster records and tracks the intended grace period before the Pod is allowed to be forcefully killed. With that forceful shutdown tracking in place, the [kubelet](#) attempts graceful shutdown.

Typically, with this graceful termination of the pod, kubelet makes requests to the container runtime to attempt to stop the containers in the pod by first sending a TERM (aka. SIGTERM) signal, with a grace period timeout, to the main process in each container. The requests to stop the containers are processed by the container runtime asynchronously. There is no guarantee to the order of processing for these requests. Many container runtimes respect the STOPSIGNAL value defined in the container image and, if different, send the container image configured STOPSIGNAL instead of TERM. Once the grace period has expired, the KILL signal is sent to any remaining processes, and the Pod is then deleted from the [API Server](#). If the kubelet or the container runtime's management service is restarted while waiting for processes to terminate, the cluster retries from the start including the full original grace period.

An example flow:

1. You use the `kubectl` tool to manually delete a specific Pod, with the default grace period (30 seconds).
2. The Pod in the API server is updated with the time beyond which the Pod is considered "dead" along with the grace period. If you use `kubectl describe` to check the Pod you're deleting, that Pod shows up as "Terminating". On the node where the Pod is running: as

soon as the kubelet sees that a Pod has been marked as terminating (a graceful shutdown duration has been set), the kubelet begins the local Pod shutdown process.

1. If one of the Pod's containers has defined a preStop [hook](#) and the terminationGracePeriodSeconds in the Pod spec is not set to 0, the kubelet runs that hook inside of the container. The default terminationGracePeriodSeconds setting is 30 seconds.

If the preStop hook is still running after the grace period expires, the kubelet requests a small, one-off grace period extension of 2 seconds.

Note:

If the `preStop` hook needs longer to complete than the default grace period allows, you must modify `terminationGracePeriodSeconds` to suit this.

2. The kubelet triggers the container runtime to send a TERM signal to process 1 inside each container.

Note:

The containers in the Pod receive the TERM signal at different times and in an arbitrary order. If the order of shutdowns matters, consider using a `preStop` hook to synchronize.

3. At the same time as the kubelet is starting graceful shutdown of the Pod, the control plane evaluates whether to remove that shutting-down Pod from EndpointSlice (and Endpoints) objects, where those objects represent a [Service](#) with a configured [selector](#). [ReplicaSets](#) and other workload resources no longer treat the shutting-down Pod as a valid, in-service replica.

Pods that shut down slowly should not continue to serve regular traffic and should start terminating and finish processing open connections. Some applications need to go beyond finishing open connections and need more graceful termination, for example, session draining and completion.

Any endpoints that represent the terminating Pods are not immediately removed from EndpointSlices, and a status indicating [terminating state](#) is exposed from the EndpointSlice API (and the legacy Endpoints API). Terminating endpoints always have their ready status as false (for backward compatibility with versions before 1.26), so load balancers will not use it for regular traffic.

If traffic draining on terminating Pod is needed, the actual readiness can be checked as a condition serving. You can find more details on how to implement connections draining in the tutorial [Pods And Endpoints Termination Flow](#)

Note:

If you don't have the EndpointSliceTerminatingCondition feature gate enabled in your cluster (the gate is on by default from Kubernetes 1.22, and locked to default in 1.26), then the Kubernetes control plane removes a Pod from any relevant EndpointSlices as soon as the Pod's

termination grace period *begins*. The behavior above is described when the feature gate `EndpointSliceTerminatingCondition` is enabled.

Note:

Beginning with Kubernetes 1.29, if your Pod includes one or more sidecar containers (init containers with an Always restart policy), the kubelet will delay sending the TERM signal to these sidecar containers until the last main container has fully terminated. The sidecar containers will be terminated in the reverse order they are defined in the Pod spec. This ensures that sidecar containers continue serving the other containers in the Pod until they are no longer needed.

Note that slow termination of a main container will also delay the termination of the sidecar containers. If the grace period expires before the termination process is complete, the Pod may enter emergency termination. In this case, all remaining containers in the Pod will be terminated simultaneously with a short grace period.

Similarly, if the Pod has a preStop hook that exceeds the termination grace period, emergency termination may occur. In general, if you have used preStop hooks to control the termination order without sidecar containers, you can now remove them and allow the kubelet to manage sidecar termination automatically.

1. When the grace period expires, the kubelet triggers forcible shutdown. The container runtime sends SIGKILL to any processes still running in any container in the Pod. The kubelet also cleans up a hidden pause container if that container runtime uses one.
2. The kubelet transitions the Pod into a terminal phase (Failed or Succeeded depending on the end state of its containers). This step is guaranteed since version 1.27.
3. The kubelet triggers forcible removal of Pod object from the API server, by setting grace period to 0 (immediate deletion).
4. The API server deletes the Pod's API object, which is then no longer visible from any client.

Forced Pod termination

Caution:

Forced deletions can be potentially disruptive for some workloads and their Pods.

By default, all deletes are graceful within 30 seconds. The `kubectl delete` command supports the `--grace-period=<seconds>` option which allows you to override the default and specify your own value.

Setting the grace period to 0 forcibly and immediately deletes the Pod from the API server. If the Pod was still running on a node, that forcible deletion triggers the kubelet to begin immediate cleanup.

Note:

You must specify an additional flag `--force` along with `--grace-period=0` in order to perform force deletions.

When a force deletion is performed, the API server does not wait for confirmation from the kubelet that the Pod has been terminated on the node it was running on. It removes the Pod in the API immediately so a new Pod can be created with the same name. On the node, Pods that are set to terminate immediately will still be given a small grace period before being force killed.

Caution:

Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.

If you need to force-delete Pods that are part of a StatefulSet, refer to the task documentation for [deleting Pods from a StatefulSet](#).

Garbage collection of Pods

For failed Pods, the API objects remain in the cluster's API until a human or [controller](#) process explicitly removes them.

The Pod garbage collector (PodGC), which is a controller in the control plane, cleans up terminated Pods (with a phase of Succeeded or Failed), when the number of Pods exceeds the configured threshold (determined by `terminated-pod-gc-threshold` in the kube-controller-manager). This avoids a resource leak as Pods are created and terminated over time.

Additionally, PodGC cleans up any Pods which satisfy any of the following conditions:

1. are orphan Pods - bound to a node which no longer exists,
2. are unscheduled terminating Pods,
3. are terminating Pods, bound to a non-ready node tainted with [node.kubernetes.io/out-of-service](https://kubernetes.io/out-of-service), when the `NodeOutOfServiceVolumeDetach` feature gate is enabled.

When the `PodDisruptionConditions` feature gate is enabled, along with cleaning up the Pods, PodGC will also mark them as failed if they are in a non-terminal phase. Also, PodGC adds a Pod disruption condition when cleaning up an orphan Pod. See [Pod disruption conditions](#) for more details.

What's next

- Get hands-on experience [attaching handlers to container lifecycle events](#).
- Get hands-on experience [configuring Liveness, Readiness and Startup Probes](#).
- Learn more about [container lifecycle hooks](#).
- Learn more about [sidecar containers](#).
- For detailed information about Pod and container status in the API, see the API reference documentation covering [status](#) for Pod.

Init Containers

This page provides an overview of init containers: specialized containers that run before app containers in a [Pod](#). Init containers can contain utilities or setup scripts not present in an app image.

You can specify init containers in the Pod specification alongside the `containers` array (which describes app containers).

In Kubernetes, a [sidecar container](#) is a container that starts before the main application container and *continues to run*. This document is about init containers: containers that run to completion during Pod initialization.

Understanding init containers

A [Pod](#) can have multiple containers running apps within it, but it can also have one or more init containers, which are run before the app containers are started.

Init containers are exactly like regular containers, except:

- Init containers always run to completion.
- Each init container must complete successfully before the next one starts.

If a Pod's init container fails, the kubelet repeatedly restarts that init container until it succeeds. However, if the Pod has a `restartPolicy` of `Never`, and an init container fails during startup of that Pod, Kubernetes treats the overall Pod as failed.

To specify an init container for a Pod, add the `initContainers` field into the [Pod specification](#), as an array of container items (similar to the `containers` field and its contents). See [Container](#) in the API reference for more details.

The status of the init containers is returned in `.status.initContainerStatuses` field as an array of the container statuses (similar to the `.status.containerStatuses` field).

Differences from regular containers

Init containers support all the fields and features of app containers, including resource limits, [volumes](#), and security settings. However, the resource requests and limits for an init container are handled differently, as documented in [Resource sharing within containers](#).

Regular init containers (in other words: excluding sidecar containers) do not support the `lifecycle`, `livenessProbe`, `readinessProbe`, or `startupProbe` fields. Init containers must run to completion before the Pod can be ready; sidecar containers continue running during a Pod's lifetime, and *do* support some probes. See [sidecar container](#) for further details about sidecar containers.

If you specify multiple init containers for a Pod, kubelet runs each init container sequentially. Each init container must succeed before the next can run. When all of the init containers have run to completion, kubelet initializes the application containers for the Pod and runs them as usual.

Differences from sidecar containers

Init containers run and complete their tasks before the main application container starts. Unlike [sidecar containers](#), init containers are not continuously running alongside the main containers.

Init containers run to completion sequentially, and the main container does not start until all the init containers have successfully completed.

init containers do not support `lifecycle`, `livenessProbe`, `readinessProbe`, or `startupProbe` whereas sidecar containers support all these [probes](#) to control their lifecycle.

Init containers share the same resources (CPU, memory, network) with the main application containers but do not interact directly with them. They can, however, use shared volumes for data exchange.

Using init containers

Because init containers have separate images from app containers, they have some advantages for start-up related code:

- Init containers can contain utilities or custom code for setup that are not present in an app image. For example, there is no need to make an image FROM another image just to use a tool like `sed`, `awk`, `python`, or `dig` during setup.
- The application image builder and deployer roles can work independently without the need to jointly build a single app image.
- Init containers can run with a different view of the filesystem than app containers in the same Pod. Consequently, they can be given access to [Secrets](#) that app containers cannot access.
- Because init containers run to completion before any app containers start, init containers offer a mechanism to block or delay app container startup until a set of preconditions are met. Once preconditions are met, all of the app containers in a Pod can start in parallel.
- Init containers can securely run utilities or custom code that would otherwise make an app container image less secure. By keeping unnecessary tools separate you can limit the attack surface of your app container image.

Examples

Here are some ideas for how to use init containers:

- Wait for a [Service](#) to be created, using a shell one-line command like:

```
for i in {1..100}; do sleep 1; if nslookup myservice; then exit 0; fi; done; exit 1
```

- Register this Pod with a remote server from the downward API with a command like:

```
curl -X POST http://$MANAGEMENT_SERVICE_HOST:$MANAGEMENT_SERVICE_PORT/register -d 'instance=$(<POD_NAME>)&ip=$(<POD_IP>)'
```

- Wait for some time before starting the app container with a command like

```
sleep 60
```

- Clone a Git repository into a [Volume](#)

- Place values into a configuration file and run a template tool to dynamically generate a
- configuration file for the main app container. For example, place the `POD_IP` value in a configuration and generate the main app configuration file using Jinja.

Init containers in use

This example defines a simple Pod that has two init containers. The first waits for `myservice`, and the second waits for `mydb`. Once both init containers complete, the Pod runs the app container from its `spec` section.

```
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
  labels:
    app.kubernetes.io/name: MyApp
spec:
  containers:
    - name: myapp-container
      image: busybox:1.28
      command: ['sh', '-c', 'echo The app is running! && sleep 3600']
  initContainers:
    - name: init-myservice
      image: busybox:1.28
      command: ['sh', '-c', "until nslookup myservice.$(cat /var/run/secrets/kubernetes.io/serviceaccount/namespace).svc.cluster.local; do echo waiting for myservice; sleep 2; done"]
    - name: init-mydb
      image: busybox:1.28
      command: ['sh', '-c', "until nslookup mydb.$(cat /var/run/secrets/kubernetes.io/serviceaccount/namespace).svc.cluster.local; do echo waiting for mydb; sleep 2; done"]
```

You can start this Pod by running:

```
kubectl apply -f myapp.yaml
```

The output is similar to this:

```
pod/myapp-pod created
```

And check on its status with:

```
kubectl get -f myapp.yaml
```

The output is similar to this:

NAME	READY	STATUS	RESTARTS	AGE
myapp-pod	0/1	Init:0/2	0	6m

or for more details:

```
kubectl describe -f myapp.yaml
```

The output is similar to this:

```

Name:      myapp-pod
Namespace: default
[...]
Labels:    app.kubernetes.io/name=MyApp
Status:    Pending
[...]
Init Containers:
  init-myservice:
  [...]
    State:      Running
  [...]
  init-mysql:
  [...]
    State:      Waiting
    Reason:     PodInitializing
    Ready:      False
  [...]
Containers:
  myapp-container:
  [...]
    State:      Waiting
    Reason:     PodInitializing
    Ready:      False
  [...]
Events:
  FirstSeen    LastSeen    Count   From              SubObjectPath              Type
Reason        Message
-----
16s          16s         1       {default-scheduler }              Normal
Scheduled     Successfully assigned myapp-pod to 172.17.4.201
16s          16s         1       {kubelet 172.17.4.201} spec.initContainers{init-myservice}
Normal        Pulling      pulling image "busybox"
13s          13s         1       {kubelet 172.17.4.201} spec.initContainers{init-myservice}
Normal        Pulled       Successfully pulled image "busybox"
13s          13s         1       {kubelet 172.17.4.201} spec.initContainers{init-myservice}
Normal        Created      Created container init-myservice
13s          13s         1       {kubelet 172.17.4.201} spec.initContainers{init-myservice}
Normal        Started      Started container init-myservice

```

To see logs for the init containers in this Pod, run:

```

kubectl logs myapp-pod -c init-myservice # Inspect the first init container
kubectl logs myapp-pod -c init-mysql     # Inspect the second init container

```

At this point, those init containers will be waiting to discover [Services](#) named mysql and myservice.

Here's a configuration you can use to make those Services appear:

```

---
apiVersion: v1
kind: Service

```



```
metadata:
  name: myservice
spec:
  ports:
  - protocol: TCP
    port: 80
    targetPort: 9376
---
apiVersion: v1
kind: Service
metadata:
  name: mydb
spec:
  ports:
  - protocol: TCP
    port: 80
    targetPort: 9377
```

To create the mydb and myservice services:

```
kubectl apply -f services.yaml
```

The output is similar to this:

```
service/myservice created
service/mydb created
```

You'll then see that those init containers complete, and that the myapp-pod Pod moves into the Running state:

```
kubectl get -f myapp.yaml
```

The output is similar to this:

NAME	READY	STATUS	RESTARTS	AGE
myapp-pod	1/1	Running	0	9m

This simple example should provide some inspiration for you to create your own init containers. [What's next](#) contains a link to a more detailed example.

Detailed behavior

During Pod startup, the kubelet delays running init containers until the networking and storage are ready. Then the kubelet runs the Pod's init containers in the order they appear in the Pod's spec.

Each init container must exit successfully before the next container starts. If a container fails to start due to the runtime or exits with failure, it is retried according to the Pod restartPolicy. However, if the Pod restartPolicy is set to Always, the init containers use restartPolicy OnFailure.

A Pod cannot be Ready until all init containers have succeeded. The ports on an init container are not aggregated under a Service. A Pod that is initializing is in the Pending state but should have a condition Initialized set to false.

If the Pod [restarts](#), or is restarted, all init containers must execute again.

Changes to the init container spec are limited to the container image field. Altering an init container image field is equivalent to restarting the Pod.

Because init containers can be restarted, retried, or re-executed, init container code should be idempotent. In particular, code that writes to files on EmptyDirs should be prepared for the possibility that an output file already exists.

Init containers have all of the fields of an app container. However, Kubernetes prohibits readinessProbe from being used because init containers cannot define readiness distinct from completion. This is enforced during validation.

Use activeDeadlineSeconds on the Pod to prevent init containers from failing forever. The active deadline includes init containers. However it is recommended to use activeDeadlineSeconds only if teams deploy their application as a Job, because activeDeadlineSeconds has an effect even after initContainer finished. The Pod which is already running correctly would be killed by activeDeadlineSeconds if you set.

The name of each app and init container in a Pod must be unique; a validation error is thrown for any container sharing a name with another.

Resource sharing within containers

Given the order of execution for init, sidecar and app containers, the following rules for resource usage apply:

- The highest of any particular resource request or limit defined on all init containers is the *effective init request/limit*. If any resource has no resource limit specified this is considered as the highest limit.
- The Pod's *effective request/limit* for a resource is the higher of:
 - the sum of all app containers request/limit for a resource
 - the effective init request/limit for a resource
- Scheduling is done based on effective requests/limits, which means init containers can reserve resources for initialization that are not used during the life of the Pod.
- The QoS (quality of service) tier of the Pod's *effective QoS tier* is the QoS tier for init containers and app containers alike.

Quota and limits are applied based on the effective Pod request and limit.

Init containers and Linux cgroups

On Linux, resource allocations for Pod level control groups (cgroups) are based on the effective Pod request and limit, the same as the scheduler.

Pod restart reasons

A Pod can restart, causing re-execution of init containers, for the following reasons:

- The Pod infrastructure container is restarted. This is uncommon and would have to be done by someone with root access to nodes.
- All containers in a Pod are terminated while restartPolicy is set to Always, forcing a restart, and the init container completion record has been lost due to [garbage collection](#).

The Pod will not be restarted when the init container image is changed, or the init container completion record has been lost due to garbage collection. This applies for Kubernetes v1.20 and later. If you are using an earlier version of Kubernetes, consult the documentation for the version you are using.

What's next

Learn more about the following:

- [Creating a Pod that has an init container](#).
- [Debug init containers](#).
- Overview of [kubelet](#) and [kubectl](#).
- [Types of probes](#): liveness, readiness, startup probe.
- [Sidecar containers](#).

Sidecar Containers

FEATURE STATE: Kubernetes v1.29 [beta]

Sidecar containers are the secondary containers that run along with the main application container within the same [Pod](#). These containers are used to enhance or to extend the functionality of the primary *app container* by providing additional services, or functionality such as logging, monitoring, security, or data synchronization, without directly altering the primary application code.

Typically, you only have one app container in a Pod. For example, if you have a web application that requires a local webserver, the local webserver is a sidecar and the web application itself is the app container.

Sidecar containers in Kubernetes

Kubernetes implements sidecar containers as a special case of [init containers](#); sidecar containers remain running after Pod startup. This document uses the term *regular init containers* to clearly refer to containers that only run during Pod startup.

Provided that your cluster has the SidecarContainers [feature gate](#) enabled (the feature is active by default since Kubernetes v1.29), you can specify a restartPolicy for containers listed in a Pod's initContainers field. These restartable *sidecar* containers are independent from other init containers and from the main application container(s) within the same pod. These can be started, stopped, or restarted without effecting the main application container and other init containers.

You can also run a Pod with multiple containers that are not marked as init or sidecar containers. This is appropriate if the containers within the Pod are required for the Pod to work overall, but you don't need to control which containers start or stop first. You could also do this if you need to support older versions of Kubernetes that don't support a container-level `restartPolicy` field.

Example application

Here's an example of a Deployment with two containers, one of which is a sidecar:

[application/deployment-sidecar.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: myapp
  labels:
    app: myapp
spec:
  replicas: 1
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
    spec:
      containers:
        - name: myapp
          image: alpine:latest
          command: ['sh', '-c', 'while true; do echo "logging" >> /opt/logs.txt; sleep 1; done']
          volumeMounts:
            - name: data
              mountPath: /opt
      initContainers:
        - name: logshipper
          image: alpine:latest
          restartPolicy: Always
          command: ['sh', '-c', 'tail -F /opt/logs.txt']
          volumeMounts:
            - name: data
              mountPath: /opt
      volumes:
        - name: data
          emptyDir: {}
```

Sidecar containers and Pod lifecycle

If an init container is created with its `restartPolicy` set to `Always`, it will start and remain running during the entire life of the Pod. This can be helpful for running supporting services separated from the main application containers.

If a `readinessProbe` is specified for this init container, its result will be used to determine the ready state of the Pod.

Since these containers are defined as init containers, they benefit from the same ordering and sequential guarantees as regular init containers, allowing you to mix sidecar containers with regular init containers for complex Pod initialization flows.

Compared to regular init containers, sidecars defined within `initContainers` continue to run after they have started. This is important when there is more than one entry inside `.spec.initContainers` for a Pod. After a sidecar-style init container is running (the kubelet has set the started status for that init container to true), the kubelet then starts the next init container from the ordered `.spec.initContainers` list. That status either becomes true because there is a process running in the container and no startup probe defined, or as a result of its `startupProbe` succeeding.

Jobs with sidecar containers

If you define a Job that uses sidecar using Kubernetes-style init containers, the sidecar container in each Pod does not prevent the Job from completing after the main container has finished.

Here's an example of a Job with two containers, one of which is a sidecar:

[application/job/job-sidecar.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
  name: myjob
spec:
  template:
    spec:
      containers:
        - name: myjob
          image: alpine:latest
          command: ['sh', '-c', 'echo "logging" > /opt/logs.txt']
          volumeMounts:
            - name: data
              mountPath: /opt
      initContainers:
        - name: logshipper
          image: alpine:latest
          restartPolicy: Always
          command: ['sh', '-c', 'tail -F /opt/logs.txt']
          volumeMounts:
            - name: data
              mountPath: /opt
      restartPolicy: Never
      volumes:
        - name: data
          emptyDir: {}
```

Differences from application containers

Sidecar containers run alongside *app containers* in the same pod. However, they do not execute the primary application logic; instead, they provide supporting functionality to the main application.

Sidecar containers have their own independent lifecycles. They can be started, stopped, and restarted independently of app containers. This means you can update, scale, or maintain sidecar containers without affecting the primary application.

Sidecar containers share the same network and storage namespaces with the primary container. This co-location allows them to interact closely and share resources.

Differences from init containers

Sidecar containers work alongside the main container, extending its functionality and providing additional services.

Sidecar containers run concurrently with the main application container. They are active throughout the lifecycle of the pod and can be started and stopped independently of the main container. Unlike [init containers](#), sidecar containers support [probes](#) to control their lifecycle.

Sidecar containers can interact directly with the main application containers, because like init containers they always share the same network, and can optionally also share volumes (filesystems).

Init containers stop before the main containers start up, so init containers cannot exchange messages with the app container in a Pod. Any data passing is one-way (for example, an init container can put information inside an emptyDir volume).

Resource sharing within containers

Given the order of execution for init, sidecar and app containers, the following rules for resource usage apply:

- The highest of any particular resource request or limit defined on all init containers is the *effective init request/limit*. If any resource has no resource limit specified this is considered as the highest limit.
- The Pod's *effective request/limit* for a resource is the sum of [pod overhead](#) and the higher of:
 - the sum of all non-init containers(app and sidecar containers) request/limit for a resource
 - the effective init request/limit for a resource
- Scheduling is done based on effective requests/limits, which means init containers can reserve resources for initialization that are not used during the life of the Pod.
- The QoS (quality of service) tier of the Pod's *effective QoS tier* is the QoS tier for all init, sidecar and app containers alike.

Quota and limits are applied based on the effective Pod request and limit.

Sidecar containers and Linux cgroups

On Linux, resource allocations for Pod level control groups (cgroups) are based on the effective Pod request and limit, the same as the scheduler.

What's next

- Read a blog post on [native sidecar containers](#).
- Read about [creating a Pod that has an init container](#).
- Learn about the [types of probes](#): liveness, readiness, startup probe.
- Learn about [pod overhead](#).

Ephemeral Containers

FEATURE STATE: Kubernetes v1.25 [stable]

This page provides an overview of ephemeral containers: a special type of container that runs temporarily in an existing [Pod](#) to accomplish user-initiated actions such as troubleshooting. You use ephemeral containers to inspect services rather than to build applications.

Understanding ephemeral containers

[Pods](#) are the fundamental building block of Kubernetes applications. Since Pods are intended to be disposable and replaceable, you cannot add a container to a Pod once it has been created. Instead, you usually delete and replace Pods in a controlled fashion using [deployments](#).

Sometimes it's necessary to inspect the state of an existing Pod, however, for example to troubleshoot a hard-to-reproduce bug. In these cases you can run an ephemeral container in an existing Pod to inspect its state and run arbitrary commands.

What is an ephemeral container?

Ephemeral containers differ from other containers in that they lack guarantees for resources or execution, and they will never be automatically restarted, so they are not appropriate for building applications. Ephemeral containers are described using the same ContainerSpec as regular containers, but many fields are incompatible and disallowed for ephemeral containers.

- Ephemeral containers may not have ports, so fields such as `ports`, `livenessProbe`, `readinessProbe` are disallowed.
- Pod resource allocations are immutable, so setting `resources` is disallowed.
- For a complete list of allowed fields, see the [EphemeralContainer reference documentation](#).

Ephemeral containers are created using a special `ephemeralcontainers` handler in the API rather than by adding them directly to `pod.spec`, so it's not possible to add an ephemeral container using `kubectl edit`.

Like regular containers, you may not change or remove an ephemeral container after you have added it to a Pod.

Note:

Ephemeral containers are not supported by [static pods](#).

Uses for ephemeral containers

Ephemeral containers are useful for interactive troubleshooting when `kubectl exec` is insufficient because a container has crashed or a container image doesn't include debugging utilities.

In particular, [distroless images](#) enable you to deploy minimal container images that reduce attack surface and exposure to bugs and vulnerabilities. Since distroless images do not include a shell or any debugging utilities, it's difficult to troubleshoot distroless images using `kubectl exec` alone.

When using ephemeral containers, it's helpful to enable [process namespace sharing](#) so you can view processes in other containers.

What's next

- Learn how to [debug pods using ephemeral containers](#).

Disruptions

This guide is for application owners who want to build highly available applications, and thus need to understand what types of disruptions can happen to Pods.

It is also for cluster administrators who want to perform automated cluster actions, like upgrading and autoscaling clusters.

Voluntary and involuntary disruptions

Pods do not disappear until someone (a person or a controller) destroys them, or there is an unavoidable hardware or system software error.

We call these unavoidable cases *involuntary disruptions* to an application. Examples are:

- a hardware failure of the physical machine backing the node
- cluster administrator deletes VM (instance) by mistake
- cloud provider or hypervisor failure makes VM disappear
- a kernel panic
- the node disappears from the cluster due to cluster network partition
- eviction of a pod due to the node being [out-of-resources](#).

Except for the out-of-resources condition, all these conditions should be familiar to most users; they are not specific to Kubernetes.

We call other cases *voluntary disruptions*. These include both actions initiated by the application owner and those initiated by a Cluster Administrator. Typical application owner actions include:

- deleting the deployment or other controller that manages the pod
- updating a deployment's pod template causing a restart
- directly deleting a pod (e.g. by accident)

Cluster administrator actions include:

- [Draining a node](#) for repair or upgrade.
- Draining a node from a cluster to scale the cluster down (learn about [Cluster Autoscaling](#)).
- Removing a pod from a node to permit something else to fit on that node.

These actions might be taken directly by the cluster administrator, or by automation run by the cluster administrator, or by your cluster hosting provider.

Ask your cluster administrator or consult your cloud provider or distribution documentation to determine if any sources of voluntary disruptions are enabled for your cluster. If none are enabled, you can skip creating Pod Disruption Budgets.

Caution:

Not all voluntary disruptions are constrained by Pod Disruption Budgets. For example, deleting deployments or pods bypasses Pod Disruption Budgets.

Dealing with disruptions

Here are some ways to mitigate involuntary disruptions:

- Ensure your pod [requests the resources](#) it needs.
- Replicate your application if you need higher availability. (Learn about running replicated [stateless](#) and [stateful](#) applications.)
- For even higher availability when running replicated applications, spread applications across racks (using [anti-affinity](#)) or across zones (if using a [multi-zone cluster](#).)

The frequency of voluntary disruptions varies. On a basic Kubernetes cluster, there are no automated voluntary disruptions (only user-triggered ones). However, your cluster administrator or hosting provider may run some additional services which cause voluntary disruptions. For example, rolling out node software updates can cause voluntary disruptions. Also, some implementations of cluster (node) autoscaling may cause voluntary disruptions to defragment and compact nodes. Your cluster administrator or hosting provider should have documented what level of voluntary disruptions, if any, to expect. Certain configuration options, such as [using PriorityClasses](#) in your pod spec can also cause voluntary (and involuntary) disruptions.

Pod disruption budgets

FEATURE STATE: Kubernetes v1.21 [stable]

Kubernetes offers features to help you run highly available applications even when you introduce frequent voluntary disruptions.

As an application owner, you can create a PodDisruptionBudget (PDB) for each application. A PDB limits the number of Pods of a replicated application that are down simultaneously from voluntary disruptions. For example, a quorum-based application would like to ensure that the number of replicas running is never brought below the number needed for a quorum. A web front end might want to ensure that the number of replicas serving load never falls below a certain percentage of the total.

Cluster managers and hosting providers should use tools which respect PodDisruptionBudgets by calling the [Eviction API](#) instead of directly deleting pods or deployments.

For example, the `kubectl drain` subcommand lets you mark a node as going out of service. When you run `kubectl drain`, the tool tries to evict all of the Pods on the Node you're taking out of service. The eviction request that `kubectl` submits on your behalf may be temporarily rejected, so the tool periodically retries all failed requests until all Pods on the target node are terminated, or until a configurable timeout is reached.

A PDB specifies the number of replicas that an application can tolerate having, relative to how many it is intended to have. For example, a Deployment which has a `.spec.replicas: 5` is supposed to have 5 pods at any given time. If its PDB allows for there to be 4 at a time, then the Eviction API will allow voluntary disruption of one (but not two) pods at a time.

The group of pods that comprise the application is specified using a label selector, the same as the one used by the application's controller (deployment, stateful-set, etc).

The "intended" number of pods is computed from the `.spec.replicas` of the workload resource that is managing those pods. The control plane discovers the owning workload resource by examining the `.metadata.ownerReferences` of the Pod.

[Involuntary disruptions](#) cannot be prevented by PDBs; however they do count against the budget.

Pods which are deleted or unavailable due to a rolling upgrade to an application do count against the disruption budget, but workload resources (such as Deployment and StatefulSet) are not limited by PDBs when doing rolling upgrades. Instead, the handling of failures during application updates is configured in the spec for the specific workload resource.

It is recommended to set `AlwaysAllow` [Unhealthy Pod Eviction Policy](#) to your PodDisruptionBudgets to support eviction of misbehaving applications during a node drain. The default behavior is to wait for the application pods to become [healthy](#) before the drain can proceed.

When a pod is evicted using the eviction API, it is gracefully [terminated](#), honoring the `terminationGracePeriodSeconds` setting in its [PodSpec](#).

PodDisruptionBudget example

Consider a cluster with 3 nodes, node-1 through node-3. The cluster is running several applications. One of them has 3 replicas initially called pod-a, pod-b, and pod-c. Another, unrelated pod without a PDB, called pod-x, is also shown. Initially, the pods are laid out as follows:

node-1	node-2	node-3
pod-a <i>available</i>	pod-b <i>available</i>	pod-c <i>available</i>
pod-x <i>available</i>		

All 3 pods are part of a deployment, and they collectively have a PDB which requires there be at least 2 of the 3 pods to be available at all times.

For example, assume the cluster administrator wants to reboot into a new kernel version to fix a bug in the kernel. The cluster administrator first tries to drain node-1 using the `kubectl drain` command. That tool tries to evict pod-a and pod-x. This succeeds immediately. Both pods go into the terminating state at the same time. This puts the cluster in this state:

node-1 <i>draining</i>	node-2	node-3
pod-a <i>terminating</i>	pod-b <i>available</i>	pod-c <i>available</i>
pod-x <i>terminating</i>		

The deployment notices that one of the pods is terminating, so it creates a replacement called pod-d. Since node-1 is cordoned, it lands on another node. Something has also created pod-y as a replacement for pod-x.

(Note: for a StatefulSet, pod-a, which would be called something like pod-0, would need to terminate completely before its replacement, which is also called pod-0 but has a different UID, could be created. Otherwise, the example applies to a StatefulSet as well.)

Now the cluster is in this state:

node-1 <i>draining</i>	node-2	node-3
pod-a <i>terminating</i>	pod-b <i>available</i>	pod-c <i>available</i>
pod-x <i>terminating</i>	pod-d <i>starting</i>	pod-y

At some point, the pods terminate, and the cluster looks like this:

node-1 <i>drained</i>	node-2	node-3
	pod-b <i>available</i>	pod-c <i>available</i>
	pod-d <i>starting</i>	pod-y

At this point, if an impatient cluster administrator tries to drain node-2 or node-3, the drain command will block, because there are only 2 available pods for the deployment, and its PDB requires at least 2. After some time passes, pod-d becomes available.

The cluster state now looks like this:

node-1 <i>drained</i>	node-2	node-3
	pod-b <i>available</i>	pod-c <i>available</i>
	pod-d <i>available</i>	pod-y

Now, the cluster administrator tries to drain node-2. The drain command will try to evict the two pods in some order, say pod-b first and then pod-d. It will succeed at evicting pod-b. But, when it tries to evict pod-d, it will be refused because that would leave only one pod available for the deployment.

The deployment creates a replacement for pod-b called pod-e. Because there are not enough resources in the cluster to schedule pod-e the drain will again block. The cluster may end up in this state:

node-1 <i>drained</i>	node-2	node-3	no node
	pod-b <i>terminating</i>	pod-c <i>available</i>	pod-e <i>pending</i>
	pod-d <i>available</i>	pod-y	

At this point, the cluster administrator needs to add a node back to the cluster to proceed with the upgrade.

You can see how Kubernetes varies the rate at which disruptions can happen, according to:

- how many replicas an application needs
- how long it takes to gracefully shutdown an instance
- how long it takes a new instance to start up
- the type of controller
- the cluster's resource capacity

Pod disruption conditions

FEATURE STATE: Kubernetes v1.26 [beta]

Note:

In order to use this behavior, you must have the PodDisruptionConditions [feature gate](#) enabled in your cluster.

When enabled, a dedicated Pod DisruptionTarget [condition](#) is added to indicate that the Pod is about to be deleted due to a [disruption](#). The reason field of the condition additionally indicates one of the following reasons for the Pod termination:

PreemptionByScheduler

Pod is due to be [preempted](#) by a scheduler in order to accommodate a new Pod with a higher priority. For more information, see [Pod priority preemption](#).

DeletionByTaintManager

Pod is due to be deleted by Taint Manager (which is part of the node lifecycle controller within kube-controller-manager) due to a NoExecute taint that the Pod does not tolerate; see [taint](#)-based evictions.

EvictionByEvictionAPI

Pod has been marked for [eviction using the Kubernetes API](#) .

DeletionByPodGC

Pod, that is bound to a no longer existing Node, is due to be deleted by [Pod garbage collection](#).

TerminationByKubelet

Pod has been terminated by the kubelet, because of either [node pressure eviction](#) or the [graceful node shutdown](#).

Note:

A Pod disruption might be interrupted. The control plane might re-attempt to continue the disruption of the same Pod, but it is not guaranteed. As a result, the DisruptionTarget condition

might be added to a Pod, but that Pod might then not actually be deleted. In such a situation, after some time, the Pod disruption condition will be cleared.

When the PodDisruptionConditions feature gate is enabled, along with cleaning up the pods, the Pod garbage collector (PodGC) will also mark them as failed if they are in a non-terminal phase (see also [Pod garbage collection](#)).

When using a Job (or CronJob), you may want to use these Pod disruption conditions as part of your Job's [Pod failure policy](#).

Separating Cluster Owner and Application Owner Roles

Often, it is useful to think of the Cluster Manager and Application Owner as separate roles with limited knowledge of each other. This separation of responsibilities may make sense in these scenarios:

- when there are many application teams sharing a Kubernetes cluster, and there is natural specialization of roles
- when third-party tools or services are used to automate cluster management

Pod Disruption Budgets support this separation of roles by providing an interface between the roles.

If you do not have such a separation of responsibilities in your organization, you may not need to use Pod Disruption Budgets.

How to perform Disruptive Actions on your Cluster

If you are a Cluster Administrator, and you need to perform a disruptive action on all the nodes in your cluster, such as a node or system software upgrade, here are some options:

- Accept downtime during the upgrade.
- Failover to another complete replica cluster.
 - No downtime, but may be costly both for the duplicated nodes and for human effort to orchestrate the switchover.
- Write disruption tolerant applications and use PDBs.
 - No downtime.
 - Minimal resource duplication.
 - Allows more automation of cluster administration.
 - Writing disruption-tolerant applications is tricky, but the work to tolerate voluntary disruptions largely overlaps with work to support autoscaling and tolerating involuntary disruptions.

What's next

- Follow steps to protect your application by [configuring a Pod Disruption Budget](#).
- Learn more about [draining nodes](#)
- Learn about [updating a deployment](#) including steps to maintain its availability during the rollout.

Pod Quality of Service Classes

This page introduces *Quality of Service (QoS) classes* in Kubernetes, and explains how Kubernetes assigns a QoS class to each Pod as a consequence of the resource constraints that you specify for the containers in that Pod. Kubernetes relies on this classification to make decisions about which Pods to evict when there are not enough available resources on a Node.

Quality of Service classes

Kubernetes classifies the Pods that you run and allocates each Pod into a specific *quality of service (QoS) class*. Kubernetes uses that classification to influence how different pods are handled. Kubernetes does this classification based on the [resource requests](#) of the [Containers](#) in that Pod, along with how those requests relate to resource limits. This is known as [Quality of Service \(QoS\)](#) class. Kubernetes assigns every Pod a QoS class based on the resource requests and limits of its component Containers. QoS classes are used by Kubernetes to decide which Pods to evict from a Node experiencing [Node Pressure](#). The possible QoS classes are Guaranteed, Burstable, and BestEffort. When a Node runs out of resources, Kubernetes will first evict BestEffort Pods running on that Node, followed by Burstable and finally Guaranteed Pods. When this eviction is due to resource pressure, only Pods exceeding resource requests are candidates for eviction.

Guaranteed

Pods that are Guaranteed have the strictest resource limits and are least likely to face eviction. They are guaranteed not to be killed until they exceed their limits or there are no lower-priority Pods that can be preempted from the Node. They may not acquire resources beyond their specified limits. These Pods can also make use of exclusive CPUs using the [static](#) CPU management policy.

Criteria

For a Pod to be given a QoS class of Guaranteed:

- Every Container in the Pod must have a memory limit and a memory request.
- For every Container in the Pod, the memory limit must equal the memory request.
- Every Container in the Pod must have a CPU limit and a CPU request.
- For every Container in the Pod, the CPU limit must equal the CPU request.

Burstable

Pods that are Burstable have some lower-bound resource guarantees based on the request, but do not require a specific limit. If a limit is not specified, it defaults to a limit equivalent to the capacity of the Node, which allows the Pods to flexibly increase their resources if resources are available. In the event of Pod eviction due to Node resource pressure, these Pods are evicted only after all BestEffort Pods are evicted. Because a Burstable Pod can include a Container that has no resource limits or requests, a Pod that is Burstable can try to use any amount of node resources.

Criteria

A Pod is given a QoS class of Burstable if:

- The Pod does not meet the criteria for QoS class Guaranteed.
- At least one Container in the Pod has a memory or CPU request or limit.

BestEffort

Pods in the BestEffort QoS class can use node resources that aren't specifically assigned to Pods in other QoS classes. For example, if you have a node with 16 CPU cores available to the kubelet, and you assign 4 CPU cores to a Guaranteed Pod, then a Pod in the BestEffort QoS class can try to use any amount of the remaining 12 CPU cores.

The kubelet prefers to evict BestEffort Pods if the node comes under resource pressure.

Criteria

A Pod has a QoS class of BestEffort if it doesn't meet the criteria for either Guaranteed or Burstable. In other words, a Pod is BestEffort only if none of the Containers in the Pod have a memory limit or a memory request, and none of the Containers in the Pod have a CPU limit or a CPU request. Containers in a Pod can request other resources (not CPU or memory) and still be classified as BestEffort.

Memory QoS with cgroup v2

FEATURE STATE: Kubernetes v1.22 [alpha]

Memory QoS uses the memory controller of cgroup v2 to guarantee memory resources in Kubernetes. Memory requests and limits of containers in pod are used to set specific interfaces `memory.min` and `memory.high` provided by the memory controller. When `memory.min` is set to memory requests, memory resources are reserved and never reclaimed by the kernel; this is how Memory QoS ensures memory availability for Kubernetes pods. And if memory limits are set in the container, this means that the system needs to limit container memory usage; Memory QoS uses `memory.high` to throttle workload approaching its memory limit, ensuring that the system is not overwhelmed by instantaneous memory allocation.

Memory QoS relies on QoS class to determine which settings to apply; however, these are different mechanisms that both provide controls over quality of service.

Some behavior is independent of QoS class

Certain behavior is independent of the QoS class assigned by Kubernetes. For example:

- Any Container exceeding a resource limit will be killed and restarted by the kubelet without affecting other Containers in that Pod.
- If a Container exceeds its resource request and the node it runs on faces resource pressure, the Pod it is in becomes a candidate for [eviction](#). If this occurs, all Containers in the Pod will be terminated. Kubernetes may create a replacement Pod, usually on a different node.

- The resource request of a Pod is equal to the sum of the resource requests of its component Containers, and the resource limit of a Pod is equal to the sum of the resource limits of its component Containers.
- The kube-scheduler does not consider QoS class when selecting which Pods to [preempt](#). Preemption can occur when a cluster does not have enough resources to run all the Pods you defined.

What's next

- Learn about [resource management for Pods and Containers](#).
- Learn about [Node-pressure eviction](#).
- Learn about [Pod priority and preemption](#).
- Learn about [Pod disruptions](#).
- Learn how to [assign memory resources to containers and pods](#).
- Learn how to [assign CPU resources to containers and pods](#).
- Learn how to [configure Quality of Service for Pods](#).

User Namespaces

FEATURE STATE: Kubernetes v1.30 [beta]

This page explains how user namespaces are used in Kubernetes pods. A user namespace isolates the user running inside the container from the one in the host.

A process running as root in a container can run as a different (non-root) user in the host; in other words, the process has full privileges for operations inside the user namespace, but is unprivileged for operations outside the namespace.

You can use this feature to reduce the damage a compromised container can do to the host or other pods in the same node. There are [several security vulnerabilities](#) rated either **HIGH** or **CRITICAL** that were not exploitable when user namespaces is active. It is expected user namespace will mitigate some future vulnerabilities too.

Before you begin

Note: This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

This is a Linux-only feature and support is needed in Linux for idmap mounts on the filesystems used. This means:

- On the node, the filesystem you use for /var/lib/kubelet/pods/, or the custom directory you configure for this, needs idmap mount support.
- All the filesystems used in the pod's volumes must support idmap mounts.

In practice this means you need at least Linux 6.3, as tmpfs started supporting idmap mounts in that version. This is usually needed as several Kubernetes features use tmpfs (the service account token that is mounted by default uses a tmpfs, Secrets use a tmpfs, etc.)

Some popular filesystems that support idmap mounts in Linux 6.3 are: btrfs, ext4, xfs, fat, tmpfs, overlayfs.

In addition, the container runtime and its underlying OCI runtime must support user namespaces. The following OCI runtimes offer support:

- [crun](#) version 1.9 or greater (it's recommend version 1.13+).

Note:

Many OCI runtimes do not include the support needed for using user namespaces in Linux pods. If you use a managed Kubernetes, or have downloaded it from packages and set it up, it's likely that nodes in your cluster use a runtime that doesn't include this support. For example, the most widely used OCI runtime is runc, and version 1.1.z of runc doesn't support all the features needed by the Kubernetes implementation of user namespaces.

If there is a newer release of runc than 1.1 available for use, check its documentation and release notes for compatibility (look for idmap mounts support in particular, because that is the missing feature).

To use user namespaces with Kubernetes, you also need to use a CRI [container runtime](#) to use this feature with Kubernetes pods:

- CRI-O: version 1.25 (and later) supports user namespaces for containers.

containerd v1.7 is not compatible with the userns support in Kubernetes v1.27 to v1.30. Kubernetes v1.25 and v1.26 used an earlier implementation that **is** compatible with containerd v1.7, in terms of userns support. If you are using a version of Kubernetes other than 1.30, check the documentation for that version of Kubernetes for the most relevant information. If there is a newer release of containerd than v1.7 available for use, also check the containerd documentation for compatibility information.

You can see the status of user namespaces support in cri-dockerd tracked in an [issue](#) on GitHub.

Introduction

User namespaces is a Linux feature that allows to map users in the container to different users in the host. Furthermore, the capabilities granted to a pod in a user namespace are valid only in the namespace and void outside of it.

A pod can opt-in to use user namespaces by setting the `pod.spec.hostUsers` field to false.

The kubelet will pick host UIDs/GIDs a pod is mapped to, and will do so in a way to guarantee that no two pods on the same node use the same mapping.

The `runAsUser`, `runAsGroup`, `fsGroup`, etc. fields in the `pod.spec` always refer to the user inside the container.

The valid UIDs/GIDs when this feature is enabled is the range 0-65535. This applies to files and processes (`runAsUser`, `runAsGroup`, etc.).

Files using a UID/GID outside this range will be seen as belonging to the overflow ID, usually 65534 (configured in `/proc/sys/kernel/overflowuid` and `/proc/sys/kernel/overflowgid`). However, it is not possible to modify those files, even by running as the 65534 user/group.

Most applications that need to run as root but don't access other host namespaces or resources, should continue to run fine without any changes needed if user namespaces is activated.

Understanding user namespaces for pods

Several container runtimes with their default configuration (like Docker Engine, containerd, CRI-O) use Linux namespaces for isolation. Other technologies exist and can be used with those runtimes too (e.g. Kata Containers uses VMs instead of Linux namespaces). This page is applicable for container runtimes using Linux namespaces for isolation.

When creating a pod, by default, several new namespaces are used for isolation: a network namespace to isolate the network of the container, a PID namespace to isolate the view of processes, etc. If a user namespace is used, this will isolate the users in the container from the users in the node.

This means containers can run as root and be mapped to a non-root user on the host. Inside the container the process will think it is running as root (and therefore tools like apt, yum, etc. work fine), while in reality the process doesn't have privileges on the host. You can verify this, for example, if you check which user the container process is running by executing `ps aux` from the host. The user `ps` shows is not the same as the user you see if you execute inside the container the command `id`.

This abstraction limits what can happen, for example, if the container manages to escape to the host. Given that the container is running as a non-privileged user on the host, it is limited what it can do to the host.

Furthermore, as users on each pod will be mapped to different non-overlapping users in the host, it is limited what they can do to other pods too.

Capabilities granted to a pod are also limited to the pod user namespace and mostly invalid out of it, some are even completely void. Here are two examples:

- `CAP_SYS_MODULE` does not have any effect if granted to a pod using user namespaces, the pod isn't able to load kernel modules.
- `CAP_SYS_ADMIN` is limited to the pod's user namespace and invalid outside of it.

Without using a user namespace a container running as root, in the case of a container breakout, has root privileges on the node. And if some capability were granted to the container, the capabilities are valid on the host too. None of this is true when we use user namespaces.

If you want to know more details about what changes when user namespaces are in use, see `man 7 user_namespaces`.

Set up a node to support user namespaces

By default, the kubelet assigns pods UIDs/GIDs above the range 0-65535, based on the assumption that the host's files and processes use UIDs/GIDs within this range, which is standard for most Linux distributions. This approach prevents any overlap between the UIDs/GIDs of the host and those of the pods.

Avoiding the overlap is important to mitigate the impact of vulnerabilities such as [CVE-2021-25741](#), where a pod can potentially read arbitrary files in the host. If the UIDs/GIDs of the pod and the host don't overlap, it is limited what a pod would be able to do: the pod UID/GID won't match the host's file owner/group.

The kubelet can use a custom range for user IDs and group IDs for pods. To configure a custom range, the node needs to have:

- A user kubelet in the system (you cannot use any other username here)
- The binary getsubids installed (part of [shadow-utils](#)) and in the PATH for the kubelet binary.
- A configuration of subordinate UIDs/GIDs for the kubelet user (see [man 5 subuid](#) and [man 5 subgid](#)).

This setting only gathers the UID/GID range configuration and does not change the user executing the kubelet.

You must follow some constraints for the subordinate ID range that you assign to the kubelet user:

- The subordinate user ID, that starts the UID range for Pods, **must** be a multiple of 65536 and must also be greater than or equal to 65536. In other words, you cannot use any ID from the range 0-65535 for Pods; the kubelet imposes this restriction to make it difficult to create an accidentally insecure configuration.
- The subordinate ID count must be a multiple of 65536
- The subordinate ID count must be at least $65536 \times \text{<maxPods>}$ where `<maxPods>` is the maximum number of pods that can run on the node.
- You must assign the same range for both user IDs and for group IDs, It doesn't matter if other users have user ID ranges that don't align with the group ID ranges.
- None of the assigned ranges should overlap with any other assignment.
- The subordinate configuration must be only one line. In other words, you can't have multiple ranges.

For example, you could define `/etc/subuid` and `/etc/subgid` to both have these entries for the kubelet user:

```
# The format is
# name:firstID:count of IDs
# where
# - firstID is 65536 (the minimum value possible)
# - count of IDs is 110 (default limit for number of) * 65536
kubelet:65536:7208960
```

Integration with Pod security admission checks

FEATURE STATE: Kubernetes v1.29 [alpha]

For Linux Pods that enable user namespaces, Kubernetes relaxes the application of [Pod Security Standards](#) in a controlled way. This behavior can be controlled by the [feature gate](#)

UserNamespacesPodSecurityStandards, which allows an early opt-in for end users. Admins have to ensure that user namespaces are enabled by all nodes within the cluster if using the feature gate.

If you enable the associated feature gate and create a Pod that uses user namespaces, the following fields won't be constrained even in contexts that enforce the *Baseline* or *Restricted* pod security standard. This behavior does not present a security concern because root inside a Pod with user namespaces actually refers to the user inside the container, that is never mapped to a privileged user on the host. Here's the list of fields that are **not** checks for Pods in those circumstances:

- `spec.securityContext.runAsNonRoot`
- `spec.containers[*].securityContext.runAsNonRoot`
- `spec.initContainers[*].securityContext.runAsNonRoot`
- `spec.ephemeralContainers[*].securityContext.runAsNonRoot`
- `spec.securityContext.runAsUser`
- `spec.containers[*].securityContext.runAsUser`
- `spec.initContainers[*].securityContext.runAsUser`
- `spec.ephemeralContainers[*].securityContext.runAsUser`

Limitations

When using a user namespace for the pod, it is disallowed to use other host namespaces. In particular, if you set `hostUsers: false` then you are not allowed to set any of:

- `hostNetwork: true`
- `hostIPC: true`
- `hostPID: true`

What's next

- Take a look at [Use a User Namespace With a Pod](#)

Downward API

There are two ways to expose Pod and container fields to a running container: environment variables, and as files that are populated by a special volume type. Together, these two ways of exposing Pod and container fields are called the downward API.

It is sometimes useful for a container to have information about itself, without being overly coupled to Kubernetes. The *downward API* allows containers to consume information about themselves or the cluster without using the Kubernetes client or API server.

An example is an existing application that assumes a particular well-known environment variable holds a unique identifier. One possibility is to wrap the application, but that is tedious and error-prone, and it violates the goal of low coupling. A better option would be to use the Pod's name as an identifier, and inject the Pod's name into the well-known environment variable.

In Kubernetes, there are two ways to expose Pod and container fields to a running container:

- as [environment variables](#)
- as [files in a downwardAPI volume](#)

Together, these two ways of exposing Pod and container fields are called the *downward API*.

Available fields

Only some Kubernetes API fields are available through the downward API. This section lists which fields you can make available.

You can pass information from available Pod-level fields using `fieldRef`. At the API level, the spec for a Pod always defines at least one [Container](#). You can pass information from available Container-level fields using `resourceFieldRef`.

Information available via `fieldRef`

For some Pod-level fields, you can provide them to a container either as an environment variable or using a downwardAPI volume. The fields available via either mechanism are:

`metadata.name`
the pod's name
`metadata.namespace`
the pod's [namespace](#)
`metadata.uid`
the pod's unique ID
`metadata.annotations['<KEY>']`
the value of the pod's [annotation](#) named <KEY> (for example, `metadata.annotations['myannotation']`)
`metadata.labels['<KEY>']`
the text value of the pod's [label](#) named <KEY> (for example, `metadata.labels['mylabel']`)

The following information is available through environment variables **but not as a downwardAPI volume fieldRef**:

`spec.serviceAccountName`
the name of the pod's [service account](#)
`spec.nodeName`
the name of the [node](#) where the Pod is executing
`status.hostIP`
the primary IP address of the node to which the Pod is assigned
`status.hostIPs`
the IP addresses is a dual-stack version of `status.hostIP`, the first is always the same as `status.hostIP`.
`status.podIP`
the pod's primary IP address (usually, its IPv4 address)
`status.podIPs`
the IP addresses is a dual-stack version of `status.podIP`, the first is always the same as `status.podIP`

The following information is available through a downwardAPI volume fieldRef, **but not as environment variables**:

metadata.labels

all of the pod's labels, formatted as label-key="escaped-label-value" with one label per line

metadata.annotations

all of the pod's annotations, formatted as annotation-key="escaped-annotation-value" with one annotation per line

Information available via resourceFieldRef

These container-level fields allow you to provide information about [requests and limits](#) for resources such as CPU and memory.

resource: limits.cpu

A container's CPU limit

resource: requests.cpu

A container's CPU request

resource: limits.memory

A container's memory limit

resource: requests.memory

A container's memory request

resource: limits.hugepages-*

A container's hugepages limit

resource: requests.hugepages-*

A container's hugepages request

resource: limits.ephemeral-storage

A container's ephemeral-storage limit

resource: requests.ephemeral-storage

A container's ephemeral-storage request

Fallback information for resource limits

If CPU and memory limits are not specified for a container, and you use the downward API to try to expose that information, then the kubelet defaults to exposing the maximum allocatable value for CPU and memory based on the [node allocatable](#) calculation.

What's next

You can read about [downwardAPI volumes](#).

You can try using the downward API to expose container- or Pod-level information:

- as [environment variables](#)
- as [files in downwardAPI volume](#)

Workload Management

Kubernetes provides several built-in APIs for declarative management of your [workloads](#) and the components of those workloads.

Ultimately, your applications run as containers inside [Pods](#); however, managing individual Pods would be a lot of effort. For example, if a Pod fails, you probably want to run a new Pod to replace it. Kubernetes can do that for you.

You use the Kubernetes API to create a workload [object](#) that represents a higher abstraction level than a Pod, and then the Kubernetes [control plane](#) automatically manages Pod objects on your behalf, based on the specification for the workload object you defined.

The built-in APIs for managing workloads are:

[Deployment](#) (and, indirectly, [ReplicaSet](#)), the most common way to run an application on your cluster. Deployment is a good fit for managing a stateless application workload on your cluster, where any Pod in the Deployment is interchangeable and can be replaced if needed. (Deployments are a replacement for the legacy [ReplicationController](#) API).

A [StatefulSet](#) lets you manage one or more Pods – all running the same application code – where the Pods rely on having a distinct identity. This is different from a Deployment where the Pods are expected to be interchangeable. The most common use for a StatefulSet is to be able to make a link between its Pods and their persistent storage. For example, you can run a StatefulSet that associates each Pod with a [PersistentVolume](#). If one of the Pods in the StatefulSet fails, Kubernetes makes a replacement Pod that is connected to the same PersistentVolume.

A [DaemonSet](#) defines Pods that provide facilities that are local to a specific [node](#); for example, a driver that lets containers on that node access a storage system. You use a DaemonSet when the driver, or other node-level service, has to run on the node where it's useful. Each Pod in a DaemonSet performs a role similar to a system daemon on a classic Unix / POSIX server. A DaemonSet might be fundamental to the operation of your cluster, such as a plugin to let that node access [cluster networking](#), it might help you to manage the node, or it could provide less essential facilities that enhance the container platform you are running. You can run DaemonSets (and their pods) across every node in your cluster, or across just a subset (for example, only install the GPU accelerator driver on nodes that have a GPU installed).

You can use a [Job](#) and / or a [CronJob](#) to define tasks that run to completion and then stop. A Job represents a one-off task, whereas each CronJob repeats according to a schedule.

Other topics in this section:

- [Automatic Cleanup for Finished Jobs](#)
- [ReplicationController](#)

Deployments

A Deployment manages a set of Pods to run an application workload, usually one that doesn't maintain state.

A *Deployment* provides declarative updates for [Pods](#) and [ReplicaSets](#).

You describe a *desired state* in a Deployment, and the Deployment [Controller](#) changes the actual state to the desired state at a controlled rate. You can define Deployments to create new ReplicaSets, or to remove existing Deployments and adopt all their resources with new Deployments.

Note:

Do not manage ReplicaSets owned by a Deployment. Consider opening an issue in the main Kubernetes repository if your use case is not covered below.

Use Case

The following are typical use cases for Deployments:

- [Create a Deployment to rollout a ReplicaSet](#). The ReplicaSet creates Pods in the background. Check the status of the rollout to see if it succeeds or not.
- [Declare the new state of the Pods](#) by updating the PodTemplateSpec of the Deployment. A new ReplicaSet is created and the Deployment manages moving the Pods from the old ReplicaSet to the new one at a controlled rate. Each new ReplicaSet updates the revision of the Deployment.
- [Rollback to an earlier Deployment revision](#) if the current state of the Deployment is not stable. Each rollback updates the revision of the Deployment.
- [Scale up the Deployment to facilitate more load](#).
- [Pause the rollout of a Deployment](#) to apply multiple fixes to its PodTemplateSpec and then resume it to start a new rollout.
- [Use the status of the Deployment](#) as an indicator that a rollout has stuck.
- [Clean up older ReplicaSets](#) that you don't need anymore.

Creating a Deployment

The following is an example of a Deployment. It creates a ReplicaSet to bring up three nginx Pods:

[controllers/nginx-deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:1.14.2
          ports:
            - containerPort: 80
```


In this example:

- A Deployment named `nginx-deployment` is created, indicated by the `.metadata.name` field. This name will become the basis for the ReplicaSets and Pods which are created later. See [Writing a Deployment Spec](#) for more details.
- The Deployment creates a ReplicaSet that creates three replicated Pods, indicated by the `.spec.replicas` field.
- The `.spec.selector` field defines how the created ReplicaSet finds which Pods to manage. In this case, you select a label that is defined in the Pod template (`app: nginx`). However, more sophisticated selection rules are possible, as long as the Pod template itself satisfies the rule.

Note:

The `.spec.selector.matchLabels` field is a map of {key,value} pairs. A single {key,value} in the matchLabels map is equivalent to an element of matchExpressions, whose key field is "key", the operator is "In", and the values array contains only "value". All of the requirements, from both matchLabels and matchExpressions, must be satisfied in order to match.

- The template field contains the following sub-fields:
 - The Pods are labeled `app: nginx` using the `.metadata.labels` field.
 - The Pod template's specification, or `.template.spec` field, indicates that the Pods run one container, `nginx`, which runs the `nginx` [Docker Hub](#) image at version 1.14.2.
 - Create one container and name it `nginx` using the `.spec.template.spec.containers[0].name` field.

Before you begin, make sure your Kubernetes cluster is up and running. Follow the steps given below to create the above Deployment:

1. Create the Deployment by running the following command:

```
kubectl apply -f https://k8s.io/examples/controllers/nginx-deployment.yaml
```

2. Run `kubectl get deployments` to check if the Deployment was created.

If the Deployment is still being created, the output is similar to the following:

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	0/3	0	0	1s

When you inspect the Deployments in your cluster, the following fields are displayed:

- NAME lists the names of the Deployments in the namespace.
- READY displays how many replicas of the application are available to your users. It follows the pattern ready/desired.
- UP-TO-DATE displays the number of replicas that have been updated to achieve the desired state.
- AVAILABLE displays how many replicas of the application are available to your users.
- AGE displays the amount of time that the application has been running.

Notice how the number of desired replicas is 3 according to `.spec.replicas` field.

3. To see the Deployment rollout status, run `kubectl rollout status deployment/nginx-deployment`.

The output is similar to:

```
Waiting for rollout to finish: 2 out of 3 new replicas have been updated...
deployment "nginx-deployment" successfully rolled out
```

4. Run the `kubectl get deployments` again a few seconds later. The output is similar to this:

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	3/3	3	3	18s

Notice that the Deployment has created all three replicas, and all replicas are up-to-date (they contain the latest Pod template) and available.

5. To see the ReplicaSet (rs) created by the Deployment, run `kubectl get rs`. The output is similar to this:

NAME	DESIRED	CURRENT	READY	AGE
nginx-deployment-75675f5897	3	3	3	18s

ReplicaSet output shows the following fields:

- NAME lists the names of the ReplicaSets in the namespace.
- DESIRED displays the desired number of *replicas* of the application, which you define when you create the Deployment. This is the *desired state*.
- CURRENT displays how many replicas are currently running.
- READY displays how many replicas of the application are available to your users.
- AGE displays the amount of time that the application has been running.

Notice that the name of the ReplicaSet is always formatted as `[DEPLOYMENT-NAME]-[HASH]`. This name will become the basis for the Pods which are created.

The HASH string is the same as the `pod-template-hash` label on the ReplicaSet.

6. To see the labels automatically generated for each Pod, run `kubectl get pods --show-labels`. The output is similar to:

NAME	READY	STATUS	RESTARTS	AGE	LABELS
nginx-deployment-75675f5897-7ci7o	1/1	Running	0	18s	app=nginx,pod-template-hash=75675f5897
nginx-deployment-75675f5897-kzszj	1/1	Running	0	18s	app=nginx,pod-template-hash=75675f5897
nginx-deployment-75675f5897-qqcnn	1/1	Running	0	18s	app=nginx,pod-template-hash=75675f5897

The created ReplicaSet ensures that there are three `nginx` Pods.

Note:

You must specify an appropriate selector and Pod template labels in a Deployment (in this case, `app: nginx`).

Do not overlap labels or selectors with other controllers (including other Deployments and StatefulSets). Kubernetes doesn't stop you from overlapping, and if multiple controllers have overlapping selectors those controllers might conflict and behave unexpectedly.

Pod-template-hash label

Caution:

Do not change this label.

The `pod-template-hash` label is added by the Deployment controller to every ReplicaSet that a Deployment creates or adopts.

This label ensures that child ReplicaSets of a Deployment do not overlap. It is generated by hashing the PodTemplate of the ReplicaSet and using the resulting hash as the label value that is added to the ReplicaSet selector, Pod template labels, and in any existing Pods that the ReplicaSet might have.

Updating a Deployment

Note:

A Deployment's rollout is triggered if and only if the Deployment's Pod template (that is, `.spec.template`) is changed, for example if the labels or container images of the template are updated. Other updates, such as scaling the Deployment, do not trigger a rollout.

Follow the steps given below to update your Deployment:

1. Let's update the `nginx` Pods to use the `nginx:1.16.1` image instead of the `nginx:1.14.2` image.

```
kubectl set image deployment.v1.apps/nginx-deployment nginx=nginx:1.16.1
```

or use the following command:

```
kubectl set image deployment/nginx-deployment nginx=nginx:1.16.1
```

where `deployment/nginx-deployment` indicates the Deployment, `nginx` indicates the Container the update will take place and `nginx:1.16.1` indicates the new image and its tag.

The output is similar to:

```
deployment.apps/nginx-deployment image updated
```

Alternatively, you can edit the Deployment and change `.spec.template.spec.containers[0].image` from `nginx:1.14.2` to `nginx:1.16.1`:

```
kubectl edit deployment/nginx-deployment
```

The output is similar to:

```
deployment.apps/nginx-deployment edited
```

2. To see the rollout status, run:

```
kubectl rollout status deployment/nginx-deployment
```

The output is similar to this:

```
Waiting for rollout to finish: 2 out of 3 new replicas have been updated...
```

or

```
deployment "nginx-deployment" successfully rolled out
```

Get more details on your updated Deployment:

- After the rollout succeeds, you can view the Deployment by running `kubectl get deployments`. The output is similar to this:

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	3/3	3	3	36s

- Run `kubectl get rs` to see that the Deployment updated the Pods by creating a new ReplicaSet and scaling it up to 3 replicas, as well as scaling down the old ReplicaSet to 0 replicas.

```
kubectl get rs
```

The output is similar to this:

NAME	DESIRED	CURRENT	READY	AGE
nginx-deployment-1564180365	3	3	3	6s
nginx-deployment-2035384211	0	0	0	36s

- Running `get pods` should now show only the new Pods:

```
kubectl get pods
```

The output is similar to this:

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-1564180365-khku8	1/1	Running	0	14s
nginx-deployment-1564180365-nacti	1/1	Running	0	14s
nginx-deployment-1564180365-z9gth	1/1	Running	0	14s

Next time you want to update these Pods, you only need to update the Deployment's Pod template again.

Deployment ensures that only a certain number of Pods are down while they are being updated. By default, it ensures that at least 75% of the desired number of Pods are up (25% max unavailable).

Deployment also ensures that only a certain number of Pods are created above the desired number of Pods. By default, it ensures that at most 125% of the desired number of Pods are up (25% max surge).

For example, if you look at the above Deployment closely, you will see that it first creates a new Pod, then deletes an old Pod, and creates another new one. It does not kill old Pods until a sufficient number of new Pods have come up, and does not create new Pods until a sufficient number of old Pods have been killed. It makes sure that at least 3 Pods are available and that at max 4 Pods in total are available. In case of a Deployment with 4 replicas, the number of Pods would be between 3 and 5.

- Get details of your Deployment:

```
kubectl describe deployments
```

The output is similar to this:

```
Name:          nginx-deployment
Namespace:     default
CreationTimestamp: Thu, 30 Nov 2017 10:56:25 +0000
Labels:        app=nginx
Annotations:    deployment.kubernetes.io/revision=2
Selector:      app=nginx
Replicas:      3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels: app=nginx
  Containers:
    nginx:
      Image:      nginx:1.16.1
      Port:       80/TCP
      Environment: <none>
      Mounts:      <none>
      Volumes:      <none>
  Conditions:
    Type           Status Reason
    ----           -
    Available      True   MinimumReplicasAvailable
    Progressing    True   NewReplicaSetAvailable
  OldReplicaSets: <none>
  NewReplicaSet:  nginx-deployment-1564180365 (3/3 replicas created)
  Events:
    Type Reason      Age From          Message
    ---  -
    Normal ScalingReplicaSet 2m deployment-controller Scaled up replica set nginx-deployment-2035384211 to 3
    Normal ScalingReplicaSet 24s deployment-controller Scaled up replica set nginx-deployment-1564180365 to 1
    Normal ScalingReplicaSet 22s deployment-controller Scaled down replica set nginx-deployment-2035384211 to 2
    Normal ScalingReplicaSet 22s deployment-controller Scaled up replica set nginx-
```

```
deployment-1564180365 to 2
Normal ScalingReplicaSet 19s deployment-controller Scaled down replica set nginx-
deployment-2035384211 to 1
Normal ScalingReplicaSet 19s deployment-controller Scaled up replica set nginx-
deployment-1564180365 to 3
Normal ScalingReplicaSet 14s deployment-controller Scaled down replica set nginx-
deployment-2035384211 to 0
```

Here you see that when you first created the Deployment, it created a ReplicaSet (nginx-deployment-2035384211) and scaled it up to 3 replicas directly. When you updated the Deployment, it created a new ReplicaSet (nginx-deployment-1564180365) and scaled it up to 1 and waited for it to come up. Then it scaled down the old ReplicaSet to 2 and scaled up the new ReplicaSet to 2 so that at least 3 Pods were available and at most 4 Pods were created at all times. It then continued scaling up and down the new and the old ReplicaSet, with the same rolling update strategy. Finally, you'll have 3 available replicas in the new ReplicaSet, and the old ReplicaSet is scaled down to 0.

Note:

Kubernetes doesn't count terminating Pods when calculating the number of availableReplicas, which must be between replicas - maxUnavailable and replicas + maxSurge. As a result, you might notice that there are more Pods than expected during a rollout, and that the total resources consumed by the Deployment is more than replicas + maxSurge until the terminationGracePeriodSeconds of the terminating Pods expires.

Rollover (aka multiple updates in-flight)

Each time a new Deployment is observed by the Deployment controller, a ReplicaSet is created to bring up the desired Pods. If the Deployment is updated, the existing ReplicaSet that controls Pods whose labels match .spec.selector but whose template does not match .spec.template are scaled down. Eventually, the new ReplicaSet is scaled to .spec.replicas and all old ReplicaSets is scaled to 0.

If you update a Deployment while an existing rollout is in progress, the Deployment creates a new ReplicaSet as per the update and start scaling that up, and rolls over the ReplicaSet that it was scaling up previously -- it will add it to its list of old ReplicaSets and start scaling it down.

For example, suppose you create a Deployment to create 5 replicas of nginx:1.14.2, but then update the Deployment to create 5 replicas of nginx:1.16.1, when only 3 replicas of nginx:1.14.2 had been created. In that case, the Deployment immediately starts killing the 3 nginx:1.14.2 Pods that it had created, and starts creating nginx:1.16.1 Pods. It does not wait for the 5 replicas of nginx:1.14.2 to be created before changing course.

Label selector updates

It is generally discouraged to make label selector updates and it is suggested to plan your selectors up front. In any case, if you need to perform a label selector update, exercise great caution and make sure you have grasped all of the implications.

Note:

In API version `apps/v1`, a Deployment's label selector is immutable after it gets created.

- Selector additions require the Pod template labels in the Deployment spec to be updated with the new label too, otherwise a validation error is returned. This change is a non-overlapping one, meaning that the new selector does not select ReplicaSets and Pods created with the old selector, resulting in orphaning all old ReplicaSets and creating a new ReplicaSet.
- Selector updates changes the existing value in a selector key -- result in the same behavior as additions.
- Selector removals removes an existing key from the Deployment selector -- do not require any changes in the Pod template labels. Existing ReplicaSets are not orphaned, and a new ReplicaSet is not created, but note that the removed label still exists in any existing Pods and ReplicaSets.

Rolling Back a Deployment

Sometimes, you may want to rollback a Deployment; for example, when the Deployment is not stable, such as crash looping. By default, all of the Deployment's rollout history is kept in the system so that you can rollback anytime you want (you can change that by modifying revision history limit).

Note:

A Deployment's revision is created when a Deployment's rollout is triggered. This means that the new revision is created if and only if the Deployment's Pod template (`.spec.template`) is changed, for example if you update the labels or container images of the template. Other updates, such as scaling the Deployment, do not create a Deployment revision, so that you can facilitate simultaneous manual- or auto-scaling. This means that when you roll back to an earlier revision, only the Deployment's Pod template part is rolled back.

- Suppose that you made a typo while updating the Deployment, by putting the image name as `nginx:1.161` instead of `nginx:1.16.1`:

```
kubectl set image deployment/nginx-deployment nginx=nginx:1.161
```

The output is similar to this:

```
deployment.apps/nginx-deployment image updated
```

- The rollout gets stuck. You can verify it by checking the rollout status:

```
kubectl rollout status deployment/nginx-deployment
```

The output is similar to this:

```
Waiting for rollout to finish: 1 out of 3 new replicas have been updated...
```

- Press Ctrl-C to stop the above rollout status watch. For more information on stuck rollouts, [read more here](#).

You see that the number of old replicas (adding the replica count from nginx-

- deployment-1564180365 and nginx-deployment-2035384211) is 3, and the number of new replicas (from nginx-deployment-3066724191) is 1.

```
kubectl get rs
```

The output is similar to this:

NAME	DESIRED	CURRENT	READY	AGE
nginx-deployment-1564180365	3	3	3	25s
nginx-deployment-2035384211	0	0	0	36s
nginx-deployment-3066724191	1	1	0	6s

- Looking at the Pods created, you see that 1 Pod created by new ReplicaSet is stuck in an image pull loop.

```
kubectl get pods
```

The output is similar to this:

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-1564180365-70iae	1/1	Running	0	25s
nginx-deployment-1564180365-jbqqo	1/1	Running	0	25s
nginx-deployment-1564180365-hysrc	1/1	Running	0	25s
nginx-deployment-3066724191-08mng	0/1	ImagePullBackOff	0	6s

Note:

The Deployment controller stops the bad rollout automatically, and stops scaling up the new ReplicaSet. This depends on the rollingUpdate parameters (maxUnavailable specifically) that you have specified. Kubernetes by default sets the value to 25%.

- Get the description of the Deployment:

```
kubectl describe deployment
```

The output is similar to this:

```
Name:      nginx-deployment
Namespace:  default
CreationTimestamp: Tue, 15 Mar 2016 14:48:04 -0700
Labels:    app=nginx
Selector:  app=nginx
Replicas:  3 desired | 1 updated | 4 total | 3 available | 1 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels: app=nginx
  Containers:
    nginx:
      Image:   nginx:1.161
      Port:    80/TCP
      Host Port: 0/TCP
```



```

Environment: <none>
Mounts:      <none>
Volumes:     <none>
Conditions:
Type          Status Reason
-----
Available     True  MinimumReplicasAvailable
Progressing   True  ReplicaSetUpdated
OldReplicaSets:  nginx-deployment-1564180365 (3/3 replicas created)
NewReplicaSet:   nginx-deployment-3066724191 (1/1 replicas created)
Events:
FirstSeen LastSeen  Count  From              SubObjectPath Type    Reason
Message
-----
1m         1m         1      {deployment-controller }      Normal    ScalingReplicaSet
Scaled up replica set nginx-deployment-2035384211 to 3
22s        22s        1      {deployment-controller }      Normal    ScalingReplicaSet
Scaled up replica set nginx-deployment-1564180365 to 1
22s        22s        1      {deployment-controller }      Normal    ScalingReplicaSet
Scaled down replica set nginx-deployment-2035384211 to 2
22s        22s        1      {deployment-controller }      Normal    ScalingReplicaSet
Scaled up replica set nginx-deployment-1564180365 to 2
21s        21s        1      {deployment-controller }      Normal    ScalingReplicaSet
Scaled down replica set nginx-deployment-2035384211 to 1
21s        21s        1      {deployment-controller }      Normal    ScalingReplicaSet
Scaled up replica set nginx-deployment-1564180365 to 3
13s        13s        1      {deployment-controller }      Normal    ScalingReplicaSet
Scaled down replica set nginx-deployment-2035384211 to 0
13s        13s        1      {deployment-controller }      Normal    ScalingReplicaSet
Scaled up replica set nginx-deployment-3066724191 to 1

```

To fix this, you need to rollback to a previous revision of Deployment that is stable.

Checking Rollout History of a Deployment

Follow the steps given below to check the rollout history:

1. First, check the revisions of this Deployment:

```
kubectl rollout history deployment/nginx-deployment
```

The output is similar to this:

```

deployments "nginx-deployment"
REVISION  CHANGE-CAUSE
1         kubectl apply --filename=https://k8s.io/examples/controllers/nginx-
deployment.yaml
2         kubectl set image deployment/nginx-deployment nginx=nginx:1.16.1
3         kubectl set image deployment/nginx-deployment nginx=nginx:1.16.1

```

CHANGE-CAUSE is copied from the Deployment annotation `kubernetes.io/change-cause` to its revisions upon creation. You can specify the CHANGE-CAUSE message by:

- Annotating the Deployment with `kubectl annotate deployment/nginx-deployment kubernetes.io/change-cause="image updated to 1.16.1"`
- Manually editing the manifest of the resource.

2. To see the details of each revision, run:

```
kubectl rollout history deployment/nginx-deployment --revision=2
```

The output is similar to this:

```
deployments "nginx-deployment" revision 2
Labels:   app=nginx
          pod-template-hash=1159050644
Annotations: kubernetes.io/change-cause=kubectl set image deployment/nginx-
deployment nginx=nginx:1.16.1
Containers:
  nginx:
    Image:   nginx:1.16.1
    Port:    80/TCP
    QoS Tier:
      cpu:      BestEffort
      memory:   BestEffort
    Environment Variables:  <none>
  No volumes.
```

Rolling Back to a Previous Revision

Follow the steps given below to rollback the Deployment from the current version to the previous version, which is version 2.

1. Now you've decided to undo the current rollout and rollback to the previous revision:

```
kubectl rollout undo deployment/nginx-deployment
```

The output is similar to this:

```
deployment.apps/nginx-deployment rolled back
```

Alternatively, you can rollback to a specific revision by specifying it with `--to-revision`:

```
kubectl rollout undo deployment/nginx-deployment --to-revision=2
```

The output is similar to this:

```
deployment.apps/nginx-deployment rolled back
```

For more details about rollout related commands, read [kubectl rollout](#).

The Deployment is now rolled back to a previous stable revision. As you can see, a `DeploymentRollback` event for rolling back to revision 2 is generated from Deployment controller.

Check if the rollback was successful and the Deployment is running as expected, run:

2. `kubectl get deployment nginx-deployment`

The output is similar to this:

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	3/3	3	3	30m

3. Get the description of the Deployment:

`kubectl describe deployment nginx-deployment`

The output is similar to this:

```
Name:          nginx-deployment
Namespace:     default
CreationTimestamp:  Sun, 02 Sep 2018 18:17:55 -0500
Labels:        app=nginx
Annotations:    deployment.kubernetes.io/revision=4
                kubernetes.io/change-cause=kubectl set image deployment/nginx-
deployment nginx=nginx:1.16.1
Selector:      app=nginx
Replicas:      3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds:  0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=nginx
  Containers:
    nginx:
      Image:   nginx:1.16.1
      Port:    80/TCP
      Host Port:  0/TCP
      Environment: <none>
      Mounts:      <none>
      Volumes:      <none>
  Conditions:
    Type           Status Reason
    ----           -
    Available       True  MinimumReplicasAvailable
    Progressing     True  NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet:  nginx-deployment-c4747d96c (3/3 replicas created)
Events:
  Type    Reason             Age   From          Message
  ----    -
  Normal  ScalingReplicaSet  12m   deployment-controller Scaled up replica set nginx-
deployment-75675f5897 to 3
  Normal  ScalingReplicaSet  11m   deployment-controller Scaled up replica set nginx-
deployment-c4747d96c to 1
  Normal  ScalingReplicaSet  11m   deployment-controller Scaled down replica set nginx-
deployment-75675f5897 to 2
  Normal  ScalingReplicaSet  11m   deployment-controller Scaled up replica set nginx-
```

```

deployment-c4747d96c to 2
Normal ScalingReplicaSet 11m deployment-controller Scaled down replica set nginx-
deployment-75675f5897 to 1
Normal ScalingReplicaSet 11m deployment-controller Scaled up replica set nginx-
deployment-c4747d96c to 3
Normal ScalingReplicaSet 11m deployment-controller Scaled down replica set nginx-
deployment-75675f5897 to 0
Normal ScalingReplicaSet 11m deployment-controller Scaled up replica set nginx-
deployment-595696685f to 1
Normal DeploymentRollback 15s deployment-controller Rolled back deployment
"nginx-deployment" to revision 2
Normal ScalingReplicaSet 15s deployment-controller Scaled down replica set nginx-
deployment-595696685f to 0

```

Scaling a Deployment

You can scale a Deployment by using the following command:

```
kubectl scale deployment/nginx-deployment --replicas=10
```

The output is similar to this:

```
deployment.apps/nginx-deployment scaled
```

Assuming [horizontal Pod autoscaling](#) is enabled in your cluster, you can set up an autoscaler for your Deployment and choose the minimum and maximum number of Pods you want to run based on the CPU utilization of your existing Pods.

```
kubectl autoscale deployment/nginx-deployment --min=10 --max=15 --cpu-percent=80
```

The output is similar to this:

```
deployment.apps/nginx-deployment scaled
```

Proportional scaling

RollingUpdate Deployments support running multiple versions of an application at the same time. When you or an autoscaler scales a RollingUpdate Deployment that is in the middle of a rollout (either in progress or paused), the Deployment controller balances the additional replicas in the existing active ReplicaSets (ReplicaSets with Pods) in order to mitigate risk. This is called *proportional scaling*.

For example, you are running a Deployment with 10 replicas, [maxSurge](#)=3, and [maxUnavailable](#)=2.

- Ensure that the 10 replicas in your Deployment are running.

```
kubectl get deploy
```

The output is similar to this:

NAME	DESIRED	CURRENT	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	10	10	10	10	50s

You update to a new image which happens to be unresolvable from inside the cluster.

- ```
kubectrl set image deployment/nginx-deployment nginx=nginx:sometag
```

The output is similar to this:

```
deployment.apps/nginx-deployment image updated
```

- The image update starts a new rollout with ReplicaSet `nginx-deployment-1989198191`, but it's blocked due to the `maxUnavailable` requirement that you mentioned above. Check out the rollout status:

```
kubectrl get rs
```

The output is similar to this:

| NAME                        | DESIRED | CURRENT | READY | AGE |
|-----------------------------|---------|---------|-------|-----|
| nginx-deployment-1989198191 | 5       | 5       | 0     | 9s  |
| nginx-deployment-618515232  | 8       | 8       | 8     | 1m  |

- Then a new scaling request for the Deployment comes along. The autoscaler increments the Deployment replicas to 15. The Deployment controller needs to decide where to add these new 5 replicas. If you weren't using proportional scaling, all 5 of them would be added in the new ReplicaSet. With proportional scaling, you spread the additional replicas across all ReplicaSets. Bigger proportions go to the ReplicaSets with the most replicas and lower proportions go to ReplicaSets with less replicas. Any leftovers are added to the ReplicaSet with the most replicas. ReplicaSets with zero replicas are not scaled up.

In our example above, 3 replicas are added to the old ReplicaSet and 2 replicas are added to the new ReplicaSet. The rollout process should eventually move all replicas to the new ReplicaSet, assuming the new replicas become healthy. To confirm this, run:

```
kubectrl get deploy
```

The output is similar to this:

| NAME             | DESIRED | CURRENT | UP-TO-DATE | AVAILABLE | AGE |
|------------------|---------|---------|------------|-----------|-----|
| nginx-deployment | 15      | 18      | 7          | 8         | 7m  |

The rollout status confirms how the replicas were added to each ReplicaSet.

```
kubectrl get rs
```

The output is similar to this:

| NAME                        | DESIRED | CURRENT | READY | AGE |
|-----------------------------|---------|---------|-------|-----|
| nginx-deployment-1989198191 | 7       | 7       | 0     | 7m  |
| nginx-deployment-618515232  | 11      | 11      | 11    | 7m  |

## Pausing and Resuming a rollout of a Deployment

When you update a Deployment, or plan to, you can pause rollouts for that Deployment before you trigger one or more updates. When you're ready to apply those changes, you resume

rollouts for the Deployment. This approach allows you to apply multiple fixes in between pausing and resuming without triggering unnecessary rollouts.

- For example, with a Deployment that was created:

Get the Deployment details:

```
kubectl get deploy
```

The output is similar to this:

| NAME  | DESIRED | CURRENT | UP-TO-DATE | AVAILABLE | AGE |
|-------|---------|---------|------------|-----------|-----|
| nginx | 3       | 3       | 3          | 3         | 1m  |

Get the rollout status:

```
kubectl get rs
```

The output is similar to this:

| NAME             | DESIRED | CURRENT | READY | AGE |
|------------------|---------|---------|-------|-----|
| nginx-2142116321 | 3       | 3       | 3     | 1m  |

- Pause by running the following command:

```
kubectl rollout pause deployment/nginx-deployment
```

The output is similar to this:

```
deployment.apps/nginx-deployment paused
```

- Then update the image of the Deployment:

```
kubectl set image deployment/nginx-deployment nginx=nginx:1.16.1
```

The output is similar to this:

```
deployment.apps/nginx-deployment image updated
```

- Notice that no new rollout started:

```
kubectl rollout history deployment/nginx-deployment
```

The output is similar to this:

```
deployments "nginx"
REVISION CHANGE-CAUSE
1 <none>
```

- Get the rollout status to verify that the existing ReplicaSet has not changed:

```
kubectl get rs
```

The output is similar to this:

| NAME             | DESIRED | CURRENT | READY | AGE |
|------------------|---------|---------|-------|-----|
| nginx-2142116321 | 3       | 3       | 3     | 2m  |

- You can make as many updates as you wish, for example, update the resources that will be used:

```
kubectl set resources deployment/nginx-deployment -c=nginx --limits=cpu=200m,memory=512Mi
```

The output is similar to this:

```
deployment.apps/nginx-deployment resource requirements updated
```

The initial state of the Deployment prior to pausing its rollout will continue its function, but new updates to the Deployment will not have any effect as long as the Deployment rollout is paused.

- Eventually, resume the Deployment rollout and observe a new ReplicaSet coming up with all the new updates:

```
kubectl rollout resume deployment/nginx-deployment
```

The output is similar to this:

```
deployment.apps/nginx-deployment resumed
```

- Watch the status of the rollout until it's done.

```
kubectl get rs -w
```

The output is similar to this:

| NAME             | DESIRED | CURRENT | READY | AGE |
|------------------|---------|---------|-------|-----|
| nginx-2142116321 | 2       | 2       | 2     | 2m  |
| nginx-3926361531 | 2       | 2       | 0     | 6s  |
| nginx-3926361531 | 2       | 2       | 1     | 18s |
| nginx-2142116321 | 1       | 2       | 2     | 2m  |
| nginx-2142116321 | 1       | 2       | 2     | 2m  |
| nginx-3926361531 | 3       | 2       | 1     | 18s |
| nginx-3926361531 | 3       | 2       | 1     | 18s |
| nginx-2142116321 | 1       | 1       | 1     | 2m  |
| nginx-3926361531 | 3       | 3       | 1     | 18s |
| nginx-3926361531 | 3       | 3       | 2     | 19s |
| nginx-2142116321 | 0       | 1       | 1     | 2m  |
| nginx-2142116321 | 0       | 1       | 1     | 2m  |
| nginx-2142116321 | 0       | 0       | 0     | 2m  |
| nginx-3926361531 | 3       | 3       | 3     | 20s |

- Get the status of the latest rollout:

```
kubectl get rs
```

The output is similar to this:

| NAME             | DESIRED | CURRENT | READY | AGE |
|------------------|---------|---------|-------|-----|
| nginx-2142116321 | 0       | 0       | 0     | 2m  |
| nginx-3926361531 | 3       | 3       | 3     | 28s |

### Note:

You cannot rollback a paused Deployment until you resume it.

## Deployment status

A Deployment enters various states during its lifecycle. It can be [progressing](#) while rolling out a new ReplicaSet, it can be [complete](#), or it can [fail to progress](#).

### Progressing Deployment

Kubernetes marks a Deployment as *progressing* when one of the following tasks is performed:

- The Deployment creates a new ReplicaSet.
- The Deployment is scaling up its newest ReplicaSet.
- The Deployment is scaling down its older ReplicaSet(s).
- New Pods become ready or available (ready for at least [MinReadySeconds](#)).

When the rollout becomes “progressing”, the Deployment controller adds a condition with the following attributes to the Deployment's `.status.conditions`:

- type: Progressing
- status: "True"
- reason: NewReplicaSetCreated | reason: FoundNewReplicaSet | reason: ReplicaSetUpdated

You can monitor the progress for a Deployment by using `kubectl rollout status`.

### Complete Deployment

Kubernetes marks a Deployment as *complete* when it has the following characteristics:

- All of the replicas associated with the Deployment have been updated to the latest version you've specified, meaning any updates you've requested have been completed.
- All of the replicas associated with the Deployment are available.
- No old replicas for the Deployment are running.

When the rollout becomes “complete”, the Deployment controller sets a condition with the following attributes to the Deployment's `.status.conditions`:

- type: Progressing
- status: "True"
- reason: NewReplicaSetAvailable

This Progressing condition will retain a status value of "True" until a new rollout is initiated. The condition holds even when availability of replicas changes (which does instead affect the Available condition).



You can check if a Deployment has completed by using `kubectl rollout status`. If the rollout completed successfully, `kubectl rollout status` returns a zero exit code.

```
kubectl rollout status deployment/nginx-deployment
```

The output is similar to this:

```
Waiting for rollout to finish: 2 of 3 updated replicas are available...
deployment "nginx-deployment" successfully rolled out
```

and the exit status from `kubectl rollout` is 0 (success):

```
echo $?
```

```
0
```

## Failed Deployment

Your Deployment may get stuck trying to deploy its newest ReplicaSet without ever completing. This can occur due to some of the following factors:

- Insufficient quota
- Readiness probe failures
- Image pull errors
- Insufficient permissions
- Limit ranges
- Application runtime misconfiguration

One way you can detect this condition is to specify a deadline parameter in your Deployment spec: ([.spec.progressDeadlineSeconds](#)). `.spec.progressDeadlineSeconds` denotes the number of seconds the Deployment controller waits before indicating (in the Deployment status) that the Deployment progress has stalled.

The following `kubectl` command sets the spec with `progressDeadlineSeconds` to make the controller report lack of progress of a rollout for a Deployment after 10 minutes:

```
kubectl patch deployment/nginx-deployment -p '{"spec":{"progressDeadlineSeconds":600}}'
```

The output is similar to this:

```
deployment.apps/nginx-deployment patched
```

Once the deadline has been exceeded, the Deployment controller adds a `DeploymentCondition` with the following attributes to the Deployment's `.status.conditions`:

- type: Progressing
- status: "False"
- reason: ProgressDeadlineExceeded

This condition can also fail early and is then set to status value of "False" due to reasons as `ReplicaSetCreateError`. Also, the deadline is not taken into account anymore once the Deployment rollout completes.

See the [Kubernetes API conventions](#) for more information on status conditions.

**Note:**

Kubernetes takes no action on a stalled Deployment other than to report a status condition with reason: ProgressDeadlineExceeded. Higher level orchestrators can take advantage of it and act accordingly, for example, rollback the Deployment to its previous version.

**Note:**

If you pause a Deployment rollout, Kubernetes does not check progress against your specified deadline. You can safely pause a Deployment rollout in the middle of a rollout and resume without triggering the condition for exceeding the deadline.

You may experience transient errors with your Deployments, either due to a low timeout that you have set or due to any other kind of error that can be treated as transient. For example, let's suppose you have insufficient quota. If you describe the Deployment you will notice the following section:

```
kubectl describe deployment nginx-deployment
```

The output is similar to this:

```
<...>
Conditions:
 Type Status Reason
 ---- -
 Available True MinimumReplicasAvailable
 Progressing True ReplicaSetUpdated
 ReplicaFailure True FailedCreate
<...>
```

If you run `kubectl get deployment nginx-deployment -o yaml`, the Deployment status is similar to this:

```
status:
 availableReplicas: 2
 conditions:
 - lastTransitionTime: 2016-10-04T12:25:39Z
 lastUpdateTime: 2016-10-04T12:25:39Z
 message: Replica set "nginx-deployment-4262182780" is progressing.
 reason: ReplicaSetUpdated
 status: "True"
 type: Progressing
 - lastTransitionTime: 2016-10-04T12:25:42Z
 lastUpdateTime: 2016-10-04T12:25:42Z
 message: Deployment has minimum availability.
 reason: MinimumReplicasAvailable
 status: "True"
 type: Available
 - lastTransitionTime: 2016-10-04T12:25:39Z
 lastUpdateTime: 2016-10-04T12:25:39Z
 message: 'Error creating: pods "nginx-deployment-4262182780-" is forbidden: exceeded quota:
 object-counts, requested: pods=1, used: pods=3, limited: pods=2'
 reason: FailedCreate
```

```
status: "True"
type: ReplicaFailure
observedGeneration: 3
replicas: 2
unavailableReplicas: 2
```

Eventually, once the Deployment progress deadline is exceeded, Kubernetes updates the status and the reason for the Progressing condition:

Conditions:

| Type           | Status | Reason                   |
|----------------|--------|--------------------------|
| Available      | True   | MinimumReplicasAvailable |
| Progressing    | False  | ProgressDeadlineExceeded |
| ReplicaFailure | True   | FailedCreate             |

You can address an issue of insufficient quota by scaling down your Deployment, by scaling down other controllers you may be running, or by increasing quota in your namespace. If you satisfy the quota conditions and the Deployment controller then completes the Deployment rollout, you'll see the Deployment's status update with a successful condition (status: "True" and reason: NewReplicaSetAvailable).

Conditions:

| Type        | Status | Reason                   |
|-------------|--------|--------------------------|
| Available   | True   | MinimumReplicasAvailable |
| Progressing | True   | NewReplicaSetAvailable   |

type: Available with status: "True" means that your Deployment has minimum availability. Minimum availability is dictated by the parameters specified in the deployment strategy. type: Progressing with status: "True" means that your Deployment is either in the middle of a rollout and it is progressing or that it has successfully completed its progress and the minimum required new replicas are available (see the Reason of the condition for the particulars - in our case reason: NewReplicaSetAvailable means that the Deployment is complete).

You can check if a Deployment has failed to progress by using `kubectl rollout status`. `kubectl rollout status` returns a non-zero exit code if the Deployment has exceeded the progression deadline.

```
kubectl rollout status deployment/nginx-deployment
```

The output is similar to this:

```
Waiting for rollout to finish: 2 out of 3 new replicas have been updated...
error: deployment "nginx" exceeded its progress deadline
```

and the exit status from `kubectl rollout` is 1 (indicating an error):

```
echo $?
```

```
1
```

## Operating on a failed deployment

All actions that apply to a complete Deployment also apply to a failed Deployment. You can scale it up/down, roll back to a previous revision, or even pause it if you need to apply multiple tweaks in the Deployment Pod template.

## Clean up Policy

You can set `.spec.revisionHistoryLimit` field in a Deployment to specify how many old ReplicaSets for this Deployment you want to retain. The rest will be garbage-collected in the background. By default, it is 10.

### Note:

Explicitly setting this field to 0, will result in cleaning up all the history of your Deployment thus that Deployment will not be able to roll back.

## Canary Deployment

If you want to roll out releases to a subset of users or servers using the Deployment, you can create multiple Deployments, one for each release, following the canary pattern described in [managing resources](#).

## Writing a Deployment Spec

As with all other Kubernetes configs, a Deployment needs `.apiVersion`, `.kind`, and `.metadata` fields. For general information about working with config files, see [deploying applications](#), configuring containers, and [using kubectl to manage resources](#) documents.

When the control plane creates new Pods for a Deployment, the `.metadata.name` of the Deployment is part of the basis for naming those Pods. The name of a Deployment must be a valid [DNS subdomain](#) value, but this can produce unexpected results for the Pod hostnames. For best compatibility, the name should follow the more restrictive rules for a [DNS label](#).

A Deployment also needs a [.spec section](#).

## Pod Template

The `.spec.template` and `.spec.selector` are the only required fields of the `.spec`.

The `.spec.template` is a [Pod template](#). It has exactly the same schema as a [Pod](#), except it is nested and does not have an `apiVersion` or `kind`.

In addition to required fields for a Pod, a Pod template in a Deployment must specify appropriate labels and an appropriate restart policy. For labels, make sure not to overlap with other controllers. See [selector](#).

Only a [.spec.template.spec.restartPolicy](#) equal to `Always` is allowed, which is the default if not specified.

## Replicas

`.spec.replicas` is an optional field that specifies the number of desired Pods. It defaults to 1.

Should you manually scale a Deployment, example via `kubectl scale deployment deployment --replicas=X`, and then you update that Deployment based on a manifest (for example: by running `kubectl apply -f deployment.yaml`), then applying that manifest overwrites the manual scaling that you previously did.

If a [HorizontalPodAutoscaler](#) (or any similar API for horizontal scaling) is managing scaling for a Deployment, don't set `.spec.replicas`.

Instead, allow the Kubernetes [control plane](#) to manage the `.spec.replicas` field automatically.

## Selector

`.spec.selector` is a required field that specifies a [label selector](#) for the Pods targeted by this Deployment.

`.spec.selector` must match `.spec.template.metadata.labels`, or it will be rejected by the API.

In API version `apps/v1`, `.spec.selector` and `.metadata.labels` do not default to `.spec.template.metadata.labels` if not set. So they must be set explicitly. Also note that `.spec.selector` is immutable after creation of the Deployment in `apps/v1`.

A Deployment may terminate Pods whose labels match the selector if their template is different from `.spec.template` or if the total number of such Pods exceeds `.spec.replicas`. It brings up new Pods with `.spec.template` if the number of Pods is less than the desired number.

### Note:

You should not create other Pods whose labels match this selector, either directly, by creating another Deployment, or by creating another controller such as a ReplicaSet or a ReplicationController. If you do so, the first Deployment thinks that it created these other Pods. Kubernetes does not stop you from doing this.

If you have multiple controllers that have overlapping selectors, the controllers will fight with each other and won't behave correctly.

## Strategy

`.spec.strategy` specifies the strategy used to replace old Pods by new ones. `.spec.strategy.type` can be "Recreate" or "RollingUpdate". "RollingUpdate" is the default value.

### Recreate Deployment

All existing Pods are killed before new ones are created when `.spec.strategy.type==Recreate`.

### Note:

This will only guarantee Pod termination previous to creation for upgrades. If you upgrade a Deployment, all Pods of the old revision will be terminated immediately. Successful removal is

awaited before any Pod of the new revision is created. If you manually delete a Pod, the lifecycle is controlled by the ReplicaSet and the replacement will be created immediately (even if the old Pod is still in a Terminating state). If you need an "at most" guarantee for your Pods, you should consider using a [StatefulSet](#).

## Rolling Update Deployment

The Deployment updates Pods in a rolling update fashion when `.spec.strategy.type==RollingUpdate`. You can specify `maxUnavailable` and `maxSurge` to control the rolling update process.

### Max Unavailable

`.spec.strategy.rollingUpdate.maxUnavailable` is an optional field that specifies the maximum number of Pods that can be unavailable during the update process. The value can be an absolute number (for example, 5) or a percentage of desired Pods (for example, 10%). The absolute number is calculated from percentage by rounding down. The value cannot be 0 if `.spec.strategy.rollingUpdate.maxSurge` is 0. The default value is 25%.

For example, when this value is set to 30%, the old ReplicaSet can be scaled down to 70% of desired Pods immediately when the rolling update starts. Once new Pods are ready, old ReplicaSet can be scaled down further, followed by scaling up the new ReplicaSet, ensuring that the total number of Pods available at all times during the update is at least 70% of the desired Pods.

### Max Surge

`.spec.strategy.rollingUpdate.maxSurge` is an optional field that specifies the maximum number of Pods that can be created over the desired number of Pods. The value can be an absolute number (for example, 5) or a percentage of desired Pods (for example, 10%). The value cannot be 0 if `MaxUnavailable` is 0. The absolute number is calculated from the percentage by rounding up. The default value is 25%.

For example, when this value is set to 30%, the new ReplicaSet can be scaled up immediately when the rolling update starts, such that the total number of old and new Pods does not exceed 130% of desired Pods. Once old Pods have been killed, the new ReplicaSet can be scaled up further, ensuring that the total number of Pods running at any time during the update is at most 130% of desired Pods.

Here are some Rolling Update Deployment examples that use the `maxUnavailable` and `maxSurge`:

- [Max Unavailable](#)
- [Max Surge](#)
- [Hybrid](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
 labels:
 app: nginx
spec:
```

```
replicas: 3
selector:
 matchLabels:
 app: nginx
template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
strategy:
 type: RollingUpdate
rollingUpdate:
 maxUnavailable: 1
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
 labels:
 app: nginx
spec:
 replicas: 3
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
 strategy:
 type: RollingUpdate
 rollingUpdate:
 maxSurge: 1
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
 labels:
 app: nginx
spec:
 replicas: 3
```

```
selector:
 matchLabels:
 app: nginx
template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
 strategy:
 type: RollingUpdate
 rollingUpdate:
 maxSurge: 1
 maxUnavailable: 1
```

## Progress Deadline Seconds

`.spec.progressDeadlineSeconds` is an optional field that specifies the number of seconds you want to wait for your Deployment to progress before the system reports back that the Deployment has [failed progressing](#) - surfaced as a condition with type: Progressing, status: "False", and reason: ProgressDeadlineExceeded in the status of the resource. The Deployment controller will keep retrying the Deployment. This defaults to 600. In the future, once automatic rollback will be implemented, the Deployment controller will roll back a Deployment as soon as it observes such a condition.

If specified, this field needs to be greater than `.spec.minReadySeconds`.

## Min Ready Seconds

`.spec.minReadySeconds` is an optional field that specifies the minimum number of seconds for which a newly created Pod should be ready without any of its containers crashing, for it to be considered available. This defaults to 0 (the Pod will be considered available as soon as it is ready). To learn more about when a Pod is considered ready, see [Container Probes](#).

## Revision History Limit

A Deployment's revision history is stored in the ReplicaSets it controls.

`.spec.revisionHistoryLimit` is an optional field that specifies the number of old ReplicaSets to retain to allow rollback. These old ReplicaSets consume resources in etcd and crowd the output of `kubectl get rs`. The configuration of each Deployment revision is stored in its ReplicaSets; therefore, once an old ReplicaSet is deleted, you lose the ability to rollback to that revision of Deployment. By default, 10 old ReplicaSets will be kept, however its ideal value depends on the frequency and stability of new Deployments.

More specifically, setting this field to zero means that all old ReplicaSets with 0 replicas will be cleaned up. In this case, a new Deployment rollout cannot be undone, since its revision history is cleaned up.



## Paused

`.spec.paused` is an optional boolean field for pausing and resuming a Deployment. The only difference between a paused Deployment and one that is not paused, is that any changes into the `PodTemplateSpec` of the paused Deployment will not trigger new rollouts as long as it is paused. A Deployment is not paused by default when it is created.

## What's next

- Learn more about [Pods](#).
- [Run a stateless application using a Deployment](#).
- Read the [Deployment](#) to understand the Deployment API.
- Read about [PodDisruptionBudget](#) and how you can use it to manage application availability during disruptions.
- Use `kubectl` to [create a Deployment](#).

# ReplicaSet

A ReplicaSet's purpose is to maintain a stable set of replica Pods running at any given time. Usually, you define a Deployment and let that Deployment manage ReplicaSets automatically.

A ReplicaSet's purpose is to maintain a stable set of replica Pods running at any given time. As such, it is often used to guarantee the availability of a specified number of identical Pods.

## How a ReplicaSet works

A ReplicaSet is defined with fields, including a selector that specifies how to identify Pods it can acquire, a number of replicas indicating how many Pods it should be maintaining, and a pod template specifying the data of new Pods it should create to meet the number of replicas criteria. A ReplicaSet then fulfills its purpose by creating and deleting Pods as needed to reach the desired number. When a ReplicaSet needs to create new Pods, it uses its Pod template.

A ReplicaSet is linked to its Pods via the Pods' [metadata.ownerReferences](#) field, which specifies what resource the current object is owned by. All Pods acquired by a ReplicaSet have their owning ReplicaSet's identifying information within their `ownerReferences` field. It's through this link that the ReplicaSet knows of the state of the Pods it is maintaining and plans accordingly.

A ReplicaSet identifies new Pods to acquire by using its selector. If there is a Pod that has no `OwnerReference` or the `OwnerReference` is not a [Controller](#) and it matches a ReplicaSet's selector, it will be immediately acquired by said ReplicaSet.

## When to use a ReplicaSet

A ReplicaSet ensures that a specified number of pod replicas are running at any given time. However, a Deployment is a higher-level concept that manages ReplicaSets and provides declarative updates to Pods along with a lot of other useful features. Therefore, we recommend using Deployments instead of directly using ReplicaSets, unless you require custom update orchestration or don't require updates at all.

This actually means that you may never need to manipulate ReplicaSet objects: use a Deployment instead, and define your application in the spec section.

## Example

[controllers/frontend.yaml](#)

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
 name: frontend
 labels:
 app: guestbook
 tier: frontend
spec:
 # modify replicas according to your case
 replicas: 3
 selector:
 matchLabels:
 tier: frontend
 template:
 metadata:
 labels:
 tier: frontend
 spec:
 containers:
 - name: php-redis
 image: us-docker.pkg.dev/google-samples/containers/gke/gb-frontend:v5
```

Saving this manifest into frontend.yaml and submitting it to a Kubernetes cluster will create the defined ReplicaSet and the Pods that it manages.

```
kubectl apply -f https://kubernetes.io/examples/controllers/frontend.yaml
```

You can then get the current ReplicaSets deployed:

```
kubectl get rs
```

And see the frontend one you created:

| NAME     | DESIRED | CURRENT | READY | AGE |
|----------|---------|---------|-------|-----|
| frontend | 3       | 3       | 3     | 6s  |

You can also check on the state of the ReplicaSet:

```
kubectl describe rs/frontend
```

And you will see output similar to:

```
Name: frontend
Namespace: default
Selector: tier=frontend
Labels: app=guestbook
```

```

 tier=frontend
Annotations: <none>
Replicas: 3 current / 3 desired
Pods Status: 3 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
 Labels: tier=frontend
 Containers:
 php-redis:
 Image: us-docker.pkg.dev/google-samples/containers/gke/gb-frontend:v5
 Port: <none>
 Host Port: <none>
 Environment: <none>
 Mounts: <none>
 Volumes: <none>
Events:
 Type Reason Age From Message
 ---- -
Normal SuccessfulCreate 13s replicaset-controller Created pod: frontend-gbgfx
Normal SuccessfulCreate 13s replicaset-controller Created pod: frontend-rwz57
Normal SuccessfulCreate 13s replicaset-controller Created pod: frontend-wkl7w

```

And lastly you can check for the Pods brought up:

```
kubectl get pods
```

You should see Pod information similar to:

| NAME           | READY | STATUS  | RESTARTS | AGE |
|----------------|-------|---------|----------|-----|
| frontend-gbgfx | 1/1   | Running | 0        | 10m |
| frontend-rwz57 | 1/1   | Running | 0        | 10m |
| frontend-wkl7w | 1/1   | Running | 0        | 10m |

You can also verify that the owner reference of these pods is set to the frontend ReplicaSet. To do this, get the yaml of one of the Pods running:

```
kubectl get pods frontend-gbgfx -o yaml
```

The output will look similar to this, with the frontend ReplicaSet's info set in the metadata's ownerReferences field:

```

apiVersion: v1
kind: Pod
metadata:
 creationTimestamp: "2024-02-28T22:30:44Z"
 generateName: frontend-
 labels:
 tier: frontend
 name: frontend-gbgfx
 namespace: default
 ownerReferences:
 - apiVersion: apps/v1
 blockOwnerDeletion: true
 controller: true
 kind: ReplicaSet

```

```
name: frontend
uid: e129deca-f864-481b-bb16-b27abfd92292
```

...

## Non-Template Pod acquisitions

While you can create bare Pods with no problems, it is strongly recommended to make sure that the bare Pods do not have labels which match the selector of one of your ReplicaSets. The reason for this is because a ReplicaSet is not limited to owning Pods specified by its template-- it can acquire other Pods in the manner specified in the previous sections.

Take the previous frontend ReplicaSet example, and the Pods specified in the following manifest:

[pods/pod-rs.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: pod1
 labels:
 tier: frontend
spec:
 containers:
 - name: hello1
 image: gcr.io/google-samples/hello-app:2.0
```

---

```
apiVersion: v1
kind: Pod
metadata:
 name: pod2
 labels:
 tier: frontend
spec:
 containers:
 - name: hello2
 image: gcr.io/google-samples/hello-app:1.0
```

As those Pods do not have a Controller (or any object) as their owner reference and match the selector of the frontend ReplicaSet, they will immediately be acquired by it.

Suppose you create the Pods after the frontend ReplicaSet has been deployed and has set up its initial Pod replicas to fulfill its replica count requirement:

```
kubectl apply -f https://kubernetes.io/examples/pods/pod-rs.yaml
```

The new Pods will be acquired by the ReplicaSet, and then immediately terminated as the ReplicaSet would be over its desired count.

Fetching the Pods:

```
kubectl get pods
```

The output shows that the new Pods are either already terminated, or in the process of being terminated:

| NAME            | READY | STATUS      | RESTARTS | AGE |
|-----------------|-------|-------------|----------|-----|
| frontend-b2zdvd | 1/1   | Running     | 0        | 10m |
| frontend-vcmts  | 1/1   | Running     | 0        | 10m |
| frontend-wtsmm  | 1/1   | Running     | 0        | 10m |
| pod1            | 0/1   | Terminating | 0        | 1s  |
| pod2            | 0/1   | Terminating | 0        | 1s  |

If you create the Pods first:

```
kubectl apply -f https://kubernetes.io/examples/pods/pod-rs.yaml
```

And then create the ReplicaSet however:

```
kubectl apply -f https://kubernetes.io/examples/controllers/frontend.yaml
```

You shall see that the ReplicaSet has acquired the Pods and has only created new ones according to its spec until the number of its new Pods and the original matches its desired count. As fetching the Pods:

```
kubectl get pods
```

Will reveal in its output:

| NAME           | READY | STATUS  | RESTARTS | AGE |
|----------------|-------|---------|----------|-----|
| frontend-hmmj2 | 1/1   | Running | 0        | 9s  |
| pod1           | 1/1   | Running | 0        | 36s |
| pod2           | 1/1   | Running | 0        | 36s |

In this manner, a ReplicaSet can own a non-homogeneous set of Pods

## Writing a ReplicaSet manifest

As with all other Kubernetes API objects, a ReplicaSet needs the `apiVersion`, `kind`, and `metadata` fields. For ReplicaSets, the `kind` is always a `ReplicaSet`.

When the control plane creates new Pods for a ReplicaSet, the `.metadata.name` of the ReplicaSet is part of the basis for naming those Pods. The name of a ReplicaSet must be a valid [DNS subdomain](#) value, but this can produce unexpected results for the Pod hostnames. For best compatibility, the name should follow the more restrictive rules for a [DNS label](#).

A ReplicaSet also needs a [.spec section](#).

## Pod Template

The `.spec.template` is a [pod template](#) which is also required to have labels in place. In our `frontend.yaml` example we had one label: `tier: frontend`. Be careful not to overlap with the selectors of other controllers, lest they try to adopt this Pod.

For the template's [restart policy](#) field, `.spec.template.spec.restartPolicy`, the only allowed value is Always, which is the default.

## Pod Selector

The `.spec.selector` field is a [label selector](#). As discussed [earlier](#) these are the labels used to identify potential Pods to acquire. In our `frontend.yaml` example, the selector was:

```
matchLabels:
 tier: frontend
```

In the `ReplicaSet`, `.spec.template.metadata.labels` must match `spec.selector`, or it will be rejected by the API.

### Note:

For 2 `ReplicaSets` specifying the same `.spec.selector` but different `.spec.template.metadata.labels` and `.spec.template.spec` fields, each `ReplicaSet` ignores the Pods created by the other `ReplicaSet`.

## Replicas

You can specify how many Pods should run concurrently by setting `.spec.replicas`. The `ReplicaSet` will create/delete its Pods to match this number.

If you do not specify `.spec.replicas`, then it defaults to 1.

## Working with ReplicaSets

### Deleting a ReplicaSet and its Pods

To delete a `ReplicaSet` and all of its Pods, use [kubectl delete](#). The [Garbage collector](#) automatically deletes all of the dependent Pods by default.

When using the REST API or the client-go library, you must set `propagationPolicy` to Background or Foreground in the `-d` option. For example:

```
kubectl proxy --port=8080
curl -X DELETE 'localhost:8080/apis/apps/v1/namespaces/default/replicasets/frontend' \
-d '{"kind":"DeleteOptions","apiVersion":"v1","propagationPolicy":"Foreground"}' \
-H "Content-Type: application/json"
```

### Deleting just a ReplicaSet

You can delete a `ReplicaSet` without affecting any of its Pods using [kubectl delete](#) with the `--cascade=orphan` option. When using the REST API or the client-go library, you must set `propagationPolicy` to Orphan. For example:

```
kubectl proxy --port=8080
curl -X DELETE 'localhost:8080/apis/apps/v1/namespaces/default/replicasets/frontend' \
-d '{"kind":"DeleteOptions","apiVersion":"v1","propagationPolicy":"Orphan"}' \
-H "Content-Type: application/json"
```

Once the original is deleted, you can create a new ReplicaSet to replace it. As long as the old and new `.spec.selector` are the same, then the new one will adopt the old Pods. However, it will not make any effort to make existing Pods match a new, different pod template. To update Pods to a new spec in a controlled way, use a [Deployment](#), as ReplicaSets do not support a rolling update directly.

## Isolating Pods from a ReplicaSet

You can remove Pods from a ReplicaSet by changing their labels. This technique may be used to remove Pods from service for debugging, data recovery, etc. Pods that are removed in this way will be replaced automatically (assuming that the number of replicas is not also changed).

## Scaling a ReplicaSet

A ReplicaSet can be easily scaled up or down by simply updating the `.spec.replicas` field. The ReplicaSet controller ensures that a desired number of Pods with a matching label selector are available and operational.

When scaling down, the ReplicaSet controller chooses which pods to delete by sorting the available pods to prioritize scaling down pods based on the following general algorithm:

1. Pending (and unschedulable) pods are scaled down first
2. If `controller.kubernetes.io/pod-deletion-cost` annotation is set, then the pod with the lower value will come first.
3. Pods on nodes with more replicas come before pods on nodes with fewer replicas.
4. If the pods' creation times differ, the pod that was created more recently comes before the older pod (the creation times are bucketed on an integer log scale when the `LogarithmicScaleDown` [feature gate](#) is enabled)

If all of the above match, then selection is random.

## Pod deletion cost

FEATURE STATE: Kubernetes v1.22 [beta]

Using the [controller.kubernetes.io/pod-deletion-cost](#) annotation, users can set a preference regarding which pods to remove first when downscaling a ReplicaSet.

The annotation should be set on the pod, the range is `[-2147483648, 2147483647]`. It represents the cost of deleting a pod compared to other pods belonging to the same ReplicaSet. Pods with lower deletion cost are preferred to be deleted before pods with higher deletion cost.

The implicit value for this annotation for pods that don't set it is 0; negative values are permitted. Invalid values will be rejected by the API server.

This feature is beta and enabled by default. You can disable it using the [feature gate](#) `PodDeletionCost` in both `kube-apiserver` and `kube-controller-manager`.

### Note:

- This is honored on a best-effort basis, so it does not offer any guarantees on pod deletion order.

- Users should avoid updating the annotation frequently, such as updating it based on a metric value, because doing so will generate a significant number of pod updates on the apiserver.

## Example Use Case

The different pods of an application could have different utilization levels. On scale down, the application may prefer to remove the pods with lower utilization. To avoid frequently updating the pods, the application should update `controller.kubernetes.io/pod-deletion-cost` once before issuing a scale down (setting the annotation to a value proportional to pod utilization level). This works if the application itself controls the down scaling; for example, the driver pod of a Spark deployment.

## ReplicaSet as a Horizontal Pod Autoscaler Target

A ReplicaSet can also be a target for [Horizontal Pod Autoscalers \(HPA\)](#). That is, a ReplicaSet can be auto-scaled by an HPA. Here is an example HPA targeting the ReplicaSet we created in the previous example.

[controllers/hpa-rs.yaml](#)

```
apiVersion: autoscaling/v1
kind: HorizontalPodAutoscaler
metadata:
 name: frontend-scaler
spec:
 scaleTargetRef:
 kind: ReplicaSet
 name: frontend
 minReplicas: 3
 maxReplicas: 10
 targetCPUUtilizationPercentage: 50
```

Saving this manifest into `hpa-rs.yaml` and submitting it to a Kubernetes cluster should create the defined HPA that autoscales the target ReplicaSet depending on the CPU usage of the replicated Pods.

```
kubectl apply -f https://k8s.io/examples/controllers/hpa-rs.yaml
```

Alternatively, you can use the `kubectl autoscale` command to accomplish the same (and it's easier!)

```
kubectl autoscale rs frontend --max=10 --min=3 --cpu-percent=50
```

## Alternatives to ReplicaSet

### Deployment (recommended)

[Deployment](#) is an object which can own ReplicaSets and update them and their Pods via declarative, server-side rolling updates. While ReplicaSets can be used independently, today they're mainly used by Deployments as a mechanism to orchestrate Pod creation, deletion and updates. When you use Deployments you don't have to worry about managing the ReplicaSets



that they create. Deployments own and manage their ReplicaSets. As such, it is recommended to use Deployments when you want ReplicaSets.

## Bare Pods

Unlike the case where a user directly created Pods, a ReplicaSet replaces Pods that are deleted or terminated for any reason, such as in the case of node failure or disruptive node maintenance, such as a kernel upgrade. For this reason, we recommend that you use a ReplicaSet even if your application requires only a single Pod. Think of it similarly to a process supervisor, only it supervises multiple Pods across multiple nodes instead of individual processes on a single node. A ReplicaSet delegates local container restarts to some agent on the node such as Kubelet.

## Job

Use a [Job](#) instead of a ReplicaSet for Pods that are expected to terminate on their own (that is, batch jobs).

## DaemonSet

Use a [DaemonSet](#) instead of a ReplicaSet for Pods that provide a machine-level function, such as machine monitoring or machine logging. These Pods have a lifetime that is tied to a machine lifetime: the Pod needs to be running on the machine before other Pods start, and are safe to terminate when the machine is otherwise ready to be rebooted/shutdown.

## ReplicationController

ReplicaSets are the successors to [ReplicationControllers](#). The two serve the same purpose, and behave similarly, except that a ReplicationController does not support set-based selector requirements as described in the [labels user guide](#). As such, ReplicaSets are preferred over ReplicationControllers

## What's next

- Learn about [Pods](#).
- Learn about [Deployments](#).
- [Run a Stateless Application Using a Deployment](#), which relies on ReplicaSets to work.
- ReplicaSet is a top-level resource in the Kubernetes REST API. Read the [ReplicaSet](#) object definition to understand the API for replica sets.
- Read about [PodDisruptionBudget](#) and how you can use it to manage application availability during disruptions.

## StatefulSets

A StatefulSet runs a group of Pods, and maintains a sticky identity for each of those Pods. This is useful for managing applications that need persistent storage or a stable, unique network identity.

StatefulSet is the workload API object used to manage stateful applications.

Manages the deployment and scaling of a set of [Pods](#), *and provides guarantees about the ordering and uniqueness* of these Pods.

Like a [Deployment](#), a StatefulSet manages Pods that are based on an identical container spec. Unlike a Deployment, a StatefulSet maintains a sticky identity for each of its Pods. These pods are created from the same spec, but are not interchangeable: each has a persistent identifier that it maintains across any rescheduling.

If you want to use storage volumes to provide persistence for your workload, you can use a StatefulSet as part of the solution. Although individual Pods in a StatefulSet are susceptible to failure, the persistent Pod identifiers make it easier to match existing volumes to the new Pods that replace any that have failed.

## Using StatefulSets

StatefulSets are valuable for applications that require one or more of the following.

- Stable, unique network identifiers.
- Stable, persistent storage.
- Ordered, graceful deployment and scaling.
- Ordered, automated rolling updates.

In the above, stable is synonymous with persistence across Pod (re)scheduling. If an application doesn't require any stable identifiers or ordered deployment, deletion, or scaling, you should deploy your application using a workload object that provides a set of stateless replicas. [Deployment](#) or [ReplicaSet](#) may be better suited to your stateless needs.

## Limitations

- The storage for a given Pod must either be provisioned by a [PersistentVolume Provisioner](#) ([examples here](#)) based on the requested *storage class*, or pre-provisioned by an admin.
- Deleting and/or scaling a StatefulSet down will *not* delete the volumes associated with the StatefulSet. This is done to ensure data safety, which is generally more valuable than an automatic purge of all related StatefulSet resources.
- StatefulSets currently require a [Headless Service](#) to be responsible for the network identity of the Pods. You are responsible for creating this Service.
- StatefulSets do not provide any guarantees on the termination of pods when a StatefulSet is deleted. To achieve ordered and graceful termination of the pods in the StatefulSet, it is possible to scale the StatefulSet down to 0 prior to deletion.
- When using [Rolling Updates](#) with the default [Pod Management Policy](#) (OrderedReady), it's possible to get into a broken state that requires [manual intervention to repair](#).

## Components

The example below demonstrates the components of a StatefulSet.

```
apiVersion: v1
kind: Service
metadata:
 name: nginx
 labels:
```

```

 app: nginx
spec:
 ports:
 - port: 80
 name: web
 clusterIP: None
 selector:
 app: nginx

apiVersion: apps/v1
kind: StatefulSet
metadata:
 name: web
spec:
 selector:
 matchLabels:
 app: nginx # has to match .spec.template.metadata.labels
 serviceName: "nginx"
 replicas: 3 # by default is 1
 minReadySeconds: 10 # by default is 0
 template:
 metadata:
 labels:
 app: nginx # has to match .spec.selector.matchLabels
 spec:
 terminationGracePeriodSeconds: 10
 containers:
 - name: nginx
 image: registry.k8s.io/nginx-slim:0.8
 ports:
 - containerPort: 80
 name: web
 volumeMounts:
 - name: www
 mountPath: /usr/share/nginx/html
 volumeClaimTemplates:
 - metadata:
 name: www
 spec:
 accessModes: ["ReadWriteOnce"]
 storageClassName: "my-storage-class"
 resources:
 requests:
 storage: 1Gi

```

### Note:

This example uses the ReadWriteOnce access mode, for simplicity. For production use, the Kubernetes project recommends using the ReadWriteOncePod access mode instead.

In the above example:

- A Headless Service, named `nginx`, is used to control the network domain.

- The StatefulSet, named web, has a Spec that indicates that 3 replicas of the nginx container will be launched in unique Pods.
- The volumeClaimTemplates will provide stable storage using [PersistentVolumes](#) provisioned by a PersistentVolume Provisioner.

The name of a StatefulSet object must be a valid [DNS label](#).

## Pod Selector

You must set the .spec.selector field of a StatefulSet to match the labels of its .spec.template.metadata.labels. Failing to specify a matching Pod Selector will result in a validation error during StatefulSet creation.

## Volume Claim Templates

You can set the .spec.volumeClaimTemplates field to create a [PersistentVolumeClaim](#). This will provide stable storage to the StatefulSet if either

- The StorageClass specified for the volume claim is set up to use [dynamic provisioning](#), or
- The cluster already contains a PersistentVolume with the correct StorageClass and sufficient available storage space.

## Minimum ready seconds

FEATURE STATE: Kubernetes v1.25 [stable]

.spec.minReadySeconds is an optional field that specifies the minimum number of seconds for which a newly created Pod should be running and ready without any of its containers crashing, for it to be considered available. This is used to check progression of a rollout when using a [Rolling Update](#) strategy. This field defaults to 0 (the Pod will be considered available as soon as it is ready). To learn more about when a Pod is considered ready, see [Container Probes](#).

## Pod Identity

StatefulSet Pods have a unique identity that consists of an ordinal, a stable network identity, and stable storage. The identity sticks to the Pod, regardless of which node it's (re)scheduled on.

## Ordinal Index

For a StatefulSet with N [replicas](#), each Pod in the StatefulSet will be assigned an integer ordinal, that is unique over the Set. By default, pods will be assigned ordinals from 0 up through N-1. The StatefulSet controller will also add a pod label with this index: apps.kubernetes.io/pod-index.

## Start ordinal

FEATURE STATE: Kubernetes v1.27 [beta]

`.spec.ordinals` is an optional field that allows you to configure the integer ordinals assigned to each Pod. It defaults to nil. You must enable the `StatefulSetStartOrdinal` [feature gate](#) to use this field. Once enabled, you can configure the following options:

- `.spec.ordinals.start`: If the `.spec.ordinals.start` field is set, Pods will be assigned ordinals from `.spec.ordinals.start` up through `.spec.ordinals.start + .spec.replicas - 1`.

## Stable Network ID

Each Pod in a StatefulSet derives its hostname from the name of the StatefulSet and the ordinal of the Pod. The pattern for the constructed hostname is `$(statefulset name)-$(ordinal)`. The example above will create three Pods named `web-0`, `web-1`, `web-2`. A StatefulSet can use a [Headless Service](#) to control the domain of its Pods. The domain managed by this Service takes the form: `$(service name).$(namespace).svc.cluster.local`, where "cluster.local" is the cluster domain. As each Pod is created, it gets a matching DNS subdomain, taking the form: `$(podname).$(governing service domain)`, where the governing service is defined by the `serviceName` field on the StatefulSet.

Depending on how DNS is configured in your cluster, you may not be able to look up the DNS name for a newly-run Pod immediately. This behavior can occur when other clients in the cluster have already sent queries for the hostname of the Pod before it was created. Negative caching (normal in DNS) means that the results of previous failed lookups are remembered and reused, even after the Pod is running, for at least a few seconds.

If you need to discover Pods promptly after they are created, you have a few options:

- Query the Kubernetes API directly (for example, using a watch) rather than relying on DNS lookups.
- Decrease the time of caching in your Kubernetes DNS provider (typically this means editing the config map for CoreDNS, which currently caches for 30 seconds).

As mentioned in the [limitations](#) section, you are responsible for creating the [Headless Service](#) responsible for the network identity of the pods.

Here are some examples of choices for Cluster Domain, Service name, StatefulSet name, and how that affects the DNS names for the StatefulSet's Pods.

| Cluster Domain | Service (ns/name) | StatefulSet (ns/name) | StatefulSet Domain              | Pod DNS                                      | Pod Hostname |
|----------------|-------------------|-----------------------|---------------------------------|----------------------------------------------|--------------|
| cluster.local  | default/nginx     | default/web           | nginx.default.svc.cluster.local | web-{0..N-1}.nginx.default.svc.cluster.local | web-{0..N-1} |
| cluster.local  | foo/nginx         | foo/web               | nginx.foo.svc.cluster.local     | web-{0..N-1}.nginx.foo.svc.cluster.local     | web-{0..N-1} |
| kube.local     | foo/nginx         | foo/web               | nginx.foo.svc.kube.local        | web-{0..N-1}.nginx.foo.svc.kube.local        | web-{0..N-1} |

### Note:

Cluster Domain will be set to `cluster.local` unless [otherwise configured](#).

## Stable Storage

For each VolumeClaimTemplate entry defined in a StatefulSet, each Pod receives one PersistentVolumeClaim. In the nginx example above, each Pod receives a single PersistentVolume with a StorageClass of my-storage-class and 1 GiB of provisioned storage. If no StorageClass is specified, then the default StorageClass will be used. When a Pod is (re)scheduled onto a node, its volumeMounts mount the PersistentVolumes associated with its PersistentVolume Claims. Note that, the PersistentVolumes associated with the Pods' PersistentVolume Claims are not deleted when the Pods, or StatefulSet are deleted. This must be done manually.

## Pod Name Label

When the StatefulSet [controller](#) creates a Pod, it adds a label, statefulset.kubernetes.io/pod-name, that is set to the name of the Pod. This label allows you to attach a Service to a specific Pod in the StatefulSet.

## Pod index label

FEATURE STATE: Kubernetes v1.28 [beta]

When the StatefulSet [controller](#) creates a Pod, the new Pod is labelled with apps.kubernetes.io/pod-index. The value of this label is the ordinal index of the Pod. This label allows you to route traffic to a particular pod index, filter logs/metrics using the pod index label, and more. Note the feature gate PodIndexLabel must be enabled for this feature, and it is enabled by default.

## Deployment and Scaling Guarantees

- For a StatefulSet with N replicas, when Pods are being deployed, they are created sequentially, in order from {0..N-1}.
- When Pods are being deleted, they are terminated in reverse order, from {N-1..0}.
- Before a scaling operation is applied to a Pod, all of its predecessors must be Running and Ready.
- Before a Pod is terminated, all of its successors must be completely shutdown.

The StatefulSet should not specify a pod.Spec.TerminationGracePeriodSeconds of 0. This practice is unsafe and strongly discouraged. For further explanation, please refer to [force deleting StatefulSet Pods](#).

When the nginx example above is created, three Pods will be deployed in the order web-0, web-1, web-2. web-1 will not be deployed before web-0 is [Running and Ready](#), and web-2 will not be deployed until web-1 is Running and Ready. If web-0 should fail, after web-1 is Running and Ready, but before web-2 is launched, web-2 will not be launched until web-0 is successfully relaunched and becomes Running and Ready.

If a user were to scale the deployed example by patching the StatefulSet such that replicas=1, web-2 would be terminated first. web-1 would not be terminated until web-2 is fully shutdown and deleted. If web-0 were to fail after web-2 has been terminated and is completely shutdown, but prior to web-1's termination, web-1 would not be terminated until web-0 is Running and Ready.

## Pod Management Policies

StatefulSet allows you to relax its ordering guarantees while preserving its uniqueness and identity guarantees via its `.spec.podManagementPolicy` field.

### OrderedReady Pod Management

OrderedReady pod management is the default for StatefulSets. It implements the behavior described [above](#).

### Parallel Pod Management

Parallel pod management tells the StatefulSet controller to launch or terminate all Pods in parallel, and to not wait for Pods to become Running and Ready or completely terminated prior to launching or terminating another Pod. This option only affects the behavior for scaling operations. Updates are not affected.

## Update strategies

A StatefulSet's `.spec.updateStrategy` field allows you to configure and disable automated rolling updates for containers, labels, resource request/limits, and annotations for the Pods in a StatefulSet. There are two possible values:

#### OnDelete

When a StatefulSet's `.spec.updateStrategy.type` is set to `OnDelete`, the StatefulSet controller will not automatically update the Pods in a StatefulSet. Users must manually delete Pods to cause the controller to create new Pods that reflect modifications made to a StatefulSet's `.spec.template`.

#### RollingUpdate

The `RollingUpdate` update strategy implements automated, rolling updates for the Pods in a StatefulSet. This is the default update strategy.

## Rolling Updates

When a StatefulSet's `.spec.updateStrategy.type` is set to `RollingUpdate`, the StatefulSet controller will delete and recreate each Pod in the StatefulSet. It will proceed in the same order as Pod termination (from the largest ordinal to the smallest), updating each Pod one at a time.

The Kubernetes control plane waits until an updated Pod is Running and Ready prior to updating its predecessor. If you have set `.spec.minReadySeconds` (see [Minimum Ready Seconds](#)), the control plane additionally waits that amount of time after the Pod turns ready, before moving on.

### Partitioned rolling updates

The `RollingUpdate` update strategy can be partitioned, by specifying a `.spec.updateStrategy.rollingUpdate.partition`. If a partition is specified, all Pods with an ordinal that is greater than or equal to the partition will be updated when the StatefulSet's `.spec.template` is updated. All Pods with an ordinal that is less than the partition will not be updated, and, even if they are deleted, they will be recreated at the previous version. If a StatefulSet's `.spec.updateStrategy.rollingUpdate.partition` is greater than its `.spec.replicas`,



updates to its `.spec.template` will not be propagated to its Pods. In most cases you will not need to use a partition, but they are useful if you want to stage an update, roll out a canary, or perform a phased roll out.

## Maximum unavailable Pods

FEATURE STATE: Kubernetes v1.24 [alpha]

You can control the maximum number of Pods that can be unavailable during an update by specifying the `.spec.updateStrategy.rollingUpdate.maxUnavailable` field. The value can be an absolute number (for example, 5) or a percentage of desired Pods (for example, 10%). Absolute number is calculated from the percentage value by rounding it up. This field cannot be 0. The default setting is 1.

This field applies to all Pods in the range 0 to `replicas - 1`. If there is any unavailable Pod in the range 0 to `replicas - 1`, it will be counted towards `maxUnavailable`.

### Note:

The `maxUnavailable` field is in Alpha stage and it is honored only by API servers that are running with the `MaxUnavailableStatefulSet` [feature gate](#) enabled.

## Forced rollback

When using [Rolling Updates](#) with the default [Pod Management Policy](#) (OrderedReady), it's possible to get into a broken state that requires manual intervention to repair.

If you update the Pod template to a configuration that never becomes Running and Ready (for example, due to a bad binary or application-level configuration error), StatefulSet will stop the rollout and wait.

In this state, it's not enough to revert the Pod template to a good configuration. Due to a [known issue](#), StatefulSet will continue to wait for the broken Pod to become Ready (which never happens) before it will attempt to revert it back to the working configuration.

After reverting the template, you must also delete any Pods that StatefulSet had already attempted to run with the bad configuration. StatefulSet will then begin to recreate the Pods using the reverted template.

## PersistentVolumeClaim retention

FEATURE STATE: Kubernetes v1.27 [beta]

The optional `.spec.persistentVolumeClaimRetentionPolicy` field controls if and how PVCs are deleted during the lifecycle of a StatefulSet. You must enable the `StatefulSetAutoDeletePVC` [feature gate](#) on the API server and the controller manager to use this field. Once enabled, there are two policies you can configure for each StatefulSet:

`whenDeleted`

configures the volume retention behavior that applies when the StatefulSet is deleted

`whenScaled`



configures the volume retention behavior that applies when the replica count of the StatefulSet is reduced; for example, when scaling down the set.

For each policy that you can configure, you can set the value to either Delete or Retain.

#### Delete

The PVCs created from the StatefulSet `volumeClaimTemplate` are deleted for each Pod affected by the policy. With the `whenDeleted` policy all PVCs from the `volumeClaimTemplate` are deleted after their Pods have been deleted. With the `whenScaled` policy, only PVCs corresponding to Pod replicas being scaled down are deleted, after their Pods have been deleted.

#### Retain (default)

PVCs from the `volumeClaimTemplate` are not affected when their Pod is deleted. This is the behavior before this new feature.

Bear in mind that these policies **only** apply when Pods are being removed due to the StatefulSet being deleted or scaled down. For example, if a Pod associated with a StatefulSet fails due to node failure, and the control plane creates a replacement Pod, the StatefulSet retains the existing PVC. The existing volume is unaffected, and the cluster will attach it to the node where the new Pod is about to launch.

The default for policies is Retain, matching the StatefulSet behavior before this new feature.

Here is an example policy.

```
apiVersion: apps/v1
kind: StatefulSet
...
spec:
 persistentVolumeClaimRetentionPolicy:
 whenDeleted: Retain
 whenScaled: Delete
...
```

The StatefulSet [controller](#) adds [owner references](#) to its PVCs, which are then deleted by the [garbage collector](#) after the Pod is terminated. This enables the Pod to cleanly unmount all volumes before the PVCs are deleted (and before the backing PV and volume are deleted, depending on the retain policy). When you set the `whenDeleted` policy to Delete, an owner reference to the StatefulSet instance is placed on all PVCs associated with that StatefulSet.

The `whenScaled` policy must delete PVCs only when a Pod is scaled down, and not when a Pod is deleted for another reason. When reconciling, the StatefulSet controller compares its desired replica count to the actual Pods present on the cluster. Any StatefulSet Pod whose id greater than the replica count is condemned and marked for deletion. If the `whenScaled` policy is Delete, the condemned Pods are first set as owners to the associated StatefulSet template PVCs, before the Pod is deleted. This causes the PVCs to be garbage collected after only the condemned Pods have terminated.

This means that if the controller crashes and restarts, no Pod will be deleted before its owner reference has been updated appropriate to the policy. If a condemned Pod is force-deleted while the controller is down, the owner reference may or may not have been set up, depending on when the controller crashed. It may take several reconcile loops to update the owner references, so some condemned Pods may have set up owner references and others may not. For this reason we recommend waiting for the controller to come back up, which will verify owner

references before terminating Pods. If that is not possible, the operator should verify the owner references on PVCs to ensure the expected objects are deleted when Pods are force-deleted.

## Replicas

`.spec.replicas` is an optional field that specifies the number of desired Pods. It defaults to 1.

Should you manually scale a deployment, example via `kubectl scale statefulset statefulset -- replicas=X`, and then you update that StatefulSet based on a manifest (for example: by running `kubectl apply -f statefulset.yaml`), then applying that manifest overwrites the manual scaling that you previously did.

If a [HorizontalPodAutoscaler](#) (or any similar API for horizontal scaling) is managing scaling for a StatefulSet, don't set `.spec.replicas`. Instead, allow the Kubernetes [control plane](#) to manage the `.spec.replicas` field automatically.

## What's next

- Learn about [Pods](#).
- Find out how to use StatefulSets
  - Follow an example of [deploying a stateful application](#).
  - Follow an example of [deploying Cassandra with Stateful Sets](#).
  - Follow an example of [running a replicated stateful application](#).
  - Learn how to [scale a StatefulSet](#).
  - Learn what's involved when you [delete a StatefulSet](#).
  - Learn how to [configure a Pod to use a volume for storage](#).
  - Learn how to [configure a Pod to use a PersistentVolume for storage](#).
- StatefulSet is a top-level resource in the Kubernetes REST API. Read the [StatefulSet](#) object definition to understand the API for stateful sets.
- Read about [PodDisruptionBudget](#) and how you can use it to manage application availability during disruptions.

## DaemonSet

A DaemonSet defines Pods that provide node-local facilities. These might be fundamental to the operation of your cluster, such as a networking helper tool, or be part of an add-on.

A *DemonSet* ensures that all (or some) Nodes run a copy of a Pod. As nodes are added to the cluster, Pods are added to them. As nodes are removed from the cluster, those Pods are garbage collected. Deleting a DaemonSet will clean up the Pods it created.

Some typical uses of a DaemonSet are:

- running a cluster storage daemon on every node
- running a logs collection daemon on every node
- running a node monitoring daemon on every node

In a simple case, one DaemonSet, covering all nodes, would be used for each type of daemon. A more complex setup might use multiple DaemonSets for a single type of daemon, but with different flags and/or different memory and cpu requests for different hardware types.

# Writing a DaemonSet Spec

## Create a DaemonSet

You can describe a DaemonSet in a YAML file. For example, the daemonset.yaml file below describes a DaemonSet that runs the fluentd-elasticsearch Docker image:

[controllers/daemonset.yaml](#)

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
 name: fluentd-elasticsearch
 namespace: kube-system
 labels:
 k8s-app: fluentd-logging
spec:
 selector:
 matchLabels:
 name: fluentd-elasticsearch
 template:
 metadata:
 labels:
 name: fluentd-elasticsearch
 spec:
 tolerations:
 # these tolerations are to have the daemonset runnable on control plane nodes
 # remove them if your control plane nodes should not run pods
 - key: node-role.kubernetes.io/control-plane
 operator: Exists
 effect: NoSchedule
 - key: node-role.kubernetes.io/master
 operator: Exists
 effect: NoSchedule
 containers:
 - name: fluentd-elasticsearch
 image: quay.io/fluentd_elasticsearch/fluentd:v2.5.2
 resources:
 limits:
 memory: 200Mi
 requests:
 cpu: 100m
 memory: 200Mi
 volumeMounts:
 - name: varlog
 mountPath: /var/log
 # it may be desirable to set a high priority class to ensure that a DaemonSet Pod
 # preempts running Pods
 # priorityClassName: important
 terminationGracePeriodSeconds: 30
 volumes:
 - name: varlog
```

```
hostPath:
 path: /var/log
```

Create a DaemonSet based on the YAML file:

```
kubectl apply -f https://k8s.io/examples/controllers/daemonset.yaml
```

## Required Fields

As with all other Kubernetes config, a DaemonSet needs `apiVersion`, `kind`, and `metadata` fields. For general information about working with config files, see [running stateless applications](#) and [object management using kubectl](#).

The name of a DaemonSet object must be a valid [DNS subdomain name](#).

A DaemonSet also needs a `.spec` section.

## Pod Template

The `.spec.template` is one of the required fields in `.spec`.

The `.spec.template` is a [pod template](#). It has exactly the same schema as a [Pod](#), except it is nested and does not have an `apiVersion` or `kind`.

In addition to required fields for a Pod, a Pod template in a DaemonSet has to specify appropriate labels (see [pod selector](#)).

A Pod Template in a DaemonSet must have a [RestartPolicy](#) equal to `Always`, or be unspecified, which defaults to `Always`.

## Pod Selector

The `.spec.selector` field is a pod selector. It works the same as the `.spec.selector` of a [Job](#).

You must specify a pod selector that matches the labels of the `.spec.template`. Also, once a DaemonSet is created, its `.spec.selector` can not be mutated. Mutating the pod selector can lead to the unintentional orphaning of Pods, and it was found to be confusing to users.

The `.spec.selector` is an object consisting of two fields:

- `matchLabels` - works the same as the `.spec.selector` of a [ReplicationController](#).
- `matchExpressions` - allows to build more sophisticated selectors by specifying key, list of values and an operator that relates the key and values.

When the two are specified the result is ANDed.

The `.spec.selector` must match the `.spec.template.metadata.labels`. Config with these two not matching will be rejected by the API.

## Running Pods on select Nodes

If you specify a `.spec.template.spec.nodeSelector`, then the DaemonSet controller will create Pods on nodes which match that [node selector](#). Likewise if you specify

a `.spec.template.spec.affinity`, then DaemonSet controller will create Pods on nodes which match that [node affinity](#). If you do not specify either, then the DaemonSet controller will create Pods on all nodes.

## How Daemon Pods are scheduled

A DaemonSet can be used to ensure that all eligible nodes run a copy of a Pod. The DaemonSet controller creates a Pod for each eligible node and adds the `spec.affinity.nodeAffinity` field of the Pod to match the target host. After the Pod is created, the default scheduler typically takes over and then binds the Pod to the target host by setting the `.spec.nodeName` field. If the new Pod cannot fit on the node, the default scheduler may preempt (evict) some of the existing Pods based on the [priority](#) of the new Pod.

### Note:

If it's important that the DaemonSet pod run on each node, it's often desirable to set the `.spec.template.spec.priorityClassName` of the DaemonSet to a [PriorityClass](#) with a higher priority to ensure that this eviction occurs.

The user can specify a different scheduler for the Pods of the DaemonSet, by setting the `.spec.template.spec.schedulerName` field of the DaemonSet.

The original node affinity specified at the `.spec.template.spec.affinity.nodeAffinity` field (if specified) is taken into consideration by the DaemonSet controller when evaluating the eligible nodes, but is replaced on the created Pod with the node affinity that matches the name of the eligible node.

```
nodeAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 nodeSelectorTerms:
 - matchFields:
 - key: metadata.name
 operator: In
 values:
 - target-host-name
```

## Taints and tolerations

The DaemonSet controller automatically adds a set of [tolerations](#) to DaemonSet Pods:

Tolerations for DaemonSet pods

| Toleration key                                   | Effect     | Details                                                                                                                                                 |
|--------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">node.kubernetes.io/not-ready</a>     | NoExecute  | DaemonSet Pods can be scheduled onto nodes that are not healthy or ready to accept Pods. Any DaemonSet Pods running on such nodes will not be evicted.  |
| <a href="#">node.kubernetes.io/unreachable</a>   | NoExecute  | DaemonSet Pods can be scheduled onto nodes that are unreachable from the node controller. Any DaemonSet Pods running on such nodes will not be evicted. |
| <a href="#">node.kubernetes.io/disk-pressure</a> | NoSchedule | DaemonSet Pods can be scheduled onto nodes with disk pressure issues.                                                                                   |
|                                                  | NoSchedule |                                                                                                                                                         |

| Toleration key                                                                                                                                    | Effect     | Details                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="https://kubernetes.io/docs/concepts/scheduling-eviction/toleration-keys/#memory-pressure">node.kubernetes.io/memory-pressure</a>         |            | DaemonSet Pods can be scheduled onto nodes with memory pressure issues.                                                                                                                 |
| <a href="https://kubernetes.io/docs/concepts/scheduling-eviction/toleration-keys/#process-pressure">node.kubernetes.io/pid-pressure</a>           | NoSchedule | DaemonSet Pods can be scheduled onto nodes with process pressure issues.                                                                                                                |
| <a href="https://kubernetes.io/docs/concepts/scheduling-eviction/toleration-keys/#unschedulable">node.kubernetes.io/unschedulable</a>             | NoSchedule | DaemonSet Pods can be scheduled onto nodes that are unschedulable.                                                                                                                      |
| <a href="https://kubernetes.io/docs/concepts/scheduling-eviction/toleration-keys/#network-unavailable">node.kubernetes.io/network-unavailable</a> | NoSchedule | <b>Only added for DaemonSet Pods that request host networking</b> , i.e., Pods having spec.hostNetwork: true. Such DaemonSet Pods can be scheduled onto nodes with unavailable network. |

You can add your own tolerations to the Pods of a DaemonSet as well, by defining these in the Pod template of the DaemonSet.

Because the DaemonSet controller sets the `node.kubernetes.io/unschedulable:NoSchedule` toleration automatically, Kubernetes can run DaemonSet Pods on nodes that are marked as *unschedulable*.

If you use a DaemonSet to provide an important node-level function, such as [cluster networking](#), it is helpful that Kubernetes places DaemonSet Pods on nodes before they are ready. For example, without that special toleration, you could end up in a deadlock situation where the node is not marked as ready because the network plugin is not running there, and at the same time the network plugin is not running on that node because the node is not yet ready.

## Communicating with Daemon Pods

Some possible patterns for communicating with Pods in a DaemonSet are:

- **Push:** Pods in the DaemonSet are configured to send updates to another service, such as a stats database. They do not have clients.
- **NodeIP and Known Port:** Pods in the DaemonSet can use a `hostPort`, so that the pods are reachable via the node IPs. Clients know the list of node IPs somehow, and know the port by convention.
- **DNS:** Create a [headless service](#) with the same pod selector, and then discover DaemonSets using the endpoints resource or retrieve multiple A records from DNS.
- **Service:** Create a service with the same Pod selector, and use the service to reach a daemon on a random node. (No way to reach specific node.)

## Updating a DaemonSet

If node labels are changed, the DaemonSet will promptly add Pods to newly matching nodes and delete Pods from newly not-matching nodes.

You can modify the Pods that a DaemonSet creates. However, Pods do not allow all fields to be updated. Also, the DaemonSet controller will use the original template the next time a node (even with the same name) is created.

You can delete a DaemonSet. If you specify `--cascade=orphan` with `kubectl`, then the Pods will be left on the nodes. If you subsequently create a new DaemonSet with the same selector, the

new DaemonSet adopts the existing Pods. If any Pods need replacing the DaemonSet replaces them according to its updateStrategy.

You can [perform a rolling update](#) on a DaemonSet.

## Alternatives to DaemonSet

### Init scripts

It is certainly possible to run daemon processes by directly starting them on a node (e.g. using init, upstartd, or systemd). This is perfectly fine. However, there are several advantages to running such processes via a DaemonSet:

- Ability to monitor and manage logs for daemons in the same way as applications.
- Same config language and tools (e.g. Pod templates, kubectl) for daemons and applications.
- Running daemons in containers with resource limits increases isolation between daemons from app containers. However, this can also be accomplished by running the daemons in a container but not in a Pod.

### Bare Pods

It is possible to create Pods directly which specify a particular node to run on. However, a DaemonSet replaces Pods that are deleted or terminated for any reason, such as in the case of node failure or disruptive node maintenance, such as a kernel upgrade. For this reason, you should use a DaemonSet rather than creating individual Pods.

### Static Pods

It is possible to create Pods by writing a file to a certain directory watched by Kubelet. These are called [static pods](#). Unlike DaemonSet, static Pods cannot be managed with kubectl or other Kubernetes API clients. Static Pods do not depend on the apiserver, making them useful in cluster bootstrapping cases. Also, static Pods may be deprecated in the future.

### Deployments

DaemonSets are similar to [Deployments](#) in that they both create Pods, and those Pods have processes which are not expected to terminate (e.g. web servers, storage servers).

Use a Deployment for stateless services, like frontends, where scaling up and down the number of replicas and rolling out updates are more important than controlling exactly which host the Pod runs on. Use a DaemonSet when it is important that a copy of a Pod always run on all or certain hosts, if the DaemonSet provides node-level functionality that allows other Pods to run correctly on that particular node.

For example, [network plugins](#) often include a component that runs as a DaemonSet. The DaemonSet component makes sure that the node where it's running has working cluster networking.



## What's next

- Learn about [Pods](#).
  - Learn about [static Pods](#), which are useful for running Kubernetes [control plane](#) components.
- Find out how to use DaemonSets
  - [Perform a rolling update on a DaemonSet](#)
  - [Perform a rollback on a DaemonSet](#) (for example, if a roll out didn't work how you expected).
- Understand [how Kubernetes assigns Pods to Nodes](#).
- Learn about [device plugins](#) and [add ons](#), which often run as DaemonSets.
- DaemonSet is a top-level resource in the Kubernetes REST API. Read the [DaemonSet](#) object definition to understand the API for daemon sets.

## Jobs

Jobs represent one-off tasks that run to completion and then stop.

A Job creates one or more Pods and will continue to retry execution of the Pods until a specified number of them successfully terminate. As pods successfully complete, the Job tracks the successful completions. When a specified number of successful completions is reached, the task (ie, Job) is complete. Deleting a Job will clean up the Pods it created. Suspending a Job will delete its active Pods until the Job is resumed again.

A simple case is to create one Job object in order to reliably run one Pod to completion. The Job object will start a new Pod if the first Pod fails or is deleted (for example due to a node hardware failure or a node reboot).

You can also use a Job to run multiple Pods in parallel.

If you want to run a Job (either a single task, or several in parallel) on a schedule, see [CronJob](#).

## Running an example Job

Here is an example Job config. It computes  $\pi$  to 2000 places and prints it out. It takes around 10s to complete.

[controllers/job.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
 name: pi
spec:
 template:
 spec:
 containers:
 - name: pi
 image: perl:5.34.0
 command: ["perl", "-Mbignum=bpi", "-wle", "print bpi(2000)"]
 restartPolicy: Never
```



```
backoffLimit: 4
```

You can run the example with this command:

```
kubectl apply -f https://kubernetes.io/examples/controllers/job.yaml
```

The output is similar to this:

```
job.batch/pi created
```

Check on the status of the Job with kubectl:

- [kubectl describe job pi](#)
- [kubectl get job pi -o yaml](#)

```
Name: pi
Namespace: default
Selector: batch.kubernetes.io/controller-uid=c9948307-e56d-4b5d-8302-ae2d7b7da67c
Labels: batch.kubernetes.io/controller-uid=c9948307-e56d-4b5d-8302-ae2d7b7da67c
 batch.kubernetes.io/job-name=pi
...
Annotations: batch.kubernetes.io/job-tracking: ""
Parallelism: 1
Completions: 1
Start Time: Mon, 02 Dec 2019 15:20:11 +0200
Completed At: Mon, 02 Dec 2019 15:21:16 +0200
Duration: 65s
Pods Statuses: 0 Running / 1 Succeeded / 0 Failed
Pod Template:
 Labels: batch.kubernetes.io/controller-uid=c9948307-e56d-4b5d-8302-ae2d7b7da67c
 batch.kubernetes.io/job-name=pi
 Containers:
 pi:
 Image: perl:5.34.0
 Port: <none>
 Host Port: <none>
 Command:
 perl
 -Mbignum=bpi
 -wle
 print bpi(2000)
 Environment: <none>
 Mounts: <none>
 Volumes: <none>
Events:
 Type Reason Age From Message
 ---- -
 Normal SuccessfulCreate 21s job-controller Created pod: pi-xf9p4
 Normal Completed 18s job-controller Job completed
```

```
apiVersion: batch/v1
```

```
kind: Job
metadata:
 annotations: batch.kubernetes.io/job-tracking: ""
 ...
 creationTimestamp: "2022-11-10T17:53:53Z"
 generation: 1
 labels:
 batch.kubernetes.io/controller-uid: 863452e6-270d-420e-9b94-53a54146c223
 batch.kubernetes.io/job-name: pi
 name: pi
 namespace: default
 resourceVersion: "4751"
 uid: 204fb678-040b-497f-9266-35ffa8716d14
spec:
 backoffLimit: 4
 completionMode: NonIndexed
 completions: 1
 parallelism: 1
 selector:
 matchLabels:
 batch.kubernetes.io/controller-uid: 863452e6-270d-420e-9b94-53a54146c223
 suspend: false
 template:
 metadata:
 creationTimestamp: null
 labels:
 batch.kubernetes.io/controller-uid: 863452e6-270d-420e-9b94-53a54146c223
 batch.kubernetes.io/job-name: pi
 spec:
 containers:
 - command:
 - perl
 - -Mbignum=bpi
 - -wle
 - print bpi(2000)
 image: perl:5.34.0
 imagePullPolicy: IfNotPresent
 name: pi
 resources: {}
 terminationMessagePath: /dev/termination-log
 terminationMessagePolicy: File
 dnsPolicy: ClusterFirst
 restartPolicy: Never
 schedulerName: default-scheduler
 securityContext: {}
 terminationGracePeriodSeconds: 30
status:
 active: 1
 ready: 0
 startTime: "2022-11-10T17:53:57Z"
 uncountedTerminatedPods: {}
```

To view completed Pods of a Job, use `kubectl get pods`.

To list all the Pods that belong to a Job in a machine readable form, you can use a command like this:

```
 pods=$(kubectl get pods --selector=batch.kubernetes.io/job-name=pi --output=jsonpath='{.items[*].metadata.name}')
 echo $pods
```

The output is similar to this:

```
pi-5rwd7
```

Here, the selector is the same as the selector for the Job. The `--output=jsonpath` option specifies an expression with the name from each Pod in the returned list.

View the standard output of one of the pods:

```
kubectl logs $pods
```

Another way to view the logs of a Job:

```
kubectl logs jobs/pi
```

The output is similar to this:

```
3.1415926535897932384626433832795028841971693993751058209749445923078164062862089986
280348253421170679821480865132823066470938446095505822317253594081284811174502841027
019385211055596446229489549303819644288109756659334461284756482337867831652712019091
456485669234603486104543266482133936072602491412737245870066063155881748815209209628
292540917153643678925903600113305305488204665213841469519415116094330572703657595919
530921861173819326117931051185480744623799627495673518857527248912279381830119491298
336733624406566430860213949463952247371907021798609437027705392171762931767523846748
184676694051320005681271452635608277857713427577896091736371787214684409012249534301
465495853710507922796892589235420199561121290219608640344181598136297747713099605187
072113499999983729780499510597317328160963185950244594553469083026425223082533446850
352619311881710100031378387528865875332083814206171776691473035982534904287554687311
595628638823537875937519577818577805321712268066130019278766111959092164201989380952
572010654858632788659361533818279682303019520353018529689957736225994138912497217752
834791315155748572424541506959508295331168617278558890750983817546374649393192550604
009277016711390098488240128583616035637076601047101819429555961989467678374494482553
797747268471040475346462080466842590694912933136770289891521047521620569660240580381
501935112533824300355876402474964732639141992726042699227967823547816360093417216412
199245863150302861829745557067498385054945885869269956909272107975093029553211653449
872027559602364806654991198818347977535663698074265425278625518184175746728909777727
938000816470600161452491921732172147723501414419735685481613611573525521334757418494
684385233239073941433345477624168625189835694855620992192221842725502542568876717904
946016534668049886272327917860857843838279679766814541009538837863609506800642251252
051173929848960841284886269456042419652850222106611863067442786220391949450471237137
869609563643719172874677646575739624138908658326459958133904780275901
```

## Writing a Job spec

As with all other Kubernetes config, a Job needs `apiVersion`, `kind`, and `metadata` fields.

When the control plane creates new Pods for a Job, the `.metadata.name` of the Job is part of the basis for naming those Pods. The name of a Job must be a valid [DNS subdomain](#) value, but this can produce unexpected results for the Pod hostnames. For best compatibility, the name should follow the more restrictive rules for a [DNS label](#). Even when the name is a DNS subdomain, the name must be no longer than 63 characters.

A Job also needs a [.spec section](#).

## Job Labels

Job labels will have `batch.kubernetes.io/` prefix for `job-name` and `controller-uid`.

## Pod Template

The `.spec.template` is the only required field of the `.spec`.

The `.spec.template` is a [pod template](#). It has exactly the same schema as a [Pod](#), except it is nested and does not have an `apiVersion` or `kind`.

In addition to required fields for a Pod, a pod template in a Job must specify appropriate labels (see [pod selector](#)) and an appropriate restart policy.

Only a [RestartPolicy](#) equal to `Never` or `OnFailure` is allowed.

## Pod selector

The `.spec.selector` field is optional. In almost all cases you should not specify it. See section [specifying your own pod selector](#).

## Parallel execution for Jobs

There are three main types of task suitable to run as a Job:

1. Non-parallel Jobs
  - normally, only one Pod is started, unless the Pod fails.
  - the Job is complete as soon as its Pod terminates successfully.
2. Parallel Jobs with a *fixed completion count*:
  - specify a non-zero positive value for `.spec.completions`.
  - the Job represents the overall task, and is complete when there are `.spec.completions` successful Pods.
  - when using `.spec.completionMode="Indexed"`, each Pod gets a different index in the range 0 to `.spec.completions-1`.
3. Parallel Jobs with a *work queue*:
  - do not specify `.spec.completions`, default to `.spec.parallelism`.
  - the Pods must coordinate amongst themselves or an external service to determine what each should work on. For example, a Pod might fetch a batch of up to N items from the work queue.
  - each Pod is independently capable of determining whether or not all its peers are done, and thus that the entire Job is done.
  - when *any* Pod from the Job terminates with success, no new Pods are created.
  - once at least one Pod has terminated with success and all Pods are terminated, then the Job is completed with success.

- once any Pod has exited with success, no other Pod should still be doing any work for this task or writing any output. They should all be in the process of exiting.

For a *non-parallel* Job, you can leave both `.spec.completions` and `.spec.parallelism` unset. When both are unset, both are defaulted to 1.

For a *fixed completion count* Job, you should set `.spec.completions` to the number of completions needed. You can set `.spec.parallelism`, or leave it unset and it will default to 1.

For a *work queue* Job, you must leave `.spec.completions` unset, and set `.spec.parallelism` to a non-negative integer.

For more information about how to make use of the different types of job, see the [job patterns](#) section.

## Controlling parallelism

The requested parallelism (`.spec.parallelism`) can be set to any non-negative value. If it is unspecified, it defaults to 1. If it is specified as 0, then the Job is effectively paused until it is increased.

Actual parallelism (number of pods running at any instant) may be more or less than requested parallelism, for a variety of reasons:

- For *fixed completion count* Jobs, the actual number of pods running in parallel will not exceed the number of remaining completions. Higher values of `.spec.parallelism` are effectively ignored.
- For *work queue* Jobs, no new Pods are started after any Pod has succeeded -- remaining Pods are allowed to complete, however.
- If the Job [Controller](#) has not had time to react.
- If the Job controller failed to create Pods for any reason (lack of `ResourceQuota`, lack of permission, etc.), then there may be fewer pods than requested.
- The Job controller may throttle new Pod creation due to excessive previous pod failures in the same Job.
- When a Pod is gracefully shut down, it takes time to stop.

## Completion mode

FEATURE STATE: `Kubernetes v1.24` [stable]

Jobs with *fixed completion count* - that is, jobs that have non null `.spec.completions` - can have a completion mode that is specified in `.spec.completionMode`:

- **NonIndexed** (default): the Job is considered complete when there have been `.spec.completions` successfully completed Pods. In other words, each Pod completion is homologous to each other. Note that Jobs that have null `.spec.completions` are implicitly **NonIndexed**.
- **Indexed**: the Pods of a Job get an associated completion index from 0 to `.spec.completions-1`. The index is available through four mechanisms:
  - The Pod annotation `batch.kubernetes.io/job-completion-index`.

- The Pod label `batch.kubernetes.io/job-completion-index` (for v1.28 and later). Note the feature gate `PodIndexLabel` must be enabled to use this label, and it is enabled by default.
- As part of the Pod hostname, following the pattern `$(job-name)-$(index)`. When you use an Indexed Job in combination with a [Service](#), Pods within the Job can use the deterministic hostnames to address each other via DNS. For more information about how to configure this, see [Job with Pod-to-Pod Communication](#).
- From the containerized task, in the environment variable `JOB_COMPLETION_INDEX`.

The Job is considered complete when there is one successfully completed Pod for each index. For more information about how to use this mode, see [Indexed Job for Parallel Processing with Static Work Assignment](#).

#### Note:

Although rare, more than one Pod could be started for the same index (due to various reasons such as node failures, kubelet restarts, or Pod evictions). In this case, only the first Pod that completes successfully will count towards the completion count and update the status of the Job. The other Pods that are running or completed for the same index will be deleted by the Job controller once they are detected.

## Handling Pod and container failures

A container in a Pod may fail for a number of reasons, such as because the process in it exited with a non-zero exit code, or the container was killed for exceeding a memory limit, etc. If this happens, and the `.spec.template.spec.restartPolicy = "OnFailure"`, then the Pod stays on the node, but the container is re-run. Therefore, your program needs to handle the case when it is restarted locally, or else specify `.spec.template.spec.restartPolicy = "Never"`. See [pod lifecycle](#) for more information on `restartPolicy`.

An entire Pod can also fail, for a number of reasons, such as when the pod is kicked off the node (node is upgraded, rebooted, deleted, etc.), or if a container of the Pod fails and the `.spec.template.spec.restartPolicy = "Never"`. When a Pod fails, then the Job controller starts a new Pod. This means that your application needs to handle the case when it is restarted in a new pod. In particular, it needs to handle temporary files, locks, incomplete output and the like caused by previous runs.

By default, each pod failure is counted towards the `.spec.backoffLimit` limit, see [pod backoff failure policy](#). However, you can customize handling of pod failures by setting the Job's [pod failure policy](#).

Additionally, you can choose to count the pod failures independently for each index of an [Indexed](#) Job by setting the `.spec.backoffLimitPerIndex` field (for more information, see [backoff limit per index](#)).

Note that even if you specify `.spec.parallelism = 1` and `.spec.completions = 1` and `.spec.template.spec.restartPolicy = "Never"`, the same program may sometimes be started twice.

If you do specify `.spec.parallelism` and `.spec.completions` both greater than 1, then there may be multiple pods running at once. Therefore, your pods must also be tolerant of concurrency.

When the [feature gates](#) `PodDisruptionConditions` and `JobPodFailurePolicy` are both enabled, and the `.spec.podFailurePolicy` field is set, the Job controller does not consider a terminating Pod (a pod that has a `.metadata.deletionTimestamp` field set) as a failure until that Pod is terminal (its `.status.phase` is `Failed` or `Succeeded`). However, the Job controller creates a replacement Pod as soon as the termination becomes apparent. Once the pod terminates, the Job controller evaluates `.backoffLimit` and `.podFailurePolicy` for the relevant Job, taking this now-terminated Pod into consideration.

If either of these requirements is not satisfied, the Job controller counts a terminating Pod as an immediate failure, even if that Pod later terminates with phase: `"Succeeded"`.

## Pod backoff failure policy

There are situations where you want to fail a Job after some amount of retries due to a logical error in configuration etc. To do so, set `.spec.backoffLimit` to specify the number of retries before considering a Job as failed. The back-off limit is set by default to 6. Failed Pods associated with the Job are recreated by the Job controller with an exponential back-off delay (10s, 20s, 40s ...) capped at six minutes.

The number of retries is calculated in two ways:

- The number of Pods with `.status.phase = "Failed"`.
- When using `restartPolicy = "OnFailure"`, the number of retries in all the containers of Pods with `.status.phase` equal to `Pending` or `Running`.

If either of the calculations reaches the `.spec.backoffLimit`, the Job is considered failed.

### Note:

If your job has `restartPolicy = "OnFailure"`, keep in mind that your Pod running the Job will be terminated once the job backoff limit has been reached. This can make debugging the Job's executable more difficult. We suggest setting `restartPolicy = "Never"` when debugging the Job or using a logging system to ensure output from failed Jobs is not lost inadvertently.

## Backoff limit per index

FEATURE STATE: Kubernetes v1.29 [beta]

### Note:

You can only configure the backoff limit per index for an [Indexed](#) Job, if you have the `JobBackoffLimitPerIndex` [feature gate](#) enabled in your cluster.

When you run an [indexed](#) Job, you can choose to handle retries for pod failures independently for each index. To do so, set the `.spec.backoffLimitPerIndex` to specify the maximal number of pod failures per index.

When the per-index backoff limit is exceeded for an index, Kubernetes considers the index as failed and adds it to the `.status.failedIndexes` field. The succeeded indexes, those with a successfully executed pods, are recorded in the `.status.completedIndexes` field, regardless of whether you set the `backoffLimitPerIndex` field.

Note that a failing index does not interrupt execution of other indexes. Once all indexes finish for a Job where you specified a backoff limit per index, if at least one of those indexes did fail, the Job controller marks the overall Job as failed, by setting the Failed condition in the status. The Job gets marked as failed even if some, potentially nearly all, of the indexes were processed successfully.

You can additionally limit the maximal number of indexes marked failed by setting the `.spec.maxFailedIndexes` field. When the number of failed indexes exceeds the `maxFailedIndexes` field, the Job controller triggers termination of all remaining running Pods for that Job. Once all pods are terminated, the entire Job is marked failed by the Job controller, by setting the Failed condition in the Job status.

Here is an example manifest for a Job that defines a `backoffLimitPerIndex`:

</controllers/job-backoff-limit-per-index-example.yaml>

```
apiVersion: batch/v1
kind: Job
metadata:
 name: job-backoff-limit-per-index-example
spec:
 completions: 10
 parallelism: 3
 completionMode: Indexed # required for the feature
 backoffLimitPerIndex: 1 # maximal number of failures per index
 maxFailedIndexes: 5 # maximal number of failed indexes before terminating the Job
 execution
 template:
 spec:
 restartPolicy: Never # required for the feature
 containers:
 - name: example
 image: python
 command:
 # The jobs fails as there is at least one failed index
 # (all even indexes fail in here), yet all indexes
 # are executed as maxFailedIndexes is not exceeded.
 - python3
 - -c
 - |
 import os, sys
 print("Hello world")
 if int(os.environ.get("JOB_COMPLETION_INDEX")) % 2 == 0:
 sys.exit(1)
```

In the example above, the Job controller allows for one restart for each of the indexes. When the total number of failed indexes exceeds 5, then the entire Job is terminated.

Once the job is finished, the Job status looks as follows:

```
kubectl get -o yaml job job-backoff-limit-per-index-example
```

```
status:
 completedIndexes: 1,3,5,7,9
 failedIndexes: 0,2,4,6,8
```



```
succeeded: 5 # 1 succeeded pod for each of 5 succeeded indexes
failed: 10 # 2 failed pods (1 retry) for each of 5 failed indexes
conditions:
- message: Job has failed indexes
 reason: FailedIndexes
 status: "True"
 type: Failed
```

Additionally, you may want to use the per-index backoff along with a [pod failure policy](#). When using per-index backoff, there is a new `FailIndex` action available which allows you to avoid unnecessary retries within an index.

## Pod failure policy

FEATURE STATE: Kubernetes v1.26 [beta]

### Note:

You can only configure a Pod failure policy for a Job if you have the `JobPodFailurePolicy` [feature gate](#) enabled in your cluster. Additionally, it is recommended to enable the `PodDisruptionConditions` feature gate in order to be able to detect and handle Pod disruption conditions in the Pod failure policy (see also: [Pod disruption conditions](#)). Both feature gates are available in Kubernetes 1.30.

A Pod failure policy, defined with the `.spec.podFailurePolicy` field, enables your cluster to handle Pod failures based on the container exit codes and the Pod conditions.

In some situations, you may want to have a better control when handling Pod failures than the control provided by the [Pod backoff failure policy](#), which is based on the Job's `.spec.backoffLimit`. These are some examples of use cases:

- To optimize costs of running workloads by avoiding unnecessary Pod restarts, you can terminate a Job as soon as one of its Pods fails with an exit code indicating a software bug.
- To guarantee that your Job finishes even if there are disruptions, you can ignore Pod failures caused by disruptions (such as [preemption](#), [API-initiated eviction](#) or [taint](#)-based eviction) so that they don't count towards the `.spec.backoffLimit` limit of retries.

You can configure a Pod failure policy, in the `.spec.podFailurePolicy` field, to meet the above use cases. This policy can handle Pod failures based on the container exit codes and the Pod conditions.

Here is a manifest for a Job that defines a `podFailurePolicy`:

</controllers/job-pod-failure-policy-example.yaml>

```
apiVersion: batch/v1
kind: Job
metadata:
 name: job-pod-failure-policy-example
spec:
 completions: 12
 parallelism: 3
```

```

template:
 spec:
 restartPolicy: Never
 containers:
 - name: main
 image: docker.io/library/bash:5
 command: ["bash"] # example command simulating a bug which triggers the FailJob
 action
 args:
 - -c
 - echo "Hello world!" && sleep 5 && exit 42
 backoffLimit: 6
 podFailurePolicy:
 rules:
 - action: FailJob
 onExitCodes:
 containerName: main # optional
 operator: In # one of: In, NotIn
 values: [42]
 - action: Ignore # one of: Ignore, FailJob, Count
 onPodConditions:
 - type: DisruptionTarget # indicates Pod disruption

```

In the example above, the first rule of the Pod failure policy specifies that the Job should be marked failed if the main container fails with the 42 exit code. The following are the rules for the main container specifically:

- an exit code of 0 means that the container succeeded
- an exit code of 42 means that the **entire Job** failed
- any other exit code represents that the container failed, and hence the entire Pod. The Pod will be re-created if the total number of restarts is below backoffLimit. If the backoffLimit is reached the **entire Job** failed.

#### Note:

Because the Pod template specifies a restartPolicy: Never, the kubelet does not restart the main container in that particular Pod.

The second rule of the Pod failure policy, specifying the Ignore action for failed Pods with condition DisruptionTarget excludes Pod disruptions from being counted towards the .spec.backoffLimit limit of retries.

#### Note:

If the Job failed, either by the Pod failure policy or Pod backoff failure policy, and the Job is running multiple Pods, Kubernetes terminates all the Pods in that Job that are still Pending or Running.

These are some requirements and semantics of the API:

- if you want to use a .spec.podFailurePolicy field for a Job, you must also define that Job's pod template with .spec.restartPolicy set to Never.

- the Pod failure policy rules you specify under `spec.podFailurePolicy.rules` are evaluated in order. Once a rule matches a Pod failure, the remaining rules are ignored. When no rule matches the Pod failure, the default handling applies.
- you may want to restrict a rule to a specific container by specifying its name `inspec.podFailurePolicy.rules[*].onExitCodes.containerName`. When not specified the rule applies to all containers. When specified, it should match one the container or `initContainer` names in the Pod template.
- you may specify the action taken when a Pod failure policy is matched by `spec.podFailurePolicy.rules[*].action`. Possible values are:
  - `FailJob`: use to indicate that the Pod's job should be marked as Failed and all running Pods should be terminated.
  - `Ignore`: use to indicate that the counter towards the `.spec.backoffLimit` should not be incremented and a replacement Pod should be created.
  - `Count`: use to indicate that the Pod should be handled in the default way. The counter towards the `.spec.backoffLimit` should be incremented.
  - `FailIndex`: use this action along with [backoff limit per index](#) to avoid unnecessary retries within the index of a failed pod.

**Note:**

When you use a `podFailurePolicy`, the job controller only matches Pods in the Failed phase. Pods with a deletion timestamp that are not in a terminal phase (Failed or Succeeded) are considered still terminating. This implies that terminating pods retain a [tracking finalizer](#) until they reach a terminal phase. Since Kubernetes 1.27, Kubelet transitions deleted pods to a terminal phase (see: [Pod Phase](#)). This ensures that deleted pods have their finalizers removed by the Job controller.

**Note:**

Starting with Kubernetes v1.28, when Pod failure policy is used, the Job controller recreates terminating Pods only once these Pods reach the terminal Failed phase. This behavior is similar to `podReplacementPolicy: Failed`. For more information, see [Pod replacement policy](#).

## Success policy

FEATURE STATE: Kubernetes v1.30 [alpha]

**Note:**

You can only configure a success policy for an Indexed Job if you have the `JobSuccessPolicy` [feature gate](#) enabled in your cluster.

When creating an Indexed Job, you can define when a Job can be declared as succeeded using a `.spec.successPolicy`, based on the pods that succeeded.

By default, a Job succeeds when the number of succeeded Pods equals `.spec.completions`. These are some situations where you might want additional control for declaring a Job succeeded:

- When running simulations with different parameters, you might not need all the simulations to succeed for the overall Job to be successful.

- When following a leader-worker pattern, only the success of the leader determines the success or failure of a Job. Examples of this are frameworks like MPI and PyTorch etc.

You can configure a success policy, in the `.spec.successPolicy` field, to meet the above use cases. This policy can handle Job success based on the succeeded pods. After the Job meets the success policy, the job controller terminates the lingering Pods. A success policy is defined by rules. Each rule can take one of the following forms:

- When you specify the `succeededIndexes` only, once all indexes specified in the `succeededIndexes` succeed, the job controller marks the Job as succeeded. The `succeededIndexes` must be a list of intervals between 0 and `.spec.completions-1`.
- When you specify the `succeededCount` only, once the number of succeeded indexes reaches the `succeededCount`, the job controller marks the Job as succeeded.
- When you specify both `succeededIndexes` and `succeededCount`, once the number of succeeded indexes from the subset of indexes specified in the `succeededIndexes` reaches the `succeededCount`, the job controller marks the Job as succeeded.

Note that when you specify multiple rules in the `.spec.successPolicy.rules`, the job controller evaluates the rules in order. Once the Job meets a rule, the job controller ignores remaining rules.

Here is a manifest for a Job with successPolicy:

</controllers/job-success-policy.yaml>

```
apiVersion: batch/v1
kind: Job
metadata:
 name: job-success
spec:
 parallelism: 10
 completions: 10
 completionMode: Indexed # Required for the success policy
 successPolicy:
 rules:
 - succeededIndexes: 0,2-3
 succeededCount: 1
 template:
 spec:
 containers:
 - name: main
 image: python
 command:
 # Provided that at least one of the Pods with 0, 2, and 3 indexes has
 # succeeded,
 # the overall Job is a success.
 - python3
 - -c
 - |
 import os, sys
 if os.environ.get("JOB_COMPLETION_INDEX") == "2":
 sys.exit(0)
 else:
 sys.exit(1)
 restartPolicy: Never
```

In the example above, both `succeededIndexes` and `succeededCount` have been specified. Therefore, the job controller will mark the Job as succeeded and terminate the lingering Pods when either of the specified indexes, 0, 2, or 3, succeed. The Job that meets the success policy gets the `SuccessCriteriaMet` condition. After the removal of the lingering Pods is issued, the Job gets the `Complete` condition.

Note that the `succeededIndexes` is represented as intervals separated by a hyphen. The number are listed in represented by the first and last element of the series, separated by a hyphen.

#### Note:

When you specify both a success policy and some terminating policies such as `.spec.backoffLimit` and `.spec.podFailurePolicy`, once the Job meets either policy, the job controller respects the terminating policy and ignores the success policy.

## Job termination and cleanup

When a Job completes, no more Pods are created, but the Pods are [usually](#) not deleted either. Keeping them around allows you to still view the logs of completed pods to check for errors, warnings, or other diagnostic output. The job object also remains after it is completed so that you can view its status. It is up to the user to delete old jobs after noting their status. Delete the job with `kubectl` (e.g. `kubectl delete jobs/pi` or `kubectl delete -f ./job.yaml`). When you delete the job using `kubectl`, all the pods it created are deleted too.

By default, a Job will run uninterrupted unless a Pod fails (`restartPolicy=Never`) or a Container exits in error (`restartPolicy=OnFailure`), at which point the Job defers to the `.spec.backoffLimit` described above. Once `.spec.backoffLimit` has been reached the Job will be marked as failed and any running Pods will be terminated.

Another way to terminate a Job is by setting an active deadline. Do this by setting the `.spec.activeDeadlineSeconds` field of the Job to a number of seconds. The `activeDeadlineSeconds` applies to the duration of the job, no matter how many Pods are created. Once a Job reaches `activeDeadlineSeconds`, all of its running Pods are terminated and the Job status will become `type: Failed` with reason: `DeadlineExceeded`.

Note that a Job's `.spec.activeDeadlineSeconds` takes precedence over its `.spec.backoffLimit`. Therefore, a Job that is retrying one or more failed Pods will not deploy additional Pods once it reaches the time limit specified by `activeDeadlineSeconds`, even if the `backoffLimit` is not yet reached.

Example:

```
apiVersion: batch/v1
kind: Job
metadata:
 name: pi-with-timeout
spec:
 backoffLimit: 5
 activeDeadlineSeconds: 100
 template:
 spec:
 containers:
 - name: pi
```

```
image: perl:5.34.0
command: ["perl", "-Mbignum=bpi", "-wle", "print bpi(2000)"]
restartPolicy: Never
```

Note that both the Job spec and the [Pod template spec](#) within the Job have an `activeDeadlineSeconds` field. Ensure that you set this field at the proper level.

Keep in mind that the `restartPolicy` applies to the Pod, and not to the Job itself: there is no automatic Job restart once the Job status is type: Failed. That is, the Job termination mechanisms activated with `.spec.activeDeadlineSeconds` and `.spec.backoffLimit` result in a permanent Job failure that requires manual intervention to resolve.

## Clean up finished jobs automatically

Finished Jobs are usually no longer needed in the system. Keeping them around in the system will put pressure on the API server. If the Jobs are managed directly by a higher level controller, such as [CronJobs](#), the Jobs can be cleaned up by CronJobs based on the specified capacity-based cleanup policy.

### TTL mechanism for finished Jobs

FEATURE STATE: Kubernetes v1.23 [stable]

Another way to clean up finished Jobs (either Complete or Failed) automatically is to use a TTL mechanism provided by a [TTL controller](#) for finished resources, by specifying the `.spec.ttlSecondsAfterFinished` field of the Job.

When the TTL controller cleans up the Job, it will delete the Job cascadingly, i.e. delete its dependent objects, such as Pods, together with the Job. Note that when the Job is deleted, its lifecycle guarantees, such as finalizers, will be honored.

For example:

```
apiVersion: batch/v1
kind: Job
metadata:
 name: pi-with-ttl
spec:
 ttlSecondsAfterFinished: 100
 template:
 spec:
 containers:
 - name: pi
 image: perl:5.34.0
 command: ["perl", "-Mbignum=bpi", "-wle", "print bpi(2000)"]
 restartPolicy: Never
```

The Job `pi-with-ttl` will be eligible to be automatically deleted, 100 seconds after it finishes.

If the field is set to 0, the Job will be eligible to be automatically deleted immediately after it finishes. If the field is unset, this Job won't be cleaned up by the TTL controller after it finishes.

## Note:

It is recommended to set `ttlSecondsAfterFinished` field because unmanaged jobs (Jobs that you created directly, and not indirectly through other workload APIs such as `CronJob`) have a default deletion policy of `orphanDependents` causing Pods created by an unmanaged Job to be left around after that Job is fully deleted. Even though the [control plane](#) eventually [garbage collects](#) the Pods from a deleted Job after they either fail or complete, sometimes those lingering pods may cause cluster performance degradation or in worst case cause the cluster to go offline due to this degradation.

You can use [LimitRanges](#) and [ResourceQuotas](#) to place a cap on the amount of resources that a particular namespace can consume.

## Job patterns

The Job object can be used to process a set of independent but related *work items*. These might be emails to be sent, frames to be rendered, files to be transcoded, ranges of keys in a NoSQL database to scan, and so on.

In a complex system, there may be multiple different sets of work items. Here we are just considering one set of work items that the user wants to manage together — a *batch job*.

There are several different patterns for parallel computation, each with strengths and weaknesses. The tradeoffs are:

- One Job object for each work item, versus a single Job object for all work items. One Job per work item creates some overhead for the user and for the system to manage large numbers of Job objects. A single Job for all work items is better for large numbers of items.
- Number of Pods created equals number of work items, versus each Pod can process multiple work items. When the number of Pods equals the number of work items, the Pods typically requires less modification to existing code and containers. Having each Pod process multiple work items is better for large numbers of items.
- Several approaches use a work queue. This requires running a queue service, and modifications to the existing program or container to make it use the work queue. Other approaches are easier to adapt to an existing containerised application.
- When the Job is associated with a [headless Service](#), you can enable the Pods within a Job to communicate with each other to collaborate in a computation.

The tradeoffs are summarized here, with columns 2 to 4 corresponding to the above tradeoffs. The pattern names are also links to examples and more detailed description.

| Pattern                                                 | Single Job object | Fewer pods than work items? | Use app unmodified? |
|---------------------------------------------------------|-------------------|-----------------------------|---------------------|
| <a href="#">Queue with Pod Per Work Item</a>            |                   |                             | sometimes           |
| <a href="#">Queue with Variable Pod Count</a>           |                   |                             |                     |
| <a href="#">Indexed Job with Static Work Assignment</a> |                   |                             |                     |
| <a href="#">Job with Pod-to-Pod Communication</a>       |                   | sometimes                   | sometimes           |
| <a href="#">Job Template Expansion</a>                  |                   |                             |                     |

When you specify completions with `.spec.completions`, each Pod created by the Job controller has an identical [spec](#). This means that all pods for a task will have the same command line and the same image, the same volumes, and (almost) the same environment variables. These patterns are different ways to arrange for pods to work on different things.

This table shows the required settings for `.spec.parallelism` and `.spec.completions` for each of the patterns. Here, W is the number of work items.

| Pattern                                                 | <code>.spec.completions</code> | <code>.spec.parallelism</code> |
|---------------------------------------------------------|--------------------------------|--------------------------------|
| <a href="#">Queue with Pod Per Work Item</a>            | W                              | any                            |
| <a href="#">Queue with Variable Pod Count</a>           | null                           | any                            |
| <a href="#">Indexed Job with Static Work Assignment</a> | W                              | any                            |
| <a href="#">Job with Pod-to-Pod Communication</a>       | W                              | W                              |
| <a href="#">Job Template Expansion</a>                  | 1                              | should be 1                    |

## Advanced usage

### Suspending a Job

FEATURE STATE: Kubernetes v1.24 [stable]

When a Job is created, the Job controller will immediately begin creating Pods to satisfy the Job's requirements and will continue to do so until the Job is complete. However, you may want to temporarily suspend a Job's execution and resume it later, or start Jobs in suspended state and have a custom controller decide later when to start them.

To suspend a Job, you can update the `.spec.suspend` field of the Job to true; later, when you want to resume it again, update it to false. Creating a Job with `.spec.suspend` set to true will create it in the suspended state.

When a Job is resumed from suspension, its `.status.startTime` field will be reset to the current time. This means that the `.spec.activeDeadlineSeconds` timer will be stopped and reset when a Job is suspended and resumed.

When you suspend a Job, any running Pods that don't have a status of Completed will be [terminated](#) with a SIGTERM signal. The Pod's graceful termination period will be honored and your Pod must handle this signal in this period. This may involve saving progress for later or undoing changes. Pods terminated this way will not count towards the Job's completions count.

An example Job definition in the suspended state can be like so:

```
kubectrl get job myjob -o yaml
```

```
apiVersion: batch/v1
kind: Job
metadata:
 name: myjob
spec:
 suspend: true
 parallelism: 1
 completions: 5
 template:
```



```
spec:
```

```
...
```

You can also toggle Job suspension by patching the Job using the command line.

Suspend an active Job:

```
kubectrl patch job/myjob --type=strategic --patch '{"spec":{"suspend":true}}'
```

Resume a suspended Job:

```
kubectrl patch job/myjob --type=strategic --patch '{"spec":{"suspend":false}}'
```

The Job's status can be used to determine if a Job is suspended or has been suspended in the past:

```
kubectrl get jobs/myjob -o yaml
```

```
apiVersion: batch/v1
kind: Job
.metadata and .spec omitted
status:
 conditions:
 - lastProbeTime: "2021-02-05T13:14:33Z"
 lastTransitionTime: "2021-02-05T13:14:33Z"
 status: "True"
 type: Suspended
 startTime: "2021-02-05T13:13:48Z"
```

The Job condition of type "Suspended" with status "True" means the Job is suspended; the `lastTransitionTime` field can be used to determine how long the Job has been suspended for. If the status of that condition is "False", then the Job was previously suspended and is now running. If such a condition does not exist in the Job's status, the Job has never been stopped.

Events are also created when the Job is suspended and resumed:

```
kubectrl describe jobs/myjob
```

```
Name: myjob
```

```
...
```

```
Events:
```

| Type   | Reason           | Age   | From           | Message                  |
|--------|------------------|-------|----------------|--------------------------|
| ----   | -----            | ----- | -----          | -----                    |
| Normal | SuccessfulCreate | 12m   | job-controller | Created pod: myjob-hlrpl |
| Normal | SuccessfulDelete | 11m   | job-controller | Deleted pod: myjob-hlrpl |
| Normal | Suspended        | 11m   | job-controller | Job suspended            |
| Normal | SuccessfulCreate | 3s    | job-controller | Created pod: myjob-jvb44 |
| Normal | Resumed          | 3s    | job-controller | Job resumed              |

The last four events, particularly the "Suspended" and "Resumed" events, are directly a result of toggling the `.spec.suspend` field. In the time between these two events, we see that no Pods were created, but Pod creation restarted as soon as the Job was resumed.

## Mutable Scheduling Directives

FEATURE STATE: Kubernetes v1.27 [stable]

In most cases, a parallel job will want the pods to run with constraints, like all in the same zone, or all either on GPU model x or y but not a mix of both.

The [suspend](#) field is the first step towards achieving those semantics. Suspend allows a custom queue controller to decide when a job should start; However, once a job is unsuspended, a custom queue controller has no influence on where the pods of a job will actually land.

This feature allows updating a Job's scheduling directives before it starts, which gives custom queue controllers the ability to influence pod placement while at the same time offloading actual pod-to-node assignment to kube-scheduler. This is allowed only for suspended Jobs that have never been unsuspended before.

The fields in a Job's pod template that can be updated are node affinity, node selector, tolerations, labels, annotations and [scheduling gates](#).

### Specifying your own Pod selector

Normally, when you create a Job object, you do not specify `.spec.selector`. The system defaulting logic adds this field when the Job is created. It picks a selector value that will not overlap with any other jobs.

However, in some cases, you might need to override this automatically set selector. To do this, you can specify the `.spec.selector` of the Job.

Be very careful when doing this. If you specify a label selector which is not unique to the pods of that Job, and which matches unrelated Pods, then pods of the unrelated job may be deleted, or this Job may count other Pods as completing it, or one or both Jobs may refuse to create Pods or run to completion. If a non-unique selector is chosen, then other controllers (e.g. ReplicationController) and their Pods may behave in unpredictable ways too. Kubernetes will not stop you from making a mistake when specifying `.spec.selector`.

Here is an example of a case when you might want to use this feature.

Say Job old is already running. You want existing Pods to keep running, but you want the rest of the Pods it creates to use a different pod template and for the Job to have a new name. You cannot update the Job because these fields are not updatable. Therefore, you delete Job old but *leave its pods running*, using `kubectrl delete jobs/old --cascade=orphan`. Before deleting it, you make a note of what selector it uses:

```
kubectrl get job old -o yaml
```

The output is similar to this:

```
kind: Job
metadata:
 name: old
 ...
spec:
 selector:
 matchLabels:
```

```
batch.kubernetes.io/controller-uid: a8f3d00d-c6d2-11e5-9f87-42010af00002
```

```
...
```

Then you create a new Job with name `new` and you explicitly specify the same selector. Since the existing Pods have label `batch.kubernetes.io/controller-uid=a8f3d00d-c6d2-11e5-9f87-42010af00002`, they are controlled by Job `new` as well.

You need to specify `manualSelector: true` in the new Job since you are not using the selector that the system normally generates for you automatically.

```
kind: Job
metadata:
 name: new
 ...
spec:
 manualSelector: true
 selector:
 matchLabels:
 batch.kubernetes.io/controller-uid: a8f3d00d-c6d2-11e5-9f87-42010af00002
 ...
```

The new Job itself will have a different uid from `a8f3d00d-c6d2-11e5-9f87-42010af00002`. Setting `manualSelector: true` tells the system that you know what you are doing and to allow this mismatch.

## Job tracking with finalizers

FEATURE STATE: Kubernetes v1.26 [stable]

The control plane keeps track of the Pods that belong to any Job and notices if any such Pod is removed from the API server. To do that, the Job controller creates Pods with the finalizer `batch.kubernetes.io/job-tracking`. The controller removes the finalizer only after the Pod has been accounted for in the Job status, allowing the Pod to be removed by other controllers or users.

### Note:

See [My pod stays terminating](#) if you observe that pods from a Job are stuck with the tracking finalizer.

## Elastic Indexed Jobs

FEATURE STATE: Kubernetes v1.27 [beta]

You can scale Indexed Jobs up or down by mutating both `.spec.parallelism` and `.spec.completions` together such that `.spec.parallelism == .spec.completions`. When the `ElasticIndexedJob` [feature gate](#) on the [API server](#) is disabled, `.spec.completions` is immutable.

Use cases for elastic Indexed Jobs include batch workloads which require scaling an indexed Job, such as MPI, Horovod, Ray, and PyTorch training jobs.

## Delayed creation of replacement pods

FEATURE STATE: Kubernetes v1.29 [beta]

### Note:

You can only set `podReplacementPolicy` on Jobs if you enable the `JobPodReplacementPolicy` [feature gate](#) (enabled by default).

By default, the Job controller recreates Pods as soon they either fail or are terminating (have a deletion timestamp). This means that, at a given time, when some of the Pods are terminating, the number of running Pods for a Job can be greater than parallelism or greater than one Pod per index (if you are using an Indexed Job).

You may choose to create replacement Pods only when the terminating Pod is fully terminal (has `status.phase: Failed`). To do this, set the `.spec.podReplacementPolicy: Failed`. The default replacement policy depends on whether the Job has a `podFailurePolicy` set. With no Pod failure policy defined for a Job, omitting the `podReplacementPolicy` field selects the `TerminatingOrFailed` replacement policy: the control plane creates replacement Pods immediately upon Pod deletion (as soon as the control plane sees that a Pod for this Job has `deletionTimestamp` set). For Jobs with a Pod failure policy set, the default `podReplacementPolicy` is `Failed`, and no other value is permitted. See [Pod failure policy](#) to learn more about Pod failure policies for Jobs.

```
kind: Job
metadata:
 name: new
 ...
spec:
 podReplacementPolicy: Failed
 ...
```

Provided your cluster has the feature gate enabled, you can inspect the `.status.terminating` field of a Job. The value of the field is the number of Pods owned by the Job that are currently terminating.

```
kubectl get jobs/myjob -o yaml
```

```
apiVersion: batch/v1
kind: Job
.metadata and .spec omitted
status:
 terminating: 3 # three Pods are terminating and have not yet reached the Failed phase
```

## Delegation of managing a Job object to external controller

FEATURE STATE: Kubernetes v1.30 [alpha]

### Note:

You can only set the `managedBy` field on Jobs if you enable the `JobManagedBy` [feature gate](#) (disabled by default).

This feature allows you to disable the built-in Job controller, for a specific Job, and delegate reconciliation of the Job to an external controller.

You indicate the controller that reconciles the Job by setting a custom value for the `spec.managedBy` field - any value other than `kubernetes.io/job-controller`. The value of the field is immutable.

**Note:**

When using this feature, make sure the controller indicated by the field is installed, otherwise the Job may not be reconciled at all.

**Note:**

When developing an external Job controller be aware that your controller needs to operate in a fashion conformant with the definitions of the API spec and status fields of the Job object.

Please review these in detail in the [Job API](#). We also recommend that you run the e2e conformance tests for the Job object to verify your implementation.

Finally, when developing an external Job controller make sure it does not use the `batch.kubernetes.io/job-tracking` finalizer, reserved for the built-in controller.

**Warning:**

If you are considering to disable the `JobManagedBy` feature gate, or to downgrade the cluster to a version without the feature gate enabled, check if there are jobs with a custom value of the `spec.managedBy` field. If there are such jobs, there is a risk that they might be reconciled by two controllers after the operation: the built-in Job controller and the external controller indicated by the field value.

## Alternatives

### Bare Pods

When the node that a Pod is running on reboots or fails, the pod is terminated and will not be restarted. However, a Job will create new Pods to replace terminated ones. For this reason, we recommend that you use a Job rather than a bare Pod, even if your application requires only a single Pod.

### Replication Controller

Jobs are complementary to [Replication Controllers](#). A Replication Controller manages Pods which are not expected to terminate (e.g. web servers), and a Job manages Pods that are expected to terminate (e.g. batch tasks).

As discussed in [Pod Lifecycle](#), Job is *only* appropriate for pods with `RestartPolicy` equal to `OnFailure` or `Never`. (Note: If `RestartPolicy` is not set, the default value is `Always`.)

## Single Job starts controller Pod

Another pattern is for a single Job to create a Pod which then creates other Pods, acting as a sort of custom controller for those Pods. This allows the most flexibility, but may be somewhat complicated to get started with and offers less integration with Kubernetes.

One example of this pattern would be a Job which starts a Pod which runs a script that in turn starts a Spark master controller (see [spark example](#)), runs a spark driver, and then cleans up.

An advantage of this approach is that the overall process gets the completion guarantee of a Job object, but maintains complete control over what Pods are created and how work is assigned to them.

## What's next

- Learn about [Pods](#).
- Read about different ways of running Jobs:
  - [Coarse Parallel Processing Using a Work Queue](#)
  - [Fine Parallel Processing Using a Work Queue](#)
  - Use an [indexed Job for parallel processing with static work assignment](#)
  - Create multiple Jobs based on a template: [Parallel Processing using Expansions](#)
- Follow the links within [Clean up finished jobs automatically](#) to learn more about how your cluster can clean up completed and / or failed tasks.
- Job is part of the Kubernetes REST API. Read the [Job](#) object definition to understand the API for jobs.
- Read about [CronJob](#), which you can use to define a series of Jobs that will run based on a schedule, similar to the UNIX tool cron.
- Practice how to configure handling of retrieable and non-retrieable pod failures using `podFailurePolicy`, based on the step-by-step [examples](#).

## Automatic Cleanup for Finished Jobs

A time-to-live mechanism to clean up old Jobs that have finished execution.

FEATURE STATE: Kubernetes v1.23 [stable]

When your Job has finished, it's useful to keep that Job in the API (and not immediately delete the Job) so that you can tell whether the Job succeeded or failed.

Kubernetes' TTL-after-finished [controller](#) provides a TTL (time to live) mechanism to limit the lifetime of Job objects that have finished execution.

## Cleanup for finished Jobs

The TTL-after-finished controller is only supported for Jobs. You can use this mechanism to clean up finished Jobs (either Complete or Failed) automatically by specifying the `.spec.ttlSecondsAfterFinished` field of a Job, as in this [example](#).

The TTL-after-finished controller assumes that a Job is eligible to be cleaned up TTL seconds after the Job has finished. The timer starts once the status condition of the Job changes to show that the Job is either Complete or Failed; once the TTL has expired, that Job becomes eligible

for [cascading](#) removal. When the TTL-after-finished controller cleans up a job, it will delete it cascadingly, that is to say it will delete its dependent objects together with it.

Kubernetes honors object lifecycle guarantees on the Job, such as waiting for [finalizers](#).

You can set the TTL seconds at any time. Here are some examples for setting the `.spec.ttlSecondsAfterFinished` field of a Job:

- Specify this field in the Job manifest, so that a Job can be cleaned up automatically some time after it finishes.
- Manually set this field of existing, already finished Jobs, so that they become eligible for cleanup.
- Use a [mutating admission webhook](#) to set this field dynamically at Job creation time. Cluster administrators can use this to enforce a TTL policy for finished jobs.
- Use a [mutating admission webhook](#) to set this field dynamically after the Job has finished, and choose different TTL values based on job status, labels. For this case, the webhook needs to detect changes to the `.status` of the Job and only set a TTL when the Job is being marked as completed.
- Write your own controller to manage the cleanup TTL for Jobs that match a particular [selector-selector](#).

## Caveats

### Updating TTL for finished Jobs

You can modify the TTL period, e.g. `.spec.ttlSecondsAfterFinished` field of Jobs, after the job is created or has finished. If you extend the TTL period after the existing `ttlSecondsAfterFinished` period has expired, Kubernetes doesn't guarantee to retain that Job, even if an update to extend the TTL returns a successful API response.

### Time skew

Because the TTL-after-finished controller uses timestamps stored in the Kubernetes jobs to determine whether the TTL has expired or not, this feature is sensitive to time skew in your cluster, which may cause the control plane to clean up Job objects at the wrong time.

Clocks aren't always correct, but the difference should be very small. Please be aware of this risk when setting a non-zero TTL.

## What's next

- Read [Clean up Jobs automatically](#)
- Refer to the [Kubernetes Enhancement Proposal](#) (KEP) for adding this mechanism.

## CronJob

A CronJob starts one-time Jobs on a repeating schedule.  
FEATURE STATE: Kubernetes v1.21 [stable]

A *CronJob* creates [Jobs](#) on a repeating schedule.

CronJob is meant for performing regular scheduled actions such as backups, report generation, and so on. One CronJob object is like one line of a *crontab* (cron table) file on a Unix system. It runs a Job periodically on a given schedule, written in [Cron](#) format.

CronJobs have limitations and idiosyncrasies. For example, in certain circumstances, a single CronJob can create multiple concurrent Jobs. See the [limitations](#) below.

When the control plane creates new Jobs and (indirectly) Pods for a CronJob, the `.metadata.name` of the CronJob is part of the basis for naming those Pods. The name of a CronJob must be a valid [DNS subdomain](#) value, but this can produce unexpected results for the Pod hostnames. For best compatibility, the name should follow the more restrictive rules for a [DNS label](#). Even when the name is a DNS subdomain, the name must be no longer than 52 characters. This is because the CronJob controller will automatically append 11 characters to the name you provide and there is a constraint that the length of a Job name is no more than 63 characters.

## Example

This example CronJob manifest prints the current time and a hello message every minute:

[application/job/cronjob.yaml](#)

```
apiVersion: batch/v1
kind: CronJob
metadata:
 name: hello
spec:
 schedule: "* * * * *"
 jobTemplate:
 spec:
 template:
 spec:
 containers:
 - name: hello
 image: busybox:1.28
 imagePullPolicy: IfNotPresent
 command:
 - /bin/sh
 - -c
 - date; echo Hello from the Kubernetes cluster
 restartPolicy: OnFailure
```

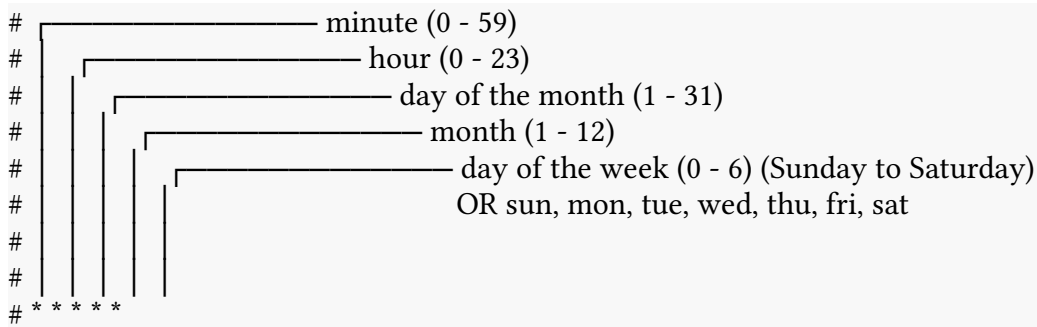
([Running Automated Tasks with a CronJob](#) takes you through this example in more detail).

## Writing a CronJob spec

### Schedule syntax

The `.spec.schedule` field is required. The value of that field follows the [Cron](#) syntax:





For example, `0 0 13 * 5` states that the task must be started every Friday at midnight, as well as on the 13th of each month at midnight.

The format also includes extended "Vixie cron" step values. As explained in the [FreeBSD manual](#):

Step values can be used in conjunction with ranges. Following a range with / <number> specifies skips of the number's value through the range. For example, 0-23/2 can be used in the hours field to specify command execution every other hour (the alternative in the V7 standard is 0,2,4,6,8,10,12,14,16,18,20,22). Steps are also permitted after an asterisk, so if you want to say "every two hours", just use \*/2.

#### Note:

A question mark (?) in the schedule has the same meaning as an asterisk \*, that is, it stands for any of available value for a given field.

Other than the standard syntax, some macros like @monthly can also be used:

| Entry                  | Description                                                | Equivalent to |
|------------------------|------------------------------------------------------------|---------------|
| @yearly (or @annually) | Run once a year at midnight of 1 January                   | 0 0 1 1 *     |
| @monthly               | Run once a month at midnight of the first day of the month | 0 0 1 * *     |
| @weekly                | Run once a week at midnight on Sunday morning              | 0 0 * * 0     |
| @daily (or @midnight)  | Run once a day at midnight                                 | 0 0 * * *     |
| @hourly                | Run once an hour at the beginning of the hour              | 0 * * * *     |

To generate CronJob schedule expressions, you can also use web tools like [crontab.guru](#).

## Job template

The `.spec.jobTemplate` defines a template for the Jobs that the CronJob creates, and it is required. It has exactly the same schema as a [Job](#), except that it is nested and does not have an `apiVersion` or `kind`. You can specify common metadata for the templated Jobs, such as [labels](#) or [annotations](#). For information about writing a Job `.spec`, see [Writing a Job Spec](#).

## Deadline for delayed Job start

The `.spec.startingDeadlineSeconds` field is optional. This field defines a deadline (in whole seconds) for starting the Job, if that Job misses its scheduled time for any reason.

After missing the deadline, the CronJob skips that instance of the Job (future occurrences are still scheduled). For example, if you have a backup Job that runs twice a day, you might allow it to start up to 8 hours late, but no later, because a backup taken any later wouldn't be useful: you would instead prefer to wait for the next scheduled run.

For Jobs that miss their configured deadline, Kubernetes treats them as failed Jobs. If you don't specify `startingDeadlineSeconds` for a CronJob, the Job occurrences have no deadline.

If the `.spec.startingDeadlineSeconds` field is set (not null), the CronJob controller measures the time between when a Job is expected to be created and now. If the difference is higher than that limit, it will skip this execution.

For example, if it is set to 200, it allows a Job to be created for up to 200 seconds after the actual schedule.

## Concurrency policy

The `.spec.concurrencyPolicy` field is also optional. It specifies how to treat concurrent executions of a Job that is created by this CronJob. The spec may specify only one of the following concurrency policies:

- Allow (default): The CronJob allows concurrently running Jobs
- Forbid: The CronJob does not allow concurrent runs; if it is time for a new Job run and the previous Job run hasn't finished yet, the CronJob skips the new Job run. Also note that when the previous Job run finishes, `.spec.startingDeadlineSeconds` is still taken into account and may result in a new Job run.
- Replace: If it is time for a new Job run and the previous Job run hasn't finished yet, the CronJob replaces the currently running Job run with a new Job run

Note that concurrency policy only applies to the Jobs created by the same CronJob. If there are multiple CronJobs, their respective Jobs are always allowed to run concurrently.

## Schedule suspension

You can suspend execution of Jobs for a CronJob, by setting the optional `.spec.suspend` field to true. The field defaults to false.

This setting does *not* affect Jobs that the CronJob has already started.

If you do set that field to true, all subsequent executions are suspended (they remain scheduled, but the CronJob controller does not start the Jobs to run the tasks) until you unsuspend the CronJob.

### Caution:

Executions that are suspended during their scheduled time count as missed Jobs. When `.spec.suspend` changes from true to false on an existing CronJob without a [starting deadline](#), the missed Jobs are scheduled immediately.

## Jobs history limits

The `.spec.successfulJobsHistoryLimit` and `.spec.failedJobsHistoryLimit` fields specify how many completed and failed Jobs should be kept. Both fields are optional.

- `.spec.successfulJobsHistoryLimit`: This field specifies the number of successful finished jobs to keep. The default value is 3. Setting this field to 0 will not keep any successful jobs.
- `.spec.failedJobsHistoryLimit`: This field specifies the number of failed finished jobs to keep. The default value is 1. Setting this field to 0 will not keep any failed jobs.

For another way to clean up Jobs automatically, see [Clean up finished Jobs automatically](#).

## Time zones

FEATURE STATE: Kubernetes v1.27 [stable]

For CronJobs with no time zone specified, the [kube-controller-manager](#) interprets schedules relative to its local time zone.

You can specify a time zone for a CronJob by setting `.spec.timeZone` to the name of a valid [time zone](#). For example, setting `.spec.timeZone: "Etc/UTC"` instructs Kubernetes to interpret the schedule relative to Coordinated Universal Time.

A time zone database from the Go standard library is included in the binaries and used as a fallback in case an external database is not available on the system.

## CronJob limitations

### Unsupported TimeZone specification

Specifying a timezone using `CRON_TZ` or `TZ` variables inside `.spec.schedule` is **not officially supported** (and never has been).

Starting with Kubernetes 1.29 if you try to set a schedule that includes `TZ` or `CRON_TZ` timezone specification, Kubernetes will fail to create the resource with a validation error. Updates to CronJobs already using `TZ` or `CRON_TZ` will continue to report a [warning](#) to the client.

### Modifying a CronJob

By design, a CronJob contains a template for *new* Jobs. If you modify an existing CronJob, the changes you make will apply to new Jobs that start to run after your modification is complete. Jobs (and their Pods) that have already started continue to run without changes. That is, the CronJob does *not* update existing Jobs, even if those remain running.

### Job creation

A CronJob creates a Job object approximately once per execution time of its schedule. The scheduling is approximate because there are certain circumstances where two Jobs might be

created, or no Job might be created. Kubernetes tries to avoid those situations, but does not completely prevent them. Therefore, the Jobs that you define should be *idempotent*.

If `startingDeadlineSeconds` is set to a large value or left unset (the default) and if `concurrencyPolicy` is set to `Allow`, the Jobs will always run at least once.

### Caution:

If `startingDeadlineSeconds` is set to a value less than 10 seconds, the CronJob may not be scheduled. This is because the CronJob controller checks things every 10 seconds.

For every CronJob, the CronJob [Controller](#) checks how many schedules it missed in the duration from its last scheduled time until now. If there are more than 100 missed schedules, then it does not start the Job and logs the error.

Cannot determine if job needs to be started. Too many missed start time (> 100). Set or decrease `.spec.startingDeadlineSeconds` or check clock skew.

It is important to note that if the `startingDeadlineSeconds` field is set (not nil), the controller counts how many missed Jobs occurred from the value of `startingDeadlineSeconds` until now rather than from the last scheduled time until now. For example, if `startingDeadlineSeconds` is 200, the controller counts how many missed Jobs occurred in the last 200 seconds.

A CronJob is counted as missed if it has failed to be created at its scheduled time. For example, if `concurrencyPolicy` is set to `Forbid` and a CronJob was attempted to be scheduled when there was a previous schedule still running, then it would count as missed.

For example, suppose a CronJob is set to schedule a new Job every one minute beginning at 08:30:00, and its `startingDeadlineSeconds` field is not set. If the CronJob controller happens to be down from 08:29:00 to 10:21:00, the Job will not start as the number of missed Jobs which missed their schedule is greater than 100.

To illustrate this concept further, suppose a CronJob is set to schedule a new Job every one minute beginning at 08:30:00, and its `startingDeadlineSeconds` is set to 200 seconds. If the CronJob controller happens to be down for the same period as the previous example (08:29:00 to 10:21:00,) the Job will still start at 10:22:00. This happens as the controller now checks how many missed schedules happened in the last 200 seconds (i.e., 3 missed schedules), rather than from the last scheduled time until now.

The CronJob is only responsible for creating Jobs that match its schedule, and the Job in turn is responsible for the management of the Pods it represents.

## What's next

- Learn about [Pods](#) and [Jobs](#), two concepts that CronJobs rely upon.
- Read about the detailed [format](#) of CronJob `.spec.schedule` fields.
- For instructions on creating and working with CronJobs, and for an example of a CronJob manifest, see [Running automated tasks with CronJobs](#).
- CronJob is part of the Kubernetes REST API. Read the [CronJob](#) API reference for more details.

# ReplicationController

Legacy API for managing workloads that can scale horizontally. Superseded by the Deployment and ReplicaSet APIs.

## Note:

A [Deployment](#) that configures a [ReplicaSet](#) is now the recommended way to set up replication.

A *ReplicationController* ensures that a specified number of pod replicas are running at any one time. In other words, a ReplicationController makes sure that a pod or a homogeneous set of pods is always up and available.

## How a ReplicationController works

If there are too many pods, the ReplicationController terminates the extra pods. If there are too few, the ReplicationController starts more pods. Unlike manually created pods, the pods maintained by a ReplicationController are automatically replaced if they fail, are deleted, or are terminated. For example, your pods are re-created on a node after disruptive maintenance such as a kernel upgrade. For this reason, you should use a ReplicationController even if your application requires only a single pod. A ReplicationController is similar to a process supervisor, but instead of supervising individual processes on a single node, the ReplicationController supervises multiple pods across multiple nodes.

ReplicationController is often abbreviated to "rc" in discussion, and as a shortcut in kubectl commands.

A simple case is to create one ReplicationController object to reliably run one instance of a Pod indefinitely. A more complex use case is to run several identical replicas of a replicated service, such as web servers.

## Running an example ReplicationController

This example ReplicationController config runs three copies of the nginx web server.

[controllers/replication.yaml](#)

```
apiVersion: v1
kind: ReplicationController
metadata:
 name: nginx
spec:
 replicas: 3
 selector:
 app: nginx
 template:
 metadata:
 name: nginx
 labels:
 app: nginx
 spec:
```

```
containers:
- name: nginx
 image: nginx
 ports:
 - containerPort: 80
```

Run the example job by downloading the example file and then running this command:

```
kubectl apply -f https://k8s.io/examples/controllers/replication.yaml
```

The output is similar to this:

```
replicationcontroller/nginx created
```

Check on the status of the ReplicationController using this command:

```
kubectl describe replicationcontrollers/nginx
```

The output is similar to this:

```
Name: nginx
Namespace: default
Selector: app=nginx
Labels: app=nginx
Annotations: <none>
Replicas: 3 current / 3 desired
Pods Status: 0 Running / 3 Waiting / 0 Succeeded / 0 Failed
Pod Template:
 Labels: app=nginx
 Containers:
 nginx:
 Image: nginx
 Port: 80/TCP
 Environment: <none>
 Mounts: <none>
 Volumes: <none>
Events:
 FirstSeen LastSeen Count From SubobjectPath Type Reason
Message

20s 20s 1 {replication-controller } Normal SuccessfulCreate
Created pod: nginx-qrm3m
20s 20s 1 {replication-controller } Normal SuccessfulCreate
Created pod: nginx-3ntk0
20s 20s 1 {replication-controller } Normal SuccessfulCreate
Created pod: nginx-4ok8v
```

Here, three pods are created, but none is running yet, perhaps because the image is being pulled. A little later, the same command may show:

```
Pods Status: 3 Running / 0 Waiting / 0 Succeeded / 0 Failed
```

To list all the pods that belong to the ReplicationController in a machine readable form, you can use a command like this:

```
 pods=$(kubectl get pods --selector=app=nginx --output=jsonpath={.items..metadata.name})
 echo $pods
```

The output is similar to this:

```
nginx-3ntk0 nginx-4ok8v nginx-qrm3m
```

Here, the selector is the same as the selector for the ReplicationController (seen in the `kubectl describe output`), and in a different form in `replication.yaml`. The `--output=jsonpath` option specifies an expression with the name from each pod in the returned list.

## Writing a ReplicationController Manifest

As with all other Kubernetes config, a ReplicationController needs `apiVersion`, `kind`, and `metadata` fields.

When the control plane creates new Pods for a ReplicationController, the `.metadata.name` of the ReplicationController is part of the basis for naming those Pods. The name of a ReplicationController must be a valid [DNS subdomain](#) value, but this can produce unexpected results for the Pod hostnames. For best compatibility, the name should follow the more restrictive rules for a [DNS label](#).

For general information about working with configuration files, see [object management](#).

A ReplicationController also needs a [.spec section](#).

### Pod Template

The `.spec.template` is the only required field of the `.spec`.

The `.spec.template` is a [pod template](#). It has exactly the same schema as a [Pod](#), except it is nested and does not have an `apiVersion` or `kind`.

In addition to required fields for a Pod, a pod template in a ReplicationController must specify appropriate labels and an appropriate restart policy. For labels, make sure not to overlap with other controllers. See [pod selector](#).

Only a [.spec.template.spec.restartPolicy](#) equal to `Always` is allowed, which is the default if not specified.

For local container restarts, ReplicationControllers delegate to an agent on the node, for example the [Kubelet](#).

### Labels on the ReplicationController

The ReplicationController can itself have labels (`.metadata.labels`). Typically, you would set these the same as the `.spec.template.metadata.labels`; if `.metadata.labels` is not specified then it defaults to `.spec.template.metadata.labels`. However, they are allowed to be different, and the `.metadata.labels` do not affect the behavior of the ReplicationController.

## Pod Selector

The `.spec.selector` field is a [label selector](#). A ReplicationController manages all the pods with labels that match the selector. It does not distinguish between pods that it created or deleted and pods that another person or process created or deleted. This allows the ReplicationController to be replaced without affecting the running pods.

If specified, the `.spec.template.metadata.labels` must be equal to the `.spec.selector`, or it will be rejected by the API. If `.spec.selector` is unspecified, it will be defaulted to `.spec.template.metadata.labels`.

Also you should not normally create any pods whose labels match this selector, either directly, with another ReplicationController, or with another controller such as Job. If you do so, the ReplicationController thinks that it created the other pods. Kubernetes does not stop you from doing this.

If you do end up with multiple controllers that have overlapping selectors, you will have to manage the deletion yourself (see [below](#)).

## Multiple Replicas

You can specify how many pods should run concurrently by setting `.spec.replicas` to the number of pods you would like to have running concurrently. The number running at any time may be higher or lower, such as if the replicas were just increased or decreased, or if a pod is gracefully shutdown, and a replacement starts early.

If you do not specify `.spec.replicas`, then it defaults to 1.

## Working with ReplicationControllers

### Deleting a ReplicationController and its Pods

To delete a ReplicationController and all its pods, use [kubectl delete](#). Kubectl will scale the ReplicationController to zero and wait for it to delete each pod before deleting the ReplicationController itself. If this kubectl command is interrupted, it can be restarted.

When using the REST API or [client library](#), you need to do the steps explicitly (scale replicas to 0, wait for pod deletions, then delete the ReplicationController).

### Deleting only a ReplicationController

You can delete a ReplicationController without affecting any of its pods.

Using kubectl, specify the `--cascade=orphan` option to [kubectl delete](#).

When using the REST API or [client library](#), you can delete the ReplicationController object.

Once the original is deleted, you can create a new ReplicationController to replace it. As long as the old and new `.spec.selector` are the same, then the new one will adopt the old pods. However, it will not make any effort to make existing pods match a new, different pod template. To update pods to a new spec in a controlled way, use a [rolling update](#).



## Isolating pods from a ReplicationController

Pods may be removed from a ReplicationController's target set by changing their labels. This technique may be used to remove pods from service for debugging and data recovery. Pods that are removed in this way will be replaced automatically (assuming that the number of replicas is not also changed).

## Common usage patterns

### Rescheduling

As mentioned above, whether you have 1 pod you want to keep running, or 1000, a ReplicationController will ensure that the specified number of pods exists, even in the event of node failure or pod termination (for example, due to an action by another control agent).

### Scaling

The ReplicationController enables scaling the number of replicas up or down, either manually or by an auto-scaling control agent, by updating the replicas field.

### Rolling updates

The ReplicationController is designed to facilitate rolling updates to a service by replacing pods one-by-one.

As explained in [#1353](#), the recommended approach is to create a new ReplicationController with 1 replica, scale the new (+1) and old (-1) controllers one by one, and then delete the old controller after it reaches 0 replicas. This predictably updates the set of pods regardless of unexpected failures.

Ideally, the rolling update controller would take application readiness into account, and would ensure that a sufficient number of pods were productively serving at any given time.

The two ReplicationControllers would need to create pods with at least one differentiating label, such as the image tag of the primary container of the pod, since it is typically image updates that motivate rolling updates.

### Multiple release tracks

In addition to running multiple releases of an application while a rolling update is in progress, it's common to run multiple releases for an extended period of time, or even continuously, using multiple release tracks. The tracks would be differentiated by labels.

For instance, a service might target all pods with tier in (frontend), environment in (prod). Now say you have 10 replicated pods that make up this tier. But you want to be able to 'canary' a new version of this component. You could set up a ReplicationController with replicas set to 9 for the bulk of the replicas, with labels tier=frontend, environment=prod, track=stable, and another ReplicationController with replicas set to 1 for the canary, with labels tier=frontend, environment=prod, track=canary. Now the service is covering both the canary and non-canary pods. But you can mess with the ReplicationControllers separately to test things out, monitor the results, etc.

## Using ReplicationControllers with Services

Multiple ReplicationControllers can sit behind a single service, so that, for example, some traffic goes to the old version, and some goes to the new version.

A ReplicationController will never terminate on its own, but it isn't expected to be as long-lived as services. Services may be composed of pods controlled by multiple ReplicationControllers, and it is expected that many ReplicationControllers may be created and destroyed over the lifetime of a service (for instance, to perform an update of pods that run the service). Both services themselves and their clients should remain oblivious to the ReplicationControllers that maintain the pods of the services.

## Writing programs for Replication

Pods created by a ReplicationController are intended to be fungible and semantically identical, though their configurations may become heterogeneous over time. This is an obvious fit for replicated stateless servers, but ReplicationControllers can also be used to maintain availability of master-elected, sharded, and worker-pool applications. Such applications should use dynamic work assignment mechanisms, such as the [RabbitMQ work queues](#), as opposed to static/one-time customization of the configuration of each pod, which is considered an anti-pattern. Any pod customization performed, such as vertical auto-sizing of resources (for example, cpu or memory), should be performed by another online controller process, not unlike the ReplicationController itself.

## Responsibilities of the ReplicationController

The ReplicationController ensures that the desired number of pods matches its label selector and are operational. Currently, only terminated pods are excluded from its count. In the future, [readiness](#) and other information available from the system may be taken into account, we may add more controls over the replacement policy, and we plan to emit events that could be used by external clients to implement arbitrarily sophisticated replacement and/or scale-down policies.

The ReplicationController is forever constrained to this narrow responsibility. It itself will not perform readiness nor liveness probes. Rather than performing auto-scaling, it is intended to be controlled by an external auto-scaler (as discussed in [#492](#)), which would change its replicas field. We will not add scheduling policies (for example, [spreading](#)) to the ReplicationController. Nor should it verify that the pods controlled match the currently specified template, as that would obstruct auto-sizing and other automated processes. Similarly, completion deadlines, ordering dependencies, configuration expansion, and other features belong elsewhere. We even plan to factor out the mechanism for bulk pod creation ([#170](#)).

The ReplicationController is intended to be a composable building-block primitive. We expect higher-level APIs and/or tools to be built on top of it and other complementary primitives for user convenience in the future. The "macro" operations currently supported by kubectl (run, scale) are proof-of-concept examples of this. For instance, we could imagine something like [Asgard](#) managing ReplicationControllers, auto-scalers, services, scheduling policies, canaries, etc.

# API Object

Replication controller is a top-level resource in the Kubernetes REST API. More details about the API object can be found at: [ReplicationController API object](#).

## Alternatives to ReplicationController

### ReplicaSet

[ReplicaSet](#) is the next-generation ReplicationController that supports the new [set-based label selector](#). It's mainly used by [Deployment](#) as a mechanism to orchestrate pod creation, deletion and updates. Note that we recommend using Deployments instead of directly using Replica Sets, unless you require custom update orchestration or don't require updates at all.

### Deployment (Recommended)

[Deployment](#) is a higher-level API object that updates its underlying Replica Sets and their Pods. Deployments are recommended if you want the rolling update functionality, because they are declarative, server-side, and have additional features.

### Bare Pods

Unlike in the case where a user directly created pods, a ReplicationController replaces pods that are deleted or terminated for any reason, such as in the case of node failure or disruptive node maintenance, such as a kernel upgrade. For this reason, we recommend that you use a ReplicationController even if your application requires only a single pod. Think of it similarly to a process supervisor, only it supervises multiple pods across multiple nodes instead of individual processes on a single node. A ReplicationController delegates local container restarts to some agent on the node, such as the kubelet.

### Job

Use a [Job](#) instead of a ReplicationController for pods that are expected to terminate on their own (that is, batch jobs).

### DaemonSet

Use a [DaemonSet](#) instead of a ReplicationController for pods that provide a machine-level function, such as machine monitoring or machine logging. These pods have a lifetime that is tied to a machine lifetime: the pod needs to be running on the machine before other pods start, and are safe to terminate when the machine is otherwise ready to be rebooted/shutdown.

## What's next

- Learn about [Pods](#).
- Learn about [Deployment](#), the replacement for ReplicationController.
- ReplicationController is part of the Kubernetes REST API. Read the [ReplicationController](#) object definition to understand the API for replication controllers.

# Autoscaling Workloads

With autoscaling, you can automatically update your workloads in one way or another. This allows your cluster to react to changes in resource demand more elastically and efficiently.

In Kubernetes, you can *scale* a workload depending on the current demand of resources. This allows your cluster to react to changes in resource demand more elastically and efficiently.

When you scale a workload, you can either increase or decrease the number of replicas managed by the workload, or adjust the resources available to the replicas in-place.

The first approach is referred to as *horizontal scaling*, while the second is referred to as *vertical scaling*.

There are manual and automatic ways to scale your workloads, depending on your use case.

## Scaling workloads manually

Kubernetes supports *manual scaling* of workloads. Horizontal scaling can be done using the kubectl CLI. For vertical scaling, you need to *patch* the resource definition of your workload.

See below for examples of both strategies.

- **Horizontal scaling:** [Running multiple instances of your app](#)
- **Vertical scaling:** [Resizing CPU and memory resources assigned to containers](#)

## Scaling workloads automatically

Kubernetes also supports *automatic scaling* of workloads, which is the focus of this page.

The concept of *Autoscaling* in Kubernetes refers to the ability to automatically update an object that manages a set of Pods (for example a [Deployment](#)).

### Scaling workloads horizontally

In Kubernetes, you can automatically scale a workload horizontally using a *HorizontalPodAutoscaler* (HPA).

It is implemented as a Kubernetes API resource and a [controller](#) and periodically adjusts the number of [replicas](#) in a workload to match observed resource utilization such as CPU or memory usage.

There is a [walkthrough tutorial](#) of configuring a HorizontalPodAutoscaler for a Deployment.

### Scaling workloads vertically

FEATURE STATE: Kubernetes v1.25 [stable]

You can automatically scale a workload vertically using a *VerticalPodAutoscaler* (VPA). Unlike the HPA, the VPA doesn't come with Kubernetes by default, but is a separate project that can be found [on GitHub](#).

Once installed, it allows you to create [CustomResourceDefinitions](#) (CRDs) for your workloads which define *how* and *when* to scale the resources of the managed replicas.

**Note:**

You will need to have the [Metrics Server](#) installed to your cluster for the HPA to work.

At the moment, the VPA can operate in four different modes:

Different modes of the VPA

| Mode     | Description                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Auto     | Currently, Recreate might change to in-place updates in the future                                                                                                                        |
| Recreate | The VPA assigns resource requests on pod creation as well as updates them on existing pods by evicting them when the requested resources differ significantly from the new recommendation |
| Initial  | The VPA only assigns resource requests on pod creation and never changes them later.                                                                                                      |
| Off      | The VPA does not automatically change the resource requirements of the pods. The recommendations are calculated and can be inspected in the VPA object.                                   |

### Requirements for in-place resizing

FEATURE STATE: Kubernetes v1.27 [alpha]

Resizing a workload in-place **without** restarting the [Pods](#) or its [Containers](#) requires Kubernetes version 1.27 or later. Additionally, the InPlaceVerticalScaling feature gate needs to be enabled.

InPlacePodVerticalScaling: Enables in-place Pod vertical scaling.

### Autoscaling based on cluster size

For workloads that need to be scaled based on the size of the cluster (for example cluster-dns or other system components), you can use the [Cluster Proportional Autoscaler](#). Just like the VPA, it is not part of the Kubernetes core, but hosted as its own project on GitHub.

The Cluster Proportional Autoscaler watches the number of schedulable [nodes](#) and cores and scales the number of replicas of the target workload accordingly.

If the number of replicas should stay the same, you can scale your workloads vertically according to the cluster size using the [Cluster Proportional Vertical Autoscaler](#). The project is **currently in beta** and can be found on GitHub.

While the Cluster Proportional Autoscaler scales the number of replicas of a workload, the Cluster Proportional Vertical Autoscaler adjusts the resource requests for a workload (for example a Deployment or DaemonSet) based on the number of nodes and/or cores in the cluster.

### Event driven Autoscaling

It is also possible to scale workloads based on events, for example using the [Kubernetes Event Driven Autoscaler \(KEDA\)](#).

KEDA is a CNCF graduated enabling you to scale your workloads based on the number of events to be processed, for example the amount of messages in a queue. There exists a wide range of adapters for different event sources to choose from.

## Autoscaling based on schedules

Another strategy for scaling your workloads is to **schedule** the scaling operations, for example in order to reduce resource consumption during off-peak hours.

Similar to event driven autoscaling, such behavior can be achieved using KEDA in conjunction with its [Cron scaler](#). The Cron scaler allows you to define schedules (and time zones) for scaling your workloads in or out.

## Scaling cluster infrastructure

If scaling workloads isn't enough to meet your needs, you can also scale your cluster infrastructure itself.

Scaling the cluster infrastructure normally means adding or removing [nodes](#). Read [cluster autoscaling](#) for more information.

## What's next

- Learn more about scaling horizontally
  - [Scale a StatefulSet](#)
  - [HorizontalPodAutoscaler Walkthrough](#)
- [Resize Container Resources In-Place](#)
- [Autoscale the DNS Service in a Cluster](#)
- Learn about [cluster autoscaling](#)

## Managing Workloads

You've deployed your application and exposed it via a Service. Now what? Kubernetes provides a number of tools to help you manage your application deployment, including scaling and updating.

## Organizing resource configurations

Many applications require multiple resources to be created, such as a Deployment along with a Service. Management of multiple resources can be simplified by grouping them together in the same file (separated by `---` in YAML). For example:

[application/nginx-app.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
 name: my-nginx-svc
labels:
 app: nginx
```

```
spec:
 type: LoadBalancer
 ports:
 - port: 80
 selector:
 app: nginx

apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
 labels:
 app: nginx
spec:
 replicas: 3
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
```

Multiple resources can be created the same way as a single resource:

```
kubectl apply -f https://k8s.io/examples/application/nginx-app.yaml
```

```
service/my-nginx-svc created
deployment.apps/my-nginx created
```

The resources will be created in the order they appear in the manifest. Therefore, it's best to specify the Service first, since that will ensure the scheduler can spread the pods associated with the Service as they are created by the controller(s), such as Deployment.

kubectl apply also accepts multiple -f arguments:

```
kubectl apply -f https://k8s.io/examples/application/nginx/nginx-svc.yaml \
-f https://k8s.io/examples/application/nginx/nginx-deployment.yaml
```

It is a recommended practice to put resources related to the same microservice or application tier into the same file, and to group all of the files associated with your application in the same directory. If the tiers of your application bind to each other using DNS, you can deploy all of the components of your stack together.

A URL can also be specified as a configuration source, which is handy for deploying directly from manifests in your source control system:

```
kubectl apply -f https://k8s.io/examples/application/nginx/nginx-deployment.yaml
```

```
deployment.apps/my-nginx created
```

If you need to define more manifests, such as adding a ConfigMap, you can do that too.

## External tools

This section lists only the most common tools used for managing workloads on Kubernetes. To see a larger list, view [Application definition and image build](#) in the [CNCF](#) Landscape.

### Helm

This item links to a third party project or product that is not part of Kubernetes itself. [More information](#)

[Helm](#) is a tool for managing packages of pre-configured Kubernetes resources. These packages are known as *Helm charts*.

### Kustomize

[Kustomize](#) traverses a Kubernetes manifest to add, remove or update configuration options. It is available both as a standalone binary and as a [native feature](#) of kubectl.

## Bulk operations in kubectl

Resource creation isn't the only operation that kubectl can perform in bulk. It can also extract resource names from configuration files in order to perform other operations, in particular to delete the same resources you created:

```
kubectl delete -f https://k8s.io/examples/application/nginx-app.yaml
```

```
deployment.apps "my-nginx" deleted
service "my-nginx-svc" deleted
```

In the case of two resources, you can specify both resources on the command line using the resource/name syntax:

```
kubectl delete deployments/my-nginx services/my-nginx-svc
```

For larger numbers of resources, you'll find it easier to specify the selector (label query) specified using `-l` or `--selector`, to filter resources by their labels:

```
kubectl delete deployment,services -l app=nginx
```

```
deployment.apps "my-nginx" deleted
service "my-nginx-svc" deleted
```

## Chaining and filtering

Because kubectl outputs resource names in the same syntax it accepts, you can chain operations using `$()` or `xargs`:



```
kubectl get $(kubectl create -f docs/concepts/cluster-administration/nginx/ -o name | grep service/)
kubectl create -f docs/concepts/cluster-administration/nginx/ -o name | grep service/ | xargs -i
kubectl get '{}'
```

The output might be similar to:

| NAME         | TYPE         | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|--------------|--------------|------------|-------------|---------|-----|
| my-nginx-svc | LoadBalancer | 10.0.0.208 | <pending>   | 80/TCP  | 0s  |

With the above commands, first you create resources under `examples/application/nginx/` and print the resources created with `-o name` output format (print each resource as `resource/name`). Then you `grep` only the Service, and then print it with [kubectl get](#).

## Recursive operations on local files

If you happen to organize your resources across several subdirectories within a particular directory, you can recursively perform the operations on the subdirectories also, by specifying `--recursive` or `-R` alongside the `--filename/-f` argument.

For instance, assume there is a directory `project/k8s/development` that holds all of the [manifests](#) needed for the development environment, organized by resource type:

```
project/k8s/development
├── configmap
│ └── my-configmap.yaml
├── deployment
│ └── my-deployment.yaml
└── pvc
 └── my-pvc.yaml
```

By default, performing a bulk operation on `project/k8s/development` will stop at the first level of the directory, not processing any subdirectories. If you had tried to create the resources in this directory using the following command, we would have encountered an error:

```
kubectl apply -f project/k8s/development
```

```
error: you must provide one or more resources by argument or filename (.json|.yaml|.yml|stdin)
```

Instead, specify the `--recursive` or `-R` command line argument along with the `--filename/-f` argument:

```
kubectl apply -f project/k8s/development --recursive
```

```
configmap/my-config created
deployment.apps/my-deployment created
persistentvolumeclaim/my-pvc created
```

The `--recursive` argument works with any operation that accepts the `--filename/-f` argument such as: `kubectl create`, `kubectl get`, `kubectl delete`, `kubectl describe`, or even `kubectl rollout`.

The `--recursive` argument also works when multiple `-f` arguments are provided:

```
kubectl apply -f project/k8s/namespaces -f project/k8s/development --recursive
```

```
namespace/development created
namespace/staging created
configmap/my-config created
deployment.apps/my-deployment created
persistentvolumeclaim/my-pvc created
```

If you're interested in learning more about kubectl, go ahead and read [Command line tool \(kubectl\)](#).

## Updating your application without an outage

At some point, you'll eventually need to update your deployed application, typically by specifying a new image or image tag. kubectl supports several update operations, each of which is applicable to different scenarios.

You can run multiple copies of your app, and use a *rollout* to gradually shift the traffic to new healthy Pods. Eventually, all the running Pods would have the new software.

This section of the page guides you through how to create and update applications with Deployments.

Let's say you were running version 1.14.2 of nginx:

```
kubectl create deployment my-nginx --image=nginx:1.14.2
```

```
deployment.apps/my-nginx created
```

Ensure that there is 1 replica:

```
kubectl scale --replicas 1 deployments/my-nginx --subresource='scale' --type='merge' -p '{"spec": {"replicas": 1}}'
```

```
deployment.apps/my-nginx scaled
```

and allow Kubernetes to add more temporary replicas during a rollout, by setting a *surge maximum* of 100%:

```
kubectl patch --type='merge' -p '{"spec":{"strategy":{"rollingUpdate":{"maxSurge": "100%" }}}}'
```

```
deployment.apps/my-nginx patched
```

To update to version 1.16.1, change `.spec.template.spec.containers[0].image` from `nginx:1.14.2` to `nginx:1.16.1` using kubectl edit:

```
kubectl edit deployment/my-nginx
Change the manifest to use the newer container image, then save your changes
```

That's it! The Deployment will declaratively update the deployed nginx application progressively behind the scene. It ensures that only a certain number of old replicas may be down while they are being updated, and only a certain number of new replicas may be created above the desired number of pods. To learn more details about how this happens, visit [Deployment](#).

You can use rollouts with DaemonSets, Deployments, or StatefulSets.

## Managing rollouts

You can use [kubectl rollout](#) to manage a progressive update of an existing application.

For example:

```
kubectl apply -f my-deployment.yaml
```

```
wait for rollout to finish
```

```
kubectl rollout status deployment/my-deployment --timeout 10m # 10 minute timeout
```

or

```
kubectl apply -f backing-stateful-component.yaml
```

```
don't wait for rollout to finish, just check the status
```

```
kubectl rollout status statefulsets/backing-stateful-component --watch=false
```

You can also pause, resume or cancel a rollout. Visit [kubectl rollout](#) to learn more.

## Canary deployments

Another scenario where multiple labels are needed is to distinguish deployments of different releases or configurations of the same component. It is common practice to deploy a *canary* of a new application release (specified via image tag in the pod template) side by side with the previous release so that the new release can receive live production traffic before fully rolling it out.

For instance, you can use a track label to differentiate different releases.

The primary, stable release would have a track label with value as stable:

```
name: frontend
replicas: 3
...
labels:
 app: guestbook
 tier: frontend
 track: stable
...
image: gb-frontend:v3
```

and then you can create a new release of the guestbook frontend that carries the track label with different value (i.e. canary), so that two sets of pods would not overlap:

```
name: frontend-canary
replicas: 1
...
labels:
 app: guestbook
 tier: frontend
 track: canary
```

```
...
image: gb-frontend:v4
```

The frontend service would span both sets of replicas by selecting the common subset of their labels (i.e. omitting the track label), so that the traffic will be redirected to both applications:

```
selector:
 app: guestbook
 tier: frontend
```

You can tweak the number of replicas of the stable and canary releases to determine the ratio of each release that will receive live production traffic (in this case, 3:1). Once you're confident, you can update the stable track to the new application release and remove the canary one.

## Updating annotations

Sometimes you would want to attach annotations to resources. Annotations are arbitrary non-identifying metadata for retrieval by API clients such as tools or libraries. This can be done with `kubectl annotate`. For example:

```
kubectl annotate pods my-nginx-v4-9gw19 description='my frontend running nginx'
kubectl get pods my-nginx-v4-9gw19 -o yaml
```

```
apiVersion: v1
kind: pod
metadata:
 annotations:
 description: my frontend running nginx
...
```

For more information, see [annotations](#) and [kubectl annotate](#).

## Scaling your application

When load on your application grows or shrinks, use `kubectl` to scale your application. For instance, to decrease the number of nginx replicas from 3 to 1, do:

```
kubectl scale deployment/my-nginx --replicas=1
```

```
deployment.apps/my-nginx scaled
```

Now you only have one pod managed by the deployment.

```
kubectl get pods -l app=nginx
```

| NAME                      | READY | STATUS  | RESTARTS | AGE |
|---------------------------|-------|---------|----------|-----|
| my-nginx-2035384211-j5fhi | 1/1   | Running | 0        | 30m |

To have the system automatically choose the number of nginx replicas as needed, ranging from 1 to 3, do:

```
This requires an existing source of container and Pod metrics
kubectl autoscale deployment/my-nginx --min=1 --max=3
```

```
horizontalpodautoscaler.autoscaling/my-nginx autoscaled
```

Now your nginx replicas will be scaled up and down as needed, automatically.

For more information, please see [kubectyl scale](#), [kubectyl autoscale](#) and [horizontal pod autoscaler](#) document.

## In-place updates of resources

Sometimes it's necessary to make narrow, non-disruptive updates to resources you've created.

### kubectyl apply

It is suggested to maintain a set of configuration files in source control (see [configuration as code](#)), so that they can be maintained and versioned along with the code for the resources they configure. Then, you can use [kubectyl apply](#) to push your configuration changes to the cluster.

This command will compare the version of the configuration that you're pushing with the previous version and apply the changes you've made, without overwriting any automated changes to properties you haven't specified.

```
kubectyl apply -f https://k8s.io/examples/application/nginx/nginx-deployment.yaml
```

```
deployment.apps/my-nginx configured
```

To learn more about the underlying mechanism, read [server-side apply](#).

### kubectyl edit

Alternatively, you may also update resources with [kubectyl edit](#):

```
kubectyl edit deployment/my-nginx
```

This is equivalent to first get the resource, edit it in text editor, and then apply the resource with the updated version:

```
kubectyl get deployment my-nginx -o yaml > /tmp/nginx.yaml
```

```
vi /tmp/nginx.yaml
```

```
do some edit, and then save the file
```

```
kubectyl apply -f /tmp/nginx.yaml
```

```
deployment.apps/my-nginx configured
```

```
rm /tmp/nginx.yaml
```

This allows you to do more significant changes more easily. Note that you can specify the editor with your EDITOR or KUBE\_EDITOR environment variables.

For more information, please see [kubectyl edit](#).

## kubectl patch

You can use [kubectl patch](#) to update API objects in place. This subcommand supports JSON patch, JSON merge patch, and strategic merge patch.

See [Update API Objects in Place Using kubectl patch](#) for more details.

## Disruptive updates

In some cases, you may need to update resource fields that cannot be updated once initialized, or you may want to make a recursive change immediately, such as to fix broken pods created by a Deployment. To change such fields, use `replace --force`, which deletes and re-creates the resource. In this case, you can modify your original configuration file:

```
kubectl replace -f https://k8s.io/examples/application/nginx/nginx-deployment.yaml --force
```

```
deployment.apps/my-nginx deleted
deployment.apps/my-nginx replaced
```

## What's next

- Learn about [how to use kubectl for application introspection and debugging](#).

# Services, Load Balancing, and Networking

Concepts and resources behind networking in Kubernetes.

## The Kubernetes network model

Every [Pod](#) in a cluster gets its own unique cluster-wide IP address. This means you do not need to explicitly create links between Pods and you almost never need to deal with mapping container ports to host ports.

This creates a clean, backwards-compatible model where Pods can be treated much like VMs or physical hosts from the perspectives of port allocation, naming, service discovery, [load balancing](#), application configuration, and migration.

Kubernetes imposes the following fundamental requirements on any networking implementation (barring any intentional network segmentation policies):

- pods can communicate with all other pods on any other [node](#) without NAT
- agents on a node (e.g. system daemons, kubelet) can communicate with all pods on that node

Note: For those platforms that support Pods running in the host network (e.g. Linux), when pods are attached to the host network of a node they can still communicate with all pods on all nodes without NAT.

This model is not only less complex overall, but it is principally compatible with the desire for Kubernetes to enable low-friction porting of apps from VMs to containers. If your job

previously ran in a VM, your VM had an IP and could talk to other VMs in your project. This is the same basic model.

Kubernetes IP addresses exist at the Pod scope - containers within a Pod share their network namespaces - including their IP address and MAC address. This means that containers within a Pod can all reach each other's ports on localhost. This also means that containers within a Pod must coordinate port usage, but this is no different from processes in a VM. This is called the "IP-per-pod" model.

How this is implemented is a detail of the particular container runtime in use.

It is possible to request ports on the Node itself which forward to your Pod (called host ports), but this is a very niche operation. How that forwarding is implemented is also a detail of the container runtime. The Pod itself is blind to the existence or non-existence of host ports.

Kubernetes networking addresses four concerns:

- Containers within a Pod [use networking to communicate](#) via loopback.
- Cluster networking provides communication between different Pods.
- The [Service](#) API lets you [expose an application running in Pods](#) to be reachable from outside your cluster.
  - [Ingress](#) provides extra functionality specifically for exposing HTTP applications, websites and APIs.
  - [Gateway API](#) is an [add-on](#) that provides an expressive, extensible, and role-oriented family of API kinds for modeling service networking.
- You can also use Services to [publish services only for consumption inside your cluster](#).

The [Connecting Applications with Services](#) tutorial lets you learn about Services and Kubernetes networking with a hands-on example.

[Cluster Networking](#) explains how to set up networking for your cluster, and also provides an overview of the technologies involved.

---

## [Service](#)

Expose an application running in your cluster behind a single outward-facing endpoint, even when the workload is split across multiple backends.

## [Ingress](#)

Make your HTTP (or HTTPS) network service available using a protocol-aware configuration mechanism, that understands web concepts like URIs, hostnames, paths, and more. The Ingress concept lets you map traffic to different backends based on rules you define via the Kubernetes API.

## [Ingress Controllers](#)

In order for an [Ingress](#) to work in your cluster, there must be an *ingress controller* running. You need to select at least one ingress controller and make sure it is set up in your cluster. This page lists common ingress controllers that you can deploy.

## [Gateway API](#)

Gateway API is a family of API kinds that provide dynamic infrastructure provisioning and advanced traffic routing.

## [EndpointSlices](#)

The EndpointSlice API is the mechanism that Kubernetes uses to let your Service scale to handle large numbers of backends, and allows the cluster to update its list of healthy backends efficiently.

## [Network Policies](#)

If you want to control traffic flow at the IP address or port level (OSI layer 3 or 4), NetworkPolicies allow you to specify rules for traffic flow within your cluster, and also between Pods and the outside world. Your cluster must use a network plugin that supports NetworkPolicy enforcement.

## [DNS for Services and Pods](#)

Your workload can discover Services within your cluster using DNS; this page explains how that works.

## [IPv4/IPv6 dual-stack](#)

Kubernetes lets you configure single-stack IPv4 networking, single-stack IPv6 networking, or dual stack networking with both network families active. This page explains how.

## [Topology Aware Routing](#)

*Topology Aware Routing* provides a mechanism to help keep network traffic within the zone where it originated. Preferring same-zone traffic between Pods in your cluster can help with reliability, performance (network latency and throughput), or cost.

## [Networking on Windows](#)

## [Service ClusterIP allocation](#)

## [Service Internal Traffic Policy](#)

If two Pods in your cluster want to communicate, and both Pods are actually running on the same node, use *Service Internal Traffic Policy* to keep network traffic within that node. Avoiding a round trip via the cluster network can help with reliability, performance (network latency and throughput), or cost.

# Service

Expose an application running in your cluster behind a single outward-facing endpoint, even when the workload is split across multiple backends.



In Kubernetes, a Service is a method for exposing a network application that is running as one or more [Pods](#) in your cluster.

A key aim of Services in Kubernetes is that you don't need to modify your existing application to use an unfamiliar service discovery mechanism. You can run code in Pods, whether this is a code designed for a cloud-native world, or an older app you've containerized. You use a Service to make that set of Pods available on the network so that clients can interact with it.

If you use a [Deployment](#) to run your app, that Deployment can create and destroy Pods dynamically. From one moment to the next, you don't know how many of those Pods are working and healthy; you might not even know what those healthy Pods are named. Kubernetes [Pods](#) are created and destroyed to match the desired state of your cluster. Pods are ephemeral resources (you should not expect that an individual Pod is reliable and durable).

Each Pod gets its own IP address (Kubernetes expects network plugins to ensure this). For a given Deployment in your cluster, the set of Pods running in one moment in time could be different from the set of Pods running that application a moment later.

This leads to a problem: if some set of Pods (call them "backends") provides functionality to other Pods (call them "frontends") inside your cluster, how do the frontends find out and keep track of which IP address to connect to, so that the frontend can use the backend part of the workload?

Enter *Services*.

## Services in Kubernetes

The Service API, part of Kubernetes, is an abstraction to help you expose groups of Pods over a network. Each Service object defines a logical set of endpoints (usually these endpoints are Pods) along with a policy about how to make those pods accessible.

For example, consider a stateless image-processing backend which is running with 3 replicas. Those replicas are fungible—frontends do not care which backend they use. While the actual Pods that compose the backend set may change, the frontend clients should not need to be aware of that, nor should they need to keep track of the set of backends themselves.

The Service abstraction enables this decoupling.

The set of Pods targeted by a Service is usually determined by a [selector](#) that you define. To learn about other ways to define Service endpoints, see [Services without selectors](#).

If your workload speaks HTTP, you might choose to use an [Ingress](#) to control how web traffic reaches that workload. Ingress is not a Service type, but it acts as the entry point for your cluster. An Ingress lets you consolidate your routing rules into a single resource, so that you can expose multiple components of your workload, running separately in your cluster, behind a single listener.

The [Gateway](#) API for Kubernetes provides extra capabilities beyond Ingress and Service. You can add Gateway to your cluster - it is a family of extension APIs, implemented using [CustomResourceDefinitions](#) - and then use these to configure access to network services that are running in your cluster.

## Cloud-native service discovery

If you're able to use Kubernetes APIs for service discovery in your application, you can query the [API server](#) for matching EndpointSlices. Kubernetes updates the EndpointSlices for a Service whenever the set of Pods in a Service changes.

For non-native applications, Kubernetes offers ways to place a network port or load balancer in between your application and the backend Pods.

Either way, your workload can use these [service discovery](#) mechanisms to find the target it wants to connect to.

## Defining a Service

A Service is an [object](#) (the same way that a Pod or a ConfigMap is an object). You can create, view or modify Service definitions using the Kubernetes API. Usually you use a tool such as kubectl to make those API calls for you.

For example, suppose you have a set of Pods that each listen on TCP port 9376 and are labelled as app.kubernetes.io/name=MyApp. You can define a Service to publish that TCP listener:

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
spec:
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - protocol: TCP
 port: 80
 targetPort: 9376
```

Applying this manifest creates a new Service named "my-service" with the default ClusterIP [service type](#). The Service targets TCP port 9376 on any Pod with the app.kubernetes.io/name: MyApp label.

Kubernetes assigns this Service an IP address (the *cluster IP*), that is used by the virtual IP address mechanism. For more details on that mechanism, read [Virtual IPs and Service Proxies](#).

The controller for that Service continuously scans for Pods that match its selector, and then makes any necessary updates to the set of EndpointSlices for the Service.

The name of a Service object must be a valid [RFC 1035 label name](#).

### Note:

A Service can map *any* incoming port to a targetPort. By default and for convenience, the targetPort is set to the same value as the port field.

## Port definitions

Port definitions in Pods have names, and you can reference these names in the `targetPort` attribute of a Service. For example, we can bind the `targetPort` of the Service to the Pod port in the following way:

```
apiVersion: v1
kind: Pod
metadata:
 name: nginx
 labels:
 app.kubernetes.io/name: proxy
spec:
 containers:
 - name: nginx
 image: nginx:stable
 ports:
 - containerPort: 80
 name: http-web-svc

apiVersion: v1
kind: Service
metadata:
 name: nginx-service
spec:
 selector:
 app.kubernetes.io/name: proxy
 ports:
 - name: name-of-service-port
 protocol: TCP
 port: 80
 targetPort: http-web-svc
```

This works even if there is a mixture of Pods in the Service using a single configured name, with the same network protocol available via different port numbers. This offers a lot of flexibility for deploying and evolving your Services. For example, you can change the port numbers that Pods expose in the next version of your backend software, without breaking clients.

The default protocol for Services is [TCP](#); you can also use any other [supported protocol](#).

Because many Services need to expose more than one port, Kubernetes supports [multiple port definitions](#) for a single Service. Each port definition can have the same protocol, or a different one.

## Services without selectors

Services most commonly abstract access to Kubernetes Pods thanks to the selector, but when used with a corresponding set of [EndpointSlices](#) objects and without a selector, the Service can abstract other kinds of backends, including ones that run outside the cluster.

For example:

- You want to have an external database cluster in production, but in your test environment you use your own databases.
- You want to point your Service to a Service in a different [Namespace](#) or on another cluster.
- You are migrating a workload to Kubernetes. While evaluating the approach, you run only a portion of your backends in Kubernetes.

In any of these scenarios you can define a Service *without* specifying a selector to match Pods. For example:

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
spec:
 ports:
 - name: http
 protocol: TCP
 port: 80
 targetPort: 9376
```

Because this Service has no selector, the corresponding EndpointSlice (and legacy Endpoints) objects are not created automatically. You can map the Service to the network address and port where it's running, by adding an EndpointSlice object manually. For example:

```
apiVersion: discovery.k8s.io/v1
kind: EndpointSlice
metadata:
 name: my-service-1 # by convention, use the name of the Service
 # as a prefix for the name of the EndpointSlice
labels:
 # You should set the "kubernetes.io/service-name" label.
 # Set its value to match the name of the Service
 kubernetes.io/service-name: my-service
addressType: IPv4
ports:
 - name: http # should match with the name of the service port defined above
 appProtocol: http
 protocol: TCP
 port: 9376
endpoints:
 - addresses:
 - "10.4.5.6"
 - addresses:
 - "10.1.2.3"
```

## Custom EndpointSlices

When you create an [EndpointSlice](#) object for a Service, you can use any name for the EndpointSlice. Each EndpointSlice in a namespace must have a unique name. You link an

EndpointSlice to a Service by setting the `kubernetes.io/service-name` [label](#) on that EndpointSlice.

**Note:**

The endpoint IPs *must not* be: loopback (127.0.0.0/8 for IPv4, ::1/128 for IPv6), or link-local (169.254.0.0/16 and 224.0.0.0/24 for IPv4, fe80::/64 for IPv6).

The endpoint IP addresses cannot be the cluster IPs of other Kubernetes Services, because [kube-proxy](#) doesn't support virtual IPs as a destination.

For an EndpointSlice that you create yourself, or in your own code, you should also pick a value to use for the label [endpointslice.kubernetes.io/managed-by](#). If you create your own controller code to manage EndpointSlices, consider using a value similar to "my-domain.example/name-of-controller". If you are using a third party tool, use the name of the tool in all-lowercase and change spaces and other punctuation to dashes (-). If people are directly using a tool such as kubectl to manage EndpointSlices, use a name that describes this manual management, such as "staff" or "cluster-admins". You should avoid using the reserved value "controller", which identifies EndpointSlices managed by Kubernetes' own control plane.

### Accessing a Service without a selector

Accessing a Service without a selector works the same as if it had a selector. In the [example](#) for a Service without a selector, traffic is routed to one of the two endpoints defined in the EndpointSlice manifest: a TCP connection to 10.1.2.3 or 10.4.5.6, on port 9376.

**Note:**

The Kubernetes API server does not allow proxying to endpoints that are not mapped to pods. Actions such as `kubectl proxy <service-name>` where the service has no selector will fail due to this constraint. This prevents the Kubernetes API server from being used as a proxy to endpoints the caller may not be authorized to access.

An ExternalName Service is a special case of Service that does not have selectors and uses DNS names instead. For more information, see the [ExternalName](#) section.

## EndpointSlices

FEATURE STATE: Kubernetes v1.21 [stable]

[EndpointSlices](#) are objects that represent a subset (a *slice*) of the backing network endpoints for a Service.

Your Kubernetes cluster tracks how many endpoints each EndpointSlice represents. If there are so many endpoints for a Service that a threshold is reached, then Kubernetes adds another empty EndpointSlice and stores new endpoint information there. By default, Kubernetes makes a new EndpointSlice once the existing EndpointSlices all contain at least 100 endpoints. Kubernetes does not make the new EndpointSlice until an extra endpoint needs to be added.

See [EndpointSlices](#) for more information about this API.

## Endpoints

In the Kubernetes API, an [Endpoints](#) (the resource kind is plural) defines a list of network endpoints, typically referenced by a Service to define which Pods the traffic can be sent to.

The EndpointSlice API is the recommended replacement for Endpoints.

### Over-capacity endpoints

Kubernetes limits the number of endpoints that can fit in a single Endpoints object. When there are over 1000 backing endpoints for a Service, Kubernetes truncates the data in the Endpoints object. Because a Service can be linked with more than one EndpointSlice, the 1000 backing endpoint limit only affects the legacy Endpoints API.

In that case, Kubernetes selects at most 1000 possible backend endpoints to store into the Endpoints object, and sets an [annotation](#) on the Endpoints: [endpoints.kubernetes.io/over-capacity: truncated](https://kubernetes.io/docs/concepts/services-networking/over-capacity-truncated/). The control plane also removes that annotation if the number of backend Pods drops below 1000.

Traffic is still sent to backends, but any load balancing mechanism that relies on the legacy Endpoints API only sends traffic to at most 1000 of the available backing endpoints.

The same API limit means that you cannot manually update an Endpoints to have more than 1000 endpoints.

## Application protocol

FEATURE STATE: Kubernetes v1.20 [stable]

The appProtocol field provides a way to specify an application protocol for each Service port. This is used as a hint for implementations to offer richer behavior for protocols that they understand. The value of this field is mirrored by the corresponding Endpoints and EndpointSlice objects.

This field follows standard Kubernetes label syntax. Valid values are one of:

- [IANA standard service names](#).
- Implementation-defined prefixed names such as mycompany.com/my-custom-protocol.
- Kubernetes-defined prefixed names:

| Protocol          | Description                                                       |
|-------------------|-------------------------------------------------------------------|
| kubernetes.io/h2c | HTTP/2 over cleartext as described in <a href="#">RFC 7540</a>    |
| kubernetes.io/ws  | WebSocket over cleartext as described in <a href="#">RFC 6455</a> |
| kubernetes.io/wss | WebSocket over TLS as described in <a href="#">RFC 6455</a>       |

## Multi-port Services

For some Services, you need to expose more than one port. Kubernetes lets you configure multiple port definitions on a Service object. When using multiple ports for a Service, you must give all of your ports names so that these are unambiguous. For example:

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
spec:
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - name: http
 protocol: TCP
 port: 80
 targetPort: 9376
 - name: https
 protocol: TCP
 port: 443
 targetPort: 9377
```

### Note:

As with Kubernetes [names](#) in general, names for ports must only contain lowercase alphanumeric characters and -. Port names must also start and end with an alphanumeric character.

For example, the names `123-abc` and `web` are valid, but `123_abc` and `-web` are not.

## Service type

For some parts of your application (for example, frontends) you may want to expose a Service onto an external IP address, one that's accessible from outside of your cluster.

Kubernetes Service types allow you to specify what kind of Service you want.

The available type values and their behaviors are:

### [ClusterIP](#)

Exposes the Service on a cluster-internal IP. Choosing this value makes the Service only reachable from within the cluster. This is the default that is used if you don't explicitly specify a type for a Service. You can expose the Service to the public internet using an [Ingress](#) or a [Gateway](#).

### [NodePort](#)

Exposes the Service on each Node's IP at a static port (the NodePort). To make the node port available, Kubernetes sets up a cluster IP address, the same as if you had requested a Service of type: ClusterIP.

### [LoadBalancer](#)

Exposes the Service externally using an external load balancer. Kubernetes does not directly offer a load balancing component; you must provide one, or you can integrate your Kubernetes cluster with a cloud provider.

### [ExternalName](#)

Maps the Service to the contents of the externalName field (for example, to the hostname `api.foo.bar.example`). The mapping configures your cluster's DNS server to return a CNAME record with that external hostname value. No proxying of any kind is set up.

The type field in the Service API is designed as nested functionality - each level adds to the previous. However there is an exception to this nested design. You can define a LoadBalancer Service by [disabling the load balancer NodePort allocation](#).

## **type: ClusterIP**

This default Service type assigns an IP address from a pool of IP addresses that your cluster has reserved for that purpose.

Several of the other types for Service build on the ClusterIP type as a foundation.

If you define a Service that has the .spec.clusterIP set to "None" then Kubernetes does not assign an IP address. See [headless Services](#) for more information.

## **Choosing your own IP address**

You can specify your own cluster IP address as part of a Service creation request. To do this, set the .spec.clusterIP field. For example, if you already have an existing DNS entry that you wish to reuse, or legacy systems that are configured for a specific IP address and difficult to re-configure.

The IP address that you choose must be a valid IPv4 or IPv6 address from within the service-cluster-ip-range CIDR range that is configured for the API server. If you try to create a Service with an invalid clusterIP address value, the API server will return a 422 HTTP status code to indicate that there's a problem.

Read [avoiding collisions](#) to learn how Kubernetes helps reduce the risk and impact of two different Services both trying to use the same IP address.

## **type: NodePort**

If you set the type field to NodePort, the Kubernetes control plane allocates a port from a range specified by --service-node-port-range flag (default: 30000-32767). Each node proxies that port (the same port number on every Node) into your Service. Your Service reports the allocated port in its .spec.ports[\*].nodePort field.

Using a NodePort gives you the freedom to set up your own load balancing solution, to configure environments that are not fully supported by Kubernetes, or even to expose one or more nodes' IP addresses directly.

For a node port Service, Kubernetes additionally allocates a port (TCP, UDP or SCTP to match the protocol of the Service). Every node in the cluster configures itself to listen on that assigned port and to forward traffic to one of the ready endpoints associated with that Service. You'll be able to contact the type: NodePort Service, from outside the cluster, by connecting to any node using the appropriate protocol (for example: TCP), and the appropriate port (as assigned to that Service).

## **Choosing your own port**

If you want a specific port number, you can specify a value in the nodePort field. The control plane will either allocate you that port or report that the API transaction failed. This means that



you need to take care of possible port collisions yourself. You also have to use a valid port number, one that's inside the range configured for NodePort use.

Here is an example manifest for a Service of type: NodePort that specifies a NodePort value (30007, in this example):

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
spec:
 type: NodePort
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - port: 80
 # By default and for convenience, the `targetPort` is set to
 # the same value as the `port` field.
 targetPort: 80
 # Optional field
 # By default and for convenience, the Kubernetes control plane
 # will allocate a port from a range (default: 30000-32767)
 nodePort: 30007
```

## Reserve Nodeport ranges to avoid collisions

FEATURE STATE: Kubernetes v1.29 [stable]

The policy for assigning ports to NodePort services applies to both the auto-assignment and the manual assignment scenarios. When a user wants to create a NodePort service that uses a specific port, the target port may conflict with another port that has already been assigned.

To avoid this problem, the port range for NodePort services is divided into two bands. Dynamic port assignment uses the upper band by default, and it may use the lower band once the upper band has been exhausted. Users can then allocate from the lower band with a lower risk of port collision.

## Custom IP address configuration for type: NodePort Services

You can set up nodes in your cluster to use a particular IP address for serving node port services. You might want to do this if each node is connected to multiple networks (for example: one network for application traffic, and another network for traffic between nodes and the control plane).

If you want to specify particular IP address(es) to proxy the port, you can set the `--nodeport-addresses` flag for kube-proxy or the equivalent `nodePortAddresses` field of the [kube-proxy configuration file](#) to particular IP block(s).

This flag takes a comma-delimited list of IP blocks (e.g. 10.0.0.0/8, 192.0.2.0/25) to specify IP address ranges that kube-proxy should consider as local to this node.

For example, if you start kube-proxy with the `--nodeport-addresses=127.0.0.0/8` flag, kube-proxy only selects the loopback interface for NodePort Services. The default for `--nodeport-addresses`

is an empty list. This means that kube-proxy should consider all available network interfaces for NodePort. (That's also compatible with earlier Kubernetes releases.)

**Note:**

This Service is visible as `<NodeIP>:spec.ports[*].nodePort` and `.spec.clusterIP:spec.ports[*].port`. If the `--nodeport-addresses` flag for kube-proxy or the equivalent field in the kube-proxy configuration file is set, `<NodeIP>` would be a filtered node IP address (or possibly IP addresses).

## **type: LoadBalancer**

On cloud providers which support external load balancers, setting the `type` field to `LoadBalancer` provisions a load balancer for your Service. The actual creation of the load balancer happens asynchronously, and information about the provisioned balancer is published in the Service's `.status.loadBalancer` field. For example:

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
spec:
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - protocol: TCP
 port: 80
 targetPort: 9376
 clusterIP: 10.0.171.239
 type: LoadBalancer
status:
 loadBalancer:
 ingress:
 - ip: 192.0.2.127
```

Traffic from the external load balancer is directed at the backend Pods. The cloud provider decides how it is load balanced.

To implement a Service of type: `LoadBalancer`, Kubernetes typically starts off by making the changes that are equivalent to you requesting a Service of type: `NodePort`. The cloud-controller-manager component then configures the external load balancer to forward traffic to that assigned node port.

You can configure a load balanced Service to [omit](#) assigning a node port, provided that the cloud provider implementation supports this.

Some cloud providers allow you to specify the `loadBalancerIP`. In those cases, the load-balancer is created with the user-specified `loadBalancerIP`. If the `loadBalancerIP` field is not specified, the load balancer is set up with an ephemeral IP address. If you specify a `loadBalancerIP` but your cloud provider does not support the feature, the `loadBalancerIP` field that you set is ignored.

**Note:**

The `spec.loadBalancerIP` field for a Service was deprecated in Kubernetes v1.24.

This field was under-specified and its meaning varies across implementations. It also cannot support dual-stack networking. This field may be removed in a future API version.

If you're integrating with a provider that supports specifying the load balancer IP address(es) for a Service via a (provider specific) annotation, you should switch to doing that.

If you are writing code for a load balancer integration with Kubernetes, avoid using this field. You can integrate with [Gateway](#) rather than Service, or you can define your own (provider specific) annotations on the Service that specify the equivalent detail.

## **Node liveness impact on load balancer traffic**

Load balancer health checks are critical to modern applications. They are used to determine which server (virtual machine, or IP address) the load balancer should dispatch traffic to. The Kubernetes APIs do not define how health checks have to be implemented for Kubernetes managed load balancers, instead it's the cloud providers (and the people implementing integration code) who decide on the behavior. Load balancer health checks are extensively used within the context of supporting the externalTrafficPolicy field for Services.

## **Load balancers with mixed protocol types**

FEATURE STATE: Kubernetes v1.26 [stable]

By default, for LoadBalancer type of Services, when there is more than one port defined, all ports must have the same protocol, and the protocol must be one which is supported by the cloud provider.

The feature gate MixedProtocolLBService (enabled by default for the kube-apiserver as of v1.24) allows the use of different protocols for LoadBalancer type of Services, when there is more than one port defined.

### **Note:**

The set of protocols that can be used for load balanced Services is defined by your cloud provider; they may impose restrictions beyond what the Kubernetes API enforces.

## **Disabling load balancer NodePort allocation**

FEATURE STATE: Kubernetes v1.24 [stable]

You can optionally disable node port allocation for a Service of type: LoadBalancer, by setting the field spec.allocateLoadBalancerNodePorts to false. This should only be used for load balancer implementations that route traffic directly to pods as opposed to using node ports. By default, spec.allocateLoadBalancerNodePorts is true and type LoadBalancer Services will continue to allocate node ports. If spec.allocateLoadBalancerNodePorts is set to false on an existing Service with allocated node ports, those node ports will **not** be de-allocated automatically. You must explicitly remove the nodePorts entry in every Service port to de-allocate those node ports.

## **Specifying class of load balancer implementation**

FEATURE STATE: Kubernetes v1.24 [stable]

For a Service with type set to LoadBalancer, the `.spec.loadBalancerClass` field enables you to use a load balancer implementation other than the cloud provider default.

By default, `.spec.loadBalancerClass` is not set and a LoadBalancer type of Service uses the cloud provider's default load balancer implementation if the cluster is configured with a cloud provider using the `--cloud-provider` component flag.

If you specify `.spec.loadBalancerClass`, it is assumed that a load balancer implementation that matches the specified class is watching for Services. Any default load balancer implementation (for example, the one provided by the cloud provider) will ignore Services that have this field set. `.spec.loadBalancerClass` can be set on a Service of type LoadBalancer only. Once set, it cannot be changed. The value of `.spec.loadBalancerClass` must be a label-style identifier, with an optional prefix such as "internal-vip" or "example.com/internal-vip". Unprefixed names are reserved for end-users.

## Specifying IPMode of load balancer status

FEATURE STATE: Kubernetes v1.30 [beta]

As a Beta feature in Kubernetes 1.30, a [feature gate](#) named `LoadBalancerIPMode` allows you to set the `.status.loadBalancer.ingress.ipMode` for a Service with type set to LoadBalancer. The `.status.loadBalancer.ingress.ipMode` specifies how the load-balancer IP behaves. It may be specified only when the `.status.loadBalancer.ingress.ip` field is also specified.

There are two possible values for `.status.loadBalancer.ingress.ipMode`: "VIP" and "Proxy". The default value is "VIP" meaning that traffic is delivered to the node with the destination set to the load-balancer's IP and port. There are two cases when setting this to "Proxy", depending on how the load-balancer from the cloud provider delivers the traffics:

- If the traffic is delivered to the node then DNATed to the pod, the destination would be set to the node's IP and node port;
- If the traffic is delivered directly to the pod, the destination would be set to the pod's IP and port.

Service implementations may use this information to adjust traffic routing.

## Internal load balancer

In a mixed environment it is sometimes necessary to route traffic from Services inside the same (virtual) network address block.

In a split-horizon DNS environment you would need two Services to be able to route both external and internal traffic to your endpoints.

To set an internal load balancer, add one of the following annotations to your Service depending on the cloud service provider you're using:

- [Default](#)
- [GCP](#)
- [AWS](#)
- [Azure](#)
- [IBM Cloud](#)
- [OpenStack](#)

- [Baidu Cloud](#)
- [Tencent Cloud](#)
- [Alibaba Cloud](#)
- [OCI](#)

Select one of the tabs.

```
metadata:
 name: my-service
 annotations:
 networking.gke.io/load-balancer-type: "Internal"
```

```
metadata:
 name: my-service
 annotations:
 service.beta.kubernetes.io/aws-load-balancer-internal: "true"
```

```
metadata:
 name: my-service
 annotations:
 service.beta.kubernetes.io/azure-load-balancer-internal: "true"
```

```
metadata:
 name: my-service
 annotations:
 service.kubernetes.io/ibm-load-balancer-cloud-provider-ip-type: "private"
```

```
metadata:
 name: my-service
 annotations:
 service.beta.kubernetes.io/openstack-internal-load-balancer: "true"
```

```
metadata:
 name: my-service
 annotations:
 service.beta.kubernetes.io/cce-load-balancer-internal-vpc: "true"
```

```
metadata:
 annotations:
 service.kubernetes.io/qcloud-loadbalancer-internal-subnetid: subnet-xxxxx
```

```
metadata:
 annotations:
 service.beta.kubernetes.io/alibaba-cloud-loadbalancer-address-type: "intranet"
```

```
metadata:
 name: my-service
 annotations:
 service.beta.kubernetes.io/oci-load-balancer-internal: true
```

## type: ExternalName

Services of type ExternalName map a Service to a DNS name, not to a typical selector such as my-service or cassandra. You specify these Services with the spec.externalName parameter.

This Service definition, for example, maps the `my-service` Service in the `prod` namespace to `my.database.example.com`:

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
 namespace: prod
spec:
 type: ExternalName
 externalName: my.database.example.com
```

**Note:**

A Service of type: `ExternalName` accepts an IPv4 address string, but treats that string as a DNS name comprised of digits, not as an IP address (the internet does not however allow such names in DNS). Services with external names that resemble IPv4 addresses are not resolved by DNS servers.

If you want to map a Service directly to a specific IP address, consider using [headless Services](#).

When looking up the host `my-service.prod.svc.cluster.local`, the cluster DNS Service returns a CNAME record with the value `my.database.example.com`. Accessing `my-service` works in the same way as other Services but with the crucial difference that redirection happens at the DNS level rather than via proxying or forwarding. Should you later decide to move your database into your cluster, you can start its Pods, add appropriate selectors or endpoints, and change the Service's type.

**Caution:**

You may have trouble using `ExternalName` for some common protocols, including HTTP and HTTPS. If you use `ExternalName` then the hostname used by clients inside your cluster is different from the name that the `ExternalName` references.

For protocols that use hostnames this difference may lead to errors or unexpected responses. HTTP requests will have a `Host:` header that the origin server does not recognize; TLS servers will not be able to provide a certificate matching the hostname that the client connected to.

## Headless Services

Sometimes you don't need load-balancing and a single Service IP. In this case, you can create what are termed *headless Services*, by explicitly specifying "None" for the cluster IP address (`.spec.clusterIP`).

You can use a headless Service to interface with other service discovery mechanisms, without being tied to Kubernetes' implementation.

For headless Services, a cluster IP is not allocated, kube-proxy does not handle these Services, and there is no load balancing or proxying done by the platform for them.

A headless Service allows a client to connect to whichever Pod it prefers, directly. Services that are headless don't configure routes and packet forwarding using [virtual IP addresses and](#)

[proxies](#); instead, headless Services report the endpoint IP addresses of the individual pods via internal DNS records, served through the cluster's [DNS service](#). To define a headless Service, you make a Service with `.spec.type` set to `ClusterIP` (which is also the default for `type`), and you additionally set `.spec.clusterIP` to `None`.

The string value `None` is a special case and is not the same as leaving the `.spec.clusterIP` field unset.

How DNS is automatically configured depends on whether the Service has selectors defined:

## With selectors

For headless Services that define selectors, the endpoints controller creates `EndpointSlices` in the Kubernetes API, and modifies the DNS configuration to return `A` or `AAAA` records (IPv4 or IPv6 addresses) that point directly to the Pods backing the Service.

## Without selectors

For headless Services that do not define selectors, the control plane does not create `EndpointSlice` objects. However, the DNS system looks for and configures either:

- DNS `CNAME` records for [type: ExternalName](#) Services.
- DNS `A` / `AAAA` records for all IP addresses of the Service's ready endpoints, for all Service types other than `ExternalName`.
  - For IPv4 endpoints, the DNS system creates `A` records.
  - For IPv6 endpoints, the DNS system creates `AAAA` records.

When you define a headless Service without a selector, the port must match the `targetPort`.

## Discovering services

For clients running inside your cluster, Kubernetes supports two primary modes of finding a Service: environment variables and DNS.

## Environment variables

When a Pod is run on a Node, the kubelet adds a set of environment variables for each active Service. It adds `{SVCNAME}_SERVICE_HOST` and `{SVCNAME}_SERVICE_PORT` variables, where the Service name is upper-cased and dashes are converted to underscores.

For example, the Service `redis-primary` which exposes TCP port 6379 and has been allocated cluster IP address 10.0.0.11, produces the following environment variables:

```
REDIS_PRIMARY_SERVICE_HOST=10.0.0.11
REDIS_PRIMARY_SERVICE_PORT=6379
REDIS_PRIMARY_PORT=tcp://10.0.0.11:6379
REDIS_PRIMARY_PORT_6379_TCP=tcp://10.0.0.11:6379
REDIS_PRIMARY_PORT_6379_TCP_PROTO=tcp
REDIS_PRIMARY_PORT_6379_TCP_PORT=6379
REDIS_PRIMARY_PORT_6379_TCP_ADDR=10.0.0.11
```



## Note:

When you have a Pod that needs to access a Service, and you are using the environment variable method to publish the port and cluster IP to the client Pods, you must create the Service *before* the client Pods come into existence. Otherwise, those client Pods won't have their environment variables populated.

If you only use DNS to discover the cluster IP for a Service, you don't need to worry about this ordering issue.

Kubernetes also supports and provides variables that are compatible with Docker Engine's "[legacy container links](#)" feature. You can read [makeLinkVariables](#) to see how this is implemented in Kubernetes.

## DNS

You can (and almost always should) set up a DNS service for your Kubernetes cluster using an [add-on](#).

A cluster-aware DNS server, such as CoreDNS, watches the Kubernetes API for new Services and creates a set of DNS records for each one. If DNS has been enabled throughout your cluster then all Pods should automatically be able to resolve Services by their DNS name.

For example, if you have a Service called `my-service` in a Kubernetes namespace `my-ns`, the control plane and the DNS Service acting together create a DNS record for `my-service.my-ns`. Pods in the `my-ns` namespace should be able to find the service by doing a name lookup for `my-service` (`my-service.my-ns` would also work).

Pods in other namespaces must qualify the name as `my-service.my-ns`. These names will resolve to the cluster IP assigned for the Service.

Kubernetes also supports DNS SRV (Service) records for named ports. If the `my-service.my-ns` Service has a port named `http` with the protocol set to TCP, you can do a DNS SRV query for `_http._tcp.my-service.my-ns` to discover the port number for `http`, as well as the IP address.

The Kubernetes DNS server is the only way to access ExternalName Services. You can find more information about ExternalName resolution in [DNS for Services and Pods](#).

## Virtual IP addressing mechanism

Read [Virtual IPs and Service Proxies](#) explains the mechanism Kubernetes provides to expose a Service with a virtual IP address.

## Traffic policies

You can set the `.spec.internalTrafficPolicy` and `.spec.externalTrafficPolicy` fields to control how Kubernetes routes traffic to healthy ("ready") backends.

See [Traffic Policies](#) for more details.



## Traffic distribution

FEATURE STATE: Kubernetes v1.30 [alpha]

The `.spec.trafficDistribution` field provides another way to influence traffic routing within a Kubernetes Service. While traffic policies focus on strict semantic guarantees, traffic distribution allows you to express *preferences* (such as routing to topologically closer endpoints). This can help optimize for performance, cost, or reliability. This optional field can be used if you have enabled the ServiceTrafficDistribution [feature gate](#) for your cluster and all of its nodes. In Kubernetes 1.30, the following field value is supported:

### PreferClose

Indicates a preference for routing traffic to endpoints that are topologically proximate to the client. The interpretation of "topologically proximate" may vary across implementations and could encompass endpoints within the same node, rack, zone, or even region. Setting this value gives implementations permission to make different tradeoffs, e.g. optimizing for proximity rather than equal distribution of load. Users should not set this value if such tradeoffs are not acceptable.

If the field is not set, the implementation will apply its default routing strategy.

See [Traffic Distribution](#) for more details

## Session stickiness

If you want to make sure that connections from a particular client are passed to the same Pod each time, you can configure session affinity based on the client's IP address. Read [session affinity](#) to learn more.

## External IPs

If there are external IPs that route to one or more cluster nodes, Kubernetes Services can be exposed on those externalIPs. When network traffic arrives into the cluster, with the external IP (as destination IP) and the port matching that Service, rules and routes that Kubernetes has configured ensure that the traffic is routed to one of the endpoints for that Service.

When you define a Service, you can specify externalIPs for any [service type](#). In the example below, the Service named "my-service" can be accessed by clients using TCP, on "198.51.100.32:80" (calculated from `.spec.externalIPs[]` and `.spec.ports[].port`).

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
spec:
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - name: http
 protocol: TCP
 port: 80
 targetPort: 49152
```

```
externalIPs:
 - 198.51.100.32
```

**Note:**

Kubernetes does not manage allocation of externalIPs; these are the responsibility of the cluster administrator.

## API Object

Service is a top-level resource in the Kubernetes REST API. You can find more details about the [Service API object](#).

## What's next

Learn more about Services and how they fit into Kubernetes:

- Follow the [Connecting Applications with Services](#) tutorial.
- Read about [Ingress](#), which exposes HTTP and HTTPS routes from outside the cluster to Services within your cluster.
- Read about [Gateway](#), an extension to Kubernetes that provides more flexibility than Ingress.

For more context, read the following:

- [Virtual IPs and Service Proxies](#)
- [EndpointSlices](#)
- [Service API reference](#)
- [EndpointSlice API reference](#)
- [Endpoint API reference \(legacy\)](#)

## Ingress

Make your HTTP (or HTTPS) network service available using a protocol-aware configuration mechanism, that understands web concepts like URIs, hostnames, paths, and more. The Ingress concept lets you map traffic to different backends based on rules you define via the Kubernetes API.

FEATURE STATE: Kubernetes v1.19 [stable]

An API object that manages external access to the services in a cluster, typically HTTP.

Ingress may provide load balancing, SSL termination and name-based virtual hosting.

**Note:**

Ingress is frozen. New features are being added to the [Gateway API](#).

# Terminology

For clarity, this guide defines the following terms:

- Node: A worker machine in Kubernetes, part of a cluster.
- Cluster: A set of Nodes that run containerized applications managed by Kubernetes. For this example, and in most common Kubernetes deployments, nodes in the cluster are not part of the public internet.
- Edge router: A router that enforces the firewall policy for your cluster. This could be a gateway managed by a cloud provider or a physical piece of hardware.
- Cluster network: A set of links, logical or physical, that facilitate communication within a cluster according to the Kubernetes [networking model](#).
- Service: A Kubernetes [Service](#) that identifies a set of Pods using [label](#) selectors. Unless mentioned otherwise, Services are assumed to have virtual IPs only routable within the cluster network.

## What is Ingress?

[Ingress](#) exposes HTTP and HTTPS routes from outside the cluster to [services](#) within the cluster. Traffic routing is controlled by rules defined on the Ingress resource.

Here is a simple example where an Ingress sends all its traffic to one Service:

[ingress-diagram](#)

Figure. Ingress

An Ingress may be configured to give Services externally-reachable URLs, load balance traffic, terminate SSL / TLS, and offer name-based virtual hosting. An [Ingress controller](#) is responsible for fulfilling the Ingress, usually with a load balancer, though it may also configure your edge router or additional frontends to help handle the traffic.

An Ingress does not expose arbitrary ports or protocols. Exposing services other than HTTP and HTTPS to the internet typically uses a service of type [Service.Type=NodePort](#) or [Service.Type=LoadBalancer](#).

## Prerequisites

You must have an [Ingress controller](#) to satisfy an Ingress. Only creating an Ingress resource has no effect.

You may need to deploy an Ingress controller such as [ingress-nginx](#). You can choose from a number of [Ingress controllers](#).

Ideally, all Ingress controllers should fit the reference specification. In reality, the various Ingress controllers operate slightly differently.

### Note:

Make sure you review your Ingress controller's documentation to understand the caveats of choosing it.

# The Ingress resource

A minimal Ingress resource example:

[service/networking/minimal-ingress.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: minimal-ingress
 annotations:
 nginx.ingress.kubernetes.io/rewrite-target: /
spec:
 ingressClassName: nginx-example
 rules:
 - http:
 paths:
 - path: /testpath
 pathType: Prefix
 backend:
 service:
 name: test
 port:
 number: 80
```

An Ingress needs `apiVersion`, `kind`, `metadata` and `spec` fields. The name of an Ingress object must be a valid [DNS subdomain name](#). For general information about working with config files, see [deploying applications](#), [configuring containers](#), [managing resources](#). Ingress frequently uses annotations to configure some options depending on the Ingress controller, an example of which is the [rewrite-target annotation](#). Different [Ingress controllers](#) support different annotations. Review the documentation for your choice of Ingress controller to learn which annotations are supported.

The [Ingress spec](#) has all the information needed to configure a load balancer or proxy server. Most importantly, it contains a list of rules matched against all incoming requests. Ingress resource only supports rules for directing HTTP(S) traffic.

If the `ingressClassName` is omitted, a [default Ingress class](#) should be defined.

There are some ingress controllers, that work without the definition of a default IngressClass. For example, the Ingress-NGINX controller can be configured with a [flag](#) `--watch-ingress-without-class`. It is [recommended](#) though, to specify the default IngressClass as shown [below](#).

## Ingress rules

Each HTTP rule contains the following information:

- An optional host. In this example, no host is specified, so the rule applies to all inbound HTTP traffic through the IP address specified. If a host is provided (for example, `foo.bar.com`), the rules apply to that host.
- A list of paths (for example, `/testpath`), each of which has an associated backend defined with a `service.name` and a `service.port.name` or `service.port.number`. Both the host and

path must match the content of an incoming request before the load balancer directs traffic to the referenced Service.

- A backend is a combination of Service and port names as described in the [Service doc](#) or a [custom resource backend](#) by way of a [CRD](#). HTTP (and HTTPS) requests to the Ingress that match the host and path of the rule are sent to the listed backend.

A defaultBackend is often configured in an Ingress controller to service any requests that do not match a path in the spec.

## DefaultBackend

An Ingress with no rules sends all traffic to a single default backend and .spec.defaultBackend is the backend that should handle requests in that case. The defaultBackend is conventionally a configuration option of the [Ingress controller](#) and is not specified in your Ingress resources. If no .spec.rules are specified, .spec.defaultBackend must be specified. If defaultBackend is not set, the handling of requests that do not match any of the rules will be up to the ingress controller (consult the documentation for your ingress controller to find out how it handles this case).

If none of the hosts or paths match the HTTP request in the Ingress objects, the traffic is routed to your default backend.

## Resource backends

A Resource backend is an ObjectRef to another Kubernetes resource within the same namespace as the Ingress object. A Resource is a mutually exclusive setting with Service, and will fail validation if both are specified. A common usage for a Resource backend is to ingress data to an object storage backend with static assets.

[service/networking/ingress-resource-backend.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: ingress-resource-backend
spec:
 defaultBackend:
 resource:
 apiGroup: k8s.example.com
 kind: StorageBucket
 name: static-assets
 rules:
 - http:
 paths:
 - path: /icons
 pathType: ImplementationSpecific
 backend:
 resource:
 apiGroup: k8s.example.com
 kind: StorageBucket
 name: icon-assets
```

After creating the Ingress above, you can view it with the following command:

```
kubectl describe ingress ingress-resource-backend
```

```
Name: ingress-resource-backend
Namespace: default
Address:
Default backend: APIGroup: k8s.example.com, Kind: StorageBucket, Name: static-assets
Rules:
 Host Path Backends
 ---- -
 *
 /icons APIGroup: k8s.example.com, Kind: StorageBucket, Name: icon-assets
Annotations: <none>
Events: <none>
```

## Path types

Each path in an Ingress is required to have a corresponding path type. Paths that do not include an explicit pathType will fail validation. There are three supported path types:

- **ImplementationSpecific:** With this path type, matching is up to the IngressClass. Implementations can treat this as a separate pathType or treat it identically to Prefix or Exact path types.
- **Exact:** Matches the URL path exactly and with case sensitivity.
- **Prefix:** Matches based on a URL path prefix split by /. Matching is case sensitive and done on a path element by element basis. A path element refers to the list of labels in the path split by the / separator. A request is a match for path *p* if every *p* is an element-wise prefix of *p* of the request path.

### Note:

If the last element of the path is a substring of the last element in request path, it is not a match (for example: /foo/bar matches /foo/bar/baz, but does not match /foo/barbaz).

## Examples

| Kind   | Path(s)   | Request path(s) | Matches?                    |
|--------|-----------|-----------------|-----------------------------|
| Prefix | /         | (all paths)     | Yes                         |
| Exact  | /foo      | /foo            | Yes                         |
| Exact  | /foo      | /bar            | No                          |
| Exact  | /foo      | /foo/           | No                          |
| Exact  | /foo/     | /foo            | No                          |
| Prefix | /foo      | /foo, /foo/     | Yes                         |
| Prefix | /foo/     | /foo, /foo/     | Yes                         |
| Prefix | /aaa/bb   | /aaa/bbb        | No                          |
| Prefix | /aaa/bbb  | /aaa/bbb        | Yes                         |
| Prefix | /aaa/bbb/ | /aaa/bbb        | Yes, ignores trailing slash |
| Prefix | /aaa/bbb  | /aaa/bbb/       | Yes, matches trailing slash |
| Prefix | /aaa/bbb  | /aaa/bbb/cc     | Yes, matches subpath        |

| Kind   | Path(s)                     | Request path(s) | Matches?                         |
|--------|-----------------------------|-----------------|----------------------------------|
| Prefix | /aaa/bbb                    | /aaa/bbbxyz     | No, does not match string prefix |
| Prefix | /, /aaa                     | /aaa/ccc        | Yes, matches /aaa prefix         |
| Prefix | /, /aaa, /aaa/bbb           | /aaa/bbb        | Yes, matches /aaa/bbb prefix     |
| Prefix | /, /aaa, /aaa/bbb           | /ccc            | Yes, matches / prefix            |
| Prefix | /aaa                        | /ccc            | No, uses default backend         |
| Mixed  | /foo (Prefix), /foo (Exact) | /foo            | Yes, prefers Exact               |

## Multiple matches

In some cases, multiple paths within an Ingress will match a request. In those cases precedence will be given first to the longest matching path. If two paths are still equally matched, precedence will be given to paths with an exact path type over prefix path type.

## Hostname wildcards

Hosts can be precise matches (for example “foo.bar.com”) or a wildcard (for example “\*.foo.com”). Precise matches require that the HTTP host header matches the host field. Wildcard matches require the HTTP host header is equal to the suffix of the wildcard rule.

| Host      | Host header     | Match?                                            |
|-----------|-----------------|---------------------------------------------------|
| *.foo.com | bar.foo.com     | Matches based on shared suffix                    |
| *.foo.com | baz.bar.foo.com | No match, wildcard only covers a single DNS label |
| *.foo.com | foo.com         | No match, wildcard only covers a single DNS label |

[service/networking/ingress-wildcard-host.yaml](#)

```

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: ingress-wildcard-host
spec:
 rules:
 - host: "foo.bar.com"
 http:
 paths:
 - pathType: Prefix
 path: "/bar"
 backend:
 service:
 name: service1
 port:
 number: 80
 - host: "*.foo.com"
 http:
 paths:
 - pathType: Prefix
 path: "/foo"
 backend:
 service:
 name: service2

```

```
port:
 number: 80
```

## Ingress class

Ingresses can be implemented by different controllers, often with different configuration. Each Ingress should specify a class, a reference to an IngressClass resource that contains additional configuration including the name of the controller that should implement the class.

[service/networking/external-lb.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
 name: external-lb
spec:
 controller: example.com/ingress-controller
 parameters:
 apiGroup: k8s.example.com
 kind: IngressParameters
 name: external-lb
```

The `.spec.parameters` field of an IngressClass lets you reference another resource that provides configuration related to that IngressClass.

The specific type of parameters to use depends on the ingress controller that you specify in the `.spec.controller` field of the IngressClass.

## IngressClass scope

Depending on your ingress controller, you may be able to use parameters that you set cluster-wide, or just for one namespace.

- [Cluster](#)
- [Namespaced](#)

The default scope for IngressClass parameters is cluster-wide.

If you set the `.spec.parameters` field and don't set `.spec.parameters.scope`, or if you set `.spec.parameters.scope` to `Cluster`, then the IngressClass refers to a cluster-scoped resource. The kind (in combination the `apiGroup`) of the parameters refers to a cluster-scoped API (possibly a custom resource), and the name of the parameters identifies a specific cluster scoped resource for that API.

For example:

```

apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
 name: external-lb-1
spec:
 controller: example.com/ingress-controller
```



```
parameters:
The parameters for this IngressClass are specified in a
ClusterIngressParameter (API group k8s.example.net) named
"external-config-1". This definition tells Kubernetes to
look for a cluster-scoped parameter resource.
scope: Cluster
apiGroup: k8s.example.net
kind: ClusterIngressParameter
name: external-config-1
```

FEATURE STATE: Kubernetes v1.23 [stable]

If you set the `.spec.parameters` field and set `.spec.parameters.scope` to `Namespace`, then the `IngressClass` refers to a namespaced-scoped resource. You must also set the `namespace` field within `.spec.parameters` to the namespace that contains the parameters you want to use.

The kind (in combination the `apiGroup`) of the parameters refers to a namespaced API (for example: `ConfigMap`), and the name of the parameters identifies a specific resource in the namespace you specified in `namespace`.

Namespace-scoped parameters help the cluster operator delegate control over the configuration (for example: load balancer settings, API gateway definition) that is used for a workload. If you used a cluster-scoped parameter then either:

- the cluster operator team needs to approve a different team's changes every time there's a new configuration change being applied.
- the cluster operator must define specific access controls, such as [RBAC](#) roles and bindings, that let the application team make changes to the cluster-scoped parameters resource.

The `IngressClass` API itself is always cluster-scoped.

Here is an example of an `IngressClass` that refers to parameters that are namespaced:

```

apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
 name: external-lb-2
spec:
 controller: example.com/ingress-controller
 parameters:
The parameters for this IngressClass are specified in an
IngressParameter (API group k8s.example.com) named "external-config",
that's in the "external-configuration" namespace.
scope: Namespace
apiGroup: k8s.example.com
kind: IngressParameter
namespace: external-configuration
name: external-config
```

## Deprecated annotation

Before the IngressClass resource and ingressClassName field were added in Kubernetes 1.18, Ingress classes were specified with a `kubernetes.io/ingress.class` annotation on the Ingress. This annotation was never formally defined, but was widely supported by Ingress controllers.

The newer ingressClassName field on Ingresses is a replacement for that annotation, but is not a direct equivalent. While the annotation was generally used to reference the name of the Ingress controller that should implement the Ingress, the field is a reference to an IngressClass resource that contains additional Ingress configuration, including the name of the Ingress controller.

## Default IngressClass

You can mark a particular IngressClass as default for your cluster. Setting the `ingressclass.kubernetes.io/is-default-class` annotation to `true` on an IngressClass resource will ensure that new Ingresses without an ingressClassName field specified will be assigned this default IngressClass.

### Caution:

If you have more than one IngressClass marked as the default for your cluster, the admission controller prevents creating new Ingress objects that don't have an ingressClassName specified. You can resolve this by ensuring that at most 1 IngressClass is marked as default in your cluster.

There are some ingress controllers, that work without the definition of a default IngressClass. For example, the Ingress-NGINX controller can be configured with a [flag](#) `--watch-ingress-without-class`. It is [recommended](#) though, to specify the default IngressClass:

[service/networking/default-ingressclass.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
 labels:
 app.kubernetes.io/component: controller
 name: nginx-example
 annotations:
 ingressclass.kubernetes.io/is-default-class: "true"
spec:
 controller: k8s.io/ingress-nginx
```

## Types of Ingress

### Ingress backed by a single Service

There are existing Kubernetes concepts that allow you to expose a single Service (see [alternatives](#)). You can also do this with an Ingress by specifying a *default backend* with no rules.

[service/networking/test-ingress.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: test-ingress
spec:
 defaultBackend:
 service:
 name: test
 port:
 number: 80
```

If you create it using `kubectl apply -f` you should be able to view the state of the Ingress you added:

```
kubectl get ingress test-ingress
```

| NAME         | CLASS       | HOSTS | ADDRESS       | PORTS | AGE |
|--------------|-------------|-------|---------------|-------|-----|
| test-ingress | external-lb | *     | 203.0.113.123 | 80    | 59s |

Where 203.0.113.123 is the IP allocated by the Ingress controller to satisfy this Ingress.

### Note:

Ingress controllers and load balancers may take a minute or two to allocate an IP address. Until that time, you often see the address listed as <pending>.

## Simple fanout

A fanout configuration routes traffic from a single IP address to more than one Service, based on the HTTP URI being requested. An Ingress allows you to keep the number of load balancers down to a minimum. For example, a setup like:

[ingress-fanout-diagram](#)

Figure. Ingress Fan Out

It would require an Ingress such as:

[service/networking/simple-fanout-example.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: simple-fanout-example
spec:
 rules:
 - host: foo.bar.com
 http:
 paths:
 - path: /foo
 pathType: Prefix
 backend:
 service:
```

```
name: service1
port:
 number: 4200
- path: /bar
 pathType: Prefix
 backend:
 service:
 name: service2
 port:
 number: 8080
```

When you create the Ingress with kubectl apply -f:

```
kubectl describe ingress simple-fanout-example
```

```
Name: simple-fanout-example
Namespace: default
Address: 178.91.123.132
Default backend: default-http-backend:80 (10.8.2.3:8080)
Rules:
 Host Path Backends
 ---- -
foo.bar.com
 /foo service1:4200 (10.8.0.90:4200)
 /bar service2:8080 (10.8.0.91:8080)
Events:
 Type Reason Age From Message
 ---- -
Normal ADD 22s loadbalancer-controller default/test
```

The Ingress controller provisions an implementation-specific load balancer that satisfies the Ingress, as long as the Services (service1, service2) exist. When it has done so, you can see the address of the load balancer at the Address field.

### Note:

Depending on the [Ingress controller](#) you are using, you may need to create a default-http-backend [Service](#).

## Name based virtual hosting

Name-based virtual hosts support routing HTTP traffic to multiple host names at the same IP address.

[ingress-namebase-diagram](#)

Figure. Ingress Name Based Virtual hosting

The following Ingress tells the backing load balancer to route requests based on the [Host header](#).

[service/networking/name-virtual-host-ingress.yaml](#)

```

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: name-virtual-host-ingress
spec:
 rules:
 - host: foo.bar.com
 http:
 paths:
 - pathType: Prefix
 path: "/"
 backend:
 service:
 name: service1
 port:
 number: 80
 - host: bar.foo.com
 http:
 paths:
 - pathType: Prefix
 path: "/"
 backend:
 service:
 name: service2
 port:
 number: 80

```

If you create an Ingress resource without any hosts defined in the rules, then any web traffic to the IP address of your Ingress controller can be matched without a name based virtual host being required.

For example, the following Ingress routes traffic requested for first.bar.com to service1, second.bar.com to service2, and any traffic whose request host header doesn't match first.bar.com and second.bar.com to service3.

[service/networking/name-virtual-host-ingress-no-third-host.yaml](#)

```

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: name-virtual-host-ingress-no-third-host
spec:
 rules:
 - host: first.bar.com
 http:
 paths:
 - pathType: Prefix
 path: "/"
 backend:
 service:
 name: service1
 port:
 number: 80

```

```

- host: second.bar.com
 http:
 paths:
 - pathType: Prefix
 path: "/"
 backend:
 service:
 name: service2
 port:
 number: 80
- http:
 paths:
 - pathType: Prefix
 path: "/"
 backend:
 service:
 name: service3
 port:
 number: 80

```

## TLS

You can secure an Ingress by specifying a [Secret](#) that contains a TLS private key and certificate. The Ingress resource only supports a single TLS port, 443, and assumes TLS termination at the ingress point (traffic to the Service and its Pods is in plaintext). If the TLS configuration section in an Ingress specifies different hosts, they are multiplexed on the same port according to the hostname specified through the SNI TLS extension (provided the Ingress controller supports SNI). The TLS secret must contain keys named `tls.crt` and `tls.key` that contain the certificate and private key to use for TLS. For example:

```

apiVersion: v1
kind: Secret
metadata:
 name: testsecret-tls
 namespace: default
data:
 tls.crt: base64 encoded cert
 tls.key: base64 encoded key
type: kubernetes.io/tls

```

Referencing this secret in an Ingress tells the Ingress controller to secure the channel from the client to the load balancer using TLS. You need to make sure the TLS secret you created came from a certificate that contains a Common Name (CN), also known as a Fully Qualified Domain Name (FQDN) for `https-example.foo.com`.

### Note:

Keep in mind that TLS will not work on the default rule because the certificates would have to be issued for all the possible sub-domains. Therefore, hosts in the `tls` section need to explicitly match the host in the rules section.

[service/networking/tls-example-ingress.yaml](#)

```

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: tls-example-ingress
spec:
 tls:
 - hosts:
 - https-example.foo.com
 secretName: testsecret-tls
 rules:
 - host: https-example.foo.com
 http:
 paths:
 - path: /
 pathType: Prefix
 backend:
 service:
 name: service1
 port:
 number: 80

```

#### Note:

There is a gap between TLS features supported by various Ingress controllers. Please refer to documentation on [nginx](#), [GCE](#), or any other platform specific Ingress controller to understand how TLS works in your environment.

## Load balancing

An Ingress controller is bootstrapped with some load balancing policy settings that it applies to all Ingress, such as the load balancing algorithm, backend weight scheme, and others. More advanced load balancing concepts (e.g. persistent sessions, dynamic weights) are not yet exposed through the Ingress. You can instead get these features through the load balancer used for a Service.

It's also worth noting that even though health checks are not exposed directly through the Ingress, there exist parallel concepts in Kubernetes such as [readiness probes](#) that allow you to achieve the same end result. Please review the controller specific documentation to see how they handle health checks (for example: [nginx](#), or [GCE](#)).

## Updating an Ingress

To update an existing Ingress to add a new Host, you can update it by editing the resource:

```
kubectl describe ingress test
```

```

Name: test
Namespace: default
Address: 178.91.123.132
Default backend: default-http-backend:80 (10.8.2.3:8080)
Rules:
 Host Path Backends

```

```

---- ---- -----
foo.bar.com
 /foo service1:80 (10.8.0.90:80)
Annotations:
 nginx.ingress.kubernetes.io/rewrite-target: /
Events:
 Type Reason Age From Message
 ---- -
Normal ADD 35s loadbalancer-controller default/test

```

```
kubectl edit ingress test
```

This pops up an editor with the existing configuration in YAML format. Modify it to include the new Host:

```

spec:
 rules:
 - host: foo.bar.com
 http:
 paths:
 - backend:
 service:
 name: service1
 port:
 number: 80
 path: /foo
 pathType: Prefix
 - host: bar.baz.com
 http:
 paths:
 - backend:
 service:
 name: service2
 port:
 number: 80
 path: /foo
 pathType: Prefix
 ..

```

After you save your changes, kubectl updates the resource in the API server, which tells the Ingress controller to reconfigure the load balancer.

Verify this:

```
kubectl describe ingress test
```

```

Name: test
Namespace: default
Address: 178.91.123.132
Default backend: default-http-backend:80 (10.8.2.3:8080)
Rules:
 Host Path Backends
 ---- -
foo.bar.com

```



```
/foo service1:80 (10.8.0.90:80)
bar.baz.com
/foo service2:80 (10.8.0.91:80)
Annotations:
 nginx.ingress.kubernetes.io/rewrite-target: /
Events:
```

| Type   | Reason | Age | From                    | Message      |
|--------|--------|-----|-------------------------|--------------|
| Normal | ADD    | 45s | loadbalancer-controller | default/test |

You can achieve the same outcome by invoking `kubectl replace -f` on a modified Ingress YAML file.

## Failing across availability zones

Techniques for spreading traffic across failure domains differ between cloud providers. Please check the documentation of the relevant [Ingress controller](#) for details.

## Alternatives

You can expose a Service in multiple ways that don't directly involve the Ingress resource:

- Use [Service.Type=LoadBalancer](#)
- Use [Service.Type=NodePort](#)

## What's next

- Learn about the [Ingress](#) API
- Learn about [Ingress controllers](#)
- [Set up Ingress on Minikube with the NGINX Controller](#)

# Ingress Controllers

In order for an [Ingress](#) to work in your cluster, there must be an *ingress controller* running. You need to select at least one ingress controller and make sure it is set up in your cluster. This page lists common ingress controllers that you can deploy.

In order for the Ingress resource to work, the cluster must have an ingress controller running.

Unlike other types of controllers which run as part of the kube-controller-manager binary, Ingress controllers are not started automatically with a cluster. Use this page to choose the ingress controller implementation that best fits your cluster.

Kubernetes as a project supports and maintains [AWS](#), [GCE](#), and [nginx](#) ingress controllers.

## Additional controllers

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are

listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

- [AKS Application Gateway Ingress Controller](#) is an ingress controller that configures the [Azure Application Gateway](#).
- [Alibaba Cloud MSE Ingress](#) is an ingress controller that configures the [Alibaba Cloud Native Gateway](#), which is also the commercial version of [Higress](#).
- [Apache APISIX ingress controller](#) is an [Apache APISIX](#)-based ingress controller.
- [Avi Kubernetes Operator](#) provides L4-L7 load-balancing using [VMware NSX Advanced Load Balancer](#).
- [BFE Ingress Controller](#) is a [BFE](#)-based ingress controller.
- [Cilium Ingress Controller](#) is an ingress controller powered by [Cilium](#).
- The [Citrix ingress controller](#) works with Citrix Application Delivery Controller.
- [Contour](#) is an [Envoy](#) based ingress controller.
- [Emissary-Ingress](#) API Gateway is an [Envoy](#)-based ingress controller.
- [EnRoute](#) is an [Envoy](#) based API gateway that can run as an ingress controller.
- [Easegress IngressController](#) is an [Easegress](#) based API gateway that can run as an ingress controller.
- F5 BIG-IP [Container Ingress Services for Kubernetes](#) lets you use an Ingress to configure F5 BIG-IP virtual servers.
- [FortiADC Ingress Controller](#) support the Kubernetes Ingress resources and allows you to manage FortiADC objects from Kubernetes
- [Gloo](#) is an open-source ingress controller based on [Envoy](#), which offers API gateway functionality.
- [HAProxy Ingress](#) is an ingress controller for [HAProxy](#).
- [Higress](#) is an [Envoy](#) based API gateway that can run as an ingress controller.
- The [HAProxy Ingress Controller for Kubernetes](#) is also an ingress controller for [HAProxy](#).
- [Istio Ingress](#) is an [Istio](#) based ingress controller.
- The [Kong Ingress Controller for Kubernetes](#) is an ingress controller driving [Kong Gateway](#).
- [Kusk Gateway](#) is an OpenAPI-driven ingress controller based on [Envoy](#).
- The [NGINX Ingress Controller for Kubernetes](#) works with the [NGINX](#) webserver (as a proxy).
- The [ngrok Kubernetes Ingress Controller](#) is an open source controller for adding secure public access to your K8s services using the [ngrok platform](#).
- The [OCI Native Ingress Controller](#) is an Ingress controller for Oracle Cloud Infrastructure which allows you to manage the [OCI Load Balancer](#).
- The [Pomerium Ingress Controller](#) is based on [Pomerium](#), which offers context-aware access policy.
- [Skipper](#) HTTP router and reverse proxy for service composition, including use cases like Kubernetes Ingress, designed as a library to build your custom proxy.
- The [Traefik Kubernetes Ingress provider](#) is an ingress controller for the [Traefik](#) proxy.
- [Tyk Operator](#) extends Ingress with Custom Resources to bring API Management capabilities to Ingress. Tyk Operator works with the Open Source Tyk Gateway & Tyk Cloud control plane.
- [Voyager](#) is an ingress controller for [HAProxy](#).
- [Wallarm Ingress Controller](#) is an Ingress Controller that provides WAAP (WAF) and API Security capabilities.

# Using multiple Ingress controllers

You may deploy any number of ingress controllers using [ingress class](#) within a cluster. Note the `.metadata.name` of your ingress class resource. When you create an ingress you would need that name to specify the `ingressClassName` field on your Ingress object (refer to [IngressSpec v1 reference](#)). `ingressClassName` is a replacement of the older [annotation method](#).

If you do not specify an `IngressClass` for an Ingress, and your cluster has exactly one `IngressClass` marked as default, then Kubernetes [applies](#) the cluster's default `IngressClass` to the Ingress. You mark an `IngressClass` as default by setting the [ingressclass.kubernetes.io/is-default-class annotation](#) on that `IngressClass`, with the string value `"true"`.

Ideally, all ingress controllers should fulfill this specification, but the various ingress controllers operate slightly differently.

## Note:

Make sure you review your ingress controller's documentation to understand the caveats of choosing it.

## What's next

- Learn more about [Ingress](#).
- [Set up Ingress on Minikube with the NGINX Controller](#).

# Gateway API

Gateway API is a family of API kinds that provide dynamic infrastructure provisioning and advanced traffic routing.

Make network services available by using an extensible, role-oriented, protocol-aware configuration mechanism. [Gateway API](#) is an [add-on](#) containing API [kinds](#) that provide dynamic infrastructure provisioning and advanced traffic routing.

## Design principles

The following principles shaped the design and architecture of Gateway API:

- **Role-oriented:** Gateway API kinds are modeled after organizational roles that are responsible for managing Kubernetes service networking:
  - **Infrastructure Provider:** Manages infrastructure that allows multiple isolated clusters to serve multiple tenants, e.g. a cloud provider.
  - **Cluster Operator:** Manages clusters and is typically concerned with policies, network access, application permissions, etc.
  - **Application Developer:** Manages an application running in a cluster and is typically concerned with application-level configuration and [Service](#) composition.
- **Portable:** Gateway API specifications are defined as [custom resources](#) and are supported by many [implementations](#).

- **Expressive:** Gateway API kinds support functionality for common traffic routing use cases such as header-based matching, traffic weighting, and others that were only possible in [Ingress](#) by using custom annotations.
- **Extensible:** Gateway allows for custom resources to be linked at various layers of the API. This makes granular customization possible at the appropriate places within the API structure.

## Resource model

Gateway API has three stable API kinds:

- **GatewayClass:** Defines a set of gateways with common configuration and managed by a controller that implements the class.
- **Gateway:** Defines an instance of traffic handling infrastructure, such as cloud load balancer.
- **HTTPRoute:** Defines HTTP-specific rules for mapping traffic from a Gateway listener to a representation of backend network endpoints. These endpoints are often represented as a [Service](#).

Gateway API is organized into different API kinds that have interdependent relationships to support the role-oriented nature of organizations. A Gateway object is associated with exactly one GatewayClass; the GatewayClass describes the gateway controller responsible for managing Gateways of this class. One or more route kinds such as HTTPRoute, are then associated to Gateways. A Gateway can filter the routes that may be attached to its listeners, forming a bidirectional trust model with routes.

The following figure illustrates the relationships of the three stable Gateway API kinds:

A figure illustrating the relationships of the three stable Gateway API kinds

### GatewayClass

Gateways can be implemented by different controllers, often with different configurations. A Gateway must reference a GatewayClass that contains the name of the controller that implements the class.

A minimal GatewayClass example:

```
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
 name: example-class
spec:
 controllerName: example.com/gateway-controller
```

In this example, a controller that has implemented Gateway API is configured to manage GatewayClasses with the controller name `example.com/gateway-controller`. Gateways of this class will be managed by the implementation's controller.

See the [GatewayClass](#) reference for a full definition of this API kind.

## Gateway

A Gateway describes an instance of traffic handling infrastructure. It defines a network endpoint that can be used for processing traffic, i.e. filtering, balancing, splitting, etc. for backends such as a Service. For example, a Gateway may represent a cloud load balancer or an in-cluster proxy server that is configured to accept HTTP traffic.

A minimal Gateway resource example:

```
apiVersion: gateway.networking.k8s.io/v1
kind: Gateway
metadata:
 name: example-gateway
spec:
 gatewayClassName: example-class
 listeners:
 - name: http
 protocol: HTTP
 port: 80
```

In this example, an instance of traffic handling infrastructure is programmed to listen for HTTP traffic on port 80. Since the addresses field is unspecified, an address or hostname is assigned to the Gateway by the implementation's controller. This address is used as a network endpoint for processing traffic of backend network endpoints defined in routes.

See the [Gateway](#) reference for a full definition of this API kind.

## HTTPRoute

The HTTPRoute kind specifies routing behavior of HTTP requests from a Gateway listener to backend network endpoints. For a Service backend, an implementation may represent the backend network endpoint as a Service IP or the backing Endpoints of the Service. An HTTPRoute represents configuration that is applied to the underlying Gateway implementation. For example, defining a new HTTPRoute may result in configuring additional traffic routes in a cloud load balancer or in-cluster proxy server.

A minimal HTTPRoute example:

```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
 name: example-httproute
spec:
 parentRefs:
 - name: example-gateway
 hostnames:
 - "www.example.com"
 rules:
 - matches:
 - path:
 type: PathPrefix
 value: /login
 backendRefs:
```

```
- name: example-svc
 port: 8080
```

In this example, HTTP traffic from Gateway `example-gateway` with the `Host:` header set to `www.example.com` and the request path specified as `/login` will be routed to Service `example-svc` on port 8080.

See the [HTTPRoute](#) reference for a full definition of this API kind.

## Request flow

Here is a simple example of HTTP traffic being routed to a Service by using a Gateway and an HTTPRoute:

A diagram that provides an example of HTTP traffic being routed to a Service by using a Gateway and an HTTPRoute

In this example, the request flow for a Gateway implemented as a reverse proxy is:

1. The client starts to prepare an HTTP request for the URL `http://www.example.com`
2. The client's DNS resolver queries for the destination name and learns a mapping to one or more IP addresses associated with the Gateway.
3. The client sends a request to the Gateway IP address; the reverse proxy receives the HTTP request and uses the `Host:` header to match a configuration that was derived from the Gateway and attached HTTPRoute.
4. Optionally, the reverse proxy can perform request header and/or path matching based on match rules of the HTTPRoute.
5. Optionally, the reverse proxy can modify the request; for example, to add or remove headers, based on filter rules of the HTTPRoute.
6. Lastly, the reverse proxy forwards the request to one or more backends.

## Conformance

Gateway API covers a broad set of features and is widely implemented. This combination requires clear conformance definitions and tests to ensure that the API provides a consistent experience wherever it is used.

See the [conformance](#) documentation to understand details such as release channels, support levels, and running conformance tests.

## Migrating from Ingress

Gateway API is the successor to the [Ingress](#) API. However, it does not include the Ingress kind. As a result, a one-time conversion from your existing Ingress resources to Gateway API resources is necessary.

Refer to the [ingress migration](#) guide for details on migrating Ingress resources to Gateway API resources.

## What's next

Instead of Gateway API resources being natively implemented by Kubernetes, the specifications are defined as [Custom Resources](#) supported by a wide range of [implementations](#). [Install](#) the Gateway API CRDs or follow the installation instructions of your selected implementation. After installing an implementation, use the [Getting Started](#) guide to help you quickly start working with Gateway API.

### Note:

Make sure to review the documentation of your selected implementation to understand any caveats.

Refer to the [API specification](#) for additional details of all Gateway API kinds.

## EndpointSlices

The EndpointSlice API is the mechanism that Kubernetes uses to let your Service scale to handle large numbers of backends, and allows the cluster to update its list of healthy backends efficiently.

FEATURE STATE: Kubernetes v1.21 [stable]

Kubernetes' *EndpointSlice* API provides a way to track network endpoints within a Kubernetes cluster. EndpointSlices offer a more scalable and extensible alternative to [Endpoints](#).

## EndpointSlice API

In Kubernetes, an EndpointSlice contains references to a set of network endpoints. The control plane automatically creates EndpointSlices for any Kubernetes Service that has a [selector](#) specified. These EndpointSlices include references to all the Pods that match the Service selector. EndpointSlices group network endpoints together by unique combinations of protocol, port number, and Service name. The name of a EndpointSlice object must be a valid [DNS subdomain name](#).

As an example, here's a sample EndpointSlice object, that's owned by the example Kubernetes Service.

```
apiVersion: discovery.k8s.io/v1
kind: EndpointSlice
metadata:
 name: example-abc
 labels:
 kubernetes.io/service-name: example
addressType: IPv4
ports:
 - name: http
 protocol: TCP
 port: 80
endpoints:
 - addresses:
```



```
- "10.1.2.3"
conditions:
 ready: true
hostname: pod-1
nodeName: node-1
zone: us-west2-a
```

By default, the control plane creates and manages EndpointSlices to have no more than 100 endpoints each. You can configure this with the `--max-endpoints-per-slice` [kube-controller-manager](#) flag, up to a maximum of 1000.

EndpointSlices can act as the source of truth for [kube-proxy](#) when it comes to how to route internal traffic.

## Address types

EndpointSlices support three address types:

- IPv4
- IPv6
- FQDN (Fully Qualified Domain Name)

Each EndpointSlice object represents a specific IP address type. If you have a Service that is available via IPv4 and IPv6, there will be at least two EndpointSlice objects (one for IPv4, and one for IPv6).

## Conditions

The EndpointSlice API stores conditions about endpoints that may be useful for consumers. The three conditions are ready, serving, and terminating.

### Ready

ready is a condition that maps to a Pod's Ready condition. A running Pod with the Ready condition set to True should have this EndpointSlice condition also set to true. For compatibility reasons, ready is NEVER true when a Pod is terminating. Consumers should refer to the serving condition to inspect the readiness of terminating Pods. The only exception to this rule is for Services with `spec.publishNotReadyAddresses` set to true. Endpoints for these Services will always have the ready condition set to true.

### Serving

FEATURE STATE: Kubernetes v1.26 [stable]

The serving condition is almost identical to the ready condition. The difference is that consumers of the EndpointSlice API should check the serving condition if they care about pod readiness while the pod is also terminating.

### Note:

Although serving is almost identical to ready, it was added to prevent breaking the existing meaning of ready. It may be unexpected for existing clients if ready could be true for



terminating endpoints, since historically terminating endpoints were never included in the Endpoints or EndpointSlice API to begin with. For this reason, `ready` is *always* false for terminating endpoints, and a new condition `Serving` was added in v1.20 so that clients can track readiness for terminating pods independent of the existing semantics for `ready`.

## Terminating

FEATURE STATE: [Kubernetes v1.22 \[beta\]](#)

`Terminating` is a condition that indicates whether an endpoint is terminating. For pods, this is any pod that has a deletion timestamp set.

## Topology information

Each endpoint within an EndpointSlice can contain relevant topology information. The topology information includes the location of the endpoint and information about the corresponding Node and zone. These are available in the following per endpoint fields on EndpointSlices:

- `nodeName` - The name of the Node this endpoint is on.
- `zone` - The zone this endpoint is in.

### Note:

In the v1 API, the per endpoint topology was effectively removed in favor of the dedicated fields `nodeName` and `zone`.

Setting arbitrary topology fields on the endpoint field of an EndpointSlice resource has been deprecated and is not supported in the v1 API. Instead, the v1 API supports setting individual `nodeName` and `zone` fields. These fields are automatically translated between API versions. For example, the value of the `"topology.kubernetes.io/zone"` key in the topology field in the v1beta1 API is accessible as the `zone` field in the v1 API.

## Management

Most often, the control plane (specifically, the endpoint slice [controller](#)) creates and manages EndpointSlice objects. There are a variety of other use cases for EndpointSlices, such as service mesh implementations, that could result in other entities or controllers managing additional sets of EndpointSlices.

To ensure that multiple entities can manage EndpointSlices without interfering with each other, Kubernetes defines the [label](#) `endpointslice.kubernetes.io/managed-by`, which indicates the entity managing an EndpointSlice. The endpoint slice controller sets `endpointslice-controller.k8s.io` as the value for this label on all EndpointSlices it manages. Other entities managing EndpointSlices should also set a unique value for this label.

## Ownership

In most use cases, EndpointSlices are owned by the Service that the endpoint slice object tracks endpoints for. This ownership is indicated by an owner reference on each EndpointSlice as well as a `kubernetes.io/service-name` label that enables simple lookups of all EndpointSlices belonging to a Service.

## EndpointSlice mirroring

In some cases, applications create custom Endpoints resources. To ensure that these applications do not need to concurrently write to both Endpoints and EndpointSlice resources, the cluster's control plane mirrors most Endpoints resources to corresponding EndpointSlices.

The control plane mirrors Endpoints resources unless:

- the Endpoints resource has a `endpointslice.kubernetes.io/skip-mirror` label set to `true`.
- the Endpoints resource has a `control-plane.alpha.kubernetes.io/leader` annotation.
- the corresponding Service resource does not exist.
- the corresponding Service resource has a non-nil selector.

Individual Endpoints resources may translate into multiple EndpointSlices. This will occur if an Endpoints resource has multiple subsets or includes endpoints with multiple IP families (IPv4 and IPv6). A maximum of 1000 addresses per subset will be mirrored to EndpointSlices.

## Distribution of EndpointSlices

Each EndpointSlice has a set of ports that applies to all endpoints within the resource. When named ports are used for a Service, Pods may end up with different target port numbers for the same named port, requiring different EndpointSlices. This is similar to the logic behind how subsets are grouped with Endpoints.

The control plane tries to fill EndpointSlices as full as possible, but does not actively rebalance them. The logic is fairly straightforward:

1. Iterate through existing EndpointSlices, remove endpoints that are no longer desired and update matching endpoints that have changed.
2. Iterate through EndpointSlices that have been modified in the first step and fill them up with any new endpoints needed.
3. If there's still new endpoints left to add, try to fit them into a previously unchanged slice and/or create new ones.

Importantly, the third step prioritizes limiting EndpointSlice updates over a perfectly full distribution of EndpointSlices. As an example, if there are 10 new endpoints to add and 2 EndpointSlices with room for 5 more endpoints each, this approach will create a new EndpointSlice instead of filling up the 2 existing EndpointSlices. In other words, a single EndpointSlice creation is preferable to multiple EndpointSlice updates.

With kube-proxy running on each Node and watching EndpointSlices, every change to an EndpointSlice becomes relatively expensive since it will be transmitted to every Node in the cluster. This approach is intended to limit the number of changes that need to be sent to every Node, even if it may result with multiple EndpointSlices that are not full.

In practice, this less than ideal distribution should be rare. Most changes processed by the EndpointSlice controller will be small enough to fit in an existing EndpointSlice, and if not, a new EndpointSlice is likely going to be necessary soon anyway. Rolling updates of Deployments also provide a natural repacking of EndpointSlices with all Pods and their corresponding endpoints getting replaced.

## Duplicate endpoints

Due to the nature of EndpointSlice changes, endpoints may be represented in more than one EndpointSlice at the same time. This naturally occurs as changes to different EndpointSlice objects can arrive at the Kubernetes client watch / cache at different times.

### Note:

Clients of the EndpointSlice API must iterate through all the existing EndpointSlices associated to a Service and build a complete list of unique network endpoints. It is important to mention that endpoints may be duplicated in different EndpointSlices.

You can find a reference implementation for how to perform this endpoint aggregation and deduplication as part of the EndpointSliceCache code within kube-proxy.

## Comparison with Endpoints

The original Endpoints API provided a simple and straightforward way of tracking network endpoints in Kubernetes. As Kubernetes clusters and [Services](#) grew to handle more traffic and to send more traffic to more backend Pods, the limitations of that original API became more visible. Most notably, those included challenges with scaling to larger numbers of network endpoints.

Since all network endpoints for a Service were stored in a single Endpoints object, those Endpoints objects could get quite large. For Services that stayed stable (the same set of endpoints over a long period of time) the impact was less noticeable; even then, some use cases of Kubernetes weren't well served.

When a Service had a lot of backend endpoints and the workload was either scaling frequently, or rolling out new changes frequently, each update to the single Endpoints object for that Service meant a lot of traffic between Kubernetes cluster components (within the control plane, and also between nodes and the API server). This extra traffic also had a cost in terms of CPU use.

With EndpointSlices, adding or removing a single Pod triggers the same *number* of updates to clients that are watching for changes, but the size of those update message is much smaller at large scale.

EndpointSlices also enabled innovation around new features such dual-stack networking and topology-aware routing.

## What's next

- Follow the [Connecting Applications with Services](#) tutorial
- Read the [API reference](#) for the EndpointSlice API
- Read the [API reference](#) for the Endpoints API

# Network Policies

If you want to control traffic flow at the IP address or port level (OSI layer 3 or 4), NetworkPolicies allow you to specify rules for traffic flow within your cluster, and also between Pods and the outside world. Your cluster must use a network plugin that supports NetworkPolicy enforcement.

If you want to control traffic flow at the IP address or port level for TCP, UDP, and SCTP protocols, then you might consider using Kubernetes NetworkPolicies for particular applications in your cluster. NetworkPolicies are an application-centric construct which allow you to specify how a [pod](#) is allowed to communicate with various network "entities" (we use the word "entity" here to avoid overloading the more common terms such as "endpoints" and "services", which have specific Kubernetes connotations) over the network. NetworkPolicies apply to a connection with a pod on one or both ends, and are not relevant to other connections.

The entities that a Pod can communicate with are identified through a combination of the following three identifiers:

1. Other pods that are allowed (exception: a pod cannot block access to itself)
2. Namespaces that are allowed
3. IP blocks (exception: traffic to and from the node where a Pod is running is always allowed, regardless of the IP address of the Pod or the node)

When defining a pod- or namespace-based NetworkPolicy, you use a [selector](#) to specify what traffic is allowed to and from the Pod(s) that match the selector.

Meanwhile, when IP-based NetworkPolicies are created, we define policies based on IP blocks (CIDR ranges).

## Prerequisites

Network policies are implemented by the [network plugin](#). To use network policies, you must be using a networking solution which supports NetworkPolicy. Creating a NetworkPolicy resource without a controller that implements it will have no effect.

## The two sorts of pod isolation

There are two sorts of isolation for a pod: isolation for egress, and isolation for ingress. They concern what connections may be established. "Isolation" here is not absolute, rather it means "some restrictions apply". The alternative, "non-isolated for \$direction", means that no restrictions apply in the stated direction. The two sorts of isolation (or not) are declared independently, and are both relevant for a connection from one pod to another.

By default, a pod is non-isolated for egress; all outbound connections are allowed. A pod is isolated for egress if there is any NetworkPolicy that both selects the pod and has "Egress" in its policyTypes; we say that such a policy applies to the pod for egress. When a pod is isolated for egress, the only allowed connections from the pod are those allowed by the egress list of some NetworkPolicy that applies to the pod for egress. Reply traffic for those allowed connections will also be implicitly allowed. The effects of those egress lists combine additively.

By default, a pod is non-isolated for ingress; all inbound connections are allowed. A pod is isolated for ingress if there is any NetworkPolicy that both selects the pod and has "Ingress" in its policyTypes; we say that such a policy applies to the pod for ingress. When a pod is isolated for ingress, the only allowed connections into the pod are those from the pod's node and those allowed by the ingress list of some NetworkPolicy that applies to the pod for ingress. Reply traffic for those allowed connections will also be implicitly allowed. The effects of those ingress lists combine additively.

Network policies do not conflict; they are additive. If any policy or policies apply to a given pod for a given direction, the connections allowed in that direction from that pod is the union of what the applicable policies allow. Thus, order of evaluation does not affect the policy result.

For a connection from a source pod to a destination pod to be allowed, both the egress policy on the source pod and the ingress policy on the destination pod need to allow the connection. If either side does not allow the connection, it will not happen.

## The NetworkPolicy resource

See the [NetworkPolicy](#) reference for a full definition of the resource.

An example NetworkPolicy might look like this:

[service/networking/networkpolicy.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: test-network-policy
 namespace: default
spec:
 podSelector:
 matchLabels:
 role: db
 policyTypes:
 - Ingress
 - Egress
 ingress:
 - from:
 - ipBlock:
 cidr: 172.17.0.0/16
 except:
 - 172.17.1.0/24
 - namespaceSelector:
 matchLabels:
 project: myproject
 - podSelector:
 matchLabels:
 role: frontend
 ports:
 - protocol: TCP
 port: 6379
 egress:
 - to:
```

```
- ipBlock:
 cidr: 10.0.0.0/24
ports:
- protocol: TCP
 port: 5978
```

**Note:**

POSTing this to the API server for your cluster will have no effect unless your chosen networking solution supports network policy.

**Mandatory Fields:** As with all other Kubernetes config, a NetworkPolicy needs apiVersion, kind, and metadata fields. For general information about working with config files, see [Configure a Pod to Use a ConfigMap](#), and [Object Management](#).

**spec:** NetworkPolicy [spec](#) has all the information needed to define a particular network policy in the given namespace.

**podSelector:** Each NetworkPolicy includes a podSelector which selects the grouping of pods to which the policy applies. The example policy selects pods with the label "role=db". An empty podSelector selects all pods in the namespace.

**policyTypes:** Each NetworkPolicy includes a policyTypes list which may include either Ingress, Egress, or both. The policyTypes field indicates whether or not the given policy applies to ingress traffic to selected pod, egress traffic from selected pods, or both. If no policyTypes are specified on a NetworkPolicy then by default Ingress will always be set and Egress will be set if the NetworkPolicy has any egress rules.

**ingress:** Each NetworkPolicy may include a list of allowed ingress rules. Each rule allows traffic which matches both the from and ports sections. The example policy contains a single rule, which matches traffic on a single port, from one of three sources, the first specified via an ipBlock, the second via a namespaceSelector and the third via a podSelector.

**egress:** Each NetworkPolicy may include a list of allowed egress rules. Each rule allows traffic which matches both the to and ports sections. The example policy contains a single rule, which matches traffic on a single port to any destination in 10.0.0.0/24.

So, the example NetworkPolicy:

1. isolates role=db pods in the default namespace for both ingress and egress traffic (if they weren't already isolated)
2. (Ingress rules) allows connections to all pods in the default namespace with the label role=db on TCP port 6379 from:
  - any pod in the default namespace with the label role=frontend
  - any pod in a namespace with the label project=myproject
  - IP addresses in the ranges 172.17.0.0–172.17.0.255 and 172.17.2.0–172.17.255.255 (ie, all of 172.17.0.0/16 except 172.17.1.0/24)
3. (Egress rules) allows connections from any pod in the default namespace with the label role=db to CIDR 10.0.0.0/24 on TCP port 5978

See the [Declare Network Policy](#) walkthrough for further examples.

## Behavior of to and from selectors

There are four kinds of selectors that can be specified in an ingress from section or egress to section:

**podSelector:** This selects particular Pods in the same namespace as the NetworkPolicy which should be allowed as ingress sources or egress destinations.

**namespaceSelector:** This selects particular namespaces for which all Pods should be allowed as ingress sources or egress destinations.

**namespaceSelector and podSelector:** A single to/from entry that specifies both namespaceSelector and podSelector selects particular Pods within particular namespaces. Be careful to use correct YAML syntax. For example:

```
...
ingress:
- from:
 - namespaceSelector:
 matchLabels:
 user: alice
 podSelector:
 matchLabels:
 role: client
...
```

This policy contains a single from element allowing connections from Pods with the label role=client in namespaces with the label user=alice. But the following policy is different:

```
...
ingress:
- from:
 - namespaceSelector:
 matchLabels:
 user: alice
 - podSelector:
 matchLabels:
 role: client
...
```

It contains two elements in the from array, and allows connections from Pods in the local Namespace with the label role=client, *or* from any Pod in any namespace with the label user=alice.

When in doubt, use `kubectl describe` to see how Kubernetes has interpreted the policy.

**ipBlock:** This selects particular IP CIDR ranges to allow as ingress sources or egress destinations. These should be cluster-external IPs, since Pod IPs are ephemeral and unpredictable.

Cluster ingress and egress mechanisms often require rewriting the source or destination IP of packets. In cases where this happens, it is not defined whether this happens before or after NetworkPolicy processing, and the behavior may be different for different combinations of network plugin, cloud provider, Service implementation, etc.

In the case of ingress, this means that in some cases you may be able to filter incoming packets based on the actual original source IP, while in other cases, the "source IP" that the NetworkPolicy acts on may be the IP of a LoadBalancer or of the Pod's node, etc.

For egress, this means that connections from pods to Service IPs that get rewritten to cluster-external IPs may or may not be subject to ipBlock-based policies.

## Default policies

By default, if no policies exist in a namespace, then all ingress and egress traffic is allowed to and from pods in that namespace. The following examples let you change the default behavior in that namespace.

### Default deny all ingress traffic

You can create a "default" ingress isolation policy for a namespace by creating a NetworkPolicy that selects all pods but does not allow any ingress traffic to those pods.

[service/networking/network-policy-default-deny-ingress.yaml](https://kubernetes.io/docs/service/networking/network-policy-default-deny-ingress.yaml)

```

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: default-deny-ingress
spec:
 podSelector: {}
 policyTypes:
 - Ingress
```

This ensures that even pods that aren't selected by any other NetworkPolicy will still be isolated for ingress. This policy does not affect isolation for egress from any pod.

### Allow all ingress traffic

If you want to allow all incoming connections to all pods in a namespace, you can create a policy that explicitly allows that.

[service/networking/network-policy-allow-all-ingress.yaml](https://kubernetes.io/docs/service/networking/network-policy-allow-all-ingress.yaml)

```

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: allow-all-ingress
spec:
 podSelector: {}
 ingress:
```



```
- {}
policyTypes:
- Ingress
```

With this policy in place, no additional policy or policies can cause any incoming connection to those pods to be denied. This policy has no effect on isolation for egress from any pod.

## Default deny all egress traffic

You can create a "default" egress isolation policy for a namespace by creating a NetworkPolicy that selects all pods but does not allow any egress traffic from those pods.

<service/networking/network-policy-default-deny-egress.yaml>

```

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: default-deny-egress
spec:
 podSelector: {}
 policyTypes:
 - Egress
```

This ensures that even pods that aren't selected by any other NetworkPolicy will not be allowed egress traffic. This policy does not change the ingress isolation behavior of any pod.

## Allow all egress traffic

If you want to allow all connections from all pods in a namespace, you can create a policy that explicitly allows all outgoing connections from pods in that namespace.

<service/networking/network-policy-allow-all-egress.yaml>

```

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: allow-all-egress
spec:
 podSelector: {}
 egress:
 - {}
 policyTypes:
 - Egress
```

With this policy in place, no additional policy or policies can cause any outgoing connection from those pods to be denied. This policy has no effect on isolation for ingress to any pod.

## Default deny all ingress and all egress traffic

You can create a "default" policy for a namespace which prevents all ingress AND egress traffic by creating the following NetworkPolicy in that namespace.

[service/networking/network-policy-default-deny-all.yaml](#)

```

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: default-deny-all
spec:
 podSelector: {}
 policyTypes:
 - Ingress
 - Egress
```

This ensures that even pods that aren't selected by any other NetworkPolicy will not be allowed ingress or egress traffic.

## Network traffic filtering

NetworkPolicy is defined for [layer 4](#) connections (TCP, UDP, and optionally SCTP). For all the other protocols, the behaviour may vary across network plugins.

### Note:

You must be using a [CNI](#) plugin that supports SCTP protocol NetworkPolicies.

When a deny all network policy is defined, it is only guaranteed to deny TCP, UDP and SCTP connections. For other protocols, such as ARP or ICMP, the behaviour is undefined. The same applies to allow rules: when a specific pod is allowed as ingress source or egress destination, it is undefined what happens with (for example) ICMP packets. Protocols such as ICMP may be allowed by some network plugins and denied by others.

## Targeting a range of ports

FEATURE STATE: Kubernetes v1.25 [stable]

When writing a NetworkPolicy, you can target a range of ports instead of a single port.

This is achievable with the usage of the endPort field, as the following example:

[service/networking/networkpolicy-multiport-egress.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: multi-port-egress
 namespace: default
spec:
 podSelector:
 matchLabels:
 role: db
 policyTypes:
 - Egress
```

```
egress:
- to:
 - ipBlock:
 cidr: 10.0.0.0/24
ports:
- protocol: TCP
 port: 32000
 endPort: 32768
```

The above rule allows any Pod with label `role=db` on the namespace `default` to communicate with any IP within the range `10.0.0.0/24` over TCP, provided that the target port is between the range `32000` and `32768`.

The following restrictions apply when using this field:

- The `endPort` field must be equal to or greater than the `port` field.
- `endPort` can only be defined if `port` is also defined.
- Both ports must be numeric.

#### Note:

Your cluster must be using a [CNI](#) plugin that supports the `endPort` field in NetworkPolicy specifications. If your [network plugin](#) does not support the `endPort` field and you specify a NetworkPolicy with that, the policy will be applied only for the single port field.

## Targeting multiple namespaces by label

In this scenario, your Egress NetworkPolicy targets more than one namespace using their label names. For this to work, you need to label the target namespaces. For example:

```
kubectl label namespace frontend namespace=frontend
kubectl label namespace backend namespace=backend
```

Add the labels under `namespaceSelector` in your NetworkPolicy document. For example:

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: egress-namespaces
spec:
 podSelector:
 matchLabels:
 app: myapp
 policyTypes:
 - Egress
 egress:
 - to:
 - namespaceSelector:
 matchExpressions:
 - key: namespace
 operator: In
 values: ["frontend", "backend"]
```

**Note:**

It is not possible to directly specify the name of the namespaces in a NetworkPolicy. You must use a namespaceSelector with matchLabels or matchExpressions to select the namespaces based on their labels.

## Targeting a Namespace by its name

The Kubernetes control plane sets an immutable label `kubernetes.io/metadata.name` on all namespaces, the value of the label is the namespace name.

While NetworkPolicy cannot target a namespace by its name with some object field, you can use the standardized label to target a specific namespace.

## Pod lifecycle

**Note:**

The following applies to clusters with a conformant networking plugin and a conformant implementation of NetworkPolicy.

When a new NetworkPolicy object is created, it may take some time for a network plugin to handle the new object. If a pod that is affected by a NetworkPolicy is created before the network plugin has completed NetworkPolicy handling, that pod may be started unprotected, and isolation rules will be applied when the NetworkPolicy handling is completed.

Once the NetworkPolicy is handled by a network plugin,

1. All newly created pods affected by a given NetworkPolicy will be isolated before they are started. Implementations of NetworkPolicy must ensure that filtering is effective throughout the Pod lifecycle, even from the very first instant that any container in that Pod is started. Because they are applied at Pod level, NetworkPolicies apply equally to init containers, sidecar containers, and regular containers.
2. Allow rules will be applied eventually after the isolation rules (or may be applied at the same time). In the worst case, a newly created pod may have no network connectivity at all when it is first started, if isolation rules were already applied, but no allow rules were applied yet.

Every created NetworkPolicy will be handled by a network plugin eventually, but there is no way to tell from the Kubernetes API when exactly that happens.

Therefore, pods must be resilient against being started up with different network connectivity than expected. If you need to make sure the pod can reach certain destinations before being started, you can use an [init container](#) to wait for those destinations to be reachable before kubelet starts the app containers.

Every NetworkPolicy will be applied to all selected pods eventually. Because the network plugin may implement NetworkPolicy in a distributed manner, it is possible that pods may see a slightly inconsistent view of network policies when the pod is first created, or when pods or policies change. For example, a newly-created pod that is supposed to be able to reach both Pod

A on Node 1 and Pod B on Node 2 may find that it can reach Pod A immediately, but cannot reach Pod B until a few seconds later.

## NetworkPolicy and hostNetwork pods

NetworkPolicy behaviour for hostNetwork pods is undefined, but it should be limited to 2 possibilities:

- The network plugin can distinguish hostNetwork pod traffic from all other traffic (including being able to distinguish traffic from different hostNetwork pods on the same node), and will apply NetworkPolicy to hostNetwork pods just like it does to pod-network pods.
- The network plugin cannot properly distinguish hostNetwork pod traffic, and so it ignores hostNetwork pods when matching podSelector and namespaceSelector. Traffic to/from hostNetwork pods is treated the same as all other traffic to/from the node IP. (This is the most common implementation.)

This applies when

1. a hostNetwork pod is selected by spec.podSelector.

```
...
spec:
 podSelector:
 matchLabels:
 role: client
...
```

2. a hostNetwork pod is selected by a podSelector or namespaceSelector in an ingress or egress rule.

```
...
ingress:
- from:
- podSelector:
 matchLabels:
 role: client
...
```

At the same time, since hostNetwork pods have the same IP addresses as the nodes they reside on, their connections will be treated as node connections. For example, you can allow traffic from a hostNetwork Pod using an ipBlock rule.

## What you can't do with network policies (at least, not yet)

As of Kubernetes 1.30, the following functionality does not exist in the NetworkPolicy API, but you might be able to implement workarounds using Operating System components (such as SELinux, OpenVSwitch, IPTables, and so on) or Layer 7 technologies (Ingress controllers, Service Mesh implementations) or admission controllers. In case you are new to network

security in Kubernetes, its worth noting that the following User Stories cannot (yet) be implemented using the NetworkPolicy API.

- Forcing internal cluster traffic to go through a common gateway (this might be best served with a service mesh or other proxy).
- Anything TLS related (use a service mesh or ingress controller for this).
- Node specific policies (you can use CIDR notation for these, but you cannot target nodes by their Kubernetes identities specifically).
- Targeting of services by name (you can, however, target pods or namespaces by their [labels](#), which is often a viable workaround).
- Creation or management of "Policy requests" that are fulfilled by a third party.
- Default policies which are applied to all namespaces or pods (there are some third party Kubernetes distributions and projects which can do this).
- Advanced policy querying and reachability tooling.
- The ability to log network security events (for example connections that are blocked or accepted).
- The ability to explicitly deny policies (currently the model for NetworkPolicies are deny by default, with only the ability to add allow rules).
- The ability to prevent loopback or incoming host traffic (Pods cannot currently block localhost access, nor do they have the ability to block access from their resident node).

## NetworkPolicy's impact on existing connections

When the set of NetworkPolicies that applies to an existing connection changes - this could happen either due to a change in NetworkPolicies or if the relevant labels of the namespaces/pods selected by the policy (both subject and peers) are changed in the middle of an existing connection - it is implementation defined as to whether the change will take effect for that existing connection or not. Example: A policy is created that leads to denying a previously allowed connection, the underlying network plugin implementation is responsible for defining if that new policy will close the existing connections or not. It is recommended not to modify policies/pods/namespaces in ways that might affect existing connections.

## What's next

- See the [Declare Network Policy](#) walkthrough for further examples.
- See more [recipes](#) for common scenarios enabled by the NetworkPolicy resource.

## DNS for Services and Pods

Your workload can discover Services within your cluster using DNS; this page explains how that works.

Kubernetes creates DNS records for Services and Pods. You can contact Services with consistent DNS names instead of IP addresses.

Kubernetes publishes information about Pods and Services which is used to program DNS. Kubelet configures Pods' DNS so that running containers can lookup Services by name rather than IP.

Services defined in the cluster are assigned DNS names. By default, a client Pod's DNS search list includes the Pod's own namespace and the cluster's default domain.

## Namespaces of Services

A DNS query may return different results based on the namespace of the Pod making it. DNS queries that don't specify a namespace are limited to the Pod's namespace. Access Services in other namespaces by specifying it in the DNS query.

For example, consider a Pod in a `test` namespace. A data Service is in the `prod` namespace.

A query for `data` returns no results, because it uses the Pod's `test` namespace.

A query for `data.prod` returns the intended result, because it specifies the namespace.

DNS queries may be expanded using the Pod's `/etc/resolv.conf`. Kubelet configures this file for each Pod. For example, a query for `just data` may be expanded to `data.test.svc.cluster.local`. The values of the search option are used to expand queries. To learn more about DNS queries, see [the resolv.conf manual page](#).

```
nameserver 10.32.0.10
search <namespace>.svc.cluster.local svc.cluster.local cluster.local
options ndots:5
```

In summary, a Pod in the `test` namespace can successfully resolve either `data.prod` or `data.prod.svc.cluster.local`.

## DNS Records

What objects get DNS records?

1. Services
2. Pods

The following sections detail the supported DNS record types and layout that is supported. Any other layout or names or queries that happen to work are considered implementation details and are subject to change without warning. For more up-to-date specification, see [Kubernetes DNS-Based Service Discovery](#).

## Services

### A/AAAA records

"Normal" (not headless) Services are assigned DNS A and/or AAAA records, depending on the IP family or families of the Service, with a name of the form `my-svc.my-namespace.svc.cluster-domain.example`. This resolves to the cluster IP of the Service.

[Headless Services](#) (without a cluster IP) Services are also assigned DNS A and/or AAAA records, with a name of the form `my-svc.my-namespace.svc.cluster-domain.example`. Unlike normal Services, this resolves to the set of IPs of all of the Pods selected by the Service. Clients are expected to consume the set or else use standard round-robin selection from the set.

## SRV records

SRV Records are created for named ports that are part of normal or headless services. For each named port, the SRV record has the form `_port-name._port-protocol.my-svc.my-namespace.svc.cluster-domain.example`. For a regular Service, this resolves to the port number and the domain name: `my-svc.my-namespace.svc.cluster-domain.example`. For a headless Service, this resolves to multiple answers, one for each Pod that is backing the Service, and contains the port number and the domain name of the Pod of the form `hostname.my-svc.my-namespace.svc.cluster-domain.example`.

## Pods

### A/AAAA records

Kube-DNS versions, prior to the implementation of the [DNS specification](#), had the following DNS resolution:

`pod-ipv4-address.my-namespace.pod.cluster-domain.example`.

For example, if a Pod in the default namespace has the IP address 172.17.0.3, and the domain name for your cluster is `cluster.local`, then the Pod has a DNS name:

`172-17-0-3.default.pod.cluster.local`.

Any Pods exposed by a Service have the following DNS resolution available:

`pod-ipv4-address.service-name.my-namespace.svc.cluster-domain.example`.

### Pod's hostname and subdomain fields

Currently when a Pod is created, its hostname (as observed from within the Pod) is the Pod's `metadata.name` value.

The Pod spec has an optional `hostname` field, which can be used to specify a different hostname. When specified, it takes precedence over the Pod's name to be the hostname of the Pod (again, as observed from within the Pod). For example, given a Pod with `spec.hostname` set to `"my-host"`, the Pod will have its hostname set to `"my-host"`.

The Pod spec also has an optional `subdomain` field which can be used to indicate that the pod is part of sub-group of the namespace. For example, a Pod with `spec.hostname` set to `"foo"`, and `spec.subdomain` set to `"bar"`, in namespace `"my-namespace"`, will have its hostname set to `"foo"` and its fully qualified domain name (FQDN) set to `"foo.bar.my-namespace.svc.cluster.local"` (once more, as observed from within the Pod).

If there exists a headless Service in the same namespace as the Pod, with the same name as the subdomain, the cluster's DNS Server also returns A and/or AAAA records for the Pod's fully qualified hostname.

Example:

```
apiVersion: v1
kind: Service
metadata:
```



```
name: busybox-subdomain
spec:
 selector:
 name: busybox
 clusterIP: None
 ports:
 - name: foo # name is not required for single-port Services
 port: 1234
```

```

apiVersion: v1
kind: Pod
metadata:
 name: busybox1
 labels:
 name: busybox
spec:
 hostname: busybox-1
 subdomain: busybox-subdomain
 containers:
 - image: busybox:1.28
 command:
 - sleep
 - "3600"
 name: busybox
```

```

apiVersion: v1
kind: Pod
metadata:
 name: busybox2
 labels:
 name: busybox
spec:
 hostname: busybox-2
 subdomain: busybox-subdomain
 containers:
 - image: busybox:1.28
 command:
 - sleep
 - "3600"
 name: busybox
```

Given the above Service "busybox-subdomain" and the Pods which set spec.subdomain to "busybox-subdomain", the first Pod will see its own FQDN as "busybox-1.busybox-subdomain.my-namespace.svc.cluster-domain.example". DNS serves A and/or AAAA records at that name, pointing to the Pod's IP. Both Pods "busybox1" and "busybox2" will have their own address records.

An [EndpointSlice](#) can specify the DNS hostname for any endpoint addresses, along with its IP.

**Note:**

Because A and AAAA records are not created for Pod names, hostname is required for the Pod's A or AAAA record to be created. A Pod with no hostname but with subdomain will only create

the A or AAAA record for the headless Service (busybox-subdomain.my-namespace.svc.cluster-domain.example), pointing to the Pods' IP addresses. Also, the Pod needs to be ready in order to have a record unless `publishNotReadyAddresses=True` is set on the Service.

## Pod's `setHostnameAsFQDN` field

FEATURE STATE: Kubernetes v1.22 [stable]

When a Pod is configured to have fully qualified domain name (FQDN), its hostname is the short hostname. For example, if you have a Pod with the fully qualified domain name `busybox-1.busybox-subdomain.my-namespace.svc.cluster-domain.example`, then by default the hostname command inside that Pod returns `busybox-1` and the `hostname --fqdn` command returns the FQDN.

When you set `setHostnameAsFQDN: true` in the Pod spec, the kubelet writes the Pod's FQDN into the hostname for that Pod's namespace. In this case, both `hostname` and `hostname --fqdn` return the Pod's FQDN.

### Note:

In Linux, the hostname field of the kernel (the `nodename` field of `struct utsname`) is limited to 64 characters.

If a Pod enables this feature and its FQDN is longer than 64 character, it will fail to start. The Pod will remain in Pending status (ContainerCreating as seen by `kubectl`) generating error events, such as `Failed to construct FQDN from Pod hostname and cluster domain, FQDN long-FQDN is too long (64 characters is the max, 70 characters requested)`. One way of improving user experience for this scenario is to create an [admission webhook controller](#) to control FQDN size when users create top level objects, for example, Deployment.

## Pod's DNS Policy

DNS policies can be set on a per-Pod basis. Currently Kubernetes supports the following Pod-specific DNS policies. These policies are specified in the `dnsPolicy` field of a Pod Spec.

- "Default": The Pod inherits the name resolution configuration from the node that the Pods run on. See [related discussion](#) for more details.
- "ClusterFirst": Any DNS query that does not match the configured cluster domain suffix, such as `"www.kubernetes.io"`, is forwarded to an upstream nameserver by the DNS server. Cluster administrators may have extra stub-domain and upstream DNS servers configured. See [related discussion](#) for details on how DNS queries are handled in those cases.
- "ClusterFirstWithHostNet": For Pods running with `hostNetwork`, you should explicitly set its DNS policy to `"ClusterFirstWithHostNet"`. Otherwise, Pods running with `hostNetwork` and `"ClusterFirst"` will fallback to the behavior of the `"Default"` policy.
  - Note: This is not supported on Windows. See [below](#) for details
- "None": It allows a Pod to ignore DNS settings from the Kubernetes environment. All DNS settings are supposed to be provided using the `dnsConfig` field in the Pod Spec. See [Pod's DNS config](#) subsection below.

**Note:**

"Default" is not the default DNS policy. If `dnsPolicy` is not explicitly specified, then "ClusterFirst" is used.

The example below shows a Pod with its DNS policy set to "ClusterFirstWithHostNet" because it has `hostNetwork` set to true.

```
apiVersion: v1
kind: Pod
metadata:
 name: busybox
 namespace: default
spec:
 containers:
 - image: busybox:1.28
 command:
 - sleep
 - "3600"
 imagePullPolicy: IfNotPresent
 name: busybox
 restartPolicy: Always
 hostNetwork: true
 dnsPolicy: ClusterFirstWithHostNet
```

## Pod's DNS Config

FEATURE STATE: Kubernetes v1.14 [stable]

Pod's DNS Config allows users more control on the DNS settings for a Pod.

The `dnsConfig` field is optional and it can work with any `dnsPolicy` settings. However, when a Pod's `dnsPolicy` is set to "None", the `dnsConfig` field has to be specified.

Below are the properties a user can specify in the `dnsConfig` field:

- `nameservers`: a list of IP addresses that will be used as DNS servers for the Pod. There can be at most 3 IP addresses specified. When the Pod's `dnsPolicy` is set to "None", the list must contain at least one IP address, otherwise this property is optional. The servers listed will be combined to the base nameservers generated from the specified DNS policy with duplicate addresses removed.
- `searches`: a list of DNS search domains for hostname lookup in the Pod. This property is optional. When specified, the provided list will be merged into the base search domain names generated from the chosen DNS policy. Duplicate domain names are removed. Kubernetes allows up to 32 search domains.
- `options`: an optional list of objects where each object may have a `name` property (required) and a `value` property (optional). The contents in this property will be merged to the options generated from the specified DNS policy. Duplicate entries are removed.

The following is an example Pod with custom DNS settings:

[service/networking/custom-dns.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 namespace: default
 name: dns-example
spec:
 containers:
 - name: test
 image: nginx
 dnsPolicy: "None"
 dnsConfig:
 nameservers:
 - 192.0.2.1 # this is an example
 searches:
 - ns1.svc.cluster-domain.example
 - my.dns.search.suffix
 options:
 - name: ndots
 value: "2"
 - name: edns0
```

When the Pod above is created, the container test gets the following contents in its `/etc/resolv.conf` file:

```
nameserver 192.0.2.1
search ns1.svc.cluster-domain.example my.dns.search.suffix
options ndots:2 edns0
```

For IPv6 setup, search path and name server should be set up like this:

```
kubect exec -it dns-example -- cat /etc/resolv.conf
```

The output is similar to this:

```
nameserver 2001:db8:30::a
search default.svc.cluster-domain.example svc.cluster-domain.example cluster-domain.example
options ndots:5
```

## DNS search domain list limits

FEATURE STATE: Kubernetes 1.28 [stable]

Kubernetes itself does not limit the DNS Config until the length of the search domain list exceeds 32 or the total length of all search domains exceeds 2048. This limit applies to the node's resolver configuration file, the Pod's DNS Config, and the merged DNS Config respectively.

### Note:

Some container runtimes of earlier versions may have their own restrictions on the number of DNS search domains. Depending on the container runtime environment, the pods with a large number of DNS search domains may get stuck in the pending state.

It is known that containerd v1.5.5 or earlier and CRI-O v1.21 or earlier have this problem.

## DNS resolution on Windows nodes

- ClusterFirstWithHostNet is not supported for Pods that run on Windows nodes. Windows treats all names with a . as a FQDN and skips FQDN resolution.
- On Windows, there are multiple DNS resolvers that can be used. As these come with slightly different behaviors, using the [Resolve-DNSName](#) powershell cmdlet for name query resolutions is recommended.
- On Linux, you have a DNS suffix list, which is used after resolution of a name as fully qualified has failed. On Windows, you can only have 1 DNS suffix, which is the DNS suffix associated with that Pod's namespace (example: mydns.svc.cluster.local). Windows can resolve FQDNs, Services, or network name which can be resolved with this single suffix. For example, a Pod spawned in the default namespace, will have the DNS suffix default.svc.cluster.local. Inside a Windows Pod, you can resolve both `kubernetes.default.svc.cluster.local` and `kubernetes`, but not the partially qualified names (`kubernetes.default` or `kubernetes.default.svc`).

## What's next

For guidance on administering DNS configurations, check [Configure DNS Service](#)

## IPv4/IPv6 dual-stack

Kubernetes lets you configure single-stack IPv4 networking, single-stack IPv6 networking, or dual stack networking with both network families active. This page explains how.

FEATURE STATE: Kubernetes v1.23 [stable]

IPv4/IPv6 dual-stack networking enables the allocation of both IPv4 and IPv6 addresses to [Pods](#) and [Services](#).

IPv4/IPv6 dual-stack networking is enabled by default for your Kubernetes cluster starting in 1.21, allowing the simultaneous assignment of both IPv4 and IPv6 addresses.

## Supported Features

IPv4/IPv6 dual-stack on your Kubernetes cluster provides the following features:

- Dual-stack Pod networking (a single IPv4 and IPv6 address assignment per Pod)
- IPv4 and IPv6 enabled Services
- Pod off-cluster egress routing (eg. the Internet) via both IPv4 and IPv6 interfaces

## Prerequisites

The following prerequisites are needed in order to utilize IPv4/IPv6 dual-stack Kubernetes clusters:

- Kubernetes 1.20 or later

For information about using dual-stack services with earlier Kubernetes versions, refer to the documentation for that version of Kubernetes.

- Provider support for dual-stack networking (Cloud provider or otherwise must be able to provide Kubernetes nodes with routable IPv4/IPv6 network interfaces)
- A [network plugin](#) that supports dual-stack networking.

## Configure IPv4/IPv6 dual-stack

To configure IPv4/IPv6 dual-stack, set dual-stack cluster network assignments:

- kube-apiserver:
  - `--service-cluster-ip-range=<IPv4 CIDR>,<IPv6 CIDR>`
- kube-controller-manager:
  - `--cluster-cidr=<IPv4 CIDR>,<IPv6 CIDR>`
  - `--service-cluster-ip-range=<IPv4 CIDR>,<IPv6 CIDR>`
  - `--node-cidr-mask-size-ipv4|--node-cidr-mask-size-ipv6` defaults to /24 for IPv4 and /64 for IPv6
- kube-proxy:
  - `--cluster-cidr=<IPv4 CIDR>,<IPv6 CIDR>`
- kubelet:
  - `--node-ip=<IPv4 IP>,<IPv6 IP>`
    - This option is required for bare metal dual-stack nodes (nodes that do not define a cloud provider with the `--cloud-provider` flag). If you are using a cloud provider and choose to override the node IPs chosen by the cloud provider, set the `--node-ip` option.
    - (The legacy built-in cloud providers do not support dual-stack `--node-ip`.)

### Note:

An example of an IPv4 CIDR: 10.244.0.0/16 (though you would supply your own address range)

An example of an IPv6 CIDR: fdXY:IJKL:MNOP:15::/64 (this shows the format but is not a valid address - see [RFC 4193](#))

## Services

You can create [Services](#) which can use IPv4, IPv6, or both.

The address family of a Service defaults to the address family of the first service cluster IP range (configured via the `--service-cluster-ip-range` flag to the kube-apiserver).

When you define a Service you can optionally configure it as dual stack. To specify the behavior you want, you set the `.spec.ipFamilyPolicy` field to one of the following values:

- SingleStack: Single-stack service. The control plane allocates a cluster IP for the Service, using the first configured service cluster IP range.
- PreferDualStack:
  - Allocates IPv4 and IPv6 cluster IPs for the Service.

- **RequireDualStack**: Allocates Service `.spec.ClusterIPs` from both IPv4 and IPv6 address ranges.
  - Selects the `.spec.ClusterIP` from the list of `.spec.ClusterIPs` based on the address family of the first element in the `.spec.ipFamilies` array.

If you would like to define which IP family to use for single stack or define the order of IP families for dual-stack, you can choose the address families by setting an optional field, `.spec.ipFamilies`, on the Service.

### Note:

The `.spec.ipFamilies` field is conditionally mutable: you can add or remove a secondary IP address family, but you cannot change the primary IP address family of an existing Service.

You can set `.spec.ipFamilies` to any of the following array values:

- ["IPv4"]
- ["IPv6"]
- ["IPv4","IPv6"] (dual stack)
- ["IPv6","IPv4"] (dual stack)

The first family you list is used for the legacy `.spec.ClusterIP` field.

## Dual-stack Service configuration scenarios

These examples demonstrate the behavior of various dual-stack Service configuration scenarios.

### Dual-stack options on new Services

1. This Service specification does not explicitly define `.spec.ipFamilyPolicy`. When you create this Service, Kubernetes assigns a cluster IP for the Service from the first configured service-cluster-ip-range and sets the `.spec.ipFamilyPolicy` to `SingleStack`. ([Services without selectors](#) and [headless Services](#) with selectors will behave in this same way.)

[service/networking/dual-stack-default-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
 labels:
 app.kubernetes.io/name: MyApp
spec:
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - protocol: TCP
 port: 80
```

2. This Service specification explicitly defines `PreferDualStack` in `.spec.ipFamilyPolicy`. When you create this Service on a dual-stack cluster, Kubernetes assigns both IPv4 and IPv6 addresses for the service. The control plane updates the `.spec` for the Service to

record the IP address assignments. The field `.spec.ClusterIPs` is the primary field, and contains both assigned IP addresses; `.spec.ClusterIP` is a secondary field with its value calculated from `.spec.ClusterIPs`.

- For the `.spec.ClusterIP` field, the control plane records the IP address that is from the same address family as the first service cluster IP range.
- On a single-stack cluster, the `.spec.ClusterIPs` and `.spec.ClusterIP` fields both only list one address.
- On a cluster with dual-stack enabled, specifying `RequireDualStack` in `.spec.ipFamilyPolicy` behaves the same as `PreferDualStack`.

[service/networking/dual-stack-preferred-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
 labels:
 app.kubernetes.io/name: MyApp
spec:
 ipFamilyPolicy: PreferDualStack
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - protocol: TCP
 port: 80
```

3. This Service specification explicitly defines IPv6 and IPv4 in `.spec.ipFamilies` as well as defining `PreferDualStack` in `.spec.ipFamilyPolicy`. When Kubernetes assigns an IPv6 and IPv4 address in `.spec.ClusterIPs`, `.spec.ClusterIP` is set to the IPv6 address because that is the first element in the `.spec.ClusterIPs` array, overriding the default.

[service/networking/dual-stack-preferred-ipfamilies-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
 labels:
 app.kubernetes.io/name: MyApp
spec:
 ipFamilyPolicy: PreferDualStack
 ipFamilies:
 - IPv6
 - IPv4
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - protocol: TCP
 port: 80
```



## Dual-stack defaults on existing Services

These examples demonstrate the default behavior when dual-stack is newly enabled on a cluster where Services already exist. (Upgrading an existing cluster to 1.21 or beyond will enable dual-stack.)

1. When dual-stack is enabled on a cluster, existing Services (whether IPv4 or IPv6) are configured by the control plane to set `.spec.ipFamilyPolicy` to `SingleStack` and set `.spec.ipFamilies` to the address family of the existing Service. The existing Service cluster IP will be stored in `.spec.ClusterIPs`.

[service/networking/dual-stack-default-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
 labels:
 app.kubernetes.io/name: MyApp
spec:
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - protocol: TCP
 port: 80
```

You can validate this behavior by using `kubectl` to inspect an existing service.

```
kubectl get svc my-service -o yaml
```

```
apiVersion: v1
kind: Service
metadata:
 labels:
 app.kubernetes.io/name: MyApp
 name: my-service
spec:
 clusterIP: 10.0.197.123
 clusterIPs:
 - 10.0.197.123
 ipFamilies:
 - IPv4
 ipFamilyPolicy: SingleStack
 ports:
 - port: 80
 protocol: TCP
 targetPort: 80
 selector:
 app.kubernetes.io/name: MyApp
 type: ClusterIP
status:
 loadBalancer: {}
```

- When dual-stack is enabled on a cluster, existing [headless Services](#) with selectors are
2. configured by the control plane to set `.spec.ipFamilyPolicy` to `SingleStack` and set `.spec.ipFamilies` to the address family of the first service cluster IP range (configured via the `--service-cluster-ip-range` flag to the kube-apiserver) even though `.spec.ClusterIP` is set to `None`.

[service/networking/dual-stack-default-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
 labels:
 app.kubernetes.io/name: MyApp
spec:
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - protocol: TCP
 port: 80
```

You can validate this behavior by using `kubectl` to inspect an existing headless service with selectors.

```
kubectl get svc my-service -o yaml
```

```
apiVersion: v1
kind: Service
metadata:
 labels:
 app.kubernetes.io/name: MyApp
 name: my-service
spec:
 clusterIP: None
 clusterIPs:
 - None
 ipFamilies:
 - IPv4
 ipFamilyPolicy: SingleStack
 ports:
 - port: 80
 protocol: TCP
 targetPort: 80
 selector:
 app.kubernetes.io/name: MyApp
```

## Switching Services between single-stack and dual-stack

Services can be changed from single-stack to dual-stack and from dual-stack to single-stack.

1. To change a Service from single-stack to dual-stack, change `.spec.ipFamilyPolicy` from `SingleStack` to `PreferDualStack` or `RequireDualStack` as desired. When you change this

Service from single-stack to dual-stack, Kubernetes assigns the missing address family so that the Service now has IPv4 and IPv6 addresses.

Edit the Service specification updating the `.spec.ipFamilyPolicy` from `SingleStack` to `PreferDualStack`.

Before:

```
spec:
 ipFamilyPolicy: SingleStack
```

After:

```
spec:
 ipFamilyPolicy: PreferDualStack
```

2. To change a Service from dual-stack to single-stack, change `.spec.ipFamilyPolicy` from `PreferDualStack` or `RequireDualStack` to `SingleStack`. When you change this Service from dual-stack to single-stack, Kubernetes retains only the first element in the `.spec.ClusterIPs` array, and sets `.spec.ClusterIP` to that IP address and sets `.spec.ipFamilies` to the address family of `.spec.ClusterIPs`.

## Headless Services without selector

For [Headless Services without selectors](#) and without `.spec.ipFamilyPolicy` explicitly set, the `.spec.ipFamilyPolicy` field defaults to `RequireDualStack`.

## Service type LoadBalancer

To provision a dual-stack load balancer for your Service:

- Set the `.spec.type` field to `LoadBalancer`
- Set `.spec.ipFamilyPolicy` field to `PreferDualStack` or `RequireDualStack`

### Note:

To use a dual-stack `LoadBalancer` type Service, your cloud provider must support IPv4 and IPv6 load balancers.

## Egress traffic

If you want to enable egress traffic in order to reach off-cluster destinations (eg. the public Internet) from a Pod that uses non-publicly routable IPv6 addresses, you need to enable the Pod to use a publicly routed IPv6 address via a mechanism such as transparent proxying or IP masquerading. The [ip-masq-agent](#) project supports IP masquerading on dual-stack clusters.

### Note:

Ensure your [CNI](#) provider supports IPv6.

## Windows support

Kubernetes on Windows does not support single-stack "IPv6-only" networking. However, dual-stack IPv4/IPv6 networking for pods and nodes with single-family services is supported.

You can use IPv4/IPv6 dual-stack networking with l2bridge networks.

### Note:

Overlay (VXLAN) networks on Windows **do not** support dual-stack networking.

You can read more about the different network modes for Windows within the [Networking on Windows](#) topic.

## What's next

- [Validate IPv4/IPv6 dual-stack](#) networking
- [Enable dual-stack networking using kubeadm](#)

## Topology Aware Routing

*Topology Aware Routing* provides a mechanism to help keep network traffic within the zone where it originated. Preferring same-zone traffic between Pods in your cluster can help with reliability, performance (network latency and throughput), or cost.

FEATURE STATE: Kubernetes v1.23 [beta]

### Note:

Prior to Kubernetes 1.27, this feature was known as *Topology Aware Hints*.

*Topology Aware Routing* adjusts routing behavior to prefer keeping traffic in the zone it originated from. In some cases this can help reduce costs or improve network performance.

## Motivation

Kubernetes clusters are increasingly deployed in multi-zone environments. *Topology Aware Routing* provides a mechanism to help keep traffic within the zone it originated from. When calculating the endpoints for a [Service](#), the EndpointSlice controller considers the topology (region and zone) of each endpoint and populates the hints field to allocate it to a zone. Cluster components such as [kube-proxy](#) can then consume those hints, and use them to influence how the traffic is routed (favoring topologically closer endpoints).

## Enabling Topology Aware Routing

### Note:

Prior to Kubernetes 1.27, this behavior was controlled using the `service.kubernetes.io/topology-aware-hints` annotation.

You can enable Topology Aware Routing for a Service by setting the `service.kubernetes.io/topology-mode` annotation to `Auto`. When there are enough endpoints available in each zone, Topology Hints will be populated on EndpointSlices to allocate individual endpoints to specific zones, resulting in traffic being routed closer to where it originated from.

## When it works best

This feature works best when:

### 1. Incoming traffic is evenly distributed

If a large proportion of traffic is originating from a single zone, that traffic could overload the subset of endpoints that have been allocated to that zone. This feature is not recommended when incoming traffic is expected to originate from a single zone.

### 2. The Service has 3 or more endpoints per zone

In a three zone cluster, this means 9 or more endpoints. If there are fewer than 3 endpoints per zone, there is a high ( $\approx 50\%$ ) probability that the EndpointSlice controller will not be able to allocate endpoints evenly and instead will fall back to the default cluster-wide routing approach.

## How It Works

The "Auto" heuristic attempts to proportionally allocate a number of endpoints to each zone. Note that this heuristic works best for Services that have a significant number of endpoints.

### EndpointSlice controller

The EndpointSlice controller is responsible for setting hints on EndpointSlices when this heuristic is enabled. The controller allocates a proportional amount of endpoints to each zone. This proportion is based on the [allocatable](#) CPU cores for nodes running in that zone. For example, if one zone had 2 CPU cores and another zone only had 1 CPU core, the controller would allocate twice as many endpoints to the zone with 2 CPU cores.

The following example shows what an EndpointSlice looks like when hints have been populated:

```
apiVersion: discovery.k8s.io/v1
kind: EndpointSlice
metadata:
 name: example-hints
 labels:
 kubernetes.io/service-name: example-svc
addressType: IPv4
ports:
 - name: http
 protocol: TCP
 port: 80
endpoints:
 - addresses:
```

```
- "10.1.2.3"
conditions:
 ready: true
hostname: pod-1
zone: zone-a
hints:
 forZones:
 - name: "zone-a"
```

## kube-proxy

The kube-proxy component filters the endpoints it routes to based on the hints set by the EndpointSlice controller. In most cases, this means that the kube-proxy is able to route traffic to endpoints in the same zone. Sometimes the controller allocates endpoints from a different zone to ensure more even distribution of endpoints between zones. This would result in some traffic being routed to other zones.

## Safeguards

The Kubernetes control plane and the kube-proxy on each node apply some safeguard rules before using Topology Aware Hints. If these don't check out, the kube-proxy selects endpoints from anywhere in your cluster, regardless of the zone.

1. **Insufficient number of endpoints:** If there are less endpoints than zones in a cluster, the controller will not assign any hints.
2. **Impossible to achieve balanced allocation:** In some cases, it will be impossible to achieve a balanced allocation of endpoints among zones. For example, if zone-a is twice as large as zone-b, but there are only 2 endpoints, an endpoint allocated to zone-a may receive twice as much traffic as zone-b. The controller does not assign hints if it can't get this "expected overload" value below an acceptable threshold for each zone. Importantly this is not based on real-time feedback. It is still possible for individual endpoints to become overloaded.
3. **One or more Nodes has insufficient information:** If any node does not have a `topology.kubernetes.io/zone` label or is not reporting a value for allocatable CPU, the control plane does not set any topology-aware endpoint hints and so kube-proxy does not filter endpoints by zone.
4. **One or more endpoints does not have a zone hint:** When this happens, the kube-proxy assumes that a transition from or to Topology Aware Hints is underway. Filtering endpoints for a Service in this state would be dangerous so the kube-proxy falls back to using all endpoints.
5. **A zone is not represented in hints:** If the kube-proxy is unable to find at least one endpoint with a hint targeting the zone it is running in, it falls back to using endpoints from all zones. This is most likely to happen as you add a new zone into your existing cluster.

## Constraints

- Topology Aware Hints are not used when `internalTrafficPolicy` is set to `Local` on a Service. It is possible to use both features in the same cluster on different Services, just not on the same Service.
- This approach will not work well for Services that have a large proportion of traffic originating from a subset of zones. Instead this assumes that incoming traffic will be roughly proportional to the capacity of the Nodes in each zone.
- The EndpointSlice controller ignores unready nodes as it calculates the proportions of each zone. This could have unintended consequences if a large portion of nodes are unready.
- The EndpointSlice controller ignores nodes with the `node-role.kubernetes.io/control-plane` or `node-role.kubernetes.io/master` label set. This could be problematic if workloads are also running on those nodes.
- The EndpointSlice controller does not take into account [tolerations](#) when deploying or calculating the proportions of each zone. If the Pods backing a Service are limited to a subset of Nodes in the cluster, this will not be taken into account.
- This may not work well with autoscaling. For example, if a lot of traffic is originating from a single zone, only the endpoints allocated to that zone will be handling that traffic. That could result in [Horizontal Pod Autoscaler](#) either not picking up on this event, or newly added pods starting in a different zone.

## Custom heuristics

Kubernetes is deployed in many different ways, there is no single heuristic for allocating endpoints to zones will work for every use case. A key goal of this feature is to enable custom heuristics to be developed if the built in heuristic does not work for your use case. The first steps to enable custom heuristics were included in the 1.27 release. This is a limited implementation that may not yet cover some relevant and plausible situations.

## What's next

- Follow the [Connecting Applications with Services](#) tutorial
- Learn about the [trafficDistribution](#) field, which is closely related to the `service.kubernetes.io/topology-mode` annotation and provides flexible options for traffic routing within Kubernetes.

## Networking on Windows

Kubernetes supports running nodes on either Linux or Windows. You can mix both kinds of node within a single cluster. This page provides an overview to networking specific to the Windows operating system.

# Container networking on Windows

Networking for Windows containers is exposed through [CNI plugins](#). Windows containers function similarly to virtual machines in regards to networking. Each container has a virtual network adapter (vNIC) which is connected to a Hyper-V virtual switch (vSwitch). The Host Networking Service (HNS) and the Host Compute Service (HCS) work together to create containers and attach container vNICs to networks. HCS is responsible for the management of containers whereas HNS is responsible for the management of networking resources such as:

- Virtual networks (including creation of vSwitches)
- Endpoints / vNICs
- Namespaces
- Policies including packet encapsulations, load-balancing rules, ACLs, and NAT rules.

The Windows HNS and vSwitch implement namespacing and can create virtual NICs as needed for a pod or container. However, many configurations such as DNS, routes, and metrics are stored in the Windows registry database rather than as files inside /etc, which is how Linux stores those configurations. The Windows registry for the container is separate from that of the host, so concepts like mapping /etc/resolv.conf from the host into a container don't have the same effect they would on Linux. These must be configured using Windows APIs run in the context of that container. Therefore CNI implementations need to call the HNS instead of relying on file mappings to pass network details into the pod or container.

## Network modes

Windows supports five different networking drivers/modes: L2bridge, L2tunnel, Overlay (Beta), Transparent, and NAT. In a heterogeneous cluster with Windows and Linux worker nodes, you need to select a networking solution that is compatible on both Windows and Linux. The following table lists the out-of-tree plugins are supported on Windows, with recommendations on when to use each CNI:

| Network Driver | Description                                                                                                                                                                                                           | Container Packet Modifications                                                             | Network Plugins                                                                               | Network Plugin Characteristics                                                                                                                                                  |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L2bridge       | Containers are attached to an external vSwitch. Containers are attached to the underlay network, although the physical network doesn't need to learn the container MACs because they are rewritten on ingress/egress. | MAC is rewritten to host MAC, IP may be rewritten to host IP using HNS OutboundNAT policy. | <a href="#">win-bridge</a> , <a href="#">Azure-CNI</a> , Flannel host-gateway uses win-bridge | win-bridge uses L2bridge network mode, connects containers to the underlay of hosts, offering best performance. Requires user-defined routes (UDR) for inter-node connectivity. |
| L2Tunnel       | This is a special case of l2bridge, but only used on Azure. All packets are sent to                                                                                                                                   | MAC rewritten, IP visible on the underlay network                                          | <a href="#">Azure-CNI</a>                                                                     | Azure-CNI allows integration of containers with Azure vNET, and allows them to leverage                                                                                         |



| Network Driver                                                     | Description                                                                                                                                                                                    | Container Packet Modifications                                                                                                                                                                                                                                                                  | Network Plugins                                                   | Network Plugin Characteristics                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                    | the virtualization host where SDN policy is applied.                                                                                                                                           |                                                                                                                                                                                                                                                                                                 |                                                                   | the set of capabilities that <a href="#">Azure Virtual Network provides</a> . For example, securely connect to Azure services or use Azure NSGs. See <a href="#">azure-cni for some examples</a>                                                                                                                                                               |
| Overlay                                                            | Containers are given a vNIC connected to an external vSwitch. Each overlay network gets its own IP subnet, defined by a custom IP prefix. The overlay network driver uses VXLAN encapsulation. | Encapsulated with an outer header.                                                                                                                                                                                                                                                              | <a href="#">win-overlay</a> , Flannel<br>VXLAN (uses win-overlay) | win-overlay should be used when virtual container networks are desired to be isolated from underlay of hosts (e.g. for security reasons). Allows for IPs to be re-used for different overlay networks (which have different VNID tags) if you are restricted on IPs in your datacenter. This option requires <a href="#">KB4489899</a> on Windows Server 2019. |
| Transparent (special use case for <a href="#">ovn-kubernetes</a> ) | Requires an external vSwitch. Containers are attached to an external vSwitch which enables intra-pod communication via logical networks (logical switches and routers).                        | Packet is encapsulated either via <a href="#">GENEVE</a> or <a href="#">STT</a> tunneling to reach pods which are not on the same host. Packets are forwarded or dropped via the tunnel metadata information supplied by the ovn network controller. NAT is done for north-south communication. | <a href="#">ovn-kubernetes</a>                                    | <a href="#">Deploy via ansible</a> . Distributed ACLs can be applied via Kubernetes policies. IPAM support. Load-balancing can be achieved without kube-proxy. NATing is done without using iptables/netsh.                                                                                                                                                    |
| NAT ( <i>not used in Kubernetes</i> )                              | Containers are given a vNIC connected to an internal vSwitch. DNS/DHCP is provided using an internal component called <a href="#">WinNAT</a>                                                   | MAC and IP is rewritten to host MAC/IP.                                                                                                                                                                                                                                                         | <a href="#">nat</a>                                               | Included here for completeness                                                                                                                                                                                                                                                                                                                                 |

As outlined above, the [Flannel CNI plugin](#) is also [supported](#) on Windows via the [VXLAN network backend](#) (**Beta support** ; delegates to win-overlay) and [host-gateway network backend](#) (stable support; delegates to win-bridge).

This plugin supports delegating to one of the reference CNI plugins (win-overlay, win-bridge), to work in conjunction with Flannel daemon on Windows (Flanneld) for automatic node subnet lease assignment and HNS network creation. This plugin reads in its own configuration file (cni.conf), and aggregates it with the environment variables from the Flanneld generated subnet.env file. It then delegates to one of the reference CNI plugins for network plumbing, and sends the correct configuration containing the node-assigned subnet to the IPAM plugin (for example: host-local).

For Node, Pod, and Service objects, the following network flows are supported for TCP/UDP traffic:

- Pod → Pod (IP)
- Pod → Pod (Name)
- Pod → Service (Cluster IP)
- Pod → Service (PQDN, but only if there are no ".")
- Pod → Service (FQDN)
- Pod → external (IP)
- Pod → external (DNS)
- Node → Pod
- Pod → Node

## IP address management (IPAM)

The following IPAM options are supported on Windows:

- [host-local](#)
- [azure-vnet-ipam](#) (for azure-cni only)
- [Windows Server IPAM](#) (fallback option if no IPAM is set)

## Load balancing and Services

A Kubernetes [Service](#) is an abstraction that defines a logical set of Pods and a means to access them over a network. In a cluster that includes Windows nodes, you can use the following types of Service:

- NodePort
- ClusterIP
- LoadBalancer
- ExternalName

Windows container networking differs in some important ways from Linux networking. The [Microsoft documentation for Windows Container Networking](#) provides additional details and background.

On Windows, you can use the following settings to configure Services and load balancing behavior:

Windows Service Settings

| Feature                         | Description                                                                                                                                                                            | Minimum Supported Windows OS build | How to enable                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Session affinity                | Ensures that connections from a particular client are passed to the same Pod each time.                                                                                                | Windows Server 2022                | Set <code>service.spec.sessionAffinity</code> to "ClientIP"                                                      |
| Direct Server Return (DSR)      | Load balancing mode where the IP address fixups and the LBNAT occurs at the container vSwitch port directly; service traffic arrives with the source IP set as the originating pod IP. | Windows Server 2019                | Set the following flags in kube-proxy: <code>--feature-gates="WinDSR=true"</code> <code>--enable-dsr=true</code> |
| Preserve-Destination            | Skips DNAT of service traffic, thereby preserving the virtual IP of the target service in packets reaching the backend Pod. Also disables node-node forwarding.                        | Windows Server, version 1903       | Set <code>"preserve-destination": "true"</code> in service annotations and enable DSR in kube-proxy.             |
| IPv4/IPv6 dual-stack networking | Native IPv4-to-IPv4 in parallel with IPv6-to-IPv6 communications to, from, and within a cluster                                                                                        | Windows Server 2019                | See <a href="#">IPv4/IPv6 dual-stack</a>                                                                         |
| Client IP preservation          | Ensures that source IP of incoming ingress traffic gets preserved. Also disables node-node forwarding.                                                                                 | Windows Server 2019                | Set <code>service.spec.externalTrafficPolicy</code> to "Local" and enable DSR in kube-proxy                      |

### Warning:

There are known issue with NodePort Services on overlay networking, if the destination node is running Windows Server 2022. To avoid the issue entirely, you can configure the service with `externalTrafficPolicy: Local`.

There are known issues with Pod to Pod connectivity on l2bridge network on Windows Server 2022 with KB5005619 or higher installed. To workaround the issue and restore Pod to Pod connectivity, you can disable the WinDSR feature in kube-proxy.

These issues require OS fixes. Please follow <https://github.com/microsoft/Windows-Containers/issues/204> for updates.

## Limitations

The following networking functionality is *not* supported on Windows nodes:

- Host networking mode

- Local NodePort access from the node itself (works for other nodes or external clients)
- More than 64 backend pods (or unique destination addresses) for a single Service
- IPv6 communication between Windows pods connected to overlay networks
- Local Traffic Policy in non-DSR mode
- Outbound communication using the ICMP protocol via the win-overlay, win-bridge, or using the Azure-CNI plugin.

Specifically, the Windows data plane ([VFP](#)) doesn't support ICMP packet transpositions, and this means:

- ICMP packets directed to destinations within the same network (such as pod to pod communication via ping) work as expected;
- TCP/UDP packets work as expected;
- ICMP packets directed to pass through a remote network (e.g. pod to external internet communication via ping) cannot be transposed and thus will not be routed back to their source;
- Since TCP/UDP packets can still be transposed, you can substitute ping <destination> with curl <destination> when debugging connectivity with the outside world.

Other limitations:

- Windows reference network plugins win-bridge and win-overlay do not implement [CNI spec](#) v0.4.0, due to a missing CHECK implementation.
- The Flannel VXLAN CNI plugin has the following limitations on Windows:
  - Node-pod connectivity is only possible for local pods with Flannel v0.12.0 (or higher).
  - Flannel is restricted to using VNI 4096 and UDP port 4789. See the official [Flannel VXLAN](#) backend docs for more details on these parameters.

## Service ClusterIP allocation

In Kubernetes, [Services](#) are an abstract way to expose an application running on a set of Pods. Services can have a cluster-scoped virtual IP address (using a Service of type: ClusterIP). Clients can connect using that virtual IP address, and Kubernetes then load-balances traffic to that Service across the different backing Pods.

### How Service ClusterIPs are allocated?

When Kubernetes needs to assign a virtual IP address for a Service, that assignment happens one of two ways:

*dynamically*

the cluster's control plane automatically picks a free IP address from within the configured IP range for type: ClusterIP Services.

*statically*

you specify an IP address of your choice, from within the configured IP range for Services.

Across your whole cluster, every Service ClusterIP must be unique. Trying to create a Service with a specific ClusterIP that has already been allocated will return an error.

# Why do you need to reserve Service Cluster IPs?

Sometimes you may want to have Services running in well-known IP addresses, so other components and users in the cluster can use them.

The best example is the DNS Service for the cluster. As a soft convention, some Kubernetes installers assign the 10th IP address from the Service IP range to the DNS service. Assuming you configured your cluster with Service IP range 10.96.0.0/16 and you want your DNS Service IP to be 10.96.0.10, you'd have to create a Service like this:

```
apiVersion: v1
kind: Service
metadata:
 labels:
 k8s-app: kube-dns
 kubernetes.io/cluster-service: "true"
 kubernetes.io/name: CoreDNS
 name: kube-dns
 namespace: kube-system
spec:
 clusterIP: 10.96.0.10
 ports:
 - name: dns
 port: 53
 protocol: UDP
 targetPort: 53
 - name: dns-tcp
 port: 53
 protocol: TCP
 targetPort: 53
 selector:
 k8s-app: kube-dns
 type: ClusterIP
```

but as it was explained before, the IP address 10.96.0.10 has not been reserved; if other Services are created before or in parallel with dynamic allocation, there is a chance they can allocate this IP, hence, you will not be able to create the DNS Service because it will fail with a conflict error.

## How can you avoid Service ClusterIP conflicts?

The allocation strategy implemented in Kubernetes to allocate ClusterIPs to Services reduces the risk of collision.

The ClusterIP range is divided, based on the formula  $\min(\max(16, \text{cidrSize} / 16), 256)$ , described as *never less than 16 or more than 256 with a graduated step between them*.

Dynamic IP assignment uses the upper band by default, once this has been exhausted it will use the lower range. This will allow users to use static allocations on the lower band with a low risk of collision.

# Examples

## Example 1

This example uses the IP address range: 10.96.0.0/24 (CIDR notation) for the IP addresses of Services.

Range Size:  $2^8 - 2 = 254$   
Band Offset:  $\min(\max(16, 256/16), 256) = \min(16, 256) = 16$   
Static band start: 10.96.0.1  
Static band end: 10.96.0.16  
Range end: 10.96.0.254

pie showData title 10.96.0.0/24 "Static" : 16 "Dynamic" : 238

*JavaScript must be [enabled](#) to view this content*

## Example 2

This example uses the IP address range: 10.96.0.0/20 (CIDR notation) for the IP addresses of Services.

Range Size:  $2^{12} - 2 = 4094$   
Band Offset:  $\min(\max(16, 4096/16), 256) = \min(256, 256) = 256$   
Static band start: 10.96.0.1  
Static band end: 10.96.1.0  
Range end: 10.96.15.254

pie showData title 10.96.0.0/20 "Static" : 256 "Dynamic" : 3838

*JavaScript must be [enabled](#) to view this content*

## Example 3

This example uses the IP address range: 10.96.0.0/16 (CIDR notation) for the IP addresses of Services.

Range Size:  $2^{16} - 2 = 65534$   
Band Offset:  $\min(\max(16, 65536/16), 256) = \min(4096, 256) = 256$   
Static band start: 10.96.0.1  
Static band ends: 10.96.1.0  
Range end: 10.96.255.254

pie showData title 10.96.0.0/16 "Static" : 256 "Dynamic" : 65278

*JavaScript must be [enabled](#) to view this content*

## What's next

- Read about [Service External Traffic Policy](#)
- Read about [Connecting Applications with Services](#)

- Read about [Services](#)

# Service Internal Traffic Policy

If two Pods in your cluster want to communicate, and both Pods are actually running on the same node, use *Service Internal Traffic Policy* to keep network traffic within that node. Avoiding a round trip via the cluster network can help with reliability, performance (network latency and throughput), or cost.

FEATURE STATE: Kubernetes v1.26 [stable]

*Service Internal Traffic Policy* enables internal traffic restrictions to only route internal traffic to endpoints within the node the traffic originated from. The "internal" traffic here refers to traffic originated from Pods in the current cluster. This can help to reduce costs and improve performance.

## Using Service Internal Traffic Policy

You can enable the internal-only traffic policy for a [Service](#), by setting its `.spec.internalTrafficPolicy` to `Local`. This tells kube-proxy to only use node local endpoints for cluster internal traffic.

### Note:

For pods on nodes with no endpoints for a given Service, the Service behaves as if it has zero endpoints (for Pods on this node) even if the service does have endpoints on other nodes.

The following example shows what a Service looks like when you set `.spec.internalTrafficPolicy` to `Local`:

```
apiVersion: v1
kind: Service
metadata:
 name: my-service
spec:
 selector:
 app.kubernetes.io/name: MyApp
 ports:
 - protocol: TCP
 port: 80
 targetPort: 9376
 internalTrafficPolicy: Local
```

## How it works

The kube-proxy filters the endpoints it routes to based on the `spec.internalTrafficPolicy` setting. When it's set to `Local`, only node local endpoints are considered. When it's `Cluster` (the default), or is not set, Kubernetes considers all endpoints.

## What's next

- Read about [Topology Aware Routing](#)
- Read about [Service External Traffic Policy](#)
- Follow the [Connecting Applications with Services](#) tutorial

## Storage

Ways to provide both long-term and temporary storage to Pods in your cluster.

---

[Volumes](#)

[Persistent Volumes](#)

[Projected Volumes](#)

[Ephemeral Volumes](#)

[Storage Classes](#)

[Volume Attributes Classes](#)

[Dynamic Volume Provisioning](#)

[Volume Snapshots](#)

[Volume Snapshot Classes](#)

[CSI Volume Cloning](#)

[Storage Capacity](#)

[Node-specific Volume Limits](#)

[Volume Health Monitoring](#)

[Windows Storage](#)

## Volumes

On-disk files in a container are ephemeral, which presents some problems for non-trivial applications when running in containers. One problem occurs when a container crashes or is stopped. Container state is not saved so all of the files that were created or modified during the lifetime of the container are lost. During a crash, kubelet restarts the container with a clean state. Another problem occurs when multiple containers are running in a Pod and need to share files. It can be challenging to setup and access a shared filesystem across all of the containers. The Kubernetes [volume](#) abstraction solves both of these problems. Familiarity with [Pods](#) is suggested.



# Background

Kubernetes supports many types of volumes. A [Pod](#) can use any number of volume types simultaneously. [Ephemeral volume](#) types have a lifetime of a pod, but [persistent volumes](#) exist beyond the lifetime of a pod. When a pod ceases to exist, Kubernetes destroys ephemeral volumes; however, Kubernetes does not destroy persistent volumes. For any kind of volume in a given pod, data is preserved across container restarts.

At its core, a volume is a directory, possibly with some data in it, which is accessible to the containers in a pod. How that directory comes to be, the medium that backs it, and the contents of it are determined by the particular volume type used.

To use a volume, specify the volumes to provide for the Pod in `.spec.volumes` and declare where to mount those volumes into containers in `.spec.containers[*].volumeMounts`. A process in a container sees a filesystem view composed from the initial contents of the [container image](#), plus volumes (if defined) mounted inside the container. The process sees a root filesystem that initially matches the contents of the container image. Any writes to within that filesystem hierarchy, if allowed, affect what that process views when it performs a subsequent filesystem access. Volumes mount at the [specified paths](#) within the image. For each container defined within a Pod, you must independently specify where to mount each volume that the container uses.

Volumes cannot mount within other volumes (but see [Using subPath](#) for a related mechanism). Also, a volume cannot contain a hard link to anything in a different volume.

## Types of volumes

Kubernetes supports several types of volumes.

### **awsElasticBlockStore (deprecated)**

In Kubernetes 1.30, all operations for the in-tree `awsElasticBlockStore` type are redirected to the `ebs.csi.aws.com` [CSI](#) driver.

The `AWSElasticBlockStore` in-tree storage driver was deprecated in the Kubernetes v1.19 release and then removed entirely in the v1.27 release.

The Kubernetes project suggests that you use the [AWS EBS](#) third party storage driver instead.

### **azureDisk (deprecated)**

In Kubernetes 1.30, all operations for the in-tree `azureDisk` type are redirected to the `disk.csi.azure.com` [CSI](#) driver.

The `AzureDisk` in-tree storage driver was deprecated in the Kubernetes v1.19 release and then removed entirely in the v1.27 release.

The Kubernetes project suggests that you use the [Azure Disk](#) third party storage driver instead.

### **azureFile (deprecated)**

FEATURE STATE: Kubernetes v1.21 [deprecated]

The `azureFile` volume type mounts a Microsoft Azure File volume (SMB 2.1 and 3.0) into a pod.

For more details, see the [azureFile volume plugin](#).

### **azureFile CSI migration**

FEATURE STATE: Kubernetes v1.26 [stable]

The CSIMigration feature for `azureFile`, when enabled, redirects all plugin operations from the existing in-tree plugin to the `file.csi.azure.com` Container Storage Interface (CSI) Driver. In order to use this feature, the [Azure File CSI Driver](#) must be installed on the cluster and the `CSIMigrationAzureFile` [feature gates](#) must be enabled.

Azure File CSI driver does not support using same volume with different fsgroups. If `CSIMigrationAzureFile` is enabled, using same volume with different fsgroups won't be supported at all.

### **azureFile CSI migration complete**

FEATURE STATE: Kubernetes v1.21 [alpha]

To disable the `azureFile` storage plugin from being loaded by the controller manager and the kubelet, set the `InTreePluginAzureFileUnregister` flag to true.

### **cephfs (deprecated)**

FEATURE STATE: Kubernetes v1.28 [deprecated]

#### **Note:**

The Kubernetes project suggests that you use the [CephFS CSI](#) third party storage driver instead.

A `cephfs` volume allows an existing CephFS volume to be mounted into your Pod. Unlike `emptyDir`, which is erased when a pod is removed, the contents of a `cephfs` volume are preserved and the volume is merely unmounted. This means that a `cephfs` volume can be pre-populated with data, and that data can be shared between pods. The `cephfs` volume can be mounted by multiple writers simultaneously.

#### **Note:**

You must have your own Ceph server running with the share exported before you can use it.

See the [CephFS example](#) for more details.

### **cinder (deprecated)**

In Kubernetes 1.30, all operations for the in-tree `cinder` type are redirected to the `cinder.csi.openstack.org` [CSI](#) driver.

The OpenStack Cinder in-tree storage driver was deprecated in the Kubernetes v1.11 release and then removed entirely in the v1.26 release.

The Kubernetes project suggests that you use the [OpenStack Cinder](#) third party storage driver instead.

## configMap

A [ConfigMap](#) provides a way to inject configuration data into pods. The data stored in a ConfigMap can be referenced in a volume of type configMap and then consumed by containerized applications running in a pod.

When referencing a ConfigMap, you provide the name of the ConfigMap in the volume. You can customize the path to use for a specific entry in the ConfigMap. The following configuration shows how to mount the log-config ConfigMap onto a Pod called configmap-pod:

```
apiVersion: v1
kind: Pod
metadata:
 name: configmap-pod
spec:
 containers:
 - name: test
 image: busybox:1.28
 command: ['sh', '-c', 'echo "The app is running!" && tail -f /dev/null']
 volumeMounts:
 - name: config-vol
 mountPath: /etc/config
 volumes:
 - name: config-vol
 configMap:
 name: log-config
 items:
 - key: log_level
 path: log_level
```

The log-config ConfigMap is mounted as a volume, and all contents stored in its log\_level entry are mounted into the Pod at path /etc/config/log\_level. Note that this path is derived from the volume's mountPath and the path keyed with log\_level.

### Note:

- You must [create a ConfigMap](#) before you can use it.
- A ConfigMap is always mounted as readOnly.
- A container using a ConfigMap as a [subPath](#) volume mount will not receive ConfigMap updates.
- Text data is exposed as files using the UTF-8 character encoding. For other character encodings, use binaryData.

## downwardAPI

A downwardAPI volume makes [downward API](#) data available to applications. Within the volume, you can find the exposed data as read-only files in plain text format.

**Note:**

A container using the downward API as a [subPath](#) volume mount does not receive updates when field values change.

See [Expose Pod Information to Containers Through Files](#) to learn more.

**emptyDir**

For a Pod that defines an `emptyDir` volume, the volume is created when the Pod is assigned to a node. As the name says, the `emptyDir` volume is initially empty. All containers in the Pod can read and write the same files in the `emptyDir` volume, though that volume can be mounted at the same or different paths in each container. When a Pod is removed from a node for any reason, the data in the `emptyDir` is deleted permanently.

**Note:**

A container crashing does *not* remove a Pod from a node. The data in an `emptyDir` volume is safe across container crashes.

Some uses for an `emptyDir` are:

- scratch space, such as for a disk-based merge sort
- checkpointing a long computation for recovery from crashes
- holding files that a content-manager container fetches while a webserver container serves the data

The `emptyDir.medium` field controls where `emptyDir` volumes are stored. By default `emptyDir` volumes are stored on whatever medium that backs the node such as disk, SSD, or network storage, depending on your environment. If you set the `emptyDir.medium` field to "Memory", Kubernetes mounts a tmpfs (RAM-backed filesystem) for you instead. While tmpfs is very fast be aware that, unlike disks, files you write count against the memory limit of the container that wrote them.

A size limit can be specified for the default medium, which limits the capacity of the `emptyDir` volume. The storage is allocated from [node ephemeral storage](#). If that is filled up from another source (for example, log files or image overlays), the `emptyDir` may run out of capacity before this limit.

**Note:**

If the `SizeMemoryBackedVolumes` [feature gate](#) is enabled, you can specify a size for memory backed volumes. If no size is specified, memory backed volumes are sized to node allocatable memory.

**emptyDir configuration example**

```
apiVersion: v1
kind: Pod
metadata:
 name: test-pd
spec:
```

```
containers:
- image: registry.k8s.io/test-webserver
 name: test-container
 volumeMounts:
 - mountPath: /cache
 name: cache-volume
volumes:
- name: cache-volume
 emptyDir:
 sizeLimit: 500Mi
```

## fc (fibre channel)

An fc volume type allows an existing fibre channel block storage volume to mount in a Pod. You can specify single or multiple target world wide names (WWNs) using the parameter `targetWWNs` in your Volume configuration. If multiple WWNs are specified, `targetWWNs` expect that those WWNs are from multi-path connections.

### Note:

You must configure FC SAN Zoning to allocate and mask those LUNs (volumes) to the target WWNs beforehand so that Kubernetes hosts can access them.

See the [fibre channel example](#) for more details.

## gcePersistentDisk (deprecated)

In Kubernetes 1.30, all operations for the in-tree `gcePersistentDisk` type are redirected to the `pd.csi.storage.gke.io` [CSI](#) driver.

The `gcePersistentDisk` in-tree storage driver was deprecated in the Kubernetes v1.17 release and then removed entirely in the v1.28 release.

The Kubernetes project suggests that you use the [Google Compute Engine Persistent Disk CSI](#) third party storage driver instead.

## gitRepo (deprecated)

### Warning:

The `gitRepo` volume type is deprecated.

To provision a Pod that has a Git repository mounted, you can mount an [emptyDir](#) volume into an [init container](#) that clones the repo using Git, then mount the [EmptyDir](#) into the Pod's container.

---

You can restrict the use of `gitRepo` volumes in your cluster using [policies](#) such as [ValidatingAdmissionPolicy](#). You can use the following Common Expression Language (CEL) expression as part of a policy to reject use of `gitRepo` volumes: `has(object.spec.volumes) || ! object.spec.volumes.exists(v, has(v.gitRepo))`.

A `gitRepo` volume is an example of a volume plugin. This plugin mounts an empty directory and clones a git repository into this directory for your Pod to use.

Here is an example of a `gitRepo` volume:

```
apiVersion: v1
kind: Pod
metadata:
 name: server
spec:
 containers:
 - image: nginx
 name: nginx
 volumeMounts:
 - mountPath: /mypath
 name: git-volume
 volumes:
 - name: git-volume
 gitRepo:
 repository: "git@somewhere:me/my-git-repository.git"
 revision: "22f1d8406d464b0c0874075539c1f2e96c253775"
```

## glusterfs (removed)

Kubernetes 1.30 does not include a `glusterfs` volume type.

The GlusterFS in-tree storage driver was deprecated in the Kubernetes v1.25 release and then removed entirely in the v1.26 release.

## hostPath

A `hostPath` volume mounts a file or directory from the host node's filesystem into your Pod. This is not something that most Pods will need, but it offers a powerful escape hatch for some applications.

### Warning:

Using the `hostPath` volume type presents many security risks. If you can avoid using a `hostPath` volume, you should. For example, define a [local PersistentVolume](#), and use that instead.

If you are restricting access to specific directories on the node using admission-time validation, that restriction is only effective when you additionally require that any mounts of that `hostPath` volume are **read only**. If you allow a read-write mount of any host path by an untrusted Pod, the containers in that Pod may be able to subvert the read-write host mount.

---

Take care when using `hostPath` volumes, whether these are mounted as read-only or as read-write, because:

- Access to the host filesystem can expose privileged system credentials (such as for the kubelet) or privileged APIs (such as the container runtime socket), that can be used for container escape or to attack other parts of the cluster.

- Pods with identical configuration (such as created from a PodTemplate) may behave differently on different nodes due to different files on the nodes.

Some uses for a hostPath are:

- running a container that needs access to node-level system components (such as a container that transfers system logs to a central location, accessing those logs using a read-only mount of /var/log)
- making a configuration file stored on the host system available read-only to a [static pod](#); unlike normal Pods, static Pods cannot access ConfigMaps

## hostPath volume types

In addition to the required path property, you can optionally specify a type for a hostPath volume.

The available values for type are:

| Value             | Behavior                                                                                                                                                               |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ""                | Empty string (default) is for backward compatibility, which means that no checks will be performed before mounting the hostPath volume.                                |
| DirectoryOrCreate | If nothing exists at the given path, an empty directory will be created there as needed with permission set to 0755, having the same group and ownership with Kubelet. |
| Directory         | A directory must exist at the given path                                                                                                                               |
| FileOrCreate      | If nothing exists at the given path, an empty file will be created there as needed with permission set to 0644, having the same group and ownership with Kubelet.      |
| File              | A file must exist at the given path                                                                                                                                    |
| Socket            | A UNIX socket must exist at the given path                                                                                                                             |
| CharDevice        | (Linux nodes only) A character device must exist at the given path                                                                                                     |
| BlockDevice       | (Linux nodes only) A block device must exist at the given path                                                                                                         |

### Caution:

The FileOrCreate mode does **not** create the parent directory of the file. If the parent directory of the mounted file does not exist, the pod fails to start. To ensure that this mode works, you can try to mount directories and files separately, as shown in the [FileOrCreate example](#) for hostPath.

Some files or directories created on the underlying hosts might only be accessible by root. You then either need to run your process as root in a [privileged container](#) or modify the file permissions on the host to be able to read from (or write to) a hostPath volume.

## hostPath configuration example

- [Linux node](#)
- [Windows node](#)

---

```

This manifest mounts /data/foo on the host as /foo inside the
single container that runs within the hostpath-example-linux Pod.
#
The mount into the container is read-only.
apiVersion: v1
kind: Pod
metadata:
 name: hostpath-example-linux
spec:
 os: { name: linux }
 nodeSelector:
 kubernetes.io/os: linux
 containers:
 - name: example-container
 image: registry.k8s.io/test-webserver
 volumeMounts:
 - mountPath: /foo
 name: example-volume
 readOnly: true
 volumes:
 - name: example-volume
 # mount /data/foo, but only if that directory already exists
 hostPath:
 path: /data/foo # directory location on host
 type: Directory # this field is optional

```

```

This manifest mounts C:\Data\foo on the host as C:\foo, inside the
single container that runs within the hostpath-example-windows Pod.
#
The mount into the container is read-only.
apiVersion: v1
kind: Pod
metadata:
 name: hostpath-example-windows
spec:
 os: { name: windows }
 nodeSelector:
 kubernetes.io/os: windows
 containers:
 - name: example-container
 image: microsoft/windowsservercore:1709
 volumeMounts:
 - name: example-volume
 mountPath: "C:\\foo"
 readOnly: true
 volumes:
 - name: example-volume
 # mount C:\Data\foo from the host, but only if that directory already exists
 hostPath:
 path: "C:\\Data\\foo" # directory location on host
 type: Directory # this field is optional

```



## hostPath FileOrCreate configuration example

The following manifest defines a Pod that mounts `/var/local/aaa` inside the single container in the Pod. If the node does not already have a path `/var/local/aaa`, the kubelet creates it as a directory and then mounts it into the Pod.

If `/var/local/aaa` already exists but is not a directory, the Pod fails. Additionally, the kubelet attempts to make a file named `/var/local/aaa/1.txt` inside that directory (as seen from the host); if something already exists at that path and isn't a regular file, the Pod fails.

Here's the example manifest:

```
apiVersion: v1
kind: Pod
metadata:
 name: test-webserver
spec:
 os: { name: linux }
 nodeSelector:
 kubernetes.io/os: linux
 containers:
 - name: test-webserver
 image: registry.k8s.io/test-webserver:latest
 volumeMounts:
 - mountPath: /var/local/aaa
 name: mydir
 - mountPath: /var/local/aaa/1.txt
 name: myfile
 volumes:
 - name: mydir
 hostPath:
 # Ensure the file directory is created.
 path: /var/local/aaa
 type: DirectoryOrCreate
 - name: myfile
 hostPath:
 path: /var/local/aaa/1.txt
 type: FileOrCreate
```

## iscsi

An `iscsi` volume allows an existing iSCSI (SCSI over IP) volume to be mounted into your Pod. Unlike `emptyDir`, which is erased when a Pod is removed, the contents of an `iscsi` volume are preserved and the volume is merely unmounted. This means that an `iscsi` volume can be pre-populated with data, and that data can be shared between pods.

### Note:

You must have your own iSCSI server running with the volume created before you can use it.

A feature of iSCSI is that it can be mounted as read-only by multiple consumers simultaneously. This means that you can pre-populate a volume with your dataset and then serve it in parallel

from as many Pods as you need. Unfortunately, iSCSI volumes can only be mounted by a single consumer in read-write mode. Simultaneous writers are not allowed.

See the [iSCSI example](#) for more details.

## local

A local volume represents a mounted local storage device such as a disk, partition or directory.

Local volumes can only be used as a statically created PersistentVolume. Dynamic provisioning is not supported.

Compared to hostPath volumes, local volumes are used in a durable and portable manner without manually scheduling pods to nodes. The system is aware of the volume's node constraints by looking at the node affinity on the PersistentVolume.

However, local volumes are subject to the availability of the underlying node and are not suitable for all applications. If a node becomes unhealthy, then the local volume becomes inaccessible by the pod. The pod using this volume is unable to run. Applications using local volumes must be able to tolerate this reduced availability, as well as potential data loss, depending on the durability characteristics of the underlying disk.

The following example shows a PersistentVolume using a local volume and nodeAffinity:

```
apiVersion: v1
kind: PersistentVolume
metadata:
 name: example-pv
spec:
 capacity:
 storage: 100Gi
 volumeMode: Filesystem
 accessModes:
 - ReadWriteOnce
 persistentVolumeReclaimPolicy: Delete
 storageClassName: local-storage
 local:
 path: /mnt/disks/ssd1
 nodeAffinity:
 required:
 nodeSelectorTerms:
 - matchExpressions:
 - key: kubernetes.io/hostname
 operator: In
 values:
 - example-node
```

You must set a PersistentVolume nodeAffinity when using local volumes. The Kubernetes scheduler uses the PersistentVolume nodeAffinity to schedule these Pods to the correct node.

PersistentVolume volumeMode can be set to "Block" (instead of the default value "Filesystem") to expose the local volume as a raw block device.

When using local volumes, it is recommended to create a StorageClass with volumeBindingMode set to WaitForFirstConsumer. For more details, see the local [StorageClass](#) example. Delaying volume binding ensures that the PersistentVolumeClaim binding decision will also be evaluated with any other node constraints the Pod may have, such as node resource requirements, node selectors, Pod affinity, and Pod anti-affinity.

An external static provisioner can be run separately for improved management of the local volume lifecycle. Note that this provisioner does not support dynamic provisioning yet. For an example on how to run an external local provisioner, see the [local volume provisioner user guide](#).

**Note:**

The local PersistentVolume requires manual cleanup and deletion by the user if the external static provisioner is not used to manage the volume lifecycle.

## nfs

An nfs volume allows an existing NFS (Network File System) share to be mounted into a Pod. Unlike emptyDir, which is erased when a Pod is removed, the contents of an nfs volume are preserved and the volume is merely unmounted. This means that an NFS volume can be pre-populated with data, and that data can be shared between pods. NFS can be mounted by multiple writers simultaneously.

```
apiVersion: v1
kind: Pod
metadata:
 name: test-pd
spec:
 containers:
 - image: registry.k8s.io/test-webserver
 name: test-container
 volumeMounts:
 - mountPath: /my-nfs-data
 name: test-volume
 volumes:
 - name: test-volume
 nfs:
 server: my-nfs-server.example.com
 path: /my-nfs-volume
 readOnly: true
```

**Note:**

You must have your own NFS server running with the share exported before you can use it.

Also note that you can't specify NFS mount options in a Pod spec. You can either set mount options server-side or use [/etc/nfsmount.conf](#). You can also mount NFS volumes via PersistentVolumes which do allow you to set mount options.

See the [NFS example](#) for an example of mounting NFS volumes with PersistentVolumes.

## **persistentVolumeClaim**

A `persistentVolumeClaim` volume is used to mount a [PersistentVolume](#) into a Pod. `PersistentVolumeClaims` are a way for users to "claim" durable storage (such as an iSCSI volume) without knowing the details of the particular cloud environment.

See the information about [PersistentVolumes](#) for more details.

## **portworxVolume (deprecated)**

FEATURE STATE: Kubernetes v1.25 [deprecated]

A `portworxVolume` is an elastic block storage layer that runs hyperconverged with Kubernetes. [Portworx](#) fingerprints storage in a server, tiers based on capabilities, and aggregates capacity across multiple servers. Portworx runs in-guest in virtual machines or on bare metal Linux nodes.

A `portworxVolume` can be dynamically created through Kubernetes or it can also be pre-provisioned and referenced inside a Pod. Here is an example Pod referencing a pre-provisioned Portworx volume:

```
apiVersion: v1
kind: Pod
metadata:
 name: test-portworx-volume-pod
spec:
 containers:
 - image: registry.k8s.io/test-webserver
 name: test-container
 volumeMounts:
 - mountPath: /mnt
 name: pxvol
 volumes:
 - name: pxvol
 # This Portworx volume must already exist.
 portworxVolume:
 volumeID: "pxvol"
 fsType: "<fs-type>"
```

### **Note:**

Make sure you have an existing `PortworxVolume` with name `pxvol` before using it in the Pod.

For more details, see the [Portworx volume](#) examples.

## **Portworx CSI migration**

FEATURE STATE: Kubernetes v1.25 [beta]

The `CSIMigration` feature for Portworx has been added but disabled by default in Kubernetes 1.23 since it's in alpha state. It has been beta now since v1.25 but it is still turned off by default. It redirects all plugin operations from the existing in-tree plugin to the `pxd.portworx.com`

Container Storage Interface (CSI) Driver. [Portworx CSI Driver](#) must be installed on the cluster. To enable the feature, set CSIMigrationPortworx=true in kube-controller-manager and kubelet.

## projected

A projected volume maps several existing volume sources into the same directory. For more details, see [projected volumes](#).

## rbd

FEATURE STATE: Kubernetes v1.28 [deprecated]

### Note:

The Kubernetes project suggests that you use the [Ceph CSI](#) third party storage driver instead, in RBD mode.

An rbd volume allows a [Rados Block Device](#) (RBD) volume to mount into your Pod. Unlike emptyDir, which is erased when a pod is removed, the contents of an rbd volume are preserved and the volume is unmounted. This means that a RBD volume can be pre-populated with data, and that data can be shared between pods.

### Note:

You must have a Ceph installation running before you can use RBD.

A feature of RBD is that it can be mounted as read-only by multiple consumers simultaneously. This means that you can pre-populate a volume with your dataset and then serve it in parallel from as many pods as you need. Unfortunately, RBD volumes can only be mounted by a single consumer in read-write mode. Simultaneous writers are not allowed.

See the [RBD example](#) for more details.

## RBD CSI migration

FEATURE STATE: Kubernetes v1.28 [deprecated]

The CSIMigration feature for RBD, when enabled, redirects all plugin operations from the existing in-tree plugin to the rbd.csi.ceph.com [CSI](#) driver. In order to use this feature, the [Ceph CSI driver](#) must be installed on the cluster and the CSIMigrationRBD [feature gate](#) must be enabled. (Note that the csiMigrationRBD flag has been removed and replaced with CSIMigrationRBD in release v1.24)

### Note:

As a Kubernetes cluster operator that administers storage, here are the prerequisites that you must complete before you attempt migration to the RBD CSI driver:

- You must install the Ceph CSI driver (rbd.csi.ceph.com), v3.5.0 or above, into your Kubernetes cluster.
- considering the clusterID field is a required parameter for CSI driver for its operations, but in-tree StorageClass has monitors field as a required parameter, a Kubernetes storage

admin has to create a clusterID based on the monitors hash ( ex:#echo -n '<monitors\_string>' | md5sum) in the CSI config map and keep the monitors under this clusterID configuration.

- Also, if the value of adminId in the in-tree Storageclass is different from admin, the adminSecretName mentioned in the in-tree Storageclass has to be patched with the base64 value of the adminId parameter value, otherwise this step can be skipped.

## secret

A secret volume is used to pass sensitive information, such as passwords, to Pods. You can store secrets in the Kubernetes API and mount them as files for use by pods without coupling to Kubernetes directly. secret volumes are backed by tmpfs (a RAM-backed filesystem) so they are never written to non-volatile storage.

### Note:

- You must create a Secret in the Kubernetes API before you can use it.
- A Secret is always mounted as readOnly.
- A container using a Secret as a [subPath](#) volume mount will not receive Secret updates.

For more details, see [Configuring Secrets](#).

## vsphereVolume (deprecated)

### Note:

The Kubernetes project recommends using the [vSphere CSI](#) out-of-tree storage driver instead.

A vsphereVolume is used to mount a vSphere VMDK volume into your Pod. The contents of a volume are preserved when it is unmounted. It supports both VMFS and VSAN datastore.

For more information, see the [vSphere volume](#) examples.

## vSphere CSI migration

FEATURE STATE: Kubernetes v1.26 [stable]

In Kubernetes 1.30, all operations for the in-tree vsphereVolume type are redirected to the csi.vsphere.vmware.com [CSI](#) driver.

[vSphere CSI driver](#) must be installed on the cluster. You can find additional advice on how to migrate in-tree vsphereVolume in VMware's documentation page [Migrating In-Tree vSphere Volumes to vSphere Container Storage lug-in](#). If vSphere CSI Driver is not installed volume operations can not be performed on the PV created with the in-tree vsphereVolume type.

You must run vSphere 7.0u2 or later in order to migrate to the vSphere CSI driver.

If you are running a version of Kubernetes other than v1.30, consult the documentation for that version of Kubernetes.

**Note:**

The following StorageClass parameters from the built-in `vsphereVolume` plugin are not supported by the vSphere CSI driver:

- `diskformat`
- `hostfailurestotolerate`
- `forceprovisioning`
- `cachereservation`
- `diskstripes`
- `objectspacereservation`
- `iopslimit`

Existing volumes created using these parameters will be migrated to the vSphere CSI driver, but new volumes created by the vSphere CSI driver will not be honoring these parameters.

**vSphere CSI migration complete**

FEATURE STATE: Kubernetes v1.19 [beta]

To turn off the `vsphereVolume` plugin from being loaded by the controller manager and the kubelet, you need to set `InTreePluginvSphereUnregister` feature flag to true. You must install a `csi.vsphere.vmware.com` [CSI](#) driver on all worker nodes.

## Using subPath

Sometimes, it is useful to share one volume for multiple uses in a single pod. The `volumeMounts[*].subPath` property specifies a sub-path inside the referenced volume instead of its root.

The following example shows how to configure a Pod with a LAMP stack (Linux Apache MySQL PHP) using a single, shared volume. This sample `subPath` configuration is not recommended for production use.

The PHP application's code and assets map to the volume's `html` folder and the MySQL database is stored in the volume's `mysql` folder. For example:

```
apiVersion: v1
kind: Pod
metadata:
 name: my-lamp-site
spec:
 containers:
 - name: mysql
 image: mysql
 env:
 - name: MYSQL_ROOT_PASSWORD
 value: "rootpasswd"
 volumeMounts:
 - mountPath: /var/lib/mysql
 name: site-data
 subPath: mysql
```

```
- name: php
 image: php:7.0-apache
 volumeMounts:
 - mountPath: /var/www/html
 name: site-data
 subPath: html
 volumes:
 - name: site-data
 persistentVolumeClaim:
 claimName: my-lamp-site-data
```

## Using subPath with expanded environment variables

FEATURE STATE: Kubernetes v1.17 [stable]

Use the `subPathExpr` field to construct `subPath` directory names from downward API environment variables. The `subPath` and `subPathExpr` properties are mutually exclusive.

In this example, a Pod uses `subPathExpr` to create a directory `pod1` within the `hostPath` volume `/var/log/pods`. The `hostPath` volume takes the Pod name from the downwardAPI. The host directory `/var/log/pods/pod1` is mounted at `/logs` in the container.

```
apiVersion: v1
kind: Pod
metadata:
 name: pod1
spec:
 containers:
 - name: container1
 env:
 - name: POD_NAME
 valueFrom:
 fieldRef:
 apiVersion: v1
 fieldPath: metadata.name
 image: busybox:1.28
 command: ["sh", "-c", "while [true]; do echo 'Hello'; sleep 10; done | tee -a /logs/hello.txt"]
 volumeMounts:
 - name: workdir1
 mountPath: /logs
 # The variable expansion uses round brackets (not curly brackets).
 subPathExpr: $(POD_NAME)
 restartPolicy: Never
 volumes:
 - name: workdir1
 hostPath:
 path: /var/log/pods
```

## Resources

The storage media (such as Disk or SSD) of an `emptyDir` volume is determined by the medium of the filesystem holding the kubelet root dir (typically `/var/lib/kubelet`). There is no limit on



how much space an `emptyDir` or `hostPath` volume can consume, and no isolation between containers or between pods.

To learn about requesting space using a resource specification, see [how to manage resources](#).

## Out-of-tree volume plugins

The out-of-tree volume plugins include [Container Storage Interface](#) (CSI), and also FlexVolume (which is deprecated). These plugins enable storage vendors to create custom storage plugins without adding their plugin source code to the Kubernetes repository.

Previously, all volume plugins were "in-tree". The "in-tree" plugins were built, linked, compiled, and shipped with the core Kubernetes binaries. This meant that adding a new storage system to Kubernetes (a volume plugin) required checking code into the core Kubernetes code repository.

Both CSI and FlexVolume allow volume plugins to be developed independent of the Kubernetes code base, and deployed (installed) on Kubernetes clusters as extensions.

For storage vendors looking to create an out-of-tree volume plugin, please refer to the [volume plugin FAQ](#).

### csi

[Container Storage Interface](#) (CSI) defines a standard interface for container orchestration systems (like Kubernetes) to expose arbitrary storage systems to their container workloads.

Please read the [CSI design proposal](#) for more information.

#### Note:

Support for CSI spec versions 0.2 and 0.3 are deprecated in Kubernetes v1.13 and will be removed in a future release.

#### Note:

CSI drivers may not be compatible across all Kubernetes releases. Please check the specific CSI driver's documentation for supported deployments steps for each Kubernetes release and a compatibility matrix.

Once a CSI compatible volume driver is deployed on a Kubernetes cluster, users may use the `csi` volume type to attach or mount the volumes exposed by the CSI driver.

A `csi` volume can be used in a Pod in three different ways:

- through a reference to a [PersistentVolumeClaim](#)
- with a [generic ephemeral volume](#)
- with a [CSI ephemeral volume](#) if the driver supports that

The following fields are available to storage administrators to configure a CSI persistent volume:

- **driver:** A string value that specifies the name of the volume driver to use. This value must correspond to the value returned in the `GetPluginInfoResponse` by the CSI driver as defined in the [CSI spec](#). It is used by Kubernetes to identify which CSI driver to call out to, and by CSI driver components to identify which PV objects belong to the CSI driver.
- **volumeHandle:** A string value that uniquely identifies the volume. This value must correspond to the value returned in the `volume.id` field of the `CreateVolumeResponse` by the CSI driver as defined in the [CSI spec](#). The value is passed as `volume_id` on all calls to the CSI volume driver when referencing the volume.
- **readOnly:** An optional boolean value indicating whether the volume is to be "ControllerPublished" (attached) as read only. Default is false. This value is passed to the CSI driver via the `readonly` field in the `ControllerPublishVolumeRequest`.
- **fsType:** If the PV's `VolumeMode` is `Filesystem` then this field may be used to specify the filesystem that should be used to mount the volume. If the volume has not been formatted and formatting is supported, this value will be used to format the volume. This value is passed to the CSI driver via the `VolumeCapability` field of `ControllerPublishVolumeRequest`, `NodeStageVolumeRequest`, and `NodePublishVolumeRequest`.
- **volumeAttributes:** A map of string to string that specifies static properties of a volume. This map must correspond to the map returned in the `volume.attributes` field of the `CreateVolumeResponse` by the CSI driver as defined in the [CSI spec](#). The map is passed to the CSI driver via the `volume_context` field in the `ControllerPublishVolumeRequest`, `NodeStageVolumeRequest`, and `NodePublishVolumeRequest`.
- **controllerPublishSecretRef:** A reference to the secret object containing sensitive information to pass to the CSI driver to complete the CSI `ControllerPublishVolume` and `ControllerUnpublishVolume` calls. This field is optional, and may be empty if no secret is required. If the Secret contains more than one secret, all secrets are passed.
- **nodeExpandSecretRef:** A reference to the secret containing sensitive information to pass to the CSI driver to complete the CSI `NodeExpandVolume` call. This field is optional, and may be empty if no secret is required. If the object contains more than one secret, all secrets are passed. When you have configured secret data for node-initiated volume expansion, the kubelet passes that data via the `NodeExpandVolume()` call to the CSI driver. In order to use the `nodeExpandSecretRef` field, your cluster should be running Kubernetes version 1.25 or later.
- If you are running Kubernetes Version 1.25 or 1.26, you must enable the [feature gate](#) named `CSINodeExpandSecret` for each kube-apiserver and for the kubelet on every node. In Kubernetes version 1.27 this feature has been enabled by default and no explicit enablement of the feature gate is required. You must also be using a CSI driver that supports or requires secret data during node-initiated storage resize operations.
- **nodePublishSecretRef:** A reference to the secret object containing sensitive information to pass to the CSI driver to complete the CSI `NodePublishVolume` call. This field is optional, and may be empty if no secret is required. If the secret object contains more than one secret, all secrets are passed.
- **nodeStageSecretRef:** A reference to the secret object containing sensitive information to pass to the CSI driver to complete the CSI `NodeStageVolume` call. This field is optional, and may be empty if no secret is required. If the Secret contains more than one secret, all secrets are passed.

## CSI raw block volume support

FEATURE STATE: Kubernetes v1.18 [stable]

Vendors with external CSI drivers can implement raw block volume support in Kubernetes workloads.

You can set up your [PersistentVolume/PersistentVolumeClaim with raw block volume support](#) as usual, without any CSI specific changes.

## CSI ephemeral volumes

FEATURE STATE: Kubernetes v1.25 [stable]

You can directly configure CSI volumes within the Pod specification. Volumes specified in this way are ephemeral and do not persist across pod restarts. See [Ephemeral Volumes](#) for more information.

For more information on how to develop a CSI driver, refer to the [kubernetes-csi documentation](#)

## Windows CSI proxy

FEATURE STATE: Kubernetes v1.22 [stable]

CSI node plugins need to perform various privileged operations like scanning of disk devices and mounting of file systems. These operations differ for each host operating system. For Linux worker nodes, containerized CSI node plugins are typically deployed as privileged containers. For Windows worker nodes, privileged operations for containerized CSI node plugins is supported using [csi-proxy](#), a community-managed, stand-alone binary that needs to be pre-installed on each Windows node.

For more details, refer to the deployment guide of the CSI plugin you wish to deploy.

## Migrating to CSI drivers from in-tree plugins

FEATURE STATE: Kubernetes v1.25 [stable]

The CSIMigration feature directs operations against existing in-tree plugins to corresponding CSI plugins (which are expected to be installed and configured). As a result, operators do not have to make any configuration changes to existing Storage Classes, PersistentVolumes or PersistentVolumeClaims (referring to in-tree plugins) when transitioning to a CSI driver that supersedes an in-tree plugin.

### Note:

Existing PVs created by a in-tree volume plugin can still be used in the future without any configuration changes, even after the migration to CSI is completed for that volume type, and even after you upgrade to a version of Kubernetes that doesn't have compiled-in support for that kind of storage.

As part of that migration, you - or another cluster administrator - **must** have installed and configured the appropriate CSI driver for that storage. The core of Kubernetes does not install that software for you.

---

After that migration, you can also define new PVCs and PVs that refer to the legacy, built-in storage integrations. Provided you have the appropriate CSI driver installed and configured, the PV creation continues to work, even for brand new volumes. The actual storage management now happens through the CSI driver.

The operations and features that are supported include: provisioning/delete, attach/detach, mount/unmount and resizing of volumes.

In-tree plugins that support CSIMigration and have a corresponding CSI driver implemented are listed in [Types of Volumes](#).

The following in-tree plugins support persistent storage on Windows nodes:

- [azureFile](#)
- [gcePersistentDisk](#)
- [vsphereVolume](#)

## **flexVolume (deprecated)**

FEATURE STATE: Kubernetes v1.23 [deprecated]

FlexVolume is an out-of-tree plugin interface that uses an exec-based model to interface with storage drivers. The FlexVolume driver binaries must be installed in a pre-defined volume plugin path on each node and in some cases the control plane nodes as well.

Pods interact with FlexVolume drivers through the flexVolume in-tree volume plugin. For more details, see the FlexVolume [README](#) document.

The following FlexVolume [plugins](#), deployed as PowerShell scripts on the host, support Windows nodes:

- [SMB](#)
- [iSCSI](#)

### **Note:**

FlexVolume is deprecated. Using an out-of-tree CSI driver is the recommended way to integrate external storage with Kubernetes.

Maintainers of FlexVolume driver should implement a CSI Driver and help to migrate users of FlexVolume drivers to CSI. Users of FlexVolume should move their workloads to use the equivalent CSI Driver.

## **Mount propagation**

Mount propagation allows for sharing volumes mounted by a container to other containers in the same pod, or even to other pods on the same node.

Mount propagation of a volume is controlled by the `mountPropagation` field in `containers[*].volumeMounts`. Its values are:

- **None** - This volume mount will not receive any subsequent mounts that are mounted to this volume or any of its subdirectories by the host. In similar fashion, no mounts created by the container will be visible on the host. This is the default mode.

This mode is equal to `rprivate` mount propagation as described in [mount\(8\)](#)

However, the CRI runtime may choose `rslave` mount propagation (i.e., `HostToContainer`) instead, when `rprivate` propagation is not applicable. `cri-dockerd` (Docker) is known to choose `rslave` mount propagation when the mount source contains the Docker daemon's root directory (`/var/lib/docker`).

- **HostToContainer** - This volume mount will receive all subsequent mounts that are mounted to this volume or any of its subdirectories.

In other words, if the host mounts anything inside the volume mount, the container will see it mounted there.

Similarly, if any Pod with `Bidirectional` mount propagation to the same volume mounts anything there, the container with `HostToContainer` mount propagation will see it.

This mode is equal to `rslave` mount propagation as described in the [mount\(8\)](#)

- **Bidirectional** - This volume mount behaves the same the `HostToContainer` mount. In addition, all volume mounts created by the container will be propagated back to the host and to all containers of all pods that use the same volume.

A typical use case for this mode is a Pod with a `FlexVolume` or `CSI` driver or a Pod that needs to mount something on the host using a `hostPath` volume.

This mode is equal to `rshared` mount propagation as described in the [mount\(8\)](#)

### **Warning:**

`Bidirectional` mount propagation can be dangerous. It can damage the host operating system and therefore it is allowed only in privileged containers. Familiarity with Linux kernel behavior is strongly recommended. In addition, any volume mounts created by containers in pods must be destroyed (unmounted) by the containers on termination.

## **Read-only mounts**

A mount can be made read-only by setting the `.spec.containers[].volumeMounts[].readOnly` field to `true`. This does not make the volume itself read-only, but that specific container will not be able to write to it. Other containers in the Pod may mount the same volume as read-write.

On Linux, read-only mounts are not recursively read-only by default. For example, consider a Pod which mounts the hosts `/mnt` as a `hostPath` volume. If there is another filesystem mounted read-write on `/mnt/<SUBMOUNT>` (such as `tmpfs`, `NFS`, or `USB` storage), the volume mounted into the container(s) will also have a writeable `/mnt/<SUBMOUNT>`, even if the mount itself was specified as read-only.

## Recursive read-only mounts

FEATURE STATE: Kubernetes v1.30 [alpha]

Recursive read-only mounts can be enabled by setting the RecursiveReadOnlyMounts [feature gate](#) for kubelet and kube-apiserver, and setting the `.spec.containers[].volumeMounts[].recursiveReadOnly` field for a pod.

The allowed values are:

- Disabled (default): no effect.
- Enabled: makes the mount recursively read-only. Needs all the following requirements to be satisfied:
  - `readOnly` is set to `true`
  - `mountPropagation` is unset, or, set to `None`
  - The host is running with Linux kernel v5.12 or later
  - The [CRI-level](#) container runtime supports recursive read-only mounts
  - The OCI-level container runtime supports recursive read-only mounts. It will fail if any of these is not true.
- IfPossible: attempts to apply Enabled, and falls back to Disabled if the feature is not supported by the kernel or the runtime class.

Example:

[storage/rro.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: rro
spec:
 volumes:
 - name: mnt
 hostPath:
 # tmpfs is mounted on /mnt/tmpfs
 path: /mnt
 containers:
 - name: busybox
 image: busybox
 args: ["sleep", "infinity"]
 volumeMounts:
 # /mnt-rro/tmpfs is not writable
 - name: mnt
 mountPath: /mnt-rro
 readOnly: true
 mountPropagation: None
 recursiveReadOnly: Enabled
 # /mnt-ro/tmpfs is writable
 - name: mnt
 mountPath: /mnt-ro
 readOnly: true
```

```
/mnt-rw/tmpfs is writable
- name: mnt
 mountPath: /mnt-rw
```

When this property is recognized by kubelet and kube-apiserver, the `.status.containerStatuses[].volumeMounts[].recursiveReadOnly` field is set to either `Enabled` or `Disabled`.

## Implementations

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

The following container runtimes are known to support recursive read-only mounts.

CRI-level:

- [containerd](#), since v2.0

OCI-level:

- [runc](#), since v1.1
- [crun](#), since v1.8.6

## What's next

Follow an example of [deploying WordPress and MySQL with Persistent Volumes](#).

# Persistent Volumes

This document describes *persistent volumes* in Kubernetes. Familiarity with [volumes](#), [StorageClasses](#) and [VolumeAttributesClasses](#) is suggested.

## Introduction

Managing storage is a distinct problem from managing compute instances. The PersistentVolume subsystem provides an API for users and administrators that abstracts details of how storage is provided from how it is consumed. To do this, we introduce two new API resources: PersistentVolume and PersistentVolumeClaim.

A *PersistentVolume* (PV) is a piece of storage in the cluster that has been provisioned by an administrator or dynamically provisioned using [Storage Classes](#). It is a resource in the cluster just like a node is a cluster resource. PVs are volume plugins like Volumes, but have a lifecycle independent of any individual Pod that uses the PV. This API object captures the details of the implementation of the storage, be that NFS, iSCSI, or a cloud-provider-specific storage system.

A *PersistentVolumeClaim* (PVC) is a request for storage by a user. It is similar to a Pod. Pods consume node resources and PVCs consume PV resources. Pods can request specific levels of resources (CPU and Memory). Claims can request specific size and access modes (e.g., they can



be mounted `ReadWriteOnce`, `ReadOnlyMany`, `ReadWriteMany`, or `ReadWriteOncePod`, see [AccessModes](#)).

While `PersistentVolumeClaims` allow a user to consume abstract storage resources, it is common that users need `PersistentVolumes` with varying properties, such as performance, for different problems. Cluster administrators need to be able to offer a variety of `PersistentVolumes` that differ in more ways than size and access modes, without exposing users to the details of how those volumes are implemented. For these needs, there is the *StorageClass* resource.

See the [detailed walkthrough with working examples](#).

## Lifecycle of a volume and claim

PVs are resources in the cluster. PVCs are requests for those resources and also act as claim checks to the resource. The interaction between PVs and PVCs follows this lifecycle:

### Provisioning

There are two ways PVs may be provisioned: statically or dynamically.

#### Static

A cluster administrator creates a number of PVs. They carry the details of the real storage, which is available for use by cluster users. They exist in the Kubernetes API and are available for consumption.

#### Dynamic

When none of the static PVs the administrator created match a user's `PersistentVolumeClaim`, the cluster may try to dynamically provision a volume specially for the PVC. This provisioning is based on `StorageClasses`: the PVC must request a [storage class](#) and the administrator must have created and configured that class for dynamic provisioning to occur. Claims that request the class `""` effectively disable dynamic provisioning for themselves.

To enable dynamic storage provisioning based on storage class, the cluster administrator needs to enable the `DefaultStorageClass` [admission controller](#) on the API server. This can be done, for example, by ensuring that `DefaultStorageClass` is among the comma-delimited, ordered list of values for the `--enable-admission-plugins` flag of the API server component. For more information on API server command-line flags, check [kube-apiserver](#) documentation.

### Binding

A user creates, or in the case of dynamic provisioning, has already created, a `PersistentVolumeClaim` with a specific amount of storage requested and with certain access modes. A control loop in the control plane watches for new PVCs, finds a matching PV (if possible), and binds them together. If a PV was dynamically provisioned for a new PVC, the loop will always bind that PV to the PVC. Otherwise, the user will always get at least what they asked for, but the volume may be in excess of what was requested. Once bound, `PersistentVolumeClaim` binds are exclusive, regardless of how they were bound. A PVC to PV



binding is a one-to-one mapping, using a ClaimRef which is a bi-directional binding between the PersistentVolume and the PersistentVolumeClaim.

Claims will remain unbound indefinitely if a matching volume does not exist. Claims will be bound as matching volumes become available. For example, a cluster provisioned with many 50Gi PVs would not match a PVC requesting 100Gi. The PVC can be bound when a 100Gi PV is added to the cluster.

## Using

Pods use claims as volumes. The cluster inspects the claim to find the bound volume and mounts that volume for a Pod. For volumes that support multiple access modes, the user specifies which mode is desired when using their claim as a volume in a Pod.

Once a user has a claim and that claim is bound, the bound PV belongs to the user for as long as they need it. Users schedule Pods and access their claimed PVs by including a persistentVolumeClaim section in a Pod's volumes block. See [Claims As Volumes](#) for more details on this.

## Storage Object in Use Protection

The purpose of the Storage Object in Use Protection feature is to ensure that PersistentVolumeClaims (PVCs) in active use by a Pod and PersistentVolume (PVs) that are bound to PVCs are not removed from the system, as this may result in data loss.

### Note:

PVC is in active use by a Pod when a Pod object exists that is using the PVC.

If a user deletes a PVC in active use by a Pod, the PVC is not removed immediately. PVC removal is postponed until the PVC is no longer actively used by any Pods. Also, if an admin deletes a PV that is bound to a PVC, the PV is not removed immediately. PV removal is postponed until the PV is no longer bound to a PVC.

You can see that a PVC is protected when the PVC's status is Terminating and the Finalizers list includes `kubernetes.io/pvc-protection`:

```
kubectl describe pvc hostpath
Name: hostpath
Namespace: default
StorageClass: example-hostpath
Status: Terminating
Volume:
Labels: <none>
Annotations: volume.beta.kubernetes.io/storage-class=example-hostpath
 volume.beta.kubernetes.io/storage-provisioner=example.com/hostpath
Finalizers: [kubernetes.io/pvc-protection]
...
```

You can see that a PV is protected when the PV's status is Terminating and the Finalizers list includes `kubernetes.io/pv-protection` too:

```
kubectl describe pv task-pv-volume
Name: task-pv-volume
Labels: type=local
Annotations: <none>
Finalizers: [kubernetes.io/pv-protection]
StorageClass: standard
Status: Terminating
Claim:
Reclaim Policy: Delete
Access Modes: RWO
Capacity: 1Gi
Message:
Source:
 Type: HostPath (bare host directory volume)
 Path: /tmp/data
 HostPathType:
Events: <none>
```

## Reclaiming

When a user is done with their volume, they can delete the PVC objects from the API that allows reclamation of the resource. The reclaim policy for a PersistentVolume tells the cluster what to do with the volume after it has been released of its claim. Currently, volumes can either be Retained, Recycled, or Deleted.

### Retain

The Retain reclaim policy allows for manual reclamation of the resource. When the PersistentVolumeClaim is deleted, the PersistentVolume still exists and the volume is considered "released". But it is not yet available for another claim because the previous claimant's data remains on the volume. An administrator can manually reclaim the volume with the following steps.

1. Delete the PersistentVolume. The associated storage asset in external infrastructure still exists after the PV is deleted.
2. Manually clean up the data on the associated storage asset accordingly.
3. Manually delete the associated storage asset.

If you want to reuse the same storage asset, create a new PersistentVolume with the same storage asset definition.

### Delete

For volume plugins that support the Delete reclaim policy, deletion removes both the PersistentVolume object from Kubernetes, as well as the associated storage asset in the external infrastructure. Volumes that were dynamically provisioned inherit the [reclaim policy of their StorageClass](#), which defaults to Delete. The administrator should configure the StorageClass according to users' expectations; otherwise, the PV must be edited or patched after it is created. See [Change the Reclaim Policy of a PersistentVolume](#).

## Recycle

### Warning:

The Recycle reclaim policy is deprecated. Instead, the recommended approach is to use dynamic provisioning.

If supported by the underlying volume plugin, the Recycle reclaim policy performs a basic scrub (`rm -rf /thevolume/*`) on the volume and makes it available again for a new claim.

However, an administrator can configure a custom recycler Pod template using the Kubernetes controller manager command line arguments as described in the [reference](#). The custom recycler Pod template must contain a volumes specification, as shown in the example below:

```
apiVersion: v1
kind: Pod
metadata:
 name: pv-recycler
 namespace: default
spec:
 restartPolicy: Never
 volumes:
 - name: vol
 hostPath:
 path: /any/path/it/will/be/replaced
 containers:
 - name: pv-recycler
 image: "registry.k8s.io/busybox"
 command: ["/bin/sh", "-c",
"test -e /scrub && rm -rf /scrub/..?* /scrub/.[*] /scrub/* && test -z \"$(ls -A /scrub)\" || exit 1"]
 volumeMounts:
 - name: vol
 mountPath: /scrub
```

However, the particular path specified in the custom recycler Pod template in the volumes part is replaced with the particular path of the volume that is being recycled.

## PersistentVolume deletion protection finalizer

FEATURE STATE: Kubernetes v1.23 [alpha]

Finalizers can be added on a PersistentVolume to ensure that PersistentVolumes having Delete reclaim policy are deleted only after the backing storage are deleted.

The newly introduced finalizers `kubernetes.io/pv-controller` and `external-provisioner.volume.kubernetes.io/finalizer` are only added to dynamically provisioned volumes.

The finalizer `kubernetes.io/pv-controller` is added to in-tree plugin volumes. The following is an example

```
kubectl describe pv pvc-74a498d6-3929-47e8-8c02-078c1ece4d78
Name: pvc-74a498d6-3929-47e8-8c02-078c1ece4d78
Labels: <none>
```

Annotations: kubernetes.io/createdby: vsphere-volume-dynamic-provisioner  
pv.kubernetes.io/bound-by-controller: yes  
pv.kubernetes.io/provisioned-by: kubernetes.io/vsphere-volume  
Finalizers: [kubernetes.io/pv-protection kubernetes.io/pv-controller]  
StorageClass: vcp-sc  
Status: Bound  
Claim: default/vcp-pvc-1  
Reclaim Policy: Delete  
Access Modes: RWO  
VolumeMode: Filesystem  
Capacity: 1Gi  
Node Affinity: <none>  
Message:  
Source:  
Type: vSphereVolume (a Persistent Disk resource in vSphere)  
VolumePath: [vsanDatastore] d49c4a62-166f-ce12-c464-020077ba5d46/kubernetes-dynamic-pvc-74a498d6-3929-47e8-8c02-078c1ece4d78.vmdk  
FSType: ext4  
StoragePolicyName: vSAN Default Storage Policy  
Events: <none>

The finalizer external-provisioner.volume.kubernetes.io/finalizer is added for CSI volumes. The following is an example:

Name: pvc-2f0bab97-85a8-4552-8044-eb8be45cf48d  
Labels: <none>  
Annotations: pv.kubernetes.io/provisioned-by: csi.vsphere.vmware.com  
Finalizers: [kubernetes.io/pv-protection external-provisioner.volume.kubernetes.io/finalizer]  
StorageClass: fast  
Status: Bound  
Claim: demo-app/nginx-logs  
Reclaim Policy: Delete  
Access Modes: RWO  
VolumeMode: Filesystem  
Capacity: 200Mi  
Node Affinity: <none>  
Message:  
Source:  
Type: CSI (a Container Storage Interface (CSI) volume source)  
Driver: csi.vsphere.vmware.com  
FSType: ext4  
VolumeHandle: 44830fa8-79b4-406b-8b58-621ba25353fd  
ReadOnly: false  
VolumeAttributes: storage.kubernetes.io/csiProvisionerIdentity=1648442357185-8081-csi.vsphere.vmware.com  
type=vSphere CNS Block Volume  
Events: <none>

When the CSIMigration{provider} feature flag is enabled for a specific in-tree volume plugin, the kubernetes.io/pv-controller finalizer is replaced by the external-provisioner.volume.kubernetes.io/finalizer finalizer.

## Reserving a PersistentVolume

The control plane can [bind PersistentVolumeClaims to matching PersistentVolumes](#) in the cluster. However, if you want a PVC to bind to a specific PV, you need to pre-bind them.

By specifying a PersistentVolume in a PersistentVolumeClaim, you declare a binding between that specific PV and PVC. If the PersistentVolume exists and has not reserved PersistentVolumeClaims through its claimRef field, then the PersistentVolume and PersistentVolumeClaim will be bound.

The binding happens regardless of some volume matching criteria, including node affinity. The control plane still checks that [storage class](#), access modes, and requested storage size are valid.

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: foo-pvc
 namespace: foo
spec:
 storageClassName: ""
Empty string must be explicitly set otherwise default StorageClass will be set
 volumeName: foo-pv
...
```

This method does not guarantee any binding privileges to the PersistentVolume. If other PersistentVolumeClaims could use the PV that you specify, you first need to reserve that storage volume. Specify the relevant PersistentVolumeClaim in the claimRef field of the PV so that other PVCs can not bind to it.

```
apiVersion: v1
kind: PersistentVolume
metadata:
 name: foo-pv
spec:
 storageClassName: ""
 claimRef:
 name: foo-pvc
 namespace: foo
...
```

This is useful if you want to consume PersistentVolumes that have their persistentVolumeReclaimPolicy set to Retain, including cases where you are reusing an existing PV.

## Expanding Persistent Volumes Claims

FEATURE STATE: Kubernetes v1.24 [stable]

Support for expanding PersistentVolumeClaims (PVCs) is enabled by default. You can expand the following types of volumes:

- azureFile (deprecated)
- [csi](#)

- flexVolume (deprecated)
- rbd (deprecated)
- portworxVolume (deprecated)

You can only expand a PVC if its storage class's `allowVolumeExpansion` field is set to `true`.

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: example-vol-default
provisioner: vendor-name.example/magicstorage
parameters:
 resturl: "http://192.168.10.100:8080"
 restuser: ""
 secretNamespace: ""
 secretName: ""
allowVolumeExpansion: true
```

To request a larger volume for a PVC, edit the PVC object and specify a larger size. This triggers expansion of the volume that backs the underlying `PersistentVolume`. A new `PersistentVolume` is never created to satisfy the claim. Instead, an existing volume is resized.

### Warning:

Directly editing the size of a `PersistentVolume` can prevent an automatic resize of that volume. If you edit the capacity of a `PersistentVolume`, and then edit the `.spec` of a matching `PersistentVolumeClaim` to make the size of the `PersistentVolumeClaim` match the `PersistentVolume`, then no storage resize happens. The Kubernetes control plane will see that the desired state of both resources matches, conclude that the backing volume size has been manually increased and that no resize is necessary.

### CSI Volume expansion

FEATURE STATE: Kubernetes v1.24 [stable]

Support for expanding CSI volumes is enabled by default but it also requires a specific CSI driver to support volume expansion. Refer to documentation of the specific CSI driver for more information.

### Resizing a volume containing a file system

You can only resize volumes containing a file system if the file system is XFS, Ext3, or Ext4.

When a volume contains a file system, the file system is only resized when a new Pod is using the `PersistentVolumeClaim` in `ReadWrite` mode. File system expansion is either done when a Pod is starting up or when a Pod is running and the underlying file system supports online expansion.

FlexVolumes (deprecated since Kubernetes v1.23) allow resize if the driver is configured with the `RequiresFSResize` capability to `true`. The FlexVolume can be resized on Pod restart.

## Resizing an in-use PersistentVolumeClaim

FEATURE STATE: Kubernetes v1.24 [stable]

In this case, you don't need to delete and recreate a Pod or deployment that is using an existing PVC. Any in-use PVC automatically becomes available to its Pod as soon as its file system has been expanded. This feature has no effect on PVCs that are not in use by a Pod or deployment. You must create a Pod that uses the PVC before the expansion can complete.

Similar to other volume types - FlexVolume volumes can also be expanded when in-use by a Pod.

### Note:

FlexVolume resize is possible only when the underlying driver supports resize.

## Recovering from Failure when Expanding Volumes

If a user specifies a new size that is too big to be satisfied by underlying storage system, expansion of PVC will be continuously retried until user or cluster administrator takes some action. This can be undesirable and hence Kubernetes provides following methods of recovering from such failures.

- [Manually with Cluster Administrator access](#)
- [By requesting expansion to smaller size](#)

If expanding underlying storage fails, the cluster administrator can manually recover the Persistent Volume Claim (PVC) state and cancel the resize requests. Otherwise, the resize requests are continuously retried by the controller without administrator intervention.

1. Mark the PersistentVolume(PV) that is bound to the PersistentVolumeClaim(PVC) with Retain reclaim policy.
2. Delete the PVC. Since PV has Retain reclaim policy - we will not lose any data when we recreate the PVC.
3. Delete the claimRef entry from PV specs, so as new PVC can bind to it. This should make the PV Available.
4. Re-create the PVC with smaller size than PV and set volumeName field of the PVC to the name of the PV. This should bind new PVC to existing PV.
5. Don't forget to restore the reclaim policy of the PV.

FEATURE STATE: Kubernetes v1.23 [alpha]

### Note:

Recovery from failing PVC expansion by users is available as an alpha feature since Kubernetes 1.23. The RecoverVolumeExpansionFailure feature must be enabled for this feature to work. Refer to the [feature gate](#) documentation for more information.

If the feature gates RecoverVolumeExpansionFailure is enabled in your cluster, and expansion has failed for a PVC, you can retry expansion with a smaller size than the previously requested value. To request a new expansion attempt with a smaller proposed size, edit .spec.resources for that PVC and choose a value that is less than the value you previously tried. This is useful if expansion to a higher value did not succeed because of capacity constraint. If that has

happened, or you suspect that it might have, you can retry expansion by specifying a size that is within the capacity limits of underlying storage provider. You can monitor status of resize operation by watching `.status.allocatedResourceStatuses` and events on the PVC.

Note that, although you can specify a lower amount of storage than what was requested previously, the new value must still be higher than `.status.capacity`. Kubernetes does not support shrinking a PVC to less than its current size.

## Types of Persistent Volumes

PersistentVolume types are implemented as plugins. Kubernetes currently supports the following plugins:

- [csi](#) - Container Storage Interface (CSI)
- [fc](#) - Fibre Channel (FC) storage
- [hostPath](#) - HostPath volume (for single node testing only; WILL NOT WORK in a multi-node cluster; consider using local volume instead)
- [iscsi](#) - iSCSI (SCSI over IP) storage
- [local](#) - local storage devices mounted on nodes.
- [nfs](#) - Network File System (NFS) storage

The following types of PersistentVolume are deprecated but still available. If you are using these volume types except for flexVolume, cephfs and rbd, please install corresponding CSI drivers.

- [awsElasticBlockStore](#) - AWS Elastic Block Store (EBS) (**migration on by default** starting v1.23)
- [azureDisk](#) - Azure Disk (**migration on by default** starting v1.23)
- [azureFile](#) - Azure File (**migration on by default** starting v1.24)
- [cephfs](#) - CephFS volume (**deprecated** starting v1.28, no migration plan, support will be removed in a future release)
- [cinder](#) - Cinder (OpenStack block storage) (**migration on by default** starting v1.21)
- [flexVolume](#) - FlexVolume (**deprecated** starting v1.23, no migration plan and no plan to remove support)
- [gcePersistentDisk](#) - GCE Persistent Disk (**migration on by default** starting v1.23)
- [portworxVolume](#) - Portworx volume (**deprecated** starting v1.25)
- [rbd](#) - Rados Block Device (RBD) volume (**deprecated** starting v1.28, no migration plan, support will be removed in a future release)
- [vsphereVolume](#) - vSphere VMDK volume (**migration on by default** starting v1.25)

Older versions of Kubernetes also supported the following in-tree PersistentVolume types:

- [photonPersistentDisk](#) - Photon controller persistent disk. (**not available** starting v1.15)
- [scaleIO](#) - ScaleIO volume. (**not available** starting v1.21)
- [flocker](#) - Flocker storage. (**not available** starting v1.25)
- [quobyte](#) - Quobyte volume. (**not available** starting v1.25)
- [storageos](#) - StorageOS volume. (**not available** starting v1.25)

## Persistent Volumes

Each PV contains a spec and status, which is the specification and status of the volume. The name of a PersistentVolume object must be a valid [DNS subdomain name](#).



```
apiVersion: v1
kind: PersistentVolume
metadata:
 name: pv0003
spec:
 capacity:
 storage: 5Gi
 volumeMode: Filesystem
 accessModes:
 - ReadWriteOnce
 persistentVolumeReclaimPolicy: Recycle
 storageClassName: slow
 mountOptions:
 - hard
 - nfsvers=4.1
 nfs:
 path: /tmp
 server: 172.17.0.2
```

### Note:

Helper programs relating to the volume type may be required for consumption of a PersistentVolume within a cluster. In this example, the PersistentVolume is of type NFS and the helper program `/sbin/mount.nfs` is required to support the mounting of NFS filesystems.

## Capacity

Generally, a PV will have a specific storage capacity. This is set using the PV's capacity attribute which is a [Quantity](#) value.

Currently, storage size is the only resource that can be set or requested. Future attributes may include IOPS, throughput, etc.

## Volume Mode

FEATURE STATE: Kubernetes v1.18 [stable]

Kubernetes supports two volumeModes of PersistentVolumes: Filesystem and Block.

volumeMode is an optional API parameter. Filesystem is the default mode used when volumeMode parameter is omitted.

A volume with volumeMode: Filesystem is *mounted* into Pods into a directory. If the volume is backed by a block device and the device is empty, Kubernetes creates a filesystem on the device before mounting it for the first time.

You can set the value of volumeMode to Block to use a volume as a raw block device. Such volume is presented into a Pod as a block device, without any filesystem on it. This mode is useful to provide a Pod the fastest possible way to access a volume, without any filesystem layer between the Pod and the volume. On the other hand, the application running in the Pod must know how to handle a raw block device. See [Raw Block Volume Support](#) for an example on how to use a volume with volumeMode: Block in a Pod.

## Access Modes

A PersistentVolume can be mounted on a host in any way supported by the resource provider. As shown in the table below, providers will have different capabilities and each PV's access modes are set to the specific modes supported by that particular volume. For example, NFS can support multiple read/write clients, but a specific NFS PV might be exported on the server as read-only. Each PV gets its own set of access modes describing that specific PV's capabilities.

The access modes are:

### ReadWriteOnce

the volume can be mounted as read-write by a single node. ReadWriteOnce access mode still can allow multiple pods to access the volume when the pods are running on the same node. For single pod access, please see ReadWriteOncePod.

### ReadOnlyMany

the volume can be mounted as read-only by many nodes.

### ReadWriteMany

the volume can be mounted as read-write by many nodes.

### ReadWriteOncePod

FEATURE STATE: Kubernetes v1.29 [stable]

the volume can be mounted as read-write by a single Pod. Use ReadWriteOncePod access mode if you want to ensure that only one pod across the whole cluster can read that PVC or write to it.

### Note:

The ReadWriteOncePod access mode is only supported for [CSI](#) volumes and Kubernetes version 1.22+. To use this feature you will need to update the following [CSI sidecars](#) to these versions or greater:

- [csi-provisioner:v3.0.0+](#)
- [csi-attacher:v3.3.0+](#)
- [csi-resizer:v1.3.0+](#)

In the CLI, the access modes are abbreviated to:

- RWO - ReadWriteOnce
- ROX - ReadOnlyMany
- RWX - ReadWriteMany
- RWOP - ReadWriteOncePod

### Note:

Kubernetes uses volume access modes to match PersistentVolumeClaims and PersistentVolumes. In some cases, the volume access modes also constrain where the PersistentVolume can be mounted. Volume access modes do **not** enforce write protection once the storage has been mounted. Even if the access modes are specified as ReadWriteOnce, ReadOnlyMany, or ReadWriteMany, they don't set any constraints on the volume. For example, even if a PersistentVolume is created as ReadOnlyMany, it is no guarantee that it will be read-only. If the access modes are specified as ReadWriteOncePod, the volume is constrained and can be mounted on only a single Pod.

**Important!** A volume can only be mounted using one access mode at a time, even if it supports many.

| Volume Plugin  | ReadWriteOnce         | ReadOnlyMany          | ReadWriteMany                      | ReadWriteOncePod      |
|----------------|-----------------------|-----------------------|------------------------------------|-----------------------|
| AzureFile      |                       |                       |                                    | -                     |
| CephFS         |                       |                       |                                    | -                     |
| CSI            | depends on the driver | depends on the driver | depends on the driver              | depends on the driver |
| FC             |                       |                       | -                                  | -                     |
| FlexVolume     |                       |                       | depends on the driver              | -                     |
| HostPath       |                       | -                     | -                                  | -                     |
| iSCSI          |                       |                       | -                                  | -                     |
| NFS            |                       |                       |                                    | -                     |
| RBD            |                       |                       | -                                  | -                     |
| VsphereVolume  |                       | -                     | - (works when Pods are collocated) | -                     |
| PortworxVolume |                       | -                     |                                    | -                     |

## Class

A PV can have a class, which is specified by setting the `storageClassName` attribute to the name of a [StorageClass](#). A PV of a particular class can only be bound to PVCs requesting that class. A PV with no `storageClassName` has no class and can only be bound to PVCs that request no particular class.

In the past, the annotation `volume.beta.kubernetes.io/storage-class` was used instead of the `storageClassName` attribute. This annotation is still working; however, it will become fully deprecated in a future Kubernetes release.

## Reclaim Policy

Current reclaim policies are:

- Retain -- manual reclamation
- Recycle -- basic scrub (`rm -rf /thevolume/*`)
- Delete -- delete the volume

For Kubernetes 1.30, only `nfs` and `hostPath` volume types support recycling.

## Mount Options

A Kubernetes administrator can specify additional mount options for when a Persistent Volume is mounted on a node.

### Note:

Not all Persistent Volume types support mount options.

The following volume types support mount options:

- azureFile
- cephfs (**deprecated** in v1.28)
- cinder (**deprecated** in v1.18)
- iscsi
- nfs
- rbd (**deprecated** in v1.28)
- vsphereVolume

Mount options are not validated. If a mount option is invalid, the mount fails.

In the past, the annotation `volume.beta.kubernetes.io/mount-options` was used instead of the `mountOptions` attribute. This annotation is still working; however, it will become fully deprecated in a future Kubernetes release.

## Node Affinity

### Note:

For most volume types, you do not need to set this field. You need to explicitly set this for [local](#) volumes.

A PV can specify node affinity to define constraints that limit what nodes this volume can be accessed from. Pods that use a PV will only be scheduled to nodes that are selected by the node affinity. To specify node affinity, set `nodeAffinity` in the `.spec` of a PV. The [PersistentVolume](#) API reference has more details on this field.

## Phase

A `PersistentVolume` will be in one of the following phases:

### Available

a free resource that is not yet bound to a claim

### Bound

the volume is bound to a claim

### Released

the claim has been deleted, but the associated storage resource is not yet reclaimed by the cluster

### Failed

the volume has failed its (automated) reclamation

You can see the name of the PVC bound to the PV using `kubectl describe persistentvolume <name>`.

## Phase transition timestamp

FEATURE STATE: Kubernetes v1.29 [beta]

The `.status` field for a `PersistentVolume` can include an alpha `lastPhaseTransitionTime` field. This field records the timestamp of when the volume last transitioned its phase. For newly created volumes the phase is set to `Pending` and `lastPhaseTransitionTime` is set to the current time.

## Note:

You need to enable the PersistentVolumeLastPhaseTransitionTime [feature gate](#) to use or see the lastPhaseTransitionTime field.

# PersistentVolumeClaims

Each PVC contains a spec and status, which is the specification and status of the claim. The name of a PersistentVolumeClaim object must be a valid [DNS subdomain name](#).

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: myclaim
spec:
 accessModes:
 - ReadWriteOnce
 volumeMode: Filesystem
 resources:
 requests:
 storage: 8Gi
 storageClassName: slow
 selector:
 matchLabels:
 release: "stable"
 matchExpressions:
 - {key: environment, operator: In, values: [dev]}
```

## Access Modes

Claims use [the same conventions as volumes](#) when requesting storage with specific access modes.

## Volume Modes

Claims use [the same convention as volumes](#) to indicate the consumption of the volume as either a filesystem or block device.

## Resources

Claims, like Pods, can request specific quantities of a resource. In this case, the request is for storage. The same [resource model](#) applies to both volumes and claims.

## Selector

Claims can specify a [label selector](#) to further filter the set of volumes. Only the volumes whose labels match the selector can be bound to the claim. The selector can consist of two fields:

- matchLabels - the volume must have a label with this value

- `matchExpressions` - a list of requirements made by specifying key, list of values, and operator that relates the key and values. Valid operators include `In`, `NotIn`, `Exists`, and `DoesNotExist`.

All of the requirements, from both `matchLabels` and `matchExpressions`, are ANDed together – they must all be satisfied in order to match.

## Class

A claim can request a particular class by specifying the name of a [StorageClass](#) using the attribute `storageClassName`. Only PVs of the requested class, ones with the same `storageClassName` as the PVC, can be bound to the PVC.

PVCs don't necessarily have to request a class. A PVC with its `storageClassName` set equal to "" is always interpreted to be requesting a PV with no class, so it can only be bound to PVs with no class (no annotation or one set equal to ""). A PVC with no `storageClassName` is not quite the same and is treated differently by the cluster, depending on whether the [DefaultStorageClass admission plugin](#) is turned on.

- If the admission plugin is turned on, the administrator may specify a default StorageClass. All PVCs that have no `storageClassName` can be bound only to PVs of that default. Specifying a default StorageClass is done by setting the annotation `storageclass.kubernetes.io/is-default-class` equal to `true` in a StorageClass object. If the administrator does not specify a default, the cluster responds to PVC creation as if the admission plugin were turned off. If more than one default StorageClass is specified, the newest default is used when the PVC is dynamically provisioned.
- If the admission plugin is turned off, there is no notion of a default StorageClass. All PVCs that have `storageClassName` set to "" can be bound only to PVs that have `storageClassName` also set to "". However, PVCs with missing `storageClassName` can be updated later once default StorageClass becomes available. If the PVC gets updated it will no longer bind to PVs that have `storageClassName` also set to "".

See [retroactive default StorageClass assignment](#) for more details.

Depending on installation method, a default StorageClass may be deployed to a Kubernetes cluster by addon manager during installation.

When a PVC specifies a selector in addition to requesting a StorageClass, the requirements are ANDed together: only a PV of the requested class and with the requested labels may be bound to the PVC.

### Note:

Currently, a PVC with a non-empty selector can't have a PV dynamically provisioned for it.

In the past, the annotation `volume.beta.kubernetes.io/storage-class` was used instead of `storageClassName` attribute. This annotation is still working; however, it won't be supported in a future Kubernetes release.

## Retroactive default StorageClass assignment

FEATURE STATE: Kubernetes v1.28 [stable]

You can create a PersistentVolumeClaim without specifying a storageClassName for the new PVC, and you can do so even when no default StorageClass exists in your cluster. In this case, the new PVC creates as you defined it, and the storageClassName of that PVC remains unset until default becomes available.

When a default StorageClass becomes available, the control plane identifies any existing PVCs without storageClassName. For the PVCs that either have an empty value for storageClassName or do not have this key, the control plane then updates those PVCs to set storageClassName to match the new default StorageClass. If you have an existing PVC where the storageClassName is "", and you configure a default StorageClass, then this PVC will not get updated.

In order to keep binding to PVs with storageClassName set to "" (while a default StorageClass is present), you need to set the storageClassName of the associated PVC to "".

This behavior helps administrators change default StorageClass by removing the old one first and then creating or setting another one. This brief window while there is no default causes PVCs without storageClassName created at that time to not have any default, but due to the retroactive default StorageClass assignment this way of changing defaults is safe.

## Claims As Volumes

Pods access storage by using the claim as a volume. Claims must exist in the same namespace as the Pod using the claim. The cluster finds the claim in the Pod's namespace and uses it to get the PersistentVolume backing the claim. The volume is then mounted to the host and into the Pod.

```
apiVersion: v1
kind: Pod
metadata:
 name: mypod
spec:
 containers:
 - name: myfrontend
 image: nginx
 volumeMounts:
 - mountPath: "/var/www/html"
 name: mypd
 volumes:
 - name: mypd
 persistentVolumeClaim:
 claimName: myclaim
```

## A Note on Namespaces

PersistentVolumes binds are exclusive, and since PersistentVolumeClaims are namespaced objects, mounting claims with "Many" modes (ROX, RWX) is only possible within one namespace.

## PersistentVolumes typed hostPath

A hostPath PersistentVolume uses a file or directory on the Node to emulate network-attached storage. See [an example of hostPath typed volume](#).

## Raw Block Volume Support

FEATURE STATE: Kubernetes v1.18 [stable]

The following volume plugins support raw block volumes, including dynamic provisioning where applicable:

- CSI
- FC (Fibre Channel)
- iSCSI
- Local volume
- OpenStack Cinder
- RBD (deprecated)
- RBD (Ceph Block Device; deprecated)
- VsphereVolume

## PersistentVolume using a Raw Block Volume

```
apiVersion: v1
kind: PersistentVolume
metadata:
 name: block-pv
spec:
 capacity:
 storage: 10Gi
 accessModes:
 - ReadWriteOnce
 volumeMode: Block
 persistentVolumeReclaimPolicy: Retain
 fc:
 targetWWNs: ["50060e801049cfd1"]
 lun: 0
 readOnly: false
```

## PersistentVolumeClaim requesting a Raw Block Volume

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: block-pvc
spec:
 accessModes:
 - ReadWriteOnce
 volumeMode: Block
 resources:
 requests:
 storage: 10Gi
```



## Pod specification adding Raw Block Device path in container

```
apiVersion: v1
kind: Pod
metadata:
 name: pod-with-block-volume
spec:
 containers:
 - name: fc-container
 image: fedora:26
 command: ["/bin/sh", "-c"]
 args: ["tail -f /dev/null"]
 volumeDevices:
 - name: data
 devicePath: /dev/xvda
 volumes:
 - name: data
 persistentVolumeClaim:
 claimName: block-pvc
```

### Note:

When adding a raw block device for a Pod, you specify the device path in the container instead of a mount path.

## Binding Block Volumes

If a user requests a raw block volume by indicating this using the `volumeMode` field in the `PersistentVolumeClaim` spec, the binding rules differ slightly from previous releases that didn't consider this mode as part of the spec. Listed is a table of possible combinations the user and admin might specify for requesting a raw block device. The table indicates if the volume will be bound or not given the combinations: Volume binding matrix for statically provisioned volumes:

| PV volumeMode | PVC volumeMode | Result  |
|---------------|----------------|---------|
| unspecified   | unspecified    | BIND    |
| unspecified   | Block          | NO BIND |
| unspecified   | Filesystem     | BIND    |
| Block         | unspecified    | NO BIND |
| Block         | Block          | BIND    |
| Block         | Filesystem     | NO BIND |
| Filesystem    | Filesystem     | BIND    |
| Filesystem    | Block          | NO BIND |
| Filesystem    | unspecified    | BIND    |

### Note:

Only statically provisioned volumes are supported for alpha release. Administrators should take care to consider these values when working with raw block devices.

# Volume Snapshot and Restore Volume from Snapshot Support

FEATURE STATE: Kubernetes v1.20 [stable]

Volume snapshots only support the out-of-tree CSI volume plugins. For details, see [Volume Snapshots](#). In-tree volume plugins are deprecated. You can read about the deprecated volume plugins in the [Volume Plugin FAQ](#).

## Create a PersistentVolumeClaim from a Volume Snapshot

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: restore-pvc
spec:
 storageClassName: csi-hostpath-sc
 dataSource:
 name: new-snapshot-test
 kind: VolumeSnapshot
 apiGroup: snapshot.storage.k8s.io
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 10Gi
```

## Volume Cloning

[Volume Cloning](#) only available for CSI volume plugins.

## Create PersistentVolumeClaim from an existing PVC

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: cloned-pvc
spec:
 storageClassName: my-csi-plugin
 dataSource:
 name: existing-src-pvc-name
 kind: PersistentVolumeClaim
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 10Gi
```

# Volume populators and data sources

FEATURE STATE: Kubernetes v1.24 [beta]

Kubernetes supports custom volume populators. To use custom volume populators, you must enable the AnyVolumeDataSource [feature gate](#) for the kube-apiserver and kube-controller-manager.

Volume populators take advantage of a PVC spec field called dataSourceRef. Unlike the dataSource field, which can only contain either a reference to another PersistentVolumeClaim or to a VolumeSnapshot, the dataSourceRef field can contain a reference to any object in the same namespace, except for core objects other than PVCs. For clusters that have the feature gate enabled, use of the dataSourceRef is preferred over dataSource.

## Cross namespace data sources

FEATURE STATE: Kubernetes v1.26 [alpha]

Kubernetes supports cross namespace volume data sources. To use cross namespace volume data sources, you must enable the AnyVolumeDataSource and CrossNamespaceVolumeDataSource [feature gates](#) for the kube-apiserver and kube-controller-manager. Also, you must enable the CrossNamespaceVolumeDataSource feature gate for the csi-provisioner.

Enabling the CrossNamespaceVolumeDataSource feature gate allows you to specify a namespace in the dataSourceRef field.

### Note:

When you specify a namespace for a volume data source, Kubernetes checks for a ReferenceGrant in the other namespace before accepting the reference. ReferenceGrant is part of the gateway.networking.k8s.io extension APIs. See [ReferenceGrant](#) in the Gateway API documentation for details. This means that you must extend your Kubernetes cluster with at least ReferenceGrant from the Gateway API before you can use this mechanism.

## Data source references

The dataSourceRef field behaves almost the same as the dataSource field. If one is specified while the other is not, the API server will give both fields the same value. Neither field can be changed after creation, and attempting to specify different values for the two fields will result in a validation error. Therefore the two fields will always have the same contents.

There are two differences between the dataSourceRef field and the dataSource field that users should be aware of:

- The dataSource field ignores invalid values (as if the field was blank) while the dataSourceRef field never ignores values and will cause an error if an invalid value is used. Invalid values are any core object (objects with no apiGroup) except for PVCs.
- The dataSourceRef field may contain different types of objects, while the dataSource field only allows PVCs and VolumeSnapshots.

When the `CrossNamespaceVolumeDataSource` feature is enabled, there are additional differences:

- The `dataSource` field only allows local objects, while the `dataSourceRef` field allows objects in any namespaces.
- When namespace is specified, `dataSource` and `dataSourceRef` are not synced.

Users should always use `dataSourceRef` on clusters that have the feature gate enabled, and fall back to `dataSource` on clusters that do not. It is not necessary to look at both fields under any circumstance. The duplicated values with slightly different semantics exist only for backwards compatibility. In particular, a mixture of older and newer controllers are able to interoperate because the fields are the same.

## Using volume populators

Volume populators are [controllers](#) that can create non-empty volumes, where the contents of the volume are determined by a Custom Resource. Users create a populated volume by referring to a Custom Resource using the `dataSourceRef` field:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: populated-pvc
spec:
 dataSourceRef:
 name: example-name
 kind: ExampleDataSource
 apiGroup: example.storage.k8s.io
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 10Gi
```

Because volume populators are external components, attempts to create a PVC that uses one can fail if not all the correct components are installed. External controllers should generate events on the PVC to provide feedback on the status of the creation, including warnings if the PVC cannot be created due to some missing component.

You can install the alpha [volume data source validator](#) controller into your cluster. That controller generates warning Events on a PVC in the case that no populator is registered to handle that kind of data source. When a suitable populator is installed for a PVC, it's the responsibility of that populator controller to report Events that relate to volume creation and issues during the process.

## Using a cross-namespace volume data source

FEATURE STATE: Kubernetes v1.26 [alpha]

Create a `ReferenceGrant` to allow the namespace owner to accept the reference. You define a populated volume by specifying a cross namespace volume data source using the `dataSourceRef` field. You must already have a valid `ReferenceGrant` in the source namespace:

```
apiVersion: gateway.networking.k8s.io/v1beta1
kind: ReferenceGrant
metadata:
 name: allow-ns1-pvc
 namespace: default
spec:
 from:
 - group: ""
 kind: PersistentVolumeClaim
 namespace: ns1
 to:
 - group: snapshot.storage.k8s.io
 kind: VolumeSnapshot
 name: new-snapshot-demo
```

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: foo-pvc
 namespace: ns1
spec:
 storageClassName: example
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 1Gi
 dataSourceRef:
 apiGroup: snapshot.storage.k8s.io
 kind: VolumeSnapshot
 name: new-snapshot-demo
 namespace: default
 volumeMode: Filesystem
```

## Writing Portable Configuration

If you're writing configuration templates or examples that run on a wide range of clusters and need persistent storage, it is recommended that you use the following pattern:

- Include PersistentVolumeClaim objects in your bundle of config (alongside Deployments, ConfigMaps, etc).
- Do not include PersistentVolume objects in the config, since the user instantiating the config may not have permission to create PersistentVolumes.
- Give the user the option of providing a storage class name when instantiating the template.
  - If the user provides a storage class name, put that value into the persistentVolumeClaim.storageClassName field. This will cause the PVC to match the right storage class if the cluster has StorageClasses enabled by the admin.
  - If the user does not provide a storage class name, leave the persistentVolumeClaim.storageClassName field as nil. This will cause a PV to be automatically provisioned for the user with the default StorageClass in the cluster.

Many cluster environments have a default StorageClass installed, or administrators can create their own default StorageClass.

- In your tooling, watch for PVCs that are not getting bound after some time and surface this to the user, as this may indicate that the cluster has no dynamic storage support (in which case the user should create a matching PV) or the cluster has no storage system (in which case the user cannot deploy config requiring PVCs).

## What's next

- Learn more about [Creating a PersistentVolume](#).
- Learn more about [Creating a PersistentVolumeClaim](#).
- Read the [Persistent Storage design document](#).

## API references

Read about the APIs described in this page:

- [PersistentVolume](#)
- [PersistentVolumeClaim](#)

# Projected Volumes

This document describes *projected volumes* in Kubernetes. Familiarity with [volumes](#) is suggested.

## Introduction

A projected volume maps several existing volume sources into the same directory.

Currently, the following types of volume sources can be projected:

- [secret](#)
- [downwardAPI](#)
- [configMap](#)
- [serviceAccountToken](#)
- [clusterTrustBundle](#)

All sources are required to be in the same namespace as the Pod. For more details, see the [all-in-one volume](#) design document.

## Example configuration with a secret, a downwardAPI, and a configMap

[pods/storage/projected-secret-downwardapi-configmap.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: volume-test
spec:
 containers:
```

```

- name: container-test
 image: busybox:1.28
 command: ["sleep", "3600"]
 volumeMounts:
 - name: all-in-one
 mountPath: "/projected-volume"
 readOnly: true
volumes:
- name: all-in-one
 projected:
 sources:
 - secret:
 name: mysecret
 items:
 - key: username
 path: my-group/my-username
 - downwardAPI:
 items:
 - path: "labels"
 fieldRef:
 fieldPath: metadata.labels
 - path: "cpu_limit"
 resourceFieldRef:
 containerName: container-test
 resource: limits.cpu
 - configMap:
 name: myconfigmap
 items:
 - key: config
 path: my-group/my-config

```

## Example configuration: secrets with a non-default permission mode set

[pods/storage/projected-secrets-nondefault-permission-mode.yaml](#)

```

apiVersion: v1
kind: Pod
metadata:
 name: volume-test
spec:
 containers:
 - name: container-test
 image: busybox:1.28
 command: ["sleep", "3600"]
 volumeMounts:
 - name: all-in-one
 mountPath: "/projected-volume"
 readOnly: true
 volumes:
 - name: all-in-one
 projected:
 sources:
 - secret:

```

```

 name: mysecret
 items:
 - key: username
 path: my-group/my-username
 - secret:
 name: mysecret2
 items:
 - key: password
 path: my-group/my-password
 mode: 511

```

Each projected volume source is listed in the spec under sources. The parameters are nearly the same with two exceptions:

- For secrets, the `secretName` field has been changed to `name` to be consistent with ConfigMap naming.
- The `defaultMode` can only be specified at the projected level and not for each volume source. However, as illustrated above, you can explicitly set the mode for each individual projection.

## serviceAccountToken projected volumes

You can inject the token for the current [service account](#) into a Pod at a specified path. For example:

[pods/storage/projected-service-account-token.yaml](#)

```

apiVersion: v1
kind: Pod
metadata:
 name: sa-token-test
spec:
 containers:
 - name: container-test
 image: busybox:1.28
 command: ["sleep", "3600"]
 volumeMounts:
 - name: token-vol
 mountPath: "/service-account"
 readOnly: true
 serviceAccountName: default
 volumes:
 - name: token-vol
 projected:
 sources:
 - serviceAccountToken:
 audience: api
 expirationSeconds: 3600
 path: token

```

The example Pod has a projected volume containing the injected service account token. Containers in this Pod can use that token to access the Kubernetes API server, authenticating with the identity of [the pod's ServiceAccount](#). The audience field contains the intended



audience of the token. A recipient of the token must identify itself with an identifier specified in the audience of the token, and otherwise should reject the token. This field is optional and it defaults to the identifier of the API server.

The `expirationSeconds` is the expected duration of validity of the service account token. It defaults to 1 hour and must be at least 10 minutes (600 seconds). An administrator can also limit its maximum value by specifying the `--service-account-max-token-expiration` option for the API server. The `path` field specifies a relative path to the mount point of the projected volume.

**Note:**

A container using a projected volume source as a [subPath](#) volume mount will not receive updates for those volume sources.

## clusterTrustBundle projected volumes

FEATURE STATE: Kubernetes v1.29 [alpha]

**Note:**

To use this feature in Kubernetes 1.30, you must enable support for ClusterTrustBundle objects with the ClusterTrustBundle [feature gate](#) and `--runtime-config=certificates.k8s.io/v1alpha1/clustertrustbundles=true` kube-apiserver flag, then enable the ClusterTrustBundleProjection feature gate.

The clusterTrustBundle projected volume source injects the contents of one or more [ClusterTrustBundle](#) objects as an automatically-updating file in the container filesystem.

ClusterTrustBundles can be selected either by [name](#) or by [signer name](#).

To select by name, use the `name` field to designate a single ClusterTrustBundle object.

To select by signer name, use the `signerName` field (and optionally the `labelSelector` field) to designate a set of ClusterTrustBundle objects that use the given signer name. If `labelSelector` is not present, then all ClusterTrustBundles for that signer are selected.

The kubelet deduplicates the certificates in the selected ClusterTrustBundle objects, normalizes the PEM representations (discarding comments and headers), reorders the certificates, and writes them into the file named by path. As the set of selected ClusterTrustBundles or their content changes, kubelet keeps the file up-to-date.

By default, the kubelet will prevent the pod from starting if the named ClusterTrustBundle is not found, or if `signerName` / `labelSelector` do not match any ClusterTrustBundles. If this behavior is not what you want, then set the optional field to true, and the pod will start up with an empty file at path.

[pods/storage/projected-clustertrustbundle.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: sa-ctb-name-test
spec:
```

```
containers:
- name: container-test
 image: busybox
 command: ["sleep", "3600"]
 volumeMounts:
 - name: token-vol
 mountPath: "/root-certificates"
 readOnly: true
serviceAccountName: default
volumes:
- name: token-vol
 projected:
 sources:
 - clusterTrustBundle:
 name: example
 path: example-roots.pem
 - clusterTrustBundle:
 signerName: "example.com/mysigner"
 labelSelector:
 matchLabels:
 version: live
 path: mysigner-roots.pem
 optional: true
```

## SecurityContext interactions

The [proposal](#) for file permission handling in projected service account volume enhancement introduced the projected files having the correct owner permissions set.

### Linux

In Linux pods that have a projected volume and RunAsUser set in the Pod [SecurityContext](#), the projected files have the correct ownership set including container user ownership.

When all containers in a pod have the same runAsUser set in their [PodSecurityContext](#) or container [SecurityContext](#), then the kubelet ensures that the contents of the serviceAccountToken volume are owned by that user, and the token file has its permission mode set to 0600.

### Note:

[Ephemeral containers](#) added to a Pod after it is created do *not* change volume permissions that were set when the pod was created.

If a Pod's serviceAccountToken volume permissions were set to 0600 because all other containers in the Pod have the same runAsUser, ephemeral containers must use the same runAsUser to be able to read the token.

### Windows

In Windows pods that have a projected volume and RunAsUsername set in the Pod [SecurityContext](#), the ownership is not enforced due to the way user accounts are managed in

Windows. Windows stores and manages local user and group accounts in a database file called Security Account Manager (SAM). Each container maintains its own instance of the SAM database, to which the host has no visibility into while the container is running. Windows containers are designed to run the user mode portion of the OS in isolation from the host, hence the maintenance of a virtual SAM database. As a result, the kubelet running on the host does not have the ability to dynamically configure host file ownership for virtualized container accounts. It is recommended that if files on the host machine are to be shared with the container then they should be placed into their own volume mount outside of C:\.

By default, the projected files will have the following ownership as shown for an example projected volume file:

```
PS C:\> Get-Acl C:\var\run\secrets\kubernetes.io\serviceaccount\..2021_08_31_22_22_18.318230061\ca.crt | Format-List
```

```
Path : Microsoft.PowerShell.Core\FileSystem::C:
\var\run\secrets\kubernetes.io\serviceaccount\..2021_08_31_22_22_18.318230061\ca.crt
Owner : BUILTIN\Administrators
Group : NT AUTHORITY\SYSTEM
Access : NT AUTHORITY\SYSTEM Allow FullControl
 BUILTIN\Administrators Allow FullControl
 BUILTIN\Users Allow ReadAndExecute, Synchronize
Audit :
Sddl : O:BAG:SYD:AI(A;ID;FA;;;SY)(A;ID;FA;;;BA)(A;ID;0x1200a9;;;BU)
```

This implies all administrator users like ContainerAdministrator will have read, write and execute access while, non-administrator users will have read and execute access.

### Note:

In general, granting the container access to the host is discouraged as it can open the door for potential security exploits.

Creating a Windows Pod with RunAsUser in its SecurityContext will result in the Pod being stuck at ContainerCreating forever. So it is advised to not use the Linux only RunAsUser option with Windows Pods.

## Ephemeral Volumes

This document describes *ephemeral volumes* in Kubernetes. Familiarity with [volumes](#) is suggested, in particular PersistentVolumeClaim and PersistentVolume.

Some applications need additional storage but don't care whether that data is stored persistently across restarts. For example, caching services are often limited by memory size and can move infrequently used data into storage that is slower than memory with little impact on overall performance.

Other applications expect some read-only input data to be present in files, like configuration data or secret keys.

*Ephemeral volumes* are designed for these use cases. Because volumes follow the Pod's lifetime and get created and deleted along with the Pod, Pods can be stopped and restarted without being limited to where some persistent volume is available.

Ephemeral volumes are specified *inline* in the Pod spec, which simplifies application deployment and management.

## Types of ephemeral volumes

Kubernetes supports several different kinds of ephemeral volumes for different purposes:

- [emptyDir](#): empty at Pod startup, with storage coming locally from the kubelet base directory (usually the root disk) or RAM
- [configMap](#), [downwardAPI](#), [secret](#): inject different kinds of Kubernetes data into a Pod
- [CSI ephemeral volumes](#): similar to the previous volume kinds, but provided by special [CSI](#) drivers which specifically [support this feature](#)
- [generic ephemeral volumes](#), which can be provided by all storage drivers that also support persistent volumes

`emptyDir`, `configMap`, `downwardAPI`, `secret` are provided as [local ephemeral storage](#). They are managed by kubelet on each node.

CSI ephemeral volumes *must* be provided by third-party CSI storage drivers.

Generic ephemeral volumes *can* be provided by third-party CSI storage drivers, but also by any other storage driver that supports dynamic provisioning. Some CSI drivers are written specifically for CSI ephemeral volumes and do not support dynamic provisioning: those then cannot be used for generic ephemeral volumes.

The advantage of using third-party drivers is that they can offer functionality that Kubernetes itself does not support, for example storage with different performance characteristics than the disk that is managed by kubelet, or injecting different data.

## CSI ephemeral volumes

FEATURE STATE: Kubernetes v1.25 [stable]

### Note:

CSI ephemeral volumes are only supported by a subset of CSI drivers. The [Kubernetes CSI Drivers list](#) shows which drivers support ephemeral volumes.

Conceptually, CSI ephemeral volumes are similar to `configMap`, `downwardAPI` and `secret` volume types: the storage is managed locally on each node and is created together with other local resources after a Pod has been scheduled onto a node. Kubernetes has no concept of rescheduling Pods anymore at this stage. Volume creation has to be unlikely to fail, otherwise Pod startup gets stuck. In particular, [storage capacity aware Pod scheduling](#) is *not* supported for these volumes. They are currently also not covered by the storage resource usage limits of a Pod, because that is something that kubelet can only enforce for storage that it manages itself.

Here's an example manifest for a Pod that uses CSI ephemeral storage:

```

kind: Pod
apiVersion: v1
metadata:
 name: my-csi-app
spec:
 containers:
 - name: my-frontend
 image: busybox:1.28
 volumeMounts:
 - mountPath: "/data"
 name: my-csi-inline-vol
 command: ["sleep", "1000000"]
 volumes:
 - name: my-csi-inline-vol
 csi:
 driver: inline.storage.kubernetes.io
 volumeAttributes:
 foo: bar

```

The volumeAttributes determine what volume is prepared by the driver. These attributes are specific to each driver and not standardized. See the documentation of each CSI driver for further instructions.

## CSI driver restrictions

CSI ephemeral volumes allow users to provide volumeAttributes directly to the CSI driver as part of the Pod spec. A CSI driver allowing volumeAttributes that are typically restricted to administrators is NOT suitable for use in an inline ephemeral volume. For example, parameters that are normally defined in the StorageClass should not be exposed to users through the use of inline ephemeral volumes.

Cluster administrators who need to restrict the CSI drivers that are allowed to be used as inline volumes within a Pod spec may do so by:

- Removing Ephemeral from volumeLifecycleModes in the CSIDriver spec, which prevents the driver from being used as an inline ephemeral volume.
- Using an [admission webhook](#) to restrict how this driver is used.

## Generic ephemeral volumes

FEATURE STATE: Kubernetes v1.23 [stable]

Generic ephemeral volumes are similar to emptyDir volumes in the sense that they provide a per-pod directory for scratch data that is usually empty after provisioning. But they may also have additional features:

- Storage can be local or network-attached.
- Volumes can have a fixed size that Pods are not able to exceed.
- Volumes may have some initial data, depending on the driver and parameters.
- Typical operations on volumes are supported assuming that the driver supports them, including [snapshotting](#), [cloning](#), [resizing](#), and [storage capacity tracking](#).

Example:

```

kind: Pod
apiVersion: v1
metadata:
 name: my-app
spec:
 containers:
 - name: my-frontend
 image: busybox:1.28
 volumeMounts:
 - mountPath: "/scratch"
 name: scratch-volume
 command: ["sleep", "1000000"]
 volumes:
 - name: scratch-volume
 ephemeral:
 volumeClaimTemplate:
 metadata:
 labels:
 type: my-frontend-volume
 spec:
 accessModes: ["ReadWriteOnce"]
 storageClassName: "scratch-storage-class"
 resources:
 requests:
 storage: 1Gi

```

## Lifecycle and PersistentVolumeClaim

The key design idea is that the [parameters for a volume claim](#) are allowed inside a volume source of the Pod. Labels, annotations and the whole set of fields for a PersistentVolumeClaim are supported. When such a Pod gets created, the ephemeral volume controller then creates an actual PersistentVolumeClaim object in the same namespace as the Pod and ensures that the PersistentVolumeClaim gets deleted when the Pod gets deleted.

That triggers volume binding and/or provisioning, either immediately if the [StorageClass](#) uses immediate volume binding or when the Pod is tentatively scheduled onto a node (WaitForFirstConsumer volume binding mode). The latter is recommended for generic ephemeral volumes because then the scheduler is free to choose a suitable node for the Pod. With immediate binding, the scheduler is forced to select a node that has access to the volume once it is available.

In terms of [resource ownership](#), a Pod that has generic ephemeral storage is the owner of the PersistentVolumeClaim(s) that provide that ephemeral storage. When the Pod is deleted, the Kubernetes garbage collector deletes the PVC, which then usually triggers deletion of the volume because the default reclaim policy of storage classes is to delete volumes. You can create quasi-ephemeral local storage using a StorageClass with a reclaim policy of retain: the storage outlives the Pod, and in this case you need to ensure that volume clean up happens separately.

While these PVCs exist, they can be used like any other PVC. In particular, they can be referenced as data source in volume cloning or snapshotting. The PVC object also holds the current status of the volume.

## PersistentVolumeClaim naming

Naming of the automatically created PVCs is deterministic: the name is a combination of the Pod name and volume name, with a hyphen (-) in the middle. In the example above, the PVC name will be my-app-scratch-volume. This deterministic naming makes it easier to interact with the PVC because one does not have to search for it once the Pod name and volume name are known.

The deterministic naming also introduces a potential conflict between different Pods (a Pod "pod-a" with volume "scratch" and another Pod with name "pod" and volume "a-scratch" both end up with the same PVC name "pod-a-scratch") and between Pods and manually created PVCs.

Such conflicts are detected: a PVC is only used for an ephemeral volume if it was created for the Pod. This check is based on the ownership relationship. An existing PVC is not overwritten or modified. But this does not resolve the conflict because without the right PVC, the Pod cannot start.

### Caution:

Take care when naming Pods and volumes inside the same namespace, so that these conflicts can't occur.

## Security

Using generic ephemeral volumes allows users to create PVCs indirectly if they can create Pods, even if they do not have permission to create PVCs directly. Cluster administrators must be aware of this. If this does not fit their security model, they should use an [admission webhook](#) that rejects objects like Pods that have a generic ephemeral volume.

The normal [namespace quota for PVCs](#) still applies, so even if users are allowed to use this new mechanism, they cannot use it to circumvent other policies.

## What's next

### Ephemeral volumes managed by kubelet

See [local ephemeral storage](#).

### CSI ephemeral volumes

- For more information on the design, see the [Ephemeral Inline CSI volumes KEP](#).
- For more information on further development of this feature, see the [enhancement tracking issue #596](#).

### Generic ephemeral volumes

- For more information on the design, see the [Generic ephemeral inline volumes KEP](#).



# Storage Classes

This document describes the concept of a StorageClass in Kubernetes. Familiarity with [volumes](#) and [persistent volumes](#) is suggested.

A StorageClass provides a way for administrators to describe the *classes* of storage they offer. Different classes might map to quality-of-service levels, or to backup policies, or to arbitrary policies determined by the cluster administrators. Kubernetes itself is unopinionated about what classes represent.

The Kubernetes concept of a storage class is similar to “profiles” in some other storage system designs.

## StorageClass objects

Each StorageClass contains the fields `provisioner`, `parameters`, and `reclaimPolicy`, which are used when a PersistentVolume belonging to the class needs to be dynamically provisioned to satisfy a PersistentVolumeClaim (PVC).

The name of a StorageClass object is significant, and is how users can request a particular class. Administrators set the name and other parameters of a class when first creating StorageClass objects.

As an administrator, you can specify a default StorageClass that applies to any PVCs that don't request a specific class. For more details, see the [PersistentVolumeClaim concept](#).

Here's an example of a StorageClass:

[storage/storageclass-low-latency.yaml](#)

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: low-latency
 annotations:
 storageclass.kubernetes.io/is-default-class: "false"
provisioner: csi-driver.example-vendor.example
reclaimPolicy: Retain # default value is Delete
allowVolumeExpansion: true
mountOptions:
 - discard # this might enable UNMAP / TRIM at the block storage layer
volumeBindingMode: WaitForFirstConsumer
parameters:
 guaranteedReadWriteLatency: "true" # provider-specific
```

## Default StorageClass

You can mark a StorageClass as the default for your cluster. For instructions on setting the default StorageClass, see [Change the default StorageClass](#).

When a PVC does not specify a `storageClassName`, the default StorageClass is used.



If you set the [storageclass.kubernetes.io/is-default-class](https://kubernetes.io/docs/concepts/storage/storage-classes/#default) annotation to true on more than one StorageClass in your cluster, and you then create a PersistentVolumeClaim with no storageClassName set, Kubernetes uses the most recently created default StorageClass.

#### Note:

You should try to only have one StorageClass in your cluster that is marked as the default. The reason that Kubernetes allows you to have multiple default StorageClasses is to allow for seamless migration.

You can create a PersistentVolumeClaim without specifying a storageClassName for the new PVC, and you can do so even when no default StorageClass exists in your cluster. In this case, the new PVC creates as you defined it, and the storageClassName of that PVC remains unset until a default becomes available.

You can have a cluster without any default StorageClass. If you don't mark any StorageClass as default (and one hasn't been set for you by, for example, a cloud provider), then Kubernetes cannot apply that defaulting for PersistentVolumeClaims that need it.

If or when a default StorageClass becomes available, the control plane identifies any existing PVCs without storageClassName. For the PVCs that either have an empty value for storageClassName or do not have this key, the control plane then updates those PVCs to set storageClassName to match the new default StorageClass. If you have an existing PVC where the storageClassName is "", and you configure a default StorageClass, then this PVC will not get updated.

In order to keep binding to PVs with storageClassName set to "" (while a default StorageClass is present), you need to set the storageClassName of the associated PVC to "".

## Provisioner

Each StorageClass has a provisioner that determines what volume plugin is used for provisioning PVs. This field must be specified.

| Volume Plugin  | Internal Provisioner | Config Example                  |
|----------------|----------------------|---------------------------------|
| AzureFile      |                      | <a href="#">Azure File</a>      |
| CephFS         | -                    | -                               |
| FC             | -                    | -                               |
| FlexVolume     | -                    | -                               |
| iSCSI          | -                    | -                               |
| Local          | -                    | <a href="#">Local</a>           |
| NFS            | -                    | <a href="#">NFS</a>             |
| PortworxVolume |                      | <a href="#">Portworx Volume</a> |
| RBD            | -                    | <a href="#">Ceph RBD</a>        |
| VsphereVolume  |                      | <a href="#">vSphere</a>         |

You are not restricted to specifying the "internal" provisioners listed here (whose names are prefixed with "kubernetes.io" and shipped alongside Kubernetes). You can also run and specify external provisioners, which are independent programs that follow a [specification](#) defined by Kubernetes. Authors of external provisioners have full discretion over where their code lives, how the provisioner is shipped, how it needs to be run, what volume plugin it uses (including

Flex), etc. The repository [kubernetes-sigs/sig-storage-lib-external-provisioner](https://github.com/kubernetes-sigs/sig-storage-lib-external-provisioner) houses a library for writing external provisioners that implements the bulk of the specification. Some external provisioners are listed under the repository [kubernetes-sigs/sig-storage-lib-external-provisioner](https://github.com/kubernetes-sigs/sig-storage-lib-external-provisioner).

For example, NFS doesn't provide an internal provisioner, but an external provisioner can be used. There are also cases when 3rd party storage vendors provide their own external provisioner.

## Reclaim policy

PersistentVolumes that are dynamically created by a StorageClass will have the [reclaim policy](#) specified in the reclaimPolicy field of the class, which can be either Delete or Retain. If no reclaimPolicy is specified when a StorageClass object is created, it will default to Delete.

PersistentVolumes that are created manually and managed via a StorageClass will have whatever reclaim policy they were assigned at creation.

## Volume expansion

PersistentVolumes can be configured to be expandable. This allows you to resize the volume by editing the corresponding PVC object, requesting a new larger amount of storage.

The following types of volumes support volume expansion, when the underlying StorageClass has the field allowVolumeExpansion set to true.

Table of Volume types and the version of Kubernetes they require

| Volume type | Required Kubernetes version for volume expansion |
|-------------|--------------------------------------------------|
| Azure File  | 1.11                                             |
| CSI         | 1.24                                             |
| FlexVolume  | 1.13                                             |
| Portworx    | 1.11                                             |
| rbd         | 1.11                                             |

### Note:

You can only use the volume expansion feature to grow a Volume, not to shrink it.

## Mount options

PersistentVolumes that are dynamically created by a StorageClass will have the mount options specified in the mountOptions field of the class.

If the volume plugin does not support mount options but mount options are specified, provisioning will fail. Mount options are **not** validated on either the class or PV. If a mount option is invalid, the PV mount fails.

# Volume binding mode

The `volumeBindingMode` field controls when [volume binding and dynamic provisioning](#) should occur. When unset, Immediate mode is used by default.

The Immediate mode indicates that volume binding and dynamic provisioning occurs once the `PersistentVolumeClaim` is created. For storage backends that are topology-constrained and not globally accessible from all Nodes in the cluster, `PersistentVolumes` will be bound or provisioned without knowledge of the Pod's scheduling requirements. This may result in unschedulable Pods.

A cluster administrator can address this issue by specifying the `WaitForFirstConsumer` mode which will delay the binding and provisioning of a `PersistentVolume` until a Pod using the `PersistentVolumeClaim` is created. `PersistentVolumes` will be selected or provisioned conforming to the topology that is specified by the Pod's scheduling constraints. These include, but are not limited to, [resource requirements](#), [node selectors](#), [pod affinity and anti-affinity](#), and [taints and tolerations](#).

The following plugins support `WaitForFirstConsumer` with dynamic provisioning:

- CSI volumes, provided that the specific CSI driver supports this

The following plugins support `WaitForFirstConsumer` with pre-created `PersistentVolume` binding:

- CSI volumes, provided that the specific CSI driver supports this
- [local](#)

## Note:

If you choose to use `WaitForFirstConsumer`, do not use `nodeName` in the Pod spec to specify node affinity. If `nodeName` is used in this case, the scheduler will be bypassed and PVC will remain in pending state.

Instead, you can use node selector for `kubernetes.io/hostname`:

```
apiVersion: v1
kind: Pod
metadata:
 name: task-pv-pod
spec:
 nodeSelector:
 kubernetes.io/hostname: kube-01
 volumes:
 - name: task-pv-storage
 persistentVolumeClaim:
 claimName: task-pv-claim
 containers:
 - name: task-pv-container
 image: nginx
 ports:
 - containerPort: 80
 name: "http-server"
```

```
volumeMounts:
 - mountPath: "/usr/share/nginx/html"
 name: task-pv-storage
```

## Allowed topologies

When a cluster operator specifies the `WaitForFirstConsumer` volume binding mode, it is no longer necessary to restrict provisioning to specific topologies in most situations. However, if still required, `allowedTopologies` can be specified.

This example demonstrates how to restrict the topology of provisioned volumes to specific zones and should be used as a replacement for the `zone` and `zones` parameters for the supported plugins.

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: standard
provisioner: kubernetes.io/example
parameters:
 type: pd-standard
volumeBindingMode: WaitForFirstConsumer
allowedTopologies:
 - matchLabelExpressions:
 - key: topology.kubernetes.io/zone
 values:
 - us-central-1a
 - us-central-1b
```

## Parameters

StorageClasses have parameters that describe volumes belonging to the storage class. Different parameters may be accepted depending on the provisioner. When a parameter is omitted, some default is used.

There can be at most 512 parameters defined for a StorageClass. The total length of the parameters object including its keys and values cannot exceed 256 KiB.

## AWS EBS

Kubernetes 1.30 does not include a `awsElasticBlockStore` volume type.

The `AWSElasticBlockStore` in-tree storage driver was deprecated in the Kubernetes v1.19 release and then removed entirely in the v1.27 release.

The Kubernetes project suggests that you use the [AWS EBS](#) out-of-tree storage driver instead.

Here is an example StorageClass for the AWS EBS CSI driver:

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
```

```
name: ebs-sc
provisioner: ebs.csi.aws.com
volumeBindingMode: WaitForFirstConsumer
parameters:
 csi.storage.k8s.io/fstype: xfs
 type: io1
 iopsPerGB: "50"
 encrypted: "true"
allowedTopologies:
- matchLabelExpressions:
 - key: topology.ebs.csi.aws.com/zone
 values:
 - us-east-2c
```

## NFS

To configure NFS storage, you can use the in-tree driver or the [NFS CSI driver for Kubernetes](#) (recommended).

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: example-nfs
provisioner: example.com/external-nfs
parameters:
 server: nfs-server.example.com
 path: /share
 readOnly: "false"
```

- **server:** Server is the hostname or IP address of the NFS server.
- **path:** Path that is exported by the NFS server.
- **readOnly:** A flag indicating whether the storage will be mounted as read only (default false).

Kubernetes doesn't include an internal NFS provisioner. You need to use an external provisioner to create a StorageClass for NFS. Here are some examples:

- [NFS Ganesha server and external provisioner](#)
- [NFS subdir external provisioner](#)

## vSphere

There are two types of provisioners for vSphere storage classes:

- [CSI provisioner](#): csi.vsphere.vmware.com
- [vCP provisioner](#): kubernetes.io/vsphere-volume

In-tree provisioners are [deprecated](#). For more information on the CSI provisioner, see [Kubernetes vSphere CSI Driver](#) and [vSphereVolume CSI migration](#).

## CSI Provisioner

The vSphere CSI StorageClass provisioner works with Tanzu Kubernetes clusters. For an example, refer to the [vSphere CSI repository](#).

## vCP Provisioner

The following examples use the VMware Cloud Provider (vCP) StorageClass provisioner.

1. Create a StorageClass with a user specified disk format.

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: fast
provisioner: kubernetes.io/vsphere-volume
parameters:
 diskformat: zeroedthick
```

diskformat: thin, zeroedthick and eagerzeroedthick. Default: "thin".

2. Create a StorageClass with a disk format on a user specified datastore.

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: fast
provisioner: kubernetes.io/vsphere-volume
parameters:
 diskformat: zeroedthick
 datastore: VSANDatastore
```

datastore: The user can also specify the datastore in the StorageClass. The volume will be created on the datastore specified in the StorageClass, which in this case is VSANDatastore. This field is optional. If the datastore is not specified, then the volume will be created on the datastore specified in the vSphere config file used to initialize the vSphere Cloud Provider.

3. Storage Policy Management inside kubernetes

- Using existing vCenter SPBM policy

One of the most important features of vSphere for Storage Management is policy based Management. Storage Policy Based Management (SPBM) is a storage policy framework that provides a single unified control plane across a broad range of data services and storage solutions. SPBM enables vSphere administrators to overcome upfront storage provisioning challenges, such as capacity planning, differentiated service levels and managing capacity headroom.

The SPBM policies can be specified in the StorageClass using the storagePolicyName parameter.

- Virtual SAN policy support inside Kubernetes

Vsphere Infrastructure (VI) Admins will have the ability to specify custom Virtual SAN Storage Capabilities during dynamic volume provisioning. You can now define storage requirements, such as performance and availability, in the form of storage capabilities during dynamic volume provisioning. The storage capability requirements are converted into a Virtual SAN policy which are then pushed down to the Virtual SAN layer when a persistent volume (virtual disk) is being created. The virtual disk is distributed across the Virtual SAN datastore to meet the requirements.

You can see [Storage Policy Based Management for dynamic provisioning of volumes](#) for more details on how to use storage policies for persistent volumes management.

There are few [vSphere examples](#) which you try out for persistent volume management inside Kubernetes for vSphere.

## Ceph RBD (deprecated)

### Note:

FEATURE STATE: Kubernetes v1.28 [deprecated]

This internal provisioner of Ceph RBD is deprecated. Please use [CephFS RBD CSI driver](#).

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: fast
provisioner: kubernetes.io/rbd
parameters:
 monitors: 10.16.153.105:6789
 adminId: kube
 adminSecretName: ceph-secret
 adminSecretNamespace: kube-system
 pool: kube
 userId: kube
 userSecretName: ceph-secret-user
 userSecretNamespace: default
 fsType: ext4
 imageFormat: "2"
 imageFeatures: "layering"
```

- **monitors:** Ceph monitors, comma delimited. This parameter is required.
- **adminId:** Ceph client ID that is capable of creating images in the pool. Default is "admin".
- **adminSecretName:** Secret Name for adminId. This parameter is required. The provided secret must have type "kubernetes.io/rbd".
- **adminSecretNamespace:** The namespace for adminSecretName. Default is "default".
- **pool:** Ceph RBD pool. Default is "rbd".
- **userId:** Ceph client ID that is used to map the RBD image. Default is the same as adminId.

- `userSecretName`: The name of Ceph Secret for `userId` to map RBD image. It must exist in the same namespace as PVCs. This parameter is required. The provided secret must have type "kubernetes.io/rbd", for example created in this way:

```
kubectl create secret generic ceph-secret --type="kubernetes.io/rbd" \
 --from-literal=key='QVFEQ1pMdFhPUnQrSmhBQUFYaERWNHJsZ3BsMmNjcDR6RFZS
T0E9PQ==' \
 --namespace=kube-system
```

- `userSecretNamespace`: The namespace for `userSecretName`.
- `fsType`: fsType that is supported by kubernetes. Default: "ext4".
- `imageFormat`: Ceph RBD image format, "1" or "2". Default is "2".
- `imageFeatures`: This parameter is optional and should only be used if you set `imageFormat` to "2". Currently supported features are layering only. Default is "", and no features are turned on.

## Azure Disk

Kubernetes 1.30 does not include a `azureDisk` volume type.

The `azureDisk` in-tree storage driver was deprecated in the Kubernetes v1.19 release and then removed entirely in the v1.27 release.

The Kubernetes project suggests that you use the [Azure Disk](#) third party storage driver instead.

## Azure File (deprecated)

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: azurefile
provisioner: kubernetes.io/azure-file
parameters:
 skuName: Standard_LRS
 location: eastus
 storageAccount: azure_storage_account_name
```

- `skuName`: Azure storage account SKU tier. Default is empty.
- `location`: Azure storage account location. Default is empty.
- `storageAccount`: Azure storage account name. Default is empty. If a storage account is not provided, all storage accounts associated with the resource group are searched to find one that matches `skuName` and `location`. If a storage account is provided, it must reside in the same resource group as the cluster, and `skuName` and `location` are ignored.
- `secretNamespace`: the namespace of the secret that contains the Azure Storage Account Name and Key. Default is the same as the Pod.
- `secretName`: the name of the secret that contains the Azure Storage Account Name and Key. Default is `azure-storage-account-<accountName>-secret`
- `readOnly`: a flag indicating whether the storage will be mounted as read only. Defaults to false which means a read/write mount. This setting will impact the `ReadOnly` setting in `VolumeMounts` as well.



During storage provisioning, a secret named by `secretName` is created for the mounting credentials. If the cluster has enabled both [RBAC](#) and [Controller Roles](#), add the create permission of resource secret for clusterrole system:controller:persistent-volume-binder.

In a multi-tenancy context, it is strongly recommended to set the value for `secretNamespace` explicitly, otherwise the storage account credentials may be read by other users.

## Portworx volume (deprecated)

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: portworx-io-priority-high
provisioner: kubernetes.io/portworx-volume
parameters:
 repl: "1"
 snap_interval: "70"
 priority_io: "high"
```

- `fs`: filesystem to be laid out: none/xfs/ext4 (default: ext4).
- `block_size`: block size in Kbytes (default: 32).
- `repl`: number of synchronous replicas to be provided in the form of replication factor 1..3 (default: 1) A string is expected here i.e. "1" and not 1.
- `priority_io`: determines whether the volume will be created from higher performance or a lower priority storage high/medium/low (default: low).
- `snap_interval`: clock/time interval in minutes for when to trigger snapshots. Snapshots are incremental based on difference with the prior snapshot, 0 disables snaps (default: 0). A string is expected here i.e. "70" and not 70.
- `aggregation_level`: specifies the number of chunks the volume would be distributed into, 0 indicates a non-aggregated volume (default: 0). A string is expected here i.e. "0" and not 0
- `ephemeral`: specifies whether the volume should be cleaned-up after unmount or should be persistent. emptyDir use case can set this value to true and persistent volumes use case such as for databases like Cassandra should set to false, true/false (default false). A string is expected here i.e. "true" and not true.

## Local

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: local-storage
provisioner: kubernetes.io/no-provisioner
volumeBindingMode: WaitForFirstConsumer
```

Local volumes do not support dynamic provisioning in Kubernetes 1.30; however a StorageClass should still be created to delay volume binding until a Pod is actually scheduled to the appropriate node. This is specified by the `WaitForFirstConsumer` volume binding mode.

Delaying volume binding allows the scheduler to consider all of a Pod's scheduling constraints when choosing an appropriate PersistentVolume for a PersistentVolumeClaim.

# Volume Attributes Classes

FEATURE STATE: Kubernetes v1.29 [alpha]

This page assumes that you are familiar with [StorageClasses](#), [volumes](#) and [PersistentVolumes](#) in Kubernetes.

A VolumeAttributesClass provides a way for administrators to describe the mutable "classes" of storage they offer. Different classes might map to different quality-of-service levels. Kubernetes itself is unopinionated about what these classes represent.

This is an alpha feature and disabled by default.

If you want to test the feature whilst it's alpha, you need to enable the VolumeAttributesClass [feature gate](#) for the kube-controller-manager and the kube-apiserver. You use the `--feature-gates` command line argument:

```
--feature-gates="...,VolumeAttributesClass=true"
```

You can also only use VolumeAttributesClasses with storage backed by [Container Storage Interface](#), and only where the relevant CSI driver implements the ModifyVolume API.

## The VolumeAttributesClass API

Each VolumeAttributesClass contains the `driverName` and `parameters`, which are used when a PersistentVolume (PV) belonging to the class needs to be dynamically provisioned or modified.

The name of a VolumeAttributesClass object is significant and is how users can request a particular class. Administrators set the name and other parameters of a class when first creating VolumeAttributesClass objects. While the name of a VolumeAttributesClass object in a PersistentVolumeClaim is mutable, the parameters in an existing class are immutable.

```
apiVersion: storage.k8s.io/v1alpha1
kind: VolumeAttributesClass
metadata:
 name: silver
driverName: pd.csi.storage.gke.io
parameters:
 provisioned-iops: "3000"
 provisioned-throughput: "50"
```

### Provisioner

Each VolumeAttributesClass has a provisioner that determines what volume plugin is used for provisioning PVs. The field `driverName` must be specified.

The feature support for VolumeAttributesClass is implemented in [kubernetes-csi/external-provisioner](#).

You are not restricted to specifying the [kubernetes-csi/external-provisioner](#). You can also run and specify external provisioners, which are independent programs that follow a specification defined by Kubernetes. Authors of external provisioners have full discretion over where their

code lives, how the provisioner is shipped, how it needs to be run, what volume plugin it uses, etc.

## Resizer

Each `VolumeAttributesClass` has a resizer that determines what volume plugin is used for modifying PVs. The field `driverName` must be specified.

The modifying volume feature support for `VolumeAttributesClass` is implemented in [kubernetes-csi/external-resizer](https://github.com/kubernetes-csi/external-resizer).

For example, a existing `PersistentVolumeClaim` is using a `VolumeAttributesClass` named `silver`:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: test-pv-claim
spec:
 ...
 volumeAttributesClassName: silver
 ...
```

A new `VolumeAttributesClass` `gold` is available in the cluster:

```
apiVersion: storage.k8s.io/v1alpha1
kind: VolumeAttributesClass
metadata:
 name: gold
driverName: pd.csi.storage.gke.io
parameters:
 iops: "4000"
 throughput: "60"
```

The end user can update the PVC with the new `VolumeAttributesClass` `gold` and apply:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: test-pv-claim
spec:
 ...
 volumeAttributesClassName: gold
 ...
```

## Parameters

`VolumeAttributeClasses` have parameters that describe volumes belonging to them. Different parameters may be accepted depending on the provisioner or the resizer. For example, the value 4000, for the parameter `iops`, and the parameter `throughput` are specific to GCE PD. When a parameter is omitted, the default is used at volume provisioning. If a user apply the PVC with a different `VolumeAttributesClass` with omitted parameters, the default value of the parameters may be used depends on the CSI driver implementation. Please refer to the related CSI driver documentation for more details.

There can be at most 512 parameters defined for a `VolumeAttributesClass`. The total length of the parameters object including its keys and values cannot exceed 256 KiB.

# Dynamic Volume Provisioning

Dynamic volume provisioning allows storage volumes to be created on-demand. Without dynamic provisioning, cluster administrators have to manually make calls to their cloud or storage provider to create new storage volumes, and then create [PersistentVolume objects](#) to represent them in Kubernetes. The dynamic provisioning feature eliminates the need for cluster administrators to pre-provision storage. Instead, it automatically provisions storage when users create [PersistentVolumeClaim objects](#).

## Background

The implementation of dynamic volume provisioning is based on the API object `StorageClass` from the API group `storage.k8s.io`. A cluster administrator can define as many `StorageClass` objects as needed, each specifying a *volume plugin* (aka *provisioner*) that provisions a volume and the set of parameters to pass to that provisioner when provisioning. A cluster administrator can define and expose multiple flavors of storage (from the same or different storage systems) within a cluster, each with a custom set of parameters. This design also ensures that end users don't have to worry about the complexity and nuances of how storage is provisioned, but still have the ability to select from multiple storage options.

More information on storage classes can be found [here](#).

## Enabling Dynamic Provisioning

To enable dynamic provisioning, a cluster administrator needs to pre-create one or more `StorageClass` objects for users. `StorageClass` objects define which provisioner should be used and what parameters should be passed to that provisioner when dynamic provisioning is invoked. The name of a `StorageClass` object must be a valid [DNS subdomain name](#).

The following manifest creates a storage class "slow" which provisions standard disk-like persistent disks.

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: slow
provisioner: kubernetes.io/gce-pd
parameters:
 type: pd-standard
```

The following manifest creates a storage class "fast" which provisions SSD-like persistent disks.

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: fast
provisioner: kubernetes.io/gce-pd
```

```
parameters:
 type: pd-ssd
```

## Using Dynamic Provisioning

Users request dynamically provisioned storage by including a storage class in their `PersistentVolumeClaim`. Before Kubernetes v1.6, this was done via the `volume.beta.kubernetes.io/storage-class` annotation. However, this annotation is deprecated since v1.9. Users now can and should instead use the `storageClassName` field of the `PersistentVolumeClaim` object. The value of this field must match the name of a `StorageClass` configured by the administrator (see [below](#)).

To select the "fast" storage class, for example, a user would create the following `PersistentVolumeClaim`:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: claim1
spec:
 accessModes:
 - ReadWriteOnce
 storageClassName: fast
 resources:
 requests:
 storage: 30Gi
```

This claim results in an SSD-like Persistent Disk being automatically provisioned. When the claim is deleted, the volume is destroyed.

## Defaulting Behavior

Dynamic provisioning can be enabled on a cluster such that all claims are dynamically provisioned if no storage class is specified. A cluster administrator can enable this behavior by:

- Marking one `StorageClass` object as *default*;
- Making sure that the [DefaultStorageClass admission controller](#) is enabled on the API server.

An administrator can mark a specific `StorageClass` as default by adding the [storageclass.kubernetes.io/is-default-class](#) annotation to it. When a default `StorageClass` exists in a cluster and a user creates a `PersistentVolumeClaim` with `storageClassName` unspecified, the `DefaultStorageClass` admission controller automatically adds the `storageClassName` field pointing to the default storage class.

Note that if you set the `storageclass.kubernetes.io/is-default-class` annotation to true on more than one `StorageClass` in your cluster, and you then create a `PersistentVolumeClaim` with no `storageClassName` set, Kubernetes uses the most recently created default `StorageClass`.

# Topology Awareness

In [Multi-Zone](#) clusters, Pods can be spread across Zones in a Region. Single-Zone storage backends should be provisioned in the Zones where Pods are scheduled. This can be accomplished by setting the [Volume Binding Mode](#).

## Volume Snapshots

In Kubernetes, a *VolumeSnapshot* represents a snapshot of a volume on a storage system. This document assumes that you are already familiar with Kubernetes [persistent volumes](#).

### Introduction

Similar to how API resources `PersistentVolume` and `PersistentVolumeClaim` are used to provision volumes for users and administrators, `VolumeSnapshotContent` and `VolumeSnapshot` API resources are provided to create volume snapshots for users and administrators.

A `VolumeSnapshotContent` is a snapshot taken from a volume in the cluster that has been provisioned by an administrator. It is a resource in the cluster just like a `PersistentVolume` is a cluster resource.

A `VolumeSnapshot` is a request for snapshot of a volume by a user. It is similar to a `PersistentVolumeClaim`.

`VolumeSnapshotClass` allows you to specify different attributes belonging to a `VolumeSnapshot`. These attributes may differ among snapshots taken from the same volume on the storage system and therefore cannot be expressed by using the same `StorageClass` of a `PersistentVolumeClaim`.

Volume snapshots provide Kubernetes users with a standardized way to copy a volume's contents at a particular point in time without creating an entirely new volume. This functionality enables, for example, database administrators to backup databases before performing edit or delete modifications.

Users need to be aware of the following when using this feature:

- API Objects `VolumeSnapshot`, `VolumeSnapshotContent`, and `VolumeSnapshotClass` are [CRDs](#), not part of the core API.
- `VolumeSnapshot` support is only available for CSI drivers.
- As part of the deployment process of `VolumeSnapshot`, the Kubernetes team provides a snapshot controller to be deployed into the control plane, and a sidecar helper container called `csi-snapshotter` to be deployed together with the CSI driver. The snapshot controller watches `VolumeSnapshot` and `VolumeSnapshotContent` objects and is responsible for the creation and deletion of `VolumeSnapshotContent` object. The sidecar `csi-snapshotter` watches `VolumeSnapshotContent` objects and triggers `CreateSnapshot` and `DeleteSnapshot` operations against a CSI endpoint.
- There is also a validating webhook server which provides tightened validation on snapshot objects. This should be installed by the Kubernetes distros along with the snapshot controller and CRDs, not CSI drivers. It should be installed in all Kubernetes clusters that has the snapshot feature enabled.

- CSI drivers may or may not have implemented the volume snapshot functionality. The CSI drivers that have provided support for volume snapshot will likely use the csi-snapshotter. See [CSI Driver documentation](#) for details.
- The CRDs and snapshot controller installations are the responsibility of the Kubernetes distribution.

## Lifecycle of a volume snapshot and volume snapshot content

VolumeSnapshotContents are resources in the cluster. VolumeSnapshots are requests for those resources. The interaction between VolumeSnapshotContents and VolumeSnapshots follow this lifecycle:

### Provisioning Volume Snapshot

There are two ways snapshots may be provisioned: pre-provisioned or dynamically provisioned.

#### Pre-provisioned

A cluster administrator creates a number of VolumeSnapshotContents. They carry the details of the real volume snapshot on the storage system which is available for use by cluster users. They exist in the Kubernetes API and are available for consumption.

#### Dynamic

Instead of using a pre-existing snapshot, you can request that a snapshot to be dynamically taken from a PersistentVolumeClaim. The [VolumeSnapshotClass](#) specifies storage provider-specific parameters to use when taking a snapshot.

### Binding

The snapshot controller handles the binding of a VolumeSnapshot object with an appropriate VolumeSnapshotContent object, in both pre-provisioned and dynamically provisioned scenarios. The binding is a one-to-one mapping.

In the case of pre-provisioned binding, the VolumeSnapshot will remain unbound until the requested VolumeSnapshotContent object is created.

### Persistent Volume Claim as Snapshot Source Protection

The purpose of this protection is to ensure that in-use [PersistentVolumeClaim](#) API objects are not removed from the system while a snapshot is being taken from it (as this may result in data loss).

While a snapshot is being taken of a PersistentVolumeClaim, that PersistentVolumeClaim is in-use. If you delete a PersistentVolumeClaim API object in active use as a snapshot source, the PersistentVolumeClaim object is not removed immediately. Instead, removal of the PersistentVolumeClaim object is postponed until the snapshot is readyToUse or aborted.



## Delete

Deletion is triggered by deleting the `VolumeSnapshot` object, and the `DeletionPolicy` will be followed. If the `DeletionPolicy` is `Delete`, then the underlying storage snapshot will be deleted along with the `VolumeSnapshotContent` object. If the `DeletionPolicy` is `Retain`, then both the underlying snapshot and `VolumeSnapshotContent` remain.

## VolumeSnapshots

Each `VolumeSnapshot` contains a spec and a status.

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshot
metadata:
 name: new-snapshot-test
spec:
 volumeSnapshotClassName: csi-hostpath-snapclass
 source:
 persistentVolumeClaimName: pvc-test
```

`persistentVolumeClaimName` is the name of the `PersistentVolumeClaim` data source for the snapshot. This field is required for dynamically provisioning a snapshot.

A volume snapshot can request a particular class by specifying the name of a [VolumeSnapshotClass](#) using the attribute `volumeSnapshotClassName`. If nothing is set, then the default class is used if available.

For pre-provisioned snapshots, you need to specify a `volumeSnapshotContentName` as the source for the snapshot as shown in the following example. The `volumeSnapshotContentName` source field is required for pre-provisioned snapshots.

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshot
metadata:
 name: test-snapshot
spec:
 source:
 volumeSnapshotContentName: test-content
```

## Volume Snapshot Contents

Each `VolumeSnapshotContent` contains a spec and status. In dynamic provisioning, the snapshot common controller creates `VolumeSnapshotContent` objects. Here is an example:

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshotContent
metadata:
 name: snapcontent-72d9a349-aacd-42d2-a240-d775650d2455
spec:
 deletionPolicy: Delete
 driver: hostpath.csi.k8s.io
 source:
```



```
volumeHandle: ee0cfb94-f8d4-11e9-b2d8-0242ac110002
sourceVolumeMode: Filesystem
volumeSnapshotClassName: csi-hostpath-snapclass
volumeSnapshotRef:
 name: new-snapshot-test
 namespace: default
 uid: 72d9a349-aacd-42d2-a240-d775650d2455
```

`volumeHandle` is the unique identifier of the volume created on the storage backend and returned by the CSI driver during the volume creation. This field is required for dynamically provisioning a snapshot. It specifies the volume source of the snapshot.

For pre-provisioned snapshots, you (as cluster administrator) are responsible for creating the `VolumeSnapshotContent` object as follows.

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshotContent
metadata:
 name: new-snapshot-content-test
spec:
 deletionPolicy: Delete
 driver: hostpath.csi.k8s.io
 source:
 snapshotHandle: 7bdd0de3-aaeb-11e8-9aae-0242ac110002
 sourceVolumeMode: Filesystem
 volumeSnapshotRef:
 name: new-snapshot-test
 namespace: default
```

`snapshotHandle` is the unique identifier of the volume snapshot created on the storage backend. This field is required for the pre-provisioned snapshots. It specifies the CSI snapshot id on the storage system that this `VolumeSnapshotContent` represents.

`sourceVolumeMode` is the mode of the volume whose snapshot is taken. The value of the `sourceVolumeMode` field can be either `Filesystem` or `Block`. If the source volume mode is not specified, Kubernetes treats the snapshot as if the source volume's mode is unknown.

`volumeSnapshotRef` is the reference of the corresponding `VolumeSnapshot`. Note that when the `VolumeSnapshotContent` is being created as a pre-provisioned snapshot, the `VolumeSnapshot` referenced in `volumeSnapshotRef` might not exist yet.

## Converting the volume mode of a Snapshot

If the `VolumeSnapshots` API installed on your cluster supports the `sourceVolumeMode` field, then the API has the capability to prevent unauthorized users from converting the mode of a volume.

To check if your cluster has capability for this feature, run the following command:

```
$ kubectl get crd volumesnapshotcontent -o yaml
```

If you want to allow users to create a `PersistentVolumeClaim` from an existing `VolumeSnapshot`, but with a different volume mode than the source, the annotation

`snapshot.storage.kubernetes.io/allow-volume-mode-change: "true"` needs to be added to the `VolumeSnapshotContent` that corresponds to the `VolumeSnapshot`.

For pre-provisioned snapshots, `spec.sourceVolumeMode` needs to be populated by the cluster administrator.

An example `VolumeSnapshotContent` resource with this feature enabled would look like:

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshotContent
metadata:
 name: new-snapshot-content-test
 annotations:
 - snapshot.storage.kubernetes.io/allow-volume-mode-change: "true"
spec:
 deletionPolicy: Delete
 driver: hostpath.csi.k8s.io
 source:
 snapshotHandle: 7bdd0de3-aaeb-11e8-9aae-0242ac110002
 sourceVolumeMode: Filesystem
 volumeSnapshotRef:
 name: new-snapshot-test
 namespace: default
```

## Provisioning Volumes from Snapshots

You can provision a new volume, pre-populated with data from a snapshot, by using the `dataSource` field in the `PersistentVolumeClaim` object.

For more details, see [Volume Snapshot and Restore Volume from Snapshot](#).

## Volume Snapshot Classes

This document describes the concept of `VolumeSnapshotClass` in Kubernetes. Familiarity with [volume snapshots](#) and [storage classes](#) is suggested.

### Introduction

Just like `StorageClass` provides a way for administrators to describe the "classes" of storage they offer when provisioning a volume, `VolumeSnapshotClass` provides a way to describe the "classes" of storage when provisioning a volume snapshot.

### The VolumeSnapshotClass Resource

Each `VolumeSnapshotClass` contains the fields `driver`, `deletionPolicy`, and `parameters`, which are used when a `VolumeSnapshot` belonging to the class needs to be dynamically provisioned.

The name of a `VolumeSnapshotClass` object is significant, and is how users can request a particular class. Administrators set the name and other parameters of a class when first creating `VolumeSnapshotClass` objects, and the objects cannot be updated once they are created.

## Note:

Installation of the CRDs is the responsibility of the Kubernetes distribution. Without the required CRDs present, the creation of a VolumeSnapshotClass fails.

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshotClass
metadata:
 name: csi-hostpath-snapclass
driver: hostpath.csi.k8s.io
deletionPolicy: Delete
parameters:
```

Administrators can specify a default VolumeSnapshotClass for VolumeSnapshots that don't request any particular class to bind to by adding the `snapshot.storage.kubernetes.io/is-default-class: "true"` annotation:

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshotClass
metadata:
 name: csi-hostpath-snapclass
 annotations:
 snapshot.storage.kubernetes.io/is-default-class: "true"
driver: hostpath.csi.k8s.io
deletionPolicy: Delete
parameters:
```

## Driver

Volume snapshot classes have a driver that determines what CSI volume plugin is used for provisioning VolumeSnapshots. This field must be specified.

## DeletionPolicy

Volume snapshot classes have a [deletionPolicy](#). It enables you to configure what happens to a VolumeSnapshotContent when the VolumeSnapshot object it is bound to is to be deleted. The deletionPolicy of a volume snapshot class can either be Retain or Delete. This field must be specified.

If the deletionPolicy is Delete, then the underlying storage snapshot will be deleted along with the VolumeSnapshotContent object. If the deletionPolicy is Retain, then both the underlying snapshot and VolumeSnapshotContent remain.

## Parameters

Volume snapshot classes have parameters that describe volume snapshots belonging to the volume snapshot class. Different parameters may be accepted depending on the driver.

# CSI Volume Cloning

This document describes the concept of cloning existing CSI Volumes in Kubernetes. Familiarity with [Volumes](#) is suggested.

## Introduction

The [CSI](#) Volume Cloning feature adds support for specifying existing [PVCs](#) in the `dataSource` field to indicate a user would like to clone a [Volume](#).

A Clone is defined as a duplicate of an existing Kubernetes Volume that can be consumed as any standard Volume would be. The only difference is that upon provisioning, rather than creating a "new" empty Volume, the back end device creates an exact duplicate of the specified Volume.

The implementation of cloning, from the perspective of the Kubernetes API, adds the ability to specify an existing PVC as a `dataSource` during new PVC creation. The source PVC must be bound and available (not in use).

Users need to be aware of the following when using this feature:

- Cloning support (`VolumePVCDDataSource`) is only available for CSI drivers.
- Cloning support is only available for dynamic provisioners.
- CSI drivers may or may not have implemented the volume cloning functionality.
- You can only clone a PVC when it exists in the same namespace as the destination PVC (source and destination must be in the same namespace).
- Cloning is supported with a different Storage Class.
  - Destination volume can be the same or a different storage class as the source.
  - Default storage class can be used and `storageClassName` omitted in the spec.
- Cloning can only be performed between two volumes that use the same `VolumeMode` setting (if you request a block mode volume, the source **MUST** also be block mode)

## Provisioning

Clones are provisioned like any other PVC with the exception of adding a `dataSource` that references an existing PVC in the same namespace.

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: clone-of-pvc-1
 namespace: myns
spec:
 accessModes:
 - ReadWriteOnce
 storageClassName: cloning
 resources:
 requests:
 storage: 5Gi
 dataSource:
```

```
kind: PersistentVolumeClaim
name: pvc-1
```

#### Note:

You must specify a capacity value for `spec.resources.requests.storage`, and the value you specify must be the same or larger than the capacity of the source volume.

The result is a new PVC with the name `clone-of-pvc-1` that has the exact same content as the specified source `pvc-1`.

## Usage

Upon availability of the new PVC, the cloned PVC is consumed the same as other PVC. It's also expected at this point that the newly created PVC is an independent object. It can be consumed, cloned, snapshotted, or deleted independently and without consideration for its original `dataSource` PVC. This also implies that the source is not linked in any way to the newly created clone, it may also be modified or deleted without affecting the newly created clone.

# Storage Capacity

Storage capacity is limited and may vary depending on the node on which a pod runs: network-attached storage might not be accessible by all nodes, or storage is local to a node to begin with.

FEATURE STATE: Kubernetes v1.24 [stable]

This page describes how Kubernetes keeps track of storage capacity and how the scheduler uses that information to [schedule Pods](#) onto nodes that have access to enough storage capacity for the remaining missing volumes. Without storage capacity tracking, the scheduler may choose a node that doesn't have enough capacity to provision a volume and multiple scheduling retries will be needed.

## Before you begin

Kubernetes v1.30 includes cluster-level API support for storage capacity tracking. To use this you must also be using a CSI driver that supports capacity tracking. Consult the documentation for the CSI drivers that you use to find out whether this support is available and, if so, how to use it. If you are not running Kubernetes v1.30, check the documentation for that version of Kubernetes.

## API

There are two API extensions for this feature:

- [CSIStorageCapacity](#) objects: these get produced by a CSI driver in the namespace where the driver is installed. Each object contains capacity information for one storage class and defines which nodes have access to that storage.
- [The CSIDriverSpec.StorageCapacity field](#): when set to true, the Kubernetes scheduler will consider storage capacity for volumes that use the CSI driver.

# Scheduling

Storage capacity information is used by the Kubernetes scheduler if:

- a Pod uses a volume that has not been created yet,
- that volume uses a [StorageClass](#) which references a CSI driver and uses WaitForFirstConsumer [volume binding mode](#), and
- the CSIDriver object for the driver has StorageCapacity set to true.

In that case, the scheduler only considers nodes for the Pod which have enough storage available to them. This check is very simplistic and only compares the size of the volume against the capacity listed in CSIStorageCapacity objects with a topology that includes the node.

For volumes with Immediate volume binding mode, the storage driver decides where to create the volume, independently of Pods that will use the volume. The scheduler then schedules Pods onto nodes where the volume is available after the volume has been created.

For [CSI ephemeral volumes](#), scheduling always happens without considering storage capacity. This is based on the assumption that this volume type is only used by special CSI drivers which are local to a node and do not need significant resources there.

## Rescheduling

When a node has been selected for a Pod with WaitForFirstConsumer volumes, that decision is still tentative. The next step is that the CSI storage driver gets asked to create the volume with a hint that the volume is supposed to be available on the selected node.

Because Kubernetes might have chosen a node based on out-dated capacity information, it is possible that the volume cannot really be created. The node selection is then reset and the Kubernetes scheduler tries again to find a node for the Pod.

## Limitations

Storage capacity tracking increases the chance that scheduling works on the first try, but cannot guarantee this because the scheduler has to decide based on potentially out-dated information. Usually, the same retry mechanism as for scheduling without any storage capacity information handles scheduling failures.

One situation where scheduling can fail permanently is when a Pod uses multiple volumes: one volume might have been created already in a topology segment which then does not have enough capacity left for another volume. Manual intervention is necessary to recover from this, for example by increasing capacity or deleting the volume that was already created.

## What's next

- For more information on the design, see the [Storage Capacity Constraints for Pod Scheduling KEP](#).

# Node-specific Volume Limits

This page describes the maximum number of volumes that can be attached to a Node for various cloud providers.

Cloud providers like Google, Amazon, and Microsoft typically have a limit on how many volumes can be attached to a Node. It is important for Kubernetes to respect those limits. Otherwise, Pods scheduled on a Node could get stuck waiting for volumes to attach.

## Kubernetes default limits

The Kubernetes scheduler has default limits on the number of volumes that can be attached to a Node:

| Cloud service                                    | Maximum volumes per Node |
|--------------------------------------------------|--------------------------|
| <a href="#">Amazon Elastic Block Store (EBS)</a> | 39                       |
| <a href="#">Google Persistent Disk</a>           | 16                       |
| <a href="#">Microsoft Azure Disk Storage</a>     | 16                       |

## Custom limits

You can change these limits by setting the value of the `KUBE_MAX_PD_VOLS` environment variable, and then starting the scheduler. CSI drivers might have a different procedure, see their documentation on how to customize their limits.

Use caution if you set a limit that is higher than the default limit. Consult the cloud provider's documentation to make sure that Nodes can actually support the limit you set.

The limit applies to the entire cluster, so it affects all Nodes.

## Dynamic volume limits

FEATURE STATE: [Kubernetes v1.17](#) [stable]

Dynamic volume limits are supported for following volume types.

- Amazon EBS
- Google Persistent Disk
- Azure Disk
- CSI

For volumes managed by in-tree volume plugins, Kubernetes automatically determines the Node type and enforces the appropriate maximum number of volumes for the node. For example:

- On [Google Compute Engine](#), up to 127 volumes can be attached to a node, [depending on the node type](#).

- For Amazon EBS disks on M5,C5,R5,T3 and Z1D instance types, Kubernetes allows only
- 25 volumes to be attached to a Node. For other instance types on [Amazon Elastic Compute Cloud \(EC2\)](#), Kubernetes allows 39 volumes to be attached to a Node.
  - On Azure, up to 64 disks can be attached to a node, depending on the node type. For more details, refer to [Sizes for virtual machines in Azure](#).
  - If a CSI storage driver advertises a maximum number of volumes for a Node (using NodeGetInfo), the [kube-scheduler](#) honors that limit. Refer to the [CSI specifications](#) for details.
  - For volumes managed by in-tree plugins that have been migrated to a CSI driver, the maximum number of volumes will be the one reported by the CSI driver.

# Volume Health Monitoring

FEATURE STATE: Kubernetes v1.21 [alpha]

[CSI](#) volume health monitoring allows CSI Drivers to detect abnormal volume conditions from the underlying storage systems and report them as events on [PVCs](#) or [Pods](#).

## Volume health monitoring

Kubernetes *volume health monitoring* is part of how Kubernetes implements the Container Storage Interface (CSI). Volume health monitoring feature is implemented in two components: an External Health Monitor controller, and the [kubelet](#).

If a CSI Driver supports Volume Health Monitoring feature from the controller side, an event will be reported on the related [PersistentVolumeClaim](#) (PVC) when an abnormal volume condition is detected on a CSI volume.

The External Health Monitor [controller](#) also watches for node failure events. You can enable node failure monitoring by setting the enable-node-watcher flag to true. When the external health monitor detects a node failure event, the controller reports an Event will be reported on the PVC to indicate that pods using this PVC are on a failed node.

If a CSI Driver supports Volume Health Monitoring feature from the node side, an Event will be reported on every Pod using the PVC when an abnormal volume condition is detected on a CSI volume. In addition, Volume Health information is exposed as Kubelet VolumeStats metrics. A new metric kubelet\_volume\_stats\_health\_status\_abnormal is added. This metric includes two labels: namespace and persistentvolumeclaim. The count is either 1 or 0. 1 indicates the volume is unhealthy, 0 indicates volume is healthy. For more information, please check [KEP](#).

### Note:

You need to enable the CSIVolumeHealth [feature gate](#) to use this feature from the node side.

## What's next

See the [CSI driver documentation](#) to find out which CSI drivers have implemented this feature.



# Windows Storage

This page provides an storage overview specific to the Windows operating system.

## Persistent storage

Windows has a layered filesystem driver to mount container layers and create a copy filesystem based on NTFS. All file paths in the container are resolved only within the context of that container.

- With Docker, volume mounts can only target a directory in the container, and not an individual file. This limitation does not apply to containerd.
- Volume mounts cannot project files or directories back to the host filesystem.
- Read-only filesystems are not supported because write access is always required for the Windows registry and SAM database. However, read-only volumes are supported.
- Volume user-masks and permissions are not available. Because the SAM is not shared between the host & container, there's no mapping between them. All permissions are resolved within the context of the container.

As a result, the following storage functionality is not supported on Windows nodes:

- Volume subpath mounts: only the entire volume can be mounted in a Windows container
- Subpath volume mounting for Secrets
- Host mount projection
- Read-only root filesystem (mapped volumes still support readOnly)
- Block device mapping
- Memory as the storage medium (for example, emptyDir.medium set to Memory)
- File system features like uid/gid; per-user Linux filesystem permissions
- Setting [secret permissions with DefaultMode](#) (due to UID/GID dependency)
- NFS based storage/volume support
- Expanding the mounted volume (resizefs)

Kubernetes [volumes](#) enable complex applications, with data persistence and Pod volume sharing requirements, to be deployed on Kubernetes. Management of persistent volumes associated with a specific storage back-end or protocol includes actions such as provisioning/de-provisioning/resizing of volumes, attaching/detaching a volume to/from a Kubernetes node and mounting/dismounting a volume to/from individual containers in a pod that needs to persist data.

Volume management components are shipped as Kubernetes volume [plugin](#). The following broad classes of Kubernetes volume plugins are supported on Windows:

- [FlexVolume plugins](#)
  - Please note that FlexVolumes have been deprecated as of 1.23
- [CSI Plugins](#)

### In-tree volume plugins

The following in-tree plugins support persistent storage on Windows nodes:

- [azureFile](#)
- [vsphereVolume](#)

# Configuration

Resources that Kubernetes provides for configuring Pods.

---

[Configuration Best Practices](#)

[ConfigMaps](#)

[Secrets](#)

[Resource Management for Pods and Containers](#)

[Organizing Cluster Access Using kubeconfig Files](#)

[Resource Management for Windows nodes](#)

## Configuration Best Practices

This document highlights and consolidates configuration best practices that are introduced throughout the user guide, Getting Started documentation, and examples.

This is a living document. If you think of something that is not on this list but might be useful to others, please don't hesitate to file an issue or submit a PR.

### General Configuration Tips

- When defining configurations, specify the latest stable API version.
- Configuration files should be stored in version control before being pushed to the cluster. This allows you to quickly roll back a configuration change if necessary. It also aids cluster re-creation and restoration.
- Write your configuration files using YAML rather than JSON. Though these formats can be used interchangeably in almost all scenarios, YAML tends to be more user-friendly.
- Group related objects into a single file whenever it makes sense. One file is often easier to manage than several. See the [guestbook-all-in-one.yaml](#) file as an example of this syntax.
- Note also that many kubectl commands can be called on a directory. For example, you can call kubectl apply on a directory of config files.
- Don't specify default values unnecessarily: simple, minimal configuration will make errors less likely.
- Put object descriptions in annotations, to allow better introspection.

## Note:

There is a breaking change introduced in the [YAML 1.2](#) boolean values specification with respect to [YAML 1.1](#). This is a known [issue](#) in Kubernetes. YAML 1.2 only recognizes **true** and **false** as valid booleans, while YAML 1.1 also accepts **yes**, **no**, **on**, and **off** as booleans. However, Kubernetes uses YAML [parsers](#) that are mostly compatible with YAML 1.1, which means that using **yes** or **no** instead of **true** or **false** in a YAML manifest may cause unexpected errors or behaviors. To avoid this issue, it is recommended to always use **true** or **false** for boolean values in YAML manifests, and to quote any strings that may be confused with booleans, such as "yes" or "no".

Besides booleans, there are additional specifications changes between YAML versions. Please refer to the [YAML Specification Changes](#) documentation for a comprehensive list.

## "Naked" Pods versus ReplicaSets, Deployments, and Jobs

- Don't use naked Pods (that is, Pods not bound to a [ReplicaSet](#) or [Deployment](#)) if you can avoid it. Naked Pods will not be rescheduled in the event of a node failure.

A Deployment, which both creates a ReplicaSet to ensure that the desired number of Pods is always available, and specifies a strategy to replace Pods (such as [RollingUpdate](#)), is almost always preferable to creating Pods directly, except for some explicit [restartPolicy: Never](#) scenarios. A [Job](#) may also be appropriate.

## Services

- Create a [Service](#) before its corresponding backend workloads (Deployments or ReplicaSets), and before any workloads that need to access it. When Kubernetes starts a container, it provides environment variables pointing to all the Services which were running when the container was started. For example, if a Service named foo exists, all containers will get the following variables in their initial environment:

```
FOO_SERVICE_HOST=<the host the Service is running on>
FOO_SERVICE_PORT=<the port the Service is running on>
```

*This does imply an ordering requirement* - any Service that a Pod wants to access must be created before the Pod itself, or else the environment variables will not be populated. DNS does not have this restriction.

- An optional (though strongly recommended) [cluster add-on](#) is a DNS server. The DNS server watches the Kubernetes API for new Services and creates a set of DNS records for each. If DNS has been enabled throughout the cluster then all Pods should be able to do name resolution of Services automatically.
- Don't specify a hostPort for a Pod unless it is absolutely necessary. When you bind a Pod to a hostPort, it limits the number of places the Pod can be scheduled, because each <hostIP, hostPort, protocol> combination must be unique. If you don't specify the hostIP and protocol explicitly, Kubernetes will use 0.0.0.0 as the default hostIP and TCP as the default protocol.

If you only need access to the port for debugging purposes, you can use the [apiserver proxy](#) or [kubectl port-forward](#).

If you explicitly need to expose a Pod's port on the node, consider using a [NodePort](#) Service before resorting to hostPort.

- Avoid using hostNetwork, for the same reasons as hostPort.
- Use [headless Services](#) (which have a ClusterIP of None) for service discovery when you don't need kube-proxy load balancing.

## Using Labels

- Define and use [labels](#) that identify **semantic attributes** of your application or Deployment, such as { app.kubernetes.io/name: MyApp, tier: frontend, phase: test, deployment: v3 }. You can use these labels to select the appropriate Pods for other resources; for example, a Service that selects all tier: frontend Pods, or all phase: test components of app.kubernetes.io/name: MyApp. See the [guestbook](#) app for examples of this approach.

A Service can be made to span multiple Deployments by omitting release-specific labels from its selector. When you need to update a running service without downtime, use a [Deployment](#).

A desired state of an object is described by a Deployment, and if changes to that spec are *applied*, the deployment controller changes the actual state to the desired state at a controlled rate.

- Use the [Kubernetes common labels](#) for common use cases. These standardized labels enrich the metadata in a way that allows tools, including kubectl and [dashboard](#), to work in an interoperable way.
- You can manipulate labels for debugging. Because Kubernetes controllers (such as ReplicaSet) and Services match to Pods using selector labels, removing the relevant labels from a Pod will stop it from being considered by a controller or from being served traffic by a Service. If you remove the labels of an existing Pod, its controller will create a new Pod to take its place. This is a useful way to debug a previously "live" Pod in a "quarantine" environment. To interactively remove or add labels, use [kubectl label](#).

## Using kubectl

- Use kubectl apply -f <directory>. This looks for Kubernetes configuration in all .yaml, .yml, and .json files in <directory> and passes it to apply.
- Use label selectors for get and delete operations instead of specific object names. See the sections on [label selectors](#) and [using labels effectively](#).
- Use kubectl create deployment and kubectl expose to quickly create single-container Deployments and Services. See [Use a Service to Access an Application in a Cluster](#) for an example.

## ConfigMaps

A ConfigMap is an API object used to store non-confidential data in key-value pairs. [Pods](#) can consume ConfigMaps as environment variables, command-line arguments, or as configuration files in a [volume](#).

A ConfigMap allows you to decouple environment-specific configuration from your [container images](#), so that your applications are easily portable.

### Caution:

ConfigMap does not provide secrecy or encryption. If the data you want to store are confidential, use a [Secret](#) rather than a ConfigMap, or use additional (third party) tools to keep your data private.

## Motivation

Use a ConfigMap for setting configuration data separately from application code.

For example, imagine that you are developing an application that you can run on your own computer (for development) and in the cloud (to handle real traffic). You write the code to look in an environment variable named DATABASE\_HOST. Locally, you set that variable to localhost. In the cloud, you set it to refer to a Kubernetes [Service](#) that exposes the database component to your cluster. This lets you fetch a container image running in the cloud and debug the exact same code locally if needed.

### Note:

A ConfigMap is not designed to hold large chunks of data. The data stored in a ConfigMap cannot exceed 1 MiB. If you need to store settings that are larger than this limit, you may want to consider mounting a volume or use a separate database or file service.

## ConfigMap object

A ConfigMap is an [API object](#) that lets you store configuration for other objects to use. Unlike most Kubernetes objects that have a spec, a ConfigMap has data and binaryData fields. These fields accept key-value pairs as their values. Both the data field and the binaryData are optional. The data field is designed to contain UTF-8 strings while the binaryData field is designed to contain binary data as base64-encoded strings.

The name of a ConfigMap must be a valid [DNS subdomain name](#).

Each key under the data or the binaryData field must consist of alphanumeric characters, -, \_ or .. The keys stored in data must not overlap with the keys in the binaryData field.

Starting from v1.19, you can add an immutable field to a ConfigMap definition to create an [immutable ConfigMap](#).

## ConfigMaps and Pods

You can write a Pod spec that refers to a ConfigMap and configures the container(s) in that Pod based on the data in the ConfigMap. The Pod and the ConfigMap must be in the same [namespace](#).

## Note:

The spec of a [static Pod](#) cannot refer to a ConfigMap or any other API objects.

Here's an example ConfigMap that has some keys with single values, and other keys where the value looks like a fragment of a configuration format.

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: game-demo
data:
 # property-like keys; each key maps to a simple value
 player_initial_lives: "3"
 ui_properties_file_name: "user-interface.properties"

 # file-like keys
 game.properties: |
 enemy.types=aliens,monsters
 player.maximum-lives=5
 user-interface.properties: |
 color.good=purple
 color.bad=yellow
 allow.textmode=true
```

There are four different ways that you can use a ConfigMap to configure a container inside a Pod:

1. Inside a container command and args
2. Environment variables for a container
3. Add a file in read-only volume, for the application to read
4. Write code to run inside the Pod that uses the Kubernetes API to read a ConfigMap

These different methods lend themselves to different ways of modeling the data being consumed. For the first three methods, the [kubelet](#) uses the data from the ConfigMap when it launches container(s) for a Pod.

The fourth method means you have to write code to read the ConfigMap and its data. However, because you're using the Kubernetes API directly, your application can subscribe to get updates whenever the ConfigMap changes, and react when that happens. By accessing the Kubernetes API directly, this technique also lets you access a ConfigMap in a different namespace.

Here's an example Pod that uses values from `game-demo` to configure a Pod:

[configmap/configure-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: configmap-demo-pod
spec:
 containers:
 - name: demo
 image: alpine
```

```

command: ["sleep", "3600"]
env:
 # Define the environment variable
 - name: PLAYER_INITIAL_LIVES # Notice that the case is different here
 # from the key name in the ConfigMap.
 valueFrom:
 configMapKeyRef:
 name: game-demo # The ConfigMap this value comes from.
 key: player_initial_lives # The key to fetch.
 - name: UI_PROPERTIES_FILE_NAME
 valueFrom:
 configMapKeyRef:
 name: game-demo
 key: ui_properties_file_name
volumeMounts:
 - name: config
 mountPath: "/config"
 readOnly: true
volumes:
 # You set volumes at the Pod level, then mount them into containers inside that Pod
 - name: config
 configMap:
 # Provide the name of the ConfigMap you want to mount.
 name: game-demo
 # An array of keys from the ConfigMap to create as files
 items:
 - key: "game.properties"
 path: "game.properties"
 - key: "user-interface.properties"
 path: "user-interface.properties"

```

A ConfigMap doesn't differentiate between single line property values and multi-line file-like values. What matters is how Pods and other objects consume those values.

For this example, defining a volume and mounting it inside the demo container as /config creates two files, /config/game.properties and /config/user-interface.properties, even though there are four keys in the ConfigMap. This is because the Pod definition specifies an items array in the volumes section. If you omit the items array entirely, every key in the ConfigMap becomes a file with the same name as the key, and you get 4 files.

## Using ConfigMaps

ConfigMaps can be mounted as data volumes. ConfigMaps can also be used by other parts of the system, without being directly exposed to the Pod. For example, ConfigMaps can hold data that other parts of the system should use for configuration.

The most common way to use ConfigMaps is to configure settings for containers running in a Pod in the same namespace. You can also use a ConfigMap separately.

For example, you might encounter [addons](#) or [operators](#) that adjust their behavior based on a ConfigMap.



## Using ConfigMaps as files from a Pod

To consume a ConfigMap in a volume in a Pod:

1. Create a ConfigMap or use an existing one. Multiple Pods can reference the same ConfigMap.
2. Modify your Pod definition to add a volume under `.spec.volumes[]`. Name the volume anything, and have a `.spec.volumes[].configMap.name` field set to reference your ConfigMap object.
3. Add a `.spec.containers[].volumeMounts[]` to each container that needs the ConfigMap. Specify `.spec.containers[].volumeMounts[].readOnly = true` and `.spec.containers[].volumeMounts[].mountPath` to an unused directory name where you would like the ConfigMap to appear.
4. Modify your image or command line so that the program looks for files in that directory. Each key in the ConfigMap data map becomes the filename under `mountPath`.

This is an example of a Pod that mounts a ConfigMap in a volume:

```
apiVersion: v1
kind: Pod
metadata:
 name: mypod
spec:
 containers:
 - name: mypod
 image: redis
 volumeMounts:
 - name: foo
 mountPath: "/etc/foo"
 readOnly: true
 volumes:
 - name: foo
 configMap:
 name: myconfigmap
```

Each ConfigMap you want to use needs to be referred to in `.spec.volumes`.

If there are multiple containers in the Pod, then each container needs its own `volumeMounts` block, but only one `.spec.volumes` is needed per ConfigMap.

### Mounted ConfigMaps are updated automatically

When a ConfigMap currently consumed in a volume is updated, projected keys are eventually updated as well. The kubelet checks whether the mounted ConfigMap is fresh on every periodic sync. However, the kubelet uses its local cache for getting the current value of the ConfigMap. The type of the cache is configurable using the `configMapAndSecretChangeDetectionStrategy` field in the [KubeletConfiguration struct](#). A ConfigMap can be either propagated by watch (default), ttl-based, or by redirecting all requests directly to the API server. As a result, the total delay from the moment when the ConfigMap is updated to the moment when new keys are projected to the Pod can be as long as the kubelet sync period + cache propagation delay, where the cache propagation delay depends on the chosen cache type (it equals to watch propagation delay, ttl of cache, or zero correspondingly).



ConfigMaps consumed as environment variables are not updated automatically and require a pod restart.

**Note:**

A container using a ConfigMap as a [subPath](#) volume mount will not receive ConfigMap updates.

## Using Configmaps as environment variables

To use a Configmap in an [environment variable](#) in a Pod:

1. For each container in your Pod specification, add an environment variable for each Configmap key that you want to use to the `env[].valueFrom.configMapKeyRef` field.
2. Modify your image and/or command line so that the program looks for values in the specified environment variables.

This is an example of defining a ConfigMap as a pod environment variable:

```
apiVersion: v1
kind: Pod
metadata:
 name: env-configmap
spec:
 containers:
 - name: envvars-test-container
 image: nginx
 env:
 - name: CONFIGMAP_USERNAME
 valueFrom:
 configMapKeyRef:
 name: myconfigmap
 key: username
```

It's important to note that the range of characters allowed for environment variable names in pods is [restricted](#). If any keys do not meet the rules, those keys are not made available to your container, though the Pod is allowed to start.

## Immutable ConfigMaps

FEATURE STATE: Kubernetes v1.21 [stable]

The Kubernetes feature *Immutable Secrets and ConfigMaps* provides an option to set individual Secrets and ConfigMaps as immutable. For clusters that extensively use ConfigMaps (at least tens of thousands of unique ConfigMap to Pod mounts), preventing changes to their data has the following advantages:

- protects you from accidental (or unwanted) updates that could cause applications outages
- improves performance of your cluster by significantly reducing load on kube-apiserver, by closing watches for ConfigMaps marked as immutable.

You can create an immutable ConfigMap by setting the `immutable` field to `true`. For example:

```
apiVersion: v1
kind: ConfigMap
metadata:
 ...
data:
 ...
immutable: true
```

Once a ConfigMap is marked as immutable, it is *not* possible to revert this change nor to mutate the contents of the data or the binaryData field. You can only delete and recreate the ConfigMap. Because existing Pods maintain a mount point to the deleted ConfigMap, it is recommended to recreate these pods.

## What's next

- Read about [Secrets](#).
- Read [Configure a Pod to Use a ConfigMap](#).
- Read about [changing a ConfigMap \(or any other Kubernetes object\)](#)
- Read [The Twelve-Factor App](#) to understand the motivation for separating code from configuration.

## Secrets

A Secret is an object that contains a small amount of sensitive data such as a password, a token, or a key. Such information might otherwise be put in a [Pod](#) specification or in a [container image](#). Using a Secret means that you don't need to include confidential data in your application code.

Because Secrets can be created independently of the Pods that use them, there is less risk of the Secret (and its data) being exposed during the workflow of creating, viewing, and editing Pods. Kubernetes, and applications that run in your cluster, can also take additional precautions with Secrets, such as avoiding writing sensitive data to nonvolatile storage.

Secrets are similar to [ConfigMaps](#) but are specifically intended to hold confidential data.

### Caution:

Kubernetes Secrets are, by default, stored unencrypted in the API server's underlying data store (etcd). Anyone with API access can retrieve or modify a Secret, and so can anyone with access to etcd. Additionally, anyone who is authorized to create a Pod in a namespace can use that access to read any Secret in that namespace; this includes indirect access such as the ability to create a Deployment.

In order to safely use Secrets, take at least the following steps:

1. [Enable Encryption at Rest](#) for Secrets.
2. [Enable or configure RBAC rules](#) with least-privilege access to Secrets.
3. Restrict Secret access to specific containers.
4. [Consider using external Secret store providers](#).

For more guidelines to manage and improve the security of your Secrets, refer to [Good practices for Kubernetes Secrets](#).

See [Information security for Secrets](#) for more details.

## Uses for Secrets

You can use Secrets for purposes such as the following:

- [Set environment variables for a container](#).
- [Provide credentials such as SSH keys or passwords to Pods](#).
- [Allow the kubelet to pull container images from private registries](#).

The Kubernetes control plane also uses Secrets; for example, [bootstrap token Secrets](#) are a mechanism to help automate node registration.

### Use case: dotfiles in a secret volume

You can make your data "hidden" by defining a key that begins with a dot. This key represents a dotfile or "hidden" file. For example, when the following Secret is mounted into a volume, secret-volume, the volume will contain a single file, called .secret-file, and the dotfile-test-container will have this file present at the path /etc/secret-volume/.secret-file.

#### Note:

Files beginning with dot characters are hidden from the output of `ls -l`; you must use `ls -la` to see them when listing directory contents.

[secret/dotfile-secret.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
 name: dotfile-secret
data:
 .secret-file: dmFsdWUtMg0KDQo=

apiVersion: v1
kind: Pod
metadata:
 name: secret-dotfiles-pod
spec:
 volumes:
 - name: secret-volume
 secret:
 secretName: dotfile-secret
 containers:
 - name: dotfile-test-container
 image: registry.k8s.io/busybox
 command:
 - ls
 - "-l"
 - "/etc/secret-volume"
```

```
volumeMounts:
 - name: secret-volume
 readOnly: true
 mountPath: "/etc/secret-volume"
```

## Use case: Secret visible to one container in a Pod

Consider a program that needs to handle HTTP requests, do some complex business logic, and then sign some messages with an HMAC. Because it has complex application logic, there might be an unnoticed remote file reading exploit in the server, which could expose the private key to an attacker.

This could be divided into two processes in two containers: a frontend container which handles user interaction and business logic, but which cannot see the private key; and a signer container that can see the private key, and responds to simple signing requests from the frontend (for example, over localhost networking).

With this partitioned approach, an attacker now has to trick the application server into doing something rather arbitrary, which may be harder than getting it to read a file.

## Alternatives to Secrets

Rather than using a Secret to protect confidential data, you can pick from alternatives.

Here are some of your options:

- If your cloud-native component needs to authenticate to another application that you know is running within the same Kubernetes cluster, you can use a [ServiceAccount](#) and its tokens to identify your client.
- There are third-party tools that you can run, either within or outside your cluster, that manage sensitive data. For example, a service that Pods access over HTTPS, that reveals a Secret if the client correctly authenticates (for example, with a ServiceAccount token).
- For authentication, you can implement a custom signer for X.509 certificates, and use [CertificateSigningRequests](#) to let that custom signer issue certificates to Pods that need them.
- You can use a [device plugin](#) to expose node-local encryption hardware to a specific Pod. For example, you can schedule trusted Pods onto nodes that provide a Trusted Platform Module, configured out-of-band.

You can also combine two or more of those options, including the option to use Secret objects themselves.

For example: implement (or deploy) an [operator](#) that fetches short-lived session tokens from an external service, and then creates Secrets based on those short-lived session tokens. Pods running in your cluster can make use of the session tokens, and operator ensures they are valid. This separation means that you can run Pods that are unaware of the exact mechanisms for issuing and refreshing those session tokens.

# Types of Secret

When creating a Secret, you can specify its type using the type field of the [Secret](#) resource, or certain equivalent kubectl command line flags (if available). The Secret type is used to facilitate programmatic handling of the Secret data.

Kubernetes provides several built-in types for some common usage scenarios. These types vary in terms of the validations performed and the constraints Kubernetes imposes on them.

| Built-in Type                       | Usage                                 |
|-------------------------------------|---------------------------------------|
| Opaque                              | arbitrary user-defined data           |
| kubernetes.io/service-account-token | ServiceAccount token                  |
| kubernetes.io/dockercfg             | serialized ~/.dockercfg file          |
| kubernetes.io/dockerconfigjson      | serialized ~/.docker/config.json file |
| kubernetes.io/basic-auth            | credentials for basic authentication  |
| kubernetes.io/ssh-auth              | credentials for SSH authentication    |
| kubernetes.io/tls                   | data for a TLS client or server       |
| bootstrap.kubernetes.io/token       | bootstrap token data                  |

You can define and use your own Secret type by assigning a non-empty string as the type value for a Secret object (an empty string is treated as an Opaque type).

Kubernetes doesn't impose any constraints on the type name. However, if you are using one of the built-in types, you must meet all the requirements defined for that type.

If you are defining a type of Secret that's for public use, follow the convention and structure the Secret type to have your domain name before the name, separated by a /. For example: cloud-hosting.example.net/cloud-api-credentials.

## Opaque Secrets

Opaque is the default Secret type if you don't explicitly specify a type in a Secret manifest. When you create a Secret using kubectl, you must use the generic subcommand to indicate an Opaque Secret type. For example, the following command creates an empty Secret of type Opaque:

```
kubectl create secret generic empty-secret
kubectl get secret empty-secret
```

The output looks like:

| NAME         | TYPE   | DATA | AGE  |
|--------------|--------|------|------|
| empty-secret | Opaque | 0    | 2m6s |

The DATA column shows the number of data items stored in the Secret. In this case, 0 means you have created an empty Secret.

## ServiceAccount token Secrets

A `kubernetes.io/service-account-token` type of Secret is used to store a token credential that identifies a [ServiceAccount](#). This is a legacy mechanism that provides long-lived ServiceAccount credentials to Pods.

In Kubernetes v1.22 and later, the recommended approach is to obtain a short-lived, automatically rotating ServiceAccount token by using the [TokenRequest](#) API instead. You can get these short-lived tokens using the following methods:

- Call the TokenRequest API either directly or by using an API client like kubectl. For example, you can use the [kubectl create token](#) command.
- Request a mounted token in a [projected volume](#) in your Pod manifest. Kubernetes creates the token and mounts it in the Pod. The token is automatically invalidated when the Pod that it's mounted in is deleted. For details, see [Launch a Pod using service account token projection](#).

### Note:

You should only create a ServiceAccount token Secret if you can't use the TokenRequest API to obtain a token, and the security exposure of persisting a non-expiring token credential in a readable API object is acceptable to you. For instructions, see [Manually create a long-lived API token for a ServiceAccount](#).

When using this Secret type, you need to ensure that the `kubernetes.io/service-account.name` annotation is set to an existing ServiceAccount name. If you are creating both the ServiceAccount and the Secret objects, you should create the ServiceAccount object first.

After the Secret is created, a Kubernetes [controller](#) fills in some other fields such as the `kubernetes.io/service-account.uid` annotation, and the token key in the data field, which is populated with an authentication token.

The following example configuration declares a ServiceAccount token Secret:

[secret/serviceaccount-token-secret.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
 name: secret-sa-sample
 annotations:
 kubernetes.io/service-account.name: "sa-name"
type: kubernetes.io/service-account-token
data:
 extra: YmFyCg==
```

After creating the Secret, wait for Kubernetes to populate the token key in the data field.

See the [ServiceAccount](#) documentation for more information on how ServiceAccounts work. You can also check the `automountServiceAccountToken` field and the `serviceAccountName` field of the [Pod](#) for information on referencing ServiceAccount credentials from within Pods.

## Docker config Secrets

If you are creating a Secret to store credentials for accessing a container image registry, you must use one of the following type values for that Secret:

- `kubernetes.io/dockercfg`: store a serialized `~/.dockercfg` which is the legacy format for configuring Docker command line. The Secret data field contains a `.dockercfg` key whose value is the content of a base64 encoded `~/.dockercfg` file.
- `kubernetes.io/dockerconfigjson`: store a serialized JSON that follows the same format rules as the `~/.docker/config.json` file, which is a new format for `~/.dockercfg`. The Secret data field must contain a `.dockerconfigjson` key for which the value is the content of a base64 encoded `~/.docker/config.json` file.

Below is an example for a `kubernetes.io/dockercfg` type of Secret:

[secret/dockercfg-secret.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
 name: secret-dockercfg
type: kubernetes.io/dockercfg
data:
 .dockercfg: |

eyJhdXRocyI6eyJodHRwczovL2V4YW1wbGUvdjEvIjpw7ImF1dGgiOiJvcGVuc2VzYW1lIn19fQo=
```

### Note:

If you do not want to perform the base64 encoding, you can choose to use the `stringData` field instead.

When you create Docker config Secrets using a manifest, the API server checks whether the expected key exists in the data field, and it verifies if the value provided can be parsed as a valid JSON. The API server doesn't validate if the JSON actually is a Docker config file.

You can also use `kubectl` to create a Secret for accessing a container registry, such as when you don't have a Docker configuration file:

```
kubectl create secret docker-registry secret-tiger-docker \
--docker-email=tiger@acme.example \
--docker-username=tiger \
--docker-password=pass1234 \
--docker-server=my-registry.example:5000
```

This command creates a Secret of type `kubernetes.io/dockerconfigjson`.

Retrieve the `.data.dockerconfigjson` field from that new Secret and decode the data:

```
kubectl get secret secret-tiger-docker -o jsonpath='{.data.*}' | base64 -d
```

The output is equivalent to the following JSON document (which is also a valid Docker configuration file):

```
{
 "auths": {
 "my-registry.example:5000": {
 "username": "tiger",
 "password": "pass1234",
 "email": "tiger@acme.example",
 "auth": "dGlnZXI6cGFzc2EyMzQ="
 }
 }
}
```

### Caution:

The auth value there is base64 encoded; it is obscured but not secret. Anyone who can read that Secret can learn the registry access bearer token.

It is suggested to use [credential providers](#) to dynamically and securely provide pull secrets on-demand.

## Basic authentication Secret

The `kubernetes.io/basic-auth` type is provided for storing credentials needed for basic authentication. When using this Secret type, the data field of the Secret must contain one of the following two keys:

- `username`: the user name for authentication
- `password`: the password or token for authentication

Both values for the above two keys are base64 encoded strings. You can alternatively provide the clear text content using the `stringData` field in the Secret manifest.

The following manifest is an example of a basic authentication Secret:

[secret/basicauth-secret.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
 name: secret-basic-auth
type: kubernetes.io/basic-auth
stringData:
 username: admin # required field for kubernetes.io/basic-auth
 password: t0p-Secret # required field for kubernetes.io/basic-auth
```

### Note:

The `stringData` field for a Secret does not work well with server-side apply.

The basic authentication Secret type is provided only for convenience. You can create an Opaque type for credentials used for basic authentication. However, using the defined and public Secret type (`kubernetes.io/basic-auth`) helps other people to understand the purpose of your Secret, and sets a convention for what key names to expect. The Kubernetes API verifies that the required keys are set for a Secret of this type.



## SSH authentication Secrets

The builtin type `kubernetes.io/ssh-auth` is provided for storing data used in SSH authentication. When using this Secret type, you will have to specify a `ssh-privatekey` key-value pair in the `data` (or `stringData`) field as the SSH credential to use.

The following manifest is an example of a Secret used for SSH public/private key authentication:

[secret/ssh-auth-secret.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
 name: secret-ssh-auth
type: kubernetes.io/ssh-auth
data:
 # the data is abbreviated in this example
 ssh-privatekey: |
 UG91cmLuZzYlRW1vdGljb24lU2N1YmE=
```

The SSH authentication Secret type is provided only for convenience. You can create an `Opaque` type for credentials used for SSH authentication. However, using the defined and public Secret type (`kubernetes.io/ssh-auth`) helps other people to understand the purpose of your Secret, and sets a convention for what key names to expect. The Kubernetes API verifies that the required keys are set for a Secret of this type.

### Caution:

SSH private keys do not establish trusted communication between an SSH client and host server on their own. A secondary means of establishing trust is needed to mitigate "man in the middle" attacks, such as a `known_hosts` file added to a `ConfigMap`.

## TLS Secrets

The `kubernetes.io/tls` Secret type is for storing a certificate and its associated key that are typically used for TLS.

One common use for TLS Secrets is to configure encryption in transit for an [Ingress](#), but you can also use it with other resources or directly in your workload. When using this type of Secret, the `tls.key` and the `tls.crt` key must be provided in the `data` (or `stringData`) field of the Secret configuration, although the API server doesn't actually validate the values for each key.

As an alternative to using `stringData`, you can use the `data` field to provide the base64 encoded certificate and private key. For details, see [Constraints on Secret names and data](#).

The following YAML contains an example config for a TLS Secret:

[secret/tls-auth-secret.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
```

name: secret-tls

type: kubernetes.io/tls

data:

# values are base64 encoded, which obscures them but does NOT provide

# any useful level of confidentiality

tls.crt: |

LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSB0tLS0tCk1JSUNVakNDQWJzQ0FnMytNQTBHQ1NxR1N  
JYjNE

UUVCQlFVQU1JR2JNUXN3Q1FZRfZRUUdFd0pLVURFT01Bd0cKQTFVRUNCTUZWRzlyZVc4e  
EVEQU9C

Z05WQkFjVEIwTm9kVzh0YTNVeEVUQVBCZ05WQkFvVENFWnlZVzVyTkVSRQpNUmd3Rmd  
ZRFZRUUx

dzlYWldKRFPYSjBJRk4xY0hCdmNuUXhHREFXQmdOVkJBTVREMFp5WVc1ck5FUkVJRmRsCll  
pQkRR

VEVqTUNFR0NTcUdTSWlZRFFFSkFSWVVM1Z3Y0c5eWRFQm1jbUZ1YXpSa1pDNWpiMjB3S  
GhjTk1U

TXcKTVRFeE1EUTFNVE01V2hjTk1UZ3dNVEV3TURRMU1UTTVXakJMTVFzd0NRWURWUV  
FHREFKS1VE

RVBNQTBHQTFVRQpDQXdHWEZSdmEzbHZNUkV3RHdZRFZRUUtEQWhHY21GdWF6UkVS  
REVZTUJZR0Ex

VUVBd3dQZDNkM0xtVjRZVzF3CmJHVXVZMjI0TUIHYU1BMEdDU3FHU0liM0RRRUJBUVVB  
QTRHSUFE

Q0JoQUo5WThFaUhmeHhNL25PbjJTbkxWHgKRHdPdEJEVDfKRjBReTliMVIKaN2YjdjaTEw  
ZjVN

Vm1UQllqMUZTVWZNOU1vejJDVVFZdW4yRFljV29IcFA4ZQpqSG1BUFVrNVd5cDJRN1ArMj  
h1bkII

QkphVGZlQ09PekZSUFY2MEdTWWUzNmFScG04L3dVVM16eGFLOGtCOWVaCmhPN3F1Tjdt  
SWQxL2pW

cTNKODhDQXdFQUFUQU5CZ2txaGtpRzl3MEJBUVVGQUFPQmdRQU1meTQzeE15OHh3QTU  
KVjF2T2NS

OEtyNWNNaSXdtbFhCUU8xeFEzazlxSGtyNFIUY1JxTVQ5WjVKTM1rWHYxK2VSAgcwTi9WM  
W5NUTRZ

RgpnWXcxbnlESnBnOTduZUV4VzQyeXVIMFIHSDYyV1hYUUhYOVNVREgrRlowVnQvRGZsd  
klVTWRj

UUFEZjM4aU9zCjlQbG1kb3YrcE0vNCs5a1h5aDhSUEkzZXZ6OS9NQ09CiotLS0tRU5EIEFUIR  
J

RkIDQVRFLS0tLS0K

# In this example, the key data is not a real PEM-encoded private key

```
tls.key: |
RXhhbXBsZSBkYXRhIGZvcjB0aGUgVExTIGNydCBmaWVsZA==
```

The TLS Secret type is provided only for convenience. You can create an Opaque type for credentials used for TLS authentication. However, using the defined and public Secret type (kubernetes.io/tls) helps ensure the consistency of Secret format in your project. The API server verifies if the required keys are set for a Secret of this type.

To create a TLS Secret using kubectl, use the tls subcommand:

```
kubectl create secret tls my-tls-secret \
--cert=path/to/cert/file \
--key=path/to/key/file
```

The public/private key pair must exist before hand. The public key certificate for --cert must be .PEM encoded and must match the given private key for --key.

## Bootstrap token Secrets

The bootstrap.kubernetes.io/token Secret type is for tokens used during the node bootstrap process. It stores tokens used to sign well-known ConfigMaps.

A bootstrap token Secret is usually created in the kube-system namespace and named in the form bootstrap-token-<token-id> where <token-id> is a 6 character string of the token ID.

As a Kubernetes manifest, a bootstrap token Secret might look like the following:

[secret/bootstrap-token-secret-base64.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
 name: bootstrap-token-5emitj
 namespace: kube-system
type: bootstrap.kubernetes.io/token
data:
 auth-extra-groups: c3lzdGVtOmJvb3RzdHJhcHBlcjM6a3ViZWZkbTpkZWZhdWx0LW5vZGUtdG9rZW4=
 expiration: MjAyMC0wOS0xM1QwND0zOT0xM00=
 token-id: NWVtaXRq
 token-secret: a3E0Z2lodnN6emduMXAwcg==
 usage-bootstrap-authentication: dHJ1ZQ==
 usage-bootstrap-signing: dHJ1ZQ==
```

A bootstrap token Secret has the following keys specified under data:

- token-id: A random 6 character string as the token identifier. Required.
- token-secret: A random 16 character string as the actual token Secret. Required.
- description: A human-readable string that describes what the token is used for. Optional.
- expiration: An absolute UTC time using [RFC3339](#) specifying when the token should be expired. Optional.
- usage-bootstrap-<usage>: A boolean flag indicating additional usage for the bootstrap token.

- `auth-extra-groups`: A comma-separated list of group names that will be authenticated as in addition to the `system:bootstrappers` group.

You can alternatively provide the values in the `stringData` field of the Secret without base64 encoding them:

[secret/bootstrap-token-secret-literal.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
 # Note how the Secret is named
 name: bootstrap-token-5emitj
 # A bootstrap token Secret usually resides in the kube-system namespace
 namespace: kube-system
type: bootstrap.kubernetes.io/token
stringData:
 auth-extra-groups: "system:bootstrappers:kubeadm:default-node-token"
 expiration: "2020-09-13T04:39:10Z"
 # This token ID is used in the name
 token-id: "5emitj"
 token-secret: "kq4gihvszzgn1p0r"
 # This token can be used for authentication
 usage-bootstrap-authentication: "true"
 # and it can be used for signing
 usage-bootstrap-signing: "true"
```

**Note:**

The `stringData` field for a Secret does not work well with server-side apply.

## Working with Secrets

### Creating a Secret

There are several options to create a Secret:

- [Use kubectl](#)
- [Use a configuration file](#)
- [Use the Kustomize tool](#)

### Constraints on Secret names and data

The name of a Secret object must be a valid [DNS subdomain name](#).

You can specify the data and/or the `stringData` field when creating a configuration file for a Secret. The `data` and the `stringData` fields are optional. The values for all keys in the `data` field have to be base64-encoded strings. If the conversion to base64 string is not desirable, you can choose to specify the `stringData` field instead, which accepts arbitrary strings as values.

The keys of `data` and `stringData` must consist of alphanumeric characters, `-`, `_` or `.`. All key-value pairs in the `stringData` field are internally merged into the `data` field. If a key appears in both the `data` and the `stringData` field, the value specified in the `stringData` field takes precedence.

## Size limit

Individual Secrets are limited to 1MiB in size. This is to discourage creation of very large Secrets that could exhaust the API server and kubelet memory. However, creation of many smaller Secrets could also exhaust memory. You can use a [resource quota](#) to limit the number of Secrets (or other resources) in a namespace.

## Editing a Secret

You can edit an existing Secret unless it is [immutable](#). To edit a Secret, use one of the following methods:

- [Use kubectl](#)
- [Use a configuration file](#)

You can also edit the data in a Secret using the [Kustomize tool](#). However, this method creates a new Secret object with the edited data.

Depending on how you created the Secret, as well as how the Secret is used in your Pods, updates to existing Secret objects are propagated automatically to Pods that use the data. For more information, refer to [Using Secrets as files from a Pod](#) section.

## Using a Secret

Secrets can be mounted as data volumes or exposed as [environment variables](#) to be used by a container in a Pod. Secrets can also be used by other parts of the system, without being directly exposed to the Pod. For example, Secrets can hold credentials that other parts of the system should use to interact with external systems on your behalf.

Secret volume sources are validated to ensure that the specified object reference actually points to an object of type Secret. Therefore, a Secret needs to be created before any Pods that depend on it.

If the Secret cannot be fetched (perhaps because it does not exist, or due to a temporary lack of connection to the API server) the kubelet periodically retries running that Pod. The kubelet also reports an Event for that Pod, including details of the problem fetching the Secret.

## Optional Secrets

When you reference a Secret in a Pod, you can mark the Secret as *optional*, such as in the following example. If an optional Secret doesn't exist, Kubernetes ignores it.

[secret/optional-secret.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: mypod
spec:
```

```
containers:
- name: mypod
 image: redis
 volumeMounts:
 - name: foo
 mountPath: "/etc/foo"
 readOnly: true
volumes:
- name: foo
 secret:
 secretName: mysecret
 optional: true
```

By default, Secrets are required. None of a Pod's containers will start until all non-optional Secrets are available.

If a Pod references a specific key in a non-optional Secret and that Secret does exist, but is missing the named key, the Pod fails during startup.

## Using Secrets as files from a Pod

If you want to access data from a Secret in a Pod, one way to do that is to have Kubernetes make the value of that Secret be available as a file inside the filesystem of one or more of the Pod's containers.

For instructions, refer to [Distribute credentials securely using Secrets](#).

When a volume contains data from a Secret, and that Secret is updated, Kubernetes tracks this and updates the data in the volume, using an eventually-consistent approach.

### Note:

A container using a Secret as a [subPath](#) volume mount does not receive automated Secret updates.

The kubelet keeps a cache of the current keys and values for the Secrets that are used in volumes for pods on that node. You can configure the way that the kubelet detects changes from the cached values. The `configMapAndSecretChangeDetectionStrategy` field in the [kubelet configuration](#) controls which strategy the kubelet uses. The default strategy is Watch.

Updates to Secrets can be either propagated by an API watch mechanism (the default), based on a cache with a defined time-to-live, or polled from the cluster API server on each kubelet synchronisation loop.

As a result, the total delay from the moment when the Secret is updated to the moment when new keys are projected to the Pod can be as long as the kubelet sync period + cache propagation delay, where the cache propagation delay depends on the chosen cache type (following the same order listed in the previous paragraph, these are: watch propagation delay, the configured cache TTL, or zero for direct polling).

## Using Secrets as environment variables

To use a Secret in an [environment variable](#) in a Pod:

1. For each container in your Pod specification, add an environment variable for each Secret key that you want to use to the `env[].valueFrom.secretKeyRef` field.
2. Modify your image and/or command line so that the program looks for values in the specified environment variables.

For instructions, refer to [Define container environment variables using Secret data](#).

It's important to note that the range of characters allowed for environment variable names in pods is [restricted](#). If any keys do not meet the rules, those keys are not made available to your container, though the Pod is allowed to start.

## Container image pull Secrets

If you want to fetch container images from a private repository, you need a way for the kubelet on each node to authenticate to that repository. You can configure *image pull Secrets* to make this possible. These Secrets are configured at the Pod level.

### Using imagePullSecrets

The `imagePullSecrets` field is a list of references to Secrets in the same namespace. You can use an `imagePullSecrets` to pass a Secret that contains a Docker (or other) image registry password to the kubelet. The kubelet uses this information to pull a private image on behalf of your Pod. See the [PodSpec API](#) for more information about the `imagePullSecrets` field.

#### Manually specifying an imagePullSecret

You can learn how to specify `imagePullSecrets` from the [container images](#) documentation.

#### Arranging for imagePullSecrets to be automatically attached

You can manually create `imagePullSecrets`, and reference these from a `ServiceAccount`. Any Pods created with that `ServiceAccount` or created with that `ServiceAccount` by default, will get their `imagePullSecrets` field set to that of the service account. See [Add ImagePullSecrets to a service account](#) for a detailed explanation of that process.

## Using Secrets with static Pods

You cannot use ConfigMaps or Secrets with [static Pods](#).

## Immutable Secrets

FEATURE STATE: Kubernetes v1.21 [stable]

Kubernetes lets you mark specific Secrets (and ConfigMaps) as *immutable*. Preventing changes to the data of an existing Secret has the following benefits:

- protects you from accidental (or unwanted) updates that could cause applications outages

- (for clusters that extensively use Secrets - at least tens of thousands of unique Secret to Pod mounts), switching to immutable Secrets improves the performance of your cluster by significantly reducing load on kube-apiserver. The kubelet does not need to maintain a [watch] on any Secrets that are marked as immutable.

## Marking a Secret as immutable

You can create an immutable Secret by setting the `immutable` field to `true`. For example,

```
apiVersion: v1
kind: Secret
metadata: ...
data: ...
immutable: true
```

You can also update any existing mutable Secret to make it immutable.

### Note:

Once a Secret or ConfigMap is marked as immutable, it is *not* possible to revert this change nor to mutate the contents of the data field. You can only delete and recreate the Secret. Existing Pods maintain a mount point to the deleted Secret - it is recommended to recreate these pods.

## Information security for Secrets

Although ConfigMap and Secret work similarly, Kubernetes applies some additional protection for Secret objects.

Secrets often hold values that span a spectrum of importance, many of which can cause escalations within Kubernetes (e.g. service account tokens) and to external systems. Even if an individual app can reason about the power of the Secrets it expects to interact with, other apps within the same namespace can render those assumptions invalid.

A Secret is only sent to a node if a Pod on that node requires it. For mounting Secrets into Pods, the kubelet stores a copy of the data into a tmpfs so that the confidential data is not written to durable storage. Once the Pod that depends on the Secret is deleted, the kubelet deletes its local copy of the confidential data from the Secret.

There may be several containers in a Pod. By default, containers you define only have access to the default ServiceAccount and its related Secret. You must explicitly define environment variables or map a volume into a container in order to provide access to any other Secret.

There may be Secrets for several Pods on the same node. However, only the Secrets that a Pod requests are potentially visible within its containers. Therefore, one Pod does not have access to the Secrets of another Pod.

## Configure least-privilege access to Secrets

To enhance the security measures around Secrets, Kubernetes provides a mechanism: you can annotate a ServiceAccount as `kubernetes.io/enforce-mountable-secrets: "true"`.

For more information, you can refer to the [documentation about this annotation](#).



## Warning:

Any containers that run with `privileged: true` on a node can access all Secrets used on that node.

## What's next

- For guidelines to manage and improve the security of your Secrets, refer to [Good practices for Kubernetes Secrets](#).
- Learn how to [manage Secrets using kubectl](#)
- Learn how to [manage Secrets using config file](#)
- Learn how to [manage Secrets using kustomize](#)
- Read the [API reference](#) for Secret

# Resource Management for Pods and Containers

When you specify a [Pod](#), you can optionally specify how much of each resource a [container](#) needs. The most common resources to specify are CPU and memory (RAM); there are others.

When you specify the resource *request* for containers in a Pod, the [kube-scheduler](#) uses this information to decide which node to place the Pod on. When you specify a resource *limit* for a container, the [kubelet](#) enforces those limits so that the running container is not allowed to use more of that resource than the limit you set. The kubelet also reserves at least the *request* amount of that system resource specifically for that container to use.

## Requests and limits

If the node where a Pod is running has enough of a resource available, it's possible (and allowed) for a container to use more resource than its request for that resource specifies. However, a container is not allowed to use more than its resource limit.

For example, if you set a `memory` request of 256 MiB for a container, and that container is in a Pod scheduled to a Node with 8GiB of memory and no other Pods, then the container can try to use more RAM.

If you set a memory limit of 4GiB for that container, the kubelet (and [container runtime](#)) enforce the limit. The runtime prevents the container from using more than the configured resource limit. For example: when a process in the container tries to consume more than the allowed amount of memory, the system kernel terminates the process that attempted the allocation, with an out of memory (OOM) error.

Limits can be implemented either reactively (the system intervenes once it sees a violation) or by enforcement (the system prevents the container from ever exceeding the limit). Different runtimes can have different ways to implement the same restrictions.

### Note:

If you specify a limit for a resource, but do not specify any request, and no admission-time mechanism has applied a default request for that resource, then Kubernetes copies the limit you specified and uses it as the requested value for the resource.

## Resource types

*CPU* and *memory* are each a *resource type*. A resource type has a base unit. CPU represents compute processing and is specified in units of [Kubernetes CPUs](#). Memory is specified in units of bytes. For Linux workloads, you can specify *huge page* resources. Huge pages are a Linux-specific feature where the node kernel allocates blocks of memory that are much larger than the default page size.

For example, on a system where the default page size is 4KiB, you could specify a limit, `hugepages-2Mi: 80Mi`. If the container tries allocating over 40 2MiB huge pages (a total of 80 MiB), that allocation fails.

### Note:

You cannot overcommit `hugepages-*` resources. This is different from the `memory` and `cpu` resources.

CPU and memory are collectively referred to as *compute resources*, or *resources*. Compute resources are measurable quantities that can be requested, allocated, and consumed. They are distinct from [API resources](#). API resources, such as Pods and [Services](#) are objects that can be read and modified through the Kubernetes API server.

## Resource requests and limits of Pod and container

For each container, you can specify resource limits and requests, including the following:

- `spec.containers[].resources.limits.cpu`
- `spec.containers[].resources.limits.memory`
- `spec.containers[].resources.limits.hugepages-<size>`
- `spec.containers[].resources.requests.cpu`
- `spec.containers[].resources.requests.memory`
- `spec.containers[].resources.requests.hugepages-<size>`

Although you can only specify requests and limits for individual containers, it is also useful to think about the overall resource requests and limits for a Pod. For a particular resource, a *Pod resource request/limit* is the sum of the resource requests/limits of that type for each container in the Pod.

## Resource units in Kubernetes

### CPU resource units

Limits and requests for CPU resources are measured in *cpu* units. In Kubernetes, 1 CPU unit is equivalent to **1 physical CPU core**, or **1 virtual core**, depending on whether the node is a physical host or a virtual machine running inside a physical machine.

Fractional requests are allowed. When you define a container with `spec.containers[].resources.requests.cpu` set to 0.5, you are requesting half as much CPU time compared to if you asked for 1.0 CPU. For CPU resource units, the [quantity](#) expression 0.1 is equivalent to the expression 100m, which can be read as "one hundred millicpu". Some people say "one hundred millicores", and this is understood to mean the same thing.

CPU resource is always specified as an absolute amount of resource, never as a relative amount. For example, 500m CPU represents the roughly same amount of computing power whether that container runs on a single-core, dual-core, or 48-core machine.

#### Note:

Kubernetes doesn't allow you to specify CPU resources with a precision finer than 1m or 0.001 CPU. To avoid accidentally using an invalid CPU quantity, it's useful to specify CPU units using the milliCPU form instead of the decimal form when using less than 1 CPU unit.

For example, you have a Pod that uses 5m or 0.005 CPU and would like to decrease its CPU resources. By using the decimal form, it's harder to spot that 0.0005 CPU is an invalid value, while by using the milliCPU form, it's easier to spot that 0.5m is an invalid value.

## Memory resource units

Limits and requests for memory are measured in bytes. You can express memory as a plain integer or as a fixed-point number using one of these [quantity](#) suffixes: E, P, T, G, M, k. You can also use the power-of-two equivalents: Ei, Pi, Ti, Gi, Mi, Ki. For example, the following represent roughly the same value:

```
128974848, 129e6, 129M, 128974848000m, 123Mi
```

Pay attention to the case of the suffixes. If you request 400m of memory, this is a request for 0.4 bytes. Someone who types that probably meant to ask for 400 mebibytes (400Mi) or 400 megabytes (400M).

## Container resources example

The following Pod has two containers. Both containers are defined with a request for 0.25 CPU and 64MiB ( $2^{26}$  bytes) of memory. Each container has a limit of 0.5 CPU and 128MiB of memory. You can say the Pod has a request of 0.5 CPU and 128 MiB of memory, and a limit of 1 CPU and 256MiB of memory.

```

apiVersion: v1
kind: Pod
metadata:
 name: frontend
spec:
 containers:
 - name: app
 image: images.my-company.example/app:v4
 resources:
 requests:
 memory: "64Mi"
```

```
 cpu: "250m"
limits:
 memory: "128Mi"
 cpu: "500m"
- name: log-aggregator
image: images.my-company.example/log-aggregator:v6
resources:
 requests:
 memory: "64Mi"
 cpu: "250m"
 limits:
 memory: "128Mi"
 cpu: "500m"
```

## How Pods with resource requests are scheduled

When you create a Pod, the Kubernetes scheduler selects a node for the Pod to run on. Each node has a maximum capacity for each of the resource types: the amount of CPU and memory it can provide for Pods. The scheduler ensures that, for each resource type, the sum of the resource requests of the scheduled containers is less than the capacity of the node. Note that although actual memory or CPU resource usage on nodes is very low, the scheduler still refuses to place a Pod on a node if the capacity check fails. This protects against a resource shortage on a node when resource usage later increases, for example, during a daily peak in request rate.

## How Kubernetes applies resource requests and limits

When the kubelet starts a container as part of a Pod, the kubelet passes that container's requests and limits for memory and CPU to the container runtime.

On Linux, the container runtime typically configures kernel [cgroups](#) that apply and enforce the limits you defined.

- The CPU limit defines a hard ceiling on how much CPU time that the container can use. During each scheduling interval (time slice), the Linux kernel checks to see if this limit is exceeded; if so, the kernel waits before allowing that cgroup to resume execution.
- The CPU request typically defines a weighting. If several different containers (cgroups) want to run on a contended system, workloads with larger CPU requests are allocated more CPU time than workloads with small requests.
- The memory request is mainly used during (Kubernetes) Pod scheduling. On a node that uses cgroups v2, the container runtime might use the memory request as a hint to set `memory.min` and `memory.low`.
- The memory limit defines a memory limit for that cgroup. If the container tries to allocate more memory than this limit, the Linux kernel out-of-memory subsystem activates and, typically, intervenes by stopping one of the processes in the container that tried to allocate memory. If that process is the container's PID 1, and the container is marked as restartable, Kubernetes restarts the container.
- The memory limit for the Pod or container can also apply to pages in memory backed volumes, such as an `emptyDir`. The kubelet tracks tmpfs `emptyDir` volumes as container memory use, rather than as local ephemeral storage.

If a container exceeds its memory request and the node that it runs on becomes short of memory overall, it is likely that the Pod the container belongs to will be [evicted](#).

A container might or might not be allowed to exceed its CPU limit for extended periods of time. However, container runtimes don't terminate Pods or containers for excessive CPU usage.

To determine whether a container cannot be scheduled or is being killed due to resource limits, see the [Troubleshooting](#) section.

## Monitoring compute & memory resource usage

The kubelet reports the resource usage of a Pod as part of the Pod [status](#).

If optional [tools for monitoring](#) are available in your cluster, then Pod resource usage can be retrieved either from the [Metrics API](#) directly or from your monitoring tools.

## Local ephemeral storage

FEATURE STATE: Kubernetes v1.25 [stable]

Nodes have local ephemeral storage, backed by locally-attached writeable devices or, sometimes, by RAM. "Ephemeral" means that there is no long-term guarantee about durability.

Pods use ephemeral local storage for scratch space, caching, and for logs. The kubelet can provide scratch space to Pods using local ephemeral storage to mount [emptyDir volumes](#) into containers.

The kubelet also uses this kind of storage to hold [node-level container logs](#), container images, and the writable layers of running containers.

### Caution:

If a node fails, the data in its ephemeral storage can be lost. Your applications cannot expect any performance SLAs (disk IOPS for example) from local ephemeral storage.

### Note:

To make the resource quota work on ephemeral-storage, two things need to be done:

- An admin sets the resource quota for ephemeral-storage in a namespace.
- A user needs to specify limits for the ephemeral-storage resource in the Pod spec.

If the user doesn't specify the ephemeral-storage resource limit in the Pod spec, the resource quota is not enforced on ephemeral-storage.

Kubernetes lets you track, reserve and limit the amount of ephemeral local storage a Pod can consume.

## Configurations for local ephemeral storage

Kubernetes supports two ways to configure local ephemeral storage on a node:

- [Single filesystem](#)
- [Two filesystems](#)

In this configuration, you place all different kinds of ephemeral local data (emptyDir volumes, writable layers, container images, logs) into one filesystem. The most effective way to configure the kubelet means dedicating this filesystem to Kubernetes (kubelet) data.

The kubelet also writes [node-level container logs](#) and treats these similarly to ephemeral local storage.

The kubelet writes logs to files inside its configured log directory (/var/log by default); and has a base directory for other locally stored data (/var/lib/kubelet by default).

Typically, both /var/lib/kubelet and /var/log are on the system root filesystem, and the kubelet is designed with that layout in mind.

Your node can have as many other filesystems, not used for Kubernetes, as you like.

You have a filesystem on the node that you're using for ephemeral data that comes from running Pods: logs, and emptyDir volumes. You can use this filesystem for other data (for example: system logs not related to Kubernetes); it can even be the root filesystem.

The kubelet also writes [node-level container logs](#) into the first filesystem, and treats these similarly to ephemeral local storage.

You also use a separate filesystem, backed by a different logical storage device. In this configuration, the directory where you tell the kubelet to place container image layers and writable layers is on this second filesystem.

The first filesystem does not hold any image layers or writable layers.

Your node can have as many other filesystems, not used for Kubernetes, as you like.

The kubelet can measure how much local storage it is using. It does this provided that you have set up the node using one of the supported configurations for local ephemeral storage.

If you have a different configuration, then the kubelet does not apply resource limits for ephemeral local storage.

**Note:**

The kubelet tracks tmpfs emptyDir volumes as container memory use, rather than as local ephemeral storage.

**Note:**

The kubelet will only track the root filesystem for ephemeral storage. OS layouts that mount a separate disk to /var/lib/kubelet or /var/lib/containers will not report ephemeral storage correctly.

## Setting requests and limits for local ephemeral storage

You can specify ephemeral-storage for managing local ephemeral storage. Each container of a Pod can specify either or both of the following:

- spec.containers[].resources.limits.ephemeral-storage

- `spec.containers[].resources.requests.ephemeral-storage`

Limits and requests for ephemeral-storage are measured in byte quantities. You can express storage as a plain integer or as a fixed-point number using one of these suffixes: E, P, T, G, M, k. You can also use the power-of-two equivalents: Ei, Pi, Ti, Gi, Mi, Ki. For example, the following quantities all represent roughly the same value:

- 128974848
- 129e6
- 129M
- 123Mi

Pay attention to the case of the suffixes. If you request 400m of ephemeral-storage, this is a request for 0.4 bytes. Someone who types that probably meant to ask for 400 mebibytes (400Mi) or 400 megabytes (400M).

In the following example, the Pod has two containers. Each container has a request of 2GiB of local ephemeral storage. Each container has a limit of 4GiB of local ephemeral storage. Therefore, the Pod has a request of 4GiB of local ephemeral storage, and a limit of 8GiB of local ephemeral storage. 500Mi of that limit could be consumed by the emptyDir volume.

```
apiVersion: v1
kind: Pod
metadata:
 name: frontend
spec:
 containers:
 - name: app
 image: images.my-company.example/app:v4
 resources:
 requests:
 ephemeral-storage: "2Gi"
 limits:
 ephemeral-storage: "4Gi"
 volumeMounts:
 - name: ephemeral
 mountPath: "/tmp"
 - name: log-aggregator
 image: images.my-company.example/log-aggregator:v6
 resources:
 requests:
 ephemeral-storage: "2Gi"
 limits:
 ephemeral-storage: "4Gi"
 volumeMounts:
 - name: ephemeral
 mountPath: "/tmp"
 volumes:
 - name: ephemeral
 emptyDir:
 sizeLimit: 500Mi
```

## How Pods with ephemeral-storage requests are scheduled

When you create a Pod, the Kubernetes scheduler selects a node for the Pod to run on. Each node has a maximum amount of local ephemeral storage it can provide for Pods. For more information, see [Node Allocatable](#).

The scheduler ensures that the sum of the resource requests of the scheduled containers is less than the capacity of the node.

## Ephemeral storage consumption management

If the kubelet is managing local ephemeral storage as a resource, then the kubelet measures storage use in:

- emptyDir volumes, except *tmpfs* emptyDir volumes
- directories holding node-level logs
- writeable container layers

If a Pod is using more ephemeral storage than you allow it to, the kubelet sets an eviction signal that triggers Pod eviction.

For container-level isolation, if a container's writable layer and log usage exceeds its storage limit, the kubelet marks the Pod for eviction.

For pod-level isolation the kubelet works out an overall Pod storage limit by summing the limits for the containers in that Pod. In this case, if the sum of the local ephemeral storage usage from all containers and also the Pod's emptyDir volumes exceeds the overall Pod storage limit, then the kubelet also marks the Pod for eviction.

### Caution:

If the kubelet is not measuring local ephemeral storage, then a Pod that exceeds its local storage limit will not be evicted for breaching local storage resource limits.

However, if the filesystem space for writeable container layers, node-level logs, or emptyDir volumes falls low, the node [taints](#) itself as short on local storage and this taint triggers eviction for any Pods that don't specifically tolerate the taint.

See the supported [configurations](#) for ephemeral local storage.

The kubelet supports different ways to measure Pod storage use:

- [Periodic scanning](#)
- [Filesystem project quota](#)

The kubelet performs regular, scheduled checks that scan each emptyDir volume, container log directory, and writeable container layer.

The scan measures how much space is used.

### Note:

In this mode, the kubelet does not track open file descriptors for deleted files.



If you (or a container) create a file inside an `emptyDir` volume, something then opens that file, and you delete the file while it is still open, then the inode for the deleted file stays until you close that file but the kubelet does not categorize the space as in use.

FEATURE STATE: Kubernetes v1.15 [alpha]

Project quotas are an operating-system level feature for managing storage use on filesystems. With Kubernetes, you can enable project quotas for monitoring storage use. Make sure that the filesystem backing the `emptyDir` volumes, on the node, provides project quota support. For example, XFS and ext4fs offer project quotas.

### Note:

Project quotas let you monitor storage use; they do not enforce limits.

Kubernetes uses project IDs starting from 1048576. The IDs in use are registered in `/etc/projects` and `/etc/projid`. If project IDs in this range are used for other purposes on the system, those project IDs must be registered in `/etc/projects` and `/etc/projid` so that Kubernetes does not use them.

Quotas are faster and more accurate than directory scanning. When a directory is assigned to a project, all files created under a directory are created in that project, and the kernel merely has to keep track of how many blocks are in use by files in that project. If a file is created and deleted, but has an open file descriptor, it continues to consume space. Quota tracking records that space accurately whereas directory scans overlook the storage used by deleted files.

If you want to use project quotas, you should:

- Enable the `LocalStorageCapacityIsolationFSQuotaMonitoring=true` [feature gate](#) using the `featureGates` field in the [kubelet configuration](#) or the `--feature-gates` command line flag.
- Ensure that the root filesystem (or optional runtime filesystem) has project quotas enabled. All XFS filesystems support project quotas. For ext4 filesystems, you need to enable the project quota tracking feature while the filesystem is not mounted.

```
For ext4, with /dev/block-device not mounted
sudo tune2fs -O project -Q prjquota /dev/block-device
```

- Ensure that the root filesystem (or optional runtime filesystem) is mounted with project quotas enabled. For both XFS and ext4fs, the mount option is named `prjquota`.

## Extended resources

Extended resources are fully-qualified resource names outside the `kubernetes.io` domain. They allow cluster operators to advertise and users to consume the non-Kubernetes-built-in resources.

There are two steps required to use Extended Resources. First, the cluster operator must advertise an Extended Resource. Second, users must request the Extended Resource in Pods.

## Managing extended resources

### Node-level extended resources

Node-level extended resources are tied to nodes.

#### Device plugin managed resources

See [Device Plugin](#) for how to advertise device plugin managed resources on each node.

#### Other resources

To advertise a new node-level extended resource, the cluster operator can submit a PATCH HTTP request to the API server to specify the available quantity in the status.capacity for a node in the cluster. After this operation, the node's status.capacity will include a new resource. The status.allocatable field is updated automatically with the new resource asynchronously by the kubelet.

Because the scheduler uses the node's status.allocatable value when evaluating Pod fitness, the scheduler only takes account of the new value after that asynchronous update. There may be a short delay between patching the node capacity with a new resource and the time when the first Pod that requests the resource can be scheduled on that node.

#### Example:

Here is an example showing how to use curl to form an HTTP request that advertises five "example.com/foo" resources on node k8s-node-1 whose master is k8s-master.

```
curl --header "Content-Type: application/json-patch+json" \
--request PATCH \
--data ' [{"op": "add", "path": "/status/capacity/example.com~1foo", "value": "5"}] ' \
http://k8s-master:8080/api/v1/nodes/k8s-node-1/status
```

#### Note:

In the preceding request, ~1 is the encoding for the character / in the patch path. The operation path value in JSON-Patch is interpreted as a JSON-Pointer. For more details, see [IETF RFC 6901, section 3](#).

### Cluster-level extended resources

Cluster-level extended resources are not tied to nodes. They are usually managed by scheduler extenders, which handle the resource consumption and resource quota.

You can specify the extended resources that are handled by scheduler extenders in [scheduler configuration](#)

#### Example:

The following configuration for a scheduler policy indicates that the cluster-level extended resource "example.com/foo" is handled by the scheduler extender.

- The scheduler sends a Pod to the scheduler extender only if the Pod requests "example.com/foo".
- The `ignoredByScheduler` field specifies that the scheduler does not check the "example.com/foo" resource in its `PodFitsResources` predicate.

```
{
 "kind": "Policy",
 "apiVersion": "v1",
 "extenders": [
 {
 "urlPrefix": "<extender-endpoint>",
 "bindVerb": "bind",
 "managedResources": [
 {
 "name": "example.com/foo",
 "ignoredByScheduler": true
 }
]
 }
]
}
```

## Consuming extended resources

Users can consume extended resources in Pod specs like CPU and memory. The scheduler takes care of the resource accounting so that no more than the available amount is simultaneously allocated to Pods.

The API server restricts quantities of extended resources to whole numbers. Examples of *valid* quantities are 3, 3000m and 3Ki. Examples of *invalid* quantities are 0.5 and 1500m (because 1500m would result in 1.5).

### Note:

Extended resources replace Opaque Integer Resources. Users can use any domain name prefix other than `kubernetes.io` which is reserved.

To consume an extended resource in a Pod, include the resource name as a key in the `spec.containers[].resources.limits` map in the container spec.

### Note:

Extended resources cannot be overcommitted, so request and limit must be equal if both are present in a container spec.

A Pod is scheduled only if all of the resource requests are satisfied, including CPU, memory and any extended resources. The Pod remains in the PENDING state as long as the resource request cannot be satisfied.

### Example:

The Pod below requests 2 CPUs and 1 "example.com/foo" (an extended resource).

```
apiVersion: v1
kind: Pod
metadata:
 name: my-pod
spec:
 containers:
 - name: my-container
 image: myimage
 resources:
 requests:
 cpu: 2
 example.com/foo: 1
 limits:
 example.com/foo: 1
```

## PID limiting

Process ID (PID) limits allow for the configuration of a kubelet to limit the number of PIDs that a given Pod can consume. See [PID Limiting](#) for information.

## Troubleshooting

### My Pods are pending with event message FailedScheduling

If the scheduler cannot find any node where a Pod can fit, the Pod remains unscheduled until a place can be found. An [Event](#) is produced each time the scheduler fails to find a place for the Pod. You can use `kubectl` to view the events for a Pod; for example:

```
kubectl describe pod frontend | grep -A 999999999 Events
```

Events:

| Type    | Reason           | Age | From              | Message                                |
|---------|------------------|-----|-------------------|----------------------------------------|
| Warning | FailedScheduling | 23s | default-scheduler | 0/42 nodes available: insufficient cpu |

In the preceding example, the Pod named "frontend" fails to be scheduled due to insufficient CPU resource on any node. Similar error messages can also suggest failure due to insufficient memory (`PodExceedsFreeMemory`). In general, if a Pod is pending with a message of this type, there are several things to try:

- Add more nodes to the cluster.
- Terminate unneeded Pods to make room for pending Pods.
- Check that the Pod is not larger than all the nodes. For example, if all the nodes have a capacity of `cpu: 1`, then a Pod with a request of `cpu: 1.1` will never be scheduled.
- Check for node taints. If most of your nodes are tainted, and the new Pod does not tolerate that taint, the scheduler only considers placements onto the remaining nodes that don't have that taint.

You can check node capacities and amounts allocated with the `kubectl describe nodes` command. For example:

```
kubectl describe nodes e2e-test-node-pool-4lw4
```

```
Name: e2e-test-node-pool-4lw4
```

```
[... lines removed for clarity ...]
```

```
Capacity:
```

```
cpu: 2
```

```
memory: 7679792Ki
```

```
Pods: 110
```

```
Allocatable:
```

```
cpu: 1800m
```

```
memory: 7474992Ki
```

```
Pods: 110
```

```
[... lines removed for clarity ...]
```

```
Non-terminated Pods: (5 in total)
```

| Namespace   | Name                                 | CPU Requests | CPU Limits | Memory Requests | Memory Limits |
|-------------|--------------------------------------|--------------|------------|-----------------|---------------|
| kube-system | fluentd-gcp-v1.38-28bv1              | 100m (5%)    | 0 (0%)     | 200Mi (2%)      | 200Mi (2%)    |
| kube-system | kube-dns-3297075139-61lj3            | 260m (13%)   | 0 (0%)     | 100Mi (1%)      | 170Mi (2%)    |
| kube-system | kube-proxy-e2e-test-...              | 100m (5%)    | 0 (0%)     | 0 (0%)          | 0 (0%)        |
| kube-system | monitoring-influxdb-grafana-v4-z1m12 | 200m (10%)   | 200m (10%) | 600Mi (8%)      | 600Mi (8%)    |
| kube-system | node-problem-detector-v0.1-fj7m3     | 20m (1%)     | 200m (10%) | 20Mi (0%)       | 100Mi (1%)    |

```
Allocated resources:
```

```
(Total limits may be over 100 percent, i.e., overcommitted.)
```

```
CPU Requests CPU Limits Memory Requests Memory Limits
```

```

```

|            |            |             |              |
|------------|------------|-------------|--------------|
| 680m (34%) | 400m (20%) | 920Mi (11%) | 1070Mi (13%) |
|------------|------------|-------------|--------------|

In the preceding output, you can see that if a Pod requests more than 1.120 CPUs or more than 6.23Gi of memory, that Pod will not fit on the node.

By looking at the “Pods” section, you can see which Pods are taking up space on the node.

The amount of resources available to Pods is less than the node capacity because system daemons use a portion of the available resources. Within the Kubernetes API, each Node has a `.status.allocatable` field (see [NodeStatus](#) for details).

The `.status.allocatable` field describes the amount of resources that are available to Pods on that node (for example: 15 virtual CPUs and 7538 MiB of memory). For more information on node allocatable resources in Kubernetes, see [Reserve Compute Resources for System Daemons](#).

You can configure [resource quotas](#) to limit the total amount of resources that a namespace can consume. Kubernetes enforces quotas for objects in particular namespace when there is a `ResourceQuota` in that namespace. For example, if you assign specific namespaces to different teams, you can add `ResourceQuotas` into those namespaces. Setting resource quotas helps to prevent one team from using so much of any resource that this over-use affects other teams.

You should also consider what access you grant to that namespace: **full** write access to a namespace allows someone with that access to remove any resource, including a configured `ResourceQuota`.

## My container is terminated

Your container might get terminated because it is resource-starved. To check whether a container is being killed because it is hitting a resource limit, call `kubectl describe pod` on the Pod of interest:

```
kubectl describe pod simmemleak-hra99
```

The output is similar to:

```
Name: simmemleak-hra99
Namespace: default
Image(s): saadali/simmemleak
Node: kubernetes-node-tf0f/10.240.216.66
Labels: name=simmemleak
Status: Running
Reason:
Message:
IP: 10.244.2.75
Containers:
 simmemleak:
 Image: saadali/simmemleak:latest
 Limits:
 cpu: 100m
 memory: 50Mi
 State: Running
 Started: Tue, 07 Jul 2019 12:54:41 -0700
 Last State: Terminated
 Reason: OOMKilled
 Exit Code: 137
 Started: Fri, 07 Jul 2019 12:54:30 -0700
 Finished: Fri, 07 Jul 2019 12:54:33 -0700
 Ready: False
 Restart Count: 5
Conditions:
 Type Status
 Ready False
Events:
 Type Reason Age From Message
 ---- -
 Normal Scheduled 42s default-scheduler Successfully assigned simmemleak-hra99 to
kubernetes-node-tf0f
 Normal Pulled 41s kubelet Container image "saadali/simmemleak:latest" already
present on machine
 Normal Created 41s kubelet Created container simmemleak
 Normal Started 40s kubelet Started container simmemleak
 Normal Killing 32s kubelet Killing container with id
ead3fb35-5cf5-44ed-9ae1-488115be66c6: Need to kill Pod
```

In the preceding example, the `Restart Count: 5` indicates that the `simmemleak` container in the Pod was terminated and restarted five times (so far). The `OOMKilled` reason shows that the container tried to use more memory than its limit.

Your next step might be to check the application code for a memory leak. If you find that the application is behaving how you expect, consider setting a higher memory limit (and possibly request) for that container.

## What's next

- Get hands-on experience [assigning Memory resources to containers and Pods](#).
- Get hands-on experience [assigning CPU resources to containers and Pods](#).
- Read how the API reference defines a [container](#) and its [resource requirements](#)
- Read about [project quotas](#) in XFS
- Read more about the [kube-scheduler configuration reference \(v1\)](#)
- Read more about [Quality of Service classes for Pods](#)

# Organizing Cluster Access Using kubeconfig Files

Use kubeconfig files to organize information about clusters, users, namespaces, and authentication mechanisms. The kubectl command-line tool uses kubeconfig files to find the information it needs to choose a cluster and communicate with the API server of a cluster.

### Note:

A file that is used to configure access to clusters is called a *kubeconfig file*. This is a generic way of referring to configuration files. It does not mean that there is a file named kubeconfig.

### Warning:

Only use kubeconfig files from trusted sources. Using a specially-crafted kubeconfig file could result in malicious code execution or file exposure. If you must use an untrusted kubeconfig file, inspect it carefully first, much as you would a shell script.

By default, kubectl looks for a file named config in the \$HOME/.kube directory. You can specify other kubeconfig files by setting the KUBECONFIG environment variable or by setting the `--kubeconfig` flag.

For step-by-step instructions on creating and specifying kubeconfig files, see [Configure Access to Multiple Clusters](#).

## Supporting multiple clusters, users, and authentication mechanisms

Suppose you have several clusters, and your users and components authenticate in a variety of ways. For example:

- A running kubelet might authenticate using certificates.
- A user might authenticate using tokens.
- Administrators might have sets of certificates that they provide to individual users.

With kubeconfig files, you can organize your clusters, users, and namespaces. You can also define contexts to quickly and easily switch between clusters and namespaces.

## Context

A *context* element in a kubeconfig file is used to group access parameters under a convenient name. Each context has three parameters: cluster, namespace, and user. By default, the kubectl command-line tool uses parameters from the *current context* to communicate with the cluster.

To choose the current context:

```
kubectl config use-context
```

## The KUBECONFIG environment variable

The KUBECONFIG environment variable holds a list of kubeconfig files. For Linux and Mac, the list is colon-delimited. For Windows, the list is semicolon-delimited. The KUBECONFIG environment variable is not required. If the KUBECONFIG environment variable doesn't exist, kubectl uses the default kubeconfig file, \$HOME/.kube/config.

If the KUBECONFIG environment variable does exist, kubectl uses an effective configuration that is the result of merging the files listed in the KUBECONFIG environment variable.

## Merging kubeconfig files

To see your configuration, enter this command:

```
kubectl config view
```

As described previously, the output might be from a single kubeconfig file, or it might be the result of merging several kubeconfig files.

Here are the rules that kubectl uses when it merges kubeconfig files:

1. If the `--kubeconfig` flag is set, use only the specified file. Do not merge. Only one instance of this flag is allowed.

Otherwise, if the KUBECONFIG environment variable is set, use it as a list of files that should be merged. Merge the files listed in the KUBECONFIG environment variable according to these rules:

- Ignore empty filenames.
- Produce errors for files with content that cannot be deserialized.
- The first file to set a particular value or map key wins.
- Never change the value or map key. Example: Preserve the context of the first file to set current-context. Example: If two files specify a red-user, use only values from the first file's red-user. Even if the second file has non-conflicting entries under red-user, discard them.

For an example of setting the KUBECONFIG environment variable, see [Setting the KUBECONFIG environment variable](#).



Otherwise, use the default kubeconfig file, `$HOME/.kube/config`, with no merging.

2. Determine the context to use based on the first hit in this chain:

1. Use the `--context` command-line flag if it exists.
2. Use the `current-context` from the merged kubeconfig files.

An empty context is allowed at this point.

3. Determine the cluster and user. At this point, there might or might not be a context. Determine the cluster and user based on the first hit in this chain, which is run twice: once for user and once for cluster:

1. Use a command-line flag if it exists: `--user` or `--cluster`.
2. If the context is non-empty, take the user or cluster from the context.

The user and cluster can be empty at this point.

4. Determine the actual cluster information to use. At this point, there might or might not be cluster information. Build each piece of the cluster information based on this chain; the first hit wins:

1. Use command line flags if they exist: `--server`, `--certificate-authority`, `--insecure-skip-tls-verify`.
2. If any cluster information attributes exist from the merged kubeconfig files, use them.
3. If there is no server location, fail.

5. Determine the actual user information to use. Build user information using the same rules as cluster information, except allow only one authentication technique per user:

1. Use command line flags if they exist: `--client-certificate`, `--client-key`, `--username`, `--password`, `--token`.
2. Use the user fields from the merged kubeconfig files.
3. If there are two conflicting techniques, fail.

6. For any information still missing, use default values and potentially prompt for authentication information.

## File references

File and path references in a kubeconfig file are relative to the location of the kubeconfig file. File references on the command line are relative to the current working directory. In `$HOME/.kube/config`, relative paths are stored relatively, and absolute paths are stored absolutely.

## Proxy

You can configure `kubectl` to use a proxy per cluster using `proxy-url` in your kubeconfig file, like this:

```
apiVersion: v1
kind: Config
```

```
clusters:
- cluster:
 proxy-url: http://proxy.example.org:3128
 server: https://k8s.example.org/k8s/clusters/c-xyyyzz
 name: development

users:
- name: developer

contexts:
- context:
 name: development
```

## What's next

- [Configure Access to Multiple Clusters](#)
- [kubectl config](#)

# Resource Management for Windows nodes

This page outlines the differences in how resources are managed between Linux and Windows.

On Linux nodes, [cgroups](#) are used as a pod boundary for resource control. Containers are created within that boundary for network, process and file system isolation. The Linux cgroup APIs can be used to gather CPU, I/O, and memory use statistics.

In contrast, Windows uses a [job object](#) per container with a system namespace filter to contain all processes in a container and provide logical isolation from the host. (Job objects are a Windows process isolation mechanism and are different from what Kubernetes refers to as a [Job](#)).

There is no way to run a Windows container without the namespace filtering in place. This means that system privileges cannot be asserted in the context of the host, and thus privileged containers are not available on Windows. Containers cannot assume an identity from the host because the Security Account Manager (SAM) is separate.

## Memory management

Windows does not have an out-of-memory process killer as Linux does. Windows always treats all user-mode memory allocations as virtual, and pagefiles are mandatory.

Windows nodes do not overcommit memory for processes. The net effect is that Windows won't reach out of memory conditions the same way Linux does, and processes page to disk instead of being subject to out of memory (OOM) termination. If memory is over-provisioned and all physical memory is exhausted, then paging can slow down performance.

## CPU management

Windows can limit the amount of CPU time allocated for different processes but cannot guarantee a minimum amount of CPU time.

On Windows, the kubelet supports a command-line flag to set the [scheduling priority](#) of the kubelet process: `--windows-priorityclass`. This flag allows the kubelet process to get more CPU time slices when compared to other processes running on the Windows host. More information on the allowable values and their meaning is available at [Windows Priority Classes](#). To ensure that running Pods do not starve the kubelet of CPU cycles, set this flag to `ABOVE_NORMAL_PRIORITY_CLASS` or above.

## Resource reservation

To account for memory and CPU used by the operating system, the container runtime, and by Kubernetes host processes such as the kubelet, you can (and should) reserve memory and CPU resources with the `--kube-reserved` and/or `--system-reserved` kubelet flags. On Windows these values are only used to calculate the node's [allocatable](#) resources.

### Caution:

As you deploy workloads, set resource memory and CPU limits on containers. This also subtracts from `NodeAllocatable` and helps the cluster-wide scheduler in determining which pods to place on which nodes.

Scheduling pods without limits may over-provision the Windows nodes and in extreme cases can cause the nodes to become unhealthy.

On Windows, a good practice is to reserve at least 2GiB of memory.

To determine how much CPU to reserve, identify the maximum pod density for each node and monitor the CPU usage of the system services running there, then choose a value that meets your workload needs.

## Security

Concepts for keeping your cloud-native workload secure.

This section of the Kubernetes documentation aims to help you learn to run workloads more securely, and about the essential aspects of keeping a Kubernetes cluster secure.

Kubernetes is based on a cloud-native architecture, and draws on advice from the [CNCF](#) about good practice for cloud native information security.

Read [Cloud Native Security and Kubernetes](#) for the broader context about how to secure your cluster and the applications that you're running on it.

# Kubernetes security mechanisms

Kubernetes includes several APIs and security controls, as well as ways to define [policies](#) that can form part of how you manage information security.

## Control plane protection

A key security mechanism for any Kubernetes cluster is to [control access to the Kubernetes API](#).

Kubernetes expects you to configure and use TLS to provide [data encryption in transit](#) within the control plane, and between the control plane and its clients. You can also enable [encryption at rest](#) for the data stored within Kubernetes control plane; this is separate from using encryption at rest for your own workloads' data, which might also be a good idea.

## Secrets

The [Secret](#) API provides basic protection for configuration values that require confidentiality.

## Workload protection

Enforce [Pod security standards](#) to ensure that Pods and their containers are isolated appropriately. You can also use [RuntimeClasses](#) to define custom isolation if you need it.

[Network policies](#) let you control network traffic between Pods, or between Pods and the network outside your cluster.

You can deploy security controls from the wider ecosystem to implement preventative or detective controls around Pods, their containers, and the images that run in them.

## Auditing

Kubernetes [audit logging](#) provides a security-relevant, chronological set of records documenting the sequence of actions in a cluster. The cluster audits the activities generated by users, by applications that use the Kubernetes API, and by the control plane itself.

## Cloud provider security

**Note:** Items on this page refer to vendors external to Kubernetes. The Kubernetes project authors aren't responsible for those third-party products or projects. To add a vendor, product or project to this list, read the [content guide](#) before submitting a change. [More information.](#)

If you are running a Kubernetes cluster on your own hardware or a different cloud provider, consult your documentation for security best practices. Here are links to some of the popular cloud providers' security documentation:

### Cloud provider security

| IaaS Provider         | Link                                                                                              |
|-----------------------|---------------------------------------------------------------------------------------------------|
| Alibaba Cloud         | <a href="https://www.alibabacloud.com/trust-center">https://www.alibabacloud.com/trust-center</a> |
| Amazon Web Services   | <a href="https://aws.amazon.com/security">https://aws.amazon.com/security</a>                     |
| Google Cloud Platform | <a href="https://cloud.google.com/security">https://cloud.google.com/security</a>                 |

| IaaS Provider               | Link                                                                                                                                              |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Huawei Cloud                | <a href="https://www.huaweicloud.com/intl/en-us/securecenter/overallsafety">https://www.huaweicloud.com/intl/en-us/securecenter/overallsafety</a> |
| IBM Cloud                   | <a href="https://www.ibm.com/cloud/security">https://www.ibm.com/cloud/security</a>                                                               |
| Microsoft Azure             | <a href="https://docs.microsoft.com/en-us/azure/security/azure-security">https://docs.microsoft.com/en-us/azure/security/azure-security</a>       |
| Oracle Cloud Infrastructure | <a href="https://www.oracle.com/security">https://www.oracle.com/security</a>                                                                     |
| VMware vSphere              | <a href="https://www.vmware.com/security/hardening-guides">https://www.vmware.com/security/hardening-guides</a>                                   |

## Policies

You can define security policies using Kubernetes-native mechanisms, such as [NetworkPolicy](#) (declarative control over network packet filtering) or [ValidatingAdmissionPolicy](#) (declarative restrictions on what changes someone can make using the Kubernetes API).

However, you can also rely on policy implementations from the wider ecosystem around Kubernetes. Kubernetes provides extension mechanisms to let those ecosystem projects implement their own policy controls on source code review, container image approval, API access controls, networking, and more.

For more information about policy mechanisms and Kubernetes, read [Policies](#).

## What's next

Learn about related Kubernetes security topics:

- [Securing your cluster](#)
- [Known vulnerabilities](#) in Kubernetes (and links to further information)
- [Data encryption in transit](#) for the control plane
- [Data encryption at rest](#)
- [Controlling Access to the Kubernetes API](#)
- [Network policies](#) for Pods
- [Secrets in Kubernetes](#)
- [Pod security standards](#)
- [RuntimeClasses](#)

Learn the context:

- [Cloud Native Security and Kubernetes](#)

Get certified:

- [Certified Kubernetes Security Specialist](#) certification and official training course.

Read more in this section:

- [Pod Security Standards](#)
- [Pod Security Admission](#)
- [Service Accounts](#)
- [Pod Security Policies](#)
- [Security For Windows Nodes](#)
- [Controlling Access to the Kubernetes API](#)

- [Role Based Access Control Good Practices](#)
- [Good practices for Kubernetes Secrets](#)
- [Multi-tenancy](#)
- [Hardening Guide - Authentication Mechanisms](#)
- [Kubernetes API Server Bypass Risks](#)
- [Linux kernel security constraints for Pods and containers](#)
- [Security Checklist](#)

# Cloud Native Security and Kubernetes

Concepts for keeping your cloud-native workload secure.

Kubernetes is based on a cloud-native architecture, and draws on advice from the [CNCF](#) about good practice for cloud native information security.

Read on through this page for an overview of how Kubernetes is designed to help you deploy a secure cloud native platform.

## Cloud native information security

The CNCF [white paper](#) on cloud native security defines security controls and practices that are appropriate to different *lifecycle phases*.

### *Develop lifecycle phase*

- Ensure the integrity of development environments.
- Design applications following good practice for information security, appropriate for your context.
- Consider end user security as part of solution design.

To achieve this, you can:

1. Adopt an architecture, such as [zero trust](#), that minimizes attack surfaces, even for internal threats.
2. Define a code review process that considers security concerns.
3. Build a *threat model* of your system or application that identifies trust boundaries. Use that to model to identify risks and to help find ways to treat those risks.
4. Incorporate advanced security automation, such as *fuzzing* and [security chaos engineering](#), where it's justified.

### *Distribute lifecycle phase*

- Ensure the security of the supply chain for container images you execute.
- Ensure the security of the supply chain for the cluster and other components that execute your application. An example of another component might be an external database that your cloud-native application uses for persistence.

To achieve this, you can:

1. Scan container images and other artifacts for known vulnerabilities.

2. Ensure that software distribution uses encryption in transit, with a chain of trust for the software source.
3. Adopt and follow processes to update dependencies when updates are available, especially in response to security announcements.
4. Use validation mechanisms such as digital certificates for supply chain assurance.
5. Subscribe to feeds and other mechanisms to alert you to security risks.
6. Restrict access to artifacts. Place container images in a [private registry](#) that only allows authorized clients to pull images.

## ***Deploy lifecycle phase***

Ensure appropriate restrictions on what can be deployed, who can deploy it, and where it can be deployed to. You can enforce measures from the *distribute* phase, such as verifying the cryptographic identity of container image artifacts.

When you deploy Kubernetes, you also set the foundation for your applications' runtime environment: a Kubernetes cluster (or multiple clusters). That IT infrastructure must provide the security guarantees that higher layers expect.

## ***Runtime lifecycle phase***

The Runtime phase comprises three critical areas: [compute](#), [access](#), and [storage](#).

### **Runtime protection: access**

The Kubernetes API is what makes your cluster work. Protecting this API is key to providing effective cluster security.

Other pages in the Kubernetes documentation have more detail about how to set up specific aspects of access control. The [security checklist](#) has a set of suggested basic checks for your cluster.

Beyond that, securing your cluster means implementing effective [authentication](#) and [authorization](#) for API access. Use [ServiceAccounts](#) to provide and manage security identities for workloads and cluster components.

Kubernetes uses TLS to protect API traffic; make sure to deploy the cluster using TLS (including for traffic between nodes and the control plane), and protect the encryption keys. If you use Kubernetes' own API for [CertificateSigningRequests](#), pay special attention to restricting misuse there.

### **Runtime protection: compute**

[Containers](#) provide two things: isolation between different applications, and a mechanism to combine those isolated applications to run on the same host computer. Those two aspects, isolation and aggregation, mean that runtime security involves trade-offs and finding an appropriate balance.

Kubernetes relies on a [container runtime](#) to actually set up and run containers. The Kubernetes project does not recommend a specific container runtime and you should make sure that the runtime(s) that you choose meet your information security needs.

To protect your compute at runtime, you can:

1. Enforce [Pod security standards](#) for applications, to help ensure they run with only the necessary privileges.
2. Run a specialized operating system on your nodes that is designed specifically for running containerized workloads. This is typically based on a read-only operating system (*immutable image*) that provides only the services essential for running containers.

Container-specific operating systems help to isolate system components and present a reduced attack surface in case of a container escape.

3. Define [ResourceQuotas](#) to fairly allocate shared resources, and use mechanisms such as [LimitRanges](#) to ensure that Pods specify their resource requirements.
4. Partition workloads across different nodes. Use [node isolation](#) mechanisms, either from Kubernetes itself or from the ecosystem, to ensure that Pods with different trust contexts are run on separate sets of nodes.
5. Use a [container runtime](#) that provides security restrictions.
6. On Linux nodes, use a Linux security module such as [AppArmor](#) or [seccomp](#).

## Runtime protection: storage

To protect storage for your cluster and the applications that run there, you can:

1. Integrate your cluster with an external storage plugin that provides encryption at rest for volumes.
2. Enable [encryption at rest](#) for API objects.
3. Protect data durability using backups. Verify that you can restore these, whenever you need to.
4. Authenticate connections between cluster nodes and any network storage they rely upon.
5. Implement data encryption within your own application.

For encryption keys, generating these within specialized hardware provides the best protection against disclosure risks. A *hardware security module* can let you perform cryptographic operations without allowing the security key to be copied elsewhere.

## Networking and security

You should also consider network security measures, such as [NetworkPolicy](#) or a [service mesh](#). Some network plugins for Kubernetes provide encryption for your cluster network, using technologies such as a virtual private network (VPN) overlay. By design, Kubernetes lets you use your own networking plugin for your cluster (if you use managed Kubernetes, the person or organization managing your cluster may have chosen a network plugin for you).

The network plugin you choose and the way you integrate it can have a strong impact on the security of information in transit.



## Observability and runtime security

Kubernetes lets you extend your cluster with extra tooling. You can set up third party solutions to help you monitor or troubleshoot your applications and the clusters they are running. You also get some basic observability features built in to Kubernetes itself. Your code running in containers can generate logs, publish metrics or provide other observability data; at deploy time, you need to make sure your cluster provides an appropriate level of protection there.

If you set up a metrics dashboard or something similar, review the chain of components that populate data into that dashboard, as well as the dashboard itself. Make sure that the whole chain is designed with enough resilience and enough integrity protection that you can rely on it even during an incident where your cluster might be degraded.

Where appropriate, deploy security measures below the level of Kubernetes itself, such as cryptographically measured boot, or authenticated distribution of time (which helps ensure the fidelity of logs and audit records).

For a high assurance environment, deploy cryptographic protections to ensure that logs are both tamper-proof and confidential.

## What's next

### Cloud native security

- CNCF [white paper](#) on cloud native security.
- CNCF [white paper](#) on good practices for securing a software supply chain.
- [Fixing the Kubernetes clusterf\\*\\*k: Understanding security from the kernel up](#) (FOSDEM 2020)
- [Kubernetes Security Best Practices](#) (Kubernetes Forum Seoul 2019)
- [Towards Measured Boot Out of the Box](#) (Linux Security Summit 2016)

### Kubernetes and information security

- [Kubernetes security](#)
- [Securing your cluster](#)
- [Data encryption in transit](#) for the control plane
- [Data encryption at rest](#)
- [Secrets in Kubernetes](#)
- [Controlling Access to the Kubernetes API](#)
- [Network policies](#) for Pods
- [Pod security standards](#)
- [RuntimeClasses](#)

## Pod Security Standards

A detailed look at the different policy levels defined in the Pod Security Standards.

The Pod Security Standards define three different *policies* to broadly cover the security spectrum. These policies are *cumulative* and range from highly-permissive to highly-restrictive. This guide outlines the requirements of each policy.

| Profile           | Description                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Privileged</b> | Unrestricted policy, providing the widest possible level of permissions. This policy allows for known privilege escalations.         |
| <b>Baseline</b>   | Minimally restrictive policy which prevents known privilege escalations. Allows the default (minimally specified) Pod configuration. |
| <b>Restricted</b> | Heavily restricted policy, following current Pod hardening best practices.                                                           |

## Profile Details

### Privileged

The ***Privileged*** policy is **purposely-open, and entirely unrestricted**. This type of policy is typically aimed at system- and infrastructure-level workloads managed by privileged, trusted users.

The Privileged policy is defined by an absence of restrictions. Allow-by-default mechanisms (such as gatekeeper) may be Privileged by default. In contrast, for a deny-by-default mechanism (such as Pod Security Policy) the Privileged policy should disable all restrictions.

### Baseline

The ***Baseline*** policy is aimed at **ease of adoption for common containerized workloads while preventing known privilege escalations**. This policy is targeted at application operators and developers of non-critical applications. The following listed controls should be enforced/disallowed:

#### Note:

In this table, wildcards (\*) indicate all elements in a list. For example, `spec.containers[*].securityContext` refers to the Security Context object for *all defined containers*. If any of the listed containers fails to meet the requirements, the entire pod will fail validation.

Baseline policy specification

| Control     | Policy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HostProcess | <p>Windows pods offer the ability to run <a href="#">HostProcess containers</a> which enables privileged access to the Windows node. Privileged access to the host is disallowed in the baseline policy.</p> <p>FEATURE STATE: Kubernetes v1.26 [stable]</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li><code>spec.securityContext.windowsOptions.hostProcess</code></li> <li><code>spec.containers[*].securityContext.windowsOptions.hostProcess</code></li> <li><code>spec.initContainers[*].securityContext.windowsOptions.hostProcess</code></li> <li><code>spec.ephemeralContainers[*].securityContext.windowsOptions.hostProcess</code></li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>Undefined/nil</li> </ul> |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <ul style="list-style-type: none"> <li>• false</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                        |
| Host<br>Namespaces       | <p>Sharing the host namespaces must be disallowed.</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.hostNetwork</li> <li>• spec.hostPID</li> <li>• spec.hostIPC</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• false</li> </ul>                                                                                                                                      |
| Privileged<br>Containers | <p>Privileged Pods disable most security mechanisms and must be disallowed.</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.containers[*].securityContext.privileged</li> <li>• spec.initContainers[*].securityContext.privileged</li> <li>• spec.ephemeralContainers[*].securityContext.privileged</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• false</li> </ul> |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Capabilities     | <p>Adding additional capabilities beyond those listed below must be disallowed.</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.containers[*].securityContext.capabilities.add</li> <li>• spec.initContainers[*].securityContext.capabilities.add</li> <li>• spec.ephemeralContainers[*].securityContext.capabilities.add</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• AUDIT_WRITE</li> <li>• CHOWN</li> <li>• DAC_OVERRIDE</li> <li>• FOWNER</li> <li>• FSETID</li> <li>• KILL</li> <li>• MKNOD</li> <li>• NET_BIND_SERVICE</li> <li>• SETFCAP</li> <li>• SETGID</li> <li>• SETPCAP</li> <li>• SETUID</li> <li>• SYS_CHROOT</li> </ul> |
| HostPath Volumes | <p>HostPath volumes must be forbidden.</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.volumes[*].hostPath</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/nil</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Host Ports       | <p>HostPorts should be disallowed entirely (recommended) or restricted to a known list</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.containers[*].ports[*].hostPort</li> <li>• spec.initContainers[*].ports[*].hostPort</li> <li>• spec.ephemeralContainers[*].ports[*].hostPort</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• Known list (not supported by the built-in <a href="#">Pod Security Admission controller</a>)</li> <li>• 0</li> </ul>                                                                                                                                                                                   |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AppArmor | <p>On supported hosts, the RuntimeDefault AppArmor profile is applied by default. The baseline policy should prevent overriding or disabling the default AppArmor profile, or restrict overrides to an allowed set of profiles.</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.securityContext.appArmorProfile.type</li> <li>• spec.containers[*].securityContext.appArmorProfile.type</li> <li>• spec.initContainers[*].securityContext.appArmorProfile.type</li> <li>• spec.ephemeralContainers[*].securityContext.appArmorProfile.type</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• RuntimeDefault</li> <li>• Localhost</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• metadata.annotations["container.apparmor.security.beta.kubernetes.io/*"]</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• runtime/default</li> <li>• localhost/*</li> </ul>                                                                                                                                                                   |
| SELinux  | <p>Setting the SELinux type is restricted, and setting a custom SELinux user or role option is forbidden.</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.securityContext.seLinuxOptions.type</li> <li>• spec.containers[*].securityContext.seLinuxOptions.type</li> <li>• spec.initContainers[*].securityContext.seLinuxOptions.type</li> <li>• spec.ephemeralContainers[*].securityContext.seLinuxOptions.type</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/""</li> <li>• container_t</li> <li>• container_init_t</li> <li>• container_kvm_t</li> </ul> <hr/> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.securityContext.seLinuxOptions.user</li> <li>• spec.containers[*].securityContext.seLinuxOptions.user</li> <li>• spec.initContainers[*].securityContext.seLinuxOptions.user</li> <li>• spec.ephemeralContainers[*].securityContext.seLinuxOptions.user</li> <li>• spec.securityContext.seLinuxOptions.role</li> <li>• spec.containers[*].securityContext.seLinuxOptions.role</li> <li>• spec.initContainers[*].securityContext.seLinuxOptions.role</li> </ul> |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <ul style="list-style-type: none"> <li>• spec.ephemeralContainers[*].securityContext.seLinuxOptions.role</li> </ul> <b>Allowed Values</b> <ul style="list-style-type: none"> <li>• Undefined/""</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| /proc Mount Type | <p>The default /proc masks are set up to reduce attack surface, and should be required.</p> <b>Restricted Fields</b> <ul style="list-style-type: none"> <li>• spec.containers[*].securityContext.procMount</li> <li>• spec.initContainers[*].securityContext.procMount</li> <li>• spec.ephemeralContainers[*].securityContext.procMount</li> </ul> <b>Allowed Values</b> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• Default</li> </ul>                                                                                                                                                                                                                                              |
| Seccomp          | <p>Seccomp profile must not be explicitly set to Unconfined.</p> <b>Restricted Fields</b> <ul style="list-style-type: none"> <li>• spec.securityContext.seccompProfile.type</li> <li>• spec.containers[*].securityContext.seccompProfile.type</li> <li>• spec.initContainers[*].securityContext.seccompProfile.type</li> <li>• spec.ephemeralContainers[*].securityContext.seccompProfile.type</li> </ul> <b>Allowed Values</b> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• RuntimeDefault</li> <li>• Localhost</li> </ul>                                                                                                                                                           |
| Sysctls          | <p>Sysctls can disable security mechanisms or affect all containers on a host, and should be disallowed except for an allowed "safe" subset. A sysctl is considered safe if it is namespaced in the container or the Pod, and it is isolated from other Pods or processes on the same Node.</p> <b>Restricted Fields</b> <ul style="list-style-type: none"> <li>• spec.securityContext.sysctls[*].name</li> </ul> <b>Allowed Values</b> <ul style="list-style-type: none"> <li>• Undefined/nil</li> <li>• kernel.shm_rmid_forced</li> <li>• net.ipv4.ip_local_port_range</li> <li>• net.ipv4.ip_unprivileged_port_start</li> <li>• net.ipv4.tcp_syncookies</li> <li>• net.ipv4.ping_group_range</li> </ul> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• <code>net.ipv4.ip_local_reserved_ports</code> (since Kubernetes 1.27)</li> <li>• <code>net.ipv4.tcp_keepalive_time</code> (since Kubernetes 1.29)</li> <li>• <code>net.ipv4.tcp_fin_timeout</code> (since Kubernetes 1.29)</li> <li>• <code>net.ipv4.tcp_keepalive_intvl</code> (since Kubernetes 1.29)</li> <li>• <code>net.ipv4.tcp_keepalive_probes</code> (since Kubernetes 1.29)</li> </ul> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Restricted

**The *Restricted* policy is aimed at enforcing current Pod hardening best practices, at the expense of some compatibility.** It is targeted at operators and developers of security-critical applications, as well as lower-trust users. The following listed controls should be enforced/disallowed:

### Note:

In this table, wildcards (\*) indicate all elements in a list. For example, `spec.containers[*].securityContext` refers to the Security Context object for *all defined containers*. If any of the listed containers fails to meet the requirements, the entire pod will fail validation.

Restricted policy specification

| Control                                      | Policy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Everything from the baseline profile.</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Volume Types                                 | <p>The restricted policy only permits the following volume types.</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• <code>spec.volumes[*]</code></li> </ul> <p><b>Allowed Values</b></p> <p>Every item in the <code>spec.volumes[*]</code> list must set one of the following fields to a non-null value:</p> <ul style="list-style-type: none"> <li>• <code>spec.volumes[*].configMap</code></li> <li>• <code>spec.volumes[*].csi</code></li> <li>• <code>spec.volumes[*].downwardAPI</code></li> <li>• <code>spec.volumes[*].emptyDir</code></li> <li>• <code>spec.volumes[*].ephemeral</code></li> <li>• <code>spec.volumes[*].persistentVolumeClaim</code></li> <li>• <code>spec.volumes[*].projected</code></li> <li>• <code>spec.volumes[*].secret</code></li> </ul> |
| Privilege Escalation (v1.8+)                 | <p>Privilege escalation (such as via <code>set-user-ID</code> or <code>set-group-ID</code> file mode) should not be allowed. <a href="#"><i>This is Linux only policy</i></a> in v1.25+ (<code>spec.os.name != windows</code>)</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• <code>spec.containers[*].securityContext.allowPrivilegeEscalation</code></li> <li>• <code>spec.initContainers[*].securityContext.allowPrivilegeEscalation</code></li> <li>• <code>spec.ephemeralContainers[*].securityContext.allowPrivilegeEscalation</code></li> </ul>                                                                                                                                                                                                                  |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• false</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Running as Non-root               | <p>Containers must be required to run as non-root users.</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.securityContext.runAsNonRoot</li> <li>• spec.containers[*].securityContext.runAsNonRoot</li> <li>• spec.initContainers[*].securityContext.runAsNonRoot</li> <li>• spec.ephemeralContainers[*].securityContext.runAsNonRoot</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• true</li> </ul> <p>The container fields may be undefined/nil if the pod-level spec.securityContext.runAsNonRoot is set to true.</p>                                                                                                                            |
| Running as Non-root user (v1.23+) | <p>Containers must not set runAsUser to 0</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.securityContext.runAsUser</li> <li>• spec.containers[*].securityContext.runAsUser</li> <li>• spec.initContainers[*].securityContext.runAsUser</li> <li>• spec.ephemeralContainers[*].securityContext.runAsUser</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• any non-zero value</li> <li>• undefined/null</li> </ul>                                                                                                                                                                                                                                   |
| Seccomp (v1.19+)                  | <p>Seccomp profile must be explicitly set to one of the allowed values. Both the Unconfined profile and the <i>absence</i> of a profile are prohibited. <a href="#">This is Linux only policy</a> in v1.25+ (<i>spec.os.name != windows</i>)</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li>• spec.securityContext.seccompProfile.type</li> <li>• spec.containers[*].securityContext.seccompProfile.type</li> <li>• spec.initContainers[*].securityContext.seccompProfile.type</li> <li>• spec.ephemeralContainers[*].securityContext.seccompProfile.type</li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• RuntimeDefault</li> <li>• Localhost</li> </ul> |



|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | The container fields may be undefined/nil if the pod-level <code>spec.securityContext.seccompProfile.type</code> field is set appropriately. Conversely, the pod-level field may be undefined/nil if <code>_all_</code> container- level fields are set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Capabilities (v1.22+) | <p>Containers must drop ALL capabilities, and are only permitted to add back the <code>NET_BIND_SERVICE</code> capability. <a href="#">This is Linux only policy</a> in v1.25+ (<code>spec.os.name != "windows"</code>)</p> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li><code>spec.containers[*].securityContext.capabilities.drop</code></li> <li><code>spec.initContainers[*].securityContext.capabilities.drop</code></li> <li><code>spec.ephemeralContainers[*].securityContext.capabilities.drop</code></li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>Any list of capabilities that includes ALL</li> </ul> <hr/> <p><b>Restricted Fields</b></p> <ul style="list-style-type: none"> <li><code>spec.containers[*].securityContext.capabilities.add</code></li> <li><code>spec.initContainers[*].securityContext.capabilities.add</code></li> <li><code>spec.ephemeralContainers[*].securityContext.capabilities.add</code></li> </ul> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>Undefined/nil</li> <li><code>NET_BIND_SERVICE</code></li> </ul> |

## Policy Instantiation

Decoupling policy definition from policy instantiation allows for a common understanding and consistent language of policies across clusters, independent of the underlying enforcement mechanism.

As mechanisms mature, they will be defined below on a per-policy basis. The methods of enforcement of individual policies are not defined here.

### Pod Security Admission Controller

- [Privileged namespace](#)
- [Baseline namespace](#)
- [Restricted namespace](#)

## Alternatives

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Other alternatives for enforcing policies are being developed in the Kubernetes ecosystem, such as:

- [Kubewarden](#)
- [Kyverno](#)
- [OPA Gatekeeper](#)

## Pod OS field

Kubernetes lets you use nodes that run either Linux or Windows. You can mix both kinds of node in one cluster. Windows in Kubernetes has some limitations and differentiators from Linux-based workloads. Specifically, many of the Pod securityContext fields [have no effect on Windows](#).

### Note:

Kubelets prior to v1.24 don't enforce the pod OS field, and if a cluster has nodes on versions earlier than v1.24 the restricted policies should be pinned to a version prior to v1.25.

## Restricted Pod Security Standard changes

Another important change, made in Kubernetes v1.25 is that the *restricted* Pod security has been updated to use the pod.spec.os.name field. Based on the OS name, certain policies that are specific to a particular OS can be relaxed for the other OS.

### OS-specific policy controls

Restrictions on the following controls are only required if .spec.os.name is not windows:

- Privilege Escalation
- Seccomp
- Linux Capabilities

## User namespaces

User Namespaces are a Linux-only feature to run workloads with increased isolation. How they work together with Pod Security Standards is described in the [documentation](#) for Pods that use user namespaces.

## FAQ

### Why isn't there a profile between privileged and baseline?

The three profiles defined here have a clear linear progression from most secure (restricted) to least secure (privileged), and cover a broad set of workloads. Privileges required above the baseline policy are typically very application specific, so we do not offer a standard profile in this niche. This is not to say that the privileged profile should always be used in this case, but that policies in this space need to be defined on a case-by-case basis.

SIG Auth may reconsider this position in the future, should a clear need for other profiles arise.

## What's the difference between a security profile and a security context?

[Security Contexts](#) configure Pods and Containers at runtime. Security contexts are defined as part of the Pod and container specifications in the Pod manifest, and represent parameters to the container runtime.

Security profiles are control plane mechanisms to enforce specific settings in the Security Context, as well as other related parameters outside the Security Context. As of July 2021, [Pod Security Policies](#) are deprecated in favor of the built-in [Pod Security Admission Controller](#).

## What about sandboxed Pods?

There is not currently an API standard that controls whether a Pod is considered sandboxed or not. Sandbox Pods may be identified by the use of a sandboxed runtime (such as gVisor or Kata Containers), but there is no standard definition of what a sandboxed runtime is.

The protections necessary for sandboxed workloads can differ from others. For example, the need to restrict privileged permissions is lessened when the workload is isolated from the underlying kernel. This allows for workloads requiring heightened permissions to still be isolated.

Additionally, the protection of sandboxed workloads is highly dependent on the method of sandboxing. As such, no single recommended profile is recommended for all sandboxed workloads.

# Pod Security Admission

An overview of the Pod Security Admission Controller, which can enforce the Pod Security Standards.

FEATURE STATE: Kubernetes v1.25 [stable]

The Kubernetes [Pod Security Standards](#) define different isolation levels for Pods. These standards let you define how you want to restrict the behavior of pods in a clear, consistent fashion.

Kubernetes offers a built-in *Pod Security* [admission controller](#) to enforce the Pod Security Standards. Pod security restrictions are applied at the [namespace](#) level when pods are created.

## Built-in Pod Security admission enforcement

This page is part of the documentation for Kubernetes v1.30. If you are running a different version of Kubernetes, consult the documentation for that release.

## Pod Security levels

Pod Security admission places requirements on a Pod's [Security Context](#) and other related fields according to the three levels defined by the [Pod Security Standards](#): privileged, baseline, and restricted. Refer to the [Pod Security Standards](#) page for an in-depth look at those requirements.

# Pod Security Admission labels for namespaces

Once the feature is enabled or the webhook is installed, you can configure namespaces to define the admission control mode you want to use for pod security in each namespace. Kubernetes defines a set of [labels](#) that you can set to define which of the predefined Pod Security Standard levels you want to use for a namespace. The label you select defines what action the [control plane](#) takes if a potential violation is detected:

Pod Security Admission modes

| Mode           | Description                                                                                                                                            |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>enforce</b> | Policy violations will cause the pod to be rejected.                                                                                                   |
| <b>audit</b>   | Policy violations will trigger the addition of an audit annotation to the event recorded in the <a href="#">audit log</a> , but are otherwise allowed. |
| <b>warn</b>    | Policy violations will trigger a user-facing warning, but are otherwise allowed.                                                                       |

A namespace can configure any or all modes, or even set a different level for different modes.

For each mode, there are two labels that determine the policy used:

```
The per-mode level label indicates which policy level to apply for the mode.
#
MODE must be one of `enforce`, `audit`, or `warn`.
LEVEL must be one of `privileged`, `baseline`, or `restricted`.
pod-security.kubernetes.io/<MODE>: <LEVEL>

Optional: per-mode version label that can be used to pin the policy to the
version that shipped with a given Kubernetes minor version (for example v1.30).
#
MODE must be one of `enforce`, `audit`, or `warn`.
VERSION must be a valid Kubernetes minor version, or `latest`.
pod-security.kubernetes.io/<MODE>-version: <VERSION>
```

Check out [Enforce Pod Security Standards with Namespace Labels](#) to see example usage.

## Workload resources and Pod templates

Pods are often created indirectly, by creating a [workload object](#) such as a [Deployment](#) or [Job](#). The workload object defines a *Pod template* and a [controller](#) for the workload resource creates Pods based on that template. To help catch violations early, both the audit and warning modes are applied to the workload resources. However, enforce mode is **not** applied to workload resources, only to the resulting pod objects.

## Exemptions

You can define *exemptions* from pod security enforcement in order to allow the creation of pods that would have otherwise been prohibited due to the policy associated with a given namespace. Exemptions can be statically configured in the [Admission Controller configuration](#).

Exemptions must be explicitly enumerated. Requests meeting exemption criteria are *ignored* by the Admission Controller (all enforce, audit and warn behaviors are skipped). Exemption dimensions include:

- **Username:** requests from users with an exempt authenticated (or impersonated) username are ignored.
- **RuntimeClassNames:** pods and [workload resources](#) specifying an exempt runtime class name are ignored.
- **Namespaces:** pods and [workload resources](#) in an exempt namespace are ignored.

### Caution:

Most pods are created by a controller in response to a [workload resource](#), meaning that exempting an end user will only exempt them from enforcement when creating pods directly, but not when creating a workload resource. Controller service accounts (such as `system:serviceaccount:kube-system:replicaset-controller`) should generally not be exempted, as doing so would implicitly exempt any user that can create the corresponding workload resource.

Updates to the following pod fields are exempt from policy checks, meaning that if a pod update request only changes these fields, it will not be denied even if the pod is in violation of the current policy level:

- Any metadata updates **except** changes to the seccomp or AppArmor annotations:
  - `seccomp.security.alpha.kubernetes.io/pod` (deprecated)
  - `container.seccomp.security.alpha.kubernetes.io/*` (deprecated)
  - `container.apparmor.security.beta.kubernetes.io/*` (deprecated)
- Valid updates to `.spec.activeDeadlineSeconds`
- Valid updates to `.spec.tolerations`

## Metrics

Here are the Prometheus metrics exposed by kube-apiserver:

- `pod_security_errors_total`: This metric indicates the number of errors preventing normal evaluation. Non-fatal errors may result in the latest restricted profile being used for enforcement.
- `pod_security_evaluations_total`: This metric indicates the number of policy evaluations that have occurred, not counting ignored or exempt requests during exporting.
- `pod_security_exemptions_total`: This metric indicates the number of exempt requests, not counting ignored or out of scope requests.

## What's next

- [Pod Security Standards](#)
- [Enforcing Pod Security Standards](#)
- [Enforce Pod Security Standards by Configuring the Built-in Admission Controller](#)
- [Enforce Pod Security Standards with Namespace Labels](#)

If you are running an older version of Kubernetes and want to upgrade to a version of Kubernetes that does not include PodSecurityPolicies, read [migrate from PodSecurityPolicy to the Built-In PodSecurity Admission Controller](#).

# Service Accounts

Learn about ServiceAccount objects in Kubernetes.

This page introduces the ServiceAccount object in Kubernetes, providing information about how service accounts work, use cases, limitations, alternatives, and links to resources for additional guidance.

## What are service accounts?

A service account is a type of non-human account that, in Kubernetes, provides a distinct identity in a Kubernetes cluster. Application Pods, system components, and entities inside and outside the cluster can use a specific ServiceAccount's credentials to identify as that ServiceAccount. This identity is useful in various situations, including authenticating to the API server or implementing identity-based security policies.

Service accounts exist as ServiceAccount objects in the API server. Service accounts have the following properties:

- **Namespaced:** Each service account is bound to a Kubernetes [namespace](#). Every namespace gets a [default ServiceAccount](#) upon creation.
- **Lightweight:** Service accounts exist in the cluster and are defined in the Kubernetes API. You can quickly create service accounts to enable specific tasks.
- **Portable:** A configuration bundle for a complex containerized workload might include service account definitions for the system's components. The lightweight nature of service accounts and the namespaced identities make the configurations portable.

Service accounts are different from user accounts, which are authenticated human users in the cluster. By default, user accounts don't exist in the Kubernetes API server; instead, the API server treats user identities as opaque data. You can authenticate as a user account using multiple methods. Some Kubernetes distributions might add custom extension APIs to represent user accounts in the API server.

Comparison between service accounts and users

| Description    | ServiceAccount                                                    | User or group                                                      |
|----------------|-------------------------------------------------------------------|--------------------------------------------------------------------|
| Location       | Kubernetes API (ServiceAccount object)                            | External                                                           |
| Access control | Kubernetes RBAC or other <a href="#">authorization mechanisms</a> | Kubernetes RBAC or other identity and access management mechanisms |
| Intended use   | Workloads, automation                                             | People                                                             |

## Default service accounts

When you create a cluster, Kubernetes automatically creates a ServiceAccount object named `default` for every namespace in your cluster. The default service accounts in each namespace get no permissions by default other than the [default API discovery permissions](#) that Kubernetes grants to all authenticated principals if role-based access control (RBAC) is enabled. If you delete the default ServiceAccount object in a namespace, the [control plane](#) replaces it with a new one.

If you deploy a Pod in a namespace, and you don't [manually assign a ServiceAccount to the Pod](#), Kubernetes assigns the default ServiceAccount for that namespace to the Pod.

## Use cases for Kubernetes service accounts

As a general guideline, you can use service accounts to provide identities in the following scenarios:

- Your Pods need to communicate with the Kubernetes API server, for example in situations such as the following:
  - Providing read-only access to sensitive information stored in Secrets.
  - Granting [cross-namespace access](#), such as allowing a Pod in namespace example to read, list, and watch for Lease objects in the kube-node-lease namespace.
- Your Pods need to communicate with an external service. For example, a workload Pod requires an identity for a commercially available cloud API, and the commercial provider allows configuring a suitable trust relationship.
- [Authenticating to a private image registry using an imagePullSecret](#).
- An external service needs to communicate with the Kubernetes API server. For example, authenticating to the cluster as part of a CI/CD pipeline.
- You use third-party security software in your cluster that relies on the ServiceAccount identity of different Pods to group those Pods into different contexts.

## How to use service accounts

To use a Kubernetes service account, you do the following:

1. Create a ServiceAccount object using a Kubernetes client like kubectl or a manifest that defines the object.
2. Grant permissions to the ServiceAccount object using an authorization mechanism such as [RBAC](#).
3. Assign the ServiceAccount object to Pods during Pod creation.

If you're using the identity from an external service, [retrieve the ServiceAccount token](#) and use it from that service instead.

For instructions, refer to [Configure Service Accounts for Pods](#).

## Grant permissions to a ServiceAccount

You can use the built-in Kubernetes [role-based access control \(RBAC\)](#) mechanism to grant the minimum permissions required by each service account. You create a *role*, which grants access, and then *bind* the role to your ServiceAccount. RBAC lets you define a minimum set of permissions so that the service account permissions follow the principle of least privilege. Pods that use that service account don't get more permissions than are required to function correctly.

For instructions, refer to [ServiceAccount permissions](#).



## Cross-namespace access using a ServiceAccount

You can use RBAC to allow service accounts in one namespace to perform actions on resources in a different namespace in the cluster. For example, consider a scenario where you have a service account and Pod in the dev namespace and you want your Pod to see Jobs running in the maintenance namespace. You could create a Role object that grants permissions to list Job objects. Then, you'd create a RoleBinding object in the maintenance namespace to bind the Role to the ServiceAccount object. Now, Pods in the dev namespace can list Job objects in the maintenance namespace using that service account.

## Assign a ServiceAccount to a Pod

To assign a ServiceAccount to a Pod, you set the `spec.serviceAccountName` field in the Pod specification. Kubernetes then automatically provides the credentials for that ServiceAccount to the Pod. In v1.22 and later, Kubernetes gets a short-lived, **automatically rotating** token using the TokenRequest API and mounts the token as a [projected volume](#).

By default, Kubernetes provides the Pod with the credentials for an assigned ServiceAccount, whether that is the default ServiceAccount or a custom ServiceAccount that you specify.

To prevent Kubernetes from automatically injecting credentials for a specified ServiceAccount or the default ServiceAccount, set the `automountServiceAccountToken` field in your Pod specification to false.

In versions earlier than 1.22, Kubernetes provides a long-lived, static token to the Pod as a Secret.

## Manually retrieve ServiceAccount credentials

If you need the credentials for a ServiceAccount to mount in a non-standard location, or for an audience that isn't the API server, use one of the following methods:

- [TokenRequest API](#) (recommended): Request a short-lived service account token from within your own *application code*. The token expires automatically and can rotate upon expiration. If you have a legacy application that is not aware of Kubernetes, you could use a sidecar container within the same pod to fetch these tokens and make them available to the application workload.
- [Token Volume Projection](#) (also recommended): In Kubernetes v1.20 and later, use the Pod specification to tell the kubelet to add the service account token to the Pod as a *projected volume*. Projected tokens expire automatically, and the kubelet rotates the token before it expires.
- [Service Account Token Secrets](#) (not recommended): You can mount service account tokens as Kubernetes Secrets in Pods. These tokens don't expire and don't rotate. In versions prior to v1.24, a permanent token was automatically created for each service account. This method is not recommended anymore, especially at scale, because of the risks associated with static, long-lived credentials. The [LegacyServiceAccountTokenNoAutoGeneration feature gate](#) (which was enabled by default from Kubernetes v1.24 to v1.26), prevented Kubernetes from automatically creating these tokens for ServiceAccounts. The feature gate is removed in v1.27, because it was elevated to GA status; you can still create indefinite service account tokens manually, but should take into account the security implications.



## Note:

For applications running outside your Kubernetes cluster, you might be considering creating a long-lived ServiceAccount token that is stored in a Secret. This allows authentication, but the Kubernetes project recommends you avoid this approach. Long-lived bearer tokens represent a security risk as, once disclosed, the token can be misused. Instead, consider using an alternative. For example, your external application can authenticate using a well-protected private key and a certificate, or using a custom mechanism such as an [authentication webhook](#) that you implement yourself.

You can also use TokenRequest to obtain short-lived tokens for your external application.

## Restricting access to Secrets

Kubernetes provides an annotation called `kubernetes.io/enforce-mountable-secrets` that you can add to your ServiceAccounts. When this annotation is applied, the ServiceAccount's secrets can only be mounted on specified types of resources, enhancing the security posture of your cluster.

You can add the annotation to a ServiceAccount using a manifest:

```
apiVersion: v1
kind: ServiceAccount
metadata:
 annotations:
 kubernetes.io/enforce-mountable-secrets: "true"
 name: my-serviceaccount
 namespace: my-namespace
```

When this annotation is set to "true", the Kubernetes control plane ensures that the Secrets from this ServiceAccount are subject to certain mounting restrictions.

1. The name of each Secret that is mounted as a volume in a Pod must appear in the `secrets` field of the Pod's ServiceAccount.
2. The name of each Secret referenced using `envFrom` in a Pod must also appear in the `secrets` field of the Pod's ServiceAccount.
3. The name of each Secret referenced using `imagePullSecrets` in a Pod must also appear in the `secrets` field of the Pod's ServiceAccount.

By understanding and enforcing these restrictions, cluster administrators can maintain a tighter security profile and ensure that secrets are accessed only by the appropriate resources.

## Authenticating service account credentials

ServiceAccounts use signed [JSON Web Tokens](#) (JWTs) to authenticate to the Kubernetes API server, and to any other system where a trust relationship exists. Depending on how the token was issued (either time-limited using a TokenRequest or using a legacy mechanism with a Secret), a ServiceAccount token might also have an expiry time, an audience, and a time after which the token *starts* being valid. When a client that is acting as a ServiceAccount tries to communicate with the Kubernetes API server, the client includes an `Authorization: Bearer <token>` header with the HTTP request. The API server checks the validity of that bearer token as follows:

1. Checks the token signature.

2. Checks whether the token has expired.
3. Checks whether object references in the token claims are currently valid.
4. Checks whether the token is currently valid.
5. Checks the audience claims.

The TokenRequest API produces *bound tokens* for a ServiceAccount. This binding is linked to the lifetime of the client, such as a Pod, that is acting as that ServiceAccount. See [Token Volume Projection](#) for an example of a bound pod service account token's JWT schema and payload.

For tokens issued using the TokenRequest API, the API server also checks that the specific object reference that is using the ServiceAccount still exists, matching by the [unique ID](#) of that object. For legacy tokens that are mounted as Secrets in Pods, the API server checks the token against the Secret.

For more information about the authentication process, refer to [Authentication](#).

## Authenticating service account credentials in your own code

If you have services of your own that need to validate Kubernetes service account credentials, you can use the following methods:

- [TokenReview API](#) (recommended)
- OIDC discovery

The Kubernetes project recommends that you use the TokenReview API, because this method invalidates tokens that are bound to API objects such as Secrets, ServiceAccounts, Pods or Nodes when those objects are deleted. For example, if you delete the Pod that contains a projected ServiceAccount token, the cluster invalidates that token immediately and a TokenReview immediately fails. If you use OIDC validation instead, your clients continue to treat the token as valid until the token reaches its expiration timestamp.

Your application should always define the audience that it accepts, and should check that the token's audiences match the audiences that the application expects. This helps to minimize the scope of the token so that it can only be used in your application and nowhere else.

## Alternatives

- Issue your own tokens using another mechanism, and then use [Webhook Token Authentication](#) to validate bearer tokens using your own validation service.
- Provide your own identities to Pods.
  - [Use the SPIFFE CSI driver plugin to provide SPIFFE SVIDs as X.509 certificate pairs to Pods.](#)

This item links to a third party project or product that is not part of Kubernetes itself. [More information](#)
  - [Use a service mesh such as Istio to provide certificates to Pods.](#)
- Authenticate from outside the cluster to the API server without using service account tokens:
  - [Configure the API server to accept OpenID Connect \(OIDC\) tokens from your identity provider.](#)

- Use service accounts or user accounts created using an external Identity and Access Management (IAM) service, such as from a cloud provider, to authenticate to your cluster.
- [Use the CertificateSigningRequest API with client certificates.](#)
- [Configure the kubelet to retrieve credentials from an image registry.](#)
- Use a Device Plugin to access a virtual Trusted Platform Module (TPM), which then allows authentication using a private key.

## What's next

- Learn how to [manage your ServiceAccounts as a cluster administrator.](#)
- Learn how to [assign a ServiceAccount to a Pod.](#)
- Read the [ServiceAccount API reference.](#)

# Pod Security Policies

## Removed feature

PodSecurityPolicy was [deprecated](#) in Kubernetes v1.21, and removed from Kubernetes in v1.25.

Instead of using PodSecurityPolicy, you can enforce similar restrictions on Pods using either or both:

- [Pod Security Admission](#)
- a 3rd party admission plugin, that you deploy and configure yourself

For a migration guide, see [Migrate from PodSecurityPolicy to the Built-In PodSecurity Admission Controller](#). For more information on the removal of this API, see [PodSecurityPolicy Deprecation: Past, Present, and Future](#).

If you are not running Kubernetes v1.30, check the documentation for your version of Kubernetes.

# Security For Windows Nodes

This page describes security considerations and best practices specific to the Windows operating system.

## Protection for Secret data on nodes

On Windows, data from Secrets are written out in clear text onto the node's local storage (as compared to using tmpfs / in-memory filesystems on Linux). As a cluster operator, you should take both of the following additional measures:

1. Use file ACLs to secure the Secrets' file location.
2. Apply volume-level encryption using [BitLocker](#).

## Container users

[RunAsUsername](#) can be specified for Windows Pods or containers to execute the container processes as specific user. This is roughly equivalent to [RunAsUser](#).

Windows containers offer two default user accounts, ContainerUser and ContainerAdministrator. The differences between these two user accounts are covered in [When to use ContainerAdmin and ContainerUser user accounts](#) within Microsoft's *Secure Windows containers* documentation.

Local users can be added to container images during the container build process.

### Note:

- [Nano Server](#) based images run as ContainerUser by default
- [Server Core](#) based images run as ContainerAdministrator by default

Windows containers can also run as Active Directory identities by utilizing [Group Managed Service Accounts](#)

## Pod-level security isolation

Linux-specific pod security context mechanisms (such as SELinux, AppArmor, Seccomp, or custom POSIX capabilities) are not supported on Windows nodes.

Privileged containers are [not supported](#) on Windows. Instead [HostProcess containers](#) can be used on Windows to perform many of the tasks performed by privileged containers on Linux.

# Controlling Access to the Kubernetes API

This page provides an overview of controlling access to the Kubernetes API.

Users access the [Kubernetes API](#) using kubectl, client libraries, or by making REST requests. Both human users and [Kubernetes service accounts](#) can be authorized for API access. When a request reaches the API, it goes through several stages, illustrated in the following diagram:

Diagram of request handling steps for Kubernetes API request

## Transport security

By default, the Kubernetes API server listens on port 6443 on the first non-localhost network interface, protected by TLS. In a typical production Kubernetes cluster, the API serves on port 443. The port can be changed with the `--secure-port`, and the listening IP address with the `--bind-address` flag.

The API server presents a certificate. This certificate may be signed using a private certificate authority (CA), or based on a public key infrastructure linked to a generally recognized CA. The certificate and corresponding private key can be set by using the `--tls-cert-file` and `--tls-private-key-file` flags.

If your cluster uses a private certificate authority, you need a copy of that CA certificate configured into your `~/.kube/config` on the client, so that you can trust the connection and be confident it was not intercepted.

Your client can present a TLS client certificate at this stage.

## Authentication

Once TLS is established, the HTTP request moves to the Authentication step. This is shown as step **1** in the diagram. The cluster creation script or cluster admin configures the API server to run one or more Authenticator modules. Authenticators are described in more detail in [Authentication](#).

The input to the authentication step is the entire HTTP request; however, it typically examines the headers and/or client certificate.

Authentication modules include client certificates, password, and plain tokens, bootstrap tokens, and JSON Web Tokens (used for service accounts).

Multiple authentication modules can be specified, in which case each one is tried in sequence, until one of them succeeds.

If the request cannot be authenticated, it is rejected with HTTP status code 401. Otherwise, the user is authenticated as a specific username, and the user name is available to subsequent steps to use in their decisions. Some authenticators also provide the group memberships of the user, while other authenticators do not.

While Kubernetes uses usernames for access control decisions and in request logging, it does not have a User object nor does it store usernames or other information about users in its API.

## Authorization

After the request is authenticated as coming from a specific user, the request must be authorized. This is shown as step **2** in the diagram.

A request must include the username of the requester, the requested action, and the object affected by the action. The request is authorized if an existing policy declares that the user has permissions to complete the requested action.

For example, if Bob has the policy below, then he can read pods only in the namespace `projectCaribou`:

```
{
 "apiVersion": "abac.authorization.kubernetes.io/v1beta1",
 "kind": "Policy",
 "spec": {
 "user": "bob",
 "namespace": "projectCaribou",
 "resource": "pods",
 "readonly": true
 }
}
```

If Bob makes the following request, the request is authorized because he is allowed to read objects in the projectCaribou namespace:

```
{
 "apiVersion": "authorization.k8s.io/v1beta1",
 "kind": "SubjectAccessReview",
 "spec": {
 "resourceAttributes": {
 "namespace": "projectCaribou",
 "verb": "get",
 "group": "unicorn.example.org",
 "resource": "pods"
 }
 }
}
```

If Bob makes a request to write (create or update) to the objects in the projectCaribou namespace, his authorization is denied. If Bob makes a request to read (get) objects in a different namespace such as projectFish, then his authorization is denied.

Kubernetes authorization requires that you use common REST attributes to interact with existing organization-wide or cloud-provider-wide access control systems. It is important to use REST formatting because these control systems might interact with other APIs besides the Kubernetes API.

Kubernetes supports multiple authorization modules, such as ABAC mode, RBAC Mode, and Webhook mode. When an administrator creates a cluster, they configure the authorization modules that should be used in the API server. If more than one authorization modules are configured, Kubernetes checks each module, and if any module authorizes the request, then the request can proceed. If all of the modules deny the request, then the request is denied (HTTP status code 403).

To learn more about Kubernetes authorization, including details about creating policies using the supported authorization modules, see [Authorization](#).

## Admission control

Admission Control modules are software modules that can modify or reject requests. In addition to the attributes available to Authorization modules, Admission Control modules can access the contents of the object that is being created or modified.

Admission controllers act on requests that create, modify, delete, or connect to (proxy) an object. Admission controllers do not act on requests that merely read objects. When multiple admission controllers are configured, they are called in order.

This is shown as step 3 in the diagram.

Unlike Authentication and Authorization modules, if any admission controller module rejects, then the request is immediately rejected.

In addition to rejecting objects, admission controllers can also set complex defaults for fields.

The available Admission Control modules are described in [Admission Controllers](#).

Once a request passes all admission controllers, it is validated using the validation routines for the corresponding API object, and then written to the object store (shown as step 4).

## Auditing

Kubernetes auditing provides a security-relevant, chronological set of records documenting the sequence of actions in a cluster. The cluster audits the activities generated by users, by applications that use the Kubernetes API, and by the control plane itself.

For more information, see [Auditing](#).

## What's next

Read more documentation on authentication, authorization and API access control:

- [Authenticating](#)
  - [Authenticating with Bootstrap Tokens](#)
- [Admission Controllers](#)
  - [Dynamic Admission Control](#)
- [Authorization](#)
  - [Role Based Access Control](#)
  - [Attribute Based Access Control](#)
  - [Node Authorization](#)
  - [Webhook Authorization](#)
- [Certificate Signing Requests](#)
  - including [CSR approval](#) and [certificate signing](#)
- Service accounts
  - [Developer guide](#)
  - [Administration](#)

You can learn about:

- how Pods can use [Secrets](#) to obtain API credentials.

## Role Based Access Control Good Practices

Principles and practices for good RBAC design for cluster operators.

Kubernetes [RBAC](#) is a key security control to ensure that cluster users and workloads have only the access to resources required to execute their roles. It is important to ensure that, when designing permissions for cluster users, the cluster administrator understands the areas where privilege escalation could occur, to reduce the risk of excessive access leading to security incidents.

The good practices laid out here should be read in conjunction with the general [RBAC documentation](#).



# General good practice

## Least privilege

Ideally, minimal RBAC rights should be assigned to users and service accounts. Only permissions explicitly required for their operation should be used. While each cluster will be different, some general rules that can be applied are :

- Assign permissions at the namespace level where possible. Use RoleBindings as opposed to ClusterRoleBindings to give users rights only within a specific namespace.
- Avoid providing wildcard permissions when possible, especially to all resources. As Kubernetes is an extensible system, providing wildcard access gives rights not just to all object types that currently exist in the cluster, but also to all object types which are created in the future.
- Administrators should not use cluster-admin accounts except where specifically needed. Providing a low privileged account with [impersonation rights](#) can avoid accidental modification of cluster resources.
- Avoid adding users to the system:masters group. Any user who is a member of this group bypasses all RBAC rights checks and will always have unrestricted superuser access, which cannot be revoked by removing RoleBindings or ClusterRoleBindings. As an aside, if a cluster is using an authorization webhook, membership of this group also bypasses that webhook (requests from users who are members of that group are never sent to the webhook)

## Minimize distribution of privileged tokens

Ideally, pods shouldn't be assigned service accounts that have been granted powerful permissions (for example, any of the rights listed under [privilege escalation risks](#)). In cases where a workload requires powerful permissions, consider the following practices:

- Limit the number of nodes running powerful pods. Ensure that any DaemonSets you run are necessary and are run with least privilege to limit the blast radius of container escapes.
- Avoid running powerful pods alongside untrusted or publicly-exposed ones. Consider using [Taints and Toleration](#), [NodeAffinity](#), or [PodAntiAffinity](#) to ensure pods don't run alongside untrusted or less-trusted Pods. Pay special attention to situations where less-trustworthy Pods are not meeting the **Restricted** Pod Security Standard.

## Hardening

Kubernetes defaults to providing access which may not be required in every cluster. Reviewing the RBAC rights provided by default can provide opportunities for security hardening. In general, changes should not be made to rights provided to system: accounts some options to harden cluster rights exist:

- Review bindings for the system:unauthenticated group and remove them where possible, as this gives access to anyone who can contact the API server at a network level.
- Avoid the default auto-mounting of service account tokens by setting automountServiceAccountToken: false. For more details, see [using default service account token](#). Setting this value for a Pod will overwrite the service account setting, workloads which require service account tokens can still mount them.



## Periodic review

It is vital to periodically review the Kubernetes RBAC settings for redundant entries and possible privilege escalations. If an attacker is able to create a user account with the same name as a deleted user, they can automatically inherit all the rights of the deleted user, especially the rights assigned to that user.

## Kubernetes RBAC - privilege escalation risks

Within Kubernetes RBAC there are a number of privileges which, if granted, can allow a user or a service account to escalate their privileges in the cluster or affect systems outside the cluster.

This section is intended to provide visibility of the areas where cluster operators should take care, to ensure that they do not inadvertently allow for more access to clusters than intended.

### Listing secrets

It is generally clear that allowing get access on Secrets will allow a user to read their contents. It is also important to note that list and watch access also effectively allow for users to reveal the Secret contents. For example, when a List response is returned (for example, via `kubectl get secrets -A -o yaml`), the response includes the contents of all Secrets.

### Workload creation

Permission to create workloads (either Pods, or [workload resources](#) that manage Pods) in a namespace implicitly grants access to many other resources in that namespace, such as Secrets, ConfigMaps, and PersistentVolumes that can be mounted in Pods. Additionally, since Pods can run as any [ServiceAccount](#), granting permission to create workloads also implicitly grants the API access levels of any service account in that namespace.

Users who can run privileged Pods can use that access to gain node access and potentially to further elevate their privileges. Where you do not fully trust a user or other principal with the ability to create suitably secure and isolated Pods, you should enforce either the **Baseline** or **Restricted** Pod Security Standard. You can use [Pod Security admission](#) or other (third party) mechanisms to implement that enforcement.

For these reasons, namespaces should be used to separate resources requiring different levels of trust or tenancy. It is still considered best practice to follow [least privilege](#) principles and assign the minimum set of permissions, but boundaries within a namespace should be considered weak.

### Persistent volume creation

If someone - or some application - is allowed to create arbitrary PersistentVolumes, that access includes the creation of hostPath volumes, which then means that a Pod would get access to the underlying host filesystem(s) on the associated node. Granting that ability is a security risk.

There are many ways a container with unrestricted access to the host filesystem can escalate privileges, including reading data from other containers, and abusing the credentials of system services, such as Kubelet.

You should only allow access to create PersistentVolume objects for:

- Users (cluster operators) that need this access for their work, and who you trust.
- The Kubernetes control plane components which creates PersistentVolumes based on PersistentVolumeClaims that are configured for automatic provisioning. This is usually setup by the Kubernetes provider or by the operator when installing a CSI driver.

Where access to persistent storage is required trusted administrators should create PersistentVolumes, and constrained users should use PersistentVolumeClaims to access that storage.

## **Access to proxy subresource of Nodes**

Users with access to the proxy sub-resource of node objects have rights to the Kubelet API, which allows for command execution on every pod on the node(s) to which they have rights. This access bypasses audit logging and admission control, so care should be taken before granting rights to this resource.

## **Escalate verb**

Generally, the RBAC system prevents users from creating clusterroles with more rights than the user possesses. The exception to this is the escalate verb. As noted in the [RBAC documentation](#), users with this right can effectively escalate their privileges.

## **Bind verb**

Similar to the escalate verb, granting users this right allows for the bypass of Kubernetes in-built protections against privilege escalation, allowing users to create bindings to roles with rights they do not already have.

## **Impersonate verb**

This verb allows users to impersonate and gain the rights of other users in the cluster. Care should be taken when granting it, to ensure that excessive permissions cannot be gained via one of the impersonated accounts.

## **CSRs and certificate issuing**

The CSR API allows for users with create rights to CSRs and update rights on certificatesigningrequests/approval where the signer is kubernetes.io/kube-apiserver-client to create new client certificates which allow users to authenticate to the cluster. Those client certificates can have arbitrary names including duplicates of Kubernetes system components. This will effectively allow for privilege escalation.

## **Token request**

Users with create rights on serviceaccounts/token can create TokenRequests to issue tokens for existing service accounts.

## Control admission webhooks

Users with control over `validatingwebhookconfigurations` or `mutatingwebhookconfigurations` can control webhooks that can read any object admitted to the cluster, and in the case of mutating webhooks, also mutate admitted objects.

## Namespace modification

Users who can perform **patch** operations on Namespace objects (through a namespaced RoleBinding to a Role with that access) can modify labels on that namespace. In clusters where Pod Security Admission is used, this may allow a user to configure the namespace for a more permissive policy than intended by the administrators. For clusters where NetworkPolicy is used, users may be set labels that indirectly allow access to services that an administrator did not intend to allow.

## Kubernetes RBAC - denial of service risks

### Object creation denial-of-service

Users who have rights to create objects in a cluster may be able to create sufficient large objects to create a denial of service condition either based on the size or number of objects, as discussed in [etcd used by Kubernetes is vulnerable to OOM attack](#). This may be specifically relevant in multi-tenant clusters if semi-trusted or untrusted users are allowed limited access to a system.

One option for mitigation of this issue would be to use [resource quotas](#) to limit the quantity of objects which can be created.

## What's next

- To learn more about RBAC, see the [RBAC documentation](#).

## Good practices for Kubernetes Secrets

Principles and practices for good Secret management for cluster administrators and application developers.

In Kubernetes, a Secret is an object that stores sensitive information, such as passwords, OAuth tokens, and SSH keys.

Secrets give you more control over how sensitive information is used and reduces the risk of accidental exposure. Secret values are encoded as base64 strings and are stored unencrypted by default, but can be configured to be [encrypted at rest](#).

A [Pod](#) can reference the Secret in a variety of ways, such as in a volume mount or as an environment variable. Secrets are designed for confidential data and [ConfigMaps](#) are designed for non-confidential data.

The following good practices are intended for both cluster administrators and application developers. Use these guidelines to improve the security of your sensitive information in Secret objects, as well as to more effectively manage your Secrets.

## Cluster administrators

This section provides good practices that cluster administrators can use to improve the security of confidential information in the cluster.

### Configure encryption at rest

By default, Secret objects are stored unencrypted in [etcd](#). You should configure encryption of your Secret data in etcd. For instructions, refer to [Encrypt Secret Data at Rest](#).

### Configure least-privilege access to Secrets

When planning your access control mechanism, such as Kubernetes [Role-based Access Control \(RBAC\)](#), consider the following guidelines for access to Secret objects. You should also follow the other guidelines in [RBAC good practices](#).

- **Components:** Restrict watch or list access to only the most privileged, system-level components. Only grant get access for Secrets if the component's normal behavior requires it.
- **Humans:** Restrict get, watch, or list access to Secrets. Only allow cluster administrators to access etcd. This includes read-only access. For more complex access control, such as restricting access to Secrets with specific annotations, consider using third-party authorization mechanisms.

#### Caution:

Granting list access to Secrets implicitly lets the subject fetch the contents of the Secrets.

A user who can create a Pod that uses a Secret can also see the value of that Secret. Even if cluster policies do not allow a user to read the Secret directly, the same user could have access to run a Pod that then exposes the Secret. You can detect or limit the impact caused by Secret data being exposed, either intentionally or unintentionally, by a user with this access. Some recommendations include:

- Use short-lived Secrets
- Implement audit rules that alert on specific events, such as concurrent reading of multiple Secrets by a single user

### Additional ServiceAccount annotations for Secret management

You can also use the `kubernetes.io/enforce-mountable-secrets` annotation on a ServiceAccount to enforce specific rules on how Secrets are used in a Pod. For more details, see the [documentation on this annotation](#).

### Improve etcd management policies

Consider wiping or shredding the durable storage used by etcd once it is no longer in use.

If there are multiple etcd instances, configure encrypted SSL/TLS communication between the instances to protect the Secret data in transit.

## Configure access to external Secrets

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

You can use third-party Secrets store providers to keep your confidential data outside your cluster and then configure Pods to access that information. The [Kubernetes Secrets Store CSI Driver](#) is a DaemonSet that lets the kubelet retrieve Secrets from external stores, and mount the Secrets as a volume into specific Pods that you authorize to access the data.

For a list of supported providers, refer to [Providers for the Secret Store CSI Driver](#).

## Developers

This section provides good practices for developers to use to improve the security of confidential data when building and deploying Kubernetes resources.

### Restrict Secret access to specific containers

If you are defining multiple containers in a Pod, and only one of those containers needs access to a Secret, define the volume mount or environment variable configuration so that the other containers do not have access to that Secret.

### Protect Secret data after reading

Applications still need to protect the value of confidential information after reading it from an environment variable or volume. For example, your application must avoid logging the secret data in the clear or transmitting it to an untrusted party.

### Avoid sharing Secret manifests

If you configure a Secret through a [manifest](#), with the secret data encoded as base64, sharing this file or checking it in to a source repository means the secret is available to everyone who can read the manifest.

#### Caution:

Base64 encoding is *not* an encryption method, it provides no additional confidentiality over plain text.

## Multi-tenancy

This page provides an overview of available configuration options and best practices for cluster multi-tenancy.

Sharing clusters saves costs and simplifies administration. However, sharing clusters also presents challenges such as security, fairness, and managing *noisy neighbors*.

Clusters can be shared in many ways. In some cases, different applications may run in the same cluster. In other cases, multiple instances of the same application may run in the same cluster, one for each end user. All these types of sharing are frequently described using the umbrella term *multi-tenancy*.

While Kubernetes does not have first-class concepts of end users or tenants, it provides several features to help manage different tenancy requirements. These are discussed below.

## Use cases

The first step to determining how to share your cluster is understanding your use case, so you can evaluate the patterns and tools available. In general, multi-tenancy in Kubernetes clusters falls into two broad categories, though many variations and hybrids are also possible.

### Multiple teams

A common form of multi-tenancy is to share a cluster between multiple teams within an organization, each of whom may operate one or more workloads. These workloads frequently need to communicate with each other, and with other workloads located on the same or different clusters.

In this scenario, members of the teams often have direct access to Kubernetes resources via tools such as `kubectl`, or indirect access through GitOps controllers or other types of release automation tools. There is often some level of trust between members of different teams, but Kubernetes policies such as RBAC, quotas, and network policies are essential to safely and fairly share clusters.

### Multiple customers

The other major form of multi-tenancy frequently involves a Software-as-a-Service (SaaS) vendor running multiple instances of a workload for customers. This business model is so strongly associated with this deployment style that many people call it "SaaS tenancy." However, a better term might be "multi-customer tenancy," since SaaS vendors may also use other deployment models, and this deployment model can also be used outside of SaaS.

In this scenario, the customers do not have access to the cluster; Kubernetes is invisible from their perspective and is only used by the vendor to manage the workloads. Cost optimization is frequently a critical concern, and Kubernetes policies are used to ensure that the workloads are strongly isolated from each other.

## Terminology

### Tenants

When discussing multi-tenancy in Kubernetes, there is no single definition for a "tenant". Rather, the definition of a tenant will vary depending on whether multi-team or multi-customer tenancy is being discussed.

In multi-team usage, a tenant is typically a team, where each team typically deploys a small number of workloads that scales with the complexity of the service. However, the definition of "team" may itself be fuzzy, as teams may be organized into higher-level divisions or subdivided into smaller teams.

By contrast, if each team deploys dedicated workloads for each new client, they are using a multi-customer model of tenancy. In this case, a "tenant" is simply a group of users who share a single workload. This may be as large as an entire company, or as small as a single team at that company.

In many cases, the same organization may use both definitions of "tenants" in different contexts. For example, a platform team may offer shared services such as security tools and databases to multiple internal "customers" and a SaaS vendor may also have multiple teams sharing a development cluster. Finally, hybrid architectures are also possible, such as a SaaS provider using a combination of per-customer workloads for sensitive data, combined with multi-tenant shared services.

## **A cluster showing coexisting tenancy models**

### **Isolation**

There are several ways to design and build multi-tenant solutions with Kubernetes. Each of these methods comes with its own set of tradeoffs that impact the isolation level, implementation effort, operational complexity, and cost of service.

A Kubernetes cluster consists of a control plane which runs Kubernetes software, and a data plane consisting of worker nodes where tenant workloads are executed as pods. Tenant isolation can be applied in both the control plane and the data plane based on organizational requirements.

The level of isolation offered is sometimes described using terms like "hard" multi-tenancy, which implies strong isolation, and "soft" multi-tenancy, which implies weaker isolation. In particular, "hard" multi-tenancy is often used to describe cases where the tenants do not trust each other, often from security and resource sharing perspectives (e.g. guarding against attacks such as data exfiltration or DoS). Since data planes typically have much larger attack surfaces, "hard" multi-tenancy often requires extra attention to isolating the data-plane, though control plane isolation also remains critical.

However, the terms "hard" and "soft" can often be confusing, as there is no single definition that will apply to all users. Rather, "hardness" or "softness" is better understood as a broad spectrum, with many different techniques that can be used to maintain different types of isolation in your clusters, based on your requirements.

In more extreme cases, it may be easier or necessary to forgo any cluster-level sharing at all and assign each tenant their dedicated cluster, possibly even running on dedicated hardware if VMs are not considered an adequate security boundary. This may be easier with managed Kubernetes clusters, where the overhead of creating and operating clusters is at least somewhat taken on by a cloud provider. The benefit of stronger tenant isolation must be evaluated against the cost and complexity of managing multiple clusters. The [Multi-cluster SIG](#) is responsible for addressing these types of use cases.

The remainder of this page focuses on isolation techniques used for shared Kubernetes clusters. However, even if you are considering dedicated clusters, it may be valuable to review these



recommendations, as it will give you the flexibility to shift to shared clusters in the future if your needs or capabilities change.

## Control plane isolation

Control plane isolation ensures that different tenants cannot access or affect each others' Kubernetes API resources.

### Namespaces

In Kubernetes, a [Namespace](#) provides a mechanism for isolating groups of API resources within a single cluster. This isolation has two key dimensions:

1. Object names within a namespace can overlap with names in other namespaces, similar to files in folders. This allows tenants to name their resources without having to consider what other tenants are doing.
2. Many Kubernetes security policies are scoped to namespaces. For example, RBAC Roles and Network Policies are namespace-scoped resources. Using RBAC, Users and Service Accounts can be restricted to a namespace.

In a multi-tenant environment, a Namespace helps segment a tenant's workload into a logical and distinct management unit. In fact, a common practice is to isolate every workload in its own namespace, even if multiple workloads are operated by the same tenant. This ensures that each workload has its own identity and can be configured with an appropriate security policy.

The namespace isolation model requires configuration of several other Kubernetes resources, networking plugins, and adherence to security best practices to properly isolate tenant workloads. These considerations are discussed below.

### Access controls

The most important type of isolation for the control plane is authorization. If teams or their workloads can access or modify each others' API resources, they can change or disable all other types of policies thereby negating any protection those policies may offer. As a result, it is critical to ensure that each tenant has the appropriate access to only the namespaces they need, and no more. This is known as the "Principle of Least Privilege."

Role-based access control (RBAC) is commonly used to enforce authorization in the Kubernetes control plane, for both users and workloads (service accounts). [Roles](#) and [RoleBindings](#) are Kubernetes objects that are used at a namespace level to enforce access control in your application; similar objects exist for authorizing access to cluster-level objects, though these are less useful for multi-tenant clusters.

In a multi-team environment, RBAC must be used to restrict tenants' access to the appropriate namespaces, and ensure that cluster-wide resources can only be accessed or modified by privileged users such as cluster administrators.

If a policy ends up granting a user more permissions than they need, this is likely a signal that the namespace containing the affected resources should be refactored into finer-grained namespaces. Namespace management tools may simplify the management of these finer-grained namespaces by applying common RBAC policies to different namespaces, while still allowing fine-grained policies where necessary.



## Quotas

Kubernetes workloads consume node resources, like CPU and memory. In a multi-tenant environment, you can use [Resource Quotas](#) to manage resource usage of tenant workloads. For the multiple teams use case, where tenants have access to the Kubernetes API, you can use resource quotas to limit the number of API resources (for example: the number of Pods, or the number of ConfigMaps) that a tenant can create. Limits on object count ensure fairness and aim to avoid *noisy neighbor* issues from affecting other tenants that share a control plane.

Resource quotas are namespaced objects. By mapping tenants to namespaces, cluster admins can use quotas to ensure that a tenant cannot monopolize a cluster's resources or overwhelm its control plane. Namespace management tools simplify the administration of quotas. In addition, while Kubernetes quotas only apply within a single namespace, some namespace management tools allow groups of namespaces to share quotas, giving administrators far more flexibility with less effort than built-in quotas.

Quotas prevent a single tenant from consuming greater than their allocated share of resources hence minimizing the “noisy neighbor” issue, where one tenant negatively impacts the performance of other tenants' workloads.

When you apply a quota to namespace, Kubernetes requires you to also specify resource requests and limits for each container. Limits are the upper bound for the amount of resources that a container can consume. Containers that attempt to consume resources that exceed the configured limits will either be throttled or killed, based on the resource type. When resource requests are set lower than limits, each container is guaranteed the requested amount but there may still be some potential for impact across workloads.

Quotas cannot protect against all kinds of resource sharing, such as network traffic. Node isolation (described below) may be a better solution for this problem.

## Data Plane Isolation

Data plane isolation ensures that pods and workloads for different tenants are sufficiently isolated.

### Network isolation

By default, all pods in a Kubernetes cluster are allowed to communicate with each other, and all network traffic is unencrypted. This can lead to security vulnerabilities where traffic is accidentally or maliciously sent to an unintended destination, or is intercepted by a workload on a compromised node.

Pod-to-pod communication can be controlled using [Network Policies](#), which restrict communication between pods using namespace labels or IP address ranges. In a multi-tenant environment where strict network isolation between tenants is required, starting with a default policy that denies communication between pods is recommended with another rule that allows all pods to query the DNS server for name resolution. With such a default policy in place, you can begin adding more permissive rules that allow for communication within a namespace. It is also recommended not to use empty label selector '{}' for namespaceSelector field in network policy definition, in case traffic need to be allowed between namespaces. This scheme can be further refined as required. Note that this only applies to pods within a single control plane;

Pods that belong to different virtual control planes cannot talk to each other via Kubernetes networking.

Namespace management tools may simplify the creation of default or common network policies. In addition, some of these tools allow you to enforce a consistent set of namespace labels across your cluster, ensuring that they are a trusted basis for your policies.

### **Warning:**

Network policies require a [CNI plugin](#) that supports the implementation of network policies. Otherwise, NetworkPolicy resources will be ignored.

More advanced network isolation may be provided by service meshes, which provide OSI Layer 7 policies based on workload identity, in addition to namespaces. These higher-level policies can make it easier to manage namespace-based multi-tenancy, especially when multiple namespaces are dedicated to a single tenant. They frequently also offer encryption using mutual TLS, protecting your data even in the presence of a compromised node, and work across dedicated or virtual clusters. However, they can be significantly more complex to manage and may not be appropriate for all users.

## **Storage isolation**

Kubernetes offers several types of volumes that can be used as persistent storage for workloads. For security and data-isolation, [dynamic volume provisioning](#) is recommended and volume types that use node resources should be avoided.

[StorageClasses](#) allow you to describe custom "classes" of storage offered by your cluster, based on quality-of-service levels, backup policies, or custom policies determined by the cluster administrators.

Pods can request storage using a [PersistentVolumeClaim](#). A PersistentVolumeClaim is a namespaced resource, which enables isolating portions of the storage system and dedicating it to tenants within the shared Kubernetes cluster. However, it is important to note that a PersistentVolume is a cluster-wide resource and has a lifecycle independent of workloads and namespaces.

For example, you can configure a separate StorageClass for each tenant and use this to strengthen isolation. If a StorageClass is shared, you should set a [reclaim policy of Delete](#) to ensure that a PersistentVolume cannot be reused across different namespaces.

## **Sandboxing containers**

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Kubernetes pods are composed of one or more containers that execute on worker nodes. Containers utilize OS-level virtualization and hence offer a weaker isolation boundary than virtual machines that utilize hardware-based virtualization.

In a shared environment, unpatched vulnerabilities in the application and system layers can be exploited by attackers for container breakouts and remote code execution that allow access to

host resources. In some applications, like a Content Management System (CMS), customers may be allowed the ability to upload and execute untrusted scripts or code. In either case, mechanisms to further isolate and protect workloads using strong isolation are desirable.

Sandboxing provides a way to isolate workloads running in a shared cluster. It typically involves running each pod in a separate execution environment such as a virtual machine or a userspace kernel. Sandboxing is often recommended when you are running untrusted code, where workloads are assumed to be malicious. Part of the reason this type of isolation is necessary is because containers are processes running on a shared kernel; they mount file systems like `/sys` and `/proc` from the underlying host, making them less secure than an application that runs on a virtual machine which has its own kernel. While controls such as `seccomp`, `AppArmor`, and `SELinux` can be used to strengthen the security of containers, it is hard to apply a universal set of rules to all workloads running in a shared cluster. Running workloads in a sandbox environment helps to insulate the host from container escapes, where an attacker exploits a vulnerability to gain access to the host system and all the processes/files running on that host.

Virtual machines and userspace kernels are 2 popular approaches to sandboxing. The following sandboxing implementations are available:

- [gVisor](#) intercepts syscalls from containers and runs them through a userspace kernel, written in Go, with limited access to the underlying host.
- [Kata Containers](#) provide a secure container runtime that allows you to run containers in a VM. The hardware virtualization available in Kata offers an added layer of security for containers running untrusted code.

## Node Isolation

Node isolation is another technique that you can use to isolate tenant workloads from each other. With node isolation, a set of nodes is dedicated to running pods from a particular tenant and co-mingling of tenant pods is prohibited. This configuration reduces the noisy tenant issue, as all pods running on a node will belong to a single tenant. The risk of information disclosure is slightly lower with node isolation because an attacker that manages to escape from a container will only have access to the containers and volumes mounted to that node.

Although workloads from different tenants are running on different nodes, it is important to be aware that the kubelet and (unless using virtual control planes) the API service are still shared services. A skilled attacker could use the permissions assigned to the kubelet or other pods running on the node to move laterally within the cluster and gain access to tenant workloads running on other nodes. If this is a major concern, consider implementing compensating controls such as `seccomp`, `AppArmor` or `SELinux` or explore using sandboxed containers or creating separate clusters for each tenant.

Node isolation is a little easier to reason about from a billing standpoint than sandboxing containers since you can charge back per node rather than per pod. It also has fewer compatibility and performance issues and may be easier to implement than sandboxing containers. For example, nodes for each tenant can be configured with taints so that only pods with the corresponding toleration can run on them. A mutating webhook could then be used to automatically add tolerations and node affinities to pods deployed into tenant namespaces so that they run on a specific set of nodes designated for that tenant.

Node isolation can be implemented using an [pod node selectors](#) or a [Virtual Kubelet](#).

## Additional Considerations

This section discusses other Kubernetes constructs and patterns that are relevant for multi-tenancy.

### API Priority and Fairness

[API priority and fairness](#) is a Kubernetes feature that allows you to assign a priority to certain pods running within the cluster. When an application calls the Kubernetes API, the API server evaluates the priority assigned to pod. Calls from pods with higher priority are fulfilled before those with a lower priority. When contention is high, lower priority calls can be queued until the server is less busy or you can reject the requests.

Using API priority and fairness will not be very common in SaaS environments unless you are allowing customers to run applications that interface with the Kubernetes API, for example, a controller.

### Quality-of-Service (QoS)

When you're running a SaaS application, you may want the ability to offer different Quality-of-Service (QoS) tiers of service to different tenants. For example, you may have freemium service that comes with fewer performance guarantees and features and a for-fee service tier with specific performance guarantees. Fortunately, there are several Kubernetes constructs that can help you accomplish this within a shared cluster, including network QoS, storage classes, and pod priority and preemption. The idea with each of these is to provide tenants with the quality of service that they paid for. Let's start by looking at networking QoS.

Typically, all pods on a node share a network interface. Without network QoS, some pods may consume an unfair share of the available bandwidth at the expense of other pods. The Kubernetes [bandwidth plugin](#) creates an [extended resource](#) for networking that allows you to use Kubernetes resources constructs, i.e. requests/limits, to apply rate limits to pods by using Linux tc queues. Be aware that the plugin is considered experimental as per the [Network Plugins](#) documentation and should be thoroughly tested before use in production environments.

For storage QoS, you will likely want to create different storage classes or profiles with different performance characteristics. Each storage profile can be associated with a different tier of service that is optimized for different workloads such IO, redundancy, or throughput. Additional logic might be necessary to allow the tenant to associate the appropriate storage profile with their workload.

Finally, there's [pod priority and preemption](#) where you can assign priority values to pods. When scheduling pods, the scheduler will try evicting pods with lower priority when there are insufficient resources to schedule pods that are assigned a higher priority. If you have a use case where tenants have different service tiers in a shared cluster e.g. free and paid, you may want to give higher priority to certain tiers using this feature.

### DNS

Kubernetes clusters include a Domain Name System (DNS) service to provide translations from names to IP addresses, for all Services and Pods. By default, the Kubernetes DNS service allows lookups across all namespaces in the cluster.

In multi-tenant environments where tenants can access pods and other Kubernetes resources, or where stronger isolation is required, it may be necessary to prevent pods from looking up services in other Namespaces. You can restrict cross-namespace DNS lookups by configuring security rules for the DNS service. For example, CoreDNS (the default DNS service for Kubernetes) can leverage Kubernetes metadata to restrict queries to Pods and Services within a namespace. For more information, read an [example](#) of configuring this within the CoreDNS documentation.

When a [Virtual Control Plane per tenant](#) model is used, a DNS service must be configured per tenant or a multi-tenant DNS service must be used. Here is an example of a [customized version of CoreDNS](#) that supports multiple tenants.

## Operators

[Operators](#) are Kubernetes controllers that manage applications. Operators can simplify the management of multiple instances of an application, like a database service, which makes them a common building block in the multi-consumer (SaaS) multi-tenancy use case.

Operators used in a multi-tenant environment should follow a stricter set of guidelines. Specifically, the Operator should:

- Support creating resources within different tenant namespaces, rather than just in the namespace in which the Operator is deployed.
- Ensure that the Pods are configured with resource requests and limits, to ensure scheduling and fairness.
- Support configuration of Pods for data-plane isolation techniques such as node isolation and sandboxed containers.

## Implementations

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information.](#)

There are two primary ways to share a Kubernetes cluster for multi-tenancy: using Namespaces (that is, a Namespace per tenant) or by virtualizing the control plane (that is, virtual control plane per tenant).

In both cases, data plane isolation, and management of additional considerations such as API Priority and Fairness, is also recommended.

Namespace isolation is well-supported by Kubernetes, has a negligible resource cost, and provides mechanisms to allow tenants to interact appropriately, such as by allowing service-to-service communication. However, it can be difficult to configure, and doesn't apply to Kubernetes resources that can't be namespaced, such as Custom Resource Definitions, Storage Classes, and Webhooks.

Control plane virtualization allows for isolation of non-namespaced resources at the cost of somewhat higher resource usage and more difficult cross-tenant sharing. It is a good option when namespace isolation is insufficient but dedicated clusters are undesirable, due to the high cost of maintaining them (especially on-prem) or due to their higher overhead and lack of

resource sharing. However, even within a virtualized control plane, you will likely see benefits by using namespaces as well.

The two options are discussed in more detail in the following sections.

## **Namespace per tenant**

As previously mentioned, you should consider isolating each workload in its own namespace, even if you are using dedicated clusters or virtualized control planes. This ensures that each workload only has access to its own resources, such as ConfigMaps and Secrets, and allows you to tailor dedicated security policies for each workload. In addition, it is a best practice to give each namespace names that are unique across your entire fleet (that is, even if they are in separate clusters), as this gives you the flexibility to switch between dedicated and shared clusters in the future, or to use multi-cluster tooling such as service meshes.

Conversely, there are also advantages to assigning namespaces at the tenant level, not just the workload level, since there are often policies that apply to all workloads owned by a single tenant. However, this raises its own problems. Firstly, this makes it difficult or impossible to customize policies to individual workloads, and secondly, it may be challenging to come up with a single level of "tenancy" that should be given a namespace. For example, an organization may have divisions, teams, and subteams - which should be assigned a namespace?

To solve this, Kubernetes provides the [Hierarchical Namespace Controller \(HNC\)](#), which allows you to organize your namespaces into hierarchies, and share certain policies and resources between them. It also helps you manage namespace labels, namespace lifecycles, and delegated management, and share resource quotas across related namespaces. These capabilities can be useful in both multi-team and multi-customer scenarios.

Other projects that provide similar capabilities and aid in managing namespaced resources are listed below.

### **Multi-team tenancy**

- [Capsule](#)
- [Kiosk](#)

### **Multi-customer tenancy**

- [Kubeplus](#)

### **Policy engines**

Policy engines provide features to validate and generate tenant configurations:

- [Kyverno](#)
- [OPA/Gatekeeper](#)

## **Virtual control plane per tenant**

Another form of control-plane isolation is to use Kubernetes extensions to provide each tenant a virtual control-plane that enables segmentation of cluster-wide API resources. [Data plane](#)



[isolation](#) techniques can be used with this model to securely manage worker nodes across tenants.

The virtual control plane based multi-tenancy model extends namespace-based multi-tenancy by providing each tenant with dedicated control plane components, and hence complete control over cluster-wide resources and add-on services. Worker nodes are shared across all tenants, and are managed by a Kubernetes cluster that is normally inaccessible to tenants. This cluster is often referred to as a *super-cluster* (or sometimes as a *host-cluster*). Since a tenant's control-plane is not directly associated with underlying compute resources it is referred to as a *virtual control plane*.

A virtual control plane typically consists of the Kubernetes API server, the controller manager, and the etcd data store. It interacts with the super cluster via a metadata synchronization controller which coordinates changes across tenant control planes and the control plane of the super-cluster.

By using per-tenant dedicated control planes, most of the isolation problems due to sharing one API server among all tenants are solved. Examples include noisy neighbors in the control plane, uncontrollable blast radius of policy misconfigurations, and conflicts between cluster scope objects such as webhooks and CRDs. Hence, the virtual control plane model is particularly suitable for cases where each tenant requires access to a Kubernetes API server and expects the full cluster manageability.

The improved isolation comes at the cost of running and maintaining an individual virtual control plane per tenant. In addition, per-tenant control planes do not solve isolation problems in the data plane, such as node-level noisy neighbors or security threats. These must still be addressed separately.

The Kubernetes [Cluster API - Nested \(CAPN\)](#) project provides an implementation of virtual control planes.

### Other implementations

- [Kamaji](#)
- [vcluster](#)

## Hardening Guide - Authentication Mechanisms

Information on authentication options in Kubernetes and their security properties.

Selecting the appropriate authentication mechanism(s) is a crucial aspect of securing your cluster. Kubernetes provides several built-in mechanisms, each with its own strengths and weaknesses that should be carefully considered when choosing the best authentication mechanism for your cluster.

In general, it is recommended to enable as few authentication mechanisms as possible to simplify user management and prevent cases where users retain access to a cluster that is no longer required.

It is important to note that Kubernetes does not have an in-built user database within the cluster. Instead, it takes user information from the configured authentication system and uses that to make authorization decisions. Therefore, to audit user access, you need to review credentials from every configured authentication source.

For production clusters with multiple users directly accessing the Kubernetes API, it is recommended to use external authentication sources such as OIDC. The internal authentication mechanisms, such as client certificates and service account tokens, described below, are not suitable for this use-case.

## X.509 client certificate authentication

Kubernetes leverages [X.509 client certificate](#) authentication for system components, such as when the Kubelet authenticates to the API Server. While this mechanism can also be used for user authentication, it might not be suitable for production use due to several restrictions:

- Client certificates cannot be individually revoked. Once compromised, a certificate can be used by an attacker until it expires. To mitigate this risk, it is recommended to configure short lifetimes for user authentication credentials created using client certificates.
- If a certificate needs to be invalidated, the certificate authority must be re-keyed, which can introduce availability risks to the cluster.
- There is no permanent record of client certificates created in the cluster. Therefore, all issued certificates must be recorded if you need to keep track of them.
- Private keys used for client certificate authentication cannot be password-protected. Anyone who can read the file containing the key will be able to make use of it.
- Using client certificate authentication requires a direct connection from the client to the API server with no intervening TLS termination points, which can complicate network architectures.
- Group data is embedded in the O value of the client certificate, which means the user's group memberships cannot be changed for the lifetime of the certificate.

## Static token file

Although Kubernetes allows you to load credentials from a [static token file](#) located on the control plane node disks, this approach is not recommended for production servers due to several reasons:

- Credentials are stored in clear text on control plane node disks, which can be a security risk.
- Changing any credential requires a restart of the API server process to take effect, which can impact availability.
- There is no mechanism available to allow users to rotate their credentials. To rotate a credential, a cluster administrator must modify the token on disk and distribute it to the users.
- There is no lockout mechanism available to prevent brute-force attacks.



## Bootstrap tokens

[Bootstrap tokens](#) are used for joining nodes to clusters and are not recommended for user authentication due to several reasons:

- They have hard-coded group memberships that are not suitable for general use, making them unsuitable for authentication purposes.
- Manually generating bootstrap tokens can lead to weak tokens that can be guessed by an attacker, which can be a security risk.
- There is no lockout mechanism available to prevent brute-force attacks, making it easier for attackers to guess or crack the token.

## ServiceAccount secret tokens

[Service account secrets](#) are available as an option to allow workloads running in the cluster to authenticate to the API server. In Kubernetes < 1.23, these were the default option, however, they are being replaced with TokenRequest API tokens. While these secrets could be used for user authentication, they are generally unsuitable for a number of reasons:

- They cannot be set with an expiry and will remain valid until the associated service account is deleted.
- The authentication tokens are visible to any cluster user who can read secrets in the namespace that they are defined in.
- Service accounts cannot be added to arbitrary groups complicating RBAC management where they are used.

## TokenRequest API tokens

The TokenRequest API is a useful tool for generating short-lived credentials for service authentication to the API server or third-party systems. However, it is not generally recommended for user authentication as there is no revocation method available, and distributing credentials to users in a secure manner can be challenging.

When using TokenRequest tokens for service authentication, it is recommended to implement a short lifespan to reduce the impact of compromised tokens.

## OpenID Connect token authentication

Kubernetes supports integrating external authentication services with the Kubernetes API using [OpenID Connect \(OIDC\)](#). There is a wide variety of software that can be used to integrate Kubernetes with an identity provider. However, when using OIDC authentication for Kubernetes, it is important to consider the following hardening measures:

- The software installed in the cluster to support OIDC authentication should be isolated from general workloads as it will run with high privileges.
- Some Kubernetes managed services are limited in the OIDC providers that can be used.
- As with TokenRequest tokens, OIDC tokens should have a short lifespan to reduce the impact of compromised tokens.

# Webhook token authentication

[Webhook token authentication](#) is another option for integrating external authentication providers into Kubernetes. This mechanism allows for an authentication service, either running inside the cluster or externally, to be contacted for an authentication decision over a webhook. It is important to note that the suitability of this mechanism will likely depend on the software used for the authentication service, and there are some Kubernetes-specific considerations to take into account.

To configure Webhook authentication, access to control plane server filesystems is required. This means that it will not be possible with Managed Kubernetes unless the provider specifically makes it available. Additionally, any software installed in the cluster to support this access should be isolated from general workloads, as it will run with high privileges.

## Authenticating proxy

Another option for integrating external authentication systems into Kubernetes is to use an [authenticating proxy](#). With this mechanism, Kubernetes expects to receive requests from the proxy with specific header values set, indicating the username and group memberships to assign for authorization purposes. It is important to note that there are specific considerations to take into account when using this mechanism.

Firstly, securely configured TLS must be used between the proxy and Kubernetes API server to mitigate the risk of traffic interception or sniffing attacks. This ensures that the communication between the proxy and Kubernetes API server is secure.

Secondly, it is important to be aware that an attacker who is able to modify the headers of the request may be able to gain unauthorized access to Kubernetes resources. As such, it is important to ensure that the headers are properly secured and cannot be tampered with.

# Kubernetes API Server Bypass Risks

Security architecture information relating to the API server and other components

The Kubernetes API server is the main point of entry to a cluster for external parties (users and services) interacting with it.

As part of this role, the API server has several key built-in security controls, such as audit logging and [admission controllers](#). However, there are ways to modify the configuration or content of the cluster that bypass these controls.

This page describes the ways in which the security controls built into the Kubernetes API server can be bypassed, so that cluster operators and security architects can ensure that these bypasses are appropriately restricted.

## Static Pods

The [kubelet](#) on each node loads and directly manages any manifests that are stored in a named directory or fetched from a specific URL as [static Pods](#) in your cluster. The API server doesn't

manage these static Pods. An attacker with write access to this location could modify the configuration of static pods loaded from that source, or could introduce new static Pods.

Static Pods are restricted from accessing other objects in the Kubernetes API. For example, you can't configure a static Pod to mount a Secret from the cluster. However, these Pods can take other security sensitive actions, such as using hostPath mounts from the underlying node.

By default, the kubelet creates a [mirror pod](#) so that the static Pods are visible in the Kubernetes API. However, if the attacker uses an invalid namespace name when creating the Pod, it will not be visible in the Kubernetes API and can only be discovered by tooling that has access to the affected host(s).

If a static Pod fails admission control, the kubelet won't register the Pod with the API server. However, the Pod still runs on the node. For more information, refer to [kubeadm issue #1541](#).

## Mitigations

- Only [enable the kubelet static Pod manifest functionality](#) if required by the node.
- If a node uses the static Pod functionality, restrict filesystem access to the static Pod manifest directory or URL to users who need the access.
- Restrict access to kubelet configuration parameters and files to prevent an attacker setting a static Pod path or URL.
- Regularly audit and centrally report all access to directories or web storage locations that host static Pod manifests and kubelet configuration files.

## The kubelet API

The kubelet provides an HTTP API that is typically exposed on TCP port 10250 on cluster worker nodes. The API might also be exposed on control plane nodes depending on the Kubernetes distribution in use. Direct access to the API allows for disclosure of information about the pods running on a node, the logs from those pods, and execution of commands in every container running on the node.

When Kubernetes cluster users have RBAC access to Node object sub-resources, that access serves as authorization to interact with the kubelet API. The exact access depends on which sub-resource access has been granted, as detailed in [kubelet authorization](#).

Direct access to the kubelet API is not subject to admission control and is not logged by Kubernetes audit logging. An attacker with direct access to this API may be able to bypass controls that detect or prevent certain actions.

The kubelet API can be configured to authenticate requests in a number of ways. By default, the kubelet configuration allows anonymous access. Most Kubernetes providers change the default to use webhook and certificate authentication. This lets the control plane ensure that the caller is authorized to access the nodes API resource or sub-resources. The default anonymous access doesn't make this assertion with the control plane.

## Mitigations

- Restrict access to sub-resources of the nodes API object using mechanisms such as [RBAC](#). Only grant this access when required, such as by monitoring services.

- Restrict access to the kubelet port. Only allow specified and trusted IP address ranges to access the port.
- Ensure that [kubelet authentication](#) is set to webhook or certificate mode.
- Ensure that the unauthenticated "read-only" Kubelet port is not enabled on the cluster.

## The etcd API

Kubernetes clusters use etcd as a datastore. The etcd service listens on TCP port 2379. The only clients that need access are the Kubernetes API server and any backup tooling that you use. Direct access to this API allows for disclosure or modification of any data held in the cluster.

Access to the etcd API is typically managed by client certificate authentication. Any certificate issued by a certificate authority that etcd trusts allows full access to the data stored inside etcd.

Direct access to etcd is not subject to Kubernetes admission control and is not logged by Kubernetes audit logging. An attacker who has read access to the API server's etcd client certificate private key (or can create a new trusted client certificate) can gain cluster admin rights by accessing cluster secrets or modifying access rules. Even without elevating their Kubernetes RBAC privileges, an attacker who can modify etcd can retrieve any API object or create new workloads inside the cluster.

Many Kubernetes providers configure etcd to use mutual TLS (both client and server verify each other's certificate for authentication). There is no widely accepted implementation of authorization for the etcd API, although the feature exists. Since there is no authorization model, any certificate with client access to etcd can be used to gain full access to etcd. Typically, etcd client certificates that are only used for health checking can also grant full read and write access.

## Mitigations

- Ensure that the certificate authority trusted by etcd is used only for the purposes of authentication to that service.
- Control access to the private key for the etcd server certificate, and to the API server's client certificate and key.
- Consider restricting access to the etcd port at a network level, to only allow access from specified and trusted IP address ranges.

## Container runtime socket

On each node in a Kubernetes cluster, access to interact with containers is controlled by the container runtime (or runtimes, if you have configured more than one). Typically, the container runtime exposes a Unix socket that the kubelet can access. An attacker with access to this socket can launch new containers or interact with running containers.

At the cluster level, the impact of this access depends on whether the containers that run on the compromised node have access to Secrets or other confidential data that an attacker could use to escalate privileges to other worker nodes or to control plane components.

## Mitigations

- Ensure that you tightly control filesystem access to container runtime sockets. When possible, restrict this access to the root user.
- Isolate the kubelet from other components running on the node, using mechanisms such as Linux kernel namespaces.
- Ensure that you restrict or forbid the use of [hostPath mounts](#) that include the container runtime socket, either directly or by mounting a parent directory. Also hostPath mounts must be set as read-only to mitigate risks of attackers bypassing directory restrictions.
- Restrict user access to nodes, and especially restrict superuser access to nodes.

# Linux kernel security constraints for Pods and containers

Overview of Linux kernel security modules and constraints that you can use to harden your Pods and containers.

This page describes some of the security features that are built into the Linux kernel that you can use in your Kubernetes workloads. To learn how to apply these features to your Pods and containers, refer to [Configure a SecurityContext for a Pod or Container](#). You should already be familiar with Linux and with the basics of Kubernetes workloads.

## Run workloads without root privileges

When you deploy a workload in Kubernetes, use the Pod specification to restrict that workload from running as the root user on the node. You can use the Pod securityContext to define the specific Linux user and group for the processes in the Pod, and explicitly restrict containers from running as root users. Setting these values in the Pod manifest takes precedence over similar values in the container image, which is especially useful if you're running images that you don't own.

### Caution:

Ensure that the user or group that you assign to the workload has the permissions required for the application to function correctly. Changing the user or group to one that doesn't have the correct permissions could lead to file access issues or failed operations.

Configuring the kernel security features on this page provides fine-grained control over the actions that processes in your cluster can take, but managing these configurations can be challenging at scale. Running containers as non-root, or in user namespaces if you need root privileges, helps to reduce the chance that you'll need to enforce your configured kernel security capabilities.

## Security features in the Linux kernel

Kubernetes lets you configure and use Linux kernel features to improve isolation and harden your containerized workloads. Common features include the following:

- **Secure computing mode (seccomp):** Filter which system calls a process can make

- **AppArmor:** Restrict the access privileges of individual programs
- **Security Enhanced Linux (SELinux):** Assign security labels to objects for more manageable security policy enforcement

To configure settings for one of these features, the operating system that you choose for your nodes must enable the feature in the kernel. For example, Ubuntu 7.10 and later enable AppArmor by default. To learn whether your OS enables a specific feature, consult the OS documentation.

You use the `securityContext` field in your Pod specification to define the constraints that apply to those processes. The `securityContext` field also supports other security settings, such as specific Linux capabilities or file access permissions using UIDs and GIDs. To learn more, refer to [Configure a SecurityContext for a Pod or Container](#).

## seccomp

Some of your workloads might need privileges to perform specific actions as the root user on your node's host machine. Linux uses *capabilities* to divide the available privileges into categories, so that processes can get the privileges required to perform specific actions without being granted all privileges. Each capability has a set of system calls (syscalls) that a process can make. seccomp lets you restrict these individual syscalls. It can be used to sandbox the privileges of a process, restricting the calls it is able to make from userspace into the kernel.

In Kubernetes, you use a *container runtime* on each node to run your containers. Example runtimes include CRI-O, Docker, or containerd. Each runtime allows only a subset of Linux capabilities by default. You can further limit the allowed syscalls individually by using a seccomp profile. Container runtimes usually include a default seccomp profile. Kubernetes lets you automatically apply seccomp profiles loaded onto a node to your Pods and containers.

### Note:

Kubernetes also has the `allowPrivilegeEscalation` setting for Pods and containers. When set to false, this prevents processes from gaining new capabilities and restricts unprivileged users from changing the applied seccomp profile to a more permissive profile.

To learn how to implement seccomp in Kubernetes, refer to [Restrict a Container's Syscalls with seccomp](#).

To learn more about seccomp, see [Seccomp BPF](#) in the Linux kernel documentation.

## Considerations for seccomp

seccomp is a low-level security configuration that you should only configure yourself if you require fine-grained control over Linux syscalls. Using seccomp, especially at scale, has the following risks:

- Configurations might break during application updates
- Attackers can still use allowed syscalls to exploit vulnerabilities
- Profile management for individual applications becomes challenging at scale

**Recommendation:** Use the default seccomp profile that's bundled with your container runtime. If you need a more isolated environment, consider using a sandbox, such as gVisor. Sandboxes solve the preceding risks with custom seccomp profiles, but require more compute



resources on your nodes and might have compatibility issues with GPUs and other specialized hardware.

## AppArmor and SELinux: policy-based mandatory access control

You can use Linux policy-based mandatory access control (MAC) mechanisms, such as AppArmor and SELinux, to harden your Kubernetes workloads.

### AppArmor

[AppArmor](#) is a Linux kernel security module that supplements the standard Linux user and group based permissions to confine programs to a limited set of resources. AppArmor can be configured for any application to reduce its potential attack surface and provide greater in-depth defense. It is configured through profiles tuned to allow the access needed by a specific program or container, such as Linux capabilities, network access, and file permissions. Each profile can be run in either enforcing mode, which blocks access to disallowed resources, or complain mode, which only reports violations.

AppArmor can help you to run a more secure deployment by restricting what containers are allowed to do, and/or provide better auditing through system logs. The container runtime that you use might ship with a default AppArmor profile, or you can use a custom profile.

To learn how to use AppArmor in Kubernetes, refer to [Restrict a Container's Access to Resources with AppArmor](#).

### SELinux

SELinux is a Linux kernel security module that lets you restrict the access that a specific *subject*, such as a process, has to the files on your system. You define security policies that apply to subjects that have specific SELinux labels. When a process that has an SELinux label attempts to access a file, the SELinux server checks whether that process' security policy allows the access and makes an authorization decision.

In Kubernetes, you can set an SELinux label in the `securityContext` field of your manifest. The specified labels are assigned to those processes. If you have configured security policies that affect those labels, the host OS kernel enforces these policies.

To learn how to use SELinux in Kubernetes, refer to [Assign SELinux labels to a container](#).

### Differences between AppArmor and SELinux

The operating system on your Linux nodes usually includes one of either AppArmor or SELinux. Both mechanisms provide similar types of protection, but have differences such as the following:

- **Configuration:** AppArmor uses profiles to define access to resources. SELinux uses policies that apply to specific labels.
- **Policy application:** In AppArmor, you define resources using file paths. SELinux uses the index node (inode) of a resource to identify the resource.

## Summary of features

The following table describes the use cases and scope of each security control. You can use all of these controls together to build a more hardened system.

Summary of Linux kernel security features

| Security feature | Description                                                                                                                                                | How to use                                                                                                                                | Example                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| seccomp          | Restrict individual kernel calls in the userspace. Reduces the likelihood that a vulnerability that uses a restricted syscall would compromise the system. | Specify a loaded seccomp profile in the Pod or container specification to apply its constraints to the processes in the Pod.              | Reject the <code>unshare</code> syscall, which was used in <a href="#">CVE-2022-0185</a> . |
| AppArmor         | Restrict program access to specific resources. Reduces the attack surface of the program. Improves audit logging.                                          | Specify a loaded AppArmor profile in the container specification.                                                                         | Restrict a read-only program from writing to any file path in the system.                  |
| SELinux          | Restrict access to resources such as files, applications, ports, and processes using labels and security policies.                                         | Specify access restrictions for specific labels. Tag processes with those labels to enforce the access restrictions related to the label. | Restrict a container from accessing files outside its own filesystem.                      |

### Note:

Mechanisms like AppArmor and SELinux can provide protection that extends beyond the container. For example, you can use SELinux to help mitigate [CVE-2019-5736](#).

## Considerations for managing custom configurations

seccomp, AppArmor, and SELinux usually have a default configuration that offers basic protections. You can also create custom profiles and policies that meet the requirements of your workloads. Managing and distributing these custom configurations at scale might be challenging, especially if you use all three features together. To help you to manage these configurations at scale, use a tool like the [Kubernetes Security Profiles Operator](#).

## Kernel-level security features and privileged containers

Kubernetes lets you specify that some trusted containers can run in *privileged* mode. Any container in a Pod can run in privileged mode to use operating system administrative capabilities that would otherwise be inaccessible. This is available for both Windows and Linux.

Privileged containers explicitly override some of the Linux kernel constraints that you might use in your workloads, as follows:

- **seccomp:** Privileged containers run as the `Unconfined` seccomp profile, overriding any seccomp profile that you specified in your manifest.
- **AppArmor:** Privileged containers ignore any applied AppArmor profiles.



- **SELinux:** Privileged containers run as the `unconfined_t` domain.

## Privileged containers

Any container in a Pod can enable *Privileged mode* if you set the `privileged: true` field in the [securityContext](#) field for the container. Privileged containers override or undo many other hardening settings such as the applied seccomp profile, AppArmor profile, or SELinux constraints. Privileged containers are given all Linux capabilities, including capabilities that they don't require. For example, a root user in a privileged container might be able to use the `CAP_SYS_ADMIN` and `CAP_NET_ADMIN` capabilities on the node, bypassing the runtime seccomp configuration and other restrictions.

In most cases, you should avoid using privileged containers, and instead grant the specific capabilities required by your container using the `capabilities` field in the `securityContext` field. Only use privileged mode if you have a capability that you can't grant with the `securityContext`. This is useful for containers that want to use operating system administrative capabilities such as manipulating the network stack or accessing hardware devices.

In Kubernetes version 1.26 and later, you can also run Windows containers in a similarly privileged mode by setting the `windowsOptions.hostProcess` flag on the `securityContext` of the Pod spec. For details and instructions, see [Create a Windows HostProcess Pod](#).

## Recommendations and best practices

- Before configuring kernel-level security capabilities, you should consider implementing network-level isolation. For more information, read the [Security Checklist](#).
- Unless necessary, run Linux workloads as non-root by setting specific user and group IDs in your Pod manifest and by specifying `runAsNonRoot: true`.

Additionally, you can run workloads in user namespaces by setting `hostUsers: false` in your Pod manifest. This lets you run containers as root users in the user namespace, but as non-root users in the host namespace on the node. This is still in early stages of development and might not have the level of support that you need. For instructions, refer to [Use a User Namespace With a Pod](#).

## What's next

- [Learn how to use AppArmor](#)
- [Learn how to use seccomp](#)
- [Learn how to use SELinux](#)

# Security Checklist

Baseline checklist for ensuring security in Kubernetes clusters.

This checklist aims at providing a basic list of guidance with links to more comprehensive documentation on each topic. It does not claim to be exhaustive and is meant to evolve.

On how to read and use this document:

- The order of topics does not reflect an order of priority.

- Some checklist items are detailed in the paragraph below the list of each section.

### Caution:

Checklists are **not** sufficient for attaining a good security posture on their own. A good security posture requires constant attention and improvement, but a checklist can be the first step on the never-ending journey towards security preparedness. Some of the recommendations in this checklist may be too restrictive or too lax for your specific security needs. Since Kubernetes security is not "one size fits all", each category of checklist items should be evaluated on its merits.

## Authentication & Authorization

- `system:masters` group is not used for user or component authentication after bootstrapping.
- The kube-controller-manager is running with `--use-service-account-credentials` enabled.
- The root certificate is protected (either an offline CA, or a managed online CA with effective access controls).
- Intermediate and leaf certificates have an expiry date no more than 3 years in the future.
- A process exists for periodic access review, and reviews occur no more than 24 months apart.
- The [Role Based Access Control Good Practices](#) are followed for guidance related to authentication and authorization.

After bootstrapping, neither users nor components should authenticate to the Kubernetes API as `system:masters`. Similarly, running all of kube-controller-manager as `system:masters` should be avoided. In fact, `system:masters` should only be used as a break-glass mechanism, as opposed to an admin user.

## Network security

- CNI plugins in-use supports network policies.
- Ingress and egress network policies are applied to all workloads in the cluster.
- Default network policies within each namespace, selecting all pods, denying everything, are in place.
- If appropriate, a service mesh is used to encrypt all communications inside of the cluster.
- The Kubernetes API, kubelet API and etcd are not exposed publicly on Internet.
- Access from the workloads to the cloud metadata API is filtered.
- Use of LoadBalancer and ExternalIPs is restricted.

A number of [Container Network Interface \(CNI\) plugins](#) provide the functionality to restrict network resources that pods may communicate with. This is most commonly done through [Network Policies](#) which provide a namespaced resource to define rules. Default network policies blocking everything egress and ingress, in each namespace, selecting all the pods, can be useful to adopt an allow list approach, ensuring that no workloads is missed.

Not all CNI plugins provide encryption in transit. If the chosen plugin lacks this feature, an alternative solution could be to use a service mesh to provide that functionality.

The etcd datastore of the control plane should have controls to limit access and not be publicly exposed on the Internet. Furthermore, mutual TLS (mTLS) should be used to communicate securely with it. The certificate authority for this should be unique to etcd.

External Internet access to the Kubernetes API server should be restricted to not expose the API publicly. Be careful as many managed Kubernetes distributions are publicly exposing the API server by default. You can then use a bastion host to access the server.

The [kubelet](#) API access should be restricted and not publicly exposed, the defaults authentication and authorization settings, when no configuration file specified with the `--config` flag, are overly permissive.

If a cloud provider is used for hosting Kubernetes, the access from pods to the cloud metadata API 169.254.169.254 should also be restricted or blocked if not needed because it may leak information.

For restricted LoadBalancer and ExternalIPs use, see [CVE-2020-8554: Man in the middle using LoadBalancer or ExternalIPs](#) and the [DenyServiceExternalIPs admission controller](#) for further information.

## Pod security

- RBAC rights to create, update, patch, delete workloads is only granted if necessary.
- Appropriate Pod Security Standards policy is applied for all namespaces and enforced.
- Memory limit is set for the workloads with a limit equal or inferior to the request.
- CPU limit might be set on sensitive workloads.
- For nodes that support it, Seccomp is enabled with appropriate syscalls profile for programs.
- For nodes that support it, AppArmor or SELinux is enabled with appropriate profile for programs.

RBAC authorization is crucial but [cannot be granular enough to have authorization on the Pods' resources](#) (or on any resource that manages Pods). The only granularity is the API verbs on the resource itself, for example, create on Pods. Without additional admission, the authorization to create these resources allows direct unrestricted access to the schedulable nodes of a cluster.

The [Pod Security Standards](#) define three different policies, privileged, baseline and restricted that limit how fields can be set in the PodSpec regarding security. These standards can be enforced at the namespace level with the new [Pod Security](#) admission, enabled by default, or by third-party admission webhook. Please note that, contrary to the removed PodSecurityPolicy admission it replaces, [Pod Security](#) admission can be easily combined with admission webhooks and external services.

Pod Security admission restricted policy, the most restrictive policy of the [Pod Security Standards](#) set, [can operate in several modes](#), warn, audit or enforce to gradually apply the most appropriate [security context](#) according to security best practices. Nevertheless, pods' [security context](#) should be separately investigated to limit the privileges and access pods may have on top of the predefined security standards, for specific use cases.

For a hands-on tutorial on [Pod Security](#), see the blog post [Kubernetes 1.23: Pod Security Graduates to Beta](#).

[Memory and CPU limits](#) should be set in order to restrict the memory and CPU resources a pod can consume on a node, and therefore prevent potential DoS attacks from malicious or breached workloads. Such policy can be enforced by an admission controller. Please note that

CPU limits will throttle usage and thus can have unintended effects on auto-scaling features or efficiency i.e. running the process in best effort with the CPU resource available.

**Caution:**

Memory limit superior to request can expose the whole node to OOM issues.

## Enabling Seccomp

Seccomp stands for secure computing mode and has been a feature of the Linux kernel since version 2.6.12. It can be used to sandbox the privileges of a process, restricting the calls it is able to make from userspace into the kernel. Kubernetes lets you automatically apply seccomp profiles loaded onto a node to your Pods and containers.

Seccomp can improve the security of your workloads by reducing the Linux kernel syscall attack surface available inside containers. The seccomp filter mode leverages BPF to create an allow or deny list of specific syscalls, named profiles.

Since Kubernetes 1.27, you can enable the use of RuntimeDefault as the default seccomp profile for all workloads. A [security tutorial](#) is available on this topic. In addition, the [Kubernetes Security Profiles Operator](#) is a project that facilitates the management and use of seccomp in clusters.

**Note:**

Seccomp is only available on Linux nodes.

## Enabling AppArmor or SELinux

### AppArmor

[AppArmor](#) is a Linux kernel security module that can provide an easy way to implement Mandatory Access Control (MAC) and better auditing through system logs. A default AppArmor profile is enforced on nodes that support it, or a custom profile can be configured. Like seccomp, AppArmor is also configured through profiles, where each profile is either running in enforcing mode, which blocks access to disallowed resources or complain mode, which only reports violations. AppArmor profiles are enforced on a per-container basis, with an annotation, allowing for processes to gain just the right privileges.

**Note:**

AppArmor is only available on Linux nodes, and enabled in [some Linux distributions](#).

### SELinux

[SELinux](#) is also a Linux kernel security module that can provide a mechanism for supporting access control security policies, including Mandatory Access Controls (MAC). SELinux labels can be assigned to containers or pods [via their securityContext section](#).

## Note:

SELinux is only available on Linux nodes, and enabled in [some Linux distributions](#).

## Logs and auditing

- Audit logs, if enabled, are protected from general access.

## Pod placement

- Pod placement is done in accordance with the tiers of sensitivity of the application.
- Sensitive applications are running isolated on nodes or with specific sandboxed runtimes.

Pods that are on different tiers of sensitivity, for example, an application pod and the Kubernetes API server, should be deployed onto separate nodes. The purpose of node isolation is to prevent an application container breakout to directly providing access to applications with higher level of sensitivity to easily pivot within the cluster. This separation should be enforced to prevent pods accidentally being deployed onto the same node. This could be enforced with the following features:

### [Node Selectors](#)

Key-value pairs, as part of the pod specification, that specify which nodes to deploy onto. These can be enforced at the namespace and cluster level with the [PodNodeSelector](#) admission controller.

### [PodTolerationRestriction](#)

An admission controller that allows administrators to restrict permitted [tolerations](#) within a namespace. Pods within a namespace may only utilize the tolerations specified on the namespace object annotation keys that provide a set of default and allowed tolerations.

### [RuntimeClass](#)

RuntimeClass is a feature for selecting the container runtime configuration. The container runtime configuration is used to run a Pod's containers and can provide more or less isolation from the host at the cost of performance overhead.

## Secrets

- ConfigMaps are not used to hold confidential data.
- Encryption at rest is configured for the Secret API.
- If appropriate, a mechanism to inject secrets stored in third-party storage is deployed and available.
- Service account tokens are not mounted in pods that don't require them.
- [Bound service account token volume](#) is in-use instead of non-expiring tokens.

Secrets required for pods should be stored within Kubernetes Secrets as opposed to alternatives such as ConfigMap. Secret resources stored within etcd should be [encrypted at rest](#).

Pods needing secrets should have these automatically mounted through volumes, preferably stored in memory like with the [emptyDir.medium option](#). Mechanism can be used to also inject secrets from third-party storages as volume, like the [Secrets Store CSI Driver](#). This should be done preferentially as compared to providing the pods service account RBAC access to secrets. This would allow adding secrets into the pod as environment variables or files. Please note that

the environment variable method might be more prone to leakage due to crash dumps in logs and the non-confidential nature of environment variable in Linux, as opposed to the permission mechanism on files.

Service account tokens should not be mounted into pods that do not require them. This can be configured by setting [automountServiceAccountToken](#) to false either within the service account to apply throughout the namespace or specifically for a pod. For Kubernetes v1.22 and above, use [Bound Service Accounts](#) for time-bound service account credentials.

## Images

- Minimize unnecessary content in container images.
- Container images are configured to be run as unprivileged user.
- References to container images are made by sha256 digests (rather than tags) or the provenance of the image is validated by verifying the image's digital signature at deploy time [via admission control](#).
- Container images are regularly scanned during creation and in deployment, and known vulnerable software is patched.

Container image should contain the bare minimum to run the program they package. Preferably, only the program and its dependencies, building the image from the minimal possible base. In particular, image used in production should not contain shells or debugging utilities, as an [ephemeral debug container](#) can be used for troubleshooting.

Build images to directly start with an unprivileged user by using the [USER instruction in Dockerfile](#). The [Security Context](#) allows a container image to be started with a specific user and group with `runAsUser` and `runAsGroup`, even if not specified in the image manifest. However, the file permissions in the image layers might make it impossible to just start the process with a new unprivileged user without image modification.

Avoid using image tags to reference an image, especially the latest tag, the image behind a tag can be easily modified in a registry. Prefer using the complete sha256 digest which is unique to the image manifest. This policy can be enforced via an [ImagePolicyWebhook](#). Image signatures can also be automatically [verified with an admission controller](#) at deploy time to validate their authenticity and integrity.

Scanning a container image can prevent critical vulnerabilities from being deployed to the cluster alongside the container image. Image scanning should be completed before deploying a container image to a cluster and is usually done as part of the deployment process in a CI/CD pipeline. The purpose of an image scan is to obtain information about possible vulnerabilities and their prevention in the container image, such as a [Common Vulnerability Scoring System \(CVSS\)](#) score. If the result of the image scans is combined with the pipeline compliance rules, only properly patched container images will end up in Production.

## Admission controllers

- An appropriate selection of admission controllers is enabled.
- A pod security policy is enforced by the Pod Security Admission or/and a webhook admission controller.
- The admission chain plugins and webhooks are securely configured.



Admission controllers can help to improve the security of the cluster. However, they can present risks themselves as they extend the API server and [should be properly secured](#).

The following lists present a number of admission controllers that could be considered to enhance the security posture of your cluster and application. It includes controllers that may be referenced in other parts of this document.

This first group of admission controllers includes plugins [enabled by default](#), consider to leave them enabled unless you know what you are doing:

#### [CertificateApproval](#)

Performs additional authorization checks to ensure the approving user has permission to approve certificate request.

#### [CertificateSigning](#)

Performs additional authorization checks to ensure the signing user has permission to sign certificate requests.

#### [CertificateSubjectRestriction](#)

Rejects any certificate request that specifies a 'group' (or 'organization attribute') of system:masters.

#### [LimitRanger](#)

Enforce the LimitRange API constraints.

#### [MutatingAdmissionWebhook](#)

Allows the use of custom controllers through webhooks, these controllers may mutate requests that it reviews.

#### [PodSecurity](#)

Replacement for Pod Security Policy, restricts security contexts of deployed Pods.

#### [ResourceQuota](#)

Enforces resource quotas to prevent over-usage of resources.

#### [ValidatingAdmissionWebhook](#)

Allows the use of custom controllers through webhooks, these controllers do not mutate requests that it reviews.

The second group includes plugin that are not enabled by default but in general availability state and recommended to improve your security posture:

#### [DenyServiceExternalIPs](#)

Rejects all net-new usage of the Service.spec.externalIPs field. This is a mitigation for [CVE-2020-8554: Man in the middle using LoadBalancer or ExternalIPs](#).

#### [NodeRestriction](#)

Restricts kubelet's permissions to only modify the pods API resources they own or the node API resource that represent themselves. It also prevents kubelet from using the node-restriction.kubernetes.io/ annotation, which can be used by an attacker with access to the kubelet's credentials to influence pod placement to the controlled node.

The third group includes plugins that are not enabled by default but could be considered for certain use cases:

#### [AlwaysPullImages](#)

Enforces the usage of the latest version of a tagged image and ensures that the deployer has permissions to use the image.

#### [ImagePolicyWebhook](#)

Allows enforcing additional controls for images through webhooks.

## What's next

- [Privilege escalation via Pod creation](#) warns you about a specific access control risk; check how you're managing that threat.
  - If you use Kubernetes RBAC, read [RBAC Good Practices](#) for further information on authorization.
- [Securing a Cluster](#) for information on protecting a cluster from accidental or malicious access.
- [Cluster Multi-tenancy guide](#) for configuration options recommendations and best practices on multi-tenancy.
- [Blog post "A Closer Look at NSA/CISA Kubernetes Hardening Guidance"](#) for complementary resource on hardening Kubernetes clusters.

## Policies

Manage security and best-practices with policies.

Kubernetes policies are configurations that manage other configurations or runtime behaviors. Kubernetes offers various forms of policies, described below:

### Apply policies using API objects

Some API objects act as policies. Here are some examples:

- [NetworkPolicies](#) can be used to restrict ingress and egress traffic for a workload.
- [LimitRanges](#) manage resource allocation constraints across different object kinds.
- [ResourceQuotas](#) limit resource consumption for a [namespace](#).

### Apply policies using admission controllers

An [admission controller](#) runs in the API server and can validate or mutate API requests. Some admission controllers act to apply policies. For example, the [AlwaysPullImages](#) admission controller modifies a new Pod to set the image pull policy to Always.

Kubernetes has several built-in admission controllers that are configurable via the API server `--enable-admission-plugins` flag.

Details on admission controllers, with the complete list of available admission controllers, are documented in a dedicated section:

- [Admission Controllers](#)

### Apply policies using ValidatingAdmissionPolicy

Validating admission policies allow configurable validation checks to be executed in the API server using the Common Expression Language (CEL). For example, a ValidatingAdmissionPolicy can be used to disallow use of the latest image tag.

A ValidatingAdmissionPolicy operates on an API request and can be used to block, audit, and warn users about non-compliant configurations.



Details on the `ValidatingAdmissionPolicy` API, with examples, are documented in a dedicated section:

- [Validating Admission Policy](#)

## Apply policies using dynamic admission control

Dynamic admission controllers (or admission webhooks) run outside the API server as separate applications that register to receive webhooks requests to perform validation or mutation of API requests.

Dynamic admission controllers can be used to apply policies on API requests and trigger other policy-based workflows. A dynamic admission controller can perform complex checks including those that require retrieval of other cluster resources and external data. For example, an image verification check can lookup data from OCI registries to validate the container image signatures and attestations.

Details on dynamic admission control are documented in a dedicated section:

- [Dynamic Admission Control](#)

## Implementations

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Dynamic Admission Controllers that act as flexible policy engines are being developed in the Kubernetes ecosystem, such as:

- [Kubewarden](#)
- [Kyverno](#)
- [OPA Gatekeeper](#)
- [Polaris](#)

## Apply policies using Kubelet configurations

Kubernetes allows configuring the Kubelet on each worker node. Some Kubelet configurations act as policies:

- [Process ID limits and reservations](#) are used to limit and reserve allocatable PIDs.
- [Node Resource Managers](#) can manage compute, memory, and device resources for latency-critical and high-throughput workloads.

## Limit Ranges

By default, containers run with unbounded [compute resources](#) on a Kubernetes cluster. Using Kubernetes [resource quotas](#), administrators (also termed *cluster operators*) can restrict consumption and creation of cluster resources (such as CPU time, memory, and persistent storage) within a specified [namespace](#). Within a namespace, a [Pod](#) can consume as much CPU

and memory as is allowed by the ResourceQuotas that apply to that namespace. As a cluster operator, or as a namespace-level administrator, you might also be concerned about making sure that a single object cannot monopolize all available resources within a namespace.

A LimitRange is a policy to constrain the resource allocations (limits and requests) that you can specify for each applicable object kind (such as Pod or [PersistentVolumeClaim](#)) in a namespace.

A *LimitRange* provides constraints that can:

- Enforce minimum and maximum compute resources usage per Pod or Container in a namespace.
- Enforce minimum and maximum storage request per [PersistentVolumeClaim](#) in a namespace.
- Enforce a ratio between request and limit for a resource in a namespace.
- Set default request/limit for compute resources in a namespace and automatically inject them to Containers at runtime.

A LimitRange is enforced in a particular namespace when there is a LimitRange object in that namespace.

The name of a LimitRange object must be a valid [DNS subdomain name](#).

## Constraints on resource limits and requests

- The administrator creates a LimitRange in a namespace.
- Users create (or try to create) objects in that namespace, such as Pods or PersistentVolumeClaims.
- First, the LimitRange admission controller applies default request and limit values for all Pods (and their containers) that do not set compute resource requirements.
- Second, the LimitRange tracks usage to ensure it does not exceed resource minimum, maximum and ratio defined in any LimitRange present in the namespace.
- If you attempt to create or update an object (Pod or PersistentVolumeClaim) that violates a LimitRange constraint, your request to the API server will fail with an HTTP status code 403 Forbidden and a message explaining the constraint that has been violated.
- If you add a LimitRange in a namespace that applies to compute-related resources such as cpu and memory, you must specify requests or limits for those values. Otherwise, the system may reject Pod creation.
- LimitRange validations occur only at Pod admission stage, not on running Pods. If you add or modify a LimitRange, the Pods that already exist in that namespace continue unchanged.
- If two or more LimitRange objects exist in the namespace, it is not deterministic which default value will be applied.

## LimitRange and admission checks for Pods

A LimitRange does **not** check the consistency of the default values it applies. This means that a default value for the *limit* that is set by LimitRange may be less than the *request* value specified for the container in the spec that a client submits to the API server. If that happens, the final Pod will not be schedulable.

For example, you define a LimitRange with this manifest:

[concepts/policy/limit-range/problematic-limit-range.yaml](#)

```
apiVersion: v1
kind: LimitRange
metadata:
 name: cpu-resource-constraint
spec:
 limits:
 - default: # this section defines default limits
 cpu: 500m
 defaultRequest: # this section defines default requests
 cpu: 500m
 max: # max and min define the limit range
 cpu: "1"
 min:
 cpu: 100m
 type: Container
```

along with a Pod that declares a CPU resource request of 700m, but not a limit:

[concepts/policy/limit-range/example-conflict-with-limitrange-cpu.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: example-conflict-with-limitrange-cpu
spec:
 containers:
 - name: demo
 image: registry.k8s.io/pause:2.0
 resources:
 requests:
 cpu: 700m
```

then that Pod will not be scheduled, failing with an error similar to:

Pod "example-conflict-with-limitrange-cpu" is invalid: spec.containers[0].resources.requests: Invalid value: "700m": must be less than or equal to cpu limit

If you set both request and limit, then that new Pod will be scheduled successfully even with the same LimitRange in place:

[concepts/policy/limit-range/example-no-conflict-with-limitrange-cpu.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: example-no-conflict-with-limitrange-cpu
spec:
 containers:
 - name: demo
 image: registry.k8s.io/pause:2.0
 resources:
 requests:
```

```
cpu: 700m
limits:
cpu: 700m
```

## Example resource constraints

Examples of policies that could be created using `LimitRange` are:

- In a 2 node cluster with a capacity of 8 GiB RAM and 16 cores, constrain Pods in a namespace to request 100m of CPU with a max limit of 500m for CPU and request 200Mi for Memory with a max limit of 600Mi for Memory.
- Define default CPU limit and request to 150m and memory default request to 300Mi for Containers started with no cpu and memory requests in their specs.

In the case where the total limits of the namespace is less than the sum of the limits of the Pods/Containers, there may be contention for resources. In this case, the Containers or Pods will not be created.

Neither contention nor changes to a `LimitRange` will affect already created resources.

## What's next

For examples on using limits, see:

- [how to configure minimum and maximum CPU constraints per namespace.](#)
- [how to configure minimum and maximum Memory constraints per namespace.](#)
- [how to configure default CPU Requests and Limits per namespace.](#)
- [how to configure default Memory Requests and Limits per namespace.](#)
- [how to configure minimum and maximum Storage consumption per namespace.](#)
- [a detailed example on configuring quota per namespace.](#)

Refer to the [LimitRanger design document](#) for context and historical information.

## Resource Quotas

When several users or teams share a cluster with a fixed number of nodes, there is a concern that one team could use more than its fair share of resources.

Resource quotas are a tool for administrators to address this concern.

A resource quota, defined by a `ResourceQuota` object, provides constraints that limit aggregate resource consumption per namespace. It can limit the quantity of objects that can be created in a namespace by type, as well as the total amount of compute resources that may be consumed by resources in that namespace.

Resource quotas work like this:

- Different teams work in different namespaces. This can be enforced with [RBAC](#).
- The administrator creates one `ResourceQuota` for each namespace.

- Users create resources (pods, services, etc.) in the namespace, and the quota system tracks usage to ensure it does not exceed hard resource limits defined in a ResourceQuota.
- If creating or updating a resource violates a quota constraint, the request will fail with HTTP status code 403 FORBIDDEN with a message explaining the constraint that would have been violated.
- If quota is enabled in a namespace for compute resources like `cpu` and `memory`, users must specify requests or limits for those values; otherwise, the quota system may reject pod creation. Hint: Use the LimitRanger admission controller to force defaults for pods that make no compute resource requirements.

See the [walkthrough](#) for an example of how to avoid this problem.

#### Note:

- For `cpu` and `memory` resources, ResourceQuotas enforce that **every** (new) pod in that namespace sets a limit for that resource. If you enforce a resource quota in a namespace for either `cpu` or `memory`, you, and other clients, **must** specify either requests or limits for that resource, for every new Pod you submit. If you don't, the control plane may reject admission for that Pod.
- For other resources: ResourceQuota works and will ignore pods in the namespace without setting a limit or request for that resource. It means that you can create a new pod without limit/request ephemeral storage if the resource quota limits the ephemeral storage of this namespace. You can use a [LimitRange](#) to automatically set a default request for these resources.

The name of a ResourceQuota object must be a valid [DNS subdomain name](#).

Examples of policies that could be created using namespaces and quotas are:

- In a cluster with a capacity of 32 GiB RAM, and 16 cores, let team A use 20 GiB and 10 cores, let B use 10GiB and 4 cores, and hold 2GiB and 2 cores in reserve for future allocation.
- Limit the "testing" namespace to using 1 core and 1GiB RAM. Let the "production" namespace use any amount.

In the case where the total capacity of the cluster is less than the sum of the quotas of the namespaces, there may be contention for resources. This is handled on a first-come-first-served basis.

Neither contention nor changes to quota will affect already created resources.

## Enabling Resource Quota

Resource Quota support is enabled by default for many Kubernetes distributions. It is enabled when the [API server](#) `--enable-admission-plugins=` flag has ResourceQuota as one of its arguments.

A resource quota is enforced in a particular namespace when there is a ResourceQuota in that namespace.

## Compute Resource Quota

You can limit the total sum of [compute resources](#) that can be requested in a given namespace.

The following resource types are supported:

| Resource Name    | Description                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------------------|
| limits.cpu       | Across all pods in a non-terminal state, the sum of CPU limits cannot exceed this value.                                  |
| limits.memory    | Across all pods in a non-terminal state, the sum of memory limits cannot exceed this value.                               |
| requests.cpu     | Across all pods in a non-terminal state, the sum of CPU requests cannot exceed this value.                                |
| requests.memory  | Across all pods in a non-terminal state, the sum of memory requests cannot exceed this value.                             |
| hugepages-<size> | Across all pods in a non-terminal state, the number of huge page requests of the specified size cannot exceed this value. |
| cpu              | Same as requests.cpu                                                                                                      |
| memory           | Same as requests.memory                                                                                                   |

## Resource Quota For Extended Resources

In addition to the resources mentioned above, in release 1.10, quota support for [extended resources](#) is added.

As overcommit is not allowed for extended resources, it makes no sense to specify both requests and limits for the same extended resource in a quota. So for extended resources, only quota items with prefix requests. is allowed for now.

Take the GPU resource as an example, if the resource name is `nvidia.com/gpu`, and you want to limit the total number of GPUs requested in a namespace to 4, you can define a quota as follows:

- `requests.nvidia.com/gpu: 4`

See [Viewing and Setting Quotas](#) for more detail information.

## Storage Resource Quota

You can limit the total sum of [storage resources](#) that can be requested in a given namespace.

In addition, you can limit consumption of storage resources based on associated storage-class.

| Resource Name          | Description                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------|
| requests.storage       | Across all persistent volume claims, the sum of storage requests cannot exceed this value.  |
| persistentvolumeclaims | The total number of <a href="#">PersistentVolumeClaims</a> that can exist in the namespace. |

| Resource Name                                                           | Description                                                                                                                                                                 |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <storage-class-name>.storageclass.storage.k8s.io/requests.storage       | Across all persistent volume claims associated with the <storage-class-name>, the sum of storage requests cannot exceed this value.                                         |
| <storage-class-name>.storageclass.storage.k8s.io/persistentvolumeclaims | Across all persistent volume claims associated with the <storage-class-name>, the total number of <a href="#">persistent volume claims</a> that can exist in the namespace. |

For example, if an operator wants to quota storage with gold storage class separate from bronze storage class, the operator can define a quota as follows:

- gold.storageclass.storage.k8s.io/requests.storage: 500Gi
- bronze.storageclass.storage.k8s.io/requests.storage: 100Gi

In release 1.8, quota support for local ephemeral storage is added as an alpha feature:

| Resource Name              | Description                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------|
| requests.ephemeral-storage | Across all pods in the namespace, the sum of local ephemeral storage requests cannot exceed this value. |
| limits.ephemeral-storage   | Across all pods in the namespace, the sum of local ephemeral storage limits cannot exceed this value.   |
| ephemeral-storage          | Same as requests.ephemeral-storage.                                                                     |

#### Note:

When using a CRI container runtime, container logs will count against the ephemeral storage quota. This can result in the unexpected eviction of pods that have exhausted their storage quotas. Refer to [Logging Architecture](#) for details.

## Object Count Quota

You can set quota for *the total number of one particular resource kind* in the Kubernetes API, using the following syntax:

- count/<resource>.<group> for resources from non-core groups
- count/<resource> for resources from the core group

Here is an example set of resources users may want to put under object count quota:

- count/persistentvolumeclaims
- count/services
- count/secrets
- count/configmaps
- count/replicationcontrollers
- count/deployments.apps
- count/replicasets.apps
- count/statefulsets.apps
- count/jobs.batch
- count/cronjobs.batch



If you define a quota this way, it applies to Kubernetes' APIs that are part of the API server, and to any custom resources backed by a CustomResourceDefinition. If you use [API aggregation](#) to add additional, custom APIs that are not defined as CustomResourceDefinitions, the core Kubernetes control plane does not enforce quota for the aggregated API. The extension API server is expected to provide quota enforcement if that's appropriate for the custom API. For example, to create a quota on a widgets custom resource in the example.com API group, use `count/widgets.example.com`.

When using such a resource quota (nearly for all object kinds), an object is charged against the quota if the object kind exists (is defined) in the control plane. These types of quotas are useful to protect against exhaustion of storage resources. For example, you may want to limit the number of Secrets in a server given their large size. Too many Secrets in a cluster can actually prevent servers and controllers from starting. You can set a quota for Jobs to protect against a poorly configured CronJob. CronJobs that create too many Jobs in a namespace can lead to a denial of service.

There is another syntax only to set the same type of quota for certain resources. The following types are supported:

| Resource Name          | Description                                                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| configmaps             | The total number of ConfigMaps that can exist in the namespace.                                                                                                              |
| persistentvolumeclaims | The total number of <a href="#">PersistentVolumeClaims</a> that can exist in the namespace.                                                                                  |
| Pods                   | The total number of Pods in a non-terminal state that can exist in the namespace. A pod is in a terminal state if <code>.status.phase</code> in (Failed, Succeeded) is true. |
| replicationcontrollers | The total number of ReplicationControllers that can exist in the namespace.                                                                                                  |
| resourcequotas         | The total number of ResourceQuotas that can exist in the namespace.                                                                                                          |
| services               | The total number of Services that can exist in the namespace.                                                                                                                |
| services.loadbalancers | The total number of Services of type LoadBalancer that can exist in the namespace.                                                                                           |
| services.nodeports     | The total number of Services of type NodePort that can exist in the namespace.                                                                                               |
| secrets                | The total number of Secrets that can exist in the namespace.                                                                                                                 |

For example, pods quota counts and enforces a maximum on the number of pods created in a single namespace that are not terminal. You might want to set a pods quota on a namespace to avoid the case where a user creates many small pods and exhausts the cluster's supply of Pod IPs.

You can find more examples on [Viewing and Setting Quotas](#).

## Quota Scopes

Each quota can have an associated set of scopes. A quota will only measure usage for a resource if it matches the intersection of enumerated scopes.

When a scope is added to the quota, it limits the number of resources it supports to those that pertain to the scope. Resources specified on the quota outside of the allowed set results in a validation error.



| Scope                     | Description                                                                     |
|---------------------------|---------------------------------------------------------------------------------|
| Terminating               | Match pods where .spec.activeDeadlineSeconds >= 0                               |
| NotTerminating            | Match pods where .spec.activeDeadlineSeconds is nil                             |
| BestEffort                | Match pods that have best effort quality of service.                            |
| NotBestEffort             | Match pods that do not have best effort quality of service.                     |
| PriorityClass             | Match pods that references the specified <a href="#">priority class</a> .       |
| CrossNamespacePodAffinity | Match pods that have cross-namespace pod <a href="#">(anti)affinity terms</a> . |

The BestEffort scope restricts a quota to tracking the following resource:

- pods

The Terminating, NotTerminating, NotBestEffort and PriorityClass scopes restrict a quota to tracking the following resources:

- pods
- cpu
- memory
- requests.cpu
- requests.memory
- limits.cpu
- limits.memory

Note that you cannot specify both the Terminating and the NotTerminating scopes in the same quota, and you cannot specify both the BestEffort and NotBestEffort scopes in the same quota either.

The scopeSelector supports the following values in the operator field:

- In
- NotIn
- Exists
- DoesNotExist

When using one of the following values as the scopeName when defining the scopeSelector, the operator must be Exists.

- Terminating
- NotTerminating
- BestEffort
- NotBestEffort

If the operator is In or NotIn, the values field must have at least one value. For example:

```
scopeSelector:
 matchExpressions:
 - scopeName: PriorityClass
 operator: In
 values:
 - middle
```

If the operator is Exists or DoesNotExist, the values field must *NOT* be specified.

## Resource Quota Per PriorityClass

FEATURE STATE: Kubernetes v1.17 [stable]

Pods can be created at a specific [priority](#). You can control a pod's consumption of system resources based on a pod's priority, by using the `scopeSelector` field in the quota spec.

A quota is matched and consumed only if `scopeSelector` in the quota spec selects the pod.

When quota is scoped for priority class using `scopeSelector` field, quota object is restricted to track only following resources:

- pods
- cpu
- memory
- ephemeral-storage
- limits.cpu
- limits.memory
- limits.ephemeral-storage
- requests.cpu
- requests.memory
- requests.ephemeral-storage

This example creates a quota object and matches it with pods at specific priorities. The example works as follows:

- Pods in the cluster have one of the three priority classes, "low", "medium", "high".
- One quota object is created for each priority.

Save the following YAML to a file `quota.yml`.

```
apiVersion: v1
kind: List
items:
- apiVersion: v1
 kind: ResourceQuota
 metadata:
 name: pods-high
 spec:
 hard:
 cpu: "1000"
 memory: 200Gi
 pods: "10"
 scopeSelector:
 matchExpressions:
 - operator: In
 scopeName: PriorityClass
 values: ["high"]
- apiVersion: v1
 kind: ResourceQuota
 metadata:
 name: pods-medium
 spec:
 hard:
```

```

 cpu: "10"
 memory: 20Gi
 pods: "10"
scopeSelector:
 matchExpressions:
 - operator : In
 scopeName: PriorityClass
 values: ["medium"]
- apiVersion: v1
 kind: ResourceQuota
 metadata:
 name: pods-low
 spec:
 hard:
 cpu: "5"
 memory: 10Gi
 pods: "10"
 scopeSelector:
 matchExpressions:
 - operator : In
 scopeName: PriorityClass
 values: ["low"]

```

Apply the YAML using kubectl create.

```
kubectl create -f ./quota.yml
```

```

resourcequota/pods-high created
resourcequota/pods-medium created
resourcequota/pods-low created

```

Verify that Used quota is 0 using kubectl describe quota.

```
kubectl describe quota
```

```

Name: pods-high
Namespace: default
Resource Used Hard

cpu 0 1k
memory 0 200Gi
pods 0 10

```

```

Name: pods-low
Namespace: default
Resource Used Hard

cpu 0 5
memory 0 10Gi
pods 0 10

```

```
Name: pods-medium
```

```
Namespace: default
Resource Used Hard

cpu 0 10
memory 0 20Gi
pods 0 10
```

Create a pod with priority "high". Save the following YAML to a file high-priority-pod.yml.

```
apiVersion: v1
kind: Pod
metadata:
 name: high-priority
spec:
 containers:
 - name: high-priority
 image: ubuntu
 command: ["/bin/sh"]
 args: ["-c", "while true; do echo hello; sleep 10;done"]
 resources:
 requests:
 memory: "10Gi"
 cpu: "500m"
 limits:
 memory: "10Gi"
 cpu: "500m"
 priorityClassName: high
```

Apply it with kubectl create.

```
kubectl create -f ./high-priority-pod.yml
```

Verify that "Used" stats for "high" priority quota, pods-high, has changed and that the other two quotas are unchanged.

```
kubectl describe quota
```

```
Name: pods-high
Namespace: default
Resource Used Hard

cpu 500m 1k
memory 10Gi 200Gi
pods 1 10
```

```
Name: pods-low
Namespace: default
Resource Used Hard

cpu 0 5
memory 0 10Gi
pods 0 10
```

```
Name: pods-medium
Namespace: default
Resource Used Hard
----- -
cpu 0 10
memory 0 20Gi
pods 0 10
```

## Cross-namespace Pod Affinity Quota

FEATURE STATE: Kubernetes v1.24 [stable]

Operators can use `CrossNamespacePodAffinity` quota scope to limit which namespaces are allowed to have pods with affinity terms that cross namespaces. Specifically, it controls which pods are allowed to set namespaces or namespaceSelector fields in pod affinity terms.

Preventing users from using cross-namespace affinity terms might be desired since a pod with anti-affinity constraints can block pods from all other namespaces from getting scheduled in a failure domain.

Using this scope operators can prevent certain namespaces (foo-ns in the example below) from having pods that use cross-namespace pod affinity by creating a resource quota object in that namespace with `CrossNamespacePodAffinity` scope and hard limit of 0:

```
apiVersion: v1
kind: ResourceQuota
metadata:
 name: disable-cross-namespace-affinity
 namespace: foo-ns
spec:
 hard:
 pods: "0"
 scopeSelector:
 matchExpressions:
 - scopeName: CrossNamespacePodAffinity
 operator: Exists
```

If operators want to disallow using namespaces and namespaceSelector by default, and only allow it for specific namespaces, they could configure `CrossNamespacePodAffinity` as a limited resource by setting the kube-apiserver flag `--admission-control-config-file` to the path of the following configuration file:

```
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
 - name: "ResourceQuota"
 configuration:
 apiVersion: apiserver.config.k8s.io/v1
 kind: ResourceQuotaConfiguration
 limitedResources:
 - resource: pods
 matchScopes:
```

```
- scopeName: CrossNamespacePodAffinity
 operator: Exists
```

With the above configuration, pods can use namespaces and namespaceSelector in pod affinity only if the namespace where they are created have a resource quota object with CrossNamespacePodAffinity scope and a hard limit greater than or equal to the number of pods using those fields.

## Requests compared to Limits

When allocating compute resources, each container may specify a request and a limit value for either CPU or memory. The quota can be configured to quota either value.

If the quota has a value specified for requests.cpu or requests.memory, then it requires that every incoming container makes an explicit request for those resources. If the quota has a value specified for limits.cpu or limits.memory, then it requires that every incoming container specifies an explicit limit for those resources.

## Viewing and Setting Quotas

Kubectrl supports creating, updating, and viewing quotas:

```
kubectrl create namespace myspace
```

```
cat <<EOF > compute-resources.yaml
apiVersion: v1
kind: ResourceQuota
metadata:
 name: compute-resources
spec:
 hard:
 requests.cpu: "1"
 requests.memory: 1Gi
 limits.cpu: "2"
 limits.memory: 2Gi
 requests.nvidia.com/gpu: 4
EOF
```

```
kubectrl create -f ./compute-resources.yaml --namespace=myspace
```

```
cat <<EOF > object-counts.yaml
apiVersion: v1
kind: ResourceQuota
metadata:
 name: object-counts
spec:
 hard:
 configmaps: "10"
 persistentvolumeclaims: "4"
 pods: "4"
 replicationcontrollers: "20"
 secrets: "10"
```

```
services: "10"
services.loadbalancers: "2"
EOF
```

```
kubectl create -f ./object-counts.yaml --namespace=myspace
```

```
kubectl get quota --namespace=myspace
```

| NAME              | AGE |
|-------------------|-----|
| compute-resources | 30s |
| object-counts     | 32s |

```
kubectl describe quota compute-resources --namespace=myspace
```

|                         |                   |      |  |
|-------------------------|-------------------|------|--|
| Name:                   | compute-resources |      |  |
| Namespace:              | myspace           |      |  |
| Resource                | Used              | Hard |  |
| -----                   | ----              | ---- |  |
| limits.cpu              | 0                 | 2    |  |
| limits.memory           | 0                 | 2Gi  |  |
| requests.cpu            | 0                 | 1    |  |
| requests.memory         | 0                 | 1Gi  |  |
| requests.nvidia.com/gpu | 0                 | 4    |  |

```
kubectl describe quota object-counts --namespace=myspace
```

|                        |               |      |  |
|------------------------|---------------|------|--|
| Name:                  | object-counts |      |  |
| Namespace:             | myspace       |      |  |
| Resource               | Used          | Hard |  |
| -----                  | ----          | ---- |  |
| configmaps             | 0             | 10   |  |
| persistentvolumeclaims | 0             | 4    |  |
| Pods                   | 0             | 4    |  |
| replicationcontrollers | 0             | 20   |  |
| secrets                | 1             | 10   |  |
| services               | 0             | 10   |  |
| services.loadbalancers | 0             | 2    |  |

Kubectl also supports object count quota for all standard namespaced resources using the syntax `count/<resource>.<group>`:

```
kubectl create namespace myspace
```

```
kubectl create quota test --hard=count/deployments.apps=2,count/replicasets.apps=4,count/pods=3,count/secrets=4 --namespace=myspace
```

```
kubectl create deployment nginx --image=nginx --namespace=myspace --replicas=2
```

```
kubectl describe quota --namespace=myspace
```

|            |         |      |  |
|------------|---------|------|--|
| Name:      | test    |      |  |
| Namespace: | myspace |      |  |
| Resource   | Used    | Hard |  |
| -----      | ----    | ---- |  |

|                        |   |   |
|------------------------|---|---|
| count/deployments.apps | 1 | 2 |
| count/pods             | 2 | 3 |
| count/replicasets.apps | 1 | 4 |
| count/secrets          | 1 | 4 |

## Quota and Cluster Capacity

ResourceQuotas are independent of the cluster capacity. They are expressed in absolute units. So, if you add nodes to your cluster, this does *not* automatically give each namespace the ability to consume more resources.

Sometimes more complex policies may be desired, such as:

- Proportionally divide total cluster resources among several teams.
- Allow each tenant to grow resource usage as needed, but have a generous limit to prevent accidental resource exhaustion.
- Detect demand from one namespace, add nodes, and increase quota.

Such policies could be implemented using ResourceQuotas as building blocks, by writing a "controller" that watches the quota usage and adjusts the quota hard limits of each namespace according to other signals.

Note that resource quota divides up aggregate cluster resources, but it creates no restrictions around nodes: pods from several namespaces may run on the same node.

## Limit Priority Class consumption by default

It may be desired that pods at a particular priority, eg. "cluster-services", should be allowed in a namespace, if and only if, a matching quota object exists.

With this mechanism, operators are able to restrict usage of certain high priority classes to a limited number of namespaces and not every namespace will be able to consume these priority classes by default.

To enforce this, kube-apiserver flag `--admission-control-config-file` should be used to pass path to the following configuration file:

```
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
- name: "ResourceQuota"
 configuration:
 apiVersion: apiserver.config.k8s.io/v1
 kind: ResourceQuotaConfiguration
 limitedResources:
 - resource: pods
 matchScopes:
 - scopeName: PriorityClass
 operator: In
 values: ["cluster-services"]
```

Then, create a resource quota object in the kube-system namespace:



[policy/priority-class-resourcequota.yaml](#)

```
apiVersion: v1
kind: ResourceQuota
metadata:
 name: pods-cluster-services
spec:
 scopeSelector:
 matchExpressions:
 - operator: In
 scopeName: PriorityClass
 values: ["cluster-services"]
```

```
kubectl apply -f https://k8s.io/examples/policy/priority-class-resourcequota.yaml -n kube-system
```

resourcequota/pods-cluster-services created

In this case, a pod creation will be allowed if:

1. the Pod's `priorityClassName` is not specified.
2. the Pod's `priorityClassName` is specified to a value other than `cluster-services`.
3. the Pod's `priorityClassName` is set to `cluster-services`, it is to be created in the `kube-system` namespace, and it has passed the resource quota check.

A Pod creation request is rejected if its `priorityClassName` is set to `cluster-services` and it is to be created in a namespace other than `kube-system`.

## What's next

- See [ResourceQuota design doc](#) for more information.
- See a [detailed example for how to use resource quota](#).
- Read [Quota support for priority class design doc](#).
- See [LimitedResources](#)

# Process ID Limits And Reservations

FEATURE STATE: Kubernetes v1.20 [stable]

Kubernetes allow you to limit the number of process IDs (PIDs) that a [Pod](#) can use. You can also reserve a number of allocatable PIDs for each [node](#) for use by the operating system and daemons (rather than by Pods).

Process IDs (PIDs) are a fundamental resource on nodes. It is trivial to hit the task limit without hitting any other resource limits, which can then cause instability to a host machine.

Cluster administrators require mechanisms to ensure that Pods running in the cluster cannot induce PID exhaustion that prevents host daemons (such as the [kubelet](#) or [kube-proxy](#), and potentially also the container runtime) from running. In addition, it is important to ensure that PIDs are limited among Pods in order to ensure they have limited impact on other workloads on the same node.

**Note:**

On certain Linux installations, the operating system sets the PIDs limit to a low default, such as 32768. Consider raising the value of `/proc/sys/kernel/pid_max`.

You can configure a kubelet to limit the number of PIDs a given Pod can consume. For example, if your node's host OS is set to use a maximum of 262144 PIDs and expect to host less than 250 Pods, one can give each Pod a budget of 1000 PIDs to prevent using up that node's overall number of available PIDs. If the admin wants to overcommit PIDs similar to CPU or memory, they may do so as well with some additional risks. Either way, a single Pod will not be able to bring the whole machine down. This kind of resource limiting helps to prevent simple fork bombs from affecting operation of an entire cluster.

Per-Pod PID limiting allows administrators to protect one Pod from another, but does not ensure that all Pods scheduled onto that host are unable to impact the node overall. Per-Pod limiting also does not protect the node agents themselves from PID exhaustion.

You can also reserve an amount of PIDs for node overhead, separate from the allocation to Pods. This is similar to how you can reserve CPU, memory, or other resources for use by the operating system and other facilities outside of Pods and their containers.

PID limiting is a an important sibling to [compute resource](#) requests and limits. However, you specify it in a different way: rather than defining a Pod's resource limit in the `.spec` for a Pod, you configure the limit as a setting on the kubelet. Pod-defined PID limits are not currently supported.

**Caution:**

This means that the limit that applies to a Pod may be different depending on where the Pod is scheduled. To make things simple, it's easiest if all Nodes use the same PID resource limits and reservations.

## Node PID limits

Kubernetes allows you to reserve a number of process IDs for the system use. To configure the reservation, use the parameter `pid=<number>` in the `--system-reserved` and `--kube-reserved` command line options to the kubelet. The value you specified declares that the specified number of process IDs will be reserved for the system as a whole and for Kubernetes system daemons respectively.

## Pod PID limits

Kubernetes allows you to limit the number of processes running in a Pod. You specify this limit at the node level, rather than configuring it as a resource limit for a particular Pod. Each Node can have a different PID limit.

To configure the limit, you can specify the command line parameter `--pod-max-pids` to the kubelet, or set `PodPidsLimit` in the kubelet [configuration file](#).

## PID based eviction

You can configure kubelet to start terminating a Pod when it is misbehaving and consuming abnormal amount of resources. This feature is called eviction. You can [Configure Out of Resource Handling](#) for various eviction signals. Use `pid.available` eviction signal to configure the threshold for number of PIDs used by Pod. You can set soft and hard eviction policies. However, even with the hard eviction policy, if the number of PIDs growing very fast, node can still get into unstable state by hitting the node PIDs limit. Eviction signal value is calculated periodically and does NOT enforce the limit.

PID limiting - per Pod and per Node sets the hard limit. Once the limit is hit, workload will start experiencing failures when trying to get a new PID. It may or may not lead to rescheduling of a Pod, depending on how workload reacts on these failures and how liveness and readiness probes are configured for the Pod. However, if limits were set correctly, you can guarantee that other Pods workload and system processes will not run out of PIDs when one Pod is misbehaving.

## What's next

- Refer to the [PID Limiting enhancement document](#) for more information.
- For historical context, read [Process ID Limiting for Stability Improvements in Kubernetes 1.14](#).
- Read [Managing Resources for Containers](#).
- Learn how to [Configure Out of Resource Handling](#).

## Node Resource Managers

In order to support latency-critical and high-throughput workloads, Kubernetes offers a suite of Resource Managers. The managers aim to co-ordinate and optimise node's resources alignment for pods configured with a specific requirement for CPUs, devices, and memory (hugepages) resources.

The main manager, the Topology Manager, is a Kubelet component that co-ordinates the overall resource management process through its [policy](#).

The configuration of individual managers is elaborated in dedicated documents:

- [CPU Manager Policies](#)
- [Device Manager](#)
- [Memory Manager Policies](#)

## Scheduling, Preemption and Eviction

In Kubernetes, scheduling refers to making sure that Pods are matched to Nodes so that the kubelet can run them. Preemption is the process of terminating Pods with lower Priority so that Pods with higher Priority can schedule on Nodes. Eviction is the process of proactively terminating one or more Pods on resource-starved Nodes.

In Kubernetes, scheduling refers to making sure that [Pods](#) are matched to [Nodes](#) so that the [kubelet](#) can run them. Preemption is the process of terminating Pods with lower [Priority](#) so that Pods with higher Priority can schedule on Nodes. Eviction is the process of terminating one or more Pods on Nodes.

## Scheduling

- [Kubernetes Scheduler](#)
- [Assigning Pods to Nodes](#)
- [Pod Overhead](#)
- [Pod Topology Spread Constraints](#)
- [Taints and Tolerations](#)
- [Scheduling Framework](#)
- [Dynamic Resource Allocation](#)
- [Scheduler Performance Tuning](#)
- [Resource Bin Packing for Extended Resources](#)
- [Pod Scheduling Readiness](#)
- [Descheduler](#)

## Pod Disruption

[Pod disruption](#) is the process by which Pods on Nodes are terminated either voluntarily or involuntarily.

Voluntary disruptions are started intentionally by application owners or cluster administrators. Involuntary disruptions are unintentional and can be triggered by unavoidable issues like Nodes running out of resources, or by accidental deletions.

- [Pod Priority and Preemption](#)
- [Node-pressure Eviction](#)
- [API-initiated Eviction](#)

## Kubernetes Scheduler

In Kubernetes, *scheduling* refers to making sure that [Pods](#) are matched to [Nodes](#) so that [Kubelet](#) can run them.

## Scheduling overview

A scheduler watches for newly created Pods that have no Node assigned. For every Pod that the scheduler discovers, the scheduler becomes responsible for finding the best Node for that Pod to run on. The scheduler reaches this placement decision taking into account the scheduling principles described below.

If you want to understand why Pods are placed onto a particular Node, or if you're planning to implement a custom scheduler yourself, this page will help you learn about scheduling.

# kube-scheduler

[kube-scheduler](#) is the default scheduler for Kubernetes and runs as part of the [control plane](#). kube-scheduler is designed so that, if you want and need to, you can write your own scheduling component and use that instead.

Kube-scheduler selects an optimal node to run newly created or not yet scheduled (unscheduled) pods. Since containers in pods - and pods themselves - can have different requirements, the scheduler filters out any nodes that don't meet a Pod's specific scheduling needs. Alternatively, the API lets you specify a node for a Pod when you create it, but this is unusual and is only done in special cases.

In a cluster, Nodes that meet the scheduling requirements for a Pod are called *feasible* nodes. If none of the nodes are suitable, the pod remains unscheduled until the scheduler is able to place it.

The scheduler finds feasible Nodes for a Pod and then runs a set of functions to score the feasible Nodes and picks a Node with the highest score among the feasible ones to run the Pod. The scheduler then notifies the API server about this decision in a process called *binding*.

Factors that need to be taken into account for scheduling decisions include individual and collective resource requirements, hardware / software / policy constraints, affinity and anti-affinity specifications, data locality, inter-workload interference, and so on.

## Node selection in kube-scheduler

kube-scheduler selects a node for the pod in a 2-step operation:

1. Filtering
2. Scoring

The *filtering* step finds the set of Nodes where it's feasible to schedule the Pod. For example, the PodFitsResources filter checks whether a candidate Node has enough available resources to meet a Pod's specific resource requests. After this step, the node list contains any suitable Nodes; often, there will be more than one. If the list is empty, that Pod isn't (yet) schedulable.

In the *scoring* step, the scheduler ranks the remaining nodes to choose the most suitable Pod placement. The scheduler assigns a score to each Node that survived filtering, basing this score on the active scoring rules.

Finally, kube-scheduler assigns the Pod to the Node with the highest ranking. If there is more than one node with equal scores, kube-scheduler selects one of these at random.

There are two supported ways to configure the filtering and scoring behavior of the scheduler:

1. [Scheduling Policies](#) allow you to configure *Predicates* for filtering and *Priorities* for scoring.
2. [Scheduling Profiles](#) allow you to configure Plugins that implement different scheduling stages, including: QueueSort, Filter, Score, Bind, Reserve, Permit, and others. You can also configure the kube-scheduler to run different profiles.

## What's next

- Read about [scheduler performance tuning](#)
- Read about [Pod topology spread constraints](#)
- Read the [reference documentation](#) for kube-scheduler
- Read the [kube-scheduler config \(v1\)](#) reference
- Learn about [configuring multiple schedulers](#)
- Learn about [topology management policies](#)
- Learn about [Pod Overhead](#)
- Learn about scheduling of Pods that use volumes in:
  - [Volume Topology Support](#)
  - [Storage Capacity Tracking](#)
  - [Node-specific Volume Limits](#)

## Assigning Pods to Nodes

You can constrain a [Pod](#) so that it is *restricted* to run on particular [node\(s\)](#), or to *prefer* to run on particular nodes. There are several ways to do this and the recommended approaches all use [label selectors](#) to facilitate the selection. Often, you do not need to set any such constraints; the [scheduler](#) will automatically do a reasonable placement (for example, spreading your Pods across nodes so as not place Pods on a node with insufficient free resources). However, there are some circumstances where you may want to control which node the Pod deploys to, for example, to ensure that a Pod ends up on a node with an SSD attached to it, or to co-locate Pods from two different services that communicate a lot into the same availability zone.

You can use any of the following methods to choose where Kubernetes schedules specific Pods:

- [nodeSelector](#) field matching against [node labels](#)
- [Affinity and anti-affinity](#)
- [nodeName](#) field
- [Pod topology spread constraints](#)

## Node labels

Like many other Kubernetes objects, nodes have [labels](#). You can [attach labels manually](#). Kubernetes also populates a [standard set of labels](#) on all nodes in a cluster.

### Note:

The value of these labels is cloud provider specific and is not guaranteed to be reliable. For example, the value of `kubernetes.io/hostname` may be the same as the node name in some environments and a different value in other environments.

## Node isolation/restriction

Adding labels to nodes allows you to target Pods for scheduling on specific nodes or groups of nodes. You can use this functionality to ensure that specific Pods only run on nodes with certain isolation, security, or regulatory properties.

If you use labels for node isolation, choose label keys that the [kubelet](#) cannot modify. This prevents a compromised node from setting those labels on itself so that the scheduler schedules workloads onto the compromised node.

The [NodeRestriction admission plugin](#) prevents the kubelet from setting or modifying labels with a `node-restriction.kubernetes.io/` prefix.

To make use of that label prefix for node isolation:

1. Ensure you are using the [Node authorizer](#) and have *enabled* the NodeRestriction admission plugin.
2. Add labels with the `node-restriction.kubernetes.io/` prefix to your nodes, and use those labels in your [node selectors](#). For example, `example.com.node-restriction.kubernetes.io/fips=true` or `example.com.node-restriction.kubernetes.io/pci-dss=true`.

## nodeSelector

`nodeSelector` is the simplest recommended form of node selection constraint. You can add the `nodeSelector` field to your Pod specification and specify the [node labels](#) you want the target node to have. Kubernetes only schedules the Pod onto nodes that have each of the labels you specify.

See [Assign Pods to Nodes](#) for more information.

## Affinity and anti-affinity

`nodeSelector` is the simplest way to constrain Pods to nodes with specific labels. Affinity and anti-affinity expands the types of constraints you can define. Some of the benefits of affinity and anti-affinity include:

- The affinity/anti-affinity language is more expressive. `nodeSelector` only selects nodes with all the specified labels. Affinity/anti-affinity gives you more control over the selection logic.
- You can indicate that a rule is *soft* or *preferred*, so that the scheduler still schedules the Pod even if it can't find a matching node.
- You can constrain a Pod using labels on other Pods running on the node (or other topological domain), instead of just node labels, which allows you to define rules for which Pods can be co-located on a node.

The affinity feature consists of two types of affinity:

- *Node affinity* functions like the `nodeSelector` field but is more expressive and allows you to specify soft rules.
- *Inter-pod affinity/anti-affinity* allows you to constrain Pods against labels on other Pods.

## Node affinity

Node affinity is conceptually similar to `nodeSelector`, allowing you to constrain which nodes your Pod can be scheduled on based on node labels. There are two types of node affinity:

- `requiredDuringSchedulingIgnoredDuringExecution`: The scheduler can't schedule the Pod unless the rule is met. This functions like `nodeSelector`, but with a more expressive syntax.
- `preferredDuringSchedulingIgnoredDuringExecution`: The scheduler tries to find a node that meets the rule. If a matching node is not available, the scheduler still schedules the Pod.

### Note:

In the preceding types, `IgnoredDuringExecution` means that if the node labels change after Kubernetes schedules the Pod, the Pod continues to run.

You can specify node affinities using the `.spec.affinity.nodeAffinity` field in your Pod spec.

For example, consider the following Pod spec:

[pods/pod-with-node-affinity.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: with-node-affinity
spec:
 affinity:
 nodeAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 nodeSelectorTerms:
 - matchExpressions:
 - key: topology.kubernetes.io/zone
 operator: In
 values:
 - antarctica-east1
 - antarctica-west1
 preferredDuringSchedulingIgnoredDuringExecution:
 - weight: 1
 preference:
 matchExpressions:
 - key: another-node-label-key
 operator: In
 values:
 - another-node-label-value
 containers:
 - name: with-node-affinity
 image: registry.k8s.io/pause:2.0
```



In this example, the following rules apply:

- The node *must* have a label with the key `topology.kubernetes.io/zone` and the value of that label *must* be either `antarctica-east1` or `antarctica-west1`.
- The node *preferably* has a label with the key `another-node-label-key` and the value `another-node-label-value`.

You can use the `operator` field to specify a logical operator for Kubernetes to use when interpreting the rules. You can use `In`, `NotIn`, `Exists`, `DoesNotExist`, `Gt` and `Lt`.

Read [Operators](#) to learn more about how these work.

`NotIn` and `DoesNotExist` allow you to define node anti-affinity behavior. Alternatively, you can use [node taints](#) to repel Pods from specific nodes.

### Note:

If you specify both `nodeSelector` and `nodeAffinity`, *both* must be satisfied for the Pod to be scheduled onto a node.

If you specify multiple terms in `nodeSelectorTerms` associated with `nodeAffinity` types, then the Pod can be scheduled onto a node if one of the specified terms can be satisfied (terms are ORed).

If you specify multiple expressions in a single `matchExpressions` field associated with a term in `nodeSelectorTerms`, then the Pod can be scheduled onto a node only if all the expressions are satisfied (expressions are ANDed).

See [Assign Pods to Nodes using Node Affinity](#) for more information.

### Node affinity weight

You can specify a weight between 1 and 100 for each instance of the `preferredDuringSchedulingIgnoredDuringExecution` affinity type. When the scheduler finds nodes that meet all the other scheduling requirements of the Pod, the scheduler iterates through every preferred rule that the node satisfies and adds the value of the weight for that expression to a sum.

The final sum is added to the score of other priority functions for the node. Nodes with the highest total score are prioritized when the scheduler makes a scheduling decision for the Pod.

For example, consider the following Pod spec:

[pods/pod-with-affinity-preferred-weight.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: with-affinity-preferred-weight
spec:
 affinity:
 nodeAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
```

```

nodeSelectorTerms:
- matchExpressions:
 - key: kubernetes.io/os
 operator: In
 values:
 - linux
preferredDuringSchedulingIgnoredDuringExecution:
- weight: 1
 preference:
 matchExpressions:
 - key: label-1
 operator: In
 values:
 - key-1
- weight: 50
 preference:
 matchExpressions:
 - key: label-2
 operator: In
 values:
 - key-2
containers:
- name: with-node-affinity
 image: registry.k8s.io/pause:2.0

```

If there are two possible nodes that match the `preferredDuringSchedulingIgnoredDuringExecution` rule, one with the `label-1:key-1` label and another with the `label-2:key-2` label, the scheduler considers the weight of each node and adds the weight to the other scores for that node, and schedules the Pod onto the node with the highest final score.

### Note:

If you want Kubernetes to successfully schedule the Pods in this example, you must have existing nodes with the `kubernetes.io/os=linux` label.

### Node affinity per scheduling profile

FEATURE STATE: Kubernetes v1.20 [beta]

When configuring multiple [scheduling profiles](#), you can associate a profile with a node affinity, which is useful if a profile only applies to a specific set of nodes. To do so, add an `addedAffinity` to the `args` field of the [NodeAffinity plugin](#) in the [scheduler configuration](#). For example:

```

apiVersion: kubescheduler.config.k8s.io/v1beta3
kind: KubeSchedulerConfiguration

profiles:
- schedulerName: default-scheduler
- schedulerName: foo-scheduler
 pluginConfig:
 - name: NodeAffinity
 args:

```

```
addedAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 nodeSelectorTerms:
 - matchExpressions:
 - key: scheduler-profile
 operator: In
 values:
 - foo
```

The `addedAffinity` is applied to all Pods that set `.spec.schedulerName` to `foo-scheduler`, in addition to the `NodeAffinity` specified in the `PodSpec`. That is, in order to match the Pod, nodes need to satisfy `addedAffinity` and the Pod's `.spec.NodeAffinity`.

Since the `addedAffinity` is not visible to end users, its behavior might be unexpected to them. Use node labels that have a clear correlation to the scheduler profile name.

#### Note:

The `DaemonSet` controller, which [creates Pods for DaemonSets](#), does not support scheduling profiles. When the `DaemonSet` controller creates Pods, the default Kubernetes scheduler places those Pods and honors any `nodeAffinity` rules in the `DaemonSet` controller.

## Inter-pod affinity and anti-affinity

Inter-pod affinity and anti-affinity allow you to constrain which nodes your Pods can be scheduled on based on the labels of **Pods** already running on that node, instead of the node labels.

Inter-pod affinity and anti-affinity rules take the form "this Pod should (or, in the case of anti-affinity, should not) run in an X if that X is already running one or more Pods that meet rule Y", where X is a topology domain like node, rack, cloud provider zone or region, or similar and Y is the rule Kubernetes tries to satisfy.

You express these rules (Y) as [label selectors](#) with an optional associated list of namespaces. Pods are namespaced objects in Kubernetes, so Pod labels also implicitly have namespaces. Any label selectors for Pod labels should specify the namespaces in which Kubernetes should look for those labels.

You express the topology domain (X) using a `topologyKey`, which is the key for the node label that the system uses to denote the domain. For examples, see [Well-Known Labels, Annotations and Taints](#).

#### Note:

Inter-pod affinity and anti-affinity require substantial amounts of processing which can slow down scheduling in large clusters significantly. We do not recommend using them in clusters larger than several hundred nodes.

**Note:**

Pod anti-affinity requires nodes to be consistently labeled, in other words, every node in the cluster must have an appropriate label matching topologyKey. If some or all nodes are missing the specified topologyKey label, it can lead to unintended behavior.

**Types of inter-pod affinity and anti-affinity**

Similar to [node affinity](#) are two types of Pod affinity and anti-affinity as follows:

- requiredDuringSchedulingIgnoredDuringExecution
- preferredDuringSchedulingIgnoredDuringExecution

For example, you could use requiredDuringSchedulingIgnoredDuringExecution affinity to tell the scheduler to co-locate Pods of two services in the same cloud provider zone because they communicate with each other a lot. Similarly, you could use preferredDuringSchedulingIgnoredDuringExecution anti-affinity to spread Pods from a service across multiple cloud provider zones.

To use inter-pod affinity, use the affinity.podAffinity field in the Pod spec. For inter-pod anti-affinity, use the affinity.podAntiAffinity field in the Pod spec.

**Scheduling a group of pods with inter-pod affinity to themselves**

If the current Pod being scheduled is the first in a series that have affinity to themselves, it is allowed to be scheduled if it passes all other affinity checks. This is determined by verifying that no other pod in the cluster matches the namespace and selector of this pod, that the pod matches its own terms, and the chosen node matches all requested topologies. This ensures that there will not be a deadlock even if all the pods have inter-pod affinity specified.

**Pod affinity example**

Consider the following Pod spec:

[pods/pod-with-pod-affinity.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: with-pod-affinity
spec:
 affinity:
 podAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 - labelSelector:
 matchExpressions:
 - key: security
 operator: In
 values:
 - S1
 topologyKey: topology.kubernetes.io/zone
 podAntiAffinity:
```

```
preferredDuringSchedulingIgnoredDuringExecution:
- weight: 100
podAffinityTerm:
 labelSelector:
 matchExpressions:
 - key: security
 operator: In
 values:
 - S2
 topologyKey: topology.kubernetes.io/zone
containers:
- name: with-pod-affinity
 image: registry.k8s.io/pause:2.0
```

This example defines one Pod affinity rule and one Pod anti-affinity rule. The Pod affinity rule uses the "hard" `requiredDuringSchedulingIgnoredDuringExecution`, while the anti-affinity rule uses the "soft" `preferredDuringSchedulingIgnoredDuringExecution`.

The affinity rule specifies that the scheduler is allowed to place the example Pod on a node only if that node belongs to a specific [zone](#) where other Pods have been labeled with `security=S1`. For instance, if we have a cluster with a designated zone, let's call it "Zone V," consisting of nodes labeled with `topology.kubernetes.io/zone=V`, the scheduler can assign the Pod to any node within Zone V, as long as there is at least one Pod within Zone V already labeled with `security=S1`. Conversely, if there are no Pods with `security=S1` labels in Zone V, the scheduler will not assign the example Pod to any node in that zone.

The anti-affinity rule specifies that the scheduler should try to avoid scheduling the Pod on a node if that node belongs to a specific [zone](#) where other Pods have been labeled with `security=S2`. For instance, if we have a cluster with a designated zone, let's call it "Zone R," consisting of nodes labeled with `topology.kubernetes.io/zone=R`, the scheduler should avoid assigning the Pod to any node within Zone R, as long as there is at least one Pod within Zone R already labeled with `security=S2`. Conversely, the anti-affinity rule does not impact scheduling into Zone R if there are no Pods with `security=S2` labels.

To get yourself more familiar with the examples of Pod affinity and anti-affinity, refer to the [design proposal](#).

You can use the `In`, `NotIn`, `Exists` and `DoesNotExist` values in the operator field for Pod affinity and anti-affinity.

Read [Operators](#) to learn more about how these work.

In principle, the `topologyKey` can be any allowed label key with the following exceptions for performance and security reasons:

- For Pod affinity and anti-affinity, an empty `topologyKey` field is not allowed in both `requiredDuringSchedulingIgnoredDuringExecution` and `preferredDuringSchedulingIgnoredDuringExecution`.
- For `requiredDuringSchedulingIgnoredDuringExecution` Pod anti-affinity rules, the admission controller `LimitPodHardAntiAffinityTopology` limits `topologyKey` to `kubernetes.io/hostname`. You can modify or disable the admission controller if you want to allow custom topologies.

In addition to `labelSelector` and `topologyKey`, you can optionally specify a list of namespaces which the `labelSelector` should match against using the `namespaces` field at the same level as `labelSelector` and `topologyKey`. If omitted or empty, `namespaces` defaults to the namespace of the Pod where the affinity/anti-affinity definition appears.

## Namespace selector

FEATURE STATE: Kubernetes v1.24 [stable]

You can also select matching namespaces using `namespaceSelector`, which is a label query over the set of namespaces. The affinity term is applied to namespaces selected by both `namespaceSelector` and the `namespaces` field. Note that an empty `namespaceSelector` (`{}`) matches all namespaces, while a null or empty `namespaces` list and null `namespaceSelector` matches the namespace of the Pod where the rule is defined.

## matchLabelKeys

FEATURE STATE: Kubernetes v1.29 [alpha]

### Note:

The `matchLabelKeys` field is an alpha-level field and is disabled by default in Kubernetes 1.30. When you want to use it, you have to enable it via the `MatchLabelKeysInPodAffinity` [feature gate](#).

Kubernetes includes an optional `matchLabelKeys` field for Pod affinity or anti-affinity. The field specifies keys for the labels that should match with the incoming Pod's labels, when satisfying the Pod (anti)affinity.

The keys are used to look up values from the pod labels; those key-value labels are combined (using AND) with the match restrictions defined using the `labelSelector` field. The combined filtering selects the set of existing pods that will be taken into Pod (anti)affinity calculation.

A common use case is to use `matchLabelKeys` with `pod-template-hash` (set on Pods managed as part of a Deployment, where the value is unique for each revision). Using `pod-template-hash` in `matchLabelKeys` allows you to target the Pods that belong to the same revision as the incoming Pod, so that a rolling upgrade won't break affinity.

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: application-server
...
spec:
 template:
 spec:
 affinity:
 podAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 - labelSelector:
 matchExpressions:
 - key: app
```

```

 operator: In
 values:
 - database
 topologyKey: topology.kubernetes.io/zone
 # Only Pods from a given rollout are taken into consideration when calculating pod
 affinity.
 # If you update the Deployment, the replacement Pods follow their own affinity rules
 # (if there are any defined in the new Pod template)
 matchLabelKeys:
 - pod-template-hash

```

## mismatchLabelKeys

FEATURE STATE: Kubernetes v1.29 [alpha]

### Note:

The `mismatchLabelKeys` field is an alpha-level field and is disabled by default in Kubernetes 1.30. When you want to use it, you have to enable it via the [MatchLabelKeysInPodAffinity feature gate](#).

Kubernetes includes an optional `mismatchLabelKeys` field for Pod affinity or anti-affinity. The field specifies keys for the labels that should **not** match with the incoming Pod's labels, when satisfying the Pod (anti)affinity.

One example use case is to ensure Pods go to the topology domain (node, zone, etc) where only Pods from the same tenant or team are scheduled in. In other words, you want to avoid running Pods from two different tenants on the same topology domain at the same time.

```

apiVersion: v1
kind: Pod
metadata:
 labels:
 # Assume that all relevant Pods have a "tenant" label set
 tenant: tenant-a
...
spec:
 affinity:
 podAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 # ensure that pods associated with this tenant land on the correct node pool
 - matchLabelKeys:
 - tenant
 topologyKey: node-pool
 podAntiAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 # ensure that pods associated with this tenant can't schedule to nodes used for another
 tenant
 - mismatchLabelKeys:
 - tenant # whatever the value of the "tenant" label for this Pod, prevent
 # scheduling to nodes in any pool where any Pod from a different
 # tenant is running.
 labelSelector:

```

```
We have to have the labelSelector which selects only Pods with the tenant label,
otherwise this Pod would hate Pods from daemonsets as well, for example,
which aren't supposed to have the tenant label.
matchExpressions:
- key: tenant
 operator: Exists
topologyKey: node-pool
```

## More practical use-cases

Inter-pod affinity and anti-affinity can be even more useful when they are used with higher level collections such as ReplicaSets, StatefulSets, Deployments, etc. These rules allow you to configure that a set of workloads should be co-located in the same defined topology; for example, preferring to place two related Pods onto the same node.

For example: imagine a three-node cluster. You use the cluster to run a web application and also an in-memory cache (such as Redis). For this example, also assume that latency between the web application and the memory cache should be as low as is practical. You could use inter-pod affinity and anti-affinity to co-locate the web servers with the cache as much as possible.

In the following example Deployment for the Redis cache, the replicas get the label `app=store`. The `podAntiAffinity` rule tells the scheduler to avoid placing multiple replicas with the `app=store` label on a single node. This creates each cache in a separate node.

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: redis-cache
spec:
 selector:
 matchLabels:
 app: store
 replicas: 3
 template:
 metadata:
 labels:
 app: store
 spec:
 affinity:
 podAntiAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 - labelSelector:
 matchExpressions:
 - key: app
 operator: In
 values:
 - store
 topologyKey: "kubernetes.io/hostname"
 containers:
 - name: redis-server
 image: redis:3.2-alpine
```



The following example Deployment for the web servers creates replicas with the label `app=web-store`. The Pod affinity rule tells the scheduler to place each replica on a node that has a Pod with the label `app=store`. The Pod anti-affinity rule tells the scheduler never to place multiple `app=web-store` servers on a single node.

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: web-server
spec:
 selector:
 matchLabels:
 app: web-store
 replicas: 3
 template:
 metadata:
 labels:
 app: web-store
 spec:
 affinity:
 podAntiAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 - labelSelector:
 matchExpressions:
 - key: app
 operator: In
 values:
 - web-store
 topologyKey: "kubernetes.io/hostname"
 podAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 - labelSelector:
 matchExpressions:
 - key: app
 operator: In
 values:
 - store
 topologyKey: "kubernetes.io/hostname"
 containers:
 - name: web-app
 image: nginx:1.16-alpine
```

Creating the two preceding Deployments results in the following cluster layout, where each web server is co-located with a cache, on three separate nodes.

| node-1             | node-2             | node-3             |
|--------------------|--------------------|--------------------|
| <i>webserver-1</i> | <i>webserver-2</i> | <i>webserver-3</i> |
| <i>cache-1</i>     | <i>cache-2</i>     | <i>cache-3</i>     |

The overall effect is that each cache instance is likely to be accessed by a single client that is running on the same node. This approach aims to minimize both skew (imbalanced load) and latency.

You might have other reasons to use Pod anti-affinity. See the [ZooKeeper tutorial](#) for an example of a StatefulSet configured with anti-affinity for high availability, using the same technique as this example.

## nodeName

`nodeName` is a more direct form of node selection than affinity or `nodeSelector`. `nodeName` is a field in the Pod spec. If the `nodeName` field is not empty, the scheduler ignores the Pod and the kubelet on the named node tries to place the Pod on that node. Using `nodeName` overrules using `nodeSelector` or affinity and anti-affinity rules.

Some of the limitations of using `nodeName` to select nodes are:

- If the named node does not exist, the Pod will not run, and in some cases may be automatically deleted.
- If the named node does not have the resources to accommodate the Pod, the Pod will fail and its reason will indicate why, for example `OutOfMemory` or `OutOfCPU`.
- Node names in cloud environments are not always predictable or stable.

### Warning:

`nodeName` is intended for use by custom schedulers or advanced use cases where you need to bypass any configured schedulers. Bypassing the schedulers might lead to failed Pods if the assigned Nodes get oversubscribed. You can use [node affinity](#) or the [nodeSelector field](#) to assign a Pod to a specific Node without bypassing the schedulers.

Here is an example of a Pod spec using the `nodeName` field:

```
apiVersion: v1
kind: Pod
metadata:
 name: nginx
spec:
 containers:
 - name: nginx
 image: nginx
 nodeName: kube-01
```

The above Pod will only run on the node `kube-01`.

## Pod topology spread constraints

You can use *topology spread constraints* to control how [Pods](#) are spread across your cluster among failure-domains such as regions, zones, nodes, or among any other topology domains that you define. You might do this to improve performance, expected availability, or overall utilization.

Read [Pod topology spread constraints](#) to learn more about how these work.

# Operators

The following are all the logical operators that you can use in the operator field for nodeAffinity and podAffinity mentioned above.

| Operator     | Behavior                                                        |
|--------------|-----------------------------------------------------------------|
| In           | The label value is present in the supplied set of strings       |
| NotIn        | The label value is not contained in the supplied set of strings |
| Exists       | A label with this key exists on the object                      |
| DoesNotExist | No label with this key exists on the object                     |

The following operators can only be used with nodeAffinity.

| Operator | Behaviour                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gt       | The field value will be parsed as an integer, and that integer is less than or equal to the integer that results from parsing the value of a label named by this selector    |
| Lt       | The field value will be parsed as an integer, and that integer is greater than or equal to the integer that results from parsing the value of a label named by this selector |

## Note:

Gt and Lt operators will not work with non-integer values. If the given value doesn't parse as an integer, the pod will fail to get scheduled. Also, Gt and Lt are not available for podAffinity.

## What's next

- Read more about [taints and tolerations](#) .
- Read the design docs for [node affinity](#) and for [inter-pod affinity/anti-affinity](#).
- Learn about how the [topology manager](#) takes part in node-level resource allocation decisions.
- Learn how to use [nodeSelector](#).
- Learn how to use [affinity and anti-affinity](#).

# Pod Overhead

FEATURE STATE: Kubernetes v1.24 [stable]

When you run a Pod on a Node, the Pod itself takes an amount of system resources. These resources are additional to the resources needed to run the container(s) inside the Pod. In Kubernetes, *Pod Overhead* is a way to account for the resources consumed by the Pod infrastructure on top of the container requests & limits.

In Kubernetes, the Pod's overhead is set at [admission](#) time according to the overhead associated with the Pod's [RuntimeClass](#).

A pod's overhead is considered in addition to the sum of container resource requests when scheduling a Pod. Similarly, the kubelet will include the Pod overhead when sizing the Pod cgroup, and when carrying out Pod eviction ranking.

# Configuring Pod overhead

You need to make sure a RuntimeClass is utilized which defines the overhead field.

## Usage example

To work with Pod overhead, you need a RuntimeClass that defines the overhead field. As an example, you could use the following RuntimeClass definition with a virtualization container runtime (in this example, Kata Containers combined with the Firecracker virtual machine monitor) that uses around 120MiB per Pod for the virtual machine and the guest OS:

```
You need to change this example to match the actual runtime name, and per-Pod
resource overhead, that the container runtime is adding in your cluster.
apiVersion: node.k8s.io/v1
kind: RuntimeClass
metadata:
 name: kata-fc
handler: kata-fc
overhead:
 podFixed:
 memory: "120Mi"
 cpu: "250m"
```

Workloads which are created which specify the kata-fc RuntimeClass handler will take the memory and cpu overheads into account for resource quota calculations, node scheduling, as well as Pod cgroup sizing.

Consider running the given example workload, test-pod:

```
apiVersion: v1
kind: Pod
metadata:
 name: test-pod
spec:
 runtimeClassName: kata-fc
 containers:
 - name: busybox-ctr
 image: busybox:1.28
 stdin: true
 tty: true
 resources:
 limits:
 cpu: 500m
 memory: 100Mi
 - name: nginx-ctr
 image: nginx
 resources:
 limits:
 cpu: 1500m
 memory: 100Mi
```

At admission time the RuntimeClass [admission controller](#) updates the workload's PodSpec to include the overhead as described in the RuntimeClass. If the PodSpec already has this field defined, the Pod will be rejected. In the given example, since only the RuntimeClass name is specified, the admission controller mutates the Pod to include an overhead.

After the RuntimeClass admission controller has made modifications, you can check the updated Pod overhead value:

```
kubectl get pod test-pod -o jsonpath='{.spec.overhead}'
```

The output is:

```
map[cpu:250m memory:120Mi]
```

If a [ResourceQuota](#) is defined, the sum of container requests as well as the overhead field are counted.

When the kube-scheduler is deciding which node should run a new Pod, the scheduler considers that Pod's overhead as well as the sum of container requests for that Pod. For this example, the scheduler adds the requests and the overhead, then looks for a node that has 2.25 CPU and 320 MiB of memory available.

Once a Pod is scheduled to a node, the kubelet on that node creates a new [cgroup](#) for the Pod. It is within this pod that the underlying container runtime will create containers.

If the resource has a limit defined for each container (Guaranteed QoS or Burstable QoS with limits defined), the kubelet will set an upper limit for the pod cgroup associated with that resource (cpu.cfs\_quota\_us for CPU and memory.limit\_in\_bytes memory). This upper limit is based on the sum of the container limits plus the overhead defined in the PodSpec.

For CPU, if the Pod is Guaranteed or Burstable QoS, the kubelet will set cpu.shares based on the sum of container requests plus the overhead defined in the PodSpec.

Looking at our example, verify the container requests for the workload:

```
kubectl get pod test-pod -o jsonpath='{.spec.containers[*].resources.limits}'
```

The total container requests are 2000m CPU and 200MiB of memory:

```
map[cpu: 500m memory:100Mi] map[cpu:1500m memory:100Mi]
```

Check this against what is observed by the node:

```
kubectl describe node | grep test-pod -B2
```

The output shows requests for 2250m CPU, and for 320MiB of memory. The requests include Pod overhead:

| Namespace | Name     | CPU Requests | CPU Limits  | Memory Requests | Memory Limits | AGE |
|-----------|----------|--------------|-------------|-----------------|---------------|-----|
| default   | test-pod | 2250m (56%)  | 2250m (56%) | 320Mi (1%)      | 320Mi (1%)    | 36m |

## Verify Pod cgroup limits

Check the Pod's memory cgroups on the node where the workload is running. In the following example, [crictl](#) is used on the node, which provides a CLI for CRI-compatible container runtimes. This is an advanced example to show Pod overhead behavior, and it is not expected that users should need to check cgroups directly on the node.

First, on the particular node, determine the Pod identifier:

```
Run this on the node where the Pod is scheduled
POD_ID="$(sudo crictl pods --name test-pod -q)"
```

From this, you can determine the cgroup path for the Pod:

```
Run this on the node where the Pod is scheduled
sudo crictl inspectp -o=json $POD_ID | grep cgroupsPath
```

The resulting cgroup path includes the Pod's pause container. The Pod level cgroup is one directory above.

```
"cgroupsPath": "/kubepods/podd7f4b509-cf94-4951-9417-
d1087c92a5b2/7ccf55aee35dd16aca4189c952d83487297f3cd760f1bbf09620e206e7d0c27a"
```

In this specific case, the pod cgroup path is `kubepods/podd7f4b509-cf94-4951-9417-d1087c92a5b2`. Verify the Pod level cgroup setting for memory:

```
Run this on the node where the Pod is scheduled.
Also, change the name of the cgroup to match the cgroup allocated for your pod.
cat /sys/fs/cgroup/memory/kubepods/podd7f4b509-cf94-4951-9417-d1087c92a5b2/
memory.limit_in_bytes
```

This is 320 MiB, as expected:

```
335544320
```

## Observability

Some `kube_pod_overhead_*` metrics are available in [kube-state-metrics](#) to help identify when Pod overhead is being utilized and to help observe stability of workloads running with a defined overhead.

## What's next

- Learn more about [RuntimeClass](#)
- Read the [PodOverhead Design](#) enhancement proposal for extra context

## Pod Scheduling Readiness

FEATURE STATE: Kubernetes v1.30 [stable]

Pods were considered ready for scheduling once created. Kubernetes scheduler does its due diligence to find nodes to place all pending Pods. However, in a real-world case, some Pods may stay in a "miss-essential-resources" state for a long period. These Pods actually churn the scheduler (and downstream integrators like Cluster AutoScaler) in an unnecessary manner.

By specifying/removing a Pod's `.spec.schedulingGates`, you can control when a Pod is ready to be considered for scheduling.

## Configuring Pod schedulingGates

The `schedulingGates` field contains a list of strings, and each string literal is perceived as a criteria that Pod should be satisfied before considered schedulable. This field can be initialized only when a Pod is created (either by the client, or mutated during admission). After creation, each `schedulingGate` can be removed in arbitrary order, but addition of a new scheduling gate is disallowed.

[pod-scheduling-gates-diagram](#)

Figure. Pod SchedulingGates

## Usage example

To mark a Pod not-ready for scheduling, you can create it with one or more scheduling gates like this:

[pods/pod-with-scheduling-gates.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: test-pod
spec:
 schedulingGates:
 - name: example.com/foo
 - name: example.com/bar
 containers:
 - name: pause
 image: registry.k8s.io/pause:3.6
```

After the Pod's creation, you can check its state using:

```
kubectl get pod test-pod
```

The output reveals it's in `SchedulingGated` state:

| NAME     | READY | STATUS          | RESTARTS | AGE |
|----------|-------|-----------------|----------|-----|
| test-pod | 0/1   | SchedulingGated | 0        | 7s  |

You can also check its `schedulingGates` field by running:

```
kubectl get pod test-pod -o jsonpath='{.spec.schedulingGates}'
```

The output is:

```
[{"name":"example.com/foo"}, {"name":"example.com/bar"}]
```

To inform scheduler this Pod is ready for scheduling, you can remove its `schedulingGates` entirely by reapplying a modified manifest:

[pods/pod-without-scheduling-gates.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: test-pod
spec:
 containers:
 - name: pause
 image: registry.k8s.io/pause:3.6
```

You can check if the `schedulingGates` is cleared by running:

```
kubectl get pod test-pod -o jsonpath='{.spec.schedulingGates}'
```

The output is expected to be empty. And you can check its latest status by running:

```
kubectl get pod test-pod -o wide
```

Given the test-pod doesn't request any CPU/memory resources, it's expected that this Pod's state get transited from previous `SchedulingGated` to `Running`:

| NAME     | READY | STATUS  | RESTARTS | AGE | IP       | NODE   |
|----------|-------|---------|----------|-----|----------|--------|
| test-pod | 1/1   | Running | 0        | 15s | 10.0.0.4 | node-2 |

## Observability

The metric `scheduler_pending_pods` comes with a new label "gated" to distinguish whether a Pod has been tried scheduling but claimed as unschedulable, or explicitly marked as not ready for scheduling. You can use `scheduler_pending_pods{queue="gated"}` to check the metric result.

## Mutable Pod scheduling directives

You can mutate scheduling directives of Pods while they have scheduling gates, with certain constraints. At a high level, you can only tighten the scheduling directives of a Pod. In other words, the updated directives would cause the Pods to only be able to be scheduled on a subset of the nodes that it would previously match. More concretely, the rules for updating a Pod's scheduling directives are as follows:

1. For `.spec.nodeSelector`, only additions are allowed. If absent, it will be allowed to be set.
2. For `spec.affinity.nodeAffinity`, if nil, then setting anything is allowed.
3. If `NodeSelectorTerms` was empty, it will be allowed to be set. If not empty, then only additions of `NodeSelectorRequirements` to `matchExpressions` or `fieldExpressions` are allowed, and no changes to existing `matchExpressions` and `fieldExpressions` will be allowed. This is because the terms in `.requiredDuringSchedulingIgnoredDuringExecution.NodeSelectorTerms`, are ORed while



the expressions in `nodeSelectorTerms[].matchExpressions` and `nodeSelectorTerms[].fieldExpressions` are ANDed.

4. For `.preferredDuringSchedulingIgnoredDuringExecution`, all updates are allowed. This is because preferred terms are not authoritative, and so policy controllers don't validate those terms.

## What's next

- Read the [PodSchedulingReadiness KEP](#) for more details

# Pod Topology Spread Constraints

You can use *topology spread constraints* to control how [Pods](#) are spread across your cluster among failure-domains such as regions, zones, nodes, and other user-defined topology domains. This can help to achieve high availability as well as efficient resource utilization.

You can set [cluster-level constraints](#) as a default, or configure topology spread constraints for individual workloads.

## Motivation

Imagine that you have a cluster of up to twenty nodes, and you want to run a [workload](#) that automatically scales how many replicas it uses. There could be as few as two Pods or as many as fifteen. When there are only two Pods, you'd prefer not to have both of those Pods run on the same node: you would run the risk that a single node failure takes your workload offline.

In addition to this basic usage, there are some advanced usage examples that enable your workloads to benefit on high availability and cluster utilization.

As you scale up and run more Pods, a different concern becomes important. Imagine that you have three nodes running five Pods each. The nodes have enough capacity to run that many replicas; however, the clients that interact with this workload are split across three different datacenters (or infrastructure zones). Now you have less concern about a single node failure, but you notice that latency is higher than you'd like, and you are paying for network costs associated with sending network traffic between the different zones.

You decide that under normal operation you'd prefer to have a similar number of replicas [scheduled](#) into each infrastructure zone, and you'd like the cluster to self-heal in the case that there is a problem.

Pod topology spread constraints offer you a declarative way to configure that.

## topologySpreadConstraints field

The Pod API includes a field, `spec.topologySpreadConstraints`. The usage of this field looks like the following:

```

apiVersion: v1
```

```
kind: Pod
metadata:
 name: example-pod
spec:
 # Configure a topology spread constraint
 topologySpreadConstraints:
 - maxSkew: <integer>
 minDomains: <integer> # optional
 topologyKey: <string>
 whenUnsatisfiable: <string>
 labelSelector: <object>
 matchLabelKeys: <list> # optional; beta since v1.27
 nodeAffinityPolicy: [Honor|Ignore] # optional; beta since v1.26
 nodeTaintsPolicy: [Honor|Ignore] # optional; beta since v1.26
 ### other Pod fields go here
```

You can read more about this field by running `kubectl explain Pod.spec.topologySpreadConstraints` or refer to the [scheduling](#) section of the API reference for Pod.

## Spread constraint definition

You can define one or multiple `topologySpreadConstraints` entries to instruct the kube-scheduler how to place each incoming Pod in relation to the existing Pods across your cluster. Those fields are:

- **maxSkew** describes the degree to which Pods may be unevenly distributed. You must specify this field and the number must be greater than zero. Its semantics differ according to the value of `whenUnsatisfiable`:
  - if you select `whenUnsatisfiable: DoNotSchedule`, then `maxSkew` defines the maximum permitted difference between the number of matching pods in the target topology and the *global minimum* (the minimum number of matching pods in an eligible domain or zero if the number of eligible domains is less than `MinDomains`). For example, if you have 3 zones with 2, 2 and 1 matching pods respectively, `MaxSkew` is set to 1 then the global minimum is 1.
  - if you select `whenUnsatisfiable: ScheduleAnyway`, the scheduler gives higher precedence to topologies that would help reduce the skew.
- **minDomains** indicates a minimum number of eligible domains. This field is optional. A domain is a particular instance of a topology. An eligible domain is a domain whose nodes match the node selector.

### Note:

Before Kubernetes v1.30, the `minDomains` field was only available if the `MinDomainsInPodTopologySpread` [feature gate](#) was enabled (default since v1.28). In older Kubernetes clusters it might be explicitly disabled or the field might not be available.

- The value of `minDomains` must be greater than 0, when specified. You can only specify `minDomains` in conjunction with `whenUnsatisfiable: DoNotSchedule`.
- When the number of eligible domains with match topology keys is less than `minDomains`, Pod topology spread treats global minimum as 0, and then the calculation of skew is performed. The global minimum is the minimum number of

- matching Pods in an eligible domain, or zero if the number of eligible domains is less than minDomains.
  - When the number of eligible domains with matching topology keys equals or is greater than minDomains, this value has no effect on scheduling.
  - If you do not specify minDomains, the constraint behaves as if minDomains is 1.
- **topologyKey** is the key of [node labels](#). Nodes that have a label with this key and identical values are considered to be in the same topology. We call each instance of a topology (in other words, a <key, value> pair) a domain. The scheduler will try to put a balanced number of pods into each domain. Also, we define an eligible domain as a domain whose nodes meet the requirements of nodeAffinityPolicy and nodeTaintsPolicy.
- **whenUnsatisfiable** indicates how to deal with a Pod if it doesn't satisfy the spread constraint:
  - DoNotSchedule (default) tells the scheduler not to schedule it.
  - ScheduleAnyway tells the scheduler to still schedule it while prioritizing nodes that minimize the skew.
- **labelSelector** is used to find matching Pods. Pods that match this label selector are counted to determine the number of Pods in their corresponding topology domain. See [Label Selectors](#) for more details.
- **matchLabelKeys** is a list of pod label keys to select the pods over which spreading will be calculated. The keys are used to lookup values from the pod labels, those key-value labels are ANDed with labelSelector to select the group of existing pods over which spreading will be calculated for the incoming pod. The same key is forbidden to exist in both matchLabelKeys and labelSelector. matchLabelKeys cannot be set when labelSelector isn't set. Keys that don't exist in the pod labels will be ignored. A null or empty list means only match against the labelSelector.

With matchLabelKeys, you don't need to update the pod.spec between different revisions. The controller/operator just needs to set different values to the same label key for different revisions. The scheduler will assume the values automatically based on matchLabelKeys. For example, if you are configuring a Deployment, you can use the label keyed with [pod-template-hash](#), which is added automatically by the Deployment controller, to distinguish between different revisions in a single Deployment.

```
topologySpreadConstraints:
 - maxSkew: 1
 topologyKey: kubernetes.io/hostname
 whenUnsatisfiable: DoNotSchedule
 labelSelector:
 matchLabels:
 app: foo
 matchLabelKeys:
 - pod-template-hash
```

#### Note:

The matchLabelKeys field is a beta-level field and enabled by default in 1.27. You can disable it by disabling the MatchLabelKeysInPodTopologySpread [feature gate](#).

**nodeAffinityPolicy** indicates how we will treat Pod's nodeAffinity/nodeSelector when

- calculating pod topology spread skew. Options are:

- Honor: only nodes matching nodeAffinity/nodeSelector are included in the calculations.
- Ignore: nodeAffinity/nodeSelector are ignored. All nodes are included in the calculations.

If this value is null, the behavior is equivalent to the Honor policy.

**Note:**

The nodeAffinityPolicy is a beta-level field and enabled by default in 1.26. You can disable it by disabling the NodeInclusionPolicyInPodTopologySpread [feature gate](#).

- **nodeTaintsPolicy** indicates how we will treat node taints when calculating pod topology spread skew. Options are:

- Honor: nodes without taints, along with tainted nodes for which the incoming pod has a toleration, are included.
- Ignore: node taints are ignored. All nodes are included.

If this value is null, the behavior is equivalent to the Ignore policy.

**Note:**

The nodeTaintsPolicy is a beta-level field and enabled by default in 1.26. You can disable it by disabling the NodeInclusionPolicyInPodTopologySpread [feature gate](#).

When a Pod defines more than one topologySpreadConstraint, those constraints are combined using a logical AND operation: the kube-scheduler looks for a node for the incoming Pod that satisfies all the configured constraints.

## Node labels

Topology spread constraints rely on node labels to identify the topology domain(s) that each [node](#) is in. For example, a node might have labels:

```
region: us-east-1
zone: us-east-1a
```

**Note:**

For brevity, this example doesn't use the [well-known](#) label keys topology.kubernetes.io/zone and topology.kubernetes.io/region. However, those registered label keys are nonetheless recommended rather than the private (unqualified) label keys region and zone that are used here.

You can't make a reliable assumption about the meaning of a private label key between different contexts.

Suppose you have a 4-node cluster with the following labels:

| NAME  | STATUS | ROLES  | AGE   | VERSION | LABELS                |
|-------|--------|--------|-------|---------|-----------------------|
| node1 | Ready  | <none> | 4m26s | v1.16.0 | node=node1,zone=zoneA |
| node2 | Ready  | <none> | 3m58s | v1.16.0 | node=node2,zone=zoneA |
| node3 | Ready  | <none> | 3m17s | v1.16.0 | node=node3,zone=zoneB |
| node4 | Ready  | <none> | 2m43s | v1.16.0 | node=node4,zone=zoneB |

Then the cluster is logically viewed as below:

```
graph TB
 subgraph "zoneB"
 n3(Node3)
 n4(Node4)
 end
 subgraph "zoneA"
 n1(Node1)
 n2(Node2)
 end
 classDef plain fill:#ddd,stroke:#fff,stroke-width:4px,color:#000;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff;
 classDef cluster fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5;
 class n1,n2,n3,n4 k8s;
 class zoneA,zoneB cluster;
```

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## Consistency

You should set the same Pod topology spread constraints on all pods in a group.

Usually, if you are using a workload controller such as a Deployment, the pod template takes care of this for you. If you mix different spread constraints then Kubernetes follows the API definition of the field; however, the behavior is more likely to become confusing and troubleshooting is less straightforward.

You need a mechanism to ensure that all the nodes in a topology domain (such as a cloud provider region) are labeled consistently. To avoid you needing to manually label nodes, most clusters automatically populate well-known labels such as `kubernetes.io/hostname`. Check whether your cluster supports this.

## Topology spread constraint examples

### Example: one topology spread constraint

Suppose you have a 4-node cluster where 3 Pods labeled `foo: bar` are located in node1, node2 and node3 respectively:

```
graph BT
 subgraph "zoneB"
 p3(Pod) --> n3(Node3)
 n4(Node4)
 end
 subgraph "zoneA"
 p1(Pod) --> n1(Node1)
 p2(Pod) --> n2(Node2)
 end
 classDef plain fill:#ddd,stroke:#fff,stroke-width:4px,color:#000;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff;
 classDef cluster fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5;
 class n1,n2,n3,n4,p1,p2,p3 k8s;
 class zoneA,zoneB cluster;
```

JavaScript must be [enabled](#) to view this content

If you want an incoming Pod to be evenly spread with existing Pods across zones, you can use a manifest similar to:

[pods/topology-spread-constraints/one-constraint.yaml](#)

```

kind: Pod
apiVersion: v1
metadata:
 name: mypod
 labels:
 foo: bar
spec:
 topologySpreadConstraints:
 - maxSkew: 1
 topologyKey: zone
 whenUnsatisfiable: DoNotSchedule
 labelSelector:
 matchLabels:
 foo: bar
 containers:
 - name: pause
 image: registry.k8s.io/pause:3.1

```

From that manifest, `topologyKey: zone` implies the even distribution will only be applied to nodes that are labeled `zone: <any value>` (nodes that don't have a zone label are skipped). The field `whenUnsatisfiable: DoNotSchedule` tells the scheduler to let the incoming Pod stay pending if the scheduler can't find a way to satisfy the constraint.

If the scheduler placed this incoming Pod into zone A, the distribution of Pods would become [3, 1]. That means the actual skew is then 2 (calculated as 3 - 1), which violates `maxSkew: 1`. To satisfy the constraints and context for this example, the incoming Pod can only be placed onto a node in zone B:

```

graph BT
 subgraph "zoneB"
 p3(Pod) --> n3(Node3)
 p4(mypod) --> n4(Node4)
 end
 subgraph "zoneA"
 p1(Pod) --> n1(Node1)
 p2(Pod) --> n2(Node2)
 end
 classDef plain fill:#ddd,stroke:#fff,stroke-width:4px,color:#000;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff;
 classDef cluster fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5;
 class n1,n2,n3,n4,p1,p2,p3 k8s;
 class p4 plain;
 class zoneA,zoneB cluster;

```

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OR

```

graph BT
 subgraph "zoneB"
 p3(Pod) --> n3(Node3)
 p4(mypod) --> n4(Node4)
 end
 subgraph "zoneA"
 p1(Pod) --> n1(Node1)
 p2(Pod) --> n2(Node2)
 end
 classDef plain fill:#ddd,stroke:#fff,stroke-width:4px,color:#000;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff;
 classDef cluster fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5;
 class n1,n2,n3,n4,p1,p2,p3 k8s;
 class p4 plain;
 class zoneA,zoneB cluster;

```

*JavaScript must be [enabled](#) to view this content*

You can tweak the Pod spec to meet various kinds of requirements:

- Change `maxSkew` to a bigger value - such as 2 - so that the incoming Pod can be placed into zone A as well.

- Change topologyKey to node so as to distribute the Pods evenly across nodes instead of zones. In the above example, if maxSkew remains 1, the incoming Pod can only be placed onto the node node4.
- Change whenUnsatisfiable: DoNotSchedule to whenUnsatisfiable: ScheduleAnyway to ensure the incoming Pod to be always schedulable (suppose other scheduling APIs are satisfied). However, it's preferred to be placed into the topology domain which has fewer matching Pods. (Be aware that this preference is jointly normalized with other internal scheduling priorities such as resource usage ratio).

## Example: multiple topology spread constraints

This builds upon the previous example. Suppose you have a 4-node cluster where 3 existing Pods labeled foo: bar are located on node1, node2 and node3 respectively:

```
graph BT
 subgraph "zoneB"
 p3(Pod) --> n3(Node3)
 n4(Node4)
 end
 subgraph "zoneA"
 p1(Pod) --> n1(Node1)
 p2(Pod) --> n2(Node2)
 end
 classDef plain fill:#ddd,stroke:#fff,stroke-width:4px,color:#000;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff;
 classDef cluster fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5;
 class n1,n2,n3,n4,p1,p2,p3 k8s;
 class p4 plain;
 class zoneA,zoneB cluster;
```

JavaScript must be [enabled](#) to view this content

You can combine two topology spread constraints to control the spread of Pods both by node and by zone:

[pods/topology-spread-constraints/two-constraints.yaml](#)

```
kind: Pod
apiVersion: v1
metadata:
 name: mypod
 labels:
 foo: bar
spec:
 topologySpreadConstraints:
 - maxSkew: 1
 topologyKey: zone
 whenUnsatisfiable: DoNotSchedule
 labelSelector:
 matchLabels:
 foo: bar
 - maxSkew: 1
 topologyKey: node
 whenUnsatisfiable: DoNotSchedule
 labelSelector:
 matchLabels:
 foo: bar
 containers:
 - name: pause
 image: registry.k8s.io/pause:3.1
```



In this case, to match the first constraint, the incoming Pod can only be placed onto nodes in zone B; while in terms of the second constraint, the incoming Pod can only be scheduled to the node node4. The scheduler only considers options that satisfy all defined constraints, so the only valid placement is onto node node4.

## Example: conflicting topology spread constraints

Multiple constraints can lead to conflicts. Suppose you have a 3-node cluster across 2 zones:

```
graph BT
 subgraph "zoneB"
 p4(Pod) --> n3(Node3)
 p5(Pod) --> n3
 end
 subgraph "zoneA"
 p1(Pod) --> n1(Node1)
 p2(Pod) --> n1
 p3(Pod) --> n2(Node2)
 end
 classDef plain fill:#ddd,stroke:#fff,stroke-width:4px,color:#000;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff;
 classDef cluster fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5;
 class n1,n2,n3,n4,p1,p2,p3,p4,p5 k8s;
 class zoneA,zoneB cluster;
```

JavaScript must be [enabled](#) to view this content

If you were to apply [two-constraints.yaml](#) (the manifest from the previous example) to **this** cluster, you would see that the Pod mypod stays in the Pending state. This happens because: to satisfy the first constraint, the Pod mypod can only be placed into zone B; while in terms of the second constraint, the Pod mypod can only schedule to node node2. The intersection of the two constraints returns an empty set, and the scheduler cannot place the Pod.

To overcome this situation, you can either increase the value of maxSkew or modify one of the constraints to use whenUnsatisfiable: ScheduleAnyway. Depending on circumstances, you might also decide to delete an existing Pod manually - for example, if you are troubleshooting why a bug-fix rollout is not making progress.

## Interaction with node affinity and node selectors

The scheduler will skip the non-matching nodes from the skew calculations if the incoming Pod has spec.nodeSelector or spec.affinity.nodeAffinity defined.

## Example: topology spread constraints with node affinity

Suppose you have a 5-node cluster ranging across zones A to C:

```
graph BT
 subgraph "zoneB"
 p3(Pod) --> n3(Node3)
 n4(Node4)
 end
 subgraph "zoneA"
 p1(Pod) --> n1(Node1)
 p2(Pod) --> n2(Node2)
 end
 classDef plain fill:#ddd,stroke:#fff,stroke-width:4px,color:#000;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff;
 classDef cluster fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5;
 class n1,n2,n3,n4,p1,p2,p3 k8s;
 class p4 plain;
 class zoneA,zoneB cluster;
```

JavaScript must be [enabled](#) to view this content

```
graph BT
 subgraph "zoneC"
 n5(Node5)
 end
 classDef plain fill:#ddd,stroke:#fff,stroke-width:4px,color:#000;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff;
 classDef cluster fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5;
 class n5 k8s;
 class zoneC cluster;
```

JavaScript must be [enabled](#) to view this content



and you know that zone C must be excluded. In this case, you can compose a manifest as below, so that Pod mypod will be placed into zone B instead of zone C. Similarly, Kubernetes also respects spec.nodeSelector.

[pods/topology-spread-constraints/one-constraint-with-nodeaffinity.yaml](#)

```
kind: Pod
apiVersion: v1
metadata:
 name: mypod
 labels:
 foo: bar
spec:
 topologySpreadConstraints:
 - maxSkew: 1
 topologyKey: zone
 whenUnsatisfiable: DoNotSchedule
 labelSelector:
 matchLabels:
 foo: bar
 affinity:
 nodeAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 nodeSelectorTerms:
 - matchExpressions:
 - key: zone
 operator: NotIn
 values:
 - zoneC
 containers:
 - name: pause
 image: registry.k8s.io/pause:3.1
```

## Implicit conventions

There are some implicit conventions worth noting here:

- Only the Pods holding the same namespace as the incoming Pod can be matching candidates.
- The scheduler bypasses any nodes that don't have any topologySpreadConstraints[\*].topologyKey present. This implies that:
  1. any Pods located on those bypassed nodes do not impact maxSkew calculation - in the above [example](#), suppose the node node1 does not have a label "zone", then the 2 Pods will be disregarded, hence the incoming Pod will be scheduled into zone A.
  2. the incoming Pod has no chances to be scheduled onto this kind of nodes - in the above example, suppose a node node5 has the **mistyped** label zone-typo: zoneC (and no zone label set). After node node5 joins the cluster, it will be bypassed and Pods for this workload aren't scheduled there.
- Be aware of what will happen if the incoming Pod's topologySpreadConstraints[\*].labelSelector doesn't match its own labels. In the above

example, if you remove the incoming Pod's labels, it can still be placed onto nodes in zone B, since the constraints are still satisfied. However, after that placement, the degree of imbalance of the cluster remains unchanged - it's still zone A having 2 Pods labeled as foo: bar, and zone B having 1 Pod labeled as foo: bar. If this is not what you expect, update the workload's topologySpreadConstraints[\*].labelSelector to match the labels in the pod template.

## Cluster-level default constraints

It is possible to set default topology spread constraints for a cluster. Default topology spread constraints are applied to a Pod if, and only if:

- It doesn't define any constraints in its .spec.topologySpreadConstraints.
- It belongs to a Service, ReplicaSet, StatefulSet or ReplicationController.

Default constraints can be set as part of the PodTopologySpread plugin arguments in a [scheduling profile](#). The constraints are specified with the same [API above](#), except that labelSelector must be empty. The selectors are calculated from the Services, ReplicaSets, StatefulSets or ReplicationControllers that the Pod belongs to.

An example configuration might look like follows:

```
apiVersion: kubescheduler.config.k8s.io/v1beta3
kind: KubeSchedulerConfiguration

profiles:
- schedulerName: default-scheduler
 pluginConfig:
 - name: PodTopologySpread
 args:
 defaultConstraints:
 - maxSkew: 1
 topologyKey: topology.kubernetes.io/zone
 whenUnsatisfiable: ScheduleAnyway
 defaultingType: List
```

## Built-in default constraints

FEATURE STATE: Kubernetes v1.24 [stable]

If you don't configure any cluster-level default constraints for pod topology spreading, then kube-scheduler acts as if you specified the following default topology constraints:

```
defaultConstraints:
- maxSkew: 3
 topologyKey: "kubernetes.io/hostname"
 whenUnsatisfiable: ScheduleAnyway
- maxSkew: 5
 topologyKey: "topology.kubernetes.io/zone"
 whenUnsatisfiable: ScheduleAnyway
```

Also, the legacy SelectorSpread plugin, which provides an equivalent behavior, is disabled by default.

## Note:

The PodTopologySpread plugin does not score the nodes that don't have the topology keys specified in the spreading constraints. This might result in a different default behavior compared to the legacy SelectorSpread plugin when using the default topology constraints.

If your nodes are not expected to have **both** `kubernetes.io/hostname` and `topology.kubernetes.io/zone` labels set, define your own constraints instead of using the Kubernetes defaults.

If you don't want to use the default Pod spreading constraints for your cluster, you can disable those defaults by setting `defaultingType` to `List` and leaving empty `defaultConstraints` in the PodTopologySpread plugin configuration:

```
apiVersion: kubescheduler.config.k8s.io/v1beta3
kind: KubeSchedulerConfiguration
```

```
profiles:
```

```
- schedulerName: default-scheduler
```

```
 pluginConfig:
```

```
 - name: PodTopologySpread
```

```
 args:
```

```
 defaultConstraints: []
```

```
 defaultingType: List
```

## Comparison with podAffinity and podAntiAffinity

In Kubernetes, [inter-Pod affinity and anti-affinity](#) control how Pods are scheduled in relation to one another - either more packed or more scattered.

**podAffinity**

attracts Pods; you can try to pack any number of Pods into qualifying topology domain(s).

**podAntiAffinity**

repels Pods. If you set this to `requiredDuringSchedulingIgnoredDuringExecution` mode then only a single Pod can be scheduled into a single topology domain; if you choose `preferredDuringSchedulingIgnoredDuringExecution` then you lose the ability to enforce the constraint.

For finer control, you can specify topology spread constraints to distribute Pods across different topology domains - to achieve either high availability or cost-saving. This can also help on rolling update workloads and scaling out replicas smoothly.

For more context, see the [Motivation](#) section of the enhancement proposal about Pod topology spread constraints.

## Known limitations

- There's no guarantee that the constraints remain satisfied when Pods are removed. For example, scaling down a Deployment may result in imbalanced Pods distribution.

You can use a tool such as the [Descheduler](#) to rebalance the Pods distribution.

Pods matched on tainted nodes are respected. See [Issue 80921](#).

- 
- The scheduler doesn't have prior knowledge of all the zones or other topology domains that a cluster has. They are determined from the existing nodes in the cluster. This could lead to a problem in autoscaled clusters, when a node pool (or node group) is scaled to zero nodes, and you're expecting the cluster to scale up, because, in this case, those topology domains won't be considered until there is at least one node in them.

You can work around this by using a cluster autoscaling tool that is aware of Pod topology spread constraints and is also aware of the overall set of topology domains.

## What's next

- The blog article [Introducing PodTopologySpread](#) explains maxSkew in some detail, as well as covering some advanced usage examples.
- Read the [scheduling](#) section of the API reference for Pod.

## Taints and Tolerations

[Node affinity](#) is a property of [Pods](#) that *attracts* them to a set of [nodes](#) (either as a preference or a hard requirement). *Taints* are the opposite -- they allow a node to repel a set of pods.

*Tolerations* are applied to pods. Tolerations allow the scheduler to schedule pods with matching taints. Tolerations allow scheduling but don't guarantee scheduling: the scheduler also [evaluates other parameters](#) as part of its function.

Taints and tolerations work together to ensure that pods are not scheduled onto inappropriate nodes. One or more taints are applied to a node; this marks that the node should not accept any pods that do not tolerate the taints.

## Concepts

You add a taint to a node using [kubectl taint](#). For example,

```
kubectl taint nodes node1 key1=value1:NoSchedule
```

places a taint on node node1. The taint has key key1, value value1, and taint effect NoSchedule. This means that no pod will be able to schedule onto node1 unless it has a matching toleration.

To remove the taint added by the command above, you can run:

```
kubectl taint nodes node1 key1=value1:NoSchedule-
```

You specify a toleration for a pod in the PodSpec. Both of the following tolerations "match" the taint created by the `kubectl taint` line above, and thus a pod with either toleration would be able to schedule onto node1:

```
tolerations:
- key: "key1"
 operator: "Equal"
```

```
value: "value1"
effect: "NoSchedule"
```

```
tolerations:
- key: "key1"
 operator: "Exists"
 effect: "NoSchedule"
```

The default Kubernetes scheduler takes taints and tolerations into account when selecting a node to run a particular Pod. However, if you manually specify the `.spec.nodeName` for a Pod, that action bypasses the scheduler; the Pod is then bound onto the node where you assigned it, even if there are NoSchedule taints on that node that you selected. If this happens and the node also has a NoExecute taint set, the kubelet will eject the Pod unless there is an appropriate tolerance set.

Here's an example of a pod that has some tolerations defined:

[pods/pod-with-toleration.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: nginx
 labels:
 env: test
spec:
 containers:
 - name: nginx
 image: nginx
 imagePullPolicy: IfNotPresent
 tolerations:
 - key: "example-key"
 operator: "Exists"
 effect: "NoSchedule"
```

The default value for operator is Equal.

A toleration "matches" a taint if the keys are the same and the effects are the same, and:

- the operator is Exists (in which case no value should be specified), or
- the operator is Equal and the values should be equal.

### **Note:**

There are two special cases:

An empty key with operator Exists matches all keys, values and effects which means this will tolerate everything.

An empty effect matches all effects with key key1.

The above example used the effect of NoSchedule. Alternatively, you can use the effect of PreferNoSchedule.

The allowed values for the effect field are:

#### NoExecute

This affects pods that are already running on the node as follows:

- Pods that do not tolerate the taint are evicted immediately
- Pods that tolerate the taint without specifying tolerationSeconds in their toleration specification remain bound forever
- Pods that tolerate the taint with a specified tolerationSeconds remain bound for the specified amount of time. After that time elapses, the node lifecycle controller evicts the Pods from the node.

#### NoSchedule

No new Pods will be scheduled on the tainted node unless they have a matching toleration. Pods currently running on the node are **not** evicted.

#### PreferNoSchedule

PreferNoSchedule is a "preference" or "soft" version of NoSchedule. The control plane will *try* to avoid placing a Pod that does not tolerate the taint on the node, but it is not guaranteed.

You can put multiple taints on the same node and multiple tolerations on the same pod. The way Kubernetes processes multiple taints and tolerations is like a filter: start with all of a node's taints, then ignore the ones for which the pod has a matching toleration; the remaining un-ignored taints have the indicated effects on the pod. In particular,

- if there is at least one un-ignored taint with effect NoSchedule then Kubernetes will not schedule the pod onto that node
- if there is no un-ignored taint with effect NoSchedule but there is at least one un-ignored taint with effect PreferNoSchedule then Kubernetes will *try* to not schedule the pod onto the node
- if there is at least one un-ignored taint with effect NoExecute then the pod will be evicted from the node (if it is already running on the node), and will not be scheduled onto the node (if it is not yet running on the node).

For example, imagine you taint a node like this

```
kubecttl taint nodes node1 key1=value1:NoSchedule
kubecttl taint nodes node1 key1=value1:NoExecute
kubecttl taint nodes node1 key2=value2:NoSchedule
```

And a pod has two tolerations:

tolerations:

```
- key: "key1"
 operator: "Equal"
 value: "value1"
 effect: "NoSchedule"
- key: "key1"
 operator: "Equal"
 value: "value1"
 effect: "NoExecute"
```

In this case, the pod will not be able to schedule onto the node, because there is no toleration matching the third taint. But it will be able to continue running if it is already running on the node when the taint is added, because the third taint is the only one of the three that is not tolerated by the pod.

Normally, if a taint with effect `NoExecute` is added to a node, then any pods that do not tolerate the taint will be evicted immediately, and pods that do tolerate the taint will never be evicted. However, a toleration with `NoExecute` effect can specify an optional `tolerationSeconds` field that dictates how long the pod will stay bound to the node after the taint is added. For example,

```
tolerations:
- key: "key1"
 operator: "Equal"
 value: "value1"
 effect: "NoExecute"
 tolerationSeconds: 3600
```

means that if this pod is running and a matching taint is added to the node, then the pod will stay bound to the node for 3600 seconds, and then be evicted. If the taint is removed before that time, the pod will not be evicted.

## Example Use Cases

Taints and tolerations are a flexible way to steer pods *away* from nodes or evict pods that shouldn't be running. A few of the use cases are

- **Dedicated Nodes:** If you want to dedicate a set of nodes for exclusive use by a particular set of users, you can add a taint to those nodes (say, `kubectl taint nodes nodename dedicated=groupName:NoSchedule`) and then add a corresponding toleration to their pods (this would be done most easily by writing a custom [admission controller](#)). The pods with the tolerations will then be allowed to use the tainted (dedicated) nodes as well as any other nodes in the cluster. If you want to dedicate the nodes to them *and* ensure they *only* use the dedicated nodes, then you should additionally add a label similar to the taint to the same set of nodes (e.g. `dedicated=groupName`), and the admission controller should additionally add a node affinity to require that the pods can only schedule onto nodes labeled with `dedicated=groupName`.
- **Nodes with Special Hardware:** In a cluster where a small subset of nodes have specialized hardware (for example GPUs), it is desirable to keep pods that don't need the specialized hardware off of those nodes, thus leaving room for later-arriving pods that do need the specialized hardware. This can be done by tainting the nodes that have the specialized hardware (e.g. `kubectl taint nodes nodename special=true:NoSchedule` or `kubectl taint nodes nodename special=true:PreferNoSchedule`) and adding a corresponding toleration to pods that use the special hardware. As in the dedicated nodes use case, it is probably easiest to apply the tolerations using a custom [admission controller](#). For example, it is recommended to use [Extended Resources](#) to represent the special hardware, taint your special hardware nodes with the extended resource name and run the [ExtendedResourceToleration](#) admission controller. Now, because the nodes are tainted, no pods without the toleration will schedule on them. But when you submit a pod that requests the extended resource, the `ExtendedResourceToleration` admission controller will automatically add the correct toleration to the pod and that pod will schedule on the special hardware nodes. This will make sure that these special hardware nodes are dedicated for pods requesting such hardware and you don't have to manually add tolerations to your pods.
- **Taint based Evictions:** A per-pod-configurable eviction behavior when there are node problems, which is described in the next section.

# Taint based Evictions

FEATURE STATE: Kubernetes v1.18 [stable]

The node controller automatically taints a Node when certain conditions are true. The following taints are built in:

- `node.kubernetes.io/not-ready`: Node is not ready. This corresponds to the `NodeCondition Ready` being "False".
- `node.kubernetes.io/unreachable`: Node is unreachable from the node controller. This corresponds to the `NodeCondition Ready` being "Unknown".
- `node.kubernetes.io/memory-pressure`: Node has memory pressure.
- `node.kubernetes.io/disk-pressure`: Node has disk pressure.
- `node.kubernetes.io/pid-pressure`: Node has PID pressure.
- `node.kubernetes.io/network-unavailable`: Node's network is unavailable.
- `node.kubernetes.io/unschedulable`: Node is unschedulable.
- `node.cloudprovider.kubernetes.io/uninitialized`: When the kubelet is started with an "external" cloud provider, this taint is set on a node to mark it as unusable. After a controller from the cloud-controller-manager initializes this node, the kubelet removes this taint.

In case a node is to be drained, the node controller or the kubelet adds relevant taints with `NoExecute` effect. This effect is added by default for the `node.kubernetes.io/not-ready` and `node.kubernetes.io/unreachable` taints. If the fault condition returns to normal, the kubelet or node controller can remove the relevant taint(s).

In some cases when the node is unreachable, the API server is unable to communicate with the kubelet on the node. The decision to delete the pods cannot be communicated to the kubelet until communication with the API server is re-established. In the meantime, the pods that are scheduled for deletion may continue to run on the partitioned node.

## Note:

The control plane limits the rate of adding new taints to nodes. This rate limiting manages the number of evictions that are triggered when many nodes become unreachable at once (for example: if there is a network disruption).

You can specify `tolerationSeconds` for a Pod to define how long that Pod stays bound to a failing or unresponsive Node.

For example, you might want to keep an application with a lot of local state bound to node for a long time in the event of network partition, hoping that the partition will recover and thus the pod eviction can be avoided. The toleration you set for that Pod might look like:

```
tolerations:
- key: "node.kubernetes.io/unreachable"
 operator: "Exists"
 effect: "NoExecute"
 tolerationSeconds: 6000
```



## Note:

Kubernetes automatically adds a toleration for `node.kubernetes.io/not-ready` and `node.kubernetes.io/unreachable` with `tolerationSeconds=300`, unless you, or a controller, set those tolerations explicitly.

These automatically-added tolerations mean that Pods remain bound to Nodes for 5 minutes after one of these problems is detected.

[DaemonSet](#) pods are created with `NoExecute` tolerations for the following taints with no `tolerationSeconds`:

- `node.kubernetes.io/unreachable`
- `node.kubernetes.io/not-ready`

This ensures that DaemonSet pods are never evicted due to these problems.

## Taint Nodes by Condition

The control plane, using the node [controller](#), automatically creates taints with a `NoSchedule` effect for [node conditions](#).

The scheduler checks taints, not node conditions, when it makes scheduling decisions. This ensures that node conditions don't directly affect scheduling. For example, if the `DiskPressure` node condition is active, the control plane adds the `node.kubernetes.io/disk-pressure` taint and does not schedule new pods onto the affected node. If the `MemoryPressure` node condition is active, the control plane adds the `node.kubernetes.io/memory-pressure` taint.

You can ignore node conditions for newly created pods by adding the corresponding Pod tolerations. The control plane also adds the `node.kubernetes.io/memory-pressure` toleration on pods that have a [QoS class](#) other than `BestEffort`. This is because Kubernetes treats pods in the `Guaranteed` or `Burstable` QoS classes (even pods with no memory request set) as if they are able to cope with memory pressure, while new `BestEffort` pods are not scheduled onto the affected node.

The DaemonSet controller automatically adds the following `NoSchedule` tolerations to all daemons, to prevent DaemonSets from breaking.

- `node.kubernetes.io/memory-pressure`
- `node.kubernetes.io/disk-pressure`
- `node.kubernetes.io/pid-pressure` (1.14 or later)
- `node.kubernetes.io/unschedulable` (1.10 or later)
- `node.kubernetes.io/network-unavailable` (*host network only*)

Adding these tolerations ensures backward compatibility. You can also add arbitrary tolerations to DaemonSets.

## What's next

- Read about [Node-pressure Eviction](#) and how you can configure it
- Read about [Pod Priority](#)

# Scheduling Framework

FEATURE STATE: Kubernetes v1.19 [stable]

The *scheduling framework* is a pluggable architecture for the Kubernetes scheduler. It consists of a set of "plugin" APIs that are compiled directly into the scheduler. These APIs allow most scheduling features to be implemented as plugins, while keeping the scheduling "core" lightweight and maintainable. Refer to the [design proposal of the scheduling framework](#) for more technical information on the design of the framework.

## Framework workflow

The Scheduling Framework defines a few extension points. Scheduler plugins register to be invoked at one or more extension points. Some of these plugins can change the scheduling decisions and some are informational only.

Each attempt to schedule one Pod is split into two phases, the **scheduling cycle** and the **binding cycle**.

## Scheduling cycle & binding cycle

The scheduling cycle selects a node for the Pod, and the binding cycle applies that decision to the cluster. Together, a scheduling cycle and binding cycle are referred to as a "scheduling context".

Scheduling cycles are run serially, while binding cycles may run concurrently.

A scheduling or binding cycle can be aborted if the Pod is determined to be unschedulable or if there is an internal error. The Pod will be returned to the queue and retried.

## Interfaces

The following picture shows the scheduling context of a Pod and the interfaces that the scheduling framework exposes.

One plugin may implement multiple interfaces to perform more complex or stateful tasks.

Some interfaces match the scheduler extension points which can be configured through [Scheduler Configuration](#).

### Scheduling framework extension points

#### PreEnqueue

These plugins are called prior to adding Pods to the internal active queue, where Pods are marked as ready for scheduling.

Only when all PreEnqueue plugins return Success, the Pod is allowed to enter the active queue. Otherwise, it's placed in the internal unschedulable Pods list, and doesn't get an Unschedulable condition.

For more details about how internal scheduler queues work, read [Scheduling queue in kube-scheduler](#).

## EnqueueExtension

EnqueueExtension is the interface where the plugin can control whether to retry scheduling of Pods rejected by this plugin, based on changes in the cluster. Plugins that implement PreEnqueue, PreFilter, Filter, Reserve or Permit should implement this interface.

## QueueingHint

FEATURE STATE: Kubernetes v1.28 [beta]

QueueingHint is a callback function for deciding whether a Pod can be requeued to the active queue or backoff queue. It's executed every time a certain kind of event or change happens in the cluster. When the QueueingHint finds that the event might make the Pod schedulable, the Pod is put into the active queue or the backoff queue so that the scheduler will retry the scheduling of the Pod.

### Note:

QueueingHint evaluation during scheduling is a beta-level feature. The v1.28 release series initially enabled the associated feature gate; however, after the discovery of an excessive memory footprint, the Kubernetes project set that feature gate to be disabled by default. In Kubernetes 1.30, this feature gate is disabled and you need to enable it manually. You can enable it via the SchedulerQueueingHints [feature gate](#).

## QueueSort

These plugins are used to sort Pods in the scheduling queue. A queue sort plugin essentially provides a Less(Pod1, Pod2) function. Only one queue sort plugin may be enabled at a time.

## PreFilter

These plugins are used to pre-process info about the Pod, or to check certain conditions that the cluster or the Pod must meet. If a PreFilter plugin returns an error, the scheduling cycle is aborted.

## Filter

These plugins are used to filter out nodes that cannot run the Pod. For each node, the scheduler will call filter plugins in their configured order. If any filter plugin marks the node as infeasible, the remaining plugins will not be called for that node. Nodes may be evaluated concurrently.

## PostFilter

These plugins are called after the Filter phase, but only when no feasible nodes were found for the pod. Plugins are called in their configured order. If any postFilter plugin marks the node as Schedulable, the remaining plugins will not be called. A typical PostFilter implementation is preemption, which tries to make the pod schedulable by preempting other Pods.

## PreScore

These plugins are used to perform "pre-scoring" work, which generates a sharable state for Score plugins to use. If a PreScore plugin returns an error, the scheduling cycle is aborted.

## Score

These plugins are used to rank nodes that have passed the filtering phase. The scheduler will call each scoring plugin for each node. There will be a well defined range of integers representing the minimum and maximum scores. After the [NormalizeScore](#) phase, the scheduler will combine node scores from all plugins according to the configured plugin weights.

## NormalizeScore

These plugins are used to modify scores before the scheduler computes a final ranking of Nodes. A plugin that registers for this extension point will be called with the [Score](#) results from the same plugin. This is called once per plugin per scheduling cycle.

For example, suppose a plugin `BlinkingLightScorer` ranks Nodes based on how many blinking lights they have.

```
func ScoreNode(_ *v1.Pod, n *v1.Node) (int, error) {
 return getBlinkingLightCount(n)
}
```

However, the maximum count of blinking lights may be small compared to `NodeScoreMax`. To fix this, `BlinkingLightScorer` should also register for this extension point.

```
func NormalizeScores(scores map[string]int) {
 highest := 0
 for _, score := range scores {
 highest = max(highest, score)
 }
 for node, score := range scores {
 scores[node] = score*NodeScoreMax/highest
 }
}
```

If any `NormalizeScore` plugin returns an error, the scheduling cycle is aborted.

### Note:

Plugins wishing to perform "pre-reserve" work should use the `NormalizeScore` extension point.

## Reserve

A plugin that implements the `Reserve` interface has two methods, namely `Reserve` and `Unreserve`, that back two informational scheduling phases called `Reserve` and `Unreserve`, respectively. Plugins which maintain runtime state (aka "stateful plugins") should use these phases to be notified by the scheduler when resources on a node are being reserved and unreserved for a given Pod.

The Reserve phase happens before the scheduler actually binds a Pod to its designated node. It exists to prevent race conditions while the scheduler waits for the bind to succeed. The Reserve method of each Reserve plugin may succeed or fail; if one Reserve method call fails, subsequent plugins are not executed and the Reserve phase is considered to have failed. If the Reserve method of all plugins succeed, the Reserve phase is considered to be successful and the rest of the scheduling cycle and the binding cycle are executed.

The Unreserve phase is triggered if the Reserve phase or a later phase fails. When this happens, the Unreserve method of **all** Reserve plugins will be executed in the reverse order of Reserve method calls. This phase exists to clean up the state associated with the reserved Pod.

### Caution:

The implementation of the Unreserve method in Reserve plugins must be idempotent and may not fail.

## Permit

*Permit* plugins are invoked at the end of the scheduling cycle for each Pod, to prevent or delay the binding to the candidate node. A permit plugin can do one of the three things:

1. **approve**

Once all Permit plugins approve a Pod, it is sent for binding.

2. **deny**

If any Permit plugin denies a Pod, it is returned to the scheduling queue. This will trigger the Unreserve phase in [Reserve plugins](#).

3. **wait** (with a timeout)

If a Permit plugin returns "wait", then the Pod is kept in an internal "waiting" Pods list, and the binding cycle of this Pod starts but directly blocks until it gets approved. If a timeout occurs, **wait** becomes **deny** and the Pod is returned to the scheduling queue, triggering the Unreserve phase in [Reserve plugins](#).

### Note:

While any plugin can access the list of "waiting" Pods and approve them (see [FrameworkHandle](#)), we expect only the permit plugins to approve binding of reserved Pods that are in "waiting" state. Once a Pod is approved, it is sent to the [PreBind](#) phase.

## PreBind

These plugins are used to perform any work required before a Pod is bound. For example, a pre-bind plugin may provision a network volume and mount it on the target node before allowing the Pod to run there.

If any PreBind plugin returns an error, the Pod is [rejected](#) and returned to the scheduling queue.

## Bind

These plugins are used to bind a Pod to a Node. Bind plugins will not be called until all PreBind plugins have completed. Each bind plugin is called in the configured order. A bind plugin may

choose whether or not to handle the given Pod. If a bind plugin chooses to handle a Pod, **the remaining bind plugins are skipped**.

## PostBind

This is an informational interface. Post-bind plugins are called after a Pod is successfully bound. This is the end of a binding cycle, and can be used to clean up associated resources.

## Plugin API

There are two steps to the plugin API. First, plugins must register and get configured, then they use the extension point interfaces. Extension point interfaces have the following form.

```
type Plugin interface {
 Name() string
}

type QueueSortPlugin interface {
 Plugin
 Less(*v1.pod, *v1.pod) bool
}

type PreFilterPlugin interface {
 Plugin
 PreFilter(context.Context, *framework.CycleState, *v1.pod) error
}

// ...
```

## Plugin configuration

You can enable or disable plugins in the scheduler configuration. If you are using Kubernetes v1.18 or later, most scheduling [plugins](#) are in use and enabled by default.

In addition to default plugins, you can also implement your own scheduling plugins and get them configured along with default plugins. You can visit [scheduler-plugins](#) for more details.

If you are using Kubernetes v1.18 or later, you can configure a set of plugins as a scheduler profile and then define multiple profiles to fit various kinds of workload. Learn more at [multiple profiles](#).

## Dynamic Resource Allocation

FEATURE STATE: Kubernetes v1.26 [alpha]

Dynamic resource allocation is an API for requesting and sharing resources between pods and containers inside a pod. It is a generalization of the persistent volumes API for generic resources. Third-party resource drivers are responsible for tracking and allocating resources, with additional support provided by Kubernetes via *structured parameters* (introduced in Kubernetes 1.30). When a driver uses structured parameters, Kubernetes handles scheduling

and resource allocation without having to communicate with the driver. Different kinds of resources support arbitrary parameters for defining requirements and initialization.

## Before you begin

Kubernetes v1.30 includes cluster-level API support for dynamic resource allocation, but it [needs to be enabled](#) explicitly. You also must install a resource driver for specific resources that are meant to be managed using this API. If you are not running Kubernetes v1.30, check the documentation for that version of Kubernetes.

## API

The resource.k8s.io/v1alpha2 [API group](#) provides these types:

### ResourceClass

Defines which resource driver handles a certain kind of resource and provides common parameters for it. ResourceClasses are created by a cluster administrator when installing a resource driver.

### ResourceClaim

Defines a particular resource instance that is required by a workload. Created by a user (lifecycle managed manually, can be shared between different Pods) or for individual Pods by the control plane based on a ResourceClaimTemplate (automatic lifecycle, typically used by just one Pod).

### ResourceClaimTemplate

Defines the spec and some metadata for creating ResourceClaims. Created by a user when deploying a workload.

### PodSchedulingContext

Used internally by the control plane and resource drivers to coordinate pod scheduling when ResourceClaims need to be allocated for a Pod.

### ResourceSlice

Used with structured parameters to publish information about resources that are available in the cluster.

### ResourceClaimParameters

Contain the parameters for a ResourceClaim which influence scheduling, in a format that is understood by Kubernetes (the "structured parameter model"). Additional parameters may be embedded in an opaque extension, for use by the vendor driver when setting up the underlying resource.

### ResourceClassParameters

Similar to ResourceClaimParameters, the ResourceClassParameters provides a type for ResourceClass parameters which is understood by Kubernetes.

Parameters for ResourceClass and ResourceClaim are stored in separate objects, typically using the type defined by a [CRD](#) that was created when installing a resource driver.

The developer of a resource driver decides whether they want to handle these parameters in their own external controller or instead rely on Kubernetes to handle them through the use of structured parameters. A custom controller provides more flexibility, but cluster autoscaling is not going to work reliably for node-local resources. Structured parameters enable cluster autoscaling, but might not satisfy all use-cases.

When a driver uses structured parameters, it is still possible to let the end-user specify parameters with vendor-specific CRDs. When doing so, the driver needs to translate those

custom parameters into the in-tree types. Alternatively, a driver may also document how to use the in-tree types directly.

The core/v1 PodSpec defines ResourceClaims that are needed for a Pod in a resourceClaims field. Entries in that list reference either a ResourceClaim or a ResourceClaimTemplate. When referencing a ResourceClaim, all Pods using this PodSpec (for example, inside a Deployment or StatefulSet) share the same ResourceClaim instance. When referencing a ResourceClaimTemplate, each Pod gets its own instance.

The resources.claims list for container resources defines whether a container gets access to these resource instances, which makes it possible to share resources between one or more containers.

Here is an example for a fictional resource driver. Two ResourceClaim objects will get created for this Pod and each container gets access to one of them.

```
apiVersion: resource.k8s.io/v1alpha2
kind: ResourceClass
name: resource.example.com
driverName: resource-driver.example.com

apiVersion: cats.resource.example.com/v1
kind: ClaimParameters
name: large-black-cat-claim-parameters
spec:
 color: black
 size: large

apiVersion: resource.k8s.io/v1alpha2
kind: ResourceClaimTemplate
metadata:
 name: large-black-cat-claim-template
spec:
 spec:
 resourceClassName: resource.example.com
 parametersRef:
 apiGroup: cats.resource.example.com
 kind: ClaimParameters
 name: large-black-cat-claim-parameters

apiVersion: v1
kind: Pod
metadata:
 name: pod-with-cats
spec:
 containers:
 - name: container0
 image: ubuntu:20.04
 command: ["sleep", "9999"]
 resources:
 claims:
 - name: cat-0
 - name: container1
```



```
image: ubuntu:20.04
command: ["sleep", "9999"]
resources:
 claims:
 - name: cat-1
resourceClaims:
- name: cat-0
 source:
 resourceClaimTemplateName: large-black-cat-claim-template
- name: cat-1
 source:
 resourceClaimTemplateName: large-black-cat-claim-template
```

## Scheduling

### Without structured parameters

In contrast to native resources (CPU, RAM) and extended resources (managed by a device plugin, advertised by kubelet), without structured parameters the scheduler has no knowledge of what dynamic resources are available in a cluster or how they could be split up to satisfy the requirements of a specific ResourceClaim. Resource drivers are responsible for that. They mark ResourceClaims as "allocated" once resources for it are reserved. This also then tells the scheduler where in the cluster a ResourceClaim is available.

ResourceClaims can get allocated as soon as they are created ("immediate allocation"), without considering which Pods will use them. The default is to delay allocation until a Pod gets scheduled which needs the ResourceClaim (i.e. "wait for first consumer").

In that mode, the scheduler checks all ResourceClaims needed by a Pod and creates a PodScheduling object where it informs the resource drivers responsible for those ResourceClaims about nodes that the scheduler considers suitable for the Pod. The resource drivers respond by excluding nodes that don't have enough of the driver's resources left. Once the scheduler has that information, it selects one node and stores that choice in the PodScheduling object. The resource drivers then allocate their ResourceClaims so that the resources will be available on that node. Once that is complete, the Pod gets scheduled.

As part of this process, ResourceClaims also get reserved for the Pod. Currently ResourceClaims can either be used exclusively by a single Pod or an unlimited number of Pods.

One key feature is that Pods do not get scheduled to a node unless all of their resources are allocated and reserved. This avoids the scenario where a Pod gets scheduled onto one node and then cannot run there, which is bad because such a pending Pod also blocks all other resources like RAM or CPU that were set aside for it.

#### **Note:**

Scheduling of pods which use ResourceClaims is going to be slower because of the additional communication that is required. Beware that this may also impact pods that don't use ResourceClaims because only one pod at a time gets scheduled, blocking API calls are made while handling a pod with ResourceClaims, and thus scheduling the next pod gets delayed.

## With structured parameters

When a driver uses structured parameters, the scheduler takes over the responsibility of allocating resources to a ResourceClaim whenever a pod needs them. It does so by retrieving the full list of available resources from ResourceSlice objects, tracking which of those resources have already been allocated to existing ResourceClaims, and then selecting from those resources that remain. The exact resources selected are subject to the constraints provided in any ResourceClaimParameters or ResourceClassParameters associated with the ResourceClaim.

The chosen resource is recorded in the ResourceClaim status together with any vendor-specific parameters, so when a pod is about to start on a node, the resource driver on the node has all the information it needs to prepare the resource.

By using structured parameters, the scheduler is able to reach a decision without communicating with any DRA resource drivers. It is also able to schedule multiple pods quickly by keeping information about ResourceClaim allocations in memory and writing this information to the ResourceClaim objects in the background while concurrently binding the pod to a node.

## Monitoring resources

The kubelet provides a gRPC service to enable discovery of dynamic resources of running Pods. For more information on the gRPC endpoints, see the [resource allocation reporting](#).

## Pre-scheduled Pods

When you - or another API client - create a Pod with `spec.nodeName` already set, the scheduler gets bypassed. If some ResourceClaim needed by that Pod does not exist yet, is not allocated or not reserved for the Pod, then the kubelet will fail to run the Pod and re-check periodically because those requirements might still get fulfilled later.

Such a situation can also arise when support for dynamic resource allocation was not enabled in the scheduler at the time when the Pod got scheduled (version skew, configuration, feature gate, etc.). kube-controller-manager detects this and tries to make the Pod runnable by triggering allocation and/or reserving the required ResourceClaims.

### Note:

This only works with resource drivers that don't use structured parameters.

It is better to avoid bypassing the scheduler because a Pod that is assigned to a node blocks normal resources (RAM, CPU) that then cannot be used for other Pods while the Pod is stuck. To make a Pod run on a specific node while still going through the normal scheduling flow, create the Pod with a node selector that exactly matches the desired node:

```
apiVersion: v1
kind: Pod
metadata:
 name: pod-with-cats
spec:
 nodeSelector:
```

```
kubernetes.io/hostname: name-of-the-intended-node
```

```
...
```

You may also be able to mutate the incoming Pod, at admission time, to unset the `.spec.nodeName` field and to use a node selector instead.

## Enabling dynamic resource allocation

Dynamic resource allocation is an *alpha feature* and only enabled when the `DynamicResourceAllocation` [feature gate](#) and the `resource.k8s.io/v1alpha2` [API group](#) are enabled. For details on that, see the `--feature-gates` and `--runtime-config` [kube-apiserver parameters](#). `kube-scheduler`, `kube-controller-manager` and `kubelet` also need the feature gate.

A quick check whether a Kubernetes cluster supports the feature is to list `ResourceClass` objects with:

```
kubectl get resourceclasses
```

If your cluster supports dynamic resource allocation, the response is either a list of `ResourceClass` objects or:

```
No resources found
```

If not supported, this error is printed instead:

```
error: the server doesn't have a resource type "resourceclasses"
```

The default configuration of `kube-scheduler` enables the "DynamicResources" plugin if and only if the feature gate is enabled and when using the v1 configuration API. Custom configurations may have to be modified to include it.

In addition to enabling the feature in the cluster, a resource driver also has to be installed. Please refer to the driver's documentation for details.

## What's next

- For more information on the design, see the [Dynamic Resource Allocation KEP](#) and the [Structured Parameters KEP](#).

# Scheduler Performance Tuning

FEATURE STATE: Kubernetes v1.14 [beta]

[kube-scheduler](#) is the Kubernetes default scheduler. It is responsible for placement of Pods on Nodes in a cluster.

Nodes in a cluster that meet the scheduling requirements of a Pod are called *feasible* Nodes for the Pod. The scheduler finds feasible Nodes for a Pod and then runs a set of functions to score the feasible Nodes, picking a Node with the highest score among the feasible ones to run the Pod. The scheduler then notifies the API server about this decision in a process called *Binding*.

This page explains performance tuning optimizations that are relevant for large Kubernetes clusters.

In large clusters, you can tune the scheduler's behaviour balancing scheduling outcomes between latency (new Pods are placed quickly) and accuracy (the scheduler rarely makes poor placement decisions).

You configure this tuning setting via kube-scheduler setting `percentageOfNodesToScore`. This `KubeSchedulerConfiguration` setting determines a threshold for scheduling nodes in your cluster.

## Setting the threshold

The `percentageOfNodesToScore` option accepts whole numeric values between 0 and 100. The value 0 is a special number which indicates that the kube-scheduler should use its compiled-in default. If you set `percentageOfNodesToScore` above 100, kube-scheduler acts as if you had set a value of 100.

To change the value, edit the [kube-scheduler configuration file](#) and then restart the scheduler. In many cases, the configuration file can be found at `/etc/kubernetes/config/kube-scheduler.yaml`.

After you have made this change, you can run

```
kubectl get pods -n kube-system | grep kube-scheduler
```

to verify that the kube-scheduler component is healthy.

## Node scoring threshold

To improve scheduling performance, the kube-scheduler can stop looking for feasible nodes once it has found enough of them. In large clusters, this saves time compared to a naive approach that would consider every node.

You specify a threshold for how many nodes are enough, as a whole number percentage of all the nodes in your cluster. The kube-scheduler converts this into an integer number of nodes. During scheduling, if the kube-scheduler has identified enough feasible nodes to exceed the configured percentage, the kube-scheduler stops searching for more feasible nodes and moves on to the [scoring phase](#).

[How the scheduler iterates over Nodes](#) describes the process in detail.

### Default threshold

If you don't specify a threshold, Kubernetes calculates a figure using a linear formula that yields 50% for a 100-node cluster and yields 10% for a 5000-node cluster. The lower bound for the automatic value is 5%.

This means that the kube-scheduler always scores at least 5% of your cluster no matter how large the cluster is, unless you have explicitly set `percentageOfNodesToScore` to be smaller than 5.

If you want the scheduler to score all nodes in your cluster, set `percentageOfNodesToScore` to 100.

## Example

Below is an example configuration that sets `percentageOfNodesToScore` to 50%.

```
apiVersion: kubescheduler.config.k8s.io/v1alpha1
kind: KubeSchedulerConfiguration
algorithmSource:
 provider: DefaultProvider
...
percentageOfNodesToScore: 50
```

## Tuning `percentageOfNodesToScore`

`percentageOfNodesToScore` must be a value between 1 and 100 with the default value being calculated based on the cluster size. There is also a hardcoded minimum value of 50 nodes.

### Note:

In clusters with less than 50 feasible nodes, the scheduler still checks all the nodes because there are not enough feasible nodes to stop the scheduler's search early.

In a small cluster, if you set a low value for `percentageOfNodesToScore`, your change will have no or little effect, for a similar reason.

If your cluster has several hundred Nodes or fewer, leave this configuration option at its default value. Making changes is unlikely to improve the scheduler's performance significantly.

An important detail to consider when setting this value is that when a smaller number of nodes in a cluster are checked for feasibility, some nodes are not sent to be scored for a given Pod. As a result, a Node which could possibly score a higher value for running the given Pod might not even be passed to the scoring phase. This would result in a less than ideal placement of the Pod.

You should avoid setting `percentageOfNodesToScore` very low so that kube-scheduler does not make frequent, poor Pod placement decisions. Avoid setting the percentage to anything below 10%, unless the scheduler's throughput is critical for your application and the score of nodes is not important. In other words, you prefer to run the Pod on any Node as long as it is feasible.

## How the scheduler iterates over Nodes

This section is intended for those who want to understand the internal details of this feature.

In order to give all the Nodes in a cluster a fair chance of being considered for running Pods, the scheduler iterates over the nodes in a round robin fashion. You can imagine that Nodes are in an array. The scheduler starts from the start of the array and checks feasibility of the nodes until it finds enough Nodes as specified by `percentageOfNodesToScore`. For the next Pod, the

scheduler continues from the point in the Node array that it stopped at when checking feasibility of Nodes for the previous Pod.

If Nodes are in multiple zones, the scheduler iterates over Nodes in various zones to ensure that Nodes from different zones are considered in the feasibility checks. As an example, consider six nodes in two zones:

Zone 1: Node 1, Node 2, Node 3, Node 4

Zone 2: Node 5, Node 6

The Scheduler evaluates feasibility of the nodes in this order:

Node 1, Node 5, Node 2, Node 6, Node 3, Node 4

After going over all the Nodes, it goes back to Node 1.

## What's next

- Check the [kube-scheduler configuration reference \(v1\)](#)

# Resource Bin Packing

In the [scheduling-plugin](#) NodeResourcesFit of kube-scheduler, there are two scoring strategies that support the bin packing of resources: MostAllocated and RequestedToCapacityRatio.

## Enabling bin packing using MostAllocated strategy

The MostAllocated strategy scores the nodes based on the utilization of resources, favoring the ones with higher allocation. For each resource type, you can set a weight to modify its influence in the node score.

To set the MostAllocated strategy for the NodeResourcesFit plugin, use a [scheduler configuration](#) similar to the following:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- pluginConfig:
 - args:
 scoringStrategy:
 resources:
 - name: cpu
 weight: 1
 - name: memory
 weight: 1
 - name: intel.com/foo
 weight: 3
 - name: intel.com/bar
 weight: 3
 type: MostAllocated
 name: NodeResourcesFit
```

To learn more about other parameters and their default configuration, see the API documentation for [NodeResourcesFitArgs](#).

## Enabling bin packing using RequestedToCapacityRatio

The RequestedToCapacityRatio strategy allows the users to specify the resources along with weights for each resource to score nodes based on the request to capacity ratio. This allows users to bin pack extended resources by using appropriate parameters to improve the utilization of scarce resources in large clusters. It favors nodes according to a configured function of the allocated resources. The behavior of the RequestedToCapacityRatio in the NodeResourcesFit score function can be controlled by the [scoringStrategy](#) field. Within the scoringStrategy field, you can configure two parameters: requestedToCapacityRatio and resources. The shape in the requestedToCapacityRatio parameter allows the user to tune the function as least requested or most requested based on utilization and score values. The resources parameter comprises both the name of the resource to be considered during scoring and its corresponding weight, which specifies the weight of each resource.

Below is an example configuration that sets the bin packing behavior for extended resources intel.com/foo and intel.com/bar using the requestedToCapacityRatio field.

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- pluginConfig:
 - args:
 scoringStrategy:
 resources:
 - name: intel.com/foo
 weight: 3
 - name: intel.com/bar
 weight: 3
 requestedToCapacityRatio:
 shape:
 - utilization: 0
 score: 0
 - utilization: 100
 score: 10
 type: RequestedToCapacityRatio
 name: NodeResourcesFit
```

Referencing the KubeSchedulerConfiguration file with the kube-scheduler flag `--config=/path/to/config/file` will pass the configuration to the scheduler.

To learn more about other parameters and their default configuration, see the API documentation for [NodeResourcesFitArgs](#).

## Tuning the score function

shape is used to specify the behavior of the RequestedToCapacityRatio function.

```
shape:
- utilization: 0
 score: 0
```

```
- utilization: 100
 score: 10
```

The above arguments give the node a score of 0 if utilization is 0% and 10 for utilization 100%, thus enabling bin packing behavior. To enable least requested the score value must be reversed as follows.

```
shape:
- utilization: 0
 score: 10
- utilization: 100
 score: 0
```

resources is an optional parameter which defaults to:

```
resources:
- name: cpu
 weight: 1
- name: memory
 weight: 1
```

It can be used to add extended resources as follows:

```
resources:
- name: intel.com/foo
 weight: 5
- name: cpu
 weight: 3
- name: memory
 weight: 1
```

The weight parameter is optional and is set to 1 if not specified. Also, the weight cannot be set to a negative value.

## Node scoring for capacity allocation

This section is intended for those who want to understand the internal details of this feature. Below is an example of how the node score is calculated for a given set of values.

Requested resources:

```
intel.com/foo : 2
memory: 256MB
cpu: 2
```

Resource weights:

```
intel.com/foo : 5
memory: 1
cpu: 3
```

FunctionShapePoint {{0, 0}, {100, 10}}

Node 1 spec:



Available:

intel.com/foo: 4  
memory: 1 GB  
cpu: 8

Used:

intel.com/foo: 1  
memory: 256MB  
cpu: 1

Node score:

```
intel.com/foo = resourceScoringFunction((2+1),4)
 = (100 - ((4-3)*100/4))
 = (100 - 25)
 = 75 # requested + used = 75% * available
 = rawScoringFunction(75)
 = 7 # floor(75/10)

memory = resourceScoringFunction((256+256),1024)
 = (100 - ((1024-512)*100/1024))
 = 50 # requested + used = 50% * available
 = rawScoringFunction(50)
 = 5 # floor(50/10)

cpu = resourceScoringFunction((2+1),8)
 = (100 - ((8-3)*100/8))
 = 37.5 # requested + used = 37.5% * available
 = rawScoringFunction(37.5)
 = 3 # floor(37.5/10)

NodeScore = ((7 * 5) + (5 * 1) + (3 * 3)) / (5 + 1 + 3)
 = 5
```

Node 2 spec:

Available:

intel.com/foo: 8  
memory: 1GB  
cpu: 8

Used:

intel.com/foo: 2  
memory: 512MB  
cpu: 6

Node score:

```
intel.com/foo = resourceScoringFunction((2+2),8)
 = (100 - ((8-4)*100/8))
 = (100 - 50)
 = 50
 = rawScoringFunction(50)
 = 5
```

```
memory = resourceScoringFunction((256+512),1024)
 = (100 -((1024-768)*100/1024))
 = 75
 = rawScoringFunction(75)
 = 7

cpu = resourceScoringFunction((2+6),8)
 = (100 -((8-8)*100/8))
 = 100
 = rawScoringFunction(100)
 = 10

NodeScore = ((5 * 5) + (7 * 1) + (10 * 3)) / (5 + 1 + 3)
 = 7
```

## What's next

- Read more about the [scheduling framework](#)
- Read more about [scheduler configuration](#)

# Pod Priority and Preemption

FEATURE STATE: Kubernetes v1.14 [stable]

[Pods](#) can have *priority*. Priority indicates the importance of a Pod relative to other Pods. If a Pod cannot be scheduled, the scheduler tries to preempt (evict) lower priority Pods to make scheduling of the pending Pod possible.

### Warning:

In a cluster where not all users are trusted, a malicious user could create Pods at the highest possible priorities, causing other Pods to be evicted/not get scheduled. An administrator can use ResourceQuota to prevent users from creating pods at high priorities.

See [limit Priority Class consumption by default](#) for details.

## How to use priority and preemption

To use priority and preemption:

1. Add one or more [PriorityClasses](#).
2. Create Pods with [priorityClassName](#) set to one of the added PriorityClasses. Of course you do not need to create the Pods directly; normally you would add priorityClassName to the Pod template of a collection object like a Deployment.

Keep reading for more information about these steps.

## Note:

Kubernetes already ships with two PriorityClasses: `system-cluster-critical` and `system-node-critical`. These are common classes and are used to [ensure that critical components are always scheduled first](#).

## PriorityClass

A PriorityClass is a non-namespaced object that defines a mapping from a priority class name to the integer value of the priority. The name is specified in the `name` field of the PriorityClass object's metadata. The value is specified in the `value` field. The higher the value, the higher the priority. The name of a PriorityClass object must be a valid [DNS subdomain name](#), and it cannot be prefixed with `system-`.

A PriorityClass object can have any 32-bit integer value smaller than or equal to 1 billion. This means that the range of values for a PriorityClass object is from -2147483648 to 1000000000 inclusive. Larger numbers are reserved for built-in PriorityClasses that represent critical system Pods. A cluster admin should create one PriorityClass object for each such mapping that they want.

PriorityClass also has two optional fields: `globalDefault` and `description`. The `globalDefault` field indicates that the value of this PriorityClass should be used for Pods without a `priorityClassName`. Only one PriorityClass with `globalDefault` set to `true` can exist in the system. If there is no PriorityClass with `globalDefault` set, the priority of Pods with no `priorityClassName` is zero.

The `description` field is an arbitrary string. It is meant to tell users of the cluster when they should use this PriorityClass.

## Notes about PodPriority and existing clusters

- If you upgrade an existing cluster without this feature, the priority of your existing Pods is effectively zero.
- Addition of a PriorityClass with `globalDefault` set to `true` does not change the priorities of existing Pods. The value of such a PriorityClass is used only for Pods created after the PriorityClass is added.
- If you delete a PriorityClass, existing Pods that use the name of the deleted PriorityClass remain unchanged, but you cannot create more Pods that use the name of the deleted PriorityClass.

## Example PriorityClass

```
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
 name: high-priority
value: 1000000
globalDefault: false
description: "This priority class should be used for XYZ service pods only."
```

# Non-preempting PriorityClass

FEATURE STATE: Kubernetes v1.24 [stable]

Pods with `preemptionPolicy: Never` will be placed in the scheduling queue ahead of lower-priority pods, but they cannot preempt other pods. A non-preempting pod waiting to be scheduled will stay in the scheduling queue, until sufficient resources are free, and it can be scheduled. Non-preempting pods, like other pods, are subject to scheduler back-off. This means that if the scheduler tries these pods and they cannot be scheduled, they will be retried with lower frequency, allowing other pods with lower priority to be scheduled before them.

Non-preempting pods may still be preempted by other, high-priority pods.

`preemptionPolicy` defaults to `PreemptLowerPriority`, which will allow pods of that `PriorityClass` to preempt lower-priority pods (as is existing default behavior). If `preemptionPolicy` is set to `Never`, pods in that `PriorityClass` will be non-preempting.

An example use case is for data science workloads. A user may submit a job that they want to be prioritized above other workloads, but do not wish to discard existing work by preempting running pods. The high priority job with `preemptionPolicy: Never` will be scheduled ahead of other queued pods, as soon as sufficient cluster resources "naturally" become free.

## Example Non-preempting PriorityClass

```
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
 name: high-priority-nonpreempting
value: 1000000
preemptionPolicy: Never
globalDefault: false
description: "This priority class will not cause other pods to be preempted."
```

## Pod priority

After you have one or more `PriorityClasses`, you can create Pods that specify one of those `PriorityClass` names in their specifications. The priority admission controller uses the `priorityClassName` field and populates the integer value of the priority. If the priority class is not found, the Pod is rejected.

The following YAML is an example of a Pod configuration that uses the `PriorityClass` created in the preceding example. The priority admission controller checks the specification and resolves the priority of the Pod to 1000000.

```
apiVersion: v1
kind: Pod
metadata:
 name: nginx
 labels:
 env: test
spec:
 containers:
```

```
- name: nginx
 image: nginx
 imagePullPolicy: IfNotPresent
 priorityClassName: high-priority
```

## Effect of Pod priority on scheduling order

When Pod priority is enabled, the scheduler orders pending Pods by their priority and a pending Pod is placed ahead of other pending Pods with lower priority in the scheduling queue. As a result, the higher priority Pod may be scheduled sooner than Pods with lower priority if its scheduling requirements are met. If such Pod cannot be scheduled, the scheduler will continue and try to schedule other lower priority Pods.

## Preemption

When Pods are created, they go to a queue and wait to be scheduled. The scheduler picks a Pod from the queue and tries to schedule it on a Node. If no Node is found that satisfies all the specified requirements of the Pod, preemption logic is triggered for the pending Pod. Let's call the pending Pod P. Preemption logic tries to find a Node where removal of one or more Pods with lower priority than P would enable P to be scheduled on that Node. If such a Node is found, one or more lower priority Pods get evicted from the Node. After the Pods are gone, P can be scheduled on the Node.

## User exposed information

When Pod P preempts one or more Pods on Node N, `nominatedNodeName` field of Pod P's status is set to the name of Node N. This field helps the scheduler track resources reserved for Pod P and also gives users information about preemptions in their clusters.

Please note that Pod P is not necessarily scheduled to the "nominated Node". The scheduler always tries the "nominated Node" before iterating over any other nodes. After victim Pods are preempted, they get their graceful termination period. If another node becomes available while scheduler is waiting for the victim Pods to terminate, scheduler may use the other node to schedule Pod P. As a result `nominatedNodeName` and `nodeName` of Pod spec are not always the same. Also, if the scheduler preempts Pods on Node N, but then a higher priority Pod than Pod P arrives, the scheduler may give Node N to the new higher priority Pod. In such a case, scheduler clears `nominatedNodeName` of Pod P. By doing this, scheduler makes Pod P eligible to preempt Pods on another Node.

## Limitations of preemption

### Graceful termination of preemption victims

When Pods are preempted, the victims get their [graceful termination period](#). They have that much time to finish their work and exit. If they don't, they are killed. This graceful termination period creates a time gap between the point that the scheduler preempts Pods and the time when the pending Pod (P) can be scheduled on the Node (N). In the meantime, the scheduler keeps scheduling other pending Pods. As victims exit or get terminated, the scheduler tries to schedule Pods in the pending queue. Therefore, there is usually a time gap between the point that scheduler preempts victims and the time that Pod P is scheduled. In order to minimize this gap, one can set graceful termination period of lower priority Pods to zero or a small number.

## PodDisruptionBudget is supported, but not guaranteed

A [PodDisruptionBudget](#) (PDB) allows application owners to limit the number of Pods of a replicated application that are down simultaneously from voluntary disruptions. Kubernetes supports PDB when preempting Pods, but respecting PDB is best effort. The scheduler tries to find victims whose PDB are not violated by preemption, but if no such victims are found, preemption will still happen, and lower priority Pods will be removed despite their PDBs being violated.

## Inter-Pod affinity on lower-priority Pods

A Node is considered for preemption only when the answer to this question is yes: "If all the Pods with lower priority than the pending Pod are removed from the Node, can the pending Pod be scheduled on the Node?"

### Note:

Preemption does not necessarily remove all lower-priority Pods. If the pending Pod can be scheduled by removing fewer than all lower-priority Pods, then only a portion of the lower-priority Pods are removed. Even so, the answer to the preceding question must be yes. If the answer is no, the Node is not considered for preemption.

If a pending Pod has inter-pod [affinity](#) to one or more of the lower-priority Pods on the Node, the inter-Pod affinity rule cannot be satisfied in the absence of those lower-priority Pods. In this case, the scheduler does not preempt any Pods on the Node. Instead, it looks for another Node. The scheduler might find a suitable Node or it might not. There is no guarantee that the pending Pod can be scheduled.

Our recommended solution for this problem is to create inter-Pod affinity only towards equal or higher priority Pods.

## Cross node preemption

Suppose a Node N is being considered for preemption so that a pending Pod P can be scheduled on N. P might become feasible on N only if a Pod on another Node is preempted. Here's an example:

- Pod P is being considered for Node N.
- Pod Q is running on another Node in the same Zone as Node N.
- Pod P has Zone-wide anti-affinity with Pod Q (topologyKey: topology.kubernetes.io/zone).
- There are no other cases of anti-affinity between Pod P and other Pods in the Zone.
- In order to schedule Pod P on Node N, Pod Q can be preempted, but scheduler does not perform cross-node preemption. So, Pod P will be deemed unschedulable on Node N.

If Pod Q were removed from its Node, the Pod anti-affinity violation would be gone, and Pod P could possibly be scheduled on Node N.

We may consider adding cross Node preemption in future versions if there is enough demand and if we find an algorithm with reasonable performance.

# Troubleshooting

Pod priority and preemption can have unwanted side effects. Here are some examples of potential problems and ways to deal with them.

## Pods are preempted unnecessarily

Preemption removes existing Pods from a cluster under resource pressure to make room for higher priority pending Pods. If you give high priorities to certain Pods by mistake, these unintentionally high priority Pods may cause preemption in your cluster. Pod priority is specified by setting the `priorityClassName` field in the Pod's specification. The integer value for priority is then resolved and populated to the `priority` field of `podSpec`.

To address the problem, you can change the `priorityClassName` for those Pods to use lower priority classes, or leave that field empty. An empty `priorityClassName` is resolved to zero by default.

When a Pod is preempted, there will be events recorded for the preempted Pod. Preemption should happen only when a cluster does not have enough resources for a Pod. In such cases, preemption happens only when the priority of the pending Pod (preemptor) is higher than the victim Pods. Preemption must not happen when there is no pending Pod, or when the pending Pods have equal or lower priority than the victims. If preemption happens in such scenarios, please file an issue.

## Pods are preempted, but the preemptor is not scheduled

When pods are preempted, they receive their requested graceful termination period, which is by default 30 seconds. If the victim Pods do not terminate within this period, they are forcibly terminated. Once all the victims go away, the preemptor Pod can be scheduled.

While the preemptor Pod is waiting for the victims to go away, a higher priority Pod may be created that fits on the same Node. In this case, the scheduler will schedule the higher priority Pod instead of the preemptor.

This is expected behavior: the Pod with the higher priority should take the place of a Pod with a lower priority.

## Higher priority Pods are preempted before lower priority pods

The scheduler tries to find nodes that can run a pending Pod. If no node is found, the scheduler tries to remove Pods with lower priority from an arbitrary node in order to make room for the pending pod. If a node with low priority Pods is not feasible to run the pending Pod, the scheduler may choose another node with higher priority Pods (compared to the Pods on the other node) for preemption. The victims must still have lower priority than the preemptor Pod.

When there are multiple nodes available for preemption, the scheduler tries to choose the node with a set of Pods with lowest priority. However, if such Pods have `PodDisruptionBudget` that would be violated if they are preempted then the scheduler may choose another node with higher priority Pods.

When multiple nodes exist for preemption and none of the above scenarios apply, the scheduler chooses a node with the lowest priority.

# Interactions between Pod priority and quality of service

Pod priority and [QoS class](#) are two orthogonal features with few interactions and no default restrictions on setting the priority of a Pod based on its QoS classes. The scheduler's preemption logic does not consider QoS when choosing preemption targets. Preemption considers Pod priority and attempts to choose a set of targets with the lowest priority. Higher-priority Pods are considered for preemption only if the removal of the lowest priority Pods is not sufficient to allow the scheduler to schedule the preemptor Pod, or if the lowest priority Pods are protected by `PodDisruptionBudget`.

The kubelet uses Priority to determine pod order for [node-pressure eviction](#). You can use the QoS class to estimate the order in which pods are most likely to get evicted. The kubelet ranks pods for eviction based on the following factors:

1. Whether the starved resource usage exceeds requests
2. Pod Priority
3. Amount of resource usage relative to requests

See [Pod selection for kubelet eviction](#) for more details.

kubelet node-pressure eviction does not evict Pods when their usage does not exceed their requests. If a Pod with lower priority is not exceeding its requests, it won't be evicted. Another Pod with higher priority that exceeds its requests may be evicted.

## What's next

- Read about using ResourceQuotas in connection with PriorityClasses: [limit Priority Class consumption by default](#)
- Learn about [Pod Disruption](#)
- Learn about [API-initiated Eviction](#)
- Learn about [Node-pressure Eviction](#)

## Node-pressure Eviction

Node-pressure eviction is the process by which the [kubelet](#) proactively terminates pods to reclaim resources on nodes.

The [kubelet](#) monitors resources like memory, disk space, and filesystem inodes on your cluster's nodes. When one or more of these resources reach specific consumption levels, the kubelet can proactively fail one or more pods on the node to reclaim resources and prevent starvation.

During a node-pressure eviction, the kubelet sets the [phase](#) for the selected pods to Failed, and terminates the Pod.

Node-pressure eviction is not the same as [API-initiated eviction](#).

The kubelet does not respect your configured [PodDisruptionBudget](#) or the pod's `terminationGracePeriodSeconds`. If you use [soft eviction thresholds](#), the kubelet respects your configured `eviction-max-pod-grace-period`. If you use [hard eviction thresholds](#), the kubelet uses a 0s grace period (immediate shutdown) for termination.



# Self healing behavior

The kubelet attempts to [reclaim node-level resources](#) before it terminates end-user pods. For example, it removes unused container images when disk resources are starved.

If the pods are managed by a [workload](#) management object (such as [StatefulSet](#) or [Deployment](#)) that replaces failed pods, the control plane (kube-controller-manager) creates new pods in place of the evicted pods.

## Self healing for static pods

If you are running a [static pod](#) on a node that is under resource pressure, the kubelet may evict that static Pod. The kubelet then tries to create a replacement, because static Pods always represent an intent to run a Pod on that node.

The kubelet takes the *priority* of the static pod into account when creating a replacement. If the static pod manifest specifies a low priority, and there are higher-priority Pods defined within the cluster's control plane, and the node is under resource pressure, the kubelet may not be able to make room for that static pod. The kubelet continues to attempt to run all static pods even when there is resource pressure on a node.

## Eviction signals and thresholds

The kubelet uses various parameters to make eviction decisions, like the following:

- Eviction signals
- Eviction thresholds
- Monitoring intervals

### Eviction signals

Eviction signals are the current state of a particular resource at a specific point in time. Kubelet uses eviction signals to make eviction decisions by comparing the signals to eviction thresholds, which are the minimum amount of the resource that should be available on the node.

On Linux, the kubelet uses the following eviction signals:

| Eviction Signal    | Description                                                                     |
|--------------------|---------------------------------------------------------------------------------|
| memory.available   | memory.available := node.status.capacity[memory] - node.stats.memory.workingSet |
| nodefs.available   | nodefs.available := node.stats.fs.available                                     |
| nodefs.inodesFree  | nodefs.inodesFree := node.stats.fs.inodesFree                                   |
| imagefs.available  | imagefs.available := node.stats.runtime.imagefs.available                       |
| imagefs.inodesFree | imagefs.inodesFree := node.stats.runtime.imagefs.inodesFree                     |
| pid.available      | pid.available := node.stats.rlimit.maxpid - node.stats.rlimit.curproc           |

In this table, the **Description** column shows how kubelet gets the value of the signal. Each signal supports either a percentage or a literal value. Kubelet calculates the percentage value relative to the total capacity associated with the signal.

The value for `memory.available` is derived from the `cgroupfs` instead of tools like `free -m`. This is important because `free -m` does not work in a container, and if users use the [node allocatable](#) feature, out of resource decisions are made local to the end user Pod part of the `cgroup` hierarchy as well as the root node. This [script](#) or [cgroupv2 script](#) reproduces the same set of steps that the kubelet performs to calculate `memory.available`. The kubelet excludes `inactive_file` (the number of bytes of file-backed memory on the inactive LRU list) from its calculation, as it assumes that memory is reclaimable under pressure.

The kubelet recognizes two specific filesystem identifiers:

1. `nodefs`: The node's main filesystem, used for local disk volumes, `emptyDir` volumes not backed by memory, log storage, and more. For example, `nodefs` contains `/var/lib/kubelet/`.
2. `imagefs`: An optional filesystem that container runtimes use to store container images and container writable layers.

Kubelet auto-discovers these filesystems and ignores other node local filesystems. Kubelet does not support other configurations.

Some kubelet garbage collection features are deprecated in favor of eviction:

| Existing Flag                                        | Rationale                                                          |
|------------------------------------------------------|--------------------------------------------------------------------|
| <code>--maximum-dead-containers</code>               | deprecated once old logs are stored outside of container's context |
| <code>--maximum-dead-containers-per-container</code> | deprecated once old logs are stored outside of container's context |
| <code>--minimum-container-ttl-duration</code>        | deprecated once old logs are stored outside of container's context |

## Eviction thresholds

You can specify custom eviction thresholds for the kubelet to use when it makes eviction decisions. You can configure [soft](#) and [hard](#) eviction thresholds.

Eviction thresholds have the form `[eviction-signal][operator][quantity]`, where:

- `eviction-signal` is the [eviction signal](#) to use.
- `operator` is the [relational operator](#) you want, such as `<` (less than).
- `quantity` is the eviction threshold amount, such as `1Gi`. The value of `quantity` must match the quantity representation used by Kubernetes. You can use either literal values or percentages (%).

For example, if a node has `10GiB` of total memory and you want trigger eviction if the available memory falls below `1GiB`, you can define the eviction threshold as either `memory.available<10%` or `memory.available<1Gi` (you cannot use both).

### Soft eviction thresholds

A soft eviction threshold pairs an eviction threshold with a required administrator-specified grace period. The kubelet does not evict pods until the grace period is exceeded. The kubelet returns an error on startup if you do not specify a grace period.

You can specify both a soft eviction threshold grace period and a maximum allowed pod termination grace period for kubelet to use during evictions. If you specify a maximum allowed

grace period and the soft eviction threshold is met, the kubelet uses the lesser of the two grace periods. If you do not specify a maximum allowed grace period, the kubelet kills evicted pods immediately without graceful termination.

You can use the following flags to configure soft eviction thresholds:

- `eviction-soft`: A set of eviction thresholds like `memory.available<1.5Gi` that can trigger pod eviction if held over the specified grace period.
- `eviction-soft-grace-period`: A set of eviction grace periods like `memory.available=1m30s` that define how long a soft eviction threshold must hold before triggering a Pod eviction.
- `eviction-max-pod-grace-period`: The maximum allowed grace period (in seconds) to use when terminating pods in response to a soft eviction threshold being met.

## Hard eviction thresholds

A hard eviction threshold has no grace period. When a hard eviction threshold is met, the kubelet kills pods immediately without graceful termination to reclaim the starved resource.

You can use the `eviction-hard` flag to configure a set of hard eviction thresholds like `memory.available<1Gi`.

The kubelet has the following default hard eviction thresholds:

- `memory.available<100Mi`
- `nodefs.available<10%`
- `imagefs.available<15%`
- `nodefs.inodesFree<5%` (Linux nodes)
- `imagefs.inodesFree<5%` (Linux nodes)

These default values of hard eviction thresholds will only be set if none of the parameters is changed. If you change the value of any parameter, then the values of other parameters will not be inherited as the default values and will be set to zero. In order to provide custom values, you should provide all the thresholds respectively.

## Eviction monitoring interval

The kubelet evaluates eviction thresholds based on its configured `housekeeping-interval`, which defaults to 10s.

## Node conditions

The kubelet reports [node conditions](#) to reflect that the node is under pressure because hard or soft eviction threshold is met, independent of configured grace periods.

The kubelet maps eviction signals to node conditions as follows:

| Node Condition | Eviction Signal               | Description                                                      |
|----------------|-------------------------------|------------------------------------------------------------------|
| MemoryPressure | <code>memory.available</code> | Available memory on the node has satisfied an eviction threshold |
| DiskPressure   |                               |                                                                  |

| Node Condition | Eviction Signal                                                               | Description                                                                                                                  |
|----------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                | nodefs.available, nodefs.inodesFree, imagefs.available, or imagefs.inodesFree | Available disk space and inodes on either the node's root filesystem or image filesystem has satisfied an eviction threshold |
| PIDPressure    | pid.available                                                                 | Available processes identifiers on the (Linux) node has fallen below an eviction threshold                                   |

The control plane also [maps](#) these node conditions to taints.

The kubelet updates the node conditions based on the configured `--node-status-update-frequency`, which defaults to 10s.

## Node condition oscillation

In some cases, nodes oscillate above and below soft eviction thresholds without holding for the defined grace periods. This causes the reported node condition to constantly switch between true and false, leading to bad eviction decisions.

To protect against oscillation, you can use the `eviction-pressure-transition-period` flag, which controls how long the kubelet must wait before transitioning a node condition to a different state. The transition period has a default value of 5m.

## Reclaiming node level resources

The kubelet tries to reclaim node-level resources before it evicts end-user pods.

When a `DiskPressure` node condition is reported, the kubelet reclaims node-level resources based on the filesystems on the node.

### With imagefs

If the node has a dedicated `imagefs` filesystem for container runtimes to use, the kubelet does the following:

- If the `nodefs` filesystem meets the eviction thresholds, the kubelet garbage collects dead pods and containers.
- If the `imagefs` filesystem meets the eviction thresholds, the kubelet deletes all unused images.

## Without imagefs

If the node only has a `nodefs` filesystem that meets eviction thresholds, the kubelet frees up disk space in the following order:

1. Garbage collect dead pods and containers
2. Delete unused images

## Pod selection for kubelet eviction

If the kubelet's attempts to reclaim node-level resources don't bring the eviction signal below the threshold, the kubelet begins to evict end-user pods.

The kubelet uses the following parameters to determine the pod eviction order:

1. Whether the pod's resource usage exceeds requests
2. [Pod Priority](#)
3. The pod's resource usage relative to requests

As a result, kubelet ranks and evicts pods in the following order:

1. `BestEffort` or `Burstable` pods where the usage exceeds requests. These pods are evicted based on their Priority and then by how much their usage level exceeds the request.
2. `Guaranteed` pods and `Burstable` pods where the usage is less than requests are evicted last, based on their Priority.

### Note:

The kubelet does not use the pod's [QoS class](#) to determine the eviction order. You can use the QoS class to estimate the most likely pod eviction order when reclaiming resources like memory. QoS classification does not apply to `EphemeralStorage` requests, so the above scenario will not apply if the node is, for example, under `DiskPressure`.

`Guaranteed` pods are guaranteed only when requests and limits are specified for all the containers and they are equal. These pods will never be evicted because of another pod's resource consumption. If a system daemon (such as kubelet and journald) is consuming more resources than were reserved via `system-reserved` or `kube-reserved` allocations, and the node only has `Guaranteed` or `Burstable` pods using less resources than requests left on it, then the kubelet must choose to evict one of these pods to preserve node stability and to limit the impact of resource starvation on other pods. In this case, it will choose to evict pods of lowest Priority first.

If you are running a [static pod](#) and want to avoid having it evicted under resource pressure, set the `priority` field for that Pod directly. Static pods do not support the `priorityClassName` field.

When the kubelet evicts pods in response to inode or process ID starvation, it uses the Pods' relative priority to determine the eviction order, because inodes and PIDs have no requests.

The kubelet sorts pods differently based on whether the node has a dedicated `imagefs` filesystem:

## With imagefs

If nodefs is triggering evictions, the kubelet sorts pods based on nodefs usage (local volumes + logs of all containers).

If imagefs is triggering evictions, the kubelet sorts pods based on the writable layer usage of all containers.

## Without imagefs

If nodefs is triggering evictions, the kubelet sorts pods based on their total disk usage (local volumes + logs & writable layer of all containers)

## Minimum eviction reclaim

In some cases, pod eviction only reclaims a small amount of the starved resource. This can lead to the kubelet repeatedly hitting the configured eviction thresholds and triggering multiple evictions.

You can use the `--eviction-minimum-reclaim` flag or a [kubelet config file](#) to configure a minimum reclaim amount for each resource. When the kubelet notices that a resource is starved, it continues to reclaim that resource until it reclaims the quantity you specify.

For example, the following configuration sets minimum reclaim amounts:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
evictionHard:
 memory.available: "500Mi"
 nodefs.available: "1Gi"
 imagefs.available: "100Gi"
evictionMinimumReclaim:
 memory.available: "0Mi"
 nodefs.available: "500Mi"
 imagefs.available: "2Gi"
```

In this example, if the `nodefs.available` signal meets the eviction threshold, the kubelet reclaims the resource until the signal reaches the threshold of 1GiB, and then continues to reclaim the minimum amount of 500MiB, until the available nodefs storage value reaches 1.5GiB.

Similarly, the kubelet tries to reclaim the `imagefs` resource until the `imagefs.available` value reaches 102Gi, representing 102 GiB of available container image storage. If the amount of storage that the kubelet could reclaim is less than 2GiB, the kubelet doesn't reclaim anything.

The default `eviction-minimum-reclaim` is 0 for all resources.

## Node out of memory behavior

If the node experiences an *out of memory* (OOM) event prior to the kubelet being able to reclaim memory, the node depends on the [oom killer](#) to respond.

The kubelet sets an `oom_score_adj` value for each container based on the QoS for the pod.

| Quality of Service | oom_score_adj                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------|
| Guaranteed         | -997                                                                                                     |
| BestEffort         | 1000                                                                                                     |
| Burstable          | $\min(\max(2, 1000 - (1000 \times \text{memoryRequestBytes}) / \text{machineMemoryCapacityBytes}), 999)$ |

### Note:

The kubelet also sets an `oom_score_adj` value of -997 for any containers in Pods that have system-node-critical [Priority](#).

If the kubelet can't reclaim memory before a node experiences OOM, the `oom_killer` calculates an `oom_score` based on the percentage of memory it's using on the node, and then adds the `oom_score_adj` to get an effective `oom_score` for each container. It then kills the container with the highest score.

This means that containers in low QoS pods that consume a large amount of memory relative to their scheduling requests are killed first.

Unlike pod eviction, if a container is OOM killed, the kubelet can restart it based on its `restartPolicy`.

## Good practices

The following sections describe good practice for eviction configuration.

### Schedulable resources and eviction policies

When you configure the kubelet with an eviction policy, you should make sure that the scheduler will not schedule pods if they will trigger eviction because they immediately induce memory pressure.

Consider the following scenario:

- Node memory capacity: 10GiB
- Operator wants to reserve 10% of memory capacity for system daemons (kernel, kubelet, etc.)
- Operator wants to evict Pods at 95% memory utilization to reduce incidence of system OOM.

For this to work, the kubelet is launched as follows:

```
--eviction-hard=memory.available<500Mi
--system-reserved=memory=1.5Gi
```

In this configuration, the `--system-reserved` flag reserves 1.5GiB of memory for the system, which is 10% of the total memory + the eviction threshold amount.

The node can reach the eviction threshold if a pod is using more than its request, or if the system is using more than 1GiB of memory, which makes the `memory.available` signal fall below 500MiB and triggers the threshold.



## DaemonSets and node-pressure eviction

Pod priority is a major factor in making eviction decisions. If you do not want the kubelet to evict pods that belong to a DaemonSet, give those pods a high enough priority by specifying a suitable `priorityClassName` in the pod spec. You can also use a lower priority, or the default, to only allow pods from that DaemonSet to run when there are enough resources.

## Known issues

The following sections describe known issues related to out of resource handling.

### kubelet may not observe memory pressure right away

By default, the kubelet polls cAdvisor to collect memory usage stats at a regular interval. If memory usage increases within that window rapidly, the kubelet may not observe `MemoryPressure` fast enough, and the OOM killer will still be invoked.

You can use the `--kernel-memcg-notification` flag to enable the memcg notification API on the kubelet to get notified immediately when a threshold is crossed.

If you are not trying to achieve extreme utilization, but a sensible measure of overcommit, a viable workaround for this issue is to use the `--kube-reserved` and `--system-reserved` flags to allocate memory for the system.

### active\_file memory is not considered as available memory

On Linux, the kernel tracks the number of bytes of file-backed memory on active least recently used (LRU) list as the `active_file` statistic. The kubelet treats `active_file` memory areas as not reclaimable. For workloads that make intensive use of block-backed local storage, including ephemeral local storage, kernel-level caches of file and block data means that many recently accessed cache pages are likely to be counted as `active_file`. If enough of these kernel block buffers are on the active LRU list, the kubelet is liable to observe this as high resource use and taint the node as experiencing memory pressure - triggering pod eviction.

For more details, see <https://github.com/kubernetes/kubernetes/issues/43916>

You can work around that behavior by setting the memory limit and memory request the same for containers likely to perform intensive I/O activity. You will need to estimate or measure an optimal memory limit value for that container.

## What's next

- Learn about [API-initiated Eviction](#)
- Learn about [Pod Priority and Preemption](#)
- Learn about [PodDisruptionBudgets](#)
- Learn about [Quality of Service](#) (QoS)
- Check out the [Eviction API](#)



# API-initiated Eviction

API-initiated eviction is the process by which you use the [Eviction API](#) to create an Eviction object that triggers graceful pod termination.

You can request eviction by calling the Eviction API directly, or programmatically using a client of the [API server](#), like the `kubectl drain` command. This creates an Eviction object, which causes the API server to terminate the Pod.

API-initiated evictions respect your configured [PodDisruptionBudgets](#) and [terminationGracePeriodSeconds](#).

Using the API to create an Eviction object for a Pod is like performing a policy-controlled [DELETE operation](#) on the Pod.

## Calling the Eviction API

You can use a [Kubernetes language client](#) to access the Kubernetes API and create an Eviction object. To do this, you POST the attempted operation, similar to the following example:

- [policy/v1](#)
- [policy/v1beta1](#)

### Note:

`policy/v1` Eviction is available in v1.22+. Use `policy/v1beta1` with prior releases.

```
{
 "apiVersion": "policy/v1",
 "kind": "Eviction",
 "metadata": {
 "name": "quux",
 "namespace": "default"
 }
}
```

### Note:

Deprecated in v1.22 in favor of `policy/v1`

```
{
 "apiVersion": "policy/v1beta1",
 "kind": "Eviction",
 "metadata": {
 "name": "quux",
 "namespace": "default"
 }
}
```

Alternatively, you can attempt an eviction operation by accessing the API using `curl` or `wget`, similar to the following example:

```
curl -v -H 'Content-type: application/json' https://your-cluster-api-endpoint.example/api/v1/namespaces/default/pods/quux/eviction -d @eviction.json
```

## How API-initiated eviction works

When you request an eviction using the API, the API server performs admission checks and responds in one of the following ways:

- 200 OK: the eviction is allowed, the Eviction subresource is created, and the Pod is deleted, similar to sending a DELETE request to the Pod URL.
- 429 Too Many Requests: the eviction is not currently allowed because of the configured [PodDisruptionBudget](#). You may be able to attempt the eviction again later. You might also see this response because of API rate limiting.
- 500 Internal Server Error: the eviction is not allowed because there is a misconfiguration, like if multiple PodDisruptionBudgets reference the same Pod.

If the Pod you want to evict isn't part of a workload that has a PodDisruptionBudget, the API server always returns 200 OK and allows the eviction.

If the API server allows the eviction, the Pod is deleted as follows:

1. The Pod resource in the API server is updated with a deletion timestamp, after which the API server considers the Pod resource to be terminated. The Pod resource is also marked with the configured grace period.
2. The [kubelet](#) on the node where the local Pod is running notices that the Pod resource is marked for termination and starts to gracefully shut down the local Pod.
3. While the kubelet is shutting the Pod down, the control plane removes the Pod from [Endpoint](#) and [EndpointSlice](#) objects. As a result, controllers no longer consider the Pod as a valid object.
4. After the grace period for the Pod expires, the kubelet forcefully terminates the local Pod.
5. The kubelet tells the API server to remove the Pod resource.
6. The API server deletes the Pod resource.

## Troubleshooting stuck evictions

In some cases, your applications may enter a broken state, where the Eviction API will only return 429 or 500 responses until you intervene. This can happen if, for example, a ReplicaSet creates pods for your application but new pods do not enter a Ready state. You may also notice this behavior in cases where the last evicted Pod had a long termination grace period.

If you notice stuck evictions, try one of the following solutions:

- Abort or pause the automated operation causing the issue. Investigate the stuck application before you restart the operation.
- Wait a while, then directly delete the Pod from your cluster control plane instead of using the Eviction API.

## What's next

- Learn how to protect your applications with a [Pod Disruption Budget](#).
- Learn about [Node-pressure Eviction](#).

- Learn about [Pod Priority and Preemption](#).

# Cluster Administration

Lower-level detail relevant to creating or administering a Kubernetes cluster.

The cluster administration overview is for anyone creating or administering a Kubernetes cluster. It assumes some familiarity with core Kubernetes [concepts](#).

## Planning a cluster

See the guides in [Setup](#) for examples of how to plan, set up, and configure Kubernetes clusters. The solutions listed in this article are called *distros*.

### Note:

Not all distros are actively maintained. Choose distros which have been tested with a recent version of Kubernetes.

Before choosing a guide, here are some considerations:

- Do you want to try out Kubernetes on your computer, or do you want to build a high-availability, multi-node cluster? Choose distros best suited for your needs.
- Will you be using a **hosted Kubernetes cluster**, such as [Google Kubernetes Engine](#), or **hosting your own cluster**?
- Will your cluster be **on-premises**, or **in the cloud (IaaS)**? Kubernetes does not directly support hybrid clusters. Instead, you can set up multiple clusters.
- **If you are configuring Kubernetes on-premises**, consider which [networking model](#) fits best.
- Will you be running Kubernetes on **"bare metal" hardware** or on **virtual machines (VMs)**?
- Do you **want to run a cluster**, or do you expect to do **active development of Kubernetes project code**? If the latter, choose an actively-developed distro. Some distros only use binary releases, but offer a greater variety of choices.
- Familiarize yourself with the [components](#) needed to run a cluster.

## Managing a cluster

- Learn how to [manage nodes](#).
  - Read about [cluster autoscaling](#).
- Learn how to set up and manage the [resource quota](#) for shared clusters.

## Securing a cluster

- [Generate Certificates](#) describes the steps to generate certificates using different tool chains.

- [Kubernetes Container Environment](#) describes the environment for Kubelet managed containers on a Kubernetes node.
- [Controlling Access to the Kubernetes API](#) describes how Kubernetes implements access control for its own API.
- [Authenticating](#) explains authentication in Kubernetes, including the various authentication options.
- [Authorization](#) is separate from authentication, and controls how HTTP calls are handled.
- [Using Admission Controllers](#) explains plug-ins which intercepts requests to the Kubernetes API server after authentication and authorization.
- [Using Sysctls in a Kubernetes Cluster](#) describes to an administrator how to use the `sysctl` command-line tool to set kernel parameters .
- [Auditing](#) describes how to interact with Kubernetes' audit logs.

## Securing the kubelet

- [Control Plane-Node communication](#)
- [TLS bootstrapping](#)
- [Kubelet authentication/authorization](#)

## Optional Cluster Services

- [DNS Integration](#) describes how to resolve a DNS name directly to a Kubernetes service.
- [Logging and Monitoring Cluster Activity](#) explains how logging in Kubernetes works and how to implement it.

# Node Shutdowns

In a Kubernetes cluster, a [node](#) can be shutdown in a planned graceful way or unexpectedly because of reasons such as a power outage or something else external. A node shutdown could lead to workload failure if the node is not drained before the shutdown. A node shutdown can be either **graceful** or **non-graceful**.

## Graceful node shutdown

FEATURE STATE: Kubernetes v1.21 [beta]

The kubelet attempts to detect node system shutdown and terminates pods running on the node.

Kubelet ensures that pods follow the normal [pod termination process](#) during the node shutdown. During node shutdown, the kubelet does not accept new Pods (even if those Pods are already bound to the node).

The Graceful node shutdown feature depends on systemd since it takes advantage of [systemd inhibitor locks](#) to delay the node shutdown with a given duration.

Graceful node shutdown is controlled with the GracefulNodeShutdown [feature gate](#) which is enabled by default in 1.21.

Note that by default, both configuration options described below, shutdownGracePeriod and shutdownGracePeriodCriticalPods are set to zero, thus not activating the graceful node shutdown functionality. To activate the feature, the two kubelet config settings should be configured appropriately and set to non-zero values.

Once systemd detects or notifies node shutdown, the kubelet sets a NotReady condition on the Node, with the reason set to "node is shutting down". The kube-scheduler honors this condition and does not schedule any Pods onto the affected node; other third-party schedulers are expected to follow the same logic. This means that new Pods won't be scheduled onto that node and therefore none will start.

The kubelet **also** rejects Pods during the PodAdmission phase if an ongoing node shutdown has been detected, so that even Pods with a [toleration](#) for node.kubernetes.io/not-ready:NoSchedule do not start there.

At the same time when kubelet is setting that condition on its Node via the API, the kubelet also begins terminating any Pods that are running locally.

During a graceful shutdown, kubelet terminates pods in two phases:

1. Terminate regular pods running on the node.
2. Terminate [critical pods](#) running on the node.

Graceful node shutdown feature is configured with two [KubeletConfiguration](#) options:

- shutdownGracePeriod:
  - Specifies the total duration that the node should delay the shutdown by. This is the total grace period for pod termination for both regular and [critical pods](#).
- shutdownGracePeriodCriticalPods:
  - Specifies the duration used to terminate [critical pods](#) during a node shutdown. This value should be less than shutdownGracePeriod.

**Note:**

There are cases when Node termination was cancelled by the system (or perhaps manually by an administrator). In either of those situations the Node will return to the Ready state. However, Pods which already started the process of termination will not be restored by kubelet and will need to be re-scheduled.

For example, if shutdownGracePeriod=30s, and shutdownGracePeriodCriticalPods=10s, kubelet will delay the node shutdown by 30 seconds. During the shutdown, the first 20 (30-10) seconds would be reserved for gracefully terminating normal pods, and the last 10 seconds would be reserved for terminating [critical pods](#).

## Note:

When pods were evicted during the graceful node shutdown, they are marked as shutdown. Running `kubectl get pods` shows the status of the evicted pods as `Terminated`. And `kubectl describe pod` indicates that the pod was evicted because of node shutdown:

Reason: Terminated  
Message: Pod was terminated in response to imminent node shutdown.

## Pod Priority based graceful node shutdown

FEATURE STATE: Kubernetes v1.24 [beta]

To provide more flexibility during graceful node shutdown around the ordering of pods during shutdown, graceful node shutdown honors the `PriorityClass` for Pods, provided that you enabled this feature in your cluster. The feature allows cluster administrators to explicitly define the ordering of pods during graceful node shutdown based on [priority classes](#).

The [Graceful Node Shutdown](#) feature, as described above, shuts down pods in two phases, non-critical pods, followed by critical pods. If additional flexibility is needed to explicitly define the ordering of pods during shutdown in a more granular way, pod priority based graceful shutdown can be used.

When graceful node shutdown honors pod priorities, this makes it possible to do graceful node shutdown in multiple phases, each phase shutting down a particular priority class of pods. The kubelet can be configured with the exact phases and shutdown time per phase.

Assuming the following custom pod [priority classes](#) in a cluster,

| Pod priority class name | Pod priority class value |
|-------------------------|--------------------------|
| custom-class-a          | 100000                   |
| custom-class-b          | 10000                    |
| custom-class-c          | 1000                     |
| regular/unset           | 0                        |

Within the [kubelet configuration](#) the settings for `shutdownGracePeriodByPodPriority` could look like:

| Pod priority class value | Shutdown period |
|--------------------------|-----------------|
| 100000                   | 10 seconds      |
| 10000                    | 180 seconds     |
| 1000                     | 120 seconds     |
| 0                        | 60 seconds      |

The corresponding kubelet config YAML configuration would be:

`shutdownGracePeriodByPodPriority:`

- priority: 100000  
shutdownGracePeriodSeconds: 10
- priority: 10000  
shutdownGracePeriodSeconds: 180
- priority: 1000

```
shutdownGracePeriodSeconds: 120
- priority: 0
shutdownGracePeriodSeconds: 60
```

The above table implies that any pod with priority value  $\geq 100000$  will get just 10 seconds to stop, any pod with value  $\geq 10000$  and  $< 100000$  will get 180 seconds to stop, any pod with value  $\geq 1000$  and  $< 10000$  will get 120 seconds to stop. Finally, all other pods will get 60 seconds to stop.

One doesn't have to specify values corresponding to all of the classes. For example, you could instead use these settings:

| Pod priority class value | Shutdown period |
|--------------------------|-----------------|
| 100000                   | 300 seconds     |
| 1000                     | 120 seconds     |
| 0                        | 60 seconds      |

In the above case, the pods with `custom-class-b` will go into the same bucket as `custom-class-c` for shutdown.

If there are no pods in a particular range, then the kubelet does not wait for pods in that priority range. Instead, the kubelet immediately skips to the next priority class value range.

If this feature is enabled and no configuration is provided, then no ordering action will be taken.

Using this feature requires enabling the `GracefulNodeShutdownBasedOnPodPriority` [feature gate](#), and setting `ShutdownGracePeriodByPodPriority` in the [kubelet config](#) to the desired configuration containing the pod priority class values and their respective shutdown periods.

#### Note:

The ability to take Pod priority into account during graceful node shutdown was introduced as an Alpha feature in Kubernetes v1.23. In Kubernetes 1.30 the feature is Beta and is enabled by default.

Metrics `graceful_shutdown_start_time_seconds` and `graceful_shutdown_end_time_seconds` are emitted under the kubelet subsystem to monitor node shutdowns.

## Non-graceful node shutdown handling

FEATURE STATE: Kubernetes v1.28 [stable]

A node shutdown action may not be detected by kubelet's Node Shutdown Manager, either because the command does not trigger the inhibitor locks mechanism used by kubelet or because of a user error, i.e., the `ShutdownGracePeriod` and `ShutdownGracePeriodCriticalPods` are not configured properly. Please refer to above section [Graceful Node Shutdown](#) for more details.

When a node is shutdown but not detected by kubelet's Node Shutdown Manager, the pods that are part of a [StatefulSet](#) will be stuck in terminating status on the shutdown node and cannot move to a new running node. This is because kubelet on the shutdown node is not available to



delete the pods so the StatefulSet cannot create a new pod with the same name. If there are volumes used by the pods, the VolumeAttachments will not be deleted from the original shutdown node so the volumes used by these pods cannot be attached to a new running node. As a result, the application running on the StatefulSet cannot function properly. If the original shutdown node comes up, the pods will be deleted by kubelet and new pods will be created on a different running node. If the original shutdown node does not come up, these pods will be stuck in terminating status on the shutdown node forever.

To mitigate the above situation, a user can manually add the taint `node.kubernetes.io/out-of-service` with either `NoExecute` or `NoSchedule` effect to a Node marking it out-of-service. If the `NodeOutOfServiceVolumeDetach` [feature gate](#) is enabled on [kube-controller-manager](#), and a Node is marked out-of-service with this taint, the pods on the node will be forcefully deleted if there are no matching tolerations on it and volume detach operations for the pods terminating on the node will happen immediately. This allows the Pods on the out-of-service node to recover quickly on a different node.

During a non-graceful shutdown, Pods are terminated in the two phases:

1. Force delete the Pods that do not have matching out-of-service tolerations.
2. Immediately perform detach volume operation for such pods.

**Note:**

- Before adding the taint `node.kubernetes.io/out-of-service`, it should be verified that the node is already in shutdown or power off state (not in the middle of restarting).
- The user is required to manually remove the out-of-service taint after the pods are moved to a new node and the user has checked that the shutdown node has been recovered since the user was the one who originally added the taint.

## Forced storage detach on timeout

In any situation where a pod deletion has not succeeded for 6 minutes, kubernetes will force detach volumes being unmounted if the node is unhealthy at that instant. Any workload still running on the node that uses a force-detached volume will cause a violation of the [CSI specification](#), which states that `ControllerUnpublishVolume` "**must** be called after all `NodeUnstageVolume` and `NodeUnpublishVolume` on the volume are called and succeed". In such circumstances, volumes on the node in question might encounter data corruption.

The forced storage detach behaviour is optional; users might opt to use the "Non-graceful node shutdown" feature instead.

Force storage detach on timeout can be disabled by setting the `disable-force-detach-on-timeout` config field in `kube-controller-manager`. Disabling the force detach on timeout feature means that a volume that is hosted on a node that is unhealthy for more than 6 minutes will not have its associated [VolumeAttachment](#) deleted.

After this setting has been applied, unhealthy pods still attached to a volumes must be recovered via the [Non-Graceful Node Shutdown](#) procedure mentioned above.

**Note:**

- Caution must be taken while using the [Non-Graceful Node Shutdown](#) procedure.
- Deviation from the steps documented above can result in data corruption.



## What's next

Learn more about the following:

- Blog: [Non-Graceful Node Shutdown](#).
- Cluster Architecture: [Nodes](#).

## Certificates

To learn how to generate certificates for your cluster, see [Certificates](#).

## Cluster Networking

Networking is a central part of Kubernetes, but it can be challenging to understand exactly how it is expected to work. There are 4 distinct networking problems to address:

1. Highly-coupled container-to-container communications: this is solved by [Pods](#) and localhost communications.
2. Pod-to-Pod communications: this is the primary focus of this document.
3. Pod-to-Service communications: this is covered by [Services](#).
4. External-to-Service communications: this is also covered by Services.

Kubernetes is all about sharing machines among applications. Typically, sharing machines requires ensuring that two applications do not try to use the same ports. Coordinating ports across multiple developers is very difficult to do at scale and exposes users to cluster-level issues outside of their control.

Dynamic port allocation brings a lot of complications to the system - every application has to take ports as flags, the API servers have to know how to insert dynamic port numbers into configuration blocks, services have to know how to find each other, etc. Rather than deal with this, Kubernetes takes a different approach.

To learn about the Kubernetes networking model, see [here](#).

## Kubernetes IP address ranges

Kubernetes clusters require to allocate non-overlapping IP addresses for Pods, Services and Nodes, from a range of available addresses configured in the following components:

- The network plugin is configured to assign IP addresses to Pods.
- The kube-apiserver is configured to assign IP addresses to Services.
- The kubelet or the cloud-controller-manager is configured to assign IP addresses to Nodes.

A figure illustrating the different network ranges in a kubernetes cluster

# Cluster networking types

Kubernetes clusters, attending to the IP families configured, can be categorized into:

- IPv4 only: The network plugin, kube-apiserver and kubelet/cloud-controller-manager are configured to assign only IPv4 addresses.
- IPv6 only: The network plugin, kube-apiserver and kubelet/cloud-controller-manager are configured to assign only IPv6 addresses.
- IPv4/IPv6 or IPv6/IPv4 [dual-stack](#):
  - The network plugin is configured to assign IPv4 and IPv6 addresses.
  - The kube-apiserver is configured to assign IPv4 and IPv6 addresses.
  - The kubelet or cloud-controller-manager is configured to assign IPv4 and IPv6 address.
  - All components must agree on the configured primary IP family.

Kubernetes clusters only consider the IP families present on the Pods, Services and Nodes objects, independently of the existing IPs of the represented objects. Per example, a server or a pod can have multiple IP addresses on its interfaces, but only the IP addresses in `node.status.addresses` or `pod.status.ip`s are considered for implementing the Kubernetes network model and defining the type of the cluster.

## How to implement the Kubernetes network model

The network model is implemented by the container runtime on each node. The most common container runtimes use [Container Network Interface](#) (CNI) plugins to manage their network and security capabilities. Many different CNI plugins exist from many different vendors. Some of these provide only basic features of adding and removing network interfaces, while others provide more sophisticated solutions, such as integration with other container orchestration systems, running multiple CNI plugins, advanced IPAM features etc.

See [this page](#) for a non-exhaustive list of networking addons supported by Kubernetes.

## What's next

The early design of the networking model and its rationale are described in more detail in the [networking design document](#). For future plans and some on-going efforts that aim to improve Kubernetes networking, please refer to the SIG-Network [KEPs](#).

## Logging Architecture

Application logs can help you understand what is happening inside your application. The logs are particularly useful for debugging problems and monitoring cluster activity. Most modern applications have some kind of logging mechanism. Likewise, container engines are designed to support logging. The easiest and most adopted logging method for containerized applications is writing to standard output and standard error streams.

However, the native functionality provided by a container engine or runtime is usually not enough for a complete logging solution.

For example, you may want to access your application's logs if a container crashes, a pod gets evicted, or a node dies.

In a cluster, logs should have a separate storage and lifecycle independent of nodes, pods, or containers. This concept is called [cluster-level logging](#).

Cluster-level logging architectures require a separate backend to store, analyze, and query logs. Kubernetes does not provide a native storage solution for log data. Instead, there are many logging solutions that integrate with Kubernetes. The following sections describe how to handle and store logs on nodes.

## Pod and container logs

Kubernetes captures logs from each container in a running Pod.

This example uses a manifest for a Pod with a container that writes text to the standard output stream, once per second.

[debug/counter-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: counter
spec:
 containers:
 - name: count
 image: busybox:1.28
 args: [/bin/sh, -c,
 'i=0; while true; do echo "$i: $(date)"; i=$((i+1)); sleep 1; done']
```

To run this pod, use the following command:

```
kubectl apply -f https://k8s.io/examples/debug/counter-pod.yaml
```

The output is:

```
pod/counter created
```

To fetch the logs, use the `kubectl logs` command, as follows:

```
kubectl logs counter
```

The output is similar to:

```
0: Fri Apr 1 11:42:23 UTC 2022
1: Fri Apr 1 11:42:24 UTC 2022
2: Fri Apr 1 11:42:25 UTC 2022
```

You can use `kubectl logs --previous` to retrieve logs from a previous instantiation of a container. If your pod has multiple containers, specify which container's logs you want to access by appending a container name to the command, with a `-c` flag, like so:

```
kubectl logs counter -c count
```

See the [kubectl logs documentation](#) for more details.

## How nodes handle container logs

### Node level logging

A container runtime handles and redirects any output generated to a containerized application's stdout and stderr streams. Different container runtimes implement this in different ways; however, the integration with the kubelet is standardized as the *CRI logging format*.

By default, if a container restarts, the kubelet keeps one terminated container with its logs. If a pod is evicted from the node, all corresponding containers are also evicted, along with their logs.

The kubelet makes logs available to clients via a special feature of the Kubernetes API. The usual way to access this is by running `kubectl logs`.

## Log rotation

FEATURE STATE: Kubernetes v1.21 [stable]

The kubelet is responsible for rotating container logs and managing the logging directory structure. The kubelet sends this information to the container runtime (using CRI), and the runtime writes the container logs to the given location.

You can configure two kubelet [configuration settings](#), `containerLogMaxSize` (default 10Mi) and `containerLogMaxFiles` (default 5), using the [kubelet configuration file](#). These settings let you configure the maximum size for each log file and the maximum number of files allowed for each container respectively.

In order to perform an efficient log rotation in clusters where the volume of the logs generated by the workload is large, kubelet also provides a mechanism to tune how the logs are rotated in terms of how many concurrent log rotations can be performed and the interval at which the logs are monitored and rotated as required. You can configure two kubelet [configuration settings](#), `containerLogMaxWorkers` and `containerLogMonitorInterval` using the [kubelet configuration file](#).

When you run `kubectl logs` as in the basic logging example, the kubelet on the node handles the request and reads directly from the log file. The kubelet returns the content of the log file.

### Note:

Only the contents of the latest log file are available through `kubectl logs`.

For example, if a Pod writes 40 MiB of logs and the kubelet rotates logs after 10 MiB, running `kubectl logs` returns at most 10MiB of data.

# System component logs

There are two types of system components: those that typically run in a container, and those components directly involved in running containers. For example:

- The kubelet and container runtime do not run in containers. The kubelet runs your containers (grouped together in [pods](#))
- The Kubernetes scheduler, controller manager, and API server run within pods (usually [static Pods](#)). The etcd component runs in the control plane, and most commonly also as a static pod. If your cluster uses kube-proxy, you typically run this as a DaemonSet.

## Log locations

The way that the kubelet and container runtime write logs depends on the operating system that the node uses:

- [Linux](#)
- [Windows](#)

On Linux nodes that use systemd, the kubelet and container runtime write to journald by default. You use journalctl to read the systemd journal; for example: journalctl -u kubelet.

If systemd is not present, the kubelet and container runtime write to .log files in the /var/log directory. If you want to have logs written elsewhere, you can indirectly run the kubelet via a helper tool, kube-log-runner, and use that tool to redirect kubelet logs to a directory that you choose.

By default, kubelet directs your container runtime to write logs into directories within /var/log/pods.

For more information on kube-log-runner, read [System Logs](#).

By default, the kubelet writes logs to files within the directory C:\var\logs (notice that this is not C:\var\log).

Although C:\var\log is the Kubernetes default location for these logs, several cluster deployment tools set up Windows nodes to log to C:\var\log\kubelet instead.

If you want to have logs written elsewhere, you can indirectly run the kubelet via a helper tool, kube-log-runner, and use that tool to redirect kubelet logs to a directory that you choose.

However, by default, kubelet directs your container runtime to write logs within the directory C:\var\log\pods.

For more information on kube-log-runner, read [System Logs](#).

For Kubernetes cluster components that run in pods, these write to files inside the /var/log directory, bypassing the default logging mechanism (the components do not write to the systemd journal). You can use Kubernetes' storage mechanisms to map persistent storage into the container that runs the component.

Kubelet allows changing the pod logs directory from default `/var/log/pods` to a custom path. This adjustment can be made by configuring the `podLogsDir` parameter in the kubelet's configuration file.

### Caution:

It's important to note that the default location `/var/log/pods` has been in use for an extended period and certain processes might implicitly assume this path. Therefore, altering this parameter must be approached with caution and at your own risk.

Another caveat to keep in mind is that the kubelet supports the location being on the same disk as `/var`. Otherwise, if the logs are on a separate filesystem from `/var`, then the kubelet will not track that filesystem's usage, potentially leading to issues if it fills up.

For details about etcd and its logs, view the [etcd documentation](#). Again, you can use Kubernetes' storage mechanisms to map persistent storage into the container that runs the component.

### Note:

If you deploy Kubernetes cluster components (such as the scheduler) to log to a volume shared from the parent node, you need to consider and ensure that those logs are rotated. **Kubernetes does not manage that log rotation.**

Your operating system may automatically implement some log rotation - for example, if you share the directory `/var/log` into a static Pod for a component, node-level log rotation treats a file in that directory the same as a file written by any component outside Kubernetes.

Some deploy tools account for that log rotation and automate it; others leave this as your responsibility.

## Cluster-level logging architectures

While Kubernetes does not provide a native solution for cluster-level logging, there are several common approaches you can consider. Here are some options:

- Use a node-level logging agent that runs on every node.
- Include a dedicated sidecar container for logging in an application pod.
- Push logs directly to a backend from within an application.

### Using a node logging agent

Using a node level logging agent

You can implement cluster-level logging by including a *node-level logging agent* on each node. The logging agent is a dedicated tool that exposes logs or pushes logs to a backend. Commonly, the logging agent is a container that has access to a directory with log files from all of the application containers on that node.

Because the logging agent must run on every node, it is recommended to run the agent as a DaemonSet.

Node-level logging creates only one agent per node and doesn't require any changes to the applications running on the node.

Containers write to stdout and stderr, but with no agreed format. A node-level agent collects these logs and forwards them for aggregation.

## Using a sidecar container with the logging agent

You can use a sidecar container in one of the following ways:

- The sidecar container streams application logs to its own stdout.
- The sidecar container runs a logging agent, which is configured to pick up logs from an application container.

### Streaming sidecar container

Sidecar container with a streaming container

By having your sidecar containers write to their own stdout and stderr streams, you can take advantage of the kubelet and the logging agent that already run on each node. The sidecar containers read logs from a file, a socket, or journald. Each sidecar container prints a log to its own stdout or stderr stream.

This approach allows you to separate several log streams from different parts of your application, some of which can lack support for writing to stdout or stderr. The logic behind redirecting logs is minimal, so it's not a significant overhead. Additionally, because stdout and stderr are handled by the kubelet, you can use built-in tools like kubectl logs.

For example, a pod runs a single container, and the container writes to two different log files using two different formats. Here's a manifest for the Pod:

[admin/logging/two-files-counter-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: counter
spec:
 containers:
 - name: count
 image: busybox:1.28
 args:
 - /bin/sh
 - -c
 - >
 i=0;
 while true;
 do
 echo "$i: $(date)" >> /var/log/1.log;
 echo "$(date) INFO $i" >> /var/log/2.log;
 i=$((i+1));
 sleep 1;
 done
```

```
volumeMounts:
- name: varlog
 mountPath: /var/log
volumes:
- name: varlog
 emptyDir: {}
```

It is not recommended to write log entries with different formats to the same log stream, even if you managed to redirect both components to the stdout stream of the container. Instead, you can create two sidecar containers. Each sidecar container could tail a particular log file from a shared volume and then redirect the logs to its own stdout stream.

Here's a manifest for a pod that has two sidecar containers:

[admin/logging/two-files-counter-pod-streaming-sidecar.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: counter
spec:
 containers:
 - name: count
 image: busybox:1.28
 args:
 - /bin/sh
 - -c
 - >
 i=0;
 while true;
 do
 echo "$i: $(date)" >> /var/log/1.log;
 echo "$(date) INFO $i" >> /var/log/2.log;
 i=$((i+1));
 sleep 1;
 done
 volumeMounts:
 - name: varlog
 mountPath: /var/log
 - name: count-log-1
 image: busybox:1.28
 args: [/bin/sh, -c, 'tail -n+1 -F /var/log/1.log']
 volumeMounts:
 - name: varlog
 mountPath: /var/log
 - name: count-log-2
 image: busybox:1.28
 args: [/bin/sh, -c, 'tail -n+1 -F /var/log/2.log']
 volumeMounts:
 - name: varlog
 mountPath: /var/log
 volumes:
```



```
- name: varlog
 emptyDir: {}
```

Now when you run this pod, you can access each log stream separately by running the following commands:

```
kubectrl logs counter count-log-1
```

The output is similar to:

```
0: Fri Apr 1 11:42:26 UTC 2022
1: Fri Apr 1 11:42:27 UTC 2022
2: Fri Apr 1 11:42:28 UTC 2022
...
```

```
kubectrl logs counter count-log-2
```

The output is similar to:

```
Fri Apr 1 11:42:29 UTC 2022 INFO 0
Fri Apr 1 11:42:30 UTC 2022 INFO 0
Fri Apr 1 11:42:31 UTC 2022 INFO 0
...
```

If you installed a node-level agent in your cluster, that agent picks up those log streams automatically without any further configuration. If you like, you can configure the agent to parse log lines depending on the source container.

Even for Pods that only have low CPU and memory usage (order of a couple of millicores for cpu and order of several megabytes for memory), writing logs to a file and then streaming them to stdout can double how much storage you need on the node. If you have an application that writes to a single file, it's recommended to set `/dev/stdout` as the destination rather than implement the streaming sidecar container approach.

Sidecar containers can also be used to rotate log files that cannot be rotated by the application itself. An example of this approach is a small container running `logrotate` periodically. However, it's more straightforward to use `stdout` and `stderr` directly, and leave rotation and retention policies to the kubelet.

### Sidecar container with a logging agent

Sidecar container with a logging agent

If the node-level logging agent is not flexible enough for your situation, you can create a sidecar container with a separate logging agent that you have configured specifically to run with your application.

#### Note:

Using a logging agent in a sidecar container can lead to significant resource consumption. Moreover, you won't be able to access those logs using `kubectrl logs` because they are not controlled by the kubelet.

Here are two example manifests that you can use to implement a sidecar container with a logging agent. The first manifest contains a [ConfigMap](#) to configure fluentd.

[admin/logging/fluentd-sidecar-config.yaml](#)

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: fluentd-config
data:
 fluentd.conf: |
 <source>
 type tail
 format none
 path /var/log/1.log
 pos_file /var/log/1.log.pos
 tag count.format1
 </source>

 <source>
 type tail
 format none
 path /var/log/2.log
 pos_file /var/log/2.log.pos
 tag count.format2
 </source>

 <match **>
 type google_cloud
 </match>
```

**Note:**

In the sample configurations, you can replace fluentd with any logging agent, reading from any source inside an application container.

The second manifest describes a pod that has a sidecar container running fluentd. The pod mounts a volume where fluentd can pick up its configuration data.

[admin/logging/two-files-counter-pod-agent-sidecar.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: counter
spec:
 containers:
 - name: count
 image: busybox:1.28
 args:
 - /bin/sh
 - -c
 - >
```

```
i=0;
while true;
do
 echo "$i: $(date)" >> /var/log/1.log;
 echo "$(date) INFO $i" >> /var/log/2.log;
 i=$((i+1));
 sleep 1;
done
volumeMounts:
- name: varlog
 mountPath: /var/log
- name: count-agent
image: registry.k8s.io/fluentd-gcp:1.30
env:
- name: FLUENTD_ARGS
 value: -c /etc/fluentd-config/fluentd.conf
volumeMounts:
- name: varlog
 mountPath: /var/log
- name: config-volume
 mountPath: /etc/fluentd-config
volumes:
- name: varlog
 emptyDir: {}
- name: config-volume
 configMap:
 name: fluentd-config
```

## Exposing logs directly from the application

Exposing logs directly from the application

Cluster-logging that exposes or pushes logs directly from every application is outside the scope of Kubernetes.

## What's next

- Read about [Kubernetes system logs](#)
- Learn about [Traces For Kubernetes System Components](#)
- Learn how to [customise the termination message](#) that Kubernetes records when a Pod fails

# Metrics For Kubernetes System Components

System component metrics can give a better look into what is happening inside them. Metrics are particularly useful for building dashboards and alerts.

Kubernetes components emit metrics in [Prometheus format](#). This format is structured plain text, designed so that people and machines can both read it.

# Metrics in Kubernetes

In most cases metrics are available on `/metrics` endpoint of the HTTP server. For components that don't expose endpoint by default, it can be enabled using `--bind-address` flag.

Examples of those components:

- [kube-controller-manager](#)
- [kube-proxy](#)
- [kube-apiserver](#)
- [kube-scheduler](#)
- [kubelet](#)

In a production environment you may want to configure [Prometheus Server](#) or some other metrics scraper to periodically gather these metrics and make them available in some kind of time series database.

Note that [kubelet](#) also exposes metrics in `/metrics/cadvisor`, `/metrics/resource` and `/metrics/probes` endpoints. Those metrics do not have the same lifecycle.

If your cluster uses [RBAC](#), reading metrics requires authorization via a user, group or ServiceAccount with a ClusterRole that allows accessing `/metrics`. For example:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: prometheus
rules:
 - nonResourceURLs:
 - "/metrics"
 verbs:
 - get
```

## Metric lifecycle

Alpha metric → Stable metric → Deprecated metric → Hidden metric → Deleted metric

Alpha metrics have no stability guarantees. These metrics can be modified or deleted at any time.

Stable metrics are guaranteed to not change. This means:

- A stable metric without a deprecated signature will not be deleted or renamed
- A stable metric's type will not be modified

Deprecated metrics are slated for deletion, but are still available for use. These metrics include an annotation about the version in which they became deprecated.

For example:

- Before deprecation

```
HELP some_counter this counts things
TYPE some_counter counter
some_counter 0
```

- After deprecation

```
HELP some_counter (Deprecated since 1.15.0) this counts things
TYPE some_counter counter
some_counter 0
```

Hidden metrics are no longer published for scraping, but are still available for use. To use a hidden metric, please refer to the [Show hidden metrics](#) section.

Deleted metrics are no longer published and cannot be used.

## Show hidden metrics

As described above, admins can enable hidden metrics through a command-line flag on a specific binary. This intends to be used as an escape hatch for admins if they missed the migration of the metrics deprecated in the last release.

The flag `show-hidden-metrics-for-version` takes a version for which you want to show metrics deprecated in that release. The version is expressed as `x.y`, where `x` is the major version, `y` is the minor version. The patch version is not needed even though a metrics can be deprecated in a patch release, the reason for that is the metrics deprecation policy runs against the minor release.

The flag can only take the previous minor version as its value. All metrics hidden in previous will be emitted if admins set the previous version to `show-hidden-metrics-for-version`. The too old version is not allowed because this violates the metrics deprecated policy.

Take metric `A` as an example, here assumed that `A` is deprecated in `1.n`. According to metrics deprecated policy, we can reach the following conclusion:

- In release `1.n`, the metric is deprecated, and it can be emitted by default.
- In release `1.n+1`, the metric is hidden by default and it can be emitted by command line `show-hidden-metrics-for-version=1.n`.
- In release `1.n+2`, the metric should be removed from the codebase. No escape hatch anymore.

If you're upgrading from release `1.12` to `1.13`, but still depend on a metric `A` deprecated in `1.12`, you should set hidden metrics via command line: `--show-hidden-metrics=1.12` and remember to remove this metric dependency before upgrading to `1.14`

## Component metrics

### kube-controller-manager metrics

Controller manager metrics provide important insight into the performance and health of the controller manager. These metrics include common Go language runtime metrics such as `go_routine` count and controller specific metrics such as `etcd` request latencies or Cloudprovider (AWS, GCE, OpenStack) API latencies that can be used to gauge the health of a cluster.

Starting from Kubernetes 1.7, detailed Cloudprovider metrics are available for storage operations for GCE, AWS, Vsphere and OpenStack. These metrics can be used to monitor health of persistent volume operations.

For example, for GCE these metrics are called:

```
cloudprovider_gce_api_request_duration_seconds { request = "instance_list"}
cloudprovider_gce_api_request_duration_seconds { request = "disk_insert"}
cloudprovider_gce_api_request_duration_seconds { request = "disk_delete"}
cloudprovider_gce_api_request_duration_seconds { request = "attach_disk"}
cloudprovider_gce_api_request_duration_seconds { request = "detach_disk"}
cloudprovider_gce_api_request_duration_seconds { request = "list_disk"}
```

## kube-scheduler metrics

FEATURE STATE: Kubernetes v1.21 [beta]

The scheduler exposes optional metrics that reports the requested resources and the desired limits of all running pods. These metrics can be used to build capacity planning dashboards, assess current or historical scheduling limits, quickly identify workloads that cannot schedule due to lack of resources, and compare actual usage to the pod's request.

The kube-scheduler identifies the resource [requests and limits](#) configured for each Pod; when either a request or limit is non-zero, the kube-scheduler reports a metrics timeseries. The time series is labelled by:

- namespace
- pod name
- the node where the pod is scheduled or an empty string if not yet scheduled
- priority
- the assigned scheduler for that pod
- the name of the resource (for example, cpu)
- the unit of the resource if known (for example, cores)

Once a pod reaches completion (has a restartPolicy of Never or OnFailure and is in the Succeeded or Failed pod phase, or has been deleted and all containers have a terminated state) the series is no longer reported since the scheduler is now free to schedule other pods to run. The two metrics are called kube\_pod\_resource\_request and kube\_pod\_resource\_limit.

The metrics are exposed at the HTTP endpoint /metrics/resources and require the same authorization as the /metrics endpoint on the scheduler. You must use the --show-hidden-metrics-for-version=1.20 flag to expose these alpha stability metrics.

## Disabling metrics

You can explicitly turn off metrics via command line flag --disabled-metrics. This may be desired if, for example, a metric is causing a performance problem. The input is a list of disabled metrics (i.e. --disabled-metrics=metric1,metric2).

## Metric cardinality enforcement

Metrics with unbounded dimensions could cause memory issues in the components they instrument. To limit resource use, you can use the `--allow-label-value` command line option to dynamically configure an allow-list of label values for a metric.

In alpha stage, the flag can only take in a series of mappings as metric label allow-list. Each mapping is of the format `<metric_name>,<label_name>=<allowed_labels>` where `<allowed_labels>` is a comma-separated list of acceptable label names.

The overall format looks like:

```
--allow-label-value <metric_name>,<label_name>='<allow_value1>, <allow_value2>...',
<metric_name2>,<label_name>='<allow_value1>, <allow_value2>...', ...
```

Here is an example:

```
--allow-label-value number_count_metric,odd_number='1,3,5',
number_count_metric,even_number='2,4,6', date_gauge_metric,weekend='Saturday,Sunday'
```

In addition to specifying this from the CLI, this can also be done within a configuration file. You can specify the path to that configuration file using the `--allow-metric-labels-manifest` command line argument to a component. Here's an example of the contents of that configuration file:

```
allow-list:
- "metric1,label2": "v1,v2,v3"
- "metric2,label1": "v1,v2,v3"
```

Additionally, the `cardinality_enforcement_unexpected_categorizations_total` meta-metric records the count of unexpected categorizations during cardinality enforcement, that is, whenever a label value is encountered that is not allowed with respect to the allow-list constraints.

## What's next

- Read about the [Prometheus text format](#) for metrics
- See the list of [stable Kubernetes metrics](#)
- Read about the [Kubernetes deprecation policy](#)

## Metrics for Kubernetes Object States

kube-state-metrics, an add-on agent to generate and expose cluster-level metrics.

The state of Kubernetes objects in the Kubernetes API can be exposed as metrics. An add-on agent called [kube-state-metrics](#) can connect to the Kubernetes API server and expose a HTTP endpoint with metrics generated from the state of individual objects in the cluster. It exposes various information about the state of objects like labels and annotations, startup and termination times, status or the phase the object currently is in. For example, containers running in pods create a `kube_pod_container_info` metric. This includes the name of the container, the name of the pod it is part of, the [namespace](#) the pod is running in, the name of

the container image, the ID of the image, the image name from the spec of the container, the ID of the running container and the ID of the pod as labels.

This item links to a third party project or product that is not part of Kubernetes itself. [More information](#)

An external component that is able and capable to scrape the endpoint of kube-state-metrics (for example via Prometheus) can now be used to enable the following use cases.

## Example: using metrics from kube-state-metrics to query the cluster state

Metric series generated by kube-state-metrics are helpful to gather further insights into the cluster, as they can be used for querying.

If you use Prometheus or another tool that uses the same query language, the following PromQL query returns the number of pods that are not ready:

```
count(kube_pod_status_ready{condition="false"}) by (namespace, pod)
```

## Example: alerting based on from kube-state-metrics

Metrics generated from kube-state-metrics also allow for alerting on issues in the cluster.

If you use Prometheus or a similar tool that uses the same alert rule language, the following alert will fire if there are pods that have been in a Terminating state for more than 5 minutes:

```
groups:
- name: Pod state
 rules:
 - alert: PodsBlockedInTerminatingState
 expr: count(kube_pod_deletion_timestamp) by (namespace, pod) *
count(kube_pod_status_reason{reason="NodeLost"} == 0) by (namespace, pod) > 0
 for: 5m
 labels:
 severity: page
 annotations:
 summary: Pod {{$labels.namespace}}/{{$labels.pod}} blocked in Terminating state.
```

## System Logs

System component logs record events happening in cluster, which can be very useful for debugging. You can configure log verbosity to see more or less detail. Logs can be as coarse-grained as showing errors within a component, or as fine-grained as showing step-by-step traces of events (like HTTP access logs, pod state changes, controller actions, or scheduler decisions).



## Warning:

In contrast to the command line flags described here, the *log output* itself does *not* fall under the Kubernetes API stability guarantees: individual log entries and their formatting may change from one release to the next!

## Klog

klog is the Kubernetes logging library. [klog](#) generates log messages for the Kubernetes system components.

Kubernetes is in the process of simplifying logging in its components. The following klog command line flags [are deprecated](#) starting with Kubernetes v1.23 and removed in Kubernetes v1.26:

- `--add-dir-header`
- `--alsologtostderr`
- `--log-backtrace-at`
- `--log-dir`
- `--log-file`
- `--log-file-max-size`
- `--logtostderr`
- `--one-output`
- `--skip-headers`
- `--skip-log-headers`
- `--stderrthreshold`

Output will always be written to stderr, regardless of the output format. Output redirection is expected to be handled by the component which invokes a Kubernetes component. This can be a POSIX shell or a tool like systemd.

In some cases, for example a distroless container or a Windows system service, those options are not available. Then the [kube-log-runner](#) binary can be used as wrapper around a Kubernetes component to redirect output. A prebuilt binary is included in several Kubernetes base images under its traditional name as `/go-runner` and as `kube-log-runner` in server and node release archives.

This table shows how kube-log-runner invocations correspond to shell redirection:

| Usage                                    | POSIX shell (such as bash)                 | kube-log-runner <options> <cmd>                                        |
|------------------------------------------|--------------------------------------------|------------------------------------------------------------------------|
| Merge stderr and stdout, write to stdout | <code>2&gt;&amp;1</code>                   | <code>kube-log-runner</code> (default behavior)                        |
| Redirect both into log file              | <code>1&gt;&gt;/tmp/log 2&gt;&amp;1</code> | <code>kube-log-runner -log-file=/tmp/log</code>                        |
| Copy into log file and to stdout         | <code>2&gt;&amp;1   tee -a /tmp/log</code> | <code>kube-log-runner -log-file=/tmp/log -also-stdout</code>           |
| Redirect only stdout into log file       | <code>&gt;/tmp/log</code>                  | <code>kube-log-runner -log-file=/tmp/log -redirect-stderr=false</code> |

## Klog output

An example of the traditional klog native format:

```
I1025 00:15:15.525108 1 httplog.go:79] GET /api/v1/namespaces/kube-system/pods/metrics-server-v0.3.1-57c75779f-9p8wg: (1.512ms) 200 [pod_nanny/v0.0.0 (linux/amd64) kubernetes/$Format 10.56.1.19:51756]
```

The message string may contain line breaks:

```
I1025 00:15:15.525108 1 example.go:79] This is a message
which has a line break.
```

## Structured Logging

FEATURE STATE: Kubernetes v1.23 [beta]

### Warning:

Migration to structured log messages is an ongoing process. Not all log messages are structured in this version. When parsing log files, you must also handle unstructured log messages.

Log formatting and value serialization are subject to change.

Structured logging introduces a uniform structure in log messages allowing for programmatic extraction of information. You can store and process structured logs with less effort and cost. The code which generates a log message determines whether it uses the traditional unstructured klog output or structured logging.

The default formatting of structured log messages is as text, with a format that is backward compatible with traditional klog:

```
<klog header> "<message>" <key1>=<value1> <key2>=<value2>" ...
```

Example:

```
I1025 00:15:15.525108 1 controller_utils.go:116] "Pod status updated" pod="kube-system/
kubedns" status="ready"
```

Strings are quoted. Other values are formatted with [%+v](#), which may cause log messages to continue on the next line [depending on the data](#).

```
I1025 00:15:15.525108 1 example.go:116] "Example" data="This is text with a line break\nand
\"quotation marks\".\" someInt=1 someFloat=0.1 someStruct={StringField: First line,
second line.}
```

## Contextual Logging

FEATURE STATE: Kubernetes v1.30 [beta]

Contextual logging builds on top of structured logging. It is primarily about how developers use logging calls: code based on that concept is more flexible and supports additional use cases as described in the [Contextual Logging KEP](#).

If developers use additional functions like `WithValues` or `WithName` in their components, then log entries contain additional information that gets passed into functions by their caller.

For Kubernetes 1.30, this is gated behind the ContextualLogging [feature gate](#) and is enabled by default. The infrastructure for this was added in 1.24 without modifying components. The [component-base/logs/example](#) command demonstrates how to use the new logging calls and how a component behaves that supports contextual logging.

```
$ cd $GOPATH/src/k8s.io/kubernetes/staging/src/k8s.io/component-base/logs/example/cmd/
$ go run . --help
...
 --feature-gates mapStringBool A set of key=value pairs that describe feature gates for
alpha/experimental features. Options are:
 AllAlpha=true|false (ALPHA - default=false)
 AllBeta=true|false (BETA - default=false)
 ContextualLogging=true|false (BETA - default=true)
$ go run . --feature-gates ContextualLogging=true
...
I0222 15:13:31.645988 197901 example.go:54] "runtime" logger="example.myname" foo="bar"
duration="1m0s"
I0222 15:13:31.646007 197901 example.go:55] "another runtime" logger="example" foo="bar"
duration="1h0m0s" duration="1m0s"
```

The logger key and foo="bar" were added by the caller of the function which logs the runtime message and duration="1m0s" value, without having to modify that function.

With contextual logging disable, WithValues and WithName do nothing and log calls go through the global klog logger. Therefore this additional information is not in the log output anymore:

```
$ go run . --feature-gates ContextualLogging=false
...
I0222 15:14:40.497333 198174 example.go:54] "runtime" duration="1m0s"
I0222 15:14:40.497346 198174 example.go:55] "another runtime" duration="1h0m0s"
duration="1m0s"
```

## JSON log format

FEATURE STATE: Kubernetes v1.19 [alpha]

### Warning:

JSON output does not support many standard klog flags. For list of unsupported klog flags, see the [Command line tool reference](#).

Not all logs are guaranteed to be written in JSON format (for example, during process start). If you intend to parse logs, make sure you can handle log lines that are not JSON as well.

Field names and JSON serialization are subject to change.

The --logging-format=json flag changes the format of logs from klog native format to JSON format. Example of JSON log format (pretty printed):

```
{
 "ts": 1580306777.04728,
 "v": 4,
```

```
"msg": "Pod status updated",
"pod":{
 "name": "nginx-1",
 "namespace": "default"
},
"status": "ready"
}
```

Keys with special meaning:

- `ts` - timestamp as Unix time (required, float)
- `v` - verbosity (only for info and not for error messages, int)
- `err` - error string (optional, string)
- `msg` - message (required, string)

List of components currently supporting JSON format:

- [kube-controller-manager](#)
- [kube-apiserver](#)
- [kube-scheduler](#)
- [kubelet](#)

## Log verbosity level

The `-v` flag controls log verbosity. Increasing the value increases the number of logged events. Decreasing the value decreases the number of logged events. Increasing verbosity settings logs increasingly less severe events. A verbosity setting of 0 logs only critical events.

## Log location

There are two types of system components: those that run in a container and those that do not run in a container. For example:

- The Kubernetes scheduler and kube-proxy run in a container.
- The kubelet and [container runtime](#) do not run in containers.

On machines with systemd, the kubelet and container runtime write to journald. Otherwise, they write to `.log` files in the `/var/log` directory. System components inside containers always write to `.log` files in the `/var/log` directory, bypassing the default logging mechanism. Similar to the container logs, you should rotate system component logs in the `/var/log` directory. In Kubernetes clusters created by the `kube-up.sh` script, log rotation is configured by the `logrotate` tool. The `logrotate` tool rotates logs daily, or once the log size is greater than 100MB.

## Log query

FEATURE STATE: Kubernetes v1.30 [beta]

To help with debugging issues on nodes, Kubernetes v1.27 introduced a feature that allows viewing logs of services running on the node. To use the feature, ensure that the `NodeLogQuery` [feature gate](#) is enabled for that node, and that the kubelet configuration options `enableSystemLogHandler` and `enableSystemLogQuery` are both set to true. On Linux the assumption is that service logs are available via journald. On Windows the assumption is that

service logs are available in the application log provider. On both operating systems, logs are also available by reading files within `/var/log/`.

Provided you are authorized to interact with node objects, you can try out this feature on all your nodes or just a subset. Here is an example to retrieve the kubelet service logs from a node:

```
Fetch kubelet logs from a node named node-1.example
kubectl get --raw "/api/v1/nodes/node-1.example/proxy/logs/?query=kubelet"
```

You can also fetch files, provided that the files are in a directory that the kubelet allows for log fetches. For example, you can fetch a log from `/var/log` on a Linux node:

```
kubectl get --raw "/api/v1/nodes/<insert-node-name-here>/proxy/logs/?query=/<insert-log-file-name-here>"
```

The kubelet uses heuristics to retrieve logs. This helps if you are not aware whether a given system service is writing logs to the operating system's native logger like `journald` or to a log file in `/var/log/`. The heuristics first checks the native logger and if that is not available attempts to retrieve the first logs from `/var/log/<servicename>` or `/var/log/<servicename>.log` or `/var/log/<servicename>/<servicename>.log`.

The complete list of options that can be used are:

| Option    | Description                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------|
| boot      | boot show messages from a specific system boot                                                    |
| pattern   | pattern filters log entries by the provided PERL-compatible regular expression                    |
| query     | query specifies services(s) or files from which to return logs (required)                         |
| sinceTime | an <a href="#">RFC3339</a> timestamp from which to show logs (inclusive)                          |
| untilTime | an <a href="#">RFC3339</a> timestamp until which to show logs (inclusive)                         |
| tailLines | specify how many lines from the end of the log to retrieve; the default is to fetch the whole log |

Example of a more complex query:

```
Fetch kubelet logs from a node named node-1.example that have the word "error"
kubectl get --raw "/api/v1/nodes/node-1.example/proxy/logs/?query=kubelet&pattern=error"
```

## What's next

- Read about the [Kubernetes Logging Architecture](#)
- Read about [Structured Logging](#)
- Read about [Contextual Logging](#)
- Read about [deprecation of klog flags](#)
- Read about the [Conventions for logging severity](#)
- Read about [Log Query](#)

# Traces For Kubernetes System Components

FEATURE STATE: Kubernetes v1.27 [beta]

System component traces record the latency of and relationships between operations in the cluster.

Kubernetes components emit traces using the [OpenTelemetry Protocol](#) with the gRPC exporter and can be collected and routed to tracing backends using an [OpenTelemetry Collector](#).

## Trace Collection

Kubernetes components have built-in gRPC exporters for OTLP to export traces, either with an OpenTelemetry Collector, or without an OpenTelemetry Collector.

For a complete guide to collecting traces and using the collector, see [Getting Started with the OpenTelemetry Collector](#). However, there are a few things to note that are specific to Kubernetes components.

By default, Kubernetes components export traces using the gRPC exporter for OTLP on the [IANA OpenTelemetry port](#), 4317. As an example, if the collector is running as a sidecar to a Kubernetes component, the following receiver configuration will collect spans and log them to standard output:

```
receivers:
 otlp:
 protocols:
 grpc:
exporters:
 # Replace this exporter with the exporter for your backend
 logging:
 logLevel: debug
service:
 pipelines:
 traces:
 receivers: [otlp]
 exporters: [logging]
```

To directly emit traces to a backend without utilizing a collector, specify the endpoint field in the Kubernetes tracing configuration file with the desired trace backend address. This method negates the need for a collector and simplifies the overall structure.

For trace backend header configuration, including authentication details, environment variables can be used with `OTEL_EXPORTER_OTLP_HEADERS`, see [OTLP Exporter Configuration](#).

Additionally, for trace resource attribute configuration such as Kubernetes cluster name, namespace, Pod name, etc., environment variables can also be used with `OTEL_RESOURCE_ATTRIBUTES`, see [OTLP Kubernetes Resource](#).

## Component traces

### kube-apiserver traces

The kube-apiserver generates spans for incoming HTTP requests, and for outgoing requests to webhooks, etcd, and re-entrant requests. It propagates the [W3C Trace Context](#) with outgoing

requests but does not make use of the trace context attached to incoming requests, as the kube-apiserver is often a public endpoint.

## Enabling tracing in the kube-apiserver

To enable tracing, provide the kube-apiserver with a tracing configuration file with `--tracing-config-file=<path-to-config>`. This is an example config that records spans for 1 in 10000 requests, and uses the default OpenTelemetry endpoint:

```
apiVersion: apiserver.config.k8s.io/v1beta1
kind: TracingConfiguration
default value
#endpoint: localhost:4317
samplingRatePerMillion: 100
```

For more information about the TracingConfiguration struct, see [API server config API \(v1beta1\)](#).

## kubelet traces

FEATURE STATE: Kubernetes v1.27 [beta]

The kubelet CRI interface and authenticated http servers are instrumented to generate trace spans. As with the apiserver, the endpoint and sampling rate are configurable. Trace context propagation is also configured. A parent span's sampling decision is always respected. A provided tracing configuration sampling rate will apply to spans without a parent. Enabled without a configured endpoint, the default OpenTelemetry Collector receiver address of "localhost:4317" is set.

## Enabling tracing in the kubelet

To enable tracing, apply the [tracing configuration](#). This is an example snippet of a kubelet config that records spans for 1 in 10000 requests, and uses the default OpenTelemetry endpoint:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
featureGates:
 KubeletTracing: true
tracing:
 # default value
 #endpoint: localhost:4317
 samplingRatePerMillion: 100
```

If the samplingRatePerMillion is set to one million (1000000), then every span will be sent to the exporter.

The kubelet in Kubernetes v1.30 collects spans from the garbage collection, pod synchronization routine as well as every gRPC method. The kubelet propagates trace context with gRPC requests so that container runtimes with trace instrumentation, such as CRI-O and containerd, can associate their exported spans with the trace context from the kubelet. The resulting traces will have parent-child links between kubelet and container runtime spans, providing helpful context when debugging node issues.

Please note that exporting spans always comes with a small performance overhead on the networking and CPU side, depending on the overall configuration of the system. If there is any issue like that in a cluster which is running with tracing enabled, then mitigate the problem by either reducing the `samplingRatePerMillion` or disabling tracing completely by removing the configuration.

## Stability

Tracing instrumentation is still under active development, and may change in a variety of ways. This includes span names, attached attributes, instrumented endpoints, etc. Until this feature graduates to stable, there are no guarantees of backwards compatibility for tracing instrumentation.

## What's next

- Read about [Getting Started with the OpenTelemetry Collector](#)
- Read about [OTLP Exporter Configuration](#)

# Proxies in Kubernetes

This page explains proxies used with Kubernetes.

## Proxies

There are several different proxies you may encounter when using Kubernetes:

1. The [kubectrl proxy](#):

- runs on a user's desktop or in a pod
- proxies from a localhost address to the Kubernetes apiserver
- client to proxy uses HTTP
- proxy to apiserver uses HTTPS
- locates apiserver
- adds authentication headers

2. The [apiserver proxy](#):

- is a bastion built into the apiserver
- connects a user outside of the cluster to cluster IPs which otherwise might not be reachable
- runs in the apiserver processes
- client to proxy uses HTTPS (or http if apiserver so configured)
- proxy to target may use HTTP or HTTPS as chosen by proxy using available information
- can be used to reach a Node, Pod, or Service
- does load balancing when used to reach a Service

3. The [kube proxy](#):

- runs on each node



- proxies UDP, TCP and SCTP
- does not understand HTTP
- provides load balancing
- is only used to reach services

#### 4. A Proxy/Load-balancer in front of apiserver(s):

- existence and implementation varies from cluster to cluster (e.g. nginx)
- sits between all clients and one or more apiservers
- acts as load balancer if there are several apiservers.

#### 5. Cloud Load Balancers on external services:

- are provided by some cloud providers (e.g. AWS ELB, Google Cloud Load Balancer)
- are created automatically when the Kubernetes service has type LoadBalancer
- usually supports UDP/TCP only
- SCTP support is up to the load balancer implementation of the cloud provider
- implementation varies by cloud provider.

Kubernetes users will typically not need to worry about anything other than the first two types. The cluster admin will typically ensure that the latter types are set up correctly.

## Requesting redirects

Proxies have replaced redirect capabilities. Redirects have been deprecated.

## API Priority and Fairness

FEATURE STATE: Kubernetes v1.29 [stable]

Controlling the behavior of the Kubernetes API server in an overload situation is a key task for cluster administrators. The [kube-apiserver](#) has some controls available (i.e. the `--max-requests-inflight` and `--max-mutating-requests-inflight` command-line flags) to limit the amount of outstanding work that will be accepted, preventing a flood of inbound requests from overloading and potentially crashing the API server, but these flags are not enough to ensure that the most important requests get through in a period of high traffic.

The API Priority and Fairness feature (APF) is an alternative that improves upon aforementioned max-inflight limitations. APF classifies and isolates requests in a more fine-grained way. It also introduces a limited amount of queuing, so that no requests are rejected in cases of very brief bursts. Requests are dispatched from queues using a fair queuing technique so that, for example, a poorly-behaved [controller](#) need not starve others (even at the same priority level).

This feature is designed to work well with standard controllers, which use informers and react to failures of API requests with exponential back-off, and other clients that also work this way.

### Caution:

Some requests classified as "long-running"—such as remote command execution or log tailing—are not subject to the API Priority and Fairness filter. This is also true for the `--max-requests-`

inflight flag without the API Priority and Fairness feature enabled. API Priority and Fairness *does* apply to **watch** requests. When API Priority and Fairness is disabled, **watch** requests are not subject to the `--max-requests-inflight` limit.

## Enabling/Disabling API Priority and Fairness

The API Priority and Fairness feature is controlled by a command-line flag and is enabled by default. See [Options](#) for a general explanation of the available kube-apiserver command-line options and how to enable and disable them. The name of the command-line option for APF is `--enable-priority-and-fairness`. This feature also involves an [API Group](#) with: (a) a stable v1 version, introduced in 1.29, and enabled by default (b) a v1beta3 version, enabled by default, and deprecated in v1.29. You can disable the API group beta version v1beta3 by adding the following command-line flags to your kube-apiserver invocation:

```
kube-apiserver \
--runtime-config=flowcontrol.apiserver.k8s.io/v1beta3=false \
...and other flags as usual
```

The command-line flag `--enable-priority-and-fairness=false` will disable the API Priority and Fairness feature.

## Concepts

There are several distinct features involved in the API Priority and Fairness feature. Incoming requests are classified by attributes of the request using *FlowSchemas*, and assigned to priority levels. Priority levels add a degree of isolation by maintaining separate concurrency limits, so that requests assigned to different priority levels cannot starve each other. Within a priority level, a fair-queuing algorithm prevents requests from different *flows* from starving each other, and allows for requests to be queued to prevent bursty traffic from causing failed requests when the average load is acceptably low.

### Priority Levels

Without APF enabled, overall concurrency in the API server is limited by the kube-apiserver flags `--max-requests-inflight` and `--max-mutating-requests-inflight`. With APF enabled, the concurrency limits defined by these flags are summed and then the sum is divided up among a configurable set of *priority levels*. Each incoming request is assigned to a single priority level, and each priority level will only dispatch as many concurrent requests as its particular limit allows.

The default configuration, for example, includes separate priority levels for leader-election requests, requests from built-in controllers, and requests from Pods. This means that an ill-behaved Pod that floods the API server with requests cannot prevent leader election or actions by the built-in controllers from succeeding.

The concurrency limits of the priority levels are periodically adjusted, allowing under-utilized priority levels to temporarily lend concurrency to heavily-utilized levels. These limits are based on nominal limits and bounds on how much concurrency a priority level may lend and how much it may borrow, all derived from the configuration objects mentioned below.

## Seats Occupied by a Request

The above description of concurrency management is the baseline story. Requests have different durations but are counted equally at any given moment when comparing against a priority level's concurrency limit. In the baseline story, each request occupies one unit of concurrency. The word "seat" is used to mean one unit of concurrency, inspired by the way each passenger on a train or aircraft takes up one of the fixed supply of seats.

But some requests take up more than one seat. Some of these are **list** requests that the server estimates will return a large number of objects. These have been found to put an exceptionally heavy burden on the server. For this reason, the server estimates the number of objects that will be returned and considers the request to take a number of seats that is proportional to that estimated number.

## Execution time tweaks for watch requests

API Priority and Fairness manages **watch** requests, but this involves a couple more excursions from the baseline behavior. The first concerns how long a **watch** request is considered to occupy its seat. Depending on request parameters, the response to a **watch** request may or may not begin with **create** notifications for all the relevant pre-existing objects. API Priority and Fairness considers a **watch** request to be done with its seat once that initial burst of notifications, if any, is over.

The normal notifications are sent in a concurrent burst to all relevant **watch** response streams whenever the server is notified of an object create/update/delete. To account for this work, API Priority and Fairness considers every write request to spend some additional time occupying seats after the actual writing is done. The server estimates the number of notifications to be sent and adjusts the write request's number of seats and seat occupancy time to include this extra work.

## Queuing

Even within a priority level there may be a large number of distinct sources of traffic. In an overload situation, it is valuable to prevent one stream of requests from starving others (in particular, in the relatively common case of a single buggy client flooding the kube-apiserver with requests, that buggy client would ideally not have much measurable impact on other clients at all). This is handled by use of a fair-queuing algorithm to process requests that are assigned the same priority level. Each request is assigned to a *flow*, identified by the name of the matching FlowSchema plus a *flow distinguisher* — which is either the requesting user, the target resource's namespace, or nothing — and the system attempts to give approximately equal weight to requests in different flows of the same priority level. To enable distinct handling of distinct instances, controllers that have many instances should authenticate with distinct usernames

After classifying a request into a flow, the API Priority and Fairness feature then may assign the request to a queue. This assignment uses a technique known as [shuffle sharding](#), which makes relatively efficient use of queues to insulate low-intensity flows from high-intensity flows.

The details of the queuing algorithm are tunable for each priority level, and allow administrators to trade off memory use, fairness (the property that independent flows will all make progress when total traffic exceeds capacity), tolerance for bursty traffic, and the added latency induced by queuing.

## Exempt requests

Some requests are considered sufficiently important that they are not subject to any of the limitations imposed by this feature. These exemptions prevent an improperly-configured flow control configuration from totally disabling an API server.

## Resources

The flow control API involves two kinds of resources. [PriorityLevelConfigurations](#) define the available priority levels, the share of the available concurrency budget that each can handle, and allow for fine-tuning queuing behavior. [FlowSchemas](#) are used to classify individual inbound requests, matching each to a single PriorityLevelConfiguration.

### PriorityLevelConfiguration

A PriorityLevelConfiguration represents a single priority level. Each PriorityLevelConfiguration has an independent limit on the number of outstanding requests, and limitations on the number of queued requests.

The nominal concurrency limit for a PriorityLevelConfiguration is not specified in an absolute number of seats, but rather in "nominal concurrency shares." The total concurrency limit for the API Server is distributed among the existing PriorityLevelConfigurations in proportion to these shares, to give each level its nominal limit in terms of seats. This allows a cluster administrator to scale up or down the total amount of traffic to a server by restarting kube-apiserver with a different value for `--max-requests-inflight` (or `--max-mutating-requests-inflight`), and all PriorityLevelConfigurations will see their maximum allowed concurrency go up (or down) by the same fraction.

#### Caution:

In the versions before v1beta3 the relevant PriorityLevelConfiguration field is named "assured concurrency shares" rather than "nominal concurrency shares". Also, in Kubernetes release 1.25 and earlier there were no periodic adjustments: the nominal/assured limits were always applied without adjustment.

The bounds on how much concurrency a priority level may lend and how much it may borrow are expressed in the PriorityLevelConfiguration as percentages of the level's nominal limit. These are resolved to absolute numbers of seats by multiplying with the nominal limit / 100.0 and rounding. The dynamically adjusted concurrency limit of a priority level is constrained to lie between (a) a lower bound of its nominal limit minus its lendable seats and (b) an upper bound of its nominal limit plus the seats it may borrow. At each adjustment the dynamic limits are derived by each priority level reclaiming any lent seats for which demand recently appeared and then jointly fairly responding to the recent seat demand on the priority levels, within the bounds just described.

#### Caution:

With the Priority and Fairness feature enabled, the total concurrency limit for the server is set to the sum of `--max-requests-inflight` and `--max-mutating-requests-inflight`. There is no longer any distinction made between mutating and non-mutating requests; if you want to treat them

separately for a given resource, make separate FlowSchemas that match the mutating and non-mutating verbs respectively.

When the volume of inbound requests assigned to a single PriorityLevelConfiguration is more than its permitted concurrency level, the type field of its specification determines what will happen to extra requests. A type of Reject means that excess traffic will immediately be rejected with an HTTP 429 (Too Many Requests) error. A type of Queue means that requests above the threshold will be queued, with the shuffle sharding and fair queuing techniques used to balance progress between request flows.

The queuing configuration allows tuning the fair queuing algorithm for a priority level. Details of the algorithm can be read in the [enhancement proposal](#), but in short:

- Increasing queues reduces the rate of collisions between different flows, at the cost of increased memory usage. A value of 1 here effectively disables the fair-queuing logic, but still allows requests to be queued.
- Increasing queueLengthLimit allows larger bursts of traffic to be sustained without dropping any requests, at the cost of increased latency and memory usage.
- Changing handSize allows you to adjust the probability of collisions between different flows and the overall concurrency available to a single flow in an overload situation.

**Note:**

A larger handSize makes it less likely for two individual flows to collide (and therefore for one to be able to starve the other), but more likely that a small number of flows can dominate the apiserver. A larger handSize also potentially increases the amount of latency that a single high-traffic flow can cause. The maximum number of queued requests possible from a single flow is  $\text{handSize} * \text{queueLengthLimit}$ .

Following is a table showing an interesting collection of shuffle sharding configurations, showing for each the probability that a given mouse (low-intensity flow) is squished by the elephants (high-intensity flows) for an illustrative collection of numbers of elephants. See <https://play.golang.org/p/Gi0PLgVHiUg> , which computes this table.

Example Shuffle Sharding Configurations

| HandSize | Queues | 1 elephant             | 4 elephants            | 16 elephants          |
|----------|--------|------------------------|------------------------|-----------------------|
| 12       | 32     | 4.428838398950118e-09  | 0.11431348830099144    | 0.9935089607656024    |
| 10       | 32     | 1.550093439632541e-08  | 0.0626479840223545     | 0.9753101519027554    |
| 10       | 64     | 6.601827268370426e-12  | 0.00045571320990370776 | 0.49999929150089345   |
| 9        | 64     | 3.6310049976037345e-11 | 0.00045501212304112273 | 0.4282314876454858    |
| 8        | 64     | 2.25929199850899e-10   | 0.0004886697053040446  | 0.35935114681123076   |
| 8        | 128    | 6.994461389026097e-13  | 3.4055790161620863e-06 | 0.02746173137155063   |
| 7        | 128    | 1.0579122850901972e-11 | 6.960839379258192e-06  | 0.02406157386340147   |
| 7        | 256    | 7.597695465552631e-14  | 6.728547142019406e-08  | 0.0006709661542533682 |
| 6        | 256    | 2.7134626662687968e-12 | 2.9516464018476436e-07 | 0.0008895654642000348 |
| 6        | 512    | 4.116062922897309e-14  | 4.982983350480894e-09  | 2.26025764343413e-05  |
| 6        | 1024   | 6.337324016514285e-16  | 8.09060164312957e-11   | 4.517408062903668e-07 |

## FlowSchema

A FlowSchema matches some inbound requests and assigns them to a priority level. Every inbound request is tested against FlowSchemas, starting with those with the numerically lowest matchingPrecedence and working upward. The first match wins.

### Caution:

Only the first matching FlowSchema for a given request matters. If multiple FlowSchemas match a single inbound request, it will be assigned based on the one with the highest matchingPrecedence. If multiple FlowSchemas with equal matchingPrecedence match the same request, the one with lexicographically smaller name will win, but it's better not to rely on this, and instead to ensure that no two FlowSchemas have the same matchingPrecedence.

A FlowSchema matches a given request if at least one of its rules matches. A rule matches if at least one of its subjects *and* at least one of its resourceRules or nonResourceRules (depending on whether the incoming request is for a resource or non-resource URL) match the request.

For the name field in subjects, and the verbs, apiGroups, resources, namespaces, and nonResourceURLs fields of resource and non-resource rules, the wildcard \* may be specified to match all values for the given field, effectively removing it from consideration.

A FlowSchema's distinguisherMethod.type determines how requests matching that schema will be separated into flows. It may be ByUser, in which one requesting user will not be able to starve other users of capacity; ByNamespace, in which requests for resources in one namespace will not be able to starve requests for resources in other namespaces of capacity; or blank (or distinguisherMethod may be omitted entirely), in which all requests matched by this FlowSchema will be considered part of a single flow. The correct choice for a given FlowSchema depends on the resource and your particular environment.

## Defaults

Each kube-apiserver maintains two sorts of APF configuration objects: mandatory and suggested.

### Mandatory Configuration Objects

The four mandatory configuration objects reflect fixed built-in guardrail behavior. This is behavior that the servers have before those objects exist, and when those objects exist their specs reflect this behavior. The four mandatory objects are as follows.

- The mandatory exempt priority level is used for requests that are not subject to flow control at all: they will always be dispatched immediately. The mandatory exempt FlowSchema classifies all requests from the system:masters group into this priority level. You may define other FlowSchemas that direct other requests to this priority level, if appropriate.
- The mandatory catch-all priority level is used in combination with the mandatory catch-all FlowSchema to make sure that every request gets some kind of classification. Typically you should not rely on this catch-all configuration, and should create your own catch-all FlowSchema and PriorityLevelConfiguration (or use the suggested global-default priority level that is installed by default) as appropriate. Because it is not expected to be

used normally, the mandatory catch-all priority level has a very small concurrency share and does not queue requests.

## Suggested Configuration Objects

The suggested FlowSchemas and PriorityLevelConfigurations constitute a reasonable default configuration. You can modify these and/or create additional configuration objects if you want. If your cluster is likely to experience heavy load then you should consider what configuration will work best.

The suggested configuration groups requests into six priority levels:

- The node-high priority level is for health updates from nodes.
- The system priority level is for non-health requests from the system:nodes group, i.e. Kubelets, which must be able to contact the API server in order for workloads to be able to schedule on them.
- The leader-election priority level is for leader election requests from built-in controllers (in particular, requests for endpoints, configmaps, or leases coming from the system:kube-controller-manager or system:kube-scheduler users and service accounts in the kube-system namespace). These are important to isolate from other traffic because failures in leader election cause their controllers to fail and restart, which in turn causes more expensive traffic as the new controllers sync their informers.
- The workload-high priority level is for other requests from built-in controllers.
- The workload-low priority level is for requests from any other service account, which will typically include all requests from controllers running in Pods.
- The global-default priority level handles all other traffic, e.g. interactive kubectl commands run by nonprivileged users.

The suggested FlowSchemas serve to steer requests into the above priority levels, and are not enumerated here.

## Maintenance of the Mandatory and Suggested Configuration Objects

Each kube-apiserver independently maintains the mandatory and suggested configuration objects, using initial and periodic behavior. Thus, in a situation with a mixture of servers of different versions there may be thrashing as long as different servers have different opinions of the proper content of these objects.

Each kube-apiserver makes an initial maintenance pass over the mandatory and suggested configuration objects, and after that does periodic maintenance (once per minute) of those objects.

For the mandatory configuration objects, maintenance consists of ensuring that the object exists and, if it does, has the proper spec. The server refuses to allow a creation or update with a spec that is inconsistent with the server's guardrail behavior.

Maintenance of suggested configuration objects is designed to allow their specs to be overridden. Deletion, on the other hand, is not respected: maintenance will restore the object. If you do not want a suggested configuration object then you need to keep it around but set its



spec to have minimal consequences. Maintenance of suggested objects is also designed to support automatic migration when a new version of the kube-apiserver is rolled out, albeit potentially with thrashing while there is a mixed population of servers.

Maintenance of a suggested configuration object consists of creating it --- with the server's suggested spec --- if the object does not exist. OTOH, if the object already exists, maintenance behavior depends on whether the kube-apiservers or the users control the object. In the former case, the server ensures that the object's spec is what the server suggests; in the latter case, the spec is left alone.

The question of who controls the object is answered by first looking for an annotation with key `apf.kubernetes.io/autoupdate-spec`. If there is such an annotation and its value is true then the kube-apiservers control the object. If there is such an annotation and its value is false then the users control the object. If neither of those conditions holds then the `metadata.generation` of the object is consulted. If that is 1 then the kube-apiservers control the object. Otherwise the users control the object. These rules were introduced in release 1.22 and their consideration of `metadata.generation` is for the sake of migration from the simpler earlier behavior. Users who wish to control a suggested configuration object should set its `apf.kubernetes.io/autoupdate-spec` annotation to false.

Maintenance of a mandatory or suggested configuration object also includes ensuring that it has an `apf.kubernetes.io/autoupdate-spec` annotation that accurately reflects whether the kube-apiservers control the object.

Maintenance also includes deleting objects that are neither mandatory nor suggested but are annotated `apf.kubernetes.io/autoupdate-spec=true`.

## Health check concurrency exemption

The suggested configuration gives no special treatment to the health check requests on kube-apiservers from their local kubelets --- which tend to use the secured port but supply no credentials. With the suggested config, these requests get assigned to the global-default `FlowSchema` and the corresponding global-default priority level, where other traffic can crowd them out.

If you add the following additional `FlowSchema`, this exempts those requests from rate limiting.

### Caution:

Making this change also allows any hostile party to then send health-check requests that match this `FlowSchema`, at any volume they like. If you have a web traffic filter or similar external security mechanism to protect your cluster's API server from general internet traffic, you can configure rules to block any health check requests that originate from outside your cluster.

[priority-and-fairness/health-for-strangers.yaml](#)

```
apiVersion: flowcontrol.apiserver.k8s.io/v1
kind: FlowSchema
metadata:
 name: health-for-strangers
spec:
 matchingPrecedence: 1000
 priorityLevelConfiguration:
 name: exempt
```



```
rules:
- nonResourceRules:
- nonResourceURLs:
 - "/healthz"
 - "/livez"
 - "/readyz"
verbs:
- "*"
subjects:
- kind: Group
 group:
 name: "system:unauthenticated"
```

## Observability

### Metrics

#### Note:

In versions of Kubernetes before v1.20, the labels `flow_schema` and `priority_level` were inconsistently named `flowSchema` and `priorityLevel`, respectively. If you're running Kubernetes versions v1.19 and earlier, you should refer to the documentation for your version.

When you enable the API Priority and Fairness feature, the kube-apiserver exports additional metrics. Monitoring these can help you determine whether your configuration is inappropriately throttling important traffic, or find poorly-behaved workloads that may be harming system health.

#### Maturity level BETA

- `apiserver_flowcontrol_rejected_requests_total` is a counter vector (cumulative since server start) of requests that were rejected, broken down by the labels `flow_schema` (indicating the one that matched the request), `priority_level` (indicating the one to which the request was assigned), and reason. The reason label will be one of the following values:
  - `queue-full`, indicating that too many requests were already queued.
  - `concurrency-limit`, indicating that the `PriorityLevelConfiguration` is configured to reject rather than queue excess requests.
  - `time-out`, indicating that the request was still in the queue when its queuing time limit expired.
  - `cancelled`, indicating that the request is not purge locked and has been ejected from the queue.
- `apiserver_flowcontrol_dispatched_requests_total` is a counter vector (cumulative since server start) of requests that began executing, broken down by `flow_schema` and `priority_level`.
- `apiserver_flowcontrol_current_inqueue_requests` is a gauge vector holding the instantaneous number of queued (not executing) requests, broken down by `priority_level` and `flow_schema`.

- `apiserver_flowcontrol_current_executing_requests` is a gauge vector holding the instantaneous number of executing (not waiting in a queue) requests, broken down by `priority_level` and `flow_schema`.
- `apiserver_flowcontrol_current_executing_seats` is a gauge vector holding the instantaneous number of occupied seats, broken down by `priority_level` and `flow_schema`.
- `apiserver_flowcontrol_request_wait_duration_seconds` is a histogram vector of how long requests spent queued, broken down by the labels `flow_schema`, `priority_level`, and `execute`. The `execute` label indicates whether the request has started executing.

**Note:**

Since each `FlowSchema` always assigns requests to a single `PriorityLevelConfiguration`, you can add the histograms for all the `FlowSchemas` for one priority level to get the effective histogram for requests assigned to that priority level.

- `apiserver_flowcontrol_nominal_limit_seats` is a gauge vector holding each priority level's nominal concurrency limit, computed from the API server's total concurrency limit and the priority level's configured nominal concurrency shares.

## Maturity level ALPHA

- `apiserver_current_inqueue_requests` is a gauge vector of recent high water marks of the number of queued requests, grouped by a label named `request_kind` whose value is `mutating` or `readOnly`. These high water marks describe the largest number seen in the one second window most recently completed. These complement the older `apiserver_current_inflight_requests` gauge vector that holds the last window's high water mark of number of requests actively being served.
- `apiserver_current_inqueue_seats` is a gauge vector of the sum over queued requests of the largest number of seats each will occupy, grouped by labels named `flow_schema` and `priority_level`.
- `apiserver_flowcontrol_read_vs_write_current_requests` is a histogram vector of observations, made at the end of every nanosecond, of the number of requests broken down by the labels `phase` (which takes on the values `waiting` and `executing`) and `request_kind` (which takes on the values `mutating` and `readOnly`). Each observed value is a ratio, between 0 and 1, of the number of requests divided by the corresponding limit on the number of requests (queue volume limit for waiting and concurrency limit for executing).
- `apiserver_flowcontrol_request_concurrency_in_use` is a gauge vector holding the instantaneous number of occupied seats, broken down by `priority_level` and `flow_schema`.
- `apiserver_flowcontrol_priority_level_request_utilization` is a histogram vector of observations, made at the end of each nanosecond, of the number of requests broken down by the labels `phase` (which takes on the values `waiting` and `executing`) and `priority_level`. Each observed value is a ratio, between 0 and 1, of a number of requests divided by the corresponding limit on the number of requests (queue volume limit for waiting and concurrency limit for executing).

- `apiserver_flowcontrol_priority_level_seat_utilization` is a histogram vector of observations, made at the end of each nanosecond, of the utilization of a priority level's concurrency limit, broken down by `priority_level`. This utilization is the fraction (number of seats occupied) / (concurrency limit). This metric considers all stages of execution (both normal and the extra delay at the end of a write to cover for the corresponding notification work) of all requests except WATCHes; for those it considers only the initial stage that delivers notifications of pre-existing objects. Each histogram in the vector is also labeled with phase: executing (there is no seat limit for the waiting phase).
- `apiserver_flowcontrol_request_queue_length_after_enqueue` is a histogram vector of queue lengths for the queues, broken down by `priority_level` and `flow_schema`, as sampled by the enqueued requests. Each request that gets queued contributes one sample to its histogram, reporting the length of the queue immediately after the request was added. Note that this produces different statistics than an unbiased survey would.

**Note:**

An outlier value in a histogram here means it is likely that a single flow (i.e., requests by one user or for one namespace, depending on configuration) is flooding the API server, and being throttled. By contrast, if one priority level's histogram shows that all queues for that priority level are longer than those for other priority levels, it may be appropriate to increase that `PriorityLevelConfiguration`'s concurrency shares.

- `apiserver_flowcontrol_request_concurrency_limit` is the same as `apiserver_flowcontrol_nominal_limit_seats`. Before the introduction of concurrency borrowing between priority levels, this was always equal to `apiserver_flowcontrol_current_limit_seats` (which did not exist as a distinct metric).
- `apiserver_flowcontrol_lower_limit_seats` is a gauge vector holding the lower bound on each priority level's dynamic concurrency limit.
- `apiserver_flowcontrol_upper_limit_seats` is a gauge vector holding the upper bound on each priority level's dynamic concurrency limit.
- `apiserver_flowcontrol_demand_seats` is a histogram vector counting observations, at the end of every nanosecond, of each priority level's ratio of (seat demand) / (nominal concurrency limit). A priority level's seat demand is the sum, over both queued requests and those in the initial phase of execution, of the maximum of the number of seats occupied in the request's initial and final execution phases.
- `apiserver_flowcontrol_demand_seats_high_watermark` is a gauge vector holding, for each priority level, the maximum seat demand seen during the last concurrency borrowing adjustment period.
- `apiserver_flowcontrol_demand_seats_average` is a gauge vector holding, for each priority level, the time-weighted average seat demand seen during the last concurrency borrowing adjustment period.
- `apiserver_flowcontrol_demand_seats_stddev` is a gauge vector holding, for each priority level, the time-weighted population standard deviation of seat demand seen during the last concurrency borrowing adjustment period.

- `apiserver_flowcontrol_demand_seats_smoothed` is a gauge vector holding, for each priority level, the smoothed enveloped seat demand determined at the last concurrency adjustment.
- `apiserver_flowcontrol_target_seats` is a gauge vector holding, for each priority level, the concurrency target going into the borrowing allocation problem.
- `apiserver_flowcontrol_seat_fair_frac` is a gauge holding the fair allocation fraction determined in the last borrowing adjustment.
- `apiserver_flowcontrol_current_limit_seats` is a gauge vector holding, for each priority level, the dynamic concurrency limit derived in the last adjustment.
- `apiserver_flowcontrol_request_execution_seconds` is a histogram vector of how long requests took to actually execute, broken down by `flow_schema` and `priority_level`.
- `apiserver_flowcontrol_watch_count_samples` is a histogram vector of the number of active WATCH requests relevant to a given write, broken down by `flow_schema` and `priority_level`.
- `apiserver_flowcontrol_work_estimated_seats` is a histogram vector of the number of estimated seats (maximum of initial and final stage of execution) associated with requests, broken down by `flow_schema` and `priority_level`.
- `apiserver_flowcontrol_request_dispatch_no_accommodation_total` is a counter vector of the number of events that in principle could have led to a request being dispatched but did not, due to lack of available concurrency, broken down by `flow_schema` and `priority_level`.
- `apiserver_flowcontrol_epoch_advance_total` is a counter vector of the number of attempts to jump a priority level's progress meter backward to avoid numeric overflow, grouped by `priority_level` and success.

## Good practices for using API Priority and Fairness

When a given priority level exceeds its permitted concurrency, requests can experience increased latency or be dropped with an HTTP 429 (Too Many Requests) error. To prevent these side effects of APF, you can modify your workload or tweak your APF settings to ensure there are sufficient seats available to serve your requests.

To detect whether requests are being rejected due to APF, check the following metrics:

- `apiserver_flowcontrol_rejected_requests_total`: the total number of requests rejected per `FlowSchema` and `PriorityLevelConfiguration`.
- `apiserver_flowcontrol_current_inqueue_requests`: the current number of requests queued per `FlowSchema` and `PriorityLevelConfiguration`.
- `apiserver_flowcontrol_request_wait_duration_seconds`: the latency added to requests waiting in queues.
- `apiserver_flowcontrol_priority_level_seat_utilization`: the seat utilization per `PriorityLevelConfiguration`.

## Workload modifications

To prevent requests from queuing and adding latency or being dropped due to APF, you can optimize your requests by:

- Reducing the rate at which requests are executed. A fewer number of requests over a fixed period will result in a fewer number of seats being needed at a given time.
- Avoid issuing a large number of expensive requests concurrently. Requests can be optimized to use fewer seats or have lower latency so that these requests hold those seats for a shorter duration. List requests can occupy more than 1 seat depending on the number of objects fetched during the request. Restricting the number of objects retrieved in a list request, for example by using pagination, will use less total seats over a shorter period. Furthermore, replacing list requests with watch requests will require lower total concurrency shares as watch requests only occupy 1 seat during its initial burst of notifications. If using streaming lists in versions 1.27 and later, watch requests will occupy the same number of seats as a list request for its initial burst of notifications because the entire state of the collection has to be streamed. Note that in both cases, a watch request will not hold any seats after this initial phase.

Keep in mind that queuing or rejected requests from APF could be induced by either an increase in the number of requests or an increase in latency for existing requests. For example, if requests that normally take 1s to execute start taking 60s, it is possible that APF will start rejecting requests because requests are occupying seats for a longer duration than normal due to this increase in latency. If APF starts rejecting requests across multiple priority levels without a significant change in workload, it is possible there is an underlying issue with control plane performance rather than the workload or APF settings.

## Priority and fairness settings

You can also modify the default FlowSchema and PriorityLevelConfiguration objects or create new objects of these types to better accommodate your workload.

APF settings can be modified to:

- Give more seats to high priority requests.
- Isolate non-essential or expensive requests that would starve a concurrency level if it was shared with other flows.

### Give more seats to high priority requests

1. If possible, the number of seats available across all priority levels for a particular kube-apiserver can be increased by increasing the values for the max-requests-inflight and max-mutating-requests-inflight flags. Alternatively, horizontally scaling the number of kube-apiserver instances will increase the total concurrency per priority level across the cluster assuming there is sufficient load balancing of requests.
2. You can create a new FlowSchema which references a PriorityLevelConfiguration with a larger concurrency level. This new PriorityLevelConfiguration could be an existing level or a new level with its own set of nominal concurrency shares. For example, a new FlowSchema could be introduced to change the PriorityLevelConfiguration for your requests from global-default to workload-low to increase the number of seats available to your user. Creating a new PriorityLevelConfiguration will reduce the number of seats designated for existing levels. Recall that editing a default FlowSchema or

PriorityLevelConfiguration will require setting the `apf.kubernetes.io/autoupdate-spec` annotation to false.

3. You can also increase the `NominalConcurrencyShares` for the `PriorityLevelConfiguration` which is serving your high priority requests. Alternatively, for versions 1.26 and later, you can increase the `LendablePercent` for competing priority levels so that the given priority level has a higher pool of seats it can borrow.

## Isolate non-essential requests from starving other flows

For request isolation, you can create a `FlowSchema` whose subject matches the user making these requests or create a `FlowSchema` that matches what the request is (corresponding to the `resourceRules`). Next, you can map this `FlowSchema` to a `PriorityLevelConfiguration` with a low share of seats.

For example, suppose list event requests from Pods running in the default namespace are using 10 seats each and execute for 1 minute. To prevent these expensive requests from impacting requests from other Pods using the existing service-accounts `FlowSchema`, you can apply the following `FlowSchema` to isolate these list calls from other requests.

Example `FlowSchema` object to isolate list event requests:

[priority-and-fairness/list-events-default-service-account.yaml](#)

```
apiVersion: flowcontrol.apiserver.k8s.io/v1
kind: FlowSchema
metadata:
 name: list-events-default-service-account
spec:
 distinguisherMethod:
 type: ByUser
 matchingPrecedence: 8000
 priorityLevelConfiguration:
 name: catch-all
 rules:
 - resourceRules:
 - apiGroups:
 - '*'
 namespaces:
 - default
 resources:
 - events
 verbs:
 - list
 subjects:
 - kind: ServiceAccount
 serviceAccount:
 name: default
 namespace: default
```

- This `FlowSchema` captures all list event calls made by the default service account in the default namespace. The matching precedence 8000 is lower than the value of 9000 used by the existing service-accounts `FlowSchema` so these list event calls will match `list-events-default-service-account` rather than `service-accounts`.

- The catch-all `PriorityLevelConfiguration` is used to isolate these requests. The catch-all priority level has a very small concurrency share and does not queue requests.

## What's next

- You can visit flow control [reference doc](#) to learn more about troubleshooting.
- For background information on design details for API priority and fairness, see the [enhancement proposal](#).
- You can make suggestions and feature requests via [SIG API Machinery](#) or the feature's [slack channel](#).

# Cluster Autoscaling

Automatically manage the nodes in your cluster to adapt to demand.

Kubernetes requires [nodes](#) in your cluster to run [pods](#). This means providing capacity for the workload Pods and for Kubernetes itself.

You can adjust the amount of resources available in your cluster automatically: *node autoscaling*. You can either change the number of nodes, or change the capacity that nodes provide. The first approach is referred to as *horizontal scaling*, while the second is referred to as *vertical scaling*.

Kubernetes can even provide multidimensional automatic scaling for nodes.

## Manual node management

You can manually manage node-level capacity, where you configure a fixed amount of nodes; you can use this approach even if the provisioning (the process to set up, manage, and decommission) for these nodes is automated.

This page is about taking the next step, and automating management of the amount of node capacity (CPU, memory, and other node resources) available in your cluster.

## Automatic horizontal scaling

### Cluster Autoscaler

You can use the [Cluster Autoscaler](#) to manage the scale of your nodes automatically. The cluster autoscaler can integrate with a cloud provider, or with Kubernetes' [cluster API](#), to achieve the actual node management that's needed.

The cluster autoscaler adds nodes when there are unschedulable Pods, and removes nodes when those nodes are empty.

### Cloud provider integrations

The [README](#) for the cluster autoscaler lists some of the cloud provider integrations that are available.

# Cost-aware multidimensional scaling

## Karpenter

[Karpenter](#) supports direct node management, via plugins that integrate with specific cloud providers, and can manage nodes for you whilst optimizing for overall cost.

Karpenter automatically launches just the right compute resources to handle your cluster's applications. It is designed to let you take full advantage of the cloud with fast and simple compute provisioning for Kubernetes clusters.

The Karpenter tool is designed to integrate with a cloud provider that provides API-driven server management, and where the price information for available servers is also available via a web API.

For example, if you start some more Pods in your cluster, the Karpenter tool might buy a new node that is larger than one of the nodes you are already using, and then shut down an existing node once the new node is in service.

## Cloud provider integrations

**Note:** Items on this page refer to vendors external to Kubernetes. The Kubernetes project authors aren't responsible for those third-party products or projects. To add a vendor, product or project to this list, read the [content guide](#) before submitting a change. [More information.](#)

There are integrations available between Karpenter's core and the following cloud providers:

- [Amazon Web Services](#)
- [Azure](#)

## Related components

### Descheduler

The [descheduler](#) can help you consolidate Pods onto a smaller number of nodes, to help with automatic scale down when the cluster has space capacity.

### Sizing a workload based on cluster size

#### Cluster proportional autoscaler

For workloads that need to be scaled based on the size of the cluster (for example cluster-dns or other system components), you can use the [Cluster Proportional Autoscaler](#).

The Cluster Proportional Autoscaler watches the number of schedulable nodes and cores, and scales the number of replicas of the target workload accordingly.



## Cluster proportional vertical autoscaler

If the number of replicas should stay the same, you can scale your workloads vertically according to the cluster size using the [Cluster Proportional Vertical Autoscaler](#). This project is in **beta** and can be found on GitHub.

While the Cluster Proportional Autoscaler scales the number of replicas of a workload, the Cluster Proportional Vertical Autoscaler adjusts the resource requests for a workload (for example a Deployment or DaemonSet) based on the number of nodes and/or cores in the cluster.

## What's next

- Read about [workload-level autoscaling](#)

# Installing Addons

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Add-ons extend the functionality of Kubernetes.

This page lists some of the available add-ons and links to their respective installation instructions. The list does not try to be exhaustive.

## Networking and Network Policy

- [ACI](#) provides integrated container networking and network security with Cisco ACI.
- [Antrea](#) operates at Layer 3/4 to provide networking and security services for Kubernetes, leveraging Open vSwitch as the networking data plane. Antrea is a [CNCF project at the Sandbox level](#).
- [Calico](#) is a networking and network policy provider. Calico supports a flexible set of networking options so you can choose the most efficient option for your situation, including non-overlay and overlay networks, with or without BGP. Calico uses the same engine to enforce network policy for hosts, pods, and (if using Istio & Envoy) applications at the service mesh layer.
- [Canal](#) unites Flannel and Calico, providing networking and network policy.
- [Cilium](#) is a networking, observability, and security solution with an eBPF-based data plane. Cilium provides a simple flat Layer 3 network with the ability to span multiple clusters in either a native routing or overlay/encapsulation mode, and can enforce network policies on L3-L7 using an identity-based security model that is decoupled from network addressing. Cilium can act as a replacement for kube-proxy; it also offers additional, opt-in observability and security features. Cilium is a [CNCF project at the Graduated level](#).
- [CNI-Genie](#) enables Kubernetes to seamlessly connect to a choice of CNI plugins, such as Calico, Canal, Flannel, or Weave. CNI-Genie is a [CNCF project at the Sandbox level](#).
- [Contiv](#) provides configurable networking (native L3 using BGP, overlay using vxlan, classic L2, and Cisco-SDN/ACI) for various use cases and a rich policy framework. Contiv

project is fully [open sourced](#). The [installer](#) provides both kubeadm and non-kubeadm based installation options.

- [Contrail](#), based on [Tungsten Fabric](#), is an open source, multi-cloud network virtualization and policy management platform. Contrail and Tungsten Fabric are integrated with orchestration systems such as Kubernetes, OpenShift, OpenStack and Mesos, and provide isolation modes for virtual machines, containers/pods and bare metal workloads.
- [Flannel](#) is an overlay network provider that can be used with Kubernetes.
- [Gateway API](#) is an open source project managed by the [SIG Network](#) community and provides an expressive, extensible, and role-oriented API for modeling service networking.
- [Knitter](#) is a plugin to support multiple network interfaces in a Kubernetes pod.
- [Multus](#) is a Multi plugin for multiple network support in Kubernetes to support all CNI plugins (e.g. Calico, Cilium, Contiv, Flannel), in addition to SRIOV, DPDK, OVS-DPDK and VPP based workloads in Kubernetes.
- [OVN-Kubernetes](#) is a networking provider for Kubernetes based on [OVN \(Open Virtual Network\)](#), a virtual networking implementation that came out of the Open vSwitch (OVS) project. OVN-Kubernetes provides an overlay based networking implementation for Kubernetes, including an OVS based implementation of load balancing and network policy.
- [Nodus](#) is an OVN based CNI controller plugin to provide cloud native based Service function chaining(SFC).
- [NSX-T](#) Container Plug-in (NCP) provides integration between VMware NSX-T and container orchestrators such as Kubernetes, as well as integration between NSX-T and container-based CaaS/PaaS platforms such as Pivotal Container Service (PKS) and OpenShift.
- [Nuage](#) is an SDN platform that provides policy-based networking between Kubernetes Pods and non-Kubernetes environments with visibility and security monitoring.
- [Romana](#) is a Layer 3 networking solution for pod networks that also supports the [NetworkPolicy](#) API.
- [Spiderpool](#) is an underlay and RDMA networking solution for Kubernetes. Spiderpool is supported on bare metal, virtual machines, and public cloud environments.
- [Weave Net](#) provides networking and network policy, will carry on working on both sides of a network partition, and does not require an external database.

## Service Discovery

- [CoreDNS](#) is a flexible, extensible DNS server which can be [installed](#) as the in-cluster DNS for pods.

## Visualization & Control

- [Dashboard](#) is a dashboard web interface for Kubernetes.
- [Weave Scope](#) is a tool for visualizing your containers, Pods, Services and more.

## Infrastructure

- [KubeVirt](#) is an add-on to run virtual machines on Kubernetes. Usually run on bare-metal clusters.
- The [node problem detector](#) runs on Linux nodes and reports system issues as either [Events](#) or [Node conditions](#).

## Instrumentation

- [kube-state-metrics](#)

## Legacy Add-ons

There are several other add-ons documented in the deprecated [cluster/addons](#) directory.

Well-maintained ones should be linked to here. PRs welcome!

## Windows in Kubernetes

Kubernetes supports nodes that run Microsoft Windows.

Kubernetes supports worker [nodes](#) running either Linux or Microsoft Windows.

This item links to a third party project or product that is not part of Kubernetes itself. [More information](#)

The CNCF and its parent the Linux Foundation take a vendor-neutral approach towards compatibility. It is possible to join your [Windows server](#) as a worker node to a Kubernetes cluster.

You can [install and set up kubectl on Windows](#) no matter what operating system you use within your cluster.

If you are using Windows nodes, you can read:

- [Networking On Windows](#)
- [Windows Storage In Kubernetes](#)
- [Resource Management for Windows Nodes](#)
- [Configure RunAsUserName for Windows Pods and Containers](#)
- [Create A Windows HostProcess Pod](#)
- [Configure Group Managed Service Accounts for Windows Pods and Containers](#)
- [Security For Windows Nodes](#)
- [Windows Debugging Tips](#)
- [Guide for Scheduling Windows Containers in Kubernetes](#)

or, for an overview, read:

- [Windows containers in Kubernetes](#)
- [Guide for Running Windows Containers in Kubernetes](#)

## Windows containers in Kubernetes

Windows applications constitute a large portion of the services and applications that run in many organizations. [Windows containers](#) provide a way to encapsulate processes and package dependencies, making it easier to use DevOps practices and follow cloud native patterns for Windows applications.

Organizations with investments in Windows-based applications and Linux-based applications don't have to look for separate orchestrators to manage their workloads, leading to increased operational efficiencies across their deployments, regardless of operating system.

## Windows nodes in Kubernetes

To enable the orchestration of Windows containers in Kubernetes, include Windows nodes in your existing Linux cluster. Scheduling Windows containers in [Pods](#) on Kubernetes is similar to scheduling Linux-based containers.

In order to run Windows containers, your Kubernetes cluster must include multiple operating systems. While you can only run the [control plane](#) on Linux, you can deploy worker nodes running either Windows or Linux.

Windows [nodes](#) are [supported](#) provided that the operating system is Windows Server 2019 or Windows Server 2022.

This document uses the term *Windows containers* to mean Windows containers with process isolation. Kubernetes does not support running Windows containers with [Hyper-V isolation](#).

## Compatibility and limitations

Some node features are only available if you use a specific [container runtime](#); others are not available on Windows nodes, including:

- HugePages: not supported for Windows containers
- Privileged containers: not supported for Windows containers. [HostProcess Containers](#) offer similar functionality.
- TerminationGracePeriod: requires containerD

Not all features of shared namespaces are supported. See [API compatibility](#) for more details.

See [Windows OS version compatibility](#) for details on the Windows versions that Kubernetes is tested against.

From an API and kubectl perspective, Windows containers behave in much the same way as Linux-based containers. However, there are some notable differences in key functionality which are outlined in this section.

## Comparison with Linux

Key Kubernetes elements work the same way in Windows as they do in Linux. This section refers to several key workload abstractions and how they map to Windows.

- [Pods](#)

A Pod is the basic building block of Kubernetes—the smallest and simplest unit in the Kubernetes object model that you create or deploy. You may not deploy Windows and Linux containers in the same Pod. All containers in a Pod are scheduled onto a single

Node where each Node represents a specific platform and architecture. The following Pod capabilities, properties and events are supported with Windows containers:

- Single or multiple containers per Pod with process isolation and volume sharing
- Pod status fields
- Readiness, liveness, and startup probes
- postStart & preStop container lifecycle hooks
- ConfigMap, Secrets: as environment variables or volumes
- emptyDir volumes
- Named pipe host mounts
- Resource limits
- OS field:

The `.spec.os.name` field should be set to `windows` to indicate that the current Pod uses Windows containers.

If you set the `.spec.os.name` field to `windows`, you must not set the following fields in the `.spec` of that Pod:

- `spec.hostPID`
- `spec.hostIPC`
- `spec.securityContext.seLinuxOptions`
- `spec.securityContext.seccompProfile`
- `spec.securityContext.fsGroup`
- `spec.securityContext.fsGroupChangePolicy`
- `spec.securityContext.sysctls`
- `spec.shareProcessNamespace`
- `spec.securityContext.runAsUser`
- `spec.securityContext.runAsGroup`
- `spec.securityContext.supplementalGroups`
- `spec.containers[*].securityContext.seLinuxOptions`
- `spec.containers[*].securityContext.seccompProfile`
- `spec.containers[*].securityContext.capabilities`
- `spec.containers[*].securityContext.readOnlyRootFilesystem`
- `spec.containers[*].securityContext.privileged`
- `spec.containers[*].securityContext.allowPrivilegeEscalation`
- `spec.containers[*].securityContext.procMount`
- `spec.containers[*].securityContext.runAsUser`
- `spec.containers[*].securityContext.runAsGroup`

In the above list, wildcards (\*) indicate all elements in a list. For example, `spec.containers[*].securityContext` refers to the SecurityContext object for all containers. If any of these fields is specified, the Pod will not be admitted by the API server.

[Workload resources](#) including:

- - ReplicaSet
  - Deployment
  - StatefulSet
  - DaemonSet
  - Job
  - CronJob
  - ReplicationController
- [Services](#) See [Load balancing and Services](#) for more details.

Pods, workload resources, and Services are critical elements to managing Windows workloads on Kubernetes. However, on their own they are not enough to enable the proper lifecycle management of Windows workloads in a dynamic cloud native environment.

- `kubectl exec`
- Pod and container metrics
- [Horizontal pod autoscaling](#)
- [Resource quotas](#)
- Scheduler preemption

## Command line options for the kubelet

Some kubelet command line options behave differently on Windows, as described below:

- The `--windows-priorityclass` lets you set the scheduling priority of the kubelet process (see [CPU resource management](#))
- The `--kube-reserved`, `--system-reserved`, and `--eviction-hard` flags update [NodeAllocatable](#)
- Eviction by using `--enforce-node-allocable` is not implemented
- Eviction by using `--eviction-hard` and `--eviction-soft` are not implemented
- When running on a Windows node the kubelet does not have memory or CPU restrictions. `--kube-reserved` and `--system-reserved` only subtract from `NodeAllocatable` and do not guarantee resource provided for workloads. See [Resource Management for Windows nodes](#) for more information.
- The MemoryPressure Condition is not implemented
- The kubelet does not take OOM eviction actions

## API compatibility

There are subtle differences in the way the Kubernetes APIs work for Windows due to the OS and container runtime. Some workload properties were designed for Linux, and fail to run on Windows.

At a high level, these OS concepts are different:

- Identity - Linux uses userID (UID) and groupID (GID) which are represented as integer types. User and group names are not canonical - they are just an alias in `/etc/groups` or `/etc/passwd` back to UID+GID. Windows uses a larger binary [security identifier](#) (SID) which is stored in the Windows Security Access Manager (SAM) database. This database is not shared between the host and containers, or between containers.

- File permissions - Windows uses an access control list based on (SIDs), whereas POSIX systems such as Linux use a bitmask based on object permissions and UID+GID, plus *optional* access control lists.
- File paths - the convention on Windows is to use \ instead of /. The Go IO libraries typically accept both and just make it work, but when you're setting a path or command line that's interpreted inside a container, \ may be needed.
- Signals - Windows interactive apps handle termination differently, and can implement one or more of these:
  - A UI thread handles well-defined messages including WM\_CLOSE.
  - Console apps handle Ctrl-C or Ctrl-break using a Control Handler.
  - Services register a Service Control Handler function that can accept SERVICE\_CONTROL\_STOP control codes.

Container exit codes follow the same convention where 0 is success, and nonzero is failure. The specific error codes may differ across Windows and Linux. However, exit codes passed from the Kubernetes components (kubelet, kube-proxy) are unchanged.

### Field compatibility for container specifications

The following list documents differences between how Pod container specifications work between Windows and Linux:

- Huge pages are not implemented in the Windows container runtime, and are not available. They require [asserting a user privilege](#) that's not configurable for containers.
- requests.cpu and requests.memory - requests are subtracted from node available resources, so they can be used to avoid overprovisioning a node. However, they cannot be used to guarantee resources in an overprovisioned node. They should be applied to all containers as a best practice if the operator wants to avoid overprovisioning entirely.
- securityContext.allowPrivilegeEscalation - not possible on Windows; none of the capabilities are hooked up
- securityContext.capabilities - POSIX capabilities are not implemented on Windows
- securityContext.privileged - Windows doesn't support privileged containers, use [HostProcess Containers](#) instead
- securityContext.procMount - Windows doesn't have a /proc filesystem
- securityContext.readOnlyRootFilesystem - not possible on Windows; write access is required for registry & system processes to run inside the container
- securityContext.runAsGroup - not possible on Windows as there is no GID support
- securityContext.runAsNonRoot - this setting will prevent containers from running as ContainerAdministrator which is the closest equivalent to a root user on Windows.
- securityContext.runAsUser - use [runAsUserName](#) instead
- securityContext.seLinuxOptions - not possible on Windows as SELinux is Linux-specific
- terminationMessagePath - this has some limitations in that Windows doesn't support mapping single files. The default value is /dev/termination-log, which does work because it does not exist on Windows by default.

### Field compatibility for Pod specifications

The following list documents differences between how Pod specifications work between Windows and Linux:

- hostIPC and hostpid - host namespace sharing is not possible on Windows
- hostNetwork - [see below](#)



- `dnsPolicy` - setting the Pod `dnsPolicy` to `ClusterFirstWithHostNet` is not supported on Windows because host networking is not provided. Pods always run with a container network.
- `podSecurityContext` [see below](#)
- `shareProcessNamespace` - this is a beta feature, and depends on Linux namespaces which are not implemented on Windows. Windows cannot share process namespaces or the container's root filesystem. Only the network can be shared.
- `terminationGracePeriodSeconds` - this is not fully implemented in Docker on Windows, see the [GitHub issue](#). The behavior today is that the `ENTRYPOINT` process is sent `CTRL_SHUTDOWN_EVENT`, then Windows waits 5 seconds by default, and finally shuts down all processes using the normal Windows shutdown behavior. The 5 second default is actually in the Windows registry [inside the container](#), so it can be overridden when the container is built.
- `volumeDevices` - this is a beta feature, and is not implemented on Windows. Windows cannot attach raw block devices to pods.
- `volumes`
  - If you define an `emptyDir` volume, you cannot set its volume source to memory.
- You cannot enable `mountPropagation` for volume mounts as this is not supported on Windows.

## Field compatibility for `hostNetwork`

FEATURE STATE: Kubernetes v1.26 [alpha]

The kubelet can now request that pods running on Windows nodes use the host's network namespace instead of creating a new pod network namespace. To enable this functionality pass `--feature-gates=WindowsHostNetwork=true` to the kubelet.

### Note:

This functionality requires a container runtime that supports this functionality.

## Field compatibility for Pod security context

Only the `securityContext.runAsNonRoot` and `securityContext.windowsOptions` from the Pod [securityContext](#) fields work on Windows.

## Node problem detector

The node problem detector (see [Monitor Node Health](#)) has preliminary support for Windows. For more information, visit the project's [GitHub page](#).

## Pause container

In a Kubernetes Pod, an infrastructure or “pause” container is first created to host the container. In Linux, the cgroups and namespaces that make up a pod need a process to maintain their continued existence; the pause process provides this. Containers that belong to the same pod, including infrastructure and worker containers, share a common network endpoint (same IPv4 and / or IPv6 address, same network port spaces). Kubernetes uses pause containers to allow for worker containers crashing or restarting without losing any of the networking configuration.



Kubernetes maintains a multi-architecture image that includes support for Windows. For Kubernetes v1.30.0 the recommended pause image is [registry.k8s.io/pause:3.6](https://registry.k8s.io/pause:3.6). The [source code](#) is available on GitHub.

Microsoft maintains a different multi-architecture image, with Linux and Windows amd64 support, that you can find as [mcr.microsoft.com/oss/kubernetes/pause:3.6](https://mcr.microsoft.com/oss/kubernetes/pause:3.6). This image is built from the same source as the Kubernetes maintained image but all of the Windows binaries are [authenticode signed](#) by Microsoft. The Kubernetes project recommends using the Microsoft maintained image if you are deploying to a production or production-like environment that requires signed binaries.

## Container runtimes

You need to install a [container runtime](#) into each node in the cluster so that Pods can run there.

The following container runtimes work with Windows:

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

### ContainerD

FEATURE STATE: Kubernetes v1.20 [stable]

You can use [ContainerD](#) 1.4.0+ as the container runtime for Kubernetes nodes that run Windows.

Learn how to [install ContainerD on a Windows node](#).

#### Note:

There is a [known limitation](#) when using GMSA with containerd to access Windows network shares, which requires a kernel patch.

### Mirantis Container Runtime

[Mirantis Container Runtime](#) (MCR) is available as a container runtime for all Windows Server 2019 and later versions.

See [Install MCR on Windows Servers](#) for more information.

## Windows OS version compatibility

On Windows nodes, strict compatibility rules apply where the host OS version must match the container base image OS version. Only Windows containers with a container operating system of Windows Server 2019 are fully supported.

For Kubernetes v1.30, operating system compatibility for Windows nodes (and Pods) is as follows:

Windows Server LTSC release  
Windows Server 2019  
Windows Server 2022  
Windows Server SAC release  
Windows Server version 20H2

The Kubernetes [version-skew policy](#) also applies.

## Hardware recommendations and considerations

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information.](#)

### Note:

The following hardware specifications outlined here should be regarded as sensible default values. They are not intended to represent minimum requirements or specific recommendations for production environments. Depending on the requirements for your workload these values may need to be adjusted.

- 64-bit processor 4 CPU cores or more, capable of supporting virtualization
- 8GB or more of RAM
- 50GB or more of free disk space

Refer to [Hardware requirements for Windows Server Microsoft documentation](#) for the most up-to-date information on minimum hardware requirements. For guidance on deciding on resources for production worker nodes refer to [Production worker nodes Kubernetes documentation](#).

To optimize system resources, if a graphical user interface is not required, it may be preferable to use a Windows Server OS installation that excludes the [Windows Desktop Experience](#) installation option, as this configuration typically frees up more system resources.

In assessing disk space for Windows worker nodes, take note that Windows container images are typically larger than Linux container images, with container image sizes ranging from [300MB to over 10GB](#) for a single image. Additionally, take note that the C: drive in Windows containers represents a virtual free size of 20GB by default, which is not the actual consumed space, but rather the disk size for which a single container can grow to occupy when using local storage on the host. See [Containers on Windows - Container Storage Documentation](#) for more detail.

## Getting help and troubleshooting

Your main source of help for troubleshooting your Kubernetes cluster should start with the [Troubleshooting](#) page.

Some additional, Windows-specific troubleshooting help is included in this section. Logs are an important element of troubleshooting issues in Kubernetes. Make sure to include them any time you seek troubleshooting assistance from other contributors. Follow the instructions in the SIG Windows [contributing guide on gathering logs](#).

## Reporting issues and feature requests

If you have what looks like a bug, or you would like to make a feature request, please follow the [SIG Windows contributing guide](#) to create a new issue. You should first search the list of issues in case it was reported previously and comment with your experience on the issue and add additional logs. SIG Windows channel on the Kubernetes Slack is also a great avenue to get some initial support and troubleshooting ideas prior to creating a ticket.

## Validating the Windows cluster operability

The Kubernetes project provides a *Windows Operational Readiness* specification, accompanied by a structured test suite. This suite is split into two sets of tests, core and extended, each containing categories aimed at testing specific areas. It can be used to validate all the functionalities of a Windows and hybrid system (mixed with Linux nodes) with full coverage.

To set up the project on a newly created cluster, refer to the instructions in the [project guide](#).

## Deployment tools

The kubeadm tool helps you to deploy a Kubernetes cluster, providing the control plane to manage the cluster it, and nodes to run your workloads.

The Kubernetes [cluster API](#) project also provides means to automate deployment of Windows nodes.

## Windows distribution channels

For a detailed explanation of Windows distribution channels see the [Microsoft documentation](#).

Information on the different Windows Server servicing channels including their support models can be found at [Windows Server servicing channels](#).

# Guide for Running Windows Containers in Kubernetes

This page provides a walkthrough for some steps you can follow to run Windows containers using Kubernetes. The page also highlights some Windows specific functionality within Kubernetes.

It is important to note that creating and deploying services and workloads on Kubernetes behaves in much the same way for Linux and Windows containers. The [kubectrl commands](#) to interface with the cluster are identical. The examples in this page are provided to jumpstart your experience with Windows containers.

## Objectives

Configure an example deployment to run Windows containers on a Windows node.

## Before you begin

You should already have access to a Kubernetes cluster that includes a worker node running Windows Server.

## Getting Started: Deploying a Windows workload

The example YAML file below deploys a simple webserver application running inside a Windows container.

Create a manifest named `win-webserver.yaml` with the contents below:

```

apiVersion: v1
kind: Service
metadata:
 name: win-webserver
 labels:
 app: win-webserver
spec:
 ports:
 # the port that this service should serve on
 - port: 80
 targetPort: 80
 selector:
 app: win-webserver
 type: NodePort

apiVersion: apps/v1
kind: Deployment
metadata:
 labels:
 app: win-webserver
 name: win-webserver
spec:
 replicas: 2
 selector:
 matchLabels:
 app: win-webserver
 template:
 metadata:
 labels:
 app: win-webserver
 name: win-webserver
 spec:
 containers:
 - name: windowswebserver
 image: mcr.microsoft.com/windows/servercore:ltsc2019
 command:
 - powershell.exe
 - -command
 -
```

```
"<#code used from https://gist.github.com/19WAS85/5424431#> ; $$listener = New-Object
System.Net.HttpListener ; $$listener.Prefixes.Add('http://*:80/') ; $$listener.Start() ; $
$callerCounts = @{} ; Write-Host('Listening at http://*:80/') ; while ($$listener.IsListening) { ;$
$context = $$listener.GetContext() ; $$requestUrl = $$context.Request.Url ; $$clientIP = $
$context.Request.RemoteEndPoint.Address ; $$response = $$context.Response ; Write-Host
" ; Write-Host('> {0}' -f $$requestUrl) ; $$count = 1 ; $$k=$$callerCounts.Get_Item($$clientIP) ; if
($$k -ne $$null) { $$count += $$k } ; $$callerCounts.Set_Item($$clientIP, $$count) ; $$ip=(Get-
NetAdapter | Get-NetIpAddress) ; $$header='<html><body><H1>Windows Container Web
Server</H1>' ; $$callerCountsString="" ; $$callerCounts.Keys | % { $$callerCountsString+=''<p>IP
{0} callerCount {1} ' -f $$ip[1].IPAddress,$$callerCounts.Item($$_) } ; $$footer='</body></html>' ;
$$content='{0}{1}{2}' -f $$header,$$callerCountsString,$$footer ; Write-Output $$content ; $
$buffer = [System.Text.Encoding]::UTF8.GetBytes($$content) ; $$response.ContentLength64 = $
$buffer.Length ; $$response.OutputStream.Write($$buffer, 0, $$buffer.Length) ; $
$response.Close() ; $$responseStatus = $$response.StatusCode ; Write-Host('< {0}' -f $
$responseStatus) } ; "
nodeSelector:
 kubernetes.io/os: windows
```

### Note:

Port mapping is also supported, but for simplicity this example exposes port 80 of the container directly to the Service.

1. Check that all nodes are healthy:

```
kubectl get nodes
```

2. Deploy the service and watch for pod updates:

```
kubectl apply -f win-webserver.yaml
kubectl get pods -o wide -w
```

When the service is deployed correctly both Pods are marked as Ready. To exit the watch command, press Ctrl+C.

3. Check that the deployment succeeded. To verify:

- Several pods listed from the Linux control plane node, use `kubectl get pods`
- Node-to-pod communication across the network, curl port 80 of your pod IPs from the Linux control plane node to check for a web server response
- Pod-to-pod communication, ping between pods (and across hosts, if you have more than one Windows node) using `kubectl exec`
- Service-to-pod communication, curl the virtual service IP (seen under `kubectl get services`) from the Linux control plane node and from individual pods
- Service discovery, curl the service name with the Kubernetes [default DNS suffix](#)
- Inbound connectivity, curl the NodePort from the Linux control plane node or machines outside of the cluster
- Outbound connectivity, curl external IPs from inside the pod using `kubectl exec`

**Note:**

Windows container hosts are not able to access the IP of services scheduled on them due to current platform limitations of the Windows networking stack. Only Windows pods are able to access service IPs.

## Observability

### Capturing logs from workloads

Logs are an important element of observability; they enable users to gain insights into the operational aspect of workloads and are a key ingredient to troubleshooting issues. Because Windows containers and workloads inside Windows containers behave differently from Linux containers, users had a hard time collecting logs, limiting operational visibility. Windows workloads for example are usually configured to log to ETW (Event Tracing for Windows) or push entries to the application event log. [LogMonitor](#), an open source tool by Microsoft, is the recommended way to monitor configured log sources inside a Windows container. LogMonitor supports monitoring event logs, ETW providers, and custom application logs, piping them to STDOUT for consumption by `kubect logs <pod>`.

Follow the instructions in the LogMonitor GitHub page to copy its binaries and configuration files to all your containers and add the necessary entypoints for LogMonitor to push your logs to STDOUT.

## Configuring container user

### Using configurable Container usernames

Windows containers can be configured to run their entypoints and processes with different usernames than the image defaults. Learn more about it [here](#).

### Managing Workload Identity with Group Managed Service Accounts

Windows container workloads can be configured to use Group Managed Service Accounts (GMSA). Group Managed Service Accounts are a specific type of Active Directory account that provide automatic password management, simplified service principal name (SPN) management, and the ability to delegate the management to other administrators across multiple servers. Containers configured with a GMSA can access external Active Directory Domain resources while carrying the identity configured with the GMSA. Learn more about configuring and using GMSA for Windows containers [here](#).

## Taints and tolerations

Users need to use some combination of [taint](#) and node selectors in order to schedule Linux and Windows workloads to their respective OS-specific nodes. The recommended approach is outlined below, with one of its main goals being that this approach should not break compatibility for existing Linux workloads.

You can (and should) set `.spec.os.name` for each Pod, to indicate the operating system that the containers in that Pod are designed for. For Pods that run Linux containers, set `.spec.os.name` to `linux`. For Pods that run Windows containers, set `.spec.os.name` to `windows`.

#### Note:

If you are running a version of Kubernetes older than 1.24, you may need to enable the `IdentifyPodOS` [feature gate](#) to be able to set a value for `.spec.pod.os`.

The scheduler does not use the value of `.spec.os.name` when assigning Pods to nodes. You should use normal Kubernetes mechanisms for [assigning pods to nodes](#) to ensure that the control plane for your cluster places pods onto nodes that are running the appropriate operating system.

The `.spec.os.name` value has no effect on the scheduling of the Windows pods, so taints and tolerations (or node selectors) are still required to ensure that the Windows pods land onto appropriate Windows nodes.

## Ensuring OS-specific workloads land on the appropriate container host

Users can ensure Windows containers can be scheduled on the appropriate host using taints and tolerations. All Kubernetes nodes running Kubernetes 1.30 have the following default labels:

- `kubernetes.io/os = [windows|linux]`
- `kubernetes.io/arch = [amd64|arm64|...]`

If a Pod specification does not specify a `nodeSelector` such as `"kubernetes.io/os": windows`, it is possible the Pod can be scheduled on any host, Windows or Linux. This can be problematic since a Windows container can only run on Windows and a Linux container can only run on Linux. The best practice for Kubernetes 1.30 is to use a `nodeSelector`.

However, in many cases users have a pre-existing large number of deployments for Linux containers, as well as an ecosystem of off-the-shelf configurations, such as community Helm charts, and programmatic Pod generation cases, such as with operators. In those situations, you may be hesitant to make the configuration change to add `nodeSelector` fields to all Pods and Pod templates. The alternative is to use taints. Because the kubelet can set taints during registration, it could easily be modified to automatically add a taint when running on Windows only.

For example: `--register-with-taints='os=windows:NoSchedule'`

By adding a taint to all Windows nodes, nothing will be scheduled on them (that includes existing Linux Pods). In order for a Windows Pod to be scheduled on a Windows node, it would need both the `nodeSelector` and the appropriate matching toleration to choose Windows.

`nodeSelector:`

```
kubernetes.io/os: windows
node.kubernetes.io/windows-build: '10.0.17763'
```

`tolerations:`

```
- key: "os"
 operator: "Equal"
```

```
value: "windows"
effect: "NoSchedule"
```

## Handling multiple Windows versions in the same cluster

The Windows Server version used by each pod must match that of the node. If you want to use multiple Windows Server versions in the same cluster, then you should set additional node labels and nodeSelector fields.

Kubernetes automatically adds a label, [node.kubernetes.io/windows-build](https://kubernetes.io/windows-build) to simplify this.

This label reflects the Windows major, minor, and build number that need to match for compatibility. Here are values used for each Windows Server version:

| Product Name        | Version    |
|---------------------|------------|
| Windows Server 2019 | 10.0.17763 |
| Windows Server 2022 | 10.0.20348 |

## Simplifying with RuntimeClass

[RuntimeClass](#) can be used to simplify the process of using taints and tolerations. A cluster administrator can create a RuntimeClass object which is used to encapsulate these taints and tolerations.

1. Save this file to runtimeClasses.yml. It includes the appropriate nodeSelector for the Windows OS, architecture, and version.

```

apiVersion: node.k8s.io/v1
kind: RuntimeClass
metadata:
 name: windows-2019
handler: example-container-runtime-handler
scheduling:
 nodeSelector:
 kubernetes.io/os: 'windows'
 kubernetes.io/arch: 'amd64'
 node.kubernetes.io/windows-build: '10.0.17763'
 tolerations:
 - effect: NoSchedule
 key: os
 operator: Equal
 value: "windows"
```

2. Run `kubectl create -f runtimeClasses.yml` using as a cluster administrator
3. Add `runtimeClassName: windows-2019` as appropriate to Pod specs

For example:

```

apiVersion: apps/v1
kind: Deployment
```



```

metadata:
 name: iis-2019
 labels:
 app: iis-2019
spec:
 replicas: 1
 template:
 metadata:
 name: iis-2019
 labels:
 app: iis-2019
 spec:
 runtimeClassName: windows-2019
 containers:
 - name: iis
 image: mcr.microsoft.com/windows/servercore/iis:windowsservercore-ltsc2019
 resources:
 limits:
 cpu: 1
 memory: 800Mi
 requests:
 cpu: .1
 memory: 300Mi
 ports:
 - containerPort: 80
 selector:
 matchLabels:
 app: iis-2019

apiVersion: v1
kind: Service
metadata:
 name: iis
spec:
 type: LoadBalancer
 ports:
 - protocol: TCP
 port: 80
 selector:
 app: iis-2019

```

## Extending Kubernetes

Different ways to change the behavior of your Kubernetes cluster.

Kubernetes is highly configurable and extensible. As a result, there is rarely a need to fork or submit patches to the Kubernetes project code.

This guide describes the options for customizing a Kubernetes cluster. It is aimed at [cluster operators](#) who want to understand how to adapt their Kubernetes cluster to the needs of their work environment. Developers who are prospective [Platform Developers](#) or Kubernetes Project

[Contributors](#) will also find it useful as an introduction to what extension points and patterns exist, and their trade-offs and limitations.

Customization approaches can be broadly divided into [configuration](#), which only involves changing command line arguments, local configuration files, or API resources; and [extensions](#), which involve running additional programs, additional network services, or both. This document is primarily about *extensions*.

## Configuration

*Configuration files* and *command arguments* are documented in the [Reference](#) section of the online documentation, with a page for each binary:

- [kube-apiserver](#)
- [kube-controller-manager](#)
- [kube-scheduler](#)
- [kubelet](#)
- [kube-proxy](#)

Command arguments and configuration files may not always be changeable in a hosted Kubernetes service or a distribution with managed installation. When they are changeable, they are usually only changeable by the cluster operator. Also, they are subject to change in future Kubernetes versions, and setting them may require restarting processes. For those reasons, they should be used only when there are no other options.

Built-in *policy APIs*, such as [ResourceQuota](#), [NetworkPolicy](#) and Role-based Access Control ([RBAC](#)), are built-in Kubernetes APIs that provide declaratively configured policy settings. APIs are typically usable even with hosted Kubernetes services and with managed Kubernetes installations. The built-in policy APIs follow the same conventions as other Kubernetes resources such as Pods. When you use a policy APIs that is [stable](#), you benefit from a [defined support policy](#) like other Kubernetes APIs. For these reasons, policy APIs are recommended over *configuration files* and *command arguments* where suitable.

## Extensions

Extensions are software components that extend and deeply integrate with Kubernetes. They adapt it to support new types and new kinds of hardware.

Many cluster administrators use a hosted or distribution instance of Kubernetes. These clusters come with extensions pre-installed. As a result, most Kubernetes users will not need to install extensions and even fewer users will need to author new ones.

### Extension patterns

Kubernetes is designed to be automated by writing client programs. Any program that reads and/or writes to the Kubernetes API can provide useful automation. *Automation* can run on the cluster or off it. By following the guidance in this doc you can write highly available and robust automation. Automation generally works with any Kubernetes cluster, including hosted clusters and managed installations.

There is a specific pattern for writing client programs that work well with Kubernetes called the [controller](#) pattern. Controllers typically read an object's .spec, possibly do things, and then update the object's .status.

A controller is a client of the Kubernetes API. When Kubernetes is the client and calls out to a remote service, Kubernetes calls this a *webhook*. The remote service is called a *webhook backend*. As with custom controllers, webhooks do add a point of failure.

#### **Note:**

Outside of Kubernetes, the term “webhook” typically refers to a mechanism for asynchronous notifications, where the webhook call serves as a one-way notification to another system or component. In the Kubernetes ecosystem, even synchronous HTTP callouts are often described as “webhooks”.

In the webhook model, Kubernetes makes a network request to a remote service. With the alternative *binary Plugin* model, Kubernetes executes a binary (program). Binary plugins are used by the kubelet (for example, [CSI storage plugins](#) and [CNI network plugins](#)), and by kubectl (see [Extend kubectl with plugins](#)).

## **Extension points**

This diagram shows the extension points in a Kubernetes cluster and the clients that access it.

Symbolic representation of seven numbered extension points for Kubernetes

Kubernetes extension points

### **Key to the figure**

1. Users often interact with the Kubernetes API using kubectl. [Plugins](#) customise the behaviour of clients. There are generic extensions that can apply to different clients, as well as specific ways to extend kubectl.
2. The API server handles all requests. Several types of extension points in the API server allow authenticating requests, or blocking them based on their content, editing content, and handling deletion. These are described in the [API Access Extensions](#) section.
3. The API server serves various kinds of *resources*. *Built-in resource kinds*, such as pods, are defined by the Kubernetes project and can't be changed. Read [API extensions](#) to learn about extending the Kubernetes API.
4. The Kubernetes scheduler [decides](#) which nodes to place pods on. There are several ways to extend scheduling, which are described in the [Scheduling extensions](#) section.
5. Much of the behavior of Kubernetes is implemented by programs called [controllers](#), that are clients of the API server. Controllers are often used in conjunction with custom resources. Read [combining new APIs with automation](#) and [Changing built-in resources](#) to learn more.
6. The kubelet runs on servers (nodes), and helps pods appear like virtual servers with their own IPs on the cluster network. [Network Plugins](#) allow for different implementations of pod networking.

You can use [Device Plugins](#) to integrate custom hardware or other special node-local facilities, and make these available to Pods running in your cluster. The kubelet includes support for working with device plugins.

The kubelet also mounts and unmounts [volume](#) for pods and their containers. You can use [Storage Plugins](#) to add support for new kinds of storage and other volume types.

### Extension point choice flowchart

If you are unsure where to start, this flowchart can help. Note that some solutions may involve several types of extensions.

Flowchart with questions about use cases and guidance for implementers. Green circles indicate yes; red circles indicate no.

Flowchart guide to select an extension approach

---

## Client extensions

Plugins for kubectl are separate binaries that add or replace the behavior of specific subcommands. The kubectl tool can also integrate with [credential plugins](#). These extensions only affect a individual user's local environment, and so cannot enforce site-wide policies.

If you want to extend the kubectl tool, read [Extend kubectl with plugins](#).

## API extensions

### Custom resource definitions

Consider adding a *Custom Resource* to Kubernetes if you want to define new controllers, application configuration objects or other declarative APIs, and to manage them using Kubernetes tools, such as kubectl.

For more about Custom Resources, see the [Custom Resources](#) concept guide.

### API aggregation layer

You can use Kubernetes' [API Aggregation Layer](#) to integrate the Kubernetes API with additional services such as for [metrics](#).

### Combining new APIs with automation

A combination of a custom resource API and a control loop is called the [controllers](#) pattern. If your controller takes the place of a human operator deploying infrastructure based on a desired state, then the controller may also be following the [operator pattern](#). The Operator pattern is used to manage specific applications; usually, these are applications that maintain state and require care in how they are managed.

You can also make your own custom APIs and control loops that manage other resources, such as storage, or to define policies (such as an access control restriction).

## Changing built-in resources

When you extend the Kubernetes API by adding custom resources, the added resources always fall into a new API Groups. You cannot replace or change existing API groups. Adding an API does not directly let you affect the behavior of existing APIs (such as Pods), whereas *API Access Extensions* do.

## API access extensions

When a request reaches the Kubernetes API Server, it is first *authenticated*, then *authorized*, and is then subject to various types of *admission control* (some requests are in fact not authenticated, and get special treatment). See [Controlling Access to the Kubernetes API](#) for more on this flow.

Each of the steps in the Kubernetes authentication / authorization flow offers extension points.

### Authentication

[Authentication](#) maps headers or certificates in all requests to a username for the client making the request.

Kubernetes has several built-in authentication methods that it supports. It can also sit behind an authenticating proxy, and it can send a token from an Authorization: header to a remote service for verification (an [authentication webhook](#)) if those don't meet your needs.

### Authorization

[Authorization](#) determines whether specific users can read, write, and do other operations on API resources. It works at the level of whole resources -- it doesn't discriminate based on arbitrary object fields.

If the built-in authorization options don't meet your needs, an [authorization webhook](#) allows calling out to custom code that makes an authorization decision.

### Dynamic admission control

After a request is authorized, if it is a write operation, it also goes through [Admission Control](#) steps. In addition to the built-in steps, there are several extensions:

- The [Image Policy webhook](#) restricts what images can be run in containers.
- To make arbitrary admission control decisions, a general [Admission webhook](#) can be used. Admission webhooks can reject creations or updates. Some admission webhooks modify the incoming request data before it is handled further by Kubernetes.

## Infrastructure extensions

### Device plugins

*Device plugins* allow a node to discover new Node resources (in addition to the builtin ones like cpu and memory) via a [Device Plugin](#).

## Storage plugins

[Container Storage Interface](#) (CSI) plugins provide a way to extend Kubernetes with supports for new kinds of volumes. The volumes can be backed by durable external storage, or provide ephemeral storage, or they might offer a read-only interface to information using a filesystem paradigm.

Kubernetes also includes support for [FlexVolume](#) plugins, which are deprecated since Kubernetes v1.23 (in favour of CSI).

FlexVolume plugins allow users to mount volume types that aren't natively supported by Kubernetes. When you run a Pod that relies on FlexVolume storage, the kubelet calls a binary plugin to mount the volume. The archived [FlexVolume](#) design proposal has more detail on this approach.

The [Kubernetes Volume Plugin FAQ for Storage Vendors](#) includes general information on storage plugins.

## Network plugins

Your Kubernetes cluster needs a *network plugin* in order to have a working Pod network and to support other aspects of the Kubernetes network model.

[Network Plugins](#) allow Kubernetes to work with different networking topologies and technologies.

## Kubelet image credential provider plugins

FEATURE STATE: Kubernetes v1.26 [stable]

Kubelet image credential providers are plugins for the kubelet to dynamically retrieve image registry credentials. The credentials are then used when pulling images from container image registries that match the configuration.

The plugins can communicate with external services or use local files to obtain credentials. This way, the kubelet does not need to have static credentials for each registry, and can support various authentication methods and protocols.

For plugin configuration details, see [Configure a kubelet image credential provider](#).

## Scheduling extensions

The scheduler is a special type of controller that watches pods, and assigns pods to nodes. The default scheduler can be replaced entirely, while continuing to use other Kubernetes components, or [multiple schedulers](#) can run at the same time.

This is a significant undertaking, and almost all Kubernetes users find they do not need to modify the scheduler.

You can control which [scheduling plugins](#) are active, or associate sets of plugins with different named [scheduler profiles](#). You can also write your own plugin that integrates with one or more of the kube-scheduler's [extension points](#).

Finally, the built in kube-scheduler component supports a [webhook](#) that permits a remote HTTP backend (scheduler extension) to filter and / or prioritize the nodes that the kube-scheduler chooses for a pod.

**Note:**

You can only affect node filtering and node prioritization with a scheduler extender webhook; other extension points are not available through the webhook integration.

## What's next

- Learn more about infrastructure extensions
  - [Device Plugins](#)
  - [Network Plugins](#)
  - CSI [storage plugins](#)
- Learn about [kubectl plugins](#)
- Learn more about [Custom Resources](#)
- Learn more about [Extension API Servers](#)
- Learn about [Dynamic admission control](#)
- Learn about the [Operator pattern](#)

# Compute, Storage, and Networking Extensions

This section covers extensions to your cluster that do not come as part as Kubernetes itself. You can use these extensions to enhance the nodes in your cluster, or to provide the network fabric that links Pods together.

- [CSI](#) and [FlexVolume](#) storage plugins

[Container Storage Interface](#) (CSI) plugins provide a way to extend Kubernetes with supports for new kinds of volumes. The volumes can be backed by durable external storage, or provide ephemeral storage, or they might offer a read-only interface to information using a filesystem paradigm.

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The [Kubernetes Volume Plugin FAQ for Storage Vendors](#) includes general information on storage plugins.

- [Device plugins](#)

Device plugins allow a node to discover new Node facilities (in addition to the built-in node resources such as cpu and memory), and provide these custom node-local facilities to Pods that request them.

- [Network plugins](#)

A network plugin allow Kubernetes to work with different networking topologies and technologies. Your Kubernetes cluster needs a *network plugin* in order to have a working Pod network and to support other aspects of the Kubernetes network model.

Kubernetes 1.30 is compatible with [CNI](#) network plugins.

## Network Plugins

Kubernetes 1.30 supports [Container Network Interface](#) (CNI) plugins for cluster networking. You must use a CNI plugin that is compatible with your cluster and that suits your needs. Different plugins are available (both open- and closed- source) in the wider Kubernetes ecosystem.

A CNI plugin is required to implement the [Kubernetes network model](#).

You must use a CNI plugin that is compatible with the [v0.4.0](#) or later releases of the CNI specification. The Kubernetes project recommends using a plugin that is compatible with the [v1.0.0](#) CNI specification (plugins can be compatible with multiple spec versions).

## Installation

A Container Runtime, in the networking context, is a daemon on a node configured to provide CRI Services for kubelet. In particular, the Container Runtime must be configured to load the CNI plugins required to implement the Kubernetes network model.

### Note:

Prior to Kubernetes 1.24, the CNI plugins could also be managed by the kubelet using the `cni-bin-dir` and `network-plugin` command-line parameters. These command-line parameters were removed in Kubernetes 1.24, with management of the CNI no longer in scope for kubelet.

See [Troubleshooting CNI plugin-related errors](#) if you are facing issues following the removal of dockershim.

For specific information about how a Container Runtime manages the CNI plugins, see the documentation for that Container Runtime, for example:

- [containerd](#)
- [CRI-O](#)

For specific information about how to install and manage a CNI plugin, see the documentation for that plugin or [networking provider](#).



# Network Plugin Requirements

## Loopback CNI

In addition to the CNI plugin installed on the nodes for implementing the Kubernetes network model, Kubernetes also requires the container runtimes to provide a loopback interface `lo`, which is used for each sandbox (pod sandboxes, vm sandboxes, ...). Implementing the loopback interface can be accomplished by re-using the [CNI loopback plugin](#), or by developing your own code to achieve this (see [this example from CRI-O](#)).

## Support hostPort

The CNI networking plugin supports `hostPort`. You can use the official [portmap](#) plugin offered by the CNI plugin team or use your own plugin with `portMapping` functionality.

If you want to enable `hostPort` support, you must specify `portMappings` capability in your `cni-conf-dir`. For example:

```
{
 "name": "k8s-pod-network",
 "cniVersion": "0.4.0",
 "plugins": [
 {
 "type": "calico",
 "log_level": "info",
 "datastore_type": "kubernetes",
 "nodename": "127.0.0.1",
 "ipam": {
 "type": "host-local",
 "subnet": "usePodCidr"
 },
 "policy": {
 "type": "k8s"
 },
 "kubernetes": {
 "kubeconfig": "/etc/cni/net.d/calico-kubeconfig"
 }
 },
 {
 "type": "portmap",
 "capabilities": {"portMappings": true},
 "externalSetMarkChain": "KUBE-MARK-MASQ"
 }
]
}
```

## Support traffic shaping

## Experimental Feature

The CNI networking plugin also supports pod ingress and egress traffic shaping. You can use the official [bandwidth](#) plugin offered by the CNI plugin team or use your own plugin with bandwidth control functionality.

If you want to enable traffic shaping support, you must add the bandwidth plugin to your CNI configuration file (default `/etc/cni/net.d`) and ensure that the binary is included in your CNI bin dir (default `/opt/cni/bin`).

```
{
 "name": "k8s-pod-network",
 "cniVersion": "0.4.0",
 "plugins": [
 {
 "type": "calico",
 "log_level": "info",
 "datastore_type": "kubernetes",
 "nodename": "127.0.0.1",
 "ipam": {
 "type": "host-local",
 "subnet": "usePodCidr"
 },
 "policy": {
 "type": "k8s"
 },
 "kubernetes": {
 "kubeconfig": "/etc/cni/net.d/calico-kubeconfig"
 }
 },
 {
 "type": "bandwidth",
 "capabilities": {"bandwidth": true}
 }
]
}
```

Now you can add the `kubernetes.io/ingress-bandwidth` and `kubernetes.io/egress-bandwidth` annotations to your Pod. For example:

```
apiVersion: v1
kind: Pod
metadata:
 annotations:
 kubernetes.io/ingress-bandwidth: 1M
 kubernetes.io/egress-bandwidth: 1M
...
```

## What's next

- Learn more about [Cluster Networking](#)
- Learn more about [Network Policies](#)
- Learn about the [Troubleshooting CNI plugin-related errors](#)

# Device Plugins

Device plugins let you configure your cluster with support for devices or resources that require vendor-specific setup, such as GPUs, NICs, FPGAs, or non-volatile main memory.

FEATURE STATE: Kubernetes v1.26 [stable]

Kubernetes provides a device plugin framework that you can use to advertise system hardware resources to the [Kubelet](#).

Instead of customizing the code for Kubernetes itself, vendors can implement a device plugin that you deploy either manually or as a [DaemonSet](#). The targeted devices include GPUs, high-performance NICs, FPGAs, InfiniBand adapters, and other similar computing resources that may require vendor specific initialization and setup.

## Device plugin registration

The kubelet exports a Registration gRPC service:

```
service Registration {
 rpc Register(RegisterRequest) returns (Empty) {}
}
```

A device plugin can register itself with the kubelet through this gRPC service. During the registration, the device plugin needs to send:

- The name of its Unix socket.
- The Device Plugin API version against which it was built.
- The ResourceName it wants to advertise. Here ResourceName needs to follow the [extended resource naming scheme](#) as vendor-domain/resourcetype. (For example, an NVIDIA GPU is advertised as nvidia.com/gpu.)

Following a successful registration, the device plugin sends the kubelet the list of devices it manages, and the kubelet is then in charge of advertising those resources to the API server as part of the kubelet node status update. For example, after a device plugin registers hardware-vendor.example/foo with the kubelet and reports two healthy devices on a node, the node status is updated to advertise that the node has 2 "Foo" devices installed and available.

Then, users can request devices as part of a Pod specification (see [container](#)). Requesting extended resources is similar to how you manage requests and limits for other resources, with the following differences:

- Extended resources are only supported as integer resources and cannot be overcommitted.
- Devices cannot be shared between containers.

## Example

Suppose a Kubernetes cluster is running a device plugin that advertises resource hardware-vendor.example/foo on certain nodes. Here is an example of a pod requesting this resource to run a demo workload:

```

apiVersion: v1
kind: Pod
metadata:
 name: demo-pod
spec:
 containers:
 - name: demo-container-1
 image: registry.k8s.io/pause:2.0
 resources:
 limits:
 hardware-vendor.example/foo: 2
#
This Pod needs 2 of the hardware-vendor.example/foo devices
and can only schedule onto a Node that's able to satisfy
that need.
#
If the Node has more than 2 of those devices available, the
remainder would be available for other Pods to use.

```

## Device plugin implementation

The general workflow of a device plugin includes the following steps:

1. Initialization. During this phase, the device plugin performs vendor-specific initialization and setup to make sure the devices are in a ready state.
2. The plugin starts a gRPC service, with a Unix socket under the host path `/var/lib/kubelet/device-plugins/`, that implements the following interfaces:

```

service DevicePlugin {
 // GetDevicePluginOptions returns options to be communicated with Device
 Manager.
 rpc GetDevicePluginOptions(Empty) returns (DevicePluginOptions) {}

 // ListAndWatch returns a stream of List of Devices
 // Whenever a Device state change or a Device disappears, ListAndWatch
 // returns the new list
 rpc ListAndWatch(Empty) returns (stream ListAndWatchResponse) {}

 // Allocate is called during container creation so that the Device
 // Plugin can run device specific operations and instruct Kubelet
 // of the steps to make the Device available in the container
 rpc Allocate(AllocateRequest) returns (AllocateResponse) {}

 // GetPreferredAllocation returns a preferred set of devices to allocate
 // from a list of available ones. The resulting preferred allocation is not
 // guaranteed to be the allocation ultimately performed by the
 // devicemanager. It is only designed to help the devicemanager make a more
 // informed allocation decision when possible.
 rpc GetPreferredAllocation(PreferredAllocationRequest) returns
 (PreferredAllocationResponse) {}
}

```

```
// PreStartContainer is called, if indicated by Device Plugin during registration phase,
// before each container start. Device plugin can run device specific operations
// such as resetting the device before making devices available to the container.
rpc PreStartContainer(PreStartContainerRequest) returns
(PreStartContainerResponse) {}
}
```

**Note:**

Plugins are not required to provide useful implementations for `GetPreferredAllocation()` or `PreStartContainer()`. Flags indicating the availability of these calls, if any, should be set in the `DevicePluginOptions` message sent back by a call to `GetDevicePluginOptions()`. The kubelet will always call `GetDevicePluginOptions()` to see which optional functions are available, before calling any of them directly.

3. The plugin registers itself with the kubelet through the Unix socket at host path `/var/lib/kubelet/device-plugins/kubelet.sock`.

**Note:**

The ordering of the workflow is important. A plugin **MUST** start serving gRPC service before registering itself with kubelet for successful registration.

4. After successfully registering itself, the device plugin runs in serving mode, during which it keeps monitoring device health and reports back to the kubelet upon any device state changes. It is also responsible for serving `Allocate` gRPC requests. During `Allocate`, the device plugin may do device-specific preparation; for example, GPU cleanup or QRNG initialization. If the operations succeed, the device plugin returns an `AllocateResponse` that contains container runtime configurations for accessing the allocated devices. The kubelet passes this information to the container runtime.

An `AllocateResponse` contains zero or more `ContainerAllocateResponse` objects. In these, the device plugin defines modifications that must be made to a container's definition to provide access to the device. These modifications include:

- [Annotations](#)
- device nodes
- environment variables
- mounts
- fully-qualified CDI device names

**Note:**

The processing of the fully-qualified CDI device names by the Device Manager requires that the `DevicePluginCDIDevices` [feature gate](#) is enabled for both the kubelet and the kube-apiserver. This was added as an alpha feature in Kubernetes v1.28 and graduated to beta in v1.29.

## Handling kubelet restarts

A device plugin is expected to detect kubelet restarts and re-register itself with the new kubelet instance. A new kubelet instance deletes all the existing Unix sockets under `/var/lib/kubelet/device-plugins` when it starts. A device plugin can monitor the deletion of its Unix socket and re-register itself upon such an event.

## Device plugin deployment

You can deploy a device plugin as a DaemonSet, as a package for your node's operating system, or manually.

The canonical directory `/var/lib/kubelet/device-plugins` requires privileged access, so a device plugin must run in a privileged security context. If you're deploying a device plugin as a DaemonSet, `/var/lib/kubelet/device-plugins` must be mounted as a [Volume](#) in the plugin's [PodSpec](#).

If you choose the DaemonSet approach you can rely on Kubernetes to: place the device plugin's Pod onto Nodes, to restart the daemon Pod after failure, and to help automate upgrades.

## API compatibility

Previously, the versioning scheme required the Device Plugin's API version to match exactly the Kubelet's version. Since the graduation of this feature to Beta in v1.12 this is no longer a hard requirement. The API is versioned and has been stable since Beta graduation of this feature. Because of this, kubelet upgrades should be seamless but there still may be changes in the API before stabilization making upgrades not guaranteed to be non-breaking.

### Note:

Although the Device Manager component of Kubernetes is a generally available feature, the *device plugin API* is not stable. For information on the device plugin API and version compatibility, read [Device Plugin API versions](#).

As a project, Kubernetes recommends that device plugin developers:

- Watch for Device Plugin API changes in the future releases.
- Support multiple versions of the device plugin API for backward/forward compatibility.

To run device plugins on nodes that need to be upgraded to a Kubernetes release with a newer device plugin API version, upgrade your device plugins to support both versions before upgrading these nodes. Taking that approach will ensure the continuous functioning of the device allocations during the upgrade.

## Monitoring device plugin resources

FEATURE STATE: Kubernetes v1.28 [stable]

In order to monitor resources provided by device plugins, monitoring agents need to be able to discover the set of devices that are in-use on the node and obtain metadata to describe which container the metric should be associated with. [Prometheus](#) metrics exposed by device

monitoring agents should follow the [Kubernetes Instrumentation Guidelines](#), identifying containers using pod, namespace, and container prometheus labels.

The kubelet provides a gRPC service to enable discovery of in-use devices, and to provide metadata for these devices:

```
// PodResourcesLister is a service provided by the kubelet that provides information about the
// node resources consumed by pods and containers on the node
service PodResourcesLister {
 rpc List(ListPodResourcesRequest) returns (ListPodResourcesResponse) {}
 rpc GetAllocatableResources(AllocatableResourcesRequest) returns
(AllocatableResourcesResponse) {}
 rpc Get(GetPodResourcesRequest) returns (GetPodResourcesResponse) {}
}
```

## List gRPC endpoint

The List endpoint provides information on resources of running pods, with details such as the id of exclusively allocated CPUs, device id as it was reported by device plugins and id of the NUMA node where these devices are allocated. Also, for NUMA-based machines, it contains the information about memory and hugepages reserved for a container.

Starting from Kubernetes v1.27, the List endpoint can provide information on resources of running pods allocated in ResourceClaims by the DynamicResourceAllocation API. To enable this feature kubelet must be started with the following flags:

```
--feature-gates=DynamicResourceAllocation=true,KubeletPodResourcesDynamicResources=true
```

```
// ListPodResourcesResponse is the response returned by List function
message ListPodResourcesResponse {
 repeated PodResources pod_resources = 1;
}

// PodResources contains information about the node resources assigned to a pod
message PodResources {
 string name = 1;
 string namespace = 2;
 repeated ContainerResources containers = 3;
}

// ContainerResources contains information about the resources assigned to a container
message ContainerResources {
 string name = 1;
 repeated ContainerDevices devices = 2;
 repeated int64 cpu_ids = 3;
 repeated ContainerMemory memory = 4;
 repeated DynamicResource dynamic_resources = 5;
}

// ContainerMemory contains information about memory and hugepages assigned to a
container
message ContainerMemory {
```

```

 string memory_type = 1;
 uint64 size = 2;
 TopologyInfo topology = 3;
}

// Topology describes hardware topology of the resource
message TopologyInfo {
 repeated NUMANode nodes = 1;
}

// NUMA representation of NUMA node
message NUMANode {
 int64 ID = 1;
}

// ContainerDevices contains information about the devices assigned to a container
message ContainerDevices {
 string resource_name = 1;
 repeated string device_ids = 2;
 TopologyInfo topology = 3;
}

// DynamicResource contains information about the devices assigned to a container by
Dynamic Resource Allocation
message DynamicResource {
 string class_name = 1;
 string claim_name = 2;
 string claim_namespace = 3;
 repeated ClaimResource claim_resources = 4;
}

// ClaimResource contains per-plugin resource information
message ClaimResource {
 repeated CDIDevice cdi_devices = 1 [(gogoproto.customname) = "CDIDevices"];
}

// CDIDevice specifies a CDI device information
message CDIDevice {
 // Fully qualified CDI device name
 // for example: vendor.com/gpu=gpudevice1
 // see more details in the CDI specification:
 // https://github.com/container-orchestrated-devices/container-device-interface/blob/main/SPEC.md
 string name = 1;
}

```

**Note:**

cpu\_ids in the ContainerResources in the List endpoint correspond to exclusive CPUs allocated to a particular container. If the goal is to evaluate CPUs that belong to the shared pool, the List



endpoint needs to be used in conjunction with the `GetAllocatableResources` endpoint as explained below:

1. Call `GetAllocatableResources` to get a list of all the allocatable CPUs
2. Call `GetCpuIds` on all `ContainerResources` in the system
3. Subtract out all of the CPUs from the `GetCpuIds` calls from the `GetAllocatableResources` call

## GetAllocatableResources gRPC endpoint

FEATURE STATE: Kubernetes v1.28 [stable]

`GetAllocatableResources` provides information on resources initially available on the worker node. It provides more information than kubelet exports to `APIServer`.

### Note:

`GetAllocatableResources` should only be used to evaluate [allocatable](#) resources on a node. If the goal is to evaluate free/unallocated resources it should be used in conjunction with the `List()` endpoint. The result obtained by `GetAllocatableResources` would remain the same unless the underlying resources exposed to kubelet change. This happens rarely but when it does (for example: hotplug/hotunplug, device health changes), client is expected to call `GetAllocatableResources` endpoint.

However, calling `GetAllocatableResources` endpoint is not sufficient in case of cpu and/or memory update and Kubelet needs to be restarted to reflect the correct resource capacity and allocatable.

```
// AllocatableResourcesResponses contains information about all the devices known by the kubelet
message AllocatableResourcesResponse {
 repeated ContainerDevices devices = 1;
 repeated int64 cpu_ids = 2;
 repeated ContainerMemory memory = 3;
}
```

`ContainerDevices` do expose the topology information declaring to which NUMA cells the device is affine. The NUMA cells are identified using an opaque integer ID, which value is consistent to what device plugins report [when they register themselves to the kubelet](#).

The gRPC service is served over a unix socket at `/var/lib/kubelet/pod-resources/kubelet.sock`. Monitoring agents for device plugin resources can be deployed as a daemon, or as a `DaemonSet`. The canonical directory `/var/lib/kubelet/pod-resources` requires privileged access, so monitoring agents must run in a privileged security context. If a device monitoring agent is running as a `DaemonSet`, `/var/lib/kubelet/pod-resources` must be mounted as a [Volume](#) in the device monitoring agent's [PodSpec](#).

### Note:

When accessing the `/var/lib/kubelet/pod-resources/kubelet.sock` from `DaemonSet` or any other app deployed as a container on the host, which is mounting socket as a volume, it is a good practice to mount directory `/var/lib/kubelet/pod-resources/` instead of the `/var/lib/kubelet/pod-`

resources/kubelet.sock. This will ensure that after kubelet restart, container will be able to re-connect to this socket.

Container mounts are managed by inode referencing the socket or directory, depending on what was mounted. When kubelet restarts, socket is deleted and a new socket is created, while directory stays untouched. So the original inode for the socket become unusable. Inode to directory will continue working.

## Get gRPC endpoint

FEATURE STATE: Kubernetes v1.27 [alpha]

The Get endpoint provides information on resources of a running Pod. It exposes information similar to those described in the List endpoint. The Get endpoint requires PodName and PodNamespace of the running Pod.

```
// GetPodResourcesRequest contains information about the pod
message GetPodResourcesRequest {
 string pod_name = 1;
 string pod_namespace = 2;
}
```

To enable this feature, you must start your kubelet services with the following flag:

```
--feature-gates=KubeletPodResourcesGet=true
```

The Get endpoint can provide Pod information related to dynamic resources allocated by the dynamic resource allocation API. To enable this feature, you must ensure your kubelet services are started with the following flags:

```
--feature-gates=KubeletPodResourcesGet=true,DynamicResourceAllocation=true,KubeletPodResourcesDynamicResources=true
```

## Device plugin integration with the Topology Manager

FEATURE STATE: Kubernetes v1.27 [stable]

The Topology Manager is a Kubelet component that allows resources to be co-ordinated in a Topology aligned manner. In order to do this, the Device Plugin API was extended to include a TopologyInfo struct.

```
message TopologyInfo {
 repeated NUMANode nodes = 1;
}

message NUMANode {
 int64 ID = 1;
}
```

Device Plugins that wish to leverage the Topology Manager can send back a populated TopologyInfo struct as part of the device registration, along with the device IDs and the health

of the device. The device manager will then use this information to consult with the Topology Manager and make resource assignment decisions.

TopologyInfo supports setting a nodes field to either nil or a list of NUMA nodes. This allows the Device Plugin to advertise a device that spans multiple NUMA nodes.

Setting TopologyInfo to nil or providing an empty list of NUMA nodes for a given device indicates that the Device Plugin does not have a NUMA affinity preference for that device.

An example TopologyInfo struct populated for a device by a Device Plugin:

```
pluginapi.Device{ID: "25102017", Health: pluginapi.Healthy,
Topology:&pluginapi.TopologyInfo{Nodes:
[]*pluginapi.NUMANode{&pluginapi.NUMANode{ID: 0,},}}}
```

## Device plugin examples

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Here are some examples of device plugin implementations:

- [Akri](#), which lets you easily expose heterogeneous leaf devices (such as IP cameras and USB devices).
- The [AMD GPU device plugin](#)
- The [generic device plugin](#) for generic Linux devices and USB devices
- The [Intel device plugins](#) for Intel GPU, FPGA, QAT, VPU, SGX, DSA, DLB and IAA devices
- The [KubeVirt device plugins](#) for hardware-assisted virtualization
- The [NVIDIA GPU device plugin for Container-Optimized OS](#)
- The [RDMA device plugin](#)
- The [SocketCAN device plugin](#)
- The [Solarflare device plugin](#)
- The [SR-IOV Network device plugin](#)
- The [Xilinx FPGA device plugins](#) for Xilinx FPGA devices

## What's next

- Learn about [scheduling GPU resources](#) using device plugins
- Learn about [advertising extended resources](#) on a node
- Learn about the [Topology Manager](#)
- Read about using [hardware acceleration for TLS ingress](#) with Kubernetes

# Extending the Kubernetes API

Custom resources are extensions of the Kubernetes API. Kubernetes provides two ways to add custom resources to your cluster:

- The [CustomResourceDefinition](#) (CRD) mechanism allows you to declaratively define a new custom API with an API group, kind, and schema that you specify. The Kubernetes control plane serves and handles the storage of your custom resource. CRDs allow you to create new types of resources for your cluster without writing and running a custom API server.
  - The [aggregation layer](#) sits behind the primary API server, which acts as a proxy. This arrangement is called API Aggregation (AA), which allows you to provide specialized implementations for your custom resources by writing and deploying your own API server. The main API server delegates requests to your API server for the custom APIs that you specify, making them available to all of its clients.
- 

## [Custom Resources](#)

## [Kubernetes API Aggregation Layer](#)

# Custom Resources

*Custom resources* are extensions of the Kubernetes API. This page discusses when to add a custom resource to your Kubernetes cluster and when to use a standalone service. It describes the two methods for adding custom resources and how to choose between them.

## Custom resources

A *resource* is an endpoint in the [Kubernetes API](#) that stores a collection of [API objects](#) of a certain kind; for example, the built-in *pods* resource contains a collection of Pod objects.

A *custom resource* is an extension of the Kubernetes API that is not necessarily available in a default Kubernetes installation. It represents a customization of a particular Kubernetes installation. However, many core Kubernetes functions are now built using custom resources, making Kubernetes more modular.

Custom resources can appear and disappear in a running cluster through dynamic registration, and cluster admins can update custom resources independently of the cluster itself. Once a custom resource is installed, users can create and access its objects using [kubectl](#), just as they do for built-in resources like *Pods*.

## Custom controllers

On their own, custom resources let you store and retrieve structured data. When you combine a custom resource with a *custom controller*, custom resources provide a true *declarative API*.

The Kubernetes [declarative API](#) enforces a separation of responsibilities. You declare the desired state of your resource. The Kubernetes controller keeps the current state of Kubernetes objects

in sync with your declared desired state. This is in contrast to an imperative API, where you *instruct* a server what to do.

You can deploy and update a custom controller on a running cluster, independently of the cluster's lifecycle. Custom controllers can work with any kind of resource, but they are especially effective when combined with custom resources. The [Operator pattern](#) combines custom resources and custom controllers. You can use custom controllers to encode domain knowledge for specific applications into an extension of the Kubernetes API.

## Should I add a custom resource to my Kubernetes cluster?

When creating a new API, consider whether to [aggregate your API with the Kubernetes cluster APIs](#) or let your API stand alone.

| Consider API aggregation if:                                                                                                                                              | Prefer a stand-alone API if:                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Your API is <a href="#">Declarative</a> .                                                                                                                                 | Your API does not fit the <a href="#">Declarative</a> model.                                                 |
| You want your new types to be readable and writable using kubectl.                                                                                                        | kubectl support is not required                                                                              |
| You want to view your new types in a Kubernetes UI, such as dashboard, alongside built-in types.                                                                          | Kubernetes UI support is not required.                                                                       |
| You are developing a new API.                                                                                                                                             | You already have a program that serves your API and works well.                                              |
| You are willing to accept the format restriction that Kubernetes puts on REST resource paths, such as API Groups and Namespaces. (See the <a href="#">API Overview</a> .) | You need to have specific REST paths to be compatible with an already defined REST API.                      |
| Your resources are naturally scoped to a cluster or namespaces of a cluster.                                                                                              | Cluster or namespace scoped resources are a poor fit; you need control over the specifics of resource paths. |
| You want to reuse <a href="#">Kubernetes API support features</a> .                                                                                                       | You don't need those features.                                                                               |

### Declarative APIs

In a Declarative API, typically:

- Your API consists of a relatively small number of relatively small objects (resources).
- The objects define configuration of applications or infrastructure.
- The objects are updated relatively infrequently.
- Humans often need to read and write the objects.
- The main operations on the objects are CRUD-y (creating, reading, updating and deleting).
- Transactions across objects are not required: the API represents a desired state, not an exact state.

Imperative APIs are not declarative. Signs that your API might not be declarative include:

- The client says "do this", and then gets a synchronous response back when it is done.
- The client says "do this", and then gets an operation ID back, and has to check a separate Operation object to determine completion of the request.
- You talk about Remote Procedure Calls (RPCs).

- Directly storing large amounts of data; for example, > a few kB per object, or > 1000s of objects.
- High bandwidth access (10s of requests per second sustained) needed.
- Store end-user data (such as images, PII, etc.) or other large-scale data processed by applications.
- The natural operations on the objects are not CRUD-y.
- The API is not easily modeled as objects.
- You chose to represent pending operations with an operation ID or an operation object.

## Should I use a ConfigMap or a custom resource?

Use a ConfigMap if any of the following apply:

- There is an existing, well-documented configuration file format, such as a `mysql.cnf` or `pom.xml`.
- You want to put the entire configuration into one key of a ConfigMap.
- The main use of the configuration file is for a program running in a Pod on your cluster to consume the file to configure itself.
- Consumers of the file prefer to consume via file in a Pod or environment variable in a pod, rather than the Kubernetes API.
- You want to perform rolling updates via Deployment, etc., when the file is updated.

### Note:

Use a [Secret](#) for sensitive data, which is similar to a ConfigMap but more secure.

Use a custom resource (CRD or Aggregated API) if most of the following apply:

- You want to use Kubernetes client libraries and CLIs to create and update the new resource.
- You want top-level support from `kubectl`; for example, `kubectl get my-object object-name`.
- You want to build new automation that watches for updates on the new object, and then CRUD other objects, or vice versa.
- You want to write automation that handles updates to the object.
- You want to use Kubernetes API conventions like `.spec`, `.status`, and `.metadata`.
- You want the object to be an abstraction over a collection of controlled resources, or a summarization of other resources.

## Adding custom resources

Kubernetes provides two ways to add custom resources to your cluster:

- CRDs are simple and can be created without any programming.
- [API Aggregation](#) requires programming, but allows more control over API behaviors like how data is stored and conversion between API versions.

Kubernetes provides these two options to meet the needs of different users, so that neither ease of use nor flexibility is compromised.

Aggregated APIs are subordinate API servers that sit behind the primary API server, which acts as a proxy. This arrangement is called [API Aggregation](#)(AA). To users, the Kubernetes API appears extended.

CRDs allow users to create new types of resources without adding another API server. You do not need to understand API Aggregation to use CRDs.

Regardless of how they are installed, the new resources are referred to as Custom Resources to distinguish them from built-in Kubernetes resources (like pods).

**Note:**

Avoid using a Custom Resource as data storage for application, end user, or monitoring data: architecture designs that store application data within the Kubernetes API typically represent a design that is too closely coupled.

Architecturally, [cloud native](#) application architectures favor loose coupling between components. If part of your workload requires a backing service for its routine operation, run that backing service as a component or consume it as an external service. This way, your workload does not rely on the Kubernetes API for its normal operation.

## CustomResourceDefinitions

The [CustomResourceDefinition](#) API resource allows you to define custom resources. Defining a CRD object creates a new custom resource with a name and schema that you specify. The Kubernetes API serves and handles the storage of your custom resource. The name of a CRD object must be a valid [DNS subdomain name](#).

This frees you from writing your own API server to handle the custom resource, but the generic nature of the implementation means you have less flexibility than with [API server aggregation](#).

Refer to the [custom controller example](#) for an example of how to register a new custom resource, work with instances of your new resource type, and use a controller to handle events.

## API server aggregation

Usually, each resource in the Kubernetes API requires code that handles REST requests and manages persistent storage of objects. The main Kubernetes API server handles built-in resources like *Pods* and *services*, and can also generically handle custom resources through [CRDs](#).

The [aggregation layer](#) allows you to provide specialized implementations for your custom resources by writing and deploying your own API server. The main API server delegates requests to your API server for the custom resources that you handle, making them available to all of its clients.

## Choosing a method for adding custom resources

CRDs are easier to use. Aggregated APIs are more flexible. Choose the method that best meets your needs.

Typically, CRDs are a good fit if:

- You have a handful of fields



- You are using the resource within your company, or as part of a small open-source project (as opposed to a commercial product)

## Comparing ease of use

CRDs are easier to create than Aggregated APIs.

| CRDs                                                                                                                                                  | Aggregated API                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Do not require programming. Users can choose any language for a CRD controller.                                                                       | Requires programming and building binary and image.                                                                  |
| No additional service to run; CRDs are handled by API server.                                                                                         | An additional service to create and that could fail.                                                                 |
| No ongoing support once the CRD is created. Any bug fixes are picked up as part of normal Kubernetes Master upgrades.                                 | May need to periodically pickup bug fixes from upstream and rebuild and update the Aggregated API server.            |
| No need to handle multiple versions of your API; for example, when you control the client for this resource, you can upgrade it in sync with the API. | You need to handle multiple versions of your API; for example, when developing an extension to share with the world. |

## Advanced features and flexibility

Aggregated APIs offer more advanced API features and customization of other features; for example, the storage layer.

| Feature          | Description                                                                                                                                                                                 | CRDs                                                                                                                                                                                                                                                                                                                                                                  | Aggregated API                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Validation       | Help users prevent errors and allow you to evolve your API independently of your clients. These features are most useful when there are many clients who can't all update at the same time. | Yes. Most validation can be specified in the CRD using <a href="#">OpenAPI v3.0 validation</a> . <a href="#">CRDValidationRatcheting</a> feature gate allows failing validations specified using OpenAPI also can be ignored if the failing part of the resource was unchanged. Any other validations supported by addition of a <a href="#">Validating Webhook</a> . | Yes, arbitrary validation checks |
| Defaulting       | See above                                                                                                                                                                                   | Yes, either via <a href="#">OpenAPI v3.0 validation</a> default keyword (GA in 1.17), or via a <a href="#">Mutating Webhook</a> (though this will not be run when reading from etcd for old objects).                                                                                                                                                                 | Yes                              |
| Multi-versioning | Allows serving the same object through two API versions. Can help ease API changes like renaming fields. Less important if you control your client versions.                                | <a href="#">Yes</a>                                                                                                                                                                                                                                                                                                                                                   | Yes                              |
| Custom Storage   | If you need storage with a different performance mode (for example, a time-series database instead of key-value store) or isolation for                                                     | No                                                                                                                                                                                                                                                                                                                                                                    | Yes                              |



| Feature               | Description                                                                                                                                                                                                                                                              | CRDs                                                                           | Aggregated API |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------|
|                       | security (for example, encryption of sensitive information, etc.)                                                                                                                                                                                                        |                                                                                |                |
| Custom Business Logic | Perform arbitrary checks or actions when creating, reading, updating or deleting an object                                                                                                                                                                               | Yes, using <a href="#">Webhooks</a> .                                          | Yes            |
| Scale Subresource     | Allows systems like HorizontalPodAutoscaler and PodDisruptionBudget interact with your new resource                                                                                                                                                                      | <a href="#">Yes</a>                                                            | Yes            |
| Status Subresource    | Allows fine-grained access control where user writes the spec section and the controller writes the status section. Allows incrementing object Generation on custom resource data mutation (requires separate spec and status sections in the resource)                  | <a href="#">Yes</a>                                                            | Yes            |
| Other Subresources    | Add operations other than CRUD, such as "logs" or "exec".                                                                                                                                                                                                                | No                                                                             | Yes            |
| strategic-merge-patch | The new endpoints support PATCH with Content-Type: application/strategic-merge-patch+json. Useful for updating objects that may be modified both locally, and by the server. For more information, see <a href="#">"Update API Objects in Place Using kubectl patch"</a> | No                                                                             | Yes            |
| Protocol Buffers      | The new resource supports clients that want to use Protocol Buffers                                                                                                                                                                                                      | No                                                                             | Yes            |
| OpenAPI Schema        | Is there an OpenAPI (swagger) schema for the types that can be dynamically fetched from the server? Is the user protected from misspelling field names by ensuring only allowed fields are set? Are types enforced (in                                                   | Yes, based on the <a href="#">OpenAPI v3.0 validation</a> schema (GA in 1.16). | Yes            |

| Feature | Description                                       | CRDs | Aggregated API |
|---------|---------------------------------------------------|------|----------------|
|         | other words, don't put an int in a string field?) |      |                |

## Common Features

When you create a custom resource, either via a CRD or an AA, you get many features for your API, compared to implementing it outside the Kubernetes platform:

| Feature                     | What it does                                                                                                      |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|
| CRUD                        | The new endpoints support CRUD basic operations via HTTP and kubectl                                              |
| Watch                       | The new endpoints support Kubernetes Watch operations via HTTP                                                    |
| Discovery                   | Clients like kubectl and dashboard automatically offer list, display, and field edit operations on your resources |
| json-patch                  | The new endpoints support PATCH with Content-Type: application/json-patch+json                                    |
| merge-patch                 | The new endpoints support PATCH with Content-Type: application/merge-patch+json                                   |
| HTTPS                       | The new endpoints uses HTTPS                                                                                      |
| Built-in Authentication     | Access to the extension uses the core API server (aggregation layer) for authentication                           |
| Built-in Authorization      | Access to the extension can reuse the authorization used by the core API server; for example, RBAC.               |
| Finalizers                  | Block deletion of extension resources until external cleanup happens.                                             |
| Admission Webhooks          | Set default values and validate extension resources during any create/update/delete operation.                    |
| UI/CLI Display              | Kubectl, dashboard can display extension resources.                                                               |
| Unset versus Empty          | Clients can distinguish unset fields from zero-valued fields.                                                     |
| Client Libraries Generation | Kubernetes provides generic client libraries, as well as tools to generate type-specific client libraries.        |
| Labels and annotations      | Common metadata across objects that tools know how to edit for core and custom resources.                         |

## Preparing to install a custom resource

There are several points to be aware of before adding a custom resource to your cluster.

### Third party code and new points of failure

While creating a CRD does not automatically add any new points of failure (for example, by causing third party code to run on your API server), packages (for example, Charts) or other installation bundles often include CRDs as well as a Deployment of third-party code that implements the business logic for a new custom resource.

Installing an Aggregated API server always involves running a new Deployment.

## Storage

Custom resources consume storage space in the same way that ConfigMaps do. Creating too many custom resources may overload your API server's storage space.

Aggregated API servers may use the same storage as the main API server, in which case the same warning applies.

## Authentication, authorization, and auditing

CRDs always use the same authentication, authorization, and audit logging as the built-in resources of your API server.

If you use RBAC for authorization, most RBAC roles will not grant access to the new resources (except the cluster-admin role or any role created with wildcard rules). You'll need to explicitly grant access to the new resources. CRDs and Aggregated APIs often come bundled with new role definitions for the types they add.

Aggregated API servers may or may not use the same authentication, authorization, and auditing as the primary API server.

## Accessing a custom resource

Kubernetes [client libraries](#) can be used to access custom resources. Not all client libraries support custom resources. The *Go* and *Python* client libraries do.

When you add a custom resource, you can access it using:

- kubectl
- The Kubernetes dynamic client.
- A REST client that you write.
- A client generated using [Kubernetes client generation tools](#) (generating one is an advanced undertaking, but some projects may provide a client along with the CRD or AA).

## Custom resource field selectors

[Field Selectors](#) let clients select custom resources based on the value of one or more resource fields.

All custom resources support the `metadata.name` and `metadata.namespace` field selectors.

Fields declared in a [CustomResourceDefinition](#) may also be used with field selectors when included in the `spec.versions[*].selectableFields` field of the [CustomResourceDefinition](#).

## Selectable fields for custom resources

FEATURE STATE: Kubernetes v1.30 [alpha]

You need to enable the `CustomResourceFieldSelectors` [feature gate](#) to use this behavior, which then applies to all `CustomResourceDefinitions` in your cluster.

The `spec.versions[*].selectableFields` field of a [CustomResourceDefinition](#) may be used to declare which other fields in a custom resource may be used in field selectors. The following example adds the `.spec.color` and `.spec.size` fields as selectable fields.

[customresourcedefinition/shirt-resource-definition.yaml](#)

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
 name: shirts.stable.example.com
spec:
 group: stable.example.com
 scope: Namespaced
 names:
 plural: shirts
 singular: shirt
 kind: Shirt
 versions:
 - name: v1
 served: true
 storage: true
 schema:
 openAPIV3Schema:
 type: object
 properties:
 spec:
 type: object
 properties:
 color:
 type: string
 size:
 type: string
 selectableFields:
 - jsonPath: .spec.color
 - jsonPath: .spec.size
 additionalPrinterColumns:
 - jsonPath: .spec.color
 name: Color
 type: string
 - jsonPath: .spec.size
 name: Size
 type: string
```

Field selectors can then be used to get only resources with a color of blue:

```
kubectl get shirts.stable.example.com --field-selector spec.color=blue
```

The output should be:

| NAME     | COLOR | SIZE |
|----------|-------|------|
| example1 | blue  | S    |
| example2 | blue  | M    |

## What's next

- Learn how to [Extend the Kubernetes API with the aggregation layer](#).
- Learn how to [Extend the Kubernetes API with CustomResourceDefinition](#).

# Kubernetes API Aggregation Layer

The aggregation layer allows Kubernetes to be extended with additional APIs, beyond what is offered by the core Kubernetes APIs. The additional APIs can either be ready-made solutions such as a [metrics server](#), or APIs that you develop yourself.

The aggregation layer is different from [Custom Resources](#), which are a way to make the [kube-apiserver](#) recognise new kinds of object.

## Aggregation layer

The aggregation layer runs in-process with the kube-apiserver. Until an extension resource is registered, the aggregation layer will do nothing. To register an API, you add an *APIService* object, which "claims" the URL path in the Kubernetes API. At that point, the aggregation layer will proxy anything sent to that API path (e.g. /apis/myextension.mycompany.io/v1/...) to the registered *APIService*.

The most common way to implement the *APIService* is to run an *extension API server* in Pod(s) that run in your cluster. If you're using the extension API server to manage resources in your cluster, the extension API server (also written as "extension-apiserver") is typically paired with one or more [controllers](#). The apiserver-builder library provides a skeleton for both extension API servers and the associated controller(s).

## Response latency

Extension API servers should have low latency networking to and from the kube-apiserver. Discovery requests are required to round-trip from the kube-apiserver in five seconds or less.

If your extension API server cannot achieve that latency requirement, consider making changes that let you meet it.

## What's next

- To get the aggregator working in your environment, [configure the aggregation layer](#).
- Then, [setup an extension api-server](#) to work with the aggregation layer.
- Read about [APIService](#) in the API reference

Alternatively: learn how to [extend the Kubernetes API using Custom Resource Definitions](#).

# Operator pattern

Operators are software extensions to Kubernetes that make use of [custom resources](#) to manage applications and their components. Operators follow Kubernetes principles, notably the [control loop](#).

## Motivation

The *operator pattern* aims to capture the key aim of a human operator who is managing a service or set of services. Human operators who look after specific applications and services have deep knowledge of how the system ought to behave, how to deploy it, and how to react if there are problems.

People who run workloads on Kubernetes often like to use automation to take care of repeatable tasks. The operator pattern captures how you can write code to automate a task beyond what Kubernetes itself provides.

## Operators in Kubernetes

Kubernetes is designed for automation. Out of the box, you get lots of built-in automation from the core of Kubernetes. You can use Kubernetes to automate deploying and running workloads, *and* you can automate how Kubernetes does that.

Kubernetes' [operator pattern](#) concept lets you extend the cluster's behaviour without modifying the code of Kubernetes itself by linking [controllers](#) to one or more custom resources. Operators are clients of the Kubernetes API that act as controllers for a [Custom Resource](#).

## An example operator

Some of the things that you can use an operator to automate include:

- deploying an application on demand
- taking and restoring backups of that application's state
- handling upgrades of the application code alongside related changes such as database schemas or extra configuration settings
- publishing a Service to applications that don't support Kubernetes APIs to discover them
- simulating failure in all or part of your cluster to test its resilience
- choosing a leader for a distributed application without an internal member election process

What might an operator look like in more detail? Here's an example:

1. A custom resource named SampleDB, that you can configure into the cluster.
2. A Deployment that makes sure a Pod is running that contains the controller part of the operator.
3. A container image of the operator code.
4. Controller code that queries the control plane to find out what SampleDB resources are configured.

5. The core of the operator is code to tell the API server how to make reality match the configured resources.
  - If you add a new SampleDB, the operator sets up PersistentVolumeClaims to provide durable database storage, a StatefulSet to run SampleDB and a Job to handle initial configuration.
  - If you delete it, the operator takes a snapshot, then makes sure that the StatefulSet and Volumes are also removed.
6. The operator also manages regular database backups. For each SampleDB resource, the operator determines when to create a Pod that can connect to the database and take backups. These Pods would rely on a ConfigMap and / or a Secret that has database connection details and credentials.
7. Because the operator aims to provide robust automation for the resource it manages, there would be additional supporting code. For this example, code checks to see if the database is running an old version and, if so, creates Job objects that upgrade it for you.

## Deploying operators

The most common way to deploy an operator is to add the Custom Resource Definition and its associated Controller to your cluster. The Controller will normally run outside of the [control plane](#), much as you would run any containerized application. For example, you can run the controller in your cluster as a Deployment.

## Using an operator

Once you have an operator deployed, you'd use it by adding, modifying or deleting the kind of resource that the operator uses. Following the above example, you would set up a Deployment for the operator itself, and then:

```
kubecttl get SampleDB # find configured databases

kubecttl edit SampleDB/example-database # manually change some settings
```

...and that's it! The operator will take care of applying the changes as well as keeping the existing service in good shape.

## Writing your own operator

If there isn't an operator in the ecosystem that implements the behavior you want, you can code your own.

You also implement an operator (that is, a Controller) using any language / runtime that can act as a [client for the Kubernetes API](#).

Following are a few libraries and tools you can use to write your own cloud native operator.

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

- [Charmed Operator Framework](#)

- [Java Operator SDK](#)
- [Kopf](#) (Kubernetes Operator Pythonic Framework)
- [kube-rs](#) (Rust)
- [kubebuilder](#)
- [KubeOps](#) (.NET operator SDK)
- [Mast](#)
- [Metacontroller](#) along with WebHooks that you implement yourself
- [Operator Framework](#)
- [shell-operator](#)

## What's next

- Read the [CNCF Operator White Paper](#).
- Learn more about [Custom Resources](#)
- Find ready-made operators on [OperatorHub.io](#) to suit your use case
- [Publish](#) your operator for other people to use
- Read [CoreOS' original article](#) that introduced the operator pattern (this is an archived version of the original article).
- Read an [article](#) from Google Cloud about best practices for building operators



- **GETTING STARTED**

- [create](#)

- [get](#)

- [run](#)

- [expose](#)

- [delete](#)

- **APP MANAGEMENT**

- [apply](#)

- [annotate](#)

- [autoscale](#)

- [debug](#)

- [diff](#)

- [edit](#)

- [kustomize](#)

- [label](#)

- [patch](#)

- [replace](#)

- [rollout](#)

- [scale](#)

- [set](#)

- [wait](#)

- **WORKING WITH APPS**

- [attach](#)

- [auth](#)

- [cp](#)

- [describe](#)

- [exec](#)

- [logs](#)

- [port-forward](#)
- [proxy](#)
- [top](#)
- [CLUSTER MANAGEMENT](#)
- [api-versions](#)
- [certificate](#)
- [cluster-info](#)
- [cordon](#)
- [drain](#)
- [taint](#)
- [uncordon](#)
- [KUBECTL SETTINGS AND USAGE](#)
- [alpha](#)
- [api-resources](#)
- [completion](#)
- [config](#)
- [explain](#)
- [options](#)
- [plugin](#)
- [version](#)

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- example

## GETTING STARTED

This section contains the most basic commands for getting a workload running on your cluster.

- `run` will start running 1 or more instances of a container image on your cluster.
- `expose` will load balance traffic across the running instances, and can create a HA proxy for accessing the containers from outside the cluster.

Once your workloads are running, you can use the commands in the [WORKING WITH APPS](#) section to inspect them.

---

## create

Create a pod using the data in pod.json

```
kubectl create -f ./pod.json
```

Create a pod based on the JSON passed into stdin

```
cat pod.json | kubectl create -f -
```

Edit the data in docker-registry.yaml in JSON then create the resource using the edited data

```
kubectl create -f docker-registry.yaml --edit -o json
```

Create a resource from a file or from stdin.

JSON and YAML formats are accepted.

### Usage

```
$ kubectl create -f FILENAME
```

### Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                    |
|-----------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| edit                        |           | false          | Edit the API resource before creating                                                                                                                                                                    |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                       |
| filename                    | f         | []             | Filename, directory, or URL to files to use to create the resource                                                                                                                                       |
| kustomize                   | k         |                | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                     |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                        |
| raw                         |           |                | Raw URI to POST to the server. Uses the transport specified by the kubeconfig file.                                                                                                                      |

| Name                 | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|----------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record               |           | false   | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                      |
| recursive            | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| save-config          |           | false   | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| selector             | l         |         | Selector (label query) to filter on, supports '=', '==', and '!='. (e.g. -l key1=value1,key2=value2)                                                                                                                                                      |
| show-managed-fields  |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template             |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate             |           | true    | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |
| windows-line-endings |           | false   | Only relevant if --edit=true. Defaults to the line ending native to your platform.                                                                                                                                                                        |

## clusterrole

Create a cluster role named "pod-reader" that allows user to perform "get", "watch" and "list" on pods

```
kubectl create clusterrole pod-reader --verb=get,list,watch --resource=pods
```

Create a cluster role named "pod-reader" with ResourceName specified

```
kubectl create clusterrole pod-reader --verb=get --resource=pods --resource-name=readablepod --resource-name=anotherpod
```

Create a cluster role named "foo" with API Group specified

```
kubectl create clusterrole foo --verb=get,list,watch --resource=rs.extensions
```

Create a cluster role named "foo" with SubResource specified

```
kubectl create clusterrole foo --verb=get,list,watch --resource=pods,pods/status
```

Create a cluster role name "foo" with NonResourceURL specified

```
kubectl create clusterrole "foo" --verb=get --non-resource-url=/logs/*
```

Create a cluster role name "monitoring" with AggregationRule specified

```
kubectl create clusterrole monitoring --aggregation-rule="rbac.example.com/aggregate-to-monitoring=true"
```

Create a cluster role.

## Usage

```
$ kubectl create clusterrole NAME --verb=verb --resource=resource.group [--resource-name=resourcename] [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aggregation-rule            |           |                | An aggregation label selector for combining ClusterRoles.                                                                                                                                                                                                 |
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                           |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| non-resource-url            |           | []             | A partial url that user should have access to.                                                                                                                                                                                                            |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| resource                    |           | []             | Resource that the rule applies to                                                                                                                                                                                                                         |
| resource-name               |           | []             | Resource in the white list that the rule applies to, repeat this flag for multiple items                                                                                                                                                                  |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |
| verb                        |           | []             | Verb that applies to the resources contained in the rule                                                                                                                                                                                                  |

## clusterrolebinding

Create a cluster role binding for user1, user2, and group1 using the cluster-admin cluster role

```
kubectl create clusterrolebinding cluster-admin --clusterrole=cluster-admin --user=user1 --user=user2 --group=group1
```

Create a cluster role binding for a particular cluster role.

### Usage

```
$ kubectl create clusterrolebinding NAME --clusterrole=NAME [--user=username] [--group=groupname] [--serviceaccount=namespace:serviceaccountname] [--dry-run=server|client|none]
```

### Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| clusterrole                 |           |                | ClusterRole this ClusterRoleBinding should reference                                                                                                                                                                                                     |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| group                       |           | []             | Groups to bind to the clusterrole                                                                                                                                                                                                                        |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| serviceaccount              |           | []             | Service accounts to bind to the clusterrole, in the format <namespace>:<name>                                                                                                                                                                            |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## configmap

Create a new config map named my-config based on folder bar

```
kubectl create configmap my-config --from-file=path/to/bar
```

Create a new config map named my-config with specified keys instead of file basenames on disk

```
kubectl create configmap my-config --from-file=key1=/path/to/bar/file1.txt --from-file=key2=/path/to/bar/file2.txt
```

Create a new config map named my-config with key1=config1 and key2=config2

```
kubectl create configmap my-config --from-literal=key1=config1 --from-literal=key2=config2
```

Create a new config map named my-config from the key=value pairs in the file

```
kubectl create configmap my-config --from-file=path/to/bar
```

Create a new config map named my-config from an env file

```
kubectl create configmap my-config --from-env-file=path/to/bar.env
```

Create a config map based on a file, directory, or specified literal value.

A single config map may package one or more key/value pairs.

When creating a config map based on a file, the key will default to the basename of the file, and the value will default to the file content. If the basename is an invalid key, you may specify an alternate key.

When creating a config map based on a directory, each file whose basename is a valid key in the directory will be packaged into the config map. Any directory entries except regular files are ignored (e.g. subdirectories, symlinks, devices, pipes, etc).

## Usage

```
$ kubectl create configmap NAME [--from-file=[key=]source] [--from-literal=key1=value1] [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                           |
|-----------------------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
| append-hash                 |           | false   | Append a hash of the configmap to its name.                                                                                                     |
| dry-run                     |           | none    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If                     |

| Name                | Shorthand | Default        | Usage                                                                                                                                                                                                                                                                                                     |
|---------------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |                | server strategy, submit server-side request without persisting the resource.                                                                                                                                                                                                                              |
| field-manager       |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                                                                        |
| from-env-file       |           |                | Specify the path to a file to read lines of key=val pairs to create a configmap (i.e. a Docker .env file).                                                                                                                                                                                                |
| from-file           |           | []             | Key file can be specified using its file path, in which case file basename will be used as configmap key, or optionally with a key and file path, in which case the given key will be used. Specifying a directory will iterate each named file in the directory whose basename is a valid configmap key. |
| from-literal        |           | []             | Specify a key and literal value to insert in configmap (i.e. mykey=somevalue)                                                                                                                                                                                                                             |
| output              | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                                                                         |
| save-config         |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                       |
| show-managed-fields |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                             |
| template            |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                 |
| validate            |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                                                                             |

## cronjob

Create a cron job

```
kubectl create cronjob my-job --image=busybox --schedule="*/1 * * * *"
```

Create a cron job with a command

```
kubectl create cronjob my-job --image=busybox --schedule="*/1 * * * *" -- date
```

Create a cron job with the specified name.

## Usage

```
$ kubectl create cronjob NAME --image=image --schedule='0/5 * * * *?' -- [COMMAND] [args...]
```



## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| image                       |           |                | Image name to run.                                                                                                                                                                                                                                        |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| restart                     |           |                | job's restart policy. supported values: OnFailure, Never                                                                                                                                                                                                  |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| schedule                    |           |                | A schedule in the Cron format the job should be run with.                                                                                                                                                                                                 |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goolang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |

## deployment

Create a deployment named my-dep that runs the busybox image

```
kubectl create deployment my-dep --image=busybox
```

Create a deployment with a command

```
kubectl create deployment my-dep --image=busybox -- date
```

Create a deployment named my-dep that runs the nginx image with 3 replicas

```
kubectl create deployment my-dep --image=nginx --replicas=3
```

Create a deployment named my-dep that runs the busybox image and expose port 5701

```
kubectl create deployment my-dep --image=busybox --port=5701
```

Create a deployment with the specified name.

## Usage

```
$ kubectl create deployment NAME --image=image -- [COMMAND] [args...]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| image                       |           | []             | Image names to run.                                                                                                                                                                                                                                       |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| port                        |           | -1             | The port that this container exposes.                                                                                                                                                                                                                     |
| replicas                    | r         | 1              | Number of replicas to create. Default is 1.                                                                                                                                                                                                               |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goolang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |

## ingress

Create a single ingress called 'simple' that directs requests to foo.com/bar to svc # svc1:8080 with a tls secret "my-cert"

```
kubectl create ingress simple --rule="foo.com/bar=svc1:8080,tls=my-cert"
```

Create a catch all ingress of "/path" pointing to service svc:port and Ingress Class as "otheringress"

```
kubectl create ingress catch-all --class=otheringress --rule="/path=svc:port"
```

Create an ingress with two annotations: ingress.annotation1 and ingress.annotations2

```
kubectl create ingress annotated --class=default --rule="foo.com/bar=svc:port" \
--annotation ingress.annotation1=foo \
--annotation ingress.annotation2=bla
```

Create an ingress with the same host and multiple paths

```
kubectl create ingress multipath --class=default \
--rule="foo.com/=svc:port" \
--rule="foo.com/admin/=svcadmin:portadmin"
```

Create an ingress with multiple hosts and the pathType as Prefix

```
kubectl create ingress ingress1 --class=default \
--rule="foo.com/path*=svc:8080" \
--rule="bar.com/admin*=svc2:http"
```

Create an ingress with TLS enabled using the default ingress certificate and different path types

```
kubectl create ingress ingtls --class=default \
--rule="foo.com/=svc:https,tls" \
--rule="foo.com/path/subpath*=othersvc:8080"
```

Create an ingress with TLS enabled using a specific secret and pathType as Prefix

```
kubectl create ingress ingsecret --class=default \
--rule="foo.com/*=svc:8080,tls=secret1"
```

Create an ingress with a default backend

```
kubectl create ingress ingdefault --class=default \
--default-backend=defaultsvc:http \
--rule="foo.com/*=svc:8080,tls=secret1"
```

Create an ingress with the specified name.

## Usage

```
$ kubectl create ingress NAME --rule=host/path=service:port[,tls[=secret]]
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                           |
|-----------------------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyang and jsonpath output formats. |
| annotation                  |           | []      | Annotation to insert in the ingress object, in the format annotation=value                                                                      |
| class                       |           |         | Ingress Class to be used                                                                                                                        |

| Name                | Shorthand | Default        | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| default-backend     |           |                | Default service for backend, in format of svcname:port                                                                                                                                                                                                    |
| dry-run             |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager       |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| output              | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| rule                |           | []             | Rule in format host/path=service:port[,tls=secretname]. Paths containing the leading character '*' are considered pathType=Prefix. tls argument is optional.                                                                                              |
| save-config         |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| show-managed-fields |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate            |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |

## job

Create a job

```
kubectl create job my-job --image=busybox
```

Create a job with a command

```
kubectl create job my-job --image=busybox -- date
```

Create a job from a cron job named "a-cronjob"

```
kubectl create job test-job --from=cronjob/a-cronjob
```

Create a job with the specified name.

## Usage

```
$ kubectl create job NAME --image=image [--from=cronjob/name] -- [COMMAND] [args...]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| from                        |           |                | The name of the resource to create a Job from (only cronjob is supported).                                                                                                                                                                               |
| image                       |           |                | Image name to run.                                                                                                                                                                                                                                       |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## namespace

Create a new namespace named my-namespace

```
kubectl create namespace my-namespace
```

Create a namespace with the specified name.

## Usage

\$ kubectl create namespace NAME [--dry-run=server|client|none]

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                           |
|-----------------------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |

| Name                | Shorthand | Default        | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dry-run             |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager       |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| output              | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| save-config         |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| show-managed-fields |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate            |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |

## ***poddisruptionbudget***

Create a pod disruption budget named my-pdb that will select all pods with the app=rails label # and require at least one of them being available at any point in time

```
kubectl create poddisruptionbudget my-pdb --selector=app=rails --min-available=1
```

Create a pod disruption budget named my-pdb that will select all pods with the app=nginx label # and require at least half of the pods selected to be available at any point in time

```
kubectl create pdb my-pdb --selector=app=nginx --min-available=50%
```

Create a pod disruption budget with the specified name, selector, and desired minimum available pods.

### **Usage**

```
$ kubectl create poddisruptionbudget NAME --selector=SELECTOR --min-available=N [--dry-run=server|client|none]
```

### **Flags**

| Name                        | Shorthand | Default | Usage |
|-----------------------------|-----------|---------|-------|
| allow-missing-template-keys |           | true    |       |

| Name                | Shorthand | Default        | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |                | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                                          |
| dry-run             |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager       |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| max-unavailable     |           |                | The maximum number or percentage of unavailable pods this budget requires.                                                                                                                                                                               |
| min-available       |           |                | The minimum number or percentage of available pods this budget requires.                                                                                                                                                                                 |
| output              | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| save-config         |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| selector            |           |                | A label selector to use for this budget. Only equality-based selector requirements are supported.                                                                                                                                                        |
| show-managed-fields |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goyaml templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate            |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## ***priorityclass***

Create a priority class named high-priority

```
kubectl create priorityclass high-priority --value=1000 --description="high priority"
```

Create a priority class named default-priority that is considered as the global default priority

```
kubectl create priorityclass default-priority --value=1000 --global-default=true --description="default priority"
```

Create a priority class named high-priority that cannot preempt pods with lower priority

```
kubectl create priorityclass high-priority --value=1000 --description="high priority" --preemption-policy="Never"
```

Create a priority class with the specified name, value, globalDefault and description.

## Usage

```
$ kubectl create priorityclass NAME --value=VALUE --global-default=BOOL [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default              | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true                 | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                                                                          |
| description                 |           |                      | description is an arbitrary string that usually provides guidelines on when this priority class should be used.                                                                                                                                           |
| dry-run                     |           | none                 | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager               |           | kubectl-create       | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| global-default              |           | false                | global-default specifies whether this PriorityClass should be considered as the default priority.                                                                                                                                                         |
| output                      | o         |                      | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| preemption-policy           |           | PreemptLowerPriority | preemption-policy is the policy for preempting pods with lower priority.                                                                                                                                                                                  |
| save-config                 |           | false                | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| show-managed-fields         |           | false                | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template                    |           |                      | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true                 | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |
| value                       |           | 0                    | the value of this priority class.                                                                                                                                                                                                                         |



## quota

Create a new resource quota named my-quota

```
kubectl create quota my-quota --hard=cpu=1,memory=1G,pods=2,services=3,replicationcontrollers=2,resourcequotas=1,secrets=5,persistentvolumeclaims=10
```

Create a new resource quota named best-effort

```
kubectl create quota best-effort --hard=pods=100 --scopes=BestEffort
```

Create a resource quota with the specified name, hard limits, and optional scopes.

## Usage

```
$ kubectl create quota NAME [--hard=key1=value1,key2=value2] [--scopes=Scope1,Scope2] [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| hard                        |           |                | A comma-delimited set of resource=quantity pairs that define a hard limit.                                                                                                                                                                               |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| scopes                      |           |                | A comma-delimited set of quota scopes that must all match each object tracked by the quota.                                                                                                                                                              |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## role

Create a role named "pod-reader" that allows user to perform "get", "watch" and "list" on pods

```
kubectl create role pod-reader --verb=get --verb=list --verb=watch --resource=pods
```

Create a role named "pod-reader" with ResourceName specified

```
kubectl create role pod-reader --verb=get --resource=pods --resource-name=readablepod --resource-name=anotherpod
```

Create a role named "foo" with API Group specified

```
kubectl create role foo --verb=get,list,watch --resource=rs.extensions
```

Create a role named "foo" with SubResource specified

```
kubectl create role foo --verb=get,list,watch --resource=pods,pods/status
```

Create a role with single rule.

## Usage

```
$ kubectl create role NAME --verb=verb --resource=resource.group/subresource [--resource-name=resourcename] [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                               |
|-----------------------------|-----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                     |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.            |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                  |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                   |
| resource                    |           | []             | Resource that the rule applies to                                                                                                                                                                                   |
| resource-name               |           | []             | Resource in the white list that the rule applies to, repeat this flag for multiple items                                                                                                                            |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |
|                             |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                       |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| show-managed-fields |           |         |                                                                                                                                                                                                                                                           |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goolang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate            |           | true    | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |
| verb                |           | []      | Verb that applies to the resources contained in the rule                                                                                                                                                                                                  |

## rolebinding

Create a role binding for user1, user2, and group1 using the admin cluster role

```
kubectl create rolebinding admin --clusterrole=admin --user=user1 --user=user2 --group=group1
```

Create a role binding for a particular role or cluster role.

### Usage

```
$ kubectl create rolebinding NAME --clusterrole=NAME|--role=NAME [--user=username] [--group=groupname] [--serviceaccount=namespace:serviceaccountname] [--dry-run=server|client|none]
```

### Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                               |
|-----------------------------|-----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                                    |
| clusterrole                 |           |                | ClusterRole this RoleBinding should reference                                                                                                                                                                       |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.            |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                  |
| group                       |           | []             | Groups to bind to the role                                                                                                                                                                                          |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                   |
| role                        |           |                | Role this RoleBinding should reference                                                                                                                                                                              |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |
| serviceaccount              |           | []             |                                                                                                                                                                                                                     |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |         | Service accounts to bind to the role, in the format <namespace>:<name>                                                                                                                                                                                   |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates <a href="http://golang.org/pkg/text/template/#pkg-overview">[http://golang.org/pkg/text/template/#pkg-overview]</a> . |
| validate            |           | true    | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## ***secret***

Create a secret using specified subcommand.

### **Usage**

```
$ kubectl create secret
```

## ***secret docker-registry***

If you don't already have a .dockercfg file, you can create a dockercfg secret directly by using:

```
kubectl create secret docker-registry my-secret --docker-server=DOCKER_REGISTRY_SERVER
--docker-username=DOCKER_USER --docker-password=DOCKER_PASSWORD --docker-
email=DOCKER_EMAIL
```

Create a new secret named my-secret from ~/.docker/config.json

```
kubectl create secret docker-registry my-secret --from-file=.dockerconfigjson=path/to/.docker/
config.json
```

Create a new secret for use with Docker registries.

Dockercfg secrets are used to authenticate against Docker registries.

When using the Docker command line to push images, you can authenticate to a given registry by running: '\$ docker login DOCKER\_REGISTRY\_SERVER --username=DOCKER\_USER --password=DOCKER\_PASSWORD --email=DOCKER\_EMAIL'.

That produces a ~/.dockercfg file that is used by subsequent 'docker push' and 'docker pull' commands to authenticate to the registry. The email address is optional.

When creating applications, you may have a Docker registry that requires authentication. In order for the nodes to pull images on your behalf, they must have the credentials. You can provide this information by creating a dockercfg secret and attaching it to your service account.

## Usage

```
$ kubectl create secret docker-registry NAME --docker-username=user --docker-
password=password --docker-email=email [--docker-server=string] [--from-file=[key=]source]
[--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default                                                               | Usage                                                                                                                                                                                                                                                                                      |
|-----------------------------|-----------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true                                                                  | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                                                                            |
| append-hash                 |           | false                                                                 | Append a hash of the secret to its name.                                                                                                                                                                                                                                                   |
| docker-email                |           |                                                                       | Email for Docker registry                                                                                                                                                                                                                                                                  |
| docker-password             |           |                                                                       | Password for Docker registry authentication                                                                                                                                                                                                                                                |
| docker-server               |           | <a href="https://index.docker.io/v1/">https://index.docker.io/v1/</a> | Server location for Docker registry                                                                                                                                                                                                                                                        |
| docker-username             |           |                                                                       | Username for Docker registry authentication                                                                                                                                                                                                                                                |
| dry-run                     |           | none                                                                  | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                                   |
| field-manager               |           | kubectl-create                                                        | Name of the manager used to track field ownership.                                                                                                                                                                                                                                         |
| from-file                   |           | []                                                                    | Key files can be specified using their file path, in which case a default name will be given to them, or optionally with a name and file path, in which case the given name will be used. Specifying a directory will iterate each named file in the directory that is a valid secret key. |
| output                      | o         |                                                                       | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                                                          |
| save-config                 |           | false                                                                 | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                        |
| show-managed-fields         |           | false                                                                 | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                              |
| template                    |           |                                                                       | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template                                                                                                                                                                                     |

| Name     | Shorthand | Default | Usage                                                                                                                                              |
|----------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|          |           |         | format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate |           | true    | If true, use a schema to validate the input before sending it                                                                                      |

## *secret generic*

Create a new secret named my-secret with keys for each file in folder bar

```
kubectl create secret generic my-secret --from-file=path/to/bar
```

Create a new secret named my-secret with specified keys instead of names on disk

```
kubectl create secret generic my-secret --from-file=ssh-privatekey=path/to/id_rsa --from-file=ssh-publickey=path/to/id_rsa.pub
```

Create a new secret named my-secret with key1=supersecret and key2=topsecret

```
kubectl create secret generic my-secret --from-literal=key1=supersecret --from-literal=key2=topsecret
```

Create a new secret named my-secret using a combination of a file and a literal

```
kubectl create secret generic my-secret --from-file=ssh-privatekey=path/to/id_rsa --from-literal=passphrase=topsecret
```

Create a new secret named my-secret from an env file

```
kubectl create secret generic my-secret --from-env-file=path/to/bar.env
```

Create a secret based on a file, directory, or specified literal value.

A single secret may package one or more key/value pairs.

When creating a secret based on a file, the key will default to the basename of the file, and the value will default to the file content. If the basename is an invalid key or you wish to choose your own, you may specify an alternate key.

When creating a secret based on a directory, each file whose basename is a valid key in the directory will be packaged into the secret. Any directory entries except regular files are ignored (e.g. subdirectories, symlinks, devices, pipes, etc).

## Usage

```
$ kubectl create generic NAME [--type=string] [--from-file=[key=]source] [--from-literal=key1=value1] [--dry-run=server|client|none]
```

## Flags

| Name | Shorthand | Default | Usage |
|------|-----------|---------|-------|
|      |           | true    |       |

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                                                      |
|-----------------------------|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           |                | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                                                                                                           |
| append-hash                 |           | false          | Append a hash of the secret to its name.                                                                                                                                                                                                                                                   |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                                   |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                                                         |
| from-env-file               |           |                | Specify the path to a file to read lines of key=val pairs to create a secret (i.e. a Docker .env file).                                                                                                                                                                                    |
| from-file                   |           | []             | Key files can be specified using their file path, in which case a default name will be given to them, or optionally with a name and file path, in which case the given name will be used. Specifying a directory will iterate each named file in the directory that is a valid secret key. |
| from-literal                |           | []             | Specify a key and literal value to insert in secret (i.e. mykey=somevalue)                                                                                                                                                                                                                 |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                                                          |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                        |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                              |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                  |
| type                        |           |                | The type of secret to create                                                                                                                                                                                                                                                               |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                                                              |

## ***secret tls***

Create a new TLS secret named tls-secret with the given key pair

```
kubectl create secret tls tls-secret --cert=path/to/tls.cert --key=path/to/tls.key
```

Create a TLS secret from the given public/private key pair.

The public/private key pair must exist beforehand. The public key certificate must be .PEM encoded and match the given private key.

## Usage

```
$ kubectl create secret tls NAME --cert=path/to/cert/file --key=path/to/key/file [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| append-hash                 |           | false          | Append a hash of the secret to its name.                                                                                                                                                                                                                 |
| cert                        |           |                | Path to PEM encoded public key certificate.                                                                                                                                                                                                              |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| key                         |           |                | Path to private key associated with given certificate.                                                                                                                                                                                                   |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## *service*

Create a service using a specified subcommand.

## Usage

```
$ kubectl create service
```

## *service clusterip*

Create a new ClusterIP service named my-cs



```
kubectl create service clusterip my-cs --tcp=5678:8080
```

Create a new ClusterIP service named my-cs (in headless mode)

```
kubectl create service clusterip my-cs --clusterip="None"
```

Create a ClusterIP service with the specified name.

## Usage

```
$ kubectl create service clusterip NAME [--tcp=<port>:<targetPort>] [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                                                                          |
| clusterip                   |           |                | Assign your own ClusterIP or set to 'None' for a 'headless' service (no loadbalancing).                                                                                                                                                                   |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| tcp                         |           | []             | Port pairs can be specified as '<port>:<targetPort>'.                                                                                                                                                                                                     |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |

## *service externalname*

Create a new ExternalName service named my-ns

```
kubectl create service externalname my-ns --external-name bar.com
```

Create an ExternalName service with the specified name.

ExternalName service references to an external DNS address instead of only pods, which will allow application authors to reference services that exist off platform, on other clusters, or locally.

## Usage

```
$ kubectl create service externalname NAME --external-name external.name [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| external-name               |           |                | External name of service                                                                                                                                                                                                                                 |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| tcp                         |           | []             | Port pairs can be specified as '<port>:<targetPort>'.                                                                                                                                                                                                    |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goyaml templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## *service loadbalancer*

Create a new LoadBalancer service named my-lbs

```
kubectl create service loadbalancer my-lbs --tcp=5678:8080
```

Create a LoadBalancer service with the specified name.

## Usage

```
$ kubectl create service loadbalancer NAME [--tcp=port:targetPort] [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| tcp                         |           | []             | Port pairs can be specified as '<port>:<targetPort>'.                                                                                                                                                                                                     |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goolang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |

## *service nodeport*

Create a new NodePort service named my-ns

```
kubectl create service nodeport my-ns --tcp=5678:8080
```

Create a NodePort service with the specified name.

## Usage

```
$ kubectl create service nodeport NAME [--tcp=port:targetPort] [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default        | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true           | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager               |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| node-port                   |           | 0              | Port used to expose the service on each node in a cluster.                                                                                                                                                                                               |
| output                      | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| save-config                 |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| show-managed-fields         |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| tcp                         |           | []             | Port pairs can be specified as '<port>:<targetPort>'.                                                                                                                                                                                                    |
| template                    |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## serviceaccount

Create a new service account named my-service-account

```
kubectl create serviceaccount my-service-account
```

Create a service account with the specified name.

## Usage

\$ kubectl create serviceaccount NAME [--dry-run=server|client|none]

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                           |
|-----------------------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |

| Name                | Shorthand | Default        | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dry-run             |           | none           | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager       |           | kubectl-create | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| output              | o         |                | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| save-config         |           | false          | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| show-managed-fields |           | false          | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate            |           | true           | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |

## get

List all pods in ps output format

```
kubectl get pods
```

List all pods in ps output format with more information (such as node name)

```
kubectl get pods -o wide
```

List a single replication controller with specified NAME in ps output format

```
kubectl get replicationcontroller web
```

List deployments in JSON output format, in the "v1" version of the "apps" API group

```
kubectl get deployments.v1.apps -o json
```

List a single pod in JSON output format

```
kubectl get -o json pod web-pod-13je7
```

List a pod identified by type and name specified in "pod.yaml" in JSON output format

```
kubectl get -f pod.yaml -o json
```

List resources from a directory with kustomization.yaml - e.g. dir/  
kustomization.yaml

```
kubectl get -k dir/
```

Return only the phase value of the specified pod

```
kubectl get -o template pod/web-pod-13je7 --template={{.status.phase}}
```

List resource information in custom columns

```
kubectl get pod test-pod -o custom-columns=CONTAINER:.spec.containers[0].name,IMAGE:.spec.containers[0].image
```

List all replication controllers and services together in ps output format

```
kubectl get rc,services
```

List one or more resources by their type and names

```
kubectl get rc/web service/frontend pods/web-pod-13je7
```

Display one or many resources.

Prints a table of the most important information about the specified resources. You can filter the list using a label selector and the --selector flag. If the desired resource type is namespaced you will only see results in your current namespace unless you pass --all-namespaces.

Uninitialized objects are not shown unless --include-uninitialized is passed.

By specifying the output as 'template' and providing a Go template as the value of the --template flag, you can filter the attributes of the fetched resources.

Use "kubectl api-resources" for a complete list of supported resources.

## Usage

```
$ kubectl get [(-o|--output=json|yaml|name|go-template|go-template-file|template|templatefile|jsonpath|jsonpath-as-json|jsonpath-file|custom-columns|custom-columns-file|wide) (TYPE[.VERSION][.GROUP] [NAME | -l label] | TYPE[.VERSION][.GROUP]/NAME ...) [flags]
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                            |
|-----------------------------|-----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| all-namespaces              | A         | false   | If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.      |
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats. |
| chunk-size                  |           | 500     | Return large lists in chunks rather than all at once. Pass 0 to disable. This flag is beta and may change in the future.                         |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| field-selector      |           |         | Selector (field query) to filter on, supports '=', '==', and '!='. (e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| filename            | f         | []      | Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| ignore-not-found    |           | false   | If the requested object does not exist the command will return exit code 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| kustomize           | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| label-columns       | L         | []      | Accepts a comma separated list of labels that are going to be presented as columns. Names are case-sensitive. You can also use multiple flag options like -L label1 -L label2...                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| no-headers          |           | false   | When using the default or custom-column output format, don't print headers (default print headers).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| output              | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file custom-columns-file custom-columns wide<br>See custom columns [ <a href="https://kubernetes.io/docs/reference/kubectl/overview/#custom-columns">https://kubernetes.io/docs/reference/kubectl/overview/#custom-columns</a> ], golang template [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ] and jsonpath template [ <a href="https://kubernetes.io/docs/reference/kubectl/jsonpath/">https://kubernetes.io/docs/reference/kubectl/jsonpath/</a> ]. |
| output-watch-events |           | false   | Output watch event objects when --watch or --watch-only is used. Existing objects are output as initial ADDED events.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| raw                 |           |         | Raw URI to request from the server. Uses the transport specified by the kubeconfig file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| selector            | l         |         | Selector (label query) to filter on, supports '=', '==', and '!='. (e.g. -l key1=value1,key2=value2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| server-print        |           | true    | If true, have the server return the appropriate table output. Supports extension APIs and CRDs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| show-kind           |           | false   | If present, list the resource type for the requested object(s).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| show-labels         |           | false   | When printing, show all labels as the last column (default hide labels column)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| sort-by             |           |         | If non-empty, sort list types using this field specification. The field specification is expressed as a JSONPath expression (e.g. '{.metadata.name}'). The field in the API resource specified by this JSONPath expression must be an integer or a string.                                                                                                                                                                                                                                                                                                                                                                                                 |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                                                   |
|                     |           | false   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Name                      | Shorthand | Default | Usage                                                                                                                            |
|---------------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------|
| use-openapi-print-columns |           |         | If true, use x-kubernetes-print-column metadata (if present) from the OpenAPI schema for displaying a resource.                  |
| watch                     | w         | false   | After listing/getting the requested object, watch for changes. Uninitialized objects are excluded if no object name is provided. |
| watch-only                |           | false   | Watch for changes to the requested object(s), without listing/getting first.                                                     |

## run

Start a nginx pod

```
kubectl run nginx --image=nginx
```

Start a hazelcast pod and let the container expose port 5701

```
kubectl run hazelcast --image=hazelcast/hazelcast --port=5701
```

Start a hazelcast pod and set environment variables "DNS\_DOMAIN=cluster" and "POD\_NAMESPACE=default" in the container

```
kubectl run hazelcast --image=hazelcast/hazelcast --env="DNS_DOMAIN=cluster" --env="POD_NAMESPACE=default"
```

Start a hazelcast pod and set labels "app=hazelcast" and "env=prod" in the container

```
kubectl run hazelcast --image=hazelcast/hazelcast --labels="app=hazelcast,env=prod"
```

Dry run; print the corresponding API objects without creating them

```
kubectl run nginx --image=nginx --dry-run=client
```

Start a nginx pod, but overload the spec with a partial set of values parsed from JSON

```
kubectl run nginx --image=nginx --overrides='{ "apiVersion": "v1", "spec": { ... } }'
```

Start a busybox pod and keep it in the foreground, don't restart it if it exits

```
kubectl run -i -t busybox --image=busybox --restart=Never
```

Start the nginx pod using the default command, but use custom arguments (arg1 .. argN) for that command

```
kubectl run nginx --image=nginx -- <arg1> <arg2> ... <argN>
```

Start the nginx pod using a different command and custom arguments

```
kubectl run nginx --image=nginx --command -- <cmd> <arg1> ... <argN>
```



Create and run a particular image in a pod.

## Usage

```
$ kubectl run NAME --image=image [--env="key=value"] [--port=port] [--dry-run=server|client] [--overrides=inline-json] [--command] -- [COMMAND] [args...]
```

## Flags

| Name                        | Shorthand | Default     | Usage                                                                                                                                                                                                                                                                   |
|-----------------------------|-----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true        | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                                         |
| annotations                 |           | []          | Annotations to apply to the pod.                                                                                                                                                                                                                                        |
| attach                      |           | false       | If true, wait for the Pod to start running, and then attach to the Pod as if 'kubectl attach ...' were called. Default false, unless '-i/--stdin' is set, in which case the default is true. With '--restart=Never' the exit code of the container process is returned. |
| cascade                     |           | background  | Must be "background", "orphan", or "foreground". Selects the deletion cascading strategy for the dependents (e.g. Pods created by a ReplicationController). Defaults to background.                                                                                     |
| command                     |           | false       | If true and extra arguments are present, use them as the 'command' field in the container, rather than the 'args' field which is the default.                                                                                                                           |
| dry-run                     |           | none        | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                |
| env                         |           | []          | Environment variables to set in the container.                                                                                                                                                                                                                          |
| expose                      |           | false       | If true, service is created for the container(s) which are run                                                                                                                                                                                                          |
| field-manager               |           | kubectl-run | Name of the manager used to track field ownership.                                                                                                                                                                                                                      |
| filename                    | f         | []          | to use to replace the resource.                                                                                                                                                                                                                                         |
| force                       |           | false       | If true, immediately remove resources from API and bypass graceful deletion. Note that immediate deletion of some resources may result in inconsistency or data loss and requires confirmation.                                                                         |
| grace-period                |           | -1          | Period of time in seconds given to the resource to terminate gracefully. Ignored if negative. Set to 1 for immediate shutdown. Can only be set to 0 when --force is true (force deletion).                                                                              |
| hostport                    |           | -1          | The host port mapping for the container port. To demonstrate a single-machine container.                                                                                                                                                                                |
| image                       |           |             | The image for the container to run.                                                                                                                                                                                                                                     |
| image-pull-policy           |           |             | The image pull policy for the container. If left empty, this value will not be specified by the client and defaulted by the server                                                                                                                                      |
| kustomize                   | k         |             |                                                                                                                                                                                                                                                                         |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                |
|---------------------|-----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |         | Process a kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                   |
| labels              | l         |         | Comma separated labels to apply to the pod(s). Will override previous values.                                                                                                                                                        |
| leave-stdin-open    |           | false   | If the pod is started in interactive mode or with stdin, leave stdin open after the first attach completes. By default, stdin will be closed after the first attach completes.                                                       |
| limits              |           |         | The resource requirement limits for this container. For example, 'cpu=200m,memory=512Mi'. Note that server side components may assign limits depending on the server configuration, such as limit ranges.                            |
| output              | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                    |
| overrides           |           |         | An inline JSON override for the generated object. If this is non-empty, it is used to override the generated object. Requires that the object supply a valid apiVersion field.                                                       |
| pod-running-timeout |           | 1m0s    | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                                                                  |
| port                |           |         | The port that this container exposes.                                                                                                                                                                                                |
| privileged          |           | false   | If true, run the container in privileged mode.                                                                                                                                                                                       |
| quiet               | q         | false   | If true, suppress prompt messages.                                                                                                                                                                                                   |
| record              |           | false   | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists. |
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                      |
| requests            |           |         | The resource requirement requests for this container. For example, 'cpu=100m,memory=256Mi'. Note that server side components may assign requests depending on the server configuration, such as limit ranges.                        |
| restart             |           | Always  | The restart policy for this Pod. Legal values [Always, OnFailure, Never].                                                                                                                                                            |
| rm                  |           | false   | If true, delete resources created in this command for attached containers.                                                                                                                                                           |
| save-config         |           | false   | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                  |
| serviceaccount      |           |         | Service account to set in the pod spec.                                                                                                                                                                                              |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                        |
| stdin               | i         | false   |                                                                                                                                                                                                                                      |

| Name     | Shorthand | Default | Usage                                                                                                                                                                                                                                                    |
|----------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |           |         | Keep stdin open on the container(s) in the pod, even if nothing is attached.                                                                                                                                                                             |
| template |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| timeout  |           | 0s      | The length of time to wait before giving up on a delete, zero means determine a timeout from the size of the object                                                                                                                                      |
| tty      | t         | false   | Allocated a TTY for each container in the pod.                                                                                                                                                                                                           |
| wait     |           | false   | If true, wait for resources to be gone before returning. This waits for finalizers.                                                                                                                                                                      |

## expose

Create a service for a replicated nginx, which serves on port 80 and connects to the containers on port 8000

```
kubectl expose rc nginx --port=80 --target-port=8000
```

Create a service for a replication controller identified by type and name specified in "nginx-controller.yaml", which serves on port 80 and connects to the containers on port 8000

```
kubectl expose -f nginx-controller.yaml --port=80 --target-port=8000
```

Create a service for a pod valid-pod, which serves on port 444 with the name "frontend"

```
kubectl expose pod valid-pod --port=444 --name=frontend
```

Create a second service based on the above service, exposing the container port 8443 as port 443 with the name "nginx-https"

```
kubectl expose service nginx --port=443 --target-port=8443 --name=nginx-https
```

Create a service for a replicated streaming application on port 4100 balancing UDP traffic and named 'video-stream'.

```
kubectl expose rc streamer --port=4100 --protocol=UDP --name=video-stream
```

Create a service for a replicated nginx using replica set, which serves on port 80 and connects to the containers on port 8000

```
kubectl expose rs nginx --port=80 --target-port=8000
```

Create a service for an nginx deployment, which serves on port 80 and connects to the containers on port 8000

```
kubectl expose deployment nginx --port=80 --target-port=8000
```

Expose a resource as a new Kubernetes service.

Looks up a deployment, service, replica set, replication controller or pod by name and uses the selector for that resource as the selector for a new service on the specified port. A deployment or replica set will be exposed as a service only if its selector is convertible to a selector that service supports, i.e. when the selector contains only the matchLabels component. Note that if no port is specified via --port and the exposed resource has multiple ports, all will be re-used by the new service. Also if no labels are specified, the new service will re-use the labels from the resource it exposes.

Possible resources include (case insensitive):

pod (po), service (svc), replicationcontroller (rc), deployment (deploy), replicaset (rs)

## Usage

```
$ kubectrl expose (-f FILENAME | TYPE NAME) [--port=port] [--protocol=TCP|UDP|SCTP] [--target-port=number-or-name] [--name=name] [--external-ip=external-ip-of-service] [--type=type]
```

## Flags

| Name                        | Shorthand | Default         | Usage                                                                                                                                                                                                                                 |
|-----------------------------|-----------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true            | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                       |
| cluster-ip                  |           |                 | ClusterIP to be assigned to the service. Leave empty to auto-allocate, or set to 'None' to create a headless service.                                                                                                                 |
| container-port              |           |                 | Synonym for --target-port                                                                                                                                                                                                             |
| dry-run                     |           | none            | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                              |
| external-ip                 |           |                 | Additional external IP address (not managed by Kubernetes) to accept for the service. If this IP is routed to a node, the service can be accessed by this IP in addition to its generated service IP.                                 |
| field-manager               |           | kubectrl-expose | Name of the manager used to track field ownership.                                                                                                                                                                                    |
| filename                    | f         | []              | Filename, directory, or URL to files identifying the resource to expose a service                                                                                                                                                     |
| generator                   |           | service/v2      | The name of the API generator to use. There are 2 generators: 'service/v1' and 'service/v2'. The only difference between them is that service port in v1 is named 'default', while it is left unnamed in v2. Default is 'service/v2'. |
| kustomize                   | k         |                 | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                  |
| labels                      | l         |                 | Labels to apply to the service created by this call.                                                                                                                                                                                  |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| load-balancer-ip    |           |         | IP to assign to the LoadBalancer. If empty, an ephemeral IP will be created and used (cloud-provider specific).                                                                                                                                           |
| name                |           |         | The name for the newly created object.                                                                                                                                                                                                                    |
| output              | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| overrides           |           |         | An inline JSON override for the generated object. If this is non-empty, it is used to override the generated object. Requires that the object supply a valid apiVersion field.                                                                            |
| port                |           |         | The port that the service should serve on. Copied from the resource being exposed, if unspecified                                                                                                                                                         |
| protocol            |           |         | The network protocol for the service to be created. Default is 'TCP'.                                                                                                                                                                                     |
| record              |           | false   | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                      |
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| save-config         |           | false   | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| selector            |           |         | A label selector to use for this service. Only equality-based selector requirements are supported. If empty (the default) infer the selector from the replication controller or replica set.)                                                             |
| session-affinity    |           |         | If non-empty, set the session affinity for the service to this; legal values: 'None', 'ClientIP'                                                                                                                                                          |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| target-port         |           |         | Name or number for the port on the container that the service should direct traffic to. Optional.                                                                                                                                                         |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| type                |           |         | Type for this service: ClusterIP, NodePort, LoadBalancer, or ExternalName. Default is 'ClusterIP'.                                                                                                                                                        |

## delete

Delete a pod using the type and name specified in pod.json

```
kubectrl delete -f ./pod.json
```

Delete resources from a directory containing kustomization.yaml - e.g. dir/  
kustomization.yaml

```
kubectrl delete -k dir
```

Delete a pod based on the type and name in the JSON passed into stdin

```
cat pod.json | kubectrl delete -f -
```

Delete pods and services with same names "baz" and "foo"

```
kubectrl delete pod,service baz foo
```

Delete pods and services with label name=myLabel

```
kubectrl delete pods,services -l name=myLabel
```

Delete a pod with minimal delay

```
kubectrl delete pod foo --now
```

Force delete a pod on a dead node

```
kubectrl delete pod foo --force
```

Delete all pods

```
kubectrl delete pods --all
```

Delete resources by file names, stdin, resources and names, or by resources and label selector.

JSON and YAML formats are accepted. Only one type of argument may be specified: file names, resources and names, or resources and label selector.

Some resources, such as pods, support graceful deletion. These resources define a default period before they are forcibly terminated (the grace period) but you may override that value with the `--grace-period` flag, or pass `--now` to set a grace-period of 1. Because these resources often represent entities in the cluster, deletion may not be acknowledged immediately. If the node hosting a pod is down or cannot reach the API server, termination may take significantly longer than the grace period. To force delete a resource, you must specify the `--force` flag. Note: only a subset of resources support graceful deletion. In absence of the support, the `--grace-period` flag is ignored.

**IMPORTANT:** Force deleting pods does not wait for confirmation that the pod's processes have been terminated, which can leave those processes running until the node detects the deletion and completes graceful deletion. If your processes use shared storage or talk to a remote API and depend on the name of the pod to identify themselves, force deleting those pods may result in multiple processes running on different machines using the same identification which may lead to data corruption or inconsistency. Only force delete pods when you are sure the pod is terminated, or if your application can tolerate multiple copies of the same pod running at once. Also, if you force delete pods, the scheduler may place new pods on those nodes before the node has released those resources and causing those pods to be evicted immediately.

Note that the delete command does NOT do resource version checks, so if someone submits an update to a resource right when you submit a delete, their update will be lost along with the rest of the resource.

## Usage

```
$ kubectl delete ([-f FILENAME] | [-k DIRECTORY] | TYPE [(NAME | -l label | --all)])
```

## Flags

| Name             | Shorthand | Default    | Usage                                                                                                                                                                                                    |
|------------------|-----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all              |           | false      | Delete all resources, including uninitialized ones, in the namespace of the specified resource types.                                                                                                    |
| all-namespaces   | A         | false      | If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.                                                              |
| cascade          |           | background | Must be "background", "orphan", or "foreground". Selects the deletion cascading strategy for the dependents (e.g. Pods created by a ReplicationController). Defaults to background.                      |
| dry-run          |           | none       | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| field-selector   |           |            | Selector (field query) to filter on, supports '=', '==', and '!=' (e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type.                  |
| filename         | f         | []         | containing the resource to delete.                                                                                                                                                                       |
| force            |           | false      | If true, immediately remove resources from API and bypass graceful deletion. Note that immediate deletion of some resources may result in inconsistency or data loss and requires confirmation.          |
| grace-period     |           | -1         | Period of time in seconds given to the resource to terminate gracefully. Ignored if negative. Set to 1 for immediate shutdown. Can only be set to 0 when --force is true (force deletion).               |
| ignore-not-found |           | false      | Treat "resource not found" as a successful delete. Defaults to "true" when --all is specified.                                                                                                           |
| kustomize        | k         |            | Process a kustomization directory. This flag can't be used together with -f or -R.                                                                                                                       |
| now              |           | false      | If true, resources are signaled for immediate shutdown (same as --grace-period=1).                                                                                                                       |
| output           | o         |            | Output mode. Use "-o name" for shorter output (resource/name).                                                                                                                                           |
| raw              |           |            | Raw URI to DELETE to the server. Uses the transport specified by the kubeconfig file.                                                                                                                    |
| recursive        | R         | false      | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                          |

| Name     | Shorthand | Default | Usage                                                                                                               |
|----------|-----------|---------|---------------------------------------------------------------------------------------------------------------------|
| selector | l         |         | Selector (label query) to filter on, not including uninitialized ones.                                              |
| timeout  |           | 0s      | The length of time to wait before giving up on a delete, zero means determine a timeout from the size of the object |
| wait     |           | true    | If true, wait for resources to be gone before returning. This waits for finalizers.                                 |

# APP MANAGEMENT

This section contains commands for creating, updating, deleting, and viewing your workloads in a Kubernetes cluster.

---

## apply

Apply the configuration in pod.json to a pod

```
kubectl apply -f ./pod.json
```

Apply resources from a directory containing kustomization.yaml - e.g. dir/  
kustomization.yaml

```
kubectl apply -k dir/
```

Apply the JSON passed into stdin to a pod

```
cat pod.json | kubectl apply -f -
```

Note: --prune is still in Alpha # Apply the configuration in manifest.yaml that matches label app=nginx and delete all other resources that are not in the file and match label app=nginx

```
kubectl apply --prune -f manifest.yaml -l app=nginx
```

Apply the configuration in manifest.yaml and delete all the other config maps that are not in the file

```
kubectl apply --prune -f manifest.yaml --all --prune-whitelist=core/v1/ConfigMap
```

Apply a configuration to a resource by file name or stdin. The resource name must be specified. This resource will be created if it doesn't exist yet. To use 'apply', always create the resource initially with either 'apply' or 'create --save-config'.

JSON and YAML formats are accepted.

Alpha Disclaimer: the --prune functionality is not yet complete. Do not use unless you are aware of what the current state is. See <https://issues.k8s.io/34274>.



## Usage

\$ kubectl apply (-f FILENAME | -k DIRECTORY)

## Flags

| Name                        | Shorthand | Default                   | Usage                                                                                                                                                                                                       |
|-----------------------------|-----------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false                     | Select all resources in the namespace of the specified resource types.                                                                                                                                      |
| allow-missing-template-keys |           | true                      | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                             |
| cascade                     |           | background                | Must be "background", "orphan", or "foreground". Selects the deletion cascading strategy for the dependents (e.g. Pods created by a ReplicationController). Defaults to background.                         |
| dry-run                     |           | none                      | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.    |
| field-manager               |           | kubectl-client-side-apply | Name of the manager used to track field ownership.                                                                                                                                                          |
| filename                    | f         | []                        | that contains the configuration to apply                                                                                                                                                                    |
| force                       |           | false                     | If true, immediately remove resources from API and bypass graceful deletion. Note that immediate deletion of some resources may result in inconsistency or data loss and requires confirmation.             |
| force-conflicts             |           | false                     | If true, server-side apply will force the changes against conflicts.                                                                                                                                        |
| grace-period                |           | -1                        | Period of time in seconds given to the resource to terminate gracefully. Ignored if negative. Set to 1 for immediate shutdown. Can only be set to 0 when --force is true (force deletion).                  |
| kustomize                   | k         |                           | Process a kustomization directory. This flag can't be used together with -f or -R.                                                                                                                          |
| openapi-patch               |           | true                      | If true, use openapi to calculate diff when the openapi presents and the resource can be found in the openapi spec. Otherwise, fall back to use baked-in types.                                             |
| output                      | o         |                           | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                           |
| overwrite                   |           | true                      | Automatically resolve conflicts between the modified and live configuration by using values from the modified configuration                                                                                 |
| prune                       |           | false                     | Automatically delete resource objects, including the uninitialized ones, that do not appear in the configs and are created by either apply or create --save-config. Should be used with either -l or --all. |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prune-whitelist     |           | []      | Overwrite the default whitelist with <group/version/kind> for --prune                                                                                                                                                                                     |
| record              |           | false   | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                      |
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| selector            | l         |         | Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2)                                                                                                                                                       |
| server-side         |           | false   | If true, apply runs in the server instead of the client.                                                                                                                                                                                                  |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| timeout             |           | 0s      | The length of time to wait before giving up on a delete, zero means determine a timeout from the size of the object                                                                                                                                       |
| validate            |           | true    | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |
| wait                |           | false   | If true, wait for resources to be gone before returning. This waits for finalizers.                                                                                                                                                                       |

## *edit-last-applied*

Edit the last-applied-configuration annotations by type/name in YAML

```
kubectl apply edit-last-applied deployment/nginx
```

Edit the last-applied-configuration annotations by file in JSON

```
kubectl apply edit-last-applied -f deploy.yaml -o json
```

Edit the latest last-applied-configuration annotations of resources from the default editor.

The edit-last-applied command allows you to directly edit any API resource you can retrieve via the command-line tools. It will open the editor defined by your KUBE\_EDITOR, or EDITOR environment variables, or fall back to 'vi' for Linux or 'notepad' for Windows. You can edit multiple objects, although changes are applied one at a time. The command accepts file names as well as command-line arguments, although the files you point to must be previously saved versions of resources.

The default format is YAML. To edit in JSON, specify "-o json".

The flag `--windows-line-endings` can be used to force Windows line endings, otherwise the default for your operating system will be used.

In the event an error occurs while updating, a temporary file will be created on disk that contains your unapplied changes. The most common error when updating a resource is another editor changing the resource on the server. When this occurs, you will have to apply your changes to the newer version of the resource, or update your temporary saved copy to include the latest resource version.

## Usage

```
$ kubectl apply edit-last-applied (RESOURCE/NAME | -f FILENAME)
```

## Flags

| Name                        | Shorthand | Default                   | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true                      | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyang and jsonpath output formats.                                                                                                          |
| field-manager               |           | kubectl-client-side-apply | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| filename                    | f         | []                        | Filename, directory, or URL to files to use to edit the resource                                                                                                                                                                                         |
| kustomize                   | k         |                           | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| output                      | o         |                           | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| record                      |           | false                     | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                     |
| recursive                   | R         | false                     | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| show-managed-fields         |           | false                     | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                    |           |                           | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goyang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| windows-line-endings        |           | false                     | Defaults to the line ending native to your platform.                                                                                                                                                                                                     |

## ***set-last-applied***

Set the last-applied-configuration of a resource to match the contents of a file

```
kubectl apply set-last-applied -f deploy.yaml
```

Execute set-last-applied against each configuration file in a directory

```
kubectl apply set-last-applied -f path/
```

Set the last-applied-configuration of a resource to match the contents of a file; will create the annotation if it does not already exist

```
kubectl apply set-last-applied -f deploy.yaml --create-annotation=true
```

Set the latest last-applied-configuration annotations by setting it to match the contents of a file. This results in the last-applied-configuration being updated as though 'kubectl apply -f' was run, without updating any other parts of the object.

### **Usage**

```
$ kubectl apply set-last-applied -f FILENAME
```

### **Flags**

| Name                        | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                                                                          |
| create-annotation           |           | false   | Will create 'last-applied-configuration' annotations if current objects doesn't have one                                                                                                                                                                  |
| dry-run                     |           | none    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| filename                    | f         | []      | Filename, directory, or URL to files that contains the last-applied-configuration annotations                                                                                                                                                             |
| output                      | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| show-managed-fields         |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template                    |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goolang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## ***view-last-applied***

View the last-applied-configuration annotations by type/name in YAML

```
kubectl apply view-last-applied deployment/nginx
```

View the last-applied-configuration annotations by file in JSON

```
kubectl apply view-last-applied -f deploy.yaml -o json
```

View the latest last-applied-configuration annotations by type/name or file.

The default output will be printed to stdout in YAML format. You can use the -o option to change the output format.

## Usage

\$ kubectl apply view-last-applied (TYPE [NAME | -l label] | TYPE/NAME | -f FILENAME)

## Flags

| Name      | Shorthand | Default | Usage                                                                                                                                           |
|-----------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| all       |           | false   | Select all resources in the namespace of the specified resource types                                                                           |
| filename  | f         | []      | Filename, directory, or URL to files that contains the last-applied-configuration annotations                                                   |
| kustomize | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| output    | o         | yaml    | Output format. Must be one of yaml json                                                                                                         |
| recursive | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory. |
| selector  | l         |         | Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2)                                             |

# annotate

Update pod 'foo' with the annotation 'description' and the value 'my frontend' # If the same annotation is set multiple times, only the last value will be applied

```
kubectl annotate pods foo description='my frontend'
```

Update a pod identified by type and name in "pod.json"

```
kubectl annotate -f pod.json description='my frontend'
```

Update pod 'foo' with the annotation 'description' and the value 'my frontend running nginx', overwriting any existing value

```
kubectl annotate --overwrite pods foo description='my frontend running nginx'
```

Update all pods in the namespace

```
kubectl annotate pods --all description='my frontend running nginx'
```

Update pod 'foo' only if the resource is unchanged from version 1

```
kubectl annotate pods foo description='my frontend running nginx' --resource-version=1
```

Update pod 'foo' by removing an annotation named 'description' if it exists # Does not require the --overwrite flag

```
kubectl annotate pods foo description-
```

Update the annotations on one or more resources.

All Kubernetes objects support the ability to store additional data with the object as annotations. Annotations are key/value pairs that can be larger than labels and include arbitrary string values such as structured JSON. Tools and system extensions may use annotations to store their own data.

Attempting to set an annotation that already exists will fail unless --overwrite is set. If --resource-version is specified and does not match the current resource version on the server the command will fail.

Use "kubectl api-resources" for a complete list of supported resources.

## Usage

```
$ kubectl annotate [--overwrite] (-f FILENAME | TYPE NAME) KEY_1=VAL_1 ...
KEY_N=VAL_N [--resource-version=version]
```

## Flags

| Name                        | Shorthand | Default          | Usage                                                                                                                                                                                                    |
|-----------------------------|-----------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false            | Select all resources, including uninitialized ones, in the namespace of the specified resource types.                                                                                                    |
| all-namespaces              | A         | false            | If true, check the specified action in all namespaces.                                                                                                                                                   |
| allow-missing-template-keys |           | true             | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |
| dry-run                     |           | none             | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| field-manager               |           | kubectl-annotate | Name of the manager used to track field ownership.                                                                                                                                                       |
| field-selector              |           |                  | Selector (field query) to filter on, supports '=', '==', and '!=' (e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type.                  |
| filename                    | f         | []               | Filename, directory, or URL to files identifying the resource to update the annotation                                                                                                                   |
| kustomize                   | k         |                  | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                     |
| list                        |           | false            | If true, display the annotations for a given resource.                                                                                                                                                   |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| local               |           | false   | If true, annotation will NOT contact api-server but run locally.                                                                                                                                                                                          |
| output              | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| overwrite           |           | false   | If true, allow annotations to be overwritten, otherwise reject annotation updates that overwrite existing annotations.                                                                                                                                    |
| record              |           | false   | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                      |
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| resource-version    |           |         | If non-empty, the annotation update will only succeed if this is the current resource-version for the object. Only valid when specifying a single resource.                                                                                               |
| selector            | l         |         | Selector (label query) to filter on, not including uninitialized ones, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2).                                                                                                                    |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## autoscale

Auto scale a deployment "foo", with the number of pods between 2 and 10, no target CPU utilization specified so a default autoscaling policy will be used

```
kubectl autoscale deployment foo --min=2 --max=10
```

Auto scale a replication controller "foo", with the number of pods between 1 and 5, target CPU utilization at 80%

```
kubectl autoscale rc foo --max=5 --cpu-percent=80
```

Creates an autoscaler that automatically chooses and sets the number of pods that run in a Kubernetes cluster.

Looks up a deployment, replica set, stateful set, or replication controller by name and creates an autoscaler that uses the given resource as a reference. An autoscaler can automatically increase or decrease number of pods deployed within the system as needed.

## Usage

```
$ kubectl autoscale (-f FILENAME | TYPE NAME | TYPE/NAME) [--min=MINPODS] --max=MAXPODS [--cpu-percent=CPU]
```

## Flags

| Name                        | Shorthand | Default           | Usage                                                                                                                                                                                                                                |
|-----------------------------|-----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true              | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                      |
| cpu-percent                 |           | -1                | The target average CPU utilization (represented as a percent of requested CPU) over all the pods. If it's not specified or negative, a default autoscaling policy will be used.                                                      |
| dry-run                     |           | none              | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                             |
| field-manager               |           | kubectl-autoscale | Name of the manager used to track field ownership.                                                                                                                                                                                   |
| filename                    | f         | []                | Filename, directory, or URL to files identifying the resource to autoscale.                                                                                                                                                          |
| kustomize                   | k         |                   | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                 |
| max                         |           | -1                | The upper limit for the number of pods that can be set by the autoscaler. Required.                                                                                                                                                  |
| min                         |           | -1                | The lower limit for the number of pods that can be set by the autoscaler. If it's not specified or negative, the server will apply a default value.                                                                                  |
| name                        |           |                   | The name for the newly created object. If not specified, the name of the input resource will be used.                                                                                                                                |
| output                      | o         |                   | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                    |
| record                      |           | false             | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists. |
| recursive                   | R         | false             | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                      |
| save-config                 |           | false             | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                  |
|                             |           | false             | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                        |



| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| show-managed-fields |           |         |                                                                                                                                                                                                                                                           |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## debug

Create an interactive debugging session in pod mypod and immediately attach to it.  
# (requires the EphemeralContainers feature to be enabled in the cluster)

```
kubectl debug mypod -it --image=busybox
```

Create a debug container named debugger using a custom automated debugging image. # (requires the EphemeralContainers feature to be enabled in the cluster)

```
kubectl debug --image=myproj/debug-tools -c debugger mypod
```

Create a copy of mypod adding a debug container and attach to it

```
kubectl debug mypod -it --image=busybox --copy-to=my-debugger
```

Create a copy of mypod changing the command of mycontainer

```
kubectl debug mypod -it --copy-to=my-debugger --container=mycontainer -- sh
```

Create a copy of mypod changing all container images to busybox

```
kubectl debug mypod --copy-to=my-debugger --set-image=*=busybox
```

Create a copy of mypod adding a debug container and changing container images

```
kubectl debug mypod -it --copy-to=my-debugger --image=debian --set-image=app=app:debug,sidecar=sidecar:debug
```

Create an interactive debugging session on a node and immediately attach to it. #  
The container will run in the host namespaces and the host's filesystem will be mounted at /host

```
kubectl debug node/mynode -it --image=busybox
```

Debug cluster resources using interactive debugging containers.

'debug' provides automation for common debugging tasks for cluster objects identified by resource and name. Pods will be used by default if no resource is specified.

The action taken by 'debug' varies depending on what resource is specified. Supported actions include:

- Workload: Create a copy of an existing pod with certain attributes changed, for example changing the image tag to a new version.
- Workload: Add an ephemeral container to an already running pod, for example to add debugging utilities without restarting the pod.
- Node: Create a new pod that runs in the node's host namespaces and can access the node's filesystem.

## Usage

```
$ kubectl debug (POD | TYPE[.VERSION].GROUP)/NAME [-- COMMAND [args...]]
```

## Flags

| Name              | Shorthand | Default | Usage                                                                                                                                                                                   |
|-------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| arguments-only    |           | false   | If specified, everything after -- will be passed to the new container as Args instead of Command.                                                                                       |
| attach            |           | false   | If true, wait for the container to start running, and then attach as if 'kubectl attach ...' were called. Default false, unless '-i/--stdin' is set, in which case the default is true. |
| container         | c         |         | Container name to use for debug container.                                                                                                                                              |
| copy-to           |           |         | Create a copy of the target Pod with this name.                                                                                                                                         |
| env               |           | []      | Environment variables to set in the container.                                                                                                                                          |
| image             |           |         | Container image to use for debug container.                                                                                                                                             |
| image-pull-policy |           |         | The image pull policy for the container. If left empty, this value will not be specified by the client and defaulted by the server.                                                     |
| quiet             | q         | false   | If true, suppress informational messages.                                                                                                                                               |
| replace           |           | false   | When used with '--copy-to', delete the original Pod.                                                                                                                                    |
| same-node         |           | false   | When used with '--copy-to', schedule the copy of target Pod on the same node.                                                                                                           |
| set-image         |           | []      | When used with '--copy-to', a list of name=image pairs for changing container images, similar to how 'kubectl set image' works.                                                         |
| share-processes   |           | true    | When used with '--copy-to', enable process namespace sharing in the copy.                                                                                                               |
| stdin             | i         | false   | Keep stdin open on the container(s) in the pod, even if nothing is attached.                                                                                                            |
| target            |           |         | When using an ephemeral container, target processes in this container name.                                                                                                             |
| tty               | t         | false   | Allocate a TTY for the debugging container.                                                                                                                                             |

## diff

Diff resources included in pod.json

```
kubectl diff -f pod.json
```

Diff file read from stdin

```
cat service.yaml | kubectl diff -f -
```

Diff configurations specified by file name or stdin between the current online configuration, and the configuration as it would be if applied.

The output is always YAML.

KUBECTL\_EXTERNAL\_DIFF environment variable can be used to select your own diff command. Users can use external commands with params too, example:  
KUBECTL\_EXTERNAL\_DIFF="colordiff -N -u"

By default, the "diff" command available in your path will be run with the "-u" (unified diff) and "-N" (treat absent files as empty) options.

Exit status: 0 No differences were found. 1 Differences were found. >1 Kubectl or diff failed with an error.

Note: KUBECTL\_EXTERNAL\_DIFF, if used, is expected to follow that convention.

## Usage

```
$ kubectl diff -f FILENAME
```

## Flags

| Name            | Shorthand | Default                   | Usage                                                                                                                                           |
|-----------------|-----------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| field-manager   |           | kubectl-client-side-apply | Name of the manager used to track field ownership.                                                                                              |
| filename        | f         | []                        | Filename, directory, or URL to files contains the configuration to diff                                                                         |
| force-conflicts |           | false                     | If true, server-side apply will force the changes against conflicts.                                                                            |
| kustomize       | k         |                           | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| recursive       | R         | false                     | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory. |
| selector        | l         |                           | Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2)                                             |
| server-side     |           | false                     | If true, apply runs in the server instead of the client.                                                                                        |

## edit

Edit the service named 'docker-registry'

```
kubectl edit svc/docker-registry
```

Use an alternative editor

```
KUBE_EDITOR="nano" kubectl edit svc/docker-registry
```

Edit the job 'myjob' in JSON using the v1 API format

```
kubectl edit job.v1.batch/myjob -o json
```

Edit the deployment 'mydeployment' in YAML and save the modified config in its annotation

```
kubectl edit deployment/mydeployment -o yaml --save-config
```

Edit a resource from the default editor.

The edit command allows you to directly edit any API resource you can retrieve via the command-line tools. It will open the editor defined by your KUBE\_EDITOR, or EDITOR environment variables, or fall back to 'vi' for Linux or 'notepad' for Windows. You can edit multiple objects, although changes are applied one at a time. The command accepts file names as well as command-line arguments, although the files you point to must be previously saved versions of resources.

Editing is done with the API version used to fetch the resource. To edit using a specific API version, fully-qualify the resource, version, and group.

The default format is YAML. To edit in JSON, specify "-o json".

The flag --windows-line-endings can be used to force Windows line endings, otherwise the default for your operating system will be used.

In the event an error occurs while updating, a temporary file will be created on disk that contains your unapplied changes. The most common error when updating a resource is another editor changing the resource on the server. When this occurs, you will have to apply your changes to the newer version of the resource, or update your temporary saved copy to include the latest resource version.

## Usage

```
$ kubectl edit (RESOURCE/NAME | -f FILENAME)
```

## Flags

| Name                        | Shorthand | Default      | Usage                                                                                                                                            |
|-----------------------------|-----------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true         | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats. |
| field-manager               |           | kubectl-edit | Name of the manager used to track field ownership.                                                                                               |
| filename                    | f         | []           | Filename, directory, or URL to files to use to edit the resource                                                                                 |

| Name                 | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|----------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kustomize            | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                      |
| output               | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| output-patch         |           | false   | Output the patch if the resource is edited.                                                                                                                                                                                                               |
| record               |           | false   | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                      |
| recursive            | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| save-config          |           | false   | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| show-managed-fields  |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template             |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate             |           | true    | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |
| windows-line-endings |           | false   | Defaults to the line ending native to your platform.                                                                                                                                                                                                      |

## kustomize

Build the current working directory

```
kubectl kustomize
```

Build some shared configuration directory

```
kubectl kustomize /home/config/production
```

Build from github

```
kubectl kustomize https://github.com/kubernetes-sigs/kustomize.git/examples/helloWorld?ref=v1.0.6
```

Build a set of KRM resources using a 'kustomization.yaml' file. The DIR argument must be a path to a directory containing 'kustomization.yaml', or a git repository URL with a path suffix specifying same with respect to the repository root. If DIR is omitted, '.' is assumed.

## Usage

\$ kubectl kustomize DIR

## Flags

| Name                   | Shorthand | Default                  | Usage                                                                                                                                                                  |
|------------------------|-----------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| as-current-user        |           | false                    | use the uid and gid of the command executor to run the function in the container                                                                                       |
| enable-alpha-plugins   |           | false                    | enable kustomize plugins                                                                                                                                               |
| enable-helm            |           | false                    | Enable use of the Helm chart inflator generator.                                                                                                                       |
| enable-managedby-label |           | false                    | enable adding app.kubernetes.io/managed-by                                                                                                                             |
| env                    | e         | []                       | a list of environment variables to be used by functions                                                                                                                |
| helm-command           |           | helm                     | helm command (path to executable)                                                                                                                                      |
| load-restrictor        |           | LoadRestrictionsRootOnly | if set to 'LoadRestrictionsNone', local kustomizations may load files from outside their root. This does, however, break the relocatability of the kustomization.      |
| mount                  |           | []                       | a list of storage options read from the filesystem                                                                                                                     |
| network                |           | false                    | enable network access for functions that declare it                                                                                                                    |
| network-name           |           | bridge                   | the docker network to run the container in                                                                                                                             |
| output                 | o         |                          | If specified, write output to this path.                                                                                                                               |
| reorder                |           | legacy                   | Reorder the resources just before output. Use 'legacy' to apply a legacy reordering (Namespaces first, Webhooks last, etc). Use 'none' to suppress a final reordering. |

## label

Update pod 'foo' with the label 'unhealthy' and the value 'true'

```
kubectl label pods foo unhealthy=true
```

Update pod 'foo' with the label 'status' and the value 'unhealthy', overwriting any existing value

```
kubectl label --overwrite pods foo status=unhealthy
```

Update all pods in the namespace

```
kubectl label pods --all status=unhealthy
```

Update a pod identified by the type and name in "pod.json"

```
kubectl label -f pod.json status=unhealthy
```

Update pod 'foo' only if the resource is unchanged from version 1

```
kubectl label pods foo status=unhealthy --resource-version=1
```

Update pod 'foo' by removing a label named 'bar' if it exists # Does not require the --overwrite flag

```
kubectl label pods foo bar-
```

Update the labels on a resource.

- A label key and value must begin with a letter or number, and may contain letters, numbers, hyphens, dots, and underscores, up to 63 characters each.
- Optionally, the key can begin with a DNS subdomain prefix and a single '/', like example.com/my-app.
- If --overwrite is true, then existing labels can be overwritten, otherwise attempting to overwrite a label will result in an error.
- If --resource-version is specified, then updates will use this resource version, otherwise the existing resource-version will be used.

## Usage

```
$ kubectl label [--overwrite] (-f FILENAME | TYPE NAME) KEY_1=VAL_1 ... KEY_N=VAL_N [--resource-version=version]
```

## Flags

| Name                        | Shorthand | Default       | Usage                                                                                                                                                                                                    |
|-----------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false         | Select all resources, including uninitialized ones, in the namespace of the specified resource types                                                                                                     |
| all-namespaces              | A         | false         | If true, check the specified action in all namespaces.                                                                                                                                                   |
| allow-missing-template-keys |           | true          | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                         |
| dry-run                     |           | none          | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| field-manager               |           | kubectl-label | Name of the manager used to track field ownership.                                                                                                                                                       |
| field-selector              |           |               | Selector (field query) to filter on, supports '=', '==', and '!='. (e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type.                 |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| filename            | f         | []      | Filename, directory, or URL to files identifying the resource to update the labels                                                                                                                                                                       |
| kustomize           | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| list                |           | false   | If true, display the labels for a given resource.                                                                                                                                                                                                        |
| local               |           | false   | If true, label will NOT contact api-server but run locally.                                                                                                                                                                                              |
| output              | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| overwrite           |           | false   | If true, allow labels to be overwritten, otherwise reject label updates that overwrite existing labels.                                                                                                                                                  |
| record              |           | false   | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                     |
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| resource-version    |           |         | If non-empty, the labels update will only succeed if this is the current resource-version for the object. Only valid when specifying a single resource.                                                                                                  |
| selector            | l         |         | Selector (label query) to filter on, not including uninitialized ones, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2).                                                                                                                   |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## patch

Partially update a node using a strategic merge patch, specifying the patch as JSON

```
kubectl patch node k8s-node-1 -p '{"spec":{"unschedulable":true}}'
```

Partially update a node using a strategic merge patch, specifying the patch as YAML

```
kubectl patch node k8s-node-1 -p '$spec:\n unschedulable: true'
```

Partially update a node identified by the type and name specified in "node.json" using strategic merge patch

```
kubectl patch -f node.json -p '{"spec":{"unschedulable":true}}'
```



Update a container's image; spec.containers[\*].name is required because it's a merge key

```
kubectl patch pod valid-pod -p '{"spec":{"containers":[{"name":"kubernetes-serve-hostname","image":"new image"}]}}'
```

Update a container's image using a JSON patch with positional arrays

```
kubectl patch pod valid-pod --type='json' -p='[{"op": "replace", "path": "/spec/containers/0/image", "value":"new image"}]'
```

Update fields of a resource using strategic merge patch, a JSON merge patch, or a JSON patch.

JSON and YAML formats are accepted.

## Usage

```
$ kubectl patch (-f FILENAME | TYPE NAME) [-p PATCH|--patch-file FILE]
```

## Flags

| Name                        | Shorthand | Default       | Usage                                                                                                                                                                                                                                |
|-----------------------------|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true          | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                                                     |
| dry-run                     |           | none          | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                             |
| field-manager               |           | kubectl-patch | Name of the manager used to track field ownership.                                                                                                                                                                                   |
| filename                    | f         | []            | Filename, directory, or URL to files identifying the resource to update                                                                                                                                                              |
| kustomize                   | k         |               | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                 |
| local                       |           | false         | If true, patch will operate on the content of the file, not the server-side resource.                                                                                                                                                |
| output                      | o         |               | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                    |
| patch                       | p         |               | The patch to be applied to the resource JSON file.                                                                                                                                                                                   |
| patch-file                  |           |               | A file containing a patch to be applied to the resource.                                                                                                                                                                             |
| record                      |           | false         | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists. |
| recursive                   | R         | false         | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                      |

| Name                | Shorthand | Default   | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| show-managed-fields |           | false     | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |           | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| type                |           | strategic | The type of patch being provided; one of [json merge strategic]                                                                                                                                                                                          |

## replace

Replace a pod using the data in pod.json

```
kubectl replace -f ./pod.json
```

Replace a pod based on the JSON passed into stdin

```
cat pod.json | kubectl replace -f -
```

Update a single-container pod's image version (tag) to v4

```
kubectl get pod mypod -o yaml | sed 's/(image: myimage\):.*$/1:v4/' | kubectl replace -f -
```

Force replace, delete and then re-create the resource

```
kubectl replace --force -f ./pod.json
```

Replace a resource by file name or stdin.

JSON and YAML formats are accepted. If replacing an existing resource, the complete resource spec must be provided. This can be obtained by

```
$ kubectl get TYPE NAME -o yaml
```

### Usage

```
$ kubectl replace -f FILENAME
```

### Flags

| Name                        | Shorthand | Default    | Usage                                                                                                                                           |
|-----------------------------|-----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true       | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |
| cascade                     |           | background | Must be "background", "orphan", or "foreground". Selects the deletion cascading strategy for the                                                |

| Name                | Shorthand | Default         | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |                 | dependents (e.g. Pods created by a ReplicationController). Defaults to background.                                                                                                                                                                       |
| dry-run             |           | none            | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager       |           | kubectl-replace | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| filename            | f         | []              | to use to replace the resource.                                                                                                                                                                                                                          |
| force               |           | false           | If true, immediately remove resources from API and bypass graceful deletion. Note that immediate deletion of some resources may result in inconsistency or data loss and requires confirmation.                                                          |
| grace-period        |           | -1              | Period of time in seconds given to the resource to terminate gracefully. Ignored if negative. Set to 1 for immediate shutdown. Can only be set to 0 when --force is true (force deletion).                                                               |
| kustomize           | k         |                 | Process a kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                       |
| output              | o         |                 | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| raw                 |           |                 | Raw URI to PUT to the server. Uses the transport specified by the kubeconfig file.                                                                                                                                                                       |
| recursive           | R         | false           | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| save-config         |           | false           | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| show-managed-fields |           | false           | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |                 | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| timeout             |           | 0s              | The length of time to wait before giving up on a delete, zero means determine a timeout from the size of the object                                                                                                                                      |
| validate            |           | true            | If true, use a schema to validate the input before sending it                                                                                                                                                                                            |
| wait                |           | false           | If true, wait for resources to be gone before returning. This waits for finalizers.                                                                                                                                                                      |

# rollout

Rollback to the previous deployment

```
kubectl rollout undo deployment/abc
```

Check the rollout status of a daemonset

```
kubectl rollout status daemonset/foo
```

Manage the rollout of a resource.

Valid resource types include:

- deployments
- daemonsets
- statefulsets

## Usage

```
$ kubectl rollout SUBCOMMAND
```

---

## history

View the rollout history of a deployment

```
kubectl rollout history deployment/abc
```

View the details of daemonset revision 3

```
kubectl rollout history daemonset/abc --revision=3
```

View previous rollout revisions and configurations.

## Usage

```
$ kubectl rollout history (TYPE NAME | TYPE/NAME) [flags]
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                           |
|-----------------------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |
| filename                    | f         | []      | Filename, directory, or URL to files identifying the resource to get from a server.                                                             |
| kustomize                   | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| output                      | o         |         |                                                                                                                                                 |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| revision            |           | 0       | See the details, including podTemplate of the revision specified                                                                                                                                                                                         |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## *pause*

Mark the nginx deployment as paused # Any current state of the deployment will continue its function; new updates # to the deployment will not have an effect as long as the deployment is paused

```
kubectl rollout pause deployment/nginx
```

Mark the provided resource as paused.

Paused resources will not be reconciled by a controller. Use "kubectl rollout resume" to resume a paused resource. Currently only deployments support being paused.

## Usage

```
$ kubectl rollout pause RESOURCE
```

## Flags

| Name                        | Shorthand | Default         | Usage                                                                                                                                           |
|-----------------------------|-----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true            | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |
| field-manager               |           | kubectl-rollout | Name of the manager used to track field ownership.                                                                                              |
| filename                    | f         | []              | Filename, directory, or URL to files identifying the resource to get from a server.                                                             |
| kustomize                   | k         |                 | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| output                      | o         |                 | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.               |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## restart

Restart a deployment

```
kubectl rollout restart deployment/nginx
```

Restart a daemon set

```
kubectl rollout restart daemonset/abc
```

Restart a resource.

Resource rollout will be restarted.

## Usage

```
$ kubectl rollout restart RESOURCE
```

## Flags

| Name                        | Shorthand | Default         | Usage                                                                                                                                            |
|-----------------------------|-----------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true            | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats. |
| field-manager               |           | kubectl-rollout | Name of the manager used to track field ownership.                                                                                               |
| filename                    | f         | []              | Filename, directory, or URL to files identifying the resource to get from a server.                                                              |
| kustomize                   | k         |                 | Process the kustomization directory. This flag can't be used together with -f or -R.                                                             |
| output                      | o         |                 | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                |
| recursive                   | R         | false           | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.  |
|                             |           | false           | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                    |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| show-managed-fields |           |         |                                                                                                                                                                                                                                                           |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## resume

Resume an already paused deployment

```
kubectl rollout resume deployment/nginx
```

Resume a paused resource.

Paused resources will not be reconciled by a controller. By resuming a resource, we allow it to be reconciled again. Currently only deployments support being resumed.

## Usage

```
$ kubectl rollout resume RESOURCE
```

## Flags

| Name                        | Shorthand | Default         | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true            | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                                                                          |
| field-manager               |           | kubectl-rollout | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| filename                    | f         | []              | Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                       |
| kustomize                   | k         |                 | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                      |
| output                      | o         |                 | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| recursive                   | R         | false           | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| show-managed-fields         |           | false           | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template                    |           |                 | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## ***status***

Watch the rollout status of a deployment

```
kubectl rollout status deployment/nginx
```

Show the status of the rollout.

By default 'rollout status' will watch the status of the latest rollout until it's done. If you don't want to wait for the rollout to finish then you can use `--watch=false`. Note that if a new rollout starts in-between, then 'rollout status' will continue watching the latest revision. If you want to pin to a specific revision and abort if it is rolled over by another revision, use `--revision=N` where N is the revision you need to watch for.

### **Usage**

```
$ kubectl rollout status (TYPE NAME | TYPE/NAME) [flags]
```

### **Flags**

| Name      | Shorthand | Default | Usage                                                                                                                                           |
|-----------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| filename  | f         | []      | Filename, directory, or URL to files identifying the resource to get from a server.                                                             |
| kustomize | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| recursive | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory. |
| revision  |           | 0       | Pin to a specific revision for showing its status. Defaults to 0 (last revision).                                                               |
| timeout   |           | 0s      | The length of time to wait before ending watch, zero means never. Any other values should contain a corresponding time unit (e.g. 1s, 2m, 3h).  |
| watch     | w         | true    | Watch the status of the rollout until it's done.                                                                                                |

## ***undo***

Roll back to the previous deployment

```
kubectl rollout undo deployment/abc
```

Roll back to daemonset revision 3

```
kubectl rollout undo daemonset/abc --to-revision=3
```

Roll back to the previous deployment with dry-run

```
kubectl rollout undo --dry-run=server deployment/abc
```

Roll back to a previous rollout.



## Usage

\$ kubectl rollout undo (TYPE NAME | TYPE/NAME) [flags]

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| filename                    | f         | []      | Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                      |
| kustomize                   | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| output                      | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| recursive                   | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| show-managed-fields         |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                    |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| to-revision                 |           | 0       | The revision to rollback to. Default to 0 (last revision).                                                                                                                                                                                               |

## scale

Scale a replica set named 'foo' to 3

```
kubectl scale --replicas=3 rs/foo
```

Scale a resource identified by type and name specified in "foo.yaml" to 3

```
kubectl scale --replicas=3 -f foo.yaml
```

If the deployment named mysql's current size is 2, scale mysql to 3

```
kubectl scale --current-replicas=2 --replicas=3 deployment/mysql
```

Scale multiple replication controllers

```
kubectl scale --replicas=5 rc/foo rc/bar rc/baz
```

Scale stateful set named 'web' to 3

```
kubectl scale --replicas=3 statefulset/web
```

Set a new size for a deployment, replica set, replication controller, or stateful set.

Scale also allows users to specify one or more preconditions for the scale action.

If `--current-replicas` or `--resource-version` is specified, it is validated before the scale is attempted, and it is guaranteed that the precondition holds true when the scale is sent to the server.

## Usage

```
$ kubectl scale [--resource-version=version] [--current-replicas=count] --replicas=COUNT (-f
FILENAME | TYPE NAME)
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                                                                                                                |
|-----------------------------|-----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false   | Select all resources in the namespace of the specified resource types                                                                                                                                                                |
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                      |
| current-replicas            |           | -1      | Precondition for current size. Requires that the current size of the resource match this value in order to scale. -1 (default) for no condition.                                                                                     |
| dry-run                     |           | none    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                             |
| filename                    | f         | []      | Filename, directory, or URL to files identifying the resource to set a new size                                                                                                                                                      |
| kustomize                   | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                 |
| output                      | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                    |
| record                      |           | false   | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists. |
| recursive                   | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                      |
| replicas                    |           | 0       | The new desired number of replicas. Required.                                                                                                                                                                                        |
| resource-version            |           |         | Precondition for resource version. Requires that the current resource version match this value in order to scale.                                                                                                                    |
| selector                    | l         |         |                                                                                                                                                                                                                                      |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |         | Selector (label query) to filter on, supports '=', '==', and '!='. (e.g. -l key1=value1,key2=value2)                                                                                                                                                      |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| timeout             |           | 0s      | The length of time to wait before giving up on a scale operation, zero means don't wait. Any other values should contain a corresponding time unit (e.g. 1s, 2m, 3h).                                                                                     |

## set

Configure application resources.

These commands help you make changes to existing application resources.

### Usage

```
$ kubectl set SUBCOMMAND
```

### env

Update deployment 'registry' with a new environment variable

```
kubectl set env deployment/registry STORAGE_DIR=/local
```

List the environment variables defined on a deployments 'sample-build'

```
kubectl set env deployment/sample-build --list
```

List the environment variables defined on all pods

```
kubectl set env pods --all --list
```

Output modified deployment in YAML, and does not alter the object on the server

```
kubectl set env deployment/sample-build STORAGE_DIR=/data -o yaml
```

Update all containers in all replication controllers in the project to have ENV=prod

```
kubectl set env rc --all ENV=prod
```

Import environment from a secret

```
kubectl set env --from=secret/mysecret deployment/myapp
```

Import environment from a config map with a prefix

```
kubectl set env --from=configmap/myconfigmap --prefix=MYSQL_ deployment/myapp
```

Import specific keys from a config map

```
kubectl set env --keys=my-example-key --from=configmap/myconfigmap deployment/myapp
```

Remove the environment variable ENV from container 'c1' in all deployment configs

```
kubectl set env deployments --all --containers="c1" ENV-
```

Remove the environment variable ENV from a deployment definition on disk and # update the deployment config on the server

```
kubectl set env -f deploy.json ENV-
```

Set some of the local shell environment into a deployment config on the server

```
env | grep RAILS_ | kubectl set env -e - deployment/registry
```

Update environment variables on a pod template.

List environment variable definitions in one or more pods, pod templates. Add, update, or remove container environment variable definitions in one or more pod templates (within replication controllers or deployment configurations). View or modify the environment variable definitions on all containers in the specified pods or pod templates, or just those that match a wildcard.

If "--env -" is passed, environment variables can be read from STDIN using the standard env syntax.

Possible resources include (case insensitive):

pod (po), replicationcontroller (rc), deployment (deploy), daemonset (ds), statefulset (sts), cronjob (cj), replicaset (rs)

## Usage

```
$ kubectl set env RESOURCE/NAME KEY_1=VAL_1 ... KEY_N=VAL_N
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                            |
|-----------------------------|-----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false   | If true, select all resources in the namespace of the specified resource types                                                                   |
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats. |
| containers                  | c         | *       | The names of containers in the selected pod templates to change - may use wildcards                                                              |
| dry-run                     |           | none    |                                                                                                                                                  |

| Name                | Shorthand | Default     | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |             | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| env                 | e         | []          | Specify a key-value pair for an environment variable to set into each container.                                                                                                                                                                          |
| field-manager       |           | kubectl-set | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| filename            | f         | []          | Filename, directory, or URL to files the resource to update the env                                                                                                                                                                                       |
| from                |           |             | The name of a resource from which to inject environment variables                                                                                                                                                                                         |
| keys                |           | []          | Comma-separated list of keys to import from specified resource                                                                                                                                                                                            |
| kustomize           | k         |             | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                      |
| list                |           | false       | If true, display the environment and any changes in the standard format. this flag will removed when we have kubectl view env.                                                                                                                            |
| local               |           | false       | If true, set env will NOT contact api-server but run locally.                                                                                                                                                                                             |
| output              | o         |             | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| overwrite           |           | true        | If true, allow environment to be overwritten, otherwise reject updates that overwrite existing environment.                                                                                                                                               |
| prefix              |           |             | Prefix to append to variable names                                                                                                                                                                                                                        |
| recursive           | R         | false       | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| resolve             |           | false       | If true, show secret or configmap references when listing variables                                                                                                                                                                                       |
| selector            | l         |             | Selector (label query) to filter on                                                                                                                                                                                                                       |
| show-managed-fields |           | false       | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |             | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## image

Set a deployment's nginx container image to 'nginx:1.9.1', and its busybox container image to 'busybox'

```
kubectl set image deployment/nginx busybox=busybox nginx=nginx:1.9.1
```

Update all deployments' and rc's nginx container's image to 'nginx:1.9.1'

```
kubectl set image deployments,rc nginx=nginx:1.9.1 --all
```

Update image of all containers of daemonset abc to 'nginx:1.9.1'

```
kubectl set image daemonset abc *=nginx:1.9.1
```

Print result (in yaml format) of updating nginx container image from local file, without hitting the server

```
kubectl set image -f path/to/file.yaml nginx=nginx:1.9.1 --local -o yaml
```

Update existing container image(s) of resources.

Possible resources include (case insensitive):

pod (po), replicationcontroller (rc), deployment (deploy), daemonset (ds), statefulset (sts), cronjob (cj), replicaset (rs)

## Usage

```
$ kubectl set image (-f FILENAME | TYPE NAME)
CONTAINER_NAME_1=CONTAINER_IMAGE_1 ...
CONTAINER_NAME_N=CONTAINER_IMAGE_N
```

## Flags

| Name                        | Shorthand | Default     | Usage                                                                                                                                                                                                                                |
|-----------------------------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false       | Select all resources, including uninitialized ones, in the namespace of the specified resource types                                                                                                                                 |
| allow-missing-template-keys |           | true        | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                                                     |
| dry-run                     |           | none        | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                             |
| field-manager               |           | kubectl-set | Name of the manager used to track field ownership.                                                                                                                                                                                   |
| filename                    | f         | []          | Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                  |
| kustomize                   | k         |             | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                 |
| local                       |           | false       | If true, set image will NOT contact api-server but run locally.                                                                                                                                                                      |
| output                      | o         |             | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                    |
| record                      |           | false       | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists. |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| recursive           | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| selector            | l         |         | Selector (label query) to filter on, not including uninitialized ones, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2)                                                                                                                     |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goolang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## resources

Set a deployments nginx container cpu limits to "200m" and memory to "512Mi"

```
kubectl set resources deployment nginx -c=nginx --limits=cpu=200m,memory=512Mi
```

Set the resource request and limits for all containers in nginx

```
kubectl set resources deployment nginx --limits=cpu=200m,memory=512Mi --requests=cpu=100m,memory=256Mi
```

Remove the resource requests for resources on containers in nginx

```
kubectl set resources deployment nginx --limits=cpu=0,memory=0 --requests=cpu=0,memory=0
```

Print the result (in yaml format) of updating nginx container limits from a local, without hitting the server

```
kubectl set resources -f path/to/file.yaml --limits=cpu=200m,memory=512Mi --local -o yaml
```

Specify compute resource requirements (CPU, memory) for any resource that defines a pod template. If a pod is successfully scheduled, it is guaranteed the amount of resource requested, but may burst up to its specified limits.

For each compute resource, if a limit is specified and a request is omitted, the request will default to the limit.

Possible resources include (case insensitive): Use "kubectl api-resources" for a complete list of supported resources..

## Usage

```
$ kubectl set resources (-f FILENAME | TYPE NAME) ([--limits=LIMITS & --requests=REQUESTS]
```

## Flags

| Name                        | Shorthand | Default     | Usage                                                                                                                                                                                                                                |
|-----------------------------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false       | Select all resources, including uninitialized ones, in the namespace of the specified resource types                                                                                                                                 |
| allow-missing-template-keys |           | true        | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                                                     |
| containers                  | c         | *           | The names of containers in the selected pod templates to change, all containers are selected by default - may use wildcards                                                                                                          |
| dry-run                     |           | none        | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                             |
| field-manager               |           | kubectl-set | Name of the manager used to track field ownership.                                                                                                                                                                                   |
| filename                    | f         | []          | Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                  |
| kustomize                   | k         |             | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                 |
| limits                      |           |             | The resource requirement requests for this container. For example, 'cpu=100m,memory=256Mi'. Note that server side components may assign requests depending on the server configuration, such as limit ranges.                        |
| local                       |           | false       | If true, set resources will NOT contact api-server but run locally.                                                                                                                                                                  |
| output                      | o         |             | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                    |
| record                      |           | false       | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists. |
| recursive                   | R         | false       | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                      |
| requests                    |           |             | The resource requirement requests for this container. For example, 'cpu=100m,memory=256Mi'. Note that server side components may assign requests depending on the server configuration, such as limit ranges.                        |
| selector                    | l         |             | Selector (label query) to filter on, not including uninitialized ones, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2)                                                                                                |
| show-managed-fields         |           | false       | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                        |
| template                    |           |             | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is                                                                                                                     |



| Name | Shorthand | Default | Usage                                                                                                                                   |
|------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------|
|      |           |         | golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## selector

Set the labels and selector before creating a deployment/service pair

```
kubectl create service clusterip my-svc --clusterip="None" -o yaml --dry-run=client | kubectl set
selector --local -f - 'environment=qa' -o yaml | kubectl create -f -
kubectl create deployment my-dep -o yaml --dry-run=client | kubectl label --local -f - environm
ent=qa -o yaml | kubectl create -f -
```

Set the selector on a resource. Note that the new selector will overwrite the old selector if the resource had one prior to the invocation of 'set selector'.

A selector must begin with a letter or number, and may contain letters, numbers, hyphens, dots, and underscores, up to 63 characters. If --resource-version is specified, then updates will use this resource version, otherwise the existing resource-version will be used. Note: currently selectors can only be set on Service objects.

## Usage

```
$ kubectl set selector (-f FILENAME | TYPE NAME) EXPRESSIONS [--resource-
version=version]
```

## Flags

| Name                        | Shorthand | Default     | Usage                                                                                                                                                                                                    |
|-----------------------------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false       | Select all resources in the namespace of the specified resource types                                                                                                                                    |
| allow-missing-template-keys |           | true        | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |
| dry-run                     |           | none        | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| field-manager               |           | kubectl-set | Name of the manager used to track field ownership.                                                                                                                                                       |
| filename                    | f         | []          | identifying the resource.                                                                                                                                                                                |
| local                       |           | false       | If true, annotation will NOT contact api-server but run locally.                                                                                                                                         |
| output                      | o         |             | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                        |
| record                      |           | false       | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set                                                                                            |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |         | to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                                                                                                                                    |
| recursive           | R         | true    | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| resource-version    |           |         | If non-empty, the selectors update will only succeed if this is the current resource-version for the object. Only valid when specifying a single resource.                                                                                                |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goolang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## serviceaccount

Set deployment nginx-deployment's service account to serviceaccount1

```
kubectl set serviceaccount deployment nginx-deployment serviceaccount1
```

Print the result (in YAML format) of updated nginx deployment with the service account from local file, without hitting the API server

```
kubectl set sa -f nginx-deployment.yaml serviceaccount1 --local --dry-run=client -o yaml
```

Update the service account of pod template resources.

Possible resources (case insensitive) can be:

replicationcontroller (rc), deployment (deploy), daemonset (ds), job, replicaset (rs), statefulset

## Usage

\$ kubectl set serviceaccount (-f FILENAME | TYPE NAME) SERVICE\_ACCOUNT

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                                                                                    |
|-----------------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false   | Select all resources, including uninitialized ones, in the namespace of the specified resource types                                                                                                     |
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                         |
| dry-run                     |           | none    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |

| Name                | Shorthand | Default     | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| field-manager       |           | kubectl-set | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| filename            | f         | []          | Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                      |
| kustomize           | k         |             | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| local               |           | false       | If true, set serviceaccount will NOT contact api-server but run locally.                                                                                                                                                                                 |
| output              | o         |             | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| record              |           | false       | Record current kubectl command in the resource annotation. If set to false, do not record the command. If set to true, record the command. If not set, default to updating the existing annotation value only if one already exists.                     |
| recursive           | R         | false       | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| show-managed-fields |           | false       | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |             | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go-lang templates <a href="http://golang.org/pkg/text/template/#pkg-overview">[http://golang.org/pkg/text/template/#pkg-overview]</a> . |

## subject

Update a cluster role binding for serviceaccount1

```
kubectl set subject clusterrolebinding admin --serviceaccount=namespace:serviceaccount1
```

Update a role binding for user1, user2, and group1

```
kubectl set subject rolebinding admin --user=user1 --user=user2 --group=group1
```

Print the result (in YAML format) of updating rolebinding subjects from a local, without hitting the server

```
kubectl create rolebinding admin --role=admin --user=admin -o yaml --dry-run=client | kubectl set subject --local -f - --user=foo -o yaml
```

Update the user, group, or service account in a role binding or cluster role binding.

## Usage

```
$ kubectl set subject (-f FILENAME | TYPE NAME) [--user=username] [--group=groupname] [--serviceaccount=namespace:serviceaccountname] [--dry-run=server|client|none]
```

## Flags

| Name                        | Shorthand | Default     | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false       | Select all resources, including uninitialized ones, in the namespace of the specified resource types                                                                                                                                                     |
| allow-missing-template-keys |           | true        | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none        | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| field-manager               |           | kubectl-set | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| filename                    | f         | []          | Filename, directory, or URL to files the resource to update the subjects                                                                                                                                                                                 |
| group                       |           | []          | Groups to bind to the role                                                                                                                                                                                                                               |
| kustomize                   | k         |             | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| local                       |           | false       | If true, set subject will NOT contact api-server but run locally.                                                                                                                                                                                        |
| output                      | o         |             | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| recursive                   | R         | false       | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| selector                    | l         |             | Selector (label query) to filter on, not including uninitialized ones, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2)                                                                                                                    |
| serviceaccount              |           | []          | Service accounts to bind to the role                                                                                                                                                                                                                     |
| show-managed-fields         |           | false       | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                    |           |             | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## wait

Wait for the pod "busybox1" to contain the status condition of type "Ready"

```
kubectrl wait --for=condition=Ready pod/busybox1
```

The default value of status condition is true; you can set it to false

```
kubectrl wait --for=condition=Ready=false pod/busybox1
```

Wait for the pod "busybox1" to be deleted, with a timeout of 60s, after having issued the "delete" command

```
kubectl delete pod/busybox1
kubectl wait --for=delete pod/busybox1 --timeout=60s
```

Experimental: Wait for a specific condition on one or many resources.

The command takes multiple resources and waits until the specified condition is seen in the Status field of every given resource.

Alternatively, the command can wait for the given set of resources to be deleted by providing the "delete" keyword as the value to the --for flag.

A successful message will be printed to stdout indicating when the specified condition has been met. You can use -o option to change to output destination.

## Usage

```
$ kubectl wait ([-f FILENAME] | resource.group/resource.name | resource.group [-l label | --all]) [--for=delete|--for condition=available]
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                                                                    |
|-----------------------------|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false   | Select all resources in the namespace of the specified resource types                                                                                                                    |
| all-namespaces              | A         | false   | If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.                                              |
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                          |
| field-selector              |           |         | Selector (field query) to filter on, supports '=', '==', and '!='. (e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type. |
| filename                    | f         | []      | identifying the resource.                                                                                                                                                                |
| for                         |           |         | The condition to wait on: [delete condition=condition-name]. The default status value of condition-name is true, you can set false with condition=condition-name=false                   |
| local                       |           | false   | If true, annotation will NOT contact api-server but run locally.                                                                                                                         |
| output                      | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                        |
| recursive                   | R         | true    | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                          |
| selector                    | l         |         | Selector (label query) to filter on, supports '=', '==', and '!='. (e.g. -l key1=value1,key2=value2)                                                                                     |
|                             |           | false   |                                                                                                                                                                                          |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                    |
|---------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| show-managed-fields |           |         | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| timeout             |           | 30s     | The length of time to wait before giving up. Zero means check once and don't wait, negative means wait for a week.                                                                                                                                       |

## WORKING WITH APPS

This section contains commands for inspecting and debugging your applications.

- `logs` will print the logs from the specified pod + container.
- `exec` can be used to get an interactive shell on a pod + container.
- `describe` will print debug information about the given resource.

### attach

Get output from running pod mypod; use the 'kubectl.kubernetes.io/default-container' annotation # for selecting the container to be attached or the first container in the pod will be chosen

```
kubectl attach mypod
```

Get output from ruby-container from pod mypod

```
kubectl attach mypod -c ruby-container
```

Switch to raw terminal mode; sends stdin to 'bash' in ruby-container from pod mypod # and sends stdout/stderr from 'bash' back to the client

```
kubectl attach mypod -c ruby-container -i -t
```

Get output from the first pod of a replica set named nginx

```
kubectl attach rs/nginx
```

Attach to a process that is already running inside an existing container.

### Usage

```
$ kubectl attach (POD | TYPE/NAME) -c CONTAINER
```

## Flags

| Name                | Shorthand | Default | Usage                                                                                                                                                                                           |
|---------------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| container           | c         |         | Container name. If omitted, use the <code>kubectl.kubernetes.io/default-container</code> annotation for selecting the container to be attached or the first container in the pod will be chosen |
| pod-running-timeout |           | 1m0s    | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                             |
| quiet               | q         | false   | Only print output from the remote session                                                                                                                                                       |
| stdin               | i         | false   | Pass stdin to the container                                                                                                                                                                     |
| tty                 | t         | false   | Stdin is a TTY                                                                                                                                                                                  |

## auth

Inspect authorization

### Usage

```
$ kubectl auth
```

### *can-i*

Check to see if I can create pods in any namespace

```
kubectl auth can-i create pods --all-namespaces
```

Check to see if I can list deployments in my current namespace

```
kubectl auth can-i list deployments.apps
```

Check to see if I can do everything in my current namespace ("\*" means all)

```
kubectl auth can-i '*' '*'
```

Check to see if I can get the job named "bar" in namespace "foo"

```
kubectl auth can-i list jobs.batch/bar -n foo
```

Check to see if I can read pod logs

```
kubectl auth can-i get pods --subresource=log
```

Check to see if I can access the URL /logs/

```
kubectl auth can-i get /logs/
```

List all allowed actions in namespace "foo"

```
kubectrl auth can-i --list --namespace=foo
```

Check whether an action is allowed.

VERB is a logical Kubernetes API verb like 'get', 'list', 'watch', 'delete', etc. TYPE is a Kubernetes resource. Shortcuts and groups will be resolved. NONRESOURCEURL is a partial URL that starts with "/". NAME is the name of a particular Kubernetes resource.

## Usage

```
$ kubectrl auth can-i VERB [TYPE | TYPE/NAME | NONRESOURCEURL]
```

## Flags

| Name           | Shorthand | Default | Usage                                                   |
|----------------|-----------|---------|---------------------------------------------------------|
| all-namespaces | A         | false   | If true, check the specified action in all namespaces.  |
| list           |           | false   | If true, prints all allowed actions.                    |
| no-headers     |           | false   | If true, prints allowed actions without headers         |
| quiet          | q         | false   | If true, suppress output and just return the exit code. |
| subresource    |           |         | SubResource such as pod/log or deployment/scale         |

## *reconcile*

Reconcile RBAC resources from a file

```
kubectrl auth reconcile -f my-rbac-rules.yaml
```

Reconciles rules for RBAC role, role binding, cluster role, and cluster role binding objects.

Missing objects are created, and the containing namespace is created for namespaced objects, if required.

Existing roles are updated to include the permissions in the input objects, and remove extra permissions if --remove-extra-permissions is specified.

Existing bindings are updated to include the subjects in the input objects, and remove extra subjects if --remove-extra-subjects is specified.

This is preferred to 'apply' for RBAC resources so that semantically-aware merging of rules and subjects is done.

## Usage

```
$ kubectrl auth reconcile -f FILENAME
```

## Flags

| Name                        | Shorthand | Default | Usage |
|-----------------------------|-----------|---------|-------|
| allow-missing-template-keys |           | true    |       |



| Name                     | Shorthand | Default | Usage                                                                                                                                                                                                                                                    |
|--------------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |           |         | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                                          |
| dry-run                  |           | none    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| filename                 | f         | []      | Filename, directory, or URL to files identifying the resource to reconcile.                                                                                                                                                                              |
| kustomize                | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| output                   | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| recursive                | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| remove-extra-permissions |           | false   | If true, removes extra permissions added to roles                                                                                                                                                                                                        |
| remove-extra-subjects    |           | false   | If true, removes extra subjects added to rolebindings                                                                                                                                                                                                    |
| show-managed-fields      |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                 |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goyaml templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## cp

!!!Important Note!!! # Requires that the 'tar' binary is present in your container # image. If 'tar' is not present, 'kubectl cp' will fail. # # For advanced use cases, such as symlinks, wildcard expansion or # file mode preservation, consider using 'kubectl exec'. # Copy /tmp/foo local file to /tmp/bar in a remote pod in namespace

```
tar cf - /tmp/foo | kubectl exec -i -n <some-namespace> <some-pod> -- tar xf - -C /tmp/bar
```

Copy /tmp/foo from a remote pod to /tmp/bar locally

```
kubectl exec -n <some-namespace> <some-pod> -- tar cf - /tmp/foo | tar xf - -C /tmp/bar
```

Copy /tmp/foo\_dir local directory to /tmp/bar\_dir in a remote pod in the default namespace

```
kubectl cp /tmp/foo_dir <some-pod>:/tmp/bar_dir
```

Copy /tmp/foo local file to /tmp/bar in a remote pod in a specific container

```
kubectl cp /tmp/foo <some-pod>:/tmp/bar -c <specific-container>
```

Copy /tmp/foo local file to /tmp/bar in a remote pod in namespace

```
kubectcl cp /tmp/foo <some-namespace>/<some-pod>:/tmp/bar
```

Copy /tmp/foo from a remote pod to /tmp/bar locally

```
kubectcl cp <some-namespace>/<some-pod>:/tmp/foo /tmp/bar
```

Copy files and directories to and from containers.

## Usage

```
$ kubectcl cp <file-spec-src> <file-spec-dest>
```

## Flags

| Name        | Shorthand | Default | Usage                                                                                                                                                                               |
|-------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| container   | c         |         | Container name. If omitted, use the kubectcl.kubernetes.io/default-container annotation for selecting the container to be attached or the first container in the pod will be chosen |
| no-preserve |           | false   | The copied file/directory's ownership and permissions will not be preserved in the container                                                                                        |

# describe

Describe a node

```
kubectcl describe nodes kubernetes-node-emt8.c.myproject.internal
```

Describe a pod

```
kubectcl describe pods/nginx
```

Describe a pod identified by type and name in "pod.json"

```
kubectcl describe -f pod.json
```

Describe all pods

```
kubectcl describe pods
```

Describe pods by label name=myLabel

```
kubectcl describe po -l name=myLabel
```

Describe all pods managed by the 'frontend' replication controller (rc-created pods  
# get the name of the rc as a prefix in the pod the name)

```
kubectcl describe pods frontend
```

Show details of a specific resource or group of resources.

Print a detailed description of the selected resources, including related resources such as events or controllers. You may select a single object by name, all objects of that type, provide a name prefix, or label selector. For example:

```
$ kubectl describe TYPE NAME_PREFIX
```

will first check for an exact match on TYPE and NAME\_PREFIX. If no such resource exists, it will output details for every resource that has a name prefixed with NAME\_PREFIX.

Use "kubectl api-resources" for a complete list of supported resources.

## Usage

```
$ kubectl describe (-f FILENAME | TYPE [NAME_PREFIX | -l label] | TYPE/NAME)
```

## Flags

| Name           | Shorthand | Default | Usage                                                                                                                                           |
|----------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| all-namespaces | A         | false   | If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.     |
| chunk-size     |           | 500     | Return large lists in chunks rather than all at once. Pass 0 to disable. This flag is beta and may change in the future.                        |
| filename       | f         | []      | Filename, directory, or URL to files containing the resource to describe                                                                        |
| kustomize      | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| recursive      | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory. |
| selector       | l         |         | Selector (label query) to filter on, supports '=', '==', and '!='. (e.g. -l key1=value1,key2=value2)                                            |
| show-events    |           | true    | If true, display events related to the described object.                                                                                        |

## exec

Get output from running the 'date' command from pod mypod, using the first container by default

```
kubectl exec mypod -- date
```

Get output from running the 'date' command in ruby-container from pod mypod

```
kubectl exec mypod -c ruby-container -- date
```

Switch to raw terminal mode; sends stdin to 'bash' in ruby-container from pod mypod # and sends stdout/stderr from 'bash' back to the client

```
kubectl exec mypod -c ruby-container -i -t -- bash -il
```

List contents of /usr from the first container of pod mypod and sort by modification time # If the command you want to execute in the pod has any flags in common (e.g. -i), # you must use two dashes (--) to separate your command's flags/arguments # Also note, do not surround your command and its flags/arguments with quotes # unless that is how you would execute it normally (i.e., do ls -t /usr, not "ls -t /usr")

```
kubectl exec mypod -i -t -- ls -t /usr
```

Get output from running 'date' command from the first pod of the deployment mydeployment, using the first container by default

```
kubectl exec deploy/mydeployment -- date
```

Get output from running 'date' command from the first pod of the service myservice, using the first container by default

```
kubectl exec svc/myservice -- date
```

Execute a command in a container.

## Usage

```
$ kubectl exec (POD | TYPE/NAME) [-c CONTAINER] [flags] -- COMMAND [args...]
```

## Flags

| Name                | Shorthand | Default | Usage                                                                                                                                                                              |
|---------------------|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| container           | c         |         | Container name. If omitted, use the kubectl.kubernetes.io/default-container annotation for selecting the container to be attached or the first container in the pod will be chosen |
| filename            | f         | []      | to use to exec into the resource                                                                                                                                                   |
| pod-running-timeout |           | 1m0s    | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                |
| quiet               | q         | false   | Only print output from the remote session                                                                                                                                          |
| stdin               | i         | false   | Pass stdin to the container                                                                                                                                                        |
| tty                 | t         | false   | Stdin is a TTY                                                                                                                                                                     |

## logs

Return snapshot logs from pod nginx with only one container

```
kubectl logs nginx
```

Return snapshot logs from pod nginx with multi containers

```
kubectl logs nginx --all-containers=true
```

Return snapshot logs from all containers in pods defined by label app=nginx

```
kubectl logs -l app=nginx --all-containers=true
```

Return snapshot of previous terminated ruby container logs from pod web-1

```
kubectrl logs -p -c ruby web-1
```

Begin streaming the logs of the ruby container in pod web-1

```
kubectrl logs -f -c ruby web-1
```

Begin streaming the logs from all containers in pods defined by label app=nginx

```
kubectrl logs -f -l app=nginx --all-containers=true
```

Display only the most recent 20 lines of output in pod nginx

```
kubectrl logs --tail=20 nginx
```

Show all logs from pod nginx written in the last hour

```
kubectrl logs --since=1h nginx
```

Show logs from a kubelet with an expired serving certificate

```
kubectrl logs --insecure-skip-tls-verify-backend nginx
```

Return snapshot logs from first container of a job named hello

```
kubectrl logs job/hello
```

Return snapshot logs from container nginx-1 of a deployment named nginx

```
kubectrl logs deployment/nginx -c nginx-1
```

Print the logs for a container in a pod or specified resource. If the pod has only one container, the container name is optional.

## Usage

```
$ kubectrl logs [-f] [-p] (POD | TYPE/NAME) [-c CONTAINER]
```

## Flags

| Name                             | Shorthand | Default | Usage                                                                                                                                                                                                                 |
|----------------------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all-containers                   |           | false   | Get all containers' logs in the pod(s).                                                                                                                                                                               |
| container                        | c         |         | Print the logs of this container                                                                                                                                                                                      |
| follow                           | f         | false   | Specify if the logs should be streamed.                                                                                                                                                                               |
| ignore-errors                    |           | false   | If watching / following pod logs, allow for any errors that occur to be non-fatal                                                                                                                                     |
| insecure-skip-tls-verify-backend |           | false   | Skip verifying the identity of the kubelet that logs are requested from. In theory, an attacker could provide invalid log content back. You might want to use this if your kubelet serving certificates have expired. |
| limit-bytes                      |           | 0       | Maximum bytes of logs to return. Defaults to no limit.                                                                                                                                                                |
|                                  |           | 5       |                                                                                                                                                                                                                       |

| Name                | Shorthand | Default | Usage                                                                                                                                 |
|---------------------|-----------|---------|---------------------------------------------------------------------------------------------------------------------------------------|
| max-log-requests    |           |         | Specify maximum number of concurrent logs to follow when using by a selector. Defaults to 5.                                          |
| pod-running-timeout |           | 20s     | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                   |
| prefix              |           | false   | Prefix each log line with the log source (pod name and container name)                                                                |
| previous            | p         | false   | If true, print the logs for the previous instance of the container in a pod if it exists.                                             |
| selector            | l         |         | Selector (label query) to filter on.                                                                                                  |
| since               |           | 0s      | Only return logs newer than a relative duration like 5s, 2m, or 3h. Defaults to all logs. Only one of since-time / since may be used. |
| since-time          |           |         | Only return logs after a specific date (RFC3339). Defaults to all logs. Only one of since-time / since may be used.                   |
| tail                |           | -1      | Lines of recent log file to display. Defaults to -1 with no selector, showing all log lines otherwise 10, if a selector is provided.  |
| timestamps          |           | false   | Include timestamps on each line in the log output                                                                                     |

## port-forward

Listen on ports 5000 and 6000 locally, forwarding data to/from ports 5000 and 6000 in the pod

```
kubectrl port-forward pod/mypod 5000 6000
```

Listen on ports 5000 and 6000 locally, forwarding data to/from ports 5000 and 6000 in a pod selected by the deployment

```
kubectrl port-forward deployment/mydeployment 5000 6000
```

Listen on port 8443 locally, forwarding to the targetPort of the service's port named "https" in a pod selected by the service

```
kubectrl port-forward service/myservice 8443:https
```

Listen on port 8888 locally, forwarding to 5000 in the pod

```
kubectrl port-forward pod/mypod 8888:5000
```

Listen on port 8888 on all addresses, forwarding to 5000 in the pod

```
kubectrl port-forward --address 0.0.0.0 pod/mypod 8888:5000
```

Listen on port 8888 on localhost and selected IP, forwarding to 5000 in the pod

```
kubectrl port-forward --address localhost,10.19.21.23 pod/mypod 8888:5000
```

Listen on a random port locally, forwarding to 5000 in the pod

```
kubectrl port-forward pod/mypod :5000
```

Forward one or more local ports to a pod.

Use resource type/name such as deployment/mydeployment to select a pod. Resource type defaults to 'pod' if omitted.

If there are multiple pods matching the criteria, a pod will be selected automatically. The forwarding session ends when the selected pod terminates, and a rerun of the command is needed to resume forwarding.

## Usage

```
$ kubectrl port-forward TYPE/NAME [options] [LOCAL_PORT:]REMOTE_PORT [...
[LOCAL_PORT_N:]REMOTE_PORT_N]
```

## Flags

| Name                | Shorthand | Default     | Usage                                                                                                                                                                                                                                           |
|---------------------|-----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| address             |           | [localhost] | Addresses to listen on (comma separated). Only accepts IP addresses or localhost as a value. When localhost is supplied, kubectrl will try to bind on both 127.0.0.1 and ::1 and will fail if neither of these addresses are available to bind. |
| pod-running-timeout |           | 1m0s        | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                                                                             |

# proxy

To proxy all of the Kubernetes API and nothing else

```
kubectrl proxy --api-prefix=/
```

To proxy only part of the Kubernetes API and also some static files # You can get pods info with 'curl localhost:8001/api/v1/pods'

```
kubectrl proxy --www=/my/files --www-prefix=/static/ --api-prefix=/api/
```

To proxy the entire Kubernetes API at a different root # You can get pods info with 'curl localhost:8001/custom/api/v1/pods'

```
kubectrl proxy --api-prefix=/custom/
```

Run a proxy to the Kubernetes API server on port 8011, serving static content from ./local/www/

```
kubectrl proxy --port=8011 --www=./local/www/
```

Run a proxy to the Kubernetes API server on an arbitrary local port # The chosen port for the server will be output to stdout

```
kubectrl proxy --port=0
```

Run a proxy to the Kubernetes API server, changing the API prefix to k8s-api # This makes e.g. the pods API available at localhost:8001/k8s-api/v1/pods/

```
kubectrl proxy --api-prefix=/k8s-api
```

Creates a proxy server or application-level gateway between localhost and the Kubernetes API server. It also allows serving static content over specified HTTP path. All incoming data enters through one port and gets forwarded to the remote Kubernetes API server port, except for the path matching the static content path.

## Usage

```
$ kubectrl proxy [--port=PORT] [--www=static-dir] [--www-prefix=prefix] [--api-prefix=prefix]
```

## Flags

| Name           | Shorthand      | Default                                                 | Usage                                                                                                                                                |
|----------------|----------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| accept-hosts   |                | <code>^localhost\$,^127.0.0.1\$,^[:1]\$\</code>         | Regular expression for hosts that the proxy should accept.                                                                                           |
| accept-paths   |                | <code>^.*\</code>                                       | Regular expression for paths that the proxy should accept.                                                                                           |
| address        |                | <code>127.0.0.1</code>                                  | The IP address on which to serve on.                                                                                                                 |
| api-prefix     |                | <code>/</code>                                          | Prefix to serve the proxied API under.                                                                                                               |
| disable-filter |                | <code>false</code>                                      | If true, disable request filtering in the proxy. This is dangerous, and can leave you vulnerable to XSRF attacks, when used with an accessible port. |
| keepalive      |                | <code>0s</code>                                         | keepalive specifies the keep-alive period for an active network connection. Set to 0 to disable keepalive.                                           |
| port           | <code>p</code> | <code>8001</code>                                       | The port on which to run the proxy. Set to 0 to pick a random port.                                                                                  |
| reject-methods |                | <code>^\$\</code>                                       | Regular expression for HTTP methods that the proxy should reject (example <code>--reject-methods='POST,PUT,PATCH'</code> ).                          |
| reject-paths   |                | <code>^/api/./pods/./exec,^/api/./pods/./attach\</code> | Regular expression for paths that the proxy should reject. Paths specified here will be rejected even accepted by <code>--accept-paths</code> .      |
| unix-socket    | <code>u</code> |                                                         | Unix socket on which to run the proxy.                                                                                                               |
| www            | <code>w</code> |                                                         | Also serve static files from the given directory under the specified prefix.                                                                         |
| www-prefix     | <code>P</code> | <code>/static/</code>                                   | Prefix to serve static files under, if static file directory is specified.                                                                           |



# top

Display Resource (CPU/Memory) usage.

The top command allows you to see the resource consumption for nodes or pods.

This command requires Metrics Server to be correctly configured and working on the server.

## Usage

```
$ kubectl top
```

---

## *node*

Show metrics for all nodes

```
kubectl top node
```

Show metrics for a given node

```
kubectl top node NODE_NAME
```

Display resource (CPU/memory) usage of nodes.

The top-node command allows you to see the resource consumption of nodes.

## Usage

```
$ kubectl top node [NAME | -l label]
```

## Flags

| Name                 | Shorthand       | Default | Usage                                                                                                              |
|----------------------|-----------------|---------|--------------------------------------------------------------------------------------------------------------------|
| no-headers           |                 | false   | If present, print output without headers                                                                           |
| selector             | <code>-l</code> |         | Selector (label query) to filter on, supports '=', '==', and '!='. (e.g. <code>-l key1=value1,key2=value2</code> ) |
| sort-by              |                 |         | If non-empty, sort nodes list using specified field. The field can be either 'cpu' or 'memory'.                    |
| use-protocol-buffers |                 | true    | Enables using protocol-buffers to access Metrics API.                                                              |

---

## *pod*

Show metrics for all pods in the default namespace

```
kubectl top pod
```

Show metrics for all pods in the given namespace

```
kubectl top pod --namespace=NAMESPACE
```

Show metrics for a given pod and its containers

```
kubectl top pod POD_NAME --containers
```

Show metrics for the pods defined by label name=myLabel

```
kubectl top pod -l name=myLabel
```

Display resource (CPU/memory) usage of pods.

The 'top pod' command allows you to see the resource consumption of pods.

Due to the metrics pipeline delay, they may be unavailable for a few minutes since pod creation.

## Usage

```
$ kubectl top pod [NAME | -l label]
```

## Flags

| Name                 | Shorthand | Default | Usage                                                                                                                                                                                    |
|----------------------|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all-namespaces       | A         | false   | If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.                                              |
| containers           |           | false   | If present, print usage of containers within a pod.                                                                                                                                      |
| field-selector       |           |         | Selector (field query) to filter on, supports '=', '==', and '!='. (e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type. |
| no-headers           |           | false   | If present, print output without headers.                                                                                                                                                |
| selector             | l         |         | Selector (label query) to filter on, supports '=', '==', and '!='. (e.g. -l key1=value1,key2=value2)                                                                                     |
| sort-by              |           |         | If non-empty, sort pods list using specified field. The field can be either 'cpu' or 'memory'.                                                                                           |
| use-protocol-buffers |           | true    | Enables using protocol-buffers to access Metrics API.                                                                                                                                    |

# CLUSTER MANAGEMENT

---

## api-versions

Print the supported API versions

```
kubectl api-versions
```

Print the supported API versions on the server, in the form of "group/version".

## Usage

\$ kubectl api-versions

---

# certificate

Modify certificate resources.

## Usage

\$ kubectl certificate SUBCOMMAND

---

### *approve*

Approve CSR 'csr-sqgzp'

kubectl certificate approve csr-sqgzp

Approve a certificate signing request.

kubectl certificate approve allows a cluster admin to approve a certificate signing request (CSR). This action tells a certificate signing controller to issue a certificate to the requestor with the attributes requested in the CSR.

SECURITY NOTICE: Depending on the requested attributes, the issued certificate can potentially grant a requester access to cluster resources or to authenticate as a requested identity. Before approving a CSR, ensure you understand what the signed certificate can do.

## Usage

\$ kubectl certificate approve (-f FILENAME | NAME)

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                           |
|-----------------------------|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |
| filename                    | f         | []      | Filename, directory, or URL to files identifying the resource to update                                                                         |
| force                       |           | false   | Update the CSR even if it is already approved.                                                                                                  |
| kustomize                   | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| output                      | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.               |
| recursive                   | R         | false   |                                                                                                                                                 |

| Name                | Shorthand | Default | Usage                                                                                                                                                                                                                                                     |
|---------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |           |         | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| show-managed-fields |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template            |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## deny

Deny CSR 'csr-sqgzp'

```
kubectl certificate deny csr-sqgzp
```

Deny a certificate signing request.

kubectl certificate deny allows a cluster admin to deny a certificate signing request (CSR). This action tells a certificate signing controller to not to issue a certificate to the requestor.

## Usage

\$ kubectl certificate deny (-f FILENAME | NAME)

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                            |
|-----------------------------|-----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats. |
| filename                    | f         | []      | Filename, directory, or URL to files identifying the resource to update                                                                          |
| force                       |           | false   | Update the CSR even if it is already denied.                                                                                                     |
| kustomize                   | k         |         | Process the kustomization directory. This flag can't be used together with -f or -R.                                                             |
| output                      | o         |         | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                |
| recursive                   | R         | false   | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.  |
| show-managed-fields         |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                    |
| template                    |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is                                 |

| Name | Shorthand | Default | Usage                                                                                                                                   |
|------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------|
|      |           |         | golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## cluster-info

Print the address of the control plane and cluster services

```
kubectl cluster-info
```

Display addresses of the control plane and services with label `kubernetes.io/cluster-service=true`. To further debug and diagnose cluster problems, use `'kubectl cluster-info dump'`.

### Usage

```
$ kubectl cluster-info
```

---

## *dump*

Dump current cluster state to stdout

```
kubectl cluster-info dump
```

Dump current cluster state to `/path/to/cluster-state`

```
kubectl cluster-info dump --output-directory=/path/to/cluster-state
```

Dump all namespaces to stdout

```
kubectl cluster-info dump --all-namespaces
```

Dump a set of namespaces to `/path/to/cluster-state`

```
kubectl cluster-info dump --namespaces default,kube-system --output-directory=/path/to/cluster-state
```

Dump cluster information out suitable for debugging and diagnosing cluster problems. By default, dumps everything to stdout. You can optionally specify a directory with `--output-directory`. If you specify a directory, Kubernetes will build a set of files in that directory. By default, only dumps things in the current namespace and 'kube-system' namespace, but you can switch to a different namespace with the `--namespaces` flag, or specify `--all-namespaces` to dump all namespaces.

The command also dumps the logs of all of the pods in the cluster; these logs are dumped into different directories based on namespace and pod name.

### Usage

```
$ kubectl cluster-info dump
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                                                                                                                                    |
|-----------------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all-namespaces              | A         | false   | If true, dump all namespaces. If true, --namespaces is ignored.                                                                                                                                                                                          |
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| namespaces                  |           | []      | A comma separated list of namespaces to dump.                                                                                                                                                                                                            |
| output                      | o         | json    | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                        |
| output-directory            |           |         | Where to output the files. If empty or '-' uses stdout, otherwise creates a directory hierarchy in that directory                                                                                                                                        |
| pod-running-timeout         |           | 20s     | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                                                                                      |
| show-managed-fields         |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| template                    |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

## cordon

Mark node "foo" as unschedulable

```
kubectcl cordon foo
```

Mark node as unschedulable.

### Usage

```
$ kubectcl cordon NODE
```

## Flags

| Name     | Shorthand | Default | Usage                                                                                                                                                                                                    |
|----------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dry-run  |           | none    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| selector | l         |         | Selector (label query) to filter on                                                                                                                                                                      |

# drain

Drain node "foo", even if there are pods not managed by a replication controller, replica set, job, daemon set or stateful set on it

```
kubectldrain foo --force
```

As above, but abort if there are pods not managed by a replication controller, replica set, job, daemon set or stateful set, and use a grace period of 15 minutes

```
kubectldrain foo --grace-period=900
```

Drain node in preparation for maintenance.

The given node will be marked unschedulable to prevent new pods from arriving. 'drain' evicts the pods if the API server supports <https://kubernetes.io/docs/concepts/workloads/pods/disruptions/> . Otherwise, it will use normal DELETE to delete the pods. The 'drain' evicts or deletes all pods except mirror pods (which cannot be deleted through the API server). If there are daemon set-managed pods, drain will not proceed without --ignore-daemonsets, and regardless it will not delete any daemon set-managed pods, because those pods would be immediately replaced by the daemon set controller, which ignores unschedulable markings. If there are any pods that are neither mirror pods nor managed by a replication controller, replica set, daemon set, stateful set, or job, then drain will not delete any pods unless you use --force. --force will also allow deletion to proceed if the managing resource of one or more pods is missing.

'drain' waits for graceful termination. You should not operate on the machine until the command completes.

When you are ready to put the node back into service, use `kubectl uncordon`, which will make the node schedulable again.

<https://kubernetes.io/images/docs/kubectldrain.svg>

## Usage

```
$ kubectldrain NODE
```

## Flags

| Name                 | Shorthand | Default | Usage                                                                                                                       |
|----------------------|-----------|---------|-----------------------------------------------------------------------------------------------------------------------------|
| chunk-size           |           | 500     | Return large lists in chunks rather than all at once. Pass 0 to disable. This flag is beta and may change in the future.    |
| delete-emptydir-data |           | false   | Continue even if there are pods using emptyDir (local data that will be deleted when the node is drained).                  |
| delete-local-data    |           | false   | Continue even if there are pods using emptyDir (local data that will be deleted when the node is drained).                  |
| disable-eviction     |           | false   | Force drain to use delete, even if eviction is supported. This will bypass checking PodDisruptionBudgets, use with caution. |
| dry-run              |           | none    |                                                                                                                             |

| Name                         | Shorthand | Default | Usage                                                                                                                                                                                                    |
|------------------------------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |           |         | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| force                        |           | false   | Continue even if there are pods not managed by a ReplicationController, ReplicaSet, Job, DaemonSet or StatefulSet.                                                                                       |
| grace-period                 |           | -1      | Period of time in seconds given to each pod to terminate gracefully. If negative, the default value specified in the pod will be used.                                                                   |
| ignore-daemonsets            |           | false   | Ignore DaemonSet-managed pods.                                                                                                                                                                           |
| ignore-errors                |           | false   | Ignore errors occurred between drain nodes in group.                                                                                                                                                     |
| pod-selector                 |           |         | Label selector to filter pods on the node                                                                                                                                                                |
| selector                     | l         |         | Selector (label query) to filter on                                                                                                                                                                      |
| skip-wait-for-delete-timeout |           | 0       | If pod DeletionTimestamp older than N seconds, skip waiting for the pod. Seconds must be greater than 0 to skip.                                                                                         |
| timeout                      |           | 0s      | The length of time to wait before giving up, zero means infinite                                                                                                                                         |

## taint

Update node 'foo' with a taint with key 'dedicated' and value 'special-user' and effect 'NoSchedule' # If a taint with that key and effect already exists, its value is replaced as specified

```
kubectl taint nodes foo dedicated=special-user:NoSchedule
```

Remove from node 'foo' the taint with key 'dedicated' and effect 'NoSchedule' if one exists

```
kubectl taint nodes foo dedicated:NoSchedule-
```

Remove from node 'foo' all the taints with key 'dedicated'

```
kubectl taint nodes foo dedicated-
```

Add a taint with key 'dedicated' on nodes having label mylabel=X

```
kubectl taint node -l myLabel=X dedicated=foo:PreferNoSchedule
```

Add to node 'foo' a taint with key 'bar' and no value

```
kubectl taint nodes foo bar:NoSchedule
```

Update the taints on one or more nodes.

- A taint consists of a key, value, and effect. As an argument here, it is expressed as key=value:effect.



- The key must begin with a letter or number, and may contain letters, numbers, hyphens, dots, and underscores, up to 253 characters.
- Optionally, the key can begin with a DNS subdomain prefix and a single '/', like example.com/my-app.
- The value is optional. If given, it must begin with a letter or number, and may contain letters, numbers, hyphens, dots, and underscores, up to 63 characters.
- The effect must be NoSchedule, PreferNoSchedule or NoExecute.
- Currently taint can only apply to node.

## Usage

```
$ kubectl taint NODE NAME KEY_1=VAL_1:TAINT_EFFECT_1 ...
KEY_N=VAL_N:TAINT_EFFECT_N
```

## Flags

| Name                        | Shorthand | Default       | Usage                                                                                                                                                                                                                                                     |
|-----------------------------|-----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all                         |           | false         | Select all nodes in the cluster                                                                                                                                                                                                                           |
| allow-missing-template-keys |           | true          | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                                                                          |
| dry-run                     |           | none          | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| field-manager               |           | kubectl-taint | Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| output                      | o         |               | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                         |
| overwrite                   |           | false         | If true, allow taints to be overwritten, otherwise reject taint updates that overwrite existing taints.                                                                                                                                                   |
| selector                    | l         |               | Selector (label query) to filter on, supports '=', '==', and '!='. (e.g. -l key1=value1,key2=value2)                                                                                                                                                      |
| show-managed-fields         |           | false         | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| template                    |           |               | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| validate                    |           | true          | If true, use a schema to validate the input before sending it                                                                                                                                                                                             |

# uncordon

Mark node "foo" as schedulable

```
kubectl uncordon foo
```

Mark node as schedulable.

Usage

\$ kubectl uncordon NODE

Flags

| Name     | Shorthand | Default | Usage                                                                                                                                                                                                    |
|----------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dry-run  |           | none    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| selector | l         |         | Selector (label query) to filter on                                                                                                                                                                      |

# KUBECTL SETTINGS AND USAGE

## alpha

These commands correspond to alpha features that are not enabled in Kubernetes clusters by default.

Usage

\$ kubectl alpha

## api-resources

Print the supported API resources

kubectl api-resources

Print the supported API resources with more information

kubectl api-resources -o wide

Print the supported API resources sorted by a column

kubectl api-resources --sort-by=name

Print the supported namespaced resources

kubectl api-resources --namespaced=true

Print the supported non-namespaced resources

kubectl api-resources --namespaced=false

Print the supported API resources with a specific APIGroup

```
kubectl api-resources --api-group=extensions
```

Print the supported API resources on the server.

## Usage

```
$ kubectl api-resources
```

## Flags

| Name       | Shorthand | Default | Usage                                                                                                     |
|------------|-----------|---------|-----------------------------------------------------------------------------------------------------------|
| api-group  |           |         | Limit to resources in the specified API group.                                                            |
| cached     |           | false   | Use the cached list of resources if available.                                                            |
| namespaced |           | true    | If false, non-namespaced resources will be returned, otherwise returning namespaced resources by default. |
| no-headers |           | false   | When using the default or custom-column output format, don't print headers (default print headers).       |
| output     | o         |         | Output format. One of: wide name.                                                                         |
| sort-by    |           |         | If non-empty, sort list of resources using specified field. The field can be either 'name' or 'kind'.     |
| verbs      |           | []      | Limit to resources that support the specified verbs.                                                      |

# completion

Installing bash completion on macOS using homebrew ## If running Bash 3.2 included with macOS

```
brew install bash-completion
```

or, if running Bash 4.1+

```
brew install bash-completion@2
```

If kubectl is installed via homebrew, this should start working immediately ## If you've installed via other means, you may need add the completion to your completion directory

```
kubectl completion bash > $(brew --prefix)/etc/bash_completion.d/kubectl
```

Installing bash completion on Linux ## If bash-completion is not installed on Linux, install the 'bash-completion' package ## via your distribution's package manager. ## Load the kubectl completion code for bash into the current shell

```
source <(kubectl completion bash)
```

Write bash completion code to a file and source it from .bash\_profile

```
kubectl completion bash > ~/.kube/completion.bash.inc
printf "
```

### Kubectl shell completion

```
source '$HOME/.kube/completion.bash.inc'
" >> $HOME/.bash_profile
source $HOME/.bash_profile
```

Load the kubectl completion code for zsh[1] into the current shell

```
source <(kubectl completion zsh)
```

Set the kubectl completion code for zsh[1] to autoload on startup

```
kubectl completion zsh > "${fpath[1]}/_kubectl"
```

Output shell completion code for the specified shell (bash or zsh). The shell code must be evaluated to provide interactive completion of kubectl commands. This can be done by sourcing it from the .bash\_profile.

Detailed instructions on how to do this are available here:

for macOS: <https://kubernetes.io/docs/tasks/tools/install-kubectl-macos/#enable-shell-autocompletion>

for linux: <https://kubernetes.io/docs/tasks/tools/install-kubectl-linux/#enable-shell-autocompletion>

for windows: <https://kubernetes.io/docs/tasks/tools/install-kubectl-windows/#enable-shell-autocompletion>

Note for zsh users: [1] zsh completions are only supported in versions of zsh >= 5.2.

## Usage

```
$ kubectl completion SHELL
```

---

# config

Modify kubeconfig files using subcommands like "kubectl config set current-context my-context"

The loading order follows these rules:

1. If the --kubeconfig flag is set, then only that file is loaded. The flag may only be set once and no merging takes place.
2. If \$KUBECONFIG environment variable is set, then it is used as a list of paths (normal path delimiting rules for your system). These paths are merged. When a value is modified, it is modified in the file that defines the stanza. When a value is created, it is created in the first file that exists. If no files in the chain exist, then it creates the last file in the list.
3. Otherwise, \${HOME}/.kube/config is used and no merging takes place.

## Usage

```
$ kubectl config SUBCOMMAND
```

---

### ***current-context***

Display the current-context

```
kubectl config current-context
```

Display the current-context.

## Usage

```
$ kubectl config current-context
```

---

### ***delete-cluster***

Delete the minikube cluster

```
kubectl config delete-cluster minikube
```

Delete the specified cluster from the kubeconfig.

## Usage

```
$ kubectl config delete-cluster NAME
```

---

### ***delete-context***

Delete the context for the minikube cluster

```
kubectl config delete-context minikube
```

Delete the specified context from the kubeconfig.

## Usage

```
$ kubectl config delete-context NAME
```

---

### ***delete-user***

Delete the minikube user

```
kubectl config delete-user minikube
```

Delete the specified user from the kubeconfig.

Usage

\$ kubectl config delete-user NAME

---

*get-clusters*

List the clusters that kubectl knows about

kubectl config get-clusters

Display clusters defined in the kubeconfig.

Usage

\$ kubectl config get-clusters

---

*get-contexts*

List all the contexts in your kubeconfig file

kubectl config get-contexts

Describe one context in your kubeconfig file

kubectl config get-contexts my-context

Display one or many contexts from the kubeconfig file.

Usage

\$ kubectl config get-contexts [(-o|--output=)name)]

Flags

| Name       | Shorthand | Default | Usage                                                                                               |
|------------|-----------|---------|-----------------------------------------------------------------------------------------------------|
| no-headers |           | false   | When using the default or custom-column output format, don't print headers (default print headers). |
| output     | o         |         | Output format. One of: name                                                                         |

---

*get-users*

List the users that kubectl knows about

kubectl config get-users

Display users defined in the kubeconfig.

## Usage

```
$ kubectl config get-users
```

---

### ***rename-context***

Rename the context 'old-name' to 'new-name' in your kubeconfig file

```
kubectl config rename-context old-name new-name
```

Renames a context from the kubeconfig file.

CONTEXT\_NAME is the context name that you want to change.

NEW\_NAME is the new name you want to set.

Note: If the context being renamed is the 'current-context', this field will also be updated.

## Usage

```
$ kubectl config rename-context CONTEXT_NAME NEW_NAME
```

---

### ***set***

Set the server field on the my-cluster cluster to <https://1.2.3.4>

```
kubectl config set clusters.my-cluster.server https://1.2.3.4
```

Set the certificate-authority-data field on the my-cluster cluster

```
kubectl config set clusters.my-cluster.certificate-authority-data $(echo "cert_data_here" | base64 -i -)
```

Set the cluster field in the my-context context to my-cluster

```
kubectl config set contexts.my-context.cluster my-cluster
```

Set the client-key-data field in the cluster-admin user using --set-raw-bytes option

```
kubectl config set users.cluster-admin.client-key-data cert_data_here --set-raw-bytes=true
```

Set an individual value in a kubeconfig file.

PROPERTY\_NAME is a dot delimited name where each token represents either an attribute name or a map key. Map keys may not contain dots.

PROPERTY\_VALUE is the new value you want to set. Binary fields such as 'certificate-authority-data' expect a base64 encoded string unless the --set-raw-bytes flag is used.

Specifying an attribute name that already exists will merge new fields on top of existing values.

## Usage

```
$ kubectl config set PROPERTY_NAME PROPERTY_VALUE
```

## Flags

| Name          | Shorthand | Default | Usage                                                                                          |
|---------------|-----------|---------|------------------------------------------------------------------------------------------------|
| set-raw-bytes |           | false   | When writing a []byte PROPERTY_VALUE, write the given string directly without base64 decoding. |

## *set-cluster*

Set only the server field on the e2e cluster entry without touching other values

```
kubectl config set-cluster e2e --server=https://1.2.3.4
```

Embed certificate authority data for the e2e cluster entry

```
kubectl config set-cluster e2e --embed-certs --certificate-authority=~/.kube/e2e/
kubernetes.ca.crt
```

Disable cert checking for the dev cluster entry

```
kubectl config set-cluster e2e --insecure-skip-tls-verify=true
```

Set custom TLS server name to use for validation for the e2e cluster entry

```
kubectl config set-cluster e2e --tls-server-name=my-cluster-name
```

Set a cluster entry in kubeconfig.

Specifying a name that already exists will merge new fields on top of existing values for those fields.

## Usage

```
$ kubectl config set-cluster NAME [--server=server] [--certificate-authority=path/to/certificate/
authority] [--insecure-skip-tls-verify=true] [--tls-server-name=example.com]
```

## Flags

| Name        | Shorthand | Default | Usage                                           |
|-------------|-----------|---------|-------------------------------------------------|
| embed-certs |           | false   | embed-certs for the cluster entry in kubeconfig |

## *set-context*

Set the user field on the gce context entry without touching other values

```
kubectl config set-context gce --user=cluster-admin
```



Set a context entry in kubeconfig.

Specifying a name that already exists will merge new fields on top of existing values for those fields.

## Usage

```
$ kubectl config set-context [NAME | --current] [--cluster=cluster_nickname] [--user=user_nickname] [--namespace=namespace]
```

## Flags

| Name    | Shorthand | Default | Usage                      |
|---------|-----------|---------|----------------------------|
| current |           | false   | Modify the current context |

## *set-credentials*

Set only the "client-key" field on the "cluster-admin" # entry, without touching other values

```
kubectl config set-credentials cluster-admin --client-key=~/.kube/admin.key
```

Set basic auth for the "cluster-admin" entry

```
kubectl config set-credentials cluster-admin --username=admin --password=uXFGweU9l35qcif
```

Embed client certificate data in the "cluster-admin" entry

```
kubectl config set-credentials cluster-admin --client-certificate=~/.kube/admin.crt --embed-certs=true
```

Enable the Google Compute Platform auth provider for the "cluster-admin" entry

```
kubectl config set-credentials cluster-admin --auth-provider=gcp
```

Enable the OpenID Connect auth provider for the "cluster-admin" entry with additional args

```
kubectl config set-credentials cluster-admin --auth-provider=oidc --auth-provider-arg=client-id=foo --auth-provider-arg=client-secret=bar
```

Remove the "client-secret" config value for the OpenID Connect auth provider for the "cluster-admin" entry

```
kubectl config set-credentials cluster-admin --auth-provider=oidc --auth-provider-arg=client-secret-
```

Enable new exec auth plugin for the "cluster-admin" entry

```
kubectl config set-credentials cluster-admin --exec-command=/path/to/the/executable --exec-api-version=client.authentication.k8s.io/v1beta1
```

Define new exec auth plugin args for the "cluster-admin" entry

```
kubectcl config set-credentials cluster-admin --exec-arg=arg1 --exec-arg=arg2
```

Create or update exec auth plugin environment variables for the "cluster-admin" entry

```
kubectcl config set-credentials cluster-admin --exec-env=key1=val1 --exec-env=key2=val2
```

Remove exec auth plugin environment variables for the "cluster-admin" entry

```
kubectcl config set-credentials cluster-admin --exec-env=var-to-remove-
```

Set a user entry in kubeconfig.

Specifying a name that already exists will merge new fields on top of existing values.

Client-certificate flags: --client-certificate=certfile --client-key=keyfile

Bearer token flags: --token=bearer\_token

Basic auth flags: --username=basic\_user --password=basic\_password

Bearer token and basic auth are mutually exclusive.

## Usage

```
$ kubectcl config set-credentials NAME [--client-certificate=path/to/certfile] [--client-key=path/to/keyfile] [--token=bearer_token] [--username=basic_user] [--password=basic_password] [--auth-provider=provider_name] [--auth-provider-arg=key=value] [--exec-command=exec_command] [--exec-api-version=exec_api_version] [--exec-arg=arg] [--exec-env=key=value]
```

## Flags

| Name              | Shorthand | Default | Usage                                                                                 |
|-------------------|-----------|---------|---------------------------------------------------------------------------------------|
| auth-provider     |           |         | Auth provider for the user entry in kubeconfig                                        |
| auth-provider-arg |           | []      | 'key=value' arguments for the auth provider                                           |
| embed-certs       |           | false   | Embed client cert/key for the user entry in kubeconfig                                |
| exec-api-version  |           |         | API version of the exec credential plugin for the user entry in kubeconfig            |
| exec-arg          |           | []      | New arguments for the exec credential plugin command for the user entry in kubeconfig |
| exec-command      |           |         | Command for the exec credential plugin for the user entry in kubeconfig               |
| exec-env          |           | []      | 'key=value' environment values for the exec credential plugin                         |

## *unset*

Unset the current-context

```
kubectrl config unset current-context
```

Unset namespace in foo context

```
kubectrl config unset contexts.foo.namespace
```

Unset an individual value in a kubeconfig file.

PROPERTY\_NAME is a dot delimited name where each token represents either an attribute name or a map key. Map keys may not contain dots.

## Usage

```
$ kubectrl config unset PROPERTY_NAME
```

---

## *use-context*

Use the context for the minikube cluster

```
kubectrl config use-context minikube
```

Set the current-context in a kubeconfig file.

## Usage

```
$ kubectrl config use-context CONTEXT_NAME
```

---

## *view*

Show merged kubeconfig settings

```
kubectrl config view
```

Show merged kubeconfig settings and raw certificate data

```
kubectrl config view --raw
```

Get the password for the e2e user

```
kubectrl config view -o jsonpath='{.users[?(@.name == "e2e")].user.password}'
```

Display merged kubeconfig settings or a specified kubeconfig file.

You can use --output jsonpath={...} to extract specific values using a jsonpath expression.

## Usage

```
$ kubectrl config view
```

## Flags

| Name                        | Shorthand | Default | Usage                                                                                                                                                                                                                                                   |
|-----------------------------|-----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allow-missing-template-keys |           | true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                         |
| flatten                     |           | false   | Flatten the resulting kubeconfig file into self-contained output (useful for creating portable kubeconfig files)                                                                                                                                        |
| merge                       |           | true    | Merge the full hierarchy of kubeconfig files                                                                                                                                                                                                            |
| minify                      |           | false   | Remove all information not used by current-context from the output                                                                                                                                                                                      |
| output                      | o         | yaml    | Output format. One of: json yaml name go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                       |
| raw                         |           | false   | Display raw byte data                                                                                                                                                                                                                                   |
| show-managed-fields         |           | false   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                           |
| template                    |           |         | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates <a href="http://golang.org/pkg/text/template/#pkg-overview">[http://golang.org/pkg/text/template/#pkg-overview]</a> . |

## explain

Get the documentation of the resource and its fields

```
kubectl explain pods
```

Get the documentation of a specific field of a resource

```
kubectl explain pods.spec.containers
```

List the fields for supported resources.

This command describes the fields associated with each supported API resource. Fields are identified via a simple JSONPath identifier:

```
<type>.<fieldName>[.<fieldName>]
```

Add the --recursive flag to display all of the fields at once without descriptions. Information about each field is retrieved from the server in OpenAPI format.

Use "kubectl api-resources" for a complete list of supported resources.

## Usage

```
$ kubectl explain RESOURCE
```

## Flags

| Name        | Shorthand | Default | Usage                                                                     |
|-------------|-----------|---------|---------------------------------------------------------------------------|
| api-version |           |         | Get different explanations for particular API version (API group/version) |
| recursive   |           | false   | Print the fields of fields (Currently only 1 level deep)                  |

## options

Print flags inherited by all commands

```
kubectl options
```

Print the list of flags inherited by all commands

### Usage

```
$ kubectl options
```

---

## plugin

Provides utilities for interacting with plugins.

Plugins provide extended functionality that is not part of the major command-line distribution. Please refer to the documentation and examples for more information about how write your own plugins.

The easiest way to discover and install plugins is via the kubernetes sub-project krew. To install krew, visit <https://krew.sigs.k8s.io/docs/user-guide/setup/install/>

### Usage

```
$ kubectl plugin [flags]
```

---

### *list*

List all available plugin files on a user's PATH.

Available plugin files are those that are: - executable - anywhere on the user's PATH - begin with "kubectl-"

### Usage

```
$ kubectl plugin list
```

## Flags

| Name      | Shorthand | Default | Usage                                                                           |
|-----------|-----------|---------|---------------------------------------------------------------------------------|
| name-only |           | false   | If true, display only the binary name of each plugin, rather than its full path |

# version

Print the client and server versions for the current context

```
kubectl version
```

Print the client and server version information for the current context.

## Usage

```
$ kubectl version
```

## Flags

| Name   | Shorthand | Default | Usage                                                    |
|--------|-----------|---------|----------------------------------------------------------|
| client |           | false   | If true, shows client version only (no server required). |
| output | o         |         | One of 'yaml' or 'json'.                                 |
| short  |           | false   | If true, print just the version number.                  |

This section of the Kubernetes documentation contains references.

## API Reference

- [Glossary](#) - a comprehensive, standardized list of Kubernetes terminology
- [Kubernetes API Reference](#)
- [One-page API Reference for Kubernetes v1.30](#)
- [Using The Kubernetes API](#) - overview of the API for Kubernetes.
- [API access control](#) - details on how Kubernetes controls API access
- [Well-Known Labels, Annotations and Taints](#)

## Officially supported client libraries

To call the Kubernetes API from a programming language, you can use [client libraries](#). Officially supported client libraries:

- [Kubernetes Go client library](#)
- [Kubernetes Python client library](#)
- [Kubernetes Java client library](#)
- [Kubernetes JavaScript client library](#)
- [Kubernetes C# client library](#)
- [Kubernetes Haskell client library](#)

## CLI

- [kubectl](#) - Main CLI tool for running commands and managing Kubernetes clusters.
  - [JSONPath](#) - Syntax guide for using [JSONPath expressions](#) with kubectl.
- [kubeadm](#) - CLI tool to easily provision a secure Kubernetes cluster.

## Components

- [kubelet](#) - The primary agent that runs on each node. The kubelet takes a set of PodSpecs and ensures that the described containers are running and healthy.
- [kube-apiserver](#) - REST API that validates and configures data for API objects such as pods, services, replication controllers.
- [kube-controller-manager](#) - Daemon that embeds the core control loops shipped with Kubernetes.
- [kube-proxy](#) - Can do simple TCP/UDP stream forwarding or round-robin TCP/UDP forwarding across a set of back-ends.
- [kube-scheduler](#) - Scheduler that manages availability, performance, and capacity.
  - [Scheduler Policies](#)

- [Scheduler Profiles](#)
- List of [ports and protocols](#) that should be open on control plane and worker nodes

## Config APIs

This section hosts the documentation for "unpublished" APIs which are used to configure kubernetes components or tools. Most of these APIs are not exposed by the API server in a RESTful way though they are essential for a user or an operator to use or manage a cluster.

- [kubeconfig \(v1\)](#)
- [kube-apiserver admission \(v1\)](#)
- [kube-apiserver configuration \(v1alpha1\)](#) and
- [kube-apiserver configuration \(v1beta1\)](#) and [kube-apiserver configuration \(v1\)](#)
- [kube-apiserver event rate limit \(v1alpha1\)](#)
- [kubelet configuration \(v1alpha1\)](#) and [kubelet configuration \(v1beta1\)](#) [kubelet configuration \(v1\)](#)
- [kubelet credential providers \(v1\)](#)
- [kube-scheduler configuration \(v1beta3\)](#) and [kube-scheduler configuration \(v1\)](#)
- [kube-controller-manager configuration \(v1alpha1\)](#)
- [kube-proxy configuration \(v1alpha1\)](#)
- [audit.k8s.io/v1 API](#)
- [Client authentication API \(v1beta1\)](#) and [Client authentication API \(v1\)](#)
- [WebhookAdmission configuration \(v1\)](#)
- [ImagePolicy API \(v1alpha1\)](#)

## Config API for kubeadm

- [v1beta3](#)
- [v1beta4](#)

## External APIs

These are the APIs defined by the Kubernetes project, but are not implemented by the core project:

- [Metrics API \(v1beta1\)](#)
- [Custom Metrics API \(v1beta2\)](#)
- [External Metrics API \(v1beta1\)](#)

## Design Docs

An archive of the design docs for Kubernetes functionality. Good starting points are [Kubernetes Architecture](#) and [Kubernetes Design Overview](#).

## API Overview

This section provides reference information for the Kubernetes API.



The REST API is the fundamental fabric of Kubernetes. All operations and communications between components, and external user commands are REST API calls that the API Server handles. Consequently, everything in the Kubernetes platform is treated as an API object and has a corresponding entry in the [API](#).

The [Kubernetes API reference](#) lists the API for Kubernetes version v1.30.

For general background information, read [The Kubernetes API](#). [Controlling Access to the Kubernetes API](#) describes how clients can authenticate to the Kubernetes API server, and how their requests are authorized.

## API versioning

The JSON and Protobuf serialization schemas follow the same guidelines for schema changes. The following descriptions cover both formats.

The API versioning and software versioning are indirectly related. The [API and release versioning proposal](#) describes the relationship between API versioning and software versioning.

Different API versions indicate different levels of stability and support. You can find more information about the criteria for each level in the [API Changes documentation](#).

Here's a summary of each level:

- Alpha:
  - The version names contain alpha (for example, v1alpha1).
  - Built-in alpha API versions are disabled by default and must be explicitly enabled in the kube-apiserver configuration to be used.
  - The software may contain bugs. Enabling a feature may expose bugs.
  - Support for an alpha API may be dropped at any time without notice.
  - The API may change in incompatible ways in a later software release without notice.
  - The software is recommended for use only in short-lived testing clusters, due to increased risk of bugs and lack of long-term support.
- Beta:
  - The version names contain beta (for example, v2beta3).
  - Built-in beta API versions are disabled by default and must be explicitly enabled in the kube-apiserver configuration to be used (**except** for beta versions of APIs introduced prior to Kubernetes 1.22, which were enabled by default).
  - Built-in beta API versions have a maximum lifetime of 9 months or 3 minor releases (whichever is longer) from introduction to deprecation, and 9 months or 3 minor releases (whichever is longer) from deprecation to removal.
  - The software is well tested. Enabling a feature is considered safe.
  - The support for a feature will not be dropped, though the details may change.
  - The schema and/or semantics of objects may change in incompatible ways in a subsequent beta or stable API version. When this happens, migration instructions

are provided. Adapting to a subsequent beta or stable API version may require editing or re-creating API objects, and may not be straightforward. The migration may require downtime for applications that rely on the feature.

- The software is not recommended for production uses. Subsequent releases may introduce incompatible changes. Use of beta API versions is required to transition to subsequent beta or stable API versions once the beta API version is deprecated and no longer served.

**Note:**

Please try beta features and provide feedback. After the features exit beta, it may not be practical to make more changes.

- Stable:
  - The version name is vX where X is an integer.
  - Stable API versions remain available for all future releases within a Kubernetes major version, and there are no current plans for a major version revision of Kubernetes that removes stable APIs.

## API groups

[API groups](#) make it easier to extend the Kubernetes API. The API group is specified in a REST path and in the `apiVersion` field of a serialized object.

There are several API groups in Kubernetes:

- The *core* (also called *legacy*) group is found at REST path `/api/v1`. The core group is not specified as part of the `apiVersion` field, for example, `apiVersion: v1`.
- The named groups are at REST path `/apis/$GROUP_NAME/$VERSION` and use `apiVersion: $GROUP_NAME/$VERSION` (for example, `apiVersion: batch/v1`). You can find the full list of supported API groups in [Kubernetes API reference](#).

## Enabling or disabling API groups

Certain resources and API groups are enabled by default. You can enable or disable them by setting `--runtime-config` on the API server. The `--runtime-config` flag accepts comma separated `<key>[=<value>]` pairs describing the runtime configuration of the API server. If the `=<value>` part is omitted, it is treated as if `=true` is specified. For example:

- to disable `batch/v1`, set `--runtime-config=batch/v1=false`
- to enable `batch/v2alpha1`, set `--runtime-config=batch/v2alpha1`
- to enable a specific version of an API, such as `storage.k8s.io/v1beta1/csistoragecapacities`, set `--runtime-config=storage.k8s.io/v1beta1/csistoragecapacities`

**Note:**

When you enable or disable groups or resources, you need to restart the API server and controller manager to pick up the `--runtime-config` changes.

## Persistence

Kubernetes stores its serialized state in terms of the API resources by writing them into [etcd](#).

## What's next

- Learn more about [API conventions](#)
- Read the design documentation for [aggregator](#)

# Kubernetes API Concepts

The Kubernetes API is a resource-based (RESTful) programmatic interface provided via HTTP. It supports retrieving, creating, updating, and deleting primary resources via the standard HTTP verbs (POST, PUT, PATCH, DELETE, GET).

For some resources, the API includes additional subresources that allow fine grained authorization (such as separate views for Pod details and log retrievals), and can accept and serve those resources in different representations for convenience or efficiency.

Kubernetes supports efficient change notifications on resources via *watches*. Kubernetes also provides consistent list operations so that API clients can effectively cache, track, and synchronize the state of resources.

You can view the [API reference](#) online, or read on to learn about the API in general.

## Kubernetes API terminology

Kubernetes generally leverages common RESTful terminology to describe the API concepts:

- A *resource type* is the name used in the URL (pods, namespaces, services)
- All resource types have a concrete representation (their object schema) which is called a *kind*
- A list of instances of a resource type is known as a *collection*
- A single instance of a resource type is called a *resource*, and also usually represents an *object*
- For some resource types, the API includes one or more *sub-resources*, which are represented as URI paths below the resource

Most Kubernetes API resource types are [objects](#) – they represent a concrete instance of a concept on the cluster, like a pod or namespace. A smaller number of API resource types are *virtual* in that they often represent operations on objects, rather than objects, such as a permission check (use a POST with a JSON-encoded body of SubjectAccessReview to the `subjectaccessreviews` resource), or the eviction sub-resource of a Pod (used to trigger [API-initiated eviction](#)).

## Object names

All objects you can create via the API have a unique object [name](#) to allow idempotent creation and retrieval, except that virtual resource types may not have unique names if they are not retrievable, or do not rely on idempotency. Within a [namespace](#), only one object of a given kind

can have a given name at a time. However, if you delete the object, you can make a new object with the same name. Some objects are not namespaced (for example: Nodes), and so their names must be unique across the whole cluster.

## API verbs

Almost all object resource types support the standard HTTP verbs - GET, POST, PUT, PATCH, and DELETE. Kubernetes also uses its own verbs, which are often written lowercase to distinguish them from HTTP verbs.

Kubernetes uses the term **list** to describe returning a [collection](#) of resources to distinguish from retrieving a single resource which is usually called a **get**. If you sent an HTTP GET request with the ?watch query parameter, Kubernetes calls this a **watch** and not a **get** (see [Efficient detection of changes](#) for more details).

For PUT requests, Kubernetes internally classifies these as either **create** or **update** based on the state of the existing object. An **update** is different from a **patch**; the HTTP verb for a **patch** is PATCH.

## Resource URIs

All resource types are either scoped by the cluster (/apis/GROUP/VERSION/\*) or to a namespace (/apis/GROUP/VERSION/namespaces/NAMESPACE/\*). A namespace-scoped resource type will be deleted when its namespace is deleted and access to that resource type is controlled by authorization checks on the namespace scope.

Note: core resources use /api instead of /apis and omit the GROUP path segment.

Examples:

- /api/v1/namespaces
- /api/v1/pods
- /api/v1/namespaces/my-namespace/pods
- /apis/apps/v1/deployments
- /apis/apps/v1/namespaces/my-namespace/deployments
- /apis/apps/v1/namespaces/my-namespace/deployments/my-deployment

You can also access collections of resources (for example: listing all Nodes). The following paths are used to retrieve collections and resources:

- Cluster-scoped resources:
  - GET /apis/GROUP/VERSION/RESOURCETYPE - return the collection of resources of the resource type
  - GET /apis/GROUP/VERSION/RESOURCETYPE/NAME - return the resource with NAME under the resource type
- Namespace-scoped resources:
  - GET /apis/GROUP/VERSION/RESOURCETYPE - return the collection of all instances of the resource type across all namespaces
  - GET /apis/GROUP/VERSION/namespaces/NAMESPACE/RESOURCETYPE - return collection of all instances of the resource type in NAMESPACE

- GET /apis/GROUP/VERSION/namespaces/NAMESPACE/RESOURCETYPE/NAME - return the instance of the resource type with NAME in NAMESPACE

Since a namespace is a cluster-scoped resource type, you can retrieve the list (“collection”) of all namespaces with GET /api/v1/namespaces and details about a particular namespace with GET /api/v1/namespaces/NAME.

- Cluster-scoped subresource: GET /apis/GROUP/VERSION/RESOURCETYPE/NAME/SUBRESOURCE
- Namespace-scoped subresource: GET /apis/GROUP/VERSION/namespaces/NAMESPACE/RESOURCETYPE/NAME/SUBRESOURCE

The verbs supported for each subresource will differ depending on the object - see the [API reference](#) for more information. It is not possible to access sub-resources across multiple resources - generally a new virtual resource type would be used if that becomes necessary.

## Efficient detection of changes

The Kubernetes API allows clients to make an initial request for an object or a collection, and then to track changes since that initial request: a **watch**. Clients can send a **list** or a **get** and then make a follow-up **watch** request.

To make this change tracking possible, every Kubernetes object has a `resourceVersion` field representing the version of that resource as stored in the underlying persistence layer. When retrieving a collection of resources (either namespace or cluster scoped), the response from the API server contains a `resourceVersion` value. The client can use that `resourceVersion` to initiate a **watch** against the API server.

When you send a **watch** request, the API server responds with a stream of changes. These changes itemize the outcome of operations (such as **create**, **delete**, and **update**) that occurred after the `resourceVersion` you specified as a parameter to the **watch** request. The overall **watch** mechanism allows a client to fetch the current state and then subscribe to subsequent changes, without missing any events.

If a client **watch** is disconnected then that client can start a new **watch** from the last returned `resourceVersion`; the client could also perform a fresh **get** / **list** request and begin again. See [Resource Version Semantics](#) for more detail.

For example:

1. List all of the pods in a given namespace.

```
GET /api/v1/namespaces/test/pods

200 OK
Content-Type: application/json

{
 "kind": "PodList",
 "apiVersion": "v1",
 "metadata": {"resourceVersion": "10245"},
 "items": [...]
}
```

- Starting from resource version 10245, receive notifications of any API operations (such as
2. **create, delete, patch or update**) that affect Pods in the *test* namespace. Each change notification is a JSON document. The HTTP response body (served as application/json) consists a series of JSON documents.

```
GET /api/v1/namespaces/test/pods?watch=1&resourceVersion=10245

200 OK
Transfer-Encoding: chunked
Content-Type: application/json

{
 "type": "ADDED",
 "object": {"kind": "Pod", "apiVersion": "v1", "metadata": {"resourceVersion": "10596", ...}, ...}
}
{
 "type": "MODIFIED",
 "object": {"kind": "Pod", "apiVersion": "v1", "metadata": {"resourceVersion": "11020", ...}, ...}
}
...
```

A given Kubernetes server will only preserve a historical record of changes for a limited time. Clusters using etcd 3 preserve changes in the last 5 minutes by default. When the requested **watch** operations fail because the historical version of that resource is not available, clients must handle the case by recognizing the status code 410 Gone, clearing their local cache, performing a new **get** or **list** operation, and starting the **watch** from the resourceVersion that was returned.

For subscribing to collections, Kubernetes client libraries typically offer some form of standard tool for this **list-then-watch** logic. (In the Go client library, this is called a Reflector and is located in the [k8s.io/client-go/tools/cache](https://github.com/kubernetes/client-go/tree/master/tools/cache) package.)

## Watch bookmarks

To mitigate the impact of short history window, the Kubernetes API provides a watch event named BOOKMARK. It is a special kind of event to mark that all changes up to a given resourceVersion the client is requesting have already been sent. The document representing the BOOKMARK event is of the type requested by the request, but only includes a `.metadata.resourceVersion` field. For example:

```
GET /api/v1/namespaces/test/pods?
watch=1&resourceVersion=10245&allowWatchBookmarks=true

200 OK
Transfer-Encoding: chunked
Content-Type: application/json

{
 "type": "ADDED",
 "object": {"kind": "Pod", "apiVersion": "v1", "metadata": {"resourceVersion": "10596", ...}, ...}
}
...
{
```

```
"type": "BOOKMARK",
"object": {"kind": "Pod", "apiVersion": "v1", "metadata": {"resourceVersion": "12746"}}
}
```

As a client, you can request BOOKMARK events by setting the `allowWatchBookmarks=true` query parameter to a **watch** request, but you shouldn't assume bookmarks are returned at any specific interval, nor can clients assume that the API server will send any BOOKMARK event even when requested.

## Streaming lists

FEATURE STATE: Kubernetes v1.27 [alpha]

On large clusters, retrieving the collection of some resource types may result in a significant increase of resource usage (primarily RAM) on the control plane. In order to alleviate its impact and simplify the user experience of the **list** + **watch** pattern, Kubernetes v1.27 introduces as an alpha feature the support for requesting the initial state (previously requested via the **list** request) as part of the **watch** request.

Provided that the WatchList [feature gate](#) is enabled, this can be achieved by specifying `sendInitialEvents=true` as query string parameter in a **watch** request. If set, the API server starts the watch stream with synthetic init events (of type ADDED) to build the whole state of all existing objects followed by a [BOOKMARK event](#) (if requested via `allowWatchBookmarks=true` option). The bookmark event includes the resource version to which is synced. After sending the bookmark event, the API server continues as for any other **watch** request.

When you set `sendInitialEvents=true` in the query string, Kubernetes also requires that you set `resourceVersionMatch` to `NotOlderThan` value. If you provided `resourceVersion` in the query string without providing a value or don't provide it at all, this is interpreted as a request for *consistent read*; the bookmark event is sent when the state is synced at least to the moment of a consistent read from when the request started to be processed. If you specify `resourceVersion` (in the query string), the bookmark event is sent when the state is synced at least to the provided resource version.

## Example

An example: you want to watch a collection of Pods. For that collection, the current resource version is 10245 and there are two pods: foo and bar. Then sending the following request (explicitly requesting *consistent read* by setting empty resource version using `resourceVersion=`) could result in the following sequence of events:

```
GET /api/v1/namespaces/test/pods?
watch=1&sendInitialEvents=true&allowWatchBookmarks=true&resourceVersion=&resourceVersionMatch=NotOlderThan

200 OK
Transfer-Encoding: chunked
Content-Type: application/json

{
 "type": "ADDED",
 "object": {"kind": "Pod", "apiVersion": "v1", "metadata": {"resourceVersion": "8467", "name":
```



```

"foo"}, ...}
}
{
 "type": "ADDED",
 "object": {"kind": "Pod", "apiVersion": "v1", "metadata": {"resourceVersion": "5726", "name":
"bar"}, ...}
}
{
 "type": "BOOKMARK",
 "object": {"kind": "Pod", "apiVersion": "v1", "metadata": {"resourceVersion": "10245"}}
}
...
<followed by regular watch stream starting from resourceVersion="10245">

```

## Response compression

FEATURE STATE: Kubernetes v1.16 [beta]

`APIResponseCompression` is an option that allows the API server to compress the responses for **get** and **list** requests, reducing the network bandwidth and improving the performance of large-scale clusters. It is enabled by default since Kubernetes 1.16 and it can be disabled by including `APIResponseCompression=false` in the `--feature-gates` flag on the API server.

API response compression can significantly reduce the size of the response, especially for large resources or [collections](#). For example, a **list** request for pods can return hundreds of kilobytes or even megabytes of data, depending on the number of pods and their attributes. By compressing the response, the network bandwidth can be saved and the latency can be reduced.

To verify if `APIResponseCompression` is working, you can send a **get** or **list** request to the API server with an `Accept-Encoding` header, and check the response size and headers. For example:

```

GET /api/v1/pods
Accept-Encoding: gzip

200 OK
Content-Type: application/json
content-encoding: gzip
...

```

The `content-encoding` header indicates that the response is compressed with `gzip`.

## Retrieving large results sets in chunks

FEATURE STATE: Kubernetes v1.29 [stable]

On large clusters, retrieving the collection of some resource types may result in very large responses that can impact the server and client. For instance, a cluster may have tens of thousands of Pods, each of which is equivalent to roughly 2 KiB of encoded JSON. Retrieving all pods across all namespaces may result in a very large response (10-20MB) and consume a large amount of server resources.



The Kubernetes API server supports the ability to break a single large collection request into many smaller chunks while preserving the consistency of the total request. Each chunk can be returned sequentially which reduces both the total size of the request and allows user-oriented clients to display results incrementally to improve responsiveness.

You can request that the API server handles a **list** by serving single collection using pages (which Kubernetes calls *chunks*). To retrieve a single collection in chunks, two query parameters `limit` and `continue` are supported on requests against collections, and a response field `continue` is returned from all **list** operations in the collection's metadata field. A client should specify the maximum results they wish to receive in each chunk with `limit` and the server will return up to `limit` resources in the result and include a `continue` value if there are more resources in the collection.

As an API client, you can then pass this `continue` value to the API server on the next request, to instruct the server to return the next page (*chunk*) of results. By continuing until the server returns an empty `continue` value, you can retrieve the entire collection.

Like a **watch** operation, a `continue` token will expire after a short amount of time (by default 5 minutes) and return a 410 Gone if more results cannot be returned. In this case, the client will need to start from the beginning or omit the `limit` parameter.

For example, if there are 1,253 pods on the cluster and you want to receive chunks of 500 pods at a time, request those chunks as follows:

1. List all of the pods on a cluster, retrieving up to 500 pods each time.

```
GET /api/v1/pods?limit=500

200 OK
Content-Type: application/json

{
 "kind": "PodList",
 "apiVersion": "v1",
 "metadata": {
 "resourceVersion": "10245",
 "continue": "ENCODED_CONTINUE_TOKEN",
 "remainingItemCount": 753,
 ...
 },
 "items": [...] // returns pods 1-500
}
```

2. Continue the previous call, retrieving the next set of 500 pods.

```
GET /api/v1/pods?limit=500&continue=ENCODED_CONTINUE_TOKEN

200 OK
Content-Type: application/json

{
 "kind": "PodList",
 "apiVersion": "v1",
 "metadata": {
```

```
"resourceVersion":"10245",
"continue": "ENCODED_CONTINUE_TOKEN_2",
"remainingItemCount": 253,
...
},
"items": [...] // returns pods 501-1000
}
```

3. Continue the previous call, retrieving the last 253 pods.

```
GET /api/v1/pods?limit=500&continue=ENCODED_CONTINUE_TOKEN_2

200 OK
Content-Type: application/json

{
 "kind": "PodList",
 "apiVersion": "v1",
 "metadata": {
 "resourceVersion":"10245",
 "continue": "", // continue token is empty because we have reached the end of the list
 ...
 },
 "items": [...] // returns pods 1001-1253
}
```

Notice that the `resourceVersion` of the collection remains constant across each request, indicating the server is showing you a consistent snapshot of the pods. Pods that are created, updated, or deleted after version 10245 would not be shown unless you make a separate **list** request without the continue token. This allows you to break large requests into smaller chunks and then perform a **watch** operation on the full set without missing any updates.

`remainingItemCount` is the number of subsequent items in the collection that are not included in this response. If the **list** request contained label or field [selectors](#) then the number of remaining items is unknown and the API server does not include a `remainingItemCount` field in its response. If the **list** is complete (either because it is not chunking, or because this is the last chunk), then there are no more remaining items and the API server does not include a `remainingItemCount` field in its response. The intended use of the `remainingItemCount` is estimating the size of a collection.

## Collections

In Kubernetes terminology, the response you get from a **list** is a *collection*. However, Kubernetes defines concrete kinds for collections of different types of resource. Collections have a kind named for the resource kind, with List appended.

When you query the API for a particular type, all items returned by that query are of that type. For example, when you **list** Services, the collection response has kind set to [ServiceList](#); each item in that collection represents a single Service. For example:

```
GET /api/v1/services
```

```
{
 "kind": "ServiceList",
 "apiVersion": "v1",
 "metadata": {
 "resourceVersion": "2947301"
 },
 "items": [
 {
 "metadata": {
 "name": "kubernetes",
 "namespace": "default",
...
 "metadata": {
 "name": "kube-dns",
 "namespace": "kube-system",
...

```

There are dozens of collection types (such as PodList, ServiceList, and NodeList) defined in the Kubernetes API. You can get more information about each collection type from the [Kubernetes API](#) documentation.

Some tools, such as kubectl, represent the Kubernetes collection mechanism slightly differently from the Kubernetes API itself. Because the output of kubectl might include the response from multiple **list** operations at the API level, kubectl represents a list of items using kind: List. For example:

```
kubectl get services -A -o yaml
```

```
apiVersion: v1
kind: List
metadata:
 resourceVersion: ""
 selfLink: ""
items:
- apiVersion: v1
 kind: Service
 metadata:
 creationTimestamp: "2021-06-03T14:54:12Z"
 labels:
 component: apiserver
 provider: kubernetes
 name: kubernetes
 namespace: default
...
- apiVersion: v1
 kind: Service
 metadata:
 annotations:
 prometheus.io/port: "9153"
 prometheus.io/scrape: "true"
 creationTimestamp: "2021-06-03T14:54:14Z"
 labels:
 k8s-app: kube-dns

```

```
kubernetes.io/cluster-service: "true"
kubernetes.io/name: CoreDNS
name: kube-dns
namespace: kube-system
```

#### Note:

Keep in mind that the Kubernetes API does not have a kind named List.

kind: List is a client-side, internal implementation detail for processing collections that might be of different kinds of object. Avoid depending on kind: List in automation or other code.

## Receiving resources as Tables

When you run `kubectl get`, the default output format is a simple tabular representation of one or more instances of a particular resource type. In the past, clients were required to reproduce the tabular and describe output implemented in `kubectl` to perform simple lists of objects. A few limitations of that approach include non-trivial logic when dealing with certain objects. Additionally, types provided by API aggregation or third party resources are not known at compile time. This means that generic implementations had to be in place for types unrecognized by a client.

In order to avoid potential limitations as described above, clients may request the Table representation of objects, delegating specific details of printing to the server. The Kubernetes API implements standard HTTP content type negotiation: passing an Accept header containing a value of `application/json;as=Table;g=meta.k8s.io;v=v1` with a GET call will request that the server return objects in the Table content type.

For example, list all of the pods on a cluster in the Table format.

```
GET /api/v1/pods
Accept: application/json;as=Table;g=meta.k8s.io;v=v1

200 OK
Content-Type: application/json

{
 "kind": "Table",
 "apiVersion": "meta.k8s.io/v1",
 ...
 "columnDefinitions": [
 ...
]
}
```

For API resource types that do not have a custom Table definition known to the control plane, the API server returns a default Table response that consists of the resource's name and `creationTimestamp` fields.

```
GET /apis/crd.example.com/v1alpha1/namespaces/default/resources

200 OK
Content-Type: application/json
```

```

...
{
 "kind": "Table",
 "apiVersion": "meta.k8s.io/v1",
 ...
 "columnDefinitions": [
 {
 "name": "Name",
 "type": "string",
 ...
 },
 {
 "name": "Created At",
 "type": "date",
 ...
 }
]
}

```

Not all API resource types support a Table response; for example, a [CustomResourceDefinitions](#) might not define field-to-table mappings, and an APIService that [extends the core Kubernetes API](#) might not serve Table responses at all. If you are implementing a client that uses the Table information and must work against all resource types, including extensions, you should make requests that specify multiple content types in the Accept header. For example:

```
Accept: application/json;as=Table;g=meta.k8s.io;v=v1, application/json
```

## Alternate representations of resources

By default, Kubernetes returns objects serialized to JSON with content type `application/json`. This is the default serialization format for the API. However, clients may request the more efficient [Protobuf representation](#) of these objects for better performance at scale. The Kubernetes API implements standard HTTP content type negotiation: passing an Accept header with a GET call will request that the server tries to return a response in your preferred media type, while sending an object in Protobuf to the server for a PUT or POST call means that you must set the Content-Type header appropriately.

The server will return a response with a Content-Type header if the requested format is supported, or the 406 Not acceptable error if none of the media types you requested are supported. All built-in resource types support the `application/json` media type.

See the Kubernetes [API reference](#) for a list of supported content types for each API.

For example:

1. List all of the pods on a cluster in Protobuf format.

```

GET /api/v1/pods
Accept: application/vnd.kubernetes.protobuf

200 OK
Content-Type: application/vnd.kubernetes.protobuf

```

```
... binary encoded PodList object
```

2. Create a pod by sending Protobuf encoded data to the server, but request a response in JSON.

```
POST /api/v1/namespaces/test/pods
Content-Type: application/vnd.kubernetes.protobuf
Accept: application/json
... binary encoded Pod object

200 OK
Content-Type: application/json

{
 "kind": "Pod",
 "apiVersion": "v1",
 ...
}
```

Not all API resource types support Protobuf; specifically, Protobuf isn't available for resources that are defined as [CustomResourceDefinitions](#) or are served via the [aggregation layer](#). As a client, if you might need to work with extension types you should specify multiple content types in the request Accept header to support fallback to JSON. For example:

```
Accept: application/vnd.kubernetes.protobuf, application/json
```

## Kubernetes Protobuf encoding

Kubernetes uses an envelope wrapper to encode Protobuf responses. That wrapper starts with a 4 byte magic number to help identify content in disk or in etcd as Protobuf (as opposed to JSON), and then is followed by a Protobuf encoded wrapper message, which describes the encoding and type of the underlying object and then contains the object.

The wrapper format is:

A four byte magic number prefix:

```
Bytes 0-3: "k8s\x00" [0x6b, 0x38, 0x73, 0x00]
```

An encoded Protobuf message with the following IDL:

```
message Unknown {
 // typeMeta should have the string values for "kind" and "apiVersion" as set on the JSON
 object
 optional TypeMeta typeMeta = 1;

 // raw will hold the complete serialized object in protobuf. See the protobuf definitions in the
 client libraries for a given kind.
 optional bytes raw = 2;

 // contentEncoding is encoding used for the raw data. Unspecified means no encoding.
 optional string contentEncoding = 3;

 // contentType is the serialization method used to serialize 'raw'. Unspecified means
```

```

application/vnd.kubernetes.protobuf and is usually
// omitted.
optional string contentType = 4;
}

message TypeMeta {
// apiVersion is the group/version for this type
optional string apiVersion = 1;
// kind is the name of the object schema. A protobuf definition should exist for this object.
optional string kind = 2;
}

```

### Note:

Clients that receive a response in `application/vnd.kubernetes.protobuf` that does not match the expected prefix should reject the response, as future versions may need to alter the serialization format in an incompatible way and will do so by changing the prefix.

## Resource deletion

When you **delete** a resource this takes place in two phases.

1. *finalization*
2. removal

```

{
 "kind": "ConfigMap",
 "apiVersion": "v1",
 "metadata": {
 "finalizers": ["url.io/neat-finalization", "other-url.io/my-finalizer"],
 "deletionTimestamp": nil,
 }
}

```

When a client first sends a **delete** to request the removal of a resource, the `.metadata.deletionTimestamp` is set to the current time. Once the `.metadata.deletionTimestamp` is set, external controllers that act on finalizers may start performing their cleanup work at any time, in any order.

Order is **not** enforced between finalizers because it would introduce significant risk of stuck `.metadata.finalizers`.

The `.metadata.finalizers` field is shared: any actor with permission can reorder it. If the finalizer list were processed in order, then this might lead to a situation in which the component responsible for the first finalizer in the list is waiting for some signal (field value, external system, or other) produced by a component responsible for a finalizer later in the list, resulting in a deadlock.

Without enforced ordering, finalizers are free to order amongst themselves and are not vulnerable to ordering changes in the list.

Once the last finalizer is removed, the resource is actually removed from etcd.

# Single resource API

The Kubernetes API verbs **get**, **create**, **update**, **patch**, **delete** and **proxy** support single resources only. These verbs with single resource support have no support for submitting multiple resources together in an ordered or unordered list or transaction.

When clients (including kubectl) act on a set of resources, the client makes a series of single-resource API requests, then aggregates the responses if needed.

By contrast, the Kubernetes API verbs **list** and **watch** allow getting multiple resources, and **deletecollection** allows deleting multiple resources.

## Field validation

Kubernetes always validates the type of fields. For example, if a field in the API is defined as a number, you cannot set the field to a text value. If a field is defined as an array of strings, you can only provide an array. Some fields allow you to omit them, other fields are required. Omitting a required field from an API request is an error.

If you make a request with an extra field, one that the cluster's control plane does not recognize, then the behavior of the API server is more complicated.

By default, the API server drops fields that it does not recognize from an input that it receives (for example, the JSON body of a PUT request).

There are two situations where the API server drops fields that you supplied in an HTTP request.

These situations are:

1. The field is unrecognized because it is not in the resource's OpenAPI schema. (One exception to this is for [CRDs](#) that explicitly choose not to prune unknown fields via x-kubernetes-preserve-unknown-fields).
2. The field is duplicated in the object.

## Validation for unrecognized or duplicate fields

FEATURE STATE: Kubernetes v1.27 [stable]

From 1.25 onward, unrecognized or duplicate fields in an object are detected via validation on the server when you use HTTP verbs that can submit data (POST, PUT, and PATCH). Possible levels of validation are Ignore, Warn (default), and Strict.

### Ignore

The API server succeeds in handling the request as it would without the erroneous fields being set, dropping all unknown and duplicate fields and giving no indication it has done so.

### Warn

(Default) The API server succeeds in handling the request, and reports a warning to the client. The warning is sent using the Warning: response header, adding one warning item for each unknown or duplicate field. For more information about warnings and the Kubernetes API, see the blog article [Warning: Helpful Warnings Ahead](#).



## Strict

The API server rejects the request with a 400 Bad Request error when it detects any unknown or duplicate fields. The response message from the API server specifies all the unknown or duplicate fields that the API server has detected.

The field validation level is set by the `fieldValidation` query parameter.

### Note:

If you submit a request that specifies an unrecognized field, and that is also invalid for a different reason (for example, the request provides a string value where the API expects an integer for a known field), then the API server responds with a 400 Bad Request error, but will not provide any information on unknown or duplicate fields (only which fatal error it encountered first).

You always receive an error response in this case, no matter what field validation level you requested.

Tools that submit requests to the server (such as `kubectl`), might set their own defaults that are different from the Warn validation level that the API server uses by default.

The `kubectl` tool uses the `--validate` flag to set the level of field validation. It accepts the values `ignore`, `warn`, and `strict` while also accepting the values `true` (equivalent to `strict`) and `false` (equivalent to `ignore`). The default validation setting for `kubectl` is `--validate=true`, which means strict server-side field validation.

When `kubectl` cannot connect to an API server with field validation (API servers prior to Kubernetes 1.27), it will fall back to using client-side validation. Client-side validation will be removed entirely in a future version of `kubectl`.

### Note:

Prior to Kubernetes 1.25 `kubectl --validate` was used to toggle client-side validation on or off as a boolean flag.

## Dry-run

FEATURE STATE: Kubernetes v1.19 [stable]

When you use HTTP verbs that can modify resources (POST, PUT, PATCH, and DELETE), you can submit your request in a *dry run* mode. Dry run mode helps to evaluate a request through the typical request stages (admission chain, validation, merge conflicts) up until persisting objects to storage. The response body for the request is as close as possible to a non-dry-run response. Kubernetes guarantees that dry-run requests will not be persisted in storage or have any other side effects.

### Make a dry-run request

Dry-run is triggered by setting the `dryRun` query parameter. This parameter is a string, working as an enum, and the only accepted values are:

[no value set]

Allow side effects. You request this with a query string such as `?dryRun` or `?dryRun&pretty=true`. The response is the final object that would have been persisted, or an error if the request could not be fulfilled.

All

Every stage runs as normal, except for the final storage stage where side effects are prevented.

When you set `?dryRun=All`, any relevant [admission controllers](#) are run, validating admission controllers check the request post-mutation, merge is performed on PATCH, fields are defaulted, and schema validation occurs. The changes are not persisted to the underlying storage, but the final object which would have been persisted is still returned to the user, along with the normal status code.

If the non-dry-run version of a request would trigger an admission controller that has side effects, the request will be failed rather than risk an unwanted side effect. All built in admission control plugins support dry-run. Additionally, admission webhooks can declare in their [configuration object](#) that they do not have side effects, by setting their `sideEffects` field to `None`.

### Note:

If a webhook actually does have side effects, then the `sideEffects` field should be set to `"NoneOnDryRun"`. That change is appropriate provided that the webhook is also be modified to understand the `DryRun` field in `AdmissionReview`, and to prevent side effects on any request marked as dry runs.

Here is an example dry-run request that uses `?dryRun=All`:

```
POST /api/v1/namespaces/test/pods?dryRun=All
Content-Type: application/json
Accept: application/json
```

The response would look the same as for non-dry-run request, but the values of some generated fields may differ.

## Generated values

Some values of an object are typically generated before the object is persisted. It is important not to rely upon the values of these fields set by a dry-run request, since these values will likely be different in dry-run mode from when the real request is made. Some of these fields are:

- `name`: if `generateName` is set, name will have a unique random name
- `creationTimestamp` / `deletionTimestamp`: records the time of creation/deletion
- `UID`: [uniquely identifies](#) the object and is randomly generated (non-deterministic)
- `resourceVersion`: tracks the persisted version of the object
- Any field set by a mutating admission controller
- For the Service resource: Ports or IP addresses that the kube-apiserver assigns to Service objects

## Dry-run authorization

Authorization for dry-run and non-dry-run requests is identical. Thus, to make a dry-run request, you must be authorized to make the non-dry-run request.

For example, to run a dry-run **patch** for a Deployment, you must be authorized to perform that **patch**. Here is an example of a rule for Kubernetes [RBAC](#) that allows patching Deployments:

```
rules:
- apiGroups: ["apps"]
 resources: ["deployments"]
 verbs: ["patch"]
```

See [Authorization Overview](#).

## Updates to existing resources

Kubernetes provides several ways to update existing objects. You can read [choosing an update mechanism](#) to learn about which approach might be best for your use case.

You can overwrite (**update**) an existing resource - for example, a ConfigMap - using an HTTP PUT. For a PUT request, it is the client's responsibility to specify the resourceVersion (taking this from the object being updated). Kubernetes uses that resourceVersion information so that the API server can detect lost updates and reject requests made by a client that is out of date with the cluster. In the event that the resource has changed (the resourceVersion the client provided is stale), the API server returns a 409 Conflict error response.

Instead of sending a PUT request, the client can send an instruction to the API server to **patch** an existing resource. A **patch** is typically appropriate if the change that the client wants to make isn't conditional on the existing data. Clients that need effective detection of lost updates should consider making their request conditional on the existing resourceVersion (either HTTP PUT or HTTP PATCH), and then handle any retries that are needed in case there is a conflict.

The Kubernetes API supports four different PATCH operations, determined by their corresponding HTTP Content-Type header:

### application/apply-patch+yaml

Server Side Apply YAML (a Kubernetes-specific extension, based on YAML). All JSON documents are valid YAML, so you can also submit JSON using this media type. See [Server Side Apply serialization](#) for more details.

To Kubernetes, this is a **create** operation if the object does not exist, or a **patch** operation if the object already exists.

### application/json-patch+json

JSON Patch, as defined in [RFC6902](#). A JSON patch is a sequence of operations that are executed on the resource; for example `{"op": "add", "path": "/a/b/c", "value": [ "foo", "bar" ]}`. To Kubernetes, this is a **patch** operation.

A **patch** using `application/json-patch+json` can include conditions to validate consistency, allowing the operation to fail if those conditions are not met (for example, to avoid a lost update).

### application/merge-patch+json

JSON Merge Patch, as defined in [RFC7386](#). A JSON Merge Patch is essentially a partial representation of the resource. The submitted JSON is combined with the current resource to create a new one, then the new one is saved.

To Kubernetes, this is a **patch** operation.

### application/strategic-merge-patch+json

Strategic Merge Patch (a Kubernetes-specific extension based on JSON). Strategic Merge Patch is a custom implementation of JSON Merge Patch. You can only use Strategic Merge Patch with built-in APIs, or with aggregated API servers that have special support for it. You cannot use application/strategic-merge-patch+json with any API defined using a [CustomResourceDefinition](#).

**Note:**

The Kubernetes *server side apply* mechanism has superseded Strategic Merge Patch.

Kubernetes' [Server Side Apply](#) feature allows the control plane to track managed fields for newly created objects. Server Side Apply provides a clear pattern for managing field conflicts, offers server-side **apply** and **update** operations, and replaces the client-side functionality of `kubectl apply`.

For Server-Side Apply, Kubernetes treats the request as a **create** if the object does not yet exist, and a **patch** otherwise. For other requests that use PATCH at the HTTP level, the logical Kubernetes operation is always **patch**.

See [Server Side Apply](#) for more details.

## Choosing an update mechanism

### HTTP PUT to replace existing resource

The **update** (HTTP PUT) operation is simple to implement and flexible, but has drawbacks:

- You need to handle conflicts where the `resourceVersion` of the object changes between your client reading it and trying to write it back. Kubernetes always detects the conflict, but you as the client author need to implement retries.
- You might accidentally drop fields if you decode an object locally (for example, using `client-go`, you could receive fields that your client does not know how to handle - and then drop them as part of your update.
- If there's a lot of contention on the object (even on a field, or set of fields, that you're not trying to edit), you might have trouble sending the update. The problem is worse for larger objects and for objects with many fields.

### HTTP PATCH using JSON Patch

A **patch** update is helpful, because:

- As you're only sending differences, you have less data to send in the PATCH request.
- You can make changes that rely on existing values, such as copying the value of a particular field into an annotation.
- Unlike with an **update** (HTTP PUT), making your change can happen right away even if there are frequent changes to unrelated fields): you usually would not need to retry.
  - You might still need to specify the `resourceVersion` (to match an existing object) if you want to be extra careful to avoid lost updates
  - It's still good practice to write in some retry logic in case of errors.
- You can use test conditions to carefully craft specific update conditions. For example, you can increment a counter without reading it if the existing value matches what you expect. You can do this with no lost update risk, even if the object has changed in other ways

since you last wrote to it. (If the test condition fails, you can fall back to reading the current value and then write back the changed number).

However:

- you need more local (client) logic to build the patch; it helps a lot if you have a library implementation of JSON Patch, or even for making a JSON Patch specifically against Kubernetes
- as the author of client software, you need to be careful when building the patch (the HTTP request body) not to drop fields (the order of operations matters)

## HTTP PATCH using Server-Side Apply

Server-Side Apply has some clear benefits:

- A single round trip: it rarely requires making a GET request first.
  - and you can still detect conflicts for unexpected changes
  - you have the option to force override a conflict, if appropriate
- Client implementations are easy to make
- You get an atomic create-or-update operation without extra effort (similar to UPSERT in some SQL dialects)

However:

- Server-Side Apply does not work at all for field changes that depend on a current value of the object
- You can only apply updates to objects. Some resources in the Kubernetes HTTP API are not objects (they do not have a `.metadata` field), and Server-Side Apply is only relevant for Kubernetes objects.

## Resource versions

Resource versions are strings that identify the server's internal version of an object. Resource versions can be used by clients to determine when objects have changed, or to express data consistency requirements when getting, listing and watching resources. Resource versions must be treated as opaque by clients and passed unmodified back to the server.

You must not assume resource versions are numeric or collatable. API clients may only compare two resource versions for equality (this means that you must not compare resource versions for greater-than or less-than relationships).

### resourceVersion fields in metadata

Clients find resource versions in resources, including the resources from the response stream for a **watch**, or when using **list** to enumerate resources.

[v1.meta/ObjectMeta](#) - The `metadata.resourceVersion` of a resource instance identifies the resource version the instance was last modified at.

[v1.meta/ListMeta](#) - The `metadata.resourceVersion` of a resource collection (the response to a **list**) identifies the resource version at which the collection was constructed.

## resourceVersion parameters in query strings

The **get**, **list**, and **watch** operations support the `resourceVersion` parameter. From version v1.19, Kubernetes API servers also support the `resourceVersionMatch` parameter on *list* requests.

The API server interprets the `resourceVersion` parameter differently depending on the operation you request, and on the value of `resourceVersion`. If you set `resourceVersionMatch` then this also affects the way matching happens.

### Semantics for get and list

For **get** and **list**, the semantics of `resourceVersion` are:

**get:**

| <code>resourceVersion</code> unset | <code>resourceVersion="0"</code> | <code>resourceVersion="{value other than 0}"</code> |
|------------------------------------|----------------------------------|-----------------------------------------------------|
| Most Recent                        | Any                              | Not older than                                      |

**list:**

From version v1.19, Kubernetes API servers support the `resourceVersionMatch` parameter on *list* requests. If you set both `resourceVersion` and `resourceVersionMatch`, the `resourceVersionMatch` parameter determines how the API server interprets `resourceVersion`.

You should always set the `resourceVersionMatch` parameter when setting `resourceVersion` on a **list** request. However, be prepared to handle the case where the API server that responds is unaware of `resourceVersionMatch` and ignores it.

Unless you have strong consistency requirements, using `resourceVersionMatch=NotOlderThan` and a known `resourceVersion` is preferable since it can achieve better performance and scalability of your cluster than leaving `resourceVersion` and `resourceVersionMatch` unset, which requires quorum read to be served.

Setting the `resourceVersionMatch` parameter without setting `resourceVersion` is not valid.

This table explains the behavior of **list** requests with various combinations of `resourceVersion` and `resourceVersionMatch`:

`resourceVersionMatch` and paging parameters for list

| <code>resourceVersionMatch</code> param        | paging params                                             | <code>resourceVersion</code> not set | <code>resourceVersion="0"</code>                | <code>resourceVersion="{value other than 0}"</code> |
|------------------------------------------------|-----------------------------------------------------------|--------------------------------------|-------------------------------------------------|-----------------------------------------------------|
| <i>unset</i>                                   | <i>limit unset</i>                                        | Most Recent                          | Any                                             | Not older than                                      |
| <i>unset</i>                                   | <i>limit=&lt;n&gt;</i> ,<br><i>continue unset</i>         | Most Recent                          | Any                                             | Exact                                               |
| <i>unset</i>                                   | <i>limit=&lt;n&gt;</i> ,<br><i>continue=&lt;token&gt;</i> | Continue Token,<br>Exact             | Invalid, treated as<br>Continue Token,<br>Exact | Invalid, HTTP 400 Request                           |
| <code>resourceVersionMatch=Exact</code>        | <i>limit unset</i>                                        | Invalid                              | Invalid                                         | Exact                                               |
| <code>resourceVersionMatch=Exact</code>        | <i>limit=&lt;n&gt;</i> ,<br><i>continue unset</i>         | Invalid                              | Invalid                                         | Exact                                               |
| <code>resourceVersionMatch=NotOlderThan</code> | <i>limit unset</i>                                        | Invalid                              | Any                                             | Not older than                                      |
| <code>resourceVersionMatch=NotOlderThan</code> |                                                           | Invalid                              | Any                                             | Not older than                                      |

| resourceVersionMatch param | paging params                | resourceVersion not set | resourceVersion="0" | resourceVersion other than 0 |
|----------------------------|------------------------------|-------------------------|---------------------|------------------------------|
|                            | limit=<n>,<br>continue unset |                         |                     |                              |

**Note:**

If your cluster's API server does not honor the `resourceVersionMatch` parameter, the behavior is the same as if you did not set it.

The meaning of the **get** and **list** semantics are:

Any

Return data at any resource version. The newest available resource version is preferred, but strong consistency is not required; data at any resource version may be served. It is possible for the request to return data at a much older resource version that the client has previously observed, particularly in high availability configurations, due to partitions or stale caches. Clients that cannot tolerate this should not use this semantic.

Most recent

Return data at the most recent resource version. The returned data must be consistent (in detail: served from etcd via a quorum read).

Not older than

Return data at least as new as the provided `resourceVersion`. The newest available data is preferred, but any data not older than the provided `resourceVersion` may be served. For **list** requests to servers that honor the `resourceVersionMatch` parameter, this guarantees that the collection's `.metadata.resourceVersion` is not older than the requested `resourceVersion`, but does not make any guarantee about the `.metadata.resourceVersion` of any of the items in that collection.

Exact

Return data at the exact resource version provided. If the provided `resourceVersion` is unavailable, the server responds with HTTP 410 "Gone". For **list** requests to servers that honor the `resourceVersionMatch` parameter, this guarantees that the collection's `.metadata.resourceVersion` is the same as the `resourceVersion` you requested in the query string. That guarantee does not apply to the `.metadata.resourceVersion` of any items within that collection.

Continue Token, Exact

Return data at the resource version of the initial paginated **list** call. The returned *continue tokens* are responsible for keeping track of the initially provided resource version for all paginated **list** calls after the initial paginated **list**.

**Note:**

When you **list** resources and receive a collection response, the response includes the [list metadata](#) of the collection as well as [object metadata](#) for each item in that collection. For individual objects found within a collection response, `.metadata.resourceVersion` tracks when that object was last updated, and not how up-to-date the object is when served.

When using `resourceVersionMatch=NotOlderThan` and `limit` is set, clients must handle HTTP 410 "Gone" responses. For example, the client might retry with a newer `resourceVersion` or fall back to `resourceVersion=""`.

When using `resourceVersionMatch=Exact` and `limit` is unset, clients must verify that the collection's `.metadata.resourceVersion` matches the requested `resourceVersion`, and handle the case where it does not. For example, the client might fall back to a request with `limit` set.

## Semantics for watch

For **watch**, the semantics of resource version are:

### watch:

resourceVersion for watch

| resourceVersion unset              | resourceVersion="0"        | resourceVersion="{value other than 0}" |
|------------------------------------|----------------------------|----------------------------------------|
| Get State and Start at Most Recent | Get State and Start at Any | Start at Exact                         |

The meaning of those **watch** semantics are:

Get State and Start at Any

### Caution:

Watches initialized this way may return arbitrarily stale data. Please review this semantic before using it, and favor the other semantics where possible.

Start a **watch** at any resource version; the most recent resource version available is preferred, but not required. Any starting resource version is allowed. It is possible for the **watch** to start at a much older resource version that the client has previously observed, particularly in high availability configurations, due to partitions or stale caches. Clients that cannot tolerate this apparent rewinding should not start a **watch** with this semantic. To establish initial state, the **watch** begins with synthetic "Added" events for all resource instances that exist at the starting resource version. All following watch events are for all changes that occurred after the resource version the **watch** started at.

Get State and Start at Most Recent

Start a **watch** at the most recent resource version, which must be consistent (in detail: served from etcd via a quorum read). To establish initial state, the **watch** begins with synthetic "Added" events of all resources instances that exist at the starting resource version. All following watch events are for all changes that occurred after the resource version the **watch** started at.

Start at Exact

Start a **watch** at an exact resource version. The watch events are for all changes after the provided resource version. Unlike "Get State and Start at Most Recent" and "Get State and Start at Any", the **watch** is not started with synthetic "Added" events for the provided resource version. The client is assumed to already have the initial state at the starting resource version since the client provided the resource version.

## "410 Gone" responses

Servers are not required to serve all older resource versions and may return a HTTP 410 (Gone) status code if a client requests a `resourceVersion` older than the server has retained. Clients must be able to tolerate 410 (Gone) responses. See [Efficient detection of changes](#) for details on how to handle 410 (Gone) responses when watching resources.



If you request a resourceVersion outside the applicable limit then, depending on whether a request is served from cache or not, the API server may reply with a 410 Gone HTTP response.

## Unavailable resource versions

Servers are not required to serve unrecognized resource versions. If you request **list** or **get** for a resource version that the API server does not recognize, then the API server may either:

- wait briefly for the resource version to become available, then timeout with a 504 (Gateway Timeout) if the provided resource versions does not become available in a reasonable amount of time;
- respond with a Retry-After response header indicating how many seconds a client should wait before retrying the request.

If you request a resource version that an API server does not recognize, the kube-apiserver additionally identifies its error responses with a "Too large resource version" message.

If you make a **watch** request for an unrecognized resource version, the API server may wait indefinitely (until the request timeout) for the resource version to become available.

# Server-Side Apply

FEATURE STATE: Kubernetes v1.22 [stable]

Kubernetes supports multiple appliers collaborating to manage the fields of a single [object](#).

Server-Side Apply provides an optional mechanism for your cluster's control plane to track changes to an object's fields. At the level of a specific resource, Server-Side Apply records and tracks information about control over the fields of that object.

Server-Side Apply helps users and [controllers](#) manage their resources through declarative configuration. Clients can create and modify [objects](#) declaratively by submitting their *fully specified intent*.

A fully specified intent is a partial object that only includes the fields and values for which the user has an opinion. That intent either creates a new object (using default values for unspecified fields), or is [combined](#), by the API server, with the existing object.

[Comparison with Client-Side Apply](#) explains how Server-Side Apply differs from the original, client-side kubectl apply implementation.

## Field management

The Kubernetes API server tracks *managed fields* for all newly created objects.

When trying to apply an object, fields that have a different value and are owned by another [manager](#) will result in a [conflict](#). This is done in order to signal that the operation might undo another collaborator's changes. Writes to objects with managed fields can be forced, in which case the value of any conflicted field will be overridden, and the ownership will be transferred.

Whenever a field's value does change, ownership moves from its current manager to the manager making the change.

Apply checks if there are any other field managers that also own the field. If the field is not owned by any other field managers, that field is set to its default value (if there is one), or otherwise is deleted from the object. The same rule applies to fields that are lists, associative lists, or maps.

For a user to manage a field, in the Server-Side Apply sense, means that the user relies on and expects the value of the field not to change. The user who last made an assertion about the value of a field will be recorded as the current field manager. This can be done by changing the field manager details explicitly using HTTP POST (**create**), PUT (**update**), or non-apply PATCH (**patch**). You can also declare and record a field manager by including a value for that field in a Server-Side Apply operation.

A Server-Side Apply **patch** request requires the client to provide its identity as a [field manager](#). When using Server-Side Apply, trying to change a field that is controlled by a different manager results in a rejected request unless the client forces an override. For details of overrides, see [Conflicts](#).

When two or more appliers set a field to the same value, they share ownership of that field. Any subsequent attempt to change the value of the shared field, by any of the appliers, results in a conflict. Shared field owners may give up ownership of a field by making a Server-Side Apply **patch** request that doesn't include that field.

Field management details are stored in a `managedFields` field that is part of an object's [metadata](#).

If you remove a field from a manifest and apply that manifest, Server-Side Apply checks if there are any other field managers that also own the field. If the field is not owned by any other field managers, it is either deleted from the live object or reset to its default value, if it has one. The same rule applies to associative list or map items.

Compared to the (legacy) [kubectrl.kubernetes.io/last-applied-configuration](https://kubernetes.io/docs/concepts/configuration/last-applied-configuration/) annotation managed by kubectrl, Server-Side Apply uses a more declarative approach, that tracks a user's (or client's) field management, rather than a user's last applied state. As a side effect of using Server-Side Apply, information about which field manager manages each field in an object also becomes available.

## Example

A simple example of an object created using Server-Side Apply could look like this:

```

apiVersion: v1
kind: ConfigMap
metadata:
 name: test-cm
 namespace: default
 labels:
 test-label: test
 managedFields:
 - manager: kubectrl
 operation: Apply # note capitalization: "Apply" (or "Update")
 apiVersion: v1
 time: "2010-10-10T0:00:00Z"
```

```
fieldsType: FieldsV1
fieldsV1:
 f:metadata:
 f:labels:
 f:test-label: {}
 f:data:
 f:key: {}
data:
 key: some value
```

That example ConfigMap object contains a single field management record in `.metadata.managedFields`. The field management record consists of basic information about the managing entity itself, plus details about the fields being managed and the relevant operation (Apply or Update). If the request that last changed that field was a Server-Side Apply **patch** then the value of operation is Apply; otherwise, it is Update.

There is another possible outcome. A client could submit an invalid request body. If the fully specified intent does not produce a valid object, the request fails.

It is however possible to change `.metadata.managedFields` through an **update**, or through a **patch** operation that does not use Server-Side Apply. Doing so is highly discouraged, but might be a reasonable option to try if, for example, the `.metadata.managedFields` get into an inconsistent state (which should not happen in normal operations).

The format of managedFields is [described](#) in the Kubernetes API reference.

### Caution:

The `.metadata.managedFields` field is managed by the API server. You should avoid updating it manually.

## Conflicts

A *conflict* is a special status error that occurs when an Apply operation tries to change a field that another manager also claims to manage. This prevents an applier from unintentionally overwriting the value set by another user. When this occurs, the applier has 3 options to resolve the conflicts:

- **Overwrite value, become sole manager:** If overwriting the value was intentional (or if the applier is an automated process like a controller) the applier should set the force query parameter to true (for kubectl apply, you use the `--force-conflicts` command line parameter), and make the request again. This forces the operation to succeed, changes the value of the field, and removes the field from all other managers' entries in managedFields.
- **Don't overwrite value, give up management claim:** If the applier doesn't care about the value of the field any more, the applier can remove it from their local model of the resource, and make a new request with that particular field omitted. This leaves the value unchanged, and causes the field to be removed from the applier's entry in managedFields.
- **Don't overwrite value, become shared manager:** If the applier still cares about the value of a field, but doesn't want to overwrite it, they can change the value of that field in their local model of the resource so as to match the value of the object on the server, and

then make a new request that takes into account that local update. Doing so leaves the value unchanged, and causes that field's management to be shared by the applier along with all other field managers that already claimed to manage it.

## Field managers

Managers identify distinct workflows that are modifying the object (especially useful on conflicts!), and can be specified through the [fieldManager](#) query parameter as part of a modifying request. When you Apply to a resource, the `fieldManager` parameter is required. For other updates, the API server infers a field manager identity from the "User-Agent:" HTTP header (if present).

When you use the `kubectl` tool to perform a Server-Side Apply operation, `kubectl` sets the manager identity to "kubectl" by default.

## Serialization

At the protocol level, Kubernetes represents Server-Side Apply message bodies as [YAML](#), with the media type `application/apply-patch+yaml`.

### Note:

Whether you are submitting JSON data or YAML data, use `application/apply-patch+yaml` as the Content-Type header value.

All JSON documents are valid YAML. However, Kubernetes has a bug where it uses a YAML parser that does not fully implement the YAML specification. Some JSON escapes may not be recognized.

The serialization is the same as for Kubernetes objects, with the exception that clients are not required to send a complete object.

Here's an example of a Server-Side Apply message body (fully specified intent):

```
{
 "apiVersion": "v1",
 "kind": "ConfigMap"
}
```

(this would make a no-change update, provided that it was sent as the body of a **patch** request to a valid `v1/configmaps` resource, and with the appropriate request Content-Type).

## Operations in scope for field management

The Kubernetes API operations where field management is considered are:

1. Server-Side Apply (HTTP PATCH, with content type `application/apply-patch+yaml`)
2. Replacing an existing object (**update** to Kubernetes; PUT at the HTTP level)

Both operations update `.metadata.managedFields`, but behave a little differently.

Unless you specify a forced override, an apply operation that encounters field-level conflicts always fails; by contrast, if you make a change using **update** that would affect a managed field, a conflict never provokes failure of the operation.

All Server-Side Apply **patch** requests are required to identify themselves by providing a `fieldManager` query parameter, while the query parameter is optional for **update** operations. Finally, when using the Apply operation you cannot define `managedFields` in the body of the request that you submit.

An example object with multiple managers could look like this:

```

apiVersion: v1
kind: ConfigMap
metadata:
 name: test-cm
 namespace: default
 labels:
 test-label: test
 managedFields:
 - manager: kubectl
 operation: Apply
 apiVersion: v1
 fields:
 f:metadata:
 f:labels:
 f:test-label: {}
 - manager: kube-controller-manager
 operation: Update
 apiVersion: v1
 time: '2019-03-30T16:00:00.000Z'
 fields:
 f:data:
 f:key: {}
data:
 key: new value
```

In this example, a second operation was run as an **update** by the manager called `kube-controller-manager`. The update request succeeded and changed a value in the `data` field, which caused that field's management to change to the `kube-controller-manager`.

If this update has instead been attempted using Server-Side Apply, the request would have failed due to conflicting ownership.

## Merge strategy

The merging strategy, implemented with Server-Side Apply, provides a generally more stable object lifecycle. Server-Side Apply tries to merge fields based on the actor who manages them instead of overruling based on values. This way multiple actors can update the same object without causing unexpected interference.

When a user sends a *fully-specified intent* object to the Server-Side Apply endpoint, the server merges it with the live object favoring the value from the request body if it is specified in both

places. If the set of items present in the applied config is not a superset of the items applied by the same user last time, each missing item not managed by any other appliers is removed. For more information about how an object's schema is used to make decisions when merging, see [sigs.k8s.io/structured-merge-diff](https://sigs.k8s.io/structured-merge-diff).

The Kubernetes API (and the Go code that implements that API for Kubernetes) allows defining *merge strategy markers*. These markers describe the merge strategy supported for fields within Kubernetes objects. For a [CustomResourceDefinition](#), you can set these markers when you define the custom resource.

| Golang marker | OpenAPI extension          | Possible values                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|----------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| //+listType   | x-kubernetes-list-type     | atomic/set/map                                 | Applicable to lists. set applies to lists that include only scalar elements. These elements must be unique. map applies to lists of nested types only. The key values (see listMapKey) must be unique in the list. atomic can apply to any list. If configured as atomic, the entire list is replaced during merge. At any point in time, a single manager owns the list. If set or map, different managers can manage entries separately. |
| //+listMapKey | x-kubernetes-list-map-keys | List of field names, e.g. ["port", "protocol"] | Only applicable when +listType=map. A list of field names whose values uniquely identify entries in the list. While there can be multiple keys, listMapKey is singular because keys need to be specified individually in the Go type. The key fields must be scalars.                                                                                                                                                                      |
| //+mapType    | x-kubernetes-map-type      | atomic/granular                                | Applicable to maps. atomic means that the map can only be entirely replaced by a single manager. granular means that the map supports separate managers updating individual fields.                                                                                                                                                                                                                                                        |
| //+structType | x-kubernetes-map-type      | atomic/granular                                | Applicable to structs; otherwise same usage and OpenAPI annotation as //+mapType.                                                                                                                                                                                                                                                                                                                                                          |

If listType is missing, the API server interprets a patchStrategy=merge marker as a listType=map and the corresponding patchMergeKey marker as a listMapKey.

The atomic list type is recursive.

(In the [Go](#) code for Kubernetes, these markers are specified as comments and code authors need not repeat them as field tags).

## Custom resources and Server-Side Apply

By default, Server-Side Apply treats custom resources as unstructured data. All keys are treated the same as struct fields, and all lists are considered atomic.

If the CustomResourceDefinition defines a [schema](#) that contains annotations as defined in the previous [Merge Strategy](#) section, these annotations will be used when merging objects of this type.

## Compatibility across topology changes

On rare occurrences, the author for a CustomResourceDefinition (CRD) or built-in may want to change the specific topology of a field in their resource, without incrementing its API version. Changing the topology of types, by upgrading the cluster or updating the CRD, has different consequences when updating existing objects. There are two categories of changes: when a field goes from map/set/granular to atomic, and the other way around.

When the listType, mapType, or structType changes from map/set/granular to atomic, the whole list, map, or struct of existing objects will end-up being owned by actors who owned an element of these types. This means that any further change to these objects would cause a conflict.

When a listType, mapType, or structType changes from atomic to map/set/granular, the API server is unable to infer the new ownership of these fields. Because of that, no conflict will be produced when objects have these fields updated. For that reason, it is not recommended to change a type from atomic to map/set/granular.

Take for example, the custom resource:

```

apiVersion: example.com/v1
kind: Foo
metadata:
 name: foo-sample
 managedFields:
 - manager: "manager-one"
 operation: Apply
 apiVersion: example.com/v1
 fields:
 f:spec:
 f:data: {}
spec:
 data:
 key1: val1
 key2: val2
```

Before spec.data gets changed from atomic to granular, manager-one owns the field spec.data, and all the fields within it (key1 and key2). When the CRD gets changed to make spec.data granular, manager-one continues to own the top-level field spec.data (meaning no other managers can delete the map called data without a conflict), but it no longer owns key1 and key2, so another manager can then modify or delete those fields without conflict.

## Using Server-Side Apply in a controller

As a developer of a controller, you can use Server-Side Apply as a way to simplify the update logic of your controller. The main differences with a read-modify-write and/or patch are the following:

- the applied object must contain all the fields that the controller cares about.
- there is no way to remove fields that haven't been applied by the controller before (controller can still send a **patch** or **update** for these use-cases).

- the object doesn't have to be read beforehand; resourceVersion doesn't have to be specified.

It is strongly recommended for controllers to always force conflicts on objects that they own and manage, since they might not be able to resolve or act on these conflicts.

## Transferring ownership

In addition to the concurrency controls provided by [conflict resolution](#), Server-Side Apply provides ways to perform coordinated field ownership transfers from users to controllers.

This is best explained by example. Let's look at how to safely transfer ownership of the replicas field from a user to a controller while enabling automatic horizontal scaling for a Deployment, using the HorizontalPodAutoscaler resource and its accompanying controller.

Say a user has defined Deployment with replicas set to the desired value:

[application/ssa/nginx-deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
 labels:
 app: nginx
spec:
 replicas: 3
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
```

And the user has created the Deployment using Server-Side Apply, like so:

```
kubectl apply -f https://k8s.io/examples/application/ssa/nginx-deployment.yaml --server-side
```

Then later, automatic scaling is enabled for the Deployment; for example:

```
kubectl autoscale deployment nginx-deployment --cpu-percent=50 --min=1 --max=10
```

Now, the user would like to remove replicas from their configuration, so they don't accidentally fight with the HorizontalPodAutoscaler (HPA) and its controller. However, there is a race: it might take some time before the HPA feels the need to adjust .spec.replicas; if the user removes .spec.replicas before the HPA writes to the field and becomes its owner, then the API server would set .spec.replicas to 1 (the default replica count for Deployment). This is not what the user wants to happen, even temporarily - it might well degrade a running workload.



There are two solutions:

- (basic) Leave replicas in the configuration; when the HPA eventually writes to that field, the system gives the user a conflict over it. At that point, it is safe to remove from the configuration.
- (more advanced) If, however, the user doesn't want to wait, for example because they want to keep the cluster legible to their colleagues, then they can take the following steps to make it safe to remove replicas from their configuration:

First, the user defines a new manifest containing only the replicas field:

```
Save this file as 'nginx-deployment-replicas-only.yaml'.
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 replicas: 3
```

**Note:**

The YAML file for SSA in this case only contains the fields you want to change. You are not supposed to provide a fully compliant Deployment manifest if you only want to modify the spec.replicas field using SSA.

The user applies that manifest using a private field manager name. In this example, the user picked handover-to-hpa:

```
kubectl apply -f nginx-deployment-replicas-only.yaml \
--server-side --field-manager=handover-to-hpa \
--validate=false
```

If the apply results in a conflict with the HPA controller, then do nothing. The conflict indicates the controller has claimed the field earlier in the process than it sometimes does.

At this point the user may remove the replicas field from their manifest:

[application/ssa/nginx-deployment-no-replicas.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
labels:
 app: nginx
spec:
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
```

```
spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
```

Note that whenever the HPA controller sets the `replicas` field to a new value, the temporary field manager will no longer own any fields and will be automatically deleted. No further clean up is required.

## Transferring ownership between managers

Field managers can transfer ownership of a field between each other by setting the field to the same value in both of their applied configurations, causing them to share ownership of the field. Once the managers share ownership of the field, one of them can remove the field from their applied configuration to give up ownership and complete the transfer to the other field manager.

## Comparison with Client-Side Apply

Server-Side Apply is meant both as a replacement for the original client-side implementation of the `kubectl apply` subcommand, and as simple and effective mechanism for [controllers](#) to enact their changes.

Compared to the last-applied annotation managed by `kubectl`, Server-Side Apply uses a more declarative approach, which tracks an object's field management, rather than a user's last applied state. This means that as a side effect of using Server-Side Apply, information about which field manager manages each field in an object also becomes available.

A consequence of the conflict detection and resolution implemented by Server-Side Apply is that an applier always has up to date field values in their local state. If they don't, they get a conflict the next time they apply. Any of the three options to resolve conflicts results in the applied configuration being an up to date subset of the object on the server's fields.

This is different from Client-Side Apply, where outdated values which have been overwritten by other users are left in an applier's local config. These values only become accurate when the user updates that specific field, if ever, and an applier has no way of knowing whether their next apply will overwrite other users' changes.

Another difference is that an applier using Client-Side Apply is unable to change the API version they are using, but Server-Side Apply supports this use case.

## Migration between client-side and server-side apply

### Upgrading from client-side apply to server-side apply

Client-side apply users who manage a resource with `kubectl apply` can start using server-side apply with the following flag.

```
kubectl apply --server-side [--dry-run=server]
```

By default, field management of the object transfers from client-side apply to `kubectl server-side apply`, without encountering conflicts.

## Caution:

Keep the `last-applied-configuration` annotation up to date. The annotation infers client-side applies managed fields. Any fields not managed by client-side apply raise conflicts.

For example, if you used `kubectl scale` to update the `replicas` field after client-side apply, then this field is not owned by client-side apply and creates conflicts on `kubectl apply --server-side`.

This behavior applies to server-side apply with the `kubectl` field manager. As an exception, you can opt-out of this behavior by specifying a different, non-default field manager, as seen in the following example. The default field manager for `kubectl` server-side apply is `kubectl`.

```
kubectl apply --server-side --field-manager=my-manager [--dry-run=server]
```

## Downgrading from server-side apply to client-side apply

If you manage a resource with `kubectl apply --server-side`, you can downgrade to client-side apply directly with `kubectl apply`.

Downgrading works because `kubectl` Server-Side Apply keeps the `last-applied-configuration` annotation up-to-date if you use `kubectl apply`.

This behavior applies to Server-Side Apply with the `kubectl` field manager. As an exception, you can opt-out of this behavior by specifying a different, non-default field manager, as seen in the following example. The default field manager for `kubectl` server-side apply is `kubectl`.

```
kubectl apply --server-side --field-manager=my-manager [--dry-run=server]
```

## API implementation

The `PATCH` verb for a resource that supports Server-Side Apply can accept the unofficial `application/apply-patch+yaml` content type. Users of Server-Side Apply can send partially specified objects as YAML as the body of a `PATCH` request to the URI of a resource. When applying a configuration, you should always include all the fields that are important to the outcome (such as a desired state) that you want to define.

All JSON messages are valid YAML. Some clients specify Server-Side Apply requests using YAML request bodies that are also valid JSON.

## Access control and permissions

Since Server-Side Apply is a type of `PATCH`, a principal (such as a Role for Kubernetes [RBAC](#)) requires the **patch** permission to edit existing resources, and also needs the **create** verb permission in order to create new resources with Server-Side Apply.

## Clearing managedFields

It is possible to strip all `managedFields` from an object by overwriting them using a **patch** (JSON Merge Patch, Strategic Merge Patch, JSON Patch), or through an **update** (HTTP PUT); in other words, through every write operation other than **apply**. This can be done by overwriting the `managedFields` field with an empty entry. Two examples are:

```
PATCH /api/v1/namespaces/default/configmaps/example-cm
Accept: application/json
Content-Type: application/merge-patch+json
```

```
{
 "metadata": {
 "managedFields": [
 {}
]
 }
}
```

```
PATCH /api/v1/namespaces/default/configmaps/example-cm
Accept: application/json
Content-Type: application/json-patch+json
If-Match: 1234567890123456789
```

```
[{"op": "replace", "path": "/metadata/managedFields", "value": [{}]}]
```

This will overwrite the managedFields with a list containing a single empty entry that then results in the managedFields being stripped entirely from the object. Note that setting the managedFields to an empty list will not reset the field. This is on purpose, so managedFields never get stripped by clients not aware of the field.

In cases where the reset operation is combined with changes to other fields than the managedFields, this will result in the managedFields being reset first and the other changes being processed afterwards. As a result the applier takes ownership of any fields updated in the same request.

#### **Note:**

Server-Side Apply does not correctly track ownership on sub-resources that don't receive the resource object type. If you are using Server-Side Apply with such a sub-resource, the changed fields may not be tracked.

## **What's next**

You can read about managedFields within the Kubernetes API reference for the [metadata](#) top level field.

## **Client Libraries**

This page contains an overview of the client libraries for using the Kubernetes API from various programming languages.

To write applications using the [Kubernetes REST API](#), you do not need to implement the API calls and request/response types yourself. You can use a client library for the programming language you are using.

Client libraries often handle common tasks such as authentication for you. Most client libraries can discover and use the Kubernetes Service Account to authenticate if the API client is

running inside the Kubernetes cluster, or can understand the [kubeconfig file](#) format to read the credentials and the API Server address.

## Officially-supported Kubernetes client libraries

The following client libraries are officially maintained by [Kubernetes SIG API Machinery](#).

| Language   | Client Library                                                                                        | Sample Programs        |
|------------|-------------------------------------------------------------------------------------------------------|------------------------|
| C          | <a href="https://github.com/kubernetes-client/c">github.com/kubernetes-client/c</a>                   | <a href="#">browse</a> |
| dotnet     | <a href="https://github.com/kubernetes-client/csharp">github.com/kubernetes-client/csharp</a>         | <a href="#">browse</a> |
| Go         | <a href="https://github.com/kubernetes/client-go/">github.com/kubernetes/client-go/</a>               | <a href="#">browse</a> |
| Haskell    | <a href="https://github.com/kubernetes-client/haskell">github.com/kubernetes-client/haskell</a>       | <a href="#">browse</a> |
| Java       | <a href="https://github.com/kubernetes-client/java">github.com/kubernetes-client/java</a>             | <a href="#">browse</a> |
| JavaScript | <a href="https://github.com/kubernetes-client/javascript">github.com/kubernetes-client/javascript</a> | <a href="#">browse</a> |
| Perl       | <a href="https://github.com/kubernetes-client/perl/">github.com/kubernetes-client/perl/</a>           | <a href="#">browse</a> |
| Python     | <a href="https://github.com/kubernetes-client/python/">github.com/kubernetes-client/python/</a>       | <a href="#">browse</a> |
| Ruby       | <a href="https://github.com/kubernetes-client/ruby/">github.com/kubernetes-client/ruby/</a>           | <a href="#">browse</a> |

## Community-maintained client libraries

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

The following Kubernetes API client libraries are provided and maintained by their authors, not the Kubernetes team.

| Language             | Client Library                                                                                                      |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Clojure              | <a href="https://github.com/yanatan16/clj-kubernetes-api">github.com/yanatan16/clj-kubernetes-api</a>               |
| DotNet               | <a href="https://github.com/tonnyeremin/kubernetes_gen">github.com/tonnyeremin/kubernetes_gen</a>                   |
| DotNet (RestSharp)   | <a href="https://github.com/masroorhasan/Kubernetes.DotNet">github.com/masroorhasan/Kubernetes.DotNet</a>           |
| Elixir               | <a href="https://github.com/obmarg/kazan">github.com/obmarg/kazan</a>                                               |
| Elixir               | <a href="https://github.com/coryodaniel/k8s">github.com/coryodaniel/k8s</a>                                         |
| Java (OSGi)          | <a href="https://bitbucket.org/amdatulabs/amdatu-kubernetes">bitbucket.org/amdatulabs/amdatu-kubernetes</a>         |
| Java (Fabric8, OSGi) | <a href="https://github.com/fabric8io/kubernetes-client">github.com/fabric8io/kubernetes-client</a>                 |
| Java                 | <a href="https://github.com/manusa/yakc">github.com/manusa/yakc</a>                                                 |
| Lisp                 | <a href="https://github.com/brendandburns/cl-k8s">github.com/brendandburns/cl-k8s</a>                               |
| Lisp                 | <a href="https://github.com/xh4/cube">github.com/xh4/cube</a>                                                       |
| Node.js (TypeScript) | <a href="https://github.com/Goyoo/node-k8s-client">github.com/Goyoo/node-k8s-client</a>                             |
| Node.js              | <a href="https://github.com/ajpauwels/easy-k8s">github.com/ajpauwels/easy-k8s</a>                                   |
| Node.js              | <a href="https://github.com/godaddy/kubernetes-client">github.com/godaddy/kubernetes-client</a>                     |
| Node.js              | <a href="https://github.com/tenxcloud/node-kubernetes-client">github.com/tenxcloud/node-kubernetes-client</a>       |
| Perl                 | <a href="https://metacpan.org/pod/Net::Kubernetes">metacpan.org/pod/Net::Kubernetes</a>                             |
| PHP                  | <a href="https://github.com/allansun/kubernetes-php-client">github.com/allansun/kubernetes-php-client</a>           |
| PHP                  | <a href="https://github.com/maclof/kubernetes-client">github.com/maclof/kubernetes-client</a>                       |
| PHP                  | <a href="https://github.com/travisghansen/kubernetes-client-php">github.com/travisghansen/kubernetes-client-php</a> |

| Language | Client Library                                                                                    |
|----------|---------------------------------------------------------------------------------------------------|
| PHP      | <a href="https://github.com/renoki-co/php-k8s">github.com/renoki-co/php-k8s</a>                   |
| Python   | <a href="https://github.com/fiaas/k8s">github.com/fiaas/k8s</a>                                   |
| Python   | <a href="https://github.com/gtsystem/lightkube">github.com/gtsystem/lightkube</a>                 |
| Python   | <a href="https://github.com/kr8s-org/kr8s">github.com/kr8s-org/kr8s</a>                           |
| Python   | <a href="https://github.com/mnubo/kubernetes-py">github.com/mnubo/kubernetes-py</a>               |
| Python   | <a href="https://github.com/tomplus/kubernetes_asyncio">github.com/tomplus/kubernetes_asyncio</a> |
| Python   | <a href="https://github.com/Frankkkkk/pykorm">github.com/Frankkkkk/pykorm</a>                     |
| Ruby     | <a href="https://github.com/abonas/kubeclient">github.com/abonas/kubeclient</a>                   |
| Ruby     | <a href="https://github.com/k8s-ruby/k8s-ruby">github.com/k8s-ruby/k8s-ruby</a>                   |
| Ruby     | <a href="https://github.com/kontena/k8s-client">github.com/kontena/k8s-client</a>                 |
| Rust     | <a href="https://github.com/kube-rs/kube">github.com/kube-rs/kube</a>                             |
| Rust     | <a href="https://github.com/ynqa/kubernetes-rust">github.com/ynqa/kubernetes-rust</a>             |
| Scala    | <a href="https://github.com/hagay3/skuber">github.com/hagay3/skuber</a>                           |
| Scala    | <a href="https://github.com/hnaderi/scala-k8s">github.com/hnaderi/scala-k8s</a>                   |
| Scala    | <a href="https://github.com/joan38/kubernetes-client">github.com/joan38/kubernetes-client</a>     |
| Swift    | <a href="https://github.com/swiftkube/client">github.com/swiftkube/client</a>                     |

# Common Expression Language in Kubernetes

The [Common Expression Language \(CEL\)](#) is used in the Kubernetes API to declare validation rules, policy rules, and other constraints or conditions.

CEL expressions are evaluated directly in the [API server](#), making CEL a convenient alternative to out-of-process mechanisms, such as webhooks, for many extensibility use cases. Your CEL expressions continue to execute so long as the control plane's API server component remains available.

## Language overview

The [CEL language](#) has a straightforward syntax that is similar to the expressions in C, C++, Java, JavaScript and Go.

CEL was designed to be embedded into applications. Each CEL "program" is a single expression that evaluates to a single value. CEL expressions are typically short "one-liners" that inline well into the string fields of Kubernetes API resources.

Inputs to a CEL program are "variables". Each Kubernetes API field that contains CEL declares in the API documentation which variables are available to use for that field. For example, in the `x-kubernetes-validations[i].rules` field of CustomResourceDefinitions, the `self` and `oldSelf` variables are available and refer to the previous and current state of the custom resource data to be validated by the CEL expression. Other Kubernetes API fields may declare different variables. See the API documentation of the API fields to learn which variables are available for that field.

Example CEL expressions:

Examples of CEL expressions and the purpose of each

| Rule                                                                                                  | Purpose                                                                           |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <code>self.minReplicas &lt;= self.replicas &amp;&amp; self.replicas &lt;= self.maxReplicas</code>     | Validate that the three fields defining replicas are ordered appropriately        |
| <code>'Available' in self.stateCounts</code>                                                          | Validate that an entry with the 'Available' key exists in a map                   |
| <code>(self.list1.size() == 0) != (self.list2.size() == 0)</code>                                     | Validate that one of two lists is non-empty, but not both                         |
| <code>self.envvars.filter(e, e.name == 'MY_ENV').all(e, e.value.matches('[a-zA-Z]*\$'))</code>        | Validate the 'value' field of a listMap entry where key field 'name' is 'MY_ENV'  |
| <code>has(self.expired) &amp;&amp; self.created + self.ttl &lt; self.expired</code>                   | Validate that 'expired' date is after a 'create' date plus a 'ttl' duration       |
| <code>self.health.startsWith('ok')</code>                                                             | Validate a 'health' string field has the prefix 'ok'                              |
| <code>self.widgets.exists(w, w.key == 'x' &amp;&amp; w.foo &lt; 10)</code>                            | Validate that the 'foo' property of a listMap item with a key 'x' is less than 10 |
| <code>type(self) == string ? self == '99%' : self == 42</code>                                        | Validate an int-or-string field for both the int and string cases                 |
| <code>self.metadata.name == 'singleton'</code>                                                        | Validate that an object's name matches a specific value (making it a singleton)   |
| <code>self.set1.all(e, !(e in self.set2))</code>                                                      | Validate that two listSets are disjoint                                           |
| <code>self.names.size() == self.details.size() &amp;&amp; self.names.all(n, n in self.details)</code> | Validate the 'details' map is keyed by the items in the 'names' listSet           |
| <code>self.details.all(key, key.matches('[a-zA-Z]*\$'))</code>                                        | Validate the keys of the 'details' map                                            |
| <code>self.details.all(key, self.details[key].matches('[a-zA-Z]*\$'))</code>                          | Validate the values of the 'details' map                                          |

## CEL options, language features, and libraries

CEL is configured with the following options, libraries and language features, introduced at the specified Kubernetes versions:

| CEL option, library or language feature        | Included                                                                                                                     |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Standard macros</a>                | <code>has</code> , <code>all</code> , <code>exists</code> , <code>exists_one</code> , <code>map</code> , <code>filter</code> |
| <a href="#">Standard functions</a>             | See <a href="#">official list of standard definitions</a>                                                                    |
| <a href="#">Homogeneous Aggregate Literals</a> |                                                                                                                              |
| <a href="#">Default UTC Time Zone</a>          |                                                                                                                              |
| <a href="#">Eagerly Validate Declarations</a>  |                                                                                                                              |



| CEL option, library or language feature              | Included                                                                                    |
|------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <a href="#">extended strings library</a> , Version 1 | charAt, indexOf, lastIndexOf, lowerAscii, upperAscii, replace, split, join, substring, trim |
| Kubernetes list library                              | See <a href="#">Kubernetes list library</a>                                                 |
| Kubernetes regex library                             | See <a href="#">Kubernetes regex library</a>                                                |
| Kubernetes URL library                               | See <a href="#">Kubernetes URL library</a>                                                  |
| Kubernetes authorizer library                        | See <a href="#">Kubernetes authorizer library</a>                                           |
| Kubernetes quantity library                          | See <a href="#">Kubernetes quantity library</a>                                             |
| CEL optional types                                   | See <a href="#">CEL optional types</a>                                                      |
| CEL CrossTypeNumericComparisons                      | See <a href="#">CEL CrossTypeNumericComparisons</a>                                         |

CEL functions, features and language settings support Kubernetes control plane rollbacks. For example, *CEL Optional Values* was introduced at Kubernetes 1.29 and so only API servers at that version or newer will accept write requests to CEL expressions that use *CEL Optional Values*. However, when a cluster is rolled back to Kubernetes 1.28 CEL expressions using "CEL Optional Values" that are already stored in API resources will continue to evaluate correctly.

## Kubernetes CEL libraries

In addition to the CEL community libraries, Kubernetes includes CEL libraries that are available everywhere CEL is used in Kubernetes.

### Kubernetes list library

The list library includes `indexOf` and `lastIndexOf`, which work similar to the strings functions of the same names. These functions either the first or last positional index of the provided element in the list.

The list library also includes `min`, `max` and `sum`. Sum is supported on all number types as well as the duration type. Min and max are supported on all comparable types.

`isSorted` is also provided as a convenience function and is supported on all comparable types.

Examples:



Examples of CEL expressions using list library functions

| CEL Expression                                                                                   | Purpose                                                   |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <code>names.isSorted()</code>                                                                    | Verify that a list of names is kept in alphabetical order |
| <code>items.map(x, x.weight).sum() == 1.0</code>                                                 | Verify that the "weights" of a list of objects sum to 1.0 |
| <code>lowPriorities.map(x, x.priority).max() &lt; highPriorities.map(x, x.priority).min()</code> | Verify that two sets of priorities do not overlap         |
| <code>names.indexOf('should-be-first') == 1</code>                                               | Require that the first name in a list if a specific value |

See the [Kubernetes List Library](#) godoc for more information.

## Kubernetes regex library

In addition to the `matches` function provided by the CEL standard library, the regex library provides `find` and `findAll`, enabling a much wider range of regex operations.

Examples:

Examples of CEL expressions using regex library functions

| CEL Expression                                                            | Purpose                                                  |
|---------------------------------------------------------------------------|----------------------------------------------------------|
| <code>"abc 123".find('[0-9]*')</code>                                     | Find the first number in a string                        |
| <code>"1, 2, 3, 4".findAll('[0-9]*').map(x, int(x)).sum() &lt; 100</code> | Verify that the numbers in a string sum to less than 100 |

See the [Kubernetes regex library](#) godoc for more information.

## Kubernetes URL library

To make it easier and safer to process URLs, the following functions have been added:

- `isURL(string)` checks if a string is a valid URL according to the [Go's net/url](#) package. The string must be an absolute URL.
- `url(string)` URL converts a string to a URL or results in an error if the string is not a valid URL.

Once parsed via the `url` function, the resulting URL object has `getScheme`, `getHost`, `getHostname`, `getPort`, `getEscapedPath` and `getQuery` accessors.

Examples:

Examples of CEL expressions using URL library functions

| CEL Expression                                                             | Purpose                                        |
|----------------------------------------------------------------------------|------------------------------------------------|
| <code>url('https://example.com:80/').getHost()</code>                      | Get the 'example.com:80' host part of the URL. |
| <code>url('https://example.com/path with spaces/').getEscapedPath()</code> | Returns '/path%20with%20spaces/'               |

See the [Kubernetes URL library](#) godoc for more information.

## Kubernetes authorizer library

For CEL expressions in the API where a variable of type `Authorizer` is available, the authorizer may be used to perform authorization checks for the principal (authenticated user) of the request.

API resource checks are performed as follows:

1. Specify the group and resource to check: `Authorizer.group(string).resource(string) ResourceCheck`
2. Optionally call any combination of the following builder functions to further narrow the authorization check. Note that these functions return the receiver type and can be chained:
  - `ResourceCheck.subresource(string) ResourceCheck`
  - `ResourceCheck.namespace(string) ResourceCheck`
  - `ResourceCheck.name(string) ResourceCheck`
1. Call `ResourceCheck.check(verb string) Decision` to perform the authorization check.
2. Call `allowed() bool` or `reason() string` to inspect the result of the authorization check.

Non-resource authorization performed are used as follows:

1. specify only a path: `Authorizer.path(string) PathCheck`
2. Call `PathCheck.check(httpVerb string) Decision` to perform the authorization check.
3. Call `allowed() bool` or `reason() string` to inspect the result of the authorization check.

To perform an authorization check for a service account:

- `Authorizer.serviceAccount(namespace string, name string) Authorizer`

Examples of CEL expressions using URL library functions

| CEL Expression                                                                                    | Purpose                                                                                                    |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>authorizer.group('').resource('pods').namespace('default').check('create').allowed()</code> | Returns true if the principal (user or service account) is allowed create pods in the 'default' namespace. |
| <code>authorizer.path('/healthz').check('get').allowed()</code>                                   | Checks if the principal (user or service account) is authorized to make HTTP GET requests to the /         |

| CEL Expression                                                                                                          | Purpose                                                            |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|                                                                                                                         | healthz API path.                                                  |
| <code>authorizer.serviceAccount('default', 'myserviceaccount').resource('deployments').check('delete').allowed()</code> | Checks if the service account is authorized to delete deployments. |

See the [Kubernetes Authz library](#) godoc for more information.

## Kubernetes quantity library

Kubernetes 1.28 adds support for manipulating quantity strings (ex 1.5G, 512k, 20Mi)

- `isQuantity(string)` checks if a string is a valid Quantity according to [Kubernetes' resource.Quantity](#).
- `quantity(string)` Quantity converts a string to a Quantity or results in an error if the string is not a valid quantity.

Once parsed via the quantity function, the resulting Quantity object has the following library of member functions:

Available member functions of a Quantity

| Member Function                              | CEL Return Value | Description                                                                                                                                                           |
|----------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>isInteger()</code>                     | bool             | returns true if and only if <code>asInteger</code> is safe to call without an error                                                                                   |
| <code>asInteger()</code>                     | int              | returns a representation of the current value as an int64 if possible or results in an error if conversion would result in overflow or loss of precision.             |
| <code>asApproximateFloat()</code>            | float            | returns a float64 representation of the quantity which may lose precision. If the value of the quantity is outside the range of a float64 +Inf/-Inf will be returned. |
| <code>sign()</code>                          | int              | Returns 1 if the quantity is positive, -1 if it is negative. 0 if it is zero                                                                                          |
| <code>add(&lt;Quantity&gt;)</code>           | Quantity         | Returns sum of two quantities                                                                                                                                         |
| <code>add(&lt;int&gt;)</code>                | Quantity         | Returns sum of quantity and an integer                                                                                                                                |
| <code>sub(&lt;Quantity&gt;)</code>           | Quantity         | Returns difference between two quantities                                                                                                                             |
| <code>sub(&lt;int&gt;)</code>                | Quantity         | Returns difference between a quantity and an integer                                                                                                                  |
| <code>isLessThan(&lt;Quantity&gt;)</code>    | bool             | Returns true if and only if the receiver is less than the operand                                                                                                     |
| <code>isGreaterThan(&lt;Quantity&gt;)</code> | bool             | Returns true if and only if the receiver is greater than the operand                                                                                                  |
| <code>compareTo(&lt;Quantity&gt;)</code>     | int              | Compares receiver to operand and returns 0 if they are equal, 1 if the receiver is greater, or -1 if the receiver is less than the operand                            |

Examples:

### Examples of CEL expressions using URL library functions

[illegible]

# Type checking

CEL is a gradually typed language.

Some Kubernetes API fields contain fully type checked CEL expressions. For example, [CustomResourceDefinitions Validation Rules](#) are fully type checked.

Some Kubernetes API fields contain partially type checked CEL expressions. A partially type checked expression is an expressions where some of the variables are statically typed but others are dynamically typed. For example, in the CEL expressions of [ValidatingAdmissionPolicies](#) the request variable is typed, but the object variable is dynamically typed. As a result, an expression containing request.name would fail type checking because the name field is not defined. However, object.name would pass type checking even when the name field is not defined for the resource kinds that object refers to, because object is dynamically typed.

The `has()` macro in CEL may be used in CEL expressions to check if a field of a dynamically typed variable is accessible before attempting to access the field's value. For example:

```
has(object.namex) ? object.namex == 'special' : request.name == 'special'
```

# Type system integration

Table showing the relationship between OpenAPIv3 types and CEL types

| OpenAPIv3 type                                     | CEL type                                                                                                                     |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 'object' with Properties                           | object / "message type" (type(<object>) evaluates to selfType<uniqueNumber>.path.to.object.from.self                         |
| 'object' with AdditionalProperties                 | map                                                                                                                          |
| 'object' with x-kubernetes-embedded-type           | object / "message type", 'apiVersion', 'kind', 'metadata.name' and 'metadata.generateName' are implicitly included in schema |
| 'object' with x-kubernetes-preserve-unknown-fields | object / "message type", unknown fields are NOT accessible in CEL expression                                                 |
| x-kubernetes-int-or-string                         | union of int or string, self.intOrString < 100    self.intOrString == '50%' evaluates to true for both 50 and "50%"          |
| 'array'                                            | list                                                                                                                         |
| 'array' with x-kubernetes-list-type=map            | list with map based Equality & unique key guarantees                                                                         |
| 'array' with x-kubernetes-list-type=set            | list with set based Equality & unique entry guarantees                                                                       |
| 'boolean'                                          | boolean                                                                                                                      |
| 'number' (all formats)                             | double                                                                                                                       |
| 'integer' (all formats)                            | int (64)                                                                                                                     |
| <i>no equivalent</i>                               | uint (64)                                                                                                                    |
| 'null'                                             | null_type                                                                                                                    |
| 'string'                                           | string                                                                                                                       |
| 'string' with format=byte (base64 encoded)         | bytes                                                                                                                        |
| 'string' with format=date                          | timestamp (google.protobuf.Timestamp)                                                                                        |
| 'string' with format=datetime                      | timestamp (google.protobuf.Timestamp)                                                                                        |
| 'string' with format=duration                      | duration (google.protobuf.Duration)                                                                                          |

Also see: [CEL types](#), [OpenAPI types](#), [Kubernetes Structural Schemas](#).

Equality comparison for arrays with x-kubernetes-list-type of set or map ignores element order. For example `[1, 2] == [2, 1]` if the arrays represent Kubernetes set values.

Concatenation on arrays with x-kubernetes-list-type use the semantics of the list type:

- set: `X + Y` performs a union where the array positions of all elements in X are preserved and non-intersecting elements in Y are appended, retaining their partial order.
- map: `X + Y` performs a merge where the array positions of all keys in X are preserved but the values are overwritten by values in Y when the key sets of X and Y intersect. Elements in Y with non-intersecting keys are appended, retaining their partial order.

# Escaping

Only Kubernetes resource property names of the form `[a-zA-Z_./][a-zA-Z0-9_./]*` are accessible from CEL. Accessible property names are escaped according to the following rules when accessed in the expression:

Table of CEL identifier escaping rules

| escape sequence              | property name equivalent             |
|------------------------------|--------------------------------------|
| <code>__underscores__</code> | <code>__</code>                      |
| <code>__dot__</code>         | <code>.</code>                       |
| <code>__dash__</code>        | <code>-</code>                       |
| <code>__slash__</code>       | <code>/</code>                       |
| <code>__{keyword}__</code>   | <a href="#">CEL RESERVED keyword</a> |

When you escape any of CEL's **RESERVED** keywords you need to match the exact property name use the underscore escaping (for example, `int` in the word `sprint` would not be escaped and nor would it need to be).

Examples on escaping:

Examples escaped CEL identifiers

| property name          | rule with escaped property name                 |
|------------------------|-------------------------------------------------|
| <code>namespace</code> | <code>self.__namespace__ &gt; 0</code>          |
| <code>x-prop</code>    | <code>self.x__dash__prop &gt; 0</code>          |
| <code>redact_d</code>  | <code>self.redact__underscores__d &gt; 0</code> |
| <code>string</code>    | <code>self.startsWith('kube')</code>            |

## Resource constraints

CEL is non-Turing complete and offers a variety of production safety controls to limit execution time. CEL's *resource constraint* features provide feedback to developers about expression complexity and help protect the API server from excessive resource consumption during evaluation. CEL's resource constraint features are used to prevent CEL evaluation from consuming excessive API server resources.

A key element of the resource constraint features is a *cost unit* that CEL defines as a way of tracking CPU utilization. Cost units are independent of system load and hardware. Cost units are also deterministic; for any given CEL expression and input data, evaluation of the expression by the CEL interpreter will always result in the same cost.

Many of CEL's core operations have fixed costs. The simplest operations, such as comparisons (e.g. `<`) have a cost of 1. Some have a higher fixed cost, for example list literal declarations have a fixed base cost of 40 cost units.

Calls to functions implemented in native code approximate cost based on the time complexity of the operation. For example: operations that use regular expressions, such as `match` and `find`, are estimated using an approximated cost of `length(regexString)*length(inputString)`. The approximated cost reflects the worst case time complexity of Go's RE2 implementation.

## Runtime cost budget

All CEL expressions evaluated by Kubernetes are constrained by a runtime cost budget. The runtime cost budget is an estimate of actual CPU utilization computed by incrementing a cost unit counter while interpreting a CEL expression. If the CEL interpreter executes too many instructions, the runtime cost budget will be exceeded, execution of the expressions will be halted, and an error will result.

Some Kubernetes resources define an additional runtime cost budget that bounds the execution of multiple expressions. If the sum total of the cost of expressions exceed the budget, execution of the expressions will be halted, and an error will result. For example the validation of a custom resource has a *per-validation* runtime cost budget for all [Validation Rules](#) evaluated to validate the custom resource.

## Estimated cost limits

For some Kubernetes resources, the API server may also check if worst case estimated running time of CEL expressions would be prohibitively expensive to execute. If so, the API server prevent the CEL expression from being written to API resources by rejecting create or update operations containing the CEL expression to the API resources. This feature offers a stronger assurance that CEL expressions written to the API resource will be evaluate at runtime without exceeding the runtime cost budget.

# Kubernetes Deprecation Policy

This document details the deprecation policy for various facets of the system.

Kubernetes is a large system with many components and many contributors. As with any such software, the feature set naturally evolves over time, and sometimes a feature may need to be removed. This could include an API, a flag, or even an entire feature. To avoid breaking existing users, Kubernetes follows a deprecation policy for aspects of the system that are slated to be removed.

## Deprecating parts of the API

Since Kubernetes is an API-driven system, the API has evolved over time to reflect the evolving understanding of the problem space. The Kubernetes API is actually a set of APIs, called "API groups", and each API group is independently versioned. [API versions](#) fall into 3 main tracks, each of which has different policies for deprecation:

| Example  | Track                            |
|----------|----------------------------------|
| v1       | GA (generally available, stable) |
| v1beta1  | Beta (pre-release)               |
| v1alpha1 | Alpha (experimental)             |

A given release of Kubernetes can support any number of API groups and any number of versions of each.

The following rules govern the deprecation of elements of the API. This includes:

- REST resources (aka API objects)
- Fields of REST resources
- Annotations on REST resources, including "beta" annotations but not including "alpha" annotations.
- Enumerated or constant values
- Component config structures

These rules are enforced between official releases, not between arbitrary commits to master or release branches.

**Rule #1: API elements may only be removed by incrementing the version of the API group.**

Once an API element has been added to an API group at a particular version, it can not be removed from that version or have its behavior significantly changed, regardless of track.

**Note:**

For historical reasons, there are 2 "monolithic" API groups - "core" (no group name) and "extensions". Resources will incrementally be moved from these legacy API groups into more domain-specific API groups.

**Rule #2: API objects must be able to round-trip between API versions in a given release without information loss, with the exception of whole REST resources that do not exist in some versions.**

For example, an object can be written as v1 and then read back as v2 and converted to v1, and the resulting v1 resource will be identical to the original. The representation in v2 might be different from v1, but the system knows how to convert between them in both directions. Additionally, any new field added in v2 must be able to round-trip to v1 and back, which means v1 might have to add an equivalent field or represent it as an annotation.

**Rule #3: An API version in a given track may not be deprecated in favor of a less stable API version.**

- GA API versions can replace beta and alpha API versions.
- Beta API versions can replace earlier beta and alpha API versions, but *may not* replace GA API versions.
- Alpha API versions can replace earlier alpha API versions, but *may not* replace GA or beta API versions.

**Rule #4a: API lifetime is determined by the API stability level**

- GA API versions may be marked as deprecated, but must not be removed within a major version of Kubernetes
- Beta API versions are deprecated no more than 9 months or 3 minor releases after introduction (whichever is longer), and are no longer served 9 months or 3 minor releases after deprecation (whichever is longer)
- Alpha API versions may be removed in any release without prior deprecation notice



This ensures beta API support covers the [maximum supported version skew of 2 releases](#), and that APIs don't stagnate on unstable beta versions, accumulating production usage that will be disrupted when support for the beta API ends.

**Note:**

There are no current plans for a major version revision of Kubernetes that removes GA APIs.

**Note:**

Until [#52185](#) is resolved, no API versions that have been persisted to storage may be removed. Serving REST endpoints for those versions may be disabled (subject to the deprecation timelines in this document), but the API server must remain capable of decoding/converting previously persisted data from storage.

**Rule #4b: The "preferred" API version and the "storage version" for a given group may not advance until after a release has been made that supports both the new version and the previous version**

Users must be able to upgrade to a new release of Kubernetes and then roll back to a previous release, without converting anything to the new API version or suffering breakages (unless they explicitly used features only available in the newer version). This is particularly evident in the stored representation of objects.

All of this is best illustrated by examples. Imagine a Kubernetes release, version X, which introduces a new API group. A new Kubernetes release is made every approximately 4 months (3 per year). The following table describes which API versions are supported in a series of subsequent releases.

| Release | API Versions                                   | Preferred/<br>Storage Version | Notes                                                                                              |
|---------|------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------|
| X       | v1alpha1                                       | v1alpha1                      |                                                                                                    |
| X+1     | v1alpha2                                       | v1alpha2                      | <ul style="list-style-type: none"><li>• v1alpha1 is removed, "action required" relnote</li></ul>   |
| X+2     | v1beta1                                        | v1beta1                       | <ul style="list-style-type: none"><li>• v1alpha2 is removed, "action required" relnote</li></ul>   |
| X+3     | v1beta2, v1beta1 (deprecated)                  | v1beta1                       | <ul style="list-style-type: none"><li>• v1beta1 is deprecated, "action required" relnote</li></ul> |
| X+4     | v1beta2, v1beta1 (deprecated)                  | v1beta2                       |                                                                                                    |
| X+5     | v1, v1beta1 (deprecated), v1beta2 (deprecated) | v1beta2                       | <ul style="list-style-type: none"><li>• v1beta2 is deprecated, "action required" relnote</li></ul> |
| X+6     | v1, v1beta2 (deprecated)                       | v1                            | <ul style="list-style-type: none"><li>• v1beta1 is removed, "action required" relnote</li></ul>    |

| Release | API Versions                                                    | Preferred/<br>Storage Version | Notes                                                                                                                                                                |
|---------|-----------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X+7     | v1, v1beta2 (deprecated)                                        | v1                            |                                                                                                                                                                      |
| X+8     | v2alpha1, v1                                                    | v1                            | <ul style="list-style-type: none"> <li>v1beta2 is removed, "action required" relnote</li> </ul>                                                                      |
| X+9     | v2alpha2, v1                                                    | v1                            | <ul style="list-style-type: none"> <li>v2alpha1 is removed, "action required" relnote</li> </ul>                                                                     |
| X+10    | v2beta1, v1                                                     | v1                            | <ul style="list-style-type: none"> <li>v2alpha2 is removed, "action required" relnote</li> </ul>                                                                     |
| X+11    | v2beta2, v2beta1 (deprecated), v1                               | v1                            | <ul style="list-style-type: none"> <li>v2beta1 is deprecated, "action required" relnote</li> </ul>                                                                   |
| X+12    | v2, v2beta2 (deprecated), v2beta1 (deprecated), v1 (deprecated) | v1                            | <ul style="list-style-type: none"> <li>v2beta2 is deprecated, "action required" relnote</li> <li>v1 is deprecated in favor of v2, but will not be removed</li> </ul> |
| X+13    | v2, v2beta1 (deprecated), v2beta2 (deprecated), v1 (deprecated) | v2                            |                                                                                                                                                                      |
| X+14    | v2, v2beta2 (deprecated), v1 (deprecated)                       | v2                            | <ul style="list-style-type: none"> <li>v2beta1 is removed, "action required" relnote</li> </ul>                                                                      |
| X+15    | v2, v1 (deprecated)                                             | v2                            | <ul style="list-style-type: none"> <li>v2beta2 is removed, "action required" relnote</li> </ul>                                                                      |

## REST resources (aka API objects)

Consider a hypothetical REST resource named Widget, which was present in API v1 in the above timeline, and which needs to be deprecated. We document and [announce](#) the deprecation in sync with release X+1. The Widget resource still exists in API version v1 (deprecated) but not in v2alpha1. The Widget resource continues to exist and function in releases up to and including X+8. Only in release X+9, when API v1 has aged out, does the Widget resource cease to exist, and the behavior get removed.

Starting in Kubernetes v1.19, making an API request to a deprecated REST API endpoint:

1. Returns a Warning header (as defined in [RFC7234, Section 5.5](#)) in the API response.
2. Adds a "k8s.io/deprecated":"true" annotation to the [audit event](#) recorded for the request.

- Sets an `apiserver_requested_deprecated_apis` gauge metric to 1 in the kube-apiserver process. The metric has labels for group, version, resource, subresource that can be joined to the `apiserver_request_total` metric, and a `removed_release` label that indicates the Kubernetes release in which the API will no longer be served. The following Prometheus query returns information about requests made to deprecated APIs which will be removed in v1.22:

```
apiserver_requested_deprecated_apis{removed_release="1.22"} * on(group,version,resource,subresource) group_right() apiserver_request_total
```

## Fields of REST resources

As with whole REST resources, an individual field which was present in API v1 must exist and function until API v1 is removed. Unlike whole resources, the v2 APIs may choose a different representation for the field, as long as it can be round-tripped. For example a v1 field named "magnitude" which was deprecated might be named "deprecatedMagnitude" in API v2. When v1 is eventually removed, the deprecated field can be removed from v2.

## Enumerated or constant values

As with whole REST resources and fields thereof, a constant value which was supported in API v1 must exist and function until API v1 is removed.

## Component config structures

Component configs are versioned and managed similar to REST resources.

## Future work

Over time, Kubernetes will introduce more fine-grained API versions, at which point these rules will be adjusted as needed.

## Deprecating a flag or CLI

The Kubernetes system is comprised of several different programs cooperating. Sometimes, a Kubernetes release might remove flags or CLI commands (collectively "CLI elements") in these programs. The individual programs naturally sort into two main groups - user-facing and admin-facing programs, which vary slightly in their deprecation policies. Unless a flag is explicitly prefixed or documented as "alpha" or "beta", it is considered GA.

CLI elements are effectively part of the API to the system, but since they are not versioned in the same way as the REST API, the rules for deprecation are as follows:

**Rule #5a: CLI elements of user-facing components (e.g. kubectl) must function after their announced deprecation for no less than:**

- **GA: 12 months or 2 releases (whichever is longer)**
- **Beta: 3 months or 1 release (whichever is longer)**
- **Alpha: 0 releases**

**Rule #5b: CLI elements of admin-facing components (e.g. kubelet) must function after their announced deprecation for no less than:**

- GA: 6 months or 1 release (whichever is longer)
- Beta: 3 months or 1 release (whichever is longer)
- Alpha: 0 releases

**Rule #5c: Command line interface (CLI) elements cannot be deprecated in favor of less stable CLI elements**

Similar to the Rule #3 for APIs, if an element of a command line interface is being replaced with an alternative implementation, such as by renaming an existing element, or by switching to use configuration sourced from a file instead of a command line argument, that recommended alternative must be of the same or higher stability level.

**Rule #6: Deprecated CLI elements must emit warnings (optionally disable) when used.**

## Deprecating a feature or behavior

Occasionally a Kubernetes release needs to deprecate some feature or behavior of the system that is not controlled by the API or CLI. In this case, the rules for deprecation are as follows:

**Rule #7: Deprecated behaviors must function for no less than 1 year after their announced deprecation.**

If the feature or behavior is being replaced with an alternative implementation that requires work to adopt the change, there should be an effort to simplify the transition whenever possible. If an alternative implementation is under Kubernetes organization control, the following rules apply:

**Rule #8: The feature or behavior must not be deprecated in favor of an alternative implementation that is less stable**

For example, a generally available feature cannot be deprecated in favor of a Beta replacement. The Kubernetes project does, however, encourage users to adopt and transitions to alternative implementations even before they reach the same maturity level. This is particularly important for exploring new use cases of a feature or getting an early feedback on the replacement.

Alternative implementations may sometimes be external tools or products, for example a feature may move from the kubelet to container runtime that is not under Kubernetes project control. In such cases, the rule cannot be applied, but there must be an effort to ensure that there is a transition path that does not compromise on components' maturity levels. In the example with container runtimes, the effort may involve trying to ensure that popular container runtimes have versions that offer the same level of stability while implementing that replacement behavior.

Deprecation rules for features and behaviors do not imply that all changes to the system are governed by this policy. These rules applies only to significant, user-visible behaviors which impact the correctness of applications running on Kubernetes or that impact the administration of Kubernetes clusters, and which are being removed entirely.

An exception to the above rule is *feature gates*. Feature gates are key=value pairs that allow for users to enable/disable experimental features.

Feature gates are intended to cover the development life cycle of a feature - they are not intended to be long-term APIs. As such, they are expected to be deprecated and removed after a feature becomes GA or is dropped.

As a feature moves through the stages, the associated feature gate evolves. The feature life cycle matched to its corresponding feature gate is:

- Alpha: the feature gate is disabled by default and can be enabled by the user.
- Beta: the feature gate is enabled by default and can be disabled by the user.
- GA: the feature gate is deprecated (see "[Deprecation](#)") and becomes non-operational.
- GA, deprecation window complete: the feature gate is removed and calls to it are no longer accepted.

## Deprecation

Features can be removed at any point in the life cycle prior to GA. When features are removed prior to GA, their associated feature gates are also deprecated.

When an invocation tries to disable a non-operational feature gate, the call fails in order to avoid unsupported scenarios that might otherwise run silently.

In some cases, removing pre-GA features requires considerable time. Feature gates can remain operational until their associated feature is fully removed, at which point the feature gate itself can be deprecated.

When removing a feature gate for a GA feature also requires considerable time, calls to feature gates may remain operational if the feature gate has no effect on the feature, and if the feature gate causes no errors.

Features intended to be disabled by users should include a mechanism for disabling the feature in the associated feature gate.

Versioning for feature gates is different from the previously discussed components, therefore the rules for deprecation are as follows:

**Rule #9: Feature gates must be deprecated when the corresponding feature they control transitions a lifecycle stage as follows. Feature gates must function for no less than:**

- Beta feature to GA: 6 months or 2 releases (whichever is longer)
- Beta feature to EOL: 3 months or 1 release (whichever is longer)
- Alpha feature to EOL: 0 releases

**Rule #10: Deprecated feature gates must respond with a warning when used. When a feature gate is deprecated it must be documented in both in the release notes and the corresponding CLI help. Both warnings and documentation must indicate whether a feature gate is non-operational.**

## Deprecating a metric

Each component of the Kubernetes control-plane exposes metrics (usually the /metrics endpoint), which are typically ingested by cluster administrators. Not all metrics are the same: some metrics are commonly used as SLIs or used to determine SLOs, these tend to have greater

import. Other metrics are more experimental in nature or are used primarily in the Kubernetes development process.

Accordingly, metrics fall under three stability classes (ALPHA, BETA, STABLE); this impacts removal of a metric during a Kubernetes release. These classes are determined by the perceived importance of the metric. The rules for deprecating and removing a metric are as follows:

**Rule #11a: Metrics, for the corresponding stability class, must function for no less than:**

- **STABLE: 4 releases or 12 months (whichever is longer)**
- **BETA: 2 releases or 8 months (whichever is longer)**
- **ALPHA: 0 releases**

**Rule #11b: Metrics, after their *announced deprecation*, must function for no less than:**

- **STABLE: 3 releases or 9 months (whichever is longer)**
- **BETA: 1 releases or 4 months (whichever is longer)**
- **ALPHA: 0 releases**

Deprecated metrics will have their description text prefixed with a deprecation notice string '(Deprecated from x.y)' and a warning log will be emitted during metric registration. Like their stable undeprecated counterparts, deprecated metrics will be automatically registered to the metrics endpoint and therefore visible.

On a subsequent release (when the metric's `deprecatedVersion` is equal to `current_kubernetes_version - 3`), a deprecated metric will become a *hidden* metric. **Unlike** their deprecated counterparts, hidden metrics will *no longer* be automatically registered to the metrics endpoint (hence hidden). However, they can be explicitly enabled through a command line flag on the binary (`--show-hidden-metrics-for-version=`). This provides cluster admins an escape hatch to properly migrate off of a deprecated metric, if they were not able to react to the earlier deprecation warnings. Hidden metrics should be deleted after one release.

## Exceptions

No policy can cover every possible situation. This policy is a living document, and will evolve over time. In practice, there will be situations that do not fit neatly into this policy, or for which this policy becomes a serious impediment. Such situations should be discussed with SIGs and project leaders to find the best solutions for those specific cases, always bearing in mind that Kubernetes is committed to being a stable system that, as much as possible, never breaks users. Exceptions will always be announced in all relevant release notes.

## Deprecated API Migration Guide

As the Kubernetes API evolves, APIs are periodically reorganized or upgraded. When APIs evolve, the old API is deprecated and eventually removed. This page contains information you need to know when migrating from deprecated API versions to newer and more stable API versions.

# Removed APIs by release

## v1.32

The **v1.32** release will stop serving the following deprecated API versions:

### Flow control resources

The **flowcontrol.apiserver.k8s.io/v1beta3** API version of FlowSchema and PriorityLevelConfiguration will no longer be served in v1.32.

- Migrate manifests and API clients to use the **flowcontrol.apiserver.k8s.io/v1** API version, available since v1.29.
- All existing persisted objects are accessible via the new API
- Notable changes in **flowcontrol.apiserver.k8s.io/v1**:
  - The PriorityLevelConfiguration spec.limited.nominalConcurrencyShares field only defaults to 30 when unspecified, and an explicit value of 0 is not changed to 30.

## v1.29

The **v1.29** release stopped serving the following deprecated API versions:

### Flow control resources

The **flowcontrol.apiserver.k8s.io/v1beta2** API version of FlowSchema and PriorityLevelConfiguration is no longer served as of v1.29.

- Migrate manifests and API clients to use the **flowcontrol.apiserver.k8s.io/v1** API version, available since v1.29, or the **flowcontrol.apiserver.k8s.io/v1beta3** API version, available since v1.26.
- All existing persisted objects are accessible via the new API
- Notable changes in **flowcontrol.apiserver.k8s.io/v1**:
  - The PriorityLevelConfiguration spec.limited.assuredConcurrencyShares field is renamed to spec.limited.nominalConcurrencyShares and only defaults to 30 when unspecified, and an explicit value of 0 is not changed to 30.
- Notable changes in **flowcontrol.apiserver.k8s.io/v1beta3**:
  - The PriorityLevelConfiguration spec.limited.assuredConcurrencyShares field is renamed to spec.limited.nominalConcurrencyShares

## v1.27

The **v1.27** release stopped serving the following deprecated API versions:

### CSIStorageCapacity

The **storage.k8s.io/v1beta1** API version of CSIStorageCapacity is no longer served as of v1.27.

- Migrate manifests and API clients to use the **storage.k8s.io/v1** API version, available since v1.24.
- All existing persisted objects are accessible via the new API
- No notable changes

## v1.26

The **v1.26** release stopped serving the following deprecated API versions:

### Flow control resources

The **flowcontrol.apiserver.k8s.io/v1beta1** API version of FlowSchema and PriorityLevelConfiguration is no longer served as of v1.26.

- Migrate manifests and API clients to use the **flowcontrol.apiserver.k8s.io/v1beta2** API version.
- All existing persisted objects are accessible via the new API
- No notable changes

### HorizontalPodAutoscaler

The **autoscaling/v2beta2** API version of HorizontalPodAutoscaler is no longer served as of v1.26.

- Migrate manifests and API clients to use the **autoscaling/v2** API version, available since v1.23.
- All existing persisted objects are accessible via the new API
- Notable changes:
  - targetAverageUtilization is replaced with target.averageUtilization and target.type: Utilization. See [Autoscaling on multiple metrics and custom metrics](#).

## v1.25

The **v1.25** release stopped serving the following deprecated API versions:

### CronJob

The **batch/v1beta1** API version of CronJob is no longer served as of v1.25.

- Migrate manifests and API clients to use the **batch/v1** API version, available since v1.21.
- All existing persisted objects are accessible via the new API
- No notable changes

### EndpointSlice

The **discovery.k8s.io/v1beta1** API version of EndpointSlice is no longer served as of v1.25.

- Migrate manifests and API clients to use the **discovery.k8s.io/v1** API version, available since v1.21.
- All existing persisted objects are accessible via the new API
- Notable changes in **discovery.k8s.io/v1**:
  - use per Endpoint nodeName field instead of deprecated topology["kubernetes.io/hostname"] field
  - use per Endpoint zone field instead of deprecated topology["topology.kubernetes.io/zone"] field
  - topology is replaced with the deprecatedTopology field which is not writable in v1



## Event

The **events.k8s.io/v1beta1** API version of Event is no longer served as of v1.25.

- Migrate manifests and API clients to use the **events.k8s.io/v1** API version, available since v1.19.
- All existing persisted objects are accessible via the new API
- Notable changes in **events.k8s.io/v1**:
  - type is limited to Normal and Warning
  - involvedObject is renamed to regarding
  - action, reason, reportingController, and reportingInstance are required when creating new **events.k8s.io/v1** Events
  - use eventTime instead of the deprecated firstTimestamp field (which is renamed to deprecatedFirstTimestamp and not permitted in new **events.k8s.io/v1** Events)
  - use series.lastObservedTime instead of the deprecated lastTimestamp field (which is renamed to deprecatedLastTimestamp and not permitted in new **events.k8s.io/v1** Events)
  - use series.count instead of the deprecated count field (which is renamed to deprecatedCount and not permitted in new **events.k8s.io/v1** Events)
  - use reportingController instead of the deprecated source.component field (which is renamed to deprecatedSource.component and not permitted in new **events.k8s.io/v1** Events)
  - use reportingInstance instead of the deprecated source.host field (which is renamed to deprecatedSource.host and not permitted in new **events.k8s.io/v1** Events)

## HorizontalPodAutoscaler

The **autoscaling/v2beta1** API version of HorizontalPodAutoscaler is no longer served as of v1.25.

- Migrate manifests and API clients to use the **autoscaling/v2** API version, available since v1.23.
- All existing persisted objects are accessible via the new API
- Notable changes:
  - targetAverageUtilization is replaced with target.averageUtilization and target.type: Utilization. See [Autoscaling on multiple metrics and custom metrics](#).

## PodDisruptionBudget

The **policy/v1beta1** API version of PodDisruptionBudget is no longer served as of v1.25.

- Migrate manifests and API clients to use the **policy/v1** API version, available since v1.21.
- All existing persisted objects are accessible via the new API
- Notable changes in **policy/v1**:
  - an empty spec.selector ({} ) written to a policy/v1 PodDisruptionBudget selects all pods in the namespace (in policy/v1beta1 an empty spec.selector selected no pods). An unset spec.selector selects no pods in either API version.

## PodSecurityPolicy

PodSecurityPolicy in the **policy/v1beta1** API version is no longer served as of v1.25, and the PodSecurityPolicy admission controller will be removed.

Migrate to [Pod Security Admission](#) or a [3rd party admission webhook](#). For a migration guide, see [Migrate from PodSecurityPolicy to the Built-In PodSecurity Admission Controller](#). For more information on the deprecation, see [PodSecurityPolicy Deprecation: Past, Present, and Future](#).

## RuntimeClass

RuntimeClass in the **node.k8s.io/v1beta1** API version is no longer served as of v1.25.

- Migrate manifests and API clients to use the **node.k8s.io/v1** API version, available since v1.20.
- All existing persisted objects are accessible via the new API
- No notable changes

## v1.22

The **v1.22** release stopped serving the following deprecated API versions:

### Webhook resources

The **admissionregistration.k8s.io/v1beta1** API version of MutatingWebhookConfiguration and ValidatingWebhookConfiguration is no longer served as of v1.22.

- Migrate manifests and API clients to use the **admissionregistration.k8s.io/v1** API version, available since v1.16.
- All existing persisted objects are accessible via the new APIs
- Notable changes:
  - `webhooks[*].failurePolicy` default changed from Ignore to Fail for v1
  - `webhooks[*].matchPolicy` default changed from Exact to Equivalent for v1
  - `webhooks[*].timeoutSeconds` default changed from 30s to 10s for v1
  - `webhooks[*].sideEffects` default value is removed, and the field made required, and only None and NoneOnDryRun are permitted for v1
  - `webhooks[*].admissionReviewVersions` default value is removed and the field made required for v1 (supported versions for AdmissionReview are v1 and v1beta1)
  - `webhooks[*].name` must be unique in the list for objects created via `admissionregistration.k8s.io/v1`

## CustomResourceDefinition

The **apiextensions.k8s.io/v1beta1** API version of CustomResourceDefinition is no longer served as of v1.22.

- Migrate manifests and API clients to use the **apiextensions.k8s.io/v1** API version, available since v1.16.
- All existing persisted objects are accessible via the new API
- Notable changes:
  - `spec.scope` is no longer defaulted to Namespaced and must be explicitly specified
  - `spec.version` is removed in v1; use `spec.versions` instead
  - `spec.validation` is removed in v1; use `spec.versions[*].schema` instead
  - `spec.subresources` is removed in v1; use `spec.versions[*].subresources` instead
  - `spec.additionalPrinterColumns` is removed in v1; use `spec.versions[*].additionalPrinterColumns` instead

- `spec.conversion.webhookClientConfig` is moved to `spec.conversion.webhook.clientConfig` in v1
- `spec.conversion.conversionReviewVersions` is moved to `spec.conversion.webhook.conversionReviewVersions` in v1
- `spec.versions[*].schema.openAPIV3Schema` is now required when creating v1 `CustomResourceDefinition` objects, and must be a [structural schema](#)
- `spec.preserveUnknownFields: true` is disallowed when creating v1 `CustomResourceDefinition` objects; it must be specified within schema definitions as `x-kubernetes-preserve-unknown-fields: true`
- In additionalPrinterColumns items, the JSONPath field was renamed to `jsonPath` in v1 (fixes [#66531](#))

## APIService

The **apiregistration.k8s.io/v1beta1** API version of APIService is no longer served as of v1.22.

- Migrate manifests and API clients to use the **apiregistration.k8s.io/v1** API version, available since v1.10.
- All existing persisted objects are accessible via the new API
- No notable changes

## TokenReview

The **authentication.k8s.io/v1beta1** API version of TokenReview is no longer served as of v1.22.

- Migrate manifests and API clients to use the **authentication.k8s.io/v1** API version, available since v1.6.
- No notable changes

## SubjectAccessReview resources

The **authorization.k8s.io/v1beta1** API version of LocalSubjectAccessReview, SelfSubjectAccessReview, SubjectAccessReview, and SelfSubjectRulesReview is no longer served as of v1.22.

- Migrate manifests and API clients to use the **authorization.k8s.io/v1** API version, available since v1.6.
- Notable changes:
  - `spec.group` was renamed to `spec.groups` in v1 (fixes [#32709](#))

## CertificateSigningRequest

The **certificates.k8s.io/v1beta1** API version of CertificateSigningRequest is no longer served as of v1.22.

- Migrate manifests and API clients to use the **certificates.k8s.io/v1** API version, available since v1.19.
- All existing persisted objects are accessible via the new API

- Notable changes in `certificates.k8s.io/v1`:
  - For API clients requesting certificates:
    - `spec.signerName` is now required (see [known Kubernetes signers](#)), and requests for `kubernetes.io/legacy-unknown` are not allowed to be created via the `certificates.k8s.io/v1` API
    - `spec.usages` is now required, may not contain duplicate values, and must only contain known usages
  - For API clients approving or signing certificates:
    - `status.conditions` may not contain duplicate types
    - `status.conditions[*].status` is now required
    - `status.certificate` must be PEM-encoded, and contain only CERTIFICATE blocks

## Lease

The **`coordination.k8s.io/v1beta1`** API version of Lease is no longer served as of v1.22.

- Migrate manifests and API clients to use the **`coordination.k8s.io/v1`** API version, available since v1.14.
- All existing persisted objects are accessible via the new API
- No notable changes

## Ingress

The **`extensions/v1beta1`** and **`networking.k8s.io/v1beta1`** API versions of Ingress is no longer served as of v1.22.

- Migrate manifests and API clients to use the **`networking.k8s.io/v1`** API version, available since v1.19.
- All existing persisted objects are accessible via the new API
- Notable changes:
  - `spec.backend` is renamed to `spec.defaultBackend`
  - The backend `serviceName` field is renamed to `service.name`
  - Numeric backend `servicePort` fields are renamed to `service.port.number`
  - String backend `servicePort` fields are renamed to `service.port.name`
  - `pathType` is now required for each specified path. Options are `Prefix`, `Exact`, and `ImplementationSpecific`. To match the undefined v1beta1 behavior, use `ImplementationSpecific`.

## IngressClass

The **`networking.k8s.io/v1beta1`** API version of IngressClass is no longer served as of v1.22.

- Migrate manifests and API clients to use the **`networking.k8s.io/v1`** API version, available since v1.19.
- All existing persisted objects are accessible via the new API
- No notable changes

## RBAC resources

The **rbac.authorization.k8s.io/v1beta1** API version of ClusterRole, ClusterRoleBinding, Role, and RoleBinding is no longer served as of v1.22.

- Migrate manifests and API clients to use the **rbac.authorization.k8s.io/v1** API version, available since v1.8.
- All existing persisted objects are accessible via the new APIs
- No notable changes

## PriorityClass

The **scheduling.k8s.io/v1beta1** API version of PriorityClass is no longer served as of v1.22.

- Migrate manifests and API clients to use the **scheduling.k8s.io/v1** API version, available since v1.14.
- All existing persisted objects are accessible via the new API
- No notable changes

## Storage resources

The **storage.k8s.io/v1beta1** API version of CSIDriver, CSINode, StorageClass, and VolumeAttachment is no longer served as of v1.22.

- Migrate manifests and API clients to use the **storage.k8s.io/v1** API version
  - CSIDriver is available in **storage.k8s.io/v1** since v1.19.
  - CSINode is available in **storage.k8s.io/v1** since v1.17
  - StorageClass is available in **storage.k8s.io/v1** since v1.6
  - VolumeAttachment is available in **storage.k8s.io/v1** v1.13
- All existing persisted objects are accessible via the new APIs
- No notable changes

## v1.16

The **v1.16** release stopped serving the following deprecated API versions:

### NetworkPolicy

The **extensions/v1beta1** API version of NetworkPolicy is no longer served as of v1.16.

- Migrate manifests and API clients to use the **networking.k8s.io/v1** API version, available since v1.8.
- All existing persisted objects are accessible via the new API

### DaemonSet

The **extensions/v1beta1** and **apps/v1beta2** API versions of DaemonSet are no longer served as of v1.16.

- Migrate manifests and API clients to use the **apps/v1** API version, available since v1.9.
- All existing persisted objects are accessible via the new API

- Notable changes:
  - `spec.templateGeneration` is removed
  - `spec.selector` is now required and immutable after creation; use the existing template labels as the selector for seamless upgrades
  - `spec.updateStrategy.type` now defaults to `RollingUpdate` (the default in `extensions/v1beta1` was `OnDelete`)

## Deployment

The **`extensions/v1beta1`**, **`apps/v1beta1`**, and **`apps/v1beta2`** API versions of `Deployment` are no longer served as of v1.16.

- Migrate manifests and API clients to use the **`apps/v1`** API version, available since v1.9.
- All existing persisted objects are accessible via the new API
- Notable changes:
  - `spec.rollbackTo` is removed
  - `spec.selector` is now required and immutable after creation; use the existing template labels as the selector for seamless upgrades
  - `spec.progressDeadlineSeconds` now defaults to 600 seconds (the default in `extensions/v1beta1` was no deadline)
  - `spec.revisionHistoryLimit` now defaults to 10 (the default in `apps/v1beta1` was 2, the default in `extensions/v1beta1` was to retain all)
  - `maxSurge` and `maxUnavailable` now default to 25% (the default in `extensions/v1beta1` was 1)

## StatefulSet

The **`apps/v1beta1`** and **`apps/v1beta2`** API versions of `StatefulSet` are no longer served as of v1.16.

- Migrate manifests and API clients to use the **`apps/v1`** API version, available since v1.9.
- All existing persisted objects are accessible via the new API
- Notable changes:
  - `spec.selector` is now required and immutable after creation; use the existing template labels as the selector for seamless upgrades
  - `spec.updateStrategy.type` now defaults to `RollingUpdate` (the default in `apps/v1beta1` was `OnDelete`)

## ReplicaSet

The **`extensions/v1beta1`**, **`apps/v1beta1`**, and **`apps/v1beta2`** API versions of `ReplicaSet` are no longer served as of v1.16.

- Migrate manifests and API clients to use the **`apps/v1`** API version, available since v1.9.
- All existing persisted objects are accessible via the new API
- Notable changes:
  - `spec.selector` is now required and immutable after creation; use the existing template labels as the selector for seamless upgrades

## PodSecurityPolicy

The **extensions/v1beta1** API version of PodSecurityPolicy is no longer served as of v1.16.

- Migrate manifests and API client to use the **policy/v1beta1** API version, available since v1.10.
- Note that the **policy/v1beta1** API version of PodSecurityPolicy will be removed in v1.25.

## What to do

### Test with deprecated APIs disabled

You can test your clusters by starting an API server with specific API versions disabled to simulate upcoming removals. Add the following flag to the API server startup arguments:

```
--runtime-config=<group>/<version>=false
```

For example:

```
--runtime-config=admissionregistration.k8s.io/v1beta1=false,apiextensions.k8s.io/v1beta1,...
```

### Locate use of deprecated APIs

Use [client warnings, metrics, and audit information available in 1.19+](#) to locate use of deprecated APIs.

### Migrate to non-deprecated APIs

- Update custom integrations and controllers to call the non-deprecated APIs
- Change YAML files to reference the non-deprecated APIs

You can use the `kubectl convert` command to automatically convert an existing object:

```
kubectl convert -f <file> --output-version <group>/<version>.
```

For example, to convert an older Deployment to `apps/v1`, you can run:

```
kubectl convert -f ./my-deployment.yaml --output-version apps/v1
```

This conversion may use non-ideal default values. To learn more about a specific resource, check the Kubernetes [API reference](#).

#### Note:

The `kubectl convert` tool is not installed by default, although in fact it once was part of `kubectl` itself. For more details, you can read the [deprecation and removal issue](#) for the built-in subcommand.

To learn how to set up `kubectl convert` on your computer, visit the page that is right for your operating system: [Linux](#), [macOS](#), or [Windows](#).

# Kubernetes API health endpoints

The Kubernetes [API server](#) provides API endpoints to indicate the current status of the API server. This page describes these API endpoints and explains how you can use them.

## API endpoints for health

The Kubernetes API server provides 3 API endpoints (healthz, livez and readyz) to indicate the current status of the API server. The healthz endpoint is deprecated (since Kubernetes v1.16), and you should use the more specific livez and readyz endpoints instead. The livez endpoint can be used with the `--livez-grace-period` [flag](#) to specify the startup duration. For a graceful shutdown you can specify the `--shutdown-delay-duration` [flag](#) with the `/readyz` endpoint. Machines that check the healthz/livez/readyz of the API server should rely on the HTTP status code. A status code 200 indicates the API server is healthy/live/ready, depending on the called endpoint. The more verbose options shown below are intended to be used by human operators to debug their cluster or understand the state of the API server.

The following examples will show how you can interact with the health API endpoints.

For all endpoints, you can use the `verbose` parameter to print out the checks and their status. This can be useful for a human operator to debug the current status of the API server, it is not intended to be consumed by a machine:

```
curl -k https://localhost:6443/livez?verbose
```

or from a remote host with authentication:

```
kubectl get --raw='/readyz?verbose'
```

The output will look like this:

```
[+]ping ok
[+]log ok
[+]etcd ok
[+]poststarthook/start-kube-apiserver-admission-initializer ok
[+]poststarthook/generic-apiserver-start-informers ok
[+]poststarthook/start-apiextensions-informers ok
[+]poststarthook/start-apiextensions-controllers ok
[+]poststarthook/crd-informer-synced ok
[+]poststarthook/bootstrap-controller ok
[+]poststarthook/rbac/bootstrap-roles ok
[+]poststarthook/scheduling/bootstrap-system-priority-classes ok
[+]poststarthook/start-cluster-authentication-info-controller ok
[+]poststarthook/start-kube-aggregator-informers ok
[+]poststarthook/apiservice-registration-controller ok
[+]poststarthook/apiservice-status-available-controller ok
[+]poststarthook/kube-apiserver-autoregistration ok
[+]autoregister-completion ok
[+]poststarthook/apiservice-openapi-controller ok
healthz check passed
```



The Kubernetes API server also supports to exclude specific checks. The query parameters can also be combined like in this example:

```
curl -k 'https://localhost:6443/readyz?verbose&exclude=etcd'
```

The output show that the etcd check is excluded:

```
[+]ping ok
[+]log ok
[+]etcd excluded: ok
[+]poststarthook/start-kube-apiserver-admission-initializer ok
[+]poststarthook/generic-apiserver-start-informers ok
[+]poststarthook/start-apiextensions-informers ok
[+]poststarthook/start-apiextensions-controllers ok
[+]poststarthook/crd-informer-synced ok
[+]poststarthook/bootstrap-controller ok
[+]poststarthook/rbac/bootstrap-roles ok
[+]poststarthook/scheduling/bootstrap-system-priority-classes ok
[+]poststarthook/start-cluster-authentication-info-controller ok
[+]poststarthook/start-kube-aggregator-informers ok
[+]poststarthook/apiservice-registration-controller ok
[+]poststarthook/apiservice-status-available-controller ok
[+]poststarthook/kube-apiserver-autoregistration ok
[+]autoregister-completion ok
[+]poststarthook/apiservice-openapi-controller ok
[+]shutdown ok
healthz check passed
```

## Individual health checks

FEATURE STATE: Kubernetes v1.30 [alpha]

Each individual health check exposes an HTTP endpoint and can be checked individually. The schema for the individual health checks is `/livez/<healthcheck-name>` or `/readyz/<healthcheck-name>`, where `livez` and `readyz` can be used to indicate if you want to check the liveness or the readiness of the API server, respectively. The `<healthcheck-name>` path can be discovered using the `verbose` flag from above and take the path between `[+]` and `ok`. These individual health checks should not be consumed by machines but can be helpful for a human operator to debug a system:

```
curl -k https://localhost:6443/livez/etcd
```

## API Access Control

For an introduction to how Kubernetes implements and controls API access, read [Controlling Access to the Kubernetes API](#).

Reference documentation:

- [Authenticating](#)
  - [Authenticating with Bootstrap Tokens](#)

- [Admission Controllers](#)
  - [Dynamic Admission Control](#)
- [Authorization](#)
  - [Role Based Access Control](#)
  - [Attribute Based Access Control](#)
  - [Node Authorization](#)
  - [Webhook Authorization](#)
- [Certificate Signing Requests](#)
  - including [CSR approval](#) and [certificate signing](#)
- Service accounts
  - [Developer guide](#)
  - [Administration](#)
- [Kubelet Authentication & Authorization](#)
  - including kubelet [TLS bootstrapping](#)

# Authenticating

This page provides an overview of authentication.

## Users in Kubernetes

All Kubernetes clusters have two categories of users: service accounts managed by Kubernetes, and normal users.

It is assumed that a cluster-independent service manages normal users in the following ways:

- an administrator distributing private keys
- a user store like Keystone or Google Accounts
- a file with a list of usernames and passwords

In this regard, *Kubernetes does not have objects which represent normal user accounts*. Normal users cannot be added to a cluster through an API call.

Even though a normal user cannot be added via an API call, any user that presents a valid certificate signed by the cluster's certificate authority (CA) is considered authenticated. In this configuration, Kubernetes determines the username from the common name field in the 'subject' of the cert (e.g., "/CN=bob"). From there, the role based access control (RBAC) sub-system would determine whether the user is authorized to perform a specific operation on a resource. For more details, refer to the normal users topic in [certificate request](#) for more details about this.

In contrast, service accounts are users managed by the Kubernetes API. They are bound to specific namespaces, and created automatically by the API server or manually through API calls. Service accounts are tied to a set of credentials stored as Secrets, which are mounted into pods allowing in-cluster processes to talk to the Kubernetes API.

API requests are tied to either a normal user or a service account, or are treated as [anonymous requests](#). This means every process inside or outside the cluster, from a human user typing kubectl on a workstation, to kubelets on nodes, to members of the control plane, must authenticate when making requests to the API server, or be treated as an anonymous user.

# Authentication strategies

Kubernetes uses client certificates, bearer tokens, or an authenticating proxy to authenticate API requests through authentication plugins. As HTTP requests are made to the API server, plugins attempt to associate the following attributes with the request:

- Username: a string which identifies the end user. Common values might be kube-admin or jane@example.com.
- UID: a string which identifies the end user and attempts to be more consistent and unique than username.
- Groups: a set of strings, each of which indicates the user's membership in a named logical collection of users. Common values might be system:masters or devops-team.
- Extra fields: a map of strings to list of strings which holds additional information authorizers may find useful.

All values are opaque to the authentication system and only hold significance when interpreted by an [authorizer](#).

You can enable multiple authentication methods at once. You should usually use at least two methods:

- service account tokens for service accounts
- at least one other method for user authentication.

When multiple authenticator modules are enabled, the first module to successfully authenticate the request short-circuits evaluation. The API server does not guarantee the order authenticators run in.

The `system:authenticated` group is included in the list of groups for all authenticated users.

Integrations with other authentication protocols (LDAP, SAML, Kerberos, alternate x509 schemes, etc) can be accomplished using an [authenticating proxy](#) or the [authentication webhook](#).

## X509 client certificates

Client certificate authentication is enabled by passing the `--client-ca-file=SOMEFILE` option to API server. The referenced file must contain one or more certificate authorities to use to validate client certificates presented to the API server. If a client certificate is presented and verified, the common name of the subject is used as the user name for the request. As of Kubernetes 1.4, client certificates can also indicate a user's group memberships using the certificate's organization fields. To include multiple group memberships for a user, include multiple organization fields in the certificate.

For example, using the openssl command line tool to generate a certificate signing request:

```
openssl req -new -key jbeda.pem -out jbeda-csr.pem -subj "/CN=jbeda/O=app1/O=app2"
```

This would create a CSR for the username "jbeda", belonging to two groups, "app1" and "app2".

See [Managing Certificates](#) for how to generate a client cert.

## Static token file

The API server reads bearer tokens from a file when given the `--token-auth-file=SOMEFILE` option on the command line. Currently, tokens last indefinitely, and the token list cannot be changed without restarting the API server.

The token file is a csv file with a minimum of 3 columns: token, user name, user uid, followed by optional group names.

### Note:

If you have more than one group the column must be double quoted e.g.

```
token,user,uid,"group1,group2,group3"
```

## Putting a bearer token in a request

When using bearer token authentication from an http client, the API server expects an Authorization header with a value of Bearer <token>. The bearer token must be a character sequence that can be put in an HTTP header value using no more than the encoding and quoting facilities of HTTP. For example: if the bearer token is 31ada4fd-adec-460c-809a-9e56ceb75269 then it would appear in an HTTP header as shown below.

```
Authorization: Bearer 31ada4fd-adec-460c-809a-9e56ceb75269
```

## Bootstrap tokens

FEATURE STATE: Kubernetes v1.18 [stable]

To allow for streamlined bootstrapping for new clusters, Kubernetes includes a dynamically-managed Bearer token type called a *Bootstrap Token*. These tokens are stored as Secrets in the kube-system namespace, where they can be dynamically managed and created. Controller Manager contains a TokenCleaner controller that deletes bootstrap tokens as they expire.

The tokens are of the form [a-z0-9]{6}.[a-z0-9]{16}. The first component is a Token ID and the second component is the Token Secret. You specify the token in an HTTP header as follows:

```
Authorization: Bearer 781292.db7bc3a58fc5f07e
```

You must enable the Bootstrap Token Authenticator with the `--enable-bootstrap-token-auth` flag on the API Server. You must enable the TokenCleaner controller via the `--controllers` flag on the Controller Manager. This is done with something like `--controllers=*,tokencleaner`. kubeadm will do this for you if you are using it to bootstrap a cluster.

The authenticator authenticates as `system:bootstrap:<Token ID>`. It is included in the `system:bootstrappers` group. The naming and groups are intentionally limited to discourage users from using these tokens past bootstrapping. The user names and group can be used (and are used by kubeadm) to craft the appropriate authorization policies to support bootstrapping a cluster.

Please see [Bootstrap Tokens](#) for in depth documentation on the Bootstrap Token authenticator and controllers along with how to manage these tokens with kubeadm.

## Service account tokens

A service account is an automatically enabled authenticator that uses signed bearer tokens to verify requests. The plugin takes two optional flags:

- `--service-account-key-file` File containing PEM-encoded x509 RSA or ECDSA private or public keys, used to verify ServiceAccount tokens. The specified file can contain multiple keys, and the flag can be specified multiple times with different files. If unspecified, `--tls-private-key-file` is used.
- `--service-account-lookup` If enabled, tokens which are deleted from the API will be revoked.

Service accounts are usually created automatically by the API server and associated with pods running in the cluster through the ServiceAccount [Admission Controller](#). Bearer tokens are mounted into pods at well-known locations, and allow in-cluster processes to talk to the API server. Accounts may be explicitly associated with pods using the `serviceAccountName` field of a PodSpec.

### Note:

`serviceAccountName` is usually omitted because this is done automatically.

```
apiVersion: apps/v1 # this apiVersion is relevant as of Kubernetes 1.9
kind: Deployment
metadata:
 name: nginx-deployment
 namespace: default
spec:
 replicas: 3
 template:
 metadata:
 # ...
 spec:
 serviceAccountName: bob-the-bot
 containers:
 - name: nginx
 image: nginx:1.14.2
```

Service account bearer tokens are perfectly valid to use outside the cluster and can be used to create identities for long standing jobs that wish to talk to the Kubernetes API. To manually create a service account, use the `kubectl create serviceaccount (NAME)` command. This creates a service account in the current namespace.

```
kubectl create serviceaccount jenkins
```

```
serviceaccount/jenkins created
```

Create an associated token:

```
kubectl create token jenkins
```

```
eyJhbGciOiJSUzI1NiIsImtp...
```

The created token is a signed JSON Web Token (JWT).

The signed JWT can be used as a bearer token to authenticate as the given service account. See [above](#) for how the token is included in a request. Normally these tokens are mounted into pods for in-cluster access to the API server, but can be used from outside the cluster as well.

Service accounts authenticate with the username `system:serviceaccount:(NAMESPACE):(SERVICEACCOUNT)`, and are assigned to the groups `system:serviceaccounts` and `system:serviceaccounts:(NAMESPACE)`.

### Warning:

Because service account tokens can also be stored in Secret API objects, any user with write access to Secrets can request a token, and any user with read access to those Secrets can authenticate as the service account. Be cautious when granting permissions to service accounts and read or write capabilities for Secrets.

## OpenID Connect Tokens

[OpenID Connect](#) is a flavor of OAuth2 supported by some OAuth2 providers, notably Microsoft Entra ID, Salesforce, and Google. The protocol's main extension of OAuth2 is an additional field returned with the access token called an [ID Token](#). This token is a JSON Web Token (JWT) with well known fields, such as a user's email, signed by the server.

To identify the user, the authenticator uses the `id_token` (not the `access_token`) from the OAuth2 [token response](#) as a bearer token. See [above](#) for how the token is included in a request.

```
sequenceDiagram
 participant user as User
 participant idp as Identity Provider
 participant kube as Kubectl
 participant api as API Server

 user->>idp: 1. Log in to IdP
 activate idp
 idp-->>user: 2. Provide access_token, id_token, and refresh_token
 deactivate idp
 activate user
 user->>kube: 3. Call Kubectl with --token being the id_token
 deactivate user
 activate kube
 kube->>api: 4. OR add tokens to .kube/config
 deactivate kube
 activate api
 api-->>kube: 5. Authorization: Bearer...
 deactivate api
 activate kube
 kube->>api: 6. Is JWT signature valid?
 deactivate kube
 activate api
 api-->>kube: 7. Has the JWT expired? (iat+exp)
 deactivate api
 activate kube
 kube->>api: 8. User authorized?
 deactivate kube
 activate api
 api-->>kube: 9. Authorized: Perform action and return result
 deactivate api
 activate kube
 kube-->>user: 10. Return result
 deactivate kube
 deactivate user
```

*JavaScript must be [enabled](#) to view this content*

1. Log in to your identity provider
2. Your identity provider will provide you with an `access_token`, `id_token` and a `refresh_token`
3. When using `kubectl`, use your `id_token` with the `--token` flag or add it directly to your `kubeconfig`
4. `kubectl` sends your `id_token` in a header called `Authorization` to the API server
5. The API server will make sure the JWT signature is valid
6. Check to make sure the `id_token` hasn't expired
  1. Perform claim and/or user validation if CEL expressions are configured with `AuthenticationConfiguration`.
7. Make sure the user is authorized
8. Once authorized the API server returns a response to `kubectl`
9. `kubectl` provides feedback to the user

Since all of the data needed to validate who you are is in the `id_token`, Kubernetes doesn't need to "phone home" to the identity provider. In a model where every request is stateless this provides a very scalable solution for authentication. It does offer a few challenges:

1. Kubernetes has no "web interface" to trigger the authentication process. There is no browser or interface to collect credentials which is why you need to authenticate to your identity provider first.
2. The `id_token` can't be revoked, it's like a certificate so it should be short-lived (only a few minutes) so it can be very annoying to have to get a new token every few minutes.
3. To authenticate to the Kubernetes dashboard, you must use the `kubectl proxy` command or a reverse proxy that injects the `id_token`.

## Configuring the API Server

### Using flags

To enable the plugin, configure the following flags on the API server:

| Parameter                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Example                                                                                                                                                                                    | Required |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <code>--oidc-issuer-url</code>      | URL of the provider that allows the API server to discover public signing keys. Only URLs that use the <code>https://</code> scheme are accepted. This is typically the provider's discovery URL, changed to have an empty path                                                                                                                                                                                                                                                                                         | If the issuer's OIDC discovery URL is <code>https://accounts.provider.example/.well-known/openid-configuration</code> , the value should be <code>https://accounts.provider.example</code> | Yes      |
| <code>--oidc-client-id</code>       | A client id that all tokens must be issued for.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | kubernetes                                                                                                                                                                                 | Yes      |
| <code>--oidc-username-claim</code>  | JWT claim to use as the user name. By default <code>sub</code> , which is expected to be a unique identifier of the end user. Admins can choose other claims, such as email or name, depending on their provider. However, claims other than email will be prefixed with the issuer URL to prevent naming clashes with other plugins.                                                                                                                                                                                   | <code>sub</code>                                                                                                                                                                           | No       |
| <code>--oidc-username-prefix</code> | Prefix prepended to username claims to prevent clashes with existing names (such as <code>system:users</code> ). For example, the value <code>oidc:</code> will create usernames like <code>oidc:jane.doe</code> . If this flag isn't provided and <code>--oidc-username-claim</code> is a value other than email the prefix defaults to <code>( Issuer URL )#</code> where <code>( Issuer URL )</code> is the value of <code>--oidc-issuer-url</code> . The value <code>-</code> can be used to disable all prefixing. | <code>oidc:</code>                                                                                                                                                                         | No       |

| Parameter                          | Description                                                                                                                                                                                                                          | Example                       | Required |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------|
| <code>--oidc-groups-claim</code>   | JWT claim to use as the user's group. If the claim is present it must be an array of strings.                                                                                                                                        | groups                        | No       |
| <code>--oidc-groups-prefix</code>  | Prefix prepended to group claims to prevent clashes with existing names (such as system: groups). For example, the value <code>oidc:</code> will create group names like <code>oidc:engineering</code> and <code>oidc:infra</code> . | oidc:                         | No       |
| <code>--oidc-required-claim</code> | A key=value pair that describes a required claim in the ID Token. If set, the claim is verified to be present in the ID Token with a matching value. Repeat this flag to specify multiple claims.                                    | claim=value                   | No       |
| <code>--oidc-ca-file</code>        | The path to the certificate for the CA that signed your identity provider's web certificate. Defaults to the host's root CAs.                                                                                                        | /etc/kubernetes/ssl/kc-ca.pem | No       |
| <code>--oidc-signing-algs</code>   | The signing algorithms accepted. Default is "RS256".                                                                                                                                                                                 | RS512                         | No       |

#### Authentication configuration from a file

FEATURE STATE: Kubernetes v1.30 [beta]

JWT Authenticator is an authenticator to authenticate Kubernetes users using JWT compliant tokens. The authenticator will attempt to parse a raw ID token, verify it's been signed by the configured issuer. The public key to verify the signature is discovered from the issuer's public endpoint using OIDC discovery.

The minimum valid JWT payload must contain the following claims:

```
{
 "iss": "https://example.com", // must match the issuer.url
 "aud": ["my-app"], // at least one of the entries in issuer.audiences must match the
 // "aud" claim in presented JWTs.
 "exp": 1234567890, // token expiration as Unix time (the number of seconds elapsed
 // since January 1, 1970 UTC)
 "<username-claim>": "user" // this is the username claim configured in the
 // claimMappings.username.claim or claimMappings.username.expression
}
```

The configuration file approach allows you to configure multiple JWT authenticators, each with a unique issuer.url and issuer.discoveryURL. The configuration file even allows you to specify [CEL](#) expressions to map claims to user attributes, and to validate claims and user information. The API server also automatically reloads the authenticators when the configuration file is modified. You can use `apiserver_authentication_config_controller_automatic_reload_last_timestamp_seconds` metric to monitor the last time the configuration was reloaded by the API server.



You must specify the path to the authentication configuration using the `--authentication-config` flag on the API server. If you want to use command line flags instead of the configuration file, those will continue to work as-is. To access the new capabilities like configuring multiple authenticators, setting multiple audiences for an issuer, switch to using the configuration file.

For Kubernetes v1.30, the structured authentication configuration file format is beta-level, and the mechanism for using that configuration is also beta. Provided you didn't specifically disable the StructuredAuthenticationConfiguration [feature gate](#) for your cluster, you can turn on structured authentication by specifying the `--authentication-config` command line argument to the kube-apiserver. An example of the structured authentication configuration file is shown below.

#### Note:

If you specify `--authentication-config` along with any of the `--oidc-*` command line arguments, this is a misconfiguration. In this situation, the API server reports an error and then immediately exits. If you want to switch to using structured authentication configuration, you have to remove the `--oidc-*` command line arguments, and use the configuration file instead.

```

#
CAUTION: this is an example configuration.
Do not use this for your own cluster!
#
apiVersion: apiserver.config.k8s.io/v1beta1
kind: AuthenticationConfiguration
list of authenticators to authenticate Kubernetes users using JWT compliant tokens.
the maximum number of allowed authenticators is 64.
jwt:
- issuer:
 # url must be unique across all authenticators.
 # url must not conflict with issuer configured in --service-account-issuer.
 url: https://example.com # Same as --oidc-issuer-url.
 # discoveryURL, if specified, overrides the URL used to fetch discovery
 # information instead of using "{url}/.well-known/openid-configuration".
 # The exact value specified is used, so "/.well-known/openid-configuration"
 # must be included in discoveryURL if needed.
 #
 # The "issuer" field in the fetched discovery information must match the "issuer.url" field
 # in the AuthenticationConfiguration and will be used to validate the "iss" claim in the
presented JWT.
 # This is for scenarios where the well-known and jwks endpoints are hosted at a different
 # location than the issuer (such as locally in the cluster).
 # discoveryURL must be different from url if specified and must be unique across all
authenticators.
 discoveryURL: https://discovery.example.com/.well-known/openid-configuration
 # PEM encoded CA certificates used to validate the connection when fetching
 # discovery information. If not set, the system verifier will be used.
 # Same value as the content of the file referenced by the --oidc-ca-file flag.
 certificateAuthority: <PEM encoded CA certificates>
 # audiences is the set of acceptable audiences the JWT must be issued to.
 # At least one of the entries must match the "aud" claim in presented JWTs.
 audiences:
```

- my-app # Same as --oidc-client-id.
- my-other-app
- # this is required to be set to "MatchAny" when multiple audiences are specified.
- audienceMatchPolicy: MatchAny
- # rules applied to validate token claims to authenticate users.
- claimValidationRules:
  - # Same as --oidc-required-claim key=value.
- claim: hd
  - requiredValue: example.com
  - # Instead of claim and requiredValue, you can use expression to validate the claim.
  - # expression is a CEL expression that evaluates to a boolean.
  - # all the expressions must evaluate to true for validation to succeed.
- expression: 'claims.hd == "example.com"'
  - # Message customizes the error message seen in the API server logs when the validation fails.
  - message: the hd claim must be set to example.com
- expression: 'claims.exp - claims.nbf <= 86400'
  - message: total token lifetime must not exceed 24 hours
- claimMappings:
  - # username represents an option for the username attribute.
  - # This is the only required attribute.
  - username:
    - # Same as --oidc-username-claim. Mutually exclusive with username.expression.
    - claim: "sub"
    - # Same as --oidc-username-prefix. Mutually exclusive with username.expression.
    - # if username.claim is set, username.prefix is required.
    - # Explicitly set it to "" if no prefix is desired.
    - prefix: ""
    - # Mutually exclusive with username.claim and username.prefix.
    - # expression is a CEL expression that evaluates to a string.
    - #
    - # 1. If username.expression uses 'claims.email', then 'claims.email\_verified' must be used in username.expression or extra[\*].valueExpression or claimValidationRules[\*].expression.
    - # An example claim validation rule expression that matches the validation automatically applied when username.claim is set to 'email' is 'claims.?email\_verified.orValue(true)'.
    - # 2. If the username asserted based on username.expression is the empty string, the
- authentication
  - # request will fail.
  - expression: 'claims.username + ":external-user"'
  - # groups represents an option for the groups attribute.
  - groups:
    - # Same as --oidc-groups-claim. Mutually exclusive with groups.expression.
    - claim: "sub"
    - # Same as --oidc-groups-prefix. Mutually exclusive with groups.expression.
    - # if groups.claim is set, groups.prefix is required.
    - # Explicitly set it to "" if no prefix is desired.
    - prefix: ""
    - # Mutually exclusive with groups.claim and groups.prefix.
    - # expression is a CEL expression that evaluates to a string or a list of strings.
    - expression: 'claims.roles.split(",")'
  - # uid represents an option for the uid attribute.
  - uid:
    - # Mutually exclusive with uid.expression.
    - claim: 'sub'

```

Mutually exclusive with uid.claim
expression is a CEL expression that evaluates to a string.
expression: 'claims.sub'
extra attributes to be added to the UserInfo object. Keys must be domain-prefix path and
must be unique.
extra:
- key: 'example.com/tenant'
 # valueExpression is a CEL expression that evaluates to a string or a list of strings.
 valueExpression: 'claims.tenant'
validation rules applied to the final user object.
userValidationRules:
expression is a CEL expression that evaluates to a boolean.
all the expressions must evaluate to true for the user to be valid.
- expression: "!user.username.startsWith('system:')"
Message customizes the error message seen in the API server logs when the validation fails.
message: 'username cannot used reserved system: prefix'
- expression: "user.groups.all(group, !group.startsWith('system:'))"
message: 'groups cannot used reserved system: prefix'

```

- Claim validation rule expression

`jwt.claimValidationRules[i].expression` represents the expression which will be evaluated by CEL. CEL expressions have access to the contents of the token payload, organized into claims CEL variable. `claims` is a map of claim names (as strings) to claim values (of any type).

- User validation rule expression

`jwt.userValidationRules[i].expression` represents the expression which will be evaluated by CEL. CEL expressions have access to the contents of `userInfo`, organized into user CEL variable. Refer to the [UserInfo](#) API documentation for the schema of user.

- Claim mapping expression

`jwt.claimMappings.username.expression`, `jwt.claimMappings.groups.expression`, `jwt.claimMappings.uid.expression` `jwt.claimMappings.extra[i].valueExpression` represents the expression which will be evaluated by CEL. CEL expressions have access to the contents of the token payload, organized into claims CEL variable. `claims` is a map of claim names (as strings) to claim values (of any type).

To learn more, see the [Documentation on CEL](#)

Here are examples of the `AuthenticationConfiguration` with different token payloads.

- [Valid token](#)
- [Fails claim validation](#)
- [Fails user validation](#)

```

apiVersion: apiserver.config.k8s.io/v1beta1
kind: AuthenticationConfiguration
jwt:
- issuer:
 url: https://example.com
 audiences:

```



```

 "user",
 "admin"
],
 "extra": {
 "example.com/tenant": "72f988bf-86f1-41af-91ab-2d7cd011db4a"
 }
}

```

apiVersion: apiserver.config.k8s.io/v1beta1

kind: AuthenticationConfiguration

jwt:

- issuer:

url: <https://example.com>

audiences:

- my-app

### claimValidationRules:

- expression: `'claims.hd == "example.com"'` # the token below does not have this claim, so validation will fail.

message: the hd claim must be set to example.com

claimMappings:

username:

```
expression: 'claims.username + ":external-user"'
```

groups:

expression: 'claims.roles.split(",")'

uid:

expression: 'claims.sub'

extra:

```
- key: 'example.com/tenant'
```

```
valueExpression: 'claims.tenant'
```

### userValidationRules:

- expression: `"!user.username.startsWith('system:')" # the expression will evaluate to true, so validation will succeed.`

```
message: 'username cannot used reserved system: prefix'
```

TOKEN=eyJhbGciOiJSUzI1NiIsImtpZCI6ImY3dF9tOEROWmFTQk1oWGw5QXZTWGhBUC04Y0JmZ0JVbFVpTG5oQkgxdXMlLCJ0eXAiOiJKV1QiZXhwLjoxNzAzMjMyOTQ5LCJpYXQiOiE3MDExMDcyMzMslmlzcyl6Imh0dHBzOi8vZXhhbXBsZS5jb20iLCJqdGkiOiI3YzMzNzk0MjgwNTU3M2NhYTJjMzBiODY4YWMMWY2U5MTBiY2UwMmRkY2JmZWJlOGMyM2I4YjVmMjdhdzdDYyODczLiwiYmJmLjoxNzAxMTA3MjMjZCJyb2xlcyl6InVzZXIsYWRTaW4iLCJzdWIiOiJhdXRoliwidGVuYW50LjoibnZmOTg4YmYtODZmMS00MWFMlTkxYWItMmQ3Y2QwMTFkyYjRhIiwidXNlcm5lbHBUiOiJmb28ifQ.TBWF2RkQHm4QQz85AYPcwLxSk-VLvQW-mNDHx7SEOSv9LVwcPYPuPajJpuqn9c\_gKq1R94QKSQ5F6UGHMILz8OfmPKmX\_00wpwwNVGeevJ79ieX2V-\_\_W56iNR5gj-i9nn6FYk5pwfVREB0l4HSlpTOmu80gbPWAXY5hlW0ztEc1JTTEEmefORHV2ge8e3jp1xGaFNy6LdJWabYuKiwd8D7Qga\_HxtKB-t0KRmnZLRs7rka\_SfqgOdSYektuxHLbiDKqhmrffGlQKXGVzUsuvfw7IGM5ZWnzgEMdzCI357obHeM3Trqpnp5WRjtB8Om7JgnCyMaJi-P3icD88iu1xnZA

---

where the token payload is:

```
{
 "aud": "kubernetes",
```

```
{
 "exp": 1703232949,
 "iat": 1701107233,
 "iss": "https://example.com",
 "jti": "7c337942807e73caa2c30c868ac0ce910bce02ddcbfebe8c23b8b5f27ad62873",
 "nbf": 1701107233,
 "roles": "user,admin",
 "sub": "auth",
 "tenant": "72f988bf-86f1-41af-91ab-2d7cd011db4a",
 "username": "foo"
}
```

The token with the above `AuthenticationConfiguration` will fail to authenticate because the `hd` claim is not set to `example.com`. The API server will return 401 Unauthorized error.

```
apiVersion: apiserver.config.k8s.io/v1beta1
kind: AuthenticationConfiguration
jwt:
- issuer:
 url: https://example.com
 audiences:
 - my-app
 claimValidationRules:
 - expression: 'claims.hd == "example.com"'
 message: the hd claim must be set to example.com
 claimMappings:
 username:
 expression: '"system:" + claims.username' # this will prefix the username with
"system:" and will fail user validation.
 groups:
 expression: 'claims.roles.split(",')
 uid:
 expression: 'claims.sub'
 extra:
 - key: 'example.com/tenant'
 valueExpression: 'claims.tenant'
 userValidationRules:
 - expression: '!user.username.startsWith('system:~)' # the username will be system:foo
and expression will evaluate to false, so validation will fail.
 message: 'username cannot used reserved system: prefix'
```

TOKEN=eyJhbGciOiJSUzI1NiIsImtpZCI6ImY3dF9tOEROWmFTQk1oWGW5QXZTWGhBUC04Y0JmZ0JVbFVpTG5oQkgxdXMiLCJ0eXAiOiJKV1QiQifQ.eyJhdWQiOiJrdWJlcm5ldGVzLiwiaXhwIjoxNzAzMjMyOTQ5LCJoZCI6ImV4YW1wbGUuY29tIiwiaWF0IjoxNzAxMTEzMjA5MTA5LCJpc3MiOiJodHRwczovL2V4YW1wbGUuY29tIiwianRpIjoieyJyViMDY1MjM3MmNmMjBlMzQ1YjZmZGZmY2RjMjE4MWY0YWZkNmYyNTlhYW10YjdlMzU4ODEyMzdKmjkyMjBiYyIsIm5iZil6MTcwMTExMzEwMSwicm9sZXMiOiJ1c2VyLGFKbWluIiwic3ViIjoieXV0aCIsInRlbmFudCI6IjcyZjk4OGJmLTg2ZjEtNDZhZi05MWFhLTIjkn2NkMDExZW50YSIsbnVzZXJuYW1lIjoieyJmZm9vIn0.fgPJBYLobo9jnbHreooBlvpgeEcSPWnKfX6dc0IvdLRBF0dCcgy91oCJeK\_aBk-8zh5AKUXoFTllnfLckPivMOJqMECA1YTrMUwt\_IVqwb116AqihfByUYIIqzmjvUbthtbpleHQm2ff0HbrUqa\_Q0uaYwgy8mD807h7sBcUMjNd215ff\_nFIHss-9zegH8GI1d9fiBf-g6zjkr1j987EP748khpqh9IxPjMJbsgg\_uH5x80YFuqgEWwq-aYJPQxxX6FatP96a2EAan7wfppGlPrT0HcBOvq5pCnudgcgfvgiojiLr\_7robQu4T1bis0W75VPEvwWtgFcLnvcOx0JWg

where the token payload is:

```
{
 "aud": "kubernetes",
 "exp": 1703232949,
 "hd": "example.com",
 "iat": 1701113101,
 "iss": "https://example.com",
 "jti": "b5b0652372cd20e345b6fdffcdc2181f4afd6f259aab4b7e35881237d29220bc",
 "nbf": 1701113101,
 "roles": "user,admin",
 "sub": "auth",
 "tenant": "72f988bf-86f1-41af-91ab-2d7cd011db4a",
 "username": "foo"
}
```

The token with the above AuthenticationConfiguration will produce the following UserInfo object:

```
{
 "username": "system:foo",
 "uid": "auth",
 "groups": [
 "user",
 "admin"
],
 "extra": {
 "example.com/tenant": "72f988bf-86f1-41af-91ab-2d7cd011db4a"
 }
}
```

which will fail user validation because the username starts with `system:`. The API server will return 401 Unauthorized error.

#### Limitations

1. Distributed claims do not work via [CEL](#) expressions.
2. Egress selector configuration is not supported for calls to `issuer.url` and `issuer.discoveryURL`.

Kubernetes does not provide an OpenID Connect Identity Provider. You can use an existing public OpenID Connect Identity Provider (such as Google, or [others](#)). Or, you can run your own Identity Provider, such as [dex](#), [Keycloak](#), CloudFoundry [UAA](#), or Tremolo Security's [OpenUnison](#).

For an identity provider to work with Kubernetes it must:

1. Support [OpenID connect discovery](#)

The public key to verify the signature is discovered from the issuer's public endpoint using OIDC discovery. If you're using the authentication configuration file, the identity provider doesn't need to publicly expose the discovery endpoint. You can host the

discovery endpoint at a different location than the issuer (such as locally in the cluster) and specify the issuer.discoveryURL in the configuration file.

2. Run in TLS with non-obsolete ciphers

3. Have a CA signed certificate (even if the CA is not a commercial CA or is self signed)

A note about requirement #3 above, requiring a CA signed certificate. If you deploy your own identity provider (as opposed to one of the cloud providers like Google or Microsoft) you **MUST** have your identity provider's web server certificate signed by a certificate with the CA flag set to TRUE, even if it is self signed. This is due to GoLang's TLS client implementation being very strict to the standards around certificate validation. If you don't have a CA handy, you can use the [gencert script](#) from the Dex team to create a simple CA and a signed certificate and key pair. Or you can use [this similar script](#) that generates SHA256 certs with a longer life and larger key size.

Setup instructions for specific systems:

- [UAA](#)
- [Dex](#)
- [OpenUnison](#)

## Using kubectl

### Option 1 - OIDC Authenticator

The first option is to use the kubectl oidc authenticator, which sets the id\_token as a bearer token for all requests and refreshes the token once it expires. After you've logged into your provider, use kubectl to add your id\_token, refresh\_token, client\_id, and client\_secret to configure the plugin.

Providers that don't return an id\_token as part of their refresh token response aren't supported by this plugin and should use "Option 2" below.

```
kubectl config set-credentials USER_NAME \
--auth-provider=oidc \
--auth-provider-arg=idp-issuer-url=(issuer url) \
--auth-provider-arg=client-id=(your client id) \
--auth-provider-arg=client-secret=(your client secret) \
--auth-provider-arg=refresh-token=(your refresh token) \
--auth-provider-arg=idp-certificate-authority=(path to your ca certificate) \
--auth-provider-arg=id-token=(your id_token)
```

As an example, running the below command after authenticating to your identity provider:

```
kubectl config set-credentials mmosley \
--auth-provider=oidc \
--auth-provider-arg=idp-issuer-url=https://oidcidp.tremolo.lan:8443/auth/idp/OidcIdP \
--auth-provider-arg=client-id=kubernetes \
--auth-provider-arg=client-secret=1db158f6-177d-4d9c-8a8b-d36869918ec5 \
--auth-provider-arg=refresh-token=q1bKLFOyUiosTfawzA93TzZIDzH2TNa2SMm0zEiPKT
UwME6BkEo6Sql5yUWVBSWpKUGphaWpxSVAfekBOZbBhaEW+VIFUeVRGcluyVF5JT4+haZ
mPsluFoFu5XkpXk5BXqHega4GAXlF+ma+vmYpFcHe5eZR+slBFpZKtQA= \

```



```
--auth-provider-arg=idp-certificate-authority=/root/ca.pem \
--auth-provider-arg=id-token=eyJraWQiOiJDTj1vaWRjaWRwLnRyZW1vbG8ubGFuLCBPVT1EZW1vLCBPPVRybWVvbG8gU2VjdXJpdHksIEw9QXJsaW5ndG9uLCBTVD1WaXJnaW5pYSwgQz1VUy1DTj1rdWJlLWNhLTEyMDIxNDc5MjEwMzYwNzMyMTUyIiwiaWF0IjoiUlMyNTYifQ.eyJpc3MiOiJodHRwczovL29pZG9pZHAudHJlbW9sby5sYW46ODQ0My9hdXRoL2lkC9PaWRjSWRQIiwiaXVkJjoia3ViZXJuZXRlcyIsImV4cCI6MTQ4MzU0OTUxMSwianRpIjoiaMm96US15TXdFcHV4WDIHZUhQdy1hZyIsImhhdCI6MTQ4MzU0OTQ1MSwibmJmIjoxNDgzNTQ5MzMxLCJzdWIiOiI0YWViMzdiYS1iNjQ1LTQ4ZmQtYWZlMC0xYTAxZWU0MWUyMTgifQ.w6p4J_6qQ1HzTG9nrEOrubxIMb9K5hzcMPxc9IxPx2K4xO9l-oFiUw93daH3m5pluP6K7eOE6txBuRVfEcpJSwlelsOsW8gb8VJcnzMS9EnZpeA0tW_p-mnkFc3VcfyXuhe5R3G7aa5d8uHv70yJ9Y3-UhjiN9EhpMdfPAoEB9fYKKkJRzF7utTTIPGrSaSU6d2pcpfYKaxIwePzEkT4DfcQthoZdy9ucNvvLoi1DIC-UocFD8HLs8LYKEqSxQvOcvnThbObJ9af71EwmuE21fO5KzMW20KtAeget1gnldOosPtz1G5EwvaQ401-RPQzPGMVBld0_zMCAwZttJ4knw
```

Which would produce the below configuration:

```
users:
- name: mmosley
 user:
 auth-provider:
 config:
 client-id: kubernetes
 client-secret: 1db158f6-177d-4d9c-8a8b-d36869918ec5
 id-token: eyJraWQiOiJDTj1vaWRjaWRwLnRyZW1vbG8ubGFuLCBPPVT1EZW1vLCBPPVRybWVvbG8gU2VjdXJpdHksIEw9QXJsaW5ndG9uLCBTVD1WaXJnaW5pYSwgQz1VUy1DTj1rdWJlLWNhLTEyMDIxNDc5MjEwMzYwNzMyMTUyIiwiaWF0IjoiUlMyNTYifQ.eyJpc3MiOiJodHRwczovL29pZG9pZHAudHJlbW9sby5sYW46ODQ0My9hdXRoL2lkC9PaWRjSWRQIiwiaXVkJjoia3ViZXJuZXRlcyIsImV4cCI6MTQ4MzU0OTUxMSwianRpIjoiaMm96US15TXdFcHV4WDIHZUhQdy1hZyIsImhhdCI6MTQ4MzU0OTQ1MSwibmJmIjoxNDgzNTQ5MzMxLCJzdWIiOiI0YWViMzdiYS1iNjQ1LTQ4ZmQtYWZlMC0xYTAxZWU0MWUyMTgifQ.w6p4J_6qQ1HzTG9nrEOrubxIMb9K5hzcMPxc9IxPx2K4xO9l-oFiUw93daH3m5pluP6K7eOE6txBuRVfEcpJSwlelsOsW8gb8VJcnzMS9EnZpeA0tW_p-mnkFc3VcfyXuhe5R3G7aa5d8uHv70yJ9Y3-UhjiN9EhpMdfPAoEB9fYKKkJRzF7utTTIPGrSaSU6d2pcpfYKaxIwePzEkT4DfcQthoZdy9ucNvvLoi1DIC-UocFD8HLs8LYKEqSxQvOcvnThbObJ9af71EwmuE21fO5KzMW20KtAeget1gnldOosPtz1G5EwvaQ401-RPQzPGMVBld0_zMCAwZttJ4knw
 idp-certificate-authority: /root/ca.pem
 idp-issuer-url: https://oidcidp.tremolo.lan:8443/auth/idp/OidcIdP
 refresh-token: q1bKLFOyUiosTfawzA93TzZIDzH2TNa2SMm0zEiPKTUwME6BkEo6Sql5yUWVBSWpKUGphaWpxSVAfekBOZbBhaEW+VIFUeVRGcluyVF5JT4+haZmPsluFoFu5XkpXk5BXq
 name: oidc
```

Once your `id_token` expires, `kubectl` will attempt to refresh your `id_token` using your `refresh_token` and `client_secret` storing the new values for the `refresh_token` and `id_token` in your `.kube/config`.

## Option 2 - Use the --token Option

The `kubectl` command lets you pass in a token using the `--token` option. Copy and paste the `id_token` into this option:

```
kubectl --token=eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJodHRwczovL21sYi50cmVtb2xvLmxhbjo4MDQzL2F1dGgvaWRwL29pZGMiLCJhdWQiOiJrdWJlcm5ldGVzliwiZXhwIjoxNDc0NTk2NjY5LCJqdGkiOiI2RDUzNXoxUEpFNjJlbnR3QXaWVYIm9RliwiaWF0IjoxNDc0NTk2MzY5LCJmYmYiOiE0NzQ1OTYyNDksInN1YiI6Im13aW5kdSIsInVzZXJfcm9sZSI6WyJ1c2VycyIsIm5ldy1uYW1lc3BhY2Utdmld2VyIl0sImVtYWlsIjoibXdpbmR1QG5vbW9yZWplZGkuY29tIn0.f2As579n9VNoaKzoF-dOQGmXkFKf1FMyNV0-va_B63jn-_n9LGSCca_6IVMP8pO-Zb4KvRqGyTP0r3HkHxYy5c81Anlh8ijarruczl-TK_yF5akjSTHFZD-0gRzlevBDiH8Q79NAr-ky0P4iIXS8lY9Vnjch5MF74Zx0c3alKJHJUnnpjIACByfF2SCaYzbWFMUNat-K1PaUk5-ujMBG7yYnr95xD-63n8CO8teGUAaEMx6zRjzfhnhbzX-ajwZLGwGUBT4WqjMs70-6a7_8gZmLZb2az1cZynkFRj2BaCkVT3A2RrjeEwZEtGXlMqKJ1_I2ulrOVsYx01_yD35-rw get nodes
```

## Webhook Token Authentication

Webhook authentication is a hook for verifying bearer tokens.

- `--authentication-token-webhook-config-file` a configuration file describing how to access the remote webhook service.
- `--authentication-token-webhook-cache-ttl` how long to cache authentication decisions. Defaults to two minutes.
- `--authentication-token-webhook-version` determines whether to use `authentication.k8s.io/v1beta1` or `authentication.k8s.io/v1` `TokenReview` objects to send/receive information from the webhook. Defaults to `v1beta1`.

The configuration file uses the [kubernetes config](#) file format. Within the file, `clusters` refers to the remote service and `users` refers to the API server webhook. An example would be:

```
Kubernetes API version
apiVersion: v1
kind of the API object
kind: Config
clusters refers to the remote service.
clusters:
- name: name-of-remote-authn-service
 cluster:
 certificate-authority: /path/to/ca.pem # CA for verifying the remote service.
 server: https://authn.example.com/authenticate # URL of remote service to query. 'https'
recommended for production.

users refers to the API server's webhook configuration.
users:
- name: name-of-api-server
 user:
 client-certificate: /path/to/cert.pem # cert for the webhook plugin to use
 client-key: /path/to/key.pem # key matching the cert

kubeconfig files require a context. Provide one for the API server.
current-context: webhook
```

```
contexts:
- context:
 cluster: name-of-remote-authn-service
 user: name-of-api-server
 name: webhook
```

When a client attempts to authenticate with the API server using a bearer token as discussed [above](#), the authentication webhook POSTs a JSON-serialized TokenReview object containing the token to the remote service.

Note that webhook API objects are subject to the same [versioning compatibility rules](#) as other Kubernetes API objects. Implementers should check the `apiVersion` field of the request to ensure correct deserialization, and **must** respond with a TokenReview object of the same version as the request.

- [authentication.k8s.io/v1](#)
- [authentication.k8s.io/v1beta1](#)

#### Note:

The Kubernetes API server defaults to sending `authentication.k8s.io/v1beta1` token reviews for backwards compatibility. To opt into receiving `authentication.k8s.io/v1` token reviews, the API server must be started with `--authentication-token-webhook-version=v1`.

```
{
 "apiVersion": "authentication.k8s.io/v1",
 "kind": "TokenReview",
 "spec": {
 # Opaque bearer token sent to the API server
 "token": "014fbff9a07c...",

 # Optional list of the audience identifiers for the server the token was presented to.
 # Audience-aware token authenticators (for example, OIDC token authenticators)
 # should verify the token was intended for at least one of the audiences in this list,
 # and return the intersection of this list and the valid audiences for the token in the response
 status.
 # This ensures the token is valid to authenticate to the server it was presented to.
 # If no audiences are provided, the token should be validated to authenticate to the
 Kubernetes API server.
 "audiences": ["https://myserver.example.com", "https://myserver.internal.example.com"]
 }
}
```

```
{
 "apiVersion": "authentication.k8s.io/v1beta1",
 "kind": "TokenReview",
 "spec": {
 # Opaque bearer token sent to the API server
 "token": "014fbff9a07c...",

 # Optional list of the audience identifiers for the server the token was presented to.
 # Audience-aware token authenticators (for example, OIDC token authenticators)
 # should verify the token was intended for at least one of the audiences in this list,
```

```

and return the intersection of this list and the valid audiences for the token in the response
status.
This ensures the token is valid to authenticate to the server it was presented to.
If no audiences are provided, the token should be validated to authenticate to the
Kubernetes API server.
"audiences": ["https://myserver.example.com", "https://myserver.internal.example.com"]
}
}

```

The remote service is expected to fill the status field of the request to indicate the success of the login. The response body's spec field is ignored and may be omitted. The remote service must return a response using the same TokenReview API version that it received. A successful validation of the bearer token would return:

- [authentication.k8s.io/v1](https://kubernetes.io/api/authentication.k8s.io/v1)
- [authentication.k8s.io/v1beta1](https://kubernetes.io/api/authentication.k8s.io/v1beta1)

```

{
 "apiVersion": "authentication.k8s.io/v1",
 "kind": "TokenReview",
 "status": {
 "authenticated": true,
 "user": {
 # Required
 "username": "janedoe@example.com",
 # Optional
 "uid": "42",
 # Optional group memberships
 "groups": ["developers", "qa"],
 # Optional additional information provided by the authenticator.
 # This should not contain confidential data, as it can be recorded in logs
 # or API objects, and is made available to admission webhooks.
 "extra": {
 "extrafield1": [
 "extravalue1",
 "extravalue2"
]
 }
 },
 # Optional list audience-aware token authenticators can return,
 # containing the audiences from the `spec.audiences` list for which the provided token was
 valid.
 # If this is omitted, the token is considered to be valid to authenticate to the Kubernetes API
 server.
 "audiences": ["https://myserver.example.com"]
 }
}

```

```

{
 "apiVersion": "authentication.k8s.io/v1beta1",
 "kind": "TokenReview",
 "status": {
 "authenticated": true,

```

```

"user": {
 # Required
 "username": "janedoe@example.com",
 # Optional
 "uid": "42",
 # Optional group memberships
 "groups": ["developers", "qa"],
 # Optional additional information provided by the authenticator.
 # This should not contain confidential data, as it can be recorded in logs
 # or API objects, and is made available to admission webhooks.
 "extra": {
 "extrafield1": [
 "extravalue1",
 "extravalue2"
]
 }
},
Optional list audience-aware token authenticators can return,
containing the audiences from the `spec.audiences` list for which the provided token was
valid.
If this is omitted, the token is considered to be valid to authenticate to the Kubernetes API
server.
"audiences": ["https://myserver.example.com"]
}
}

```

An unsuccessful request would return:

- [authentication.k8s.io/v1](https://kubernetes.io/api/authentication.k8s.io/v1)
- [authentication.k8s.io/v1beta1](https://kubernetes.io/api/authentication.k8s.io/v1beta1)

```

{
 "apiVersion": "authentication.k8s.io/v1",
 "kind": "TokenReview",
 "status": {
 "authenticated": false,
 # Optionally include details about why authentication failed.
 # If no error is provided, the API will return a generic Unauthorized message.
 # The error field is ignored when authenticated=true.
 "error": "Credentials are expired"
 }
}

```

```

{
 "apiVersion": "authentication.k8s.io/v1beta1",
 "kind": "TokenReview",
 "status": {
 "authenticated": false,
 # Optionally include details about why authentication failed.
 # If no error is provided, the API will return a generic Unauthorized message.
 # The error field is ignored when authenticated=true.
 "error": "Credentials are expired"
 }
}

```

```
}
}
```

## Authenticating Proxy

The API server can be configured to identify users from request header values, such as X-Remote-User. It is designed for use in combination with an authenticating proxy, which sets the request header value.

- `--requestheader-username-headers` Required, case-insensitive. Header names to check, in order, for the user identity. The first header containing a value is used as the username.
- `--requestheader-group-headers` 1.6+. Optional, case-insensitive. "X-Remote-Group" is suggested. Header names to check, in order, for the user's groups. All values in all specified headers are used as group names.
- `--requestheader-extra-headers-prefix` 1.6+. Optional, case-insensitive. "X-Remote-Extra-" is suggested. Header prefixes to look for to determine extra information about the user (typically used by the configured authorization plugin). Any headers beginning with any of the specified prefixes have the prefix removed. The remainder of the header name is lowercased and [percent-decoded](#) and becomes the extra key, and the header value is the extra value.

### Note:

Prior to 1.11.3 (and 1.10.7, 1.9.11), the extra key could only contain characters which were [legal in HTTP header labels](#).

For example, with this configuration:

```
--requestheader-username-headers=X-Remote-User
--requestheader-group-headers=X-Remote-Group
--requestheader-extra-headers-prefix=X-Remote-Extra-
```

this request:

```
GET / HTTP/1.1
X-Remote-User: fido
X-Remote-Group: dogs
X-Remote-Group: dachshunds
X-Remote-Extra-Acme.com%2Fproject: some-project
X-Remote-Extra-Scopes: openid
X-Remote-Extra-Scopes: profile
```

would result in this user info:

```
name: fido
groups:
- dogs
- dachshunds
extra:
 acme.com/project:
 - some-project
scopes:
```

- openid
- profile

In order to prevent header spoofing, the authenticating proxy is required to present a valid client certificate to the API server for validation against the specified CA before the request headers are checked. WARNING: do **not** reuse a CA that is used in a different context unless you understand the risks and the mechanisms to protect the CA's usage.

- `--requestheader-client-ca-file` Required. PEM-encoded certificate bundle. A valid client certificate must be presented and validated against the certificate authorities in the specified file before the request headers are checked for user names.
- `--requestheader-allowed-names` Optional. List of Common Name values (CNs). If set, a valid client certificate with a CN in the specified list must be presented before the request headers are checked for user names. If empty, any CN is allowed.

## Anonymous requests

When enabled, requests that are not rejected by other configured authentication methods are treated as anonymous requests, and given a username of `system:anonymous` and a group of `system:unauthenticated`.

For example, on a server with token authentication configured, and anonymous access enabled, a request providing an invalid bearer token would receive a 401 Unauthorized error. A request providing no bearer token would be treated as an anonymous request.

In 1.5.1-1.5.x, anonymous access is disabled by default, and can be enabled by passing the `--anonymous-auth=true` option to the API server.

In 1.6+, anonymous access is enabled by default if an authorization mode other than `AlwaysAllow` is used, and can be disabled by passing the `--anonymous-auth=false` option to the API server. Starting in 1.6, the ABAC and RBAC authorizers require explicit authorization of the `system:anonymous` user or the `system:unauthenticated` group, so legacy policy rules that grant access to the `*` user or `*` group do not include anonymous users.

## User impersonation

A user can act as another user through impersonation headers. These let requests manually override the user info a request authenticates as. For example, an admin could use this feature to debug an authorization policy by temporarily impersonating another user and seeing if a request was denied.

Impersonation requests first authenticate as the requesting user, then switch to the impersonated user info.

- A user makes an API call with their credentials *and* impersonation headers.
- API server authenticates the user.
- API server ensures the authenticated users have impersonation privileges.
- Request user info is replaced with impersonation values.
- Request is evaluated, authorization acts on impersonated user info.

The following HTTP headers can be used to performing an impersonation request:

- `Impersonate-User`: The username to act as.

- **Impersonate-Group:** A group name to act as. Can be provided multiple times to set multiple groups. Optional. Requires "Impersonate-User".
- **Impersonate-Extra-( extra name ):** A dynamic header used to associate extra fields with the user. Optional. Requires "Impersonate-User". In order to be preserved consistently, ( extra name ) must be lower-case, and any characters which aren't [legal in HTTP header labels](#) MUST be utf8 and [percent-encoded](#).
- **Impersonate-Uid:** A unique identifier that represents the user being impersonated. Optional. Requires "Impersonate-User". Kubernetes does not impose any format requirements on this string.

**Note:**

Prior to 1.11.3 (and 1.10.7, 1.9.11), ( extra name ) could only contain characters which were [legal in HTTP header labels](#).

**Note:**

Impersonate-Uid is only available in versions 1.22.0 and higher.

An example of the impersonation headers used when impersonating a user with groups:

```
Impersonate-User: jane.doe@example.com
Impersonate-Group: developers
Impersonate-Group: admins
```

An example of the impersonation headers used when impersonating a user with a UID and extra fields:

```
Impersonate-User: jane.doe@example.com
Impersonate-Extra-dn: cn=jane,ou=engineers,dc=example,dc=com
Impersonate-Extra-acme.com%2Fproject: some-project
Impersonate-Extra-scopes: view
Impersonate-Extra-scopes: development
Impersonate-Uid: 06f6ce97-e2c5-4ab8-7ba5-7654dd08d52b
```

When using `kubectl` set the `--as` flag to configure the `Impersonate-User` header, set the `--as-group` flag to configure the `Impersonate-Group` header.

```
kubectl drain mynode
```

Error from server (Forbidden): User "clark" cannot get nodes at the cluster scope. (get nodes mynode)

Set the `--as` and `--as-group` flag:

```
kubectl drain mynode --as=superman --as-group=system:masters
```

```
node/mynode cordoned
node/mynode drained
```

**Note:**

`kubectl` cannot impersonate extra fields or UIDs.



To impersonate a user, group, user identifier (UID) or extra fields, the impersonating user must have the ability to perform the "impersonate" verb on the kind of attribute being impersonated ("user", "group", "uid", etc.). For clusters that enable the RBAC authorization plugin, the following ClusterRole encompasses the rules needed to set user and group impersonation headers:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: impersonator
rules:
- apiGroups: [""]
 resources: ["users", "groups", "serviceaccounts"]
 verbs: ["impersonate"]
```

For impersonation, extra fields and impersonated UIDs are both under the "authentication.k8s.io" apiGroup. Extra fields are evaluated as sub-resources of the resource "userextras". To allow a user to use impersonation headers for the extra field "scopes" and for UIDs, a user should be granted the following role:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: scopes-and-uid-impersonator
rules:
Can set "Impersonate-Extra-scopes" header and the "Impersonate-Uid" header.
- apiGroups: ["authentication.k8s.io"]
 resources: ["userextras/scopes", "uids"]
 verbs: ["impersonate"]
```

The values of impersonation headers can also be restricted by limiting the set of resourceNames a resource can take.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: limited-impersonator
rules:
Can impersonate the user "jane.doe@example.com"
- apiGroups: [""]
 resources: ["users"]
 verbs: ["impersonate"]
 resourceNames: ["jane.doe@example.com"]

Can impersonate the groups "developers" and "admins"
- apiGroups: [""]
 resources: ["groups"]
 verbs: ["impersonate"]
 resourceNames: ["developers", "admins"]

Can impersonate the extras field "scopes" with the values "view" and "development"
- apiGroups: ["authentication.k8s.io"]
 resources: ["userextras/scopes"]
```

```
verbs: ["impersonate"]
resourceNames: ["view", "development"]

Can impersonate the uid "06f6ce97-e2c5-4ab8-7ba5-7654dd08d52b"
- apiGroups: ["authentication.k8s.io"]
 resources: ["uids"]
 verbs: ["impersonate"]
 resourceNames: ["06f6ce97-e2c5-4ab8-7ba5-7654dd08d52b"]
```

#### Note:

Impersonating a user or group allows you to perform any action as if you were that user or group; for that reason, impersonation is not namespace scoped. If you want to allow impersonation using Kubernetes RBAC, this requires using a ClusterRole and a ClusterRoleBinding, not a Role and RoleBinding.

## client-go credential plugins

FEATURE STATE: Kubernetes v1.22 [stable]

k8s.io/client-go and tools using it such as kubectl and kubelet are able to execute an external command to receive user credentials.

This feature is intended for client side integrations with authentication protocols not natively supported by k8s.io/client-go (LDAP, Kerberos, OAuth2, SAML, etc.). The plugin implements the protocol specific logic, then returns opaque credentials to use. Almost all credential plugin use cases require a server side component with support for the [webhook token authenticator](#) to interpret the credential format produced by the client plugin.

#### Note:

Earlier versions of kubectl included built-in support for authenticating to AKS and GKE, but this is no longer present.

### Example use case

In a hypothetical use case, an organization would run an external service that exchanges LDAP credentials for user specific, signed tokens. The service would also be capable of responding to [webhook token authenticator](#) requests to validate the tokens. Users would be required to install a credential plugin on their workstation.

To authenticate against the API:

- The user issues a kubectl command.
- Credential plugin prompts the user for LDAP credentials, exchanges credentials with external service for a token.
- Credential plugin returns token to client-go, which uses it as a bearer token against the API server.
- API server uses the [webhook token authenticator](#) to submit a TokenReview to the external service.
- External service verifies the signature on the token and returns the user's username and groups.

## Configuration

Credential plugins are configured through [kubectl config files](#) as part of the user fields.

- [client.authentication.k8s.io/v1](#)
- [client.authentication.k8s.io/v1beta1](#)

```
apiVersion: v1
kind: Config
users:
- name: my-user
 user:
 exec:
 # Command to execute. Required.
 command: "example-client-go-exec-plugin"

 # API version to use when decoding the ExecCredentials resource. Required.
 #
 # The API version returned by the plugin MUST match the version listed here.
 #
 # To integrate with tools that support multiple versions (such as client.authentication.k8s.io/
v1beta1),

set an environment variable, pass an argument to the tool that indicates which version the
exec plugin expects,
or read the version from the ExecCredential object in the KUBERNETES_EXEC_INFO
environment variable.
 apiVersion: "client.authentication.k8s.io/v1"

 # Environment variables to set when executing the plugin. Optional.
 env:
 - name: "FOO"
 value: "bar"

 # Arguments to pass when executing the plugin. Optional.
 args:
 - "arg1"
 - "arg2"

 # Text shown to the user when the executable doesn't seem to be present. Optional.
 installHint: |
 example-client-go-exec-plugin is required to authenticate
 to the current cluster. It can be installed:

 On macOS: brew install example-client-go-exec-plugin

 On Ubuntu: apt-get install example-client-go-exec-plugin

 On Fedora: dnf install example-client-go-exec-plugin

 ...

 # Whether or not to provide cluster information, which could potentially contain
```

```

very large CA data, to this exec plugin as a part of the KUBERNETES_EXEC_INFO
environment variable.
provideClusterInfo: true

The contract between the exec plugin and the standard input I/O stream. If the
contract cannot be satisfied, this plugin will not be run and an error will be
returned. Valid values are "Never" (this exec plugin never uses standard input),
"IfAvailable" (this exec plugin wants to use standard input if it is available),
or "Always" (this exec plugin requires standard input to function). Required.
interactiveMode: Never

clusters:
- name: my-cluster
 cluster:
 server: "https://172.17.4.100:6443"
 certificate-authority: "/etc/kubernetes/ca.pem"
 extensions:
 - name: client.authentication.k8s.io/exec # reserved extension name for per cluster exec
 config
 extension:
 arbitrary: config
 this: can be provided via the KUBERNETES_EXEC_INFO environment variable upon
 setting provideClusterInfo
 you: ["can", "put", "anything", "here"]
 contexts:
 - name: my-cluster
 context:
 cluster: my-cluster
 user: my-user
 current-context: my-cluster

```

```

apiVersion: v1
kind: Config
users:
- name: my-user
 user:
 exec:
 # Command to execute. Required.
 command: "example-client-go-exec-plugin"

 # API version to use when decoding the ExecCredentials resource. Required.
 #
 # The API version returned by the plugin MUST match the version listed here.
 #
 # To integrate with tools that support multiple versions (such as client.authentication.k8s.io/
v1),

set an environment variable, pass an argument to the tool that indicates which version the
exec plugin expects,
or read the version from the ExecCredential object in the KUBERNETES_EXEC_INFO
environment variable.
 apiVersion: "client.authentication.k8s.io/v1beta1"

 # Environment variables to set when executing the plugin. Optional.

```

```

env:
- name: "FOO"
 value: "bar"

Arguments to pass when executing the plugin. Optional.
args:
- "arg1"
- "arg2"

Text shown to the user when the executable doesn't seem to be present. Optional.
installHint: |
 example-client-go-exec-plugin is required to authenticate
 to the current cluster. It can be installed:

 On macOS: brew install example-client-go-exec-plugin

 On Ubuntu: apt-get install example-client-go-exec-plugin

 On Fedora: dnf install example-client-go-exec-plugin

 ...

Whether or not to provide cluster information, which could potentially contain
very large CA data, to this exec plugin as a part of the KUBERNETES_EXEC_INFO
environment variable.
provideClusterInfo: true

The contract between the exec plugin and the standard input I/O stream. If the
contract cannot be satisfied, this plugin will not be run and an error will be
returned. Valid values are "Never" (this exec plugin never uses standard input),
"IfAvailable" (this exec plugin wants to use standard input if it is available),
or "Always" (this exec plugin requires standard input to function). Optional.
Defaults to "IfAvailable".
interactiveMode: Never

clusters:
- name: my-cluster
 cluster:
 server: "https://172.17.4.100:6443"
 certificate-authority: "/etc/kubernetes/ca.pem"
 extensions:
 - name: client.authentication.k8s.io/exec # reserved extension name for per cluster exec
 config
 extension:
 arbitrary: config
 this: can be provided via the KUBERNETES_EXEC_INFO environment variable upon
 setting provideClusterInfo
 you: ["can", "put", "anything", "here"]
 contexts:
 - name: my-cluster
 context:
 cluster: my-cluster
 user: my-user
 current-context: my-cluster

```

Relative command paths are interpreted as relative to the directory of the config file. If KUBECONFIG is set to /home/jane/kubeconfig and the exec command is ./bin/example-client-go-exec-plugin, the binary /home/jane/bin/example-client-go-exec-plugin is executed.

```
- name: my-user
 user:
 exec:
 # Path relative to the directory of the kubeconfig
 command: "./bin/example-client-go-exec-plugin"
 apiVersion: "client.authentication.k8s.io/v1"
 interactiveMode: Never
```

## Input and output formats

The executed command prints an ExecCredential object to stdout. k8s.io/client-go authenticates against the Kubernetes API using the returned credentials in the status. The executed command is passed an ExecCredential object as input via the KUBERNETES\_EXEC\_INFO environment variable. This input contains helpful information like the expected API version of the returned ExecCredential object and whether or not the plugin can use stdin to interact with the user.

When run from an interactive session (i.e., a terminal), stdin can be exposed directly to the plugin. Plugins should use the spec.interactive field of the input ExecCredential object from the KUBERNETES\_EXEC\_INFO environment variable in order to determine if stdin has been provided. A plugin's stdin requirements (i.e., whether stdin is optional, strictly required, or never used in order for the plugin to run successfully) is declared via the user.exec.interactiveMode field in the [kubeconfig](#) (see table below for valid values). The user.exec.interactiveMode field is optional in client.authentication.k8s.io/v1beta1 and required in client.authentication.k8s.io/v1.

interactiveMode values

| interactiveMode Value | Meaning                                                                                                                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Never                 | This exec plugin never needs to use standard input, and therefore the exec plugin will be run regardless of whether standard input is available for user input.                                                                                                                                                              |
| IfAvailable           | This exec plugin would like to use standard input if it is available, but can still operate if standard input is not available. Therefore, the exec plugin will be run regardless of whether stdin is available for user input. If standard input is available for user input, then it will be provided to this exec plugin. |
| Always                | This exec plugin requires standard input in order to run, and therefore the exec plugin will only be run if standard input is available for user input. If standard input is not available for user input, then the exec plugin will not be run and an error will be returned by the exec plugin runner.                     |

To use bearer token credentials, the plugin returns a token in the status of the [ExecCredential](#)

- [client.authentication.k8s.io/v1](#)
- [client.authentication.k8s.io/v1beta1](#)

```
{
 "apiVersion": "client.authentication.k8s.io/v1",
 "kind": "ExecCredential",
```

```

"status": {
 "token": "my-bearer-token"
}
}

```

```

{
 "apiVersion": "client.authentication.k8s.io/v1beta1",
 "kind": "ExecCredential",
 "status": {
 "token": "my-bearer-token"
 }
}

```

Alternatively, a PEM-encoded client certificate and key can be returned to use TLS client auth. If the plugin returns a different certificate and key on a subsequent call, k8s.io/client-go will close existing connections with the server to force a new TLS handshake.

If specified, clientKeyData and clientCertificateData must both be present.

clientCertificateData may contain additional intermediate certificates to send to the server.

- [client.authentication.k8s.io/v1](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.17/#execcredential-v1-clientauthentication-k8s-io)
- [client.authentication.k8s.io/v1beta1](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.17/#execcredential-v1beta1-clientauthentication-k8s-io)

```

{
 "apiVersion": "client.authentication.k8s.io/v1",
 "kind": "ExecCredential",
 "status": {
 "clientCertificateData": "-----BEGIN CERTIFICATE-----\n...\n-----END CERTIFICATE-----",
 "clientKeyData": "-----BEGIN RSA PRIVATE KEY-----\n...\n-----END RSA PRIVATE KEY-----"
 }
}

```

```

{
 "apiVersion": "client.authentication.k8s.io/v1beta1",
 "kind": "ExecCredential",
 "status": {
 "clientCertificateData": "-----BEGIN CERTIFICATE-----\n...\n-----END CERTIFICATE-----",
 "clientKeyData": "-----BEGIN RSA PRIVATE KEY-----\n...\n-----END RSA PRIVATE KEY-----"
 }
}

```

Optionally, the response can include the expiry of the credential formatted as a [RFC 3339](https://tools.ietf.org/html/rfc3339) timestamp.

Presence or absence of an expiry has the following impact:

- If an expiry is included, the bearer token and TLS credentials are cached until the expiry time is reached, or if the server responds with a 401 HTTP status code, or when the process exits.
  - If an expiry is omitted, the bearer token and TLS credentials are cached until the server responds with a 401 HTTP status code or until the process exits.
- [client.authentication.k8s.io/v1](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.17/#execcredential-v1-clientauthentication-k8s-io)

- [client.authentication.k8s.io/v1beta1](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.17/#execcredential-v1-client.authentication.k8s.io)

```
{
 "apiVersion": "client.authentication.k8s.io/v1",
 "kind": "ExecCredential",
 "status": {
 "token": "my-bearer-token",
 "expirationTimestamp": "2018-03-05T17:30:20-08:00"
 }
}
```

```
{
 "apiVersion": "client.authentication.k8s.io/v1beta1",
 "kind": "ExecCredential",
 "status": {
 "token": "my-bearer-token",
 "expirationTimestamp": "2018-03-05T17:30:20-08:00"
 }
}
```

To enable the exec plugin to obtain cluster-specific information, set `provideClusterInfo` on the `user.exec` field in the [kubeconfig](#). The plugin will then be supplied this cluster-specific information in the `KUBERNETES_EXEC_INFO` environment variable. Information from this environment variable can be used to perform cluster-specific credential acquisition logic. The following `ExecCredential` manifest describes a cluster information sample.

- [client.authentication.k8s.io/v1](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.17/#execcredential-v1-client.authentication.k8s.io)
- [client.authentication.k8s.io/v1beta1](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.17/#execcredential-v1-client.authentication.k8s.io)

```
{
 "apiVersion": "client.authentication.k8s.io/v1",
 "kind": "ExecCredential",
 "spec": {
 "cluster": {
 "server": "https://172.17.4.100:6443",
 "certificate-authority-data": "LS0t...",
 "config": {
 "arbitrary": "config",
 "this": "can be provided via the KUBERNETES_EXEC_INFO environment variable upon setting provideClusterInfo",
 "you": ["can", "put", "anything", "here"]
 }
 },
 "interactive": true
 }
}
```

```
{
 "apiVersion": "client.authentication.k8s.io/v1beta1",
 "kind": "ExecCredential",
 "spec": {
 "cluster": {
 "server": "https://172.17.4.100:6443",
 "certificate-authority-data": "LS0t...",

```



```

 "config": {
 "arbitrary": "config",
 "this": "can be provided via the KUBERNETES_EXEC_INFO environment variable upon
setting provideClusterInfo",
 "you": ["can", "put", "anything", "here"]
 }
 },
 "interactive": true
}
}

```

## API access to authentication information for a client

FEATURE STATE: Kubernetes v1.28 [stable]

If your cluster has the API enabled, you can use the `SelfSubjectReview` API to find out how your Kubernetes cluster maps your authentication information to identify you as a client. This works whether you are authenticating as a user (typically representing a real person) or as a `ServiceAccount`.

`SelfSubjectReview` objects do not have any configurable fields. On receiving a request, the Kubernetes API server fills the status with the user attributes and returns it to the user.

Request example (the body would be a `SelfSubjectReview`):

POST `/apis/authentication.k8s.io/v1/selfsubjectreviews`

```

{
 "apiVersion": "authentication.k8s.io/v1",
 "kind": "SelfSubjectReview"
}

```

Response example:

```

{
 "apiVersion": "authentication.k8s.io/v1",
 "kind": "SelfSubjectReview",
 "status": {
 "userInfo": {
 "name": "jane.doe",
 "uid": "b6c7cfd4-f166-11ec-8ea0-0242ac120002",
 "groups": [
 "viewers",
 "editors",
 "system:authenticated"
],
 "extra": {
 "provider_id": ["token.company.example"]
 }
 }
 }
}

```

For convenience, the `kubectl auth whoami` command is present. Executing this command will produce the following output (yet different user attributes will be shown):

- Simple output example

| ATTRIBUTE | VALUE                  |
|-----------|------------------------|
| Username  | jane.doe               |
| Groups    | [system:authenticated] |

- Complex example including extra attributes

| ATTRIBUTE       | VALUE                                    |
|-----------------|------------------------------------------|
| Username        | jane.doe                                 |
| UID             | b79dbf30-0c6a-11ed-861d-0242ac120002     |
| Groups          | [students teachers system:authenticated] |
| Extra: skills   | [reading learning]                       |
| Extra: subjects | [math sports]                            |

By providing the output flag, it is also possible to print the JSON or YAML representation of the result:

- [JSON](#)
- [YAML](#)

```
{
 "apiVersion": "authentication.k8s.io/v1",
 "kind": "SelfSubjectReview",
 "status": {
 "userInfo": {
 "username": "jane.doe",
 "uid": "b79dbf30-0c6a-11ed-861d-0242ac120002",
 "groups": [
 "students",
 "teachers",
 "system:authenticated"
],
 "extra": {
 "skills": [
 "reading",
 "learning"
],
 "subjects": [
 "math",
 "sports"
]
 }
 }
 }
}
```

```
apiVersion: authentication.k8s.io/v1
kind: SelfSubjectReview
status:
 userInfo:
```

```
username: jane.doe
uid: b79dbf30-0c6a-11ed-861d-0242ac120002
groups:
- students
- teachers
- system:authenticated
extra:
skills:
- reading
- learning
subjects:
- math
- sports
```

This feature is extremely useful when a complicated authentication flow is used in a Kubernetes cluster, for example, if you use [webhook token authentication](#) or [authenticating proxy](#).

#### Note:

The Kubernetes API server fills the `userInfo` after all authentication mechanisms are applied, including [impersonation](#). If you, or an authentication proxy, make a `SelfSubjectReview` using impersonation, you see the user details and properties for the user that was impersonated.

By default, all authenticated users can create `SelfSubjectReview` objects when the `APISelfSubjectReview` feature is enabled. It is allowed by the `system:basic-user` cluster role.

#### Note:

You can only make `SelfSubjectReview` requests if:

- the `APISelfSubjectReview` [feature gate](#) is enabled for your cluster (not needed for Kubernetes 1.30, but older Kubernetes versions might not offer this feature gate, or might default it to be off)
- (if you are running a version of Kubernetes older than v1.28) the API server for your cluster has the `authentication.k8s.io/v1alpha1` or `authentication.k8s.io/v1beta1` [API group](#) enabled.

## What's next

- Read the [client authentication reference \(v1beta1\)](#)
- Read the [client authentication reference \(v1\)](#)

# Authenticating with Bootstrap Tokens

FEATURE STATE: Kubernetes v1.18 [stable]

Bootstrap tokens are a simple bearer token that is meant to be used when creating new clusters or joining new nodes to an existing cluster. It was built to support [kubeadm](#), but can be used in other contexts for users that wish to start clusters without `kubeadm`. It is also built to work, via RBAC policy, with the [Kubelet TLS Bootstrapping](#) system.

# Bootstrap Tokens Overview

Bootstrap Tokens are defined with a specific type (`bootstrap.kubernetes.io/token`) of secrets that lives in the `kube-system` namespace. These Secrets are then read by the Bootstrap Authenticator in the API Server. Expired tokens are removed with the TokenCleaner controller in the Controller Manager. The tokens are also used to create a signature for a specific ConfigMap used in a "discovery" process through a BootstrapSigner controller.

## Token Format

Bootstrap Tokens take the form of `abcdef.0123456789abcdef`. More formally, they must match the regular expression `[a-z0-9]{6}\.[a-z0-9]{16}`.

The first part of the token is the "Token ID" and is considered public information. It is used when referring to a token without leaking the secret part used for authentication. The second part is the "Token Secret" and should only be shared with trusted parties.

## Enabling Bootstrap Token Authentication

The Bootstrap Token authenticator can be enabled using the following flag on the API server:

```
--enable-bootstrap-token-auth
```

When enabled, bootstrapping tokens can be used as bearer token credentials to authenticate requests against the API server.

```
Authorization: Bearer 07401b.f395accd246ae52d
```

Tokens authenticate as the username `system:bootstrap:<token id>` and are members of the group `system:bootstrappers`. Additional groups may be specified in the token's Secret.

Expired tokens can be deleted automatically by enabling the `tokencleaner` controller on the controller manager.

```
--controllers=*,tokencleaner
```

## Bootstrap Token Secret Format

Each valid token is backed by a secret in the `kube-system` namespace. You can find the full design doc [here](#).

Here is what the secret looks like.

```
apiVersion: v1
kind: Secret
metadata:
 # Name MUST be of form "bootstrap-token-<token id>"
 name: bootstrap-token-07401b
 namespace: kube-system
Type MUST be 'bootstrap.kubernetes.io/token'
```

```
type: bootstrap.kubernetes.io/token
stringData:
 # Human readable description. Optional.
 description: "The default bootstrap token generated by 'kubeadm init'."

 # Token ID and secret. Required.
 token-id: 07401b
 token-secret: f395accd246ae52d

 # Expiration. Optional.
 expiration: 2017-03-10T03:22:11Z

 # Allowed usages.
 usage-bootstrap-authentication: "true"
 usage-bootstrap-signing: "true"

 # Extra groups to authenticate the token as. Must start with "system:bootstrappers:"
 auth-extra-groups: system:bootstrappers:worker,system:bootstrappers:ingress
```

The type of the secret must be `bootstrap.kubernetes.io/token` and the name must be `bootstrap-token-<token id>`. It must also exist in the `kube-system` namespace.

The `usage-bootstrap-*` members indicate what this secret is intended to be used for. A value must be set to `true` to be enabled.

- `usage-bootstrap-authentication` indicates that the token can be used to authenticate to the API server as a bearer token.
- `usage-bootstrap-signing` indicates that the token may be used to sign the cluster-info ConfigMap as described below.

The `expiration` field controls the expiry of the token. Expired tokens are rejected when used for authentication and ignored during ConfigMap signing. The expiry value is encoded as an absolute UTC time using RFC3339. Enable the `tokencleaner` controller to automatically delete expired tokens.

## Token Management with kubeadm

You can use the `kubeadm` tool to manage tokens on a running cluster. See the [kubeadm token docs](#) for details.

## ConfigMap Signing

In addition to authentication, the tokens can be used to sign a ConfigMap. This is used early in a cluster bootstrap process before the client trusts the API server. The signed ConfigMap can be authenticated by the shared token.

Enable ConfigMap signing by enabling the `bootstrapsigner` controller on the Controller Manager.

```
--controllers=*,bootstrapsigner
```

The ConfigMap that is signed is cluster-info in the kube-public namespace. The typical flow is that a client reads this ConfigMap while unauthenticated and ignoring TLS errors. It then validates the payload of the ConfigMap by looking at a signature embedded in the ConfigMap.

The ConfigMap may look like this:

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: cluster-info
 namespace: kube-public
data:
 jws-kubeconfig-07401b: eyJhbGciOiJIUzI1NiIsImtpZCI6IjA3NDAxYiJ9..tYefbo6zDNo40MQE07aZcQX2m3EB2rO3NuXtxVMYm9U
 kubeconfig: |
 apiVersion: v1
 clusters:
 - cluster:
 certificate-authority-data: <really long certificate data>
 server: https://10.138.0.2:6443
 name: ""
 contexts: []
 current-context: ""
 kind: Config
 preferences: {}
 users: []
```

The kubeconfig member of the ConfigMap is a config file with only the cluster information filled out. The key thing being communicated here is the certificate-authority-data. This may be expanded in the future.

The signature is a JWS signature using the "detached" mode. To validate the signature, the user should encode the kubeconfig payload according to JWS rules (base64 encoded while discarding any trailing =). That encoded payload is then used to form a whole JWS by inserting it between the 2 dots. You can verify the JWS using the HS256 scheme (HMAC-SHA256) with the full token (e.g. 07401b.f395accd246ae52d) as the shared secret. Users *must* verify that HS256 is used.

### Warning:

Any party with a bootstrapping token can create a valid signature for that token. When using ConfigMap signing it's discouraged to share the same token with many clients, since a compromised client can potentially man-in-the middle another client relying on the signature to bootstrap TLS trust.

Consult the [kubeadm implementation details](#) section for more information.

## Authorization

Details of Kubernetes authorization mechanisms and supported authorization modes.

Kubernetes authorization takes place following [authentication](#). Usually, a client making a request must be authenticated (logged in) before its request can be allowed; however, Kubernetes also allows anonymous requests in some circumstances.

For an overview of how authorization fits into the wider context of API access control, read [Controlling Access to the Kubernetes API](#).

## Authorization verdicts

Kubernetes authorization of API requests takes place within the API server. The API server evaluates all of the request attributes against all policies, potentially also consulting external services, and then allows or denies the request.

All parts of an API request must be allowed by some authorization mechanism in order to proceed. In other words: access is denied by default.

### Note:

Access controls and policies that depend on specific fields of specific kinds of objects are handled by [admission controllers](#).

Kubernetes admission control happens after authorization has completed (and, therefore, only when the authorization decision was to allow the request).

When multiple [authorization modules](#) are configured, each is checked in sequence. If any authorizer *approves* or *denies* a request, that decision is immediately returned and no other authorizer is consulted. If all modules have *no opinion* on the request, then the request is denied. An overall deny verdict means that the API server rejects the request and responds with an HTTP 403 (Forbidden) status.

## Request attributes used in authorization

Kubernetes reviews only the following API request attributes:

- **user** - The user string provided during authentication.
- **group** - The list of group names to which the authenticated user belongs.
- **extra** - A map of arbitrary string keys to string values, provided by the authentication layer.
- **API** - Indicates whether the request is for an API resource.
- **Request path** - Path to miscellaneous non-resource endpoints like /api or /healthz.
- **API request verb** - API verbs like get, list, create, update, patch, watch, delete, and deletecollection are used for resource requests. To determine the request verb for a resource API endpoint, see [request verbs and authorization](#).
- **HTTP request verb** - Lowercased HTTP methods like get, post, put, and delete are used for non-resource requests.
- **Resource** - The ID or name of the resource that is being accessed (for resource requests only) -- For resource requests using get, update, patch, and delete verbs, you must provide the resource name.
- **Subresource** - The subresource that is being accessed (for resource requests only).
- **Namespace** - The namespace of the object that is being accessed (for namespaced resource requests only).

- **API group** - The [API Group](#) being accessed (for resource requests only). An empty string designates the *core* [API group](#).

## Request verbs and authorization

### Non-resource requests

Requests to endpoints other than `/api/v1/...` or `/apis/<group>/<version>/...` are considered *non-resource requests*, and use the lower-cased HTTP method of the request as the verb. For example, making a GET request using HTTP to endpoints such as `/api` or `/healthz` would use **get** as the verb.

### Resource requests

To determine the request verb for a resource API endpoint, Kubernetes maps the HTTP verb used and considers whether or not the request acts on an individual resource or on a collection of resources:

| HTTP verb | request verb                                                                                                                                                                       |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POST      | <b>create</b>                                                                                                                                                                      |
| GET, HEAD | <b>get</b> (for individual resources), <b>list</b> (for collections, including full object content), <b>watch</b> (for watching an individual resource or collection of resources) |
| PUT       | <b>update</b>                                                                                                                                                                      |
| PATCH     | <b>patch</b>                                                                                                                                                                       |
| DELETE    | <b>delete</b> (for individual resources), <b>deletecollection</b> (for collections)                                                                                                |

### Caution:

+The **get**, **list** and **watch** verbs can all return the full details of a resource. In terms of access to the returned data they are equivalent. For example, **list** on secrets will reveal the **data** attributes of any returned resources.

Kubernetes sometimes checks authorization for additional permissions using specialized verbs. For example:

- Special cases of [authentication](#)
  - **impersonate** verb on users, groups, and serviceaccounts in the core API group, and the userextras in the authentication.k8s.io API group.
- [Authorization of CertificateSigningRequests](#)
  - **approve** verb for CertificateSigningRequests, and **update** for revisions to existing approvals
- [RBAC](#)
  - **bind** and **escalate** verbs on roles and clusterroles resources in the rbac.authorization.k8s.io API group.



# Authorization context

Kubernetes expects attributes that are common to REST API requests. This means that Kubernetes authorization works with existing organization-wide or cloud-provider-wide access control systems which may handle other APIs besides the Kubernetes API.

## Authorization modes

The Kubernetes API server may authorize a request using one of several authorization modes:

### AlwaysAllow

This mode allows all requests, which brings [security risks](#). Use this authorization mode only if you do not require authorization for your API requests (for example, for testing).

### AlwaysDeny

This mode blocks all requests. Use this authorization mode only for testing.

### ABAC ([attribute-based access control](#))

Kubernetes ABAC mode defines an access control paradigm whereby access rights are granted to users through the use of policies which combine attributes together.

The policies can use any type of attributes (user attributes, resource attributes, object, environment attributes, etc).

### RBAC ([role-based access control](#))

Kubernetes RBAC is a method of regulating access to computer or network resources based on the roles of individual users within an enterprise. In this context, access is the ability of an individual user to perform a specific task, such as view, create, or modify a file.

In this mode, Kubernetes uses the `rbac.authorization.k8s.io` API group to drive authorization decisions, allowing you to dynamically configure permission policies through the Kubernetes API.

### Node

A special-purpose authorization mode that grants permissions to kubelets based on the pods they are scheduled to run. To learn more about the Node authorization mode, see [Node Authorization](#).

### Webhook

Kubernetes [webhook mode](#) for authorization makes a synchronous HTTP callout, blocking the request until the remote HTTP service responds to the query. You can write your own software to handle the callout, or use solutions from the ecosystem.

## Warning:

Enabling the AlwaysAllow mode bypasses authorization; do not use this on a cluster where you do not trust **all** potential API clients, including the workloads that you run.

Authorization mechanisms typically return either a *deny* or *no opinion* result; see [authorization verdicts](#) for more on this. Activating the AlwaysAllow means that if all other authorizers return “no opinion”, the request is allowed. For example, `--authorization-mode=AlwaysAllow,RBAC` has the same effect as `--authorization-mode=AlwaysAllow` because Kubernetes RBAC does not provide negative (deny) access rules.

You should not use the AlwaysAllow mode on a Kubernetes cluster where the API server is reachable from the public internet.

## Authorization mode configuration

You can configure the Kubernetes API server's authorizer chain using either [command line arguments](#) only or, as a beta feature, using a [configuration file](#).

You have to pick one of the two configuration approaches; setting both `--authorization-config` path and configuring an authorization webhook using the `--authorization-mode` and `--authorization-webhook-*` command line arguments is not allowed. If you try this, the API server reports an error message during startup, then exits immediately.

## Command line authorization mode configuration

FEATURE STATE: Kubernetes v1.8 [stable]

You can use the following modes:

- `--authorization-mode=ABAC` (Attribute-based access control mode)
- `--authorization-mode=RBAC` (Role-based access control mode)
- `--authorization-mode=Node` (Node authorizer)
- `--authorization-mode=Webhook` (Webhook authorization mode)
- `--authorization-mode=AlwaysAllow` (always allows requests; carries [security risks](#))
- `--authorization-mode=AlwaysDeny` (always denies requests)

You can choose more than one authorization mode; for example: `--authorization-mode=Node,Webhook`

Kubernetes checks authorization modules based on the order that you specify them on the API server's command line, so an earlier module has higher priority to allow or deny a request.

You cannot combine the `--authorization-mode` command line argument with the `--authorization-config` command line argument used for [configuring authorization using a local file](#).

For more information on command line arguments to the API server, read the [kube-apiserver reference](#).

## Configuring the API Server using an authorization config file

FEATURE STATE: Kubernetes v1.30 [beta]

As a beta feature, Kubernetes lets you configure authorization chains that can include multiple webhooks. The authorization items in that chain can have well-defined parameters that validate requests in a particular order, offering you fine-grained control, such as explicit Deny on failures.

The configuration file approach even allows you to specify [CEL](#) rules to pre-filter requests before they are dispatched to webhooks, helping you to prevent unnecessary invocations. The

API server also automatically reloads the authorizer chain when the configuration file is modified.

You specify the path to the authorization configuration using the `--authorization-config` command line argument.

If you want to use command line arguments instead of a configuration file, that's also a valid and supported approach. Some authorization capabilities (for example: multiple webhooks, webhook failure policy, and pre-filter rules) are only available if you use an authorization configuration file.

### Example configuration

```

#
DO NOT USE THE CONFIG AS IS. THIS IS AN EXAMPLE.
#
apiVersion: apiserver.config.k8s.io/v1beta1
kind: AuthorizationConfiguration
authorizers:
- type: Webhook
 # Name used to describe the authorizer
 # This is explicitly used in monitoring machinery for metrics
 # Note:
 # - Validation for this field is similar to how K8s labels are validated today.
 # Required, with no default
 name: webhook
 webhook:
 # The duration to cache 'authorized' responses from the webhook
 # authorizer.
 # Same as setting `--authorization-webhook-cache-authorized-ttl` flag
 # Default: 5m0s
 authorizedTTL: 30s
 # The duration to cache 'unauthorized' responses from the webhook
 # authorizer.
 # Same as setting `--authorization-webhook-cache-unauthorized-ttl` flag
 # Default: 30s
 unauthorizedTTL: 30s
 # Timeout for the webhook request
 # Maximum allowed is 30s.
 # Required, with no default.
 timeout: 3s
 # The API version of the authorization.k8s.io SubjectAccessReview to
 # send to and expect from the webhook.
 # Same as setting `--authorization-webhook-version` flag
 # Required, with no default
 # Valid values: v1beta1, v1
 subjectAccessReviewVersion: v1
 # MatchConditionSubjectAccessReviewVersion specifies the SubjectAccessReview
 # version the CEL expressions are evaluated against
 # Valid values: v1
 # Required, no default value
 matchConditionSubjectAccessReviewVersion: v1
```

```

Controls the authorization decision when a webhook request fails to
complete or returns a malformed response or errors evaluating
matchConditions.
Valid values:
- NoOpinion: continue to subsequent authorizers to see if one of
them allows the request
- Deny: reject the request without consulting subsequent authorizers
Required, with no default.
failurePolicy: Deny
connectionInfo:
 # Controls how the webhook should communicate with the server.
 # Valid values:
 # - KubeConfig: use the file specified in kubeConfigFile to locate the
 # server.
 # - InClusterConfig: use the in-cluster configuration to call the
 # SubjectAccessReview API hosted by kube-apiserver. This mode is not
 # allowed for kube-apiserver.
 type: KubeConfig
 # Path to KubeConfigFile for connection info
 # Required, if connectionInfo.Type is KubeConfig
 kubeConfigFile: /kube-system-authz-webhook.yaml
 # matchConditions is a list of conditions that must be met for a request to be sent to this
 # webhook. An empty list of matchConditions matches all requests.
 # There are a maximum of 64 match conditions allowed.
 #
 # The exact matching logic is (in order):
 # 1. If at least one matchCondition evaluates to FALSE, then the webhook is skipped.
 # 2. If ALL matchConditions evaluate to TRUE, then the webhook is called.
 # 3. If at least one matchCondition evaluates to an error (but none are FALSE):
 # - If failurePolicy=Deny, then the webhook rejects the request
 # - If failurePolicy=NoOpinion, then the error is ignored and the webhook is skipped
matchConditions:
expression represents the expression which will be evaluated by CEL. Must evaluate to
bool.
CEL expressions have access to the contents of the SubjectAccessReview in v1 version.
If version specified by subjectAccessReviewVersion in the request variable is v1beta1,
the contents would be converted to the v1 version before evaluating the CEL expression.
#
Documentation on CEL: https://kubernetes.io/docs/reference/using-api/cel/
#
only send resource requests to the webhook
- expression: has(request.resourceAttributes)
only intercept requests to kube-system
- expression: request.resourceAttributes.namespace == 'kube-system'
don't intercept requests from kube-system service accounts
- expression: !('system:serviceaccounts:kube-system' in request.user.groups)
- type: Node
 name: node
- type: RBAC
 name: rbac
- type: Webhook
 name: in-cluster-authorizer
webhook:

```

```
authorizedTTL: 5m
unauthorizedTTL: 30s
timeout: 3s
subjectAccessReviewVersion: v1
failurePolicy: NoOpinion
connectionInfo:
 type: InClusterConfig
```

When configuring the authorizer chain using a configuration file, make sure all the control plane nodes have the same file contents. Take a note of the API server configuration when upgrading / downgrading your clusters. For example, if upgrading from Kubernetes 1.29 to Kubernetes 1.30, you would need to make sure the config file is in a format that Kubernetes 1.30 can understand, before you upgrade the cluster. If you downgrade to 1.29, you would need to set the configuration appropriately.

## Authorization configuration and reloads

Kubernetes reloads the authorization configuration file when the API server observes a change to the file, and also on a 60 second schedule if no change events were observed.

### Note:

You must ensure that all non-webhook authorizer types remain unchanged in the file on reload.

A reload **must not** add or remove Node or RBAC authorizers (they can be reordered, but cannot be added or removed).

## Privilege escalation via workload creation or edits

Users who can create/edit pods in a namespace, either directly or through an object that enables indirect [workload management](#), may be able to escalate their privileges in that namespace. The potential routes to privilege escalation include Kubernetes [API extensions](#) and their associated [controllers](#).

### Caution:

As a cluster administrator, use caution when granting access to create or edit workloads. Some details of how these can be misused are documented in [escalation paths](#).

## Escalation paths

There are different ways that an attacker or untrustworthy user could gain additional privilege within a namespace, if you allow them to run arbitrary Pods in that namespace:

- Mounting arbitrary Secrets in that namespace
  - Can be used to access confidential information meant for other workloads
  - Can be used to obtain a more privileged ServiceAccount's service account token
- Using arbitrary ServiceAccounts in that namespace
  - Can perform Kubernetes API actions as another workload (impersonation)
  - Can perform any privileged actions that ServiceAccount has

- Mounting or using ConfigMaps meant for other workloads in that namespace
  - Can be used to obtain information meant for other workloads, such as database host names.
- Mounting volumes meant for other workloads in that namespace
  - Can be used to obtain information meant for other workloads, and change it.

### Caution:

As a system administrator, you should be cautious when deploying CustomResourceDefinitions that let users make changes to the above areas. These may open privilege escalations paths. Consider the consequences of this kind of change when deciding on your authorization controls.

## Checking API access

kubectl provides the `auth can-i` subcommand for quickly querying the API authorization layer. The command uses the `SelfSubjectAccessReview` API to determine if the current user can perform a given action, and works regardless of the authorization mode used.

```
kubectl auth can-i create deployments --namespace dev
```

The output is similar to this:

```
yes
```

```
kubectl auth can-i create deployments --namespace prod
```

The output is similar to this:

```
no
```

Administrators can combine this with [user impersonation](#) to determine what action other users can perform.

```
kubectl auth can-i list secrets --namespace dev --as dave
```

The output is similar to this:

```
no
```

Similarly, to check whether a ServiceAccount named `dev-sa` in Namespace `dev` can list Pods in the Namespace `target`:

```
kubectl auth can-i list pods \
 --namespace target \
 --as system:serviceaccount:dev:dev-sa
```

The output is similar to this:

```
yes
```

`SelfSubjectAccessReview` is part of the `authorization.k8s.io` API group, which exposes the API server authorization to external services. Other resources in this group include:

### SubjectAccessReview

Access review for any user, not only the current one. Useful for delegating authorization decisions to the API server. For example, the kubelet and extension API servers use this to determine user access to their own APIs.

### LocalSubjectAccessReview

Like SubjectAccessReview but restricted to a specific namespace.

### SelfSubjectRulesReview

A review which returns the set of actions a user can perform within a namespace. Useful for users to quickly summarize their own access, or for UIs to hide/show actions.

These APIs can be queried by creating normal Kubernetes resources, where the response status field of the returned object is the result of the query. For example:

```
kubectl create -f - -o yaml << EOF
apiVersion: authorization.k8s.io/v1
kind: SelfSubjectAccessReview
spec:
 resourceAttributes:
 group: apps
 resource: deployments
 verb: create
 namespace: dev
EOF
```

The generated SelfSubjectAccessReview is similar to:

```
apiVersion: authorization.k8s.io/v1
kind: SelfSubjectAccessReview
metadata:
 creationTimestamp: null
spec:
 resourceAttributes:
 group: apps
 resource: deployments
 namespace: dev
 verb: create
status:
 allowed: true
 denied: false
```

## What's next

- To learn more about Authentication, see [Authentication](#).
- For an overview, read [Controlling Access to the Kubernetes API](#).
- To learn more about Admission Control, see [Using Admission Controllers](#).
- Read more about [Common Expression Language in Kubernetes](#).

## Using RBAC Authorization

Role-based access control (RBAC) is a method of regulating access to computer or network resources based on the roles of individual users within your organization.

RBAC authorization uses the `rbac.authorization.k8s.io` [API group](#) to drive authorization decisions, allowing you to dynamically configure policies through the Kubernetes API.

To enable RBAC, start the [API server](#) with the `--authorization-mode` flag set to a comma-separated list that includes RBAC; for example:

```
kube-apiserver --authorization-mode=Example,RBAC --other-options --more-options
```

## API objects

The RBAC API declares four kinds of Kubernetes object: *Role*, *ClusterRole*, *RoleBinding* and *ClusterRoleBinding*. You can describe or amend the RBAC [objects](#) using tools such as `kubectl`, just like any other Kubernetes object.

### Caution:

These objects, by design, impose access restrictions. If you are making changes to a cluster as you learn, see [privilege escalation prevention and bootstrapping](#) to understand how those restrictions can prevent you making some changes.

## Role and ClusterRole

An RBAC *Role* or *ClusterRole* contains rules that represent a set of permissions. Permissions are purely additive (there are no "deny" rules).

A Role always sets permissions within a particular [namespace](#); when you create a Role, you have to specify the namespace it belongs in.

ClusterRole, by contrast, is a non-namespaced resource. The resources have different names (Role and ClusterRole) because a Kubernetes object always has to be either namespaced or not namespaced; it can't be both.

ClusterRoles have several uses. You can use a ClusterRole to:

1. define permissions on namespaced resources and be granted access within individual namespace(s)
2. define permissions on namespaced resources and be granted access across all namespaces
3. define permissions on cluster-scoped resources

If you want to define a role within a namespace, use a Role; if you want to define a role cluster-wide, use a ClusterRole.

### Role example

Here's an example Role in the "default" namespace that can be used to grant read access to [pods](#):

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
 namespace: default
 name: pod-reader
```



rules:

```
- apiGroups: [""] # "" indicates the core API group
resources: ["pods"]
verbs: ["get", "watch", "list"]
```

## ClusterRole example

A ClusterRole can be used to grant the same permissions as a Role. Because ClusterRoles are cluster-scoped, you can also use them to grant access to:

- cluster-scoped resources (like [nodes](#))
- non-resource endpoints (like `/healthz`)
- namespaced resources (like Pods), across all namespaces

For example: you can use a ClusterRole to allow a particular user to run `kubectl get pods --all-namespaces`

Here is an example of a ClusterRole that can be used to grant read access to [secrets](#) in any particular namespace, or across all namespaces (depending on how it is [bound](#)):

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 # "namespace" omitted since ClusterRoles are not namespaced
 name: secret-reader
rules:
- apiGroups: [""]
 #
 # at the HTTP level, the name of the resource for accessing Secret
 # objects is "secrets"
 resources: ["secrets"]
 verbs: ["get", "watch", "list"]
```

The name of a Role or a ClusterRole object must be a valid [path segment name](#).

## RoleBinding and ClusterRoleBinding

A role binding grants the permissions defined in a role to a user or set of users. It holds a list of *subjects* (users, groups, or service accounts), and a reference to the role being granted. A RoleBinding grants permissions within a specific namespace whereas a ClusterRoleBinding grants that access cluster-wide.

A RoleBinding may reference any Role in the same namespace. Alternatively, a RoleBinding can reference a ClusterRole and bind that ClusterRole to the namespace of the RoleBinding. If you want to bind a ClusterRole to all the namespaces in your cluster, you use a ClusterRoleBinding.

The name of a RoleBinding or ClusterRoleBinding object must be a valid [path segment name](#).

## RoleBinding examples

Here is an example of a RoleBinding that grants the "pod-reader" Role to the user "jane" within the "default" namespace. This allows "jane" to read pods in the "default" namespace.

```
apiVersion: rbac.authorization.k8s.io/v1
This role binding allows "jane" to read pods in the "default" namespace.
You need to already have a Role named "pod-reader" in that namespace.
kind: RoleBinding
metadata:
 name: read-pods
 namespace: default
subjects:
You can specify more than one "subject"
- kind: User
 name: jane # "name" is case sensitive
 apiGroup: rbac.authorization.k8s.io
roleRef:
"roleRef" specifies the binding to a Role / ClusterRole
kind: Role #this must be Role or ClusterRole
name: pod-reader # this must match the name of the Role or ClusterRole you wish to bind to
apiGroup: rbac.authorization.k8s.io
```

A RoleBinding can also reference a ClusterRole to grant the permissions defined in that ClusterRole to resources inside the RoleBinding's namespace. This kind of reference lets you define a set of common roles across your cluster, then reuse them within multiple namespaces.

For instance, even though the following RoleBinding refers to a ClusterRole, "dave" (the subject, case sensitive) will only be able to read Secrets in the "development" namespace, because the RoleBinding's namespace (in its metadata) is "development".

```
apiVersion: rbac.authorization.k8s.io/v1
This role binding allows "dave" to read secrets in the "development" namespace.
You need to already have a ClusterRole named "secret-reader".
kind: RoleBinding
metadata:
 name: read-secrets
 #
 # The namespace of the RoleBinding determines where the permissions are granted.
 # This only grants permissions within the "development" namespace.
 namespace: development
subjects:
- kind: User
 name: dave # Name is case sensitive
 apiGroup: rbac.authorization.k8s.io
roleRef:
 kind: ClusterRole
 name: secret-reader
 apiGroup: rbac.authorization.k8s.io
```

## ClusterRoleBinding example

To grant permissions across a whole cluster, you can use a ClusterRoleBinding. The following ClusterRoleBinding allows any user in the group "manager" to read secrets in any namespace.

```
apiVersion: rbac.authorization.k8s.io/v1
This cluster role binding allows anyone in the "manager" group to read secrets in any
namespace.
kind: ClusterRoleBinding
metadata:
 name: read-secrets-global
subjects:
- kind: Group
 name: manager # Name is case sensitive
 apiGroup: rbac.authorization.k8s.io
roleRef:
 kind: ClusterRole
 name: secret-reader
 apiGroup: rbac.authorization.k8s.io
```

After you create a binding, you cannot change the Role or ClusterRole that it refers to. If you try to change a binding's roleRef, you get a validation error. If you do want to change the roleRef for a binding, you need to remove the binding object and create a replacement.

There are two reasons for this restriction:

1. Making roleRef immutable allows granting someone update permission on an existing binding object, so that they can manage the list of subjects, without being able to change the role that is granted to those subjects.
2. A binding to a different role is a fundamentally different binding. Requiring a binding to be deleted/recreated in order to change the roleRef ensures the full list of subjects in the binding is intended to be granted the new role (as opposed to enabling or accidentally modifying only the roleRef without verifying all of the existing subjects should be given the new role's permissions).

The `kubectl auth reconcile` command-line utility creates or updates a manifest file containing RBAC objects, and handles deleting and recreating binding objects if required to change the role they refer to. See [command usage and examples](#) for more information.

## Referring to resources

In the Kubernetes API, most resources are represented and accessed using a string representation of their object name, such as pods for a Pod. RBAC refers to resources using exactly the same name that appears in the URL for the relevant API endpoint. Some Kubernetes APIs involve a *subresource*, such as the logs for a Pod. A request for a Pod's logs looks like:

```
GET /api/v1/namespaces/{namespace}/pods/{name}/log
```

In this case, pods is the namespaced resource for Pod resources, and log is a subresource of pods. To represent this in an RBAC role, use a slash (/) to delimit the resource and subresource. To allow a subject to read pods and also access the log subresource for each of those Pods, you write:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
 namespace: default
 name: pod-and-pod-logs-reader
rules:
- apiGroups: [""]
 resources: ["pods", "pods/log"]
 verbs: ["get", "list"]
```

You can also refer to resources by name for certain requests through the `resourceNames` list. When specified, requests can be restricted to individual instances of a resource. Here is an example that restricts its subject to only get or update a [ConfigMap](#) named my-configmap:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
 namespace: default
 name: configmap-updater
rules:
- apiGroups: [""]
 #
 # at the HTTP level, the name of the resource for accessing ConfigMap
 # objects is "configmaps"
 resources: ["configmaps"]
 resourceNames: ["my-configmap"]
 verbs: ["update", "get"]
```

#### **Note:**

You cannot restrict create or delete requests by their resource name. For create, this limitation is because the name of the new object may not be known at authorization time. If you restrict list or watch by `resourceName`, clients must include a `metadata.name` field selector in their list or watch request that matches the specified `resourceName` in order to be authorized. For example, `kubectl get configmaps --field-selector=metadata.name=my-configmap`

Rather than referring to individual resources, `apiGroups`, and verbs, you can use the wildcard `*` symbol to refer to all such objects. For nonResourceURLs, you can use the wildcard `*` as a suffix glob match. For `resourceNames`, an empty set means that everything is allowed. Here is an example that allows access to perform any current and future action on all current and future resources in the example.com API group. This is similar to the built-in cluster-admin role.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
 namespace: default
 name: example.com-superuser # DO NOT USE THIS ROLE, IT IS JUST AN EXAMPLE
rules:
- apiGroups: ["example.com"]
 resources: ["*"]
 verbs: ["*"]
```

### Caution:

Using wildcards in resource and verb entries could result in overly permissive access being granted to sensitive resources. For instance, if a new resource type is added, or a new subresource is added, or a new custom verb is checked, the wildcard entry automatically grants access, which may be undesirable. The [principle of least privilege](#) should be employed, using specific resources and verbs to ensure only the permissions required for the workload to function correctly are applied.

## Aggregated ClusterRoles

You can *aggregate* several ClusterRoles into one combined ClusterRole. A controller, running as part of the cluster control plane, watches for ClusterRole objects with an aggregationRule set. The aggregationRule defines a label [selector](#) that the controller uses to match other ClusterRole objects that should be combined into the rules field of this one.

### Caution:

The control plane overwrites any values that you manually specify in the rules field of an aggregate ClusterRole. If you want to change or add rules, do so in the ClusterRole objects that are selected by the aggregationRule.

Here is an example aggregated ClusterRole:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: monitoring
aggregationRule:
 clusterRoleSelectors:
 - matchLabels:
 rbac.example.com/aggregate-to-monitoring: "true"
rules: [] # The control plane automatically fills in the rules
```

If you create a new ClusterRole that matches the label selector of an existing aggregated ClusterRole, that change triggers adding the new rules into the aggregated ClusterRole. Here is an example that adds rules to the "monitoring" ClusterRole, by creating another ClusterRole labeled rbac.example.com/aggregate-to-monitoring: true.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: monitoring-endpoints
labels:
 rbac.example.com/aggregate-to-monitoring: "true"
When you create the "monitoring-endpoints" ClusterRole,
the rules below will be added to the "monitoring" ClusterRole.
rules:
- apiGroups: [""]
 resources: ["services", "endpointslices", "pods"]
 verbs: ["get", "list", "watch"]
```

The [default user-facing roles](#) use ClusterRole aggregation. This lets you, as a cluster administrator, include rules for custom resources, such as those served by [CustomResourceDefinitions](#) or aggregated API servers, to extend the default roles.

For example: the following ClusterRoles let the "admin" and "edit" default roles manage the custom resource named CronTab, whereas the "view" role can perform only read actions on CronTab resources. You can assume that CronTab objects are named "crontabs" in URLs as seen by the API server.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: aggregate-cron-tabs-edit
 labels:
 # Add these permissions to the "admin" and "edit" default roles.
 rbac.authorization.k8s.io/aggregate-to-admin: "true"
 rbac.authorization.k8s.io/aggregate-to-edit: "true"
rules:
- apiGroups: ["stable.example.com"]
 resources: ["crontabs"]
 verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]

kind: ClusterRole
apiVersion: rbac.authorization.k8s.io/v1
metadata:
 name: aggregate-cron-tabs-view
 labels:
 # Add these permissions to the "view" default role.
 rbac.authorization.k8s.io/aggregate-to-view: "true"
rules:
- apiGroups: ["stable.example.com"]
 resources: ["crontabs"]
 verbs: ["get", "list", "watch"]
```

## Role examples

The following examples are excerpts from Role or ClusterRole objects, showing only the rules section.

Allow reading "pods" resources in the core [API Group](#):

```
rules:
- apiGroups: [""]
 #
 # at the HTTP level, the name of the resource for accessing Pod
 # objects is "pods"
 resources: ["pods"]
 verbs: ["get", "list", "watch"]
```

Allow reading/writing Deployments (at the HTTP level: objects with "deployments" in the resource part of their URL) in the "apps" API groups:

rules:

```
- apiGroups: ["apps"]
#
at the HTTP level, the name of the resource for accessing Deployment
objects is "deployments"
resources: ["deployments"]
verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
```

Allow reading Pods in the core API group, as well as reading or writing Job resources in the "batch" API group:

rules:

```
- apiGroups: [""]
#
at the HTTP level, the name of the resource for accessing Pod
objects is "pods"
resources: ["pods"]
verbs: ["get", "list", "watch"]
- apiGroups: ["batch"]
#
at the HTTP level, the name of the resource for accessing Job
objects is "jobs"
resources: ["jobs"]
verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
```

Allow reading a ConfigMap named "my-config" (must be bound with a RoleBinding to limit to a single ConfigMap in a single namespace):

rules:

```
- apiGroups: [""]
#
at the HTTP level, the name of the resource for accessing ConfigMap
objects is "configmaps"
resources: ["configmaps"]
resourceNames: ["my-config"]
verbs: ["get"]
```

Allow reading the resource "nodes" in the core group (because a Node is cluster-scoped, this must be in a ClusterRole bound with a ClusterRoleBinding to be effective):

rules:

```
- apiGroups: [""]
#
at the HTTP level, the name of the resource for accessing Node
objects is "nodes"
resources: ["nodes"]
verbs: ["get", "list", "watch"]
```

Allow GET and POST requests to the non-resource endpoint `/healthz` and all subpaths (must be in a ClusterRole bound with a ClusterRoleBinding to be effective):

rules:

```
- nonResourceURLs: ["/healthz", "/healthz/*"] # '*' in a nonResourceURL is a suffix glob match
verbs: ["get", "post"]
```

## Referring to subjects

A RoleBinding or ClusterRoleBinding binds a role to subjects. Subjects can be groups, users or [ServiceAccounts](#).

Kubernetes represents usernames as strings. These can be: plain names, such as "alice"; email-style names, like "bob@example.com"; or numeric user IDs represented as a string. It is up to you as a cluster administrator to configure the [authentication modules](#) so that authentication produces usernames in the format you want.

### Caution:

The prefix `system:` is reserved for Kubernetes system use, so you should ensure that you don't have users or groups with names that start with `system:` by accident. Other than this special prefix, the RBAC authorization system does not require any format for usernames.

In Kubernetes, Authenticator modules provide group information. Groups, like users, are represented as strings, and that string has no format requirements, other than that the prefix `system:` is reserved.

[ServiceAccounts](#) have names prefixed with `system:serviceaccount:`, and belong to groups that have names prefixed with `system:serviceaccounts:`.

### Note:

- `system:serviceaccount:` (singular) is the prefix for service account usernames.
- `system:serviceaccounts:` (plural) is the prefix for service account groups.

## RoleBinding examples

The following examples are RoleBinding excerpts that only show the subjects section.

For a user named `alice@example.com`:

```
subjects:
- kind: User
 name: "alice@example.com"
 apiGroup: rbac.authorization.k8s.io
```

For a group named `frontend-admins`:

```
subjects:
- kind: Group
 name: "frontend-admins"
 apiGroup: rbac.authorization.k8s.io
```

For the default service account in the "kube-system" namespace:

```
subjects:
- kind: ServiceAccount
 name: default
 namespace: kube-system
```



For all service accounts in the "qa" namespace:

```
subjects:
- kind: Group
 name: system:serviceaccounts:qa
 apiGroup: rbac.authorization.k8s.io
```

For all service accounts in any namespace:

```
subjects:
- kind: Group
 name: system:serviceaccounts
 apiGroup: rbac.authorization.k8s.io
```

For all authenticated users:

```
subjects:
- kind: Group
 name: system:authenticated
 apiGroup: rbac.authorization.k8s.io
```

For all unauthenticated users:

```
subjects:
- kind: Group
 name: system:unauthenticated
 apiGroup: rbac.authorization.k8s.io
```

For all users:

```
subjects:
- kind: Group
 name: system:authenticated
 apiGroup: rbac.authorization.k8s.io
- kind: Group
 name: system:unauthenticated
 apiGroup: rbac.authorization.k8s.io
```

## Default roles and role bindings

API servers create a set of default ClusterRole and ClusterRoleBinding objects. Many of these are system: prefixed, which indicates that the resource is directly managed by the cluster control plane. All of the default ClusterRoles and ClusterRoleBindings are labeled with `kubernetes.io/bootstrapping=rbac-defaults`.

### Caution:

Take care when modifying ClusterRoles and ClusterRoleBindings with names that have a `system:` prefix. Modifications to these resources can result in non-functional clusters.

## Auto-reconciliation

At each start-up, the API server updates default cluster roles with any missing permissions, and updates default cluster role bindings with any missing subjects. This allows the cluster to repair accidental modifications, and helps to keep roles and role bindings up-to-date as permissions and subjects change in new Kubernetes releases.

To opt out of this reconciliation, set the `rbac.authorization.kubernetes.io/autoupdate` annotation on a default cluster role or rolebinding to false. Be aware that missing default permissions and subjects can result in non-functional clusters.

Auto-reconciliation is enabled by default if the RBAC authorizer is active.

## API discovery roles

Default role bindings authorize unauthenticated and authenticated users to read API information that is deemed safe to be publicly accessible (including CustomResourceDefinitions). To disable anonymous unauthenticated access, add `--anonymous-auth=false` to the API server configuration.

To view the configuration of these roles via `kubectl` run:

```
kubectl get clusterroles system:discovery -o yaml
```

### Note:

If you edit that ClusterRole, your changes will be overwritten on API server restart via [auto-reconciliation](#). To avoid that overwriting, either do not manually edit the role, or disable auto-reconciliation.

Kubernetes RBAC API discovery roles

| Default ClusterRole              | Default ClusterRoleBinding                                           | Description                                                                                                                                                                                   |
|----------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>system:basic-user</b>         | <b>system:authenticated</b> group                                    | Allows a user read-only access to basic information about themselves. Prior to v1.14, this role was also bound to <code>system:unauthenticated</code> by default.                             |
| <b>system:discovery</b>          | <b>system:authenticated</b> group                                    | Allows read-only access to API discovery endpoints needed to discover and negotiate an API level. Prior to v1.14, this role was also bound to <code>system:unauthenticated</code> by default. |
| <b>system:public-info-viewer</b> | <b>system:authenticated</b> and <b>system:unauthenticated</b> groups | Allows read-only access to non-sensitive information about the cluster. Introduced in Kubernetes v1.14.                                                                                       |

## User-facing roles

Some of the default ClusterRoles are not `system:` prefixed. These are intended to be user-facing roles. They include super-user roles (cluster-admin), roles intended to be granted cluster-wide using ClusterRoleBindings, and roles intended to be granted within particular namespaces using RoleBindings (admin, edit, view).

User-facing ClusterRoles use [ClusterRole aggregation](#) to allow admins to include rules for custom resources on these ClusterRoles. To add rules to the admin, edit, or view roles, create a ClusterRole with one or more of the following labels:

metadata:

labels:

rbac.authorization.k8s.io/aggregate-to-admin: "true"

rbac.authorization.k8s.io/aggregate-to-edit: "true"

rbac.authorization.k8s.io/aggregate-to-view: "true"

| Default ClusterRole  | Default ClusterRoleBinding  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>cluster-admin</b> | <b>system:masters</b> group | Allows super-user access to perform any action on any resource. When used in a <b>ClusterRoleBinding</b> , it gives full control over every resource in the cluster and in all namespaces. When used in a <b>RoleBinding</b> , it gives full control over every resource in the role binding's namespace, including the namespace itself.                                                                                                                                                                                                                                                                     |
| <b>admin</b>         | None                        | <p>Allows admin access, intended to be granted within a namespace using a <b>RoleBinding</b>.</p> <p>If used in a <b>RoleBinding</b>, allows read/write access to most resources in a namespace, including the ability to create roles and role bindings within the namespace. This role does not allow write access to resource quota or to the namespace itself. This role also does not allow write access to EndpointSlices (or Endpoints) in clusters created using Kubernetes v1.22+. More information is available in the <a href="#">"Write Access for EndpointSlices and Endpoints" section</a>.</p> |
| <b>edit</b>          | None                        | <p>Allows read/write access to most objects in a namespace.</p> <p>This role does not allow viewing or modifying roles or role bindings. However, this role allows accessing Secrets and running Pods as any ServiceAccount in the namespace, so it can be used to gain the API access levels of any ServiceAccount in the namespace. This role also does not allow write access to EndpointSlices (or Endpoints) in clusters created using Kubernetes v1.22+. More information is available in the <a href="#">"Write Access for EndpointSlices and Endpoints" section</a>.</p>                              |
| <b>view</b>          | None                        | <p>Allows read-only access to see most objects in a namespace. It does not allow viewing roles or role bindings.</p> <p>This role does not allow viewing Secrets, since reading the contents of Secrets enables access to ServiceAccount credentials in the namespace, which would allow API access as any ServiceAccount in the namespace (a form of privilege escalation).</p>                                                                                                                                                                                                                              |

## Core component roles

| Default ClusterRole                         | Default ClusterRoleBinding                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>system:kube-scheduler</code>          | <code>system:kube-scheduler</code> user          | Allows access to the resources required by the <a href="#">scheduler</a> component.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>system:volume-scheduler</code>        | <code>system:kube-scheduler</code> user          | Allows access to the volume resources required by the kube-scheduler component.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>system:kube-controller-manager</code> | <code>system:kube-controller-manager</code> user | Allows access to the resources required by the <a href="#">controller manager</a> component. The permissions required by individual controllers are detailed in the <a href="#">controller roles</a> .                                                                                                                                                                                                                                                                                                                                       |
| <code>system:node</code>                    | None                                             | <p>Allows access to resources required by the kubelet, <b>including read access to all secrets, and write access to all pod status objects.</b></p> <p>You should use the <a href="#">Node authorizer</a> and <a href="#">NodeRestriction admission plugin</a> instead of the <code>system:node</code> role, and allow granting API access to kubelets based on the Pods scheduled to run on them.</p> <p>The <code>system:node</code> role only exists for compatibility with Kubernetes clusters upgraded from versions prior to v1.8.</p> |
| <code>system:node-proxier</code>            | <code>system:kube-proxy</code> user              | Allows access to the resources required by the <a href="#">kube-proxy</a> component.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Other component roles

| Default ClusterRole                       | Default ClusterRoleBinding                                                      | Description                                                                                                                                         |
|-------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>system:auth-delegator</code>        | None                                                                            | Allows delegated authentication and authorization checks. This is commonly used by add-on API servers for unified authentication and authorization. |
| <code>system:heapster</code>              | None                                                                            | Role for the <a href="#">Heapster</a> component (deprecated).                                                                                       |
| <code>system:kube-aggregator</code>       | None                                                                            | Role for the <a href="#">kube-aggregator</a> component.                                                                                             |
| <code>system:kube-dns</code>              | <code>kube-dns</code> service account in the <code>kube-system</code> namespace | Role for the <a href="#">kube-dns</a> component.                                                                                                    |
| <code>system:kubelet-api-admin</code>     | None                                                                            | Allows full access to the kubelet API.                                                                                                              |
| <code>system:node-bootstrapper</code>     | None                                                                            | Allows access to the resources required to perform <a href="#">kubelet TLS bootstrapping</a> .                                                      |
| <code>system:node-problem-detector</code> | None                                                                            | Role for the <a href="#">node-problem-detector</a> component.                                                                                       |

| Default ClusterRole                         | Default ClusterRoleBinding     | Description                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>system:persistent-volume-provisioner</b> | None                           | Allows access to the resources required by most <a href="#">dynamic volume provisioners</a> .                                                                                                                                                                                                                                                             |
| <b>system:monitoring</b>                    | <b>system:monitoring</b> group | Allows read access to control-plane monitoring endpoints (i.e. <a href="#">kube-apiserver</a> liveness and readiness endpoints (/healthz, /livez, /readyz), the individual health-check endpoints (/healthz/*, /livez/*, /readyz/*), and /metrics). Note that individual health check endpoints and the metric endpoint may expose sensitive information. |

## Roles for built-in controllers

The Kubernetes [controller manager](#) runs [controllers](#) that are built in to the Kubernetes control plane. When invoked with `--use-service-account-credentials`, kube-controller-manager starts each controller using a separate service account. Corresponding roles exist for each built-in controller, prefixed with `system:controller:`. If the controller manager is not started with `--use-service-account-credentials`, it runs all control loops using its own credential, which must be granted all the relevant roles. These roles include:

- system:controller:attachdetach-controller
- system:controller:certificate-controller
- system:controller:clusterrole-aggregation-controller
- system:controller:cronjob-controller
- system:controller:daemon-set-controller
- system:controller:deployment-controller
- system:controller:disruption-controller
- system:controller:endpoint-controller
- system:controller:expand-controller
- system:controller:generic-garbage-collector
- system:controller:horizontal-pod-autoscaler
- system:controller:job-controller
- system:controller:namespace-controller
- system:controller:node-controller
- system:controller:persistent-volume-binder
- system:controller:pod-garbage-collector
- system:controller:pv-protection-controller
- system:controller:pvc-protection-controller
- system:controller:replicaset-controller
- system:controller:replication-controller
- system:controller:resourcequota-controller
- system:controller:root-ca-cert-publisher
- system:controller:route-controller
- system:controller:service-account-controller
- system:controller:service-controller
- system:controller:statefulset-controller
- system:controller:ttl-controller

# Privilege escalation prevention and bootstrapping

The RBAC API prevents users from escalating privileges by editing roles or role bindings. Because this is enforced at the API level, it applies even when the RBAC authorizer is not in use.

## Restrictions on role creation or update

You can only create/update a role if at least one of the following things is true:

1. You already have all the permissions contained in the role, at the same scope as the object being modified (cluster-wide for a ClusterRole, within the same namespace or cluster-wide for a Role).
2. You are granted explicit permission to perform the `escalate` verb on the `roles` or `clusterroles` resource in the `rbac.authorization.k8s.io` API group.

For example, if user-1 does not have the ability to list Secrets cluster-wide, they cannot create a ClusterRole containing that permission. To allow a user to create/update roles:

1. Grant them a role that allows them to create/update Role or ClusterRole objects, as desired.
2. Grant them permission to include specific permissions in the roles they create/update:
  - implicitly, by giving them those permissions (if they attempt to create or modify a Role or ClusterRole with permissions they themselves have not been granted, the API request will be forbidden)
  - or explicitly allow specifying any permission in a Role or ClusterRole by giving them permission to perform the `escalate` verb on `roles` or `clusterroles` resources in the `rbac.authorization.k8s.io` API group

## Restrictions on role binding creation or update

You can only create/update a role binding if you already have all the permissions contained in the referenced role (at the same scope as the role binding) *or* if you have been authorized to perform the `bind` verb on the referenced role. For example, if user-1 does not have the ability to list Secrets cluster-wide, they cannot create a ClusterRoleBinding to a role that grants that permission. To allow a user to create/update role bindings:

1. Grant them a role that allows them to create/update RoleBinding or ClusterRoleBinding objects, as desired.
2. Grant them permissions needed to bind a particular role:
  - implicitly, by giving them the permissions contained in the role.
  - explicitly, by giving them permission to perform the `bind` verb on the particular Role (or ClusterRole).

For example, this ClusterRole and RoleBinding would allow user-1 to grant other users the `admin`, `edit`, and `view` roles in the namespace `user-1-namespace`:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: role-grantor
rules:
- apiGroups: ["rbac.authorization.k8s.io"]
```

```

resources: ["rolebindings"]
verbs: ["create"]
- apiGroups: ["rbac.authorization.k8s.io"]
resources: ["clusterroles"]
verbs: ["bind"]
omit resourceNames to allow binding any ClusterRole
resourceNames: ["admin","edit","view"]

apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
 name: role-grantor-binding
 namespace: user-1-namespace
roleRef:
 apiGroup: rbac.authorization.k8s.io
 kind: ClusterRole
 name: role-grantor
subjects:
- apiGroup: rbac.authorization.k8s.io
 kind: User
 name: user-1

```

When bootstrapping the first roles and role bindings, it is necessary for the initial user to grant permissions they do not yet have. To bootstrap initial roles and role bindings:

- Use a credential with the "system:masters" group, which is bound to the "cluster-admin" super-user role by the default bindings.

## Command-line utilities

### kubectl create role

Creates a Role object defining permissions within a single namespace. Examples:

- Create a Role named "pod-reader" that allows users to perform get, watch and list on pods:

```
kubectl create role pod-reader --verb=get --verb=list --verb=watch --resource=pods
```

- Create a Role named "pod-reader" with resourceNames specified:

```
kubectl create role pod-reader --verb=get --resource=pods --resource-name=readablepod
--resource-name=anotherpod
```

- Create a Role named "foo" with apiGroups specified:

```
kubectl create role foo --verb=get,list,watch --resource=replicasets.apps
```

- Create a Role named "foo" with subresource permissions:

```
kubectl create role foo --verb=get,list,watch --resource=pods,pods/status
```

- Create a Role named "my-component-lease-holder" with permissions to get/update a resource with a specific name:

```
kubectl create role my-component-lease-holder --verb=get,list,watch,update --resource=lease --resource-name=my-component
```

## kubectl create clusterrole

Creates a ClusterRole. Examples:

- Create a ClusterRole named "pod-reader" that allows user to perform get, watch and list on pods:

```
kubectl create clusterrole pod-reader --verb=get,list,watch --resource=pods
```

- Create a ClusterRole named "pod-reader" with resourceNames specified:

```
kubectl create clusterrole pod-reader --verb=get --resource=pods --resource-name=readablepod --resource-name=anotherpod
```

- Create a ClusterRole named "foo" with apiGroups specified:

```
kubectl create clusterrole foo --verb=get,list,watch --resource=replicasets.apps
```

- Create a ClusterRole named "foo" with subresource permissions:

```
kubectl create clusterrole foo --verb=get,list,watch --resource=pods,pods/status
```

- Create a ClusterRole named "foo" with nonResourceURL specified:

```
kubectl create clusterrole "foo" --verb=get --non-resource-url=/logs/*
```

- Create a ClusterRole named "monitoring" with an aggregationRule specified:

```
kubectl create clusterrole monitoring --aggregation-rule="rbac.example.com/aggregate-to-monitoring=true"
```

## kubectl create rolebinding

Grants a Role or ClusterRole within a specific namespace. Examples:

- Within the namespace "acme", grant the permissions in the "admin" ClusterRole to a user named "bob":

```
kubectl create rolebinding bob-admin-binding --clusterrole=admin --user=bob --namespace=acme
```

- Within the namespace "acme", grant the permissions in the "view" ClusterRole to the service account in the namespace "acme" named "myapp":

```
kubectl create rolebinding myapp-view-binding --clusterrole=view --serviceaccount=acme:myapp --namespace=acme
```

- Within the namespace "acme", grant the permissions in the "view" ClusterRole to a service account in the namespace "myappnamespace" named "myapp":



```
kubectl create rolebinding myappnamespace-myapp-view-binding --clusterrole=view --serviceaccount=myappnamespace:myapp --namespace=acme
```

## kubectl create clusterrolebinding

Grants a ClusterRole across the entire cluster (all namespaces). Examples:

- Across the entire cluster, grant the permissions in the "cluster-admin" ClusterRole to a user named "root":

```
kubectl create clusterrolebinding root-cluster-admin-binding --clusterrole=cluster-admin --user=root
```

- Across the entire cluster, grant the permissions in the "system:node-proxier" ClusterRole to a user named "system:kube-proxy":

```
kubectl create clusterrolebinding kube-proxy-binding --clusterrole=system:node-proxier --user=system:kube-proxy
```

- Across the entire cluster, grant the permissions in the "view" ClusterRole to a service account named "myapp" in the namespace "acme":

```
kubectl create clusterrolebinding myapp-view-binding --clusterrole=view --serviceaccount=acme:myapp
```

## kubectl auth reconcile

Creates or updates rbac.authorization.k8s.io/v1 API objects from a manifest file.

Missing objects are created, and the containing namespace is created for namespaced objects, if required.

Existing roles are updated to include the permissions in the input objects, and remove extra permissions if `--remove-extra-permissions` is specified.

Existing bindings are updated to include the subjects in the input objects, and remove extra subjects if `--remove-extra-subjects` is specified.

Examples:

- Test applying a manifest file of RBAC objects, displaying changes that would be made:

```
kubectl auth reconcile -f my-rbac-rules.yaml --dry-run=client
```

- Apply a manifest file of RBAC objects, preserving any extra permissions (in roles) and any extra subjects (in bindings):

```
kubectl auth reconcile -f my-rbac-rules.yaml
```

- Apply a manifest file of RBAC objects, removing any extra permissions (in roles) and any extra subjects (in bindings):

```
kubectl auth reconcile -f my-rbac-rules.yaml --remove-extra-subjects --remove-extra-permissions
```

# ServiceAccount permissions

Default RBAC policies grant scoped permissions to control-plane components, nodes, and controllers, but grant *no permissions* to service accounts outside the kube-system namespace (beyond discovery permissions given to all authenticated users).

This allows you to grant particular roles to particular ServiceAccounts as needed. Fine-grained role bindings provide greater security, but require more effort to administrate. Broader grants can give unnecessary (and potentially escalating) API access to ServiceAccounts, but are easier to administrate.

In order from most secure to least secure, the approaches are:

1. Grant a role to an application-specific service account (best practice)

This requires the application to specify a `serviceAccountName` in its pod spec, and for the service account to be created (via the API, application manifest, `kubectl create serviceaccount`, etc.).

For example, grant read-only permission within "my-namespace" to the "my-sa" service account:

```
kubectl create rolebinding my-sa-view \
 --clusterrole=view \
 --serviceaccount=my-namespace:my-sa \
 --namespace=my-namespace
```

2. Grant a role to the "default" service account in a namespace

If an application does not specify a `serviceAccountName`, it uses the "default" service account.

**Note:**

Permissions given to the "default" service account are available to any pod in the namespace that does not specify a `serviceAccountName`.

For example, grant read-only permission within "my-namespace" to the "default" service account:

```
kubectl create rolebinding default-view \
 --clusterrole=view \
 --serviceaccount=my-namespace:default \
 --namespace=my-namespace
```

Many [add-ons](#) run as the "default" service account in the kube-system namespace. To allow those add-ons to run with super-user access, grant cluster-admin permissions to the "default" service account in the kube-system namespace.

**Caution:**

Enabling this means the kube-system namespace contains Secrets that grant super-user access to your cluster's API.

```
kubectl create clusterrolebinding add-on-cluster-admin \
 --clusterrole=cluster-admin \
 --serviceaccount=kube-system:default
```

### 3. Grant a role to all service accounts in a namespace

If you want all applications in a namespace to have a role, no matter what service account they use, you can grant a role to the service account group for that namespace.

For example, grant read-only permission within "my-namespace" to all service accounts in that namespace:

```
kubectl create rolebinding serviceaccounts-view \
 --clusterrole=view \
 --group=system:serviceaccounts:my-namespace \
 --namespace=my-namespace
```

### 4. Grant a limited role to all service accounts cluster-wide (discouraged)

If you don't want to manage permissions per-namespace, you can grant a cluster-wide role to all service accounts.

For example, grant read-only permission across all namespaces to all service accounts in the cluster:

```
kubectl create clusterrolebinding serviceaccounts-view \
 --clusterrole=view \
 --group=system:serviceaccounts
```

### 5. Grant super-user access to all service accounts cluster-wide (strongly discouraged)

If you don't care about partitioning permissions at all, you can grant super-user access to all service accounts.

#### **Warning:**

This allows any application full access to your cluster, and also grants any user with read access to Secrets (or the ability to create any pod) full access to your cluster.

```
kubectl create clusterrolebinding serviceaccounts-cluster-admin \
 --clusterrole=cluster-admin \
 --group=system:serviceaccounts
```

## Write access for EndpointSlices and Endpoints

Kubernetes clusters created before Kubernetes v1.22 include write access to EndpointSlices (and Endpoints) in the aggregated "edit" and "admin" roles. As a mitigation for [CVE-2021-25740](#), this access is not part of the aggregated roles in clusters that you create using Kubernetes v1.22 or later.

Existing clusters that have been upgraded to Kubernetes v1.22 will not be subject to this change. The [CVE announcement](#) includes guidance for restricting this access in existing clusters.

If you want new clusters to retain this level of access in the aggregated roles, you can create the following ClusterRole:

[access/endpoints-aggregated.yaml](#)

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 annotations:
 kubernetes.io/description: |-
 Add endpoints write permissions to the edit and admin roles. This was
 removed by default in 1.22 because of CVE-2021-25740. See
 https://issue.k8s.io/103675. This can allow writers to direct LoadBalancer
 or Ingress implementations to expose backend IPs that would not otherwise
 be accessible, and can circumvent network policies or security controls
 intended to prevent/isolate access to those backends.
 EndpointSlices were never included in the edit or admin roles, so there
 is nothing to restore for the EndpointSlice API.
 labels:
 rbac.authorization.k8s.io/aggregate-to-edit: "true"
name: custom:aggregate-to-edit:endpoints # you can change this if you wish
rules:
- apiGroups: [""]
 resources: ["endpoints"]
 verbs: ["create", "delete", "deletecollection", "patch", "update"]
```

## Upgrading from ABAC

Clusters that originally ran older Kubernetes versions often used permissive ABAC policies, including granting full API access to all service accounts.

Default RBAC policies grant scoped permissions to control-plane components, nodes, and controllers, but grant *no permissions* to service accounts outside the kube-system namespace (beyond discovery permissions given to all authenticated users).

While far more secure, this can be disruptive to existing workloads expecting to automatically receive API permissions. Here are two approaches for managing this transition:

### Parallel authorizers

Run both the RBAC and ABAC authorizers, and specify a policy file that contains the [legacy ABAC policy](#):

```
--authorization-mode=...,RBAC,ABAC --authorization-policy-file=mypolicy.json
```

To explain that first command line option in detail: if earlier authorizers, such as Node, deny a request, then the RBAC authorizer attempts to authorize the API request. If RBAC also denies that API request, the ABAC authorizer is then run. This means that any request allowed by *either* the RBAC or ABAC policies is allowed.

When the kube-apiserver is run with a log level of 5 or higher for the RBAC component (`--vmodule=rbac*=5` or `--v=5`), you can see RBAC denials in the API server log (prefixed with

RBAC). You can use that information to determine which roles need to be granted to which users, groups, or service accounts.

Once you have [granted roles to service accounts](#) and workloads are running with no RBAC denial messages in the server logs, you can remove the ABAC authorizer.

## Permissive RBAC permissions

You can replicate a permissive ABAC policy using RBAC role bindings.

### Warning:

The following policy allows **ALL** service accounts to act as cluster administrators. Any application running in a container receives service account credentials automatically, and could perform any action against the API, including viewing secrets and modifying permissions. This is not a recommended policy.

```
kubectl create clusterrolebinding permissive-binding \
--clusterrole=cluster-admin \
--user=admin \
--user=kubelet \
--group=system:serviceaccounts
```

After you have transitioned to use RBAC, you should adjust the access controls for your cluster to ensure that these meet your information security needs.

# Using Node Authorization

Node authorization is a special-purpose authorization mode that specifically authorizes API requests made by kubelets.

## Overview

The Node authorizer allows a kubelet to perform API operations. This includes:

Read operations:

- services
- endpoints
- nodes
- pods
- secrets, configmaps, persistent volume claims and persistent volumes related to pods bound to the kubelet's node

Write operations:

- nodes and node status (enable the NodeRestriction admission plugin to limit a kubelet to modify its own node)
- pods and pod status (enable the NodeRestriction admission plugin to limit a kubelet to modify pods bound to itself)
- events

Auth-related operations:

- read/write access to the [CertificateSigningRequests API](#) for TLS bootstrapping
- the ability to create TokenReviews and SubjectAccessReviews for delegated authentication/authorization checks

In future releases, the node authorizer may add or remove permissions to ensure kubelets have the minimal set of permissions required to operate correctly.

In order to be authorized by the Node authorizer, kubelets must use a credential that identifies them as being in the system:nodes group, with a username of system:node:<nodeName>. This group and user name format match the identity created for each kubelet as part of [kubelet TLS bootstrapping](#).

The value of <nodeName> **must** match precisely the name of the node as registered by the kubelet. By default, this is the host name as provided by hostname, or overridden via the [kubelet option](#) `--hostname-override`. However, when using the `--cloud-provider` kubelet option, the specific hostname may be determined by the cloud provider, ignoring the local hostname and the `--hostname-override` option. For specifics about how the kubelet determines the hostname, see the [kubelet options reference](#).

To enable the Node authorizer, start the apiserver with `--authorization-mode=Node`.

To limit the API objects kubelets are able to write, enable the [NodeRestriction](#) admission plugin by starting the apiserver with `--enable-admission-plugins=...,NodeRestriction,...`

## Migration considerations

### Kubelets outside the system:nodes group

Kubelets outside the system:nodes group would not be authorized by the Node authorization mode, and would need to continue to be authorized via whatever mechanism currently authorizes them. The node admission plugin would not restrict requests from these kubelets.

### Kubelets with undifferentiated usernames

In some deployments, kubelets have credentials that place them in the system:nodes group, but do not identify the particular node they are associated with, because they do not have a username in the system:node:... format. These kubelets would not be authorized by the Node authorization mode, and would need to continue to be authorized via whatever mechanism currently authorizes them.

The NodeRestriction admission plugin would ignore requests from these kubelets, since the default node identifier implementation would not consider that a node identity.

## Webhook Mode

A WebHook is an HTTP callback: an HTTP POST that occurs when something happens; a simple event-notification via HTTP POST. A web application implementing WebHooks will POST a message to a URL when certain things happen.

When specified, mode Webhook causes Kubernetes to query an outside REST service when determining user privileges.

## Configuration File Format

Mode Webhook requires a file for HTTP configuration, specify by the `--authorization-webhook-config-file=SOME_FILENAME` flag.

The configuration file uses the [kubeconfig](#) file format. Within the file "users" refers to the API Server webhook and "clusters" refers to the remote service.

A configuration example which uses HTTPS client auth:

```
Kubernetes API version
apiVersion: v1
kind of the API object
kind: Config
clusters refers to the remote service.
clusters:
- name: name-of-remote-authz-service
 cluster:
 # CA for verifying the remote service.
 certificate-authority: /path/to/ca.pem
 # URL of remote service to query. Must use 'https'. May not include parameters.
 server: https://authz.example.com/authorize

users refers to the API Server's webhook configuration.
users:
- name: name-of-api-server
 user:
 client-certificate: /path/to/cert.pem # cert for the webhook plugin to use
 client-key: /path/to/key.pem # key matching the cert

kubeconfig files require a context. Provide one for the API Server.
current-context: webhook
contexts:
- context:
 cluster: name-of-remote-authz-service
 user: name-of-api-server
 name: webhook
```

## Request Payloads

When faced with an authorization decision, the API Server POSTs a JSON- serialized `authorization.k8s.io/v1beta1 SubjectAccessReview` object describing the action. This object contains fields describing the user attempting to make the request, and either details about the resource being accessed or requests attributes.

Note that webhook API objects are subject to the same [versioning compatibility rules](#) as other Kubernetes API objects. Implementers should be aware of looser compatibility promises for beta objects and check the "apiVersion" field of the request to ensure correct deserialization.

Additionally, the API Server must enable the `authorization.k8s.io/v1beta1` API extensions group (`--runtime-config=authorization.k8s.io/v1beta1=true`).

An example request body:

```
{
 "apiVersion": "authorization.k8s.io/v1beta1",
 "kind": "SubjectAccessReview",
 "spec": {
 "resourceAttributes": {
 "namespace": "kittensandponies",
 "verb": "get",
 "group": "unicorn.example.org",
 "resource": "pods"
 },
 "user": "jane",
 "group": [
 "group1",
 "group2"
]
 }
}
```

The remote service is expected to fill the `status` field of the request and respond to either allow or disallow access. The response body's `spec` field is ignored and may be omitted. A permissive response would return:

```
{
 "apiVersion": "authorization.k8s.io/v1beta1",
 "kind": "SubjectAccessReview",
 "status": {
 "allowed": true
 }
}
```

For disallowing access there are two methods.

The first method is preferred in most cases, and indicates the authorization webhook does not allow, or has "no opinion" about the request, but if other authorizers are configured, they are given a chance to allow the request. If there are no other authorizers, or none of them allow the request, the request is forbidden. The webhook would return:

```
{
 "apiVersion": "authorization.k8s.io/v1beta1",
 "kind": "SubjectAccessReview",
 "status": {
 "allowed": false,
 "reason": "user does not have read access to the namespace"
 }
}
```

The second method denies immediately, short-circuiting evaluation by other configured authorizers. This should only be used by webhooks that have detailed knowledge of the full authorizer configuration of the cluster. The webhook would return:



```
{
 "apiVersion": "authorization.k8s.io/v1beta1",
 "kind": "SubjectAccessReview",
 "status": {
 "allowed": false,
 "denied": true,
 "reason": "user does not have read access to the namespace"
 }
}
```

Access to non-resource paths are sent as:

```
{
 "apiVersion": "authorization.k8s.io/v1beta1",
 "kind": "SubjectAccessReview",
 "spec": {
 "nonResourceAttributes": {
 "path": "/debug",
 "verb": "get"
 },
 "user": "jane",
 "group": [
 "group1",
 "group2"
]
 }
}
```

Non-resource paths include: `/api`, `/apis`, `/metrics`, `/logs`, `/debug`, `/healthz`, `/livez`, `/openapi/v2`, `/readyz`, and `/version`. Clients require access to `/api`, `/api/*`, `/apis`, `/apis/*`, and `/version` to discover what resources and versions are present on the server. Access to other non-resource paths can be disallowed without restricting access to the REST api.

For further documentation refer to the `authorization.v1beta1` API objects and [webhook.go](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.16/#subjectaccessreview-authorization-k8s-io-v1beta1).

## Using ABAC Authorization

Attribute-based access control (ABAC) defines an access control paradigm whereby access rights are granted to users through the use of policies which combine attributes together.

### Policy File Format

To enable ABAC mode, specify `--authorization-policy-file=SOME_FILENAME` and `--authorization-mode=ABAC` on startup.

The file format is [one JSON object per line](#). There should be no enclosing list or map, only one map per line.

Each line is a "policy object", where each such object is a map with the following properties:

- Versioning properties:
  - `apiVersion`, type string; valid values are "abac.authorization.kubernetes.io/v1beta1". Allows versioning and conversion of the policy format.
  - `kind`, type string; valid values are "Policy". Allows versioning and conversion of the policy format.
- `spec` property set to a map with the following properties:
  - Subject-matching properties:
    - `user`, type string; the user-string from `--token-auth-file`. If you specify `user`, it must match the username of the authenticated user.
    - `group`, type string; if you specify `group`, it must match one of the groups of the authenticated user. `system:authenticated` matches all authenticated requests. `system:unauthenticated` matches all unauthenticated requests.
  - Resource-matching properties:
    - `apiGroup`, type string; an API group.
      - Ex: `apps`, `networking.k8s.io`
      - Wildcard: `*` matches all API groups.
    - `namespace`, type string; a namespace.
      - Ex: `kube-system`
      - Wildcard: `*` matches all resource requests.
    - `resource`, type string; a resource type
      - Ex: `pods`, `deployments`
      - Wildcard: `*` matches all resource requests.
  - Non-resource-matching properties:
    - `nonResourcePath`, type string; non-resource request paths.
      - Ex: `/version` or `/apis`
      - Wildcard:
        - `*` matches all non-resource requests.
        - `/foo/*` matches all subpaths of `/foo/`.
  - `readOnly`, type boolean, when true, means that the Resource-matching policy only applies to get, list, and watch operations, Non-resource-matching policy only applies to get operation.

#### Note:

An unset property is the same as a property set to the zero value for its type (e.g. empty string, 0, false). However, unset should be preferred for readability.

In the future, policies may be expressed in a JSON format, and managed via a REST interface.

## Authorization Algorithm

A request has attributes which correspond to the properties of a policy object.

When a request is received, the attributes are determined. Unknown attributes are set to the zero value of its type (e.g. empty string, 0, false).

A property set to `"**"` will match any value of the corresponding attribute.

The tuple of attributes is checked for a match against every policy in the policy file. If at least one line matches the request attributes, then the request is authorized (but may fail later validation).

To permit any authenticated user to do something, write a policy with the group property set to "system:authenticated".

To permit any unauthenticated user to do something, write a policy with the group property set to "system:unauthenticated".

To permit a user to do anything, write a policy with the apiGroup, namespace, resource, and nonResourcePath properties set to "\*".

## Kubectl

Kubectl uses the /api and /apis endpoints of apiserver to discover served resource types, and validates objects sent to the API by create/update operations using schema information located at /openapi/v2.

When using ABAC authorization, those special resources have to be explicitly exposed via the nonResourcePath property in a policy (see [examples](#) below):

- /api, /api/\*, /apis, and /apis/\* for API version negotiation.
- /version for retrieving the server version via kubectl version.
- /swaggerapi/\* for create/update operations.

To inspect the HTTP calls involved in a specific kubectl operation you can turn up the verbosity:

```
kubectl --v=8 version
```

## Examples

1. Alice can do anything to all resources:

```
{"apiVersion": "abac.authorization.kubernetes.io/v1beta1", "kind": "Policy", "spec": {"user": "alice", "namespace": "*", "resource": "*", "apiGroup": "*"}}
```

2. The kubelet can read any pods:

```
{"apiVersion": "abac.authorization.kubernetes.io/v1beta1", "kind": "Policy", "spec": {"user": "kubelet", "namespace": "*", "resource": "pods", "readonly": true}}
```

3. The kubelet can read and write events:

```
{"apiVersion": "abac.authorization.kubernetes.io/v1beta1", "kind": "Policy", "spec": {"user": "kubelet", "namespace": "*", "resource": "events"}}
```

4. Bob can just read pods in namespace "projectCaribou":

```
{"apiVersion": "abac.authorization.kubernetes.io/v1beta1", "kind": "Policy", "spec": {"user": "bob", "namespace": "projectCaribou", "resource": "pods", "readonly": true}}
```

Anyone can make read-only requests to all non-resource paths:

5.

```
{"apiVersion": "abac.authorization.kubernetes.io/v1beta1", "kind": "Policy", "spec":
 {"group": "system:authenticated", "readonly": true, "nonResourcePath": "**"}}
{"apiVersion": "abac.authorization.kubernetes.io/v1beta1", "kind": "Policy", "spec": {"group
 ": "system:unauthenticated", "readonly": true, "nonResourcePath": "**"}}
```

[Complete file example](#)

## A quick note on service accounts

Every service account has a corresponding ABAC username, and that service account's username is generated according to the naming convention:

```
system:serviceaccount:<namespace>:<serviceaccountname>
```

Creating a new namespace leads to the creation of a new service account in the following format:

```
system:serviceaccount:<namespace>:default
```

For example, if you wanted to grant the default service account (in the kube-system namespace) full privilege to the API using ABAC, you would add this line to your policy file:

```
{"apiVersion": "abac.authorization.kubernetes.io/v1beta1", "kind": "Policy", "spec": {"user": "system:se
rviceaccount:kube-system:default", "namespace": "**", "resource": "**", "apiGroup": "**"}}
```

The apiserver will need to be restarted to pick up the new policy lines.

# Admission Controllers Reference

This page provides an overview of Admission Controllers.

## What are they?

An *admission controller* is a piece of code that intercepts requests to the Kubernetes API server prior to persistence of the object, but after the request is authenticated and authorized.

Admission controllers may be *validating*, *mutating*, or both. Mutating controllers may modify objects related to the requests they admit; validating controllers may not.

Admission controllers limit requests to create, delete, modify objects. Admission controllers can also block custom verbs, such as a request connect to a Pod via an API server proxy. Admission controllers do *not* (and cannot) block requests to read (**get**, **watch** or **list**) objects.

The admission controllers in Kubernetes 1.30 consist of the [list](#) below, are compiled into the kube-apiserver binary, and may only be configured by the cluster administrator. In that list, there are two special controllers: MutatingAdmissionWebhook and ValidatingAdmissionWebhook. These execute the mutating and validating (respectively) [admission control webhooks](#) which are configured in the API.

## Admission control phases

The admission control process proceeds in two phases. In the first phase, mutating admission controllers are run. In the second phase, validating admission controllers are run. Note again that some of the controllers are both.

If any of the controllers in either phase reject the request, the entire request is rejected immediately and an error is returned to the end-user.

Finally, in addition to sometimes mutating the object in question, admission controllers may sometimes have side effects, that is, mutate related resources as part of request processing. Incrementing quota usage is the canonical example of why this is necessary. Any such side-effect needs a corresponding reclamation or reconciliation process, as a given admission controller does not know for sure that a given request will pass all of the other admission controllers.

## Why do I need them?

Several important features of Kubernetes require an admission controller to be enabled in order to properly support the feature. As a result, a Kubernetes API server that is not properly configured with the right set of admission controllers is an incomplete server and will not support all the features you expect.

## How do I turn on an admission controller?

The Kubernetes API server flag `enable-admission-plugins` takes a comma-delimited list of admission control plugins to invoke prior to modifying objects in the cluster. For example, the following command line enables the `NamespaceLifecycle` and the `LimitRanger` admission control plugins:

```
kube-apiserver --enable-admission-plugins=NamespaceLifecycle,LimitRanger ...
```

### Note:

Depending on the way your Kubernetes cluster is deployed and how the API server is started, you may need to apply the settings in different ways. For example, you may have to modify the systemd unit file if the API server is deployed as a systemd service, you may modify the manifest file for the API server if Kubernetes is deployed in a self-hosted way.

## How do I turn off an admission controller?

The Kubernetes API server flag `disable-admission-plugins` takes a comma-delimited list of admission control plugins to be disabled, even if they are in the list of plugins enabled by default.

```
kube-apiserver --disable-admission-plugins=PodNodeSelector,AlwaysDeny ...
```

## Which plugins are enabled by default?

To see which admission plugins are enabled:

```
kube-apiserver -h | grep enable-admission-plugins
```

In Kubernetes 1.30, the default ones are:

CertificateApproval, CertificateSigning, CertificateSubjectRestriction, DefaultIngressClass, DefaultStorageClass, DefaultTolerationSeconds, LimitRanger, MutatingAdmissionWebhook, NamespaceLifecycle, PersistentVolumeClaimResize, PodSecurity, Priority, ResourceQuota, RuntimeClass, ServiceAccount, StorageObjectInUseProtection, TaintNodesByCondition, ValidatingAdmissionPolicy, ValidatingAdmissionWebhook

#### Note:

The [ValidatingAdmissionPolicy](#) admission plugin is enabled by default, but is only active if you enable the ValidatingAdmissionPolicy [feature gate](#) **and** the admissionregistration.k8s.io/v1alpha1 API.

## What does each admission controller do?

### AlwaysAdmit

FEATURE STATE: Kubernetes v1.13 [deprecated]

**Type:** Validating.

This admission controller allows all pods into the cluster. It is **deprecated** because its behavior is the same as if there were no admission controller at all.

### AlwaysDeny

FEATURE STATE: Kubernetes v1.13 [deprecated]

**Type:** Validating.

Rejects all requests. AlwaysDeny is **deprecated** as it has no real meaning.

### AlwaysPullImages

**Type:** Mutating and Validating.

This admission controller modifies every new Pod to force the image pull policy to Always. This is useful in a multitenant cluster so that users can be assured that their private images can only be used by those who have the credentials to pull them. Without this admission controller, once an image has been pulled to a node, any pod from any user can use it by knowing the image's name (assuming the Pod is scheduled onto the right node), without any authorization check against the image. When this admission controller is enabled, images are always pulled prior to starting containers, which means valid credentials are required.

### CertificateApproval

**Type:** Validating.

This admission controller observes requests to approve `CertificateSigningRequest` resources and performs additional authorization checks to ensure the approving user has permission to **approve** certificate requests with the `spec.signerName` requested on the `CertificateSigningRequest` resource.

See [Certificate Signing Requests](#) for more information on the permissions required to perform different actions on `CertificateSigningRequest` resources.

## CertificateSigning

**Type:** Validating.

This admission controller observes updates to the `status.certificate` field of `CertificateSigningRequest` resources and performs an additional authorization checks to ensure the signing user has permission to **sign** certificate requests with the `spec.signerName` requested on the `CertificateSigningRequest` resource.

See [Certificate Signing Requests](#) for more information on the permissions required to perform different actions on `CertificateSigningRequest` resources.

## CertificateSubjectRestriction

**Type:** Validating.

This admission controller observes creation of `CertificateSigningRequest` resources that have a `spec.signerName` of `kubernetes.io/kube-apiserver-client`. It rejects any request that specifies a 'group' (or 'organization attribute') of `system:masters`.

## DefaultIngressClass

**Type:** Mutating.

This admission controller observes creation of `Ingress` objects that do not request any specific ingress class and automatically adds a default ingress class to them. This way, users that do not request any special ingress class do not need to care about them at all and they will get the default one.

This admission controller does not do anything when no default ingress class is configured. When more than one ingress class is marked as default, it rejects any creation of `Ingress` with an error and an administrator must revisit their `IngressClass` objects and mark only one as default (with the annotation `"ingressclass.kubernetes.io/is-default-class"`). This admission controller ignores any `Ingress` updates; it acts only on creation.

See the [Ingress](#) documentation for more about ingress classes and how to mark one as default.

## DefaultStorageClass

**Type:** Mutating.

This admission controller observes creation of `PersistentVolumeClaim` objects that do not request any specific storage class and automatically adds a default storage class to them. This way, users that do not request any special storage class do not need to care about them at all and they will get the default one.

This admission controller does not do anything when no default storage class is configured. When more than one storage class is marked as default, it rejects any creation of PersistentVolumeClaim with an error and an administrator must revisit their StorageClass objects and mark only one as default. This admission controller ignores any PersistentVolumeClaim updates; it acts only on creation.

See [persistent volume](#) documentation about persistent volume claims and storage classes and how to mark a storage class as default.

## DefaultTolerationSeconds

**Type:** Mutating.

This admission controller sets the default forgiveness toleration for pods to tolerate the taints notready:NoExecute and unreachable:NoExecute based on the k8s-apiserver input parameters default-not-ready-toleration-seconds and default-unreachable-toleration-seconds if the pods don't already have toleration for taints node.kubernetes.io/not-ready:NoExecute or node.kubernetes.io/unreachable:NoExecute. The default value for default-not-ready-toleration-seconds and default-unreachable-toleration-seconds is 5 minutes.

## DenyServiceExternalIPs

**Type:** Validating.

This admission controller rejects all net-new usage of the Service field externalIPs. This feature is very powerful (allows network traffic interception) and not well controlled by policy. When enabled, users of the cluster may not create new Services which use externalIPs and may not add new values to externalIPs on existing Service objects. Existing uses of externalIPs are not affected, and users may remove values from externalIPs on existing Service objects.

Most users do not need this feature at all, and cluster admins should consider disabling it. Clusters that do need to use this feature should consider using some custom policy to manage usage of it.

This admission controller is disabled by default.

## EventRateLimit

FEATURE STATE: Kubernetes v1.13 [alpha]

**Type:** Validating.

This admission controller mitigates the problem where the API server gets flooded by requests to store new Events. The cluster admin can specify event rate limits by:

- Enabling the EventRateLimit admission controller;
- Referencing an EventRateLimit configuration file from the file provided to the API server's command line flag --admission-control-config-file:

```
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
- name: EventRateLimit
```



```
path: eventconfig.yaml
```

```
...
```

There are four types of limits that can be specified in the configuration:

- Server: All Event requests (creation or modifications) received by the API server share a single bucket.
- Namespace: Each namespace has a dedicated bucket.
- User: Each user is allocated a bucket.
- SourceAndObject: A bucket is assigned by each combination of source and involved object of the event.

Below is a sample eventconfig.yaml for such a configuration:

```
apiVersion: eventratelimit.admission.k8s.io/v1alpha1
kind: Configuration
limits:
- type: Namespace
 qps: 50
 burst: 100
 cacheSize: 2000
- type: User
 qps: 10
 burst: 50
```

See the [EventRateLimit Config API \(v1alpha1\)](#) for more details.

This admission controller is disabled by default.

## ExtendedResourceToleration

**Type:** Mutating.

This plug-in facilitates creation of dedicated nodes with extended resources. If operators want to create dedicated nodes with extended resources (like GPUs, FPGAs etc.), they are expected to [taint the node](#) with the extended resource name as the key. This admission controller, if enabled, automatically adds tolerations for such taints to pods requesting extended resources, so users don't have to manually add these tolerations.

This admission controller is disabled by default.

## ImagePolicyWebhook

**Type:** Validating.

The ImagePolicyWebhook admission controller allows a backend webhook to make admission decisions.

This admission controller is disabled by default.

## Configuration file format

ImagePolicyWebhook uses a configuration file to set options for the behavior of the backend. This file may be json or yaml and has the following format:

```
imagePolicy:
 kubeConfigFile: /path/to/kubeconfig/for/backend
 # time in s to cache approval
 allowTTL: 50
 # time in s to cache denial
 denyTTL: 50
 # time in ms to wait between retries
 retryBackoff: 500
 # determines behavior if the webhook backend fails
 defaultAllow: true
```

Reference the ImagePolicyWebhook configuration file from the file provided to the API server's command line flag `--admission-control-config-file`:

```
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
 - name: ImagePolicyWebhook
 path: imagepolicyconfig.yaml
...
```

Alternatively, you can embed the configuration directly in the file:

```
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
 - name: ImagePolicyWebhook
 configuration:
 imagePolicy:
 kubeConfigFile: <path-to-kubeconfig-file>
 allowTTL: 50
 denyTTL: 50
 retryBackoff: 500
 defaultAllow: true
```

The ImagePolicyWebhook config file must reference a [kubeconfig](#) formatted file which sets up the connection to the backend. It is required that the backend communicate over TLS.

The kubeconfig file's `cluster` field must point to the remote service, and the `user` field must contain the returned authorizer.

```
clusters refers to the remote service.
clusters:
 - name: name-of-remote-imagepolicy-service
 cluster:
 certificate-authority: /path/to/ca.pem # CA for verifying the remote service.
 server: https://images.example.com/policy # URL of remote service to query. Must use 'https'.
```

```
users refers to the API server's webhook configuration.
users:
- name: name-of-api-server
 user:
 client-certificate: /path/to/cert.pem # cert for the webhook admission controller to use
 client-key: /path/to/key.pem # key matching the cert
```

For additional HTTP configuration, refer to the [kubecfg](#) documentation.

## Request payloads

When faced with an admission decision, the API Server POSTs a JSON serialized `imagepolicy.k8s.io/v1alpha1 ImageReview` object describing the action. This object contains fields describing the containers being admitted, as well as any pod annotations that match `*.image-policy.k8s.io/*`.

### Note:

The webhook API objects are subject to the same versioning compatibility rules as other Kubernetes API objects. Implementers should be aware of looser compatibility promises for alpha objects and check the `apiVersion` field of the request to ensure correct deserialization. Additionally, the API Server must enable the `imagepolicy.k8s.io/v1alpha1` API extensions group (`--runtime-config=imagepolicy.k8s.io/v1alpha1=true`).

An example request body:

```
{
 "apiVersion": "imagepolicy.k8s.io/v1alpha1",
 "kind": "ImageReview",
 "spec": {
 "containers": [
 {
 "image": "myrepo/myimage:v1"
 },
 {
 "image": "myrepo/
myimage@sha256:beb6bd6a68f114c1dc2ea4b28db81bdf91de202a9014972bec5e4d9171d90ed"
 }
],
 "annotations": {
 "mycluster.image-policy.k8s.io/ticket-1234": "break-glass"
 },
 "namespace": "mynamespace"
 }
}
```

The remote service is expected to fill the `status` field of the request and respond to either allow or disallow access. The response body's `spec` field is ignored, and may be omitted. A permissive response would return:

```
{
 "apiVersion": "imagepolicy.k8s.io/v1alpha1",
 "kind": "ImageReview",
```

```
"status": {
 "allowed": true
}
```

To disallow access, the service would return:

```
{
 "apiVersion": "imagepolicy.k8s.io/v1alpha1",
 "kind": "ImageReview",
 "status": {
 "allowed": false,
 "reason": "image currently blacklisted"
 }
}
```

For further documentation refer to the [imagepolicy.v1alpha1 API](#).

## Extending with Annotations

All annotations on a Pod that match `*.image-policy.k8s.io/*` are sent to the webhook. Sending annotations allows users who are aware of the image policy backend to send extra information to it, and for different backends implementations to accept different information.

Examples of information you might put here are:

- request to "break glass" to override a policy, in case of emergency.
- a ticket number from a ticket system that documents the break-glass request
- provide a hint to the policy server as to the imageID of the image being provided, to save it a lookup

In any case, the annotations are provided by the user and are not validated by Kubernetes in any way.

## LimitPodHardAntiAffinityTopology

**Type:** Validating.

This admission controller denies any pod that defines AntiAffinity topology key other than `kubernetes.io/hostname` in `requiredDuringSchedulingRequiredDuringExecution`.

This admission controller is disabled by default.

## LimitRanger

**Type:** Mutating and Validating.

This admission controller will observe the incoming request and ensure that it does not violate any of the constraints enumerated in the `LimitRange` object in a Namespace. If you are using `LimitRange` objects in your Kubernetes deployment, you **MUST** use this admission controller to enforce those constraints. `LimitRanger` can also be used to apply default resource requests to Pods that don't specify any; currently, the default `LimitRanger` applies a 0.1 CPU requirement to all Pods in the default namespace.

See the [LimitRange API reference](#) and the [example of LimitRange](#) for more details.

## MutatingAdmissionWebhook

**Type:** Mutating.

This admission controller calls any mutating webhooks which match the request. Matching webhooks are called in serial; each one may modify the object if it desires.

This admission controller (as implied by the name) only runs in the mutating phase.

If a webhook called by this has side effects (for example, decrementing quota) it *must* have a reconciliation system, as it is not guaranteed that subsequent webhooks or validating admission controllers will permit the request to finish.

If you disable the MutatingAdmissionWebhook, you must also disable the MutatingWebhookConfiguration object in the admissionregistration.k8s.io/v1 group/version via the `--runtime-config` flag, both are on by default.

### Use caution when authoring and installing mutating webhooks

- Users may be confused when the objects they try to create are different from what they get back.
- Built in control loops may break when the objects they try to create are different when read back.
  - Setting originally unset fields is less likely to cause problems than overwriting fields set in the original request. Avoid doing the latter.
- Future changes to control loops for built-in resources or third-party resources may break webhooks that work well today. Even when the webhook installation API is finalized, not all possible webhook behaviors will be guaranteed to be supported indefinitely.

## NamespaceAutoProvision

**Type:** Mutating.

This admission controller examines all incoming requests on namespaced resources and checks if the referenced namespace does exist. It creates a namespace if it cannot be found. This admission controller is useful in deployments that do not want to restrict creation of a namespace prior to its usage.

## NamespaceExists

**Type:** Validating.

This admission controller checks all requests on namespaced resources other than Namespace itself. If the namespace referenced from a request doesn't exist, the request is rejected.

## NamespaceLifecycle

**Type:** Validating.

This admission controller enforces that a Namespace that is undergoing termination cannot have new objects created in it, and ensures that requests in a non-existent Namespace are rejected. This admission controller also prevents deletion of three system reserved namespaces default, kube-system, kube-public.

A Namespace deletion kicks off a sequence of operations that remove all objects (pods, services, etc.) in that namespace. In order to enforce integrity of that process, we strongly recommend running this admission controller.

## NodeRestriction

**Type:** Validating.

This admission controller limits the Node and Pod objects a kubelet can modify. In order to be limited by this admission controller, kubelets must use credentials in the system:nodes group, with a username in the form system:node:<nodeName>. Such kubelets will only be allowed to modify their own Node API object, and only modify Pod API objects that are bound to their node. kubelets are not allowed to update or remove taints from their Node API object.

The NodeRestriction admission plugin prevents kubelets from deleting their Node API object, and enforces kubelet modification of labels under the kubernetes.io/ or k8s.io/ prefixes as follows:

- **Prevents** kubelets from adding/removing/updating labels with a node-restriction.kubernetes.io/ prefix. This label prefix is reserved for administrators to label their Node objects for workload isolation purposes, and kubelets will not be allowed to modify labels with that prefix.
- **Allows** kubelets to add/remove/update these labels and label prefixes:
  - kubernetes.io/hostname
  - kubernetes.io/arch
  - kubernetes.io/os
  - beta.kubernetes.io/instance-type
  - node.kubernetes.io/instance-type
  - failure-domain.beta.kubernetes.io/region (deprecated)
  - failure-domain.beta.kubernetes.io/zone (deprecated)
  - topology.kubernetes.io/region
  - topology.kubernetes.io/zone
  - kubelet.kubernetes.io/-prefixed labels
  - node.kubernetes.io/-prefixed labels

Use of any other labels under the kubernetes.io or k8s.io prefixes by kubelets is reserved, and may be disallowed or allowed by the NodeRestriction admission plugin in the future.

Future versions may add additional restrictions to ensure kubelets have the minimal set of permissions required to operate correctly.

## OwnerReferencesPermissionEnforcement

**Type:** Validating.

This admission controller protects the access to the metadata.ownerReferences of an object so that only users with **delete** permission to the object can change it. This admission controller also protects the access to metadata.ownerReferences[x].blockOwnerDeletion of an object, so

that only users with **update** permission to the finalizers subresource of the referenced *owner* can change it.

## PersistentVolumeClaimResize

FEATURE STATE: Kubernetes v1.24 [stable]

**Type:** Validating.

This admission controller implements additional validations for checking incoming PersistentVolumeClaim resize requests.

Enabling the PersistentVolumeClaimResize admission controller is recommended. This admission controller prevents resizing of all claims by default unless a claim's StorageClass explicitly enables resizing by setting allowVolumeExpansion to true.

For example: all PersistentVolumeClaims created from the following StorageClass support volume expansion:

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: gluster-vol-default
provisioner: kubernetes.io/glusterfs
parameters:
 resturl: "http://192.168.10.100:8080"
 restuser: ""
 secretNamespace: ""
 secretName: ""
allowVolumeExpansion: true
```

For more information about persistent volume claims, see [PersistentVolumeClaims](#).

## PersistentVolumeLabel

FEATURE STATE: Kubernetes v1.13 [deprecated]

**Type:** Mutating.

This admission controller automatically attaches region or zone labels to PersistentVolumes as defined by the cloud provider (for example, Azure or GCP). It helps ensure the Pods and the PersistentVolumes mounted are in the same region and/or zone. If the admission controller doesn't support automatic labelling your PersistentVolumes, you may need to add the labels manually to prevent pods from mounting volumes from a different zone. PersistentVolumeLabel is **deprecated** as labeling for persistent volumes has been taken over by the [cloud-controller-manager](#).

This admission controller is disabled by default.

## PodNodeSelector

FEATURE STATE: Kubernetes v1.5 [alpha]

**Type:** Validating.

This admission controller defaults and limits what node selectors may be used within a namespace by reading a namespace annotation and a global configuration.

This admission controller is disabled by default.

### Configuration file format

PodNodeSelector uses a configuration file to set options for the behavior of the backend. Note that the configuration file format will move to a versioned file in a future release. This file may be json or yaml and has the following format:

```
podNodeSelectorPluginConfig:
 clusterDefaultNodeSelector: name-of-node-selector
 namespace1: name-of-node-selector
 namespace2: name-of-node-selector
```

Reference the PodNodeSelector configuration file from the file provided to the API server's command line flag `--admission-control-config-file`:

```
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
- name: PodNodeSelector
 path: podnodeselector.yaml
...
```

### Configuration Annotation Format

PodNodeSelector uses the annotation key `scheduler.alpha.kubernetes.io/node-selector` to assign node selectors to namespaces.

```
apiVersion: v1
kind: Namespace
metadata:
 annotations:
 scheduler.alpha.kubernetes.io/node-selector: name-of-node-selector
 name: namespace3
```

### Internal Behavior

This admission controller has the following behavior:

1. If the Namespace has an annotation with a key `scheduler.alpha.kubernetes.io/node-selector`, use its value as the node selector.
2. If the namespace lacks such an annotation, use the `clusterDefaultNodeSelector` defined in the PodNodeSelector plugin configuration file as the node selector.
3. Evaluate the pod's node selector against the namespace node selector for conflicts. Conflicts result in rejection.
4. Evaluate the pod's node selector against the namespace-specific allowed selector defined the plugin configuration file. Conflicts result in rejection.



**Note:**

PodNodeSelector allows forcing pods to run on specifically labeled nodes. Also see the PodTolerationRestriction admission plugin, which allows preventing pods from running on specifically tainted nodes.

## PodSecurity

FEATURE STATE: Kubernetes v1.25 [stable]

**Type:** Validating.

The PodSecurity admission controller checks new Pods before they are admitted, determines if it should be admitted based on the requested security context and the restrictions on permitted [Pod Security Standards](#) for the namespace that the Pod would be in.

See the [Pod Security Admission](#) documentation for more information.

PodSecurity replaced an older admission controller named PodSecurityPolicy.

## PodTolerationRestriction

FEATURE STATE: Kubernetes v1.7 [alpha]

**Type:** Mutating and Validating.

The PodTolerationRestriction admission controller verifies any conflict between tolerations of a pod and the tolerations of its namespace. It rejects the pod request if there is a conflict. It then merges the tolerations annotated on the namespace into the tolerations of the pod. The resulting tolerations are checked against a list of allowed tolerations annotated to the namespace. If the check succeeds, the pod request is admitted otherwise it is rejected.

If the namespace of the pod does not have any associated default tolerations or allowed tolerations annotated, the cluster-level default tolerations or cluster-level list of allowed tolerations are used instead if they are specified.

Tolerations to a namespace are assigned via the scheduler.alpha.kubernetes.io/defaultTolerations annotation key. The list of allowed tolerations can be added via the scheduler.alpha.kubernetes.io/tolerationsWhitelist annotation key.

Example for namespace annotations:

```
apiVersion: v1
kind: Namespace
metadata:
 name: apps-that-need-nodes-exclusively
 annotations:
 scheduler.alpha.kubernetes.io/defaultTolerations: '[{"operator": "Exists", "effect": "NoSchedule", "key": "dedicated-node"}]'
 scheduler.alpha.kubernetes.io/tolerationsWhitelist: '[{"operator": "Exists", "effect": "NoSchedule", "key": "dedicated-node"}]'
```

This admission controller is disabled by default.

## Priority

**Type:** Mutating and Validating.

The priority admission controller uses the `priorityClassName` field and populates the integer value of the priority. If the priority class is not found, the Pod is rejected.

## ResourceQuota

**Type:** Validating.

This admission controller will observe the incoming request and ensure that it does not violate any of the constraints enumerated in the ResourceQuota object in a Namespace. If you are using ResourceQuota objects in your Kubernetes deployment, you **MUST** use this admission controller to enforce quota constraints.

See the [ResourceQuota API reference](#) and the [example of Resource Quota](#) for more details.

## RuntimeClass

**Type:** Mutating and Validating.

If you define a RuntimeClass with [Pod overhead](#) configured, this admission controller checks incoming Pods. When enabled, this admission controller rejects any Pod create requests that have the overhead already set. For Pods that have a RuntimeClass configured and selected in their `.spec`, this admission controller sets `.spec.overhead` in the Pod based on the value defined in the corresponding RuntimeClass.

See also [Pod Overhead](#) for more information.

## ServiceAccount

**Type:** Mutating and Validating.

This admission controller implements automation for [serviceAccounts](#). The Kubernetes project strongly recommends enabling this admission controller. You should enable this admission controller if you intend to make any use of Kubernetes ServiceAccount objects.

Regarding the annotation `kubernetes.io/enforce-mountable-secrets`: While the annotation's name suggests it only concerns the mounting of Secrets, its enforcement also extends to other ways Secrets are used in the context of a Pod. Therefore, it is crucial to ensure that all the referenced secrets are correctly specified in the ServiceAccount.

## StorageObjectInUseProtection

**Type:** Mutating.

The StorageObjectInUseProtection plugin adds the `kubernetes.io/pvc-protection` or `kubernetes.io/pv-protection` finalizers to newly created Persistent Volume Claims (PVCs) or Persistent Volumes (PV). In case a user deletes a PVC or PV the PVC or PV is not removed until the finalizer is removed from the PVC or PV by PVC or PV Protection Controller. Refer to the [Storage Object in Use Protection](#) for more detailed information.

## TaintNodesByCondition

**Type:** Mutating.

This admission controller [taints](#) newly created Nodes as NotReady and NoSchedule. That tainting avoids a race condition that could cause Pods to be scheduled on new Nodes before their taints were updated to accurately reflect their reported conditions.

## ValidatingAdmissionPolicy

**Type:** Validating.

[This admission controller](#) implements the CEL validation for incoming matched requests. It is enabled when both feature gate validatingadmissionpolicy and admissionregistration.k8s.io/v1alpha1 group/version are enabled. If any of the ValidatingAdmissionPolicy fails, the request fails.

## ValidatingAdmissionWebhook

**Type:** Validating.

This admission controller calls any validating webhooks which match the request. Matching webhooks are called in parallel; if any of them rejects the request, the request fails. This admission controller only runs in the validation phase; the webhooks it calls may not mutate the object, as opposed to the webhooks called by the MutatingAdmissionWebhook admission controller.

If a webhook called by this has side effects (for example, decrementing quota) it *must* have a reconciliation system, as it is not guaranteed that subsequent webhooks or other validating admission controllers will permit the request to finish.

If you disable the ValidatingAdmissionWebhook, you must also disable the ValidatingWebhookConfiguration object in the admissionregistration.k8s.io/v1 group/version via the `--runtime-config` flag.

## Is there a recommended set of admission controllers to use?

Yes. The recommended admission controllers are enabled by default (shown [here](#)), so you do not need to explicitly specify them. You can enable additional admission controllers beyond the default set using the `--enable-admission-plugins` flag (**order doesn't matter**).

## Dynamic Admission Control

In addition to [compiled-in admission plugins](#), admission plugins can be developed as extensions and run as webhooks configured at runtime. This page describes how to build, configure, use, and monitor admission webhooks.

# What are admission webhooks?

Admission webhooks are HTTP callbacks that receive admission requests and do something with them. You can define two types of admission webhooks, [validating admission webhook](#) and [mutating admission webhook](#). Mutating admission webhooks are invoked first, and can modify objects sent to the API server to enforce custom defaults. After all object modifications are complete, and after the incoming object is validated by the API server, validating admission webhooks are invoked and can reject requests to enforce custom policies.

## Note:

Admission webhooks that need to guarantee they see the final state of the object in order to enforce policy should use a validating admission webhook, since objects can be modified after being seen by mutating webhooks.

## Experimenting with admission webhooks

Admission webhooks are essentially part of the cluster control-plane. You should write and deploy them with great caution. Please read the [user guides](#) for instructions if you intend to write/deploy production-grade admission webhooks. In the following, we describe how to quickly experiment with admission webhooks.

### Prerequisites

- Ensure that MutatingAdmissionWebhook and ValidatingAdmissionWebhook admission controllers are enabled. [Here](#) is a recommended set of admission controllers to enable in general.
- Ensure that the admissionregistration.k8s.io/v1 API is enabled.

### Write an admission webhook server

Please refer to the implementation of the [admission webhook server](#) that is validated in a Kubernetes e2e test. The webhook handles the AdmissionReview request sent by the API servers, and sends back its decision as an AdmissionReview object in the same version it received.

See the [webhook request](#) section for details on the data sent to webhooks.

See the [webhook response](#) section for the data expected from webhooks.

The example admission webhook server leaves the ClientAuth field [empty](#), which defaults to NoClientCert. This means that the webhook server does not authenticate the identity of the clients, supposedly API servers. If you need mutual TLS or other ways to authenticate the clients, see how to [authenticate API servers](#).

### Deploy the admission webhook service

The webhook server in the e2e test is deployed in the Kubernetes cluster, via the [deployment API](#). The test also creates a [service](#) as the front-end of the webhook server. See [code](#).

You may also deploy your webhooks outside of the cluster. You will need to update your webhook configurations accordingly.

## Configure admission webhooks on the fly

You can dynamically configure what resources are subject to what admission webhooks via [ValidatingWebhookConfiguration](#) or [MutatingWebhookConfiguration](#).

The following is an example ValidatingWebhookConfiguration, a mutating webhook configuration is similar. See the [webhook configuration](#) section for details about each config field.

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
metadata:
 name: "pod-policy.example.com"
webhooks:
- name: "pod-policy.example.com"
 rules:
 - apiGroups: [""]
 apiVersions: ["v1"]
 operations: ["CREATE"]
 resources: ["pods"]
 scope: "Namespaced"
 clientConfig:
 service:
 namespace: "example-namespace"
 name: "example-service"
 caBundle: <CA_BUNDLE>
 admissionReviewVersions: ["v1"]
 sideEffects: None
 timeoutSeconds: 5
```

### Note:

You must replace the <CA\_BUNDLE> in the above example by a valid CA bundle which is a PEM-encoded (field value is Base64 encoded) CA bundle for validating the webhook's server certificate.

The scope field specifies if only cluster-scoped resources ("Cluster") or namespace-scoped resources ("Namespaced") will match this rule. "\*" means that there are no scope restrictions.

### Note:

When using clientConfig.service, the server cert must be valid for <svc\_name>.<svc\_namespace>.svc.

### Note:

Default timeout for a webhook call is 10 seconds, You can set the timeout and it is encouraged to use a short timeout for webhooks. If the webhook call times out, the request is handled according to the webhook's failure policy.

When an API server receives a request that matches one of the rules, the API server sends an admissionReview request to webhook as specified in the clientConfig.

After you create the webhook configuration, the system will take a few seconds to honor the new configuration.

## Authenticate API servers

If your admission webhooks require authentication, you can configure the API servers to use basic auth, bearer token, or a cert to authenticate itself to the webhooks. There are three steps to complete the configuration.

- When starting the API server, specify the location of the admission control configuration file via the --admission-control-config-file flag.
- In the admission control configuration file, specify where the MutatingAdmissionWebhook controller and ValidatingAdmissionWebhook controller should read the credentials. The credentials are stored in kubeConfig files (yes, the same schema that's used by kubectl), so the field name is kubeConfigFile. Here is an example admission control configuration file:
- [apiserver.config.k8s.io/v1](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.17/#apiserverconfig-k8s-io-v1)
- [apiserver.k8s.io/v1alpha1](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.17/#apiserverconfig-k8s-io-v1alpha1)

```
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
- name: ValidatingAdmissionWebhook
 configuration:
 apiVersion: apiserver.config.k8s.io/v1
 kind: WebhookAdmissionConfiguration
 kubeConfigFile: "<path-to-kubeconfig-file>"
- name: MutatingAdmissionWebhook
 configuration:
 apiVersion: apiserver.config.k8s.io/v1
 kind: WebhookAdmissionConfiguration
 kubeConfigFile: "<path-to-kubeconfig-file>"
```

```
Deprecated in v1.17 in favor of apiserver.config.k8s.io/v1
apiVersion: apiserver.k8s.io/v1alpha1
kind: AdmissionConfiguration
plugins:
- name: ValidatingAdmissionWebhook
 configuration:
 # Deprecated in v1.17 in favor of apiserver.config.k8s.io/v1,
kind=WebhookAdmissionConfiguration
 apiVersion: apiserver.config.k8s.io/v1alpha1
 kind: WebhookAdmission
 kubeConfigFile: "<path-to-kubeconfig-file>"
- name: MutatingAdmissionWebhook
 configuration:
 # Deprecated in v1.17 in favor of apiserver.config.k8s.io/v1,
kind=WebhookAdmissionConfiguration
```

```
apiVersion: apiserver.config.k8s.io/v1alpha1
kind: WebhookAdmission
kubeConfigFile: "<path-to-kubeconfig-file>"
```

For more information about AdmissionConfiguration, see the [AdmissionConfiguration \(v1\) reference](#). See the [webhook configuration](#) section for details about each config field.

In the kubeConfig file, provide the credentials:

```
apiVersion: v1
kind: Config
users:
name should be set to the DNS name of the service or the host (including port) of the URL the
webhook is configured to speak to.
If a non-443 port is used for services, it must be included in the name when configuring 1.16+
API servers.
#
For a webhook configured to speak to a service on the default port (443), specify the DNS
name of the service:
- name: webhook1.ns1.svc
user: ...
#
For a webhook configured to speak to a service on non-default port (e.g. 8443), specify the
DNS name and port of the service in 1.16+:
- name: webhook1.ns1.svc:8443
user: ...
and optionally create a second stanza using only the DNS name of the service for
compatibility with 1.15 API servers:
- name: webhook1.ns1.svc
user: ...
#
For webhooks configured to speak to a URL, match the host (and port) specified in the
webhook's URL. Examples:
A webhook with `url: https://www.example.com`:
- name: www.example.com
user: ...
#
A webhook with `url: https://www.example.com:443`:
- name: www.example.com:443
user: ...
#
A webhook with `url: https://www.example.com:8443`:
- name: www.example.com:8443
user: ...
#
- name: 'webhook1.ns1.svc'
 user:
 client-certificate-data: "<pem encoded certificate>"
 client-key-data: "<pem encoded key>"
The `name` supports using * to wildcard-match prefixing segments.
- name: '*.webhook-company.org'
 user:
 password: "<password>"
```

```
username: "<name>"
'*' is the default match.
- name: '*'
user:
 token: "<token>"
```

Of course you need to set up the webhook server to handle these authentication requests.

## Webhook request and response

### Request

Webhooks are sent as POST requests, with Content-Type: application/json, with an AdmissionReview API object in the admission.k8s.io API group serialized to JSON as the body.

Webhooks can specify what versions of AdmissionReview objects they accept with the admissionReviewVersions field in their configuration:

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 admissionReviewVersions: ["v1", "v1beta1"]
```

admissionReviewVersions is a required field when creating webhook configurations. Webhooks are required to support at least one AdmissionReview version understood by the current and previous API server.

API servers send the first AdmissionReview version in the admissionReviewVersions list they support. If none of the versions in the list are supported by the API server, the configuration will not be allowed to be created. If an API server encounters a webhook configuration that was previously created and does not support any of the AdmissionReview versions the API server knows how to send, attempts to call to the webhook will fail and be subject to the [failure policy](#).

This example shows the data contained in an AdmissionReview object for a request to update the scale subresource of an apps/v1 Deployment:

```
apiVersion: admission.k8s.io/v1
kind: AdmissionReview
request:
 # Random uid uniquely identifying this admission call
 uid: 705ab4f5-6393-11e8-b7cc-42010a800002

 # Fully-qualified group/version/kind of the incoming object
 kind:
 group: autoscaling
 version: v1
 kind: Scale

 # Fully-qualified group/version/kind of the resource being modified
 resource:
 group: apps
 version: v1
```



```
resource: deployments

subresource, if the request is to a subresource
subResource: scale

Fully-qualified group/version/kind of the incoming object in the original request to the API
server.
This only differs from `kind` if the webhook specified `matchPolicy: Equivalent` and the
original request to the API server was converted to a version the webhook registered for.
requestKind:
 group: autoscaling
 version: v1
 kind: Scale

Fully-qualified group/version/kind of the resource being modified in the original request to
the API server.
This only differs from `resource` if the webhook specified `matchPolicy: Equivalent` and the
original request to the API server was converted to a version the webhook registered for.
requestResource:
 group: apps
 version: v1
 resource: deployments

subresource, if the request is to a subresource

This only differs from `subResource` if the webhook specified `matchPolicy: Equivalent` and
the
original request to the API server was converted to a version the webhook registered for.
requestSubResource: scale

Name of the resource being modified
name: my-deployment

Namespace of the resource being modified, if the resource is namespaced (or is a Namespace
object)
namespace: my-namespace

operation can be CREATE, UPDATE, DELETE, or CONNECT
operation: UPDATE

userInfo:
 # Username of the authenticated user making the request to the API server
 username: admin

 # UID of the authenticated user making the request to the API server
 uid: 014fbff9a07c

 # Group memberships of the authenticated user making the request to the API server
 groups:
 - system:authenticated
 - my-admin-group
 # Arbitrary extra info associated with the user making the request to the API server.
 # This is populated by the API server authentication layer and should be included
```

```

if any SubjectAccessReview checks are performed by the webhook.
extra:
 some-key:
 - some-value1
 - some-value2

object is the new object being admitted.
It is null for DELETE operations.
object:
 apiVersion: autoscaling/v1
 kind: Scale

oldObject is the existing object.
It is null for CREATE and CONNECT operations.
oldObject:
 apiVersion: autoscaling/v1
 kind: Scale

options contains the options for the operation being admitted, like meta.k8s.io/v1
CreateOptions, UpdateOptions, or DeleteOptions.
It is null for CONNECT operations.
options:
 apiVersion: meta.k8s.io/v1
 kind: UpdateOptions

dryRun indicates the API request is running in dry run mode and will not be persisted.
Webhooks with side effects should avoid actuating those side effects when dryRun is true.
See http://k8s.io/docs/reference/using-api/api-concepts/#make-a-dry-run-request for more
details.
dryRun: False

```

## Response

Webhooks respond with a 200 HTTP status code, Content-Type: application/json, and a body containing an AdmissionReview object (in the same version they were sent), with the response stanza populated, serialized to JSON.

At a minimum, the response stanza must contain the following fields:

- uid, copied from the request.uid sent to the webhook
- allowed, either set to true or false

Example of a minimal response from a webhook to allow a request:

```

{
 "apiVersion": "admission.k8s.io/v1",
 "kind": "AdmissionReview",
 "response": {
 "uid": "<value from request.uid>",
 "allowed": true
 }
}

```

Example of a minimal response from a webhook to forbid a request:

```
{
 "apiVersion": "admission.k8s.io/v1",
 "kind": "AdmissionReview",
 "response": {
 "uid": "<value from request.uid>",
 "allowed": false
 }
}
```

When rejecting a request, the webhook can customize the http code and message returned to the user using the status field. The specified status object is returned to the user. See the [API documentation](#) for details about the status type. Example of a response to forbid a request, customizing the HTTP status code and message presented to the user:

```
{
 "apiVersion": "admission.k8s.io/v1",
 "kind": "AdmissionReview",
 "response": {
 "uid": "<value from request.uid>",
 "allowed": false,
 "status": {
 "code": 403,
 "message": "You cannot do this because it is Tuesday and your name starts with A"
 }
 }
}
```

When allowing a request, a mutating admission webhook may optionally modify the incoming object as well. This is done using the patch and patchType fields in the response. The only currently supported patchType is JSONPatch. See [JSON patch](#) documentation for more details. For patchType: JSONPatch, the patch field contains a base64-encoded array of JSON patch operations.

As an example, a single patch operation that would set spec.replicas would be [{"op": "add", "path": "/spec/replicas", "value": 3}]

Base64-encoded, this would be

W3sib3AiOiAiYWRkIiwgInBhdGgiOiAiL3NwZWVvcmljYXMiLCBldmFsdWUiOiAiAzfV0=

So a webhook response to add that label would be:

```
{
 "apiVersion": "admission.k8s.io/v1",
 "kind": "AdmissionReview",
 "response": {
 "uid": "<value from request.uid>",
 "allowed": true,
 "patchType": "JSONPatch",
 "patch": "W3sib3AiOiAiYWRkIiwgInBhdGgiOiAiL3NwZWVvcmljYXMiLCBldmFsdWUiOiAiAzfV0="
 }
}
```

Admission webhooks can optionally return warning messages that are returned to the requesting client in HTTP Warning headers with a warning code of 299. Warnings can be sent with allowed or rejected admission responses.

If you're implementing a webhook that returns a warning:

- Don't include a "Warning:" prefix in the message
- Use warning messages to describe problems the client making the API request should correct or be aware of
- Limit warnings to 120 characters if possible

### Caution:

Individual warning messages over 256 characters may be truncated by the API server before being returned to clients. If more than 4096 characters of warning messages are added (from all sources), additional warning messages are ignored.

```
{
 "apiVersion": "admission.k8s.io/v1",
 "kind": "AdmissionReview",
 "response": {
 "uid": "<value from request.uid>",
 "allowed": true,
 "warnings": [
 "duplicate envvar entries specified with name MY_ENV",
 "memory request less than 4MB specified for container mycontainer, which will not start successfully"
]
 }
}
```

## Webhook configuration

To register admission webhooks, create `MutatingWebhookConfiguration` or `ValidatingWebhookConfiguration` API objects. The name of a `MutatingWebhookConfiguration` or a `ValidatingWebhookConfiguration` object must be a valid [DNS subdomain name](#).

Each configuration can contain one or more webhooks. If multiple webhooks are specified in a single configuration, each must be given a unique name. This is required in order to make resulting audit logs and metrics easier to match up to active configurations.

Each webhook defines the following things.

### Matching requests: rules

Each webhook must specify a list of rules used to determine if a request to the API server should be sent to the webhook. Each rule specifies one or more operations, apiGroups, apiVersions, and resources, and a resource scope:

- operations lists one or more operations to match. Can be "CREATE", "UPDATE", "DELETE", "CONNECT", or "\*" to match all.

- `apiGroups` lists one or more API groups to match. `""` is the core API group. `""` matches all API groups.
- `apiVersions` lists one or more API versions to match. `""` matches all API versions.
- `resources` lists one or more resources to match.
  - `""` matches all resources, but not subresources.
  - `"/"` matches all resources and subresources.
  - `"pods/"` matches all subresources of pods.
  - `"/status"` matches all status subresources.
- `scope` specifies a scope to match. Valid values are `"Cluster"`, `"Namespaced"`, and `""`. Subresources match the scope of their parent resource. Default is `""`.
  - `"Cluster"` means that only cluster-scoped resources will match this rule (Namespace API objects are cluster-scoped).
  - `"Namespaced"` means that only namespaced resources will match this rule.
  - `""` means that there are no scope restrictions.

If an incoming request matches one of the specified operations, groups, versions, resources, and scope for any of a webhook's rules, the request is sent to the webhook.

Here are other examples of rules that could be used to specify which resources should be intercepted.

Match CREATE or UPDATE requests to `apps/v1` and `apps/v1beta1` deployments and replicaset:

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
...
webhooks:
- name: my-webhook.example.com
 rules:
 - operations: ["CREATE", "UPDATE"]
 apiGroups: ["apps"]
 apiVersions: ["v1", "v1beta1"]
 resources: ["deployments", "replicasets"]
 scope: "Namespaced"
...
```

Match create requests for all resources (but not subresources) in all API groups and versions:

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 rules:
 - operations: ["CREATE"]
 apiGroups: [""]
 apiVersions: [""]
 resources: [""]
 scope: ""
```

Match update requests for all status subresources in all API groups and versions:

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 rules:
 - operations: ["UPDATE"]
 apiGroups: ["*"]
 apiVersions: ["*"]
 resources: ["*/status"]
 scope: "*"

```

## Matching requests: objectSelector

Webhooks may optionally limit which requests are intercepted based on the labels of the objects they would be sent, by specifying an `objectSelector`. If specified, the `objectSelector` is evaluated against both the object and `oldObject` that would be sent to the webhook, and is considered to match if either object matches the selector.

A null object (`oldObject` in the case of create, or `newObject` in the case of delete), or an object that cannot have labels (like a `DeploymentRollback` or a `PodProxyOptions` object) is not considered to match.

Use the object selector only if the webhook is opt-in, because end users may skip the admission webhook by setting the labels.

This example shows a mutating webhook that would match a CREATE of any resource (but not subresources) with the label `foo: bar`:

```
apiVersion: admissionregistration.k8s.io/v1
kind: MutatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 objectSelector:
 matchLabels:
 foo: bar
 rules:
 - operations: ["CREATE"]
 apiGroups: ["*"]
 apiVersions: ["*"]
 resources: ["*"]
 scope: "*"

```

See [labels concept](#) for more examples of label selectors.

## Matching requests: namespaceSelector

Webhooks may optionally limit which requests for namespaced resources are intercepted, based on the labels of the containing namespace, by specifying a `namespaceSelector`.

The `namespaceSelector` decides whether to run the webhook on a request for a namespaced resource (or a `Namespace` object), based on whether the namespace's labels match the selector.

If the object itself is a namespace, the matching is performed on `object.metadata.labels`. If the object is a cluster scoped resource other than a Namespace, `namespaceSelector` has no effect.

This example shows a mutating webhook that matches a CREATE of any namespaced resource inside a namespace that does not have a "runlevel" label of "0" or "1":

```
apiVersion: admissionregistration.k8s.io/v1
kind: MutatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 namespaceSelector:
 matchExpressions:
 - key: runlevel
 operator: NotIn
 values: ["0", "1"]
 rules:
 - operations: ["CREATE"]
 apiGroups: ["*"]
 apiVersions: ["*"]
 resources: ["*"]
 scope: "Namespaced"
```

This example shows a validating webhook that matches a CREATE of any namespaced resource inside a namespace that is associated with the "environment" of "prod" or "staging":

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 namespaceSelector:
 matchExpressions:
 - key: environment
 operator: In
 values: ["prod", "staging"]
 rules:
 - operations: ["CREATE"]
 apiGroups: ["*"]
 apiVersions: ["*"]
 resources: ["*"]
 scope: "Namespaced"
```

See [labels concept](#) for more examples of label selectors.

## Matching requests: matchPolicy

API servers can make objects available via multiple API groups or versions.

For example, if a webhook only specified a rule for some API groups/versions (like `apiGroups: ["apps"]`, `apiVersions: ["v1", "v1beta1"]`), and a request was made to modify the resource via another API group/version (like `extensions/v1beta1`), the request would not be sent to the webhook.

The `matchPolicy` lets a webhook define how its rules are used to match incoming requests. Allowed values are `Exact` or `Equivalent`.

- `Exact` means a request should be intercepted only if it exactly matches a specified rule.
- `Equivalent` means a request should be intercepted if modifies a resource listed in rules, even via another API group or version.

In the example given above, the webhook that only registered for `apps/v1` could use `matchPolicy`:

- `matchPolicy: Exact` would mean the `extensions/v1beta1` request would not be sent to the webhook
- `matchPolicy: Equivalent` means the `extensions/v1beta1` request would be sent to the webhook (with the objects converted to a version the webhook had specified: `apps/v1`)

Specifying `Equivalent` is recommended, and ensures that webhooks continue to intercept the resources they expect when upgrades enable new versions of the resource in the API server.

When a resource stops being served by the API server, it is no longer considered equivalent to other versions of that resource that are still served. For example, `extensions/v1beta1` deployments were first deprecated and then removed (in Kubernetes v1.16).

Since that removal, a webhook with a `apiGroups:["extensions"]`, `apiVersions:["v1beta1"]`, `resources:["deployments"]` rule does not intercept deployments created via `apps/v1` APIs. For that reason, webhooks should prefer registering for stable versions of resources.

This example shows a validating webhook that intercepts modifications to deployments (no matter the API group or version), and is always sent an `apps/v1` Deployment object:

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 matchPolicy: Equivalent
 rules:
 - operations: ["CREATE","UPDATE","DELETE"]
 apiGroups: ["apps"]
 apiVersions: ["v1"]
 resources: ["deployments"]
 scope: "Namespaced"
```

The `matchPolicy` for an admission webhooks defaults to `Equivalent`.

## Matching requests: `matchConditions`

FEATURE STATE: Kubernetes v1.30 [stable]

You can define *match conditions* for webhooks if you need fine-grained request filtering. These conditions are useful if you find that match rules, objectSelectors and namespaceSelectors still doesn't provide the filtering you want over when to call out over HTTP. Match conditions are [CEL expressions](#). All match conditions must evaluate to true for the webhook to be called.

Here is an example illustrating a few different uses for match conditions:



```

apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 matchPolicy: Equivalent
 rules:
 - operations: ['CREATE','UPDATE']
 apiGroups: ['*']
 apiVersions: ['*']
 resources: ['*']
 failurePolicy: 'Ignore' # Fail-open (optional)
 sideEffects: None
 clientConfig:
 service:
 namespace: my-namespace
 name: my-webhook
 caBundle: '<omitted>'
 # You can have up to 64 matchConditions per webhook
 matchConditions:
 - name: 'exclude-leases' # Each match condition must have a unique name
 expression: '!(request.resource.group == "coordination.k8s.io" &&
request.resource.resource == "leases")' # Match non-lease resources.
 - name: 'exclude-kubelet-requests'
 expression: '!("system:nodes" in request.userInfo.groups)' # Match requests made by non-
node users.
 - name: 'rbac' # Skip RBAC requests, which are handled by the second webhook.
 expression: 'request.resource.group != "rbac.authorization.k8s.io"'

 # This example illustrates the use of the 'authorizer'. The authorization check is more
expensive
 # than a simple expression, so in this example it is scoped to only RBAC requests by using a
second
webhook. Both webhooks can be served by the same endpoint.
- name: rbac.my-webhook.example.com
 matchPolicy: Equivalent
 rules:
 - operations: ['CREATE','UPDATE']
 apiGroups: ['rbac.authorization.k8s.io']
 apiVersions: ['*']
 resources: ['*']
 failurePolicy: 'Fail' # Fail-closed (the default)
 sideEffects: None
 clientConfig:
 service:
 namespace: my-namespace
 name: my-webhook
 caBundle: '<omitted>'
 # You can have up to 64 matchConditions per webhook
 matchConditions:
 - name: 'breakglass'
 # Skip requests made by users authorized to 'breakglass' on this webhook.
 # The 'breakglass' API verb does not need to exist outside this check.
 expression: '!'

```

```
authorizer.group("admissionregistration.k8s.io").resource("validatingwebhookconfigurations").name("my-webhook.example.com").check("breakglass").allowed()'
```

### Note:

You can define up to 64 elements in the `matchConditions` field per webhook.

Match conditions have access to the following CEL variables:

- `object` - The object from the incoming request. The value is null for DELETE requests. The object version may be converted based on the [matchPolicy](#).
- `oldObject` - The existing object. The value is null for CREATE requests.
- `request` - The request portion of the [AdmissionReview](#), excluding `object` and `oldObject`.
- `authorizer` - A CEL Authorizer. May be used to perform authorization checks for the principal (authenticated user) of the request. See [Authz](#) in the Kubernetes CEL library documentation for more details.
- `authorizer.requestResource` - A shortcut for an authorization check configured with the request resource (group, resource, (subresource), namespace, name).

For more information on CEL expressions, refer to the [Common Expression Language in Kubernetes reference](#).

In the event of an error evaluating a match condition the webhook is never called. Whether to reject the request is determined as follows:

1. If **any** match condition evaluated to false (regardless of other errors), the API server skips the webhook.
2. Otherwise:
  - for [failurePolicy: Fail](#), reject the request (without calling the webhook).
  - for [failurePolicy: Ignore](#), proceed with the request but skip the webhook.

## Contacting the webhook

Once the API server has determined a request should be sent to a webhook, it needs to know how to contact the webhook. This is specified in the `clientConfig` stanza of the webhook configuration.

Webhooks can either be called via a URL or a service reference, and can optionally include a custom CA bundle to use to verify the TLS connection.

### URL

`url` gives the location of the webhook, in standard URL form (`scheme://host:port/path`).

The host should not refer to a service running in the cluster; use a service reference by specifying the `service` field instead. The host might be resolved via external DNS in some API servers (e.g., kube-apiserver cannot resolve in-cluster DNS as that would be a layering violation). host may also be an IP address.

Please note that using `localhost` or `127.0.0.1` as a host is risky unless you take great care to run this webhook on all hosts which run an API server which might need to make calls to this webhook. Such installations are likely to be non-portable or not readily run in a new cluster.

The scheme must be "https"; the URL must begin with "https://".

Attempting to use a user or basic auth (for example user:password@) is not allowed. Fragments (#...) and query parameters (?...) are also not allowed.

Here is an example of a mutating webhook configured to call a URL (and expects the TLS certificate to be verified using system trust roots, so does not specify a caBundle):

```
apiVersion: admissionregistration.k8s.io/v1
kind: MutatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 clientConfig:
 url: "https://my-webhook.example.com:9443/my-webhook-path"
```

## Service reference

The service stanza inside clientConfig is a reference to the service for this webhook. If the webhook is running within the cluster, then you should use service instead of url. The service namespace and name are required. The port is optional and defaults to 443. The path is optional and defaults to "/".

Here is an example of a mutating webhook configured to call a service on port "1234" at the subpath "/my-path", and to verify the TLS connection against the ServerName my-service-name.my-service-namespace.svc using a custom CA bundle:

```
apiVersion: admissionregistration.k8s.io/v1
kind: MutatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 clientConfig:
 caBundle: <CA_BUNDLE>
 service:
 namespace: my-service-namespace
 name: my-service-name
 path: /my-path
 port: 1234
```

## Note:

You must replace the <CA\_BUNDLE> in the above example by a valid CA bundle which is a PEM-encoded CA bundle for validating the webhook's server certificate.

## Side effects

Webhooks typically operate only on the content of the AdmissionReview sent to them. Some webhooks, however, make out-of-band changes as part of processing admission requests.

Webhooks that make out-of-band changes ("side effects") must also have a reconciliation mechanism (like a controller) that periodically determines the actual state of the world, and adjusts the out-of-band data modified by the admission webhook to reflect reality. This is because a call to an admission webhook does not guarantee the admitted object will be

persisted as is, or at all. Later webhooks can modify the content of the object, a conflict could be encountered while writing to storage, or the server could power off before persisting the object.

Additionally, webhooks with side effects must skip those side-effects when `dryRun: true` admission requests are handled. A webhook must explicitly indicate that it will not have side-effects when run with `dryRun`, or the dry-run request will not be sent to the webhook and the API request will fail instead.

Webhooks indicate whether they have side effects using the `sideEffects` field in the webhook configuration:

- `None`: calling the webhook will have no side effects.
- `NoneOnDryRun`: calling the webhook will possibly have side effects, but if a request with `dryRun: true` is sent to the webhook, the webhook will suppress the side effects (the webhook is `dryRun`-aware).

Here is an example of a validating webhook indicating it has no side effects on `dryRun: true` requests:

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 sideEffects: NoneOnDryRun
```

## Timeouts

Because webhooks add to API request latency, they should evaluate as quickly as possible. `timeoutSeconds` allows configuring how long the API server should wait for a webhook to respond before treating the call as a failure.

If the timeout expires before the webhook responds, the webhook call will be ignored or the API call will be rejected based on the [failure policy](#).

The timeout value must be between 1 and 30 seconds.

Here is an example of a validating webhook with a custom timeout of 2 seconds:

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 timeoutSeconds: 2
```

The timeout for an admission webhook defaults to 10 seconds.

## Reinvocation policy

A single ordering of mutating admissions plugins (including webhooks) does not work for all cases (see <https://issue.k8s.io/64333> as an example). A mutating webhook can add a new sub-structure to the object (like adding a container to a pod), and other mutating plugins which have already run may have opinions on those new structures (like setting an `imagePullPolicy` on all containers).

To allow mutating admission plugins to observe changes made by other plugins, built-in mutating admission plugins are re-run if a mutating webhook modifies an object, and mutating webhooks can specify a `reinvocationPolicy` to control whether they are reinvoked as well.

`reinvocationPolicy` may be set to `Never` or `IfNeeded`. It defaults to `Never`.

- `Never`: the webhook must not be called more than once in a single admission evaluation.
- `IfNeeded`: the webhook may be called again as part of the admission evaluation if the object being admitted is modified by other admission plugins after the initial webhook call.

The important elements to note are:

- The number of additional invocations is not guaranteed to be exactly one.
- If additional invocations result in further modifications to the object, webhooks are not guaranteed to be invoked again.
- Webhooks that use this option may be reordered to minimize the number of additional invocations.
- To validate an object after all mutations are guaranteed complete, use a validating admission webhook instead (recommended for webhooks with side-effects).

Here is an example of a mutating webhook opting into being re-invoked if later admission plugins modify the object:

```
apiVersion: admissionregistration.k8s.io/v1
kind: MutatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 reinvocationPolicy: IfNeeded
```

Mutating webhooks must be [idempotent](#), able to successfully process an object they have already admitted and potentially modified. This is true for all mutating admission webhooks, since any change they can make in an object could already exist in the user-provided object, but it is essential for webhooks that opt into reinvocation.

## Failure policy

`failurePolicy` defines how unrecognized errors and timeout errors from the admission webhook are handled. Allowed values are `Ignore` or `Fail`.

- `Ignore` means that an error calling the webhook is ignored and the API request is allowed to continue.
- `Fail` means that an error calling the webhook causes the admission to fail and the API request to be rejected.

Here is a mutating webhook configured to reject an API request if errors are encountered calling the admission webhook:

```
apiVersion: admissionregistration.k8s.io/v1
kind: MutatingWebhookConfiguration
webhooks:
- name: my-webhook.example.com
 failurePolicy: Fail
```

The default failurePolicy for an admission webhooks is Fail.

## Monitoring admission webhooks

The API server provides ways to monitor admission webhook behaviors. These monitoring mechanisms help cluster admins to answer questions like:

1. Which mutating webhook mutated the object in a API request?
2. What change did the mutating webhook applied to the object?
3. Which webhooks are frequently rejecting API requests? What's the reason for a rejection?

## Mutating webhook auditing annotations

Sometimes it's useful to know which mutating webhook mutated the object in a API request, and what change did the webhook apply.

The Kubernetes API server performs [auditing](#) on each mutating webhook invocation. Each invocation generates an auditing annotation capturing if a request object is mutated by the invocation, and optionally generates an annotation capturing the applied patch from the webhook admission response. The annotations are set in the audit event for given request on given stage of its execution, which is then pre-processed according to a certain policy and written to a backend.

The audit level of a event determines which annotations get recorded:

- At Metadata audit level or higher, an annotation with key `mutation.webhook.admission.k8s.io/round_{round idx}_index_{order idx}` gets logged with JSON payload indicating a webhook gets invoked for given request and whether it mutated the object or not.

For example, the following annotation gets recorded for a webhook being reinvoked. The webhook is ordered the third in the mutating webhook chain, and didn't mutated the request object during the invocation.

```
the audit event recorded
{
 "kind": "Event",
 "apiVersion": "audit.k8s.io/v1",
 "annotations": {
 "mutation.webhook.admission.k8s.io/round_1_index_2": "{\"configuration\":\"my-mutating-webhook-configuration.example.com\",\"webhook\":\"my-webhook.example.com\",\"mutated\": false}"
 # other annotations
 ...
 }
 # other fields
 ...
}
```

```
the annotation value deserialized
{
 "configuration": "my-mutating-webhook-configuration.example.com",
 "webhook": "my-webhook.example.com",
 "mutated": false
}
```

The following annotation gets recorded for a webhook being invoked in the first round. The webhook is ordered the first in the mutating webhook chain, and mutated the request object during the invocation.

```
the audit event recorded
{
 "kind": "Event",
 "apiVersion": "audit.k8s.io/v1",
 "annotations": {
 "mutation.webhook.admission.k8s.io/round_0_index_0": "{\"configuration\":\"my-mutating-webhook-configuration.example.com\",\"webhook\":\"my-webhook-always-mutate.example.com\",\"mutated\": true}"
 # other annotations
 ...
 }
 # other fields
 ...
}
```

```
the annotation value deserialized
{
 "configuration": "my-mutating-webhook-configuration.example.com",
 "webhook": "my-webhook-always-mutate.example.com",
 "mutated": true
}
```

- At Request audit level or higher, an annotation with key `patch.webhook.admission.k8s.io/round_{round idx}_index_{order idx}` gets logged with JSON payload indicating a webhook gets invoked for given request and what patch gets applied to the request object.

For example, the following annotation gets recorded for a webhook being reinvoked. The webhook is ordered the fourth in the mutating webhook chain, and responded with a JSON patch which got applied to the request object.

```
the audit event recorded
{
 "kind": "Event",
 "apiVersion": "audit.k8s.io/v1",
 "annotations": {
 "patch.webhook.admission.k8s.io/round_1_index_3": "{\"configuration\":\"my-other-mutating-webhook-configuration.example.com\",\"webhook\":\"my-webhook-always-mutate.example.com\",\"patch\":{\"op\":\"add\",\"path\":\"/data/mutation-stage\",\"value\":\"yes\"}},\"patchType\":\"JSONPatch\"}"
 # other annotations
 ...
 }
}
```

```

other fields
...
}

the annotation value deserialized
{
 "configuration": "my-other-mutating-webhook-configuration.example.com",
 "webhook": "my-webhook-always-mutate.example.com",
 "patchType": "JSONPatch",
 "patch": [
 {
 "op": "add",
 "path": "/data/mutation-stage",
 "value": "yes"
 }
]
}

```

## Admission webhook metrics

The API server exposes Prometheus metrics from the `/metrics` endpoint, which can be used for monitoring and diagnosing API server status. The following metrics record status related to admission webhooks.

### API server admission webhook rejection count

Sometimes it's useful to know which admission webhooks are frequently rejecting API requests, and the reason for a rejection.

The API server exposes a Prometheus counter metric recording admission webhook rejections. The metrics are labelled to identify the causes of webhook rejection(s):

- `name`: the name of the webhook that rejected a request.
- `operation`: the operation type of the request, can be one of CREATE, UPDATE, DELETE and CONNECT.
- `type`: the admission webhook type, can be one of admit and validating.
- `error_type`: identifies if an error occurred during the webhook invocation that caused the rejection. Its value can be one of:
  - `calling_webhook_error`: unrecognized errors or timeout errors from the admission webhook happened and the webhook's [Failure policy](#) is set to Fail.
  - `no_error`: no error occurred. The webhook rejected the request with `allowed: false` in the admission response. The metrics label `rejection_code` records the `.status.code` set in the admission response.
  - `apiserver_internal_error`: an API server internal error happened.
- `rejection_code`: the HTTP status code set in the admission response when a webhook rejected a request.

Example of the rejection count metrics:



```
HELP apiserver_admission_webhook_rejection_count [ALPHA] Admission webhook
rejection count, identified by name and broken out for each admission type (validating or
admit) and operation. Additional labels specify an error type (calling_webhook_error or
apiserver_internal_error if an error occurred; no_error otherwise) and optionally a non-zero
rejection code if the webhook rejects the request with an HTTP status code (honored by the
apiserver when the code is greater or equal to 400). Codes greater than 600 are truncated to 600,
to keep the metrics cardinality bounded.
TYPE apiserver_admission_webhook_rejection_count counter
apiserver_admission_webhook_rejection_count{error_type="calling_webhook_error",name="always-timeout-
webhook.example.com",operation="CREATE",rejection_code="0",type="validating"} 1
apiserver_admission_webhook_rejection_count{error_type="calling_webhook_error",name="in
valid-admission-response-
webhook.example.com",operation="CREATE",rejection_code="0",type="validating"} 1
apiserver_admission_webhook_rejection_count{error_type="no_error",name="deny-unwanted-
configmap-data.example.com",operation="CREATE",rejection_code="400",type="validating"} 13
```

## Best practices and warnings

### Idempotence

An idempotent mutating admission webhook is able to successfully process an object it has already admitted and potentially modified. The admission can be applied multiple times without changing the result beyond the initial application.

#### Example of idempotent mutating admission webhooks:

1. For a CREATE pod request, set the field `.spec.securityContext.runAsNonRoot` of the pod to true, to enforce security best practices.
2. For a CREATE pod request, if the field `.spec.containers[].resources.limits` of a container is not set, set default resource limits.
3. For a CREATE pod request, inject a sidecar container with name `foo-sidecar` if no container with the name `foo-sidecar` already exists.

In the cases above, the webhook can be safely reinvoked, or admit an object that already has the fields set.

#### Example of non-idempotent mutating admission webhooks:

1. For a CREATE pod request, inject a sidecar container with name `foo-sidecar` suffixed with the current timestamp (e.g. `foo-sidecar-19700101-000000`).
2. For a CREATE/UPDATE pod request, reject if the pod has label `"env"` set, otherwise add an `"env": "prod"` label to the pod.
3. For a CREATE pod request, blindly append a sidecar container named `foo-sidecar` without looking to see if there is already a `foo-sidecar` container in the pod.

In the first case above, reinvoking the webhook can result in the same sidecar being injected multiple times to a pod, each time with a different container name. Similarly the webhook can inject duplicated containers if the sidecar already exists in a user-provided pod.

In the second case above, reinvoking the webhook will result in the webhook failing on its own output.

In the third case above, reinvoking the webhook will result in duplicated containers in the pod spec, which makes the request invalid and rejected by the API server.

## **Intercepting all versions of an object**

It is recommended that admission webhooks should always intercept all versions of an object by setting `.webhooks[].matchPolicy` to `Equivalent`. It is also recommended that admission webhooks should prefer registering for stable versions of resources. Failure to intercept all versions of an object can result in admission policies not being enforced for requests in certain versions. See [Matching requests: matchPolicy](#) for examples.

## **Availability**

It is recommended that admission webhooks should evaluate as quickly as possible (typically in milliseconds), since they add to API request latency. It is encouraged to use a small timeout for webhooks. See [Timeouts](#) for more detail.

It is recommended that admission webhooks should leverage some format of load-balancing, to provide high availability and performance benefits. If a webhook is running within the cluster, you can run multiple webhook backends behind a service to leverage the load-balancing that service supports.

## **Guaranteeing the final state of the object is seen**

Admission webhooks that need to guarantee they see the final state of the object in order to enforce policy should use a validating admission webhook, since objects can be modified after being seen by mutating webhooks.

For example, a mutating admission webhook is configured to inject a sidecar container with name "foo-sidecar" on every CREATE pod request. If the sidecar *must* be present, a validating admission webhook should also be configured to intercept CREATE pod requests, and validate that a container with name "foo-sidecar" with the expected configuration exists in the to-be-created object.

## **Avoiding deadlocks in self-hosted webhooks**

A webhook running inside the cluster might cause deadlocks for its own deployment if it is configured to intercept resources required to start its own pods.

For example, a mutating admission webhook is configured to admit CREATE pod requests only if a certain label is set in the pod (e.g. "env": "prod"). The webhook server runs in a deployment which doesn't set the "env" label. When a node that runs the webhook server pods becomes unhealthy, the webhook deployment will try to reschedule the pods to another node. However the requests will get rejected by the existing webhook server since the "env" label is unset, and the migration cannot happen.

It is recommended to exclude the namespace where your webhook is running with a [namespaceSelector](#).

## Side effects

It is recommended that admission webhooks should avoid side effects if possible, which means the webhooks operate only on the content of the AdmissionReview sent to them, and do not make out-of-band changes. The `.webhooks[].sideEffects` field should be set to `None` if a webhook doesn't have any side effect.

If side effects are required during the admission evaluation, they must be suppressed when processing an AdmissionReview object with `dryRun` set to `true`, and the `.webhooks[].sideEffects` field should be set to `NoneOnDryRun`. See [Side effects](#) for more detail.

## Avoiding operating on the kube-system namespace

The `kube-system` namespace contains objects created by the Kubernetes system, e.g. service accounts for the control plane components, pods like `kube-dns`. Accidentally mutating or rejecting requests in the `kube-system` namespace may cause the control plane components to stop functioning or introduce unknown behavior. If your admission webhooks don't intend to modify the behavior of the Kubernetes control plane, exclude the `kube-system` namespace from being intercepted using a [namespaceSelector](#).

# Managing Service Accounts

A *ServiceAccount* provides an identity for processes that run in a Pod.

A process inside a Pod can use the identity of its associated service account to authenticate to the cluster's API server.

For an introduction to service accounts, read [configure service accounts](#).

This task guide explains some of the concepts behind ServiceAccounts. The guide also explains how to obtain or revoke tokens that represent ServiceAccounts.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To be able to follow these steps exactly, ensure you have a namespace named `examplens`. If you don't, create one by running:

```
kubectl create namespace examplens
```

# User accounts versus service accounts

Kubernetes distinguishes between the concept of a user account and a service account for a number of reasons:

- User accounts are for humans. Service accounts are for application processes, which (for Kubernetes) run in containers that are part of pods.
- User accounts are intended to be global: names must be unique across all namespaces of a cluster. No matter what namespace you look at, a particular username that represents a user represents the same user. In Kubernetes, service accounts are namespaced: two different namespaces can contain ServiceAccounts that have identical names.
- Typically, a cluster's user accounts might be synchronised from a corporate database, where new user account creation requires special privileges and is tied to complex business processes. By contrast, service account creation is intended to be more lightweight, allowing cluster users to create service accounts for specific tasks on demand. Separating ServiceAccount creation from the steps to onboard human users makes it easier for workloads to follow the principle of least privilege.
- Auditing considerations for humans and service accounts may differ; the separation makes that easier to achieve.
- A configuration bundle for a complex system may include definition of various service accounts for components of that system. Because service accounts can be created without many constraints and have namespaced names, such configuration is usually portable.

## Bound service account tokens

ServiceAccount tokens can be bound to API objects that exist in the kube-apiserver. This can be used to tie the validity of a token to the existence of another API object. Supported object types are as follows:

- Pod (used for projected volume mounts, see below)
- Secret (can be used to allow revoking a token by deleting the Secret)
- Node (in v1.30, creating new node-bound tokens is alpha, using existing node-bound tokens is beta)

When a token is bound to an object, the object's `metadata.name` and `metadata.uid` are stored as extra 'private claims' in the issued JWT.

When a bound token is presented to the kube-apiserver, the service account authenticator will extract and verify these claims. If the referenced object no longer exists (or its `metadata.uid` does not match), the request will not be authenticated.

## Additional metadata in Pod bound tokens

FEATURE STATE: Kubernetes v1.30 [beta]

When a service account token is bound to a Pod object, additional metadata is also embedded into the token that indicates the value of the bound pod's `spec.nodeName` field, and the uid of that Node, if available.

This node information is **not** verified by the kube-apiserver when the token is used for authentication. It is included so integrators do not have to fetch Pod or Node API objects to check the associated Node name and uid when inspecting a JWT.

## Verifying and inspecting private claims

The TokenReview API can be used to verify and extract private claims from a token:

1. First, assume you have a pod named test-pod and a service account named my-sa.
2. Create a token that is bound to this Pod:

```
kubectl create token my-sa --bound-object-kind="Pod" --bound-object-name="test-pod"
```

1. Copy this token into a new file named tokenreview.yaml:

```
apiVersion: authentication.k8s.io/v1
kind: TokenReview
spec:
 token: <token from step 2>
```

1. Submit this resource to the apiserver for review:

```
kubectl create -o yaml -f tokenreview.yaml # we use '-o yaml' so we can inspect the output
```

You should see an output like below:

```
apiVersion: authentication.k8s.io/v1
kind: TokenReview
metadata:
 creationTimestamp: null
spec:
 token: <token>
status:
 audiences:
 - https://kubernetes.default.svc.cluster.local
 authenticated: true
 user:
 extra:
 authentication.kubernetes.io/credential-id:
 - JTI=7ee52be0-9045-4653-aa5e-0da57b8dccdc
 authentication.kubernetes.io/node-name:
 - kind-control-plane
 authentication.kubernetes.io/node-uid:
 - 497e9d9a-47aa-4930-b0f6-9f2fb574c8c6
 authentication.kubernetes.io/pod-name:
 - test-pod
 authentication.kubernetes.io/pod-uid:
 - e87dbbd6-3d7e-45db-aafb-72b24627dff5
 groups:
 - system:serviceaccounts
 - system:serviceaccounts:default
 - system:authenticated
 uid: f8b4161b-2e2b-11e9-86b7-2afc33b31a7e
 username: system:serviceaccount:default:my-sa
```

## Note:

Despite using `kubectl create -f` to create this resource, and defining it similar to other resource types in Kubernetes, `TokenReview` is a special type and the `kube-apiserver` does not actually persist the `TokenReview` object into `etcd`. Hence `kubectl get tokenreview` is not a valid command.

## Bound service account token volume mechanism

FEATURE STATE: Kubernetes v1.22 [stable]

By default, the Kubernetes control plane (specifically, the [ServiceAccount admission controller](#)) adds a [projected volume](#) to Pods, and this volume includes a token for Kubernetes API access.

Here's an example of how that looks for a launched Pod:

```
...
- name: kube-api-access-<random-suffix>
 projected:
 sources:
 - serviceAccountToken:
 path: token # must match the path the app expects
 - configMap:
 items:
 - key: ca.crt
 path: ca.crt
 name: kube-root-ca.crt
 - downwardAPI:
 items:
 - fieldRef:
 apiVersion: v1
 fieldPath: metadata.namespace
 path: namespace
```

That manifest snippet defines a projected volume that consists of three sources. In this case, each source also represents a single path within that volume. The three sources are:

1. A `serviceAccountToken` source, that contains a token that the kubelet acquires from `kube-apiserver`. The kubelet fetches time-bound tokens using the `TokenRequest` API. A token served for a `TokenRequest` expires either when the pod is deleted or after a defined lifespan (by default, that is 1 hour). The kubelet also refreshes that token before the token expires. The token is bound to the specific Pod and has the `kube-apiserver` as its audience. This mechanism superseded an earlier mechanism that added a volume based on a `Secret`, where the `Secret` represented the `ServiceAccount` for the Pod, but did not expire.
2. A `configMap` source. The `ConfigMap` contains a bundle of certificate authority data. Pods can use these certificates to make sure that they are connecting to your cluster's `kube-apiserver` (and not to middlebox or an accidentally misconfigured peer).
3. A `downwardAPI` source that looks up the name of the namespace containing the Pod, and makes that name information available to application code running inside the Pod.

Any container within the Pod that mounts this particular volume can access the above information.

**Note:**

There is no specific mechanism to invalidate a token issued via `TokenRequest`. If you no longer trust a bound service account token for a Pod, you can delete that Pod. Deleting a Pod expires its bound service account tokens.

## Manual Secret management for ServiceAccounts

Versions of Kubernetes before v1.22 automatically created credentials for accessing the Kubernetes API. This older mechanism was based on creating token Secrets that could then be mounted into running Pods.

In more recent versions, including Kubernetes v1.30, API credentials are [obtained directly](#) using the [TokenRequest](#) API, and are mounted into Pods using a projected volume. The tokens obtained using this method have bounded lifetimes, and are automatically invalidated when the Pod they are mounted into is deleted.

You can still [manually create](#) a Secret to hold a service account token; for example, if you need a token that never expires.

Once you manually create a Secret and link it to a ServiceAccount, the Kubernetes control plane automatically populates the token into that Secret.

**Note:**

Although the manual mechanism for creating a long-lived ServiceAccount token exists, using [TokenRequest](#) to obtain short-lived API access tokens is recommended instead.

## Auto-generated legacy ServiceAccount token clean up

Before version 1.24, Kubernetes automatically generated Secret-based tokens for ServiceAccounts. To distinguish between automatically generated tokens and manually created ones, Kubernetes checks for a reference from the ServiceAccount's `secrets` field. If the Secret is referenced in the `secrets` field, it is considered an auto-generated legacy token. Otherwise, it is considered a manually created legacy token. For example:

```
apiVersion: v1
kind: ServiceAccount
metadata:
 name: build-robot
 namespace: default
secrets:
 - name: build-robot-secret # usually NOT present for a manually generated token
```

Beginning from version 1.29, legacy ServiceAccount tokens that were generated automatically will be marked as invalid if they remain unused for a certain period of time (set to default at one year). Tokens that continue to be unused for this defined period (again, by default, one year) will subsequently be purged by the control plane.

If users use an invalidated auto-generated token, the token validator will

1. add an audit annotation for the key-value pair `authentication.k8s.io/legacy-token-invalidated: <secret name>/<namespace>`,
2. increment the `invalid_legacy_auto_token_uses_total` metric count,
3. update the Secret label `kubernetes.io/legacy-token-last-used` with the new date,
4. return an error indicating that the token has been invalidated.

When receiving this validation error, users can update the Secret to remove the `kubernetes.io/legacy-token-invalid-since` label to temporarily allow use of this token.

Here's an example of an auto-generated legacy token that has been marked with the `kubernetes.io/legacy-token-last-used` and `kubernetes.io/legacy-token-invalid-since` labels:

```
apiVersion: v1
kind: Secret
metadata:
 name: build-robot-secret
 namespace: default
labels:
 kubernetes.io/legacy-token-last-used: 2022-10-24
 kubernetes.io/legacy-token-invalid-since: 2023-10-25
annotations:
 kubernetes.io/service-account.name: build-robot
type: kubernetes.io/service-account-token
```

## Control plane details

### ServiceAccount controller

A ServiceAccount controller manages the ServiceAccounts inside namespaces, and ensures a ServiceAccount named "default" exists in every active namespace.

### Token controller

The service account token controller runs as part of `kube-controller-manager`. This controller acts asynchronously. It:

- watches for ServiceAccount deletion and deletes all corresponding ServiceAccount token Secrets.
- watches for ServiceAccount token Secret addition, and ensures the referenced ServiceAccount exists, and adds a token to the Secret if needed.
- watches for Secret deletion and removes a reference from the corresponding ServiceAccount if needed.

You must pass a service account private key file to the token controller in the `kube-controller-manager` using the `--service-account-private-key-file` flag. The private key is used to sign generated service account tokens. Similarly, you must pass the corresponding public key to the `kube-apiserver` using the `--service-account-key-file` flag. The public key will be used to verify the tokens during authentication.



## ServiceAccount admission controller

The modification of pods is implemented via a plugin called an [Admission Controller](#). It is part of the API server. This admission controller acts synchronously to modify pods as they are created. When this plugin is active (and it is by default on most distributions), then it does the following when a Pod is created:

1. If the pod does not have a `.spec.serviceAccountName` set, the admission controller sets the name of the ServiceAccount for this incoming Pod to default.
2. The admission controller ensures that the ServiceAccount referenced by the incoming Pod exists. If there is no ServiceAccount with a matching name, the admission controller rejects the incoming Pod. That check applies even for the default ServiceAccount.
3. Provided that neither the ServiceAccount's `automountServiceAccountToken` field nor the Pod's `automountServiceAccountToken` field is set to false:
  - the admission controller mutates the incoming Pod, adding an extra [volume](#) that contains a token for API access.
  - the admission controller adds a `volumeMount` to each container in the Pod, skipping any containers that already have a volume mount defined for the path `/var/run/secrets/kubernetes.io/serviceaccount`. For Linux containers, that volume is mounted at `/var/run/secrets/kubernetes.io/serviceaccount`; on Windows nodes, the mount is at the equivalent path.
4. If the spec of the incoming Pod doesn't already contain any `imagePullSecrets`, then the admission controller adds `imagePullSecrets`, copying them from the ServiceAccount.

## Legacy ServiceAccount token tracking controller

FEATURE STATE: Kubernetes v1.28 [stable]

This controller generates a ConfigMap called `kube-system/kube-apiserver-legacy-service-account-token-tracking` in the `kube-system` namespace. The ConfigMap records the timestamp when legacy service account tokens began to be monitored by the system.

## Legacy ServiceAccount token cleaner

FEATURE STATE: Kubernetes v1.30 [stable]

The legacy ServiceAccount token cleaner runs as part of the `kube-controller-manager` and checks every 24 hours to see if any auto-generated legacy ServiceAccount token has not been used in a *specified amount of time*. If so, the cleaner marks those tokens as invalid.

The cleaner works by first checking the ConfigMap created by the control plane (provided that `LegacyServiceAccountTokenTracking` is enabled). If the current time is a *specified amount of time* after the date in the ConfigMap, the cleaner then loops through the list of Secrets in the cluster and evaluates each Secret that has the type `kubernetes.io/service-account-token`.

If a Secret meets all of the following conditions, the cleaner marks it as invalid:

- The Secret is auto-generated, meaning that it is bi-directionally referenced by a ServiceAccount.
- The Secret is not currently mounted by any pods.
- The Secret has not been used in a *specified amount of time* since it was created or since it was last used.

The cleaner marks a Secret invalid by adding a label called `kubernetes.io/legacy-token-invalid-since` to the Secret, with the current date as the value. If an invalid Secret is not used in a *specified amount of time*, the cleaner will delete it.

#### Note:

All the *specified amount of time* above defaults to one year. The cluster administrator can configure this value through the `--legacy-service-account-token-clean-up-period` command line argument for the `kube-controller-manager` component.

## TokenRequest API

FEATURE STATE: Kubernetes v1.22 [stable]

You use the [TokenRequest](#) subresource of a ServiceAccount to obtain a time-bound token for that ServiceAccount. You don't need to call this to obtain an API token for use within a container, since the kubelet sets this up for you using a *projected volume*.

If you want to use the TokenRequest API from `kubectl`, see [Manually create an API token for a ServiceAccount](#).

The Kubernetes control plane (specifically, the ServiceAccount admission controller) adds a projected volume to Pods, and the kubelet ensures that this volume contains a token that lets containers authenticate as the right ServiceAccount.

(This mechanism superseded an earlier mechanism that added a volume based on a Secret, where the Secret represented the ServiceAccount for the Pod but did not expire.)

Here's an example of how that looks for a launched Pod:

```
...
- name: kube-api-access-<random-suffix>
 projected:
 defaultMode: 420 # decimal equivalent of octal 0644
 sources:
 - serviceAccountToken:
 expirationSeconds: 3607
 path: token
 - configMap:
 items:
 - key: ca.crt
 path: ca.crt
 name: kube-root-ca.crt
 - downwardAPI:
 items:
 - fieldRef:
 apiVersion: v1
 fieldPath: metadata.namespace
 path: namespace
```

That manifest snippet defines a projected volume that combines information from three sources:

1. A serviceAccountToken source, that contains a token that the kubelet acquires from kube-apiserver. The kubelet fetches time-bound tokens using the TokenRequest API. A token served for a TokenRequest expires either when the pod is deleted or after a defined lifespan (by default, that is 1 hour). The token is bound to the specific Pod and has the kube-apiserver as its audience.
2. A configMap source. The ConfigMap contains a bundle of certificate authority data. Pods can use these certificates to make sure that they are connecting to your cluster's kube-apiserver (and not to middlebox or an accidentally misconfigured peer).
3. A downwardAPI source. This downwardAPI volume makes the name of the namespace containing the Pod available to application code running inside the Pod.

Any container within the Pod that mounts this volume can access the above information.

## Create additional API tokens

### Caution:

Only create long-lived API tokens if the [token request](#) mechanism is not suitable. The token request mechanism provides time-limited tokens; because these expire, they represent a lower risk to information security.

To create a non-expiring, persisted API token for a ServiceAccount, create a Secret of type `kubernetes.io/service-account-token` with an annotation referencing the ServiceAccount. The control plane then generates a long-lived token and updates that Secret with that generated token data.

Here is a sample manifest for such a Secret:

[secret/serviceaccount/mysecretname.yaml](#)

```
apiVersion: v1
kind: Secret
type: kubernetes.io/service-account-token
metadata:
 name: mysecretname
 annotations:
 kubernetes.io/service-account.name: myserviceaccount
```

To create a Secret based on this example, run:

```
kubectl -n examplens create -f https://k8s.io/examples/secret/serviceaccount/
mysecretname.yaml
```

To see the details for that Secret, run:

```
kubectl -n examplens describe secret mysecretname
```

The output is similar to:

```
Name: mysecretname
Namespace: examplens
Labels: <none>
Annotations: kubernetes.io/service-account.name=myserviceaccount
 kubernetes.io/service-account.uid=8a85c4c4-8483-11e9-bc42-526af7764f64

Type: kubernetes.io/service-account-token

Data
====
ca.crt: 1362 bytes
namespace: 9 bytes
token: ...
```

If you launch a new Pod into the examplens namespace, it can use the myserviceaccount service-account-token Secret that you just created.

### Caution:

Do not reference manually created Secrets in the secrets field of a ServiceAccount. Or the manually created Secrets will be cleaned if it is not used for a long time. Please refer to [auto-generated legacy ServiceAccount token clean up](#).

## Delete/invalidate a ServiceAccount token

If you know the name of the Secret that contains the token you want to remove:

```
kubectl delete secret name-of-secret
```

Otherwise, first find the Secret for the ServiceAccount.

```
This assumes that you already have a namespace named 'examplens'
kubectl -n examplens get serviceaccount/example-automated-thing -o yaml
```

The output is similar to:

```
apiVersion: v1
kind: ServiceAccount
metadata:
 annotations:
 kubectl.kubernetes.io/last-applied-configuration: |
 {"apiVersion":"v1","kind":"ServiceAccount","metadata":{"annotations":{},"name":"example-automated-thing","namespace":"examplens"}}
 creationTimestamp: "2019-07-21T07:07:07Z"
 name: example-automated-thing
 namespace: examplens
 resourceVersion: "777"
 selfLink: /api/v1/namespaces/examplens/serviceaccounts/example-automated-thing
 uid: f23fd170-66f2-4697-b049-e1e266b7f835
secrets:
 - name: example-automated-thing-token-zyxwv
```

Then, delete the Secret you now know the name of:

```
kubectl -n exemplens delete secret/example-automated-thing-token-zyxwv
```

## Clean up

If you created a namespace `exemplens` to experiment with, you can remove it:

```
kubectl delete namespace exemplens
```

## What's next

- Read more details about [projected volumes](#).

# Certificates and Certificate Signing Requests

Kubernetes certificate and trust bundle APIs enable automation of [X.509](#) credential provisioning by providing a programmatic interface for clients of the Kubernetes API to request and obtain X.509 [certificates](#) from a Certificate Authority (CA).

There is also experimental (alpha) support for distributing [trust bundles](#).

## Certificate signing requests

FEATURE STATE: Kubernetes v1.19 [stable]

A `CertificateSigningRequest` (CSR) resource is used to request that a certificate be signed by a denoted signer, after which the request may be approved or denied before finally being signed.

### Request signing process

The `CertificateSigningRequest` resource type allows a client to ask for an X.509 certificate be issued, based on a signing request. The `CertificateSigningRequest` object includes a PEM-encoded PKCS#10 signing request in the `spec.request` field. The `CertificateSigningRequest` denotes the signer (the recipient that the request is being made to) using the `spec.signerName` field. Note that `spec.signerName` is a required key after API version `certificates.k8s.io/v1`. In Kubernetes v1.22 and later, clients may optionally set the `spec.expirationSeconds` field to request a particular lifetime for the issued certificate. The minimum valid value for this field is 600, i.e. ten minutes.

Once created, a `CertificateSigningRequest` must be approved before it can be signed. Depending on the signer selected, a `CertificateSigningRequest` may be automatically approved by a [controller](#). Otherwise, a `CertificateSigningRequest` must be manually approved either via the REST API (or `client-go`) or by running `kubectl certificate approve`. Likewise, a `CertificateSigningRequest` may also be denied, which tells the configured signer that it must not sign the request.

For certificates that have been approved, the next step is signing. The relevant signing controller first validates that the signing conditions are met and then creates a certificate. The signing controller then updates the `CertificateSigningRequest`, storing the new certificate into

the `status.certificate` field of the existing `CertificateSigningRequest` object. The `status.certificate` field is either empty or contains a X.509 certificate, encoded in PEM format. The `CertificateSigningRequest` `status.certificate` field is empty until the signer does this.

Once the `status.certificate` field has been populated, the request has been completed and clients can now fetch the signed certificate PEM data from the `CertificateSigningRequest` resource. The signers can instead deny certificate signing if the approval conditions are not met.

In order to reduce the number of old `CertificateSigningRequest` resources left in a cluster, a garbage collection controller runs periodically. The garbage collection removes `CertificateSigningRequests` that have not changed state for some duration:

- Approved requests: automatically deleted after 1 hour
- Denied requests: automatically deleted after 1 hour
- Failed requests: automatically deleted after 1 hour
- Pending requests: automatically deleted after 24 hours
- All requests: automatically deleted after the issued certificate has expired

## Certificate signing authorization

To allow creating a `CertificateSigningRequest` and retrieving any `CertificateSigningRequest`:

- Verbs: create, get, list, watch, group: certificates.k8s.io, resource: certificatesigningrequests

For example:

[access/certificate-signing-request/clusterrole-create.yaml](#)

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: csr-creator
rules:
- apiGroups:
 - certificates.k8s.io
 resources:
 - certificatesigningrequests
 verbs:
 - create
 - get
 - list
 - watch
```

To allow approving a `CertificateSigningRequest`:

- Verbs: get, list, watch, group: certificates.k8s.io, resource: certificatesigningrequests
- Verbs: update, group: certificates.k8s.io, resource: certificatesigningrequests/approval
- Verbs: approve, group: certificates.k8s.io, resource: signers, resourceName: <signerNameDomain>/<signerNamePath> or <signerNameDomain>/\*

For example:

[access/certificate-signing-request/clusterrole-approve.yaml](#)

```

apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: csr-approver
rules:
- apiGroups:
 - certificates.k8s.io
 resources:
 - certificatesigningrequests
 verbs:
 - get
 - list
 - watch
- apiGroups:
 - certificates.k8s.io
 resources:
 - certificatesigningrequests/approval
 verbs:
 - update
- apiGroups:
 - certificates.k8s.io
 resources:
 - signers
 resourceNames:
 - example.com/my-signer-name
example.com/* can be used to authorize for all signers in the 'example.com' domain
verbs:
- approve

```

To allow signing a CertificateSigningRequest:

- Verbs: get, list, watch, group: certificates.k8s.io, resource: certificatesigningrequests
- Verbs: update, group: certificates.k8s.io, resource: certificatesigningrequests/status
- Verbs: sign, group: certificates.k8s.io, resource: signers, resourceName: <signerNameDomain>/<signerNamePath> or <signerNameDomain>/\*

[access/certificate-signing-request/clusterrole-sign.yaml](#)

```

apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: csr-signer
rules:
- apiGroups:
 - certificates.k8s.io
 resources:
 - certificatesigningrequests
 verbs:
 - get
 - list
 - watch
- apiGroups:

```

```
- certificates.k8s.io
resources:
- certificatesigningrequests/status
verbs:
- update
- apiGroups:
- certificates.k8s.io
resources:
- signers
resourceNames:
- example.com/my-signer-name
example.com/* can be used to authorize for all signers in the 'example.com' domain
verbs:
- sign
```

## Signers

Signers abstractly represent the entity or entities that might sign, or have signed, a security certificate.

Any signer that is made available for outside a particular cluster should provide information about how the signer works, so that consumers can understand what that means for CertificateSigningRequests and (if enabled) [ClusterTrustBundles](#).

This includes:

1. **Trust distribution:** how trust anchors (CA certificates or certificate bundles) are distributed.
2. **Permitted subjects:** any restrictions on and behavior when a disallowed subject is requested.
3. **Permitted x509 extensions:** including IP subjectAltNames, DNS subjectAltNames, Email subjectAltNames, URI subjectAltNames etc, and behavior when a disallowed extension is requested.
4. **Permitted key usages / extended key usages:** any restrictions on and behavior when usages different than the signer-determined usages are specified in the CSR.
5. **Expiration/certificate lifetime:** whether it is fixed by the signer, configurable by the admin, determined by the CSR spec.expirationSeconds field, etc and the behavior when the signer-determined expiration is different from the CSR spec.expirationSeconds field.
6. **CA bit allowed/disallowed:** and behavior if a CSR contains a request a for a CA certificate when the signer does not permit it.

Commonly, the status.certificate field of a CertificateSigningRequest contains a single PEM-encoded X.509 certificate once the CSR is approved and the certificate is issued. Some signers store multiple certificates into the status.certificate field. In that case, the documentation for the signer should specify the meaning of additional certificates; for example, this might be the certificate plus intermediates to be presented during TLS handshakes.

If you want to make the *trust anchor* (root certificate) available, this should be done separately from a CertificateSigningRequest and its status.certificate field. For example, you could use a ClusterTrustBundle.

The PKCS#10 signing request format does not have a standard mechanism to specify a certificate expiration or lifetime. The expiration or lifetime therefore has to be set through the spec.expirationSeconds field of the CSR object. The built-in signers use the



ClusterSigningDuration configuration option, which defaults to 1 year, (the --cluster-signing-duration command-line flag of the kube-controller-manager) as the default when no spec.expirationSeconds is specified. When spec.expirationSeconds is specified, the minimum of spec.expirationSeconds and ClusterSigningDuration is used.

**Note:**

The spec.expirationSeconds field was added in Kubernetes v1.22. Earlier versions of Kubernetes do not honor this field. Kubernetes API servers prior to v1.22 will silently drop this field when the object is created.

## Kubernetes signers

Kubernetes provides built-in signers that each have a well-known signerName:

1. `kubernetes.io/kube-apiserver-client`: signs certificates that will be honored as client certificates by the API server. Never auto-approved by [kube-controller-manager](#).
  1. Trust distribution: signed certificates must be honored as client certificates by the API server. The CA bundle is not distributed by any other means.
  2. Permitted subjects - no subject restrictions, but approvers and signers may choose not to approve or sign. Certain subjects like cluster-admin level users or groups vary between distributions and installations, but deserve additional scrutiny before approval and signing. The CertificateSubjectRestriction admission plugin is enabled by default to restrict system:masters, but it is often not the only cluster-admin subject in a cluster.
  3. Permitted x509 extensions - honors subjectAltName and key usage extensions and discards other extensions.
  4. Permitted key usages - must include ["client auth"]. Must not include key usages beyond ["digital signature", "key encipherment", "client auth"].
  5. Expiration/certificate lifetime - for the kube-controller-manager implementation of this signer, set to the minimum of the --cluster-signing-duration option or, if specified, the spec.expirationSeconds field of the CSR object.
  6. CA bit allowed/disallowed - not allowed.
2. `kubernetes.io/kube-apiserver-client-kubelet`: signs client certificates that will be honored as client certificates by the API server. May be auto-approved by [kube-controller-manager](#).
  1. Trust distribution: signed certificates must be honored as client certificates by the API server. The CA bundle is not distributed by any other means.
  2. Permitted subjects - organizations are exactly ["system:nodes"], common name starts with "system:node:".
  3. Permitted x509 extensions - honors key usage extensions, forbids subjectAltName extensions and drops other extensions.
  4. Permitted key usages - ["key encipherment", "digital signature", "client auth"] or ["digital signature", "client auth"].
  5. Expiration/certificate lifetime - for the kube-controller-manager implementation of this signer, set to the minimum of the --cluster-signing-duration option or, if specified, the spec.expirationSeconds field of the CSR object.
  6. CA bit allowed/disallowed - not allowed.

- kubernetes.io/kubelet-serving: signs serving certificates that are honored as a valid
3. kubelet serving certificate by the API server, but has no other guarantees. Never auto-approved by [kube-controller-manager](#).
    1. Trust distribution: signed certificates must be honored by the API server as valid to terminate connections to a kubelet. The CA bundle is not distributed by any other means.
    2. Permitted subjects - organizations are exactly ["system:nodes"], common name starts with "system:node:".
    3. Permitted x509 extensions - honors key usage and DNSName/IPAddress subjectAltName extensions, forbids EmailAddress and URI subjectAltName extensions, drops other extensions. At least one DNS or IP subjectAltName must be present.
    4. Permitted key usages - ["key encipherment", "digital signature", "server auth"] or ["digital signature", "server auth"].
    5. Expiration/certificate lifetime - for the kube-controller-manager implementation of this signer, set to the minimum of the --cluster-signing-duration option or, if specified, the spec.expirationSeconds field of the CSR object.
    6. CA bit allowed/disallowed - not allowed.
  4. kubernetes.io/legacy-unknown: has no guarantees for trust at all. Some third-party distributions of Kubernetes may honor client certificates signed by it. The stable CertificateSigningRequest API (version certificates.k8s.io/v1 and later) does not allow to set the signerName as kubernetes.io/legacy-unknown. Never auto-approved by [kube-controller-manager](#).
    1. Trust distribution: None. There is no standard trust or distribution for this signer in a Kubernetes cluster.
    2. Permitted subjects - any
    3. Permitted x509 extensions - honors subjectAltName and key usage extensions and discards other extensions.
    4. Permitted key usages - any
    5. Expiration/certificate lifetime - for the kube-controller-manager implementation of this signer, set to the minimum of the --cluster-signing-duration option or, if specified, the spec.expirationSeconds field of the CSR object.
    6. CA bit allowed/disallowed - not allowed.

The kube-controller-manager implements [control plane signing](#) for each of the built in signers. Failures for all of these are only reported in kube-controller-manager logs.

**Note:**

The spec.expirationSeconds field was added in Kubernetes v1.22. Earlier versions of Kubernetes do not honor this field. Kubernetes API servers prior to v1.22 will silently drop this field when the object is created.

Distribution of trust happens out of band for these signers. Any trust outside of those described above are strictly coincidental. For instance, some distributions may honor kubernetes.io/legacy-unknown as client certificates for the kube-apiserver, but this is not a standard. None of these usages are related to ServiceAccount token secrets .data[ca.crt] in any way. That CA bundle is only guaranteed to verify a connection to the API server using the default service (kubernetes.default.svc).

## Custom signers

You can also introduce your own custom signer, which should have a similar prefixed name but using your own domain name. For example, if you represent an open source project that uses the domain `open-fictional.example` then you might use `issuer.open-fictional.example/service-mesh` as a signer name.

A custom signer uses the Kubernetes API to issue a certificate. See [API-based signers](#).

## Signing

### Control plane signer

The Kubernetes control plane implements each of the [Kubernetes signers](#), as part of the kube-controller-manager.

#### Note:

Prior to Kubernetes v1.18, the kube-controller-manager would sign any CSRs that were marked as approved.

#### Note:

The `spec.expirationSeconds` field was added in Kubernetes v1.22. Earlier versions of Kubernetes do not honor this field. Kubernetes API servers prior to v1.22 will silently drop this field when the object is created.

### API-based signers

Users of the REST API can sign CSRs by submitting an UPDATE request to the status subresource of the CSR to be signed.

As part of this request, the `status.certificate` field should be set to contain the signed certificate. This field contains one or more PEM-encoded certificates.

All PEM blocks must have the "CERTIFICATE" label, contain no headers, and the encoded data must be a BER-encoded ASN.1 Certificate structure as described in [section 4 of RFC5280](#).

Example certificate content:

```
-----BEGIN CERTIFICATE-----
MIIDgjCCAmqgAwIBAgIUC1N1EJ4Qnsd322BhDPRwmg3b/oAwDQYJKoZIhvcNAQEL
BQAwXDELMAkGA1UEBhMCeHgxCjAIBgNVBAGMAXgxGjAIBgNVBACMAXgxGjAIBgNV
BAoMAXgxGjAIBgNVBAsMAXgxGjAIBgNVBAMMAmNhMRAwDgYJKoZIhvcNAQkBFgF4
MB4XDTIwMDcwNjIyMDcwMFoXDTI1MDcwNTIyMDcwMFowNzEVMBMGGA1UEChMMc3lz
dGVtOm5vZGVzMR4wHAYDVQQDEhVzeXN0ZW06bm9kZToxMjc0MC4wLjEwggEiMA0G
CSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQDne5X2eQ1JcLZkKvhzCR4Hx19+ZmU3
+e1zfOywLdoQxrPi+o4hVsUH3q0y52BMA7u1yehHDRSaq9u62cmi5ekgXhXHHzGmm
kmW5n0itRECV3SFsSm2DSghRKf0mm6iTYHWDHhZUXKdm9lPPWoSOxoR5oqOsm3JEh
Q7Et13wrvTJqBMJo1GTwQuF+HYOku0NF/DLqbZlcpI08yQKyrBgYz2uO51/oNp8a
sTCsV4OUfyHhx2BBLUo4g4SptHFySTBwlpRWBnSjZPOhmN74JcpTLB4J5f4iEeA7
```

```
2QytZfADckG4wVkhH3C2EJUmRtFIBVirwDn39GXkSGlnvnMgF3uLZ6zNagMBAAGj
YTBfMA4GA1UdDwEB/wQEAwIFoDATBgNVHSUEDDAKBggrBgEFBQcDAjAMBgNVHRMB
Af8EAjAAMB0GA1UdDgQWBBTREl2hW54lkQBDeVCd2f2VSlB1DALBgNVHREEBDAC
ggAwDQYJKoZIhvcNAQELBQADggEBABpZjuIKTq8pCaX8dMEGPWtAykgLsTcD2jYr
L0/TCrqmuaaliUa42jQTt2OVsVP/L8ofFunj/KjpQU0bvKJPLMRKtmxbhXuQCQi1
qCRkp8o93mHvEz3mTUN+D1cfQ2fpsBENLnpS0F4G/JyY2Vrh19/X8+mImMEK5eOy
o0BMby7byUj98WmcUvNCiXbC6F45QTmkwEhMqWns0JZQY+/XeDhEcg+lJvz9Eyo2
aGgPsy1o3DpyXnyfJWAWMhOz7cikS5X2adesbgI86PhEHBXPIJ1v13ZdfCExmdd
M1fLPhLyR54fGaY+7/X8P9AZzPefAkwiseXwe9ii6/a08vWoiE4=
-----END CERTIFICATE-----
```

Non-PEM content may appear before or after the CERTIFICATE PEM blocks and is unvalidated, to allow for explanatory text as described in [section 5.2 of RFC7468](#).

When encoded in JSON or YAML, this field is base-64 encoded. A CertificateSigningRequest containing the example certificate above would look like this:

```
apiVersion: certificates.k8s.io/v1
kind: CertificateSigningRequest
...
status:
 certificate: "LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JS..."
```

## Approval or rejection

Before a [signer](#) issues a certificate based on a CertificateSigningRequest, the signer typically checks that the issuance for that CSR has been *approved*.

### Control plane automated approval

The kube-controller-manager ships with a built-in approver for certificates with a signerName of kubernetes.io/kube-apiserver-client-kubelet that delegates various permissions on CSRs for node credentials to authorization. The kube-controller-manager POSTs SubjectAccessReview resources to the API server in order to check authorization for certificate approval.

### Approval or rejection using kubectl

A Kubernetes administrator (with appropriate permissions) can manually approve (or deny) CertificateSigningRequests by using the kubectl certificate approve and kubectl certificate deny commands.

To approve a CSR with kubectl:

```
kubectl certificate approve <certificate-signing-request-name>
```

Likewise, to deny a CSR:

```
kubectl certificate deny <certificate-signing-request-name>
```

## Approval or rejection using the Kubernetes API

Users of the REST API can approve CSRs by submitting an UPDATE request to the approval subresource of the CSR to be approved. For example, you could write an [operator](#) that watches for a particular kind of CSR and then sends an UPDATE to approve them.

When you make an approval or rejection request, set either the Approved or Denied status condition based on the state you determine:

For Approved CSRs:

```
apiVersion: certificates.k8s.io/v1
kind: CertificateSigningRequest
...
status:
 conditions:
 - lastUpdateTime: "2020-02-08T11:37:35Z"
 lastTransitionTime: "2020-02-08T11:37:35Z"
 message: Approved by my custom approver controller
 reason: ApprovedByMyPolicy # You can set this to any string
 type: Approved
```

For Denied CSRs:

```
apiVersion: certificates.k8s.io/v1
kind: CertificateSigningRequest
...
status:
 conditions:
 - lastUpdateTime: "2020-02-08T11:37:35Z"
 lastTransitionTime: "2020-02-08T11:37:35Z"
 message: Denied by my custom approver controller
 reason: DeniedByMyPolicy # You can set this to any string
 type: Denied
```

It's usual to set status.conditions.reason to a machine-friendly reason code using TitleCase; this is a convention but you can set it to anything you like. If you want to add a note for human consumption, use the status.conditions.message field.

## Cluster trust bundles

FEATURE STATE: Kubernetes v1.27 [alpha]

### Note:

In Kubernetes 1.30, you must enable the ClusterTrustBundle [feature gate](#) and the certificates.k8s.io/v1alpha1 [API group](#) in order to use this API.

A ClusterTrustBundles is a cluster-scoped object for distributing X.509 trust anchors (root certificates) to workloads within the cluster. They're designed to work well with the [signer](#) concept from CertificateSigningRequests.

ClusterTrustBundles can be used in two modes: [signer-linked](#) and [signer-unlinked](#).

## Common properties and validation

All ClusterTrustBundle objects have strong validation on the contents of their trustBundle field. That field must contain one or more X.509 certificates, DER-serialized, each wrapped in a PEM CERTIFICATE block. The certificates must parse as valid X.509 certificates.

Esoteric PEM features like inter-block data and intra-block headers are either rejected during object validation, or can be ignored by consumers of the object. Additionally, consumers are allowed to reorder the certificates in the bundle with their own arbitrary but stable ordering.

ClusterTrustBundle objects should be considered world-readable within the cluster. If your cluster uses [RBAC](#) authorization, all ServiceAccounts have a default grant that allows them to **get**, **list**, and **watch** all ClusterTrustBundle objects. If you use your own authorization mechanism and you have enabled ClusterTrustBundles in your cluster, you should set up an equivalent rule to make these objects public within the cluster, so that they work as intended.

If you do not have permission to list cluster trust bundles by default in your cluster, you can impersonate a service account you have access to in order to see available ClusterTrustBundles:

```
kubectl get clustertrustbundles --as='system:serviceaccount:mynamespace:default'
```

## Signer-linked ClusterTrustBundles

Signer-linked ClusterTrustBundles are associated with a *signer name*, like this:

```
apiVersion: certificates.k8s.io/v1alpha1
kind: ClusterTrustBundle
metadata:
 name: example.com:mysigner:foo
spec:
 signerName: example.com/mysigner
 trustBundle: "<... PEM data ...>"
```

These ClusterTrustBundles are intended to be maintained by a signer-specific controller in the cluster, so they have several security features:

- To create or update a signer-linked ClusterTrustBundle, you must be permitted to **attest** on the signer (custom authorization verb attest, API group certificates.k8s.io; resource path signers). You can configure authorization for the specific resource name `<signerNameDomain>/<signerNamePath>` or match a pattern such as `<signerNameDomain>/*`.
- Signer-linked ClusterTrustBundles **must** be named with a prefix derived from their spec.signerName field. Slashes (/) are replaced with colons (:), and a final colon is appended. This is followed by an arbitrary name. For example, the signer example.com/mysigner can be linked to a ClusterTrustBundle example.com:mysigner:<arbitrary-name>.

Signer-linked ClusterTrustBundles will typically be consumed in workloads by a combination of a [field selector](#) on the signer name, and a separate [label selector](#).

## Signer-unlinked ClusterTrustBundles

Signer-unlinked ClusterTrustBundles have an empty spec.signerName field, like this:



```
apiVersion: certificates.k8s.io/v1alpha1
kind: ClusterTrustBundle
metadata:
 name: foo
spec:
 # no signerName specified, so the field is blank
 trustBundle: "<... PEM data ...>"
```

They are primarily intended for cluster configuration use cases. Each signer-unlinked ClusterTrustBundle is an independent object, in contrast to the customary grouping behavior of signer-linked ClusterTrustBundles.

Signer-unlinked ClusterTrustBundles have no attest verb requirement. Instead, you control access to them directly using the usual mechanisms, such as role-based access control.

To distinguish them from signer-linked ClusterTrustBundles, the names of signer-unlinked ClusterTrustBundles **must not** contain a colon (:).

## Accessing ClusterTrustBundles from pods

FEATURE STATE: Kubernetes v1.29 [alpha]

The contents of ClusterTrustBundles can be injected into the container filesystem, similar to ConfigMaps and Secrets. See the [clusterTrustBundle projected volume source](#) for more details.

## How to issue a certificate for a user

A few steps are required in order to get a normal user to be able to authenticate and invoke an API. First, this user must have a certificate issued by the Kubernetes cluster, and then present that certificate to the Kubernetes API.

### Create private key

The following scripts show how to generate PKI private key and CSR. It is important to set CN and O attribute of the CSR. CN is the name of the user and O is the group that this user will belong to. You can refer to [RBAC](#) for standard groups.

```
openssl genrsa -out myuser.key 2048
openssl req -new -key myuser.key -out myuser.csr -subj "/CN=myuser"
```

### Create a CertificateSigningRequest

Create a CertificateSigningRequest and submit it to a Kubernetes Cluster via kubectl. Below is a script to generate the CertificateSigningRequest.

```
cat <<EOF | kubectl apply -f -
apiVersion: certificates.k8s.io/v1
kind: CertificateSigningRequest
metadata:
 name: myuser
spec:
 request:
```

```
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSBRSRVFVRVNULS0tLS0KTUJQ1ZqQ0NBVDRDQVFBd0V
URVBNQTBHQTFRUF3d0dZVzVuWld4aE1JSUJJakFOQmdrcWhraUc5dzBCQVFFRgpBQU9D
QVE4QU1JSUJDZ0tDQVFFQTByczhJTHRhdTYxakx2dHhWTTJSVIRWMDNHw1JTWWw0dWI
uVWo4RElaWjBOCnR2MUZtRVFSd3VoaUZsOFZzcWl0Qm0wMUFSMkNJVXBGd2ZzSjZ4MXF3
ckJzVkhZbGlBNVhwRVpZM3ExcGswSDQKM3Z3aGJlK1o2MVNrVHF5SVBYUwUwrTWM5T1Ns
bm0xb0R2N0NtSkZNMUIMRVI3QTVGZnZKOEdFRjJ6dHB0aUJFMwpub1dtdHNZb3JuT2wzc2l
HQ2ZGZzR4Zmd4eW8ybmlneFNVekl1bXNnVm9PM2tT0x1RVF6cXpkakJ3TFJXbWIEcklmMX
BMWnoyalVnald4UkhCM1gyWnVvV1d1T09PZnpXM01LaE8ybHEvZi9DdS8wYk83c0x0Mct3U
2ZMSU91TFcKcW90blZtRmxMMytqTy82WDNDKzBERHk5aUtwbXJjVDBnWGZLemE1dHJRSU
RBUUFcb0FBd0RRWUpLb1pJaHZjTgpBUUVMQlFBRGdnRUJBR05WdmVIOGR4ZzNvK21VeVR
kbnFjVmQ1N24zSkExdnZEU1JWREkyQTZ1eXN3ZFp1L1BVckkwZXpZWV0RVNnSk1IRmQy
cVVMjNuNVJsSXJ3R0xuUXFISUh5VStWWHhsdnZsRnpNOVpEWlSTmU3QlJvYXgKQVlEdUI
5STZXT3FYbkFvczFqRmxNUG5NbFpqdU5kSGxpT1BjTU1oNndLaTZZZFhpVStHYTJ2RUVLY01j
SVUyRgpnU2djUWdMYTk0aEpacGk3ZnNMdm1OQUxoT045UHdNMGM1dVJVeV4T0dGMU0tC
bWRSeEgvaUNOS2JKYjFRQm1HCckwYitEUEdaTktXTU0xMzhIQXdoV0tkNjVoVHdYOWl4V3Z
HMkh4TG1WQzg0L1BHT0tWQW9FNkpsYWVHdTIQVmkKdj1OSjVaZlZrcXdCd0hKbzZXdk9xV
lA3SVFjZmg3d0drWm89Ci0tLS0tRU5EIEENFUIRJRklDQVRFIJFUVVFU1QtLS0tLQo=
signerName: kubernetes.io/kube-apiserver-client
expirationSeconds: 86400 # one day
usages:
- client auth
EOF
```

Some points to note:

- usages has to be 'client auth'
- expirationSeconds could be made longer (i.e. 864000 for ten days) or shorter (i.e. 3600 for one hour)
- request is the base64 encoded value of the CSR file content. You can get the content using this command:

```
cat myuser.csr | base64 | tr -d "\n"
```

## Approve the CertificateSigningRequest

Use kubectl to create a CSR and approve it.

Get the list of CSRs:

```
kubectl get csr
```

Approve the CSR:

```
kubectl certificate approve myuser
```

## Get the certificate

Retrieve the certificate from the CSR:

```
kubectl get csr/myuser -o yaml
```

The certificate value is in Base64-encoded format under status.certificate.



Export the issued certificate from the CertificateSigningRequest.

```
kubectl get csr myuser -o jsonpath='{.status.certificate}' | base64 -d > myuser.crt
```

## Create Role and RoleBinding

With the certificate created it is time to define the Role and RoleBinding for this user to access Kubernetes cluster resources.

This is a sample command to create a Role for this new user:

```
kubectl create role developer --verb=create --verb=get --verb=list --verb=update --verb=delete --resource=pods
```

This is a sample command to create a RoleBinding for this new user:

```
kubectl create rolebinding developer-binding-myuser --role=developer --user=myuser
```

## Add to kubeconfig

The last step is to add this user into the kubeconfig file.

First, you need to add new credentials:

```
kubectl config set-credentials myuser --client-key=myuser.key --client-certificate=myuser.crt --embed-certs=true
```

Then, you need to add the context:

```
kubectl config set-context myuser --cluster=kubernetes --user=myuser
```

To test it, change the context to myuser:

```
kubectl config use-context myuser
```

## What's next

- Read [Manage TLS Certificates in a Cluster](#)
- View the source code for the kube-controller-manager built in [signer](#)
- View the source code for the kube-controller-manager built in [approver](#)
- For details of X.509 itself, refer to [RFC 5280](#) section 3.1
- For information on the syntax of PKCS#10 certificate signing requests, refer to [RFC 2986](#)
- Read about the ClusterTrustBundle API:
  - [ClusterTrustBundle API reference](#)

# Mapping PodSecurityPolicies to Pod Security Standards

The tables below enumerate the configuration parameters on PodSecurityPolicy objects, whether the field mutates and/or validates pods, and how the configuration values map to the [Pod Security Standards](#).

For each applicable parameter, the allowed values for the [Baseline](#) and [Restricted](#) profiles are listed. Anything outside the allowed values for those profiles would fall under the [Privileged](#) profile. "No opinion" means all values are allowed under all Pod Security Standards.

For a step-by-step migration guide, see [Migrate from PodSecurityPolicy to the Built-In PodSecurity Admission Controller](#).

## PodSecurityPolicy Spec

The fields enumerated in this table are part of the PodSecurityPolicySpec, which is specified under the .spec field path.

Mapping PodSecurityPolicySpec fields to Pod Security Standards

| PodSecurityPolicySpec    | Type                  | Pod Security Standards Equivalent                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| privileged               | Validating            | <b>Baseline &amp; Restricted:</b> false / undefined / nil                                                                                                                                                                                                                                                                                                                                                           |
| defaultAddCapabilities   | Mutating & Validating | Requirements match allowedCapabilities below.                                                                                                                                                                                                                                                                                                                                                                       |
| allowedCapabilities      | Validating            | <b>Baseline:</b> subset of <ul style="list-style-type: none"><li>• AUDIT_WRITE</li><li>• CHOWN</li><li>• DAC_OVERRIDE</li><li>• FOWNER</li><li>• FSETID</li><li>• KILL</li><li>• MKNOD</li><li>• NET_BIND_SERVICE</li><li>• SETFCAP</li><li>• SETGID</li><li>• SETPCAP</li><li>• SETUID</li><li>• SYS_CHROOT</li></ul> <b>Restricted:</b> empty / undefined / nil OR a list containing <i>only</i> NET_BIND_SERVICE |
| requiredDropCapabilities | Mutating & Validating | <b>Baseline:</b> no opinion<br><b>Restricted:</b> must include ALL                                                                                                                                                                                                                                                                                                                                                  |
| volumes                  | Validating            | <b>Baseline:</b> anything except <ul style="list-style-type: none"><li>• hostPath</li><li>• *</li></ul> <b>Restricted:</b> subset of <ul style="list-style-type: none"><li>• configMap</li><li>• csi</li></ul>                                                                                                                                                                                                      |

|                                 |                                   |                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                                   | <ul style="list-style-type: none"> <li>• downwardAPI</li> <li>• emptyDir</li> <li>• ephemeral</li> <li>• persistentVolumeClaim</li> <li>• projected</li> <li>• secret</li> </ul>                                                                                                                                                                 |
| hostNetwork                     | Validating                        | <b>Baseline &amp; Restricted:</b> false / undefined / nil                                                                                                                                                                                                                                                                                        |
| hostPorts                       | Validating                        | <b>Baseline &amp; Restricted:</b> undefined / nil / empty                                                                                                                                                                                                                                                                                        |
| hostPID                         | Validating                        | <b>Baseline &amp; Restricted:</b> false / undefined / nil                                                                                                                                                                                                                                                                                        |
| hostIPC                         | Validating                        | <b>Baseline &amp; Restricted:</b> false / undefined / nil                                                                                                                                                                                                                                                                                        |
| seLinux                         | Mutating & Validating             | <b>Baseline &amp; Restricted:</b> seLinux.rule is MustRunAs, with the following options <ul style="list-style-type: none"> <li>• user is unset ("" / undefined / nil)</li> <li>• role is unset ("" / undefined / nil)</li> <li>• type is unset or one of: container_t, container_init_t, container_kvm_t</li> <li>• level is anything</li> </ul> |
| runAsUser                       | Mutating & Validating             | <b>Baseline:</b> Anything<br><b>Restricted:</b> rule is MustRunAsNonRoot                                                                                                                                                                                                                                                                         |
| runAsGroup                      | Mutating (MustRunAs) & Validating | <i>No opinion</i>                                                                                                                                                                                                                                                                                                                                |
| supplementalGroups              | Mutating & Validating             | <i>No opinion</i>                                                                                                                                                                                                                                                                                                                                |
| fsGroup                         | Mutating & Validating             | <i>No opinion</i>                                                                                                                                                                                                                                                                                                                                |
| readOnlyRootFilesystem          | Mutating & Validating             | <i>No opinion</i>                                                                                                                                                                                                                                                                                                                                |
| defaultAllowPrivilegeEscalation | Mutating                          | <i>No opinion (non-validating)</i>                                                                                                                                                                                                                                                                                                               |
| allowPrivilegeEscalation        | Mutating & Validating             | <i>Only mutating if set to false</i><br><b>Baseline:</b> No opinion<br><b>Restricted:</b> false                                                                                                                                                                                                                                                  |
| allowedHostPaths                | Validating                        | <i>No opinion (volumes takes precedence)</i>                                                                                                                                                                                                                                                                                                     |
| allowedFlexVolumes              | Validating                        | <i>No opinion (volumes takes precedence)</i>                                                                                                                                                                                                                                                                                                     |
| allowedCSIDrivers               | Validating                        | <i>No opinion (volumes takes precedence)</i>                                                                                                                                                                                                                                                                                                     |
| allowedUnsafeSysctls            | Validating                        |                                                                                                                                                                                                                                                                                                                                                  |

|                                                   |            |                                                                          |
|---------------------------------------------------|------------|--------------------------------------------------------------------------|
|                                                   |            | <b>Baseline &amp; Restricted:</b> undefined / nil / empty                |
| forbiddenSysctls                                  | Validating | <i>No opinion</i>                                                        |
| allowedProcMountTypes<br>( <i>alpha feature</i> ) | Validating | <b>Baseline &amp; Restricted:</b> ["Default"] OR undefined / nil / empty |
| runtimeClass<br>.defaultRuntimeClassName          | Mutating   | <i>No opinion</i>                                                        |
| runtimeClass<br>.allowedRuntimeClassNames         | Validating | <i>No opinion</i>                                                        |

## PodSecurityPolicy annotations

The [annotations](#) enumerated in this table can be specified under .metadata.annotations on the PodSecurityPolicy object.

Mapping PodSecurityPolicy annotations to Pod Security Standards

| PSP Annotation                                           | Type       | Pod Security Standards Equivalent                                                                                                                                                                                                     |
|----------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| seccomp.security.alpha.kubernetes.io/defaultProfileName  | Mutating   | <i>No opinion</i>                                                                                                                                                                                                                     |
| seccomp.security.alpha.kubernetes.io/allowedProfileNames | Validating | <b>Baseline:</b> "runtime/default," ( <i>Trailing comma to allow unset</i> )<br><b>Restricted:</b> "runtime/default" ( <i>No trailing comma</i> )<br><i>localhost/* values are also permitted for both Baseline &amp; Restricted.</i> |
| apparmor.security.beta.kubernetes.io/defaultProfileName  | Mutating   | <i>No opinion</i>                                                                                                                                                                                                                     |
| apparmor.security.beta.kubernetes.io/allowedProfileNames | Validating | <b>Baseline:</b> "runtime/default," ( <i>Trailing comma to allow unset</i> )<br><b>Restricted:</b> "runtime/default" ( <i>No trailing comma</i> )<br><i>localhost/* values are also permitted for both Baseline &amp; Restricted.</i> |

## Kubelet authentication/authorization

### Overview

A kubelet's HTTPS endpoint exposes APIs which give access to data of varying sensitivity, and allow you to perform operations with varying levels of power on the node and within containers.

This document describes how to authenticate and authorize access to the kubelet's HTTPS endpoint.

## Kubelet authentication

By default, requests to the kubelet's HTTPS endpoint that are not rejected by other configured authentication methods are treated as anonymous requests, and given a username of `system:anonymous` and a group of `system:unauthenticated`.

To disable anonymous access and send 401 Unauthorized responses to unauthenticated requests:

- start the kubelet with the `--anonymous-auth=false` flag

To enable X509 client certificate authentication to the kubelet's HTTPS endpoint:

- start the kubelet with the `--client-ca-file` flag, providing a CA bundle to verify client certificates with
- start the apiserver with `--kubelet-client-certificate` and `--kubelet-client-key` flags
- see the [apiserver authentication documentation](#) for more details

To enable API bearer tokens (including service account tokens) to be used to authenticate to the kubelet's HTTPS endpoint:

- ensure the `authentication.k8s.io/v1beta1` API group is enabled in the API server
- start the kubelet with the `--authentication-token-webhook` and `--kubeconfig` flags
- the kubelet calls the `TokenReview` API on the configured API server to determine user information from bearer tokens

## Kubelet authorization

Any request that is successfully authenticated (including an anonymous request) is then authorized. The default authorization mode is `AlwaysAllow`, which allows all requests.

There are many possible reasons to subdivide access to the kubelet API:

- anonymous auth is enabled, but anonymous users' ability to call the kubelet API should be limited
- bearer token auth is enabled, but arbitrary API users' (like service accounts) ability to call the kubelet API should be limited
- client certificate auth is enabled, but only some of the client certificates signed by the configured CA should be allowed to use the kubelet API

To subdivide access to the kubelet API, delegate authorization to the API server:

- ensure the `authorization.k8s.io/v1beta1` API group is enabled in the API server
- start the kubelet with the `--authorization-mode=Webhook` and the `--kubeconfig` flags
- the kubelet calls the `SubjectAccessReview` API on the configured API server to determine whether each request is authorized

The kubelet authorizes API requests using the same [request attributes](#) approach as the apiserver.

The verb is determined from the incoming request's HTTP verb:

| HTTP verb | request verb |
|-----------|--------------|
| POST      | create       |
| GET, HEAD | get          |
| PUT       | update       |
| PATCH     | patch        |
| DELETE    | delete       |

The resource and subresource is determined from the incoming request's path:

| Kubelet API       | resource | subresource |
|-------------------|----------|-------------|
| /stats/*          | nodes    | stats       |
| /metrics/*        | nodes    | metrics     |
| /logs/*           | nodes    | log         |
| /spec/*           | nodes    | spec        |
| <i>all others</i> | nodes    | proxy       |

The namespace and API group attributes are always an empty string, and the resource name is always the name of the kubelet's Node API object.

When running in this mode, ensure the user identified by the `--kubelet-client-certificate` and `--kubelet-client-key` flags passed to the apiserver is authorized for the following attributes:

- verb=\*, resource=nodes, subresource=proxy
- verb=\*, resource=nodes, subresource=stats
- verb=\*, resource=nodes, subresource=log
- verb=\*, resource=nodes, subresource=spec
- verb=\*, resource=nodes, subresource=metrics

## TLS bootstrapping

In a Kubernetes cluster, the components on the worker nodes - kubelet and kube-proxy - need to communicate with Kubernetes control plane components, specifically kube-apiserver. In order to ensure that communication is kept private, not interfered with, and ensure that each component of the cluster is talking to another trusted component, we strongly recommend using client TLS certificates on nodes.

The normal process of bootstrapping these components, especially worker nodes that need certificates so they can communicate safely with kube-apiserver, can be a challenging process as it is often outside of the scope of Kubernetes and requires significant additional work. This in turn, can make it challenging to initialize or scale a cluster.

In order to simplify the process, beginning in version 1.4, Kubernetes introduced a certificate request and signing API. The proposal can be found [here](#).

This document describes the process of node initialization, how to set up TLS client certificate bootstrapping for kubelets, and how it works.

# Initialization process

When a worker node starts up, the kubelet does the following:

1. Look for its kubeconfig file
2. Retrieve the URL of the API server and credentials, normally a TLS key and signed certificate from the kubeconfig file
3. Attempt to communicate with the API server using the credentials.

Assuming that the kube-apiserver successfully validates the kubelet's credentials, it will treat the kubelet as a valid node, and begin to assign pods to it.

Note that the above process depends upon:

- Existence of a key and certificate on the local host in the kubeconfig
- The certificate having been signed by a Certificate Authority (CA) trusted by the kube-apiserver

All of the following are responsibilities of whoever sets up and manages the cluster:

1. Creating the CA key and certificate
2. Distributing the CA certificate to the control plane nodes, where kube-apiserver is running
3. Creating a key and certificate for each kubelet; strongly recommended to have a unique one, with a unique CN, for each kubelet
4. Signing the kubelet certificate using the CA key
5. Distributing the kubelet key and signed certificate to the specific node on which the kubelet is running

The TLS Bootstrapping described in this document is intended to simplify, and partially or even completely automate, steps 3 onwards, as these are the most common when initializing or scaling a cluster.

## Bootstrap initialization

In the bootstrap initialization process, the following occurs:

1. kubelet begins
2. kubelet sees that it does *not* have a kubeconfig file
3. kubelet searches for and finds a bootstrap-kubeconfig file
4. kubelet reads its bootstrap file, retrieving the URL of the API server and a limited usage "token"
5. kubelet connects to the API server, authenticates using the token
6. kubelet now has limited credentials to create and retrieve a certificate signing request (CSR)
7. kubelet creates a CSR for itself with the signerName set to kubernetes.io/kube-apiserver-client-kubelet
8. CSR is approved in one of two ways:
  - If configured, kube-controller-manager automatically approves the CSR
  - If configured, an outside process, possibly a person, approves the CSR using the Kubernetes API or via kubectl
9. Certificate is created for the kubelet
10. Certificate is issued to the kubelet

11. kubelet retrieves the certificate
12. kubelet creates a proper kubeconfig with the key and signed certificate
13. kubelet begins normal operation
14. Optional: if configured, kubelet automatically requests renewal of the certificate when it is close to expiry
15. The renewed certificate is approved and issued, either automatically or manually, depending on configuration.

The rest of this document describes the necessary steps to configure TLS Bootstrapping, and its limitations.

## Configuration

To configure for TLS bootstrapping and optional automatic approval, you must configure options on the following components:

- kube-apiserver
- kube-controller-manager
- kubelet
- in-cluster resources: ClusterRoleBinding and potentially ClusterRole

In addition, you need your Kubernetes Certificate Authority (CA).

## Certificate Authority

As without bootstrapping, you will need a Certificate Authority (CA) key and certificate. As without bootstrapping, these will be used to sign the kubelet certificate. As before, it is your responsibility to distribute them to control plane nodes.

For the purposes of this document, we will assume these have been distributed to control plane nodes at `/var/lib/kubernetes/ca.pem` (certificate) and `/var/lib/kubernetes/ca-key.pem` (key). We will refer to these as "Kubernetes CA certificate and key".

All Kubernetes components that use these certificates - kubelet, kube-apiserver, kube-controller-manager - assume the key and certificate to be PEM-encoded.

## kube-apiserver configuration

The kube-apiserver has several requirements to enable TLS bootstrapping:

- Recognizing CA that signs the client certificate
- Authenticating the bootstrapping kubelet to the `system:bootstrappers` group
- Authorize the bootstrapping kubelet to create a certificate signing request (CSR)

### Recognizing client certificates

This is normal for all client certificate authentication. If not already set, add the `--client-ca-file=FILENAME` flag to the kube-apiserver command to enable client certificate authentication, referencing a certificate authority bundle containing the signing certificate, for example `--client-ca-file=/var/lib/kubernetes/ca.pem`.



## Initial bootstrap authentication

In order for the bootstrapping kubelet to connect to kube-apiserver and request a certificate, it must first authenticate to the server. You can use any [authenticator](#) that can authenticate the kubelet.

While any authentication strategy can be used for the kubelet's initial bootstrap credentials, the following two authenticators are recommended for ease of provisioning.

1. [Bootstrap Tokens](#)
2. [Token authentication file](#)

Using bootstrap tokens is a simpler and more easily managed method to authenticate kubelets, and does not require any additional flags when starting kube-apiserver.

Whichever method you choose, the requirement is that the kubelet be able to authenticate as a user with the rights to:

1. create and retrieve CSRs
2. be automatically approved to request node client certificates, if automatic approval is enabled.

A kubelet authenticating using bootstrap tokens is authenticated as a user in the group `system:bootstrappers`, which is the standard method to use.

As this feature matures, you should ensure tokens are bound to a Role Based Access Control (RBAC) policy which limits requests (using the [bootstrap token](#)) strictly to client requests related to certificate provisioning. With RBAC in place, scoping the tokens to a group allows for great flexibility. For example, you could disable a particular bootstrap group's access when you are done provisioning the nodes.

### Bootstrap tokens

Bootstrap tokens are described in detail [here](#). These are tokens that are stored as secrets in the Kubernetes cluster, and then issued to the individual kubelet. You can use a single token for an entire cluster, or issue one per worker node.

The process is two-fold:

1. Create a Kubernetes secret with the token ID, secret and scope(s).
2. Issue the token to the kubelet

From the kubelet's perspective, one token is like another and has no special meaning. From the kube-apiserver's perspective, however, the bootstrap token is special. Due to its type, namespace and name, kube-apiserver recognizes it as a special token, and grants anyone authenticating with that token special bootstrap rights, notably treating them as a member of the `system:bootstrappers` group. This fulfills a basic requirement for TLS bootstrapping.

The details for creating the secret are available [here](#).

If you want to use bootstrap tokens, you must enable it on kube-apiserver with the flag:

```
--enable-bootstrap-token-auth=true
```

## Token authentication file

kube-apiserver has the ability to accept tokens as authentication. These tokens are arbitrary but should represent at least 128 bits of entropy derived from a secure random number generator (such as `/dev/urandom` on most modern Linux systems). There are multiple ways you can generate a token. For example:

```
head -c 16 /dev/urandom | od -An -t x | tr -d ' '
```

This will generate tokens that look like `02b50b05283e98dd0fd71db496ef01e8`.

The token file should look like the following example, where the first three values can be anything and the quoted group name should be as depicted:

```
02b50b05283e98dd0fd71db496ef01e8,kubelet-bootstrap,10001,"system:bootstrappers"
```

Add the `--token-auth-file=FILENAME` flag to the kube-apiserver command (in your systemd unit file perhaps) to enable the token file. See docs [here](#) for further details.

## Authorize kubelet to create CSR

Now that the bootstrapping node is *authenticated* as part of the `system:bootstrappers` group, it needs to be *authorized* to create a certificate signing request (CSR) as well as retrieve it when done. Fortunately, Kubernetes ships with a ClusterRole with precisely these (and only these) permissions, `system:node-bootstrapper`.

To do this, you only need to create a ClusterRoleBinding that binds the `system:bootstrappers` group to the cluster role `system:node-bootstrapper`.

```
enable bootstrapping nodes to create CSR
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
 name: create-csrs-for-bootstrapping
subjects:
- kind: Group
 name: system:bootstrappers
 apiGroup: rbac.authorization.k8s.io
roleRef:
 kind: ClusterRole
 name: system:node-bootstrapper
 apiGroup: rbac.authorization.k8s.io
```

## kube-controller-manager configuration

While the apiserver receives the requests for certificates from the kubelet and authenticates those requests, the controller-manager is responsible for issuing actual signed certificates.

The controller-manager performs this function via a certificate-issuing control loop. This takes the form of a [cfssl](#) local signer using assets on disk. Currently, all certificates issued have one year validity and a default set of key usages.

In order for the controller-manager to sign certificates, it needs the following:

- access to the "Kubernetes CA key and certificate" that you created and distributed
- enabling CSR signing

## Access to key and certificate

As described earlier, you need to create a Kubernetes CA key and certificate, and distribute it to the control plane nodes. These will be used by the controller-manager to sign the kubelet certificates.

Since these signed certificates will, in turn, be used by the kubelet to authenticate as a regular kubelet to kube-apiserver, it is important that the CA provided to the controller-manager at this stage also be trusted by kube-apiserver for authentication. This is provided to kube-apiserver with the flag `--client-ca-file=FILENAME` (for example, `--client-ca-file=/var/lib/kubernetes/ca.pem`), as described in the kube-apiserver configuration section.

To provide the Kubernetes CA key and certificate to kube-controller-manager, use the following flags:

```
--cluster-signing-cert-file="/etc/path/to/kubernetes/ca/ca.crt" --cluster-signing-key-file="/etc/path/to/kubernetes/ca/ca.key"
```

For example:

```
--cluster-signing-cert-file="/var/lib/kubernetes/ca.pem" --cluster-signing-key-file="/var/lib/kubernetes/ca-key.pem"
```

The validity duration of signed certificates can be configured with flag:

```
--cluster-signing-duration
```

## Approval

In order to approve CSRs, you need to tell the controller-manager that it is acceptable to approve them. This is done by granting RBAC permissions to the correct group.

There are two distinct sets of permissions:

- `nodeclient`: If a node is creating a new certificate for a node, then it does not have a certificate yet. It is authenticating using one of the tokens listed above, and thus is part of the group `system:bootstrappers`.
- `selfnodeclient`: If a node is renewing its certificate, then it already has a certificate (by definition), which it uses continuously to authenticate as part of the group `system:nodes`.

To enable the kubelet to request and receive a new certificate, create a `ClusterRoleBinding` that binds the group in which the bootstrapping node is a member `system:bootstrappers` to the `ClusterRole` that grants it permission, `system:certificates.k8s.io:certificatesigningrequests:nodeclient`:

```
Approve all CSRs for the group "system:bootstrappers"
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
```

```
name: auto-approve-csrs-for-group
subjects:
- kind: Group
 name: system:bootstrappers
 apiGroup: rbac.authorization.k8s.io
roleRef:
 kind: ClusterRole
 name: system:certificates.k8s.io:certificatesigningrequests:nodeclient
 apiGroup: rbac.authorization.k8s.io
```

To enable the kubelet to renew its own client certificate, create a ClusterRoleBinding that binds the group in which the fully functioning node is a member system:nodes to the ClusterRole that grants it permission, system:certificates.k8s.io:certificatesigningrequests:selfnodeclient:

```
Approve renewal CSRs for the group "system:nodes"
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
 name: auto-approve-renewals-for-nodes
subjects:
- kind: Group
 name: system:nodes
 apiGroup: rbac.authorization.k8s.io
roleRef:
 kind: ClusterRole
 name: system:certificates.k8s.io:certificatesigningrequests:selfnodeclient
 apiGroup: rbac.authorization.k8s.io
```

The csrapproving controller that ships as part of [kube-controller-manager](#) and is enabled by default. The controller uses the [SubjectAccessReview API](#) to determine if a given user is authorized to request a CSR, then approves based on the authorization outcome. To prevent conflicts with other approvers, the built-in approver doesn't explicitly deny CSRs. It only ignores unauthorized requests. The controller also prunes expired certificates as part of garbage collection.

## kubelet configuration

Finally, with the control plane nodes properly set up and all of the necessary authentication and authorization in place, we can configure the kubelet.

The kubelet requires the following configuration to bootstrap:

- A path to store the key and certificate it generates (optional, can use default)
- A path to a kubeconfig file that does not yet exist; it will place the bootstrapped config file here
- A path to a bootstrap kubeconfig file to provide the URL for the server and bootstrap credentials, e.g. a bootstrap token
- Optional: instructions to rotate certificates

The bootstrap kubeconfig should be in a path available to the kubelet, for example /var/lib/kubelet/bootstrap-kubeconfig.

Its format is identical to a normal kubeconfig file. A sample file might look as follows:

```

apiVersion: v1
kind: Config
clusters:
- cluster:
 certificate-authority: /var/lib/kubernetes/ca.pem
 server: https://my.server.example.com:6443
 name: bootstrap
contexts:
- context:
 cluster: bootstrap
 user: kubelet-bootstrap
 name: bootstrap
current-context: bootstrap
preferences: {}
users:
- name: kubelet-bootstrap
 user:
 token: 07401b.f395accd246ae52d

```

The important elements to note are:

- `certificate-authority`: path to a CA file, used to validate the server certificate presented by kube-apiserver
- `server`: URL to kube-apiserver
- `token`: the token to use

The format of the token does not matter, as long as it matches what kube-apiserver expects. In the above example, we used a bootstrap token. As stated earlier, *any* valid authentication method can be used, not only tokens.

Because the bootstrap kubeconfig is a standard kubeconfig, you can use `kubect` to generate it. To create the above example file:

```

kubect config --kubeconfig=/var/lib/kubelet/bootstrap-kubeconfig set-cluster bootstrap --
server='https://my.server.example.com:6443' --certificate-authority=/var/lib/kubernetes/ca.pem
kubect config --kubeconfig=/var/lib/kubelet/bootstrap-kubeconfig set-credentials kubelet-
bootstrap --token=07401b.f395accd246ae52d
kubect config --kubeconfig=/var/lib/kubelet/bootstrap-kubeconfig set-context bootstrap --
user=kubelet-bootstrap --cluster=bootstrap
kubect config --kubeconfig=/var/lib/kubelet/bootstrap-kubeconfig use-context bootstrap

```

To indicate to the kubelet to use the bootstrap kubeconfig, use the following kubelet flag:

```

--bootstrap-kubeconfig="/var/lib/kubelet/bootstrap-kubeconfig" --kubeconfig="/var/lib/kubelet/
kubeconfig"

```

When starting the kubelet, if the file specified via `--kubeconfig` does not exist, the bootstrap kubeconfig specified via `--bootstrap-kubeconfig` is used to request a client certificate from the API server. On approval of the certificate request and receipt back by the kubelet, a kubeconfig file referencing the generated key and obtained certificate is written to the path specified by `--kubeconfig`. The certificate and key file will be placed in the directory specified by `--cert-dir`.

## Client and serving certificates

All of the above relate to kubelet *client* certificates, specifically, the certificates a kubelet uses to authenticate to kube-apiserver.

A kubelet also can use *serving* certificates. The kubelet itself exposes an https endpoint for certain features. To secure these, the kubelet can do one of:

- use provided key and certificate, via the `--tls-private-key-file` and `--tls-cert-file` flags
- create self-signed key and certificate, if a key and certificate are not provided
- request serving certificates from the cluster server, via the CSR API

The client certificate provided by TLS bootstrapping is signed, by default, for client auth only, and thus cannot be used as serving certificates, or server auth.

However, you *can* enable its server certificate, at least partially, via certificate rotation.

## Certificate rotation

Kubernetes v1.8 and higher kubelet implements features for enabling rotation of its client and/or serving certificates. Note, rotation of serving certificate is a **beta** feature and requires the `RotateKubeletServerCertificate` feature flag on the kubelet (enabled by default).

You can configure the kubelet to rotate its client certificates by creating new CSRs as its existing credentials expire. To enable this feature, use the `rotateCertificates` field of [kubelet configuration file](#) or pass the following command line argument to the kubelet (deprecated):

```
--rotate-certificates
```

Enabling `RotateKubeletServerCertificate` causes the kubelet **both** to request a serving certificate after bootstrapping its client credentials **and** to rotate that certificate. To enable this behavior, use the field `serverTLSBootstrap` of the [kubelet configuration file](#) or pass the following command line argument to the kubelet (deprecated):

```
--rotate-server-certificates
```

### Note:

The CSR approving controllers implemented in core Kubernetes do not approve node *serving* certificates for [security reasons](#). To use `RotateKubeletServerCertificate` operators need to run a custom approving controller, or manually approve the serving certificate requests.

A deployment-specific approval process for kubelet serving certificates should typically only approve CSRs which:

1. are requested by nodes (ensure the `spec.username` field is of the form `system:node:<nodeName>` and `spec.groups` contains `system:nodes`)
2. request usages for a serving certificate (ensure `spec.usages` contains `server auth`, optionally contains `digital signature` and `key encipherment`, and contains no other usages)
3. only have IP and DNS `subjectAltNames` that belong to the requesting node, and have no URI and Email `subjectAltNames` (parse the x509 Certificate Signing Request in `spec.request` to verify `subjectAltNames`)

## Other authenticating components

All of TLS bootstrapping described in this document relates to the kubelet. However, other components may need to communicate directly with kube-apiserver. Notable is kube-proxy, which is part of the Kubernetes node components and runs on every node, but may also include other components such as monitoring or networking.

Like the kubelet, these other components also require a method of authenticating to kube-apiserver. You have several options for generating these credentials:

- The old way: Create and distribute certificates the same way you did for kubelet before TLS bootstrapping
- DaemonSet: Since the kubelet itself is loaded on each node, and is sufficient to start base services, you can run kube-proxy and other node-specific services not as a standalone process, but rather as a daemonset in the kube-system namespace. Since it will be in-cluster, you can give it a proper service account with appropriate permissions to perform its activities. This may be the simplest way to configure such services.

## kubectl approval

CSRs can be approved outside of the approval flows built into the controller manager.

The signing controller does not immediately sign all certificate requests. Instead, it waits until they have been flagged with an "Approved" status by an appropriately-privileged user. This flow is intended to allow for automated approval handled by an external approval controller or the approval controller implemented in the core controller-manager. However cluster administrators can also manually approve certificate requests using kubectl. An administrator can list CSRs with `kubectl get csr` and describe one in detail with `kubectl describe csr <name>`. An administrator can approve or deny a CSR with `kubectl certificate approve <name>` and `kubectl certificate deny <name>`.

## Validating Admission Policy

FEATURE STATE: Kubernetes v1.30 [stable]

This page provides an overview of Validating Admission Policy.

### What is Validating Admission Policy?

Validating admission policies offer a declarative, in-process alternative to validating admission webhooks.

Validating admission policies use the Common Expression Language (CEL) to declare the validation rules of a policy. Validation admission policies are highly configurable, enabling policy authors to define policies that can be parameterized and scoped to resources as needed by cluster administrators.

# What Resources Make a Policy

A policy is generally made up of three resources:

- The `ValidatingAdmissionPolicy` describes the abstract logic of a policy (think: "this policy makes sure a particular label is set to a particular value").
- A `ValidatingAdmissionPolicyBinding` links the above resources together and provides scoping. If you only want to require an `owner` label to be set for Pods, the binding is where you would specify this restriction.
- A parameter resource provides information to a `ValidatingAdmissionPolicy` to make it a concrete statement (think "the owner label must be set to something that ends in `.company.com`"). A native type such as `ConfigMap` or a CRD defines the schema of a parameter resource. `ValidatingAdmissionPolicy` objects specify what Kind they are expecting for their parameter resource.

At least a `ValidatingAdmissionPolicy` and a corresponding `ValidatingAdmissionPolicyBinding` must be defined for a policy to have an effect.

If a `ValidatingAdmissionPolicy` does not need to be configured via parameters, simply leave `spec.paramKind` in `ValidatingAdmissionPolicy` not specified.

## Before you begin

- Ensure the `ValidatingAdmissionPolicy` [feature gate](#) is enabled.
- Ensure that the `admissionregistration.k8s.io/v1beta1` API is enabled.

## Getting Started with Validating Admission Policy

Validating Admission Policy is part of the cluster control-plane. You should write and deploy them with great caution. The following describes how to quickly experiment with Validating Admission Policy.

### Creating a ValidatingAdmissionPolicy

The following is an example of a `ValidatingAdmissionPolicy`.

[validatingadmissionpolicy/basic-example-policy.yaml](#)

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
metadata:
 name: "demo-policy.example.com"
spec:
 failurePolicy: Fail
 matchConstraints:
 resourceRules:
 - apiGroups: ["apps"]
 apiVersions: ["v1"]
 operations: ["CREATE", "UPDATE"]
```



```
resources: ["deployments"]
validations:
- expression: "object.spec.replicas <= 5"
```

spec.validations contains CEL expressions which use the [Common Expression Language \(CEL\)](#) to validate the request. If an expression evaluates to false, the validation check is enforced according to the spec.failurePolicy field.

### Note:

You can quickly test CEL expressions in [CEL Playground](#).

To configure a validating admission policy for use in a cluster, a binding is required. The following is an example of a ValidatingAdmissionPolicyBinding.:

[validatingadmissionpolicy/basic-example-binding.yaml](#)

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicyBinding
metadata:
 name: "demo-binding-test.example.com"
spec:
 policyName: "demo-policy.example.com"
 validationActions: [Deny]
 matchResources:
 namespaceSelector:
 matchLabels:
 environment: test
```

When trying to create a deployment with replicas set not satisfying the validation expression, an error will return containing message:

```
ValidatingAdmissionPolicy 'demo-policy.example.com' with binding 'demo-binding-test.example.com' denied request: failed expression: object.spec.replicas <= 5
```

The above provides a simple example of using ValidatingAdmissionPolicy without a parameter configured.

### Validation actions

Each ValidatingAdmissionPolicyBinding must specify one or more validationActions to declare how validations of a policy are enforced.

The supported validationActions are:

- Deny: Validation failure results in a denied request.
- Warn: Validation failure is reported to the request client as a [warning](#).
- Audit: Validation failure is included in the audit event for the API request.

For example, to both warn clients about a validation failure and to audit the validation failures, use:

```
validationActions: [Warn, Audit]
```

Deny and Warn may not be used together since this combination needlessly duplicates the validation failure both in the API response body and the HTTP warning headers.

A validation that evaluates to false is always enforced according to these actions. Failures defined by the failurePolicy are enforced according to these actions only if the failurePolicy is set to Fail (or not specified), otherwise the failures are ignored.

See [Audit Annotations: validation failures](#) for more details about the validation failure audit annotation.

## Parameter resources

Parameter resources allow a policy configuration to be separate from its definition. A policy can define paramKind, which outlines GVK of the parameter resource, and then a policy binding ties a policy by name (via policyName) to a particular parameter resource via paramRef.

If parameter configuration is needed, the following is an example of a ValidatingAdmissionPolicy with parameter configuration.

[validatingadmissionpolicy/policy-with-param.yaml](#)

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
metadata:
 name: "replicalimit-policy.example.com"
spec:
 failurePolicy: Fail
 paramKind:
 apiVersion: rules.example.com/v1
 kind: ReplicaLimit
 matchConstraints:
 resourceRules:
 - apiGroups: ["apps"]
 apiVersions: ["v1"]
 operations: ["CREATE", "UPDATE"]
 resources: ["deployments"]
 validations:
 - expression: "object.spec.replicas <= params.maxReplicas"
 reason: Invalid
```

The spec.paramKind field of the ValidatingAdmissionPolicy specifies the kind of resources used to parameterize this policy. For this example, it is configured by ReplicaLimit custom resources. Note in this example how the CEL expression references the parameters via the CEL params variable, e.g. params.maxReplicas. spec.matchConstraints specifies what resources this policy is designed to validate. Note that the native types such like ConfigMap could also be used as parameter reference.

The spec.validations fields contain CEL expressions. If an expression evaluates to false, the validation check is enforced according to the spec.failurePolicy field.

The validating admission policy author is responsible for providing the ReplicaLimit parameter CRD.

To configure an validating admission policy for use in a cluster, a binding and parameter resource are created. The following is an example of a ValidatingAdmissionPolicyBinding that uses a **cluster-wide** param - the same param will be used to validate every resource request that matches the binding:

[validatingadmissionpolicy/binding-with-param.yaml](#)

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicyBinding
metadata:
 name: "replicalimit-binding-test.example.com"
spec:
 policyName: "replicalimit-policy.example.com"
 validationActions: [Deny]
 paramRef:
 name: "replica-limit-test.example.com"
 namespace: "default"
 matchResources:
 namespaceSelector:
 matchLabels:
 environment: test
```

Notice this binding applies a parameter to the policy for all resources which are in the test environment.

The parameter resource could be as following:

[validatingadmissionpolicy/replicalimit-param.yaml](#)

```
apiVersion: rules.example.com/v1
kind: ReplicaLimit
metadata:
 name: "replica-limit-test.example.com"
 namespace: "default"
maxReplicas: 3
```

This policy parameter resource limits deployments to a max of 3 replicas.

An admission policy may have multiple bindings. To bind all other environments to have a maxReplicas limit of 100, create another ValidatingAdmissionPolicyBinding:

[validatingadmissionpolicy/binding-with-param-prod.yaml](#)

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicyBinding
metadata:
 name: "replicalimit-binding-nontest"
spec:
 policyName: "replicalimit-policy.example.com"
 validationActions: [Deny]
 paramRef:
 name: "replica-limit-prod.example.com"
 namespace: "default"
 matchResources:
```

```
namespaceSelector:
 matchExpressions:
 - key: environment
 operator: NotIn
 values:
 - test
```

Notice this binding applies a different parameter to resources which are not in the test environment.

And have a parameter resource:

[validatingadmissionpolicy/replicalimit-param-prod.yaml](#)

```
apiVersion: rules.example.com/v1
kind: ReplicaLimit
metadata:
 name: "replica-limit-prod.example.com"
maxReplicas: 100
```

For each admission request, the API server evaluates CEL expressions of each (policy, binding, param) combination that match the request. For a request to be admitted it must pass **all** evaluations.

If multiple bindings match the request, the policy will be evaluated for each, and they must all pass evaluation for the policy to be considered passed.

If multiple parameters match a single binding, the policy rules will be evaluated for each param, and they too must all pass for the binding to be considered passed. Bindings can have overlapping match criteria. The policy is evaluated for each matching binding-parameter combination. A policy may even be evaluated multiple times if multiple bindings match it, or a single binding that matches multiple parameters.

The params object representing a parameter resource will not be set if a parameter resource has not been bound, so for policies requiring a parameter resource, it can be useful to add a check to ensure one has been bound. A parameter resource will not be bound and params will be null if paramKind of the policy, or paramRef of the binding are not specified.

For the use cases require parameter configuration, we recommend to add a param check in spec.validations[0].expression:

```
- expression: "params != null"
 message: "params missing but required to bind to this policy"
```

## Optional parameters

It can be convenient to be able to have optional parameters as part of a parameter resource, and only validate them if present. CEL provides has(), which checks if the key passed to it exists. CEL also implements Boolean short-circuiting. If the first half of a logical OR evaluates to true, it won't evaluate the other half (since the result of the entire OR will be true regardless).

Combining the two, we can provide a way to validate optional parameters:

```
!has(params.optionalNumber) || (params.optionalNumber >= 5 && params.optionalNumber <= 10)
```

Here, we first check that the optional parameter is present with `!has(params.optionalNumber)`.

- If `optionalNumber` hasn't been defined, then the expression short-circuits since `!has(params.optionalNumber)` will evaluate to true.
- If `optionalNumber` has been defined, then the latter half of the CEL expression will be evaluated, and `optionalNumber` will be checked to ensure that it contains a value between 5 and 10 inclusive.

## Per-namespace Parameters

As the author of a `ValidatingAdmissionPolicy` and its `ValidatingAdmissionPolicyBinding`, you can choose to specify cluster-wide, or per-namespace parameters. If you specify a namespace for the binding's `paramRef`, the control plane only searches for parameters in that namespace.

However, if `namespace` is not specified in the `ValidatingAdmissionPolicyBinding`, the API server can search for relevant parameters in the namespace that a request is against. For example, if you make a request to modify a `ConfigMap` in the default namespace and there is a relevant `ValidatingAdmissionPolicyBinding` with no namespace set, then the API server looks for a parameter object in default. This design enables policy configuration that depends on the namespace of the resource being manipulated, for more fine-tuned control.

## Parameter selector

In addition to specify a parameter in a binding by name, you may choose instead to specify label selector, such that all resources of the policy's `paramKind`, and the `param`'s namespace (if applicable) that match the label selector are selected for evaluation. See [selector](#) for more information on how label selectors match resources.

If multiple parameters are found to meet the condition, the policy's rules are evaluated for each parameter found and the results will be ANDed together.

If `namespace` is provided, only objects of the `paramKind` in the provided namespace are eligible for selection. Otherwise, when `namespace` is empty and `paramKind` is namespace-scoped, the namespace used in the request being admitted will be used.

## Authorization checks

We introduced the authorization check for parameter resources. User is expected to have read access to the resources referenced by `paramKind` in `ValidatingAdmissionPolicy` and `paramRef` in `ValidatingAdmissionPolicyBinding`.

Note that if a resource in `paramKind` fails resolving via the restmapper, read access to all resources of groups is required.

## Failure Policy

`failurePolicy` defines how mis-configurations and CEL expressions evaluating to error from the admission policy are handled. Allowed values are Ignore or Fail.

- Ignore means that an error calling the `ValidatingAdmissionPolicy` is ignored and the API request is allowed to continue.
- Fail means that an error calling the `ValidatingAdmissionPolicy` causes the admission to fail and the API request to be rejected.

Note that the `failurePolicy` is defined inside `ValidatingAdmissionPolicy`:

[validatingadmissionpolicy/failure-policy-ignore.yaml](#)

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
spec:
...
failurePolicy: Ignore # The default is "Fail"
validations:
- expression: "object.spec.xyz == params.x"
```

## Validation Expression

`spec.validations[i].expression` represents the expression which will be evaluated by CEL. To learn more, see the [CEL language specification](#) CEL expressions have access to the contents of the Admission request/response, organized into CEL variables as well as some other useful variables:

- 'object' - The object from the incoming request. The value is null for DELETE requests.
- 'oldObject' - The existing object. The value is null for CREATE requests.
- 'request' - Attributes of the [admission request](#).
- 'params' - Parameter resource referred to by the policy binding being evaluated. The value is null if `ParamKind` is not specified.
- `namespaceObject` - The namespace, as a Kubernetes resource, that the incoming object belongs to. The value is null if the incoming object is cluster-scoped.
- `authorizer` - A CEL Authorizer. May be used to perform authorization checks for the principal (authenticated user) of the request. See [Authz](#) in the Kubernetes CEL library documentation for more details.
- `authorizer.requestResource` - A shortcut for an authorization check configured with the request resource (group, resource, (subresource), namespace, name).

The `apiVersion`, `kind`, `metadata.name` and `metadata.generateName` are always accessible from the root of the object. No other metadata properties are accessible.

Equality on arrays with list type of 'set' or 'map' ignores element order, i.e. `[1, 2] == [2, 1]`. Concatenation on arrays with `x-kubernetes-list-type` use the semantics of the list type:

- 'set': `X + Y` performs a union where the array positions of all elements in `X` are preserved and non-intersecting elements in `Y` are appended, retaining their partial order.
- 'map': `X + Y` performs a merge where the array positions of all keys in `X` are preserved but the values are overwritten by values in `Y` when the key sets of `X` and `Y` intersect. Elements in `Y` with non-intersecting keys are appended, retaining their partial order.

## Validation expression examples

| Expression                                                                                                  | Purpose                                                                                        |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <code>object.minReplicas &lt;= object.replicas &amp;&amp; object.replicas &lt;= object.maxReplicas</code>   | Validate that the three fields defining replicas are ordered appropriately                     |
| <code>'Available' in object.stateCounts</code>                                                              | Validate that an entry with the 'Available' key exists in a map                                |
| <code>(size(object.list1) == 0) != (size(object.list2) == 0)</code>                                         | Validate that one of two lists is non-empty, but not both                                      |
| <code>!('MY_KEY' in object.map1)    object['MY_KEY'].matches('^[a-zA-Z]*\$')</code>                         | Validate the value of a map for a specific key, if it is in the map                            |
| <code>object.envvars.filter(e, e.name == 'MY_ENV').all(e, e.value.matches('^[a-zA-Z]*\$'))</code>           | Validate the 'value' field of a listMap entry where key field 'name' is 'MY_ENV'               |
| <code>has(object.expired) &amp;&amp; object.created + object.ttl &lt; object.expired</code>                 | Validate that 'expired' date is after a 'create' date plus a 'ttl' duration                    |
| <code>object.health.startsWith('ok')</code>                                                                 | Validate a 'health' string field has the prefix 'ok'                                           |
| <code>object.widgets.exists(w, w.key == 'x' &amp;&amp; w.foo &lt; 10)</code>                                | Validate that the 'foo' property of a listMap item with a key 'x' is less than 10              |
| <code>type(object) == string ? object == '100%' : object == 1000</code>                                     | Validate an int-or-string field for both the int and string cases                              |
| <code>object.metadata.name.startsWith(object.prefix)</code>                                                 | Validate that an object's name has the prefix of another field value                           |
| <code>object.set1.all(e, !(e in object.set2))</code>                                                        | Validate that two listSets are disjoint                                                        |
| <code>size(object.names) == size(object.details) &amp;&amp; object.names.all(n, n in object.details)</code> | Validate the 'details' map is keyed by the items in the 'names' listSet                        |
| <code>size(object.clusters.filter(c, c.name == object.primary)) == 1</code>                                 | Validate that the 'primary' property has one and only one occurrence in the 'clusters' listMap |

Read [Supported evaluation on CEL](#) for more information about CEL rules.

`spec.validation[i].reason` represents a machine-readable description of why this validation failed. If this is the first validation in the list to fail, this reason, as well as the corresponding HTTP response code, are used in the HTTP response to the client. The currently supported reasons are: Unauthorized, Forbidden, Invalid, RequestEntityTooLarge. If not set, `StatusReasonInvalid` is used in the response to the client.

## Matching requests: matchConditions

You can define *match conditions* for a `ValidatingAdmissionPolicy` if you need fine-grained request filtering. These conditions are useful if you find that match rules, objectSelectors and namespaceSelectors still doesn't provide the filtering you want. Match conditions are [CEL expressions](#). All match conditions must evaluate to true for the resource to be evaluated.

Here is an example illustrating a few different uses for match conditions:

[access/validating-admission-policy-match-conditions.yaml](#)



```

apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
metadata:
 name: "demo-policy.example.com"
spec:
 failurePolicy: Fail
 matchConstraints:
 resourceRules:
 - apiGroups: ["*"]
 apiVersions: ["*"]
 operations: ["CREATE", "UPDATE"]
 resources: ["*"]
 matchConditions:
 - name: 'exclude-leases' # Each match condition must have a unique name
 expression: '!(request.resource.group == "coordination.k8s.io" && request.resource.resource
 == "leases")' # Match non-lease resources.
 - name: 'exclude-kubelet-requests'
 expression: '!("system:nodes" in request.userInfo.groups)' # Match requests made by non-
 node users.
 - name: 'rbac' # Skip RBAC requests.
 expression: 'request.resource.group != "rbac.authorization.k8s.io"'
 validations:
 - expression: "!object.metadata.name.contains('demo') || object.metadata.namespace ==
 'demo'"

```

Match conditions have access to the same CEL variables as validation expressions.

In the event of an error evaluating a match condition the policy is not evaluated. Whether to reject the request is determined as follows:

1. If **any** match condition evaluated to false (regardless of other errors), the API server skips the policy.
2. Otherwise:
  - for [failurePolicy: Fail](#), reject the request (without evaluating the policy).
  - for [failurePolicy: Ignore](#), proceed with the request but skip the policy.

## Audit annotations

auditAnnotations may be used to include audit annotations in the audit event of the API request.

For example, here is an admission policy with an audit annotation:

[access/validating-admission-policy-audit-annotation.yaml](#)

```

apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
metadata:
 name: "demo-policy.example.com"
spec:
 failurePolicy: Fail
 matchConstraints:

```



```

resourceRules:
- apiGroups: ["apps"]
 apiVersions: ["v1"]
 operations: ["CREATE", "UPDATE"]
 resources: ["deployments"]
validations:
- expression: "object.spec.replicas > 50"
 messageExpression: "Deployment spec.replicas set to ' + string(object.spec.replicas)"
auditAnnotations:
- key: "high-replica-count"
 valueExpression: "Deployment spec.replicas set to ' + string(object.spec.replicas)"

```

When an API request is validated with this admission policy, the resulting audit event will look like:

```

the audit event recorded
{
 "kind": "Event",
 "apiVersion": "audit.k8s.io/v1",
 "annotations": {
 "demo-policy.example.com/high-replica-count": "Deployment spec.replicas set to 128"
 # other annotations
 ...
 }
 # other fields
 ...
}

```

In this example the annotation will only be included if the `spec.replicas` of the Deployment is more than 50, otherwise the CEL expression evaluates to null and the annotation will not be included.

Note that audit annotation keys are prefixed by the name of the ValidatingAdmissionWebhook and a /. If another admission controller, such as an admission webhook, uses the exact same audit annotation key, the value of the first admission controller to include the audit annotation will be included in the audit event and all other values will be ignored.

## Message expression

To return a more friendly message when the policy rejects a request, we can use a CEL expression to composite a message with `spec.validations[i].messageExpression`. Similar to the validation expression, a message expression has access to `object`, `oldObject`, `request`, `params`, and `namespaceObject`. Unlike validations, message expression must evaluate to a string.

For example, to better inform the user of the reason of denial when the policy refers to a parameter, we can have the following validation:

[access/deployment-replicas-policy.yaml](#)

```

apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
metadata:
 name: "deploy-replica-policy.example.com"

```

```
spec:
 paramKind:
 apiVersion: rules.example.com/v1
 kind: ReplicaLimit
 matchConstraints:
 resourceRules:
 - apiGroups: ["apps"]
 apiVersions: ["v1"]
 operations: ["CREATE", "UPDATE"]
 resources: ["deployments"]
 validations:
 - expression: "object.spec.replicas <= params.maxReplicas"
 messageExpression: "'object.spec.replicas must be no greater than ' +
string(params.maxReplicas)"
 reason: Invalid
```

After creating a params object that limits the replicas to 3 and setting up the binding, when we try to create a deployment with 5 replicas, we will receive the following message.

```
$ kubectl create deploy --image=nginx nginx --replicas=5
error: failed to create deployment: deployments.apps "nginx" is forbidden:
ValidatingAdmissionPolicy 'deploy-replica-policy.example.com' with binding 'demo-binding-
test.example.com' denied request: object.spec.replicas must be no greater than 3
```

This is more informative than a static message of "too many replicas".

The message expression takes precedence over the static message defined in `spec.validations[i].message` if both are defined. However, if the message expression fails to evaluate, the static message will be used instead. Additionally, if the message expression evaluates to a multi-line string, the evaluation result will be discarded and the static message will be used if present. Note that static message is validated against multi-line strings.

## Type checking

When a policy definition is created or updated, the validation process parses the expressions it contains and reports any syntax errors, rejecting the definition if any errors are found. Afterward, the referred variables are checked for type errors, including missing fields and type confusion, against the matched types of `spec.matchConstraints`. The result of type checking can be retrieved from `status.typeChecking`. The presence of `status.typeChecking` indicates the completion of type checking, and an empty `status.typeChecking` means that no errors were detected.

For example, given the following policy definition:

[validatingadmissionpolicy/typechecking.yaml](#)

```
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
metadata:
 name: "deploy-replica-policy.example.com"
spec:
 matchConstraints:
```

```

resourceRules:
- apiGroups: ["apps"]
 apiVersions: ["v1"]
 operations: ["CREATE", "UPDATE"]
 resources: ["deployments"]
validations:
- expression: "object.replicas > 1" # should be "object.spec.replicas > 1"
 message: "must be replicated"
 reason: Invalid

```

The status will yield the following information:

```

status:
 typeChecking:
 expressionWarnings:
 - fieldRef: spec.validations[0].expression
 warning: |-
 apps/v1, Kind=Deployment: ERROR: <input>:1:7: undefined field 'replicas'
 | object.replicas > 1
 |^

```

If multiple resources are matched in `spec.matchConstraints`, all of matched resources will be checked against. For example, the following policy definition

[validatingadmissionpolicy/typechecking-multiple-match.yaml](#)

```

apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
metadata:
 name: "replica-policy.example.com"
spec:
 matchConstraints:
 resourceRules:
 - apiGroups: ["apps"]
 apiVersions: ["v1"]
 operations: ["CREATE", "UPDATE"]
 resources: ["deployments", "replicasets"]
 validations:
 - expression: "object.replicas > 1" # should be "object.spec.replicas > 1"
 message: "must be replicated"
 reason: Invalid

```

will have multiple types and type checking result of each type in the warning message.

```

status:
 typeChecking:
 expressionWarnings:
 - fieldRef: spec.validations[0].expression
 warning: |-
 apps/v1, Kind=Deployment: ERROR: <input>:1:7: undefined field 'replicas'
 | object.replicas > 1
 |^
 apps/v1, Kind=ReplicaSet: ERROR: <input>:1:7: undefined field 'replicas'

```

```
| object.replicas > 1
|^
```

Type Checking has the following limitation:

- No wildcard matching. If `spec.matchConstraints.resourceRules` contains "\*" in any of `apiGroups`, `apiVersions` or `resources`, the types that "\*" matches will not be checked.
- The number of matched types is limited to 10. This is to prevent a policy that manually specifying too many types. to consume excessive computing resources. In the order of ascending group, version, and then resource, 11th combination and beyond are ignored.
- Type Checking does not affect the policy behavior in any way. Even if the type checking detects errors, the policy will continue to evaluate. If errors do occur during evaluate, the failure policy will decide its outcome.
- Type Checking does not apply to CRDs, including matched CRD types and reference of `paramKind`. The support for CRDs will come in future release.

## Variable composition

If an expression grows too complicated, or part of the expression is reusable and computationally expensive to evaluate, you can extract some part of the expressions into variables. A variable is a named expression that can be referred later in variables in other expressions.

```
spec:
 variables:
 - name: foo
 expression: "'foo' in object.spec.metadata.labels ? object.spec.metadata.labels['foo'] : 'default'"
 validations:
 - expression: variables.foo == 'bar'
```

A variable is lazily evaluated when it is first referred. Any error that occurs during the evaluation will be reported during the evaluation of the referring expression. Both the result and potential error are memorized and count only once towards the runtime cost.

The order of variables are important because a variable can refer to other variables that are defined before it. This ordering prevents circular references.

The following is a more complex example of enforcing that image repo names match the environment defined in its namespace.

[access/image-matches-namespace-environment.policy.yaml](#)

```
This policy enforces that all containers of a deployment has the image repo match the
environment label of its namespace.
Except for "exempt" deployments, or any containers that do not belong to the "example.com"
organization (e.g. common sidecars).
For example, if the namespace has a label of {"environment": "staging"}, all container images
must be either staging.example.com/*
or do not contain "example.com" at all, unless the deployment has {"exempt": "true"} label.
apiVersion: admissionregistration.k8s.io/v1
kind: ValidatingAdmissionPolicy
metadata:
 name: "image-matches-namespace-environment.policy.example.com"
spec:
```

```

failurePolicy: Fail
matchConstraints:
 resourceRules:
 - apiGroups: ["apps"]
 apiVersions: ["v1"]
 operations: ["CREATE", "UPDATE"]
 resources: ["deployments"]
 variables:
 - name: environment
 expression: "'environment' in namespaceObject.metadata.labels ?
namespaceObject.metadata.labels['environment'] : 'prod'"
 - name: exempt
 expression: "'exempt' in object.metadata.labels && object.metadata.labels['exempt'] == 'true'"
 - name: containers
 expression: "object.spec.template.spec.containers"
 - name: containersToCheck
 expression: "variables.containers.filter(c, c.image.contains('example.com/'))"
 validations:
 - expression: "variables.exempt || variables.containersToCheck.all(c,
c.image.startsWith(variables.environment + ':'))"
 messageExpression: "'only ' + variables.environment + ' images are allowed in namespace ' +
namespaceObject.metadata.name"

```

With the policy bound to the namespace default, which is labeled `environment: prod`, the following attempt to create a deployment would be rejected.

```
kubectl create deploy --image=dev.example.com/nginx invalid
```

The error message is similar to this.

```

error: failed to create deployment: deployments.apps "invalid" is forbidden:
ValidatingAdmissionPolicy 'image-matches-namespace-environment.policy.example.com' with
binding 'demo-binding-test.example.com' denied request: only prod images are allowed in
namespace default

```

## Well-Known Labels, Annotations and Taints

Kubernetes reserves all labels and annotations in the `kubernetes.io` and `k8s.io` namespaces.

This document serves both as a reference to the values and as a coordination point for assigning values.

### Labels, annotations and taints used on API objects

#### `apf.kubernetes.io/autoupdate-spec`

Type: Annotation

Example: `apf.kubernetes.io/autoupdate-spec: "true"`

Used on: [FlowSchema and PriorityLevelConfiguration Objects](#)

If this annotation is set to true on a FlowSchema or PriorityLevelConfiguration, the spec for that object is managed by the kube-apiserver. If the API server does not recognize an APF object, and you annotate it for automatic update, the API server deletes the entire object. Otherwise, the API server does not manage the object spec. For more details, read [Maintenance of the Mandatory and Suggested Configuration Objects](#).

## **app.kubernetes.io/component**

Type: Label

Example: `app.kubernetes.io/component: "database"`

Used on: All Objects (typically used on [workload resources](#)).

The component within the application architecture.

One of the [recommended labels](#).

## **app.kubernetes.io/created-by (deprecated)**

Type: Label

Example: `app.kubernetes.io/created-by: "controller-manager"`

Used on: All Objects (typically used on [workload resources](#)).

The controller/user who created this resource.

### **Note:**

Starting from v1.9, this label is deprecated.

## **app.kubernetes.io/instance**

Type: Label

Example: `app.kubernetes.io/instance: "mysql-abcxyz"`

Used on: All Objects (typically used on [workload resources](#)).

A unique name identifying the instance of an application. To assign a non-unique name, use [app.kubernetes.io/name](#).

One of the [recommended labels](#).

## **app.kubernetes.io/managed-by**

Type: Label

Example: `app.kubernetes.io/managed-by: "helm"`

Used on: All Objects (typically used on [workload resources](#)).

The tool being used to manage the operation of an application.

One of the [recommended labels](#).

## **app.kubernetes.io/name**

Type: Label

Example: `app.kubernetes.io/name: "mysql"`

Used on: All Objects (typically used on [workload resources](#)).

The name of the application.

One of the [recommended labels](#).

## **app.kubernetes.io/part-of**

Type: Label

Example: `app.kubernetes.io/part-of: "wordpress"`

Used on: All Objects (typically used on [workload resources](#)).

The name of a higher-level application this object is part of.

One of the [recommended labels](#).

## **app.kubernetes.io/version**

Type: Label

Example: `app.kubernetes.io/version: "5.7.21"`

Used on: All Objects (typically used on [workload resources](#)).

The current version of the application.

Common forms of values include:

- [semantic version](#)
- the Git [revision hash](#) for the source code.

One of the [recommended labels](#).

## **applyset.kubernetes.io/additional-namespaces (alpha)**

Type: Annotation

Example: `applyset.kubernetes.io/additional-namespaces: "namespace1,namespace2"`

Used on: Objects being used as ApplySet parents.

Use of this annotation is Alpha. For Kubernetes version 1.30, you can use this annotation on Secrets, ConfigMaps, or custom resources if the [CustomResourceDefinition](#) defining them has the `applyset.kubernetes.io/is-parent-type` label.

Part of the specification used to implement [ApplySet-based pruning in kubectl](#). This annotation is applied to the parent object used to track an ApplySet to extend the scope of the ApplySet beyond the parent object's own namespace (if any). The value is a comma-separated list of the names of namespaces other than the parent's namespace in which objects are found.

## **applyset.kubernetes.io/contains-group-resources (alpha)**

Type: Annotation

Example: `applyset.kubernetes.io/contains-group-resources: "certificates.cert-manager.io,configmaps,deployments.apps,secrets,services"`

Used on: Objects being used as ApplySet parents.

Use of this annotation is Alpha. For Kubernetes version 1.30, you can use this annotation on Secrets, ConfigMaps, or custom resources if the CustomResourceDefinition defining them has the `applyset.kubernetes.io/is-parent-type` label.

Part of the specification used to implement [ApplySet-based pruning in kubectl](#). This annotation is applied to the parent object used to track an ApplySet to optimize listing of ApplySet member objects. It is optional in the ApplySet specification, as tools can perform discovery or use a different optimization. However, as of Kubernetes version 1.30, it is required by kubectl. When present, the value of this annotation must be a comma separated list of the group-kinds, in the fully-qualified name format, i.e. `<resource>.<group>`.

## **applyset.kubernetes.io/id (alpha)**

Type: Label

Example: `applyset.kubernetes.io/id: "applyset-0eFHV8ySq7XoShsGvyWFQD3s96yqwHmzc4e0HR1dsY-v1"`

Used on: Objects being used as ApplySet parents.

Use of this label is Alpha. For Kubernetes version 1.30, you can use this label on Secrets, ConfigMaps, or custom resources if the CustomResourceDefinition defining them has the `applyset.kubernetes.io/is-parent-type` label.

Part of the specification used to implement [ApplySet-based pruning in kubectl](#). This label is what makes an object an ApplySet parent object. Its value is the unique ID of the ApplySet, which is derived from the identity of the parent object itself. This ID **must** be the base64 encoding (using the URL safe encoding of RFC4648) of the hash of the group-kind-name-namespace of the object it is on, in the form:

`<base64(sha256(<name>.<namespace>.<kind>.<group>))>`. There is no relation between the value of this label and object UID.

## **applyset.kubernetes.io/is-parent-type (alpha)**

Type: Label



Example: `applyset.kubernetes.io/is-parent-type: "true"`

Used on: Custom Resource Definition (CRD)

Use of this label is Alpha. Part of the specification used to implement [ApplySet-based pruning in kubectl](#). You can set this label on a CustomResourceDefinition (CRD) to identify the custom resource type it defines (not the CRD itself) as an allowed parent for an ApplySet. The only permitted value for this label is "true"; if you want to mark a CRD as not being a valid parent for ApplySets, omit this label.

## **applyset.kubernetes.io/part-of (alpha)**

Type: Label

Example: `applyset.kubernetes.io/part-of: "applyset-0eFHV8ySqP7XoShsGvyWFQD3s96yqwHmzc4e0HR1dsY-v1"`

Used on: All objects.

Use of this label is Alpha. Part of the specification used to implement [ApplySet-based pruning in kubectl](#). This label is what makes an object a member of an ApplySet. The value of the label **must** match the value of the `applyset.kubernetes.io/id` label on the parent object.

## **applyset.kubernetes.io/tooling (alpha)**

Type: Annotation

Example: `applyset.kubernetes.io/tooling: "kubectl/v1.30"`

Used on: Objects being used as ApplySet parents.

Use of this annotation is Alpha. For Kubernetes version 1.30, you can use this annotation on Secrets, ConfigMaps, or custom resources if the CustomResourceDefinition defining them has the `applyset.kubernetes.io/is-parent-type` label.

Part of the specification used to implement [ApplySet-based pruning in kubectl](#). This annotation is applied to the parent object used to track an ApplySet to indicate which tooling manages that ApplySet. Tooling should refuse to mutate ApplySets belonging to other tools. The value must be in the format `<toolname>/<semver>`.

## **apps.kubernetes.io/pod-index (beta)**

Type: Label

Example: `apps.kubernetes.io/pod-index: "0"`

Used on: Pod

When a StatefulSet controller creates a Pod for the StatefulSet, it sets this label on that Pod. The value of the label is the ordinal index of the pod being created.

See [Pod Index Label](#) in the StatefulSet topic for more details. Note the [PodIndexLabel](#) feature gate must be enabled for this label to be added to pods.

## **cluster-autoscaler.kubernetes.io/safe-to-evict**

Type: Annotation

Example: `cluster-autoscaler.kubernetes.io/safe-to-evict: "true"`

Used on: Pod

When this annotation is set to "true", the cluster autoscaler is allowed to evict a Pod even if other rules would normally prevent that. The cluster autoscaler never evicts Pods that have this annotation explicitly set to "false"; you could set that on an important Pod that you want to keep running. If this annotation is not set then the cluster autoscaler follows its Pod-level behavior.

## **config.kubernetes.io/local-config**

Type: Annotation

Example: `config.kubernetes.io/local-config: "true"`

Used on: All objects

This annotation is used in manifests to mark an object as local configuration that should not be submitted to the Kubernetes API.

A value of "true" for this annotation declares that the object is only consumed by client-side tooling and should not be submitted to the API server.

A value of "false" can be used to declare that the object should be submitted to the API server even when it would otherwise be assumed to be local.

This annotation is part of the Kubernetes Resource Model (KRM) Functions Specification, which is used by Kustomize and similar third-party tools. For example, Kustomize removes objects with this annotation from its final build output.

## **container.apparmor.security.beta.kubernetes.io/\* (deprecated)**

Type: Annotation

Example: `container.apparmor.security.beta.kubernetes.io/my-container: my-custom-profile`

Used on: Pods

This annotation allows you to specify the AppArmor security profile for a container within a Kubernetes pod. As of Kubernetes v1.30, this should be set with the `appArmorProfile` field instead. To learn more, see the [AppArmor](#) tutorial. The tutorial illustrates using AppArmor to restrict a container's abilities and access.

The profile specified dictates the set of rules and restrictions that the containerized process must adhere to. This helps enforce security policies and isolation for your containers.

## **internal.config.kubernetes.io/\* (reserved prefix)**

Type: Annotation

Used on: All objects

This prefix is reserved for internal use by tools that act as orchestrators in accordance with the Kubernetes Resource Model (KRM) Functions Specification. Annotations with this prefix are internal to the orchestration process and are not persisted to the manifests on the filesystem. In other words, the orchestrator tool should set these annotations when reading files from the local filesystem and remove them when writing the output of functions back to the filesystem.

A KRM function **must not** modify annotations with this prefix, unless otherwise specified for a given annotation. This enables orchestrator tools to add additional internal annotations, without requiring changes to existing functions.

## **internal.config.kubernetes.io/path**

Type: Annotation

Example: `internal.config.kubernetes.io/path: "relative/file/path.yaml"`

Used on: All objects

This annotation records the slash-delimited, OS-agnostic, relative path to the manifest file the object was loaded from. The path is relative to a fixed location on the filesystem, determined by the orchestrator tool.

This annotation is part of the Kubernetes Resource Model (KRM) Functions Specification, which is used by Kustomize and similar third-party tools.

A KRM Function **should not** modify this annotation on input objects unless it is modifying the referenced files. A KRM Function **may** include this annotation on objects it generates.

## **internal.config.kubernetes.io/index**

Type: Annotation

Example: `internal.config.kubernetes.io/index: "2"`

Used on: All objects

This annotation records the zero-indexed position of the YAML document that contains the object within the manifest file the object was loaded from. Note that YAML documents are separated by three dashes (---) and can each contain one object. When this annotation is not specified, a value of 0 is implied.

This annotation is part of the Kubernetes Resource Model (KRM) Functions Specification, which is used by Kustomize and similar third-party tools.

A KRM Function **should not** modify this annotation on input objects unless it is modifying the referenced files. A KRM Function **may** include this annotation on objects it generates.

## **kubernetes.io/arch**

Type: Label

Example: `kubernetes.io/arch: "amd64"`

Used on: Node

The Kubelet populates this with `runtime.GOARCH` as defined by Go. This can be handy if you are mixing ARM and x86 nodes.

## **kubernetes.io/os**

Type: Label

Example: `kubernetes.io/os: "linux"`

Used on: Node, Pod

For nodes, the kubelet populates this with `runtime.GOOS` as defined by Go. This can be handy if you are mixing operating systems in your cluster (for example: mixing Linux and Windows nodes).

You can also set this label on a Pod. Kubernetes allows you to set any value for this label; if you use this label, you should nevertheless set it to the Go `runtime.GOOS` string for the operating system that this Pod actually works with.

When the `kubernetes.io/os` label value for a Pod does not match the label value on a Node, the kubelet on the node will not admit the Pod. However, this is not taken into account by the kube-scheduler. Alternatively, the kubelet refuses to run a Pod where you have specified a Pod OS, if this isn't the same as the operating system for the node where that kubelet is running. Just look for [Pods OS](#) for more details.

## **kubernetes.io/metadata.name**

Type: Label

Example: `kubernetes.io/metadata.name: "mynamespace"`

Used on: Namespaces

The Kubernetes API server (part of the [control plane](#)) sets this label on all namespaces. The label value is set to the name of the namespace. You can't change this label's value.

This is useful if you want to target a specific namespace with a label [selector](#).

## **kubernetes.io/limit-ranger**

Type: Annotation

Example: `kubernetes.io/limit-ranger: "LimitRanger plugin set: cpu, memory request for container nginx; cpu, memory limit for container nginx"`

Used on: Pod

Kubernetes by default doesn't provide any resource limit, that means unless you explicitly define limits, your container can consume unlimited CPU and memory. You can define a default request or default limit for pods. You do this by creating a `LimitRange` in the relevant namespace. Pods deployed after you define a `LimitRange` will have these limits applied to them.

The annotation `kubernetes.io/limit-ranger` records that resource defaults were specified for the Pod, and they were applied successfully. For more details, read about [LimitRanges](#).

## **kubernetes.io/config.hash**

Type: Annotation

Example: `kubernetes.io/config.hash: "df7cc47f8477b6b1226d7d23a904867b"`

Used on: Pod

When the kubelet creates a static Pod based on a given manifest, it attaches this annotation to the static Pod. The value of the annotation is the UID of the Pod. Note that the kubelet also sets the `.spec.nodeName` to the current node name as if the Pod was scheduled to the node.

## **kubernetes.io/config.mirror**

Type: Annotation

Example: `kubernetes.io/config.mirror: "df7cc47f8477b6b1226d7d23a904867b"`

Used on: Pod

For a static Pod created by the kubelet on a node, a [mirror Pod](#) is created on the API server. The kubelet adds an annotation to indicate that this Pod is actually a mirror Pod. The annotation value is copied from the [kubernetes.io/config.hash](#) annotation, which is the UID of the Pod.

When updating a Pod with this annotation set, the annotation cannot be changed or removed. If a Pod doesn't have this annotation, it cannot be added during a Pod update.

## **kubernetes.io/config.source**

Type: Annotation

Example: `kubernetes.io/config.source: "file"`

Used on: Pod

This annotation is added by the kubelet to indicate where the Pod comes from. For static Pods, the annotation value could be one of `file` or `http` depending on where the Pod manifest is located. For a Pod created on the API server and then scheduled to the current node, the annotation value is `api`.

## **kubernetes.io/config.seen**

Type: Annotation

Example: `kubernetes.io/config.seen: "2023-10-27T04:04:56.011314488Z"`

Used on: Pod

When the kubelet sees a Pod for the first time, it may add this annotation to the Pod with a value of current timestamp in the RFC3339 format.

## **addonmanager.kubernetes.io/mode**

Type: Label

Example: `addonmanager.kubernetes.io/mode: "Reconcile"`

Used on: All objects

To specify how an add-on should be managed, you can use the `addonmanager.kubernetes.io/mode` label. This label can have one of three values: `Reconcile`, `EnsureExists`, or `Ignore`.

- `Reconcile`: Addon resources will be periodically reconciled with the expected state. If there are any differences, the add-on manager will recreate, reconfigure or delete the resources as needed. This is the default mode if no label is specified.
- `EnsureExists`: Addon resources will be checked for existence only but will not be modified after creation. The add-on manager will create or re-create the resources when there is no instance of the resource with that name.
- `Ignore`: Addon resources will be ignored. This mode is useful for add-ons that are not compatible with the add-on manager or that are managed by another controller.

For more details, see [Add-on manager](#).

## **beta.kubernetes.io/arch (deprecated)**

Type: Label

This label has been deprecated. Please use [kubernetes.io/arch](#) instead.

## **beta.kubernetes.io/os (deprecated)**

Type: Label

This label has been deprecated. Please use [kubernetes.io/os](#) instead.

## **kube-aggregator.kubernetes.io/automanaged**

Type: Label

Example: `kube-aggregator.kubernetes.io/automanaged: "onstart"`

Used on: APIService

The kube-apiserver sets this label on any APIService object that the API server has created automatically. The label marks how the control plane should manage that APIService. You should not add, modify, or remove this label by yourself.

### **Note:**

Automanaged APIService objects are deleted by kube-apiserver when it has no built-in or custom resource API corresponding to the API group/version of the APIService.

There are two possible values:

- `onstart`: The APIService should be reconciled when an API server starts up, but not otherwise.
- `true`: The API server should reconcile this APIService continuously.

## **service.alpha.kubernetes.io/tolerate-unready-endpoints (deprecated)**

Type: Annotation

Used on: StatefulSet

This annotation on a Service denotes if the Endpoints controller should go ahead and create Endpoints for unready Pods. Endpoints of these Services retain their DNS records and continue receiving traffic for the Service from the moment the kubelet starts all containers in the pod and marks it *Running*, til the kubelet stops all containers and deletes the pod from the API server.

## **kubernetes.io/hostname**

Type: Label

Example: `kubernetes.io/hostname: "ip-172-20-114-199.ec2.internal"`

Used on: Node

The Kubelet populates this label with the hostname of the node. Note that the hostname can be changed from the "actual" hostname by passing the `--hostname-override` flag to the kubelet.

This label is also used as part of the topology hierarchy. See [topology.kubernetes.io/zone](https://kubernetes.io/docs/concepts/topology/topology-hierarchy/) for more information.

## **kubernetes.io/change-cause**

Type: Annotation

Example: `kubernetes.io/change-cause: "kubectl edit --record deployment foo"`

Used on: All Objects

This annotation is a best guess at why something was changed.

It is populated when adding `--record` to a `kubectl` command that may change an object.

## **kubernetes.io/description**

Type: Annotation

Example: `kubernetes.io/description: "Description of K8s object."`

Used on: All Objects

This annotation is used for describing specific behaviour of given object.

## kubernetes.io/enforce-mountable-secrets

Type: Annotation

Example: `kubernetes.io/enforce-mountable-secrets: "true"`

Used on: ServiceAccount

The value for this annotation must be **true** to take effect. When you set this annotation to "true", Kubernetes enforces the following rules for Pods running as this ServiceAccount:

1. Secrets mounted as volumes must be listed in the ServiceAccount's `secrets` field.
2. Secrets referenced in `envFrom` for containers (including sidecar containers and init containers) must also be listed in the ServiceAccount's `secrets` field. If any container in a Pod references a Secret not listed in the ServiceAccount's `secrets` field (and even if the reference is marked as optional), then the Pod will fail to start, and an error indicating the non-compliant secret reference will be generated.
3. Secrets referenced in a Pod's `imagePullSecrets` must be present in the ServiceAccount's `imagePullSecrets` field, the Pod will fail to start, and an error indicating the non-compliant image pull secret reference will be generated.

When you create or update a Pod, these rules are checked. If a Pod doesn't follow them, it won't start and you'll see an error message. If a Pod is already running and you change the `kubernetes.io/enforce-mountable-secrets` annotation to true, or you edit the associated ServiceAccount to remove the reference to a Secret that the Pod is already using, the Pod continues to run.

## node.kubernetes.io/exclude-from-external-load-balancers

Type: Label

Example: `node.kubernetes.io/exclude-from-external-load-balancers`

Used on: Node

Kubernetes automatically enables the `ServiceNodeExclusion` feature gate on the clusters it creates. With this feature gate enabled on a cluster, you can add labels to particular worker nodes to exclude them from the list of backend servers. The following command can be used to exclude a worker node from the list of backend servers in a backend set:

```
kubectll label nodes <node-name> node.kubernetes.io/exclude-from-external-load-balancers=true
```

## controller.kubernetes.io/pod-deletion-cost

Type: Annotation

Example: `controller.kubernetes.io/pod-deletion-cost: "10"`

Used on: Pod

This annotation is used to set [Pod Deletion Cost](#) which allows users to influence ReplicaSet downscaling order. The annotation value parses into an `int32` type.



## **cluster-autoscaler.kubernetes.io/enable-ds-eviction**

Type: Annotation

Example: cluster-autoscaler.kubernetes.io/enable-ds-eviction: "true"

Used on: Pod

This annotation controls whether a DaemonSet pod should be evicted by a ClusterAutoscaler. This annotation needs to be specified on DaemonSet pods in a DaemonSet manifest. When this annotation is set to "true", the ClusterAutoscaler is allowed to evict a DaemonSet Pod, even if other rules would normally prevent that. To disallow the ClusterAutoscaler from evicting DaemonSet pods, you can set this annotation to "false" for important DaemonSet pods. If this annotation is not set, then the ClusterAutoscaler follows its overall behavior (i.e evict the DaemonSets based on its configuration).

### **Note:**

This annotation only impacts DaemonSet Pods.

## **kubernetes.io/ingress-bandwidth**

Type: Annotation

Example: kubernetes.io/ingress-bandwidth: 10M

Used on: Pod

You can apply quality-of-service traffic shaping to a pod and effectively limit its available bandwidth. Ingress traffic to a Pod is handled by shaping queued packets to effectively handle data. To limit the bandwidth on a Pod, write an object definition JSON file and specify the data traffic speed using kubernetes.io/ingress-bandwidth annotation. The unit used for specifying ingress rate is bits per second, as a [Quantity](#). For example, 10M means 10 megabits per second.

### **Note:**

Ingress traffic shaping annotation is an experimental feature. If you want to enable traffic shaping support, you must add the bandwidth plugin to your CNI configuration file (default /etc/cni/net.d) and ensure that the binary is included in your CNI bin dir (default /opt/cni/bin).

## **kubernetes.io/egress-bandwidth**

Type: Annotation

Example: kubernetes.io/egress-bandwidth: 10M

Used on: Pod

Egress traffic from a Pod is handled by policing, which simply drops packets in excess of the configured rate. The limits you place on a Pod do not affect the bandwidth of other Pods. To limit the bandwidth on a Pod, write an object definition JSON file and specify the data traffic speed using kubernetes.io/egress-bandwidth annotation. The unit used for specifying egress rate is bits per second, as a [Quantity](#). For example, 10M means 10 megabits per second.

**Note:**

Egress traffic shaping annotation is an experimental feature. If you want to enable traffic shaping support, you must add the bandwidth plugin to your CNI configuration file (default /etc/cni/net.d) and ensure that the binary is included in your CNI bin dir (default /opt/cni/bin).

**beta.kubernetes.io/instance-type (deprecated)**

Type: Label

**Note:**

Starting in v1.17, this label is deprecated in favor of [node.kubernetes.io/instance-type](https://kubernetes.io/docs/concepts/scheduling/kubernetes-scheduler-configuration/#node-labels).

**node.kubernetes.io/instance-type**

Type: Label

Example: node.kubernetes.io/instance-type: "m3.medium"

Used on: Node

The Kubelet populates this with the instance type as defined by the cloud provider. This will be set only if you are using a cloud provider. This setting is handy if you want to target certain workloads to certain instance types, but typically you want to rely on the Kubernetes scheduler to perform resource-based scheduling. You should aim to schedule based on properties rather than on instance types (for example: require a GPU, instead of requiring a g2.2xlarge).

**failure-domain.beta.kubernetes.io/region (deprecated)**

Type: Label

**Note:**

Starting in v1.17, this label is deprecated in favor of [topology.kubernetes.io/region](https://kubernetes.io/docs/concepts/scheduling/kubernetes-scheduler-configuration/#node-labels).

**failure-domain.beta.kubernetes.io/zone (deprecated)**

Type: Label

**Note:**

Starting in v1.17, this label is deprecated in favor of [topology.kubernetes.io/zone](https://kubernetes.io/docs/concepts/scheduling/kubernetes-scheduler-configuration/#node-labels).

**pv.kubernetes.io/bind-completed**

Type: Annotation

Example: pv.kubernetes.io/bind-completed: "yes"

Used on: PersistentVolumeClaim

When this annotation is set on a PersistentVolumeClaim (PVC), that indicates that the lifecycle of the PVC has passed through initial binding setup. When present, that information changes how the control plane interprets the state of PVC objects. The value of this annotation does not matter to Kubernetes.

## **pv.kubernetes.io/bound-by-controller**

Type: Annotation

Example: `pv.kubernetes.io/bound-by-controller: "yes"`

Used on: PersistentVolume, PersistentVolumeClaim

If this annotation is set on a PersistentVolume or PersistentVolumeClaim, it indicates that a storage binding (PersistentVolume → PersistentVolumeClaim, or PersistentVolumeClaim → PersistentVolume) was installed by the [controller](#). If the annotation isn't set, and there is a storage binding in place, the absence of that annotation means that the binding was done manually. The value of this annotation does not matter.

## **pv.kubernetes.io/provisioned-by**

Type: Annotation

Example: `pv.kubernetes.io/provisioned-by: "kubernetes.io/rbd"`

Used on: PersistentVolume

This annotation is added to a PersistentVolume(PV) that has been dynamically provisioned by Kubernetes. Its value is the name of volume plugin that created the volume. It serves both users (to show where a PV comes from) and Kubernetes (to recognize dynamically provisioned PVs in its decisions).

## **pv.kubernetes.io/migrated-to**

Type: Annotation

Example: `pv.kubernetes.io/migrated-to: pd.csi.storage.gke.io`

Used on: PersistentVolume, PersistentVolumeClaim

It is added to a PersistentVolume(PV) and PersistentVolumeClaim(PVC) that is supposed to be dynamically provisioned/deleted by its corresponding CSI driver through the CSIMigration feature gate. When this annotation is set, the Kubernetes components will "stand-down" and the external-provisioner will act on the objects.

## **statefulset.kubernetes.io/pod-name**

Type: Label

Example: `statefulset.kubernetes.io/pod-name: "mystatefulset-7"`

Used on: Pod

When a StatefulSet controller creates a Pod for the StatefulSet, the control plane sets this label on that Pod. The value of the label is the name of the Pod being created.

See [Pod Name Label](#) in the StatefulSet topic for more details.

## **scheduler.alpha.kubernetes.io/node-selector**

Type: Annotation

Example: scheduler.alpha.kubernetes.io/node-selector: "name-of-node-selector"

Used on: Namespace

The [PodNodeSelector](#) uses this annotation key to assign node selectors to pods in namespaces.

## **topology.kubernetes.io/region**

Type: Label

Example: topology.kubernetes.io/region: "us-east-1"

Used on: Node, PersistentVolume

See [topology.kubernetes.io/zone](#).

## **topology.kubernetes.io/zone**

Type: Label

Example: topology.kubernetes.io/zone: "us-east-1c"

Used on: Node, PersistentVolume

**On Node:** The kubelet or the external cloud-controller-manager populates this with the information from the cloud provider. This will be set only if you are using a cloud provider. However, you can consider setting this on nodes if it makes sense in your topology.

**On PersistentVolume:** topology-aware volume provisioners will automatically set node affinity constraints on a PersistentVolume.

A zone represents a logical failure domain. It is common for Kubernetes clusters to span multiple zones for increased availability. While the exact definition of a zone is left to infrastructure implementations, common properties of a zone include very low network latency within a zone, no-cost network traffic within a zone, and failure independence from other zones. For example, nodes within a zone might share a network switch, but nodes in different zones should not.

A region represents a larger domain, made up of one or more zones. It is uncommon for Kubernetes clusters to span multiple regions. While the exact definition of a zone or region is left to infrastructure implementations, common properties of a region include higher network latency between them than within them, non-zero cost for network traffic between them, and failure independence from other zones or regions. For example, nodes within a region might share power infrastructure (e.g. a UPS or generator), but nodes in different regions typically would not.

Kubernetes makes a few assumptions about the structure of zones and regions:

1. regions and zones are hierarchical: zones are strict subsets of regions and no zone can be in 2 regions
2. zone names are unique across regions; for example region "africa-east-1" might be comprised of zones "africa-east-1a" and "africa-east-1b"

It should be safe to assume that topology labels do not change. Even though labels are strictly mutable, consumers of them can assume that a given node is not going to be moved between zones without being destroyed and recreated.

Kubernetes can use this information in various ways. For example, the scheduler automatically tries to spread the Pods in a ReplicaSet across nodes in a single-zone cluster (to reduce the impact of node failures, see [kubernetes.io/hostname](https://kubernetes.io/hostname)). With multiple-zone clusters, this spreading behavior also applies to zones (to reduce the impact of zone failures). This is achieved via *SelectorSpreadPriority*.

*SelectorSpreadPriority* is a best effort placement. If the zones in your cluster are heterogeneous (for example: different numbers of nodes, different types of nodes, or different pod resource requirements), this placement might prevent equal spreading of your Pods across zones. If desired, you can use homogeneous zones (same number and types of nodes) to reduce the probability of unequal spreading.

The scheduler (through the *VolumeZonePredicate* predicate) also will ensure that Pods, that claim a given volume, are only placed into the same zone as that volume. Volumes cannot be attached across zones.

If *PersistentVolumeLabel* does not support automatic labeling of your *PersistentVolumes*, you should consider adding the labels manually (or adding support for *PersistentVolumeLabel*). With *PersistentVolumeLabel*, the scheduler prevents Pods from mounting volumes in a different zone. If your infrastructure doesn't have this constraint, you don't need to add the zone labels to the volumes at all.

## **volume.beta.kubernetes.io/storage-provisioner (deprecated)**

Type: Annotation

Example: `volume.beta.kubernetes.io/storage-provisioner: "k8s.io/minikube-hostpath"`

Used on: *PersistentVolumeClaim*

This annotation has been deprecated since v1.23. See [volume.kubernetes.io/storage-provisioner](https://volume.kubernetes.io/storage-provisioner).

## **volume.beta.kubernetes.io/storage-class (deprecated)**

Type: Annotation

Example: `volume.beta.kubernetes.io/storage-class: "example-class"`

Used on: *PersistentVolume*, *PersistentVolumeClaim*

This annotation can be used for *PersistentVolume*(PV) or *PersistentVolumeClaim*(PVC) to specify the name of [StorageClass](#). When both the `storageClassName` attribute and the

`volume.beta.kubernetes.io/storage-class` annotation are specified, the annotation `volume.beta.kubernetes.io/storage-class` takes precedence over the `storageClassName` attribute.

This annotation has been deprecated. Instead, set the [storageClassName field](#) for the `PersistentVolumeClaim` or `PersistentVolume`.

## **`volume.beta.kubernetes.io/mount-options` (deprecated)**

Type: Annotation

Example : `volume.beta.kubernetes.io/mount-options: "ro,soft"`

Used on: `PersistentVolume`

A Kubernetes administrator can specify additional [mount options](#) for when a `PersistentVolume` is mounted on a node.

## **`volume.kubernetes.io/storage-provisioner`**

Type: Annotation

Used on: `PersistentVolumeClaim`

This annotation is added to a PVC that is supposed to be dynamically provisioned. Its value is the name of a volume plugin that is supposed to provision a volume for this PVC.

## **`volume.kubernetes.io/selected-node`**

Type: Annotation

Used on: `PersistentVolumeClaim`

This annotation is added to a PVC that is triggered by a scheduler to be dynamically provisioned. Its value is the name of the selected node.

## **`volumes.kubernetes.io/controller-managed-attach-detach`**

Type: Annotation

Used on: Node

If a node has the annotation `volumes.kubernetes.io/controller-managed-attach-detach`, its storage attach and detach operations are being managed by the *volume attach/detach* [controller](#).

The value of the annotation isn't important.

## **`node.kubernetes.io/windows-build`**

Type: Label

Example: `node.kubernetes.io/windows-build: "10.0.17763"`

Used on: Node

When the kubelet is running on Microsoft Windows, it automatically labels its Node to record the version of Windows Server in use.

The label's value is in the format "MajorVersion.MinorVersion.BuildNumber".

## **service.kubernetes.io/headless**

Type: Label

Example: `service.kubernetes.io/headless: ""`

Used on: Endpoints

The control plane adds this label to an Endpoints object when the owning Service is headless. To learn more, read [Headless Services](#).

## **service.kubernetes.io/topology-aware-hints (deprecated)**

Example: `service.kubernetes.io/topology-aware-hints: "Auto"`

Used on: Service

This annotation was used for enabling *topology aware hints* on Services. Topology aware hints have since been renamed: the concept is now called [topology aware routing](#). Setting the annotation to Auto, on a Service, configured the Kubernetes control plane to add topology hints on EndpointSlices associated with that Service. You can also explicitly set the annotation to Disabled.

If you are running a version of Kubernetes older than 1.30, check the documentation for that Kubernetes version to see how topology aware routing works in that release.

There are no other valid values for this annotation. If you don't want topology aware hints for a Service, don't add this annotation.

## **service.kubernetes.io/topology-mode**

Type: Annotation

Example: `service.kubernetes.io/topology-mode: Auto`

Used on: Service

This annotation provides a way to define how Services handle network topology; for example, you can configure a Service so that Kubernetes prefers keeping traffic between a client and server within a single topology zone. In some cases this can help reduce costs or improve network performance.

See [Topology Aware Routing](#) for more details.

## **kubernetes.io/service-name**

Type: Label

Example: `kubernetes.io/service-name: "my-website"`

Used on: EndpointSlice

Kubernetes associates [EndpointSlices](#) with [Services](#) using this label.

This label records the [name](#) of the Service that the EndpointSlice is backing. All EndpointSlices should have this label set to the name of their associated Service.

## **kubernetes.io/service-account.name**

Type: Annotation

Example: kubernetes.io/service-account.name: "sa-name"

Used on: Secret

This annotation records the [name](#) of the ServiceAccount that the token (stored in the Secret of type kubernetes.io/service-account-token) represents.

## **kubernetes.io/service-account.uid**

Type: Annotation

Example: kubernetes.io/service-account.uid: da68f9c6-9d26-11e7-b84e-002dc52800da

Used on: Secret

This annotation records the [unique ID](#) of the ServiceAccount that the token (stored in the Secret of type kubernetes.io/service-account-token) represents.

## **kubernetes.io/legacy-token-last-used**

Type: Label

Example: kubernetes.io/legacy-token-last-used: 2022-10-24

Used on: Secret

The control plane only adds this label to Secrets that have the type kubernetes.io/service-account-token. The value of this label records the date (ISO 8601 format, UTC time zone) when the control plane last saw a request where the client authenticated using the service account token.

If a legacy token was last used before the cluster gained the feature (added in Kubernetes v1.26), then the label isn't set.

## **kubernetes.io/legacy-token-invalid-since**

Type: Label

Example: kubernetes.io/legacy-token-invalid-since: 2023-10-27

Used on: Secret



The control plane automatically adds this label to auto-generated Secrets that have the type `kubernetes.io/service-account-token`. This label marks the Secret-based token as invalid for authentication. The value of this label records the date (ISO 8601 format, UTC time zone) when the control plane detects that the auto-generated Secret has not been used for a specified duration (defaults to one year).

### **endpointslice.kubernetes.io/managed-by**

Type: Label

Example: `endpointslice.kubernetes.io/managed-by: endpointslice-controller.k8s.io`

Used on: EndpointSlices

The label is used to indicate the controller or entity that manages the EndpointSlice. This label aims to enable different EndpointSlice objects to be managed by different controllers or entities within the same cluster.

### **endpointslice.kubernetes.io/skip-mirror**

Type: Label

Example: `endpointslice.kubernetes.io/skip-mirror: "true"`

Used on: Endpoints

The label can be set to `"true"` on an Endpoints resource to indicate that the EndpointSliceMirroring controller should not mirror this resource with EndpointSlices.

### **service.kubernetes.io/service-proxy-name**

Type: Label

Example: `service.kubernetes.io/service-proxy-name: "foo-bar"`

Used on: Service

The kube-proxy has this label for custom proxy, which delegates service control to custom proxy.

### **experimental.windows.kubernetes.io/isolation-type (deprecated)**

Type: Annotation

Example: `experimental.windows.kubernetes.io/isolation-type: "hyperv"`

Used on: Pod

The annotation is used to run Windows containers with Hyper-V isolation.

**Note:**

Starting from v1.20, this annotation is deprecated. Experimental Hyper-V support was removed in 1.21.

**ingressclass.kubernetes.io/is-default-class**

Type: Annotation

Example: `ingressclass.kubernetes.io/is-default-class: "true"`

Used on: IngressClass

When a IngressClass resource has this annotation set to "true", new Ingress resource without a class specified will be assigned this default class.

**kubernetes.io/ingress.class (deprecated)**

Type: Annotation

Used on: Ingress

**Note:**

Starting in v1.18, this annotation is deprecated in favor of `spec.ingressClassName`.

**storageclass.kubernetes.io/is-default-class**

Type: Annotation

Example: `storageclass.kubernetes.io/is-default-class: "true"`

Used on: StorageClass

When a single StorageClass resource has this annotation set to "true", new PersistentVolumeClaim resource without a class specified will be assigned this default class.

**alpha.kubernetes.io/provided-node-ip (alpha)**

Type: Annotation

Example: `alpha.kubernetes.io/provided-node-ip: "10.0.0.1"`

Used on: Node

The kubelet can set this annotation on a Node to denote its configured IPv4 and/or IPv6 address.

When kubelet is started with the `--cloud-provider` flag set to any value (includes both external and legacy in-tree cloud providers), it sets this annotation on the Node to denote an IP address set from the command line flag (`--node-ip`). This IP is verified with the cloud provider as valid by the cloud-controller-manager.

## **batch.kubernetes.io/job-completion-index**

Type: Annotation, Label

Example: batch.kubernetes.io/job-completion-index: "3"

Used on: Pod

The Job controller in the kube-controller-manager sets this as a label and annotation for Pods created with Indexed [completion mode](#).

Note the [PodIndexLabel](#) feature gate must be enabled for this to be added as a pod **label**, otherwise it will just be an annotation.

## **batch.kubernetes.io/cronjob-scheduled-timestamp**

Type: Annotation

Example: batch.kubernetes.io/cronjob-scheduled-timestamp: "2016-05-19T03:00:00-07:00"

Used on: Jobs and Pods controlled by CronJobs

This annotation is used to record the original (expected) creation timestamp for a Job, when that Job is part of a CronJob. The control plane sets the value to that timestamp in RFC3339 format. If the Job belongs to a CronJob with a timezone specified, then the timestamp is in that timezone. Otherwise, the timestamp is in controller-manager's local time.

## **kubectrl.kubernetes.io/default-container**

Type: Annotation

Example: kubectrl.kubernetes.io/default-container: "front-end-app"

The value of the annotation is the container name that is default for this Pod. For example, `kubectrl logs` or `kubectrl exec` without `-c` or `--container` flag will use this default container.

## **kubectrl.kubernetes.io/default-logs-container (deprecated)**

Type: Annotation

Example: kubectrl.kubernetes.io/default-logs-container: "front-end-app"

The value of the annotation is the container name that is the default logging container for this Pod. For example, `kubectrl logs` without `-c` or `--container` flag will use this default container.

### **Note:**

This annotation is deprecated. You should use the [kubectrl.kubernetes.io/default-container](#) annotation instead. Kubernetes versions 1.25 and newer ignore this annotation.

## **kubectrl.kubernetes.io/last-applied-configuration**

Type: Annotation

Example: *see following snippet*

```
kubectl.kubernetes.io/last-applied-configuration: >
{"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":
{"name":"example","namespace":"default"},"spec":{"selector":{"matchLabels":{"app.kubernetes.io/
name":"foo"},"template":{"metadata":{"labels":{"app.kubernetes.io/name":"foo"},"spec":
{"containers":[{"image":"container-registry.example/foo-bar:1.42","name":"foo-bar","ports":
[{"containerPort":42}]}}}}}}}
```

Used on: all objects

The kubectl command line tool uses this annotation as a legacy mechanism to track changes. That mechanism has been superseded by [Server-side apply](#).

## **endpoints.kubernetes.io/over-capacity**

Type: Annotation

Example: endpoints.kubernetes.io/over-capacity:truncated

Used on: Endpoints

The [control plane](#) adds this annotation to an [Endpoints](#) object if the associated [Service](#) has more than 1000 backing endpoints. The annotation indicates that the Endpoints object is over capacity and the number of endpoints has been truncated to 1000.

If the number of backend endpoints falls below 1000, the control plane removes this annotation.

## **endpoints.kubernetes.io/last-change-trigger-time**

Type: Annotation

Example: endpoints.kubernetes.io/last-change-trigger-time: "2023-07-20T04:45:21Z"

Used on: Endpoints

This annotation set to an [Endpoints](#) object that represents the timestamp (The timestamp is stored in RFC 3339 date-time string format. For example, '2018-10-22T19:32:52.1Z'). This is timestamp of the last change in some Pod or Service object, that triggered the change to the Endpoints object.

## **control-plane.alpha.kubernetes.io/leader (deprecated)**

Type: Annotation

```
Example: control-plane.alpha.kubernetes.io/
leader={"holderIdentity":"controller-0","leaseDurationSeconds":
15,"acquireTime":"2023-01-19T13:12:57Z","renewTime":"2023-01-19T13:13:54Z","leaderTransitions
":1}
```

Used on: Endpoints

The [control plane](#) previously set annotation on an [Endpoints](#) object. This annotation provided the following detail:

- Who is the current leader.
- The time when the current leadership was acquired.
- The duration of the lease (of the leadership) in seconds.
- The time the current lease (the current leadership) should be renewed.
- The number of leadership transitions that happened in the past.

Kubernetes now uses [Leases](#) to manage leader assignment for the Kubernetes control plane.

## **batch.kubernetes.io/job-tracking (deprecated)**

Type: Annotation

Example: batch.kubernetes.io/job-tracking: ""

Used on: Jobs

The presence of this annotation on a Job used to indicate that the control plane is [tracking the Job status using finalizers](#). Adding or removing this annotation no longer has an effect (Kubernetes v1.27 and later) All Jobs are tracked with finalizers.

## **job-name (deprecated)**

Type: Label

Example: job-name: "pi"

Used on: Jobs and Pods controlled by Jobs

### **Note:**

Starting from Kubernetes 1.27, this label is deprecated. Kubernetes 1.27 and newer ignore this label and use the prefixed job-name label.

## **controller-uid (deprecated)**

Type: Label

Example: controller-uid: "\$UID"

Used on: Jobs and Pods controlled by Jobs

### **Note:**

Starting from Kubernetes 1.27, this label is deprecated. Kubernetes 1.27 and newer ignore this label and use the prefixed controller-uid label.

## **batch.kubernetes.io/job-name**

Type: Label

Example: `batch.kubernetes.io/job-name: "pi"`

Used on: Jobs and Pods controlled by Jobs

This label is used as a user-friendly way to get Pods corresponding to a Job. The job-name comes from the name of the Job and allows for an easy way to get Pods corresponding to the Job.

## **batch.kubernetes.io/controller-uid**

Type: Label

Example: `batch.kubernetes.io/controller-uid: "$UID"`

Used on: Jobs and Pods controlled by Jobs

This label is used as a programmatic way to get all Pods corresponding to a Job. The controller-uid is a unique identifier that gets set in the selector field so the Job controller can get all the corresponding Pods.

## **scheduler.alpha.kubernetes.io/defaultTolerations**

Type: Annotation

Example: `scheduler.alpha.kubernetes.io/defaultTolerations: '[{"operator": "Equal", "value": "value1", "effect": "NoSchedule", "key": "dedicated-node"}]'`

Used on: Namespace

This annotation requires the [PodTolerationRestriction](#) admission controller to be enabled. This annotation key allows assigning tolerations to a namespace and any new pods created in this namespace would get these tolerations added.

## **scheduler.alpha.kubernetes.io/tolerationsWhitelist**

Type: Annotation

Example: `scheduler.alpha.kubernetes.io/tolerationsWhitelist: '[{"operator": "Exists", "effect": "NoSchedule", "key": "dedicated-node"}]'`

Used on: Namespace

This annotation is only useful when the (Alpha) [PodTolerationRestriction](#) admission controller is enabled. The annotation value is a JSON document that defines a list of allowed tolerations for the namespace it annotates. When you create a Pod or modify its tolerations, the API server checks the tolerations to see if they are mentioned in the allow list. The pod is admitted only if the check succeeds.

## **scheduler.alpha.kubernetes.io/preferAvoidPods (deprecated)**

Type: Annotation

Used on: Node

This annotation requires the [NodePreferAvoidPods scheduling plugin](#) to be enabled. The plugin is deprecated since Kubernetes 1.22. Use [Taints and Tolerations](#) instead.

## **node.kubernetes.io/not-ready**

Type: Taint

Example: `node.kubernetes.io/not-ready: "NoExecute"`

Used on: Node

The Node controller detects whether a Node is ready by monitoring its health and adds or removes this taint accordingly.

## **node.kubernetes.io/unreachable**

Type: Taint

Example: `node.kubernetes.io/unreachable: "NoExecute"`

Used on: Node

The Node controller adds the taint to a Node corresponding to the [NodeCondition](#) Ready being Unknown.

## **node.kubernetes.io/unschedulable**

Type: Taint

Example: `node.kubernetes.io/unschedulable: "NoSchedule"`

Used on: Node

The taint will be added to a node when initializing the node to avoid race condition.

## **node.kubernetes.io/memory-pressure**

Type: Taint

Example: `node.kubernetes.io/memory-pressure: "NoSchedule"`

Used on: Node

The kubelet detects memory pressure based on `memory.available` and `allocatableMemory.available` observed on a Node. The observed values are then compared to the corresponding thresholds that can be set on the kubelet to determine if the Node condition and taint should be added/removed.

## **node.kubernetes.io/disk-pressure**

Type: Taint

Example: `node.kubernetes.io/disk-pressure : "NoSchedule"`

Used on: Node

The kubelet detects disk pressure based on `imagefs.available`, `imagefs.inodesFree`, `nodefs.available` and `nodefs.inodesFree` (Linux only) observed on a Node. The observed values are then compared to the corresponding thresholds that can be set on the kubelet to determine if the Node condition and taint should be added/removed.

## **node.kubernetes.io/network-unavailable**

Type: Taint

Example: `node.kubernetes.io/network-unavailable: "NoSchedule"`

Used on: Node

This is initially set by the kubelet when the cloud provider used indicates a requirement for additional network configuration. Only when the route on the cloud is configured properly will the taint be removed by the cloud provider.

## **node.kubernetes.io/pid-pressure**

Type: Taint

Example: `node.kubernetes.io/pid-pressure: "NoSchedule"`

Used on: Node

The kubelet checks D-value of the size of `/proc/sys/kernel/pid_max` and the PIDs consumed by Kubernetes on a node to get the number of available PIDs that referred to as the `pid.available` metric. The metric is then compared to the corresponding threshold that can be set on the kubelet to determine if the node condition and taint should be added/removed.

## **node.kubernetes.io/out-of-service**

Type: Taint

Example: `node.kubernetes.io/out-of-service:NoExecute`

Used on: Node

A user can manually add the taint to a Node marking it out-of-service. If the `NodeOutOfServiceVolumeDetach` [feature gate](#) is enabled on kube-controller-manager, and a Node is marked out-of-service with this taint, the Pods on the node will be forcefully deleted if there are no matching tolerations on it and volume detach operations for the Pods terminating on the node will happen immediately. This allows the Pods on the out-of-service node to recover quickly on a different node.

### **Caution:**

Refer to [Non-graceful node shutdown](#) for further details about when and how to use this taint.



## **node.cloudprovider.kubernetes.io/uninitialized**

Type: Taint

Example: node.cloudprovider.kubernetes.io/uninitialized: "NoSchedule"

Used on: Node

Sets this taint on a Node to mark it as unusable, when kubelet is started with the "external" cloud provider, until a controller from the cloud-controller-manager initializes this Node, and then removes the taint.

## **node.cloudprovider.kubernetes.io/shutdown**

Type: Taint

Example: node.cloudprovider.kubernetes.io/shutdown: "NoSchedule"

Used on: Node

If a Node is in a cloud provider specified shutdown state, the Node gets tainted accordingly with node.cloudprovider.kubernetes.io/shutdown and the taint effect of NoSchedule.

## **feature.node.kubernetes.io/\***

Type: Label

Example: feature.node.kubernetes.io/network-sriov.capable: "true"

Used on: Node

These labels are used by the Node Feature Discovery (NFD) component to advertise features on a node. All built-in labels use the feature.node.kubernetes.io label namespace and have the format feature.node.kubernetes.io/<feature-name>: "true". NFD has many extension points for creating vendor and application-specific labels. For details, see the [customization guide](#).

## **nfd.node.kubernetes.io/master.version**

Type: Annotation

Example: nfd.node.kubernetes.io/master.version: "v0.6.0"

Used on: Node

For node(s) where the Node Feature Discovery (NFD) [master](#) is scheduled, this annotation records the version of the NFD master. It is used for informative use only.

## **nfd.node.kubernetes.io/worker.version**

Type: Annotation

Example: nfd.node.kubernetes.io/worker.version: "v0.4.0"

Used on: Nodes

This annotation records the version for a Node Feature Discovery's [worker](#) if there is one running on a node. It's used for informative use only.

### **nfd.node.kubernetes.io/feature-labels**

Type: Annotation

Example: `nfd.node.kubernetes.io/feature-labels: "cpu-cpuid.ADX,cpu-cpuid.AESNI,cpu-hardware_multithreading,kernel-version.full"`

Used on: Nodes

This annotation records a comma-separated list of node feature labels managed by [Node Feature Discovery](#) (NFD). NFD uses this for an internal mechanism. You should not edit this annotation yourself.

### **nfd.node.kubernetes.io/extended-resources**

Type: Annotation

Example: `nfd.node.kubernetes.io/extended-resources: "accelerator.acme.example/q500,example.com/coprocessor-fx5"`

Used on: Nodes

This annotation records a comma-separated list of [extended resources](#) managed by [Node Feature Discovery](#) (NFD). NFD uses this for an internal mechanism. You should not edit this annotation yourself.

### **nfd.node.kubernetes.io/node-name**

Type: Label

Example: `nfd.node.kubernetes.io/node-name: node-1`

Used on: Nodes

It specifies which node the NodeFeature object is targeting. Creators of NodeFeature objects must set this label and consumers of the objects are supposed to use the label for filtering features designated for a certain node.

#### **Note:**

These Node Feature Discovery (NFD) labels or annotations only apply to the nodes where NFD is running. To learn more about NFD and its components go to its official [documentation](#).

### **service.beta.kubernetes.io/aws-load-balancer-access-log-emit-interval (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-access-log-emit-interval: "5"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures the load balancer for a Service based on this annotation. The value determines how often the load balancer writes log entries. For example, if you set the value to 5, the log writes occur 5 seconds apart.

### **service.beta.kubernetes.io/aws-load-balancer-access-log-enabled (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-access-log-enabled: "false"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures the load balancer for a Service based on this annotation. Access logging is enabled if you set the annotation to "true".

### **service.beta.kubernetes.io/aws-load-balancer-access-log-s3-bucket-name (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-access-log-enabled: example`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures the load balancer for a Service based on this annotation. The load balancer writes logs to an S3 bucket with the name you specify.

### **service.beta.kubernetes.io/aws-load-balancer-access-log-s3-bucket-prefix (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-access-log-enabled: "/example"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures the load balancer for a Service based on this annotation. The load balancer writes log objects with the prefix that you specify.

### **service.beta.kubernetes.io/aws-load-balancer-additional-resource-tags (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-additional-resource-tags: "Environment=demo,Project=example"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures tags (an AWS concept) for a load balancer based on the comma-separated key/value pairs in the value of this annotation.

## **service.beta.kubernetes.io/aws-load-balancer-alpn-policy (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-alpn-policy: HTTP2Optional`

Used on: Service

The [AWS load balancer controller](#) uses this annotation. See [annotations](#) in the AWS load balancer controller documentation.

## **service.beta.kubernetes.io/aws-load-balancer-attributes (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-attributes: "deletion_protection.enabled=true"`

Used on: Service

The [AWS load balancer controller](#) uses this annotation. See [annotations](#) in the AWS load balancer controller documentation.

## **service.beta.kubernetes.io/aws-load-balancer-backend-protocol (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-backend-protocol: tcp`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures the load balancer listener based on the value of this annotation.

## **service.beta.kubernetes.io/aws-load-balancer-connection-draining-enabled (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-connection-draining-enabled: "false"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures the load balancer based on this annotation. The load balancer's connection draining setting depends on the value you set.

## **service.beta.kubernetes.io/aws-load-balancer-connection-draining-timeout (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-connection-draining-timeout: "60"`

Used on: Service

If you configure [connection draining](#) for a Service of type: `LoadBalancer`, and you use the AWS cloud, the integration configures the draining period based on this annotation. The value you set determines the draining timeout in seconds.

## **service.beta.kubernetes.io/aws-load-balancer-ip-address-type (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-ip-address-type: ipv4`

Used on: Service

The [AWS load balancer controller](#) uses this annotation. See [annotations](#) in the AWS load balancer controller documentation.

## **service.beta.kubernetes.io/aws-load-balancer-connection-idle-timeout (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-connection-idle-timeout: "60"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The load balancer has a configured idle timeout period (in seconds) that applies to its connections. If no data has been sent or received by the time that the idle timeout period elapses, the load balancer closes the connection.

## **service.beta.kubernetes.io/aws-load-balancer-cross-zone-load-balancing-enabled (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-cross-zone-load-balancing-enabled: "true"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. If you set this annotation to "true", each load balancer node distributes requests evenly across the registered targets in all enabled [availability zones](#). If you disable cross-zone load balancing, each load balancer node distributes requests evenly across the registered targets in its availability zone only.

## **service.beta.kubernetes.io/aws-load-balancer-eip-allocations (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-eip-allocations: "eipalloc-01bcdef23bcdef456,eipalloc-def1234abc4567890"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The value is a comma-separated list of elastic IP address allocation IDs.

This annotation is only relevant for Services of type: LoadBalancer, where the load balancer is an AWS Network Load Balancer.

## **service.beta.kubernetes.io/aws-load-balancer-extra-security-groups (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-extra-security-groups: "sg-12abcd3456,sg-34dcba6543"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The annotation value is a comma-separated list of extra AWS VPC security groups to configure for the load balancer.

## **service.beta.kubernetes.io/aws-load-balancer-healthcheck-healthy-threshold (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-healthcheck-healthy-threshold: "3"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The annotation value specifies the number of successive successful health checks required for a backend to be considered healthy for traffic.

## **service.beta.kubernetes.io/aws-load-balancer-healthcheck-interval (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-healthcheck-interval: "30"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The annotation value specifies the interval, in seconds, between health check probes made by the load balancer.

## **service.beta.kubernetes.io/aws-load-balancer-healthcheck-path (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-healthcheck-path: /healthcheck`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The annotation value determines the path part of the URL that is used for HTTP health checks.

## **service.beta.kubernetes.io/aws-load-balancer-healthcheck-port (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-healthcheck-port: "24"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The annotation value determines which port the load balancer connects to when performing health checks.

## **service.beta.kubernetes.io/aws-load-balancer-healthcheck-protocol (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-healthcheck-protocol: TCP`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The annotation value determines how the load balancer checks the health of backend targets.

## **service.beta.kubernetes.io/aws-load-balancer-healthcheck-timeout (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-healthcheck-timeout: "3"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The annotation value specifies the number of seconds before a probe that hasn't yet succeeded is automatically treated as having failed.

## **service.beta.kubernetes.io/aws-load-balancer-healthcheck-unhealthy-threshold (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-healthcheck-unhealthy-threshold: "3"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. The annotation value specifies the number of successive unsuccessful health checks required for a backend to be considered unhealthy for traffic.

## **service.beta.kubernetes.io/aws-load-balancer-internal (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-internal: "true"`

Used on: Service

The cloud controller manager integration with AWS elastic load balancing configures a load balancer based on this annotation. When you set this annotation to "true", the integration configures an internal load balancer.

If you use the [AWS load balancer controller](#), see [service.beta.kubernetes.io/aws-load-balancer-scheme](#).

## **service.beta.kubernetes.io/aws-load-balancer-manage-backend-security-group-rules (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-manage-backend-security-group-rules: "true"`

Used on: Service

The [AWS load balancer controller](#) uses this annotation. See [annotations](#) in the AWS load balancer controller documentation.

### **service.beta.kubernetes.io/aws-load-balancer-name (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-name: my-elb`

Used on: Service

If you set this annotation on a Service, and you also annotate that Service with `service.beta.kubernetes.io/aws-load-balancer-type: "external"`, and you use the [AWS load balancer controller](#) in your cluster, then the AWS load balancer controller sets the name of that load balancer to the value you set for *this* annotation.

See [annotations](#) in the AWS load balancer controller documentation.

### **service.beta.kubernetes.io/aws-load-balancer-nlb-target-type (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-nlb-target-type: "true"`

Used on: Service

The [AWS load balancer controller](#) uses this annotation. See [annotations](#) in the AWS load balancer controller documentation.

### **service.beta.kubernetes.io/aws-load-balancer-private-ipv4-addresses (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-private-ipv4-addresses: "198.51.100.0,198.51.100.64"`

Used on: Service

The [AWS load balancer controller](#) uses this annotation. See [annotations](#) in the AWS load balancer controller documentation.

### **service.beta.kubernetes.io/aws-load-balancer-proxy-protocol (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-proxy-protocol: ""`

Used on: Service

The official Kubernetes integration with AWS elastic load balancing configures a load balancer based on this annotation. The only permitted value is `""`, which indicates that the load balancer should wrap TCP connections to the backend Pod with the PROXY protocol.

### **service.beta.kubernetes.io/aws-load-balancer-scheme (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-scheme: internal`

Used on: Service



The [AWS load balancer controller](#) uses this annotation. See [annotations](#) in the AWS load balancer controller documentation.

### **service.beta.kubernetes.io/aws-load-balancer-security-groups (deprecated)**

Example: `service.beta.kubernetes.io/aws-load-balancer-security-groups: "sg-53fae93f,sg-8725gr62r"`

Used on: Service

The AWS load balancer controller uses this annotation to specify a comma separated list of security groups you want to attach to an AWS load balancer. Both name and ID of security are supported where name matches a Name tag, not the groupName attribute.

When this annotation is added to a Service, the load-balancer controller attaches the security groups referenced by the annotation to the load balancer. If you omit this annotation, the AWS load balancer controller automatically creates a new security group and attaches it to the load balancer.

#### **Note:**

Kubernetes v1.27 and later do not directly set or read this annotation. However, the AWS load balancer controller (part of the Kubernetes project) does still use the `service.beta.kubernetes.io/aws-load-balancer-security-groups` annotation.

### **service.beta.kubernetes.io/load-balancer-source-ranges (deprecated)**

Example: `service.beta.kubernetes.io/load-balancer-source-ranges: "192.0.2.0/25"`

Used on: Service

The [AWS load balancer controller](#) uses this annotation. You should set `.spec.loadBalancerSourceRanges` for the Service instead.

### **service.beta.kubernetes.io/aws-load-balancer-ssl-cert (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-ssl-cert: "arn:aws:acm:us-east-1:123456789012:certificate/12345678-1234-1234-1234-123456789012"`

Used on: Service

The official integration with AWS elastic load balancing configures TLS for a Service of type: `LoadBalancer` based on this annotation. The value of the annotation is the AWS Resource Name (ARN) of the X.509 certificate that the load balancer listener should use.

(The TLS protocol is based on an older technology that abbreviates to SSL.)

### **service.beta.kubernetes.io/aws-load-balancer-ssl-negotiation-policy (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-ssl-negotiation-policy: ELBSecurityPolicy-TLS-1-2-2017-01`

The official integration with AWS elastic load balancing configures TLS for a Service of type: LoadBalancer based on this annotation. The value of the annotation is the name of an AWS policy for negotiating TLS with a client peer.

### **service.beta.kubernetes.io/aws-load-balancer-ssl-ports (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-ssl-ports: "**"`

The official integration with AWS elastic load balancing configures TLS for a Service of type: LoadBalancer based on this annotation. The value of the annotation is either "\*\*", which means that all the load balancer's ports should use TLS, or it is a comma separated list of port numbers.

### **service.beta.kubernetes.io/aws-load-balancer-subnets (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-subnets: "private-a,private-b"`

Kubernetes' official integration with AWS uses this annotation to configure a load balancer and determine in which AWS availability zones to deploy the managed load balancing service. The value is either a comma separated list of subnet names, or a comma separated list of subnet IDs.

### **service.beta.kubernetes.io/aws-load-balancer-target-group-attributes (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-target-group-attributes: "stickiness.enabled=true,stickiness.type=source_ip"`

Used on: Service

The [AWS load balancer controller](#) uses this annotation. See [annotations](#) in the AWS load balancer controller documentation.

### **service.beta.kubernetes.io/aws-load-balancer-target-node-labels (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-target-node-labels: "kubernetes.io/os=Linux,topology.kubernetes.io/region=us-east-2"`

Kubernetes' official integration with AWS uses this annotation to determine which nodes in your cluster should be considered as valid targets for the load balancer.

### **service.beta.kubernetes.io/aws-load-balancer-type (beta)**

Example: `service.beta.kubernetes.io/aws-load-balancer-type: external`

Kubernetes' official integrations with AWS use this annotation to determine whether the AWS cloud provider integration should manage a Service of type: LoadBalancer.

There are two permitted values:

`nlb`

the cloud controller manager configures a Network Load Balancer

`external`

the cloud controller manager does not configure any load balancer

If you deploy a Service of type: `LoadBalancer` on AWS, and you don't set any `service.beta.kubernetes.io/aws-load-balancer-type` annotation, the AWS integration deploys a classic Elastic Load Balancer. This behavior, with no annotation present, is the default unless you specify otherwise.

When you set this annotation to `external` on a Service of type: `LoadBalancer`, and your cluster has a working deployment of the AWS Load Balancer controller, then the AWS Load Balancer controller attempts to deploy a load balancer based on the Service specification.

#### Caution:

Do not modify or add the `service.beta.kubernetes.io/aws-load-balancer-type` annotation on an existing Service object. See the AWS documentation on this topic for more details.

### **`service.beta.kubernetes.io/azure-load-balancer-disable-tcp-reset` (deprecated)**

Example: `service.beta.kubernetes.io/azure-load-balancer-disable-tcp-reset: "false"`

Used on: Service

This annotation only works for Azure standard load balancer backed service. This annotation is used on the Service to specify whether the load balancer should disable or enable TCP reset on idle timeout. If enabled, it helps applications to behave more predictably, to detect the termination of a connection, remove expired connections and initiate new connections. You can set the value to be either true or false.

See [Load Balancer TCP Reset](#) for more information.

#### Note:

This annotation is deprecated.

### **`pod-security.kubernetes.io/enforce`**

Type: Label

Example: `pod-security.kubernetes.io/enforce: "baseline"`

Used on: Namespace

Value **must** be one of `privileged`, `baseline`, or `restricted` which correspond to [Pod Security Standard](#) levels. Specifically, the `enforce` label *prohibits* the creation of any Pod in the labeled Namespace which does not meet the requirements outlined in the indicated level.

See [Enforcing Pod Security at the Namespace Level](#) for more information.

### **`pod-security.kubernetes.io/enforce-version`**

Type: Label

Example: `pod-security.kubernetes.io/enforce-version: "1.30"`

Used on: Namespace

Value **must** be latest or a valid Kubernetes version in the format v<major>.<minor>. This determines the version of the [Pod Security Standard](#) policies to apply when validating a Pod.

See [Enforcing Pod Security at the Namespace Level](#) for more information.

## **pod-security.kubernetes.io/audit**

Type: Label

Example: pod-security.kubernetes.io/audit: "baseline"

Used on: Namespace

Value **must** be one of privileged, baseline, or restricted which correspond to [Pod Security Standard](#) levels. Specifically, the audit label does not prevent the creation of a Pod in the labeled Namespace which does not meet the requirements outlined in the indicated level, but adds an this annotation to the Pod.

See [Enforcing Pod Security at the Namespace Level](#) for more information.

## **pod-security.kubernetes.io/audit-version**

Type: Label

Example: pod-security.kubernetes.io/audit-version: "1.30"

Used on: Namespace

Value **must** be latest or a valid Kubernetes version in the format v<major>.<minor>. This determines the version of the [Pod Security Standard](#) policies to apply when validating a Pod.

See [Enforcing Pod Security at the Namespace Level](#) for more information.

## **pod-security.kubernetes.io/warn**

Type: Label

Example: pod-security.kubernetes.io/warn: "baseline"

Used on: Namespace

Value **must** be one of privileged, baseline, or restricted which correspond to [Pod Security Standard](#) levels. Specifically, the warn label does not prevent the creation of a Pod in the labeled Namespace which does not meet the requirements outlined in the indicated level, but returns a warning to the user after doing so. Note that warnings are also displayed when creating or updating objects that contain Pod templates, such as Deployments, Jobs, StatefulSets, etc.

See [Enforcing Pod Security at the Namespace Level](#) for more information.

## **pod-security.kubernetes.io/warn-version**

Type: Label

Example: `pod-security.kubernetes.io/warn-version: "1.30"`

Used on: Namespace

Value **must** be latest or a valid Kubernetes version in the format `v<major>.<minor>`. This determines the version of the [Pod Security Standard](#) policies to apply when validating a submitted Pod. Note that warnings are also displayed when creating or updating objects that contain Pod templates, such as Deployments, Jobs, StatefulSets, etc.

See [Enforcing Pod Security at the Namespace Level](#) for more information.

## **`rbac.authorization.kubernetes.io/autoupdate`**

Type: Annotation

Example: `rbac.authorization.kubernetes.io/autoupdate: "false"`

Used on: ClusterRole, ClusterRoleBinding, Role, RoleBinding

When this annotation is set to `"true"` on default RBAC objects created by the API server, they are automatically updated at server start to add missing permissions and subjects (extra permissions and subjects are left in place). To prevent autoupdating a particular role or rolebinding, set this annotation to `"false"`. If you create your own RBAC objects and set this annotation to `"false"`, `kubectl auth reconcile` (which allows reconciling arbitrary RBAC objects in a [manifest](#)) respects this annotation and does not automatically add missing permissions and subjects.

## **`kubernetes.io/psp (deprecated)`**

Type: Annotation

Example: `kubernetes.io/psp: restricted`

Used on: Pod

This annotation was only relevant if you were using [PodSecurityPolicy](#) objects. Kubernetes v1.30 does not support the PodSecurityPolicy API.

When the PodSecurityPolicy admission controller admitted a Pod, the admission controller modified the Pod to have this annotation. The value of the annotation was the name of the PodSecurityPolicy that was used for validation.

## **`seccomp.security.alpha.kubernetes.io/pod (non-functional)`**

Type: Annotation

Used on: Pod

Kubernetes before v1.25 allowed you to configure seccomp behavior using this annotation. See [Restrict a Container's Syscalls with seccomp](#) to learn the supported way to specify seccomp restrictions for a Pod.

## **container.seccomp.security.alpha.kubernetes.io/[NAME] (non-functional)**

Type: Annotation

Used on: Pod

Kubernetes before v1.25 allowed you to configure seccomp behavior using this annotation. See [Restrict a Container's Syscalls with seccomp](#) to learn the supported way to specify seccomp restrictions for a Pod.

## **snapshot.storage.kubernetes.io/allow-volume-mode-change**

Type: Annotation

Example: `snapshot.storage.kubernetes.io/allow-volume-mode-change: "true"`

Used on: VolumeSnapshotContent

Value can either be true or false. This determines whether a user can modify the mode of the source volume when a PersistentVolumeClaim is being created from a VolumeSnapshot.

Refer to [Converting the volume mode of a Snapshot](#) and the [Kubernetes CSI Developer Documentation](#) for more information.

## **scheduler.alpha.kubernetes.io/critical-pod (deprecated)**

Type: Annotation

Example: `scheduler.alpha.kubernetes.io/critical-pod: ""`

Used on: Pod

This annotation lets Kubernetes control plane know about a Pod being a critical Pod so that the descheduler will not remove this Pod.

### **Note:**

Starting in v1.16, this annotation was removed in favor of [Pod Priority](#).

## **Annotations used for audit**

- [authorization.k8s.io/decision](#)
- [authorization.k8s.io/reason](#)
- [insecure-sha1.invalid-cert.kubernetes.io/\\$hostname](#)
- [missing-san.invalid-cert.kubernetes.io/\\$hostname](#)
- [pod-security.kubernetes.io/audit-violations](#)
- [pod-security.kubernetes.io/enforce-policy](#)
- [pod-security.kubernetes.io/exempt](#)
- [validation.policy.admission.k8s.io/validation\\_failure](#)

See more details on [Audit Annotations](#).

# kubeadm

## **kubeadm.alpha.kubernetes.io/cri-socket**

Type: Annotation

Example: kubeadm.alpha.kubernetes.io/cri-socket: unix:///run/containerd/container.sock

Used on: Node

Annotation that kubeadm uses to preserve the CRI socket information given to kubeadm at init/join time for later use. kubeadm annotates the Node object with this information. The annotation remains "alpha", since ideally this should be a field in KubeletConfiguration instead.

## **kubeadm.kubernetes.io/etcd.advertise-client-urls**

Type: Annotation

Example: kubeadm.kubernetes.io/etcd.advertise-client-urls: https://172.17.0.18:2379

Used on: Pod

Annotation that kubeadm places on locally managed etcd Pods to keep track of a list of URLs where etcd clients should connect to. This is used mainly for etcd cluster health check purposes.

## **kubeadm.kubernetes.io/kube-apiserver.advertise-address.endpoint**

Type: Annotation

Example: kubeadm.kubernetes.io/kube-apiserver.advertise-address.endpoint: https://172.17.0.18:6443

Used on: Pod

Annotation that kubeadm places on locally managed kube-apiserver Pods to keep track of the exposed advertise address/port endpoint for that API server instance.

## **kubeadm.kubernetes.io/component-config.hash**

Type: Annotation

Example: kubeadm.kubernetes.io/component-config.hash:  
2c26b46b68ffc68ff99b453c1d30413413422d706483bfa0f98a5e886266e7ae

Used on: ConfigMap

Annotation that kubeadm places on ConfigMaps that it manages for configuring components. It contains a hash (SHA-256) used to determine if the user has applied settings different from the kubeadm defaults for a particular component.

## **node-role.kubernetes.io/control-plane**

Type: Label

Used on: Node

A marker label to indicate that the node is used to run control plane components. The kubeadm tool applies this label to the control plane nodes that it manages. Other cluster management tools typically also set this taint.

You can label control plane nodes with this label to make it easier to schedule Pods only onto these nodes, or to avoid running Pods on the control plane. If this label is set, the [EndpointSlice controller](#) ignores that node while calculating Topology Aware Hints.

## **node-role.kubernetes.io/control-plane**

Type: Taint

Example: node-role.kubernetes.io/control-plane:NoSchedule

Used on: Node

Taint that kubeadm applies on control plane nodes to restrict placing Pods and allow only specific pods to schedule on them.

If this Taint is applied, control plane nodes allow only critical workloads to be scheduled onto them. You can manually remove this taint with the following command on a specific node.

```
kubecttl taint nodes <node-name> node-role.kubernetes.io/control-plane:NoSchedule-
```

## **node-role.kubernetes.io/master (deprecated)**

Type: Taint

Used on: Node

Example: node-role.kubernetes.io/master:NoSchedule

Taint that kubeadm previously applied on control plane nodes to allow only critical workloads to schedule on them. Replaced by the [node-role.kubernetes.io/control-plane](#) taint. kubeadm no longer sets or uses this deprecated taint.

# **Audit Annotations**

This page serves as a reference for the audit annotations of the kubernetes.io namespace. These annotations apply to Event object from API group audit.k8s.io.

### **Note:**

The following annotations are not used within the Kubernetes API. When you [enable auditing](#) in your cluster, audit event data is written using Event from API group audit.k8s.io. The annotations apply to audit events. Audit events are different from objects in the [Event API](#) (API group events.k8s.io).



## pod-security.kubernetes.io/exempt

Example: pod-security.kubernetes.io/exempt: namespace

Value **must** be one of user, namespace, or runtimeClass which correspond to [Pod Security Exemption](#) dimensions. This annotation indicates on which dimension was based the exemption from the PodSecurity enforcement.

## pod-security.kubernetes.io/enforce-policy

Example: pod-security.kubernetes.io/enforce-policy: restricted:latest

Value **must** be privileged:<version>, baseline:<version>, restricted:<version> which correspond to [Pod Security Standard](#) levels accompanied by a version which **must** be latest or a valid Kubernetes version in the format v<MAJOR>.<MINOR>. This annotations informs about the enforcement level that allowed or denied the pod during PodSecurity admission.

See [Pod Security Standards](#) for more information.

## pod-security.kubernetes.io/audit-violations

Example: pod-security.kubernetes.io/audit-violations: would violate PodSecurity "restricted:latest": allowPrivilegeEscalation != false (container "example" must set securityContext.allowPrivilegeEscalation=false), ...

Value details an audit policy violation, it contains the [Pod Security Standard](#) level that was transgressed as well as the specific policies on the fields that were violated from the PodSecurity enforcement.

See [Pod Security Standards](#) for more information.

## authorization.k8s.io/decision

Example: authorization.k8s.io/decision: "forbid"

This annotation indicates whether or not a request was authorized in Kubernetes audit logs.

See [Auditing](#) for more information.

## authorization.k8s.io/reason

Example: authorization.k8s.io/reason: "Human-readable reason for the decision"

This annotation gives reason for the [decision](#) in Kubernetes audit logs.

See [Auditing](#) for more information.

## **missing-san.invalid-cert.kubernetes.io/\$hostname**

Example:

```
missing-san.invalid-cert.kubernetes.io/example-svc.example-namespace.svc: "relies on a legacy Common Name field instead of the SAN extension for subject validation"
```

Used by Kubernetes version v1.24 and later

This annotation indicates a webhook or aggregated API server is using an invalid certificate that is missing subjectAltNames. Support for these certificates was disabled by default in Kubernetes 1.19, and removed in Kubernetes 1.23.

Requests to endpoints using these certificates will fail. Services using these certificates should replace them as soon as possible to avoid disruption when running in Kubernetes 1.23+ environments.

There's more information about this in the Go documentation: [X.509 CommonName deprecation](#).

## **insecure-sha1.invalid-cert.kubernetes.io/\$hostname**

Example: `insecure-sha1.invalid-cert.kubernetes.io/example-svc.example-namespace.svc: "uses an insecure SHA-1 signature"`

Used by Kubernetes version v1.24 and later

This annotation indicates a webhook or aggregated API server is using an insecure certificate signed with a SHA-1 hash. Support for these insecure certificates is disabled by default in Kubernetes 1.24, and will be removed in a future release.

Services using these certificates should replace them as soon as possible, to ensure connections are secured properly and to avoid disruption in future releases.

There's more information about this in the Go documentation: [Rejecting SHA-1 certificates](#).

## **validation.policy.admission.k8s.io/validation\_failure**

Example: `validation.policy.admission.k8s.io/validation_failure: [{"message": "Invalid value", "policy": "policy.example.com", "binding": "policybinding.example.com", "expressionIndex": "1", "validationActions": ["Audit"]}]"`

Used by Kubernetes version v1.27 and later.

This annotation indicates that a admission policy validation evaluated to false for an API request, or that the validation resulted in an error while the policy was configured with failurePolicy: Fail.

The value of the annotation is a JSON object. The message in the JSON provides the message about the validation failure.

The `policy`, `binding` and `expressionIndex` in the JSON identifies the name of the `ValidatingAdmissionPolicy`, the name of the `ValidatingAdmissionPolicyBinding` and the index in the policy validations of the CEL expressions that failed, respectively.

The `validationActions` shows what actions were taken for this validation failure. See [Validating Admission Policy](#) for more details about `validationActions`.

# Kubernetes API

Kubernetes' API is the application that serves Kubernetes functionality through a RESTful interface and stores the state of the cluster.

Kubernetes resources and "records of intent" are all stored as API objects, and modified via RESTful calls to the API. The API allows configuration to be managed in a declarative way. Users can interact with the Kubernetes API directly, or via tools like `kubectl`. The core Kubernetes API is flexible and can also be extended to support custom resources.

---

[Workload Resources](#)

[Service Resources](#)

[Config and Storage Resources](#)

[Authentication Resources](#)

[Authorization Resources](#)

[Policy Resources](#)

[Extend Resources](#)

[Cluster Resources](#)

[Common Definitions](#)

[Other Resources](#)

[Common Parameters](#)

## Workload Resources

---

[Pod](#)

Pod is a collection of containers that can run on a host.

### [PodTemplate](#)

PodTemplate describes a template for creating copies of a predefined pod.

### [ReplicationController](#)

ReplicationController represents the configuration of a replication controller.

### [ReplicaSet](#)

ReplicaSet ensures that a specified number of pod replicas are running at any given time.

### [Deployment](#)

Deployment enables declarative updates for Pods and ReplicaSets.

### [StatefulSet](#)

StatefulSet represents a set of pods with consistent identities.

### [ControllerRevision](#)

ControllerRevision implements an immutable snapshot of state data.

### [DaemonSet](#)

DemonSet represents the configuration of a daemon set.

### [Job](#)

Job represents the configuration of a single job.

### [CronJob](#)

CronJob represents the configuration of a single cron job.

### [HorizontalPodAutoscaler](#)

configuration of a horizontal pod autoscaler.

### [HorizontalPodAutoscaler](#)

HorizontalPodAutoscaler is the configuration for a horizontal pod autoscaler, which automatically manages the replica count of any resource implementing the scale subresource based on the metrics specified.

### [PriorityClass](#)

PriorityClass defines mapping from a priority class name to the priority integer value.

### [PodSchedulingContext v1alpha2](#)

PodSchedulingContext objects hold information that is needed to schedule a Pod with ResourceClaims that use "WaitForFirstConsumer" allocation mode.

### [ResourceClaim v1alpha2](#)

ResourceClaim describes which resources are needed by a resource consumer.

### [ResourceClaimTemplate v1alpha2](#)

ResourceClaimTemplate is used to produce ResourceClaim objects.

### [ResourceClass v1alpha2](#)

ResourceClass is used by administrators to influence how resources are allocated.

## Pod

Pod is a collection of containers that can run on a host.

apiVersion: v1

```
import "k8s.io/api/core/v1"
```

## Pod

Pod is a collection of containers that can run on a host. This resource is created by clients and scheduled onto hosts.

- 
- **apiVersion:** v1
  - **kind:** Pod
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([PodSpec](#))

Specification of the desired behavior of the pod. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([PodStatus](#))

Most recently observed status of the pod. This data may not be up to date. Populated by the system. Read-only. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

# PodSpec

PodSpec is a description of a pod.

---

## Containers

- **containers** ([][Container](#)), required

*Patch strategy: merge on key name*

List of containers belonging to the pod. Containers cannot currently be added or removed. There must be at least one container in a Pod. Cannot be updated.

- **initContainers** ([][Container](#))

*Patch strategy: merge on key name*

List of initialization containers belonging to the pod. Init containers are executed in order prior to containers being started. If any init container fails, the pod is considered to have failed and is handled according to its restartPolicy. The name for an init container or normal container must be unique among all containers. Init containers may not have Lifecycle actions, Readiness probes, Liveness probes, or Startup probes. The resourceRequirements of an init container are taken into account during scheduling by finding the highest request/limit for each resource type, and then using the max of that value or the sum of the normal containers. Limits are applied to init containers in a similar fashion. Init containers cannot currently be added or removed. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/workloads/pods/init-containers/>

- **ephemeralContainers** ([][EphemeralContainer](#))

*Patch strategy: merge on key name*

List of ephemeral containers run in this pod. Ephemeral containers may be run in an existing pod to perform user-initiated actions such as debugging. This list cannot be specified when creating a pod, and it cannot be modified by updating the pod spec. In order to add an ephemeral container to an existing pod, use the pod's ephemeralcontainers subresource.

- **imagePullSecrets** ([][LocalObjectReference](#))

*Patch strategy: merge on key name*

ImagePullSecrets is an optional list of references to secrets in the same namespace to use for pulling any of the images used by this PodSpec. If specified, these secrets will be passed to individual puller implementations for them to use. More info: <https://kubernetes.io/docs/concepts/containers/images#specifying-imagepullsecrets-on-a-pod>

- **enableServiceLinks** (boolean)

EnableServiceLinks indicates whether information about services should be injected into pod's environment variables, matching the syntax of Docker links. Optional: Defaults to true.

## • **os** (PodOS)

Specifies the OS of the containers in the pod. Some pod and container fields are restricted if this is set.

If the OS field is set to linux, the following fields must be unset: - securityContext.windowsOptions

If the OS field is set to windows, following fields must be unset: - spec.hostPID - spec.hostIPC - spec.hostUsers - spec.securityContext.seLinuxOptions - spec.securityContext.seccompProfile - spec.securityContext.fsGroup - spec.securityContext.fsGroupChangePolicy - spec.securityContext.sysctl - spec.shareProcessNamespace - spec.securityContext.runAsUser - spec.securityContext.runAsGroup - spec.securityContext.supplementalGroups - spec.containers[].securityContext.seLinuxOptions - spec.containers[].securityContext.seccompProfile - spec.containers[].securityContext.capabilities - spec.containers[].securityContext.readOnlyRootFilesystem - spec.containers[].securityContext.privileged - spec.containers[].securityContext.allowPrivilegeEscalation - spec.containers[].securityContext.procMount - spec.containers[].securityContext.runAsUser - spec.containers[\*].securityContext.runAsGroup

*PodOS defines the OS parameters of a pod.*

- **os.name** (string), required

Name is the name of the operating system. The currently supported values are linux and windows. Additional value may be defined in future and can be one of: <https://github.com/opencontainers/runtime-spec/blob/master/config.md#platform-specific-configuration> Clients should expect to handle additional values and treat unrecognized values in this field as os: null

## Volumes

- **volumes** ([][Volume](#))

*Patch strategies: retainKeys, merge on key name*

List of volumes that can be mounted by containers belonging to the pod. More info: <https://kubernetes.io/docs/concepts/storage/volumes>

## Scheduling

- **nodeSelector** (map[string]string)

NodeSelector is a selector which must be true for the pod to fit on a node. Selector which must match a node's labels for the pod to be scheduled on that node. More info: <https://kubernetes.io/docs/concepts/configuration/assign-pod-node/>

- **nodeName** (string)

nodeName is a request to schedule this pod onto a specific node. If it is non-empty, the scheduler simply schedules this pod onto that node, assuming that it fits resource requirements.

- **affinity** (Affinity)

If specified, the pod's scheduling constraints

*Affinity is a group of affinity scheduling rules.*

- **affinity.nodeName** ([NodeAffinity](#))

Describes node affinity scheduling rules for the pod.

- **affinity.podAffinity** ([PodAffinity](#))

Describes pod affinity scheduling rules (e.g. co-locate this pod in the same node, zone, etc. as some other pod(s)).

- **affinity.podAntiAffinity** ([PodAntiAffinity](#))

Describes pod anti-affinity scheduling rules (e.g. avoid putting this pod in the same node, zone, etc. as some other pod(s)).

- **tolerations** ([]Toleration)

If specified, the pod's tolerations.

*The pod this Toleration is attached to tolerates any taint that matches the triple <key,value,effect> using the matching operator .*

- **tolerations.key** (string)

Key is the taint key that the toleration applies to. Empty means match all taint keys. If the key is empty, operator must be Exists; this combination means to match all values and all keys.

- **tolerations.operator** (string)

Operator represents a key's relationship to the value. Valid operators are Exists and Equal. Defaults to Equal. Exists is equivalent to wildcard for value, so that a pod can tolerate all taints of a particular category.

- **tolerations.value** (string)

Value is the taint value the toleration matches to. If the operator is Exists, the value should be empty, otherwise just a regular string.

- **tolerations.effect** (string)

Effect indicates the taint effect to match. Empty means match all taint effects. When specified, allowed values are NoSchedule, PreferNoSchedule and NoExecute.

- **tolerations.tolerationSeconds** (int64)



TolerationSeconds represents the period of time the toleration (which must be of effect NoExecute, otherwise this field is ignored) tolerates the taint. By default, it is not set, which means tolerate the taint forever (do not evict). Zero and negative values will be treated as 0 (evict immediately) by the system.

- **schedulerName** (string)

If specified, the pod will be dispatched by specified scheduler. If not specified, the pod will be dispatched by default scheduler.

- **runtimeClassName** (string)

RuntimeClassName refers to a RuntimeClass object in the node.k8s.io group, which should be used to run this pod. If no RuntimeClass resource matches the named class, the pod will not be run. If unset or empty, the "legacy" RuntimeClass will be used, which is an implicit class with an empty definition that uses the default runtime handler. More info: <https://git.k8s.io/enhancements/keps/sig-node/585-runtime-class>

- **priorityClassName** (string)

If specified, indicates the pod's priority. "system-node-critical" and "system-cluster-critical" are two special keywords which indicate the highest priorities with the former being the highest priority. Any other name must be defined by creating a PriorityClass object with that name. If not specified, the pod priority will be default or zero if there is no default.

- **priority** (int32)

The priority value. Various system components use this field to find the priority of the pod. When Priority Admission Controller is enabled, it prevents users from setting this field. The admission controller populates this field from PriorityClassName. The higher the value, the higher the priority.

- **preemptionPolicy** (string)

PreemptionPolicy is the Policy for preempting pods with lower priority. One of Never, PreemptLowerPriority. Defaults to PreemptLowerPriority if unset.

- **topologySpreadConstraints** ([]TopologySpreadConstraint)

*Patch strategy: merge on key topologyKey*

*Map: unique values on keys topologyKey, whenUnsatisfiable will be kept during a merge*

TopologySpreadConstraints describes how a group of pods ought to spread across topology domains. Scheduler will schedule pods in a way which abides by the constraints. All topologySpreadConstraints are ANDed.

*TopologySpreadConstraint specifies how to spread matching pods among the given topology.*

- **topologySpreadConstraints.maxSkew** (int32), required

MaxSkew describes the degree to which pods may be unevenly distributed. When `whenUnsatisfiable=DoNotSchedule`, it is the maximum permitted difference between the number of matching pods in the target topology and the global

minimum. The global minimum is the minimum number of matching pods in an eligible domain or zero if the number of eligible domains is less than MinDomains. For example, in a 3-zone cluster, MaxSkew is set to 1, and pods with the same labelSelector spread as 2/2/1: In this case, the global minimum is 1. | zone1 | zone2 | zone3 || P P | P P | P | - if MaxSkew is 1, incoming pod can only be scheduled to zone3 to become 2/2/2; scheduling it onto zone1(zone2) would make the ActualSkew(3-1) on zone1(zone2) violate MaxSkew(1). - if MaxSkew is 2, incoming pod can be scheduled onto any zone. When whenUnsatisfiable=ScheduleAnyway, it is used to give higher precedence to topologies that satisfy it. It's a required field. Default value is 1 and 0 is not allowed.

- **topologySpreadConstraints.topologyKey** (string), required

TopologyKey is the key of node labels. Nodes that have a label with this key and identical values are considered to be in the same topology. We consider each <key, value> as a "bucket", and try to put balanced number of pods into each bucket. We define a domain as a particular instance of a topology. Also, we define an eligible domain as a domain whose nodes meet the requirements of nodeAffinityPolicy and nodeTaintsPolicy. e.g. If TopologyKey is "kubernetes.io/hostname", each Node is a domain of that topology. And, if TopologyKey is "topology.kubernetes.io/zone", each zone is a domain of that topology. It's a required field.

- **topologySpreadConstraints.whenUnsatisfiable** (string), required

WhenUnsatisfiable indicates how to deal with a pod if it doesn't satisfy the spread constraint. - DoNotSchedule (default) tells the scheduler not to schedule it. - ScheduleAnyway tells the scheduler to schedule the pod in any location, but giving higher precedence to topologies that would help reduce the skew. A constraint is considered "Unsatisfiable" for an incoming pod if and only if every possible node assignment for that pod would violate "MaxSkew" on some topology. For example, in a 3-zone cluster, MaxSkew is set to 1, and pods with the same labelSelector spread as 3/1/1: | zone1 | zone2 | zone3 || P P P | P | P | If WhenUnsatisfiable is set to DoNotSchedule, incoming pod can only be scheduled to zone2(zone3) to become 3/2/1(3/1/2) as ActualSkew(2-1) on zone2(zone3) satisfies MaxSkew(1). In other words, the cluster can still be imbalanced, but scheduler won't make it *more* imbalanced. It's a required field.

- **topologySpreadConstraints.labelSelector** ([LabelSelector](#))

LabelSelector is used to find matching pods. Pods that match this label selector are counted to determine the number of pods in their corresponding topology domain.

- **topologySpreadConstraints.matchLabelKeys** ([]string)

*Atomic: will be replaced during a merge*

MatchLabelKeys is a set of pod label keys to select the pods over which spreading will be calculated. The keys are used to lookup values from the incoming pod labels, those key-value labels are ANDed with labelSelector to select the group of existing pods over which spreading will be calculated for the incoming pod. The same key is forbidden to exist in both MatchLabelKeys and LabelSelector. MatchLabelKeys cannot be set when LabelSelector isn't set. Keys that don't exist in the incoming pod labels will be ignored. A null or empty list means only match against labelSelector.

This is a beta field and requires the MatchLabelKeysInPodTopologySpread feature gate to be enabled (enabled by default).

- **topologySpreadConstraints.minDomains** (int32)

MinDomains indicates a minimum number of eligible domains. When the number of eligible domains with matching topology keys is less than minDomains, Pod Topology Spread treats "global minimum" as 0, and then the calculation of Skew is performed. And when the number of eligible domains with matching topology keys equals or greater than minDomains, this value has no effect on scheduling. As a result, when the number of eligible domains is less than minDomains, scheduler won't schedule more than maxSkew Pods to those domains. If value is nil, the constraint behaves as if MinDomains is equal to 1. Valid values are integers greater than 0. When value is not nil, WhenUnsatisfiable must be DoNotSchedule.

For example, in a 3-zone cluster, MaxSkew is set to 2, MinDomains is set to 5 and pods with the same labelSelector spread as 2/2/2: | zone1 | zone2 | zone3 | | P P | P P | P P | The number of domains is less than 5(MinDomains), so "global minimum" is treated as 0. In this situation, new pod with the same labelSelector cannot be scheduled, because computed skew will be 3(3 - 0) if new Pod is scheduled to any of the three zones, it will violate MaxSkew.

This is a beta field and requires the MinDomainsInPodTopologySpread feature gate to be enabled (enabled by default).

- **topologySpreadConstraints.nodeAffinityPolicy** (string)

NodeAffinityPolicy indicates how we will treat Pod's nodeAffinity/nodeSelector when calculating pod topology spread skew. Options are: - Honor: only nodes matching nodeAffinity/nodeSelector are included in the calculations. - Ignore: nodeAffinity/nodeSelector are ignored. All nodes are included in the calculations.

If this value is nil, the behavior is equivalent to the Honor policy. This is a beta-level feature default enabled by the NodeInclusionPolicyInPodTopologySpread feature flag.

- **topologySpreadConstraints.nodeTaintsPolicy** (string)

NodeTaintsPolicy indicates how we will treat node taints when calculating pod topology spread skew. Options are: - Honor: nodes without taints, along with tainted nodes for which the incoming pod has a toleration, are included. - Ignore: node taints are ignored. All nodes are included.

If this value is nil, the behavior is equivalent to the Ignore policy. This is a beta-level feature default enabled by the NodeInclusionPolicyInPodTopologySpread feature flag.

- **overhead** (map[string][Quantity](#))

Overhead represents the resource overhead associated with running a pod for a given RuntimeClass. This field will be autopopulated at admission time by the RuntimeClass admission controller. If the RuntimeClass admission controller is enabled, overhead must not be set in Pod create requests. The RuntimeClass admission controller will reject Pod create requests which have the overhead already set. If RuntimeClass is configured and

selected in the PodSpec, Overhead will be set to the value defined in the corresponding RuntimeClass, otherwise it will remain unset and treated as zero. More info: <https://git.k8s.io/enhancements/keps/sig-node/688-pod-overhead/README.md>

## Lifecycle

- **restartPolicy** (string)

Restart policy for all containers within the pod. One of Always, OnFailure, Never. In some contexts, only a subset of those values may be permitted. Default to Always. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle/#restart-policy>

- **terminationGracePeriodSeconds** (int64)

Optional duration in seconds the pod needs to terminate gracefully. May be decreased in delete request. Value must be non-negative integer. The value zero indicates stop immediately via the kill signal (no opportunity to shut down). If this value is nil, the default grace period will be used instead. The grace period is the duration in seconds after the processes running in the pod are sent a termination signal and the time when the processes are forcibly halted with a kill signal. Set this value longer than the expected cleanup time for your process. Defaults to 30 seconds.

- **activeDeadlineSeconds** (int64)

Optional duration in seconds the pod may be active on the node relative to StartTime before the system will actively try to mark it failed and kill associated containers. Value must be a positive integer.

- **readinessGates** ([]PodReadinessGate)

If specified, all readiness gates will be evaluated for pod readiness. A pod is ready when all its containers are ready AND all conditions specified in the readiness gates have status equal to "True" More info: <https://git.k8s.io/enhancements/keps/sig-network/580-pod-readiness-gates>

*PodReadinessGate contains the reference to a pod condition*

- **readinessGates.conditionType** (string), required

ConditionType refers to a condition in the pod's condition list with matching type.

## Hostname and Name resolution

- **hostname** (string)

Specifies the hostname of the Pod If not specified, the pod's hostname will be set to a system-defined value.

- **setHostnameAsFQDN** (boolean)

If true the pod's hostname will be configured as the pod's FQDN, rather than the leaf name (the default). In Linux containers, this means setting the FQDN in the hostname field of the kernel (the nodename field of struct utsname). In Windows containers, this means setting the registry value of hostname for the registry key

HKEY\_LOCAL\_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters to FQDN. If a pod does not have FQDN, this has no effect. Default to false.

- **subdomain** (string)

If specified, the fully qualified Pod hostname will be "<hostname>.<subdomain>.<pod namespace>.svc.<cluster domain>". If not specified, the pod will not have a domainname at all.

- **hostAliases** ([]HostAlias)

*Patch strategy: merge on key ip*

HostAliases is an optional list of hosts and IPs that will be injected into the pod's hosts file if specified. This is only valid for non-hostNetwork pods.

*HostAlias holds the mapping between IP and hostnames that will be injected as an entry in the pod's hosts file.*

- **hostAliases.hostnames** ([]string)

Hostnames for the above IP address.

- **hostAliases.ip** (string)

IP address of the host file entry.

- **dnsConfig** (PodDNSConfig)

Specifies the DNS parameters of a pod. Parameters specified here will be merged to the generated DNS configuration based on DNSPolicy.

*PodDNSConfig defines the DNS parameters of a pod in addition to those generated from DNSPolicy.*

- **dnsConfig.nameservers** ([]string)

A list of DNS name server IP addresses. This will be appended to the base nameservers generated from DNSPolicy. Duplicated nameservers will be removed.

- **dnsConfig.options** ([]PodDNSConfigOption)

A list of DNS resolver options. This will be merged with the base options generated from DNSPolicy. Duplicated entries will be removed. Resolution options given in Options will override those that appear in the base DNSPolicy.

*PodDNSConfigOption defines DNS resolver options of a pod.*

- **dnsConfig.options.name** (string)

Required.

- **dnsConfig.options.value** (string)

- **dnsConfig.searches** ([]string)

A list of DNS search domains for host-name lookup. This will be appended to the base search paths generated from DNSPolicy. Duplicated search paths will be removed.

- **dnsPolicy** (string)

Set DNS policy for the pod. Defaults to "ClusterFirst". Valid values are 'ClusterFirstWithHostNet', 'ClusterFirst', 'Default' or 'None'. DNS parameters given in DNSConfig will be merged with the policy selected with DNSPolicy. To have DNS options set along with hostNetwork, you have to specify DNS policy explicitly to 'ClusterFirstWithHostNet'.

## Hosts namespaces

- **hostNetwork** (boolean)

Host networking requested for this pod. Use the host's network namespace. If this option is set, the ports that will be used must be specified. Default to false.

- **hostPID** (boolean)

Use the host's pid namespace. Optional: Default to false.

- **hostIPC** (boolean)

Use the host's ipc namespace. Optional: Default to false.

- **shareProcessNamespace** (boolean)

Share a single process namespace between all of the containers in a pod. When this is set containers will be able to view and signal processes from other containers in the same pod, and the first process in each container will not be assigned PID 1. HostPID and ShareProcessNamespace cannot both be set. Optional: Default to false.

## Service account

- **serviceAccountName** (string)

ServiceAccountName is the name of the ServiceAccount to use to run this pod. More info: <https://kubernetes.io/docs/tasks/configure-pod-container/configure-service-account/>

- **automountServiceAccountToken** (boolean)

AutomountServiceAccountToken indicates whether a service account token should be automatically mounted.

## Security context

- **securityContext** (PodSecurityContext)

SecurityContext holds pod-level security attributes and common container settings. Optional: Defaults to empty. See type description for default values of each field.

*PodSecurityContext holds pod-level security attributes and common container settings. Some fields are also present in container.securityContext. Field values of container.securityContext take precedence over field values of PodSecurityContext.*

- **securityContext.runAsUser** (int64)

The UID to run the entrypoint of the container process. Defaults to user specified in image metadata if unspecified. May also be set in SecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence for that container. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.runAsNonRoot** (boolean)

Indicates that the container must run as a non-root user. If true, the Kubelet will validate the image at runtime to ensure that it does not run as UID 0 (root) and fail to start the container if it does. If unset or false, no such validation will be performed. May also be set in SecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence.

- **securityContext.runAsGroup** (int64)

The GID to run the entrypoint of the container process. Uses runtime default if unset. May also be set in SecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence for that container. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.supplementalGroups** ([]int64)

A list of groups applied to the first process run in each container, in addition to the container's primary GID, the fsGroup (if specified), and group memberships defined in the container image for the uid of the container process. If unspecified, no additional groups are added to any container. Note that group memberships defined in the container image for the uid of the container process are still effective, even if they are not included in this list. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.fsGroup** (int64)

A special supplemental group that applies to all containers in a pod. Some volume types allow the Kubelet to change the ownership of that volume to be owned by the pod:

1. The owning GID will be the FSGroup
2. The setgid bit is set (new files created in the volume will be owned by FSGroup)
3. The permission bits are OR'd with rw-rw----

If unset, the Kubelet will not modify the ownership and permissions of any volume. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.fsGroupChangePolicy** (string)

fsGroupChangePolicy defines behavior of changing ownership and permission of the volume before being exposed inside Pod. This field will only apply to volume types which support fsGroup based ownership(and permissions). It will have no

effect on ephemeral volume types such as: secret, configmaps and emptydir. Valid values are "OnRootMismatch" and "Always". If not specified, "Always" is used. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.seccompProfile** (SeccompProfile)

The seccomp options to use by the containers in this pod. Note that this field cannot be set when spec.os.name is windows.

*SeccompProfile defines a pod/container's seccomp profile settings. Only one profile source may be set.*

- **securityContext.seccompProfile.type** (string), required

type indicates which kind of seccomp profile will be applied. Valid options are:

Localhost - a profile defined in a file on the node should be used.

RuntimeDefault - the container runtime default profile should be used.

Unconfined - no profile should be applied.

- **securityContext.seccompProfile.localhostProfile** (string)

localhostProfile indicates a profile defined in a file on the node should be used. The profile must be preconfigured on the node to work. Must be a descending path, relative to the kubelet's configured seccomp profile location. Must be set if type is "Localhost". Must NOT be set for any other type.

- **securityContext.seLinuxOptions** (SELinuxOptions)

The SELinux context to be applied to all containers. If unspecified, the container runtime will allocate a random SELinux context for each container. May also be set in SecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence for that container. Note that this field cannot be set when spec.os.name is windows.

*SELinuxOptions are the labels to be applied to the container*

- **securityContext.seLinuxOptions.level** (string)

Level is SELinux level label that applies to the container.

- **securityContext.seLinuxOptions.role** (string)

Role is a SELinux role label that applies to the container.

- **securityContext.seLinuxOptions.type** (string)

Type is a SELinux type label that applies to the container.

- **securityContext.seLinuxOptions.user** (string)

User is a SELinux user label that applies to the container.

- **securityContext.sysctls** ([]Sysctl)



Sysctls hold a list of namespaced sysctls used for the pod. Pods with unsupported sysctls (by the container runtime) might fail to launch. Note that this field cannot be set when spec.os.name is windows.

*Sysctl defines a kernel parameter to be set*

- **securityContext.sysctls.name** (string), required

Name of a property to set

- **securityContext.sysctls.value** (string), required

Value of a property to set

- **securityContext.windowsOptions** (WindowsSecurityContextOptions)

The Windows specific settings applied to all containers. If unspecified, the options within a container's SecurityContext will be used. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is linux.

*WindowsSecurityContextOptions contain Windows-specific options and credentials.*

- **securityContext.windowsOptions.gmsaCredentialSpec** (string)

GMSACredentialSpec is where the GMSA admission webhook (<https://github.com/kubernetes-sigs/windows-gmsa>) inlines the contents of the GMSA credential spec named by the GMSACredentialSpecName field.

- **securityContext.windowsOptions.gmsaCredentialSpecName** (string)

GMSACredentialSpecName is the name of the GMSA credential spec to use.

- **securityContext.windowsOptions.hostProcess** (boolean)

HostProcess determines if a container should be run as a 'Host Process' container. All of a Pod's containers must have the same effective HostProcess value (it is not allowed to have a mix of HostProcess containers and non-HostProcess containers). In addition, if HostProcess is true then HostNetwork must also be set to true.

- **securityContext.windowsOptions.runAsUserName** (string)

The UserName in Windows to run the entrypoint of the container process. Defaults to the user specified in image metadata if unspecified. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence.

## Alpha level

- **hostUsers** (boolean)

Use the host's user namespace. Optional: Default to true. If set to true or not present, the pod will be run in the host user namespace, useful for when the pod needs a feature only available to the host user namespace, such as loading a kernel module with

CAP\_SYS\_MODULE. When set to false, a new users is created for the pod. Setting false is useful for mitigating container breakout vulnerabilities even allowing users to run their containers as root without actually having root privileges on the host. This field is alpha-level and is only honored by servers that enable the UserNamespacesSupport feature.

- **resourceClaims** ([]PodResourceClaim)

*Patch strategies: retainKeys, merge on key name*

*Map: unique values on key name will be kept during a merge*

ResourceClaims defines which ResourceClaims must be allocated and reserved before the Pod is allowed to start. The resources will be made available to those containers which consume them by name.

This is an alpha field and requires enabling the DynamicResourceAllocation feature gate.

This field is immutable.

*PodResourceClaim references exactly one ResourceClaim through a ClaimSource. It adds a name to it that uniquely identifies the ResourceClaim inside the Pod. Containers that need access to the ResourceClaim reference it with this name.*

- **resourceClaims.name** (string), required

Name uniquely identifies this resource claim inside the pod. This must be a DNS\_LABEL.

- **resourceClaims.source** (ClaimSource)

Source describes where to find the ResourceClaim.

\*ClaimSource describes a reference to a ResourceClaim.

Exactly one of these fields should be set. Consumers of this type must treat an empty object as if it has an unknown value.\*

- **resourceClaims.source.resourceClaimName** (string)

ResourceClaimName is the name of a ResourceClaim object in the same namespace as this pod.

- **resourceClaims.source.resourceClaimTemplateName** (string)

ResourceClaimTemplateName is the name of a ResourceClaimTemplate object in the same namespace as this pod.

The template will be used to create a new ResourceClaim, which will be bound to this pod. When this pod is deleted, the ResourceClaim will also be deleted. The pod name and resource name, along with a generated component, will be used to form a unique name for the ResourceClaim, which will be recorded in pod.status.resourceClaimStatuses.

This field is immutable and no changes will be made to the corresponding ResourceClaim by the control plane after creating the ResourceClaim.

• **schedulingGates** ([]PodSchedulingGate)

*Patch strategy: merge on key name*

*Map: unique values on key name will be kept during a merge*

SchedulingGates is an opaque list of values that if specified will block scheduling the pod. If schedulingGates is not empty, the pod will stay in the SchedulingGated state and the scheduler will not attempt to schedule the pod.

SchedulingGates can only be set at pod creation time, and be removed only afterwards.

This is a beta feature enabled by the PodSchedulingReadiness feature gate.

*PodSchedulingGate is associated to a Pod to guard its scheduling.*

- **schedulingGates.name** (string), required

Name of the scheduling gate. Each scheduling gate must have a unique name field.

## Deprecated

- **serviceAccount** (string)

DeprecatedServiceAccount is a depreciated alias for ServiceAccountName. Deprecated: Use serviceAccountName instead.

## Container

A single application container that you want to run within a pod.

---

- **name** (string), required

Name of the container specified as a DNS\_LABEL. Each container in a pod must have a unique name (DNS\_LABEL). Cannot be updated.

## Image

- **image** (string)

Container image name. More info: <https://kubernetes.io/docs/concepts/containers/images>  
This field is optional to allow higher level config management to default or override container images in workload controllers like Deployments and StatefulSets.

- **imagePullPolicy** (string)

Image pull policy. One of Always, Never, IfNotPresent. Defaults to Always if :latest tag is specified, or IfNotPresent otherwise. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/containers/images#updating-images>

## Entrypoint

- **command** ([]string)

Entrypoint array. Not executed within a shell. The container image's ENTRYPOINT is used if this is not provided. Variable references \$(VAR\_NAME) are expanded using the container's environment. If a variable cannot be resolved, the reference in the input string will be unchanged. Double \$\$ are reduced to a single \$, which allows for escaping the \$(VAR\_NAME) syntax: i.e. "\$\$(VAR\_NAME)" will produce the string literal "\$ (VAR\_NAME)". Escaped references will never be expanded, regardless of whether the variable exists or not. Cannot be updated. More info: <https://kubernetes.io/docs/tasks/inject-data-application/define-command-argument-container/#running-a-command-in-a-shell>

- **args** ([]string)

Arguments to the entrypoint. The container image's CMD is used if this is not provided. Variable references \$(VAR\_NAME) are expanded using the container's environment. If a variable cannot be resolved, the reference in the input string will be unchanged. Double \$ are reduced to a single \$, which allows for escaping the \$(VAR\_NAME) syntax: i.e. "\$\$ (VAR\_NAME)" will produce the string literal "\$(VAR\_NAME)". Escaped references will never be expanded, regardless of whether the variable exists or not. Cannot be updated. More info: <https://kubernetes.io/docs/tasks/inject-data-application/define-command-argument-container/#running-a-command-in-a-shell>

- **workingDir** (string)

Container's working directory. If not specified, the container runtime's default will be used, which might be configured in the container image. Cannot be updated.

## Ports

- **ports** ([]ContainerPort)

*Patch strategy: merge on key containerPort*

*Map: unique values on keys containerPort, protocol will be kept during a merge*

List of ports to expose from the container. Not specifying a port here DOES NOT prevent that port from being exposed. Any port which is listening on the default "0.0.0.0" address inside a container will be accessible from the network. Modifying this array with strategic merge patch may corrupt the data. For more information See <https://github.com/kubernetes/kubernetes/issues/108255>. Cannot be updated.

*ContainerPort represents a network port in a single container.*

- **ports.containerPort** (int32), required

Number of port to expose on the pod's IP address. This must be a valid port number, 0 < x < 65536.

- **ports.hostIP** (string)

What host IP to bind the external port to.

**ports.hostPort** (int32)

- Number of port to expose on the host. If specified, this must be a valid port number,  $0 < x < 65536$ . If HostNetwork is specified, this must match ContainerPort. Most containers do not need this.

- **ports.name** (string)

If specified, this must be an IANA\_SVC\_NAME and unique within the pod. Each named port in a pod must have a unique name. Name for the port that can be referred to by services.

- **ports.protocol** (string)

Protocol for port. Must be UDP, TCP, or SCTP. Defaults to "TCP".

## Environment variables

- **env** ([]EnvVar)

*Patch strategy: merge on key name*

List of environment variables to set in the container. Cannot be updated.

*EnvVar represents an environment variable present in a Container.*

- **env.name** (string), required

Name of the environment variable. Must be a C\_IDENTIFIER.

- **env.value** (string)

Variable references  $\$(VAR\_NAME)$  are expanded using the previously defined environment variables in the container and any service environment variables. If a variable cannot be resolved, the reference in the input string will be unchanged. Double  $\$$  are reduced to a single  $\$$ , which allows for escaping the  $\$(VAR\_NAME)$  syntax: i.e.  $\$\$(VAR\_NAME)$  will produce the string literal  $\$(VAR\_NAME)$ . Escaped references will never be expanded, regardless of whether the variable exists or not. Defaults to "".

- **env.valueFrom** (EnvVarSource)

Source for the environment variable's value. Cannot be used if value is not empty.

*EnvVarSource represents a source for the value of an EnvVar.*

- **env.valueFrom.configMapKeyRef** (ConfigMapKeySelector)

Selects a key of a ConfigMap.

*Selects a key from a ConfigMap.*

- **env.valueFrom.configMapKeyRef.key** (string), required

The key to select.

▪ **env.valueFrom.configMapKeyRef.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

▪ **env.valueFrom.configMapKeyRef.optional** (boolean)

Specify whether the ConfigMap or its key must be defined

▪ **env.valueFrom.fieldRef** ([ObjectFieldSelector](#))

Selects a field of the pod: supports metadata.name, metadata.namespace, metadata.labels['\<KEY>'], metadata.annotations['\<KEY>'], spec.nodeName, spec.serviceAccountName, status.hostIP, status.podIP, status.podIPs.

▪ **env.valueFrom.resourceFieldRef** ([ResourceFieldSelector](#))

Selects a resource of the container: only resources limits and requests (limits.cpu, limits.memory, limits.ephemeral-storage, requests.cpu, requests.memory and requests.ephemeral-storage) are currently supported.

▪ **env.valueFrom.secretKeyRef** (SecretKeySelector)

Selects a key of a secret in the pod's namespace

*SecretKeySelector selects a key of a Secret.*

▪ **env.valueFrom.secretKeyRef.key** (string), required

The key of the secret to select from. Must be a valid secret key.

▪ **env.valueFrom.secretKeyRef.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

▪ **env.valueFrom.secretKeyRef.optional** (boolean)

Specify whether the Secret or its key must be defined

• **envFrom** ([]EnvFromSource)

List of sources to populate environment variables in the container. The keys defined within a source must be a C\_IDENTIFIER. All invalid keys will be reported as an event when the container is starting. When a key exists in multiple sources, the value associated with the last source will take precedence. Values defined by an Env with a duplicate key will take precedence. Cannot be updated.

*EnvFromSource represents the source of a set of ConfigMaps*

◦ **envFrom.configMapRef** (ConfigMapEnvSource)

The ConfigMap to select from

\*ConfigMapEnvSource selects a ConfigMap to populate the environment variables with.

The contents of the target ConfigMap's Data field will represent the key-value pairs as environment variables.\*

- **envFrom.configMapRef.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **envFrom.configMapRef.optional** (boolean)

Specify whether the ConfigMap must be defined

- **envFrom.prefix** (string)

An optional identifier to prepend to each key in the ConfigMap. Must be a C\_IDENTIFIER.

- **envFrom.secretRef** (SecretEnvSource)

The Secret to select from

\*SecretEnvSource selects a Secret to populate the environment variables with.

The contents of the target Secret's Data field will represent the key-value pairs as environment variables.\*

- **envFrom.secretRef.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **envFrom.secretRef.optional** (boolean)

Specify whether the Secret must be defined

## Volumes

- **volumeMounts** ([]VolumeMount)

*Patch strategy: merge on key mountPath*

Pod volumes to mount into the container's filesystem. Cannot be updated.

*VolumeMount describes a mounting of a Volume within a container.*

- **volumeMounts.mountPath** (string), required

Path within the container at which the volume should be mounted. Must not contain ':

- **volumeMounts.name** (string), required

This must match the Name of a Volume.

- **volumeMounts.mountPropagation** (string)

mountPropagation determines how mounts are propagated from the host to container and the other way around. When not set, MountPropagationNone is used. This field is beta in 1.10.

- **volumeMounts.readOnly** (boolean)

Mounted read-only if true, read-write otherwise (false or unspecified). Defaults to false.

- **volumeMounts.subPath** (string)

Path within the volume from which the container's volume should be mounted. Defaults to "" (volume's root).

- **volumeMounts.subPathExpr** (string)

Expanded path within the volume from which the container's volume should be mounted. Behaves similarly to SubPath but environment variable references \$ (VAR\_NAME) are expanded using the container's environment. Defaults to "" (volume's root). SubPathExpr and SubPath are mutually exclusive.

- **volumeDevices** ([]VolumeDevice)

*Patch strategy: merge on key devicePath*

volumeDevices is the list of block devices to be used by the container.

*volumeDevice describes a mapping of a raw block device within a container.*

- **volumeDevices.devicePath** (string), required

devicePath is the path inside of the container that the device will be mapped to.

- **volumeDevices.name** (string), required

name must match the name of a persistentVolumeClaim in the pod

## Resources

- **resources** (ResourceRequirements)

Compute Resources required by this container. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/>

*ResourceRequirements describes the compute resource requirements.*

- **resources.claims** ([]ResourceClaim)

*Map: unique values on key name will be kept during a merge*

Claims lists the names of resources, defined in spec.resourceClaims, that are used by this container.

This is an alpha field and requires enabling the DynamicResourceAllocation feature gate.



This field is immutable. It can only be set for containers.

*ResourceClaim references one entry in PodSpec.ResourceClaims.*

- **resources.claims.name** (string), required

Name must match the name of one entry in pod.spec.resourceClaims of the Pod where this field is used. It makes that resource available inside a container.

- **resources.limits** (map[string][Quantity](https://kubernetes.io/docs/concepts/containers/quantity/))

Limits describes the maximum amount of compute resources allowed. More info: <https://kubernetes.io/docs/concepts/containers/quantity/>

- **resources.requests** (map[string][Quantity](https://kubernetes.io/docs/concepts/containers/quantity/))

Requests describes the minimum amount of compute resources required. If Requests is omitted for a container, it defaults to Limits if that is explicitly specified, otherwise to an implementation-defined value. Requests cannot exceed Limits. More info: <https://kubernetes.io/docs/concepts/containers/quantity/>

- **resizePolicy** ([]ContainerResizePolicy)

*Atomic: will be replaced during a merge*

Resources resize policy for the container.

*ContainerResizePolicy represents resource resize policy for the container.*

- **resizePolicy.resourceName** (string), required

Name of the resource to which this resource resize policy applies. Supported values: cpu, memory.

- **resizePolicy.restartPolicy** (string), required

Restart policy to apply when specified resource is resized. If not specified, it defaults to NotRequired.

## Lifecycle

- **lifecycle** (Lifecycle)

Actions that the management system should take in response to container lifecycle events. Cannot be updated.

*Lifecycle describes actions that the management system should take in response to container lifecycle events. For the PostStart and PreStop lifecycle handlers, management of the container blocks until the action is complete, unless the container process fails, in which case the handler is aborted.*

- **lifecycle.postStart** ([LifecycleHandler](https://kubernetes.io/docs/concepts/containers/lifecycle-hooks/))

PostStart is called immediately after a container is created. If the handler fails, the container is terminated and restarted according to its restart policy. Other management of the container blocks until the hook completes. More info: <https://kubernetes.io/docs/concepts/containers/container-lifecycle-hooks/#container-hooks>

- **lifecycle.preStop** ([LifecycleHandler](#))

PreStop is called immediately before a container is terminated due to an API request or management event such as liveness/startup probe failure, preemption, resource contention, etc. The handler is not called if the container crashes or exits. The Pod's termination grace period countdown begins before the PreStop hook is executed. Regardless of the outcome of the handler, the container will eventually terminate within the Pod's termination grace period (unless delayed by finalizers). Other management of the container blocks until the hook completes or until the termination grace period is reached. More info: <https://kubernetes.io/docs/concepts/containers/container-lifecycle-hooks/#container-hooks>

- **terminationMessagePath** (string)

Optional: Path at which the file to which the container's termination message will be written is mounted into the container's filesystem. Message written is intended to be brief final status, such as an assertion failure message. Will be truncated by the node if greater than 4096 bytes. The total message length across all containers will be limited to 12kb. Defaults to /dev/termination-log. Cannot be updated.

- **terminationMessagePolicy** (string)

Indicate how the termination message should be populated. File will use the contents of terminationMessagePath to populate the container status message on both success and failure. FallbackToLogsOnError will use the last chunk of container log output if the termination message file is empty and the container exited with an error. The log output is limited to 2048 bytes or 80 lines, whichever is smaller. Defaults to File. Cannot be updated.

- **livenessProbe** ([Probe](#))

Periodic probe of container liveness. Container will be restarted if the probe fails. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#container-probes>

- **readinessProbe** ([Probe](#))

Periodic probe of container service readiness. Container will be removed from service endpoints if the probe fails. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#container-probes>

- **startupProbe** ([Probe](#))

StartupProbe indicates that the Pod has successfully initialized. If specified, no other probes are executed until this completes successfully. If this probe fails, the Pod will be restarted, just as if the livenessProbe failed. This can be used to provide different probe parameters at the beginning of a Pod's lifecycle, when it might take a long time to load

data or warm a cache, than during steady-state operation. This cannot be updated. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#container-probes>

- **restartPolicy** (string)

RestartPolicy defines the restart behavior of individual containers in a pod. This field may only be set for init containers, and the only allowed value is "Always". For non-init containers or when this field is not specified, the restart behavior is defined by the Pod's restart policy and the container type. Setting the RestartPolicy as "Always" for the init container will have the following effect: this init container will be continually restarted on exit until all regular containers have terminated. Once all regular containers have completed, all init containers with restartPolicy "Always" will be shut down. This lifecycle differs from normal init containers and is often referred to as a "sidecar" container. Although this init container still starts in the init container sequence, it does not wait for the container to complete before proceeding to the next init container. Instead, the next init container starts immediately after this init container is started, or after any startupProbe has successfully completed.

## Security Context

- **securityContext** (SecurityContext)

SecurityContext defines the security options the container should be run with. If set, the fields of SecurityContext override the equivalent fields of PodSecurityContext. More info: <https://kubernetes.io/docs/tasks/configure-pod-container/security-context/>

*SecurityContext holds security configuration that will be applied to a container. Some fields are present in both SecurityContext and PodSecurityContext. When both are set, the values in SecurityContext take precedence.*

- **securityContext.runAsUser** (int64)

The UID to run the entrypoint of the container process. Defaults to user specified in image metadata if unspecified. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.runAsNonRoot** (boolean)

Indicates that the container must run as a non-root user. If true, the Kubelet will validate the image at runtime to ensure that it does not run as UID 0 (root) and fail to start the container if it does. If unset or false, no such validation will be performed. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence.

- **securityContext.runAsGroup** (int64)

The GID to run the entrypoint of the container process. Uses runtime default if unset. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.readOnlyRootFilesystem** (boolean)

Whether this container has a read-only root filesystem. Default is false. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.procMount** (string)

procMount denotes the type of proc mount to use for the containers. The default is DefaultProcMount which uses the container runtime defaults for readonly paths and masked paths. This requires the ProcMountType feature flag to be enabled. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.privileged** (boolean)

Run container in privileged mode. Processes in privileged containers are essentially equivalent to root on the host. Defaults to false. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.allowPrivilegeEscalation** (boolean)

AllowPrivilegeEscalation controls whether a process can gain more privileges than its parent process. This bool directly controls if the no\_new\_privs flag will be set on the container process. AllowPrivilegeEscalation is true always when the container is: 1) run as Privileged 2) has CAP\_SYS\_ADMIN Note that this field cannot be set when spec.os.name is windows.

- **securityContext.capabilities** (Capabilities)

The capabilities to add/drop when running containers. Defaults to the default set of capabilities granted by the container runtime. Note that this field cannot be set when spec.os.name is windows.

*Adds and removes POSIX capabilities from running containers.*

- **securityContext.capabilities.add** ([]string)

Added capabilities

- **securityContext.capabilities.drop** ([]string)

Removed capabilities

- **securityContext.seccompProfile** (SeccompProfile)

The seccomp options to use by this container. If seccomp options are provided at both the pod & container level, the container options override the pod options. Note that this field cannot be set when spec.os.name is windows.

*SeccompProfile defines a pod/container's seccomp profile settings. Only one profile source may be set.*

- **securityContext.seccompProfile.type** (string), required

type indicates which kind of seccomp profile will be applied. Valid options are:

Localhost - a profile defined in a file on the node should be used.  
RuntimeDefault - the container runtime default profile should be used.  
Unconfined - no profile should be applied.

- **securityContext.seccompProfile.localhostProfile** (string)

localhostProfile indicates a profile defined in a file on the node should be used. The profile must be preconfigured on the node to work. Must be a descending path, relative to the kubelet's configured seccomp profile location. Must be set if type is "Localhost". Must NOT be set for any other type.

- **securityContext.seLinuxOptions** (SELinuxOptions)

The SELinux context to be applied to the container. If unspecified, the container runtime will allocate a random SELinux context for each container. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is windows.

*SELinuxOptions are the labels to be applied to the container*

- **securityContext.seLinuxOptions.level** (string)

Level is SELinux level label that applies to the container.

- **securityContext.seLinuxOptions.role** (string)

Role is a SELinux role label that applies to the container.

- **securityContext.seLinuxOptions.type** (string)

Type is a SELinux type label that applies to the container.

- **securityContext.seLinuxOptions.user** (string)

User is a SELinux user label that applies to the container.

- **securityContext.windowsOptions** (WindowsSecurityContextOptions)

The Windows specific settings applied to all containers. If unspecified, the options from the PodSecurityContext will be used. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is linux.

*WindowsSecurityContextOptions contain Windows-specific options and credentials.*

- **securityContext.windowsOptions.gmsaCredentialSpec** (string)

GMSACredentialSpec is where the GMSA admission webhook (<https://github.com/kubernetes-sigs/windows-gmsa>) inlines the contents of the GMSA credential spec named by the GMSACredentialSpecName field.

- **securityContext.windowsOptions.gmsaCredentialSpecName** (string)

GMSACredentialSpecName is the name of the GMSA credential spec to use.

- **securityContext.windowsOptions.hostProcess** (boolean)

HostProcess determines if a container should be run as a 'Host Process' container. All of a Pod's containers must have the same effective HostProcess value (it is not allowed to have a mix of HostProcess containers and non-HostProcess containers). In addition, if HostProcess is true then HostNetwork must also be set to true.

- **securityContext.windowsOptions.runAsUserName** (string)

The UserName in Windows to run the entrypoint of the container process. Defaults to the user specified in image metadata if unspecified. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence.

## Debugging

- **stdin** (boolean)

Whether this container should allocate a buffer for stdin in the container runtime. If this is not set, reads from stdin in the container will always result in EOF. Default is false.

- **stdinOnce** (boolean)

Whether the container runtime should close the stdin channel after it has been opened by a single attach. When stdin is true the stdin stream will remain open across multiple attach sessions. If stdinOnce is set to true, stdin is opened on container start, is empty until the first client attaches to stdin, and then remains open and accepts data until the client disconnects, at which time stdin is closed and remains closed until the container is restarted. If this flag is false, a container processes that reads from stdin will never receive an EOF. Default is false

- **tty** (boolean)

Whether this container should allocate a TTY for itself, also requires 'stdin' to be true. Default is false.

## EphemeralContainer

An EphemeralContainer is a temporary container that you may add to an existing Pod for user-initiated activities such as debugging. Ephemeral containers have no resource or scheduling guarantees, and they will not be restarted when they exit or when a Pod is removed or restarted. The kubelet may evict a Pod if an ephemeral container causes the Pod to exceed its resource allocation.

To add an ephemeral container, use the ephemeralcontainers subresource of an existing Pod. Ephemeral containers may not be removed or restarted.

---

- **name** (string), required

Name of the ephemeral container specified as a DNS\_LABEL. This name must be unique among all containers, init containers and ephemeral containers.

- **targetContainerName** (string)

If set, the name of the container from PodSpec that this ephemeral container targets. The ephemeral container will be run in the namespaces (IPC, PID, etc) of this container. If not set then the ephemeral container uses the namespaces configured in the Pod spec.

The container runtime must implement support for this feature. If the runtime does not support namespace targeting then the result of setting this field is undefined.

## Image

- **image** (string)

Container image name. More info: <https://kubernetes.io/docs/concepts/containers/images>

- **imagePullPolicy** (string)

Image pull policy. One of Always, Never, IfNotPresent. Defaults to Always if :latest tag is specified, or IfNotPresent otherwise. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/containers/images#updating-images>

## Entrypoint

- **command** ([]string)

Entrypoint array. Not executed within a shell. The image's ENTRYPOINT is used if this is not provided. Variable references \$(VAR\_NAME) are expanded using the container's environment. If a variable cannot be resolved, the reference in the input string will be unchanged. Double \$\$ are reduced to a single \$, which allows for escaping the \$(VAR\_NAME) syntax: i.e. "\$\$(VAR\_NAME)" will produce the string literal "\$ (VAR\_NAME)". Escaped references will never be expanded, regardless of whether the variable exists or not. Cannot be updated. More info: <https://kubernetes.io/docs/tasks/inject-data-application/define-command-argument-container/#running-a-command-in-a-shell>

- **args** ([]string)

Arguments to the entrypoint. The image's CMD is used if this is not provided. Variable references \$(VAR\_NAME) are expanded using the container's environment. If a variable cannot be resolved, the reference in the input string will be unchanged. Double \$\$ are reduced to a single \$, which allows for escaping the \$(VAR\_NAME) syntax: i.e. "\$\$(VAR\_NAME)" will produce the string literal "\$(VAR\_NAME)". Escaped references will never be expanded, regardless of whether the variable exists or not. Cannot be updated. More info: <https://kubernetes.io/docs/tasks/inject-data-application/define-command-argument-container/#running-a-command-in-a-shell>

- **workingDir** (string)

Container's working directory. If not specified, the container runtime's default will be used, which might be configured in the container image. Cannot be updated.

## Environment variables

- **env** ([]EnvVar)

*Patch strategy: merge on key name*

List of environment variables to set in the container. Cannot be updated.

*EnvVar represents an environment variable present in a Container.*

- **env.name** (string), required

Name of the environment variable. Must be a C\_IDENTIFIER.

- **env.value** (string)

Variable references \$(VAR\_NAME) are expanded using the previously defined environment variables in the container and any service environment variables. If a variable cannot be resolved, the reference in the input string will be unchanged. Double \$\$ are reduced to a single \$, which allows for escaping the \$(VAR\_NAME) syntax: i.e. "\$\$(VAR\_NAME)" will produce the string literal "\$(VAR\_NAME)". Escaped references will never be expanded, regardless of whether the variable exists or not. Defaults to "".

- **env.valueFrom** (EnvVarSource)

Source for the environment variable's value. Cannot be used if value is not empty.

*EnvVarSource represents a source for the value of an EnvVar.*

- **env.valueFrom.configMapKeyRef** (ConfigMapKeySelector)

Selects a key of a ConfigMap.

*Selects a key from a ConfigMap.*

- **env.valueFrom.configMapKeyRef.key** (string), required

The key to select.

- **env.valueFrom.configMapKeyRef.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **env.valueFrom.configMapKeyRef.optional** (boolean)

Specify whether the ConfigMap or its key must be defined

- **env.valueFrom.fieldRef** ([ObjectFieldSelector](#))

Selects a field of the pod: supports metadata.name, metadata.namespace, metadata.labels['<KEY>'], metadata.annotations['<KEY>'], spec.nodeName, spec.serviceAccountName, status.hostIP, status.podIP, status.podIPs.



▪ **env.valueFrom.resourceFieldRef** ([ResourceFieldSelector](#))

Selects a resource of the container: only resources limits and requests (limits.cpu, limits.memory, limits.ephemeral-storage, requests.cpu, requests.memory and requests.ephemeral-storage) are currently supported.

▪ **env.valueFrom.secretKeyRef** (SecretKeySelector)

Selects a key of a secret in the pod's namespace

*SecretKeySelector selects a key of a Secret.*

▪ **env.valueFrom.secretKeyRef.key** (string), required

The key of the secret to select from. Must be a valid secret key.

▪ **env.valueFrom.secretKeyRef.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

▪ **env.valueFrom.secretKeyRef.optional** (boolean)

Specify whether the Secret or its key must be defined

• **envFrom** ([]EnvFromSource)

List of sources to populate environment variables in the container. The keys defined within a source must be a C\_IDENTIFIER. All invalid keys will be reported as an event when the container is starting. When a key exists in multiple sources, the value associated with the last source will take precedence. Values defined by an Env with a duplicate key will take precedence. Cannot be updated.

*EnvFromSource represents the source of a set of ConfigMaps*

◦ **envFrom.configMapRef** (ConfigMapEnvSource)

The ConfigMap to select from

\*ConfigMapEnvSource selects a ConfigMap to populate the environment variables with.

The contents of the target ConfigMap's Data field will represent the key-value pairs as environment variables.\*

▪ **envFrom.configMapRef.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

▪ **envFrom.configMapRef.optional** (boolean)

Specify whether the ConfigMap must be defined

◦ **envFrom.prefix** (string)

An optional identifier to prepend to each key in the ConfigMap. Must be a C\_IDENTIFIER.

- **envFrom.secretRef** (SecretEnvSource)

The Secret to select from

\*SecretEnvSource selects a Secret to populate the environment variables with.

The contents of the target Secret's Data field will represent the key-value pairs as environment variables.\*

- **envFrom.secretRef.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **envFrom.secretRef.optional** (boolean)

Specify whether the Secret must be defined

## Volumes

- **volumeMounts** ([]VolumeMount)

*Patch strategy: merge on key mountPath*

Pod volumes to mount into the container's filesystem. Subpath mounts are not allowed for ephemeral containers. Cannot be updated.

*VolumeMount describes a mounting of a Volume within a container.*

- **volumeMounts.mountPath** (string), required

Path within the container at which the volume should be mounted. Must not contain ':

- **volumeMounts.name** (string), required

This must match the Name of a Volume.

- **volumeMounts.mountPropagation** (string)

mountPropagation determines how mounts are propagated from the host to container and the other way around. When not set, MountPropagationNone is used. This field is beta in 1.10.

- **volumeMounts.readOnly** (boolean)

Mounted read-only if true, read-write otherwise (false or unspecified). Defaults to false.

- **volumeMounts.subPath** (string)

Path within the volume from which the container's volume should be mounted. Defaults to "" (volume's root).

- **volumeMounts.subPathExpr** (string)

Expanded path within the volume from which the container's volume should be mounted. Behaves similarly to SubPath but environment variable references \$ (VAR\_NAME) are expanded using the container's environment. Defaults to "" (volume's root). SubPathExpr and SubPath are mutually exclusive.

- **volumeDevices** ([]VolumeDevice)

*Patch strategy: merge on key devicePath*

volumeDevices is the list of block devices to be used by the container.

*volumeDevice describes a mapping of a raw block device within a container.*

- **volumeDevices.devicePath** (string), required

devicePath is the path inside of the container that the device will be mapped to.

- **volumeDevices.name** (string), required

name must match the name of a persistentVolumeClaim in the pod

## Resources

- **resizePolicy** ([]ContainerResizePolicy)

*Atomic: will be replaced during a merge*

Resources resize policy for the container.

*ContainerResizePolicy represents resource resize policy for the container.*

- **resizePolicy.resourceName** (string), required

Name of the resource to which this resource resize policy applies. Supported values: cpu, memory.

- **resizePolicy.restartPolicy** (string), required

Restart policy to apply when specified resource is resized. If not specified, it defaults to NotRequired.

## Lifecycle

- **terminationMessagePath** (string)

Optional: Path at which the file to which the container's termination message will be written is mounted into the container's filesystem. Message written is intended to be brief final status, such as an assertion failure message. Will be truncated by the node if greater than 4096 bytes. The total message length across all containers will be limited to 12kb. Defaults to /dev/termination-log. Cannot be updated.

- **terminationMessagePolicy** (string)

Indicate how the termination message should be populated. File will use the contents of terminationMessagePath to populate the container status message on both success and failure. FallbackToLogsOnError will use the last chunk of container log output if the termination message file is empty and the container exited with an error. The log output is limited to 2048 bytes or 80 lines, whichever is smaller. Defaults to File. Cannot be updated.

- **restartPolicy** (string)

Restart policy for the container to manage the restart behavior of each container within a pod. This may only be set for init containers. You cannot set this field on ephemeral containers.

## Debugging

- **stdin** (boolean)

Whether this container should allocate a buffer for stdin in the container runtime. If this is not set, reads from stdin in the container will always result in EOF. Default is false.

- **stdinOnce** (boolean)

Whether the container runtime should close the stdin channel after it has been opened by a single attach. When stdin is true the stdin stream will remain open across multiple attach sessions. If stdinOnce is set to true, stdin is opened on container start, is empty until the first client attaches to stdin, and then remains open and accepts data until the client disconnects, at which time stdin is closed and remains closed until the container is restarted. If this flag is false, a container processes that reads from stdin will never receive an EOF. Default is false

- **tty** (boolean)

Whether this container should allocate a TTY for itself, also requires 'stdin' to be true. Default is false.

## Security context

- **securityContext** (SecurityContext)

Optional: SecurityContext defines the security options the ephemeral container should be run with. If set, the fields of SecurityContext override the equivalent fields of PodSecurityContext.

*SecurityContext holds security configuration that will be applied to a container. Some fields are present in both SecurityContext and PodSecurityContext. When both are set, the values in SecurityContext take precedence.*

- **securityContext.runAsUser** (int64)

The UID to run the entrypoint of the container process. Defaults to user specified in image metadata if unspecified. May also be set in PodSecurityContext. If set in both

SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.runAsNonRoot** (boolean)

Indicates that the container must run as a non-root user. If true, the Kubelet will validate the image at runtime to ensure that it does not run as UID 0 (root) and fail to start the container if it does. If unset or false, no such validation will be performed. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence.

- **securityContext.runAsGroup** (int64)

The GID to run the entrypoint of the container process. Uses runtime default if unset. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.readOnlyRootFilesystem** (boolean)

Whether this container has a read-only root filesystem. Default is false. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.procMount** (string)

procMount denotes the type of proc mount to use for the containers. The default is DefaultProcMount which uses the container runtime defaults for readonly paths and masked paths. This requires the ProcMountType feature flag to be enabled. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.privileged** (boolean)

Run container in privileged mode. Processes in privileged containers are essentially equivalent to root on the host. Defaults to false. Note that this field cannot be set when spec.os.name is windows.

- **securityContext.allowPrivilegeEscalation** (boolean)

AllowPrivilegeEscalation controls whether a process can gain more privileges than its parent process. This bool directly controls if the no\_new\_privs flag will be set on the container process. AllowPrivilegeEscalation is true always when the container is: 1) run as Privileged 2) has CAP\_SYS\_ADMIN Note that this field cannot be set when spec.os.name is windows.

- **securityContext.capabilities** (Capabilities)

The capabilities to add/drop when running containers. Defaults to the default set of capabilities granted by the container runtime. Note that this field cannot be set when spec.os.name is windows.

*Adds and removes POSIX capabilities from running containers.*

- **securityContext.capabilities.add** ([]string)

Added capabilities

- **securityContext.capabilities.drop** ([]string)

- Removed capabilities

- **securityContext.seccompProfile** (SeccompProfile)

The seccomp options to use by this container. If seccomp options are provided at both the pod & container level, the container options override the pod options. Note that this field cannot be set when spec.os.name is windows.

*SeccompProfile defines a pod/container's seccomp profile settings. Only one profile source may be set.*

- **securityContext.seccompProfile.type** (string), required

- type indicates which kind of seccomp profile will be applied. Valid options are:

- Localhost - a profile defined in a file on the node should be used.

- RuntimeDefault - the container runtime default profile should be used.

- Unconfined - no profile should be applied.

- **securityContext.seccompProfile.localhostProfile** (string)

- localhostProfile indicates a profile defined in a file on the node should be used. The profile must be preconfigured on the node to work. Must be a descending path, relative to the kubelet's configured seccomp profile location. Must be set if type is "Localhost". Must NOT be set for any other type.

- **securityContext.seLinuxOptions** (SELinuxOptions)

The SELinux context to be applied to the container. If unspecified, the container runtime will allocate a random SELinux context for each container. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is windows.

*SELinuxOptions are the labels to be applied to the container*

- **securityContext.seLinuxOptions.level** (string)

- Level is SELinux level label that applies to the container.

- **securityContext.seLinuxOptions.role** (string)

- Role is a SELinux role label that applies to the container.

- **securityContext.seLinuxOptions.type** (string)

- Type is a SELinux type label that applies to the container.

- **securityContext.seLinuxOptions.user** (string)

- User is a SELinux user label that applies to the container.

**securityContext.windowsOptions** (WindowsSecurityContextOptions)

◦

The Windows specific settings applied to all containers. If unspecified, the options from the PodSecurityContext will be used. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence. Note that this field cannot be set when spec.os.name is linux.

*WindowsSecurityContextOptions contain Windows-specific options and credentials.*

▪ **securityContext.windowsOptions.gmsaCredentialSpec** (string)

GMSACredentialSpec is where the GMSA admission webhook (<https://github.com/kubernetes-sigs/windows-gmsa>) inlines the contents of the GMSA credential spec named by the GMSACredentialSpecName field.

▪ **securityContext.windowsOptions.gmsaCredentialSpecName** (string)

GMSACredentialSpecName is the name of the GMSA credential spec to use.

▪ **securityContext.windowsOptions.hostProcess** (boolean)

HostProcess determines if a container should be run as a 'Host Process' container. All of a Pod's containers must have the same effective HostProcess value (it is not allowed to have a mix of HostProcess containers and non-HostProcess containers). In addition, if HostProcess is true then HostNetwork must also be set to true.

▪ **securityContext.windowsOptions.runAsUserName** (string)

The UserName in Windows to run the entrypoint of the container process. Defaults to the user specified in image metadata if unspecified. May also be set in PodSecurityContext. If set in both SecurityContext and PodSecurityContext, the value specified in SecurityContext takes precedence.

## Not allowed

• **ports** ([]ContainerPort)

*Patch strategy: merge on key containerPort*

*Map: unique values on keys containerPort, protocol will be kept during a merge*

Ports are not allowed for ephemeral containers.

*ContainerPort represents a network port in a single container.*

◦ **ports.containerPort** (int32), required

Number of port to expose on the pod's IP address. This must be a valid port number,  $0 < x < 65536$ .

◦ **ports.hostIP** (string)

What host IP to bind the external port to.

**ports.hostPort** (int32)

- Number of port to expose on the host. If specified, this must be a valid port number,  $0 < x < 65536$ . If HostNetwork is specified, this must match ContainerPort. Most containers do not need this.

- **ports.name** (string)

If specified, this must be an IANA\_SVC\_NAME and unique within the pod. Each named port in a pod must have a unique name. Name for the port that can be referred to by services.

- **ports.protocol** (string)

Protocol for port. Must be UDP, TCP, or SCTP. Defaults to "TCP".

- **resources** (ResourceRequirements)

Resources are not allowed for ephemeral containers. Ephemeral containers use spare resources already allocated to the pod.

*ResourceRequirements describes the compute resource requirements.*

- **resources.claims** ([]ResourceClaim)

*Map: unique values on key name will be kept during a merge*

Claims lists the names of resources, defined in spec.resourceClaims, that are used by this container.

This is an alpha field and requires enabling the DynamicResourceAllocation feature gate.

This field is immutable. It can only be set for containers.

*ResourceClaim references one entry in PodSpec.ResourceClaims.*

- **resources.claims.name** (string), required

Name must match the name of one entry in pod.spec.resourceClaims of the Pod where this field is used. It makes that resource available inside a container.

- **resources.limits** (map[string][Quantity](#))

Limits describes the maximum amount of compute resources allowed. More info: <https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/>

- **resources.requests** (map[string][Quantity](#))

Requests describes the minimum amount of compute resources required. If Requests is omitted for a container, it defaults to Limits if that is explicitly specified, otherwise to an implementation-defined value. Requests cannot exceed Limits. More info: <https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/>



## **lifecycle** (Lifecycle)

- Lifecycle is not allowed for ephemeral containers.

*Lifecycle describes actions that the management system should take in response to container lifecycle events. For the PostStart and PreStop lifecycle handlers, management of the container blocks until the action is complete, unless the container process fails, in which case the handler is aborted.*

- **lifecycle.postStart** ([LifecycleHandler](#))

PostStart is called immediately after a container is created. If the handler fails, the container is terminated and restarted according to its restart policy. Other management of the container blocks until the hook completes. More info: <https://kubernetes.io/docs/concepts/containers/container-lifecycle-hooks/#container-hooks>

- **lifecycle.preStop** ([LifecycleHandler](#))

PreStop is called immediately before a container is terminated due to an API request or management event such as liveness/startup probe failure, preemption, resource contention, etc. The handler is not called if the container crashes or exits. The Pod's termination grace period countdown begins before the PreStop hook is executed. Regardless of the outcome of the handler, the container will eventually terminate within the Pod's termination grace period (unless delayed by finalizers). Other management of the container blocks until the hook completes or until the termination grace period is reached. More info: <https://kubernetes.io/docs/concepts/containers/container-lifecycle-hooks/#container-hooks>

- **livenessProbe** ([Probe](#))

Probes are not allowed for ephemeral containers.

- **readinessProbe** ([Probe](#))

Probes are not allowed for ephemeral containers.

- **startupProbe** ([Probe](#))

Probes are not allowed for ephemeral containers.

## **LifecycleHandler**

LifecycleHandler defines a specific action that should be taken in a lifecycle hook. One and only one of the fields, except TCPSocket must be specified.

---

- **exec** (ExecAction)

Exec specifies the action to take.

*ExecAction describes a "run in container" action.*

- **exec.command** ([]string)

Command is the command line to execute inside the container, the working directory for the command is root ('/') in the container's filesystem. The command is simply exec'd, it is not run inside a shell, so traditional shell instructions ('|', etc) won't work. To use a shell, you need to explicitly call out to that shell. Exit status of 0 is treated as live/healthy and non-zero is unhealthy.

- **httpGet** (HTTPGetAction)

HTTPGet specifies the http request to perform.

*HTTPGetAction describes an action based on HTTP Get requests.*

- **httpGet.port** (IntOrString), required

Name or number of the port to access on the container. Number must be in the range 1 to 65535. Name must be an IANA\_SVC\_NAME.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **httpGet.host** (string)

Host name to connect to, defaults to the pod IP. You probably want to set "Host" in httpHeaders instead.

- **httpGet.httpHeaders** ([]HTTPHeader)

Custom headers to set in the request. HTTP allows repeated headers.

*HTTPHeader describes a custom header to be used in HTTP probes*

- **httpGet.httpHeaders.name** (string), required

The header field name. This will be canonicalized upon output, so case-variant names will be understood as the same header.

- **httpGet.httpHeaders.value** (string), required

The header field value

- **httpGet.path** (string)

Path to access on the HTTP server.

- **httpGet.scheme** (string)

Scheme to use for connecting to the host. Defaults to HTTP.

- **tcpSocket** (TCPSocketAction)

Deprecated. TCPSocket is NOT supported as a LifecycleHandler and kept for the backward compatibility. There are no validation of this field and lifecycle hooks will fail in runtime when tcp handler is specified.

*TCPSocketAction describes an action based on opening a socket*

- **tcpSocket.port** (IntOrString), required

Number or name of the port to access on the container. Number must be in the range 1 to 65535. Name must be an IANA\_SVC\_NAME.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **tcpSocket.host** (string)

Optional: Host name to connect to, defaults to the pod IP.

## NodeAffinity

Node affinity is a group of node affinity scheduling rules.

---

- **preferredDuringSchedulingIgnoredDuringExecution** ([]PreferredSchedulingTerm)

The scheduler will prefer to schedule pods to nodes that satisfy the affinity expressions specified by this field, but it may choose a node that violates one or more of the expressions. The node that is most preferred is the one with the greatest sum of weights, i.e. for each node that meets all of the scheduling requirements (resource request, requiredDuringScheduling affinity expressions, etc.), compute a sum by iterating through the elements of this field and adding "weight" to the sum if the node matches the corresponding matchExpressions; the node(s) with the highest sum are the most preferred.

*An empty preferred scheduling term matches all objects with implicit weight 0 (i.e. it's a no-op). A null preferred scheduling term matches no objects (i.e. is also a no-op).*

- **preferredDuringSchedulingIgnoredDuringExecution.preference** (NodeSelectorTerm), required

A node selector term, associated with the corresponding weight.

*A null or empty node selector term matches no objects. The requirements of them are ANDed. The TopologySelectorTerm type implements a subset of the NodeSelectorTerm.*

- **preferredDuringSchedulingIgnoredDuringExecution.preference.matchExpressions** ([][NodeSelectorRequirement](#))

A list of node selector requirements by node's labels.

- **preferredDuringSchedulingIgnoredDuringExecution.preference.matchFields** ([][NodeSelectorRequirement](#))

A list of node selector requirements by node's fields.

- **preferredDuringSchedulingIgnoredDuringExecution.weight** (int32), required

Weight associated with matching the corresponding nodeSelectorTerm, in the range 1-100.

- **requiredDuringSchedulingIgnoredDuringExecution** (NodeSelector)

If the affinity requirements specified by this field are not met at scheduling time, the pod will not be scheduled onto the node. If the affinity requirements specified by this field cease to be met at some point during pod execution (e.g. due to an update), the system may or may not try to eventually evict the pod from its node.

*A node selector represents the union of the results of one or more label queries over a set of nodes; that is, it represents the OR of the selectors represented by the node selector terms.*

- **requiredDuringSchedulingIgnoredDuringExecution.nodeSelectorTerms** ([]NodeSelectorTerm), required

Required. A list of node selector terms. The terms are ORed.

*A null or empty node selector term matches no objects. The requirements of them are ANDed. The TopologySelectorTerm type implements a subset of the NodeSelectorTerm.*

- **requiredDuringSchedulingIgnoredDuringExecution.nodeSelectorTerms.matchExpressions** ([]NodeSelectorRequirement)

A list of node selector requirements by node's labels.

- **requiredDuringSchedulingIgnoredDuringExecution.nodeSelectorTerms.matchFields** ([]NodeSelectorRequirement)

A list of node selector requirements by node's fields.

## PodAffinity

Pod affinity is a group of inter pod affinity scheduling rules.

---

- **preferredDuringSchedulingIgnoredDuringExecution** ([]WeightedPodAffinityTerm)

The scheduler will prefer to schedule pods to nodes that satisfy the affinity expressions specified by this field, but it may choose a node that violates one or more of the expressions. The node that is most preferred is the one with the greatest sum of weights, i.e. for each node that meets all of the scheduling requirements (resource request, requiredDuringScheduling affinity expressions, etc.), compute a sum by iterating through the elements of this field and adding "weight" to the sum if the node has pods which matches the corresponding podAffinityTerm; the node(s) with the highest sum are the most preferred.

*The weights of all of the matched WeightedPodAffinityTerm fields are added per-node to find the most preferred node(s)*

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm** (PodAffinityTerm), required

Required. A pod affinity term, associated with the corresponding weight.

*Defines a set of pods (namely those matching the labelSelector relative to the given namespace(s)) that this pod should be co-located (affinity) or not co-located (anti-affinity) with, where co-located is defined as running on a node whose value of the label with key matches that of any node on which a pod of the set of pods is running*

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm.topologyKey** (string), required

This pod should be co-located (affinity) or not co-located (anti-affinity) with the pods matching the labelSelector in the specified namespaces, where co-located is defined as running on a node whose value of the label with key topologyKey matches that of any node on which any of the selected pods is running. Empty topologyKey is not allowed.

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm.labelSelector** ([LabelSelector](#))

A label query over a set of resources, in this case pods.

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm.namespaces** ([LabelSelector](#))

A label query over the set of namespaces that the term applies to. The term is applied to the union of the namespaces selected by this field and the ones listed in the namespaces field. null selector and null or empty namespaces list means "this pod's namespace". An empty selector ({}) matches all namespaces.

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm.namespaces** ([]string)

namespaces specifies a static list of namespace names that the term applies to. The term is applied to the union of the namespaces listed in this field and the ones selected by namespaceSelector. null or empty namespaces list and null namespaceSelector means "this pod's namespace".

- **preferredDuringSchedulingIgnoredDuringExecution.weight** (int32), required

weight associated with matching the corresponding podAffinityTerm, in the range 1-100.

- **requiredDuringSchedulingIgnoredDuringExecution** ([]PodAffinityTerm)

If the affinity requirements specified by this field are not met at scheduling time, the pod will not be scheduled onto the node. If the affinity requirements specified by this field cease to be met at some point during pod execution (e.g. due to a pod label update), the system may or may not try to eventually evict the pod from its node. When there are multiple elements, the lists of nodes corresponding to each podAffinityTerm are intersected, i.e. all terms must be satisfied.

*Defines a set of pods (namely those matching the labelSelector relative to the given namespace(s)) that this pod should be co-located (affinity) or not co-located (anti-affinity)*

with, where co-located is defined as running on a node whose value of the label with key matches that of any node on which a pod of the set of pods is running

- **requiredDuringSchedulingIgnoredDuringExecution.topologyKey** (string), required

This pod should be co-located (affinity) or not co-located (anti-affinity) with the pods matching the labelSelector in the specified namespaces, where co-located is defined as running on a node whose value of the label with key topologyKey matches that of any node on which any of the selected pods is running. Empty topologyKey is not allowed.

- **requiredDuringSchedulingIgnoredDuringExecution.labelSelector** ([LabelSelector](#))

A label query over a set of resources, in this case pods.

- **requiredDuringSchedulingIgnoredDuringExecution.namespaceSelector** ([LabelSelector](#))

A label query over the set of namespaces that the term applies to. The term is applied to the union of the namespaces selected by this field and the ones listed in the namespaces field. null selector and null or empty namespaces list means "this pod's namespace". An empty selector ({} ) matches all namespaces.

- **requiredDuringSchedulingIgnoredDuringExecution.namespaces** ([]string)

namespaces specifies a static list of namespace names that the term applies to. The term is applied to the union of the namespaces listed in this field and the ones selected by namespaceSelector. null or empty namespaces list and null namespaceSelector means "this pod's namespace".

## PodAntiAffinity

Pod anti affinity is a group of inter pod anti affinity scheduling rules.

---

- **preferredDuringSchedulingIgnoredDuringExecution** ([]WeightedPodAffinityTerm)

The scheduler will prefer to schedule pods to nodes that satisfy the anti-affinity expressions specified by this field, but it may choose a node that violates one or more of the expressions. The node that is most preferred is the one with the greatest sum of weights, i.e. for each node that meets all of the scheduling requirements (resource request, requiredDuringScheduling anti-affinity expressions, etc.), compute a sum by iterating through the elements of this field and adding "weight" to the sum if the node has pods which matches the corresponding podAffinityTerm; the node(s) with the highest sum are the most preferred.

*The weights of all of the matched WeightedPodAffinityTerm fields are added per-node to find the most preferred node(s)*

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm** (PodAffinityTerm), required

Required. A pod affinity term, associated with the corresponding weight.

*Defines a set of pods (namely those matching the labelSelector relative to the given namespace(s)) that this pod should be co-located (affinity) or not co-located (anti-affinity) with, where co-located is defined as running on a node whose value of the label with key matches that of any node on which a pod of the set of pods is running*

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm.topologyKey** (string), required

This pod should be co-located (affinity) or not co-located (anti-affinity) with the pods matching the labelSelector in the specified namespaces, where co-located is defined as running on a node whose value of the label with key topologyKey matches that of any node on which any of the selected pods is running. Empty topologyKey is not allowed.

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm.labelSelector** ([LabelSelector](#))

A label query over a set of resources, in this case pods.

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm.namespaces** ([LabelSelector](#))

A label query over the set of namespaces that the term applies to. The term is applied to the union of the namespaces selected by this field and the ones listed in the namespaces field. null selector and null or empty namespaces list means "this pod's namespace". An empty selector ({} ) matches all namespaces.

- **preferredDuringSchedulingIgnoredDuringExecution.podAffinityTerm.namespaces** ([]string)

namespaces specifies a static list of namespace names that the term applies to. The term is applied to the union of the namespaces listed in this field and the ones selected by namespaceSelector. null or empty namespaces list and null namespaceSelector means "this pod's namespace".

- **preferredDuringSchedulingIgnoredDuringExecution.weight** (int32), required

weight associated with matching the corresponding podAffinityTerm, in the range 1-100.

- **requiredDuringSchedulingIgnoredDuringExecution** ([]PodAffinityTerm)

If the anti-affinity requirements specified by this field are not met at scheduling time, the pod will not be scheduled onto the node. If the anti-affinity requirements specified by this field cease to be met at some point during pod execution (e.g. due to a pod label update), the system may or may not try to eventually evict the pod from its node. When there are multiple elements, the lists of nodes corresponding to each podAffinityTerm are intersected, i.e. all terms must be satisfied.

*Defines a set of pods (namely those matching the labelSelector relative to the given namespace(s)) that this pod should be co-located (affinity) or not co-located (anti-affinity)*

with, where co-located is defined as running on a node whose value of the label with key matches that of any node on which a pod of the set of pods is running

- **requiredDuringSchedulingIgnoredDuringExecution.topologyKey** (string), required

This pod should be co-located (affinity) or not co-located (anti-affinity) with the pods matching the labelSelector in the specified namespaces, where co-located is defined as running on a node whose value of the label with key topologyKey matches that of any node on which any of the selected pods is running. Empty topologyKey is not allowed.

- **requiredDuringSchedulingIgnoredDuringExecution.labelSelector** ([LabelSelector](#))

A label query over a set of resources, in this case pods.

- **requiredDuringSchedulingIgnoredDuringExecution.namespaceSelector** ([LabelSelector](#))

A label query over the set of namespaces that the term applies to. The term is applied to the union of the namespaces selected by this field and the ones listed in the namespaces field. null selector and null or empty namespaces list means "this pod's namespace". An empty selector ({} ) matches all namespaces.

- **requiredDuringSchedulingIgnoredDuringExecution.namespaces** ([]string)

namespaces specifies a static list of namespace names that the term applies to. The term is applied to the union of the namespaces listed in this field and the ones selected by namespaceSelector. null or empty namespaces list and null namespaceSelector means "this pod's namespace".

## Probe

Probe describes a health check to be performed against a container to determine whether it is alive or ready to receive traffic.

- 
- **exec** (ExecAction)

Exec specifies the action to take.

*ExecAction describes a "run in container" action.*

- **exec.command** ([]string)

Command is the command line to execute inside the container, the working directory for the command is root ('/') in the container's filesystem. The command is simply exec'd, it is not run inside a shell, so traditional shell instructions ('|', etc) won't work. To use a shell, you need to explicitly call out to that shell. Exit status of 0 is treated as live/healthy and non-zero is unhealthy.

- **httpGet** (HTTPGetAction)



HTTPGet specifies the http request to perform.

*HTTPGetAction describes an action based on HTTP Get requests.*

- **httpGet.port** (IntOrString), required

Name or number of the port to access on the container. Number must be in the range 1 to 65535. Name must be an IANA\_SVC\_NAME.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **httpGet.host** (string)

Host name to connect to, defaults to the pod IP. You probably want to set "Host" in httpHeaders instead.

- **httpGet.httpHeaders** ([]HTTPHeader)

Custom headers to set in the request. HTTP allows repeated headers.

*HTTPHeader describes a custom header to be used in HTTP probes*

- **httpGet.httpHeaders.name** (string), required

The header field name. This will be canonicalized upon output, so case-variant names will be understood as the same header.

- **httpGet.httpHeaders.value** (string), required

The header field value

- **httpGet.path** (string)

Path to access on the HTTP server.

- **httpGet.scheme** (string)

Scheme to use for connecting to the host. Defaults to HTTP.

- **tcpSocket** (TCPSocketAction)

TCPSocket specifies an action involving a TCP port.

*TCPSocketAction describes an action based on opening a socket*

- **tcpSocket.port** (IntOrString), required

Number or name of the port to access on the container. Number must be in the range 1 to 65535. Name must be an IANA\_SVC\_NAME.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

**tcpSocket.host** (string)

◦

Optional: Host name to connect to, defaults to the pod IP.

- **initialDelaySeconds** (int32)

Number of seconds after the container has started before liveness probes are initiated.

More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#container-probes>

- **terminationGracePeriodSeconds** (int64)

Optional duration in seconds the pod needs to terminate gracefully upon probe failure.

The grace period is the duration in seconds after the processes running in the pod are sent a termination signal and the time when the processes are forcibly halted with a kill signal. Set this value longer than the expected cleanup time for your process. If this value is nil, the pod's terminationGracePeriodSeconds will be used. Otherwise, this value overrides the value provided by the pod spec. Value must be non-negative integer. The value zero indicates stop immediately via the kill signal (no opportunity to shut down). This is a beta field and requires enabling ProbeTerminationGracePeriod feature gate. Minimum value is 1. spec.terminationGracePeriodSeconds is used if unset.

- **periodSeconds** (int32)

How often (in seconds) to perform the probe. Default to 10 seconds. Minimum value is 1.

- **timeoutSeconds** (int32)

Number of seconds after which the probe times out. Defaults to 1 second. Minimum value is 1. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#container-probes>

- **failureThreshold** (int32)

Minimum consecutive failures for the probe to be considered failed after having succeeded. Defaults to 3. Minimum value is 1.

- **successThreshold** (int32)

Minimum consecutive successes for the probe to be considered successful after having failed. Defaults to 1. Must be 1 for liveness and startup. Minimum value is 1.

- **grpc** (GRPCAction)

GRPC specifies an action involving a GRPC port.

\*\*

- **grpc.port** (int32), required

Port number of the gRPC service. Number must be in the range 1 to 65535.

- **grpc.service** (string)

Service is the name of the service to place in the gRPC HealthCheckRequest (see <https://github.com/grpc/grpc/blob/master/doc/health-checking.md>).

If this is not specified, the default behavior is defined by gRPC.

## PodStatus

PodStatus represents information about the status of a pod. Status may trail the actual state of a system, especially if the node that hosts the pod cannot contact the control plane.

---

- **nominatedNodeName** (string)

nominatedNodeName is set only when this pod preempts other pods on the node, but it cannot be scheduled right away as preemption victims receive their graceful termination periods. This field does not guarantee that the pod will be scheduled on this node. Scheduler may decide to place the pod elsewhere if other nodes become available sooner. Scheduler may also decide to give the resources on this node to a higher priority pod that is created after preemption. As a result, this field may be different than PodSpec.nodeName when the pod is scheduled.

- **hostIP** (string)

hostIP holds the IP address of the host to which the pod is assigned. Empty if the pod has not started yet. A pod can be assigned to a node that has a problem in kubelet which in turns mean that HostIP will not be updated even if there is a node is assigned to pod

- **hostIPs** ([]HostIP)

*Patch strategy: merge on key ip*

*Atomic: will be replaced during a merge*

hostIPs holds the IP addresses allocated to the host. If this field is specified, the first entry must match the hostIP field. This list is empty if the pod has not started yet. A pod can be assigned to a node that has a problem in kubelet which in turns means that HostIPs will not be updated even if there is a node is assigned to this pod.

*HostIP represents a single IP address allocated to the host.*

- **hostIPs.ip** (string)

IP is the IP address assigned to the host

- **startTime** (Time)

RFC 3339 date and time at which the object was acknowledged by the Kubelet. This is before the Kubelet pulled the container image(s) for the pod.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **phase** (string)

The phase of a Pod is a simple, high-level summary of where the Pod is in its lifecycle. The conditions array, the reason and message fields, and the individual container status arrays contain more detail about the pod's status. There are five possible phase values:

**Pending:** The pod has been accepted by the Kubernetes system, but one or more of the container images has not been created. This includes time before being scheduled as well as time spent downloading images over the network, which could take a while. **Running:** The pod has been bound to a node, and all of the containers have been created. At least one container is still running, or is in the process of starting or restarting. **Succeeded:** All containers in the pod have terminated in success, and will not be restarted. **Failed:** All containers in the pod have terminated, and at least one container has terminated in failure. The container either exited with non-zero status or was terminated by the system. **Unknown:** For some reason the state of the pod could not be obtained, typically due to an error in communicating with the host of the pod.

More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#pod-phase>

- **message** (string)

A human readable message indicating details about why the pod is in this condition.

- **reason** (string)

A brief CamelCase message indicating details about why the pod is in this state. e.g. 'Evicted'

- **podIP** (string)

podIP address allocated to the pod. Routable at least within the cluster. Empty if not yet allocated.

- **podIPs** ([]PodIP)

*Patch strategy: merge on key ip*

podIPs holds the IP addresses allocated to the pod. If this field is specified, the 0th entry must match the podIP field. Pods may be allocated at most 1 value for each of IPv4 and IPv6. This list is empty if no IPs have been allocated yet.

*PodIP represents a single IP address allocated to the pod.*

- **podIPs.ip** (string)

IP is the IP address assigned to the pod

- **conditions** ([]PodCondition)

*Patch strategy: merge on key type*

Current service state of pod. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#pod-conditions>

*PodCondition contains details for the current condition of this pod.*

- **conditions.status** (string), required

Status is the status of the condition. Can be True, False, Unknown. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#pod-conditions>

- **conditions.type** (string), required

Type is the type of the condition. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#pod-conditions>

- **conditions.lastProbeTime** (Time)

Last time we probed the condition.

*Time is a wrapper around time. Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.lastTransitionTime** (Time)

Last time the condition transitioned from one status to another.

*Time is a wrapper around time. Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

Human-readable message indicating details about last transition.

- **conditions.reason** (string)

Unique, one-word, CamelCase reason for the condition's last transition.

- **qosClass** (string)

The Quality of Service (QOS) classification assigned to the pod based on resource requirements See PodQOSClass type for available QOS classes More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-qos/#quality-of-service-classes>

- **initContainerStatuses** ([]ContainerStatus)

The list has one entry per init container in the manifest. The most recent successful init container will have ready = true, the most recently started container will have startTime set. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#pod-and-container-status>

*ContainerStatus contains details for the current status of this container.*

- **containerStatuses** ([]ContainerStatus)

The list has one entry per container in the manifest. More info: <https://kubernetes.io/docs/concepts/workloads/pods/pod-lifecycle#pod-and-container-status>

*ContainerStatus contains details for the current status of this container.*

- **ephemeralContainerStatuses** ([]ContainerStatus)

Status for any ephemeral containers that have run in this pod.

*ContainerStatus contains details for the current status of this container.*

- **resourceClaimStatuses** ([]PodResourceClaimStatus)

*Patch strategies: retainKeys, merge on key name*

*Map: unique values on key name will be kept during a merge*

Status of resource claims.

*PodResourceClaimStatus is stored in the PodStatus for each PodResourceClaim which references a ResourceClaimTemplate. It stores the generated name for the corresponding ResourceClaim.*

- **resourceClaimStatuses.name** (string), required

Name uniquely identifies this resource claim inside the pod. This must match the name of an entry in pod.spec.resourceClaims, which implies that the string must be a DNS\_LABEL.

- **resourceClaimStatuses.resourceClaimName** (string)

ResourceClaimName is the name of the ResourceClaim that was generated for the Pod in the namespace of the Pod. If this is unset, then generating a ResourceClaim was not necessary. The pod.spec.resourceClaims entry can be ignored in this case.

- **resize** (string)

Status of resources resize desired for pod's containers. It is empty if no resources resize is pending. Any changes to container resources will automatically set this to "Proposed"

## PodList

PodList is a list of Pods.

- 
- **items** ([]Pod), required

List of pods. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md>

- **apiVersion** (string)

APIVersion defines the versioned schema of this representation of an object. Servers should convert recognized schemas to the latest internal value, and may reject unrecognized values. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources>

- **kind** (string)

Kind is a string value representing the REST resource this object represents. Servers may infer this from the endpoint the client submits requests to. Cannot be updated. In

CamelCase. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

## Operations

---

### get read the specified Pod

#### HTTP Request

GET /api/v1/namespaces/{namespace}/pods/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([Pod](#)): OK

401: Unauthorized

### get read ephemeralcontainers of the specified Pod

#### HTTP Request

GET /api/v1/namespaces/{namespace}/pods/{name}/ephemeralcontainers

#### Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Pod](#)): OK

401: Unauthorized

## get read log of the specified Pod

### HTTP Request

GET /api/v1/namespaces/{namespace}/pods/{name}/log

### Parameters

- **name** (*in path*): string, required

name of the Pod

- **namespace** (*in path*): string, required

[namespace](#)

- **container** (*in query*): string

The container for which to stream logs. Defaults to only container if there is one container in the pod.

- **follow** (*in query*): boolean

Follow the log stream of the pod. Defaults to false.

- **insecureSkipTLSVerifyBackend** (*in query*): boolean

`insecureSkipTLSVerifyBackend` indicates that the apiserver should not confirm the validity of the serving certificate of the backend it is connecting to. This will make the HTTPS connection between the apiserver and the backend insecure. This means the apiserver cannot verify the log data it is receiving came from the real kubelet. If the kubelet is configured to verify the apiserver's TLS credentials, it does not mean the connection to the real kubelet is vulnerable to a man in the middle attack (e.g. an attacker could not intercept the actual log data coming from the real kubelet).

- **limitBytes** (*in query*): integer

If set, the number of bytes to read from the server before terminating the log output. This may not display a complete final line of logging, and may return slightly more or slightly less than the specified limit.



- **pretty** (*in query*): string

- [pretty](#)

- **previous** (*in query*): boolean

Return previous terminated container logs. Defaults to false.

- **sinceSeconds** (*in query*): integer

A relative time in seconds before the current time from which to show logs. If this value precedes the time a pod was started, only logs since the pod start will be returned. If this value is in the future, no logs will be returned. Only one of sinceSeconds or sinceTime may be specified.

- **tailLines** (*in query*): integer

If set, the number of lines from the end of the logs to show. If not specified, logs are shown from the creation of the container or sinceSeconds or sinceTime

- **timestamps** (*in query*): boolean

If true, add an RFC3339 or RFC3339Nano timestamp at the beginning of every line of log output. Defaults to false.

## Response

200 (string): OK

401: Unauthorized

## get read status of the specified Pod

### HTTP Request

GET /api/v1/namespaces/{namespace}/pods/{name}/status

### Parameters

- **name** (*in path*): string, required

name of the Pod

- **namespace** (*in path*): string, required

- [namespace](#)

- **pretty** (*in query*): string

- [pretty](#)

## Response

200 ([Pod](#)): OK

401: Unauthorized

## list list or watch objects of kind Pod

### HTTP Request

GET /api/v1/namespaces/{namespace}/pods

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([PodList](#)): OK

401: Unauthorized

## list list or watch objects of kind Pod

### HTTP Request

GET /api/v1/pods

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([PodList](#)): OK

401: Unauthorized

## create create a Pod

### HTTP Request

POST /api/v1/namespaces/{namespace}/pods

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Pod](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Pod](#)): OK

201 ([Pod](#)): Created

202 ([Pod](#)): Accepted

401: Unauthorized

## update replace the specified Pod

### HTTP Request

PUT /api/v1/namespaces/{namespace}/pods/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Pod](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Pod](#)): OK

201 ([Pod](#)): Created

401: Unauthorized

## update replace ephemeralcontainers of the specified Pod

### HTTP Request

PUT /api/v1/namespaces/{namespace}/pods/{name}/ephemeralcontainers

## Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Pod](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Pod](#)): OK

201 ([Pod](#)): Created

401: Unauthorized

## update replace status of the specified Pod

### HTTP Request

PUT /api/v1/namespaces/{namespace}/pods/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Pod](#), required

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Pod](#)): OK

201 ([Pod](#)): Created

401: Unauthorized

## patch partially update the specified Pod

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/pods/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)

- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Pod](#)): OK

201 ([Pod](#)): Created

401: Unauthorized

## patch partially update ephemeralcontainers of the specified Pod

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/pods/{name}/ephemeralcontainers

### Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)



## Response

200 ([Pod](#)): OK

201 ([Pod](#)): Created

401: Unauthorized

## patch partially update status of the specified Pod

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/pods/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Pod](#)): OK

201 ([Pod](#)): Created

401: Unauthorized

## delete delete a Pod

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/pods/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Pod
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([Pod](#)): OK

202 ([Pod](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Pod

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/pods

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)

- **body:** [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# PodTemplate

PodTemplate describes a template for creating copies of a predefined pod.

apiVersion: v1

import "k8s.io/api/core/v1"

## PodTemplate

PodTemplate describes a template for creating copies of a predefined pod.

---

- **apiVersion:** v1
- **kind:** PodTemplate
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **template** ([PodTemplateSpec](#))

Template defines the pods that will be created from this pod template. <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## PodTemplateSpec

PodTemplateSpec describes the data a pod should have when created from a template

---

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([PodSpec](#))

Specification of the desired behavior of the pod. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## PodTemplateList

PodTemplateList is a list of PodTemplates.

---

- **apiVersion:** v1
- **kind:** PodTemplateList

- **metadata** ([ListMeta](#))  
Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>
- **items** ([\[\]PodTemplate](#)), required  
List of pod templates

## Operations

---

### get read the specified PodTemplate

#### HTTP Request

GET /api/v1/namespaces/{namespace}/podtemplates/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the PodTemplate
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([PodTemplate](#)): OK

401: Unauthorized

### list list or watch objects of kind PodTemplate

#### HTTP Request

GET /api/v1/namespaces/{namespace}/podtemplates

#### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([PodTemplateList](#)): OK

401: Unauthorized

## list or watch objects of kind PodTemplate

### HTTP Request

GET /api/v1/podtemplates

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([PodTemplateList](#)): OK

401: Unauthorized

## create create a PodTemplate

### HTTP Request

POST /api/v1/namespaces/{namespace}/podtemplates

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [PodTemplate](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([PodTemplate](#)): OK

201 ([PodTemplate](#)): Created

202 ([PodTemplate](#)): Accepted

401: Unauthorized

## update replace the specified PodTemplate

### HTTP Request

PUT /api/v1/namespaces/{namespace}/podtemplates/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PodTemplate
- **namespace** (*in path*): string, required



[namespace](#)

- **body**: [PodTemplate](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PodTemplate](#)): OK

201 ([PodTemplate](#)): Created

401: Unauthorized

## patch partially update the specified PodTemplate

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/podtemplates/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PodTemplate
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodTemplate](#)): OK

201 ([PodTemplate](#)): Created

401: Unauthorized

## delete delete a PodTemplate

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/podtemplates/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PodTemplate
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([PodTemplate](#)): OK

202 ([PodTemplate](#)): Accepted

401: Unauthorized

## deletecollection delete collection of PodTemplate

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/podtemplates

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ReplicationController

ReplicationController represents the configuration of a replication controller.

apiVersion: v1

import "k8s.io/api/core/v1"

## ReplicationController

ReplicationController represents the configuration of a replication controller.

- 
- **apiVersion**: v1
  - **kind**: ReplicationController
  - **metadata** ([ObjectMeta](#))

If the Labels of a ReplicationController are empty, they are defaulted to be the same as the Pod(s) that the replication controller manages. Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([ReplicationControllerSpec](#))

Spec defines the specification of the desired behavior of the replication controller. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([ReplicationControllerStatus](#))

Status is the most recently observed status of the replication controller. This data may be out of date by some window of time. Populated by the system. Read-only. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## ReplicationControllerSpec

ReplicationControllerSpec is the specification of a replication controller.

---

- **selector** (map[string]string)

Selector is a label query over pods that should match the Replicas count. If Selector is empty, it is defaulted to the labels present on the Pod template. Label keys and values that must match in order to be controlled by this replication controller, if empty defaulted to labels on Pod template. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/#label-selectors>

- **template** ([PodTemplateSpec](#))

Template is the object that describes the pod that will be created if insufficient replicas are detected. This takes precedence over a TemplateRef. The only allowed template.spec.restartPolicy value is "Always". More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller#pod-template>

- **replicas** (int32)

Replicas is the number of desired replicas. This is a pointer to distinguish between explicit zero and unspecified. Defaults to 1. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller#what-is-a-replicationcontroller>

- **minReadySeconds** (int32)

Minimum number of seconds for which a newly created pod should be ready without any of its container crashing, for it to be considered available. Defaults to 0 (pod will be considered available as soon as it is ready)

## ReplicationControllerStatus

ReplicationControllerStatus represents the current status of a replication controller.

---

- **replicas** (int32), required

Replicas is the most recently observed number of replicas. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller#what-is-a-replicationcontroller>

- **availableReplicas** (int32)

The number of available replicas (ready for at least minReadySeconds) for this replication controller.

**readyReplicas** (int32)

- The number of ready replicas for this replication controller.

• **fullyLabeledReplicas** (int32)

The number of pods that have labels matching the labels of the pod template of the replication controller.

• **conditions** ([]ReplicationControllerCondition)

*Patch strategy: merge on key type*

Represents the latest available observations of a replication controller's current state.

*ReplicationControllerCondition describes the state of a replication controller at a certain point.*

- **conditions.status** (string), required

Status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

Type of replication controller condition.

- **conditions.lastTransitionTime** (Time)

The last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

A human readable message indicating details about the transition.

- **conditions.reason** (string)

The reason for the condition's last transition.

• **observedGeneration** (int64)

ObservedGeneration reflects the generation of the most recently observed replication controller.

## ReplicationControllerList

ReplicationControllerList is a collection of replication controllers.

---

• **apiVersion:** v1

• **kind:** ReplicationControllerList

**metadata** ([ListMeta](#))

- Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]ReplicationController](#)), required

List of replication controllers. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller>

## Operations

---

### get read the specified ReplicationController

#### HTTP Request

GET /api/v1/namespaces/{namespace}/replicationcontrollers/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ReplicationController
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ReplicationController](#)): OK

401: Unauthorized

### get read status of the specified ReplicationController

#### HTTP Request

GET /api/v1/namespaces/{namespace}/replicationcontrollers/{name}/status

#### Parameters

- **name** (*in path*): string, required  
name of the ReplicationController

- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ReplicationController](#)): OK

401: Unauthorized

## list list or watch objects of kind ReplicationController

### HTTP Request

GET /api/v1/namespaces/{namespace}/replicationcontrollers

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)



- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ReplicationControllerList](#)): OK

401: Unauthorized

## list list or watch objects of kind ReplicationController

## HTTP Request

GET /api/v1/replicationcontrollers

## Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ReplicationControllerList](#)): OK

401: Unauthorized

## create create a ReplicationController

### HTTP Request

POST /api/v1/namespaces/{namespace}/replicationcontrollers

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ReplicationController](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ReplicationController](#)): OK

201 ([ReplicationController](#)): Created

202 ([ReplicationController](#)): Accepted

401: Unauthorized

## update replace the specified ReplicationController

## HTTP Request

PUT /api/v1/namespaces/{namespace}/replicationcontrollers/{name}

## Parameters

- **name** (*in path*): string, required  
name of the ReplicationController
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ReplicationController](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ReplicationController](#)): OK

201 ([ReplicationController](#)): Created

401: Unauthorized

## update replace status of the specified ReplicationController

### HTTP Request

PUT /api/v1/namespaces/{namespace}/replicationcontrollers/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ReplicationController
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ReplicationController](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ReplicationController](#)): OK

201 ([ReplicationController](#)): Created

401: Unauthorized

## patch partially update the specified ReplicationController

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/replicationcontrollers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ReplicationController

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ReplicationController](#)): OK

201 ([ReplicationController](#)): Created

401: Unauthorized

## patch partially update status of the specified ReplicationController

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/replicationcontrollers/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ReplicationController
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ReplicationController](#)): OK

201 ([ReplicationController](#)): Created

401: Unauthorized

## delete delete a ReplicationController

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/replicationcontrollers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ReplicationController
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ReplicationController

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/replicationcontrollers

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **continue** (*in query*): string

[continue](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ReplicaSet

ReplicaSet ensures that a specified number of pod replicas are running at any given time.

apiVersion: apps/v1

import "k8s.io/api/apps/v1"

## ReplicaSet

ReplicaSet ensures that a specified number of pod replicas are running at any given time.

- 
- **apiVersion**: apps/v1
  - **kind**: ReplicaSet
  - **metadata** ([ObjectMeta](#))

If the Labels of a ReplicaSet are empty, they are defaulted to be the same as the Pod(s) that the ReplicaSet manages. Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([ReplicaSetSpec](#))



Spec defines the specification of the desired behavior of the ReplicaSet. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([ReplicaSetStatus](#))

Status is the most recently observed status of the ReplicaSet. This data may be out of date by some window of time. Populated by the system. Read-only. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## ReplicaSetSpec

ReplicaSetSpec is the specification of a ReplicaSet.

---

- **selector** ([LabelSelector](#)), required

Selector is a label query over pods that should match the replica count. Label keys and values that must match in order to be controlled by this replica set. It must match the pod template's labels. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/#label-selectors>

- **template** ([PodTemplateSpec](#))

Template is the object that describes the pod that will be created if insufficient replicas are detected. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller#pod-template>

- **replicas** (int32)

Replicas is the number of desired replicas. This is a pointer to distinguish between explicit zero and unspecified. Defaults to 1. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller/#what-is-a-replicationcontroller>

- **minReadySeconds** (int32)

Minimum number of seconds for which a newly created pod should be ready without any of its container crashing, for it to be considered available. Defaults to 0 (pod will be considered available as soon as it is ready)

## ReplicaSetStatus

ReplicaSetStatus represents the current status of a ReplicaSet.

---

- **replicas** (int32), required

Replicas is the most recently observed number of replicas. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller/#what-is-a-replicationcontroller>

- **availableReplicas** (int32)

The number of available replicas (ready for at least minReadySeconds) for this replica set.

- **readyReplicas** (int32)

readyReplicas is the number of pods targeted by this ReplicaSet with a Ready Condition.

- **fullyLabeledReplicas** (int32)

The number of pods that have labels matching the labels of the pod template of the replicaset.

- **conditions** ([]ReplicaSetCondition)

*Patch strategy: merge on key type*

Represents the latest available observations of a replica set's current state.

*ReplicaSetCondition describes the state of a replica set at a certain point.*

- **conditions.status** (string), required

Status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

Type of replica set condition.

- **conditions.lastTransitionTime** (Time)

The last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

A human readable message indicating details about the transition.

- **conditions.reason** (string)

The reason for the condition's last transition.

- **observedGeneration** (int64)

ObservedGeneration reflects the generation of the most recently observed ReplicaSet.

## ReplicaSetList

ReplicaSetList is a collection of ReplicaSets.

---

- **apiVersion:** apps/v1

- **kind:** ReplicaSetList

**metadata** ([ListMeta](#))

- Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]ReplicaSet](#)), required

List of ReplicaSets. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller>

## Operations

---

### get read the specified ReplicaSet

#### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/replicasets/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ReplicaSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ReplicaSet](#)): OK

401: Unauthorized

### get read status of the specified ReplicaSet

#### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/replicasets/{name}/status

#### Parameters

- **name** (*in path*): string, required  
name of the ReplicaSet

- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ReplicaSet](#)): OK

401: Unauthorized

## list list or watch objects of kind ReplicaSet

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/replicasets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ReplicaSetList](#)): OK

401: Unauthorized

## list list or watch objects of kind ReplicaSet

### HTTP Request

GET /apis/apps/v1/replicasets

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ReplicaSetList](#)): OK

401: Unauthorized

## create create a ReplicaSet

### HTTP Request

POST /apis/apps/v1/namespaces/{namespace}/replicasets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ReplicaSet](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ReplicaSet](#)): OK

201 ([ReplicaSet](#)): Created

202 ([ReplicaSet](#)): Accepted

401: Unauthorized

## update replace the specified ReplicaSet

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/replicasets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ReplicaSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ReplicaSet](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ReplicaSet](#)): OK

201 ([ReplicaSet](#)): Created

401: Unauthorized

## update replace status of the specified ReplicaSet

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/replicasets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ReplicaSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ReplicaSet](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ReplicaSet](#)): OK

201 ([ReplicaSet](#)): Created

401: Unauthorized

## patch partially update the specified ReplicaSet

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/replicasets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ReplicaSet



- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ReplicaSet](#)): OK

201 ([ReplicaSet](#)): Created

401: Unauthorized

## patch partially update status of the specified ReplicaSet

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/replicasets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ReplicaSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ReplicaSet](#)): OK

201 ([ReplicaSet](#)): Created

401: Unauthorized

## delete delete a ReplicaSet

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/replicasets/{name}

### Parameters

- **name** (*in path*): string, required

name of the ReplicaSet

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ReplicaSet

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/replicasets

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **continue** (*in query*): string

[continue](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Deployment

Deployment enables declarative updates for Pods and ReplicaSets.

apiVersion: apps/v1

import "k8s.io/api/apps/v1"

## Deployment

Deployment enables declarative updates for Pods and ReplicaSets.

- 
- **apiVersion**: apps/v1
  - **kind**: Deployment
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([DeploymentSpec](#))

Specification of the desired behavior of the Deployment.

**status** ([DeploymentStatus](#))

- Most recently observed status of the Deployment.

## DeploymentSpec

DeploymentSpec is the specification of the desired behavior of the Deployment.

---

- **selector** ([LabelSelector](#)), required

Label selector for pods. Existing ReplicaSets whose pods are selected by this will be the ones affected by this deployment. It must match the pod template's labels.

- **template** ([PodTemplateSpec](#)), required

Template describes the pods that will be created. The only allowed `template.spec.restartPolicy` value is "Always".

- **replicas** (int32)

Number of desired pods. This is a pointer to distinguish between explicit zero and not specified. Defaults to 1.

- **minReadySeconds** (int32)

Minimum number of seconds for which a newly created pod should be ready without any of its container crashing, for it to be considered available. Defaults to 0 (pod will be considered available as soon as it is ready)

- **strategy** (DeploymentStrategy)

*Patch strategy: retainKeys*

The deployment strategy to use to replace existing pods with new ones.

*DeploymentStrategy describes how to replace existing pods with new ones.*

- **strategy.type** (string)

Type of deployment. Can be "Recreate" or "RollingUpdate". Default is RollingUpdate.

- **strategy.rollingUpdate** (RollingUpdateDeployment)

Rolling update config params. Present only if DeploymentStrategyType = RollingUpdate.

*Spec to control the desired behavior of rolling update.*

- **strategy.rollingUpdate.maxSurge** (IntOrString)

The maximum number of pods that can be scheduled above the desired number of pods. Value can be an absolute number (ex: 5) or a percentage of desired pods (ex: 10%). This can not be 0 if MaxUnavailable is 0. Absolute

number is calculated from percentage by rounding up. Defaults to 25%. Example: when this is set to 30%, the new ReplicaSet can be scaled up immediately when the rolling update starts, such that the total number of old and new pods do not exceed 130% of desired pods. Once old pods have been killed, new ReplicaSet can be scaled up further, ensuring that total number of pods running at any time during the update is at most 130% of desired pods.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **strategy.rollingUpdate.maxUnavailable** (IntOrString)

The maximum number of pods that can be unavailable during the update. Value can be an absolute number (ex: 5) or a percentage of desired pods (ex: 10%). Absolute number is calculated from percentage by rounding down. This can not be 0 if MaxSurge is 0. Defaults to 25%. Example: when this is set to 30%, the old ReplicaSet can be scaled down to 70% of desired pods immediately when the rolling update starts. Once new pods are ready, old ReplicaSet can be scaled down further, followed by scaling up the new ReplicaSet, ensuring that the total number of pods available at all times during the update is at least 70% of desired pods.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **revisionHistoryLimit** (int32)

The number of old ReplicaSets to retain to allow rollback. This is a pointer to distinguish between explicit zero and not specified. Defaults to 10.

- **progressDeadlineSeconds** (int32)

The maximum time in seconds for a deployment to make progress before it is considered to be failed. The deployment controller will continue to process failed deployments and a condition with a ProgressDeadlineExceeded reason will be surfaced in the deployment status. Note that progress will not be estimated during the time a deployment is paused. Defaults to 600s.

- **paused** (boolean)

Indicates that the deployment is paused.

## DeploymentStatus

DeploymentStatus is the most recently observed status of the Deployment.

---

- **replicas** (int32)

Total number of non-terminated pods targeted by this deployment (their labels match the selector).

- **availableReplicas** (int32)

Total number of available pods (ready for at least minReadySeconds) targeted by this deployment.

- **readyReplicas** (int32)

readyReplicas is the number of pods targeted by this Deployment with a Ready Condition.

- **unavailableReplicas** (int32)

Total number of unavailable pods targeted by this deployment. This is the total number of pods that are still required for the deployment to have 100% available capacity. They may either be pods that are running but not yet available or pods that still have not been created.

- **updatedReplicas** (int32)

Total number of non-terminated pods targeted by this deployment that have the desired template spec.

- **collisionCount** (int32)

Count of hash collisions for the Deployment. The Deployment controller uses this field as a collision avoidance mechanism when it needs to create the name for the newest ReplicaSet.

- **conditions** ([]DeploymentCondition)

*Patch strategy: merge on key type*

Represents the latest available observations of a deployment's current state.

*DeploymentCondition describes the state of a deployment at a certain point.*

- **conditions.status** (string), required

Status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

Type of deployment condition.

- **conditions.lastTransitionTime** (Time)

Last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.lastUpdateTime** (Time)

The last time this condition was updated.

*Time is a wrapper around time. Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

A human readable message indicating details about the transition.

- **conditions.reason** (string)

The reason for the condition's last transition.

- **observedGeneration** (int64)

The generation observed by the deployment controller.

## DeploymentList

DeploymentList is a list of Deployments.

---

- **apiVersion:** apps/v1

- **kind:** DeploymentList

- **metadata** ([ListMeta](#))

Standard list metadata.

- **items** ([\[\]Deployment](#)), required

Items is the list of Deployments.

## Operations

---

### get read the specified Deployment

#### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/deployments/{name}

#### Parameters

- **name** (*in path*): string, required

name of the Deployment

- **namespace** (*in path*): string, required



[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Deployment](#)): OK

401: Unauthorized

## get read status of the specified Deployment

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/deployments/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Deployment
- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Deployment](#)): OK

401: Unauthorized

## list list or watch objects of kind Deployment

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/deployments

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([DeploymentList](#)): OK

401: Unauthorized

## list or watch objects of kind Deployment

### HTTP Request

GET /apis/apps/v1/deployments

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([DeploymentList](#)): OK

401: Unauthorized

## create create a Deployment

### HTTP Request

POST /apis/apps/v1/namespaces/{namespace}/deployments

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Deployment](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Deployment](#)): OK

201 ([Deployment](#)): Created

202 ([Deployment](#)): Accepted

401: Unauthorized

## update replace the specified Deployment

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/deployments/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Deployment
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Deployment](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Deployment](#)): OK

201 ([Deployment](#)): Created

401: Unauthorized

## update replace status of the specified Deployment

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/deployments/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Deployment
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Deployment](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Deployment](#)): OK

201 ([Deployment](#)): Created

401: Unauthorized

## patch partially update the specified Deployment

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/deployments/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Deployment
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Deployment](#)): OK

201 ([Deployment](#)): Created

401: Unauthorized

## patch partially update status of the specified Deployment

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/deployments/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Deployment
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Deployment](#)): OK

201 ([Deployment](#)): Created

401: Unauthorized

## delete delete a Deployment

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/deployments/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Deployment
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Deployment

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/deployments

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)



- **body:** [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# StatefulSet

StatefulSet represents a set of pods with consistent identities.

apiVersion: apps/v1

import "k8s.io/api/apps/v1"

## StatefulSet

StatefulSet represents a set of pods with consistent identities. Identities are defined as:

- Network: A single stable DNS and hostname.
- Storage: As many VolumeClaims as requested.

The StatefulSet guarantees that a given network identity will always map to the same storage identity.

- 
- **apiVersion:** apps/v1
  - **kind:** StatefulSet
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([StatefulSetSpec](#))

Spec defines the desired identities of pods in this set.

- **status** ([StatefulSetStatus](#))

Status is the current status of Pods in this StatefulSet. This data may be out of date by some window of time.

## StatefulSetSpec

A StatefulSetSpec is the specification of a StatefulSet.

- 
- **serviceName** (string), required

serviceName is the name of the service that governs this StatefulSet. This service must exist before the StatefulSet, and is responsible for the network identity of the set. Pods get DNS/hostnames that follow the pattern: pod-specific-string.serviceName.default.svc.cluster.local where "pod-specific-string" is managed by the StatefulSet controller.

- **selector** ([LabelSelector](#)), required

selector is a label query over pods that should match the replica count. It must match the pod template's labels. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/#label-selectors>

- **template** ([PodTemplateSpec](#)), required

template is the object that describes the pod that will be created if insufficient replicas are detected. Each pod stamped out by the StatefulSet will fulfill this Template, but have a unique identity from the rest of the StatefulSet. Each pod will be named with the format <statefulsetname>-<podindex>. For example, a pod in a StatefulSet named "web" with index number "3" would be named "web-3". The only allowed template.spec.restartPolicy value is "Always".

- **replicas** (int32)

replicas is the desired number of replicas of the given Template. These are replicas in the sense that they are instantiations of the same Template, but individual replicas also have a consistent identity. If unspecified, defaults to 1.

- **updateStrategy** (StatefulSetUpdateStrategy)

updateStrategy indicates the StatefulSetUpdateStrategy that will be employed to update Pods in the StatefulSet when a revision is made to Template.

*StatefulSetUpdateStrategy indicates the strategy that the StatefulSet controller will use to perform updates. It includes any additional parameters necessary to perform the update for the indicated strategy.*

- **updateStrategy.type** (string)

Type indicates the type of the StatefulSetUpdateStrategy. Default is RollingUpdate.

- **updateStrategy.rollingUpdate** (RollingUpdateStatefulSetStrategy)

RollingUpdate is used to communicate parameters when Type is RollingUpdateStatefulSetStrategyType.

*RollingUpdateStatefulSetStrategy is used to communicate parameter for RollingUpdateStatefulSetStrategyType.*

- **updateStrategy.rollingUpdate.maxUnavailable** (IntOrString)

The maximum number of pods that can be unavailable during the update. Value can be an absolute number (ex: 5) or a percentage of desired pods (ex: 10%). Absolute number is calculated from percentage by rounding up. This can not be 0. Defaults to 1. This field is alpha-level and is only honored by servers that enable the MaxUnavailableStatefulSet feature. The field applies to all pods in the range 0 to Replicas-1. That means if there is any unavailable pod in the range 0 to Replicas-1, it will be counted towards MaxUnavailable.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

▪ **updateStrategy.rollingUpdate.partition** (int32)

Partition indicates the ordinal at which the StatefulSet should be partitioned for updates. During a rolling update, all pods from ordinal Replicas-1 to Partition are updated. All pods from ordinal Partition-1 to 0 remain untouched. This is helpful in being able to do a canary based deployment. The default value is 0.

• **podManagementPolicy** (string)

podManagementPolicy controls how pods are created during initial scale up, when replacing pods on nodes, or when scaling down. The default policy is OrderedReady, where pods are created in increasing order (pod-0, then pod-1, etc) and the controller will wait until each pod is ready before continuing. When scaling down, the pods are removed in the opposite order. The alternative policy is Parallel which will create pods in parallel to match the desired scale without waiting, and on scale down will delete all pods at once.

• **revisionHistoryLimit** (int32)

revisionHistoryLimit is the maximum number of revisions that will be maintained in the StatefulSet's revision history. The revision history consists of all revisions not represented by a currently applied StatefulSetSpec version. The default value is 10.

• **volumeClaimTemplates** ([][PersistentVolumeClaim](#))

volumeClaimTemplates is a list of claims that pods are allowed to reference. The StatefulSet controller is responsible for mapping network identities to claims in a way that maintains the identity of a pod. Every claim in this list must have at least one matching (by name) volumeMount in one container in the template. A claim in this list takes precedence over any volumes in the template, with the same name.

• **minReadySeconds** (int32)

Minimum number of seconds for which a newly created pod should be ready without any of its container crashing for it to be considered available. Defaults to 0 (pod will be considered available as soon as it is ready)

• **persistentVolumeClaimRetentionPolicy**  
(StatefulSetPersistentVolumeClaimRetentionPolicy)

persistentVolumeClaimRetentionPolicy describes the lifecycle of persistent volume claims created from volumeClaimTemplates. By default, all persistent volume claims are created as needed and retained until manually deleted. This policy allows the lifecycle to be altered, for example by deleting persistent volume claims when their stateful set is deleted, or when their pod is scaled down. This requires the StatefulSetAutoDeletePVC feature gate to be enabled, which is alpha. +optional

*StatefulSetPersistentVolumeClaimRetentionPolicy describes the policy used for PVCs created from the StatefulSet VolumeClaimTemplates.*

◦ **persistentVolumeClaimRetentionPolicy.whenDeleted** (string)

WhenDeleted specifies what happens to PVCs created from StatefulSet VolumeClaimTemplates when the StatefulSet is deleted. The default policy of Retain

causes PVCs to not be affected by StatefulSet deletion. The Delete policy causes those PVCs to be deleted.

- **persistentVolumeClaimRetentionPolicy.whenScaled** (string)

WhenScaled specifies what happens to PVCs created from StatefulSet VolumeClaimTemplates when the StatefulSet is scaled down. The default policy of Retain causes PVCs to not be affected by a scaledown. The Delete policy causes the associated PVCs for any excess pods above the replica count to be deleted.

- **ordinals** (StatefulSetOrdinals)

ordinals controls the numbering of replica indices in a StatefulSet. The default ordinals behavior assigns a "0" index to the first replica and increments the index by one for each additional replica requested. Using the ordinals field requires the StatefulSetStartOrdinal feature gate to be enabled, which is beta.

*StatefulSetOrdinals describes the policy used for replica ordinal assignment in this StatefulSet.*

- **ordinals.start** (int32)

start is the number representing the first replica's index. It may be used to number replicas from an alternate index (eg: 1-indexed) over the default 0-indexed names, or to orchestrate progressive movement of replicas from one StatefulSet to another. If set, replica indices will be in the range: [.spec.ordinals.start, .spec.ordinals.start + .spec.replicas). If unset, defaults to 0. Replica indices will be in the range: [0, .spec.replicas).

## StatefulSetStatus

StatefulSetStatus represents the current state of a StatefulSet.

---

- **replicas** (int32), required

replicas is the number of Pods created by the StatefulSet controller.

- **readyReplicas** (int32)

readyReplicas is the number of pods created for this StatefulSet with a Ready Condition.

- **currentReplicas** (int32)

currentReplicas is the number of Pods created by the StatefulSet controller from the StatefulSet version indicated by currentRevision.

- **updatedReplicas** (int32)

updatedReplicas is the number of Pods created by the StatefulSet controller from the StatefulSet version indicated by updateRevision.

- **availableReplicas** (int32)

Total number of available pods (ready for at least minReadySeconds) targeted by this statefulset.

- **collisionCount** (int32)

collisionCount is the count of hash collisions for the StatefulSet. The StatefulSet controller uses this field as a collision avoidance mechanism when it needs to create the name for the newest ControllerRevision.

- **conditions** ([]StatefulSetCondition)

*Patch strategy: merge on key type*

Represents the latest available observations of a statefulset's current state.

*StatefulSetCondition describes the state of a statefulset at a certain point.*

- **conditions.status** (string), required

Status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

Type of statefulset condition.

- **conditions.lastTransitionTime** (Time)

Last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

A human readable message indicating details about the transition.

- **conditions.reason** (string)

The reason for the condition's last transition.

- **currentRevision** (string)

currentRevision, if not empty, indicates the version of the StatefulSet used to generate Pods in the sequence [0,currentReplicas).

- **updateRevision** (string)

updateRevision, if not empty, indicates the version of the StatefulSet used to generate Pods in the sequence [replicas-updatedReplicas,replicas)

- **observedGeneration** (int64)

observedGeneration is the most recent generation observed for this StatefulSet. It corresponds to the StatefulSet's generation, which is updated on mutation by the API Server.

# StatefulSetList

StatefulSetList is a collection of StatefulSets.

---

- **apiVersion:** apps/v1
- **kind:** StatefulSetList
- **metadata** ([ListMeta](#))

Standard list's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]StatefulSet](#)), required

Items is the list of stateful sets.

## Operations

---

### get read the specified StatefulSet

#### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/statefulsets/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the StatefulSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([StatefulSet](#)): OK

401: Unauthorized

## get read status of the specified StatefulSet

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/statefulsets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the StatefulSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([StatefulSet](#)): OK

401: Unauthorized

## list list or watch objects of kind StatefulSet

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/statefulsets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)



- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([StatefulSetList](#)): OK

401: Unauthorized

## list list or watch objects of kind StatefulSet

### HTTP Request

GET /apis/apps/v1/statefulsets

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)

- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([StatefulSetList](#)): OK

401: Unauthorized

## create create a StatefulSet

### HTTP Request

POST /apis/apps/v1/namespaces/{namespace}/statefulsets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [StatefulSet](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([StatefulSet](#)): OK

201 ([StatefulSet](#)): Created

202 ([StatefulSet](#)): Accepted

401: Unauthorized

## update replace the specified StatefulSet

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/statefulsets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the StatefulSet
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [StatefulSet](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([StatefulSet](#)): OK

201 ([StatefulSet](#)): Created

401: Unauthorized

## update replace status of the specified StatefulSet

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/statefulsets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the StatefulSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [StatefulSet](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([StatefulSet](#)): OK

201 ([StatefulSet](#)): Created

401: Unauthorized

## patch partially update the specified StatefulSet

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/statefulsets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the StatefulSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([StatefulSet](#)): OK

201 ([StatefulSet](#)): Created

401: Unauthorized

## patch partially update status of the specified StatefulSet

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/statefulsets/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the StatefulSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([StatefulSet](#)): OK

201 ([StatefulSet](#)): Created

401: Unauthorized

## delete delete a StatefulSet

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/statefulsets/{name}

## Parameters

- **name** (*in path*): string, required  
name of the StatefulSet
- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [DeleteOptions](#)
- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## **deletecollection** delete collection of StatefulSet

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/statefulsets

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [DeleteOptions](#)

- **continue** (*in query*): string

[continue](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ControllerRevision

ControllerRevision implements an immutable snapshot of state data.

apiVersion: apps/v1

import "k8s.io/api/apps/v1"



# ControllerRevision

ControllerRevision implements an immutable snapshot of state data. Clients are responsible for serializing and deserializing the objects that contain their internal state. Once a ControllerRevision has been successfully created, it can not be updated. The API Server will fail validation of all requests that attempt to mutate the Data field. ControllerRevisions may, however, be deleted. Note that, due to its use by both the DaemonSet and StatefulSet controllers for update and rollback, this object is beta. However, it may be subject to name and representation changes in future releases, and clients should not depend on its stability. It is primarily for internal use by controllers.

---

- **apiVersion:** apps/v1
- **kind:** ControllerRevision
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **revision** (int64), required

Revision indicates the revision of the state represented by Data.

- **data** (RawExtension)

Data is the serialized representation of the state.

\*RawExtension is used to hold extensions in external versions.

To use this, make a field which has RawExtension as its type in your external, versioned struct, and Object in your internal struct. You also need to register your various plugin types.

// Internal package:

```
type MyAPIObject struct { runtime.TypeMeta json:",inline" MyPlugin runtime.Object
json:"myPlugin" }
```

```
type PluginA struct { AOption string json:"aOption" }
```

// External package:

```
type MyAPIObject struct { runtime.TypeMeta json:",inline" MyPlugin
runtime.RawExtension json:"myPlugin" }
```

```
type PluginA struct { AOption string json:"aOption" }
```

// On the wire, the JSON will look something like this:

```
{ "kind": "MyAPIObject", "apiVersion": "v1", "myPlugin": { "kind": "PluginA", "aOption": "foo",
}, }
```

So what happens? Decode first uses json or yaml to unmarshal the serialized data into your external MyAPIObject. That causes the raw JSON to be stored, but not unpacked. The next step is to copy (using pkg/conversion) into the internal struct. The runtime package's DefaultScheme has conversion functions installed which will unpack the JSON stored in RawExtension, turning it into the correct object type, and storing it in the Object. (TODO: In the case where the object is of an unknown type, a runtime.Unknown object will be created and stored.)\*

## ControllerRevisionList

ControllerRevisionList is a resource containing a list of ControllerRevision objects.

---

- **apiVersion:** apps/v1
- **kind:** ControllerRevisionList
- **metadata** ([ListMeta](#))

More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]ControllerRevision](#)), required

Items is the list of ControllerRevisions

## Operations

---

### get read the specified ControllerRevision

#### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/controllerrevisions/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ControllerRevision
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ControllerRevision](#)): OK

401: Unauthorized

## list list or watch objects of kind ControllerRevision

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/controllerrevisions

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ControllerRevisionList](#)): OK

401: Unauthorized

## list list or watch objects of kind ControllerRevision

## HTTP Request

GET /apis/apps/v1/controllerrevisions

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ControllerRevisionList](#)): OK

401: Unauthorized

## create create a ControllerRevision

### HTTP Request

POST /apis/apps/v1/namespaces/{namespace}/controllerrevisions

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [ControllerRevision](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ControllerRevision](#)): OK

201 ([ControllerRevision](#)): Created

202 ([ControllerRevision](#)): Accepted

401: Unauthorized

## update replace the specified ControllerRevision

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/controllerrevisions/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ControllerRevision
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ControllerRevision](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ControllerRevision](#)): OK

201 ([ControllerRevision](#)): Created

401: Unauthorized

## patch partially update the specified ControllerRevision

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/controllerrevisions/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ControllerRevision

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ControllerRevision](#)): OK

201 ([ControllerRevision](#)): Created

401: Unauthorized

## delete delete a ControllerRevision

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/controllerrevisions/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ControllerRevision
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ControllerRevision

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/controllerrevisions

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **continue** (*in query*): string

[continue](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string



[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# DaemonSet

DaemonSet represents the configuration of a daemon set.

apiVersion: apps/v1

import "k8s.io/api/apps/v1"

## DaemonSet

DaemonSet represents the configuration of a daemon set.

---

- **apiVersion**: apps/v1
- **kind**: DaemonSet

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([DaemonSetSpec](#))

The desired behavior of this daemon set. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([DaemonSetStatus](#))

The current status of this daemon set. This data may be out of date by some window of time. Populated by the system. Read-only. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## DaemonSetSpec

DaemonSetSpec is the specification of a daemon set.

---

- **selector** ([LabelSelector](#)), required

A label query over pods that are managed by the daemon set. Must match in order to be controlled. It must match the pod template's labels. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/#label-selectors>

- **template** ([PodTemplateSpec](#)), required

An object that describes the pod that will be created. The DaemonSet will create exactly one copy of this pod on every node that matches the template's node selector (or on every node if no node selector is specified). The only allowed template.spec.restartPolicy value is "Always". More info: <https://kubernetes.io/docs/concepts/workloads/controllers/replicationcontroller#pod-template>

- **minReadySeconds** (int32)

The minimum number of seconds for which a newly created DaemonSet pod should be ready without any of its container crashing, for it to be considered available. Defaults to 0 (pod will be considered available as soon as it is ready).

- **updateStrategy** ([DaemonSetUpdateStrategy](#))

An update strategy to replace existing DaemonSet pods with new pods.

*DaemonSetUpdateStrategy is a struct used to control the update strategy for a DaemonSet.*

- **updateStrategy.type** (string)

Type of daemon set update. Can be "RollingUpdate" or "OnDelete". Default is RollingUpdate.

- **updateStrategy.rollingUpdate** ([RollingUpdateDaemonSet](#))

Rolling update config params. Present only if type = "RollingUpdate".

*Spec to control the desired behavior of daemon set rolling update.*

- **updateStrategy.rollingUpdate.maxSurge** (IntOrString)

The maximum number of nodes with an existing available DaemonSet pod that can have an updated DaemonSet pod during during an update. Value can be an absolute number (ex: 5) or a percentage of desired pods (ex: 10%). This can not be 0 if MaxUnavailable is 0. Absolute number is calculated from percentage by rounding up to a minimum of 1. Default value is 0. Example: when this is set to 30%, at most 30% of the total number of nodes that should be running the daemon pod (i.e. status.desiredNumberScheduled) can have their a new pod created before the old pod is marked as deleted. The update starts by launching new pods on 30% of nodes. Once an updated pod is available (Ready for at least minReadySeconds) the old DaemonSet pod on that node is marked deleted. If the old pod becomes unavailable for any reason (Ready transitions to false, is evicted, or is drained) an updated pod is immediately created on that node without considering surge limits. Allowing surge implies the possibility that the resources consumed by the daemonset on any given node can double if the readiness check fails, and so resource intensive daemonsets should take into account that they may cause evictions during disruption.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **updateStrategy.rollingUpdate.maxUnavailable** (IntOrString)

The maximum number of DaemonSet pods that can be unavailable during the update. Value can be an absolute number (ex: 5) or a percentage of total number of DaemonSet pods at the start of the update (ex: 10%). Absolute number is calculated from percentage by rounding up. This cannot be 0 if MaxSurge is 0 Default value is 1. Example: when this is set to 30%, at most 30% of the total number of nodes that should be running the daemon pod (i.e. status.desiredNumberScheduled) can have their pods stopped for an update at any given time. The update starts by stopping at most 30% of those DaemonSet pods and then brings up new DaemonSet pods in their place. Once the new pods are available, it then proceeds onto other DaemonSet pods, thus ensuring that at least 70% of original number of DaemonSet pods are available at all times during the update.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **revisionHistoryLimit** (int32)

The number of old history to retain to allow rollback. This is a pointer to distinguish between explicit zero and not specified. Defaults to 10.

# DaemonSetStatus

DaemonSetStatus represents the current status of a daemon set.

---

- **numberReady** (int32), required

numberReady is the number of nodes that should be running the daemon pod and have one or more of the daemon pod running with a Ready Condition.

- **numberAvailable** (int32)

The number of nodes that should be running the daemon pod and have one or more of the daemon pod running and available (ready for at least spec.minReadySeconds)

- **numberUnavailable** (int32)

The number of nodes that should be running the daemon pod and have none of the daemon pod running and available (ready for at least spec.minReadySeconds)

- **numberMisscheduled** (int32), required

The number of nodes that are running the daemon pod, but are not supposed to run the daemon pod. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/daemonset/>

- **desiredNumberScheduled** (int32), required

The total number of nodes that should be running the daemon pod (including nodes correctly running the daemon pod). More info: <https://kubernetes.io/docs/concepts/workloads/controllers/daemonset/>

- **currentNumberScheduled** (int32), required

The number of nodes that are running at least 1 daemon pod and are supposed to run the daemon pod. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/daemonset/>

- **updatedNumberScheduled** (int32)

The total number of nodes that are running updated daemon pod

- **collisionCount** (int32)

Count of hash collisions for the DaemonSet. The DaemonSet controller uses this field as a collision avoidance mechanism when it needs to create the name for the newest ControllerRevision.

- **conditions** ([]DaemonSetCondition)

*Patch strategy: merge on key type*

Represents the latest available observations of a DaemonSet's current state.

*DaemonSetCondition describes the state of a DaemonSet at a certain point.*

- **conditions.status** (string), required

Status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

Type of DaemonSet condition.

- **conditions.lastTransitionTime** (Time)

Last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

A human readable message indicating details about the transition.

- **conditions.reason** (string)

The reason for the condition's last transition.

- **observedGeneration** (int64)

The most recent generation observed by the daemon set controller.

## DaemonSetList

DaemonSetList is a collection of daemon sets.

- 
- **apiVersion**: apps/v1

- **kind**: DaemonSetList

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]DaemonSet](#)), required

A list of daemon sets.

## Operations

---

## get read the specified DaemonSet

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/daemonsets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the DaemonSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([DaemonSet](#)): OK

401: Unauthorized

## get read status of the specified DaemonSet

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/daemonsets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the DaemonSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([DaemonSet](#)): OK

401: Unauthorized

## list list or watch objects of kind DaemonSet

### HTTP Request

GET /apis/apps/v1/namespaces/{namespace}/daemonsets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([DaemonSetList](#)): OK

401: Unauthorized

## list list or watch objects of kind DaemonSet

### HTTP Request

GET /apis/apps/v1/daemonsets

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)



- **watch** (*in query*): boolean

- [watch](#)

## Response

200 ([DaemonSetList](#)): OK

401: Unauthorized

## create create a DaemonSet

### HTTP Request

POST /apis/apps/v1/namespaces/{namespace}/daemonsets

### Parameters

- **namespace** (*in path*): string, required

- [namespace](#)

- **body**: [DaemonSet](#), required

- **dryRun** (*in query*): string

- [dryRun](#)

- **fieldManager** (*in query*): string

- [fieldManager](#)

- **fieldValidation** (*in query*): string

- [fieldValidation](#)

- **pretty** (*in query*): string

- [pretty](#)

## Response

200 ([DaemonSet](#)): OK

201 ([DaemonSet](#)): Created

202 ([DaemonSet](#)): Accepted

401: Unauthorized

## update replace the specified DaemonSet

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/daemonsets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the DaemonSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DaemonSet](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([DaemonSet](#)): OK

201 ([DaemonSet](#)): Created

401: Unauthorized

## update replace status of the specified DaemonSet

### HTTP Request

PUT /apis/apps/v1/namespaces/{namespace}/daemonsets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the DaemonSet

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DaemonSet](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([DaemonSet](#)): OK

201 ([DaemonSet](#)): Created

401: Unauthorized

## patch partially update the specified DaemonSet

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/daemonsets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the DaemonSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([DaemonSet](#)): OK

201 ([DaemonSet](#)): Created

401: Unauthorized

## patch partially update status of the specified DaemonSet

### HTTP Request

PATCH /apis/apps/v1/namespaces/{namespace}/daemonsets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the DaemonSet
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([DaemonSet](#)): OK

201 ([DaemonSet](#)): Created

401: Unauthorized

## delete delete a DaemonSet

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/daemonsets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the DaemonSet
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of DaemonSet

### HTTP Request

DELETE /apis/apps/v1/namespaces/{namespace}/daemonsets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Job

Job represents the configuration of a single job.

apiVersion: batch/v1

import "k8s.io/api/batch/v1"

# Job

Job represents the configuration of a single job.

- 
- **apiVersion**: batch/v1
  - **kind**: Job
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([JobSpec](#))

Specification of the desired behavior of a job. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([JobStatus](#))

Current status of a job. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## JobSpec

JobSpec describes how the job execution will look like.

---

## Replicas

- **template** ([PodTemplateSpec](#)), required

Describes the pod that will be created when executing a job. The only allowed `template.spec.restartPolicy` values are "Never" or "OnFailure". More info: <https://kubernetes.io/docs/concepts/workloads/controllers/jobs-run-to-completion/>

- **parallelism** (int32)

Specifies the maximum desired number of pods the job should run at any given time. The actual number of pods running in steady state will be less than this number when  $((\text{.spec.completions} - \text{.status.successful}) < \text{.spec.parallelism})$ , i.e. when the work left to do is less than max parallelism. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/jobs-run-to-completion/>

## Lifecycle

- **completions** (int32)

Specifies the desired number of successfully finished pods the job should be run with. Setting to null means that the success of any pod signals the success of all pods, and allows parallelism to have any positive value. Setting to 1 means that parallelism is limited to 1 and the success of that pod signals the success of the job. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/jobs-run-to-completion/>

- **completionMode** (string)

`completionMode` specifies how Pod completions are tracked. It can be `NonIndexed` (default) or `Indexed`.

`NonIndexed` means that the Job is considered complete when there have been `.spec.completions` successfully completed Pods. Each Pod completion is homologous to each other.

`Indexed` means that the Pods of a Job get an associated completion index from 0 to  $(\text{.spec.completions} - 1)$ , available in the annotation `batch.kubernetes.io/job-completion-index`. The Job is considered complete when there is one successfully completed Pod for each index. When value is `Indexed`, `.spec.completions` must be specified and `.spec.parallelism` must be less than or equal to  $10^5$ . In addition, The Pod name takes the form `$(job-name)-$(index)-$(random-string)`, the Pod hostname takes the form `$(job-name)-$(index)`.

More completion modes can be added in the future. If the Job controller observes a mode that it doesn't recognize, which is possible during upgrades due to version skew, the controller skips updates for the Job.

- **backoffLimit** (int32)

Specifies the number of retries before marking this job failed. Defaults to 6

- **activeDeadlineSeconds** (int64)



Specifies the duration in seconds relative to the start`Time` that the job may be continuously active before the system tries to terminate it; value must be positive integer. If a Job is suspended (at creation or through an update), this timer will effectively be stopped and reset when the Job is resumed again.

- **ttlSecondsAfterFinished** (int32)

ttlSecondsAfterFinished limits the lifetime of a Job that has finished execution (either Complete or Failed). If this field is set, ttlSecondsAfterFinished after the Job finishes, it is eligible to be automatically deleted. When the Job is being deleted, its lifecycle guarantees (e.g. finalizers) will be honored. If this field is unset, the Job won't be automatically deleted. If this field is set to zero, the Job becomes eligible to be deleted immediately after it finishes.

- **suspend** (boolean)

suspend specifies whether the Job controller should create Pods or not. If a Job is created with suspend set to true, no Pods are created by the Job controller. If a Job is suspended after creation (i.e. the flag goes from false to true), the Job controller will delete all active Pods associated with this Job. Users must design their workload to gracefully handle this. Suspending a Job will reset the Start`Time` field of the Job, effectively resetting the ActiveDeadlineSeconds timer too. Defaults to false.

## Selector

- **selector** ([LabelSelector](#))

A label query over pods that should match the pod count. Normally, the system sets this field for you. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/#label-selectors>

- **manualSelector** (boolean)

manualSelector controls generation of pod labels and pod selectors. Leave manualSelector unset unless you are certain what you are doing. When false or unset, the system pick labels unique to this job and appends those labels to the pod template. When true, the user is responsible for picking unique labels and specifying the selector. Failure to pick a unique label may cause this and other jobs to not function correctly. However, You may see manualSelector=true in jobs that were created with the old extensions/v1beta1 API. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/jobs-run-to-completion/#specifying-your-own-pod-selector>

## Beta level

- **podFailurePolicy** (PodFailurePolicy)

Specifies the policy of handling failed pods. In particular, it allows to specify the set of actions and conditions which need to be satisfied to take the associated action. If empty, the default behaviour applies - the counter of failed pods, represented by the jobs's .status.failed field, is incremented and it is checked against the backoffLimit. This field cannot be used in combination with restartPolicy=OnFailure.

This field is beta-level. It can be used when the JobPodFailurePolicy feature gate is enabled (enabled by default).

*PodFailurePolicy describes how failed pods influence the backoffLimit.*

- **podFailurePolicy.rules** ([]PodFailurePolicyRule), required

*Atomic: will be replaced during a merge*

A list of pod failure policy rules. The rules are evaluated in order. Once a rule matches a Pod failure, the remaining of the rules are ignored. When no rule matches the Pod failure, the default handling applies - the counter of pod failures is incremented and it is checked against the backoffLimit. At most 20 elements are allowed.

*PodFailurePolicyRule describes how a pod failure is handled when the requirements are met. One of onExitCodes and onPodConditions, but not both, can be used in each rule.*

- **podFailurePolicy.rules.action** (string), required

Specifies the action taken on a pod failure when the requirements are satisfied. Possible values are:

- FailJob: indicates that the pod's job is marked as Failed and all running pods are terminated.
- FailIndex: indicates that the pod's index is marked as Failed and will not be restarted. This value is alpha-level. It can be used when the JobBackoffLimitPerIndex feature gate is enabled (disabled by default).
- Ignore: indicates that the counter towards the .backoffLimit is not incremented and a replacement pod is created.
- Count: indicates that the pod is handled in the default way - the counter towards the .backoffLimit is incremented. Additional values are considered to be added in the future. Clients should react to an unknown action by skipping the rule.

- **podFailurePolicy.rules.onPodConditions** ([]PodFailurePolicyOnPodConditionsPattern), required

*Atomic: will be replaced during a merge*

Represents the requirement on the pod conditions. The requirement is represented as a list of pod condition patterns. The requirement is satisfied if at least one pattern matches an actual pod condition. At most 20 elements are allowed.

*PodFailurePolicyOnPodConditionsPattern describes a pattern for matching an actual pod condition type.*

- **podFailurePolicy.rules.onPodConditions.status** (string), required

Specifies the required Pod condition status. To match a pod condition it is required that the specified status equals the pod condition status. Defaults to True.

- **podFailurePolicy.rules.onPodConditions.type** (string), required  
Specifies the required Pod condition type. To match a pod condition it is required that specified type equals the pod condition type.
- **podFailurePolicy.rules.onExitCodes**  
(PodFailurePolicyOnExitCodesRequirement)

Represents the requirement on the container exit codes.

*PodFailurePolicyOnExitCodesRequirement describes the requirement for handling a failed pod based on its container exit codes. In particular, it lookups the .state.terminated.exitCode for each app container and init container status, represented by the .status.containerStatuses and .status.initContainerStatuses fields in the Pod status, respectively. Containers completed with success (exit code 0) are excluded from the requirement check.*

- **podFailurePolicy.rules.onExitCodes.operator** (string), required

Represents the relationship between the container exit code(s) and the specified values. Containers completed with success (exit code 0) are excluded from the requirement check. Possible values are:

- In: the requirement is satisfied if at least one container exit code (might be multiple if there are multiple containers not restricted by the 'containerName' field) is in the set of specified values.
- NotIn: the requirement is satisfied if at least one container exit code (might be multiple if there are multiple containers not restricted by the 'containerName' field) is not in the set of specified values. Additional values are considered to be added in the future. Clients should react to an unknown operator by assuming the requirement is not satisfied.

- **podFailurePolicy.rules.onExitCodes.values** ([]int32), required

*Set: unique values will be kept during a merge*

Specifies the set of values. Each returned container exit code (might be multiple in case of multiple containers) is checked against this set of values with respect to the operator. The list of values must be ordered and must not contain duplicates. Value '0' cannot be used for the In operator. At least one element is required. At most 255 elements are allowed.

- **podFailurePolicy.rules.onExitCodes.containerName** (string)

Restricts the check for exit codes to the container with the specified name. When null, the rule applies to all containers. When specified, it should match one the container or initContainer names in the pod template.

## Alpha level

- **backoffLimitPerIndex** (int32)

Specifies the limit for the number of retries within an index before marking this index as failed. When enabled the number of failures per index is kept in the pod's `batch.kubernetes.io/job-index-failure-count` annotation. It can only be set when Job's `completionMode=Indexed`, and the Pod's restart policy is `Never`. The field is immutable. This field is alpha-level. It can be used when the `JobBackoffLimitPerIndex` feature gate is enabled (disabled by default).

- **maxFailedIndexes** (int32)

Specifies the maximal number of failed indexes before marking the Job as failed, when `backoffLimitPerIndex` is set. Once the number of failed indexes exceeds this number the entire Job is marked as Failed and its execution is terminated. When left as null the job continues execution of all of its indexes and is marked with the Complete Job condition. It can only be specified when `backoffLimitPerIndex` is set. It can be null or up to completions. It is required and must be less than or equal to  $10^4$  when is completions greater than  $10^5$ . This field is alpha-level. It can be used when the `JobBackoffLimitPerIndex` feature gate is enabled (disabled by default).

- **podReplacementPolicy** (string)

`podReplacementPolicy` specifies when to create replacement Pods. Possible values are: - `TerminatingOrFailed` means that we recreate pods when they are terminating (has a `metadata.deletionTimestamp`) or failed.

- Failed means to wait until a previously created Pod is fully terminated (has phase Failed or Succeeded) before creating a replacement Pod.

When using `podFailurePolicy`, Failed is the the only allowed value. `TerminatingOrFailed` and Failed are allowed values when `podFailurePolicy` is not in use. This is an alpha field. Enable `JobPodReplacementPolicy` to be able to use this field.

## JobStatus

JobStatus represents the current state of a Job.

---

- **startTime** (Time)

Represents time when the job controller started processing a job. When a Job is created in the suspended state, this field is not set until the first time it is resumed. This field is reset every time a Job is resumed from suspension. It is represented in RFC3339 form and is in UTC.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **completionTime** (Time)

Represents time when the job was completed. It is not guaranteed to be set in happens-before order across separate operations. It is represented in RFC3339 form and is in UTC. The completion time is only set when the job finishes successfully.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

**active** (int32)

- The number of pending and running pods.

• **failed** (int32)

The number of pods which reached phase Failed.

• **succeeded** (int32)

The number of pods which reached phase Succeeded.

• **completedIndexes** (string)

completedIndexes holds the completed indexes when .spec.completionMode = "Indexed" in a text format. The indexes are represented as decimal integers separated by commas. The numbers are listed in increasing order. Three or more consecutive numbers are compressed and represented by the first and last element of the series, separated by a hyphen. For example, if the completed indexes are 1, 3, 4, 5 and 7, they are represented as "1,3-5,7".

• **conditions** ([]JobCondition)

*Patch strategy: merge on key type*

*Atomic: will be replaced during a merge*

The latest available observations of an object's current state. When a Job fails, one of the conditions will have type "Failed" and status true. When a Job is suspended, one of the conditions will have type "Suspended" and status true; when the Job is resumed, the status of this condition will become false. When a Job is completed, one of the conditions will have type "Complete" and status true. More info: <https://kubernetes.io/docs/concepts/workloads/controllers/jobs-run-to-completion/>

*JobCondition describes current state of a job.*

- **conditions.status** (string), required

Status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

Type of job condition, Complete or Failed.

- **conditions.lastProbeTime** (Time)

Last time the condition was checked.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.lastTransitionTime** (Time)

Last time the condition transit from one status to another.

*Time is a wrapper around time. Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

Human readable message indicating details about last transition.

- **conditions.reason** (string)

(brief) reason for the condition's last transition.

- **uncountedTerminatedPods** (UncountedTerminatedPods)

uncountedTerminatedPods holds the UIDs of Pods that have terminated but the job controller hasn't yet accounted for in the status counters.

The job controller creates pods with a finalizer. When a pod terminates (succeeded or failed), the controller does three steps to account for it in the job status:

1. Add the pod UID to the arrays in this field.
2. Remove the pod finalizer.
3. Remove the pod UID from the arrays while increasing the corresponding counter.

Old jobs might not be tracked using this field, in which case the field remains null.

*UncountedTerminatedPods holds UIDs of Pods that have terminated but haven't been accounted in Job status counters.*

- **uncountedTerminatedPods.failed** ([]string)

*Set: unique values will be kept during a merge*

failed holds UIDs of failed Pods.

- **uncountedTerminatedPods.succeeded** ([]string)

*Set: unique values will be kept during a merge*

succeeded holds UIDs of succeeded Pods.

## Beta level

- **ready** (int32)

The number of pods which have a Ready condition.

This field is beta-level. The job controller populates the field when the feature gate JobReadyPods is enabled (enabled by default).

## Alpha level

- **failedIndexes** (string)

FailedIndexes holds the failed indexes when backoffLimitPerIndex=true. The indexes are represented in the text format analogous as for the completedIndexes field, ie. they are

kept as decimal integers separated by commas. The numbers are listed in increasing order. Three or more consecutive numbers are compressed and represented by the first and last element of the series, separated by a hyphen. For example, if the failed indexes are 1, 3, 4, 5 and 7, they are represented as "1,3-5,7". This field is alpha-level. It can be used when the `JobBackoffLimitPerIndex` feature gate is enabled (disabled by default).

- **terminating** (int32)

The number of pods which are terminating (in phase Pending or Running and have a `deletionTimestamp`).

This field is alpha-level. The job controller populates the field when the feature gate `JobPodReplacementPolicy` is enabled (disabled by default).

## JobList

JobList is a collection of jobs.

---

- **apiVersion**: batch/v1

- **kind**: JobList

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]Job](#)), required

items is the list of Jobs.

## Operations

---

### get read the specified Job

#### HTTP Request

GET /apis/batch/v1/namespaces/{namespace}/jobs/{name}

#### Parameters

- **name** (*in path*): string, required

name of the Job

- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Job](#)): OK

401: Unauthorized

## get read status of the specified Job

### HTTP Request

GET /apis/batch/v1/namespaces/{namespace}/jobs/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Job
- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Job](#)): OK

401: Unauthorized

## list list or watch objects of kind Job

### HTTP Request

GET /apis/batch/v1/namespaces/{namespace}/jobs

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)

- **continue** (*in query*): string



[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([JobList](#)): OK

401: Unauthorized

## list list or watch objects of kind Job

## HTTP Request

GET /apis/batch/v1/jobs

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([JobList](#)): OK

401: Unauthorized

## create create a Job

## HTTP Request

POST /apis/batch/v1/namespaces/{namespace}/jobs

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Job](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Job](#)): OK

201 ([Job](#)): Created

202 ([Job](#)): Accepted

401: Unauthorized

## update replace the specified Job

### HTTP Request

PUT /apis/batch/v1/namespaces/{namespace}/jobs/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Job
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Job](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Job](#)): OK

201 ([Job](#)): Created

401: Unauthorized

## update replace status of the specified Job

### HTTP Request

PUT /apis/batch/v1/namespaces/{namespace}/jobs/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Job
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Job](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Job](#)): OK

201 ([Job](#)): Created

401: Unauthorized

## patch partially update the specified Job

### HTTP Request

PATCH /apis/batch/v1/namespaces/{namespace}/jobs/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Job
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Job](#)): OK

201 ([Job](#)): Created

401: Unauthorized

## patch partially update status of the specified Job

### HTTP Request

PATCH /apis/batch/v1/namespaces/{namespace}/jobs/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Job
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Job](#)): OK

201 ([Job](#)): Created

401: Unauthorized

## delete delete a Job

### HTTP Request

DELETE /apis/batch/v1/namespaces/{namespace}/jobs/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Job
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Job

### HTTP Request

DELETE /apis/batch/v1/namespaces/{namespace}/jobs

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# CronJob

CronJob represents the configuration of a single cron job.



apiVersion: batch/v1

import "k8s.io/api/batch/v1"

## CronJob

CronJob represents the configuration of a single cron job.

---

- **apiVersion:** batch/v1
- **kind:** CronJob
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([CronJobSpec](#))

Specification of the desired behavior of a cron job, including the schedule. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([CronJobStatus](#))

Current status of a cron job. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## CronJobSpec

CronJobSpec describes how the job execution will look like and when it will actually run.

---

- **jobTemplate** ([JobTemplateSpec](#)), required

Specifies the job that will be created when executing a CronJob.

*JobTemplateSpec describes the data a Job should have when created from a template*

- **jobTemplate.metadata** ([ObjectMeta](#))

Standard object's metadata of the jobs created from this template. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **jobTemplate.spec** ([JobSpec](#))

Specification of the desired behavior of the job. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **schedule** (string), required

The schedule in Cron format, see <https://en.wikipedia.org/wiki/Cron>.

- **timeZone** (string)

The time zone name for the given schedule, see [https://en.wikipedia.org/wiki/List of tz database time zones](https://en.wikipedia.org/wiki/List_of_tz_database_time_zones). If not specified, this will default to the time zone of the kube-controller-manager process. The set of valid time zone names and the time zone offset is loaded from the system-wide time zone database by the API server during CronJob validation and the controller manager during execution. If no system-wide time zone database can be found a bundled version of the database is used instead. If the time zone name becomes invalid during the lifetime of a CronJob or due to a change in host configuration, the controller will stop creating new new Jobs and will create a system event with the reason UnknownTimeZone. More information can be found in <https://kubernetes.io/docs/concepts/workloads/controllers/cron-jobs/#time-zones>

- **concurrencyPolicy** (string)

Specifies how to treat concurrent executions of a Job. Valid values are:

- "Allow" (default): allows CronJobs to run concurrently; - "Forbid": forbids concurrent runs, skipping next run if previous run hasn't finished yet; - "Replace": cancels currently running job and replaces it with a new one

- **startingDeadlineSeconds** (int64)

Optional deadline in seconds for starting the job if it misses scheduled time for any reason. Missed jobs executions will be counted as failed ones.

- **suspend** (boolean)

This flag tells the controller to suspend subsequent executions, it does not apply to already started executions. Defaults to false.

- **successfulJobsHistoryLimit** (int32)

The number of successful finished jobs to retain. Value must be non-negative integer. Defaults to 3.

- **failedJobsHistoryLimit** (int32)

The number of failed finished jobs to retain. Value must be non-negative integer. Defaults to 1.

## CronJobStatus

CronJobStatus represents the current state of a cron job.

---

- **active** ([][ObjectReference](#))

*Atomic: will be replaced during a merge*

A list of pointers to currently running jobs.

- **lastScheduleTime** (Time)

Information when was the last time the job was successfully scheduled.

*Time is a wrapper around time. Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **lastSuccessfulTime** (Time)

Information when was the last time the job successfully completed.

*Time is a wrapper around time. Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

## CronJobList

CronJobList is a collection of cron jobs.

---

- **apiVersion:** batch/v1

- **kind:** CronJobList

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]CronJob](#)), required

items is the list of CronJobs.

## Operations

---

### get read the specified CronJob

#### HTTP Request

GET /apis/batch/v1/namespaces/{namespace}/cronjobs/{name}

#### Parameters

- **name** (*in path*): string, required

name of the CronJob

- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([CronJob](#)): OK

401: Unauthorized

## get read status of the specified CronJob

### HTTP Request

GET /apis/batch/v1/namespaces/{namespace}/cronjobs/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the CronJob
- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([CronJob](#)): OK

401: Unauthorized

## list list or watch objects of kind CronJob

### HTTP Request

GET /apis/batch/v1/namespaces/{namespace}/cronjobs

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([CronJobList](#)): OK

401: Unauthorized

## list list or watch objects of kind CronJob

## HTTP Request

GET /apis/batch/v1/cronjobs

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([CronJobList](#)): OK

401: Unauthorized

## create create a CronJob

### HTTP Request

POST /apis/batch/v1/namespaces/{namespace}/cronjobs

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [CronJob](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CronJob](#)): OK

201 ([CronJob](#)): Created

202 ([CronJob](#)): Accepted

401: Unauthorized

## update replace the specified CronJob

## HTTP Request

PUT /apis/batch/v1/namespaces/{namespace}/cronjobs/{name}

## Parameters

- **name** (*in path*): string, required  
name of the CronJob
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [CronJob](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([CronJob](#)): OK

201 ([CronJob](#)): Created

401: Unauthorized

## update replace status of the specified CronJob

### HTTP Request

PUT /apis/batch/v1/namespaces/{namespace}/cronjobs/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the CronJob
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [CronJob](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string



[pretty](#)

## Response

200 ([CronJob](#)): OK

201 ([CronJob](#)): Created

401: Unauthorized

## patch partially update the specified CronJob

### HTTP Request

PATCH /apis/batch/v1/namespaces/{namespace}/cronjobs/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CronJob
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CronJob](#)): OK

201 ([CronJob](#)): Created

401: Unauthorized

## patch partially update status of the specified CronJob

### HTTP Request

PATCH /apis/batch/v1/namespaces/{namespace}/cronjobs/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the CronJob
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([CronJob](#)): OK

201 ([CronJob](#)): Created

401: Unauthorized

## delete delete a CronJob

### HTTP Request

DELETE /apis/batch/v1/namespaces/{namespace}/cronjobs/{name}

## Parameters

- **name** (*in path*): string, required  
name of the CronJob
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of CronJob

### HTTP Request

DELETE /apis/batch/v1/namespaces/{namespace}/cronjobs

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# HorizontalPodAutoscaler

configuration of a horizontal pod autoscaler.

apiVersion: autoscaling/v1

import "k8s.io/api/autoscaling/v1"

## HorizontalPodAutoscaler

configuration of a horizontal pod autoscaler.

---

- **apiVersion:** autoscaling/v1
- **kind:** HorizontalPodAutoscaler
- **metadata** ([ObjectMeta](#))

Standard object metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([HorizontalPodAutoscalerSpec](#))

spec defines the behaviour of autoscaler. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>.

- **status** ([HorizontalPodAutoscalerStatus](#))

status is the current information about the autoscaler.

## HorizontalPodAutoscalerSpec

specification of a horizontal pod autoscaler.

---

- **maxReplicas** (int32), required

maxReplicas is the upper limit for the number of pods that can be set by the autoscaler; cannot be smaller than MinReplicas.

- **scaleTargetRef** (CrossVersionObjectReference), required

reference to scaled resource; horizontal pod autoscaler will learn the current resource consumption and will set the desired number of pods by using its Scale subresource.

*CrossVersionObjectReference contains enough information to let you identify the referred resource.*

- **scaleTargetRef.kind** (string), required

kind is the kind of the referent; More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **scaleTargetRef.name** (string), required

name is the name of the referent; More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

**scaleTargetRef.apiVersion** (string)

- apiVersion is the API version of the referent

- **minReplicas** (int32)

minReplicas is the lower limit for the number of replicas to which the autoscaler can scale down. It defaults to 1 pod. minReplicas is allowed to be 0 if the alpha feature gate HPAScaleToZero is enabled and at least one Object or External metric is configured. Scaling is active as long as at least one metric value is available.

- **targetCPUUtilizationPercentage** (int32)

targetCPUUtilizationPercentage is the target average CPU utilization (represented as a percentage of requested CPU) over all the pods; if not specified the default autoscaling policy will be used.

## HorizontalPodAutoscalerStatus

current status of a horizontal pod autoscaler

---

- **currentReplicas** (int32), required

currentReplicas is the current number of replicas of pods managed by this autoscaler.

- **desiredReplicas** (int32), required

desiredReplicas is the desired number of replicas of pods managed by this autoscaler.

- **currentCPUUtilizationPercentage** (int32)

currentCPUUtilizationPercentage is the current average CPU utilization over all pods, represented as a percentage of requested CPU, e.g. 70 means that an average pod is using now 70% of its requested CPU.

- **lastScaleTime** (Time)

lastScaleTime is the last time the HorizontalPodAutoscaler scaled the number of pods; used by the autoscaler to control how often the number of pods is changed.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **observedGeneration** (int64)

observedGeneration is the most recent generation observed by this autoscaler.

## HorizontalPodAutoscalerList

list of horizontal pod autoscaler objects.

---

- **apiVersion**: autoscaling/v1

- **kind:** HorizontalPodAutoscalerList
- **metadata** ([ListMeta](#))  
Standard list metadata.
- **items** ([\[\]HorizontalPodAutoscaler](#)), required  
items is the list of horizontal pod autoscaler objects.

## Operations

---

### get read the specified HorizontalPodAutoscaler

#### HTTP Request

GET /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([HorizontalPodAutoscaler](#)): OK

401: Unauthorized

### get read status of the specified HorizontalPodAutoscaler

#### HTTP Request

GET /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers/{name}/status

#### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler

- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

401: Unauthorized

## list list or watch objects of kind HorizontalPodAutoscaler

### HTTP Request

GET /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)



- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([HorizontalPodAutoscalerList](#)): OK

401: Unauthorized

## list list or watch objects of kind HorizontalPodAutoscaler

## HTTP Request

GET /apis/autoscaling/v1/horizontalpodautoscalers

## Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([HorizontalPodAutoscalerList](#)): OK

401: Unauthorized

## create create a HorizontalPodAutoscaler

### HTTP Request

POST /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [HorizontalPodAutoscaler](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

202 ([HorizontalPodAutoscaler](#)): Accepted

401: Unauthorized

## update replace the specified HorizontalPodAutoscaler

## HTTP Request

PUT /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers/{name}

## Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [HorizontalPodAutoscaler](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

401: Unauthorized

## update replace status of the specified HorizontalPodAutoscaler

### HTTP Request

PUT /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [HorizontalPodAutoscaler](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

401: Unauthorized

## patch partially update the specified HorizontalPodAutoscaler

### HTTP Request

PATCH /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

401: Unauthorized

## patch partially update status of the specified HorizontalPodAutoscaler

### HTTP Request

PATCH /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

401: Unauthorized

## delete delete a HorizontalPodAutoscaler

### HTTP Request

DELETE /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of HorizontalPodAutoscaler

### HTTP Request

DELETE /apis/autoscaling/v1/namespaces/{namespace}/horizontalpodautoscalers

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **continue** (*in query*): string

[continue](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# HorizontalPodAutoscaler

HorizontalPodAutoscaler is the configuration for a horizontal pod autoscaler, which automatically manages the replica count of any resource implementing the scale subresource based on the metrics specified.

apiVersion: autoscaling/v2

import "k8s.io/api/autoscaling/v2"

## HorizontalPodAutoscaler

HorizontalPodAutoscaler is the configuration for a horizontal pod autoscaler, which automatically manages the replica count of any resource implementing the scale subresource based on the metrics specified.

- 
- **apiVersion**: autoscaling/v2
  - **kind**: HorizontalPodAutoscaler
  - **metadata** ([ObjectMeta](#))

metadata is the standard object metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>



**spec** ([HorizontalPodAutoscalerSpec](#))

- spec is the specification for the behaviour of the autoscaler. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>.

• **status** ([HorizontalPodAutoscalerStatus](#))

status is the current information about the autoscaler.

## HorizontalPodAutoscalerSpec

HorizontalPodAutoscalerSpec describes the desired functionality of the HorizontalPodAutoscaler.

---

• **maxReplicas** (int32), required

maxReplicas is the upper limit for the number of replicas to which the autoscaler can scale up. It cannot be less than minReplicas.

• **scaleTargetRef** (CrossVersionObjectReference), required

scaleTargetRef points to the target resource to scale, and is used to the pods for which metrics should be collected, as well as to actually change the replica count.

*CrossVersionObjectReference contains enough information to let you identify the referred resource.*

- **scaleTargetRef.kind** (string), required

kind is the kind of the referent; More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **scaleTargetRef.name** (string), required

name is the name of the referent; More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **scaleTargetRef.apiVersion** (string)

apiVersion is the API version of the referent

• **minReplicas** (int32)

minReplicas is the lower limit for the number of replicas to which the autoscaler can scale down. It defaults to 1 pod. minReplicas is allowed to be 0 if the alpha feature gate HPAScaleToZero is enabled and at least one Object or External metric is configured. Scaling is active as long as at least one metric value is available.

• **behavior** (HorizontalPodAutoscalerBehavior)

behavior configures the scaling behavior of the target in both Up and Down directions (scaleUp and scaleDown fields respectively). If not set, the default HPAScalingRules for scale up and scale down are used.

*HorizontalPodAutoscalerBehavior configures the scaling behavior of the target in both Up and Down directions (scaleUp and scaleDown fields respectively).*

- **behavior.scaleDown** (HPAScalingRules)

scaleDown is scaling policy for scaling Down. If not set, the default value is to allow to scale down to minReplicas pods, with a 300 second stabilization window (i.e., the highest recommendation for the last 300sec is used).

*HPAScalingRules configures the scaling behavior for one direction. These Rules are applied after calculating DesiredReplicas from metrics for the HPA. They can limit the scaling velocity by specifying scaling policies. They can prevent flapping by specifying the stabilization window, so that the number of replicas is not set instantly, instead, the safest value from the stabilization window is chosen.*

- **behavior.scaleDown.policies** ([]HPAScalingPolicy)

*Atomic: will be replaced during a merge*

policies is a list of potential scaling policies which can be used during scaling. At least one policy must be specified, otherwise the HPAScalingRules will be discarded as invalid

*HPAScalingPolicy is a single policy which must hold true for a specified past interval.*

- **behavior.scaleDown.policies.type** (string), required

type is used to specify the scaling policy.

- **behavior.scaleDown.policies.value** (int32), required

value contains the amount of change which is permitted by the policy. It must be greater than zero

- **behavior.scaleDown.policies.periodSeconds** (int32), required

periodSeconds specifies the window of time for which the policy should hold true. PeriodSeconds must be greater than zero and less than or equal to 1800 (30 min).

- **behavior.scaleDown.selectPolicy** (string)

selectPolicy is used to specify which policy should be used. If not set, the default value Max is used.

- **behavior.scaleDown.stabilizationWindowSeconds** (int32)

stabilizationWindowSeconds is the number of seconds for which past recommendations should be considered while scaling up or scaling down. StabilizationWindowSeconds must be greater than or equal to zero and less than or equal to 3600 (one hour). If not set, use the default values: - For scale up: 0 (i.e. no stabilization is done). - For scale down: 300 (i.e. the stabilization window is 300 seconds long).

## **behavior.scaleUp** (HPAScalingRules)

- scaleUp is scaling policy for scaling Up. If not set, the default value is the higher of:

- increase no more than 4 pods per 60 seconds
- double the number of pods per 60 seconds No stabilization is used.

*HPAScalingRules configures the scaling behavior for one direction. These Rules are applied after calculating DesiredReplicas from metrics for the HPA. They can limit the scaling velocity by specifying scaling policies. They can prevent flapping by specifying the stabilization window, so that the number of replicas is not set instantly, instead, the safest value from the stabilization window is chosen.*

### ▪ **behavior.scaleUp.policies** ([]HPAScalingPolicy)

*Atomic: will be replaced during a merge*

policies is a list of potential scaling policies which can be used during scaling. At least one policy must be specified, otherwise the HPAScalingRules will be discarded as invalid

*HPAScalingPolicy is a single policy which must hold true for a specified past interval.*

#### ▪ **behavior.scaleUp.policies.type** (string), required

type is used to specify the scaling policy.

#### ▪ **behavior.scaleUp.policies.value** (int32), required

value contains the amount of change which is permitted by the policy. It must be greater than zero

#### ▪ **behavior.scaleUp.policies.periodSeconds** (int32), required

periodSeconds specifies the window of time for which the policy should hold true. PeriodSeconds must be greater than zero and less than or equal to 1800 (30 min).

### ▪ **behavior.scaleUp.selectPolicy** (string)

selectPolicy is used to specify which policy should be used. If not set, the default value Max is used.

### ▪ **behavior.scaleUp.stabilizationWindowSeconds** (int32)

stabilizationWindowSeconds is the number of seconds for which past recommendations should be considered while scaling up or scaling down. StabilizationWindowSeconds must be greater than or equal to zero and less than or equal to 3600 (one hour). If not set, use the default values: - For scale up: 0 (i.e. no stabilization is done). - For scale down: 300 (i.e. the stabilization window is 300 seconds long).

## • **metrics** ([]MetricSpec)

*Atomic: will be replaced during a merge*

metrics contains the specifications for which to use to calculate the desired replica count (the maximum replica count across all metrics will be used). The desired replica count is calculated multiplying the ratio between the target value and the current value by the current number of pods. Ergo, metrics used must decrease as the pod count is increased, and vice-versa. See the individual metric source types for more information about how each type of metric must respond. If not set, the default metric will be set to 80% average CPU utilization.

*MetricSpec specifies how to scale based on a single metric (only type and one other matching field should be set at once).*

- **metrics.type** (string), required

type is the type of metric source. It should be one of "ContainerResource", "External", "Object", "Pods" or "Resource", each mapping to a matching field in the object. Note: "ContainerResource" type is available on when the feature-gate HPAContainerMetrics is enabled

- **metrics.containerResource** (ContainerResourceMetricSource)

containerResource refers to a resource metric (such as those specified in requests and limits) known to Kubernetes describing a single container in each pod of the current scale target (e.g. CPU or memory). Such metrics are built in to Kubernetes, and have special scaling options on top of those available to normal per-pod metrics using the "pods" source. This is an alpha feature and can be enabled by the HPAContainerMetrics feature flag.

*ContainerResourceMetricSource indicates how to scale on a resource metric known to Kubernetes, as specified in requests and limits, describing each pod in the current scale target (e.g. CPU or memory). The values will be averaged together before being compared to the target. Such metrics are built in to Kubernetes, and have special scaling options on top of those available to normal per-pod metrics using the "pods" source. Only one "target" type should be set.*

- **metrics.containerResource.container** (string), required

container is the name of the container in the pods of the scaling target

- **metrics.containerResource.name** (string), required

name is the name of the resource in question.

- **metrics.containerResource.target** (MetricTarget), required

target specifies the target value for the given metric

*MetricTarget defines the target value, average value, or average utilization of a specific metric*

- **metrics.containerResource.target.type** (string), required

type represents whether the metric type is Utilization, Value, or AverageValue

- **metrics.containerResource.target.averageUtilization** (int32)

averageUtilization is the target value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods. Currently only valid for Resource metric source type

- **metrics.containerResource.target.averageValue** ([Quantity](#))

averageValue is the target value of the average of the metric across all relevant pods (as a quantity)

- **metrics.containerResource.target.value** ([Quantity](#))

value is the target value of the metric (as a quantity).

- **metrics.external** (ExternalMetricSource)

external refers to a global metric that is not associated with any Kubernetes object. It allows autoscaling based on information coming from components running outside of cluster (for example length of queue in cloud messaging service, or QPS from loadbalancer running outside of cluster).

*ExternalMetricSource indicates how to scale on a metric not associated with any Kubernetes object (for example length of queue in cloud messaging service, or QPS from loadbalancer running outside of cluster).*

- **metrics.external.metric** (MetricIdentifier), required

metric identifies the target metric by name and selector

*MetricIdentifier defines the name and optionally selector for a metric*

- **metrics.external.metric.name** (string), required

name is the name of the given metric

- **metrics.external.metric.selector** ([LabelSelector](#))

selector is the string-encoded form of a standard kubernetes label selector for the given metric When set, it is passed as an additional parameter to the metrics server for more specific metrics scoping. When unset, just the metricName will be used to gather metrics.

- **metrics.external.target** (MetricTarget), required

target specifies the target value for the given metric

*MetricTarget defines the target value, average value, or average utilization of a specific metric*

- **metrics.external.target.type** (string), required

type represents whether the metric type is Utilization, Value, or AverageValue

- **metrics.external.target.averageUtilization** (int32)

averageUtilization is the target value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods. Currently only valid for Resource metric source type

- **metrics.external.target.averageValue** ([Quantity](#))

averageValue is the target value of the average of the metric across all relevant pods (as a quantity)

- **metrics.external.target.value** ([Quantity](#))

value is the target value of the metric (as a quantity).

- **metrics.object** (ObjectMetricSource)

object refers to a metric describing a single kubernetes object (for example, hits-per-second on an Ingress object).

*ObjectMetricSource indicates how to scale on a metric describing a kubernetes object (for example, hits-per-second on an Ingress object).*

- **metrics.object.describedObject** (CrossVersionObjectReference), required

describedObject specifies the descriptions of a object,such as kind,name apiVersion

*CrossVersionObjectReference contains enough information to let you identify the referred resource.*

- **metrics.object.describedObject.kind** (string), required

kind is the kind of the referent; More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **metrics.object.describedObject.name** (string), required

name is the name of the referent; More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **metrics.object.describedObject.apiVersion** (string)

apiVersion is the API version of the referent

- **metrics.object.metric** (MetricIdentifier), required

metric identifies the target metric by name and selector

*MetricIdentifier defines the name and optionally selector for a metric*

- **metrics.object.metric.name** (string), required

name is the name of the given metric

▪ **metrics.object.metric.selector** ([LabelSelector](#))

selector is the string-encoded form of a standard kubernetes label selector for the given metric When set, it is passed as an additional parameter to the metrics server for more specific metrics scoping. When unset, just the metricName will be used to gather metrics.

▪ **metrics.object.target** (MetricTarget), required

target specifies the target value for the given metric

*MetricTarget defines the target value, average value, or average utilization of a specific metric*

▪ **metrics.object.target.type** (string), required

type represents whether the metric type is Utilization, Value, or AverageValue

▪ **metrics.object.target.averageUtilization** (int32)

averageUtilization is the target value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods. Currently only valid for Resource metric source type

▪ **metrics.object.target.averageValue** ([Quantity](#))

averageValue is the target value of the average of the metric across all relevant pods (as a quantity)

▪ **metrics.object.target.value** ([Quantity](#))

value is the target value of the metric (as a quantity).

◦ **metrics.pods** (PodsMetricSource)

pods refers to a metric describing each pod in the current scale target (for example, transactions-processed-per-second). The values will be averaged together before being compared to the target value.

*PodsMetricSource indicates how to scale on a metric describing each pod in the current scale target (for example, transactions-processed-per-second). The values will be averaged together before being compared to the target value.*

▪ **metrics.pods.metric** (MetricIdentifier), required

metric identifies the target metric by name and selector

*MetricIdentifier defines the name and optionally selector for a metric*

▪ **metrics.pods.metric.name** (string), required

name is the name of the given metric

▪ **metrics.pods.metric.selector** ([LabelSelector](#))

selector is the string-encoded form of a standard kubernetes label selector for the given metric When set, it is passed as an additional parameter to the metrics server for more specific metrics scoping. When unset, just the metricName will be used to gather metrics.

▪ **metrics.pods.target** (MetricTarget), required

target specifies the target value for the given metric

*MetricTarget defines the target value, average value, or average utilization of a specific metric*

▪ **metrics.pods.target.type** (string), required

type represents whether the metric type is Utilization, Value, or AverageValue

▪ **metrics.pods.target.averageUtilization** (int32)

averageUtilization is the target value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods. Currently only valid for Resource metric source type

▪ **metrics.pods.target.averageValue** ([Quantity](#))

averageValue is the target value of the average of the metric across all relevant pods (as a quantity)

▪ **metrics.pods.target.value** ([Quantity](#))

value is the target value of the metric (as a quantity).

◦ **metrics.resource** (ResourceMetricSource)

resource refers to a resource metric (such as those specified in requests and limits) known to Kubernetes describing each pod in the current scale target (e.g. CPU or memory). Such metrics are built in to Kubernetes, and have special scaling options on top of those available to normal per-pod metrics using the "pods" source.

*ResourceMetricSource indicates how to scale on a resource metric known to Kubernetes, as specified in requests and limits, describing each pod in the current scale target (e.g. CPU or memory). The values will be averaged together before being compared to the target. Such metrics are built in to Kubernetes, and have special scaling options on top of those available to normal per-pod metrics using the "pods" source. Only one "target" type should be set.*

▪ **metrics.resource.name** (string), required

name is the name of the resource in question.

▪ **metrics.resource.target** (MetricTarget), required



target specifies the target value for the given metric

*MetricTarget defines the target value, average value, or average utilization of a specific metric*

- **metrics.resource.target.type** (string), required

type represents whether the metric type is Utilization, Value, or AverageValue

- **metrics.resource.target.averageUtilization** (int32)

averageUtilization is the target value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods. Currently only valid for Resource metric source type

- **metrics.resource.target.averageValue** ([Quantity](#))

averageValue is the target value of the average of the metric across all relevant pods (as a quantity)

- **metrics.resource.target.value** ([Quantity](#))

value is the target value of the metric (as a quantity).

## HorizontalPodAutoscalerStatus

HorizontalPodAutoscalerStatus describes the current status of a horizontal pod autoscaler.

---

- **desiredReplicas** (int32), required

desiredReplicas is the desired number of replicas of pods managed by this autoscaler, as last calculated by the autoscaler.

- **conditions** ([]HorizontalPodAutoscalerCondition)

*Patch strategy: merge on key type*

*Map: unique values on key type will be kept during a merge*

conditions is the set of conditions required for this autoscaler to scale its target, and indicates whether or not those conditions are met.

*HorizontalPodAutoscalerCondition describes the state of a HorizontalPodAutoscaler at a certain point.*

- **conditions.status** (string), required

status is the status of the condition (True, False, Unknown)

- **conditions.type** (string), required

type describes the current condition

**conditions.lastTransitionTime** (Time)

- lastTransitionTime is the last time the condition transitioned from one status to another

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

message is a human-readable explanation containing details about the transition

- **conditions.reason** (string)

reason is the reason for the condition's last transition.

- **currentMetrics** ([]MetricStatus)

*Atomic: will be replaced during a merge*

currentMetrics is the last read state of the metrics used by this autoscaler.

*MetricStatus describes the last-read state of a single metric.*

- **currentMetrics.type** (string), required

type is the type of metric source. It will be one of "ContainerResource", "External", "Object", "Pods" or "Resource", each corresponds to a matching field in the object. Note: "ContainerResource" type is available on when the feature-gate HPAContainerMetrics is enabled

- **currentMetrics.containerResource** (ContainerResourceMetricStatus)

container resource refers to a resource metric (such as those specified in requests and limits) known to Kubernetes describing a single container in each pod in the current scale target (e.g. CPU or memory). Such metrics are built in to Kubernetes, and have special scaling options on top of those available to normal per-pod metrics using the "pods" source.

*ContainerResourceMetricStatus indicates the current value of a resource metric known to Kubernetes, as specified in requests and limits, describing a single container in each pod in the current scale target (e.g. CPU or memory). Such metrics are built in to Kubernetes, and have special scaling options on top of those available to normal per-pod metrics using the "pods" source.*

- **currentMetrics.containerResource.container** (string), required

container is the name of the container in the pods of the scaling target

- **currentMetrics.containerResource.current** (MetricValueStatus), required

current contains the current value for the given metric

*MetricValueStatus holds the current value for a metric*

- **currentMetrics.containerResource.current.averageUtilization** (int32)

currentAverageUtilization is the current value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods.

- **currentMetrics.containerResource.current.averageValue** ([Quantity](#))

averageValue is the current value of the average of the metric across all relevant pods (as a quantity)

- **currentMetrics.containerResource.current.value** ([Quantity](#))

value is the current value of the metric (as a quantity).

- **currentMetrics.containerResource.name** (string), required

name is the name of the resource in question.

- **currentMetrics.external** (ExternalMetricStatus)

external refers to a global metric that is not associated with any Kubernetes object. It allows autoscaling based on information coming from components running outside of cluster (for example length of queue in cloud messaging service, or QPS from loadbalancer running outside of cluster).

*ExternalMetricStatus indicates the current value of a global metric not associated with any Kubernetes object.*

- **currentMetrics.external.current** (MetricValueStatus), required

current contains the current value for the given metric

*MetricValueStatus holds the current value for a metric*

- **currentMetrics.external.current.averageUtilization** (int32)

currentAverageUtilization is the current value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods.

- **currentMetrics.external.current.averageValue** ([Quantity](#))

averageValue is the current value of the average of the metric across all relevant pods (as a quantity)

- **currentMetrics.external.current.value** ([Quantity](#))

value is the current value of the metric (as a quantity).

- **currentMetrics.external.metric** (MetricIdentifier), required

metric identifies the target metric by name and selector

*MetricIdentifier defines the name and optionally selector for a metric*

- **currentMetrics.external.metric.name** (string), required

name is the name of the given metric

- **currentMetrics.external.metric.selector** ([LabelSelector](#))

selector is the string-encoded form of a standard kubernetes label selector for the given metric When set, it is passed as an additional parameter to the metrics server for more specific metrics scoping. When unset, just the metricName will be used to gather metrics.

- **currentMetrics.object** (ObjectMetricStatus)

object refers to a metric describing a single kubernetes object (for example, hits-per-second on an Ingress object).

*ObjectMetricStatus indicates the current value of a metric describing a kubernetes object (for example, hits-per-second on an Ingress object).*

- **currentMetrics.object.current** (MetricValueStatus), required

current contains the current value for the given metric

*MetricValueStatus holds the current value for a metric*

- **currentMetrics.object.current.averageUtilization** (int32)

currentAverageUtilization is the current value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods.

- **currentMetrics.object.current.averageValue** ([Quantity](#))

averageValue is the current value of the average of the metric across all relevant pods (as a quantity)

- **currentMetrics.object.current.value** ([Quantity](#))

value is the current value of the metric (as a quantity).

- **currentMetrics.object.describedObject** (CrossVersionObjectReference), required

DescribedObject specifies the descriptions of a object,such as kind,name apiVersion

*CrossVersionObjectReference contains enough information to let you identify the referred resource.*

- **currentMetrics.object.describedObject.kind** (string), required

kind is the kind of the referent; More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **currentMetrics.object.describedObject.name** (string), required

name is the name of the referent; More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **currentMetrics.object.describedObject.apiVersion** (string)

apiVersion is the API version of the referent

- **currentMetrics.object.metric** (MetricIdentifier), required

metric identifies the target metric by name and selector

*MetricIdentifier defines the name and optionally selector for a metric*

- **currentMetrics.object.metric.name** (string), required

name is the name of the given metric

- **currentMetrics.object.metric.selector** ([LabelSelector](#))

selector is the string-encoded form of a standard kubernetes label selector for the given metric When set, it is passed as an additional parameter to the metrics server for more specific metrics scoping. When unset, just the metricName will be used to gather metrics.

- **currentMetrics.pods** (PodsMetricStatus)

pods refers to a metric describing each pod in the current scale target (for example, transactions-processed-per-second). The values will be averaged together before being compared to the target value.

*PodsMetricStatus indicates the current value of a metric describing each pod in the current scale target (for example, transactions-processed-per-second).*

- **currentMetrics.pods.current** (MetricValueStatus), required

current contains the current value for the given metric

*MetricValueStatus holds the current value for a metric*

- **currentMetrics.pods.current.averageUtilization** (int32)

currentAverageUtilization is the current value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods.

- **currentMetrics.pods.current.averageValue** ([Quantity](#))

averageValue is the current value of the average of the metric across all relevant pods (as a quantity)

- **currentMetrics.pods.current.value** ([Quantity](#))

- value is the current value of the metric (as a quantity).

- **currentMetrics.pods.metric** (MetricIdentifier), required

- metric identifies the target metric by name and selector

- MetricIdentifier defines the name and optionally selector for a metric*

- **currentMetrics.pods.metric.name** (string), required

- name is the name of the given metric

- **currentMetrics.pods.metric.selector** ([LabelSelector](#))

- selector is the string-encoded form of a standard kubernetes label selector for the given metric When set, it is passed as an additional parameter to the metrics server for more specific metrics scoping. When unset, just the metricName will be used to gather metrics.

- **currentMetrics.resource** (ResourceMetricStatus)

- resource refers to a resource metric (such as those specified in requests and limits) known to Kubernetes describing each pod in the current scale target (e.g. CPU or memory). Such metrics are built in to Kubernetes, and have special scaling options on top of those available to normal per-pod metrics using the "pods" source.

- ResourceMetricStatus indicates the current value of a resource metric known to Kubernetes, as specified in requests and limits, describing each pod in the current scale target (e.g. CPU or memory). Such metrics are built in to Kubernetes, and have special scaling options on top of those available to normal per-pod metrics using the "pods" source.*

- **currentMetrics.resource.current** (MetricValueStatus), required

- current contains the current value for the given metric

- MetricValueStatus holds the current value for a metric*

- **currentMetrics.resource.current.averageUtilization** (int32)

- currentAverageUtilization is the current value of the average of the resource metric across all relevant pods, represented as a percentage of the requested value of the resource for the pods.

- **currentMetrics.resource.current.averageValue** ([Quantity](#))

- averageValue is the current value of the average of the metric across all relevant pods (as a quantity)

- **currentMetrics.resource.current.value** ([Quantity](#))

- value is the current value of the metric (as a quantity).

- **currentMetrics.resource.name** (string), required  
name is the name of the resource in question.
- **currentReplicas** (int32)  
  
currentReplicas is current number of replicas of pods managed by this autoscaler, as last seen by the autoscaler.
- **lastScaleTime** (Time)  
  
lastScaleTime is the last time the HorizontalPodAutoscaler scaled the number of pods, used by the autoscaler to control how often the number of pods is changed.  
  
*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*
- **observedGeneration** (int64)  
  
observedGeneration is the most recent generation observed by this autoscaler.

## HorizontalPodAutoscalerList

HorizontalPodAutoscalerList is a list of horizontal pod autoscaler objects.

---

- **apiVersion**: autoscaling/v2
- **kind**: HorizontalPodAutoscalerList
- **metadata** ([ListMeta](#))  
  
metadata is the standard list metadata.
- **items** ([\[\]HorizontalPodAutoscaler](#)), required  
  
items is the list of horizontal pod autoscaler objects.

## Operations

---

### get read the specified HorizontalPodAutoscaler

#### HTTP Request

GET /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers/{name}

#### Parameters

- **name** (*in path*): string, required  
  
name of the HorizontalPodAutoscaler

- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

401: Unauthorized

## get read status of the specified HorizontalPodAutoscaler

### HTTP Request

GET /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

401: Unauthorized

## list list or watch objects of kind HorizontalPodAutoscaler

### HTTP Request

GET /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)



- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([HorizontalPodAutoscalerList](#)): OK

401: Unauthorized

**list** list or watch objects of kind HorizontalPodAutoscaler

## HTTP Request

GET /apis/autoscaling/v2/horizontalpodautoscalers

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([HorizontalPodAutoscalerList](#)): OK

401: Unauthorized

## create create a HorizontalPodAutoscaler

### HTTP Request

POST /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [HorizontalPodAutoscaler](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([HorizontalPodAutoscaler](#)): OK  
201 ([HorizontalPodAutoscaler](#)): Created  
202 ([HorizontalPodAutoscaler](#)): Accepted  
401: Unauthorized

## update replace the specified HorizontalPodAutoscaler

### HTTP Request

PUT /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [HorizontalPodAutoscaler](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

401: Unauthorized

## update replace status of the specified HorizontalPodAutoscaler

### HTTP Request

PUT /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [HorizontalPodAutoscaler](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

401: Unauthorized

## patch partially update the specified HorizontalPodAutoscaler

### HTTP Request

PATCH /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

401: Unauthorized

## patch partially update status of the specified HorizontalPodAutoscaler

### HTTP Request

PATCH /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([HorizontalPodAutoscaler](#)): OK

201 ([HorizontalPodAutoscaler](#)): Created

401: Unauthorized

## delete delete a HorizontalPodAutoscaler

### HTTP Request

DELETE /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the HorizontalPodAutoscaler
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of HorizontalPodAutoscaler

### HTTP Request

DELETE /apis/autoscaling/v2/namespaces/{namespace}/horizontalpodautoscalers

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)

- **body:** [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized



# PriorityClass

PriorityClass defines mapping from a priority class name to the priority integer value.

apiVersion: scheduling.k8s.io/v1

import "k8s.io/api/scheduling/v1"

## PriorityClass

PriorityClass defines mapping from a priority class name to the priority integer value. The value can be any valid integer.

---

- **apiVersion:** scheduling.k8s.io/v1

- **kind:** PriorityClass

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **value** (int32), required

value represents the integer value of this priority class. This is the actual priority that pods receive when they have the name of this class in their pod spec.

- **description** (string)

description is an arbitrary string that usually provides guidelines on when this priority class should be used.

- **globalDefault** (boolean)

globalDefault specifies whether this PriorityClass should be considered as the default priority for pods that do not have any priority class. Only one PriorityClass can be marked as globalDefault. However, if more than one PriorityClasses exists with their globalDefault field set to true, the smallest value of such global default PriorityClasses will be used as the default priority.

- **preemptionPolicy** (string)

preemptionPolicy is the Policy for preempting pods with lower priority. One of Never, PreemptLowerPriority. Defaults to PreemptLowerPriority if unset.

## PriorityClassList

PriorityClassList is a collection of priority classes.

---

- **apiVersion:** scheduling.k8s.io/v1

**kind:** PriorityClassList

- 
- **metadata** ([ListMeta](#))

Standard list metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([][PriorityClass](#)), required

items is the list of PriorityClasses

## Operations

---

### get read the specified PriorityClass

#### HTTP Request

GET /apis/scheduling.k8s.io/v1/priorityclasses/{name}

#### Parameters

- **name** (*in path*): string, required

name of the PriorityClass

- **pretty** (*in query*): string

[pretty](#)

#### Response

200 ([PriorityClass](#)): OK

401: Unauthorized

### list list or watch objects of kind PriorityClass

#### HTTP Request

GET /apis/scheduling.k8s.io/v1/priorityclasses

#### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([PriorityClassList](#)): OK

401: Unauthorized

## create create a PriorityClass

### HTTP Request

POST /apis/scheduling.k8s.io/v1/priorityclasses

### Parameters

- **body**: [PriorityClass](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PriorityClass](#)): OK

201 ([PriorityClass](#)): Created

202 ([PriorityClass](#)): Accepted

401: Unauthorized

## update replace the specified PriorityClass

### HTTP Request

PUT /apis/scheduling.k8s.io/v1/priorityclasses/{name}

### Parameters

- **name** (*in path*): string, required

name of the PriorityClass

- **body**: [PriorityClass](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PriorityClass](#)): OK

201 ([PriorityClass](#)): Created

401: Unauthorized

## patch partially update the specified PriorityClass

### HTTP Request

PATCH /apis/scheduling.k8s.io/v1/priorityclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PriorityClass
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PriorityClass](#)): OK

201 ([PriorityClass](#)): Created

401: Unauthorized

## delete delete a PriorityClass

### HTTP Request

DELETE /apis/scheduling.k8s.io/v1/priorityclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PriorityClass
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of PriorityClass

### HTTP Request

DELETE /apis/scheduling.k8s.io/v1/priorityclasses

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# PodSchedulingContext v1alpha2

PodSchedulingContext objects hold information that is needed to schedule a Pod with ResourceClaims that use "WaitForFirstConsumer" allocation mode.

apiVersion: resource.k8s.io/v1alpha2

```
import "k8s.io/api/resource/v1alpha2"
```

## PodSchedulingContext

PodSchedulingContext objects hold information that is needed to schedule a Pod with ResourceClaims that use "WaitForFirstConsumer" allocation mode.

This is an alpha type and requires enabling the DynamicResourceAllocation feature gate.

---

- **apiVersion:** resource.k8s.io/v1alpha2

- **kind:** PodSchedulingContext

- **metadata** ([ObjectMeta](#))

Standard object metadata

- **spec** ([PodSchedulingContextSpec](#)), required

Spec describes where resources for the Pod are needed.

- **status** ([PodSchedulingContextStatus](#))

Status describes where resources for the Pod can be allocated.

## PodSchedulingContextSpec

PodSchedulingContextSpec describes where resources for the Pod are needed.

---

- **potentialNodes** ([]string)

*Set: unique values will be kept during a merge*

PotentialNodes lists nodes where the Pod might be able to run.

The size of this field is limited to 128. This is large enough for many clusters. Larger clusters may need more attempts to find a node that suits all pending resources. This may get increased in the future, but not reduced.

- **selectedNode** (string)

SelectedNode is the node for which allocation of ResourceClaims that are referenced by the Pod and that use "WaitForFirstConsumer" allocation is to be attempted.

## PodSchedulingContextStatus

PodSchedulingContextStatus describes where resources for the Pod can be allocated.

---

- **resourceClaims** ([]ResourceClaimSchedulingStatus)



*Map: unique values on key name will be kept during a merge*

ResourceClaims describes resource availability for each pod.spec.resourceClaim entry where the corresponding ResourceClaim uses "WaitForFirstConsumer" allocation mode.

*ResourceClaimSchedulingStatus contains information about one particular ResourceClaim with "WaitForFirstConsumer" allocation mode.*

- **resourceClaims.name** (string)

Name matches the pod.spec.resourceClaims[\*].Name field.

- **resourceClaims.unsuitableNodes** ([]string)

*Set: unique values will be kept during a merge*

UnsuitableNodes lists nodes that the ResourceClaim cannot be allocated for.

The size of this field is limited to 128, the same as for PodSchedulingSpec.PotentialNodes. This may get increased in the future, but not reduced.

## PodSchedulingContextList

PodSchedulingContextList is a collection of Pod scheduling objects.

- 
- **apiVersion:** resource.k8s.io/v1alpha2
  - **kind:** PodSchedulingContextList
  - **metadata** ([ListMeta](#))
    - Standard list metadata
  - **items** ([][PodSchedulingContext](#)), required

Items is the list of PodSchedulingContext objects.

## Operations

---

### get read the specified PodSchedulingContext

#### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts/{name}

#### Parameters

- **name** (*in path*): string, required

name of the PodSchedulingContext

- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PodSchedulingContext](#)): OK

401: Unauthorized

## get read status of the specified PodSchedulingContext

### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts/{name}/status

### Parameters

- **name** (*in path*): string, required

name of the PodSchedulingContext

- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PodSchedulingContext](#)): OK

401: Unauthorized

## list list or watch objects of kind PodSchedulingContext

### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([PodSchedulingContextList](#)): OK

401: Unauthorized

## list list or watch objects of kind PodSchedulingContext

### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/podschedulingcontexts

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([PodSchedulingContextList](#)): OK

401: Unauthorized

## create create a PodSchedulingContext

### HTTP Request

POST /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [PodSchedulingContext](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodSchedulingContext](#)): OK

201 ([PodSchedulingContext](#)): Created

202 ([PodSchedulingContext](#)): Accepted

401: Unauthorized

## update replace the specified PodSchedulingContext

### HTTP Request

PUT /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts/{name}

## Parameters

- **name** (*in path*): string, required  
name of the PodSchedulingContext
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [PodSchedulingContext](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodSchedulingContext](#)): OK

201 ([PodSchedulingContext](#)): Created

401: Unauthorized

## update replace status of the specified PodSchedulingContext

### HTTP Request

PUT /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the PodSchedulingContext
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [PodSchedulingContext](#), required

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodSchedulingContext](#)): OK

201 ([PodSchedulingContext](#)): Created

401: Unauthorized

## patch partially update the specified PodSchedulingContext

### HTTP Request

PATCH /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PodSchedulingContext
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)

- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodSchedulingContext](#)): OK

201 ([PodSchedulingContext](#)): Created

401: Unauthorized

## patch partially update status of the specified PodSchedulingContext

### HTTP Request

PATCH /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PodSchedulingContext
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)



## Response

200 ([PodSchedulingContext](#)): OK

201 ([PodSchedulingContext](#)): Created

401: Unauthorized

## delete delete a PodSchedulingContext

### HTTP Request

DELETE /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PodSchedulingContext
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([PodSchedulingContext](#)): OK

202 ([PodSchedulingContext](#)): Accepted

401: Unauthorized

## deletecollection delete collection of PodSchedulingContext

### HTTP Request

DELETE /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/podschedulingcontexts

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ResourceClaim v1alpha2

ResourceClaim describes which resources are needed by a resource consumer.

apiVersion: resource.k8s.io/v1alpha2

import "k8s.io/api/resource/v1alpha2"

## ResourceClaim

ResourceClaim describes which resources are needed by a resource consumer. Its status tracks whether the resource has been allocated and what the resulting attributes are.

This is an alpha type and requires enabling the DynamicResourceAllocation feature gate.

- 
- **apiVersion**: resource.k8s.io/v1alpha2

- **kind**: ResourceClaim

- **metadata** ([ObjectMeta](#))

Standard object metadata

- **spec** ([ResourceClaimSpec](#)), required

Spec describes the desired attributes of a resource that then needs to be allocated. It can only be set once when creating the ResourceClaim.

- **status** ([ResourceClaimStatus](#))

Status describes whether the resource is available and with which attributes.

## ResourceClaimSpec

ResourceClaimSpec defines how a resource is to be allocated.

- 
- **resourceClassName** (string), required

ResourceClassName references the driver and additional parameters via the name of a ResourceClass that was created as part of the driver deployment.

- **allocationMode** (string)

Allocation can start immediately or when a Pod wants to use the resource. "WaitForFirstConsumer" is the default.

- **parametersRef** (ResourceClaimParametersReference)

ParametersRef references a separate object with arbitrary parameters that will be used by the driver when allocating a resource for the claim.

The object must be in the same namespace as the ResourceClaim.

*ResourceClaimParametersReference contains enough information to let you locate the parameters for a ResourceClaim. The object must be in the same namespace as the ResourceClaim.*

- **parametersRef.kind** (string), required

Kind is the type of resource being referenced. This is the same value as in the parameter object's metadata, for example "ConfigMap".

- **parametersRef.name** (string), required

Name is the name of resource being referenced.

- **parametersRef.apiGroup** (string)

APIGroup is the group for the resource being referenced. It is empty for the core API. This matches the group in the APIVersion that is used when creating the resources.

## ResourceClaimStatus

ResourceClaimStatus tracks whether the resource has been allocated and what the resulting attributes are.

---

- **allocation** (AllocationResult)

Allocation is set by the resource driver once a resource or set of resources has been allocated successfully. If this is not specified, the resources have not been allocated yet.

*AllocationResult contains attributes of an allocated resource.*

- **allocation.availableOnNodes** (NodeSelector)

This field will get set by the resource driver after it has allocated the resource to inform the scheduler where it can schedule Pods using the ResourceClaim.

Setting this field is optional. If null, the resource is available everywhere.

*A node selector represents the union of the results of one or more label queries over a set of nodes; that is, it represents the OR of the selectors represented by the node selector terms.*

- **allocation.availableOnNodes.nodeSelectorTerms** ([]NodeSelectorTerm), required

Required. A list of node selector terms. The terms are ORed.

*A null or empty node selector term matches no objects. The requirements of them are ANDed. The TopologySelectorTerm type implements a subset of the NodeSelectorTerm.*

- **allocation.availableOnNodes.nodeSelectorTerms.matchExpressions** ([]NodeSelectorRequirement)

A list of node selector requirements by node's labels.

- **allocation.availableOnNodes.nodeSelectorTerms.matchFields** ([]NodeSelectorRequirement)

A list of node selector requirements by node's fields.

- **allocation.resourceHandles** ([]ResourceHandle)

*Atomic: will be replaced during a merge*

ResourceHandles contain the state associated with an allocation that should be maintained throughout the lifetime of a claim. Each ResourceHandle contains data that should be passed to a specific kubelet plugin once it lands on a node. This data is returned by the driver after a successful allocation and is opaque to Kubernetes. Driver documentation may explain to users how to interpret this data if needed.

Setting this field is optional. It has a maximum size of 32 entries. If null (or empty), it is assumed this allocation will be processed by a single kubelet plugin with no ResourceHandle data attached. The name of the kubelet plugin invoked will match the DriverName set in the ResourceClaimStatus this AllocationResult is embedded in.

*ResourceHandle holds opaque resource data for processing by a specific kubelet plugin.*

- **allocation.resourceHandles.data** (string)

Data contains the opaque data associated with this ResourceHandle. It is set by the controller component of the resource driver whose name matches the DriverName set in the ResourceClaimStatus this ResourceHandle is embedded in. It is set at allocation time and is intended for processing by the kubelet plugin whose name matches the DriverName set in this ResourceHandle.

The maximum size of this field is 16KiB. This may get increased in the future, but not reduced.

- **allocation.resourceHandles.driverName** (string)

DriverName specifies the name of the resource driver whose kubelet plugin should be invoked to process this ResourceHandle's data once it lands on a node. This may differ from the DriverName set in ResourceClaimStatus this ResourceHandle is embedded in.

- **allocation.shareable** (boolean)

Shareable determines whether the resource supports more than one consumer at a time.

- **deallocationRequested** (boolean)

DeallocationRequested indicates that a ResourceClaim is to be deallocated.

The driver then must deallocate this claim and reset the field together with clearing the Allocation field.

While DeallocationRequested is set, no new consumers may be added to ReservedFor.

- **driverName** (string)

DriverName is a copy of the driver name from the ResourceClass at the time when allocation started.

- **reservedFor** ([]ResourceClaimConsumerReference)

*Map: unique values on key uid will be kept during a merge*

ReservedFor indicates which entities are currently allowed to use the claim. A Pod which references a ResourceClaim which is not reserved for that Pod will not be started.

There can be at most 32 such reservations. This may get increased in the future, but not reduced.

*ResourceClaimConsumerReference contains enough information to let you locate the consumer of a ResourceClaim. The user must be a resource in the same namespace as the ResourceClaim.*

- **reservedFor.name** (string), required

Name is the name of resource being referenced.

- **reservedFor.resource** (string), required

Resource is the type of resource being referenced, for example "pods".

- **reservedFor.uid** (string), required

UID identifies exactly one incarnation of the resource.

- **reservedFor.apiGroup** (string)

APIGroup is the group for the resource being referenced. It is empty for the core API. This matches the group in the APIVersion that is used when creating the resources.

# ResourceClaimList

ResourceClaimList is a collection of claims.

---

- **apiVersion:** resource.k8s.io/v1alpha2
- **kind:** ResourceClaimList
- **metadata** ([ListMeta](#))  
Standard list metadata
- **items** ([\[\]ResourceClaim](#)), required  
Items is the list of resource claims.

## Operations

---

### get read the specified ResourceClaim

#### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ResourceClaim](#)): OK

401: Unauthorized

### get read status of the specified ResourceClaim

#### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the ResourceClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClaim](#)): OK

401: Unauthorized

## list or watch objects of kind ResourceClaim

## HTTP Request

GET /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string



[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ResourceClaimList](#)): OK

401: Unauthorized

## list list or watch objects of kind ResourceClaim

### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/resourceclaims

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ResourceClaimList](#)): OK

401: Unauthorized

## create create a ResourceClaim

### HTTP Request

POST /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [ResourceClaim](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClaim](#)): OK

201 ([ResourceClaim](#)): Created

202 ([ResourceClaim](#)): Accepted

401: Unauthorized

## update replace the specified ResourceClaim

### HTTP Request

PUT /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ResourceClaim](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClaim](#)): OK

201 ([ResourceClaim](#)): Created

401: Unauthorized

## update replace status of the specified ResourceClaim

### HTTP Request

PUT /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ResourceClaim](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClaim](#)): OK

201 ([ResourceClaim](#)): Created

401: Unauthorized

## patch partially update the specified ResourceClaim

### HTTP Request

PATCH /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ResourceClaim](#)): OK

201 ([ResourceClaim](#)): Created

401: Unauthorized

## patch partially update status of the specified ResourceClaim

### HTTP Request

PATCH /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the ResourceClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClaim](#)): OK

201 ([ResourceClaim](#)): Created

401: Unauthorized

## delete delete a ResourceClaim

### HTTP Request

DELETE /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaim
- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [DeleteOptions](#)
- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([ResourceClaim](#)): OK

202 ([ResourceClaim](#)): Accepted

401: Unauthorized

## **deletecollection** delete collection of ResourceClaim

### HTTP Request

DELETE /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaims

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [DeleteOptions](#)

- **continue** (*in query*): string

[continue](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ResourceClaimTemplate v1alpha2

ResourceClaimTemplate is used to produce ResourceClaim objects.

```
apiVersion: resource.k8s.io/v1alpha2
```

```
import "k8s.io/api/resource/v1alpha2"
```

## ResourceClaimTemplate

ResourceClaimTemplate is used to produce ResourceClaim objects.



- 
- **apiVersion:** resource.k8s.io/v1alpha2

- **kind:** ResourceClaimTemplate

- **metadata** ([ObjectMeta](#))

Standard object metadata

- **spec** ([ResourceClaimTemplateSpec](#)), required

Describes the ResourceClaim that is to be generated.

This field is immutable. A ResourceClaim will get created by the control plane for a Pod when needed and then not get updated anymore.

## ResourceClaimTemplateSpec

ResourceClaimTemplateSpec contains the metadata and fields for a ResourceClaim.

---

- **spec** ([ResourceClaimSpec](#)), required

Spec for the ResourceClaim. The entire content is copied unchanged into the ResourceClaim that gets created from this template. The same fields as in a ResourceClaim are also valid here.

- **metadata** ([ObjectMeta](#))

ObjectMeta may contain labels and annotations that will be copied into the PVC when creating it. No other fields are allowed and will be rejected during validation.

## ResourceClaimTemplateList

ResourceClaimTemplateList is a collection of claim templates.

---

- **apiVersion:** resource.k8s.io/v1alpha2

- **kind:** ResourceClaimTemplateList

- **metadata** ([ListMeta](#))

Standard list metadata

- **items** ([\[\]ResourceClaimTemplate](#)), required

Items is the list of resource claim templates.

## Operations

---

## get read the specified ResourceClaimTemplate

### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaimtemplates/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaimTemplate
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ResourceClaimTemplate](#)): OK

401: Unauthorized

## list list or watch objects of kind ResourceClaimTemplate

### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaimtemplates

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)

- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ResourceClaimTemplateList](#)): OK

401: Unauthorized

## list list or watch objects of kind ResourceClaimTemplate

### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/resourceclaimtemplates

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)

- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ResourceClaimTemplateList](#)): OK

401: Unauthorized

## create create a ResourceClaimTemplate

### HTTP Request

POST /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaimtemplates

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ResourceClaimTemplate](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ResourceClaimTemplate](#)): OK

201 ([ResourceClaimTemplate](#)): Created

202 ([ResourceClaimTemplate](#)): Accepted

401: Unauthorized

## update replace the specified ResourceClaimTemplate

### HTTP Request

PUT /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaimtemplates/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaimTemplate
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ResourceClaimTemplate](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)

- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClaimTemplate](#)): OK

201 ([ResourceClaimTemplate](#)): Created

401: Unauthorized

## patch partially update the specified ResourceClaimTemplate

### HTTP Request

PATCH /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaimtemplates/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaimTemplate
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClaimTemplate](#)): OK

201 ([ResourceClaimTemplate](#)): Created

401: Unauthorized

## delete delete a ResourceClaimTemplate

### HTTP Request

DELETE /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaimtemplates/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClaimTemplate
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([ResourceClaimTemplate](#)): OK

202 ([ResourceClaimTemplate](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ResourceClaimTemplate

### HTTP Request

DELETE /apis/resource.k8s.io/v1alpha2/namespaces/{namespace}/resourceclaimtemplates

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean



[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ResourceClass v1alpha2

ResourceClass is used by administrators to influence how resources are allocated.

apiVersion: resource.k8s.io/v1alpha2

import "k8s.io/api/resource/v1alpha2"

## ResourceClass

ResourceClass is used by administrators to influence how resources are allocated.

This is an alpha type and requires enabling the DynamicResourceAllocation feature gate.

- 
- **apiVersion**: resource.k8s.io/v1alpha2

- **kind**: ResourceClass

- **metadata** ([ObjectMeta](#))

Standard object metadata

- **driverName** (string), required

DriverName defines the name of the dynamic resource driver that is used for allocation of a ResourceClaim that uses this class.

Resource drivers have a unique name in forward domain order (acme.example.com).

- **parametersRef** (ResourceClassParametersReference)

ParametersRef references an arbitrary separate object that may hold parameters that will be used by the driver when allocating a resource that uses this class. A dynamic resource driver can distinguish between parameters stored here and those stored in ResourceClaimSpec.

*ResourceClassParametersReference contains enough information to let you locate the parameters for a ResourceClass.*

- **parametersRef.kind** (string), required

Kind is the type of resource being referenced. This is the same value as in the parameter object's metadata.

- **parametersRef.name** (string), required

Name is the name of resource being referenced.

- **parametersRef.apiGroup** (string)

APIGroup is the group for the resource being referenced. It is empty for the core API. This matches the group in the APIVersion that is used when creating the resources.

- **parametersRef.namespace** (string)

Namespace that contains the referenced resource. Must be empty for cluster-scoped resources and non-empty for namespaced resources.

- **suitableNodes** (NodeSelector)

Only nodes matching the selector will be considered by the scheduler when trying to find a Node that fits a Pod when that Pod uses a ResourceClaim that has not been allocated yet.

Setting this field is optional. If null, all nodes are candidates.

*A node selector represents the union of the results of one or more label queries over a set of nodes; that is, it represents the OR of the selectors represented by the node selector terms.*

- **suitableNodes.nodeSelectorTerms** ([]NodeSelectorTerm), required

Required. A list of node selector terms. The terms are ORed.

*A null or empty node selector term matches no objects. The requirements of them are ANDed. The TopologySelectorTerm type implements a subset of the NodeSelectorTerm.*

- **suitableNodes.nodeSelectorTerms.matchExpressions**  
([]NodeSelectorRequirement)

A list of node selector requirements by node's labels.

- **suitableNodes.nodeSelectorTerms.matchFields**  
([]NodeSelectorRequirement)

A list of node selector requirements by node's fields.

## ResourceClassList

ResourceClassList is a collection of classes.

- 
- **apiVersion:** resource.k8s.io/v1alpha2
  - **kind:** ResourceClassList

- **metadata** ([ListMeta](#))
- **items** ([\[\]ResourceClass](#)), required  
Items is the list of resource classes.

## Operations

---

### get read the specified ResourceClass

#### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/resourceclasses/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ResourceClass
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ResourceClass](#)): OK

401: Unauthorized

### list list or watch objects of kind ResourceClass

#### HTTP Request

GET /apis/resource.k8s.io/v1alpha2/resourceclasses

#### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ResourceClassList](#)): OK

401: Unauthorized

## create create a ResourceClass

### HTTP Request

POST /apis/resource.k8s.io/v1alpha2/resourceclasses

### Parameters

- **body**: [ResourceClass](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClass](#)): OK

201 ([ResourceClass](#)): Created

202 ([ResourceClass](#)): Accepted

401: Unauthorized

## update replace the specified ResourceClass

### HTTP Request

PUT /apis/resource.k8s.io/v1alpha2/resourceclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClass
- **body**: [ResourceClass](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClass](#)): OK

201 ([ResourceClass](#)): Created

401: Unauthorized

## patch partially update the specified ResourceClass

### HTTP Request

PATCH /apis/resource.k8s.io/v1alpha2/resourceclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClass
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceClass](#)): OK

201 ([ResourceClass](#)): Created

401: Unauthorized

## delete delete a ResourceClass

### HTTP Request

DELETE /apis/resource.k8s.io/v1alpha2/resourceclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceClass
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([ResourceClass](#)): OK

202 ([ResourceClass](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ResourceClass

### HTTP Request

DELETE /apis/resource.k8s.io/v1alpha2/resourceclasses

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Service Resources

---



## Service

Service is a named abstraction of software service (for example, mysql) consisting of local port (for example 3306) that the proxy listens on, and the selector that determines which pods will answer requests sent through the proxy.

## Endpoints

Endpoints is a collection of endpoints that implement the actual service.

## EndpointSlice

EndpointSlice represents a subset of the endpoints that implement a service.

## Ingress

Ingress is a collection of rules that allow inbound connections to reach the endpoints defined by a backend.

## IngressClass

IngressClass represents the class of the Ingress, referenced by the Ingress Spec.

# Service

Service is a named abstraction of software service (for example, mysql) consisting of local port (for example 3306) that the proxy listens on, and the selector that determines which pods will answer requests sent through the proxy.

apiVersion: v1

import "k8s.io/api/core/v1"

# Service

Service is a named abstraction of software service (for example, mysql) consisting of local port (for example 3306) that the proxy listens on, and the selector that determines which pods will answer requests sent through the proxy.

- 
- **apiVersion:** v1
  - **kind:** Service
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([ServiceSpec](#))

Spec defines the behavior of a service. <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([ServiceStatus](#))

Most recently observed status of the service. Populated by the system. Read-only. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## ServiceSpec

ServiceSpec describes the attributes that a user creates on a service.

---

- **selector** (map[string]string)

Route service traffic to pods with label keys and values matching this selector. If empty or not present, the service is assumed to have an external process managing its endpoints, which Kubernetes will not modify. Only applies to types ClusterIP, NodePort, and LoadBalancer. Ignored if type is ExternalName. More info: <https://kubernetes.io/docs/concepts/services-networking/service/>

- **ports** ([]ServicePort)

*Patch strategy: merge on key port*

*Map: unique values on keys port, protocol will be kept during a merge*

The list of ports that are exposed by this service. More info: <https://kubernetes.io/docs/concepts/services-networking/service/#virtual-ips-and-service-proxies>

*ServicePort contains information on service's port.*

- **ports.port** (int32), required

The port that will be exposed by this service.

- **ports.targetPort** (IntOrString)

Number or name of the port to access on the pods targeted by the service. Number must be in the range 1 to 65535. Name must be an IANA\_SVC\_NAME. If this is a string, it will be looked up as a named port in the target Pod's container ports. If this is not specified, the value of the 'port' field is used (an identity map). This field is ignored for services with clusterIP=None, and should be omitted or set equal to the 'port' field. More info: <https://kubernetes.io/docs/concepts/services-networking/service/#defining-a-service>

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **ports.protocol** (string)

The IP protocol for this port. Supports "TCP", "UDP", and "SCTP". Default is TCP.

**ports.name** (string)

- The name of this port within the service. This must be a DNS\_LABEL. All ports within a ServiceSpec must have unique names. When considering the endpoints for a Service, this must match the 'name' field in the EndpointPort. Optional if only one ServicePort is defined on this service.

◦ **ports.nodePort** (int32)

The port on each node on which this service is exposed when type is NodePort or LoadBalancer. Usually assigned by the system. If a value is specified, in-range, and not in use it will be used, otherwise the operation will fail. If not specified, a port will be allocated if this Service requires one. If this field is specified when creating a Service which does not need it, creation will fail. This field will be wiped when updating a Service to no longer need it (e.g. changing type from NodePort to ClusterIP). More info: <https://kubernetes.io/docs/concepts/services-networking/service/#type-nodeport>

◦ **ports.appProtocol** (string)

The application protocol for this port. This is used as a hint for implementations to offer richer behavior for protocols that they understand. This field follows standard Kubernetes label syntax. Valid values are either:

- Un-prefixed protocol names - reserved for IANA standard service names (as per RFC-6335 and <https://www.iana.org/assignments/service-names>).
- Kubernetes-defined prefixed names:
  - 'kubernetes.io/h2c' - HTTP/2 over cleartext as described in <https://www.rfc-editor.org/rfc/rfc7540>
  - 'kubernetes.io/ws' - WebSocket over cleartext as described in <https://www.rfc-editor.org/rfc/rfc6455>
  - 'kubernetes.io/wss' - WebSocket over TLS as described in <https://www.rfc-editor.org/rfc/rfc6455>
- Other protocols should use implementation-defined prefixed names such as mycompany.com/my-custom-protocol.

• **type** (string)

type determines how the Service is exposed. Defaults to ClusterIP. Valid options are ExternalName, ClusterIP, NodePort, and LoadBalancer. "ClusterIP" allocates a cluster-internal IP address for load-balancing to endpoints. Endpoints are determined by the selector or if that is not specified, by manual construction of an Endpoints object or EndpointSlice objects. If clusterIP is "None", no virtual IP is allocated and the endpoints are published as a set of endpoints rather than a virtual IP. "NodePort" builds on ClusterIP and allocates a port on every node which routes to the same endpoints as the clusterIP. "LoadBalancer" builds on NodePort and creates an external load-balancer (if supported in the current cloud) which routes to the same endpoints as the clusterIP. "ExternalName" aliases this service to the specified externalName. Several other fields do not apply to ExternalName services. More info: <https://kubernetes.io/docs/concepts/services-networking/service/#publishing-services-service-types>

- **ipFamilies** ([]string)

*Atomic: will be replaced during a merge*

IPFamilies is a list of IP families (e.g. IPv4, IPv6) assigned to this service. This field is usually assigned automatically based on cluster configuration and the ipFamilyPolicy field. If this field is specified manually, the requested family is available in the cluster, and ipFamilyPolicy allows it, it will be used; otherwise creation of the service will fail. This field is conditionally mutable: it allows for adding or removing a secondary IP family, but it does not allow changing the primary IP family of the Service. Valid values are "IPv4" and "IPv6". This field only applies to Services of types ClusterIP, NodePort, and LoadBalancer, and does apply to "headless" services. This field will be wiped when updating a Service to type ExternalName.

This field may hold a maximum of two entries (dual-stack families, in either order). These families must correspond to the values of the clusterIPs field, if specified. Both clusterIPs and ipFamilies are governed by the ipFamilyPolicy field.

- **ipFamilyPolicy** (string)

IPFamilyPolicy represents the dual-stack-ness requested or required by this Service. If there is no value provided, then this field will be set to SingleStack. Services can be "SingleStack" (a single IP family), "PreferDualStack" (two IP families on dual-stack configured clusters or a single IP family on single-stack clusters), or "RequireDualStack" (two IP families on dual-stack configured clusters, otherwise fail). The ipFamilies and clusterIPs fields depend on the value of this field. This field will be wiped when updating a service to type ExternalName.

- **clusterIP** (string)

clusterIP is the IP address of the service and is usually assigned randomly. If an address is specified manually, is in-range (as per system configuration), and is not in use, it will be allocated to the service; otherwise creation of the service will fail. This field may not be changed through updates unless the type field is also being changed to ExternalName (which requires this field to be blank) or the type field is being changed from ExternalName (in which case this field may optionally be specified, as describe above). Valid values are "None", empty string (""), or a valid IP address. Setting this to "None" makes a "headless service" (no virtual IP), which is useful when direct endpoint connections are preferred and proxying is not required. Only applies to types ClusterIP, NodePort, and LoadBalancer. If this field is specified when creating a Service of type ExternalName, creation will fail. This field will be wiped when updating a Service to type ExternalName. More info: <https://kubernetes.io/docs/concepts/services-networking/service/#virtual-ips-and-service-proxies>

- **clusterIPs** ([]string)

*Atomic: will be replaced during a merge*

ClusterIPs is a list of IP addresses assigned to this service, and are usually assigned randomly. If an address is specified manually, is in-range (as per system configuration), and is not in use, it will be allocated to the service; otherwise creation of the service will fail. This field may not be changed through updates unless the type field is also being changed to ExternalName (which requires this field to be empty) or the type field is being changed from ExternalName (in which case this field may optionally be specified, as

describe above). Valid values are "None", empty string (""), or a valid IP address. Setting this to "None" makes a "headless service" (no virtual IP), which is useful when direct endpoint connections are preferred and proxying is not required. Only applies to types ClusterIP, NodePort, and LoadBalancer. If this field is specified when creating a Service of type ExternalName, creation will fail. This field will be wiped when updating a Service to type ExternalName. If this field is not specified, it will be initialized from the clusterIP field. If this field is specified, clients must ensure that clusterIPs[0] and clusterIP have the same value.

This field may hold a maximum of two entries (dual-stack IPs, in either order). These IPs must correspond to the values of the ipFamilies field. Both clusterIPs and ipFamilies are governed by the ipFamilyPolicy field. More info: <https://kubernetes.io/docs/concepts/services-networking/service/#virtual-ips-and-service-proxies>

- **externalIPs** ([]string)

externalIPs is a list of IP addresses for which nodes in the cluster will also accept traffic for this service. These IPs are not managed by Kubernetes. The user is responsible for ensuring that traffic arrives at a node with this IP. A common example is external load-balancers that are not part of the Kubernetes system.

- **sessionAffinity** (string)

Supports "ClientIP" and "None". Used to maintain session affinity. Enable client IP based session affinity. Must be ClientIP or None. Defaults to None. More info: <https://kubernetes.io/docs/concepts/services-networking/service/#virtual-ips-and-service-proxies>

- **loadBalancerIP** (string)

Only applies to Service Type: LoadBalancer. This feature depends on whether the underlying cloud-provider supports specifying the loadBalancerIP when a load balancer is created. This field will be ignored if the cloud-provider does not support the feature. Deprecated: This field was under-specified and its meaning varies across implementations. Using it is non-portable and it may not support dual-stack. Users are encouraged to use implementation-specific annotations when available.

- **loadBalancerSourceRanges** ([]string)

If specified and supported by the platform, this will restrict traffic through the cloud-provider load-balancer will be restricted to the specified client IPs. This field will be ignored if the cloud-provider does not support the feature." More info: <https://kubernetes.io/docs/tasks/access-application-cluster/create-external-load-balancer/>

- **loadBalancerClass** (string)

loadBalancerClass is the class of the load balancer implementation this Service belongs to. If specified, the value of this field must be a label-style identifier, with an optional prefix, e.g. "internal-vip" or "example.com/internal-vip". Unprefixed names are reserved for end-users. This field can only be set when the Service type is 'LoadBalancer'. If not set, the default load balancer implementation is used, today this is typically done through the cloud provider integration, but should apply for any default implementation. If set, it is assumed that a load balancer implementation is watching for Services with a matching class. Any default load balancer implementation (e.g. cloud providers) should ignore

Services that set this field. This field can only be set when creating or updating a Service to type 'LoadBalancer'. Once set, it can not be changed. This field will be wiped when a service is updated to a non 'LoadBalancer' type.

- **externalName** (string)

externalName is the external reference that discovery mechanisms will return as an alias for this service (e.g. a DNS CNAME record). No proxying will be involved. Must be a lowercase RFC-1123 hostname (<https://tools.ietf.org/html/rfc1123>) and requires type to be "ExternalName".

- **externalTrafficPolicy** (string)

externalTrafficPolicy describes how nodes distribute service traffic they receive on one of the Service's "externally-facing" addresses (NodePorts, ExternalIPs, and LoadBalancer IPs). If set to "Local", the proxy will configure the service in a way that assumes that external load balancers will take care of balancing the service traffic between nodes, and so each node will deliver traffic only to the node-local endpoints of the service, without masquerading the client source IP. (Traffic mistakenly sent to a node with no endpoints will be dropped.) The default value, "Cluster", uses the standard behavior of routing to all endpoints evenly (possibly modified by topology and other features). Note that traffic sent to an External IP or LoadBalancer IP from within the cluster will always get "Cluster" semantics, but clients sending to a NodePort from within the cluster may need to take traffic policy into account when picking a node.

- **internalTrafficPolicy** (string)

InternalTrafficPolicy describes how nodes distribute service traffic they receive on the ClusterIP. If set to "Local", the proxy will assume that pods only want to talk to endpoints of the service on the same node as the pod, dropping the traffic if there are no local endpoints. The default value, "Cluster", uses the standard behavior of routing to all endpoints evenly (possibly modified by topology and other features).

- **healthCheckNodePort** (int32)

healthCheckNodePort specifies the healthcheck nodePort for the service. This only applies when type is set to LoadBalancer and externalTrafficPolicy is set to Local. If a value is specified, is in-range, and is not in use, it will be used. If not specified, a value will be automatically allocated. External systems (e.g. load-balancers) can use this port to determine if a given node holds endpoints for this service or not. If this field is specified when creating a Service which does not need it, creation will fail. This field will be wiped when updating a Service to no longer need it (e.g. changing type). This field cannot be updated once set.

- **publishNotReadyAddresses** (boolean)

publishNotReadyAddresses indicates that any agent which deals with endpoints for this Service should disregard any indications of ready/not-ready. The primary use case for setting this field is for a StatefulSet's Headless Service to propagate SRV DNS records for its Pods for the purpose of peer discovery. The Kubernetes controllers that generate Endpoints and EndpointSlice resources for Services interpret this to mean that all endpoints are considered "ready" even if the Pods themselves are not. Agents which consume only Kubernetes generated endpoints through the Endpoints or EndpointSlice resources can safely assume this behavior.

### **sessionAffinityConfig** (SessionAffinityConfig)

- sessionAffinityConfig contains the configurations of session affinity.

*SessionAffinityConfig represents the configurations of session affinity.*

- **sessionAffinityConfig.clientIP** (ClientIPConfig)

clientIP contains the configurations of Client IP based session affinity.

*ClientIPConfig represents the configurations of Client IP based session affinity.*

- **sessionAffinityConfig.clientIP.timeoutSeconds** (int32)

timeoutSeconds specifies the seconds of ClientIP type session sticky time.

The value must be >0 && <=86400(for 1 day) if ServiceAffinity == "ClientIP".

Default value is 10800(for 3 hours).

- **allocateLoadBalancerNodePorts** (boolean)

allocateLoadBalancerNodePorts defines if NodePorts will be automatically allocated for services with type LoadBalancer. Default is "true". It may be set to "false" if the cluster load-balancer does not rely on NodePorts. If the caller requests specific NodePorts (by specifying a value), those requests will be respected, regardless of this field. This field may only be set for services with type LoadBalancer and will be cleared if the type is changed to any other type.

## ServiceStatus

ServiceStatus represents the current status of a service.

---

- **conditions** ([]Condition)

*Patch strategy: merge on key type*

*Map: unique values on key type will be kept during a merge*

Current service state

*Condition contains details for one aspect of the current state of this API Resource.*

- **conditions.lastTransitionTime** (Time), required

lastTransitionTime is the last time the condition transitioned from one status to another. This should be when the underlying condition changed. If that is not known, then using the time when the API field changed is acceptable.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string), required

message is a human readable message indicating details about the transition. This may be an empty string.

- **conditions.reason** (string), required

reason contains a programmatic identifier indicating the reason for the condition's last transition. Producers of specific condition types may define expected values and meanings for this field, and whether the values are considered a guaranteed API. The value should be a CamelCase string. This field may not be empty.

- **conditions.status** (string), required

status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

type of condition in CamelCase or in foo.example.com/CamelCase.

- **conditions.observedGeneration** (int64)

observedGeneration represents the .metadata.generation that the condition was set based upon. For instance, if .metadata.generation is currently 12, but the .status.conditions[x].observedGeneration is 9, the condition is out of date with respect to the current state of the instance.

- **loadBalancer** (LoadBalancerStatus)

LoadBalancer contains the current status of the load-balancer, if one is present.

*LoadBalancerStatus represents the status of a load-balancer.*

- **loadBalancer.ingress** ([]LoadBalancerIngress)

Ingress is a list containing ingress points for the load-balancer. Traffic intended for the service should be sent to these ingress points.

*LoadBalancerIngress represents the status of a load-balancer ingress point: traffic intended for the service should be sent to an ingress point.*

- **loadBalancer.ingress.hostname** (string)

Hostname is set for load-balancer ingress points that are DNS based (typically AWS load-balancers)

- **loadBalancer.ingress.ip** (string)

IP is set for load-balancer ingress points that are IP based (typically GCE or OpenStack load-balancers)

- **loadBalancer.ingress.ipMode** (string)

IPMode specifies how the load-balancer IP behaves, and may only be specified when the ip field is specified. Setting this to "VIP" indicates that traffic is delivered to the node with the destination set to the load-balancer's IP and port. Setting this to "Proxy" indicates that traffic is delivered to the



node or pod with the destination set to the node's IP and node port or the pod's IP and port. Service implementations may use this information to adjust traffic routing.

- **loadBalancer.ingress.ports** ([]PortStatus)

*Atomic: will be replaced during a merge*

Ports is a list of records of service ports If used, every port defined in the service should have an entry in it

\*\*

- **loadBalancer.ingress.ports.port** (int32), required

Port is the port number of the service port of which status is recorded here

- **loadBalancer.ingress.ports.protocol** (string), required

Protocol is the protocol of the service port of which status is recorded here The supported values are: "TCP", "UDP", "SCTP"

- **loadBalancer.ingress.ports.error** (string)

Error is to record the problem with the service port The format of the error shall comply with the following rules: - built-in error values shall be specified in this file and those shall use CamelCase names

- cloud provider specific error values must have names that comply with the format foo.example.com/CamelCase.

## ServiceList

ServiceList holds a list of services.

- 
- **apiVersion:** v1
  - **kind:** ServiceList
  - **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([][Service](#)), required

List of services

## Operations

---

## get read the specified Service

### HTTP Request

GET /api/v1/namespaces/{namespace}/services/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Service
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Service](#)): OK

401: Unauthorized

## get read status of the specified Service

### HTTP Request

GET /api/v1/namespaces/{namespace}/services/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Service
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Service](#)): OK

401: Unauthorized

## list list or watch objects of kind Service

### HTTP Request

GET /api/v1/namespaces/{namespace}/services

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ServiceList](#)): OK

401: Unauthorized

## list list or watch objects of kind Service

## HTTP Request

GET /api/v1/services

## Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

- **watch** (*in query*): boolean

- [watch](#)

## Response

200 ([ServiceList](#)): OK

401: Unauthorized

## create create a Service

### HTTP Request

POST /api/v1/namespaces/{namespace}/services

### Parameters

- **namespace** (*in path*): string, required

- [namespace](#)

- **body**: [Service](#), required

- **dryRun** (*in query*): string

- [dryRun](#)

- **fieldManager** (*in query*): string

- [fieldManager](#)

- **fieldValidation** (*in query*): string

- [fieldValidation](#)

- **pretty** (*in query*): string

- [pretty](#)

## Response

200 ([Service](#)): OK

201 ([Service](#)): Created

202 ([Service](#)): Accepted

401: Unauthorized

## update replace the specified Service

### HTTP Request

PUT /api/v1/namespaces/{namespace}/services/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Service
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Service](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Service](#)): OK

201 ([Service](#)): Created

401: Unauthorized

## update replace status of the specified Service

### HTTP Request

PUT /api/v1/namespaces/{namespace}/services/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Service

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Service](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Service](#)): OK

201 ([Service](#)): Created

401: Unauthorized

## patch partially update the specified Service

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/services/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Service
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Service](#)): OK

201 ([Service](#)): Created

401: Unauthorized

## patch partially update status of the specified Service

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/services/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Service
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean



[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Service](#)): OK

201 ([Service](#)): Created

401: Unauthorized

## delete delete a Service

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/services/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Service
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Service](#)): OK

202 ([Service](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Service

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/services

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)

• **sendInitialEvents** (*in query*): boolean

• [sendInitialEvents](#)

• **timeoutSeconds** (*in query*): integer

• [timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Endpoints

Endpoints is a collection of endpoints that implement the actual service.

apiVersion: v1

```
import "k8s.io/api/core/v1"
```

## Endpoints

Endpoints is a collection of endpoints that implement the actual service. Example:

```
Name: "mysvc",
Subsets: [
 {
 Addresses: [{"ip": "10.10.1.1"}, {"ip": "10.10.2.2"}],
 Ports: [{"name": "a", "port": 8675}, {"name": "b", "port": 309}]
 },
 {
 Addresses: [{"ip": "10.10.3.3"}],
 Ports: [{"name": "a", "port": 93}, {"name": "b", "port": 76}]
 },
]
```

---

• **apiVersion**: v1

• **kind**: Endpoints

• **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

• **subsets** ([])EndpointSubset)

The set of all endpoints is the union of all subsets. Addresses are placed into subsets according to the IPs they share. A single address with multiple ports, some of which are

ready and some of which are not (because they come from different containers) will result in the address being displayed in different subsets for the different ports. No address will appear in both `Addresses` and `NotReadyAddresses` in the same subset. Sets of addresses and ports that comprise a service.

\*`EndpointSubset` is a group of addresses with a common set of ports. The expanded set of endpoints is the Cartesian product of `Addresses` x `Ports`. For example, given:

```
{ Addresses: [{"ip": "10.10.1.1"}, {"ip": "10.10.2.2"}], Ports: [{"name": "a", "port": 8675}, {"name": "b", "port": 309}] }
```

The resulting set of endpoints can be viewed as:

a: [ 10.10.1.1:8675, 10.10.2.2:8675 ], b: [ 10.10.1.1:309, 10.10.2.2:309 ]\*

- **subsets.addresses** ([]`EndpointAddress`)

IP addresses which offer the related ports that are marked as ready. These endpoints should be considered safe for load balancers and clients to utilize.

*EndpointAddress is a tuple that describes single IP address.*

- **subsets.addresses.ip** (string), required

The IP of this endpoint. May not be loopback (127.0.0.0/8 or ::1), link-local (169.254.0.0/16 or fe80::/10), or link-local multicast (224.0.0.0/24 or ff02::/16).

- **subsets.addresses.hostname** (string)

The Hostname of this endpoint

- **subsets.addresses.nodeName** (string)

Optional: Node hosting this endpoint. This can be used to determine endpoints local to a node.

- **subsets.addresses.targetRef** ([ObjectReference](#))

Reference to object providing the endpoint.

- **subsets.notReadyAddresses** ([]`EndpointAddress`)

IP addresses which offer the related ports but are not currently marked as ready because they have not yet finished starting, have recently failed a readiness check, or have recently failed a liveness check.

*EndpointAddress is a tuple that describes single IP address.*

- **subsets.notReadyAddresses.ip** (string), required

The IP of this endpoint. May not be loopback (127.0.0.0/8 or ::1), link-local (169.254.0.0/16 or fe80::/10), or link-local multicast (224.0.0.0/24 or ff02::/16).

- **subsets.notReadyAddresses.hostname** (string)

The Hostname of this endpoint

- **subsets.notReadyAddresses.nodeName** (string)

- Optional: Node hosting this endpoint. This can be used to determine endpoints local to a node.

- **subsets.notReadyAddresses.targetRef** ([ObjectReference](#))

- Reference to object providing the endpoint.

- **subsets.ports** ([]EndpointPort)

Port numbers available on the related IP addresses.

*EndpointPort is a tuple that describes a single port.*

- **subsets.ports.port** (int32), required

- The port number of the endpoint.

- **subsets.ports.protocol** (string)

- The IP protocol for this port. Must be UDP, TCP, or SCTP. Default is TCP.

- **subsets.ports.name** (string)

- The name of this port. This must match the 'name' field in the corresponding ServicePort. Must be a DNS\_LABEL. Optional only if one port is defined.

- **subsets.ports.appProtocol** (string)

- The application protocol for this port. This is used as a hint for implementations to offer richer behavior for protocols that they understand. This field follows standard Kubernetes label syntax. Valid values are either:

- Un-prefixed protocol names - reserved for IANA standard service names (as per RFC-6335 and <https://www.iana.org/assignments/service-names>).
- Kubernetes-defined prefixed names:
  - 'kubernetes.io/h2c' - HTTP/2 over cleartext as described in <https://www.rfc-editor.org/rfc/rfc7540>
  - 'kubernetes.io/ws' - WebSocket over cleartext as described in <https://www.rfc-editor.org/rfc/rfc6455>
  - 'kubernetes.io/wss' - WebSocket over TLS as described in <https://www.rfc-editor.org/rfc/rfc6455>
- Other protocols should use implementation-defined prefixed names such as mycompany.com/my-custom-protocol.

## EndpointsList

EndpointsList is a list of endpoints.

- 
- **apiVersion:** v1
  - **kind:** EndpointsList
  - **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]Endpoints](#)), required

List of endpoints.

## Operations

---

### get read the specified Endpoints

#### HTTP Request

GET /api/v1/namespaces/{namespace}/endpoints/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the Endpoints
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([Endpoints](#)): OK

401: Unauthorized

### list list or watch objects of kind Endpoints

#### HTTP Request

GET /api/v1/namespaces/{namespace}/endpoints

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([EndpointsList](#)): OK

401: Unauthorized

## list list or watch objects of kind Endpoints

### HTTP Request

GET /api/v1/endpoints

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)



## Response

200 ([EndpointsList](#)): OK

401: Unauthorized

## create create Endpoints

### HTTP Request

POST /api/v1/namespaces/{namespace}/endpoints

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Endpoints](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Endpoints](#)): OK

201 ([Endpoints](#)): Created

202 ([Endpoints](#)): Accepted

401: Unauthorized

## update replace the specified Endpoints

### HTTP Request

PUT /api/v1/namespaces/{namespace}/endpoints/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Endpoints
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Endpoints](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Endpoints](#)): OK

201 ([Endpoints](#)): Created

401: Unauthorized

## patch partially update the specified Endpoints

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/endpoints/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Endpoints
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Endpoints](#)): OK

201 ([Endpoints](#)): Created

401: Unauthorized

## delete delete Endpoints

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/endpoints/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Endpoints
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)

- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Endpoints

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/endpoints

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)

- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# EndpointSlice

EndpointSlice represents a subset of the endpoints that implement a service.

apiVersion: discovery.k8s.io/v1

import "k8s.io/api/discovery/v1"

## EndpointSlice

EndpointSlice represents a subset of the endpoints that implement a service. For a given service there may be multiple EndpointSlice objects, selected by labels, which must be joined to produce the full set of endpoints.

- 
- **apiVersion**: discovery.k8s.io/v1
  - **kind**: EndpointSlice
  - **metadata** ([ObjectMeta](#))

Standard object's metadata.

**addressType** (string), required

- addressType specifies the type of address carried by this EndpointSlice. All addresses in this slice must be the same type. This field is immutable after creation. The following address types are currently supported: \* IPv4: Represents an IPv4 Address. \* IPv6: Represents an IPv6 Address. \* FQDN: Represents a Fully Qualified Domain Name.

- **endpoints** ([]Endpoint), required

*Atomic: will be replaced during a merge*

endpoints is a list of unique endpoints in this slice. Each slice may include a maximum of 1000 endpoints.

*Endpoint represents a single logical "backend" implementing a service.*

- **endpoints.addresses** ([]string), required

*Set: unique values will be kept during a merge*

addresses of this endpoint. The contents of this field are interpreted according to the corresponding EndpointSlice addressType field. Consumers must handle different types of addresses in the context of their own capabilities. This must contain at least one address but no more than 100. These are all assumed to be fungible and clients may choose to only use the first element. Refer to: <https://issue.k8s.io/106267>

- **endpoints.conditions** (EndpointConditions)

conditions contains information about the current status of the endpoint.

*EndpointConditions represents the current condition of an endpoint.*

- **endpoints.conditions.ready** (boolean)

ready indicates that this endpoint is prepared to receive traffic, according to whatever system is managing the endpoint. A nil value indicates an unknown state. In most cases consumers should interpret this unknown state as ready. For compatibility reasons, ready should never be "true" for terminating endpoints, except when the normal readiness behavior is being explicitly overridden, for example when the associated Service has set the publishNotReadyAddresses flag.

- **endpoints.conditions.serving** (boolean)

serving is identical to ready except that it is set regardless of the terminating state of endpoints. This condition should be set to true for a ready endpoint that is terminating. If nil, consumers should defer to the ready condition.

- **endpoints.conditions.terminating** (boolean)

terminating indicates that this endpoint is terminating. A nil value indicates an unknown state. Consumers should interpret this unknown state to mean that the endpoint is not terminating.

**endpoints.deprecatedTopology** (map[string]string)

- deprecatedTopology contains topology information part of the v1beta1 API. This field is deprecated, and will be removed when the v1beta1 API is removed (no sooner than kubernetes v1.24). While this field can hold values, it is not writable through the v1 API, and any attempts to write to it will be silently ignored. Topology information can be found in the zone and nodeName fields instead.

- **endpoints.hints** (EndpointHints)

hints contains information associated with how an endpoint should be consumed.

*EndpointHints provides hints describing how an endpoint should be consumed.*

- **endpoints.hints.forZones** ([]ForZone)

*Atomic: will be replaced during a merge*

forZones indicates the zone(s) this endpoint should be consumed by to enable topology aware routing.

*ForZone provides information about which zones should consume this endpoint.*

- **endpoints.hints.forZones.name** (string), required

name represents the name of the zone.

- **endpoints.hostname** (string)

hostname of this endpoint. This field may be used by consumers of endpoints to distinguish endpoints from each other (e.g. in DNS names). Multiple endpoints which use the same hostname should be considered fungible (e.g. multiple A values in DNS). Must be lowercase and pass DNS Label (RFC 1123) validation.

- **endpoints.nodeName** (string)

nodeName represents the name of the Node hosting this endpoint. This can be used to determine endpoints local to a Node.

- **endpoints.targetRef** ([ObjectReference](#))

targetRef is a reference to a Kubernetes object that represents this endpoint.

- **endpoints.zone** (string)

zone is the name of the Zone this endpoint exists in.

- **ports** ([]EndpointPort)

*Atomic: will be replaced during a merge*

ports specifies the list of network ports exposed by each endpoint in this slice. Each port must have a unique name. When ports is empty, it indicates that there are no defined ports. When a port is defined with a nil port value, it indicates "all ports". Each slice may include a maximum of 100 ports.

*EndpointPort represents a Port used by an EndpointSlice*

- **ports.port** (int32)

port represents the port number of the endpoint. If this is not specified, ports are not restricted and must be interpreted in the context of the specific consumer.

- **ports.protocol** (string)

protocol represents the IP protocol for this port. Must be UDP, TCP, or SCTP. Default is TCP.

- **ports.name** (string)

name represents the name of this port. All ports in an EndpointSlice must have a unique name. If the EndpointSlice is derived from a Kubernetes service, this corresponds to the Service.ports[].name. Name must either be an empty string or pass DNS\_LABEL validation: \* must be no more than 63 characters long. \* must consist of lower case alphanumeric characters or '-'. \* must start and end with an alphanumeric character. Default is empty string.

- **ports.appProtocol** (string)

The application protocol for this port. This is used as a hint for implementations to offer richer behavior for protocols that they understand. This field follows standard Kubernetes label syntax. Valid values are either:

- Un-prefixed protocol names - reserved for IANA standard service names (as per RFC-6335 and <https://www.iana.org/assignments/service-names>).
- Kubernetes-defined prefixed names:
  - 'kubernetes.io/h2c' - HTTP/2 over cleartext as described in <https://www.rfc-editor.org/rfc/rfc7540>
  - 'kubernetes.io/ws' - WebSocket over cleartext as described in <https://www.rfc-editor.org/rfc/rfc6455>
  - 'kubernetes.io/wss' - WebSocket over TLS as described in <https://www.rfc-editor.org/rfc/rfc6455>
- Other protocols should use implementation-defined prefixed names such as mycompany.com/my-custom-protocol.

## EndpointSliceList

EndpointSliceList represents a list of endpoint slices

- 
- **apiVersion:** discovery.k8s.io/v1
  - **kind:** EndpointSliceList
  - **metadata** ([ListMeta](#))

Standard list metadata.



- **items** ([[EndpointSlice](#)]), required  
items is the list of endpoint slices

## Operations

---

### get read the specified EndpointSlice

#### HTTP Request

GET /apis/discovery.k8s.io/v1/namespaces/{namespace}/endpointslices/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the EndpointSlice
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([EndpointSlice](#)): OK

401: Unauthorized

### list list or watch objects of kind EndpointSlice

#### HTTP Request

GET /apis/discovery.k8s.io/v1/namespaces/{namespace}/endpointslices

#### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([EndpointSliceList](#)): OK

401: Unauthorized

## list list or watch objects of kind EndpointSlice

## HTTP Request

GET /apis/discovery.k8s.io/v1/endpointslices

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([EndpointSliceList](#)): OK

401: Unauthorized

## create create an EndpointSlice

### HTTP Request

POST /apis/discovery.k8s.io/v1/namespaces/{namespace}/endpointslices

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [EndpointSlice](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([EndpointSlice](#)): OK

201 ([EndpointSlice](#)): Created

202 ([EndpointSlice](#)): Accepted

401: Unauthorized

## update replace the specified EndpointSlice

### HTTP Request

PUT /apis/discovery.k8s.io/v1/namespaces/{namespace}/endpointslices/{name}

## Parameters

- **name** (*in path*): string, required  
name of the EndpointSlice
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [EndpointSlice](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([EndpointSlice](#)): OK

201 ([EndpointSlice](#)): Created

401: Unauthorized

## patch partially update the specified EndpointSlice

### HTTP Request

PATCH /apis/discovery.k8s.io/v1/namespaces/{namespace}/endpointslices/{name}

### Parameters

- **name** (*in path*): string, required  
name of the EndpointSlice
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([EndpointSlice](#)): OK

201 ([EndpointSlice](#)): Created

401: Unauthorized

## delete delete an EndpointSlice

### HTTP Request

DELETE /apis/discovery.k8s.io/v1/namespaces/{namespace}/endpointslices/{name}

### Parameters

- **name** (*in path*): string, required  
name of the EndpointSlice
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of EndpointSlice

### HTTP Request

DELETE /apis/discovery.k8s.io/v1/namespaces/{namespace}/endpointslices

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)

• **sendInitialEvents** (*in query*): boolean

• [sendInitialEvents](#)

• **timeoutSeconds** (*in query*): integer

• [timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Ingress

Ingress is a collection of rules that allow inbound connections to reach the endpoints defined by a backend.

apiVersion: networking.k8s.io/v1

import "k8s.io/api/networking/v1"

## Ingress

Ingress is a collection of rules that allow inbound connections to reach the endpoints defined by a backend. An Ingress can be configured to give services externally-reachable urls, load balance traffic, terminate SSL, offer name based virtual hosting etc.

---

• **apiVersion**: networking.k8s.io/v1

• **kind**: Ingress

• **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

• **spec** ([IngressSpec](#))

spec is the desired state of the Ingress. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

• **status** ([IngressStatus](#))

status is the current state of the Ingress. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>



# IngressSpec

IngressSpec describes the Ingress the user wishes to exist.

---

- **defaultBackend** ([IngressBackend](#))

defaultBackend is the backend that should handle requests that don't match any rule. If Rules are not specified, DefaultBackend must be specified. If DefaultBackend is not set, the handling of requests that do not match any of the rules will be up to the Ingress controller.

- **ingressClassName** (string)

ingressClassName is the name of an IngressClass cluster resource. Ingress controller implementations use this field to know whether they should be serving this Ingress resource, by a transitive connection (controller -> IngressClass -> Ingress resource). Although the `kubernetes.io/ingress.class` annotation (simple constant name) was never formally defined, it was widely supported by Ingress controllers to create a direct binding between Ingress controller and Ingress resources. Newly created Ingress resources should prefer using the field. However, even though the annotation is officially deprecated, for backwards compatibility reasons, ingress controllers should still honor that annotation if present.

- **rules** ([]IngressRule)

*Atomic: will be replaced during a merge*

rules is a list of host rules used to configure the Ingress. If unspecified, or no rule matches, all traffic is sent to the default backend.

*IngressRule represents the rules mapping the paths under a specified host to the related backend services. Incoming requests are first evaluated for a host match, then routed to the backend associated with the matching IngressRuleValue.*

- **rules.host** (string)

host is the fully qualified domain name of a network host, as defined by RFC 3986. Note the following deviations from the "host" part of the URI as defined in RFC 3986: 1. IPs are not allowed. Currently an IngressRuleValue can only apply to the IP in the Spec of the parent Ingress. 2. The `:` delimiter is not respected because ports are not allowed. Currently the port of an Ingress is implicitly `:80` for http and `:443` for https. Both these may change in the future. Incoming requests are matched against the host before the IngressRuleValue. If the host is unspecified, the Ingress routes all traffic based on the specified IngressRuleValue.

host can be "precise" which is a domain name without the terminating dot of a network host (e.g. "foo.bar.com") or "wildcard", which is a domain name prefixed with a single wildcard label (e.g. ".foo.com"). The wildcard character `"` must appear by itself as the first DNS label and matches only a single label. You cannot have a wildcard label by itself (e.g. Host == "\*"). Requests will be matched against the Host field in the following way: 1. If host is precise, the request matches this rule if the http host header is equal to Host. 2. If host is a wildcard, then the request matches

this rule if the http host header is to equal to the suffix (removing the first label) of the wildcard rule.

- **rules.http** (HTTPIngressRuleValue)

*HTTPIngressRuleValue is a list of http selectors pointing to backends. In the example: http:/// -> backend where where parts of the url correspond to RFC 3986, this resource will be used to match against everything after the last '/' and before the first '?' or '#'.*

- **rules.http.paths** ([]HTTPIngressPath), required

*Atomic: will be replaced during a merge*

paths is a collection of paths that map requests to backends.

*HTTPIngressPath associates a path with a backend. Incoming urls matching the path are forwarded to the backend.*

- **rules.http.paths.backend** ([IngressBackend](#)), required

backend defines the referenced service endpoint to which the traffic will be forwarded to.

- **rules.http.paths.pathType** (string), required

pathType determines the interpretation of the path matching.

PathType can be one of the following values: \* Exact: Matches the URL path exactly. \* Prefix: Matches based on a URL path prefix split by '/'. Matching is done on a path element by element basis. A path element refers is the list of labels in the path split by the '/' separator. A request is a match for path p if every p is an element-wise prefix of p of the request path. Note that if the last element of the path is a substring of the last element in request path, it is not a match (e.g. /foo/bar matches /foo/bar/baz, but does not match /foo/barbaz).

- ImplementationSpecific: Interpretation of the Path matching is up to the IngressClass. Implementations can treat this as a separate PathType or treat it identically to Prefix or Exact path types. Implementations are required to support all path types.

- **rules.http.paths.path** (string)

path is matched against the path of an incoming request. Currently it can contain characters disallowed from the conventional "path" part of a URL as defined by RFC 3986. Paths must begin with a '/' and must be present when using PathType with value "Exact" or "Prefix".

- **tls** ([]IngressTLS)

*Atomic: will be replaced during a merge*

tls represents the TLS configuration. Currently the Ingress only supports a single TLS port, 443. If multiple members of this list specify different hosts, they will be multiplexed on the same port according to the hostname specified through the SNI TLS extension, if the ingress controller fulfilling the ingress supports SNI.

*IngressTLS describes the transport layer security associated with an ingress.*

- **tls.hosts** ([]string)

*Atomic: will be replaced during a merge*

hosts is a list of hosts included in the TLS certificate. The values in this list must match the name/s used in the tlsSecret. Defaults to the wildcard host setting for the loadbalancer controller fulfilling this Ingress, if left unspecified.

- **tls.secretName** (string)

secretName is the name of the secret used to terminate TLS traffic on port 443. Field is left optional to allow TLS routing based on SNI hostname alone. If the SNI host in a listener conflicts with the "Host" header field used by an IngressRule, the SNI host is used for termination and value of the "Host" header is used for routing.

## IngressBackend

IngressBackend describes all endpoints for a given service and port.

---

- **resource** ([TypedLocalObjectReference](#))

resource is an ObjectRef to another Kubernetes resource in the namespace of the Ingress object. If resource is specified, a service.Name and service.Port must not be specified. This is a mutually exclusive setting with "Service".

- **service** (IngressServiceBackend)

service references a service as a backend. This is a mutually exclusive setting with "Resource".

*IngressServiceBackend references a Kubernetes Service as a Backend.*

- **service.name** (string), required

name is the referenced service. The service must exist in the same namespace as the Ingress object.

- **service.port** (ServiceBackendPort)

port of the referenced service. A port name or port number is required for a IngressServiceBackend.

*ServiceBackendPort is the service port being referenced.*

- **service.port.name** (string)

name is the name of the port on the Service. This is a mutually exclusive setting with "Number".

- **service.port.number** (int32)

number is the numerical port number (e.g. 80) on the Service. This is a mutually exclusive setting with "Name".

## IngressStatus

IngressStatus describe the current state of the Ingress.

---

- **loadBalancer** (IngressLoadBalancerStatus)

loadBalancer contains the current status of the load-balancer.

*IngressLoadBalancerStatus represents the status of a load-balancer.*

- **loadBalancer.ingress** ([]IngressLoadBalancerIngress)

ingress is a list containing ingress points for the load-balancer.

*IngressLoadBalancerIngress represents the status of a load-balancer ingress point.*

- **loadBalancer.ingress.hostname** (string)

hostname is set for load-balancer ingress points that are DNS based.

- **loadBalancer.ingress.ip** (string)

ip is set for load-balancer ingress points that are IP based.

- **loadBalancer.ingress.ports** ([]IngressPortStatus)

*Atomic: will be replaced during a merge*

ports provides information about the ports exposed by this LoadBalancer.

*IngressPortStatus represents the error condition of a service port*

- **loadBalancer.ingress.ports.port** (int32), required

port is the port number of the ingress port.

- **loadBalancer.ingress.ports.protocol** (string), required

protocol is the protocol of the ingress port. The supported values are: "TCP", "UDP", "SCTP"

- **loadBalancer.ingress.ports.error** (string)

error is to record the problem with the service port The format of the error shall comply with the following rules: - built-in error values shall be specified in this file and those shall use CamelCase names

- cloud provider specific error values must have names that comply with the format foo.example.com/CamelCase.

# IngressList

IngressList is a collection of Ingress.

---

- **items** ([\[\]Ingress](#)), required

items is the list of Ingress.

- **apiVersion** (string)

APIVersion defines the versioned schema of this representation of an object. Servers should convert recognized schemas to the latest internal value, and may reject unrecognized values. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources>

- **kind** (string)

Kind is a string value representing the REST resource this object represents. Servers may infer this from the endpoint the client submits requests to. Cannot be updated. In CamelCase. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **metadata** ([ListMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

## Operations

---

### get read the specified Ingress

#### HTTP Request

GET /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the Ingress
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Ingress](#)): OK

401: Unauthorized

## get read status of the specified Ingress

### HTTP Request

GET /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Ingress
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Ingress](#)): OK

401: Unauthorized

## list list or watch objects of kind Ingress

### HTTP Request

GET /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)

- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([IngressList](#)): OK

401: Unauthorized

## list list or watch objects of kind Ingress

### HTTP Request

GET /apis/networking.k8s.io/v1/ingresses

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)

- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([IngressList](#)): OK

401: Unauthorized

## create create an Ingress

### HTTP Request

POST /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses



## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Ingress](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Ingress](#)): OK

201 ([Ingress](#)): Created

202 ([Ingress](#)): Accepted

401: Unauthorized

## update replace the specified Ingress

### HTTP Request

PUT /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Ingress
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Ingress](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Ingress](#)): OK

201 ([Ingress](#)): Created

401: Unauthorized

## update replace status of the specified Ingress

### HTTP Request

PUT /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Ingress
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Ingress](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Ingress](#)): OK

201 ([Ingress](#)): Created

401: Unauthorized

## patch partially update the specified Ingress

### HTTP Request

PATCH /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Ingress
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Ingress](#)): OK

201 ([Ingress](#)): Created

401: Unauthorized

## patch partially update status of the specified Ingress

### HTTP Request

PATCH /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Ingress
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Ingress](#)): OK

201 ([Ingress](#)): Created

401: Unauthorized

## delete delete an Ingress

### HTTP Request

DELETE /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Ingress
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Ingress

### HTTP Request

DELETE /apis/networking.k8s.io/v1/namespaces/{namespace}/ingresses

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# IngressClass

IngressClass represents the class of the Ingress, referenced by the Ingress Spec.

apiVersion: networking.k8s.io/v1

import "k8s.io/api/networking/v1"

## IngressClass

IngressClass represents the class of the Ingress, referenced by the Ingress Spec. The `ingressclass.kubernetes.io/is-default-class` annotation can be used to indicate that an IngressClass should be considered default. When a single IngressClass resource has this annotation set to true, new Ingress resources without a class specified will be assigned this default class.

- 
- **apiVersion:** networking.k8s.io/v1

- **kind:** IngressClass

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([IngressClassSpec](#))

spec is the desired state of the IngressClass. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## IngressClassSpec

IngressClassSpec provides information about the class of an Ingress.

- 
- **controller** (string)

controller refers to the name of the controller that should handle this class. This allows for different "flavors" that are controlled by the same controller. For example, you may have different parameters for the same implementing controller. This should be specified as a domain-prefixed path no more than 250 characters in length, e.g. "acme.io/ingress-controller". This field is immutable.

- **parameters** ([IngressClassParametersReference](#))

parameters is a link to a custom resource containing additional configuration for the controller. This is optional if the controller does not require extra parameters.

*IngressClassParametersReference identifies an API object. This can be used to specify a cluster or namespace-scoped resource.*

- **parameters.kind** (string), required

kind is the type of resource being referenced.

- **parameters.name** (string), required

name is the name of resource being referenced.

- **parameters.apiGroup** (string)

apiGroup is the group for the resource being referenced. If APIGroup is not specified, the specified Kind must be in the core API group. For any other third-party types, APIGroup is required.

- **parameters.namespace** (string)

namespace is the namespace of the resource being referenced. This field is required when scope is set to "Namespace" and must be unset when scope is set to "Cluster".

- **parameters.scope** (string)

scope represents if this refers to a cluster or namespace scoped resource. This may be set to "Cluster" (default) or "Namespace".

## IngressClassList

IngressClassList is a collection of IngressClasses.

---

- **apiVersion:** networking.k8s.io/v1

- **kind:** IngressClassList

- **metadata** ([ListMeta](#))

Standard list metadata.

- **items** ([\[\]IngressClass](#)), required

items is the list of IngressClasses.

## Operations

---

### get read the specified IngressClass

#### HTTP Request

GET /apis/networking.k8s.io/v1/ingressclasses/{name}

#### Parameters

- **name** (*in path*): string, required

name of the IngressClass

- **pretty** (*in query*): string



[pretty](#)

## Response

200 ([IngressClass](#)): OK

401: Unauthorized

## list list or watch objects of kind IngressClass

## HTTP Request

GET /apis/networking.k8s.io/v1/ingressclasses

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([IngressClassList](#)): OK

401: Unauthorized

## create create an IngressClass

### HTTP Request

POST /apis/networking.k8s.io/v1/ingressclasses

### Parameters

- **body**: [IngressClass](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([IngressClass](#)): OK

201 ([IngressClass](#)): Created

202 ([IngressClass](#)): Accepted

401: Unauthorized

## update replace the specified IngressClass

### HTTP Request

PUT /apis/networking.k8s.io/v1/ingressclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the IngressClass
- **body**: [IngressClass](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([IngressClass](#)): OK

201 ([IngressClass](#)): Created

401: Unauthorized

## patch partially update the specified IngressClass

### HTTP Request

PATCH /apis/networking.k8s.io/v1/ingressclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the IngressClass
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([IngressClass](#)): OK

201 ([IngressClass](#)): Created

401: Unauthorized

## delete delete an IngressClass

### HTTP Request

DELETE /apis/networking.k8s.io/v1/ingressclasses/{name}

### Parameters

- **name** (*in path*): string, required

name of the IngressClass

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of IngressClass

### HTTP Request

DELETE /apis/networking.k8s.io/v1/ingressclasses

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Config and Storage Resources

---

## [ConfigMap](#)

ConfigMap holds configuration data for pods to consume.

## [Secret](#)

Secret holds secret data of a certain type.

## [Volume](#)

Volume represents a named volume in a pod that may be accessed by any container in the pod.

## [PersistentVolumeClaim](#)

PersistentVolumeClaim is a user's request for and claim to a persistent volume.

## [PersistentVolume](#)

PersistentVolume (PV) is a storage resource provisioned by an administrator.

## [StorageClass](#)

StorageClass describes the parameters for a class of storage for which PersistentVolumes can be dynamically provisioned.

## [VolumeAttachment](#)

VolumeAttachment captures the intent to attach or detach the specified volume to/from the specified node.

## [CSIDriver](#)

CSIDriver captures information about a Container Storage Interface (CSI) volume driver deployed on the cluster.

## [CSINode](#)

CSINode holds information about all CSI drivers installed on a node.

## [CSIStorageCapacity](#)

CSIStorageCapacity stores the result of one CSI GetCapacity call.

# ConfigMap

ConfigMap holds configuration data for pods to consume.

```
apiVersion: v1
```

```
import "k8s.io/api/core/v1"
```

# ConfigMap

ConfigMap holds configuration data for pods to consume.

---

- **apiVersion:** v1
- **kind:** ConfigMap
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **binaryData** (map[string][]byte)

BinaryData contains the binary data. Each key must consist of alphanumeric characters, '-', '\_' or '.'. BinaryData can contain byte sequences that are not in the UTF-8 range. The keys stored in BinaryData must not overlap with the ones in the Data field, this is enforced during validation process. Using this field will require 1.10+ apiserver and kubelet.

- **data** (map[string]string)

Data contains the configuration data. Each key must consist of alphanumeric characters, '-', '\_' or '.'. Values with non-UTF-8 byte sequences must use the BinaryData field. The keys stored in Data must not overlap with the keys in the BinaryData field, this is enforced during validation process.

- **immutable** (boolean)

Immutable, if set to true, ensures that data stored in the ConfigMap cannot be updated (only object metadata can be modified). If not set to true, the field can be modified at any time. Defaulted to nil.

## ConfigMapList

ConfigMapList is a resource containing a list of ConfigMap objects.

---

- **apiVersion:** v1
- **kind:** ConfigMapList
- **metadata** ([ListMeta](#))

More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]ConfigMap](#)), required

Items is the list of ConfigMaps.

## Operations

---

### get read the specified ConfigMap

#### HTTP Request

GET /api/v1/namespaces/{namespace}/configmaps/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ConfigMap
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)



## Response

200 ([ConfigMap](#)): OK

401: Unauthorized

## list list or watch objects of kind ConfigMap

### HTTP Request

GET /api/v1/namespaces/{namespace}/configmaps

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ConfigMapList](#)): OK

401: Unauthorized

## list list or watch objects of kind ConfigMap

## HTTP Request

GET /api/v1/configmaps

## Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ConfigMapList](#)): OK

401: Unauthorized

## create create a ConfigMap

### HTTP Request

POST /api/v1/namespaces/{namespace}/configmaps

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ConfigMap](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ConfigMap](#)): OK

201 ([ConfigMap](#)): Created

202 ([ConfigMap](#)): Accepted

401: Unauthorized

## update replace the specified ConfigMap

### HTTP Request

PUT /api/v1/namespaces/{namespace}/configmaps/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ConfigMap
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ConfigMap](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ConfigMap](#)): OK

201 ([ConfigMap](#)): Created

401: Unauthorized

## patch partially update the specified ConfigMap

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/configmaps/{name}

## Parameters

- **name** (*in path*): string, required  
name of the ConfigMap
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ConfigMap](#)): OK

201 ([ConfigMap](#)): Created

401: Unauthorized

## delete delete a ConfigMap

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/configmaps/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ConfigMap
- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [DeleteOptions](#)
- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ConfigMap

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/configmaps

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [DeleteOptions](#)

- **continue** (*in query*): string

[continue](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Secret

Secret holds secret data of a certain type.

```
apiVersion: v1
```

```
import "k8s.io/api/core/v1"
```

# Secret

Secret holds secret data of a certain type. The total bytes of the values in the Data field must be less than MaxSecretSize bytes.

---

- **apiVersion:** v1
- **kind:** Secret
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **data** (map[string][]byte)

Data contains the secret data. Each key must consist of alphanumeric characters, '-', '\_' or '.'. The serialized form of the secret data is a base64 encoded string, representing the arbitrary (possibly non-string) data value here. Described in <https://tools.ietf.org/html/rfc4648#section-4>

- **immutable** (boolean)

Immutable, if set to true, ensures that data stored in the Secret cannot be updated (only object metadata can be modified). If not set to true, the field can be modified at any time. Defaulted to nil.

- **stringData** (map[string]string)

stringData allows specifying non-binary secret data in string form. It is provided as a write-only input field for convenience. All keys and values are merged into the data field on write, overwriting any existing values. The stringData field is never output when reading from the API.

- **type** (string)

Used to facilitate programmatic handling of secret data. More info: <https://kubernetes.io/docs/concepts/configuration/secret/#secret-types>

# SecretList

SecretList is a list of Secret.

---

- **apiVersion:** v1
- **kind:** SecretList
- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>



**items** ([[Secret](#)]), required

- Items is a list of secret objects. More info: <https://kubernetes.io/docs/concepts/configuration/secret>

## Operations

---

### get read the specified Secret

#### HTTP Request

GET /api/v1/namespaces/{namespace}/secrets/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the Secret
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([Secret](#)): OK

401: Unauthorized

### list list or watch objects of kind Secret

#### HTTP Request

GET /api/v1/namespaces/{namespace}/secrets

#### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([SecretList](#)): OK

401: Unauthorized

## list list or watch objects of kind Secret

## HTTP Request

GET /api/v1/secrets

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([SecretList](#)): OK

401: Unauthorized

## create create a Secret

### HTTP Request

POST /api/v1/namespaces/{namespace}/secrets

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Secret](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Secret](#)): OK

201 ([Secret](#)): Created

202 ([Secret](#)): Accepted

401: Unauthorized

## update replace the specified Secret

### HTTP Request

PUT /api/v1/namespaces/{namespace}/secrets/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Secret
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Secret](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Secret](#)): OK

201 ([Secret](#)): Created

401: Unauthorized

## patch partially update the specified Secret

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/secrets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Secret
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Secret](#)): OK

201 ([Secret](#)): Created

401: Unauthorized

## delete delete a Secret

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/secrets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Secret
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Secret

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/secrets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Volume

Volume represents a named volume in a pod that may be accessed by any container in the pod.

```
import "k8s.io/api/core/v1"
```

## Volume

Volume represents a named volume in a pod that may be accessed by any container in the pod.

- 
- **name** (string), required

name of the volume. Must be a DNS\_LABEL and unique within the pod. More info:  
<https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

## Exposed Persistent volumes

- **persistentVolumeClaim** (PersistentVolumeClaimVolumeSource)

`persistentVolumeClaimVolumeSource` represents a reference to a `PersistentVolumeClaim` in the same namespace. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#persistentvolumeclaims>

*PersistentVolumeClaimVolumeSource references the user's PVC in the same namespace. This volume finds the bound PV and mounts that volume for the pod. A PersistentVolumeClaimVolumeSource is, essentially, a wrapper around another type of volume that is owned by someone else (the system).*

- **persistentVolumeClaim.claimName** (string), required

`claimName` is the name of a `PersistentVolumeClaim` in the same namespace as the pod using this volume. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#persistentvolumeclaims>

- **persistentVolumeClaim.readOnly** (boolean)

`readOnly` Will force the `ReadOnly` setting in `VolumeMounts`. Default false.



## Projections

- **configMap** (ConfigMapVolumeSource)

configMap represents a configMap that should populate this volume

\*Adapts a ConfigMap into a volume.

The contents of the target ConfigMap's Data field will be presented in a volume as files using the keys in the Data field as the file names, unless the items element is populated with specific mappings of keys to paths. ConfigMap volumes support ownership management and SELinux relabeling.\*

- **configMap.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **configMap.optional** (boolean)

optional specify whether the ConfigMap or its keys must be defined

- **configMap.defaultMode** (int32)

defaultMode is optional: mode bits used to set permissions on created files by default. Must be an octal value between 0000 and 0777 or a decimal value between 0 and 511. YAML accepts both octal and decimal values, JSON requires decimal values for mode bits. Defaults to 0644. Directories within the path are not affected by this setting. This might be in conflict with other options that affect the file mode, like fsGroup, and the result can be other mode bits set.

- **configMap.items** ([][KeyToPath](#))

items if unspecified, each key-value pair in the Data field of the referenced ConfigMap will be projected into the volume as a file whose name is the key and content is the value. If specified, the listed keys will be projected into the specified paths, and unlisted keys will not be present. If a key is specified which is not present in the ConfigMap, the volume setup will error unless it is marked optional. Paths must be relative and may not contain the '..' path or start with '..'.

- **secret** (SecretVolumeSource)

secret represents a secret that should populate this volume. More info: <https://kubernetes.io/docs/concepts/storage/volumes#secret>

\*Adapts a Secret into a volume.

The contents of the target Secret's Data field will be presented in a volume as files using the keys in the Data field as the file names. Secret volumes support ownership management and SELinux relabeling.\*

- **secret.secretName** (string)

secretName is the name of the secret in the pod's namespace to use. More info: <https://kubernetes.io/docs/concepts/storage/volumes#secret>

**secret.optional** (boolean)

- optional field specify whether the Secret or its keys must be defined
- **secret.defaultMode** (int32)

defaultMode is Optional: mode bits used to set permissions on created files by default. Must be an octal value between 0000 and 0777 or a decimal value between 0 and 511. YAML accepts both octal and decimal values, JSON requires decimal values for mode bits. Defaults to 0644. Directories within the path are not affected by this setting. This might be in conflict with other options that affect the file mode, like fsGroup, and the result can be other mode bits set.

- **secret.items** ([][KeyToPath](#))

items If unspecified, each key-value pair in the Data field of the referenced Secret will be projected into the volume as a file whose name is the key and content is the value. If specified, the listed keys will be projected into the specified paths, and unlisted keys will not be present. If a key is specified which is not present in the Secret, the volume setup will error unless it is marked optional. Paths must be relative and may not contain the '..' path or start with '..'.

- **downwardAPI** (DownwardAPIVolumeSource)

downwardAPI represents downward API about the pod that should populate this volume

*DownwardAPIVolumeSource represents a volume containing downward API info. Downward API volumes support ownership management and SELinux relabeling.*

- **downwardAPI.defaultMode** (int32)

Optional: mode bits to use on created files by default. Must be a Optional: mode bits used to set permissions on created files by default. Must be an octal value between 0000 and 0777 or a decimal value between 0 and 511. YAML accepts both octal and decimal values, JSON requires decimal values for mode bits. Defaults to 0644. Directories within the path are not affected by this setting. This might be in conflict with other options that affect the file mode, like fsGroup, and the result can be other mode bits set.

- **downwardAPI.items** ([][DownwardAPIVolumeFile](#))

Items is a list of downward API volume file

- **projected** (ProjectedVolumeSource)

projected items for all in one resources secrets, configmaps, and downward API

*Represents a projected volume source*

- **projected.defaultMode** (int32)

defaultMode are the mode bits used to set permissions on created files by default. Must be an octal value between 0000 and 0777 or a decimal value between 0 and 511. YAML accepts both octal and decimal values, JSON requires decimal values for mode bits. Directories within the path are not affected by this setting. This might be

in conflict with other options that affect the file mode, like fsGroup, and the result can be other mode bits set.

- **projected.sources** ([]VolumeProjection)

sources is the list of volume projections

*Projection that may be projected along with other supported volume types*

- **projected.sources.configMap** (ConfigMapProjection)

configMap information about the configMap data to project

\*Adapts a ConfigMap into a projected volume.

The contents of the target ConfigMap's Data field will be presented in a projected volume as files using the keys in the Data field as the file names, unless the items element is populated with specific mappings of keys to paths. Note that this is identical to a configmap volume source without the default mode.\*

- **projected.sources.configMap.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **projected.sources.configMap.optional** (boolean)

optional specify whether the ConfigMap or its keys must be defined

- **projected.sources.configMap.items** ([][KeyToPath](#))

items if unspecified, each key-value pair in the Data field of the referenced ConfigMap will be projected into the volume as a file whose name is the key and content is the value. If specified, the listed keys will be projected into the specified paths, and unlisted keys will not be present. If a key is specified which is not present in the ConfigMap, the volume setup will error unless it is marked optional. Paths must be relative and may not contain the '..' path or start with '..'.

- **projected.sources.downwardAPI** (DownwardAPIProjection)

downwardAPI information about the downwardAPI data to project

*Represents downward API info for projecting into a projected volume. Note that this is identical to a downwardAPI volume source without the default mode.*

- **projected.sources.downwardAPI.items**  
([][DownwardAPIVolumeFile](#))

Items is a list of DownwardAPIVolume file

- **projected.sources.secret** (SecretProjection)

secret information about the secret data to project

\*Adapts a secret into a projected volume.

The contents of the target Secret's Data field will be presented in a projected volume as files using the keys in the Data field as the file names. Note that this is identical to a secret volume source without the default mode.\*

- **projected.sources.secret.name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **projected.sources.secret.optional** (boolean)

optional field specify whether the Secret or its key must be defined

- **projected.sources.secret.items** ([][KeyToPath](#))

items if unspecified, each key-value pair in the Data field of the referenced Secret will be projected into the volume as a file whose name is the key and content is the value. If specified, the listed keys will be projected into the specified paths, and unlisted keys will not be present. If a key is specified which is not present in the Secret, the volume setup will error unless it is marked optional. Paths must be relative and may not contain the '..' path or start with '..'.

- **projected.sources.serviceAccountToken**  
(ServiceAccountTokenProjection)

serviceAccountToken is information about the serviceAccountToken data to project

*ServiceAccountTokenProjection represents a projected service account token volume. This projection can be used to insert a service account token into the pods runtime filesystem for use against APIs (Kubernetes API Server or otherwise).*

- **projected.sources.serviceAccountToken.path** (string), required

path is the path relative to the mount point of the file to project the token into.

- **projected.sources.serviceAccountToken.audience** (string)

audience is the intended audience of the token. A recipient of a token must identify itself with an identifier specified in the audience of the token, and otherwise should reject the token. The audience defaults to the identifier of the apiserver.

- **projected.sources.serviceAccountToken.expirationSeconds**  
(int64)

expirationSeconds is the requested duration of validity of the service account token. As the token approaches expiration, the kubelet volume plugin will proactively rotate the service account token. The kubelet will start trying to rotate the token if the token is older than 80 percent

of its time to live or if the token is older than 24 hours. Defaults to 1 hour and must be at least 10 minutes.

## Local / Temporary Directory

- **emptyDir** (EmptyDirVolumeSource)

emptyDir represents a temporary directory that shares a pod's lifetime. More info: <https://kubernetes.io/docs/concepts/storage/volumes#emptydir>

*Represents an empty directory for a pod. Empty directory volumes support ownership management and SELinux relabeling.*

- **emptyDir.medium** (string)

medium represents what type of storage medium should back this directory. The default is "" which means to use the node's default medium. Must be an empty string (default) or Memory. More info: <https://kubernetes.io/docs/concepts/storage/volumes#emptydir>

- **emptyDir.sizeLimit** ([Quantity](#))

sizeLimit is the total amount of local storage required for this EmptyDir volume. The size limit is also applicable for memory medium. The maximum usage on memory medium EmptyDir would be the minimum value between the SizeLimit specified here and the sum of memory limits of all containers in a pod. The default is nil which means that the limit is undefined. More info: <https://kubernetes.io/docs/concepts/storage/volumes#emptydir>

- **hostPath** (HostPathVolumeSource)

hostPath represents a pre-existing file or directory on the host machine that is directly exposed to the container. This is generally used for system agents or other privileged things that are allowed to see the host machine. Most containers will NOT need this. More info: <https://kubernetes.io/docs/concepts/storage/volumes#hostpath>

*Represents a host path mapped into a pod. Host path volumes do not support ownership management or SELinux relabeling.*

- **hostPath.path** (string), required

path of the directory on the host. If the path is a symlink, it will follow the link to the real path. More info: <https://kubernetes.io/docs/concepts/storage/volumes#hostpath>

- **hostPath.type** (string)

type for HostPath Volume Defaults to "" More info: <https://kubernetes.io/docs/concepts/storage/volumes#hostpath>

## Persistent volumes

- **awsElasticBlockStore** (AWSElasticBlockStoreVolumeSource)

`awsElasticBlockStore` represents an AWS Disk resource that is attached to a kubelet's host machine and then exposed to the pod. More info: <https://kubernetes.io/docs/concepts/storage/volumes#awselasticblockstore>

\*Represents a Persistent Disk resource in AWS.

An AWS EBS disk must exist before mounting to a container. The disk must also be in the same AWS zone as the kubelet. An AWS EBS disk can only be mounted as read/write once. AWS EBS volumes support ownership management and SELinux relabeling.\*

- **`awsElasticBlockStore.volumeID`** (string), required

`volumeID` is unique ID of the persistent disk resource in AWS (Amazon EBS volume). More info: <https://kubernetes.io/docs/concepts/storage/volumes#awselasticblockstore>

- **`awsElasticBlockStore.fsType`** (string)

`fsType` is the filesystem type of the volume that you want to mount. Tip: Ensure that the filesystem type is supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://kubernetes.io/docs/concepts/storage/volumes#awselasticblockstore>

- **`awsElasticBlockStore.partition`** (int32)

`partition` is the partition in the volume that you want to mount. If omitted, the default is to mount by volume name. Examples: For volume `/dev/sda1`, you specify the partition as "1". Similarly, the volume partition for `/dev/sda` is "0" (or you can leave the property empty).

- **`awsElasticBlockStore.readOnly`** (boolean)

`readOnly` value true will force the `readOnly` setting in `VolumeMounts`. More info: <https://kubernetes.io/docs/concepts/storage/volumes#awselasticblockstore>

- **`azureDisk`** (AzureDiskVolumeSource)

`azureDisk` represents an Azure Data Disk mount on the host and bind mount to the pod.

*AzureDisk represents an Azure Data Disk mount on the host and bind mount to the pod.*

- **`azureDisk.diskName`** (string), required

`diskName` is the Name of the data disk in the blob storage

- **`azureDisk.diskURI`** (string), required

`diskURI` is the URI of data disk in the blob storage

- **`azureDisk.cachingMode`** (string)

`cachingMode` is the Host Caching mode: None, Read Only, Read Write.

- **`azureDisk.fsType`** (string)

fsType is Filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **azureDisk.kind** (string)

kind expected values are Shared: multiple blob disks per storage account Dedicated: single blob disk per storage account Managed: azure managed data disk (only in managed availability set). defaults to shared

- **azureDisk.readOnly** (boolean)

readOnly Defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

- **azureFile** (AzureFileVolumeSource)

azureFile represents an Azure File Service mount on the host and bind mount to the pod.

*AzureFile represents an Azure File Service mount on the host and bind mount to the pod.*

- **azureFile.secretName** (string), required

secretName is the name of secret that contains Azure Storage Account Name and Key

- **azureFile.shareName** (string), required

shareName is the azure share Name

- **azureFile.readOnly** (boolean)

readOnly defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

- **cephfs** (CephFSVolumeSource)

cephFS represents a Ceph FS mount on the host that shares a pod's lifetime

*Represents a Ceph Filesystem mount that lasts the lifetime of a pod Cephfs volumes do not support ownership management or SELinux relabeling.*

- **cephfs.monitors** ([]string), required

monitors is Required: Monitors is a collection of Ceph monitors More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

- **cephfs.path** (string)

path is Optional: Used as the mounted root, rather than the full Ceph tree, default is /

- **cephfs.readOnly** (boolean)



readOnly is Optional: Defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts. More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

- **cephfs.secretFile** (string)

secretFile is Optional: SecretFile is the path to key ring for User, default is /etc/ceph/user.secret More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

- **cephfs.secretRef** ([LocalObjectReference](#))

secretRef is Optional: SecretRef is reference to the authentication secret for User, default is empty. More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

- **cephfs.user** (string)

user is optional: User is the rados user name, default is admin More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

- **cinder** (CinderVolumeSource)

cinder represents a cinder volume attached and mounted on kubelets host machine. More info: <https://examples.k8s.io/mysql-cinder-pd/README.md>

*Represents a cinder volume resource in Openstack. A Cinder volume must exist before mounting to a container. The volume must also be in the same region as the kubelet. Cinder volumes support ownership management and SELinux relabeling.*

- **cinder.volumeID** (string), required

volumeID used to identify the volume in cinder. More info: <https://examples.k8s.io/mysql-cinder-pd/README.md>

- **cinder.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://examples.k8s.io/mysql-cinder-pd/README.md>

- **cinder.readOnly** (boolean)

readOnly defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts. More info: <https://examples.k8s.io/mysql-cinder-pd/README.md>

- **cinder.secretRef** ([LocalObjectReference](#))

secretRef is optional: points to a secret object containing parameters used to connect to OpenStack.

- **csi** (CSIVolumeSource)



csi (Container Storage Interface) represents ephemeral storage that is handled by certain external CSI drivers (Beta feature).

*Represents a source location of a volume to mount, managed by an external CSI driver*

- **csi.driver** (string), required

driver is the name of the CSI driver that handles this volume. Consult with your admin for the correct name as registered in the cluster.

- **csi.fsType** (string)

fsType to mount. Ex. "ext4", "xfs", "ntfs". If not provided, the empty value is passed to the associated CSI driver which will determine the default filesystem to apply.

- **csi.nodePublishSecretRef** ([LocalObjectReference](#))

nodePublishSecretRef is a reference to the secret object containing sensitive information to pass to the CSI driver to complete the CSI NodePublishVolume and NodeUnpublishVolume calls. This field is optional, and may be empty if no secret is required. If the secret object contains more than one secret, all secret references are passed.

- **csi.readOnly** (boolean)

readOnly specifies a read-only configuration for the volume. Defaults to false (read/write).

- **csi.volumeAttributes** (map[string]string)

volumeAttributes stores driver-specific properties that are passed to the CSI driver. Consult your driver's documentation for supported values.

- **ephemeral** (EphemeralVolumeSource)

ephemeral represents a volume that is handled by a cluster storage driver. The volume's lifecycle is tied to the pod that defines it - it will be created before the pod starts, and deleted when the pod is removed.

Use this if: a) the volume is only needed while the pod runs, b) features of normal volumes like restoring from snapshot or capacity tracking are needed, c) the storage driver is specified through a storage class, and d) the storage driver supports dynamic volume provisioning through a PersistentVolumeClaim (see EphemeralVolumeSource for more information on the connection between this volume type and PersistentVolumeClaim).

Use PersistentVolumeClaim or one of the vendor-specific APIs for volumes that persist for longer than the lifecycle of an individual pod.

Use CSI for light-weight local ephemeral volumes if the CSI driver is meant to be used that way - see the documentation of the driver for more information.

A pod can use both types of ephemeral volumes and persistent volumes at the same time.

*Represents an ephemeral volume that is handled by a normal storage driver.*

- **ephemeral.volumeClaimTemplate** ([PersistentVolumeClaimTemplate](#))

Will be used to create a stand-alone PVC to provision the volume. The pod in which this EphemeralVolumeSource is embedded will be the owner of the PVC, i.e. the PVC will be deleted together with the pod. The name of the PVC will be \<pod name>-\<volume name> where \<volume name> is the name from the PodSpec.Volumes array entry. Pod validation will reject the pod if the concatenated name is not valid for a PVC (for example, too long).

An existing PVC with that name that is not owned by the pod will *not* be used for the pod to avoid using an unrelated volume by mistake. Starting the pod is then blocked until the unrelated PVC is removed. If such a pre-created PVC is meant to be used by the pod, the PVC has to be updated with an owner reference to the pod once the pod exists. Normally this should not be necessary, but it may be useful when manually reconstructing a broken cluster.

This field is read-only and no changes will be made by Kubernetes to the PVC after it has been created.

Required, must not be nil.

*PersistentVolumeClaimTemplate is used to produce PersistentVolumeClaim objects as part of an EphemeralVolumeSource.*

- **ephemeral.volumeClaimTemplate.spec** ([PersistentVolumeClaimSpec](#)), required

The specification for the PersistentVolumeClaim. The entire content is copied unchanged into the PVC that gets created from this template. The same fields as in a PersistentVolumeClaim are also valid here.

- **ephemeral.volumeClaimTemplate.metadata** ([ObjectMeta](#))

May contain labels and annotations that will be copied into the PVC when creating it. No other fields are allowed and will be rejected during validation.

- **fc** ([FCVolumeSource](#))

fc represents a Fibre Channel resource that is attached to a kubelet's host machine and then exposed to the pod.

*Represents a Fibre Channel volume. Fibre Channel volumes can only be mounted as read/write once. Fibre Channel volumes support ownership management and SELinux relabeling.*

- **fc.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **fc.lun** (int32)

lun is Optional: FC target lun number

◦ **fc.readOnly** (boolean)

readOnly is Optional: Defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

◦ **fc.targetWWNs** ([]string)

targetWWNs is Optional: FC target worldwide names (WWNs)

◦ **fc.wwids** ([]string)

wwids Optional: FC volume world wide identifiers (wwids) Either wwids or combination of targetWWNs and lun must be set, but not both simultaneously.

• **flexVolume** (FlexVolumeSource)

flexVolume represents a generic volume resource that is provisioned/attached using an exec based plugin.

*FlexVolume represents a generic volume resource that is provisioned/attached using an exec based plugin.*

◦ **flexVolume.driver** (string), required

driver is the name of the driver to use for this volume.

◦ **flexVolume.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". The default filesystem depends on FlexVolume script.

◦ **flexVolume.options** (map[string]string)

options is Optional: this field holds extra command options if any.

◦ **flexVolume.readOnly** (boolean)

readOnly is Optional: defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

◦ **flexVolume.secretRef** ([LocalObjectReference](#))

secretRef is Optional: secretRef is reference to the secret object containing sensitive information to pass to the plugin scripts. This may be empty if no secret object is specified. If the secret object contains more than one secret, all secrets are passed to the plugin scripts.

• **flocker** (FlockerVolumeSource)

flocker represents a Flocker volume attached to a kubelet's host machine. This depends on the Flocker control service being running

*Represents a Flocker volume mounted by the Flocker agent. One and only one of `datasetName` and `datasetUUID` should be set. Flocker volumes do not support ownership management or SELinux relabeling.*

- **flocker.datasetName** (string)

`datasetName` is Name of the dataset stored as metadata -> name on the dataset for Flocker should be considered as deprecated

- **flocker.datasetUUID** (string)

`datasetUUID` is the UUID of the dataset. This is unique identifier of a Flocker dataset

- **gcePersistentDisk** (GCEPersistentDiskVolumeSource)

`gcePersistentDisk` represents a GCE Disk resource that is attached to a kubelet's host machine and then exposed to the pod. More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

\*Represents a Persistent Disk resource in Google Compute Engine.

A GCE PD must exist before mounting to a container. The disk must also be in the same GCE project and zone as the kubelet. A GCE PD can only be mounted as read/write once or read-only many times. GCE PDs support ownership management and SELinux relabeling.\*

- **gcePersistentDisk.pdName** (string), required

`pdName` is unique name of the PD resource in GCE. Used to identify the disk in GCE. More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

- **gcePersistentDisk.fsType** (string)

`fsType` is filesystem type of the volume that you want to mount. Tip: Ensure that the filesystem type is supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

- **gcePersistentDisk.partition** (int32)

`partition` is the partition in the volume that you want to mount. If omitted, the default is to mount by volume name. Examples: For volume `/dev/sda1`, you specify the partition as "1". Similarly, the volume partition for `/dev/sda` is "0" (or you can leave the property empty). More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

- **gcePersistentDisk.readOnly** (boolean)

`readOnly` here will force the `ReadOnly` setting in `VolumeMounts`. Defaults to false. More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

- **glusterfs** (GlusterfsVolumeSource)

glusterfs represents a Glusterfs mount on the host that shares a pod's lifetime. More info: <https://examples.k8s.io/volumes/glusterfs/README.md>

*Represents a Glusterfs mount that lasts the lifetime of a pod. Glusterfs volumes do not support ownership management or SELinux relabeling.*

- **glusterfs.endpoints** (string), required

endpoints is the endpoint name that details Glusterfs topology. More info: <https://examples.k8s.io/volumes/glusterfs/README.md#create-a-pod>

- **glusterfs.path** (string), required

path is the Glusterfs volume path. More info: <https://examples.k8s.io/volumes/glusterfs/README.md#create-a-pod>

- **glusterfs.readOnly** (boolean)

readOnly here will force the Glusterfs volume to be mounted with read-only permissions. Defaults to false. More info: <https://examples.k8s.io/volumes/glusterfs/README.md#create-a-pod>

- **iscsi** (ISCSIVolumeSource)

iscsi represents an iSCSI Disk resource that is attached to a kubelet's host machine and then exposed to the pod. More info: <https://examples.k8s.io/volumes/iscsi/README.md>

*Represents an iSCSI disk. iSCSI volumes can only be mounted as read/write once. iSCSI volumes support ownership management and SELinux relabeling.*

- **iscsi.iqn** (string), required

iqn is the target iSCSI Qualified Name.

- **iscsi.lun** (int32), required

lun represents iSCSI Target Lun number.

- **iscsi.targetPortal** (string), required

targetPortal is iSCSI Target Portal. The Portal is either an IP or ip\_addr:port if the port is other than default (typically TCP ports 860 and 3260).

- **iscsi.chapAuthDiscovery** (boolean)

chapAuthDiscovery defines whether support iSCSI Discovery CHAP authentication

- **iscsi.chapAuthSession** (boolean)

chapAuthSession defines whether support iSCSI Session CHAP authentication

- **iscsi.fsType** (string)

fsType is the filesystem type of the volume that you want to mount. Tip: Ensure that the filesystem type is supported by the host operating system. Examples:

"ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://kubernetes.io/docs/concepts/storage/volumes#iscsi>

- **iscsi.initiatorName** (string)

initiatorName is the custom iSCSI Initiator Name. If initiatorName is specified with iscsiInterface simultaneously, new iSCSI interface <target portal>:<volume name> will be created for the connection.

- **iscsi.iscsiInterface** (string)

iscsiInterface is the interface Name that uses an iSCSI transport. Defaults to 'default' (tcp).

- **iscsi.portals** ([]string)

portals is the iSCSI Target Portal List. The portal is either an IP or ip\_addr:port if the port is other than default (typically TCP ports 860 and 3260).

- **iscsi.readOnly** (boolean)

readOnly here will force the ReadOnly setting in VolumeMounts. Defaults to false.

- **iscsi.secretRef** ([LocalObjectReference](#))

secretRef is the CHAP Secret for iSCSI target and initiator authentication

- **nfs** (NFSVolumeSource)

nfs represents an NFS mount on the host that shares a pod's lifetime More info: <https://kubernetes.io/docs/concepts/storage/volumes#nfs>

*Represents an NFS mount that lasts the lifetime of a pod. NFS volumes do not support ownership management or SELinux relabeling.*

- **nfs.path** (string), required

path that is exported by the NFS server. More info: <https://kubernetes.io/docs/concepts/storage/volumes#nfs>

- **nfs.server** (string), required

server is the hostname or IP address of the NFS server. More info: <https://kubernetes.io/docs/concepts/storage/volumes#nfs>

- **nfs.readOnly** (boolean)

readOnly here will force the NFS export to be mounted with read-only permissions. Defaults to false. More info: <https://kubernetes.io/docs/concepts/storage/volumes#nfs>

- **photonPersistentDisk** (PhotonPersistentDiskVolumeSource)

photonPersistentDisk represents a PhotonController persistent disk attached and mounted on kubelets host machine

*Represents a Photon Controller persistent disk resource.*

- **photonPersistentDisk.pdID** (string), required

pdID is the ID that identifies Photon Controller persistent disk

- **photonPersistentDisk.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **portworxVolume** (PortworxVolumeSource)

portworxVolume represents a portworx volume attached and mounted on kubelets host machine

*PortworxVolumeSource represents a Portworx volume resource.*

- **portworxVolume.volumeID** (string), required

volumeID uniquely identifies a Portworx volume

- **portworxVolume.fsType** (string)

fsType represents the filesystem type to mount Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs". Implicitly inferred to be "ext4" if unspecified.

- **portworxVolume.readOnly** (boolean)

readOnly defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

- **quobyte** (QuobyteVolumeSource)

quobyte represents a Quobyte mount on the host that shares a pod's lifetime

*Represents a Quobyte mount that lasts the lifetime of a pod. Quobyte volumes do not support ownership management or SELinux relabeling.*

- **quobyte.registry** (string), required

registry represents a single or multiple Quobyte Registry services specified as a string as host:port pair (multiple entries are separated with commas) which acts as the central registry for volumes

- **quobyte.volume** (string), required

volume is a string that references an already created Quobyte volume by name.

- **quobyte.group** (string)

group to map volume access to Default is no group

- **quobyte.readOnly** (boolean)

readOnly here will force the Quobyte volume to be mounted with read-only permissions. Defaults to false.

- **quobyte.tenant** (string)

tenant owning the given Quobyte volume in the Backend Used with dynamically provisioned Quobyte volumes, value is set by the plugin

- **quobyte.user** (string)

user to map volume access to Defaults to serviceaccount user

- **rbd** (RBDVolumeSource)

rbd represents a Rados Block Device mount on the host that shares a pod's lifetime. More info: <https://examples.k8s.io/volumes/rbd/README.md>

*Represents a Rados Block Device mount that lasts the lifetime of a pod. RBD volumes support ownership management and SELinux relabeling.*

- **rbd.image** (string), required

image is the rados image name. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

- **rbd.monitors** ([]string), required

monitors is a collection of Ceph monitors. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

- **rbd.fsType** (string)

fsType is the filesystem type of the volume that you want to mount. Tip: Ensure that the filesystem type is supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://kubernetes.io/docs/concepts/storage/volumes#rbd>

- **rbd.keyring** (string)

keyring is the path to key ring for RBDUser. Default is /etc/ceph/keyring. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

- **rbd.pool** (string)

pool is the rados pool name. Default is rbd. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

- **rbd.readOnly** (boolean)

readOnly here will force the ReadOnly setting in VolumeMounts. Defaults to false. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

- **rbd.secretRef** ([LocalObjectReference](#))



secretRef is name of the authentication secret for RBDUser. If provided overrides keyring. Default is nil. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

- **rbd.user** (string)

user is the rados user name. Default is admin. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

- **scaleIO** (ScaleIOVolumeSource)

scaleIO represents a ScaleIO persistent volume attached and mounted on Kubernetes nodes.

*ScaleIOVolumeSource represents a persistent ScaleIO volume*

- **scaleIO.gateway** (string), required

gateway is the host address of the ScaleIO API Gateway.

- **scaleIO.secretRef** ([LocalObjectReference](#)), required

secretRef references to the secret for ScaleIO user and other sensitive information. If this is not provided, Login operation will fail.

- **scaleIO.system** (string), required

system is the name of the storage system as configured in ScaleIO.

- **scaleIO.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Default is "xfs".

- **scaleIO.protectionDomain** (string)

protectionDomain is the name of the ScaleIO Protection Domain for the configured storage.

- **scaleIO.readOnly** (boolean)

readOnly Defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

- **scaleIO.sslEnabled** (boolean)

sslEnabled Flag enable/disable SSL communication with Gateway, default false

- **scaleIO.storageMode** (string)

storageMode indicates whether the storage for a volume should be ThickProvisioned or ThinProvisioned. Default is ThinProvisioned.

- **scaleIO.storagePool** (string)

storagePool is the ScaleIO Storage Pool associated with the protection domain.

**scaleIO.volumeName** (string)

- volumeName is the name of a volume already created in the ScaleIO system that is associated with this volume source.

- **storageos** (StorageOSVolumeSource)

storageOS represents a StorageOS volume attached and mounted on Kubernetes nodes.

*Represents a StorageOS persistent volume resource.*

- **storageos.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **storageos.readOnly** (boolean)

readOnly defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

- **storageos.secretRef** ([LocalObjectReference](#))

secretRef specifies the secret to use for obtaining the StorageOS API credentials. If not specified, default values will be attempted.

- **storageos.volumeName** (string)

volumeName is the human-readable name of the StorageOS volume. Volume names are only unique within a namespace.

- **storageos.volumeNamespace** (string)

volumeNamespace specifies the scope of the volume within StorageOS. If no namespace is specified then the Pod's namespace will be used. This allows the Kubernetes name scoping to be mirrored within StorageOS for tighter integration. Set VolumeName to any name to override the default behaviour. Set to "default" if you are not using namespaces within StorageOS. Namespaces that do not pre-exist within StorageOS will be created.

- **vsphereVolume** (VsphereVirtualDiskVolumeSource)

vsphereVolume represents a vSphere volume attached and mounted on kubelets host machine

*Represents a vSphere volume resource.*

- **vsphereVolume.volumePath** (string), required

volumePath is the path that identifies vSphere volume vmdk

- **vsphereVolume.fsType** (string)

fsType is filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **vsphereVolume.storagePolicyID** (string)

storagePolicyID is the storage Policy Based Management (SPBM) profile ID associated with the StoragePolicyName.

- **vsphereVolume.storagePolicyName** (string)

storagePolicyName is the storage Policy Based Management (SPBM) profile name.

## Deprecated

- **gitRepo** (GitRepoVolumeSource)

gitRepo represents a git repository at a particular revision. DEPRECATED: GitRepo is deprecated. To provision a container with a git repo, mount an EmptyDir into an InitContainer that clones the repo using git, then mount the EmptyDir into the Pod's container.

\*Represents a volume that is populated with the contents of a git repository. Git repo volumes do not support ownership management. Git repo volumes support SELinux relabeling.

DEPRECATED: GitRepo is deprecated. To provision a container with a git repo, mount an EmptyDir into an InitContainer that clones the repo using git, then mount the EmptyDir into the Pod's container.\*

- **gitRepo.repository** (string), required

repository is the URL

- **gitRepo.directory** (string)

directory is the target directory name. Must not contain or start with '..'. If '' is supplied, the volume directory will be the git repository. Otherwise, if specified, the volume will contain the git repository in the subdirectory with the given name.

- **gitRepo.revision** (string)

revision is the commit hash for the specified revision.

## DownwardAPIVolumeFile

DownwardAPIVolumeFile represents information to create the file containing the pod field

---

- **path** (string), required

Required: Path is the relative path name of the file to be created. Must not be absolute or contain the '..' path. Must be utf-8 encoded. The first item of the relative path must not start with '..'

**fieldRef** ([ObjectFieldSelector](#))

- Required: Selects a field of the pod: only annotations, labels, name and namespace are supported.
- **mode** (int32)  
  
Optional: mode bits used to set permissions on this file, must be an octal value between 0000 and 0777 or a decimal value between 0 and 511. YAML accepts both octal and decimal values, JSON requires decimal values for mode bits. If not specified, the volume defaultMode will be used. This might be in conflict with other options that affect the file mode, like fsGroup, and the result can be other mode bits set.
- **resourceFieldRef** ([ResourceFieldSelector](#))  
  
Selects a resource of the container: only resources limits and requests (limits.cpu, limits.memory, requests.cpu and requests.memory) are currently supported.

## KeyToPath

Maps a string key to a path within a volume.

---

- **key** (string), required  
  
key is the key to project.
- **path** (string), required  
  
path is the relative path of the file to map the key to. May not be an absolute path. May not contain the path element '..'. May not start with the string '..'.
- **mode** (int32)  
  
mode is Optional: mode bits used to set permissions on this file. Must be an octal value between 0000 and 0777 or a decimal value between 0 and 511. YAML accepts both octal and decimal values, JSON requires decimal values for mode bits. If not specified, the volume defaultMode will be used. This might be in conflict with other options that affect the file mode, like fsGroup, and the result can be other mode bits set.

## PersistentVolumeClaim

PersistentVolumeClaim is a user's request for and claim to a persistent volume.

apiVersion: v1

import "k8s.io/api/core/v1"

## PersistentVolumeClaim

PersistentVolumeClaim is a user's request for and claim to a persistent volume

- 
- **apiVersion:** v1
  - **kind:** PersistentVolumeClaim
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([PersistentVolumeClaimSpec](#))

spec defines the desired characteristics of a volume requested by a pod author. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#persistentvolumeclaims>

- **status** ([PersistentVolumeClaimStatus](#))

status represents the current information/status of a persistent volume claim. Read-only. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#persistentvolumeclaims>

## PersistentVolumeClaimSpec

PersistentVolumeClaimSpec describes the common attributes of storage devices and allows a Source for provider-specific attributes

---

- **accessModes** ([]string)

accessModes contains the desired access modes the volume should have. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#access-modes-1>

- **selector** ([LabelSelector](#))

selector is a label query over volumes to consider for binding.

- **resources** (ResourceRequirements)

resources represents the minimum resources the volume should have. If RecoverVolumeExpansionFailure feature is enabled users are allowed to specify resource requirements that are lower than previous value but must still be higher than capacity recorded in the status field of the claim. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#resources>

*ResourceRequirements describes the compute resource requirements.*

- **resources.claims** ([]ResourceClaim)

*Map: unique values on key name will be kept during a merge*

Claims lists the names of resources, defined in spec.resourceClaims, that are used by this container.

This is an alpha field and requires enabling the `DynamicResourceAllocation` feature gate.

This field is immutable. It can only be set for containers.

*ResourceClaim references one entry in PodSpec.ResourceClaims.*

- **resources.claims.name** (string), required

Name must match the name of one entry in `pod.spec.resourceClaims` of the Pod where this field is used. It makes that resource available inside a container.

- **resources.limits** (map[string][Quantity](#))

Limits describes the maximum amount of compute resources allowed. More info: <https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/>

- **resources.requests** (map[string][Quantity](#))

Requests describes the minimum amount of compute resources required. If Requests is omitted for a container, it defaults to Limits if that is explicitly specified, otherwise to an implementation-defined value. Requests cannot exceed Limits. More info: <https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/>

- **volumeName** (string)

volumeName is the binding reference to the PersistentVolume backing this claim.

- **storageClassName** (string)

storageClassName is the name of the StorageClass required by the claim. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#class-1>

- **volumeMode** (string)

volumeMode defines what type of volume is required by the claim. Value of Filesystem is implied when not included in claim spec.

## Beta level

- **dataSource** ([TypedLocalObjectReference](#))

dataSource field can be used to specify either: \* An existing VolumeSnapshot object (snapshot.storage.k8s.io/VolumeSnapshot) \* An existing PVC (PersistentVolumeClaim) If the provisioner or an external controller can support the specified data source, it will create a new volume based on the contents of the specified data source. When the AnyVolumeDataSource feature gate is enabled, dataSource contents will be copied to dataSourceRef, and dataSourceRef contents will be copied to dataSource when dataSourceRef.namespace is not specified. If the namespace is specified, then dataSourceRef will not be copied to dataSource.

- **dataSourceRef** (TypedObjectReference)

`dataSourceRef` specifies the object from which to populate the volume with data, if a non-empty volume is desired. This may be any object from a non-empty API group (non core object) or a `PersistentVolumeClaim` object. When this field is specified, volume binding will only succeed if the type of the specified object matches some installed volume populator or dynamic provisioner. This field will replace the functionality of the `dataSource` field and as such if both fields are non-empty, they must have the same value. For backwards compatibility, when namespace isn't specified in `dataSourceRef`, both fields (`dataSource` and `dataSourceRef`) will be set to the same value automatically if one of them is empty and the other is non-empty. When namespace is specified in `dataSourceRef`, `dataSource` isn't set to the same value and must be empty. There are three important differences between `dataSource` and `dataSourceRef`: \* While `dataSource` only allows two specific types of objects, `dataSourceRef` allows any non-core object, as well as `PersistentVolumeClaim` objects.

- While `dataSource` ignores disallowed values (dropping them), `dataSourceRef` preserves all values, and generates an error if a disallowed value is specified.
- While `dataSource` only allows local objects, `dataSourceRef` allows objects in any namespaces. (Beta) Using this field requires the `AnyVolumeDataSource` feature gate to be enabled. (Alpha) Using the namespace field of `dataSourceRef` requires the `CrossNamespaceVolumeDataSource` feature gate to be enabled.

\*\*

- **`dataSourceRef.kind`** (string), required

Kind is the type of resource being referenced

- **`dataSourceRef.name`** (string), required

Name is the name of resource being referenced

- **`dataSourceRef.apiGroup`** (string)

APIGroup is the group for the resource being referenced. If APIGroup is not specified, the specified Kind must be in the core API group. For any other third-party types, APIGroup is required.

- **`dataSourceRef.namespace`** (string)

Namespace is the namespace of resource being referenced Note that when a namespace is specified, a `gateway.networking.k8s.io/ReferenceGrant` object is required in the referent namespace to allow that namespace's owner to accept the reference. See the `ReferenceGrant` documentation for details. (Alpha) This field requires the `CrossNamespaceVolumeDataSource` feature gate to be enabled.

## PersistentVolumeClaimStatus

`PersistentVolumeClaimStatus` is the current status of a persistent volume claim.

- 
- **`accessModes`** ([]string)

`accessModes` contains the actual access modes the volume backing the PVC has. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#access-modes-1>

## **allocatedResourceStatuses** (map[string]string)

allocatedResourceStatuses stores status of resource being resized for the given PVC. Key names follow standard Kubernetes label syntax. Valid values are either: \* Un-prefixed keys: - storage - the capacity of the volume. \* Custom resources must use implementation-defined prefixed names such as "example.com/my-custom-resource". Apart from above values - keys that are unprefixed or have kubernetes.io prefix are considered reserved and hence may not be used.

ClaimResourceStatus can be in any of following states: - ControllerResizeInProgress: State set when resize controller starts resizing the volume in control-plane. - ControllerResizeFailed: State set when resize has failed in resize controller with a terminal error. - NodeResizePending: State set when resize controller has finished resizing the volume but further resizing of volume is needed on the node. - NodeResizeInProgress: State set when kubelet starts resizing the volume. - NodeResizeFailed: State set when resizing has failed in kubelet with a terminal error. Transient errors don't set NodeResizeFailed. For example: if expanding a PVC for more capacity - this field can be one of the following states: - pvc.status.allocatedResourceStatus['storage'] = "ControllerResizeInProgress" - pvc.status.allocatedResourceStatus['storage'] = "ControllerResizeFailed" - pvc.status.allocatedResourceStatus['storage'] = "NodeResizePending" - pvc.status.allocatedResourceStatus['storage'] = "NodeResizeInProgress" - pvc.status.allocatedResourceStatus['storage'] = "NodeResizeFailed" When this field is not set, it means that no resize operation is in progress for the given PVC.

A controller that receives PVC update with previously unknown resourceName or ClaimResourceStatus should ignore the update for the purpose it was designed. For example - a controller that only is responsible for resizing capacity of the volume, should ignore PVC updates that change other valid resources associated with PVC.

This is an alpha field and requires enabling RecoverVolumeExpansionFailure feature.

## • **allocatedResources** (map[string>[Quantity](#))

allocatedResources tracks the resources allocated to a PVC including its capacity. Key names follow standard Kubernetes label syntax. Valid values are either: \* Un-prefixed keys: - storage - the capacity of the volume. \* Custom resources must use implementation-defined prefixed names such as "example.com/my-custom-resource". Apart from above values - keys that are unprefixed or have kubernetes.io prefix are considered reserved and hence may not be used.

Capacity reported here may be larger than the actual capacity when a volume expansion operation is requested. For storage quota, the larger value from allocatedResources and PVC.spec.resources is used. If allocatedResources is not set, PVC.spec.resources alone is used for quota calculation. If a volume expansion capacity request is lowered, allocatedResources is only lowered if there are no expansion operations in progress and if the actual volume capacity is equal or lower than the requested capacity.

A controller that receives PVC update with previously unknown resourceName should ignore the update for the purpose it was designed. For example - a controller that only is responsible for resizing capacity of the volume, should ignore PVC updates that change other valid resources associated with PVC.

This is an alpha field and requires enabling RecoverVolumeExpansionFailure feature.



• **capacity** (map[string][Quantity](#))

• capacity represents the actual resources of the underlying volume.

• **conditions** ([]PersistentVolumeClaimCondition)

*Patch strategy: merge on key type*

conditions is the current Condition of persistent volume claim. If underlying persistent volume is being resized then the Condition will be set to 'ResizeStarted'.

*PersistentVolumeClaimCondition contains details about state of pvc*

◦ **conditions.status** (string), required

◦ **conditions.type** (string), required

◦ **conditions.lastProbeTime** (Time)

lastProbeTime is the time we probed the condition.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

◦ **conditions.lastTransitionTime** (Time)

lastTransitionTime is the time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

◦ **conditions.message** (string)

message is the human-readable message indicating details about last transition.

◦ **conditions.reason** (string)

reason is a unique, this should be a short, machine understandable string that gives the reason for condition's last transition. If it reports "ResizeStarted" that means the underlying persistent volume is being resized.

• **phase** (string)

phase represents the current phase of PersistentVolumeClaim.

## PersistentVolumeClaimList

PersistentVolumeClaimList is a list of PersistentVolumeClaim items.

---

• **apiVersion**: v1

**kind:** PersistentVolumeClaimList

- 
- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]PersistentVolumeClaim](#)), required

items is a list of persistent volume claims. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#persistentvolumeclaims>

## Operations

---

### get read the specified PersistentVolumeClaim

#### HTTP Request

GET /api/v1/namespaces/{namespace}/persistentvolumeclaims/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolumeClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([PersistentVolumeClaim](#)): OK

401: Unauthorized

### get read status of the specified PersistentVolumeClaim

#### HTTP Request

GET /api/v1/namespaces/{namespace}/persistentvolumeclaims/{name}/status

#### Parameters

- **name** (*in path*): string, required

name of the PersistentVolumeClaim

- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PersistentVolumeClaim](#)): OK

401: Unauthorized

## list list or watch objects of kind PersistentVolumeClaim

### HTTP Request

GET /api/v1/namespaces/{namespace}/persistentvolumeclaims

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([PersistentVolumeClaimList](#)): OK

401: Unauthorized

## list list or watch objects of kind PersistentVolumeClaim

## HTTP Request

GET /api/v1/persistentvolumeclaims

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([PersistentVolumeClaimList](#)): OK

401: Unauthorized

## create create a PersistentVolumeClaim

### HTTP Request

POST /api/v1/namespaces/{namespace}/persistentvolumeclaims

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [PersistentVolumeClaim](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PersistentVolumeClaim](#)): OK

201 ([PersistentVolumeClaim](#)): Created

202 ([PersistentVolumeClaim](#)): Accepted

401: Unauthorized

## update replace the specified PersistentVolumeClaim

### HTTP Request

PUT /api/v1/namespaces/{namespace}/persistentvolumeclaims/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolumeClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [PersistentVolumeClaim](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PersistentVolumeClaim](#)): OK

201 ([PersistentVolumeClaim](#)): Created

401: Unauthorized

## update replace status of the specified PersistentVolumeClaim

### HTTP Request

PUT /api/v1/namespaces/{namespace}/persistentvolumeclaims/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolumeClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [PersistentVolumeClaim](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([PersistentVolumeClaim](#)): OK

201 ([PersistentVolumeClaim](#)): Created

401: Unauthorized

## patch partially update the specified PersistentVolumeClaim

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/persistentvolumeclaims/{name}

## Parameters

- **name** (*in path*): string, required  
name of the PersistentVolumeClaim
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PersistentVolumeClaim](#)): OK

201 ([PersistentVolumeClaim](#)): Created

401: Unauthorized

## patch partially update status of the specified PersistentVolumeClaim

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/persistentvolumeclaims/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the PersistentVolumeClaim
- **namespace** (*in path*): string, required



[namespace](#)

- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PersistentVolumeClaim](#)): OK

201 ([PersistentVolumeClaim](#)): Created

401: Unauthorized

## delete delete a PersistentVolumeClaim

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/persistentvolumeclaims/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolumeClaim
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([PersistentVolumeClaim](#)): OK

202 ([PersistentVolumeClaim](#)): Accepted

401: Unauthorized

## deletecollection delete collection of PersistentVolumeClaim

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/persistentvolumeclaims

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)

- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# PersistentVolume

PersistentVolume (PV) is a storage resource provisioned by an administrator.

apiVersion: v1

import "k8s.io/api/core/v1"

## PersistentVolume

PersistentVolume (PV) is a storage resource provisioned by an administrator. It is analogous to a node. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes>

- 
- **apiVersion**: v1
  - **kind**: PersistentVolume
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([PersistentVolumeSpec](#))

spec defines a specification of a persistent volume owned by the cluster. Provisioned by an administrator. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#persistent-volumes>

- **status** ([PersistentVolumeStatus](#))

status represents the current information/status for the persistent volume. Populated by the system. Read-only. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#persistent-volumes>

## PersistentVolumeSpec

PersistentVolumeSpec is the specification of a persistent volume.

---

- **accessModes** ([]string)

accessModes contains all ways the volume can be mounted. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#access-modes>

- **capacity** (map[string][Quantity](#))

capacity is the description of the persistent volume's resources and capacity. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#capacity>

- **claimRef** ([ObjectReference](#))

claimRef is part of a bi-directional binding between PersistentVolume and PersistentVolumeClaim. Expected to be non-nil when bound. claim.VolumeName is the authoritative bind between PV and PVC. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#binding>

- **mountOptions** ([]string)

mountOptions is the list of mount options, e.g. ["ro", "soft"]. Not validated - mount will simply fail if one is invalid. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes/#mount-options>

- **nodeAffinity** (VolumeNodeAffinity)

nodeAffinity defines constraints that limit what nodes this volume can be accessed from. This field influences the scheduling of pods that use this volume.

*VolumeNodeAffinity defines constraints that limit what nodes this volume can be accessed from.*

- **nodeAffinity.required** (NodeSelector)

required specifies hard node constraints that must be met.

*A node selector represents the union of the results of one or more label queries over a set of nodes; that is, it represents the OR of the selectors represented by the node selector terms.*

- **nodeAffinity.required.nodeSelectorTerms** ([]NodeSelectorTerm), required

Required. A list of node selector terms. The terms are ORed.

*A null or empty node selector term matches no objects. The requirements of them are ANDed. The TopologySelectorTerm type implements a subset of the NodeSelectorTerm.*

- **nodeAffinity.required.nodeSelectorTerms.matchExpressions**  
([]NodeSelectorRequirement)

A list of node selector requirements by node's labels.

- **nodeAffinity.required.nodeSelectorTerms.matchFields**  
([]NodeSelectorRequirement)

A list of node selector requirements by node's fields.

- **persistentVolumeReclaimPolicy** (string)

`persistentVolumeReclaimPolicy` defines what happens to a persistent volume when released from its claim. Valid options are Retain (default for manually created PersistentVolumes), Delete (default for dynamically provisioned PersistentVolumes), and Recycle (deprecated). Recycle must be supported by the volume plugin underlying this PersistentVolume. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#reclaiming>

- **storageClassName** (string)

`storageClassName` is the name of StorageClass to which this persistent volume belongs. Empty value means that this volume does not belong to any StorageClass.

- **volumeMode** (string)

`volumeMode` defines if a volume is intended to be used with a formatted filesystem or to remain in raw block state. Value of Filesystem is implied when not included in spec.

## Local

- **hostPath** (HostPathVolumeSource)

`hostPath` represents a directory on the host. Provisioned by a developer or tester. This is useful for single-node development and testing only! On-host storage is not supported in any way and WILL NOT WORK in a multi-node cluster. More info: <https://kubernetes.io/docs/concepts/storage/volumes#hostpath>

*Represents a host path mapped into a pod. Host path volumes do not support ownership management or SELinux relabeling.*

- **hostPath.path** (string), required

path of the directory on the host. If the path is a symlink, it will follow the link to the real path. More info: <https://kubernetes.io/docs/concepts/storage/volumes#hostpath>

- **hostPath.type** (string)

type for HostPath Volume Defaults to "" More info: <https://kubernetes.io/docs/concepts/storage/volumes#hostpath>

- **local** (LocalVolumeSource)

local represents directly-attached storage with node affinity

*Local represents directly-attached storage with node affinity (Beta feature)*

- **local.path** (string), required

path of the full path to the volume on the node. It can be either a directory or block device (disk, partition, ...).

- **local.fsType** (string)

fsType is the filesystem type to mount. It applies only when the Path is a block device. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". The default value is to auto-select a filesystem if unspecified.

## Persistent volumes

- **awsElasticBlockStore** (AWSElasticBlockStoreVolumeSource)

awsElasticBlockStore represents an AWS Disk resource that is attached to a kubelet's host machine and then exposed to the pod. More info: <https://kubernetes.io/docs/concepts/storage/volumes#awselasticblockstore>

\*Represents a Persistent Disk resource in AWS.

An AWS EBS disk must exist before mounting to a container. The disk must also be in the same AWS zone as the kubelet. An AWS EBS disk can only be mounted as read/write once. AWS EBS volumes support ownership management and SELinux relabeling.\*

- **awsElasticBlockStore.volumeID** (string), required

volumeID is unique ID of the persistent disk resource in AWS (Amazon EBS volume). More info: <https://kubernetes.io/docs/concepts/storage/volumes#awselasticblockstore>

- **awsElasticBlockStore.fsType** (string)

fsType is the filesystem type of the volume that you want to mount. Tip: Ensure that the filesystem type is supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://kubernetes.io/docs/concepts/storage/volumes#awselasticblockstore>

- **awsElasticBlockStore.partition** (int32)

partition is the partition in the volume that you want to mount. If omitted, the default is to mount by volume name. Examples: For volume /dev/sda1, you specify the partition as "1". Similarly, the volume partition for /dev/sda is "0" (or you can leave the property empty).

- **awsElasticBlockStore.readOnly** (boolean)

readOnly value true will force the readOnly setting in VolumeMounts. More info: <https://kubernetes.io/docs/concepts/storage/volumes#awselasticblockstore>

- **azureDisk** (AzureDiskVolumeSource)

azureDisk represents an Azure Data Disk mount on the host and bind mount to the pod.

*AzureDisk represents an Azure Data Disk mount on the host and bind mount to the pod.*

- **azureDisk.diskName** (string), required

diskName is the Name of the data disk in the blob storage

- **azureDisk.diskURI** (string), required

diskURI is the URI of data disk in the blob storage

- **azureDisk.cachingMode** (string)

cachingMode is the Host Caching mode: None, Read Only, Read Write.

- **azureDisk.fsType** (string)

fsType is Filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **azureDisk.kind** (string)

kind expected values are Shared: multiple blob disks per storage account Dedicated: single blob disk per storage account Managed: azure managed data disk (only in managed availability set). defaults to shared

- **azureDisk.readOnly** (boolean)

readOnly Defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

- **azureFile** (AzureFilePersistentVolumeSource)

azureFile represents an Azure File Service mount on the host and bind mount to the pod.

*AzureFile represents an Azure File Service mount on the host and bind mount to the pod.*

- **azureFile.secretName** (string), required

secretName is the name of secret that contains Azure Storage Account Name and Key

**azureFile.shareName** (string), required

- shareName is the azure Share Name

◦ **azureFile.readOnly** (boolean)

readOnly defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

◦ **azureFile.secretNamespace** (string)

secretNamespace is the namespace of the secret that contains Azure Storage Account Name and Key default is the same as the Pod

- **cephfs** (CephFSPersistentVolumeSource)

cephFS represents a Ceph FS mount on the host that shares a pod's lifetime

*Represents a Ceph Filesystem mount that lasts the lifetime of a pod Cephfs volumes do not support ownership management or SELinux relabeling.*

◦ **cephfs.monitors** ([]string), required

monitors is Required: Monitors is a collection of Ceph monitors More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

◦ **cephfs.path** (string)

path is Optional: Used as the mounted root, rather than the full Ceph tree, default is /

◦ **cephfs.readOnly** (boolean)

readOnly is Optional: Defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts. More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

◦ **cephfs.secretFile** (string)

secretFile is Optional: SecretFile is the path to key ring for User, default is /etc/ceph/user.secret More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

◦ **cephfs.secretRef** (SecretReference)

secretRef is Optional: SecretRef is reference to the authentication secret for User, default is empty. More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **cephfs.secretRef.name** (string)

name is unique within a namespace to reference a secret resource.



- **cephfs.secretRef.namespace** (string)

- namespace defines the space within which the secret name must be unique.

- **cephfs.user** (string)

- user is Optional: User is the rados user name, default is admin More info: <https://examples.k8s.io/volumes/cephfs/README.md#how-to-use-it>

- **cinder** (CinderPersistentVolumeSource)

cinder represents a cinder volume attached and mounted on kubelets host machine. More info: <https://examples.k8s.io/mysql-cinder-pd/README.md>

*Represents a cinder volume resource in Openstack. A Cinder volume must exist before mounting to a container. The volume must also be in the same region as the kubelet. Cinder volumes support ownership management and SELinux relabeling.*

- **cinder.volumeID** (string), required

- volumeID used to identify the volume in cinder. More info: <https://examples.k8s.io/mysql-cinder-pd/README.md>

- **cinder.fsType** (string)

- fsType Filesystem type to mount. Must be a filesystem type supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://examples.k8s.io/mysql-cinder-pd/README.md>

- **cinder.readOnly** (boolean)

- readOnly is Optional: Defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts. More info: <https://examples.k8s.io/mysql-cinder-pd/README.md>

- **cinder.secretRef** (SecretReference)

- secretRef is Optional: points to a secret object containing parameters used to connect to OpenStack.

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **cinder.secretRef.name** (string)

- name is unique within a namespace to reference a secret resource.

- **cinder.secretRef.namespace** (string)

- namespace defines the space within which the secret name must be unique.

- **csi** (CSIPersistentVolumeSource)

csi represents storage that is handled by an external CSI driver (Beta feature).

*Represents storage that is managed by an external CSI volume driver (Beta feature)*

- **csi.driver** (string), required

driver is the name of the driver to use for this volume. Required.

- **csi.volumeHandle** (string), required

volumeHandle is the unique volume name returned by the CSI volume plugin's CreateVolume to refer to the volume on all subsequent calls. Required.

- **csi.controllerExpandSecretRef** (SecretReference)

controllerExpandSecretRef is a reference to the secret object containing sensitive information to pass to the CSI driver to complete the CSI ControllerExpandVolume call. This field is optional, and may be empty if no secret is required. If the secret object contains more than one secret, all secrets are passed.

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **csi.controllerExpandSecretRef.name** (string)

name is unique within a namespace to reference a secret resource.

- **csi.controllerExpandSecretRef.namespace** (string)

namespace defines the space within which the secret name must be unique.

- **csi.controllerPublishSecretRef** (SecretReference)

controllerPublishSecretRef is a reference to the secret object containing sensitive information to pass to the CSI driver to complete the CSI ControllerPublishVolume and ControllerUnpublishVolume calls. This field is optional, and may be empty if no secret is required. If the secret object contains more than one secret, all secrets are passed.

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **csi.controllerPublishSecretRef.name** (string)

name is unique within a namespace to reference a secret resource.

- **csi.controllerPublishSecretRef.namespace** (string)

namespace defines the space within which the secret name must be unique.

- **csi.fsType** (string)

fsType to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs".

- **csi.nodeExpandSecretRef** (SecretReference)

`nodeExpandSecretRef` is a reference to the secret object containing sensitive information to pass to the CSI driver to complete the `CSI NodeExpandVolume` call. This is a beta field which is enabled default by `CSINodeExpandSecret` feature gate. This field is optional, may be omitted if no secret is required. If the secret object contains more than one secret, all secrets are passed.

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **`csi.nodeExpandSecretRef.name`** (string)

name is unique within a namespace to reference a secret resource.

- **`csi.nodeExpandSecretRef.namespace`** (string)

namespace defines the space within which the secret name must be unique.

- **`csi.nodePublishSecretRef`** (SecretReference)

`nodePublishSecretRef` is a reference to the secret object containing sensitive information to pass to the CSI driver to complete the `CSI NodePublishVolume` and `NodeUnpublishVolume` calls. This field is optional, and may be empty if no secret is required. If the secret object contains more than one secret, all secrets are passed.

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **`csi.nodePublishSecretRef.name`** (string)

name is unique within a namespace to reference a secret resource.

- **`csi.nodePublishSecretRef.namespace`** (string)

namespace defines the space within which the secret name must be unique.

- **`csi.nodeStageSecretRef`** (SecretReference)

`nodeStageSecretRef` is a reference to the secret object containing sensitive information to pass to the CSI driver to complete the `CSI NodeStageVolume` and `NodeUnstageVolume` calls. This field is optional, and may be empty if no secret is required. If the secret object contains more than one secret, all secrets are passed.

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **`csi.nodeStageSecretRef.name`** (string)

name is unique within a namespace to reference a secret resource.

- **`csi.nodeStageSecretRef.namespace`** (string)

namespace defines the space within which the secret name must be unique.

- **`csi.readOnly`** (boolean)

readOnly value to pass to ControllerPublishVolumeRequest. Defaults to false (read/write).

- **csi.volumeAttributes** (map[string]string)

volumeAttributes of the volume to publish.

- **fc** (FCVolumeSource)

fc represents a Fibre Channel resource that is attached to a kubelet's host machine and then exposed to the pod.

*Represents a Fibre Channel volume. Fibre Channel volumes can only be mounted as read/write once. Fibre Channel volumes support ownership management and SELinux relabeling.*

- **fc.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **fc.lun** (int32)

lun is Optional: FC target lun number

- **fc.readOnly** (boolean)

readOnly is Optional: Defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

- **fc.targetWWNs** ([]string)

targetWWNs is Optional: FC target worldwide names (WWNs)

- **fc.wwids** ([]string)

wwids Optional: FC volume world wide identifiers (wwids) Either wwids or combination of targetWWNs and lun must be set, but not both simultaneously.

- **flexVolume** (FlexPersistentVolumeSource)

flexVolume represents a generic volume resource that is provisioned/attached using an exec based plugin.

*FlexPersistentVolumeSource represents a generic persistent volume resource that is provisioned/attached using an exec based plugin.*

- **flexVolume.driver** (string), required

driver is the name of the driver to use for this volume.

- **flexVolume.fsType** (string)

fsType is the Filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". The default filesystem depends on FlexVolume script.

**flexVolume.options** (map[string]string)

- options is Optional: this field holds extra command options if any.

◦ **flexVolume.readOnly** (boolean)

readOnly is Optional: defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

◦ **flexVolume.secretRef** (SecretReference)

secretRef is Optional: SecretRef is reference to the secret object containing sensitive information to pass to the plugin scripts. This may be empty if no secret object is specified. If the secret object contains more than one secret, all secrets are passed to the plugin scripts.

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

▪ **flexVolume.secretRef.name** (string)

name is unique within a namespace to reference a secret resource.

▪ **flexVolume.secretRef.namespace** (string)

namespace defines the space within which the secret name must be unique.

• **flocker** (FlockerVolumeSource)

flocker represents a Flocker volume attached to a kubelet's host machine and exposed to the pod for its usage. This depends on the Flocker control service being running

*Represents a Flocker volume mounted by the Flocker agent. One and only one of datasetName and datasetUUID should be set. Flocker volumes do not support ownership management or SELinux relabeling.*

◦ **flocker.datasetName** (string)

datasetName is Name of the dataset stored as metadata -> name on the dataset for Flocker should be considered as deprecated

◦ **flocker.datasetUUID** (string)

datasetUUID is the UUID of the dataset. This is unique identifier of a Flocker dataset

• **gcePersistentDisk** (GCEPersistentDiskVolumeSource)

gcePersistentDisk represents a GCE Disk resource that is attached to a kubelet's host machine and then exposed to the pod. Provisioned by an admin. More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

\*Represents a Persistent Disk resource in Google Compute Engine.

A GCE PD must exist before mounting to a container. The disk must also be in the same GCE project and zone as the kubelet. A GCE PD can only be mounted as read/write once or read-only many times. GCE PDs support ownership management and SELinux relabeling.\*

- **gcePersistentDisk.pdName** (string), required

pdName is unique name of the PD resource in GCE. Used to identify the disk in GCE. More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

- **gcePersistentDisk.fsType** (string)

fsType is filesystem type of the volume that you want to mount. Tip: Ensure that the filesystem type is supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

- **gcePersistentDisk.partition** (int32)

partition is the partition in the volume that you want to mount. If omitted, the default is to mount by volume name. Examples: For volume /dev/sda1, you specify the partition as "1". Similarly, the volume partition for /dev/sda is "0" (or you can leave the property empty). More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

- **gcePersistentDisk.readOnly** (boolean)

readOnly here will force the ReadOnly setting in VolumeMounts. Defaults to false. More info: <https://kubernetes.io/docs/concepts/storage/volumes#gcepersistentdisk>

- **glusterfs** (GlusterfsPersistentVolumeSource)

glusterfs represents a Glusterfs volume that is attached to a host and exposed to the pod. Provisioned by an admin. More info: <https://examples.k8s.io/volumes/glusterfs/README.md>

*Represents a Glusterfs mount that lasts the lifetime of a pod. Glusterfs volumes do not support ownership management or SELinux relabeling.*

- **glusterfs.endpoints** (string), required

endpoints is the endpoint name that details Glusterfs topology. More info: <https://examples.k8s.io/volumes/glusterfs/README.md#create-a-pod>

- **glusterfs.path** (string), required

path is the Glusterfs volume path. More info: <https://examples.k8s.io/volumes/glusterfs/README.md#create-a-pod>

- **glusterfs.endpointsNamespace** (string)

endpointsNamespace is the namespace that contains Glusterfs endpoint. If this field is empty, the EndpointNamespace defaults to the same namespace as the bound

PVC. More info: <https://examples.k8s.io/volumes/glusterfs/README.md#create-a-pod>

- **glusterfs.readOnly** (boolean)

readOnly here will force the Glusterfs volume to be mounted with read-only permissions. Defaults to false. More info: <https://examples.k8s.io/volumes/glusterfs/README.md#create-a-pod>

- **iscsi** (ISCSIPersistentVolumeSource)

iscsi represents an iSCSI Disk resource that is attached to a kubelet's host machine and then exposed to the pod. Provisioned by an admin.

*ISCSIPersistentVolumeSource represents an iSCSI disk. iSCSI volumes can only be mounted as read/write once. iSCSI volumes support ownership management and SELinux relabeling.*

- **iscsi.iqn** (string), required

iqn is Target iSCSI Qualified Name.

- **iscsi.lun** (int32), required

lun is iSCSI Target Lun number.

- **iscsi.targetPortal** (string), required

targetPortal is iSCSI Target Portal. The Portal is either an IP or ip\_addr:port if the port is other than default (typically TCP ports 860 and 3260).

- **iscsi.chapAuthDiscovery** (boolean)

chapAuthDiscovery defines whether support iSCSI Discovery CHAP authentication

- **iscsi.chapAuthSession** (boolean)

chapAuthSession defines whether support iSCSI Session CHAP authentication

- **iscsi.fsType** (string)

fsType is the filesystem type of the volume that you want to mount. Tip: Ensure that the filesystem type is supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://kubernetes.io/docs/concepts/storage/volumes#iscsi>

- **iscsi.initiatorName** (string)

initiatorName is the custom iSCSI Initiator Name. If initiatorName is specified with iscsiInterface simultaneously, new iSCSI interface <target portal>:<volume name> will be created for the connection.

- **iscsi.iscsiInterface** (string)

iscsiInterface is the interface Name that uses an iSCSI transport. Defaults to 'default' (tcp).

**iscsi.portals** ([]string)

- portals is the iSCSI Target Portal List. The Portal is either an IP or ip\_addr:port if the port is other than default (typically TCP ports 860 and 3260).

◦ **iscsi.readOnly** (boolean)

readOnly here will force the ReadOnly setting in VolumeMounts. Defaults to false.

◦ **iscsi.secretRef** (SecretReference)

secretRef is the CHAP Secret for iSCSI target and initiator authentication

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

▪ **iscsi.secretRef.name** (string)

name is unique within a namespace to reference a secret resource.

▪ **iscsi.secretRef.namespace** (string)

namespace defines the space within which the secret name must be unique.

• **nfs** (NFSVolumeSource)

nfs represents an NFS mount on the host. Provisioned by an admin. More info: <https://kubernetes.io/docs/concepts/storage/volumes#nfs>

*Represents an NFS mount that lasts the lifetime of a pod. NFS volumes do not support ownership management or SELinux relabeling.*

◦ **nfs.path** (string), required

path that is exported by the NFS server. More info: <https://kubernetes.io/docs/concepts/storage/volumes#nfs>

◦ **nfs.server** (string), required

server is the hostname or IP address of the NFS server. More info: <https://kubernetes.io/docs/concepts/storage/volumes#nfs>

◦ **nfs.readOnly** (boolean)

readOnly here will force the NFS export to be mounted with read-only permissions. Defaults to false. More info: <https://kubernetes.io/docs/concepts/storage/volumes#nfs>

• **photonPersistentDisk** (PhotonPersistentDiskVolumeSource)

photonPersistentDisk represents a PhotonController persistent disk attached and mounted on kubelets host machine

*Represents a Photon Controller persistent disk resource.*

◦ **photonPersistentDisk.pdID** (string), required



pdID is the ID that identifies Photon Controller persistent disk

- **photonPersistentDisk.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **portworxVolume** (PortworxVolumeSource)

portworxVolume represents a portworx volume attached and mounted on kubelets host machine

*PortworxVolumeSource represents a Portworx volume resource.*

- **portworxVolume.volumeID** (string), required

volumeID uniquely identifies a Portworx volume

- **portworxVolume.fsType** (string)

fsType represents the filesystem type to mount Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs". Implicitly inferred to be "ext4" if unspecified.

- **portworxVolume.readOnly** (boolean)

readOnly defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.

- **quobyte** (QuobyteVolumeSource)

quobyte represents a Quobyte mount on the host that shares a pod's lifetime

*Represents a Quobyte mount that lasts the lifetime of a pod. Quobyte volumes do not support ownership management or SELinux relabeling.*

- **quobyte.registry** (string), required

registry represents a single or multiple Quobyte Registry services specified as a string as host:port pair (multiple entries are separated with commas) which acts as the central registry for volumes

- **quobyte.volume** (string), required

volume is a string that references an already created Quobyte volume by name.

- **quobyte.group** (string)

group to map volume access to Default is no group

- **quobyte.readOnly** (boolean)

readOnly here will force the Quobyte volume to be mounted with read-only permissions. Defaults to false.

**quobyte.tenant** (string)

- tenant owning the given Quobyte volume in the Backend Used with dynamically provisioned Quobyte volumes, value is set by the plugin

◦ **quobyte.user** (string)

user to map volume access to Defaults to serviceaccount user

• **rbd** (RBDPersistentVolumeSource)

rbd represents a Rados Block Device mount on the host that shares a pod's lifetime. More info: <https://examples.k8s.io/volumes/rbd/README.md>

*Represents a Rados Block Device mount that lasts the lifetime of a pod. RBD volumes support ownership management and SELinux relabeling.*

◦ **rbd.image** (string), required

image is the rados image name. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

◦ **rbd.monitors** ([]string), required

monitors is a collection of Ceph monitors. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

◦ **rbd.fsType** (string)

fsType is the filesystem type of the volume that you want to mount. Tip: Ensure that the filesystem type is supported by the host operating system. Examples: "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified. More info: <https://kubernetes.io/docs/concepts/storage/volumes#rbd>

◦ **rbd.keyring** (string)

keyring is the path to key ring for RBDUser. Default is /etc/ceph/keyring. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

◦ **rbd.pool** (string)

pool is the rados pool name. Default is rbd. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

◦ **rbd.readOnly** (boolean)

readOnly here will force the ReadOnly setting in VolumeMounts. Defaults to false. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

◦ **rbd.secretRef** (SecretReference)

secretRef is name of the authentication secret for RBDUser. If provided overrides keyring. Default is nil. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **rbd.secretRef.name** (string)

name is unique within a namespace to reference a secret resource.

- **rbd.secretRef.namespace** (string)

namespace defines the space within which the secret name must be unique.

- **rbd.user** (string)

user is the rados user name. Default is admin. More info: <https://examples.k8s.io/volumes/rbd/README.md#how-to-use-it>

- **scaleIO** (ScaleIOPersistentVolumeSource)

scaleIO represents a ScaleIO persistent volume attached and mounted on Kubernetes nodes.

*ScaleIOPersistentVolumeSource represents a persistent ScaleIO volume*

- **scaleIO.gateway** (string), required

gateway is the host address of the ScaleIO API Gateway.

- **scaleIO.secretRef** (SecretReference), required

secretRef references to the secret for ScaleIO user and other sensitive information. If this is not provided, Login operation will fail.

*SecretReference represents a Secret Reference. It has enough information to retrieve secret in any namespace*

- **scaleIO.secretRef.name** (string)

name is unique within a namespace to reference a secret resource.

- **scaleIO.secretRef.namespace** (string)

namespace defines the space within which the secret name must be unique.

- **scaleIO.system** (string), required

system is the name of the storage system as configured in ScaleIO.

- **scaleIO.fsType** (string)

fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Default is "xfs"

- **scaleIO.protectionDomain** (string)

protectionDomain is the name of the ScaleIO Protection Domain for the configured storage.

- **scaleIO.readOnly** (boolean)
  - readOnly defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.
- **scaleIO.sslEnabled** (boolean)
  - sslEnabled is the flag to enable/disable SSL communication with Gateway, default false
- **scaleIO.storageMode** (string)
  - storageMode indicates whether the storage for a volume should be ThickProvisioned or ThinProvisioned. Default is ThinProvisioned.
- **scaleIO.storagePool** (string)
  - storagePool is the ScaleIO Storage Pool associated with the protection domain.
- **scaleIO.volumeName** (string)
  - volumeName is the name of a volume already created in the ScaleIO system that is associated with this volume source.
- **storageos** (StorageOSPersistentVolumeSource)
  - storageOS represents a StorageOS volume that is attached to the kubelet's host machine and mounted into the pod More info: <https://examples.k8s.io/volumes/storageos/README.md>
  - *Represents a StorageOS persistent volume resource.*
  - **storageos.fsType** (string)
    - fsType is the filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.
  - **storageos.readOnly** (boolean)
    - readOnly defaults to false (read/write). ReadOnly here will force the ReadOnly setting in VolumeMounts.
  - **storageos.secretRef** ([ObjectReference](#))
    - secretRef specifies the secret to use for obtaining the StorageOS API credentials. If not specified, default values will be attempted.
  - **storageos.volumeName** (string)
    - volumeName is the human-readable name of the StorageOS volume. Volume names are only unique within a namespace.
  - **storageos.volumeNamespace** (string)

`volumeNamespace` specifies the scope of the volume within StorageOS. If no namespace is specified then the Pod's namespace will be used. This allows the Kubernetes name scoping to be mirrored within StorageOS for tighter integration. Set `VolumeName` to any name to override the default behaviour. Set to "default" if you are not using namespaces within StorageOS. Namespaces that do not pre-exist within StorageOS will be created.

- **vsphereVolume** (VsphereVirtualDiskVolumeSource)

`vsphereVolume` represents a vSphere volume attached and mounted on kubelets host machine

*Represents a vSphere volume resource.*

- **vsphereVolume.volumePath** (string), required

`volumePath` is the path that identifies vSphere volume vmdk

- **vsphereVolume.fsType** (string)

`fsType` is filesystem type to mount. Must be a filesystem type supported by the host operating system. Ex. "ext4", "xfs", "ntfs". Implicitly inferred to be "ext4" if unspecified.

- **vsphereVolume.storagePolicyID** (string)

`storagePolicyID` is the storage Policy Based Management (SPBM) profile ID associated with the `StoragePolicyName`.

- **vsphereVolume.storagePolicyName** (string)

`storagePolicyName` is the storage Policy Based Management (SPBM) profile name.

## PersistentVolumeStatus

`PersistentVolumeStatus` is the current status of a persistent volume.

---

- **lastPhaseTransitionTime** (Time)

`lastPhaseTransitionTime` is the time the phase transitioned from one to another and automatically resets to current time everytime a volume phase transitions. This is an alpha field and requires enabling `PersistentVolumeLastPhaseTransitionTime` feature.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **message** (string)

`message` is a human-readable message indicating details about why the volume is in this state.

- **phase** (string)

phase indicates if a volume is available, bound to a claim, or released by a claim. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#phase>

- **reason** (string)

reason is a brief CamelCase string that describes any failure and is meant for machine parsing and tidy display in the CLI.

## PersistentVolumeList

PersistentVolumeList is a list of PersistentVolume items.

---

- **apiVersion**: v1
- **kind**: PersistentVolumeList
- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]PersistentVolume](#)), required

items is a list of persistent volumes. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes>

## Operations

---

### get read the specified PersistentVolume

#### HTTP Request

GET /api/v1/persistentvolumes/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolume
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([PersistentVolume](#)): OK

401: Unauthorized

## get read status of the specified PersistentVolume

### HTTP Request

GET /api/v1/persistentvolumes/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolume
- **pretty** (*in query*): string

[pretty](#)

### Response

200 ([PersistentVolume](#)): OK

401: Unauthorized

## list list or watch objects of kind PersistentVolume

### HTTP Request

GET /api/v1/persistentvolumes

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)

- **continue** (*in query*): string  
[continue](#)

- **fieldSelector** (*in query*): string  
[fieldSelector](#)

- **labelSelector** (*in query*): string  
[labelSelector](#)

- **limit** (*in query*): integer  
[limit](#)

- **pretty** (*in query*): string  
[pretty](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([PersistentVolumeList](#)): OK

401: Unauthorized

## create create a PersistentVolume

### HTTP Request

POST /api/v1/persistentvolumes

### Parameters

- **body**: [PersistentVolume](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)



## Response

200 ([PersistentVolume](#)): OK

201 ([PersistentVolume](#)): Created

202 ([PersistentVolume](#)): Accepted

401: Unauthorized

## update replace the specified PersistentVolume

### HTTP Request

PUT /api/v1/persistentvolumes/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolume
- **body**: [PersistentVolume](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PersistentVolume](#)): OK

201 ([PersistentVolume](#)): Created

401: Unauthorized

## update replace status of the specified PersistentVolume

### HTTP Request

PUT /api/v1/persistentvolumes/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the PersistentVolume
- **body**: [PersistentVolume](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PersistentVolume](#)): OK

201 ([PersistentVolume](#)): Created

401: Unauthorized

## patch partially update the specified PersistentVolume

### HTTP Request

PATCH /api/v1/persistentvolumes/{name}

## Parameters

- **name** (*in path*): string, required  
name of the PersistentVolume
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PersistentVolume](#)): OK

201 ([PersistentVolume](#)): Created

401: Unauthorized

## patch partially update status of the specified PersistentVolume

### HTTP Request

PATCH /api/v1/persistentvolumes/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolume
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PersistentVolume](#)): OK

201 ([PersistentVolume](#)): Created

401: Unauthorized

## delete delete a PersistentVolume

### HTTP Request

DELETE /api/v1/persistentvolumes/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PersistentVolume
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([PersistentVolume](#)): OK

202 ([PersistentVolume](#)): Accepted

401: Unauthorized

## deletecollection delete collection of PersistentVolume

### HTTP Request

DELETE /api/v1/persistentvolumes

## Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

# StorageClass

StorageClass describes the parameters for a class of storage for which PersistentVolumes can be dynamically provisioned.

apiVersion: storage.k8s.io/v1

```
import "k8s.io/api/storage/v1"
```

## StorageClass

StorageClass describes the parameters for a class of storage for which PersistentVolumes can be dynamically provisioned.

StorageClasses are non-namespaced; the name of the storage class according to etcd is in ObjectMeta.Name.

- 
- **apiVersion:** storage.k8s.io/v1

- **kind:** StorageClass

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **provisioner** (string), required

provisioner indicates the type of the provisioner.

- **allowVolumeExpansion** (boolean)

allowVolumeExpansion shows whether the storage class allow volume expand.

- **allowedTopologies** ([]TopologySelectorTerm)

*Atomic: will be replaced during a merge*

allowedTopologies restrict the node topologies where volumes can be dynamically provisioned. Each volume plugin defines its own supported topology specifications. An empty TopologySelectorTerm list means there is no topology restriction. This field is only honored by servers that enable the VolumeScheduling feature.

*A topology selector term represents the result of label queries. A null or empty topology selector term matches no objects. The requirements of them are ANDed. It provides a subset of functionality as NodeSelectorTerm. This is an alpha feature and may change in the future.*

- **allowedTopologies.matchLabelExpressions**  
([]TopologySelectorLabelRequirement)

A list of topology selector requirements by labels.

*A topology selector requirement is a selector that matches given label. This is an alpha feature and may change in the future.*

- **allowedTopologies.matchLabelExpressions.key** (string), required

The label key that the selector applies to.

- **allowedTopologies.matchLabelExpressions.values** ([]string), required

An array of string values. One value must match the label to be selected.  
Each entry in Values is ORed.

- **mountOptions** ([]string)

mountOptions controls the mountOptions for dynamically provisioned PersistentVolumes of this storage class. e.g. ["ro", "soft"]. Not validated - mount of the PVs will simply fail if one is invalid.

- **parameters** (map[string]string)

parameters holds the parameters for the provisioner that should create volumes of this storage class.

- **reclaimPolicy** (string)

reclaimPolicy controls the reclaimPolicy for dynamically provisioned PersistentVolumes of this storage class. Defaults to Delete.

- **volumeBindingMode** (string)

volumeBindingMode indicates how PersistentVolumeClaims should be provisioned and bound. When unset, VolumeBindingImmediate is used. This field is only honored by servers that enable the VolumeScheduling feature.

## StorageClassList

StorageClassList is a collection of storage classes.

- 
- **apiVersion:** storage.k8s.io/v1
  - **kind:** StorageClassList
  - **metadata** ([ListMeta](#))

Standard list metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([][StorageClass](#)), required

items is the list of StorageClasses

# Operations

---

## get read the specified StorageClass

### HTTP Request

GET /apis/storage.k8s.io/v1/storageclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the StorageClass
- **pretty** (*in query*): string

[pretty](#)

### Response

200 ([StorageClass](#)): OK

401: Unauthorized

## list list or watch objects of kind StorageClass

### HTTP Request

GET /apis/storage.k8s.io/v1/storageclasses

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)



- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([StorageClassList](#)): OK

401: Unauthorized

## create create a StorageClass

### HTTP Request

POST /apis/storage.k8s.io/v1/storageclasses

### Parameters

- **body**: [StorageClass](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([StorageClass](#)): OK

201 ([StorageClass](#)): Created

202 ([StorageClass](#)): Accepted

401: Unauthorized

## update replace the specified StorageClass

## HTTP Request

PUT /apis/storage.k8s.io/v1/storageclasses/{name}

## Parameters

- **name** (*in path*): string, required  
name of the StorageClass
- **body**: [StorageClass](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([StorageClass](#)): OK

201 ([StorageClass](#)): Created

401: Unauthorized

## patch partially update the specified StorageClass

### HTTP Request

PATCH /apis/storage.k8s.io/v1/storageclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the StorageClass
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([StorageClass](#)): OK

201 ([StorageClass](#)): Created

401: Unauthorized

## delete delete a StorageClass

### HTTP Request

DELETE /apis/storage.k8s.io/v1/storageclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the StorageClass

- **body:** [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([StorageClass](#)): OK

202 ([StorageClass](#)): Accepted

401: Unauthorized

## deletecollection delete collection of StorageClass

### HTTP Request

DELETE /apis/storage.k8s.io/v1/storageclasses

### Parameters

- **body:** [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# VolumeAttachment

VolumeAttachment captures the intent to attach or detach the specified volume to/from the specified node.

apiVersion: storage.k8s.io/v1

```
import "k8s.io/api/storage/v1"
```

## VolumeAttachment

VolumeAttachment captures the intent to attach or detach the specified volume to/from the specified node.

VolumeAttachment objects are non-namespaced.

- 
- **apiVersion:** storage.k8s.io/v1

- **kind:** VolumeAttachment

- **metadata** ([ObjectMeta](#))

Standard object metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([VolumeAttachmentSpec](#)), required

spec represents specification of the desired attach/detach volume behavior. Populated by the Kubernetes system.

- **status** ([VolumeAttachmentStatus](#))

status represents status of the VolumeAttachment request. Populated by the entity completing the attach or detach operation, i.e. the external-attacher.

## VolumeAttachmentSpec

VolumeAttachmentSpec is the specification of a VolumeAttachment request.

---

- **attacher** (string), required

attacher indicates the name of the volume driver that MUST handle this request. This is the name returned by GetPluginName().

- **nodeName** (string), required

nodeName represents the node that the volume should be attached to.

- **source** (VolumeAttachmentSource), required

source represents the volume that should be attached.

*VolumeAttachmentSource represents a volume that should be attached. Right now only PersistentVolumes can be attached via external attacher, in future we may allow also inline volumes in pods. Exactly one member can be set.*

- **source.inlineVolumeSpec** ([PersistentVolumeSpec](#))

inlineVolumeSpec contains all the information necessary to attach a persistent volume defined by a pod's inline VolumeSource. This field is populated only for the CSIMigration feature. It contains translated fields from a pod's inline VolumeSource to a PersistentVolumeSpec. This field is beta-level and is only honored by servers that enabled the CSIMigration feature.

- **source.persistentVolumeName** (string)

persistentVolumeName represents the name of the persistent volume to attach.

# VolumeAttachmentStatus

VolumeAttachmentStatus is the status of a VolumeAttachment request.

---

- **attached** (boolean), required

attached indicates the volume is successfully attached. This field must only be set by the entity completing the attach operation, i.e. the external-attacher.

- **attachError** (VolumeError)

attachError represents the last error encountered during attach operation, if any. This field must only be set by the entity completing the attach operation, i.e. the external-attacher.

*VolumeError captures an error encountered during a volume operation.*

- **attachError.message** (string)

message represents the error encountered during Attach or Detach operation. This string may be logged, so it should not contain sensitive information.

- **attachError.time** (Time)

time represents the time the error was encountered.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **attachmentMetadata** (map[string]string)

attachmentMetadata is populated with any information returned by the attach operation, upon successful attach, that must be passed into subsequent WaitForAttach or Mount calls. This field must only be set by the entity completing the attach operation, i.e. the external-attacher.

- **detachError** (VolumeError)

detachError represents the last error encountered during detach operation, if any. This field must only be set by the entity completing the detach operation, i.e. the external-attacher.

*VolumeError captures an error encountered during a volume operation.*

- **detachError.message** (string)

message represents the error encountered during Attach or Detach operation. This string may be logged, so it should not contain sensitive information.

- **detachError.time** (Time)

time represents the time the error was encountered.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

## VolumeAttachmentList

VolumeAttachmentList is a collection of VolumeAttachment objects.

---

- **apiVersion:** storage.k8s.io/v1

- **kind:** VolumeAttachmentList

- **metadata** ([ListMeta](#))

Standard list metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]VolumeAttachment](#)), required

items is the list of VolumeAttachments

## Operations

---

### get read the specified VolumeAttachment

#### HTTP Request

GET /apis/storage.k8s.io/v1/volumeattachments/{name}

#### Parameters

- **name** (*in path*): string, required

name of the VolumeAttachment

- **pretty** (*in query*): string

[pretty](#)

#### Response

200 ([VolumeAttachment](#)): OK

401: Unauthorized



## get read status of the specified VolumeAttachment

### HTTP Request

GET /apis/storage.k8s.io/v1/volumeattachments/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the VolumeAttachment
- **pretty** (*in query*): string

[pretty](#)

### Response

200 ([VolumeAttachment](#)): OK

401: Unauthorized

## list list or watch objects of kind VolumeAttachment

### HTTP Request

GET /apis/storage.k8s.io/v1/volumeattachments

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)

- **continue** (*in query*): string  
[continue](#)

- **fieldSelector** (*in query*): string  
[fieldSelector](#)

- **labelSelector** (*in query*): string  
[labelSelector](#)

- **limit** (*in query*): integer  
[limit](#)

- **pretty** (*in query*): string  
[pretty](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([VolumeAttachmentList](#)): OK

401: Unauthorized

## create create a VolumeAttachment

### HTTP Request

POST /apis/storage.k8s.io/v1/volumeattachments

### Parameters

- **body**: [VolumeAttachment](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([VolumeAttachment](#)): OK

201 ([VolumeAttachment](#)): Created

202 ([VolumeAttachment](#)): Accepted

401: Unauthorized

## update replace the specified VolumeAttachment

### HTTP Request

PUT /apis/storage.k8s.io/v1/volumeattachments/{name}

### Parameters

- **name** (*in path*): string, required  
name of the VolumeAttachment
- **body**: [VolumeAttachment](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([VolumeAttachment](#)): OK

201 ([VolumeAttachment](#)): Created

401: Unauthorized

## update replace status of the specified VolumeAttachment

### HTTP Request

PUT /apis/storage.k8s.io/v1/volumeattachments/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the VolumeAttachment
- **body**: [VolumeAttachment](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([VolumeAttachment](#)): OK

201 ([VolumeAttachment](#)): Created

401: Unauthorized

## patch partially update the specified VolumeAttachment

### HTTP Request

PATCH /apis/storage.k8s.io/v1/volumeattachments/{name}

## Parameters

- **name** (*in path*): string, required  
name of the VolumeAttachment
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([VolumeAttachment](#)): OK

201 ([VolumeAttachment](#)): Created

401: Unauthorized

## patch partially update status of the specified VolumeAttachment

### HTTP Request

PATCH /apis/storage.k8s.io/v1/volumeattachments/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the VolumeAttachment
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([VolumeAttachment](#)): OK

201 ([VolumeAttachment](#)): Created

401: Unauthorized

## delete delete a VolumeAttachment

### HTTP Request

DELETE /apis/storage.k8s.io/v1/volumeattachments/{name}

### Parameters

- **name** (*in path*): string, required  
name of the VolumeAttachment
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([VolumeAttachment](#)): OK

202 ([VolumeAttachment](#)): Accepted

401: Unauthorized

## deletecollection delete collection of VolumeAttachment

### HTTP Request

DELETE /apis/storage.k8s.io/v1/volumeattachments

## Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# CSIDriver

CSIDriver captures information about a Container Storage Interface (CSI) volume driver deployed on the cluster.

apiVersion: storage.k8s.io/v1

```
import "k8s.io/api/storage/v1"
```

## CSIDriver

CSIDriver captures information about a Container Storage Interface (CSI) volume driver deployed on the cluster. Kubernetes attach detach controller uses this object to determine whether attach is required. Kubelet uses this object to determine whether pod information needs to be passed on mount. CSIDriver objects are non-namespaced.

- 
- **apiVersion:** storage.k8s.io/v1
  - **kind:** CSIDriver
  - **metadata** ([ObjectMeta](#))

Standard object metadata. metadata.Name indicates the name of the CSI driver that this object refers to; it MUST be the same name returned by the CSI GetPluginName() call for that driver. The driver name must be 63 characters or less, beginning and ending with an alphanumeric character ([a-z0-9A-Z]) with dashes (-), dots (.), and alphanumerics between. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([CSIDriverSpec](#)), required

spec represents the specification of the CSI Driver.

## CSIDriverSpec

CSIDriverSpec is the specification of a CSIDriver.

- 
- **attachRequired** (boolean)

attachRequired indicates this CSI volume driver requires an attach operation (because it implements the CSI ControllerPublishVolume() method), and that the Kubernetes attach detach controller should call the attach volume interface which checks the volumeattachment status and waits until the volume is attached before proceeding to mounting. The CSI external-attacher coordinates with CSI volume driver and updates the volumeattachment status when the attach operation is complete. If the CSIDriverRegistry feature gate is enabled and the value is specified to false, the attach operation will be skipped. Otherwise the attach operation will be called.



This field is immutable.

- **fsGroupPolicy** (string)

fsGroupPolicy defines if the underlying volume supports changing ownership and permission of the volume before being mounted. Refer to the specific FSGroupPolicy values for additional details.

This field is immutable.

Defaults to `ReadWriteOnceWithFSType`, which will examine each volume to determine if Kubernetes should modify ownership and permissions of the volume. With the default policy the defined fsGroup will only be applied if a fstype is defined and the volume's access mode contains `ReadWriteOnce`.

- **podInfoOnMount** (boolean)

podInfoOnMount indicates this CSI volume driver requires additional pod information (like podName, podUID, etc.) during mount operations, if set to true. If set to false, pod information will not be passed on mount. Default is false.

The CSI driver specifies podInfoOnMount as part of driver deployment. If true, Kubelet will pass pod information as VolumeContext in the CSI NodePublishVolume() calls. The CSI driver is responsible for parsing and validating the information passed in as VolumeContext.

The following VolumeContext will be passed if podInfoOnMount is set to true. This list might grow, but the prefix will be used. "csi.storage.k8s.io/pod.name": pod.Name "csi.storage.k8s.io/pod.namespace": pod.Namespace "csi.storage.k8s.io/pod.uid": string(pod.UID) "csi.storage.k8s.io/ephemeral": "true" if the volume is an ephemeral inline volume defined by a CSIVolumeSource, otherwise "false"

"csi.storage.k8s.io/ephemeral" is a new feature in Kubernetes 1.16. It is only required for drivers which support both the "Persistent" and "Ephemeral" VolumeLifecycleMode. Other drivers can leave pod info disabled and/or ignore this field. As Kubernetes 1.15 doesn't support this field, drivers can only support one mode when deployed on such a cluster and the deployment determines which mode that is, for example via a command line parameter of the driver.

This field is immutable.

- **requiresRepublish** (boolean)

requiresRepublish indicates the CSI driver wants NodePublishVolume being periodically called to reflect any possible change in the mounted volume. This field defaults to false.

Note: After a successful initial NodePublishVolume call, subsequent calls to NodePublishVolume should only update the contents of the volume. New mount points will not be seen by a running container.

- **seLinuxMount** (boolean)

seLinuxMount specifies if the CSI driver supports "-o context" mount option.

When "true", the CSI driver must ensure that all volumes provided by this CSI driver can be mounted separately with different `-o` context options. This is typical for storage backends that provide volumes as filesystems on block devices or as independent shared volumes. Kubernetes will call `NodeStage / NodePublish` with `"-o context=xyz"` mount option when mounting a `ReadWriteOncePod` volume used in Pod that has explicitly set SELinux context. In the future, it may be expanded to other volume `AccessModes`. In any case, Kubernetes will ensure that the volume is mounted only with a single SELinux context.

When "false", Kubernetes won't pass any special SELinux mount options to the driver. This is typical for volumes that represent subdirectories of a bigger shared filesystem.

Default is "false".

- **storageCapacity** (boolean)

`storageCapacity` indicates that the CSI volume driver wants pod scheduling to consider the storage capacity that the driver deployment will report by creating `CSIStorageCapacity` objects with capacity information, if set to true.

The check can be enabled immediately when deploying a driver. In that case, provisioning new volumes with late binding will pause until the driver deployment has published some suitable `CSIStorageCapacity` object.

Alternatively, the driver can be deployed with the field unset or false and it can be flipped later when storage capacity information has been published.

This field was immutable in Kubernetes  $\leq$  1.22 and now is mutable.

- **tokenRequests** (`[]TokenRequest`)

*Atomic: will be replaced during a merge*

`tokenRequests` indicates the CSI driver needs pods' service account tokens it is mounting volume for to do necessary authentication. Kubelet will pass the tokens in `VolumeContext` in the CSI `NodePublishVolume` calls. The CSI driver should parse and validate the following `VolumeContext`: `"csi.storage.k8s.io/serviceAccount.tokens": { "<audience>": { "token": <token>, "expirationTimestamp": <expiration timestamp in RFC3339>, }, ... }`

Note: Audience in each `TokenRequest` should be different and at most one token is empty string. To receive a new token after expiry, `RequiresRepublish` can be used to trigger `NodePublishVolume` periodically.

*TokenRequest contains parameters of a service account token.*

- **tokenRequests.audience** (string), required

audience is the intended audience of the token in `"TokenRequestSpec"`. It will default to the audiences of kube apiserver.

- **tokenRequests.expirationSeconds** (int64)

expirationSeconds is the duration of validity of the token in `"TokenRequestSpec"`. It has the same default value of `"ExpirationSeconds"` in `"TokenRequestSpec"`.

**volumeLifecycleModes** ([]string)

- *Set: unique values will be kept during a merge*

volumeLifecycleModes defines what kind of volumes this CSI volume driver supports. The default if the list is empty is "Persistent", which is the usage defined by the CSI specification and implemented in Kubernetes via the usual PV/PVC mechanism.

The other mode is "Ephemeral". In this mode, volumes are defined inline inside the pod spec with CSIVolumeSource and their lifecycle is tied to the lifecycle of that pod. A driver has to be aware of this because it is only going to get a NodePublishVolume call for such a volume.

For more information about implementing this mode, see <https://kubernetes-csi.github.io/docs/ephemeral-local-volumes.html> A driver can support one or more of these modes and more modes may be added in the future.

This field is beta. This field is immutable.

## CSIDriverList

CSIDriverList is a collection of CSIDriver objects.

- 
- **apiVersion:** storage.k8s.io/v1
  - **kind:** CSIDriverList
  - **metadata** ([ListMeta](#))

Standard list metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([][CSIDriver](#)), required
- items is the list of CSIDriver

## Operations

---

### get read the specified CSIDriver

#### HTTP Request

GET /apis/storage.k8s.io/v1/csidrivers/{name}

#### Parameters

- **name** (*in path*): string, required
- name of the CSIDriver

- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CSIDriver](#)): OK

401: Unauthorized

## list list or watch objects of kind CSIDriver

### HTTP Request

GET /apis/storage.k8s.io/v1/csidrivers

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([CSIDriverList](#)): OK

401: Unauthorized

## create create a CSIDriver

### HTTP Request

POST /apis/storage.k8s.io/v1/csidrivers

### Parameters

- **body**: [CSIDriver](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CSIDriver](#)): OK

201 ([CSIDriver](#)): Created

202 ([CSIDriver](#)): Accepted

401: Unauthorized

## update replace the specified CSIDriver

### HTTP Request

PUT /apis/storage.k8s.io/v1/csdrivers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CSIDriver
- **body**: [CSIDriver](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([CSIDriver](#)): OK

201 ([CSIDriver](#)): Created

401: Unauthorized

## patch partially update the specified CSIDriver

### HTTP Request

PATCH /apis/storage.k8s.io/v1/csdrivers/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CSIDriver
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([CSIDriver](#)): OK

201 ([CSIDriver](#)): Created

401: Unauthorized

## delete delete a CSIDriver

### HTTP Request

DELETE /apis/storage.k8s.io/v1/csidrivers/{name}

### Parameters

- **name** (*in path*): string, required

name of the CSIDriver

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([CSIDriver](#)): OK

202 ([CSIDriver](#)): Accepted

401: Unauthorized

## deletecollection delete collection of CSIDriver

### HTTP Request

DELETE /apis/storage.k8s.io/v1/csidrivers

### Parameters

- **body:** [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string



[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# CSINode

CSINode holds information about all CSI drivers installed on a node.

apiVersion: storage.k8s.io/v1

import "k8s.io/api/storage/v1"

## CSINode

CSINode holds information about all CSI drivers installed on a node. CSI drivers do not need to create the CSINode object directly. As long as they use the node-driver-registrar sidecar container, the kubelet will automatically populate the CSINode object for the CSI driver as part of kubelet plugin registration. CSINode has the same name as a node. If the object is missing, it means either there are no CSI Drivers available on the node, or the Kubelet version is low enough that it doesn't create this object. CSINode has an OwnerReference that points to the corresponding node object.

- 
- **apiVersion**: storage.k8s.io/v1
  - **kind**: CSINode
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. metadata.name must be the Kubernetes node name.

- **spec** ([CSINodeSpec](#)), required

spec is the specification of CSINode

# CSINodeSpec

CSINodeSpec holds information about the specification of all CSI drivers installed on a node

---

- **drivers** ([]CSINodeDriver), required

*Patch strategy: merge on key name*

drivers is a list of information of all CSI Drivers existing on a node. If all drivers in the list are uninstalled, this can become empty.

*CSINodeDriver holds information about the specification of one CSI driver installed on a node*

- **drivers.name** (string), required

name represents the name of the CSI driver that this object refers to. This MUST be the same name returned by the CSI GetPluginName() call for that driver.

- **drivers.nodeID** (string), required

nodeID of the node from the driver point of view. This field enables Kubernetes to communicate with storage systems that do not share the same nomenclature for nodes. For example, Kubernetes may refer to a given node as "node1", but the storage system may refer to the same node as "nodeA". When Kubernetes issues a command to the storage system to attach a volume to a specific node, it can use this field to refer to the node name using the ID that the storage system will understand, e.g. "nodeA" instead of "node1". This field is required.

- **drivers.allocatable** (VolumeNodeResources)

allocatable represents the volume resources of a node that are available for scheduling. This field is beta.

*VolumeNodeResources is a set of resource limits for scheduling of volumes.*

- **drivers.allocatable.count** (int32)

count indicates the maximum number of unique volumes managed by the CSI driver that can be used on a node. A volume that is both attached and mounted on a node is considered to be used once, not twice. The same rule applies for a unique volume that is shared among multiple pods on the same node. If this field is not specified, then the supported number of volumes on this node is unbounded.

- **drivers.topologyKeys** ([]string)

topologyKeys is the list of keys supported by the driver. When a driver is initialized on a cluster, it provides a set of topology keys that it understands (e.g. "company.com/zone", "company.com/region"). When a driver is initialized on a node, it provides the same topology keys along with values. Kubelet will expose these topology keys as labels on its own node object. When Kubernetes does topology aware provisioning, it can use this list to determine which labels it should

retrieve from the node object and pass back to the driver. It is possible for different nodes to use different topology keys. This can be empty if driver does not support topology.

## CSINodeList

CSINodeList is a collection of CSINode objects.

---

- **apiVersion:** storage.k8s.io/v1
- **kind:** CSINodeList
- **metadata** ([ListMeta](#))

Standard list metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]CSINode](#)), required

items is the list of CSINode

## Operations

---

### get read the specified CSINode

#### HTTP Request

GET /apis/storage.k8s.io/v1/csinodes/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the CSINode
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([CSINode](#)): OK

401: Unauthorized

## list list or watch objects of kind CSINode

### HTTP Request

GET /apis/storage.k8s.io/v1/csinodes

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([CSINodeList](#)): OK

401: Unauthorized

## create create a CSINode

### HTTP Request

POST /apis/storage.k8s.io/v1/csinodes

### Parameters

- **body**: [CSINode](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CSINode](#)): OK

201 ([CSINode](#)): Created

202 ([CSINode](#)): Accepted

401: Unauthorized

## update replace the specified CSINode

### HTTP Request

PUT /apis/storage.k8s.io/v1/csinodes/{name}

### Parameters

- **name** (*in path*): string, required

name of the CSINode

- **body**: [CSINode](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CSINode](#)): OK

201 ([CSINode](#)): Created

401: Unauthorized

## patch partially update the specified CSINode

### HTTP Request

PATCH /apis/storage.k8s.io/v1/csinodes/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CSINode
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)

- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CSINode](#)): OK

201 ([CSINode](#)): Created

401: Unauthorized

## delete delete a CSINode

### HTTP Request

DELETE /apis/storage.k8s.io/v1/csinodes/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CSINode
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([CSINode](#)): OK

202 ([CSINode](#)): Accepted

401: Unauthorized

## deletecollection delete collection of CSINode

### HTTP Request

DELETE /apis/storage.k8s.io/v1/csinodes

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer



[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# CSIStorageCapacity

CSIStorageCapacity stores the result of one CSI GetCapacity call.

apiVersion: storage.k8s.io/v1

```
import "k8s.io/api/storage/v1"
```

## CSIStorageCapacity

CSIStorageCapacity stores the result of one CSI GetCapacity call. For a given StorageClass, this describes the available capacity in a particular topology segment. This can be used when considering where to instantiate new PersistentVolumes.

For example this can express things like: - StorageClass "standard" has "1234 GiB" available in "topology.kubernetes.io/zone=us-east1" - StorageClass "localssd" has "10 GiB" available in "kubernetes.io/hostname=knode-abc123"

The following three cases all imply that no capacity is available for a certain combination: - no object exists with suitable topology and storage class name - such an object exists, but the capacity is unset - such an object exists, but the capacity is zero

The producer of these objects can decide which approach is more suitable.

They are consumed by the kube-scheduler when a CSI driver opts into capacity-aware scheduling with CSIDriverSpec.StorageCapacity. The scheduler compares the MaximumVolumeSize against the requested size of pending volumes to filter out unsuitable nodes. If MaximumVolumeSize is unset, it falls back to a comparison against the less precise Capacity. If that is also unset, the scheduler assumes that capacity is insufficient and tries some other node.

- 
- **apiVersion:** storage.k8s.io/v1
  - **kind:** CSIStorageCapacity
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. The name has no particular meaning. It must be a DNS subdomain (dots allowed, 253 characters). To ensure that there are no conflicts with other CSI drivers on the cluster, the recommendation is to use csisc-**<uuid>**, a generated name, or a reverse-domain name which ends with the unique CSI driver name.

Objects are namespaced.

More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **storageClassName** (string), required

storageClassName represents the name of the StorageClass that the reported capacity applies to. It must meet the same requirements as the name of a StorageClass object (non-empty, DNS subdomain). If that object no longer exists, the CSIStorageCapacity object is obsolete and should be removed by its creator. This field is immutable.

- **capacity** ([Quantity](#))

capacity is the value reported by the CSI driver in its GetCapacityResponse for a GetCapacityRequest with topology and parameters that match the previous fields.

The semantic is currently (CSI spec 1.2) defined as: The available capacity, in bytes, of the storage that can be used to provision volumes. If not set, that information is currently unavailable.

- **maximumVolumeSize** ([Quantity](#))

maximumVolumeSize is the value reported by the CSI driver in its GetCapacityResponse for a GetCapacityRequest with topology and parameters that match the previous fields.

This is defined since CSI spec 1.4.0 as the largest size that may be used in a CreateVolumeRequest.capacity\_range.required\_bytes field to create a volume with the same parameters as those in GetCapacityRequest. The corresponding value in the Kubernetes API is ResourceRequirements.Requests in a volume claim.

- **nodeTopology** ([LabelSelector](#))

nodeTopology defines which nodes have access to the storage for which capacity was reported. If not set, the storage is not accessible from any node in the cluster. If empty, the storage is accessible from all nodes. This field is immutable.

## CSIStorageCapacityList

CSIStorageCapacityList is a collection of CSIStorageCapacity objects.

- 
- **apiVersion**: storage.k8s.io/v1
  - **kind**: CSIStorageCapacityList
  - **metadata** ([ListMeta](#))

Standard list metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]CSIStorageCapacity](#)), required

*Map: unique values on key name will be kept during a merge*

items is the list of CSIStorageCapacity objects.

# Operations

---

## get read the specified CSIStorageCapacity

### HTTP Request

GET /apis/storage.k8s.io/v1/namespaces/{namespace}/csistoragecapacities/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CSIStorageCapacity
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([CSIStorageCapacity](#)): OK

401: Unauthorized

## list list or watch objects of kind CSIStorageCapacity

### HTTP Request

GET /apis/storage.k8s.io/v1/namespaces/{namespace}/csistoragecapacities

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)

- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([CSISStorageCapacityList](#)): OK

401: Unauthorized

## list list or watch objects of kind CSISStorageCapacity

### HTTP Request

GET /apis/storage.k8s.io/v1/csistoragecapacities

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)

- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([CSIStorageCapacityList](#)): OK

401: Unauthorized

## create create a CSIStorageCapacity

### HTTP Request

POST /apis/storage.k8s.io/v1/namespaces/{namespace}/csistoragecapacities

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)

- **body:** [CSIStorageCapacity](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CSIStorageCapacity](#)): OK

201 ([CSIStorageCapacity](#)): Created

202 ([CSIStorageCapacity](#)): Accepted

401: Unauthorized

## update replace the specified CSIStorageCapacity

### HTTP Request

PUT /apis/storage.k8s.io/v1/namespaces/{namespace}/csistoragecapacities/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CSIStorageCapacity
- **namespace** (*in path*): string, required  
[namespace](#)
- **body:** [CSIStorageCapacity](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CSISStorageCapacity](#)): OK

201 ([CSISStorageCapacity](#)): Created

401: Unauthorized

## patch partially update the specified CSISStorageCapacity

### HTTP Request

PATCH /apis/storage.k8s.io/v1/namespaces/{namespace}/csistoragecapacities/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CSISStorageCapacity
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CSIStorageCapacity](#)): OK

201 ([CSIStorageCapacity](#)): Created

401: Unauthorized

## delete delete a CSIStorageCapacity

### HTTP Request

DELETE /apis/storage.k8s.io/v1/namespaces/{namespace}/csistoragecapacities/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CSIStorageCapacity
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized



## deletecollection delete collection of CSISStorageCapacity

### HTTP Request

DELETE /apis/storage.k8s.io/v1/namespaces/{namespace}/csistoragecapacities

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Authentication Resources

---

## [ServiceAccount](#)

ServiceAccount binds together: \* a name, understood by users, and perhaps by peripheral systems, for an identity \* a principal that can be authenticated and authorized \* a set of secrets.

## [TokenRequest](#)

TokenRequest requests a token for a given service account.

## [TokenReview](#)

TokenReview attempts to authenticate a token to a known user.

## [CertificateSigningRequest](#)

CertificateSigningRequest objects provide a mechanism to obtain x509 certificates by submitting a certificate signing request, and having it asynchronously approved and issued.

## [ClusterTrustBundle v1alpha1](#)

ClusterTrustBundle is a cluster-scoped container for X.

## [SelfSubjectReview](#)

SelfSubjectReview contains the user information that the kube-apiserver has about the user making this request.

# ServiceAccount

ServiceAccount binds together: \* a name, understood by users, and perhaps by peripheral systems, for an identity \* a principal that can be authenticated and authorized \* a set of secrets.

apiVersion: v1

```
import "k8s.io/api/core/v1"
```

## ServiceAccount

ServiceAccount binds together: \* a name, understood by users, and perhaps by peripheral systems, for an identity \* a principal that can be authenticated and authorized \* a set of secrets

---

- **apiVersion:** v1
- **kind:** ServiceAccount
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **automountServiceAccountToken** (boolean)

AutomountServiceAccountToken indicates whether pods running as this service account should have an API token automatically mounted. Can be overridden at the pod level.

- **imagePullSecrets** ([\[\]LocalObjectReference](#))

ImagePullSecrets is a list of references to secrets in the same namespace to use for pulling any images in pods that reference this ServiceAccount. ImagePullSecrets are distinct from Secrets because Secrets can be mounted in the pod, but ImagePullSecrets are only accessed by the kubelet. More info: <https://kubernetes.io/docs/concepts/containers/images/#specifying-imagepullsecrets-on-a-pod>

- **secrets** ([\[\]ObjectReference](#))

*Patch strategy: merge on key name*

Secrets is a list of the secrets in the same namespace that pods running using this ServiceAccount are allowed to use. Pods are only limited to this list if this service account has a "kubernetes.io/enforce-mountable-secrets" annotation set to "true". This field should not be used to find auto-generated service account token secrets for use outside of pods. Instead, tokens can be requested directly using the TokenRequest API, or service account token secrets can be manually created. More info: <https://kubernetes.io/docs/concepts/configuration/secret>

## ServiceAccountList

ServiceAccountList is a list of ServiceAccount objects

---

- **apiVersion:** v1
- **kind:** ServiceAccountList
- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]ServiceAccount](#)), required

List of ServiceAccounts. More info: <https://kubernetes.io/docs/tasks/configure-pod-container/configure-service-account/>

## Operations

---

### get read the specified ServiceAccount

#### HTTP Request

GET /api/v1/namespaces/{namespace}/serviceaccounts/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ServiceAccount
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ServiceAccount](#)): OK

401: Unauthorized

### list list or watch objects of kind ServiceAccount

#### HTTP Request

GET /api/v1/namespaces/{namespace}/serviceaccounts

#### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ServiceAccountList](#)): OK

401: Unauthorized

## list or watch objects of kind ServiceAccount

### HTTP Request

GET /api/v1/serviceaccounts

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ServiceAccountList](#)): OK

401: Unauthorized

## create create a ServiceAccount

### HTTP Request

POST /api/v1/namespaces/{namespace}/serviceaccounts

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ServiceAccount](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ServiceAccount](#)): OK

201 ([ServiceAccount](#)): Created

202 ([ServiceAccount](#)): Accepted

401: Unauthorized

## update replace the specified ServiceAccount

### HTTP Request

PUT /api/v1/namespaces/{namespace}/serviceaccounts/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ServiceAccount
- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [ServiceAccount](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ServiceAccount](#)): OK

201 ([ServiceAccount](#)): Created

401: Unauthorized

## patch partially update the specified ServiceAccount

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/serviceaccounts/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ServiceAccount
- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)



- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ServiceAccount](#)): OK

201 ([ServiceAccount](#)): Created

401: Unauthorized

## delete delete a ServiceAccount

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/serviceaccounts/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ServiceAccount
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([ServiceAccount](#)): OK

202 ([ServiceAccount](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ServiceAccount

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/serviceaccounts

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# TokenRequest

TokenRequest requests a token for a given service account.

apiVersion: authentication.k8s.io/v1

import "k8s.io/api/authentication/v1"

## TokenRequest

TokenRequest requests a token for a given service account.

- 
- **apiVersion**: authentication.k8s.io/v1
  - **kind**: TokenRequest
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([TokenRequestSpec](#)), required

Spec holds information about the request being evaluated

- **status** ([TokenRequestStatus](#))

Status is filled in by the server and indicates whether the token can be authenticated.

# TokenRequestSpec

TokenRequestSpec contains client provided parameters of a token request.

---

- **audiences** ([]string), required

Audiences are the intended audiences of the token. A recipient of a token must identify itself with an identifier in the list of audiences of the token, and otherwise should reject the token. A token issued for multiple audiences may be used to authenticate against any of the audiences listed but implies a high degree of trust between the target audiences.

- **boundObjectRef** (BoundObjectReference)

BoundObjectRef is a reference to an object that the token will be bound to. The token will only be valid for as long as the bound object exists. NOTE: The API server's TokenReview endpoint will validate the BoundObjectRef, but other audiences may not. Keep ExpirationSeconds small if you want prompt revocation.

*BoundObjectReference is a reference to an object that a token is bound to.*

- **boundObjectRef.apiVersion** (string)

API version of the referent.

- **boundObjectRef.kind** (string)

Kind of the referent. Valid kinds are 'Pod' and 'Secret'.

- **boundObjectRef.name** (string)

Name of the referent.

- **boundObjectRef.uid** (string)

UID of the referent.

- **expirationSeconds** (int64)

ExpirationSeconds is the requested duration of validity of the request. The token issuer may return a token with a different validity duration so a client needs to check the 'expiration' field in a response.

# TokenRequestStatus

TokenRequestStatus is the result of a token request.

---

- **expirationTimestamp** (Time), required

ExpirationTimestamp is the time of expiration of the returned token.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **token** (string), required

Token is the opaque bearer token.

## Operations

---

### create create token of a ServiceAccount

#### HTTP Request

POST /api/v1/namespaces/{namespace}/serviceaccounts/{name}/token

#### Parameters

- **name** (*in path*): string, required  
name of the TokenRequest
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [TokenRequest](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([TokenRequest](#)): OK

201 ([TokenRequest](#)): Created

202 ([TokenRequest](#)): Accepted

401: Unauthorized

# TokenReview

TokenReview attempts to authenticate a token to a known user.

apiVersion: authentication.k8s.io/v1

```
import "k8s.io/api/authentication/v1"
```

## TokenReview

TokenReview attempts to authenticate a token to a known user. Note: TokenReview requests may be cached by the webhook token authenticator plugin in the kube-apiserver.

---

- **apiVersion:** authentication.k8s.io/v1

- **kind:** TokenReview

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([TokenReviewSpec](#)), required

Spec holds information about the request being evaluated

- **status** ([TokenReviewStatus](#))

Status is filled in by the server and indicates whether the request can be authenticated.

## TokenReviewSpec

TokenReviewSpec is a description of the token authentication request.

---

- **audiences** ([]string)

Audiences is a list of the identifiers that the resource server presented with the token identifies as. Audience-aware token authenticators will verify that the token was intended for at least one of the audiences in this list. If no audiences are provided, the audience will default to the audience of the Kubernetes apiserver.

- **token** (string)

Token is the opaque bearer token.

## TokenReviewStatus

TokenReviewStatus is the result of the token authentication request.

---

- **audiences** ([]string)

Audiences are audience identifiers chosen by the authenticator that are compatible with both the TokenReview and token. An identifier is any identifier in the intersection of the TokenReviewSpec audiences and the token's audiences. A client of the TokenReview API that sets the spec.audiences field should validate that a compatible audience identifier is returned in the status.audiences field to ensure that the TokenReview server is audience aware. If a TokenReview returns an empty status.audience field where status.authenticated is "true", the token is valid against the audience of the Kubernetes API server.

- **authenticated** (boolean)

Authenticated indicates that the token was associated with a known user.

- **error** (string)

Error indicates that the token couldn't be checked

- **user** (UserInfo)

User is the UserInfo associated with the provided token.

*UserInfo holds the information about the user needed to implement the user.Info interface.*

- **user.extra** (map[string][]string)

Any additional information provided by the authenticator.

- **user.groups** ([]string)

The names of groups this user is a part of.

- **user.uid** (string)

A unique value that identifies this user across time. If this user is deleted and another user by the same name is added, they will have different UIDs.

- **user.username** (string)

The name that uniquely identifies this user among all active users.

## Operations

---

### create create a TokenReview

#### HTTP Request

POST /apis/authentication.k8s.io/v1/tokenreviews

## Parameters

- **body**: [TokenReview](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([TokenReview](#)): OK

201 ([TokenReview](#)): Created

202 ([TokenReview](#)): Accepted

401: Unauthorized

# CertificateSigningRequest

CertificateSigningRequest objects provide a mechanism to obtain x509 certificates by submitting a certificate signing request, and having it asynchronously approved and issued.

apiVersion: certificates.k8s.io/v1

import "k8s.io/api/certificates/v1"

## CertificateSigningRequest

CertificateSigningRequest objects provide a mechanism to obtain x509 certificates by submitting a certificate signing request, and having it asynchronously approved and issued.

Kubelets use this API to obtain:

1. client certificates to authenticate to kube-apiserver (with the "kubernetes.io/kube-apiserver-client-kubelet" signerName).
2. serving certificates for TLS endpoints kube-apiserver can connect to securely (with the "kubernetes.io/kubelet-serving" signerName).



This API can be used to request client certificates to authenticate to kube-apiserver (with the "kubernetes.io/kube-apiserver-client" signerName), or to obtain certificates from custom non-Kubernetes signers.

---

- **apiVersion:** certificates.k8s.io/v1
- **kind:** CertificateSigningRequest
- **metadata** ([ObjectMeta](#))
- **spec** ([CertificateSigningRequestSpec](#)), required

spec contains the certificate request, and is immutable after creation. Only the request, signerName, expirationSeconds, and usages fields can be set on creation. Other fields are derived by Kubernetes and cannot be modified by users.

- **status** ([CertificateSigningRequestStatus](#))

status contains information about whether the request is approved or denied, and the certificate issued by the signer, or the failure condition indicating signer failure.

## CertificateSigningRequestSpec

CertificateSigningRequestSpec contains the certificate request.

---

- **request** ([]byte), required

*Atomic: will be replaced during a merge*

request contains an x509 certificate signing request encoded in a "CERTIFICATE REQUEST" PEM block. When serialized as JSON or YAML, the data is additionally base64-encoded.

- **signerName** (string), required

signerName indicates the requested signer, and is a qualified name.

List/watch requests for CertificateSigningRequests can filter on this field using a "spec.signerName=NAME" fieldSelector.

Well-known Kubernetes signers are:

1. "kubernetes.io/kube-apiserver-client": issues client certificates that can be used to authenticate to kube-apiserver. Requests for this signer are never auto-approved by kube-controller-manager, can be issued by the "csrsigning" controller in kube-controller-manager.
2. "kubernetes.io/kube-apiserver-client-kubelet": issues client certificates that kubelets use to authenticate to kube-apiserver. Requests for this signer can be auto-approved by the "csrapproving" controller in kube-controller-manager, and can be issued by the "csrsigning" controller in kube-controller-manager.
3. "kubernetes.io/kubelet-serving" issues serving certificates that kubelets use to serve TLS endpoints, which kube-apiserver can connect to securely. Requests for this

signer are never auto-approved by kube-controller-manager, and can be issued by the "csrsigning" controller in kube-controller-manager.

More details are available at <https://k8s.io/docs/reference/access-authn-authz/certificate-signing-requests/#kubernetes-signers>

Custom signerNames can also be specified. The signer defines:

1. Trust distribution: how trust (CA bundles) are distributed.
2. Permitted subjects: and behavior when a disallowed subject is requested.
3. Required, permitted, or forbidden x509 extensions in the request (including whether subjectAltNames are allowed, which types, restrictions on allowed values) and behavior when a disallowed extension is requested.
4. Required, permitted, or forbidden key usages / extended key usages.
5. Expiration/certificate lifetime: whether it is fixed by the signer, configurable by the admin.
6. Whether or not requests for CA certificates are allowed.

- **expirationSeconds** (int32)

expirationSeconds is the requested duration of validity of the issued certificate. The certificate signer may issue a certificate with a different validity duration so a client must check the delta between the notBefore and notAfter fields in the issued certificate to determine the actual duration.

The v1.22+ in-tree implementations of the well-known Kubernetes signers will honor this field as long as the requested duration is not greater than the maximum duration they will honor per the --cluster-signing-duration CLI flag to the Kubernetes controller manager.

Certificate signers may not honor this field for various reasons:

1. Old signer that is unaware of the field (such as the in-tree implementations prior to v1.22)
2. Signer whose configured maximum is shorter than the requested duration
3. Signer whose configured minimum is longer than the requested duration

The minimum valid value for expirationSeconds is 600, i.e. 10 minutes.

- **extra** (map[string]string)

extra contains extra attributes of the user that created the CertificateSigningRequest. Populated by the API server on creation and immutable.

- **groups** ([]string)

*Atomic: will be replaced during a merge*

groups contains group membership of the user that created the CertificateSigningRequest. Populated by the API server on creation and immutable.

- **uid** (string)

uid contains the uid of the user that created the CertificateSigningRequest. Populated by the API server on creation and immutable.

• **usages** ([]string)

*Atomic: will be replaced during a merge*

usages specifies a set of key usages requested in the issued certificate.

Requests for TLS client certificates typically request: "digital signature", "key encipherment", "client auth".

Requests for TLS serving certificates typically request: "key encipherment", "digital signature", "server auth".

Valid values are: "signing", "digital signature", "content commitment", "key encipherment", "key agreement", "data encipherment", "cert sign", "crl sign", "encipher only", "decipher only", "any", "server auth", "client auth", "code signing", "email protection", "s/mime", "ipsec end system", "ipsec tunnel", "ipsec user", "timestamping", "ocsp signing", "microsoft sgc", "netscape sgc"

• **username** (string)

username contains the name of the user that created the CertificateSigningRequest. Populated by the API server on creation and immutable.

## CertificateSigningRequestStatus

CertificateSigningRequestStatus contains conditions used to indicate approved/denied/failed status of the request, and the issued certificate.

---

• **certificate** ([]byte)

*Atomic: will be replaced during a merge*

certificate is populated with an issued certificate by the signer after an Approved condition is present. This field is set via the /status subresource. Once populated, this field is immutable.

If the certificate signing request is denied, a condition of type "Denied" is added and this field remains empty. If the signer cannot issue the certificate, a condition of type "Failed" is added and this field remains empty.

Validation requirements:

1. certificate must contain one or more PEM blocks.
2. All PEM blocks must have the "CERTIFICATE" label, contain no headers, and the encoded data must be a BER-encoded ASN.1 Certificate structure as described in section 4 of RFC5280.
3. Non-PEM content may appear before or after the "CERTIFICATE" PEM blocks and is unvalidated, to allow for explanatory text as described in section 5.2 of RFC7468.

If more than one PEM block is present, and the definition of the requested spec.signerName does not indicate otherwise, the first block is the issued certificate, and subsequent blocks should be treated as intermediate certificates and presented in TLS handshakes.

The certificate is encoded in PEM format.

When serialized as JSON or YAML, the data is additionally base64-encoded, so it consists of:

```
base64(
-----BEGIN CERTIFICATE-----
...
-----END CERTIFICATE-----
)
```

- **conditions** ([]CertificateSigningRequestCondition)

*Map: unique values on key type will be kept during a merge*

conditions applied to the request. Known conditions are "Approved", "Denied", and "Failed".

*CertificateSigningRequestCondition describes a condition of a CertificateSigningRequest object*

- **conditions.status** (string), required

status of the condition, one of True, False, Unknown. Approved, Denied, and Failed conditions may not be "False" or "Unknown".

- **conditions.type** (string), required

type of the condition. Known conditions are "Approved", "Denied", and "Failed".

An "Approved" condition is added via the /approval subresource, indicating the request was approved and should be issued by the signer.

A "Denied" condition is added via the /approval subresource, indicating the request was denied and should not be issued by the signer.

A "Failed" condition is added via the /status subresource, indicating the signer failed to issue the certificate.

Approved and Denied conditions are mutually exclusive. Approved, Denied, and Failed conditions cannot be removed once added.

Only one condition of a given type is allowed.

- **conditions.lastTransitionTime** (Time)

lastTransitionTime is the time the condition last transitioned from one status to another. If unset, when a new condition type is added or an existing condition's status is changed, the server defaults this to the current time.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.lastUpdateTime** (Time)

lastUpdateTime is the time of the last update to this condition

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

message contains a human readable message with details about the request state

- **conditions.reason** (string)

reason indicates a brief reason for the request state

## CertificateSigningRequestList

CertificateSigningRequestList is a collection of CertificateSigningRequest objects

---

- **apiVersion:** certificates.k8s.io/v1
- **kind:** CertificateSigningRequestList
- **metadata** ([ListMeta](#))
- **items** ([][CertificateSigningRequest](#)), required

items is a collection of CertificateSigningRequest objects

## Operations

---

### get read the specified CertificateSigningRequest

#### HTTP Request

GET /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([CertificateSigningRequest](#)): OK

401: Unauthorized

## get read approval of the specified CertificateSigningRequest

### HTTP Request

GET /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}/approval

### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([CertificateSigningRequest](#)): OK

401: Unauthorized

## get read status of the specified CertificateSigningRequest

### HTTP Request

GET /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([CertificateSigningRequest](#)): OK

401: Unauthorized

## list list or watch objects of kind CertificateSigningRequest

### HTTP Request

GET /apis/certificates.k8s.io/v1/certificatesigningrequests

## Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([CertificateSigningRequestList](#)): OK

401: Unauthorized

## create create a CertificateSigningRequest

### HTTP Request

POST /apis/certificates.k8s.io/v1/certificatesigningrequests

### Parameters

- **body**: [CertificateSigningRequest](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([CertificateSigningRequest](#)): OK

201 ([CertificateSigningRequest](#)): Created

202 ([CertificateSigningRequest](#)): Accepted

401: Unauthorized

## update replace the specified CertificateSigningRequest

### HTTP Request

PUT /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **body**: [CertificateSigningRequest](#), required
- **dryRun** (*in query*): string  
[dryRun](#)



- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CertificateSigningRequest](#)): OK

201 ([CertificateSigningRequest](#)): Created

401: Unauthorized

## update replace approval of the specified CertificateSigningRequest

### HTTP Request

PUT /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}/approval

### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **body**: [CertificateSigningRequest](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CertificateSigningRequest](#)): OK

201 ([CertificateSigningRequest](#)): Created

401: Unauthorized

## update replace status of the specified CertificateSigningRequest

### HTTP Request

PUT /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **body**: [CertificateSigningRequest](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([CertificateSigningRequest](#)): OK

201 ([CertificateSigningRequest](#)): Created

401: Unauthorized

## patch partially update the specified CertificateSigningRequest

### HTTP Request

PATCH /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}

### Parameters

- **name** (*in path*): string, required

name of the CertificateSigningRequest

- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CertificateSigningRequest](#)): OK

201 ([CertificateSigningRequest](#)): Created

401: Unauthorized

## patch partially update approval of the specified CertificateSigningRequest

### HTTP Request

PATCH /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}/approval

### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CertificateSigningRequest](#)): OK

201 ([CertificateSigningRequest](#)): Created

401: Unauthorized

## patch partially update status of the specified CertificateSigningRequest

### HTTP Request

PATCH /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CertificateSigningRequest](#)): OK

201 ([CertificateSigningRequest](#)): Created

401: Unauthorized

## delete delete a CertificateSigningRequest

### HTTP Request

DELETE /apis/certificates.k8s.io/v1/certificatesigningrequests/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CertificateSigningRequest
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of CertificateSigningRequest

### HTTP Request

DELETE /apis/certificates.k8s.io/v1/certificatesigningrequests

## Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

# ClusterTrustBundle v1alpha1

ClusterTrustBundle is a cluster-scoped container for X.

apiVersion: certificates.k8s.io/v1alpha1

```
import "k8s.io/api/certificates/v1alpha1"
```

## ClusterTrustBundle

ClusterTrustBundle is a cluster-scoped container for X.509 trust anchors (root certificates).

ClusterTrustBundle objects are considered to be readable by any authenticated user in the cluster, because they can be mounted by pods using the clusterTrustBundle projection. All service accounts have read access to ClusterTrustBundles by default. Users who only have namespace-level access to a cluster can read ClusterTrustBundles by impersonating a serviceaccount that they have access to.

It can be optionally associated with a particular assigner, in which case it contains one valid set of trust anchors for that signer. Signers may have multiple associated ClusterTrustBundles; each is an independent set of trust anchors for that signer. Admission control is used to enforce that only users with permissions on the signer can create or modify the corresponding bundle.

---

- **apiVersion:** certificates.k8s.io/v1alpha1

- **kind:** ClusterTrustBundle

- **metadata** ([ObjectMeta](#))

metadata contains the object metadata.

- **spec** ([ClusterTrustBundleSpec](#)), required

spec contains the signer (if any) and trust anchors.

## ClusterTrustBundleSpec

ClusterTrustBundleSpec contains the signer and trust anchors.

---

- **trustBundle** (string), required

trustBundle contains the individual X.509 trust anchors for this bundle, as PEM bundle of PEM-wrapped, DER-formatted X.509 certificates.

The data must consist only of PEM certificate blocks that parse as valid X.509 certificates. Each certificate must include a basic constraints extension with the CA bit set. The API server will reject objects that contain duplicate certificates, or that use PEM block headers.

Users of ClusterTrustBundles, including Kubelet, are free to reorder and deduplicate certificate blocks in this file according to their own logic, as well as to drop PEM block headers and inter-block data.

- **signerName** (string)

signerName indicates the associated signer, if any.

In order to create or update a ClusterTrustBundle that sets signerName, you must have the following cluster-scoped permission: group=certificates.k8s.io resource=signers resourceName=<the signer name> verb=attest.

If signerName is not empty, then the ClusterTrustBundle object must be named with the signer name as a prefix (translating slashes to colons). For example, for the signer name example.com/foo, valid ClusterTrustBundle object names include example.com:foo:abc and example.com:foo:v1.

If signerName is empty, then the ClusterTrustBundle object's name must not have such a prefix.

List/watch requests for ClusterTrustBundles can filter on this field using a spec.signerName=NAME field selector.

## ClusterTrustBundleList

ClusterTrustBundleList is a collection of ClusterTrustBundle objects

---

- **apiVersion**: certificates.k8s.io/v1alpha1
- **kind**: ClusterTrustBundleList
- **metadata** ([ListMeta](#))

metadata contains the list metadata.

- **items** ([\[\]ClusterTrustBundle](#)), required

items is a collection of ClusterTrustBundle objects

## Operations

---

### get read the specified ClusterTrustBundle

#### HTTP Request

GET /apis/certificates.k8s.io/v1alpha1/clustertrustbundles/{name}

#### Parameters

- **name** (*in path*): string, required



name of the ClusterTrustBundle

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ClusterTrustBundle](#)): OK

401: Unauthorized

## list or watch objects of kind ClusterTrustBundle

### HTTP Request

GET /apis/certificates.k8s.io/v1alpha1/clustertrustbundles

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ClusterTrustBundleList](#)): OK

401: Unauthorized

## create create a ClusterTrustBundle

### HTTP Request

POST /apis/certificates.k8s.io/v1alpha1/clustertrustbundles

### Parameters

- **body**: [ClusterTrustBundle](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ClusterTrustBundle](#)): OK

201 ([ClusterTrustBundle](#)): Created

202 ([ClusterTrustBundle](#)): Accepted

401: Unauthorized

## update replace the specified ClusterTrustBundle

### HTTP Request

PUT /apis/certificates.k8s.io/v1alpha1/clustertrustbundles/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ClusterTrustBundle
- **body**: [ClusterTrustBundle](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ClusterTrustBundle](#)): OK

201 ([ClusterTrustBundle](#)): Created

401: Unauthorized

## patch partially update the specified ClusterTrustBundle

### HTTP Request

PATCH /apis/certificates.k8s.io/v1alpha1/clustertrustbundles/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ClusterTrustBundle
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ClusterTrustBundle](#)): OK

201 ([ClusterTrustBundle](#)): Created

401: Unauthorized

## delete delete a ClusterTrustBundle

### HTTP Request

DELETE /apis/certificates.k8s.io/v1alpha1/clustertrustbundles/{name}

### Parameters

- **name** (*in path*): string, required

name of the ClusterTrustBundle

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ClusterTrustBundle

### HTTP Request

DELETE /apis/certificates.k8s.io/v1alpha1/clustertrustbundles

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# SelfSubjectReview

SelfSubjectReview contains the user information that the kube-apiserver has about the user making this request.

apiVersion: authentication.k8s.io/v1

```
import "k8s.io/api/authentication/v1"
```

## SelfSubjectReview

SelfSubjectReview contains the user information that the kube-apiserver has about the user making this request. When using impersonation, users will receive the user info of the user being impersonated. If impersonation or request header authentication is used, any extra keys will have their case ignored and returned as lowercase.

- 
- **apiVersion**: authentication.k8s.io/v1
  - **kind**: SelfSubjectReview
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **status** ([SelfSubjectReviewStatus](#))

Status is filled in by the server with the user attributes.

# SelfSubjectReviewStatus

SelfSubjectReviewStatus is filled by the kube-apiserver and sent back to a user.

---

- **userInfo** (UserInfo)

User attributes of the user making this request.

*UserInfo holds the information about the user needed to implement the user.Info interface.*

- **userInfo.extra** (map[string][]string)

Any additional information provided by the authenticator.

- **userInfo.groups** ([]string)

The names of groups this user is a part of.

- **userInfo.uid** (string)

A unique value that identifies this user across time. If this user is deleted and another user by the same name is added, they will have different UIDs.

- **userInfo.username** (string)

The name that uniquely identifies this user among all active users.

## Operations

---

### create create a SelfSubjectReview

#### HTTP Request

POST /apis/authentication.k8s.io/v1/selfsubjectreviews

#### Parameters

- **body**: [SelfSubjectReview](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

**pretty** (*in query*): string

•

[pretty](#)

## Response

200 ([SelfSubjectReview](#)): OK

201 ([SelfSubjectReview](#)): Created

202 ([SelfSubjectReview](#)): Accepted

401: Unauthorized

# Authorization Resources

---

## [LocalSubjectAccessReview](#)

LocalSubjectAccessReview checks whether or not a user or group can perform an action in a given namespace.

## [SelfSubjectAccessReview](#)

SelfSubjectAccessReview checks whether or the current user can perform an action.

## [SelfSubjectRulesReview](#)

SelfSubjectRulesReview enumerates the set of actions the current user can perform within a namespace.

## [SubjectAccessReview](#)

SubjectAccessReview checks whether or not a user or group can perform an action.

## [ClusterRole](#)

ClusterRole is a cluster level, logical grouping of PolicyRules that can be referenced as a unit by a RoleBinding or ClusterRoleBinding.

## [ClusterRoleBinding](#)

ClusterRoleBinding references a ClusterRole, but not contain it.

## [Role](#)

Role is a namespaced, logical grouping of PolicyRules that can be referenced as a unit by a RoleBinding.



## [RoleBinding](#)

RoleBinding references a role, but does not contain it.

# LocalSubjectAccessReview

LocalSubjectAccessReview checks whether or not a user or group can perform an action in a given namespace.

apiVersion: authorization.k8s.io/v1

import "k8s.io/api/authorization/v1"

## LocalSubjectAccessReview

LocalSubjectAccessReview checks whether or not a user or group can perform an action in a given namespace. Having a namespace scoped resource makes it much easier to grant namespace scoped policy that includes permissions checking.

---

- **apiVersion:** authorization.k8s.io/v1

- **kind:** LocalSubjectAccessReview

- **metadata** ([ObjectMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([SubjectAccessReviewSpec](#)), required

Spec holds information about the request being evaluated. spec.namespace must be equal to the namespace you made the request against. If empty, it is defaulted.

- **status** ([SubjectAccessReviewStatus](#))

Status is filled in by the server and indicates whether the request is allowed or not

## Operations

---

### create create a LocalSubjectAccessReview

#### HTTP Request

POST /apis/authorization.k8s.io/v1/namespaces/{namespace}/localsubjectaccessreviews

#### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [LocalSubjectAccessReview](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([LocalSubjectAccessReview](#)): OK

201 ([LocalSubjectAccessReview](#)): Created

202 ([LocalSubjectAccessReview](#)): Accepted

401: Unauthorized

# SelfSubjectAccessReview

SelfSubjectAccessReview checks whether or the current user can perform an action.

apiVersion: authorization.k8s.io/v1

import "k8s.io/api/authorization/v1"

## SelfSubjectAccessReview

SelfSubjectAccessReview checks whether or the current user can perform an action. Not filling in a spec.namespace means "in all namespaces". Self is a special case, because users should always be able to check whether they can perform an action

- 
- **apiVersion**: authorization.k8s.io/v1
  - **kind**: SelfSubjectAccessReview
  - **metadata** ([ObjectMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([SelfSubjectAccessReviewSpec](#)), required
  - Spec holds information about the request being evaluated. user and groups must be empty
- **status** ([SubjectAccessReviewStatus](#))
  - Status is filled in by the server and indicates whether the request is allowed or not

## SelfSubjectAccessReviewSpec

SelfSubjectAccessReviewSpec is a description of the access request. Exactly one of ResourceAuthorizationAttributes and NonResourceAuthorizationAttributes must be set

---

- **nonResourceAttributes** (NonResourceAttributes)

NonResourceAttributes describes information for a non-resource access request

*NonResourceAttributes includes the authorization attributes available for non-resource requests to the Authorizer interface*

- **nonResourceAttributes.path** (string)

Path is the URL path of the request

- **nonResourceAttributes.verb** (string)

Verb is the standard HTTP verb

- **resourceAttributes** (ResourceAttributes)

ResourceAuthorizationAttributes describes information for a resource access request

*ResourceAttributes includes the authorization attributes available for resource requests to the Authorizer interface*

- **resourceAttributes.group** (string)

Group is the API Group of the Resource. "\*" means all.

- **resourceAttributes.name** (string)

Name is the name of the resource being requested for a "get" or deleted for a "delete". "" (empty) means all.

- **resourceAttributes.namespace** (string)

Namespace is the namespace of the action being requested. Currently, there is no distinction between no namespace and all namespaces "" (empty) is defaulted for LocalSubjectAccessReviews "" (empty) is empty for cluster-scoped resources "" (empty) means "all" for namespace scoped resources from a SubjectAccessReview or SelfSubjectAccessReview

- **resourceAttributes.resource** (string)

Resource is one of the existing resource types. "\*" means all.

- **resourceAttributes.subresource** (string)

Subresource is one of the existing resource types. "" means none.

- **resourceAttributes.verb** (string)

Verb is a kubernetes resource API verb, like: get, list, watch, create, update, delete, proxy. "\*" means all.

- **resourceAttributes.version** (string)

Version is the API Version of the Resource. "\*" means all.

## Operations

---

### create create a SelfSubjectAccessReview

#### HTTP Request

POST /apis/authorization.k8s.io/v1/selfsubjectaccessreviews

#### Parameters

- **body**: [SelfSubjectAccessReview](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([SelfSubjectAccessReview](#)): OK

201 ([SelfSubjectAccessReview](#)): Created

202 ([SelfSubjectAccessReview](#)): Accepted

401: Unauthorized

# SelfSubjectRulesReview

SelfSubjectRulesReview enumerates the set of actions the current user can perform within a namespace.

apiVersion: authorization.k8s.io/v1

import "k8s.io/api/authorization/v1"

## SelfSubjectRulesReview

SelfSubjectRulesReview enumerates the set of actions the current user can perform within a namespace. The returned list of actions may be incomplete depending on the server's authorization mode, and any errors experienced during the evaluation. SelfSubjectRulesReview should be used by UIs to show/hide actions, or to quickly let an end user reason about their permissions. It should NOT Be used by external systems to drive authorization decisions as this raises confused deputy, cache lifetime/revocation, and correctness concerns.

SubjectAccessReview, and LocalAccessReview are the correct way to defer authorization decisions to the API server.

---

- **apiVersion:** authorization.k8s.io/v1

- **kind:** SelfSubjectRulesReview

- **metadata** ([ObjectMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([SelfSubjectRulesReviewSpec](#)), required

Spec holds information about the request being evaluated.

- **status** (SubjectRulesReviewStatus)

Status is filled in by the server and indicates the set of actions a user can perform.

*SubjectRulesReviewStatus contains the result of a rules check. This check can be incomplete depending on the set of authorizers the server is configured with and any errors experienced during evaluation. Because authorization rules are additive, if a rule appears in a list it's safe to assume the subject has that permission, even if that list is incomplete.*

- **status.incomplete** (boolean), required

Incomplete is true when the rules returned by this call are incomplete. This is most commonly encountered when an authorizer, such as an external authorizer, doesn't support rules evaluation.

- **status.nonResourceRules** ([]NonResourceRule), required

NonResourceRules is the list of actions the subject is allowed to perform on non-resources. The list ordering isn't significant, may contain duplicates, and possibly be incomplete.

*NonResourceRule holds information that describes a rule for the non-resource*

- **status.nonResourceRules.verbs** ([]string), required

Verb is a list of kubernetes non-resource API verbs, like: get, post, put, delete, patch, head, options. "\*" means all.

- **status.nonResourceRules.nonResourceURLs** ([]string)

NonResourceURLs is a set of partial urls that a user should have access to. s are allowed, but only as the full, final step in the path. "" means all.

- **status.resourceRules** ([]ResourceRule), required

ResourceRules is the list of actions the subject is allowed to perform on resources. The list ordering isn't significant, may contain duplicates, and possibly be incomplete.

*ResourceRule is the list of actions the subject is allowed to perform on resources. The list ordering isn't significant, may contain duplicates, and possibly be incomplete.*

- **status.resourceRules.verbs** ([]string), required

Verb is a list of kubernetes resource API verbs, like: get, list, watch, create, update, delete, proxy. "\*" means all.

- **status.resourceRules.apiGroups** ([]string)

APIGroups is the name of the APIGroup that contains the resources. If multiple API groups are specified, any action requested against one of the enumerated resources in any API group will be allowed. "\*" means all.

- **status.resourceRules.resourceNames** ([]string)

ResourceNames is an optional white list of names that the rule applies to. An empty set means that everything is allowed. "\*" means all.

- **status.resourceRules.resources** ([]string)

Resources is a list of resources this rule applies to. "" means all in the specified apiGroups. "/foo" represents the subresource 'foo' for all resources in the specified apiGroups.

- **status.evaluationError** (string)

EvaluationError can appear in combination with Rules. It indicates an error occurred during rule evaluation, such as an authorizer that doesn't support rule evaluation, and that ResourceRules and/or NonResourceRules may be incomplete.

# SelfSubjectRulesReviewSpec

SelfSubjectRulesReviewSpec defines the specification for SelfSubjectRulesReview.

---

- **namespace** (string)

Namespace to evaluate rules for. Required.

## Operations

---

### create create a SelfSubjectRulesReview

#### HTTP Request

POST /apis/authorization.k8s.io/v1/selfsubjectrulesreviews

#### Parameters

- **body**: [SelfSubjectRulesReview](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([SelfSubjectRulesReview](#)): OK

201 ([SelfSubjectRulesReview](#)): Created

202 ([SelfSubjectRulesReview](#)): Accepted

401: Unauthorized

# SubjectAccessReview

SubjectAccessReview checks whether or not a user or group can perform an action.

apiVersion: authorization.k8s.io/v1

import "k8s.io/api/authorization/v1"

## SubjectAccessReview

SubjectAccessReview checks whether or not a user or group can perform an action.

---

- **apiVersion:** authorization.k8s.io/v1

- **kind:** SubjectAccessReview

- **metadata** ([ObjectMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([SubjectAccessReviewSpec](#)), required

Spec holds information about the request being evaluated

- **status** ([SubjectAccessReviewStatus](#))

Status is filled in by the server and indicates whether the request is allowed or not

## SubjectAccessReviewSpec

SubjectAccessReviewSpec is a description of the access request. Exactly one of ResourceAuthorizationAttributes and NonResourceAuthorizationAttributes must be set

---

- **extra** (map[string][]string)

Extra corresponds to the user.Info.GetExtra() method from the authenticator. Since that is input to the authorizer it needs a reflection here.

- **groups** ([]string)

Groups is the groups you're testing for.

- **nonResourceAttributes** (NonResourceAttributes)

NonResourceAttributes describes information for a non-resource access request

*NonResourceAttributes includes the authorization attributes available for non-resource requests to the Authorizer interface*

- **nonResourceAttributes.path** (string)



Path is the URL path of the request

- **nonResourceAttributes.verb** (string)

Verb is the standard HTTP verb

- **resourceAttributes** (ResourceAttributes)

ResourceAuthorizationAttributes describes information for a resource access request

*ResourceAttributes includes the authorization attributes available for resource requests to the Authorizer interface*

- **resourceAttributes.group** (string)

Group is the API Group of the Resource. "\*" means all.

- **resourceAttributes.name** (string)

Name is the name of the resource being requested for a "get" or deleted for a "delete". "" (empty) means all.

- **resourceAttributes.namespace** (string)

Namespace is the namespace of the action being requested. Currently, there is no distinction between no namespace and all namespaces "" (empty) is defaulted for LocalSubjectAccessReviews "" (empty) is empty for cluster-scoped resources "" (empty) means "all" for namespace scoped resources from a SubjectAccessReview or SelfSubjectAccessReview

- **resourceAttributes.resource** (string)

Resource is one of the existing resource types. "\*" means all.

- **resourceAttributes.subresource** (string)

Subresource is one of the existing resource types. "" means none.

- **resourceAttributes.verb** (string)

Verb is a kubernetes resource API verb, like: get, list, watch, create, update, delete, proxy. "\*" means all.

- **resourceAttributes.version** (string)

Version is the API Version of the Resource. "\*" means all.

- **uid** (string)

UID information about the requesting user.

- **user** (string)

User is the user you're testing for. If you specify "User" but not "Groups", then is it interpreted as "What if User were not a member of any groups"

# SubjectAccessReviewStatus

SubjectAccessReviewStatus

---

- **allowed** (boolean), required

Allowed is required. True if the action would be allowed, false otherwise.

- **denied** (boolean)

Denied is optional. True if the action would be denied, otherwise false. If both allowed is false and denied is false, then the authorizer has no opinion on whether to authorize the action. Denied may not be true if Allowed is true.

- **evaluationError** (string)

EvaluationError is an indication that some error occurred during the authorization check. It is entirely possible to get an error and be able to continue determine authorization status in spite of it. For instance, RBAC can be missing a role, but enough roles are still present and bound to reason about the request.

- **reason** (string)

Reason is optional. It indicates why a request was allowed or denied.

## Operations

---

### create create a SubjectAccessReview

#### HTTP Request

POST /apis/authorization.k8s.io/v1/subjectaccessreviews

#### Parameters

- **body**: [SubjectAccessReview](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([SubjectAccessReview](#)): OK

201 ([SubjectAccessReview](#)): Created

202 ([SubjectAccessReview](#)): Accepted

401: Unauthorized

# ClusterRole

ClusterRole is a cluster level, logical grouping of PolicyRules that can be referenced as a unit by a RoleBinding or ClusterRoleBinding.

apiVersion: rbac.authorization.k8s.io/v1

```
import "k8s.io/api/rbac/v1"
```

## ClusterRole

ClusterRole is a cluster level, logical grouping of PolicyRules that can be referenced as a unit by a RoleBinding or ClusterRoleBinding.

---

- **apiVersion:** rbac.authorization.k8s.io/v1

- **kind:** ClusterRole

- **metadata** ([ObjectMeta](#))

Standard object's metadata.

- **aggregationRule** (AggregationRule)

AggregationRule is an optional field that describes how to build the Rules for this ClusterRole. If AggregationRule is set, then the Rules are controller managed and direct changes to Rules will be stomped by the controller.

*AggregationRule describes how to locate ClusterRoles to aggregate into the ClusterRole*

- **aggregationRule.clusterRoleSelectors** ([][LabelSelector](#))

ClusterRoleSelectors holds a list of selectors which will be used to find ClusterRoles and create the rules. If any of the selectors match, then the ClusterRole's permissions will be added

- **rules** ([]PolicyRule)

Rules holds all the PolicyRules for this ClusterRole

*PolicyRule holds information that describes a policy rule, but does not contain information about who the rule applies to or which namespace the rule applies to.*

- **rules.apiGroups** ([]string)

APIGroups is the name of the APIGroup that contains the resources. If multiple API groups are specified, any action requested against one of the enumerated resources in any API group will be allowed. "" represents the core API group and "\*" represents all API groups.

- **rules.resources** ([]string)

Resources is a list of resources this rule applies to. "\*" represents all resources.

- **rules.verbs** ([]string), required

Verbs is a list of Verbs that apply to ALL the ResourceKinds contained in this rule. "\*" represents all verbs.

- **rules.resourceNames** ([]string)

ResourceNames is an optional white list of names that the rule applies to. An empty set means that everything is allowed.

- **rules.nonResourceURLs** ([]string)

NonResourceURLs is a set of partial urls that a user should have access to. \*s are allowed, but only as the full, final step in the path Since non-resource URLs are not namespaced, this field is only applicable for ClusterRoles referenced from a ClusterRoleBinding. Rules can either apply to API resources (such as "pods" or "secrets") or non-resource URL paths (such as "/api"), but not both.

## ClusterRoleList

ClusterRoleList is a collection of ClusterRoles

---

- **apiVersion:** rbac.authorization.k8s.io/v1

- **kind:** ClusterRoleList

- **metadata** ([ListMeta](#))

Standard object's metadata.

- **items** ([][ClusterRole](#)), required

Items is a list of ClusterRoles

## Operations

---

## get read the specified ClusterRole

### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/clusterroles/{name}

### Parameters

- **name** (*in path*): string, required

name of the ClusterRole

- **pretty** (*in query*): string

[pretty](#)

### Response

200 ([ClusterRole](#)): OK

401: Unauthorized

## list list or watch objects of kind ClusterRole

### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/clusterroles

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ClusterRoleList](#)): OK

401: Unauthorized

## create create a ClusterRole

### HTTP Request

POST /apis/rbac.authorization.k8s.io/v1/clusterroles

### Parameters

- **body**: [ClusterRole](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ClusterRole](#)): OK

201 ([ClusterRole](#)): Created

202 ([ClusterRole](#)): Accepted

401: Unauthorized

## update replace the specified ClusterRole

### HTTP Request

PUT /apis/rbac.authorization.k8s.io/v1/clusterroles/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ClusterRole
- **body**: [ClusterRole](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ClusterRole](#)): OK

201 ([ClusterRole](#)): Created

401: Unauthorized

## patch partially update the specified ClusterRole

### HTTP Request

PATCH /apis/rbac.authorization.k8s.io/v1/clusterroles/{name}

## Parameters

- **name** (*in path*): string, required  
name of the ClusterRole
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ClusterRole](#)): OK

201 ([ClusterRole](#)): Created

401: Unauthorized

## delete delete a ClusterRole

### HTTP Request

DELETE /apis/rbac.authorization.k8s.io/v1/clusterroles/{name}

## Parameters

- **name** (*in path*): string, required  
name of the ClusterRole
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)



- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ClusterRole

### HTTP Request

DELETE /apis/rbac.authorization.k8s.io/v1/clusterroles

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)

- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ClusterRoleBinding

ClusterRoleBinding references a ClusterRole, but not contain it.

apiVersion: rbac.authorization.k8s.io/v1

```
import "k8s.io/api/rbac/v1"
```

## ClusterRoleBinding

ClusterRoleBinding references a ClusterRole, but not contain it. It can reference a ClusterRole in the global namespace, and adds who information via Subject.

- 
- **apiVersion**: rbac.authorization.k8s.io/v1
  - **kind**: ClusterRoleBinding
  - **metadata** ([ObjectMeta](#))  
Standard object's metadata.
  - **roleRef** (RoleRef), required

RoleRef can only reference a ClusterRole in the global namespace. If the RoleRef cannot be resolved, the Authorizer must return an error. This field is immutable.

*RoleRef contains information that points to the role being used*

- **roleRef.apiGroup** (string), required

APIGroup is the group for the resource being referenced

- **roleRef.kind** (string), required

Kind is the type of resource being referenced

- **roleRef.name** (string), required

Name is the name of resource being referenced

- **subjects** ([]Subject)

Subjects holds references to the objects the role applies to.

*Subject contains a reference to the object or user identities a role binding applies to. This can either hold a direct API object reference, or a value for non-objects such as user and group names.*

- **subjects.kind** (string), required

Kind of object being referenced. Values defined by this API group are "User", "Group", and "ServiceAccount". If the Authorizer does not recognize the kind value, the Authorizer should report an error.

- **subjects.name** (string), required

Name of the object being referenced.

- **subjects.apiGroup** (string)

APIGroup holds the API group of the referenced subject. Defaults to "" for ServiceAccount subjects. Defaults to "rbac.authorization.k8s.io" for User and Group subjects.

- **subjects.namespace** (string)

Namespace of the referenced object. If the object kind is non-namespace, such as "User" or "Group", and this value is not empty the Authorizer should report an error.

## ClusterRoleBindingList

ClusterRoleBindingList is a collection of ClusterRoleBindings

- 
- **apiVersion:** rbac.authorization.k8s.io/v1
  - **kind:** ClusterRoleBindingList

- **metadata** ([ListMeta](#))
- Standard object's metadata.
- **items** ([\[\]ClusterRoleBinding](#)), required  
Items is a list of ClusterRoleBindings

## Operations

---

### get read the specified ClusterRoleBinding

#### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/clusterrolebindings/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ClusterRoleBinding
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ClusterRoleBinding](#)): OK

401: Unauthorized

### list list or watch objects of kind ClusterRoleBinding

#### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/clusterrolebindings

#### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ClusterRoleBindingList](#)): OK

401: Unauthorized

## create create a ClusterRoleBinding

### HTTP Request

POST /apis/rbac.authorization.k8s.io/v1/clusterrolebindings

### Parameters

- **body**: [ClusterRoleBinding](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ClusterRoleBinding](#)): OK

201 ([ClusterRoleBinding](#)): Created

202 ([ClusterRoleBinding](#)): Accepted

401: Unauthorized

## update replace the specified ClusterRoleBinding

### HTTP Request

PUT /apis/rbac.authorization.k8s.io/v1/clusterrolebindings/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ClusterRoleBinding
- **body**: [ClusterRoleBinding](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ClusterRoleBinding](#)): OK

201 ([ClusterRoleBinding](#)): Created

401: Unauthorized

## patch partially update the specified ClusterRoleBinding

### HTTP Request

PATCH /apis/rbac.authorization.k8s.io/v1/clusterrolebindings/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ClusterRoleBinding
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ClusterRoleBinding](#)): OK

201 ([ClusterRoleBinding](#)): Created

401: Unauthorized

## delete delete a ClusterRoleBinding

### HTTP Request

DELETE /apis/rbac.authorization.k8s.io/v1/clusterrolebindings/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ClusterRoleBinding
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ClusterRoleBinding

### HTTP Request

DELETE /apis/rbac.authorization.k8s.io/v1/clusterrolebindings

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string



[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Role

Role is a namespaced, logical grouping of PolicyRules that can be referenced as a unit by a RoleBinding.

apiVersion: rbac.authorization.k8s.io/v1

```
import "k8s.io/api/rbac/v1"
```

## Role

Role is a namespaced, logical grouping of PolicyRules that can be referenced as a unit by a RoleBinding.

---

- **apiVersion:** rbac.authorization.k8s.io/v1

- **kind:** Role

- **metadata** ([ObjectMeta](#))

Standard object's metadata.

- **rules** ([]PolicyRule)

Rules holds all the PolicyRules for this Role

*PolicyRule holds information that describes a policy rule, but does not contain information about who the rule applies to or which namespace the rule applies to.*

- **rules.apiGroups** ([]string)

APIGroups is the name of the APIGroup that contains the resources. If multiple API groups are specified, any action requested against one of the enumerated resources in any API group will be allowed. "" represents the core API group and "\*" represents all API groups.

- **rules.resources** ([]string)

Resources is a list of resources this rule applies to. "\*" represents all resources.

- **rules.verbs** ([]string), required

Verbs is a list of Verbs that apply to ALL the ResourceKinds contained in this rule. "\*" represents all verbs.

- **rules.resourceNames** ([]string)

ResourceNames is an optional white list of names that the rule applies to. An empty set means that everything is allowed.

- **rules.nonResourceURLs** ([]string)

NonResourceURLs is a set of partial urls that a user should have access to. \*s are allowed, but only as the full, final step in the path Since non-resource URLs are not namespaced, this field is only applicable for ClusterRoles referenced from a ClusterRoleBinding. Rules can either apply to API resources (such as "pods" or "secrets") or non-resource URL paths (such as "/api"), but not both.

# RoleList

RoleList is a collection of Roles

---

- **apiVersion:** rbac.authorization.k8s.io/v1
- **kind:** RoleList
- **metadata** ([ListMeta](#))  
Standard object's metadata.
- **items** ([\[\]Role](#)), required  
Items is a list of Roles

## Operations

---

### get read the specified Role

#### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/roles/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the Role
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([Role](#)): OK

401: Unauthorized

### list list or watch objects of kind Role

#### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/roles

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([RoleList](#)): OK

401: Unauthorized

## list list or watch objects of kind Role

### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/roles

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([RoleList](#)): OK

401: Unauthorized

## create create a Role

### HTTP Request

POST /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/roles

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Role](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Role](#)): OK

201 ([Role](#)): Created

202 ([Role](#)): Accepted

401: Unauthorized

## update replace the specified Role

### HTTP Request

PUT /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/roles/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Role
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Role](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Role](#)): OK

201 ([Role](#)): Created

401: Unauthorized

## patch partially update the specified Role

### HTTP Request

PATCH /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/roles/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Role
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Role](#)): OK

201 ([Role](#)): Created

401: Unauthorized

## delete delete a Role

### HTTP Request

DELETE /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/roles/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Role
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)



- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Role

### HTTP Request

DELETE /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/roles

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)

- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# RoleBinding

RoleBinding references a role, but does not contain it.

apiVersion: rbac.authorization.k8s.io/v1

```
import "k8s.io/api/rbac/v1"
```

## RoleBinding

RoleBinding references a role, but does not contain it. It can reference a Role in the same namespace or a ClusterRole in the global namespace. It adds who information via Subjects and namespace information by which namespace it exists in. RoleBindings in a given namespace only have effect in that namespace.

- 
- **apiVersion**: rbac.authorization.k8s.io/v1
  - **kind**: RoleBinding
  - **metadata** ([ObjectMeta](#))

Standard object's metadata.

**roleRef** (RoleRef), required

- RoleRef can reference a Role in the current namespace or a ClusterRole in the global namespace. If the RoleRef cannot be resolved, the Authorizer must return an error. This field is immutable.

*RoleRef contains information that points to the role being used*

- **roleRef.apiGroup** (string), required

APIGroup is the group for the resource being referenced

- **roleRef.kind** (string), required

Kind is the type of resource being referenced

- **roleRef.name** (string), required

Name is the name of resource being referenced

- **subjects** ([]Subject)

Subjects holds references to the objects the role applies to.

*Subject contains a reference to the object or user identities a role binding applies to. This can either hold a direct API object reference, or a value for non-objects such as user and group names.*

- **subjects.kind** (string), required

Kind of object being referenced. Values defined by this API group are "User", "Group", and "ServiceAccount". If the Authorizer does not recognize the kind value, the Authorizer should report an error.

- **subjects.name** (string), required

Name of the object being referenced.

- **subjects.apiGroup** (string)

APIGroup holds the API group of the referenced subject. Defaults to "" for ServiceAccount subjects. Defaults to "rbac.authorization.k8s.io" for User and Group subjects.

- **subjects.namespace** (string)

Namespace of the referenced object. If the object kind is non-namespace, such as "User" or "Group", and this value is not empty the Authorizer should report an error.

## RoleBindingList

RoleBindingList is a collection of RoleBindings

- 
- **apiVersion:** rbac.authorization.k8s.io/v1
  - **kind:** RoleBindingList
  - **metadata** ([ListMeta](#))  
Standard object's metadata.
  - **items** ([\[\]RoleBinding](#)), required  
Items is a list of RoleBindings

## Operations

---

### get read the specified RoleBinding

#### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/rolebindings/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the RoleBinding
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([RoleBinding](#)): OK

401: Unauthorized

### list list or watch objects of kind RoleBinding

#### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/rolebindings

#### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([RoleBindingList](#)): OK

401: Unauthorized

## list list or watch objects of kind RoleBinding

### HTTP Request

GET /apis/rbac.authorization.k8s.io/v1/rolebindings

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([RoleBindingList](#)): OK

401: Unauthorized

## create create a RoleBinding

### HTTP Request

POST /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/rolebindings

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [RoleBinding](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([RoleBinding](#)): OK

201 ([RoleBinding](#)): Created

202 ([RoleBinding](#)): Accepted

401: Unauthorized

## update replace the specified RoleBinding

### HTTP Request

PUT /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/rolebindings/{name}

## Parameters

- **name** (*in path*): string, required  
name of the RoleBinding
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [RoleBinding](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([RoleBinding](#)): OK

201 ([RoleBinding](#)): Created

401: Unauthorized

## patch partially update the specified RoleBinding

### HTTP Request

PATCH /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/rolebindings/{name}

## Parameters

- **name** (*in path*): string, required  
name of the RoleBinding
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required



- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([RoleBinding](#)): OK

201 ([RoleBinding](#)): Created

401: Unauthorized

## delete delete a RoleBinding

### HTTP Request

DELETE /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/rolebindings/{name}

### Parameters

- **name** (*in path*): string, required  
name of the RoleBinding
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)

- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of RoleBinding

### HTTP Request

DELETE /apis/rbac.authorization.k8s.io/v1/namespaces/{namespace}/rolebindings

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)

- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Policy Resources

---

## [LimitRange](#)

LimitRange sets resource usage limits for each kind of resource in a Namespace.

## [ResourceQuota](#)

ResourceQuota sets aggregate quota restrictions enforced per namespace.

## [NetworkPolicy](#)

NetworkPolicy describes what network traffic is allowed for a set of Pods.

## [PodDisruptionBudget](#)

PodDisruptionBudget is an object to define the max disruption that can be caused to a collection of pods.

## [IPAddress v1alpha1](#)

IPAddress represents a single IP of a single IP Family.

# LimitRange

LimitRange sets resource usage limits for each kind of resource in a Namespace.

apiVersion: v1

import "k8s.io/api/core/v1"

# LimitRange

LimitRange sets resource usage limits for each kind of resource in a Namespace.

---

- **apiVersion:** v1
- **kind:** LimitRange
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([LimitRangeSpec](#))

Spec defines the limits enforced. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

# LimitRangeSpec

LimitRangeSpec defines a min/max usage limit for resources that match on kind.

---

- **limits** ([]LimitRangeItem), required

Limits is the list of LimitRangeItem objects that are enforced.

*LimitRangeItem defines a min/max usage limit for any resource that matches on kind.*

- **limits.type** (string), required

Type of resource that this limit applies to.

- **limits.default** (map[string][Quantity](#))

Default resource requirement limit value by resource name if resource limit is omitted.

- **limits.defaultRequest** (map[string][Quantity](#))

DefaultRequest is the default resource requirement request value by resource name if resource request is omitted.

- **limits.max** (map[string][Quantity](#))

Max usage constraints on this kind by resource name.

- **limits.maxLimitRequestRatio** (map[string][Quantity](#))

MaxLimitRequestRatio if specified, the named resource must have a request and limit that are both non-zero where limit divided by request is less than or equal to the enumerated value; this represents the max burst for the named resource.

- **limits.min** (map[string][Quantity](#))

Min usage constraints on this kind by resource name.

## LimitRangeList

LimitRangeList is a list of LimitRange items.

- 
- **apiVersion:** v1
  - **kind:** LimitRangeList
  - **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([][LimitRange](#)), required

Items is a list of LimitRange objects. More info: <https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/>

## Operations

---

### get read the specified LimitRange

#### HTTP Request

GET /api/v1/namespaces/{namespace}/limitranges/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the LimitRange
- **namespace** (*in path*): string, required

[namespace](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([LimitRange](#)): OK

401: Unauthorized

## list list or watch objects of kind LimitRange

### HTTP Request

GET /api/v1/namespaces/{namespace}/limitranges

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([LimitRangeList](#)): OK

401: Unauthorized

## list list or watch objects of kind LimitRange

## HTTP Request

GET /api/v1/limitranges

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([LimitRangeList](#)): OK

401: Unauthorized

## create create a LimitRange

### HTTP Request

POST /api/v1/namespaces/{namespace}/limitranges

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [LimitRange](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)



## Response

200 ([LimitRange](#)): OK

201 ([LimitRange](#)): Created

202 ([LimitRange](#)): Accepted

401: Unauthorized

## update replace the specified LimitRange

### HTTP Request

PUT /api/v1/namespaces/{namespace}/limitranges/{name}

### Parameters

- **name** (*in path*): string, required  
name of the LimitRange
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [LimitRange](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([LimitRange](#)): OK

201 ([LimitRange](#)): Created

401: Unauthorized

## patch partially update the specified LimitRange

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/limitranges/{name}

### Parameters

- **name** (*in path*): string, required  
name of the LimitRange
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([LimitRange](#)): OK

201 ([LimitRange](#)): Created

401: Unauthorized

## delete delete a LimitRange

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/limitranges/{name}

## Parameters

- **name** (*in path*): string, required  
name of the LimitRange
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of LimitRange

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/limitranges

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ResourceQuota

ResourceQuota sets aggregate quota restrictions enforced per namespace.

apiVersion: v1

import "k8s.io/api/core/v1"

## ResourceQuota

ResourceQuota sets aggregate quota restrictions enforced per namespace

---

- **apiVersion:** v1
- **kind:** ResourceQuota
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([ResourceQuotaSpec](#))

Spec defines the desired quota. <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([ResourceQuotaStatus](#))

Status defines the actual enforced quota and its current usage. <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## ResourceQuotaSpec

ResourceQuotaSpec defines the desired hard limits to enforce for Quota.

---

- **hard** (map[string][Quantity](#))

hard is the set of desired hard limits for each named resource. More info: <https://kubernetes.io/docs/concepts/policy/resource-quotas/>

- **scopeSelector** (ScopeSelector)

scopeSelector is also a collection of filters like scopes that must match each object tracked by a quota but expressed using ScopeSelectorOperator in combination with possible values. For a resource to match, both scopes AND scopeSelector (if specified in spec), must be matched.

*A scope selector represents the AND of the selectors represented by the scoped-resource selector requirements.*

- **scopeSelector.matchExpressions** ([]ScopedResourceSelectorRequirement)

A list of scope selector requirements by scope of the resources.

*A scoped-resource selector requirement is a selector that contains values, a scope name, and an operator that relates the scope name and values.*

- **scopeSelector.matchExpressions.operator** (string), required

Represents a scope's relationship to a set of values. Valid operators are In, NotIn, Exists, DoesNotExist.

- **scopeSelector.matchExpressions.scopeName** (string), required

The name of the scope that the selector applies to.

- **scopeSelector.matchExpressions.values** ([]string)

An array of string values. If the operator is In or NotIn, the values array must be non-empty. If the operator is Exists or DoesNotExist, the values array must be empty. This array is replaced during a strategic merge patch.

- **scopes** ([]string)

A collection of filters that must match each object tracked by a quota. If not specified, the quota matches all objects.

## ResourceQuotaStatus

ResourceQuotaStatus defines the enforced hard limits and observed use.

---

- **hard** (map[string][Quantity](#))

Hard is the set of enforced hard limits for each named resource. More info: <https://kubernetes.io/docs/concepts/policy/resource-quotas/>

- **used** (map[string][Quantity](#))

Used is the current observed total usage of the resource in the namespace.

## ResourceQuotaList

ResourceQuotaList is a list of ResourceQuota items.

---

- **apiVersion**: v1
- **kind**: ResourceQuotaList
- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([][ResourceQuota](#)), required

Items is a list of ResourceQuota objects. More info: <https://kubernetes.io/docs/concepts/policy/resource-quotas/>

## Operations

---

### get read the specified ResourceQuota

#### HTTP Request

GET /api/v1/namespaces/{namespace}/resourcequotas/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ResourceQuota
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ResourceQuota](#)): OK

401: Unauthorized

### get read status of the specified ResourceQuota

#### HTTP Request

GET /api/v1/namespaces/{namespace}/resourcequotas/{name}/status

#### Parameters

- **name** (*in path*): string, required  
name of the ResourceQuota
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceQuota](#)): OK

401: Unauthorized

## list list or watch objects of kind ResourceQuota

### HTTP Request

GET /api/v1/namespaces/{namespace}/resourcequotas

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)



- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ResourceQuotaList](#)): OK

401: Unauthorized

## list list or watch objects of kind ResourceQuota

## HTTP Request

GET /api/v1/resourcequotas

## Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ResourceQuotaList](#)): OK

401: Unauthorized

## create create a ResourceQuota

### HTTP Request

POST /api/v1/namespaces/{namespace}/resourcequotas

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ResourceQuota](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceQuota](#)): OK

201 ([ResourceQuota](#)): Created

202 ([ResourceQuota](#)): Accepted

401: Unauthorized

## update replace the specified ResourceQuota

### HTTP Request

PUT /api/v1/namespaces/{namespace}/resourcequotas/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceQuota
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ResourceQuota](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ResourceQuota](#)): OK

201 ([ResourceQuota](#)): Created

401: Unauthorized

## update replace status of the specified ResourceQuota

### HTTP Request

PUT /api/v1/namespaces/{namespace}/resourcequotas/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the ResourceQuota
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [ResourceQuota](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceQuota](#)): OK

201 ([ResourceQuota](#)): Created

401: Unauthorized

## patch partially update the specified ResourceQuota

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/resourcequotas/{name}

## Parameters

- **name** (*in path*): string, required  
name of the ResourceQuota
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required

- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceQuota](#)): OK

201 ([ResourceQuota](#)): Created

401: Unauthorized

## patch partially update status of the specified ResourceQuota

### HTTP Request

PATCH /api/v1/namespaces/{namespace}/resourcequotas/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ResourceQuota
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ResourceQuota](#)): OK

201 ([ResourceQuota](#)): Created

401: Unauthorized

## delete delete a ResourceQuota

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/resourcequotas/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ResourceQuota
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([ResourceQuota](#)): OK

202 ([ResourceQuota](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ResourceQuota

### HTTP Request

DELETE /api/v1/namespaces/{namespace}/resourcequotas

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# NetworkPolicy

NetworkPolicy describes what network traffic is allowed for a set of Pods.

apiVersion: networking.k8s.io/v1

import "k8s.io/api/networking/v1"

## NetworkPolicy

NetworkPolicy describes what network traffic is allowed for a set of Pods

- 
- **apiVersion:** networking.k8s.io/v1
  - **kind:** NetworkPolicy
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([NetworkPolicySpec](#))

spec represents the specification of the desired behavior for this NetworkPolicy.

## NetworkPolicySpec

NetworkPolicySpec provides the specification of a NetworkPolicy



- 
- **podSelector** ([LabelSelector](#)), required

podSelector selects the pods to which this NetworkPolicy object applies. The array of ingress rules is applied to any pods selected by this field. Multiple network policies can select the same set of pods. In this case, the ingress rules for each are combined additively. This field is NOT optional and follows standard label selector semantics. An empty podSelector matches all pods in this namespace.

- **policyTypes** ([]string)

policyTypes is a list of rule types that the NetworkPolicy relates to. Valid options are ["Ingress"], ["Egress"], or ["Ingress", "Egress"]. If this field is not specified, it will default based on the existence of ingress or egress rules; policies that contain an egress section are assumed to affect egress, and all policies (whether or not they contain an ingress section) are assumed to affect ingress. If you want to write an egress-only policy, you must explicitly specify policyTypes [ "Egress" ]. Likewise, if you want to write a policy that specifies that no egress is allowed, you must specify a policyTypes value that include "Egress" (since such a policy would not include an egress section and would otherwise default to just [ "Ingress" ]). This field is beta-level in 1.8

- **ingress** ([]NetworkPolicyIngressRule)

ingress is a list of ingress rules to be applied to the selected pods. Traffic is allowed to a pod if there are no NetworkPolicies selecting the pod (and cluster policy otherwise allows the traffic), OR if the traffic source is the pod's local node, OR if the traffic matches at least one ingress rule across all of the NetworkPolicy objects whose podSelector matches the pod. If this field is empty then this NetworkPolicy does not allow any traffic (and serves solely to ensure that the pods it selects are isolated by default)

*NetworkPolicyIngressRule describes a particular set of traffic that is allowed to the pods matched by a NetworkPolicySpec's podSelector. The traffic must match both ports and from.*

- **ingress.from** ([]NetworkPolicyPeer)

from is a list of sources which should be able to access the pods selected for this rule. Items in this list are combined using a logical OR operation. If this field is empty or missing, this rule matches all sources (traffic not restricted by source). If this field is present and contains at least one item, this rule allows traffic only if the traffic matches at least one item in the from list.

*NetworkPolicyPeer describes a peer to allow traffic to/from. Only certain combinations of fields are allowed*

- **ingress.from.ipBlock** (IPBlock)

ipBlock defines policy on a particular IPBlock. If this field is set then neither of the other fields can be.

*IPBlock describes a particular CIDR (Ex. "192.168.1.0/24","2001:db8::/64") that is allowed to the pods matched by a NetworkPolicySpec's podSelector. The except entry describes CIDRs that should not be included within this rule.*

- **ingress.from.ipBlock.cidr** (string), required

cidr is a string representing the IPBlock Valid examples are "192.168.1.0/24" or "2001:db8::/64"

- **ingress.from.ipBlock.except** ([]string)

except is a slice of CIDRs that should not be included within an IPBlock Valid examples are "192.168.1.0/24" or "2001:db8::/64" Except values will be rejected if they are outside the cidr range

- **ingress.from.namespaceSelector** ([LabelSelector](#))

namespaceSelector selects namespaces using cluster-scoped labels. This field follows standard label selector semantics; if present but empty, it selects all namespaces.

If podSelector is also set, then the NetworkPolicyPeer as a whole selects the pods matching podSelector in the namespaces selected by namespaceSelector. Otherwise it selects all pods in the namespaces selected by namespaceSelector.

- **ingress.from.podSelector** ([LabelSelector](#))

podSelector is a label selector which selects pods. This field follows standard label selector semantics; if present but empty, it selects all pods.

If namespaceSelector is also set, then the NetworkPolicyPeer as a whole selects the pods matching podSelector in the Namespaces selected by NamespaceSelector. Otherwise it selects the pods matching podSelector in the policy's own namespace.

- **ingress.ports** ([]NetworkPolicyPort)

ports is a list of ports which should be made accessible on the pods selected for this rule. Each item in this list is combined using a logical OR. If this field is empty or missing, this rule matches all ports (traffic not restricted by port). If this field is present and contains at least one item, then this rule allows traffic only if the traffic matches at least one port in the list.

*NetworkPolicyPort describes a port to allow traffic on*

- **ingress.ports.port** (IntOrString)

port represents the port on the given protocol. This can either be a numerical or named port on a pod. If this field is not provided, this matches all port names and numbers. If present, only traffic on the specified protocol AND port will be matched.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **ingress.ports.endPort** (int32)

endPort indicates that the range of ports from port to endPort if set, inclusive, should be allowed by the policy. This field cannot be defined if the port field is not defined or if the port field is defined as a named (string) port. The endPort must be equal or greater than port.

- **ingress.ports.protocol** (string)

protocol represents the protocol (TCP, UDP, or SCTP) which traffic must match. If not specified, this field defaults to TCP.

- **egress** ([]NetworkPolicyEgressRule)

egress is a list of egress rules to be applied to the selected pods. Outgoing traffic is allowed if there are no NetworkPolicies selecting the pod (and cluster policy otherwise allows the traffic), OR if the traffic matches at least one egress rule across all of the NetworkPolicy objects whose podSelector matches the pod. If this field is empty then this NetworkPolicy limits all outgoing traffic (and serves solely to ensure that the pods it selects are isolated by default). This field is beta-level in 1.8

*NetworkPolicyEgressRule describes a particular set of traffic that is allowed out of pods matched by a NetworkPolicySpec's podSelector. The traffic must match both ports and to. This type is beta-level in 1.8*

- **egress.to** ([]NetworkPolicyPeer)

to is a list of destinations for outgoing traffic of pods selected for this rule. Items in this list are combined using a logical OR operation. If this field is empty or missing, this rule matches all destinations (traffic not restricted by destination). If this field is present and contains at least one item, this rule allows traffic only if the traffic matches at least one item in the to list.

*NetworkPolicyPeer describes a peer to allow traffic to/from. Only certain combinations of fields are allowed*

- **egress.to.ipBlock** (IPBlock)

ipBlock defines policy on a particular IPBlock. If this field is set then neither of the other fields can be.

*IPBlock describes a particular CIDR (Ex. "192.168.1.0/24","2001:db8::/64") that is allowed to the pods matched by a NetworkPolicySpec's podSelector. The except entry describes CIDRs that should not be included within this rule.*

- **egress.to.ipBlock.cidr** (string), required

cidr is a string representing the IPBlock Valid examples are "192.168.1.0/24" or "2001:db8::/64"

- **egress.to.ipBlock.except** ([]string)

except is a slice of CIDRs that should not be included within an IPBlock Valid examples are "192.168.1.0/24" or "2001:db8::/64" Except values will be rejected if they are outside the cidr range

- **egress.to.namespaceSelector** ([LabelSelector](#))

namespaceSelector selects namespaces using cluster-scoped labels. This field follows standard label selector semantics; if present but empty, it selects all namespaces.

If podSelector is also set, then the NetworkPolicyPeer as a whole selects the pods matching podSelector in the namespaces selected by namespaceSelector. Otherwise it selects all pods in the namespaces selected by namespaceSelector.

- **egress.to.podSelector** ([LabelSelector](#))

podSelector is a label selector which selects pods. This field follows standard label selector semantics; if present but empty, it selects all pods.

If namespaceSelector is also set, then the NetworkPolicyPeer as a whole selects the pods matching podSelector in the Namespaces selected by NamespaceSelector. Otherwise it selects the pods matching podSelector in the policy's own namespace.

- **egress.ports** ([]NetworkPolicyPort)

ports is a list of destination ports for outgoing traffic. Each item in this list is combined using a logical OR. If this field is empty or missing, this rule matches all ports (traffic not restricted by port). If this field is present and contains at least one item, then this rule allows traffic only if the traffic matches at least one port in the list.

*NetworkPolicyPort describes a port to allow traffic on*

- **egress.ports.port** (IntOrString)

port represents the port on the given protocol. This can either be a numerical or named port on a pod. If this field is not provided, this matches all port names and numbers. If present, only traffic on the specified protocol AND port will be matched.

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **egress.ports.endPort** (int32)

endPort indicates that the range of ports from port to endPort if set, inclusive, should be allowed by the policy. This field cannot be defined if the port field is not defined or if the port field is defined as a named (string) port. The endPort must be equal or greater than port.

- **egress.ports.protocol** (string)

protocol represents the protocol (TCP, UDP, or SCTP) which traffic must match. If not specified, this field defaults to TCP.

# NetworkPolicyList

NetworkPolicyList is a list of NetworkPolicy objects.

---

- **apiVersion:** networking.k8s.io/v1
- **kind:** NetworkPolicyList
- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]NetworkPolicy](#)), required
- items is a list of schema objects.

## Operations

---

### get read the specified NetworkPolicy

#### HTTP Request

GET /apis/networking.k8s.io/v1/namespaces/{namespace}/networkpolicies/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the NetworkPolicy
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([NetworkPolicy](#)): OK

401: Unauthorized

## list list or watch objects of kind NetworkPolicy

### HTTP Request

GET /apis/networking.k8s.io/v1/namespaces/{namespace}/networkpolicies

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([NetworkPolicyList](#)): OK

401: Unauthorized

## list list or watch objects of kind NetworkPolicy

### HTTP Request

GET /apis/networking.k8s.io/v1/networkpolicies

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

• **watch** (*in query*): boolean

• [watch](#)

## Response

200 ([NetworkPolicyList](#)): OK

401: Unauthorized

## create create a NetworkPolicy

### HTTP Request

POST /apis/networking.k8s.io/v1/namespaces/{namespace}/networkpolicies

### Parameters

• **namespace** (*in path*): string, required

[namespace](#)

• **body**: [NetworkPolicy](#), required

• **dryRun** (*in query*): string

[dryRun](#)

• **fieldManager** (*in query*): string

[fieldManager](#)

• **fieldValidation** (*in query*): string

[fieldValidation](#)

• **pretty** (*in query*): string

[pretty](#)

## Response

200 ([NetworkPolicy](#)): OK

201 ([NetworkPolicy](#)): Created

202 ([NetworkPolicy](#)): Accepted

401: Unauthorized



## update replace the specified NetworkPolicy

### HTTP Request

PUT /apis/networking.k8s.io/v1/namespaces/{namespace}/networkpolicies/{name}

### Parameters

- **name** (*in path*): string, required  
name of the NetworkPolicy
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [NetworkPolicy](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([NetworkPolicy](#)): OK

201 ([NetworkPolicy](#)): Created

401: Unauthorized

## patch partially update the specified NetworkPolicy

### HTTP Request

PATCH /apis/networking.k8s.io/v1/namespaces/{namespace}/networkpolicies/{name}

### Parameters

- **name** (*in path*): string, required  
name of the NetworkPolicy

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([NetworkPolicy](#)): OK

201 ([NetworkPolicy](#)): Created

401: Unauthorized

## delete delete a NetworkPolicy

### HTTP Request

DELETE /apis/networking.k8s.io/v1/namespaces/{namespace}/networkpolicies/{name}

### Parameters

- **name** (*in path*): string, required  
name of the NetworkPolicy
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of NetworkPolicy

### HTTP Request

DELETE /apis/networking.k8s.io/v1/namespaces/{namespace}/networkpolicies

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [DeleteOptions](#)

- **continue** (*in query*): string

[continue](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# PodDisruptionBudget

PodDisruptionBudget is an object to define the max disruption that can be caused to a collection of pods.

apiVersion: policy/v1

```
import "k8s.io/api/policy/v1"
```

## PodDisruptionBudget

PodDisruptionBudget is an object to define the max disruption that can be caused to a collection of pods

- 
- **apiVersion**: policy/v1

**kind:** PodDisruptionBudget

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([PodDisruptionBudgetSpec](#))

Specification of the desired behavior of the PodDisruptionBudget.

- **status** ([PodDisruptionBudgetStatus](#))

Most recently observed status of the PodDisruptionBudget.

## PodDisruptionBudgetSpec

PodDisruptionBudgetSpec is a description of a PodDisruptionBudget.

---

- **maxUnavailable** (IntOrString)

An eviction is allowed if at most "maxUnavailable" pods selected by "selector" are unavailable after the eviction, i.e. even in absence of the evicted pod. For example, one can prevent all voluntary evictions by specifying 0. This is a mutually exclusive setting with "minAvailable".

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **minAvailable** (IntOrString)

An eviction is allowed if at least "minAvailable" pods selected by "selector" will still be available after the eviction, i.e. even in the absence of the evicted pod. So for example you can prevent all voluntary evictions by specifying "100%".

*IntOrString is a type that can hold an int32 or a string. When used in JSON or YAML marshalling and unmarshalling, it produces or consumes the inner type. This allows you to have, for example, a JSON field that can accept a name or number.*

- **selector** ([LabelSelector](#))

Label query over pods whose evictions are managed by the disruption budget. A null selector will match no pods, while an empty ({} ) selector will select all pods within the namespace.

- **unhealthyPodEvictionPolicy** (string)

UnhealthyPodEvictionPolicy defines the criteria for when unhealthy pods should be considered for eviction. Current implementation considers healthy pods, as pods that have status.conditions item with type="Ready",status="True".

Valid policies are `IfHealthyBudget` and `AlwaysAllow`. If no policy is specified, the default behavior will be used, which corresponds to the `IfHealthyBudget` policy.

`IfHealthyBudget` policy means that running pods (`status.phase="Running"`), but not yet healthy can be evicted only if the guarded application is not disrupted (`status.currentHealthy` is at least equal to `status.desiredHealthy`). Healthy pods will be subject to the PDB for eviction.

`AlwaysAllow` policy means that all running pods (`status.phase="Running"`), but not yet healthy are considered disrupted and can be evicted regardless of whether the criteria in a PDB is met. This means perspective running pods of a disrupted application might not get a chance to become healthy. Healthy pods will be subject to the PDB for eviction.

Additional policies may be added in the future. Clients making eviction decisions should disallow eviction of unhealthy pods if they encounter an unrecognized policy in this field.

This field is beta-level. The eviction API uses this field when the feature gate `PDBUnhealthyPodEvictionPolicy` is enabled (enabled by default).

## PodDisruptionBudgetStatus

`PodDisruptionBudgetStatus` represents information about the status of a `PodDisruptionBudget`. Status may trail the actual state of a system.

---

- **currentHealthy** (int32), required  
current number of healthy pods
- **desiredHealthy** (int32), required  
minimum desired number of healthy pods
- **disruptionsAllowed** (int32), required  
Number of pod disruptions that are currently allowed.
- **expectedPods** (int32), required  
total number of pods counted by this disruption budget
- **conditions** ([]Condition)

*Patch strategy: merge on key type*

*Map: unique values on key type will be kept during a merge*

Conditions contain conditions for PDB. The disruption controller sets the `DisruptionAllowed` condition. The following are known values for the reason field (additional reasons could be added in the future): - `SyncFailed`: The controller

encountered an error and wasn't able to compute the number of allowed disruptions. Therefore no disruptions are allowed and the status of the condition will be False.

- **InsufficientPods**: The number of pods are either at or below the number required by the PodDisruptionBudget. No disruptions are allowed and the status of the condition will be False.
- **SufficientPods**: There are more pods than required by the PodDisruptionBudget. The condition will be True, and the number of allowed disruptions are provided by the disruptionsAllowed property.

*Condition contains details for one aspect of the current state of this API Resource.*

- **conditions.lastTransitionTime** (Time), required

lastTransitionTime is the last time the condition transitioned from one status to another. This should be when the underlying condition changed. If that is not known, then using the time when the API field changed is acceptable.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string), required

message is a human readable message indicating details about the transition. This may be an empty string.

- **conditions.reason** (string), required

reason contains a programmatic identifier indicating the reason for the condition's last transition. Producers of specific condition types may define expected values and meanings for this field, and whether the values are considered a guaranteed API. The value should be a CamelCase string. This field may not be empty.

- **conditions.status** (string), required

status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

type of condition in CamelCase or in foo.example.com/CamelCase.

- **conditions.observedGeneration** (int64)

observedGeneration represents the .metadata.generation that the condition was set based upon. For instance, if .metadata.generation is currently 12, but the .status.conditions[x].observedGeneration is 9, the condition is out of date with respect to the current state of the instance.

- **disruptedPods** (map[string]Time)

DisruptedPods contains information about pods whose eviction was processed by the API server eviction subresource handler but has not yet been observed by the PodDisruptionBudget controller. A pod will be in this map from the time when the API server processed the eviction request to the time when the pod is seen by PDB controller

as having been marked for deletion (or after a timeout). The key in the map is the name of the pod and the value is the time when the API server processed the eviction request. If the deletion didn't occur and a pod is still there it will be removed from the list automatically by PodDisruptionBudget controller after some time. If everything goes smooth this map should be empty for the most of the time. Large number of entries in the map may indicate problems with pod deletions.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **observedGeneration** (int64)

Most recent generation observed when updating this PDB status. DisruptionsAllowed and other status information is valid only if observedGeneration equals to PDB's object generation.

## PodDisruptionBudgetList

PodDisruptionBudgetList is a collection of PodDisruptionBudgets.

---

- **apiVersion:** policy/v1
- **kind:** PodDisruptionBudgetList
- **metadata** ([ListMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([][PodDisruptionBudget](#)), required

Items is a list of PodDisruptionBudgets

## Operations

---

### get read the specified PodDisruptionBudget

#### HTTP Request

GET /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the PodDisruptionBudget
- **namespace** (*in path*): string, required  
[namespace](#)



- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodDisruptionBudget](#)): OK

401: Unauthorized

## get read status of the specified PodDisruptionBudget

### HTTP Request

GET /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PodDisruptionBudget
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodDisruptionBudget](#)): OK

401: Unauthorized

## list list or watch objects of kind PodDisruptionBudget

### HTTP Request

GET /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)

- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([PodDisruptionBudgetList](#)): OK

401: Unauthorized

**list list or watch objects of kind PodDisruptionBudget**

## HTTP Request

GET /apis/policy/v1/poddisruptionbudgets

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([PodDisruptionBudgetList](#)): OK

401: Unauthorized

## create create a PodDisruptionBudget

### HTTP Request

POST /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [PodDisruptionBudget](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([PodDisruptionBudget](#)): OK

201 ([PodDisruptionBudget](#)): Created

202 ([PodDisruptionBudget](#)): Accepted

401: Unauthorized

## update replace the specified PodDisruptionBudget

### HTTP Request

PUT /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PodDisruptionBudget
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [PodDisruptionBudget](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PodDisruptionBudget](#)): OK

201 ([PodDisruptionBudget](#)): Created

401: Unauthorized

## update replace status of the specified PodDisruptionBudget

### HTTP Request

PUT /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PodDisruptionBudget
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [PodDisruptionBudget](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodDisruptionBudget](#)): OK

201 ([PodDisruptionBudget](#)): Created

401: Unauthorized

## patch partially update the specified PodDisruptionBudget

### HTTP Request

PATCH /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PodDisruptionBudget
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodDisruptionBudget](#)): OK

201 ([PodDisruptionBudget](#)): Created

401: Unauthorized

## patch partially update status of the specified PodDisruptionBudget

### HTTP Request

PATCH /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PodDisruptionBudget
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PodDisruptionBudget](#)): OK

201 ([PodDisruptionBudget](#)): Created

401: Unauthorized

## delete delete a PodDisruptionBudget

### HTTP Request

DELETE /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PodDisruptionBudget
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of PodDisruptionBudget

### HTTP Request

DELETE /apis/policy/v1/namespaces/{namespace}/poddisruptionbudgets

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)



- **body:** [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# IPAddress v1alpha1

IPAddress represents a single IP of a single IP Family.

apiVersion: networking.k8s.io/v1alpha1

```
import "k8s.io/api/networking/v1alpha1"
```

## IPAddress

IPAddress represents a single IP of a single IP Family. The object is designed to be used by APIs that operate on IP addresses. The object is used by the Service core API for allocation of IP addresses. An IP address can be represented in different formats, to guarantee the uniqueness of the IP, the name of the object is the IP address in canonical format, four decimal digits separated by dots suppressing leading zeros for IPv4 and the representation defined by RFC 5952 for IPv6. Valid: 192.168.1.5 or 2001:db8::1 or 2001:db8:aaaa:bbbb:cccc:dddd:eeee:1 Invalid: 10.01.2.3 or 2001:db8:0:0:0::1

- 
- **apiVersion:** networking.k8s.io/v1alpha1

- **kind:** IPAddress

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([IPAddressSpec](#))

spec is the desired state of the IPAddress. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## IPAddressSpec

IPAddressSpec describe the attributes in an IP Address.

- 
- **parentRef** ([ParentReference](#))

ParentRef references the resource that an IPAddress is attached to. An IPAddress must reference a parent object.

*ParentReference describes a reference to a parent object.*

- **parentRef.group** (string)

Group is the group of the object being referenced.

- **parentRef.name** (string)

Name is the name of the object being referenced.

- **parentRef.namespace** (string)  
Namespace is the namespace of the object being referenced.
- **parentRef.resource** (string)  
Resource is the resource of the object being referenced.
- **parentRef.uid** (string)  
UID is the uid of the object being referenced.

## IPAddressList

IPAddressList contains a list of IPAddress.

- 
- **apiVersion:** networking.k8s.io/v1alpha1
  - **kind:** IPAddressList
  - **metadata** ([ListMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]IPAddress](#)), required  
items is the list of IPAddresses.

## Operations

---

### get read the specified IPAddress

#### HTTP Request

GET /apis/networking.k8s.io/v1alpha1/ipaddresses/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the IPAddress
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([IPAddress](#)): OK

401: Unauthorized

## list list or watch objects of kind IPAddress

### HTTP Request

GET /apis/networking.k8s.io/v1alpha1/ipaddresses

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([IPAddressList](#)): OK

401: Unauthorized

## create create an IPAddress

### HTTP Request

POST /apis/networking.k8s.io/v1alpha1/ipaddresses

### Parameters

- **body**: [IPAddress](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([IPAddress](#)): OK

201 ([IPAddress](#)): Created

202 ([IPAddress](#)): Accepted

401: Unauthorized

## update replace the specified IPAddress

### HTTP Request

PUT /apis/networking.k8s.io/v1alpha1/ipaddresses/{name}

### Parameters

- **name** (*in path*): string, required

name of the IPAddress

- **body:** [IPAddress](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([IPAddress](#)): OK

201 ([IPAddress](#)): Created

401: Unauthorized

## patch partially update the specified IPAddress

### HTTP Request

PATCH /apis/networking.k8s.io/v1alpha1/ipaddresses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the IPAddress
- **body:** [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)

- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([IPAddress](#)): OK

201 ([IPAddress](#)): Created

401: Unauthorized

## delete delete an IPAddress

### HTTP Request

DELETE /apis/networking.k8s.io/v1alpha1/ipaddresses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the IPAddress
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of IPAddress

### HTTP Request

DELETE /apis/networking.k8s.io/v1alpha1/ipaddresses

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer



[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Extend Resources

---

## [CustomResourceDefinition](#)

CustomResourceDefinition represents a resource that should be exposed on the API server.

## [MutatingWebhookConfiguration](#)

MutatingWebhookConfiguration describes the configuration of and admission webhook that accept or reject and may change the object.

## [ValidatingWebhookConfiguration](#)

ValidatingWebhookConfiguration describes the configuration of and admission webhook that accept or reject and object without changing it.

## [ValidatingAdmissionPolicy v1beta1](#)

ValidatingAdmissionPolicy describes the definition of an admission validation policy that accepts or rejects an object without changing it.

# CustomResourceDefinition

CustomResourceDefinition represents a resource that should be exposed on the API server.

```
apiVersion: apiextensions.k8s.io/v1
```

```
import "k8s.io/apiextensions-apiserver/pkg/apis/apiextensions/v1"
```

## CustomResourceDefinition

CustomResourceDefinition represents a resource that should be exposed on the API server. Its name MUST be in the format `<.spec.name>.<.spec.group>`.

---

- **apiVersion:** apiextensions.k8s.io/v1
- **kind:** CustomResourceDefinition
- **metadata** ([ObjectMeta](#))

Standard object's metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([CustomResourceDefinitionSpec](#)), required

spec describes how the user wants the resources to appear

- **status** ([CustomResourceDefinitionStatus](#))

status indicates the actual state of the CustomResourceDefinition

## CustomResourceDefinitionSpec

CustomResourceDefinitionSpec describes how a user wants their resource to appear

---

- **group** (string), required

group is the API group of the defined custom resource. The custom resources are served under `/apis/<group>/...`. Must match the name of the CustomResourceDefinition (in the form `<names.plural>.<group>`).

- **names** ([CustomResourceDefinitionNames](#)), required

names specify the resource and kind names for the custom resource.

*CustomResourceDefinitionNames indicates the names to serve this CustomResourceDefinition*

- **names.kind** (string), required

kind is the serialized kind of the resource. It is normally CamelCase and singular. Custom resource instances will use this value as the kind attribute in API calls.

- **names.plural** (string), required

plural is the plural name of the resource to serve. The custom resources are served under `/apis/<group>/<version>/.../<plural>`. Must match the name of the CustomResourceDefinition (in the form `<names.plural>.<group>`). Must be all lowercase.

- **names.categories** ([]string)

categories is a list of grouped resources this custom resource belongs to (e.g. 'all'). This is published in API discovery documents, and used by clients to support invocations like `kubectl get all`.

- **names.listKind** (string)

listKind is the serialized kind of the list for this resource. Defaults to "kindList".

- **names.shortNames** ([]string)

shortNames are short names for the resource, exposed in API discovery documents, and used by clients to support invocations like `kubectl get <shortname>`. It must be all lowercase.

- **names.singular** (string)

singular is the singular name of the resource. It must be all lowercase. Defaults to lowercased kind.

- **scope** (string), required

scope indicates whether the defined custom resource is cluster- or namespace-scoped. Allowed values are Cluster and Namespaced.

- **versions** ([]CustomResourceDefinitionVersion), required

versions is the list of all API versions of the defined custom resource. Version names are used to compute the order in which served versions are listed in API discovery. If the version string is "kube-like", it will sort above non "kube-like" version strings, which are ordered lexicographically. "Kube-like" versions start with a "v", then are followed by a number (the major version), then optionally the string "alpha" or "beta" and another number (the minor version). These are sorted first by GA > beta > alpha (where GA is a version with no suffix such as beta or alpha), and then by comparing major version, then minor version. An example sorted list of versions: v10, v2, v1, v11beta2, v10beta3, v3beta1, v12alpha1, v11alpha2, foo1, foo10.

*CustomResourceDefinitionVersion describes a version for CRD.*

- **versions.name** (string), required

name is the version name, e.g. "v1", "v2beta1", etc. The custom resources are served under this version at `/apis/<group>/<version>/...` if served is true.

- **versions.served** (boolean), required

served is a flag enabling/disabling this version from being served via REST APIs

- **versions.storage** (boolean), required

storage indicates this version should be used when persisting custom resources to storage. There must be exactly one version with storage=true.

- **versions.additionalPrinterColumns** ([]CustomResourceColumnDefinition)

additionalPrinterColumns specifies additional columns returned in Table output. See <https://kubernetes.io/docs/reference/using-api/api-concepts/#receiving-resources-as-tables> for details. If no columns are specified, a single column displaying the age of the custom resource is used.

*CustomResourceColumnDefinition specifies a column for server side printing.*

- **versions.additionalPrinterColumns.jsonPath** (string), required

jsonPath is a simple JSON path (i.e. with array notation) which is evaluated against each custom resource to produce the value for this column.

- **versions.additionalPrinterColumns.name** (string), required
  - name is a human readable name for the column.
- **versions.additionalPrinterColumns.type** (string), required
  - type is an OpenAPI type definition for this column. See <https://github.com/OAI/OpenAPI-Specification/blob/master/versions/2.0.md#data-types> for details.
- **versions.additionalPrinterColumns.description** (string)
  - description is a human readable description of this column.
- **versions.additionalPrinterColumns.format** (string)
  - format is an optional OpenAPI type definition for this column. The 'name' format is applied to the primary identifier column to assist in clients identifying column is the resource name. See <https://github.com/OAI/OpenAPI-Specification/blob/master/versions/2.0.md#data-types> for details.
- **versions.additionalPrinterColumns.priority** (int32)
  - priority is an integer defining the relative importance of this column compared to others. Lower numbers are considered higher priority. Columns that may be omitted in limited space scenarios should be given a priority greater than 0.
- **versions.deprecated** (boolean)
  - deprecated indicates this version of the custom resource API is deprecated. When set to true, API requests to this version receive a warning header in the server response. Defaults to false.
- **versions.deprecationWarning** (string)
  - deprecationWarning overrides the default warning returned to API clients. May only be set when deprecated is true. The default warning indicates this version is deprecated and recommends use of the newest served version of equal or greater stability, if one exists.
- **versions.schema** (CustomResourceValidation)
  - schema describes the schema used for validation, pruning, and defaulting of this version of the custom resource.
  - *CustomResourceValidation is a list of validation methods for CustomResources.*
  - **versions.schema.openAPIV3Schema** ([JSONSchemaProps](#))
    - openAPIV3Schema is the OpenAPI v3 schema to use for validation and pruning.
- **versions.subresources** (CustomResourceSubresources)

subresources specify what subresources this version of the defined custom resource have.

*CustomResourceSubresources* defines the status and scale subresources for *CustomResources*.

- **versions.subresources.scale** (*CustomResourceSubresourceScale*)

scale indicates the custom resource should serve a `/scale` subresource that returns an autoscaling/v1 Scale object.

*CustomResourceSubresourceScale* defines how to serve the scale subresource for *CustomResources*.

- **versions.subresources.scale.specReplicasPath** (string), required

specReplicasPath defines the JSON path inside of a custom resource that corresponds to Scale spec.replicas. Only JSON paths without the array notation are allowed. Must be a JSON Path under .spec. If there is no value under the given path in the custom resource, the `/scale` subresource will return an error on GET.

- **versions.subresources.scale.statusReplicasPath** (string), required

statusReplicasPath defines the JSON path inside of a custom resource that corresponds to Scale status.replicas. Only JSON paths without the array notation are allowed. Must be a JSON Path under .status. If there is no value under the given path in the custom resource, the status.replicas value in the `/scale` subresource will default to 0.

- **versions.subresources.scale.labelSelectorPath** (string)

labelSelectorPath defines the JSON path inside of a custom resource that corresponds to Scale status.selector. Only JSON paths without the array notation are allowed. Must be a JSON Path under .status or .spec. Must be set to work with HorizontalPodAutoscaler. The field pointed by this JSON path must be a string field (not a complex selector struct) which contains a serialized label selector in string form. More info: <https://kubernetes.io/docs/tasks/access-kubernetes-api/custom-resources/custom-resource-definitions#scale-subresource> If there is no value under the given path in the custom resource, the status.selector value in the `/scale` subresource will default to the empty string.

- **versions.subresources.status** (*CustomResourceSubresourceStatus*)

status indicates the custom resource should serve a `/status` subresource. When enabled: 1. requests to the custom resource primary endpoint ignore changes to the status stanza of the object. 2. requests to the custom resource `/status` subresource ignore changes to anything other than the status stanza of the object.

*CustomResourceSubresourceStatus* defines how to serve the status subresource for *CustomResources*. Status is represented by the `.status` JSON path inside of a *CustomResource*. When set, \* exposes a `/status` subresource for the custom

*resource \* PUT requests to the /status subresource take a custom resource object, and ignore changes to anything except the status stanza \* PUT/POST/PATCH requests to the custom resource ignore changes to the status stanza*

- **conversion** (CustomResourceConversion)

conversion defines conversion settings for the CRD.

*CustomResourceConversion describes how to convert different versions of a CR.*

- **conversion.strategy** (string), required

strategy specifies how custom resources are converted between versions. Allowed values are: - "None": The converter only change the apiVersion and would not touch any other field in the custom resource. - "Webhook": API Server will call to an external webhook to do the conversion. Additional information is needed for this option. This requires spec.preserveUnknownFields to be false, and spec.conversion.webhook to be set.

- **conversion.webhook** (WebhookConversion)

webhook describes how to call the conversion webhook. Required when strategy is set to "Webhook".

*WebhookConversion describes how to call a conversion webhook*

- **conversion.webhook.conversionReviewVersions** ([]string), required

conversionReviewVersions is an ordered list of preferred ConversionReview versions the Webhook expects. The API server will use the first version in the list which it supports. If none of the versions specified in this list are supported by API server, conversion will fail for the custom resource. If a persisted Webhook configuration specifies allowed versions and does not include any versions known to the API Server, calls to the webhook will fail.

- **conversion.webhook.clientConfig** (WebhookClientConfig)

clientConfig is the instructions for how to call the webhook if strategy is Webhook.

*WebhookClientConfig contains the information to make a TLS connection with the webhook.*

- **conversion.webhook.clientConfig.caBundle** ([]byte)

caBundle is a PEM encoded CA bundle which will be used to validate the webhook's server certificate. If unspecified, system trust roots on the apiserver are used.

- **conversion.webhook.clientConfig.service** (ServiceReference)

service is a reference to the service for this webhook. Either service or url must be specified.

If the webhook is running within the cluster, then you should use service.

*ServiceReference holds a reference to Service.legacy.k8s.io*

- **conversion.webhook.clientConfig.service.name** (string), required

name is the name of the service. Required

- **conversion.webhook.clientConfig.service.namespace** (string), required

namespace is the namespace of the service. Required

- **conversion.webhook.clientConfig.service.path** (string)

path is an optional URL path at which the webhook will be contacted.

- **conversion.webhook.clientConfig.service.port** (int32)

port is an optional service port at which the webhook will be contacted. port should be a valid port number (1-65535, inclusive). Defaults to 443 for backward compatibility.

- **conversion.webhook.clientConfig.url** (string)

url gives the location of the webhook, in standard URL form (scheme://host:port/path). Exactly one of url or service must be specified.

The host should not refer to a service running in the cluster; use the service field instead. The host might be resolved via external DNS in some apiservers (e.g., kube-apiserver cannot resolve in-cluster DNS as that would be a layering violation). host may also be an IP address.

Please note that using localhost or 127.0.0.1 as a host is risky unless you take great care to run this webhook on all hosts which run an apiserver which might need to make calls to this webhook. Such installs are likely to be non-portable, i.e., not easy to turn up in a new cluster.

The scheme must be "https"; the URL must begin with "https://".

A path is optional, and if present may be any string permissible in a URL. You may use the path to pass an arbitrary string to the webhook, for example, a cluster identifier.

Attempting to use a user or basic auth e.g. "user:password@" is not allowed. Fragments ("#...") and query parameters ("?...") are not allowed, either.

- **preserveUnknownFields** (boolean)

preserveUnknownFields indicates that object fields which are not specified in the OpenAPI schema should be preserved when persisting to storage. apiVersion, kind, metadata and known fields inside metadata are always preserved. This field is deprecated in favor of setting x-preserve-unknown-fields to true in spec.versions[\*].schema.openAPIV3Schema. See <https://kubernetes.io/docs/tasks/extend-kubernetes/custom-resources/custom-resource-definitions/#field-pruning> for details.

## JSONSchemaProps

JSONSchemaProps is a JSON-Schema following Specification Draft 4 (<http://json-schema.org/>).

---

- **\$ref** (string)
- **\$schema** (string)
- **additionalItems** (JSONSchemaPropsOrBool)

*JSONSchemaPropsOrBool represents JSONSchemaProps or a boolean value. Defaults to true for the boolean property.*

- **additionalProperties** (JSONSchemaPropsOrBool)

*JSONSchemaPropsOrBool represents JSONSchemaProps or a boolean value. Defaults to true for the boolean property.*

- **allOf** ([][JSONSchemaProps](#))
- **anyOf** ([][JSONSchemaProps](#))
- **default** (JSON)

default is a default value for undefined object fields. Defaulting is a beta feature under the CustomResourceDefaulting feature gate. Defaulting requires spec.preserveUnknownFields to be false.

*JSON represents any valid JSON value. These types are supported: bool, int64, float64, string, []interface{}, map[string]interface{} and nil.*

- **definitions** (map[string][JSONSchemaProps](#))
- **dependencies** (map[string]JSONSchemaPropsOrStringArray)

*JSONSchemaPropsOrStringArray represents a JSONSchemaProps or a string array.*

- **description** (string)
- **enum** ([]JSON)

*JSON represents any valid JSON value. These types are supported: bool, int64, float64, string, []interface{}, map[string]interface{} and nil.*

- **example** (JSON)



*JSON represents any valid JSON value. These types are supported: bool, int64, float64, string, []interface{}, map[string]interface{} and nil.*

- **exclusiveMaximum** (boolean)
- **exclusiveMinimum** (boolean)
- **externalDocs** (ExternalDocumentation)

*ExternalDocumentation allows referencing an external resource for extended documentation.*

- **externalDocs.description** (string)
- **externalDocs.url** (string)

- **format** (string)

format is an OpenAPI v3 format string. Unknown formats are ignored. The following formats are validated:

- **bsonobjectid**: a bson object ID, i.e. a 24 characters hex string - **uri**: an URI as parsed by Golang net/url.ParseRequestURI - **email**: an email address as parsed by Golang net/mail.ParseAddress - **hostname**: a valid representation for an Internet host name, as defined by RFC 1034, section 3.1 [RFC1034]. - **ipv4**: an IPv4 IP as parsed by Golang net.ParseIP - **ipv6**: an IPv6 IP as parsed by Golang net.ParseIP - **cidr**: a CIDR as parsed by Golang net.ParseCIDR - **mac**: a MAC address as parsed by Golang net.ParseMAC - **uuid**: an UUID that allows uppercase defined by the regex `(?i)^[0-9a-f]{8}-?[0-9a-f]{4}-?[0-9a-f]{4}-?[0-9a-f]{12}$` - **uuid3**: an UUID3 that allows uppercase defined by the regex `(?i)^[0-9a-f]{8}-?[0-9a-f]{4}-?3[0-9a-f]{3}-?[0-9a-f]{4}-?[0-9a-f]{12}$` - **uuid4**: an UUID4 that allows uppercase defined by the regex `(?i)^[0-9a-f]{8}-?[0-9a-f]{4}-?4[0-9a-f]{3}-?[89ab][0-9a-f]{3}-?[0-9a-f]{12}$` - **uuid5**: an UUID5 that allows uppercase defined by the regex `(?i)^[0-9a-f]{8}-?[0-9a-f]{4}-?5[0-9a-f]{3}-?[89ab][0-9a-f]{3}-?[0-9a-f]{12}$` - **isbn**: an ISBN10 or ISBN13 number string like "0321751043" or "978-0321751041" - **isbn10**: an ISBN10 number string like "0321751043" - **isbn13**: an ISBN13 number string like "978-0321751041" - **creditcard**: a credit card number defined by the regex `^(?:4[0-9]{12}(?:[0-9]{3})?|5[1-5][0-9]{14}|6(?:011|5[0-9][0-9])[0-9]{12}|3[47][0-9]{13}|3(?:0[0-5]||[68][0-9])[0-9]{11})(?:2131|1800|35\d{3})\d{11}$` with any non digit characters mixed in - **ssn**: a U.S. social security number following the regex `^\d{3}[- ]?\d{2}[- ]?\d{4}$` - **hexcolor**: an hexadecimal color code like "#FFFFFF" following the regex `^#?([0-9a-fA-F]{3}|[0-9a-fA-F]{6})$` - **rgbcolor**: an RGB color code like rgb like "rgb(255,255,255)" - **byte**: base64 encoded binary data - **password**: any kind of string - **date**: a date string like "2006-01-02" as defined by full-date in RFC3339 - **duration**: a duration string like "22 ns" as parsed by Golang time.ParseDuration or compatible with Scala duration format - **datetime**: a date time string like "2014-12-15T19:30:20.000Z" as defined by date-time in RFC3339.

- **id** (string)
- **items** (JSONSchemaPropsOrArray)

*JSONSchemaPropsOrArray represents a value that can either be a JSONSchemaProps or an array of JSONSchemaProps. Mainly here for serialization purposes.*

- **maxItems** (int64)
- **maxLength** (int64)
- **maxProperties** (int64)
- **maximum** (double)
- **minItems** (int64)
- **minLength** (int64)
- **minProperties** (int64)
- **minimum** (double)
- **multipleOf** (double)
- **not** ([JSONSchemaProps](#))
- **nullable** (boolean)
- **oneOf** ([\[\]JSONSchemaProps](#))
- **pattern** (string)
- **patternProperties** (map[string][JSONSchemaProps](#))
- **properties** (map[string][JSONSchemaProps](#))
- **required** ([\[\]string](#))
- **title** (string)
- **type** (string)
- **uniqueItems** (boolean)
- **x-kubernetes-embedded-resource** (boolean)

x-kubernetes-embedded-resource defines that the value is an embedded Kubernetes runtime.Object, with TypeMeta and ObjectMeta. The type must be object. It is allowed to further restrict the embedded object. kind, apiVersion and metadata are validated automatically. x-kubernetes-preserve-unknown-fields is allowed to be true, but does not have to be if the object is fully specified (up to kind, apiVersion, metadata).

- **x-kubernetes-int-or-string** (boolean)

x-kubernetes-int-or-string specifies that this value is either an integer or a string. If this is true, an empty type is allowed and type as child of anyOf is permitted if following one of the following patterns:

1. anyOf:
  - type: integer
  - type: string

2. allOf:

- anyOf:
  - type: integer
  - type: string
- ... zero or more

- **x-kubernetes-list-map-keys** ([]string)

x-kubernetes-list-map-keys annotates an array with the x-kubernetes-list-type map by specifying the keys used as the index of the map.

This tag MUST only be used on lists that have the "x-kubernetes-list-type" extension set to "map". Also, the values specified for this attribute must be a scalar typed field of the child structure (no nesting is supported).

The properties specified must either be required or have a default value, to ensure those properties are present for all list items.

- **x-kubernetes-list-type** (string)

x-kubernetes-list-type annotates an array to further describe its topology. This extension must only be used on lists and may have 3 possible values:

1. atomic: the list is treated as a single entity, like a scalar. Atomic lists will be entirely replaced when updated. This extension may be used on any type of list (struct, scalar, ...).
2. set: Sets are lists that must not have multiple items with the same value. Each value must be a scalar, an object with x-kubernetes-map-type atomic or an array with x-kubernetes-list-type atomic.
3. map: These lists are like maps in that their elements have a non-index key used to identify them. Order is preserved upon merge. The map tag must only be used on a list with elements of type object. Defaults to atomic for arrays.

- **x-kubernetes-map-type** (string)

x-kubernetes-map-type annotates an object to further describe its topology. This extension must only be used when type is object and may have 2 possible values:

1. granular: These maps are actual maps (key-value pairs) and each fields are independent from each other (they can each be manipulated by separate actors). This is the default behaviour for all maps.
2. atomic: the list is treated as a single entity, like a scalar. Atomic maps will be entirely replaced when updated.

- **x-kubernetes-preserve-unknown-fields** (boolean)

x-kubernetes-preserve-unknown-fields stops the API server decoding step from pruning fields which are not specified in the validation schema. This affects fields recursively, but switches back to normal pruning behaviour if nested properties or additionalProperties are specified in the schema. This can either be true or undefined. False is forbidden.

- **x-kubernetes-validations** ([]ValidationRule)

*Patch strategy: merge on key rule*

*Map: unique values on key rule will be kept during a merge*

x-kubernetes-validations describes a list of validation rules written in the CEL expression language. This field is an alpha-level. Using this field requires the feature gate CustomResourceValidationExpressions to be enabled.

*ValidationRule describes a validation rule written in the CEL expression language.*

- **x-kubernetes-validations.rule** (string), required

Rule represents the expression which will be evaluated by CEL. ref: <https://github.com/google/cel-spec> The Rule is scoped to the location of the x-kubernetes-validations extension in the schema. The self variable in the CEL expression is bound to the scoped value. Example: - Rule scoped to the root of a resource with a status subresource: {"rule": "self.status.actual <= self.spec.maxDesired"}

If the Rule is scoped to an object with properties, the accessible properties of the object are field selectable via self.field and field presence can be checked via has(self.field). Null valued fields are treated as absent fields in CEL expressions. If the Rule is scoped to an object with additionalProperties (i.e. a map) the value of the map are accessible via self[mapKey], map containment can be checked via mapKey in self and all entries of the map are accessible via CEL macros and functions such as self.all(...). If the Rule is scoped to an array, the elements of the array are accessible via self[i] and also by macros and functions. If the Rule is scoped to a scalar, self is bound to the scalar value. Examples: - Rule scoped to a map of objects: {"rule": "self.components['Widget'].priority < 10"} - Rule scoped to a list of integers: {"rule": "self.values.all(value, value >= 0 && value < 100)"} - Rule scoped to a string value: {"rule": "self.startsWith('kube')"}

The apiVersion, kind, metadata.name and metadata.generateName are always accessible from the root of the object and from any x-kubernetes-embedded-resource annotated objects. No other metadata properties are accessible.

Unknown data preserved in custom resources via x-kubernetes-preserve-unknown-fields is not accessible in CEL expressions. This includes: - Unknown field values that are preserved by object schemas with x-kubernetes-preserve-unknown-fields. - Object properties where the property schema is of an "unknown type". An "unknown type" is recursively defined as:

- A schema with no type and x-kubernetes-preserve-unknown-fields set to true
- An array where the items schema is of an "unknown type"
- An object where the additionalProperties schema is of an "unknown type"

Only property names of the form [a-zA-Z\_./][a-zA-Z0-9\_./]\* are accessible.

Accessible property names are escaped according to the following rules when accessed in the expression: - " **escapes to 'underscores'** - ' **escapes to 'dot'** - ' **escapes to 'dash'** - '/' **escapes to 'slash'** - **Property names that exactly match a CEL RESERVED keyword escape to '{keyword}\_\_'**. The keywords are: "true", "false", "null", "in", "as", "break", "const", "continue", "else", "for", "function", "if", "import", "let", "loop", "package", "namespace", "return". Examples:

- Rule accessing a property named "namespace": {"rule": "self.namespace > 0"}
- Rule accessing a property named "x-prop": {"rule": "self.x\_\_dash\_\_prop > 0"}

- Rule accessing a property named "redact\_\_d": {"rule": "self.redact\_\_underscores\_\_d > 0"}

Equality on arrays with x-kubernetes-list-type of 'set' or 'map' ignores element order, i.e. [1, 2] == [2, 1]. Concatenation on arrays with x-kubernetes-list-type use the semantics of the list type:

- 'set':  $X + Y$  performs a union where the array positions of all elements in  $X$  are preserved and non-intersecting elements in  $Y$  are appended, retaining their partial order.
- 'map':  $X + Y$  performs a merge where the array positions of all keys in  $X$  are preserved but the values are overwritten by values in  $Y$  when the key sets of  $X$  and  $Y$  intersect. Elements in  $Y$  with non-intersecting keys are appended, retaining their partial order.

◦ **x-kubernetes-validations.fieldPath** (string)

fieldPath represents the field path returned when the validation fails. It must be a relative JSON path (i.e. with array notation) scoped to the location of this x-kubernetes-validations extension in the schema and refer to an existing field. e.g. when validation checks if a specific attribute foo under a map testMap, the fieldPath could be set to .testMap.foo If the validation checks two lists must have unique attributes, the fieldPath could be set to either of the list: e.g. .testList It does not support list numeric index. It supports child operation to refer to an existing field currently. Refer to [JSONPath support in Kubernetes](#) for more info. Numeric index of array is not supported. For field name which contains special characters, use ['specialName'] to refer the field name. e.g. for attribute foo.34\$ appears in a list testList, the fieldPath could be set to .testList['foo.34\$']

◦ **x-kubernetes-validations.message** (string)

Message represents the message displayed when validation fails. The message is required if the Rule contains line breaks. The message must not contain line breaks. If unset, the message is "failed rule: {Rule}". e.g. "must be a URL with the host matching spec.host"

◦ **x-kubernetes-validations.messageExpression** (string)

MessageExpression declares a CEL expression that evaluates to the validation failure message that is returned when this rule fails. Since messageExpression is used as a failure message, it must evaluate to a string. If both message and messageExpression are present on a rule, then messageExpression will be used if validation fails. If messageExpression results in a runtime error, the runtime error is logged, and the validation failure message is produced as if the messageExpression field were unset. If messageExpression evaluates to an empty string, a string with only spaces, or a string that contains line breaks, then the validation failure message will also be produced as if the messageExpression field were unset, and the fact that messageExpression produced an empty string/string with only spaces/string with line breaks will be logged. messageExpression has access to all the same variables as the rule; the only difference is the return type. Example: "x must be less than max (" + string(self.max) + ")"

◦ **x-kubernetes-validations.reason** (string)

reason provides a machine-readable validation failure reason that is returned to the caller when a request fails this validation rule. The HTTP status code returned to the caller will match the reason of the first failed validation rule. The currently supported reasons are: "FieldValueInvalid", "FieldValueForbidden", "FieldValueRequired", "FieldValueDuplicate". If not set, default to use "FieldValueInvalid". All future added reasons must be accepted by clients when reading this value and unknown reasons should be treated as FieldValueInvalid.

## CustomResourceDefinitionStatus

CustomResourceDefinitionStatus indicates the state of the CustomResourceDefinition

---

- **acceptedNames** (CustomResourceDefinitionNames)

acceptedNames are the names that are actually being used to serve discovery. They may be different than the names in spec.

*CustomResourceDefinitionNames indicates the names to serve this CustomResourceDefinition*

- **acceptedNames.kind** (string), required

kind is the serialized kind of the resource. It is normally CamelCase and singular. Custom resource instances will use this value as the kind attribute in API calls.

- **acceptedNames.plural** (string), required

plural is the plural name of the resource to serve. The custom resources are served under /apis/<group>/<version>/.../<plural>. Must match the name of the CustomResourceDefinition (in the form <names.plural>.<group>). Must be all lowercase.

- **acceptedNames.categories** ([]string)

categories is a list of grouped resources this custom resource belongs to (e.g. 'all'). This is published in API discovery documents, and used by clients to support invocations like kubectl get all.

- **acceptedNames.listKind** (string)

listKind is the serialized kind of the list for this resource. Defaults to "kindList".

- **acceptedNames.shortNames** ([]string)

shortNames are short names for the resource, exposed in API discovery documents, and used by clients to support invocations like kubectl get <shortname>. It must be all lowercase.

- **acceptedNames.singular** (string)

singular is the singular name of the resource. It must be all lowercase. Defaults to lowercased kind.

- **conditions** ([]CustomResourceDefinitionCondition)

*Map: unique values on key type will be kept during a merge*

conditions indicate state for particular aspects of a CustomResourceDefinition

*CustomResourceDefinitionCondition contains details for the current condition of this pod.*

- **conditions.status** (string), required

status is the status of the condition. Can be True, False, Unknown.

- **conditions.type** (string), required

type is the type of the condition. Types include Established, NamesAccepted and Terminating.

- **conditions.lastTransitionTime** (Time)

lastTransitionTime last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

message is a human-readable message indicating details about last transition.

- **conditions.reason** (string)

reason is a unique, one-word, CamelCase reason for the condition's last transition.

- **storedVersions** ([]string)

storedVersions lists all versions of CustomResources that were ever persisted. Tracking these versions allows a migration path for stored versions in etcd. The field is mutable so a migration controller can finish a migration to another version (ensuring no old objects are left in storage), and then remove the rest of the versions from this list. Versions may not be removed from spec.versions while they exist in this list.

## CustomResourceDefinitionList

CustomResourceDefinitionList is a list of CustomResourceDefinition objects.

---

- **items** ([][CustomResourceDefinition](#)), required

items list individual CustomResourceDefinition objects

- **apiVersion** (string)

APIVersion defines the versioned schema of this representation of an object. Servers should convert recognized schemas to the latest internal value, and may reject unrecognized values. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources>

**kind** (string)

- Kind is a string value representing the REST resource this object represents. Servers may infer this from the endpoint the client submits requests to. Cannot be updated. In CamelCase. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **metadata** ([ListMeta](#))

Standard object's metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

## Operations

---

### get read the specified CustomResourceDefinition

#### HTTP Request

GET /apis/apiextensions.k8s.io/v1/customresourcedefinitions/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the CustomResourceDefinition
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([CustomResourceDefinition](#)): OK

401: Unauthorized

### get read status of the specified CustomResourceDefinition

#### HTTP Request

GET /apis/apiextensions.k8s.io/v1/customresourcedefinitions/{name}/status

#### Parameters

- **name** (*in path*): string, required  
name of the CustomResourceDefinition
- **pretty** (*in query*): string



[pretty](#)

## Response

200 ([CustomResourceDefinition](#)): OK

401: Unauthorized

## list list or watch objects of kind CustomResourceDefinition

## HTTP Request

GET /apis/apiextensions.k8s.io/v1/customresourcedefinitions

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([CustomResourceDefinitionList](#)): OK

401: Unauthorized

## create create a CustomResourceDefinition

### HTTP Request

POST /apis/apiextensions.k8s.io/v1/customresourcedefinitions

### Parameters

- **body**: [CustomResourceDefinition](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([CustomResourceDefinition](#)): OK

201 ([CustomResourceDefinition](#)): Created

202 ([CustomResourceDefinition](#)): Accepted

401: Unauthorized

## update replace the specified CustomResourceDefinition

### HTTP Request

PUT /apis/apiextensions.k8s.io/v1/customresourcedefinitions/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CustomResourceDefinition
- **body**: [CustomResourceDefinition](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([CustomResourceDefinition](#)): OK

201 ([CustomResourceDefinition](#)): Created

401: Unauthorized

## update replace status of the specified CustomResourceDefinition

### HTTP Request

PUT /apis/apiextensions.k8s.io/v1/customresourcedefinitions/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the CustomResourceDefinition
- **body**: [CustomResourceDefinition](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([CustomResourceDefinition](#)): OK

201 ([CustomResourceDefinition](#)): Created

401: Unauthorized

## patch partially update the specified CustomResourceDefinition

### HTTP Request

PATCH /apis/apiextensions.k8s.io/v1/customresourcedefinitions/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CustomResourceDefinition

- **body**: [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([CustomResourceDefinition](#)): OK

201 ([CustomResourceDefinition](#)): Created

401: Unauthorized

## patch partially update status of the specified CustomResourceDefinition

### HTTP Request

PATCH /apis/apiextensions.k8s.io/v1/customresourcedefinitions/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the CustomResourceDefinition
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([CustomResourceDefinition](#)): OK

201 ([CustomResourceDefinition](#)): Created

401: Unauthorized

## delete delete a CustomResourceDefinition

### HTTP Request

DELETE /apis/apiextensions.k8s.io/v1/customresourcedefinitions/{name}

### Parameters

- **name** (*in path*): string, required  
name of the CustomResourceDefinition
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of CustomResourceDefinition

### HTTP Request

DELETE /apis/apiextensions.k8s.io/v1/customresourcedefinitions

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# MutatingWebhookConfiguration

MutatingWebhookConfiguration describes the configuration of and admission webhook that accept or reject and may change the object.

apiVersion: admissionregistration.k8s.io/v1

```
import "k8s.io/api/admissionregistration/v1"
```

## MutatingWebhookConfiguration

MutatingWebhookConfiguration describes the configuration of and admission webhook that accept or reject and may change the object.

---

- **apiVersion:** admissionregistration.k8s.io/v1
- **kind:** MutatingWebhookConfiguration
- **metadata** ([ObjectMeta](#))

Standard object metadata; More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>.

- **webhooks** ([]MutatingWebhook)

*Patch strategy: merge on key name*

Webhooks is a list of webhooks and the affected resources and operations.

*MutatingWebhook describes an admission webhook and the resources and operations it applies to.*

- **webhooks.admissionReviewVersions** ([]string), required

AdmissionReviewVersions is an ordered list of preferred AdmissionReview versions the Webhook expects. API server will try to use first version in the list which it supports. If none of the versions specified in this list supported by API server, validation will fail for this object. If a persisted webhook configuration specifies allowed versions and does not include any versions known to the API Server, calls to the webhook will fail and be subject to the failure policy.

- **webhooks.clientConfig** (WebhookClientConfig), required

ClientConfig defines how to communicate with the hook. Required

*WebhookClientConfig contains the information to make a TLS connection with the webhook*

- **webhooks.clientConfig.caBundle** ([]byte)

caBundle is a PEM encoded CA bundle which will be used to validate the webhook's server certificate. If unspecified, system trust roots on the apiserver are used.

- **webhooks.clientConfig.service** (ServiceReference)

service is a reference to the service for this webhook. Either service or url must be specified.

If the webhook is running within the cluster, then you should use service.



*ServiceReference holds a reference to Service.legacy.k8s.io*

- **webhooks.clientConfig.service.name** (string), required

name is the name of the service. Required

- **webhooks.clientConfig.service.namespace** (string), required

namespace is the namespace of the service. Required

- **webhooks.clientConfig.service.path** (string)

path is an optional URL path which will be sent in any request to this service.

- **webhooks.clientConfig.service.port** (int32)

If specified, the port on the service that hosting webhook. Default to 443 for backward compatibility. port should be a valid port number (1-65535, inclusive).

- **webhooks.clientConfig.url** (string)

url gives the location of the webhook, in standard URL form (scheme://host:port/path). Exactly one of url or service must be specified.

The host should not refer to a service running in the cluster; use the service field instead. The host might be resolved via external DNS in some apiservers (e.g., kube-apiserver cannot resolve in-cluster DNS as that would be a layering violation). host may also be an IP address.

Please note that using localhost or 127.0.0.1 as a host is risky unless you take great care to run this webhook on all hosts which run an apiserver which might need to make calls to this webhook. Such installs are likely to be non-portable, i.e., not easy to turn up in a new cluster.

The scheme must be "https"; the URL must begin with "https://".

A path is optional, and if present may be any string permissible in a URL. You may use the path to pass an arbitrary string to the webhook, for example, a cluster identifier.

Attempting to use a user or basic auth e.g. "user:password@" is not allowed. Fragments ("#...") and query parameters ("?...") are not allowed, either.

- **webhooks.name** (string), required

The name of the admission webhook. Name should be fully qualified, e.g., imagepolicy.kubernetes.io, where "imagepolicy" is the name of the webhook, and kubernetes.io is the name of the organization. Required.

- **webhooks.sideEffects** (string), required

SideEffects states whether this webhook has side effects. Acceptable values are: None, NoneOnDryRun (webhooks created via v1beta1 may also specify Some or

Unknown). Webhooks with side effects MUST implement a reconciliation system, since a request may be rejected by a future step in the admission chain and the side effects therefore need to be undone. Requests with the dryRun attribute will be auto-rejected if they match a webhook with sideEffects == Unknown or Some.

- **webhooks.failurePolicy** (string)

FailurePolicy defines how unrecognized errors from the admission endpoint are handled - allowed values are Ignore or Fail. Defaults to Fail.

- **webhooks.matchConditions** ([]MatchCondition)

*Patch strategy: merge on key name*

*Map: unique values on key name will be kept during a merge*

MatchConditions is a list of conditions that must be met for a request to be sent to this webhook. Match conditions filter requests that have already been matched by the rules, namespaceSelector, and objectSelector. An empty list of matchConditions matches all requests. There are a maximum of 64 match conditions allowed.

The exact matching logic is (in order):

1. If ANY matchCondition evaluates to FALSE, the webhook is skipped.
2. If ALL matchConditions evaluate to TRUE, the webhook is called.
3. If any matchCondition evaluates to an error (but none are FALSE):
  - If failurePolicy=Fail, reject the request
  - If failurePolicy=Ignore, the error is ignored and the webhook is skipped

This is a beta feature and managed by the AdmissionWebhookMatchConditions feature gate.

*MatchCondition represents a condition which must be fulfilled for a request to be sent to a webhook.*

- **webhooks.matchConditions.expression** (string), required

Expression represents the expression which will be evaluated by CEL. Must evaluate to bool. CEL expressions have access to the contents of the AdmissionRequest and Authorizer, organized into CEL variables:

'object' - The object from the incoming request. The value is null for DELETE requests. 'oldObject' - The existing object. The value is null for CREATE requests. 'request' - Attributes of the admission request (/pkg/apis/admission/types.go#AdmissionRequest). 'authorizer' - A CEL Authorizer. May be used to perform authorization checks for the principal (user or service account) of the request. See <https://pkg.go.dev/k8s.io/apiserver/pkg/cel/library#Authz> 'authorizer.requestResource' - A CEL ResourceCheck constructed from the 'authorizer' and configured with the request resource. Documentation on CEL: <https://kubernetes.io/docs/reference/using-api/cel/>

Required.

- **webhooks.matchConditions.name** (string), required

Name is an identifier for this match condition, used for strategic merging of MatchConditions, as well as providing an identifier for logging purposes. A good name should be descriptive of the associated expression. Name must be a qualified name consisting of alphanumeric characters, '-', '.' or '\_', and must start and end with an alphanumeric character (e.g. 'MyName', or 'my.name', or '123-abc', regex used for validation is '([A-Za-z0-9](-|\.|\_)?[A-Za-z0-9])' with an optional DNS subdomain prefix and '/' (e.g. 'example.com/MyName')

Required.

- **webhooks.matchPolicy** (string)

matchPolicy defines how the "rules" list is used to match incoming requests. Allowed values are "Exact" or "Equivalent".

- **Exact:** match a request only if it exactly matches a specified rule. For example, if deployments can be modified via apps/v1, apps/v1beta1, and extensions/v1beta1, but "rules" only included apiGroups: ["apps"], apiVersions: ["v1"], resources: ["deployments"], a request to apps/v1beta1 or extensions/v1beta1 would not be sent to the webhook.
- **Equivalent:** match a request if modifies a resource listed in rules, even via another API group or version. For example, if deployments can be modified via apps/v1, apps/v1beta1, and extensions/v1beta1, and "rules" only included apiGroups: ["apps"], apiVersions: ["v1"], resources: ["deployments"], a request to apps/v1beta1 or extensions/v1beta1 would be converted to apps/v1 and sent to the webhook.

Defaults to "Equivalent"

- **webhooks.namespaceSelector** ([LabelSelector](#))

NamespaceSelector decides whether to run the webhook on an object based on whether the namespace for that object matches the selector. If the object itself is a namespace, the matching is performed on object.metadata.labels. If the object is another cluster scoped resource, it never skips the webhook.

For example, to run the webhook on any objects whose namespace is not associated with "runlevel" of "0" or "1"; you will set the selector as follows: "namespaceSelector": { "matchExpressions": [ { "key": "runlevel", "operator": "NotIn", "values": [ "0", "1" ] } ] }

If instead you want to only run the webhook on any objects whose namespace is associated with the "environment" of "prod" or "staging"; you will set the selector as follows: "namespaceSelector": { "matchExpressions": [ { "key": "environment", "operator": "In", "values": [ "prod", "staging" ] } ] }

See <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/> for more examples of label selectors.

Default to the empty LabelSelector, which matches everything.

- **webhooks.objectSelector** ([LabelSelector](#))

ObjectSelector decides whether to run the webhook based on if the object has matching labels. objectSelector is evaluated against both the oldObject and newObject that would be sent to the webhook, and is considered to match if either object matches the selector. A null object (oldObject in the case of create, or newObject in the case of delete) or an object that cannot have labels (like a DeploymentRollback or a PodProxyOptions object) is not considered to match. Use the object selector only if the webhook is opt-in, because end users may skip the admission webhook by setting the labels. Default to the empty LabelSelector, which matches everything.

- **webhooks.reinvocationPolicy** (string)

reinvocationPolicy indicates whether this webhook should be called multiple times as part of a single admission evaluation. Allowed values are "Never" and "IfNeeded".

Never: the webhook will not be called more than once in a single admission evaluation.

IfNeeded: the webhook will be called at least one additional time as part of the admission evaluation if the object being admitted is modified by other admission plugins after the initial webhook call. Webhooks that specify this option *must* be idempotent, able to process objects they previously admitted. Note: \* the number of additional invocations is not guaranteed to be exactly one. \* if additional invocations result in further modifications to the object, webhooks are not guaranteed to be invoked again. \* webhooks that use this option may be reordered to minimize the number of additional invocations. \* to validate an object after all mutations are guaranteed complete, use a validating admission webhook instead.

Defaults to "Never".

- **webhooks.rules** ([]RuleWithOperations)

Rules describes what operations on what resources/subresources the webhook cares about. The webhook cares about an operation if it matches *any* Rule. However, in order to prevent ValidatingAdmissionWebhooks and MutatingAdmissionWebhooks from putting the cluster in a state which cannot be recovered from without completely disabling the plugin, ValidatingAdmissionWebhooks and MutatingAdmissionWebhooks are never called on admission requests for ValidatingWebhookConfiguration and MutatingWebhookConfiguration objects.

*RuleWithOperations is a tuple of Operations and Resources. It is recommended to make sure that all the tuple expansions are valid.*

- **webhooks.rules.apiGroups** ([]string)

*Atomic: will be replaced during a merge*

APIGroups is the API groups the resources belong to. '' is all groups. If '' is present, the length of the slice must be one. Required.

- **webhooks.rules.apiVersions** ([]string)

*Atomic: will be replaced during a merge*

APIVersions is the API versions the resources belong to. '' is all versions. If '' is present, the length of the slice must be one. Required.

- **webhooks.rules.operations** ([]string)

*Atomic: will be replaced during a merge*

Operations is the operations the admission hook cares about - CREATE, UPDATE, DELETE, CONNECT or \* for all of those operations and any future admission operations that are added. If '\*' is present, the length of the slice must be one. Required.

- **webhooks.rules.resources** ([]string)

*Atomic: will be replaced during a merge*

Resources is a list of resources this rule applies to.

For example: 'pods' means pods. 'pods/log' means the log subresource of pods. '' means all resources, but not subresources. 'pods/' means all subresources of pods. '/scale' means all scale subresources. '/' means all resources and their subresources.

If wildcard is present, the validation rule will ensure resources do not overlap with each other.

Depending on the enclosing object, subresources might not be allowed. Required.

- **webhooks.rules.scope** (string)

scope specifies the scope of this rule. Valid values are "Cluster", "Namespaced", and "" "Cluster" means that only cluster-scoped resources will match this rule. Namespace API objects are cluster-scoped. "Namespaced" means that only namespaced resources will match this rule. "" means that there are no scope restrictions. Subresources match the scope of their parent resource. Default is "".

- **webhooks.timeoutSeconds** (int32)

TimeoutSeconds specifies the timeout for this webhook. After the timeout passes, the webhook call will be ignored or the API call will fail based on the failure policy. The timeout value must be between 1 and 30 seconds. Default to 10 seconds.

## MutatingWebhookConfigurationList

MutatingWebhookConfigurationList is a list of MutatingWebhookConfiguration.

- 
- **apiVersion:** admissionregistration.k8s.io/v1
  - **kind:** MutatingWebhookConfigurationList
  - **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]MutatingWebhookConfiguration](#)), required

List of MutatingWebhookConfiguration.

## Operations

---

### get read the specified MutatingWebhookConfiguration

#### HTTP Request

GET /apis/admissionregistration.k8s.io/v1/mutatingwebhookconfigurations/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the MutatingWebhookConfiguration
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([MutatingWebhookConfiguration](#)): OK

401: Unauthorized

### list list or watch objects of kind MutatingWebhookConfiguration

#### HTTP Request

GET /apis/admissionregistration.k8s.io/v1/mutatingwebhookconfigurations

#### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)

- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([MutatingWebhookConfigurationList](#)): OK

401: Unauthorized

## create create a MutatingWebhookConfiguration

### HTTP Request

POST /apis/admissionregistration.k8s.io/v1/mutatingwebhookconfigurations

### Parameters

- **body**: [MutatingWebhookConfiguration](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([MutatingWebhookConfiguration](#)): OK

201 ([MutatingWebhookConfiguration](#)): Created

202 ([MutatingWebhookConfiguration](#)): Accepted

401: Unauthorized

## update replace the specified MutatingWebhookConfiguration

### HTTP Request

PUT /apis/admissionregistration.k8s.io/v1/mutatingwebhookconfigurations/{name}

### Parameters

- **name** (*in path*): string, required  
name of the MutatingWebhookConfiguration
- **body**: [MutatingWebhookConfiguration](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([MutatingWebhookConfiguration](#)): OK



201 ([MutatingWebhookConfiguration](#)): Created

401: Unauthorized

## patch partially update the specified MutatingWebhookConfiguration

### HTTP Request

PATCH /apis/admissionregistration.k8s.io/v1/mutatingwebhookconfigurations/{name}

### Parameters

- **name** (*in path*): string, required  
name of the MutatingWebhookConfiguration
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([MutatingWebhookConfiguration](#)): OK

201 ([MutatingWebhookConfiguration](#)): Created

401: Unauthorized

## delete delete a MutatingWebhookConfiguration

### HTTP Request

DELETE /apis/admissionregistration.k8s.io/v1/mutatingwebhookconfigurations/{name}

## Parameters

- **name** (*in path*): string, required  
name of the MutatingWebhookConfiguration
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of MutatingWebhookConfiguration

### HTTP Request

DELETE /apis/admissionregistration.k8s.io/v1/mutatingwebhookconfigurations

## Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ValidatingWebhookConfiguration

ValidatingWebhookConfiguration describes the configuration of and admission webhook that accept or reject and object without changing it.

apiVersion: admissionregistration.k8s.io/v1

import "k8s.io/api/admissionregistration/v1"

# ValidatingWebhookConfiguration

ValidatingWebhookConfiguration describes the configuration of and admission webhook that accept or reject and object without changing it.

---

- **apiVersion:** admissionregistration.k8s.io/v1
- **kind:** ValidatingWebhookConfiguration
- **metadata** ([ObjectMeta](#))

Standard object metadata; More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>.

- **webhooks** ([]ValidatingWebhook)

*Patch strategy: merge on key name*

Webhooks is a list of webhooks and the affected resources and operations.

*ValidatingWebhook describes an admission webhook and the resources and operations it applies to.*

- **webhooks.admissionReviewVersions** ([]string), required

AdmissionReviewVersions is an ordered list of preferred AdmissionReview versions the Webhook expects. API server will try to use first version in the list which it supports. If none of the versions specified in this list supported by API server, validation will fail for this object. If a persisted webhook configuration specifies allowed versions and does not include any versions known to the API Server, calls to the webhook will fail and be subject to the failure policy.

- **webhooks.clientConfig** (WebhookClientConfig), required

ClientConfig defines how to communicate with the hook. Required

*WebhookClientConfig contains the information to make a TLS connection with the webhook*

- **webhooks.clientConfig.caBundle** ([]byte)

caBundle is a PEM encoded CA bundle which will be used to validate the webhook's server certificate. If unspecified, system trust roots on the apiserver are used.

- **webhooks.clientConfig.service** (ServiceReference)

service is a reference to the service for this webhook. Either service or url must be specified.

If the webhook is running within the cluster, then you should use service.

*ServiceReference holds a reference to Service.legacy.k8s.io*

- **webhooks.clientConfig.service.name** (string), required

name is the name of the service. Required

- **webhooks.clientConfig.service.namespace** (string), required

namespace is the namespace of the service. Required

- **webhooks.clientConfig.service.path** (string)

path is an optional URL path which will be sent in any request to this service.

- **webhooks.clientConfig.service.port** (int32)

If specified, the port on the service that hosting webhook. Default to 443 for backward compatibility. port should be a valid port number (1-65535, inclusive).

- **webhooks.clientConfig.url** (string)

url gives the location of the webhook, in standard URL form (scheme://host:port/path). Exactly one of url or service must be specified.

The host should not refer to a service running in the cluster; use the service field instead. The host might be resolved via external DNS in some apiservers (e.g., kube-apiserver cannot resolve in-cluster DNS as that would be a layering violation). host may also be an IP address.

Please note that using localhost or 127.0.0.1 as a host is risky unless you take great care to run this webhook on all hosts which run an apiserver which might need to make calls to this webhook. Such installs are likely to be non-portable, i.e., not easy to turn up in a new cluster.

The scheme must be "https"; the URL must begin with "https://".

A path is optional, and if present may be any string permissible in a URL. You may use the path to pass an arbitrary string to the webhook, for example, a cluster identifier.

Attempting to use a user or basic auth e.g. "user:password@" is not allowed. Fragments ("#...") and query parameters ("?...") are not allowed, either.

- **webhooks.name** (string), required

The name of the admission webhook. Name should be fully qualified, e.g., imagepolicy.kubernetes.io, where "imagepolicy" is the name of the webhook, and kubernetes.io is the name of the organization. Required.

- **webhooks.sideEffects** (string), required

SideEffects states whether this webhook has side effects. Acceptable values are: None, NoneOnDryRun (webhooks created via v1beta1 may also specify Some or

Unknown). Webhooks with side effects MUST implement a reconciliation system, since a request may be rejected by a future step in the admission chain and the side effects therefore need to be undone. Requests with the dryRun attribute will be auto-rejected if they match a webhook with sideEffects == Unknown or Some.

- **webhooks.failurePolicy** (string)

FailurePolicy defines how unrecognized errors from the admission endpoint are handled - allowed values are Ignore or Fail. Defaults to Fail.

- **webhooks.matchConditions** ([]MatchCondition)

*Patch strategy: merge on key name*

*Map: unique values on key name will be kept during a merge*

MatchConditions is a list of conditions that must be met for a request to be sent to this webhook. Match conditions filter requests that have already been matched by the rules, namespaceSelector, and objectSelector. An empty list of matchConditions matches all requests. There are a maximum of 64 match conditions allowed.

The exact matching logic is (in order):

1. If ANY matchCondition evaluates to FALSE, the webhook is skipped.
2. If ALL matchConditions evaluate to TRUE, the webhook is called.
3. If any matchCondition evaluates to an error (but none are FALSE):
  - If failurePolicy=Fail, reject the request
  - If failurePolicy=Ignore, the error is ignored and the webhook is skipped

This is a beta feature and managed by the AdmissionWebhookMatchConditions feature gate.

*MatchCondition represents a condition which must be fulfilled for a request to be sent to a webhook.*

- **webhooks.matchConditions.expression** (string), required

Expression represents the expression which will be evaluated by CEL. Must evaluate to bool. CEL expressions have access to the contents of the AdmissionRequest and Authorizer, organized into CEL variables:

'object' - The object from the incoming request. The value is null for DELETE requests. 'oldObject' - The existing object. The value is null for CREATE requests. 'request' - Attributes of the admission request (/pkg/apis/admission/types.go#AdmissionRequest). 'authorizer' - A CEL Authorizer. May be used to perform authorization checks for the principal (user or service account) of the request. See <https://pkg.go.dev/k8s.io/apiserver/pkg/cel/library#Authz> 'authorizer.requestResource' - A CEL ResourceCheck constructed from the 'authorizer' and configured with the request resource. Documentation on CEL: <https://kubernetes.io/docs/reference/using-api/cel/>

Required.

- **webhooks.matchConditions.name** (string), required

Name is an identifier for this match condition, used for strategic merging of MatchConditions, as well as providing an identifier for logging purposes. A good name should be descriptive of the associated expression. Name must be a qualified name consisting of alphanumeric characters, '-', '.' or '\_', and must start and end with an alphanumeric character (e.g. 'MyName', or 'my.name', or '123-abc', regex used for validation is '([A-Za-z0-9](-|\.|\_)?[A-Za-z0-9])' with an optional DNS subdomain prefix and '/' (e.g. 'example.com/MyName')

Required.

- **webhooks.matchPolicy** (string)

matchPolicy defines how the "rules" list is used to match incoming requests. Allowed values are "Exact" or "Equivalent".

- **Exact:** match a request only if it exactly matches a specified rule. For example, if deployments can be modified via apps/v1, apps/v1beta1, and extensions/v1beta1, but "rules" only included apiGroups: ["apps"], apiVersions: ["v1"], resources: ["deployments"], a request to apps/v1beta1 or extensions/v1beta1 would not be sent to the webhook.
- **Equivalent:** match a request if modifies a resource listed in rules, even via another API group or version. For example, if deployments can be modified via apps/v1, apps/v1beta1, and extensions/v1beta1, and "rules" only included apiGroups: ["apps"], apiVersions: ["v1"], resources: ["deployments"], a request to apps/v1beta1 or extensions/v1beta1 would be converted to apps/v1 and sent to the webhook.

Defaults to "Equivalent"

- **webhooks.namespaceSelector** ([LabelSelector](#))

NamespaceSelector decides whether to run the webhook on an object based on whether the namespace for that object matches the selector. If the object itself is a namespace, the matching is performed on object.metadata.labels. If the object is another cluster scoped resource, it never skips the webhook.

For example, to run the webhook on any objects whose namespace is not associated with "runlevel" of "0" or "1"; you will set the selector as follows: "namespaceSelector": { "matchExpressions": [ { "key": "runlevel", "operator": "NotIn", "values": [ "0", "1" ] } ] }

If instead you want to only run the webhook on any objects whose namespace is associated with the "environment" of "prod" or "staging"; you will set the selector as follows: "namespaceSelector": { "matchExpressions": [ { "key": "environment", "operator": "In", "values": [ "prod", "staging" ] } ] }

See <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels> for more examples of label selectors.

Default to the empty LabelSelector, which matches everything.

- **webhooks.objectSelector** ([LabelSelector](#))

ObjectSelector decides whether to run the webhook based on if the object has matching labels. objectSelector is evaluated against both the oldObject and newObject that would be sent to the webhook, and is considered to match if either object matches the selector. A null object (oldObject in the case of create, or newObject in the case of delete) or an object that cannot have labels (like a DeploymentRollback or a PodProxyOptions object) is not considered to match. Use the object selector only if the webhook is opt-in, because end users may skip the admission webhook by setting the labels. Default to the empty LabelSelector, which matches everything.

- **webhooks.rules** ([]RuleWithOperations)

Rules describes what operations on what resources/subresources the webhook cares about. The webhook cares about an operation if it matches *any* Rule. However, in order to prevent ValidatingAdmissionWebhooks and MutatingAdmissionWebhooks from putting the cluster in a state which cannot be recovered from without completely disabling the plugin, ValidatingAdmissionWebhooks and MutatingAdmissionWebhooks are never called on admission requests for ValidatingWebhookConfiguration and MutatingWebhookConfiguration objects.

*RuleWithOperations is a tuple of Operations and Resources. It is recommended to make sure that all the tuple expansions are valid.*

- **webhooks.rules.apiGroups** ([]string)

*Atomic: will be replaced during a merge*

APIGroups is the API groups the resources belong to. '' is all groups. If '' is present, the length of the slice must be one. Required.

- **webhooks.rules.apiVersions** ([]string)

*Atomic: will be replaced during a merge*

APIVersions is the API versions the resources belong to. '' is all versions. If '' is present, the length of the slice must be one. Required.

- **webhooks.rules.operations** ([]string)

*Atomic: will be replaced during a merge*

Operations is the operations the admission hook cares about - CREATE, UPDATE, DELETE, CONNECT or \* for all of those operations and any future admission operations that are added. If '\*' is present, the length of the slice must be one. Required.

- **webhooks.rules.resources** ([]string)

*Atomic: will be replaced during a merge*

Resources is a list of resources this rule applies to.

For example: 'pods' means pods. 'pods/log' means the log subresource of pods. '' means all resources, but not subresources. 'pods/' means all subresources



of pods. *'/scale' means all scale subresources. '/' means all resources and their subresources.*

If wildcard is present, the validation rule will ensure resources do not overlap with each other.

Depending on the enclosing object, subresources might not be allowed.  
Required.

- **webhooks.rules.scope** (string)

scope specifies the scope of this rule. Valid values are "Cluster", "Namespaced", and "" *"Cluster" means that only cluster-scoped resources will match this rule. Namespace API objects are cluster-scoped. "Namespaced" means that only namespaced resources will match this rule. "" means that there are no scope restrictions. Subresources match the scope of their parent resource. Default is "".*

- **webhooks.timeoutSeconds** (int32)

TimeoutSeconds specifies the timeout for this webhook. After the timeout passes, the webhook call will be ignored or the API call will fail based on the failure policy. The timeout value must be between 1 and 30 seconds. Default to 10 seconds.

## ValidatingWebhookConfigurationList

ValidatingWebhookConfigurationList is a list of ValidatingWebhookConfiguration.

---

- **apiVersion:** admissionregistration.k8s.io/v1
- **kind:** ValidatingWebhookConfigurationList
- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]ValidatingWebhookConfiguration](#)), required

List of ValidatingWebhookConfiguration.

## Operations

---

### get read the specified ValidatingWebhookConfiguration

#### HTTP Request

GET /apis/admissionregistration.k8s.io/v1/validatingwebhookconfigurations/{name}

## Parameters

- **name** (*in path*): string, required  
name of the ValidatingWebhookConfiguration
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ValidatingWebhookConfiguration](#)): OK

401: Unauthorized

## list list or watch objects of kind ValidatingWebhookConfiguration

## HTTP Request

GET /apis/admissionregistration.k8s.io/v1/validatingwebhookconfigurations

## Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ValidatingWebhookConfigurationList](#)): OK

401: Unauthorized

## create create a ValidatingWebhookConfiguration

### HTTP Request

POST /apis/admissionregistration.k8s.io/v1/validatingwebhookconfigurations

### Parameters

- **body**: [ValidatingWebhookConfiguration](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ValidatingWebhookConfiguration](#)): OK

201 ([ValidatingWebhookConfiguration](#)): Created

202 ([ValidatingWebhookConfiguration](#)): Accepted

401: Unauthorized

## **update replace the specified ValidatingWebhookConfiguration**

### **HTTP Request**

PUT /apis/admissionregistration.k8s.io/v1/validatingwebhookconfigurations/{name}

### **Parameters**

- **name** (*in path*): string, required  
name of the ValidatingWebhookConfiguration
- **body**: [ValidatingWebhookConfiguration](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### **Response**

200 ([ValidatingWebhookConfiguration](#)): OK

201 ([ValidatingWebhookConfiguration](#)): Created

401: Unauthorized

## **patch partially update the specified ValidatingWebhookConfiguration**

### **HTTP Request**

PATCH /apis/admissionregistration.k8s.io/v1/validatingwebhookconfigurations/{name}

### **Parameters**

- **name** (*in path*): string, required  
name of the ValidatingWebhookConfiguration

- **body:** [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ValidatingWebhookConfiguration](#)): OK

201 ([ValidatingWebhookConfiguration](#)): Created

401: Unauthorized

## delete delete a ValidatingWebhookConfiguration

### HTTP Request

DELETE /apis/admissionregistration.k8s.io/v1/validatingwebhookconfigurations/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ValidatingWebhookConfiguration
- **body:** [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ValidatingWebhookConfiguration

### HTTP Request

DELETE /apis/admissionregistration.k8s.io/v1/validatingwebhookconfigurations

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# ValidatingAdmissionPolicy v1beta1

ValidatingAdmissionPolicy describes the definition of an admission validation policy that accepts or rejects an object without changing it.

apiVersion: admissionregistration.k8s.io/v1beta1

import "k8s.io/api/admissionregistration/v1beta1"

## ValidatingAdmissionPolicy

ValidatingAdmissionPolicy describes the definition of an admission validation policy that accepts or rejects an object without changing it.

- 
- **apiVersion**: admissionregistration.k8s.io/v1beta1
  - **kind**: ValidatingAdmissionPolicy
  - **metadata** ([ObjectMeta](#))

Standard object metadata; More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>.

- **spec** (ValidatingAdmissionPolicySpec)

Specification of the desired behavior of the ValidatingAdmissionPolicy.

*ValidatingAdmissionPolicySpec is the specification of the desired behavior of the AdmissionPolicy.*

- **spec.auditAnnotations** ([]AuditAnnotation)

*Atomic: will be replaced during a merge*

auditAnnotations contains CEL expressions which are used to produce audit annotations for the audit event of the API request. validations and auditAnnotations may not both be empty; a least one of validations or auditAnnotations is required.

*AuditAnnotation describes how to produce an audit annotation for an API request.*

- **spec.auditAnnotations.key** (string), required

key specifies the audit annotation key. The audit annotation keys of a ValidatingAdmissionPolicy must be unique. The key must be a qualified name ([A-Za-z0-9][-A-Za-z0-9\_.]\*) no more than 63 bytes in length.

The key is combined with the resource name of the ValidatingAdmissionPolicy to construct an audit annotation key: "{ValidatingAdmissionPolicy name}/{key}".

If an admission webhook uses the same resource name as this ValidatingAdmissionPolicy and the same audit annotation key, the annotation key will be identical. In this case, the first annotation written with the key will be included in the audit event and all subsequent annotations with the same key will be discarded.

Required.

- **spec.auditAnnotations.valueExpression** (string), required

valueExpression represents the expression which is evaluated by CEL to produce an audit annotation value. The expression must evaluate to either a string or null value. If the expression evaluates to a string, the audit annotation is included with the string value. If the expression evaluates to null or empty string the audit annotation will be omitted. The valueExpression may be no longer than 5kb in length. If the result of the valueExpression is more than 10kb in length, it will be truncated to 10kb.

If multiple ValidatingAdmissionPolicyBinding resources match an API request, then the valueExpression will be evaluated for each binding. All unique values produced by the valueExpressions will be joined together in a comma-separated list.

Required.

- **spec.failurePolicy** (string)

failurePolicy defines how to handle failures for the admission policy. Failures can occur from CEL expression parse errors, type check errors, runtime errors and invalid or mis-configured policy definitions or bindings.



A policy is invalid if `spec.paramKind` refers to a non-existent Kind. A binding is invalid if `spec.paramRef.name` refers to a non-existent resource.

`failurePolicy` does not define how validations that evaluate to false are handled.

When `failurePolicy` is set to Fail, `ValidatingAdmissionPolicyBinding` `validationActions` define how failures are enforced.

Allowed values are Ignore or Fail. Defaults to Fail.

- **`spec.matchConditions`** (`[]MatchCondition`)

*Patch strategy: merge on key name*

*Map: unique values on key name will be kept during a merge*

`MatchConditions` is a list of conditions that must be met for a request to be validated. Match conditions filter requests that have already been matched by the rules, `namespaceSelector`, and `objectSelector`. An empty list of `matchConditions` matches all requests. There are a maximum of 64 match conditions allowed.

If a parameter object is provided, it can be accessed via the `params` handle in the same manner as validation expressions.

The exact matching logic is (in order):

1. If ANY `matchCondition` evaluates to FALSE, the policy is skipped.
2. If ALL `matchConditions` evaluate to TRUE, the policy is evaluated.
3. If any `matchCondition` evaluates to an error (but none are FALSE):
  - If `failurePolicy=Fail`, reject the request
  - If `failurePolicy=Ignore`, the policy is skipped

*MatchCondition represents a condition which must be fulfilled for a request to be sent to a webhook.*

- **`spec.matchConditions.expression`** (string), required

Expression represents the expression which will be evaluated by CEL. Must evaluate to bool. CEL expressions have access to the contents of the `AdmissionRequest` and `Authorizer`, organized into CEL variables:

'object' - The object from the incoming request. The value is null for DELETE requests. 'oldObject' - The existing object. The value is null for CREATE requests. 'request' - Attributes of the admission request (`/pkg/apis/admission/types.go#AdmissionRequest`). 'authorizer' - A CEL Authorizer. May be used to perform authorization checks for the principal (user or service account) of the request. See <https://pkg.go.dev/k8s.io/apiserver/pkg/cel/library#Authz> 'authorizer.requestResource' - A CEL ResourceCheck constructed from the 'authorizer' and configured with the request resource. Documentation on CEL: <https://kubernetes.io/docs/reference/using-api/cel/>

Required.

- **`spec.matchConditions.name`** (string), required

Name is an identifier for this match condition, used for strategic merging of MatchConditions, as well as providing an identifier for logging purposes. A good name should be descriptive of the associated expression. Name must be a qualified name consisting of alphanumeric characters, '-', '\_' or '.', and must start and end with an alphanumeric character (e.g. 'MyName', or 'my.name', or '123-abc', regex used for validation is '([A-Za-z0-9](-|\_|\.)\*)?[A-Za-z0-9]') with an optional DNS subdomain prefix and '/' (e.g. 'example.com/MyName')

Required.

- **spec.matchConstraints** (MatchResources)

MatchConstraints specifies what resources this policy is designed to validate. The AdmissionPolicy cares about a request if it matches *all* Constraints. However, in order to prevent clusters from being put into an unstable state that cannot be recovered from via the API ValidatingAdmissionPolicy cannot match ValidatingAdmissionPolicy and ValidatingAdmissionPolicyBinding. Required.

*MatchResources decides whether to run the admission control policy on an object based on whether it meets the match criteria. The exclude rules take precedence over include rules (if a resource matches both, it is excluded)*

- **spec.matchConstraints.excludeResourceRules**  
([]NamedRuleWithOperations)

*Atomic: will be replaced during a merge*

ExcludeResourceRules describes what operations on what resources/subresources the ValidatingAdmissionPolicy should not care about. The exclude rules take precedence over include rules (if a resource matches both, it is excluded)

*NamedRuleWithOperations is a tuple of Operations and Resources with ResourceNames.*

- **spec.matchConstraints.excludeResourceRules.apiGroups**  
([]string)

*Atomic: will be replaced during a merge*

APIGroups is the API groups the resources belong to. '' is all groups. If '' is present, the length of the slice must be one. Required.

- **spec.matchConstraints.excludeResourceRules.apiVersions**  
([]string)

*Atomic: will be replaced during a merge*

APIVersions is the API versions the resources belong to. '' is all versions. If '' is present, the length of the slice must be one. Required.

- **spec.matchConstraints.excludeResourceRules.operations**  
([]string)

*Atomic: will be replaced during a merge*

Operations is the operations the admission hook cares about - CREATE, UPDATE, DELETE, CONNECT or \* for all of those operations and any future admission operations that are added. If '\*' is present, the length of the slice must be one. Required.

- **spec.matchConstraints.excludeResourceRules.resourceNames** ([]string)

*Atomic: will be replaced during a merge*

ResourceNames is an optional white list of names that the rule applies to. An empty set means that everything is allowed.

- **spec.matchConstraints.excludeResourceRules.resources** ([]string)

*Atomic: will be replaced during a merge*

Resources is a list of resources this rule applies to.

For example: 'pods' means pods. 'pods/log' means the log subresource of pods. '' means all resources, but not subresources. 'pods/' means all subresources of pods. '/scale' means all scale subresources. '/' means all resources and their subresources.

If wildcard is present, the validation rule will ensure resources do not overlap with each other.

Depending on the enclosing object, subresources might not be allowed. Required.

- **spec.matchConstraints.excludeResourceRules.scope** (string)

scope specifies the scope of this rule. Valid values are "Cluster", "Namespaced", and "" *"Cluster" means that only cluster-scoped resources will match this rule. Namespace API objects are cluster-scoped. "Namespaced" means that only namespaced resources will match this rule. "" means that there are no scope restrictions. Subresources match the scope of their parent resource. Default is ""*.

- **spec.matchConstraints.matchPolicy** (string)

matchPolicy defines how the "MatchResources" list is used to match incoming requests. Allowed values are "Exact" or "Equivalent".

- **Exact:** match a request only if it exactly matches a specified rule. For example, if deployments can be modified via apps/v1, apps/v1beta1, and extensions/v1beta1, but "rules" only included apiGroups:["apps"], apiVersions:["v1"], resources: ["deployments"], a request to apps/v1beta1 or extensions/v1beta1 would not be sent to the ValidatingAdmissionPolicy.
- **Equivalent:** match a request if modifies a resource listed in rules, even via another API group or version. For example, if deployments can be modified via apps/v1, apps/v1beta1, and extensions/v1beta1, and

"rules" only included `apiGroups:["apps"], apiVersions:["v1"], resources: ["deployments"],` a request to `apps/v1beta1` or `extensions/v1beta1` would be converted to `apps/v1` and sent to the `ValidatingAdmissionPolicy`.

Defaults to "Equivalent"

- **`spec.matchConstraints.namespaceSelector`** ([LabelSelector](#))

`NamespaceSelector` decides whether to run the admission control policy on an object based on whether the namespace for that object matches the selector. If the object itself is a namespace, the matching is performed on `object.metadata.labels`. If the object is another cluster scoped resource, it never skips the policy.

For example, to run the webhook on any objects whose namespace is not associated with "runlevel" of "0" or "1"; you will set the selector as follows: `"namespaceSelector": { "matchExpressions": [ { "key": "runlevel", "operator": "NotIn", "values": [ "0", "1" ] } ] }`

If instead you want to only run the policy on any objects whose namespace is associated with the "environment" of "prod" or "staging"; you will set the selector as follows: `"namespaceSelector": { "matchExpressions": [ { "key": "environment", "operator": "In", "values": [ "prod", "staging" ] } ] }`

See <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/> for more examples of label selectors.

Default to the empty `LabelSelector`, which matches everything.

- **`spec.matchConstraints.objectSelector`** ([LabelSelector](#))

`ObjectSelector` decides whether to run the validation based on if the object has matching labels. `objectSelector` is evaluated against both the `oldObject` and `newObject` that would be sent to the cel validation, and is considered to match if either object matches the selector. A null object (`oldObject` in the case of create, or `newObject` in the case of delete) or an object that cannot have labels (like a `DeploymentRollback` or a `PodProxyOptions` object) is not considered to match. Use the object selector only if the webhook is opt-in, because end users may skip the admission webhook by setting the labels. Default to the empty `LabelSelector`, which matches everything.

- **`spec.matchConstraints.resourceRules`** (`[]NamedRuleWithOperations`)

*Atomic: will be replaced during a merge*

`ResourceRules` describes what operations on what resources/subresources the `ValidatingAdmissionPolicy` matches. The policy cares about an operation if it matches *any* Rule.

*NamedRuleWithOperations is a tuple of Operations and Resources with ResourceNames.*

- **`spec.matchConstraints.resourceRules.apiGroups`** (`[]string`)

*Atomic: will be replaced during a merge*

APIGroups is the API groups the resources belong to. '' is all groups. If '' is present, the length of the slice must be one. Required.

▪ **spec.matchConstraints.resourceRules.apiVersions** ([]string)

*Atomic: will be replaced during a merge*

APIVersions is the API versions the resources belong to. '' is all versions. If '' is present, the length of the slice must be one. Required.

▪ **spec.matchConstraints.resourceRules.operations** ([]string)

*Atomic: will be replaced during a merge*

Operations is the operations the admission hook cares about - CREATE, UPDATE, DELETE, CONNECT or \* for all of those operations and any future admission operations that are added. If '\*' is present, the length of the slice must be one. Required.

▪ **spec.matchConstraints.resourceRules.resourceNames** ([]string)

*Atomic: will be replaced during a merge*

ResourceNames is an optional white list of names that the rule applies to. An empty set means that everything is allowed.

▪ **spec.matchConstraints.resourceRules.resources** ([]string)

*Atomic: will be replaced during a merge*

Resources is a list of resources this rule applies to.

For example: 'pods' means pods. 'pods/log' means the log subresource of pods. '' means all resources, but not subresources. 'pods/' means all subresources of pods. '/scale' means all scale subresources. '/' means all resources and their subresources.

If wildcard is present, the validation rule will ensure resources do not overlap with each other.

Depending on the enclosing object, subresources might not be allowed. Required.

▪ **spec.matchConstraints.resourceRules.scope** (string)

scope specifies the scope of this rule. Valid values are "Cluster", "Namespaced", and "" "Cluster" means that only cluster-scoped resources will match this rule. Namespace API objects are cluster-scoped. "Namespaced" means that only namespaced resources will match this rule. "" means that there are no scope restrictions. Subresources match the scope of their parent resource. Default is "".

◦ **spec.paramKind** (ParamKind)

ParamKind specifies the kind of resources used to parameterize this policy. If absent, there are no parameters for this policy and the param CEL variable will not be provided to validation expressions. If ParamKind refers to a non-existent kind, this policy definition is mis-configured and the FailurePolicy is applied. If paramKind is specified but paramRef is unset in ValidatingAdmissionPolicyBinding, the params variable will be null.

*ParamKind is a tuple of Group Kind and Version.*

- **spec.paramKind.apiVersion** (string)

APIVersion is the API group version the resources belong to. In format of "group/version". Required.

- **spec.paramKind.kind** (string)

Kind is the API kind the resources belong to. Required.

- **spec.validations** ([]Validation)

*Atomic: will be replaced during a merge*

Validations contain CEL expressions which is used to apply the validation. Validations and AuditAnnotations may not both be empty; a minimum of one Validations or AuditAnnotations is required.

*Validation specifies the CEL expression which is used to apply the validation.*

- **spec.validations.expression** (string), required

Expression represents the expression which will be evaluated by CEL. ref: <https://github.com/google/cel-spec> CEL expressions have access to the contents of the API request/response, organized into CEL variables as well as some other useful variables:

- 'object' - The object from the incoming request. The value is null for DELETE requests. - 'oldObject' - The existing object. The value is null for CREATE requests. - 'request' - Attributes of the API request(ref). - 'params' - Parameter resource referred to by the policy binding being evaluated. Only populated if the policy has a ParamKind. - 'namespaceObject' - The namespace object that the incoming object belongs to. The value is null for cluster-scoped resources. - 'variables' - Map of composited variables, from its name to its lazily evaluated value. For example, a variable named 'foo' can be accessed as 'variables.foo'.
- 'authorizer' - A CEL Authorizer. May be used to perform authorization checks for the principal (user or service account) of the request. See <https://pkg.go.dev/k8s.io/apiserver/pkg/cel/library#Authz>
- 'authorizer.requestResource' - A CEL ResourceCheck constructed from the 'authorizer' and configured with the request resource.

The apiVersion, kind, metadata.name and metadata.generateName are always accessible from the root of the object. No other metadata properties are accessible.

Only property names of the form `[a-zA-Z_./][a-zA-Z0-9_./]*` are accessible. Accessible property names are escaped according to the following rules when accessed in the expression: - `"` escapes to `'underscores'` - `'` escapes to `'dot'` - `-` escapes to `'dash'` - `/` escapes to `'slash'` - **Property names that exactly match a CEL RESERVED keyword escape to `{keyword}__`**. The keywords are: `"true"`, `"false"`, `"null"`, `"in"`, `"as"`, `"break"`, `"const"`, `"continue"`, `"else"`, `"for"`, `"function"`, `"if"`, `"import"`, `"let"`, `"loop"`, `"package"`, `"namespace"`, `"return"`. Examples:

- Expression accessing a property named `"namespace"`: `{"Expression": "object.namespace > 0"}`
- Expression accessing a property named `"x-prop"`: `{"Expression": "object.x__dash__prop > 0"}`
- Expression accessing a property named `"redact__d"`: `{"Expression": "object.redact__underscores__d > 0"}`

Equality on arrays with list type of `'set'` or `'map'` ignores element order, i.e. `[1, 2] == [2, 1]`. Concatenation on arrays with `x-kubernetes-list-type` use the semantics of the list type:

- `'set'`: `X + Y` performs a union where the array positions of all elements in `X` are preserved and non-intersecting elements in `Y` are appended, retaining their partial order.
- `'map'`: `X + Y` performs a merge where the array positions of all keys in `X` are preserved but the values are overwritten by values in `Y` when the key sets of `X` and `Y` intersect. Elements in `Y` with non-intersecting keys are appended, retaining their partial order. Required.

- **`spec.validations.message`** (string)

Message represents the message displayed when validation fails. The message is required if the Expression contains line breaks. The message must not contain line breaks. If unset, the message is `"failed rule: {Rule}"`. e.g. `"must be a URL with the host matching spec.host"` If the Expression contains line breaks. Message is required. The message must not contain line breaks. If unset, the message is `"failed Expression: {Expression}"`.

- **`spec.validations.messageExpression`** (string)

`messageExpression` declares a CEL expression that evaluates to the validation failure message that is returned when this rule fails. Since `messageExpression` is used as a failure message, it must evaluate to a string. If both `message` and `messageExpression` are present on a validation, then `messageExpression` will be used if validation fails. If `messageExpression` results in a runtime error, the runtime error is logged, and the validation failure message is produced as if the `messageExpression` field were unset. If `messageExpression` evaluates to an empty string, a string with only spaces, or a string that contains line breaks, then the validation failure message will also be produced as if the `messageExpression` field were unset, and the fact that `messageExpression` produced an empty string/string with only spaces/string with line breaks will be logged. `messageExpression` has access to all the same variables as the expression except for `'authorizer'` and `'authorizer.requestResource'`. Example: `"object.x must be less than max (" + string(params.max) + ")"`

- **spec.validations.reason** (string)

Reason represents a machine-readable description of why this validation failed. If this is the first validation in the list to fail, this reason, as well as the corresponding HTTP response code, are used in the HTTP response to the client. The currently supported reasons are: "Unauthorized", "Forbidden", "Invalid", "RequestEntityTooLarge". If not set, StatusReasonInvalid is used in the response to the client.

- **spec.variables** ([]Variable)

*Patch strategy: merge on key name*

*Map: unique values on key name will be kept during a merge*

Variables contain definitions of variables that can be used in composition of other expressions. Each variable is defined as a named CEL expression. The variables defined here will be available under variables in other expressions of the policy except MatchConditions because MatchConditions are evaluated before the rest of the policy.

The expression of a variable can refer to other variables defined earlier in the list but not those after. Thus, Variables must be sorted by the order of first appearance and acyclic.

*Variable is the definition of a variable that is used for composition. A variable is defined as a named expression.*

- **spec.variables.expression** (string), required

Expression is the expression that will be evaluated as the value of the variable. The CEL expression has access to the same identifiers as the CEL expressions in Validation.

- **spec.variables.name** (string), required

Name is the name of the variable. The name must be a valid CEL identifier and unique among all variables. The variable can be accessed in other expressions through variables. For example, if name is "foo", the variable will be available as variables.foo

- **status** (ValidatingAdmissionPolicyStatus)

The status of the ValidatingAdmissionPolicy, including warnings that are useful to determine if the policy behaves in the expected way. Populated by the system. Read-only.

*ValidatingAdmissionPolicyStatus represents the status of an admission validation policy.*

- **status.conditions** ([]Condition)

*Map: unique values on key type will be kept during a merge*

The conditions represent the latest available observations of a policy's current state.



*Condition contains details for one aspect of the current state of this API Resource.*

- **status.conditions.lastTransitionTime** (Time), required

lastTransitionTime is the last time the condition transitioned from one status to another. This should be when the underlying condition changed. If that is not known, then using the time when the API field changed is acceptable.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **status.conditions.message** (string), required

message is a human readable message indicating details about the transition. This may be an empty string.

- **status.conditions.reason** (string), required

reason contains a programmatic identifier indicating the reason for the condition's last transition. Producers of specific condition types may define expected values and meanings for this field, and whether the values are considered a guaranteed API. The value should be a CamelCase string. This field may not be empty.

- **status.conditions.status** (string), required

status of the condition, one of True, False, Unknown.

- **status.conditions.type** (string), required

type of condition in CamelCase or in foo.example.com/CamelCase.

- **status.conditions.observedGeneration** (int64)

observedGeneration represents the .metadata.generation that the condition was set based upon. For instance, if .metadata.generation is currently 12, but the .status.conditions[x].observedGeneration is 9, the condition is out of date with respect to the current state of the instance.

- **status.observedGeneration** (int64)

The generation observed by the controller.

- **status.typeChecking** (TypeChecking)

The results of type checking for each expression. Presence of this field indicates the completion of the type checking.

*TypeChecking contains results of type checking the expressions in the ValidatingAdmissionPolicy*

- **status.typeChecking.expressionWarnings** ([]ExpressionWarning)

*Atomic: will be replaced during a merge*

The type checking warnings for each expression.

*ExpressionWarning* is a warning information that targets a specific expression.

- **status.typeChecking.expressionWarnings.fieldRef** (string), required

The path to the field that refers the expression. For example, the reference to the expression of the first item of validations is "spec.validations[0].expression"

- **status.typeChecking.expressionWarnings.warning** (string), required

The content of type checking information in a human-readable form. Each line of the warning contains the type that the expression is checked against, followed by the type check error from the compiler.

## ValidatingAdmissionPolicyList

ValidatingAdmissionPolicyList is a list of ValidatingAdmissionPolicy.

---

- **apiVersion** (string)

APIVersion defines the versioned schema of this representation of an object. Servers should convert recognized schemas to the latest internal value, and may reject unrecognized values. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources>

- **items** ([\[\]ValidatingAdmissionPolicy](#))

List of ValidatingAdmissionPolicy.

- **kind** (string)

Kind is a string value representing the REST resource this object represents. Servers may infer this from the endpoint the client submits requests to. Cannot be updated. In CamelCase. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

## ValidatingAdmissionPolicyBinding

ValidatingAdmissionPolicyBinding binds the ValidatingAdmissionPolicy with parameterized resources. ValidatingAdmissionPolicyBinding and parameter CRDs together define how cluster administrators configure policies for clusters.

For a given admission request, each binding will cause its policy to be evaluated N times, where N is 1 for policies/bindings that don't use params, otherwise N is the number of parameters selected by the binding.

The CEL expressions of a policy must have a computed CEL cost below the maximum CEL budget. Each evaluation of the policy is given an independent CEL cost budget. Adding/removing policies, bindings, or params can not affect whether a given (policy, binding, param) combination is within its own CEL budget.

---

- **apiVersion** (string)

APIVersion defines the versioned schema of this representation of an object. Servers should convert recognized schemas to the latest internal value, and may reject unrecognized values. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources>

- **kind** (string)

Kind is a string value representing the REST resource this object represents. Servers may infer this from the endpoint the client submits requests to. Cannot be updated. In CamelCase. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **metadata** ([ObjectMeta](#))

Standard object metadata; More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>.

- **spec** ([ValidatingAdmissionPolicyBindingSpec](#))

Specification of the desired behavior of the [ValidatingAdmissionPolicyBinding](#).

*[ValidatingAdmissionPolicyBindingSpec](#) is the specification of the [ValidatingAdmissionPolicyBinding](#).*

- **spec.matchResources** ([MatchResources](#))

[MatchResources](#) declares what resources match this binding and will be validated by it. Note that this is intersected with the policy's [matchConstraints](#), so only requests that are matched by the policy can be selected by this. If this is unset, all resources matched by the policy are validated by this binding. When [resourceRules](#) is unset, it does not constrain resource matching. If a resource is matched by the other fields of this object, it will be validated. Note that this differs from [ValidatingAdmissionPolicy matchConstraints](#), where [resourceRules](#) are required.

*[MatchResources](#) decides whether to run the admission control policy on an object based on whether it meets the match criteria. The exclude rules take precedence over include rules (if a resource matches both, it is excluded)*

- **spec.matchResources.excludeResourceRules**  
([\[\]NamedRuleWithOperations](#))

*Atomic: will be replaced during a merge*

ExcludeResourceRules describes what operations on what resources/subresources the ValidatingAdmissionPolicy should not care about. The exclude rules take precedence over include rules (if a resource matches both, it is excluded)

*NamedRuleWithOperations is a tuple of Operations and Resources with ResourceNames.*

- **spec.matchResources.excludeResourceRules.apiGroups** ([]string)

*Atomic: will be replaced during a merge*

APIGroups is the API groups the resources belong to. '' is all groups. If '' is present, the length of the slice must be one. Required.

- **spec.matchResources.excludeResourceRules.apiVersions** ([]string)

*Atomic: will be replaced during a merge*

APIVersions is the API versions the resources belong to. '' is all versions. If '' is present, the length of the slice must be one. Required.

- **spec.matchResources.excludeResourceRules.operations** ([]string)

*Atomic: will be replaced during a merge*

Operations is the operations the admission hook cares about - CREATE, UPDATE, DELETE, CONNECT or \* for all of those operations and any future admission operations that are added. If '\*' is present, the length of the slice must be one. Required.

- **spec.matchResources.excludeResourceRules.resourceNames** ([]string)

*Atomic: will be replaced during a merge*

ResourceNames is an optional white list of names that the rule applies to. An empty set means that everything is allowed.

- **spec.matchResources.excludeResourceRules.resources** ([]string)

*Atomic: will be replaced during a merge*

Resources is a list of resources this rule applies to.

For example: 'pods' means pods. 'pods/log' means the log subresource of pods. '' means all resources, but not subresources. 'pods/' means all subresources of pods. '/scale' means all scale subresources. '/' means all resources and their subresources.

If wildcard is present, the validation rule will ensure resources do not overlap with each other.

Depending on the enclosing object, subresources might not be allowed. Required.

- **spec.matchResources.excludeResourceRules.scope** (string)

scope specifies the scope of this rule. Valid values are "Cluster", "Namespaced", and "" *"Cluster" means that only cluster-scoped resources will match this rule. Namespace API objects are cluster-scoped. "Namespaced" means that only namespaced resources will match this rule.* "" means that there are no scope restrictions. Subresources match the scope of their parent resource. Default is "".

- **spec.matchResources.matchPolicy** (string)

matchPolicy defines how the "MatchResources" list is used to match incoming requests. Allowed values are "Exact" or "Equivalent".

- **Exact:** match a request only if it exactly matches a specified rule. For example, if deployments can be modified via apps/v1, apps/v1beta1, and extensions/v1beta1, but "rules" only included apiGroups:["apps"], apiVersions:["v1"], resources: ["deployments"], a request to apps/v1beta1 or extensions/v1beta1 would not be sent to the ValidatingAdmissionPolicy.
- **Equivalent:** match a request if modifies a resource listed in rules, even via another API group or version. For example, if deployments can be modified via apps/v1, apps/v1beta1, and extensions/v1beta1, and "rules" only included apiGroups:["apps"], apiVersions:["v1"], resources: ["deployments"], a request to apps/v1beta1 or extensions/v1beta1 would be converted to apps/v1 and sent to the ValidatingAdmissionPolicy.

Defaults to "Equivalent"

- **spec.matchResources.namespaceSelector** ([LabelSelector](#))

NamespaceSelector decides whether to run the admission control policy on an object based on whether the namespace for that object matches the selector. If the object itself is a namespace, the matching is performed on object.metadata.labels. If the object is another cluster scoped resource, it never skips the policy.

For example, to run the webhook on any objects whose namespace is not associated with "runlevel" of "0" or "1"; you will set the selector as follows: "namespaceSelector": { "matchExpressions": [ { "key": "runlevel", "operator": "NotIn", "values": [ "0", "1" ] } ] }

If instead you want to only run the policy on any objects whose namespace is associated with the "environment" of "prod" or "staging"; you will set the selector as follows: "namespaceSelector": { "matchExpressions": [ { "key": "environment", "operator": "In", "values": [ "prod", "staging" ] } ] }

See <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/> for more examples of label selectors.

Default to the empty LabelSelector, which matches everything.

- **spec.matchResources.objectSelector** ([LabelSelector](#))

ObjectSelector decides whether to run the validation based on if the object has matching labels. objectSelector is evaluated against both the oldObject and newObject that would be sent to the cel validation, and is considered to match if either object matches the selector. A null object (oldObject in the case of create, or newObject in the case of delete) or an object that cannot have labels (like a DeploymentRollback or a PodProxyOptions object) is not considered to match. Use the object selector only if the webhook is opt-in, because end users may skip the admission webhook by setting the labels. Default to the empty LabelSelector, which matches everything.

- **spec.matchResources.resourceRules** ([]NamedRuleWithOperations)

*Atomic: will be replaced during a merge*

ResourceRules describes what operations on what resources/subresources the ValidatingAdmissionPolicy matches. The policy cares about an operation if it matches *any* Rule.

*NamedRuleWithOperations is a tuple of Operations and Resources with ResourceNames.*

- **spec.matchResources.resourceRules.apiGroups** ([]string)

*Atomic: will be replaced during a merge*

APIGroups is the API groups the resources belong to. '' is all groups. If '' is present, the length of the slice must be one. Required.

- **spec.matchResources.resourceRules.apiVersions** ([]string)

*Atomic: will be replaced during a merge*

APIVersions is the API versions the resources belong to. '' is all versions. If '' is present, the length of the slice must be one. Required.

- **spec.matchResources.resourceRules.operations** ([]string)

*Atomic: will be replaced during a merge*

Operations is the operations the admission hook cares about - CREATE, UPDATE, DELETE, CONNECT or \* for all of those operations and any future admission operations that are added. If '\*' is present, the length of the slice must be one. Required.

- **spec.matchResources.resourceRules.resourceNames** ([]string)

*Atomic: will be replaced during a merge*

ResourceNames is an optional white list of names that the rule applies to. An empty set means that everything is allowed.

▪ **spec.matchResources.resourceRules.resources** ([]string)

*Atomic: will be replaced during a merge*

Resources is a list of resources this rule applies to.

For example: 'pods' means pods. 'pods/log' means the log subresource of pods. '' means all resources, but not subresources. 'pods/' means all subresources of pods. '/scale' means all scale subresources. '/' means all resources and their subresources.

If wildcard is present, the validation rule will ensure resources do not overlap with each other.

Depending on the enclosing object, subresources might not be allowed. Required.

▪ **spec.matchResources.resourceRules.scope** (string)

scope specifies the scope of this rule. Valid values are "Cluster", "Namespaced", and "" *"Cluster" means that only cluster-scoped resources will match this rule. Namespace API objects are cluster-scoped. "Namespaced" means that only namespaced resources will match this rule. "" means that there are no scope restrictions. Subresources match the scope of their parent resource. Default is "".*

◦ **spec.paramRef** (ParamRef)

paramRef specifies the parameter resource used to configure the admission control policy. It should point to a resource of the type specified in ParamKind of the bound ValidatingAdmissionPolicy. If the policy specifies a ParamKind and the resource referred to by ParamRef does not exist, this binding is considered mis-configured and the FailurePolicy of the ValidatingAdmissionPolicy applied. If the policy does not specify a ParamKind then this field is ignored, and the rules are evaluated without a param.

*ParamRef describes how to locate the params to be used as input to expressions of rules applied by a policy binding.*

▪ **spec.paramRef.name** (string)

name is the name of the resource being referenced.

One of name or selector must be set, but name and selector are mutually exclusive properties. If one is set, the other must be unset.

A single parameter used for all admission requests can be configured by setting the name field, leaving selector blank, and setting namespace if paramKind is namespace-scoped.

▪ **spec.paramRef.namespace** (string)

namespace is the namespace of the referenced resource. Allows limiting the search for params to a specific namespace. Applies to both name and selector fields.

A per-namespace parameter may be used by specifying a namespace-scoped paramKind in the policy and leaving this field empty.

- If paramKind is cluster-scoped, this field MUST be unset. Setting this field results in a configuration error.
- If paramKind is namespace-scoped, the namespace of the object being evaluated for admission will be used when this field is left unset. Take care that if this is left empty the binding must not match any cluster-scoped resources, which will result in an error.

▪ **spec.paramRef.parameterNotFoundAction** (string)

parameterNotFoundAction controls the behavior of the binding when the resource exists, and name or selector is valid, but there are no parameters matched by the binding. If the value is set to Allow, then no matched parameters will be treated as successful validation by the binding. If set to Deny, then no matched parameters will be subject to the failurePolicy of the policy.

Allowed values are Allow or Deny

Required

▪ **spec.paramRef.selector** ([LabelSelector](#))

selector can be used to match multiple param objects based on their labels. Supply selector: {} to match all resources of the ParamKind.

If multiple params are found, they are all evaluated with the policy expressions and the results are ANDed together.

One of name or selector must be set, but name and selector are mutually exclusive properties. If one is set, the other must be unset.

◦ **spec.policyName** (string)

PolicyName references a ValidatingAdmissionPolicy name which the ValidatingAdmissionPolicyBinding binds to. If the referenced resource does not exist, this binding is considered invalid and will be ignored Required.

◦ **spec.validationActions** ([]string)

*Set: unique values will be kept during a merge*

validationActions declares how Validations of the referenced ValidatingAdmissionPolicy are enforced. If a validation evaluates to false it is always enforced according to these actions.

Failures defined by the ValidatingAdmissionPolicy's FailurePolicy are enforced according to these actions only if the FailurePolicy is set to Fail, otherwise the failures are ignored. This includes compilation errors, runtime errors and misconfigurations of the policy.



validationActions is declared as a set of action values. Order does not matter. validationActions may not contain duplicates of the same action.

The supported actions values are:

"Deny" specifies that a validation failure results in a denied request.

"Warn" specifies that a validation failure is reported to the request client in HTTP Warning headers, with a warning code of 299. Warnings can be sent both for allowed or denied admission responses.

"Audit" specifies that a validation failure is included in the published audit event for the request. The audit event will contain a validation.policy.admission.k8s.io/validation\_failure audit annotation with a value containing the details of the validation failures, formatted as a JSON list of objects, each with the following fields: - message: The validation failure message string - policy: The resource name of the ValidatingAdmissionPolicy - binding: The resource name of the ValidatingAdmissionPolicyBinding - expressionIndex: The index of the failed validations in the ValidatingAdmissionPolicy - validationActions: The enforcement actions enacted for the validation failure Example audit annotation:

```
"validation.policy.admission.k8s.io/validation_failure": "[{"message": "Invalid value",
{"policy": "policy.example.com", {"binding": "policybinding.example.com",
{"expressionIndex": "1", {"validationActions": ["Audit"]}]"]"
```

Clients should expect to handle additional values by ignoring any values not recognized.

"Deny" and "Warn" may not be used together since this combination needlessly duplicates the validation failure both in the API response body and the HTTP warning headers.

Required.

## Operations

---

### get read the specified ValidatingAdmissionPolicy

#### HTTP Request

GET /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ValidatingAdmissionPolicy
- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ValidatingAdmissionPolicy](#)): OK

401: Unauthorized

## get read status of the specified ValidatingAdmissionPolicy

### HTTP Request

GET /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ValidatingAdmissionPolicy
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ValidatingAdmissionPolicy](#)): OK

401: Unauthorized

## list list or watch objects of kind ValidatingAdmissionPolicy

### HTTP Request

GET /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)

- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([ValidatingAdmissionPolicyList](#)): OK

401: Unauthorized

## create create a ValidatingAdmissionPolicy

### HTTP Request

POST /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies

### Parameters

- **body**: [ValidatingAdmissionPolicy](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ValidatingAdmissionPolicy](#)): OK

201 ([ValidatingAdmissionPolicy](#)): Created

202 ([ValidatingAdmissionPolicy](#)): Accepted

401: Unauthorized

## update replace the specified ValidatingAdmissionPolicy

### HTTP Request

PUT /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ValidatingAdmissionPolicy
- **body**: [ValidatingAdmissionPolicy](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ValidatingAdmissionPolicy](#)): OK

201 ([ValidatingAdmissionPolicy](#)): Created

401: Unauthorized

## update replace status of the specified ValidatingAdmissionPolicy

### HTTP Request

PUT /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ValidatingAdmissionPolicy
- **body**: [ValidatingAdmissionPolicy](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ValidatingAdmissionPolicy](#)): OK

201 ([ValidatingAdmissionPolicy](#)): Created

401: Unauthorized

## patch partially update the specified ValidatingAdmissionPolicy

### HTTP Request

PATCH /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ValidatingAdmissionPolicy
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ValidatingAdmissionPolicy](#)): OK

201 ([ValidatingAdmissionPolicy](#)): Created

401: Unauthorized

## patch partially update status of the specified ValidatingAdmissionPolicy

### HTTP Request

PATCH /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the ValidatingAdmissionPolicy
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([ValidatingAdmissionPolicy](#)): OK

201 ([ValidatingAdmissionPolicy](#)): Created

401: Unauthorized

## delete delete a ValidatingAdmissionPolicy

### HTTP Request

DELETE /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ValidatingAdmissionPolicy

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ValidatingAdmissionPolicy

### HTTP Request

DELETE /apis/admissionregistration.k8s.io/v1beta1/validatingadmissionpolicies

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer



[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Cluster Resources

---

## [Node](#)

Node is a worker node in Kubernetes.

## [Namespace](#)

Namespace provides a scope for Names.

## [Event](#)

Event is a report of an event somewhere in the cluster.

## [APIService](#)

APIService represents a server for a particular GroupVersion.

## [Lease](#)

Lease defines a lease concept.

## [RuntimeClass](#)

RuntimeClass defines a class of container runtime supported in the cluster.

## [FlowSchema v1beta3](#)

FlowSchema defines the schema of a group of flows.

## [PriorityLevelConfiguration v1beta3](#)

PriorityLevelConfiguration represents the configuration of a priority level.

## [Binding](#)

Binding ties one object to another; for example, a pod is bound to a node by a scheduler.

## [ComponentStatus](#)

ComponentStatus (and ComponentStatusList) holds the cluster validation info.

## [ClusterCIDR v1alpha1](#)

ClusterCIDR represents a single configuration for per-Node Pod CIDR allocations when the MultiCIDRRangeAllocator is enabled (see the config for kube-controller-manager).

# Node

Node is a worker node in Kubernetes.

apiVersion: v1

```
import "k8s.io/api/core/v1"
```

# Node

Node is a worker node in Kubernetes. Each node will have a unique identifier in the cache (i.e. in etcd).

- 
- **apiVersion:** v1
  - **kind:** Node
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([NodeSpec](#))

Spec defines the behavior of a node. <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([NodeStatus](#))

Most recently observed status of the node. Populated by the system. Read-only. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

# NodeSpec

NodeSpec describes the attributes that a node is created with.

- 
- **configSource** (NodeConfigSource)

Deprecated: Previously used to specify the source of the node's configuration for the DynamicKubeletConfig feature. This feature is removed.

*NodeConfigSource specifies a source of node configuration. Exactly one subfield (excluding metadata) must be non-nil. This API is deprecated since 1.22*

- **configSource.configMap** (ConfigMapNodeConfigSource)

ConfigMap is a reference to a Node's ConfigMap

*ConfigMapNodeConfigSource contains the information to reference a ConfigMap as a config source for the Node. This API is deprecated since 1.22: <https://git.k8s.io/enhancements/keps/sig-node/281-dynamic-kubelet-configuration>*

- **configSource.configMap.kubeletConfigKey** (string), required

KubeletConfigKey declares which key of the referenced ConfigMap corresponds to the KubeletConfiguration structure. This field is required in all cases.

- **configSource.configMap.name** (string), required

Name is the metadata.name of the referenced ConfigMap. This field is required in all cases.

- **configSource.configMap.namespace** (string), required

Namespace is the metadata.namespace of the referenced ConfigMap. This field is required in all cases.

- **configSource.configMap.resourceVersion** (string)

ResourceVersion is the metadata.ResourceVersion of the referenced ConfigMap. This field is forbidden in Node.Spec, and required in Node.Status.

- **configSource.configMap.uid** (string)

UID is the metadata.UID of the referenced ConfigMap. This field is forbidden in Node.Spec, and required in Node.Status.

- **externalID** (string)

Deprecated. Not all kubelets will set this field. Remove field after 1.13. see: <https://issues.k8s.io/61966>

- **podCIDR** (string)

PodCIDR represents the pod IP range assigned to the node.

- **podCIDRs** ([]string)

podCIDRs represents the IP ranges assigned to the node for usage by Pods on that node. If this field is specified, the 0th entry must match the podCIDR field. It may contain at most 1 value for each of IPv4 and IPv6.

- **providerID** (string)

ID of the node assigned by the cloud provider in the format: <ProviderName>://<ProviderSpecificNodeID>

- **taints** ([]Taint)

If specified, the node's taints.

*The node this Taint is attached to has the "effect" on any pod that does not tolerate the Taint.*

- **taints.effect** (string), required

Required. The effect of the taint on pods that do not tolerate the taint. Valid effects are NoSchedule, PreferNoSchedule and NoExecute.

- **taints.key** (string), required

Required. The taint key to be applied to a node.

- **taints.timeAdded** (Time)

TimeAdded represents the time at which the taint was added. It is only written for NoExecute taints.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **taints.value** (string)

The taint value corresponding to the taint key.

- **unschedulable** (boolean)

Unschedulable controls node schedulability of new pods. By default, node is schedulable. More info: <https://kubernetes.io/docs/concepts/nodes/node/#manual-node-administration>

## NodeStatus

NodeStatus is information about the current status of a node.

---

- **addresses** ([]NodeAddress)

*Patch strategy: merge on key type*

List of addresses reachable to the node. Queried from cloud provider, if available. More info: <https://kubernetes.io/docs/concepts/nodes/node/#addresses> Note: This field is declared as mergeable, but the merge key is not sufficiently unique, which can cause data corruption when it is merged. Callers should instead use a full-replacement patch. See <https://pr.k8s.io/79391> for an example. Consumers should assume that addresses can change during the lifetime of a Node. However, there are some exceptions where this

may not be possible, such as Pods that inherit a Node's address in its own status or consumers of the downward API (status.hostIP).

*NodeAddress contains information for the node's address.*

- **addresses.address** (string), required

The node address.

- **addresses.type** (string), required

Node address type, one of Hostname, ExternalIP or InternalIP.

- **allocatable** (map[string][Quantity](#))

Allocatable represents the resources of a node that are available for scheduling. Defaults to Capacity.

- **capacity** (map[string][Quantity](#))

Capacity represents the total resources of a node. More info: <https://kubernetes.io/docs/concepts/storage/persistent-volumes#capacity>

- **conditions** ([]NodeCondition)

*Patch strategy: merge on key type*

Conditions is an array of current observed node conditions. More info: <https://kubernetes.io/docs/concepts/nodes/node/#condition>

*NodeCondition contains condition information for a node.*

- **conditions.status** (string), required

Status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

Type of node condition.

- **conditions.lastHeartbeatTime** (Time)

Last time we got an update on a given condition.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.lastTransitionTime** (Time)

Last time the condition transit from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)
  - Human readable message indicating details about last transition.
- **conditions.reason** (string)
  - (brief) reason for the condition's last transition.
- **config** (NodeConfigStatus)
 

Status of the config assigned to the node via the dynamic Kubelet config feature.

*NodeConfigStatus describes the status of the config assigned by Node.Spec.ConfigSource.*

  - **config.active** (NodeConfigSource)
 

Active reports the checkpointed config the node is actively using. Active will represent either the current version of the Assigned config, or the current LastKnownGood config, depending on whether attempting to use the Assigned config results in an error.

*NodeConfigSource specifies a source of node configuration. Exactly one subfield (excluding metadata) must be non-nil. This API is deprecated since 1.22*

    - **config.active.configMap** (ConfigMapNodeConfigSource)
 

ConfigMap is a reference to a Node's ConfigMap

*ConfigMapNodeConfigSource contains the information to reference a ConfigMap as a config source for the Node. This API is deprecated since 1.22: <https://git.k8s.io/enhancements/keps/sig-node/281-dynamic-kubelet-configuration>*

      - **config.active.configMap.kubeletConfigKey** (string), required
 

KubeletConfigKey declares which key of the referenced ConfigMap corresponds to the KubeletConfiguration structure This field is required in all cases.
      - **config.active.configMap.name** (string), required
 

Name is the metadata.name of the referenced ConfigMap. This field is required in all cases.
      - **config.active.configMap.namespace** (string), required
 

Namespace is the metadata.namespace of the referenced ConfigMap. This field is required in all cases.
      - **config.active.configMap.resourceVersion** (string)
 

ResourceVersion is the metadata.ResourceVersion of the referenced ConfigMap. This field is forbidden in Node.Spec, and required in Node.Status.

▪ **config.active.configMap.uid** (string)

UID is the metadata.UID of the referenced ConfigMap. This field is forbidden in Node.Spec, and required in Node.Status.

◦ **config.assigned** (NodeConfigSource)

Assigned reports the checkpointed config the node will try to use. When Node.Spec.ConfigSource is updated, the node checkpoints the associated config payload to local disk, along with a record indicating intended config. The node refers to this record to choose its config checkpoint, and reports this record in Assigned. Assigned only updates in the status after the record has been checkpointed to disk. When the Kubelet is restarted, it tries to make the Assigned config the Active config by loading and validating the checkpointed payload identified by Assigned.

*NodeConfigSource specifies a source of node configuration. Exactly one subfield (excluding metadata) must be non-nil. This API is deprecated since 1.22*

▪ **config.assigned.configMap** (ConfigMapNodeConfigSource)

ConfigMap is a reference to a Node's ConfigMap

*ConfigMapNodeConfigSource contains the information to reference a ConfigMap as a config source for the Node. This API is deprecated since 1.22: <https://git.k8s.io/enhancements/keps/sig-node/281-dynamic-kubelet-configuration>*

▪ **config.assigned.configMap.kubeletConfigKey** (string), required

KubeletConfigKey declares which key of the referenced ConfigMap corresponds to the KubeletConfiguration structure. This field is required in all cases.

▪ **config.assigned.configMap.name** (string), required

Name is the metadata.name of the referenced ConfigMap. This field is required in all cases.

▪ **config.assigned.configMap.namespace** (string), required

Namespace is the metadata.namespace of the referenced ConfigMap. This field is required in all cases.

▪ **config.assigned.configMap.resourceVersion** (string)

ResourceVersion is the metadata.ResourceVersion of the referenced ConfigMap. This field is forbidden in Node.Spec, and required in Node.Status.

▪ **config.assigned.configMap.uid** (string)

UID is the metadata.UID of the referenced ConfigMap. This field is forbidden in Node.Spec, and required in Node.Status.

**config.error** (string)

◦

Error describes any problems reconciling the Spec.ConfigSource to the Active config. Errors may occur, for example, attempting to checkpoint Spec.ConfigSource to the local Assigned record, attempting to checkpoint the payload associated with Spec.ConfigSource, attempting to load or validate the Assigned config, etc. Errors may occur at different points while syncing config. Earlier errors (e.g. download or checkpointing errors) will not result in a rollback to LastKnownGood, and may resolve across Kubelet retries. Later errors (e.g. loading or validating a checkpointed config) will result in a rollback to LastKnownGood. In the latter case, it is usually possible to resolve the error by fixing the config assigned in Spec.ConfigSource. You can find additional information for debugging by searching the error message in the Kubelet log. Error is a human-readable description of the error state; machines can check whether or not Error is empty, but should not rely on the stability of the Error text across Kubelet versions.

◦ **config.lastKnownGood** (NodeConfigSource)

LastKnownGood reports the checkpointed config the node will fall back to when it encounters an error attempting to use the Assigned config. The Assigned config becomes the LastKnownGood config when the node determines that the Assigned config is stable and correct. This is currently implemented as a 10-minute soak period starting when the local record of Assigned config is updated. If the Assigned config is Active at the end of this period, it becomes the LastKnownGood. Note that if Spec.ConfigSource is reset to nil (use local defaults), the LastKnownGood is also immediately reset to nil, because the local default config is always assumed good. You should not make assumptions about the node's method of determining config stability and correctness, as this may change or become configurable in the future.

*NodeConfigSource specifies a source of node configuration. Exactly one subfield (excluding metadata) must be non-nil. This API is deprecated since 1.22*

▪ **config.lastKnownGood.configMap** (ConfigMapNodeConfigSource)

ConfigMap is a reference to a Node's ConfigMap

*ConfigMapNodeConfigSource contains the information to reference a ConfigMap as a config source for the Node. This API is deprecated since 1.22: <https://git.k8s.io/enhancements/keps/sig-node/281-dynamic-kubelet-configuration>*

▪ **config.lastKnownGood.configMap.kubeletConfigKey** (string), required

KubeletConfigKey declares which key of the referenced ConfigMap corresponds to the KubeletConfiguration structure. This field is required in all cases.

▪ **config.lastKnownGood.configMap.name** (string), required

Name is the metadata.name of the referenced ConfigMap. This field is required in all cases.

▪ **config.lastKnownGood.configMap.namespace** (string), required



Namespace is the metadata.namespace of the referenced ConfigMap. This field is required in all cases.

- **config.lastKnownGood.configMap.resourceVersion** (string)

ResourceVersion is the metadata.ResourceVersion of the referenced ConfigMap. This field is forbidden in Node.Spec, and required in Node.Status.

- **config.lastKnownGood.configMap.uid** (string)

UID is the metadata.UID of the referenced ConfigMap. This field is forbidden in Node.Spec, and required in Node.Status.

- **daemonEndpoints** (NodeDaemonEndpoints)

Endpoints of daemons running on the Node.

*NodeDaemonEndpoints lists ports opened by daemons running on the Node.*

- **daemonEndpoints.kubeletEndpoint** (DaemonEndpoint)

Endpoint on which Kubelet is listening.

*DaemonEndpoint contains information about a single Daemon endpoint.*

- **daemonEndpoints.kubeletEndpoint.Port** (int32), required

Port number of the given endpoint.

- **images** ([]ContainerImage)

List of container images on this node

*Describe a container image*

- **images.names** ([]string)

Names by which this image is known. e.g. ["kubernetes.example/hyperkube:v1.0.7", "cloud-vendor.registry.example/cloud-vendor/hyperkube:v1.0.7"]

- **images.sizeBytes** (int64)

The size of the image in bytes.

- **nodeInfo** (NodeSystemInfo)

Set of ids/uuids to uniquely identify the node. More info: <https://kubernetes.io/docs/concepts/nodes/node/#info>

*NodeSystemInfo is a set of ids/uuids to uniquely identify the node.*

- **nodeInfo.architecture** (string), required

The Architecture reported by the node

- **nodeInfo.bootID** (string), required
  - Boot ID reported by the node.
- **nodeInfo.containerRuntimeVersion** (string), required
  - ContainerRuntime Version reported by the node through runtime remote API (e.g. containerd://1.4.2).
- **nodeInfo.kernelVersion** (string), required
  - Kernel Version reported by the node from 'uname -r' (e.g. 3.16.0-0.bpo.4-amd64).
- **nodeInfo.kubeProxyVersion** (string), required
  - KubeProxy Version reported by the node.
- **nodeInfo.kubeletVersion** (string), required
  - Kubelet Version reported by the node.
- **nodeInfo.machineID** (string), required
  - MachineID reported by the node. For unique machine identification in the cluster this field is preferred. Learn more from man(5) machine-id: <http://man7.org/linux/man-pages/man5/machine-id.5.html>
- **nodeInfo.operatingSystem** (string), required
  - The Operating System reported by the node
- **nodeInfo.osImage** (string), required
  - OS Image reported by the node from /etc/os-release (e.g. Debian GNU/Linux 7 (wheezy)).
- **nodeInfo.systemUUID** (string), required
  - SystemUUID reported by the node. For unique machine identification MachineID is preferred. This field is specific to Red Hat hosts [https://access.redhat.com/documentation/en-us/red\\_hat\\_subscription\\_management/1/html/rhsm/uuid](https://access.redhat.com/documentation/en-us/red_hat_subscription_management/1/html/rhsm/uuid)
- **phase** (string)
  - NodePhase is the recently observed lifecycle phase of the node. More info: <https://kubernetes.io/docs/concepts/nodes/node/#phase> The field is never populated, and now is deprecated.
- **volumesAttached** ([]AttachedVolume)
  - List of volumes that are attached to the node.
  - *AttachedVolume describes a volume attached to a node*
  - **volumesAttached.devicePath** (string), required

DevicePath represents the device path where the volume should be available

- **volumesAttached.name** (string), required

Name of the attached volume

- **volumesInUse** ([]string)

List of attachable volumes in use (mounted) by the node.

## NodeList

NodeList is the whole list of all Nodes which have been registered with master.

---

- **apiVersion**: v1

- **kind**: NodeList

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([][Node](#)), required

List of nodes

## Operations

---

### get read the specified Node

#### HTTP Request

GET /api/v1/nodes/{name}

#### Parameters

- **name** (*in path*): string, required

name of the Node

- **pretty** (*in query*): string

[pretty](#)

#### Response

200 ([Node](#)): OK

401: Unauthorized

## get read status of the specified Node

### HTTP Request

GET /api/v1/nodes/{name}/status

### Parameters

- **name** (*in path*): string, required

name of the Node

- **pretty** (*in query*): string

[pretty](#)

### Response

200 ([Node](#)): OK

401: Unauthorized

## list list or watch objects of kind Node

### HTTP Request

GET /api/v1/nodes

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([NodeList](#)): OK

401: Unauthorized

## create create a Node

### HTTP Request

POST /api/v1/nodes

### Parameters

- **body**: [Node](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Node](#)): OK

201 ([Node](#)): Created

202 ([Node](#)): Accepted

401: Unauthorized

## update replace the specified Node

### HTTP Request

PUT /api/v1/nodes/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Node
- **body**: [Node](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Node](#)): OK

201 ([Node](#)): Created

401: Unauthorized

## update replace status of the specified Node

### HTTP Request

PUT /api/v1/nodes/{name}/status

## Parameters

- **name** (*in path*): string, required  
name of the Node
- **body**: [Node](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Node](#)): OK

201 ([Node](#)): Created

401: Unauthorized

## patch partially update the specified Node

### HTTP Request

PATCH /api/v1/nodes/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Node
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Node](#)): OK

201 ([Node](#)): Created

401: Unauthorized

## patch partially update status of the specified Node

### HTTP Request

PATCH /api/v1/nodes/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Node
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)



## Response

200 ([Node](#)): OK

201 ([Node](#)): Created

401: Unauthorized

## delete delete a Node

### HTTP Request

DELETE /api/v1/nodes/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Node
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Node

### HTTP Request

DELETE /api/v1/nodes

## Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Namespace

Namespace provides a scope for Names.

apiVersion: v1

```
import "k8s.io/api/core/v1"
```

## Namespace

Namespace provides a scope for Names. Use of multiple namespaces is optional.

---

- **apiVersion:** v1
- **kind:** Namespace
- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([NamespaceSpec](#))

Spec defines the behavior of the Namespace. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([NamespaceStatus](#))

Status describes the current status of a Namespace. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## NamespaceSpec

NamespaceSpec describes the attributes on a Namespace.

---

- **finalizers** ([]string)

Finalizers is an opaque list of values that must be empty to permanently remove object from storage. More info: <https://kubernetes.io/docs/tasks/administer-cluster/namespaces/>

## NamespaceStatus

NamespaceStatus is information about the current status of a Namespace.

---

- **conditions** ([]NamespaceCondition)

*Patch strategy: merge on key type*

Represents the latest available observations of a namespace's current state.

*NamespaceCondition contains details about state of namespace.*

- **conditions.status** (string), required

Status of the condition, one of True, False, Unknown.

- **conditions.type** (string), required

Type of namespace controller condition.

- **conditions.lastTransitionTime** (Time)

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

- **conditions.reason** (string)

- **phase** (string)

Phase is the current lifecycle phase of the namespace. More info: <https://kubernetes.io/docs/tasks/administer-cluster/namespaces/>

## NamespaceList

NamespaceList is a list of Namespaces.

---

- **apiVersion:** v1
- **kind:** NamespaceList
- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]Namespace](#)), required

Items is the list of Namespace objects in the list. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/namespaces/>

## Operations

---

## get read the specified Namespace

### HTTP Request

GET /api/v1/namespaces/{name}

### Parameters

- **name** (*in path*): string, required

name of the Namespace

- **pretty** (*in query*): string

[pretty](#)

### Response

200 ([Namespace](#)): OK

401: Unauthorized

## get read status of the specified Namespace

### HTTP Request

GET /api/v1/namespaces/{name}/status

### Parameters

- **name** (*in path*): string, required

name of the Namespace

- **pretty** (*in query*): string

[pretty](#)

### Response

200 ([Namespace](#)): OK

401: Unauthorized

## list list or watch objects of kind Namespace

### HTTP Request

GET /api/v1/namespaces

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([NamespaceList](#)): OK

401: Unauthorized

## create create a Namespace

### HTTP Request

POST /api/v1/namespaces

### Parameters

- **body**: [Namespace](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Namespace](#)): OK

201 ([Namespace](#)): Created

202 ([Namespace](#)): Accepted

401: Unauthorized

## update replace the specified Namespace

### HTTP Request

PUT /api/v1/namespaces/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Namespace
- **body**: [Namespace](#), required
- **dryRun** (*in query*): string  
[dryRun](#)

- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Namespace](#)): OK

201 ([Namespace](#)): Created

401: Unauthorized

## update replace finalize of the specified Namespace

### HTTP Request

PUT /api/v1/namespaces/{name}/finalize

### Parameters

- **name** (*in path*): string, required  
name of the Namespace
- **body**: [Namespace](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Namespace](#)): OK



201 ([Namespace](#)): Created

401: Unauthorized

## update replace status of the specified Namespace

### HTTP Request

PUT /api/v1/namespaces/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Namespace
- **body**: [Namespace](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([Namespace](#)): OK

201 ([Namespace](#)): Created

401: Unauthorized

## patch partially update the specified Namespace

### HTTP Request

PATCH /api/v1/namespaces/{name}

### Parameters

- **name** (*in path*): string, required

name of the Namespace

- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Namespace](#)): OK

201 ([Namespace](#)): Created

401: Unauthorized

## patch partially update status of the specified Namespace

### HTTP Request

PATCH /api/v1/namespaces/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the Namespace
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Namespace](#)): OK

201 ([Namespace](#)): Created

401: Unauthorized

## delete delete a Namespace

### HTTP Request

DELETE /api/v1/namespaces/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Namespace
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

# Event

Event is a report of an event somewhere in the cluster.

apiVersion: events.k8s.io/v1

import "k8s.io/api/events/v1"

## Event

Event is a report of an event somewhere in the cluster. It generally denotes some state change in the system. Events have a limited retention time and triggers and messages may evolve with time. Event consumers should not rely on the timing of an event with a given Reason reflecting a consistent underlying trigger, or the continued existence of events with that Reason. Events should be treated as informative, best-effort, supplemental data.

---

- **apiVersion:** events.k8s.io/v1

- **kind:** Event

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **eventTime** (MicroTime), required

eventTime is the time when this Event was first observed. It is required.

*MicroTime is version of Time with microsecond level precision.*

- **action** (string)

action is what action was taken/failed regarding to the regarding object. It is machine-readable. This field cannot be empty for new Events and it can have at most 128 characters.

- **deprecatedCount** (int32)

deprecatedCount is the deprecated field assuring backward compatibility with core.v1 Event type.

- **deprecatedFirstTimestamp** (Time)

deprecatedFirstTimestamp is the deprecated field assuring backward compatibility with core.v1 Event type.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **deprecatedLastTimestamp** (Time)

deprecatedLastTimestamp is the deprecated field assuring backward compatibility with core.v1 Event type.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **deprecatedSource** (EventSource)

deprecatedSource is the deprecated field assuring backward compatibility with core.v1 Event type.

*EventSource contains information for an event.*

- **deprecatedSource.component** (string)

Component from which the event is generated.

- **deprecatedSource.host** (string)

Node name on which the event is generated.

- **note** (string)

note is a human-readable description of the status of this operation. Maximal length of the note is 1kB, but libraries should be prepared to handle values up to 64kB.

- **reason** (string)

reason is why the action was taken. It is human-readable. This field cannot be empty for new Events and it can have at most 128 characters.

- **regarding** ([ObjectReference](#))

regarding contains the object this Event is about. In most cases it's an Object reporting controller implements, e.g. ReplicaSetController implements ReplicaSets and this event is emitted because it acts on some changes in a ReplicaSet object.

- **related** ([ObjectReference](#))

related is the optional secondary object for more complex actions. E.g. when regarding object triggers a creation or deletion of related object.

- **reportingController** (string)

reportingController is the name of the controller that emitted this Event, e.g. kubernetes.io/kubelet. This field cannot be empty for new Events.

- **reportingInstance** (string)

reportingInstance is the ID of the controller instance, e.g. kubelet-xyzf. This field cannot be empty for new Events and it can have at most 128 characters.

- **series** (EventSeries)

series is data about the Event series this event represents or nil if it's a singleton Event.

*EventSeries contain information on series of events, i.e. thing that was/is happening continuously for some time. How often to update the EventSeries is up to the event reporters. The default event reporter in "k8s.io/client-go/tools/events/event\_broadcaster.go" shows how this struct is updated on heartbeats and can guide customized reporter implementations.*

- **series.count** (int32), required

count is the number of occurrences in this series up to the last heartbeat time.

- **series.lastObservedTime** (MicroTime), required

lastObservedTime is the time when last Event from the series was seen before last heartbeat.

*MicroTime is version of Time with microsecond level precision.*

- **type** (string)

type is the type of this event (Normal, Warning), new types could be added in the future. It is machine-readable. This field cannot be empty for new Events.

## EventList

EventList is a list of Event objects.

---

- **apiVersion**: events.k8s.io/v1

- **kind**: EventList

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]Event](#)), required

items is a list of schema objects.

## Operations

---

### get read the specified Event

#### HTTP Request

GET /apis/events.k8s.io/v1/namespaces/{namespace}/events/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Event
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Event](#)): OK

401: Unauthorized

## list list or watch objects of kind Event

### HTTP Request

GET /apis/events.k8s.io/v1/namespaces/{namespace}/events

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([EventList](#)): OK

401: Unauthorized

## list list or watch objects of kind Event

### HTTP Request

GET /apis/events.k8s.io/v1/events

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer



[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([EventList](#)): OK

401: Unauthorized

## create create an Event

### HTTP Request

POST /apis/events.k8s.io/v1/namespaces/{namespace}/events

### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Event](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Event](#)): OK

201 ([Event](#)): Created

202 ([Event](#)): Accepted

401: Unauthorized

## update replace the specified Event

### HTTP Request

PUT /apis/events.k8s.io/v1/namespaces/{namespace}/events/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Event
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Event](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Event](#)): OK

201 ([Event](#)): Created

401: Unauthorized

## patch partially update the specified Event

### HTTP Request

PATCH /apis/events.k8s.io/v1/namespaces/{namespace}/events/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Event
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Event](#)): OK

201 ([Event](#)): Created

401: Unauthorized

## delete delete an Event

### HTTP Request

DELETE /apis/events.k8s.io/v1/namespaces/{namespace}/events/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Event
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

### Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Event

### HTTP Request

DELETE /apis/events.k8s.io/v1/namespaces/{namespace}/events

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)

- **body:** [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# APIService

APIService represents a server for a particular GroupVersion.

apiVersion: apiregistration.k8s.io/v1

```
import "k8s.io/kube-aggregator/pkg/apis/apiregistration/v1"
```

## APIService

APIService represents a server for a particular GroupVersion. Name must be "version.group".

---

- **apiVersion:** apiregistration.k8s.io/v1

- **kind:** APIService

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([APIServiceSpec](#))

Spec contains information for locating and communicating with a server

- **status** ([APIServiceStatus](#))

Status contains derived information about an API server

## APIServiceSpec

APIServiceSpec contains information for locating and communicating with a server. Only https is supported, though you are able to disable certificate verification.

---

- **groupPriorityMinimum** (int32), required

GroupPriorityMinimum is the priority this group should have at least. Higher priority means that the group is preferred by clients over lower priority ones. Note that other versions of this group might specify even higher GroupPriorityMinimum values such that the whole group gets a higher priority. The primary sort is based on GroupPriorityMinimum, ordered highest number to lowest (20 before 10). The secondary sort is based on the alphabetical comparison of the name of the object. (v1.bar before v1.foo) We'd recommend something like: \*.k8s.io (except extensions) at 18000 and PaaSes (OpenShift, Deis) are recommended to be in the 2000s

- **versionPriority** (int32), required

VersionPriority controls the ordering of this API version inside of its group. Must be greater than zero. The primary sort is based on VersionPriority, ordered highest to lowest (20 before 10). Since it's inside of a group, the number can be small, probably in the 10s.

In case of equal version priorities, the version string will be used to compute the order inside a group. If the version string is "kube-like", it will sort above non "kube-like" version strings, which are ordered lexicographically. "Kube-like" versions start with a "v", then are followed by a number (the major version), then optionally the string "alpha" or "beta" and another number (the minor version). These are sorted first by GA > beta > alpha (where GA is a version with no suffix such as beta or alpha), and then by comparing major version, then minor version. An example sorted list of versions: v10, v2, v1, v11beta2, v10beta3, v3beta1, v12alpha1, v11alpha2, foo1, foo10.

- **caBundle** ([]byte)

*Atomic: will be replaced during a merge*

CABundle is a PEM encoded CA bundle which will be used to validate an API server's serving certificate. If unspecified, system trust roots on the apiserver are used.

- **group** (string)

Group is the API group name this server hosts

- **insecureSkipTLSVerify** (boolean)

InsecureSkipTLSVerify disables TLS certificate verification when communicating with this server. This is strongly discouraged. You should use the CABundle instead.

- **service** (ServiceReference)

Service is a reference to the service for this API server. It must communicate on port 443. If the Service is nil, that means the handling for the API groupversion is handled locally on this server. The call will simply delegate to the normal handler chain to be fulfilled.

*ServiceReference holds a reference to Service.legacy.k8s.io*

- **service.name** (string)

Name is the name of the service

- **service.namespace** (string)

Namespace is the namespace of the service

- **service.port** (int32)

If specified, the port on the service that hosting webhook. Default to 443 for backward compatibility. port should be a valid port number (1-65535, inclusive).

- **version** (string)

Version is the API version this server hosts. For example, "v1"

## APIServiceStatus

APIServiceStatus contains derived information about an API server

- 
- **conditions** ([]APIServiceCondition)

*Patch strategy: merge on key type*

*Map: unique values on key type will be kept during a merge*

Current service state of apiService.

*APIServiceCondition describes the state of an APIService at a particular point*

- **conditions.status** (string), required

Status is the status of the condition. Can be True, False, Unknown.

- **conditions.type** (string), required

Type is the type of the condition.

- **conditions.lastTransitionTime** (Time)

Last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

Human-readable message indicating details about last transition.

- **conditions.reason** (string)

Unique, one-word, CamelCase reason for the condition's last transition.

## APIServiceList

APIServiceList is a list of APIService objects.

- 
- **apiVersion**: apiregistration.k8s.io/v1

- **kind**: APIServiceList

- **metadata** ([ListMeta](#))

Standard list metadata More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([][APIService](#)), required

Items is the list of APIService



# Operations

---

## get read the specified APIService

### HTTP Request

GET /apis/apiregistration.k8s.io/v1/apiservices/{name}

### Parameters

- **name** (*in path*): string, required  
name of the APIService
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([APIService](#)): OK

401: Unauthorized

## get read status of the specified APIService

### HTTP Request

GET /apis/apiregistration.k8s.io/v1/apiservices/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the APIService
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([APIService](#)): OK

401: Unauthorized

## list list or watch objects of kind APIService

### HTTP Request

GET /apis/apiregistration.k8s.io/v1/apiservices

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([APIServiceList](#)): OK

401: Unauthorized

## create create an APIService

### HTTP Request

POST /apis/apiregistration.k8s.io/v1/apiservices

### Parameters

- **body**: [APIService](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([APIService](#)): OK

201 ([APIService](#)): Created

202 ([APIService](#)): Accepted

401: Unauthorized

## update replace the specified APIService

### HTTP Request

PUT /apis/apiregistration.k8s.io/v1/apiservices/{name}

### Parameters

- **name** (*in path*): string, required

name of the APIService

- **body**: [APIService](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([APIService](#)): OK

201 ([APIService](#)): Created

401: Unauthorized

## update replace status of the specified APIService

### HTTP Request

PUT /apis/apiregistration.k8s.io/v1/apiservices/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the APIService
- **body**: [APIService](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)

- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([APIService](#)): OK

201 ([APIService](#)): Created

401: Unauthorized

## patch partially update the specified APIService

### HTTP Request

PATCH /apis/apiregistration.k8s.io/v1/apiservices/{name}

### Parameters

- **name** (*in path*): string, required  
name of the APIService
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([APIService](#)): OK

201 ([APIService](#)): Created

401: Unauthorized

## patch partially update status of the specified APIService

### HTTP Request

PATCH /apis/apiregistration.k8s.io/v1/apiservices/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the APIService
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([APIService](#)): OK

201 ([APIService](#)): Created

401: Unauthorized

## delete delete an APIService

### HTTP Request

DELETE /apis/apiregistration.k8s.io/v1/apiservices/{name}

### Parameters

- **name** (*in path*): string, required  
name of the APIService

- **body:** [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of APIService

### HTTP Request

DELETE /apis/apiregistration.k8s.io/v1/apiservices

### Parameters

- **body:** [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Lease

Lease defines a lease concept.

apiVersion: coordination.k8s.io/v1

import "k8s.io/api/coordination/v1"

## Lease

Lease defines a lease concept.

- 
- **apiVersion**: coordination.k8s.io/v1
  - **kind**: Lease



- **metadata** ([ObjectMeta](#))

More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([LeaseSpec](#))

spec contains the specification of the Lease. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## LeaseSpec

LeaseSpec is a specification of a Lease.

---

- **acquireTime** (MicroTime)

acquireTime is a time when the current lease was acquired.

*MicroTime is version of Time with microsecond level precision.*

- **holderIdentity** (string)

holderIdentity contains the identity of the holder of a current lease.

- **leaseDurationSeconds** (int32)

leaseDurationSeconds is a duration that candidates for a lease need to wait to force acquire it. This is measure against time of last observed renewTime.

- **leaseTransitions** (int32)

leaseTransitions is the number of transitions of a lease between holders.

- **renewTime** (MicroTime)

renewTime is a time when the current holder of a lease has last updated the lease.

*MicroTime is version of Time with microsecond level precision.*

## LeaseList

LeaseList is a list of Lease objects.

---

- **apiVersion**: coordination.k8s.io/v1

- **kind**: LeaseList

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([[Lease](#)]), required  
items is a list of schema objects.

## Operations

---

### get read the specified Lease

#### HTTP Request

GET /apis/coordination.k8s.io/v1/namespaces/{namespace}/leases/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the Lease
- **namespace** (*in path*): string, required  
[namespace](#)
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([Lease](#)): OK

401: Unauthorized

### list list or watch objects of kind Lease

#### HTTP Request

GET /apis/coordination.k8s.io/v1/namespaces/{namespace}/leases

#### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([LeaseList](#)): OK

401: Unauthorized

## list list or watch objects of kind Lease

## HTTP Request

GET /apis/coordination.k8s.io/v1/leases

## Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([LeaseList](#)): OK

401: Unauthorized

## create create a Lease

## HTTP Request

POST /apis/coordination.k8s.io/v1/namespaces/{namespace}/leases

## Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Lease](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Lease](#)): OK

201 ([Lease](#)): Created

202 ([Lease](#)): Accepted

401: Unauthorized

## update replace the specified Lease

### HTTP Request

PUT /apis/coordination.k8s.io/v1/namespaces/{namespace}/leases/{name}

## Parameters

- **name** (*in path*): string, required  
name of the Lease
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [Lease](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Lease](#)): OK

201 ([Lease](#)): Created

401: Unauthorized

## patch partially update the specified Lease

### HTTP Request

PATCH /apis/coordination.k8s.io/v1/namespaces/{namespace}/leases/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Lease
- **namespace** (*in path*): string, required

[namespace](#)

- **body**: [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([Lease](#)): OK

201 ([Lease](#)): Created

401: Unauthorized

## delete delete a Lease

### HTTP Request

DELETE /apis/coordination.k8s.io/v1/namespaces/{namespace}/leases/{name}

### Parameters

- **name** (*in path*): string, required  
name of the Lease
- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of Lease

### HTTP Request

DELETE /apis/coordination.k8s.io/v1/namespaces/{namespace}/leases

### Parameters

- **namespace** (*in path*): string, required  
[namespace](#)
- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)



- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# RuntimeClass

RuntimeClass defines a class of container runtime supported in the cluster.

apiVersion: node.k8s.io/v1

import "k8s.io/api/node/v1"

## RuntimeClass

RuntimeClass defines a class of container runtime supported in the cluster. The RuntimeClass is used to determine which container runtime is used to run all containers in a pod.

RuntimeClasses are manually defined by a user or cluster provisioner, and referenced in the PodSpec. The Kubelet is responsible for resolving the RuntimeClassName reference before running the pod. For more details, see <https://kubernetes.io/docs/concepts/containers/runtime-class/>

- 
- **apiVersion**: node.k8s.io/v1
  - **kind**: RuntimeClass
  - **metadata** ([ObjectMeta](#))

More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **handler** (string), required

handler specifies the underlying runtime and configuration that the CRI implementation will use to handle pods of this class. The possible values are specific to the node & CRI configuration. It is assumed that all handlers are available on every node, and handlers of the same name are equivalent on every node. For example, a handler called "runc" might specify that the runc OCI runtime (using native Linux containers) will be used to run the containers in a pod. The Handler must be lowercase, conform to the DNS Label (RFC 1123) requirements, and is immutable.

- **overhead** (Overhead)

overhead represents the resource overhead associated with running a pod for a given RuntimeClass. For more details, see <https://kubernetes.io/docs/concepts/scheduling-eviction/pod-overhead/>

*Overhead structure represents the resource overhead associated with running a pod.*

- **overhead.podFixed** (map[string][Quantity](#))

podFixed represents the fixed resource overhead associated with running a pod.

- **scheduling** (Scheduling)

scheduling holds the scheduling constraints to ensure that pods running with this RuntimeClass are scheduled to nodes that support it. If scheduling is nil, this RuntimeClass is assumed to be supported by all nodes.

*Scheduling specifies the scheduling constraints for nodes supporting a RuntimeClass.*

- **scheduling.nodeSelector** (map[string]string)

nodeSelector lists labels that must be present on nodes that support this RuntimeClass. Pods using this RuntimeClass can only be scheduled to a node matched by this selector. The RuntimeClass nodeSelector is merged with a pod's existing nodeSelector. Any conflicts will cause the pod to be rejected in admission.

- **scheduling.tolerations** ([]Toleration)

*Atomic: will be replaced during a merge*

tolerations are appended (excluding duplicates) to pods running with this RuntimeClass during admission, effectively unioning the set of nodes tolerated by the pod and the RuntimeClass.

*The pod this Toleration is attached to tolerates any taint that matches the triple <key,value,effect> using the matching operator .*

- **scheduling.tolerations.key** (string)

Key is the taint key that the toleration applies to. Empty means match all taint keys. If the key is empty, operator must be Exists; this combination means to match all values and all keys.

- **scheduling.tolerations.operator** (string)

Operator represents a key's relationship to the value. Valid operators are Exists and Equal. Defaults to Equal. Exists is equivalent to wildcard for value, so that a pod can tolerate all taints of a particular category.

- **scheduling.tolerations.value** (string)

Value is the taint value the toleration matches to. If the operator is Exists, the value should be empty, otherwise just a regular string.

- **scheduling.tolerations.effect** (string)

Effect indicates the taint effect to match. Empty means match all taint effects. When specified, allowed values are NoSchedule, PreferNoSchedule and NoExecute.

- **scheduling.tolerations.tolerationSeconds** (int64)

TolerationSeconds represents the period of time the toleration (which must be of effect NoExecute, otherwise this field is ignored) tolerates the taint. By default, it is not set, which means tolerate the taint forever (do not evict). Zero and negative values will be treated as 0 (evict immediately) by the system.

## RuntimeClassList

RuntimeClassList is a list of RuntimeClass objects.

---

- **apiVersion**: node.k8s.io/v1

- **kind**: RuntimeClassList

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]RuntimeClass](#)), required

items is a list of schema objects.

## Operations

---

### get read the specified RuntimeClass

#### HTTP Request

GET /apis/node.k8s.io/v1/runtimeclasses/{name}

#### Parameters

- **name** (*in path*): string, required

name of the RuntimeClass

- **pretty** (*in query*): string

[pretty](#)

#### Response

200 ([RuntimeClass](#)): OK

401: Unauthorized

## list list or watch objects of kind RuntimeClass

### HTTP Request

GET /apis/node.k8s.io/v1/runtimeclasses

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([RuntimeClassList](#)): OK

401: Unauthorized

## create create a RuntimeClass

### HTTP Request

POST /apis/node.k8s.io/v1/runtimeclasses

### Parameters

- **body**: [RuntimeClass](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([RuntimeClass](#)): OK

201 ([RuntimeClass](#)): Created

202 ([RuntimeClass](#)): Accepted

401: Unauthorized

## update replace the specified RuntimeClass

### HTTP Request

PUT /apis/node.k8s.io/v1/runtimeclasses/{name}

### Parameters

- **name** (*in path*): string, required

name of the RuntimeClass

- **body**: [RuntimeClass](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([RuntimeClass](#)): OK

201 ([RuntimeClass](#)): Created

401: Unauthorized

## patch partially update the specified RuntimeClass

### HTTP Request

PATCH /apis/node.k8s.io/v1/runtimeclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the RuntimeClass
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)

- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([RuntimeClass](#)): OK

201 ([RuntimeClass](#)): Created

401: Unauthorized

## delete delete a RuntimeClass

### HTTP Request

DELETE /apis/node.k8s.io/v1/runtimeclasses/{name}

### Parameters

- **name** (*in path*): string, required  
name of the RuntimeClass
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of RuntimeClass

### HTTP Request

DELETE /apis/node.k8s.io/v1/runtimeclasses

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer



[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# FlowSchema v1beta3

FlowSchema defines the schema of a group of flows.

apiVersion: flowcontrol.apiserver.k8s.io/v1beta3

import "k8s.io/api/flowcontrol/v1beta3"

## FlowSchema

FlowSchema defines the schema of a group of flows. Note that a flow is made up of a set of inbound API requests with similar attributes and is identified by a pair of strings: the name of the FlowSchema and a "flow distinguisher".

---

- **apiVersion:** flowcontrol.apiserver.k8s.io/v1beta3

- **kind:** FlowSchema

- **metadata** ([ObjectMeta](#))

metadata is the standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([FlowSchemaSpec](#))

spec is the specification of the desired behavior of a FlowSchema. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([FlowSchemaStatus](#))

status is the current status of a FlowSchema. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## FlowSchemaSpec

FlowSchemaSpec describes how the FlowSchema's specification looks like.

---

- **priorityLevelConfiguration** (PriorityLevelConfigurationReference), required

`priorityLevelConfiguration` should reference a `PriorityLevelConfiguration` in the cluster. If the reference cannot be resolved, the `FlowSchema` will be ignored and marked as invalid in its status. Required.

*`PriorityLevelConfigurationReference` contains information that points to the "request-priority" being used.*

- **`priorityLevelConfiguration.name`** (string), required

`name` is the name of the priority level configuration being referenced Required.

- **`distinguisherMethod`** (`FlowDistinguisherMethod`)

`distinguisherMethod` defines how to compute the flow distinguisher for requests that match this schema. `nil` specifies that the distinguisher is disabled and thus will always be the empty string.

*`FlowDistinguisherMethod` specifies the method of a flow distinguisher.*

- **`distinguisherMethod.type`** (string), required

`type` is the type of flow distinguisher method The supported types are "ByUser" and "ByNamespace". Required.

- **`matchingPrecedence`** (int32)

`matchingPrecedence` is used to choose among the `FlowSchemas` that match a given request. The chosen `FlowSchema` is among those with the numerically lowest (which we take to be logically highest) `MatchingPrecedence`. Each `MatchingPrecedence` value must be ranged in [1,10000]. Note that if the precedence is not specified, it will be set to 1000 as default.

- **`rules`** (`[]PolicyRulesWithSubjects`)

*Atomic: will be replaced during a merge*

`rules` describes which requests will match this flow schema. This `FlowSchema` matches a request if and only if at least one member of `rules` matches the request. if it is an empty slice, there will be no requests matching the `FlowSchema`.

*`PolicyRulesWithSubjects` prescribes a test that applies to a request to an apiserver. The test considers the subject making the request, the verb being requested, and the resource to be acted upon. This `PolicyRulesWithSubjects` matches a request if and only if both (a) at least one member of `subjects` matches the request and (b) at least one member of `resourceRules` or `nonResourceRules` matches the request.*

- **`rules.subjects`** (`[]Subject`), required

*Atomic: will be replaced during a merge*

`subjects` is the list of normal user, serviceaccount, or group that this rule cares about. There must be at least one member in this slice. A slice that includes both the `system:authenticated` and `system:unauthenticated` user groups matches every request. Required.

*Subject matches the originator of a request, as identified by the request authentication system. There are three ways of matching an originator; by user, group, or service account.*

- **rules.subjects.kind** (string), required

kind indicates which one of the other fields is non-empty. Required

- **rules.subjects.group** (GroupSubject)

group matches based on user group name.

*GroupSubject holds detailed information for group-kind subject.*

- **rules.subjects.group.name** (string), required

name is the user group that matches, or "\*" to match all user groups. See <https://github.com/kubernetes/apiserver/blob/master/pkg/authentication/user/user.go> for some well-known group names. Required.

- **rules.subjects.serviceAccount** (ServiceAccountSubject)

serviceAccount matches ServiceAccounts.

*ServiceAccountSubject holds detailed information for service-account-kind subject.*

- **rules.subjects.serviceAccount.name** (string), required

name is the name of matching ServiceAccount objects, or "\*" to match regardless of name. Required.

- **rules.subjects.serviceAccount.namespace** (string), required

namespace is the namespace of matching ServiceAccount objects. Required.

- **rules.subjects.user** (UserSubject)

user matches based on username.

*UserSubject holds detailed information for user-kind subject.*

- **rules.subjects.user.name** (string), required

name is the username that matches, or "\*" to match all usernames. Required.

- **rules.nonResourceRules** ([]NonResourcePolicyRule)

*Atomic: will be replaced during a merge*

nonResourceRules is a list of NonResourcePolicyRules that identify matching requests according to their verb and the target non-resource URL.

*NonResourcePolicyRule is a predicate that matches non-resource requests according to their verb and the target non-resource URL. A NonResourcePolicyRule matches a request if and only if both (a) at least one member of verbs matches the request and (b) at least one member of nonResourceURLs matches the request.*

- **rules.nonResourceRules.nonResourceURLs** ([]string), required

*Set: unique values will be kept during a merge*

nonResourceURLs is a set of url prefixes that a user should have access to and may not be empty. For example:

- `"/healthz"` is legal
- `"/hea*"` is illegal
- `"/hea"` is legal but matches nothing
- `"/hea/*"` also matches nothing
- `"/healthz/"` matches all per-component health checks. `""` matches all non-resource urls. if it is present, it must be the only entry. Required.

- **rules.nonResourceRules.verbs** ([]string), required

*Set: unique values will be kept during a merge*

verbs is a list of matching verbs and may not be empty. `""` matches all verbs. If it is present, it must be the only entry. Required.

- **rules.resourceRules** ([]ResourcePolicyRule)

*Atomic: will be replaced during a merge*

resourceRules is a slice of ResourcePolicyRules that identify matching requests according to their verb and the target resource. At least one of resourceRules and nonResourceRules has to be non-empty.

*ResourcePolicyRule is a predicate that matches some resource requests, testing the request's verb and the target resource. A ResourcePolicyRule matches a resource request if and only if: (a) at least one member of verbs matches the request, (b) at least one member of apiGroups matches the request, (c) at least one member of resources matches the request, and (d) either (d1) the request does not specify a namespace (i.e., Namespace=="") and clusterScope is true or (d2) the request specifies a namespace and least one member of namespaces matches the request's namespace.*

- **rules.resourceRules.apiGroups** ([]string), required

*Set: unique values will be kept during a merge*

apiGroups is a list of matching API groups and may not be empty. `""` matches all API groups and, if present, must be the only entry. Required.

- **rules.resourceRules.resources** ([]string), required

*Set: unique values will be kept during a merge*

resources is a list of matching resources (i.e., lowercase and plural) with, if desired, subresource. For example, [ `"services"`, `"nodes/status"` ]. This list may

not be empty. "\*" matches all resources and, if present, must be the only entry. Required.

- **rules.resourceRules.verbs** ([]string), required

*Set: unique values will be kept during a merge*

verbs is a list of matching verbs and may not be empty. "\*" matches all verbs and, if present, must be the only entry. Required.

- **rules.resourceRules.clusterScope** (boolean)

clusterScope indicates whether to match requests that do not specify a namespace (which happens either because the resource is not namespaced or the request targets all namespaces). If this field is omitted or false then the namespaces field must contain a non-empty list.

- **rules.resourceRules.namespaces** ([]string)

*Set: unique values will be kept during a merge*

namespaces is a list of target namespaces that restricts matches. A request that specifies a target namespace matches only if either (a) this list contains that target namespace or (b) this list contains "". *Note that "" matches any specified namespace but does not match a request that does not specify a namespace (see the clusterScope field for that).* This list may be empty, but only if clusterScope is true.

## FlowSchemaStatus

FlowSchemaStatus represents the current state of a FlowSchema.

---

- **conditions** ([]FlowSchemaCondition)

*Patch strategy: merge on key type*

*Map: unique values on key type will be kept during a merge*

conditions is a list of the current states of FlowSchema.

*FlowSchemaCondition describes conditions for a FlowSchema.*

- **conditions.lastTransitionTime** (Time)

lastTransitionTime is the last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)

`message` is a human-readable message indicating details about last transition.

- **conditions.reason** (string)

`reason` is a unique, one-word, CamelCase reason for the condition's last transition.

- **conditions.status** (string)

`status` is the status of the condition. Can be True, False, Unknown. Required.

- **conditions.type** (string)

`type` is the type of the condition. Required.

## FlowSchemaList

FlowSchemaList is a list of FlowSchema objects.

- 
- **apiVersion:** flowcontrol.apiserver.k8s.io/v1beta3
  - **kind:** FlowSchemaList
  - **metadata** ([ListMeta](#))

`metadata` is the standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **items** ([\[\]FlowSchema](#)), required

`items` is a list of FlowSchemas.

## Operations

---

### get read the specified FlowSchema

#### HTTP Request

GET /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the FlowSchema
- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([FlowSchema](#)): OK

401: Unauthorized

## get read status of the specified FlowSchema

### HTTP Request

GET /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas/{name}/status

### Parameters

- **name** (*in path*): string, required

name of the FlowSchema

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([FlowSchema](#)): OK

401: Unauthorized

## list list or watch objects of kind FlowSchema

### HTTP Request

GET /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas

### Parameters

- **allowWatchBookmarks** (*in query*): boolean

[allowWatchBookmarks](#)

- **continue** (*in query*): string

[continue](#)

- **fieldSelector** (*in query*): string

[fieldSelector](#)

- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([FlowSchemaList](#)): OK

401: Unauthorized

## create create a FlowSchema

### HTTP Request

POST /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas

### Parameters

- **body**: [FlowSchema](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string



[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([FlowSchema](#)): OK

201 ([FlowSchema](#)): Created

202 ([FlowSchema](#)): Accepted

401: Unauthorized

## update replace the specified FlowSchema

### HTTP Request

PUT /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas/{name}

### Parameters

- **name** (*in path*): string, required

name of the FlowSchema

- **body**: [FlowSchema](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([FlowSchema](#)): OK

201 ([FlowSchema](#)): Created

401: Unauthorized

## update replace status of the specified FlowSchema

### HTTP Request

PUT /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the FlowSchema
- **body**: [FlowSchema](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([FlowSchema](#)): OK

201 ([FlowSchema](#)): Created

401: Unauthorized

## patch partially update the specified FlowSchema

### HTTP Request

PATCH /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas/{name}

### Parameters

- **name** (*in path*): string, required  
name of the FlowSchema
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([FlowSchema](#)): OK

201 ([FlowSchema](#)): Created

401: Unauthorized

## patch partially update status of the specified FlowSchema

### HTTP Request

PATCH /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas/{name}/status

### Parameters

- **name** (*in path*): string, required

name of the FlowSchema

- **body**: [Patch](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([FlowSchema](#)): OK

201 ([FlowSchema](#)): Created

401: Unauthorized

## delete delete a FlowSchema

### HTTP Request

DELETE /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas/{name}

### Parameters

- **name** (*in path*): string, required  
name of the FlowSchema
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of FlowSchema

### HTTP Request

DELETE /apis/flowcontrol.apiserver.k8s.io/v1beta3/flowschemas

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# PriorityLevelConfiguration v1beta3

PriorityLevelConfiguration represents the configuration of a priority level.

apiVersion: flowcontrol.apiserver.k8s.io/v1beta3

import "k8s.io/api/flowcontrol/v1beta3"

## PriorityLevelConfiguration

PriorityLevelConfiguration represents the configuration of a priority level.

- 
- **apiVersion:** flowcontrol.apiserver.k8s.io/v1beta3
  - **kind:** PriorityLevelConfiguration
  - **metadata** ([ObjectMeta](#))

metadata is the standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([PriorityLevelConfigurationSpec](#))

spec is the specification of the desired behavior of a "request-priority". More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

- **status** ([PriorityLevelConfigurationStatus](#))

status is the current status of a "request-priority". More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## PriorityLevelConfigurationSpec

PriorityLevelConfigurationSpec specifies the configuration of a priority level.

- 
- **type** (string), required

type indicates whether this priority level is subject to limitation on request execution. A value of "Exempt" means that requests of this priority level are not subject to a limit (and thus are never queued) and do not detract from the capacity made available to other

priority levels. A value of "Limited" means that (a) requests of this priority level *are* subject to limits and (b) some of the server's limited capacity is made available exclusively to this priority level. Required.

- **exempt** (ExemptPriorityLevelConfiguration)

`exempt` specifies how requests are handled for an exempt priority level. This field **MUST** be empty if type is "Limited". This field **MAY** be non-empty if type is "Exempt". If empty and type is "Exempt" then the default values for ExemptPriorityLevelConfiguration apply.

*ExemptPriorityLevelConfiguration describes the configurable aspects of the handling of exempt requests. In the mandatory exempt configuration object the values in the fields here can be modified by authorized users, unlike the rest of the spec.*

- **exempt.lendablePercent** (int32)

`lendablePercent` prescribes the fraction of the level's NominalCL that can be borrowed by other priority levels. This value of this field must be between 0 and 100, inclusive, and it defaults to 0. The number of seats that other levels can borrow from this level, known as this level's LendableConcurrencyLimit (LendableCL), is defined as follows.

$$\text{LendableCL}(i) = \text{round}(\text{NominalCL}(i) * \text{lendablePercent}(i)/100.0)$$

- **exempt.nominalConcurrencyShares** (int32)

`nominalConcurrencyShares` (NCS) contributes to the computation of the NominalConcurrencyLimit (NominalCL) of this level. This is the number of execution seats nominally reserved for this priority level. This **DOES NOT** limit the dispatching from this priority level but affects the other priority levels through the borrowing mechanism. The server's concurrency limit (ServerCL) is divided among all the priority levels in proportion to their NCS values:

$$\text{NominalCL}(i) = \text{ceil}(\text{ServerCL} * \text{NCS}(i) / \text{sum\_ncs}) \quad \text{sum\_ncs} = \text{sum}[\text{priority level } k] \text{NCS}(k)$$

Bigger numbers mean a larger nominal concurrency limit, at the expense of every other priority level. This field has a default value of zero.

- **limited** (LimitedPriorityLevelConfiguration)

`limited` specifies how requests are handled for a Limited priority level. This field must be non-empty if and only if type is "Limited".

\*LimitedPriorityLevelConfiguration specifies how to handle requests that are subject to limits. It addresses two issues:

- How are requests for this priority level limited?
  - What should be done with requests that exceed the limit?\*
  - **limited.borrowingLimitPercent** (int32)

`borrowingLimitPercent`, if present, configures a limit on how many seats this priority level can borrow from other priority levels. The limit is known as this

level's `BorrowingConcurrencyLimit` (`BorrowingCL`) and is a limit on the total number of seats that this level may borrow at any one time. This field holds the ratio of that limit to the level's nominal concurrency limit. When this field is non-nil, it must hold a non-negative integer and the limit is calculated as follows.

$$\text{BorrowingCL}(i) = \text{round}(\text{NominalCL}(i) * \text{borrowingLimitPercent}(i)/100.0)$$

The value of this field can be more than 100, implying that this priority level can borrow a number of seats that is greater than its own nominal concurrency limit (`NominalCL`). When this field is left nil, the limit is effectively infinite.

- **limited.lendablePercent** (int32)

`lendablePercent` prescribes the fraction of the level's `NominalCL` that can be borrowed by other priority levels. The value of this field must be between 0 and 100, inclusive, and it defaults to 0. The number of seats that other levels can borrow from this level, known as this level's `LendableConcurrencyLimit` (`LendableCL`), is defined as follows.

$$\text{LendableCL}(i) = \text{round}(\text{NominalCL}(i) * \text{lendablePercent}(i)/100.0)$$

- **limited.limitResponse** (`LimitResponse`)

`limitResponse` indicates what to do with requests that can not be executed right now

*LimitResponse defines how to handle requests that can not be executed right now.*

- **limited.limitResponse.type** (string), required

`type` is "Queue" or "Reject". "Queue" means that requests that can not be executed upon arrival are held in a queue until they can be executed or a queuing limit is reached. "Reject" means that requests that can not be executed upon arrival are rejected. Required.

- **limited.limitResponse.queuing** (`QueuingConfiguration`)

`queuing` holds the configuration parameters for queuing. This field may be non-empty only if `type` is "Queue".

*QueuingConfiguration holds the configuration parameters for queuing*

- **limited.limitResponse.queuing.handSize** (int32)

`handSize` is a small positive number that configures the shuffle sharding of requests into queues. When enqueueing a request at this priority level the request's flow identifier (a string pair) is hashed and the hash value is used to shuffle the list of queues and deal a hand of the size specified here. The request is put into one of the shortest queues in that hand. `handSize` must be no larger than queues, and should be significantly smaller (so that a few heavy flows do not saturate most of the queues). See the user-facing documentation for more extensive guidance on setting this field. This field has a default value of 8.



- **limited.limitResponse.queueing.queueLengthLimit** (int32)

queueLengthLimit is the maximum number of requests allowed to be waiting in a given queue of this priority level at a time; excess requests are rejected. This value must be positive. If not specified, it will be defaulted to 50.

- **limited.limitResponse.queueing.queues** (int32)

queues is the number of queues for this priority level. The queues exist independently at each apiserver. The value must be positive. Setting it to 1 effectively precludes shufflesharding and thus makes the distinguisher method of associated flow schemas irrelevant. This field has a default value of 64.

- **limited.nominalConcurrencyShares** (int32)

nominalConcurrencyShares (NCS) contributes to the computation of the NominalConcurrencyLimit (NominalCL) of this level. This is the number of execution seats available at this priority level. This is used both for requests dispatched from this priority level as well as requests dispatched from other priority levels borrowing seats from this level. The server's concurrency limit (ServerCL) is divided among the Limited priority levels in proportion to their NCS values:

$$\text{NominalCL}(i) = \text{ceil}(\text{ServerCL} * \text{NCS}(i) / \text{sum\_ncs}) \quad \text{sum\_ncs} = \sum[\text{priority level } k] \text{NCS}(k)$$

Bigger numbers mean a larger nominal concurrency limit, at the expense of every other priority level. This field has a default value of 30.

## PriorityLevelConfigurationStatus

PriorityLevelConfigurationStatus represents the current state of a "request-priority".

---

- **conditions** ([]PriorityLevelConfigurationCondition)

*Patch strategy: merge on key type*

*Map: unique values on key type will be kept during a merge*

conditions is the current state of "request-priority".

*PriorityLevelConfigurationCondition defines the condition of priority level.*

- **conditions.lastTransitionTime** (Time)

lastTransitionTime is the last time the condition transitioned from one status to another.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **conditions.message** (string)  
message is a human-readable message indicating details about last transition.
- **conditions.reason** (string)  
reason is a unique, one-word, CamelCase reason for the condition's last transition.
- **conditions.status** (string)  
status is the status of the condition. Can be True, False, Unknown. Required.
- **conditions.type** (string)  
type is the type of the condition. Required.

## PriorityLevelConfigurationList

PriorityLevelConfigurationList is a list of PriorityLevelConfiguration objects.

---

- **apiVersion**: flowcontrol.apiserver.k8s.io/v1beta3
- **kind**: PriorityLevelConfigurationList
- **metadata** ([ListMeta](#))  
metadata is the standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>
- **items** ([\[\]PriorityLevelConfiguration](#)), required  
items is a list of request-priorities.

## Operations

---

### get read the specified PriorityLevelConfiguration

#### HTTP Request

GET /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the PriorityLevelConfiguration
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PriorityLevelConfiguration](#)): OK

401: Unauthorized

## get read status of the specified PriorityLevelConfiguration

### HTTP Request

GET /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PriorityLevelConfiguration
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([PriorityLevelConfiguration](#)): OK

401: Unauthorized

## list list or watch objects of kind PriorityLevelConfiguration

### HTTP Request

GET /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)

- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)
- **watch** (*in query*): boolean  
[watch](#)

## Response

200 ([PriorityLevelConfigurationList](#)): OK

401: Unauthorized

## create create a PriorityLevelConfiguration

### HTTP Request

POST /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations

### Parameters

- **body**: [PriorityLevelConfiguration](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PriorityLevelConfiguration](#)): OK

201 ([PriorityLevelConfiguration](#)): Created

202 ([PriorityLevelConfiguration](#)): Accepted

401: Unauthorized

## update replace the specified PriorityLevelConfiguration

### HTTP Request

PUT /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PriorityLevelConfiguration
- **body**: [PriorityLevelConfiguration](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PriorityLevelConfiguration](#)): OK

201 ([PriorityLevelConfiguration](#)): Created

401: Unauthorized

## update replace status of the specified PriorityLevelConfiguration

### HTTP Request

PUT /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PriorityLevelConfiguration
- **body**: [PriorityLevelConfiguration](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([PriorityLevelConfiguration](#)): OK

201 ([PriorityLevelConfiguration](#)): Created

401: Unauthorized

## patch partially update the specified PriorityLevelConfiguration

### HTTP Request

PATCH /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PriorityLevelConfiguration
- **body**: [Patch](#), required
- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string

[fieldValidation](#)

- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PriorityLevelConfiguration](#)): OK

201 ([PriorityLevelConfiguration](#)): Created

401: Unauthorized

## patch partially update status of the specified PriorityLevelConfiguration

### HTTP Request

PATCH /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations/{name}/status

### Parameters

- **name** (*in path*): string, required  
name of the PriorityLevelConfiguration
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean

[force](#)

- **pretty** (*in query*): string

[pretty](#)

## Response

200 ([PriorityLevelConfiguration](#)): OK

201 ([PriorityLevelConfiguration](#)): Created

401: Unauthorized

## delete delete a PriorityLevelConfiguration

### HTTP Request

DELETE /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations/{name}

### Parameters

- **name** (*in path*): string, required  
name of the PriorityLevelConfiguration

- **body**: [DeleteOptions](#)

- **dryRun** (*in query*): string

[dryRun](#)

- **gracePeriodSeconds** (*in query*): integer

[gracePeriodSeconds](#)

- **pretty** (*in query*): string

[pretty](#)

- **propagationPolicy** (*in query*): string

[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized



## deletecollection delete collection of PriorityLevelConfiguration

### HTTP Request

DELETE /apis/flowcontrol.apiserver.k8s.io/v1beta3/prioritylevelconfigurations

### Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Binding

Binding ties one object to another; for example, a pod is bound to a node by a scheduler.

apiVersion: v1

```
import "k8s.io/api/core/v1"
```

## Binding

Binding ties one object to another; for example, a pod is bound to a node by a scheduler.  
Deprecated in 1.7, please use the bindings subresource of pods instead.

- 
- **apiVersion:** v1
  - **kind:** Binding
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **target** ([ObjectReference](#)), required

The target object that you want to bind to the standard object.

## Operations

---

### create create a Binding

#### HTTP Request

POST /api/v1/namespaces/{namespace}/bindings

#### Parameters

- **namespace** (*in path*): string, required

[namespace](#)

- **body:** [Binding](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Binding](#)): OK

201 ([Binding](#)): Created

202 ([Binding](#)): Accepted

401: Unauthorized

## create create binding of a Pod

### HTTP Request

POST /api/v1/namespaces/{namespace}/pods/{name}/binding

### Parameters

- **name** (*in path*): string, required  
name of the Binding
- **namespace** (*in path*): string, required  
[namespace](#)
- **body:** [Binding](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([Binding](#)): OK

201 ([Binding](#)): Created

202 ([Binding](#)): Accepted

401: Unauthorized

# ComponentStatus

ComponentStatus (and ComponentStatusList) holds the cluster validation info.

apiVersion: v1

import "k8s.io/api/core/v1"

## ComponentStatus

ComponentStatus (and ComponentStatusList) holds the cluster validation info. Deprecated: This API is deprecated in v1.19+

- 
- **apiVersion:** v1
  - **kind:** ComponentStatus
  - **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **conditions** ([]ComponentCondition)

*Patch strategy: merge on key type*

List of component conditions observed

*Information about the condition of a component.*

- **conditions.status** (string), required

Status of the condition for a component. Valid values for "Healthy": "True", "False", or "Unknown".

- **conditions.type** (string), required  
Type of condition for a component. Valid value: "Healthy"
- **conditions.error** (string)  
Condition error code for a component. For example, a health check error code.
- **conditions.message** (string)  
Message about the condition for a component. For example, information about a health check.

## ComponentStatusList

Status of all the conditions for the component as a list of ComponentStatus objects. Deprecated: This API is deprecated in v1.19+

- 
- **apiVersion:** v1
  - **kind:** ComponentStatusList
  - **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **items** ([\[\]ComponentStatus](#)), required  
List of ComponentStatus objects.

## Operations

---

### get read the specified ComponentStatus

#### HTTP Request

GET /api/v1/componentstatuses/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ComponentStatus
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ComponentStatus](#)): OK

401: Unauthorized

## list list objects of kind ComponentStatus

### HTTP Request

GET /api/v1/componentstatuses

### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

**watch** (*in query*): boolean

•

[watch](#)

## Response

200 ([ComponentStatusList](#)): OK

401: Unauthorized

# ClusterCIDR v1alpha1

ClusterCIDR represents a single configuration for per-Node Pod CIDR allocations when the MultiCIDRRangeAllocator is enabled (see the config for kube-controller-manager).

apiVersion: networking.k8s.io/v1alpha1

```
import "k8s.io/api/networking/v1alpha1"
```

## ClusterCIDR

ClusterCIDR represents a single configuration for per-Node Pod CIDR allocations when the MultiCIDRRangeAllocator is enabled (see the config for kube-controller-manager). A cluster may have any number of ClusterCIDR resources, all of which will be considered when allocating a CIDR for a Node. A ClusterCIDR is eligible to be used for a given Node when the node selector matches the node in question and has free CIDRs to allocate. In case of multiple matching ClusterCIDR resources, the allocator will attempt to break ties using internal heuristics, but any ClusterCIDR whose node selector matches the Node may be used.

---

- **apiVersion:** networking.k8s.io/v1alpha1

- **kind:** ClusterCIDR

- **metadata** ([ObjectMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

- **spec** ([ClusterCIDRSpec](#))

spec is the desired state of the ClusterCIDR. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## ClusterCIDRSpec

ClusterCIDRSpec defines the desired state of ClusterCIDR.

---

- **perNodeHostBits** (int32), required

perNodeHostBits defines the number of host bits to be configured per node. A subnet mask determines how much of the address is used for network bits and host bits. For example an IPv4 address of 192.168.0.0/24, splits the address into 24 bits for the network portion and 8 bits for the host portion. To allocate 256 IPs, set this field to 8 (a /24 mask for IPv4 or a /120 for IPv6). Minimum value is 4 (16 IPs). This field is immutable.

- **ipv4** (string)

ipv4 defines an IPv4 IP block in CIDR notation(e.g. "10.0.0.0/8"). At least one of ipv4 and ipv6 must be specified. This field is immutable.

- **ipv6** (string)

ipv6 defines an IPv6 IP block in CIDR notation(e.g. "2001:db8::/64"). At least one of ipv4 and ipv6 must be specified. This field is immutable.

- **nodeSelector** (NodeSelector)

nodeSelector defines which nodes the config is applicable to. An empty or nil nodeSelector selects all nodes. This field is immutable.

*A node selector represents the union of the results of one or more label queries over a set of nodes; that is, it represents the OR of the selectors represented by the node selector terms.*

- **nodeSelector.nodeSelectorTerms** ([]NodeSelectorTerm), required

Required. A list of node selector terms. The terms are ORed.

*A null or empty node selector term matches no objects. The requirements of them are ANDed. The TopologySelectorTerm type implements a subset of the NodeSelectorTerm.*

- **nodeSelector.nodeSelectorTerms.matchExpressions**  
([][NodeSelectorRequirement](#))

A list of node selector requirements by node's labels.

- **nodeSelector.nodeSelectorTerms.matchFields**  
([][NodeSelectorRequirement](#))

A list of node selector requirements by node's fields.

## ClusterCIDRList

ClusterCIDRList contains a list of ClusterCIDR.

- 
- **apiVersion**: networking.k8s.io/v1alpha1
  - **kind**: ClusterCIDRList
  - **metadata** ([ListMeta](#))

Standard object's metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>



- **items** ([][ClusterCIDR](#)), required  
items is the list of ClusterCIDRs.

## Operations

---

### get read the specified ClusterCIDR

#### HTTP Request

GET /apis/networking.k8s.io/v1alpha1/clustercidrs/{name}

#### Parameters

- **name** (*in path*): string, required  
name of the ClusterCIDR
- **pretty** (*in query*): string  
[pretty](#)

#### Response

200 ([ClusterCIDR](#)): OK

401: Unauthorized

### list list or watch objects of kind ClusterCIDR

#### HTTP Request

GET /apis/networking.k8s.io/v1alpha1/clustercidrs

#### Parameters

- **allowWatchBookmarks** (*in query*): boolean  
[allowWatchBookmarks](#)
- **continue** (*in query*): string  
[continue](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)
- **labelSelector** (*in query*): string

[labelSelector](#)

- **limit** (*in query*): integer

[limit](#)

- **pretty** (*in query*): string

[pretty](#)

- **resourceVersion** (*in query*): string

[resourceVersion](#)

- **resourceVersionMatch** (*in query*): string

[resourceVersionMatch](#)

- **sendInitialEvents** (*in query*): boolean

[sendInitialEvents](#)

- **timeoutSeconds** (*in query*): integer

[timeoutSeconds](#)

- **watch** (*in query*): boolean

[watch](#)

## Response

200 ([ClusterCIDRList](#)): OK

401: Unauthorized

## create create a ClusterCIDR

### HTTP Request

POST /apis/networking.k8s.io/v1alpha1/clustercidrs

### Parameters

- **body**: [ClusterCIDR](#), required

- **dryRun** (*in query*): string

[dryRun](#)

- **fieldManager** (*in query*): string

[fieldManager](#)

- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ClusterCIDR](#)): OK

201 ([ClusterCIDR](#)): Created

202 ([ClusterCIDR](#)): Accepted

401: Unauthorized

## update replace the specified ClusterCIDR

### HTTP Request

PUT /apis/networking.k8s.io/v1alpha1/clustercidrs/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ClusterCIDR
- **body**: [ClusterCIDR](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **pretty** (*in query*): string  
[pretty](#)

## Response

200 ([ClusterCIDR](#)): OK

201 ([ClusterCIDR](#)): Created

401: Unauthorized

## patch partially update the specified ClusterCIDR

### HTTP Request

PATCH /apis/networking.k8s.io/v1alpha1/clustercidrs/{name}

### Parameters

- **name** (*in path*): string, required  
name of the ClusterCIDR
- **body**: [Patch](#), required
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldManager** (*in query*): string  
[fieldManager](#)
- **fieldValidation** (*in query*): string  
[fieldValidation](#)
- **force** (*in query*): boolean  
[force](#)
- **pretty** (*in query*): string  
[pretty](#)

### Response

200 ([ClusterCIDR](#)): OK

201 ([ClusterCIDR](#)): Created

401: Unauthorized

## delete delete a ClusterCIDR

### HTTP Request

DELETE /apis/networking.k8s.io/v1alpha1/clustercidrs/{name}

## Parameters

- **name** (*in path*): string, required  
name of the ClusterCIDR
- **body**: [DeleteOptions](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)

## Response

200 ([Status](#)): OK

202 ([Status](#)): Accepted

401: Unauthorized

## deletecollection delete collection of ClusterCIDR

### HTTP Request

DELETE /apis/networking.k8s.io/v1alpha1/clustercidrs

## Parameters

- **body**: [DeleteOptions](#)
- **continue** (*in query*): string  
[continue](#)
- **dryRun** (*in query*): string  
[dryRun](#)
- **fieldSelector** (*in query*): string  
[fieldSelector](#)

- **gracePeriodSeconds** (*in query*): integer  
[gracePeriodSeconds](#)
- **labelSelector** (*in query*): string  
[labelSelector](#)
- **limit** (*in query*): integer  
[limit](#)
- **pretty** (*in query*): string  
[pretty](#)
- **propagationPolicy** (*in query*): string  
[propagationPolicy](#)
- **resourceVersion** (*in query*): string  
[resourceVersion](#)
- **resourceVersionMatch** (*in query*): string  
[resourceVersionMatch](#)
- **sendInitialEvents** (*in query*): boolean  
[sendInitialEvents](#)
- **timeoutSeconds** (*in query*): integer  
[timeoutSeconds](#)

## Response

200 ([Status](#)): OK

401: Unauthorized

# Common Definitions

---

## [DeleteOptions](#)

DeleteOptions may be provided when deleting an API object.

## [LabelSelector](#)

A label selector is a label query over a set of resources.

### [ListMeta](#)

ListMeta describes metadata that synthetic resources must have, including lists and various status objects.

### [LocalObjectReference](#)

LocalObjectReference contains enough information to let you locate the referenced object inside the same namespace.

### [NodeSelectorRequirement](#)

A node selector requirement is a selector that contains values, a key, and an operator that relates the key and values.

### [ObjectFieldSelector](#)

ObjectFieldSelector selects an APIVersioned field of an object.

### [ObjectMeta](#)

ObjectMeta is metadata that all persisted resources must have, which includes all objects users must create.

### [ObjectReference](#)

ObjectReference contains enough information to let you inspect or modify the referred object.

### [Patch](#)

Patch is provided to give a concrete name and type to the Kubernetes PATCH request body.

### [Quantity](#)

Quantity is a fixed-point representation of a number.

### [ResourceFieldSelector](#)

ResourceFieldSelector represents container resources (cpu, memory) and their output format.

### [Status](#)

Status is a return value for calls that don't return other objects.

### [TypedLocalObjectReference](#)

TypedLocalObjectReference contains enough information to let you locate the typed referenced object inside the same namespace.

# DeleteOptions

DeleteOptions may be provided when deleting an API object.

```
import "k8s.io/apimachinery/pkg/apis/meta/v1"
```

DeleteOptions may be provided when deleting an API object.

---

- **apiVersion** (string)

APIVersion defines the versioned schema of this representation of an object. Servers should convert recognized schemas to the latest internal value, and may reject unrecognized values. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources>

- **dryRun** ([]string)

When present, indicates that modifications should not be persisted. An invalid or unrecognized dryRun directive will result in an error response and no further processing of the request. Valid values are: - All: all dry run stages will be processed

- **gracePeriodSeconds** (int64)

The duration in seconds before the object should be deleted. Value must be non-negative integer. The value zero indicates delete immediately. If this value is nil, the default grace period for the specified type will be used. Defaults to a per object value if not specified. zero means delete immediately.

- **kind** (string)

Kind is a string value representing the REST resource this object represents. Servers may infer this from the endpoint the client submits requests to. Cannot be updated. In CamelCase. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **orphanDependents** (boolean)

Deprecated: please use the PropagationPolicy, this field will be deprecated in 1.7. Should the dependent objects be orphaned. If true/false, the "orphan" finalizer will be added to/ removed from the object's finalizers list. Either this field or PropagationPolicy may be set, but not both.

- **preconditions** (Preconditions)

Must be fulfilled before a deletion is carried out. If not possible, a 409 Conflict status will be returned.

*Preconditions must be fulfilled before an operation (update, delete, etc.) is carried out.*

- **preconditions.resourceVersion** (string)

Specifies the target ResourceVersion



**preconditions.uid** (string)

- Specifies the target UID.

- **propagationPolicy** (string)

Whether and how garbage collection will be performed. Either this field or OrphanDependents may be set, but not both. The default policy is decided by the existing finalizer set in the metadata.finalizers and the resource-specific default policy. Acceptable values are: 'Orphan' - orphan the dependents; 'Background' - allow the garbage collector to delete the dependents in the background; 'Foreground' - a cascading policy that deletes all dependents in the foreground.

## LabelSelector

A label selector is a label query over a set of resources.

```
import "k8s.io/apimachinery/pkg/apis/meta/v1"
```

A label selector is a label query over a set of resources. The result of matchLabels and matchExpressions are ANDed. An empty label selector matches all objects. A null label selector matches no objects.

---

- **matchExpressions** ([]LabelSelectorRequirement)

matchExpressions is a list of label selector requirements. The requirements are ANDed.

*A label selector requirement is a selector that contains values, a key, and an operator that relates the key and values.*

- **matchExpressions.key** (string), required

key is the label key that the selector applies to.

- **matchExpressions.operator** (string), required

operator represents a key's relationship to a set of values. Valid operators are In, NotIn, Exists and DoesNotExist.

- **matchExpressions.values** ([]string)

values is an array of string values. If the operator is In or NotIn, the values array must be non-empty. If the operator is Exists or DoesNotExist, the values array must be empty. This array is replaced during a strategic merge patch.

- **matchLabels** (map[string]string)

matchLabels is a map of {key,value} pairs. A single {key,value} in the matchLabels map is equivalent to an element of matchExpressions, whose key field is "key", the operator is "In", and the values array contains only "value". The requirements are ANDed.

# ListMeta

ListMeta describes metadata that synthetic resources must have, including lists and various status objects.

```
import "k8s.io/apimachinery/pkg/apis/meta/v1"
```

ListMeta describes metadata that synthetic resources must have, including lists and various status objects. A resource may have only one of {ObjectMeta, ListMeta}.

---

- **continue** (string)

continue may be set if the user set a limit on the number of items returned, and indicates that the server has more data available. The value is opaque and may be used to issue another request to the endpoint that served this list to retrieve the next set of available objects. Continuing a consistent list may not be possible if the server configuration has changed or more than a few minutes have passed. The resourceVersion field returned when using this continue value will be identical to the value in the first response, unless you have received this token from an error message.

- **remainingItemCount** (int64)

remainingItemCount is the number of subsequent items in the list which are not included in this list response. If the list request contained label or field selectors, then the number of remaining items is unknown and the field will be left unset and omitted during serialization. If the list is complete (either because it is not chunking or because this is the last chunk), then there are no more remaining items and this field will be left unset and omitted during serialization. Servers older than v1.15 do not set this field. The intended use of the remainingItemCount is *estimating* the size of a collection. Clients should not rely on the remainingItemCount to be set or to be exact.

- **resourceVersion** (string)

String that identifies the server's internal version of this object that can be used by clients to determine when objects have changed. Value must be treated as opaque by clients and passed unmodified back to the server. Populated by the system. Read-only. More info:

<https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#concurrency-control-and-consistency>

- **selfLink** (string)

Deprecated: selfLink is a legacy read-only field that is no longer populated by the system.

## LocalObjectReference

LocalObjectReference contains enough information to let you locate the referenced object inside the same namespace.

```
import "k8s.io/api/core/v1"
```

LocalObjectReference contains enough information to let you locate the referenced object inside the same namespace.

---

- **name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

## NodeSelectorRequirement

A node selector requirement is a selector that contains values, a key, and an operator that relates the key and values.

```
import "k8s.io/api/core/v1"
```

A node selector requirement is a selector that contains values, a key, and an operator that relates the key and values.

---

- **key** (string), required

The label key that the selector applies to.

- **operator** (string), required

Represents a key's relationship to a set of values. Valid operators are In, NotIn, Exists, DoesNotExist. Gt, and Lt.

- **values** ([]string)

An array of string values. If the operator is In or NotIn, the values array must be non-empty. If the operator is Exists or DoesNotExist, the values array must be empty. If the operator is Gt or Lt, the values array must have a single element, which will be interpreted as an integer. This array is replaced during a strategic merge patch.

## ObjectFieldSelector

ObjectFieldSelector selects an APIVersioned field of an object.

```
import "k8s.io/api/core/v1"
```

ObjectFieldSelector selects an APIVersioned field of an object.

---

- **fieldPath** (string), required

Path of the field to select in the specified API version.

- **apiVersion** (string)

Version of the schema the FieldPath is written in terms of, defaults to "v1".

# ObjectMeta

ObjectMeta is metadata that all persisted resources must have, which includes all objects users must create.

```
import "k8s.io/apimachinery/pkg/apis/meta/v1"
```

ObjectMeta is metadata that all persisted resources must have, which includes all objects users must create.

---

- **name** (string)

Name must be unique within a namespace. Is required when creating resources, although some resources may allow a client to request the generation of an appropriate name automatically. Name is primarily intended for creation idempotence and configuration definition. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names#names>

- **generateName** (string)

GenerateName is an optional prefix, used by the server, to generate a unique name ONLY IF the Name field has not been provided. If this field is used, the name returned to the client will be different than the name passed. This value will also be combined with a unique suffix. The provided value has the same validation rules as the Name field, and may be truncated by the length of the suffix required to make the value unique on the server.

If this field is specified and the generated name exists, the server will return a 409.

Applied only if Name is not specified. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#idempotency>

- **namespace** (string)

Namespace defines the space within which each name must be unique. An empty namespace is equivalent to the "default" namespace, but "default" is the canonical representation. Not all objects are required to be scoped to a namespace - the value of this field for those objects will be empty.

Must be a DNS\_LABEL. Cannot be updated. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/namespaces>

- **labels** (map[string]string)

Map of string keys and values that can be used to organize and categorize (scope and select) objects. May match selectors of replication controllers and services. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels>

- **annotations** (map[string]string)

Annotations is an unstructured key value map stored with a resource that may be set by external tools to store and retrieve arbitrary metadata. They are not queryable and should

be preserved when modifying objects. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/annotations>

## System

- **finalizers** ([]string)

Must be empty before the object is deleted from the registry. Each entry is an identifier for the responsible component that will remove the entry from the list. If the `deletionTimestamp` of the object is non-nil, entries in this list can only be removed. Finalizers may be processed and removed in any order. Order is NOT enforced because it introduces significant risk of stuck finalizers. `finalizers` is a shared field, any actor with permission can reorder it. If the finalizer list is processed in order, then this can lead to a situation in which the component responsible for the first finalizer in the list is waiting for a signal (field value, external system, or other) produced by a component responsible for a finalizer later in the list, resulting in a deadlock. Without enforced ordering finalizers are free to order amongst themselves and are not vulnerable to ordering changes in the list.

- **managedFields** ([]ManagedFieldsEntry)

`ManagedFields` maps workflow-id and version to the set of fields that are managed by that workflow. This is mostly for internal housekeeping, and users typically shouldn't need to set or understand this field. A workflow can be the user's name, a controller's name, or the name of a specific apply path like "ci-cd". The set of fields is always in the version that the workflow used when modifying the object.

*ManagedFieldsEntry is a workflow-id, a FieldSet and the group version of the resource that the fieldset applies to.*

- **managedFields.apiVersion** (string)

`APIVersion` defines the version of this resource that this field set applies to. The format is "group/version" just like the top-level `APIVersion` field. It is necessary to track the version of a field set because it cannot be automatically converted.

- **managedFields.fieldsType** (string)

`FieldsType` is the discriminator for the different fields format and version. There is currently only one possible value: "FieldsV1"

- **managedFields.fieldsV1** (FieldsV1)

`FieldsV1` holds the first JSON version format as described in the "FieldsV1" type.

\*`FieldsV1` stores a set of fields in a data structure like a Trie, in JSON format.

Each key is either a '.' representing the field itself, and will always map to an empty set, or a string representing a sub-field or item. The string will follow one of these four formats: 'f:', where is the name of a field in a struct, or key in a map 'v:', where is the exact json formatted value of a list item 'i:', where is position of a item in a list 'k:', where is a map of a list item's key fields to their unique values If a key maps to an empty Fields value, the field that key represents is part of the set.

The exact format is defined in [sigs.k8s.io/structured-merge-diff](https://sigs.k8s.io/structured-merge-diff)\*

- **managedFields.manager** (string)

Manager is an identifier of the workflow managing these fields.

- **managedFields.operation** (string)

Operation is the type of operation which lead to this ManagedFieldsEntry being created. The only valid values for this field are 'Apply' and 'Update'.

- **managedFields.subresource** (string)

Subresource is the name of the subresource used to update that object, or empty string if the object was updated through the main resource. The value of this field is used to distinguish between managers, even if they share the same name. For example, a status update will be distinct from a regular update using the same manager name. Note that the APIVersion field is not related to the Subresource field and it always corresponds to the version of the main resource.

- **managedFields.time** (Time)

Time is the timestamp of when the ManagedFields entry was added. The timestamp will also be updated if a field is added, the manager changes any of the owned fields value or removes a field. The timestamp does not update when a field is removed from the entry because another manager took it over.

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **ownerReferences** ([]OwnerReference)

*Patch strategy: merge on key uid*

List of objects depended by this object. If ALL objects in the list have been deleted, this object will be garbage collected. If this object is managed by a controller, then an entry in this list will point to this controller, with the controller field set to true. There cannot be more than one managing controller.

*OwnerReference contains enough information to let you identify an owning object. An owning object must be in the same namespace as the dependent, or be cluster-scoped, so there is no namespace field.*

- **ownerReferences.apiVersion** (string), required

API version of the referent.

- **ownerReferences.kind** (string), required

Kind of the referent. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **ownerReferences.name** (string), required

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names#names>

- **ownerReferences.uid** (string), required

UID of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names#uids>

- **ownerReferences.blockOwnerDeletion** (boolean)

If true, AND if the owner has the "foregroundDeletion" finalizer, then the owner cannot be deleted from the key-value store until this reference is removed. See <https://kubernetes.io/docs/concepts/architecture/garbage-collection/#foreground-deletion> for how the garbage collector interacts with this field and enforces the foreground deletion. Defaults to false. To set this field, a user needs "delete" permission of the owner, otherwise 422 (Unprocessable Entity) will be returned.

- **ownerReferences.controller** (boolean)

If true, this reference points to the managing controller.

## Read-only

- **creationTimestamp** (Time)

CreationTimestamp is a timestamp representing the server time when this object was created. It is not guaranteed to be set in happens-before order across separate operations. Clients may not set this value. It is represented in RFC3339 form and is in UTC.

Populated by the system. Read-only. Null for lists. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

*Time is a wrapper around time.Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **deletionGracePeriodSeconds** (int64)

Number of seconds allowed for this object to gracefully terminate before it will be removed from the system. Only set when deletionTimestamp is also set. May only be shortened. Read-only.

- **deletionTimestamp** (Time)

DeletionTimestamp is RFC 3339 date and time at which this resource will be deleted. This field is set by the server when a graceful deletion is requested by the user, and is not directly settable by a client. The resource is expected to be deleted (no longer visible from resource lists, and not reachable by name) after the time in this field, once the finalizers list is empty. As long as the finalizers list contains items, deletion is blocked. Once the deletionTimestamp is set, this value may not be unset or be set further into the future, although it may be shortened or the resource may be deleted prior to this time. For example, a user may request that a pod is deleted in 30 seconds. The Kubelet will react by sending a graceful termination signal to the containers in the pod. After that 30 seconds, the Kubelet will send a hard termination signal (SIGKILL) to the container and after cleanup, remove the pod from the API. In the presence of network partitions, this object

may still exist after this timestamp, until an administrator or automated process can determine the resource is fully terminated. If not set, graceful deletion of the object has not been requested.

Populated by the system when a graceful deletion is requested. Read-only. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata>

*Time is a wrapper around time. Time which supports correct marshaling to YAML and JSON. Wrappers are provided for many of the factory methods that the time package offers.*

- **generation** (int64)

A sequence number representing a specific generation of the desired state. Populated by the system. Read-only.

- **resourceVersion** (string)

An opaque value that represents the internal version of this object that can be used by clients to determine when objects have changed. May be used for optimistic concurrency, change detection, and the watch operation on a resource or set of resources. Clients must treat these values as opaque and passed unmodified back to the server. They may only be valid for a particular resource or set of resources.

Populated by the system. Read-only. Value must be treated as opaque by clients and . More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#concurrency-control-and-consistency>

- **selfLink** (string)

Deprecated: selfLink is a legacy read-only field that is no longer populated by the system.

- **uid** (string)

UID is the unique in time and space value for this object. It is typically generated by the server on successful creation of a resource and is not allowed to change on PUT operations.

Populated by the system. Read-only. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names#uids>

## ObjectReference

ObjectReference contains enough information to let you inspect or modify the referred object.

```
import "k8s.io/api/core/v1"
```

ObjectReference contains enough information to let you inspect or modify the referred object.

---

- **apiVersion** (string)

API version of the referent.



**fieldPath** (string)

- 

If referring to a piece of an object instead of an entire object, this string should contain a valid JSON/Go field access statement, such as `desiredState.manifest.containers[2]`. For example, if the object reference is to a container within a pod, this would take on a value like: `"spec.containers{name}"` (where "name" refers to the name of the container that triggered the event) or if no container name is specified `"spec.containers[2]"` (container with index 2 in this pod). This syntax is chosen only to have some well-defined way of referencing a part of an object.

- **kind** (string)

Kind of the referent. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **name** (string)

Name of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#names>

- **namespace** (string)

Namespace of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/namespaces/>

- **resourceVersion** (string)

Specific resourceVersion to which this reference is made, if any. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#concurrency-control-and-consistency>

- **uid** (string)

UID of the referent. More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names/#uids>

## Patch

Patch is provided to give a concrete name and type to the Kubernetes PATCH request body.

```
import "k8s.io/apimachinery/pkg/apis/meta/v1"
```

Patch is provided to give a concrete name and type to the Kubernetes PATCH request body.

---

## Quantity

Quantity is a fixed-point representation of a number.

```
import "k8s.io/apimachinery/pkg/api/resource"
```

Quantity is a fixed-point representation of a number. It provides convenient marshaling/unmarshaling in JSON and YAML, in addition to String() and AsInt64() accessors.

The serialization format is:

(Note that \<suffix> may be empty, from the "" case in \<decimalSI>.)

```
\<digit> ::= 0 | 1 | ... | 9 \<digits> ::= \<digit> | \<digit>\<digits> \<number> ::=
\<digits> | \<digits>.\<digits> | \<digits>. | \<digits> \<sign> ::= "+" | "-"
\<signedNumber> ::= \<number> | \<sign>\<number> \<suffix> ::= \<binarySI> |
\<decimalExponent> | \<decimalSI> \<binarySI> ::= Ki | Mi | Gi | Ti | Pi | Ei
```

(International System of units; See: <http://physics.nist.gov/cuu/Units/binary.html>)

```
\<decimalSI> ::= m | "" | k | M | G | T | P | E
```

(Note that 1024 = 1Ki but 1000 = 1k; I didn't choose the capitalization.)

```
\<decimalExponent> ::= "e" \<signedNumber> | "E" \<signedNumber> ``
```

No matter which of the three exponent forms is used, no quantity may represent a number greater than  $2^{63}-1$  in magnitude, nor may it have more than 3 decimal places. Numbers larger or more precise will be capped or rounded up. (E.g.: 0.1m will rounded up to 1m.) This may be extended in the future if we require larger or smaller quantities.

When a Quantity is parsed from a string, it will remember the type of suffix it had, and will use the same type again when it is serialized.

Before serializing, Quantity will be put in "canonical form". This means that Exponent/suffix will be adjusted up or down (with a corresponding increase or decrease in Mantissa) such that:

- No precision is lost
- No fractional digits will be emitted
- The exponent (or suffix) is as large as possible.

The sign will be omitted unless the number is negative.

Examples:

- 1.5 will be serialized as "1500m" - 1.5Gi will be serialized as "1536Mi"

Note that the quantity will NEVER be internally represented by a floating point number. That is the whole point of this exercise.

Non-canonical values will still parse as long as they are well formed, but will be re-emitted in their canonical form. (So always use canonical form, or don't diff.)

This format is intended to make it difficult to use these numbers without writing some sort of special handling code in the hopes that that will cause implementors to also use a fixed point implementation.

<hr>

# ResourceFieldSelector

ResourceFieldSelector represents container resources (cpu, memory) and their output format.

```
import "k8s.io/api/core/v1"
```

ResourceFieldSelector represents container resources (cpu, memory) and their output format

---

- **resource** (string), required

Required: resource to select

- **containerName** (string)

Container name: required for volumes, optional for env vars

- **divisor** ([Quantity](#))

Specifies the output format of the exposed resources, defaults to "1"

## Status

Status is a return value for calls that don't return other objects.

```
import "k8s.io/apimachinery/pkg/apis/meta/v1"
```

Status is a return value for calls that don't return other objects.

---

- **apiVersion** (string)

APIVersion defines the versioned schema of this representation of an object. Servers should convert recognized schemas to the latest internal value, and may reject unrecognized values. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources>

- **code** (int32)

Suggested HTTP return code for this status, 0 if not set.

- **details** (StatusDetails)

Extended data associated with the reason. Each reason may define its own extended details. This field is optional and the data returned is not guaranteed to conform to any schema except that defined by the reason type.

*StatusDetails is a set of additional properties that MAY be set by the server to provide additional information about a response. The Reason field of a Status object defines what attributes will be set. Clients must ignore fields that do not match the defined type of each attribute, and should assume that any attribute may be empty, invalid, or under defined.*

- **details.causes** ([]StatusCause)

The Causes array includes more details associated with the StatusReason failure. Not all StatusReasons may provide detailed causes.

*StatusCause provides more information about an api.Status failure, including cases when multiple errors are encountered.*

- **details.causes.field** (string)

The field of the resource that has caused this error, as named by its JSON serialization. May include dot and postfix notation for nested attributes. Arrays are zero-indexed. Fields may appear more than once in an array of causes due to fields having multiple errors. Optional.

Examples: "name" - the field "name" on the current resource "items[0].name" - the field "name" on the first array entry in "items"

- **details.causes.message** (string)

A human-readable description of the cause of the error. This field may be presented as-is to a reader.

- **details.causes.reason** (string)

A machine-readable description of the cause of the error. If this value is empty there is no information available.

- **details.group** (string)

The group attribute of the resource associated with the status StatusReason.

- **details.kind** (string)

The kind attribute of the resource associated with the status StatusReason. On some operations may differ from the requested resource Kind. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **details.name** (string)

The name attribute of the resource associated with the status StatusReason (when there is a single name which can be described).

- **details.retryAfterSeconds** (int32)

If specified, the time in seconds before the operation should be retried. Some errors may indicate the client must take an alternate action - for those errors this field may indicate how long to wait before taking the alternate action.

- **details.uid** (string)

UID of the resource. (when there is a single resource which can be described). More info: <https://kubernetes.io/docs/concepts/overview/working-with-objects/names#uids>

- **kind** (string)

Kind is a string value representing the REST resource this object represents. Servers may infer this from the endpoint the client submits requests to. Cannot be updated. In CamelCase. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **message** (string)

A human-readable description of the status of this operation.

- **metadata** ([ListMeta](#))

Standard list metadata. More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds>

- **reason** (string)

A machine-readable description of why this operation is in the "Failure" status. If this value is empty there is no information available. A Reason clarifies an HTTP status code but does not override it.

- **status** (string)

Status of the operation. One of: "Success" or "Failure". More info: <https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status>

## TypedLocalObjectReference

TypedLocalObjectReference contains enough information to let you locate the typed referenced object inside the same namespace.

```
import "k8s.io/api/core/v1"
```

TypedLocalObjectReference contains enough information to let you locate the typed referenced object inside the same namespace.

- 
- **kind** (string), required

Kind is the type of resource being referenced

- **name** (string), required

Name is the name of resource being referenced

- **apiGroup** (string)

APIGroup is the group for the resource being referenced. If APIGroup is not specified, the specified Kind must be in the core API group. For any other third-party types, APIGroup is required.

## Other Resources

---

# ValidatingAdmissionPolicyBindingList v1beta1

apiVersion: admissionregistration.k8s.io/v1beta1

import "k8s.io/api/admissionregistration/v1beta1"

## Common Parameters

### allowWatchBookmarks

allowWatchBookmarks requests watch events with type "BOOKMARK". Servers that do not implement bookmarks may ignore this flag and bookmarks are sent at the server's discretion. Clients should not assume bookmarks are returned at any specific interval, nor may they assume the server will send any BOOKMARK event during a session. If this is not a watch, this field is ignored.

---

### continue

The continue option should be set when retrieving more results from the server. Since this value is server defined, clients may only use the continue value from a previous query result with identical query parameters (except for the value of continue) and the server may reject a continue value it does not recognize. If the specified continue value is no longer valid whether due to expiration (generally five to fifteen minutes) or a configuration change on the server, the server will respond with a 410 ResourceExpired error together with a continue token. If the client needs a consistent list, it must restart their list without the continue field. Otherwise, the client may send another list request with the token received with the 410 error, the server will respond with a list starting from the next key, but from the latest snapshot, which is inconsistent from the previous list results - objects that are created, modified, or deleted after the first list request will be included in the response, as long as their keys are after the "next key".

This field is not supported when watch is true. Clients may start a watch from the last resourceVersion value returned by the server and not miss any modifications.

---

### dryRun

When present, indicates that modifications should not be persisted. An invalid or unrecognized dryRun directive will result in an error response and no further processing of the request. Valid values are: - All: all dry run stages will be processed

---

## fieldManager

fieldManager is a name associated with the actor or entity that is making these changes. The value must be less than or 128 characters long, and only contain printable characters, as defined by <https://pkg.go.dev/unicode#IsPrint>.

---

## fieldSelector

A selector to restrict the list of returned objects by their fields. Defaults to everything.

---

## fieldValidation

fieldValidation instructs the server on how to handle objects in the request (POST/PUT/PATCH) containing unknown or duplicate fields. Valid values are: - Ignore: This will ignore any unknown fields that are silently dropped from the object, and will ignore all but the last duplicate field that the decoder encounters. This is the default behavior prior to v1.23. - Warn: This will send a warning via the standard warning response header for each unknown field that is dropped from the object, and for each duplicate field that is encountered. The request will still succeed if there are no other errors, and will only persist the last of any duplicate fields. This is the default in v1.23+ - Strict: This will fail the request with a BadRequest error if any unknown fields would be dropped from the object, or if any duplicate fields are present. The error returned from the server will contain all unknown and duplicate fields encountered.

---

## force

Force is going to "force" Apply requests. It means user will re-acquire conflicting fields owned by other people. Force flag must be unset for non-apply patch requests.

---

## gracePeriodSeconds

The duration in seconds before the object should be deleted. Value must be non-negative integer. The value zero indicates delete immediately. If this value is nil, the default grace period for the specified type will be used. Defaults to a per object value if not specified. zero means delete immediately.

---

## labelSelector

A selector to restrict the list of returned objects by their labels. Defaults to everything.

---

## limit

limit is a maximum number of responses to return for a list call. If more items exist, the server will set the continue field on the list metadata to a value that can be used with the same initial query to retrieve the next set of results. Setting a limit may return fewer than the requested amount of items (up to zero items) in the event all requested objects are filtered out and clients should only use the presence of the continue field to determine whether more results are available. Servers may choose not to support the limit argument and will return all of the available results. If limit is specified and the continue field is empty, clients may assume that no more results are available. This field is not supported if watch is true.

The server guarantees that the objects returned when using continue will be identical to issuing a single list call without a limit - that is, no objects created, modified, or deleted after the first request is issued will be included in any subsequent continued requests. This is sometimes referred to as a consistent snapshot, and ensures that a client that is using limit to receive smaller chunks of a very large result can ensure they see all possible objects. If objects are updated during a chunked list the version of the object that was present at the time the first list result was calculated is returned.

---

## namespace

object name and auth scope, such as for teams and projects

---

## pretty

If 'true', then the output is pretty printed.

---

## propagationPolicy

Whether and how garbage collection will be performed. Either this field or OrphanDependents may be set, but not both. The default policy is decided by the existing finalizer set in the metadata.finalizers and the resource-specific default policy. Acceptable values are: 'Orphan' - orphan the dependents; 'Background' - allow the garbage collector to delete the dependents in the background; 'Foreground' - a cascading policy that deletes all dependents in the foreground.

---

## resourceVersion

resourceVersion sets a constraint on what resource versions a request may be served from. See <https://kubernetes.io/docs/reference/using-api/api-concepts/#resource-versions> for details.

Defaults to unset

---



## resourceVersionMatch

resourceVersionMatch determines how resourceVersion is applied to list calls. It is highly recommended that resourceVersionMatch be set for list calls where resourceVersion is set See <https://kubernetes.io/docs/reference/using-api/api-concepts/#resource-versions> for details.

Defaults to unset

---

## sendInitialEvents

sendInitialEvents=true may be set together with watch=true. In that case, the watch stream will begin with synthetic events to produce the current state of objects in the collection. Once all such events have been sent, a synthetic "Bookmark" event will be sent. The bookmark will report the ResourceVersion (RV) corresponding to the set of objects, and be marked with "k8s.io/initial-events-end": "true" annotation. Afterwards, the watch stream will proceed as usual, sending watch events corresponding to changes (subsequent to the RV) to objects watched.

When sendInitialEvents option is set, we require resourceVersionMatch option to also be set. The semantic of the watch request is as following: - resourceVersionMatch = NotOlderThan is interpreted as "data at least as new as the provided resourceVersion" and the bookmark event is send when the state is synced to a resourceVersion at least as fresh as the one provided by the ListOptions. If resourceVersion is unset, this is interpreted as "consistent read" and the bookmark event is send when the state is synced at least to the moment when request started being processed.

- resourceVersionMatch set to any other value or unset Invalid error is returned.

Defaults to true if resourceVersion="" or resourceVersion="0" (for backward compatibility reasons) and to false otherwise.

---

## timeoutSeconds

Timeout for the list/watch call. This limits the duration of the call, regardless of any activity or inactivity.

---

## watch

Watch for changes to the described resources and return them as a stream of add, update, and remove notifications. Specify resourceVersion.

---

## Instrumentation

---

## [CRI Pod & Container Metrics](#)

Collection of Pod & Container metrics via the CRI.

## [Node metrics data](#)

Mechanisms for accessing metrics at node, volume, pod and container level, as seen by the kubelet.

## [Kubernetes Metrics Reference](#)

Details of the metric data that Kubernetes components export.

# Kubernetes Component SLI Metrics

FEATURE STATE: Kubernetes v1.29 [stable]

By default, Kubernetes 1.30 publishes Service Level Indicator (SLI) metrics for each Kubernetes component binary. This metric endpoint is exposed on the serving HTTPS port of each component, at the path `/metrics/slis`. The ComponentSLIs [feature gate](#) defaults to enabled for each Kubernetes component as of v1.27.

## SLI Metrics

With SLI metrics enabled, each Kubernetes component exposes two metrics, labeled per healthcheck:

- a gauge (which represents the current state of the healthcheck)
- a counter (which records the cumulative counts observed for each healthcheck state)

You can use the metric information to calculate per-component availability statistics. For example, the API server checks the health of etcd. You can work out and report how available or unavailable etcd has been - as reported by its client, the API server.

The prometheus gauge data looks like this:

```
HELP kubernetes_healthcheck [ALPHA] This metric records the result of a single healthcheck.
TYPE kubernetes_healthcheck gauge
kubernetes_healthcheck{name="autoregister-completion",type="healthz"} 1
kubernetes_healthcheck{name="autoregister-completion",type="readyz"} 1
kubernetes_healthcheck{name="etcd",type="healthz"} 1
kubernetes_healthcheck{name="etcd",type="readyz"} 1
kubernetes_healthcheck{name="etcd-readiness",type="readyz"} 1
kubernetes_healthcheck{name="informer-sync",type="readyz"} 1
kubernetes_healthcheck{name="log",type="healthz"} 1
kubernetes_healthcheck{name="log",type="readyz"} 1
kubernetes_healthcheck{name="ping",type="healthz"} 1
kubernetes_healthcheck{name="ping",type="readyz"} 1
```

While the counter data looks like this:

```
HELP kubernetes_healthchecks_total [ALPHA] This metric records the results of all
healthcheck.
TYPE kubernetes_healthchecks_total counter
kubernetes_healthchecks_total{name="autoregister-completion",status="error",type="readyz"} 1
kubernetes_healthchecks_total{name="autoregister-completion",status="success",type="healthz"} 15
kubernetes_healthchecks_total{name="autoregister-completion",status="success",type="readyz"}
14
kubernetes_healthchecks_total{name="etcd",status="success",type="healthz"} 15
kubernetes_healthchecks_total{name="etcd",status="success",type="readyz"} 15
kubernetes_healthchecks_total{name="etcd-readiness",status="success",type="readyz"} 15
kubernetes_healthchecks_total{name="informer-sync",status="error",type="readyz"} 1
kubernetes_healthchecks_total{name="informer-sync",status="success",type="readyz"} 14
kubernetes_healthchecks_total{name="log",status="success",type="healthz"} 15
kubernetes_healthchecks_total{name="log",status="success",type="readyz"} 15
kubernetes_healthchecks_total{name="ping",status="success",type="healthz"} 15
kubernetes_healthchecks_total{name="ping",status="success",type="readyz"} 15
```

## Using this data

The component SLIs metrics endpoint is intended to be scraped at a high frequency. Scraping at a high frequency means that you end up with greater granularity of the gauge's signal, which can be then used to calculate SLOs. The `/metrics/slis` endpoint provides the raw data necessary to calculate an availability SLO for the respective Kubernetes component.

# CRI Pod & Container Metrics

Collection of Pod & Container metrics via the CRI.  
FEATURE STATE: Kubernetes v1.23 [alpha]

The [kubelet](#) collects pod and container metrics via [cAdvisor](#). As an alpha feature, Kubernetes lets you configure the collection of pod and container metrics via the [Container Runtime Interface](#) (CRI). You must enable the `PodAndContainerStatsFromCRI` [feature gate](#) and use a compatible CRI implementation (containerd  $\geq$  1.6.0, CRI-O  $\geq$  1.23.0) to use the CRI based collection mechanism.

## CRI Pod & Container Metrics

With `PodAndContainerStatsFromCRI` enabled, the kubelet polls the underlying container runtime for pod and container stats instead of inspecting the host system directly using cAdvisor. The benefits of relying on the container runtime for this information as opposed to direct collection with cAdvisor include:

- Potential improved performance if the container runtime already collects this information during normal operations. In this case, the data can be re-used instead of being aggregated again by the kubelet.
- It further decouples the kubelet and the container runtime allowing collection of metrics for container runtimes that don't run processes directly on the host with kubelet where they are observable by cAdvisor (for example: container runtimes that use virtualization).

# Node metrics data

Mechanisms for accessing metrics at node, volume, pod and container level, as seen by the kubelet.

The [kubelet](#) gathers metric statistics at the node, volume, pod and container level, and emits this information in the [Summary API](#).

You can send a proxied request to the stats summary API via the Kubernetes API server.

Here is an example of a Summary API request for a node named minikube:

```
kubectl get --raw "/api/v1/nodes/minikube/proxy/stats/summary"
```

Here is the same API call using curl:

```
You need to run "kubectl proxy" first
Change 8080 to the port that "kubectl proxy" assigns
curl http://localhost:8080/api/v1/nodes/minikube/proxy/stats/summary
```

## Note:

Beginning with metrics-server 0.6.x, metrics-server queries the /metrics/resource kubelet endpoint, and not /stats/summary.

## Summary metrics API source

By default, Kubernetes fetches node summary metrics data using an embedded [cAdvisor](#) that runs within the kubelet. If you enable the PodAndContainerStatsFromCRI [feature gate](#) in your cluster, and you use a container runtime that supports statistics access via [Container Runtime Interface](#) (CRI), then the kubelet [fetches Pod- and container-level metric data using CRI](#), and not via cAdvisor.

## What's next

The task pages for [Troubleshooting Clusters](#) discuss how to use a metrics pipeline that rely on these data.

# Kubernetes Metrics Reference

Details of the metric data that Kubernetes components export.

## Metrics (v1.30)

This page details the metrics that different Kubernetes components export. You can query the metrics endpoint for these components using an HTTP scrape, and fetch the current metrics data in Prometheus format.

## List of Stable Kubernetes Metrics

Stable metrics observe strict API contracts and no labels can be added or removed from stable metrics during their lifetime.

`apiserver_admission_controller_admission_duration_seconds`

Admission controller latency histogram in seconds, identified by name and broken out for each operation and API resource and type (validate or admit).

- Stability Level:STABLE
- Type: Histogram
- Labels:nameoperationrejectedtype

`apiserver_admission_step_admission_duration_seconds`

Admission sub-step latency histogram in seconds, broken out for each operation and API resource and step type (validate or admit).

- Stability Level:STABLE
- Type: Histogram
- Labels:operationrejectedtype

`apiserver_admission_webhook_admission_duration_seconds`

Admission webhook latency histogram in seconds, identified by name and broken out for each operation and API resource and type (validate or admit).

- Stability Level:STABLE
- Type: Histogram
- Labels:nameoperationrejectedtype

`apiserver_current_inflight_requests`

Maximal number of currently used inflight request limit of this apiserver per request kind in last second.

- Stability Level:STABLE
- Type: Gauge
- Labels:request\_kind

`apiserver_longrunning_requests`

Gauge of all active long-running apiserver requests broken out by verb, group, version, resource, scope and component. Not all requests are tracked this way.

- Stability Level:STABLE
- Type: Gauge
- Labels:componentgroupresourcescopesubresourceverbversion

`apiserver_requested_deprecated_apis`

Gauge of deprecated APIs that have been requested, broken out by API group, version, resource, subresource, and removed\_release.

- Stability Level:STABLE
- Type: Gauge
- Labels:groupremoved\_releaseresourcesubresourceversion

`apiserver_response_sizes`

Response size distribution in bytes for each group, version, verb, resource, subresource, scope and component.

- Stability Level:STABLE
- Type: Histogram
- Labels:componentgroupresourcescopesubresourceverbversion

apiserver\_storage\_objects

Number of stored objects at the time of last check split by kind. In case of a fetching error, the value will be -1.

- Stability Level:STABLE
- Type: Gauge
- Labels:resource

apiserver\_storage\_size\_bytes

Size of the storage database file physically allocated in bytes.

- Stability Level:STABLE
- Type: Custom
- Labels:storage\_cluster\_id

container\_cpu\_usage\_seconds\_total

Cumulative cpu time consumed by the container in core-seconds

- Stability Level:STABLE
- Type: Custom
- Labels:containerpodnamespace

container\_memory\_working\_set\_bytes

Current working set of the container in bytes

- Stability Level:STABLE
- Type: Custom
- Labels:containerpodnamespace

container\_start\_time\_seconds

Start time of the container since unix epoch in seconds

- Stability Level:STABLE
- Type: Custom
- Labels:containerpodnamespace

cronjob\_controller\_job\_creation\_skew\_duration\_seconds

Time between when a cronjob is scheduled to be run, and when the corresponding job is created

- Stability Level:STABLE
- Type: Histogram

job\_controller\_job\_pods\_finished\_total

The number of finished Pods that are fully tracked

- Stability Level:STABLE
- Type: Counter

- Labels:completion\_moderesult

job\_controller\_job\_sync\_duration\_seconds

The time it took to sync a job

- Stability Level:STABLE
- Type: Histogram
- Labels:actioncompletion\_moderesult

job\_controller\_job\_syncs\_total

The number of job syncs

- Stability Level:STABLE
- Type: Counter
- Labels:actioncompletion\_moderesult

job\_controller\_jobs\_finished\_total

The number of finished jobs

- Stability Level:STABLE
- Type: Counter
- Labels:completion\_modereasonresult

kube\_pod\_resource\_limit

Resources limit for workloads on the cluster, broken down by pod. This shows the resource usage the scheduler and kubelet expect per pod for resources along with the unit for the resource if any.

- Stability Level:STABLE
- Type: Custom
- Labels:namespacepodnodeschedulerpriorityresourceunit

kube\_pod\_resource\_request

Resources requested by workloads on the cluster, broken down by pod. This shows the resource usage the scheduler and kubelet expect per pod for resources along with the unit for the resource if any.

- Stability Level:STABLE
- Type: Custom
- Labels:namespacepodnodeschedulerpriorityresourceunit

kubernetes\_healthcheck

This metric records the result of a single healthcheck.

- Stability Level:STABLE
- Type: Gauge
- Labels:nametype

kubernetes\_healthchecks\_total

This metric records the results of all healthcheck.

- Stability Level:STABLE
- Type: Counter
- Labels:nameinstancetype

node\_collector\_evictions\_total

Number of Node evictions that happened since current instance of NodeController started.

- Stability Level:STABLE
- Type: Counter
- Labels:zone

node\_cpu\_usage\_seconds\_total

Cumulative cpu time consumed by the node in core-seconds

- Stability Level:STABLE
- Type: Custom

node\_memory\_working\_set\_bytes

Current working set of the node in bytes

- Stability Level:STABLE
- Type: Custom

pod\_cpu\_usage\_seconds\_total

Cumulative cpu time consumed by the pod in core-seconds

- Stability Level:STABLE
- Type: Custom
- Labels:podnamespace

pod\_memory\_working\_set\_bytes

Current working set of the pod in bytes

- Stability Level:STABLE
- Type: Custom
- Labels:podnamespace

resource\_scrape\_error

1 if there was an error while getting container metrics, 0 otherwise

- Stability Level:STABLE
- Type: Custom

scheduler\_framework\_extension\_point\_duration\_seconds

Latency for running all plugins of a specific extension point.

- Stability Level:STABLE
- Type: Histogram
- Labels:extension\_pointprofilestatus

scheduler\_pending\_pods

Number of pending pods, by the queue type. 'active' means number of pods in activeQ; 'backoff' means number of pods in backoffQ; 'unschedulable' means number of pods in unschedulablePods that the scheduler attempted to schedule and failed; 'gated' is the number of unschedulable pods that the scheduler never attempted to schedule because they are gated.

- Stability Level:STABLE
- Type: Gauge
- Labels:queue



scheduler\_pod\_scheduling\_attempts

Number of attempts to successfully schedule a pod.

- Stability Level:STABLE
- Type: Histogram

scheduler\_pod\_scheduling\_duration\_seconds

E2e latency for a pod being scheduled which may include multiple scheduling attempts.

- Stability Level:STABLE
- Type: Histogram
- Labels:attempts
- Deprecated Versions:1.29.0

scheduler\_preemption\_attempts\_total

Total preemption attempts in the cluster till now

- Stability Level:STABLE
- Type: Counter

scheduler\_preemption\_victims

Number of selected preemption victims

- Stability Level:STABLE
- Type: Histogram

scheduler\_queue\_incoming\_pods\_total

Number of pods added to scheduling queues by event and queue type.

- Stability Level:STABLE
- Type: Counter
- Labels:eventqueue

scheduler\_schedule\_attempts\_total

Number of attempts to schedule pods, by the result. 'unschedulable' means a pod could not be scheduled, while 'error' means an internal scheduler problem.

- Stability Level:STABLE
- Type: Counter
- Labels:profileresult

scheduler\_scheduling\_attempt\_duration\_seconds

Scheduling attempt latency in seconds (scheduling algorithm + binding)

- Stability Level:STABLE
- Type: Histogram
- Labels:profileresult

## List of Beta Kubernetes Metrics

Beta metrics observe a looser API contract than its stable counterparts. No labels can be removed from beta metrics during their lifetime, however, labels can be added while the metric is in the beta stage. This offers the assurance that beta metrics will honor existing dashboards and alerts, while allowing for amendments in the future.

apiserver\_flowcontrol\_current\_executing\_requests

Number of requests in initial (for a WATCH) or any (for a non-WATCH) execution stage in the API Priority and Fairness subsystem

- Stability Level: BETA
- Type: Gauge
- Labels: flow\_schemapriority\_level

apiserver\_flowcontrol\_current\_executing\_seats

Concurrency (number of seats) occupied by the currently executing (initial stage for a WATCH, any stage otherwise) requests in the API Priority and Fairness subsystem

- Stability Level: BETA
- Type: Gauge
- Labels: flow\_schemapriority\_level

apiserver\_flowcontrol\_current\_inqueue\_requests

Number of requests currently pending in queues of the API Priority and Fairness subsystem

- Stability Level: BETA
- Type: Gauge
- Labels: flow\_schemapriority\_level

apiserver\_flowcontrol\_dispatched\_requests\_total

Number of requests executed by API Priority and Fairness subsystem

- Stability Level: BETA
- Type: Counter
- Labels: flow\_schemapriority\_level

apiserver\_flowcontrol\_nominal\_limit\_seats

Nominal number of execution seats configured for each priority level

- Stability Level: BETA
- Type: Gauge
- Labels: priority\_level

apiserver\_flowcontrol\_rejected\_requests\_total

Number of requests rejected by API Priority and Fairness subsystem

- Stability Level: BETA
- Type: Counter
- Labels: flow\_schemapriority\_levelreason

apiserver\_flowcontrol\_request\_wait\_duration\_seconds

Length of time a request spent waiting in its queue

- Stability Level: BETA
- Type: Histogram
- Labels: executeflow\_schemapriority\_level

disabled\_metrics\_total

The count of disabled metrics.

- Stability Level: BETA

- Type: Counter

hidden\_metrics\_total

The count of hidden metrics.

- Stability Level: BETA
- Type: Counter

kubernetes\_feature\_enabled

This metric records the data about the stage and enablement of a k8s feature.

- Stability Level: BETA
- Type: Gauge
- Labels: name, stage

registered\_metrics\_total

The count of registered metrics broken by stability level and deprecation version.

- Stability Level: BETA
- Type: Counter
- Labels: deprecated\_version, stability\_level

scheduler\_pod\_scheduling\_sli\_duration\_seconds

E2e latency for a pod being scheduled, from the time the pod enters the scheduling queue and might involve multiple scheduling attempts.

- Stability Level: BETA
- Type: Histogram
- Labels: attempts

## List of Alpha Kubernetes Metrics

Alpha metrics do not have any API guarantees. These metrics must be used at your own risk, subsequent versions of Kubernetes may remove these metrics altogether, or mutate the API in such a way that breaks existing dashboards and alerts.

aggregator\_discovery\_aggregation\_count\_total

Counter of number of times discovery was aggregated

- Stability Level: ALPHA
- Type: Counter

aggregator\_openapi\_v2\_regeneration\_count

Counter of OpenAPI v2 spec regeneration count broken down by causing API Service name and reason.

- Stability Level: ALPHA
- Type: Counter
- Labels: apiservice, reason

aggregator\_openapi\_v2\_regeneration\_duration

Gauge of OpenAPI v2 spec regeneration duration in seconds.

- Stability Level: ALPHA

- Type: Gauge
- Labels:reason

aggregator\_unavailable\_apiservice

Gauge of APIServices which are marked as unavailable broken down by APIService name.

- Stability Level:ALPHA
- Type: Custom
- Labels:name

aggregator\_unavailable\_apiservice\_total

Counter of APIServices which are marked as unavailable broken down by APIService name and reason.

- Stability Level:ALPHA
- Type: Counter
- Labels:namereason

apiextensions\_apiserver\_validation\_ratcheting\_seconds

Time for comparison of old to new for the purposes of CRDValidationRatcheting during an UPDATE in seconds.

- Stability Level:ALPHA
- Type: Histogram

apiextensions\_openapi\_v2\_regeneration\_count

Counter of OpenAPI v2 spec regeneration count broken down by causing CRD name and reason.

- Stability Level:ALPHA
- Type: Counter
- Labels:crdreason

apiextensions\_openapi\_v3\_regeneration\_count

Counter of OpenAPI v3 spec regeneration count broken down by group, version, causing CRD and reason.

- Stability Level:ALPHA
- Type: Counter
- Labels:crdgroupreasonversion

apiserver\_admission\_match\_condition\_evaluation\_errors\_total

Admission match condition evaluation errors count, identified by name of resource containing the match condition and broken out for each kind containing matchConditions (webhook or policy), operation and admission type (validate or admit).

- Stability Level:ALPHA
- Type: Counter
- Labels:kindnameoperationtype

apiserver\_admission\_match\_condition\_evaluation\_seconds

Admission match condition evaluation time in seconds, identified by name and broken out for each kind containing matchConditions (webhook or policy), operation and type (validate or admit).

- Stability Level:ALPHA
- Type: Histogram
- Labels:kindnameoperationtype

apiserver\_admission\_match\_condition\_exclusions\_total

Admission match condition evaluation exclusions count, identified by name of resource containing the match condition and broken out for each kind containing matchConditions (webhook or policy), operation and admission type (validate or admit).

- Stability Level:ALPHA
- Type: Counter
- Labels:kindnameoperationtype

apiserver\_admission\_step\_admission\_duration\_seconds\_summary

Admission sub-step latency summary in seconds, broken out for each operation and API resource and step type (validate or admit).

- Stability Level:ALPHA
- Type: Summary
- Labels:operationrejectedtype

apiserver\_admission\_webhook\_fail\_open\_count

Admission webhook fail open count, identified by name and broken out for each admission type (validating or mutating).

- Stability Level:ALPHA
- Type: Counter
- Labels:nametype

apiserver\_admission\_webhook\_rejection\_count

Admission webhook rejection count, identified by name and broken out for each admission type (validating or admit) and operation. Additional labels specify an error type (calling\_webhook\_error or apiserver\_internal\_error if an error occurred; no\_error otherwise) and optionally a non-zero rejection code if the webhook rejects the request with an HTTP status code (honored by the apiserver when the code is greater or equal to 400). Codes greater than 600 are truncated to 600, to keep the metrics cardinality bounded.

- Stability Level:ALPHA
- Type: Counter
- Labels:error\_typeoperationrejection\_codetype

apiserver\_admission\_webhook\_request\_total

Admission webhook request total, identified by name and broken out for each admission type (validating or mutating) and operation. Additional labels specify whether the request was rejected or not and an HTTP status code. Codes greater than 600 are truncated to 600, to keep the metrics cardinality bounded.

- Stability Level:ALPHA
- Type: Counter
- Labels:codenameoperationrejectedtype

apiserver\_audit\_error\_total

Counter of audit events that failed to be audited properly. Plugin identifies the plugin affected by the error.

- Stability Level:ALPHA
- Type: Counter
- Labels:plugin

apiserver\_audit\_event\_total

Counter of audit events generated and sent to the audit backend.

- Stability Level:ALPHA
- Type: Counter

apiserver\_audit\_level\_total

Counter of policy levels for audit events (1 per request).

- Stability Level:ALPHA
- Type: Counter
- Labels:level

apiserver\_audit\_requests\_rejected\_total

Counter of apiserver requests rejected due to an error in audit logging backend.

- Stability Level:ALPHA
- Type: Counter

apiserver\_authentication\_config\_controller\_automatic\_reload\_last\_timestamp\_seconds

Timestamp of the last automatic reload of authentication configuration split by status and apiserver identity.

- Stability Level:ALPHA
- Type: Gauge
- Labels:apiserver\_id\_hashstatus

apiserver\_authentication\_config\_controller\_automatic\_reloads\_total

Total number of automatic reloads of authentication configuration split by status and apiserver identity.

- Stability Level:ALPHA
- Type: Counter
- Labels:apiserver\_id\_hashstatus

apiserver\_authentication\_jwt\_authenticator\_latency\_seconds

Latency of jwt authentication operations in seconds. This is the time spent authenticating a token for cache miss only (i.e. when the token is not found in the cache).

- Stability Level:ALPHA
- Type: Histogram
- Labels:jwt\_issuer\_hashresult

apiserver\_authorization\_decisions\_total

Total number of terminal decisions made by an authorizer split by authorizer type, name, and decision.

- Stability Level:ALPHA
- Type: Counter
- Labels:decisionnametype

apiserver\_authorization\_match\_condition\_evaluation\_errors\_total

Total number of errors when an authorization webhook encounters a match condition error split by authorizer type and name.

- Stability Level:ALPHA
- Type: Counter
- Labels:nametype

apiserver\_authorization\_match\_condition\_evaluation\_seconds

Authorization match condition evaluation time in seconds, split by authorizer type and name.

- Stability Level:ALPHA
- Type: Histogram
- Labels:nametype

apiserver\_authorization\_match\_condition\_exclusions\_total

Total number of exclusions when an authorization webhook is skipped because match conditions exclude it.

- Stability Level:ALPHA
- Type: Counter
- Labels:nametype

apiserver\_authorization\_webhook\_duration\_seconds

Request latency in seconds.

- Stability Level:ALPHA
- Type: Histogram
- Labels:nameresult

apiserver\_authorization\_webhook\_evaluations\_fail\_open\_total

NoOpinion results due to webhook timeout or error.

- Stability Level:ALPHA
- Type: Counter
- Labels:nameresult

apiserver\_authorization\_webhook\_evaluations\_total

Round-trips to authorization webhooks.

- Stability Level:ALPHA
- Type: Counter
- Labels:nameresult

apiserver\_cache\_list\_fetched\_objects\_total

Number of objects read from watch cache in the course of serving a LIST request

- Stability Level:ALPHA

- Type: Counter
- Labels:indexresource\_prefix

apiserver\_cache\_list\_returned\_objects\_total

Number of objects returned for a LIST request from watch cache

- Stability Level:ALPHA
- Type: Counter
- Labels:resource\_prefix

apiserver\_cache\_list\_total

Number of LIST requests served from watch cache

- Stability Level:ALPHA
- Type: Counter
- Labels:indexresource\_prefix

apiserver\_cel\_compilation\_duration\_seconds

CEL compilation time in seconds.

- Stability Level:ALPHA
- Type: Histogram

apiserver\_cel\_evaluation\_duration\_seconds

CEL evaluation time in seconds.

- Stability Level:ALPHA
- Type: Histogram

apiserver\_certificates\_registry\_csr\_honored\_duration\_total

Total number of issued CSRs with a requested duration that was honored, sliced by signer (only kubernetes.io signer names are specifically identified)

- Stability Level:ALPHA
- Type: Counter
- Labels:signerName

apiserver\_certificates\_registry\_csr\_requested\_duration\_total

Total number of issued CSRs with a requested duration, sliced by signer (only kubernetes.io signer names are specifically identified)

- Stability Level:ALPHA
- Type: Counter
- Labels:signerName

apiserver\_client\_certificate\_expiration\_seconds

Distribution of the remaining lifetime on the certificate used to authenticate a request.

- Stability Level:ALPHA
- Type: Histogram

apiserver\_clusterip\_repair\_ip\_errors\_total



Number of errors detected on clusterips by the repair loop broken down by type of error: leak, repair, full, outOfRange, duplicate, unknown, invalid

- Stability Level:ALPHA
- Type: Counter
- Labels:type

apiserver\_clusterip\_repair\_reconcile\_errors\_total

Number of reconciliation failures on the clusterip repair reconcile loop

- Stability Level:ALPHA
- Type: Counter

apiserver\_conversion\_webhook\_duration\_seconds

Conversion webhook request latency

- Stability Level:ALPHA
- Type: Histogram
- Labels:failure\_typeresult

apiserver\_conversion\_webhook\_request\_total

Counter for conversion webhook requests with success/failure and failure error type

- Stability Level:ALPHA
- Type: Counter
- Labels:failure\_typeresult

apiserver\_crd\_conversion\_webhook\_duration\_seconds

CRD webhook conversion duration in seconds

- Stability Level:ALPHA
- Type: Histogram
- Labels:crd\_namefrom\_versionsucceededto\_version

apiserver\_current\_inqueue\_requests

Maximal number of queued requests in this apiserver per request kind in last second.

- Stability Level:ALPHA
- Type: Gauge
- Labels:request\_kind

apiserver\_delegated\_authn\_request\_duration\_seconds

Request latency in seconds. Broken down by status code.

- Stability Level:ALPHA
- Type: Histogram
- Labels:code

apiserver\_delegated\_authn\_request\_total

Number of HTTP requests partitioned by status code.

- Stability Level:ALPHA
- Type: Counter
- Labels:code

apiserver\_delegated\_authz\_request\_duration\_seconds  
Request latency in seconds. Broken down by status code.

- Stability Level:ALPHA
- Type: Histogram
- Labels:code

apiserver\_delegated\_authz\_request\_total  
Number of HTTP requests partitioned by status code.

- Stability Level:ALPHA
- Type: Counter
- Labels:code

apiserver\_egress\_dialer\_dial\_duration\_seconds  
Dial latency histogram in seconds, labeled by the protocol (http-connect or grpc), transport (tcp or uds)

- Stability Level:ALPHA
- Type: Histogram
- Labels:protocoltransport

apiserver\_egress\_dialer\_dial\_failure\_count  
Dial failure count, labeled by the protocol (http-connect or grpc), transport (tcp or uds), and stage (connect or proxy). The stage indicates at which stage the dial failed

- Stability Level:ALPHA
- Type: Counter
- Labels:protocolstagetransport

apiserver\_egress\_dialer\_dial\_start\_total  
Dial starts, labeled by the protocol (http-connect or grpc) and transport (tcp or uds).

- Stability Level:ALPHA
- Type: Counter
- Labels:protocoltransport

apiserver\_encryption\_config\_controller\_automatic\_reload\_failures\_total  
Total number of failed automatic reloads of encryption configuration split by apiserver identity.

- Stability Level:ALPHA
- Type: Counter
- Labels:apiserver\_id\_hash
- Deprecated Versions:1.30.0

apiserver\_encryption\_config\_controller\_automatic\_reload\_success\_total  
Total number of successful automatic reloads of encryption configuration split by apiserver identity.

- Stability Level:ALPHA
- Type: Counter
- Labels:apiserver\_id\_hash
- Deprecated Versions:1.30.0

apiserver\_envelope\_encryption\_dek\_cache\_fill\_percent

Percent of the cache slots currently occupied by cached DEKs.

- Stability Level:ALPHA
- Type: Gauge

apiserver\_envelope\_encryption\_dek\_cache\_inter\_arrival\_time\_seconds  
Time (in seconds) of inter arrival of transformation requests.

- Stability Level:ALPHA
- Type: Histogram
- Labels:transformation\_type

apiserver\_envelope\_encryption\_dek\_source\_cache\_size  
Number of records in data encryption key (DEK) source cache. On a restart, this value is an approximation of the number of decrypt RPC calls the server will make to the KMS plugin.

- Stability Level:ALPHA
- Type: Gauge
- Labels:provider\_name

apiserver\_envelope\_encryption\_invalid\_key\_id\_from\_status\_total  
Number of times an invalid keyID is returned by the Status RPC call split by error.

- Stability Level:ALPHA
- Type: Counter
- Labels:errorprovider\_name

apiserver\_envelope\_encryption\_key\_id\_hash\_last\_timestamp\_seconds  
The last time in seconds when a keyID was used.

- Stability Level:ALPHA
- Type: Gauge
- Labels:apiserver\_id\_hashkey\_id\_hashprovider\_namestransformation\_type

apiserver\_envelope\_encryption\_key\_id\_hash\_status\_last\_timestamp\_seconds  
The last time in seconds when a keyID was returned by the Status RPC call.

- Stability Level:ALPHA
- Type: Gauge
- Labels:apiserver\_id\_hashkey\_id\_hashprovider\_name

apiserver\_envelope\_encryption\_key\_id\_hash\_total  
Number of times a keyID is used split by transformation type, provider, and apiserver identity.

- Stability Level:ALPHA
- Type: Counter
- Labels:apiserver\_id\_hashkey\_id\_hashprovider\_namestransformation\_type

apiserver\_envelope\_encryption\_kms\_operations\_latency\_seconds  
KMS operation duration with gRPC error code status total.

- Stability Level:ALPHA
- Type: Histogram
- Labels:grpc\_status\_codemethod\_nameprovider\_name

apiserver\_flowcontrol\_current\_inqueue\_seats

Number of seats currently pending in queues of the API Priority and Fairness subsystem

- Stability Level:ALPHA
- Type: Gauge
- Labels:flow\_schemapriority\_level

apiserver\_flowcontrol\_current\_limit\_seats

current derived number of execution seats available to each priority level

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_current\_r

R(time of last change)

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_demand\_seats

Observations, at the end of every nanosecond, of (the number of seats each priority level could use) / (nominal number of seats for that level)

- Stability Level:ALPHA
- Type: TimingRatioHistogram
- Labels:priority\_level

apiserver\_flowcontrol\_demand\_seats\_average

Time-weighted average, over last adjustment period, of demand\_seats

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_demand\_seats\_high\_watermark

High watermark, over last adjustment period, of demand\_seats

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_demand\_seats\_smoothed

Smoothed seat demands

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_demand\_seats\_stddev

Time-weighted standard deviation, over last adjustment period, of demand\_seats

- Stability Level:ALPHA
- Type: Gauge

- Labels:priority\_level

apiserver\_flowcontrol\_dispatch\_r  
R(time of last dispatch)

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_epoch\_advance\_total  
Number of times the queueset's progress meter jumped backward

- Stability Level:ALPHA
- Type: Counter
- Labels:priority\_levelsuccess

apiserver\_flowcontrol\_latest\_s  
S(most recently dispatched request)

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_lower\_limit\_seats  
Configured lower bound on number of execution seats available to each priority level

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_next\_discounted\_s\_bounds  
min and max, over queues, of S(oldest waiting request in queue) - estimated work in progress

- Stability Level:ALPHA
- Type: Gauge
- Labels:boundpriority\_level

apiserver\_flowcontrol\_next\_s\_bounds  
min and max, over queues, of S(oldest waiting request in queue)

- Stability Level:ALPHA
- Type: Gauge
- Labels:boundpriority\_level

apiserver\_flowcontrol\_priority\_level\_request\_utilization  
Observations, at the end of every nanosecond, of number of requests (as a fraction of the relevant limit) waiting or in any stage of execution (but only initial stage for WATCHes)

- Stability Level:ALPHA
- Type: TimingRatioHistogram
- Labels:phasepriority\_level

apiserver\_flowcontrol\_priority\_level\_seat\_utilization

Observations, at the end of every nanosecond, of utilization of seats for any stage of execution (but only initial stage for WATCHes)

- Stability Level:ALPHA
- Type: TimingRatioHistogram
- Labels:priority\_level
- Const Labels:phase:executing

apiserver\_flowcontrol\_read\_vs\_write\_current\_requests

Observations, at the end of every nanosecond, of the number of requests (as a fraction of the relevant limit) waiting or in regular stage of execution

- Stability Level:ALPHA
- Type: TimingRatioHistogram
- Labels:phaserequest\_kind

apiserver\_flowcontrol\_request\_concurrency\_in\_use

Concurrency (number of seats) occupied by the currently executing (initial stage for a WATCH, any stage otherwise) requests in the API Priority and Fairness subsystem

- Stability Level:ALPHA
- Type: Gauge
- Labels:flow\_schemapriority\_level
- Deprecated Versions:1.31.0

apiserver\_flowcontrol\_request\_concurrency\_limit

Nominal number of execution seats configured for each priority level

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level
- Deprecated Versions:1.30.0

apiserver\_flowcontrol\_request\_dispatch\_no\_accommodation\_total

Number of times a dispatch attempt resulted in a non accommodation due to lack of available seats

- Stability Level:ALPHA
- Type: Counter
- Labels:flow\_schemapriority\_level

apiserver\_flowcontrol\_request\_execution\_seconds

Duration of initial stage (for a WATCH) or any (for a non-WATCH) stage of request execution in the API Priority and Fairness subsystem

- Stability Level:ALPHA
- Type: Histogram
- Labels:flow\_schemapriority\_leveltype

apiserver\_flowcontrol\_request\_queue\_length\_after\_enqueue

Length of queue in the API Priority and Fairness subsystem, as seen by each request after it is enqueued

- Stability Level:ALPHA

- Type: Histogram
- Labels:flow\_schemapriority\_level

apiserver\_flowcontrol\_seat\_fair\_frac

Fair fraction of server's concurrency to allocate to each priority level that can use it

- Stability Level:ALPHA
- Type: Gauge

apiserver\_flowcontrol\_target\_seats

Seat allocation targets

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_upper\_limit\_seats

Configured upper bound on number of execution seats available to each priority level

- Stability Level:ALPHA
- Type: Gauge
- Labels:priority\_level

apiserver\_flowcontrol\_watch\_count\_samples

count of watchers for mutating requests in API Priority and Fairness

- Stability Level:ALPHA
- Type: Histogram
- Labels:flow\_schemapriority\_level

apiserver\_flowcontrol\_work\_estimated\_seats

Number of estimated seats (maximum of initial and final seats) associated with requests in API Priority and Fairness

- Stability Level:ALPHA
- Type: Histogram
- Labels:flow\_schemapriority\_level

apiserver\_init\_events\_total

Counter of init events processed in watch cache broken by resource type.

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_kube\_aggregator\_x509\_insecure\_sha1\_total

Counts the number of requests to servers with insecure SHA1 signatures in their serving certificate OR the number of connection failures due to the insecure SHA1 signatures (either/or, based on the runtime environment)

- Stability Level:ALPHA
- Type: Counter

apiserver\_kube\_aggregator\_x509\_missing\_san\_total

Counts the number of requests to servers missing SAN extension in their serving certificate OR the number of connection failures due to the lack of x509 certificate SAN extension missing (either/or, based on the runtime environment)

- Stability Level:ALPHA
- Type: Counter

apiserver\_nodeport\_repair\_port\_errors\_total

Number of errors detected on ports by the repair loop broken down by type of error: leak, repair, full, outOfRange, duplicate, unknown

- Stability Level:ALPHA
- Type: Counter
- Labels:type

apiserver\_request\_aborts\_total

Number of requests which apiserver aborted possibly due to a timeout, for each group, version, verb, resource, subresource and scope

- Stability Level:ALPHA
- Type: Counter
- Labels:groupresourcescopesubresourceverbversion

apiserver\_request\_body\_size\_bytes

Apiserver request body size in bytes broken out by resource and verb.

- Stability Level:ALPHA
- Type: Histogram
- Labels:resourceverb

apiserver\_request\_filter\_duration\_seconds

Request filter latency distribution in seconds, for each filter type

- Stability Level:ALPHA
- Type: Histogram
- Labels:filter

apiserver\_request\_post\_timeout\_total

Tracks the activity of the request handlers after the associated requests have been timed out by the apiserver

- Stability Level:ALPHA
- Type: Counter
- Labels:sourcestatus

apiserver\_request\_sli\_duration\_seconds

Response latency distribution (not counting webhook duration and priority & fairness queue wait times) in seconds for each verb, group, version, resource, subresource, scope and component.

- Stability Level:ALPHA
- Type: Histogram
- Labels:componentgroupresourcescopesubresourceverbversion

apiserver\_request\_slo\_duration\_seconds



Response latency distribution (not counting webhook duration and priority & fairness queue wait times) in seconds for each verb, group, version, resource, subresource, scope and component.

- Stability Level:ALPHA
- Type: Histogram
- Labels:componentgroupresourcescopesubresourceverbversion
- Deprecated Versions:1.27.0

apiserver\_request\_terminations\_total

Number of requests which apiserver terminated in self-defense.

- Stability Level:ALPHA
- Type: Counter
- Labels:codecomponentgroupresourcescopesubresourceverbversion

apiserver\_request\_timestamp\_comparison\_time

Time taken for comparison of old vs new objects in UPDATE or PATCH requests

- Stability Level:ALPHA
- Type: Histogram
- Labels:code\_path

apiserver\_rerouted\_request\_total

Total number of requests that were proxied to a peer kube apiserver because the local apiserver was not capable of serving it

- Stability Level:ALPHA
- Type: Counter
- Labels:code

apiserver\_selfrequest\_total

Counter of apiserver self-requests broken out for each verb, API resource and subresource.

- Stability Level:ALPHA
- Type: Counter
- Labels:resourcesubresourceverb

apiserver\_storage\_data\_key\_generation\_duration\_seconds

Latencies in seconds of data encryption key(DEK) generation operations.

- Stability Level:ALPHA
- Type: Histogram

apiserver\_storage\_data\_key\_generation\_failures\_total

Total number of failed data encryption key(DEK) generation operations.

- Stability Level:ALPHA
- Type: Counter

apiserver\_storage\_db\_total\_size\_in\_bytes

Total size of the storage database file physically allocated in bytes.

- Stability Level:ALPHA
- Type: Gauge

- Labels:endpoint
- Deprecated Versions:1.28.0

apiserver\_storage\_decode\_errors\_total

Number of stored object decode errors split by object type

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_storage\_envelope\_transformation\_cache\_misses\_total

Total number of cache misses while accessing key decryption key(KEK).

- Stability Level:ALPHA
- Type: Counter

apiserver\_storage\_events\_received\_total

Number of etcd events received split by kind.

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_storage\_list\_evaluated\_objects\_total

Number of objects tested in the course of serving a LIST request from storage

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_storage\_list\_fetched\_objects\_total

Number of objects read from storage in the course of serving a LIST request

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_storage\_list\_returned\_objects\_total

Number of objects returned for a LIST request from storage

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_storage\_list\_total

Number of LIST requests served from storage

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_storage\_transformation\_duration\_seconds

Latencies in seconds of value transformation operations.

- Stability Level:ALPHA

- Type: Histogram
- Labels: transformation\_type transformer\_prefix

apiserver\_storage\_transformation\_operations\_total

Total number of transformations. Successful transformation will have a status 'OK' and a varied status string when the transformation fails. This status and transformation\_type fields may be used for alerting on encryption/decryption failure using transformation\_type from\_storage for decryption and to\_storage for encryption

- Stability Level: ALPHA
- Type: Counter
- Labels: status transformation\_type transformer\_prefix

apiserver\_stream\_translator\_requests\_total

Total number of requests that were handled by the StreamTranslatorProxy, which processes streaming RemoteCommand/V5

- Stability Level: ALPHA
- Type: Counter
- Labels: code

apiserver\_terminated\_watchers\_total

Counter of watchers closed due to unresponsiveness broken by resource type.

- Stability Level: ALPHA
- Type: Counter
- Labels: resource

apiserver\_tls\_handshake\_errors\_total

Number of requests dropped with 'TLS handshake error from' error

- Stability Level: ALPHA
- Type: Counter

apiserver\_validating\_admission\_policy\_check\_duration\_seconds

Validation admission latency for individual validation expressions in seconds, labeled by policy and further including binding, state and enforcement action taken.

- Stability Level: ALPHA
- Type: Histogram
- Labels: enforcement\_action policy policy\_binding state

apiserver\_validating\_admission\_policy\_check\_total

Validation admission policy check total, labeled by policy and further identified by binding, enforcement action taken, and state.

- Stability Level: ALPHA
- Type: Counter
- Labels: enforcement\_action policy policy\_binding state

apiserver\_validating\_admission\_policy\_definition\_total

Validation admission policy count total, labeled by state and enforcement action.

- Stability Level: ALPHA
- Type: Counter

- Labels:enforcement\_actionstate

apiserver\_watch\_cache\_events\_dispatched\_total

Counter of events dispatched in watch cache broken by resource type.

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_watch\_cache\_events\_received\_total

Counter of events received in watch cache broken by resource type.

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_watch\_cache\_initializations\_total

Counter of watch cache initializations broken by resource type.

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

apiserver\_watch\_cache\_read\_wait\_seconds

Histogram of time spent waiting for a watch cache to become fresh.

- Stability Level:ALPHA
- Type: Histogram
- Labels:resource

apiserver\_watch\_events\_sizes

Watch event size distribution in bytes

- Stability Level:ALPHA
- Type: Histogram
- Labels:groupkindversion

apiserver\_watch\_events\_total

Number of events sent in watch clients

- Stability Level:ALPHA
- Type: Counter
- Labels:groupkindversion

apiserver\_watch\_list\_duration\_seconds

Response latency distribution in seconds for watch list requests broken by group, version, resource and scope.

- Stability Level:ALPHA
- Type: Histogram
- Labels:groupresourcescopeversion

apiserver\_webhooks\_x509\_insecure\_sha1\_total

Counts the number of requests to servers with insecure SHA1 signatures in their serving certificate OR the number of connection failures due to the insecure SHA1 signatures (either/or, based on the runtime environment)

- Stability Level:ALPHA
- Type: Counter

apiserver\_webhooks\_x509\_missing\_san\_total

Counts the number of requests to servers missing SAN extension in their serving certificate OR the number of connection failures due to the lack of x509 certificate SAN extension missing (either/or, based on the runtime environment)

- Stability Level:ALPHA
- Type: Counter

attach\_detach\_controller\_attachdetach\_controller\_forced\_detaches

Number of times the A/D Controller performed a forced detach

- Stability Level:ALPHA
- Type: Counter
- Labels:reason

attachdetach\_controller\_total\_volumes

Number of volumes in A/D Controller

- Stability Level:ALPHA
- Type: Custom
- Labels:plugin\_namestate

authenticated\_user\_requests

Counter of authenticated requests broken out by username.

- Stability Level:ALPHA
- Type: Counter
- Labels:username

authentication\_attempts

Counter of authenticated attempts.

- Stability Level:ALPHA
- Type: Counter
- Labels:result

authentication\_duration\_seconds

Authentication duration in seconds broken out by result.

- Stability Level:ALPHA
- Type: Histogram
- Labels:result

authentication\_token\_cache\_active\_fetch\_count

- Stability Level:ALPHA
- Type: Gauge
- Labels:status

authentication\_token\_cache\_fetch\_total

- Stability Level:ALPHA
- Type: Counter
- Labels:status

authentication\_token\_cache\_request\_duration\_seconds

- Stability Level:ALPHA
- Type: Histogram
- Labels:status

authentication\_token\_cache\_request\_total

- Stability Level:ALPHA
- Type: Counter
- Labels:status

authorization\_attempts\_total

Counter of authorization attempts broken down by result. It can be either 'allowed', 'denied', 'no-opinion' or 'error'.

- Stability Level:ALPHA
- Type: Counter
- Labels:result

authorization\_duration\_seconds

Authorization duration in seconds broken out by result.

- Stability Level:ALPHA
- Type: Histogram
- Labels:result

cloud\_provider\_webhook\_request\_duration\_seconds

Request latency in seconds. Broken down by status code.

- Stability Level:ALPHA
- Type: Histogram
- Labels:codewebhook

cloud\_provider\_webhook\_request\_total

Number of HTTP requests partitioned by status code.

- Stability Level:ALPHA
- Type: Counter
- Labels:codewebhook

cloudprovider\_gce\_api\_request\_duration\_seconds

Latency of a GCE API call

- Stability Level:ALPHA
- Type: Histogram
- Labels:regionrequestversionzone

cloudprovider\_gce\_api\_request\_errors

Number of errors for an API call

- Stability Level:ALPHA
- Type: Counter
- Labels:regionrequestversionzone

container\_swap\_usage\_bytes

Current amount of the container swap usage in bytes. Reported only on non-windows systems

- Stability Level:ALPHA
- Type: Custom
- Labels:containerpodnamespace

csi\_operations\_seconds

Container Storage Interface operation duration with gRPC error code status total

- Stability Level:ALPHA
- Type: Histogram
- Labels:driver\_namegrpc\_status\_codemethod\_namemigrated

endpoint\_slice\_controller\_changes

Number of EndpointSlice changes

- Stability Level:ALPHA
- Type: Counter
- Labels:operation

endpoint\_slice\_controller\_desired\_endpoint\_slices

Number of EndpointSlices that would exist with perfect endpoint allocation

- Stability Level:ALPHA
- Type: Gauge

endpoint\_slice\_controller\_endpoints\_added\_per\_sync

Number of endpoints added on each Service sync

- Stability Level:ALPHA
- Type: Histogram

endpoint\_slice\_controller\_endpoints\_desired

Number of endpoints desired

- Stability Level:ALPHA
- Type: Gauge

endpoint\_slice\_controller\_endpoints\_removed\_per\_sync

Number of endpoints removed on each Service sync

- Stability Level:ALPHA
- Type: Histogram

endpoint\_slice\_controller\_endpointslices\_changed\_per\_sync

Number of EndpointSlices changed on each Service sync

- Stability Level:ALPHA

- Type: Histogram
- Labels: topologytraffic\_distribution

endpoint\_slice\_controller\_num\_endpoint\_slices  
Number of EndpointSlices

- Stability Level: ALPHA
- Type: Gauge

endpoint\_slice\_controller\_services\_count\_by\_traffic\_distribution  
Number of Services using some specific trafficDistribution

- Stability Level: ALPHA
- Type: Gauge
- Labels: traffic\_distribution

endpoint\_slice\_controller\_syncs  
Number of EndpointSlice syncs

- Stability Level: ALPHA
- Type: Counter
- Labels: result

endpoint\_slice\_mirroring\_controller\_addresses\_skipped\_per\_sync  
Number of addresses skipped on each Endpoints sync due to being invalid or exceeding MaxEndpointsPerSubset

- Stability Level: ALPHA
- Type: Histogram

endpoint\_slice\_mirroring\_controller\_changes  
Number of EndpointSlice changes

- Stability Level: ALPHA
- Type: Counter
- Labels: operation

endpoint\_slice\_mirroring\_controller\_desired\_endpoint\_slices  
Number of EndpointSlices that would exist with perfect endpoint allocation

- Stability Level: ALPHA
- Type: Gauge

endpoint\_slice\_mirroring\_controller\_endpoints\_added\_per\_sync  
Number of endpoints added on each Endpoints sync

- Stability Level: ALPHA
- Type: Histogram

endpoint\_slice\_mirroring\_controller\_endpoints\_desired  
Number of endpoints desired

- Stability Level: ALPHA
- Type: Gauge



endpoint\_slice\_mirroring\_controller\_endpoints\_removed\_per\_sync  
Number of endpoints removed on each Endpoints sync

- Stability Level:ALPHA
- Type: Histogram

endpoint\_slice\_mirroring\_controller\_endpoints\_sync\_duration  
Duration of syncEndpoints() in seconds

- Stability Level:ALPHA
- Type: Histogram

endpoint\_slice\_mirroring\_controller\_endpoints\_updated\_per\_sync  
Number of endpoints updated on each Endpoints sync

- Stability Level:ALPHA
- Type: Histogram

endpoint\_slice\_mirroring\_controller\_num\_endpoint\_slices  
Number of EndpointSlices

- Stability Level:ALPHA
- Type: Gauge

ephemeral\_volume\_controller\_create\_failures\_total  
Number of PersistenVolumeClaims creation requests

- Stability Level:ALPHA
- Type: Counter

ephemeral\_volume\_controller\_create\_total  
Number of PersistenVolumeClaims creation requests

- Stability Level:ALPHA
- Type: Counter

etcd\_bookmark\_counts  
Number of etcd bookmarks (progress notify events) split by kind.

- Stability Level:ALPHA
- Type: Gauge
- Labels:resource

etcd\_lease\_object\_counts  
Number of objects attached to a single etcd lease.

- Stability Level:ALPHA
- Type: Histogram

etcd\_request\_duration\_seconds  
Etd request latency in seconds for each operation and object type.

- Stability Level:ALPHA
- Type: Histogram
- Labels:operationtype

etcd\_request\_errors\_total

Etcd failed request counts for each operation and object type.

- Stability Level:ALPHA
- Type: Counter
- Labels:operationtype

etcd\_requests\_total

Etcd request counts for each operation and object type.

- Stability Level:ALPHA
- Type: Counter
- Labels:operationtype

etcd\_version\_info

Etcd server's binary version

- Stability Level:ALPHA
- Type: Gauge
- Labels:binary\_version

field\_validation\_request\_duration\_seconds

Response latency distribution in seconds for each field validation value

- Stability Level:ALPHA
- Type: Histogram
- Labels:field\_validation

force\_cleaned\_failed\_volume\_operation\_errors\_total

The number of volumes that failed force cleanup after their reconstruction failed during kubelet startup.

- Stability Level:ALPHA
- Type: Counter

force\_cleaned\_failed\_volume\_operations\_total

The number of volumes that were force cleaned after their reconstruction failed during kubelet startup. This includes both successful and failed cleanups.

- Stability Level:ALPHA
- Type: Counter

garbagecollector\_controller\_resources\_sync\_error\_total

Number of garbage collector resources sync errors

- Stability Level:ALPHA
- Type: Counter

get\_token\_count

Counter of total Token() requests to the alternate token source

- Stability Level:ALPHA
- Type: Counter

get\_token\_fail\_count

Counter of failed Token() requests to the alternate token source

- Stability Level:ALPHA
- Type: Counter

horizontal\_pod\_autoscaler\_controller\_metric\_computation\_duration\_seconds

The time(seconds) that the HPA controller takes to calculate one metric. The label 'action' should be either 'scale\_down', 'scale\_up', or 'none'. The label 'error' should be either 'spec', 'internal', or 'none'. The label 'metric\_type' corresponds to HPA.spec.metrics[\*].type

- Stability Level:ALPHA
- Type: Histogram
- Labels:actionerrormetric\_type

horizontal\_pod\_autoscaler\_controller\_metric\_computation\_total

Number of metric computations. The label 'action' should be either 'scale\_down', 'scale\_up', or 'none'. Also, the label 'error' should be either 'spec', 'internal', or 'none'. The label 'metric\_type' corresponds to HPA.spec.metrics[\*].type

- Stability Level:ALPHA
- Type: Counter
- Labels:actionerrormetric\_type

horizontal\_pod\_autoscaler\_controller\_reconciliation\_duration\_seconds

The time(seconds) that the HPA controller takes to reconcile once. The label 'action' should be either 'scale\_down', 'scale\_up', or 'none'. Also, the label 'error' should be either 'spec', 'internal', or 'none'. Note that if both spec and internal errors happen during a reconciliation, the first one to occur is reported in `error` label.

- Stability Level:ALPHA
- Type: Histogram
- Labels:actionerror

horizontal\_pod\_autoscaler\_controller\_reconciliations\_total

Number of reconciliations of HPA controller. The label 'action' should be either 'scale\_down', 'scale\_up', or 'none'. Also, the label 'error' should be either 'spec', 'internal', or 'none'. Note that if both spec and internal errors happen during a reconciliation, the first one to occur is reported in `error` label.

- Stability Level:ALPHA
- Type: Counter
- Labels:actionerror

job\_controller\_job\_finished\_indexes\_total

`The number of finished indexes. Possible values for the, status label are: "succeeded", "failed". Possible values for the, backoffLimit label are: "perIndex" and "global"`

- Stability Level:ALPHA
- Type: Counter
- Labels:backoffLimitstatus

job\_controller\_job\_pods\_creation\_total

`The number of Pods created by the Job controller labelled with a reason for the Pod creation., This metric also distinguishes between Pods created using different PodReplacementPolicy

settings., Possible values of the "reason" label are:, "new", "recreate\_terminating\_or\_failed", "recreate\_failed"., Possible values of the "status" label are:, "succeeded", "failed".`

- Stability Level:ALPHA
- Type: Counter
- Labels:reasonstatus

job\_controller\_jobs\_by\_external\_controller\_total  
The number of Jobs managed by an external controller

- Stability Level:ALPHA
- Type: Counter
- Labels:controller\_name

job\_controller\_pod\_failures\_handled\_by\_failure\_policy\_total  
`The number of failed Pods handled by failure policy with, respect to the failure policy action applied based on the matched, rule. Possible values of the action label correspond to the, possible values for the failure policy rule action, which are:, "FailJob", "Ignore" and "Count".`

- Stability Level:ALPHA
- Type: Counter
- Labels:action

job\_controller\_terminated\_pods\_tracking\_finalizer\_total  
`The number of terminated pods (phase=Failed|Succeeded), that have the finalizer batch.kubernetes.io/job-tracking, The event label can be "add" or "delete".`

- Stability Level:ALPHA
- Type: Counter
- Labels:event

kube\_apiserver\_clusterip\_allocator\_allocated\_ips  
Gauge measuring the number of allocated IPs for Services

- Stability Level:ALPHA
- Type: Gauge
- Labels:cidr

kube\_apiserver\_clusterip\_allocator\_allocation\_errors\_total  
Number of errors trying to allocate Cluster IPs

- Stability Level:ALPHA
- Type: Counter
- Labels:cidrscope

kube\_apiserver\_clusterip\_allocator\_allocation\_total  
Number of Cluster IPs allocations

- Stability Level:ALPHA
- Type: Counter
- Labels:cidrscope

kube\_apiserver\_clusterip\_allocator\_available\_ips

Gauge measuring the number of available IPs for Services

- Stability Level:ALPHA
- Type: Gauge
- Labels:cidr

kube\_apiserver\_nodeport\_allocator\_allocated\_ports

Gauge measuring the number of allocated NodePorts for Services

- Stability Level:ALPHA
- Type: Gauge

kube\_apiserver\_nodeport\_allocator\_available\_ports

Gauge measuring the number of available NodePorts for Services

- Stability Level:ALPHA
- Type: Gauge

kube\_apiserver\_pod\_logs\_backend\_tls\_failure\_total

Total number of requests for pods/logs that failed due to kubelet server TLS verification

- Stability Level:ALPHA
- Type: Counter

kube\_apiserver\_pod\_logs\_insecure\_backend\_total

Total number of requests for pods/logs sliced by usage type: enforce\_tls, skip\_tls\_allowed, skip\_tls\_denied

- Stability Level:ALPHA
- Type: Counter
- Labels:usage

kube\_apiserver\_pod\_logs\_pods\_logs\_backend\_tls\_failure\_total

Total number of requests for pods/logs that failed due to kubelet server TLS verification

- Stability Level:ALPHA
- Type: Counter
- Deprecated Versions:1.27.0

kube\_apiserver\_pod\_logs\_pods\_logs\_insecure\_backend\_total

Total number of requests for pods/logs sliced by usage type: enforce\_tls, skip\_tls\_allowed, skip\_tls\_denied

- Stability Level:ALPHA
- Type: Counter
- Labels:usage
- Deprecated Versions:1.27.0

kubelet\_active\_pods

The number of pods the kubelet considers active and which are being considered when admitting new pods. static is true if the pod is not from the apiserver.

- Stability Level:ALPHA
- Type: Gauge
- Labels:static

kubelet\_certificate\_manager\_client\_expiration\_renew\_errors

Counter of certificate renewal errors.

- Stability Level:ALPHA
- Type: Counter

kubelet\_certificate\_manager\_client\_ttl\_seconds

Gauge of the TTL (time-to-live) of the Kubelet's client certificate. The value is in seconds until certificate expiry (negative if already expired). If client certificate is invalid or unused, the value will be +INF.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_certificate\_manager\_server\_rotation\_seconds

Histogram of the number of seconds the previous certificate lived before being rotated.

- Stability Level:ALPHA
- Type: Histogram

kubelet\_certificate\_manager\_server\_ttl\_seconds

Gauge of the shortest TTL (time-to-live) of the Kubelet's serving certificate. The value is in seconds until certificate expiry (negative if already expired). If serving certificate is invalid or unused, the value will be +INF.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_cgroup\_manager\_duration\_seconds

Duration in seconds for cgroup manager operations. Broken down by method.

- Stability Level:ALPHA
- Type: Histogram
- Labels:operation\_type

kubelet\_container\_log\_filesystem\_used\_bytes

Bytes used by the container's logs on the filesystem.

- Stability Level:ALPHA
- Type: Custom
- Labels:uidnamespacepodcontainer

kubelet\_containers\_per\_pod\_count

The number of containers per pod.

- Stability Level:ALPHA
- Type: Histogram

kubelet\_cpu\_manager\_pinning\_errors\_total

The number of cpu core allocations which required pinning failed.

- Stability Level:ALPHA
- Type: Counter

kubelet\_cpu\_manager\_pinning\_requests\_total

The number of cpu core allocations which required pinning.

- Stability Level:ALPHA
- Type: Counter

kubelet\_credential\_provider\_plugin\_duration

Duration of execution in seconds for credential provider plugin

- Stability Level:ALPHA
- Type: Histogram
- Labels:plugin\_name

kubelet\_credential\_provider\_plugin\_errors

Number of errors from credential provider plugin

- Stability Level:ALPHA
- Type: Counter
- Labels:plugin\_name

kubelet\_desired\_pods

The number of pods the kubelet is being instructed to run. static is true if the pod is not from the apiserver.

- Stability Level:ALPHA
- Type: Gauge
- Labels:static

kubelet\_device\_plugin\_alloc\_duration\_seconds

Duration in seconds to serve a device plugin Allocation request. Broken down by resource name.

- Stability Level:ALPHA
- Type: Histogram
- Labels:resource\_name

kubelet\_device\_plugin\_registration\_total

Cumulative number of device plugin registrations. Broken down by resource name.

- Stability Level:ALPHA
- Type: Counter
- Labels:resource\_name

kubelet\_evented\_pleg\_connection\_error\_count

The number of errors encountered during the establishment of streaming connection with the CRI runtime.

- Stability Level:ALPHA
- Type: Counter

kubelet\_evented\_pleg\_connection\_latency\_seconds

The latency of streaming connection with the CRI runtime, measured in seconds.

- Stability Level:ALPHA
- Type: Histogram

kubelet\_evented\_pleg\_connection\_success\_count

The number of times a streaming client was obtained to receive CRI Events.

- Stability Level:ALPHA
- Type: Counter

kubelet\_eviction\_stats\_age\_seconds

Time between when stats are collected, and when pod is evicted based on those stats by eviction signal

- Stability Level:ALPHA
- Type: Histogram
- Labels:eviction\_signal

kubelet\_evictions

Cumulative number of pod evictions by eviction signal

- Stability Level:ALPHA
- Type: Counter
- Labels:eviction\_signal

kubelet\_graceful\_shutdown\_end\_time\_seconds

Last graceful shutdown start time since unix epoch in seconds

- Stability Level:ALPHA
- Type: Gauge

kubelet\_graceful\_shutdown\_start\_time\_seconds

Last graceful shutdown start time since unix epoch in seconds

- Stability Level:ALPHA
- Type: Gauge

kubelet\_http\_inflight\_requests

Number of the inflight http requests

- Stability Level:ALPHA
- Type: Gauge
- Labels:long\_runningmethodpathserver\_type

kubelet\_http\_requests\_duration\_seconds

Duration in seconds to serve http requests

- Stability Level:ALPHA
- Type: Histogram
- Labels:long\_runningmethodpathserver\_type

kubelet\_http\_requests\_total

Number of the http requests received since the server started

- Stability Level:ALPHA
- Type: Counter
- Labels:long\_runningmethodpathserver\_type

kubelet\_image\_garbage\_collected\_total



Total number of images garbage collected by the kubelet, whether through disk usage or image age.

- Stability Level:ALPHA
- Type: Counter
- Labels:reason

kubelet\_image\_pull\_duration\_seconds

Duration in seconds to pull an image.

- Stability Level:ALPHA
- Type: Histogram
- Labels:image\_size\_in\_bytes

kubelet\_lifecycle\_handler\_http\_fallbacks\_total

The number of times lifecycle handlers successfully fell back to http from https.

- Stability Level:ALPHA
- Type: Counter

kubelet\_managed\_ephemeral\_containers

Current number of ephemeral containers in pods managed by this kubelet.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_memory\_manager\_pinning\_errors\_total

The number of memory pages allocations which required pinning that failed.

- Stability Level:ALPHA
- Type: Counter

kubelet\_memory\_manager\_pinning\_requests\_total

The number of memory pages allocations which required pinning.

- Stability Level:ALPHA
- Type: Counter

kubelet\_mirror\_pods

The number of mirror pods the kubelet will try to create (one per admitted static pod)

- Stability Level:ALPHA
- Type: Gauge

kubelet\_node\_name

The node's name. The count is always 1.

- Stability Level:ALPHA
- Type: Gauge
- Labels:node

kubelet\_node\_startup\_duration\_seconds

Duration in seconds of node startup in total.

- Stability Level:ALPHA

- Type: Gauge

kubelet\_node\_startup\_post\_registration\_duration\_seconds  
Duration in seconds of node startup after registration.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_node\_startup\_pre\_kubelet\_duration\_seconds  
Duration in seconds of node startup before kubelet starts.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_node\_startup\_pre\_registration\_duration\_seconds  
Duration in seconds of node startup before registration.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_node\_startup\_registration\_duration\_seconds  
Duration in seconds of node startup during registration.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_orphan\_pod\_cleaned\_volumes  
The total number of orphaned Pods whose volumes were cleaned in the last periodic sweep.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_orphan\_pod\_cleaned\_volumes\_errors  
The number of orphaned Pods whose volumes failed to be cleaned in the last periodic sweep.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_orphaned\_runtime\_pods\_total  
Number of pods that have been detected in the container runtime without being already known to the pod worker. This typically indicates the kubelet was restarted while a pod was force deleted in the API or in the local configuration, which is unusual.

- Stability Level:ALPHA
- Type: Counter

kubelet\_pleg\_discard\_events  
The number of discard events in PLEG.

- Stability Level:ALPHA
- Type: Counter

kubelet\_pleg\_last\_seen\_seconds

Timestamp in seconds when PLEG was last seen active.

- Stability Level:ALPHA
- Type: Gauge

kubelet\_pleg\_relist\_duration\_seconds

Duration in seconds for relisting pods in PLEG.

- Stability Level:ALPHA
- Type: Histogram

kubelet\_pleg\_relist\_interval\_seconds

Interval in seconds between relisting in PLEG.

- Stability Level:ALPHA
- Type: Histogram

kubelet\_pod\_resources\_endpoint\_errors\_get

Number of requests to the PodResource Get endpoint which returned error. Broken down by server api version.

- Stability Level:ALPHA
- Type: Counter
- Labels:server\_api\_version

kubelet\_pod\_resources\_endpoint\_errors\_get\_allocatable

Number of requests to the PodResource GetAllocatableResources endpoint which returned error. Broken down by server api version.

- Stability Level:ALPHA
- Type: Counter
- Labels:server\_api\_version

kubelet\_pod\_resources\_endpoint\_errors\_list

Number of requests to the PodResource List endpoint which returned error. Broken down by server api version.

- Stability Level:ALPHA
- Type: Counter
- Labels:server\_api\_version

kubelet\_pod\_resources\_endpoint\_requests\_get

Number of requests to the PodResource Get endpoint. Broken down by server api version.

- Stability Level:ALPHA
- Type: Counter
- Labels:server\_api\_version

kubelet\_pod\_resources\_endpoint\_requests\_get\_allocatable

Number of requests to the PodResource GetAllocatableResources endpoint. Broken down by server api version.

- Stability Level:ALPHA
- Type: Counter
- Labels:server\_api\_version

kubelet\_pod\_resources\_endpoint\_requests\_list

Number of requests to the PodResource List endpoint. Broken down by server api version.

- Stability Level:ALPHA
- Type: Counter
- Labels:server\_api\_version

kubelet\_pod\_resources\_endpoint\_requests\_total

Cumulative number of requests to the PodResource endpoint. Broken down by server api version.

- Stability Level:ALPHA
- Type: Counter
- Labels:server\_api\_version

kubelet\_pod\_start\_duration\_seconds

Duration in seconds from kubelet seeing a pod for the first time to the pod starting to run

- Stability Level:ALPHA
- Type: Histogram

kubelet\_pod\_start\_sli\_duration\_seconds

Duration in seconds to start a pod, excluding time to pull images and run init containers, measured from pod creation timestamp to when all its containers are reported as started and observed via watch

- Stability Level:ALPHA
- Type: Histogram

kubelet\_pod\_start\_total\_duration\_seconds

Duration in seconds to start a pod since creation, including time to pull images and run init containers, measured from pod creation timestamp to when all its containers are reported as started and observed via watch

- Stability Level:ALPHA
- Type: Histogram

kubelet\_pod\_status\_sync\_duration\_seconds

Duration in seconds to sync a pod status update. Measures time from detection of a change to pod status until the API is successfully updated for that pod, even if multiple intervening changes to pod status occur.

- Stability Level:ALPHA
- Type: Histogram

kubelet\_pod\_worker\_duration\_seconds

Duration in seconds to sync a single pod. Broken down by operation type: create, update, or sync

- Stability Level:ALPHA
- Type: Histogram
- Labels:operation\_type

kubelet\_pod\_worker\_start\_duration\_seconds

Duration in seconds from kubelet seeing a pod to starting a worker.

- Stability Level:ALPHA
- Type: Histogram

kubelet\_preemptions

Cumulative number of pod preemptions by preemption resource

- Stability Level:ALPHA
- Type: Counter
- Labels:preemption\_signal

kubelet\_restarted\_pods\_total

Number of pods that have been restarted because they were deleted and recreated with the same UID while the kubelet was watching them (common for static pods, extremely uncommon for API pods)

- Stability Level:ALPHA
- Type: Counter
- Labels:static

kubelet\_run\_podsandbox\_duration\_seconds

Duration in seconds of the run\_podsandbox operations. Broken down by RuntimeClass.Handler.

- Stability Level:ALPHA
- Type: Histogram
- Labels:runtime\_handler

kubelet\_run\_podsandbox\_errors\_total

Cumulative number of the run\_podsandbox operation errors by RuntimeClass.Handler.

- Stability Level:ALPHA
- Type: Counter
- Labels:runtime\_handler

kubelet\_running\_containers

Number of containers currently running

- Stability Level:ALPHA
- Type: Gauge
- Labels:container\_state

kubelet\_running\_pods

Number of pods that have a running pod sandbox

- Stability Level:ALPHA
- Type: Gauge

kubelet\_runtime\_operations\_duration\_seconds

Duration in seconds of runtime operations. Broken down by operation type.

- Stability Level:ALPHA
- Type: Histogram
- Labels:operation\_type

kubelet\_runtime\_operations\_errors\_total

Cumulative number of runtime operation errors by operation type.

- Stability Level:ALPHA
- Type: Counter
- Labels:operation\_type

kubelet\_runtime\_operations\_total

Cumulative number of runtime operations by operation type.

- Stability Level:ALPHA
- Type: Counter
- Labels:operation\_type

kubelet\_server\_expiration\_renew\_errors

Counter of certificate renewal errors.

- Stability Level:ALPHA
- Type: Counter

kubelet\_sleep\_action\_terminated\_early\_total

The number of times lifecycle sleep handler got terminated before it finishes

- Stability Level:ALPHA
- Type: Counter

kubelet\_started\_containers\_errors\_total

Cumulative number of errors when starting containers

- Stability Level:ALPHA
- Type: Counter
- Labels:codecontainer\_type

kubelet\_started\_containers\_total

Cumulative number of containers started

- Stability Level:ALPHA
- Type: Counter
- Labels:container\_type

kubelet\_started\_host\_process\_containers\_errors\_total

Cumulative number of errors when starting hostprocess containers. This metric will only be collected on Windows.

- Stability Level:ALPHA
- Type: Counter
- Labels:codecontainer\_type

kubelet\_started\_host\_process\_containers\_total

Cumulative number of hostprocess containers started. This metric will only be collected on Windows.

- Stability Level:ALPHA
- Type: Counter
- Labels:container\_type

kubelet\_started\_pods\_errors\_total  
Cumulative number of errors when starting pods

- Stability Level:ALPHA
- Type: Counter

kubelet\_started\_pods\_total  
Cumulative number of pods started

- Stability Level:ALPHA
- Type: Counter

kubelet\_topology\_manager\_admission\_duration\_ms  
Duration in milliseconds to serve a pod admission request.

- Stability Level:ALPHA
- Type: Histogram

kubelet\_topology\_manager\_admission\_errors\_total  
The number of admission request failures where resources could not be aligned.

- Stability Level:ALPHA
- Type: Counter

kubelet\_topology\_manager\_admission\_requests\_total  
The number of admission requests where resources have to be aligned.

- Stability Level:ALPHA
- Type: Counter

kubelet\_volume\_metric\_collection\_duration\_seconds  
Duration in seconds to calculate volume stats

- Stability Level:ALPHA
- Type: Histogram
- Labels:metric\_source

kubelet\_volume\_stats\_available\_bytes  
Number of available bytes in the volume

- Stability Level:ALPHA
- Type: Custom
- Labels:namespacepersistentvolumeclaim

kubelet\_volume\_stats\_capacity\_bytes  
Capacity in bytes of the volume

- Stability Level:ALPHA
- Type: Custom
- Labels:namespacepersistentvolumeclaim

kubelet\_volume\_stats\_health\_status\_abnormal

Abnormal volume health status. The count is either 1 or 0. 1 indicates the volume is unhealthy, 0 indicates volume is healthy

- Stability Level:ALPHA
- Type: Custom
- Labels:namespacepersistentvolumeclaim

kubelet\_volume\_stats\_inodes

Maximum number of inodes in the volume

- Stability Level:ALPHA
- Type: Custom
- Labels:namespacepersistentvolumeclaim

kubelet\_volume\_stats\_inodes\_free

Number of free inodes in the volume

- Stability Level:ALPHA
- Type: Custom
- Labels:namespacepersistentvolumeclaim

kubelet\_volume\_stats\_inodes\_used

Number of used inodes in the volume

- Stability Level:ALPHA
- Type: Custom
- Labels:namespacepersistentvolumeclaim

kubelet\_volume\_stats\_used\_bytes

Number of used bytes in the volume

- Stability Level:ALPHA
- Type: Custom
- Labels:namespacepersistentvolumeclaim

kubelet\_working\_pods

Number of pods the kubelet is actually running, broken down by lifecycle phase, whether the pod is desired, orphaned, or runtime only (also orphaned), and whether the pod is static. An orphaned pod has been removed from local configuration or force deleted in the API and consumes resources that are not otherwise visible.

- Stability Level:ALPHA
- Type: Gauge
- Labels:configlifecyclestatic

kubeproxy\_network\_programming\_duration\_seconds

In Cluster Network Programming Latency in seconds

- Stability Level:ALPHA
- Type: Histogram

kubeproxy\_proxy\_healthz\_total

Cumulative proxy healthz HTTP status

- Stability Level:ALPHA



- Type: Counter
- Labels:code

kubeproxy\_proxy\_livez\_total  
Cumulative proxy livez HTTP status

- Stability Level:ALPHA
- Type: Counter
- Labels:code

kubeproxy\_sync\_full\_proxy\_rules\_duration\_seconds  
SyncProxyRules latency in seconds for full resyncs

- Stability Level:ALPHA
- Type: Histogram

kubeproxy\_sync\_partial\_proxy\_rules\_duration\_seconds  
SyncProxyRules latency in seconds for partial resyncs

- Stability Level:ALPHA
- Type: Histogram

kubeproxy\_sync\_proxy\_rules\_duration\_seconds  
SyncProxyRules latency in seconds

- Stability Level:ALPHA
- Type: Histogram

kubeproxy\_sync\_proxy\_rules\_endpoint\_changes\_pending  
Pending proxy rules Endpoint changes

- Stability Level:ALPHA
- Type: Gauge

kubeproxy\_sync\_proxy\_rules\_endpoint\_changes\_total  
Cumulative proxy rules Endpoint changes

- Stability Level:ALPHA
- Type: Counter

kubeproxy\_sync\_proxy\_rules\_iptables\_last  
Number of iptables rules written by kube-proxy in last sync

- Stability Level:ALPHA
- Type: Gauge
- Labels:table

kubeproxy\_sync\_proxy\_rules\_iptables\_partial\_restore\_failures\_total  
Cumulative proxy iptables partial restore failures

- Stability Level:ALPHA
- Type: Counter

kubeproxy\_sync\_proxy\_rules\_iptables\_restore\_failures\_total

## Cumulative proxy iptables restore failures

- Stability Level:ALPHA
- Type: Counter

## kubeproxy\_sync\_proxy\_rules\_iptables\_total

Total number of iptables rules owned by kube-proxy

- Stability Level:ALPHA
- Type: Gauge
- Labels:table

## kubeproxy\_sync\_proxy\_rules\_last\_queued\_timestamp\_seconds

The last time a sync of proxy rules was queued

- Stability Level:ALPHA
- Type: Gauge

## kubeproxy\_sync\_proxy\_rules\_last\_timestamp\_seconds

The last time proxy rules were successfully synced

- Stability Level:ALPHA
- Type: Gauge

## kubeproxy\_sync\_proxy\_rules\_no\_local\_endpoints\_total

Number of services with a Local traffic policy and no endpoints

- Stability Level:ALPHA
- Type: Gauge
- Labels:traffic\_policy

## kubeproxy\_sync\_proxy\_rules\_service\_changes\_pending

Pending proxy rules Service changes

- Stability Level:ALPHA
- Type: Gauge

## kubeproxy\_sync\_proxy\_rules\_service\_changes\_total

Cumulative proxy rules Service changes

- Stability Level:ALPHA
- Type: Counter

## kubernetes\_build\_info

A metric with a constant '1' value labeled by major, minor, git version, git commit, git tree state, build date, Go version, and compiler from which Kubernetes was built, and platform on which it is running.

- Stability Level:ALPHA
- Type: Gauge
- Labels:build\_datecompilergit\_commitgit\_tree\_stategit\_versiongo\_versionmajorminormapplatform

## leader\_election\_master\_status

Gauge of if the reporting system is master of the relevant lease, 0 indicates backup, 1 indicates master. 'name' is the string used to identify the lease. Please make sure to group by name.

- Stability Level:ALPHA
- Type: Gauge
- Labels:name

leader\_election\_slowpath\_total

Total number of slow path exercised in renewing leader leases. 'name' is the string used to identify the lease. Please make sure to group by name.

- Stability Level:ALPHA
- Type: Counter
- Labels:name

node\_authorizer\_graph\_actions\_duration\_seconds

Histogram of duration of graph actions in node authorizer.

- Stability Level:ALPHA
- Type: Histogram
- Labels:operation

node\_collector\_unhealthy\_nodes\_in\_zone

Gauge measuring number of not Ready Nodes per zones.

- Stability Level:ALPHA
- Type: Gauge
- Labels:zone

node\_collector\_update\_all\_nodes\_health\_duration\_seconds

Duration in seconds for NodeController to update the health of all nodes.

- Stability Level:ALPHA
- Type: Histogram

node\_collector\_update\_node\_health\_duration\_seconds

Duration in seconds for NodeController to update the health of a single node.

- Stability Level:ALPHA
- Type: Histogram

node\_collector\_zone\_health

Gauge measuring percentage of healthy nodes per zone.

- Stability Level:ALPHA
- Type: Gauge
- Labels:zone

node\_collector\_zone\_size

Gauge measuring number of registered Nodes per zones.

- Stability Level:ALPHA
- Type: Gauge
- Labels:zone

node\_controller\_cloud\_provider\_taint\_removal\_delay\_seconds

Number of seconds after node creation when NodeController removed the cloud-provider taint of a single node.

- Stability Level:ALPHA
- Type: Histogram

node\_controller\_initial\_node\_sync\_delay\_seconds

Number of seconds after node creation when NodeController finished the initial synchronization of a single node.

- Stability Level:ALPHA
- Type: Histogram

node\_ipam\_controller\_cidrset\_allocation\_tries\_per\_request

Number of endpoints added on each Service sync

- Stability Level:ALPHA
- Type: Histogram
- Labels:clusterCIDR

node\_ipam\_controller\_cidrset\_cidrs\_allocations\_total

Counter measuring total number of CIDR allocations.

- Stability Level:ALPHA
- Type: Counter
- Labels:clusterCIDR

node\_ipam\_controller\_cidrset\_cidrs\_releases\_total

Counter measuring total number of CIDR releases.

- Stability Level:ALPHA
- Type: Counter
- Labels:clusterCIDR

node\_ipam\_controller\_cidrset\_usage\_cidrs

Gauge measuring percentage of allocated CIDRs.

- Stability Level:ALPHA
- Type: Gauge
- Labels:clusterCIDR

node\_ipam\_controller\_cirdset\_max\_cidrs

Maximum number of CIDRs that can be allocated.

- Stability Level:ALPHA
- Type: Gauge
- Labels:clusterCIDR

node\_swap\_usage\_bytes

Current swap usage of the node in bytes. Reported only on non-windows systems

- Stability Level:ALPHA
- Type: Custom

number\_of\_l4\_ilbs

Number of L4 ILBs

- Stability Level:ALPHA
- Type: Gauge
- Labels:feature

plugin\_manager\_total\_plugins

Number of plugins in Plugin Manager

- Stability Level:ALPHA
- Type: Custom
- Labels:socket\_pathstate

pod\_gc\_collector\_force\_delete\_pod\_errors\_total

Number of errors encountered when forcefully deleting the pods since the Pod GC Controller started.

- Stability Level:ALPHA
- Type: Counter
- Labels:namespace

pod\_gc\_collector\_force\_delete\_pods\_total

Number of pods that are being forcefully deleted since the Pod GC Controller started.

- Stability Level:ALPHA
- Type: Counter
- Labels:namespace

pod\_security\_errors\_total

Number of errors preventing normal evaluation. Non-fatal errors may result in the latest restricted profile being used for evaluation.

- Stability Level:ALPHA
- Type: Counter
- Labels:fatalrequest\_operationresourcesubresource

pod\_security\_evaluations\_total

Number of policy evaluations that occurred, not counting ignored or exempt requests.

- Stability Level:ALPHA
- Type: Counter
- Labels:decisionmodepolicy\_levelpolicy\_versionrequest\_operationresourcesubresource

pod\_security\_exemptions\_total

Number of exempt requests, not counting ignored or out of scope requests.

- Stability Level:ALPHA
- Type: Counter
- Labels:request\_operationresourcesubresource

pod\_swap\_usage\_bytes

Current amount of the pod swap usage in bytes. Reported only on non-windows systems

- Stability Level:ALPHA

- Type: Custom
- Labels: podnamespace

prober\_probe\_duration\_seconds

Duration in seconds for a probe response.

- Stability Level: ALPHA
- Type: Histogram
- Labels: containernamespacepodprobe\_type

prober\_probe\_total

Cumulative number of a liveness, readiness or startup probe for a container by result.

- Stability Level: ALPHA
- Type: Counter
- Labels: containernamespacepodpod\_uidprobe\_typeresult

pv\_collector\_bound\_pv\_count

Gauge measuring number of persistent volume currently bound

- Stability Level: ALPHA
- Type: Custom
- Labels: storage\_class

pv\_collector\_bound\_pvc\_count

Gauge measuring number of persistent volume claim currently bound

- Stability Level: ALPHA
- Type: Custom
- Labels: namespace

pv\_collector\_total\_pv\_count

Gauge measuring total number of persistent volumes

- Stability Level: ALPHA
- Type: Custom
- Labels: plugin\_namevolume\_mode

pv\_collector\_unbound\_pv\_count

Gauge measuring number of persistent volume currently unbound

- Stability Level: ALPHA
- Type: Custom
- Labels: storage\_class

pv\_collector\_unbound\_pvc\_count

Gauge measuring number of persistent volume claim currently unbound

- Stability Level: ALPHA
- Type: Custom
- Labels: namespace

reconstruct\_volume\_operations\_errors\_total

The number of volumes that failed reconstruction from the operating system during kubelet startup.

- Stability Level:ALPHA
- Type: Counter

reconstruct\_volume\_operations\_total

The number of volumes that were attempted to be reconstructed from the operating system during kubelet startup. This includes both successful and failed reconstruction.

- Stability Level:ALPHA
- Type: Counter

replicaset\_controller\_sorting\_deletion\_age\_ratio

The ratio of chosen deleted pod's ages to the current youngest pod's age (at the time). Should be <2. The intent of this metric is to measure the rough efficacy of the LogarithmicScaleDown feature gate's effect on the sorting (and deletion) of pods when a replicaset scales down. This only considers Ready pods when calculating and reporting.

- Stability Level:ALPHA
- Type: Histogram

resourceclaim\_controller\_create\_attempts\_total

Number of ResourceClaims creation requests

- Stability Level:ALPHA
- Type: Counter

resourceclaim\_controller\_create\_failures\_total

Number of ResourceClaims creation request failures

- Stability Level:ALPHA
- Type: Counter

rest\_client\_dns\_resolution\_duration\_seconds

DNS resolver latency in seconds. Broken down by host.

- Stability Level:ALPHA
- Type: Histogram
- Labels:host

rest\_client\_exec\_plugin\_call\_total

Number of calls to an exec plugin, partitioned by the type of event encountered (no\_error, plugin\_execution\_error, plugin\_not\_found\_error, client\_internal\_error) and an optional exit code. The exit code will be set to 0 if and only if the plugin call was successful.

- Stability Level:ALPHA
- Type: Counter
- Labels:call\_statuscode

rest\_client\_exec\_plugin\_certificate\_rotation\_age

Histogram of the number of seconds the last auth exec plugin client certificate lived before being rotated. If auth exec plugin client certificates are unused, histogram will contain no data.

- Stability Level:ALPHA

- Type: Histogram

rest\_client\_exec\_plugin\_ttl\_seconds

Gauge of the shortest TTL (time-to-live) of the client certificate(s) managed by the auth exec plugin. The value is in seconds until certificate expiry (negative if already expired). If auth exec plugins are unused or manage no TLS certificates, the value will be +INF.

- Stability Level:ALPHA
- Type: Gauge

rest\_client\_rate\_limiter\_duration\_seconds

Client side rate limiter latency in seconds. Broken down by verb, and host.

- Stability Level:ALPHA
- Type: Histogram
- Labels:hostverb

rest\_client\_request\_duration\_seconds

Request latency in seconds. Broken down by verb, and host.

- Stability Level:ALPHA
- Type: Histogram
- Labels:hostverb

rest\_client\_request\_retries\_total

Number of request retries, partitioned by status code, verb, and host.

- Stability Level:ALPHA
- Type: Counter
- Labels:codehostverb

rest\_client\_request\_size\_bytes

Request size in bytes. Broken down by verb and host.

- Stability Level:ALPHA
- Type: Histogram
- Labels:hostverb

rest\_client\_requests\_total

Number of HTTP requests, partitioned by status code, method, and host.

- Stability Level:ALPHA
- Type: Counter
- Labels:codehostmethod

rest\_client\_response\_size\_bytes

Response size in bytes. Broken down by verb and host.

- Stability Level:ALPHA
- Type: Histogram
- Labels:hostverb

rest\_client\_transport\_cache\_entries



Number of transport entries in the internal cache.

- Stability Level:ALPHA
- Type: Gauge

rest\_client\_transport\_create\_calls\_total

Number of calls to get a new transport, partitioned by the result of the operation hit: obtained from the cache, miss: created and added to the cache, uncacheable: created and not cached

- Stability Level:ALPHA
- Type: Counter
- Labels:result

retroactive\_storageclass\_errors\_total

Total number of failed retroactive StorageClass assignments to persistent volume claim

- Stability Level:ALPHA
- Type: Counter

retroactive\_storageclass\_total

Total number of retroactive StorageClass assignments to persistent volume claim

- Stability Level:ALPHA
- Type: Counter

root\_ca\_cert\_publisher\_sync\_duration\_seconds

Number of namespace syncs happened in root ca cert publisher.

- Stability Level:ALPHA
- Type: Histogram
- Labels:code

root\_ca\_cert\_publisher\_sync\_total

Number of namespace syncs happened in root ca cert publisher.

- Stability Level:ALPHA
- Type: Counter
- Labels:code

running\_managed\_controllers

Indicates where instances of a controller are currently running

- Stability Level:ALPHA
- Type: Gauge
- Labels:managername

scheduler\_goroutines

Number of running goroutines split by the work they do such as binding.

- Stability Level:ALPHA
- Type: Gauge
- Labels:operation

scheduler\_permit\_wait\_duration\_seconds

Duration of waiting on permit.

- Stability Level:ALPHA
- Type: Histogram
- Labels:result

scheduler\_plugin\_evaluation\_total

Number of attempts to schedule pods by each plugin and the extension point (available only in PreFilter, Filter, PreScore, and Score).

- Stability Level:ALPHA
- Type: Counter
- Labels:extension\_pointpluginprofile

scheduler\_plugin\_execution\_duration\_seconds

Duration for running a plugin at a specific extension point.

- Stability Level:ALPHA
- Type: Histogram
- Labels:extension\_pointpluginstatus

scheduler\_scheduler\_cache\_size

Number of nodes, pods, and assumed (bound) pods in the scheduler cache.

- Stability Level:ALPHA
- Type: Gauge
- Labels:type

scheduler\_scheduling\_algorithm\_duration\_seconds

Scheduling algorithm latency in seconds

- Stability Level:ALPHA
- Type: Histogram

scheduler\_unschedulable\_pods

The number of unschedulable pods broken down by plugin name. A pod will increment the gauge for all plugins that caused it to not schedule and so this metric have meaning only when broken down by plugin.

- Stability Level:ALPHA
- Type: Gauge
- Labels:pluginprofile

scheduler\_volume\_binder\_cache\_requests\_total

Total number for request volume binding cache

- Stability Level:ALPHA
- Type: Counter
- Labels:operation

scheduler\_volume\_scheduling\_stage\_error\_total

Volume scheduling stage error count

- Stability Level:ALPHA
- Type: Counter

- Labels:operation

scrape\_error

1 if there was an error while getting container metrics, 0 otherwise

- Stability Level:ALPHA
- Type: Custom
- Deprecated Versions:1.29.0

service\_controller\_loadbalancer\_sync\_total

A metric counting the amount of times any load balancer has been configured, as an effect of service/node changes on the cluster

- Stability Level:ALPHA
- Type: Counter

service\_controller\_nodesync\_error\_total

A metric counting the amount of times any load balancer has been configured and errored, as an effect of node changes on the cluster

- Stability Level:ALPHA
- Type: Counter

service\_controller\_nodesync\_latency\_seconds

A metric measuring the latency for nodesync which updates loadbalancer hosts on cluster node updates.

- Stability Level:ALPHA
- Type: Histogram

service\_controller\_update\_loadbalancer\_host\_latency\_seconds

A metric measuring the latency for updating each load balancer hosts.

- Stability Level:ALPHA
- Type: Histogram

serviceaccount\_invalid\_legacy\_auto\_token\_uses\_total

Cumulative invalid auto-generated legacy tokens used

- Stability Level:ALPHA
- Type: Counter

serviceaccount\_legacy\_auto\_token\_uses\_total

Cumulative auto-generated legacy tokens used

- Stability Level:ALPHA
- Type: Counter

serviceaccount\_legacy\_manual\_token\_uses\_total

Cumulative manually created legacy tokens used

- Stability Level:ALPHA
- Type: Counter

serviceaccount\_legacy\_tokens\_total

Cumulative legacy service account tokens used

- Stability Level:ALPHA
- Type: Counter

serviceaccount\_stale\_tokens\_total

Cumulative stale projected service account tokens used

- Stability Level:ALPHA
- Type: Counter

serviceaccount\_valid\_tokens\_total

Cumulative valid projected service account tokens used

- Stability Level:ALPHA
- Type: Counter

storage\_count\_attachable\_volumes\_in\_use

Measure number of volumes in use

- Stability Level:ALPHA
- Type: Custom
- Labels:nodevolume\_plugin

storage\_operation\_duration\_seconds

Storage operation duration

- Stability Level:ALPHA
- Type: Histogram
- Labels:migratedoperation\_namestatusvolume\_plugin

taint\_eviction\_controller\_pod\_deletion\_duration\_seconds

Latency, in seconds, between the time when a taint effect has been activated for the Pod and its deletion via TaintEvictionController.

- Stability Level:ALPHA
- Type: Histogram

taint\_eviction\_controller\_pod\_deletions\_total

Total number of Pods deleted by TaintEvictionController since its start.

- Stability Level:ALPHA
- Type: Counter

ttl\_after\_finished\_controller\_job\_deletion\_duration\_seconds

The time it took to delete the job since it became eligible for deletion

- Stability Level:ALPHA
- Type: Histogram

volume\_manager\_selinux\_container\_errors\_total

Number of errors when kubelet cannot compute SELinux context for a container. Kubelet can't start such a Pod then and it will retry, therefore value of this metric may not represent the actual nr. of containers.

- Stability Level:ALPHA
- Type: Gauge
- Labels:access\_mode

volume\_manager\_selinux\_container\_warnings\_total

Number of errors when kubelet cannot compute SELinux context for a container that are ignored. They will become real errors when SELinuxMountReadWriteOncePod feature is expanded to all volume access modes.

- Stability Level:ALPHA
- Type: Gauge
- Labels:access\_mode

volume\_manager\_selinux\_pod\_context\_mismatch\_errors\_total

Number of errors when a Pod defines different SELinux contexts for its containers that use the same volume. Kubelet can't start such a Pod then and it will retry, therefore value of this metric may not represent the actual nr. of Pods.

- Stability Level:ALPHA
- Type: Gauge
- Labels:access\_mode

volume\_manager\_selinux\_pod\_context\_mismatch\_warnings\_total

Number of errors when a Pod defines different SELinux contexts for its containers that use the same volume. They are not errors yet, but they will become real errors when SELinuxMountReadWriteOncePod feature is expanded to all volume access modes.

- Stability Level:ALPHA
- Type: Gauge
- Labels:access\_mode

volume\_manager\_selinux\_volume\_context\_mismatch\_errors\_total

Number of errors when a Pod uses a volume that is already mounted with a different SELinux context than the Pod needs. Kubelet can't start such a Pod then and it will retry, therefore value of this metric may not represent the actual nr. of Pods.

- Stability Level:ALPHA
- Type: Gauge
- Labels:access\_modevolume\_plugin

volume\_manager\_selinux\_volume\_context\_mismatch\_warnings\_total

Number of errors when a Pod uses a volume that is already mounted with a different SELinux context than the Pod needs. They are not errors yet, but they will become real errors when SELinuxMountReadWriteOncePod feature is expanded to all volume access modes.

- Stability Level:ALPHA
- Type: Gauge
- Labels:access\_modevolume\_plugin

volume\_manager\_selinux\_volumes\_admitted\_total

Number of volumes whose SELinux context was fine and will be mounted with mount -o context option.

- Stability Level:ALPHA
- Type: Gauge
- Labels:access\_modevolume\_plugin

volume\_manager\_total\_volumes

Number of volumes in Volume Manager

- Stability Level:ALPHA
- Type: Custom
- Labels:plugin\_namestate

volume\_operation\_total\_errors

Total volume operation errors

- Stability Level:ALPHA
- Type: Counter
- Labels:operation\_nameplugin\_name

volume\_operation\_total\_seconds

Storage operation end to end duration in seconds

- Stability Level:ALPHA
- Type: Histogram
- Labels:operation\_nameplugin\_name

watch\_cache\_capacity

Total capacity of watch cache broken by resource type.

- Stability Level:ALPHA
- Type: Gauge
- Labels:resource

watch\_cache\_capacity\_decrease\_total

Total number of watch cache capacity decrease events broken by resource type.

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

watch\_cache\_capacity\_increase\_total

Total number of watch cache capacity increase events broken by resource type.

- Stability Level:ALPHA
- Type: Counter
- Labels:resource

workqueue\_adds\_total

Total number of adds handled by workqueue

- Stability Level:ALPHA
- Type: Counter
- Labels:name

workqueue\_depth  
Current depth of workqueue

- Stability Level:ALPHA
- Type: Gauge
- Labels:name

workqueue\_longest\_running\_processor\_seconds  
How many seconds has the longest running processor for workqueue been running.

- Stability Level:ALPHA
- Type: Gauge
- Labels:name

workqueue\_queue\_duration\_seconds  
How long in seconds an item stays in workqueue before being requested.

- Stability Level:ALPHA
- Type: Histogram
- Labels:name

workqueue\_retries\_total  
Total number of retries handled by workqueue

- Stability Level:ALPHA
- Type: Counter
- Labels:name

workqueue\_unfinished\_work\_seconds  
How many seconds of work has done that is in progress and hasn't been observed by work\_duration. Large values indicate stuck threads. One can deduce the number of stuck threads by observing the rate at which this increases.

- Stability Level:ALPHA
- Type: Gauge
- Labels:name

workqueue\_work\_duration\_seconds  
How long in seconds processing an item from workqueue takes.

- Stability Level:ALPHA
- Type: Histogram
- Labels:name

# Kubernetes Issues and Security

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[Kubernetes Issue Tracker](#)

[Kubernetes Security and Disclosure Information](#)

[Official CVE Feed](#)

# Kubernetes Issue Tracker

To report a security issue, please follow the [Kubernetes security disclosure process](#).

Work on Kubernetes code and public issues are tracked using [GitHub Issues](#).

- Official [list of known CVEs](#) (security vulnerabilities) that have been announced by the [Security Response Committee](#)
- [CVE-related GitHub issues](#)

Security-related announcements are sent to the [kubernetes-security-announce@googlegroups.com](mailto:kubernetes-security-announce@googlegroups.com) mailing list.

## Kubernetes Security and Disclosure Information

This page describes Kubernetes security and disclosure information.

### Security Announcements

Join the [kubernetes-security-announce](#) group for emails about security and major API announcements.

### Report a Vulnerability

We're extremely grateful for security researchers and users that report vulnerabilities to the Kubernetes Open Source Community. All reports are thoroughly investigated by a set of community volunteers.

To make a report, submit your vulnerability to the [Kubernetes bug bounty program](#). This allows triage and handling of the vulnerability with standardized response times.

You can also email the private [security@kubernetes.io](mailto:security@kubernetes.io) list with the security details and the details expected for [all Kubernetes bug reports](#).

You may encrypt your email to this list using the GPG keys of the [Security Response Committee members](#). Encryption using GPG is NOT required to make a disclosure.

### When Should I Report a Vulnerability?

- You think you discovered a potential security vulnerability in Kubernetes
- You are unsure how a vulnerability affects Kubernetes
- You think you discovered a vulnerability in another project that Kubernetes depends on
  - For projects with their own vulnerability reporting and disclosure process, please report it directly there



## When Should I NOT Report a Vulnerability?

- You need help tuning Kubernetes components for security
- You need help applying security related updates
- Your issue is not security related

## Security Vulnerability Response

Each report is acknowledged and analyzed by Security Response Committee members within 3 working days. This will set off the [Security Release Process](#).

Any vulnerability information shared with Security Response Committee stays within Kubernetes project and will not be disseminated to other projects unless it is necessary to get the issue fixed.

As the security issue moves from triage, to identified fix, to release planning we will keep the reporter updated.

## Public Disclosure Timing

A public disclosure date is negotiated by the Kubernetes Security Response Committee and the bug submitter. We prefer to fully disclose the bug as soon as possible once a user mitigation is available. It is reasonable to delay disclosure when the bug or the fix is not yet fully understood, the solution is not well-tested, or for vendor coordination. The timeframe for disclosure is from immediate (especially if it's already publicly known) to a few weeks. For a vulnerability with a straightforward mitigation, we expect report date to disclosure date to be on the order of 7 days. The Kubernetes Security Response Committee holds the final say when setting a disclosure date.

## Official CVE Feed

FEATURE STATE: Kubernetes v1.27 [beta]

This is a community maintained list of official CVEs announced by the Kubernetes Security Response Committee. See [Kubernetes Security and Disclosure Information](#) for more details.

The Kubernetes project publishes a programmatically accessible feed of published security issues in [JSON feed](#) and [RSS feed](#) formats. You can access it by executing the following commands:

- [JSON feed](#)
- [RSS feed](#)

[Link to JSON format](#)

```
curl -Lv https://k8s.io/docs/reference/issues-security/official-cve-feed/index.json
```

[Link to RSS format](#)

```
curl -Lv https://k8s.io/docs/reference/issues-security/official-cve-feed/feed.xml
```

Official Kubernetes CVE List (last updated: 24 May 2024 19:08:31 UTC)

| CVE ID                                                           | Issue Summary                                                                                                                              | CVE GitHub Issue URL    |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <a href="#">CVE-2024-3744</a>                                    | azure-file-csi-driver discloses service account tokens in logs                                                                             | <a href="#">#124759</a> |
| <a href="#">CVE-2024-3177</a>                                    | Bypassing mountable secrets policy imposed by the ServiceAccount admission plugin                                                          | <a href="#">#124336</a> |
| <a href="#">CVE-2023-5528</a>                                    | Insufficient input sanitization in in-tree storage plugin leads to privilege escalation on Windows nodes                                   | <a href="#">#121879</a> |
| <a href="#">CVE-2023-3955</a>                                    | Insufficient input sanitization on Windows nodes leads to privilege escalation                                                             | <a href="#">#119595</a> |
| <a href="#">CVE-2023-3893</a>                                    | Insufficient input sanitization on kubernetes-csi-proxy leads to privilege escalation                                                      | <a href="#">#119594</a> |
| <a href="#">CVE-2023-3676</a>                                    | Insufficient input sanitization on Windows nodes leads to privilege escalation                                                             | <a href="#">#119339</a> |
| <a href="#">CVE-2023-2431</a>                                    | Bypass of seccomp profile enforcement                                                                                                      | <a href="#">#118690</a> |
| <a href="#">CVE-2023-2727</a> ,<br><a href="#">CVE-2023-2728</a> | Bypassing policies imposed by the ImagePolicyWebhook and bypassing mountable secrets policy imposed by the ServiceAccount admission plugin | <a href="#">#118640</a> |
| <a href="#">CVE-2023-2878</a>                                    | secrets-store-csi-driver discloses service account tokens in logs                                                                          | <a href="#">#118419</a> |
| <a href="#">CVE-2022-3294</a>                                    | Node address isn't always verified when proxying                                                                                           | <a href="#">#113757</a> |
| <a href="#">CVE-2022-3162</a>                                    | Unauthorized read of Custom Resources                                                                                                      | <a href="#">#113756</a> |
| <a href="#">CVE-2022-3172</a>                                    | Aggregated API server can cause clients to be redirected (SSRF)                                                                            | <a href="#">#112513</a> |
| <a href="#">CVE-2021-25749</a>                                   | `runAsNonRoot` logic bypass for Windows containers                                                                                         | <a href="#">#112192</a> |
| <a href="#">CVE-2021-25741</a>                                   | Symlink Exchange Can Allow Host Filesystem Access                                                                                          | <a href="#">#104980</a> |
| <a href="#">CVE-2021-25737</a>                                   | Holes in EndpointSlice Validation Enable Host Network Hijack                                                                               | <a href="#">#102106</a> |
| <a href="#">CVE-2021-3121</a>                                    | Processes may panic upon receipt of malicious protobuf messages                                                                            | <a href="#">#101435</a> |
| <a href="#">CVE-2021-25735</a>                                   | Validating Admission Webhook does not observe some previous fields                                                                         | <a href="#">#100096</a> |
| <a href="#">CVE-2020-8554</a>                                    | Man in the middle using LoadBalancer or ExternalIPs                                                                                        | <a href="#">#97076</a>  |
| <a href="#">CVE-2020-8566</a>                                    | Ceph RBD adminSecrets exposed in logs when loglevel >= 4                                                                                   | <a href="#">#95624</a>  |
| <a href="#">CVE-2020-8565</a>                                    | Incomplete fix for CVE-2019-11250 allows for token leak in logs when logLevel >= 9                                                         | <a href="#">#95623</a>  |
| <a href="#">CVE-2020-8564</a>                                    | Docker config secrets leaked when file is malformed and log level >= 4                                                                     | <a href="#">#95622</a>  |
| <a href="#">CVE-2020-8563</a>                                    | Secret leaks in kube-controller-manager when using vSphere provider                                                                        | <a href="#">#95621</a>  |
| <a href="#">CVE-2020-8557</a>                                    | Node disk DOS by writing to container /etc/hosts                                                                                           | <a href="#">#93032</a>  |
| <a href="#">CVE-2020-8559</a>                                    | Privilege escalation from compromised node to cluster                                                                                      | <a href="#">#92914</a>  |
| <a href="#">CVE-2020-8558</a>                                    | Node setting allows for neighboring hosts to bypass localhost boundary                                                                     | <a href="#">#92315</a>  |
| <a href="#">CVE-2020-8555</a>                                    | Half-Blind SSRF in kube-controller-manager                                                                                                 | <a href="#">#91542</a>  |
| <a href="#">CVE-2020-10749</a>                                   |                                                                                                                                            | <a href="#">#91507</a>  |

| CVE ID                           | Issue Summary                                                                                                             | CVE GitHub Issue URL   |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------|
|                                  | IPv4 only clusters susceptible to MitM attacks via IPv6 rogue router advertisements                                       |                        |
| <a href="#">CVE-2019-11254</a>   | kube-apiserver Denial of Service vulnerability from malicious YAML payloads                                               | <a href="#">#89535</a> |
| <a href="#">CVE-2020-8552</a>    | apiserver DoS (oom)                                                                                                       | <a href="#">#89378</a> |
| <a href="#">CVE-2020-8551</a>    | Kubelet DoS via API                                                                                                       | <a href="#">#89377</a> |
| <a href="#">CVE-2019-11251</a>   | kubectl cp symlink vulnerability                                                                                          | <a href="#">#87773</a> |
| <a href="#">CVE-2018-1002102</a> | Unvalidated redirect                                                                                                      | <a href="#">#85867</a> |
| <a href="#">CVE-2019-11255</a>   | CSI volume snapshot, cloning and resizing features can result in unauthorized volume data access or mutation              | <a href="#">#85233</a> |
| <a href="#">CVE-2019-11253</a>   | Kubernetes API Server JSON/YAML parsing vulnerable to resource exhaustion attack                                          | <a href="#">#83253</a> |
| <a href="#">CVE-2019-11250</a>   | Bearer tokens are revealed in logs                                                                                        | <a href="#">#81114</a> |
| <a href="#">CVE-2019-11248</a>   | /debug/pprof exposed on kubelet's healthz port                                                                            | <a href="#">#81023</a> |
| <a href="#">CVE-2019-11249</a>   | Incomplete fixes for CVE-2019-1002101 and CVE-2019-11246, kubectl cp potential directory traversal                        | <a href="#">#80984</a> |
| <a href="#">CVE-2019-11247</a>   | API server allows access to custom resources via wrong scope                                                              | <a href="#">#80983</a> |
| <a href="#">CVE-2019-11245</a>   | container uid changes to root after first restart or if image is already pulled to the node                               | <a href="#">#78308</a> |
| <a href="#">CVE-2019-11243</a>   | rest.AnonymousClientConfig() does not remove the serviceaccount credentials from config created by rest.InClusterConfig() | <a href="#">#76797</a> |
| <a href="#">CVE-2019-11244</a>   | `kubectl:-http-cache=<world-accessible dir>` creates world-writeable cached schema files                                  | <a href="#">#76676</a> |
| <a href="#">CVE-2019-1002100</a> | json-patch requests can exhaust apiserver resources                                                                       | <a href="#">#74534</a> |
| <a href="#">CVE-2018-1002105</a> | proxy request handling in kube-apiserver can leave vulnerable TCP connections                                             | <a href="#">#71411</a> |
| <a href="#">CVE-2018-1002101</a> | smb mount security issue                                                                                                  | <a href="#">#65750</a> |
| <a href="#">CVE-2018-1002100</a> | Kubectl copy doesn't check for paths outside of it's destination directory.                                               | <a href="#">#61297</a> |
| <a href="#">CVE-2017-1002102</a> | atomic writer volume handling allows arbitrary file deletion in host filesystem                                           | <a href="#">#60814</a> |
| <a href="#">CVE-2017-1002101</a> | subpath volume mount handling allows arbitrary file access in host filesystem                                             | <a href="#">#60813</a> |
| <a href="#">CVE-2017-1002100</a> | Azure PV should be Private scope not Container scope                                                                      | <a href="#">#47611</a> |
| <a href="#">CVE-2017-1000056</a> | PodSecurityPolicy admission plugin authorizes incorrectly                                                                 | <a href="#">#43459</a> |

This feed is auto-refreshing with a noticeable but small lag (minutes to hours) from the time a CVE is announced to the time it is accessible in this feed.

The source of truth of this feed is a set of GitHub Issues, filtered by a controlled and restricted label official-cve-feed. The raw data is stored in a Google Cloud Bucket which is writable only by a small number of trusted members of the Community.

# Node Reference Information

This section contains the following reference topics about nodes:

- the kubelet's [checkpoint API](#)
- a list of [Articles on dockershim Removal and on Using CRI-compatible Runtimes](#)
- [Kubelet Device Manager API Versions](#)
- [Node Labels Populated By The Kubelet](#)
- [Node .status information](#)

You can also read node reference details from elsewhere in the Kubernetes documentation, including:

- [Node Metrics Data](#).
- [CRI Pod & Container Metrics](#).

## Kubelet Checkpoint API

FEATURE STATE: Kubernetes v1.25 [alpha]

Checkpointing a container is the functionality to create a stateful copy of a running container. Once you have a stateful copy of a container, you could move it to a different computer for debugging or similar purposes.

If you move the checkpointed container data to a computer that's able to restore it, that restored container continues to run at exactly the same point it was checkpointed. You can also inspect the saved data, provided that you have suitable tools for doing so.

Creating a checkpoint of a container might have security implications. Typically a checkpoint contains all memory pages of all processes in the checkpointed container. This means that everything that used to be in memory is now available on the local disk. This includes all private data and possibly keys used for encryption. The underlying CRI implementations (the container runtime on that node) should create the checkpoint archive to be only accessible by the root user. It is still important to remember if the checkpoint archive is transferred to another system all memory pages will be readable by the owner of the checkpoint archive.

## Operations

### post checkpoint the specified container

Tell the kubelet to checkpoint a specific container from the specified Pod.

Consult the [Kubelet authentication/authorization reference](#) for more information about how access to the kubelet checkpoint interface is controlled.

The kubelet will request a checkpoint from the underlying [CRI](#) implementation. In the checkpoint request the kubelet will specify the name of the checkpoint archive as `checkpoint-<podFullName>-<containerName>-<timestamp>.tar` and also request to store the checkpoint archive in the checkpoints directory below its root directory (as defined by `--root-dir`). This defaults to `/var/lib/kubelet/checkpoints`.

The checkpoint archive is in *tar* format, and could be listed using an implementation of [tar](#). The contents of the archive depend on the underlying CRI implementation (the container runtime on that node).

## HTTP Request

POST /checkpoint/{namespace}/{pod}/{container}

## Parameters

- **namespace** (*in path*): string, required

[Namespace](#)

- **pod** (*in path*): string, required

[Pod](#)

- **container** (*in path*): string, required

[Container](#)

- **timeout** (*in query*): integer

Timeout in seconds to wait until the checkpoint creation is finished. If zero or no timeout is specified the default [CRI](#) timeout value will be used. Checkpoint creation time depends directly on the used memory of the container. The more memory a container uses the more time is required to create the corresponding checkpoint.

## Response

200: OK

401: Unauthorized

404: Not Found (if the ContainerCheckpoint feature gate is disabled)

404: Not Found (if the specified namespace, pod or container cannot be found)

500: Internal Server Error (if the CRI implementation encounter an error during checkpointing (see error message for further details))

500: Internal Server Error (if the CRI implementation does not implement the checkpoint CRI API (see error message for further details))

# Articles on dockershim Removal and on Using CRI-compatible Runtimes

This is a list of articles and other pages that are either about the Kubernetes' deprecation and removal of *dockershim*, or about using CRI-compatible container runtimes, in connection with that removal.

## Kubernetes project

- Kubernetes blog: [Dockershim Removal FAQ](#) (originally published 2020/12/02)
- Kubernetes blog: [Updated: Dockershim Removal FAQ](#) (updated published 2022/02/17)
- Kubernetes blog: [Kubernetes is Moving on From Dockershim: Commitments and Next Steps](#) (published 2022/01/07)
- Kubernetes blog: [Dockershim removal is coming. Are you ready?](#) (published 2021/11/12)
- Kubernetes documentation: [Migrating from dockershim](#)
- Kubernetes documentation: [Container Runtimes](#)
- Kubernetes enhancement proposal: [KEP-2221: Removing dockershim from kubelet](#)
- Kubernetes enhancement proposal issue: [Removing dockershim from kubelet](#) (*k/enhancements#2221*)

You can provide feedback via the GitHub issue [Dockershim removal feedback & issues](#). (*k/kubernetes/#106917*)

## External sources

- Amazon Web Services EKS documentation: [Amazon EKS is ending support for Dockershim](#)
- CNCF conference video: [Lessons Learned Migrating Kubernetes from Docker to containerd Runtime](#) (Ana Caylin, at KubeCon Europe 2019)
- Docker.com blog: [What developers need to know about Docker, Docker Engine, and Kubernetes v1.20](#) (published 2020/12/04)
- "Google Open Source" channel on YouTube: [Learn Kubernetes with Google - Migrating from Dockershim to Containerd](#)
- Microsoft Apps on Azure blog: [Dockershim deprecation and AKS](#) (published 2022/01/21)
- Mirantis blog: [The Future of Dockershim is cri-dockerd](#) (published 2021/04/21)
- Mirantis: [Mirantis/cri-dockerd](#) Official Documentation

- Tripwire: [How Dockershim's Forthcoming Deprecation Affects Your Kubernetes](#)  
• (published 2021/07/01)

# Node Labels Populated By The Kubelet

Kubernetes [nodes](#) come pre-populated with a standard set of [labels](#).

You can also set your own labels on nodes, either through the kubelet configuration or using the Kubernetes API.

## Preset labels

The preset labels that Kubernetes sets on nodes are:

- [kubernetes.io/arch](#)
- [kubernetes.io/hostname](#)
- [kubernetes.io/os](#)
- [node.kubernetes.io/instance-type](#) (if known to the kubelet – Kubernetes may not have this information to set the label)
- [topology.kubernetes.io/region](#) (if known to the kubelet – Kubernetes may not have this information to set the label)
- [topology.kubernetes.io/zone](#) (if known to the kubelet – Kubernetes may not have this information to set the label)

### Note:

The value of these labels is cloud provider specific and is not guaranteed to be reliable. For example, the value of `kubernetes.io/hostname` may be the same as the node name in some environments and a different value in other environments.

## What's next

- See [Well-Known Labels, Annotations and Taints](#) for a list of common labels.
- Learn how to [add a label to a node](#).

# Kubelet Configuration Directory Merging

When using the kubelet's `--config-dir` flag to specify a drop-in directory for configuration, there is some specific behavior on how different types are merged.

Here are some examples of how different data types behave during configuration merging:

## Structure Fields

There are two types of structure fields in a YAML structure: singular (or a scalar type) and embedded (structures that contain scalar types). The configuration merging process handles the overriding of singular and embedded struct fields to create a resulting kubelet configuration.

For instance, you may want a baseline kubelet configuration for all nodes, but you may want to customize the address and authorization fields. This can be done as follows:

Main kubelet configuration file contents:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
port: 20250
authorization:
 mode: Webhook
 webhook:
 cacheAuthorizedTTL: "5m"
 cacheUnauthorizedTTL: "30s"
serializeImagePulls: false
address: "192.168.0.1"
```

Contents of a file in --config-dir directory:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
authorization:
 mode: AlwaysAllow
 webhook:
 cacheAuthorizedTTL: "8m"
 cacheUnauthorizedTTL: "45s"
address: "192.168.0.8"
```

The resulting configuration will be as follows:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
port: 20250
serializeImagePulls: false
authorization:
 mode: AlwaysAllow
 webhook:
 cacheAuthorizedTTL: "8m"
 cacheUnauthorizedTTL: "45s"
address: "192.168.0.8"
```

## Lists

You can override the slices/lists values of the kubelet configuration. However, the entire list gets overridden during the merging process. For example, you can override the clusterDNS list as follows:

Main kubelet configuration file contents:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
port: 20250
serializeImagePulls: false
clusterDNS:
```



```
- "192.168.0.9"
- "192.168.0.8"
```

Contents of a file in `--config-dir` directory:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
clusterDNS:
- "192.168.0.2"
- "192.168.0.3"
- "192.168.0.5"
```

The resulting configuration will be as follows:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
port: 20250
serializeImagePulls: false
clusterDNS:
- "192.168.0.2"
- "192.168.0.3"
- "192.168.0.5"
```

## Maps, including Nested Structures

Individual fields in maps, regardless of their value types (boolean, string, etc.), can be selectively overridden. However, for `map[string][]string`, the entire list associated with a specific field gets overridden. Let's understand this better with an example, particularly on fields like `featureGates` and `staticPodURLHeader`:

Main kubelet configuration file contents:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
port: 20250
serializeImagePulls: false
featureGates:
 AllAlpha: false
 MemoryQoS: true
staticPodURLHeader:
 kubelet-api-support:
 - "Authorization: 234APSDFA"
 - "X-Custom-Header: 123"
 custom-static-pod:
 - "Authorization: 223EWRWER"
 - "X-Custom-Header: 456"
```

Contents of a file in `--config-dir` directory:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
featureGates:
 MemoryQoS: false
 KubeletTracing: true
```

```
DynamicResourceAllocation: true
staticPodURLHeader:
 custom-static-pod:
 - "Authorization: 223EWRWER"
 - "X-Custom-Header: 345"
```

The resulting configuration will be as follows:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
port: 20250
serializeImagePulls: false
featureGates:
 AllAlpha: false
 MemoryQoS: false
 KubeletTracing: true
 DynamicResourceAllocation: true
staticPodURLHeader:
 kubelet-api-support:
 - "Authorization: 234APSDFA"
 - "X-Custom-Header: 123"
 custom-static-pod:
 - "Authorization: 223EWRWER"
 - "X-Custom-Header: 345"
```

## Kubelet Device Manager API Versions

This page provides details of version compatibility between the Kubernetes [device plugin API](#), and different versions of Kubernetes itself.

### Compatibility matrix

|                 | v1alpha1 | v1beta1 |
|-----------------|----------|---------|
| Kubernetes 1.21 | -        |         |
| Kubernetes 1.22 | -        |         |
| Kubernetes 1.23 | -        |         |
| Kubernetes 1.24 | -        |         |
| Kubernetes 1.25 | -        |         |
| Kubernetes 1.26 | -        |         |

Key:

- Exactly the same features / API objects in both device plugin API and the Kubernetes version.
- + The device plugin API has features or API objects that may not be present in the Kubernetes cluster, either because the device plugin API has added additional new API calls, or that the server has removed an old API call. However, everything they have in common (most other APIs) will work. Note that alpha APIs may vanish or change significantly between one minor release and the next.

- - The Kubernetes cluster has features the device plugin API can't use, either because server has added additional API calls, or that device plugin API has removed an old API call. However, everything they share in common (most APIs) will work.

## Node Status

The status of a [node](#) in Kubernetes is a critical aspect of managing a Kubernetes cluster. In this article, we'll cover the basics of monitoring and maintaining node status to ensure a healthy and stable cluster.

### Node status fields

A Node's status contains the following information:

- [Addresses](#)
- [Conditions](#)
- [Capacity and Allocatable](#)
- [Info](#)

You can use `kubectl` to view a Node's status and other details:

```
kubectl describe node <insert-node-name-here>
```

Each section of the output is described below.

### Addresses

The usage of these fields varies depending on your cloud provider or bare metal configuration.

- **HostName:** The hostname as reported by the node's kernel. Can be overridden via the `kubelet --hostname-override` parameter.
- **ExternalIP:** Typically the IP address of the node that is externally routable (available from outside the cluster).
- **InternalIP:** Typically the IP address of the node that is routable only within the cluster.

### Conditions

The conditions field describes the status of all Running nodes. Examples of conditions include:

Node conditions, and a description of when each condition applies.

| Node Condition | Description                                                                                                                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ready          | True if the node is healthy and ready to accept pods, False if the node is not healthy and is not accepting pods, and Unknown if the node controller has not heard from the node in the last node-monitor-grace-period (default is 40 seconds) |
| DiskPressure   | True if pressure exists on the disk size—that is, if the disk capacity is low; otherwise False                                                                                                                                                 |
| MemoryPressure | True if pressure exists on the node memory—that is, if the node memory is low; otherwise False                                                                                                                                                 |

| Node Condition     | Description                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------|
| PIDPressure        | True if pressure exists on the processes—that is, if there are too many processes on the node; otherwise False |
| NetworkUnavailable | True if the network for the node is not correctly configured, otherwise False                                  |

### Note:

If you use command-line tools to print details of a cordoned Node, the Condition includes `SchedulingDisabled`. `SchedulingDisabled` is not a Condition in the Kubernetes API; instead, cordoned nodes are marked `Unschedulable` in their spec.

In the Kubernetes API, a node's condition is represented as part of the `.status` of the Node resource. For example, the following JSON structure describes a healthy node:

```
"conditions": [
{
 "type": "Ready",
 "status": "True",
 "reason": "KubeletReady",
 "message": "kubelet is posting ready status",
 "lastHeartbeatTime": "2019-06-05T18:38:35Z",
 "lastTransitionTime": "2019-06-05T11:41:27Z"
}
]
```

When problems occur on nodes, the Kubernetes control plane automatically creates [taints](#) that match the conditions affecting the node. An example of this is when the status of the Ready condition remains Unknown or False for longer than the kube-controller-manager's `NodeMonitorGracePeriod`, which defaults to 40 seconds. This will cause either an `node.kubernetes.io/unreachable` taint, for an Unknown status, or a `node.kubernetes.io/not-ready` taint, for a False status, to be added to the Node.

These taints affect pending pods as the scheduler takes the Node's taints into consideration when assigning a pod to a Node. Existing pods scheduled to the node may be evicted due to the application of `NoExecute` taints. Pods may also have [tolerations](#) that let them schedule to and continue running on a Node even though it has a specific taint.

See [Taint Based Evictions](#) and [Taint Nodes by Condition](#) for more details.

## Capacity and Allocatable

Describes the resources available on the node: CPU, memory, and the maximum number of pods that can be scheduled onto the node.

The fields in the capacity block indicate the total amount of resources that a Node has. The allocatable block indicates the amount of resources on a Node that is available to be consumed by normal Pods.

You may read more about capacity and allocatable resources while learning how to [reserve compute resources](#) on a Node.

## Info

Describes general information about the node, such as kernel version, Kubernetes version (kubelet and kube-proxy version), container runtime details, and which operating system the node uses. The kubelet gathers this information from the node and publishes it into the Kubernetes API.

## Heartbeats

Heartbeats, sent by Kubernetes nodes, help your cluster determine the availability of each node, and to take action when failures are detected.

For nodes there are two forms of heartbeats:

- updates to the `.status` of a Node
- [Lease](#) objects within the kube-node-lease [namespace](#). Each Node has an associated Lease object.

Compared to updates to `.status` of a Node, a Lease is a lightweight resource. Using Leases for heartbeats reduces the performance impact of these updates for large clusters.

The kubelet is responsible for creating and updating the `.status` of Nodes, and for updating their related Leases.

- The kubelet updates the node's `.status` either when there is change in status or if there has been no update for a configured interval. The default interval for `.status` updates to Nodes is 5 minutes, which is much longer than the 40 second default timeout for unreachable nodes.
- The kubelet creates and then updates its Lease object every 10 seconds (the default update interval). Lease updates occur independently from updates to the Node's `.status`. If the Lease update fails, the kubelet retries, using exponential backoff that starts at 200 milliseconds and capped at 7 seconds.

## Networking Reference

This section of the Kubernetes documentation provides reference details of Kubernetes networking.

---

[Protocols for Services](#)

[Ports and Protocols](#)

[Virtual IPs and Service Proxies](#)

## Protocols for Services

If you configure a [Service](#), you can select from any network protocol that Kubernetes supports.

Kubernetes supports the following protocols with Services:

- [SCTP](#)
- [TCP](#) (*the default*)
- [UDP](#)

When you define a Service, you can also specify the [application protocol](#) that it uses.

This document details some special cases, all of them typically using TCP as a transport protocol:

- [HTTP](#) and [HTTPS](#)
- [PROXY protocol](#)
- [TLS](#) termination at the load balancer

## Supported protocols

There are 3 valid values for the protocol of a port for a Service:

### SCTP

FEATURE STATE: Kubernetes v1.20 [stable]

When using a network plugin that supports SCTP traffic, you can use SCTP for most Services. For type: LoadBalancer Services, SCTP support depends on the cloud provider offering this facility. (Most do not).

SCTP is not supported on nodes that run Windows.

### Support for multihomed SCTP associations

The support of multihomed SCTP associations requires that the CNI plugin can support the assignment of multiple interfaces and IP addresses to a Pod.

NAT for multihomed SCTP associations requires special logic in the corresponding kernel modules.

### TCP

You can use TCP for any kind of Service, and it's the default network protocol.

### UDP

You can use UDP for most Services. For type: LoadBalancer Services, UDP support depends on the cloud provider offering this facility.

# Special cases

## HTTP

If your cloud provider supports it, you can use a Service in LoadBalancer mode to configure a load balancer outside of your Kubernetes cluster, in a special mode where your cloud provider's load balancer implements HTTP / HTTPS reverse proxying, with traffic forwarded to the backend endpoints for that Service.

Typically, you set the protocol for the Service to TCP and add an [annotation](#) (usually specific to your cloud provider) that configures the load balancer to handle traffic at the HTTP level. This configuration might also include serving HTTPS (HTTP over TLS) and reverse-proxying plain HTTP to your workload.

### Note:

You can also use an [Ingress](#) to expose HTTP/HTTPS Services.

You might additionally want to specify that the [application protocol](#) of the connection is http or https. Use http if the session from the load balancer to your workload is HTTP without TLS, and use https if the session from the load balancer to your workload uses TLS encryption.

## PROXY protocol

If your cloud provider supports it, you can use a Service set to type: LoadBalancer to configure a load balancer outside of Kubernetes itself, that will forward connections wrapped with the [PROXY protocol](#).

The load balancer then sends an initial series of octets describing the incoming connection, similar to this example (PROXY protocol v1):

```
PROXY TCP4 192.0.2.202 10.0.42.7 12345 7\r\n
```

The data after the proxy protocol preamble are the original data from the client. When either side closes the connection, the load balancer also triggers a connection close and sends any remaining data where feasible.

Typically, you define a Service with the protocol to TCP. You also set an annotation, specific to your cloud provider, that configures the load balancer to wrap each incoming connection in the PROXY protocol.

## TLS

If your cloud provider supports it, you can use a Service set to type: LoadBalancer as a way to set up external reverse proxying, where the connection from client to load balancer is TLS encrypted and the load balancer is the TLS server peer. The connection from the load balancer to your workload can also be TLS, or might be plain text. The exact options available to you depend on your cloud provider or custom Service implementation.

Typically, you set the protocol to TCP and set an annotation (usually specific to your cloud provider) that configures the load balancer to act as a TLS server. You would configure the TLS

identity (as server, and possibly also as a client that connects to your workload) using mechanisms that are specific to your cloud provider.

## Ports and Protocols

When running Kubernetes in an environment with strict network boundaries, such as on-premises datacenter with physical network firewalls or Virtual Networks in Public Cloud, it is useful to be aware of the ports and protocols used by Kubernetes components.

### Control plane

| Protocol | Direction | Port Range | Purpose                 | Used By              |
|----------|-----------|------------|-------------------------|----------------------|
| TCP      | Inbound   | 6443       | Kubernetes API server   | All                  |
| TCP      | Inbound   | 2379-2380  | etcd server client API  | kube-apiserver, etcd |
| TCP      | Inbound   | 10250      | Kubelet API             | Self, Control plane  |
| TCP      | Inbound   | 10259      | kube-scheduler          | Self                 |
| TCP      | Inbound   | 10257      | kube-controller-manager | Self                 |

Although etcd ports are included in control plane section, you can also host your own etcd cluster externally or on custom ports.

### Worker node(s)

| Protocol | Direction | Port Range  | Purpose            | Used By              |
|----------|-----------|-------------|--------------------|----------------------|
| TCP      | Inbound   | 10250       | Kubelet API        | Self, Control plane  |
| TCP      | Inbound   | 10256       | kube-proxy         | Self, Load balancers |
| TCP      | Inbound   | 30000-32767 | NodePort Services† | All                  |

† Default port range for [NodePort Services](#).

All default port numbers can be overridden. When custom ports are used those ports need to be open instead of defaults mentioned here.

One common example is API server port that is sometimes switched to 443. Alternatively, the default port is kept as is and API server is put behind a load balancer that listens on 443 and routes the requests to API server on the default port.

## Virtual IPs and Service Proxies

Every [node](#) in a Kubernetes [cluster](#) runs a [kube-proxy](#) (unless you have deployed your own alternative component in place of kube-proxy).

The kube-proxy component is responsible for implementing a *virtual IP* mechanism for [Services](#) of type other than [ExternalName](#). Each instance of kube-proxy watches the Kubernetes [control plane](#) for the addition and removal of Service and EndpointSlice [objects](#). For each Service, kube-proxy calls appropriate APIs (depending on the kube-proxy mode) to configure the node to capture traffic to the Service's clusterIP and port, and redirect that traffic to one of the Service's endpoints (usually a Pod, but possibly an arbitrary user-provided IP address). A control loop



ensures that the rules on each node are reliably synchronized with the Service and EndpointSlice state as indicated by the API server.

### **Virtual IP mechanism for Services, using iptables mode**

A question that pops up every now and then is why Kubernetes relies on proxying to forward inbound traffic to backends. What about other approaches? For example, would it be possible to configure DNS records that have multiple A values (or AAAA for IPv6), and rely on round-robin name resolution?

There are a few reasons for using proxying for Services:

- There is a long history of DNS implementations not respecting record TTLs, and caching the results of name lookups after they should have expired.
- Some apps do DNS lookups only once and cache the results indefinitely.
- Even if apps and libraries did proper re-resolution, the low or zero TTLs on the DNS records could impose a high load on DNS that then becomes difficult to manage.

Later in this page you can read about how various kube-proxy implementations work. Overall, you should note that, when running kube-proxy, kernel level rules may be modified (for example, iptables rules might get created), which won't get cleaned up, in some cases until you reboot. Thus, running kube-proxy is something that should only be done by an administrator which understands the consequences of having a low level, privileged network proxying service on a computer. Although the kube-proxy executable supports a cleanup function, this function is not an official feature and thus is only available to use as-is.

Some of the details in this reference refer to an example: the backend [Pods](#) for a stateless image-processing workloads, running with three replicas. Those replicas are fungible—frontends do not care which backend they use. While the actual Pods that compose the backend set may change, the frontend clients should not need to be aware of that, nor should they need to keep track of the set of backends themselves.

## **Proxy modes**

The kube-proxy starts up in different modes, which are determined by its configuration.

On Linux nodes, the available modes for kube-proxy are:

### [iptables](#)

A mode where the kube-proxy configures packet forwarding rules using iptables.

### [ipvs](#)

a mode where the kube-proxy configures packet forwarding rules using ipvs.

### [nftables](#)

a mode where the kube-proxy configures packet forwarding rules using nftables.

There is only one mode available for kube-proxy on Windows:

### [kernel-space](#)

a mode where the kube-proxy configures packet forwarding rules in the Windows kernel

## **iptables proxy mode**

*This proxy mode is only available on Linux nodes.*

In this mode, kube-proxy configures packet forwarding rules using the iptables API of the kernel netfilter subsystem. For each endpoint, it installs iptables rules which, by default, select a backend Pod at random.

## Example

As an example, consider the image processing application described [earlier](#) in the page. When the backend Service is created, the Kubernetes control plane assigns a virtual IP address, for example 10.0.0.1. For this example, assume that the Service port is 1234. All of the kube-proxy instances in the cluster observe the creation of the new Service.

When kube-proxy on a node sees a new Service, it installs a series of iptables rules which redirect from the virtual IP address to more iptables rules, defined per Service. The per-Service rules link to further rules for each backend endpoint, and the per- endpoint rules redirect traffic (using destination NAT) to the backends.

When a client connects to the Service's virtual IP address the iptables rule kicks in. A backend is chosen (either based on session affinity or randomly) and packets are redirected to the backend without rewriting the client IP address.

This same basic flow executes when traffic comes in through a node-port or through a load-balancer, though in those cases the client IP address does get altered.

## Optimizing iptables mode performance

In iptables mode, kube-proxy creates a few iptables rules for every Service, and a few iptables rules for each endpoint IP address. In clusters with tens of thousands of Pods and Services, this means tens of thousands of iptables rules, and kube-proxy may take a long time to update the rules in the kernel when Services (or their EndpointSlices) change. You can adjust the syncing behavior of kube-proxy via options in the [iptables section](#) of the kube-proxy [configuration file](#) (which you specify via kube-proxy --config <path>):

```
...
iptables:
 minSyncPeriod: 1s
 syncPeriod: 30s
...
```

### minSyncPeriod

The minSyncPeriod parameter sets the minimum duration between attempts to resynchronize iptables rules with the kernel. If it is 0s, then kube-proxy will always immediately synchronize the rules every time any Service or Endpoint changes. This works fine in very small clusters, but it results in a lot of redundant work when lots of things change in a small time period. For example, if you have a Service backed by a [Deployment](#) with 100 pods, and you delete the Deployment, then with minSyncPeriod: 0s, kube-proxy would end up removing the Service's endpoints from the iptables rules one by one, for a total of 100 updates. With a larger minSyncPeriod, multiple Pod deletion events would get aggregated together, so kube-proxy might instead end up making, say, 5 updates, each removing 20 endpoints, which will be much more efficient in terms of CPU, and result in the full set of changes being synchronized faster.

The larger the value of `minSyncPeriod`, the more work that can be aggregated, but the downside is that each individual change may end up waiting up to the full `minSyncPeriod` before being processed, meaning that the iptables rules spend more time being out-of-sync with the current API server state.

The default value of 1s should work well in most clusters, but in very large clusters it may be necessary to set it to a larger value. Especially, if kube-proxy's `sync_proxy_rules_duration_seconds` metric indicates an average time much larger than 1 second, then bumping up `minSyncPeriod` may make updates more efficient.

### Updating legacy `minSyncPeriod` configuration

Older versions of kube-proxy updated all the rules for all Services on every sync; this led to performance issues (update lag) in large clusters, and the recommended solution was to set a larger `minSyncPeriod`. Since Kubernetes v1.28, the iptables mode of kube-proxy uses a more minimal approach, only making updates where Services or EndpointSlices have actually changed.

If you were previously overriding `minSyncPeriod`, you should try removing that override and letting kube-proxy use the default value (1s) or at least a smaller value than you were using before upgrading.

If you are not running kube-proxy from Kubernetes 1.30, check the behavior and associated advice for the version that you are actually running.

### `syncPeriod`

The `syncPeriod` parameter controls a handful of synchronization operations that are not directly related to changes in individual Services and EndpointSlices. In particular, it controls how quickly kube-proxy notices if an external component has interfered with kube-proxy's iptables rules. In large clusters, kube-proxy also only performs certain cleanup operations once every `syncPeriod` to avoid unnecessary work.

For the most part, increasing `syncPeriod` is not expected to have much impact on performance, but in the past, it was sometimes useful to set it to a very large value (eg, 1h). This is no longer recommended, and is likely to hurt functionality more than it improves performance.

## IPVS proxy mode

*This proxy mode is only available on Linux nodes.*

In ipvs mode, kube-proxy uses the kernel IPVS and iptables APIs to create rules to redirect traffic from Service IPs to endpoint IPs.

The IPVS proxy mode is based on netfilter hook function that is similar to iptables mode, but uses a hash table as the underlying data structure and works in the kernel space. That means kube-proxy in IPVS mode redirects traffic with lower latency than kube-proxy in iptables mode, with much better performance when synchronizing proxy rules. Compared to the iptables proxy mode, IPVS mode also supports a higher throughput of network traffic.

IPVS provides more options for balancing traffic to backend Pods; these are:

- rr (Round Robin): Traffic is equally distributed amongst the backing servers.

- **wrr** (Weighted Round Robin): Traffic is routed to the backing servers based on the weights of the servers. Servers with higher weights receive new connections and get more requests than servers with lower weights.
- **lc** (Least Connection): More traffic is assigned to servers with fewer active connections.
- **wlc** (Weighted Least Connection): More traffic is routed to servers with fewer connections relative to their weights, that is, connections divided by weight.
- **lblc** (Locality based Least Connection): Traffic for the same IP address is sent to the same backing server if the server is not overloaded and available; otherwise the traffic is sent to servers with fewer connections, and keep it for future assignment.
- **lblcr** (Locality Based Least Connection with Replication): Traffic for the same IP address is sent to the server with least connections. If all the backing servers are overloaded, it picks up one with fewer connections and add it to the target set. If the target set has not changed for the specified time, the most loaded server is removed from the set, in order to avoid high degree of replication.
- **sh** (Source Hashing): Traffic is sent to a backing server by looking up a statically assigned hash table based on the source IP addresses.
- **dh** (Destination Hashing): Traffic is sent to a backing server by looking up a statically assigned hash table based on their destination addresses.
- **sed** (Shortest Expected Delay): Traffic forwarded to a backing server with the shortest expected delay. The expected delay is  $(C + 1) / U$  if sent to a server, where  $C$  is the number of connections on the server and  $U$  is the fixed service rate (weight) of the server.
- **nq** (Never Queue): Traffic is sent to an idle server if there is one, instead of waiting for a fast one; if all servers are busy, the algorithm falls back to the sed behavior.

#### **Note:**

To run kube-proxy in IPVS mode, you must make IPVS available on the node before starting kube-proxy.

When kube-proxy starts in IPVS proxy mode, it verifies whether IPVS kernel modules are available. If the IPVS kernel modules are not detected, then kube-proxy exits with an error.

### **Virtual IP address mechanism for Services, using IPVS mode**

#### **nftables proxy mode**

FEATURE STATE: Kubernetes v1.29 [alpha]

*This proxy mode is only available on Linux nodes.*

In this mode, kube-proxy configures packet forwarding rules using the nftables API of the kernel netfilter subsystem. For each endpoint, it installs nftables rules which, by default, select a backend Pod at random.

The nftables API is the successor to the iptables API, and although it is designed to provide better performance and scalability than iptables, the kube-proxy nftables mode is still under heavy development as of 1.30 and is not necessarily expected to outperform the other Linux modes at this time.

## kernel-space proxy mode

*This proxy mode is only available on Windows nodes.*

The kube-proxy configures packet filtering rules in the Windows *Virtual Filtering Platform* (VFP), an extension to Windows vSwitch. These rules process encapsulated packets within the node-level virtual networks, and rewrite packets so that the destination IP address (and layer 2 information) is correct for getting the packet routed to the correct destination. The Windows VFP is analogous to tools such as Linux nftables or iptables. The Windows VFP extends the *Hyper-V Switch*, which was initially implemented to support virtual machine networking.

When a Pod on a node sends traffic to a virtual IP address, and the kube-proxy selects a Pod on a different node as the load balancing target, the kernel-space proxy mode rewrites that packet to be destined to the target backend Pod. The Windows *Host Networking Service* (HNS) ensures that packet rewriting rules are configured so that the return traffic appears to come from the virtual IP address and not the specific backend Pod.

## Direct server return for kernel-space mode

FEATURE STATE: Kubernetes v1.14 [alpha]

As an alternative to the basic operation, a node that hosts the backend Pod for a Service can apply the packet rewriting directly, rather than placing this burden on the node where the client Pod is running. This is called *direct server return*.

To use this, you must run kube-proxy with the `--enable-dsr` command line argument **and** enable the WinDSR [feature gate](#).

Direct server return also optimizes the case for Pod return traffic even when both Pods are running on the same node.

## Session affinity

In these proxy models, the traffic bound for the Service's IP:Port is proxied to an appropriate backend without the clients knowing anything about Kubernetes or Services or Pods.

If you want to make sure that connections from a particular client are passed to the same Pod each time, you can select the session affinity based on the client's IP addresses by setting `.spec.sessionAffinity` to `ClientIP` for a Service (the default is `None`).

## Session stickiness timeout

You can also set the maximum session sticky time by setting `.spec.sessionAffinityConfig.clientIP.timeoutSeconds` appropriately for a Service. (the default value is 10800, which works out to be 3 hours).

## Note:

On Windows, setting the maximum session sticky time for Services is not supported.

## IP address assignment to Services

Unlike Pod IP addresses, which actually route to a fixed destination, Service IPs are not actually answered by a single host. Instead, kube-proxy uses packet processing logic (such as Linux iptables) to define *virtual* IP addresses which are transparently redirected as needed.

When clients connect to the VIP, their traffic is automatically transported to an appropriate endpoint. The environment variables and DNS for Services are actually populated in terms of the Service's virtual IP address (and port).

## Avoiding collisions

One of the primary philosophies of Kubernetes is that you should not be exposed to situations that could cause your actions to fail through no fault of your own. For the design of the Service resource, this means not making you choose your own IP address if that choice might collide with someone else's choice. That is an isolation failure.

In order to allow you to choose an IP address for your Services, we must ensure that no two Services can collide. Kubernetes does that by allocating each Service its own IP address from within the service-cluster-ip-range CIDR range that is configured for the [API Server](#).

## IP address allocation tracking

To ensure each Service receives a unique IP address, an internal allocator atomically updates a global allocation map in [etcd](#) prior to creating each Service. The map object must exist in the registry for Services to get IP address assignments, otherwise creations will fail with a message indicating an IP address could not be allocated.

In the control plane, a background controller is responsible for creating that map (needed to support migrating from older versions of Kubernetes that used in-memory locking). Kubernetes also uses controllers to check for invalid assignments (for example: due to administrator intervention) and for cleaning up allocated IP addresses that are no longer used by any Services.

## IP address allocation tracking using the Kubernetes API

FEATURE STATE: Kubernetes v1.27 [alpha]

If you enable the MultiCIDRServiceAllocator [feature gate](#) and the [networking.k8s.io/v1alpha1 API group](#), the control plane replaces the existing etcd allocator with a revised implementation that uses IPAddress and ServiceCIDR objects instead of an internal global allocation map. Each cluster IP address associated to a Service then references an IPAddress object.

Enabling the feature gate also replaces a background controller with an alternative that handles the IPAddress objects and supports migration from the old allocator model. Kubernetes 1.30 does not support migrating from IPAddress objects to the internal allocation map.

One of the main benefits of the revised allocator is that it removes the size limitations for the IP address range that can be used for the cluster IP address of Services. With MultiCIDRServiceAllocator enabled, there are no limitations for IPv4, and for IPv6 you can use IP address netmasks that are a /64 or smaller (as opposed to /108 with the legacy implementation).

Making IP address allocations available via the API means that you as a cluster administrator can allow users to inspect the IP addresses assigned to their Services. Kubernetes extensions, such as the [Gateway API](#), can use the IPAddress API to extend Kubernetes' inherent networking capabilities.

Here is a brief example of a user querying for IP addresses:

```
kubectl get services
```

| NAME       | TYPE      | CLUSTER-IP      | EXTERNAL-IP | PORT(S) | AGE  |
|------------|-----------|-----------------|-------------|---------|------|
| kubernetes | ClusterIP | 2001:db8:1:2::1 | <none>      | 443/TCP | 3d1h |

```
kubectl get ipaddresses
```

| NAME            | PARENTREF                     |
|-----------------|-------------------------------|
| 2001:db8:1:2::1 | services/default/kubernetes   |
| 2001:db8:1:2::a | services/kube-system/kube-dns |

Kubernetes also allow users to dynamically define the available IP ranges for Services using ServiceCIDR objects. During bootstrap, a default ServiceCIDR object named kubernetes is created from the value of the --service-cluster-ip-range command line argument to kube-apiserver:

```
kubectl get servicecidrs
```

| NAME       | CIDRS        | AGE |
|------------|--------------|-----|
| kubernetes | 10.96.0.0/28 | 17m |

Users can create or delete new ServiceCIDR objects to manage the available IP ranges for Services:

```
cat <<'EOF' | kubectl apply -f -
apiVersion: networking.k8s.io/v1alpha1
kind: ServiceCIDR
metadata:
 name: newservicecidr
spec:
 cidrs:
 - 10.96.0.0/24
EOF
```

```
servicecidr.networking.k8s.io/newcidr1 created
```

```
kubectl get servicecidrs
```

| NAME           | CIDRS        | AGE |
|----------------|--------------|-----|
| kubernetes     | 10.96.0.0/28 | 17m |
| newservicecidr | 10.96.0.0/24 | 7m  |

## IP address ranges for Service virtual IP addresses

FEATURE STATE: Kubernetes v1.26 [stable]

Kubernetes divides the ClusterIP range into two bands, based on the size of the configured service-cluster-ip-range by using the following formula  $\min(\max(16, \text{cidrSize} / 16), 256)$ . That formula paraphrases as *never less than 16 or more than 256, with a graduated step function between them*.

Kubernetes prefers to allocate dynamic IP addresses to Services by choosing from the upper band, which means that if you want to assign a specific IP address to a type: ClusterIP Service, you should manually assign an IP address from the **lower** band. That approach reduces the risk of a conflict over allocation.

## Traffic policies

You can set the `.spec.internalTrafficPolicy` and `.spec.externalTrafficPolicy` fields to control how Kubernetes routes traffic to healthy (“ready”) backends.

### Internal traffic policy

FEATURE STATE: Kubernetes v1.26 [stable]

You can set the `.spec.internalTrafficPolicy` field to control how traffic from internal sources is routed. Valid values are Cluster and Local. Set the field to Cluster to route internal traffic to all ready endpoints and Local to only route to ready node-local endpoints. If the traffic policy is Local and there are no node-local endpoints, traffic is dropped by kube-proxy.

### External traffic policy

You can set the `.spec.externalTrafficPolicy` field to control how traffic from external sources is routed. Valid values are Cluster and Local. Set the field to Cluster to route external traffic to all ready endpoints and Local to only route to ready node-local endpoints. If the traffic policy is Local and there are no node-local endpoints, the kube-proxy does not forward any traffic for the relevant Service.

If Cluster is specified all nodes are eligible load balancing targets *as long as* the node is not being deleted and kube-proxy is healthy. In this mode: load balancer health checks are configured to target the service proxy's readiness port and path. In the case of kube-proxy this evaluates to: `$_NODE_IP:10256/healthz`. kube-proxy will return either an HTTP code 200 or 503. kube-proxy's load balancer health check endpoint returns 200 if:

1. kube-proxy is healthy, meaning:
  - it's able to progress programming the network and isn't timing out while doing so (the timeout is defined to be:  $2 \times \text{iptables.syncPeriod}$ ); and
2. the node is not being deleted (there is no deletion timestamp set for the Node).

The reason why kube-proxy returns 503 and marks the node as not eligible when it's being deleted, is because kube-proxy supports connection draining for terminating nodes. A couple of important things occur from the point of view of a Kubernetes-managed load balancer when a node *is being* / *is* deleted.



While deleting:

- kube-proxy will start failing its readiness probe and essentially mark the node as not eligible for load balancer traffic. The load balancer health check failing causes load balancers which support connection draining to allow existing connections to terminate, and block new connections from establishing.

When deleted:

- The service controller in the Kubernetes cloud controller manager removes the node from the referenced set of eligible targets. Removing any instance from the load balancer's set of backend targets immediately terminates all connections. This is also the reason kube-proxy first fails the health check while the node is deleting.

It's important to note for Kubernetes vendors that if any vendor configures the kube-proxy readiness probe as a liveness probe: that kube-proxy will start restarting continuously when a node is deleting until it has been fully deleted. kube-proxy exposes a `/livez` path which, as opposed to the `/healthz` one, does **not** consider the Node's deleting state and only its progress programming the network. `/livez` is therefore the recommended path for anyone looking to define a `LivenessProbe` for kube-proxy.

Users deploying kube-proxy can inspect both the readiness / liveness state by evaluating the metrics: `proxy_livez_total` / `proxy_healthz_total`. Both metrics publish two series, one with the 200 label and one with the 503 one.

For Local Services: kube-proxy will return 200 if

1. kube-proxy is healthy/ready, and
2. has a local endpoint on the node in question.

Node deletion does **not** have an impact on kube-proxy's return code for what concerns load balancer health checks. The reason for this is: deleting nodes could end up causing an ingress outage should all endpoints simultaneously be running on said nodes.

The Kubernetes project recommends that cloud provider integration code configures load balancer health checks that target the service proxy's `healthz` port. If you are using or implementing your own virtual IP implementation, that people can use instead of kube-proxy, you should set up a similar health checking port with logic that matches the kube-proxy implementation.

## Traffic to terminating endpoints

FEATURE STATE: Kubernetes v1.28 [stable]

If the `ProxyTerminatingEndpoints` [feature gate](#) is enabled in kube-proxy and the traffic policy is Local, that node's kube-proxy uses a more complicated algorithm to select endpoints for a Service. With the feature enabled, kube-proxy checks if the node has local endpoints and whether or not all the local endpoints are marked as terminating. If there are local endpoints and **all** of them are terminating, then kube-proxy will forward traffic to those terminating endpoints. Otherwise, kube-proxy will always prefer forwarding traffic to endpoints that are not terminating.

This forwarding behavior for terminating endpoints exist to allow NodePort and LoadBalancer Services to gracefully drain connections when using `externalTrafficPolicy: Local`.

As a deployment goes through a rolling update, nodes backing a load balancer may transition from N to 0 replicas of that deployment. In some cases, external load balancers can send traffic to a node with 0 replicas in between health check probes. Routing traffic to terminating endpoints ensures that Node's that are scaling down Pods can gracefully receive and drain traffic to those terminating Pods. By the time the Pod completes termination, the external load balancer should have seen the node's health check failing and fully removed the node from the backend pool.

## Traffic Distribution

FEATURE STATE: Kubernetes v1.30 [alpha]

The `spec.trafficDistribution` field within a Kubernetes Service allows you to express preferences for how traffic should be routed to Service endpoints. Implementations like kube-proxy use the `spec.trafficDistribution` field as a guideline. The behavior associated with a given preference may subtly differ between implementations.

### PreferClose with kube-proxy

For kube-proxy, this means prioritizing sending traffic to endpoints within the same zone as the client. The EndpointSlice controller updates EndpointSlices with hints to communicate this preference, which kube-proxy then uses for routing decisions. If a client's zone does not have any available endpoints, traffic will be routed cluster-wide for that client.

In the absence of any value for `trafficDistribution`, the default routing strategy for kube-proxy is to distribute traffic to any endpoint in the cluster.

### Comparison with `service.kubernetes.io/topology-mode: Auto`

The `trafficDistribution` field with `PreferClose` and the `service.kubernetes.io/topology-mode: Auto` annotation both aim to prioritize same-zone traffic. However, there are key differences in their approaches:

- `service.kubernetes.io/topology-mode: Auto`: Attempts to distribute traffic proportionally across zones based on allocatable CPU resources. This heuristic includes safeguards (such as the [fallback behavior](#) for small numbers of endpoints) and could lead to the feature being disabled in certain scenarios for load-balancing reasons. This approach sacrifices some predictability in favor of potential load balancing.
- `trafficDistribution: PreferClose`: This approach aims to be slightly simpler and more predictable: "If there are endpoints in the zone, they will receive all traffic for that zone, if there are no endpoints in a zone, the traffic will be distributed to other zones". While the approach may offer more predictability, it does mean that you are in control of managing a [potential overload](#).

If the `service.kubernetes.io/topology-mode` annotation is set to `Auto`, it will take precedence over `trafficDistribution`. (The annotation may be deprecated in the future in favour of the `trafficDistribution` field).

## Interaction with Traffic Policies

When compared to the `trafficDistribution` field, the traffic policy fields (`externalTrafficPolicy` and `internalTrafficPolicy`) are meant to offer a stricter traffic locality requirements. Here's how `trafficDistribution` interacts with them:

- **Precedence of Traffic Policies:** For a given Service, if a traffic policy (`externalTrafficPolicy` or `internalTrafficPolicy`) is set to `Local`, it takes precedence over `trafficDistribution: PreferClose` for the corresponding traffic type (external or internal, respectively).
- **trafficDistribution Influence:** For a given Service, if a traffic policy (`externalTrafficPolicy` or `internalTrafficPolicy`) is set to `Cluster` (the default), or if the fields are not set, then `trafficDistribution: PreferClose` guides the routing behavior for the corresponding traffic type (external or internal, respectively). This means that an attempt will be made to route traffic to an endpoint that is in the same zone as the client.

## Considerations for using traffic distribution control

- **Increased Probability of Overloaded Endpoints:** The `PreferClose` heuristic will attempt to route traffic to the closest healthy endpoints instead of spreading that traffic evenly across all endpoints. If you do not have a sufficient number of endpoints within a zone, they may become overloaded. This is especially likely if incoming traffic is not proportionally distributed across zones. To mitigate this, consider the following strategies:
  - [Pod Topology Spread Constraints](#): Use Pod Topology Spread Constraints to distribute your pods more evenly across zones.
  - **Zone-specific Deployments:** If you expect to see skewed traffic patterns, create a separate Deployment for each zone. This approach allows the separate workloads to scale independently. There are also workload management addons available from the ecosystem, outside the Kubernetes project itself, that can help here.
- **Implementation-specific behavior:** Each dataplane implementation may handle this field slightly differently. If you're using an implementation other than kube-proxy, refer the documentation specific to that implementation to understand how this field is being handled.

## What's next

To learn more about Services, read [Connecting Applications with Services](#).

You can also:

- Read about [Services](#) as a concept
- Read about [Ingresses](#) as a concept
- Read the [API reference](#) for the Service API

## Setup tools

---

# Kubeadm

Kubeadm is a tool built to provide `kubeadm init` and `kubeadm join` as best-practice "fast paths" for creating Kubernetes clusters.

kubeadm performs the actions necessary to get a minimum viable cluster up and running. By design, it cares only about bootstrapping, not about provisioning machines. Likewise, installing various nice-to-have addons, like the Kubernetes Dashboard, monitoring solutions, and cloud-specific addons, is not in scope.

Instead, we expect higher-level and more tailored tooling to be built on top of kubeadm, and ideally, using kubeadm as the basis of all deployments will make it easier to create conformant clusters.

## How to install

To install kubeadm, see the [installation guide](#).

## What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node
- [kubeadm join](#) to bootstrap a Kubernetes worker node and join it to the cluster
- [kubeadm upgrade](#) to upgrade a Kubernetes cluster to a newer version
- [kubeadm config](#) if you initialized your cluster using kubeadm v1.7.x or lower, to configure your cluster for kubeadm upgrade
- [kubeadm token](#) to manage tokens for kubeadm join
- [kubeadm reset](#) to revert any changes made to this host by `kubeadm init` or `kubeadm join`
- [kubeadm certs](#) to manage Kubernetes certificates
- [kubeadm kubeconfig](#) to manage kubeconfig files
- [kubeadm version](#) to print the kubeadm version
- [kubeadm alpha](#) to preview a set of features made available for gathering feedback from the community

## kubeadm init

This command initializes a Kubernetes control-plane node.

Run this command in order to set up the Kubernetes control plane

### Synopsis

Run this command in order to set up the Kubernetes control plane

The "init" command executes the following phases:

|           |                        |
|-----------|------------------------|
| preflight | Run pre-flight checks  |
| certs     | Certificate generation |

|                             |                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------|
| /ca                         | Generate the self-signed Kubernetes CA to provision identities for other Kubernetes components       |
| /apiserver                  | Generate the certificate for serving the Kubernetes API                                              |
| /apiserver-kubelet-client   | Generate the certificate for the API server to connect to kubelet                                    |
| /front-proxy-ca             | Generate the self-signed CA to provision identities for front proxy                                  |
| /front-proxy-client         | Generate the certificate for the front proxy client                                                  |
| /etcd-ca                    | Generate the self-signed CA to provision identities for etcd                                         |
| /etcd-server                | Generate the certificate for serving etcd                                                            |
| /etcd-peer                  | Generate the certificate for etcd nodes to communicate with each other                               |
| /etcd-healthcheck-client    | Generate the certificate for liveness probes to healthcheck etcd                                     |
| /apiserver-etcd-client      | Generate the certificate the apiserver uses to access etcd                                           |
| /sa                         | Generate a private key for signing service account tokens along with its public key                  |
| kubeconfig                  | Generate all kubeconfig files necessary to establish the control plane and the admin kubeconfig file |
| /admin                      | Generate a kubeconfig file for the admin to use and for kubeadm itself                               |
| /super-admin                | Generate a kubeconfig file for the super-admin                                                       |
| /kubelet                    | Generate a kubeconfig file for the kubelet to use *only* for cluster bootstrapping purposes          |
| /controller-manager         | Generate a kubeconfig file for the controller manager to use                                         |
| /scheduler                  | Generate a kubeconfig file for the scheduler to use                                                  |
| etcd                        | Generate static Pod manifest file for local etcd                                                     |
| /local                      | Generate the static Pod manifest file for a local, single-node local etcd instance                   |
| control-plane               | Generate all static Pod manifest files necessary to establish the control plane                      |
| /apiserver                  | Generates the kube-apiserver static Pod manifest                                                     |
| /controller-manager         | Generates the kube-controller-manager static Pod manifest                                            |
| /scheduler                  | Generates the kube-scheduler static Pod manifest                                                     |
| kubelet-start               | Write kubelet settings and (re)start the kubelet                                                     |
| upload-config               | Upload the kubeadm and kubelet configuration to a ConfigMap                                          |
| /kubeadm                    | Upload the kubeadm ClusterConfiguration to a ConfigMap                                               |
| /kubelet                    | Upload the kubelet component config to a ConfigMap                                                   |
| upload-certs                | Upload certificates to kubeadm-certs                                                                 |
| mark-control-plane          | Mark a node as a control-plane                                                                       |
| bootstrap-token             | Generates bootstrap tokens used to join a node to a cluster                                          |
| kubelet-finalize            | Updates settings relevant to the kubelet after TLS bootstrap                                         |
| /experimental-cert-rotation | Enable kubelet client certificate rotation                                                           |
| addon                       | Install required addons for passing conformance tests                                                |
| /coredns                    | Install the CoreDNS addon to a Kubernetes cluster                                                    |
| /kube-proxy                 | Install the kube-proxy addon to a Kubernetes cluster                                                 |
| show-join-command           | Show the join command for control-plane and worker node                                              |

kubeadm init [flags]

## Options

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                   |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |

|                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <code>--apiserver-bind-port int32</code>                                                                                                                                                                                                                                                                                                                                                                           | Default: 6443                  |
| Port for the API Server to bind to.                                                                                                                                                                                                                                                                                                                                                                                |                                |
| <code>--apiserver-cert-extra-sans strings</code>                                                                                                                                                                                                                                                                                                                                                                   |                                |
| Optional extra Subject Alternative Names (SANs) to use for the API Server serving certificate. Can be both IP addresses and DNS names.                                                                                                                                                                                                                                                                             |                                |
| <code>--cert-dir string</code>                                                                                                                                                                                                                                                                                                                                                                                     | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates.                                                                                                                                                                                                                                                                                                                                                                 |                                |
| <code>--certificate-key string</code>                                                                                                                                                                                                                                                                                                                                                                              |                                |
| Key used to encrypt the control-plane certificates in the kubeadm-certs Secret. The certificate key is a hex encoded string that is an AES key of size 32 bytes.                                                                                                                                                                                                                                                   |                                |
| <code>--config string</code>                                                                                                                                                                                                                                                                                                                                                                                       |                                |
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                              |                                |
| <code>--control-plane-endpoint string</code>                                                                                                                                                                                                                                                                                                                                                                       |                                |
| Specify a stable IP address or DNS name for the control plane.                                                                                                                                                                                                                                                                                                                                                     |                                |
| <code>--cri-socket string</code>                                                                                                                                                                                                                                                                                                                                                                                   |                                |
| Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket.                                                                                                                                                                                                                       |                                |
| <code>--dry-run</code>                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                           |                                |
| <code>--feature-gates string</code>                                                                                                                                                                                                                                                                                                                                                                                |                                |
| A set of key=value pairs that describe feature gates for various features. Options are:<br>EtdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |                                |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                            |                                |
| help for init                                                                                                                                                                                                                                                                                                                                                                                                      |                                |
| <code>--ignore-preflight-errors strings</code>                                                                                                                                                                                                                                                                                                                                                                     |                                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --image-repository string    Default: "registry.k8s.io"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --kubernetes-version string    Default: "stable-1"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Choose a specific Kubernetes version for the control plane.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --node-name string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Specify the node name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |
| --pod-network-cidr string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Specify range of IP addresses for the pod network. If set, the control plane will automatically allocate CIDRs for every node.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| --service-cidr string    Default: "10.96.0.0/12"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Use alternative range of IP address for service VIPs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --service-dns-domain string    Default: "cluster.local"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Use alternative domain for services, e.g. "myorg.internal".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --skip-certificate-key-print                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Don't print the key used to encrypt the control-plane certificates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| --skip-phases strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| List of phases to be skipped                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --skip-token-print                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Skip printing of the default bootstrap token generated by 'kubeadm init'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --token string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The token to use for establishing bidirectional trust between nodes and control-plane nodes. The format is [a-z0-9]{6}.[a-z0-9]{16} - e.g. abcdef.0123456789abcdef |
| --token-ttl duration    Default: 24h0m0s                                                                                                                           |
| The duration before the token is automatically deleted (e.g. 1s, 2m, 3h). If set to '0', the token will never expire                                               |
| --upload-certs                                                                                                                                                     |
| Upload control-plane certificates to the kubeadm-certs Secret.                                                                                                     |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## Init workflow

kubeadm init bootstraps a Kubernetes control-plane node by executing the following steps:

1. Runs a series of pre-flight checks to validate the system state before making changes. Some checks only trigger warnings, others are considered errors and will exit kubeadm until the problem is corrected or the user specifies `--ignore-preflight-errors=<list-of-errors>`.
2. Generates a self-signed CA to set up identities for each component in the cluster. The user can provide their own CA cert and/or key by dropping it in the cert directory configured via `--cert-dir` (`/etc/kubernetes/pki` by default). The APIServer certs will have additional SAN entries for any `--apiserver-cert-extra-sans` arguments, lowercased if necessary.
3. Writes kubeconfig files in `/etc/kubernetes/` for the kubelet, the controller-manager and the scheduler to use to connect to the API server, each with its own identity. Also additional kubeconfig files are written, for kubeadm as administrative entity (`admin.conf`) and for a super admin user that can bypass RBAC (`super-admin.conf`).
4. Generates static Pod manifests for the API server, controller-manager and scheduler. In case an external etcd is not provided, an additional static Pod manifest is generated for etcd.

Static Pod manifests are written to `/etc/kubernetes/manifests`; the kubelet watches this directory for Pods to create on startup.

Once control plane Pods are up and running, the kubeadm init sequence can continue.

5. Apply labels and taints to the control-plane node so that no additional workloads will run there.



- Generates the token that additional nodes can use to register themselves with a control-
6. plane in the future. Optionally, the user can provide a token via `--token`, as described in the [kubeadm token](#) docs.
  7. Makes all the necessary configurations for allowing node joining with the [Bootstrap Tokens](#) and [TLS Bootstrap](#) mechanism:
    - Write a ConfigMap for making available all the information required for joining, and set up related RBAC access rules.
    - Let Bootstrap Tokens access the CSR signing API.
    - Configure auto-approval for new CSR requests.

See [kubeadm join](#) for additional info.

8. Installs a DNS server (CoreDNS) and the kube-proxy addon components via the API server. In Kubernetes version 1.11 and later CoreDNS is the default DNS server. Please note that although the DNS server is deployed, it will not be scheduled until CNI is installed.

### **Warning:**

kube-dns usage with kubeadm is deprecated as of v1.18 and is removed in v1.21.

## **Using init phases with kubeadm**

Kubeadm allows you to create a control-plane node in phases using the `kubeadm init phase` command.

To view the ordered list of phases and sub-phases you can call `kubeadm init --help`. The list will be located at the top of the help screen and each phase will have a description next to it. Note that by calling `kubeadm init` all of the phases and sub-phases will be executed in this exact order.

Some phases have unique flags, so if you want to have a look at the list of available options add `--help`, for example:

```
sudo kubeadm init phase control-plane controller-manager --help
```

You can also use `--help` to see the list of sub-phases for a certain parent phase:

```
sudo kubeadm init phase control-plane --help
```

`kubeadm init` also exposes a flag called `--skip-phases` that can be used to skip certain phases. The flag accepts a list of phase names and the names can be taken from the above ordered list.

An example:

```
sudo kubeadm init phase control-plane all --config=configfile.yaml
sudo kubeadm init phase etcd local --config=configfile.yaml
you can now modify the control plane and etcd manifest files
sudo kubeadm init --skip-phases=control-plane,etcd --config=configfile.yaml
```

What this example would do is write the manifest files for the control plane and etcd in `/etc/kubernetes/manifests` based on the configuration in `configfile.yaml`. This allows you to modify the files and then skip these phases using `--skip-phases`. By calling the last command you will create a control plane node with the custom manifest files.

FEATURE STATE: Kubernetes v1.22 [beta]

Alternatively, you can use the `skipPhases` field under `InitConfiguration`.

## Using kubeadm init with a configuration file

### Caution:

The config file is still considered beta and may change in future versions.

It's possible to configure `kubeadm init` with a configuration file instead of command line flags, and some more advanced features may only be available as configuration file options. This file is passed using the `--config` flag and it must contain a `ClusterConfiguration` structure and optionally more structures separated by `---`. Mixing `--config` with others flags may not be allowed in some cases.

The default configuration can be printed out using the [kubeadm config print](#) command.

If your configuration is not using the latest version it is **recommended** that you migrate using the [kubeadm config migrate](#) command.

For more information on the fields and usage of the configuration you can navigate to our [API reference page](#).

## Using kubeadm init with feature gates

Kubeadm supports a set of feature gates that are unique to kubeadm and can only be applied during cluster creation with `kubeadm init`. These features can control the behavior of the cluster. Feature gates are removed after a feature graduates to GA.

To pass a feature gate you can either use the `--feature-gates` flag for `kubeadm init`, or you can add items into the `featureGates` field when you pass a [configuration file](#) using `--config`.

Passing [feature gates for core Kubernetes components](#) directly to kubeadm is not supported. Instead, it is possible to pass them by [Customizing components with the kubeadm API](#).

List of feature gates:

kubeadm feature gates

| Feature                          | Default | Alpha | Beta | GA |
|----------------------------------|---------|-------|------|----|
| EtcdLearnerMode                  | true    | 1.27  | 1.29 | -  |
| PublicKeysECDSA                  | false   | 1.19  | -    | -  |
| RootlessControlPlane             | false   | 1.22  | -    | -  |
| WaitForAllControlPlaneComponents | false   | 1.30  | -    | -  |

## Note:

Once a feature gate goes GA its value becomes locked to true by default.

Feature gate descriptions:

### EtcdLearnerMode

With this feature gate enabled, when joining a new control plane node, a new etcd member will be created as a learner and promoted to a voting member only after the etcd data are fully aligned.

### PublicKeysECDSA

Can be used to create a cluster that uses ECDSA certificates instead of the default RSA algorithm. Renewal of existing ECDSA certificates is also supported using `kubeadm certs renew`, but you cannot switch between the RSA and ECDSA algorithms on the fly or during upgrades.

### RootlessControlPlane

Setting this flag configures the kubeadm deployed control plane component static Pod containers for kube-apiserver, kube-controller-manager, kube-scheduler and etcd to run as non-root users. If the flag is not set, those components run as root. You can change the value of this feature gate before you upgrade to a newer version of Kubernetes.

### WaitForAllControlPlaneComponents

With this feature gate enabled kubeadm will wait for all control plane components (kube-apiserver, kube-controller-manager, kube-scheduler) on a control plane node to report status 200 on their `/healthz` endpoints. These checks are performed on `https://127.0.0.1:PORT/healthz`, where PORT is taken from `--secure-port` of a component. If you specify custom `--secure-port` values in the kubeadm configuration they will be respected. Without the feature gate enabled, kubeadm will only wait for the kube-apiserver on a control plane node to become ready. The wait process starts right after the kubelet on the host is started by kubeadm. You are advised to enable this feature gate in case you wish to observe a ready state from all control plane components during the kubeadm init or kubeadm join command execution.

List of deprecated feature gates:

kubeadm deprecated feature gates

| Feature                         | Default |
|---------------------------------|---------|
| UpgradeAddonsBeforeControlPlane | false   |

Feature gate descriptions:

### UpgradeAddonsBeforeControlPlane

This is as a **disabled** feature gate that was introduced for Kubernetes v1.28, in order to allow reactivating a legacy and deprecated behavior during cluster upgrade. For kubeadm versions prior to v1.28, kubeadm upgrades cluster addons (including CoreDNS and kube-proxy) immediately during kubeadm upgrade apply, regardless of whether there are other control plane instances that have not been upgraded. This may cause compatibility problems. Since v1.28, kubeadm defaults to a mode that always checks whether all the control plane instances have been upgraded before starting to upgrade the addons. This behavior is applied to both kubeadm upgrade apply and kubeadm upgrade node. kubeadm determines whether a control plane instance has been upgraded by checking whether the image of the kube-apiserver Pod has been upgraded. You must perform control plane instances upgrade sequentially or at least ensure that the last control plane instance upgrade is not started until all the other control plane instances have been

upgraded completely, and the addons upgrade will be performed after the last control plane instance is upgraded. The deprecated UpgradeAddonsBeforeControlPlane feature gate gives you a chance to keep the old upgrade behavior. You should not need this old behavior; if you do, you should consider changing your cluster or upgrade processes, as this feature gate will be removed in a future release.

List of removed feature gates:

kubeadm removed feature gates

| Feature                     | Alpha | Beta | GA   | Removed |
|-----------------------------|-------|------|------|---------|
| IPv6DualStack               | 1.16  | 1.21 | 1.23 | 1.24    |
| UnversionedKubeletConfigMap | 1.22  | 1.23 | 1.25 | 1.26    |

Feature gate descriptions:

IPv6DualStack

This flag helps to configure components dual stack when the feature is in progress. For more details on Kubernetes dual-stack support see [Dual-stack support with kubeadm](#).

UnversionedKubeletConfigMap

This flag controls the name of the [ConfigMap](#) where kubeadm stores kubelet configuration data. With this flag not specified or set to true, the ConfigMap is named kubelet-config. If you set this flag to false, the name of the ConfigMap includes the major and minor version for Kubernetes (for example: kubelet-config-1.30). Kubeadm ensures that RBAC rules for reading and writing that ConfigMap are appropriate for the value you set. When kubeadm writes this ConfigMap (during kubeadm init or kubeadm upgrade apply), kubeadm respects the value of UnversionedKubeletConfigMap. When reading that ConfigMap (during kubeadm join, kubeadm reset, kubeadm upgrade ...), kubeadm attempts to use unversioned ConfigMap name first; if that does not succeed, kubeadm falls back to using the legacy (versioned) name for that ConfigMap.

## Adding kube-proxy parameters

For information about kube-proxy parameters in the kubeadm configuration see:

- [kube-proxy reference](#)

For information about enabling IPVS mode with kubeadm see:

- [IPVS](#)

## Passing custom flags to control plane components

For information about passing flags to control plane components see:

- [control-plane-flags](#)

## Running kubeadm without an Internet connection

For running kubeadm without an Internet connection you have to pre-pull the required control-plane images.

You can list and pull the images using the kubeadm config images sub-command:

```
kubeadm config images list
kubeadm config images pull
```

You can pass `--config` to the above commands with a [kubeadm configuration file](#) to control the `kubernetesVersion` and `imageRepository` fields.

All default `registry.k8s.io` images that kubeadm requires support multiple architectures.

## Using custom images

By default, kubeadm pulls images from `registry.k8s.io`. If the requested Kubernetes version is a CI label (such as `ci/latest`) `gcr.io/k8s-staging-ci-images` is used.

You can override this behavior by using [kubeadm with a configuration file](#). Allowed customization are:

- To provide `kubernetesVersion` which affects the version of the images.
- To provide an alternative `imageRepository` to be used instead of `registry.k8s.io`.
- To provide a specific `imageRepository` and `imageTag` for `etcd` or `CoreDNS`.

Image paths between the default `registry.k8s.io` and a custom repository specified using `imageRepository` may differ for backwards compatibility reasons. For example, one image might have a subpath at `registry.k8s.io/subpath/image`, but be defaulted to `my.customrepository.io/image` when using a custom repository.

To ensure you push the images to your custom repository in paths that kubeadm can consume, you must:

- Pull images from the defaults paths at `registry.k8s.io` using `kubeadm config images {list|pull}`.
- Push images to the paths from `kubeadm config images list --config=config.yaml`, where `config.yaml` contains the custom `imageRepository`, and/or `imageTag` for `etcd` and `CoreDNS`.
- Pass the same `config.yaml` to `kubeadm init`.

## Custom sandbox (pause) images

To set a custom image for these you need to configure this in your [container runtime](#) to use the image. Consult the documentation for your container runtime to find out how to change this setting; for selected container runtimes, you can also find advice within the [Container Runtimes](#) topic.

## Uploading control-plane certificates to the cluster

By adding the flag `--upload-certs` to `kubeadm init` you can temporary upload the control-plane certificates to a Secret in the cluster. Please note that this Secret will expire automatically after 2 hours. The certificates are encrypted using a 32byte key that can be specified using `--certificate-key`. The same key can be used to download the certificates when additional control-plane nodes are joining, by passing `--control-plane` and `--certificate-key` to `kubeadm join`.

The following phase command can be used to re-upload the certificates after expiration:

```
kubeadm init phase upload-certs --upload-certs --config=SOME_YAML_FILE
```

**Note:**

A predefined `certificateKey` can be provided in `InitConfiguration` when passing the [configuration file](#) with `--config`.

If a predefined certificate key is not passed to `kubeadm init` and `kubeadm init` phase upload-certs a new key will be generated automatically.

The following command can be used to generate a new key on demand:

```
kubeadm certs certificate-key
```

## Certificate management with kubeadm

For detailed information on certificate management with kubeadm see [Certificate Management with kubeadm](#). The document includes information about using external CA, custom certificates and certificate renewal.

## Managing the kubeadm drop-in file for the kubelet

The kubeadm package ships with a configuration file for running the kubelet by systemd. Note that the kubeadm CLI never touches this drop-in file. This drop-in file is part of the kubeadm DEB/RPM package.

For further information, see [Managing the kubeadm drop-in file for systemd](#).

## Use kubeadm with CRI runtimes

By default kubeadm attempts to detect your container runtime. For more details on this detection, see the [kubeadm CRI installation guide](#).

## Setting the node name

By default, kubeadm assigns a node name based on a machine's host address. You can override this setting with the `--node-name` flag. The flag passes the appropriate [--hostname-override](#) value to the kubelet.

Be aware that overriding the hostname can [interfere with cloud providers](#).

## Automating kubeadm

Rather than copying the token you obtained from `kubeadm init` to each node, as in the [basic kubeadm tutorial](#), you can parallelize the token distribution for easier automation. To implement this automation, you must know the IP address that the control-plane node will have after it is started, or use a DNS name or an address of a load balancer.

1. Generate a token. This token must have the form `<6 character string>.<16 character string>`. More formally, it must match the regex: `[a-z0-9]{6}\.[a-z0-9]{16}`.

kubeadm can generate a token for you:

```
kubeadm token generate
```

- Start both the control-plane node and the worker nodes concurrently with this token. As
2. they come up they should find each other and form the cluster. The same `--token` argument can be used on both `kubeadm init` and `kubeadm join`.
  3. Similar can be done for `--certificate-key` when joining additional control-plane nodes. The key can be generated using:

```
kubeadm certs certificate-key
```

Once the cluster is up, you can use the `/etc/kubernetes/admin.conf` file from a control-plane node to talk to the cluster with administrator credentials or [Generating kubeconfig files for additional users](#).

Note that this style of bootstrap has some relaxed security guarantees because it does not allow the root CA hash to be validated with `--discovery-token-ca-cert-hash` (since it's not generated when the nodes are provisioned). For details, see the [kubeadm join](#).

## What's next

- [kubeadm init phase](#) to understand more about kubeadm init phases
- [kubeadm join](#) to bootstrap a Kubernetes worker node and join it to the cluster
- [kubeadm upgrade](#) to upgrade a Kubernetes cluster to a newer version
- [kubeadm reset](#) to revert any changes made to this host by kubeadm init or kubeadm join

# kubeadm join

This command initializes a Kubernetes worker node and joins it to the cluster.

Run this on any machine you wish to join an existing cluster

## Synopsis

When joining a kubeadm initialized cluster, we need to establish bidirectional trust. This is split into discovery (having the Node trust the Kubernetes Control Plane) and TLS bootstrap (having the Kubernetes Control Plane trust the Node).

There are 2 main schemes for discovery. The first is to use a shared token along with the IP address of the API server. The second is to provide a file - a subset of the standard kubeconfig file. The discovery/kubeconfig file supports token, client-go authentication plugins ("exec"), "tokenFile", and "authProvider". This file can be a local file or downloaded via an HTTPS URL. The forms are `kubeadm join --discovery-token abcdef.1234567890abcdef 1.2.3.4:6443`, `kubeadm join --discovery-file path/to/file.conf`, or `kubeadm join --discovery-file https://url/file.conf`. Only one form can be used. If the discovery information is loaded from a URL, HTTPS must be used. Also, in that case the host installed CA bundle is used to verify the connection.

If you use a shared token for discovery, you should also pass the `--discovery-token-ca-cert-hash` flag to validate the public key of the root certificate authority (CA) presented by the Kubernetes Control Plane. The value of this flag is specified as "`<hash-type>:<hex-encoded-value>`", where the supported hash type is "sha256". The hash is calculated over the bytes of the Subject Public Key Info (SPKI) object (as in RFC7469). This value is available in the output of "kubeadm init" or

can be calculated using standard tools. The `--discovery-token-ca-cert-hash` flag may be repeated multiple times to allow more than one public key.

If you cannot know the CA public key hash ahead of time, you can pass the `--discovery-token-unsafe-skip-ca-verification` flag to disable this verification. This weakens the kubeadm security model since other nodes can potentially impersonate the Kubernetes Control Plane.

The TLS bootstrap mechanism is also driven via a shared token. This is used to temporarily authenticate with the Kubernetes Control Plane to submit a certificate signing request (CSR) for a locally created key pair. By default, kubeadm will set up the Kubernetes Control Plane to automatically approve these signing requests. This token is passed in with the `--tls-bootstrap-token abcdef.1234567890abcdef` flag.

Often times the same token is used for both parts. In this case, the `--token` flag can be used instead of specifying each token individually.

The "join [api-server-endpoint]" command executes the following phases:

```
preflight Run join pre-flight checks
control-plane-prepare Prepare the machine for serving a control plane
/download-certs [EXPERIMENTAL] Download certificates shared among control-plane
nodes from the kubeadm-certs Secret
/certs Generate the certificates for the new control plane components
/kubeconfig Generate the kubeconfig for the new control plane components
/control-plane Generate the manifests for the new control plane components
kubelet-start Write kubelet settings, certificates and (re)start the kubelet
control-plane-join Join a machine as a control plane instance
/etc Add a new local etcd member
/update-status Register the new control-plane node into the ClusterStatus maintained in
the kubeadm-config ConfigMap (DEPRECATED)
/mark-control-plane Mark a node as a control-plane
wait-control-plane EXPERIMENTAL: Wait for the control plane to start
```

```
kubeadm join [api-server-endpoint] [flags]
```

## Options

|                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--apiserver-advertise-address</code> string                                                                                                                            |
| If the node should host a new control plane instance, the IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| <code>--apiserver-bind-port</code> int32    Default: 6443                                                                                                                    |
| If the node should host a new control plane instance, the port for the API Server to bind to.                                                                                |
| <code>--certificate-key</code> string                                                                                                                                        |
| Use this key to decrypt the certificate secrets uploaded by init. The certificate key is a hex encoded string that is an AES key of size 32 bytes.                           |
| <code>--config</code> string                                                                                                                                                 |



|                                               |                                                                                                                                                                                                                                                                            |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | Path to a kubeadm configuration file.                                                                                                                                                                                                                                      |
| --control-plane                               |                                                                                                                                                                                                                                                                            |
|                                               | Create a new control plane instance on this node                                                                                                                                                                                                                           |
| --cri-socket string                           |                                                                                                                                                                                                                                                                            |
|                                               | Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket.                                                                               |
| --discovery-file string                       |                                                                                                                                                                                                                                                                            |
|                                               | For file-based discovery, a file or URL from which to load cluster information.                                                                                                                                                                                            |
| --discovery-token string                      |                                                                                                                                                                                                                                                                            |
|                                               | For token-based discovery, the token used to validate cluster information fetched from the API server.                                                                                                                                                                     |
| --discovery-token-ca-cert-hash strings        |                                                                                                                                                                                                                                                                            |
|                                               | For token-based discovery, validate that the root CA public key matches this hash (format: "<type>:<value>").                                                                                                                                                              |
| --discovery-token-unsafe-skip-ca-verification |                                                                                                                                                                                                                                                                            |
|                                               | For token-based discovery, allow joining without --discovery-token-ca-cert-hash pinning.                                                                                                                                                                                   |
| --dry-run                                     |                                                                                                                                                                                                                                                                            |
|                                               | Don't apply any changes; just output what would be done.                                                                                                                                                                                                                   |
| -h, --help                                    |                                                                                                                                                                                                                                                                            |
|                                               | help for join                                                                                                                                                                                                                                                              |
| --ignore-preflight-errors strings             |                                                                                                                                                                                                                                                                            |
|                                               | A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                                                                                                                                     |
| --node-name string                            |                                                                                                                                                                                                                                                                            |
|                                               | Specify the node name.                                                                                                                                                                                                                                                     |
| --patches string                              |                                                                                                                                                                                                                                                                            |
|                                               | Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". |

|                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |
| --skip-phases strings                                                                                                                                                                                                                                                                                                |
| List of phases to be skipped                                                                                                                                                                                                                                                                                         |
| --tls-bootstrap-token string                                                                                                                                                                                                                                                                                         |
| Specify the token used to temporarily authenticate with the Kubernetes Control Plane while joining the node.                                                                                                                                                                                                         |
| --token string                                                                                                                                                                                                                                                                                                       |
| Use this token for both discovery-token and tls-bootstrap-token when those values are not provided.                                                                                                                                                                                                                  |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## The join workflow

kubeadm join bootstraps a Kubernetes worker node or a control-plane node and adds it to the cluster. This action consists of the following steps for worker nodes:

1. kubeadm downloads necessary cluster information from the API server. By default, it uses the bootstrap token and the CA key hash to verify the authenticity of that data. The root CA can also be discovered directly via a file or URL.
2. Once the cluster information is known, kubelet can start the TLS bootstrapping process.

The TLS bootstrap uses the shared token to temporarily authenticate with the Kubernetes API server to submit a certificate signing request (CSR); by default the control plane signs this CSR request automatically.

3. Finally, kubeadm configures the local kubelet to connect to the API server with the definitive identity assigned to the node.

For control-plane nodes additional steps are performed:

1. Downloading certificates shared among control-plane nodes from the cluster (if explicitly requested by the user).
2. Generating control-plane component manifests, certificates and kubeconfig.
3. Adding new local etcd member.

## Using join phases with kubeadm

Kubeadm allows you join a node to the cluster in phases using `kubeadm join phase`.

To view the ordered list of phases and sub-phases you can call `kubeadm join --help`. The list will be located at the top of the help screen and each phase will have a description next to it. Note that by calling `kubeadm join` all of the phases and sub-phases will be executed in this exact order.

Some phases have unique flags, so if you want to have a look at the list of available options add `--help`, for example:

```
kubeadm join phase kubelet-start --help
```

Similar to the [kubeadm init phase](#) command, `kubeadm join phase` allows you to skip a list of phases using the `--skip-phases` flag.

For example:

```
sudo kubeadm join --skip-phases=preflight --config=config.yaml
```

FEATURE STATE: Kubernetes v1.22 [beta]

Alternatively, you can use the `skipPhases` field in `JoinConfiguration`.

## Discovering what cluster CA to trust

The kubeadm discovery has several options, each with security tradeoffs. The right method for your environment depends on how you provision nodes and the security expectations you have about your network and node lifecycles.

### Token-based discovery with CA pinning

This is the default mode in kubeadm. In this mode, kubeadm downloads the cluster configuration (including root CA) and validates it using the token as well as validating that the root CA public key matches the provided hash and that the API server certificate is valid under the root CA.

The CA key hash has the format `sha256:<hex_encoded_hash>`. By default, the hash value is printed at the end of the `kubeadm init` command or in the output from the `kubeadm token create --print-join-command` command. It is in a standard format (see [RFC7469](#)) and can also be calculated by 3rd party tools or provisioning systems. For example, using the OpenSSL CLI:

```
openssl x509 -pubkey -in /etc/kubernetes/pki/ca.crt | openssl rsa -pubin -outform der 2>/dev/null | openssl dgst -sha256 -hex | sed 's/^.* //'
```

### Example kubeadm join commands:

For worker nodes:

```
kubeadm join --discovery-token abcdef.1234567890abcdef --discovery-token-ca-cert-hash sha256:1234..cdef 1.2.3.4:6443
```

For control-plane nodes:

```
kubeadm join --discovery-token abcdef.1234567890abcdef --discovery-token-ca-cert-hash sha256:1234..cdef --control-plane 1.2.3.4:6443
```

You can also call join for a control-plane node with `--certificate-key` to copy certificates to this node, if the `kubeadm init` command was called with `--upload-certs`.

#### **Advantages:**

- Allows bootstrapping nodes to securely discover a root of trust for the control-plane node even if other worker nodes or the network are compromised.
- Convenient to execute manually since all of the information required fits into a single `kubeadm join` command.

#### **Disadvantages:**

- The CA hash is not normally known until the control-plane node has been provisioned, which can make it more difficult to build automated provisioning tools that use `kubeadm`. By generating your CA in beforehand, you may work around this limitation.

### **Token-based discovery without CA pinning**

This mode relies only on the symmetric token to sign (HMAC-SHA256) the discovery information that establishes the root of trust for the control-plane. To use the mode the joining nodes must skip the hash validation of the CA public key, using `--discovery-token-unsafe-skip-ca-verification`. You should consider using one of the other modes if possible.

#### **Example kubeadm join command:**

```
kubeadm join --token abcdef.1234567890abcdef --discovery-token-unsafe-skip-ca-verification 1.2.3.4:6443
```

#### **Advantages:**

- Still protects against many network-level attacks.
- The token can be generated ahead of time and shared with the control-plane node and worker nodes, which can then bootstrap in parallel without coordination. This allows it to be used in many provisioning scenarios.

#### **Disadvantages:**

- If an attacker is able to steal a bootstrap token via some vulnerability, they can use that token (along with network-level access) to impersonate the control-plane node to other bootstrapping nodes. This may or may not be an appropriate tradeoff in your environment.

### **File or HTTPS-based discovery**

This provides an out-of-band way to establish a root of trust between the control-plane node and bootstrapping nodes. Consider using this mode if you are building automated provisioning using `kubeadm`. The format of the discovery file is a regular Kubernetes [kubeconfig](#) file.

In case the discovery file does not contain credentials, the TLS discovery token will be used.

## Example kubeadm join commands:

- `kubeadm join --discovery-file path/to/file.conf` (local file)
- `kubeadm join --discovery-file https://url/file.conf` (remote HTTPS URL)

## Advantages:

- Allows bootstrapping nodes to securely discover a root of trust for the control-plane node even if the network or other worker nodes are compromised.

## Disadvantages:

- Requires that you have some way to carry the discovery information from the control-plane node to the bootstrapping nodes. If the discovery file contains credentials you must keep it secret and transfer it over a secure channel. This might be possible with your cloud provider or provisioning tool.

## Use of custom kubelet credentials with kubeadm join

To allow kubeadm join to use predefined kubelet credentials and skip client TLS bootstrap and CSR approval for a new node:

1. From a working control plane node in the cluster that has `/etc/kubernetes/pki/ca.key` execute `kubeadm kubeconfig user --org system:nodes --client-name system:node:$NODE > kubelet.conf`. `$NODE` must be set to the name of the new node.
2. Modify the resulted `kubelet.conf` manually to adjust the cluster name and the server endpoint, or run `kubeadm kubeconfig user --config` (it accepts `InitConfiguration`).

If your cluster does not have the `ca.key` file, you must sign the embedded certificates in the `kubelet.conf` externally. For additional information, see [PKI certificates and requirements](#) and [Certificate Management with kubeadm](#).

1. Copy the resulting `kubelet.conf` to `/etc/kubernetes/kubelet.conf` on the new node.
2. Execute `kubeadm join` with the flag `--ignore-preflight-errors=FileAvailable--etc-kubernetes-kubelet.conf` on the new node.

## Securing your installation even more

The defaults for kubeadm may not work for everyone. This section documents how to tighten up a kubeadm installation at the cost of some usability.

### Turning off auto-approval of node client certificates

By default, there is a CSR auto-approver enabled that basically approves any client certificate request for a kubelet when a Bootstrap Token was used when authenticating. If you don't want the cluster to automatically approve kubelet client certs, you can turn it off by executing this command:

```
kubectrl delete clusterrolebinding kubeadm:node-autoapprove-bootstrap
```

After that, `kubeadm join` will block until the admin has manually approved the CSR in flight:

1. Using `kubectl get csr`, you can see that the original CSR is in the Pending state.

```
kubectl get csr
```

The output is similar to this:

| NAME                                                 | AGE | REQUESTOR | CONDITION |
|------------------------------------------------------|-----|-----------|-----------|
| node-csr-c69HXe7aYcqkS1bKmH4faEnHAWxn6i2bHZ2mD04jZyQ | 18s |           |           |
| system:bootstrap:878f07                              |     | Pending   |           |

2. `kubectl certificate approve` allows the admin to approve CSR. This action tells a certificate signing controller to issue a certificate to the requestor with the attributes requested in the CSR.

```
kubectl certificate approve node-csr-c69HXe7aYcqkS1bKmH4faEnHAWxn6i2bHZ2mD04jZyQ
```

The output is similar to this:

```
certificatesigningrequest "node-csr-c69HXe7aYcqkS1bKmH4faEnHAWxn6i2bHZ2mD04jZyQ" approved
```

3. This would change the CRS resource to Active state.

```
kubectl get csr
```

The output is similar to this:

| NAME                                                 | AGE | REQUESTOR       | CONDITION |
|------------------------------------------------------|-----|-----------------|-----------|
| node-csr-c69HXe7aYcqkS1bKmH4faEnHAWxn6i2bHZ2mD04jZyQ | 1m  |                 |           |
| system:bootstrap:878f07                              |     | Approved,Issued |           |

This forces the workflow that `kubeadm join` will only succeed if `kubectl certificate approve` has been run.

## Turning off public access to the cluster-info ConfigMap

In order to achieve the joining flow using the token as the only piece of validation information, a ConfigMap with some data needed for validation of the control-plane node's identity is exposed publicly by default. While there is no private data in this ConfigMap, some users might wish to turn it off regardless. Doing so will disable the ability to use the `--discovery-token` flag of the `kubeadm join` flow. Here are the steps to do so:

- Fetch the cluster-info file from the API Server:

```
kubectl -n kube-public get cm cluster-info -o jsonpath='{.data.kubeconfig}' | tee cluster-info.yaml
```

The output is similar to this:

```
apiVersion: v1
kind: Config
clusters:
```

```
- cluster:
 certificate-authority-data: <ca-cert>
 server: https://<ip>:<port>
 name: ""
contexts: []
current-context: ""
preferences: {}
users: []
```

- Use the cluster-info.yaml file as an argument to `kubeadm join --discovery-file`.
- Turn off public access to the cluster-info ConfigMap:

```
kubectrl -n kube-public delete rolebinding kubeadm:bootstrap-signer-clusterinfo
```

These commands should be run after `kubeadm init` but before `kubeadm join`.

## Using kubeadm join with a configuration file

### Caution:

The config file is still considered beta and may change in future versions.

It's possible to configure `kubeadm join` with a configuration file instead of command line flags, and some more advanced features may only be available as configuration file options. This file is passed using the `--config` flag and it must contain a `JoinConfiguration` structure. Mixing `--config` with others flags may not be allowed in some cases.

The default configuration can be printed out using the [kubeadm config print](#) command.

If your configuration is not using the latest version it is **recommended** that you migrate using the [kubeadm config migrate](#) command.

For more information on the fields and usage of the configuration you can navigate to our [API reference](#).

## What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node.
- [kubeadm token](#) to manage tokens for `kubeadm join`.
- [kubeadm reset](#) to revert any changes made to this host by `kubeadm init` or `kubeadm join`.

# kubeadm upgrade

`kubeadm upgrade` is a user-friendly command that wraps complex upgrading logic behind one command, with support for both planning an upgrade and actually performing it.

## kubeadm upgrade guidance

The steps for performing an upgrade using `kubeadm` are outlined in [this document](#). For older versions of `kubeadm`, please refer to older documentation sets of the Kubernetes website.

You can use `kubeadm upgrade diff` to see the changes that would be applied to static pod manifests.

In Kubernetes v1.15.0 and later, `kubeadm upgrade apply` and `kubeadm upgrade node` will also automatically renew the kubeadm managed certificates on this node, including those stored in kubeconfig files. To opt-out, it is possible to pass the flag `--certificate-renewal=false`. For more details about certificate renewal see the [certificate management documentation](#).

#### Note:

The commands `kubeadm upgrade apply` and `kubeadm upgrade plan` have a legacy `--config` flag which makes it possible to reconfigure the cluster, while performing planning or upgrade of that particular control-plane node. Please be aware that the upgrade workflow was not designed for this scenario and there are reports of unexpected results.

## kubeadm upgrade plan

Check which versions are available to upgrade to and validate whether your current cluster is upgradeable.

### Synopsis

Check which versions are available to upgrade to and validate whether your current cluster is upgradeable. This command can only run on the control plane nodes where the kubeconfig file "admin.conf" exists. To skip the internet check, pass in the optional [version] parameter.

```
kubeadm upgrade plan [version] [flags]
```

### Options

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--allow-experimental-upgrades</code>                                                                                                      |
| Show unstable versions of Kubernetes as an upgrade alternative and allow upgrading to an alpha/beta/release candidate versions of Kubernetes.   |
| <code>--allow-missing-template-keys</code> Default: true                                                                                        |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
| <code>--allow-release-candidate-upgrades</code>                                                                                                 |
| Show release candidate versions of Kubernetes as an upgrade alternative and allow upgrading to a release candidate versions of Kubernetes.      |
| <code>--config</code> string                                                                                                                    |
| Path to a kubeadm configuration file.                                                                                                           |
| <code>-o, --experimental-output</code> string    Default: "text"                                                                                |



|                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output format. One of: text json yaml go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                                                                                                                                                                                   |
| --feature-gates string                                                                                                                                                                                                                                                                                                                                                                                              |
| A set of key=value pairs that describe feature gates for various features. Options are:<br>EtcdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                          |
| help for plan                                                                                                                                                                                                                                                                                                                                                                                                       |
| --ignore-preflight-errors strings                                                                                                                                                                                                                                                                                                                                                                                   |
| A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                                                                                                                                                                                                                                                                              |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf"                                                                                                                                                                                                                                                                                                                                                        |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                                                                                                                                                                                                                                        |
| --print-config                                                                                                                                                                                                                                                                                                                                                                                                      |
| Specifies whether the configuration file that will be used in the upgrade should be printed or not.                                                                                                                                                                                                                                                                                                                 |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                               |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                       |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm upgrade apply

Upgrade your Kubernetes cluster to the specified version

### Synopsis

Upgrade your Kubernetes cluster to the specified version

kubeadm upgrade apply [version]

## Options

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --allow-experimental-upgrades         | Show unstable versions of Kubernetes as an upgrade alternative and allow upgrading to an alpha/beta/release candidate versions of Kubernetes.                                                                                                                                                                                                                                                                       |
| --allow-release-candidate-upgrades    | Show release candidate versions of Kubernetes as an upgrade alternative and allow upgrading to a release candidate versions of Kubernetes.                                                                                                                                                                                                                                                                          |
| --certificate-renewal   Default: true | Perform the renewal of certificates used by component changed during upgrades.                                                                                                                                                                                                                                                                                                                                      |
| --config string                       | Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                               |
| --dry-run                             | Do not change any state, just output what actions would be performed.                                                                                                                                                                                                                                                                                                                                               |
| --etcd-upgrade   Default: true        | Perform the upgrade of etcd.                                                                                                                                                                                                                                                                                                                                                                                        |
| --feature-gates string                | A set of key=value pairs that describe feature gates for various features. Options are:<br>EtcdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |
| -f, --force                           | Force upgrading although some requirements might not be met. This also implies non-interactive mode.                                                                                                                                                                                                                                                                                                                |
| -h, --help                            | help for apply                                                                                                                                                                                                                                                                                                                                                                                                      |
| --ignore-preflight-errors strings     |                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |
| --print-config                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Specifies whether the configuration file that will be used in the upgrade should be printed or not.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -y, --yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Perform the upgrade and do not prompt for confirmation (non-interactive mode).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm upgrade diff

Show what differences would be applied to existing static pod manifests. See also: kubeadm upgrade apply --dry-run

### Synopsis

Show what differences would be applied to existing static pod manifests. See also: kubeadm upgrade apply --dry-run

kubeadm upgrade diff [version] [flags]

### Options

|                                                                                          |
|------------------------------------------------------------------------------------------|
| --api-server-manifest string    Default: "/etc/kubernetes/manifests/kube-apiserver.yaml" |
|------------------------------------------------------------------------------------------|

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| path to API server manifest                                                                                                                                  |
| --config string                                                                                                                                              |
| Path to a kubeadm configuration file.                                                                                                                        |
| -c, --context-lines int    Default: 3                                                                                                                        |
| How many lines of context in the diff                                                                                                                        |
| --controller-manager-manifest string    Default: "/etc/kubernetes/manifests/kube-controller-manager.yaml"                                                    |
| path to controller manifest                                                                                                                                  |
| -h, --help                                                                                                                                                   |
| help for diff                                                                                                                                                |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf"                                                                                                 |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| --scheduler-manifest string    Default: "/etc/kubernetes/manifests/kube-scheduler.yaml"                                                                      |
| path to scheduler manifest                                                                                                                                   |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm upgrade node

Upgrade commands for a node in the cluster

### Synopsis

Upgrade commands for a node in the cluster

The "node" command executes the following phases:

```

preflight Run upgrade node pre-flight checks
control-plane Upgrade the control plane instance deployed on this node, if any
kubelet-config Upgrade the kubelet configuration for this node

```

## Options

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <code>--certificate-renewal</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Default: true |
| Perform the renewal of certificates used by component changed during upgrades.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
| <code>--config</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | string        |
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| <code>--dry-run</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| Do not change any state, just output the actions that would be performed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| <code>--etcd-upgrade</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Default: true |
| Perform the upgrade of etcd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| help for node                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| <code>--ignore-preflight-errors</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | strings       |
| A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| <code>--kubeconfig</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | string        |
| Default: "/etc/kubernetes/admin.conf"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <code>--patches</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | string        |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |               |
| <code>--skip-phases</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | strings       |
| List of phases to be skipped                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs string</code>                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## What's next

- [kubeadm config](#) if you initialized your cluster using kubeadm v1.7.x or lower, to configure your cluster for kubeadm upgrade

# kubeadm config

During `kubeadm init`, kubeadm uploads the `ClusterConfiguration` object to your cluster in a `ConfigMap` called `kubeadm-config` in the `kube-system` namespace. This configuration is then read during `kubeadm join`, `kubeadm reset` and `kubeadm upgrade`.

You can use `kubeadm config print` to print the default static configuration that kubeadm uses for `kubeadm init` and `kubeadm join`.

### Note:

The output of the command is meant to serve as an example. You must manually edit the output of this command to adapt to your setup. Remove the fields that you are not certain about and kubeadm will try to default them on runtime by examining the host.

For more information on `init` and `join` navigate to [Using kubeadm init with a configuration file](#) or [Using kubeadm join with a configuration file](#).

For more information on using the kubeadm configuration API navigate to [Customizing components with the kubeadm API](#).

You can use `kubeadm config migrate` to convert your old configuration files that contain a deprecated API version to a newer, supported API version.

`kubeadm config validate` can be used for validating a configuration file.

`kubeadm config images list` and `kubeadm config images pull` can be used to list and pull the images that kubeadm requires.

## kubeadm config print

Print configuration

### Synopsis

This command prints configurations for subcommands provided. For details, see: <https://pkg.go.dev/k8s.io/kubernetes/cmd/kubeadm/app/apis/kubeadm#section-directories>

kubeadm config print [flags]

Options

|                |
|----------------|
| -h, --help     |
| help for print |

Options inherited from parent commands

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --kubeconfig string                                                                                                                                          | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |
| --rootfs string                                                                                                                                              |                                       |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |                                       |

kubeadm config print init-defaults

Print default init configuration, that can be used for 'kubeadm init'

Synopsis

This command prints objects such as the default init configuration that is used for 'kubeadm init'.

Note that sensitive values like the Bootstrap Token fields are replaced with placeholder values like "abcdef.0123456789abcdef" in order to pass validation but not perform the real computation for creating a token.

kubeadm config print init-defaults [flags]

Options

|                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --component-configs strings                                                                                                                                                                                              |
| A comma-separated list for component config API objects to print the default values for. Available values: [KubeProxyConfiguration KubeletConfiguration]. If this flag is not set, no component configs will be printed. |
| -h, --help                                                                                                                                                                                                               |
| help for init-defaults                                                                                                                                                                                                   |

## Options inherited from parent commands

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <code>--kubeconfig string</code>                                                                                                                             | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |
| <code>--rootfs string</code>                                                                                                                                 |                                       |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |                                       |

## kubeadm config print join-defaults

Print default join configuration, that can be used for 'kubeadm join'

### Synopsis

This command prints objects such as the default join configuration that is used for 'kubeadm join'.

Note that sensitive values like the Bootstrap Token fields are replaced with placeholder values like "abcdef.0123456789abcdef" in order to pass validation but not perform the real computation for creating a token.

kubeadm config print join-defaults [flags]

### Options

|                         |
|-------------------------|
| <code>-h, --help</code> |
| help for join-defaults  |

## Options inherited from parent commands

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <code>--kubeconfig string</code>                                                                                                                             | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |
| <code>--rootfs string</code>                                                                                                                                 |                                       |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |                                       |

## kubeadm config migrate

Read an older version of the kubeadm configuration API types from a file, and output the similar config object for the newer version



## Synopsis

This command lets you convert configuration objects of older versions to the latest supported version, locally in the CLI tool without ever touching anything in the cluster. In this version of kubeadm, the following API versions are supported:

- kubeadm.k8s.io/v1beta3

Further, kubeadm can only write out config of version "kubeadm.k8s.io/v1beta3", but read both types. So regardless of what version you pass to the `--old-config` parameter here, the API object will be read, deserialized, defaulted, converted, validated, and re-serialized when written to stdout or `--new-config` if specified.

In other words, the output of this command is what kubeadm actually would read internally if you submitted this file to "kubeadm init"

```
kubeadm config migrate [flags]
```

## Options

|                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--allow-experimental-api</code>                                                                                                     |
| Allow migration to experimental, unreleased APIs.                                                                                         |
| <code>-h, --help</code>                                                                                                                   |
| help for migrate                                                                                                                          |
| <code>--new-config string</code>                                                                                                          |
| Path to the resulting equivalent kubeadm config file using the new API version. Optional, if not specified output will be sent to STDOUT. |
| <code>--old-config string</code>                                                                                                          |
| Path to the kubeadm config file that is using an old API version and should be converted. This flag is mandatory.                         |

## Options inherited from parent commands

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--kubeconfig string</code> Default: "/etc/kubernetes/admin.conf"                                                                                       |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| <code>--rootfs string</code>                                                                                                                                 |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |

# kubeadm config validate

Read a file containing the kubeadm configuration API and report any validation problems

## Synopsis

This command lets you validate a kubeadm configuration API file and report any warnings and errors. If there are no errors the exit status will be zero, otherwise it will be non-zero. Any unmarshaling problems such as unknown API fields will trigger errors. Unknown API versions and fields with invalid values will also trigger errors. Any other errors or warnings may be reported depending on contents of the input file.

In this version of kubeadm, the following API versions are supported:

- kubeadm.k8s.io/v1beta3

```
kubeadm config validate [flags]
```

## Options

|                                                    |
|----------------------------------------------------|
| --allow-experimental-api                           |
| Allow validation of experimental, unreleased APIs. |
| --config string                                    |
| Path to a kubeadm configuration file.              |
| -h, --help                                         |
| help for validate                                  |

## Options inherited from parent commands

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --kubeconfig string                                                                                                                                          | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |
| --rootfs string                                                                                                                                              |                                       |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |                                       |

# kubeadm config images list

Print a list of images kubeadm will use. The configuration file is used in case any images or image repositories are customized

## Synopsis

Print a list of images kubeadm will use. The configuration file is used in case any images or image repositories are customized

```
kubeadm config images list [flags]
```

## Options

|                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| --allow-missing-template-keys                                                                                                                                                                                                                                                                                                                                                                                       | Default: true              |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                                                                                                                                                                                     |                            |
| --config string                                                                                                                                                                                                                                                                                                                                                                                                     |                            |
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                               |                            |
| -o, --experimental-output string                                                                                                                                                                                                                                                                                                                                                                                    | Default: "text"            |
| Output format. One of: text json yaml go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                                                                                                                                                                                                                                                                                   |                            |
| --feature-gates string                                                                                                                                                                                                                                                                                                                                                                                              |                            |
| A set of key=value pairs that describe feature gates for various features. Options are:<br>EtcdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |                            |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                          |                            |
| help for list                                                                                                                                                                                                                                                                                                                                                                                                       |                            |
| --image-repository string                                                                                                                                                                                                                                                                                                                                                                                           | Default: "registry.k8s.io" |
| Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                       |                            |
| --kubernetes-version string                                                                                                                                                                                                                                                                                                                                                                                         | Default: "stable-1"        |
| Choose a specific Kubernetes version for the control plane.                                                                                                                                                                                                                                                                                                                                                         |                            |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                               |                            |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                       |                            |

## Options inherited from parent commands

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <code>--kubeconfig</code> string                                                                                                                             | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |
| <code>--rootfs</code> string                                                                                                                                 |                                       |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |                                       |

## kubeadm config images pull

Pull images used by kubeadm

### Synopsis

Pull images used by kubeadm

```
kubeadm config images pull [flags]
```

### Options

|                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <code>--config</code> string                                                                                                                                                                                                                                                                                                                                                                                        |                            |
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                               |                            |
| <code>--cri-socket</code> string                                                                                                                                                                                                                                                                                                                                                                                    |                            |
| Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket.                                                                                                                                                                                                                        |                            |
| <code>--feature-gates</code> string                                                                                                                                                                                                                                                                                                                                                                                 |                            |
| A set of key=value pairs that describe feature gates for various features. Options are:<br>EtcdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |                            |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                             |                            |
| help for pull                                                                                                                                                                                                                                                                                                                                                                                                       |                            |
| <code>--image-repository</code> string                                                                                                                                                                                                                                                                                                                                                                              | Default: "registry.k8s.io" |
| Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                       |                            |
| <code>--kubernetes-version</code> string                                                                                                                                                                                                                                                                                                                                                                            | Default: "stable-1"        |

Choose a specific Kubernetes version for the control plane.

## Options inherited from parent commands

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <code>--kubeconfig string</code>                                                                                                                             | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |
| <code>--rootfs string</code>                                                                                                                                 |                                       |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |                                       |

## What's next

- [kubeadm upgrade](#) to upgrade a Kubernetes cluster to a newer version

# kubeadm reset

Performs a best effort revert of changes made by `kubeadm init` or `kubeadm join`.

Performs a best effort revert of changes made to this host by 'kubeadm init' or 'kubeadm join'

## Synopsis

Performs a best effort revert of changes made to this host by 'kubeadm init' or 'kubeadm join'

The "reset" command executes the following phases:

|                    |                             |
|--------------------|-----------------------------|
| preflight          | Run reset pre-flight checks |
| remove-etcd-member | Remove a local etcd member. |
| cleanup-node       | Run cleanup node.           |

`kubeadm reset [flags]`

## Options

|                                                                                                  |                                |
|--------------------------------------------------------------------------------------------------|--------------------------------|
| <code>--cert-dir string</code>                                                                   | Default: "/etc/kubernetes/pki" |
| The path to the directory where the certificates are stored. If specified, clean this directory. |                                |
| <code>--cleanup-tmp-dir</code>                                                                   |                                |
| Cleanup the "/etc/kubernetes/tmp" directory                                                      |                                |
| <code>--config string</code>                                                                     |                                |

|                                                                           |                                                                                                                                                                                              |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                           | Path to a kubeadm configuration file.                                                                                                                                                        |
| <code>--cri-socket</code> string                                          |                                                                                                                                                                                              |
|                                                                           | Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| <code>--dry-run</code>                                                    |                                                                                                                                                                                              |
|                                                                           | Don't apply any changes; just output what would be done.                                                                                                                                     |
| <code>-f, --force</code>                                                  |                                                                                                                                                                                              |
|                                                                           | Reset the node without prompting for confirmation.                                                                                                                                           |
| <code>-h, --help</code>                                                   |                                                                                                                                                                                              |
|                                                                           | help for reset                                                                                                                                                                               |
| <code>--ignore-preflight-errors</code> strings                            |                                                                                                                                                                                              |
|                                                                           | A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                                                       |
| <code>--kubeconfig</code> string    Default: "/etc/kubernetes/admin.conf" |                                                                                                                                                                                              |
|                                                                           | The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                 |
| <code>--skip-phases</code> strings                                        |                                                                                                                                                                                              |
|                                                                           | List of phases to be skipped                                                                                                                                                                 |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs</code> string                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## Reset workflow

kubeadm reset is responsible for cleaning up a node local file system from files that were created using the kubeadm init or kubeadm join commands. For control-plane nodes reset also removes the local stacked etcd member of this node from the etcd cluster.

kubeadm reset phase can be used to execute the separate phases of the above workflow. To skip a list of phases you can use the `--skip-phases` flag, which works in a similar way to the kubeadm join and kubeadm init phase runners.

## External etcd clean up

`kubeadm reset` will not delete any etcd data if external etcd is used. This means that if you run `kubeadm init` again using the same etcd endpoints, you will see state from previous clusters.

To wipe etcd data it is recommended you use a client like `etcdctl`, such as:

```
etcdctl del "" --prefix
```

See the [etcd documentation](#) for more information.

## Graceful kube-apiserver shutdown

If you have your kube-apiserver configured with the `--shutdown-delay-duration` flag, you can run the following commands to attempt a graceful shutdown for the running API server Pod, before you run `kubeadm reset`:

```
yq eval -i '.spec.containers[0].command = []' /etc/kubernetes/manifests/kube-apiserver.yaml
timeout 60 sh -c 'while pgrep kube-apiserver >/dev/null; do sleep 1; done' || true
```

## What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node
- [kubeadm join](#) to bootstrap a Kubernetes worker node and join it to the cluster

# kubeadm token

Bootstrap tokens are used for establishing bidirectional trust between a node joining the cluster and a control-plane node, as described in [authenticating with bootstrap tokens](#).

`kubeadm init` creates an initial token with a 24-hour TTL. The following commands allow you to manage such a token and also to create and manage new ones.

## kubeadm token create

Create bootstrap tokens on the server

### Synopsis

This command will create a bootstrap token for you. You can specify the usages for this token, the "time to live" and an optional human friendly description.

The `[token]` is the actual token to write. This should be a securely generated random token of the form `"[a-z0-9]{6}.[a-z0-9]{16}"`. If no `[token]` is given, `kubeadm` will generate a random token instead.

```
kubeadm token create [token]
```

## Options

|                                                                                                                                                                                                                                  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| --certificate-key string                                                                                                                                                                                                         |  |
| When used together with '--print-join-command', print the full 'kubeadm join' flag needed to join the cluster as a control-plane. To create a new certificate key you must use 'kubeadm init phase upload-certs --upload-certs'. |  |
| --config string                                                                                                                                                                                                                  |  |
| Path to a kubeadm configuration file.                                                                                                                                                                                            |  |
| --description string                                                                                                                                                                                                             |  |
| A human friendly description of how this token is used.                                                                                                                                                                          |  |
| --groups strings    Default: "system:bootstrappers:kubeadm:default-node-token"                                                                                                                                                   |  |
| Extra groups that this token will authenticate as when used for authentication. Must match "\Asystem:bootstrappers:[a-z0-9:-]{0,255}[a-z0-9]\z"                                                                                  |  |
| -h, --help                                                                                                                                                                                                                       |  |
| help for create                                                                                                                                                                                                                  |  |
| --print-join-command                                                                                                                                                                                                             |  |
| Instead of printing only the token, print the full 'kubeadm join' flag needed to join the cluster using the token.                                                                                                               |  |
| --ttl duration    Default: 24h0m0s                                                                                                                                                                                               |  |
| The duration before the token is automatically deleted (e.g. 1s, 2m, 3h). If set to '0', the token will never expire                                                                                                             |  |
| --usages strings    Default: "signing,authentication"                                                                                                                                                                            |  |
| Describes the ways in which this token can be used. You can pass --usages multiple times or provide a comma separated list of options. Valid options: [signing,authentication]                                                   |  |

## Options inherited from parent commands

|                                                              |  |
|--------------------------------------------------------------|--|
| --dry-run                                                    |  |
| Whether to enable dry-run mode or not                        |  |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf" |  |



|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| --rootfs string                                                                                                                                              |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |

## kubeadm token delete

Delete bootstrap tokens on the server

### Synopsis

This command will delete a list of bootstrap tokens for you.

The [token-value] is the full Token of the form "[a-z0-9]{6}.[a-z0-9]{16}" or the Token ID of the form "[a-z0-9]{6}" to delete.

```
kubeadm token delete [token-value] ...
```

### Options

|                 |
|-----------------|
| -h, --help      |
| help for delete |

### Options inherited from parent commands

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --dry-run                                                                                                                                                    |
| Whether to enable dry-run mode or not                                                                                                                        |
| --kubeconfig string   Default: "/etc/kubernetes/admin.conf"                                                                                                  |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| --rootfs string                                                                                                                                              |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |

## kubeadm token generate

Generate and print a bootstrap token, but do not create it on the server

## Synopsis

This command will print out a randomly-generated bootstrap token that can be used with the "init" and "join" commands.

You don't have to use this command in order to generate a token. You can do so yourself as long as it is in the format "[a-z0-9]{6}.[a-z0-9]{16}". This command is provided for convenience to generate tokens in the given format.

You can also use "kubeadm init" without specifying a token and it will generate and print one for you.

```
kubeadm token generate [flags]
```

## Options

|                   |
|-------------------|
| -h, --help        |
| help for generate |

## Options inherited from parent commands

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --dry-run                                                                                                                                                    |
| Whether to enable dry-run mode or not                                                                                                                        |
| --kubeconfig string   Default: "/etc/kubernetes/admin.conf"                                                                                                  |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| --rootfs string                                                                                                                                              |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |

## kubeadm token list

List bootstrap tokens on the server

## Synopsis

This command will list all bootstrap tokens for you.

```
kubeadm token list [flags]
```

## Options

|                                               |
|-----------------------------------------------|
| --allow-missing-template-keys   Default: true |
|-----------------------------------------------|

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |
| -o, --experimental-output string    Default: "text"                                                                                             |
| Output format. One of: text json yaml go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.               |
| -h, --help                                                                                                                                      |
| help for list                                                                                                                                   |
| --show-managed-fields                                                                                                                           |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                   |

## Options inherited from parent commands

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --dry-run                                                                                                                                                    |
| Whether to enable dry-run mode or not                                                                                                                        |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf"                                                                                                 |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| --rootfs string                                                                                                                                              |
| [EXPERIMENTAL] The path to the 'real' host root filesystem.                                                                                                  |

## What's next

- [kubeadm join](#) to bootstrap a Kubernetes worker node and join it to the cluster

# kubeadm version

This command prints the version of kubeadm.

Print the version of kubeadm

## Synopsis

Print the version of kubeadm

```
kubeadm version [flags]
```

## Options

|                                                                 |
|-----------------------------------------------------------------|
| -h, --help                                                      |
| help for version                                                |
| -o, --output string                                             |
| Output format; available options are 'yaml', 'json' and 'short' |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

# kubeadm alpha

### Caution:

kubeadm alpha provides a preview of a set of features made available for gathering feedback from the community. Please try it out and give us feedback!

Currently there are no experimental commands under kubeadm alpha.

## What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node
- [kubeadm join](#) to connect a node to the cluster
- [kubeadm reset](#) to revert any changes made to this host by kubeadm init or kubeadm join

# kubeadm certs

kubeadm certs provides utilities for managing certificates. For more details on how these commands can be used, see [Certificate Management with kubeadm](#).

## kubeadm certs

A collection of operations for operating Kubernetes certificates.

- [overview](#)

Commands related to handling kubernetes certificates

## Synopsis

Commands related to handling kubernetes certificates

```
kubeadm certs [flags]
```

## Options

|                |
|----------------|
| -h, --help     |
| help for certs |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm certs renew

You can renew all Kubernetes certificates using the all subcommand or renew them selectively. For more details see [Manual certificate renewal](#).

- [renew](#)
- [all](#)
- [admin.conf](#)
- [apiserver-etcd-client](#)
- [apiserver-kubelet-client](#)
- [apiserver](#)
- [controller-manager.conf](#)
- [etcd-healthcheck-client](#)
- [etcd-peer](#)
- [etcd-server](#)
- [front-proxy-client](#)
- [scheduler.conf](#)
- [super-admin.conf](#)

Renew certificates for a Kubernetes cluster

## Synopsis

This command is not meant to be run on its own. See list of available subcommands.

```
kubeadm certs renew [flags]
```

## Options

|            |
|------------|
| -h, --help |
|------------|

|                |
|----------------|
| help for renew |
|----------------|

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew all available certificates

## Synopsis

Renew all known certificates necessary to run the control plane. Renewals are run unconditionally, regardless of expiration date. Renewals can also be run individually for more control.

```
kubeadm certs renew all [flags]
```

## Options

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --cert-dir string                                                                                                                                            | Default: "/etc/kubernetes/pki"        |
| The path where to save the certificates                                                                                                                      |                                       |
| --config string                                                                                                                                              |                                       |
| Path to a kubeadm configuration file.                                                                                                                        |                                       |
| -h, --help                                                                                                                                                   |                                       |
| help for all                                                                                                                                                 |                                       |
| --kubeconfig string                                                                                                                                          | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate embedded in the kubeconfig file for the admin to use and for kubeadm itself

## Synopsis

Renew the certificate embedded in the kubeconfig file for the admin to use and for kubeadm itself.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

```
kubeadm certs renew admin.conf [flags]
```

## Options

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <code>--cert-dir</code> string                                                                                                                               | Default: "/etc/kubernetes/pki"        |
| The path where to save the certificates                                                                                                                      |                                       |
| <code>--config</code> string                                                                                                                                 |                                       |
| Path to a kubeadm configuration file.                                                                                                                        |                                       |
| <code>-h, --help</code>                                                                                                                                      |                                       |
| help for admin.conf                                                                                                                                          |                                       |
| <code>--kubeconfig</code> string                                                                                                                             | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs</code> string                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate the apiserver uses to access etcd

## Synopsis

Renew the certificate the apiserver uses to access etcd.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

```
kubeadm certs renew apiserver-etcd-client [flags]
```

Options

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --cert-dir string                                                                                                                                            | Default: "/etc/kubernetes/pki"        |
| The path where to save the certificates                                                                                                                      |                                       |
| --config string                                                                                                                                              |                                       |
| Path to a kubeadm configuration file.                                                                                                                        |                                       |
| -h, --help                                                                                                                                                   |                                       |
| help for apiserver-etcd-client                                                                                                                               |                                       |
| --kubeconfig string                                                                                                                                          | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |

Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate for the API server to connect to kubelet

Synopsis

Renew the certificate for the API server to connect to kubelet.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.



After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

```
kubeadm certs renew apiserver-kubelet-client [flags]
```

## Options

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <code>--cert-dir</code> string                                                                                                                               | Default: "/etc/kubernetes/pki"        |
| The path where to save the certificates                                                                                                                      |                                       |
| <code>--config</code> string                                                                                                                                 |                                       |
| Path to a kubeadm configuration file.                                                                                                                        |                                       |
| <code>-h, --help</code>                                                                                                                                      |                                       |
| help for apiserver-kubelet-client                                                                                                                            |                                       |
| <code>--kubeconfig</code> string                                                                                                                             | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs</code> string                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate for serving the Kubernetes API

## Synopsis

Renew the certificate for serving the Kubernetes API.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

```
kubeadm certs renew apiserver [flags]
```

## Options

|                                                                                                                                                              |                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <code>--cert-dir string</code>                                                                                                                               | Default: <code>"/etc/kubernetes/pki"</code>        |
| The path where to save the certificates                                                                                                                      |                                                    |
| <code>--config string</code>                                                                                                                                 |                                                    |
| Path to a kubeadm configuration file.                                                                                                                        |                                                    |
| <code>-h, --help</code>                                                                                                                                      |                                                    |
| help for apiserver                                                                                                                                           |                                                    |
| <code>--kubeconfig string</code>                                                                                                                             | Default: <code>"/etc/kubernetes/admin.conf"</code> |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                                    |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs string</code>                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate embedded in the kubeconfig file for the controller manager to use

## Synopsis

Renew the certificate embedded in the kubeconfig file for the controller manager to use.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

```
kubeadm certs renew controller-manager.conf [flags]
```

## Options

|                                         |                                             |
|-----------------------------------------|---------------------------------------------|
| <code>--cert-dir string</code>          | Default: <code>"/etc/kubernetes/pki"</code> |
| The path where to save the certificates |                                             |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--config string</code>                                                                                                                                 |
| Path to a kubeadm configuration file.                                                                                                                        |
| <code>-h, --help</code>                                                                                                                                      |
| help for controller-manager.conf                                                                                                                             |
| <code>--kubeconfig string</code> Default: <code>"/etc/kubernetes/admin.conf"</code>                                                                          |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs string</code>                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate for liveness probes to healthcheck etcd

## Synopsis

Renew the certificate for liveness probes to healthcheck etcd.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

kubeadm certs renew etcd-healthcheck-client [flags]

## Options

|                                                                            |
|----------------------------------------------------------------------------|
| <code>--cert-dir string</code> Default: <code>"/etc/kubernetes/pki"</code> |
| The path where to save the certificates                                    |
| <code>--config string</code>                                               |
| Path to a kubeadm configuration file.                                      |
| <code>-h, --help</code>                                                    |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| help for etcd-healthcheck-client                                                                                                                             |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf"                                                                                                 |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate for etcd nodes to communicate with each other

## Synopsis

Renew the certificate for etcd nodes to communicate with each other.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

```
kubeadm certs renew etcd-peer [flags]
```

## Options

|                                                              |
|--------------------------------------------------------------|
| --cert-dir string    Default: "/etc/kubernetes/pki"          |
| The path where to save the certificates                      |
| --config string                                              |
| Path to a kubeadm configuration file.                        |
| -h, --help                                                   |
| help for etcd-peer                                           |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf" |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Options inherited from parent commands

|                 |
|-----------------|
| --rootfs string |
|-----------------|

|                                                             |
|-------------------------------------------------------------|
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-------------------------------------------------------------|

Renew the certificate for serving etcd

## Synopsis

Renew the certificate for serving etcd.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

kubeadm certs renew etcd-server [flags]

## Options

|                   |                                |
|-------------------|--------------------------------|
| --cert-dir string | Default: "/etc/kubernetes/pki" |
|-------------------|--------------------------------|

|                                         |
|-----------------------------------------|
| The path where to save the certificates |
|-----------------------------------------|

|                 |
|-----------------|
| --config string |
|-----------------|

|                                       |
|---------------------------------------|
| Path to a kubeadm configuration file. |
|---------------------------------------|

|            |
|------------|
| -h, --help |
|------------|

|                      |
|----------------------|
| help for etcd-server |
|----------------------|

|                     |                                       |
|---------------------|---------------------------------------|
| --kubeconfig string | Default: "/etc/kubernetes/admin.conf" |
|---------------------|---------------------------------------|

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate for the front proxy client

## Synopsis

Renew the certificate for the front proxy client.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

```
kubeadm certs renew front-proxy-client [flags]
```

## Options

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --cert-dir string                                                                                                                                            | Default: "/etc/kubernetes/pki"        |
| The path where to save the certificates                                                                                                                      |                                       |
| --config string                                                                                                                                              |                                       |
| Path to a kubeadm configuration file.                                                                                                                        |                                       |
| -h, --help                                                                                                                                                   |                                       |
| help for front-proxy-client                                                                                                                                  |                                       |
| --kubeconfig string                                                                                                                                          | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate embedded in the kubeconfig file for the scheduler manager to use

Synopsis

Renew the certificate embedded in the kubeconfig file for the scheduler manager to use.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

kubeadm certs renew scheduler.conf [flags]

Options

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --cert-dir string                                                                                                                                            | Default: "/etc/kubernetes/pki"        |
| The path where to save the certificates                                                                                                                      |                                       |
| --config string                                                                                                                                              |                                       |
| Path to a kubeadm configuration file.                                                                                                                        |                                       |
| -h, --help                                                                                                                                                   |                                       |
| help for scheduler.conf                                                                                                                                      |                                       |
| --kubeconfig string                                                                                                                                          | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |

Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Renew the certificate embedded in the kubeconfig file for the super-admin

Synopsis

Renew the certificate embedded in the kubeconfig file for the super-admin.

Renewals run unconditionally, regardless of certificate expiration date; extra attributes such as SANs will be based on the existing file/certificates, there is no need to resupply them.

Renewal by default tries to use the certificate authority in the local PKI managed by kubeadm; as alternative it is possible to use K8s certificate API for certificate renewal, or as a last option, to generate a CSR request.

After renewal, in order to make changes effective, is required to restart control-plane components and eventually re-distribute the renewed certificate in case the file is used elsewhere.

```
kubeadm certs renew super-admin.conf [flags]
```

Options

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --cert-dir string                                                                                                                                            | Default: "/etc/kubernetes/pki"        |
| The path where to save the certificates                                                                                                                      |                                       |
| --config string                                                                                                                                              |                                       |
| Path to a kubeadm configuration file.                                                                                                                        |                                       |
| -h, --help                                                                                                                                                   |                                       |
| help for super-admin.conf                                                                                                                                    |                                       |
| --kubeconfig string                                                                                                                                          | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |

Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

kubeadm certs certificate-key

This command can be used to generate a new control-plane certificate key. The key can be passed as --certificate-key to [kubeadm init](#) and [kubeadm join](#) to enable the automatic copy of certificates when joining additional control-plane nodes.

- [certificate-key](#)

Generate certificate keys



## Synopsis

This command will print out a secure randomly-generated certificate key that can be used with the "init" command.

You can also use "kubeadm init --upload-certs" without specifying a certificate key and it will generate and print one for you.

```
kubeadm certs certificate-key [flags]
```

## Options

|                          |
|--------------------------|
| -h, --help               |
| help for certificate-key |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm certs check-expiration

This command checks expiration for the certificates in the local PKI managed by kubeadm. For more details see [Check certificate expiration](#).

- [check-expiration](#)

Check certificates expiration for a Kubernetes cluster

## Synopsis

Checks expiration for the certificates in the local PKI managed by kubeadm.

```
kubeadm certs check-expiration [flags]
```

## Options

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| --allow-missing-template-keys    Default: true                                                                                                  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
| --cert-dir string    Default: "/etc/kubernetes/pki"                                                                                             |
| The path where to save the certificates                                                                                                         |
| --config string                                                                                                                                 |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a kubeadm configuration file.                                                                                                                        |
| -o, --experimental-output string    Default: "text"                                                                                                          |
| Output format. One of: text json yaml go-template go-template-file template templatefile jsonpath jsonpath-as-json jsonpath-file.                            |
| -h, --help                                                                                                                                                   |
| help for check-expiration                                                                                                                                    |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf"                                                                                                 |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| --show-managed-fields                                                                                                                                        |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm certs generate-csr

This command can be used to generate keys and CSRs for all control-plane certificates and kubeconfig files. The user can then sign the CSRs with a CA of their choice. To read more information on how to use the command see [Signing certificate signing requests \(CSR\) generated by kubeadm](#).

- [generate-csr](#)

Generate keys and certificate signing requests

### Synopsis

Generates keys and certificate signing requests (CSRs) for all the certificates required to run the control plane. This command also generates partial kubeconfig files with private key data in the "users > user > client-key-data" field, and for each kubeconfig file an accompanying ".csr" file is created.

This command is designed for use in [Kubeadm External CA Mode](#). It generates CSRs which you can then submit to your external certificate authority for signing.

The PEM encoded signed certificates should then be saved alongside the key files, using ".crt" as the file extension, or in the case of kubeconfig files, the PEM encoded signed certificate should

be base64 encoded and added to the kubeconfig file in the "users > user > client-certificate-data" field.

```
kubeadm certs generate-csr [flags]
```

Examples

```
The following command will generate keys and CSRs for all control-plane certificates and kubeconfig files:
kubeadm certs generate-csr --kubeconfig-dir /tmp/etc-k8s --cert-dir /tmp/etc-k8s/pki
```

Options

|                                                       |                                             |
|-------------------------------------------------------|---------------------------------------------|
| --cert-dir string                                     |                                             |
|                                                       | The path where to save the certificates     |
| --config string                                       |                                             |
|                                                       | Path to a kubeadm configuration file.       |
| -h, --help                                            |                                             |
|                                                       | help for generate-csr                       |
| --kubeconfig-dir string    Default: "/etc/kubernetes" |                                             |
|                                                       | The path where to save the kubeconfig file. |

Options inherited from parent commands

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| --rootfs string |                                                             |
|                 | [EXPERIMENTAL] The path to the 'real' host root filesystem. |

What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node
- [kubeadm join](#) to connect a node to the cluster
- [kubeadm reset](#) to revert any changes made to this host by kubeadm init or kubeadm join

kubeadm init phase

kubeadm init phase enables you to invoke atomic steps of the bootstrap process. Hence, you can let kubeadm do some of the work and you can fill in the gaps if you wish to apply customization.

kubeadm init phase is consistent with the [kubeadm init workflow](#), and behind the scene both use the same code.

# kubeadm init phase preflight

Using this command you can execute preflight checks on a control-plane node.

- [preflight](#)

Run pre-flight checks

## Synopsis

Run pre-flight checks for kubeadm init.

```
kubeadm init phase preflight [flags]
```

## Examples

```
Run pre-flight checks for kubeadm init using a config file.
kubeadm init phase preflight --config kubeadm-config.yaml
```

## Options

|                                   |                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --config string                   |                                                                                                                                                                                              |
|                                   | Path to a kubeadm configuration file.                                                                                                                                                        |
| --cri-socket string               |                                                                                                                                                                                              |
|                                   | Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| --dry-run                         |                                                                                                                                                                                              |
|                                   | Don't apply any changes; just output what would be done.                                                                                                                                     |
| -h, --help                        |                                                                                                                                                                                              |
|                                   | help for preflight                                                                                                                                                                           |
| --ignore-preflight-errors strings |                                                                                                                                                                                              |
|                                   | A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                                                       |
| --image-repository string         | Default: "registry.k8s.io"                                                                                                                                                                   |
|                                   | Choose a container registry to pull control plane images from                                                                                                                                |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm init phase kubelet-start

This phase will write the kubelet configuration file and environment file and then start the kubelet.

- [kubelet-start](#)

Write kubelet settings and (re)start the kubelet

### Synopsis

Write a file with KubeletConfiguration and an environment file with node specific kubelet settings, and then (re)start kubelet.

```
kubeadm init phase kubelet-start [flags]
```

### Examples

```
Writes a dynamic environment file with kubelet flags from a InitConfiguration file.
kubeadm init phase kubelet-start --config config.yaml
```

### Options

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --config string                                                                                                                                                                              |
| Path to a kubeadm configuration file.                                                                                                                                                        |
| --cri-socket string                                                                                                                                                                          |
| Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| --dry-run                                                                                                                                                                                    |
| Don't apply any changes; just output what would be done.                                                                                                                                     |
| -h, --help                                                                                                                                                                                   |
| help for kubelet-start                                                                                                                                                                       |
| --image-repository string   Default: "registry.k8s.io"                                                                                                                                       |
| Choose a container registry to pull control plane images from                                                                                                                                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --node-name string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Specify the node name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm init phase certs

Can be used to create all required certificates by kubeadm.

- [certs](#)
- [all](#)
- [ca](#)
- [apiserver](#)
- [apiserver-kubelet-client](#)
- [front-proxy-ca](#)
- [front-proxy-client](#)
- [etcd-ca](#)
- [etcd-server](#)
- [etcd-peer](#)
- [healthcheck-client](#)
- [apiserver-etcd-client](#)
- [sa](#)

Certificate generation

## Synopsis

This command is not meant to be run on its own. See list of available subcommands.

```
kubeadm init phase certs [flags]
```

## Options

|            |
|------------|
| -h, --help |
|------------|

|                |
|----------------|
| help for certs |
|----------------|

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate all certificates

## Synopsis

Generate all certificates

kubeadm init phase certs all [flags]

## Options

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                                   |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used.                 |
| --apiserver-cert-extra-sans strings                                                                                                    |
| Optional extra Subject Alternative Names (SANs) to use for the API Server serving certificate. Can be both IP addresses and DNS names. |
| --cert-dir string   Default: "/etc/kubernetes/pki"                                                                                     |
| The path where to save and store the certificates.                                                                                     |
| --config string                                                                                                                        |
| Path to a kubeadm configuration file.                                                                                                  |
| --control-plane-endpoint string                                                                                                        |
| Specify a stable IP address or DNS name for the control plane.                                                                         |
| --dry-run                                                                                                                              |
| Don't apply any changes; just output what would be done.                                                                               |
| -h, --help                                                                                                                             |
| help for all                                                                                                                           |
| --kubernetes-version string   Default: "stable-1"                                                                                      |

|                                                             |
|-------------------------------------------------------------|
| Choose a specific Kubernetes version for the control plane. |
| --service-cidr string    Default: "10.96.0.0/12"            |
| Use alternative range of IP address for service VIPs.       |
| --service-dns-domain string    Default: "cluster.local"     |
| Use alternative domain for services, e.g. "myorg.internal". |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the self-signed Kubernetes CA to provision identities for other Kubernetes components

## Synopsis

Generate the self-signed Kubernetes CA to provision identities for other Kubernetes components, and save them into ca.crt and ca.key files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

```
kubeadm init phase certs ca [flags]
```

## Options

|                                                          |
|----------------------------------------------------------|
| --cert-dir string    Default: "/etc/kubernetes/pki"      |
| The path where to save and store the certificates.       |
| --config string                                          |
| Path to a kubeadm configuration file.                    |
| --dry-run                                                |
| Don't apply any changes; just output what would be done. |
| -h, --help                                               |
| help for ca                                              |
| --kubernetes-version string    Default: "stable-1"       |



|                                                             |
|-------------------------------------------------------------|
| Choose a specific Kubernetes version for the control plane. |
|-------------------------------------------------------------|

## Options inherited from parent commands

|                 |
|-----------------|
| --rootfs string |
|-----------------|

|                                                             |
|-------------------------------------------------------------|
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-------------------------------------------------------------|

Generate the certificate for serving the Kubernetes API

## Synopsis

Generate the certificate for serving the Kubernetes API, and save them into apiserver.crt and apiserver.key files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

kubeadm init phase certs apiserver [flags]

## Options

|                                      |
|--------------------------------------|
| --apiserver-advertise-address string |
|--------------------------------------|

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
|------------------------------------------------------------------------------------------------------------------------|

|                                     |
|-------------------------------------|
| --apiserver-cert-extra-sans strings |
|-------------------------------------|

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| Optional extra Subject Alternative Names (SANs) to use for the API Server serving certificate. Can be both IP addresses and DNS names. |
|----------------------------------------------------------------------------------------------------------------------------------------|

|                                                    |
|----------------------------------------------------|
| --cert-dir string   Default: "/etc/kubernetes/pki" |
|----------------------------------------------------|

|                                                    |
|----------------------------------------------------|
| The path where to save and store the certificates. |
|----------------------------------------------------|

|                 |
|-----------------|
| --config string |
|-----------------|

|                                       |
|---------------------------------------|
| Path to a kubeadm configuration file. |
|---------------------------------------|

|                                 |
|---------------------------------|
| --control-plane-endpoint string |
|---------------------------------|

|                                                                |
|----------------------------------------------------------------|
| Specify a stable IP address or DNS name for the control plane. |
|----------------------------------------------------------------|

|           |
|-----------|
| --dry-run |
|-----------|

|                                                          |
|----------------------------------------------------------|
| Don't apply any changes; just output what would be done. |
|----------------------------------------------------------|

|            |
|------------|
| -h, --help |
|------------|

|                                                             |
|-------------------------------------------------------------|
| help for apiserver                                          |
| --kubernetes-version string   Default: "stable-1"           |
| Choose a specific Kubernetes version for the control plane. |
| --service-cidr string   Default: "10.96.0.0/12"             |
| Use alternative range of IP address for service VIPs.       |
| --service-dns-domain string   Default: "cluster.local"      |
| Use alternative domain for services, e.g. "myorg.internal". |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the certificate for the API server to connect to kubelet

## Synopsis

Generate the certificate for the API server to connect to kubelet, and save them into apiserver-kubelet-client.crt and apiserver-kubelet-client.key files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

kubeadm init phase certs apiserver-kubelet-client [flags]

## Options

|                                                          |
|----------------------------------------------------------|
| --cert-dir string   Default: "/etc/kubernetes/pki"       |
| The path where to save and store the certificates.       |
| --config string                                          |
| Path to a kubeadm configuration file.                    |
| --dry-run                                                |
| Don't apply any changes; just output what would be done. |
| -h, --help                                               |
| help for apiserver-kubelet-client                        |

|                                                             |                     |
|-------------------------------------------------------------|---------------------|
| --kubernetes-version string                                 | Default: "stable-1" |
| Choose a specific Kubernetes version for the control plane. |                     |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the self-signed CA to provision identities for front proxy

## Synopsis

Generate the self-signed CA to provision identities for front proxy, and save them into front-proxy-ca.crt and front-proxy-ca.key files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

kubeadm init phase certs front-proxy-ca [flags]

## Options

|                                                             |                                |
|-------------------------------------------------------------|--------------------------------|
| --cert-dir string                                           | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates.          |                                |
| --config string                                             |                                |
| Path to a kubeadm configuration file.                       |                                |
| --dry-run                                                   |                                |
| Don't apply any changes; just output what would be done.    |                                |
| -h, --help                                                  |                                |
| help for front-proxy-ca                                     |                                |
| --kubernetes-version string                                 | Default: "stable-1"            |
| Choose a specific Kubernetes version for the control plane. |                                |

## Options inherited from parent commands

|                 |
|-----------------|
| --rootfs string |
|-----------------|

|                                                             |
|-------------------------------------------------------------|
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-------------------------------------------------------------|

Generate the certificate for the front proxy client

## Synopsis

Generate the certificate for the front proxy client, and save them into front-proxy-client.crt and front-proxy-client.key files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

```
kubeadm init phase certs front-proxy-client [flags]
```

## Options

|                                                             |                                |
|-------------------------------------------------------------|--------------------------------|
| --cert-dir string                                           | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates.          |                                |
| --config string                                             |                                |
| Path to a kubeadm configuration file.                       |                                |
| --dry-run                                                   |                                |
| Don't apply any changes; just output what would be done.    |                                |
| -h, --help                                                  |                                |
| help for front-proxy-client                                 |                                |
| --kubernetes-version string                                 | Default: "stable-1"            |
| Choose a specific Kubernetes version for the control plane. |                                |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the self-signed CA to provision identities for etcd

## Synopsis

Generate the self-signed CA to provision identities for etcd, and save them into etcd/ca.crt and etcd/ca.key files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

```
kubeadm init phase certs etcd-ca [flags]
```

## Options

|                                                             |                                |
|-------------------------------------------------------------|--------------------------------|
| --cert-dir string                                           | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates.          |                                |
| --config string                                             |                                |
| Path to a kubeadm configuration file.                       |                                |
| --dry-run                                                   |                                |
| Don't apply any changes; just output what would be done.    |                                |
| -h, --help                                                  |                                |
| help for etcd-ca                                            |                                |
| --kubernetes-version string                                 | Default: "stable-1"            |
| Choose a specific Kubernetes version for the control plane. |                                |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the certificate for serving etcd

## Synopsis

Generate the certificate for serving etcd, and save them into etcd/server.crt and etcd/server.key files.

Default SANs are localhost, 127.0.0.1, 127.0.0.1, ::1

If both files already exist, kubeadm skips the generation step and existing files will be used.

```
kubeadm init phase certs etcd-server [flags]
```

## Options

|                   |                                |
|-------------------|--------------------------------|
| --cert-dir string | Default: "/etc/kubernetes/pki" |
|-------------------|--------------------------------|

|                             |                                                             |
|-----------------------------|-------------------------------------------------------------|
|                             | The path where to save and store the certificates.          |
| --config string             |                                                             |
|                             | Path to a kubeadm configuration file.                       |
| --dry-run                   |                                                             |
|                             | Don't apply any changes; just output what would be done.    |
| -h, --help                  |                                                             |
|                             | help for etcd-server                                        |
| --kubernetes-version string | Default: "stable-1"                                         |
|                             | Choose a specific Kubernetes version for the control plane. |

## Options inherited from parent commands

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| --rootfs string |                                                             |
|                 | [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the certificate for etcd nodes to communicate with each other

## Synopsis

Generate the certificate for etcd nodes to communicate with each other, and save them into etcd/peer.crt and etcd/peer.key files.

Default SANs are localhost, 127.0.0.1, 127.0.0.1, ::1

If both files already exist, kubeadm skips the generation step and existing files will be used.

```
kubeadm init phase certs etcd-peer [flags]
```

## Options

|                   |                                                    |
|-------------------|----------------------------------------------------|
| --cert-dir string | Default: "/etc/kubernetes/pki"                     |
|                   | The path where to save and store the certificates. |
| --config string   |                                                    |
|                   | Path to a kubeadm configuration file.              |
| --dry-run         |                                                    |

|                                                             |
|-------------------------------------------------------------|
| Don't apply any changes; just output what would be done.    |
| -h, --help                                                  |
| help for etcd-peer                                          |
| --kubernetes-version string    Default: "stable-1"          |
| Choose a specific Kubernetes version for the control plane. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the certificate for liveness probes to healthcheck etcd

## Synopsis

Generate the certificate for liveness probes to healthcheck etcd, and save them into etcd/healthcheck-client.crt and etcd/healthcheck-client.key files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

```
kubeadm init phase certs etcd-healthcheck-client [flags]
```

## Options

|                                                             |
|-------------------------------------------------------------|
| --cert-dir string    Default: "/etc/kubernetes/pki"         |
| The path where to save and store the certificates.          |
| --config string                                             |
| Path to a kubeadm configuration file.                       |
| --dry-run                                                   |
| Don't apply any changes; just output what would be done.    |
| -h, --help                                                  |
| help for etcd-healthcheck-client                            |
| --kubernetes-version string    Default: "stable-1"          |
| Choose a specific Kubernetes version for the control plane. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the certificate the apiserver uses to access etcd

## Synopsis

Generate the certificate the apiserver uses to access etcd, and save them into apiserver-etcd-client.crt and apiserver-etcd-client.key files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

kubeadm init phase certs apiserver-etcd-client [flags]

## Options

|                                                             |                                |
|-------------------------------------------------------------|--------------------------------|
| --cert-dir string                                           | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates.          |                                |
| --config string                                             |                                |
| Path to a kubeadm configuration file.                       |                                |
| --dry-run                                                   |                                |
| Don't apply any changes; just output what would be done.    |                                |
| -h, --help                                                  |                                |
| help for apiserver-etcd-client                              |                                |
| --kubernetes-version string                                 | Default: "stable-1"            |
| Choose a specific Kubernetes version for the control plane. |                                |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate a private key for signing service account tokens along with its public key



## Synopsis

Generate the private key for signing service account tokens along with its public key, and save them into sa.key and sa.pub files.

If both files already exist, kubeadm skips the generation step and existing files will be used.

```
kubeadm init phase certs sa [flags]
```

## Options

|                                                    |                                |
|----------------------------------------------------|--------------------------------|
| --cert-dir string                                  | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates. |                                |
| -h, --help                                         |                                |
| help for sa                                        |                                |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm init phase kubeconfig

You can create all required kubeconfig files by calling the all subcommand or call them individually.

- [kubeconfig](#)
- [all](#)
- [admin](#)
- [kubelet](#)
- [controller-manager](#)
- [scheduler](#)
- [super-admin](#)

Generate all kubeconfig files necessary to establish the control plane and the admin kubeconfig file

## Synopsis

This command is not meant to be run on its own. See list of available subcommands.

```
kubeadm init phase kubeconfig [flags]
```

## Options

|                     |
|---------------------|
| -h, --help          |
| help for kubeconfig |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate all kubeconfig files

## Synopsis

Generate all kubeconfig files

kubeadm init phase kubeconfig all [flags]

## Options

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                   |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| --apiserver-bind-port int32    Default: 6443                                                                           |
| Port for the API Server to bind to.                                                                                    |
| --cert-dir string    Default: "/etc/kubernetes/pki"                                                                    |
| The path where to save and store the certificates.                                                                     |
| --config string                                                                                                        |
| Path to a kubeadm configuration file.                                                                                  |
| --control-plane-endpoint string                                                                                        |
| Specify a stable IP address or DNS name for the control plane.                                                         |
| --dry-run                                                                                                              |
| Don't apply any changes; just output what would be done.                                                               |
| -h, --help                                                                                                             |

|                                                             |
|-------------------------------------------------------------|
| help for all                                                |
| --kubeconfig-dir string    Default: "/etc/kubernetes"       |
| The path where to save the kubeconfig file.                 |
| --kubernetes-version string    Default: "stable-1"          |
| Choose a specific Kubernetes version for the control plane. |
| --node-name string                                          |
| Specify the node name.                                      |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate a kubeconfig file for the admin to use and for kubeadm itself

## Synopsis

Generate the kubeconfig file for the admin and for kubeadm itself, and save it to admin.conf file.

```
kubeadm init phase kubeconfig admin [flags]
```

## Options

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                   |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| --apiserver-bind-port int32    Default: 6443                                                                           |
| Port for the API Server to bind to.                                                                                    |
| --cert-dir string    Default: "/etc/kubernetes/pki"                                                                    |
| The path where to save and store the certificates.                                                                     |
| --config string                                                                                                        |
| Path to a kubeadm configuration file.                                                                                  |
| --control-plane-endpoint string                                                                                        |

|                                                                |
|----------------------------------------------------------------|
| Specify a stable IP address or DNS name for the control plane. |
| --dry-run                                                      |
| Don't apply any changes; just output what would be done.       |
| -h, --help                                                     |
| help for admin                                                 |
| --kubeconfig-dir string    Default: "/etc/kubernetes"          |
| The path where to save the kubeconfig file.                    |
| --kubernetes-version string    Default: "stable-1"             |
| Choose a specific Kubernetes version for the control plane.    |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate a kubeconfig file for the kubelet to use *only* for cluster bootstrapping purposes

## Synopsis

Generate the kubeconfig file for the kubelet to use and save it to kubelet.conf file.

Please note that this should *only* be used for cluster bootstrapping purposes. After your control plane is up, you should request all kubelet credentials from the CSR API.

```
kubeadm init phase kubeconfig kubelet [flags]
```

## Options

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                   |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| --apiserver-bind-port int32    Default: 6443                                                                           |
| Port for the API Server to bind to.                                                                                    |
| --cert-dir string    Default: "/etc/kubernetes/pki"                                                                    |

|                                                                 |                                                                |
|-----------------------------------------------------------------|----------------------------------------------------------------|
|                                                                 | The path where to save and store the certificates.             |
| <code>--config string</code>                                    |                                                                |
|                                                                 | Path to a kubeadm configuration file.                          |
| <code>--control-plane-endpoint string</code>                    |                                                                |
|                                                                 | Specify a stable IP address or DNS name for the control plane. |
| <code>--dry-run</code>                                          |                                                                |
|                                                                 | Don't apply any changes; just output what would be done.       |
| <code>-h, --help</code>                                         |                                                                |
|                                                                 | help for kubelet                                               |
| <code>--kubeconfig-dir string</code> Default: "/etc/kubernetes" |                                                                |
|                                                                 | The path where to save the kubeconfig file.                    |
| <code>--kubernetes-version string</code> Default: "stable-1"    |                                                                |
|                                                                 | Choose a specific Kubernetes version for the control plane.    |
| <code>--node-name string</code>                                 |                                                                |
|                                                                 | Specify the node name.                                         |

## Options inherited from parent commands

|                              |                                                             |
|------------------------------|-------------------------------------------------------------|
| <code>--rootfs string</code> |                                                             |
|                              | [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate a kubeconfig file for the controller manager to use

## Synopsis

Generate the kubeconfig file for the controller manager to use and save it to controller-manager.conf file

kubeadm init phase kubeconfig controller-manager [flags]

## Options

|                                                   |
|---------------------------------------------------|
| <code>--apiserver-advertise-address string</code> |
|---------------------------------------------------|

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| <code>--apiserver-bind-port int32</code> Default: 6443                                                                 |
| Port for the API Server to bind to.                                                                                    |
| <code>--cert-dir string</code> Default: "/etc/kubernetes/pki"                                                          |
| The path where to save and store the certificates.                                                                     |
| <code>--config string</code>                                                                                           |
| Path to a kubeadm configuration file.                                                                                  |
| <code>--control-plane-endpoint string</code>                                                                           |
| Specify a stable IP address or DNS name for the control plane.                                                         |
| <code>--dry-run</code>                                                                                                 |
| Don't apply any changes; just output what would be done.                                                               |
| <code>-h, --help</code>                                                                                                |
| help for controller-manager                                                                                            |
| <code>--kubeconfig-dir string</code> Default: "/etc/kubernetes"                                                        |
| The path where to save the kubeconfig file.                                                                            |
| <code>--kubernetes-version string</code> Default: "stable-1"                                                           |
| Choose a specific Kubernetes version for the control plane.                                                            |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs string</code>                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate a kubeconfig file for the scheduler to use

## Synopsis

Generate the kubeconfig file for the scheduler to use and save it to scheduler.conf file.

kubeadm init phase kubeconfig scheduler [flags]

## Options

|                                      |                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string |                                                                                                                        |
|                                      | The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| --apiserver-bind-port int32          | Default: 6443                                                                                                          |
|                                      | Port for the API Server to bind to.                                                                                    |
| --cert-dir string                    | Default: "/etc/kubernetes/pki"                                                                                         |
|                                      | The path where to save and store the certificates.                                                                     |
| --config string                      |                                                                                                                        |
|                                      | Path to a kubeadm configuration file.                                                                                  |
| --control-plane-endpoint string      |                                                                                                                        |
|                                      | Specify a stable IP address or DNS name for the control plane.                                                         |
| --dry-run                            |                                                                                                                        |
|                                      | Don't apply any changes; just output what would be done.                                                               |
| -h, --help                           |                                                                                                                        |
|                                      | help for scheduler                                                                                                     |
| --kubeconfig-dir string              | Default: "/etc/kubernetes"                                                                                             |
|                                      | The path where to save the kubeconfig file.                                                                            |
| --kubernetes-version string          | Default: "stable-1"                                                                                                    |
|                                      | Choose a specific Kubernetes version for the control plane.                                                            |

## Options inherited from parent commands

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| --rootfs string |                                                             |
|                 | [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate a kubeconfig file for the super-admin

## Synopsis

Generate a kubeconfig file for the super-admin, and save it to super-admin.conf file.

```
kubeadm init phase kubeconfig super-admin [flags]
```

## Options

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| <code>--apiserver-advertise-address</code> string                                                                      |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| <code>--apiserver-bind-port</code> int32    Default: 6443                                                              |
| Port for the API Server to bind to.                                                                                    |
| <code>--cert-dir</code> string    Default: "/etc/kubernetes/pki"                                                       |
| The path where to save and store the certificates.                                                                     |
| <code>--config</code> string                                                                                           |
| Path to a kubeadm configuration file.                                                                                  |
| <code>--control-plane-endpoint</code> string                                                                           |
| Specify a stable IP address or DNS name for the control plane.                                                         |
| <code>--dry-run</code>                                                                                                 |
| Don't apply any changes; just output what would be done.                                                               |
| <code>-h, --help</code>                                                                                                |
| help for super-admin                                                                                                   |
| <code>--kubeconfig-dir</code> string    Default: "/etc/kubernetes"                                                     |
| The path where to save the kubeconfig file.                                                                            |
| <code>--kubernetes-version</code> string    Default: "stable-1"                                                        |
| Choose a specific Kubernetes version for the control plane.                                                            |

## Options inherited from parent commands

|                              |
|------------------------------|
| <code>--rootfs</code> string |
|------------------------------|



|                                                             |
|-------------------------------------------------------------|
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-------------------------------------------------------------|

## kubeadm init phase control-plane

Using this phase you can create all required static Pod files for the control plane components.

- [control-plane](#)
- [all](#)
- [apiserver](#)
- [controller-manager](#)
- [scheduler](#)

Generate all static Pod manifest files necessary to establish the control plane

### Synopsis

This command is not meant to be run on its own. See list of available subcommands.

```
kubeadm init phase control-plane [flags]
```

### Options

|            |
|------------|
| -h, --help |
|------------|

|                        |
|------------------------|
| help for control-plane |
|------------------------|

### Options inherited from parent commands

|                 |
|-----------------|
| --rootfs string |
|-----------------|

|                                                             |
|-------------------------------------------------------------|
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-------------------------------------------------------------|

Generate all static Pod manifest files

### Synopsis

Generate all static Pod manifest files

```
kubeadm init phase control-plane all [flags]
```

### Examples

```
Generates all static Pod manifest files for control plane components,
functionally equivalent to what is generated by kubeadm init.
kubeadm init phase control-plane all
```

```
Generates all static Pod manifest files using options read from a configuration file.
kubeadm init phase control-plane all --config config.yaml
```

## Options

|                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--apiserver-advertise-address</code> string                                                                                                                                                                                                                                                                                                                                                                   |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used.                                                                                                                                                                                                                                                                                              |
| <code>--apiserver-bind-port</code> int32    Default: 6443                                                                                                                                                                                                                                                                                                                                                           |
| Port for the API Server to bind to.                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>--apiserver-extra-args</code> <comma-separated 'key=value' pairs>                                                                                                                                                                                                                                                                                                                                             |
| A set of extra flags to pass to the API Server or override default ones in form of <flagname>=<value>                                                                                                                                                                                                                                                                                                               |
| <code>--cert-dir</code> string    Default: "/etc/kubernetes/pki"                                                                                                                                                                                                                                                                                                                                                    |
| The path where to save and store the certificates.                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--config</code> string                                                                                                                                                                                                                                                                                                                                                                                        |
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                               |
| <code>--control-plane-endpoint</code> string                                                                                                                                                                                                                                                                                                                                                                        |
| Specify a stable IP address or DNS name for the control plane.                                                                                                                                                                                                                                                                                                                                                      |
| <code>--controller-manager-extra-args</code> <comma-separated 'key=value' pairs>                                                                                                                                                                                                                                                                                                                                    |
| A set of extra flags to pass to the Controller Manager or override default ones in form of <flagname>=<value>                                                                                                                                                                                                                                                                                                       |
| <code>--dry-run</code>                                                                                                                                                                                                                                                                                                                                                                                              |
| Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                            |
| <code>--feature-gates</code> string                                                                                                                                                                                                                                                                                                                                                                                 |
| A set of key=value pairs that describe feature gates for various features. Options are:<br>EtcdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                             |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| help for all                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --image-repository string    Default: "registry.k8s.io"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --kubernetes-version string    Default: "stable-1"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Choose a specific Kubernetes version for the control plane.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubenetconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |
| --pod-network-cidr string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Specify range of IP addresses for the pod network. If set, the control plane will automatically allocate CIDRs for every node.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| --scheduler-extra-args <comma-separated 'key=value' pairs>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| A set of extra flags to pass to the Scheduler or override default ones in form of <flagname>=<value>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| --service-cidr string    Default: "10.96.0.0/12"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Use alternative range of IP address for service VIPs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generates the kube-apiserver static Pod manifest

## Synopsis

Generates the kube-apiserver static Pod manifest

kubeadm init phase control-plane apiserver [flags]

## Options

|                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <code>--apiserver-advertise-address</code> string                                                                                                                                                                                                                                                                                                                                                                   |  |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used.                                                                                                                                                                                                                                                                                              |  |
| <code>--apiserver-bind-port</code> int32    Default: 6443                                                                                                                                                                                                                                                                                                                                                           |  |
| Port for the API Server to bind to.                                                                                                                                                                                                                                                                                                                                                                                 |  |
| <code>--apiserver-extra-args</code> <comma-separated 'key=value' pairs>                                                                                                                                                                                                                                                                                                                                             |  |
| A set of extra flags to pass to the API Server or override default ones in form of <flagname>=<value>                                                                                                                                                                                                                                                                                                               |  |
| <code>--cert-dir</code> string    Default: "/etc/kubernetes/pki"                                                                                                                                                                                                                                                                                                                                                    |  |
| The path where to save and store the certificates.                                                                                                                                                                                                                                                                                                                                                                  |  |
| <code>--config</code> string                                                                                                                                                                                                                                                                                                                                                                                        |  |
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                               |  |
| <code>--control-plane-endpoint</code> string                                                                                                                                                                                                                                                                                                                                                                        |  |
| Specify a stable IP address or DNS name for the control plane.                                                                                                                                                                                                                                                                                                                                                      |  |
| <code>--dry-run</code>                                                                                                                                                                                                                                                                                                                                                                                              |  |
| Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                            |  |
| <code>--feature-gates</code> string                                                                                                                                                                                                                                                                                                                                                                                 |  |
| A set of key=value pairs that describe feature gates for various features. Options are:<br>EtcdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |  |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                             |  |
| help for apiserver                                                                                                                                                                                                                                                                                                                                                                                                  |  |
| <code>--image-repository</code> string    Default: "registry.k8s.io"                                                                                                                                                                                                                                                                                                                                                |  |
| Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                       |  |
| <code>--kubernetes-version</code> string    Default: "stable-1"                                                                                                                                                                                                                                                                                                                                                     |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Choose a specific Kubernetes version for the control plane.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>--patches string</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |
| <b>--service-cidr string</b> Default: "10.96.0.0/12"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Use alternative range of IP address for service VIPs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <b>--rootfs string</b>                                      |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generates the kube-controller-manager static Pod manifest

## Synopsis

Generates the kube-controller-manager static Pod manifest

kubeadm init phase control-plane controller-manager [flags]

## Options

|                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|
| <b>--cert-dir string</b> Default: "/etc/kubernetes/pki"                                                       |
| The path where to save and store the certificates.                                                            |
| <b>--config string</b>                                                                                        |
| Path to a kubeadm configuration file.                                                                         |
| <b>--controller-manager-extra-args &lt;comma-separated 'key=value' pairs&gt;</b>                              |
| A set of extra flags to pass to the Controller Manager or override default ones in form of <flagname>=<value> |
| <b>--dry-run</b>                                                                                              |
| Don't apply any changes; just output what would be done.                                                      |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| help for controller-manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --image-repository string   Default: "registry.k8s.io"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --kubernetes-version string   Default: "stable-1"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Choose a specific Kubernetes version for the control plane.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |
| --pod-network-cidr string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Specify range of IP addresses for the pod network. If set, the control plane will automatically allocate CIDRs for every node.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generates the kube-scheduler static Pod manifest

## Synopsis

Generates the kube-scheduler static Pod manifest

kubeadm init phase control-plane scheduler [flags]

## Options

|                                                    |
|----------------------------------------------------|
| --cert-dir string   Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates. |
| --config string                                    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --dry-run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| help for scheduler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| --image-repository string    Default: "registry.k8s.io"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --kubernetes-version string    Default: "stable-1"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Choose a specific Kubernetes version for the control plane.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubenetconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |
| --scheduler-extra-args <comma-separated 'key=value' pairs>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| A set of extra flags to pass to the Scheduler or override default ones in form of <flagname>=<value>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm init phase etcd

Use the following phase to create a local etcd instance based on a static Pod file.

- [etcd](#)
- [local](#)

Generate static Pod manifest file for local etcd

## Synopsis

This command is not meant to be run on its own. See list of available subcommands.

```
kubeadm init phase etcd [flags]
```

## Options

|               |
|---------------|
| -h, --help    |
| help for etcd |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the static Pod manifest file for a local, single-node local etcd instance

## Synopsis

Generate the static Pod manifest file for a local, single-node local etcd instance

```
kubeadm init phase etcd local [flags]
```

## Examples

```
Generates the static Pod manifest file for etcd, functionally
equivalent to what is generated by kubeadm init.
kubeadm init phase etcd local

Generates the static Pod manifest file for etcd using options
read from a configuration file.
kubeadm init phase etcd local --config config.yaml
```

## Options

|                                                    |                                |
|----------------------------------------------------|--------------------------------|
| --cert-dir string                                  | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates. |                                |
| --config string                                    |                                |
| Path to a kubeadm configuration file.              |                                |
| --dry-run                                          |                                |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| help for local                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| --image-repository string    Default: "registry.k8s.io"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm init phase upload-config

You can use this command to upload the kubeadm configuration to your cluster. Alternatively, you can use [kubeadm config](#).

- [upload-config](#)
- [all](#)
- [kubeadm](#)
- [kubelet](#)

Upload the kubeadm and kubelet configuration to a ConfigMap

## Synopsis

This command is not meant to be run on its own. See list of available subcommands.

```
kubeadm init phase upload-config [flags]
```

## Options

|            |
|------------|
| -h, --help |
|------------|

|                        |
|------------------------|
| help for upload-config |
|------------------------|

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Upload all configuration to a config map

## Synopsis

Upload all configuration to a config map

kubeadm init phase upload-config all [flags]

## Options

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --config string                                                                                                                                                                              |
| Path to a kubeadm configuration file.                                                                                                                                                        |
| --cri-socket string                                                                                                                                                                          |
| Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| --dry-run                                                                                                                                                                                    |
| Don't apply any changes; just output what would be done.                                                                                                                                     |
| -h, --help                                                                                                                                                                                   |
| help for all                                                                                                                                                                                 |
| --kubeconfig string   Default: "/etc/kubernetes/admin.conf"                                                                                                                                  |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                 |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Upload the kubeadm ClusterConfiguration to a ConfigMap

## Synopsis

Upload the kubeadm ClusterConfiguration to a ConfigMap called kubeadm-config in the kube-system namespace. This enables correct configuration of system components and a seamless user experience when upgrading.

Alternatively, you can use kubeadm config.

```
kubeadm init phase upload-config kubeadm [flags]
```

## Examples

```
upload the configuration of your cluster
kubeadm init phase upload-config --config=myConfig.yaml
```

## Options

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --config string                                                                                                                                                                              |
| Path to a kubeadm configuration file.                                                                                                                                                        |
| --cri-socket string                                                                                                                                                                          |
| Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| --dry-run                                                                                                                                                                                    |
| Don't apply any changes; just output what would be done.                                                                                                                                     |
| -h, --help                                                                                                                                                                                   |
| help for kubeadm                                                                                                                                                                             |
| --kubeconfig string   Default: "/etc/kubernetes/admin.conf"                                                                                                                                  |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                 |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Upload the kubelet component config to a ConfigMap

## Synopsis

Upload the kubelet configuration extracted from the kubeadm InitConfiguration object to a kubelet-config ConfigMap in the cluster

```
kubeadm init phase upload-config kubelet [flags]
```

## Examples

```
Upload the kubelet configuration from the kubeadm Config file to a ConfigMap in the cluster.
```

```
kubeadm init phase upload-config kubelet --config kubeadm.yaml
```

## Options

|                     |                                                                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --config string     |                                                                                                                                                                                              |
|                     | Path to a kubeadm configuration file.                                                                                                                                                        |
| --cri-socket string |                                                                                                                                                                                              |
|                     | Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| --dry-run           |                                                                                                                                                                                              |
|                     | Don't apply any changes; just output what would be done.                                                                                                                                     |
| -h, --help          |                                                                                                                                                                                              |
|                     | help for kubelet                                                                                                                                                                             |
| --kubeconfig string | Default: "/etc/kubernetes/admin.conf"                                                                                                                                                        |
|                     | The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                 |

## Options inherited from parent commands

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| --rootfs string |                                                             |
|                 | [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm init phase upload-certs

Use the following phase to upload control-plane certificates to the cluster. By default the certs and encryption key expire after two hours.

- [upload-certs](#)

Upload certificates to kubeadm-certs

Synopsis

Upload control plane certificates to the kubeadm-certs Secret

kubeadm init phase upload-certs [flags]

Options

|                                                             |                                                                                                                                                                  |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --certificate-key string                                    | Key used to encrypt the control-plane certificates in the kubeadm-certs Secret. The certificate key is a hex encoded string that is an AES key of size 32 bytes. |
| --config string                                             | Path to a kubeadm configuration file.                                                                                                                            |
| --dry-run                                                   | Don't apply any changes; just output what would be done.                                                                                                         |
| -h, --help                                                  | help for upload-certs                                                                                                                                            |
| --kubeconfig string   Default: "/etc/kubernetes/admin.conf" | The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.     |
| --skip-certificate-key-print                                | Don't print the key used to encrypt the control-plane certificates.                                                                                              |
| --upload-certs                                              | Upload control-plane certificates to the kubeadm-certs Secret.                                                                                                   |

Options inherited from parent commands

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| --rootfs string | [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-----------------|-------------------------------------------------------------|

# kubeadm init phase mark-control-plane

Use the following phase to label and taint the node as a control plane node.

- [mark-control-plane](#)

Mark a node as a control-plane

## Synopsis

Mark a node as a control-plane

```
kubeadm init phase mark-control-plane [flags]
```

## Examples

```
Applies control-plane label and taint to the current node, functionally equivalent to what
executed by kubeadm init.
kubeadm init phase mark-control-plane --config config.yaml

Applies control-plane label and taint to a specific node
kubeadm init phase mark-control-plane --node-name myNode
```

## Options

|                                                          |
|----------------------------------------------------------|
| --config string                                          |
| Path to a kubeadm configuration file.                    |
| --dry-run                                                |
| Don't apply any changes; just output what would be done. |
| -h, --help                                               |
| help for mark-control-plane                              |
| --node-name string                                       |
| Specify the node name.                                   |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

# kubeadm init phase bootstrap-token

Use the following phase to configure bootstrap tokens.

- [bootstrap-token](#)

Generates bootstrap tokens used to join a node to a cluster

## Synopsis

Bootstrap tokens are used for establishing bidirectional trust between a node joining the cluster and a control-plane node.

This command makes all the configurations required to make bootstrap tokens works and then creates an initial token.

```
kubeadm init phase bootstrap-token [flags]
```

## Examples

```
Make all the bootstrap token configurations and create an initial token, functionally
equivalent to what generated by kubeadm init.
kubeadm init phase bootstrap-token
```

## Options

|                     |                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --config string     |                                                                                                                                                              |
|                     | Path to a kubeadm configuration file.                                                                                                                        |
| --dry-run           |                                                                                                                                                              |
|                     | Don't apply any changes; just output what would be done.                                                                                                     |
| -h, --help          |                                                                                                                                                              |
|                     | help for bootstrap-token                                                                                                                                     |
| --kubeconfig string | Default: "/etc/kubernetes/admin.conf"                                                                                                                        |
|                     | The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| --skip-token-print  |                                                                                                                                                              |
|                     | Skip printing of the default bootstrap token generated by 'kubeadm init'.                                                                                    |

## Options inherited from parent commands

|                 |
|-----------------|
| --rootfs string |
|-----------------|

|                                                             |
|-------------------------------------------------------------|
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-------------------------------------------------------------|

## kubeadm init phase kubelet-finalize

Use the following phase to update settings relevant to the kubelet after TLS bootstrap. You can use the all subcommand to run all kubelet-finalize phases.

- [kubelet-finalize](#)
- [kubelet-finalize-all](#)
- [kubelet-finalize-cert-rotation](#)

Updates settings relevant to the kubelet after TLS bootstrap

### Synopsis

Updates settings relevant to the kubelet after TLS bootstrap

```
kubeadm init phase kubelet-finalize [flags]
```

### Examples

```
Updates settings relevant to the kubelet after TLS bootstrap"
kubeadm init phase kubelet-finalize all --config
```

### Options

|                           |
|---------------------------|
| -h, --help                |
| help for kubelet-finalize |

### Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Run all kubelet-finalize phases

### Synopsis

Run all kubelet-finalize phases

```
kubeadm init phase kubelet-finalize all [flags]
```



## Examples

```
Updates settings relevant to the kubelet after TLS bootstrap"
kubeadm init phase kubelet-finalize all --config
```

## Options

|                                                          |                                |
|----------------------------------------------------------|--------------------------------|
| --cert-dir string                                        | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates.       |                                |
| --config string                                          |                                |
| Path to a kubeadm configuration file.                    |                                |
| --dry-run                                                |                                |
| Don't apply any changes; just output what would be done. |                                |
| -h, --help                                               |                                |
| help for all                                             |                                |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Enable kubelet client certificate rotation

## Synopsis

Enable kubelet client certificate rotation

```
kubeadm init phase kubelet-finalize experimental-cert-rotation [flags]
```

## Options

|                                                    |                                |
|----------------------------------------------------|--------------------------------|
| --cert-dir string                                  | Default: "/etc/kubernetes/pki" |
| The path where to save and store the certificates. |                                |
| --config string                                    |                                |
| Path to a kubeadm configuration file.              |                                |
| --dry-run                                          |                                |

|                                                          |
|----------------------------------------------------------|
| Don't apply any changes; just output what would be done. |
| -h, --help                                               |
| help for experimental-cert-rotation                      |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm init phase addon

You can install all the available addons with the `all` subcommand, or install them selectively.

- [addon](#)
- [all](#)
- [coredns](#)
- [kube-proxy](#)

Install required addons for passing conformance tests

## Synopsis

This command is not meant to be run on its own. See list of available subcommands.

```
kubeadm init phase addon [flags]
```

## Options

|                |
|----------------|
| -h, --help     |
| help for addon |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Install all the addons

## Synopsis

Install all the addons

kubeadm init phase addon all [flags]

## Options

|                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                         | The IP address the API Server will advertise it's listening on. If not set the default network interface will be used.                                                                                                                                                                                                                                                                                              |
| --apiserver-bind-port int32    Default: 6443                 | Port for the API Server to bind to.                                                                                                                                                                                                                                                                                                                                                                                 |
| --config string                                              | Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                               |
| --control-plane-endpoint string                              | Specify a stable IP address or DNS name for the control plane.                                                                                                                                                                                                                                                                                                                                                      |
| --dry-run                                                    | Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                            |
| --feature-gates string                                       | A set of key=value pairs that describe feature gates for various features. Options are:<br>EtcdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |
| -h, --help                                                   | help for all                                                                                                                                                                                                                                                                                                                                                                                                        |
| --image-repository string    Default: "registry.k8s.io"      | Choose a container registry to pull control plane images from                                                                                                                                                                                                                                                                                                                                                       |
| --kubeconfig string    Default: "/etc/kubernetes/admin.conf" | The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                                                                                                                                                                                                                                        |
| --kubernetes-version string    Default: "stable-1"           | Choose a specific Kubernetes version for the control plane.                                                                                                                                                                                                                                                                                                                                                         |
| --pod-network-cidr string                                    |                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|
| Specify range of IP addresses for the pod network. If set, the control plane will automatically allocate CIDRs for every node. |
| --service-cidr string    Default: "10.96.0.0/12"                                                                               |
| Use alternative range of IP address for service VIPs.                                                                          |
| --service-dns-domain string    Default: "cluster.local"                                                                        |
| Use alternative domain for services, e.g. "myorg.internal".                                                                    |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Install the CoreDNS addon to a Kubernetes cluster

## Synopsis

Install the CoreDNS addon components via the API server. Please note that although the DNS server is deployed, it will not be scheduled until CNI is installed.

```
kubeadm init phase addon coredns [flags]
```

## Options

|                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --config string                                                                                                                                                                                                                                                                                                                                                                                                     |
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                               |
| --dry-run                                                                                                                                                                                                                                                                                                                                                                                                           |
| Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                            |
| --feature-gates string                                                                                                                                                                                                                                                                                                                                                                                              |
| A set of key=value pairs that describe feature gates for various features. Options are:<br>EtcdLearnerMode=true false (BETA - default=true)<br>PublicKeysECDSA=true false (DEPRECATED - default=false)<br>RootlessControlPlane=true false (ALPHA - default=false)<br>UpgradeAddonsBeforeControlPlane=true false (DEPRECATED - default=false)<br>WaitForAllControlPlaneComponents=true false (ALPHA - default=false) |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                          |
| help for coredns                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                                                                                                                                              |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --image-repository string                                                                                                                                    | Default: "registry.k8s.io"            |
| Choose a container registry to pull control plane images from                                                                                                |                                       |
| --kubeconfig string                                                                                                                                          | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |                                       |
| --kubernetes-version string                                                                                                                                  | Default: "stable-1"                   |
| Choose a specific Kubernetes version for the control plane.                                                                                                  |                                       |
| --print-manifest                                                                                                                                             |                                       |
| Print the addon manifests to STDOUT instead of installing them                                                                                               |                                       |
| --service-cidr string                                                                                                                                        | Default: "10.96.0.0/12"               |
| Use alternative range of IP address for service VIPs.                                                                                                        |                                       |
| --service-dns-domain string                                                                                                                                  | Default: "cluster.local"              |
| Use alternative domain for services, e.g. "myorg.internal".                                                                                                  |                                       |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Install the kube-proxy addon to a Kubernetes cluster

## Synopsis

Install the kube-proxy addon components via the API server.

```
kubeadm init phase addon kube-proxy [flags]
```

## Options

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                   |
| The IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| --apiserver-bind-port int32     Default: 6443                                                                          |
| Port for the API Server to bind to.                                                                                    |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--config string</code>                                                                                                                                 |
| Path to a kubeadm configuration file.                                                                                                                        |
| <code>--control-plane-endpoint string</code>                                                                                                                 |
| Specify a stable IP address or DNS name for the control plane.                                                                                               |
| <code>--dry-run</code>                                                                                                                                       |
| Don't apply any changes; just output what would be done.                                                                                                     |
| <code>-h, --help</code>                                                                                                                                      |
| help for kube-proxy                                                                                                                                          |
| <code>--image-repository string</code> Default: "registry.k8s.io"                                                                                            |
| Choose a container registry to pull control plane images from                                                                                                |
| <code>--kubeconfig string</code> Default: "/etc/kubernetes/admin.conf"                                                                                       |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |
| <code>--kubernetes-version string</code> Default: "stable-1"                                                                                                 |
| Choose a specific Kubernetes version for the control plane.                                                                                                  |
| <code>--pod-network-cidr string</code>                                                                                                                       |
| Specify range of IP addresses for the pod network. If set, the control plane will automatically allocate CIDRs for every node.                               |
| <code>--print-manifest</code>                                                                                                                                |
| Print the addon manifests to STDOUT instead of installing them                                                                                               |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs string</code>                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

For more details on each field in the v1beta3 configuration you can navigate to our [API reference pages](#).

## What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node
- [kubeadm join](#) to connect a node to the cluster
- [kubeadm reset](#) to revert any changes made to this host by kubeadm init or kubeadm join
- [kubeadm alpha](#) to try experimental functionality

## kubeadm join phase

kubeadm join phase enables you to invoke atomic steps of the join process. Hence, you can let kubeadm do some of the work and you can fill in the gaps if you wish to apply customization.

kubeadm join phase is consistent with the [kubeadm join workflow](#), and behind the scene both use the same code.

## kubeadm join phase

- [phase](#)

Use this command to invoke single phase of the join workflow

### Synopsis

Use this command to invoke single phase of the join workflow

### Options

|                |
|----------------|
| -h, --help     |
| help for phase |

### Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm join phase preflight

Using this phase you can execute preflight checks on a joining node.

- [preflight](#)

Run join pre-flight checks

## Synopsis

Run pre-flight checks for kubeadm join.

```
kubeadm join phase preflight [api-server-endpoint] [flags]
```

## Examples

```
Run join pre-flight checks using a config file.
kubeadm join phase preflight --config kubeadm-config.yaml
```

## Options

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--apiserver-advertise-address</code> string                                                                                                                                            |
| If the node should host a new control plane instance, the IP address the API Server will advertise it's listening on. If not set the default network interface will be used.                 |
| <code>--apiserver-bind-port</code> int32    Default: 6443                                                                                                                                    |
| If the node should host a new control plane instance, the port for the API Server to bind to.                                                                                                |
| <code>--certificate-key</code> string                                                                                                                                                        |
| Use this key to decrypt the certificate secrets uploaded by init. The certificate key is a hex encoded string that is an AES key of size 32 bytes.                                           |
| <code>--config</code> string                                                                                                                                                                 |
| Path to a kubeadm configuration file.                                                                                                                                                        |
| <code>--control-plane</code>                                                                                                                                                                 |
| Create a new control plane instance on this node                                                                                                                                             |
| <code>--cri-socket</code> string                                                                                                                                                             |
| Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| <code>--discovery-file</code> string                                                                                                                                                         |
| For file-based discovery, a file or URL from which to load cluster information.                                                                                                              |
| <code>--discovery-token</code> string                                                                                                                                                        |
| For token-based discovery, the token used to validate cluster information fetched from the API server.                                                                                       |
| <code>--discovery-token-ca-cert-hash</code> strings                                                                                                                                          |



|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| For token-based discovery, validate that the root CA public key matches this hash (format: "<type>:<value>").                          |
| --discovery-token-unsafe-skip-ca-verification                                                                                          |
| For token-based discovery, allow joining without --discovery-token-ca-cert-hash pinning.                                               |
| --dry-run                                                                                                                              |
| Don't apply any changes; just output what would be done.                                                                               |
| -h, --help                                                                                                                             |
| help for preflight                                                                                                                     |
| --ignore-preflight-errors strings                                                                                                      |
| A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks. |
| --node-name string                                                                                                                     |
| Specify the node name.                                                                                                                 |
| --tls-bootstrap-token string                                                                                                           |
| Specify the token used to temporarily authenticate with the Kubernetes Control Plane while joining the node.                           |
| --token string                                                                                                                         |
| Use this token for both discovery-token and tls-bootstrap-token when those values are not provided.                                    |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm join phase control-plane-prepare

Using this phase you can prepare a node for serving a control-plane.

- [control-plane-prepare](#)
- [all](#)
- [download-certs](#)
- [certs](#)

- [kubeconfig](#)
- [control-plane](#)

Prepare the machine for serving a control plane

## Synopsis

Prepare the machine for serving a control plane

```
kubeadm join phase control-plane-prepare [flags]
```

## Examples

```
Prepares the machine for serving a control plane
kubeadm join phase control-plane-prepare all
```

## Options

|                                |
|--------------------------------|
| -h, --help                     |
| help for control-plane-prepare |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Prepare the machine for serving a control plane

## Synopsis

Prepare the machine for serving a control plane

```
kubeadm join phase control-plane-prepare all [api-server-endpoint] [flags]
```

## Options

|                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                                                                         |
| If the node should host a new control plane instance, the IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| --apiserver-bind-port int32   Default: 6443                                                                                                                                  |
| If the node should host a new control plane instance, the port for the API Server to bind to.                                                                                |
| --certificate-key string                                                                                                                                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use this key to decrypt the certificate secrets uploaded by init. The certificate key is a hex encoded string that is an AES key of size 32 bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| --config string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --control-plane                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Create a new control plane instance on this node                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| --discovery-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| For file-based discovery, a file or URL from which to load cluster information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --discovery-token string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| For token-based discovery, the token used to validate cluster information fetched from the API server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --discovery-token-ca-cert-hash strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| For token-based discovery, validate that the root CA public key matches this hash (format: "<type>:<value>").                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --discovery-token-unsafe-skip-ca-verification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| For token-based discovery, allow joining without --discovery-token-ca-cert-hash pinning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --dry-run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| help for all                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --node-name string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Specify the node name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubenetconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --tls-bootstrap-token string                                                                                 |
| Specify the token used to temporarily authenticate with the Kubernetes Control Plane while joining the node. |
| --token string                                                                                               |
| Use this token for both discovery-token and tls-bootstrap-token when those values are not provided.          |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

[EXPERIMENTAL] Download certificates shared among control-plane nodes from the kubeadm-certs Secret

## Synopsis

[EXPERIMENTAL] Download certificates shared among control-plane nodes from the kubeadm-certs Secret

kubeadm join phase control-plane-prepare download-certs [api-server-endpoint] [flags]

## Options

|                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------|
| --certificate-key string                                                                                                                           |
| Use this key to decrypt the certificate secrets uploaded by init. The certificate key is a hex encoded string that is an AES key of size 32 bytes. |
| --config string                                                                                                                                    |
| Path to a kubeadm configuration file.                                                                                                              |
| --control-plane                                                                                                                                    |
| Create a new control plane instance on this node                                                                                                   |
| --discovery-file string                                                                                                                            |
| For file-based discovery, a file or URL from which to load cluster information.                                                                    |
| --discovery-token string                                                                                                                           |
| For token-based discovery, the token used to validate cluster information fetched from the API server.                                             |

|                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------|
| <code>--discovery-token-ca-cert-hash</code> strings                                                                           |
| For token-based discovery, validate that the root CA public key matches this hash (format: "<type>:<value>").                 |
| <code>--discovery-token-unsafe-skip-ca-verification</code>                                                                    |
| For token-based discovery, allow joining without <code>--discovery-token-ca-cert-hash</code> pinning.                         |
| <code>--dry-run</code>                                                                                                        |
| Don't apply any changes; just output what would be done.                                                                      |
| <code>-h, --help</code>                                                                                                       |
| help for download-certs                                                                                                       |
| <code>--tls-bootstrap-token</code> string                                                                                     |
| Specify the token used to temporarily authenticate with the Kubernetes Control Plane while joining the node.                  |
| <code>--token</code> string                                                                                                   |
| Use this token for both <code>discovery-token</code> and <code>tls-bootstrap-token</code> when those values are not provided. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs</code> string                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the certificates for the new control plane components

## Synopsis

Generate the certificates for the new control plane components

```
kubeadm join phase control-plane-prepare certs [api-server-endpoint] [flags]
```

## Options

|                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--apiserver-advertise-address</code> string                                                                                                                            |
| If the node should host a new control plane instance, the IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| <code>--config</code> string                                                                                                                                                 |

|                                                            |                                                                                                                                          |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | Path to a kubeadm configuration file.                                                                                                    |
| <code>--control-plane</code>                               |                                                                                                                                          |
|                                                            | Create a new control plane instance on this node                                                                                         |
| <code>--discovery-file</code> string                       |                                                                                                                                          |
|                                                            | For file-based discovery, a file or URL from which to load cluster information.                                                          |
| <code>--discovery-token</code> string                      |                                                                                                                                          |
|                                                            | For token-based discovery, the token used to validate cluster information fetched from the API server.                                   |
| <code>--discovery-token-ca-cert-hash</code> strings        |                                                                                                                                          |
|                                                            | For token-based discovery, validate that the root CA public key matches this hash (format: " <code>&lt;type&gt;:&lt;value&gt;</code> "). |
| <code>--discovery-token-unsafe-skip-ca-verification</code> |                                                                                                                                          |
|                                                            | For token-based discovery, allow joining without <code>--discovery-token-ca-cert-hash</code> pinning.                                    |
| <code>--dry-run</code>                                     |                                                                                                                                          |
|                                                            | Don't apply any changes; just output what would be done.                                                                                 |
| <code>-h, --help</code>                                    |                                                                                                                                          |
|                                                            | help for certs                                                                                                                           |
| <code>--node-name</code> string                            |                                                                                                                                          |
|                                                            | Specify the node name.                                                                                                                   |
| <code>--tls-bootstrap-token</code> string                  |                                                                                                                                          |
|                                                            | Specify the token used to temporarily authenticate with the Kubernetes Control Plane while joining the node.                             |
| <code>--token</code> string                                |                                                                                                                                          |
|                                                            | Use this token for both <code>discovery-token</code> and <code>tls-bootstrap-token</code> when those values are not provided.            |

## Options inherited from parent commands

|                              |
|------------------------------|
| <code>--rootfs</code> string |
|------------------------------|

[EXPERIMENTAL] The path to the 'real' host root filesystem.

Generate the kubeconfig for the new control plane components

## Synopsis

Generate the kubeconfig for the new control plane components

kubeadm join phase control-plane-prepare kubeconfig [api-server-endpoint] [flags]

## Options

|                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------|
| --certificate-key string                                                                                                                           |
| Use this key to decrypt the certificate secrets uploaded by init. The certificate key is a hex encoded string that is an AES key of size 32 bytes. |
| --config string                                                                                                                                    |
| Path to a kubeadm configuration file.                                                                                                              |
| --control-plane                                                                                                                                    |
| Create a new control plane instance on this node                                                                                                   |
| --discovery-file string                                                                                                                            |
| For file-based discovery, a file or URL from which to load cluster information.                                                                    |
| --discovery-token string                                                                                                                           |
| For token-based discovery, the token used to validate cluster information fetched from the API server.                                             |
| --discovery-token-ca-cert-hash strings                                                                                                             |
| For token-based discovery, validate that the root CA public key matches this hash (format: "<type>:<value>").                                      |
| --discovery-token-unsafe-skip-ca-verification                                                                                                      |
| For token-based discovery, allow joining without --discovery-token-ca-cert-hash pinning.                                                           |
| --dry-run                                                                                                                                          |
| Don't apply any changes; just output what would be done.                                                                                           |
| -h, --help                                                                                                                                         |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| help for kubeconfig                                                                                          |
| --tls-bootstrap-token string                                                                                 |
| Specify the token used to temporarily authenticate with the Kubernetes Control Plane while joining the node. |
| --token string                                                                                               |
| Use this token for both discovery-token and tls-bootstrap-token when those values are not provided.          |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Generate the manifests for the new control plane components

## Synopsis

Generate the manifests for the new control plane components

kubeadm join phase control-plane-prepare control-plane [flags]

## Options

|                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                                                                         |
| If the node should host a new control plane instance, the IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| --apiserver-bind-port int32   Default: 6443                                                                                                                                  |
| If the node should host a new control plane instance, the port for the API Server to bind to.                                                                                |
| --config string                                                                                                                                                              |
| Path to a kubeadm configuration file.                                                                                                                                        |
| --control-plane                                                                                                                                                              |
| Create a new control plane instance on this node                                                                                                                             |
| --dry-run                                                                                                                                                                    |
| Don't apply any changes; just output what would be done.                                                                                                                     |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| help for control-plane                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm join phase kubelet-start

Using this phase you can write the kubelet settings, certificates and (re)start the kubelet.

- [kubelet-start](#)

Write kubelet settings, certificates and (re)start the kubelet

### Synopsis

Write a file with KubeletConfiguration and an environment file with node specific kubelet settings, and then (re)start kubelet.

```
kubeadm join phase kubelet-start [api-server-endpoint] [flags]
```

### Options

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --config string                                                                                                                                                                              |
| Path to a kubeadm configuration file.                                                                                                                                                        |
| --cri-socket string                                                                                                                                                                          |
| Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| --discovery-file string                                                                                                                                                                      |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | For file-based discovery, a file or URL from which to load cluster information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|  | <code>--discovery-token string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  | For token-based discovery, the token used to validate cluster information fetched from the API server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | <code>--discovery-token-ca-cert-hash strings</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|  | For token-based discovery, validate that the root CA public key matches this hash (format: " <code>&lt;type&gt;:&lt;value&gt;</code> ").                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | <code>--discovery-token-unsafe-skip-ca-verification</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|  | For token-based discovery, allow joining without <code>--discovery-token-ca-cert-hash</code> pinning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  | <code>--dry-run</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  | help for kubelet-start                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | <code>--node-name string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|  | Specify the node name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | <code>--patches string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | Path to a directory that contains files named " <code>target[suffix][+patchtype].extension</code> ". For example, " <code>kube-apiserver0+merge.yaml</code> " or just " <code>etcd.json</code> ". " <code>target</code> " can be one of " <code>kube-apiserver</code> ", " <code>kube-controller-manager</code> ", " <code>kube-scheduler</code> ", " <code>etcd</code> ", " <code>kubeletconfiguration</code> ". " <code>patchtype</code> " can be one of " <code>strategic</code> ", " <code>merge</code> " or " <code>json</code> " and they match the patch formats supported by kubectl. The default " <code>patchtype</code> " is " <code>strategic</code> ". " <code>extension</code> " must be either " <code>json</code> " or " <code>yaml</code> ". " <code>suffix</code> " is an optional string that can be used to determine which patches are applied first alpha-numerically. |
|  | <code>--tls-bootstrap-token string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | Specify the token used to temporarily authenticate with the Kubernetes Control Plane while joining the node.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|  | <code>--token string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  | Use this token for both <code>discovery-token</code> and <code>tls-bootstrap-token</code> when those values are not provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm join phase control-plane-join

Using this phase you can join a node as a control-plane instance.

- [control-plane-join](#)
- [all](#)
- [etcd](#)
- [update-status](#)
- [mark-control-plane](#)

Join a machine as a control plane instance

### Synopsis

Join a machine as a control plane instance

```
kubeadm join phase control-plane-join [flags]
```

### Examples

```
Joins a machine as a control plane instance
kubeadm join phase control-plane-join all
```

### Options

|                             |
|-----------------------------|
| -h, --help                  |
| help for control-plane-join |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Join a machine as a control plane instance

### Synopsis

Join a machine as a control plane instance

```
kubeadm join phase control-plane-join all [flags]
```

## Options

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | If the node should host a new control plane instance, the IP address the API Server will advertise it's listening on. If not set the default network interface will be used.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --config string                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --control-plane                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Create a new control plane instance on this node                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| --dry-run                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| -h, --help                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | help for all                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --node-name string                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Specify the node name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --patches string                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubenetconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

## Options inherited from parent commands

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| --rootfs string |                                                             |
|                 | [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Add a new local etcd member

## Synopsis

Add a new local etcd member

kubeadm join phase control-plane-join etcd [flags]

## Options

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | If the node should host a new control plane instance, the IP address the API Server will advertise it's listening on. If not set the default network interface will be used.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --config string                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Path to a kubeadm configuration file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --control-plane                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Create a new control plane instance on this node                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| --dry-run                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Don't apply any changes; just output what would be done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| -h, --help                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | help for etcd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --node-name string                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Specify the node name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --patches string                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubenetconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

## Options inherited from parent commands

|                 |                                                             |
|-----------------|-------------------------------------------------------------|
| --rootfs string |                                                             |
|                 | [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Register the new control-plane node into the ClusterStatus maintained in the kubeadm-config ConfigMap (DEPRECATED)

## Synopsis

Register the new control-plane node into the ClusterStatus maintained in the kubeadm-config ConfigMap (DEPRECATED)

kubeadm join phase control-plane-join update-status [flags]

Options

|                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --apiserver-advertise-address string                                                                                                                                         |
| If the node should host a new control plane instance, the IP address the API Server will advertise it's listening on. If not set the default network interface will be used. |
| --config string                                                                                                                                                              |
| Path to a kubeadm configuration file.                                                                                                                                        |
| --control-plane                                                                                                                                                              |
| Create a new control plane instance on this node                                                                                                                             |
| -h, --help                                                                                                                                                                   |
| help for update-status                                                                                                                                                       |
| --node-name string                                                                                                                                                           |
| Specify the node name.                                                                                                                                                       |

Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Mark a node as a control-plane

Synopsis

Mark a node as a control-plane

kubeadm join phase control-plane-join mark-control-plane [flags]

Options

|                                                  |
|--------------------------------------------------|
| --config string                                  |
| Path to a kubeadm configuration file.            |
| --control-plane                                  |
| Create a new control plane instance on this node |

|                                                          |
|----------------------------------------------------------|
| --dry-run                                                |
| Don't apply any changes; just output what would be done. |
| -h, --help                                               |
| help for mark-control-plane                              |
| --node-name string                                       |
| Specify the node name.                                   |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node
- [kubeadm join](#) to connect a node to the cluster
- [kubeadm reset](#) to revert any changes made to this host by kubeadm init or kubeadm join
- [kubeadm alpha](#) to try experimental functionality

# kubeadm kubeconfig

kubeadm kubeconfig provides utilities for managing kubeconfig files.

For examples on how to use kubeadm kubeconfig user see [Generating kubeconfig files for additional users](#).

## kubeadm kubeconfig

- [overview](#)

Kubeconfig file utilities

### Synopsis

Kubeconfig file utilities.

### Options

|            |
|------------|
| -h, --help |
|------------|

|                     |
|---------------------|
| help for kubeconfig |
|---------------------|

## Options inherited from parent commands

|                 |
|-----------------|
| --rootfs string |
|-----------------|

|                                                             |
|-------------------------------------------------------------|
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-------------------------------------------------------------|

## kubeadm kubeconfig user

This command can be used to output a kubeconfig file for an additional user.

- [user](#)

Output a kubeconfig file for an additional user

### Synopsis

Output a kubeconfig file for an additional user.

```
kubeadm kubeconfig user [flags]
```

### Examples

```
Output a kubeconfig file for an additional user named foo
kubeadm kubeconfig user --client-name=foo
```

```
Output a kubeconfig file for an additional user named foo using a kubeadm config file bar
kubeadm kubeconfig user --client-name=foo --config=bar
```

### Options

|                      |
|----------------------|
| --client-name string |
|----------------------|

|                                                                                |
|--------------------------------------------------------------------------------|
| The name of user. It will be used as the CN if client certificates are created |
|--------------------------------------------------------------------------------|

|                 |
|-----------------|
| --config string |
|-----------------|

|                                       |
|---------------------------------------|
| Path to a kubeadm configuration file. |
|---------------------------------------|

|            |
|------------|
| -h, --help |
|------------|

|               |
|---------------|
| help for user |
|---------------|

|               |
|---------------|
| --org strings |
|---------------|



|                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------|
| The organizations of the client certificate. It will be used as the O if client certificates are created          |
| --token string                                                                                                    |
| The token that should be used as the authentication mechanism for this kubeconfig, instead of client certificates |
| --validity-period duration    Default: 8760h0m0s                                                                  |
| The validity period of the client certificate. It is an offset from the current time.                             |

### Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm reset phase

kubeadm reset phase enables you to invoke atomic steps of the node reset process. Hence, you can let kubeadm do some of the work and you can fill in the gaps if you wish to apply customization.

kubeadm reset phase is consistent with the [kubeadm reset workflow](#), and behind the scene both use the same code.

## kubeadm reset phase

- [phase](#)

Use this command to invoke single phase of the reset workflow

### Synopsis

Use this command to invoke single phase of the reset workflow

### Options

|                |
|----------------|
| -h, --help     |
| help for phase |

### Options inherited from parent commands

|                 |
|-----------------|
| --rootfs string |
|-----------------|

|                                                             |
|-------------------------------------------------------------|
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |
|-------------------------------------------------------------|

## kubeadm reset phase preflight

Using this phase you can execute preflight checks on a node that is being reset.

- [preflight](#)

Run reset pre-flight checks

### Synopsis

Run pre-flight checks for kubeadm reset.

```
kubeadm reset phase preflight [flags]
```

### Options

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| --dry-run                                                                                                                              |
| Don't apply any changes; just output what would be done.                                                                               |
| -f, --force                                                                                                                            |
| Reset the node without prompting for confirmation.                                                                                     |
| -h, --help                                                                                                                             |
| help for preflight                                                                                                                     |
| --ignore-preflight-errors strings                                                                                                      |
| A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks. |

### Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm reset phase remove-etcd-member

Using this phase you can remove this control-plane node's etcd member from the etcd cluster.

- [remove-etcd-member](#)

Remove a local etcd member.

## Synopsis

Remove a local etcd member for a control plane node.

```
kubeadm reset phase remove-etcd-member [flags]
```

## Options

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--dry-run</code>                                                                                                                                       |
| Don't apply any changes; just output what would be done.                                                                                                     |
| <code>-h, --help</code>                                                                                                                                      |
| help for remove-etcd-member                                                                                                                                  |
| <code>--kubeconfig string</code> Default: <code>"/etc/kubernetes/admin.conf"</code>                                                                          |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs string</code>                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## kubeadm reset phase cleanup-node

Using this phase you can perform cleanup on this node.

- [cleanup-node](#)

Run cleanup node.

## Synopsis

Run cleanup node.

```
kubeadm reset phase cleanup-node [flags]
```

## Options

|                                                                                                  |
|--------------------------------------------------------------------------------------------------|
| <code>--cert-dir string</code> Default: <code>"/etc/kubernetes/pki"</code>                       |
| The path to the directory where the certificates are stored. If specified, clean this directory. |

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--cleanup-tmp-dir</code>                                                                                                                                                               |
| Cleanup the <code>"/etc/kubernetes/tmp"</code> directory                                                                                                                                     |
| <code>--cri-socket string</code>                                                                                                                                                             |
| Path to the CRI socket to connect. If empty kubeadm will try to auto-detect this value; use this option only if you have more than one CRI installed or if you have non-standard CRI socket. |
| <code>--dry-run</code>                                                                                                                                                                       |
| Don't apply any changes; just output what would be done.                                                                                                                                     |
| <code>-h, --help</code>                                                                                                                                                                      |
| help for cleanup-node                                                                                                                                                                        |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs string</code>                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node
- [kubeadm join](#) to connect a node to the cluster
- [kubeadm reset](#) to revert any changes made to this host by `kubeadm init` or `kubeadm join`
- [kubeadm alpha](#) to try experimental functionality

# kubeadm upgrade phase

In v1.15.0, kubeadm introduced preliminary support for kubeadm upgrade node phases. Phases for other kubeadm upgrade sub-commands such as `apply`, could be added in the following releases.

## kubeadm upgrade node phase

Using this phase you can choose to execute the separate steps of the upgrade of secondary control-plane or worker nodes. Please note that `kubeadm upgrade apply` still has to be called on a primary control-plane node.

- [phase](#)
- [preflight](#)
- [control-plane](#)
- [kubelet-config](#)

Use this command to invoke single phase of the node workflow

## Synopsis

Use this command to invoke single phase of the node workflow

## Options

|                |
|----------------|
| -h, --help     |
| help for phase |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Run upgrade node pre-flight checks

## Synopsis

Run pre-flight checks for kubeadm upgrade node.

kubeadm upgrade node phase preflight [flags]

## Options

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| -h, --help                                                                                                                             |
| help for preflight                                                                                                                     |
| --ignore-preflight-errors strings                                                                                                      |
| A list of checks whose errors will be shown as warnings. Example: 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Upgrade the control plane instance deployed on this node, if any

## Synopsis

Upgrade the control plane instance deployed on this node, if any

kubeadm upgrade node phase control-plane [flags]

Options

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| --certificate-renewal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Default: true                         |
| Perform the renewal of certificates used by component changed during upgrades.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                       |
| --dry-run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       |
| Do not change any state, just output the actions that would be performed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       |
| --etcd-upgrade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Default: true                         |
| Perform the upgrade of etcd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                       |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                       |
| help for control-plane                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                       |
| --kubeconfig string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Default: "/etc/kubernetes/admin.conf" |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                       |
| --patches string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                       |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |                                       |

Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| --rootfs string                                             |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

Upgrade the kubelet configuration for this node

Synopsis

Download the kubelet configuration from the kubelet-config ConfigMap stored in the cluster

kubeadm upgrade node phase kubelet-config [flags]

## Options

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--dry-run</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Do not change any state, just output the actions that would be performed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| help for kubelet-config                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>--kubeconfig string</code> Default: "/etc/kubernetes/admin.conf"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| The kubeconfig file to use when talking to the cluster. If the flag is not set, a set of standard locations can be searched for an existing kubeconfig file.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>--patches string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

## Options inherited from parent commands

|                                                             |
|-------------------------------------------------------------|
| <code>--rootfs string</code>                                |
| [EXPERIMENTAL] The path to the 'real' host root filesystem. |

## What's next

- [kubeadm init](#) to bootstrap a Kubernetes control-plane node
- [kubeadm join](#) to connect a node to the cluster
- [kubeadm reset](#) to revert any changes made to this host by kubeadm init or kubeadm join
- [kubeadm upgrade](#) to upgrade a kubeadm node
- [kubeadm alpha](#) to try experimental functionality

## Implementation details

FEATURE STATE: Kubernetes v1.10 [stable]

kubeadm init and kubeadm join together provide a nice user experience for creating a bare Kubernetes cluster from scratch, that aligns with the best-practices. However, it might not be obvious *how* kubeadm does that.

This document provides additional details on what happens under the hood, with the aim of sharing knowledge on the best practices for a Kubernetes cluster.

## Core design principles

The cluster that kubeadm init and kubeadm join set up should be:

- **Secure:** It should adopt latest best-practices like:
  - enforcing RBAC
  - using the Node Authorizer
  - using secure communication between the control plane components
  - using secure communication between the API server and the kubelets
  - lock-down the kubelet API
  - locking down access to the API for system components like the kube-proxy and CoreDNS
  - locking down what a Bootstrap Token can access
- **User-friendly:** The user should not have to run anything more than a couple of commands:
  - kubeadm init
  - export KUBECONFIG=/etc/kubernetes/admin.conf
  - kubectl apply -f <network-of-choice.yaml>
  - kubeadm join --token <token> <endpoint>:<port>
- **Extendable:**
  - It should *not* favor any particular network provider. Configuring the cluster network is out-of-scope
  - It should provide the possibility to use a config file for customizing various parameters

## Constants and well-known values and paths

In order to reduce complexity and to simplify development of higher level tools that build on top of kubeadm, it uses a limited set of constant values for well-known paths and file names.

The Kubernetes directory /etc/kubernetes is a constant in the application, since it is clearly the given path in a majority of cases, and the most intuitive location; other constant paths and file names are:

- /etc/kubernetes/manifests as the path where the kubelet should look for static Pod manifests. Names of static Pod manifests are:
  - etcd.yaml
  - kube-apiserver.yaml
  - kube-controller-manager.yaml
  - kube-scheduler.yaml
- /etc/kubernetes/ as the path where kubeconfig files with identities for control plane components are stored. Names of kubeconfig files are:
  - kubelet.conf (bootstrap-kubelet.conf during TLS bootstrap)
  - controller-manager.conf
  - scheduler.conf
  - admin.conf for the cluster admin and kubeadm itself



- super-admin.conf for the cluster super-admin that can bypass RBAC
- Names of certificates and key files:
  - ca.crt, ca.key for the Kubernetes certificate authority
  - apiserver.crt, apiserver.key for the API server certificate
  - apiserver-kubelet-client.crt, apiserver-kubelet-client.key for the client certificate used by the API server to connect to the kubelets securely
  - sa.pub, sa.key for the key used by the controller manager when signing ServiceAccount
  - front-proxy-ca.crt, front-proxy-ca.key for the front proxy certificate authority
  - front-proxy-client.crt, front-proxy-client.key for the front proxy client

## kubeadm init workflow internal design

The kubeadm init consists of a sequence of atomic work tasks to perform, as described in the kubeadm init [internal workflow](#).

The [kubeadm init phase](#) command allows users to invoke each task individually, and ultimately offers a reusable and composable API/toolbox that can be used by other Kubernetes bootstrap tools, by any IT automation tool or by an advanced user for creating custom clusters.

### Preflight checks

Kubeadm executes a set of preflight checks before starting the init, with the aim to verify preconditions and avoid common cluster startup problems. The user can skip specific preflight checks or all of them with the --ignore-preflight-errors option.

- [Warning] if the Kubernetes version to use (specified with the --kubernetes-version flag) is at least one minor version higher than the kubeadm CLI version.
- Kubernetes system requirements:
  - if running on linux:
    - [Error] if Kernel is older than the minimum required version
    - [Error] if required cgroups subsystem aren't set up
- [Error] if the CRI endpoint does not answer
- [Error] if user is not root
- [Error] if the machine hostname is not a valid DNS subdomain
- [Warning] if the host name cannot be reached via network lookup
- [Error] if kubelet version is lower than the minimum kubelet version supported by kubeadm (current minor -1)
- [Error] if kubelet version is at least one minor higher than the required controlplane version (unsupported version skew)
- [Warning] if kubelet service does not exist or if it is disabled
- [Warning] if firewalld is active
- [Error] if API server bindPort or ports 10250/10251/10252 are used
- [Error] if /etc/kubernetes/manifest folder already exists and it is not empty
- [Error] if swap is on
- [Error] if conntrack, ip, iptables, mount, nsenter commands are not present in the command path
- [Warning] if ebtables, ethtool, socat, tc, touch, crictl commands are not present in the command path

- [Warning] if extra arg flags for API server, controller manager, scheduler contains some invalid options
- [Warning] if connection to `https://API.AdvertiseAddress:API.BindPort` goes through proxy
- [Warning] if connection to services subnet goes through proxy (only first address checked)
- [Warning] if connection to Pods subnet goes through proxy (only first address checked)
- If external etcd is provided:
  - [Error] if etcd version is older than the minimum required version
  - [Error] if etcd certificates or keys are specified, but not provided
- If external etcd is NOT provided (and thus local etcd will be installed):
  - [Error] if ports 2379 is used
  - [Error] if `EtcdataDir` folder already exists and it is not empty
- If authorization mode is ABAC:
  - [Error] if `abac_policy.json` does not exist
- If authorization mode is WebHook
  - [Error] if `webhook_authz.conf` does not exist

#### Note:

Preflight checks can be invoked individually with the [kubeadm init phase preflight](#) command.

## Generate the necessary certificates

Kubeadm generates certificate and private key pairs for different purposes:

- A self signed certificate authority for the Kubernetes cluster saved into `ca.crt` file and `ca.key` private key file
- A serving certificate for the API server, generated using `ca.crt` as the CA, and saved into `apiserver.crt` file with its private key `apiserver.key`. This certificate should contain the following alternative names:
  - The Kubernetes service's internal clusterIP (the first address in the services CIDR, e.g. 10.96.0.1 if service subnet is 10.96.0.0/12)
  - Kubernetes DNS names, e.g. `kubernetes.default.svc.cluster.local` if `--service-dns-domain` flag value is `cluster.local`, plus default DNS names `kubernetes.default.svc`, `kubernetes.default`, `kubernetes`
  - The node-name
  - The `--apiserver-advertise-address`
  - Additional alternative names specified by the user
- A client certificate for the API server to connect to the kubelets securely, generated using `ca.crt` as the CA and saved into `apiserver-kubelet-client.crt` file with its private key `apiserver-kubelet-client.key`. This certificate should be in the `system:masters` organization
- A private key for signing ServiceAccount Tokens saved into `sa.key` file along with its public key `sa.pub`
- A certificate authority for the front proxy saved into `front-proxy-ca.crt` file with its key `front-proxy-ca.key`

- A client certificate for the front proxy client, generated using front-proxy-ca.crt as the CA and saved into front-proxy-client.crt file with its private key front-proxy-client.key

Certificates are stored by default in /etc/kubernetes/pki, but this directory is configurable using the `--cert-dir` flag.

Please note that:

1. If a given certificate and private key pair both exist, and their content is evaluated to be compliant with the above specs, the existing files will be used and the generation phase for the given certificate will be skipped. This means the user can, for example, copy an existing CA to /etc/kubernetes/pki/ca.{crt,key}, and then kubeadm will use those files for signing the rest of the certs. See also [using custom certificates](#)
2. For the CA, it is possible to provide the ca.crt file but not the ca.key file. If all other certificates and kubeconfig files are already in place, kubeadm recognizes this condition and activates the ExternalCA, which also implies the csr signer controller in controller-manager won't be started
3. If kubeadm is running in [external CA mode](#); all the certificates must be provided by the user, because kubeadm cannot generate them by itself
4. In case kubeadm is executed in the `--dry-run` mode, certificate files are written in a temporary folder
5. Certificate generation can be invoked individually with the [kubeadm init phase certs all](#) command

## Generate kubeconfig files for control plane components

Kubeadm generates kubeconfig files with identities for control plane components:

- A kubeconfig file for the kubelet to use during TLS bootstrap - /etc/kubernetes/bootstrap-kubelet.conf. Inside this file, there is a bootstrap-token or embedded client certificates for authenticating this node with the cluster.

This client certificate should:

- Be in the system:nodes organization, as required by the [Node Authorization](#) module
- Have the Common Name (CN) system:node:<hostname-lowercased>
- A kubeconfig file for controller-manager, /etc/kubernetes/controller-manager.conf; inside this file is embedded a client certificate with controller-manager identity. This client certificate should have the CN system:kube-controller-manager, as defined by default [RBAC core components roles](#)
- A kubeconfig file for scheduler, /etc/kubernetes/scheduler.conf; inside this file is embedded a client certificate with scheduler identity. This client certificate should have the CN system:kube-scheduler, as defined by default [RBAC core components roles](#)

Additionally, a kubeconfig file for kubeadm as an administrative entity is generated and stored in /etc/kubernetes/admin.conf. This file includes a certificate with Subject: O = kubeadm:cluster-admins, CN = kubernet-admin. kubeadm:cluster-admins is a group managed by kubeadm. It is bound to the cluster-admin ClusterRole during kubeadm init, by using the super-admin.conf file, which does not require RBAC. This admin.conf file must remain on control plane nodes and should not be shared with additional users.

During kubeadm init another kubeconfig file is generated and stored in `/etc/kubernetes/super-admin.conf`. This file includes a certificate with Subject: O = system:masters, CN = kubernetes-super-admin. system:masters is a superuser group that bypasses RBAC and makes super-admin.conf useful in case of an emergency where a cluster is locked due to RBAC misconfiguration. The super-admin.conf file must be stored in a safe location and should not be shared with additional users.

See [RBAC user facing role bindings](#) for additional information on RBAC and built-in ClusterRoles and groups.

Please note that:

1. ca.crt certificate is embedded in all the kubeconfig files.
2. If a given kubeconfig file exists, and its content is evaluated as compliant with the above specs, the existing file will be used and the generation phase for the given kubeconfig will be skipped
3. If kubeadm is running in [ExternalCA mode](#), all the required kubeconfig must be provided by the user as well, because kubeadm cannot generate any of them by itself
4. In case kubeadm is executed in the `--dry-run` mode, kubeconfig files are written in a temporary folder
5. Generation of kubeconfig files can be invoked individually with the [kubeadm init phase kubeconfig all](#) command

## Generate static Pod manifests for control plane components

Kubeadm writes static Pod manifest files for control plane components to `/etc/kubernetes/manifests`. The kubelet watches this directory for Pods to be created on startup.

Static Pod manifests share a set of common properties:

- All static Pods are deployed on kube-system namespace
- All static Pods get tier:control-plane and component:{component-name} labels
- All static Pods use the system-node-critical priority class
- `hostNetwork: true` is set on all static Pods to allow control plane startup before a network is configured; as a consequence:
  - The address that the controller-manager and the scheduler use to refer to the API server is 127.0.0.1
  - If the etcd server is set up locally, the etcd-server address will be set to 127.0.0.1:2379
- Leader election is enabled for both the controller-manager and the scheduler
- Controller-manager and the scheduler will reference kubeconfig files with their respective, unique identities
- All static Pods get any extra flags specified by the user as described in [passing custom arguments to control plane components](#)
- All static Pods get any extra Volumes specified by the user (Host path)

Please note that:

1. All images will be pulled from registry.k8s.io by default. See [using custom images](#) for customizing the image repository
2. In case kubeadm is executed in the `--dry-run` mode, static Pod files are written in a temporary folder
3. Static Pod manifest generation for control plane components can be invoked individually with the [kubeadm init phase control-plane all](#) command

## API server

The static Pod manifest for the API server is affected by the following parameters provided by the users:

- The `apiserver-advertise-address` and `apiserver-bind-port` to bind to; if not provided, those values default to the IP address of the default network interface on the machine and port 6443
- The `service-cluster-ip-range` to use for services
- If an external etcd server is specified, the `etcd-servers` address and related TLS settings (`etcd-cafile`, `etcd-certfile`, `etcd-keyfile`); if an external etcd server is not provided, a local etcd will be used (via host network)
- If a cloud provider is specified, the corresponding `--cloud-provider` parameter is configured together with the `--cloud-config` path if such file exists (this is experimental, alpha and will be removed in a future version)

Other API server flags that are set unconditionally are:

- `--insecure-port=0` to avoid insecure connections to the api server
- `--enable-bootstrap-token-auth=true` to enable the BootstrapTokenAuthenticator authentication module. See [TLS Bootstrapping](#) for more details
- `--allow-privileged` to true (required e.g. by kube proxy)
- `--requestheader-client-ca-file` to `front-proxy-ca.crt`
- `--enable-admission-plugins` to:
  - [NamespaceLifecycle](#) e.g. to avoid deletion of system reserved namespaces
  - [LimitRanger](#) and [ResourceQuota](#) to enforce limits on namespaces
  - [ServiceAccount](#) to enforce service account automation
  - [PersistentVolumeLabel](#) attaches region or zone labels to PersistentVolumes as defined by the cloud provider (This admission controller is deprecated and will be removed in a future version. It is not deployed by kubeadm by default with v1.9 onwards when not explicitly opting into using gce or aws as cloud providers)
  - [DefaultStorageClass](#) to enforce default storage class on PersistentVolumeClaim objects
  - [DefaultTolerationSeconds](#)
  - [NodeRestriction](#) to limit what a kubelet can modify (e.g. only pods on this node)
- `--kubelet-preferred-address-types` to `InternalIP,ExternalIP,Hostname`; this makes kubect logs and other API server-kubelet communication work in environments where the hostnames of the nodes aren't resolvable

Flags for using certificates generated in previous steps:

- - `--client-ca-file` to `ca.crt`
  - `--tls-cert-file` to `apiserver.crt`
  - `--tls-private-key-file` to `apiserver.key`
  - `--kubelet-client-certificate` to `apiserver-kubelet-client.crt`
  - `--kubelet-client-key` to `apiserver-kubelet-client.key`
  - `--service-account-key-file` to `sa.pub`
  - `--requestheader-client-ca-file` to `front-proxy-ca.crt`
  - `--proxy-client-cert-file` to `front-proxy-client.crt`
  - `--proxy-client-key-file` to `front-proxy-client.key`
- Other flags for securing the front proxy ([API Aggregation](#)) communications:
  - `--requestheader-username-headers=X-Remote-User`
  - `--requestheader-group-headers=X-Remote-Group`
  - `--requestheader-extra-headers-prefix=X-Remote-Extra-`
  - `--requestheader-allowed-names=front-proxy-client`

## Controller manager

The static Pod manifest for the controller manager is affected by following parameters provided by the users:

- If kubeadm is invoked specifying a `--pod-network-cidr`, the subnet manager feature required for some CNI network plugins is enabled by setting:
  - `--allocate-node-cidrs=true`
  - `--cluster-cidr` and `--node-cidr-mask-size` flags according to the given CIDR
- If a cloud provider is specified, the corresponding `--cloud-provider` is specified together with the `--cloud-config` path if such configuration file exists (this is experimental, alpha and will be removed in a future version)

Other flags that are set unconditionally are:

- `--controllers` enabling all the default controllers plus BootstrapSigner and TokenCleaner controllers for TLS bootstrap. See [TLS Bootstrapping](#) for more details.
- `--use-service-account-credentials` to `true`
- Flags for using certificates generated in previous steps:
  - `--root-ca-file` to `ca.crt`
  - `--cluster-signing-cert-file` to `ca.crt`, if External CA mode is disabled, otherwise to `""`
  - `--cluster-signing-key-file` to `ca.key`, if External CA mode is disabled, otherwise to `""`
  - `--service-account-private-key-file` to `sa.key`

## Scheduler

The static Pod manifest for the scheduler is not affected by parameters provided by the users.

## Generate static Pod manifest for local etcd

If you specified an external etcd, this step will be skipped, otherwise kubeadm generates a static Pod manifest file for creating a local etcd instance running in a Pod with following attributes:

- listen on localhost:2379 and use HostNetwork=true
- make a hostPath mount out from the dataDir to the host's filesystem
- Any extra flags specified by the user

Please note that:

1. The etcd container image will be pulled from registry.gcr.io by default. See [using custom images](#) for customizing the image repository.
2. If you run kubeadm in --dry-run mode, the etcd static Pod manifest is written into a temporary folder.
3. You can directly invoke static Pod manifest generation for local etcd, using the [kubeadm init phase etcd local](#) command.

## Wait for the control plane to come up

kubeadm waits (upto 4m0s) until localhost:6443/healthz (kube-apiserver liveness) returns ok. However, in order to detect deadlock conditions, kubeadm fails fast if localhost:10255/healthz (kubelet liveness) or localhost:10255/healthz/syncloop (kubelet readiness) don't return ok within 40s and 60s respectively.

kubeadm relies on the kubelet to pull the control plane images and run them properly as static Pods. After the control plane is up, kubeadm completes the tasks described in following paragraphs.

## Save the kubeadm ClusterConfiguration in a ConfigMap for later reference

kubeadm saves the configuration passed to kubeadm init in a ConfigMap named kubeadm-config under kube-system namespace.

This will ensure that kubeadm actions executed in future (e.g kubeadm upgrade) will be able to determine the actual/current cluster state and make new decisions based on that data.

Please note that:

1. Before saving the ClusterConfiguration, sensitive information like the token is stripped from the configuration
2. Upload of control plane node configuration can be invoked individually with the command [kubeadm init phase upload-config](#).

## Mark the node as control-plane

As soon as the control plane is available, kubeadm executes the following actions:

- Labels the node as control-plane with node-role.kubernetes.io/control-plane=""
- Taints the node with node-role.kubernetes.io/control-plane:NoSchedule



Please note that the phase to mark the control-plane phase can be invoked individually with the [kubeadm init phase mark-control-plane](#) command.

## Configure TLS-Bootstrapping for node joining

Kubeadm uses [Authenticating with Bootstrap Tokens](#) for joining new nodes to an existing cluster; for more details see also [design proposal](#).

kubeadm init ensures that everything is properly configured for this process, and this includes following steps as well as setting API server and controller flags as already described in previous paragraphs.

### Note:

TLS bootstrapping for nodes can be configured with the command [kubeadm init phase bootstrap-token](#), executing all the configuration steps described in following paragraphs; alternatively, each step can be invoked individually.

### Create a bootstrap token

kubeadm init creates a first bootstrap token, either generated automatically or provided by the user with the --token flag; as documented in bootstrap token specification, token should be saved as a secret with name bootstrap-token-<token-id> under kube-system namespace.

Please note that:

1. The default token created by kubeadm init will be used to validate temporary user during TLS bootstrap process; those users will be member of system:bootstrappers:kubeadm:default-node-token group
2. The token has a limited validity, default 24 hours (the interval may be changed with the --token-ttl flag)
3. Additional tokens can be created with the [kubeadm token](#) command, that provide other useful functions for token management as well.

### Allow joining nodes to call CSR API

Kubeadm ensures that users in system:bootstrappers:kubeadm:default-node-token group are able to access the certificate signing API.

This is implemented by creating a ClusterRoleBinding named kubeadm:kubelet-bootstrap between the group above and the default RBAC role system:node-bootstrapper.

### Set up auto approval for new bootstrap tokens

Kubeadm ensures that the Bootstrap Token will get its CSR request automatically approved by the csrapprover controller.

This is implemented by creating ClusterRoleBinding named kubeadm:node-autoapprove-bootstrap between the system:bootstrappers:kubeadm:default-node-token group and the default role system:certificates.k8s.io:certificatesigningrequests:nodeclient.



The role `system:certificates.k8s.io:certificatesigningrequests:nodeclient` should be created as well, granting POST permission to `/apis/certificates.k8s.io/certificatesigningrequests/nodeclient`.

## Set up nodes certificate rotation with auto approval

Kubeadm ensures that certificate rotation is enabled for nodes, and that a new certificate request for nodes will get its CSR request automatically approved by the `csrapprover` controller.

This is implemented by creating `ClusterRoleBinding` named `kubeadm:node-autoapprove-certificate-rotation` between the `system:nodes` group and the default role `system:certificates.k8s.io:certificatesigningrequests:selfnodeclient`.

## Create the public cluster-info ConfigMap

This phase creates the `cluster-info` `ConfigMap` in the `kube-public` namespace.

Additionally, it creates a `Role` and a `RoleBinding` granting access to the `ConfigMap` for unauthenticated users (i.e. users in RBAC group `system:unauthenticated`).

### Note:

The access to the `cluster-info` `ConfigMap` *is not* rate-limited. This may or may not be a problem if you expose your cluster's API server to the internet; worst-case scenario here is a DoS attack where an attacker uses all the in-flight requests the `kube-apiserver` can handle to serve the `cluster-info` `ConfigMap`.

## Install addons

Kubeadm installs the internal DNS server and the `kube-proxy` addon components via the API server.

### Note:

This phase can be invoked individually with the command [kubeadm init phase addon all](#).

### proxy

A `ServiceAccount` for `kube-proxy` is created in the `kube-system` namespace; then `kube-proxy` is deployed as a `DaemonSet`:

- The credentials (`ca.crt` and `token`) to the control plane come from the `ServiceAccount`
- The location (URL) of the API server comes from a `ConfigMap`
- The `kube-proxy` `ServiceAccount` is bound to the privileges in the `system:node-proxier` `ClusterRole`

### DNS

- The `CoreDNS` service is named `kube-dns`. This is done to prevent any interruption in service when the user is switching the cluster DNS from `kube-dns` to `CoreDNS` through the `--config` method described [here](#).

A ServiceAccount for CoreDNS is created in the kube-system namespace.

- 
- The coredns ServiceAccount is bound to the privileges in the system:coredns ClusterRole

In Kubernetes version 1.21, support for using kube-dns with kubeadm was removed. You can use CoreDNS with kubeadm even when the related Service is named kube-dns.

## kubeadm join phases internal design

Similarly to kubeadm init, also kubeadm join internal workflow consists of a sequence of atomic work tasks to perform.

This is split into discovery (having the Node trust the Kubernetes Master) and TLS bootstrap (having the Kubernetes Master trust the Node).

see [Authenticating with Bootstrap Tokens](#) or the corresponding [design proposal](#).

### Preflight checks

kubeadm executes a set of preflight checks before starting the join, with the aim to verify preconditions and avoid common cluster startup problems.

Please note that:

1. kubeadm join preflight checks are basically a subset of kubeadm init preflight checks
2. Starting from 1.24, kubeadm uses crictl to communicate to all known CRI endpoints.
3. Starting from 1.9, kubeadm provides support for joining nodes running on Windows; in that case, linux specific controls are skipped.
4. In any case the user can skip specific preflight checks (or eventually all preflight checks) with the --ignore-preflight-errors option.

### Discovery cluster-info

There are 2 main schemes for discovery. The first is to use a shared token along with the IP address of the API server. The second is to provide a file (that is a subset of the standard kubeconfig file).

#### Shared token discovery

If kubeadm join is invoked with --discovery-token, token discovery is used; in this case the node basically retrieves the cluster CA certificates from the cluster-info ConfigMap in the kube-public namespace.

In order to prevent "man in the middle" attacks, several steps are taken:

- First, the CA certificate is retrieved via insecure connection (this is possible because kubeadm init is granted access to cluster-info users for system:unauthenticated)
- Then the CA certificate goes through following validation steps:
  - Basic validation: using the token ID against a JWT signature
  - Pub key validation: using provided --discovery-token-ca-cert-hash. This value is available in the output of kubeadm init or can be calculated using standard tools

(the hash is calculated over the bytes of the Subject Public Key Info (SPKI) object as in RFC7469). The `--discovery-token-ca-cert-hash` flag may be repeated multiple times to allow more than one public key.

- As an additional validation, the CA certificate is retrieved via secure connection and then compared with the CA retrieved initially

**Note:**

Pub key validation can be skipped passing `--discovery-token-unsafe-skip-ca-verification` flag; This weakens the kubeadm security model since others can potentially impersonate the Kubernetes Master.

## File/https discovery

If `kubeadm join` is invoked with `--discovery-file`, file discovery is used; this file can be a local file or downloaded via an HTTPS URL; in case of HTTPS, the host installed CA bundle is used to verify the connection.

With file discovery, the cluster CA certificate is provided into the file itself; in fact, the discovery file is a kubeconfig file with only server and certificate-authority-data attributes set, as described in the [kubeadm join](#) reference doc; when the connection with the cluster is established, kubeadm tries to access the cluster-info ConfigMap, and if available, uses it.

## TLS Bootstrap

Once the cluster info is known, the file `bootstrap-kubelet.conf` is written, thus allowing kubelet to do TLS Bootstrapping.

The TLS bootstrap mechanism uses the shared token to temporarily authenticate with the Kubernetes API server to submit a certificate signing request (CSR) for a locally created key pair.

The request is then automatically approved and the operation completes saving `ca.crt` file and `kubelet.conf` file to be used by the kubelet for joining the cluster, while `bootstrap-kubelet.conf` is deleted.

**Note:**

- The temporary authentication is validated against the token saved during the `kubeadm init` process (or with additional tokens created with `kubeadm token` command)
- The temporary authentication resolves to a user member of `system:bootstrappers:kubeadm:default-node-token` group which was granted access to the CSR api during the `kubeadm init` process
- The automatic CSR approval is managed by the `csrapprover` controller, according to the configuration present in the `kubeadm init` process

## Command line tool (kubectl)

Kubernetes provides a command line tool for communicating with a Kubernetes cluster's [control plane](#), using the Kubernetes API.

This tool is named `kubectl`.

For configuration, `kubectl` looks for a file named `config` in the `$HOME/.kube` directory. You can specify other [kubeconfig](#) files by setting the `KUBECONFIG` environment variable or by setting the `--kubeconfig` flag.

This overview covers `kubectl` syntax, describes the command operations, and provides common examples. For details about each command, including all the supported flags and subcommands, see the [kubectl](#) reference documentation.

For installation instructions, see [Installing kubectl](#); for a quick guide, see the [cheat sheet](#). If you're used to using the docker command-line tool, [kubectl for Docker Users](#) explains some equivalent commands for Kubernetes.

## Syntax

Use the following syntax to run `kubectl` commands from your terminal window:

```
kubectl [command] [TYPE] [NAME] [flags]
```

where `command`, `TYPE`, `NAME`, and `flags` are:

- `command`: Specifies the operation that you want to perform on one or more resources, for example `create`, `get`, `describe`, `delete`.
- `TYPE`: Specifies the [resource type](#). Resource types are case-insensitive and you can specify the singular, plural, or abbreviated forms. For example, the following commands produce the same output:

```
kubectl get pod pod1
kubectl get pods pod1
kubectl get po pod1
```

- `NAME`: Specifies the name of the resource. Names are case-sensitive. If the name is omitted, details for all resources are displayed, for example `kubectl get pods`.

When performing an operation on multiple resources, you can specify each resource by type and name or specify one or more files:

- To specify resources by type and name:
  - To group resources if they are all the same type: `TYPE1 name1 name2 name<#>`.  
Example: `kubectl get pod example-pod1 example-pod2`
  - To specify multiple resource types individually: `TYPE1/name1 TYPE1/name2 TYPE2/name3 TYPE<#>/name<#>`.  
Example: `kubectl get pod/example-pod1 replicationcontroller/example-rc1`
- To specify resources with one or more files: `-f file1 -f file2 -f file<#>`
  - [Use YAML rather than JSON](#) since YAML tends to be more user-friendly, especially for configuration files.  
Example: `kubectl get -f ./pod.yaml`

- flags: Specifies optional flags. For example, you can use the `-s` or `--server` flags to specify the address and port of the Kubernetes API server.

### Caution:

Flags that you specify from the command line override default values and any corresponding environment variables.

If you need help, run `kubectl help` from the terminal window.

## In-cluster authentication and namespace overrides

By default `kubectl` will first determine if it is running within a pod, and thus in a cluster. It starts by checking for the `KUBERNETES_SERVICE_HOST` and `KUBERNETES_SERVICE_PORT` environment variables and the existence of a service account token file at `/var/run/secrets/kubernetes.io/serviceaccount/token`. If all three are found in-cluster authentication is assumed.

To maintain backwards compatibility, if the `POD_NAMESPACE` environment variable is set during in-cluster authentication it will override the default namespace from the service account token. Any manifests or tools relying on namespace defaulting will be affected by this.

### POD\_NAMESPACE environment variable

If the `POD_NAMESPACE` environment variable is set, cli operations on namespaced resources will default to the variable value. For example, if the variable is set to `seattle`, `kubectl get pods` would return pods in the `seattle` namespace. This is because pods are a namespaced resource, and no namespace was provided in the command. Review the output of `kubectl api-resources` to determine if a resource is namespaced.

Explicit use of `--namespace <value>` overrides this behavior.

### How kubectl handles ServiceAccount tokens

If:

- there is Kubernetes service account token file mounted at `/var/run/secrets/kubernetes.io/serviceaccount/token`, and
- the `KUBERNETES_SERVICE_HOST` environment variable is set, and
- the `KUBERNETES_SERVICE_PORT` environment variable is set, and
- you don't explicitly specify a namespace on the `kubectl` command line

then `kubectl` assumes it is running in your cluster. The `kubectl` tool looks up the namespace of that `ServiceAccount` (this is the same as the namespace of the `Pod`) and acts against that namespace. This is different from what happens outside of a cluster; when `kubectl` runs outside a cluster and you don't specify a namespace, the `kubectl` command acts against the namespace set for the current context in your client configuration. To change the default namespace for your `kubectl` you can use the following command:

```
kubectl config set-context --current --namespace=<namespace-name>
```

# Operations

The following table includes short descriptions and the general syntax for all of the kubectl operations:

| Operation     | Syntax                                                                                                                                        | Description                                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| alpha         | kubectl alpha SUBCOMMAND [flags]                                                                                                              | List the available commands that correspond to alpha features, which are not enabled in Kubernetes clusters by default.                               |
| annotate      | kubectl annotate (-f FILENAME   TYPE NAME   TYPE/NAME) KEY_1=VAL_1 ... KEY_N=VAL_N [--overwrite] [--all] [--resource-version=version] [flags] | Add or update the annotations of one or more resources.                                                                                               |
| api-resources | kubectl api-resources [flags]                                                                                                                 | List the API resources that are available.                                                                                                            |
| api-versions  | kubectl api-versions [flags]                                                                                                                  | List the API versions that are available.                                                                                                             |
| apply         | kubectl apply -f FILENAME [flags]                                                                                                             | Apply a configuration change to a resource from a file or stdin.                                                                                      |
| attach        | kubectl attach POD -c CONTAINER [-i] [-t] [flags]                                                                                             | Attach to a running container either to view the output stream or interact with the container (stdin).                                                |
| auth          | kubectl auth [flags] [options]                                                                                                                | Inspect authorization.                                                                                                                                |
| autoscale     | kubectl autoscale (-f FILENAME   TYPE NAME   TYPE/NAME) [--min=MINPODS] --max=MAXPODS [--cpu-percent=CPU] [flags]                             | Automatically scale the set of pods that are managed by a replication controller.                                                                     |
| certificate   | kubectl certificate SUBCOMMAND [options]                                                                                                      | Modify certificate resources.                                                                                                                         |
| cluster-info  | kubectl cluster-info [flags]                                                                                                                  | Display endpoint information about the master and services in the cluster.                                                                            |
| completion    | kubectl completion SHELL [options]                                                                                                            | Output shell completion code for the specified shell (bash or zsh).                                                                                   |
| config        | kubectl config SUBCOMMAND [flags]                                                                                                             | Modifies kubeconfig files. See the individual subcommands for details.                                                                                |
| convert       | kubectl convert -f FILENAME [options]                                                                                                         | Convert config files between different API versions. Both YAML and JSON formats are accepted. Note - requires kubectl-convert plugin to be installed. |
| cordon        | kubectl cordon NODE [options]                                                                                                                 | Mark node as unschedulable.                                                                                                                           |
| cp            | kubectl cp <file-spec-src> <file-spec-dest> [options]                                                                                         | Copy files and directories to and from containers.                                                                                                    |
| create        | kubectl create -f FILENAME [flags]                                                                                                            | Create one or more resources from a file or stdin.                                                                                                    |

| Operation | Syntax                                                                                                                                                                                            | Description                                                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delete    | kubectl delete (-f FILENAME   TYPE [NAME   /NAME   -l label   --all]) [flags]                                                                                                                     | Delete resources either from a file, stdin, or specifying label selectors, names, resource selectors, or resources.                                                                                                                                       |
| describe  | kubectl describe (-f FILENAME   TYPE [NAME_PREFIX   /NAME   -l label]) [flags]                                                                                                                    | Display the detailed state of one or more resources.                                                                                                                                                                                                      |
| diff      | kubectl diff -f FILENAME [flags]                                                                                                                                                                  | Diff file or stdin against live configuration.                                                                                                                                                                                                            |
| drain     | kubectl drain NODE [options]                                                                                                                                                                      | Drain node in preparation for maintenance.                                                                                                                                                                                                                |
| edit      | kubectl edit (-f FILENAME   TYPE NAME   TYPE/NAME) [flags]                                                                                                                                        | Edit and update the definition of one or more resources on the server by using the default editor.                                                                                                                                                        |
| events    | kubectl events                                                                                                                                                                                    | List events                                                                                                                                                                                                                                               |
| exec      | kubectl exec POD [-c CONTAINER] [-i] [-t] [flags] [-- COMMAND [args...]]                                                                                                                          | Execute a command against a container in a pod.                                                                                                                                                                                                           |
| explain   | kubectl explain TYPE [--recursive=false] [flags]                                                                                                                                                  | Get documentation of various resources. For instance pods, nodes, services, etc.                                                                                                                                                                          |
| expose    | kubectl expose (-f FILENAME   TYPE NAME   TYPE/NAME) [--port=port] [--protocol=TCP UDP] [--target-port=number-or-name] [--name=name] [--external-ip=external-ip-of-service] [--type=type] [flags] | Expose a replication controller, service, or pod as a new Kubernetes service.                                                                                                                                                                             |
| get       | kubectl get (-f FILENAME   TYPE [NAME   /NAME   -l label]) [--watch] [--sort-by=FIELD] [[-o   --output]=OUTPUT_FORMAT] [flags]                                                                    | List one or more resources.                                                                                                                                                                                                                               |
| kustomize | kubectl kustomize <dir> [flags] [options]                                                                                                                                                         | List a set of API resources generated from instructions in a kustomization.yaml file. The argument must be the path to the directory containing the file, or a git repository URL with a path suffix specifying same with respect to the repository root. |
| label     | kubectl label (-f FILENAME   TYPE NAME   TYPE/NAME) KEY_1=VAL_1 ... KEY_N=VAL_N [--overwrite] [--all] [--resource-version=version] [flags]                                                        | Add or update the labels of one or more resources.                                                                                                                                                                                                        |
| logs      | kubectl logs POD [-c CONTAINER] [--follow] [flags]                                                                                                                                                | Print the logs for a container in a pod.                                                                                                                                                                                                                  |
| options   | kubectl options                                                                                                                                                                                   | List of global command-line options, which apply to all commands.                                                                                                                                                                                         |
| patch     | kubectl patch (-f FILENAME   TYPE NAME   TYPE/NAME) --patch PATCH [flags]                                                                                                                         | Update one or more fields of a resource by using the strategic merge patch process.                                                                                                                                                                       |
| plugin    | kubectl plugin [flags] [options]                                                                                                                                                                  |                                                                                                                                                                                                                                                           |

| Operation    | Syntax                                                                                                                                                  | Description                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|              |                                                                                                                                                         | Provides utilities for interacting with plugins.                                                          |
| port-forward | kubectl port-forward POD<br>[LOCAL_PORT:]REMOTE_PORT [...<br>[LOCAL_PORT_N:]REMOTE_PORT_N] [flags]                                                      | Forward one or more local ports to a pod.                                                                 |
| proxy        | kubectl proxy [--port=PORT] [--www=static-dir]<br>[--www-prefix=prefix] [--api-prefix=prefix]<br>[flags]                                                | Run a proxy to the Kubernetes API server.                                                                 |
| replace      | kubectl replace -f FILENAME                                                                                                                             | Replace a resource from a file or stdin.                                                                  |
| rollout      | kubectl rollout SUBCOMMAND [options]                                                                                                                    | Manage the rollout of a resource. Valid resource types include: deployments, daemonsets and statefulsets. |
| run          | kubectl run NAME --image=image [--env="key=value"] [--port=port] [--dry-run=server client none] [--overrides=inline-json] [flags]                       | Run a specified image on the cluster.                                                                     |
| scale        | kubectl scale (-f FILENAME   TYPE NAME   TYPE/NAME) --replicas=COUNT [--resource-version=version] [--current-replicas=count] [flags]                    | Update the size of the specified replication controller.                                                  |
| set          | kubectl set SUBCOMMAND [options]                                                                                                                        | Configure application resources.                                                                          |
| taint        | kubectl taint NODE NAME<br>KEY_1=VAL_1:TAINT_EFFECT_1 ...<br>KEY_N=VAL_N:TAINT_EFFECT_N [options]                                                       | Update the taints on one or more nodes.                                                                   |
| top          | kubectl top (POD   NODE) [flags] [options]                                                                                                              | Display Resource (CPU/Memory/Storage) usage of pod or node.                                               |
| uncordon     | kubectl uncordon NODE [options]                                                                                                                         | Mark node as schedulable.                                                                                 |
| version      | kubectl version [--client] [flags]                                                                                                                      | Display the Kubernetes version running on the client and server.                                          |
| wait         | kubectl wait ([-f FILENAME]   resource.group/resource.name   resource.group [(-l label   --all)])<br>[--for=delete --for condition=available] [options] | Experimental: Wait for a specific condition on one or many resources.                                     |

To learn more about command operations, see the [kubectl](#) reference documentation.

## Resource types

The following table includes a list of all the supported resource types and their abbreviated aliases.

(This output can be retrieved from `kubectl api-resources`, and was accurate as of Kubernetes 1.25.0)

| NAME              | SHORTNAMES | APIVERSION | NAMESPACED | KIND            |
|-------------------|------------|------------|------------|-----------------|
| bindings          |            | v1         | true       | Binding         |
| componentstatuses | cs         | v1         | false      | ComponentStatus |
| configmaps        | cm         | v1         | true       | ConfigMap       |



| NAME                            | SHORTNAMES | APIVERSION                           | NAMESPACED | KIND                           |
|---------------------------------|------------|--------------------------------------|------------|--------------------------------|
| endpoints                       | ep         | v1                                   | true       | Endpoints                      |
| events                          | ev         | v1                                   | true       | Event                          |
| limitranges                     | limits     | v1                                   | true       | LimitRange                     |
| namespaces                      | ns         | v1                                   | false      | Namespace                      |
| nodes                           | no         | v1                                   | false      | Node                           |
| persistentvolumeclaims          | pvc        | v1                                   | true       | PersistentVolumeClaim          |
| persistentvolumes               | pv         | v1                                   | false      | PersistentVolume               |
| Pods                            | po         | v1                                   | true       | Pod                            |
| podtemplates                    |            | v1                                   | true       | PodTemplate                    |
| replicationcontrollers          | rc         | v1                                   | true       | ReplicationController          |
| resourcequotas                  | quota      | v1                                   | true       | ResourceQuota                  |
| secrets                         |            | v1                                   | true       | Secret                         |
| serviceaccounts                 | sa         | v1                                   | true       | ServiceAccount                 |
| services                        | svc        | v1                                   | true       | Service                        |
| mutatingwebhookconfigurations   |            | admissionregistration.k8s.io/v1      | false      | MutatingWebhookConfiguration   |
| validatingwebhookconfigurations |            | admissionregistration.k8s.io/v1      | false      | ValidatingWebhookConfiguration |
| customresourcedefinitions       | crd,crds   | apiextensions.k8s.io/v1              | false      | CustomResourceDefinition       |
| apiservices                     |            | apiregistration.k8s.io/v1            | false      | APIService                     |
| controllerrevisions             |            | apps/v1                              | true       | ControllerRevision             |
| daemonsets                      | ds         | apps/v1                              | true       | DaemonSet                      |
| deployments                     | deploy     | apps/v1                              | true       | Deployment                     |
| replicasets                     | rs         | apps/v1                              | true       | ReplicaSet                     |
| statefulsets                    | sts        | apps/v1                              | true       | StatefulSet                    |
| tokenreviews                    |            | authentication.k8s.io/v1             | false      | TokenReview                    |
| localsubjectaccessreviews       |            | authorization.k8s.io/v1              | true       | LocalSubjectAccessReview       |
| selfsubjectaccessreviews        |            | authorization.k8s.io/v1              | false      | SelfSubjectAccessReview        |
| selfsubjectrulesreviews         |            | authorization.k8s.io/v1              | false      | SelfSubjectRulesReview         |
| subjectaccessreviews            |            | authorization.k8s.io/v1              | false      | SubjectAccessReview            |
| horizontalpodautoscalers        | hpa        | autoscaling/v2                       | true       | HorizontalPodAutoscaler        |
| cronjobs                        | cj         | batch/v1                             | true       | CronJob                        |
| jobs                            |            | batch/v1                             | true       | Job                            |
| certificatesigningrequests      | csr        | certificates.k8s.io/v1               | false      | CertificateSigningRequest      |
| leases                          |            | coordination.k8s.io/v1               | true       | Lease                          |
| endpointslices                  |            | discovery.k8s.io/v1                  | true       | EndpointSlice                  |
| events                          | ev         | events.k8s.io/v1                     | true       | Event                          |
| flowschemas                     |            | flowcontrol.apiserver.k8s.io/v1beta2 | false      | FlowSchema                     |
| prioritylevelconfigurations     |            | flowcontrol.apiserver.k8s.io/v1beta2 | false      | PriorityLevelConfiguration     |
| ingressclasses                  |            | networking.k8s.io/v1                 | false      | IngressClass                   |
| ingresses                       | ing        | networking.k8s.io/v1                 | true       | Ingress                        |
| networkpolicies                 | netpol     | networking.k8s.io/v1                 | true       | NetworkPolicy                  |
| runtimeclasses                  |            | node.k8s.io/v1                       | false      | RuntimeClass                   |

| NAME                 | SHORTNAMES | APIVERSION                   | NAMESPACED | KIND        |
|----------------------|------------|------------------------------|------------|-------------|
| poddisruptionbudgets | pdb        | policy/v1                    | true       | PodDisrup   |
| podsecuritypolicies  | psp        | policy/v1beta1               | false      | PodSecurit  |
| clusterrolebindings  |            | rbac.authorization.k8s.io/v1 | false      | ClusterRol  |
| clusterroles         |            | rbac.authorization.k8s.io/v1 | false      | ClusterRol  |
| rolebindings         |            | rbac.authorization.k8s.io/v1 | true       | RoleBindin  |
| roles                |            | rbac.authorization.k8s.io/v1 | true       | Role        |
| priorityclasses      | pc         | scheduling.k8s.io/v1         | false      | PriorityCla |
| csidrivers           |            | storage.k8s.io/v1            | false      | CSIDriver   |
| csinodes             |            | storage.k8s.io/v1            | false      | CSINode     |
| csistoragecapacities |            | storage.k8s.io/v1            | true       | CSISStorage |
| storageclasses       | sc         | storage.k8s.io/v1            | false      | StorageCla  |
| volumeattachments    |            | storage.k8s.io/v1            | false      | VolumeAtt   |

## Output options

Use the following sections for information about how you can format or sort the output of certain commands. For details about which commands support the various output options, see the [kubectl](#) reference documentation.

### Formatting output

The default output format for all `kubectl` commands is the human readable plain-text format. To output details to your terminal window in a specific format, you can add either the `-o` or `--output` flags to a supported `kubectl` command.

### Syntax

```
kubectl [command] [TYPE] [NAME] -o <output_format>
```

Depending on the `kubectl` operation, the following output formats are supported:

| Output format                                        | Description                                                                                                    |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>-o custom-columns=&lt;spec&gt;</code>          | Print a table using a comma separated list of <a href="#">custom columns</a> .                                 |
| <code>-o custom-columns-file=&lt;filename&gt;</code> | Print a table using the <a href="#">custom columns</a> template in the <code>&lt;filename&gt;</code> file.     |
| <code>-o json</code>                                 | Output a JSON formatted API object.                                                                            |
| <code>-o jsonpath=&lt;template&gt;</code>            | Print the fields defined in a <a href="#">jsonpath</a> expression.                                             |
| <code>-o jsonpath-file=&lt;filename&gt;</code>       | Print the fields defined by the <a href="#">jsonpath</a> expression in the <code>&lt;filename&gt;</code> file. |
| <code>-o name</code>                                 | Print only the resource name and nothing else.                                                                 |
| <code>-o wide</code>                                 | Output in the plain-text format with any additional information. For pods, the node name is included.          |
| <code>-o yaml</code>                                 | Output a YAML formatted API object.                                                                            |

## Example

In this example, the following command outputs the details for a single pod as a YAML formatted object:

```
kubectl get pod web-pod-13je7 -o yaml
```

Remember: See the [kubectl](#) reference documentation for details about which output format is supported by each command.

## Custom columns

To define custom columns and output only the details that you want into a table, you can use the custom-columns option. You can choose to define the custom columns inline or use a template file: `-o custom-columns=<spec>` or `-o custom-columns-file=<filename>`.

### Examples

Inline:

```
kubectl get pods <pod-name> -o custom-columns=NAME:.metadata.name,RSRC:.metadata.resourceVersion
```

Template file:

```
kubectl get pods <pod-name> -o custom-columns-file=template.txt
```

where the template.txt file contains:

```
NAME RSRC
metadata.name metadata.resourceVersion
```

The result of running either command is similar to:

```
NAME RSRC
submit-queue 610995
```

## Server-side columns

kubectl supports receiving specific column information from the server about objects. This means that for any given resource, the server will return columns and rows relevant to that resource, for the client to print. This allows for consistent human-readable output across clients used against the same cluster, by having the server encapsulate the details of printing.

This feature is enabled by default. To disable it, add the `--server-print=false` flag to the kubectl get command.

### Examples

To print information about the status of a pod, use a command like the following:

```
kubectl get pods <pod-name> --server-print=false
```

The output is similar to:

```
NAME AGE
pod-name 1m
```

## Sorting list objects

To output objects to a sorted list in your terminal window, you can add the `--sort-by` flag to a supported `kubectl` command. Sort your objects by specifying any numeric or string field with the `--sort-by` flag. To specify a field, use a [jsonpath](#) expression.

### Syntax

```
kubectl [command] [TYPE] [NAME] --sort-by=<jsonpath_exp>
```

### Example

To print a list of pods sorted by name, you run:

```
kubectl get pods --sort-by=.metadata.name
```

## Examples: Common operations

Use the following set of examples to help you familiarize yourself with running the commonly used `kubectl` operations:

`kubectl apply` - Apply or Update a resource from a file or stdin.

```
Create a service using the definition in example-service.yaml.
kubectl apply -f example-service.yaml
```

```
Create a replication controller using the definition in example-controller.yaml.
kubectl apply -f example-controller.yaml
```

```
Create the objects that are defined in any .yaml, .yml, or .json file within the <directory>
directory.
kubectl apply -f <directory>
```

`kubectl get` - List one or more resources.

```
List all pods in plain-text output format.
kubectl get pods
```

```
List all pods in plain-text output format and include additional information (such as node
name).
kubectl get pods -o wide
```

```
List the replication controller with the specified name in plain-text output format. Tip: You
can shorten and replace the 'replicationcontroller' resource type with the alias 'rc'.
kubectl get replicationcontroller <rc-name>
```

```
List all replication controllers and services together in plain-text output format.
```

```
kubectl get rc,services
```

# List all daemon sets in plain-text output format.

```
kubectl get ds
```

# List all pods running on node server01

```
kubectl get pods --field-selector=spec.nodeName=server01
```

**kubectl describe** - Display detailed state of one or more resources, including the uninitialized ones by default.

# Display the details of the node with name <node-name>.

```
kubectl describe nodes <node-name>
```

# Display the details of the pod with name <pod-name>.

```
kubectl describe pods/<pod-name>
```

# Display the details of all the pods that are managed by the replication controller named <rc-name>.

# Remember: Any pods that are created by the replication controller get prefixed with the name of the replication controller.

```
kubectl describe pods <rc-name>
```

# Describe all pods

```
kubectl describe pods
```

### **Note:**

The **kubectl get** command is usually used for retrieving one or more resources of the same resource type. It features a rich set of flags that allows you to customize the output format using the **-o** or **--output** flag, for example. You can specify the **-w** or **--watch** flag to start watching updates to a particular object. The **kubectl describe** command is more focused on describing the many related aspects of a specified resource. It may invoke several API calls to the API server to build a view for the user. For example, the **kubectl describe node** command retrieves not only the information about the node, but also a summary of the pods running on it, the events generated for the node etc.

**kubectl delete** - Delete resources either from a file, stdin, or specifying label selectors, names, resource selectors, or resources.

# Delete a pod using the type and name specified in the pod.yaml file.

```
kubectl delete -f pod.yaml
```

# Delete all the pods and services that have the label '<label-key>=<label-value>'.

```
kubectl delete pods,services -l <label-key>=<label-value>
```

# Delete all pods, including uninitialized ones.

```
kubectl delete pods --all
```

**kubectl exec** - Execute a command against a container in a pod.

# Get output from running 'date' from pod <pod-name>. By default, output is from the first container.

```
kubectl exec <pod-name> -- date
```

```
Get output from running 'date' in container <container-name> of pod <pod-name>.
kubectl exec <pod-name> -c <container-name> -- date
```

```
Get an interactive TTY and run /bin/bash from pod <pod-name>. By default, output is from
the first container.
kubectl exec -ti <pod-name> -- /bin/bash
```

`kubectl logs` - Print the logs for a container in a pod.

```
Return a snapshot of the logs from pod <pod-name>.
kubectl logs <pod-name>
```

```
Start streaming the logs from pod <pod-name>. This is similar to the 'tail -f' Linux command.
kubectl logs -f <pod-name>
```

`kubectl diff` - View a diff of the proposed updates to a cluster.

```
Diff resources included in "pod.json".
kubectl diff -f pod.json
```

```
Diff file read from stdin.
cat service.yaml | kubectl diff -f -
```

## Examples: Creating and using plugins

Use the following set of examples to help you familiarize yourself with writing and using `kubectl` plugins:

```
create a simple plugin in any language and name the resulting executable file
so that it begins with the prefix "kubectl-"
cat ./kubectl-hello
```

```
#!/bin/sh
```

```
this plugin prints the words "hello world"
echo "hello world"
```

With a plugin written, let's make it executable:

```
chmod a+x ./kubectl-hello
```

```
and move it to a location in our PATH
sudo mv ./kubectl-hello /usr/local/bin
sudo chown root:root /usr/local/bin
```

```
You have now created and "installed" a kubectl plugin.
You can begin using this plugin by invoking it from kubectl as if it were a regular command
kubectl hello
```

```
hello world
```

```
You can "uninstall" a plugin, by removing it from the folder in your
$PATH where you placed it
sudo rm /usr/local/bin/kubectl-hello
```

In order to view all of the plugins that are available to kubectl, use the kubectl plugin list subcommand:

```
kubectl plugin list
```

The output is similar to:

The following kubectl-compatible plugins are available:

```
/usr/local/bin/kubectl-hello
/usr/local/bin/kubectl-foo
/usr/local/bin/kubectl-bar
```

kubectl plugin list also warns you about plugins that are not executable, or that are shadowed by other plugins; for example:

```
sudo chmod -x /usr/local/bin/kubectl-foo # remove execute permission
kubectl plugin list
```

The following kubectl-compatible plugins are available:

```
/usr/local/bin/kubectl-hello
/usr/local/bin/kubectl-foo
- warning: /usr/local/bin/kubectl-foo identified as a plugin, but it is not executable
/usr/local/bin/kubectl-bar
```

```
error: one plugin warning was found
```

You can think of plugins as a means to build more complex functionality on top of the existing kubectl commands:

```
cat ./kubectl-whoami
```

The next few examples assume that you already made kubectl-whoami have the following contents:

```
#!/bin/bash

this plugin makes use of the `kubectl config` command in order to output
information about the current user, based on the currently selected context
kubectl config view --template='{{ range .contexts }}{{ if eq .name "$(kubectl config current-
context)" }}Current user: {{ printf "%s\n" .context.user }}{{ end }}{{ end }}'
```

Running the above command gives you an output containing the user for the current context in your KUBECONFIG file:

```
make the file executable
sudo chmod +x ./kubectl-whoami
```

```
and move it into your PATH
```

```
sudo mv ./kubectl-whoami /usr/local/bin
```

```
kubectl whoami
```

```
Current user: plugins-user
```

## What's next

- Read the [kubectl reference documentation](#):
  - the [kubectl command reference](#)
  - the [command line arguments](#) reference
- Learn about [kubectl usage conventions](#)
- Read about [JSONPath support](#) in kubectl
- Read about how to [extend kubectl with plugins](#)
  - To find out more about plugins, take a look at the [example CLI plugin](#).

# Introduction to kubectl

kubectl is the Kubernetes cli version of a swiss army knife, and can do many things.

While this Book is focused on using kubectl to declaratively manage applications in Kubernetes, it also covers other kubectl functions.

## Command Families

Most kubectl commands typically fall into one of a few categories:

| Type                            | Used For                                | Description                                                                        |
|---------------------------------|-----------------------------------------|------------------------------------------------------------------------------------|
| Declarative Resource Management | Deployment and operations (e.g. GitOps) | Declaratively manage Kubernetes workloads using resource configuration             |
| Imperative Resource Management  | Development Only                        | Run commands to manage Kubernetes workloads using Command Line arguments and flags |
| Printing Workload State         | Debugging                               | Print information about workloads                                                  |
| Interacting with Containers     | Debugging                               | Exec, attach, cp, logs                                                             |
| Cluster Management              | Cluster operations                      | Drain and cordon Nodes                                                             |

## Declarative Application Management

The preferred approach for managing resources is through declarative files called resource configuration used with the kubectl *Apply* command. This command reads a local (or remote) file structure and modifies cluster state to reflect the declared intent.

### Apply

Apply is the preferred mechanism for managing resources in a Kubernetes cluster.



## Printing State about Workloads

Users will need to view workload state.

- Printing summarize state and information about resources
- Printing complete state and information about resources
- Printing specific fields from resources
- Query resources matching labels

## Debugging Workloads

kubectl supports debugging by providing commands for:

- Printing Container logs
- Printing cluster events
- Exec or attaching to a Container
- Copying files from Containers in the cluster to a user's filesystem

## Cluster Management

On occasion, users may need to perform operations to the Nodes of cluster. kubectl supports commands to drain workloads from a Node so that it can be decommissioned or debugged.

## Porcelain

Users may find using resource configuration overly verbose for *development* and prefer to work with the cluster *imperatively* with a shell-like workflow. kubectl offers porcelain commands for generating and modifying resources.

- Generating + creating resources such as Deployments, StatefulSets, Services, ConfigMaps, etc.
- Setting fields on resources
- Editing (live) resources in a text editor

### Porcelain for Dev Only

Porcelain commands are time saving for experimenting with workloads in a dev cluster, but shouldn't be used for production.

## kubectl Quick Reference

This page contains a list of commonly used kubectl commands and flags.

### Note:

These instructions are for Kubernetes v1.30. To check the version, use the `kubectl version` command.

# Kubectl autocomplete

## BASH

```
source <(kubectl completion bash) # set up autocomplete in bash into the current shell, bash-
completion package should be installed first.
echo "source <(kubectl completion bash)" >> ~/.bashrc
add autocomplete permanently to your bash shell.
```

You can also use a shorthand alias for kubectl that also works with completion:

```
alias k=kubectl
complete -o default -F __start_kubectl k
```

## ZSH

```
source <(kubectl completion zsh) # set up autocomplete in zsh into the current shell
echo '[[$commands[kubectl]]] && source <(kubectl completion zsh)' >> ~/.zshrc # add
autocomplete permanently to your zsh shell
```

## FISH

### Note:

Requires kubectl version 1.23 or above.

```
echo 'kubectl completion fish | source' > ~/.config/fish/completions/kubectl.fish && source
~/.config/fish/completions/kubectl.fish
```

### A note on --all-namespaces

Appending --all-namespaces happens frequently enough that you should be aware of the shorthand for --all-namespaces:

```
kubectl -A
```

## Kubectl context and configuration

Set which Kubernetes cluster kubectl communicates with and modifies configuration information. See [Authenticating Across Clusters with kubeconfig](#) documentation for detailed config file information.

```
kubectl config view # Show Merged kubeconfig settings.
```

```
use multiple kubeconfig files at the same time and view merged config
KUBECONFIG=~/.kube/config:~/.kube/kubconfig2
```

```
kubectl config view
```

```
Show merged kubeconfig settings and raw certificate data and exposed secrets
kubectl config view --raw
```

```
get the password for the e2e user
kubectl config view -o jsonpath='{.users[?(@.name == "e2e")].user.password}'

get the certificate for the e2e user
kubectl config view --raw -o jsonpath='{.users[?(@.name == "e2e")].user.client-certificate-data}' |
base64 -d

kubectl config view -o jsonpath='{.users[].name}' # display the first user
kubectl config view -o jsonpath='{.users[*].name}' # get a list of users
kubectl config get-contexts # display list of contexts
kubectl config get-contexts -o name # get all context names
kubectl config current-context # display the current-context
kubectl config use-context my-cluster-name # set the default context to my-cluster-name

kubectl config set-cluster my-cluster-name # set a cluster entry in the kubeconfig

configure the URL to a proxy server to use for requests made by this client in the kubeconfig
kubectl config set-cluster my-cluster-name --proxy-url=my-proxy-url

add a new user to your kubeconfig that supports basic auth
kubectl config set-credentials kubeuser/foo.kubernetes.com --username=kubeuser --
password=kubepassword

permanently save the namespace for all subsequent kubectl commands in that context.
kubectl config set-context --current --namespace=ggckad-s2

set a context utilizing a specific username and namespace.
kubectl config set-context gce --user=cluster-admin --namespace=foo \
&& kubectl config use-context gce

kubectl config unset users.foo # delete user foo

short alias to set/show context/namespace (only works for bash and bash-compatible shells,
current context to be set before using kn to set namespace)
alias kx='f() { ["$1"] && kubectl config use-context $1 || kubectl config current-context ; } ; f'
alias kn='f() { ["$1"] && kubectl config set-context --current --namespace $1 || kubectl config
view --minify | grep namespace | cut -d" " -f6 ; } ; f'
```

## Kubectl apply

apply manages applications through files defining Kubernetes resources. It creates and updates resources in a cluster through running kubectl apply. This is the recommended way of managing Kubernetes applications on production. See [Kubectl Book](#).

## Creating objects

Kubernetes manifests can be defined in YAML or JSON. The file extension .yaml, .yml, and .json can be used.

```
kubectl apply -f ./my-manifest.yaml # create resource(s)
kubectl apply -f ./my1.yaml -f ./my2.yaml # create from multiple files
```

```

kubectl apply -f ./dir # create resource(s) in all manifest files in dir
kubectl apply -f https://example.com/manifest.yaml # create resource(s) from url (Note: this is
an example domain and does not contain a valid manifest)
kubectl create deployment nginx --image=nginx # start a single instance of nginx

create a Job which prints "Hello World"
kubectl create job hello --image=busybox:1.28 -- echo "Hello World"

create a CronJob that prints "Hello World" every minute
kubectl create cronjob hello --image=busybox:1.28 --schedule="*/1 * * * *" -- echo "Hello
World"

kubectl explain pods # get the documentation for pod manifests

Create multiple YAML objects from stdin
kubectl apply -f - <<EOF
apiVersion: v1
kind: Pod
metadata:
 name: busybox-sleep
spec:
 containers:
 - name: busybox
 image: busybox:1.28
 args:
 - sleep
 - "1000000"

apiVersion: v1
kind: Pod
metadata:
 name: busybox-sleep-less
spec:
 containers:
 - name: busybox
 image: busybox:1.28
 args:
 - sleep
 - "1000"
EOF

Create a secret with several keys
kubectl apply -f - <<EOF
apiVersion: v1
kind: Secret
metadata:
 name: mysecret
type: Opaque
data:
 password: $(echo -n "s33msi4" | base64 -w0)
 username: $(echo -n "jane" | base64 -w0)
EOF

```

## Viewing and finding resources

```
Get commands with basic output
kubectl get services # List all services in the namespace
kubectl get pods --all-namespaces # List all pods in all namespaces
kubectl get pods -o wide # List all pods in the current namespace, with more
 # details
kubectl get deployment my-dep # List a particular deployment
kubectl get pods # List all pods in the namespace
kubectl get pod my-pod -o yaml # Get a pod's YAML

Describe commands with verbose output
kubectl describe nodes my-node
kubectl describe pods my-pod

List Services Sorted by Name
kubectl get services --sort-by=.metadata.name

List pods Sorted by Restart Count
kubectl get pods --sort-by='.status.containerStatuses[0].restartCount'

List PersistentVolumes sorted by capacity
kubectl get pv --sort-by=.spec.capacity.storage

Get the version label of all pods with label app=cassandra
kubectl get pods --selector=app=cassandra -o \
 jsonpath='{.items[*].metadata.labels.version}'

Retrieve the value of a key with dots, e.g. 'ca.crt'
kubectl get configmap myconfig \
 -o jsonpath='{.data.ca\.crt}'

Retrieve a base64 encoded value with dashes instead of underscores.
kubectl get secret my-secret --template='{{index .data "key-name-with-dashes"}}'

Get all worker nodes (use a selector to exclude results that have a label
named 'node-role.kubernetes.io/control-plane')
kubectl get node --selector='!node-role.kubernetes.io/control-plane'

Get all running pods in the namespace
kubectl get pods --field-selector=status.phase=Running

Get ExternalIPs of all nodes
kubectl get nodes -o jsonpath='{.items[*].status.addresses[?(@.type=="ExternalIP")].address}'

List Names of Pods that belong to Particular RC
"jq" command useful for transformations that are too complex for jsonpath, it can be found at
https://jqlang.github.io/jq/
sel=${$(kubectl get rc my-rc --output=json | jq -j '.spec.selector | to_entries | .[] | "\(.key)=\'
(.value),\'")%?'}
echo $(kubectl get pods --selector=$sel --output=jsonpath='{.items..metadata.name}')
```

```

Show labels for all pods (or any other Kubernetes object that supports labelling)
kubectl get pods --show-labels

Check which nodes are ready
JSONPATH='{range .items[*]}{@.metadata.name}:{range @.status.conditions[*]}{@.type}
={@.status};{end}{end}' \
&& kubectl get nodes -o jsonpath="$JSONPATH" | grep "Ready=True"

Check which nodes are ready with custom-columns
kubectl get node -o custom-columns='NODE_NAME:.metadata.name,STATUS:.status.conditions
[?(@.type=="Ready")].status'

Output decoded secrets without external tools
kubectl get secret my-secret -o go-template='{{range $k,$v := .data}}{{"### "}}{{$k}}{{"\n"}}{{ $v|
base64decode}}{{"\n\n"}}{{end}}'

List all Secrets currently in use by a pod
kubectl get pods -o json | jq '.items[].spec.containers[].env[]?.valueFrom.secretKeyRef.name' |
grep -v null | sort | uniq

List all containerIDs of initContainer of all pods
Helpful when cleaning up stopped containers, while avoiding removal of initContainers.
kubectl get pods --all-namespaces -o jsonpath='{range .items[*].status.initContainerStatuses[*]}
{.containerID}{"\n"}{end}' | cut -d/ -f3

List Events sorted by timestamp
kubectl get events --sort-by=.metadata.creationTimestamp

List all warning events
kubectl events --types=Warning

Compares the current state of the cluster against the state that the cluster would be in if the
manifest was applied.
kubectl diff -f ./my-manifest.yaml

Produce a period-delimited tree of all keys returned for nodes
Helpful when locating a key within a complex nested JSON structure
kubectl get nodes -o json | jq -c 'paths|join(".")'

Produce a period-delimited tree of all keys returned for pods, etc
kubectl get pods -o json | jq -c 'paths|join(".")'

Produce ENV for all pods, assuming you have a default container for the pods, default
namespace and the `env` command is supported.
Helpful when running any supported command across all pods, not just `env`
for pod in $(kubectl get po --output=jsonpath={.items..metadata.name}); do echo $pod &&
kubectl exec -it $pod -- env; done

Get a deployment's status subresource
kubectl get deployment nginx-deployment --subresource=status

```

## Updating resources

```
kubectl set image deployment/frontend www=image:v2 # Rolling update "www"
containers of "frontend" deployment, updating the image
kubectl rollout history deployment/frontend # Check the history of deployments
including the revision
kubectl rollout undo deployment/frontend # Rollback to the previous deployment
kubectl rollout undo deployment/frontend --to-revision=2 # Rollback to a specific revision
kubectl rollout status -w deployment/frontend # Watch rolling update status of
"frontend" deployment until completion
kubectl rollout restart deployment/frontend # Rolling restart of the "frontend"
deployment

cat pod.json | kubectl replace -f - # Replace a pod based on the JSON passed
into stdin

Force replace, delete and then re-create the resource. Will cause a service outage.
kubectl replace --force -f ./pod.json

Create a service for a replicated nginx, which serves on port 80 and connects to the containers
on port 8000
kubectl expose rc nginx --port=80 --target-port=8000

Update a single-container pod's image version (tag) to v4
kubectl get pod mypod -o yaml | sed 's/\(image: myimage\):.*$/1:v4/' | kubectl replace -f -

kubectl label pods my-pod new-label=awesome # Add a Label
kubectl label pods my-pod new-label- # Remove a label
kubectl label pods my-pod new-label=new-value --overwrite # Overwrite an existing value
kubectl annotate pods my-pod icon-url=http://goo.gl/XXBTWq # Add an annotation
kubectl annotate pods my-pod icon-url- # Remove annotation
kubectl autoscale deployment foo --min=2 --max=10 # Auto scale a deployment "foo"
```

## Patching resources

```
Partially update a node
kubectl patch node k8s-node-1 -p '{"spec":{"unschedulable":true}}'

Update a container's image; spec.containers[*].name is required because it's a merge key
kubectl patch pod valid-pod -p '{"spec":{"containers":[{"name":"kubernetes-serve-
hostname","image":"new image"}]}}'

Update a container's image using a json patch with positional arrays
kubectl patch pod valid-pod --type=json -p='[{"op": "replace", "path": "/spec/containers/0/
image", "value":"new image"}]'

Disable a deployment livenessProbe using a json patch with positional arrays
kubectl patch deployment valid-deployment --type json -p='[{"op": "remove", "path": "/spec/
template/spec/containers/0/livenessProbe"}]'

Add a new element to a positional array
```

```
kubectl patch sa default --type='json' -p='[{"op": "add", "path": "/secrets/1", "value": {"name": "whatever" } }]'
```

# Update a deployment's replica count by patching its scale subresource

```
kubectl patch deployment nginx-deployment --subresource='scale' --type='merge' -p '{"spec": {"replicas":2}}'
```

## Editing resources

Edit any API resource in your preferred editor.

```
kubectl edit svc/docker-registry # Edit the service named docker-registry
```

```
KUBE_EDITOR="nano" kubectl edit svc/docker-registry # Use an alternative editor
```

## Scaling resources

```
kubectl scale --replicas=3 rs/foo # Scale a replicaset named 'foo' to 3
```

```
kubectl scale --replicas=3 -f foo.yaml # Scale a resource specified in "foo.yaml" to 3
```

```
kubectl scale --current-replicas=2 --replicas=3 deployment/mysql # If the deployment named mysql's current size is 2, scale mysql to 3
```

```
kubectl scale --replicas=5 rc/foo rc/bar rc/baz # Scale multiple replication controllers
```

## Deleting resources

```
kubectl delete -f ./pod.json # Delete a pod using the type and name specified in pod.json
```

```
kubectl delete pod unwanted --now # Delete a pod with no grace period
```

```
kubectl delete pod,service baz foo # Delete pods and services with same names "baz" and "foo"
```

```
kubectl delete pods,services -l name=myLabel # Delete pods and services with label name=myLabel
```

```
kubectl -n my-ns delete pod,svc --all # Delete all pods and services in namespace my-ns,
```

# Delete all pods matching the awk pattern1 or pattern2

```
kubectl get pods -n mynamespace --no-headers=true | awk '/pattern1|pattern2/{print $1}' | xargs kubectl delete -n mynamespace pod
```

## Interacting with running Pods

```
kubectl logs my-pod # dump pod logs (stdout)
```

```
kubectl logs -l name=myLabel # dump pod logs, with label name=myLabel (stdout)
```

```
kubectl logs my-pod --previous # dump pod logs (stdout) for a previous instantiation of a container
```

```
kubectl logs my-pod -c my-container # dump pod container logs (stdout, multi-container case)
```

```
kubectl logs -l name=myLabel -c my-container # dump pod container logs, with label name=myLabel (stdout)
```

```
kubectl logs my-pod -c my-container --previous # dump pod container logs (stdout, multi-
```



```

container case) for a previous instantiation of a container
kubectl logs -f my-pod # stream pod logs (stdout)
kubectl logs -f my-pod -c my-container # stream pod container logs (stdout, multi-
container case)
kubectl logs -f -l name=myLabel --all-containers # stream all pods logs with label
name=myLabel (stdout)
kubectl run -i --tty busybox --image=busybox:1.28 -- sh # Run pod as interactive shell
kubectl run nginx --image=nginx -n mynamespace # Start a single instance of nginx pod in
the namespace of mynamespace
kubectl run nginx --image=nginx --dry-run=client -o yaml > pod.yaml
Generate spec for running pod nginx and write it into a file
called pod.yaml
kubectl attach my-pod -i # Attach to Running Container
kubectl port-forward my-pod 5000:6000
Listen on port 5000 on the local machine and forward to port 6000 on my-pod
kubectl exec my-pod -- ls / # Run command in existing pod (1 container case)
kubectl exec --stdin --tty my-pod -- /bin/sh # Interactive shell access to a running pod (1
container case)
kubectl exec my-pod -c my-container -- ls / # Run command in existing pod (multi-
container case)
kubectl debug my-pod -it --image=busybox:1.28 # Create an interactive debugging session
within existing pod and immediately attach to it
kubectl debug node/my-node -it --image=busybox:1.28 # Create an interactive debugging
session on a node and immediately attach to it
kubectl top pod # Show metrics for all pods in the default namespace
kubectl top pod POD_NAME --containers # Show metrics for a given pod and its
containers
kubectl top pod POD_NAME --sort-by=cpu # Show metrics for a given pod and sort it
by 'cpu' or 'memory'

```

## Copying files and directories to and from containers

```

kubectl cp /tmp/foo_dir my-pod:/tmp/bar_dir # Copy /tmp/foo_dir local directory to /
tmp/bar_dir in a remote pod in the current namespace
kubectl cp /tmp/foo my-pod:/tmp/bar -c my-container # Copy /tmp/foo local file to /tmp/bar
in a remote pod in a specific container
kubectl cp /tmp/foo my-namespace/my-pod:/tmp/bar # Copy /tmp/foo local file to /tmp/bar
in a remote pod in namespace my-namespace
kubectl cp my-namespace/my-pod:/tmp/foo /tmp/bar # Copy /tmp/foo from a remote pod
to /tmp/bar locally

```

### Note:

kubectl cp requires that the 'tar' binary is present in your container image. If 'tar' is not present, kubectl cp will fail. For advanced use cases, such as symlinks, wildcard expansion or file mode preservation consider using kubectl exec.

```

tar cf - /tmp/foo | kubectl exec -i -n my-namespace my-pod -- tar xf - -C /tmp/bar #
Copy /tmp/foo local file to /tmp/bar in a remote pod in namespace my-namespace
kubectl exec -n my-namespace my-pod -- tar cf - /tmp/foo | tar xf - -C /tmp/bar # Copy /tmp/
foo from a remote pod to /tmp/bar locally

```

## Interacting with Deployments and Services

```
kubectl logs deploy/my-deployment # dump Pod logs for a Deployment (single-
container case)
kubectl logs deploy/my-deployment -c my-container # dump Pod logs for a Deployment
(multi-container case)

kubectl port-forward svc/my-service 5000 # listen on local port 5000 and forward to
port 5000 on Service backend
kubectl port-forward svc/my-service 5000:my-service-port # listen on local port 5000 and
forward to Service target port with name <my-service-port>

kubectl port-forward deploy/my-deployment 5000:6000 # listen on local port 5000 and
forward to port 6000 on a Pod created by <my-deployment>
kubectl exec deploy/my-deployment -- ls # run command in first Pod and first
container in Deployment (single- or multi-container cases)
```

## Interacting with Nodes and cluster

```
kubectl cordon my-node # Mark my-node as unschedulable
kubectl drain my-node # Drain my-node in preparation for
maintenance
kubectl uncordon my-node # Mark my-node as schedulable
kubectl top node # Show metrics for all nodes
kubectl top node my-node # Show metrics for a given node
kubectl cluster-info # Display addresses of the master and
services
kubectl cluster-info dump # Dump current cluster state to stdout
kubectl cluster-info dump --output-directory=/path/to/cluster-state # Dump current cluster
state to /path/to/cluster-state

View existing taints on which exist on current nodes.
kubectl get nodes -o='custom-
columns=NodeName:.metadata.name,TaintKey:.spec.taints[*].key,TaintValue:.spec.taints[*].valu
e,TaintEffect:.spec.taints[*].effect'

If a taint with that key and effect already exists, its value is replaced as specified.
kubectl taint nodes foo dedicated=special-user:NoSchedule
```

## Resource types

List all supported resource types along with their shortnames, [API group](#), whether they are [namespaced](#), and [kind](#):

```
kubectl api-resources
```

Other operations for exploring API resources:

```
kubectl api-resources --namespaced=true # All namespaced resources
kubectl api-resources --namespaced=false # All non-namespaced resources
kubectl api-resources -o name # All resources with simple output (only the resource
name)
```

```
kubectl api-resources -o wide # All resources with expanded (aka "wide") output
kubectl api-resources --verbs=list,get
All resources that support the "list" and "get" request verbs
kubectl api-resources --api-group=extensions # All resources in the "extensions" API group
```

## Formatting output

To output details to your terminal window in a specific format, add the `-o` (or `--output`) flag to a supported `kubectl` command.

| Output format                                        | Description                                                                                                   |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>-o=custom-columns=&lt;spec&gt;</code>          | Print a table using a comma separated list of custom columns                                                  |
| <code>-o=custom-columns-file=&lt;filename&gt;</code> | Print a table using the custom columns template in the <code>&lt;filename&gt;</code> file                     |
| <code>-o=go-template=&lt;template&gt;</code>         | Print the fields defined in a <a href="#">golang template</a>                                                 |
| <code>-o=go-template-file=&lt;filename&gt;</code>    | Print the fields defined by the <a href="#">golang template</a> in the <code>&lt;filename&gt;</code> file     |
| <code>-o=json</code>                                 | Output a JSON formatted API object                                                                            |
| <code>-o=jsonpath=&lt;template&gt;</code>            | Print the fields defined in a <a href="#">jsonpath</a> expression                                             |
| <code>-o=jsonpath-file=&lt;filename&gt;</code>       | Print the fields defined by the <a href="#">jsonpath</a> expression in the <code>&lt;filename&gt;</code> file |
| <code>-o=name</code>                                 | Print only the resource name and nothing else                                                                 |
| <code>-o=wide</code>                                 | Output in the plain-text format with any additional information, and for pods, the node name is included      |
| <code>-o=yaml</code>                                 | Output a YAML formatted API object                                                                            |

Examples using `-o=custom-columns`:

```
All images running in a cluster
kubectl get pods -A -o=custom-columns='DATA:spec.containers[*].image'

All images running in namespace: default, grouped by Pod
kubectl get pods --namespace default --output=custom-columns="NAME:.metadata.name,IMAGE:.spec.containers[*].image"

All images excluding "registry.k8s.io/coredns:1.6.2"
kubectl get pods -A -o=custom-columns='DATA:spec.containers[?(@.image!="registry.k8s.io/coredns:1.6.2")].image'

All fields under metadata regardless of name
kubectl get pods -A -o=custom-columns='DATA:metadata.*'
```

More examples in the [kubectl reference documentation](#).

## Kubectl output verbosity and debugging

Kubectl verbosity is controlled with the `-v` or `--v` flags followed by an integer representing the log level. General Kubernetes logging conventions and the associated log levels are described [here](#).

| Verbosity | Description                                                                                                                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --v=0     | Generally useful for this to <i>always</i> be visible to a cluster operator.                                                                                                                      |
| --v=1     | A reasonable default log level if you don't want verbosity.                                                                                                                                       |
| --v=2     | Useful steady state information about the service and important log messages that may correlate to significant changes in the system. This is the recommended default log level for most systems. |
| --v=3     | Extended information about changes.                                                                                                                                                               |
| --v=4     | Debug level verbosity.                                                                                                                                                                            |
| --v=5     | Trace level verbosity.                                                                                                                                                                            |
| --v=6     | Display requested resources.                                                                                                                                                                      |
| --v=7     | Display HTTP request headers.                                                                                                                                                                     |
| --v=8     | Display HTTP request contents.                                                                                                                                                                    |
| --v=9     | Display HTTP request contents without truncation of contents.                                                                                                                                     |

## What's next

- Read the [kubectl overview](#) and learn about [JsonPath](#).
- See [kubectl](#) options.
- Also read [kubectl Usage Conventions](#) to understand how to use kubectl in reusable scripts.
- See more community [kubectl cheatsheets](#).

# kubectl reference

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[kubectl](#)

[kubectl annotate](#)

[kubectl api-resources](#)

[kubectl api-versions](#)

[kubectl apply](#)

[kubectl attach](#)

[kubectl auth](#)

[kubectl autoscale](#)

[kubectl certificate](#)

[kubectl cluster-info](#)

[kubectl completion](#)

[kubectl config](#)

[kubectl cordon](#)

[kubectl cp](#)

[kubectl create](#)

[kubectl debug](#)

[kubectl delete](#)

[kubectl describe](#)

[kubectl diff](#)

[kubectl drain](#)

[kubectl edit](#)

[kubectl events](#)

[kubectl exec](#)

[kubectl explain](#)

[kubectl expose](#)

[kubectl get](#)

[kubectl kustomize](#)

[kubectl label](#)

[kubectl logs](#)

[kubectl options](#)

[kubectl patch](#)

[kubectl plugin](#)

[kubectl port-forward](#)

[kubectl proxy](#)

[kubectl replace](#)

[kubectl rollout](#)

[kubectl run](#)

[kubectl scale](#)

[kubectl set](#)

[kubectl taint](#)

[kubectl top](#)

[kubectl uncordon](#)

[kubectl version](#)

[kubectl wait](#)

# kubectl

## Synopsis

kubectl controls the Kubernetes cluster manager.

Find more information at: <https://kubernetes.io/docs/reference/kubectl/>

kubectl [flags]

## Options

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| -h, --help                                                                                                                                                      |
| help for kubectl                                                                                                                                                |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Require server version to match client version                                                                                                                                                    |
| <code>-n, --namespace string</code>                                                                                                                                                               |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| <code>--password string</code>                                                                                                                                                                    |
| Password for basic authentication to the API server                                                                                                                                               |
| <code>--profile string</code> Default: "none"                                                                                                                                                     |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| <code>--profile-output string</code> Default: "profile.pprof"                                                                                                                                     |
| Name of the file to write the profile to                                                                                                                                                          |
| <code>--request-timeout string</code> Default: "0"                                                                                                                                                |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl annotate](#) - Update the annotations on a resource
- [kubectl api-resources](#) - Print the supported API resources on the server
- [kubectl api-versions](#) - Print the supported API versions on the server, in the form of "group/version"
- [kubectl apply](#) - Apply a configuration to a resource by file name or stdin
- [kubectl attach](#) - Attach to a running container
- [kubectl auth](#) - Inspect authorization
- [kubectl autoscale](#) - Auto-scale a deployment, replica set, stateful set, or replication controller
- [kubectl certificate](#) - Modify certificate resources
- [kubectl cluster-info](#) - Display cluster information
- [kubectl completion](#) - Output shell completion code for the specified shell (bash, zsh, fish, or powershell)
- [kubectl config](#) - Modify kubeconfig files
- [kubectl cordon](#) - Mark node as unschedulable
- [kubectl cp](#) - Copy files and directories to and from containers

- [kubectl create](#) - Create a resource from a file or from stdin
- [kubectl debug](#) - Create debugging sessions for troubleshooting workloads and nodes
- [kubectl delete](#) - Delete resources by file names, stdin, resources and names, or by resources and label selector
- [kubectl describe](#) - Show details of a specific resource or group of resources
- [kubectl diff](#) - Diff the live version against a would-be applied version
- [kubectl drain](#) - Drain node in preparation for maintenance
- [kubectl edit](#) - Edit a resource on the server
- [kubectl events](#) - List events
- [kubectl exec](#) - Execute a command in a container
- [kubectl explain](#) - Get documentation for a resource
- [kubectl expose](#) - Take a replication controller, service, deployment or pod and expose it as a new Kubernetes service
- [kubectl get](#) - Display one or many resources
- [kubectl kustomize](#) - Build a kustomization target from a directory or URL
- [kubectl label](#) - Update the labels on a resource
- [kubectl logs](#) - Print the logs for a container in a pod
- [kubectl options](#) - Print the list of flags inherited by all commands
- [kubectl patch](#) - Update fields of a resource
- [kubectl plugin](#) - Provides utilities for interacting with plugins
- [kubectl port-forward](#) - Forward one or more local ports to a pod
- [kubectl proxy](#) - Run a proxy to the Kubernetes API server
- [kubectl replace](#) - Replace a resource by file name or stdin
- [kubectl rollout](#) - Manage the rollout of a resource
- [kubectl run](#) - Run a particular image on the cluster
- [kubectl scale](#) - Set a new size for a deployment, replica set, or replication controller
- [kubectl set](#) - Set specific features on objects
- [kubectl taint](#) - Update the taints on one or more nodes
- [kubectl top](#) - Display resource (CPU/memory) usage
- [kubectl unordon](#) - Mark node as schedulable
- [kubectl version](#) - Print the client and server version information
- [kubectl wait](#) - Experimental: Wait for a specific condition on one or many resources

# kubectl annotate

## Synopsis

Update the annotations on one or more resources.

All Kubernetes objects support the ability to store additional data with the object as annotations. Annotations are key/value pairs that can be larger than labels and include arbitrary string values such as structured JSON. Tools and system extensions may use annotations to store their own data.

Attempting to set an annotation that already exists will fail unless `--overwrite` is set. If `--resource-version` is specified and does not match the current resource version on the server the command will fail.

Use "kubectl api-resources" for a complete list of supported resources.

kubectl annotate [--overwrite] (-f FILENAME | TYPE NAME) KEY\_1=VAL\_1 ... KEY\_N=VAL\_N  
[--resource-version=version]

## Examples

```
Update pod 'foo' with the annotation 'description' and the value 'my frontend'
If the same annotation is set multiple times, only the last value will be applied
kubectl annotate pods foo description='my frontend'

Update a pod identified by type and name in "pod.json"
kubectl annotate -f pod.json description='my frontend'

Update pod 'foo' with the annotation 'description' and the value 'my frontend running nginx',
overwriting any existing value
kubectl annotate --overwrite pods foo description='my frontend running nginx'

Update all pods in the namespace
kubectl annotate pods --all description='my frontend running nginx'

Update pod 'foo' only if the resource is unchanged from version 1
kubectl annotate pods foo description='my frontend running nginx' --resource-version=1

Update pod 'foo' by removing an annotation named 'description' if it exists
Does not require the --overwrite flag
kubectl annotate pods foo description-
```

## Options

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --all                                                                                                                                                                                                    |
| Select all resources, in the namespace of the specified resource types.                                                                                                                                  |
| -A, --all-namespaces                                                                                                                                                                                     |
| If true, check the specified action in all namespaces.                                                                                                                                                   |
| --allow-missing-template-keys    Default: true                                                                                                                                                           |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                          |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                        |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| --field-manager string    Default: "kubectl-annotate"                                                                                                                                                    |
| Name of the manager used to track field ownership.                                                                                                                                                       |

|                           |                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --field-selector string   |                                                                                                                                                                                         |
|                           | Selector (field query) to filter on, supports '=', '==', and '!='.(e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type. |
| -f, --filename strings    |                                                                                                                                                                                         |
|                           | Filename, directory, or URL to files identifying the resource to update the annotation                                                                                                  |
| -h, --help                |                                                                                                                                                                                         |
|                           | help for annotate                                                                                                                                                                       |
| -k, --kustomize string    |                                                                                                                                                                                         |
|                           | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                    |
| --list                    |                                                                                                                                                                                         |
|                           | If true, display the annotations for a given resource.                                                                                                                                  |
| --local                   |                                                                                                                                                                                         |
|                           | If true, annotation will NOT contact api-server but run locally.                                                                                                                        |
| -o, --output string       |                                                                                                                                                                                         |
|                           | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                            |
| --overwrite               |                                                                                                                                                                                         |
|                           | If true, allow annotations to be overwritten, otherwise reject annotation updates that overwrite existing annotations.                                                                  |
| -R, --recursive           |                                                                                                                                                                                         |
|                           | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                         |
| --resource-version string |                                                                                                                                                                                         |
|                           | If non-empty, the annotation update will only succeed if this is the current resource-version for the object. Only valid when specifying a single resource.                             |
| -l, --selector string     |                                                                                                                                                                                         |
|                           | Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.              |
| --show-managed-fields     |                                                                                                                                                                                         |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --as string                                                                                                                                                                                                                                               |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                              |
| --as-group strings                                                                                                                                                                                                                                        |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                                                                                                             |
| --as-uid string                                                                                                                                                                                                                                           |
| UID to impersonate for the operation.                                                                                                                                                                                                                     |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                                                                                                                        |
| Default cache directory                                                                                                                                                                                                                                   |
| --certificate-authority string                                                                                                                                                                                                                            |
| Path to a cert file for the certificate authority                                                                                                                                                                                                         |
| --client-certificate string                                                                                                                                                                                                                               |
| Path to a client certificate file for TLS                                                                                                                                                                                                                 |
| --client-key string                                                                                                                                                                                                                                       |
| Path to a client key file for TLS                                                                                                                                                                                                                         |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                                                                                                                         |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                                                                                                                      |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                                                                                                                        |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                                                                                                                      |
| --cluster string                                                                                                                                                                                                                                          |
| The name of the kubeconfig cluster to use                                                                                                                                                                                                                 |
| --context string                                                                                                                                                                                                                                          |

|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | The name of the kubeconfig context to use                                                                                                                                                         |
| --default-not-ready-toleration-seconds int   | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>-s, --server string</code>                                                                                                  |
| The address and port of the Kubernetes API server                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                       |
| database name                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                               |
| database host:port                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                     |
| database password                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                              |
| use secure connection with database                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                       |
| table name                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                         |
| database username                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| <code>--token string</code>                                                                                                       |
| Bearer token for authentication to the API server                                                                                 |
| <code>--user string</code>                                                                                                        |
| The name of the kubeconfig user to use                                                                                            |
| <code>--username string</code>                                                                                                    |
| Username for basic authentication to the API server                                                                               |
| <code>--version version[=true]</code>                                                                                             |

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>--version, --version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
|----------------------------------------------------------------------------------------------------------------------------------------|

|                                   |
|-----------------------------------|
| <code>--warnings-as-errors</code> |
|-----------------------------------|

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl api-resources

## Synopsis

Print the supported API resources on the server.

```
kubectl api-resources [flags]
```

## Examples

```
Print the supported API resources
kubectl api-resources
```

```
Print the supported API resources with more information
kubectl api-resources -o wide
```

```
Print the supported API resources sorted by a column
kubectl api-resources --sort-by=name
```

```
Print the supported namespaced resources
kubectl api-resources --namespaced=true
```

```
Print the supported non-namespaced resources
kubectl api-resources --namespaced=false
```

```
Print the supported API resources with a specific APIGroup
kubectl api-resources --api-group=rbac.authorization.k8s.io
```

## Options

|                                 |
|---------------------------------|
| <code>--api-group string</code> |
|---------------------------------|

|                                                |
|------------------------------------------------|
| Limit to resources in the specified API group. |
|------------------------------------------------|

|                       |
|-----------------------|
| <code>--cached</code> |
|-----------------------|



|                                         |                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|
|                                         | Use the cached list of resources if available.                                                            |
| <code>--categories strings</code>       |                                                                                                           |
|                                         | Limit to resources that belong to the specified categories.                                               |
| <code>-h, --help</code>                 |                                                                                                           |
|                                         | help for api-resources                                                                                    |
| <code>--namespaced</code> Default: true |                                                                                                           |
|                                         | If false, non-namespaced resources will be returned, otherwise returning namespaced resources by default. |
| <code>--no-headers</code>               |                                                                                                           |
|                                         | When using the default or custom-column output format, don't print headers (default print headers).       |
| <code>-o, --output string</code>        |                                                                                                           |
|                                         | Output format. One of: (wide, name).                                                                      |
| <code>--sort-by string</code>           |                                                                                                           |
|                                         | If non-empty, sort list of resources using specified field. The field can be either 'name' or 'kind'.     |
| <code>--verbs strings</code>            |                                                                                                           |
|                                         | Limit to resources that support the specified verbs.                                                      |

|                                                               |                                                                                                              |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>--as string</code>                                      |                                                                                                              |
|                                                               | Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| <code>--as-group strings</code>                               |                                                                                                              |
|                                                               | Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| <code>--as-uid string</code>                                  |                                                                                                              |
|                                                               | UID to impersonate for the operation.                                                                        |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache" |                                                                                                              |
|                                                               | Default cache directory                                                                                      |
| <code>--certificate-authority string</code>                   |                                                                                                              |

|                                                                                                                     |                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                     | Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                         |                                                                                                                                                                 |
|                                                                                                                     | Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                 |                                                                                                                                                                 |
|                                                                                                                     | Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |                                                                                                                                                                 |
|                                                                                                                     | CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |                                                                                                                                                                 |
|                                                                                                                     | CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                    |                                                                                                                                                                 |
|                                                                                                                     | The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                    |                                                                                                                                                                 |
|                                                                                                                     | The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                          |                                                                                                                                                                 |
|                                                                                                                     | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                        |                                                                                                                                                                 |
|                                                                                                                     | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                               |                                                                                                                                                                 |
|                                                                                                                     | If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                          |                                                                                                                                                                 |
|                                                                                                                     | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                 |                                                                                                                                                                 |
|                                                                                                                     | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                              |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Require server version to match client version                                                                                                                                                    |
| <code>-n, --namespace string</code>                                                                                                                                                               |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| <code>--password string</code>                                                                                                                                                                    |
| Password for basic authentication to the API server                                                                                                                                               |
| <code>--profile string</code> Default: "none"                                                                                                                                                     |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| <code>--profile-output string</code> Default: "profile.pprof"                                                                                                                                     |
| Name of the file to write the profile to                                                                                                                                                          |
| <code>--request-timeout string</code> Default: "0"                                                                                                                                                |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl api-versions

## Synopsis

Print the supported API versions on the server, in the form of "group/version".

kubectl api-versions

# Examples

```
Print the supported API versions
kubectl api-versions
```

# Options

|                       |
|-----------------------|
| -h, --help            |
| help for api-versions |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--cluster string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl apply

## Synopsis

Apply a configuration to a resource by file name or stdin. The resource name must be specified. This resource will be created if it doesn't exist yet. To use 'apply', always create the resource initially with either 'apply' or 'create --save-config'.

JSON and YAML formats are accepted.

Alpha Disclaimer: the --prune functionality is not yet complete. Do not use unless you are aware of what the current state is. See <https://issues.k8s.io/34274>.

```
kubectl apply (-f FILENAME | -k DIRECTORY)
```

## Examples

```
Apply the configuration in pod.json to a pod
kubectl apply -f ./pod.json

Apply resources from a directory containing kustomization.yaml - e.g. dir/
kustomization.yaml
kubectl apply -k dir/

Apply the JSON passed into stdin to a pod
cat pod.json | kubectl apply -f -

Apply the configuration from all files that end with '.json'
kubectl apply -f '*.json'

Note: --prune is still in Alpha
Apply the configuration in manifest.yaml that matches label app=nginx and delete all other
resources that are not in the file and match label app=nginx
```



```
kubectl apply --prune -f manifest.yaml -l app=nginx
```

# Apply the configuration in manifest.yaml and delete all the other config maps that are not in the file

```
kubectl apply --prune -f manifest.yaml --all --prune-allowlist=core/v1/ConfigMap
```

## Options

|                                                                                                                                                                                                          |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <code>--all</code>                                                                                                                                                                                       |  |
| Select all resources in the namespace of the specified resource types.                                                                                                                                   |  |
| <code>--allow-missing-template-keys</code> Default: true                                                                                                                                                 |  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                          |  |
| <code>--cascade string[="background"]</code> Default: "background"                                                                                                                                       |  |
| Must be "background", "orphan", or "foreground". Selects the deletion cascading strategy for the dependents (e.g. Pods created by a ReplicationController). Defaults to background.                      |  |
| <code>--dry-run string[="unchanged"]</code> Default: "none"                                                                                                                                              |  |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |  |
| <code>--field-manager string</code> Default: "kubectl-client-side-apply"                                                                                                                                 |  |
| Name of the manager used to track field ownership.                                                                                                                                                       |  |
| <code>-f, --filename strings</code>                                                                                                                                                                      |  |
| The files that contain the configurations to apply.                                                                                                                                                      |  |
| <code>--force</code>                                                                                                                                                                                     |  |
| If true, immediately remove resources from API and bypass graceful deletion. Note that immediate deletion of some resources may result in inconsistency or data loss and requires confirmation.          |  |
| <code>--force-conflicts</code>                                                                                                                                                                           |  |
| If true, server-side apply will force the changes against conflicts.                                                                                                                                     |  |
| <code>--grace-period int</code> Default: -1                                                                                                                                                              |  |
| Period of time in seconds given to the resource to terminate gracefully. Ignored if negative. Set to 1 for immediate shutdown. Can only be set to 0 when --force is true (force deletion).               |  |

|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -h, --help                                                                                                                                                                 |
| help for apply                                                                                                                                                             |
| -k, --kustomize string                                                                                                                                                     |
| Process a kustomization directory. This flag can't be used together with -f or -R.                                                                                         |
| --openapi-patch    Default: true                                                                                                                                           |
| If true, use openapi to calculate diff when the openapi presents and the resource can be found in the openapi spec. Otherwise, fall back to use baked-in types.            |
| -o, --output string                                                                                                                                                        |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                               |
| --overwrite    Default: true                                                                                                                                               |
| Automatically resolve conflicts between the modified and live configuration by using values from the modified configuration                                                |
| --prune                                                                                                                                                                    |
| Automatically delete resource objects, that do not appear in the configs and are created by either apply or create --save-config. Should be used with either -l or --all.  |
| --prune-allowlist strings                                                                                                                                                  |
| Overwrite the default allowlist with <group/version/kind> for --prune                                                                                                      |
| -R, --recursive                                                                                                                                                            |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                            |
| -l, --selector string                                                                                                                                                      |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints. |
| --server-side                                                                                                                                                              |
| If true, apply runs in the server instead of the client.                                                                                                                   |
| --show-managed-fields                                                                                                                                                      |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                              |
| --template string                                                                                                                                                          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| --timeout duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| The length of time to wait before giving up on a delete, zero means determine a timeout from the size of the object                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --wait                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| If true, wait for resources to be gone before returning. This waits for finalizers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string   Default: "none"</p>                                                                                                                                                                |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string   Default: "profile.pprof"</p>                                                                                                                                                |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string   Default: "0"</p>                                                                                                                                                           |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration   Default: 1m0s</p>                                                                                                                                         |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string   Default: "cadvisor"</p>                                                                                                                                                  |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string   Default: "localhost:8086"</p>                                                                                                                                          |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string   Default: "root"</p>                                                                                                                                                |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string   Default: "stats"</p>                                                                                                                                                  |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string   Default: "root"</p>                                                                                                                                                    |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl apply edit-last-applied](#) - Edit latest last-applied-configuration annotations of a resource/object
- [kubectl apply set-last-applied](#) - Set the last-applied-configuration annotation on a live object to match the contents of a file
- [kubectl apply view-last-applied](#) - View the latest last-applied-configuration annotations of a resource/object

# kubectl apply edit-last-applied

## Synopsis

Edit the latest last-applied-configuration annotations of resources from the default editor.

The edit-last-applied command allows you to directly edit any API resource you can retrieve via the command-line tools. It will open the editor defined by your KUBE\_EDITOR, or EDITOR environment variables, or fall back to 'vi' for Linux or 'notepad' for Windows. You can edit multiple objects, although changes are applied one at a time. The command accepts file names as well as command-line arguments, although the files you point to must be previously saved versions of resources.

The default format is YAML. To edit in JSON, specify "-o json".

The flag `--windows-line-endings` can be used to force Windows line endings, otherwise the default for your operating system will be used.

In the event an error occurs while updating, a temporary file will be created on disk that contains your unapplied changes. The most common error when updating a resource is another editor changing the resource on the server. When this occurs, you will have to apply your changes to the newer version of the resource, or update your temporary saved copy to include the latest resource version.

```
kubectl apply edit-last-applied (RESOURCE/NAME | -f FILENAME)
```

## Examples

```
Edit the last-applied-configuration annotations by type/name in YAML
kubectl apply edit-last-applied deployment/nginx

Edit the last-applied-configuration annotations by file in JSON
kubectl apply edit-last-applied -f deploy.yaml -o json
```

## Options

|                                                                                                                                                 |                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <code>--allow-missing-template-keys</code>                                                                                                      | Default: true                                  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |                                                |
| <code>--field-manager</code>                                                                                                                    | string    Default: "kubectl-client-side-apply" |
| Name of the manager used to track field ownership.                                                                                              |                                                |
| <code>-f, --filename</code>                                                                                                                     | strings                                        |
| Filename, directory, or URL to files to use to edit the resource                                                                                |                                                |
| <code>-h, --help</code>                                                                                                                         |                                                |
| help for edit-last-applied                                                                                                                      |                                                |
| <code>-k, --kustomize</code>                                                                                                                    | string                                         |
| Process the kustomization directory. This flag can't be used together with <code>-f</code> or <code>-R</code> .                                 |                                                |
| <code>-o, --output</code>                                                                                                                       | string                                         |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).    |                                                |
| <code>-R, --recursive</code>                                                                                                                    |                                                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                            |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --windows-line-endings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Defaults to the line ending native to your platform.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |



|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl apply](#) - Apply a configuration to a resource by file name or stdin

# kubectl apply set-last-applied

## Synopsis

Set the latest last-applied-configuration annotations by setting it to match the contents of a file. This results in the last-applied-configuration being updated as though 'kubectl apply -f<file> ' was run, without updating any other parts of the object.

```
kubectl apply set-last-applied -f FILENAME
```

## Examples

```
Set the last-applied-configuration of a resource to match the contents of a file
kubectl apply set-last-applied -f deploy.yaml
```

```
Execute set-last-applied against each configuration file in a directory
kubectl apply set-last-applied -f path/
```

```
Set the last-applied-configuration of a resource to match the contents of a file; will create the
annotation if it does not already exist
kubectl apply set-last-applied -f deploy.yaml --create-annotation=true
```

## Options

|                                                                                                                                                                                                                                                          |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| --allow-missing-template-keys                                                                                                                                                                                                                            | Default: true   |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                                          |                 |
| --create-annotation                                                                                                                                                                                                                                      |                 |
| Will create 'last-applied-configuration' annotations if current objects doesn't have one                                                                                                                                                                 |                 |
| --dry-run string[="unchanged"]                                                                                                                                                                                                                           | Default: "none" |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |                 |
| -f, --filename strings                                                                                                                                                                                                                                   |                 |
| Filename, directory, or URL to files that contains the last-applied-configuration annotations                                                                                                                                                            |                 |
| -h, --help                                                                                                                                                                                                                                               |                 |
| help for set-last-applied                                                                                                                                                                                                                                |                 |
| -o, --output string                                                                                                                                                                                                                                      |                 |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                             |                 |
| --show-managed-fields                                                                                                                                                                                                                                    |                 |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |                 |
| --template string                                                                                                                                                                                                                                        |                 |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goyaml templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |                 |
|                                                                                                                                                                                                                                                          |                 |
| --as string                                                                                                                                                                                                                                              |                 |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |

## See Also

- [kubectl apply](#) - Apply a configuration to a resource by file name or stdin

# kubectl apply view-last-applied

## Synopsis

View the latest last-applied-configuration annotations by type/name or file.

The default output will be printed to stdout in YAML format. You can use the -o option to change the output format.

kubectl apply view-last-applied (TYPE [NAME | -l label] | TYPE/NAME | -f FILENAME)

## Examples

```
View the last-applied-configuration annotations by type/name in YAML
kubectl apply view-last-applied deployment/nginx

View the last-applied-configuration annotations by file in JSON
kubectl apply view-last-applied -f deploy.yaml -o json
```

## Options

|                        |                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------|
| --all                  |                                                                                               |
|                        | Select all resources in the namespace of the specified resource types                         |
| -f, --filename strings |                                                                                               |
|                        | Filename, directory, or URL to files that contains the last-applied-configuration annotations |
| -h, --help             |                                                                                               |
|                        | help for view-last-applied                                                                    |
| -k, --kustomize string |                                                                                               |
|                        | Process the kustomization directory. This flag can't be used together with -f or -R.          |
| -o, --output string    | Default: "yaml"                                                                               |
|                        | Output format. Must be one of (yaml, json)                                                    |
| -R, --recursive        |                                                                                               |



|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                            |
| -l, --selector string                                                                                                                                                      |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints. |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |
| --cluster string                                                                                                   |
| The name of the kubeconfig cluster to use                                                                          |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl apply](#) - Apply a configuration to a resource by file name or stdin

# kubectl attach

## Synopsis

Attach to a process that is already running inside an existing container.

```
kubectl attach (POD | TYPE/NAME) -c CONTAINER
```

## Examples

```
Get output from running pod mypod; use the 'kubectl.kubernetes.io/default-container'
annotation
for selecting the container to be attached or the first container in the pod will be chosen
kubectl attach mypod

Get output from ruby-container from pod mypod
kubectl attach mypod -c ruby-container

Switch to raw terminal mode; sends stdin to 'bash' in ruby-container from pod mypod
and sends stdout/stderr from 'bash' back to the client
kubectl attach mypod -c ruby-container -i -t

Get output from the first pod of a replica set named nginx
kubectl attach rs/nginx
```

## Options

|                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -c, --container string                                                                                                                                                             |
| Container name. If omitted, use the kubectl.kubernetes.io/default-container annotation for selecting the container to be attached or the first container in the pod will be chosen |

|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| -h, --help                                                                                          |
| help for attach                                                                                     |
| --pod-running-timeout duration    Default: 1m0s                                                     |
| The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running |
| -q, --quiet                                                                                         |
| Only print output from the remote session                                                           |
| -i, --stdin                                                                                         |
| Pass stdin to the container                                                                         |
| -t, --tty                                                                                           |
| Stdin is a TTY                                                                                      |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |

|                                                                                                                                                                 |                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code>                                                                                                          | Default: 130.211.0.0/22,35.191.0.0/16                                 |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |                                                                       |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code>                                                                                                            | Default: 130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |                                                                       |
| <code>--cluster string</code>                                                                                                                                   |                                                                       |
| The name of the kubeconfig cluster to use                                                                                                                       |                                                                       |
| <code>--context string</code>                                                                                                                                   |                                                                       |
| The name of the kubeconfig context to use                                                                                                                       |                                                                       |
| <code>--default-not-ready-toleration-seconds int</code>                                                                                                         | Default: 300                                                          |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |                                                                       |
| <code>--default-unreachable-toleration-seconds int</code>                                                                                                       | Default: 300                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |                                                                       |
| <code>--disable-compression</code>                                                                                                                              |                                                                       |
| If true, opt-out of response compression for all requests to the server                                                                                         |                                                                       |
| <code>--insecure-skip-tls-verify</code>                                                                                                                         |                                                                       |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |                                                                       |
| <code>--kubeconfig string</code>                                                                                                                                |                                                                       |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |                                                                       |
| <code>--match-server-version</code>                                                                                                                             |                                                                       |
| Require server version to match client version                                                                                                                  |                                                                       |
| <code>-n, --namespace string</code>                                                                                                                             |                                                                       |
| If present, the namespace scope for this CLI request                                                                                                            |                                                                       |
| <code>--password string</code>                                                                                                                                  |                                                                       |
| Password for basic authentication to the API server                                                                                                             |                                                                       |

|                                                                                                                                                                                                   |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --profile string                                                                                                                                                                                  | Default: "none"           |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |                           |
| --profile-output string                                                                                                                                                                           | Default: "profile.pprof"  |
| Name of the file to write the profile to                                                                                                                                                          |                           |
| --request-timeout string                                                                                                                                                                          | Default: "0"              |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |                           |
| -s, --server string                                                                                                                                                                               |                           |
| The address and port of the Kubernetes API server                                                                                                                                                 |                           |
| --storage-driver-buffer-duration duration                                                                                                                                                         | Default: 1m0s             |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |                           |
| --storage-driver-db string                                                                                                                                                                        | Default: "cadvisor"       |
| database name                                                                                                                                                                                     |                           |
| --storage-driver-host string                                                                                                                                                                      | Default: "localhost:8086" |
| database host:port                                                                                                                                                                                |                           |
| --storage-driver-password string                                                                                                                                                                  | Default: "root"           |
| database password                                                                                                                                                                                 |                           |
| --storage-driver-secure                                                                                                                                                                           |                           |
| use secure connection with database                                                                                                                                                               |                           |
| --storage-driver-table string                                                                                                                                                                     | Default: "stats"          |
| table name                                                                                                                                                                                        |                           |
| --storage-driver-user string                                                                                                                                                                      | Default: "root"           |
| database username                                                                                                                                                                                 |                           |
| --tls-server-name string                                                                                                                                                                          |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |                           |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --token string                                                                                               |
| Bearer token for authentication to the API server                                                            |
| --user string                                                                                                |
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl auth

## Synopsis

Inspect authorization.

```
kubectl auth [flags]
```

## Options

|               |
|---------------|
| -h, --help    |
| help for auth |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |



|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                   |
| --as-uid string                                                                                                                                                 |
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                              |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                               |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                              |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                       |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                     |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | If true, opt-out of response compression for all requests to the server                                                                                                                           |
|  | <code>--insecure-skip-tls-verify</code>                                                                                                                                                           |
|  | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
|  | <code>--kubeconfig string</code>                                                                                                                                                                  |
|  | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
|  | <code>--match-server-version</code>                                                                                                                                                               |
|  | Require server version to match client version                                                                                                                                                    |
|  | <code>-n, --namespace string</code>                                                                                                                                                               |
|  | If present, the namespace scope for this CLI request                                                                                                                                              |
|  | <code>--password string</code>                                                                                                                                                                    |
|  | Password for basic authentication to the API server                                                                                                                                               |
|  | <code>--profile string</code> Default: "none"                                                                                                                                                     |
|  | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
|  | <code>--profile-output string</code> Default: "profile.pprof"                                                                                                                                     |
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | <code>--request-timeout string</code> Default: "0"                                                                                                                                                |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | <code>-s, --server string</code>                                                                                                                                                                  |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
|  | database name                                                                                                                                                                                     |
|  | <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database host:port                                                                                                           |
| --storage-driver-password string    Default: "root"                                                                          |
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl auth can-i](#) - Check whether an action is allowed

- [kubectl auth reconcile](#) - Reconciles rules for RBAC role, role binding, cluster role, and cluster role binding objects
- [kubectl auth whoami](#) - Experimental: Check self subject attributes

# kubectl auth can-i

## Synopsis

Check whether an action is allowed.

VERB is a logical Kubernetes API verb like 'get', 'list', 'watch', 'delete', etc. TYPE is a Kubernetes resource. Shortcuts and groups will be resolved. NONRESOURCEURL is a partial URL that starts with "/". NAME is the name of a particular Kubernetes resource. This command pairs nicely with impersonation. See --as global flag.

```
kubectl auth can-i VERB [TYPE | TYPE/NAME | NONRESOURCEURL]
```

## Examples

```
Check to see if I can create pods in any namespace
kubectl auth can-i create pods --all-namespaces

Check to see if I can list deployments in my current namespace
kubectl auth can-i list deployments.apps

Check to see if service account "foo" of namespace "dev" can list pods
in the namespace "prod".
You must be allowed to use impersonation for the global option "--as".
kubectl auth can-i list pods --as=system:serviceaccount:dev:foo -n prod

Check to see if I can do everything in my current namespace ("*" means all)
kubectl auth can-i '*' '*'

Check to see if I can get the job named "bar" in namespace "foo"
kubectl auth can-i list jobs.batch/bar -n foo

Check to see if I can read pod logs
kubectl auth can-i get pods --subresource=log

Check to see if I can access the URL /logs/
kubectl auth can-i get /logs/

List all allowed actions in namespace "foo"
kubectl auth can-i --list --namespace=foo
```

## Options

|                      |
|----------------------|
| -A, --all-namespaces |
|----------------------|

|                                                         |
|---------------------------------------------------------|
| If true, check the specified action in all namespaces.  |
| -h, --help                                              |
| help for can-i                                          |
| --list                                                  |
| If true, prints all allowed actions.                    |
| --no-headers                                            |
| If true, prints allowed actions without headers         |
| -q, --quiet                                             |
| If true, suppress output and just return the exit code. |
| --subresource string                                    |
| SubResource such as pod/log or deployment/scale         |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string    Default: "none"</p>                                                                                                                                                               |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string    Default: "profile.pprof"</p>                                                                                                                                               |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string    Default: "0"</p>                                                                                                                                                          |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration    Default: 1m0s</p>                                                                                                                                        |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string    Default: "cadvisor"</p>                                                                                                                                                 |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string    Default: "localhost:8086"</p>                                                                                                                                         |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string    Default: "root"</p>                                                                                                                                               |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string    Default: "stats"</p>                                                                                                                                                 |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string    Default: "root"</p>                                                                                                                                                   |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl auth](#) - Inspect authorization

# kubectl auth reconcile

## Synopsis

Reconciles rules for RBAC role, role binding, cluster role, and cluster role binding objects.

Missing objects are created, and the containing namespace is created for namespaced objects, if required.

Existing roles are updated to include the permissions in the input objects, and remove extra permissions if --remove-extra-permissions is specified.

Existing bindings are updated to include the subjects in the input objects, and remove extra subjects if --remove-extra-subjects is specified.

This is preferred to 'apply' for RBAC resources so that semantically-aware merging of rules and subjects is done.

```
kubectl auth reconcile -f FILENAME
```



# Examples

```
Reconcile RBAC resources from a file
kubectl auth reconcile -f my-rbac-rules.yaml
```

# Options

|                                                                                                                                                                                                          |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| --allow-missing-template-keys                                                                                                                                                                            | Default: true   |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                         |                 |
| --dry-run string[="unchanged"]                                                                                                                                                                           | Default: "none" |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |                 |
| -f, --filename strings                                                                                                                                                                                   |                 |
| Filename, directory, or URL to files identifying the resource to reconcile.                                                                                                                              |                 |
| -h, --help                                                                                                                                                                                               |                 |
| help for reconcile                                                                                                                                                                                       |                 |
| -k, --kustomize string                                                                                                                                                                                   |                 |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                     |                 |
| -o, --output string                                                                                                                                                                                      |                 |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                             |                 |
| -R, --recursive                                                                                                                                                                                          |                 |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                          |                 |
| --remove-extra-permissions                                                                                                                                                                               |                 |
| If true, removes extra permissions added to roles                                                                                                                                                        |                 |
| --remove-extra-subjects                                                                                                                                                                                  |                 |
| If true, removes extra subjects added to rolebindings                                                                                                                                                    |                 |
| --show-managed-fields                                                                                                                                                                                    |                 |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --as string                                                                                                                                                                                                                                               |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                              |
| --as-group strings                                                                                                                                                                                                                                        |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                                                                                                             |
| --as-uid string                                                                                                                                                                                                                                           |
| UID to impersonate for the operation.                                                                                                                                                                                                                     |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                                                                                                                        |
| Default cache directory                                                                                                                                                                                                                                   |
| --certificate-authority string                                                                                                                                                                                                                            |
| Path to a cert file for the certificate authority                                                                                                                                                                                                         |
| --client-certificate string                                                                                                                                                                                                                               |
| Path to a client certificate file for TLS                                                                                                                                                                                                                 |
| --client-key string                                                                                                                                                                                                                                       |
| Path to a client key file for TLS                                                                                                                                                                                                                         |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                                                                                                                         |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                                                                                                                      |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                                                                                                                        |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                                                                                                                      |
| --cluster string                                                                                                                                                                                                                                          |
| The name of the kubeconfig cluster to use                                                                                                                                                                                                                 |
| --context string                                                                                                                                                                                                                                          |

|                                              |                                                                                                                                                                                                   |
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|                                              | The name of the kubeconfig context to use                                                                                                                                                         |
| --default-not-ready-toleration-seconds int   | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>-s, --server string</code>                                                                                                  |
| The address and port of the Kubernetes API server                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                       |
| database name                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                               |
| database host:port                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                     |
| database password                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                              |
| use secure connection with database                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                       |
| table name                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                         |
| database username                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| <code>--token string</code>                                                                                                       |
| Bearer token for authentication to the API server                                                                                 |
| <code>--user string</code>                                                                                                        |
| The name of the kubeconfig user to use                                                                                            |
| <code>--username string</code>                                                                                                    |
| Username for basic authentication to the API server                                                                               |
| <code>--version version[=true]</code>                                                                                             |

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>--version, --version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
|----------------------------------------------------------------------------------------------------------------------------------------|

|                                   |
|-----------------------------------|
| <code>--warnings-as-errors</code> |
|-----------------------------------|

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl auth](#) - Inspect authorization

# kubectl auth whoami

## Synopsis

Experimental: Check who you are and your attributes (groups, extra).

This command is helpful to get yourself aware of the current user attributes, especially when dynamic authentication, e.g., token webhook, auth proxy, or OIDC provider, is enabled in the Kubernetes cluster.

`kubectl auth whoami`

## Examples

# Get your subject attributes.

`kubectl auth whoami`

# Get your subject attributes in JSON format.

`kubectl auth whoami -o json`

## Options

|                                                          |
|----------------------------------------------------------|
| <code>--allow-missing-template-keys</code> Default: true |
|----------------------------------------------------------|

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
|-------------------------------------------------------------------------------------------------------------------------------------------------|

|                         |
|-------------------------|
| <code>-h, --help</code> |
|-------------------------|

|                 |
|-----------------|
| help for whoami |
|-----------------|

|                                  |
|----------------------------------|
| <code>-o, --output string</code> |
|----------------------------------|

|                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------|
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file). |
|----------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--show-managed-fields</code>                                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                         |
| <code>--template string</code>                                                                                                                                                                                                                                                        |
| Template string or path to template file to use when <code>-o=go-template</code> , <code>-o=go-template-file</code> . The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------|
| <code>--as string</code>                                                                                                      |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                  |
| <code>--as-group strings</code>                                                                                               |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                 |
| <code>--as-uid string</code>                                                                                                  |
| UID to impersonate for the operation.                                                                                         |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                                       |
| <code>--certificate-authority string</code>                                                                                   |
| Path to a cert file for the certificate authority                                                                             |
| <code>--client-certificate string</code>                                                                                      |
| Path to a client certificate file for TLS                                                                                     |
| <code>--client-key string</code>                                                                                              |
| Path to a client key file for TLS                                                                                             |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                          |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                          |
| <code>--cluster string</code>                                                                                                 |
| The name of the kubeconfig cluster to use                                                                                     |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |



|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl auth](#) - Inspect authorization

# kubectl autoscale

## Synopsis

Creates an autoscaler that automatically chooses and sets the number of pods that run in a Kubernetes cluster.

Looks up a deployment, replica set, stateful set, or replication controller by name and creates an autoscaler that uses the given resource as a reference. An autoscaler can automatically increase or decrease number of pods deployed within the system as needed.

```
kubectl autoscale (-f FILENAME | TYPE NAME | TYPE/NAME) [--min=MINPODS] --max=MAXPODS [--cpu-percent=CPU]
```

## Examples

```
Auto scale a deployment "foo", with the number of pods between 2 and 10, no target CPU utilization specified so a default autoscaling policy will be used
```

```
kubectl autoscale deployment foo --min=2 --max=10
```

```
Auto scale a replication controller "foo", with the number of pods between 1 and 5, target CPU utilization at 80%
```

```
kubectl autoscale rc foo --max=5 --cpu-percent=80
```

## Options

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| --allow-missing-template-keys    Default: true                                                                                                  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
| --cpu-percent int32    Default: -1                                                                                                              |

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The target average CPU utilization (represented as a percent of requested CPU) over all the pods. If it's not specified or negative, a default autoscaling policy will be used.                          |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                        |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| --field-manager string    Default: "kubectl-autoscale"                                                                                                                                                   |
| Name of the manager used to track field ownership.                                                                                                                                                       |
| -f, --filename strings                                                                                                                                                                                   |
| Filename, directory, or URL to files identifying the resource to autoscale.                                                                                                                              |
| -h, --help                                                                                                                                                                                               |
| help for autoscale                                                                                                                                                                                       |
| -k, --kustomize string                                                                                                                                                                                   |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                     |
| --max int32    Default: -1                                                                                                                                                                               |
| The upper limit for the number of pods that can be set by the autoscaler. Required.                                                                                                                      |
| --min int32    Default: -1                                                                                                                                                                               |
| The lower limit for the number of pods that can be set by the autoscaler. If it's not specified or negative, the server will apply a default value.                                                      |
| --name string                                                                                                                                                                                            |
| The name for the newly created object. If not specified, the name of the input resource will be used.                                                                                                    |
| -o, --output string                                                                                                                                                                                      |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                             |
| -R, --recursive                                                                                                                                                                                          |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                          |
| --save-config                                                                                                                                                                                            |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|                                           |                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                  | Default: "0"                                                                                                                                                                                      |
|                                           | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                       |                                                                                                                                                                                                   |
|                                           | The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration | Default: 1m0s                                                                                                                                                                                     |
|                                           | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string                | Default: "cadvisor"                                                                                                                                                                               |
|                                           | database name                                                                                                                                                                                     |
| --storage-driver-host string              | Default: "localhost:8086"                                                                                                                                                                         |
|                                           | database host:port                                                                                                                                                                                |
| --storage-driver-password string          | Default: "root"                                                                                                                                                                                   |
|                                           | database password                                                                                                                                                                                 |
| --storage-driver-secure                   |                                                                                                                                                                                                   |
|                                           | use secure connection with database                                                                                                                                                               |
| --storage-driver-table string             | Default: "stats"                                                                                                                                                                                  |
|                                           | table name                                                                                                                                                                                        |
| --storage-driver-user string              | Default: "root"                                                                                                                                                                                   |
|                                           | database username                                                                                                                                                                                 |
| --tls-server-name string                  |                                                                                                                                                                                                   |
|                                           | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                            |                                                                                                                                                                                                   |
|                                           | Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                             |                                                                                                                                                                                                   |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl certificate

## Synopsis

Modify certificate resources.

kubectl certificate SUBCOMMAND

## Options

|                      |
|----------------------|
| -h, --help           |
| help for certificate |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |

|                                                                                                                                                                 |                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| --cache-dir string                                                                                                                                              | Default: "\$HOME/.kube/cache"                                         |
| Default cache directory                                                                                                                                         |                                                                       |
| --certificate-authority string                                                                                                                                  |                                                                       |
| Path to a cert file for the certificate authority                                                                                                               |                                                                       |
| --client-certificate string                                                                                                                                     |                                                                       |
| Path to a client certificate file for TLS                                                                                                                       |                                                                       |
| --client-key string                                                                                                                                             |                                                                       |
| Path to a client key file for TLS                                                                                                                               |                                                                       |
| --cloud-provider-gce-l7lb-src-cidrs cidrs                                                                                                                       | Default: 130.211.0.0/22,35.191.0.0/16                                 |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |                                                                       |
| --cloud-provider-gce-lb-src-cidrs cidrs                                                                                                                         | Default: 130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |                                                                       |
| --cluster string                                                                                                                                                |                                                                       |
| The name of the kubeconfig cluster to use                                                                                                                       |                                                                       |
| --context string                                                                                                                                                |                                                                       |
| The name of the kubeconfig context to use                                                                                                                       |                                                                       |
| --default-not-ready-toleration-seconds int                                                                                                                      | Default: 300                                                          |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |                                                                       |
| --default-unreachable-toleration-seconds int                                                                                                                    | Default: 300                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |                                                                       |
| --disable-compression                                                                                                                                           |                                                                       |
| If true, opt-out of response compression for all requests to the server                                                                                         |                                                                       |
| --insecure-skip-tls-verify                                                                                                                                      |                                                                       |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |                                                                       |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubect](#)l - kubectl controls the Kubernetes cluster manager
- [kubectl certificate approve](#) - Approve a certificate signing request
- [kubectl certificate deny](#) - Deny a certificate signing request

# kubectl certificate approve

## Synopsis

Approve a certificate signing request.

kubectl certificate approve allows a cluster admin to approve a certificate signing request (CSR). This action tells a certificate signing controller to issue a certificate to the requester with the attributes requested in the CSR.

SECURITY NOTICE: Depending on the requested attributes, the issued certificate can potentially grant a requester access to cluster resources or to authenticate as a requested identity. Before approving a CSR, ensure you understand what the signed certificate can do.

kubectl certificate approve (-f FILENAME | NAME)

## Examples

```
Approve CSR 'csr-sqgzp'
kubectl certificate approve csr-sqgzp
```

## Options

|                                                                                                                                                 |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| --allow-missing-template-keys                                                                                                                   | Default: true |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |               |
| -f, --filename strings                                                                                                                          |               |
| Filename, directory, or URL to files identifying the resource to update                                                                         |               |
| --force                                                                                                                                         |               |
| Update the CSR even if it is already approved.                                                                                                  |               |
| -h, --help                                                                                                                                      |               |
| help for approve                                                                                                                                |               |
| -k, --kustomize string                                                                                                                          |               |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |               |
| -o, --output string                                                                                                                             |               |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).    |               |
| -R, --recursive                                                                                                                                 |               |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory. |               |
| --show-managed-fields                                                                                                                           |               |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --as string                                                                                                                                                                                                                                               |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                              |
| --as-group strings                                                                                                                                                                                                                                        |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                                                                                                             |
| --as-uid string                                                                                                                                                                                                                                           |
| UID to impersonate for the operation.                                                                                                                                                                                                                     |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                                                                                                                        |
| Default cache directory                                                                                                                                                                                                                                   |
| --certificate-authority string                                                                                                                                                                                                                            |
| Path to a cert file for the certificate authority                                                                                                                                                                                                         |
| --client-certificate string                                                                                                                                                                                                                               |
| Path to a client certificate file for TLS                                                                                                                                                                                                                 |
| --client-key string                                                                                                                                                                                                                                       |
| Path to a client key file for TLS                                                                                                                                                                                                                         |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                                                                                                                         |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                                                                                                                      |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                                                                                                                        |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                                                                                                                      |
| --cluster string                                                                                                                                                                                                                                          |
| The name of the kubeconfig cluster to use                                                                                                                                                                                                                 |
| --context string                                                                                                                                                                                                                                          |

|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | The name of the kubeconfig context to use                                                                                                                                                         |
| --default-not-ready-toleration-seconds int   | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>-s, --server string</code>                                                                                                  |
| The address and port of the Kubernetes API server                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                       |
| database name                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                               |
| database host:port                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                     |
| database password                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                              |
| use secure connection with database                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                       |
| table name                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                         |
| database username                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| <code>--token string</code>                                                                                                       |
| Bearer token for authentication to the API server                                                                                 |
| <code>--user string</code>                                                                                                        |
| The name of the kubeconfig user to use                                                                                            |
| <code>--username string</code>                                                                                                    |
| Username for basic authentication to the API server                                                                               |
| <code>--version version[=true]</code>                                                                                             |

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>--version, --version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
| <code>--warnings-as-errors</code>                                                                                                      |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                   |

## See Also

- [kubectl certificate](#) - Modify certificate resources

# kubectl certificate deny

## Synopsis

Deny a certificate signing request.

`kubectl certificate deny` allows a cluster admin to deny a certificate signing request (CSR). This action tells a certificate signing controller to not to issue a certificate to the requester.

`kubectl certificate deny (-f FILENAME | NAME)`

## Examples

```
Deny CSR 'csr-sqgzp'
kubectl certificate deny csr-sqgzp
```

## Options

|                                                                                                                                                 |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <code>--allow-missing-template-keys</code>                                                                                                      | Default: true |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |               |
| <code>-f, --filename</code>                                                                                                                     | strings       |
| Filename, directory, or URL to files identifying the resource to update                                                                         |               |
| <code>--force</code>                                                                                                                            |               |
| Update the CSR even if it is already denied.                                                                                                    |               |
| <code>-h, --help</code>                                                                                                                         |               |
| help for deny                                                                                                                                   |               |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -k, --kustomize string                                                                                                                                                                                                                                    |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                      |
| -o, --output string                                                                                                                                                                                                                                       |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                              |
| -R, --recursive                                                                                                                                                                                                                                           |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| --show-managed-fields                                                                                                                                                                                                                                     |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |



|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string   Default: "none"</p>                                                                                                                                                                |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string   Default: "profile.pprof"</p>                                                                                                                                                |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string   Default: "0"</p>                                                                                                                                                           |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration   Default: 1m0s</p>                                                                                                                                         |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string   Default: "cadvisor"</p>                                                                                                                                                  |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string   Default: "localhost:8086"</p>                                                                                                                                          |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string   Default: "root"</p>                                                                                                                                                |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string   Default: "stats"</p>                                                                                                                                                  |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string   Default: "root"</p>                                                                                                                                                    |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl certificate](#) - Modify certificate resources

# kubectl cluster-info

## Synopsis

Display addresses of the control plane and services with label `kubernetes.io/cluster-service=true`. To further debug and diagnose cluster problems, use `'kubectl cluster-info dump'`.

```
kubectl cluster-info [flags]
```

## Examples

```
Print the address of the control plane and cluster services
kubectl cluster-info
```

## Options

```
-h, --help
```

|                       |
|-----------------------|
| help for cluster-info |
|-----------------------|

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |
| --cluster string                                                                                                   |
| The name of the kubeconfig cluster to use                                                                          |
| --context string                                                                                                   |
| The name of the kubeconfig context to use                                                                          |
| --default-not-ready-toleration-seconds int   Default: 300                                                          |

|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                          |                                                                                                                                                                                                   |
|                                              | The address and port of the Kubernetes API server                                                                                                                                                 |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl cluster-info dump](#) - Dump relevant information for debugging and diagnosis

# kubectl cluster-info dump

## Synopsis

Dump cluster information out suitable for debugging and diagnosing cluster problems. By default, dumps everything to stdout. You can optionally specify a directory with `--output-directory`. If you specify a directory, Kubernetes will build a set of files in that directory. By default, only dumps things in the current namespace and 'kube-system' namespace, but you can switch to a different namespace with the `--namespaces` flag, or specify `--all-namespaces` to dump all namespaces.

The command also dumps the logs of all of the pods in the cluster; these logs are dumped into different directories based on namespace and pod name.

```
kubectl cluster-info dump [flags]
```

## Examples

```
Dump current cluster state to stdout
kubectl cluster-info dump

Dump current cluster state to /path/to/cluster-state
kubectl cluster-info dump --output-directory=/path/to/cluster-state

Dump all namespaces to stdout
kubectl cluster-info dump --all-namespaces

Dump a set of namespaces to /path/to/cluster-state
kubectl cluster-info dump --namespaces default,kube-system --output-directory=/path/to/cluster-state
```

## Options

|                                                                 |
|-----------------------------------------------------------------|
| -A, --all-namespaces                                            |
| If true, dump all namespaces. If true, --namespaces is ignored. |
| --allow-missing-template-keys    Default: true                  |

|                                |                                                                                                                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| -h, --help                     |                                                                                                                                                                                                                                                          |
|                                | help for dump                                                                                                                                                                                                                                            |
| --namespaces strings           |                                                                                                                                                                                                                                                          |
|                                | A comma separated list of namespaces to dump.                                                                                                                                                                                                            |
| -o, --output string            | Default: "json"                                                                                                                                                                                                                                          |
|                                | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                             |
| --output-directory string      |                                                                                                                                                                                                                                                          |
|                                | Where to output the files. If empty or '-' uses stdout, otherwise creates a directory hierarchy in that directory                                                                                                                                        |
| --pod-running-timeout duration | Default: 20s                                                                                                                                                                                                                                             |
|                                | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                                                                                      |
| --show-managed-fields          |                                                                                                                                                                                                                                                          |
|                                | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string              |                                                                                                                                                                                                                                                          |
|                                | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                    |                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| --as string        |                                                                                                              |
|                    | Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings |                                                                                                              |
|                    | Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string    |                                                                                                              |
|                    | UID to impersonate for the operation.                                                                        |
| --cache-dir string | Default: "\$HOME/.kube/cache"                                                                                |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |



|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
|  | --match-server-version                                                                                                                                                                            |
|  | Require server version to match client version                                                                                                                                                    |
|  | -n, --namespace string                                                                                                                                                                            |
|  | If present, the namespace scope for this CLI request                                                                                                                                              |
|  | --password string                                                                                                                                                                                 |
|  | Password for basic authentication to the API server                                                                                                                                               |
|  | --profile string   Default: "none"                                                                                                                                                                |
|  | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
|  | --profile-output string   Default: "profile.pprof"                                                                                                                                                |
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | --request-timeout string   Default: "0"                                                                                                                                                           |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | -s, --server string                                                                                                                                                                               |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
|  | database name                                                                                                                                                                                     |
|  | --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
|  | database host:port                                                                                                                                                                                |
|  | --storage-driver-password string   Default: "root"                                                                                                                                                |
|  | database password                                                                                                                                                                                 |
|  | --storage-driver-secure                                                                                                                                                                           |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl cluster-info](#) - Display cluster information

# kubectl completion

## Synopsis

Output shell completion code for the specified shell (bash, zsh, fish, or powershell). The shell code must be evaluated to provide interactive completion of kubectl commands. This can be done by sourcing it from the .bash\_profile.

Detailed instructions on how to do this are available here:

```
for macOS:
https://kubernetes.io/docs/tasks/tools/install-kubectl-macos/#enable-shell-autocompletion

for linux:
https://kubernetes.io/docs/tasks/tools/install-kubectl-linux/#enable-shell-autocompletion

for windows:
https://kubernetes.io/docs/tasks/tools/install-kubectl-windows/#enable-shell-autocompletion
```

Note for zsh users: [1] zsh completions are only supported in versions of zsh >= 5.2.

kubectl completion SHELL

## Examples

```
Installing bash completion on macOS using homebrew
If running Bash 3.2 included with macOS
brew install bash-completion
or, if running Bash 4.1+
brew install bash-completion@2
If kubectl is installed via homebrew, this should start working immediately
If you've installed via other means, you may need add the completion to your completion
directory
kubectl completion bash > $(brew --prefix)/etc/bash_completion.d/kubectl
```

```
Installing bash completion on Linux
If bash-completion is not installed on Linux, install the 'bash-completion' package
via your distribution's package manager.
Load the kubectl completion code for bash into the current shell
source <(kubectl completion bash)
Write bash completion code to a file and source it from .bash_profile
kubectl completion bash > ~/.kube/completion.bash.inc
printf "
kubectl shell completion
source '$HOME/.kube/completion.bash.inc'
" >> $HOME/.bash_profile
source $HOME/.bash_profile
```

```
Load the kubectl completion code for zsh[1] into the current shell
source <(kubectl completion zsh)
Set the kubectl completion code for zsh[1] to autoload on startup
kubectl completion zsh > "${fpath[1]}/_kubectl"
```

```
Load the kubectl completion code for fish[2] into the current shell
kubectl completion fish | source
To load completions for each session, execute once:
kubectl completion fish > ~/.config/fish/completions/kubectl.fish
```

```
Load the kubectl completion code for powershell into the current shell
kubectl completion powershell | Out-String | Invoke-Expression
Set kubectl completion code for powershell to run on startup
Save completion code to a script and execute in the profile
kubectl completion powershell > $HOME\.kube\completion.ps1
Add-Content $PROFILE "$HOME\.kube\completion.ps1"
Execute completion code in the profile
Add-Content $PROFILE "if (Get-Command kubectl -ErrorAction SilentlyContinue) {
kubectl completion powershell | Out-String | Invoke-Expression
}"
Add completion code directly to the $PROFILE script
kubectl completion powershell >> $PROFILE
```

## Options

|                     |
|---------------------|
| -h, --help          |
| help for completion |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                            |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --tls-server-name string                                                                                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Bearer token for authentication to the API server                                                            |
| --user string                                                                                                |
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectrl](#) - kubectrl controls the Kubernetes cluster manager

# kubectrl config

## Synopsis

Modify kubeconfig files using subcommands like "kubectrl config set current-context my-context".

The loading order follows these rules:

1. If the --kubeconfig flag is set, then only that file is loaded. The flag may only be set once and no merging takes place.
2. If \$KUBECONFIG environment variable is set, then it is used as a list of paths (normal path delimiting rules for your system). These paths are merged. When a value is modified, it is modified in the file that defines the stanza. When a value is created, it is created in the first file that exists. If no files in the chain exist, then it creates the last file in the list.
3. Otherwise, \${HOME}/.kube/config is used and no merging takes place.

kubectrl config SUBCOMMAND

## Options

|            |
|------------|
| -h, --help |
|------------|

|                                  |
|----------------------------------|
| help for config                  |
| --kubeconfig string              |
| use a particular kubeconfig file |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |
| --cluster string                                                                                                   |
| The name of the kubeconfig cluster to use                                                                          |
| --context string                                                                                                   |



|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | The name of the kubeconfig context to use                                                                                                                                                         |
| --default-not-ready-toleration-seconds int   | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                          |                                                                                                                                                                                                   |
|                                              | The address and port of the Kubernetes API server                                                                                                                                                 |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl config current-context](#) - Display the current-context
- [kubectl config delete-cluster](#) - Delete the specified cluster from the kubeconfig
- [kubectl config delete-context](#) - Delete the specified context from the kubeconfig
- [kubectl config delete-user](#) - Delete the specified user from the kubeconfig
- [kubectl config get-clusters](#) - Display clusters defined in the kubeconfig
- [kubectl config get-contexts](#) - Describe one or many contexts
- [kubectl config get-users](#) - Display users defined in the kubeconfig
- [kubectl config rename-context](#) - Rename a context from the kubeconfig file
- [kubectl config set](#) - Set an individual value in a kubeconfig file
- [kubectl config set-cluster](#) - Set a cluster entry in kubeconfig
- [kubectl config set-context](#) - Set a context entry in kubeconfig
- [kubectl config set-credentials](#) - Set a user entry in kubeconfig
- [kubectl config unset](#) - Unset an individual value in a kubeconfig file
- [kubectl config use-context](#) - Set the current-context in a kubeconfig file
- [kubectl config view](#) - Display merged kubeconfig settings or a specified kubeconfig file

# kubectl config current-context

## Synopsis

Display the current-context.

```
kubectl config current-context [flags]
```

## Examples

```
Display the current-context
kubectl config current-context
```

## Options

|                          |
|--------------------------|
| -h, --help               |
| help for current-context |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |

|                                                                                                                               |                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--as-group strings</code>                                                                                               |                                                                                                                                                                 |
|                                                                                                                               | Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                   |
| <code>--as-uid string</code>                                                                                                  |                                                                                                                                                                 |
|                                                                                                                               | UID to impersonate for the operation.                                                                                                                           |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                                 |                                                                                                                                                                 |
|                                                                                                                               | Default cache directory                                                                                                                                         |
| <code>--certificate-authority string</code>                                                                                   |                                                                                                                                                                 |
|                                                                                                                               | Path to a cert file for the certificate authority                                                                                                               |
| <code>--client-certificate string</code>                                                                                      |                                                                                                                                                                 |
|                                                                                                                               | Path to a client certificate file for TLS                                                                                                                       |
| <code>--client-key string</code>                                                                                              |                                                                                                                                                                 |
|                                                                                                                               | Path to a client key file for TLS                                                                                                                               |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                  |                                                                                                                                                                 |
|                                                                                                                               | CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |                                                                                                                                                                 |
|                                                                                                                               | CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| <code>--cluster string</code>                                                                                                 |                                                                                                                                                                 |
|                                                                                                                               | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                                                                                 |                                                                                                                                                                 |
|                                                                                                                               | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300                                                          |                                                                                                                                                                 |
|                                                                                                                               | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300                                                        |                                                                                                                                                                 |
|                                                                                                                               | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                                                                            |                                                                                                                                                                 |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | If true, opt-out of response compression for all requests to the server                                                                                                                           |
|  | --insecure-skip-tls-verify                                                                                                                                                                        |
|  | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
|  | --kubeconfig string                                                                                                                                                                               |
|  | use a particular kubeconfig file                                                                                                                                                                  |
|  | --match-server-version                                                                                                                                                                            |
|  | Require server version to match client version                                                                                                                                                    |
|  | -n, --namespace string                                                                                                                                                                            |
|  | If present, the namespace scope for this CLI request                                                                                                                                              |
|  | --password string                                                                                                                                                                                 |
|  | Password for basic authentication to the API server                                                                                                                                               |
|  | --profile string   Default: "none"                                                                                                                                                                |
|  | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
|  | --profile-output string   Default: "profile.pprof"                                                                                                                                                |
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | --request-timeout string   Default: "0"                                                                                                                                                           |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | -s, --server string                                                                                                                                                                               |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
|  | database name                                                                                                                                                                                     |
|  | --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database host:port                                                                                                           |
| --storage-driver-password string    Default: "root"                                                                          |
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl config](#) - Modify kubeconfig files

# kubectl config delete-cluster

## Synopsis

Delete the specified cluster from the kubeconfig.

```
kubectl config delete-cluster NAME
```

## Examples

```
Delete the minikube cluster
kubectl config delete-cluster minikube
```

## Options

|                         |
|-------------------------|
| -h, --help              |
| help for delete-cluster |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| use a particular kubeconfig file                                                                                                                                |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |



|                                                                                                                                                                                                   |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --profile string                                                                                                                                                                                  | Default: "none"           |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |                           |
| --profile-output string                                                                                                                                                                           | Default: "profile.pprof"  |
| Name of the file to write the profile to                                                                                                                                                          |                           |
| --request-timeout string                                                                                                                                                                          | Default: "0"              |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |                           |
| -s, --server string                                                                                                                                                                               |                           |
| The address and port of the Kubernetes API server                                                                                                                                                 |                           |
| --storage-driver-buffer-duration duration                                                                                                                                                         | Default: 1m0s             |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |                           |
| --storage-driver-db string                                                                                                                                                                        | Default: "cadvisor"       |
| database name                                                                                                                                                                                     |                           |
| --storage-driver-host string                                                                                                                                                                      | Default: "localhost:8086" |
| database host:port                                                                                                                                                                                |                           |
| --storage-driver-password string                                                                                                                                                                  | Default: "root"           |
| database password                                                                                                                                                                                 |                           |
| --storage-driver-secure                                                                                                                                                                           |                           |
| use secure connection with database                                                                                                                                                               |                           |
| --storage-driver-table string                                                                                                                                                                     | Default: "stats"          |
| table name                                                                                                                                                                                        |                           |
| --storage-driver-user string                                                                                                                                                                      | Default: "root"           |
| database username                                                                                                                                                                                 |                           |
| --tls-server-name string                                                                                                                                                                          |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |                           |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --token string                                                                                               |
| Bearer token for authentication to the API server                                                            |
| --user string                                                                                                |
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config delete-context

## Synopsis

Delete the specified context from the kubeconfig.

```
kubectl config delete-context NAME
```

## Examples

```
Delete the context for the minikube cluster
kubectl config delete-context minikube
```

## Options

|                         |
|-------------------------|
| -h, --help              |
| help for delete-context |

|             |
|-------------|
| --as string |
|-------------|

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| use a particular kubeconfig file                                                                                                                                                                  |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |

# See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config delete-user

## Synopsis

Delete the specified user from the kubeconfig.

```
kubectl config delete-user NAME
```

## Examples

```
Delete the minikube user
kubectl config delete-user minikube
```

## Options

|                      |
|----------------------|
| -h, --help           |
| help for delete-user |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| use a particular kubeconfig file                                                                                                                                |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubecttl config](#) - Modify kubeconfig files

# kubectl config get-clusters

## Synopsis

Display clusters defined in the kubeconfig.

```
kubectl config get-clusters [flags]
```

## Examples

```
List the clusters that kubectl knows about
kubectl config get-clusters
```

# Options

|                       |
|-----------------------|
| -h, --help            |
| help for get-clusters |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |
| --cluster string                                                                                                   |
| The name of the kubeconfig cluster to use                                                                          |
| --context string                                                                                                   |

|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | The name of the kubeconfig context to use                                                                                                                                                         |
| --default-not-ready-toleration-seconds int   | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | use a particular kubeconfig file                                                                                                                                                                  |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>-s, --server string</code>                                                                                                  |
| The address and port of the Kubernetes API server                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                       |
| database name                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                               |
| database host:port                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                     |
| database password                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                              |
| use secure connection with database                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                       |
| table name                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                         |
| database username                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| <code>--token string</code>                                                                                                       |
| Bearer token for authentication to the API server                                                                                 |
| <code>--user string</code>                                                                                                        |
| The name of the kubeconfig user to use                                                                                            |
| <code>--username string</code>                                                                                                    |
| Username for basic authentication to the API server                                                                               |
| <code>--version version[=true]</code>                                                                                             |

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>--version, --version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
|----------------------------------------------------------------------------------------------------------------------------------------|

|                                   |
|-----------------------------------|
| <code>--warnings-as-errors</code> |
|-----------------------------------|

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config get-contexts

## Synopsis

Display one or many contexts from the kubeconfig file.

```
kubectl config get-contexts [(-o|--output=)name)]
```

## Examples

```
List all the contexts in your kubeconfig file
kubectl config get-contexts
```

```
Describe one context in your kubeconfig file
kubectl config get-contexts my-context
```

## Options

|                         |
|-------------------------|
| <code>-h, --help</code> |
|-------------------------|

|                       |
|-----------------------|
| help for get-contexts |
|-----------------------|

|                           |
|---------------------------|
| <code>--no-headers</code> |
|---------------------------|

|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| When using the default or custom-column output format, don't print headers (default print headers). |
|-----------------------------------------------------------------------------------------------------|

|                                  |
|----------------------------------|
| <code>-o, --output string</code> |
|----------------------------------|

|                                |
|--------------------------------|
| Output format. One of: (name). |
|--------------------------------|

|                          |
|--------------------------|
| <code>--as string</code> |
|--------------------------|

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| use a particular kubeconfig file                                                                                                                                                                  |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |



# See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config get-users

## Synopsis

Display users defined in the kubeconfig.

```
kubectl config get-users [flags]
```

## Examples

```
List the users that kubectl knows about
kubectl config get-users
```

## Options

|                    |
|--------------------|
| -h, --help         |
| help for get-users |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| use a particular kubeconfig file                                                                                                                                |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config rename-context

## Synopsis

Renames a context from the kubeconfig file.

CONTEXT\_NAME is the context name that you want to change.

NEW\_NAME is the new name you want to set.

Note: If the context being renamed is the 'current-context', this field will also be updated.

```
kubectl config rename-context CONTEXT_NAME NEW_NAME
```

# Examples

```
Rename the context 'old-name' to 'new-name' in your kubeconfig file
kubectl config rename-context old-name new-name
```

# Options

|                         |
|-------------------------|
| -h, --help              |
| help for rename-context |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |

|                                                              |                                                                                                                                                                 |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --cluster string                                             |                                                                                                                                                                 |
|                                                              | The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                             |                                                                                                                                                                 |
|                                                              | The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300   |                                                                                                                                                                 |
|                                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300 |                                                                                                                                                                 |
|                                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                        |                                                                                                                                                                 |
|                                                              | If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                   |                                                                                                                                                                 |
|                                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                          |                                                                                                                                                                 |
|                                                              | use a particular kubeconfig file                                                                                                                                |
| --match-server-version                                       |                                                                                                                                                                 |
|                                                              | Require server version to match client version                                                                                                                  |
| -n, --namespace string                                       |                                                                                                                                                                 |
|                                                              | If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                            |                                                                                                                                                                 |
|                                                              | Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                          |                                                                                                                                                                 |
|                                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                              | Name of the file to write the profile to                                                                                                                        |
| --request-timeout string    Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                                                                  |
| <code>--version version[=true]</code>                                                                                                                |
| <code>--version</code> , <code>--version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
| <code>--warnings-as-errors</code>                                                                                                                    |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                                 |

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config set

## Synopsis

Set an individual value in a kubeconfig file.

PROPERTY\_NAME is a dot delimited name where each token represents either an attribute name or a map key. Map keys may not contain dots.

PROPERTY\_VALUE is the new value you want to set. Binary fields such as 'certificate-authority-data' expect a base64 encoded string unless the `--set-raw-bytes` flag is used.

Specifying an attribute name that already exists will merge new fields on top of existing values.

```
kubectl config set PROPERTY_NAME PROPERTY_VALUE
```

## Examples

```
Set the server field on the my-cluster cluster to https://1.2.3.4
kubectl config set clusters.my-cluster.server https://1.2.3.4

Set the certificate-authority-data field on the my-cluster cluster
kubectl config set clusters.my-cluster.certificate-authority-data $(echo "cert_data_here" |
base64 -i -)

Set the cluster field in the my-context context to my-cluster
kubectl config set contexts.my-context.cluster my-cluster

Set the client-key-data field in the cluster-admin user using --set-raw-bytes option
kubectl config set users.cluster-admin.client-key-data cert_data_here --set-raw-bytes=true
```



# Options

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| -h, --help                                                                                     |
| help for set                                                                                   |
| --set-raw-bytes tristate[=true]                                                                |
| When writing a []byte PROPERTY_VALUE, write the given string directly without base64 decoding. |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |
| --cluster string                                                                                                   |

|                                              |                                                                                                                                                                 |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                             |                                                                                                                                                                 |
|                                              | The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int   | Default: 300                                                                                                                                                    |
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                    |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                        |                                                                                                                                                                 |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                   |                                                                                                                                                                 |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                          |                                                                                                                                                                 |
|                                              | use a particular kubeconfig file                                                                                                                                |
| --match-server-version                       |                                                                                                                                                                 |
|                                              | Require server version to match client version                                                                                                                  |
| -n, --namespace string                       |                                                                                                                                                                 |
|                                              | If present, the namespace scope for this CLI request                                                                                                            |
| --password string                            |                                                                                                                                                                 |
|                                              | Password for basic authentication to the API server                                                                                                             |
| --profile string                             | Default: "none"                                                                                                                                                 |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                        |
|                                              | Name of the file to write the profile to                                                                                                                        |
| --request-timeout string                     | Default: "0"                                                                                                                                                    |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                                                                  |
| <code>--version version[=true]</code>                                                                                                                |
| <code>--version</code> , <code>--version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
| <code>--warnings-as-errors</code>                                                                                                                    |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                                 |

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config set-cluster

## Synopsis

Set a cluster entry in kubeconfig.

Specifying a name that already exists will merge new fields on top of existing values for those fields.

```
kubectl config set-cluster NAME [--server=server] [--certificate-authority=path/to/certificate/authority] [--insecure-skip-tls-verify=true] [--tls-server-name=example.com]
```

## Examples

```
Set only the server field on the e2e cluster entry without touching other values
kubectl config set-cluster e2e --server=https://1.2.3.4

Embed certificate authority data for the e2e cluster entry
kubectl config set-cluster e2e --embed-certs --certificate-authority=~/.kube/e2e/
kubernetes.ca.crt

Disable cert checking for the e2e cluster entry
kubectl config set-cluster e2e --insecure-skip-tls-verify=true

Set the custom TLS server name to use for validation for the e2e cluster entry
kubectl config set-cluster e2e --tls-server-name=my-cluster-name

Set the proxy URL for the e2e cluster entry
kubectl config set-cluster e2e --proxy-url=https://1.2.3.4
```

## Options

|                                                                        |
|------------------------------------------------------------------------|
| --certificate-authority string                                         |
| Path to certificate-authority file for the cluster entry in kubeconfig |
| --embed-certs tristate[=true]                                          |
| embed-certs for the cluster entry in kubeconfig                        |
| -h, --help                                                             |
| help for set-cluster                                                   |
| --insecure-skip-tls-verify tristate[=true]                             |
| insecure-skip-tls-verify for the cluster entry in kubeconfig           |
| --proxy-url string                                                     |
| proxy-url for the cluster entry in kubeconfig                          |
| --server string                                                        |
| server for the cluster entry in kubeconfig                             |
| --tls-server-name string                                               |
| tls-server-name for the cluster entry in kubeconfig                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --client-certificate string                                                                                  |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --kubeconfig string                                                                                                                                             |
| use a particular kubeconfig file                                                                                                                                |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string    Default: "none"                                                                                                                                                               |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --token string                                                                                                                                                                                    |
| Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                                                                                                                                                                                     |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config set-context

## Synopsis

Set a context entry in kubeconfig.

Specifying a name that already exists will merge new fields on top of existing values for those fields.

```
kubectl config set-context [NAME | --current] [--cluster=cluster_nickname] [--user=user_nickname] [--namespace=namespace]
```

## Examples

```
Set the user field on the gce context entry without touching other values
kubectl config set-context gce --user=cluster-admin
```

## Options

|                                             |
|---------------------------------------------|
| --cluster string                            |
| cluster for the context entry in kubeconfig |
| --current                                   |
| Modify the current context                  |



|                                               |
|-----------------------------------------------|
| -h, --help                                    |
| help for set-context                          |
| --namespace string                            |
| namespace for the context entry in kubeconfig |
| --user string                                 |
| user for the context entry in kubeconfig      |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |

|                                                                           |                                                                                                                                                                                                   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--context string</code>                                             |                                                                                                                                                                                                   |
|                                                                           | The name of the kubeconfig context to use                                                                                                                                                         |
| <code>--default-not-ready-toleration-seconds int    Default: 300</code>   |                                                                                                                                                                                                   |
|                                                                           | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| <code>--default-unreachable-toleration-seconds int    Default: 300</code> |                                                                                                                                                                                                   |
|                                                                           | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| <code>--disable-compression</code>                                        |                                                                                                                                                                                                   |
|                                                                           | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| <code>--insecure-skip-tls-verify</code>                                   |                                                                                                                                                                                                   |
|                                                                           | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| <code>--kubeconfig string</code>                                          |                                                                                                                                                                                                   |
|                                                                           | use a particular kubeconfig file                                                                                                                                                                  |
| <code>--match-server-version</code>                                       |                                                                                                                                                                                                   |
|                                                                           | Require server version to match client version                                                                                                                                                    |
| <code>--password string</code>                                            |                                                                                                                                                                                                   |
|                                                                           | Password for basic authentication to the API server                                                                                                                                               |
| <code>--profile string    Default: "none"</code>                          |                                                                                                                                                                                                   |
|                                                                           | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| <code>--profile-output string    Default: "profile.pprof"</code>          |                                                                                                                                                                                                   |
|                                                                           | Name of the file to write the profile to                                                                                                                                                          |
| <code>--request-timeout string    Default: "0"</code>                     |                                                                                                                                                                                                   |
|                                                                           | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                          |                                                                                                                                                                                                   |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| The address and port of the Kubernetes API server                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

Treat warnings received from the server as errors and exit with a non-zero exit code

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config set-credentials

## Synopsis

Set a user entry in kubeconfig.

Specifying a name that already exists will merge new fields on top of existing values.

Client-certificate flags:

--client-certificate=certfile --client-key=keyfile

Bearer token flags:

--token=bearer\_token

Basic auth flags:

--username=basic\_user --password=basic\_password

Bearer token and basic auth are mutually exclusive.

```
kubectl config set-credentials NAME [--client-certificate=path/to/certfile] [--client-key=path/to/keyfile] [--token=bearer_token] [--username=basic_user] [--password=basic_password] [--auth-provider=provider_name] [--auth-provider-arg=key=value] [--exec-command=exec_command] [--exec-api-version=exec_api_version] [--exec-arg=arg] [--exec-env=key=value]
```

## Examples

```
Set only the "client-key" field on the "cluster-admin"
entry, without touching other values
kubectl config set-credentials cluster-admin --client-key=~/.kube/admin.key

Set basic auth for the "cluster-admin" entry
kubectl config set-credentials cluster-admin --username=admin --password=uXFGweU9l35qcif

Embed client certificate data in the "cluster-admin" entry
kubectl config set-credentials cluster-admin --client-certificate=~/.kube/admin.crt --embed-certs=true

Enable the Google Compute Platform auth provider for the "cluster-admin" entry
kubectl config set-credentials cluster-admin --auth-provider=gcp

Enable the OpenID Connect auth provider for the "cluster-admin" entry with additional
arguments
```

```

kubectrl config set-credentials cluster-admin --auth-provider=oidc --auth-provider-arg=client-id=foo --auth-provider-arg=client-secret=bar

Remove the "client-secret" config value for the OpenID Connect auth provider for the "cluster-admin" entry
kubectrl config set-credentials cluster-admin --auth-provider=oidc --auth-provider-arg=client-secret-

Enable new exec auth plugin for the "cluster-admin" entry
kubectrl config set-credentials cluster-admin --exec-command=/path/to/the/executable --exec-api-version=client.authentication.k8s.io/v1beta1

Enable new exec auth plugin for the "cluster-admin" entry with interactive mode
kubectrl config set-credentials cluster-admin --exec-command=/path/to/the/executable --exec-api-version=client.authentication.k8s.io/v1beta1 --exec-interactive-mode=Never

Define new exec auth plugin arguments for the "cluster-admin" entry
kubectrl config set-credentials cluster-admin --exec-arg=arg1 --exec-arg=arg2

Create or update exec auth plugin environment variables for the "cluster-admin" entry
kubectrl config set-credentials cluster-admin --exec-env=key1=val1 --exec-env=key2=val2

Remove exec auth plugin environment variables for the "cluster-admin" entry
kubectrl config set-credentials cluster-admin --exec-env=var-to-remove-

```

## Options

|                               |                                                                            |
|-------------------------------|----------------------------------------------------------------------------|
| --auth-provider string        |                                                                            |
|                               | Auth provider for the user entry in kubeconfig                             |
| --auth-provider-arg strings   |                                                                            |
|                               | 'key=value' arguments for the auth provider                                |
| --client-certificate string   |                                                                            |
|                               | Path to client-certificate file for the user entry in kubeconfig           |
| --client-key string           |                                                                            |
|                               | Path to client-key file for the user entry in kubeconfig                   |
| --embed-certs tristate[=true] |                                                                            |
|                               | Embed client cert/key for the user entry in kubeconfig                     |
| --exec-api-version string     |                                                                            |
|                               | API version of the exec credential plugin for the user entry in kubeconfig |

|                                                                                       |
|---------------------------------------------------------------------------------------|
| --exec-arg strings                                                                    |
| New arguments for the exec credential plugin command for the user entry in kubeconfig |
| --exec-command string                                                                 |
| Command for the exec credential plugin for the user entry in kubeconfig               |
| --exec-env strings                                                                    |
| 'key=value' environment values for the exec credential plugin                         |
| --exec-interactive-mode string                                                        |
| InteractiveMode of the exec credentials plugin for the user entry in kubeconfig       |
| --exec-provide-cluster-info tristate[=true]                                           |
| ProvideClusterInfo of the exec credentials plugin for the user entry in kubeconfig    |
| -h, --help                                                                            |
| help for set-credentials                                                              |
| --password string                                                                     |
| password for the user entry in kubeconfig                                             |
| --token string                                                                        |
| token for the user entry in kubeconfig                                                |
| --username string                                                                     |
| username for the user entry in kubeconfig                                             |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| use a particular kubeconfig file                                                                                                                                |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --profile string    Default: "none"                                                                                                                                                               |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --tls-server-name string                                                                                                                                                                          |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config unset

## Synopsis

Unset an individual value in a kubeconfig file.

PROPERTY\_NAME is a dot delimited name where each token represents either an attribute name or a map key. Map keys may not contain dots.

```
kubectl config unset PROPERTY_NAME
```

## Examples

```
Unset the current-context
kubectl config unset current-context

Unset namespace in foo context
kubectl config unset contexts.foo.namespace
```

## Options

|                |
|----------------|
| -h, --help     |
| help for unset |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |

|                                                                                                                                                                                                   |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| --default-unreachable-toleration-seconds int                                                                                                                                                      | Default: 300             |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |                          |
| --disable-compression                                                                                                                                                                             |                          |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |                          |
| --insecure-skip-tls-verify                                                                                                                                                                        |                          |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |                          |
| --kubeconfig string                                                                                                                                                                               |                          |
| use a particular kubeconfig file                                                                                                                                                                  |                          |
| --match-server-version                                                                                                                                                                            |                          |
| Require server version to match client version                                                                                                                                                    |                          |
| -n, --namespace string                                                                                                                                                                            |                          |
| If present, the namespace scope for this CLI request                                                                                                                                              |                          |
| --password string                                                                                                                                                                                 |                          |
| Password for basic authentication to the API server                                                                                                                                               |                          |
| --profile string                                                                                                                                                                                  | Default: "none"          |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |                          |
| --profile-output string                                                                                                                                                                           | Default: "profile.pprof" |
| Name of the file to write the profile to                                                                                                                                                          |                          |
| --request-timeout string                                                                                                                                                                          | Default: "0"             |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |                          |
| -s, --server string                                                                                                                                                                               |                          |
| The address and port of the Kubernetes API server                                                                                                                                                 |                          |
| --storage-driver-buffer-duration duration                                                                                                                                                         | Default: 1m0s            |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

Treat warnings received from the server as errors and exit with a non-zero exit code

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config use-context

## Synopsis

Set the current-context in a kubeconfig file.

```
kubectl config use-context CONTEXT_NAME
```

## Examples

```
Use the context for the minikube cluster
kubectl config use-context minikube
```

## Options

|                      |
|----------------------|
| -h, --help           |
| help for use-context |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |

|  |                                                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Path to a cert file for the certificate authority                                                                                                               |
|  | --client-certificate string                                                                                                                                     |
|  | Path to a client certificate file for TLS                                                                                                                       |
|  | --client-key string                                                                                                                                             |
|  | Path to a client key file for TLS                                                                                                                               |
|  | --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
|  | CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
|  | --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
|  | CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
|  | --cluster string                                                                                                                                                |
|  | The name of the kubeconfig cluster to use                                                                                                                       |
|  | --context string                                                                                                                                                |
|  | The name of the kubeconfig context to use                                                                                                                       |
|  | --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
|  | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
|  | --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
|  | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
|  | --disable-compression                                                                                                                                           |
|  | If true, opt-out of response compression for all requests to the server                                                                                         |
|  | --insecure-skip-tls-verify                                                                                                                                      |
|  | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
|  | --kubeconfig string                                                                                                                                             |
|  | use a particular kubeconfig file                                                                                                                                |
|  | --match-server-version                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Require server version to match client version                                                                                                                                                    |
| <code>-n, --namespace string</code>                                                                                                                                                               |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| <code>--password string</code>                                                                                                                                                                    |
| Password for basic authentication to the API server                                                                                                                                               |
| <code>--profile string</code> Default: "none"                                                                                                                                                     |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| <code>--profile-output string</code> Default: "profile.pprof"                                                                                                                                     |
| Name of the file to write the profile to                                                                                                                                                          |
| <code>--request-timeout string</code> Default: "0"                                                                                                                                                |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl config view

## Synopsis

Display merged kubeconfig settings or a specified kubeconfig file.

You can use --output jsonpath={...} to extract specific values using a jsonpath expression.

kubectl config view [flags]



## Examples

```
Show merged kubeconfig settings
kubectl config view

Show merged kubeconfig settings, raw certificate data, and exposed secrets
kubectl config view --raw

Get the password for the e2e user
kubectl config view -o jsonpath='{.users[?(@.name == "e2e")].user.password}'
```

## Options

|                                                                                                                                                 |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| --allow-missing-template-keys                                                                                                                   | Default: true   |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |                 |
| --flatten                                                                                                                                       |                 |
| Flatten the resulting kubeconfig file into self-contained output (useful for creating portable kubeconfig files)                                |                 |
| -h, --help                                                                                                                                      |                 |
| help for view                                                                                                                                   |                 |
| --merge tristate[=true]                                                                                                                         | Default: true   |
| Merge the full hierarchy of kubeconfig files                                                                                                    |                 |
| --minify                                                                                                                                        |                 |
| Remove all information not used by current-context from the output                                                                              |                 |
| -o, --output string                                                                                                                             | Default: "yaml" |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).    |                 |
| --raw                                                                                                                                           |                 |
| Display raw byte data and sensitive data                                                                                                        |                 |
| --show-managed-fields                                                                                                                           |                 |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                   |                 |
| --template string                                                                                                                               |                 |

Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [<http://golang.org/pkg/text/template/#pkg-overview>].

--as string

Username to impersonate for the operation. User could be a regular user or a service account in a namespace.

--as-group strings

Group to impersonate for the operation, this flag can be repeated to specify multiple groups.

--as-uid string

UID to impersonate for the operation.

--cache-dir string Default: "\$HOME/.kube/cache"

Default cache directory

--certificate-authority string

Path to a cert file for the certificate authority

--client-certificate string

Path to a client certificate file for TLS

--client-key string

Path to a client key file for TLS

--cloud-provider-gce-l7lb-src-cidrs cidrs Default: 130.211.0.0/22,35.191.0.0/16

CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks

--cloud-provider-gce-lb-src-cidrs cidrs Default:  
130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16

CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks

--cluster string

The name of the kubeconfig cluster to use

--context string

The name of the kubeconfig context to use

--default-not-ready-toleration-seconds int Default: 300

|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | use a particular kubeconfig file                                                                                                                                                                  |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                          |                                                                                                                                                                                                   |
|                                              | The address and port of the Kubernetes API server                                                                                                                                                 |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl config](#) - Modify kubeconfig files

# kubectl cordon

## Synopsis

Mark node as unschedulable.

```
kubectl cordon NODE
```

## Examples

```
Mark node "foo" as unschedulable
kubectl cordon foo
```

## Options

|                                                   |
|---------------------------------------------------|
| --dry-run string[="unchanged"]    Default: "none" |
|---------------------------------------------------|

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|            |
|------------|
| -h, --help |
|------------|

|                 |
|-----------------|
| help for cordon |
|-----------------|

|                       |
|-----------------------|
| -l, --selector string |
|-----------------------|

|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|             |
|-------------|
| --as string |
|-------------|

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
|--------------------------------------------------------------------------------------------------------------|

|                    |
|--------------------|
| --as-group strings |
|--------------------|

|                                                                                               |
|-----------------------------------------------------------------------------------------------|
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups. |
|-----------------------------------------------------------------------------------------------|

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--as-uid string</code>                                                                                                                                    |
| UID to impersonate for the operation.                                                                                                                           |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                                                                   |
| Default cache directory                                                                                                                                         |
| <code>--certificate-authority string</code>                                                                                                                     |
| Path to a cert file for the certificate authority                                                                                                               |
| <code>--client-certificate string</code>                                                                                                                        |
| Path to a client certificate file for TLS                                                                                                                       |
| <code>--client-key string</code>                                                                                                                                |
| Path to a client key file for TLS                                                                                                                               |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                                                    |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                   |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| <code>--cluster string</code>                                                                                                                                   |
| The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                                                                                                                   |
| The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300                                                                                            |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300                                                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                                                                                                              |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                                                                                                         |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager



# kubectl cp

## Synopsis

Copy files and directories to and from containers.

```
kubectl cp <file-spec-src> <file-spec-dest>
```

## Examples

```
!!!Important Note!!!
Requires that the 'tar' binary is present in your container
image. If 'tar' is not present, 'kubectl cp' will fail.
#
For advanced use cases, such as symlinks, wildcard expansion or
file mode preservation, consider using 'kubectl exec'.

Copy /tmp/foo local file to /tmp/bar in a remote pod in namespace <some-namespace>
tar cf - /tmp/foo | kubectl exec -i -n <some-namespace> <some-pod> -- tar xf - -C /tmp/bar

Copy /tmp/foo from a remote pod to /tmp/bar locally
kubectl exec -n <some-namespace> <some-pod> -- tar cf - /tmp/foo | tar xf - -C /tmp/bar

Copy /tmp/foo_dir local directory to /tmp/bar_dir in a remote pod in the default namespace
kubectl cp /tmp/foo_dir <some-pod>:/tmp/bar_dir

Copy /tmp/foo local file to /tmp/bar in a remote pod in a specific container
kubectl cp /tmp/foo <some-pod>:/tmp/bar -c <specific-container>

Copy /tmp/foo local file to /tmp/bar in a remote pod in namespace <some-namespace>
kubectl cp /tmp/foo <some-namespace>:<some-pod>:/tmp/bar

Copy /tmp/foo from a remote pod to /tmp/bar locally
kubectl cp <some-namespace>:<some-pod>:/tmp/foo /tmp/bar
```

## Options

|                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -c, --container string                                                                                                                                                             |
| Container name. If omitted, use the kubectl.kubernetes.io/default-container annotation for selecting the container to be attached or the first container in the pod will be chosen |
| -h, --help                                                                                                                                                                         |
| help for cp                                                                                                                                                                        |
| --no-preserve                                                                                                                                                                      |
| The copied file/directory's ownership and permissions will not be preserved in the container                                                                                       |

|                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --retries int                                                                                                                                                       |
| Set number of retries to complete a copy operation from a container. Specify 0 to disable or any negative value for infinite retrying. The default is 0 (no retry). |
| --as string                                                                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                        |
| --as-group strings                                                                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                       |
| --as-uid string                                                                                                                                                     |
| UID to impersonate for the operation.                                                                                                                               |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                                  |
| Default cache directory                                                                                                                                             |
| --certificate-authority string                                                                                                                                      |
| Path to a cert file for the certificate authority                                                                                                                   |
| --client-certificate string                                                                                                                                         |
| Path to a client certificate file for TLS                                                                                                                           |
| --client-key string                                                                                                                                                 |
| Path to a client key file for TLS                                                                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                                   |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                                  |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                                |
| --cluster string                                                                                                                                                    |
| The name of the kubeconfig cluster to use                                                                                                                           |
| --context string                                                                                                                                                    |
| The name of the kubeconfig context to use                                                                                                                           |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                                                        |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                                                      |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string    Default: "none"                                                                                                                                                               |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| The address and port of the Kubernetes API server                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>--version, --version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
|----------------------------------------------------------------------------------------------------------------------------------------|

|                                   |
|-----------------------------------|
| <code>--warnings-as-errors</code> |
|-----------------------------------|

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl create

## Synopsis

Create a resource from a file or from stdin.

JSON and YAML formats are accepted.

```
kubectl create -f FILENAME
```

## Examples

```
Create a pod using the data in pod.json
kubectl create -f ./pod.json
```

```
Create a pod based on the JSON passed into stdin
cat pod.json | kubectl create -f -
```

```
Edit the data in registry.yaml in JSON then create the resource using the edited data
kubectl create -f registry.yaml --edit -o json
```

## Options

|                                                          |
|----------------------------------------------------------|
| <code>--allow-missing-template-keys</code> Default: true |
|----------------------------------------------------------|

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
|-------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                             |
|-------------------------------------------------------------|
| <code>--dry-run string[="unchanged"]</code> Default: "none" |
|-------------------------------------------------------------|

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                     |
|---------------------|
| <code>--edit</code> |
|---------------------|

|                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Edit the API resource before creating                                                                                                                                                                               |
| --field-manager string    Default: "kubectl-create"                                                                                                                                                                 |
| Name of the manager used to track field ownership.                                                                                                                                                                  |
| -f, --filename strings                                                                                                                                                                                              |
| Filename, directory, or URL to files to use to create the resource                                                                                                                                                  |
| -h, --help                                                                                                                                                                                                          |
| help for create                                                                                                                                                                                                     |
| -k, --kustomize string                                                                                                                                                                                              |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                |
| -o, --output string                                                                                                                                                                                                 |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                        |
| --raw string                                                                                                                                                                                                        |
| Raw URI to POST to the server. Uses the transport specified by the kubeconfig file.                                                                                                                                 |
| -R, --recursive                                                                                                                                                                                                     |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                     |
| --save-config                                                                                                                                                                                                       |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |
| -l, --selector string                                                                                                                                                                                               |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                          |
| --show-managed-fields                                                                                                                                                                                               |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                       |
| --template string                                                                                                                                                                                                   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --windows-line-endings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Only relevant if --edit=true. Defaults to the line ending native to your platform.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                           |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |



|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --tls-server-name string                                                                                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Bearer token for authentication to the API server                                                            |
| --user string                                                                                                |
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl create clusterrole](#) - Create a cluster role
- [kubectl create clusterrolebinding](#) - Create a cluster role binding for a particular cluster role
- [kubectl create configmap](#) - Create a config map from a local file, directory or literal value
- [kubectl create cronjob](#) - Create a cron job with the specified name
- [kubectl create deployment](#) - Create a deployment with the specified name
- [kubectl create ingress](#) - Create an ingress with the specified name
- [kubectl create job](#) - Create a job with the specified name
- [kubectl create namespace](#) - Create a namespace with the specified name
- [kubectl create poddisruptionbudget](#) - Create a pod disruption budget with the specified name
- [kubectl create priorityclass](#) - Create a priority class with the specified name
- [kubectl create quota](#) - Create a quota with the specified name
- [kubectl create role](#) - Create a role with single rule
- [kubectl create rolebinding](#) - Create a role binding for a particular role or cluster role
- [kubectl create secret](#) - Create a secret using a specified subcommand
- [kubectl create service](#) - Create a service using a specified subcommand
- [kubectl create serviceaccount](#) - Create a service account with the specified name
- [kubectl create token](#) - Request a service account token

# kubectl create clusterrole

## Synopsis

Create a cluster role.

```
kubectl create clusterrole NAME --verb=verb --resource=resource.group [--resource-
name=resourcename] [--dry-run=server|client|none]
```

## Examples

```
Create a cluster role named "pod-reader" that allows user to perform "get", "watch" and "list"
on pods
kubectl create clusterrole pod-reader --verb=get,list,watch --resource=pods

Create a cluster role named "pod-reader" with ResourceName specified
kubectl create clusterrole pod-reader --verb=get --resource=pods --resource-
name=readablepod --resource-name=anotherpod

Create a cluster role named "foo" with API Group specified
kubectl create clusterrole foo --verb=get,list,watch --resource=rs.apps

Create a cluster role named "foo" with SubResource specified
kubectl create clusterrole foo --verb=get,list,watch --resource=pods,pods/status

Create a cluster role name "foo" with NonResourceURL specified
kubectl create clusterrole "foo" --verb=get --non-resource-url=/logs/*

Create a cluster role name "monitoring" with AggregationRule specified
kubectl create clusterrole monitoring --aggregation-rule="rbac.example.com/aggregate-to-
monitoring=true"
```

## Options

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --aggregation-rule <comma-separated 'key=value' pairs>                                                                                                                                                   |
| An aggregation label selector for combining ClusterRoles.                                                                                                                                                |
| --allow-missing-template-keys    Default: true                                                                                                                                                           |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                         |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                        |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| --field-manager string    Default: "kubectl-create"                                                                                                                                                      |
| Name of the manager used to track field ownership.                                                                                                                                                       |
| -h, --help                                                                                                                                                                                               |
| help for clusterrole                                                                                                                                                                                     |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --non-resource-url strings                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | A partial url that user should have access to.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| -o, --output string                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --resource strings                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | Resource that the rule applies to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| --resource-name strings                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | Resource in the white list that the rule applies to, repeat this flag for multiple items                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --save-config                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --show-managed-fields                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --template string                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                                               |
| --validate string[="strict"]    Default: "strict" |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | <p>Must be one of: strict (or true), warn, ignore (or false).</p> <p>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.</p> <p>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.</p> <p>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.</p> |
| --verb strings                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                   | Verb that applies to the resources contained in the rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --as string                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create clusterrolebinding

## Synopsis

Create a cluster role binding for a particular cluster role.

```
kubectl create clusterrolebinding NAME --clusterrole=NAME [--user=username] [--group=groupname] [--serviceaccount=namespace:serviceaccountname] [--dry-run=server|client|none]
```

## Examples

```
Create a cluster role binding for user1, user2, and group1 using the cluster-admin cluster role
kubectl create clusterrolebinding cluster-admin --clusterrole=cluster-admin --user=user1 --user=user2 --group=group1
```

## Options

|                                                                                                                                                                                                          |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                            | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyang and jsonpath output formats.                                                          |                           |
| --clusterrole string                                                                                                                                                                                     |                           |
| ClusterRole this ClusterRoleBinding should reference                                                                                                                                                     |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                           | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |                           |
| --field-manager string                                                                                                                                                                                   | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                       |                           |
| --group strings                                                                                                                                                                                          |                           |
| Groups to bind to the clusterrole. The flag can be repeated to add multiple groups.                                                                                                                      |                           |
| -h, --help                                                                                                                                                                                               |                           |
| help for clusterrolebinding                                                                                                                                                                              |                           |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-o, --output string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>--save-config</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>--serviceaccount strings</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Service accounts to bind to the clusterrole, in the format <namespace>:<name>. The flag can be repeated to add multiple service accounts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>--show-managed-fields</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>--template string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                            |
| <code>--user strings</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Username to bind to the clusterrole. The flag can be repeated to add multiple users.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>--validate string[="strict"]</code> Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| <code>--as string</code>                                                                                     |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| <code>--as-group strings</code>                                                                              |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| <code>--as-uid string</code>                                                                                 |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                             |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubect create](#) - Create a resource from a file or from stdin

# kubect create configmap

## Synopsis

Create a config map based on a file, directory, or specified literal value.

A single config map may package one or more key/value pairs.

When creating a config map based on a file, the key will default to the basename of the file, and the value will default to the file content. If the basename is an invalid key, you may specify an alternate key.

When creating a config map based on a directory, each file whose basename is a valid key in the directory will be packaged into the config map. Any directory entries except regular files are ignored (e.g. subdirectories, symlinks, devices, pipes, etc).

```
kubectl create configmap NAME [--from-file=[key=]source] [--from-literal=key1=value1] [--dry-run=server|client|none]
```

## Examples

```
Create a new config map named my-config based on folder bar
kubectl create configmap my-config --from-file=path/to/bar

Create a new config map named my-config with specified keys instead of file basenames on disk
kubectl create configmap my-config --from-file=key1=/path/to/bar/file1.txt --from-file=key2=/path/to/bar/file2.txt

Create a new config map named my-config with key1=config1 and key2=config2
kubectl create configmap my-config --from-literal=key1=config1 --from-literal=key2=config2

Create a new config map named my-config from the key=value pairs in the file
kubectl create configmap my-config --from-file=path/to/bar

Create a new config map named my-config from an env file
kubectl create configmap my-config --from-env-file=path/to/foo.env --from-env-file=path/to/bar.env
```

## Options

|                                                                                                                                                                                                          |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                            | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |                           |
| --append-hash                                                                                                                                                                                            |                           |
| Append a hash of the configmap to its name.                                                                                                                                                              |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                           | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |                           |
| --field-manager string                                                                                                                                                                                   | Default: "kubectl-create" |

|                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of the manager used to track field ownership.                                                                                                                                                                                                                                                        |
| --from-env-file strings                                                                                                                                                                                                                                                                                   |
| Specify the path to a file to read lines of key=val pairs to create a configmap.                                                                                                                                                                                                                          |
| --from-file strings                                                                                                                                                                                                                                                                                       |
| Key file can be specified using its file path, in which case file basename will be used as configmap key, or optionally with a key and file path, in which case the given key will be used. Specifying a directory will iterate each named file in the directory whose basename is a valid configmap key. |
| --from-literal strings                                                                                                                                                                                                                                                                                    |
| Specify a key and literal value to insert in configmap (i.e. mykey=somevalue)                                                                                                                                                                                                                             |
| -h, --help                                                                                                                                                                                                                                                                                                |
| help for configmap                                                                                                                                                                                                                                                                                        |
| -o, --output string                                                                                                                                                                                                                                                                                       |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                              |
| --save-config                                                                                                                                                                                                                                                                                             |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                       |
| --show-managed-fields                                                                                                                                                                                                                                                                                     |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                 |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                         |

Must be one of: strict (or true), warn, ignore (or false).  
 "true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.  
 "warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.  
 "false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.        |
| --as-group strings                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                       |
| --as-uid string                                                                                                     |
| UID to impersonate for the operation.                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                             |
| --certificate-authority string                                                                                      |
| Path to a cert file for the certificate authority                                                                   |
| --client-certificate string                                                                                         |
| Path to a client certificate file for TLS                                                                           |
| --client-key string                                                                                                 |
| Path to a client key file for TLS                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                |
| --cluster string                                                                                                    |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |



|                                                                                                                                                                                                   |
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| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                                                                  |
| <code>--version version[=true]</code>                                                                                                                |
| <code>--version</code> , <code>--version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
| <code>--warnings-as-errors</code>                                                                                                                    |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                                 |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create cronjob

## Synopsis

Create a cron job with the specified name.

```
kubectl create cronjob NAME --image=image --schedule='0/5 * * * ?' -- [COMMAND] [args...]
[flags]
```

## Examples

```
Create a cron job
kubectl create cronjob my-job --image=busybox --schedule="*/1 * * * *"

Create a cron job with a command
kubectl create cronjob my-job --image=busybox --schedule="*/1 * * * *" -- date
```

## Options

|                                                                                                                                                                                                          |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <code>--allow-missing-template-keys</code>                                                                                                                                                               | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |                           |
| <code>--dry-run string[="unchanged"]</code>                                                                                                                                                              | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |                           |
| <code>--field-manager string</code>                                                                                                                                                                      | Default: "kubectl-create" |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| Name of the manager used to track field ownership.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| help for cronjob                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --image string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Image name to run.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -o, --output string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --restart string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| job's restart policy. supported values: OnFailure, Never                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --save-config                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --schedule string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| A schedule in the Cron format the job should be run with.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                                               |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>Must be one of: strict (or true), warn, ignore (or false).</p> <p>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.</p> <p>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.</p> <p>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.</p> |
| --as string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
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| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create deployment

## Synopsis

Create a deployment with the specified name.

```
kubectl create deployment NAME --image=image -- [COMMAND] [args...]
```

## Examples

```
Create a deployment named my-dep that runs the busybox image
kubectl create deployment my-dep --image=busybox

Create a deployment with a command
kubectl create deployment my-dep --image=busybox -- date

Create a deployment named my-dep that runs the nginx image with 3 replicas
kubectl create deployment my-dep --image=nginx --replicas=3

Create a deployment named my-dep that runs the busybox image and expose port 5701
kubectl create deployment my-dep --image=busybox --port=5701

Create a deployment named my-dep that runs multiple containers
kubectl create deployment my-dep --image=busybox:latest --image=ubuntu:latest --
image=nginx
```

## Options

|                                                                                                                                                                                                          |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                            | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                         |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                           | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |                           |
| --field-manager string                                                                                                                                                                                   | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                       |                           |
| -h, --help                                                                                                                                                                                               |                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| help for deployment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --image strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Image names to run. A deployment can have multiple images set for multi-container pod.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| -o, --output string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --port int32    Default: -1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| The containerPort that this deployment exposes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -r, --replicas int32    Default: 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Number of replicas to create. Default is 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| --save-config                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                            |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --as string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



|                                                                                                                               |                                                                                                                                                                 |
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| <code>--as-group strings</code>                                                                                               |                                                                                                                                                                 |
|                                                                                                                               | Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                   |
| <code>--as-uid string</code>                                                                                                  |                                                                                                                                                                 |
|                                                                                                                               | UID to impersonate for the operation.                                                                                                                           |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                                 |                                                                                                                                                                 |
|                                                                                                                               | Default cache directory                                                                                                                                         |
| <code>--certificate-authority string</code>                                                                                   |                                                                                                                                                                 |
|                                                                                                                               | Path to a cert file for the certificate authority                                                                                                               |
| <code>--client-certificate string</code>                                                                                      |                                                                                                                                                                 |
|                                                                                                                               | Path to a client certificate file for TLS                                                                                                                       |
| <code>--client-key string</code>                                                                                              |                                                                                                                                                                 |
|                                                                                                                               | Path to a client key file for TLS                                                                                                                               |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                  |                                                                                                                                                                 |
|                                                                                                                               | CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |                                                                                                                                                                 |
|                                                                                                                               | CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| <code>--cluster string</code>                                                                                                 |                                                                                                                                                                 |
|                                                                                                                               | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                                                                                 |                                                                                                                                                                 |
|                                                                                                                               | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300                                                          |                                                                                                                                                                 |
|                                                                                                                               | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300                                                        |                                                                                                                                                                 |
|                                                                                                                               | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                                                                            |                                                                                                                                                                 |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | If true, opt-out of response compression for all requests to the server                                                                                                                           |
|  | --insecure-skip-tls-verify                                                                                                                                                                        |
|  | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
|  | --kubeconfig string                                                                                                                                                                               |
|  | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
|  | --match-server-version                                                                                                                                                                            |
|  | Require server version to match client version                                                                                                                                                    |
|  | -n, --namespace string                                                                                                                                                                            |
|  | If present, the namespace scope for this CLI request                                                                                                                                              |
|  | --password string                                                                                                                                                                                 |
|  | Password for basic authentication to the API server                                                                                                                                               |
|  | --profile string   Default: "none"                                                                                                                                                                |
|  | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
|  | --profile-output string   Default: "profile.pprof"                                                                                                                                                |
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | --request-timeout string   Default: "0"                                                                                                                                                           |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | -s, --server string                                                                                                                                                                               |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
|  | database name                                                                                                                                                                                     |
|  | --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database host:port                                                                                                           |
| --storage-driver-password string    Default: "root"                                                                          |
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl create](#) - Create a resource from a file or from stdin

# kubectl create ingress

## Synopsis

Create an ingress with the specified name.

```
kubectl create ingress NAME --rule=host/path=service:port[,tls[=secret]]
```

## Examples

```
Create a single ingress called 'simple' that directs requests to foo.com/bar to svc
svc1:8080 with a TLS secret "my-cert"
kubectl create ingress simple --rule="foo.com/bar=svc1:8080,tls=my-cert"
```

```
Create a catch all ingress of "/"path" pointing to service svc:port and Ingress Class as
"otheringress"
kubectl create ingress catch-all --class=otheringress --rule="/path=svc:port"
```

```
Create an ingress with two annotations: ingress.annotation1 and ingress.annotations2
kubectl create ingress annotated --class=default --rule="foo.com/bar=svc:port" \
--annotation ingress.annotation1=foo \
--annotation ingress.annotation2=bla
```

```
Create an ingress with the same host and multiple paths
kubectl create ingress multipath --class=default \
--rule="foo.com/=svc:port" \
--rule="foo.com/admin/=svcadmin:portadmin"
```

```
Create an ingress with multiple hosts and the pathType as Prefix
kubectl create ingress ingress1 --class=default \
--rule="foo.com/path*=svc:8080" \
--rule="bar.com/admin*=svc2:http"
```

```
Create an ingress with TLS enabled using the default ingress certificate and different path
types
kubectl create ingress ingtls --class=default \
--rule="foo.com/=svc:https,tls" \
--rule="foo.com/path/subpath*=othersvc:8080"
```

```
Create an ingress with TLS enabled using a specific secret and pathType as Prefix
kubectl create ingress ingsecret --class=default \
--rule="foo.com/*=svc:8080,tls=secret1"
```

```
Create an ingress with a default backend
kubectl create ingress ingdefault --class=default \
--default-backend=defaultsvc:http \
--rule="foo.com/*=svc:8080,tls=secret1"
```

# Options

|                                                                                                                                                                                                                     |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                                       | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                                    |                           |
| --annotation strings                                                                                                                                                                                                |                           |
| Annotation to insert in the ingress object, in the format annotation=value                                                                                                                                          |                           |
| --class string                                                                                                                                                                                                      |                           |
| Ingress Class to be used                                                                                                                                                                                            |                           |
| --default-backend string                                                                                                                                                                                            |                           |
| Default service for backend, in format of svcname:port                                                                                                                                                              |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                                      | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.            |                           |
| --field-manager string                                                                                                                                                                                              | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                  |                           |
| -h, --help                                                                                                                                                                                                          |                           |
| help for ingress                                                                                                                                                                                                    |                           |
| -o, --output string                                                                                                                                                                                                 |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                        |                           |
| --rule strings                                                                                                                                                                                                      |                           |
| Rule in format host/path=service:port[,tls=secretname]. Paths containing the leading character '*' are considered pathType=Prefix. tls argument is optional.                                                        |                           |
| --save-config                                                                                                                                                                                                       |                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |                           |
| --show-managed-fields                                                                                                                                                                                               |                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                           |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --tls-server-name string                                                                                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                                                                                                                                                                                    |



|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Bearer token for authentication to the API server                                                            |
| --user string                                                                                                |
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create job

## Synopsis

Create a job with the specified name.

```
kubectl create job NAME --image=image [--from=cronjob/name] -- [COMMAND] [args...]
```

## Examples

```
Create a job
kubectl create job my-job --image=busybox

Create a job with a command
kubectl create job my-job --image=busybox -- date

Create a job from a cron job named "a-cronjob"
kubectl create job test-job --from=cronjob/a-cronjob
```

## Options

|                               |               |
|-------------------------------|---------------|
| --allow-missing-template-keys | Default: true |
|-------------------------------|---------------|

|                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                                                                                                |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                                                                                                                                              |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                                                                       |
| --field-manager string    Default: "kubectl-create"                                                                                                                                                                                                                                                                            |
| Name of the manager used to track field ownership.                                                                                                                                                                                                                                                                             |
| --from string                                                                                                                                                                                                                                                                                                                  |
| The name of the resource to create a Job from (only cronjob is supported).                                                                                                                                                                                                                                                     |
| -h, --help                                                                                                                                                                                                                                                                                                                     |
| help for job                                                                                                                                                                                                                                                                                                                   |
| --image string                                                                                                                                                                                                                                                                                                                 |
| Image name to run.                                                                                                                                                                                                                                                                                                             |
| -o, --output string                                                                                                                                                                                                                                                                                                            |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                   |
| --save-config                                                                                                                                                                                                                                                                                                                  |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                            |
| --show-managed-fields                                                                                                                                                                                                                                                                                                          |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                  |
| --template string                                                                                                                                                                                                                                                                                                              |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                       |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                              |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not. |

"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.  
"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |
| --cluster string                                                                                                   |
| The name of the kubeconfig cluster to use                                                                          |
| --context string                                                                                                   |
| The name of the kubeconfig context to use                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                                                        |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                                                      |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string    Default: "none"                                                                                                                                                               |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| The address and port of the Kubernetes API server                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>--version, --version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
|----------------------------------------------------------------------------------------------------------------------------------------|

|                                   |
|-----------------------------------|
| <code>--warnings-as-errors</code> |
|-----------------------------------|

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create namespace

## Synopsis

Create a namespace with the specified name.

```
kubectl create namespace NAME [--dry-run=server|client|none]
```

## Examples

```
Create a new namespace named my-namespace
kubectl create namespace my-namespace
```

## Options

|                                                          |
|----------------------------------------------------------|
| <code>--allow-missing-template-keys</code> Default: true |
|----------------------------------------------------------|

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
|-------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                             |
|-------------------------------------------------------------|
| <code>--dry-run string[="unchanged"]</code> Default: "none" |
|-------------------------------------------------------------|

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                               |
|---------------------------------------------------------------|
| <code>--field-manager string</code> Default: "kubectl-create" |
|---------------------------------------------------------------|

|                                                    |
|----------------------------------------------------|
| Name of the manager used to track field ownership. |
|----------------------------------------------------|

|                         |
|-------------------------|
| <code>-h, --help</code> |
|-------------------------|

|                    |
|--------------------|
| help for namespace |
|--------------------|

|                                  |
|----------------------------------|
| <code>-o, --output string</code> |
|----------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --save-config                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                            |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --as string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --as-group strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --as-uid string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| UID to impersonate for the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Default cache directory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| --certificate-authority string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Path to a cert file for the certificate authority                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--client-certificate string</code>                                                                                                                        |
| Path to a client certificate file for TLS                                                                                                                       |
| <code>--client-key string</code>                                                                                                                                |
| Path to a client key file for TLS                                                                                                                               |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                                                    |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                      |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| <code>--cluster string</code>                                                                                                                                   |
| The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                                                                                                                   |
| The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300                                                                                            |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300                                                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                                                                                                              |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                                                                                                         |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                                                                                                                |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                                                                                                             |
| Require server version to match client version                                                                                                                  |



|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |

|                                                                                                                              |
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| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubecttl create](#) - Create a resource from a file or from stdin

# kubectl create poddisruptionbudget

## Synopsis

Create a pod disruption budget with the specified name, selector, and desired minimum available pods.

```
kubectl create poddisruptionbudget NAME --selector=SELECTOR --min-available=N [--dry-run=server|client|none]
```

## Examples

```
Create a pod disruption budget named my-pdb that will select all pods with the app=rails label
and require at least one of them being available at any point in time
```

```
kubectl create poddisruptionbudget my-pdb --selector=app=rails --min-available=1

Create a pod disruption budget named my-pdb that will select all pods with the app=nginx
label
and require at least half of the pods selected to be available at any point in time
kubectl create pdb my-pdb --selector=app=nginx --min-available=50%
```

## Options

|                                                                                                                                                                                                                     |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                                       | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                     |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                                      | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.            |                           |
| --field-manager string                                                                                                                                                                                              | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                  |                           |
| -h, --help                                                                                                                                                                                                          |                           |
| help for poddisruptionbudget                                                                                                                                                                                        |                           |
| --max-unavailable string                                                                                                                                                                                            |                           |
| The maximum number or percentage of unavailable pods this budget requires.                                                                                                                                          |                           |
| --min-available string                                                                                                                                                                                              |                           |
| The minimum number or percentage of available pods this budget requires.                                                                                                                                            |                           |
| -o, --output string                                                                                                                                                                                                 |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                        |                           |
| --save-config                                                                                                                                                                                                       |                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |                           |
| --selector string                                                                                                                                                                                                   |                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A label selector to use for this budget. Only equality-based selector requirements are supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                                              |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>Must be one of: strict (or true), warn, ignore (or false).</p> <p>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.</p> <p>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.</p> <p>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.</p> |

|                                                                                                              |
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| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |

|                                                                                                                                                                 |
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| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                          |
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| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string   Default: "none"</p>                                                                                                                                                                |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string   Default: "profile.pprof"</p>                                                                                                                                                |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string   Default: "0"</p>                                                                                                                                                           |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration   Default: 1m0s</p>                                                                                                                                         |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string   Default: "cadvisor"</p>                                                                                                                                                  |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string   Default: "localhost:8086"</p>                                                                                                                                          |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string   Default: "root"</p>                                                                                                                                                |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string   Default: "stats"</p>                                                                                                                                                  |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string   Default: "root"</p>                                                                                                                                                    |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |

|                                                                                                                              |
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| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl create](#) - Create a resource from a file or from stdin

# kubectrl create priorityclass

## Synopsis

Create a priority class with the specified name, value, globalDefault and description.

```
kubectrl create priorityclass NAME --value=VALUE --global-default=BOOL [--dry-run=server|client|none]
```

## Examples

```
Create a priority class named high-priority
kubectrl create priorityclass high-priority --value=1000 --description="high priority"

Create a priority class named default-priority that is considered as the global default priority
kubectrl create priorityclass default-priority --value=1000 --global-default=true --description="default priority"

Create a priority class named high-priority that cannot preempt pods with lower priority
```

```
kubectcl create priorityclass high-priority --value=1000 --description="high priority" --preemption-policy="Never"
```

## Options

|                                                                                                                                                                                                                      |                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| --allow-missing-template-keys                                                                                                                                                                                        | Default: true                   |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                      |                                 |
| --description string                                                                                                                                                                                                 |                                 |
| description is an arbitrary string that usually provides guidelines on when this priority class should be used.                                                                                                      |                                 |
| --dry-run string[="unchanged"]                                                                                                                                                                                       | Default: "none"                 |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.             |                                 |
| --field-manager string                                                                                                                                                                                               | Default: "kubectcl-create"      |
| Name of the manager used to track field ownership.                                                                                                                                                                   |                                 |
| --global-default                                                                                                                                                                                                     |                                 |
| global-default specifies whether this PriorityClass should be considered as the default priority.                                                                                                                    |                                 |
| -h, --help                                                                                                                                                                                                           |                                 |
| help for priorityclass                                                                                                                                                                                               |                                 |
| -o, --output string                                                                                                                                                                                                  |                                 |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                         |                                 |
| --preemption-policy string                                                                                                                                                                                           | Default: "PreemptLowerPriority" |
| preemption-policy is the policy for preempting pods with lower priority.                                                                                                                                             |                                 |
| --save-config                                                                                                                                                                                                        |                                 |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectcl apply on this object in the future. |                                 |
| --show-managed-fields                                                                                                                                                                                                |                                 |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --value int32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| the value of this priority class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |

|                                                                                                                                                                 |
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| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                          |
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| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string   Default: "none"</p>                                                                                                                                                                |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string   Default: "profile.pprof"</p>                                                                                                                                                |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string   Default: "0"</p>                                                                                                                                                           |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration   Default: 1m0s</p>                                                                                                                                         |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string   Default: "cadvisor"</p>                                                                                                                                                  |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string   Default: "localhost:8086"</p>                                                                                                                                          |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string   Default: "root"</p>                                                                                                                                                |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string   Default: "stats"</p>                                                                                                                                                  |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string   Default: "root"</p>                                                                                                                                                    |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl create](#) - Create a resource from a file or from stdin

# kubectrl create quota

## Synopsis

Create a resource quota with the specified name, hard limits, and optional scopes.

```
kubectrl create quota NAME [--hard=key1=value1,key2=value2] [--scopes=Scope1,Scope2] [--dry-run=server|client|none]
```

## Examples

```
Create a new resource quota named my-quota
kubectrl create quota my-quota --
hard=cpu=1,memory=1G,pods=2,services=3,replicationcontrollers=2,resourcequotas=1,secrets=
5,persistentvolumeclaims=10
```

```
Create a new resource quota named best-effort
kubectrl create quota best-effort --hard=pods=100 --scopes=BestEffort
```

# Options

|                                                                                                                                                                                                                                                          |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                                                                            | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                                                                           | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |                           |
| --field-manager string                                                                                                                                                                                                                                   | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                                                       |                           |
| --hard string                                                                                                                                                                                                                                            |                           |
| A comma-delimited set of resource=quantity pairs that define a hard limit.                                                                                                                                                                               |                           |
| -h, --help                                                                                                                                                                                                                                               |                           |
| help for quota                                                                                                                                                                                                                                           |                           |
| -o, --output string                                                                                                                                                                                                                                      |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                             |                           |
| --save-config                                                                                                                                                                                                                                            |                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                      |                           |
| --scopes string                                                                                                                                                                                                                                          |                           |
| A comma-delimited set of quota scopes that must all match each object tracked by the quota.                                                                                                                                                              |                           |
| --show-managed-fields                                                                                                                                                                                                                                    |                           |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |                           |
| --template string                                                                                                                                                                                                                                        |                           |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |                           |
| --validate string[="strict"]                                                                                                                                                                                                                             | Default: "strict"         |

Must be one of: strict (or true), warn, ignore (or false).  
 "true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.  
 "warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.  
 "false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.        |
| --as-group strings                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                       |
| --as-uid string                                                                                                     |
| UID to impersonate for the operation.                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                             |
| --certificate-authority string                                                                                      |
| Path to a cert file for the certificate authority                                                                   |
| --client-certificate string                                                                                         |
| Path to a client certificate file for TLS                                                                           |
| --client-key string                                                                                                 |
| Path to a client key file for TLS                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                |
| --cluster string                                                                                                    |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
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| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |



|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                                                                  |
| <code>--version version[=true]</code>                                                                                                                |
| <code>--version</code> , <code>--version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
| <code>--warnings-as-errors</code>                                                                                                                    |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                                 |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create role

## Synopsis

Create a role with single rule.

```
kubectl create role NAME --verb=verb --resource=resource.group/subresource [--resource-name=resourcename] [--dry-run=server|client|none]
```

## Examples

```
Create a role named "pod-reader" that allows user to perform "get", "watch" and "list" on pods
kubectl create role pod-reader --verb=get --verb=list --verb=watch --resource=pods

Create a role named "pod-reader" with ResourceName specified
kubectl create role pod-reader --verb=get --resource=pods --resource-name=readablepod --resource-name=anotherpod

Create a role named "foo" with API Group specified
kubectl create role foo --verb=get,list,watch --resource=rs.apps

Create a role named "foo" with SubResource specified
kubectl create role foo --verb=get,list,watch --resource=pods,pods/status
```

## Options

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--allow-missing-template-keys</code> Default: true                                                                                        |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |

|                                                                                                                                                                                                                                                           |                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <code>--dry-run string[="unchanged"]</code>                                                                                                                                                                                                               | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |                           |
| <code>--field-manager string</code>                                                                                                                                                                                                                       | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                                                        |                           |
| <code>-h, --help</code>                                                                                                                                                                                                                                   |                           |
| help for role                                                                                                                                                                                                                                             |                           |
| <code>-o, --output string</code>                                                                                                                                                                                                                          |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                              |                           |
| <code>--resource strings</code>                                                                                                                                                                                                                           |                           |
| Resource that the rule applies to                                                                                                                                                                                                                         |                           |
| <code>--resource-name strings</code>                                                                                                                                                                                                                      |                           |
| Resource in the white list that the rule applies to, repeat this flag for multiple items                                                                                                                                                                  |                           |
| <code>--save-config</code>                                                                                                                                                                                                                                |                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |                           |
| <code>--show-managed-fields</code>                                                                                                                                                                                                                        |                           |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |                           |
| <code>--template string</code>                                                                                                                                                                                                                            |                           |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |                           |
| <code>--validate string[="strict"]</code>                                                                                                                                                                                                                 | Default: "strict"         |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --verb strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Verb that applies to the resources contained in the rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | --request-timeout string    Default: "0"                                                                                                                                                          |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | -s, --server string                                                                                                                                                                               |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
|  | database name                                                                                                                                                                                     |
|  | --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
|  | database host:port                                                                                                                                                                                |
|  | --storage-driver-password string    Default: "root"                                                                                                                                               |
|  | database password                                                                                                                                                                                 |
|  | --storage-driver-secure                                                                                                                                                                           |
|  | use secure connection with database                                                                                                                                                               |
|  | --storage-driver-table string    Default: "stats"                                                                                                                                                 |
|  | table name                                                                                                                                                                                        |
|  | --storage-driver-user string    Default: "root"                                                                                                                                                   |
|  | database username                                                                                                                                                                                 |
|  | --tls-server-name string                                                                                                                                                                          |
|  | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
|  | --token string                                                                                                                                                                                    |
|  | Bearer token for authentication to the API server                                                                                                                                                 |
|  | --user string                                                                                                                                                                                     |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create rolebinding

## Synopsis

Create a role binding for a particular role or cluster role.

```
kubectl create rolebinding NAME --clusterrole=NAME|--role=NAME [--user=username] [--group=groupname] [--serviceaccount=namespace:serviceaccountname] [--dry-run=server|client|none]
```

## Examples

```
Create a role binding for user1, user2, and group1 using the admin cluster role
kubectl create rolebinding admin --clusterrole=admin --user=user1 --user=user2 --group=group1
```

```
Create a role binding for serviceaccount monitoring:sa-dev using the admin role
kubectl create rolebinding admin-binding --role=admin --serviceaccount=monitoring:sa-dev
```

## Options

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| --allow-missing-template-keys    Default: true                                                                                                  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
| --clusterrole string                                                                                                                            |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ClusterRole this RoleBinding should reference                                                                                                                                                                                                             |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                                                                         |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                  |
| --field-manager string    Default: "kubectl-create"                                                                                                                                                                                                       |
| Name of the manager used to track field ownership.                                                                                                                                                                                                        |
| --group strings                                                                                                                                                                                                                                           |
| Groups to bind to the role. The flag can be repeated to add multiple groups.                                                                                                                                                                              |
| -h, --help                                                                                                                                                                                                                                                |
| help for rolebinding                                                                                                                                                                                                                                      |
| -o, --output string                                                                                                                                                                                                                                       |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                              |
| --role string                                                                                                                                                                                                                                             |
| Role this RoleBinding should reference                                                                                                                                                                                                                    |
| --save-config                                                                                                                                                                                                                                             |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                       |
| --serviceaccount strings                                                                                                                                                                                                                                  |
| Service accounts to bind to the role, in the format <namespace>:<name>. The flag can be repeated to add multiple service accounts.                                                                                                                        |
| --show-managed-fields                                                                                                                                                                                                                                     |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --user strings                                                                                                                                                                                                                                            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p> Usernames to bind to the role. The flag can be repeated to add multiple users. </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p> --validate string[="strict"]    Default: "strict" </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p> Must be one of: strict (or true), warn, ignore (or false).<br/> "true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br/> "warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br/> "false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. </p> |
| <p> --as string </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p> Username to impersonate for the operation. User could be a regular user or a service account in a namespace. </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p> --as-group strings </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p> Group to impersonate for the operation, this flag can be repeated to specify multiple groups. </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p> --as-uid string </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p> UID to impersonate for the operation. </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p> --cache-dir string    Default: "\$HOME/.kube/cache" </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p> Default cache directory </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p> --certificate-authority string </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p> Path to a cert file for the certificate authority </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p> --client-certificate string </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p> Path to a client certificate file for TLS </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p> --client-key string </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p> Path to a client key file for TLS </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p> --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16 </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p> CIDRs opened in GCE firewall for L7 LB traffic proxy &amp; health checks </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p> --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br/> 130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --tls-server-name string                                                                                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                                                                                                                                                                                    |
| Bearer token for authentication to the API server                                                                                                                                                 |
| --username string                                                                                                                                                                                 |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create secret

## Synopsis

Create a secret with specified type.

A docker-registry type secret is for accessing a container registry.

A generic type secret indicate an Opaque secret type.

A tls type secret holds TLS certificate and its associated key.

kubectl create secret (docker-registry | generic | tls)

## Options

|                 |
|-----------------|
| -h, --help      |
| help for secret |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |

|                                                                                                                                                                 |
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| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                             |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |

|                                                                                                                                                                                                   |
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| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin
- [kubectl create secret docker-registry](#) - Create a secret for use with a Docker registry
- [kubectl create secret generic](#) - Create a secret from a local file, directory, or literal value
- [kubectl create secret tls](#) - Create a TLS secret

# kubectl create secret docker-registry

## Synopsis

Create a new secret for use with Docker registries.

Dockercfg secrets are used to authenticate against Docker registries.

When using the Docker command line to push images, you can authenticate to a given registry by running:

```
'$ docker login DOCKER_REGISTRY_SERVER --username=DOCKER_USER --password=DOCKER_PASSWORD --email=DOCKER_EMAIL'.
```

That produces a ~/.dockercfg file that is used by subsequent 'docker push' and 'docker pull' commands to authenticate to the registry. The email address is optional.

When creating applications, you may have a Docker registry that requires authentication. In order for the nodes to pull images on your behalf, they must have the credentials. You can provide this information by creating a dockercfg secret and attaching it to your service account.

```
kubectl create secret docker-registry NAME --docker-username=user --docker-password=password --docker-email=email [--docker-server=string] [--from-file=[key=]source] [--dry-run=server|client|none]
```

## Examples

```
If you do not already have a .dockercfg file, create a dockercfg secret directly
kubectl create secret docker-registry my-secret --docker-server=DOCKER_REGISTRY_SERVER --docker-username=DOCKER_USER --docker-password=DOCKER_PASSWORD --docker-email=DOCKER_EMAIL

Create a new secret named my-secret from ~/.docker/config.json
kubectl create secret docker-registry my-secret --from-file=.dockerconfigjson=path/to/.docker/config.json
```

## Options

|                                                                                                                                                  |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <code>--allow-missing-template-keys</code>                                                                                                       | Default: true |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats. |               |
| <code>--append-hash</code>                                                                                                                       |               |
| Append a hash of the secret to its name.                                                                                                         |               |
| <code>--docker-email</code>                                                                                                                      | string        |

|                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Email for Docker registry                                                                                                                                                                                                                                                                  |
| --docker-password string                                                                                                                                                                                                                                                                   |
| Password for Docker registry authentication                                                                                                                                                                                                                                                |
| --docker-server string   Default: "https://index.docker.io/v1/"                                                                                                                                                                                                                            |
| Server location for Docker registry                                                                                                                                                                                                                                                        |
| --docker-username string                                                                                                                                                                                                                                                                   |
| Username for Docker registry authentication                                                                                                                                                                                                                                                |
| --dry-run string[="unchanged"]   Default: "none"                                                                                                                                                                                                                                           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                                   |
| --field-manager string   Default: "kubectl-create"                                                                                                                                                                                                                                         |
| Name of the manager used to track field ownership.                                                                                                                                                                                                                                         |
| --from-file strings                                                                                                                                                                                                                                                                        |
| Key files can be specified using their file path, in which case a default name will be given to them, or optionally with a name and file path, in which case the given name will be used. Specifying a directory will iterate each named file in the directory that is a valid secret key. |
| -h, --help                                                                                                                                                                                                                                                                                 |
| help for docker-registry                                                                                                                                                                                                                                                                   |
| -o, --output string                                                                                                                                                                                                                                                                        |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                               |
| --save-config                                                                                                                                                                                                                                                                              |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                        |
| --show-managed-fields                                                                                                                                                                                                                                                                      |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                              |
| --template string                                                                                                                                                                                                                                                                          |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --as string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --as-group strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --as-uid string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| UID to impersonate for the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Default cache directory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| --certificate-authority string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Path to a cert file for the certificate authority                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --client-certificate string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Path to a client certificate file for TLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --client-key string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Path to a client key file for TLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|                                           |                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                  | Default: "0"                                                                                                                                                                                      |
|                                           | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                       |                                                                                                                                                                                                   |
|                                           | The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration | Default: 1m0s                                                                                                                                                                                     |
|                                           | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string                | Default: "cadvisor"                                                                                                                                                                               |
|                                           | database name                                                                                                                                                                                     |
| --storage-driver-host string              | Default: "localhost:8086"                                                                                                                                                                         |
|                                           | database host:port                                                                                                                                                                                |
| --storage-driver-password string          | Default: "root"                                                                                                                                                                                   |
|                                           | database password                                                                                                                                                                                 |
| --storage-driver-secure                   |                                                                                                                                                                                                   |
|                                           | use secure connection with database                                                                                                                                                               |
| --storage-driver-table string             | Default: "stats"                                                                                                                                                                                  |
|                                           | table name                                                                                                                                                                                        |
| --storage-driver-user string              | Default: "root"                                                                                                                                                                                   |
|                                           | database username                                                                                                                                                                                 |
| --tls-server-name string                  |                                                                                                                                                                                                   |
|                                           | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                            |                                                                                                                                                                                                   |
|                                           | Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                             |                                                                                                                                                                                                   |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl create secret](#) - Create a secret using a specified subcommand

# kubectl create secret generic

## Synopsis

Create a secret based on a file, directory, or specified literal value.

A single secret may package one or more key/value pairs.

When creating a secret based on a file, the key will default to the basename of the file, and the value will default to the file content. If the basename is an invalid key or you wish to chose your own, you may specify an alternate key.

When creating a secret based on a directory, each file whose basename is a valid key in the directory will be packaged into the secret. Any directory entries except regular files are ignored (e.g. subdirectories, symlinks, devices, pipes, etc).

```
kubectl create secret generic NAME [--type=string] [--from-file=[key=]source] [--from-literal=key1=value1] [--dry-run=server|client|none]
```

## Examples

```
Create a new secret named my-secret with keys for each file in folder bar
kubectl create secret generic my-secret --from-file=path/to/bar

Create a new secret named my-secret with specified keys instead of names on disk
kubectl create secret generic my-secret --from-file=ssh-privatekey=path/to/id_rsa --from-file=ssh-publickey=path/to/id_rsa.pub

Create a new secret named my-secret with key1=supersecret and key2=topsecret
```

```
kubectl create secret generic my-secret --from-literal=key1=supersecret --from-literal=key2=topsecret

Create a new secret named my-secret using a combination of a file and a literal
kubectl create secret generic my-secret --from-file=ssh-privatekey=path/to/id_rsa --from-literal=passphrase=topsecret

Create a new secret named my-secret from env files
kubectl create secret generic my-secret --from-env-file=path/to/foo.env --from-env-file=path/to/bar.env
```

## Options

|                                                                                                                                                                                                                                                                                            |                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <code>--allow-missing-template-keys</code>                                                                                                                                                                                                                                                 | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                                                            |                           |
| <code>--append-hash</code>                                                                                                                                                                                                                                                                 |                           |
| Append a hash of the secret to its name.                                                                                                                                                                                                                                                   |                           |
| <code>--dry-run string[="unchanged"]</code>                                                                                                                                                                                                                                                | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                                   |                           |
| <code>--field-manager string</code>                                                                                                                                                                                                                                                        | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                                                                                         |                           |
| <code>--from-env-file strings</code>                                                                                                                                                                                                                                                       |                           |
| Specify the path to a file to read lines of key=val pairs to create a secret.                                                                                                                                                                                                              |                           |
| <code>--from-file strings</code>                                                                                                                                                                                                                                                           |                           |
| Key files can be specified using their file path, in which case a default name will be given to them, or optionally with a name and file path, in which case the given name will be used. Specifying a directory will iterate each named file in the directory that is a valid secret key. |                           |
| <code>--from-literal strings</code>                                                                                                                                                                                                                                                        |                           |
| Specify a key and literal value to insert in secret (i.e. mykey=somevalue)                                                                                                                                                                                                                 |                           |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                    |                           |
| help for generic                                                                                                                                                                                                                                                                           |                           |
| <code>-o, --output string</code>                                                                                                                                                                                                                                                           |                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --save-config                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                            |
| --type string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| The type of secret to create                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --as string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --as-group strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --as-uid string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| UID to impersonate for the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Default cache directory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--certificate-authority string</code>                                                                                                                     |
| Path to a cert file for the certificate authority                                                                                                               |
| <code>--client-certificate string</code>                                                                                                                        |
| Path to a client certificate file for TLS                                                                                                                       |
| <code>--client-key string</code>                                                                                                                                |
| Path to a client key file for TLS                                                                                                                               |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                                                    |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                   |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| <code>--cluster string</code>                                                                                                                                   |
| The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                                                                                                                   |
| The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300                                                                                            |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300                                                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                                                                                                              |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                                                                                                         |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                                                                                                                |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |

|                                                                      |                                                                                                                                                                                                   |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--match-server-version</code>                                  |                                                                                                                                                                                                   |
|                                                                      | Require server version to match client version                                                                                                                                                    |
| <code>-n, --namespace string</code>                                  |                                                                                                                                                                                                   |
|                                                                      | If present, the namespace scope for this CLI request                                                                                                                                              |
| <code>--password string</code>                                       |                                                                                                                                                                                                   |
|                                                                      | Password for basic authentication to the API server                                                                                                                                               |
| <code>--profile string</code> Default: "none"                        |                                                                                                                                                                                                   |
|                                                                      | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| <code>--profile-output string</code> Default: "profile.pprof"        |                                                                                                                                                                                                   |
|                                                                      | Name of the file to write the profile to                                                                                                                                                          |
| <code>--request-timeout string</code> Default: "0"                   |                                                                                                                                                                                                   |
|                                                                      | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                     |                                                                                                                                                                                                   |
|                                                                      | The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s |                                                                                                                                                                                                   |
|                                                                      | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"          |                                                                                                                                                                                                   |
|                                                                      | database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"  |                                                                                                                                                                                                   |
|                                                                      | database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"        |                                                                                                                                                                                                   |
|                                                                      | database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                 |                                                                                                                                                                                                   |
|                                                                      | use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"          |                                                                                                                                                                                                   |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl create secret](#) - Create a secret using a specified subcommand

# kubectl create secret tls

## Synopsis

Create a TLS secret from the given public/private key pair.

The public/private key pair must exist beforehand. The public key certificate must be .PEM encoded and match the given private key.

```
kubectl create secret tls NAME --cert=path/to/cert/file --key=path/to/key/file [--dry-run=server|client|none]
```

## Examples

```
Create a new TLS secret named tls-secret with the given key pair
kubectl create secret tls tls-secret --cert=path/to/tls.crt --key=path/to/tls.key
```

## Options

|                                                                                                                                                                                                                     |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <code>--allow-missing-template-keys</code>                                                                                                                                                                          | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                     |                           |
| <code>--append-hash</code>                                                                                                                                                                                          |                           |
| Append a hash of the secret to its name.                                                                                                                                                                            |                           |
| <code>--cert string</code>                                                                                                                                                                                          |                           |
| Path to PEM encoded public key certificate.                                                                                                                                                                         |                           |
| <code>--dry-run string[="unchanged"]</code>                                                                                                                                                                         | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.            |                           |
| <code>--field-manager string</code>                                                                                                                                                                                 | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                  |                           |
| <code>-h, --help</code>                                                                                                                                                                                             |                           |
| help for tls                                                                                                                                                                                                        |                           |
| <code>--key string</code>                                                                                                                                                                                           |                           |
| Path to private key associated with given certificate.                                                                                                                                                              |                           |
| <code>-o, --output string</code>                                                                                                                                                                                    |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                        |                           |
| <code>--save-config</code>                                                                                                                                                                                          |                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |                           |
| <code>--show-managed-fields</code>                                                                                                                                                                                  |                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                           |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |

|                                                                                                                                                                                                   |
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| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --tls-server-name string                                                                                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                                                                                                                                                                                    |

|                                                                                                              |
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| Bearer token for authentication to the API server                                                            |
| --user string                                                                                                |
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl create secret](#) - Create a secret using a specified subcommand

# kubectl create service

## Synopsis

Create a service using a specified subcommand.

```
kubectl create service [flags]
```

## Options

|                  |
|------------------|
| -h, --help       |
| help for service |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |

|                                                                                                                                                                 |
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| <code>--as-uid string</code>                                                                                                                                    |
| UID to impersonate for the operation.                                                                                                                           |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                                                                   |
| Default cache directory                                                                                                                                         |
| <code>--certificate-authority string</code>                                                                                                                     |
| Path to a cert file for the certificate authority                                                                                                               |
| <code>--client-certificate string</code>                                                                                                                        |
| Path to a client certificate file for TLS                                                                                                                       |
| <code>--client-key string</code>                                                                                                                                |
| Path to a client key file for TLS                                                                                                                               |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                                                    |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                   |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| <code>--cluster string</code>                                                                                                                                   |
| The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                                                                                                                   |
| The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300                                                                                            |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300                                                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                                                                                                              |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                                                                                                         |

|                                                                                                                                                                                                   |
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| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin
- [kubectl create service clusterip](#) - Create a ClusterIP service
- [kubectl create service externalname](#) - Create an ExternalName service
- [kubectl create service loadbalancer](#) - Create a LoadBalancer service
- [kubectl create service nodeport](#) - Create a NodePort service

# kubectl create service clusterip

## Synopsis

Create a ClusterIP service with the specified name.

```
kubectl create service clusterip NAME [--tcp=<port>:<targetPort>] [--dry-run=server|client|none]
```

## Examples

```
Create a new ClusterIP service named my-cs
kubectl create service clusterip my-cs --tcp=5678:8080

Create a new ClusterIP service named my-cs (in headless mode)
kubectl create service clusterip my-cs --clusterip="None"
```

## Options

|                                                                                                                                                                                                          |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                            | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                         |                           |
| --clusterip string                                                                                                                                                                                       |                           |
| Assign your own ClusterIP or set to 'None' for a 'headless' service (no loadbalancing).                                                                                                                  |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                           | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |                           |
| --field-manager string                                                                                                                                                                                   | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                       |                           |
| -h, --help                                                                                                                                                                                               |                           |
| help for clusterip                                                                                                                                                                                       |                           |
| -o, --output string                                                                                                                                                                                      |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                             |                           |
| --save-config                                                                                                                                                                                            |                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --tcp strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Port pairs can be specified as '<port>:<targetPort>'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |

|                                                                                                                                                                 |
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| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl create service](#) - Create a service using a specified subcommand

# kubectl create service externalname

## Synopsis

Create an ExternalName service with the specified name.

ExternalName service references to an external DNS address instead of only pods, which will allow application authors to reference services that exist off platform, on other clusters, or locally.

```
kubectl create service externalname NAME --external-name external.name [--dry-run=server|client|none]
```

## Examples

```
Create a new ExternalName service named my-ns
kubectl create service externalname my-ns --external-name bar.com
```

## Options

|                                                                                                                                                                                                                     |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                                       | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                     |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                                      | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.            |                           |
| --external-name string                                                                                                                                                                                              |                           |
| External name of service                                                                                                                                                                                            |                           |
| --field-manager string                                                                                                                                                                                              | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                  |                           |
| -h, --help                                                                                                                                                                                                          |                           |
| help for externalname                                                                                                                                                                                               |                           |
| -o, --output string                                                                                                                                                                                                 |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                        |                           |
| --save-config                                                                                                                                                                                                       |                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |                           |
| --show-managed-fields                                                                                                                                                                                               |                           |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                       |                           |
| --tcp strings                                                                                                                                                                                                       |                           |
| Port pairs can be specified as '<port>:<targetPort>'.                                                                                                                                                               |                           |
| --template string                                                                                                                                                                                                   |                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --as string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --as-group strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --as-uid string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| UID to impersonate for the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Default cache directory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| --certificate-authority string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Path to a cert file for the certificate authority                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --client-certificate string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Path to a client certificate file for TLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --client-key string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Path to a client key file for TLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|                                           |                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                  | Default: "0"                                                                                                                                                                                      |
|                                           | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                       |                                                                                                                                                                                                   |
|                                           | The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration | Default: 1m0s                                                                                                                                                                                     |
|                                           | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string                | Default: "cadvisor"                                                                                                                                                                               |
|                                           | database name                                                                                                                                                                                     |
| --storage-driver-host string              | Default: "localhost:8086"                                                                                                                                                                         |
|                                           | database host:port                                                                                                                                                                                |
| --storage-driver-password string          | Default: "root"                                                                                                                                                                                   |
|                                           | database password                                                                                                                                                                                 |
| --storage-driver-secure                   |                                                                                                                                                                                                   |
|                                           | use secure connection with database                                                                                                                                                               |
| --storage-driver-table string             | Default: "stats"                                                                                                                                                                                  |
|                                           | table name                                                                                                                                                                                        |
| --storage-driver-user string              | Default: "root"                                                                                                                                                                                   |
|                                           | database username                                                                                                                                                                                 |
| --tls-server-name string                  |                                                                                                                                                                                                   |
|                                           | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                            |                                                                                                                                                                                                   |
|                                           | Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                             |                                                                                                                                                                                                   |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl create service](#) - Create a service using a specified subcommand

# kubectl create service loadbalancer

## Synopsis

Create a LoadBalancer service with the specified name.

```
kubectl create service loadbalancer NAME [--tcp=port:targetPort] [--dry-run=server|client|none]
```

## Examples

```
Create a new LoadBalancer service named my-lbs
kubectl create service loadbalancer my-lbs --tcp=5678:8080
```

## Options

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --allow-missing-template-keys    Default: true                                                                                                                                                           |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                        |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --field-manager string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                           |
| help for loadbalancer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |
| -o, --output string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                           |
| --save-config                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                           |
| --tcp strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                           |
| Port pairs can be specified as '<port>:<targetPort>'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                                               |                           |
| --validate string[="strict"]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Default: "strict"         |
| <p>Must be one of: strict (or true), warn, ignore (or false).</p> <p>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.</p> <p>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.</p> <p>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.</p> |                           |
| --as string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                           |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                           |
| --as-group strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                   |
| --as-uid string                                                                                                                                                 |
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                              |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                               |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                              |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                       |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                     |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | If true, opt-out of response compression for all requests to the server                                                                                                                           |
|  | <code>--insecure-skip-tls-verify</code>                                                                                                                                                           |
|  | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
|  | <code>--kubeconfig string</code>                                                                                                                                                                  |
|  | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
|  | <code>--match-server-version</code>                                                                                                                                                               |
|  | Require server version to match client version                                                                                                                                                    |
|  | <code>-n, --namespace string</code>                                                                                                                                                               |
|  | If present, the namespace scope for this CLI request                                                                                                                                              |
|  | <code>--password string</code>                                                                                                                                                                    |
|  | Password for basic authentication to the API server                                                                                                                                               |
|  | <code>--profile string</code> Default: "none"                                                                                                                                                     |
|  | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
|  | <code>--profile-output string</code> Default: "profile.pprof"                                                                                                                                     |
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | <code>--request-timeout string</code> Default: "0"                                                                                                                                                |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | <code>-s, --server string</code>                                                                                                                                                                  |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
|  | database name                                                                                                                                                                                     |
|  | <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database host:port                                                                                                           |
| --storage-driver-password string    Default: "root"                                                                          |
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl create service](#) - Create a service using a specified subcommand

# kubectl create service nodeport

## Synopsis

Create a NodePort service with the specified name.

```
kubectl create service nodeport NAME [--tcp=port:targetPort] [--dry-run=server|client|none]
```

## Examples

```
Create a new NodePort service named my-ns
kubectl create service nodeport my-ns --tcp=5678:8080
```

## Options

|                                                                                                                                                                                                                     |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --allow-missing-template-keys                                                                                                                                                                                       | Default: true             |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyang and jsonpath output formats.                                                                     |                           |
| --dry-run string[="unchanged"]                                                                                                                                                                                      | Default: "none"           |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.            |                           |
| --field-manager string                                                                                                                                                                                              | Default: "kubectl-create" |
| Name of the manager used to track field ownership.                                                                                                                                                                  |                           |
| -h, --help                                                                                                                                                                                                          |                           |
| help for nodeport                                                                                                                                                                                                   |                           |
| --node-port int                                                                                                                                                                                                     |                           |
| Port used to expose the service on each node in a cluster.                                                                                                                                                          |                           |
| -o, --output string                                                                                                                                                                                                 |                           |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                        |                           |
| --save-config                                                                                                                                                                                                       |                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |                           |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--show-managed-fields</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>--tcp strings</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Port pairs can be specified as '<port>:<targetPort>'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--template string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>--validate string[="strict"]</code> Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>Must be one of: strict (or true), warn, ignore (or false).</p> <p>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.</p> <p>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.</p> <p>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.</p> |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| <code>--as string</code>                                                                                     |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| <code>--as-group strings</code>                                                                              |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| <code>--as-uid string</code>                                                                                 |
| UID to impersonate for the operation.                                                                        |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                |
| Default cache directory                                                                                      |
| <code>--certificate-authority string</code>                                                                  |
| Path to a cert file for the certificate authority                                                            |
| <code>--client-certificate string</code>                                                                     |
| Path to a client certificate file for TLS                                                                    |
| <code>--client-key string</code>                                                                             |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string    Default: "none"</p>                                                                                                                                                               |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string    Default: "profile.pprof"</p>                                                                                                                                               |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string    Default: "0"</p>                                                                                                                                                          |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration    Default: 1m0s</p>                                                                                                                                        |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string    Default: "cadvisor"</p>                                                                                                                                                 |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string    Default: "localhost:8086"</p>                                                                                                                                         |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string    Default: "root"</p>                                                                                                                                               |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string    Default: "stats"</p>                                                                                                                                                 |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string    Default: "root"</p>                                                                                                                                                   |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl create service](#) - Create a service using a specified subcommand

# kubectl create serviceaccount

## Synopsis

Create a service account with the specified name.

```
kubectl create serviceaccount NAME [--dry-run=server|client|none]
```

## Examples

```
Create a new service account named my-service-account
kubectl create serviceaccount my-service-account
```

## Options

|                                                |
|------------------------------------------------|
| --allow-missing-template-keys    Default: true |
|------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| --field-manager string    Default: "kubectl-create"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Name of the manager used to track field ownership.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| help for serviceaccount                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -o, --output string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --save-config                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --show-managed-fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --template string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                            |
| --validate string[="strict"]    Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| --as string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |



## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl create token

## Synopsis

Request a service account token.

```
kubectl create token SERVICE_ACCOUNT_NAME
```

## Examples

```
Request a token to authenticate to the kube-apiserver as the service account "myapp" in the
current namespace
kubectl create token myapp

Request a token for a service account in a custom namespace
kubectl create token myapp --namespace myns

Request a token with a custom expiration
kubectl create token myapp --duration 10m

Request a token with a custom audience
kubectl create token myapp --audience https://example.com

Request a token bound to an instance of a Secret object
kubectl create token myapp --bound-object-kind Secret --bound-object-name mysecret

Request a token bound to an instance of a Secret object with a specific UID
kubectl create token myapp --bound-object-kind Secret --bound-object-name mysecret --
bound-object-uid 0d4691ed-659b-4935-a832-355f77ee47cc
```

## Options

|                                                                                                                                                                                    |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| --allow-missing-template-keys                                                                                                                                                      | Default: true |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                    |               |
| --audience strings                                                                                                                                                                 |               |
| Audience of the requested token. If unset, defaults to requesting a token for use with the Kubernetes API server. May be repeated to request a token valid for multiple audiences. |               |
| --bound-object-kind string                                                                                                                                                         |               |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind of an object to bind the token to. Supported kinds are Pod, Secret. If set, --bound-object-name must be provided.                                                                                                                                    |
| --bound-object-name string                                                                                                                                                                                                                                |
| Name of an object to bind the token to. The token will expire when the object is deleted. Requires --bound-object-kind.                                                                                                                                   |
| --bound-object-uid string                                                                                                                                                                                                                                 |
| UID of an object to bind the token to. Requires --bound-object-kind and --bound-object-name. If unset, the UID of the existing object is used.                                                                                                            |
| --duration duration                                                                                                                                                                                                                                       |
| Requested lifetime of the issued token. If not set or if set to 0, the lifetime will be determined by the server automatically. The server may return a token with a longer or shorter lifetime.                                                          |
| -h, --help                                                                                                                                                                                                                                                |
| help for token                                                                                                                                                                                                                                            |
| -o, --output string                                                                                                                                                                                                                                       |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                              |
| --show-managed-fields                                                                                                                                                                                                                                     |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
|  | --match-server-version                                                                                                                                                                            |
|  | Require server version to match client version                                                                                                                                                    |
|  | -n, --namespace string                                                                                                                                                                            |
|  | If present, the namespace scope for this CLI request                                                                                                                                              |
|  | --password string                                                                                                                                                                                 |
|  | Password for basic authentication to the API server                                                                                                                                               |
|  | --profile string   Default: "none"                                                                                                                                                                |
|  | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
|  | --profile-output string   Default: "profile.pprof"                                                                                                                                                |
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | --request-timeout string   Default: "0"                                                                                                                                                           |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | -s, --server string                                                                                                                                                                               |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
|  | database name                                                                                                                                                                                     |
|  | --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
|  | database host:port                                                                                                                                                                                |
|  | --storage-driver-password string   Default: "root"                                                                                                                                                |
|  | database password                                                                                                                                                                                 |
|  | --storage-driver-secure                                                                                                                                                                           |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl create](#) - Create a resource from a file or from stdin

# kubectl debug

## Synopsis

Debug cluster resources using interactive debugging containers.

'debug' provides automation for common debugging tasks for cluster objects identified by resource and name. Pods will be used by default if no resource is specified.

The action taken by 'debug' varies depending on what resource is specified. Supported actions include:

- Workload: Create a copy of an existing pod with certain attributes changed, for example changing the image tag to a new version.
- Workload: Add an ephemeral container to an already running pod, for example to add debugging utilities without restarting the pod.
- Node: Create a new pod that runs in the node's host namespaces and can access the node's filesystem.

```
kubectl debug (POD | TYPE[.VERSION].GROUP)/NAME) [-- COMMAND [args...]]
```

## Examples

```
Create an interactive debugging session in pod mypod and immediately attach to it.
kubectl debug mypod -it --image=busybox

Create an interactive debugging session for the pod in the file pod.yaml and immediately
attach to it.
(requires the EphemeralContainers feature to be enabled in the cluster)
kubectl debug -f pod.yaml -it --image=busybox

Create a debug container named debugger using a custom automated debugging image.
kubectl debug --image=myproj/debug-tools -c debugger mypod

Create a copy of mypod adding a debug container and attach to it
kubectl debug mypod -it --image=busybox --copy-to=my-debugger

Create a copy of mypod changing the command of mycontainer
kubectl debug mypod -it --copy-to=my-debugger --container=mycontainer -- sh

Create a copy of mypod changing all container images to busybox
kubectl debug mypod --copy-to=my-debugger --set-image=*=busybox

Create a copy of mypod adding a debug container and changing container images
kubectl debug mypod -it --copy-to=my-debugger --image=debian --set-
image=app=app:debug,sidecar=sidecar:debug

Create an interactive debugging session on a node and immediately attach to it.
The container will run in the host namespaces and the host's filesystem will be mounted at /
host
kubectl debug node/mynode -it --image=busybox
```

## Options

| --arguments-only                                                                                  |
|---------------------------------------------------------------------------------------------------|
| If specified, everything after -- will be passed to the new container as Args instead of Command. |
| --attach                                                                                          |

|                            |                                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | If true, wait for the container to start running, and then attach as if 'kubectl attach ...' were called. Default false, unless '-i/--stdin' is set, in which case the default is true. |
| -c, --container string     |                                                                                                                                                                                         |
|                            | Container name to use for debug container.                                                                                                                                              |
| --copy-to string           |                                                                                                                                                                                         |
|                            | Create a copy of the target Pod with this name.                                                                                                                                         |
| --env stringToString       | Default: []                                                                                                                                                                             |
|                            | Environment variables to set in the container.                                                                                                                                          |
| -f, --filename strings     |                                                                                                                                                                                         |
|                            | identifying the resource to debug                                                                                                                                                       |
| -h, --help                 |                                                                                                                                                                                         |
|                            | help for debug                                                                                                                                                                          |
| --image string             |                                                                                                                                                                                         |
|                            | Container image to use for debug container.                                                                                                                                             |
| --image-pull-policy string |                                                                                                                                                                                         |
|                            | The image pull policy for the container. If left empty, this value will not be specified by the client and defaulted by the server.                                                     |
| --profile string           | Default: "legacy"                                                                                                                                                                       |
|                            | Options are "legacy", "general", "baseline", "netadmin", "restricted" or "sysadmin".                                                                                                    |
| -q, --quiet                |                                                                                                                                                                                         |
|                            | If true, suppress informational messages.                                                                                                                                               |
| --replace                  |                                                                                                                                                                                         |
|                            | When used with '--copy-to', delete the original Pod.                                                                                                                                    |
| --same-node                |                                                                                                                                                                                         |
|                            | When used with '--copy-to', schedule the copy of target Pod on the same node.                                                                                                           |
| --set-image stringToString | Default: []                                                                                                                                                                             |

|                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------|
| When used with '--copy-to', a list of name=image pairs for changing container images, similar to how 'kubectl set image' works. |
| --share-processes    Default: true                                                                                              |
| When used with '--copy-to', enable process namespace sharing in the copy.                                                       |
| -i, --stdin                                                                                                                     |
| Keep stdin open on the container(s) in the pod, even if nothing is attached.                                                    |
| --target string                                                                                                                 |
| When using an ephemeral container, target processes in this container name.                                                     |
| -t, --tty                                                                                                                       |
| Allocate a TTY for the debugging container.                                                                                     |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                           |



|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|                                           |                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                  | Default: "0"                                                                                                                                                                                      |
|                                           | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                       |                                                                                                                                                                                                   |
|                                           | The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration | Default: 1m0s                                                                                                                                                                                     |
|                                           | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string                | Default: "cadvisor"                                                                                                                                                                               |
|                                           | database name                                                                                                                                                                                     |
| --storage-driver-host string              | Default: "localhost:8086"                                                                                                                                                                         |
|                                           | database host:port                                                                                                                                                                                |
| --storage-driver-password string          | Default: "root"                                                                                                                                                                                   |
|                                           | database password                                                                                                                                                                                 |
| --storage-driver-secure                   |                                                                                                                                                                                                   |
|                                           | use secure connection with database                                                                                                                                                               |
| --storage-driver-table string             | Default: "stats"                                                                                                                                                                                  |
|                                           | table name                                                                                                                                                                                        |
| --storage-driver-user string              | Default: "root"                                                                                                                                                                                   |
|                                           | database username                                                                                                                                                                                 |
| --tls-server-name string                  |                                                                                                                                                                                                   |
|                                           | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                            |                                                                                                                                                                                                   |
|                                           | Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                             |                                                                                                                                                                                                   |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl delete

## Synopsis

Delete resources by file names, stdin, resources and names, or by resources and label selector.

JSON and YAML formats are accepted. Only one type of argument may be specified: file names, resources and names, or resources and label selector.

Some resources, such as pods, support graceful deletion. These resources define a default period before they are forcibly terminated (the grace period) but you may override that value with the --grace-period flag, or pass --now to set a grace-period of 1. Because these resources often represent entities in the cluster, deletion may not be acknowledged immediately. If the node hosting a pod is down or cannot reach the API server, termination may take significantly longer than the grace period. To force delete a resource, you must specify the --force flag. Note: only a subset of resources support graceful deletion. In absence of the support, the --grace-period flag is ignored.

**IMPORTANT:** Force deleting pods does not wait for confirmation that the pod's processes have been terminated, which can leave those processes running until the node detects the deletion and completes graceful deletion. If your processes use shared storage or talk to a remote API and depend on the name of the pod to identify themselves, force deleting those pods may result in multiple processes running on different machines using the same identification which may lead to data corruption or inconsistency. Only force delete pods when you are sure the pod is terminated, or if your application can tolerate multiple copies of the same pod running at once. Also, if you force delete pods, the scheduler may place new pods on those nodes before the node has released those resources and causing those pods to be evicted immediately.

Note that the delete command does NOT do resource version checks, so if someone submits an update to a resource right when you submit a delete, their update will be lost along with the rest of the resource.

After a CustomResourceDefinition is deleted, invalidation of discovery cache may take up to 6 hours. If you don't want to wait, you might want to run "kubectl api-resources" to refresh the discovery cache.

```
kubectl delete ([-f FILENAME] | [-k DIRECTORY] | TYPE [(NAME | -l label | --all)])
```

## Examples

```
Delete a pod using the type and name specified in pod.json
kubectl delete -f ./pod.json

Delete resources from a directory containing kustomization.yaml - e.g. dir/
kustomization.yaml
kubectl delete -k dir

Delete resources from all files that end with '.json'
kubectl delete -f '*.json'

Delete a pod based on the type and name in the JSON passed into stdin
cat pod.json | kubectl delete -f -

Delete pods and services with same names "baz" and "foo"
kubectl delete pod,service baz foo

Delete pods and services with label name=myLabel
kubectl delete pods,services -l name=myLabel

Delete a pod with minimal delay
kubectl delete pod foo --now

Force delete a pod on a dead node
kubectl delete pod foo --force

Delete all pods
kubectl delete pods --all
```

## Options

|                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------|
| --all                                                                                                                                       |
| Delete all resources, in the namespace of the specified resource types.                                                                     |
| -A, --all-namespaces                                                                                                                        |
| If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace. |

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--cascade string[="background"]</code> Default: "background"                                                                                                                                       |
| Must be "background", "orphan", or "foreground". Selects the deletion cascading strategy for the dependents (e.g. Pods created by a ReplicationController). Defaults to background.                      |
| <code>--dry-run string[="unchanged"]</code> Default: "none"                                                                                                                                              |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| <code>--field-selector string</code>                                                                                                                                                                     |
| Selector (field query) to filter on, supports '=', '==', and '!='.(e.g. <code>--field-selector key1=value1,key2=value2</code> ). The server only supports a limited number of field queries per type.    |
| <code>-f, --filename strings</code>                                                                                                                                                                      |
| containing the resource to delete.                                                                                                                                                                       |
| <code>--force</code>                                                                                                                                                                                     |
| If true, immediately remove resources from API and bypass graceful deletion. Note that immediate deletion of some resources may result in inconsistency or data loss and requires confirmation.          |
| <code>--grace-period int</code> Default: -1                                                                                                                                                              |
| Period of time in seconds given to the resource to terminate gracefully. Ignored if negative. Set to 1 for immediate shutdown. Can only be set to 0 when <code>--force</code> is true (force deletion).  |
| <code>-h, --help</code>                                                                                                                                                                                  |
| help for delete                                                                                                                                                                                          |
| <code>--ignore-not-found</code>                                                                                                                                                                          |
| Treat "resource not found" as a successful delete. Defaults to "true" when <code>--all</code> is specified.                                                                                              |
| <code>-i, --interactive</code>                                                                                                                                                                           |
| If true, delete resource only when user confirms.                                                                                                                                                        |
| <code>-k, --kustomize string</code>                                                                                                                                                                      |
| Process a kustomization directory. This flag can't be used together with <code>-f</code> or <code>-R</code> .                                                                                            |
| <code>--now</code>                                                                                                                                                                                       |
| If true, resources are signaled for immediate shutdown (same as <code>--grace-period=1</code> ).                                                                                                         |

|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -o, --output string                                                                                                                                                        |
| Output mode. Use "-o name" for shorter output (resource/name).                                                                                                             |
| --raw string                                                                                                                                                               |
| Raw URI to DELETE to the server. Uses the transport specified by the kubeconfig file.                                                                                      |
| -R, --recursive                                                                                                                                                            |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                            |
| -l, --selector string                                                                                                                                                      |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints. |
| --timeout duration                                                                                                                                                         |
| The length of time to wait before giving up on a delete, zero means determine a timeout from the size of the object                                                        |
| --wait     Default: true                                                                                                                                                   |
| If true, wait for resources to be gone before returning. This waits for finalizers.                                                                                        |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string     Default: "\$HOME/.kube/cache"                                                         |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl](#) - kubectrl controls the Kubernetes cluster manager

# kubectrl describe

## Synopsis

Show details of a specific resource or group of resources.

Print a detailed description of the selected resources, including related resources such as events or controllers. You may select a single object by name, all objects of that type, provide a name prefix, or label selector. For example:

```
$ kubectrl describe TYPE NAME_PREFIX
```

will first check for an exact match on TYPE and NAME\_PREFIX. If no such resource exists, it will output details for every resource that has a name prefixed with NAME\_PREFIX.

Use "kubectrl api-resources" for a complete list of supported resources.

kubectl describe (-f FILENAME | TYPE [NAME\_PREFIX | -l label] | TYPE/NAME)

# Examples

```
Describe a node
kubectl describe nodes kubernetes-node-emt8.c.myproject.internal

Describe a pod
kubectl describe pods/nginx

Describe a pod identified by type and name in "pod.json"
kubectl describe -f pod.json

Describe all pods
kubectl describe pods

Describe pods by label name=myLabel
kubectl describe pods -l name=myLabel

Describe all pods managed by the 'frontend' replication controller
(rc-created pods get the name of the rc as a prefix in the pod name)
kubectl describe pods frontend
```

# Options

|                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------|
| -A, --all-namespaces                                                                                                                        |
| If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace. |
| --chunk-size int    Default: 500                                                                                                            |
| Return large lists in chunks rather than all at once. Pass 0 to disable. This flag is beta and may change in the future.                    |
| -f, --filename strings                                                                                                                      |
| Filename, directory, or URL to files containing the resource to describe                                                                    |
| -h, --help                                                                                                                                  |
| help for describe                                                                                                                           |
| -k, --kustomize string                                                                                                                      |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                        |
| -R, --recursive                                                                                                                             |

|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                            |
| -l, --selector string                                                                                                                                                      |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints. |
| --show-events    Default: true                                                                                                                                             |
| If true, display events related to the described object.                                                                                                                   |

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.        |
| --as-group strings                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                       |
| --as-uid string                                                                                                     |
| UID to impersonate for the operation.                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                             |
| --certificate-authority string                                                                                      |
| Path to a cert file for the certificate authority                                                                   |
| --client-certificate string                                                                                         |
| Path to a client certificate file for TLS                                                                           |
| --client-key string                                                                                                 |
| Path to a client key file for TLS                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--cluster string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectrl](#) - kubectrl controls the Kubernetes cluster manager

# kubectrl diff

## Synopsis

Diff configurations specified by file name or stdin between the current online configuration, and the configuration as it would be if applied.

The output is always YAML.

KUBECTL\_EXTERNAL\_DIFF environment variable can be used to select your own diff command. Users can use external commands with params too, example:  
KUBECTL\_EXTERNAL\_DIFF="colordiff -N -u"

By default, the "diff" command available in your path will be run with the "-u" (unified diff) and "-N" (treat absent files as empty) options.

Exit status: 0 No differences were found. 1 Differences were found. >1 Kubectrl or diff failed with an error.

Note: KUBECTL\_EXTERNAL\_DIFF, if used, is expected to follow that convention.

```
kubectrl diff -f FILENAME
```

## Examples

```
Diff resources included in pod.json
kubectrl diff -f pod.json

Diff file read from stdin
cat service.yaml | kubectrl diff -f -
```

# Options

|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --concurrency int    Default: 1                                                                                                                                            |
| Number of objects to process in parallel when diffing against the live version. Larger number = faster, but more memory, I/O and CPU over that shorter period of time.     |
| --field-manager string    Default: "kubectl-client-side-apply"                                                                                                             |
| Name of the manager used to track field ownership.                                                                                                                         |
| -f, --filename strings                                                                                                                                                     |
| Filename, directory, or URL to files contains the configuration to diff                                                                                                    |
| --force-conflicts                                                                                                                                                          |
| If true, server-side apply will force the changes against conflicts.                                                                                                       |
| -h, --help                                                                                                                                                                 |
| help for diff                                                                                                                                                              |
| -k, --kustomize string                                                                                                                                                     |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                       |
| --prune                                                                                                                                                                    |
| Include resources that would be deleted by pruning. Can be used with -l and default shows all resources would be pruned                                                    |
| --prune-allowlist strings                                                                                                                                                  |
| Overwrite the default allowlist with <group/version/kind> for --prune                                                                                                      |
| -R, --recursive                                                                                                                                                            |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                            |
| -l, --selector string                                                                                                                                                      |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints. |
| --server-side                                                                                                                                                              |
| If true, apply runs in the server instead of the client.                                                                                                                   |
| --show-managed-fields                                                                                                                                                      |

If true, include managed fields in the diff.

--as string

Username to impersonate for the operation. User could be a regular user or a service account in a namespace.

--as-group strings

Group to impersonate for the operation, this flag can be repeated to specify multiple groups.

--as-uid string

UID to impersonate for the operation.

--cache-dir string Default: "\$HOME/.kube/cache"

Default cache directory

--certificate-authority string

Path to a cert file for the certificate authority

--client-certificate string

Path to a client certificate file for TLS

--client-key string

Path to a client key file for TLS

--cloud-provider-gce-l7lb-src-cidrs cidrs Default: 130.211.0.0/22,35.191.0.0/16

CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks

--cloud-provider-gce-lb-src-cidrs cidrs Default:  
130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16

CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks

--cluster string

The name of the kubeconfig cluster to use

--context string

The name of the kubeconfig context to use

--default-not-ready-toleration-seconds int Default: 300



|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                          |                                                                                                                                                                                                   |
|                                              | The address and port of the Kubernetes API server                                                                                                                                                 |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

Treat warnings received from the server as errors and exit with a non-zero exit code

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl drain

## Synopsis

Drain node in preparation for maintenance.

The given node will be marked unschedulable to prevent new pods from arriving. 'drain' evicts the pods if the API server supports <https://kubernetes.io/docs/concepts/workloads/pods/disruptions/> eviction <https://kubernetes.io/docs/concepts/workloads/pods/disruptions/> . Otherwise, it will use normal DELETE to delete the pods. The 'drain' evicts or deletes all pods except mirror pods (which cannot be deleted through the API server). If there are daemon set-managed pods, drain will not proceed without --ignore-daemonsets, and regardless it will not delete any daemon set-managed pods, because those pods would be immediately replaced by the daemon set controller, which ignores unschedulable markings. If there are any pods that are neither mirror pods nor managed by a replication controller, replica set, daemon set, stateful set, or job, then drain will not delete any pods unless you use --force. --force will also allow deletion to proceed if the managing resource of one or more pods is missing.

'drain' waits for graceful termination. You should not operate on the machine until the command completes.

When you are ready to put the node back into service, use `kubectl uncordon`, which will make the node schedulable again.

[https://kubernetes.io/images/docs/kubectl\\_drain.svg](https://kubernetes.io/images/docs/kubectl_drain.svg) Workflow[https://kubernetes.io/images/docs/kubectl\\_drain.svg](https://kubernetes.io/images/docs/kubectl_drain.svg)

```
kubectl drain NODE
```

## Examples

```
Drain node "foo", even if there are pods not managed by a replication controller, replica set, job, daemon set, or stateful set on it
kubectl drain foo --force
```

```
As above, but abort if there are pods not managed by a replication controller, replica set, job, daemon set, or stateful set, and use a grace period of 15 minutes
kubectl drain foo --grace-period=900
```

# Options

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--chunk-size int</code> Default: 500                                                                                                                                                               |
| Return large lists in chunks rather than all at once. Pass 0 to disable. This flag is beta and may change in the future.                                                                                 |
| <code>--delete-emptydir-data</code>                                                                                                                                                                      |
| Continue even if there are pods using emptyDir (local data that will be deleted when the node is drained).                                                                                               |
| <code>--disable-eviction</code>                                                                                                                                                                          |
| Force drain to use delete, even if eviction is supported. This will bypass checking PodDisruptionBudgets, use with caution.                                                                              |
| <code>--dry-run string[="unchanged"]</code> Default: "none"                                                                                                                                              |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| <code>--force</code>                                                                                                                                                                                     |
| Continue even if there are pods that do not declare a controller.                                                                                                                                        |
| <code>--grace-period int</code> Default: -1                                                                                                                                                              |
| Period of time in seconds given to each pod to terminate gracefully. If negative, the default value specified in the pod will be used.                                                                   |
| <code>-h, --help</code>                                                                                                                                                                                  |
| help for drain                                                                                                                                                                                           |
| <code>--ignore-daemonsets</code>                                                                                                                                                                         |
| Ignore DaemonSet-managed pods.                                                                                                                                                                           |
| <code>--pod-selector string</code>                                                                                                                                                                       |
| Label selector to filter pods on the node                                                                                                                                                                |
| <code>-l, --selector string</code>                                                                                                                                                                       |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                               |
| <code>--skip-wait-for-delete-timeout int</code>                                                                                                                                                          |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| If pod DeletionTimestamp older than N seconds, skip waiting for the pod. Seconds must be greater than 0 to skip.   |
| --timeout duration                                                                                                 |
| The length of time to wait before giving up, zero means infinite                                                   |
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |
| --cluster string                                                                                                   |
| The name of the kubeconfig cluster to use                                                                          |
| --context string                                                                                                   |

|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | The name of the kubeconfig context to use                                                                                                                                                         |
| --default-not-ready-toleration-seconds int   | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>-s, --server string</code>                                                                                                  |
| The address and port of the Kubernetes API server                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                       |
| database name                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                               |
| database host:port                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                     |
| database password                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                              |
| use secure connection with database                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                       |
| table name                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                         |
| database username                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| <code>--token string</code>                                                                                                       |
| Bearer token for authentication to the API server                                                                                 |
| <code>--user string</code>                                                                                                        |
| The name of the kubeconfig user to use                                                                                            |
| <code>--username string</code>                                                                                                    |
| Username for basic authentication to the API server                                                                               |
| <code>--version version[=true]</code>                                                                                             |

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>--version, --version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
| <code>--warnings-as-errors</code>                                                                                                      |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                   |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl edit

## Synopsis

Edit a resource from the default editor.

The edit command allows you to directly edit any API resource you can retrieve via the command-line tools. It will open the editor defined by your KUBE\_EDITOR, or EDITOR environment variables, or fall back to 'vi' for Linux or 'notepad' for Windows. When attempting to open the editor, it will first attempt to use the shell that has been defined in the 'SHELL' environment variable. If this is not defined, the default shell will be used, which is '/bin/bash' for Linux or 'cmd' for Windows.

You can edit multiple objects, although changes are applied one at a time. The command accepts file names as well as command-line arguments, although the files you point to must be previously saved versions of resources.

Editing is done with the API version used to fetch the resource. To edit using a specific API version, fully-qualify the resource, version, and group.

The default format is YAML. To edit in JSON, specify "-o json".

The flag --windows-line-endings can be used to force Windows line endings, otherwise the default for your operating system will be used.

In the event an error occurs while updating, a temporary file will be created on disk that contains your unapplied changes. The most common error when updating a resource is another editor changing the resource on the server. When this occurs, you will have to apply your changes to the newer version of the resource, or update your temporary saved copy to include the latest resource version.

```
kubectl edit (RESOURCE/NAME | -f FILENAME)
```

## Examples

```
Edit the service named 'registry'
kubectl edit svc/registry
```



```
Use an alternative editor
KUBE_EDITOR="nano" kubectl edit svc/registry

Edit the job 'myjob' in JSON using the v1 API format
kubectl edit job.v1.batch/myjob -o json

Edit the deployment 'mydeployment' in YAML and save the modified config in its annotation
kubectl edit deployment/mydeployment -o yaml --save-config

Edit the 'status' subresource for the 'mydeployment' deployment
kubectl edit deployment mydeployment --subresource='status'
```

## Options

|                                                                                                                                                              |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <code>--allow-missing-template-keys</code>                                                                                                                   | Default: true           |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.              |                         |
| <code>--field-manager</code> string                                                                                                                          | Default: "kubectl-edit" |
| Name of the manager used to track field ownership.                                                                                                           |                         |
| <code>-f, --filename</code> strings                                                                                                                          |                         |
| Filename, directory, or URL to files to use to edit the resource                                                                                             |                         |
| <code>-h, --help</code>                                                                                                                                      |                         |
| help for edit                                                                                                                                                |                         |
| <code>-k, --kustomize</code> string                                                                                                                          |                         |
| Process the kustomization directory. This flag can't be used together with <code>-f</code> or <code>-R</code> .                                              |                         |
| <code>-o, --output</code> string                                                                                                                             |                         |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                 |                         |
| <code>--output-patch</code>                                                                                                                                  |                         |
| Output the patch if the resource is edited.                                                                                                                  |                         |
| <code>-R, --recursive</code>                                                                                                                                 |                         |
| Process the directory used in <code>-f, --filename</code> recursively. Useful when you want to manage related manifests organized within the same directory. |                         |
| <code>--save-config</code>                                                                                                                                   |                         |

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>--show-managed-fields</code>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                             | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>--subresource string</code>                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                             | If specified, edit will operate on the subresource of the requested object. Must be one of [status]. This flag is beta and may change in the future.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>--template string</code>                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                             | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                           |
| <code>--validate string[="strict"]</code> Default: "strict" |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                             | Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.<br>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields. |
| <code>--windows-line-endings</code>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                             | Defaults to the line ending native to your platform.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                               |                                                                                                              |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>--as string</code>                                      |                                                                                                              |
|                                                               | Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| <code>--as-group strings</code>                               |                                                                                                              |
|                                                               | Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| <code>--as-uid string</code>                                  |                                                                                                              |
|                                                               | UID to impersonate for the operation.                                                                        |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache" |                                                                                                              |
|                                                               | Default cache directory                                                                                      |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--certificate-authority string</code>                                                                                                                     |
| Path to a cert file for the certificate authority                                                                                                               |
| <code>--client-certificate string</code>                                                                                                                        |
| Path to a client certificate file for TLS                                                                                                                       |
| <code>--client-key string</code>                                                                                                                                |
| Path to a client key file for TLS                                                                                                                               |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                                                    |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                   |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| <code>--cluster string</code>                                                                                                                                   |
| The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                                                                                                                   |
| The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300                                                                                            |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300                                                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                                                                                                              |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                                                                                                         |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                                                                                                                |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |

|                                                                      |                                                                                                                                                                                                   |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--match-server-version</code>                                  |                                                                                                                                                                                                   |
|                                                                      | Require server version to match client version                                                                                                                                                    |
| <code>-n, --namespace string</code>                                  |                                                                                                                                                                                                   |
|                                                                      | If present, the namespace scope for this CLI request                                                                                                                                              |
| <code>--password string</code>                                       |                                                                                                                                                                                                   |
|                                                                      | Password for basic authentication to the API server                                                                                                                                               |
| <code>--profile string</code> Default: "none"                        |                                                                                                                                                                                                   |
|                                                                      | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| <code>--profile-output string</code> Default: "profile.pprof"        |                                                                                                                                                                                                   |
|                                                                      | Name of the file to write the profile to                                                                                                                                                          |
| <code>--request-timeout string</code> Default: "0"                   |                                                                                                                                                                                                   |
|                                                                      | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                     |                                                                                                                                                                                                   |
|                                                                      | The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s |                                                                                                                                                                                                   |
|                                                                      | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"          |                                                                                                                                                                                                   |
|                                                                      | database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"  |                                                                                                                                                                                                   |
|                                                                      | database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"        |                                                                                                                                                                                                   |
|                                                                      | database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                 |                                                                                                                                                                                                   |
|                                                                      | use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"          |                                                                                                                                                                                                   |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl events

## Synopsis

Display events.

Prints a table of the most important information about events. You can request events for a namespace, for all namespace, or filtered to only those pertaining to a specified resource.

```
kubectl events [(-o|--output=)json|yaml|name|go-template|go-template-file|template|templatefile|
jsonpath|jsonpath-as-json|jsonpath-file] [--for TYPE/NAME] [--watch] [--
types=Normal,Warning]
```

# Examples

```
List recent events in the default namespace
kubectl events

List recent events in all namespaces
kubectl events --all-namespaces

List recent events for the specified pod, then wait for more events and list them as they arrive
kubectl events --for pod/web-pod-13je7 --watch

List recent events in YAML format
kubectl events -oyaml

List recent only events of type 'Warning' or 'Normal'
kubectl events --types=Warning,Normal
```

# Options

|                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------|
| -A, --all-namespaces                                                                                                                             |
| If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.      |
| --allow-missing-template-keys    Default: true                                                                                                   |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats. |
| --chunk-size int    Default: 500                                                                                                                 |
| Return large lists in chunks rather than all at once. Pass 0 to disable. This flag is beta and may change in the future.                         |
| --for string                                                                                                                                     |
| Filter events to only those pertaining to the specified resource.                                                                                |
| -h, --help                                                                                                                                       |
| help for events                                                                                                                                  |
| --no-headers                                                                                                                                     |
| When using the default output format, don't print headers.                                                                                       |
| -o, --output string                                                                                                                              |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                             |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --types strings                                                                                                                                                                                                                                          |
| Output only events of given types.                                                                                                                                                                                                                       |
| -w, --watch                                                                                                                                                                                                                                              |
| After listing the requested events, watch for more events.                                                                                                                                                                                               |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                            |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |



|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --tls-server-name string                                                                                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Bearer token for authentication to the API server                                                            |
| --user string                                                                                                |
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubect](#)l - kubectl controls the Kubernetes cluster manager

# kubectl exec

## Synopsis

Execute a command in a container.

```
kubectl exec (POD | TYPE/NAME) [-c CONTAINER] [flags] -- COMMAND [args...]
```

## Examples

```
Get output from running the 'date' command from pod mypod, using the first container by default
```

```
kubectl exec mypod -- date
```

```
Get output from running the 'date' command in ruby-container from pod mypod
```

```
kubectl exec mypod -c ruby-container -- date
```

```
Switch to raw terminal mode; sends stdin to 'bash' in ruby-container from pod mypod
```

```
and sends stdout/stderr from 'bash' back to the client
```

```
kubectl exec mypod -c ruby-container -i -t -- bash -il
```

```
List contents of /usr from the first container of pod mypod and sort by modification time
```

```
If the command you want to execute in the pod has any flags in common (e.g. -i),
```

```
you must use two dashes (--) to separate your command's flags/arguments
```

```
Also note, do not surround your command and its flags/arguments with quotes
unless that is how you would execute it normally (i.e., do ls -t /usr, not "ls -t /usr")
kubectl exec mypod -i -t -- ls -t /usr
```

```
Get output from running 'date' command from the first pod of the deployment
mydeployment, using the first container by default
kubectl exec deploy/mydeployment -- date
```

```
Get output from running 'date' command from the first pod of the service myservice, using
the first container by default
kubectl exec svc/myservice -- date
```

## Options

|                                                 |                                                                                                                                                                                    |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -c, --container string                          | Container name. If omitted, use the kubectl.kubernetes.io/default-container annotation for selecting the container to be attached or the first container in the pod will be chosen |
| -f, --filename strings                          | to use to exec into the resource                                                                                                                                                   |
| -h, --help                                      | help for exec                                                                                                                                                                      |
| --pod-running-timeout duration    Default: 1m0s | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                |
| -q, --quiet                                     | Only print output from the remote session                                                                                                                                          |
| -i, --stdin                                     | Pass stdin to the container                                                                                                                                                        |
| -t, --tty                                       | Stdin is a TTY                                                                                                                                                                     |
| --as string                                     | Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                       |
| --as-group strings                              |                                                                                                                                                                                    |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                   |
| --as-uid string                                                                                                                                                 |
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                              |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                               |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                              |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                       |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                     |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | If true, opt-out of response compression for all requests to the server                                                                                                                           |
|  | <code>--insecure-skip-tls-verify</code>                                                                                                                                                           |
|  | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
|  | <code>--kubeconfig string</code>                                                                                                                                                                  |
|  | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
|  | <code>--match-server-version</code>                                                                                                                                                               |
|  | Require server version to match client version                                                                                                                                                    |
|  | <code>-n, --namespace string</code>                                                                                                                                                               |
|  | If present, the namespace scope for this CLI request                                                                                                                                              |
|  | <code>--password string</code>                                                                                                                                                                    |
|  | Password for basic authentication to the API server                                                                                                                                               |
|  | <code>--profile string</code> Default: "none"                                                                                                                                                     |
|  | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
|  | <code>--profile-output string</code> Default: "profile.pprof"                                                                                                                                     |
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | <code>--request-timeout string</code> Default: "0"                                                                                                                                                |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | <code>-s, --server string</code>                                                                                                                                                                  |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
|  | database name                                                                                                                                                                                     |
|  | <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database host:port                                                                                                           |
| --storage-driver-password string    Default: "root"                                                                          |
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl explain

## Synopsis

Describe fields and structure of various resources.

This command describes the fields associated with each supported API resource. Fields are identified via a simple JSONPath identifier:

```
<type>.<fieldName>[.<fieldName>]
```

Information about each field is retrieved from the server in OpenAPI format.

Use "kubectl api-resources" for a complete list of supported resources.

```
kubectl explain TYPE [--recursive=FALSE|TRUE] [--api-version=api-version-group] [--output=plaintext|plaintext-openapiv2]
```

## Examples

```
Get the documentation of the resource and its fields
kubectl explain pods

Get all the fields in the resource
kubectl explain pods --recursive

Get the explanation for deployment in supported api versions
kubectl explain deployments --api-version=apps/v1

Get the documentation of a specific field of a resource
kubectl explain pods.spec.containers

Get the documentation of resources in different format
kubectl explain deployment --output=plaintext-openapiv2
```

## Options

| --api-version string                                                                      |
|-------------------------------------------------------------------------------------------|
| Use given api-version (group/version) of the resource.                                    |
| -h, --help                                                                                |
| help for explain                                                                          |
| --output string   Default: "plaintext"                                                    |
| Format in which to render the schema. Valid values are: (plaintext, plaintext-openapiv2). |
| --recursive                                                                               |

When true, print the name of all the fields recursively. Otherwise, print the available fields with their description.

--as string

Username to impersonate for the operation. User could be a regular user or a service account in a namespace.

--as-group strings

Group to impersonate for the operation, this flag can be repeated to specify multiple groups.

--as-uid string

UID to impersonate for the operation.

--cache-dir string Default: "\$HOME/.kube/cache"

Default cache directory

--certificate-authority string

Path to a cert file for the certificate authority

--client-certificate string

Path to a client certificate file for TLS

--client-key string

Path to a client key file for TLS

--cloud-provider-gce-l7lb-src-cidrs cidrs Default: 130.211.0.0/22,35.191.0.0/16

CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks

--cloud-provider-gce-lb-src-cidrs cidrs Default:  
130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16

CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks

--cluster string

The name of the kubeconfig cluster to use

--context string

The name of the kubeconfig context to use

--default-not-ready-toleration-seconds int Default: 300



|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                          |                                                                                                                                                                                                   |
|                                              | The address and port of the Kubernetes API server                                                                                                                                                 |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

Treat warnings received from the server as errors and exit with a non-zero exit code

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl expose

## Synopsis

Expose a resource as a new Kubernetes service.

Looks up a deployment, service, replica set, replication controller or pod by name and uses the selector for that resource as the selector for a new service on the specified port. A deployment or replica set will be exposed as a service only if its selector is convertible to a selector that service supports, i.e. when the selector contains only the matchLabels component. Note that if no port is specified via `--port` and the exposed resource has multiple ports, all will be re-used by the new service. Also if no labels are specified, the new service will re-use the labels from the resource it exposes.

Possible resources include (case insensitive):

pod (po), service (svc), replicationcontroller (rc), deployment (deploy), replicaset (rs)

```
kubectl expose (-f FILENAME | TYPE NAME) [--port=port] [--protocol=TCP|UDP|SCTP] [--target-port=number-or-name] [--name=name] [--external-ip=external-ip-of-service] [--type=type]
```

## Examples

```
Create a service for a replicated nginx, which serves on port 80 and connects to the containers on port 8000
```

```
kubectl expose rc nginx --port=80 --target-port=8000
```

```
Create a service for a replication controller identified by type and name specified in "nginx-controller.yaml", which serves on port 80 and connects to the containers on port 8000
```

```
kubectl expose -f nginx-controller.yaml --port=80 --target-port=8000
```

```
Create a service for a pod valid-pod, which serves on port 444 with the name "frontend"
```

```
kubectl expose pod valid-pod --port=444 --name=frontend
```

```
Create a second service based on the above service, exposing the container port 8443 as port 443 with the name "nginx-https"
```

```
kubectl expose service nginx --port=443 --target-port=8443 --name=nginx-https
```

```
Create a service for a replicated streaming application on port 4100 balancing UDP traffic and named 'video-stream'.
```

```
kubectl expose rc streamer --port=4100 --protocol=UDP --name=video-stream
```

```
Create a service for a replicated nginx using replica set, which serves on port 80 and
connects to the containers on port 8000
kubectl expose rs nginx --port=80 --target-port=8000

Create a service for an nginx deployment, which serves on port 80 and connects to the
containers on port 8000
kubectl expose deployment nginx --port=80 --target-port=8000
```

## Options

|                                                                                                                                                                                                          |                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <code>--allow-missing-template-keys</code>                                                                                                                                                               | Default: true                        |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                          |                                      |
| <code>--cluster-ip</code>                                                                                                                                                                                | string                               |
| ClusterIP to be assigned to the service. Leave empty to auto-allocate, or set to 'None' to create a headless service.                                                                                    |                                      |
| <code>--dry-run</code>                                                                                                                                                                                   | string[="unchanged"] Default: "none" |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |                                      |
| <code>--external-ip</code>                                                                                                                                                                               | string                               |
| Additional external IP address (not managed by Kubernetes) to accept for the service. If this IP is routed to a node, the service can be accessed by this IP in addition to its generated service IP.    |                                      |
| <code>--field-manager</code>                                                                                                                                                                             | string Default: "kubectl-expose"     |
| Name of the manager used to track field ownership.                                                                                                                                                       |                                      |
| <code>-f, --filename</code>                                                                                                                                                                              | strings                              |
| Filename, directory, or URL to files identifying the resource to expose a service                                                                                                                        |                                      |
| <code>-h, --help</code>                                                                                                                                                                                  |                                      |
| help for expose                                                                                                                                                                                          |                                      |
| <code>-k, --kustomize</code>                                                                                                                                                                             | string                               |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                     |                                      |
| <code>-l, --labels</code>                                                                                                                                                                                | string                               |

|  |                                                                                                                                                                                                                     |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Labels to apply to the service created by this call.                                                                                                                                                                |
|  | --load-balancer-ip string                                                                                                                                                                                           |
|  | IP to assign to the LoadBalancer. If empty, an ephemeral IP will be created and used (cloud-provider specific).                                                                                                     |
|  | --name string                                                                                                                                                                                                       |
|  | The name for the newly created object.                                                                                                                                                                              |
|  | -o, --output string                                                                                                                                                                                                 |
|  | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                        |
|  | --override-type string   Default: "merge"                                                                                                                                                                           |
|  | The method used to override the generated object: json, merge, or strategic.                                                                                                                                        |
|  | --overrides string                                                                                                                                                                                                  |
|  | An inline JSON override for the generated object. If this is non-empty, it is used to override the generated object. Requires that the object supply a valid apiVersion field.                                      |
|  | --port string                                                                                                                                                                                                       |
|  | The port that the service should serve on. Copied from the resource being exposed, if unspecified                                                                                                                   |
|  | --protocol string                                                                                                                                                                                                   |
|  | The network protocol for the service to be created. Default is 'TCP'.                                                                                                                                               |
|  | -R, --recursive                                                                                                                                                                                                     |
|  | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                     |
|  | --save-config                                                                                                                                                                                                       |
|  | If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |
|  | --selector string                                                                                                                                                                                                   |
|  | A label selector to use for this service. Only equality-based selector requirements are supported. If empty (the default) infer the selector from the replication controller or replica set.)                       |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --session-affinity string                                                                                                                                                                                                                                |
| If non-empty, set the session affinity for the service to this; legal values: 'None', 'ClientIP'                                                                                                                                                         |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --target-port string                                                                                                                                                                                                                                     |
| Name or number for the port on the container that the service should direct traffic to.<br>Optional.                                                                                                                                                     |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --type string                                                                                                                                                                                                                                            |
| Type for this service: ClusterIP, NodePort, LoadBalancer, or ExternalName. Default is 'ClusterIP'.                                                                                                                                                       |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string   Default: "none"</p>                                                                                                                                                                |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string   Default: "profile.pprof"</p>                                                                                                                                                |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string   Default: "0"</p>                                                                                                                                                           |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration   Default: 1m0s</p>                                                                                                                                         |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string   Default: "cadvisor"</p>                                                                                                                                                  |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string   Default: "localhost:8086"</p>                                                                                                                                          |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string   Default: "root"</p>                                                                                                                                                |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string   Default: "stats"</p>                                                                                                                                                  |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string   Default: "root"</p>                                                                                                                                                    |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl](#) - kubectrl controls the Kubernetes cluster manager

# kubectrl get

## Synopsis

Display one or many resources.

Prints a table of the most important information about the specified resources. You can filter the list using a label selector and the --selector flag. If the desired resource type is namespaced you will only see results in your current namespace unless you pass --all-namespaces.

By specifying the output as 'template' and providing a Go template as the value of the --template flag, you can filter the attributes of the fetched resources.

Use "kubectrl api-resources" for a complete list of supported resources.

```
kubectrl get [(-o|--output=)json|yaml|name|go-template|go-template-file|template|templatefile|
jsonpath|jsonpath-as-json|jsonpath-file|custom-columns|custom-columns-file|wide]
(TYPE[.VERSION][.GROUP] [NAME | -l label] | TYPE[.VERSION][.GROUP]/NAME ...) [flags]
```

## Examples

```
List all pods in ps output format
kubectl get pods

List all pods in ps output format with more information (such as node name)
kubectl get pods -o wide

List a single replication controller with specified NAME in ps output format
kubectl get replicationcontroller web

List deployments in JSON output format, in the "v1" version of the "apps" API group
kubectl get deployments.v1.apps -o json

List a single pod in JSON output format
kubectl get -o json pod web-pod-13je7

List a pod identified by type and name specified in "pod.yaml" in JSON output format
kubectl get -f pod.yaml -o json

List resources from a directory with kustomization.yaml - e.g. dir/kustomization.yaml
kubectl get -k dir/

Return only the phase value of the specified pod
kubectl get -o template pod/web-pod-13je7 --template={{.status.phase}}

List resource information in custom columns
kubectl get pod test-pod -o custom-
columns=CONTAINER:.spec.containers[0].name,IMAGE:.spec.containers[0].image

List all replication controllers and services together in ps output format
kubectl get rc,services

List one or more resources by their type and names
kubectl get rc/web service/frontend pods/web-pod-13je7

List the 'status' subresource for a single pod
kubectl get pod web-pod-13je7 --subresource status
```

## Options

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| -A, --all-namespaces                                                                                                                            |
| If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.     |
| --allow-missing-template-keys    Default: true                                                                                                  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <code>--chunk-size int</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Default: 500 |
| Return large lists in chunks rather than all at once. Pass 0 to disable. This flag is beta and may change in the future.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |
| <code>--field-selector string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |
| Selector (field query) to filter on, supports '=', '==', and '!='.(e.g. <code>--field-selector key1=value1,key2=value2</code> ). The server only supports a limited number of field queries per type.                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |
| <code>-f, --filename strings</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |              |
| help for get                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |
| <code>--ignore-not-found</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |
| If the requested object does not exist the command will return exit code 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |
| <code>-k, --kustomize string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| Process the kustomization directory. This flag can't be used together with <code>-f</code> or <code>-R</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |
| <code>-L, --label-columns strings</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |
| Accepts a comma separated list of labels that are going to be presented as columns. Names are case-sensitive. You can also use multiple flag options like <code>-L label1 -L label2...</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |
| <code>--no-headers</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |
| When using the default or custom-column output format, don't print headers (default print headers).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| <code>-o, --output string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file, custom-columns, custom-columns-file, wide). See custom columns [ <a href="https://kubernetes.io/docs/reference/kubectl/#custom-columns">https://kubernetes.io/docs/reference/kubectl/#custom-columns</a> ], golang template [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ] and jsonpath template [ <a href="https://kubernetes.io/docs/reference/kubectl/jsonpath/">https://kubernetes.io/docs/reference/kubectl/jsonpath/</a> ]. |              |
| <code>--output-watch-events</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |
| Output watch event objects when <code>--watch</code> or <code>--watch-only</code> is used. Existing objects are output as initial ADDED events.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |
| <code>--raw string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |

|                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raw URI to request from the server. Uses the transport specified by the kubeconfig file.                                                                                                                                                                   |
| -R, --recursive                                                                                                                                                                                                                                            |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                            |
| -l, --selector string                                                                                                                                                                                                                                      |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                                 |
| --server-print    Default: true                                                                                                                                                                                                                            |
| If true, have the server return the appropriate table output. Supports extension APIs and CRDs.                                                                                                                                                            |
| --show-kind                                                                                                                                                                                                                                                |
| If present, list the resource type for the requested object(s).                                                                                                                                                                                            |
| --show-labels                                                                                                                                                                                                                                              |
| When printing, show all labels as the last column (default hide labels column)                                                                                                                                                                             |
| --show-managed-fields                                                                                                                                                                                                                                      |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                              |
| --sort-by string                                                                                                                                                                                                                                           |
| If non-empty, sort list types using this field specification. The field specification is expressed as a JSONPath expression (e.g. '{.metadata.name}'). The field in the API resource specified by this JSONPath expression must be an integer or a string. |
| --subresource string                                                                                                                                                                                                                                       |
| If specified, gets the subresource of the requested object. Must be one of [status scale]. This flag is beta and may change in the future.                                                                                                                 |
| --template string                                                                                                                                                                                                                                          |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].   |
| -w, --watch                                                                                                                                                                                                                                                |
| After listing/getting the requested object, watch for changes.                                                                                                                                                                                             |
| --watch-only                                                                                                                                                                                                                                               |

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| Watch for changes to the requested object(s), without listing/getting first.                                        |
| --as string                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.        |
| --as-group strings                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                       |
| --as-uid string                                                                                                     |
| UID to impersonate for the operation.                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                             |
| --certificate-authority string                                                                                      |
| Path to a cert file for the certificate authority                                                                   |
| --client-certificate string                                                                                         |
| Path to a client certificate file for TLS                                                                           |
| --client-key string                                                                                                 |
| Path to a client key file for TLS                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                |
| --cluster string                                                                                                    |
| The name of the kubeconfig cluster to use                                                                           |
| --context string                                                                                                    |
| The name of the kubeconfig context to use                                                                           |
| --default-not-ready-toleration-seconds int    Default: 300                                                          |

|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                          |                                                                                                                                                                                                   |
|                                              | The address and port of the Kubernetes API server                                                                                                                                                 |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl kustomize

## Synopsis

Build a set of KRM resources using a 'kustomization.yaml' file. The DIR argument must be a path to a directory containing 'kustomization.yaml', or a git repository URL with a path suffix specifying same with respect to the repository root. If DIR is omitted, '.' is assumed.

```
kubectl kustomize DIR [flags]
```

## Examples

```
Build the current working directory
kubectl kustomize

Build some shared configuration directory
kubectl kustomize /home/config/production

Build from github
kubectl kustomize https://github.com/kubernetes-sigs/kustomize.git/examples/helloWorld?
ref=v1.0.6
```

## Options

|                                                                                  |
|----------------------------------------------------------------------------------|
| --as-current-user                                                                |
| use the uid and gid of the command executor to run the function in the container |
| --enable-alpha-plugins                                                           |
| enable kustomize plugins                                                         |
| --enable-helm                                                                    |
| Enable use of the Helm chart inflator generator.                                 |
| -e, --env strings                                                                |
| a list of environment variables to be used by functions                          |



|                                                                                                                                                                   |                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| --helm-command string                                                                                                                                             | Default: "helm"                     |
| helm command (path to executable)                                                                                                                                 |                                     |
| -h, --help                                                                                                                                                        |                                     |
| help for kustomize                                                                                                                                                |                                     |
| --load-restrictor string                                                                                                                                          | Default: "LoadRestrictionsRootOnly" |
| if set to 'LoadRestrictionsNone', local kustomizations may load files from outside their root. This does, however, break the relocatability of the kustomization. |                                     |
| --mount strings                                                                                                                                                   |                                     |
| a list of storage options read from the filesystem                                                                                                                |                                     |
| --network                                                                                                                                                         |                                     |
| enable network access for functions that declare it                                                                                                               |                                     |
| --network-name string                                                                                                                                             | Default: "bridge"                   |
| the docker network to run the container in                                                                                                                        |                                     |
| -o, --output string                                                                                                                                               |                                     |
| If specified, write output to this path.                                                                                                                          |                                     |

|                                                                                                              |                               |
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| --as string                                                                                                  |                               |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |                               |
| --as-group strings                                                                                           |                               |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |                               |
| --as-uid string                                                                                              |                               |
| UID to impersonate for the operation.                                                                        |                               |
| --cache-dir string                                                                                           | Default: "\$HOME/.kube/cache" |
| Default cache directory                                                                                      |                               |
| --certificate-authority string                                                                               |                               |
| Path to a cert file for the certificate authority                                                            |                               |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--client-certificate string</code>                                                                                                                        |
| Path to a client certificate file for TLS                                                                                                                       |
| <code>--client-key string</code>                                                                                                                                |
| Path to a client key file for TLS                                                                                                                               |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                                                                    |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                      |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| <code>--cluster string</code>                                                                                                                                   |
| The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                                                                                                                   |
| The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300                                                                                            |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300                                                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                                                                                                              |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                                                                                                         |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                                                                                                                |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                                                                                                             |
| Require server version to match client version                                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl label

## Synopsis

Update the labels on a resource.

- A label key and value must begin with a letter or number, and may contain letters, numbers, hyphens, dots, and underscores, up to 63 characters each.
- Optionally, the key can begin with a DNS subdomain prefix and a single '/', like example.com/my-app.
- If --overwrite is true, then existing labels can be overwritten, otherwise attempting to overwrite a label will result in an error.
- If --resource-version is specified, then updates will use this resource version, otherwise the existing resource-version will be used.

kubectl label [--overwrite] (-f FILENAME | TYPE NAME) KEY\_1=VAL\_1 ... KEY\_N=VAL\_N [--resource-version=version]

## Examples

```
Update pod 'foo' with the label 'unhealthy' and the value 'true'
kubectl label pods foo unhealthy=true

Update pod 'foo' with the label 'status' and the value 'unhealthy', overwriting any existing value
kubectl label --overwrite pods foo status=unhealthy

Update all pods in the namespace
kubectl label pods --all status=unhealthy

Update a pod identified by the type and name in "pod.json"
kubectl label -f pod.json status=unhealthy

Update pod 'foo' only if the resource is unchanged from version 1
kubectl label pods foo status=unhealthy --resource-version=1

Update pod 'foo' by removing a label named 'bar' if it exists
Does not require the --overwrite flag
kubectl label pods foo bar-
```

## Options

|                                                    |                                                                                                                                                                                                          |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --all                                              |                                                                                                                                                                                                          |
|                                                    | Select all resources, in the namespace of the specified resource types                                                                                                                                   |
| -A, --all-namespaces                               |                                                                                                                                                                                                          |
|                                                    | If true, check the specified action in all namespaces.                                                                                                                                                   |
| --allow-missing-template-keys    Default: true     |                                                                                                                                                                                                          |
|                                                    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |
| --dry-run string[="unchanged"]    Default: "none"  |                                                                                                                                                                                                          |
|                                                    | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| --field-manager string    Default: "kubectl-label" |                                                                                                                                                                                                          |
|                                                    | Name of the manager used to track field ownership.                                                                                                                                                       |
| --field-selector string                            |                                                                                                                                                                                                          |

|                                                                                                                                                                                         |
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| Selector (field query) to filter on, supports '=', '==', and '!='.(e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type. |
| -f, --filename strings                                                                                                                                                                  |
| Filename, directory, or URL to files identifying the resource to update the labels                                                                                                      |
| -h, --help                                                                                                                                                                              |
| help for label                                                                                                                                                                          |
| -k, --kustomize string                                                                                                                                                                  |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                    |
| --list                                                                                                                                                                                  |
| If true, display the labels for a given resource.                                                                                                                                       |
| --local                                                                                                                                                                                 |
| If true, label will NOT contact api-server but run locally.                                                                                                                             |
| -o, --output string                                                                                                                                                                     |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                            |
| --overwrite                                                                                                                                                                             |
| If true, allow labels to be overwritten, otherwise reject label updates that overwrite existing labels.                                                                                 |
| -R, --recursive                                                                                                                                                                         |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                         |
| --resource-version string                                                                                                                                                               |
| If non-empty, the labels update will only succeed if this is the current resource-version for the object. Only valid when specifying a single resource.                                 |
| -l, --selector string                                                                                                                                                                   |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.              |
| --show-managed-fields                                                                                                                                                                   |

|                                                                                                                                                                                                                                                           |
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| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --as string                                                                                                                                                                                                                                               |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                                                                                                              |
| --as-group strings                                                                                                                                                                                                                                        |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                                                                                                             |
| --as-uid string                                                                                                                                                                                                                                           |
| UID to impersonate for the operation.                                                                                                                                                                                                                     |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                                                                                                                        |
| Default cache directory                                                                                                                                                                                                                                   |
| --certificate-authority string                                                                                                                                                                                                                            |
| Path to a cert file for the certificate authority                                                                                                                                                                                                         |
| --client-certificate string                                                                                                                                                                                                                               |
| Path to a client certificate file for TLS                                                                                                                                                                                                                 |
| --client-key string                                                                                                                                                                                                                                       |
| Path to a client key file for TLS                                                                                                                                                                                                                         |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                                                                                                                         |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                                                                                                                      |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                                                                                                                        |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                                                                                                                      |
| --cluster string                                                                                                                                                                                                                                          |
| The name of the kubeconfig cluster to use                                                                                                                                                                                                                 |
| --context string                                                                                                                                                                                                                                          |

|                                              |                                                                                                                                                                                                   |
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|                                              | The name of the kubeconfig context to use                                                                                                                                                         |
| --default-not-ready-toleration-seconds int   | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |



|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>-s, --server string</code>                                                                                                  |
| The address and port of the Kubernetes API server                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                       |
| database name                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                               |
| database host:port                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                     |
| database password                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                              |
| use secure connection with database                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                       |
| table name                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                         |
| database username                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| <code>--token string</code>                                                                                                       |
| Bearer token for authentication to the API server                                                                                 |
| <code>--user string</code>                                                                                                        |
| The name of the kubeconfig user to use                                                                                            |
| <code>--username string</code>                                                                                                    |
| Username for basic authentication to the API server                                                                               |
| <code>--version version[=true]</code>                                                                                             |

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>--version, --version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
| <code>--warnings-as-errors</code>                                                                                                      |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                   |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl logs

## Synopsis

Print the logs for a container in a pod or specified resource. If the pod has only one container, the container name is optional.

```
kubectl logs [-f] [-p] (POD | TYPE/NAME) [-c CONTAINER]
```

## Examples

```
Return snapshot logs from pod nginx with only one container
kubectl logs nginx
```

```
Return snapshot logs from pod nginx with multi containers
kubectl logs nginx --all-containers=true
```

```
Return snapshot logs from all containers in pods defined by label app=nginx
kubectl logs -l app=nginx --all-containers=true
```

```
Return snapshot of previous terminated ruby container logs from pod web-1
kubectl logs -p -c ruby web-1
```

```
Begin streaming the logs of the ruby container in pod web-1
kubectl logs -f -c ruby web-1
```

```
Begin streaming the logs from all containers in pods defined by label app=nginx
kubectl logs -f -l app=nginx --all-containers=true
```

```
Display only the most recent 20 lines of output in pod nginx
kubectl logs --tail=20 nginx
```

```
Show all logs from pod nginx written in the last hour
kubectl logs --since=1h nginx
```

```
Show logs from a kubelet with an expired serving certificate
kubectl logs --insecure-skip-tls-verify-backend nginx
```

```
Return snapshot logs from first container of a job named hello
kubectl logs job/hello
```

```
Return snapshot logs from container nginx-1 of a deployment named nginx
kubectl logs deployment/nginx -c nginx-1
```

## Options

|                                    |                                                                                                                                                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --all-containers                   |                                                                                                                                                                                                                       |
|                                    | Get all containers' logs in the pod(s).                                                                                                                                                                               |
| -c, --container string             |                                                                                                                                                                                                                       |
|                                    | Print the logs of this container                                                                                                                                                                                      |
| -f, --follow                       |                                                                                                                                                                                                                       |
|                                    | Specify if the logs should be streamed.                                                                                                                                                                               |
| -h, --help                         |                                                                                                                                                                                                                       |
|                                    | help for logs                                                                                                                                                                                                         |
| --ignore-errors                    |                                                                                                                                                                                                                       |
|                                    | If watching / following pod logs, allow for any errors that occur to be non-fatal                                                                                                                                     |
| --insecure-skip-tls-verify-backend |                                                                                                                                                                                                                       |
|                                    | Skip verifying the identity of the kubelet that logs are requested from. In theory, an attacker could provide invalid log content back. You might want to use this if your kubelet serving certificates have expired. |
| --limit-bytes int                  |                                                                                                                                                                                                                       |
|                                    | Maximum bytes of logs to return. Defaults to no limit.                                                                                                                                                                |
| --max-log-requests int             | Default: 5                                                                                                                                                                                                            |
|                                    | Specify maximum number of concurrent logs to follow when using by a selector. Defaults to 5.                                                                                                                          |
| --pod-running-timeout duration     | Default: 20s                                                                                                                                                                                                          |
|                                    | The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                                                   |
| --prefix                           |                                                                                                                                                                                                                       |

|                                                                                                                                                                                          |
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| Prefix each log line with the log source (pod name and container name)                                                                                                                   |
| <code>-p, --previous</code>                                                                                                                                                              |
| If true, print the logs for the previous instance of the container in a pod if it exists.                                                                                                |
| <code>-l, --selector string</code>                                                                                                                                                       |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. <code>-l key1=value1,key2=value2</code> ). Matching objects must satisfy all of the specified label constraints. |
| <code>--since duration</code>                                                                                                                                                            |
| Only return logs newer than a relative duration like 5s, 2m, or 3h. Defaults to all logs. Only one of since-time / since may be used.                                                    |
| <code>--since-time string</code>                                                                                                                                                         |
| Only return logs after a specific date (RFC3339). Defaults to all logs. Only one of since-time / since may be used.                                                                      |
| <code>--tail int</code> Default: -1                                                                                                                                                      |
| Lines of recent log file to display. Defaults to -1 with no selector, showing all log lines otherwise 10, if a selector is provided.                                                     |
| <code>--timestamps</code>                                                                                                                                                                |
| Include timestamps on each line in the log output                                                                                                                                        |

|                                                                                                              |
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| <code>--as string</code>                                                                                     |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| <code>--as-group strings</code>                                                                              |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| <code>--as-uid string</code>                                                                                 |
| UID to impersonate for the operation.                                                                        |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                |
| Default cache directory                                                                                      |
| <code>--certificate-authority string</code>                                                                  |

|                                                                                                                     |                                                                                                                                                                 |
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|                                                                                                                     | Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                         |                                                                                                                                                                 |
|                                                                                                                     | Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                 |                                                                                                                                                                 |
|                                                                                                                     | Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |                                                                                                                                                                 |
|                                                                                                                     | CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |                                                                                                                                                                 |
|                                                                                                                     | CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                    |                                                                                                                                                                 |
|                                                                                                                     | The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                    |                                                                                                                                                                 |
|                                                                                                                     | The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                          |                                                                                                                                                                 |
|                                                                                                                     | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                        |                                                                                                                                                                 |
|                                                                                                                     | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                               |                                                                                                                                                                 |
|                                                                                                                     | If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                          |                                                                                                                                                                 |
|                                                                                                                     | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                 |                                                                                                                                                                 |
|                                                                                                                     | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                              |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Require server version to match client version                                                                                                                                                    |
| <code>-n, --namespace string</code>                                                                                                                                                               |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| <code>--password string</code>                                                                                                                                                                    |
| Password for basic authentication to the API server                                                                                                                                               |
| <code>--profile string</code> Default: "none"                                                                                                                                                     |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| <code>--profile-output string</code> Default: "profile.pprof"                                                                                                                                     |
| Name of the file to write the profile to                                                                                                                                                          |
| <code>--request-timeout string</code> Default: "0"                                                                                                                                                |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl options

## Synopsis

Print the list of flags inherited by all commands

kubectl options [flags]

# Examples

```
Print flags inherited by all commands
kubectl options
```

# Options

|                  |
|------------------|
| -h, --help       |
| help for options |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |



|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--cluster string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectrl](#) - kubectrl controls the Kubernetes cluster manager

# kubectrl patch

## Synopsis

Update fields of a resource using strategic merge patch, a JSON merge patch, or a JSON patch.

JSON and YAML formats are accepted.

Note: Strategic merge patch is not supported for custom resources.

```
kubectrl patch (-f FILENAME | TYPE NAME) [-p PATCH|--patch-file FILE]
```

## Examples

```
Partially update a node using a strategic merge patch, specifying the patch as JSON
kubectrl patch node k8s-node-1 -p '{"spec":{"unschedulable":true}}'

Partially update a node using a strategic merge patch, specifying the patch as YAML
kubectrl patch node k8s-node-1 -p '$spec:\n unschedulable: true'

Partially update a node identified by the type and name specified in "node.json" using
strategic merge patch
kubectrl patch -f node.json -p '{"spec":{"unschedulable":true}}'

Update a container's image; spec.containers[*].name is required because it's a merge key
kubectrl patch pod valid-pod -p '{"spec":{"containers":[{"name":"kubernetes-serve-
hostname","image":"new image"}]}}'

Update a container's image using a JSON patch with positional arrays
kubectrl patch pod valid-pod --type='json' -p='[{"op": "replace", "path": "/spec/containers/0/
image", "value":"new image"}]'
```

```
Update a deployment's replicas through the 'scale' subresource using a merge patch
```

```
kubectl patch deployment nginx-deployment --subresource='scale' --type='merge' -p '{"spec": {"replicas":2}}'
```

## Options

|                                                                                                                                                                                                          |                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| --allow-missing-template-keys                                                                                                                                                                            | Default: true            |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |                          |
| --dry-run string[="unchanged"]                                                                                                                                                                           | Default: "none"          |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |                          |
| --field-manager string                                                                                                                                                                                   | Default: "kubectl-patch" |
| Name of the manager used to track field ownership.                                                                                                                                                       |                          |
| -f, --filename strings                                                                                                                                                                                   |                          |
| Filename, directory, or URL to files identifying the resource to update                                                                                                                                  |                          |
| -h, --help                                                                                                                                                                                               |                          |
| help for patch                                                                                                                                                                                           |                          |
| -k, --kustomize string                                                                                                                                                                                   |                          |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                     |                          |
| --local                                                                                                                                                                                                  |                          |
| If true, patch will operate on the content of the file, not the server-side resource.                                                                                                                    |                          |
| -o, --output string                                                                                                                                                                                      |                          |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                             |                          |
| -p, --patch string                                                                                                                                                                                       |                          |
| The patch to be applied to the resource JSON file.                                                                                                                                                       |                          |
| --patch-file string                                                                                                                                                                                      |                          |
| A file containing a patch to be applied to the resource.                                                                                                                                                 |                          |
| -R, --recursive                                                                                                                                                                                          |                          |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| --show-managed-fields                                                                                                                                                                                                                                     |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --subresource string                                                                                                                                                                                                                                      |
| If specified, patch will operate on the subresource of the requested object. Must be one of [status scale]. This flag is beta and may change in the future.                                                                                               |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --type string   Default: "strategic"                                                                                                                                                                                                                      |
| The type of patch being provided; one of [json merge strategic]                                                                                                                                                                                           |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |
| Path to a client key file for TLS                                                                            |

|                                                                                                                                                                 |                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code>                                                                                                          | Default: 130.211.0.0/22,35.191.0.0/16                                 |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |                                                                       |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code>                                                                                                            | Default: 130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |                                                                       |
| <code>--cluster string</code>                                                                                                                                   |                                                                       |
| The name of the kubeconfig cluster to use                                                                                                                       |                                                                       |
| <code>--context string</code>                                                                                                                                   |                                                                       |
| The name of the kubeconfig context to use                                                                                                                       |                                                                       |
| <code>--default-not-ready-toleration-seconds int</code>                                                                                                         | Default: 300                                                          |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |                                                                       |
| <code>--default-unreachable-toleration-seconds int</code>                                                                                                       | Default: 300                                                          |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |                                                                       |
| <code>--disable-compression</code>                                                                                                                              |                                                                       |
| If true, opt-out of response compression for all requests to the server                                                                                         |                                                                       |
| <code>--insecure-skip-tls-verify</code>                                                                                                                         |                                                                       |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |                                                                       |
| <code>--kubeconfig string</code>                                                                                                                                |                                                                       |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |                                                                       |
| <code>--match-server-version</code>                                                                                                                             |                                                                       |
| Require server version to match client version                                                                                                                  |                                                                       |
| <code>-n, --namespace string</code>                                                                                                                             |                                                                       |
| If present, the namespace scope for this CLI request                                                                                                            |                                                                       |
| <code>--password string</code>                                                                                                                                  |                                                                       |
| Password for basic authentication to the API server                                                                                                             |                                                                       |

|                                                                                                                                                                                                   |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --profile string                                                                                                                                                                                  | Default: "none"           |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |                           |
| --profile-output string                                                                                                                                                                           | Default: "profile.pprof"  |
| Name of the file to write the profile to                                                                                                                                                          |                           |
| --request-timeout string                                                                                                                                                                          | Default: "0"              |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |                           |
| -s, --server string                                                                                                                                                                               |                           |
| The address and port of the Kubernetes API server                                                                                                                                                 |                           |
| --storage-driver-buffer-duration duration                                                                                                                                                         | Default: 1m0s             |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |                           |
| --storage-driver-db string                                                                                                                                                                        | Default: "cadvisor"       |
| database name                                                                                                                                                                                     |                           |
| --storage-driver-host string                                                                                                                                                                      | Default: "localhost:8086" |
| database host:port                                                                                                                                                                                |                           |
| --storage-driver-password string                                                                                                                                                                  | Default: "root"           |
| database password                                                                                                                                                                                 |                           |
| --storage-driver-secure                                                                                                                                                                           |                           |
| use secure connection with database                                                                                                                                                               |                           |
| --storage-driver-table string                                                                                                                                                                     | Default: "stats"          |
| table name                                                                                                                                                                                        |                           |
| --storage-driver-user string                                                                                                                                                                      | Default: "root"           |
| database username                                                                                                                                                                                 |                           |
| --tls-server-name string                                                                                                                                                                          |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |                           |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --token string                                                                                               |
| Bearer token for authentication to the API server                                                            |
| --user string                                                                                                |
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectrl](#) - kubectrl controls the Kubernetes cluster manager

# kubectrl plugin

## Synopsis

Provides utilities for interacting with plugins.

Plugins provide extended functionality that is not part of the major command-line distribution. Please refer to the documentation and examples for more information about how write your own plugins.

The easiest way to discover and install plugins is via the kubernetes sub-project krew. To install krew, visit <https://krew.sigs.k8s.io/docs/user-guide/setup/install/> krew.sigs.k8s.io <https://krew.sigs.k8s.io/docs/user-guide/setup/install/>

kubectrl plugin [flags]

## Options

|                 |
|-----------------|
| -h, --help      |
| help for plugin |



|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                          |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                           |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                          |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                   |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |

|                                                                                                                                                                                                   |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| --default-unreachable-toleration-seconds int                                                                                                                                                      | Default: 300             |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |                          |
| --disable-compression                                                                                                                                                                             |                          |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |                          |
| --insecure-skip-tls-verify                                                                                                                                                                        |                          |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |                          |
| --kubeconfig string                                                                                                                                                                               |                          |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |                          |
| --match-server-version                                                                                                                                                                            |                          |
| Require server version to match client version                                                                                                                                                    |                          |
| -n, --namespace string                                                                                                                                                                            |                          |
| If present, the namespace scope for this CLI request                                                                                                                                              |                          |
| --password string                                                                                                                                                                                 |                          |
| Password for basic authentication to the API server                                                                                                                                               |                          |
| --profile string                                                                                                                                                                                  | Default: "none"          |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |                          |
| --profile-output string                                                                                                                                                                           | Default: "profile.pprof" |
| Name of the file to write the profile to                                                                                                                                                          |                          |
| --request-timeout string                                                                                                                                                                          | Default: "0"             |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |                          |
| -s, --server string                                                                                                                                                                               |                          |
| The address and port of the Kubernetes API server                                                                                                                                                 |                          |
| --storage-driver-buffer-duration duration                                                                                                                                                         | Default: 1m0s            |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl plugin list](#) - List all visible plugin executables on a user's PATH

# kubectl plugin list

## Synopsis

List all available plugin files on a user's PATH.

Available plugin files are those that are: - executable - anywhere on the user's PATH - begin with "kubectl-"

```
kubectl plugin list [flags]
```

## Examples

```
List all available plugins
kubectl plugin list
```

## Options

|                                                                                 |
|---------------------------------------------------------------------------------|
| -h, --help                                                                      |
| help for list                                                                   |
| --name-only                                                                     |
| If true, display only the binary name of each plugin, rather than its full path |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                             |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl plugin](#) - Provides utilities for interacting with plugins

# kubectl port-forward

## Synopsis

Forward one or more local ports to a pod.

Use resource type/name such as deployment/mydeployment to select a pod. Resource type defaults to 'pod' if omitted.

If there are multiple pods matching the criteria, a pod will be selected automatically. The forwarding session ends when the selected pod terminates, and a rerun of the command is needed to resume forwarding.

```
kubectl port-forward TYPE/NAME [options] [LOCAL_PORT:]REMOTE_PORT [...
[LOCAL_PORT_N:]REMOTE_PORT_N]
```

## Examples

```
Listen on ports 5000 and 6000 locally, forwarding data to/from ports 5000 and 6000 in the
pod
```

```
kubectl port-forward pod/mypod 5000 6000
```

```
Listen on ports 5000 and 6000 locally, forwarding data to/from ports 5000 and 6000 in a pod
selected by the deployment
```

```
kubectl port-forward deployment/mydeployment 5000 6000
```

```
Listen on port 8443 locally, forwarding to the targetPort of the service's port named "https"
in a pod selected by the service
```

```
kubectl port-forward service/myservice 8443:https
```

```
Listen on port 8888 locally, forwarding to 5000 in the pod
```

```
kubectl port-forward pod/mypod 8888:5000
```

```
Listen on port 8888 on all addresses, forwarding to 5000 in the pod
```

```
kubectl port-forward --address 0.0.0.0 pod/mypod 8888:5000
```

```
Listen on port 8888 on localhost and selected IP, forwarding to 5000 in the pod
```

```
kubectl port-forward --address localhost,10.19.21.23 pod/mypod 8888:5000
```

```
Listen on a random port locally, forwarding to 5000 in the pod
```

```
kubectl port-forward pod/mypod :5000
```

## Options

|                   |                      |
|-------------------|----------------------|
| --address strings | Default: "localhost" |
|-------------------|----------------------|

|                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Addresses to listen on (comma separated). Only accepts IP addresses or localhost as a value. When localhost is supplied, kubectl will try to bind on both 127.0.0.1 and ::1 and will fail if neither of these addresses are available to bind. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| -h, --help                                                                                          |
| help for port-forward                                                                               |
| --pod-running-timeout duration    Default: 1m0s                                                     |
| The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running |

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.        |
| --as-group strings                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                       |
| --as-uid string                                                                                                     |
| UID to impersonate for the operation.                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                             |
| --certificate-authority string                                                                                      |
| Path to a cert file for the certificate authority                                                                   |
| --client-certificate string                                                                                         |
| Path to a client certificate file for TLS                                                                           |
| --client-key string                                                                                                 |
| Path to a client key file for TLS                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                |
| --cluster string                                                                                                    |
| The name of the kubeconfig cluster to use                                                                           |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl proxy

## Synopsis

Creates a proxy server or application-level gateway between localhost and the Kubernetes API server. It also allows serving static content over specified HTTP path. All incoming data enters through one port and gets forwarded to the remote Kubernetes API server port, except for the path matching the static content path.

```
kubectl proxy [--port=PORT] [--www=static-dir] [--www-prefix=prefix] [--api-prefix=prefix]
```

## Examples

```
To proxy all of the Kubernetes API and nothing else
kubectl proxy --api-prefix=/

To proxy only part of the Kubernetes API and also some static files
You can get pods info with 'curl localhost:8001/api/v1/pods'
kubectl proxy --www=/my/files --www-prefix=/static/ --api-prefix=/api/

To proxy the entire Kubernetes API at a different root
You can get pods info with 'curl localhost:8001/custom/api/v1/pods'
kubectl proxy --api-prefix=/custom/

Run a proxy to the Kubernetes API server on port 8011, serving static content from ./local/
www/
kubectl proxy --port=8011 --www=./local/www/

Run a proxy to the Kubernetes API server on an arbitrary local port
The chosen port for the server will be output to stdout
kubectl proxy --port=0
```

```
Run a proxy to the Kubernetes API server, changing the API prefix to k8s-api
This makes e.g. the pods API available at localhost:8001/k8s-api/v1/pods/
kubectl proxy --api-prefix=/k8s-api
```

## Options

|                                                                                                                                                      |                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <code>--accept-hosts string</code>                                                                                                                   | Default: <code>"^localhost\$,^127\.0\.0\.1\$,^\[::1\]\$"</code>        |
| Regular expression for hosts that the proxy should accept.                                                                                           |                                                                        |
| <code>--accept-paths string</code>                                                                                                                   | Default: <code>"^.*"</code>                                            |
| Regular expression for paths that the proxy should accept.                                                                                           |                                                                        |
| <code>--address string</code>                                                                                                                        | Default: <code>"127.0.0.1"</code>                                      |
| The IP address on which to serve on.                                                                                                                 |                                                                        |
| <code>--api-prefix string</code>                                                                                                                     | Default: <code>"/"</code>                                              |
| Prefix to serve the proxied API under.                                                                                                               |                                                                        |
| <code>--append-server-path</code>                                                                                                                    |                                                                        |
| If true, enables automatic path appending of the kube context server path to each request.                                                           |                                                                        |
| <code>--disable-filter</code>                                                                                                                        |                                                                        |
| If true, disable request filtering in the proxy. This is dangerous, and can leave you vulnerable to XSRF attacks, when used with an accessible port. |                                                                        |
| <code>-h, --help</code>                                                                                                                              |                                                                        |
| help for proxy                                                                                                                                       |                                                                        |
| <code>--keepalive duration</code>                                                                                                                    |                                                                        |
| keepalive specifies the keep-alive period for an active network connection. Set to 0 to disable keepalive.                                           |                                                                        |
| <code>-p, --port int</code>                                                                                                                          | Default: <code>8001</code>                                             |
| The port on which to run the proxy. Set to 0 to pick a random port.                                                                                  |                                                                        |
| <code>--reject-methods string</code>                                                                                                                 | Default: <code>"^\$"</code>                                            |
| Regular expression for HTTP methods that the proxy should reject (example <code>--reject-methods='POST,PUT,PATCH'</code> ).                          |                                                                        |
| <code>--reject-paths string</code>                                                                                                                   | Default: <code>"^/api/*/pods/*/exec,<br/>^/api/*/pods/*/attach"</code> |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| Regular expression for paths that the proxy should reject. Paths specified here will be rejected even accepted by --accept-paths. |
| -u, --unix-socket string                                                                                                          |
| Unix socket on which to run the proxy.                                                                                            |
| -w, --www string                                                                                                                  |
| Also serve static files from the given directory under the specified prefix.                                                      |
| -P, --www-prefix string   Default: "/static/"                                                                                     |
| Prefix to serve static files under, if static file directory is specified.                                                        |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|                                           |                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                  | Default: "0"                                                                                                                                                                                      |
|                                           | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                       |                                                                                                                                                                                                   |
|                                           | The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration | Default: 1m0s                                                                                                                                                                                     |
|                                           | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string                | Default: "cadvisor"                                                                                                                                                                               |
|                                           | database name                                                                                                                                                                                     |
| --storage-driver-host string              | Default: "localhost:8086"                                                                                                                                                                         |
|                                           | database host:port                                                                                                                                                                                |
| --storage-driver-password string          | Default: "root"                                                                                                                                                                                   |
|                                           | database password                                                                                                                                                                                 |
| --storage-driver-secure                   |                                                                                                                                                                                                   |
|                                           | use secure connection with database                                                                                                                                                               |
| --storage-driver-table string             | Default: "stats"                                                                                                                                                                                  |
|                                           | table name                                                                                                                                                                                        |
| --storage-driver-user string              | Default: "root"                                                                                                                                                                                   |
|                                           | database username                                                                                                                                                                                 |
| --tls-server-name string                  |                                                                                                                                                                                                   |
|                                           | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                            |                                                                                                                                                                                                   |
|                                           | Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                             |                                                                                                                                                                                                   |



|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl replace

## Synopsis

Replace a resource by file name or stdin.

JSON and YAML formats are accepted. If replacing an existing resource, the complete resource spec must be provided. This can be obtained by

```
$ kubectl get TYPE NAME -o yaml
```

```
kubectl replace -f FILENAME
```

## Examples

```
Replace a pod using the data in pod.json
kubectl replace -f ./pod.json
```

```
Replace a pod based on the JSON passed into stdin
cat pod.json | kubectl replace -f -
```

```
Update a single-container pod's image version (tag) to v4
kubectl get pod mypod -o yaml | sed 's/^(image: myimage\):.*$/\1:v4/' | kubectl replace -f -
```

```
Force replace, delete and then re-create the resource
kubectl replace --force -f ./pod.json
```

# Options

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--allow-missing-template-keys</code> Default: true                                                                                                                                                 |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |
| <code>--cascade string[="background"]</code> Default: "background"                                                                                                                                       |
| Must be "background", "orphan", or "foreground". Selects the deletion cascading strategy for the dependents (e.g. Pods created by a ReplicationController). Defaults to background.                      |
| <code>--dry-run string[="unchanged"]</code> Default: "none"                                                                                                                                              |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| <code>--field-manager string</code> Default: "kubectl-replace"                                                                                                                                           |
| Name of the manager used to track field ownership.                                                                                                                                                       |
| <code>-f, --filename strings</code>                                                                                                                                                                      |
| The files that contain the configurations to replace.                                                                                                                                                    |
| <code>--force</code>                                                                                                                                                                                     |
| If true, immediately remove resources from API and bypass graceful deletion. Note that immediate deletion of some resources may result in inconsistency or data loss and requires confirmation.          |
| <code>--grace-period int</code> Default: -1                                                                                                                                                              |
| Period of time in seconds given to the resource to terminate gracefully. Ignored if negative. Set to 1 for immediate shutdown. Can only be set to 0 when --force is true (force deletion).               |
| <code>-h, --help</code>                                                                                                                                                                                  |
| help for replace                                                                                                                                                                                         |
| <code>-k, --kustomize string</code>                                                                                                                                                                      |
| Process a kustomization directory. This flag can't be used together with -f or -R.                                                                                                                       |
| <code>-o, --output string</code>                                                                                                                                                                         |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                             |
| <code>--raw string</code>                                                                                                                                                                                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raw URI to PUT to the server. Uses the transport specified by the kubeconfig file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-R, --recursive</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Process the directory used in <code>-f, --filename</code> recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>--save-config</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform <code>kubectl apply</code> on this object in the future.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>--show-managed-fields</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| If true, keep the <code>managedFields</code> when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>--subresource string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| If specified, replace will operate on the subresource of the requested object. Must be one of <code>[status scale]</code> . This flag is beta and may change in the future.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>--template string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Template string or path to template file to use when <code>-o=go-template</code> , <code>-o=go-template-file</code> . The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>--timeout duration</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| The length of time to wait before giving up on a delete, zero means determine a timeout from the size of the object                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--validate string[="strict"]</code> Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>Must be one of: <code>strict</code> (or <code>true</code>), <code>warn</code>, <code>ignore</code> (or <code>false</code>).</p> <p>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if <code>ServerSideFieldValidation</code> is enabled on the api-server, but will fall back to less reliable client-side validation if not.</p> <p>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise.</p> <p>"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.</p> |
| <code>--wait</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| If true, wait for resources to be gone before returning. This waits for finalizers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--as string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl rollout

## Synopsis

Manage the rollout of one or many resources.

Valid resource types include:

- deployments
- daemonsets
- statefulsets

kubectl rollout SUBCOMMAND

## Examples

```
Rollback to the previous deployment
kubectl rollout undo deployment/abc

Check the rollout status of a daemonset
kubectl rollout status daemonset/foo

Restart a deployment
kubectl rollout restart deployment/abc

Restart deployments with the 'app=nginx' label
kubectl rollout restart deployment --selector=app=nginx
```

## Options

|                  |
|------------------|
| -h, --help       |
| help for rollout |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                             |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |



|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl rollout history](#) - View rollout history
- [kubectl rollout pause](#) - Mark the provided resource as paused
- [kubectl rollout restart](#) - Restart a resource
- [kubectl rollout resume](#) - Resume a paused resource
- [kubectl rollout status](#) - Show the status of the rollout
- [kubectl rollout undo](#) - Undo a previous rollout

# kubectl rollout history

## Synopsis

View previous rollout revisions and configurations.

```
kubectl rollout history (TYPE NAME | TYPE/NAME) [flags]
```

## Examples

```
View the rollout history of a deployment
kubectl rollout history deployment/abc

View the details of daemonset revision 3
kubectl rollout history daemonset/abc --revision=3
```

## Options

|                                                                                                                                                  |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| --allow-missing-template-keys                                                                                                                    | Default: true |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats. |               |
| -f, --filename strings                                                                                                                           |               |
| Filename, directory, or URL to files identifying the resource to get from a server.                                                              |               |
| -h, --help                                                                                                                                       |               |
| help for history                                                                                                                                 |               |
| -k, --kustomize string                                                                                                                           |               |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                             |               |
| -o, --output string                                                                                                                              |               |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).     |               |
| -R, --recursive                                                                                                                                  |               |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.  |               |
| --revision int                                                                                                                                   |               |
| See the details, including podTemplate of the revision specified                                                                                 |               |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -l, --selector string                                                                                                                                                                                                                                    |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                               |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string    Default: "cadvisor"                                                                                                                                                 |
| database name                                                                                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                                                                                         |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                                                                                               |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                                                                                 |
| table name                                                                                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                                                                                   |
| database username                                                                                                                                                                                 |
| --tls-server-name string                                                                                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                                                                                                                                                                                    |
| Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                                                                                                                                                                                     |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl rollout](#) - Manage the rollout of a resource

# kubectl rollout pause

## Synopsis

Mark the provided resource as paused.

Paused resources will not be reconciled by a controller. Use "kubectl rollout resume" to resume a paused resource. Currently only deployments support being paused.

```
kubectl rollout pause RESOURCE
```

## Examples

```
Mark the nginx deployment as paused
Any current state of the deployment will continue its function; new updates
to the deployment will not have an effect as long as the deployment is paused
kubectl rollout pause deployment/nginx
```

## Options

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| --allow-missing-template-keys    Default: true                                                                                                  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats. |
| --field-manager string    Default: "kubectl-rollout"                                                                                            |

|                        |                                                                                                                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| -f, --filename strings |                                                                                                                                                                                                                                                          |
|                        | Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                      |
| -h, --help             |                                                                                                                                                                                                                                                          |
|                        | help for pause                                                                                                                                                                                                                                           |
| -k, --kustomize string |                                                                                                                                                                                                                                                          |
|                        | Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| -o, --output string    |                                                                                                                                                                                                                                                          |
|                        | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                             |
| -R, --recursive        |                                                                                                                                                                                                                                                          |
|                        | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| -l, --selector string  |                                                                                                                                                                                                                                                          |
|                        | Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                               |
| --show-managed-fields  |                                                                                                                                                                                                                                                          |
|                        | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string      |                                                                                                                                                                                                                                                          |
|                        | Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                    |                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| --as string        |                                                                                                              |
|                    | Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings |                                                                                                              |
|                    | Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string    |                                                                                                              |



|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                             |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl rollout](#) - Manage the rollout of a resource

# kubectl rollout restart

## Synopsis

Restart a resource.

Resource rollout will be restarted.

kubectl rollout restart RESOURCE

## Examples

```
Restart all deployments in test-namespace namespace
kubectl rollout restart deployment -n test-namespace

Restart a deployment
kubectl rollout restart deployment/nginx

Restart a daemon set
kubectl rollout restart daemonset/abc

Restart deployments with the app=nginx label
kubectl rollout restart deployment --selector=app=nginx
```

## Options

|                                                                                                                                                  |                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| --allow-missing-template-keys                                                                                                                    | Default: true              |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats. |                            |
| --field-manager string                                                                                                                           | Default: "kubectl-rollout" |
| Name of the manager used to track field ownership.                                                                                               |                            |
| -f, --filename strings                                                                                                                           |                            |
| Filename, directory, or URL to files identifying the resource to get from a server.                                                              |                            |
| -h, --help                                                                                                                                       |                            |
| help for restart                                                                                                                                 |                            |
| -k, --kustomize string                                                                                                                           |                            |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                             |                            |
| -o, --output string                                                                                                                              |                            |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                              |
| -R, --recursive                                                                                                                                                                                                                                           |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| -l, --selector string                                                                                                                                                                                                                                     |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                                |
| --show-managed-fields                                                                                                                                                                                                                                     |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |
| Path to a client certificate file for TLS                                                                    |
| --client-key string                                                                                          |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string   Default: "none"</p>                                                                                                                                                                |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string   Default: "profile.pprof"</p>                                                                                                                                                |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string   Default: "0"</p>                                                                                                                                                           |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration   Default: 1m0s</p>                                                                                                                                         |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string   Default: "cadvisor"</p>                                                                                                                                                  |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string   Default: "localhost:8086"</p>                                                                                                                                          |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string   Default: "root"</p>                                                                                                                                                |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string   Default: "stats"</p>                                                                                                                                                  |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string   Default: "root"</p>                                                                                                                                                    |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl rollout](#) - Manage the rollout of a resource

# kubectrl rollout resume

## Synopsis

Resume a paused resource.

Paused resources will not be reconciled by a controller. By resuming a resource, we allow it to be reconciled again. Currently only deployments support being resumed.

```
kubectrl rollout resume RESOURCE
```

## Examples

```
Resume an already paused deployment
kubectrl rollout resume deployment/nginx
```



# Options

|                                                                                                                                                                                                                                                           |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| --allow-missing-template-keys    Default: true                                                                                                                                                                                                            |  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                                                                          |  |
| --field-manager string    Default: "kubectrl-rollout"                                                                                                                                                                                                     |  |
| Name of the manager used to track field ownership.                                                                                                                                                                                                        |  |
| -f, --filename strings                                                                                                                                                                                                                                    |  |
| Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                       |  |
| -h, --help                                                                                                                                                                                                                                                |  |
| help for resume                                                                                                                                                                                                                                           |  |
| -k, --kustomize string                                                                                                                                                                                                                                    |  |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                      |  |
| -o, --output string                                                                                                                                                                                                                                       |  |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                              |  |
| -R, --recursive                                                                                                                                                                                                                                           |  |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |  |
| -l, --selector string                                                                                                                                                                                                                                     |  |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                                |  |
| --show-managed-fields                                                                                                                                                                                                                                     |  |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |  |
| --template string                                                                                                                                                                                                                                         |  |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is goolang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |  |
| --as string                                                                                                                                                                                                                                               |  |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |

## See Also

- [kubectl rollout](#) - Manage the rollout of a resource

# kubectl rollout status

## Synopsis

Show the status of the rollout.

By default 'rollout status' will watch the status of the latest rollout until it's done. If you don't want to wait for the rollout to finish then you can use `--watch=false`. Note that if a new rollout starts in-between, then 'rollout status' will continue watching the latest revision. If you want to pin to a specific revision and abort if it is rolled over by another revision, use `--revision=N` where N is the revision you need to watch for.

```
kubectl rollout status (TYPE NAME | TYPE/NAME) [flags]
```

## Examples

```
Watch the rollout status of a deployment
kubectl rollout status deployment/nginx
```

## Options

|                        |                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| -f, --filename strings |                                                                                                                                                 |
|                        | Filename, directory, or URL to files identifying the resource to get from a server.                                                             |
| -h, --help             |                                                                                                                                                 |
|                        | help for status                                                                                                                                 |
| -k, --kustomize string |                                                                                                                                                 |
|                        | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| -R, --recursive        |                                                                                                                                                 |
|                        | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory. |
| --revision int         |                                                                                                                                                 |
|                        | Pin to a specific revision for showing its status. Defaults to 0 (last revision).                                                               |
| -l, --selector string  |                                                                                                                                                 |

|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints. |
| --timeout duration                                                                                                                                                         |
| The length of time to wait before ending watch, zero means never. Any other values should contain a corresponding time unit (e.g. 1s, 2m, 3h).                             |
| -w, --watch     Default: true                                                                                                                                              |
| Watch the status of the rollout until it's done.                                                                                                                           |

|                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                          |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.         |
| --as-group strings                                                                                                   |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                        |
| --as-uid string                                                                                                      |
| UID to impersonate for the operation.                                                                                |
| --cache-dir string     Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                              |
| --certificate-authority string                                                                                       |
| Path to a cert file for the certificate authority                                                                    |
| --client-certificate string                                                                                          |
| Path to a client certificate file for TLS                                                                            |
| --client-key string                                                                                                  |
| Path to a client key file for TLS                                                                                    |
| --cloud-provider-gce-l7lb-src-cidrs cidrs     Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                 |
| --cloud-provider-gce-lb-src-cidrs cidrs     Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                 |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--cluster string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | <code>-s, --server string</code>                                                                                                                                                                  |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
|  | database name                                                                                                                                                                                     |
|  | <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
|  | database host:port                                                                                                                                                                                |
|  | <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
|  | database password                                                                                                                                                                                 |
|  | <code>--storage-driver-secure</code>                                                                                                                                                              |
|  | use secure connection with database                                                                                                                                                               |
|  | <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
|  | table name                                                                                                                                                                                        |
|  | <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
|  | database username                                                                                                                                                                                 |
|  | <code>--tls-server-name string</code>                                                                                                                                                             |
|  | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
|  | <code>--token string</code>                                                                                                                                                                       |
|  | Bearer token for authentication to the API server                                                                                                                                                 |
|  | <code>--user string</code>                                                                                                                                                                        |
|  | The name of the kubeconfig user to use                                                                                                                                                            |
|  | <code>--username string</code>                                                                                                                                                                    |



|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                                                                  |
| <code>--version version[=true]</code>                                                                                                                |
| <code>--version</code> , <code>--version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |
| <code>--warnings-as-errors</code>                                                                                                                    |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                                 |

## See Also

- [kubectl rollout](#) - Manage the rollout of a resource

# kubectl rollout undo

## Synopsis

Roll back to a previous rollout.

```
kubectl rollout undo (TYPE NAME | TYPE/NAME) [flags]
```

## Examples

```
Roll back to the previous deployment
kubectl rollout undo deployment/abc

Roll back to daemonset revision 3
kubectl rollout undo daemonset/abc --to-revision=3

Roll back to the previous deployment with dry-run
kubectl rollout undo --dry-run=server deployment/abc
```

## Options

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--allow-missing-template-keys</code> Default: true                                                                                                                                                 |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |
| <code>--dry-run string[="unchanged"]</code> Default: "none"                                                                                                                                              |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -f, --filename strings                                                                                                                                                                                                                                    |
| Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                       |
| -h, --help                                                                                                                                                                                                                                                |
| help for undo                                                                                                                                                                                                                                             |
| -k, --kustomize string                                                                                                                                                                                                                                    |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                      |
| -o, --output string                                                                                                                                                                                                                                       |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                              |
| -R, --recursive                                                                                                                                                                                                                                           |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                           |
| -l, --selector string                                                                                                                                                                                                                                     |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                                |
| --show-managed-fields                                                                                                                                                                                                                                     |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                             |
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --to-revision int                                                                                                                                                                                                                                         |
| The revision to rollback to. Default to 0 (last revision).                                                                                                                                                                                                |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                             |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl rollout](#) - Manage the rollout of a resource

# kubectl run

## Synopsis

Create and run a particular image in a pod.

```
kubectl run NAME --image=image [--env="key=value"] [--port=port] [--dry-run=server|client]
[--overrides=inline-json] [--command] -- [COMMAND] [args...]
```

## Examples

```
Start a nginx pod
kubectl run nginx --image=nginx

Start a hazelcast pod and let the container expose port 5701
kubectl run hazelcast --image=hazelcast/hazelcast --port=5701

Start a hazelcast pod and set environment variables "DNS_DOMAIN=cluster" and
"POD_NAMESPACE=default" in the container
kubectl run hazelcast --image=hazelcast/hazelcast --env="DNS_DOMAIN=cluster" --
env="POD_NAMESPACE=default"

Start a hazelcast pod and set labels "app=hazelcast" and "env=prod" in the container
kubectl run hazelcast --image=hazelcast/hazelcast --labels="app=hazelcast,env=prod"

Dry run; print the corresponding API objects without creating them
kubectl run nginx --image=nginx --dry-run=client

Start a nginx pod, but overload the spec with a partial set of values parsed from JSON
kubectl run nginx --image=nginx --overrides='{ "apiVersion": "v1", "spec": { ... } }'

Start a busybox pod and keep it in the foreground, don't restart it if it exits
kubectl run -i -t busybox --image=busybox --restart=Never

Start the nginx pod using the default command, but use custom arguments (arg1 .. argN) for
that command
kubectl run nginx --image=nginx -- <arg1> <arg2> ... <argN>

Start the nginx pod using a different command and custom arguments
kubectl run nginx --image=nginx --command -- <cmd> <arg1> ... <argN>
```

## Options

|                                                                                                                                                 |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <code>--allow-missing-template-keys</code>                                                                                                      | Default: true |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats. |               |
| <code>--annotations</code>                                                                                                                      | strings       |

|                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotations to apply to the pod.                                                                                                                                                                                                                                        |
| --attach                                                                                                                                                                                                                                                                |
| If true, wait for the Pod to start running, and then attach to the Pod as if 'kubectl attach ...' were called. Default false, unless '-i/--stdin' is set, in which case the default is true. With '--restart=Never' the exit code of the container process is returned. |
| --command                                                                                                                                                                                                                                                               |
| If true and extra arguments are present, use them as the 'command' field in the container, rather than the 'args' field which is the default.                                                                                                                           |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                                                                                       |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                |
| --env strings                                                                                                                                                                                                                                                           |
| Environment variables to set in the container.                                                                                                                                                                                                                          |
| --expose --port                                                                                                                                                                                                                                                         |
| If true, create a ClusterIP service associated with the pod. Requires --port.                                                                                                                                                                                           |
| --field-manager string    Default: "kubectl-run"                                                                                                                                                                                                                        |
| Name of the manager used to track field ownership.                                                                                                                                                                                                                      |
| -h, --help                                                                                                                                                                                                                                                              |
| help for run                                                                                                                                                                                                                                                            |
| --image string                                                                                                                                                                                                                                                          |
| The image for the container to run.                                                                                                                                                                                                                                     |
| --image-pull-policy string                                                                                                                                                                                                                                              |
| The image pull policy for the container. If left empty, this value will not be specified by the client and defaulted by the server.                                                                                                                                     |
| -l, --labels string                                                                                                                                                                                                                                                     |
| Comma separated labels to apply to the pod. Will override previous values.                                                                                                                                                                                              |
| --leave-stdin-open                                                                                                                                                                                                                                                      |

|                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If the pod is started in interactive mode or with stdin, leave stdin open after the first attach completes. By default, stdin will be closed after the first attach completes.                                      |
| <code>-o, --output string</code>                                                                                                                                                                                    |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                        |
| <code>--override-type string</code> Default: "merge"                                                                                                                                                                |
| The method used to override the generated object: json, merge, or strategic.                                                                                                                                        |
| <code>--overrides string</code>                                                                                                                                                                                     |
| An inline JSON override for the generated object. If this is non-empty, it is used to override the generated object. Requires that the object supply a valid apiVersion field.                                      |
| <code>--pod-running-timeout duration</code> Default: 1m0s                                                                                                                                                           |
| The length of time (like 5s, 2m, or 3h, higher than zero) to wait until at least one pod is running                                                                                                                 |
| <code>--port string</code>                                                                                                                                                                                          |
| The port that this container exposes.                                                                                                                                                                               |
| <code>--privileged</code>                                                                                                                                                                                           |
| If true, run the container in privileged mode.                                                                                                                                                                      |
| <code>-q, --quiet</code>                                                                                                                                                                                            |
| If true, suppress prompt messages.                                                                                                                                                                                  |
| <code>--restart string</code> Default: "Always"                                                                                                                                                                     |
| The restart policy for this Pod. Legal values [Always, OnFailure, Never].                                                                                                                                           |
| <code>--rm</code>                                                                                                                                                                                                   |
| If true, delete the pod after it exits. Only valid when attaching to the container, e.g. with '--attach' or with '-i/--stdin'.                                                                                      |
| <code>--save-config</code>                                                                                                                                                                                          |
| If true, the configuration of current object will be saved in its annotation. Otherwise, the annotation will be unchanged. This flag is useful when you want to perform kubectl apply on this object in the future. |
| <code>--show-managed-fields</code>                                                                                                                                                                                  |



|                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                         |
| <code>-i, --stdin</code>                                                                                                                                                                                                                                                              |
| Keep stdin open on the container in the pod, even if nothing is attached.                                                                                                                                                                                                             |
| <code>--template string</code>                                                                                                                                                                                                                                                        |
| Template string or path to template file to use when <code>-o=go-template</code> , <code>-o=go-template-file</code> . The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| <code>-t, --tty</code>                                                                                                                                                                                                                                                                |
| Allocate a TTY for the container in the pod.                                                                                                                                                                                                                                          |

|                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|
| <code>--as string</code>                                                                                                   |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.               |
| <code>--as-group strings</code>                                                                                            |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                              |
| <code>--as-uid string</code>                                                                                               |
| UID to impersonate for the operation.                                                                                      |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                              |
| Default cache directory                                                                                                    |
| <code>--certificate-authority string</code>                                                                                |
| Path to a cert file for the certificate authority                                                                          |
| <code>--client-certificate string</code>                                                                                   |
| Path to a client certificate file for TLS                                                                                  |
| <code>--client-key string</code>                                                                                           |
| Path to a client key file for TLS                                                                                          |
| <code>--cloud-provider-gce-l7lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,35.191.0.0/16                               |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                       |
| <code>--cloud-provider-gce-lb-src-cidrs cidrs</code> Default: 130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |

|                                           |                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                  | Default: "0"                                                                                                                                                                                      |
|                                           | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                       |                                                                                                                                                                                                   |
|                                           | The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration | Default: 1m0s                                                                                                                                                                                     |
|                                           | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string                | Default: "cadvisor"                                                                                                                                                                               |
|                                           | database name                                                                                                                                                                                     |
| --storage-driver-host string              | Default: "localhost:8086"                                                                                                                                                                         |
|                                           | database host:port                                                                                                                                                                                |
| --storage-driver-password string          | Default: "root"                                                                                                                                                                                   |
|                                           | database password                                                                                                                                                                                 |
| --storage-driver-secure                   |                                                                                                                                                                                                   |
|                                           | use secure connection with database                                                                                                                                                               |
| --storage-driver-table string             | Default: "stats"                                                                                                                                                                                  |
|                                           | table name                                                                                                                                                                                        |
| --storage-driver-user string              | Default: "root"                                                                                                                                                                                   |
|                                           | database username                                                                                                                                                                                 |
| --tls-server-name string                  |                                                                                                                                                                                                   |
|                                           | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                            |                                                                                                                                                                                                   |
|                                           | Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                             |                                                                                                                                                                                                   |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl scale

## Synopsis

Set a new size for a deployment, replica set, replication controller, or stateful set.

Scale also allows users to specify one or more preconditions for the scale action.

If --current-replicas or --resource-version is specified, it is validated before the scale is attempted, and it is guaranteed that the precondition holds true when the scale is sent to the server.

```
kubectl scale [--resource-version=version] [--current-replicas=count] --replicas=COUNT (-f
FILENAME | TYPE NAME)
```

## Examples

```
Scale a replica set named 'foo' to 3
kubectl scale --replicas=3 rs/foo

Scale a resource identified by type and name specified in "foo.yaml" to 3
kubectl scale --replicas=3 -f foo.yaml

If the deployment named mysql's current size is 2, scale mysql to 3
kubectl scale --current-replicas=2 --replicas=3 deployment/mysql

Scale multiple replication controllers
kubectl scale --replicas=5 rc/example1 rc/example2 rc/example3
```

```
Scale stateful set named 'web' to 3
kubectl scale --replicas=3 statefulset/web
```

## Options

|                                                                                                                                                                                                          |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <code>--all</code>                                                                                                                                                                                       |  |
| Select all resources in the namespace of the specified resource types                                                                                                                                    |  |
| <code>--allow-missing-template-keys</code> Default: true                                                                                                                                                 |  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                          |  |
| <code>--current-replicas</code> int    Default: -1                                                                                                                                                       |  |
| Precondition for current size. Requires that the current size of the resource match this value in order to scale. -1 (default) for no condition.                                                         |  |
| <code>--dry-run</code> string[="unchanged"]    Default: "none"                                                                                                                                           |  |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |  |
| <code>-f, --filename</code> strings                                                                                                                                                                      |  |
| Filename, directory, or URL to files identifying the resource to set a new size                                                                                                                          |  |
| <code>-h, --help</code>                                                                                                                                                                                  |  |
| help for scale                                                                                                                                                                                           |  |
| <code>-k, --kustomize</code> string                                                                                                                                                                      |  |
| Process the kustomization directory. This flag can't be used together with <code>-f</code> or <code>-R</code> .                                                                                          |  |
| <code>-o, --output</code> string                                                                                                                                                                         |  |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                             |  |
| <code>-R, --recursive</code>                                                                                                                                                                             |  |
| Process the directory used in <code>-f, --filename</code> recursively. Useful when you want to manage related manifests organized within the same directory.                                             |  |
| <code>--replicas</code> int                                                                                                                                                                              |  |
| The new desired number of replicas. Required.                                                                                                                                                            |  |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--resource-version string</code>                                                                                                                                                                                                                   |
| Precondition for resource version. Requires that the current resource version match this value in order to scale.                                                                                                                                        |
| <code>-l, --selector string</code>                                                                                                                                                                                                                       |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                               |
| <code>--show-managed-fields</code>                                                                                                                                                                                                                       |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| <code>--template string</code>                                                                                                                                                                                                                           |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| <code>--timeout duration</code>                                                                                                                                                                                                                          |
| The length of time to wait before giving up on a scale operation, zero means don't wait. Any other values should contain a corresponding time unit (e.g. 1s, 2m, 3h).                                                                                    |

|                                                                                                              |
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| <code>--as string</code>                                                                                     |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| <code>--as-group strings</code>                                                                              |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| <code>--as-uid string</code>                                                                                 |
| UID to impersonate for the operation.                                                                        |
| <code>--cache-dir string</code> Default: "\$HOME/.kube/cache"                                                |
| Default cache directory                                                                                      |
| <code>--certificate-authority string</code>                                                                  |
| Path to a cert file for the certificate authority                                                            |
| <code>--client-certificate string</code>                                                                     |
| Path to a client certificate file for TLS                                                                    |
| <code>--client-key string</code>                                                                             |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Password for basic authentication to the API server</p>                                                                                                                                               |
| <p>--profile string   Default: "none"</p>                                                                                                                                                                |
| <p>Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)</p>                                                                                                             |
| <p>--profile-output string   Default: "profile.pprof"</p>                                                                                                                                                |
| <p>Name of the file to write the profile to</p>                                                                                                                                                          |
| <p>--request-timeout string   Default: "0"</p>                                                                                                                                                           |
| <p>The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests.</p> |
| <p>-s, --server string</p>                                                                                                                                                                               |
| <p>The address and port of the Kubernetes API server</p>                                                                                                                                                 |
| <p>--storage-driver-buffer-duration duration   Default: 1m0s</p>                                                                                                                                         |
| <p>Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction</p>                                                                 |
| <p>--storage-driver-db string   Default: "cadvisor"</p>                                                                                                                                                  |
| <p>database name</p>                                                                                                                                                                                     |
| <p>--storage-driver-host string   Default: "localhost:8086"</p>                                                                                                                                          |
| <p>database host:port</p>                                                                                                                                                                                |
| <p>--storage-driver-password string   Default: "root"</p>                                                                                                                                                |
| <p>database password</p>                                                                                                                                                                                 |
| <p>--storage-driver-secure</p>                                                                                                                                                                           |
| <p>use secure connection with database</p>                                                                                                                                                               |
| <p>--storage-driver-table string   Default: "stats"</p>                                                                                                                                                  |
| <p>table name</p>                                                                                                                                                                                        |
| <p>--storage-driver-user string   Default: "root"</p>                                                                                                                                                    |
| <p>database username</p>                                                                                                                                                                                 |
| <p>--tls-server-name string</p>                                                                                                                                                                          |



|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl](#) - kubectrl controls the Kubernetes cluster manager

# kubectrl set

## Synopsis

Configure application resources.

These commands help you make changes to existing application resources.

kubectrl set SUBCOMMAND

## Options

|              |
|--------------|
| -h, --help   |
| help for set |

|             |
|-------------|
| --as string |
|-------------|

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl set env](#) - Update environment variables on a pod template
- [kubectl set image](#) - Update the image of a pod template
- [kubectl set resources](#) - Update resource requests/limits on objects with pod templates
- [kubectl set selector](#) - Set the selector on a resource
- [kubectl set serviceaccount](#) - Update the service account of a resource
- [kubectl set subject](#) - Update the user, group, or service account in a role binding or cluster role binding

# kubectl set env

## Synopsis

Update environment variables on a pod template.

List environment variable definitions in one or more pods, pod templates. Add, update, or remove container environment variable definitions in one or more pod templates (within replication controllers or deployment configurations). View or modify the environment variable definitions on all containers in the specified pods or pod templates, or just those that match a wildcard.

If "--env -" is passed, environment variables can be read from STDIN using the standard env syntax.

Possible resources include (case insensitive):

pod (po), replicationcontroller (rc), deployment (deploy), daemonset (ds), statefulset (sts), cronjob (cj), replicaset (rs)

```
kubectl set env RESOURCE/NAME KEY_1=VAL_1 ... KEY_N=VAL_N
```

## Examples

```
Update deployment 'registry' with a new environment variable
kubectl set env deployment/registry STORAGE_DIR=/local

List the environment variables defined on a deployments 'sample-build'
kubectl set env deployment/sample-build --list

List the environment variables defined on all pods
kubectl set env pods --all --list

Output modified deployment in YAML, and does not alter the object on the server
kubectl set env deployment/sample-build STORAGE_DIR=/data -o yaml

Update all containers in all replication controllers in the project to have ENV=prod
kubectl set env rc --all ENV=prod
```

```
Import environment from a secret
kubectl set env --from=secret/mysecret deployment/myapp

Import environment from a config map with a prefix
kubectl set env --from=configmap/myconfigmap --prefix=MYSQL_ deployment/myapp

Import specific keys from a config map
kubectl set env --keys=my-example-key --from=configmap/myconfigmap deployment/myapp

Remove the environment variable ENV from container 'c1' in all deployment configs
kubectl set env deployments --all --containers="c1" ENV-

Remove the environment variable ENV from a deployment definition on disk and
update the deployment config on the server
kubectl set env -f deploy.json ENV-

Set some of the local shell environment into a deployment config on the server
env | grep RAILS_ | kubectl set env -e - deployment/registry
```

## Options

|                                             |                                                                                                                                                                                                          |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--all</code>                          |                                                                                                                                                                                                          |
|                                             | If true, select all resources in the namespace of the specified resource types                                                                                                                           |
| <code>--allow-missing-template-keys</code>  | Default: true                                                                                                                                                                                            |
|                                             | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to go lang and jsonpath output formats.                                                         |
| <code>-c, --containers string</code>        | Default: ""                                                                                                                                                                                              |
|                                             | The names of containers in the selected pod templates to change - may use wildcards                                                                                                                      |
| <code>--dry-run string[="unchanged"]</code> | Default: "none"                                                                                                                                                                                          |
|                                             | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| <code>-e, --env strings</code>              |                                                                                                                                                                                                          |
|                                             | Specify a key-value pair for an environment variable to set into each container.                                                                                                                         |
| <code>--field-manager string</code>         | Default: "kubectl-set"                                                                                                                                                                                   |
|                                             | Name of the manager used to track field ownership.                                                                                                                                                       |
| <code>-f, --filename strings</code>         |                                                                                                                                                                                                          |
|                                             | Filename, directory, or URL to files the resource to update the env                                                                                                                                      |

|                              |                                                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| --from string                |                                                                                                                                                 |
|                              | The name of a resource from which to inject environment variables                                                                               |
| -h, --help                   |                                                                                                                                                 |
|                              | help for env                                                                                                                                    |
| --keys strings               |                                                                                                                                                 |
|                              | Comma-separated list of keys to import from specified resource                                                                                  |
| -k, --kustomize string       |                                                                                                                                                 |
|                              | Process the kustomization directory. This flag can't be used together with -f or -R.                                                            |
| --list                       |                                                                                                                                                 |
|                              | If true, display the environment and any changes in the standard format. this flag will removed when we have kubectl view env.                  |
| --local                      |                                                                                                                                                 |
|                              | If true, set env will NOT contact api-server but run locally.                                                                                   |
| -o, --output string          |                                                                                                                                                 |
|                              | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).    |
| --overwrite    Default: true |                                                                                                                                                 |
|                              | If true, allow environment to be overwritten, otherwise reject updates that overwrite existing environment.                                     |
| --prefix string              |                                                                                                                                                 |
|                              | Prefix to append to variable names                                                                                                              |
| -R, --recursive              |                                                                                                                                                 |
|                              | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory. |
| --resolve                    |                                                                                                                                                 |
|                              | If true, show secret or configmap references when listing variables                                                                             |
| -l, --selector string        |                                                                                                                                                 |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                               |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |



|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--cluster string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl set](#) - Set specific features on objects

# kubectl set image

## Synopsis

Update existing container image(s) of resources.

Possible resources include (case insensitive):

pod (po), replicationcontroller (rc), deployment (deploy), daemonset (ds), statefulset (sts), cronjob (cj), replicaset (rs)

```
kubectl set image (-f FILENAME | TYPE NAME)
CONTAINER_NAME_1=CONTAINER_IMAGE_1 ...
CONTAINER_NAME_N=CONTAINER_IMAGE_N
```

## Examples

```
Set a deployment's nginx container image to 'nginx:1.9.1', and its busybox container image to 'busybox'
kubectl set image deployment/nginx busybox=busybox nginx=nginx:1.9.1

Update all deployments' and rc's nginx container's image to 'nginx:1.9.1'
kubectl set image deployments,rc nginx=nginx:1.9.1 --all

Update image of all containers of daemonset abc to 'nginx:1.9.1'
kubectl set image daemonset abc *=nginx:1.9.1

Print result (in yaml format) of updating nginx container image from local file, without hitting the server
kubectl set image -f path/to/file.yaml nginx=nginx:1.9.1 --local -o yaml
```

# Options

|                                                             |                                                                                                                                                                                                          |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--all</code>                                          |                                                                                                                                                                                                          |
|                                                             | Select all resources, in the namespace of the specified resource types                                                                                                                                   |
| <code>--allow-missing-template-keys</code> Default: true    |                                                                                                                                                                                                          |
|                                                             | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                          |
| <code>--dry-run string[="unchanged"]</code> Default: "none" |                                                                                                                                                                                                          |
|                                                             | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| <code>--field-manager string</code> Default: "kubectl-set"  |                                                                                                                                                                                                          |
|                                                             | Name of the manager used to track field ownership.                                                                                                                                                       |
| <code>-f, --filename strings</code>                         |                                                                                                                                                                                                          |
|                                                             | Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                      |
| <code>-h, --help</code>                                     |                                                                                                                                                                                                          |
|                                                             | help for image                                                                                                                                                                                           |
| <code>-k, --kustomize string</code>                         |                                                                                                                                                                                                          |
|                                                             | Process the kustomization directory. This flag can't be used together with <code>-f</code> or <code>-R</code> .                                                                                          |
| <code>--local</code>                                        |                                                                                                                                                                                                          |
|                                                             | If true, set image will NOT contact api-server but run locally.                                                                                                                                          |
| <code>-o, --output string</code>                            |                                                                                                                                                                                                          |
|                                                             | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                             |
| <code>-R, --recursive</code>                                |                                                                                                                                                                                                          |
|                                                             | Process the directory used in <code>-f, --filename</code> recursively. Useful when you want to manage related manifests organized within the same directory.                                             |
| <code>-l, --selector string</code>                          |                                                                                                                                                                                                          |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                               |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                               |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--cluster string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl set](#) - Set specific features on objects

# kubectl set resources

## Synopsis

Specify compute resource requirements (CPU, memory) for any resource that defines a pod template. If a pod is successfully scheduled, it is guaranteed the amount of resource requested, but may burst up to its specified limits.

For each compute resource, if a limit is specified and a request is omitted, the request will default to the limit.

Possible resources include (case insensitive): Use "kubectl api-resources" for a complete list of supported resources..

```
kubectl set resources (-f FILENAME | TYPE NAME) ([--limits=LIMITS & --requests=REQUESTS]
```

## Examples

```
Set a deployments nginx container cpu limits to "200m" and memory to "512Mi"
kubectl set resources deployment nginx -c=nginx --limits=cpu=200m,memory=512Mi

Set the resource request and limits for all containers in nginx
kubectl set resources deployment nginx --limits=cpu=200m,memory=512Mi --requests=cpu=100m,memory=256Mi

Remove the resource requests for resources on containers in nginx
kubectl set resources deployment nginx --limits=cpu=0,memory=0 --requests=cpu=0,memory=0

Print the result (in yaml format) of updating nginx container limits from a local, without hitting the server
kubectl set resources -f path/to/file.yaml --limits=cpu=200m,memory=512Mi --local -o yaml
```



# Options

|                                                                                                                                                                                                               |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <code>--all</code>                                                                                                                                                                                            |  |
| Select all resources, in the namespace of the specified resource types                                                                                                                                        |  |
| <code>--allow-missing-template-keys</code> Default: true                                                                                                                                                      |  |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                               |  |
| <code>-c, --containers string</code> Default: ""                                                                                                                                                              |  |
| The names of containers in the selected pod templates to change, all containers are selected by default - may use wildcards                                                                                   |  |
| <code>--dry-run string[="unchanged"]</code> Default: "none"                                                                                                                                                   |  |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.      |  |
| <code>--field-manager string</code> Default: "kubectl-set"                                                                                                                                                    |  |
| Name of the manager used to track field ownership.                                                                                                                                                            |  |
| <code>-f, --filename strings</code>                                                                                                                                                                           |  |
| Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                           |  |
| <code>-h, --help</code>                                                                                                                                                                                       |  |
| help for resources                                                                                                                                                                                            |  |
| <code>-k, --kustomize string</code>                                                                                                                                                                           |  |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                          |  |
| <code>--limits string</code>                                                                                                                                                                                  |  |
| The resource requirement requests for this container. For example, 'cpu=100m,memory=256Mi'. Note that server side components may assign requests depending on the server configuration, such as limit ranges. |  |
| <code>--local</code>                                                                                                                                                                                          |  |
| If true, set resources will NOT contact api-server but run locally.                                                                                                                                           |  |
| <code>-o, --output string</code>                                                                                                                                                                              |  |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                             |
| -R, --recursive                                                                                                                                                                                                                                          |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| --requests string                                                                                                                                                                                                                                        |
| The resource requirement requests for this container. For example, 'cpu=100m,memory=256Mi'. Note that server side components may assign requests depending on the server configuration, such as limit ranges.                                            |
| -l, --selector string                                                                                                                                                                                                                                    |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                               |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectrl set](#) - Set specific features on objects

# kubectrl set selector

## Synopsis

Set the selector on a resource. Note that the new selector will overwrite the old selector if the resource had one prior to the invocation of 'set selector'.

A selector must begin with a letter or number, and may contain letters, numbers, hyphens, dots, and underscores, up to 63 characters. If --resource-version is specified, then updates will use this resource version, otherwise the existing resource-version will be used. Note: currently selectors can only be set on Service objects.

```
kubectrl set selector (-f FILENAME | TYPE NAME) EXPRESSIONS [--resource-version=version]
```

## Examples

```
Set the labels and selector before creating a deployment/service pair
kubectl create service clusterip my-svc --clusterip="None" -o yaml --dry-run=client | kubectl
set selector --local -f - 'environment=qa' -o yaml | kubectl create -f -
kubectl create deployment my-dep -o yaml --dry-run=client | kubectl label --local -f -
environment=qa -o yaml | kubectl create -f -
```

## Options

|                                                   |                                                                                                                                                                                                          |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --all                                             | Select all resources in the namespace of the specified resource types                                                                                                                                    |
| --allow-missing-template-keys    Default: true    | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                          |
| --dry-run string[="unchanged"]    Default: "none" | Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| --field-manager string    Default: "kubectl-set"  | Name of the manager used to track field ownership.                                                                                                                                                       |
| -f, --filename strings                            | identifying the resource.                                                                                                                                                                                |
| -h, --help                                        | help for selector                                                                                                                                                                                        |
| --local                                           | If true, annotation will NOT contact api-server but run locally.                                                                                                                                         |
| -o, --output string                               | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                             |
| -R, --recursive    Default: true                  | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                          |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --resource-version string                                                                                                                                                                                                                                |
| If non-empty, the selectors update will only succeed if this is the current resource-version for the object. Only valid when specifying a single resource.                                                                                               |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                        |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.       |
| --as-group strings                                                                                                 |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                      |
| --as-uid string                                                                                                    |
| UID to impersonate for the operation.                                                                              |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                            |
| --certificate-authority string                                                                                     |
| Path to a cert file for the certificate authority                                                                  |
| --client-certificate string                                                                                        |
| Path to a client certificate file for TLS                                                                          |
| --client-key string                                                                                                |
| Path to a client key file for TLS                                                                                  |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                               |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| --profile-output string    Default: "profile.pprof"                                                                                                             |



|                                           |                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                  | Default: "0"                                                                                                                                                                                      |
|                                           | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                       |                                                                                                                                                                                                   |
|                                           | The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration | Default: 1m0s                                                                                                                                                                                     |
|                                           | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string                | Default: "cadvisor"                                                                                                                                                                               |
|                                           | database name                                                                                                                                                                                     |
| --storage-driver-host string              | Default: "localhost:8086"                                                                                                                                                                         |
|                                           | database host:port                                                                                                                                                                                |
| --storage-driver-password string          | Default: "root"                                                                                                                                                                                   |
|                                           | database password                                                                                                                                                                                 |
| --storage-driver-secure                   |                                                                                                                                                                                                   |
|                                           | use secure connection with database                                                                                                                                                               |
| --storage-driver-table string             | Default: "stats"                                                                                                                                                                                  |
|                                           | table name                                                                                                                                                                                        |
| --storage-driver-user string              | Default: "root"                                                                                                                                                                                   |
|                                           | database username                                                                                                                                                                                 |
| --tls-server-name string                  |                                                                                                                                                                                                   |
|                                           | Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| --token string                            |                                                                                                                                                                                                   |
|                                           | Bearer token for authentication to the API server                                                                                                                                                 |
| --user string                             |                                                                                                                                                                                                   |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| The name of the kubeconfig user to use                                                                       |
| --username string                                                                                            |
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl set](#) - Set specific features on objects

# kubectl set serviceaccount

## Synopsis

Update the service account of pod template resources.

Possible resources (case insensitive) can be:

replicationcontroller (rc), deployment (deploy), daemonset (ds), job, replicaset (rs), statefulset

```
kubectl set serviceaccount (-f FILENAME | TYPE NAME) SERVICE_ACCOUNT
```

## Examples

```
Set deployment nginx-deployment's service account to serviceaccount1
kubectl set serviceaccount deployment nginx-deployment serviceaccount1

Print the result (in YAML format) of updated nginx deployment with the service account
from local file, without hitting the API server
kubectl set sa -f nginx-deployment.yaml serviceaccount1 --local --dry-run=client -o yaml
```

## Options

|                                                                        |
|------------------------------------------------------------------------|
| --all                                                                  |
| Select all resources, in the namespace of the specified resource types |
| --allow-missing-template-keys    Default: true                         |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                          |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                                                                        |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                 |
| --field-manager string    Default: "kubectl-set"                                                                                                                                                                                                         |
| Name of the manager used to track field ownership.                                                                                                                                                                                                       |
| -f, --filename strings                                                                                                                                                                                                                                   |
| Filename, directory, or URL to files identifying the resource to get from a server.                                                                                                                                                                      |
| -h, --help                                                                                                                                                                                                                                               |
| help for serviceaccount                                                                                                                                                                                                                                  |
| -k, --kustomize string                                                                                                                                                                                                                                   |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| --local                                                                                                                                                                                                                                                  |
| If true, set serviceaccount will NOT contact api-server but run locally.                                                                                                                                                                                 |
| -o, --output string                                                                                                                                                                                                                                      |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                             |
| -R, --recursive                                                                                                                                                                                                                                          |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --as string                                                                                                                                                                                                                                              |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                                                                           |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs   Default: 130.211.0.0/22,35.191.0.0/16                                                                            |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs   Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                           |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int   Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --default-unreachable-toleration-seconds int   Default: 300                                                                                                  |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |

|                                                                                                                              |                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| --storage-driver-db string                                                                                                   | Default: "cadvisor"       |
| database name                                                                                                                |                           |
| --storage-driver-host string                                                                                                 | Default: "localhost:8086" |
| database host:port                                                                                                           |                           |
| --storage-driver-password string                                                                                             | Default: "root"           |
| database password                                                                                                            |                           |
| --storage-driver-secure                                                                                                      |                           |
| use secure connection with database                                                                                          |                           |
| --storage-driver-table string                                                                                                | Default: "stats"          |
| table name                                                                                                                   |                           |
| --storage-driver-user string                                                                                                 | Default: "root"           |
| database username                                                                                                            |                           |
| --tls-server-name string                                                                                                     |                           |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |                           |
| --token string                                                                                                               |                           |
| Bearer token for authentication to the API server                                                                            |                           |
| --user string                                                                                                                |                           |
| The name of the kubeconfig user to use                                                                                       |                           |
| --username string                                                                                                            |                           |
| Username for basic authentication to the API server                                                                          |                           |
| --version version[=true]                                                                                                     |                           |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |                           |
| --warnings-as-errors                                                                                                         |                           |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |                           |

## See Also

- [kubect set](#) - Set specific features on objects

# kubect set subject

## Synopsis

Update the user, group, or service account in a role binding or cluster role binding.

```
kubect set subject (-f FILENAME | TYPE NAME) [--user=username] [--group=groupname] [--serviceaccount=namespace:serviceaccountname] [--dry-run=server|client|none]
```

## Examples

```
Update a cluster role binding for serviceaccount1
kubect set subject clusterrolebinding admin --serviceaccount=namespace:serviceaccount1

Update a role binding for user1, user2, and group1
kubect set subject rolebinding admin --user=user1 --user=user2 --group=group1

Print the result (in YAML format) of updating rolebinding subjects from a local, without hitting the server
kubect create rolebinding admin --role=admin --user=admin -o yaml --dry-run=client |
kubect set subject --local -f - --user=foo -o yaml
```

## Options

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --all                                                                                                                                                                                                    |
| Select all resources, in the namespace of the specified resource types                                                                                                                                   |
| --allow-missing-template-keys    Default: true                                                                                                                                                           |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goolang and jsonpath output formats.                                                         |
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                        |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| --field-manager string    Default: "kubectl-set"                                                                                                                                                         |
| Name of the manager used to track field ownership.                                                                                                                                                       |
| -f, --filename strings                                                                                                                                                                                   |

|                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filename, directory, or URL to files the resource to update the subjects                                                                                                                                                                                 |
| --group strings                                                                                                                                                                                                                                          |
| Groups to bind to the role                                                                                                                                                                                                                               |
| -h, --help                                                                                                                                                                                                                                               |
| help for subject                                                                                                                                                                                                                                         |
| -k, --kustomize string                                                                                                                                                                                                                                   |
| Process the kustomization directory. This flag can't be used together with -f or -R.                                                                                                                                                                     |
| --local                                                                                                                                                                                                                                                  |
| If true, set subject will NOT contact api-server but run locally.                                                                                                                                                                                        |
| -o, --output string                                                                                                                                                                                                                                      |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                             |
| -R, --recursive                                                                                                                                                                                                                                          |
| Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                          |
| -l, --selector string                                                                                                                                                                                                                                    |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                               |
| --serviceaccount strings                                                                                                                                                                                                                                 |
| Service accounts to bind to the role                                                                                                                                                                                                                     |
| --show-managed-fields                                                                                                                                                                                                                                    |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                            |
| --template string                                                                                                                                                                                                                                        |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --user strings                                                                                                                                                                                                                                           |
| Username to bind to the role                                                                                                                                                                                                                             |



|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                          |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                           |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                          |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                   |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                                                      |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                                                                                                                                                                             |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                                                                                                                                                                        |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string    Default: "none"                                                                                                                                                               |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string    Default: "profile.pprof"                                                                                                                                               |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string    Default: "0"                                                                                                                                                          |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                                                                                        |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cAdvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                              |

## See Also

- [kubectl set](#) - Set specific features on objects

# kubectl taint

## Synopsis

Update the taints on one or more nodes.

- A taint consists of a key, value, and effect. As an argument here, it is expressed as key=value:effect.
- The key must begin with a letter or number, and may contain letters, numbers, hyphens, dots, and underscores, up to 253 characters.
- Optionally, the key can begin with a DNS subdomain prefix and a single '/', like example.com/my-app.
- The value is optional. If given, it must begin with a letter or number, and may contain letters, numbers, hyphens, dots, and underscores, up to 63 characters.
- The effect must be NoSchedule, PreferNoSchedule or NoExecute.
- Currently taint can only apply to node.

```
kubectl taint NODE NAME KEY_1=VAL_1:TAINT_EFFECT_1 ...
KEY_N=VAL_N:TAINT_EFFECT_N
```

## Examples

```
Update node 'foo' with a taint with key 'dedicated' and value 'special-user' and effect
'NoSchedule'
If a taint with that key and effect already exists, its value is replaced as specified
kubectl taint nodes foo dedicated=special-user:NoSchedule

Remove from node 'foo' the taint with key 'dedicated' and effect 'NoSchedule' if one exists
kubectl taint nodes foo dedicated:NoSchedule-

Remove from node 'foo' all the taints with key 'dedicated'
kubectl taint nodes foo dedicated-

Add a taint with key 'dedicated' on nodes having label myLabel=X
kubectl taint node -l myLabel=X dedicated=foo:PreferNoSchedule

Add to node 'foo' a taint with key 'bar' and no value
kubectl taint nodes foo bar:NoSchedule
```

## Options

|                                 |
|---------------------------------|
| --all                           |
| Select all nodes in the cluster |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--allow-missing-template-keys</code> Default: true                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to golang and jsonpath output formats.                                                                                                                                                                                                                                                                                                                                                                   |
| <code>--dry-run string[="unchanged"]</code> Default: "none"                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource.                                                                                                                                                                                                                                                                                                          |
| <code>--field-manager string</code> Default: "kubectl-taint"                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Name of the manager used to track field ownership.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-h, --help</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| help for taint                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-o, --output string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                                                                                                                                                                                                                                                      |
| <code>--overwrite</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| If true, allow taints to be overwritten, otherwise reject taint updates that overwrite existing taints.                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-l, --selector string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                                                                                                                                                                                                                                                                                                                                        |
| <code>--show-managed-fields</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>--template string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is golang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ].                                                                                                                                                                                                                                                          |
| <code>--validate string[="strict"]</code> Default: "strict"                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Must be one of: strict (or true), warn, ignore (or false).<br>"true" or "strict" will use a schema to validate the input and fail the request if invalid. It will perform server side validation if ServerSideFieldValidation is enabled on the api-server, but will fall back to less reliable client-side validation if not.<br>"warn" will warn about unknown or duplicate fields without blocking the request if server-side field validation is enabled on the API server, and behave as "ignore" otherwise. |

"false" or "ignore" will not perform any schema validation, silently dropping any unknown or duplicate fields.

--as string

Username to impersonate for the operation. User could be a regular user or a service account in a namespace.

--as-group strings

Group to impersonate for the operation, this flag can be repeated to specify multiple groups.

--as-uid string

UID to impersonate for the operation.

--cache-dir string Default: "\$HOME/.kube/cache"

Default cache directory

--certificate-authority string

Path to a cert file for the certificate authority

--client-certificate string

Path to a client certificate file for TLS

--client-key string

Path to a client key file for TLS

--cloud-provider-gce-l7lb-src-cidrs cidrs Default: 130.211.0.0/22,35.191.0.0/16

CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks

--cloud-provider-gce-lb-src-cidrs cidrs Default:  
130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16

CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks

--cluster string

The name of the kubeconfig cluster to use

--context string

The name of the kubeconfig context to use

--default-not-ready-toleration-seconds int Default: 300

|                                              |                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                      |
| --default-unreachable-toleration-seconds int | Default: 300                                                                                                                                                                                      |
|                                              | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |
| --disable-compression                        |                                                                                                                                                                                                   |
|                                              | If true, opt-out of response compression for all requests to the server                                                                                                                           |
| --insecure-skip-tls-verify                   |                                                                                                                                                                                                   |
|                                              | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                          |                                                                                                                                                                                                   |
|                                              | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                       |                                                                                                                                                                                                   |
|                                              | Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                       |                                                                                                                                                                                                   |
|                                              | If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                            |                                                                                                                                                                                                   |
|                                              | Password for basic authentication to the API server                                                                                                                                               |
| --profile string                             | Default: "none"                                                                                                                                                                                   |
|                                              | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string                      | Default: "profile.pprof"                                                                                                                                                                          |
|                                              | Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string                     | Default: "0"                                                                                                                                                                                      |
|                                              | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                          |                                                                                                                                                                                                   |
|                                              | The address and port of the Kubernetes API server                                                                                                                                                 |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| --storage-driver-buffer-duration duration    Default: 1m0s                                                                        |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |



|                                                                                      |
|--------------------------------------------------------------------------------------|
| Treat warnings received from the server as errors and exit with a non-zero exit code |
|--------------------------------------------------------------------------------------|

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl top

## Synopsis

Display resource (CPU/memory) usage.

The top command allows you to see the resource consumption for nodes or pods.

This command requires Metrics Server to be correctly configured and working on the server.

kubectl top [flags]

## Options

|              |
|--------------|
| -h, --help   |
| help for top |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                                           |
| Default cache directory                                                                                      |
| --certificate-authority string                                                                               |
| Path to a cert file for the certificate authority                                                            |
| --client-certificate string                                                                                  |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |
| database password                                                                                                                                                                                 |
| --storage-driver-secure                                                                                                                                                                           |
| use secure connection with database                                                                                                                                                               |
| --storage-driver-table string   Default: "stats"                                                                                                                                                  |
| table name                                                                                                                                                                                        |
| --storage-driver-user string   Default: "root"                                                                                                                                                    |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
- [kubectl top node](#) - Display resource (CPU/memory) usage of nodes
- [kubectl top pod](#) - Display resource (CPU/memory) usage of pods

# kubectl top node

## Synopsis

Display resource (CPU/memory) usage of nodes.

The top-node command allows you to see the resource consumption of nodes.

```
kubectl top node [NAME | -l label]
```

# Examples

```
Show metrics for all nodes
kubectl top node

Show metrics for a given node
kubectl top node NODE_NAME
```

# Options

|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -h, --help                                                                                                                                                                 |
| help for node                                                                                                                                                              |
| --no-headers                                                                                                                                                               |
| If present, print output without headers                                                                                                                                   |
| -l, --selector string                                                                                                                                                      |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints. |
| --show-capacity                                                                                                                                                            |
| Print node resources based on Capacity instead of Allocatable(default) of the nodes.                                                                                       |
| --sort-by string                                                                                                                                                           |
| If non-empty, sort nodes list using specified field. The field can be either 'cpu' or 'memory'.                                                                            |
| --use-protocol-buffers    Default: true                                                                                                                                    |
| Enables using protocol-buffers to access Metrics API.                                                                                                                      |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |
| UID to impersonate for the operation.                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                          |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |

|  |                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
|  | --match-server-version                                                                                                                                                                            |
|  | Require server version to match client version                                                                                                                                                    |
|  | -n, --namespace string                                                                                                                                                                            |
|  | If present, the namespace scope for this CLI request                                                                                                                                              |
|  | --password string                                                                                                                                                                                 |
|  | Password for basic authentication to the API server                                                                                                                                               |
|  | --profile string   Default: "none"                                                                                                                                                                |
|  | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
|  | --profile-output string   Default: "profile.pprof"                                                                                                                                                |
|  | Name of the file to write the profile to                                                                                                                                                          |
|  | --request-timeout string   Default: "0"                                                                                                                                                           |
|  | The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
|  | -s, --server string                                                                                                                                                                               |
|  | The address and port of the Kubernetes API server                                                                                                                                                 |
|  | --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
|  | Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
|  | --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
|  | database name                                                                                                                                                                                     |
|  | --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
|  | database host:port                                                                                                                                                                                |
|  | --storage-driver-password string   Default: "root"                                                                                                                                                |
|  | database password                                                                                                                                                                                 |
|  | --storage-driver-secure                                                                                                                                                                           |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl top](#) - Display resource (CPU/memory) usage

# kubectl top pod

## Synopsis

Display resource (CPU/memory) usage of pods.

The 'top pod' command allows you to see the resource consumption of pods.



Due to the metrics pipeline delay, they may be unavailable for a few minutes since pod creation.

```
kubecttl top pod [NAME | -l label]
```

## Examples

```
Show metrics for all pods in the default namespace
kubecttl top pod

Show metrics for all pods in the given namespace
kubecttl top pod --namespace=NAMESPACE

Show metrics for a given pod and its containers
kubecttl top pod POD_NAME --containers

Show metrics for the pods defined by label name=myLabel
kubecttl top pod -l name=myLabel
```

## Options

|                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -A, --all-namespaces                                                                                                                                                                    |
| If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.                                             |
| --containers                                                                                                                                                                            |
| If present, print usage of containers within a pod.                                                                                                                                     |
| --field-selector string                                                                                                                                                                 |
| Selector (field query) to filter on, supports '=', '==', and '!='.(e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type. |
| -h, --help                                                                                                                                                                              |
| help for pod                                                                                                                                                                            |
| --no-headers                                                                                                                                                                            |
| If present, print output without headers.                                                                                                                                               |
| -l, --selector string                                                                                                                                                                   |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.              |
| --sort-by string                                                                                                                                                                        |

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| If non-empty, sort pods list using specified field. The field can be either 'cpu' or 'memory'. |
| --sum                                                                                          |
| Print the sum of the resource usage                                                            |
| --use-protocol-buffers    Default: true                                                        |
| Enables using protocol-buffers to access Metrics API.                                          |

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.        |
| --as-group strings                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                       |
| --as-uid string                                                                                                     |
| UID to impersonate for the operation.                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                             |
| --certificate-authority string                                                                                      |
| Path to a cert file for the certificate authority                                                                   |
| --client-certificate string                                                                                         |
| Path to a client certificate file for TLS                                                                           |
| --client-key string                                                                                                 |
| Path to a client key file for TLS                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                |
| --cluster string                                                                                                    |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl top](#) - Display resource (CPU/memory) usage

# kubectl uncordon

## Synopsis

Mark node as schedulable.

kubectl uncordon NODE

## Examples

```
Mark node "foo" as schedulable
kubectl uncordon foo
```

## Options

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --dry-run string[="unchanged"]    Default: "none"                                                                                                                                                        |
| Must be "none", "server", or "client". If client strategy, only print the object that would be sent, without sending it. If server strategy, submit server-side request without persisting the resource. |
| -h, --help                                                                                                                                                                                               |
| help for uncordon                                                                                                                                                                                        |
| -l, --selector string                                                                                                                                                                                    |
| Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2). Matching objects must satisfy all of the specified label constraints.                               |

|                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.                                                 |
| --as-group strings                                                                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                                                                |
| --as-uid string                                                                                                                                              |
| UID to impersonate for the operation.                                                                                                                        |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                          |
| Default cache directory                                                                                                                                      |
| --certificate-authority string                                                                                                                               |
| Path to a cert file for the certificate authority                                                                                                            |
| --client-certificate string                                                                                                                                  |
| Path to a client certificate file for TLS                                                                                                                    |
| --client-key string                                                                                                                                          |
| Path to a client key file for TLS                                                                                                                            |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                           |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                         |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                          |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                         |
| --cluster string                                                                                                                                             |
| The name of the kubeconfig cluster to use                                                                                                                    |
| --context string                                                                                                                                             |
| The name of the kubeconfig context to use                                                                                                                    |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                   |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration. |

|                                                                                                                                                                                                   |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| --default-unreachable-toleration-seconds int                                                                                                                                                      | Default: 300             |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                   |                          |
| --disable-compression                                                                                                                                                                             |                          |
| If true, opt-out of response compression for all requests to the server                                                                                                                           |                          |
| --insecure-skip-tls-verify                                                                                                                                                                        |                          |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |                          |
| --kubeconfig string                                                                                                                                                                               |                          |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |                          |
| --match-server-version                                                                                                                                                                            |                          |
| Require server version to match client version                                                                                                                                                    |                          |
| -n, --namespace string                                                                                                                                                                            |                          |
| If present, the namespace scope for this CLI request                                                                                                                                              |                          |
| --password string                                                                                                                                                                                 |                          |
| Password for basic authentication to the API server                                                                                                                                               |                          |
| --profile string                                                                                                                                                                                  | Default: "none"          |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |                          |
| --profile-output string                                                                                                                                                                           | Default: "profile.pprof" |
| Name of the file to write the profile to                                                                                                                                                          |                          |
| --request-timeout string                                                                                                                                                                          | Default: "0"             |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |                          |
| -s, --server string                                                                                                                                                                               |                          |
| The address and port of the Kubernetes API server                                                                                                                                                 |                          |
| --storage-driver-buffer-duration duration                                                                                                                                                         | Default: 1m0s            |

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction |
| --storage-driver-db string    Default: "cadvisor"                                                                                 |
| database name                                                                                                                     |
| --storage-driver-host string    Default: "localhost:8086"                                                                         |
| database host:port                                                                                                                |
| --storage-driver-password string    Default: "root"                                                                               |
| database password                                                                                                                 |
| --storage-driver-secure                                                                                                           |
| use secure connection with database                                                                                               |
| --storage-driver-table string    Default: "stats"                                                                                 |
| table name                                                                                                                        |
| --storage-driver-user string    Default: "root"                                                                                   |
| database username                                                                                                                 |
| --tls-server-name string                                                                                                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used      |
| --token string                                                                                                                    |
| Bearer token for authentication to the API server                                                                                 |
| --user string                                                                                                                     |
| The name of the kubeconfig user to use                                                                                            |
| --username string                                                                                                                 |
| Username for basic authentication to the API server                                                                               |
| --version version[=true]                                                                                                          |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                      |
| --warnings-as-errors                                                                                                              |



Treat warnings received from the server as errors and exit with a non-zero exit code

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl version

## Synopsis

Print the client and server version information for the current context.

```
kubectl version [flags]
```

## Examples

```
Print the client and server versions for the current context
kubectl version
```

## Options

|                                                          |
|----------------------------------------------------------|
| --client                                                 |
| If true, shows client version only (no server required). |
| -h, --help                                               |
| help for version                                         |
| -o, --output string                                      |
| One of 'yaml' or 'json'.                                 |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                  |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace. |
| --as-group strings                                                                                           |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                |
| --as-uid string                                                                                              |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UID to impersonate for the operation.                                                                                                                           |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                                                             |
| Default cache directory                                                                                                                                         |
| --certificate-authority string                                                                                                                                  |
| Path to a cert file for the certificate authority                                                                                                               |
| --client-certificate string                                                                                                                                     |
| Path to a client certificate file for TLS                                                                                                                       |
| --client-key string                                                                                                                                             |
| Path to a client key file for TLS                                                                                                                               |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| --disable-compression                                                                                                                                           |
| If true, opt-out of response compression for all requests to the server                                                                                         |
| --insecure-skip-tls-verify                                                                                                                                      |

|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                                                                |
| --kubeconfig string                                                                                                                                                                               |
| Path to the kubeconfig file to use for CLI requests.                                                                                                                                              |
| --match-server-version                                                                                                                                                                            |
| Require server version to match client version                                                                                                                                                    |
| -n, --namespace string                                                                                                                                                                            |
| If present, the namespace scope for this CLI request                                                                                                                                              |
| --password string                                                                                                                                                                                 |
| Password for basic authentication to the API server                                                                                                                                               |
| --profile string   Default: "none"                                                                                                                                                                |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                                                             |
| --profile-output string   Default: "profile.pprof"                                                                                                                                                |
| Name of the file to write the profile to                                                                                                                                                          |
| --request-timeout string   Default: "0"                                                                                                                                                           |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| -s, --server string                                                                                                                                                                               |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| --storage-driver-buffer-duration duration   Default: 1m0s                                                                                                                                         |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| --storage-driver-db string   Default: "cadvisor"                                                                                                                                                  |
| database name                                                                                                                                                                                     |
| --storage-driver-host string   Default: "localhost:8086"                                                                                                                                          |
| database host:port                                                                                                                                                                                |
| --storage-driver-password string   Default: "root"                                                                                                                                                |

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| database password                                                                                                            |
| --storage-driver-secure                                                                                                      |
| use secure connection with database                                                                                          |
| --storage-driver-table string    Default: "stats"                                                                            |
| table name                                                                                                                   |
| --storage-driver-user string    Default: "root"                                                                              |
| database username                                                                                                            |
| --tls-server-name string                                                                                                     |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used |
| --token string                                                                                                               |
| Bearer token for authentication to the API server                                                                            |
| --user string                                                                                                                |
| The name of the kubeconfig user to use                                                                                       |
| --username string                                                                                                            |
| Username for basic authentication to the API server                                                                          |
| --version version[=true]                                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                 |
| --warnings-as-errors                                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl wait

## Synopsis

Experimental: Wait for a specific condition on one or many resources.

The command takes multiple resources and waits until the specified condition is seen in the Status field of every given resource.

Alternatively, the command can wait for the given set of resources to be deleted by providing the "delete" keyword as the value to the --for flag.

A successful message will be printed to stdout indicating when the specified condition has been met. You can use -o option to change to output destination.

```
kubectl wait ([-f FILENAME] | resource.group/resource.name | resource.group [--label label | --all])
[--for=delete|--for condition=available|--for=jsonpath='{.}'[=value]]
```

## Examples

```
Wait for the pod "busybox1" to contain the status condition of type "Ready"
kubectl wait --for=condition=Ready pod/busybox1

The default value of status condition is true; you can wait for other targets after an equal
delimiter (compared after Unicode simple case folding, which is a more general form of case-
insensitivity)
kubectl wait --for=condition=Ready=false pod/busybox1

Wait for the pod "busybox1" to contain the status phase to be "Running"
kubectl wait --for=jsonpath='{.status.phase}'=Running pod/busybox1

Wait for pod "busybox1" to be Ready
kubectl wait --for=jsonpath='{.status.conditions[?(@.type=="Ready")].status}=True' pod/
busybox1

Wait for the service "loadbalancer" to have ingress.
kubectl wait --for=jsonpath='{.status.loadBalancer.ingress}' service/loadbalancer

Wait for the pod "busybox1" to be deleted, with a timeout of 60s, after having issued the
"delete" command
kubectl delete pod/busybox1
kubectl wait --for=delete pod/busybox1 --timeout=60s
```

## Options

|                                                                       |
|-----------------------------------------------------------------------|
| --all                                                                 |
| Select all resources in the namespace of the specified resource types |
| -A, --all-namespaces                                                  |

|  |                                                                                                                                                                                                                                                                                            |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | If present, list the requested object(s) across all namespaces. Namespace in current context is ignored even if specified with --namespace.                                                                                                                                                |
|  | --allow-missing-template-keys    Default: true                                                                                                                                                                                                                                             |
|  | If true, ignore any errors in templates when a field or map key is missing in the template. Only applies to goyaml and jsonpath output formats.                                                                                                                                            |
|  | --field-selector string                                                                                                                                                                                                                                                                    |
|  | Selector (field query) to filter on, supports '=', '==', and '!='.(e.g. --field-selector key1=value1,key2=value2). The server only supports a limited number of field queries per type.                                                                                                    |
|  | -f, --filename strings                                                                                                                                                                                                                                                                     |
|  | identifying the resource.                                                                                                                                                                                                                                                                  |
|  | --for string                                                                                                                                                                                                                                                                               |
|  | The condition to wait on: [delete condition=condition-name[=condition-value] jsonpath='{JSONPath expression}'=[JSONPath value]]. The default condition-value is true. Condition values are compared after Unicode simple case folding, which is a more general form of case-insensitivity. |
|  | -h, --help                                                                                                                                                                                                                                                                                 |
|  | help for wait                                                                                                                                                                                                                                                                              |
|  | --local                                                                                                                                                                                                                                                                                    |
|  | If true, annotation will NOT contact api-server but run locally.                                                                                                                                                                                                                           |
|  | -o, --output string                                                                                                                                                                                                                                                                        |
|  | Output format. One of: (json, yaml, name, go-template, go-template-file, template, templatefile, jsonpath, jsonpath-as-json, jsonpath-file).                                                                                                                                               |
|  | -R, --recursive    Default: true                                                                                                                                                                                                                                                           |
|  | Process the directory used in -f, --filename recursively. Useful when you want to manage related manifests organized within the same directory.                                                                                                                                            |
|  | -l, --selector string                                                                                                                                                                                                                                                                      |
|  | Selector (label query) to filter on, supports '=', '==', and '!='.(e.g. -l key1=value1,key2=value2)                                                                                                                                                                                        |
|  | --show-managed-fields                                                                                                                                                                                                                                                                      |
|  | If true, keep the managedFields when printing objects in JSON or YAML format.                                                                                                                                                                                                              |

|                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --template string                                                                                                                                                                                                                                         |
| Template string or path to template file to use when -o=go-template, -o=go-template-file. The template format is go lang templates [ <a href="http://golang.org/pkg/text/template/#pkg-overview">http://golang.org/pkg/text/template/#pkg-overview</a> ]. |
| --timeout duration    Default: 30s                                                                                                                                                                                                                        |
| The length of time to wait before giving up. Zero means check once and don't wait, negative means wait for a week.                                                                                                                                        |

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| --as string                                                                                                         |
| Username to impersonate for the operation. User could be a regular user or a service account in a namespace.        |
| --as-group strings                                                                                                  |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups.                       |
| --as-uid string                                                                                                     |
| UID to impersonate for the operation.                                                                               |
| --cache-dir string    Default: "\$HOME/.kube/cache"                                                                 |
| Default cache directory                                                                                             |
| --certificate-authority string                                                                                      |
| Path to a cert file for the certificate authority                                                                   |
| --client-certificate string                                                                                         |
| Path to a client certificate file for TLS                                                                           |
| --client-key string                                                                                                 |
| Path to a client key file for TLS                                                                                   |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                  |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16 |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                |
| --cluster string                                                                                                    |

|                                                                        |                                                                                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                        | The name of the kubeconfig cluster to use                                                                                                                       |
| <code>--context string</code>                                          |                                                                                                                                                                 |
|                                                                        | The name of the kubeconfig context to use                                                                                                                       |
| <code>--default-not-ready-toleration-seconds int</code> Default: 300   |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| <code>--default-unreachable-toleration-seconds int</code> Default: 300 |                                                                                                                                                                 |
|                                                                        | Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| <code>--disable-compression</code>                                     |                                                                                                                                                                 |
|                                                                        | If true, opt-out of response compression for all requests to the server                                                                                         |
| <code>--insecure-skip-tls-verify</code>                                |                                                                                                                                                                 |
|                                                                        | If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| <code>--kubeconfig string</code>                                       |                                                                                                                                                                 |
|                                                                        | Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| <code>--match-server-version</code>                                    |                                                                                                                                                                 |
|                                                                        | Require server version to match client version                                                                                                                  |
| <code>-n, --namespace string</code>                                    |                                                                                                                                                                 |
|                                                                        | If present, the namespace scope for this CLI request                                                                                                            |
| <code>--password string</code>                                         |                                                                                                                                                                 |
|                                                                        | Password for basic authentication to the API server                                                                                                             |
| <code>--profile string</code> Default: "none"                          |                                                                                                                                                                 |
|                                                                        | Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |
| <code>--profile-output string</code> Default: "profile.pprof"          |                                                                                                                                                                 |
|                                                                        | Name of the file to write the profile to                                                                                                                        |
| <code>--request-timeout string</code> Default: "0"                     |                                                                                                                                                                 |



|                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |
| <code>-s, --server string</code>                                                                                                                                                                  |
| The address and port of the Kubernetes API server                                                                                                                                                 |
| <code>--storage-driver-buffer-duration duration</code> Default: 1m0s                                                                                                                              |
| Writes in the storage driver will be buffered for this duration, and committed to the non memory backends as a single transaction                                                                 |
| <code>--storage-driver-db string</code> Default: "cadvisor"                                                                                                                                       |
| database name                                                                                                                                                                                     |
| <code>--storage-driver-host string</code> Default: "localhost:8086"                                                                                                                               |
| database host:port                                                                                                                                                                                |
| <code>--storage-driver-password string</code> Default: "root"                                                                                                                                     |
| database password                                                                                                                                                                                 |
| <code>--storage-driver-secure</code>                                                                                                                                                              |
| use secure connection with database                                                                                                                                                               |
| <code>--storage-driver-table string</code> Default: "stats"                                                                                                                                       |
| table name                                                                                                                                                                                        |
| <code>--storage-driver-user string</code> Default: "root"                                                                                                                                         |
| database username                                                                                                                                                                                 |
| <code>--tls-server-name string</code>                                                                                                                                                             |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |
| <code>--token string</code>                                                                                                                                                                       |
| Bearer token for authentication to the API server                                                                                                                                                 |
| <code>--user string</code>                                                                                                                                                                        |
| The name of the kubeconfig user to use                                                                                                                                                            |
| <code>--username string</code>                                                                                                                                                                    |

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| Username for basic authentication to the API server                                                          |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --warnings-as-errors                                                                                         |
| Treat warnings received from the server as errors and exit with a non-zero exit code                         |

## See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager

# kubectl

## Synopsis

kubectl controls the Kubernetes cluster manager.

Find more information in [Command line tool](#) (kubectl).

kubectl [flags]

## Options

|                                                                                               |
|-----------------------------------------------------------------------------------------------|
| --add-dir-header                                                                              |
| If true, adds the file directory to the header of the log messages                            |
| --alsologtostderr                                                                             |
| log to standard error as well as files                                                        |
| --as string                                                                                   |
| Username to impersonate for the operation                                                     |
| --as-group stringArray                                                                        |
| Group to impersonate for the operation, this flag can be repeated to specify multiple groups. |
| --azure-container-registry-config string                                                      |
| Path to the file containing Azure container registry configuration information.               |
| --cache-dir string   Default: "\$HOME/.kube/cache"                                            |
| Default cache directory                                                                       |
| --certificate-authority string                                                                |
| Path to a cert file for the certificate authority                                             |
| --client-certificate string                                                                   |
| Path to a client certificate file for TLS                                                     |
| --client-key string                                                                           |
| Path to a client key file for TLS                                                             |

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                              |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                            |
| --cloud-provider-gce-lb-src-cidrs cidrs    Default:<br>130.211.0.0/22,209.85.152.0/22,209.85.204.0/22,35.191.0.0/16                                             |
| CIDRs opened in GCE firewall for L4 LB traffic proxy & health checks                                                                                            |
| --cluster string                                                                                                                                                |
| The name of the kubeconfig cluster to use                                                                                                                       |
| --context string                                                                                                                                                |
| The name of the kubeconfig context to use                                                                                                                       |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                      |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.    |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                    |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration. |
| -h, --help                                                                                                                                                      |
| help for kubectl                                                                                                                                                |
| --insecure-skip-tls-verify                                                                                                                                      |
| If true, the server's certificate will not be checked for validity. This will make your HTTPS connections insecure                                              |
| --kubeconfig string                                                                                                                                             |
| Path to the kubeconfig file to use for CLI requests.                                                                                                            |
| --log-backtrace-at traceLocation    Default: :0                                                                                                                 |
| when logging hits line file:N, emit a stack trace                                                                                                               |
| --log-dir string                                                                                                                                                |
| If non-empty, write log files in this directory                                                                                                                 |
| --log-file string                                                                                                                                               |
| If non-empty, use this log file                                                                                                                                 |
| --log-file-max-size uint    Default: 1800                                                                                                                       |
| Defines the maximum size a log file can grow to. Unit is megabytes. If the value is 0, the maximum file size is unlimited.                                      |
| --log-flush-frequency duration    Default: 5s                                                                                                                   |
| Maximum number of seconds between log flushes                                                                                                                   |
| --logtostderr    Default: true                                                                                                                                  |
| log to standard error instead of files                                                                                                                          |
| --match-server-version                                                                                                                                          |
| Require server version to match client version                                                                                                                  |
| -n, --namespace string                                                                                                                                          |
| If present, the namespace scope for this CLI request                                                                                                            |
| --one-output                                                                                                                                                    |
| If true, only write logs to their native severity level (vs also writing to each lower severity level)                                                          |
| --password string                                                                                                                                               |
| Password for basic authentication to the API server                                                                                                             |
| --profile string    Default: "none"                                                                                                                             |
| Name of profile to capture. One of (none cpu heap goroutine threadcreate block mutex)                                                                           |

|                                                                                                                                                                                                   |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| --profile-output string                                                                                                                                                                           | Default: "profile.pprof" |
| Name of the file to write the profile to                                                                                                                                                          |                          |
| --request-timeout string                                                                                                                                                                          | Default: "0"             |
| The length of time to wait before giving up on a single server request. Non-zero values should contain a corresponding time unit (e.g. 1s, 2m, 3h). A value of zero means don't timeout requests. |                          |
| -s, --server string                                                                                                                                                                               |                          |
| The address and port of the Kubernetes API server                                                                                                                                                 |                          |
| --skip-headers                                                                                                                                                                                    |                          |
| If true, avoid header prefixes in the log messages                                                                                                                                                |                          |
| --skip-log-headers                                                                                                                                                                                |                          |
| If true, avoid headers when opening log files                                                                                                                                                     |                          |
| --stderrthreshold severity                                                                                                                                                                        | Default: 2               |
| logs at or above this threshold go to stderr                                                                                                                                                      |                          |
| --tls-server-name string                                                                                                                                                                          |                          |
| Server name to use for server certificate validation. If it is not provided, the hostname used to contact the server is used                                                                      |                          |
| --token string                                                                                                                                                                                    |                          |
| Bearer token for authentication to the API server                                                                                                                                                 |                          |
| --user string                                                                                                                                                                                     |                          |
| The name of the kubeconfig user to use                                                                                                                                                            |                          |
| --username string                                                                                                                                                                                 |                          |
| Username for basic authentication to the API server                                                                                                                                               |                          |
| -v, --v Level                                                                                                                                                                                     |                          |
| number for the log level verbosity                                                                                                                                                                |                          |
| --version version[=true]                                                                                                                                                                          |                          |
| Print version information and quit                                                                                                                                                                |                          |
| --vmodule moduleSpec                                                                                                                                                                              |                          |
| comma-separated list of pattern=N settings for file-filtered logging                                                                                                                              |                          |
| --warnings-as-errors                                                                                                                                                                              |                          |
| Treat warnings received from the server as errors and exit with a non-zero exit code                                                                                                              |                          |

## Environment variables

|                                                                                                                                                |  |
|------------------------------------------------------------------------------------------------------------------------------------------------|--|
| KUBECONFIG                                                                                                                                     |  |
| Path to the kubectl configuration ("kubeconfig") file. Default: "\$HOME/.kube/config"                                                          |  |
| KUBECTL_COMMAND_HEADERS                                                                                                                        |  |
| When set to false, turns off extra HTTP headers detailing invoked kubectl command (Kubernetes version v1.22 or later)                          |  |
| KUBECTL_DEBUG_CUSTOM_PROFILE                                                                                                                   |  |
| When set to true, custom flag will be enabled in kubectl debug. This flag is used to customize the pre-defined profiles.                       |  |
| KUBECTL_EXPLAIN_OPENAPIV3                                                                                                                      |  |
| Toggles whether calls to `kubectl explain` use the new OpenAPIv3 data source available. OpenAPIv3 is enabled by default since Kubernetes 1.24. |  |
| KUBECTL_ENABLE_CMD_SHADOW                                                                                                                      |  |

|                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| When set to true, external plugins can be used as subcommands for builtin commands if subcommand does not exist. In alpha stage, this feature can only be used for create command(e.g. <code>kubectl create networkpolicy</code> ).                         |
| KUBECTL_PORT_FORWARD_WEBSOCKETS                                                                                                                                                                                                                             |
| When set to true, the <code>kubectl port-forward</code> command will attempt to stream using the websockets protocol. If the upgrade to websockets fails, the commands will fallback to use the current SPDY protocol.                                      |
| KUBECTL_REMOTE_COMMAND_WEBSOCKETS                                                                                                                                                                                                                           |
| When set to true, the <code>kubectl exec</code> , <code>cp</code> , and <code>attach</code> commands will attempt to stream using the websockets protocol. If the upgrade to websockets fails, the commands will fallback to use the current SPDY protocol. |

## See Also

- [kubectl annotate](#) - Update the annotations on a resource
- [kubectl api-resources](#) - Print the supported API resources on the server
- [kubectl api-versions](#) - Print the supported API versions on the server, in the form of "group/version"
- [kubectl apply](#) - Apply a configuration to a resource by filename or stdin
- [kubectl attach](#) - Attach to a running container
- [kubectl auth](#) - Inspect authorization
- [kubectl autoscale](#) - Auto-scale a Deployment, ReplicaSet, or ReplicationController
- [kubectl certificate](#) - Modify certificate resources.
- [kubectl cluster-info](#) - Display cluster info
- [kubectl completion](#) - Output shell completion code for the specified shell (bash or zsh)
- [kubectl config](#) - Modify kubeconfig files
- [kubectl cordon](#) - Mark node as unschedulable
- [kubectl cp](#) - Copy files and directories to and from containers.
- [kubectl create](#) - Create a resource from a file or from stdin.
- [kubectl debug](#) - Create debugging sessions for troubleshooting workloads and nodes
- [kubectl delete](#) - Delete resources by filenames, stdin, resources and names, or by resources and label selector
- [kubectl describe](#) - Show details of a specific resource or group of resources
- [kubectl diff](#) - Diff live version against would-be applied version
- [kubectl drain](#) - Drain node in preparation for maintenance
- [kubectl edit](#) - Edit a resource on the server
- [kubectl events](#) - List events
- [kubectl exec](#) - Execute a command in a container
- [kubectl explain](#) - Documentation of resources
- [kubectl expose](#) - Take a replication controller, service, deployment or pod and expose it as a new Kubernetes Service
- [kubectl get](#) - Display one or many resources
- [kubectl kustomize](#) - Build a kustomization target from a directory or a remote url.
- [kubectl label](#) - Update the labels on a resource
- [kubectl logs](#) - Print the logs for a container in a pod
- [kubectl options](#) - Print the list of flags inherited by all commands
- [kubectl patch](#) - Update field(s) of a resource
- [kubectl plugin](#) - Provides utilities for interacting with plugins.
- [kubectl port-forward](#) - Forward one or more local ports to a pod
- [kubectl proxy](#) - Run a proxy to the Kubernetes API server
- [kubectl replace](#) - Replace a resource by filename or stdin

- [kubectl rollout](#) - Manage the rollout of a resource
- [kubectl run](#) - Run a particular image on the cluster
- [kubectl scale](#) - Set a new size for a Deployment, ReplicaSet or Replication Controller
- [kubectl set](#) - Set specific features on objects
- [kubectl taint](#) - Update the taints on one or more nodes
- [kubectl top](#) - Display Resource (CPU/Memory/Storage) usage.
- [kubectl uncordon](#) - Mark node as schedulable
- [kubectl version](#) - Print the client and server version information
- [kubectl wait](#) - Experimental: Wait for a specific condition on one or many resources.

## JSONPath Support

Kubectl supports JSONPath template.

JSONPath template is composed of JSONPath expressions enclosed by curly braces {}. Kubectl uses JSONPath expressions to filter on specific fields in the JSON object and format the output. In addition to the original JSONPath template syntax, the following functions and syntax are valid:

1. Use double quotes to quote text inside JSONPath expressions.
2. Use the range, end operators to iterate lists.
3. Use negative slice indices to step backwards through a list. Negative indices do not "wrap around" a list and are valid as long as `-index + listLength >= 0`.

### Note:

- The \$ operator is optional since the expression always starts from the root object by default.
- The result object is printed as its String() function.

Given the JSON input:

```
{
 "kind": "List",
 "items": [
 {
 "kind": "None",
 "metadata": {
 "name": "127.0.0.1",
 "labels": {
 "kubernetes.io/hostname": "127.0.0.1"
 }
 }
 },
 {
 "status": {
 "capacity": { "cpu": "4" },
 "addresses": [{ "type": "LegacyHostIP", "address": "127.0.0.1" }]
 }
 },
 {
 "kind": "None",
 "metadata": { "name": "127.0.0.2" },

```

```

 "status":{
 "capacity":{"cpu":"8"},
 "addresses":[
 {"type": "LegacyHostIP", "address":"127.0.0.2"},
 {"type": "another", "address":"127.0.0.3"}
]
 }
 },
 "users":[
 {
 "name": "myself",
 "user": {}
 },
 {
 "name": "e2e",
 "user": {"username": "admin", "password": "secret"}
 }
]
}

```

| Function         | Description                  | Example                                                    | Result                                                   |
|------------------|------------------------------|------------------------------------------------------------|----------------------------------------------------------|
| text             | the plain text               | kind is {kind}                                             | kind is List                                             |
| @                | the current object           | {@}                                                        | the same as input                                        |
| . or []          | child operator               | {kind}, {[kind']} or {[name.type']}                        | List                                                     |
| ..               | recursive descent            | {..name}                                                   | 127.0.0.1 127.0.0.2<br>myself e2e                        |
| *                | wildcard. Get all objects    | {items[*].metadata.name}                                   | [127.0.0.1<br>127.0.0.2]                                 |
| [start:end:step] | subscript operator           | {users[0].name}                                            | myself                                                   |
| [,]              | union operator               | {items[*]['metadata.name', 'status.capacity']}             | 127.0.0.1 127.0.0.2<br>map[cpu:4]<br>map[cpu:8]          |
| ?()              | filter                       | {users[?(@.name=="e2e")].user.password}                    | secret                                                   |
| range, end       | iterate list                 | {range items[*]}[{metadata.name}, {status.capacity}] {end} | [127.0.0.1,<br>map[cpu:4]]<br>[127.0.0.2,<br>map[cpu:8]] |
| "                | quote interpreted string     | {range items[*]}{metadata.name}{\t}{end}                   | 127.0.0.1 127.0.0.2                                      |
| \                | escape termination character | {items[0].metadata.labels.kubernetes.io/hostname}          | 127.0.0.1                                                |

Examples using kubectl and JSONPath expressions:

```

kubectl get pods -o json
kubectl get pods -o=jsonpath='{@}'

```

```
kubectl get pods -o=jsonpath='{.items[0]}'
kubectl get pods -o=jsonpath='{.items[0].metadata.name}'
kubectl get pods -o=jsonpath="{.items[*]['metadata.name', 'status.capacity']}"
kubectl get pods -o=jsonpath='{range .items[*]}{.metadata.name}{\t}{.status.startTime}{\n}{end}'
kubectl get pods -o=jsonpath='{.items[0].metadata.labels.kubernetes\.io/hostname}'
```

#### Note:

On Windows, you must *double* quote any JSONPath template that contains spaces (not single quote as shown above for bash). This in turn means that you must use a single quote or escaped double quote around any literals in the template. For example:

```
kubectl get pods -o=jsonpath="{range .items[*]}{.metadata.name}{\t}{.status.startTime}{\n}{end}"
kubectl get pods -o=jsonpath="{range .items[*]}{.metadata.name}{\"\\t\"}{.status.startTime}{\\\"\\n\\\"}{end}"
```

#### Note:

JSONPath regular expressions are not supported. If you want to match using regular expressions, you can use a tool such as jq.

```
kubectl does not support regular expressions for JSONpath output
The following command does not work
kubectl get pods -o jsonpath='{.items[?(@.metadata.name=~/^test$/)].metadata.name}'

The following command achieves the desired result
kubectl get pods -o json | jq -r '.items[] | select(.metadata.name | test("test-")).metadata.name'
```

## kubectl for Docker Users

You can use the Kubernetes command line tool kubectl to interact with the API Server. Using kubectl is straightforward if you are familiar with the Docker command line tool. However, there are a few differences between the Docker commands and the kubectl commands. The following sections show a Docker sub-command and describe the equivalent kubectl command.

### docker run

To run an nginx Deployment and expose the Deployment, see [kubectl create deployment](#).  
docker:

```
docker run -d --restart=always -e DOMAIN=cluster --name nginx-app -p 80:80 nginx
```

```
55c103fa129692154a7652490236fee9be47d70a8dd562281ae7d2f9a339a6db
```

```
docker ps
```

| CONTAINER ID | IMAGE | COMMAND | CREATED |
|--------------|-------|---------|---------|
| STATUS       | PORTS | NAMES   |         |



```
55c103fa1296 nginx "nginx -g 'daemon of..." 9 seconds ago Up 9 seconds
0.0.0.0:80->80/tcp nginx-app
```

kubectl:

```
start the pod running nginx
kubectl create deployment --image=nginx nginx-app
```

```
deployment.apps/nginx-app created
```

```
add env to nginx-app
kubectl set env deployment/nginx-app DOMAIN=cluster
```

```
deployment.apps/nginx-app env updated
```

### Note:

kubectl commands print the type and name of the resource created or mutated, which can then be used in subsequent commands. You can expose a new Service after a Deployment is created.

```
expose a port through with a service
kubectl expose deployment nginx-app --port=80 --name=nginx-http
```

```
service "nginx-http" exposed
```

By using kubectl, you can create a [Deployment](#) to ensure that N pods are running nginx, where N is the number of replicas stated in the spec and defaults to 1. You can also create a [service](#) with a selector that matches the pod labels. For more information, see [Use a Service to Access an Application in a Cluster](#).

By default images run in the background, similar to `docker run -d ...`. To run things in the foreground, use [kubectl run](#) to create pod:

```
kubectl run [-i] [--tty] --attach <name> --image=<image>
```

Unlike `docker run ...`, if you specify `--attach`, then you attach stdin, stdout and stderr. You cannot control which streams are attached (`docker -a ...`). To detach from the container, you can type the escape sequence `Ctrl+P` followed by `Ctrl+Q`.

## docker ps

To list what is currently running, see [kubectl get](#).

docker:

```
docker ps -a
```

| CONTAINER ID | IMAGE              | COMMAND                  | CREATED              |
|--------------|--------------------|--------------------------|----------------------|
| STATUS       | PORTS              | NAMES                    |                      |
| 14636241935f | ubuntu:16.04       | "echo test"              | 5 seconds ago        |
| ago          | cocky_fermi        |                          | Exited (0) 5 seconds |
| 55c103fa1296 | nginx              | "nginx -g 'daemon of..." | About a minute ago   |
| minute       | 0.0.0.0:80->80/tcp | nginx-app                | Up About a           |

kubectl:

```
kubectl get po
```

| NAME                      | READY | STATUS    | RESTARTS | AGE |
|---------------------------|-------|-----------|----------|-----|
| nginx-app-8df569cb7-4gd89 | 1/1   | Running   | 0        | 3m  |
| ubuntu                    | 0/1   | Completed | 0        | 20s |

## docker attach

To attach a process that is already running in a container, see [kubectl attach](#).

docker:

```
docker ps
```

| CONTAINER ID       | IMAGE     | COMMAND                   | CREATED       |
|--------------------|-----------|---------------------------|---------------|
| 55c103fa1296       | nginx     | "nginx -g 'daemon of...'" | 5 minutes ago |
| 0.0.0.0:80->80/tcp | nginx-app |                           | Up 5 minutes  |

```
docker attach 55c103fa1296
```

...

kubectl:

```
kubectl get pods
```

| NAME            | READY | STATUS  | RESTARTS | AGE |
|-----------------|-------|---------|----------|-----|
| nginx-app-5jyvm | 1/1   | Running | 0        | 10m |

```
kubectl attach -it nginx-app-5jyvm
```

...

To detach from the container, you can type the escape sequence Ctrl+P followed by Ctrl+Q.

## docker exec

To execute a command in a container, see [kubectl exec](#).

docker:

```
docker ps
```

| CONTAINER ID       | IMAGE     | COMMAND                   | CREATED       |
|--------------------|-----------|---------------------------|---------------|
| 55c103fa1296       | nginx     | "nginx -g 'daemon of...'" | 6 minutes ago |
| 0.0.0.0:80->80/tcp | nginx-app |                           | Up 6 minutes  |

```
docker exec 55c103fa1296 cat /etc/hostname
```

```
55c103fa1296
```

kubectl:

```
kubectl get po
```

| NAME            | READY | STATUS  | RESTARTS | AGE |
|-----------------|-------|---------|----------|-----|
| nginx-app-5jyvm | 1/1   | Running | 0        | 10m |

```
kubectl exec nginx-app-5jyvm -- cat /etc/hostname
```

```
nginx-app-5jyvm
```

To use interactive commands.

docker:

```
docker exec -ti 55c103fa1296 /bin/sh
exit
```

kubectl:

```
kubectl exec -ti nginx-app-5jyvm -- /bin/sh
exit
```

For more information, see [Get a Shell to a Running Container](#).

## docker logs

To follow stdout/stderr of a process that is running, see [kubectl logs](#).

docker:

```
docker logs -f a9e
```

```
192.168.9.1 - - [14/Jul/2015:01:04:02 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.35.0" "-"
192.168.9.1 - - [14/Jul/2015:01:04:03 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.35.0" "-"
```

kubectl:

```
kubectl logs -f nginx-app-zibvs
```

```
10.240.63.110 - - [14/Jul/2015:01:09:01 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.26.0" "-"
10.240.63.110 - - [14/Jul/2015:01:09:02 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.26.0" "-"
```

There is a slight difference between pods and containers; by default pods do not terminate if their processes exit. Instead the pods restart the process. This is similar to the docker run option `--restart=always` with one major difference. In docker, the output for each invocation of the process is concatenated, but for Kubernetes, each invocation is separate. To see the output from a previous run in Kubernetes, do this:

```
kubectl logs --previous nginx-app-zibvs
```

```
10.240.63.110 - - [14/Jul/2015:01:09:01 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.26.0" "-"
10.240.63.110 - - [14/Jul/2015:01:09:02 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.26.0" "-"
```

For more information, see [Logging Architecture](#).

## docker stop and docker rm

To stop and delete a running process, see [kubectl delete](#).

docker:

```
docker ps
```

| CONTAINER ID        | IMAGE     | COMMAND               | CREATED      | STATUS      |
|---------------------|-----------|-----------------------|--------------|-------------|
| PORTS               | NAMES     |                       |              |             |
| a9ec34d98787        | nginx     | "nginx -g 'daemon of" | 22 hours ago | Up 22 hours |
| 0.0.0.0:80->80/tcp, | nginx-app |                       |              |             |

```
docker stop a9ec34d98787
```

```
a9ec34d98787
```

```
docker rm a9ec34d98787
```

```
a9ec34d98787
```

kubectl:

```
kubectl get deployment nginx-app
```

| NAME      | READY | UP-TO-DATE | AVAILABLE | AGE |
|-----------|-------|------------|-----------|-----|
| nginx-app | 1/1   | 1          | 1         | 2m  |

```
kubectl get po -l app=nginx-app
```

| NAME                       | READY | STATUS  | RESTARTS | AGE |
|----------------------------|-------|---------|----------|-----|
| nginx-app-2883164633-aklf7 | 1/1   | Running | 0        | 2m  |

```
kubectl delete deployment nginx-app
```

```
deployment "nginx-app" deleted
```

```
kubectl get po -l app=nginx-app
```

```
Return nothing
```

### Note:

When you use kubectl, you don't delete the pod directly. You have to first delete the Deployment that owns the pod. If you delete the pod directly, the Deployment recreates the pod.

## docker login

There is no direct analog of docker login in kubectl. If you are interested in using Kubernetes with a private registry, see [Using a Private Registry](#).

## docker version

To get the version of client and server, see [kubectl version](#).

docker:

```
docker version
```

```
Client version: 1.7.0
Client API version: 1.19
Go version (client): go1.4.2
Git commit (client): 0baf609
OS/Arch (client): linux/amd64
Server version: 1.7.0
Server API version: 1.19
Go version (server): go1.4.2
Git commit (server): 0baf609
OS/Arch (server): linux/amd64
```

kubectl:

```
kubectl version
```

```
Client Version: version.Info{Major:"1", Minor:"6", GitVersion:"v1.6.9+a3d1dfa6f4335",
GitCommit:"9b77fed11a9843ce3780f70dd251e92901c43072", GitTreeState:"dirty",
BuildDate:"2017-08-29T20:32:58Z", OpenPaasKubernetesVersion:"v1.03.02", GoVersion:"go1.7.5",
Compiler:"gc", Platform:"linux/amd64"}
Server Version: version.Info{Major:"1", Minor:"6", GitVersion:"v1.6.9+a3d1dfa6f4335",
GitCommit:"9b77fed11a9843ce3780f70dd251e92901c43072", GitTreeState:"dirty",
BuildDate:"2017-08-29T20:32:58Z", OpenPaasKubernetesVersion:"v1.03.02", GoVersion:"go1.7.5",
Compiler:"gc", Platform:"linux/amd64"}
```

## docker info

To get miscellaneous information about the environment and configuration, see [kubectl cluster-info](#).

docker:

```
docker info
```

```
Containers: 40
Images: 168
Storage Driver: aufs
Root Dir: /usr/local/google/docker/aufs
Backing Filesystem: extfs
Dirs: 248
Dirperm1 Supported: false
Execution Driver: native-0.2
Logging Driver: json-file
Kernel Version: 3.13.0-53-generic
Operating System: Ubuntu 14.04.2 LTS
CPUs: 12
```

```
Total Memory: 31.32 GiB
Name: k8s-is-fun.mtv.corp.google.com
ID: ADUV:GCYR:B3VJ:HMPO:LNPQ:KD5S:YKFQ:76VN:IANZ:7TFV:ZBF4:BYJO
WARNING: No swap limit support
```

kubectl:

kubectl cluster-info

```
Kubernetes master is running at https://203.0.113.141
KubeDNS is running at https://203.0.113.141/api/v1/namespaces/kube-system/services/kube-dns/proxy
kubernetes-dashboard is running at https://203.0.113.141/api/v1/namespaces/kube-system/services/kubernetes-dashboard/proxy
Grafana is running at https://203.0.113.141/api/v1/namespaces/kube-system/services/monitoring-grafana/proxy
Heapster is running at https://203.0.113.141/api/v1/namespaces/kube-system/services/monitoring-heapster/proxy
InfluxDB is running at https://203.0.113.141/api/v1/namespaces/kube-system/services/monitoring-influxdb/proxy
```

## kubectl Usage Conventions

Recommended usage conventions for kubectl.

### Using kubectl in Reusable Scripts

For a stable output in a script:

- Request one of the machine-oriented output forms, such as `-o name`, `-o json`, `-o yaml`, `-o go-template`, or `-o jsonpath`.
- Fully-qualify the version. For example, `jobs.v1.batch/myjob`. This will ensure that kubectl does not use its default version that can change over time.
- Don't rely on context, preferences, or other implicit states.

### Subresources

- You can use the `--subresource` beta flag for kubectl commands like `get`, `patch`, `edit` and `replace` to fetch and update subresources for all resources that support them. Currently, only the status and scale subresources are supported.
  - For kubectl `edit`, the scale subresource is not supported. If you use `--subresource` with kubectl `edit` and specify scale as the subresource, the command will error out.
- The API contract against a subresource is identical to a full resource. While updating the status subresource to a new value, keep in mind that the subresource could be potentially reconciled by a controller to a different value.

# Best Practices

## kubectl run

For `kubectl run` to satisfy infrastructure as code:

- Tag the image with a version-specific tag and don't move that tag to a new version. For example, use `:v1234`, `v1.2.3`, `r03062016-1-4`, rather than `:latest` (For more information, see [Best Practices for Configuration](#)).
- Check in the script for an image that is heavily parameterized.
- Switch to configuration files checked into source control for features that are needed, but not expressible via `kubectl run` flags.

You can use the `--dry-run=client` flag to preview the object that would be sent to your cluster, without really submitting it.

## kubectl apply

- You can use `kubectl apply` to create or update resources. For more information about using `kubectl apply` to update resources, see [Kubectl Book](#).

# Component tools

---

[Feature Gates](#)

[Feature Gates \(removed\)](#)

[kubelet](#)

[kube-apiserver](#)

[kube-controller-manager](#)

[kube-proxy](#)

[kube-scheduler](#)

# Feature Gates

This page contains an overview of the various feature gates an administrator can specify on different Kubernetes components.

See [feature stages](#) for an explanation of the stages for a feature.

# Overview

Feature gates are a set of key=value pairs that describe Kubernetes features. You can turn these features on or off using the `--feature-gates` command line flag on each Kubernetes component.

Each Kubernetes component lets you enable or disable a set of feature gates that are relevant to that component. Use `-h` flag to see a full set of feature gates for all components. To set feature gates for a component, such as kubelet, use the `--feature-gates` flag assigned to a list of feature pairs:

```
--feature-gates=...,GracefulNodeShutdown=true
```

The following tables are a summary of the feature gates that you can set on different Kubernetes components.

- The "Since" column contains the Kubernetes release when a feature is introduced or its release stage is changed.
- The "Until" column, if not empty, contains the last Kubernetes release in which you can still use a feature gate.
- If a feature is in the Alpha or Beta state, you can find the feature listed in the [Alpha/Beta feature gate table](#).
- If a feature is stable you can find all stages for that feature listed in the [Graduated/Deprecated feature gate table](#).
- The [Graduated/Deprecated feature gate table](#) also lists deprecated and withdrawn features.

## Note:

For a reference to old feature gates that are removed, please refer to [feature gates removed](#).

## Feature gates for Alpha or Beta features

Feature gates for features in Alpha or Beta states

| Feature                       | Default | Stage | Since | Until |
|-------------------------------|---------|-------|-------|-------|
| AnyVolumeDataSource           | false   | Alpha | 1.18  | 1.23  |
| AnyVolumeDataSource           | true    | Beta  | 1.24  | –     |
| APIResponseCompression        | false   | Alpha | 1.7   | 1.15  |
| APIResponseCompression        | true    | Beta  | 1.16  | –     |
| APIServerIdentity             | false   | Alpha | 1.20  | 1.25  |
| APIServerIdentity             | true    | Beta  | 1.26  | –     |
| APIServerTracing              | false   | Alpha | 1.22  | 1.26  |
| APIServerTracing              | true    | Beta  | 1.27  | –     |
| AppArmor                      | true    | Beta  | 1.4   | –     |
| CloudControllerManagerWebhook | false   | Alpha | 1.27  | –     |
| ClusterTrustBundle            | false   | Alpha | 1.27  | –     |
| ClusterTrustBundleProjection  | false   | Alpha | 1.29  | –     |
| ComponentSLIs                 | false   | Alpha | 1.26  | 1.26  |
| ComponentSLIs                 | true    | Beta  | 1.27  | –     |
| ConsistentListFromCache       | false   | Alpha | 1.28  | –     |



| Feature                                | Default | Stage | Since | Until |
|----------------------------------------|---------|-------|-------|-------|
| ContainerCheckpoint                    | false   | Alpha | 1.25  | –     |
| ContextualLogging                      | false   | Alpha | 1.24  | –     |
| ContextualLogging                      | true    | Beta  | 1.30  | –     |
| CPUManagerPolicyAlphaOptions           | false   | Alpha | 1.23  | –     |
| CPUManagerPolicyBetaOptions            | true    | Beta  | 1.23  | –     |
| CPUManagerPolicyOptions                | false   | Alpha | 1.22  | 1.22  |
| CPUManagerPolicyOptions                | true    | Beta  | 1.23  | –     |
| CRDValidationRatcheting                | false   | Alpha | 1.28  | 1.29  |
| CRDValidationRatcheting                | true    | Beta  | 1.30  | –     |
| CronJobsScheduledAnnotation            | true    | Beta  | 1.28  | –     |
| CrossNamespaceVolumeDataSource         | false   | Alpha | 1.26  | –     |
| CSIMigrationPortworx                   | false   | Alpha | 1.23  | 1.24  |
| CSIMigrationPortworx                   | false   | Beta  | 1.25  | –     |
| CSIVolumeHealth                        | false   | Alpha | 1.21  | –     |
| CustomCPUCFSQuotaPeriod                | false   | Alpha | 1.12  | –     |
| CustomResourceFieldSelectors           | false   | Alpha | 1.30  | –     |
| DevicePluginCDIDevices                 | false   | Alpha | 1.28  | 1.28  |
| DevicePluginCDIDevices                 | true    | Beta  | 1.29  | –     |
| DisableCloudProviders                  | false   | Alpha | 1.22  | 1.28  |
| DisableCloudProviders                  | true    | Beta  | 1.29  | –     |
| DisableKubeletCloudCredentialProviders | false   | Alpha | 1.23  | 1.28  |
| DisableKubeletCloudCredentialProviders | true    | Beta  | 1.29  | –     |
| DisableNodeKubeProxyVersion            | false   | Alpha | 1.29  | –     |
| DynamicResourceAllocation              | false   | Alpha | 1.26  | –     |
| ElasticIndexedJob                      | true    | Beta  | 1.27  | –     |
| EventedPLEG                            | false   | Alpha | 1.26  | 1.26  |
| EventedPLEG                            | false   | Beta  | 1.27  | –     |
| GracefulNodeShutdown                   | false   | Alpha | 1.20  | 1.20  |
| GracefulNodeShutdown                   | true    | Beta  | 1.21  | –     |
| GracefulNodeShutdownBasedOnPodPriority | false   | Alpha | 1.23  | 1.23  |
| GracefulNodeShutdownBasedOnPodPriority | true    | Beta  | 1.24  | –     |
| HonorPVReclaimPolicy                   | false   | Alpha | 1.23  | –     |
| HPAScaleToZero                         | false   | Alpha | 1.16  | –     |
| ImageMaximumGCAge                      | false   | Alpha | 1.29  | 1.29  |
| ImageMaximumGCAge                      | true    | Beta  | 1.30  | –     |
| InPlacePodVerticalScaling              | false   | Alpha | 1.27  | –     |
| InTreePluginAWSUnregister              | false   | Alpha | 1.21  | –     |
| InTreePluginAzureDiskUnregister        | false   | Alpha | 1.21  | –     |
| InTreePluginAzureFileUnregister        | false   | Alpha | 1.21  | –     |
| InTreePluginGCEUnregister              | false   | Alpha | 1.21  | –     |
| InTreePluginOpenStackUnregister        | false   | Alpha | 1.21  | –     |
| InTreePluginPortworxUnregister         | false   | Alpha | 1.23  | –     |
| InTreePluginvSphereUnregister          | false   | Alpha | 1.21  | –     |
| JobBackoffLimitPerIndex                | false   | Alpha | 1.28  | 1.28  |

| Feature                                        | Default | Stage | Since | Until |
|------------------------------------------------|---------|-------|-------|-------|
| JobBackoffLimitPerIndex                        | true    | Beta  | 1.29  | –     |
| JobManagedBy                                   | false   | Alpha | 1.30  | –     |
| JobPodFailurePolicy                            | false   | Alpha | 1.25  | 1.25  |
| JobPodFailurePolicy                            | true    | Beta  | 1.26  | –     |
| JobPodReplacementPolicy                        | false   | Alpha | 1.28  | 1.28  |
| JobPodReplacementPolicy                        | true    | Beta  | 1.29  | –     |
| JobSuccessPolicy                               | false   | Alpha | 1.30  | –     |
| KubeletCgroupDriverFromCRI                     | false   | Alpha | 1.28  | –     |
| KubeletInUserNamespace                         | false   | Alpha | 1.22  | –     |
| KubeletPodResourcesDynamicResources            | false   | Alpha | 1.27  | –     |
| KubeletPodResourcesGet                         | false   | Alpha | 1.27  | –     |
| KubeletSeparateDiskGC                          | false   | Alpha | 1.29  | –     |
| KubeletTracing                                 | false   | Alpha | 1.25  | 1.26  |
| KubeletTracing                                 | true    | Beta  | 1.27  | –     |
| KubeProxyDrainingTerminatingNodes              | false   | Alpha | 1.28  | 1.30  |
| KubeProxyDrainingTerminatingNodes              | true    | Beta  | 1.30  | –     |
| LoadBalancerIPMode                             | false   | Alpha | 1.29  | 1.30  |
| LoadBalancerIPMode                             | true    | Beta  | 1.30  | –     |
| LocalStorageCapacityIsolationFSQuotaMonitoring | false   | Alpha | 1.15  | –     |
| LogarithmicScaleDown                           | false   | Alpha | 1.21  | 1.21  |
| LogarithmicScaleDown                           | true    | Beta  | 1.22  | –     |
| LoggingAlphaOptions                            | false   | Alpha | 1.24  | –     |
| LoggingBetaOptions                             | true    | Beta  | 1.24  | –     |
| MatchLabelKeysInPodAffinity                    | false   | Alpha | 1.29  | –     |
| MatchLabelKeysInPodTopologySpread              | false   | Alpha | 1.25  | 1.26  |
| MatchLabelKeysInPodTopologySpread              | true    | Beta  | 1.27  | –     |
| MaxUnavailableStatefulSet                      | false   | Alpha | 1.24  | –     |
| MemoryManager                                  | false   | Alpha | 1.21  | 1.21  |
| MemoryManager                                  | true    | Beta  | 1.22  | –     |
| MemoryQoS                                      | false   | Alpha | 1.22  | –     |
| MultiCIDRServiceAllocator                      | false   | Alpha | 1.27  | –     |
| NameGenerationRetries                          | false   | Alpha | 1.30  | –     |
| NFTablesProxyMode                              | false   | Alpha | 1.29  | –     |
| NodeInclusionPolicyInPodTopologySpread         | false   | Alpha | 1.25  | 1.25  |
| NodeInclusionPolicyInPodTopologySpread         | true    | Beta  | 1.26  | –     |
| NodeLogQuery                                   | false   | Alpha | 1.27  | 1.29  |
| NodeLogQuery                                   | false   | Beta  | 1.30  | –     |
| NodeSwap                                       | false   | Alpha | 1.22  | 1.27  |
| NodeSwap                                       | false   | Beta  | 1.28  | –     |
| OpenAPIEnums                                   | false   | Alpha | 1.23  | 1.23  |
| OpenAPIEnums                                   | true    | Beta  | 1.24  | –     |
| PDBUnhealthyPodEvictionPolicy                  | false   | Alpha | 1.26  | 1.26  |
| PDBUnhealthyPodEvictionPolicy                  | true    | Beta  | 1.27  | –     |
| PersistentVolumeLastPhaseTransitionTime        | false   | Alpha | 1.28  | 1.28  |

| Feature                                  | Default | Stage | Since | Until |
|------------------------------------------|---------|-------|-------|-------|
| PersistentVolumeLastPhaseTransitionTime  | true    | Beta  | 1.29  | –     |
| PodAndContainerStatsFromCRI              | false   | Alpha | 1.23  | –     |
| PodDeletionCost                          | false   | Alpha | 1.21  | 1.21  |
| PodDeletionCost                          | true    | Beta  | 1.22  | –     |
| PodDisruptionConditions                  | false   | Alpha | 1.25  | 1.25  |
| PodDisruptionConditions                  | true    | Beta  | 1.26  | –     |
| PodIndexLabel                            | true    | Beta  | 1.28  | –     |
| PodLifecycleSleepAction                  | false   | Alpha | 1.29  | 1.29  |
| PodLifecycleSleepAction                  | true    | Beta  | 1.30  | –     |
| PodReadyToStartContainersCondition       | false   | Alpha | 1.28  | 1.28  |
| PodReadyToStartContainersCondition       | true    | Beta  | 1.29  | –     |
| PortForwardWebsockets                    | false   | Alpha | 1.30  | –     |
| ProcMountType                            | false   | Alpha | 1.12  | –     |
| QOSReserved                              | false   | Alpha | 1.11  | –     |
| RecoverVolumeExpansionFailure            | false   | Alpha | 1.23  | –     |
| RecursiveReadOnlyMounts                  | false   | Alpha | 1.30  | –     |
| RelaxedEnvironmentVariableValidation     | false   | Alpha | 1.30  | –     |
| RotateKubeletServerCertificate           | false   | Alpha | 1.7   | 1.11  |
| RotateKubeletServerCertificate           | true    | Beta  | 1.12  | –     |
| RuntimeClassInImageCriApi                | false   | Alpha | 1.29  | –     |
| SchedulerQueueingHints                   | true    | Beta  | 1.28  | 1.28  |
| SchedulerQueueingHints                   | false   | Beta  | 1.29  | –     |
| SELinuxMount                             | false   | Alpha | 1.30  | –     |
| SELinuxMountReadWriteOncePod             | false   | Alpha | 1.25  | 1.26  |
| SELinuxMountReadWriteOncePod             | false   | Beta  | 1.27  | 1.27  |
| SELinuxMountReadWriteOncePod             | true    | Beta  | 1.28  | –     |
| SeparateTaintEvictionController          | true    | Beta  | 1.29  | –     |
| ServiceAccountTokenJTI                   | false   | Alpha | 1.29  | 1.29  |
| ServiceAccountTokenJTI                   | true    | Beta  | 1.30  | –     |
| ServiceAccountTokenNodeBinding           | false   | Alpha | 1.29  | –     |
| ServiceAccountTokenNodeBindingValidation | false   | Alpha | 1.29  | 1.29  |
| ServiceAccountTokenNodeBindingValidation | true    | Beta  | 1.30  | –     |
| ServiceAccountTokenPodNodeInfo           | false   | Alpha | 1.29  | 1.29  |
| ServiceAccountTokenPodNodeInfo           | true    | Beta  | 1.30  | –     |
| ServiceTrafficDistribution               | false   | Alpha | 1.30  | –     |
| SidecarContainers                        | false   | Alpha | 1.28  | 1.28  |
| SidecarContainers                        | true    | Beta  | 1.29  | –     |
| SizeMemoryBackedVolumes                  | false   | Alpha | 1.20  | 1.21  |
| SizeMemoryBackedVolumes                  | true    | Beta  | 1.22  | –     |
| StatefulSetAutoDeletePVC                 | false   | Alpha | 1.23  | 1.26  |
| StatefulSetAutoDeletePVC                 | true    | Beta  | 1.27  | –     |
| StatefulSetStartOrdinal                  | false   | Alpha | 1.26  | 1.26  |
| StatefulSetStartOrdinal                  | true    | Beta  | 1.27  | –     |
| StorageVersionAPI                        | false   | Alpha | 1.20  | –     |

| Feature                               | Default | Stage | Since | Until |
|---------------------------------------|---------|-------|-------|-------|
| StorageVersionHash                    | false   | Alpha | 1.14  | 1.14  |
| StorageVersionHash                    | true    | Beta  | 1.15  | –     |
| StorageVersionMigrator                | false   | Alpha | 1.30  | 1.32  |
| StructuredAuthenticationConfiguration | false   | Alpha | 1.29  | 1.29  |
| StructuredAuthenticationConfiguration | true    | Beta  | 1.30  | –     |
| StructuredAuthorizationConfiguration  | false   | Alpha | 1.29  | 1.29  |
| StructuredAuthorizationConfiguration  | true    | Beta  | 1.30  | –     |
| TopologyAwareHints                    | false   | Alpha | 1.21  | 1.22  |
| TopologyAwareHints                    | false   | Beta  | 1.23  | 1.23  |
| TopologyAwareHints                    | true    | Beta  | 1.24  | –     |
| TopologyManagerPolicyAlphaOptions     | false   | Alpha | 1.26  | –     |
| TopologyManagerPolicyBetaOptions      | false   | Beta  | 1.26  | 1.27  |
| TopologyManagerPolicyBetaOptions      | true    | Beta  | 1.28  | –     |
| TopologyManagerPolicyOptions          | false   | Alpha | 1.26  | 1.27  |
| TopologyManagerPolicyOptions          | true    | Beta  | 1.28  | –     |
| TranslateStreamCloseWebsocketRequests | true    | Beta  | 1.30  | –     |
| UnauthenticatedHTTP2DOSMitigation     | false   | Beta  | 1.28  | 1.28  |
| UnauthenticatedHTTP2DOSMitigation     | true    | Beta  | 1.29  | –     |
| UnknownVersionInteroperabilityProxy   | false   | Alpha | 1.28  | –     |
| UserNamespacesPodSecurityStandards    | false   | Alpha | 1.29  | –     |
| UserNamespacesSupport                 | false   | Alpha | 1.28  | 1.29  |
| UserNamespacesSupport                 | false   | Beta  | 1.30  | –     |
| VolumeAttributesClass                 | false   | Alpha | 1.29  | –     |
| VolumeCapacityPriority                | false   | Alpha | 1.21  | –     |
| WatchList                             | false   | Alpha | 1.27  | –     |
| WindowsHostNetwork                    | true    | Alpha | 1.26  | –     |
| WinDSR                                | false   | Alpha | 1.14  | –     |
| WinOverlay                            | false   | Alpha | 1.14  | 1.19  |
| WinOverlay                            | true    | Beta  | 1.20  | –     |
| ZeroLimitedNominalConcurrencyShares   | false   | Beta  | 1.29  | –     |

## Feature gates for graduated or deprecated features

### Feature Gates for Graduated or Deprecated Features

| Feature                         | Default | Stage      | Since | Until |
|---------------------------------|---------|------------|-------|-------|
| AdmissionWebhookMatchConditions | false   | Alpha      | 1.27  | 1.27  |
| AdmissionWebhookMatchConditions | true    | Beta       | 1.28  | 1.29  |
| AdmissionWebhookMatchConditions | true    | GA         | 1.30  | –     |
| AggregatedDiscoveryEndpoint     | false   | Alpha      | 1.26  | 1.26  |
| AggregatedDiscoveryEndpoint     | true    | Beta       | 1.27  | 1.29  |
| AggregatedDiscoveryEndpoint     | true    | GA         | 1.30  | –     |
| AllowServiceLBStatusOnNonLB     | false   | Deprecated | 1.29  | –     |
| APIListChunking                 | false   | Alpha      | 1.8   | 1.8   |
| APIListChunking                 | true    | Beta       | 1.9   | 1.28  |
| APIListChunking                 | true    | GA         | 1.29  | –     |

| Feature                                   | Default | Stage      | Since | Until |
|-------------------------------------------|---------|------------|-------|-------|
| APIPriorityAndFairness                    | false   | Alpha      | 1.18  | 1.19  |
| APIPriorityAndFairness                    | true    | Beta       | 1.20  | 1.28  |
| APIPriorityAndFairness                    | true    | GA         | 1.29  | –     |
| CloudDualStackNodeIPs                     | false   | Alpha      | 1.27  | 1.28  |
| CloudDualStackNodeIPs                     | true    | Beta       | 1.29  | 1.29  |
| CloudDualStackNodeIPs                     | true    | GA         | 1.30  | –     |
| ConsistentHTTPGetHandlers                 | true    | GA         | 1.25  | –     |
| CPUManager                                | false   | Alpha      | 1.8   | 1.9   |
| CPUManager                                | true    | Beta       | 1.10  | 1.25  |
| CPUManager                                | true    | GA         | 1.26  | –     |
| CSIMigrationRBD                           | false   | Alpha      | 1.23  | 1.27  |
| CSIMigrationRBD                           | false   | Deprecated | 1.28  | –     |
| CSINodeExpandSecret                       | false   | Alpha      | 1.25  | 1.26  |
| CSINodeExpandSecret                       | true    | Beta       | 1.27  | 1.28  |
| CSINodeExpandSecret                       | true    | GA         | 1.29  | –     |
| CustomResourceValidationExpressions       | false   | Alpha      | 1.23  | 1.24  |
| CustomResourceValidationExpressions       | true    | Beta       | 1.25  | 1.28  |
| CustomResourceValidationExpressions       | true    | GA         | 1.29  | –     |
| DefaultHostNetworkHostPortsInPodTemplates | false   | Deprecated | 1.28  | –     |
| EfficientWatchResumption                  | false   | Alpha      | 1.20  | 1.20  |
| EfficientWatchResumption                  | true    | Beta       | 1.21  | 1.23  |
| EfficientWatchResumption                  | true    | GA         | 1.24  | –     |
| ExecProbeTimeout                          | true    | GA         | 1.20  | –     |
| HPAContainerMetrics                       | false   | Alpha      | 1.20  | 1.26  |
| HPAContainerMetrics                       | true    | Beta       | 1.27  | 1.29  |
| HPAContainerMetrics                       | true    | GA         | 1.30  | –     |
| InTreePluginRBDUnregister                 | false   | Alpha      | 1.23  | 1.27  |
| InTreePluginRBDUnregister                 | false   | Deprecated | 1.28  | –     |
| JobReadyPods                              | false   | Alpha      | 1.23  | 1.23  |
| JobReadyPods                              | true    | Beta       | 1.24  | 1.28  |
| JobReadyPods                              | true    | GA         | 1.29  | –     |
| KMSv1                                     | true    | Deprecated | 1.28  | 1.28  |
| KMSv1                                     | false   | Deprecated | 1.29  | –     |
| KMSv2                                     | false   | Alpha      | 1.25  | 1.26  |
| KMSv2                                     | true    | Beta       | 1.27  | 1.28  |
| KMSv2                                     | true    | GA         | 1.29  | –     |
| KMSv2KDF                                  | false   | Beta       | 1.28  | 1.28  |
| KMSv2KDF                                  | true    | GA         | 1.29  | –     |
| LegacyServiceAccountTokenCleanUp          | false   | Alpha      | 1.28  | 1.28  |
| LegacyServiceAccountTokenCleanUp          | true    | Beta       | 1.29  | 1.29  |
| LegacyServiceAccountTokenCleanUp          | true    | GA         | 1.30  | –     |
| MinDomainsInPodTopologySpread             | false   | Alpha      | 1.24  | 1.24  |
| MinDomainsInPodTopologySpread             | false   | Beta       | 1.25  | 1.26  |
| MinDomainsInPodTopologySpread             | true    | Beta       | 1.27  | 1.29  |

| <b>Feature</b>                 | <b>Default</b> | <b>Stage</b> | <b>Since</b> | <b>Until</b> |
|--------------------------------|----------------|--------------|--------------|--------------|
| MinDomainsInPodTopologySpread  | true           | GA           | 1.30         | –            |
| NewVolumeManagerReconstruction | false          | Beta         | 1.27         | 1.27         |
| NewVolumeManagerReconstruction | true           | Beta         | 1.28         | 1.29         |
| NewVolumeManagerReconstruction | true           | GA           | 1.30         | –            |
| NodeOutOfServiceVolumeDetach   | false          | Alpha        | 1.24         | 1.25         |
| NodeOutOfServiceVolumeDetach   | true           | Beta         | 1.26         | 1.27         |
| NodeOutOfServiceVolumeDetach   | true           | GA           | 1.28         | –            |
| PodHostIPs                     | false          | Alpha        | 1.28         | 1.28         |
| PodHostIPs                     | true           | Beta         | 1.29         | 1.30         |
| PodHostIPs                     | true           | GA           | 1.30         | –            |
| PodSchedulingReadiness         | false          | Alpha        | 1.26         | 1.26         |
| PodSchedulingReadiness         | true           | Beta         | 1.27         | 1.29         |
| PodSchedulingReadiness         | true           | GA           | 1.30         | –            |
| ReadWriteOncePod               | false          | Alpha        | 1.22         | 1.26         |
| ReadWriteOncePod               | true           | Beta         | 1.27         | 1.28         |
| ReadWriteOncePod               | true           | GA           | 1.29         | –            |
| RemainingItemCount             | false          | Alpha        | 1.15         | 1.15         |
| RemainingItemCount             | true           | Beta         | 1.16         | 1.28         |
| RemainingItemCount             | true           | GA           | 1.29         | –            |
| ServerSideApply                | false          | Alpha        | 1.14         | 1.15         |
| ServerSideApply                | true           | Beta         | 1.16         | 1.21         |
| ServerSideApply                | true           | GA           | 1.22         | –            |
| ServerSideFieldValidation      | false          | Alpha        | 1.23         | 1.24         |
| ServerSideFieldValidation      | true           | Beta         | 1.25         | 1.26         |
| ServerSideFieldValidation      | true           | GA           | 1.27         | –            |
| ServiceNodePortStaticSubrange  | false          | Alpha        | 1.27         | 1.27         |
| ServiceNodePortStaticSubrange  | true           | Beta         | 1.28         | 1.28         |
| ServiceNodePortStaticSubrange  | true           | GA           | 1.29         | –            |
| SkipReadOnlyValidationGCE      | false          | Alpha        | 1.28         | 1.28         |
| SkipReadOnlyValidationGCE      | true           | Deprecated   | 1.29         | –            |
| StableLoadBalancerNodeSet      | true           | Beta         | 1.27         | 1.29         |
| StableLoadBalancerNodeSet      | true           | GA           | 1.30         | –            |
| ValidatingAdmissionPolicy      | false          | Alpha        | 1.26         | 1.27         |
| ValidatingAdmissionPolicy      | false          | Beta         | 1.28         | 1.29         |
| ValidatingAdmissionPolicy      | true           | GA           | 1.30         | –            |
| WatchBookmark                  | false          | Alpha        | 1.15         | 1.15         |
| WatchBookmark                  | true           | Beta         | 1.16         | 1.16         |
| WatchBookmark                  | true           | GA           | 1.17         | –            |

# Using a feature

## Feature stages

A feature can be in *Alpha*, *Beta* or *GA* stage. An *Alpha* feature means:

- Disabled by default.
- Might be buggy. Enabling the feature may expose bugs.
- Support for feature may be dropped at any time without notice.
- The API may change in incompatible ways in a later software release without notice.
- Recommended for use only in short-lived testing clusters, due to increased risk of bugs and lack of long-term support.

A *Beta* feature means:

- Usually enabled by default. Beta API groups are [disabled by default](#).
- The feature is well tested. Enabling the feature is considered safe.
- Support for the overall feature will not be dropped, though details may change.
- The schema and/or semantics of objects may change in incompatible ways in a subsequent beta or stable release. When this happens, we will provide instructions for migrating to the next version. This may require deleting, editing, and re-creating API objects. The editing process may require some thought. This may require downtime for applications that rely on the feature.
- Recommended for only non-business-critical uses because of potential for incompatible changes in subsequent releases. If you have multiple clusters that can be upgraded independently, you may be able to relax this restriction.

### Note:

Please do try *Beta* features and give feedback on them! After they exit beta, it may not be practical for us to make more changes.

A *General Availability* (GA) feature is also referred to as a *stable* feature. It means:

- The feature is always enabled; you cannot disable it.
- The corresponding feature gate is no longer needed.
- Stable versions of features will appear in released software for many subsequent versions.

## List of feature gates

Each feature gate is designed for enabling/disabling a specific feature.

- AdmissionWebhookMatchConditions: Enable [match conditions](#) on mutating & validating admission webhooks.
- AggregatedDiscoveryEndpoint: Enable a single HTTP endpoint `/discovery/<version>` which supports native HTTP caching with ETags containing all APIResources known to the API server.
- AllowServiceLBStatusOnNonLB: Enables `.status.ingress.loadBalancer` to be set on Services of types other than LoadBalancer.

- AnyVolumeDataSource: Enable use of any custom resource as the DataSource of a [PVC](#).
- 
- APIListChunking: Enable the API clients to retrieve (LIST or GET) resources from API server in chunks.
- APIPriorityAndFairness: Enable managing request concurrency with prioritization and fairness at each server. (Renamed from RequestManagement)
- APIResponseCompression: Compress the API responses for LIST or GET requests.
- APIServerIdentity: Assign each API server an ID in a cluster, using a [Lease](#).
- APIServerTracing: Add support for distributed tracing in the API server. See [Traces for Kubernetes System Components](#) for more details.
- AppArmor: Enable use of AppArmor mandatory access control for Pods running on Linux nodes. See [AppArmor Tutorial](#) for more details.
- CloudControllerManagerWebhook: Enable webhooks in cloud controller manager.
- CloudDualStackNodeIPs: Enables dual-stack kubelet --node-ip with external cloud providers. See [Configure IPv4/IPv6 dual-stack](#) for more details.
- ClusterTrustBundle: Enable ClusterTrustBundle objects and kubelet integration.
- ClusterTrustBundleProjection: [clusterTrustBundle projected volume sources](#).
- ComponentSLIs: Enable the /metrics/slis endpoint on Kubernetes components like kubelet, kube-scheduler, kube-proxy, kube-controller-manager, cloud-controller-manager allowing you to scrape health check metrics.
- ConsistentHTTPGetHandlers: Normalize HTTP get URL and Header passing for lifecycle handlers with probes.
- ConsistentListFromCache: Allow the API server to serve consistent lists from cache.
- ContainerCheckpoint: Enables the kubelet checkpoint API. See [Kubelet Checkpoint API](#) for more details.
- ContextualLogging: Enables extra details in log output of Kubernetes components that support contextual logging.
- CPUManager: Enable container level CPU affinity support, see [CPU Management Policies](#).
- CPUManagerPolicyAlphaOptions: This allows fine-tuning of CPUManager policies, experimental, Alpha-quality options This feature gate guards *a group* of CPUManager options whose quality level is alpha. This feature gate will never graduate to beta or stable.
- CPUManagerPolicyBetaOptions: This allows fine-tuning of CPUManager policies, experimental, Beta-quality options This feature gate guards *a group* of CPUManager options whose quality level is beta. This feature gate will never graduate to stable.
- CPUManagerPolicyOptions: Allow fine-tuning of CPUManager policies.



- **CRDValidationRatcheting**: Enable updates to custom resources to contain violations of their OpenAPI schema if the offending portions of the resource update did not change. See [Validation Ratcheting](#) for more details.
- **CronJobsScheduledAnnotation**: Set the scheduled job time as an [annotation](#) on Jobs that were created on behalf of a CronJob.
- **CrossNamespaceVolumeDataSource**: Enable the usage of cross namespace volume data source to allow you to specify a source namespace in the dataSourceRef field of a PersistentVolumeClaim.
- **CSIMigrationPortworx**: Enables shims and translation logic to route volume operations from the Portworx in-tree plugin to Portworx CSI plugin. Requires Portworx CSI driver to be installed and configured in the cluster.
- **CSIMigrationRBD**: Enables shims and translation logic to route volume operations from the RBD in-tree plugin to Ceph RBD CSI plugin. Requires CSIMigration and csiMigrationRBD feature flags enabled and Ceph CSI plugin installed and configured in the cluster. This flag has been deprecated in favor of the InTreePluginRBDUnregister feature flag which prevents the registration of in-tree RBD plugin.
- **CSINodeExpandSecret**: Enable passing secret authentication data to a CSI driver for use during a NodeExpandVolume CSI operation.
- **CSIVolumeHealth**: Enable support for CSI volume health monitoring on node.
- **CustomCPUCFSQuotaPeriod**: Enable nodes to change cpuCFSQuotaPeriod in [kubelet config](#).
- **CustomResourceFieldSelectors**: Enable selectableFields in the [CustomResourceDefinition](#) API to allow filtering of custom resource **list**, **watch** and **deletecollection** requests.
- **CustomResourceValidationExpressions**: Enable expression language validation in CRD which will validate customer resource based on validation rules written in the x-kubernetes-validations extension.
- **DefaultHostNetworkHostPortsInPodTemplates**:

This feature gate controls the point at which a default value for `.spec.containers[*].ports[*].hostPort` is assigned, for Pods using `hostNetwork: true`. The default since Kubernetes v1.28 is to only set a default value in Pods.

Enabling this means a default will be assigned even to the `.spec` of an embedded [PodTemplate](#) (for example, in a Deployment), which is the way that older releases of Kubernetes worked. You should migrate your code so that it does not rely on the legacy behavior.

- **DevicePluginCDIDevices**: Enable support to CDI device IDs in the [Device Plugin](#) API.
- **DisableCloudProviders**: Disables any functionality in kube-apiserver, kube-controller-manager and kubelet related to the `--cloud-provider` component flag.
- **DisableKubeletCloudCredentialProviders**: Disable the in-tree functionality in kubelet to authenticate to a cloud provider container registry for image pull credentials.

- DisableNodeKubeProxyVersion: Disable setting the kubeProxyVersion field of the Node.
- DynamicResourceAllocation: Enables support for resources with custom parameters and a lifecycle that is independent of a Pod.
- EfficientWatchResumption: Allows for storage-originated bookmark (progress notify) events to be delivered to the users. This is only applied to watch operations.
- ElasticIndexedJob: Enables Indexed Jobs to be scaled up or down by mutating both spec.completions and spec.parallelism together such that spec.completions == spec.parallelism. See docs on [elastic Indexed Jobs](#) for more details.
- EventedPLEG: Enable support for the kubelet to receive container life cycle events from the [container runtime](#) via an extension to [CRI](#). (PLEG is an abbreviation for “Pod lifecycle event generator”). For this feature to be useful, you also need to enable support for container lifecycle events in each container runtime running in your cluster. If the container runtime does not announce support for container lifecycle events then the kubelet automatically switches to the legacy generic PLEG mechanism, even if you have this feature gate enabled.
- ExecProbeTimeout: Ensure kubelet respects exec probe timeouts. This feature gate exists in case any of your existing workloads depend on a now-corrected fault where Kubernetes ignored exec probe timeouts. See [readiness probes](#).
- GracefulNodeShutdown: Enables support for graceful shutdown in kubelet. During a system shutdown, kubelet will attempt to detect the shutdown event and gracefully terminate pods running on the node. See [Graceful Node Shutdown](#) for more details.
- GracefulNodeShutdownBasedOnPodPriority: Enables the kubelet to check Pod priorities when shutting down a node gracefully.
- HonorPVReclaimPolicy: Honor persistent volume reclaim policy when it is Delete irrespective of PV-PVC deletion ordering. For more details, check the [PersistentVolume deletion protection finalizer](#) documentation.
- HPAContainerMetrics: Allow [HorizontalPodAutoscalers](#) to scale based on metrics from individual containers within target pods.
- HPAScaleToZero: Enables setting minReplicas to 0 for HorizontalPodAutoscaler resources when using custom or external metrics.
- ImageMaximumGCAge: Enables the kubelet configuration field imageMaximumGCAge, allowing an administrator to specify the age after which an image will be garbage collected.
- InPlacePodVerticalScaling: Enables in-place Pod vertical scaling.
- InTreePluginAWSUnregister: Stops registering the aws-ebs in-tree plugin in kubelet and volume controllers.
- InTreePluginAzureDiskUnregister: Stops registering the azuredisk in-tree plugin in kubelet and volume controllers.
- InTreePluginAzureFileUnregister: Stops registering the azurefile in-tree plugin in kubelet and volume controllers.

- `InTreePluginGCEUnregister`: Stops registering the gce-pd in-tree plugin in kubelet and volume controllers.
- `InTreePluginOpenStackUnregister`: Stops registering the OpenStack cinder in-tree plugin in kubelet and volume controllers.
- `InTreePluginPortworxUnregister`: Stops registering the Portworx in-tree plugin in kubelet and volume controllers.
- `InTreePluginRBDUnregister`: Stops registering the RBD in-tree plugin in kubelet and volume controllers.
- `InTreePluginvSphereUnregister`: Stops registering the vSphere in-tree plugin in kubelet and volume controllers.
- `JobBackoffLimitPerIndex`: Allows specifying the maximal number of pod retries per index in Indexed jobs.
- `JobManagedBy`: Allows to delegate reconciliation of a Job object to an external controller.
- `JobPodFailurePolicy`: Allow users to specify handling of pod failures based on container exit codes and pod conditions.
- `JobPodReplacementPolicy`: Allows you to specify pod replacement for terminating pods in a [Job](#)
- `JobReadyPods`: Enables tracking the number of Pods that have a Ready [condition](#). The count of Ready pods is recorded in the [status](#) of a [Job](#) status.
- `JobSuccessPolicy`: Allow users to specify when a Job can be declared as succeeded based on the set of succeeded pods.
- `KMSv1`: Enables KMS v1 API for encryption at rest. See [Using a KMS Provider for data encryption](#) for more details.
- `KMSv2`: Enables KMS v2 API for encryption at rest. See [Using a KMS Provider for data encryption](#) for more details.
- `KMSv2KDF`: Enables KMS v2 to generate single use data encryption keys. See [Using a KMS Provider for data encryption](#) for more details. If the KMSv2 feature gate is not enabled in your cluster, the value of the KMSv2KDF feature gate has no effect.
- `KubeletCgroupDriverFromCRI`: Enable detection of the kubelet cgroup driver configuration option from the [CRI](#). You can use this feature gate on nodes with a kubelet that supports the feature gate and where there is a CRI container runtime that supports the RuntimeConfig CRI call. If both CRI and kubelet support this feature, the kubelet ignores the cgroupDriver configuration setting (or deprecated `--cgroup-driver` command line argument). If you enable this feature gate and the container runtime doesn't support it, the kubelet falls back to using the driver configured using the cgroupDriver configuration setting. See [Configuring a cgroup driver](#) for more details.
- `KubeletInUserNamespace`: Enables support for running kubelet in a [user namespace](#). See [Running Kubernetes Node Components as a Non-root User](#).

- **KubeletPodResourcesDynamicResources:** Extend the kubelet's pod resources gRPC endpoint to include resources allocated in ResourceClaims via DynamicResourceAllocation API. See [resource allocation reporting](#) for more details. with information about the allocatable resources, enabling clients to properly track the free compute resources on a node.
- **KubeletPodResourcesGet:** Enable the Get gRPC endpoint on kubelet's for Pod resources. This API augments the [resource allocation reporting](#).
- **KubeletSeparateDiskGC:** Enable kubelet to garbage collect container images and containers even when those are on a separate filesystem.
- **KubeletTracing:** Add support for distributed tracing in the kubelet. When enabled, kubelet CRI interface and authenticated http servers are instrumented to generate OpenTelemetry trace spans. See [Traces for Kubernetes System Components](#) for more details.
- **KubeProxyDrainingTerminatingNodes:** Implement connection draining for terminating nodes for externalTrafficPolicy: Cluster services.
- **LegacyServiceAccountTokenCleanUp:** Enable cleaning up Secret-based [service account tokens](#) when they are not used in a specified time (default to be one year).
- **LoadBalancerIPMode:** Allows setting ipMode for Services where type is set to LoadBalancer. See [Specifying IPMode of load balancer status](#) for more information.
- **LocalStorageCapacityIsolationFSQuotaMonitoring:** When LocalStorageCapacityIsolation is enabled for [local ephemeral storage](#) and the backing filesystem for [emptyDir volumes](#) supports project quotas and they are enabled, use project quotas to monitor [emptyDir volume](#) storage consumption rather than filesystem walk for better performance and accuracy.
- **LogarithmicScaleDown:** Enable semi-random selection of pods to evict on controller scaledown based on logarithmic bucketing of pod timestamps.
- **LoggingAlphaOptions:** Allow fine-tuning of experimental, alpha-quality logging options.
- **LoggingBetaOptions:** Allow fine-tuning of experimental, beta-quality logging options.
- **MatchLabelKeysInPodAffinity:** Enable the matchLabelKeys and mismatchLabelKeys field for [pod \(anti\)affinity](#).
- **MatchLabelKeysInPodTopologySpread:** Enable the matchLabelKeys field for [Pod topology spread constraints](#).
- **MaxUnavailableStatefulSet:** Enables setting the maxUnavailable field for the [rolling update strategy](#) of a StatefulSet. The field specifies the maximum number of Pods that can be unavailable during the update.
- **MemoryManager:** Allows setting memory affinity for a container based on NUMA topology.
- **MemoryQoS:** Enable memory protection and usage throttle on pod / container using cgroup v2 memory controller.

- MinDomainsInPodTopologySpread: Enable minDomains in [Pod topology spread constraints](#).
- MultiCIDRServiceAllocator: Track IP address allocations for Service cluster IPs using IPAddress objects.
- NameGenerationRetries: Enables retrying of object creation when the [API server](#) is expected to generate a [name](#). When this feature is enabled, requests using generateName are retried automatically in case the control plane detects a name conflict with an existing object, up to a limit of 8 total attempts.
- NewVolumeManagerReconstruction:
 

Enables improved discovery of mounted volumes during kubelet startup. Since the associated code had been significantly refactored, Kubernetes versions 1.25 to 1.29 allowed you to opt-out in case the kubelet got stuck at the startup, or did not unmount volumes from terminated Pods.

This refactoring was behind the SELinuxMountReadWriteOncePod feature gate in Kubernetes releases 1.25 and 1.26.
- NFTablesProxyMode: Allow running kube-proxy with in [nftables mode](#).
- NodeInclusionPolicyInPodTopologySpread: Enable using nodeAffinityPolicy and nodeTaintsPolicy in [Pod topology spread constraints](#) when calculating pod topology spread skew.
- NodeLogQuery: Enables querying logs of node services using the /logs endpoint.
- NodeOutOfServiceVolumeDetach: When a Node is marked out-of-service using the node.kubernetes.io/out-of-service taint, Pods on the node will be forcefully deleted if they can not tolerate this taint, and the volume detach operations for Pods terminating on the node will happen immediately. The deleted Pods can recover quickly on different nodes.
- NodeSwap: Enable the kubelet to allocate swap memory for Kubernetes workloads on a node. Must be used with KubeletConfiguration.failSwapOn set to false. For more details, please see [swap memory](#)
- OpenAPIEnums: Enables populating "enum" fields of OpenAPI schemas in the spec returned from the API server.
- PDBUnhealthyPodEvictionPolicy: Enables the unhealthyPodEvictionPolicy field of a PodDisruptionBudget. This specifies when unhealthy pods should be considered for eviction. Please see [Unhealthy Pod Eviction Policy](#) for more details.
- PersistentVolumeLastPhaseTransitionTime: Adds a new field to PersistentVolume which holds a timestamp of when the volume last transitioned its phase.
- PodAndContainerStatsFromCRI: Configure the kubelet to gather container and pod stats from the CRI container runtime rather than gathering them from cAdvisor. As of 1.26, this also includes gathering metrics from CRI and emitting them over /metrics/cadvisor (rather than having cAdvisor emit them directly).
- PodDeletionCost: Enable the [Pod Deletion Cost](#) feature which allows users to influence ReplicaSet downscaling order.

- **PodDisruptionConditions:** Enables support for appending a dedicated pod condition indicating that the pod is being deleted due to a disruption.
- **PodHostIPs:** Enable the status.hostIPs field for pods and the [downward API](#). The field lets you expose host IP addresses to workloads.
- **PodIndexLabel:** Enables the Job controller and StatefulSet controller to add the pod index as a label when creating new pods. See [Job completion mode docs](#) and [StatefulSet pod index label docs](#) for more details.
- **PodLifecycleSleepAction:** Enables the sleep action in Container lifecycle hooks.
- **PodReadyToStartContainersCondition:**

Enable the kubelet to mark the [PodReadyToStartContainers](#) condition on pods.

This feature gate was previously known as PodHasNetworkCondition, and the associated condition was named PodHasNetwork.

- **PodSchedulingReadiness:** Enable setting schedulingGates field to control a Pod's [scheduling readiness](#).
- **PortForwardWebsockets:** Allow WebSocket streaming of the portforward sub-protocol (port-forward) from clients requesting version v2 (v2.portforward.k8s.io) of the sub-protocol.
- **ProcMountType:** Enables control over the type proc mounts for containers by setting the procMount field of a SecurityContext.
- **QOSReserved:** Allows resource reservations at the QoS level preventing pods at lower QoS levels from bursting into resources requested at higher QoS levels (memory only for now).
- **ReadWriteOncePod:** Enables the usage of ReadWriteOncePod PersistentVolume access mode.
- **RecoverVolumeExpansionFailure:** Enables users to edit their PVCs to smaller sizes so as they can recover from previously issued volume expansion failures. See [Recovering from Failure when Expanding Volumes](#) for more details.
- **RecursiveReadOnlyMounts:** Enables support for recursive read-only mounts. For more details, see [read-only mounts](#).
- **RelaxedEnvironmentVariableValidation:** Allow almost all printable ASCII characters in environment variables.
- **RemainingItemCount:** Allow the API servers to show a count of remaining items in the response to a [chunking list request](#).
- **RotateKubeletServerCertificate:** Enable the rotation of the server TLS certificate on the kubelet. See [kubelet configuration](#) for more details.
- **RuntimeClassInImageCriApi:** Enables images to be pulled based on the [runtime class](#) of the pods that reference them.



SchedulerQueueingHints: Enables [the scheduler's queueing hints enhancement](#), which

- benefits to reduce the useless requeueing. The scheduler retries scheduling pods if something changes in the cluster that could make the pod scheduled. Queueing hints are internal signals that allow the scheduler to filter the changes in the cluster that are relevant to the unscheduled pod, based on previous scheduling attempts.

- SELinuxMount:

Speeds up container startup by allowing kubelet to mount volumes for a Pod directly with the correct SELinux label instead of changing each file on the volumes recursively. It widens the performance improvements behind the SELinuxMountReadWriteOncePod feature gate by extending the implementation to all volumes.

Enabling the SELinuxMount feature gate requires the feature gate SELinuxMountReadWriteOncePod to be enabled.

- SELinuxMountReadWriteOncePod: Speeds up container startup by allowing kubelet to mount volumes for a Pod directly with the correct SELinux label instead of changing each file on the volumes recursively. The initial implementation focused on ReadWriteOncePod volumes.
- SeparateTaintEvictionController: Enables running TaintEvictionController, that performs [Taint-based Evictions](#), in a controller separated from NodeLifecycleController. When this feature is enabled, users can optionally disable Taint-based Eviction setting the `--controllers=-taint-eviction-controller` flag on the kube-controller-manager.
- ServerSideApply: Enables the [Sever Side Apply \(SSA\)](#) feature on the API Server.
- ServerSideFieldValidation: Enables server-side field validation. This means the validation of resource schema is performed at the API server side rather than the client side (for example, the `kubectl create` or `kubectl apply` command line).
- ServiceAccountTokenJTI: Controls whether JTIs (UUIDs) are embedded into generated service account tokens, and whether these JTIs are recorded into the Kubernetes audit log for future requests made by these tokens.
- ServiceAccountTokenNodeBinding: Controls whether the apiserver allows binding service account tokens to Node objects.
- ServiceAccountTokenNodeBindingValidation: Controls whether the apiserver will validate a Node reference in service account tokens.
- ServiceAccountTokenPodNodeInfo: Controls whether the apiserver embeds the node name and uid for the associated node when issuing service account tokens bound to Pod objects.
- ServiceNodePortStaticSubrange: Enables the use of different port allocation strategies for NodePort Services. For more details, see [reserve NodePort ranges to avoid collisions](#).
- ServiceTrafficDistribution: Allows usage of the optional `spec.trafficDistribution` field in Services. The field offers a way to express preferences for how traffic is distributed to Service endpoints.

- SidecarContainers: Allow setting the restartPolicy of an init container to Always so that
- the container becomes a sidecar container (restartable init containers). See [Sidecar containers and restartPolicy](#) for more details.
  - SizeMemoryBackedVolumes: Enable kubelets to determine the size limit for memory-backed volumes (mainly emptyDir volumes).
  - SkipReadOnlyValidationGCE: Skip validation for GCE, will enable in the next version.
  - StableLoadBalancerNodeSet: Enables less load balancer re-configurations by the service controller (KCCM) as an effect of changing node state.
  - StatefulSetAutoDeletePVC: Allows the use of the optional .spec.persistentVolumeClaimRetentionPolicy field, providing control over the deletion of PVCs in a StatefulSet's lifecycle. See [PersistentVolumeClaim retention](#) for more details.
  - StatefulSetStartOrdinal: Allow configuration of the start ordinal in a StatefulSet. See [Start ordinal](#) for more details.
  - StorageVersionAPI: Enable the [storage version API](#).
  - StorageVersionHash: Allow API servers to expose the storage version hash in the discovery.
  - StorageVersionMigrator: Enables storage version migration. See [Migrate Kubernetes Objects Using Storage Version Migration](#) for more details.
  - StructuredAuthenticationConfiguration: Enable [structured authentication configuration](#) for the API server.
  - StructuredAuthorizationConfiguration: Enable structured authorization configuration, so that cluster administrators can specify more than one [authorization webhook](#) in the API server handler chain.
  - TopologyAwareHints: Enables topology aware routing based on topology hints in EndpointSlices. See [Topology Aware Hints](#) for more details.
  - TopologyManagerPolicyAlphaOptions: Allow fine-tuning of topology manager policies, experimental, Alpha-quality options. This feature gate guards *a group* of topology manager options whose quality level is alpha. This feature gate will never graduate to beta or stable.
  - TopologyManagerPolicyBetaOptions: Allow fine-tuning of topology manager policies, experimental, Beta-quality options. This feature gate guards *a group* of topology manager options whose quality level is beta. This feature gate will never graduate to stable.
  - TopologyManagerPolicyOptions: Enable [fine-tuning](#) of topology manager policies.
  - TranslateStreamCloseWebsocketRequests: Allow WebSocket streaming of the remote command sub-protocol (exec, cp, attach) from clients requesting version 5 (v5) of the sub-protocol.



- **UnauthenticatedHTTP2DOSMitigation**: Enables HTTP/2 Denial of Service (DoS) mitigations for unauthenticated clients. Kubernetes v1.28.0 through v1.28.2 do not include this feature gate.
- **UnknownVersionInteroperabilityProxy**: Proxy resource requests to the correct peer kube-apiserver when multiple kube-apiservers exist at varied versions. See [Mixed version proxy](#) for more information.
- **UserNamespacesPodSecurityStandards**: Enable Pod Security Standards policies relaxation for pods that run with namespaces. You must set the value of this feature gate consistently across all nodes in your cluster, and you must also enable **UserNamespacesSupport** to use this feature.
- **UserNamespacesSupport**: Enable user namespace support for Pods.
- **ValidatingAdmissionPolicy**: Enable [ValidatingAdmissionPolicy](#) support for CEL validations be used in Admission Control.
- **VolumeAttributesClass**: Enable support for VolumeAttributesClasses. See [Volume Attributes Classes](#) for more information.
- **VolumeCapacityPriority**: Enable support for prioritizing nodes in different topologies based on available PV capacity.
- **WatchBookmark**: Enable support for watch bookmark events.
- **WatchList**: Enable support for [streaming initial state of objects in watch requests](#).
- **WindowsHostNetwork**: Enables support for joining Windows containers to a hosts' network namespace.
- **WinDSR**: Allows kube-proxy to create DSR loadbalancers for Windows.
- **WinOverlay**: Allows kube-proxy to run in overlay mode for Windows.
- **ZeroLimitedNominalConcurrencyShares**: Allow [Priority & Fairness](#) in the API server to use a zero value for the `nominalConcurrencyShares` field of the `limited` section of a priority level.

## What's next

- The [deprecation policy](#) for Kubernetes explains the project's approach to removing features and components.
- Since Kubernetes 1.24, new beta APIs are not enabled by default. When enabling a beta feature, you will also need to enable any associated API resources. For example, to enable a particular resource like `storage.k8s.io/v1beta1/csistoragecapacities`, set `--runtime-config=storage.k8s.io/v1beta1/csistoragecapacities`. See [API Versioning](#) for more details on the command line flags.

# Feature Gates (removed)

This page contains list of feature gates that have been removed. The information on this page is for reference. A removed feature gate is different from a GA'ed or deprecated one in that a removed one is no longer recognized as a valid feature gate. However, a GA'ed or a deprecated feature gate is still recognized by the corresponding Kubernetes components although they are unable to cause any behavior differences in a cluster.

For feature gates that are still recognized by the Kubernetes components, please refer to the [Alpha/Beta feature gate table](#) or the [Graduated/Deprecated feature gate table](#)

## Feature gates that are removed

In the following table:

- The "From" column contains the Kubernetes release when a feature is introduced or its release stage is changed.
- The "To" column, if not empty, contains the last Kubernetes release in which you can still use a feature gate. If the feature stage is either "Deprecated" or "GA", the "To" column is the Kubernetes release when the feature is removed.

Feature Gates Removed

| Feature                        | Default | Stage      | From | To   |
|--------------------------------|---------|------------|------|------|
| Accelerators                   | false   | Alpha      | 1.6  | 1.10 |
| Accelerators                   | –       | Deprecated | 1.11 | 1.11 |
| AdvancedAuditing               | false   | Alpha      | 1.7  | 1.7  |
| AdvancedAuditing               | true    | Beta       | 1.8  | 1.11 |
| AdvancedAuditing               | true    | GA         | 1.12 | 1.27 |
| AffinityInAnnotations          | false   | Alpha      | 1.6  | 1.7  |
| AffinityInAnnotations          | –       | Deprecated | 1.8  | 1.8  |
| AllowExtTrafficLocalEndpoints  | false   | Beta       | 1.4  | 1.6  |
| AllowExtTrafficLocalEndpoints  | true    | GA         | 1.7  | 1.9  |
| AllowInsecureBackendProxy      | true    | Beta       | 1.17 | 1.20 |
| AllowInsecureBackendProxy      | true    | GA         | 1.21 | 1.25 |
| APISelfSubjectReview           | false   | Alpha      | 1.26 | 1.26 |
| APISelfSubjectReview           | true    | Beta       | 1.27 | 1.27 |
| APISelfSubjectReview           | true    | GA         | 1.28 | 1.29 |
| AttachVolumeLimit              | false   | Alpha      | 1.11 | 1.11 |
| AttachVolumeLimit              | true    | Beta       | 1.12 | 1.16 |
| AttachVolumeLimit              | true    | GA         | 1.17 | 1.21 |
| BalanceAttachedNodeVolumes     | false   | Alpha      | 1.11 | 1.21 |
| BalanceAttachedNodeVolumes     | false   | Deprecated | 1.22 | 1.22 |
| BlockVolume                    | false   | Alpha      | 1.9  | 1.12 |
| BlockVolume                    | true    | Beta       | 1.13 | 1.17 |
| BlockVolume                    | true    | GA         | 1.18 | 1.21 |
| BoundServiceAccountTokenVolume | false   | Alpha      | 1.13 | 1.20 |
| BoundServiceAccountTokenVolume | true    | Beta       | 1.21 | 1.21 |

| Feature                          | Default | Stage      | From | To   |
|----------------------------------|---------|------------|------|------|
| BoundServiceAccountTokenVolume   | true    | GA         | 1.22 | 1.23 |
| ConfigurableFSGroupPolicy        | false   | Alpha      | 1.18 | 1.19 |
| ConfigurableFSGroupPolicy        | true    | Beta       | 1.20 | 1.22 |
| ConfigurableFSGroupPolicy        | true    | GA         | 1.23 | 1.25 |
| ControllerManagerLeaderMigration | false   | Alpha      | 1.21 | 1.21 |
| ControllerManagerLeaderMigration | true    | Beta       | 1.22 | 1.23 |
| ControllerManagerLeaderMigration | true    | GA         | 1.24 | 1.26 |
| CRIOContainerLogRotation         | false   | Alpha      | 1.10 | 1.10 |
| CRIOContainerLogRotation         | true    | Beta       | 1.11 | 1.20 |
| CRIOContainerLogRotation         | true    | GA         | 1.21 | 1.22 |
| CronJobControllerV2              | false   | Alpha      | 1.20 | 1.20 |
| CronJobControllerV2              | true    | Beta       | 1.21 | 1.21 |
| CronJobControllerV2              | true    | GA         | 1.22 | 1.23 |
| CronJobTimeZone                  | false   | Alpha      | 1.24 | 1.24 |
| CronJobTimeZone                  | true    | Beta       | 1.25 | 1.26 |
| CronJobTimeZone                  | true    | GA         | 1.27 | 1.28 |
| CSIBlockVolume                   | false   | Alpha      | 1.11 | 1.13 |
| CSIBlockVolume                   | true    | Beta       | 1.14 | 1.17 |
| CSIBlockVolume                   | true    | GA         | 1.18 | 1.21 |
| CSIDriverRegistry                | false   | Alpha      | 1.12 | 1.13 |
| CSIDriverRegistry                | true    | Beta       | 1.14 | 1.17 |
| CSIDriverRegistry                | true    | GA         | 1.18 | 1.21 |
| CSIInlineVolume                  | false   | Alpha      | 1.15 | 1.15 |
| CSIInlineVolume                  | true    | Beta       | 1.16 | 1.24 |
| CSIInlineVolume                  | true    | GA         | 1.25 | 1.26 |
| CSIMigration                     | false   | Alpha      | 1.14 | 1.16 |
| CSIMigration                     | true    | Beta       | 1.17 | 1.24 |
| CSIMigration                     | true    | GA         | 1.25 | 1.26 |
| CSIMigrationAWS                  | false   | Alpha      | 1.14 | 1.16 |
| CSIMigrationAWS                  | false   | Beta       | 1.17 | 1.22 |
| CSIMigrationAWS                  | true    | Beta       | 1.23 | 1.24 |
| CSIMigrationAWS                  | true    | GA         | 1.25 | 1.26 |
| CSIMigrationAWSComplete          | false   | Alpha      | 1.17 | 1.20 |
| CSIMigrationAWSComplete          | –       | Deprecated | 1.21 | 1.21 |
| CSIMigrationAzureDisk            | false   | Alpha      | 1.15 | 1.18 |
| CSIMigrationAzureDisk            | false   | Beta       | 1.19 | 1.22 |
| CSIMigrationAzureDisk            | true    | Beta       | 1.23 | 1.23 |
| CSIMigrationAzureDisk            | true    | GA         | 1.24 | 1.26 |
| CSIMigrationAzureDiskComplete    | false   | Alpha      | 1.17 | 1.20 |
| CSIMigrationAzureDiskComplete    | –       | Deprecated | 1.21 | 1.21 |
| CSIMigrationAzureFile            | false   | Alpha      | 1.15 | 1.20 |
| CSIMigrationAzureFile            | false   | Beta       | 1.21 | 1.23 |
| CSIMigrationAzureFile            | true    | Beta       | 1.24 | 1.25 |
| CSIMigrationAzureFile            | true    | GA         | 1.26 | 1.29 |

| <b>Feature</b>                | <b>Default</b> | <b>Stage</b> | <b>From</b> | <b>To</b> |
|-------------------------------|----------------|--------------|-------------|-----------|
| CSIMigrationAzureFileComplete | false          | Alpha        | 1.17        | 1.20      |
| CSIMigrationAzureFileComplete | –              | Deprecated   | 1.21        | 1.21      |
| CSIMigrationGCE               | false          | Alpha        | 1.14        | 1.16      |
| CSIMigrationGCE               | false          | Beta         | 1.17        | 1.22      |
| CSIMigrationGCE               | true           | Beta         | 1.23        | 1.24      |
| CSIMigrationGCE               | true           | GA           | 1.25        | 1.27      |
| CSIMigrationGCEComplete       | false          | Alpha        | 1.17        | 1.20      |
| CSIMigrationGCEComplete       | –              | Deprecated   | 1.21        | 1.21      |
| CSIMigrationOpenStack         | false          | Alpha        | 1.14        | 1.17      |
| CSIMigrationOpenStack         | true           | Beta         | 1.18        | 1.23      |
| CSIMigrationOpenStack         | true           | GA           | 1.24        | 1.25      |
| CSIMigrationOpenStackComplete | false          | Alpha        | 1.17        | 1.20      |
| CSIMigrationOpenStackComplete | –              | Deprecated   | 1.21        | 1.21      |
| CSIMigrationvSphere           | false          | Alpha        | 1.18        | 1.18      |
| CSIMigrationvSphere           | false          | Beta         | 1.19        | 1.24      |
| CSIMigrationvSphere           | true           | Beta         | 1.25        | 1.25      |
| CSIMigrationvSphere           | true           | GA           | 1.26        | 1.28      |
| CSIMigrationvSphereComplete   | false          | Beta         | 1.19        | 1.21      |
| CSIMigrationvSphereComplete   | –              | Deprecated   | 1.22        | 1.22      |
| CSINodeInfo                   | false          | Alpha        | 1.12        | 1.13      |
| CSINodeInfo                   | true           | Beta         | 1.14        | 1.16      |
| CSINodeInfo                   | true           | GA           | 1.17        | 1.22      |
| CSIPersistentVolume           | false          | Alpha        | 1.9         | 1.9       |
| CSIPersistentVolume           | true           | Beta         | 1.10        | 1.12      |
| CSIPersistentVolume           | true           | GA           | 1.13        | 1.16      |
| CSIServiceAccountToken        | false          | Alpha        | 1.20        | 1.20      |
| CSIServiceAccountToken        | true           | Beta         | 1.21        | 1.21      |
| CSIServiceAccountToken        | true           | GA           | 1.22        | 1.24      |
| CSIStorageCapacity            | false          | Alpha        | 1.19        | 1.20      |
| CSIStorageCapacity            | true           | Beta         | 1.21        | 1.23      |
| CSIStorageCapacity            | true           | GA           | 1.24        | 1.27      |
| CSIVolumeFSGroupPolicy        | false          | Alpha        | 1.19        | 1.19      |
| CSIVolumeFSGroupPolicy        | true           | Beta         | 1.20        | 1.22      |
| CSIVolumeFSGroupPolicy        | true           | GA           | 1.23        | 1.25      |
| CSRDuration                   | true           | Beta         | 1.22        | 1.23      |
| CSRDuration                   | true           | GA           | 1.24        | 1.25      |
| CustomPodDNS                  | false          | Alpha        | 1.9         | 1.9       |
| CustomPodDNS                  | true           | Beta         | 1.10        | 1.13      |
| CustomPodDNS                  | true           | GA           | 1.14        | 1.16      |
| CustomResourceDefaulting      | false          | Alpha        | 1.15        | 1.15      |
| CustomResourceDefaulting      | true           | Beta         | 1.16        | 1.16      |
| CustomResourceDefaulting      | true           | GA           | 1.17        | 1.18      |
| CustomResourcePublishOpenAPI  | false          | Alpha        | 1.14        | 1.14      |
| CustomResourcePublishOpenAPI  | true           | Beta         | 1.15        | 1.15      |

| Feature                          | Default | Stage      | From | To   |
|----------------------------------|---------|------------|------|------|
| CustomResourcePublishOpenAPI     | true    | GA         | 1.16 | 1.18 |
| CustomResourceSubresources       | false   | Alpha      | 1.10 | 1.10 |
| CustomResourceSubresources       | true    | Beta       | 1.11 | 1.15 |
| CustomResourceSubresources       | true    | GA         | 1.16 | 1.18 |
| CustomResourceValidation         | false   | Alpha      | 1.8  | 1.8  |
| CustomResourceValidation         | true    | Beta       | 1.9  | 1.15 |
| CustomResourceValidation         | true    | GA         | 1.16 | 1.18 |
| CustomResourceWebhookConversion  | false   | Alpha      | 1.13 | 1.14 |
| CustomResourceWebhookConversion  | true    | Beta       | 1.15 | 1.15 |
| CustomResourceWebhookConversion  | true    | GA         | 1.16 | 1.18 |
| DaemonSetUpdateSurge             | false   | Alpha      | 1.21 | 1.21 |
| DaemonSetUpdateSurge             | true    | Beta       | 1.22 | 1.24 |
| DaemonSetUpdateSurge             | true    | GA         | 1.25 | 1.28 |
| DefaultPodTopologySpread         | false   | Alpha      | 1.19 | 1.19 |
| DefaultPodTopologySpread         | true    | Beta       | 1.20 | 1.23 |
| DefaultPodTopologySpread         | true    | GA         | 1.24 | 1.25 |
| DelegateFSGroupToCSIDriver       | false   | Alpha      | 1.22 | 1.22 |
| DelegateFSGroupToCSIDriver       | true    | Beta       | 1.23 | 1.25 |
| DelegateFSGroupToCSIDriver       | true    | GA         | 1.26 | 1.27 |
| DevicePlugins                    | false   | Alpha      | 1.8  | 1.9  |
| DevicePlugins                    | true    | Beta       | 1.10 | 1.25 |
| DevicePlugins                    | true    | GA         | 1.26 | 1.27 |
| DisableAcceleratorUsageMetrics   | false   | Alpha      | 1.19 | 1.19 |
| DisableAcceleratorUsageMetrics   | true    | Beta       | 1.20 | 1.24 |
| DisableAcceleratorUsageMetrics   | true    | GA         | 1.25 | 1.27 |
| DownwardAPIHugePages             | false   | Alpha      | 1.20 | 1.20 |
| DownwardAPIHugePages             | false   | Beta       | 1.21 | 1.21 |
| DownwardAPIHugePages             | true    | Beta       | 1.22 | 1.26 |
| DownwardAPIHugePages             | true    | GA         | 1.27 | 1.28 |
| DryRun                           | false   | Alpha      | 1.12 | 1.12 |
| DryRun                           | true    | Beta       | 1.13 | 1.18 |
| DryRun                           | true    | GA         | 1.19 | 1.27 |
| DynamicAuditing                  | false   | Alpha      | 1.13 | 1.18 |
| DynamicAuditing                  | –       | Deprecated | 1.19 | 1.19 |
| DynamicKubeletConfig             | false   | Alpha      | 1.4  | 1.10 |
| DynamicKubeletConfig             | true    | Beta       | 1.11 | 1.21 |
| DynamicKubeletConfig             | false   | Deprecated | 1.22 | 1.25 |
| DynamicProvisioningScheduling    | false   | Alpha      | 1.11 | 1.11 |
| DynamicProvisioningScheduling    | –       | Deprecated | 1.12 | –    |
| DynamicVolumeProvisioning        | true    | Alpha      | 1.3  | 1.7  |
| DynamicVolumeProvisioning        | true    | GA         | 1.8  | 1.12 |
| EnableAggregatedDiscoveryTimeout | true    | Deprecated | 1.16 | 1.17 |
| EnableEquivalenceClassCache      | false   | Alpha      | 1.8  | 1.12 |
| EnableEquivalenceClassCache      | –       | Deprecated | 1.13 | 1.23 |

| <b>Feature</b>                          | <b>Default</b> | <b>Stage</b> | <b>From</b> | <b>To</b> |
|-----------------------------------------|----------------|--------------|-------------|-----------|
| EndpointSlice                           | false          | Alpha        | 1.16        | 1.16      |
| EndpointSlice                           | false          | Beta         | 1.17        | 1.17      |
| EndpointSlice                           | true           | Beta         | 1.18        | 1.20      |
| EndpointSlice                           | true           | GA           | 1.21        | 1.24      |
| EndpointSliceNodeName                   | false          | Alpha        | 1.20        | 1.20      |
| EndpointSliceNodeName                   | true           | GA           | 1.21        | 1.24      |
| EndpointSliceProxying                   | false          | Alpha        | 1.18        | 1.18      |
| EndpointSliceProxying                   | true           | Beta         | 1.19        | 1.21      |
| EndpointSliceProxying                   | true           | GA           | 1.22        | 1.24      |
| EndpointSliceTerminatingCondition       | false          | Alpha        | 1.20        | 1.21      |
| EndpointSliceTerminatingCondition       | true           | Beta         | 1.22        | 1.25      |
| EndpointSliceTerminatingCondition       | true           | GA           | 1.26        | 1.27      |
| EphemeralContainers                     | false          | Alpha        | 1.16        | 1.22      |
| EphemeralContainers                     | true           | Beta         | 1.23        | 1.24      |
| EphemeralContainers                     | true           | GA           | 1.25        | 1.26      |
| EvenPodsSpread                          | false          | Alpha        | 1.16        | 1.17      |
| EvenPodsSpread                          | true           | Beta         | 1.18        | 1.18      |
| EvenPodsSpread                          | true           | GA           | 1.19        | 1.21      |
| ExpandCSIVolumes                        | false          | Alpha        | 1.14        | 1.15      |
| ExpandCSIVolumes                        | true           | Beta         | 1.16        | 1.23      |
| ExpandCSIVolumes                        | true           | GA           | 1.24        | 1.26      |
| ExpandedDNSConfig                       | false          | Alpha        | 1.22        | 1.25      |
| ExpandedDNSConfig                       | true           | Beta         | 1.26        | 1.27      |
| ExpandedDNSConfig                       | true           | GA           | 1.28        | 1.29      |
| ExpandInUsePersistentVolumes            | false          | Alpha        | 1.11        | 1.14      |
| ExpandInUsePersistentVolumes            | true           | Beta         | 1.15        | 1.23      |
| ExpandInUsePersistentVolumes            | true           | GA           | 1.24        | 1.26      |
| ExpandPersistentVolumes                 | false          | Alpha        | 1.8         | 1.10      |
| ExpandPersistentVolumes                 | true           | Beta         | 1.11        | 1.23      |
| ExpandPersistentVolumes                 | true           | GA           | 1.24        | 1.26      |
| ExperimentalCriticalPodAnnotation       | false          | Alpha        | 1.5         | 1.12      |
| ExperimentalCriticalPodAnnotation       | false          | Deprecated   | 1.13        | 1.16      |
| ExperimentalHostUserNamespaceDefaulting | false          | Beta         | 1.5         | 1.27      |
| ExperimentalHostUserNamespaceDefaulting | false          | Deprecated   | 1.28        | 1.29      |
| ExternalPolicyForExternalIP             | true           | GA           | 1.18        | 1.22      |
| GCERegionalPersistentDisk               | true           | Beta         | 1.10        | 1.12      |
| GCERegionalPersistentDisk               | true           | GA           | 1.13        | 1.16      |
| GenericEphemeralVolume                  | false          | Alpha        | 1.19        | 1.20      |
| GenericEphemeralVolume                  | true           | Beta         | 1.21        | 1.22      |
| GenericEphemeralVolume                  | true           | GA           | 1.23        | 1.24      |
| GRPCContainerProbe                      | false          | Alpha        | 1.23        | 1.23      |
| GRPCContainerProbe                      | true           | Beta         | 1.24        | 1.26      |
| GRPCContainerProbe                      | true           | GA           | 1.27        | 1.28      |
| HugePages                               | false          | Alpha        | 1.8         | 1.9       |

| Feature                            | Default | Stage      | From | To   |
|------------------------------------|---------|------------|------|------|
| HugePages                          | true    | Beta       | 1.10 | 1.13 |
| HugePages                          | true    | GA         | 1.14 | 1.16 |
| HugePageStorageMediumSize          | false   | Alpha      | 1.18 | 1.18 |
| HugePageStorageMediumSize          | true    | Beta       | 1.19 | 1.21 |
| HugePageStorageMediumSize          | true    | GA         | 1.22 | 1.24 |
| HyperVContainer                    | false   | Alpha      | 1.10 | 1.19 |
| HyperVContainer                    | false   | Deprecated | 1.20 | 1.20 |
| IdentifyPodOS                      | false   | Alpha      | 1.23 | 1.23 |
| IdentifyPodOS                      | true    | Beta       | 1.24 | 1.24 |
| IdentifyPodOS                      | true    | GA         | 1.25 | 1.26 |
| ImmutableEphemeralVolumes          | false   | Alpha      | 1.18 | 1.18 |
| ImmutableEphemeralVolumes          | true    | Beta       | 1.19 | 1.20 |
| ImmutableEphemeralVolumes          | true    | GA         | 1.21 | 1.24 |
| IndexedJob                         | false   | Alpha      | 1.21 | 1.21 |
| IndexedJob                         | true    | Beta       | 1.22 | 1.23 |
| IndexedJob                         | true    | GA         | 1.24 | 1.25 |
| IngressClassNamespacedParams       | false   | Alpha      | 1.21 | 1.21 |
| IngressClassNamespacedParams       | true    | Beta       | 1.22 | 1.22 |
| IngressClassNamespacedParams       | true    | GA         | 1.23 | 1.24 |
| Initializers                       | false   | Alpha      | 1.7  | 1.13 |
| Initializers                       | –       | Deprecated | 1.14 | 1.14 |
| IPTablesOwnershipCleanup           | false   | Alpha      | 1.25 | 1.26 |
| IPTablesOwnershipCleanup           | true    | Beta       | 1.27 | 1.27 |
| IPTablesOwnershipCleanup           | true    | GA         | 1.28 | 1.29 |
| IPv6DualStack                      | false   | Alpha      | 1.15 | 1.20 |
| IPv6DualStack                      | true    | Beta       | 1.21 | 1.22 |
| IPv6DualStack                      | true    | GA         | 1.23 | 1.24 |
| JobMutableNodeSchedulingDirectives | true    | Beta       | 1.23 | 1.26 |
| JobMutableNodeSchedulingDirectives | true    | GA         | 1.27 | 1.28 |
| JobTrackingWithFinalizers          | false   | Alpha      | 1.22 | 1.22 |
| JobTrackingWithFinalizers          | false   | Beta       | 1.23 | 1.24 |
| JobTrackingWithFinalizers          | true    | Beta       | 1.25 | 1.25 |
| JobTrackingWithFinalizers          | true    | GA         | 1.26 | 1.28 |
| KubeletConfigFile                  | false   | Alpha      | 1.8  | 1.9  |
| KubeletConfigFile                  | –       | Deprecated | 1.10 | 1.10 |
| KubeletCredentialProviders         | false   | Alpha      | 1.20 | 1.23 |
| KubeletCredentialProviders         | true    | Beta       | 1.24 | 1.25 |
| KubeletCredentialProviders         | true    | GA         | 1.26 | 1.28 |
| KubeletPluginsWatcher              | false   | Alpha      | 1.11 | 1.11 |
| KubeletPluginsWatcher              | true    | Beta       | 1.12 | 1.12 |
| KubeletPluginsWatcher              | true    | GA         | 1.13 | 1.16 |
| KubeletPodResources                | false   | Alpha      | 1.13 | 1.14 |
| KubeletPodResources                | true    | Beta       | 1.15 | 1.27 |
| KubeletPodResources                | true    | GA         | 1.28 | 1.29 |

| Feature                                   | Default | Stage      | From | To   |
|-------------------------------------------|---------|------------|------|------|
| KubeletPodResourcesGetAllocatable         | false   | Alpha      | 1.21 | 1.22 |
| KubeletPodResourcesGetAllocatable         | true    | Beta       | 1.23 | 1.27 |
| KubeletPodResourcesGetAllocatable         | true    | GA         | 1.28 | 1.29 |
| LegacyNodeRoleBehavior                    | false   | Alpha      | 1.16 | 1.18 |
| LegacyNodeRoleBehavior                    | true    | Beta       | 1.19 | 1.20 |
| LegacyNodeRoleBehavior                    | false   | GA         | 1.21 | 1.22 |
| LegacyServiceAccountTokenNoAutoGeneration | true    | Beta       | 1.24 | 1.25 |
| LegacyServiceAccountTokenNoAutoGeneration | true    | GA         | 1.26 | 1.28 |
| LegacyServiceAccountTokenTracking         | false   | Alpha      | 1.26 | 1.26 |
| LegacyServiceAccountTokenTracking         | true    | Beta       | 1.27 | 1.27 |
| LegacyServiceAccountTokenTracking         | true    | GA         | 1.28 | 1.29 |
| LocalStorageCapacityIsolation             | false   | Alpha      | 1.7  | 1.9  |
| LocalStorageCapacityIsolation             | true    | Beta       | 1.10 | 1.24 |
| LocalStorageCapacityIsolation             | true    | GA         | 1.25 | 1.26 |
| MinimizeIPTablesRestore                   | false   | Alpha      | 1.26 | 1.26 |
| MinimizeIPTablesRestore                   | true    | Beta       | 1.27 | 1.27 |
| MinimizeIPTablesRestore                   | true    | GA         | 1.28 | 1.29 |
| MixedProtocolLBService                    | false   | Alpha      | 1.20 | 1.23 |
| MixedProtocolLBService                    | true    | Beta       | 1.24 | 1.25 |
| MixedProtocolLBService                    | true    | GA         | 1.26 | 1.27 |
| MountContainers                           | false   | Alpha      | 1.9  | 1.16 |
| MountContainers                           | false   | Deprecated | 1.17 | 1.17 |
| MountPropagation                          | false   | Alpha      | 1.8  | 1.9  |
| MountPropagation                          | true    | Beta       | 1.10 | 1.11 |
| MountPropagation                          | true    | GA         | 1.12 | 1.14 |
| MultiCIDRRangeAllocator                   | false   | Alpha      | 1.25 | 1.28 |
| NamespaceDefaultLabelName                 | true    | Beta       | 1.21 | 1.21 |
| NamespaceDefaultLabelName                 | true    | GA         | 1.22 | 1.23 |
| NetworkPolicyEndPort                      | false   | Alpha      | 1.21 | 1.21 |
| NetworkPolicyEndPort                      | true    | Beta       | 1.22 | 1.24 |
| NetworkPolicyEndPort                      | true    | GA         | 1.25 | 1.26 |
| NetworkPolicyStatus                       | false   | Alpha      | 1.24 | 1.27 |
| NodeDisruptionExclusion                   | false   | Alpha      | 1.16 | 1.18 |
| NodeDisruptionExclusion                   | true    | Beta       | 1.19 | 1.20 |
| NodeDisruptionExclusion                   | true    | GA         | 1.21 | 1.22 |
| NodeLease                                 | false   | Alpha      | 1.12 | 1.13 |
| NodeLease                                 | true    | Beta       | 1.14 | 1.16 |
| NodeLease                                 | true    | GA         | 1.17 | 1.23 |
| NonPreemptingPriority                     | false   | Alpha      | 1.15 | 1.18 |
| NonPreemptingPriority                     | true    | Beta       | 1.19 | 1.23 |
| NonPreemptingPriority                     | true    | GA         | 1.24 | 1.25 |
| OpenAPIV3                                 | false   | Alpha      | 1.23 | 1.23 |
| OpenAPIV3                                 | true    | Beta       | 1.24 | 1.26 |
| OpenAPIV3                                 | true    | GA         | 1.27 | 1.28 |



| <b>Feature</b>               | <b>Default</b> | <b>Stage</b> | <b>From</b> | <b>To</b> |
|------------------------------|----------------|--------------|-------------|-----------|
| PersistentLocalVolumes       | false          | Alpha        | 1.7         | 1.9       |
| PersistentLocalVolumes       | true           | Beta         | 1.10        | 1.13      |
| PersistentLocalVolumes       | true           | GA           | 1.14        | 1.16      |
| PodAffinityNamespaceSelector | false          | Alpha        | 1.21        | 1.21      |
| PodAffinityNamespaceSelector | true           | Beta         | 1.22        | 1.23      |
| PodAffinityNamespaceSelector | true           | GA           | 1.24        | 1.25      |
| PodDisruptionBudget          | false          | Alpha        | 1.3         | 1.4       |
| PodDisruptionBudget          | true           | Beta         | 1.5         | 1.20      |
| PodDisruptionBudget          | true           | GA           | 1.21        | 1.25      |
| PodHasNetworkCondition       | false          | Alpha        | 1.25        | 1.27      |
| PodOverhead                  | false          | Alpha        | 1.16        | 1.17      |
| PodOverhead                  | true           | Beta         | 1.18        | 1.23      |
| PodOverhead                  | true           | GA           | 1.24        | 1.25      |
| PodPriority                  | false          | Alpha        | 1.8         | 1.10      |
| PodPriority                  | true           | Beta         | 1.11        | 1.13      |
| PodPriority                  | true           | GA           | 1.14        | 1.18      |
| PodReadinessGates            | false          | Alpha        | 1.11        | 1.11      |
| PodReadinessGates            | true           | Beta         | 1.12        | 1.13      |
| PodReadinessGates            | true           | GA           | 1.14        | 1.16      |
| PodSecurity                  | false          | Alpha        | 1.22        | 1.22      |
| PodSecurity                  | true           | Beta         | 1.23        | 1.24      |
| PodSecurity                  | true           | GA           | 1.25        | 1.27      |
| PodShareProcessNamespace     | false          | Alpha        | 1.10        | 1.11      |
| PodShareProcessNamespace     | true           | Beta         | 1.12        | 1.16      |
| PodShareProcessNamespace     | true           | GA           | 1.17        | 1.19      |
| PreferNominatedNode          | false          | Alpha        | 1.21        | 1.21      |
| PreferNominatedNode          | true           | Beta         | 1.22        | 1.23      |
| PreferNominatedNode          | true           | GA           | 1.24        | 1.25      |
| ProbeTerminationGracePeriod  | false          | Alpha        | 1.21        | 1.21      |
| ProbeTerminationGracePeriod  | false          | Beta         | 1.22        | 1.24      |
| ProbeTerminationGracePeriod  | true           | Beta         | 1.25        | 1.27      |
| ProbeTerminationGracePeriod  | true           | GA           | 1.28        | 1.28      |
| ProxyTerminatingEndpoints    | false          | Alpha        | 1.22        | 1.25      |
| ProxyTerminatingEndpoints    | true           | Beta         | 1.26        | 1.27      |
| ProxyTerminatingEndpoints    | true           | GA           | 1.28        | 1.29      |
| PVCProtection                | false          | Alpha        | 1.9         | 1.9       |
| PVCProtection                | –              | Deprecated   | 1.10        | 1.10      |
| ReadOnlyAPIDataVolumes       | true           | Beta         | 1.8         | 1.9       |
| ReadOnlyAPIDataVolumes       | –              | GA           | 1.10        | 1.10      |
| RemoveSelfLink               | false          | Alpha        | 1.16        | 1.19      |
| RemoveSelfLink               | true           | Beta         | 1.20        | 1.23      |
| RemoveSelfLink               | true           | GA           | 1.24        | 1.29      |
| RequestManagement            | false          | Alpha        | 1.15        | 1.16      |
| RequestManagement            | –              | Deprecated   | 1.17        | 1.17      |

| Feature                        | Default | Stage      | From | To   |
|--------------------------------|---------|------------|------|------|
| ResourceLimitsPriorityFunction | false   | Alpha      | 1.9  | 1.18 |
| ResourceLimitsPriorityFunction | –       | Deprecated | 1.19 | 1.19 |
| ResourceQuotaScopeSelectors    | false   | Alpha      | 1.11 | 1.11 |
| ResourceQuotaScopeSelectors    | true    | Beta       | 1.12 | 1.16 |
| ResourceQuotaScopeSelectors    | true    | GA         | 1.17 | 1.18 |
| RetroactiveDefaultStorageClass | false   | Alpha      | 1.25 | 1.25 |
| RetroactiveDefaultStorageClass | true    | Beta       | 1.26 | 1.27 |
| RetroactiveDefaultStorageClass | true    | GA         | 1.28 | 1.28 |
| RootCAConfigMap                | false   | Alpha      | 1.13 | 1.19 |
| RootCAConfigMap                | true    | Beta       | 1.20 | 1.20 |
| RootCAConfigMap                | true    | GA         | 1.21 | 1.22 |
| RotateKubeletClientCertificate | true    | Beta       | 1.8  | 1.18 |
| RotateKubeletClientCertificate | true    | GA         | 1.19 | 1.21 |
| RunAsGroup                     | true    | Beta       | 1.14 | 1.20 |
| RunAsGroup                     | true    | GA         | 1.21 | 1.22 |
| RuntimeClass                   | false   | Alpha      | 1.12 | 1.13 |
| RuntimeClass                   | true    | Beta       | 1.14 | 1.19 |
| RuntimeClass                   | true    | GA         | 1.20 | 1.24 |
| ScheduleDaemonSetPods          | false   | Alpha      | 1.11 | 1.11 |
| ScheduleDaemonSetPods          | true    | Beta       | 1.12 | 1.16 |
| ScheduleDaemonSetPods          | true    | GA         | 1.17 | 1.18 |
| SCTPSupport                    | false   | Alpha      | 1.12 | 1.18 |
| SCTPSupport                    | true    | Beta       | 1.19 | 1.19 |
| SCTPSupport                    | true    | GA         | 1.20 | 1.22 |
| SeccompDefault                 | false   | Alpha      | 1.22 | 1.24 |
| SeccompDefault                 | true    | Beta       | 1.25 | 1.26 |
| SeccompDefault                 | true    | GA         | 1.27 | 1.28 |
| SecurityContextDeny            | false   | Alpha      | 1.27 | 1.29 |
| SelectorIndex                  | false   | Alpha      | 1.18 | 1.18 |
| SelectorIndex                  | true    | Beta       | 1.19 | 1.19 |
| SelectorIndex                  | true    | GA         | 1.20 | 1.25 |
| ServiceAccountIssuerDiscovery  | false   | Alpha      | 1.18 | 1.19 |
| ServiceAccountIssuerDiscovery  | true    | Beta       | 1.20 | 1.20 |
| ServiceAccountIssuerDiscovery  | true    | GA         | 1.21 | 1.23 |
| ServiceAppProtocol             | false   | Alpha      | 1.18 | 1.18 |
| ServiceAppProtocol             | true    | Beta       | 1.19 | 1.19 |
| ServiceAppProtocol             | true    | GA         | 1.20 | 1.22 |
| ServiceInternalTrafficPolicy   | false   | Alpha      | 1.21 | 1.21 |
| ServiceInternalTrafficPolicy   | true    | Beta       | 1.22 | 1.25 |
| ServiceInternalTrafficPolicy   | true    | GA         | 1.26 | 1.27 |
| ServiceIPStaticSubrange        | false   | Alpha      | 1.24 | 1.24 |
| ServiceIPStaticSubrange        | true    | Beta       | 1.25 | 1.25 |
| ServiceIPStaticSubrange        | true    | GA         | 1.26 | 1.27 |
| ServiceLBNodePortControl       | false   | Alpha      | 1.20 | 1.21 |

| Feature                      | Default | Stage      | From | To   |
|------------------------------|---------|------------|------|------|
| ServiceLBNodePortControl     | true    | Beta       | 1.22 | 1.23 |
| ServiceLBNodePortControl     | true    | GA         | 1.24 | 1.25 |
| ServiceLoadBalancerClass     | false   | Alpha      | 1.21 | 1.21 |
| ServiceLoadBalancerClass     | true    | Beta       | 1.22 | 1.23 |
| ServiceLoadBalancerClass     | true    | GA         | 1.24 | 1.25 |
| ServiceLoadBalancerFinalizer | false   | Alpha      | 1.15 | 1.15 |
| ServiceLoadBalancerFinalizer | true    | Beta       | 1.16 | 1.16 |
| ServiceLoadBalancerFinalizer | true    | GA         | 1.17 | 1.20 |
| ServiceNodeExclusion         | false   | Alpha      | 1.8  | 1.18 |
| ServiceNodeExclusion         | true    | Beta       | 1.19 | 1.20 |
| ServiceNodeExclusion         | true    | GA         | 1.21 | 1.22 |
| ServiceTopology              | false   | Alpha      | 1.17 | 1.19 |
| ServiceTopology              | false   | Deprecated | 1.20 | 1.22 |
| SetHostnameAsFQDN            | false   | Alpha      | 1.19 | 1.19 |
| SetHostnameAsFQDN            | true    | Beta       | 1.20 | 1.21 |
| SetHostnameAsFQDN            | true    | GA         | 1.22 | 1.24 |
| StartupProbe                 | false   | Alpha      | 1.16 | 1.17 |
| StartupProbe                 | true    | Beta       | 1.18 | 1.19 |
| StartupProbe                 | true    | GA         | 1.20 | 1.23 |
| StatefulSetMinReadySeconds   | false   | Alpha      | 1.22 | 1.22 |
| StatefulSetMinReadySeconds   | true    | Beta       | 1.23 | 1.24 |
| StatefulSetMinReadySeconds   | true    | GA         | 1.25 | 1.26 |
| StorageObjectInUseProtection | true    | Beta       | 1.10 | 1.10 |
| StorageObjectInUseProtection | true    | GA         | 1.11 | 1.24 |
| StreamingProxyRedirects      | false   | Beta       | 1.5  | 1.5  |
| StreamingProxyRedirects      | true    | Beta       | 1.6  | 1.17 |
| StreamingProxyRedirects      | true    | Deprecated | 1.18 | 1.21 |
| StreamingProxyRedirects      | false   | Deprecated | 1.22 | 1.24 |
| SupportIPVSProxyMode         | false   | Alpha      | 1.8  | 1.8  |
| SupportIPVSProxyMode         | false   | Beta       | 1.9  | 1.9  |
| SupportIPVSProxyMode         | true    | Beta       | 1.10 | 1.10 |
| SupportIPVSProxyMode         | true    | GA         | 1.11 | 1.20 |
| SupportNodePidsLimit         | false   | Alpha      | 1.14 | 1.14 |
| SupportNodePidsLimit         | true    | Beta       | 1.15 | 1.19 |
| SupportNodePidsLimit         | true    | GA         | 1.20 | 1.23 |
| SupportPodPidsLimit          | false   | Alpha      | 1.10 | 1.13 |
| SupportPodPidsLimit          | true    | Beta       | 1.14 | 1.19 |
| SupportPodPidsLimit          | true    | GA         | 1.20 | 1.23 |
| SuspendJob                   | false   | Alpha      | 1.21 | 1.21 |
| SuspendJob                   | true    | Beta       | 1.22 | 1.23 |
| SuspendJob                   | true    | GA         | 1.24 | 1.25 |
| Sysctls                      | true    | Beta       | 1.11 | 1.20 |
| Sysctls                      | true    | GA         | 1.21 | 1.22 |
| TaintBasedEvictions          | false   | Alpha      | 1.6  | 1.12 |

| Feature                            | Default | Stage      | From | To   |
|------------------------------------|---------|------------|------|------|
| TaintBasedEvictions                | true    | Beta       | 1.13 | 1.17 |
| TaintBasedEvictions                | true    | GA         | 1.18 | 1.20 |
| TaintNodesByCondition              | false   | Alpha      | 1.8  | 1.11 |
| TaintNodesByCondition              | true    | Beta       | 1.12 | 1.16 |
| TaintNodesByCondition              | true    | GA         | 1.17 | 1.18 |
| TokenRequest                       | false   | Alpha      | 1.10 | 1.11 |
| TokenRequest                       | true    | Beta       | 1.12 | 1.19 |
| TokenRequest                       | true    | GA         | 1.20 | 1.21 |
| TokenRequestProjection             | false   | Alpha      | 1.11 | 1.11 |
| TokenRequestProjection             | true    | Beta       | 1.12 | 1.19 |
| TokenRequestProjection             | true    | GA         | 1.20 | 1.21 |
| TopologyManager                    | false   | Alpha      | 1.16 | 1.17 |
| TopologyManager                    | true    | Beta       | 1.18 | 1.26 |
| TopologyManager                    | true    | GA         | 1.27 | 1.28 |
| TTLAfterFinished                   | false   | Alpha      | 1.12 | 1.20 |
| TTLAfterFinished                   | true    | Beta       | 1.21 | 1.22 |
| TTLAfterFinished                   | true    | GA         | 1.23 | 1.24 |
| UserNamespacesStatelessPodsSupport | false   | Alpha      | 1.25 | 1.27 |
| ValidateProxyRedirects             | false   | Alpha      | 1.12 | 1.13 |
| ValidateProxyRedirects             | true    | Beta       | 1.14 | 1.21 |
| ValidateProxyRedirects             | true    | Deprecated | 1.22 | 1.24 |
| VolumePVCDDataSource               | false   | Alpha      | 1.15 | 1.15 |
| VolumePVCDDataSource               | true    | Beta       | 1.16 | 1.17 |
| VolumePVCDDataSource               | true    | GA         | 1.18 | 1.21 |
| VolumeScheduling                   | false   | Alpha      | 1.9  | 1.9  |
| VolumeScheduling                   | true    | Beta       | 1.10 | 1.12 |
| VolumeScheduling                   | true    | GA         | 1.13 | 1.16 |
| VolumeSnapshotDataSource           | false   | Alpha      | 1.12 | 1.16 |
| VolumeSnapshotDataSource           | true    | Beta       | 1.17 | 1.19 |
| VolumeSnapshotDataSource           | true    | GA         | 1.20 | 1.22 |
| VolumeSubpath                      | true    | GA         | 1.10 | 1.24 |
| VolumeSubpathEnvExpansion          | false   | Alpha      | 1.14 | 1.14 |
| VolumeSubpathEnvExpansion          | true    | Beta       | 1.15 | 1.16 |
| VolumeSubpathEnvExpansion          | true    | GA         | 1.17 | 1.24 |
| WarningHeaders                     | true    | Beta       | 1.19 | 1.21 |
| WarningHeaders                     | true    | GA         | 1.22 | 1.24 |
| WindowsEndpointSliceProxying       | false   | Alpha      | 1.19 | 1.20 |
| WindowsEndpointSliceProxying       | true    | Beta       | 1.21 | 1.21 |
| WindowsEndpointSliceProxying       | true    | GA         | 1.22 | 1.24 |
| WindowsGMSA                        | false   | Alpha      | 1.14 | 1.15 |
| WindowsGMSA                        | true    | Beta       | 1.16 | 1.17 |
| WindowsGMSA                        | true    | GA         | 1.18 | 1.20 |
| WindowsHostProcessContainers       | false   | Alpha      | 1.22 | 1.22 |
| WindowsHostProcessContainers       | true    | Beta       | 1.23 | 1.25 |

| Feature                      | Default | Stage | From | To   |
|------------------------------|---------|-------|------|------|
| WindowsHostProcessContainers | true    | GA    | 1.26 | 1.27 |
| WindowsRunAsUserName         | false   | Alpha | 1.16 | 1.16 |
| WindowsRunAsUserName         | true    | Beta  | 1.17 | 1.17 |
| WindowsRunAsUserName         | true    | GA    | 1.18 | 1.20 |

## Descriptions for removed feature gates

- **Accelerators**: Provided an early form of plugin to enable Nvidia GPU support when using Docker Engine; no longer available. See [Device Plugins](#) for an alternative.
- **AdvancedAuditing**: Enable [advanced auditing](#)
- **AffinityInAnnotations**: Enable setting [Pod affinity or anti-affinity](#).
- **AllowExtTrafficLocalEndpoints**: Enable a service to route external requests to node local endpoints.
- **AllowInsecureBackendProxy**: Enable the users to skip TLS verification of kubelets on Pod log requests.
- **APISelfSubjectReview**: Activate the SelfSubjectReview API which allows users to see the requesting subject's authentication information. See [API access to authentication information for a client](#) for more details.
- **AttachVolumeLimit**: Enable volume plugins to report limits on number of volumes that can be attached to a node. See [dynamic volume limits](#) for more details.
- **BalanceAttachedNodeVolumes**: Include volume count on node to be considered for balanced resource allocation while scheduling. A node which has closer CPU, memory utilization, and volume count is favored by the scheduler while making decisions.
- **BlockVolume**: Enable the definition and consumption of raw block devices in Pods. See [Raw Block Volume Support](#) for more details.
- **BoundServiceAccountTokenVolume**:

Migrate ServiceAccount volumes to use a projected volume consisting of a ServiceAccountTokenVolumeProjection. Cluster admins can use metric `serviceaccount_stale_tokens_total` to monitor workloads that are depending on the extended tokens. If there are no such workloads, turn off extended tokens by starting kube-apiserver with flag `--service-account-extend-token-expiration=false`.

Check [Bound Service Account Tokens](#) for more details.

- **ConfigurableFSGroupPolicy**: Allows user to configure volume permission change policy for fsGroups when mounting a volume in a Pod. See [Configure volume permission and ownership change policy for Pods](#) for more details.
- **ControllerManagerLeaderMigration**: Enables Leader Migration for [kube-controller-manager](#) and [cloud-controller-manager](#) which allows a cluster operator to live migrate controllers from the kube-controller-manager into an external controller-manager (e.g. the cloud-controller-manager) in an HA cluster without downtime.

- CRIContainerLogRotation: Enable container log rotation for CRI container runtime. The default max size of a log file is 10MB and the default max number of log files allowed for a container is 5. These values can be configured in the kubelet config. See [logging at node level](#) for more details.
- CronJobControllerV2: Use an alternative implementation of the [CronJob](#) controller. Otherwise, version 1 of the same controller is selected.
  - CronJobTimeZone: Allow the use of the timeZone optional field in [CronJobs](#)
  - CSIBlockVolume: Enable external CSI volume drivers to support block storage. See [csi raw block volume support](#) for more details.
  - CSIDriverRegistry: Enable all logic related to the CSIDriver API object in csi.storage.k8s.io.
  - CSIInlineVolume: Enable CSI Inline volumes support for pods.
  - CSIMigration: Enables shims and translation logic to route volume operations from in-tree plugins to corresponding pre-installed CSI plugins
  - CSIMigrationAWS: Enables shims and translation logic to route volume operations from the AWS-EBS in-tree plugin to EBS CSI plugin. Supports falling back to in-tree EBS plugin for mount operations to nodes that have the feature disabled or that do not have EBS CSI plugin installed and configured. Does not support falling back for provision operations, for those the CSI plugin must be installed and configured.
  - CSIMigrationAWSComplete: Stops registering the EBS in-tree plugin in kubelet and volume controllers and enables shims and translation logic to route volume operations from the AWS-EBS in-tree plugin to EBS CSI plugin. Requires CSIMigration and CSIMigrationAWS feature flags enabled and EBS CSI plugin installed and configured on all nodes in the cluster. This flag has been deprecated in favor of the InTreePluginAWSUnregister feature flag which prevents the registration of in-tree EBS plugin.
  - CSIMigrationAzureDisk: Enables shims and translation logic to route volume operations from the Azure-Disk in-tree plugin to AzureDisk CSI plugin. Supports falling back to in-tree AzureDisk plugin for mount operations to nodes that have the feature disabled or that do not have AzureDisk CSI plugin installed and configured. Does not support falling back for provision operations, for those the CSI plugin must be installed and configured. Requires CSIMigration feature flag enabled.
  - CSIMigrationAzureDiskComplete: Stops registering the Azure-Disk in-tree plugin in kubelet and volume controllers and enables shims and translation logic to route volume operations from the Azure-Disk in-tree plugin to AzureDisk CSI plugin. Requires CSIMigration and CSIMigrationAzureDisk feature flags enabled and AzureDisk CSI plugin installed and configured on all nodes in the cluster. This flag has been deprecated in favor of the InTreePluginAzureDiskUnregister feature flag which prevents the registration of in-tree AzureDisk plugin.
  - CSIMigrationAzureFile: Enables shims and translation logic to route volume operations from the Azure-File in-tree plugin to AzureFile CSI plugin. Supports falling back to in-tree AzureFile plugin for mount operations to nodes that have the feature disabled or that do not have AzureFile CSI plugin installed and configured. Does not support falling back

for provision operations, for those the CSI plugin must be installed and configured. Requires CSIMigration feature flag enabled.

- **CSIMigrationAzureFileComplete**: Stops registering the Azure-File in-tree plugin in kubelet and volume controllers and enables shims and translation logic to route volume operations from the Azure-File in-tree plugin to AzureFile CSI plugin. Requires CSIMigration and CSIMigrationAzureFile feature flags enabled and AzureFile CSI plugin installed and configured on all nodes in the cluster. This flag has been deprecated in favor of the InTreePluginAzureFileUnregister feature flag which prevents the registration of in-tree AzureFile plugin.
- **CSIMigrationGCE**: Enables shims and translation logic to route volume operations from the GCE-PD in-tree plugin to PD CSI plugin. Supports falling back to in-tree GCE plugin for mount operations to nodes that have the feature disabled or that do not have PD CSI plugin installed and configured. Does not support falling back for provision operations, for those the CSI plugin must be installed and configured. Requires CSIMigration feature flag enabled.
- **CSIMigrationGCEComplete**: Stops registering the GCE-PD in-tree plugin in kubelet and volume controllers and enables shims and translation logic to route volume operations from the GCE-PD in-tree plugin to PD CSI plugin. Requires CSIMigration and CSIMigrationGCE feature flags enabled and PD CSI plugin installed and configured on all nodes in the cluster. This flag has been deprecated in favor of the InTreePluginGCEUnregister feature flag which prevents the registration of in-tree GCE PD plugin.
- **CSIMigrationOpenStack**: Enables shims and translation logic to route volume operations from the Cinder in-tree plugin to Cinder CSI plugin. Supports falling back to in-tree Cinder plugin for mount operations to nodes that have the feature disabled or that do not have Cinder CSI plugin installed and configured. Does not support falling back for provision operations, for those the CSI plugin must be installed and configured. Requires CSIMigration feature flag enabled.
- **CSIMigrationOpenStackComplete**: Stops registering the Cinder in-tree plugin in kubelet and volume controllers and enables shims and translation logic to route volume operations from the Cinder in-tree plugin to Cinder CSI plugin. Requires CSIMigration and CSIMigrationOpenStack feature flags enabled and Cinder CSI plugin installed and configured on all nodes in the cluster. This flag has been deprecated in favor of the InTreePluginOpenStackUnregister feature flag which prevents the registration of in-tree openstack cinder plugin.
- **CSIMigrationvSphere**: Enables shims and translation logic to route volume operations from the vSphere in-tree plugin to vSphere CSI plugin. Supports falling back to in-tree vSphere plugin for mount operations to nodes that have the feature disabled or that do not have vSphere CSI plugin installed and configured. Does not support falling back for provision operations, for those the CSI plugin must be installed and configured. Requires CSIMigration feature flag enabled.
- **CSIMigrationvSphereComplete**: Stops registering the vSphere in-tree plugin in kubelet and volume controllers and enables shims and translation logic to route volume operations from the vSphere in-tree plugin to vSphere CSI plugin. Requires CSIMigration and CSIMigrationvSphere feature flags enabled and vSphere CSI plugin installed and configured on all nodes in the cluster. This flag has been deprecated in favor of the

InTreePluginvSphereUnregister feature flag which prevents the registration of in-tree vsphere plugin.

- CSINodeInfo: Enable all logic related to the CSINodeInfo API object in `csi.storage.k8s.io`.
- CSIPersistentVolume: Enable discovering and mounting volumes provisioned through a [CSI \(Container Storage Interface\)](#) compatible volume plugin.
- CSIServiceAccountToken: Enable CSI drivers to receive the pods' service account token that they mount volumes for. See [Token Requests](#).
- CSIStorageCapacity: Enables CSI drivers to publish storage capacity information and the Kubernetes scheduler to use that information when scheduling pods. See [Storage Capacity](#). Check the [csi volume type](#) documentation for more details.
- CSIVolumeFSGroupPolicy: Allows CSIDrivers to use the fsGroupPolicy field. This field controls whether volumes created by a CSIDriver support volume ownership and permission modifications when these volumes are mounted.
- CSRDuration: Allows clients to request a duration for certificates issued via the Kubernetes CSR API.
- CustomPodDNS: Enable customizing the DNS settings for a Pod using its `dnsConfig` property. Check [Pod's DNS Config](#) for more details.
- CustomResourceDefaulting: Enable CRD support for default values in OpenAPI v3 validation schemas.
- CustomResourcePublishOpenAPI: Enables publishing of CRD OpenAPI specs.
- CustomResourceSubresources: Enable `/status` and `/scale` subresources on resources created from [CustomResourceDefinition](#).
- CustomResourceValidation: Enable schema based validation on resources created from [CustomResourceDefinition](#).
- CustomResourceWebhookConversion: Enable webhook-based conversion on resources created from [CustomResourceDefinition](#).
- DaemonSetUpdateSurge: Enables the DaemonSet workloads to maintain availability during update per node. See [Perform a Rolling Update on a DaemonSet](#).
- DefaultPodTopologySpread: Enables the use of PodTopologySpread scheduling plugin to do [default spreading](#).
- DelegateFSGroupToCSIDriver: If supported by the CSI driver, delegates the role of applying fsGroup from a Pod's securityContext to the driver by passing fsGroup through the NodeStageVolume and NodePublishVolume CSI calls.
- DevicePlugins: Enable the [device-plugins](#) based resource provisioning on nodes.
- DisableAcceleratorUsageMetrics: [Disable accelerator metrics collected by the kubelet](#).
- DownwardAPIHugePages: Enables usage of hugepages in [downward API](#).



- DryRun: Enable server-side [dry run](#) requests so that validation, merging, and mutation can be tested without committing.
- DynamicAuditing: Used to enable dynamic auditing before v1.19.
- DynamicKubeletConfig: Enable the dynamic configuration of kubelet. The feature is no longer supported outside of supported skew policy. The feature gate was removed from kubelet in 1.24.
- DynamicProvisioningScheduling: Extend the default scheduler to be aware of volume topology and handle PV provisioning. This feature was superseded by the VolumeScheduling feature in v1.12.
- DynamicVolumeProvisioning: Enable the [dynamic provisioning](#) of persistent volumes to Pods.
- EnableAggregatedDiscoveryTimeout: Enable the five second timeout on aggregated discovery calls.
- EnableEquivalenceClassCache: Enable the scheduler to cache equivalence of nodes when scheduling Pods.
- EndpointSlice: Enables EndpointSlices for more scalable and extensible network endpoints. See [Enabling EndpointSlices](#).
- EndpointSliceNodeName: Enables EndpointSlice nodeName field.
- EndpointSliceProxying: When enabled, kube-proxy running on Linux will use EndpointSlices as the primary data source instead of Endpoints, enabling scalability and performance improvements. See [Enabling Endpoint Slices](#).
- EndpointSliceTerminatingCondition: Enables EndpointSlice terminating and serving condition fields.
- EphemeralContainers: Enable the ability to add [ephemeral containers](#) to running Pods.
- EvenPodsSpread: Enable pods to be scheduled evenly across topology domains. See [Pod Topology Spread Constraints](#).
- ExpandCSIVolumes: Enable the expanding of CSI volumes.
- ExpandedDNSConfig: Enable kubelet and kube-apiserver to allow more DNS search paths and longer list of DNS search paths. This feature requires container runtime support(Containerd: v1.5.6 or higher, CRI-O: v1.22 or higher). See [Expanded DNS Configuration](#).
- ExpandInUsePersistentVolumes: Enable expanding in-use PVCs. See [Resizing an in-use PersistentVolumeClaim](#).
- ExpandPersistentVolumes: Enable the expanding of persistent volumes. See [Expanding Persistent Volumes Claims](#).
- ExperimentalCriticalPodAnnotation: Enable annotating specific pods as *critical* so that their [scheduling is guaranteed](#). This feature is deprecated by Pod Priority and Preemption as of v1.13.

- **ExperimentalHostUserNamespaceDefaulting**: Enabling the defaulting user namespace to host. This is for containers that are using other host namespaces, host mounts, or containers that are privileged or using specific non-namespaced capabilities (e.g. MKNODE, SYS\_MODULE etc.). This should only be enabled if user namespace remapping is enabled in the Docker daemon.
- **ExternalPolicyForExternalIP**: Fix a bug where ExternalTrafficPolicy is not applied to Service ExternalIPs.
- **GCERegionalPersistentDisk**: Enable the regional PD feature on GCE.
- **GenericEphemeralVolume**: Enables ephemeral, inline volumes that support all features of normal volumes (can be provided by third-party storage vendors, storage capacity tracking, restore from snapshot, etc.). See [Ephemeral Volumes](#).
- **GRPCContainerProbe**: Enables the gRPC probe method for {Liveness,Readiness,Startup} Probe. See [Configure Liveness, Readiness and Startup Probes](#).
- **HugePages**: Enable the allocation and consumption of pre-allocated [huge pages](#).
- **HugePageStorageMediumSize**: Enable support for multiple sizes pre-allocated [huge pages](#).
- **HyperVContainer**: Enable [Hyper-V isolation](#) for Windows containers.
- **IdentifyPodOS**: Allows the Pod OS field to be specified. This helps in identifying the OS of the pod authoritatively during the API server admission time.
- **ImmutableEphemeralVolumes**: Allows for marking individual Secrets and ConfigMaps as immutable for better safety and performance.
- **IndexedJob**: Allows the [Job](#) controller to manage Pod completions per completion index.
- **IngressClassNamespacedParams**: Allow namespace-scoped parameters reference in IngressClass resource. This feature adds two fields - Scope and Namespace to IngressClass.spec.parameters.
- **Initializers**: Allow asynchronous coordination of object creation using the Initializers admission plugin.
- **IPTablesOwnershipCleanup**: This causes kubelet to no longer create legacy iptables rules.
- **IPv6DualStack**: Enable [dual stack](#) support for IPv6.
- **JobMutableNodeSchedulingDirectives**: Allows updating node scheduling directives in the pod template of [Job](#).
- **JobTrackingWithFinalizers**: Enables tracking [Job](#) completions without relying on Pods remaining in the cluster indefinitely. The Job controller uses Pod finalizers and a field in the Job status to keep track of the finished Pods to count towards completion.
- **KubeletConfigFile**: Enable loading kubelet configuration from a file specified using a config file. See [setting kubelet parameters via a config file](#) for more details.

- **KubeletCredentialProviders:** Enable kubelet exec credential providers for image pull credentials.
- **KubeletPluginsWatcher:** Enable probe-based plugin watcher utility to enable kubelet to discover plugins such as [CSI volume drivers](#).
- **KubeletPodResources:** Enable the kubelet's pod resources gRPC endpoint. See [Support Device Monitoring](#) for more details.
- **KubeletPodResourcesGetAllocatable:** Enable the kubelet's pod resources GetAllocatableResources functionality. This API augments the [resource allocation reporting](#)
- **LegacyNodeRoleBehavior:** When disabled, legacy behavior in service load balancers and node disruption will ignore the node-role.kubernetes.io/master label in favor of the feature-specific labels provided by NodeDisruptionExclusion and ServiceNodeExclusion.
- **LegacyServiceAccountTokenNoAutoGeneration:** Stop auto-generation of Secret-based [service account tokens](#).
- **LegacyServiceAccountTokenTracking:** Track usage of Secret-based [service account tokens](#).
- **LocalStorageCapacityIsolation:** Enable the consumption of [local ephemeral storage](#) and also the sizeLimit property of an [emptyDir volume](#).
- **MinimizeIPTablesRestore:** Enables new performance improvement logics in the kube-proxy iptables mode.
- **MixedProtocolLBService:** Enable using different protocols in the same LoadBalancer type Service instance.
- **MountContainers:** Enable using utility containers on host as the volume mounter.
- **MountPropagation:** Enable sharing volume mounted by one container to other containers or pods. For more details, please see [mount propagation](#).
- **MultiCIDRRangeAllocator:** Enables the MultiCIDR range allocator.
- **NamespaceDefaultLabelName:** Configure the API Server to set an immutable [label](#) kubernetes.io/metadata.name on all namespaces, containing the namespace name.
- **NetworkPolicyEndPort:** Allows you to define ports in a [NetworkPolicy](#) rule as a range of port numbers.
- **NetworkPolicyStatus:** Enable the status subresource for NetworkPolicy objects.
- **NodeDisruptionExclusion:** Enable use of the Node label node.kubernetes.io/exclude-disruption which prevents nodes from being evacuated during zone failures.
- **NodeLease:** Enable the new Lease API to report node heartbeats, which could be used as a node health signal.
- **NonPreemptingPriority:** Enable preemptionPolicy field for PriorityClass and Pod.

- OpenAPIV3: Enables the API server to publish OpenAPI v3.
- PersistentLocalVolumes: Enable the usage of local volume type in Pods. Pod affinity has to be specified if requesting a local volume.
- PodAffinityNamespaceSelector: Enable the [Pod Affinity Namespace Selector](#) and [CrossNamespacePodAffinity](#) quota scope features.
- PodDisruptionBudget: Enable the [PodDisruptionBudget](#) feature.
- PodHasNetworkCondition: Enable the kubelet to mark the [PodHasNetwork](#) condition on pods. This was renamed to PodReadyToStartContainersCondition in 1.28.
- PodOverhead: Enable the [PodOverhead](#) feature to account for pod overheads.
- PodPriority: Enable the descheduling and preemption of Pods based on their [priorities](#).
- PodReadinessGates: Enable the setting of PodReadinessGate field for extending Pod readiness evaluation. See [Pod readiness gate](#) for more details.
- PodSecurity: Enables the PodSecurity admission plugin.
- PodShareProcessNamespace: Enable the setting of shareProcessNamespace in a Pod for sharing a single process namespace between containers running in a pod. More details can be found in [Share Process Namespace between Containers in a Pod](#).
- PreferNominatedNode: This flag tells the scheduler whether the nominated nodes will be checked first before looping through all the other nodes in the cluster.
- ProbeTerminationGracePeriod: Enable [setting probe-level terminationGracePeriodSeconds](#) on pods. See the [enhancement proposal](#) for more details.
- ProxyTerminatingEndpoints: Enable the kube-proxy to handle terminating endpoints when ExternalTrafficPolicy=Local.
- PVCProtection: Enable the prevention of a PersistentVolumeClaim (PVC) from being deleted when it is still used by any Pod.
- ReadOnlyAPIDataVolumes:
 

Set [configMap](#), [secret](#), [downwardAPI](#) and [projected volumes](#) to be mounted read-only.

Since Kubernetes v1.10, these volume types are always read-only and you cannot opt out.
- RemoveSelfLink: Sets the .metadata.selfLink field to blank (empty string) for all objects and collections. This field has been deprecated since the Kubernetes v1.16 release. When this feature is enabled, the .metadata.selfLink field remains part of the Kubernetes API, but is always unset.
- RequestManagement: Enables managing request concurrency with prioritization and fairness at each API server. Deprecated by APIPriorityAndFairness since 1.17.
- ResourceLimitsPriorityFunction: Enable a scheduler priority function that assigns a lowest possible score of 1 to a node that satisfies at least one of the input Pod's cpu and memory limits. The intent is to break ties between nodes with same scores.

ResourceQuotaScopeSelectors: Enable resource quota scope selectors.

- 
- RetroactiveDefaultStorageClass: Allow assigning StorageClass to unbound PVCs retroactively.
- RootCAConfigMap: Configure the kube-controller-manager to publish a [ConfigMap](#) named kube-root-ca.crt to every namespace. This ConfigMap contains a CA bundle used for verifying connections to the kube-apiserver. See [Bound Service Account Tokens](#) for more details.
- RotateKubeletClientCertificate: Enable the rotation of the client TLS certificate on the kubelet. See [kubelet configuration](#) for more details.
- RunAsGroup: Enable control over the primary group ID set on the init processes of containers.
- RuntimeClass: Enable the [RuntimeClass](#) feature for selecting container runtime configurations.
- ScheduleDaemonSetPods: Enable DaemonSet Pods to be scheduled by the default scheduler instead of the DaemonSet controller.
- SCTPSupport: Enables the *SCTP* protocol value in Pod, Service, Endpoints, EndpointSlice, and NetworkPolicy definitions.
- SeccompDefault: Enables the use of RuntimeDefault as the default seccomp profile for all workloads. The seccomp profile is specified in the securityContext of a Pod and/or a Container.
- SecurityContextDeny: This gate signals that the SecurityContextDeny admission controller is deprecated.
- SelectorIndex: Allows label and field based indexes in API server watch cache to accelerate list operations.
- ServiceAccountIssuerDiscovery: Enable OIDC discovery endpoints (issuer and JWKS URLs) for the service account issuer in the API server. See [Configure Service Accounts for Pods](#) for more details.
- ServiceAppProtocol: Enables the appProtocol field on Services and Endpoints.
- ServiceInternalTrafficPolicy: Enables the internalTrafficPolicy field on Services
- ServiceIPStaticSubrange: Enables a strategy for Services ClusterIP allocations, whereby the ClusterIP range is subdivided. Dynamic allocated ClusterIP addresses will be allocated preferentially from the upper range allowing users to assign static ClusterIPs from the lower range with a low risk of collision. See [Avoiding collisions](#) for more details.
- ServiceLBNodePortControl: Enables the allocateLoadBalancerNodePorts field on Services.
- ServiceLoadBalancerClass: Enables the loadBalancerClass field on Services. See [Specifying class of load balancer implementation](#) for more details.
- ServiceLoadBalancerFinalizer: Enable finalizer protection for Service load balancers.

- **ServiceNodeExclusion:** Enable the exclusion of nodes from load balancers created by a cloud provider. A node is eligible for exclusion if labelled with "node.kubernetes.io/exclude-from-external-load-balancers".
- **ServiceTopology:** Enable service to route traffic based upon the Node topology of the cluster.
- **SetHostnameAsFQDN:** Enable the ability of setting Fully Qualified Domain Name(FQDN) as the hostname of a pod. See [Pod's setHostnameAsFQDN field](#).
- **StartupProbe:** Enable the [startup](#) probe in the kubelet.
- **StatefulSetMinReadySeconds:** Allows minReadySeconds to be respected by the StatefulSet controller.
- **StorageObjectInUseProtection:** Postpone the deletion of PersistentVolume or PersistentVolumeClaim objects if they are still being used.
- **StreamingProxyRedirects:** Instructs the API server to intercept (and follow) redirects from the backend (kubelet) for streaming requests. Examples of streaming requests include the exec, attach and port-forward requests.
- **SupportIPVSProxyMode:** Enable providing in-cluster service load balancing using IPVS. See [service proxies](#) for more details.
- **SupportNodePidsLimit:** Enable the support to limiting PIDs on the Node. The parameter pid=<number> in the --system-reserved and --kube-reserved options can be specified to ensure that the specified number of process IDs will be reserved for the system as a whole and for Kubernetes system daemons respectively.
- **SupportPodPidsLimit:** Enable the support to limiting PIDs in Pods.
- **SuspendJob:** Enable support to suspend and resume Jobs. For more details, see [the Jobs docs](#).
- **Sysctls:** Enable support for namespaced kernel parameters (sysctls) that can be set for each pod. See [sysctls](#) for more details.
- **TaintBasedEvictions:** Enable evicting pods from nodes based on taints on Nodes and tolerations on Pods. See [taints and tolerations](#) for more details.
- **TaintNodesByCondition:** Enable automatic tainting nodes based on [node conditions](#).
- **TokenRequest:** Enable the TokenRequest endpoint on service account resources.
- **TokenRequestProjection:** Enable the injection of service account tokens into a Pod through a [projected volume](#).
- **TopologyManager:** Enable a mechanism to coordinate fine-grained hardware resource assignments for different components in Kubernetes. See [Control Topology Management Policies on a node](#).
- **TTLAfterFinished:** Allow a [TTL controller](#) to clean up resources after they finish execution.



- `UserNamespacesStatelessPodsSupport`: Enable user namespace support for stateless Pods. This feature gate was superseded by the `UserNamespacesSupport` feature gate in the Kubernetes v1.28 release.
- `ValidateProxyRedirects`: This flag controls whether the API server should validate that redirects are only followed to the same host. Only used if the `StreamingProxyRedirects` flag is enabled.
- `VolumePVCDDataSource`: Enable support for specifying an existing PVC as a `DataSource`.
- `VolumeScheduling`: Enable volume topology aware scheduling and make the `PersistentVolumeClaim` (PVC) binding aware of scheduling decisions. It also enables the usage of [local](#) volume type when used together with the `PersistentLocalVolumes` feature gate.
- `VolumeSnapshotDataSource`: Enable volume snapshot data source support.
- `VolumeSubpath`: Allow mounting a subpath of a volume in a container.
- `VolumeSubpathEnvExpansion`: Enable `subPathExpr` field for expanding environment variables into a `subPath`.
- `WarningHeaders`: Allow sending warning headers in API responses.
- `WindowsEndpointSliceProxying`: When enabled, kube-proxy running on Windows will use `EndpointSlices` as the primary data source instead of `Endpoints`, enabling scalability and performance improvements. See [Enabling Endpoint Slices](#).
- `WindowsGMSA`: Enables passing of GMSA credential specs from pods to container runtimes.
- `WindowsHostProcessContainers`: Enables support for Windows `HostProcess` containers.
- `WindowsRunAsUserName`: Enable support for running applications in Windows containers with as a non-default user. See [Configuring RunAsUserName](#) for more details.

# kubelet

## Synopsis

The kubelet is the primary "node agent" that runs on each node. It can register the node with the apiserver using one of: the hostname; a flag to override the hostname; or specific logic for a cloud provider.

The kubelet works in terms of a `PodSpec`. A `PodSpec` is a YAML or JSON object that describes a pod. The kubelet takes a set of `PodSpecs` that are provided through various mechanisms (primarily through the apiserver) and ensures that the containers described in those `PodSpecs` are running and healthy. The kubelet doesn't manage containers which were not created by Kubernetes.

Other than from a PodSpec from the apiserver, there are two ways that a container manifest can be provided to the kubelet.

- File: Path passed as a flag on the command line. Files under this path will be monitored periodically for updates. The monitoring period is 20s by default and is configurable via a flag.
- HTTP endpoint: HTTP endpoint passed as a parameter on the command line. This endpoint is checked every 20 seconds (also configurable with a flag).

kubelet [flags]

## Options

|                                                                                                                                                                      |                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| --address string                                                                                                                                                     | Default: 0.0.0.0              |
| The IP address for the kubelet to serve on (set to 0.0.0.0 or :: for listening on all interfaces and IP address family)                                              |                               |
| --allowed-unsafe-sysctls strings                                                                                                                                     |                               |
| Comma-separated whitelist of unsafe sysctls or unsafe sysctl patterns (ending in *). Use these at your own risk                                                      |                               |
| --anonymous-auth                                                                                                                                                     | Default: true                 |
| Enables anonymous requests to the kubelet server. Requests that are not rejected by another authentication method are accepted                                       |                               |
| --authentication-token-webhook                                                                                                                                       |                               |
| Use the TokenReview API to determine authentication for bearer tokens. (DEPRECATED: This parameter should be removed in the future)                                  |                               |
| --authentication-token-webhook-cache-ttl duration                                                                                                                    | Default: 2m0s                 |
| The duration to cache responses from the webhook token authenticator. (DEPRECATED: This parameter should be removed in the future)                                   |                               |
| --authorization-mode string                                                                                                                                          | Default: AlwaysAllow          |
| Authorization mode for kubelet server. Valid options are "AlwaysAllow" or "Webhook". Webhook mode uses the TokenReview API                                           |                               |
| --authorization-webhook-cache-authorized-ttl duration                                                                                                                | Default: 5m0s                 |
| The duration to cache 'authorized' responses from the webhook authorizer. (DEPRECATED: This parameter should be removed in the future)                               |                               |
| --authorization-webhook-cache-unauthorized-ttl duration                                                                                                              | Default: 30s                  |
| The duration to cache 'unauthorized' responses from the webhook authorizer. (DEPRECATED: This parameter should be removed in the future)                             |                               |
| --bootstrap-kubeconfig string                                                                                                                                        |                               |
| Path to a kubeconfig file that will be used to get client certificate for kubelet. If the file specified by --kubeconfig is not found, kubelet will use this file    |                               |
| --cert-dir string                                                                                                                                                    | Default: /var/lib/kubelet/pki |
| The directory where the TLS certs are located. If --tls-cert-file and --tls-private-key-file are provided, this flag is ignored                                      |                               |
| --cgroup-driver string                                                                                                                                               | Default: cgroupfs             |
| Driver that the kubelet uses to manipulate cgroups on the host. Possible values: "cgroupfs", "systemd". (DEPRECATED: This parameter should be removed in the future) |                               |
| --cgroup-root string                                                                                                                                                 | Default: "                    |
| Optional root cgroup to use for pods. This is handled by the container runtime on a best effort basis. Default is empty string                                       |                               |
| --cgroups-per-qos                                                                                                                                                    | Default: true                 |
| Enable creation of QoS cgroup hierarchy, if true, top level QoS and pod cgroups are created. (DEPRECATED: This parameter should be removed in the future)            |                               |
| --client-ca-file string                                                                                                                                              |                               |
| If set, any request presenting a client certificate signed by one of the authorities in the client-ca-file is authorized to get a bearer token                       |                               |
| --cloud-config string                                                                                                                                                |                               |
| The path to the cloud provider configuration file. Empty string for no configuration file. (DEPRECATED: This parameter should be removed in the future)              |                               |
| --cloud-provider string                                                                                                                                              |                               |
| The provider for cloud services. Set to empty string for running with no cloud provider. Set to 'external' for non-cloud providers                                   |                               |
| --cluster-dns strings                                                                                                                                                |                               |



|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| Comma-separated list of DNS server IP address. This value is used for containers DNS server in case of Pods        |
| <b>Note:</b> all DNS servers appearing in the list MUST serve the same set of records otherwise name resolution w  |
| --cluster-domain string                                                                                            |
| Domain for this cluster. If set, kubelet will configure all containers to search this domain in addition to the h  |
| --config string                                                                                                    |
| The kubelet will load its initial configuration from this file. The path may be absolute or relative; relative pat |
| --config-dir string   Default: "                                                                                   |
| Path to a directory to specify drop-ins, allows the user to optionally specify additional configs to overwrite v   |
| <b>Note:</b> Set the 'KUBELET_CONFIG_DROPIN_DIR_ALPHA' environment variable to specify the directory.              |
| --container-log-max-files int32   Default: 5                                                                       |
| <Warning: Beta feature> Set the maximum number of container log files that can be present for a container.         |
| --container-log-max-size string   Default: 10Mi                                                                    |
| <Warning: Beta feature> Set the maximum size (e.g. 10Mi) of container log file before it is rotated. (DEPREC       |
| --container-runtime-endpoint string   Default: "unix:///run/containerd/containerd.sock"                            |
| The endpoint of remote runtime service. UNIX domain sockets are supported on Linux, while 'npipe' and 'tcp         |
| --contention-profiling                                                                                             |
| Enable block profiling, if profiling is enabled. (DEPRECATED: This parameter should be set via the config fil      |
| --cpu-cfs-quota   Default: true                                                                                    |
| Enable CPU CFS quota enforcement for containers that specify CPU limits. (DEPRECATED: This parameter               |
| --cpu-cfs-quota-period duration   Default: 100ms                                                                   |
| Sets CPU CFS quota period value, cpu.cfs_period_us, defaults to Linux Kernel default. (DEPRECATED: This            |
| --cpu-manager-policy string   Default: none                                                                        |
| The CPU manager policy to use. Possible values: "none", "static". (DEPRECATED: This parameter should be s          |
| --cpu-manager-policy-options string                                                                                |
| A set of 'key=value' CPU manager policy options to use, to fine tune their behaviour. If not supplied, keep th     |
| --cpu-manager-reconcile-period duration   Default: 10s                                                             |
| <Warning: Alpha feature> CPU manager reconciliation period. Examples: "10s", or "1m". If not supplied, defa        |
| --enable-controller-attach-detach   Default: true                                                                  |
| Enables the Attach/Detach controller to manage attachment/detachment of volumes scheduled to this node,            |
| --enable-debugging-handlers   Default: true                                                                        |
| Enables server endpoints for log collection and local running of containers and commands. (DEPRECATED:             |
| --enable-server   Default: true                                                                                    |
| Enable the kubelet's server. (DEPRECATED: This parameter should be set via the config file specified by the        |
| --enforce-node-allocatable strings   Default: pods                                                                 |
| A comma separated list of levels of node allocatable enforcement to be enforced by kubelet. Acceptable optio       |
| --event-burst int32   Default: 100                                                                                 |
| Maximum size of a bursty event records, temporarily allows event records to burst to this number, while stil       |
| --event-qps int32   Default: 50                                                                                    |
| QPS to limit event creations. The number must be >= 0. If 0 will use default QPS (50). (DEPRECATED: This p         |
| --eviction-hard strings   Default: imagefs.available<15%,memory.available<100Mi,nodefs.available<10%               |
| A set of eviction thresholds (e.g. "memory.available<1Gi") that if met would trigger a pod eviction. On a Linu     |
| --eviction-max-pod-grace-period int32                                                                              |
| Maximum allowed grace period (in seconds) to use when terminating pods in response to a soft eviction thro         |
| --eviction-minimum-reclaim strings                                                                                 |
| A set of minimum reclaims (e.g. "imagefs.available=2Gi") that describes the minimum amount of resource th          |
| --eviction-pressure-transition-period duration   Default: 5m0s                                                     |

|                                                                                                                                                                     |
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| Duration for which the kubelet has to wait before transitioning out of an eviction pressure condition. (DEPRECATED)                                                 |
| --eviction-soft strings                                                                                                                                             |
| A set of eviction thresholds (e.g. "memory.available<1.5Gi") that if met over a corresponding grace period will trigger an eviction                                 |
| --eviction-soft-grace-period strings                                                                                                                                |
| A set of eviction grace periods (e.g. "memory.available=1m30s") that correspond to how long a soft eviction threshold can be violated before triggering an eviction |
| --exit-on-lock-contention                                                                                                                                           |
| Whether kubelet should exit upon lock-file contention.                                                                                                              |
| --experimental-allocatable-ignore-eviction Default: false                                                                                                           |
| When set to true, hard eviction thresholds will be ignored while calculating node allocatable. See <a href="#">here</a> for more details                            |
| --experimental-mounter-path string Default: mount                                                                                                                   |
| [Experimental] Path of mounter binary. Leave empty to use the default mount. (DEPRECATED: will be removed in 1.15)                                                  |
| --fail-swap-on Default: true                                                                                                                                        |
| Makes the kubelet fail to start if swap is enabled on the node. (DEPRECATED: This parameter should be set to false)                                                 |
| --feature-gates <A list of 'key=true/false' pairs>                                                                                                                  |
| A set of key=value pairs that describe feature gates for alpha/experimental features. Options are:                                                                  |
| APIResponseCompression=true false (BETA - default=true)                                                                                                             |
| APIServerIdentity=true false (BETA - default=true)                                                                                                                  |
| APIServerTracing=true false (BETA - default=true)                                                                                                                   |
| APIServingWithRoutine=true false (BETA - default=true)                                                                                                              |
| AllAlpha=true false (ALPHA - default=false)                                                                                                                         |
| AllBeta=true false (BETA - default=false)                                                                                                                           |
| AnyVolumeDataSource=true false (BETA - default=true)                                                                                                                |
| AppArmor=true false (BETA - default=true)                                                                                                                           |
| AppArmorFields=true false (BETA - default=true)                                                                                                                     |
| CPUManagerPolicyAlphaOptions=true false (ALPHA - default=false)                                                                                                     |
| CPUManagerPolicyBetaOptions=true false (BETA - default=true)                                                                                                        |
| CPUManagerPolicyOptions=true false (BETA - default=true)                                                                                                            |
| CRDValidationRatcheting=true false (BETA - default=true)                                                                                                            |
| CSIMigrationPortworx=true false (BETA - default=false)                                                                                                              |
| CSIVolumeHealth=true false (ALPHA - default=false)                                                                                                                  |
| CloudControllerManagerWebhook=true false (ALPHA - default=false)                                                                                                    |
| ClusterTrustBundle=true false (ALPHA - default=false)                                                                                                               |
| ClusterTrustBundleProjection=true false (ALPHA - default=false)                                                                                                     |
| ComponentSLIs=true false (BETA - default=true)                                                                                                                      |
| ConsistentListFromCache=true false (ALPHA - default=false)                                                                                                          |
| ContainerCheckpoint=true false (BETA - default=true)                                                                                                                |
| ContextualLogging=true false (BETA - default=true)                                                                                                                  |
| CronJobsScheduledAnnotation=true false (BETA - default=true)                                                                                                        |
| CrossNamespaceVolumeDataSource=true false (ALPHA - default=false)                                                                                                   |
| CustomCPUCFSQuotaPeriod=true false (ALPHA - default=false)                                                                                                          |
| CustomResourceFieldSelectors=true false (ALPHA - default=false)                                                                                                     |
| DevicePluginCDIDevices=true false (BETA - default=true)                                                                                                             |
| DisableCloudProviders=true false (BETA - default=true)                                                                                                              |
| DisableKubeletCloudCredentialProviders=true false (BETA - default=true)                                                                                             |
| DisableNodeKubeProxyVersion=true false (ALPHA - default=false)                                                                                                      |
| DynamicResourceAllocation=true false (ALPHA - default=false)                                                                                                        |
| ElasticIndexedJob=true false (BETA - default=true)                                                                                                                  |
| EventedPLEG=true false (ALPHA - default=false)                                                                                                                      |
| GracefulNodeShutdown=true false (BETA - default=true)                                                                                                               |
| GracefulNodeShutdownBasedOnPodPriority=true false (BETA - default=true)                                                                                             |

HPAScaleToZero=true|false (ALPHA - default=false)  
HonorPVReclaimPolicy=true|false (ALPHA - default=false)  
ImageMaximumGCAge=true|false (BETA - default=true)  
InPlacePodVerticalScaling=true|false (ALPHA - default=false)  
InTreePluginAWSUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureDiskUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureFileUnregister=true|false (ALPHA - default=false)  
InTreePluginGCEUnregister=true|false (ALPHA - default=false)  
InTreePluginOpenStackUnregister=true|false (ALPHA - default=false)  
InTreePluginPortworxUnregister=true|false (ALPHA - default=false)  
InTreePluginvSphereUnregister=true|false (ALPHA - default=false)  
InformerResourceVersion=true|false (ALPHA - default=false)  
JobBackoffLimitPerIndex=true|false (BETA - default=true)  
JobManagedBy=true|false (ALPHA - default=false)  
JobPodFailurePolicy=true|false (BETA - default=true)  
JobPodReplacementPolicy=true|false (BETA - default=true)  
JobSuccessPolicy=true|false (ALPHA - default=false)  
KubeProxyDrainingTerminatingNodes=true|false (BETA - default=true)  
KubeletCgroupDriverFromCRI=true|false (ALPHA - default=false)  
KubeletInUserNamespace=true|false (ALPHA - default=false)  
KubeletPodResourcesDynamicResources=true|false (ALPHA - default=false)  
KubeletPodResourcesGet=true|false (ALPHA - default=false)  
KubeletSeparateDiskGC=true|false (ALPHA - default=false)  
KubeletTracing=true|false (BETA - default=true)  
LoadBalancerIPMode=true|false (BETA - default=true)  
LocalStorageCapacityIsolationFSQuotaMonitoring=true|false (ALPHA - default=false)  
LogarithmicScaleDown=true|false (BETA - default=true)  
LoggingAlphaOptions=true|false (ALPHA - default=false)  
LoggingBetaOptions=true|false (BETA - default=true)  
MatchLabelKeysInPodAffinity=true|false (ALPHA - default=false)  
MatchLabelKeysInPodTopologySpread=true|false (BETA - default=true)  
MaxUnavailableStatefulSet=true|false (ALPHA - default=false)  
MemoryManager=true|false (BETA - default=true)  
MemoryQoS=true|false (ALPHA - default=false)  
MultiCIDRServiceAllocator=true|false (ALPHA - default=false)  
MutatingAdmissionPolicy=true|false (ALPHA - default=false)  
NFTablesProxyMode=true|false (ALPHA - default=false)  
NodeInclusionPolicyInPodTopologySpread=true|false (BETA - default=true)  
NodeLogQuery=true|false (BETA - default=false)  
NodeSwap=true|false (BETA - default=true)  
OpenAPIEnums=true|false (BETA - default=true)  
PDBUnhealthyPodEvictionPolicy=true|false (BETA - default=true)  
PersistentVolumeLastPhaseTransitionTime=true|false (BETA - default=true)  
PodAndContainerStatsFromCRI=true|false (ALPHA - default=false)  
PodDeletionCost=true|false (BETA - default=true)  
PodDisruptionConditions=true|false (BETA - default=true)  
PodIndexLabel=true|false (BETA - default=true)  
PodLifecycleSleepAction=true|false (BETA - default=true)  
PodReadyToStartContainersCondition=true|false (BETA - default=true)  
PortForwardWebsockets=true|false (ALPHA - default=false)  
ProcMountType=true|false (ALPHA - default=false)  
QOSReserved=true|false (ALPHA - default=false)  
RecoverVolumeExpansionFailure=true|false (ALPHA - default=false)

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| RecursiveReadOnlyMounts=true false (ALPHA - default=false)                                                                                                                                                                                                                       |
| RelaxedEnvironmentVariableValidation=true false (ALPHA - default=false)                                                                                                                                                                                                          |
| RetryGenerateName=true false (ALPHA - default=false)                                                                                                                                                                                                                             |
| RotateKubeletServerCertificate=true false (BETA - default=true)                                                                                                                                                                                                                  |
| RuntimeClassInImageCriApi=true false (ALPHA - default=false)                                                                                                                                                                                                                     |
| SELinuxMount=true false (ALPHA - default=false)                                                                                                                                                                                                                                  |
| SELinuxMountReadWriteOncePod=true false (BETA - default=true)                                                                                                                                                                                                                    |
| SchedulerQueueingHints=true false (BETA - default=false)                                                                                                                                                                                                                         |
| SeparateCacheWatchRPC=true false (BETA - default=true)                                                                                                                                                                                                                           |
| SeparateTaintEvictionController=true false (BETA - default=true)                                                                                                                                                                                                                 |
| ServiceAccountTokenJTI=true false (BETA - default=true)                                                                                                                                                                                                                          |
| ServiceAccountTokenNodeBinding=true false (ALPHA - default=false)                                                                                                                                                                                                                |
| ServiceAccountTokenNodeBindingValidation=true false (BETA - default=true)                                                                                                                                                                                                        |
| ServiceAccountTokenPodNodeInfo=true false (BETA - default=true)                                                                                                                                                                                                                  |
| ServiceTrafficDistribution=true false (ALPHA - default=false)                                                                                                                                                                                                                    |
| SidecarContainers=true false (BETA - default=true)                                                                                                                                                                                                                               |
| SizeMemoryBackedVolumes=true false (BETA - default=true)                                                                                                                                                                                                                         |
| StatefulSetAutoDeletePVC=true false (BETA - default=true)                                                                                                                                                                                                                        |
| StatefulSetStartOrdinal=true false (BETA - default=true)                                                                                                                                                                                                                         |
| StorageNamespaceIndex=true false (BETA - default=true)                                                                                                                                                                                                                           |
| StorageVersionAPI=true false (ALPHA - default=false)                                                                                                                                                                                                                             |
| StorageVersionHash=true false (BETA - default=true)                                                                                                                                                                                                                              |
| StorageVersionMigrator=true false (ALPHA - default=false)                                                                                                                                                                                                                        |
| StructuredAuthenticationConfiguration=true false (BETA - default=true)                                                                                                                                                                                                           |
| StructuredAuthorizationConfiguration=true false (BETA - default=true)                                                                                                                                                                                                            |
| TopologyAwareHints=true false (BETA - default=true)                                                                                                                                                                                                                              |
| TopologyManagerPolicyAlphaOptions=true false (ALPHA - default=false)                                                                                                                                                                                                             |
| TopologyManagerPolicyBetaOptions=true false (BETA - default=true)                                                                                                                                                                                                                |
| TopologyManagerPolicyOptions=true false (BETA - default=true)                                                                                                                                                                                                                    |
| TranslateStreamCloseWebsocketRequests=true false (BETA - default=true)                                                                                                                                                                                                           |
| UnauthenticatedHTTP2DOSMitigation=true false (BETA - default=true)                                                                                                                                                                                                               |
| UnknownVersionInteroperabilityProxy=true false (ALPHA - default=false)                                                                                                                                                                                                           |
| UserNamespacesPodSecurityStandards=true false (ALPHA - default=false)                                                                                                                                                                                                            |
| UserNamespacesSupport=true false (BETA - default=false)                                                                                                                                                                                                                          |
| VolumeAttributesClass=true false (ALPHA - default=false)                                                                                                                                                                                                                         |
| VolumeCapacityPriority=true false (ALPHA - default=false)                                                                                                                                                                                                                        |
| WatchFromStorageWithoutResourceVersion=true false (BETA - default=false)                                                                                                                                                                                                         |
| WatchList=true false (ALPHA - default=false)                                                                                                                                                                                                                                     |
| WatchListClient=true false (BETA - default=false)                                                                                                                                                                                                                                |
| WinDSR=true false (ALPHA - default=false)                                                                                                                                                                                                                                        |
| WinOverlay=true false (BETA - default=true)                                                                                                                                                                                                                                      |
| WindowsHostNetwork=true false (ALPHA - default=true)                                                                                                                                                                                                                             |
| (DEPRECATED: This parameter should be set via the config file specified by the kubelet's --config flag. See <a href="#">kubernetes.io/docs/reference/command-line-tools-reference/kubelet-config-file-v1</a> )                                                                   |
| --file-check-frequency duration    Default: 20s                                                                                                                                                                                                                                  |
| Duration between checking config files for new data. (DEPRECATED: This parameter should be set via the config file specified by the kubelet's --config flag. See <a href="#">kubernetes.io/docs/reference/command-line-tools-reference/kubelet-config-file-v1</a> )              |
| --hairpin-mode string    Default: promiscuous-bridge                                                                                                                                                                                                                             |
| How should the kubelet setup hairpin NAT. This allows endpoints of a Service to load balance back to themselves.                                                                                                                                                                 |
| --healthz-bind-address string    Default: 127.0.0.1                                                                                                                                                                                                                              |
| The IP address for the healthz server to serve on (set to "0.0.0.0" or "::" for listening in all interfaces and IP families)                                                                                                                                                     |
| --healthz-port int32    Default: 10248                                                                                                                                                                                                                                           |
| The port of the localhost healthz endpoint (set to 0 to disable). (DEPRECATED: This parameter should be set via the config file specified by the kubelet's --config flag. See <a href="#">kubernetes.io/docs/reference/command-line-tools-reference/kubelet-config-file-v1</a> ) |

|                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------|
| -h, --help                                                                                                                         |
| help for kubelet                                                                                                                   |
| --hostname-override string                                                                                                         |
| If non-empty, will use this string as identification instead of the actual hostname. If --cloud-provider is set, the               |
| --http-check-frequency duration    Default: 20s                                                                                    |
| Duration between checking HTTP for new data. (DEPRECATED: This parameter should be set via the config file)                        |
| --image-credential-provider-bin-dir string                                                                                         |
| The path to the directory where credential provider plugin binaries are located.                                                   |
| --image-credential-provider-config string                                                                                          |
| The path to the credential provider plugin config file.                                                                            |
| --image-gc-high-threshold int32    Default: 85                                                                                     |
| The percent of disk usage after which image garbage collection is always run. Values must be within the range [1, 100]             |
| --image-gc-low-threshold int32    Default: 80                                                                                      |
| The percent of disk usage before which image garbage collection is never run. Lowest disk usage to garbage collect                 |
| --image-service-endpoint string                                                                                                    |
| The endpoint of remote image service. If not specified, it will be the same with --container-runtime-endpoint                      |
| --keep-terminated-pod-volumes                                                                                                      |
| Keep terminated pod volumes mounted to the node after the pod terminates. Can be useful for debugging volumes                      |
| --kernel-memcg-notification                                                                                                        |
| If enabled, the kubelet will integrate with the kernel memcg notification to determine if memory eviction threshold                |
| --kube-api-burst int32    Default: 100                                                                                             |
| Burst to use while talking with kubernetes API server. The number must be >= 0. If 0 will use default burst (100)                  |
| --kube-api-content-type string    Default: application/vnd.kubernetes.protobuf                                                     |
| Content type of requests sent to apiserver. (DEPRECATED: This parameter should be set via the config file)                         |
| --kube-api-qps int32    Default: 50                                                                                                |
| QPS to use while talking with kubernetes API server. The number must be >= 0. If 0 will use default QPS (50)                       |
| --kube-reserved strings    Default: <None>                                                                                         |
| A set of <resource name>=<resource quantity> (e.g. "cpu=200m,memory=500Mi,ephemeral-storage=1Gi,pid=1000")                         |
| --kube-reserved-cgroup string    Default: "                                                                                        |
| Absolute name of the top level cgroup that is used to manage kubernetes components for which compute resource                      |
| --kubeconfig string                                                                                                                |
| Path to a kubeconfig file, specifying how to connect to the API server. Providing --kubeconfig enables API server                  |
| --kubelet-cgroups string                                                                                                           |
| Optional absolute name of cgroups to create and run the kubelet in. (DEPRECATED: This parameter should be set via the config file) |
| --local-storage-capacity-isolation>    Default: true                                                                               |
| If true, local ephemeral storage isolation is enabled. Otherwise, local storage isolation feature will be disabled                 |
| --lock-file string                                                                                                                 |
| <Warning: Alpha feature> The path to file for kubelet to use as a lock file.                                                       |
| --log-flush-frequency duration    Default: 5s                                                                                      |
| Maximum number of seconds between log flushes.                                                                                     |
| --log-json-info-buffer-size string    Default: '0'                                                                                 |
| [Alpha] In JSON format with split output streams, the info messages can be buffered for a while to increase performance            |
| --log-json-split-stream                                                                                                            |
| [Alpha] In JSON format, write error messages to stderr and info messages to stdout. The default is to write all                    |
| --log-text-info-buffer-size string    Default: '0'                                                                                 |

|                                                                                                                                                                                 |
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| [Alpha] In text format with split output streams, the info messages can be buffered for a while to increase performance.                                                        |
| --log-text-split-stream                                                                                                                                                         |
| [Alpha] In text format, write error messages to stderr and info messages to stdout. The default is to write a single log message to stderr.                                     |
| --logging-format string    Default: text                                                                                                                                        |
| Sets the log format. Permitted formats: "json" (gated by LoggingBetaOptions, "text"). (DEPRECATED: This parameter should be set via the config file)                            |
| --make-iptables-util-chains    Default: true                                                                                                                                    |
| If true, kubelet will ensure iptables utility rules are present on host. (DEPRECATED: This parameter should be set via the config file)                                         |
| --manifest-url string                                                                                                                                                           |
| URL for accessing additional Pod specifications to run. (DEPRECATED: This parameter should be set via the config file)                                                          |
| --manifest-url-header strings                                                                                                                                                   |
| Comma-separated list of HTTP headers to use when accessing the URL provided to --manifest-url. Multiple headers can be specified.                                               |
| --max-open-files int    Default: 1000000                                                                                                                                        |
| Number of files that can be opened by kubelet process. (DEPRECATED: This parameter should be set via the config file)                                                           |
| --max-pods int32    Default: 110                                                                                                                                                |
| Number of Pods that can run on this kubelet. (DEPRECATED: This parameter should be set via the config file)                                                                     |
| --maximum-dead-containers int32    Default: -1                                                                                                                                  |
| Maximum number of old instances of containers to retain globally. Each container takes up some disk space.                                                                      |
| --maximum-dead-containers-per-container int32    Default: 1                                                                                                                     |
| Maximum number of old instances to retain per container. Each container takes up some disk space. (DEPRECATED: This parameter should be set via the config file)                |
| --memory-manager-policy string    Default: None                                                                                                                                 |
| Memory Manager policy to use. Possible values: "None", "Static". (DEPRECATED: This parameter should be set via the config file)                                                 |
| --minimum-container-ttl-duration duration                                                                                                                                       |
| Minimum age for a finished container before it is garbage collected. Examples: "300ms", "10s" or "2h45m". (DEPRECATED: This parameter should be set via the config file)        |
| --minimum-image-ttl-duration duration    Default: 2m0s                                                                                                                          |
| Minimum age for an unused image before it is garbage collected. Examples: "300ms", "10s" or "2h45m". (DEPRECATED: This parameter should be set via the config file)             |
| --node-ip string                                                                                                                                                                |
| IP address (or comma-separated dual-stack IP addresses) of the node. If unset, kubelet will use the node's default IP address.                                                  |
| --node-labels <key=value pairs>                                                                                                                                                 |
| <Warning: Alpha feature>Labels to add when registering the node in the cluster. Labels must be key=value pairs.                                                                 |
| --node-status-max-images int32    Default: 50                                                                                                                                   |
| The maximum number of images to report in node.status.images. If -1 is specified, no cap will be applied. (DEPRECATED: This parameter should be set via the config file)        |
| --node-status-update-frequency duration    Default: 10s                                                                                                                         |
| Specifies how often kubelet posts node status to master. <b>Note:</b> be cautious when changing the constant, it may affect the cluster's health.                               |
| --oom-score-adj int32    Default: -999                                                                                                                                          |
| The oom-score-adj value for kubelet process. Values must be within the range [-1000, 1000]. (DEPRECATED: This parameter should be set via the config file)                      |
| --pod-cidr string                                                                                                                                                               |
| The CIDR to use for pod IP addresses, only used in standalone mode. In cluster mode, this is obtained from the kube-apiserver.                                                  |
| --pod-infra-container-image string    Default: registry.k8s.io/pause:3.9                                                                                                        |
| Specified image will not be pruned by the image garbage collector. CRI implementations have their own container image garbage collection.                                       |
| --pod-manifest-path string                                                                                                                                                      |
| Path to the directory containing static pod files to run, or the path to a single static pod file. Files starting with "kubeletconfig-" are reserved for kubelet configuration. |
| --pod-max-pids int    Default: -1                                                                                                                                               |
| Set the maximum number of processes per pod. If -1, the kubelet defaults to the node allocatable PID capacity.                                                                  |
| --pods-per-core int32                                                                                                                                                           |
| Number of Pods per core that can run on this kubelet. The total number of pods on this kubelet cannot exceed the number of cores on the node.                                   |

|                                                                                                                                                                                                 |                           |
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| --port int32                                                                                                                                                                                    | Default: 10250            |
| The port for the kubelet to serve on. (DEPRECATED: This parameter should be set via the config file specified by --config)                                                                      |                           |
| --protect-kernel-defaults                                                                                                                                                                       |                           |
| Default kubelet behaviour for kernel tuning. If set, kubelet errors if any of kernel tunables is different than kernel default                                                                  |                           |
| --provider-id string                                                                                                                                                                            |                           |
| Unique identifier for identifying the node in a machine database, i.e cloud provider.                                                                                                           |                           |
| --qos-reserved string                                                                                                                                                                           |                           |
| <Warning: Alpha feature> A set of <resource name>=<percentage> (e.g. "memory=50%") pairs that describe reserved resources for QoS                                                               |                           |
| --read-only-port int32                                                                                                                                                                          | Default: 10255            |
| The read-only port for the kubelet to serve on with no authentication/authorization (set to 0 to disable). (DEPRECATED: This parameter should be set via the config file specified by --config) |                           |
| --register-node                                                                                                                                                                                 | Default: true             |
| Register the node with the API server. If --kubeconfig is not provided, this flag is irrelevant, as the kubelet will attempt to register with the default endpoint                              |                           |
| --register-schedulable                                                                                                                                                                          | Default: true             |
| Register the node as schedulable. Won't have any effect if --register-node is false. (DEPRECATED: will be removed in 1.10)                                                                      |                           |
| --register-with-taints string                                                                                                                                                                   |                           |
| Register the node with the given list of taints (comma separated <key>=<value>:<effect>). No-op if --register-node is false                                                                     |                           |
| --registry-burst int32                                                                                                                                                                          | Default: 10               |
| Maximum size of a bursty pulls, temporarily allows pulls to burst to this number, while still not exceeding --registry-qps                                                                      |                           |
| --registry-qps int32                                                                                                                                                                            | Default: 5                |
| If > 0, limit registry pull QPS to this value. If 0, unlimited. (DEPRECATED: This parameter should be set via the config file specified by --config)                                            |                           |
| --reserved-cpus string                                                                                                                                                                          |                           |
| A comma-separated list of CPUs or CPU ranges that are reserved for system and kubernetes usage. This specification is in the same format as the cgroup v1 cpu.shares value                      |                           |
| --reserved-memory string                                                                                                                                                                        |                           |
| A comma-separated list of memory reservations for NUMA nodes. (e.g. "--reserved-memory 0:memory=1Gi,1:memory=2Gi")                                                                              |                           |
| --resolv-conf string                                                                                                                                                                            | Default: /etc/resolv.conf |
| Resolver configuration file used as the basis for the container DNS resolution configuration. (DEPRECATED: This parameter should be set via the config file specified by --config)              |                           |
| --root-dir string                                                                                                                                                                               | Default: /var/lib/kubelet |
| Directory path for managing kubelet files (volume mounts, etc).                                                                                                                                 |                           |
| --rotate-certificates                                                                                                                                                                           |                           |
| Auto rotate the kubelet client certificates by requesting new certificates from the kube-apiserver when the certificates are close to expiration                                                |                           |
| --rotate-server-certificates                                                                                                                                                                    |                           |
| Auto-request and rotate the kubelet serving certificates by requesting new certificates from the kube-apiserver when the certificates are close to expiration                                   |                           |
| --runonce                                                                                                                                                                                       |                           |
| If true, exit after spawning pods from local manifests or remote urls. Exclusive with --enable-server (DEPRECATED: This parameter should be set via the config file specified by --config)      |                           |
| --runtime-cgroups string                                                                                                                                                                        |                           |
| Optional absolute name of cgroups to create and run the runtime in.                                                                                                                             |                           |
| --runtime-request-timeout duration                                                                                                                                                              | Default: 2m0s             |
| Timeout of all runtime requests except long running request - pull, logs, exec and attach. When timeout expires, the request is cancelled                                                       |                           |
| --seccomp-default                                                                                                                                                                               |                           |
| Enable the use of RuntimeDefault as the default seccomp profile for all workloads.                                                                                                              |                           |
| --serialize-image-pulls                                                                                                                                                                         | Default: true             |
| Pull images one at a time. We recommend *not* changing the default value on nodes that run docker daemon                                                                                        |                           |
| --streaming-connection-idle-timeout duration                                                                                                                                                    | Default: 4h0m0s           |
| Maximum time a streaming connection can be idle before the connection is automatically closed. 0 indicates no timeout                                                                           |                           |
| --sync-frequency duration                                                                                                                                                                       | Default: 1m0s             |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| Max period between synchronizing running containers and config. (DEPRECATED: This parameter should be set via the config file specified by the kubelet's --config flag. See <a href="#">kubelet documentation</a> .)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| --system-cgroups string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Optional absolute name of cgroups in which to place all non-kernel processes that are not already inside a cgroup.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --system-reserved string   Default: <none>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| A set of <resource name>=<resource quantity> (e.g. "cpu=200m,memory=500Mi,ephemeral-storage=1Gi,pid=1000000")                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --system-reserved-cgroup string   Default: "                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Absolute name of the top level cgroup that is used to manage non-kubernetes components for which compute resources are reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --tls-cert-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| File containing x509 certificate used for serving HTTPS (with intermediate certs, if any, concatenated after the main cert).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --tls-cipher-suites string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Comma-separated list of cipher suites for the server. If omitted, the default Go cipher suites will be used. Preferred values: TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384, TLS_CHACHA20_POLY1305_SHA256, TLS_RSA_WITH_AES_256_GCM_SHA384 Insecure values: TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256, TLS_ECDHE_ECDSA_WITH_RC4_128_SHA, TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256, TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384, TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256, TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384, TLS_RSA_WITH_3DES_EDE_CBC_SHA, TLS_RSA_WITH_AES_128_CBC_SHA256, TLS_RSA_WITH_AES_256_CBC_SHA384, TLS_RSA_WITH_AES_128_GCM_SHA256, TLS_RSA_WITH_AES_256_GCM_SHA384 (DEPRECATED: This parameter should be set via the config file specified by the kubelet's --config flag. See <a href="#">kubelet documentation</a> .) |
| --tls-min-version string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Minimum TLS version supported. Possible values: "VersionTLS10", "VersionTLS11", "VersionTLS12", "VersionTLS13"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --tls-private-key-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| File containing x509 private key matching --tls-cert-file. (DEPRECATED: This parameter should be set via the config file specified by the kubelet's --config flag. See <a href="#">kubelet documentation</a> .)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| --topology-manager-policy string   Default: 'none'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Topology Manager policy to use. Possible values: "none", "best-effort", "restricted", "single-numa-node". (DEPRECATED: This parameter should be set via the config file specified by the kubelet's --config flag. See <a href="#">kubelet documentation</a> .)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --topology-manager-policy-options string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| A set of <key>=<value> topology manager policy options to use, to fine tune their behaviour. If not supplied, the default values will be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --topology-manager-scope string   Default: container                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Scope to which topology hints are applied. Topology manager collects hints from hint providers and applies them to pods in the specified scope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| -v, --v Level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Number for the log level verbosity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --version version[=true]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Print version information and quit; --version=vX.Y.Z... sets the reported version.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --vmodule <A list of 'pattern=N' strings>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Comma-separated list of pattern=N settings for file-filtered logging (only works for text log format).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --volume-plugin-dir string   Default: /usr/libexec/kubernetes/kubelet-plugins/volume/exec/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| The full path of the directory in which to search for additional third party volume plugins. (DEPRECATED: This parameter should be set via the config file specified by the kubelet's --config flag. See <a href="#">kubelet documentation</a> .)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| --volume-stats-aggr-period duration   Default: 1m0s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Specifies interval for kubelet to calculate and cache the volume disk usage for all pods and volumes. To disable, set to 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

# kube-apiserver

## Synopsis

The Kubernetes API server validates and configures data for the api objects which include pods, services, replicationcontrollers, and others. The API Server services REST operations and provides the frontend to the cluster's shared state through which all other components interact.

kube-apiserver [flags]



# Options

|                                                                                                                                                                                                                                                                                                                            |  |
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| <code>--admission-control-config-file</code> string                                                                                                                                                                                                                                                                        |  |
| File with admission control configuration.                                                                                                                                                                                                                                                                                 |  |
| <code>--advertise-address</code> string                                                                                                                                                                                                                                                                                    |  |
| The IP address on which to advertise the apiserver to members of the cluster. This address must be reachable by the rest of the cluster. If blank, the <code>--bind-address</code> will be used. If <code>--bind-address</code> is unspecified, the host's default interface will be used.                                 |  |
| <code>--aggregator-reject-forwarding-redirect</code> Default: true                                                                                                                                                                                                                                                         |  |
| Aggregator reject forwarding redirect response back to client.                                                                                                                                                                                                                                                             |  |
| <code>--allow-metric-labels</code> stringToString    Default: []                                                                                                                                                                                                                                                           |  |
| The map from metric-label to value allow-list of this label. The key's format is <code>&lt;MetricName&gt;,&lt;LabelName&gt;</code> . The value's format is <code>&lt;allowed_value&gt;,&lt;allowed_value&gt;...e.g. metric1,label1='v1,v2,v3', metric1,label2='v1,v2,v3' metric2,label1='v1,v2,v3'</code> .                |  |
| <code>--allow-metric-labels-manifest</code> string                                                                                                                                                                                                                                                                         |  |
| The path to the manifest file that contains the allow-list mapping. The format of the file is the same as the flag <code>--allow-metric-labels</code> . Note that the flag <code>--allow-metric-labels</code> will override the manifest file.                                                                             |  |
| <code>--allow-privileged</code>                                                                                                                                                                                                                                                                                            |  |
| If true, allow privileged containers. [default=false]                                                                                                                                                                                                                                                                      |  |
| <code>--anonymous-auth</code> Default: true                                                                                                                                                                                                                                                                                |  |
| Enables anonymous requests to the secure port of the API server. Requests that are not rejected by another authentication method are treated as anonymous requests. Anonymous requests have a username of <code>system:anonymous</code> , and a group name of <code>system:unauthenticated</code> .                        |  |
| <code>--api-audiences</code> strings                                                                                                                                                                                                                                                                                       |  |
| Identifiers of the API. The service account token authenticator will validate that tokens used against the API are bound to at least one of these audiences. If the <code>--service-account-issuer</code> flag is configured and this flag is not, this field defaults to a single element list containing the issuer URL. |  |
| <code>--audit-log-batch-buffer-size</code> int    Default: 10000                                                                                                                                                                                                                                                           |  |
| The size of the buffer to store events before batching and writing. Only used in batch mode.                                                                                                                                                                                                                               |  |
| <code>--audit-log-batch-max-size</code> int    Default: 1                                                                                                                                                                                                                                                                  |  |

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|                                                          | The maximum size of a batch. Only used in batch mode.                                                                                                                                                                   |
| <code>--audit-log-batch-max-wait duration</code>         |                                                                                                                                                                                                                         |
|                                                          | The amount of time to wait before force writing the batch that hadn't reached the max size. Only used in batch mode.                                                                                                    |
| <code>--audit-log-batch-throttle-burst int</code>        |                                                                                                                                                                                                                         |
|                                                          | Maximum number of requests sent at the same moment if ThrottleQPS was not utilized before. Only used in batch mode.                                                                                                     |
| <code>--audit-log-batch-throttle-enable</code>           |                                                                                                                                                                                                                         |
|                                                          | Whether batching throttling is enabled. Only used in batch mode.                                                                                                                                                        |
| <code>--audit-log-batch-throttle-qps float</code>        |                                                                                                                                                                                                                         |
|                                                          | Maximum average number of batches per second. Only used in batch mode.                                                                                                                                                  |
| <code>--audit-log-compress</code>                        |                                                                                                                                                                                                                         |
|                                                          | If set, the rotated log files will be compressed using gzip.                                                                                                                                                            |
| <code>--audit-log-format string</code> Default: "json"   |                                                                                                                                                                                                                         |
|                                                          | Format of saved audits. "legacy" indicates 1-line text format for each event. "json" indicates structured json format. Known formats are legacy,json.                                                                   |
| <code>--audit-log-maxage int</code>                      |                                                                                                                                                                                                                         |
|                                                          | The maximum number of days to retain old audit log files based on the timestamp encoded in their filename.                                                                                                              |
| <code>--audit-log-maxbackup int</code>                   |                                                                                                                                                                                                                         |
|                                                          | The maximum number of old audit log files to retain. Setting a value of 0 will mean there's no restriction on the number of files.                                                                                      |
| <code>--audit-log-maxsize int</code>                     |                                                                                                                                                                                                                         |
|                                                          | The maximum size in megabytes of the audit log file before it gets rotated.                                                                                                                                             |
| <code>--audit-log-mode string</code> Default: "blocking" |                                                                                                                                                                                                                         |
|                                                          | Strategy for sending audit events. Blocking indicates sending events should block server responses. Batch causes the backend to buffer and write events asynchronously. Known modes are batch,blocking,blocking-strict. |
| <code>--audit-log-path string</code>                     |                                                                                                                                                                                                                         |

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| If set, all requests coming to the apiserver will be logged to this file. '-' means standard out.                                                                                                                            |
| --audit-log-truncate-enabled                                                                                                                                                                                                 |
| Whether event and batch truncating is enabled.                                                                                                                                                                               |
| --audit-log-truncate-max-batch-size int    Default: 10485760                                                                                                                                                                 |
| Maximum size of the batch sent to the underlying backend. Actual serialized size can be several hundreds of bytes greater. If a batch exceeds this limit, it is split into several batches of smaller size.                  |
| --audit-log-truncate-max-event-size int    Default: 102400                                                                                                                                                                   |
| Maximum size of the audit event sent to the underlying backend. If the size of an event is greater than this number, first request and response are removed, and if this doesn't reduce the size enough, event is discarded. |
| --audit-log-version string    Default: "audit.k8s.io/v1"                                                                                                                                                                     |
| API group and version used for serializing audit events written to log.                                                                                                                                                      |
| --audit-policy-file string                                                                                                                                                                                                   |
| Path to the file that defines the audit policy configuration.                                                                                                                                                                |
| --audit-webhook-batch-buffer-size int    Default: 10000                                                                                                                                                                      |
| The size of the buffer to store events before batching and writing. Only used in batch mode.                                                                                                                                 |
| --audit-webhook-batch-max-size int    Default: 400                                                                                                                                                                           |
| The maximum size of a batch. Only used in batch mode.                                                                                                                                                                        |
| --audit-webhook-batch-max-wait duration    Default: 30s                                                                                                                                                                      |
| The amount of time to wait before force writing the batch that hadn't reached the max size. Only used in batch mode.                                                                                                         |
| --audit-webhook-batch-throttle-burst int    Default: 15                                                                                                                                                                      |
| Maximum number of requests sent at the same moment if ThrottleQPS was not utilized before. Only used in batch mode.                                                                                                          |
| --audit-webhook-batch-throttle-enable    Default: true                                                                                                                                                                       |
| Whether batching throttling is enabled. Only used in batch mode.                                                                                                                                                             |
| --audit-webhook-batch-throttle-qps float    Default: 10                                                                                                                                                                      |

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| Maximum average number of batches per second. Only used in batch mode.                                                                                                                                                                                                                    |
| --audit-webhook-config-file string                                                                                                                                                                                                                                                        |
| Path to a kubeconfig formatted file that defines the audit webhook configuration.                                                                                                                                                                                                         |
| --audit-webhook-initial-backoff duration    Default: 10s                                                                                                                                                                                                                                  |
| The amount of time to wait before retrying the first failed request.                                                                                                                                                                                                                      |
| --audit-webhook-mode string    Default: "batch"                                                                                                                                                                                                                                           |
| Strategy for sending audit events. Blocking indicates sending events should block server responses. Batch causes the backend to buffer and write events asynchronously. Known modes are batch,blocking,blocking-strict.                                                                   |
| --audit-webhook-truncate-enabled                                                                                                                                                                                                                                                          |
| Whether event and batch truncating is enabled.                                                                                                                                                                                                                                            |
| --audit-webhook-truncate-max-batch-size int    Default: 10485760                                                                                                                                                                                                                          |
| Maximum size of the batch sent to the underlying backend. Actual serialized size can be several hundreds of bytes greater. If a batch exceeds this limit, it is split into several batches of smaller size.                                                                               |
| --audit-webhook-truncate-max-event-size int    Default: 102400                                                                                                                                                                                                                            |
| Maximum size of the audit event sent to the underlying backend. If the size of an event is greater than this number, first request and response are removed, and if this doesn't reduce the size enough, event is discarded.                                                              |
| --audit-webhook-version string    Default: "audit.k8s.io/v1"                                                                                                                                                                                                                              |
| API group and version used for serializing audit events written to webhook.                                                                                                                                                                                                               |
| --authentication-config string                                                                                                                                                                                                                                                            |
| File with Authentication Configuration to configure the JWT Token authenticator. Note: This feature is in Alpha since v1.29.--feature-gate=StructuredAuthenticationConfiguration=true needs to be set for enabling this feature.This feature is mutually exclusive with the oidc-* flags. |
| --authentication-token-webhook-cache-ttl duration    Default: 2m0s                                                                                                                                                                                                                        |
| The duration to cache responses from the webhook token authenticator.                                                                                                                                                                                                                     |
| --authentication-token-webhook-config-file string                                                                                                                                                                                                                                         |
| File with webhook configuration for token authentication in kubeconfig format. The API server will query the remote service to determine authentication for bearer tokens.                                                                                                                |

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| --authentication-token-webhook-version string                                                                                                                                                                                                                                                                                                               | Default: "v1beta1"             |
| The API version of the authentication.k8s.io TokenReview to send to and expect from the webhook.                                                                                                                                                                                                                                                            |                                |
| --authorization-config string                                                                                                                                                                                                                                                                                                                               |                                |
| File with Authorization Configuration to configure the authorizer chain.Note: This feature is in Alpha since v1.29.--feature-gate=StructuredAuthorizationConfiguration=true feature flag needs to be set to true for enabling the functionality.This feature is mutually exclusive with the other --authorization-mode and --authorization-webhook-* flags. |                                |
| --authorization-mode strings                                                                                                                                                                                                                                                                                                                                |                                |
| Ordered list of plug-ins to do authorization on secure port. Defaults to AlwaysAllow if --authorization-config is not used. Comma-delimited list of: AlwaysAllow,AlwaysDeny,ABAC,Webhook,RBAC,Node.                                                                                                                                                         |                                |
| --authorization-policy-file string                                                                                                                                                                                                                                                                                                                          |                                |
| File with authorization policy in json line by line format, used with --authorization-mode=ABAC, on the secure port.                                                                                                                                                                                                                                        |                                |
| --authorization-webhook-cache-authorized-ttl duration                                                                                                                                                                                                                                                                                                       | Default: 5m0s                  |
| The duration to cache 'authorized' responses from the webhook authorizer.                                                                                                                                                                                                                                                                                   |                                |
| --authorization-webhook-cache-unauthorized-ttl duration                                                                                                                                                                                                                                                                                                     | Default: 30s                   |
| The duration to cache 'unauthorized' responses from the webhook authorizer.                                                                                                                                                                                                                                                                                 |                                |
| --authorization-webhook-config-file string                                                                                                                                                                                                                                                                                                                  |                                |
| File with webhook configuration in kubeconfig format, used with --authorization-mode=Webhook. The API server will query the remote service to determine access on the API server's secure port.                                                                                                                                                             |                                |
| --authorization-webhook-version string                                                                                                                                                                                                                                                                                                                      | Default: "v1beta1"             |
| The API version of the authorization.k8s.io SubjectAccessReview to send to and expect from the webhook.                                                                                                                                                                                                                                                     |                                |
| --bind-address string                                                                                                                                                                                                                                                                                                                                       | Default: 0.0.0.0               |
| The IP address on which to listen for the --secure-port port. The associated interface(s) must be reachable by the rest of the cluster, and by CLI/web clients. If blank or an unspecified address (0.0.0.0 or ::), all interfaces and IP address families will be used.                                                                                    |                                |
| --cert-dir string                                                                                                                                                                                                                                                                                                                                           | Default: "/var/run/kubernetes" |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The directory where the TLS certs are located. If --tls-cert-file and --tls-private-key-file are provided, this flag will be ignored.                                                                                                                                                                                                                                                                                                                                                |
| --client-ca-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| If set, any request presenting a client certificate signed by one of the authorities in the client-ca-file is authenticated with an identity corresponding to the CommonName of the client certificate.                                                                                                                                                                                                                                                                              |
| --cloud-provider-gce-l7lb-src-cidrs cidrs    Default: 130.211.0.0/22,35.191.0.0/16                                                                                                                                                                                                                                                                                                                                                                                                   |
| CIDRs opened in GCE firewall for L7 LB traffic proxy & health checks                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --contention-profiling                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Enable block profiling, if profiling is enabled                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| --cors-allowed-origins strings                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| List of allowed origins for CORS, comma separated. An allowed origin can be a regular expression to support subdomain matching. If this list is empty CORS will not be enabled. Please ensure each expression matches the entire hostname by anchoring to the start with '^' or including the '/' prefix, and by anchoring to the end with '\$' or including the ':' port separator suffix. Examples of valid expressions are '//example.com(: \$)' and '^https://example.com(: \$)' |
| --debug-socket-path string                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Use an unprotected (no authn/authz) unix-domain socket for profiling with the given path                                                                                                                                                                                                                                                                                                                                                                                             |
| --default-not-ready-toleration-seconds int    Default: 300                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Indicates the tolerationSeconds of the toleration for notReady:NoExecute that is added by default to every pod that does not already have such a toleration.                                                                                                                                                                                                                                                                                                                         |
| --default-unreachable-toleration-seconds int    Default: 300                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Indicates the tolerationSeconds of the toleration for unreachable:NoExecute that is added by default to every pod that does not already have such a toleration.                                                                                                                                                                                                                                                                                                                      |
| --delete-collection-workers int    Default: 1                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Number of workers spawned for DeleteCollection call. These are used to speed up namespace cleanup.                                                                                                                                                                                                                                                                                                                                                                                   |
| --disable-admission-plugins strings                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| admission plugins that should be disabled although they are in the default enabled plugins list (NamespaceLifecycle, LimitRanger, ServiceAccount, TaintNodesByCondition, PodSecurity, Priority, DefaultTolerationSeconds, DefaultStorageClass, StorageObjectInUseProtection, PersistentVolumeClaimResize, RuntimeClass, CertificateApproval, CertificateSigning,                                                                                                                     |

ClusterTrustBundleAttest, CertificateSubjectRestriction, DefaultIngressClass, MutatingAdmissionWebhook, ValidatingAdmissionPolicy, ValidatingAdmissionWebhook, ResourceQuota). Comma-delimited list of admission plugins: AlwaysAdmit, AlwaysDeny, AlwaysPullImages, CertificateApproval, CertificateSigning, CertificateSubjectRestriction, ClusterTrustBundleAttest, DefaultIngressClass, DefaultStorageClass, DefaultTolerationSeconds, DenyServiceExternalIPs, EventRateLimit, ExtendedResourceToleration, ImagePolicyWebhook, LimitPodHardAntiAffinityTopology, LimitRanger, MutatingAdmissionWebhook, NamespaceAutoProvision, NamespaceExists, NamespaceLifecycle, NodeRestriction, OwnerReferencesPermissionEnforcement, PersistentVolumeClaimResize, PersistentVolumeLabel, PodNodeSelector, PodSecurity, PodTolerationRestriction, Priority, ResourceQuota, RuntimeClass, ServiceAccount, StorageObjectInUseProtection, TaintNodesByCondition, ValidatingAdmissionPolicy, ValidatingAdmissionWebhook. The order of plugins in this flag does not matter.

#### --disabled-metrics strings

This flag provides an escape hatch for misbehaving metrics. You must provide the fully qualified metric name in order to disable it. Disclaimer: disabling metrics is higher in precedence than showing hidden metrics.

#### --egress-selector-config-file string

File with apiserver egress selector configuration.

#### --enable-admission-plugins strings

admission plugins that should be enabled in addition to default enabled ones (NamespaceLifecycle, LimitRanger, ServiceAccount, TaintNodesByCondition, PodSecurity, Priority, DefaultTolerationSeconds, DefaultStorageClass, StorageObjectInUseProtection, PersistentVolumeClaimResize, RuntimeClass, CertificateApproval, CertificateSigning, ClusterTrustBundleAttest, CertificateSubjectRestriction, DefaultIngressClass, MutatingAdmissionWebhook, ValidatingAdmissionPolicy, ValidatingAdmissionWebhook, ResourceQuota). Comma-delimited list of admission plugins: AlwaysAdmit, AlwaysDeny, AlwaysPullImages, CertificateApproval, CertificateSigning, CertificateSubjectRestriction, ClusterTrustBundleAttest, DefaultIngressClass, DefaultStorageClass, DefaultTolerationSeconds, DenyServiceExternalIPs, EventRateLimit, ExtendedResourceToleration, ImagePolicyWebhook, LimitPodHardAntiAffinityTopology, LimitRanger, MutatingAdmissionWebhook, NamespaceAutoProvision, NamespaceExists, NamespaceLifecycle, NodeRestriction, OwnerReferencesPermissionEnforcement, PersistentVolumeClaimResize, PersistentVolumeLabel, PodNodeSelector, PodSecurity, PodTolerationRestriction, Priority, ResourceQuota, RuntimeClass, ServiceAccount, StorageObjectInUseProtection, TaintNodesByCondition, ValidatingAdmissionPolicy, ValidatingAdmissionWebhook. The order of plugins in this flag does not matter.

#### --enable-aggregator-routing

Turns on aggregator routing requests to endpoints IP rather than cluster IP.

#### --enable-bootstrap-token-auth

|                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enable to allow secrets of type 'bootstrap.kubernetes.io/token' in the 'kube-system' namespace to be used for TLS bootstrapping authentication.                                                                                                    |
| --enable-garbage-collector    Default: true                                                                                                                                                                                                        |
| Enables the generic garbage collector. MUST be synced with the corresponding flag of the kube-controller-manager.                                                                                                                                  |
| --enable-priority-and-fairness    Default: true                                                                                                                                                                                                    |
| If true, replace the max-in-flight handler with an enhanced one that queues and dispatches with priority and fairness                                                                                                                              |
| --encryption-provider-config string                                                                                                                                                                                                                |
| The file containing configuration for encryption providers to be used for storing secrets in etcd                                                                                                                                                  |
| --encryption-provider-config-automatic-reload                                                                                                                                                                                                      |
| Determines if the file set by --encryption-provider-config should be automatically reloaded if the disk contents change. Setting this to true disables the ability to uniquely identify distinct KMS plugins via the API server healthz endpoints. |
| --endpoint-reconciler-type string    Default: "lease"                                                                                                                                                                                              |
| Use an endpoint reconciler (master-count, lease, none) master-count is deprecated, and will be removed in a future version.                                                                                                                        |
| --etcd-cafile string                                                                                                                                                                                                                               |
| SSL Certificate Authority file used to secure etcd communication.                                                                                                                                                                                  |
| --etcd-certfile string                                                                                                                                                                                                                             |
| SSL certification file used to secure etcd communication.                                                                                                                                                                                          |
| --etcd-compaction-interval duration    Default: 5m0s                                                                                                                                                                                               |
| The interval of compaction requests. If 0, the compaction request from apiserver is disabled.                                                                                                                                                      |
| --etcd-count-metric-poll-period duration    Default: 1m0s                                                                                                                                                                                          |
| Frequency of polling etcd for number of resources per type. 0 disables the metric collection.                                                                                                                                                      |
| --etcd-db-metric-poll-interval duration    Default: 30s                                                                                                                                                                                            |
| The interval of requests to poll etcd and update metric. 0 disables the metric collection                                                                                                                                                          |
| --etcd-healthcheck-timeout duration    Default: 2s                                                                                                                                                                                                 |



|                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                             | The timeout to use when checking etcd health.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>--etcd-keyfile string</code>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                             | SSL key file used to secure etcd communication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>--etcd-prefix string</code> Default: <code>"/registry"</code>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                             | The prefix to prepend to all resource paths in etcd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>--etcd-readycheck-timeout duration</code> Default: <code>2s</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                             | The timeout to use when checking etcd readiness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>--etcd-servers strings</code>                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                             | List of etcd servers to connect with (scheme://ip:port), comma separated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--etcd-servers-overrides strings</code>                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                             | Per-resource etcd servers overrides, comma separated. The individual override format: group/resource#servers, where servers are URLs, semicolon separated. Note that this applies only to resources compiled into this server binary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>--event-ttl duration</code> Default: <code>1h0m0s</code>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                             | Amount of time to retain events.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>--external-hostname string</code>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                             | The hostname to use when generating externalized URLs for this master (e.g. Swagger API Docs or OpenID Discovery).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>--feature-gates &lt;comma-separated 'key=True False' pairs&gt;</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                             | <p>A set of key=value pairs that describe feature gates for alpha/experimental features. Options are:</p> <p>APIResponseCompression=true false (BETA - default=true)</p> <p>APIServerIdentity=true false (BETA - default=true)</p> <p>APIServerTracing=true false (BETA - default=true)</p> <p>APIServingWithRoutine=true false (BETA - default=true)</p> <p>AllAlpha=true false (ALPHA - default=false)</p> <p>AllBeta=true false (BETA - default=false)</p> <p>AnyVolumeDataSource=true false (BETA - default=true)</p> <p>AppArmor=true false (BETA - default=true)</p> <p>AppArmorFields=true false (BETA - default=true)</p> <p>CPUManagerPolicyAlphaOptions=true false (ALPHA - default=false)</p> <p>CPUManagerPolicyBetaOptions=true false (BETA - default=true)</p> <p>CPUManagerPolicyOptions=true false (BETA - default=true)</p> <p>CRDValidationRatcheting=true false (BETA - default=true)</p> <p>CSIMigrationPortworx=true false (BETA - default=false)</p> |

CSIVolumeHealth=true|false (ALPHA - default=false)  
CloudControllerManagerWebhook=true|false (ALPHA - default=false)  
ClusterTrustBundle=true|false (ALPHA - default=false)  
ClusterTrustBundleProjection=true|false (ALPHA - default=false)  
ComponentSLIs=true|false (BETA - default=true)  
ConsistentListFromCache=true|false (ALPHA - default=false)  
ContainerCheckpoint=true|false (BETA - default=true)  
ContextualLogging=true|false (BETA - default=true)  
CronJobsScheduledAnnotation=true|false (BETA - default=true)  
CrossNamespaceVolumeDataSource=true|false (ALPHA - default=false)  
CustomCPUCFSQuotaPeriod=true|false (ALPHA - default=false)  
CustomResourceFieldSelectors=true|false (ALPHA - default=false)  
DevicePluginCDIDevices=true|false (BETA - default=true)  
DisableCloudProviders=true|false (BETA - default=true)  
DisableKubeletCloudCredentialProviders=true|false (BETA - default=true)  
DisableNodeKubeProxyVersion=true|false (ALPHA - default=false)  
DynamicResourceAllocation=true|false (ALPHA - default=false)  
ElasticIndexedJob=true|false (BETA - default=true)  
EventedPLEG=true|false (ALPHA - default=false)  
GracefulNodeShutdown=true|false (BETA - default=true)  
GracefulNodeShutdownBasedOnPodPriority=true|false (BETA - default=true)  
HPAScaleToZero=true|false (ALPHA - default=false)  
HonorPVReclaimPolicy=true|false (ALPHA - default=false)  
ImageMaximumGCAge=true|false (BETA - default=true)  
InPlacePodVerticalScaling=true|false (ALPHA - default=false)  
InTreePluginAWSUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureDiskUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureFileUnregister=true|false (ALPHA - default=false)  
InTreePluginGCEUnregister=true|false (ALPHA - default=false)  
InTreePluginOpenStackUnregister=true|false (ALPHA - default=false)  
InTreePluginPortworxUnregister=true|false (ALPHA - default=false)  
InTreePluginvSphereUnregister=true|false (ALPHA - default=false)  
InformerResourceVersion=true|false (ALPHA - default=false)  
JobBackoffLimitPerIndex=true|false (BETA - default=true)  
JobManagedBy=true|false (ALPHA - default=false)  
JobPodFailurePolicy=true|false (BETA - default=true)  
JobPodReplacementPolicy=true|false (BETA - default=true)  
JobSuccessPolicy=true|false (ALPHA - default=false)  
KubeProxyDrainingTerminatingNodes=true|false (BETA - default=true)  
KubeletCgroupDriverFromCRI=true|false (ALPHA - default=false)  
KubeletInUserNamespace=true|false (ALPHA - default=false)  
KubeletPodResourcesDynamicResources=true|false (ALPHA - default=false)  
KubeletPodResourcesGet=true|false (ALPHA - default=false)  
KubeletSeparateDiskGC=true|false (ALPHA - default=false)  
KubeletTracing=true|false (BETA - default=true)  
LoadBalancerIPMode=true|false (BETA - default=true)  
LocalStorageCapacityIsolationFSQuotaMonitoring=true|false (ALPHA - default=false)  
LogarithmicScaleDown=true|false (BETA - default=true)  
LoggingAlphaOptions=true|false (ALPHA - default=false)  
LoggingBetaOptions=true|false (BETA - default=true)  
MatchLabelKeysInPodAffinity=true|false (ALPHA - default=false)  
MatchLabelKeysInPodTopologySpread=true|false (BETA - default=true)  
MaxUnavailableStatefulSet=true|false (ALPHA - default=false)

MemoryManager=true|false (BETA - default=true)  
MemoryQoS=true|false (ALPHA - default=false)  
MultiCIDRServiceAllocator=true|false (ALPHA - default=false)  
MutatingAdmissionPolicy=true|false (ALPHA - default=false)  
NFTablesProxyMode=true|false (ALPHA - default=false)  
NodeInclusionPolicyInPodTopologySpread=true|false (BETA - default=true)  
NodeLogQuery=true|false (BETA - default=false)  
NodeSwap=true|false (BETA - default=true)  
OpenAPIEnums=true|false (BETA - default=true)  
PDBUnhealthyPodEvictionPolicy=true|false (BETA - default=true)  
PersistentVolumeLastPhaseTransitionTime=true|false (BETA - default=true)  
PodAndContainerStatsFromCRI=true|false (ALPHA - default=false)  
PodDeletionCost=true|false (BETA - default=true)  
PodDisruptionConditions=true|false (BETA - default=true)  
PodIndexLabel=true|false (BETA - default=true)  
PodLifecycleSleepAction=true|false (BETA - default=true)  
PodReadyToStartContainersCondition=true|false (BETA - default=true)  
PortForwardWebsockets=true|false (ALPHA - default=false)  
ProcMountType=true|false (ALPHA - default=false)  
QOSReserved=true|false (ALPHA - default=false)  
RecoverVolumeExpansionFailure=true|false (ALPHA - default=false)  
RecursiveReadOnlyMounts=true|false (ALPHA - default=false)  
RelaxedEnvironmentVariableValidation=true|false (ALPHA - default=false)  
RetryGenerateName=true|false (ALPHA - default=false)  
RotateKubeletServerCertificate=true|false (BETA - default=true)  
RuntimeClassInImageCriApi=true|false (ALPHA - default=false)  
SELinuxMount=true|false (ALPHA - default=false)  
SELinuxMountReadWriteOncePod=true|false (BETA - default=true)  
SchedulerQueueingHints=true|false (BETA - default=false)  
SeparateCacheWatchRPC=true|false (BETA - default=true)  
SeparateTaintEvictionController=true|false (BETA - default=true)  
ServiceAccountTokenJTI=true|false (BETA - default=true)  
ServiceAccountTokenNodeBinding=true|false (ALPHA - default=false)  
ServiceAccountTokenNodeBindingValidation=true|false (BETA - default=true)  
ServiceAccountTokenPodNodeInfo=true|false (BETA - default=true)  
ServiceTrafficDistribution=true|false (ALPHA - default=false)  
SidecarContainers=true|false (BETA - default=true)  
SizeMemoryBackedVolumes=true|false (BETA - default=true)  
StatefulSetAutoDeletePVC=true|false (BETA - default=true)  
StatefulSetStartOrdinal=true|false (BETA - default=true)  
StorageNamespaceIndex=true|false (BETA - default=true)  
StorageVersionAPI=true|false (ALPHA - default=false)  
StorageVersionHash=true|false (BETA - default=true)  
StorageVersionMigrator=true|false (ALPHA - default=false)  
StructuredAuthenticationConfiguration=true|false (BETA - default=true)  
StructuredAuthorizationConfiguration=true|false (BETA - default=true)  
TopologyAwareHints=true|false (BETA - default=true)  
TopologyManagerPolicyAlphaOptions=true|false (ALPHA - default=false)  
TopologyManagerPolicyBetaOptions=true|false (BETA - default=true)  
TopologyManagerPolicyOptions=true|false (BETA - default=true)  
TranslateStreamCloseWebsocketRequests=true|false (BETA - default=true)  
UnauthenticatedHTTP2DOSMitigation=true|false (BETA - default=true)  
UnknownVersionInteroperabilityProxy=true|false (ALPHA - default=false)

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| UserNamespacesPodSecurityStandards=true false (ALPHA - default=false)<br>UserNamespacesSupport=true false (BETA - default=false)<br>VolumeAttributesClass=true false (ALPHA - default=false)<br>VolumeCapacityPriority=true false (ALPHA - default=false)<br>WatchFromStorageWithoutResourceVersion=true false (BETA - default=false)<br>WatchList=true false (ALPHA - default=false)<br>WatchListClient=true false (BETA - default=false)<br>WinDSR=true false (ALPHA - default=false)<br>WinOverlay=true false (BETA - default=true)<br>WindowsHostNetwork=true false (ALPHA - default=true) |
| --goaway-chance float                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| To prevent HTTP/2 clients from getting stuck on a single apiserver, randomly close a connection (GOAWAY). The client's other in-flight requests won't be affected, and the client will reconnect, likely landing on a different apiserver after going through the load balancer again. This argument sets the fraction of requests that will be sent a GOAWAY. Clusters with single apiservers, or which don't use a load balancer, should NOT enable this. Min is 0 (off), Max is .02 (1/50 requests); .001 (1/1000) is a recommended starting point.                                         |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| help for kube-apiserver                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --http2-max-streams-per-connection int                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| The limit that the server gives to clients for the maximum number of streams in an HTTP/2 connection. Zero means to use golang's default.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| --kubelet-certificate-authority string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Path to a cert file for the certificate authority.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| --kubelet-client-certificate string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Path to a client cert file for TLS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| --kubelet-client-key string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Path to a client key file for TLS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| --kubelet-preferred-address-types strings    Default:<br>"Hostname,InternalDNS,InternalIP,ExternalDNS,ExternalIP"                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| List of the preferred NodeAddressTypes to use for kubelet connections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| --kubelet-timeout duration    Default: 5s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Timeout for kubelet operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| --kubernetes-service-node-port int                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| <p>If non-zero, the Kubernetes master service (which apiserver creates/maintains) will be of type NodePort, using this as the value of the port. If zero, the Kubernetes master service will be of type ClusterIP.</p>                                                                                                                                                                   |
| <p>--lease-reuse-duration-seconds int    Default: 60</p>                                                                                                                                                                                                                                                                                                                                 |
| <p>The time in seconds that each lease is reused. A lower value could avoid large number of objects reusing the same lease. Notice that a too small value may cause performance problems at storage layer.</p>                                                                                                                                                                           |
| <p>--livez-grace-period duration</p>                                                                                                                                                                                                                                                                                                                                                     |
| <p>This option represents the maximum amount of time it should take for apiserver to complete its startup sequence and become live. From apiserver's start time to when this amount of time has elapsed, /livez will assume that unfinished post-start hooks will complete successfully and therefore return true.</p>                                                                   |
| <p>--log-flush-frequency duration    Default: 5s</p>                                                                                                                                                                                                                                                                                                                                     |
| <p>Maximum number of seconds between log flushes</p>                                                                                                                                                                                                                                                                                                                                     |
| <p>--log-text-info-buffer-size quantity</p>                                                                                                                                                                                                                                                                                                                                              |
| <p>[Alpha] In text format with split output streams, the info messages can be buffered for a while to increase performance. The default value of zero bytes disables buffering. The size can be specified as number of bytes (512), multiples of 1000 (1K), multiples of 1024 (2Ki), or powers of those (3M, 4G, 5Mi, 6Gi). Enable the LoggingAlphaOptions feature gate to use this.</p> |
| <p>--log-text-split-stream</p>                                                                                                                                                                                                                                                                                                                                                           |
| <p>[Alpha] In text format, write error messages to stderr and info messages to stdout. The default is to write a single stream to stdout. Enable the LoggingAlphaOptions feature gate to use this.</p>                                                                                                                                                                                   |
| <p>--logging-format string    Default: "text"</p>                                                                                                                                                                                                                                                                                                                                        |
| <p>Sets the log format. Permitted formats: "text".</p>                                                                                                                                                                                                                                                                                                                                   |
| <p>--max-connection-bytes-per-sec int</p>                                                                                                                                                                                                                                                                                                                                                |
| <p>If non-zero, throttle each user connection to this number of bytes/sec. Currently only applies to long-running requests.</p>                                                                                                                                                                                                                                                          |
| <p>--max-mutating-requests-inflight int    Default: 200</p>                                                                                                                                                                                                                                                                                                                              |
| <p>This and --max-requests-inflight are summed to determine the server's total concurrency limit (which must be positive) if --enable-priority-and-fairness is true. Otherwise, this flag limits the maximum number of mutating requests in flight, or a zero value disables the limit completely.</p>                                                                                   |
| <p>--max-requests-inflight int    Default: 400</p>                                                                                                                                                                                                                                                                                                                                       |

|                                                                                                                                                                                                                                                                                                                                        |
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| <p>This and --max-mutating-requests-inflight are summed to determine the server's total concurrency limit (which must be positive) if --enable-priority-and-fairness is true. Otherwise, this flag limits the maximum number of non-mutating requests in flight, or a zero value disables the limit completely.</p>                    |
| <p>--min-request-timeout int    Default: 1800</p>                                                                                                                                                                                                                                                                                      |
| <p>An optional field indicating the minimum number of seconds a handler must keep a request open before timing it out. Currently only honored by the watch request handler, which picks a randomized value above this number as the connection timeout, to spread out load.</p>                                                        |
| <p>--oidc-ca-file string</p>                                                                                                                                                                                                                                                                                                           |
| <p>If set, the OpenID server's certificate will be verified by one of the authorities in the oidc-ca-file, otherwise the host's root CA set will be used.</p>                                                                                                                                                                          |
| <p>--oidc-client-id string</p>                                                                                                                                                                                                                                                                                                         |
| <p>The client ID for the OpenID Connect client, must be set if oidc-issuer-url is set.</p>                                                                                                                                                                                                                                             |
| <p>--oidc-groups-claim string</p>                                                                                                                                                                                                                                                                                                      |
| <p>If provided, the name of a custom OpenID Connect claim for specifying user groups. The claim value is expected to be a string or array of strings. This flag is experimental, please see the authentication documentation for further details.</p>                                                                                  |
| <p>--oidc-groups-prefix string</p>                                                                                                                                                                                                                                                                                                     |
| <p>If provided, all groups will be prefixed with this value to prevent conflicts with other authentication strategies.</p>                                                                                                                                                                                                             |
| <p>--oidc-issuer-url string</p>                                                                                                                                                                                                                                                                                                        |
| <p>The URL of the OpenID issuer, only HTTPS scheme will be accepted. If set, it will be used to verify the OIDC JSON Web Token (JWT).</p>                                                                                                                                                                                              |
| <p>--oidc-required-claim &lt;comma-separated 'key=value' pairs&gt;</p>                                                                                                                                                                                                                                                                 |
| <p>A key=value pair that describes a required claim in the ID Token. If set, the claim is verified to be present in the ID Token with a matching value. Repeat this flag to specify multiple claims.</p>                                                                                                                               |
| <p>--oidc-signing-algs strings    Default: "RS256"</p>                                                                                                                                                                                                                                                                                 |
| <p>Comma-separated list of allowed JOSE asymmetric signing algorithms. JWTs with a supported 'alg' header values are: RS256, RS384, RS512, ES256, ES384, ES512, PS256, PS384, PS512. Values are defined by RFC 7518 <a href="https://tools.ietf.org/html/rfc7518#section-3.1">https://tools.ietf.org/html/rfc7518#section-3.1</a>.</p> |
| <p>--oidc-username-claim string    Default: "sub"</p>                                                                                                                                                                                                                                                                                  |

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| <p>The OpenID claim to use as the user name. Note that claims other than the default ('sub') is not guaranteed to be unique and immutable. This flag is experimental, please see the authentication documentation for further details.</p>                                                                                                                                                       |
| <p>--oidc-username-prefix string</p>                                                                                                                                                                                                                                                                                                                                                             |
| <p>If provided, all usernames will be prefixed with this value. If not provided, username claims other than 'email' are prefixed by the issuer URL to avoid clashes. To skip any prefixing, provide the value '-'.</p>                                                                                                                                                                           |
| <p>--peer-advertise-ip string</p>                                                                                                                                                                                                                                                                                                                                                                |
| <p>If set and the UnknownVersionInteroperabilityProxy feature gate is enabled, this IP will be used by peer kube-apiservers to proxy requests to this kube-apiserver when the request cannot be handled by the peer due to version skew between the kube-apiservers. This flag is only used in clusters configured with multiple kube-apiservers for high availability.</p>                      |
| <p>--peer-advertise-port string</p>                                                                                                                                                                                                                                                                                                                                                              |
| <p>If set and the UnknownVersionInteroperabilityProxy feature gate is enabled, this port will be used by peer kube-apiservers to proxy requests to this kube-apiserver when the request cannot be handled by the peer due to version skew between the kube-apiservers. This flag is only used in clusters configured with multiple kube-apiservers for high availability.</p>                    |
| <p>--peer-ca-file string</p>                                                                                                                                                                                                                                                                                                                                                                     |
| <p>If set and the UnknownVersionInteroperabilityProxy feature gate is enabled, this file will be used to verify serving certificates of peer kube-apiservers. This flag is only used in clusters configured with multiple kube-apiservers for high availability.</p>                                                                                                                             |
| <p>--permit-address-sharing</p>                                                                                                                                                                                                                                                                                                                                                                  |
| <p>If true, SO_REUSEADDR will be used when binding the port. This allows binding to wildcard IPs like 0.0.0.0 and specific IPs in parallel, and it avoids waiting for the kernel to release sockets in TIME_WAIT state. [default=false]</p>                                                                                                                                                      |
| <p>--permit-port-sharing</p>                                                                                                                                                                                                                                                                                                                                                                     |
| <p>If true, SO_REUSEPORT will be used when binding the port, which allows more than one instance to bind on the same address and port. [default=false]</p>                                                                                                                                                                                                                                       |
| <p>--profiling   Default: true</p>                                                                                                                                                                                                                                                                                                                                                               |
| <p>Enable profiling via web interface host:port/debug/pprof/</p>                                                                                                                                                                                                                                                                                                                                 |
| <p>--proxy-client-cert-file string</p>                                                                                                                                                                                                                                                                                                                                                           |
| <p>Client certificate used to prove the identity of the aggregator or kube-apiserver when it must call out during a request. This includes proxying requests to a user api-server and calling out to webhook admission plugins. It is expected that this cert includes a signature from the CA in the --requestheader-client-ca-file flag. That CA is published in the 'extension-apiserver-</p> |

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| authentication' configmap in the kube-system namespace. Components receiving calls from kube-aggregator should use that CA to perform their half of the mutual TLS verification.                                                                                                                                                                                                                                                                                                                                                                                               |
| --proxy-client-key-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Private key for the client certificate used to prove the identity of the aggregator or kube-apiserver when it must call out during a request. This includes proxying requests to a user api-server and calling out to webhook admission plugins.                                                                                                                                                                                                                                                                                                                               |
| --request-timeout duration    Default: 1m0s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| An optional field indicating the duration a handler must keep a request open before timing it out. This is the default request timeout for requests but may be overridden by flags such as --min-request-timeout for specific types of requests.                                                                                                                                                                                                                                                                                                                               |
| --requestheader-allowed-names strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| List of client certificate common names to allow to provide usernames in headers specified by --requestheader-username-headers. If empty, any client certificate validated by the authorities in --requestheader-client-ca-file is allowed.                                                                                                                                                                                                                                                                                                                                    |
| --requestheader-client-ca-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Root certificate bundle to use to verify client certificates on incoming requests before trusting usernames in headers specified by --requestheader-username-headers. WARNING: generally do not depend on authorization being already done for incoming requests.                                                                                                                                                                                                                                                                                                              |
| --requestheader-extra-headers-prefix strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| List of request header prefixes to inspect. X-Remote-Extra- is suggested.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| --requestheader-group-headers strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| List of request headers to inspect for groups. X-Remote-Group is suggested.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --requestheader-username-headers strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| List of request headers to inspect for usernames. X-Remote-User is common.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| --runtime-config <comma-separated 'key=value' pairs>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| A set of key=value pairs that enable or disable built-in APIs. Supported options are:<br>v1=true false for the core API group<br><group>/<version>=true false for a specific API group and version (e.g. apps/v1=true)<br>api/all=true false controls all API versions<br>api/ga=true false controls all API versions of the form v[0-9]+<br>api/beta=true false controls all API versions of the form v[0-9]+beta[0-9]+<br>api/alpha=true false controls all API versions of the form v[0-9]+alpha[0-9]+<br>api/legacy is deprecated, and will be removed in a future version |
| --secure-port int    Default: 6443                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



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| The port on which to serve HTTPS with authentication and authorization. It cannot be switched off with 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| --service-account-extend-token-expiration    Default: true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Turns on projected service account expiration extension during token generation, which helps safe transition from legacy token to bound service account token feature. If this flag is enabled, admission injected tokens would be extended up to 1 year to prevent unexpected failure during transition, ignoring value of service-account-max-token-expiration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| --service-account-issuer strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Identifier of the service account token issuer. The issuer will assert this identifier in "iss" claim of issued tokens. This value is a string or URI. If this option is not a valid URI per the OpenID Discovery 1.0 spec, the ServiceAccountIssuerDiscovery feature will remain disabled, even if the feature gate is set to true. It is highly recommended that this value comply with the OpenID spec: <a href="https://openid.net/specs/openid-connect-discovery-1_0.html">https://openid.net/specs/openid-connect-discovery-1_0.html</a> . In practice, this means that service-account-issuer must be an https URL. It is also highly recommended that this URL be capable of serving OpenID discovery documents at {service-account-issuer}/.well-known/openid-configuration. When this flag is specified multiple times, the first is used to generate tokens and all are used to determine which issuers are accepted. |
| --service-account-jwks-uri string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Overrides the URI for the JSON Web Key Set in the discovery doc served at /.well-known/openid-configuration. This flag is useful if the discovery doc and key set are served to relying parties from a URL other than the API server's external (as auto-detected or overridden with external-hostname).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| --service-account-key-file strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| File containing PEM-encoded x509 RSA or ECDSA private or public keys, used to verify ServiceAccount tokens. The specified file can contain multiple keys, and the flag can be specified multiple times with different files. If unspecified, --tls-private-key-file is used. Must be specified when --service-account-signing-key-file is provided                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --service-account-lookup    Default: true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| If true, validate ServiceAccount tokens exist in etcd as part of authentication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --service-account-max-token-expiration duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| The maximum validity duration of a token created by the service account token issuer. If an otherwise valid TokenRequest with a validity duration larger than this value is requested, a token will be issued with a validity duration of this value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| --service-account-signing-key-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Path to the file that contains the current private key of the service account token issuer. The issuer will sign issued ID tokens with this private key.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| --service-cluster-ip-range string                                               | A CIDR notation IP range from which to assign service cluster IPs. This must not overlap with any IP ranges assigned to nodes or pods. Max of two dual-stack CIDRs is allowed.                                                                                                                                                                                                                                  |
| --service-node-port-range <a string in the form 'N1-N2'>   Default: 30000-32767 | A port range to reserve for services with NodePort visibility. This must not overlap with the ephemeral port range on nodes. Example: '30000-32767'. Inclusive at both ends of the range.                                                                                                                                                                                                                       |
| --show-hidden-metrics-for-version string                                        | The previous version for which you want to show hidden metrics. Only the previous minor version is meaningful, other values will not be allowed. The format is <major>.<minor>, e.g.: '1.16'. The purpose of this format is make sure you have the opportunity to notice if the next release hides additional metrics, rather than being surprised when they are permanently removed in the release after that. |
| --shutdown-delay-duration duration                                              | Time to delay the termination. During that time the server keeps serving requests normally. The endpoints /healthz and /livez will return success, but /readyz immediately returns failure. Graceful termination starts after this delay has elapsed. This can be used to allow load balancer to stop sending traffic to this server.                                                                           |
| --shutdown-send-retry-after                                                     | If true the HTTP Server will continue listening until all non long running request(s) in flight have been drained, during this window all incoming requests will be rejected with a status code 429 and a 'Retry-After' response header, in addition 'Connection: close' response header is set in order to tear down the TCP connection when idle.                                                             |
| --shutdown-watch-termination-grace-period duration                              | This option, if set, represents the maximum amount of grace period the apiserver will wait for active watch request(s) to drain during the graceful server shutdown window.                                                                                                                                                                                                                                     |
| --storage-backend string                                                        | The storage backend for persistence. Options: 'etcd3' (default).                                                                                                                                                                                                                                                                                                                                                |
| --storage-media-type string   Default: "application/vnd.kubernetes.protobuf"    | The media type to use to store objects in storage. Some resources or storage backends may only support a specific media type and will ignore this setting. Supported media types: [application/json, application/yaml, application/vnd.kubernetes.protobuf]                                                                                                                                                     |
| --strict-transport-security-directives strings                                  | List of directives for HSTS, comma separated. If this list is empty, then HSTS directives will not be added. Example: 'max-age=31536000,includeSubDomains,preload'                                                                                                                                                                                                                                              |

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| --tls-cert-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>File containing the default x509 Certificate for HTTPS. (CA cert, if any, concatenated after server cert). If HTTPS serving is enabled, and --tls-cert-file and --tls-private-key-file are not provided, a self-signed certificate and key are generated for the public address and saved to the directory specified by --cert-dir.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| --tls-cipher-suites strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p>Comma-separated list of cipher suites for the server. If omitted, the default Go cipher suites will be used.</p> <p>Preferred values: TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384, TLS_CHACHA20_POLY1305_SHA256, TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA, TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256, TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA, TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384, TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305, TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256, TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256, TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384, TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305, TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256.</p> <p>Insecure values: TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256, TLS_ECDHE_ECDSA_WITH_RC4_128_SHA, TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256, TLS_ECDHE_RSA_WITH_RC4_128_SHA, TLS_RSA_WITH_3DES_EDE_CBC_SHA, TLS_RSA_WITH_AES_128_CBC_SHA, TLS_RSA_WITH_AES_128_CBC_SHA256, TLS_RSA_WITH_AES_128_GCM_SHA256, TLS_RSA_WITH_AES_256_CBC_SHA, TLS_RSA_WITH_AES_256_GCM_SHA384, TLS_RSA_WITH_RC4_128_SHA.</p> |
| --tls-min-version string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p>Minimum TLS version supported. Possible values: VersionTLS10, VersionTLS11, VersionTLS12, VersionTLS13</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --tls-private-key-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>File containing the default x509 private key matching --tls-cert-file.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --tls-sni-cert-key string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| A pair of x509 certificate and private key file paths, optionally suffixed with a list of domain patterns which are fully qualified domain names, possibly with prefixed wildcard segments. The domain patterns also allow IP addresses, but IPs should only be used if the apiserver has visibility to the IP address requested by a client. If no domain patterns are provided, the names of the certificate are extracted. Non-wildcard matches trump over wildcard matches, explicit domain patterns trump over extracted names. For multiple key/certificate pairs, use the --tls-sni-cert-key multiple times. Examples: "example.crt,example.key" or "foo.crt,foo.key:*.foo.com,foo.com".                                |
| --token-auth-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| If set, the file that will be used to secure the secure port of the API server via token authentication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --tracing-config-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| File with apiserver tracing configuration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -v, --v int                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| number for the log level verbosity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| --version version[=true]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --vmodule pattern=N,...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| comma-separated list of pattern=N settings for file-filtered logging (only works for text log format)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --watch-cache     Default: true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Enable watch caching in the apiserver                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --watch-cache-sizes strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Watch cache size settings for some resources (pods, nodes, etc.), comma separated. The individual setting format: resource[.group]#size, where resource is lowercase plural (no version), group is omitted for resources of apiVersion v1 (the legacy core API) and included for others, and size is a number. This option is only meaningful for resources built into the apiserver, not ones defined by CRDs or aggregated from external servers, and is only consulted if the watch-cache is enabled. The only meaningful size setting to supply here is zero, which means to disable watch caching for the associated resource; all non-zero values are equivalent and mean to not disable watch caching for that resource |

# kube-controller-manager

## Synopsis

The Kubernetes controller manager is a daemon that embeds the core control loops shipped with Kubernetes. In applications of robotics and automation, a control loop is a non-terminating loop that regulates the state of the system. In Kubernetes, a controller is a control loop that watches the shared state of the cluster through the apiserver and makes changes attempting to move the current state towards the desired state. Examples of controllers that ship with Kubernetes today are the replication controller, endpoints controller, namespace controller, and serviceaccounts controller.

kube-controller-manager [flags]

## Options

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| <code>--allocate-node-cidrs</code>                                                                                                                                                                                                                      |
| Should CIDRs for Pods be allocated and set on the cloud provider.                                                                                                                                                                                       |
| <code>--allow-metric-labels</code> stringToString    Default: []                                                                                                                                                                                        |
| The map from metric-label to value allow-list of this label. The key's format is <MetricName>,<LabelName>. The value's format is <allowed_value>,<allowed_value>...e.g. metric1,label1='v1,v2,v3', metric1,label2='v1,v2,v3' metric2,label1='v1,v2,v3'. |
| <code>--allow-metric-labels-manifest</code> string                                                                                                                                                                                                      |
| The path to the manifest file that contains the allow-list mapping. The format of the file is the same as the flag <code>--allow-metric-labels</code> . Note that the flag <code>--allow-metric-labels</code> will override the manifest file.          |
| <code>--attach-detach-reconcile-sync-period</code> duration    Default: 1m0s                                                                                                                                                                            |
| The reconciler sync wait time between volume attach detach. This duration must be larger than one second, and increasing this value from the default may allow for volumes to be mismatched with pods.                                                  |
| <code>--authentication-kubeconfig</code> string                                                                                                                                                                                                         |
| kubeconfig file pointing at the 'core' kubernetes server with enough rights to create tokenreviews.authentication.k8s.io. This is optional. If empty, all token requests are considered to be anonymous and no client CA is looked up in the cluster.   |
| <code>--authentication-skip-lookup</code>                                                                                                                                                                                                               |
| If false, the authentication-kubeconfig will be used to lookup missing authentication configuration from the cluster.                                                                                                                                   |
| <code>--authentication-token-webhook-cache-ttl</code> duration    Default: 10s                                                                                                                                                                          |

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| The duration to cache responses from the webhook token authenticator.                                                                                                                                                                                                    |
| --authentication-tolerate-lookup-failure                                                                                                                                                                                                                                 |
| If true, failures to look up missing authentication configuration from the cluster are not considered fatal. Note that this can result in authentication that treats all requests as anonymous.                                                                          |
| --authorization-always-allow-paths strings    Default: "/healthz,/readyz,/livez"                                                                                                                                                                                         |
| A list of HTTP paths to skip during authorization, i.e. these are authorized without contacting the 'core' kubernetes server.                                                                                                                                            |
| --authorization-kubeconfig string                                                                                                                                                                                                                                        |
| kubeconfig file pointing at the 'core' kubernetes server with enough rights to create subjectaccessreviews.authorization.k8s.io. This is optional. If empty, all requests not skipped by authorization are forbidden.                                                    |
| --authorization-webhook-cache-authorized-ttl duration    Default: 10s                                                                                                                                                                                                    |
| The duration to cache 'authorized' responses from the webhook authorizer.                                                                                                                                                                                                |
| --authorization-webhook-cache-unauthorized-ttl duration    Default: 10s                                                                                                                                                                                                  |
| The duration to cache 'unauthorized' responses from the webhook authorizer.                                                                                                                                                                                              |
| --bind-address string    Default: 0.0.0.0                                                                                                                                                                                                                                |
| The IP address on which to listen for the --secure-port port. The associated interface(s) must be reachable by the rest of the cluster, and by CLI/web clients. If blank or an unspecified address (0.0.0.0 or ::), all interfaces and IP address families will be used. |
| --cert-dir string                                                                                                                                                                                                                                                        |
| The directory where the TLS certs are located. If --tls-cert-file and --tls-private-key-file are provided, this flag will be ignored.                                                                                                                                    |
| --cidr-allocator-type string    Default: "RangeAllocator"                                                                                                                                                                                                                |
| Type of CIDR allocator to use                                                                                                                                                                                                                                            |
| --client-ca-file string                                                                                                                                                                                                                                                  |
| If set, any request presenting a client certificate signed by one of the authorities in the client-ca-file is authenticated with an identity corresponding to the CommonName of the client certificate.                                                                  |
| --cloud-config string                                                                                                                                                                                                                                                    |

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| The path to the cloud provider configuration file. Empty string for no configuration file.                                                                                                                    |
| --cloud-provider string                                                                                                                                                                                       |
| The provider for cloud services. Empty string for no provider.                                                                                                                                                |
| --cluster-cidr string                                                                                                                                                                                         |
| CIDR Range for Pods in cluster. Requires --allocate-node-cidrs to be true                                                                                                                                     |
| --cluster-name string    Default: "kubernetes"                                                                                                                                                                |
| The instance prefix for the cluster.                                                                                                                                                                          |
| --cluster-signing-cert-file string                                                                                                                                                                            |
| Filename containing a PEM-encoded X509 CA certificate used to issue cluster-scoped certificates. If specified, no more specific --cluster-signing-* flag may be specified.                                    |
| --cluster-signing-duration duration    Default: 8760h0m0s                                                                                                                                                     |
| The max length of duration signed certificates will be given. Individual CSRs may request shorter certs by setting spec.expirationSeconds.                                                                    |
| --cluster-signing-key-file string                                                                                                                                                                             |
| Filename containing a PEM-encoded RSA or ECDSA private key used to sign cluster-scoped certificates. If specified, no more specific --cluster-signing-* flag may be specified.                                |
| --cluster-signing-kube-apiserver-client-cert-file string                                                                                                                                                      |
| Filename containing a PEM-encoded X509 CA certificate used to issue certificates for the kubernetes.io/kube-apiserver-client signer. If specified, --cluster-signing-{cert,key}-file must not be set.         |
| --cluster-signing-kube-apiserver-client-key-file string                                                                                                                                                       |
| Filename containing a PEM-encoded RSA or ECDSA private key used to sign certificates for the kubernetes.io/kube-apiserver-client signer. If specified, --cluster-signing-{cert,key}-file must not be set.     |
| --cluster-signing-kubelet-client-cert-file string                                                                                                                                                             |
| Filename containing a PEM-encoded X509 CA certificate used to issue certificates for the kubernetes.io/kube-apiserver-client-kubelet signer. If specified, --cluster-signing-{cert,key}-file must not be set. |
| --cluster-signing-kubelet-client-key-file string                                                                                                                                                              |

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| Filename containing a PEM-encoded RSA or ECDSA private key used to sign certificates for the kubernetes.io/kube-apiserver-client-kubelet signer. If specified, --cluster-signing-{cert,key}-file must not be set. |
| --cluster-signing-kubelet-serving-cert-file string                                                                                                                                                                |
| Filename containing a PEM-encoded X509 CA certificate used to issue certificates for the kubernetes.io/kubelet-serving signer. If specified, --cluster-signing-{cert,key}-file must not be set.                   |
| --cluster-signing-kubelet-serving-key-file string                                                                                                                                                                 |
| Filename containing a PEM-encoded RSA or ECDSA private key used to sign certificates for the kubernetes.io/kubelet-serving signer. If specified, --cluster-signing-{cert,key}-file must not be set.               |
| --cluster-signing-legacy-unknown-cert-file string                                                                                                                                                                 |
| Filename containing a PEM-encoded X509 CA certificate used to issue certificates for the kubernetes.io/legacy-unknown signer. If specified, --cluster-signing-{cert,key}-file must not be set.                    |
| --cluster-signing-legacy-unknown-key-file string                                                                                                                                                                  |
| Filename containing a PEM-encoded RSA or ECDSA private key used to sign certificates for the kubernetes.io/legacy-unknown signer. If specified, --cluster-signing-{cert,key}-file must not be set.                |
| --concurrent-cron-job-syncs int32    Default: 5                                                                                                                                                                   |
| The number of cron job objects that are allowed to sync concurrently. Larger number = more responsive jobs, but more CPU (and network) load                                                                       |
| --concurrent-deployment-syncs int32    Default: 5                                                                                                                                                                 |
| The number of deployment objects that are allowed to sync concurrently. Larger number = more responsive deployments, but more CPU (and network) load                                                              |
| --concurrent-endpoint-syncs int32    Default: 5                                                                                                                                                                   |
| The number of endpoint syncing operations that will be done concurrently. Larger number = faster endpoint updating, but more CPU (and network) load                                                               |
| --concurrent-ephemeralvolume-syncs int32    Default: 5                                                                                                                                                            |
| The number of ephemeral volume syncing operations that will be done concurrently. Larger number = faster ephemeral volume updating, but more CPU (and network) load                                               |
| --concurrent-gc-syncs int32    Default: 20                                                                                                                                                                        |



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| The number of garbage collector workers that are allowed to sync concurrently.                                                                                                                        |
| --concurrent-horizontal-pod-autoscaler-syncs int32    Default: 5                                                                                                                                      |
| The number of horizontal pod autoscaler objects that are allowed to sync concurrently. Larger number = more responsive horizontal pod autoscaler objects processing, but more CPU (and network) load. |
| --concurrent-job-syncs int32    Default: 5                                                                                                                                                            |
| The number of job objects that are allowed to sync concurrently. Larger number = more responsive jobs, but more CPU (and network) load                                                                |
| --concurrent-namespace-syncs int32    Default: 10                                                                                                                                                     |
| The number of namespace objects that are allowed to sync concurrently. Larger number = more responsive namespace termination, but more CPU (and network) load                                         |
| --concurrent-rc-syncs int32    Default: 5                                                                                                                                                             |
| The number of replication controllers that are allowed to sync concurrently. Larger number = more responsive replica management, but more CPU (and network) load                                      |
| --concurrent-replicaset-syncs int32    Default: 5                                                                                                                                                     |
| The number of replica sets that are allowed to sync concurrently. Larger number = more responsive replica management, but more CPU (and network) load                                                 |
| --concurrent-resource-quota-syncs int32    Default: 5                                                                                                                                                 |
| The number of resource quotas that are allowed to sync concurrently. Larger number = more responsive quota management, but more CPU (and network) load                                                |
| --concurrent-service-endpoint-syncs int32    Default: 5                                                                                                                                               |
| The number of service endpoint syncing operations that will be done concurrently. Larger number = faster endpoint slice updating, but more CPU (and network) load. Defaults to 5.                     |
| --concurrent-service-syncs int32    Default: 1                                                                                                                                                        |
| The number of services that are allowed to sync concurrently. Larger number = more responsive service management, but more CPU (and network) load                                                     |
| --concurrent-serviceaccount-token-syncs int32    Default: 5                                                                                                                                           |
| The number of service account token objects that are allowed to sync concurrently. Larger number = more responsive token generation, but more CPU (and network) load                                  |
| --concurrent-statefulset-syncs int32    Default: 5                                                                                                                                                    |

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| The number of statefulset objects that are allowed to sync concurrently. Larger number = more responsive statefulsets, but more CPU (and network) load                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --concurrent-ttl-after-finished-syncs int32    Default: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| The number of ttl-after-finished-controller workers that are allowed to sync concurrently.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| --concurrent-validating-admission-policy-status-syncs int32    Default: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| The number of ValidatingAdmissionPolicyStatusController workers that are allowed to sync concurrently.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --configure-cloud-routes    Default: true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Should CIDRs allocated by allocate-node-cidrs be configured on the cloud provider.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --contention-profiling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Enable block profiling, if profiling is enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| --controller-start-interval duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Interval between starting controller managers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| --controllers strings    Default: ""                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>A list of controllers to enable. '*' enables all on-by-default controllers, 'foo' enables the controller named 'foo', '-foo' disables the controller named 'foo'.</p> <p>All controllers: bootstrap-signer-controller, certificatesigningrequest-approving-controller, certificatesigningrequest-cleaner-controller, certificatesigningrequest-signing-controller, cloud-node-lifecycle-controller, clusterrole-aggregation-controller, cronjob-controller, daemonset-controller, deployment-controller, disruption-controller, endpoints-controller, endpointslice-controller, endpointslice-mirroring-controller, ephemeral-volume-controller, garbage-collector-controller, horizontal-pod-autoscaler-controller, job-controller, legacy-serviceaccount-token-cleaner-controller, namespace-controller, node-ipam-controller, node-lifecycle-controller, node-route-controller, persistentvolume-attach-detach-controller, persistentvolume-binder-controller, persistentvolume-expander-controller, persistentvolume-protection-controller, persistentvolumeclaim-protection-controller, pod-garbage-collector-controller, replicaset-controller, replicationcontroller-controller, resourceclaim-controller, resourcequota-controller, root-ca-certificate-publisher-controller, service-cidr-controller, service-lb-controller, serviceaccount-controller, serviceaccount-token-controller, statefulset-controller, storage-version-migrator-controller, storageversion-garbage-collector-controller, taint-eviction-controller, token-cleaner-controller, ttl-after-finished-controller, ttl-controller, validatingadmissionpolicy-status-controller</p> <p>Disabled-by-default controllers: bootstrap-signer-controller, token-cleaner-controller</p> |
| --disable-attach-detach-reconcile-sync                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Disable volume attach detach reconciler sync. Disabling this may cause volumes to be mismatched with pods. Use wisely.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| <code>--disable-force-detach-on-timeout</code>                                                                                                                                                                                                                                                                                              |
| Prevent force detaching volumes based on maximum unmount time and node status. If this flag is set to true, the non-graceful node shutdown feature must be used to recover from node failure. See <a href="https://k8s.io/docs/storage-disable-force-detach-on-timeout/">https://k8s.io/docs/storage-disable-force-detach-on-timeout/</a> . |
| <code>--disabled-metrics strings</code>                                                                                                                                                                                                                                                                                                     |
| This flag provides an escape hatch for misbehaving metrics. You must provide the fully qualified metric name in order to disable it. Disclaimer: disabling metrics is higher in precedence than showing hidden metrics.                                                                                                                     |
| <code>--enable-dynamic-provisioning</code> Default: true                                                                                                                                                                                                                                                                                    |
| Enable dynamic provisioning for environments that support it.                                                                                                                                                                                                                                                                               |
| <code>--enable-garbage-collector</code> Default: true                                                                                                                                                                                                                                                                                       |
| Enables the generic garbage collector. MUST be synced with the corresponding flag of the kube-apiserver.                                                                                                                                                                                                                                    |
| <code>--enable-hostpath-provisioner</code>                                                                                                                                                                                                                                                                                                  |
| Enable HostPath PV provisioning when running without a cloud provider. This allows testing and development of provisioning features. HostPath provisioning is not supported in any way, won't work in a multi-node cluster, and should not be used for anything other than testing or development.                                          |
| <code>--enable-leader-migration</code>                                                                                                                                                                                                                                                                                                      |
| Whether to enable controller leader migration.                                                                                                                                                                                                                                                                                              |
| <code>--endpoint-updates-batch-period duration</code>                                                                                                                                                                                                                                                                                       |
| The length of endpoint updates batching period. Processing of pod changes will be delayed by this duration to join them with potential upcoming updates and reduce the overall number of endpoints updates. Larger number = higher endpoint programming latency, but lower number of endpoints revision generated                           |
| <code>--endpointslice-updates-batch-period duration</code>                                                                                                                                                                                                                                                                                  |
| The length of endpoint slice updates batching period. Processing of pod changes will be delayed by this duration to join them with potential upcoming updates and reduce the overall number of endpoints updates. Larger number = higher endpoint programming latency, but lower number of endpoints revision generated                     |
| <code>--external-cloud-volume-plugin string</code>                                                                                                                                                                                                                                                                                          |

The plugin to use when cloud provider is set to external. Can be empty, should only be set when cloud-provider is external. Currently used to allow node-ipam-controller, persistentvolume-binder-controller, persistentvolume-expander-controller and attach-detach-controller to work for in tree cloud providers.

--feature-gates <comma-separated 'key=True|False' pairs>

A set of key=value pairs that describe feature gates for alpha/experimental features. Options are:

APIResponseCompression=true|false (BETA - default=true)  
APIServerIdentity=true|false (BETA - default=true)  
APIServerTracing=true|false (BETA - default=true)  
APIServingWithRoutine=true|false (BETA - default=true)  
AllAlpha=true|false (ALPHA - default=false)  
AllBeta=true|false (BETA - default=false)  
AnyVolumeDataSource=true|false (BETA - default=true)  
AppArmor=true|false (BETA - default=true)  
AppArmorFields=true|false (BETA - default=true)  
CPUManagerPolicyAlphaOptions=true|false (ALPHA - default=false)  
CPUManagerPolicyBetaOptions=true|false (BETA - default=true)  
CPUManagerPolicyOptions=true|false (BETA - default=true)  
CRDValidationRatcheting=true|false (BETA - default=true)  
CSIMigrationPortworx=true|false (BETA - default=false)  
CSIVolumeHealth=true|false (ALPHA - default=false)  
CloudControllerManagerWebhook=true|false (ALPHA - default=false)  
ClusterTrustBundle=true|false (ALPHA - default=false)  
ClusterTrustBundleProjection=true|false (ALPHA - default=false)  
ComponentSLIs=true|false (BETA - default=true)  
ConsistentListFromCache=true|false (ALPHA - default=false)  
ContainerCheckpoint=true|false (BETA - default=true)  
ContextualLogging=true|false (BETA - default=true)  
CronJobsScheduledAnnotation=true|false (BETA - default=true)  
CrossNamespaceVolumeDataSource=true|false (ALPHA - default=false)  
CustomCPUCFSQuotaPeriod=true|false (ALPHA - default=false)  
CustomResourceFieldSelectors=true|false (ALPHA - default=false)  
DevicePluginCDIDevices=true|false (BETA - default=true)  
DisableCloudProviders=true|false (BETA - default=true)  
DisableKubeletCloudCredentialProviders=true|false (BETA - default=true)  
DisableNodeKubeProxyVersion=true|false (ALPHA - default=false)  
DynamicResourceAllocation=true|false (ALPHA - default=false)  
ElasticIndexedJob=true|false (BETA - default=true)  
EventedPLEG=true|false (ALPHA - default=false)  
GracefulNodeShutdown=true|false (BETA - default=true)  
GracefulNodeShutdownBasedOnPodPriority=true|false (BETA - default=true)  
HPAScaleToZero=true|false (ALPHA - default=false)  
HonorPVReclaimPolicy=true|false (ALPHA - default=false)  
ImageMaximumGCAge=true|false (BETA - default=true)  
InPlacePodVerticalScaling=true|false (ALPHA - default=false)  
InTreePluginAWSUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureDiskUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureFileUnregister=true|false (ALPHA - default=false)  
InTreePluginGCEUnregister=true|false (ALPHA - default=false)

InTreePluginOpenStackUnregister=true|false (ALPHA - default=false)  
InTreePluginPortworxUnregister=true|false (ALPHA - default=false)  
InTreePluginvSphereUnregister=true|false (ALPHA - default=false)  
InformerResourceVersion=true|false (ALPHA - default=false)  
JobBackoffLimitPerIndex=true|false (BETA - default=true)  
JobManagedBy=true|false (ALPHA - default=false)  
JobPodFailurePolicy=true|false (BETA - default=true)  
JobPodReplacementPolicy=true|false (BETA - default=true)  
JobSuccessPolicy=true|false (ALPHA - default=false)  
KubeProxyDrainingTerminatingNodes=true|false (BETA - default=true)  
KubeletCgroupDriverFromCRI=true|false (ALPHA - default=false)  
KubeletInUserNamespace=true|false (ALPHA - default=false)  
KubeletPodResourcesDynamicResources=true|false (ALPHA - default=false)  
KubeletPodResourcesGet=true|false (ALPHA - default=false)  
KubeletSeparateDiskGC=true|false (ALPHA - default=false)  
KubeletTracing=true|false (BETA - default=true)  
LoadBalancerIPMode=true|false (BETA - default=true)  
LocalStorageCapacityIsolationFSQuotaMonitoring=true|false (ALPHA - default=false)  
LogarithmicScaleDown=true|false (BETA - default=true)  
LoggingAlphaOptions=true|false (ALPHA - default=false)  
LoggingBetaOptions=true|false (BETA - default=true)  
MatchLabelKeysInPodAffinity=true|false (ALPHA - default=false)  
MatchLabelKeysInPodTopologySpread=true|false (BETA - default=true)  
MaxUnavailableStatefulSet=true|false (ALPHA - default=false)  
MemoryManager=true|false (BETA - default=true)  
MemoryQoS=true|false (ALPHA - default=false)  
MultiCIDRServiceAllocator=true|false (ALPHA - default=false)  
MutatingAdmissionPolicy=true|false (ALPHA - default=false)  
NFTablesProxyMode=true|false (ALPHA - default=false)  
NodeInclusionPolicyInPodTopologySpread=true|false (BETA - default=true)  
NodeLogQuery=true|false (BETA - default=false)  
NodeSwap=true|false (BETA - default=true)  
OpenAPIEnums=true|false (BETA - default=true)  
PDBUnhealthyPodEvictionPolicy=true|false (BETA - default=true)  
PersistentVolumeLastPhaseTransitionTime=true|false (BETA - default=true)  
PodAndContainerStatsFromCRI=true|false (ALPHA - default=false)  
PodDeletionCost=true|false (BETA - default=true)  
PodDisruptionConditions=true|false (BETA - default=true)  
PodIndexLabel=true|false (BETA - default=true)  
PodLifecycleSleepAction=true|false (BETA - default=true)  
PodReadyToStartContainersCondition=true|false (BETA - default=true)  
PortForwardWebsockets=true|false (ALPHA - default=false)  
ProcMountType=true|false (ALPHA - default=false)  
QOSReserved=true|false (ALPHA - default=false)  
RecoverVolumeExpansionFailure=true|false (ALPHA - default=false)  
RecursiveReadOnlyMounts=true|false (ALPHA - default=false)  
RelaxedEnvironmentVariableValidation=true|false (ALPHA - default=false)  
RetryGenerateName=true|false (ALPHA - default=false)  
RotateKubeletServerCertificate=true|false (BETA - default=true)  
RuntimeClassInImageCriApi=true|false (ALPHA - default=false)  
SELinuxMount=true|false (ALPHA - default=false)  
SELinuxMountReadWriteOncePod=true|false (BETA - default=true)  
SchedulerQueueingHints=true|false (BETA - default=false)

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| SeparateCacheWatchRPC=true false (BETA - default=true)<br>SeparateTaintEvictionController=true false (BETA - default=true)<br>ServiceAccountTokenJTI=true false (BETA - default=true)<br>ServiceAccountTokenNodeBinding=true false (ALPHA - default=false)<br>ServiceAccountTokenNodeBindingValidation=true false (BETA - default=true)<br>ServiceAccountTokenPodNodeInfo=true false (BETA - default=true)<br>ServiceTrafficDistribution=true false (ALPHA - default=false)<br>SidecarContainers=true false (BETA - default=true)<br>SizeMemoryBackedVolumes=true false (BETA - default=true)<br>StatefulSetAutoDeletePVC=true false (BETA - default=true)<br>StatefulSetStartOrdinal=true false (BETA - default=true)<br>StorageNamespaceIndex=true false (BETA - default=true)<br>StorageVersionAPI=true false (ALPHA - default=false)<br>StorageVersionHash=true false (BETA - default=true)<br>StorageVersionMigrator=true false (ALPHA - default=false)<br>StructuredAuthenticationConfiguration=true false (BETA - default=true)<br>StructuredAuthorizationConfiguration=true false (BETA - default=true)<br>TopologyAwareHints=true false (BETA - default=true)<br>TopologyManagerPolicyAlphaOptions=true false (ALPHA - default=false)<br>TopologyManagerPolicyBetaOptions=true false (BETA - default=true)<br>TopologyManagerPolicyOptions=true false (BETA - default=true)<br>TranslateStreamCloseWebsocketRequests=true false (BETA - default=true)<br>UnauthenticatedHTTP2DOSMitigation=true false (BETA - default=true)<br>UnknownVersionInteroperabilityProxy=true false (ALPHA - default=false)<br>UserNamespacesPodSecurityStandards=true false (ALPHA - default=false)<br>UserNamespacesSupport=true false (BETA - default=false)<br>VolumeAttributesClass=true false (ALPHA - default=false)<br>VolumeCapacityPriority=true false (ALPHA - default=false)<br>WatchFromStorageWithoutResourceVersion=true false (BETA - default=false)<br>WatchList=true false (ALPHA - default=false)<br>WatchListClient=true false (BETA - default=false)<br>WinDSR=true false (ALPHA - default=false)<br>WinOverlay=true false (BETA - default=true)<br>WindowsHostNetwork=true false (ALPHA - default=true) |                                                                 |
| --flex-volume-plugin-dir string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Default: "/usr/libexec/kubernetes/kubelet-plugins/volume/exec/" |
| Full path of the directory in which the flex volume plugin should search for additional third party volume plugins.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                 |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                 |
| help for kube-controller-manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                 |
| --horizontal-pod-autoscaler-cpu-initialization-period duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Default: 5m0s                                                   |
| The period after pod start when CPU samples might be skipped.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |
| --horizontal-pod-autoscaler-downscale-stabilization duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Default: 5m0s                                                   |

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| The period for which autoscaler will look backwards and not scale down below any recommendation it made during that period.                                                                                                                                                                                                                                                             |
| --horizontal-pod-autoscaler-initial-readiness-delay duration    Default: 30s                                                                                                                                                                                                                                                                                                            |
| The period after pod start during which readiness changes will be treated as initial readiness.                                                                                                                                                                                                                                                                                         |
| --horizontal-pod-autoscaler-sync-period duration    Default: 15s                                                                                                                                                                                                                                                                                                                        |
| The period for syncing the number of pods in horizontal pod autoscaler.                                                                                                                                                                                                                                                                                                                 |
| --horizontal-pod-autoscaler-tolerance float    Default: 0.1                                                                                                                                                                                                                                                                                                                             |
| The minimum change (from 1.0) in the desired-to-actual metrics ratio for the horizontal pod autoscaler to consider scaling.                                                                                                                                                                                                                                                             |
| --http2-max-streams-per-connection int                                                                                                                                                                                                                                                                                                                                                  |
| The limit that the server gives to clients for the maximum number of streams in an HTTP/2 connection. Zero means to use golang's default.                                                                                                                                                                                                                                               |
| --kube-api-burst int32    Default: 30                                                                                                                                                                                                                                                                                                                                                   |
| Burst to use while talking with kubernetes apiserver.                                                                                                                                                                                                                                                                                                                                   |
| --kube-api-content-type string    Default: "application/vnd.kubernetes.protobuf"                                                                                                                                                                                                                                                                                                        |
| Content type of requests sent to apiserver.                                                                                                                                                                                                                                                                                                                                             |
| --kube-api-qps float    Default: 20                                                                                                                                                                                                                                                                                                                                                     |
| QPS to use while talking with kubernetes apiserver.                                                                                                                                                                                                                                                                                                                                     |
| --kubeconfig string                                                                                                                                                                                                                                                                                                                                                                     |
| Path to kubeconfig file with authorization and master location information (the master location can be overridden by the master flag).                                                                                                                                                                                                                                                  |
| --large-cluster-size-threshold int32    Default: 50                                                                                                                                                                                                                                                                                                                                     |
| Number of nodes from which node-lifecycle-controller treats the cluster as large for the eviction logic purposes. --secondary-node-eviction-rate is implicitly overridden to 0 for clusters this size or smaller. Notice: If nodes reside in multiple zones, this threshold will be considered as zone node size threshold for each zone to determine node eviction rate independently. |
| --leader-elect    Default: true                                                                                                                                                                                                                                                                                                                                                         |
| Start a leader election client and gain leadership before executing the main loop. Enable this when running replicated components for high availability.                                                                                                                                                                                                                                |

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| <code>--leader-elect-lease-duration</code> duration    Default: 15s                                                                                                                                                                                                                                                                        |
| The duration that non-leader candidates will wait after observing a leadership renewal until attempting to acquire leadership of a led but unrenewed leader slot. This is effectively the maximum duration that a leader can be stopped before it is replaced by another candidate. This is only applicable if leader election is enabled. |
| <code>--leader-elect-renew-deadline</code> duration    Default: 10s                                                                                                                                                                                                                                                                        |
| The interval between attempts by the acting master to renew a leadership slot before it stops leading. This must be less than the lease duration. This is only applicable if leader election is enabled.                                                                                                                                   |
| <code>--leader-elect-resource-lock</code> string    Default: "leases"                                                                                                                                                                                                                                                                      |
| The type of resource object that is used for locking during leader election. Supported options are 'leases', 'endpointsleases' and 'configmapsleases'.                                                                                                                                                                                     |
| <code>--leader-elect-resource-name</code> string    Default: "kube-controller-manager"                                                                                                                                                                                                                                                     |
| The name of resource object that is used for locking during leader election.                                                                                                                                                                                                                                                               |
| <code>--leader-elect-resource-namespace</code> string    Default: "kube-system"                                                                                                                                                                                                                                                            |
| The namespace of resource object that is used for locking during leader election.                                                                                                                                                                                                                                                          |
| <code>--leader-elect-retry-period</code> duration    Default: 2s                                                                                                                                                                                                                                                                           |
| The duration the clients should wait between attempting acquisition and renewal of a leadership. This is only applicable if leader election is enabled.                                                                                                                                                                                    |
| <code>--leader-migration-config</code> string                                                                                                                                                                                                                                                                                              |
| Path to the config file for controller leader migration, or empty to use the value that reflects default configuration of the controller manager. The config file should be of type LeaderMigrationConfiguration, group controllermanager.config.k8s.io, version v1alpha1.                                                                 |
| <code>--legacy-service-account-token-clean-up-period</code> duration    Default: 8760h0m0s                                                                                                                                                                                                                                                 |
| The period of time since the last usage of an legacy service account token before it can be deleted.                                                                                                                                                                                                                                       |
| <code>--log-flush-frequency</code> duration    Default: 5s                                                                                                                                                                                                                                                                                 |
| Maximum number of seconds between log flushes                                                                                                                                                                                                                                                                                              |
| <code>--log-text-info-buffer-size</code> quantity                                                                                                                                                                                                                                                                                          |



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| <p>[Alpha] In text format with split output streams, the info messages can be buffered for a while to increase performance. The default value of zero bytes disables buffering. The size can be specified as number of bytes (512), multiples of 1000 (1K), multiples of 1024 (2Ki), or powers of those (3M, 4G, 5Mi, 6Gi). Enable the LoggingAlphaOptions feature gate to use this.</p> |
| <p>--log-text-split-stream</p>                                                                                                                                                                                                                                                                                                                                                           |
| <p>[Alpha] In text format, write error messages to stderr and info messages to stdout. The default is to write a single stream to stdout. Enable the LoggingAlphaOptions feature gate to use this.</p>                                                                                                                                                                                   |
| <p>--logging-format string    Default: "text"</p>                                                                                                                                                                                                                                                                                                                                        |
| <p>Sets the log format. Permitted formats: "text".</p>                                                                                                                                                                                                                                                                                                                                   |
| <p>--master string</p>                                                                                                                                                                                                                                                                                                                                                                   |
| <p>The address of the Kubernetes API server (overrides any value in kubeconfig).</p>                                                                                                                                                                                                                                                                                                     |
| <p>--max-endpoints-per-slice int32    Default: 100</p>                                                                                                                                                                                                                                                                                                                                   |
| <p>The maximum number of endpoints that will be added to an EndpointSlice. More endpoints per slice will result in less endpoint slices, but larger resources. Defaults to 100.</p>                                                                                                                                                                                                      |
| <p>--min-resync-period duration    Default: 12h0m0s</p>                                                                                                                                                                                                                                                                                                                                  |
| <p>The resync period in reflectors will be random between MinResyncPeriod and 2*MinResyncPeriod.</p>                                                                                                                                                                                                                                                                                     |
| <p>--mirroring-concurrent-service-endpoint-syncs int32    Default: 5</p>                                                                                                                                                                                                                                                                                                                 |
| <p>The number of service endpoint syncing operations that will be done concurrently by the endpointslice-mirroring-controller. Larger number = faster endpoint slice updating, but more CPU (and network) load. Defaults to 5.</p>                                                                                                                                                       |
| <p>--mirroring-endpointslice-updates-batch-period duration</p>                                                                                                                                                                                                                                                                                                                           |
| <p>The length of EndpointSlice updates batching period for endpointslice-mirroring-controller. Processing of EndpointSlice changes will be delayed by this duration to join them with potential upcoming updates and reduce the overall number of EndpointSlice updates. Larger number = higher endpoint programming latency, but lower number of endpoints revision generated</p>       |
| <p>--mirroring-max-endpoints-per-subset int32    Default: 1000</p>                                                                                                                                                                                                                                                                                                                       |
| <p>The maximum number of endpoints that will be added to an EndpointSlice by the endpointslice-mirroring-controller. More endpoints per slice will result in less endpoint slices, but larger resources. Defaults to 100.</p>                                                                                                                                                            |
| <p>--namespace-sync-period duration    Default: 5m0s</p>                                                                                                                                                                                                                                                                                                                                 |

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| The period for syncing namespace life-cycle updates                                                                                                                                                                                  |
| --node-cidr-mask-size int32                                                                                                                                                                                                          |
| Mask size for node cidr in cluster. Default is 24 for IPv4 and 64 for IPv6.                                                                                                                                                          |
| --node-cidr-mask-size-ipv4 int32                                                                                                                                                                                                     |
| Mask size for IPv4 node cidr in dual-stack cluster. Default is 24.                                                                                                                                                                   |
| --node-cidr-mask-size-ipv6 int32                                                                                                                                                                                                     |
| Mask size for IPv6 node cidr in dual-stack cluster. Default is 64.                                                                                                                                                                   |
| --node-eviction-rate float    Default: 0.1                                                                                                                                                                                           |
| Number of nodes per second on which pods are deleted in case of node failure when a zone is healthy (see --unhealthy-zone-threshold for definition of healthy/unhealthy). Zone refers to entire cluster in non-multizone clusters.   |
| --node-monitor-grace-period duration    Default: 40s                                                                                                                                                                                 |
| Amount of time which we allow running Node to be unresponsive before marking it unhealthy. Must be N times more than kubelet's nodeStatusUpdateFrequency, where N means number of retries allowed for kubelet to post node status.   |
| --node-monitor-period duration    Default: 5s                                                                                                                                                                                        |
| The period for syncing NodeStatus in cloud-node-lifecycle-controller.                                                                                                                                                                |
| --node-startup-grace-period duration    Default: 1m0s                                                                                                                                                                                |
| Amount of time which we allow starting Node to be unresponsive before marking it unhealthy.                                                                                                                                          |
| --permit-address-sharing                                                                                                                                                                                                             |
| If true, SO_REUSEADDR will be used when binding the port. This allows binding to wildcard IPs like 0.0.0.0 and specific IPs in parallel, and it avoids waiting for the kernel to release sockets in TIME_WAIT state. [default=false] |
| --permit-port-sharing                                                                                                                                                                                                                |
| If true, SO_REUSEPORT will be used when binding the port, which allows more than one instance to bind on the same address and port. [default=false]                                                                                  |
| --profiling    Default: true                                                                                                                                                                                                         |
| Enable profiling via web interface host:port/debug/pprof/                                                                                                                                                                            |
| --pv-recycler-increment-timeout-nfs int32    Default: 30                                                                                                                                                                             |

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| the increment of time added per Gi to ActiveDeadlineSeconds for an NFS scrubber pod                                                                                                                                                                               |
| --pv-recycler-minimum-timeout-hostpath int32    Default: 60                                                                                                                                                                                                       |
| The minimum ActiveDeadlineSeconds to use for a HostPath Recycler pod. This is for development and testing only and will not work in a multi-node cluster.                                                                                                         |
| --pv-recycler-minimum-timeout-nfs int32    Default: 300                                                                                                                                                                                                           |
| The minimum ActiveDeadlineSeconds to use for an NFS Recycler pod                                                                                                                                                                                                  |
| --pv-recycler-pod-template-filepath-hostpath string                                                                                                                                                                                                               |
| The file path to a pod definition used as a template for HostPath persistent volume recycling. This is for development and testing only and will not work in a multi-node cluster.                                                                                |
| --pv-recycler-pod-template-filepath-nfs string                                                                                                                                                                                                                    |
| The file path to a pod definition used as a template for NFS persistent volume recycling                                                                                                                                                                          |
| --pv-recycler-timeout-increment-hostpath int32    Default: 30                                                                                                                                                                                                     |
| the increment of time added per Gi to ActiveDeadlineSeconds for a HostPath scrubber pod. This is for development and testing only and will not work in a multi-node cluster.                                                                                      |
| --pvclaimbinder-sync-period duration    Default: 15s                                                                                                                                                                                                              |
| The period for syncing persistent volumes and persistent volume claims                                                                                                                                                                                            |
| --requestheader-allowed-names strings                                                                                                                                                                                                                             |
| List of client certificate common names to allow to provide usernames in headers specified by --requestheader-username-headers. If empty, any client certificate validated by the authorities in --requestheader-client-ca-file is allowed.                       |
| --requestheader-client-ca-file string                                                                                                                                                                                                                             |
| Root certificate bundle to use to verify client certificates on incoming requests before trusting usernames in headers specified by --requestheader-username-headers. WARNING: generally do not depend on authorization being already done for incoming requests. |
| --requestheader-extra-headers-prefix strings    Default: "x-remote-extra-"                                                                                                                                                                                        |
| List of request header prefixes to inspect. X-Remote-Extra- is suggested.                                                                                                                                                                                         |
| --requestheader-group-headers strings    Default: "x-remote-group"                                                                                                                                                                                                |
| List of request headers to inspect for groups. X-Remote-Group is suggested.                                                                                                                                                                                       |
| --requestheader-username-headers strings    Default: "x-remote-user"                                                                                                                                                                                              |

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| List of request headers to inspect for usernames. X-Remote-User is common.                                                                                                                                                                                                                                                                                                                                      |
| --resource-quota-sync-period duration    Default: 5m0s                                                                                                                                                                                                                                                                                                                                                          |
| The period for syncing quota usage status in the system                                                                                                                                                                                                                                                                                                                                                         |
| --root-ca-file string                                                                                                                                                                                                                                                                                                                                                                                           |
| If set, this root certificate authority will be included in service account's token secret. This must be a valid PEM-encoded CA bundle.                                                                                                                                                                                                                                                                         |
| --route-reconciliation-period duration    Default: 10s                                                                                                                                                                                                                                                                                                                                                          |
| The period for reconciling routes created for Nodes by cloud provider.                                                                                                                                                                                                                                                                                                                                          |
| --secondary-node-eviction-rate float    Default: 0.01                                                                                                                                                                                                                                                                                                                                                           |
| Number of nodes per second on which pods are deleted in case of node failure when a zone is unhealthy (see --unhealthy-zone-threshold for definition of healthy/unhealthy). Zone refers to entire cluster in non-multizone clusters. This value is implicitly overridden to 0 if the cluster size is smaller than --large-cluster-size-threshold.                                                               |
| --secure-port int    Default: 10257                                                                                                                                                                                                                                                                                                                                                                             |
| The port on which to serve HTTPS with authentication and authorization. If 0, don't serve HTTPS at all.                                                                                                                                                                                                                                                                                                         |
| --service-account-private-key-file string                                                                                                                                                                                                                                                                                                                                                                       |
| Filename containing a PEM-encoded private RSA or ECDSA key used to sign service account tokens.                                                                                                                                                                                                                                                                                                                 |
| --service-cluster-ip-range string                                                                                                                                                                                                                                                                                                                                                                               |
| CIDR Range for Services in cluster. Requires --allocate-node-cidrs to be true                                                                                                                                                                                                                                                                                                                                   |
| --show-hidden-metrics-for-version string                                                                                                                                                                                                                                                                                                                                                                        |
| The previous version for which you want to show hidden metrics. Only the previous minor version is meaningful, other values will not be allowed. The format is <major>.<minor>, e.g.: '1.16'. The purpose of this format is make sure you have the opportunity to notice if the next release hides additional metrics, rather than being surprised when they are permanently removed in the release after that. |
| --terminated-pod-gc-threshold int32    Default: 12500                                                                                                                                                                                                                                                                                                                                                           |
| Number of terminated pods that can exist before the terminated pod garbage collector starts deleting terminated pods. If <= 0, the terminated pod garbage collector is disabled.                                                                                                                                                                                                                                |
| --tls-cert-file string                                                                                                                                                                                                                                                                                                                                                                                          |

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| File containing the default x509 Certificate for HTTPS. (CA cert, if any, concatenated after server cert). If HTTPS serving is enabled, and --tls-cert-file and --tls-private-key-file are not provided, a self-signed certificate and key are generated for the public address and saved to the directory specified by --cert-dir.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| --tls-cipher-suites strings<br><p>Comma-separated list of cipher suites for the server. If omitted, the default Go cipher suites will be used.</p> <p>Preferred values: TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384, TLS_CHACHA20_POLY1305_SHA256, TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA, TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256, TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA, TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384, TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305, TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256, TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256, TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384, TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305, TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256.</p> <p>Insecure values: TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256, TLS_ECDHE_ECDSA_WITH_RC4_128_SHA, TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256, TLS_ECDHE_RSA_WITH_RC4_128_SHA, TLS_RSA_WITH_3DES_EDE_CBC_SHA, TLS_RSA_WITH_AES_128_CBC_SHA, TLS_RSA_WITH_AES_128_CBC_SHA256, TLS_RSA_WITH_AES_128_GCM_SHA256, TLS_RSA_WITH_AES_256_CBC_SHA, TLS_RSA_WITH_AES_256_GCM_SHA384, TLS_RSA_WITH_RC4_128_SHA.</p> |
| --tls-min-version string<br><p>Minimum TLS version supported. Possible values: VersionTLS10, VersionTLS11, VersionTLS12, VersionTLS13</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| --tls-private-key-file string<br><p>File containing the default x509 private key matching --tls-cert-file.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --tls-sni-cert-key string<br><p>A pair of x509 certificate and private key file paths, optionally suffixed with a list of domain patterns which are fully qualified domain names, possibly with prefixed wildcard segments. The domain patterns also allow IP addresses, but IPs should only be used if the apiserver has visibility to the IP address requested by a client. If no domain patterns are provided, the names of the certificate are extracted. Non-wildcard matches trump over wildcard matches, explicit domain patterns trump over extracted names. For multiple key/certificate pairs, use the --tls-sni-cert-key multiple times. Examples: "example.crt,example.key" or "foo.crt,foo.key:*.foo.com,foo.com".</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| <code>--unhealthy-zone-threshold float</code>                                                                                                        | Default: 0.55 |
| Fraction of Nodes in a zone which needs to be not Ready (minimum 3) for zone to be treated as unhealthy.                                             |               |
| <code>--use-service-account-credentials</code>                                                                                                       |               |
| If true, use individual service account credentials for each controller.                                                                             |               |
| <code>-v, --v int</code>                                                                                                                             |               |
| number for the log level verbosity                                                                                                                   |               |
| <code>--version version[=true]</code>                                                                                                                |               |
| <code>--version</code> , <code>--version=raw</code> prints version information and quits; <code>--version=vX.Y.Z...</code> sets the reported version |               |
| <code>--vmodule pattern=N,...</code>                                                                                                                 |               |
| comma-separated list of pattern=N settings for file-filtered logging (only works for text log format)                                                |               |

# kube-proxy

## Synopsis

The Kubernetes network proxy runs on each node. This reflects services as defined in the Kubernetes API on each node and can do simple TCP, UDP, and SCTP stream forwarding or round robin TCP, UDP, and SCTP forwarding across a set of backends. Service cluster IPs and ports are currently found through Docker-links-compatible environment variables specifying ports opened by the service proxy. There is an optional addon that provides cluster DNS for these cluster IPs. The user must create a service with the apiserver API to configure the proxy.

kube-proxy [flags]

## Options

|                                                                                         |                  |
|-----------------------------------------------------------------------------------------|------------------|
| <code>--add_dir_header</code>                                                           |                  |
| If true, adds the file directory to the header of the log messages                      |                  |
| <code>--alsologtostderr</code>                                                          |                  |
| log to standard error as well as files (no effect when <code>-logtostderr=true</code> ) |                  |
| <code>--bind-address string</code>                                                      | Default: 0.0.0.0 |

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| Overrides kube-proxy's idea of what its node's primary IP is. Note that the name is a historical artifact, and kube-proxy does not actually bind any sockets to this IP. This parameter is ignored if a config file is specified by --config.                                                                                                                         |
| --bind-address-hard-fail                                                                                                                                                                                                                                                                                                                                              |
| If true kube-proxy will treat failure to bind to a port as fatal and exit                                                                                                                                                                                                                                                                                             |
| --cleanup                                                                                                                                                                                                                                                                                                                                                             |
| If true cleanup iptables and ipvs rules and exit.                                                                                                                                                                                                                                                                                                                     |
| --cluster-cidr string                                                                                                                                                                                                                                                                                                                                                 |
| The CIDR range of the pods in the cluster. (For dual-stack clusters, this can be a comma-separated dual-stack pair of CIDR ranges.). When --detect-local-mode is set to ClusterCIDR, kube-proxy will consider traffic to be local if its source IP is in this range. (Otherwise it is not used.) This parameter is ignored if a config file is specified by --config. |
| --config string                                                                                                                                                                                                                                                                                                                                                       |
| The path to the configuration file.                                                                                                                                                                                                                                                                                                                                   |
| --config-sync-period duration    Default: 15m0s                                                                                                                                                                                                                                                                                                                       |
| How often configuration from the apiserver is refreshed. Must be greater than 0.                                                                                                                                                                                                                                                                                      |
| --conntrack-max-per-core int32    Default: 32768                                                                                                                                                                                                                                                                                                                      |
| Maximum number of NAT connections to track per CPU core (0 to leave the limit as-is and ignore conntrack-min).                                                                                                                                                                                                                                                        |
| --conntrack-min int32    Default: 131072                                                                                                                                                                                                                                                                                                                              |
| Minimum number of conntrack entries to allocate, regardless of conntrack-max-per-core (set conntrack-max-per-core=0 to leave the limit as-is).                                                                                                                                                                                                                        |
| --conntrack-tcp-be-liberal                                                                                                                                                                                                                                                                                                                                            |
| Enable liberal mode for tracking TCP packets by setting nf_conntrack_tcp_be_liberal to 1                                                                                                                                                                                                                                                                              |
| --conntrack-tcp-timeout-close-wait duration    Default: 1h0m0s                                                                                                                                                                                                                                                                                                        |
| NAT timeout for TCP connections in the CLOSE_WAIT state                                                                                                                                                                                                                                                                                                               |
| --conntrack-tcp-timeout-established duration    Default: 24h0m0s                                                                                                                                                                                                                                                                                                      |
| Idle timeout for established TCP connections (0 to leave as-is)                                                                                                                                                                                                                                                                                                       |
| --conntrack-udp-timeout duration                                                                                                                                                                                                                                                                                                                                      |

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| Idle timeout for UNREPLIED UDP connections (0 to leave as-is)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| --conntrack-udp-timeout-stream duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Idle timeout for ASSURED UDP connections (0 to leave as-is)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| --detect-local-mode LocalMode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Mode to use to detect local traffic. This parameter is ignored if a config file is specified by --config.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| --feature-gates <comma-separated 'key=True False' pairs>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>A set of key=value pairs that describe feature gates for alpha/experimental features. Options are:</p> <p>APIResponseCompression=true false (BETA - default=true)</p> <p>APIServerIdentity=true false (BETA - default=true)</p> <p>APIServerTracing=true false (BETA - default=true)</p> <p>APIServingWithRoutine=true false (BETA - default=true)</p> <p>AllAlpha=true false (ALPHA - default=false)</p> <p>AllBeta=true false (BETA - default=false)</p> <p>AnyVolumeDataSource=true false (BETA - default=true)</p> <p>AppArmor=true false (BETA - default=true)</p> <p>AppArmorFields=true false (BETA - default=true)</p> <p>CPUManagerPolicyAlphaOptions=true false (ALPHA - default=false)</p> <p>CPUManagerPolicyBetaOptions=true false (BETA - default=true)</p> <p>CPUManagerPolicyOptions=true false (BETA - default=true)</p> <p>CRDValidationRatcheting=true false (BETA - default=true)</p> <p>CSIMigrationPortworx=true false (BETA - default=false)</p> <p>CSIVolumeHealth=true false (ALPHA - default=false)</p> <p>CloudControllerManagerWebhook=true false (ALPHA - default=false)</p> <p>ClusterTrustBundle=true false (ALPHA - default=false)</p> <p>ClusterTrustBundleProjection=true false (ALPHA - default=false)</p> <p>ComponentSLIs=true false (BETA - default=true)</p> <p>ConsistentListFromCache=true false (ALPHA - default=false)</p> <p>ContainerCheckpoint=true false (BETA - default=true)</p> <p>ContextualLogging=true false (BETA - default=true)</p> <p>CronJobsScheduledAnnotation=true false (BETA - default=true)</p> <p>CrossNamespaceVolumeDataSource=true false (ALPHA - default=false)</p> <p>CustomCPUCFSQuotaPeriod=true false (ALPHA - default=false)</p> <p>CustomResourceFieldSelectors=true false (ALPHA - default=false)</p> <p>DevicePluginCDIDevices=true false (BETA - default=true)</p> <p>DisableCloudProviders=true false (BETA - default=true)</p> <p>DisableKubeletCloudCredentialProviders=true false (BETA - default=true)</p> <p>DisableNodeKubeProxyVersion=true false (ALPHA - default=false)</p> <p>DynamicResourceAllocation=true false (ALPHA - default=false)</p> <p>ElasticIndexedJob=true false (BETA - default=true)</p> <p>EventedPLEG=true false (ALPHA - default=false)</p> <p>GracefulNodeShutdown=true false (BETA - default=true)</p> <p>GracefulNodeShutdownBasedOnPodPriority=true false (BETA - default=true)</p> <p>HPAScaleToZero=true false (ALPHA - default=false)</p> <p>HonorPVReclaimPolicy=true false (ALPHA - default=false)</p> |



ImageMaximumGCAge=true|false (BETA - default=true)  
InPlacePodVerticalScaling=true|false (ALPHA - default=false)  
InTreePluginAWSUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureDiskUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureFileUnregister=true|false (ALPHA - default=false)  
InTreePluginGCEUnregister=true|false (ALPHA - default=false)  
InTreePluginOpenStackUnregister=true|false (ALPHA - default=false)  
InTreePluginPortworxUnregister=true|false (ALPHA - default=false)  
InTreePluginvSphereUnregister=true|false (ALPHA - default=false)  
InformerResourceVersion=true|false (ALPHA - default=false)  
JobBackoffLimitPerIndex=true|false (BETA - default=true)  
JobManagedBy=true|false (ALPHA - default=false)  
JobPodFailurePolicy=true|false (BETA - default=true)  
JobPodReplacementPolicy=true|false (BETA - default=true)  
JobSuccessPolicy=true|false (ALPHA - default=false)  
KubeProxyDrainingTerminatingNodes=true|false (BETA - default=true)  
KubeletCgroupDriverFromCRI=true|false (ALPHA - default=false)  
KubeletInUserNamespace=true|false (ALPHA - default=false)  
KubeletPodResourcesDynamicResources=true|false (ALPHA - default=false)  
KubeletPodResourcesGet=true|false (ALPHA - default=false)  
KubeletSeparateDiskGC=true|false (ALPHA - default=false)  
KubeletTracing=true|false (BETA - default=true)  
LoadBalancerIPMode=true|false (BETA - default=true)  
LocalStorageCapacityIsolationFSQuotaMonitoring=true|false (ALPHA - default=false)  
LogarithmicScaleDown=true|false (BETA - default=true)  
LoggingAlphaOptions=true|false (ALPHA - default=false)  
LoggingBetaOptions=true|false (BETA - default=true)  
MatchLabelKeysInPodAffinity=true|false (ALPHA - default=false)  
MatchLabelKeysInPodTopologySpread=true|false (BETA - default=true)  
MaxUnavailableStatefulSet=true|false (ALPHA - default=false)  
MemoryManager=true|false (BETA - default=true)  
MemoryQoS=true|false (ALPHA - default=false)  
MultiCIDRServiceAllocator=true|false (ALPHA - default=false)  
MutatingAdmissionPolicy=true|false (ALPHA - default=false)  
NFTablesProxyMode=true|false (ALPHA - default=false)  
NodeInclusionPolicyInPodTopologySpread=true|false (BETA - default=true)  
NodeLogQuery=true|false (BETA - default=false)  
NodeSwap=true|false (BETA - default=true)  
OpenAPIEnums=true|false (BETA - default=true)  
PDBUnhealthyPodEvictionPolicy=true|false (BETA - default=true)  
PersistentVolumeLastPhaseTransitionTime=true|false (BETA - default=true)  
PodAndContainerStatsFromCRI=true|false (ALPHA - default=false)  
PodDeletionCost=true|false (BETA - default=true)  
PodDisruptionConditions=true|false (BETA - default=true)  
PodIndexLabel=true|false (BETA - default=true)  
PodLifecycleSleepAction=true|false (BETA - default=true)  
PodReadyToStartContainersCondition=true|false (BETA - default=true)  
PortForwardWebsockets=true|false (ALPHA - default=false)  
ProcMountType=true|false (ALPHA - default=false)  
QOSReserved=true|false (ALPHA - default=false)  
RecoverVolumeExpansionFailure=true|false (ALPHA - default=false)  
RecursiveReadOnlyMounts=true|false (ALPHA - default=false)  
RelaxedEnvironmentVariableValidation=true|false (ALPHA - default=false)

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| <p> RetryGenerateName=true false (ALPHA - default=false)<br/> RotateKubeletServerCertificate=true false (BETA - default=true)<br/> RuntimeClassInImageCriApi=true false (ALPHA - default=false)<br/> SELinuxMount=true false (ALPHA - default=false)<br/> SELinuxMountReadWriteOncePod=true false (BETA - default=true)<br/> SchedulerQueueingHints=true false (BETA - default=false)<br/> SeparateCacheWatchRPC=true false (BETA - default=true)<br/> SeparateTaintEvictionController=true false (BETA - default=true)<br/> ServiceAccountTokenJTI=true false (BETA - default=true)<br/> ServiceAccountTokenNodeBinding=true false (ALPHA - default=false)<br/> ServiceAccountTokenNodeBindingValidation=true false (BETA - default=true)<br/> ServiceAccountTokenPodNodeInfo=true false (BETA - default=true)<br/> ServiceTrafficDistribution=true false (ALPHA - default=false)<br/> SidecarContainers=true false (BETA - default=true)<br/> SizeMemoryBackedVolumes=true false (BETA - default=true)<br/> StatefulSetAutoDeletePVC=true false (BETA - default=true)<br/> StatefulSetStartOrdinal=true false (BETA - default=true)<br/> StorageNamespaceIndex=true false (BETA - default=true)<br/> StorageVersionAPI=true false (ALPHA - default=false)<br/> StorageVersionHash=true false (BETA - default=true)<br/> StorageVersionMigrator=true false (ALPHA - default=false)<br/> StructuredAuthenticationConfiguration=true false (BETA - default=true)<br/> StructuredAuthorizationConfiguration=true false (BETA - default=true)<br/> TopologyAwareHints=true false (BETA - default=true)<br/> TopologyManagerPolicyAlphaOptions=true false (ALPHA - default=false)<br/> TopologyManagerPolicyBetaOptions=true false (BETA - default=true)<br/> TopologyManagerPolicyOptions=true false (BETA - default=true)<br/> TranslateStreamCloseWebsocketRequests=true false (BETA - default=true)<br/> UnauthenticatedHTTP2DOSMitigation=true false (BETA - default=true)<br/> UnknownVersionInteroperabilityProxy=true false (ALPHA - default=false)<br/> UserNamespacesPodSecurityStandards=true false (ALPHA - default=false)<br/> UserNamespacesSupport=true false (BETA - default=false)<br/> VolumeAttributesClass=true false (ALPHA - default=false)<br/> VolumeCapacityPriority=true false (ALPHA - default=false)<br/> WatchFromStorageWithoutResourceVersion=true false (BETA - default=false)<br/> WatchList=true false (ALPHA - default=false)<br/> WatchListClient=true false (BETA - default=false)<br/> WinDSR=true false (ALPHA - default=false)<br/> WinOverlay=true false (BETA - default=true)<br/> WindowsHostNetwork=true false (ALPHA - default=true)<br/> This parameter is ignored if a config file is specified by --config. </p> |
| --healthz-bind-address ipport    Default: 0.0.0.0:10256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p> The IP address and port for the health check server to serve on, defaulting to "0.0.0.0:10256" (if --bind-address is unset or IPv4), or "[::]:10256" (if --bind-address is IPv6). Set empty to disable. This parameter is ignored if a config file is specified by --config. </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| -h, --help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| help for kube-proxy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --hostname-override string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| If non-empty, will be used as the name of the Node that kube-proxy is running on. If unset, the node name is assumed to be the same as the node's hostname.                                                                            |
| --init-only                                                                                                                                                                                                                            |
| If true, perform any initialization steps that must be done with full root privileges, and then exit. After doing this, you can run kube-proxy again with only the CAP_NET_ADMIN capability.                                           |
| --iptables-localhost-nodeports    Default: true                                                                                                                                                                                        |
| If false, kube-proxy will disable the legacy behavior of allowing NodePort services to be accessed via localhost. (Applies only to iptables mode and IPv4; localhost NodePorts are never allowed with other proxy modes or with IPv6.) |
| --iptables-masquerade-bit int32    Default: 14                                                                                                                                                                                         |
| If using the iptables or ipvs proxy mode, the bit of the fwmark space to mark packets requiring SNAT with. Must be within the range [0, 31].                                                                                           |
| --iptables-min-sync-period duration    Default: 1s                                                                                                                                                                                     |
| The minimum period between iptables rule resyncs (e.g. '5s', '1m', '2h22m'). A value of 0 means every Service or EndpointSlice change will result in an immediate iptables resync.                                                     |
| --iptables-sync-period duration    Default: 30s                                                                                                                                                                                        |
| An interval (e.g. '5s', '1m', '2h22m') indicating how frequently various re-synchronizing and cleanup operations are performed. Must be greater than 0.                                                                                |
| --ipvs-exclude-cidrs strings                                                                                                                                                                                                           |
| A comma-separated list of CIDRs which the ipvs proxier should not touch when cleaning up IPVS rules.                                                                                                                                   |
| --ipvs-min-sync-period duration                                                                                                                                                                                                        |
| The minimum period between IPVS rule resyncs (e.g. '5s', '1m', '2h22m'). A value of 0 means every Service or EndpointSlice change will result in an immediate IPVS resync.                                                             |
| --ipvs-scheduler string                                                                                                                                                                                                                |
| The ipvs scheduler type when proxy mode is ipvs                                                                                                                                                                                        |
| --ipvs-strict-arp                                                                                                                                                                                                                      |
| Enable strict ARP by setting arp_ignore to 1 and arp_announce to 2                                                                                                                                                                     |
| --ipvs-sync-period duration    Default: 30s                                                                                                                                                                                            |

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| An interval (e.g. '5s', '1m', '2h22m') indicating how frequently various re-synchronizing and cleanup operations are performed. Must be greater than 0.                                                                                                                                                                                                                           |
| <code>--ipvs-tcp-timeout duration</code>                                                                                                                                                                                                                                                                                                                                          |
| The timeout for idle IPVS TCP connections, 0 to leave as-is. (e.g. '5s', '1m', '2h22m').                                                                                                                                                                                                                                                                                          |
| <code>--ipvs-tcpfin-timeout duration</code>                                                                                                                                                                                                                                                                                                                                       |
| The timeout for IPVS TCP connections after receiving a FIN packet, 0 to leave as-is. (e.g. '5s', '1m', '2h22m').                                                                                                                                                                                                                                                                  |
| <code>--ipvs-udp-timeout duration</code>                                                                                                                                                                                                                                                                                                                                          |
| The timeout for IPVS UDP packets, 0 to leave as-is. (e.g. '5s', '1m', '2h22m').                                                                                                                                                                                                                                                                                                   |
| <code>--kube-api-burst int32</code> Default: 10                                                                                                                                                                                                                                                                                                                                   |
| Burst to use while talking with kubernetes apiserver                                                                                                                                                                                                                                                                                                                              |
| <code>--kube-api-content-type string</code> Default: "application/vnd.kubernetes.protobuf"                                                                                                                                                                                                                                                                                        |
| Content type of requests sent to apiserver.                                                                                                                                                                                                                                                                                                                                       |
| <code>--kube-api-qps float</code> Default: 5                                                                                                                                                                                                                                                                                                                                      |
| QPS to use while talking with kubernetes apiserver                                                                                                                                                                                                                                                                                                                                |
| <code>--kubeconfig string</code>                                                                                                                                                                                                                                                                                                                                                  |
| Path to kubeconfig file with authorization information (the master location can be overridden by the master flag).                                                                                                                                                                                                                                                                |
| <code>--log-flush-frequency duration</code> Default: 5s                                                                                                                                                                                                                                                                                                                           |
| Maximum number of seconds between log flushes                                                                                                                                                                                                                                                                                                                                     |
| <code>--log-text-info-buffer-size quantity</code>                                                                                                                                                                                                                                                                                                                                 |
| [Alpha] In text format with split output streams, the info messages can be buffered for a while to increase performance. The default value of zero bytes disables buffering. The size can be specified as number of bytes (512), multiples of 1000 (1K), multiples of 1024 (2Ki), or powers of those (3M, 4G, 5Mi, 6Gi). Enable the LoggingAlphaOptions feature gate to use this. |
| <code>--log-text-split-stream</code>                                                                                                                                                                                                                                                                                                                                              |
| [Alpha] In text format, write error messages to stderr and info messages to stdout. The default is to write a single stream to stdout. Enable the LoggingAlphaOptions feature gate to use this.                                                                                                                                                                                   |
| <code>--log_backtrace_at &lt;a string in the form 'file:N'&gt;</code> Default: :0                                                                                                                                                                                                                                                                                                 |

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| when logging hits line file:N, emit a stack trace                                                                                                                                                                                                                                                                                             |
| --log_dir string                                                                                                                                                                                                                                                                                                                              |
| If non-empty, write log files in this directory (no effect when -logtostderr=true)                                                                                                                                                                                                                                                            |
| --log_file string                                                                                                                                                                                                                                                                                                                             |
| If non-empty, use this log file (no effect when -logtostderr=true)                                                                                                                                                                                                                                                                            |
| --log_file_max_size uint   Default: 1800                                                                                                                                                                                                                                                                                                      |
| Defines the maximum size a log file can grow to (no effect when -logtostderr=true). Unit is megabytes. If the value is 0, the maximum file size is unlimited.                                                                                                                                                                                 |
| --logging-format string   Default: "text"                                                                                                                                                                                                                                                                                                     |
| Sets the log format. Permitted formats: "text".                                                                                                                                                                                                                                                                                               |
| --logtostderr   Default: true                                                                                                                                                                                                                                                                                                                 |
| log to standard error instead of files                                                                                                                                                                                                                                                                                                        |
| --masquerade-all                                                                                                                                                                                                                                                                                                                              |
| If using the iptables or ipvs proxy mode, SNAT all traffic sent via Service cluster IPs. This may be required with some CNI plugins.                                                                                                                                                                                                          |
| --master string                                                                                                                                                                                                                                                                                                                               |
| The address of the Kubernetes API server (overrides any value in kubeconfig)                                                                                                                                                                                                                                                                  |
| --metrics-bind-address ipport   Default: 127.0.0.1:10249                                                                                                                                                                                                                                                                                      |
| The IP address and port for the metrics server to serve on, defaulting to "127.0.0.1:10249" (if --bind-address is unset or IPv4), or "[::1]:10249" (if --bind-address is IPv6). (Set to "0.0.0.0:10249" / "[::]:10249" to bind on all interfaces.) Set empty to disable. This parameter is ignored if a config file is specified by --config. |
| --nodeport-addresses strings                                                                                                                                                                                                                                                                                                                  |
| A list of CIDR ranges that contain valid node IPs. If set, connections to NodePort services will only be accepted on node IPs in one of the indicated ranges. If unset, NodePort connections will be accepted on all local IPs. This parameter is ignored if a config file is specified by --config.                                          |
| --one_output                                                                                                                                                                                                                                                                                                                                  |
| If true, only write logs to their native severity level (vs also writing to each lower severity level; no effect when -logtostderr=true)                                                                                                                                                                                                      |

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| <code>--oom-score-adj int32</code> Default: -999                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| The oom-score-adj value for kube-proxy process. Values must be within the range [-1000, 1000]. This parameter is ignored if a config file is specified by <code>--config</code> .                                                                                                                                                                                                                                                                                                                  |
| <code>--pod-bridge-interface string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| A bridge interface name. When <code>--detect-local-mode</code> is set to <code>BridgeInterface</code> , kube-proxy will consider traffic to be local if it originates from this bridge.                                                                                                                                                                                                                                                                                                            |
| <code>--pod-interface-name-prefix string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| An interface name prefix. When <code>--detect-local-mode</code> is set to <code>InterfaceNamePrefix</code> , kube-proxy will consider traffic to be local if it originates from any interface whose name begins with this prefix.                                                                                                                                                                                                                                                                  |
| <code>--profiling</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| If true enables profiling via web interface on <code>/debug/pprof</code> handler. This parameter is ignored if a config file is specified by <code>--config</code> .                                                                                                                                                                                                                                                                                                                               |
| <code>--proxy-mode ProxyMode</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Which proxy mode to use: on Linux this can be 'iptables' (default) or 'ipvs'. On Windows the only supported value is 'kernel-space'. This parameter is ignored if a config file is specified by <code>--config</code> .                                                                                                                                                                                                                                                                            |
| <code>--show-hidden-metrics-for-version string</code>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| The previous version for which you want to show hidden metrics. Only the previous minor version is meaningful, other values will not be allowed. The format is <major>.<minor>, e.g.: '1.16'. The purpose of this format is make sure you have the opportunity to notice if the next release hides additional metrics, rather than being surprised when they are permanently removed in the release after that. This parameter is ignored if a config file is specified by <code>--config</code> . |
| <code>--skip_headers</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| If true, avoid header prefixes in the log messages                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>--skip_log_headers</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| If true, avoid headers when opening log files (no effect when <code>-logtostderr=true</code> )                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>--stderrthreshold int</code> Default: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| logs at or above this threshold go to stderr when writing to files and stderr (no effect when <code>-logtostderr=true</code> or <code>-alsologtostderr=true</code> )                                                                                                                                                                                                                                                                                                                               |
| <code>-v, --v int</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| number for the log level verbosity                                                                           |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --vmodule pattern=N,...                                                                                      |
| comma-separated list of pattern=N settings for file-filtered logging (only works for text log format)        |
| --write-config-to string                                                                                     |
| If set, write the default configuration values to this file and exit.                                        |

# kube-scheduler

## Synopsis

The Kubernetes scheduler is a control plane process which assigns Pods to Nodes. The scheduler determines which Nodes are valid placements for each Pod in the scheduling queue according to constraints and available resources. The scheduler then ranks each valid Node and binds the Pod to a suitable Node. Multiple different schedulers may be used within a cluster; kube-scheduler is the reference implementation. See [scheduling](#) for more information about scheduling and the kube-scheduler component.

kube-scheduler [flags]

## Options

|                                                                                                                                                                                                                                                         |
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| --allow-metric-labels stringToString    Default: []                                                                                                                                                                                                     |
| The map from metric-label to value allow-list of this label. The key's format is <MetricName>,<LabelName>. The value's format is <allowed_value>,<allowed_value>...e.g. metric1,label1='v1,v2,v3', metric1,label2='v1,v2,v3' metric2,label1='v1,v2,v3'. |
| --allow-metric-labels-manifest string                                                                                                                                                                                                                   |
| The path to the manifest file that contains the allow-list mapping. The format of the file is the same as the flag --allow-metric-labels. Note that the flag --allow-metric-labels will override the manifest file.                                     |
| --authentication-kubeconfig string                                                                                                                                                                                                                      |

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| kubeconfig file pointing at the 'core' kubernetes server with enough rights to create tokenreviews.authentication.k8s.io. This is optional. If empty, all token requests are considered to be anonymous and no client CA is looked up in the cluster.                    |
| --authentication-skip-lookup                                                                                                                                                                                                                                             |
| If false, the authentication-kubeconfig will be used to lookup missing authentication configuration from the cluster.                                                                                                                                                    |
| --authentication-token-webhook-cache-ttl duration    Default: 10s                                                                                                                                                                                                        |
| The duration to cache responses from the webhook token authenticator.                                                                                                                                                                                                    |
| --authentication-tolerate-lookup-failure    Default: true                                                                                                                                                                                                                |
| If true, failures to look up missing authentication configuration from the cluster are not considered fatal. Note that this can result in authentication that treats all requests as anonymous.                                                                          |
| --authorization-always-allow-paths strings    Default: "/healthz,/readyz,/livez"                                                                                                                                                                                         |
| A list of HTTP paths to skip during authorization, i.e. these are authorized without contacting the 'core' kubernetes server.                                                                                                                                            |
| --authorization-kubeconfig string                                                                                                                                                                                                                                        |
| kubeconfig file pointing at the 'core' kubernetes server with enough rights to create subjectaccessreviews.authorization.k8s.io. This is optional. If empty, all requests not skipped by authorization are forbidden.                                                    |
| --authorization-webhook-cache-authorized-ttl duration    Default: 10s                                                                                                                                                                                                    |
| The duration to cache 'authorized' responses from the webhook authorizer.                                                                                                                                                                                                |
| --authorization-webhook-cache-unauthorized-ttl duration    Default: 10s                                                                                                                                                                                                  |
| The duration to cache 'unauthorized' responses from the webhook authorizer.                                                                                                                                                                                              |
| --bind-address string    Default: 0.0.0.0                                                                                                                                                                                                                                |
| The IP address on which to listen for the --secure-port port. The associated interface(s) must be reachable by the rest of the cluster, and by CLI/web clients. If blank or an unspecified address (0.0.0.0 or ::), all interfaces and IP address families will be used. |
| --cert-dir string                                                                                                                                                                                                                                                        |
| The directory where the TLS certs are located. If --tls-cert-file and --tls-private-key-file are provided, this flag will be ignored.                                                                                                                                    |
| --client-ca-file string                                                                                                                                                                                                                                                  |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If set, any request presenting a client certificate signed by one of the authorities in the client-ca-file is authenticated with an identity corresponding to the CommonName of the client certificate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --config string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| The path to the configuration file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| --contention-profiling    Default: true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| DEPRECATED: enable block profiling, if profiling is enabled. This parameter is ignored if a config file is specified in --config.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| --disabled-metrics strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| This flag provides an escape hatch for misbehaving metrics. You must provide the fully qualified metric name in order to disable it. Disclaimer: disabling metrics is higher in precedence than showing hidden metrics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| --feature-gates <comma-separated 'key=True False' pairs>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>A set of key=value pairs that describe feature gates for alpha/experimental features. Options are:</p> <p>APIResponseCompression=true false (BETA - default=true)</p> <p>APIServerIdentity=true false (BETA - default=true)</p> <p>APIServerTracing=true false (BETA - default=true)</p> <p>APIServingWithRoutine=true false (BETA - default=true)</p> <p>AllAlpha=true false (ALPHA - default=false)</p> <p>AllBeta=true false (BETA - default=false)</p> <p>AnyVolumeDataSource=true false (BETA - default=true)</p> <p>AppArmor=true false (BETA - default=true)</p> <p>AppArmorFields=true false (BETA - default=true)</p> <p>CPUManagerPolicyAlphaOptions=true false (ALPHA - default=false)</p> <p>CPUManagerPolicyBetaOptions=true false (BETA - default=true)</p> <p>CPUManagerPolicyOptions=true false (BETA - default=true)</p> <p>CRDValidationRatcheting=true false (BETA - default=true)</p> <p>CSIMigrationPortworx=true false (BETA - default=false)</p> <p>CSIVolumeHealth=true false (ALPHA - default=false)</p> <p>CloudControllerManagerWebhook=true false (ALPHA - default=false)</p> <p>ClusterTrustBundle=true false (ALPHA - default=false)</p> <p>ClusterTrustBundleProjection=true false (ALPHA - default=false)</p> <p>ComponentSLIs=true false (BETA - default=true)</p> <p>ConsistentListFromCache=true false (ALPHA - default=false)</p> <p>ContainerCheckpoint=true false (BETA - default=true)</p> <p>ContextualLogging=true false (BETA - default=true)</p> <p>CronJobsScheduledAnnotation=true false (BETA - default=true)</p> <p>CrossNamespaceVolumeDataSource=true false (ALPHA - default=false)</p> <p>CustomCPUCFSQuotaPeriod=true false (ALPHA - default=false)</p> <p>CustomResourceFieldSelectors=true false (ALPHA - default=false)</p> <p>DevicePluginCDIDevices=true false (BETA - default=true)</p> <p>DisableCloudProviders=true false (BETA - default=true)</p> |

DisableKubeletCloudCredentialProviders=true|false (BETA - default=true)  
DisableNodeKubeProxyVersion=true|false (ALPHA - default=false)  
DynamicResourceAllocation=true|false (ALPHA - default=false)  
ElasticIndexedJob=true|false (BETA - default=true)  
EventedPLEG=true|false (ALPHA - default=false)  
GracefulNodeShutdown=true|false (BETA - default=true)  
GracefulNodeShutdownBasedOnPodPriority=true|false (BETA - default=true)  
HPAScaleToZero=true|false (ALPHA - default=false)  
HonorPVReclaimPolicy=true|false (ALPHA - default=false)  
ImageMaximumGCAge=true|false (BETA - default=true)  
InPlacePodVerticalScaling=true|false (ALPHA - default=false)  
InTreePluginAWSUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureDiskUnregister=true|false (ALPHA - default=false)  
InTreePluginAzureFileUnregister=true|false (ALPHA - default=false)  
InTreePluginGCEUnregister=true|false (ALPHA - default=false)  
InTreePluginOpenStackUnregister=true|false (ALPHA - default=false)  
InTreePluginPortworxUnregister=true|false (ALPHA - default=false)  
InTreePluginvSphereUnregister=true|false (ALPHA - default=false)  
InformerResourceVersion=true|false (ALPHA - default=false)  
JobBackoffLimitPerIndex=true|false (BETA - default=true)  
JobManagedBy=true|false (ALPHA - default=false)  
JobPodFailurePolicy=true|false (BETA - default=true)  
JobPodReplacementPolicy=true|false (BETA - default=true)  
JobSuccessPolicy=true|false (ALPHA - default=false)  
KubeProxyDrainingTerminatingNodes=true|false (BETA - default=true)  
KubeletCgroupDriverFromCRI=true|false (ALPHA - default=false)  
KubeletInUserNamespace=true|false (ALPHA - default=false)  
KubeletPodResourcesDynamicResources=true|false (ALPHA - default=false)  
KubeletPodResourcesGet=true|false (ALPHA - default=false)  
KubeletSeparateDiskGC=true|false (ALPHA - default=false)  
KubeletTracing=true|false (BETA - default=true)  
LoadBalancerIPMode=true|false (BETA - default=true)  
LocalStorageCapacityIsolationFSQuotaMonitoring=true|false (ALPHA - default=false)  
LogarithmicScaleDown=true|false (BETA - default=true)  
LoggingAlphaOptions=true|false (ALPHA - default=false)  
LoggingBetaOptions=true|false (BETA - default=true)  
MatchLabelKeysInPodAffinity=true|false (ALPHA - default=false)  
MatchLabelKeysInPodTopologySpread=true|false (BETA - default=true)  
MaxUnavailableStatefulSet=true|false (ALPHA - default=false)  
MemoryManager=true|false (BETA - default=true)  
MemoryQoS=true|false (ALPHA - default=false)  
MultiCIDRServiceAllocator=true|false (ALPHA - default=false)  
MutatingAdmissionPolicy=true|false (ALPHA - default=false)  
NFTablesProxyMode=true|false (ALPHA - default=false)  
NodeInclusionPolicyInPodTopologySpread=true|false (BETA - default=true)  
NodeLogQuery=true|false (BETA - default=false)  
NodeSwap=true|false (BETA - default=true)  
OpenAPIEnums=true|false (BETA - default=true)  
PDBUnhealthyPodEvictionPolicy=true|false (BETA - default=true)  
PersistentVolumeLastPhaseTransitionTime=true|false (BETA - default=true)  
PodAndContainerStatsFromCRI=true|false (ALPHA - default=false)  
PodDeletionCost=true|false (BETA - default=true)  
PodDisruptionConditions=true|false (BETA - default=true)

PodIndexLabel=true|false (BETA - default=true)  
PodLifecycleSleepAction=true|false (BETA - default=true)  
PodReadyToStartContainersCondition=true|false (BETA - default=true)  
PortForwardWebsockets=true|false (ALPHA - default=false)  
ProcMountType=true|false (ALPHA - default=false)  
QOSReserved=true|false (ALPHA - default=false)  
RecoverVolumeExpansionFailure=true|false (ALPHA - default=false)  
RecursiveReadOnlyMounts=true|false (ALPHA - default=false)  
RelaxedEnvironmentVariableValidation=true|false (ALPHA - default=false)  
RetryGenerateName=true|false (ALPHA - default=false)  
RotateKubeletServerCertificate=true|false (BETA - default=true)  
RuntimeClassInImageCriApi=true|false (ALPHA - default=false)  
SELinuxMount=true|false (ALPHA - default=false)  
SELinuxMountReadWriteOncePod=true|false (BETA - default=true)  
SchedulerQueueingHints=true|false (BETA - default=false)  
SeparateCacheWatchRPC=true|false (BETA - default=true)  
SeparateTaintEvictionController=true|false (BETA - default=true)  
ServiceAccountTokenJTI=true|false (BETA - default=true)  
ServiceAccountTokenNodeBinding=true|false (ALPHA - default=false)  
ServiceAccountTokenNodeBindingValidation=true|false (BETA - default=true)  
ServiceAccountTokenPodNodeInfo=true|false (BETA - default=true)  
ServiceTrafficDistribution=true|false (ALPHA - default=false)  
SidecarContainers=true|false (BETA - default=true)  
SizeMemoryBackedVolumes=true|false (BETA - default=true)  
StatefulSetAutoDeletePVC=true|false (BETA - default=true)  
StatefulSetStartOrdinal=true|false (BETA - default=true)  
StorageNamespaceIndex=true|false (BETA - default=true)  
StorageVersionAPI=true|false (ALPHA - default=false)  
StorageVersionHash=true|false (BETA - default=true)  
StorageVersionMigrator=true|false (ALPHA - default=false)  
StructuredAuthenticationConfiguration=true|false (BETA - default=true)  
StructuredAuthorizationConfiguration=true|false (BETA - default=true)  
TopologyAwareHints=true|false (BETA - default=true)  
TopologyManagerPolicyAlphaOptions=true|false (ALPHA - default=false)  
TopologyManagerPolicyBetaOptions=true|false (BETA - default=true)  
TopologyManagerPolicyOptions=true|false (BETA - default=true)  
TranslateStreamCloseWebsocketRequests=true|false (BETA - default=true)  
UnauthenticatedHTTP2DOSMitigation=true|false (BETA - default=true)  
UnknownVersionInteroperabilityProxy=true|false (ALPHA - default=false)  
UserNamespacesPodSecurityStandards=true|false (ALPHA - default=false)  
UserNamespacesSupport=true|false (BETA - default=false)  
VolumeAttributesClass=true|false (ALPHA - default=false)  
VolumeCapacityPriority=true|false (ALPHA - default=false)  
WatchFromStorageWithoutResourceVersion=true|false (BETA - default=false)  
WatchList=true|false (ALPHA - default=false)  
WatchListClient=true|false (BETA - default=false)  
WinDSR=true|false (ALPHA - default=false)  
WinOverlay=true|false (BETA - default=true)  
WindowsHostNetwork=true|false (ALPHA - default=true)

-h, --help

|                                                                                                                                                                                                                                                                                                                                            |
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| help for kube-scheduler                                                                                                                                                                                                                                                                                                                    |
| --http2-max-streams-per-connection int                                                                                                                                                                                                                                                                                                     |
| The limit that the server gives to clients for the maximum number of streams in an HTTP/2 connection. Zero means to use golang's default.                                                                                                                                                                                                  |
| --kube-api-burst int32    Default: 100                                                                                                                                                                                                                                                                                                     |
| DEPRECATED: burst to use while talking with kubernetes apiserver. This parameter is ignored if a config file is specified in --config.                                                                                                                                                                                                     |
| --kube-api-content-type string    Default: "application/vnd.kubernetes.protobuf"                                                                                                                                                                                                                                                           |
| DEPRECATED: content type of requests sent to apiserver. This parameter is ignored if a config file is specified in --config.                                                                                                                                                                                                               |
| --kube-api-qps float    Default: 50                                                                                                                                                                                                                                                                                                        |
| DEPRECATED: QPS to use while talking with kubernetes apiserver. This parameter is ignored if a config file is specified in --config.                                                                                                                                                                                                       |
| --kubeconfig string                                                                                                                                                                                                                                                                                                                        |
| DEPRECATED: path to kubeconfig file with authorization and master location information. This parameter is ignored if a config file is specified in --config.                                                                                                                                                                               |
| --leader-elect    Default: true                                                                                                                                                                                                                                                                                                            |
| Start a leader election client and gain leadership before executing the main loop. Enable this when running replicated components for high availability.                                                                                                                                                                                   |
| --leader-elect-lease-duration duration    Default: 15s                                                                                                                                                                                                                                                                                     |
| The duration that non-leader candidates will wait after observing a leadership renewal until attempting to acquire leadership of a led but unrenewed leader slot. This is effectively the maximum duration that a leader can be stopped before it is replaced by another candidate. This is only applicable if leader election is enabled. |
| --leader-elect-renew-deadline duration    Default: 10s                                                                                                                                                                                                                                                                                     |
| The interval between attempts by the acting master to renew a leadership slot before it stops leading. This must be less than the lease duration. This is only applicable if leader election is enabled.                                                                                                                                   |
| --leader-elect-resource-lock string    Default: "leases"                                                                                                                                                                                                                                                                                   |
| The type of resource object that is used for locking during leader election. Supported options are 'leases', 'endpointsleases' and 'configmapsleases'.                                                                                                                                                                                     |
| --leader-elect-resource-name string    Default: "kube-scheduler"                                                                                                                                                                                                                                                                           |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | The name of resource object that is used for locking during leader election.                                                                                                                                                                                                                                                                                                      |
| --leader-elect-resource-namespace string          | Default: "kube-system"                                                                                                                                                                                                                                                                                                                                                            |
|                                                   | The namespace of resource object that is used for locking during leader election.                                                                                                                                                                                                                                                                                                 |
| --leader-elect-retry-period duration              | Default: 2s                                                                                                                                                                                                                                                                                                                                                                       |
|                                                   | The duration the clients should wait between attempting acquisition and renewal of a leadership. This is only applicable if leader election is enabled.                                                                                                                                                                                                                           |
| --log-flush-frequency duration                    | Default: 5s                                                                                                                                                                                                                                                                                                                                                                       |
|                                                   | Maximum number of seconds between log flushes                                                                                                                                                                                                                                                                                                                                     |
| --log-text-info-buffer-size quantity              |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                   | [Alpha] In text format with split output streams, the info messages can be buffered for a while to increase performance. The default value of zero bytes disables buffering. The size can be specified as number of bytes (512), multiples of 1000 (1K), multiples of 1024 (2Ki), or powers of those (3M, 4G, 5Mi, 6Gi). Enable the LoggingAlphaOptions feature gate to use this. |
| --log-text-split-stream                           |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                   | [Alpha] In text format, write error messages to stderr and info messages to stdout. The default is to write a single stream to stdout. Enable the LoggingAlphaOptions feature gate to use this.                                                                                                                                                                                   |
| --logging-format string                           | Default: "text"                                                                                                                                                                                                                                                                                                                                                                   |
|                                                   | Sets the log format. Permitted formats: "text".                                                                                                                                                                                                                                                                                                                                   |
| --master string                                   |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                   | The address of the Kubernetes API server (overrides any value in kubeconfig)                                                                                                                                                                                                                                                                                                      |
| --permit-address-sharing                          |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                   | If true, SO_REUSEADDR will be used when binding the port. This allows binding to wildcard IPs like 0.0.0.0 and specific IPs in parallel, and it avoids waiting for the kernel to release sockets in TIME_WAIT state. [default=false]                                                                                                                                              |
| --permit-port-sharing                             |                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                   | If true, SO_REUSEPORT will be used when binding the port, which allows more than one instance to bind on the same address and port. [default=false]                                                                                                                                                                                                                               |
| --pod-max-in-unschedulable-pods-duration duration | Default: 5m0s                                                                                                                                                                                                                                                                                                                                                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| DEPRECATED: the maximum time a pod can stay in unschedulablePods. If a pod stays in unschedulablePods for longer than this value, the pod will be moved from unschedulablePods to backoffQ or activeQ. This flag is deprecated and will be removed in a future version.                                                                                                                                         |
| --profiling    Default: true                                                                                                                                                                                                                                                                                                                                                                                    |
| DEPRECATED: enable profiling via web interface host:port/debug/pprof/. This parameter is ignored if a config file is specified in --config.                                                                                                                                                                                                                                                                     |
| --requestheader-allowed-names strings                                                                                                                                                                                                                                                                                                                                                                           |
| List of client certificate common names to allow to provide usernames in headers specified by --requestheader-username-headers. If empty, any client certificate validated by the authorities in --requestheader-client-ca-file is allowed.                                                                                                                                                                     |
| --requestheader-client-ca-file string                                                                                                                                                                                                                                                                                                                                                                           |
| Root certificate bundle to use to verify client certificates on incoming requests before trusting usernames in headers specified by --requestheader-username-headers. WARNING: generally do not depend on authorization being already done for incoming requests.                                                                                                                                               |
| --requestheader-extra-headers-prefix strings    Default: "x-remote-extra-"                                                                                                                                                                                                                                                                                                                                      |
| List of request header prefixes to inspect. X-Remote-Extra- is suggested.                                                                                                                                                                                                                                                                                                                                       |
| --requestheader-group-headers strings    Default: "x-remote-group"                                                                                                                                                                                                                                                                                                                                              |
| List of request headers to inspect for groups. X-Remote-Group is suggested.                                                                                                                                                                                                                                                                                                                                     |
| --requestheader-username-headers strings    Default: "x-remote-user"                                                                                                                                                                                                                                                                                                                                            |
| List of request headers to inspect for usernames. X-Remote-User is common.                                                                                                                                                                                                                                                                                                                                      |
| --secure-port int    Default: 10259                                                                                                                                                                                                                                                                                                                                                                             |
| The port on which to serve HTTPS with authentication and authorization. If 0, don't serve HTTPS at all.                                                                                                                                                                                                                                                                                                         |
| --show-hidden-metrics-for-version string                                                                                                                                                                                                                                                                                                                                                                        |
| The previous version for which you want to show hidden metrics. Only the previous minor version is meaningful, other values will not be allowed. The format is <major>.<minor>, e.g.: '1.16'. The purpose of this format is make sure you have the opportunity to notice if the next release hides additional metrics, rather than being surprised when they are permanently removed in the release after that. |
| --tls-cert-file string                                                                                                                                                                                                                                                                                                                                                                                          |
| File containing the default x509 Certificate for HTTPS. (CA cert, if any, concatenated after server cert). If HTTPS serving is enabled, and --tls-cert-file and --tls-private-key-file are not                                                                                                                                                                                                                  |

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| provided, a self-signed certificate and key are generated for the public address and saved to the directory specified by --cert-dir.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| --tls-cipher-suites strings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p>Comma-separated list of cipher suites for the server. If omitted, the default Go cipher suites will be used.</p> <p>Preferred values: TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384, TLS_CHACHA20_POLY1305_SHA256, TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA, TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256, TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA, TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384, TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305, TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256, TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256, TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384, TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305, TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256.</p> <p>Insecure values: TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256, TLS_ECDHE_ECDSA_WITH_RC4_128_SHA, TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA, TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256, TLS_ECDHE_RSA_WITH_RC4_128_SHA, TLS_RSA_WITH_3DES_EDE_CBC_SHA, TLS_RSA_WITH_AES_128_CBC_SHA, TLS_RSA_WITH_AES_128_CBC_SHA256, TLS_RSA_WITH_AES_128_GCM_SHA256, TLS_RSA_WITH_AES_256_CBC_SHA, TLS_RSA_WITH_AES_256_GCM_SHA384, TLS_RSA_WITH_RC4_128_SHA.</p> |
| --tls-min-version string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Minimum TLS version supported. Possible values: VersionTLS10, VersionTLS11, VersionTLS12, VersionTLS13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --tls-private-key-file string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| File containing the default x509 private key matching --tls-cert-file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| --tls-sni-cert-key string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>A pair of x509 certificate and private key file paths, optionally suffixed with a list of domain patterns which are fully qualified domain names, possibly with prefixed wildcard segments. The domain patterns also allow IP addresses, but IPs should only be used if the apiserver has visibility to the IP address requested by a client. If no domain patterns are provided, the names of the certificate are extracted. Non-wildcard matches trump over wildcard matches, explicit domain patterns trump over extracted names. For multiple key/certificate pairs, use the --tls-sni-cert-key multiple times. Examples: "example.crt,example.key" or "foo.crt,foo.key:*.foo.com,foo.com".</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| -v, --v int                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| number for the log level verbosity                                                                           |
| --version version[=true]                                                                                     |
| --version, --version=raw prints version information and quits; --version=vX.Y.Z... sets the reported version |
| --vmodule pattern=N,...                                                                                      |
| comma-separated list of pattern=N settings for file-filtered logging (only works for text log format)        |
| --write-config-to string                                                                                     |
| If set, write the configuration values to this file and exit.                                                |

## Debug cluster

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### [Flow control](#)

## Flow control

API Priority and Fairness controls the behavior of the Kubernetes API server in an overload situation. You can find more information about it in the [API Priority and Fairness](#) documentation.

## Diagnostics

Every HTTP response from an API server with the priority and fairness feature enabled has two extra headers: X-Kubernetes-PF-FlowSchema-UID and X-Kubernetes-PF-PriorityLevel-UID, noting the flow schema that matched the request and the priority level to which it was assigned, respectively. The API objects' names are not included in these headers (to avoid revealing details in case the requesting user does not have permission to view them). When debugging, you can use a command such as:

```
kubectl get flowschemas -o custom-columns="uid:{metadata.uid},name:{metadata.name}"
kubectl get prioritylevelconfigurations -o custom-columns="uid:{metadata.uid},name:{metadata.name}"
```

to get a mapping of UIDs to names for both FlowSchemas and PriorityLevelConfigurations.

## Debug endpoints

With the `APIPriorityAndFairness` feature enabled, the `kube-apiserver` serves the following additional paths at its HTTP(S) ports.



You need to ensure you have permissions to access these endpoints. You don't have to do anything if you are using admin. Permissions can be granted if needed following the [RBAC](#) doc to access `/debug/api_priority_and_fairness/` by specifying `nonResourceURLs`.

- `/debug/api_priority_and_fairness/dump_priority_levels` - a listing of all the priority levels and the current state of each. You can fetch like this:

```
kubectl get --raw /debug/api_priority_and_fairness/dump_priority_levels
```

The output will be in CSV and similar to this:

```
PriorityLevelName, ActiveQueues, IsIdle, IsQuiescing, WaitingRequests,
ExecutingRequests, DispatchedRequests, RejectedRequests, TimedoutRequests,
CancelledRequests
catch-all, 0, true, false, 0, 0, 1, 0,
0, 0
exempt, 0, true, false, 0, 0, 0, 0,
0, 0
global-default, 0, true, false, 0, 0, 46, 0,
0, 0
leader-election, 0, true, false, 0, 0, 4, 0,
0, 0
node-high, 0, true, false, 0, 0, 34, 0,
0, 0
system, 0, true, false, 0, 0, 48, 0,
0, 0
workload-high, 0, true, false, 0, 0, 500,
0, 0, 0
workload-low, 0, true, false, 0, 0, 0, 0,
0, 0
```

Explanation for selected column names:

- `IsQuiescing` indicates if this priority level will be removed when its queues have been drained.

- `/debug/api_priority_and_fairness/dump_queues` - a listing of all the queues and their current state. You can fetch like this:

```
kubectl get --raw /debug/api_priority_and_fairness/dump_queues
```

The output will be in CSV and similar to this:

```
PriorityLevelName, Index, PendingRequests, ExecutingRequests, SeatsInUse,
NextDispatchR, InitialSeatsSum, MaxSeatsSum, TotalWorkSum
workload-low, 14, 27, 0, 0, 77.64342019ss, 270,
270, 0.81000000ss
workload-low, 74, 26, 0, 0, 76.95387841ss, 260,
260, 0.78000000ss
...
leader-election, 0, 0, 0, 0, 5088.87053833ss, 0, 0,
0.00000000ss
leader-election, 1, 0, 0, 0, 0.00000000ss, 0, 0,
0.00000000ss
```

```
...
workload-high, 0, 0, 0, 0, 0.00000000ss, 0, 0,
0.00000000ss
workload-high, 1, 0, 0, 0, 1119.44936475ss, 0,
0, 0.00000000ss
```

Explanation for selected column names:

- NextDispatchR: The R progress meter reading, in units of seat-seconds, at which the next request will be dispatched.
- InitialSeatsSum: The sum of InitialSeats associated with all requests in a given queue.
- MaxSeatsSum: The sum of MaxSeats associated with all requests in a given queue.
- TotalWorkSum: The sum of total work, in units of seat-seconds, of all waiting requests in a given queue.

Note: seat-second (abbreviate as ss) is a measure of work, in units of seat-seconds, in the APF world.

- `/debug/api_priority_and_fairness/dump_requests` - a listing of all the requests including requests waiting in a queue and requests being executing. You can fetch like this:

```
kubectl get --raw /debug/api_priority_and_fairness/dump_requests
```

The output will be in CSV and similar to this:

```
PriorityLevelName, FlowSchemaName, QueueIndex, RequestIndexInQueue,
FlowDistinguisher, ArriveTime, InitialSeats, FinalSeats,
AdditionalLatency, StartTime
exempt, exempt, -1, -1, ,
2023-07-15T04:51:25.596404345Z, 1, 0, 0s,
2023-07-15T04:51:25.596404345Z
workload-low, service-accounts, 14, 0,
system:serviceaccount:default:loadtest, 2023-07-18T00:12:51.386556253Z, 10, 0,
0s, 0001-01-01T00:00:00Z
workload-low, service-accounts, 14, 1,
system:serviceaccount:default:loadtest, 2023-07-18T00:12:51.487092539Z, 10, 0,
0s, 0001-01-01T00:00:00Z
```

You can get a more detailed listing with a command like this:

```
kubectl get --raw '/debug/api_priority_and_fairness/dump_requests?
includeRequestDetails=1'
```

The output will be in CSV and similar to this:

```
PriorityLevelName, FlowSchemaName, QueueIndex, RequestIndexInQueue,
FlowDistinguisher, ArriveTime, InitialSeats, FinalSeats,
AdditionalLatency, StartTime, Username, Verb,
APIPath, Namespace, Name, APIVersion, Resource, SubResource
exempt, exempt, -1, -1, ,
2023-07-15T04:51:25.596404345Z, 1, 0, 0s,
2023-07-15T04:51:25.596404345Z, system:serviceaccount:system:admin, list, /api/v1/
namespaces/kube-stress/configmaps, kube-stress, , v1, configmaps,
```

```
workload-low, service-accounts, 14, 0,
system:serviceaccount:default:loadtest, 2023-07-18T00:13:08.986534842Z, 10, 0,
0s, 0001-01-01T00:00:00Z, system:serviceaccount:default:loadtest, list, /
api/v1/namespaces/kube-stress/configmaps, kube-stress, , v1, configmaps,
workload-low, service-accounts, 14, 1,
system:serviceaccount:default:loadtest, 2023-07-18T00:13:09.086476021Z, 10, 0,
0s, 0001-01-01T00:00:00Z, system:serviceaccount:default:loadtest, list, /
api/v1/namespaces/kube-stress/configmaps, kube-stress, , v1, configmaps,
```

Explanation for selected column names:

- **QueueIndex**: The index of the queue. It will be -1 for priority levels without queues.
- **RequestIndexInQueue**: The index in the queue for a given request. It will be -1 for executing requests.
- **InitialSeats**: The number of seats will be occupied during the initial (normal) stage of execution of the request.
- **FinalSeats**: The number of seats will be occupied during the final stage of request execution, accounting for the associated WATCH notifications.
- **AdditionalLatency**: The extra time taken during the final stage of request execution. FinalSeats will be occupied during this time period. It does not mean any latency that a user will observe.
- **StartTime**: The time a request starts to execute. It will be 0001-01-01T00:00:00Z for queued requests.

## Debug logging

At `-v=3` or more verbosity, the API server outputs an `httplog` line for every request in the API server log, and it includes the following attributes.

- `apf_fs`: the name of the flow schema to which the request was classified.
- `apf_pl`: the name of the priority level for that flow schema.
- `apf_iseats`: the number of seats determined for the initial (normal) stage of execution of the request.
- `apf_fseats`: the number of seats determined for the final stage of execution (accounting for the associated watch notifications) of the request.
- `apf_additionalLatency`: the duration of the final stage of execution of the request.

At higher levels of verbosity there will be log lines exposing details of how APF handled the request, primarily for debugging purposes.

## Response headers

APF adds the following two headers to each HTTP response message. They won't appear in the audit log. They can be viewed from the client side. For client using `klog`, use verbosity `-v=8` or higher to view these headers.

- `X-Kubernetes-PF-FlowSchema-UID` holds the UID of the FlowSchema object to which the corresponding request was classified.
- `X-Kubernetes-PF-PriorityLevel-UID` holds the UID of the PriorityLevelConfiguration object associated with that FlowSchema.

## What's next

For background information on design details for API priority and fairness, see the [enhancement proposal](#).

# Configuration APIs

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[Client Authentication \(v1\)](#)

[Client Authentication \(v1beta1\)](#)

[Event Rate Limit Configuration \(v1alpha1\)](#)

[Image Policy API \(v1alpha1\)](#)

[kube-apiserver Admission \(v1\)](#)

[kube-apiserver Audit Configuration \(v1\)](#)

[kube-apiserver Configuration \(v1\)](#)

[kube-apiserver Configuration \(v1alpha1\)](#)

[kube-apiserver Configuration \(v1beta1\)](#)

[kube-controller-manager Configuration \(v1alpha1\)](#)

[kube-proxy Configuration \(v1alpha1\)](#)

[kube-scheduler Configuration \(v1\)](#)

[kubeadm Configuration \(v1beta3\)](#)

[kubeadm Configuration \(v1beta4\)](#)

[kubeconfig \(v1\)](#)

[Kubelet Configuration \(v1\)](#)

[Kubelet Configuration \(v1alpha1\)](#)

[Kubelet Configuration \(v1beta1\)](#)

[Kubelet CredentialProvider \(v1\)](#)

[WebhookAdmission Configuration \(v1\)](#)

# Client Authentication (v1)

## Resource Types

- [ExecCredential](#)

## ExecCredential

ExecCredential is used by exec-based plugins to communicate credentials to HTTP transports.

| Field                                                        | Description                                                                                                   |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                         | client.authentication.k8s.io/v1                                                                               |
| kind<br>string                                               | ExecCredential                                                                                                |
| spec <b>[Required]</b><br><a href="#">ExecCredentialSpec</a> | Spec holds information passed to the plugin by the transport.                                                 |
| status<br><a href="#">ExecCredentialStatus</a>               | Status is filled in by the plugin and holds the credentials that the transport should use to contact the API. |

## Cluster

Appears in:

- [ExecCredentialSpec](#)

Cluster contains information to allow an exec plugin to communicate with the kubernetes cluster being authenticated to.

To ensure that this struct contains everything someone would need to communicate with a kubernetes cluster (just like they would via a kubeconfig), the fields should shadow "k8s.io/client-go/tools/clientcmd/api/v1".Cluster, with the exception of CertificateAuthority, since CA data will always be passed to the plugin as bytes.

| Field                              | Description                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| server <b>[Required]</b><br>string | Server is the address of the kubernetes cluster (https://hostname:port).                                                                                                               |
| tls-server-name<br>string          | TLSServerName is passed to the server for SNI and is used in the client to check server certificates against. If ServerName is empty, the hostname used to contact the server is used. |

| Field                                                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| insecure-skip-tls-verify<br>bool                                                                                          | InsecureSkipTLSVerify skips the validity check for the server's certificate. This will make your HTTPS connections insecure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| certificate-authority-data<br>[]byte                                                                                      | CAData contains PEM-encoded certificate authority certificates. If empty, system roots should be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| proxy-url<br>string                                                                                                       | ProxyURL is the URL to the proxy to be used for all requests to this cluster.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| disable-compression<br>bool                                                                                               | DisableCompression allows client to opt-out of response compression for all requests to the server. This is useful to speed up requests (specifically lists) when client-server network bandwidth is ample, by saving time on compression (server-side) and decompression (client-side): <a href="https://github.com/kubernetes/kubernetes/issues/112296">https://github.com/kubernetes/kubernetes/issues/112296</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| config<br><a href="https://k8s.io/apimachinery/pkg/runtime.RawExtension">k8s.io/apimachinery/pkg/runtime.RawExtension</a> | <p>Config holds additional config data that is specific to the exec plugin with regards to the cluster being authenticated to.</p> <p>This data is sourced from the clientcmd Cluster object's extensions[client.authentication.k8s.io/exec] field:</p> <p>clusters:</p> <ul style="list-style-type: none"> <li>• name: my-cluster cluster: ... extensions: <ul style="list-style-type: none"> <li>◦ name: client.authentication.k8s.io/exec # reserved extension name for per cluster exec config extension: audience: 06e3fbd18de8 # arbitrary config</li> </ul> </li> </ul> <p>In some environments, the user config may be exactly the same across many clusters (i.e. call this exec plugin) minus some details that are specific to each cluster such as the audience. This field allows the per cluster config to be directly specified with the cluster info. Using this field to store secret data is not recommended as one of the prime benefits of exec plugins is that no secrets need to be stored directly in the kubeconfig.</p> |

## ExecCredentialSpec

Appears in:

- [ExecCredential](#)

ExecCredentialSpec holds request and runtime specific information provided by the transport.

| Field                                    | Description                                                                                                                                                                                                                                                         |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cluster<br><a href="#">Cluster</a>       | Cluster contains information to allow an exec plugin to communicate with the kubernetes cluster being authenticated to. Note that Cluster is non-nil only when provideClusterInfo is set to true in the exec provider config (i.e., ExecConfig.ProvideClusterInfo). |
| interactive<br><b>[Required]</b><br>bool | Interactive declares whether stdin has been passed to this exec plugin.                                                                                                                                                                                             |

## ExecCredentialStatus

Appears in:

- [ExecCredential](#)

ExecCredentialStatus holds credentials for the transport to use.

Token and ClientKeyData are sensitive fields. This data should only be transmitted in-memory between client and exec plugin process. Exec plugin itself should at least be protected via file permissions.

| Field                                                | Description                                                                |
|------------------------------------------------------|----------------------------------------------------------------------------|
| expirationTimestamp<br><a href="#">meta/v1.Time</a>  | ExpirationTimestamp indicates a time when the provided credentials expire. |
| token <b>[Required]</b><br>string                    | Token is a bearer token used by the client for request authentication.     |
| clientCertificateData<br><b>[Required]</b><br>string | PEM-encoded client TLS certificates (including intermediates, if any).     |
| clientKeyData <b>[Required]</b><br>string            | PEM-encoded private key for the above certificate.                         |

## Client Authentication (v1beta1)

### Resource Types

- [ExecCredential](#)

## ExecCredential

ExecCredential is used by exec-based plugins to communicate credentials to HTTP transports.

| Field                                                        | Description                                                                                                   |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                         | client.authentication.k8s.io/v1beta1                                                                          |
| kind<br>string                                               | ExecCredential                                                                                                |
| spec <b>[Required]</b><br><a href="#">ExecCredentialSpec</a> | Spec holds information passed to the plugin by the transport.                                                 |
| status<br><a href="#">ExecCredentialStatus</a>               | Status is filled in by the plugin and holds the credentials that the transport should use to contact the API. |

## Cluster

### Appears in:

- [ExecCredentialSpec](#)

Cluster contains information to allow an exec plugin to communicate with the kubernetes cluster being authenticated to.

To ensure that this struct contains everything someone would need to communicate with a kubernetes cluster (just like they would via a kubeconfig), the fields should shadow "k8s.io/client-go/tools/clientcmd/api/v1".Cluster, with the exception of CertificateAuthority, since CA data will always be passed to the plugin as bytes.

| Field                                | Description                                                                                                                                                                            |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| server <b>[Required]</b><br>string   | Server is the address of the kubernetes cluster (https://hostname:port).                                                                                                               |
| tls-server-name<br>string            | TLSServerName is passed to the server for SNI and is used in the client to check server certificates against. If ServerName is empty, the hostname used to contact the server is used. |
| insecure-skip-tls-verify<br>bool     | InsecureSkipTLSVerify skips the validity check for the server's certificate. This will make your HTTPS connections insecure.                                                           |
| certificate-authority-data<br>[]byte | CAData contains PEM-encoded certificate authority certificates. If empty, system roots should be used.                                                                                 |
| proxy-url<br>string                  | ProxyURL is the URL to the proxy to be used for all requests to this cluster.                                                                                                          |
| disable-compression<br>bool          | DisableCompression allows client to opt-out of response compression for all requests to the server. This is useful to speed up requests                                                |



| Field                                                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                           | (specifically lists) when client-server network bandwidth is ample, by saving time on compression (server-side) and decompression (client-side): <a href="https://github.com/kubernetes/kubernetes/issues/112296">https://github.com/kubernetes/kubernetes/issues/112296</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| config<br><a href="https://k8s.io/apimachinery/pkg/runtime.RawExtension">k8s.io/apimachinery/pkg/runtime.RawExtension</a> | <p>Config holds additional config data that is specific to the exec plugin with regards to the cluster being authenticated to.</p> <p>This data is sourced from the clientcmd Cluster object's extensions[client.authentication.k8s.io/exec] field:</p> <p>clusters:</p> <ul style="list-style-type: none"> <li>• name: my-cluster cluster: ... extensions:               <ul style="list-style-type: none"> <li>◦ name: client.authentication.k8s.io/exec # reserved extension name for per cluster exec config extension: audience: 06e3fbd18de8 # arbitrary config</li> </ul> </li> </ul> <p>In some environments, the user config may be exactly the same across many clusters (i.e. call this exec plugin) minus some details that are specific to each cluster such as the audience. This field allows the per cluster config to be directly specified with the cluster info. Using this field to store secret data is not recommended as one of the prime benefits of exec plugins is that no secrets need to be stored directly in the kubeconfig.</p> |

## ExecCredentialSpec

Appears in:

- [ExecCredential](#)

ExecCredentialSpec holds request and runtime specific information provided by the transport.

| Field                                    | Description                                                                                                                                                                                                                                                         |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cluster<br><a href="#">Cluster</a>       | Cluster contains information to allow an exec plugin to communicate with the kubernetes cluster being authenticated to. Note that Cluster is non-nil only when provideClusterInfo is set to true in the exec provider config (i.e., ExecConfig.ProvideClusterInfo). |
| interactive<br><b>[Required]</b><br>bool | Interactive declares whether stdin has been passed to this exec plugin.                                                                                                                                                                                             |

## ExecCredentialStatus

Appears in:

- [ExecCredential](#)

ExecCredentialStatus holds credentials for the transport to use.

Token and ClientKeyData are sensitive fields. This data should only be transmitted in-memory between client and exec plugin process. Exec plugin itself should at least be protected via file permissions.

| Field                                               | Description                                                                |
|-----------------------------------------------------|----------------------------------------------------------------------------|
| expirationTimestamp<br><a href="#">meta/v1.Time</a> | ExpirationTimestamp indicates a time when the provided credentials expire. |
| token <b>[Required]</b><br>string                   | Token is a bearer token used by the client for request authentication.     |
| clientCertificateData <b>[Required]</b><br>string   | PEM-encoded client TLS certificates (including intermediates, if any).     |
| clientKeyData <b>[Required]</b><br>string           | PEM-encoded private key for the above certificate.                         |

## Event Rate Limit Configuration (v1alpha1)

### Resource Types

- [Configuration](#)

### Configuration

Configuration provides configuration for the EventRateLimit admission controller.

| Field                                               | Description                                                                                                                                                                                    |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                | eventratelimit.admission.k8s.io/v1alpha1                                                                                                                                                       |
| kind<br>string                                      | Configuration                                                                                                                                                                                  |
| limits <b>[Required]</b><br><a href="#">[]Limit</a> | limits are the limits to place on event queries received. Limits can be placed on events received server-wide, per namespace, per user, and per source+object. At least one limit is required. |

### Limit

Appears in:

- [Configuration](#)

Limit is the configuration for a particular limit type

| Field                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type<br>[Required]<br><a href="#">LimitType</a> | type is the type of limit to which this configuration applies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| qps<br>[Required]<br>int32                      | qps is the number of event queries per second that are allowed for this type of limit. The qps and burst fields are used together to determine if a particular event query is accepted. The qps determines how many queries are accepted once the burst amount of queries has been exhausted.                                                                                                                                                                                                                                                                                               |
| burst<br>[Required]<br>int32                    | burst is the burst number of event queries that are allowed for this type of limit. The qps and burst fields are used together to determine if a particular event query is accepted. The burst determines the maximum size of the allowance granted for a particular bucket. For example, if the burst is 10 and the qps is 3, then the admission control will accept 10 queries before blocking any queries. Every second, 3 more queries will be allowed. If some of that allowance is not used, then it will roll over to the next second, until the maximum allowance of 10 is reached. |
| cacheSize<br>int32                              | <p>cacheSize is the size of the LRU cache for this type of limit. If a bucket is evicted from the cache, then the allowance for that bucket is reset. If more queries are later received for an evicted bucket, then that bucket will re-enter the cache with a clean slate, giving that bucket a full allowance of burst queries.</p> <p>The default cache size is 4096.</p> <p>If limitType is 'server', then cacheSize is ignored.</p>                                                                                                                                                   |

## LimitType

(Alias of string)

Appears in:

- [Limit](#)

LimitType is the type of the limit (e.g., per-namespace)

## Image Policy API (v1alpha1)

### Resource Types

- [ImageReview](#)

## ImageReview

ImageReview checks if the set of images in a pod are allowed.

| Field                                                       | Description                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                        | imagepolicy.k8s.io/v1alpha1                                                                                                                                                                                                                                                                                                           |
| kind<br>string                                              | ImageReview                                                                                                                                                                                                                                                                                                                           |
| metadata<br><a href="#">meta/v1.ObjectMeta</a>              | Standard object's metadata. More info: <a href="https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata">https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata</a><br><br>Refer to the Kubernetes API documentation for the fields of the metadata field. |
| spec [ <b>Required</b> ]<br><a href="#">ImageReviewSpec</a> | Spec holds information about the pod being evaluated                                                                                                                                                                                                                                                                                  |
| status<br><a href="#">ImageReviewStatus</a>                 | Status is filled in by the backend and indicates whether the pod should be allowed.                                                                                                                                                                                                                                                   |

## ImageReviewContainerSpec

Appears in:

- [ImageReviewSpec](#)

ImageReviewContainerSpec is a description of a container within the pod creation request.

| Field           | Description                                                     |
|-----------------|-----------------------------------------------------------------|
| image<br>string | This can be in the form image:tag or image@SHA:012345679abcdef. |

## ImageReviewSpec

Appears in:

- [ImageReview](#)

ImageReviewSpec is a description of the pod creation request.

| Field                                                    | Description                                                                                     |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| containers<br><a href="#">[]ImageReviewContainerSpec</a> | Containers is a list of a subset of the information in each container of the Pod being created. |
| annotations<br>map[string]string                         |                                                                                                 |

| Field               | Description                                                                                                                                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | Annotations is a list of key-value pairs extracted from the Pod's annotations. It only includes keys which match the pattern *.image-policy.k8s.io/*. It is up to each webhook backend to determine how to interpret these annotations, if at all. |
| namespace<br>string | Namespace is the namespace the pod is being created in.                                                                                                                                                                                            |

## ImageReviewStatus

Appears in:

- [ImageReview](#)

ImageReviewStatus is the result of the review for the pod creation request.

| Field                                 | Description                                                                                                                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allowed<br>[Required]<br>bool         | Allowed indicates that all images were allowed to be run.                                                                                                                                                          |
| reason<br>string                      | Reason should be empty unless Allowed is false in which case it may contain a short description of what is wrong. Kubernetes may truncate excessively long errors when displaying to the user.                     |
| auditAnnotations<br>map[string]string | AuditAnnotations will be added to the attributes object of the admission controller request using 'AddAnnotation'. The keys should be prefix-less (i.e., the admission controller will add an appropriate prefix). |

# kube-apiserver Admission (v1)

## Resource Types

- [AdmissionReview](#)

## AdmissionReview

AdmissionReview describes an admission review request/response.

| Field                | Description         |
|----------------------|---------------------|
| apiVersion<br>string | admission.k8s.io/v1 |
| kind<br>string       | AdmissionReview     |

| Field                                         | Description                                                   |
|-----------------------------------------------|---------------------------------------------------------------|
| request<br><a href="#">AdmissionRequest</a>   | Request describes the attributes for the admission request.   |
| response<br><a href="#">AdmissionResponse</a> | Response describes the attributes for the admission response. |

## AdmissionRequest

Appears in:

- [AdmissionReview](#)

AdmissionRequest describes the admission.Attributes for the admission request.

| Field                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uid <b>[Required]</b><br><a href="#">k8s.io/apimachinery/pkg/types.UID</a> | UID is an identifier for the individual request/response. It allows us to distinguish instances of requests which are otherwise identical (parallel requests, requests when earlier requests did not modify etc) The UID is meant to track the round trip (request/response) between the KAS and the WebHook, not the user request. It is suitable for correlating log entries between the webhook and apiserver, for either auditing or debugging.                                                                                                                                                                                                                                                                                                                                  |
| kind <b>[Required]</b><br><a href="#">meta/v1.GroupVersionKind</a>         | Kind is the fully-qualified type of object being submitted (for example, v1.Pod or autoscaling.v1.Scale)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| resource <b>[Required]</b><br><a href="#">meta/v1.GroupVersionResource</a> | Resource is the fully-qualified resource being requested (for example, v1.pods)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| subResource<br>string                                                      | SubResource is the subresource being requested, if any (for example, "status" or "scale")                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| requestKind<br><a href="#">meta/v1.GroupVersionKind</a>                    | <p>RequestKind is the fully-qualified type of the original API request (for example, v1.Pod or autoscaling.v1.Scale). If this is specified and differs from the value in "kind", an equivalent match and conversion was performed.</p> <p>For example, if deployments can be modified via apps/v1 and apps/v1beta1, and a webhook registered a rule of apiGroups:["apps"], apiVersions:["v1"], resources: ["deployments"] and matchPolicy: Equivalent, an API request to apps/v1beta1 deployments would be converted and sent to the webhook with kind: {group:"apps", version:"v1", kind:"Deployment"} (matching the rule the webhook registered for), and requestKind: {group:"apps", version:"v1beta1", kind:"Deployment"} (indicating the kind of the original API request).</p> |

| Field                                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                          | See documentation for the "matchPolicy" field in the webhook configuration type for more details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| requestResource<br><a href="#">meta/v1.GroupVersionResource</a>          | <p>RequestResource is the fully-qualified resource of the original API request (for example, v1.pods). If this is specified and differs from the value in "resource", an equivalent match and conversion was performed.</p> <p>For example, if deployments can be modified via apps/v1 and apps/v1beta1, and a webhook registered a rule of apiGroups:["apps"], apiVersions:["v1"], resources: ["deployments"] and matchPolicy: Equivalent, an API request to apps/v1beta1 deployments would be converted and sent to the webhook with resource: {group:"apps", version:"v1", resource:"deployments"} (matching the resource the webhook registered for), and requestResource: {group:"apps", version:"v1beta1", resource:"deployments"} (indicating the resource of the original API request).</p> <p>See documentation for the "matchPolicy" field in the webhook configuration type.</p> |
| requestSubResource<br>string                                             | RequestSubResource is the name of the subresource of the original API request, if any (for example, "status" or "scale") If this is specified and differs from the value in "subResource", an equivalent match and conversion was performed. See documentation for the "matchPolicy" field in the webhook configuration type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| name<br>string                                                           | Name is the name of the object as presented in the request. On a CREATE operation, the client may omit name and rely on the server to generate the name. If that is the case, this field will contain an empty string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| namespace<br>string                                                      | Namespace is the namespace associated with the request (if any).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| operation <b>[Required]</b><br><a href="#">Operation</a>                 | Operation is the operation being performed. This may be different than the operation requested. e.g. a patch can result in either a CREATE or UPDATE Operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| userInfo <b>[Required]</b><br><a href="#">authentication/v1.UserInfo</a> | UserInfo is information about the requesting user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| object<br><a href="#">k8s.io/apimachinery/pkg/runtime.RawExtension</a>   | Object is the object from the incoming request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Field                                                                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oldObject<br><a href="https://k8s.io/apimachinery/pkg/runtime.RawExtension">k8s.io/apimachinery/pkg/runtime.RawExtension</a> | OldObject is the existing object. Only populated for DELETE and UPDATE requests.                                                                                                                                                                                                                                                                                                                          |
| dryRun<br>bool                                                                                                               | DryRun indicates that modifications will definitely not be persisted for this request. Defaults to false.                                                                                                                                                                                                                                                                                                 |
| options<br><a href="https://k8s.io/apimachinery/pkg/runtime.RawExtension">k8s.io/apimachinery/pkg/runtime.RawExtension</a>   | Options is the operation option structure of the operation being performed. e.g. meta.k8s.io/v1.DeleteOptions or meta.k8s.io/v1.CreateOptions. This may be different than the options the caller provided. e.g. for a patch request the performed Operation might be a CREATE, in which case the Options will a meta.k8s.io/v1.CreateOptions even though the caller provided meta.k8s.io/v1.PatchOptions. |

## AdmissionResponse

Appears in:

- [AdmissionReview](#)

AdmissionResponse describes an admission response.

| Field                                                                                                       | Description                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uid [Required]<br><a href="https://k8s.io/apimachinery/pkg/types.UID">k8s.io/apimachinery/pkg/types.UID</a> | UID is an identifier for the individual request/response. This must be copied over from the corresponding AdmissionRequest.                                                                                                                                                                                                 |
| allowed [Required]<br>bool                                                                                  | Allowed indicates whether or not the admission request was permitted.                                                                                                                                                                                                                                                       |
| status<br><a href="https://meta/v1.Status">meta/v1.Status</a>                                               | Result contains extra details into why an admission request was denied. This field IS NOT consulted in any way if "Allowed" is "true".                                                                                                                                                                                      |
| patch<br>[]byte                                                                                             | The patch body. Currently we only support "JSONPatch" which implements RFC 6902.                                                                                                                                                                                                                                            |
| patchType<br><a href="#">PatchType</a>                                                                      | The type of Patch. Currently we only allow "JSONPatch".                                                                                                                                                                                                                                                                     |
| auditAnnotations<br>map[string]string                                                                       | AuditAnnotations is an unstructured key value map set by remote admission controller (e.g. error=image-blacklisted). MutatingAdmissionWebhook and ValidatingAdmissionWebhook admission controller will prefix the keys with admission webhook name (e.g. imagepolicy.example.com/error=image-blacklisted). AuditAnnotations |



| Field                | Description                                                                                                                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | will be provided by the admission webhook to add additional context to the audit log for this request.                                                                                                                                                                                                         |
| warnings<br>[]string | warnings is a list of warning messages to return to the requesting API client. Warning messages describe a problem the client making the API request should correct or be aware of. Limit warnings to 120 characters if possible. Warnings over 256 characters and large numbers of warnings may be truncated. |

## Operation

(Alias of string)

**Appears in:**

- [AdmissionRequest](#)

Operation is the type of resource operation being checked for admission control

## PatchType

(Alias of string)

**Appears in:**

- [AdmissionResponse](#)

PatchType is the type of patch being used to represent the mutated object

# kube-apiserver Audit Configuration (v1)

## Resource Types

- [Event](#)
- [EventList](#)
- [Policy](#)
- [PolicyList](#)

## Event

**Appears in:**

- [EventList](#)

Event captures all the information that can be included in an API audit log.

| Field                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                             | audit.k8s.io/v1                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| kind<br>string                                                                   | Event                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| level [ <b>Required</b> ]<br><a href="#">Level</a>                               | AuditLevel at which event was generated                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| auditID [ <b>Required</b> ]<br><a href="#">k8s.io/apimachinery/pkg/types.UID</a> | Unique audit ID, generated for each request.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| stage [ <b>Required</b> ]<br><a href="#">Stage</a>                               | Stage of the request handling when this event instance was generated.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| requestURI [ <b>Required</b> ]<br>string                                         | RequestURI is the request URI as sent by the client to a server.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| verb [ <b>Required</b> ]<br>string                                               | Verb is the kubernetes verb associated with the request. For non-resource requests, this is the lower-cased HTTP method.                                                                                                                                                                                                                                                                                                                                                                      |
| user [ <b>Required</b> ]<br><a href="#">authentication/v1.UserInfo</a>           | Authenticated user information.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| impersonatedUser<br><a href="#">authentication/v1.UserInfo</a>                   | Impersonated user information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| sourceIPs<br>[]string                                                            | Source IPs, from where the request originated and intermediate proxies. The source IPs are listed from (in order): <ul style="list-style-type: none"> <li>1. X-Forwarded-For request header IPs</li> <li>2. X-Real-Ip header, if not present in the X-Forwarded-For list</li> <li>3. The remote address for the connection, if it doesn't match the last IP in the list up to here (X-Forwarded-For or X-Real-Ip). Note: All but the last IP can be arbitrarily set by the client.</li> </ul> |
| userAgent<br>string                                                              | UserAgent records the user agent string reported by the client. Note that the UserAgent is provided by the client, and must not be trusted.                                                                                                                                                                                                                                                                                                                                                   |
| objectRef<br><a href="#">ObjectReference</a>                                     | Object reference this request is targeted at. Does not apply for List-type requests, or non-resource requests.                                                                                                                                                                                                                                                                                                                                                                                |

| Field                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| responseStatus<br><a href="#">meta/v1.Status</a>                          | The response status, populated even when the ResponseObject is not a Status type. For successful responses, this will only include the Code and StatusSuccess. For non-status type error responses, this will be auto-populated with the error Message.                                                                                                                                                                                                                                                                                          |
| requestObject<br><a href="#">k8s.io/apimachinery/pkg/runtime.Unknown</a>  | API object from the request, in JSON format. The RequestObject is recorded as-is in the request (possibly re-encoded as JSON), prior to version conversion, defaulting, admission or merging. It is an external versioned object type, and may not be a valid object on its own. Omitted for non-resource requests. Only logged at Request Level and higher.                                                                                                                                                                                     |
| responseObject<br><a href="#">k8s.io/apimachinery/pkg/runtime.Unknown</a> | API object returned in the response, in JSON. The ResponseObject is recorded after conversion to the external type, and serialized as JSON. Omitted for non-resource requests. Only logged at Response Level.                                                                                                                                                                                                                                                                                                                                    |
| requestReceivedTimestamp<br><a href="#">meta/v1.MicroTime</a>             | Time the request reached the apiserver.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| stageTimestamp<br><a href="#">meta/v1.MicroTime</a>                       | Time the request reached current audit stage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| annotations<br>map[string]string                                          | Annotations is an unstructured key value map stored with an audit event that may be set by plugins invoked in the request serving chain, including authentication, authorization and admission plugins. Note that these annotations are for the audit event, and do not correspond to the metadata.annotations of the submitted object. Keys should uniquely identify the informing component to avoid name collisions (e.g. podsecuritypolicy.admission.k8s.io/policy). Values should be short. Annotations are included in the Metadata level. |

## EventList

EventList is a list of audit Events.

| Field                                              | Description              |
|----------------------------------------------------|--------------------------|
| apiVersion<br>string                               | audit.k8s.io/v1          |
| kind<br>string                                     | EventList                |
| metadata<br><a href="#">meta/v1.ListMeta</a>       | No description provided. |
| items <b>[Required]</b><br><a href="#">[]Event</a> | No description provided. |

# Policy

Appears in:

- [PolicyList](#)

Policy defines the configuration of audit logging, and the rules for how different request categories are logged.

| Field                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                    | audit.k8s.io/v1                                                                                                                                                                                                                                                                                                                                                                                                               |
| kind<br>string                                          | Policy                                                                                                                                                                                                                                                                                                                                                                                                                        |
| metadata<br><a href="#">meta/v1.ObjectMeta</a>          | ObjectMeta is included for interoperability with API infrastructure.<br><br>Refer to the Kubernetes API documentation for the fields of the metadata field.                                                                                                                                                                                                                                                                   |
| rules <b>[Required]</b><br><a href="#">[]PolicyRule</a> | Rules specify the audit Level a request should be recorded at. A request may match multiple rules, in which case the FIRST matching rule is used. The default audit level is None, but can be overridden by a catch-all rule at the end of the list. PolicyRules are strictly ordered.                                                                                                                                        |
| omitStages<br><a href="#">[]Stage</a>                   | OmitStages is a list of stages for which no events are created. Note that this can also be specified per rule in which case the union of both are omitted.                                                                                                                                                                                                                                                                    |
| omitManagedFields<br>bool                               | OmitManagedFields indicates whether to omit the managed fields of the request and response bodies from being written to the API audit log. This is used as a global default - a value of 'true' will omit the managed fields, otherwise the managed fields will be included in the API audit log. Note that this can also be specified per rule in which case the value specified in a rule will override the global default. |

## PolicyList

PolicyList is a list of audit Policies.

| Field                                        | Description              |
|----------------------------------------------|--------------------------|
| apiVersion<br>string                         | audit.k8s.io/v1          |
| kind<br>string                               | PolicyList               |
| metadata<br><a href="#">meta/v1.ListMeta</a> | No description provided. |
|                                              | No description provided. |

| Field                                               | Description |
|-----------------------------------------------------|-------------|
| items <b>[Required]</b><br><a href="#">[]Policy</a> |             |

## GroupResources

Appears in:

- [PolicyRule](#)

GroupResources represents resource kinds in an API group.

| Field                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| group<br>string                           | Group is the name of the API group that contains the resources. The empty string represents the core API group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| resources<br><a href="#">[]string</a>     | <p>Resources is a list of resources this rule applies to.</p> <p>For example:</p> <ul style="list-style-type: none"> <li>• pods matches pods.</li> <li>• pods/log matches the log subresource of pods.</li> <li>• * matches all resources and their subresources.</li> <li>• pods/* matches all subresources of pods.</li> <li>• */scale matches all scale subresources.</li> </ul> <p>If wildcard is present, the validation rule will ensure resources do not overlap with each other.</p> <p>An empty list implies all resources and subresources in this API groups apply.</p> |
| resourceNames<br><a href="#">[]string</a> | <p>ResourceNames is a list of resource instance names that the policy matches. Using this field requires Resources to be specified. An empty list implies that every instance of the resource is matched.</p>                                                                                                                                                                                                                                                                                                                                                                      |

## Level

(Alias of string)

Appears in:

- [Event](#)
- [PolicyRule](#)

Level defines the amount of information logged during auditing

# ObjectReference

Appears in:

- [Event](#)

ObjectReference contains enough information to let you inspect or modify the referred object.

| Field                                                                                            | Description                                                                                                              |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| resource<br>string                                                                               | No description provided.                                                                                                 |
| namespace<br>string                                                                              | No description provided.                                                                                                 |
| name<br>string                                                                                   | No description provided.                                                                                                 |
| uid<br><a href="https://k8s.io/apimachinery/pkg/types/UID">k8s.io/apimachinery/pkg/types/UID</a> | No description provided.                                                                                                 |
| apiGroup<br>string                                                                               | APIGroup is the name of the API group that contains the referred object. The empty string represents the core API group. |
| apiVersion<br>string                                                                             | APIVersion is the version of the API group that contains the referred object.                                            |
| resourceVersion<br>string                                                                        | No description provided.                                                                                                 |
| subresource<br>string                                                                            | No description provided.                                                                                                 |

# PolicyRule

Appears in:

- [Policy](#)

PolicyRule maps requests based off metadata to an audit Level. Requests must match the rules of every field (an intersection of rules).

| Field                                              | Description                                                                                    |
|----------------------------------------------------|------------------------------------------------------------------------------------------------|
| level [ <b>Required</b> ]<br><a href="#">Level</a> | The Level that requests matching this rule are recorded at.                                    |
| users<br>[]string                                  | The users (by authenticated user name) this rule applies to. An empty list implies every user. |

| Field                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| userGroups<br>[]string                        | The user groups this rule applies to. A user is considered matching if it is a member of any of the UserGroups. An empty list implies every user group.                                                                                                                                                                                                                                                                                                                                                                                                             |
| verbs<br>[]string                             | The verbs that match this rule. An empty list implies every verb.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| resources<br><a href="#">[]GroupResources</a> | Resources that this rule matches. An empty list implies all kinds in all API groups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| namespaces<br>[]string                        | Namespaces that this rule matches. The empty string "" matches non-namespaced resources. An empty list implies every namespace.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| nonResourceURLs<br>[]string                   | NonResourceURLs is a set of URL paths that should be audited. *s are allowed, but only as the full, final step in the path. Examples: <ul style="list-style-type: none"> <li>• /metrics - Log requests for apiserver metrics</li> <li>• /healthz* - Log all health checks</li> </ul>                                                                                                                                                                                                                                                                                |
| omitStages<br><a href="#">[]Stage</a>         | OmitStages is a list of stages for which no events are created. Note that this can also be specified policy wide in which case the union of both are omitted. An empty list means no restrictions will apply.                                                                                                                                                                                                                                                                                                                                                       |
| omitManagedFields<br>bool                     | OmitManagedFields indicates whether to omit the managed fields of the request and response bodies from being written to the API audit log. <ul style="list-style-type: none"> <li>• a value of 'true' will drop the managed fields from the API audit log</li> <li>• a value of 'false' indicates that the managed fields should be included in the API audit log Note that the value, if specified, in this rule will override the global default If a value is not specified then the global default specified in Policy.OmitManagedFields will stand.</li> </ul> |

## Stage

(Alias of string)

**Appears in:**

- [Event](#)
- [Policy](#)
- [PolicyRule](#)

Stage defines the stages in request handling that audit events may be generated.

# kube-apiserver Configuration (v1)

Package v1 is the v1 version of the API.

## Resource Types

- [AdmissionConfiguration](#)
- [EncryptionConfiguration](#)

## AdmissionConfiguration

AdmissionConfiguration provides versioned configuration for admission controllers.

| Field                                                     | Description                                                             |
|-----------------------------------------------------------|-------------------------------------------------------------------------|
| apiVersion<br>string                                      | apiserver.config.k8s.io/v1                                              |
| kind<br>string                                            | AdmissionConfiguration                                                  |
| plugins<br><a href="#">[]AdmissionPluginConfiguration</a> | Plugins allows specifying a configuration per admission control plugin. |

## EncryptionConfiguration

EncryptionConfiguration stores the complete configuration for encryption providers. It also allows the use of wildcards to specify the resources that should be encrypted. Use '\*.<group>' to encrypt all resources within a group or '\*' to encrypt all resources. '\*' can be used to encrypt all resource in the core group. '\*' will encrypt all resources, even custom resources that are added after API server start. Use of wildcards that overlap within the same resource list or across multiple entries are not allowed since part of the configuration would be ineffective. Resource lists are processed in order, with earlier lists taking precedence.

Example:

```
kind: EncryptionConfiguration
apiVersion: apiserver.config.k8s.io/v1
resources:
- resources:
 - events
 providers:
 - identity: {} # do not encrypt events even though '*' is specified below
- resources:
 - secrets
 - configmaps
 - pandas.awesome.bears.example
 providers:
 - aescbc:
 keys:
```



```

- name: key1
 secret: c2VjcmV0IGlzIHNIY3VyZQ==
- resources:
 - '*.apps'
providers:
- aescbc:
 keys:
 - name: key2
 secret: c2VjcmV0IGlzIHNIY3VyZSwgb3IgaXMgaXQ/Cg==
- resources:
 - '**'
providers:
- aescbc:
 keys:
 - name: key3
 secret: c2VjcmV0IGlzIHNIY3VyZSwgSSB0aGluaw==

```

| Field                                                                  | Description                                                                             |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| apiVersion<br>string                                                   | apiserver.config.k8s.io/v1                                                              |
| kind<br>string                                                         | EncryptionConfiguration                                                                 |
| resources <b>[Required]</b><br><a href="#">[]ResourceConfiguration</a> | resources is a list containing resources, and their corresponding encryption providers. |

## AESConfiguration

Appears in:

- [ProviderConfiguration](#)

AESConfiguration contains the API configuration for an AES transformer.

| Field                                              | Description                                                                                                                                          |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| keys<br><b>[Required]</b><br><a href="#">[]Key</a> | keys is a list of keys to be used for creating the AES transformer. Each key has to be 32 bytes long for AES-CBC and 16, 24 or 32 bytes for AES-GCM. |

## AdmissionPluginConfiguration

Appears in:

- [AdmissionConfiguration](#)

AdmissionPluginConfiguration provides the configuration for a single plug-in.

| Field                                                                                                                  | Description                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name <b>[Required]</b><br>string                                                                                       | Name is the name of the admission controller. It must match the registered admission plugin name.                                                                      |
| path<br>string                                                                                                         | Path is the path to a configuration file that contains the plugin's configuration                                                                                      |
| configuration<br><a href="https://k8s.io/apimachinery/pkg/runtime.Unknown">k8s.io/apimachinery/pkg/runtime.Unknown</a> | Configuration is an embedded configuration object to be used as the plugin's configuration. If present, it will be used instead of the path to the configuration file. |

## IdentityConfiguration

Appears in:

- [ProviderConfiguration](#)

IdentityConfiguration is an empty struct to allow identity transformer in provider configuration.

## KMSConfiguration

Appears in:

- [ProviderConfiguration](#)

KMSConfiguration contains the name, cache size and path to configuration file for a KMS based envelope transformer.

| Field                                       | Description                                                                                                                                                                                    |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                        | apiVersion of KeyManagementService                                                                                                                                                             |
| name<br><b>[Required]</b><br>string         | name is the name of the KMS plugin to be used.                                                                                                                                                 |
| cachesize<br>int32                          | cachesize is the maximum number of secrets which are cached in memory. The default value is 1000. Set to a negative value to disable caching. This field is only allowed for KMS v1 providers. |
| endpoint<br><b>[Required]</b><br>string     | endpoint is the gRPC server listening address, for example "unix:///var/run/kms-provider.sock".                                                                                                |
| timeout<br><a href="#">meta/v1.Duration</a> | timeout for gRPC calls to kms-plugin (ex. 5s). The default is 3 seconds.                                                                                                                       |

## Key

Appears in:

- [AESConfiguration](#)
- [SecretboxConfiguration](#)

Key contains name and secret of the provided key for a transformer.

| Field                              | Description                                                        |
|------------------------------------|--------------------------------------------------------------------|
| name <b>[Required]</b><br>string   | name is the name of the key to be used while storing data to disk. |
| secret <b>[Required]</b><br>string | secret is the actual key, encoded in base64.                       |

## ProviderConfiguration

Appears in:

- [ResourceConfiguration](#)

ProviderConfiguration stores the provided configuration for an encryption provider.

| Field                                                                 | Description                                                                                            |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| aesgcm <b>[Required]</b><br><a href="#">AESConfiguration</a>          | aesgcm is the configuration for the AES-GCM transformer.                                               |
| aescbc <b>[Required]</b><br><a href="#">AESConfiguration</a>          | aescbc is the configuration for the AES-CBC transformer.                                               |
| secretbox <b>[Required]</b><br><a href="#">SecretboxConfiguration</a> | secretbox is the configuration for the Secretbox based transformer.                                    |
| identity <b>[Required]</b><br><a href="#">IdentityConfiguration</a>   | identity is the (empty) configuration for the identity transformer.                                    |
| kms <b>[Required]</b><br><a href="#">KMSConfiguration</a>             | kms contains the name, cache size and path to configuration file for a KMS based envelope transformer. |

## ResourceConfiguration

Appears in:

- [EncryptionConfiguration](#)

ResourceConfiguration stores per resource configuration.

| Field                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resources <b>[Required]</b><br>[]string                                | resources is a list of kubernetes resources which have to be encrypted. The resource names are derived from resource or resource.group of the group/version/resource. eg: pandas.awesome.bears.example is a custom resource with 'group': awesome.bears.example, 'resource': pandas. Use '*' to encrypt all resources and '*.<group>' to encrypt all resources in a specific group. eg: '*.awesome.bears.example' will encrypt all resources in the group 'awesome.bears.example'. eg: '*' will encrypt all resources in the core group (such as pods, configmaps, etc). |
| providers <b>[Required]</b><br><a href="#">[]ProviderConfiguration</a> | providers is a list of transformers to be used for reading and writing the resources to disk. eg: aesgcm, aescbc, secretbox, identity, kms.                                                                                                                                                                                                                                                                                                                                                                                                                              |

## SecretboxConfiguration

Appears in:

- [ProviderConfiguration](#)

SecretboxConfiguration contains the API configuration for an Secretbox transformer.

| Field                                           | Description                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| keys <b>[Required]</b><br><a href="#">[]Key</a> | keys is a list of keys to be used for creating the Secretbox transformer. Each key has to be 32 bytes long. |

## kube-apiserver Configuration (v1alpha1)

Package v1alpha1 is the v1alpha1 version of the API.

### Resource Types

- [AdmissionConfiguration](#)
- [AuthenticationConfiguration](#)
- [AuthorizationConfiguration](#)
- [EgressSelectorConfiguration](#)
- [TracingConfiguration](#)

### TracingConfiguration

Appears in:

- [KubeletConfiguration](#)
- [TracingConfiguration](#)

TracingConfiguration provides versioned configuration for OpenTelemetry tracing clients.

| Field                           | Description                                                                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| endpoint<br>string              | Endpoint of the collector this component will report traces to. The connection is insecure, and does not currently support TLS. Recommended is unset, and endpoint is the otel grpc default, localhost:4317. |
| samplingRatePerMillion<br>int32 | SamplingRatePerMillion is the number of samples to collect per million spans. Recommended is unset. If unset, sampler respects its parent span's sampling rate, but otherwise never samples.                 |

## AdmissionConfiguration

AdmissionConfiguration provides versioned configuration for admission controllers.

| Field                                                     | Description                                                             |
|-----------------------------------------------------------|-------------------------------------------------------------------------|
| apiVersion<br>string                                      | apiserver.k8s.io/v1alpha1                                               |
| kind<br>string                                            | AdmissionConfiguration                                                  |
| plugins<br><a href="#">[]AdmissionPluginConfiguration</a> | Plugins allows specifying a configuration per admission control plugin. |

## AuthenticationConfiguration

AuthenticationConfiguration provides versioned configuration for authentication.

| Field                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                 | apiserver.k8s.io/v1alpha1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| kind<br>string                                       | AuthenticationConfiguration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| jwt [Required]<br><a href="#">[]JWTAuthenticator</a> | jwt is a list of authenticator to authenticate Kubernetes users using JWT compliant tokens. The authenticator will attempt to parse a raw ID token, verify it's been signed by the configured issuer. The public key to verify the signature is discovered from the issuer's public endpoint using OIDC discovery. For an incoming token, each JWT authenticator will be attempted in the order in which it is specified in this list. Note however that other authenticators may run before or after the JWT authenticators. The specific position of JWT authenticators in relation to other authenticators is neither defined nor stable across releases. Since each JWT authenticator must have a unique issuer URL, at most one JWT authenticator will attempt to cryptographically validate the token. |

| Field | Description                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | The minimum valid JWT payload must contain the following claims: {<br>"iss": "https://issuer.example.com", "aud": ["audience"], "exp": 1234567890,<br>"<username claim>": "username" } |

## AuthorizationConfiguration

| Field                                                                       | Description                                                                                                                                                         |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                        | apiserver.k8s.io/v1alpha1                                                                                                                                           |
| kind<br>string                                                              | AuthorizationConfiguration                                                                                                                                          |
| authorizers <b>[Required]</b><br><a href="#">[] AuthorizerConfiguration</a> | Authorizers is an ordered list of authorizers to authorize requests against. This is similar to the --authorization-modes kube-apiserver flag Must be at least one. |

## EgressSelectorConfiguration

EgressSelectorConfiguration provides versioned configuration for egress selector clients.

| Field                                                                       | Description                                                                  |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| apiVersion<br>string                                                        | apiserver.k8s.io/v1alpha1                                                    |
| kind<br>string                                                              | EgressSelectorConfiguration                                                  |
| egressSelections<br><b>[Required]</b><br><a href="#">[] EgressSelection</a> | connectionServices contains a list of egress selection client configurations |

## TracingConfiguration

TracingConfiguration provides versioned configuration for tracing clients.

| Field                                                                             | Description                                                                                                                   |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                              | apiserver.k8s.io/v1alpha1                                                                                                     |
| kind<br>string                                                                    | TracingConfiguration                                                                                                          |
| TracingConfiguration<br><b>[Required]</b><br><a href="#">TracingConfiguration</a> | (Members of TracingConfiguration are embedded into this type.)<br><br>Embed the component config tracing configuration struct |

## AdmissionPluginConfiguration

Appears in:

- [AdmissionConfiguration](#)

AdmissionPluginConfiguration provides the configuration for a single plug-in.

| Field                                                                                                                  | Description                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name <b>[Required]</b><br>string                                                                                       | Name is the name of the admission controller. It must match the registered admission plugin name.                                                                      |
| path<br>string                                                                                                         | Path is the path to a configuration file that contains the plugin's configuration                                                                                      |
| configuration<br><a href="https://k8s.io/apimachinery/pkg/runtime.Unknown">k8s.io/apimachinery/pkg/runtime.Unknown</a> | Configuration is an embedded configuration object to be used as the plugin's configuration. If present, it will be used instead of the path to the configuration file. |

## AudienceMatchPolicyType

(Alias of string)

Appears in:

- [Issuer](#)

AudienceMatchPolicyType is a set of valid values for issuer.audienceMatchPolicy

## AuthorizerConfiguration

Appears in:

- [AuthorizationConfiguration](#)

| Field                            | Description                                                                                                                                                                                                                     |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type <b>[Required]</b><br>string | Type refers to the type of the authorizer "Webhook" is supported in the generic API server Other API servers may support additional authorizer types like Node, RBAC, ABAC, etc.                                                |
| name <b>[Required]</b><br>string | Name used to describe the webhook This is explicitly used in monitoring machinery for metrics Note: Names must be DNS1123 labels like myauthorizername or subdomains like myauthorizer.example.domain Required, with no default |

| Field                                                             | Description                                                                                                                         |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| webhook <b>[Required]</b><br><a href="#">WebhookConfiguration</a> | Webhook defines the configuration for a Webhook authorizer Must be defined when Type=Webhook Must not be defined when Type!=Webhook |

## ClaimMappings

Appears in:

- [JWTAuthenticator](#)

ClaimMappings provides the configuration for claim mapping

| Field                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| username <b>[Required]</b><br><a href="#">PrefixedClaimOrExpression</a> | <p>username represents an option for the username attribute. The claim's value must be a singular string. Same as the --oidc-username-claim and --oidc-username-prefix flags. If username.expression is set, the expression must produce a string value. If username.expression uses 'claims.email', then 'claims.email_verified' must be used in username.expression or extra[*].valueExpression or claimValidationRules[*].expression. An example claim validation rule expression that matches the validation automatically applied when username.claim is set to 'email' is 'claims.?email_verified.orValue(true)'.</p> <p>In the flag based approach, the --oidc-username-claim and --oidc-username-prefix are optional. If --oidc-username-claim is not set, the default value is "sub". For the authentication config, there is no defaulting for claim or prefix. The claim and prefix must be set explicitly. For claim, if --oidc-username-claim was not set with legacy flag approach, configure username.claim="sub" in the authentication config. For prefix: (1) --oidc-username-prefix="-", no prefix was added to the username. For the same behavior using authentication config, set username.prefix="" (2) --oidc-username-prefix="" and --oidc-username-claim != "email", prefix was "&lt;value of --oidc-issuer-url&gt;#". For the same behavior using authentication config, set username.prefix="&lt;value of issuer.url&gt;#" (3) --oidc-username-prefix="&lt;value&gt;". For the same behavior using authentication config, set username.prefix="&lt;value&gt;"</p> |
| groups<br><a href="#">PrefixedClaimOrExpression</a>                     | groups represents an option for the groups attribute. The claim's value must be a string or string array claim. If groups.claim is set, the prefix must be specified (and can be the empty string). If groups.expression is set, the expression must produce a string or string array value. "", [], and null values are treated as the group mapping not being present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



| Field                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uid<br><a href="#">ClaimOrExpression</a> | uid represents an option for the uid attribute. Claim must be a singular string claim. If uid.expression is set, the expression must produce a string value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| extra<br><a href="#">[]ExtraMapping</a>  | <p>extra represents an option for the extra attribute. expression must produce a string or string array value. If the value is empty, the extra mapping will not be present.</p> <p>hard-coded extra key/value</p> <ul style="list-style-type: none"> <li>key: "foo" valueExpression: "bar" This will result in an extra attribute - foo: ["bar"]</li> </ul> <p>hard-coded key, value copying claim value</p> <ul style="list-style-type: none"> <li>key: "foo" valueExpression: "claims.some_claim" This will result in an extra attribute - foo: [value of some_claim]</li> </ul> <p>hard-coded key, value derived from claim value</p> <ul style="list-style-type: none"> <li>key: "admin" valueExpression: '(has(claims.is_admin) &amp;&amp; claims.is_admin) ? "true":""' This will result in:</li> <li>if is_admin claim is present and true, extra attribute - admin: ["true"]</li> <li>if is_admin claim is present and false or is_admin claim is not present, no extra attribute will be added</li> </ul> |

## ClaimOrExpression

Appears in:

- [ClaimMappings](#)

ClaimOrExpression provides the configuration for a single claim or expression.

| Field                | Description                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| claim<br>string      | claim is the JWT claim to use. Either claim or expression must be set. Mutually exclusive with expression.                                                                                                                                                                                                                                                                                                                         |
| expression<br>string | <p>expression represents the expression which will be evaluated by CEL.</p> <p>CEL expressions have access to the contents of the token claims, organized into CEL variable:</p> <ul style="list-style-type: none"> <li>'claims' is a map of claim names to claim values. For example, a variable named 'sub' can be accessed as 'claims.sub'. Nested claims can be accessed using dot notation, e.g. 'claims.foo.bar'.</li> </ul> |

| Field | Description                                                                                                                                                                       |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a><br><br>Mutually exclusive with claim. |

## ClaimValidationRule

Appears in:

- [JWTAuthenticator](#)

ClaimValidationRule provides the configuration for a single claim validation rule.

| Field                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| claim<br>string         | claim is the name of a required claim. Same as --oidc-required-claim flag. Only string claim keys are supported. Mutually exclusive with expression and message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| requiredValue<br>string | requiredValue is the value of a required claim. Same as --oidc-required-claim flag. Only string claim values are supported. If claim is set and requiredValue is not set, the claim must be present with a value set to the empty string. Mutually exclusive with expression and message.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| expression<br>string    | <p>expression represents the expression which will be evaluated by CEL. Must produce a boolean.</p> <p>CEL expressions have access to the contents of the token claims, organized into CEL variable:</p> <ul style="list-style-type: none"> <li>• 'claims' is a map of claim names to claim values. For example, a variable named 'sub' can be accessed as 'claims.sub'. Nested claims can be accessed using dot notation, e.g. 'claims.foo.bar'. Must return true for the validation to pass.</li> </ul> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a><br/><br/>Mutually exclusive with claim and requiredValue.</p> |
| message<br>string       | message customizes the returned error message when expression returns false. message is a literal string. Mutually exclusive with claim and requiredValue.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Connection

Appears in:

- [EgressSelection](#)

Connection provides the configuration for a single egress selection client.

| Field                                                       | Description                                                                                                                                         |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| proxyProtocol<br>[Required]<br><a href="#">ProtocolType</a> | Protocol is the protocol used to connect from client to the konnectivity server.                                                                    |
| transport<br><a href="#">Transport</a>                      | Transport defines the transport configurations we use to dial to the konnectivity server. This is required if ProxyProtocol is HTTPConnect or GRPC. |

## EgressSelection

Appears in:

- [EgressSelectorConfiguration](#)

EgressSelection provides the configuration for a single egress selection client.

| Field                                                  | Description                                                                                                                                                                                   |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name<br>[Required]<br>string                           | name is the name of the egress selection. Currently supported values are "controlplane", "master", "etcd" and "cluster" The "master" egress selector is deprecated in favor of "controlplane" |
| connection<br>[Required]<br><a href="#">Connection</a> | connection is the exact information used to configure the egress selection                                                                                                                    |

## ExtraMapping

Appears in:

- [ClaimMappings](#)

ExtraMapping provides the configuration for a single extra mapping.

| Field                                   | Description                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| key [Required]<br>string                | key is a string to use as the extra attribute key. key must be a domain-prefix path (e.g. example.org/foo). All characters before the first "/" must be a valid subdomain as defined by RFC 1123. All characters trailing the first "/" must be valid HTTP Path characters as defined by RFC 3986. key must be lowercase. Required to be unique. |
| valueExpression<br>[Required]<br>string | valueExpression is a CEL expression to extract extra attribute value. valueExpression must produce a string or string array value. "", [], and null values are treated as the extra mapping not being present. Empty string values contained within a string array are filtered out.                                                             |

| Field | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <p>CEL expressions have access to the contents of the token claims, organized into CEL variable:</p> <ul style="list-style-type: none"> <li>'claims' is a map of claim names to claim values. For example, a variable named 'sub' can be accessed as 'claims.sub'. Nested claims can be accessed using dot notation, e.g. 'claims.foo.bar'.</li> </ul> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a></p> |

## Issuer

Appears in:

- [JWTAuthenticator](#)

Issuer provides the configuration for an external provider's specific settings.

| Field                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| url <b>[Required]</b><br>string | <p>url points to the issuer URL in a format https://url or https://url/path. This must match the "iss" claim in the presented JWT, and the issuer returned from discovery. Same value as the --oidc-issuer-url flag. Discovery information is fetched from "{url}/.well-known/openid-configuration" unless overridden by discoveryURL. Required to be unique across all JWT authenticators. Note that egress selection configuration is not used for this network connection.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| discoveryURL<br>string          | <p>discoveryURL, if specified, overrides the URL used to fetch discovery information instead of using "{url}/.well-known/openid-configuration". The exact value specified is used, so "/.well-known/openid-configuration" must be included in discoveryURL if needed.</p> <p>The "issuer" field in the fetched discovery information must match the "issuer.url" field in the AuthenticationConfiguration and will be used to validate the "iss" claim in the presented JWT. This is for scenarios where the well-known and jwks endpoints are hosted at a different location than the issuer (such as locally in the cluster).</p> <p>Example: A discovery url that is exposed using kubernetes service 'oidc' in namespace 'oidc-namespace' and discovery information is available at '/.well-known/openid-configuration'. discoveryURL: "https://oidc.oidc-namespace/.well-known/openid-configuration" certificateAuthority is used to verify the TLS connection and the hostname on the leaf certificate must be set to 'oidc.oidc-namespace'.</p> <p>curl https://oidc.oidc-namespace/.well-known/openid-configuration (.discoveryURL field) { issuer: "https://oidc.example.com" (.url field) }</p> |

| Field                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | discoveryURL must be different from url. Required to be unique across all JWT authenticators. Note that egress selection configuration is not used for this network connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| certificateAuthority string                                 | certificateAuthority contains PEM-encoded certificate authority certificates used to validate the connection when fetching discovery information. If unset, the system verifier is used. Same value as the content of the file referenced by the --oidc-ca-file flag.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| audiences [ <b>Required</b> ] []string                      | audiences is the set of acceptable audiences the JWT must be issued to. At least one of the entries must match the "aud" claim in presented JWTs. Same value as the --oidc-client-id flag (though this field supports an array). Required to be non-empty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| audienceMatchPolicy <a href="#">AudienceMatchPolicyType</a> | <p>audienceMatchPolicy defines how the "audiences" field is used to match the "aud" claim in the presented JWT. Allowed values are:</p> <ol style="list-style-type: none"> <li>1. "MatchAny" when multiple audiences are specified and</li> <li>2. empty (or unset) or "MatchAny" when a single audience is specified.</li> </ol> <ul style="list-style-type: none"> <li>• MatchAny: the "aud" claim in the presented JWT must match at least one of the entries in the "audiences" field. For example, if "audiences" is ["foo", "bar"], the "aud" claim in the presented JWT must contain either "foo" or "bar" (and may contain both).</li> <li>• "": The match policy can be empty (or unset) when a single audience is specified in the "audiences" field. The "aud" claim in the presented JWT must contain the single audience (and may contain others).</li> </ul> <p>For more nuanced audience validation, use claimValidationRules. example: claimValidationRule[].expression: 'sets.equivalent(claims.aud, ["bar", "foo", "baz"])' to require an exact match.</p> |

## JWTAuthenticator

Appears in:

- [AuthenticationConfiguration](#)

JWTAuthenticator provides the configuration for a single JWT authenticator.

| Field                                                               | Description                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| issuer <b>[Required]</b><br><a href="#">Issuer</a>                  | issuer contains the basic OIDC provider connection options.                                                                                                                                                                                                                                                                                                        |
| claimValidationRules<br><a href="#">[]ClaimValidationRule</a>       | claimValidationRules are rules that are applied to validate token claims to authenticate users.                                                                                                                                                                                                                                                                    |
| claimMappings<br><b>[Required]</b><br><a href="#">ClaimMappings</a> | claimMappings points claims of a token to be treated as user attributes.                                                                                                                                                                                                                                                                                           |
| userValidationRules<br><a href="#">[]UserValidationRule</a>         | userValidationRules are rules that are applied to final user before completing authentication. These allow invariants to be applied to incoming identities such as preventing the use of the system: prefix that is commonly used by Kubernetes components. The validation rules are logically ANDed together and must all return true for the validation to pass. |

## PrefixedClaimOrExpression

Appears in:

- [ClaimMappings](#)

PrefixedClaimOrExpression provides the configuration for a single prefixed claim or expression.

| Field             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| claim string      | claim is the JWT claim to use. Mutually exclusive with expression.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| prefix string     | prefix is prepended to claim's value to prevent clashes with existing names. prefix needs to be set if claim is set and can be the empty string. Mutually exclusive with expression.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| expression string | <p>expression represents the expression which will be evaluated by CEL.</p> <p>CEL expressions have access to the contents of the token claims, organized into CEL variable:</p> <ul style="list-style-type: none"> <li>• 'claims' is a map of claim names to claim values. For example, a variable named 'sub' can be accessed as 'claims.sub'. Nested claims can be accessed using dot notation, e.g. 'claims.foo.bar'.</li> </ul> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a></p> <p>Mutually exclusive with claim and prefix.</p> |

## ProtocolType

(Alias of string)

**Appears in:**

- [Connection](#)

ProtocolType is a set of valid values for Connection.ProtocolType

## TCPTransport

**Appears in:**

- [Transport](#)

TCPTransport provides the information to connect to konnectivity server via TCP

| Field                                  | Description                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------|
| url<br>[Required]<br>string            | URL is the location of the konnectivity server to connect to. As an example it might be "https://127.0.0.1:8131" |
| tlsConfig<br><a href="#">TLSConfig</a> | TLSConfig is the config needed to use TLS when connecting to konnectivity server                                 |

## TLSConfig

**Appears in:**

- [TCPTransport](#)

TLSConfig provides the authentication information to connect to konnectivity server Only used with TCPTransport

| Field                | Description                                                                                                                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| caBundle<br>string   | caBundle is the file location of the CA to be used to determine trust with the konnectivity server. Must be absent/empty if TCPTransport.URL is prefixed with http:// If absent while TCPTransport.URL is prefixed with https://, default to system trust roots. |
| clientKey<br>string  | clientKey is the file location of the client key to be used in mTLS handshakes with the konnectivity server. Must be absent/empty if TCPTransport.URL is prefixed with http:// Must be configured if TCPTransport.URL is prefixed with https://                  |
| clientCert<br>string |                                                                                                                                                                                                                                                                  |

| Field | Description                                                                                                                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | clientCert is the file location of the client certificate to be used in mTLS handshakes with the konnectivity server. Must be absent/empty if TCPTransport.URL is prefixed with http:// Must be configured if TCPTransport.URL is prefixed with https:// |

## Transport

Appears in:

- [Connection](#)

Transport defines the transport configurations we use to dial to the konnectivity server

| Field                               | Description                                                                                                                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tcp<br><a href="#">TCPTransport</a> | TCP is the TCP configuration for communicating with the konnectivity server via TCP ProxyProtocol of GRPC is not supported with TCP transport at the moment Requires at least one of TCP or UDS to be set |
| uds<br><a href="#">UDSTransport</a> | UDS is the UDS configuration for communicating with the konnectivity server via UDS Requires at least one of TCP or UDS to be set                                                                         |

## UDSTransport

Appears in:

- [Transport](#)

UDSTransport provides the information to connect to konnectivity server via UDS

| Field                           | Description                                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| udsName<br>[Required]<br>string | UDSName is the name of the unix domain socket to connect to konnectivity server This does not use a unix:// prefix. (Eg: /etc/srv/kubernetes/konnectivity-server/konnectivity-server.socket) |

## UserValidationRule

Appears in:

- [JWTAuthenticator](#)

UserValidationRule provides the configuration for a single user info validation rule.



| Field                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| expression<br>[Required]<br>string | <p>expression represents the expression which will be evaluated by CEL. Must return true for the validation to pass.</p> <p>CEL expressions have access to the contents of UserInfo, organized into CEL variable:</p> <ul style="list-style-type: none"> <li>'user' - authentication.k8s.io/v1, Kind=UserInfo object Refer to <a href="https://github.com/kubernetes/api/blob/release-1.28/authentication/v1/types.go#L105-L122">https://github.com/kubernetes/api/blob/release-1.28/authentication/v1/types.go#L105-L122</a> for the definition. API documentation: <a href="https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.28/#userinfo-v1-authentication-k8s-io">https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.28/#userinfo-v1-authentication-k8s-io</a></li> </ul> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a></p> |
| message<br>string                  | <p>message customizes the returned error message when rule returns false. message is a literal string.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## WebhookConfiguration

Appears in:

- [AuthorizerConfiguration](#)

| Field                                                          | Description                                                                                                                                                                                                       |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| authorizedTTL [Required]<br><a href="#">meta/v1.Duration</a>   | The duration to cache 'authorized' responses from the webhook authorizer. Same as setting --authorization-webhook-cache-authorized-ttl flag Default: 5m0s                                                         |
| unauthorizedTTL [Required]<br><a href="#">meta/v1.Duration</a> | The duration to cache 'unauthorized' responses from the webhook authorizer. Same as setting --authorization-webhook-cache-unauthorized-ttl flag Default: 30s                                                      |
| timeout [Required]<br><a href="#">meta/v1.Duration</a>         | Timeout for the webhook request Maximum allowed value is 30s. Required, no default value.                                                                                                                         |
| subjectAccessReviewVersion [Required]<br>string                | The API version of the authorization.k8s.io SubjectAccessReview to send to and expect from the webhook. Same as setting --authorization-webhook-version flag Valid values: v1beta1, v1 Required, no default value |

| Field                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| matchConditionSubjectAccessReviewVersion<br><b>[Required]</b><br>string      | MatchConditionSubjectAccessReviewVersion specifies the SubjectAccessReview version the CEL expressions are evaluated against Valid values: v1 Required, no default value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| failurePolicy <b>[Required]</b><br>string                                    | Controls the authorization decision when a webhook request fails to complete or returns a malformed response or errors evaluating matchConditions. Valid values: <ul style="list-style-type: none"> <li>• NoOpinion: continue to subsequent authorizers to see if one of them allows the request</li> <li>• Deny: reject the request without consulting subsequent authorizers Required, with no default.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   |
| connectionInfo <b>[Required]</b><br><a href="#">WebhookConnectionInfo</a>    | ConnectionInfo defines how we talk to the webhook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| matchConditions <b>[Required]</b><br><a href="#">[]WebhookMatchCondition</a> | matchConditions is a list of conditions that must be met for a request to be sent to this webhook. An empty list of matchConditions matches all requests. There are a maximum of 64 match conditions allowed.<br><br>The exact matching logic is (in order): <ol style="list-style-type: none"> <li>1. If at least one matchCondition evaluates to FALSE, then the webhook is skipped.</li> <li>2. If ALL matchConditions evaluate to TRUE, then the webhook is called.</li> <li>3. If at least one matchCondition evaluates to an error (but none are FALSE): <ul style="list-style-type: none"> <li>◦ If failurePolicy=Deny, then the webhook rejects the request</li> <li>◦ If failurePolicy=NoOpinion, then the error is ignored and the webhook is skipped</li> </ul> </li> </ol> |

## WebhookConnectionInfo

Appears in:

- [WebhookConfiguration](#)

| Field                                      | Description                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type <b>[Required]</b><br>string           | Controls how the webhook should communicate with the server. Valid values: <ul style="list-style-type: none"> <li>KubeConfigFile: use the file specified in kubeConfigFile to locate the server.</li> <li>InClusterConfig: use the in-cluster configuration to call the SubjectAccessReview API hosted by kube-apiserver. This mode is not allowed for kube-apiserver.</li> </ul> |
| kubeConfigFile <b>[Required]</b><br>string | Path to KubeConfigFile for connection info Required, if connectionInfo.Type is KubeConfig                                                                                                                                                                                                                                                                                         |

## WebhookMatchCondition

Appears in:

- [WebhookConfiguration](#)

| Field                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| expression <b>[Required]</b><br>string | expression represents the expression which will be evaluated by CEL. Must evaluate to bool. CEL expressions have access to the contents of the SubjectAccessReview in v1 version. If version specified by subjectAccessReviewVersion in the request variable is v1beta1, the contents would be converted to the v1 version before evaluating the CEL expression.<br><br>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a> |

## kube-apiserver Configuration (v1beta1)

Package v1beta1 is the v1beta1 version of the API.

### Resource Types

- [AuthenticationConfiguration](#)
- [AuthorizationConfiguration](#)
- [EgressSelectorConfiguration](#)
- [TracingConfiguration](#)

## TracingConfiguration

Appears in:

- [KubeletConfiguration](#)

[TracingConfiguration](#)

- [TracingConfiguration](#)

TracingConfiguration provides versioned configuration for OpenTelemetry tracing clients.

| Field                           | Description                                                                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| endpoint<br>string              | Endpoint of the collector this component will report traces to. The connection is insecure, and does not currently support TLS. Recommended is unset, and endpoint is the otel grpc default, localhost:4317. |
| samplingRatePerMillion<br>int32 | SamplingRatePerMillion is the number of samples to collect per million spans. Recommended is unset. If unset, sampler respects its parent span's sampling rate, but otherwise never samples.                 |

## AuthenticationConfiguration

AuthenticationConfiguration provides versioned configuration for authentication.

| Field                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                 | apiserver.k8s.io/v1beta1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| kind<br>string                                       | AuthenticationConfiguration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| jwt [Required]<br><a href="#">[]JWTAuthenticator</a> | <p>jwt is a list of authenticator to authenticate Kubernetes users using JWT compliant tokens. The authenticator will attempt to parse a raw ID token, verify it's been signed by the configured issuer. The public key to verify the signature is discovered from the issuer's public endpoint using OIDC discovery. For an incoming token, each JWT authenticator will be attempted in the order in which it is specified in this list. Note however that other authenticators may run before or after the JWT authenticators. The specific position of JWT authenticators in relation to other authenticators is neither defined nor stable across releases. Since each JWT authenticator must have a unique issuer URL, at most one JWT authenticator will attempt to cryptographically validate the token.</p> <p>The minimum valid JWT payload must contain the following claims: {<br/>"iss": "https://issuer.example.com", "aud": ["audience"], "exp": 1234567890,<br/>"&lt;username claim&gt;": "username" }</p> |

## AuthorizationConfiguration

| Field                | Description              |
|----------------------|--------------------------|
| apiVersion<br>string | apiserver.k8s.io/v1beta1 |

| Field                                                                      | Description                                                                                                                                                         |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kind<br>string                                                             | AuthorizationConfiguration                                                                                                                                          |
| authorizers <b>[Required]</b><br><a href="#">[]AuthorizerConfiguration</a> | Authorizers is an ordered list of authorizers to authorize requests against. This is similar to the --authorization-modes kube-apiserver flag Must be at least one. |

## EgressSelectorConfiguration

EgressSelectorConfiguration provides versioned configuration for egress selector clients.

| Field                                                                   | Description                                                                  |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------|
| apiVersion<br>string                                                    | apiserver.k8s.io/v1beta1                                                     |
| kind<br>string                                                          | EgressSelectorConfiguration                                                  |
| egressSelections <b>[Required]</b><br><a href="#">[]EgressSelection</a> | connectionServices contains a list of egress selection client configurations |

## TracingConfiguration

TracingConfiguration provides versioned configuration for tracing clients.

| Field                                                                          | Description                                                                                                                   |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                           | apiserver.k8s.io/v1beta1                                                                                                      |
| kind<br>string                                                                 | TracingConfiguration                                                                                                          |
| TracingConfiguration <b>[Required]</b><br><a href="#">TracingConfiguration</a> | (Members of TracingConfiguration are embedded into this type.)<br><br>Embed the component config tracing configuration struct |

## AudienceMatchPolicyType

(Alias of string)

Appears in:

- [Issuer](#)

AudienceMatchPolicyType is a set of valid values for issuer.audienceMatchPolicy

# AuthorizerConfiguration

Appears in:

- [AuthorizationConfiguration](#)

| Field                                                             | Description                                                                                                                                                                                                                     |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type <b>[Required]</b><br>string                                  | Type refers to the type of the authorizer "Webhook" is supported in the generic API server Other API servers may support additional authorizer types like Node, RBAC, ABAC, etc.                                                |
| name <b>[Required]</b><br>string                                  | Name used to describe the webhook This is explicitly used in monitoring machinery for metrics Note: Names must be DNS1123 labels like myauthorizername or subdomains like myauthorizer.example.domain Required, with no default |
| webhook <b>[Required]</b><br><a href="#">WebhookConfiguration</a> | Webhook defines the configuration for a Webhook authorizer Must be defined when Type=Webhook Must not be defined when Type!=Webhook                                                                                             |

# ClaimMappings

Appears in:

- [JWTAuthenticator](#)

ClaimMappings provides the configuration for claim mapping

| Field                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| username <b>[Required]</b><br><a href="#">PrefixedClaimOrExpression</a> | <p>username represents an option for the username attribute. The claim's value must be a singular string. Same as the --oidc-username-claim and --oidc-username-prefix flags. If username.expression is set, the expression must produce a string value. If username.expression uses 'claims.email', then 'claims.email_verified' must be used in username.expression or extra[*].valueExpression or claimValidationRules[*].expression. An example claim validation rule expression that matches the validation automatically applied when username.claim is set to 'email' is 'claims.?email_verified.orValue(true)'.</p> <p>In the flag based approach, the --oidc-username-claim and --oidc-username-prefix are optional. If --oidc-username-claim is not set, the default value is "sub". For the authentication config, there is no defaulting for claim or prefix. The claim and prefix must be set explicitly. For claim, if --oidc-username-claim was not set with legacy flag approach, configure username.claim="sub" in the authentication config. For prefix: (1) --oidc-username-prefix="-", no</p> |

| Field                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | prefix was added to the username. For the same behavior using authentication config, set username.prefix="" (2) --oidc-username-prefix="" and --oidc-username-claim != "email", prefix was "<value of --oidc-issuer-url>#". For the same behavior using authentication config, set username.prefix="<value of issuer.url>#" (3) --oidc-username-prefix="<value>". For the same behavior using authentication config, set username.prefix="<value>"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| groups<br><a href="#">PrefixedClaimOrExpression</a> | groups represents an option for the groups attribute. The claim's value must be a string or string array claim. If groups.claim is set, the prefix must be specified (and can be the empty string). If groups.expression is set, the expression must produce a string or string array value. "", [], and null values are treated as the group mapping not being present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| uid<br><a href="#">ClaimOrExpression</a>            | uid represents an option for the uid attribute. Claim must be a singular string claim. If uid.expression is set, the expression must produce a string value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| extra<br><a href="#">[]ExtraMapping</a>             | <p>extra represents an option for the extra attribute. expression must produce a string or string array value. If the value is empty, the extra mapping will not be present.</p> <p>hard-coded extra key/value</p> <ul style="list-style-type: none"> <li>• key: "foo" valueExpression: "'bar'" This will result in an extra attribute - foo: ["bar"]</li> </ul> <p>hard-coded key, value copying claim value</p> <ul style="list-style-type: none"> <li>• key: "foo" valueExpression: "claims.some_claim" This will result in an extra attribute - foo: [value of some_claim]</li> </ul> <p>hard-coded key, value derived from claim value</p> <ul style="list-style-type: none"> <li>• key: "admin" valueExpression: '(has(claims.is_admin) &amp;&amp; claims.is_admin) ? "true":""' This will result in:</li> <li>• if is_admin claim is present and true, extra attribute - admin: ["true"]</li> <li>• if is_admin claim is present and false or is_admin claim is not present, no extra attribute will be added</li> </ul> |

## ClaimOrExpression

Appears in:

- [ClaimMappings](#)

ClaimOrExpression provides the configuration for a single claim or expression.

| Field                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| claim<br>string      | claim is the JWT claim to use. Either claim or expression must be set. Mutually exclusive with expression.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| expression<br>string | <p>expression represents the expression which will be evaluated by CEL.</p> <p>CEL expressions have access to the contents of the token claims, organized into CEL variable:</p> <ul style="list-style-type: none"><li>'claims' is a map of claim names to claim values. For example, a variable named 'sub' can be accessed as 'claims.sub'. Nested claims can be accessed using dot notation, e.g. 'claims.foo.bar'.</li></ul> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a></p> <p>Mutually exclusive with claim.</p> |

## ClaimValidationRule

Appears in:

- [JWTAuthenticator](#)

ClaimValidationRule provides the configuration for a single claim validation rule.

| Field                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| claim<br>string         | claim is the name of a required claim. Same as --oidc-required-claim flag. Only string claim keys are supported. Mutually exclusive with expression and message.                                                                                                                                                                                                                                                                                                                                      |
| requiredValue<br>string | requiredValue is the value of a required claim. Same as --oidc-required-claim flag. Only string claim values are supported. If claim is set and requiredValue is not set, the claim must be present with a value set to the empty string. Mutually exclusive with expression and message.                                                                                                                                                                                                             |
| expression<br>string    | <p>expression represents the expression which will be evaluated by CEL. Must produce a boolean.</p> <p>CEL expressions have access to the contents of the token claims, organized into CEL variable:</p> <ul style="list-style-type: none"><li>'claims' is a map of claim names to claim values. For example, a variable named 'sub' can be accessed as 'claims.sub'. Nested claims can be accessed using dot notation, e.g. 'claims.foo.bar'. Must return true for the validation to pass.</li></ul> |



| Field             | Description                                                                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a><br>Mutually exclusive with claim and requiredValue. |
| message<br>string | message customizes the returned error message when expression returns false. message is a literal string. Mutually exclusive with claim and requiredValue.                                      |

## Connection

Appears in:

- [EgressSelection](#)

Connection provides the configuration for a single egress selection client.

| Field                                                       | Description                                                                                                                                         |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| proxyProtocol<br>[Required]<br><a href="#">ProtocolType</a> | Protocol is the protocol used to connect from client to the konnectivity server.                                                                    |
| transport<br><a href="#">Transport</a>                      | Transport defines the transport configurations we use to dial to the konnectivity server. This is required if ProxyProtocol is HTTPConnect or GRPC. |

## EgressSelection

Appears in:

- [EgressSelectorConfiguration](#)

EgressSelection provides the configuration for a single egress selection client.

| Field                                                  | Description                                                                                                                                                                                   |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name<br>[Required]<br>string                           | name is the name of the egress selection. Currently supported values are "controlplane", "master", "etcd" and "cluster" The "master" egress selector is deprecated in favor of "controlplane" |
| connection<br>[Required]<br><a href="#">Connection</a> | connection is the exact information used to configure the egress selection                                                                                                                    |

# ExtraMapping

Appears in:

- [ClaimMappings](#)

ExtraMapping provides the configuration for a single extra mapping.

| Field                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| key <b>[Required]</b><br>string             | key is a string to use as the extra attribute key. key must be a domain-prefix path (e.g. example.org/foo). All characters before the first "/" must be a valid subdomain as defined by RFC 1123. All characters trailing the first "/" must be valid HTTP Path characters as defined by RFC 3986. key must be lowercase. Required to be unique.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| valueExpression <b>[Required]</b><br>string | <p>valueExpression is a CEL expression to extract extra attribute value. valueExpression must produce a string or string array value. "", [], and null values are treated as the extra mapping not being present. Empty string values contained within a string array are filtered out.</p> <p>CEL expressions have access to the contents of the token claims, organized into CEL variable:</p> <ul style="list-style-type: none"><li>• 'claims' is a map of claim names to claim values. For example, a variable named 'sub' can be accessed as 'claims.sub'. Nested claims can be accessed using dot notation, e.g. 'claims.foo.bar'.</li></ul> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a></p> |

# Issuer

Appears in:

- [JWTAuthenticator](#)

Issuer provides the configuration for an external provider's specific settings.

| Field                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| url <b>[Required]</b><br>string | url points to the issuer URL in a format https://url or https://url/path. This must match the "iss" claim in the presented JWT, and the issuer returned from discovery. Same value as the --oidc-issuer-url flag. Discovery information is fetched from "{url}/.well-known/openid-configuration" unless overridden by discoveryURL. Required to be unique across all JWT authenticators. Note that egress selection configuration is not used for this network connection. |
| discoveryURL<br>string          | discoveryURL, if specified, overrides the URL used to fetch discovery information instead of using "{url}/.well-known/openid-                                                                                                                                                                                                                                                                                                                                              |

| Field                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                             | <p>configuration". The exact value specified is used, so <code>"/.well-known/openid-configuration"</code> must be included in <code>discoveryURL</code> if needed.</p> <p>The "issuer" field in the fetched discovery information must match the "issuer.url" field in the <code>AuthenticationConfiguration</code> and will be used to validate the "iss" claim in the presented JWT. This is for scenarios where the well-known and jwks endpoints are hosted at a different location than the issuer (such as locally in the cluster).</p> <p>Example: A discovery url that is exposed using kubernetes service 'oidc' in namespace 'oidc-namespace' and discovery information is available at <code>"/.well-known/openid-configuration"</code>. <code>discoveryURL</code>: <code>"https://oidc.oidc-namespace/.well-known/openid-configuration"</code> <code>certificateAuthority</code> is used to verify the TLS connection and the hostname on the leaf certificate must be set to <code>'oidc.oidc-namespace'</code>.</p> <pre>curl https://oidc.oidc-namespace/.well-known/openid-configuration (.discoveryURL field) { issuer: "https://oidc.example.com" (.url field) }</pre> <p><code>discoveryURL</code> must be different from <code>url</code>. Required to be unique across all JWT authenticators. Note that egress selection configuration is not used for this network connection.</p> |
| <code>certificateAuthority</code><br>string                                 | <code>certificateAuthority</code> contains PEM-encoded certificate authority certificates used to validate the connection when fetching discovery information. If unset, the system verifier is used. Same value as the content of the file referenced by the <code>--oidc-ca-file</code> flag.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>audiences</code> <b>[Required]</b><br>[]string                        | <code>audiences</code> is the set of acceptable audiences the JWT must be issued to. At least one of the entries must match the "aud" claim in presented JWTs. Same value as the <code>--oidc-client-id</code> flag (though this field supports an array). Required to be non-empty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>audienceMatchPolicy</code><br><a href="#">AudienceMatchPolicyType</a> | <p><code>audienceMatchPolicy</code> defines how the "audiences" field is used to match the "aud" claim in the presented JWT. Allowed values are:</p> <ol style="list-style-type: none"> <li>1. "MatchAny" when multiple audiences are specified and</li> <li>2. empty (or unset) or "MatchAny" when a single audience is specified.</li> </ol> <ul style="list-style-type: none"> <li>• MatchAny: the "aud" claim in the presented JWT must match at least one of the entries in the "audiences" field. For example, if "audiences" is <code>["foo", "bar"]</code>, the "aud" claim in the presented JWT must contain either "foo" or "bar" (and may contain both).</li> <li>• "": The match policy can be empty (or unset) when a single audience is specified in the "audiences" field. The "aud" claim</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Field | Description                                                                                                                                                                                                                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <p>in the presented JWT must contain the single audience (and may contain others).</p> <p>For more nuanced audience validation, use <code>claimValidationRules</code>. example: <code>claimValidationRule[].expression: 'sets.equivalent(claims.aud, ["bar", "foo", "baz"])</code> to require an exact match.</p> |

## JWTAuthenticator

Appears in:

- [AuthenticationConfiguration](#)

JWTAuthenticator provides the configuration for a single JWT authenticator.

| Field                                                                 | Description                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>issuer [Required]</b><br><a href="#">Issuer</a>                    | issuer contains the basic OIDC provider connection options.                                                                                                                                                                                                                                                                                                        |
| <b>claimValidationRules []</b><br><a href="#">ClaimValidationRule</a> | claimValidationRules are rules that are applied to validate token claims to authenticate users.                                                                                                                                                                                                                                                                    |
| <b>claimMappings [Required]</b><br><a href="#">ClaimMappings</a>      | claimMappings points claims of a token to be treated as user attributes.                                                                                                                                                                                                                                                                                           |
| <b>userValidationRules []</b><br><a href="#">UserValidationRule</a>   | userValidationRules are rules that are applied to final user before completing authentication. These allow invariants to be applied to incoming identities such as preventing the use of the system: prefix that is commonly used by Kubernetes components. The validation rules are logically ANDed together and must all return true for the validation to pass. |

## PrefixedClaimOrExpression

Appears in:

- [ClaimMappings](#)

PrefixedClaimOrExpression provides the configuration for a single prefixed claim or expression.

| Field         | Description                                                        |
|---------------|--------------------------------------------------------------------|
| claim string  | claim is the JWT claim to use. Mutually exclusive with expression. |
| prefix string |                                                                    |

| Field                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | prefix is prepended to claim's value to prevent clashes with existing names. prefix needs to be set if claim is set and can be the empty string. Mutually exclusive with expression.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| expression<br>string | <p>expression represents the expression which will be evaluated by CEL.</p> <p>CEL expressions have access to the contents of the token claims, organized into CEL variable:</p> <ul style="list-style-type: none"> <li>'claims' is a map of claim names to claim values. For example, a variable named 'sub' can be accessed as 'claims.sub'. Nested claims can be accessed using dot notation, e.g. 'claims.foo.bar'.</li> </ul> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a></p> <p>Mutually exclusive with claim and prefix.</p> |

## ProtocolType

(Alias of string)

Appears in:

- [Connection](#)

ProtocolType is a set of valid values for Connection.ProtocolType

## TCPTransport

Appears in:

- [Transport](#)

TCPTransport provides the information to connect to konnectivity server via TCP

| Field                                  | Description                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------|
| url<br>[Required]<br>string            | URL is the location of the konnectivity server to connect to. As an example it might be "https://127.0.0.1:8131" |
| tlsConfig<br><a href="#">TLSConfig</a> | TLSConfig is the config needed to use TLS when connecting to konnectivity server                                 |

## TLSConfig

Appears in:

- [TCPTransport](#)

TLSConfig provides the authentication information to connect to konnectivity server Only used with TCPTransport

| Field                | Description                                                                                                                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| caBundle<br>string   | caBundle is the file location of the CA to be used to determine trust with the konnectivity server. Must be absent/empty if TCPTransport.URL is prefixed with http:// If absent while TCPTransport.URL is prefixed with https://, default to system trust roots. |
| clientKey<br>string  | clientKey is the file location of the client key to be used in mtls handshakes with the konnectivity server. Must be absent/empty if TCPTransport.URL is prefixed with http:// Must be configured if TCPTransport.URL is prefixed with https://                  |
| clientCert<br>string | clientCert is the file location of the client certificate to be used in mtls handshakes with the konnectivity server. Must be absent/empty if TCPTransport.URL is prefixed with http:// Must be configured if TCPTransport.URL is prefixed with https://         |

## Transport

Appears in:

- [Connection](#)

Transport defines the transport configurations we use to dial to the konnectivity server

| Field                               | Description                                                                                                                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tcp<br><a href="#">TCPTransport</a> | TCP is the TCP configuration for communicating with the konnectivity server via TCP ProxyProtocol of GRPC is not supported with TCP transport at the moment Requires at least one of TCP or UDS to be set |
| uds<br><a href="#">UDSTransport</a> | UDS is the UDS configuration for communicating with the konnectivity server via UDS Requires at least one of TCP or UDS to be set                                                                         |

## UDSTransport

Appears in:

- [Transport](#)

UDSTransport provides the information to connect to konnectivity server via UDS

| Field                           | Description                                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| udsName<br>[Required]<br>string | UDSName is the name of the unix domain socket to connect to konnectivity server This does not use a unix:// prefix. (Eg: /etc/srv/kubernetes/konnectivity-server/konnectivity-server.socket) |

## UserValidationRule

Appears in:

- [JWTAuthenticator](#)

UserValidationRule provides the configuration for a single user info validation rule.

| Field                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| expression<br>[Required]<br>string | <p>expression represents the expression which will be evaluated by CEL. Must return true for the validation to pass.</p> <p>CEL expressions have access to the contents of UserInfo, organized into CEL variable:</p> <ul style="list-style-type: none"><li>• 'user' - authentication.k8s.io/v1, Kind=UserInfo object Refer to <a href="https://github.com/kubernetes/api/blob/release-1.28/authentication/v1/types.go#L105-L122">https://github.com/kubernetes/api/blob/release-1.28/authentication/v1/types.go#L105-L122</a> for the definition. API documentation: <a href="https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.28/#userinfo-v1-authentication-k8s-io">https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.28/#userinfo-v1-authentication-k8s-io</a></li></ul> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a></p> |
| message<br>string                  | message customizes the returned error message when rule returns false. message is a literal string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## WebhookConfiguration

Appears in:

- [AuthorizerConfiguration](#)

| Field                                                        | Description                                                                                                                                                  |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| authorizedTTL [Required]<br><a href="#">meta/v1.Duration</a> | The duration to cache 'authorized' responses from the webhook authorizer. Same as setting --authorization-webhook-cache-authorized-ttl flag<br>Default: 5m0s |

| Field                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| unauthorizedTTL <b>[Required]</b><br><a href="#">meta/v1.Duration</a>        | The duration to cache 'unauthorized' responses from the webhook authorizer. Same as setting --authorization-webhook-cache-unauthorized-ttl flag Default: 30s                                                                                                                                                                                                                                                                                                                              |
| timeout <b>[Required]</b><br><a href="#">meta/v1.Duration</a>                | Timeout for the webhook request Maximum allowed value is 30s. Required, no default value.                                                                                                                                                                                                                                                                                                                                                                                                 |
| subjectAccessReviewVersion <b>[Required]</b><br>string                       | The API version of the authorization.k8s.io SubjectAccessReview to send to and expect from the webhook. Same as setting --authorization-webhook-version flag Valid values: v1beta1, v1 Required, no default value                                                                                                                                                                                                                                                                         |
| matchConditionSubjectAccessReviewVersion <b>[Required]</b><br>string         | MatchConditionSubjectAccessReviewVersion specifies the SubjectAccessReview version the CEL expressions are evaluated against Valid values: v1 Required, no default value                                                                                                                                                                                                                                                                                                                  |
| failurePolicy <b>[Required]</b><br>string                                    | Controls the authorization decision when a webhook request fails to complete or returns a malformed response or errors evaluating matchConditions. Valid values: <ul style="list-style-type: none"> <li>• NoOpinion: continue to subsequent authorizers to see if one of them allows the request</li> <li>• Deny: reject the request without consulting subsequent authorizers</li> </ul> Required, with no default.                                                                      |
| connectionInfo <b>[Required]</b><br><a href="#">WebhookConnectionInfo</a>    | ConnectionInfo defines how we talk to the webhook                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| matchConditions <b>[Required]</b><br><a href="#">[]WebhookMatchCondition</a> | <p>matchConditions is a list of conditions that must be met for a request to be sent to this webhook. An empty list of matchConditions matches all requests. There are a maximum of 64 match conditions allowed.</p> <p>The exact matching logic is (in order):</p> <ol style="list-style-type: none"> <li>1. If at least one matchCondition evaluates to FALSE, then the webhook is skipped.</li> <li>2. If ALL matchConditions evaluate to TRUE, then the webhook is called.</li> </ol> |



| Field | Description                                                                                                                                                                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | 3. If at least one matchCondition evaluates to an error (but none are FALSE): <ul style="list-style-type: none"> <li>◦ If failurePolicy=Deny, then the webhook rejects the request</li> <li>◦ If failurePolicy=NoOpinion, then the error is ignored and the webhook is skipped</li> </ul> |

## WebhookConnectionInfo

Appears in:

- [WebhookConfiguration](#)

| Field                                      | Description                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type <b>[Required]</b><br>string           | Controls how the webhook should communicate with the server. Valid values: <ul style="list-style-type: none"> <li>• KubeConfigFile: use the file specified in kubeConfigFile to locate the server.</li> <li>• InClusterConfig: use the in-cluster configuration to call the SubjectAccessReview API hosted by kube-apiserver. This mode is not allowed for kube-apiserver.</li> </ul> |
| kubeConfigFile <b>[Required]</b><br>string | Path to KubeConfigFile for connection info Required, if connectionInfo.Type is KubeConfig                                                                                                                                                                                                                                                                                             |

## WebhookMatchCondition

Appears in:

- [WebhookConfiguration](#)

| Field                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| expression <b>[Required]</b><br>string | <p>expression represents the expression which will be evaluated by CEL. Must evaluate to bool. CEL expressions have access to the contents of the SubjectAccessReview in v1 version. If version specified by subjectAccessReviewVersion in the request variable is v1beta1, the contents would be converted to the v1 version before evaluating the CEL expression.</p> <p>Documentation on CEL: <a href="https://kubernetes.io/docs/reference/using-api/cel/">https://kubernetes.io/docs/reference/using-api/cel/</a></p> |

# kube-controller-manager Configuration (v1alpha1)

## Resource Types

- [CloudControllerManagerConfiguration](#)
- [LeaderMigrationConfiguration](#)
- [KubeControllerManagerConfiguration](#)

## NodeControllerConfiguration

Appears in:

- [CloudControllerManagerConfiguration](#)

NodeControllerConfiguration contains elements describing NodeController.

| Field                                      | Description                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------|
| ConcurrentNodeSyncs<br>[Required]<br>int32 | ConcurrentNodeSyncs is the number of workers concurrently synchronizing nodes |

## ServiceControllerConfiguration

Appears in:

- [CloudControllerManagerConfiguration](#)
- [KubeControllerManagerConfiguration](#)

ServiceControllerConfiguration contains elements describing ServiceController.

| Field                                         | Description                                                                                                                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentServiceSyncs<br>[Required]<br>int32 | concurrentServiceSyncs is the number of services that are allowed to sync concurrently. Larger number = more responsive service management, but more CPU (and network) load. |

## CloudControllerManagerConfiguration

CloudControllerManagerConfiguration contains elements describing cloud-controller manager.

| Field                | Description                                   |
|----------------------|-----------------------------------------------|
| apiVersion<br>string | cloudcontrollermanager.config.k8s.io/v1alpha1 |
|                      | CloudControllerManagerConfiguration           |

| Field                                                                                 | Description                                                                                                                                |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| kind<br>string                                                                        |                                                                                                                                            |
| Generic <b>[Required]</b><br><a href="#">GenericControllerManagerConfiguration</a>    | Generic holds configuration for a generic controller-manager                                                                               |
| KubeCloudShared <b>[Required]</b><br><a href="#">KubeCloudSharedConfiguration</a>     | KubeCloudSharedConfiguration holds configuration for shared related features both in cloud controller manager and kube-controller manager. |
| NodeController <b>[Required]</b><br><a href="#">NodeControllerConfiguration</a>       | NodeController holds configuration for node controller related features.                                                                   |
| ServiceController <b>[Required]</b><br><a href="#">ServiceControllerConfiguration</a> | ServiceControllerConfiguration holds configuration for ServiceController related features.                                                 |
| NodeStatusUpdateFrequency <b>[Required]</b><br><a href="#">meta/v1.Duration</a>       | NodeStatusUpdateFrequency is the frequency at which the controller updates nodes' status                                                   |
| Webhook <b>[Required]</b><br><a href="#">WebhookConfiguration</a>                     | Webhook is the configuration for cloud-controller-manager hosted webhooks                                                                  |

## CloudProviderConfiguration

Appears in:

- [KubeCloudSharedConfiguration](#)

CloudProviderConfiguration contains basically elements about cloud provider.

| Field                                       | Description                                                           |
|---------------------------------------------|-----------------------------------------------------------------------|
| Name <b>[Required]</b><br>string            | Name is the provider for cloud services.                              |
| CloudConfigFile <b>[Required]</b><br>string | cloudConfigFile is the path to the cloud provider configuration file. |

## KubeCloudSharedConfiguration

Appears in:

- [CloudControllerManagerConfiguration](#)

## [KubeControllerManagerConfiguration](#)

KubeCloudSharedConfiguration contains elements shared by both kube-controller manager and cloud-controller manager, but not genericconfig.

| Field                                                                           | Description                                                                                                                                                                               |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CloudProvider <b>[Required]</b><br><a href="#">CloudProviderConfiguration</a>   | CloudProviderConfiguration holds configuration for CloudProvider related features.                                                                                                        |
| ExternalCloudVolumePlugin <b>[Required]</b><br>string                           | externalCloudVolumePlugin specifies the plugin to use when cloudProvider is "external". It is currently used by the in repo cloud providers to handle node and volume control in the KCM. |
| UseServiceAccountCredentials <b>[Required]</b><br>bool                          | useServiceAccountCredentials indicates whether controllers should be run with individual service account credentials.                                                                     |
| AllowUntaggedCloud <b>[Required]</b><br>bool                                    | run with untagged cloud instances                                                                                                                                                         |
| RouteReconciliationPeriod <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | routeReconciliationPeriod is the period for reconciling routes created for Nodes by cloud provider..                                                                                      |
| NodeMonitorPeriod <b>[Required]</b><br><a href="#">meta/v1.Duration</a>         | nodeMonitorPeriod is the period for syncing NodeStatus in NodeController.                                                                                                                 |
| ClusterName <b>[Required]</b><br>string                                         | clusterName is the instance prefix for the cluster.                                                                                                                                       |
| ClusterCIDR <b>[Required]</b><br>string                                         | clusterCIDR is CIDR Range for Pods in cluster.                                                                                                                                            |
| AllocateNodeCIDRs <b>[Required]</b><br>bool                                     | AllocateNodeCIDRs enables CIDRs for Pods to be allocated and, if ConfigureCloudRoutes is true, to be set on the cloud provider.                                                           |
| CIDRAllocatorType <b>[Required]</b><br>string                                   | CIDRAllocatorType determines what kind of pod CIDR allocator will be used.                                                                                                                |
| ConfigureCloudRoutes <b>[Required]</b><br>bool                                  | configureCloudRoutes enables CIDRs allocated with allocateNodeCIDRs to be configured on the cloud provider.                                                                               |
| NodeSyncPeriod <b>[Required]</b><br><a href="#">meta/v1.Duration</a>            | nodeSyncPeriod is the period for syncing nodes from cloudprovider. Longer periods will result in fewer calls to                                                                           |

| Field | Description                                                     |
|-------|-----------------------------------------------------------------|
|       | cloud provider, but may delay addition of new nodes to cluster. |

## WebhookConfiguration

Appears in:

- [CloudControllerManagerConfiguration](#)

WebhookConfiguration contains configuration related to cloud-controller-manager hosted webhooks

| Field                                       | Description                                                                                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Webhooks<br>[ <b>Required</b> ]<br>[]string | Webhooks is the list of webhooks to enable or disable '*' means "all enabled by default webhooks" 'foo' means "enable 'foo'" '-foo' means "disable 'foo'" first item for a particular name wins |

## LeaderMigrationConfiguration

Appears in:

- [GenericControllerManagerConfiguration](#)

LeaderMigrationConfiguration provides versioned configuration for all migrating leader locks.

| Field                                                                                    | Description                                                                                                  |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                                     | controllermanager.config.k8s.io/v1alpha1                                                                     |
| kind<br>string                                                                           | LeaderMigrationConfiguration                                                                                 |
| leaderName [ <b>Required</b> ]<br>string                                                 | LeaderName is the name of the leader election resource that protects the migration E.g. 1-20-KCM-to-1-21-CCM |
| resourceLock [ <b>Required</b> ]<br>string                                               | ResourceLock indicates the resource object type that will be used to lock Should be "leases" or "endpoints"  |
| controllerLeaders [ <b>Required</b> ]<br><a href="#">[]ControllerLeaderConfiguration</a> | ControllerLeaders contains a list of migrating leader lock configurations                                    |

## ControllerLeaderConfiguration

Appears in:

- [LeaderMigrationConfiguration](#)

ControllerLeaderConfiguration provides the configuration for a migrating leader lock.

| Field                             | Description                                                                                                                                                                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name<br>[Required]<br>string      | Name is the name of the controller being migrated E.g. service-controller, route-controller, cloud-node-controller, etc                                                                                                                       |
| component<br>[Required]<br>string | Component is the name of the component in which the controller should be running. E.g. kube-controller-manager, cloud-controller-manager, etc Or '*' meaning the controller can be run under any component that participates in the migration |

## GenericControllerManagerConfiguration

Appears in:

- [CloudControllerManagerConfiguration](#)
- [KubeControllerManagerConfiguration](#)

GenericControllerManagerConfiguration holds configuration for a generic controller-manager.

| Field                                                                        | Description                                                                                                                                      |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Port [Required]<br>int32                                                     | port is the port that the controller-manager's http service runs on.                                                                             |
| Address [Required]<br>string                                                 | address is the IP address to serve on (set to 0.0.0.0 for all interfaces).                                                                       |
| MinResyncPeriod [Required]<br><a href="#">meta/v1.Duration</a>               | minResyncPeriod is the resync period in reflectors; will be random between minResyncPeriod and 2*minResyncPeriod.                                |
| ClientConnection [Required]<br><a href="#">ClientConnectionConfiguration</a> | ClientConnection specifies the kubeconfig file and client connection settings for the proxy server to use when communicating with the apiserver. |
| ControllerStartInterval<br>[Required]<br><a href="#">meta/v1.Duration</a>    | How long to wait between starting controller managers                                                                                            |

| Field                                                                               | Description                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LeaderElection [ <b>Required</b> ]<br><a href="#">LeaderElectionConfiguration</a>   | leaderElection defines the configuration of leader election client.                                                                                                                                      |
| Controllers [ <b>Required</b> ]<br>[]string                                         | Controllers is the list of controllers to enable or disable '*' means "all enabled by default controllers" 'foo' means "enable 'foo'" '-foo' means "disable 'foo'" first item for a particular name wins |
| Debugging [ <b>Required</b> ]<br><a href="#">DebuggingConfiguration</a>             | DebuggingConfiguration holds configuration for Debugging related features.                                                                                                                               |
| LeaderMigrationEnabled [ <b>Required</b> ]<br>bool                                  | LeaderMigrationEnabled indicates whether Leader Migration should be enabled for the controller manager.                                                                                                  |
| LeaderMigration [ <b>Required</b> ]<br><a href="#">LeaderMigrationConfiguration</a> | LeaderMigration holds the configuration for Leader Migration.                                                                                                                                            |

## KubeControllerManagerConfiguration

KubeControllerManagerConfiguration contains elements describing kube-controller manager.

| Field                                                                                             | Description                                                                                                                            |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                                              | kubecontrollermanager.config.k8s.io/v1alpha1                                                                                           |
| kind<br>string                                                                                    | KubeControllerManagerConfiguration                                                                                                     |
| Generic [ <b>Required</b> ]<br><a href="#">GenericControllerManagerConfiguration</a>              | Generic holds configuration for a generic controller manager                                                                           |
| KubeCloudShared [ <b>Required</b> ]<br><a href="#">KubeCloudSharedConfiguration</a>               | KubeCloudSharedConfiguration holds configuration shared related features both in cloud controller manager and kube-controller manager. |
| AttachDetachController [ <b>Required</b> ]<br><a href="#">AttachDetachControllerConfiguration</a> | AttachDetachControllerConfiguration holds configuration for AttachDetachController related features.                                   |
| CSRSigningController [ <b>Required</b> ]<br><a href="#">CSRSigningControllerConfiguration</a>     | CSRSigningControllerConfiguration holds configuration for CSRSigningController related features.                                       |

| Field                                                                                                                 | Description                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| DaemonSetController [ <b>Required</b> ]<br><a href="#">DaemonSetControllerConfiguration</a>                           | DaemonSetControllerConfiguration holds configuration for DaemonSetController related features.                           |
| DeploymentController [ <b>Required</b> ]<br><a href="#">DeploymentControllerConfiguration</a>                         | DeploymentControllerConfiguration holds configuration for DeploymentController related features.                         |
| StatefulSetController [ <b>Required</b> ]<br><a href="#">StatefulSetControllerConfiguration</a>                       | StatefulSetControllerConfiguration holds configuration for StatefulSetController related features.                       |
| DeprecatedController [ <b>Required</b> ]<br><a href="#">DeprecatedControllerConfiguration</a>                         | DeprecatedControllerConfiguration holds configuration for some deprecated features.                                      |
| EndpointController [ <b>Required</b> ]<br><a href="#">EndpointControllerConfiguration</a>                             | EndpointControllerConfiguration holds configuration for EndpointController related features.                             |
| EndpointSliceController [ <b>Required</b> ]<br><a href="#">EndpointSliceControllerConfiguration</a>                   | EndpointSliceControllerConfiguration holds configuration for EndpointSliceController related features.                   |
| EndpointSliceMirroringController [ <b>Required</b> ]<br><a href="#">EndpointSliceMirroringControllerConfiguration</a> | EndpointSliceMirroringControllerConfiguration holds configuration for EndpointSliceMirroringController related features. |
| EphemeralVolumeController [ <b>Required</b> ]<br><a href="#">EphemeralVolumeControllerConfiguration</a>               | EphemeralVolumeControllerConfiguration holds configuration for EphemeralVolumeController related features.               |
| GarbageCollectorController [ <b>Required</b> ]<br><a href="#">GarbageCollectorControllerConfiguration</a>             | GarbageCollectorControllerConfiguration holds configuration for GarbageCollectorController related features.             |
| HPAController [ <b>Required</b> ]<br><a href="#">HPAControllerConfiguration</a>                                       | HPAControllerConfiguration holds configuration for HPAController related features.                                       |
| JobController [ <b>Required</b> ]<br><a href="#">JobControllerConfiguration</a>                                       | JobControllerConfiguration holds configuration for JobController related features.                                       |
| CronJobController [ <b>Required</b> ]<br><a href="#">CronJobControllerConfiguration</a>                               | CronJobControllerConfiguration holds configuration for CronJobController related features.                               |



| Field                                                                                                                 | Description                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| LegacySATokenCleaner [ <b>Required</b> ]<br><a href="#">LegacySATokenCleanerConfiguration</a>                         | LegacySATokenCleanerConfiguration holds configuration for LegacySATokenCleaner related features.                         |
| NamespaceController [ <b>Required</b> ]<br><a href="#">NamespaceControllerConfiguration</a>                           | NamespaceControllerConfiguration holds configuration for NamespaceController related features.                           |
| NodeIPAMController [ <b>Required</b> ]<br><a href="#">NodeIPAMControllerConfiguration</a>                             | NodeIPAMControllerConfiguration holds configuration for NodeIPAMController related features.                             |
| NodeLifecycleController [ <b>Required</b> ]<br><a href="#">NodeLifecycleControllerConfiguration</a>                   | NodeLifecycleControllerConfiguration holds configuration for NodeLifecycleController related features.                   |
| PersistentVolumeBinderController [ <b>Required</b> ]<br><a href="#">PersistentVolumeBinderControllerConfiguration</a> | PersistentVolumeBinderControllerConfiguration holds configuration for PersistentVolumeBinderController related features. |
| PodGCController [ <b>Required</b> ]<br><a href="#">PodGCControllerConfiguration</a>                                   | PodGCControllerConfiguration holds configuration for PodGCController related features.                                   |
| ReplicaSetController [ <b>Required</b> ]<br><a href="#">ReplicaSetControllerConfiguration</a>                         | ReplicaSetControllerConfiguration holds configuration for ReplicaSet related features.                                   |
| ReplicationController [ <b>Required</b> ]<br><a href="#">ReplicationControllerConfiguration</a>                       | ReplicationControllerConfiguration holds configuration for ReplicationController related features.                       |
| ResourceQuotaController [ <b>Required</b> ]<br><a href="#">ResourceQuotaControllerConfiguration</a>                   | ResourceQuotaControllerConfiguration holds configuration for ResourceQuotaController related features.                   |
| SAController [ <b>Required</b> ]<br><a href="#">SAControllerConfiguration</a>                                         | SAControllerConfiguration holds configuration for ServiceAccountController related features.                             |
| ServiceController [ <b>Required</b> ]<br><a href="#">ServiceControllerConfiguration</a>                               | ServiceControllerConfiguration holds configuration for ServiceController related features.                               |
| TTLAfterFinishedController [ <b>Required</b> ]<br><a href="#">TTLAfterFinishedControllerConfiguration</a>             | TTLAfterFinishedControllerConfiguration holds configuration for TTLAfterFinishedController related features.             |

| Field                                                                                                                                 | Description                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| ValidatingAdmissionPolicyStatusController <b>[Required]</b><br><a href="#">ValidatingAdmissionPolicyStatusControllerConfiguration</a> | ValidatingAdmissionPolicyStatusControllerConfiguration holds configuration for ValidatingAdmissionPolicyStatusController related features. |

## AttachDetachControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

AttachDetachControllerConfiguration contains elements describing AttachDetachController.

| Field                                                                          | Description                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DisableAttachDetachReconcilerSync <b>[Required]</b><br>bool                    | Reconciler runs a periodic loop to reconcile the desired state of the with the actual state of the world by triggering attach detach operations. This flag enables or disables reconcile. Is false by default, and thus enabled. |
| ReconcilerSyncLoopPeriod <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | ReconcilerSyncLoopPeriod is the amount of time the reconciler sync states loop wait between successive executions. Is set to 60 sec by default.                                                                                  |
| disableForceDetachOnTimeout <b>[Required]</b><br>bool                          | DisableForceDetachOnTimeout disables force detach when the maximum unmount time is exceeded. Is false by default, and thus force detach on unmount is enabled.                                                                   |

## CSRSigningConfiguration

Appears in:

- [CSRSigningControllerConfiguration](#)

CSRSigningConfiguration holds information about a particular CSR signer

| Field                                | Description                                                                                          |
|--------------------------------------|------------------------------------------------------------------------------------------------------|
| CertFile <b>[Required]</b><br>string | certFile is the filename containing a PEM-encoded X509 CA certificate used to issue certificates     |
| KeyFile <b>[Required]</b><br>string  | keyFile is the filename containing a PEM-encoded RSA or ECDSA private key used to issue certificates |

## CSRSigningControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

CSRSigningControllerConfiguration contains elements describing CSRSigningController.

| Field                                                                                               | Description                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ClusterSigningCertFile <b>[Required]</b><br>string                                                  | clusterSigningCertFile is the filename containing a PEM-encoded X509 CA certificate used to issue cluster-scoped certificates                                        |
| ClusterSigningKeyFile <b>[Required]</b><br>string                                                   | clusterSigningKeyFile is the filename containing a PEM-encoded RSA or ECDSA private key used to issue cluster-scoped certificates                                    |
| KubeletServingSignerConfiguration <b>[Required]</b><br><a href="#">CSRSigningConfiguration</a>      | kubeletServingSignerConfiguration holds the certificate and key used to issue certificates for the kubernetes.io/kubelet-serving signer                              |
| KubeletClientSignerConfiguration <b>[Required]</b><br><a href="#">CSRSigningConfiguration</a>       | kubeletClientSignerConfiguration holds the certificate and key used to issue certificates for the kubernetes.io/kube-apiserver-client-kubelet                        |
| KubeAPIServerClientSignerConfiguration <b>[Required]</b><br><a href="#">CSRSigningConfiguration</a> | kubeAPIServerClientSignerConfiguration holds the certificate and key used to issue certificates for the kubernetes.io/kube-apiserver-client                          |
| LegacyUnknownSignerConfiguration <b>[Required]</b><br><a href="#">CSRSigningConfiguration</a>       | legacyUnknownSignerConfiguration holds the certificate and key used to issue certificates for the kubernetes.io/legacy-unknown                                       |
| ClusterSigningDuration <b>[Required]</b><br><a href="#">meta/v1.Duration</a>                        | clusterSigningDuration is the max length of duration signed certificates will be given. Individual CSRs may request shorter certs by setting spec.expirationSeconds. |

## CronJobControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

CronJobControllerConfiguration contains elements describing CronJobController.

| Field                                                | Description                                                                                                                                                       |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentCronJobSyncs<br><b>[Required]</b><br>int32 | concurrentCronJobSyncs is the number of job objects that are allowed to sync concurrently. Larger number = more responsive jobs, but more CPU (and network) load. |

## DaemonSetControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

DaemonSetControllerConfiguration contains elements describing DaemonSetController.

| Field                                                  | Description                                                                                                                                                                    |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentDaemonSetSyncs<br><b>[Required]</b><br>int32 | concurrentDaemonSetSyncs is the number of daemonset objects that are allowed to sync concurrently. Larger number = more responsive daemonset, but more CPU (and network) load. |

## DeploymentControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

DeploymentControllerConfiguration contains elements describing DeploymentController.

| Field                                                   | Description                                                                                                                                                                        |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentDeploymentSyncs<br><b>[Required]</b><br>int32 | concurrentDeploymentSyncs is the number of deployment objects that are allowed to sync concurrently. Larger number = more responsive deployments, but more CPU (and network) load. |

## DeprecatedControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

DeprecatedControllerConfiguration contains elements be deprecated.

# EndpointControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

EndpointControllerConfiguration contains elements describing EndpointController.

| Field                                                                               | Description                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentEndpointSyncs<br><b>[Required]</b><br>int32                               | concurrentEndpointSyncs is the number of endpoint syncing operations that will be done concurrently. Larger number = faster endpoint updating, but more CPU (and network) load.                                                                  |
| EndpointUpdatesBatchPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | EndpointUpdatesBatchPeriod describes the length of endpoint updates batching period. Processing of pod changes will be delayed by this duration to join them with potential upcoming updates and reduce the overall number of endpoints updates. |

# EndpointSliceControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

EndpointSliceControllerConfiguration contains elements describing EndpointSliceController.

| Field                                                                               | Description                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentServiceEndpointSyncs<br><b>[Required]</b><br>int32                        | concurrentServiceEndpointSyncs is the number of service endpoint syncing operations that will be done concurrently. Larger number = faster endpoint slice updating, but more CPU (and network) load.                                             |
| MaxEndpointsPerSlice <b>[Required]</b><br>int32                                     | maxEndpointsPerSlice is the maximum number of endpoints that will be added to an EndpointSlice. More endpoints per slice will result in fewer and larger endpoint slices, but larger resources.                                                  |
| EndpointUpdatesBatchPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | EndpointUpdatesBatchPeriod describes the length of endpoint updates batching period. Processing of pod changes will be delayed by this duration to join them with potential upcoming updates and reduce the overall number of endpoints updates. |

# EndpointSliceMirroringControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

EndpointSliceMirroringControllerConfiguration contains elements describing EndpointSliceMirroringController.

| Field                                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MirroringConcurrentServiceEndpointSyncs<br><b>[Required]</b><br>int32                        | mirroringConcurrentServiceEndpointSyncs is the number of service endpoint syncing operations that will be done concurrently. Larger number = faster endpoint slice updating, but more CPU (and network) load.                                                                                                                                                                                                     |
| MirroringMaxEndpointsPerSubset<br><b>[Required]</b><br>int32                                 | mirroringMaxEndpointsPerSubset is the maximum number of endpoints that will be mirrored to an EndpointSlice for an EndpointSubset.                                                                                                                                                                                                                                                                                |
| MirroringEndpointUpdatesBatchPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | mirroringEndpointUpdatesBatchPeriod can be used to batch EndpointSlice updates. All updates triggered by EndpointSlice changes will be delayed by up to 'mirroringEndpointUpdatesBatchPeriod'. If other addresses in the same Endpoints resource change in that period, they will be batched to a single EndpointSlice update. Default 0 value means that each Endpoints update triggers an EndpointSlice update. |

# EphemeralVolumeControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

EphemeralVolumeControllerConfiguration contains elements describing EphemeralVolumeController.

| Field                                                        | Description                                                                                                                                                                                                  |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentEphemeralVolumeSyncs<br><b>[Required]</b><br>int32 | ConcurrentEphemeralVolumeSyncseSyncs is the number of ephemeral volume syncing operations that will be done concurrently. Larger number = faster ephemeral volume updating, but more CPU (and network) load. |

# GarbageCollectorControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

GarbageCollectorControllerConfiguration contains elements describing GarbageCollectorController.

| Field                                                               | Description                                                                                                                                                          |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EnableGarbageCollector<br>[Required]<br>bool                        | enables the generic garbage collector. MUST be synced with the corresponding flag of the kube-apiserver. WARNING: the generic garbage collector is an alpha feature. |
| ConcurrentGCSyncs<br>[Required]<br>int32                            | concurrentGCSyncs is the number of garbage collector workers that are allowed to sync concurrently.                                                                  |
| GCIgnoredResources<br>[Required]<br><a href="#">[]GroupResource</a> | gcIgnoredResources is the list of GroupResources that garbage collection should ignore.                                                                              |

# GroupResource

Appears in:

- [GarbageCollectorControllerConfiguration](#)

GroupResource describes an group resource.

| Field                         | Description                                            |
|-------------------------------|--------------------------------------------------------|
| Group [Required]<br>string    | group is the group portion of the GroupResource.       |
| Resource [Required]<br>string | resource is the resource portion of the GroupResource. |

# HPAControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

HPAControllerConfiguration contains elements describing HPAController.

| Field                                                      | Description                                                                                  |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| ConcurrentHorizontalPodAutoscalerSyncs [Required]<br>int32 | ConcurrentHorizontalPodAutoscalerSyncs is the number of HPA objects that are allowed to sync |

| Field                                                                                                     | Description                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                           | concurrently. Larger number = more responsive HPA processing, but more CPU (and network) load.                                                                                                                                                                                                |
| HorizontalPodAutoscalerSyncPeriod <b>[Required]</b><br><a href="#">meta/v1.Duration</a>                   | HorizontalPodAutoscalerSyncPeriod is the period for syncing the number of pods in horizontal pod autoscaler.                                                                                                                                                                                  |
| HorizontalPodAutoscalerUpscaleForbiddenWindow <b>[Required]</b><br><a href="#">meta/v1.Duration</a>       | HorizontalPodAutoscalerUpscaleForbiddenWindow is a period after which next upscale allowed.                                                                                                                                                                                                   |
| HorizontalPodAutoscalerDownscaleStabilizationWindow <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | HorizontalPodAutoscalerDownscaleStabilizationWindow is a period for which autoscaler will look backward and not scale down below any recommendation it made during that period.                                                                                                               |
| HorizontalPodAutoscalerDownscaleForbiddenWindow <b>[Required]</b><br><a href="#">meta/v1.Duration</a>     | HorizontalPodAutoscalerDownscaleForbiddenWindow is a period after which next downscale allowed.                                                                                                                                                                                               |
| HorizontalPodAutoscalerTolerance <b>[Required]</b><br>float64                                             | HorizontalPodAutoscalerTolerance is the tolerance when resource usage suggests upscaling/downscaling.                                                                                                                                                                                         |
| HorizontalPodAutoscalerCPUInitializationPeriod <b>[Required]</b><br><a href="#">meta/v1.Duration</a>      | HorizontalPodAutoscalerCPUInitializationPeriod is a period after pod start when CPU samples might be skipped.                                                                                                                                                                                 |
| HorizontalPodAutoscalerInitialReadinessDelay <b>[Required]</b><br><a href="#">meta/v1.Duration</a>        | HorizontalPodAutoscalerInitialReadinessDelay is a period after pod start during which readiness changes are treated as readiness being set for the first time. The effect of this is that HPA will disregard CPU samples from unready pods that had last readiness change during that period. |

## JobControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

JobControllerConfiguration contains elements describing JobController.

| Field                                         | Description |
|-----------------------------------------------|-------------|
| ConcurrentJobSyncs <b>[Required]</b><br>int32 |             |



| Field | Description                                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | concurrentJobSyncs is the number of job objects that are allowed to sync concurrently. Larger number = more responsive jobs, but more CPU (and network) load. |

## LegacySATokenCleanerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

LegacySATokenCleanerConfiguration contains elements describing LegacySATokenCleaner

| Field                                                           | Description                                                                                                                   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| CleanUpPeriod<br>[Required]<br><a href="#">meta/v1.Duration</a> | CleanUpPeriod is the period of time since the last usage of an auto-generated service account token before it can be deleted. |

## NamespaceControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

NamespaceControllerConfiguration contains elements describing NamespaceController.

| Field                                                                 | Description                                                                                        |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| NamespaceSyncPeriod<br>[Required]<br><a href="#">meta/v1.Duration</a> | namespaceSyncPeriod is the period for syncing namespace life-cycle updates.                        |
| ConcurrentNamespaceSyncs<br>[Required]<br>int32                       | concurrentNamespaceSyncs is the number of namespace objects that are allowed to sync concurrently. |

## NodeIPAMControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

NodeIPAMControllerConfiguration contains elements describing NodeIpamController.

| Field                            | Description                                        |
|----------------------------------|----------------------------------------------------|
| ServiceCIDR [Required]<br>string | serviceCIDR is CIDR Range for Services in cluster. |

| Field                                               | Description                                                                                                                                                           |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SecondaryServiceCIDR<br><b>[Required]</b><br>string | secondaryServiceCIDR is CIDR Range for Services in cluster. This is used in dual stack clusters. SecondaryServiceCIDR must be of different IP family than ServiceCIDR |
| NodeCIDRMaskSize<br><b>[Required]</b><br>int32      | NodeCIDRMaskSize is the mask size for node cidr in cluster.                                                                                                           |
| NodeCIDRMaskSizeIPv4<br><b>[Required]</b><br>int32  | NodeCIDRMaskSizeIPv4 is the mask size for node cidr in dual-stack cluster.                                                                                            |
| NodeCIDRMaskSizeIPv6<br><b>[Required]</b><br>int32  | NodeCIDRMaskSizeIPv6 is the mask size for node cidr in dual-stack cluster.                                                                                            |

## NodeLifecycleControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

NodeLifecycleControllerConfiguration contains elements describing NodeLifecycleController.

| Field                                                                           | Description                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NodeEvictionRate<br><b>[Required]</b><br>float32                                | nodeEvictionRate is the number of nodes per second on which pods are deleted in case of node failure when a zone is healthy                                                                                                                                        |
| SecondaryNodeEvictionRate<br><b>[Required]</b><br>float32                       | secondaryNodeEvictionRate is the number of nodes per second on which pods are deleted in case of node failure when a zone is unhealthy                                                                                                                             |
| NodeStartupGracePeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | nodeStartupGracePeriod is the amount of time which we allow starting a node to be unresponsive before marking it unhealthy.                                                                                                                                        |
| NodeMonitorGracePeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | nodeMonitorGracePeriod is the amount of time which we allow a running node to be unresponsive before marking it unhealthy. Must be N times more than kubelet's nodeStatusUpdateFrequency, where N means number of retries allowed for kubelet to post node status. |
| PodEvictionTimeout<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a>     | podEvictionTimeout is the grace period for deleting pods on failed nodes.                                                                                                                                                                                          |

| Field                                            | Description                                                                                                                                                            |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LargeClusterSizeThreshold<br>[Required]<br>int32 | secondaryNodeEvictionRate is implicitly overridden to 0 for clusters smaller than or equal to largeClusterSizeThreshold                                                |
| UnhealthyZoneThreshold<br>[Required]<br>float32  | Zone is treated as unhealthy in nodeEvictionRate and secondaryNodeEvictionRate when at least unhealthyZoneThreshold (no less than 3) of Nodes in the zone are NotReady |

## PersistentVolumeBinderControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

PersistentVolumeBinderControllerConfiguration contains elements describing PersistentVolumeBinderController.

| Field                                                                     | Description                                                                                                                 |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| PVClaimBinderSyncPeriod<br>[Required]<br><a href="#">meta/v1.Duration</a> | pvClaimBinderSyncPeriod is the period for syncing persistent volumes and persistent volume claims.                          |
| VolumeConfiguration [Required]<br><a href="#">VolumeConfiguration</a>     | volumeConfiguration holds configuration for volume related features.                                                        |
| VolumeHostCIDRDenylist<br>[Required]<br>[]string                          | DEPRECATED: VolumeHostCIDRDenylist is a list of CIDRs that should not be reachable by the controller from plugins.          |
| VolumeHostAllowLocalLoopback<br>[Required]<br>bool                        | DEPRECATED: VolumeHostAllowLocalLoopback indicates if local loopback hosts (127.0.0.1, etc) should be allowed from plugins. |

## PersistentVolumeRecyclerConfiguration

Appears in:

- [VolumeConfiguration](#)

PersistentVolumeRecyclerConfiguration contains elements describing persistent volume plugins.

| Field                                                   | Description                                                                                                                                                                                                       |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MaximumRetry <b>[Required]</b><br>int32                 | maximumRetry is number of retries the PV recycler will execute on failure to recycle PV.                                                                                                                          |
| MinimumTimeoutNFS <b>[Required]</b><br>int32            | minimumTimeoutNFS is the minimum ActiveDeadlineSeconds to use for an NFS Recycler pod.                                                                                                                            |
| PodTemplateFilePathNFS <b>[Required]</b><br>string      | podTemplateFilePathNFS is the file path to a pod definition used as a template for NFS persistent volume recycling                                                                                                |
| IncrementTimeoutNFS <b>[Required]</b><br>int32          | incrementTimeoutNFS is the increment of time added per Gi to ActiveDeadlineSeconds for an NFS scrubber pod.                                                                                                       |
| PodTemplateFilePathHostPath <b>[Required]</b><br>string | podTemplateFilePathHostPath is the file path to a pod definition used as a template for HostPath persistent volume recycling. This is for development and testing only and will not work in a multi-node cluster. |
| MinimumTimeoutHostPath <b>[Required]</b><br>int32       | minimumTimeoutHostPath is the minimum ActiveDeadlineSeconds to use for a HostPath Recycler pod. This is for development and testing only and will not work in a multi-node cluster.                               |
| IncrementTimeoutHostPath <b>[Required]</b><br>int32     | incrementTimeoutHostPath is the increment of time added per Gi to ActiveDeadlineSeconds for a HostPath scrubber pod. This is for development and testing only and will not work in a multi-node cluster.          |

## PodGCControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

PodGCControllerConfiguration contains elements describing PodGCController.

| Field                                               | Description                                                                                                                                                                                                      |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TerminatedPodGCThreshold <b>[Required]</b><br>int32 | terminatedPodGCThreshold is the number of terminated pods that can exist before the terminated pod garbage collector starts deleting terminated pods. If <= 0, the terminated pod garbage collector is disabled. |

## ReplicaSetControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

ReplicaSetControllerConfiguration contains elements describing ReplicaSetController.

| Field                                           | Description                                                                                                                                                                 |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentRSSyncs<br><b>[Required]</b><br>int32 | concurrentRSSyncs is the number of replica sets that are allowed to sync concurrently. Larger number = more responsive replica management, but more CPU (and network) load. |

## ReplicationControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

ReplicationControllerConfiguration contains elements describing ReplicationController.

| Field                                           | Description                                                                                                                                                                            |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentRCSyncs<br><b>[Required]</b><br>int32 | concurrentRCSyncs is the number of replication controllers that are allowed to sync concurrently. Larger number = more responsive replica management, but more CPU (and network) load. |

## ResourceQuotaControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

ResourceQuotaControllerConfiguration contains elements describing ResourceQuotaController.

| Field                                                                            | Description                                                                                                                                                                             |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ResourceQuotaSyncPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | resourceQuotaSyncPeriod is the period for syncing quota usage status in the system.                                                                                                     |
| ConcurrentResourceQuotaSyncs<br><b>[Required]</b><br>int32                       | concurrentResourceQuotaSyncs is the number of resource quotas that are allowed to sync concurrently. Larger number = more responsive quota management, but more CPU (and network) load. |

## SAControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

SAControllerConfiguration contains elements describing ServiceAccountController.

| Field                                         | Description                                                                                                                                  |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| ServiceAccountKeyFile<br>[Required]<br>string | serviceAccountKeyFile is the filename containing a PEM-encoded private RSA key used to sign service account tokens.                          |
| ConcurrentSATokenSyncs<br>[Required]<br>int32 | concurrentSATokenSyncs is the number of service account token syncing operations that will be done concurrently.                             |
| RootCAFile [Required]<br>string               | rootCAFile is the root certificate authority will be included in service account's token secret. This must be a valid PEM-encoded CA bundle. |

## StatefulSetControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

StatefulSetControllerConfiguration contains elements describing StatefulSetController.

| Field                                             | Description                                                                                                                                                                           |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentStatefulSetSyncs<br>[Required]<br>int32 | concurrentStatefulSetSyncs is the number of statefulset objects that are allowed to sync concurrently. Larger number = more responsive statefulsets, but more CPU (and network) load. |

## TTLAfterFinishedControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

TTLAfterFinishedControllerConfiguration contains elements describing TTLAfterFinishedController.

| Field                                     | Description                                                                                                     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| ConcurrentTTLSyncs<br>[Required]<br>int32 | concurrentTTLSyncs is the number of TTL-after-finished collector workers that are allowed to sync concurrently. |

# ValidatingAdmissionPolicyStatusControllerConfiguration

Appears in:

- [KubeControllerManagerConfiguration](#)

ValidatingAdmissionPolicyStatusControllerConfiguration contains elements describing ValidatingAdmissionPolicyStatusController.

| Field                                               | Description                                                                                                                                                                                  |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ConcurrentPolicySyncs<br><b>[Required]</b><br>int32 | ConcurrentPolicySyncs is the number of policy objects that are allowed to sync concurrently. Larger number = quicker type checking, but more CPU (and network) load. The default value is 5. |

## VolumeConfiguration

Appears in:

- [PersistentVolumeBinderControllerConfiguration](#)

VolumeConfiguration contains *all* enumerated flags meant to configure all volume plugins. From this config, the controller-manager binary will create many instances of volume.VolumeConfig, each containing only the configuration needed for that plugin which are then passed to the appropriate plugin. The ControllerManager binary is the only part of the code which knows what plugins are supported and which flags correspond to each plugin.

| Field                                                                                                               | Description                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EnableHostPathProvisioning<br><b>[Required]</b><br>bool                                                             | enableHostPathProvisioning enables HostPath PV provisioning when running without a cloud provider. This allows testing and development of provisioning features. HostPath provisioning is not supported in any way, won't work in a multi-node cluster, and should not be used for anything other than testing or development. |
| EnableDynamicProvisioning<br><b>[Required]</b><br>bool                                                              | enableDynamicProvisioning enables the provisioning of volumes when running within an environment that supports dynamic provisioning. Defaults to true.                                                                                                                                                                         |
| PersistentVolumeRecyclerConfiguration<br><b>[Required]</b><br><a href="#">PersistentVolumeRecyclerConfiguration</a> | persistentVolumeRecyclerConfiguration holds configuration for persistent volume plugins.                                                                                                                                                                                                                                       |

| Field                                           | Description                                                                                                                               |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| FlexVolumePluginDir <b>[Required]</b><br>string | volumePluginDir is the full path of the directory in which the flex volume plugin should search for additional third party volume plugins |

# kube-proxy Configuration (v1alpha1)

## Resource Types

- [KubeProxyConfiguration](#)

## ClientConnectionConfiguration

Appears in:

- [KubeProxyConfiguration](#)
- [KubeSchedulerConfiguration](#)
- [GenericControllerManagerConfiguration](#)

ClientConnectionConfiguration contains details for constructing a client.

| Field                                             | Description                                                                                                                                                                                                                      |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kubeconfig<br><b>[Required]</b><br>string         | kubeconfig is the path to a KubeConfig file.                                                                                                                                                                                     |
| acceptContentTypes<br><b>[Required]</b><br>string | acceptContentTypes defines the Accept header sent by clients when connecting to a server, overriding the default value of 'application/json'. This field will control all connections to the server used by a particular client. |
| contentType<br><b>[Required]</b><br>string        | contentType is the content type used when sending data to the server from this client.                                                                                                                                           |
| qps <b>[Required]</b><br>float32                  | qps controls the number of queries per second allowed for this connection.                                                                                                                                                       |
| burst <b>[Required]</b><br>int32                  | burst allows extra queries to accumulate when a client is exceeding its rate.                                                                                                                                                    |



# DebuggingConfiguration

Appears in:

- [KubeSchedulerConfiguration](#)
- [GenericControllerManagerConfiguration](#)

DebuggingConfiguration holds configuration for Debugging related features.

| Field                                               | Description                                                                       |
|-----------------------------------------------------|-----------------------------------------------------------------------------------|
| enableProfiling <b>[Required]</b><br>bool           | enableProfiling enables profiling via web interface<br>host:port/debug/pprof/     |
| enableContentionProfiling <b>[Required]</b><br>bool | enableContentionProfiling enables block profiling, if<br>enableProfiling is true. |

# LeaderElectionConfiguration

Appears in:

- [KubeSchedulerConfiguration](#)
- [GenericControllerManagerConfiguration](#)

LeaderElectionConfiguration defines the configuration of leader election clients for components that can run with leader election enabled.

| Field                                                               | Description                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| leaderElect <b>[Required]</b><br>bool                               | leaderElect enables a leader election client to gain leadership before<br>executing the main loop. Enable this when running replicated<br>components for high availability.                                                                                                                                                                                             |
| leaseDuration <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | leaseDuration is the duration that non-leader candidates will wait after<br>observing a leadership renewal until attempting to acquire leadership of<br>a led but unrenewed leader slot. This is effectively the maximum<br>duration that a leader can be stopped before it is replaced by another<br>candidate. This is only applicable if leader election is enabled. |
| renewDeadline <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | renewDeadline is the interval between attempts by the acting master to<br>renew a leadership slot before it stops leading. This must be less than or<br>equal to the lease duration. This is only applicable if leader election is<br>enabled.                                                                                                                          |

| Field                                                                | Description                                                                                                                                                            |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| retryPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | retryPeriod is the duration the clients should wait between attempting acquisition and renewal of a leadership. This is only applicable if leader election is enabled. |
| resourceLock<br><b>[Required]</b><br>string                          | resourceLock indicates the resource object type that will be used to lock during leader election cycles.                                                               |
| resourceName<br><b>[Required]</b><br>string                          | resourceName indicates the name of resource object that will be used to lock during leader election cycles.                                                            |
| resourceNamespace<br><b>[Required]</b><br>string                     | resourceName indicates the namespace of resource object that will be used to lock during leader election cycles.                                                       |

## KubeProxyConfiguration

KubeProxyConfiguration contains everything necessary to configure the Kubernetes proxy server.

| Field                                                                               | Description                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                                | kubeproxy.config.k8s.io/v1alpha1                                                                                                                                              |
| kind<br>string                                                                      | KubeProxyConfiguration                                                                                                                                                        |
| featureGates <b>[Required]</b><br>map[string]bool                                   | featureGates is a map of feature names to bools that enable or disable alpha/experimental features.                                                                           |
| clientConnection <b>[Required]</b><br><a href="#">ClientConnectionConfiguration</a> | clientConnection specifies the kubeconfig file and client connection settings for the proxy server to use when communicating with the apiserver.                              |
| logging <b>[Required]</b><br><a href="#">LoggingConfiguration</a>                   | logging specifies the options of logging. Refer to <a href="#">Logs Options</a> for more information.                                                                         |
| hostnameOverride <b>[Required]</b><br>string                                        | hostnameOverride, if non-empty, will be used as the name of the Node that kube-proxy is running on. If unset, the node name is assumed to be the same as the node's hostname. |
| bindAddress <b>[Required]</b><br>string                                             | bindAddress can be used to override kube-proxy's idea of what its node's primary IP is. Note that the name is a                                                               |

| Field                                                                          | Description                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                | historical artifact, and kube-proxy does not actually bind any sockets to this IP.                                                                                                                                                                                 |
| healthzBindAddress <b>[Required]</b><br>string                                 | healthzBindAddress is the IP address and port for the health check server to serve on, defaulting to "0.0.0.0:10256" (if bindAddress is unset or IPv4), or "[::]:10256" (if bindAddress is IPv6).                                                                  |
| metricsBindAddress <b>[Required]</b><br>string                                 | metricsBindAddress is the IP address and port for the metrics server to serve on, defaulting to "127.0.0.1:10249" (if bindAddress is unset or IPv4), or "[::1]:10249" (if bindAddress is IPv6). (Set to "0.0.0.0:10249" / "[::]:10249" to bind on all interfaces.) |
| bindAddressHardFail <b>[Required]</b><br>bool                                  | bindAddressHardFail, if true, tells kube-proxy to treat failure to bind to a port as fatal and exit                                                                                                                                                                |
| enableProfiling <b>[Required]</b><br>bool                                      | enableProfiling enables profiling via web interface on / debug/pprof handler. Profiling handlers will be handled by metrics server.                                                                                                                                |
| showHiddenMetricsForVersion <b>[Required]</b><br>string                        | showHiddenMetricsForVersion is the version for which you want to show hidden metrics.                                                                                                                                                                              |
| mode <b>[Required]</b><br><a href="#">ProxyMode</a>                            | mode specifies which proxy mode to use.                                                                                                                                                                                                                            |
| iptables <b>[Required]</b><br><a href="#">KubeProxyIPTablesConfiguration</a>   | iptables contains iptables-related configuration options.                                                                                                                                                                                                          |
| ipvs <b>[Required]</b><br><a href="#">KubeProxyIPVSConfiguration</a>           | ipvs contains ipvs-related configuration options.                                                                                                                                                                                                                  |
| nftables <b>[Required]</b><br><a href="#">KubeProxyNFTablesConfiguration</a>   | nftables contains nftables-related configuration options.                                                                                                                                                                                                          |
| winkernel <b>[Required]</b><br><a href="#">KubeProxyWinkernelConfiguration</a> | winkernel contains winkernel-related configuration options.                                                                                                                                                                                                        |
| detectLocalMode <b>[Required]</b><br><a href="#">LocalMode</a>                 | detectLocalMode determines mode to use for detecting local traffic, defaults to LocalModeClusterCIDR                                                                                                                                                               |
| detectLocal <b>[Required]</b><br><a href="#">DetectLocalConfiguration</a>      | detectLocal contains optional configuration settings related to DetectLocalMode.                                                                                                                                                                                   |

| Field                                                                          | Description                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clusterCIDR <b>[Required]</b><br>string                                        | clusterCIDR is the CIDR range of the pods in the cluster. (For dual-stack clusters, this can be a comma-separated dual-stack pair of CIDR ranges.). When DetectLocalMode is set to LocalModeClusterCIDR, kube-proxy will consider traffic to be local if its source IP is in this range. (Otherwise it is not used.) |
| nodePortAddresses <b>[Required]</b><br>[]string                                | nodePortAddresses is a list of CIDR ranges that contain valid node IPs. If set, connections to NodePort services will only be accepted on node IPs in one of the indicated ranges. If unset, NodePort connections will be accepted on all local IPs.                                                                 |
| oomScoreAdj <b>[Required]</b><br>int32                                         | oomScoreAdj is the oom-score-adj value for kube-proxy process. Values must be within the range [-1000, 1000]                                                                                                                                                                                                         |
| conntrack <b>[Required]</b><br><a href="#">KubeProxyConntrackConfiguration</a> | conntrack contains conntrack-related configuration options.                                                                                                                                                                                                                                                          |
| configSyncPeriod <b>[Required]</b><br><a href="#">meta/v1.Duration</a>         | configSyncPeriod is how often configuration from the apiserver is refreshed. Must be greater than 0.                                                                                                                                                                                                                 |
| portRange <b>[Required]</b><br>string                                          | portRange was previously used to configure the userspace proxy, but is now unused.                                                                                                                                                                                                                                   |

## DetectLocalConfiguration

Appears in:

- [KubeProxyConfiguration](#)

DetectLocalConfiguration contains optional settings related to DetectLocalMode option

| Field                                           | Description                                                                                                                                                                                                                        |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bridgeInterface <b>[Required]</b><br>string     | bridgeInterface is a bridge interface name. When DetectLocalMode is set to LocalModeBridgeInterface, kube-proxy will consider traffic to be local if it originates from this bridge.                                               |
| interfaceNamePrefix <b>[Required]</b><br>string | interfaceNamePrefix is an interface name prefix. When DetectLocalMode is set to LocalModeInterfaceNamePrefix, kube-proxy will consider traffic to be local if it originates from any interface whose name begins with this prefix. |

# KubeProxyConntrackConfiguration

Appears in:

- [KubeProxyConfiguration](#)

KubeProxyConntrackConfiguration contains conntrack settings for the Kubernetes proxy server.

| Field                                                                       | Description                                                                                                                                                                    |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| maxPerCore <b>[Required]</b><br>int32                                       | maxPerCore is the maximum number of NAT connections to track per CPU core (0 to leave the limit as-is and ignore min).                                                         |
| min <b>[Required]</b><br>int32                                              | min is the minimum value of connect-tracking records to allocate, regardless of maxPerCore (set maxPerCore=0 to leave the limit as-is).                                        |
| tcpEstablishedTimeout <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | tcpEstablishedTimeout is how long an idle TCP connection will be kept open (e.g. '2s'). Must be greater than 0 to set.                                                         |
| tcpCloseWaitTimeout <b>[Required]</b><br><a href="#">meta/v1.Duration</a>   | tcpCloseWaitTimeout is how long an idle conntrack entry in CLOSE_WAIT state will remain in the conntrack table. (e.g. '60s'). Must be greater than 0 to set.                   |
| tcpBeLiberal <b>[Required]</b><br>bool                                      | tcpBeLiberal, if true, kube-proxy will configure conntrack to run in liberal mode for TCP connections and packets with out-of-window sequence numbers won't be marked INVALID. |
| udpTimeout <b>[Required]</b><br><a href="#">meta/v1.Duration</a>            | udpTimeout is how long an idle UDP conntrack entry in UNREPLIED state will remain in the conntrack table (e.g. '30s'). Must be greater than 0 to set.                          |
| udpStreamTimeout <b>[Required]</b><br><a href="#">meta/v1.Duration</a>      | udpStreamTimeout is how long an idle UDP conntrack entry in ASSURED state will remain in the conntrack table (e.g. '300s'). Must be greater than 0 to set.                     |

# KubeProxyIPTablesConfiguration

Appears in:

- [KubeProxyConfiguration](#)

KubeProxyIPTablesConfiguration contains iptables-related configuration details for the Kubernetes proxy server.

| Field                                                                  | Description                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| masqueradeBit<br><b>[Required]</b><br>int32                            | masqueradeBit is the bit of the iptables fwmark space to use for SNAT if using the iptables or ipvs proxy mode. Values must be within the range [0, 31].                                                                                                       |
| masqueradeAll<br><b>[Required]</b><br>bool                             | masqueradeAll tells kube-proxy to SNAT all traffic sent to Service cluster IPs, when using the iptables or ipvs proxy mode. This may be required with some CNI plugins.                                                                                        |
| localhostNodePorts<br><b>[Required]</b><br>bool                        | localhostNodePorts, if false, tells kube-proxy to disable the legacy behavior of allowing NodePort services to be accessed via localhost. (Applies only to iptables mode and IPv4; localhost NodePorts are never allowed with other proxy modes or with IPv6.) |
| syncPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a>    | syncPeriod is an interval (e.g. '5s', '1m', '2h22m') indicating how frequently various re-synchronizing and cleanup operations are performed. Must be greater than 0.                                                                                          |
| minSyncPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | minSyncPeriod is the minimum period between iptables rule resyncs (e.g. '5s', '1m', '2h22m'). A value of 0 means every Service or EndpointSlice change will result in an immediate iptables resync.                                                            |

## KubeProxyIPVSConfiguration

Appears in:

- [KubeProxyConfiguration](#)

KubeProxyIPVSConfiguration contains ipvs-related configuration details for the Kubernetes proxy server.

| Field                                                                  | Description                                                                                                                                                                                 |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| syncPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a>    | syncPeriod is an interval (e.g. '5s', '1m', '2h22m') indicating how frequently various re-synchronizing and cleanup operations are performed. Must be greater than 0.                       |
| minSyncPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | minSyncPeriod is the minimum period between IPVS rule resyncs (e.g. '5s', '1m', '2h22m'). A value of 0 means every Service or EndpointSlice change will result in an immediate IPVS resync. |
| scheduler<br><b>[Required]</b><br>string                               | scheduler is the IPVS scheduler to use                                                                                                                                                      |

| Field                                                                  | Description                                                                                                                                                           |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| excludeCIDRs<br><b>[Required]</b><br>[]string                          | excludeCIDRs is a list of CIDRs which the ipvs proxier should not touch when cleaning up ipvs services.                                                               |
| strictARP<br><b>[Required]</b><br>bool                                 | strictARP configures arp_ignore and arp_announce to avoid answering ARP queries from kube-ipvs0 interface                                                             |
| tcpTimeout<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a>    | tcpTimeout is the timeout value used for idle IPVS TCP sessions. The default value is 0, which preserves the current timeout value on the system.                     |
| tcpFinTimeout<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | tcpFinTimeout is the timeout value used for IPVS TCP sessions after receiving a FIN. The default value is 0, which preserves the current timeout value on the system. |
| udpTimeout<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a>    | udpTimeout is the timeout value used for IPVS UDP packets. The default value is 0, which preserves the current timeout value on the system.                           |

## KubeProxyNFTablesConfiguration

Appears in:

- [KubeProxyConfiguration](#)

KubeProxyNFTablesConfiguration contains nftables-related configuration details for the Kubernetes proxy server.

| Field                                                                  | Description                                                                                                                                                                                         |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| masqueradeBit<br><b>[Required]</b><br>int32                            | masqueradeBit is the bit of the iptables fwmark space to use for SNAT if using the nftables proxy mode. Values must be within the range [0, 31].                                                    |
| masqueradeAll<br><b>[Required]</b><br>bool                             | masqueradeAll tells kube-proxy to SNAT all traffic sent to Service cluster IPs, when using the nftables mode. This may be required with some CNI plugins.                                           |
| syncPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a>    | syncPeriod is an interval (e.g. '5s', '1m', '2h22m') indicating how frequently various re-synchronizing and cleanup operations are performed. Must be greater than 0.                               |
| minSyncPeriod<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | minSyncPeriod is the minimum period between iptables rule resyncs (e.g. '5s', '1m', '2h22m'). A value of 0 means every Service or EndpointSlice change will result in an immediate iptables resync. |

# KubeProxyWinkernelConfiguration

Appears in:

- [KubeProxyConfiguration](#)

KubeProxyWinkernelConfiguration contains Windows/HNS settings for the Kubernetes proxy server.

| Field                                             | Description                                                                                            |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| networkName [ <b>Required</b> ]<br>string         | networkName is the name of the network kube-proxy will use to create endpoints and policies            |
| sourceVip [ <b>Required</b> ]<br>string           | sourceVip is the IP address of the source VIP endpoint used for NAT when loadbalancing                 |
| enableDSR [ <b>Required</b> ]<br>bool             | enableDSR tells kube-proxy whether HNS policies should be created with DSR                             |
| rootHnsEndpointName [ <b>Required</b> ]<br>string | rootHnsEndpointName is the name of hnsendpoint that is attached to l2bridge for root network namespace |
| forwardHealthCheckVip [ <b>Required</b> ]<br>bool | forwardHealthCheckVip forwards service VIP for health check port on Windows                            |

## LocalMode

(Alias of string)

Appears in:

- [KubeProxyConfiguration](#)

LocalMode represents modes to detect local traffic from the node

## ProxyMode

(Alias of string)

Appears in:

- [KubeProxyConfiguration](#)

ProxyMode represents modes used by the Kubernetes proxy server.



Currently, two modes of proxy are available on Linux platforms: 'iptables' and 'ipvs'. One mode of proxy is available on Windows platforms: 'kernelSpace'.

If the proxy mode is unspecified, the best-available proxy mode will be used (currently this is iptables on Linux and kernelSpace on Windows). If the selected proxy mode cannot be used (due to lack of kernel support, missing userspace components, etc) then kube-proxy will exit with an error.

# kube-scheduler Configuration (v1)

## Resource Types

- [DefaultPreemptionArgs](#)
- [InterPodAffinityArgs](#)
- [KubeSchedulerConfiguration](#)
- [NodeAffinityArgs](#)
- [NodeResourcesBalancedAllocationArgs](#)
- [NodeResourcesFitArgs](#)
- [PodTopologySpreadArgs](#)
- [VolumeBindingArgs](#)

## ClientConnectionConfiguration

Appears in:

- [KubeSchedulerConfiguration](#)

ClientConnectionConfiguration contains details for constructing a client.

| Field                                      | Description                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kubeconfig<br>[Required]<br>string         | kubeconfig is the path to a KubeConfig file.                                                                                                                                                                                     |
| acceptContentTypes<br>[Required]<br>string | acceptContentTypes defines the Accept header sent by clients when connecting to a server, overriding the default value of 'application/json'. This field will control all connections to the server used by a particular client. |
| contentType<br>[Required]<br>string        | contentType is the content type used when sending data to the server from this client.                                                                                                                                           |
| qps [Required]<br>float32                  | qps controls the number of queries per second allowed for this connection.                                                                                                                                                       |
| burst [Required]<br>int32                  | burst allows extra queries to accumulate when a client is exceeding its rate.                                                                                                                                                    |

# DebuggingConfiguration

Appears in:

- [KubeSchedulerConfiguration](#)

DebuggingConfiguration holds configuration for Debugging related features.

| Field                                               | Description                                                                       |
|-----------------------------------------------------|-----------------------------------------------------------------------------------|
| enableProfiling <b>[Required]</b><br>bool           | enableProfiling enables profiling via web interface<br>host:port/debug/pprof/     |
| enableContentionProfiling <b>[Required]</b><br>bool | enableContentionProfiling enables block profiling, if<br>enableProfiling is true. |

# LeaderElectionConfiguration

Appears in:

- [KubeSchedulerConfiguration](#)

LeaderElectionConfiguration defines the configuration of leader election clients for components that can run with leader election enabled.

| Field                                                               | Description                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| leaderElect <b>[Required]</b><br>bool                               | leaderElect enables a leader election client to gain leadership before<br>executing the main loop. Enable this when running replicated<br>components for high availability.                                                                                                                                                                                             |
| leaseDuration <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | leaseDuration is the duration that non-leader candidates will wait after<br>observing a leadership renewal until attempting to acquire leadership of<br>a led but unrenewed leader slot. This is effectively the maximum<br>duration that a leader can be stopped before it is replaced by another<br>candidate. This is only applicable if leader election is enabled. |
| renewDeadline <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | renewDeadline is the interval between attempts by the acting master to<br>renew a leadership slot before it stops leading. This must be less than or<br>equal to the lease duration. This is only applicable if leader election is<br>enabled.                                                                                                                          |
| retryPeriod <b>[Required]</b><br><a href="#">meta/v1.Duration</a>   | retryPeriod is the duration the clients should wait between attempting<br>acquisition and renewal of a leadership. This is only applicable if leader<br>election is enabled.                                                                                                                                                                                            |

| Field                                            | Description                                                                                                      |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| resourceLock<br><b>[Required]</b><br>string      | resourceLock indicates the resource object type that will be used to lock during leader election cycles.         |
| resourceName<br><b>[Required]</b><br>string      | resourceName indicates the name of resource object that will be used to lock during leader election cycles.      |
| resourceNamespace<br><b>[Required]</b><br>string | resourceName indicates the namespace of resource object that will be used to lock during leader election cycles. |

## DefaultPreemptionArgs

DefaultPreemptionArgs holds arguments used to configure the DefaultPreemption plugin.

| Field                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                      | kubescheduler.config.k8s.io/v1                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| kind<br>string                                            | DefaultPreemptionArgs                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| minCandidateNodesPercentage<br><b>[Required]</b><br>int32 | MinCandidateNodesPercentage is the minimum number of candidates to shortlist when dry running preemption as a percentage of number of nodes. Must be in the range [0, 100]. Defaults to 10% of the cluster size if unspecified.                                                                                                                                                                                                                                                |
| minCandidateNodesAbsolute<br><b>[Required]</b><br>int32   | MinCandidateNodesAbsolute is the absolute minimum number of candidates to shortlist. The likely number of candidates enumerated for dry running preemption is given by the formula: numCandidates = max(numNodes * minCandidateNodesPercentage, minCandidateNodesAbsolute)<br>We say "likely" because there are other factors such as PDB violations that play a role in the number of candidates shortlisted. Must be at least 0 nodes. Defaults to 100 nodes if unspecified. |

## InterPodAffinityArgs

InterPodAffinityArgs holds arguments used to configure the InterPodAffinity plugin.

| Field                | Description                    |
|----------------------|--------------------------------|
| apiVersion<br>string | kubescheduler.config.k8s.io/v1 |
| kind<br>string       | InterPodAffinityArgs           |

| Field                                                        | Description                                                                                                                                                                                   |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hardPodAffinityWeight <b>[Required]</b><br>int32             | HardPodAffinityWeight is the scoring weight for existing pods with a matching hard affinity to the incoming pod.                                                                              |
| ignorePreferredTermsOfExistingPods <b>[Required]</b><br>bool | IgnorePreferredTermsOfExistingPods configures the scheduler to ignore existing pods' preferred affinity rules when scoring candidate nodes, unless the incoming pod has inter-pod affinities. |

## KubeSchedulerConfiguration

KubeSchedulerConfiguration configures a scheduler

| Field                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                                | kubescheduler.config.k8s.io/v1                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| kind<br>string                                                                      | KubeSchedulerConfiguration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| parallelism <b>[Required]</b><br>int32                                              | Parallelism defines the amount of parallelism in algorithms for scheduling a Pods. Must be greater than 0. Defaults to 16                                                                                                                                                                                                                                                                                                                                                                                   |
| leaderElection <b>[Required]</b><br><a href="#">LeaderElectionConfiguration</a>     | LeaderElection defines the configuration of leader election client.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| clientConnection <b>[Required]</b><br><a href="#">ClientConnectionConfiguration</a> | ClientConnection specifies the kubeconfig file and client connection settings for the proxy server to use when communicating with the apiserver.                                                                                                                                                                                                                                                                                                                                                            |
| DebuggingConfiguration <b>[Required]</b><br><a href="#">DebuggingConfiguration</a>  | (Members of DebuggingConfiguration are embedded into this type.)<br><br>DebuggingConfiguration holds configuration for Debugging related features TODO: We might wanna make this a substruct like Debugging<br>componentbaseconfigv1alpha1.DebuggingConfiguration                                                                                                                                                                                                                                           |
| percentageOfNodesToScore <b>[Required]</b><br>int32                                 | PercentageOfNodesToScore is the percentage of all nodes that once found feasible for running a pod, the scheduler stops its search for more feasible nodes in the cluster. This helps improve scheduler's performance. Scheduler always tries to find at least "minFeasibleNodesToFind" feasible nodes no matter what the value of this flag is. Example: if the cluster size is 500 nodes and the value of this flag is 30, then scheduler stops finding further feasible nodes once it finds 150 feasible |

| Field                                                                | Description                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                      | ones. When the value is 0, default percentage (5%--50% based on the size of the cluster) of the nodes will be scored. It is overridden by profile level PercentageofNodesToScore.                                                                                                                              |
| podInitialBackoffSeconds<br><b>[Required]</b><br>int64               | PodInitialBackoffSeconds is the initial backoff for unschedulable pods. If specified, it must be greater than 0. If this value is null, the default value (1s) will be used.                                                                                                                                   |
| podMaxBackoffSeconds<br><b>[Required]</b><br>int64                   | PodMaxBackoffSeconds is the max backoff for unschedulable pods. If specified, it must be greater than podInitialBackoffSeconds. If this value is null, the default value (10s) will be used.                                                                                                                   |
| profiles <b>[Required]</b><br><a href="#">[]KubeSchedulerProfile</a> | Profiles are scheduling profiles that kube-scheduler supports. Pods can choose to be scheduled under a particular profile by setting its associated scheduler name. Pods that don't specify any scheduler name are scheduled with the "default-scheduler" profile, if present here.                            |
| extenders <b>[Required]</b><br><a href="#">[]Extender</a>            | Extenders are the list of scheduler extenders, each holding the values of how to communicate with the extender. These extenders are shared by all scheduler profiles.                                                                                                                                          |
| delayCacheUntilActive<br><b>[Required]</b><br>bool                   | DelayCacheUntilActive specifies when to start caching. If this is true and leader election is enabled, the scheduler will wait to fill informer caches until it is the leader. Doing so will have slower failover with the benefit of lower memory overhead while waiting to become leader. Defaults to false. |

## NodeAffinityArgs

NodeAffinityArgs holds arguments to configure the NodeAffinity plugin.

| Field                                                 | Description                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                  | kubescheduler.config.k8s.io/v1                                                                                                                                                                                                                                                                                                                                             |
| kind<br>string                                        | NodeAffinityArgs                                                                                                                                                                                                                                                                                                                                                           |
| addedAffinity<br><a href="#">core/v1.NodeAffinity</a> | AddedAffinity is applied to all Pods additionally to the NodeAffinity specified in the PodSpec. That is, Nodes need to satisfy AddedAffinity AND .spec.NodeAffinity. AddedAffinity is empty by default (all Nodes match). When AddedAffinity is used, some Pods with affinity requirements that match a specific Node (such as Daemonset Pods) might remain unschedulable. |

## NodeResourcesBalancedAllocationArgs

NodeResourcesBalancedAllocationArgs holds arguments used to configure NodeResourcesBalancedAllocation plugin.

| Field                                                              | Description                                                                  |
|--------------------------------------------------------------------|------------------------------------------------------------------------------|
| apiVersion<br>string                                               | kubescheduler.config.k8s.io/v1                                               |
| kind<br>string                                                     | NodeResourcesBalancedAllocationArgs                                          |
| resources<br>[ <b>Required</b> ]<br><a href="#">[]ResourceSpec</a> | Resources to be managed, the default is "cpu" and "memory" if not specified. |

## NodeResourcesFitArgs

NodeResourcesFitArgs holds arguments used to configure the NodeResourcesFit plugin.

| Field                                                                     | Description                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                      | kubescheduler.config.k8s.io/v1                                                                                                                                                                                                                                                                                                     |
| kind<br>string                                                            | NodeResourcesFitArgs                                                                                                                                                                                                                                                                                                               |
| ignoredResources<br>[ <b>Required</b> ]<br>[]string                       | IgnoredResources is the list of resources that NodeResources fit filter should ignore. This doesn't apply to scoring.                                                                                                                                                                                                              |
| ignoredResourceGroups<br>[ <b>Required</b> ]<br>[]string                  | IgnoredResourceGroups defines the list of resource groups that NodeResources fit filter should ignore. e.g. if group is ["example.com"], it will ignore all resource names that begin with "example.com", such as "example.com/aaa" and "example.com/bbb". A resource group name can't contain '/'. This doesn't apply to scoring. |
| scoringStrategy<br>[ <b>Required</b> ]<br><a href="#">ScoringStrategy</a> | ScoringStrategy selects the node resource scoring strategy. The default strategy is LeastAllocated with an equal "cpu" and "memory" weight.                                                                                                                                                                                        |

## PodTopologySpreadArgs

PodTopologySpreadArgs holds arguments used to configure the PodTopologySpread plugin.

| Field                | Description                    |
|----------------------|--------------------------------|
| apiVersion<br>string | kubescheduler.config.k8s.io/v1 |
| kind<br>string       | PodTopologySpreadArgs          |

| Field                                                                    | Description                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| defaultConstraints<br><a href="#">[]core/v1.TopologySpreadConstraint</a> | DefaultConstraints defines topology spread constraints to be applied to Pods that don't define any in pod.spec.topologySpreadConstraints. .defaultConstraints[*].labelSelector must be empty, as they are deduced from the Pod's membership to Services, ReplicationControllers, ReplicaSets or StatefulSets. When empty, .defaultingType must be "List". |
| defaultingType<br><a href="#">PodTopologySpreadConstraintsDefaulting</a> | <p>DefaultingType determines how .defaultConstraints are deduced. one of "System" or "List".</p> <ul style="list-style-type: none"> <li>• "System": Use kubernetes defined constraints that spread Pods among Nodes and Zones.</li> <li>• "List": Use constraints defined in .defaultConstraints.</li> </ul> <p>Defaults to "System".</p>                 |

## VolumeBindingArgs

VolumeBindingArgs holds arguments used to configure the VolumeBinding plugin.

| Field                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                             | kubescheduler.config.k8s.io/v1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| kind<br>string                                   | VolumeBindingArgs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| bindTimeoutSeconds<br><b>[Required]</b><br>int64 | BindTimeoutSeconds is the timeout in seconds in volume binding operation. Value must be non-negative integer. The value zero indicates no waiting. If this value is nil, the default value (600) will be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| shape<br><a href="#">[]UtilizationShapePoint</a> | <p>Shape specifies the points defining the score function shape, which is used to score nodes based on the utilization of statically provisioned PVs. The utilization is calculated by dividing the total requested storage of the pod by the total capacity of feasible PVs on each node. Each point contains utilization (ranges from 0 to 100) and its associated score (ranges from 0 to 10). You can turn the priority by specifying different scores for different utilization numbers. The default shape points are:</p> <ol style="list-style-type: none"> <li>1. 0 for 0 utilization</li> <li>2. 10 for 100 utilization</li> </ol> <p>All points must be sorted in increasing order by utilization.</p> |

# Extender

Appears in:

- [KubeSchedulerConfiguration](#)

Extender holds the parameters used to communicate with the extender. If a verb is unspecified/empty, it is assumed that the extender chose not to provide that extension.

| Field                                                             | Description                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| urlPrefix <b>[Required]</b><br>string                             | URLPrefix at which the extender is available                                                                                                                                                                                                                                                   |
| filterVerb <b>[Required]</b><br>string                            | Verb for the filter call, empty if not supported. This verb is appended to the URLPrefix when issuing the filter call to extender.                                                                                                                                                             |
| preemptVerb <b>[Required]</b><br>string                           | Verb for the preempt call, empty if not supported. This verb is appended to the URLPrefix when issuing the preempt call to extender.                                                                                                                                                           |
| prioritizeVerb <b>[Required]</b><br>string                        | Verb for the prioritize call, empty if not supported. This verb is appended to the URLPrefix when issuing the prioritize call to extender.                                                                                                                                                     |
| weight <b>[Required]</b><br>int64                                 | The numeric multiplier for the node scores that the prioritize call generates. The weight should be a positive integer                                                                                                                                                                         |
| bindVerb <b>[Required]</b><br>string                              | Verb for the bind call, empty if not supported. This verb is appended to the URLPrefix when issuing the bind call to extender. If this method is implemented by the extender, it is the extender's responsibility to bind the pod to apiserver. Only one extender can implement this function. |
| enableHTTPS <b>[Required]</b><br>bool                             | EnableHTTPS specifies whether https should be used to communicate with the extender                                                                                                                                                                                                            |
| tlsConfig <b>[Required]</b><br><a href="#">ExtenderTLSConfig</a>  | TLSConfig specifies the transport layer security config                                                                                                                                                                                                                                        |
| httpTimeout <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | HTTPTimeout specifies the timeout duration for a call to the extender. Filter timeout fails the scheduling of the pod. Prioritize timeout is ignored, k8s/other extenders priorities are used to select the node.                                                                              |



| Field                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nodeCacheCapable<br><b>[Required]</b><br>bool                 | NodeCacheCapable specifies that the extender is capable of caching node information, so the scheduler should only send minimal information about the eligible nodes assuming that the extender already cached full details of all nodes in the cluster                                                                                                                                                                                                                                                                        |
| managedResources<br><a href="#">[]ExtenderManagedResource</a> | ManagedResources is a list of extended resources that are managed by this extender. <ul style="list-style-type: none"> <li>• A pod will be sent to the extender on the Filter, Prioritize and Bind (if the extender is the binder) phases iff the pod requests at least one of the extended resources in this list. If empty or unspecified, all pods will be sent to this extender.</li> <li>• If IgnoredByScheduler is set to true for a resource, kube-scheduler will skip checking the resource in predicates.</li> </ul> |
| ignorable <b>[Required]</b><br>bool                           | Ignorable specifies if the extender is ignorable, i.e. scheduling should not fail when the extender returns an error or is not reachable.                                                                                                                                                                                                                                                                                                                                                                                     |

## ExtenderManagedResource

Appears in:

- [Extender](#)

ExtenderManagedResource describes the arguments of extended resources managed by an extender.

| Field                                           | Description                                                                                               |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| name <b>[Required]</b><br>string                | Name is the extended resource name.                                                                       |
| ignoredByScheduler<br><b>[Required]</b><br>bool | IgnoredByScheduler indicates whether kube-scheduler should ignore this resource when applying predicates. |

## ExtenderTLSConfig

Appears in:

- [Extender](#)

ExtenderTLSConfig contains settings to enable TLS with extender

| Field                                     | Description                                                                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| insecure<br><b>[Required]</b><br>bool     | Server should be accessed without verifying the TLS certificate. For testing only.                                                                                                  |
| serverName<br><b>[Required]</b><br>string | ServerName is passed to the server for SNI and is used in the client to check server certificates against. If ServerName is empty, the hostname used to contact the server is used. |
| certFile<br><b>[Required]</b><br>string   | Server requires TLS client certificate authentication                                                                                                                               |
| keyFile<br><b>[Required]</b><br>string    | Server requires TLS client certificate authentication                                                                                                                               |
| caFile<br><b>[Required]</b><br>string     | Trusted root certificates for server                                                                                                                                                |
| certData<br><b>[Required]</b><br>[]byte   | CertData holds PEM-encoded bytes (typically read from a client certificate file). CertData takes precedence over CertFile                                                           |
| keyData<br><b>[Required]</b><br>[]byte    | KeyData holds PEM-encoded bytes (typically read from a client certificate key file). KeyData takes precedence over KeyFile                                                          |
| caData<br><b>[Required]</b><br>[]byte     | CAData holds PEM-encoded bytes (typically read from a root certificates bundle). CAData takes precedence over CAFile                                                                |

## KubeSchedulerProfile

Appears in:

- [KubeSchedulerConfiguration](#)

KubeSchedulerProfile is a scheduling profile.

| Field                                                  | Description                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| schedulerName <b>[Required]</b><br>string              | SchedulerName is the name of the scheduler associated to this profile. If SchedulerName matches with the pod's "spec.schedulerName", then the pod is scheduled with this profile.                                                                                                                                         |
| percentageOfNodesToScore<br><b>[Required]</b><br>int32 | PercentageOfNodesToScore is the percentage of all nodes that once found feasible for running a pod, the scheduler stops its search for more feasible nodes in the cluster. This helps improve scheduler's performance. Scheduler always tries to find at least "minFeasibleNodesToFind" feasible nodes no matter what the |

| Field                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                  | value of this flag is. Example: if the cluster size is 500 nodes and the value of this flag is 30, then scheduler stops finding further feasible nodes once it finds 150 feasible ones. When the value is 0, default percentage (5%--50% based on the size of the cluster) of the nodes will be scored. It will override global PercentageOfNodesToScore. If it is empty, global PercentageOfNodesToScore will be used.                                                                                   |
| plugins <b>[Required]</b><br><a href="#">Plugins</a>             | Plugins specify the set of plugins that should be enabled or disabled. Enabled plugins are the ones that should be enabled in addition to the default plugins. Disabled plugins are any of the default plugins that should be disabled. When no enabled or disabled plugin is specified for an extension point, default plugins for that extension point will be used if there is any. If a QueueSort plugin is specified, the same QueueSort Plugin and PluginConfig must be specified for all profiles. |
| pluginConfig <b>[Required]</b><br><a href="#">[]PluginConfig</a> | PluginConfig is an optional set of custom plugin arguments for each plugin. Omitting config args for a plugin is equivalent to using the default config for that plugin.                                                                                                                                                                                                                                                                                                                                  |

## Plugin

Appears in:

- [PluginSet](#)

Plugin specifies a plugin name and its weight when applicable. Weight is used only for Score plugins.

| Field                             | Description                                                       |
|-----------------------------------|-------------------------------------------------------------------|
| name <b>[Required]</b><br>string  | Name defines the name of plugin                                   |
| weight <b>[Required]</b><br>int32 | Weight defines the weight of plugin, only used for Score plugins. |

## PluginConfig

Appears in:

- [KubeSchedulerProfile](#)

PluginConfig specifies arguments that should be passed to a plugin at the time of initialization. A plugin that is invoked at multiple extension points is initialized once. Args can have arbitrary structure. It is up to the plugin to process these Args.

| Field                                                                                                                                     | Description                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| name <b>[Required]</b><br>string                                                                                                          | Name defines the name of plugin being configured                                                                   |
| args <b>[Required]</b><br><a href="https://k8s.io/apimachinery/pkg/runtime.RawExtension">k8s.io/apimachinery/pkg/runtime.RawExtension</a> | Args defines the arguments passed to the plugins at the time of initialization. Args can have arbitrary structure. |

## PluginSet

Appears in:

- [Plugins](#)

PluginSet specifies enabled and disabled plugins for an extension point. If an array is empty, missing, or nil, default plugins at that extension point will be used.

| Field                                                  | Description                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| enabled <b>[Required]</b><br><a href="#">[]Plugin</a>  | Enabled specifies plugins that should be enabled in addition to default plugins. If the default plugin is also configured in the scheduler config file, the weight of plugin will be overridden accordingly. These are called after default plugins and in the same order specified here. |
| disabled <b>[Required]</b><br><a href="#">[]Plugin</a> | Disabled specifies default plugins that should be disabled. When all default plugins need to be disabled, an array containing only one "*" should be provided.                                                                                                                            |

## Plugins

Appears in:

- [KubeSchedulerProfile](#)

Plugins include multiple extension points. When specified, the list of plugins for a particular extension point are the only ones enabled. If an extension point is omitted from the config, then the default set of plugins is used for that extension point. Enabled plugins are called in the order specified here, after default plugins. If they need to be invoked before default plugins, default plugins must be disabled and re-enabled here in desired order.

| Field                                                     | Description                                                                                        |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| preEnqueue <b>[Required]</b><br><a href="#">PluginSet</a> | PreEnqueue is a list of plugins that should be invoked before adding pods to the scheduling queue. |
| queueSort <b>[Required]</b><br><a href="#">PluginSet</a>  | QueueSort is a list of plugins that should be invoked when sorting pods in the scheduling queue.   |

| Field                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| preFilter<br>[Required]<br><a href="#">PluginSet</a>  | PreFilter is a list of plugins that should be invoked at "PreFilter" extension point of the scheduling framework.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| filter<br>[Required]<br><a href="#">PluginSet</a>     | Filter is a list of plugins that should be invoked when filtering out nodes that cannot run the Pod.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| postFilter<br>[Required]<br><a href="#">PluginSet</a> | PostFilter is a list of plugins that are invoked after filtering phase, but only when no feasible nodes were found for the pod.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| preScore<br>[Required]<br><a href="#">PluginSet</a>   | PreScore is a list of plugins that are invoked before scoring.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| score<br>[Required]<br><a href="#">PluginSet</a>      | Score is a list of plugins that should be invoked when ranking nodes that have passed the filtering phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| reserve<br>[Required]<br><a href="#">PluginSet</a>    | Reserve is a list of plugins invoked when reserving/unreserving resources after a node is assigned to run the pod.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| permit<br>[Required]<br><a href="#">PluginSet</a>     | Permit is a list of plugins that control binding of a Pod. These plugins can prevent or delay binding of a Pod.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| preBind<br>[Required]<br><a href="#">PluginSet</a>    | PreBind is a list of plugins that should be invoked before a pod is bound.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| bind<br>[Required]<br><a href="#">PluginSet</a>       | Bind is a list of plugins that should be invoked at "Bind" extension point of the scheduling framework. The scheduler call these plugins in order. Scheduler skips the rest of these plugins as soon as one returns success.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| postBind<br>[Required]<br><a href="#">PluginSet</a>   | PostBind is a list of plugins that should be invoked after a pod is successfully bound.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| multiPoint<br>[Required]<br><a href="#">PluginSet</a> | <p>MultiPoint is a simplified config section to enable plugins for all valid extension points. Plugins enabled through MultiPoint will automatically register for every individual extension point the plugin has implemented. Disabling a plugin through MultiPoint disables that behavior. The same is true for disabling "*" through MultiPoint (no default plugins will be automatically registered). Plugins can still be disabled through their individual extension points.</p> <p>In terms of precedence, plugin config follows this basic hierarchy</p> <ol style="list-style-type: none"> <li>1. Specific extension points</li> <li>2. Explicitly configured MultiPoint plugins</li> </ol> |

| Field | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | 3. The set of default plugins, as MultiPoint plugins This implies that a higher precedence plugin will run first and overwrite any settings within MultiPoint. Explicitly user-configured plugins also take a higher precedence over default plugins. Within this hierarchy, an Enabled setting takes precedence over Disabled. For example, if a plugin is set in both multiPoint.Enabled and multiPoint.Disabled, the plugin will be enabled. Similarly, including multiPoint.Disabled = '*' and multiPoint.Enabled = pluginA will still register that specific plugin through MultiPoint. This follows the same behavior as all other extension point configurations. |

## PodTopologySpreadConstraintsDefaulting

(Alias of string)

Appears in:

- [PodTopologySpreadArgs](#)

PodTopologySpreadConstraintsDefaulting defines how to set default constraints for the PodTopologySpread plugin.

## RequestedToCapacityRatioParam

Appears in:

- [ScoringStrategy](#)

RequestedToCapacityRatioParam define RequestedToCapacityRatio parameters

| Field                                                                | Description                                                    |
|----------------------------------------------------------------------|----------------------------------------------------------------|
| shape [ <b>Required</b> ]<br><a href="#">[]UtilizationShapePoint</a> | Shape is a list of points defining the scoring function shape. |

## ResourceSpec

Appears in:

- [NodeResourcesBalancedAllocationArgs](#)
- [ScoringStrategy](#)

ResourceSpec represents a single resource.

| Field                              | Description           |
|------------------------------------|-----------------------|
| name [ <b>Required</b> ]<br>string | Name of the resource. |

| Field                               | Description             |
|-------------------------------------|-------------------------|
| weight [ <b>Required</b> ]<br>int64 | Weight of the resource. |

## ScoringStrategy

Appears in:

- [NodeResourcesFitArgs](#)

ScoringStrategy define ScoringStrategyType for node resource plugin

| Field                                                                                            | Description                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type [ <b>Required</b> ]<br><a href="#">ScoringStrategyType</a>                                  | Type selects which strategy to run.                                                                                                                                                                            |
| resources [ <b>Required</b> ]<br><a href="#">[]ResourceSpec</a>                                  | Resources to consider when scoring. The default resource set includes "cpu" and "memory" with an equal weight. Allowed weights go from 1 to 100. Weight defaults to 1 if not specified or explicitly set to 0. |
| requestedToCapacityRatio<br>[ <b>Required</b> ]<br><a href="#">RequestedToCapacityRatioParam</a> | Arguments specific to RequestedToCapacityRatio strategy.                                                                                                                                                       |

## ScoringStrategyType

(Alias of string)

Appears in:

- [ScoringStrategy](#)

ScoringStrategyType the type of scoring strategy used in NodeResourcesFit plugin.

## UtilizationShapePoint

Appears in:

- [VolumeBindingArgs](#)
- [RequestedToCapacityRatioParam](#)

UtilizationShapePoint represents single point of priority function shape.

| Field                                       | Description                                                                       |
|---------------------------------------------|-----------------------------------------------------------------------------------|
| utilization<br>[ <b>Required</b> ]<br>int32 | Utilization (x axis). Valid values are 0 to 100. Fully utilized node maps to 100. |

| Field                     | Description                                                             |
|---------------------------|-------------------------------------------------------------------------|
| score [Required]<br>int32 | Score assigned to given utilization (y axis). Valid values are 0 to 10. |

# kubeadm Configuration (v1beta3)

## Overview

Package v1beta3 defines the v1beta3 version of the kubeadm configuration file format. This version improves on the v1beta2 format by fixing some minor issues and adding a few new fields.

A list of changes since v1beta2:

- The deprecated "ClusterConfiguration.useHyperKubeImage" field has been removed. Kubeadm no longer supports the hyperkube image.
- The "ClusterConfiguration.DNS.Type" field has been removed since CoreDNS is the only supported DNS server type by kubeadm.
- Include "datapolicy" tags on the fields that hold secrets. This would result in the field values to be omitted when API structures are printed with klog.
- Add "InitConfiguration.SkipPhases", "JoinConfiguration.SkipPhases" to allow skipping a list of phases during kubeadm init/join command execution.
- Add "InitConfiguration.NodeRegistration.ImagePullPolicy" and "JoinConfiguration.NodeRegistration.ImagePullPolicy" to allow specifying the images pull policy during kubeadm "init" and "join". The value must be one of "Always", "Never" or "IfNotPresent". "IfNotPresent" is the default, which has been the existing behavior prior to this addition.
- Add "InitConfiguration.Patches.Directory", "JoinConfiguration.Patches.Directory" to allow the user to configure a directory from which to take patches for components deployed by kubeadm.
- Move the BootstrapToken\* API and related utilities out of the "kubeadm" API group to a new group "bootstraptoken". The kubeadm API version v1beta3 no longer contains the BootstrapToken\* structures.

Migration from old kubeadm config versions

- kubeadm v1.15.x and newer can be used to migrate from v1beta1 to v1beta2.
- kubeadm v1.22.x and newer no longer support v1beta1 and older APIs, but can be used to migrate v1beta2 to v1beta3.
- kubeadm v1.27.x and newer no longer support v1beta2 and older APIs,

## Basics

The preferred way to configure kubeadm is to pass an YAML configuration file with the `--config` option. Some of the configuration options defined in the kubeadm config file are also available as command line flags, but only the most common/simple use case are supported with this approach.

A kubeadm config file could contain multiple configuration types separated using three dashes (`---`).



kubeadm supports the following configuration types:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: InitConfiguration

apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration

apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration

apiVersion: kubeproxy.config.k8s.io/v1alpha1
kind: KubeProxyConfiguration

apiVersion: kubeadm.k8s.io/v1beta3
kind: JoinConfiguration
```

To print the defaults for "init" and "join" actions use the following commands:

```
kubeadm config print init-defaults
kubeadm config print join-defaults
```

The list of configuration types that must be included in a configuration file depends by the action you are performing (init or join) and by the configuration options you are going to use (defaults or advanced customization).

If some configuration types are not provided, or provided only partially, kubeadm will use default values; defaults provided by kubeadm includes also enforcing consistency of values across components when required (e.g. --cluster-cidr flag on controller manager and clusterCIDR on kube-proxy).

Users are always allowed to override default values, with the only exception of a small subset of setting with relevance for security (e.g. enforce authorization-mode Node and RBAC on api server).

If the user provides a configuration types that is not expected for the action you are performing, kubeadm will ignore those types and print a warning.

## Kubeadm init configuration types

When executing kubeadm init with the --config option, the following configuration types could be used: InitConfiguration, ClusterConfiguration, KubeProxyConfiguration, KubeletConfiguration, but only one between InitConfiguration and ClusterConfiguration is mandatory.

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: InitConfiguration
bootstrapTokens:
 ...
nodeRegistration:
 ...
```

The InitConfiguration type should be used to configure runtime settings, that in case of kubeadm init are the configuration of the bootstrap token and all the setting which are specific to the node where kubeadm is executed, including:

- NodeRegistration, that holds fields that relate to registering the new node to the cluster; use it to customize the node name, the CRI socket to use or any other settings that should apply to this node only (e.g. the node ip).
- LocalAPIEndpoint, that represents the endpoint of the instance of the API server to be deployed on this node; use it e.g. to customize the API server advertise address.

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
networking:
 ...
etcd:
 ...
apiServer:
 extraArgs:
 ...
 extraVolumes:
 ...
...
```

The ClusterConfiguration type should be used to configure cluster-wide settings, including settings for:

- networking that holds configuration for the networking topology of the cluster; use it e.g. to customize Pod subnet or services subnet.
- etcd: use it e.g. to customize the local etcd or to configure the API server for using an external etcd cluster.
- kube-apiserver, kube-scheduler, kube-controller-manager configurations; use it to customize control-plane components by adding customized setting or overriding kubeadm default settings.

```
apiVersion: kubeproxy.config.k8s.io/v1alpha1
kind: KubeProxyConfiguration
...
```

The KubeProxyConfiguration type should be used to change the configuration passed to kube-proxy instances deployed in the cluster. If this object is not provided or provided only partially, kubeadm applies defaults.

See <https://kubernetes.io/docs/reference/command-line-tools-reference/kube-proxy/> or <https://pkg.go.dev/k8s.io/kube-proxy/config/v1alpha1#KubeProxyConfiguration> for kube-proxy official documentation.

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
...
```

The KubeletConfiguration type should be used to change the configurations that will be passed to all kubelet instances deployed in the cluster. If this object is not provided or provided only partially, kubeadm applies defaults.

See <https://kubernetes.io/docs/reference/command-line-tools-reference/kubelet/> or <https://pkg.go.dev/k8s.io/kubelet/config/v1beta1#KubeletConfiguration> for kubelet official documentation.

Here is a fully populated example of a single YAML file containing multiple configuration types to be used during a kubeadm init run.

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: InitConfiguration
bootstrapTokens:
- token: "9a08jv.c0izixklcxtmnze7"
 description: "kubeadm bootstrap token"
 ttl: "24h"
- token: "783bde.3f89s0fje9f38fhf"
 description: "another bootstrap token"
 usages:
 - authentication
 - signing
 groups:
 - system:bootstrappers:kubeadm:default-node-token
nodeRegistration:
 name: "ec2-10-100-0-1"
 criSocket: "/var/run/dockershim.sock"
 taints:
 - key: "kubeadmNode"
 value: "someValue"
 effect: "NoSchedule"
kubeletExtraArgs:
 v: 4
ignorePreflightErrors:
- IsPrivilegedUser
imagePullPolicy: "IfNotPresent"
localAPIEndpoint:
 advertiseAddress: "10.100.0.1"
 bindPort: 6443
certificateKey: "e6a2eb8581237ab72a4f494f30285ec12a9694d750b9785706a83bfcbbbd2204"
skipPhases:
- addon/kube-proxy

apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
etcd:
 # one of local or external
 local:
 imageRepository: "registry.k8s.io"
 imageTag: "3.2.24"
 dataDir: "/var/lib/etcd"
 extraArgs:
 listen-client-urls: "http://10.100.0.1:2379"
```

```
serverCertSANs:
 - "ec2-10-100-0-1.compute-1.amazonaws.com"
peerCertSANs:
 - "10.100.0.1"
external:
endpoints:
- "10.100.0.1:2379"
- "10.100.0.2:2379"
caFile: "/etc/kubernetes/pki/etcd/etcd-ca.crt"
certFile: "/etc/kubernetes/pki/etcd/etcd.crt"
keyFile: "/etc/kubernetes/pki/etcd/etcd.key"
networking:
 serviceSubnet: "10.96.0.0/16"
 podSubnet: "10.244.0.0/24"
 dnsDomain: "cluster.local"
kubernetesVersion: "v1.21.0"
controlPlaneEndpoint: "10.100.0.1:6443"
apiServer:
 extraArgs:
 authorization-mode: "Node,RBAC"
 extraVolumes:
 - name: "some-volume"
 hostPath: "/etc/some-path"
 mountPath: "/etc/some-pod-path"
 readOnly: false
 pathType: File
 certSANs:
 - "10.100.1.1"
 - "ec2-10-100-0-1.compute-1.amazonaws.com"
 timeoutForControlPlane: 4m0s
controllerManager:
 extraArgs:
 "node-cidr-mask-size": "20"
 extraVolumes:
 - name: "some-volume"
 hostPath: "/etc/some-path"
 mountPath: "/etc/some-pod-path"
 readOnly: false
 pathType: File
scheduler:
 extraArgs:
 bind-address: "10.100.0.1"
 extraVolumes:
 - name: "some-volume"
 hostPath: "/etc/some-path"
 mountPath: "/etc/some-pod-path"
 readOnly: false
 pathType: File
certificatesDir: "/etc/kubernetes/pki"
imageRepository: "registry.k8s.io"
clusterName: "example-cluster"

apiVersion: kubelet.config.k8s.io/v1beta1
```

```
kind: KubeletConfiguration
kubelet specific options here

apiVersion: kubeproxy.config.k8s.io/v1alpha1
kind: KubeProxyConfiguration
kube-proxy specific options here
```

## Kubeadm join configuration types

When executing kubeadm join with the `--config` option, the JoinConfiguration type should be provided.

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: JoinConfiguration
...
```

The JoinConfiguration type should be used to configure runtime settings, that in case of kubeadm join are the discovery method used for accessing the cluster info and all the setting which are specific to the node where kubeadm is executed, including:

- `nodeRegistration`, that holds fields that relate to registering the new node to the cluster; use it to customize the node name, the CRI socket to use or any other settings that should apply to this node only (e.g. the node ip).
- `apiEndpoint`, that represents the endpoint of the instance of the API server to be eventually deployed on this node.

## Resource Types

- [ClusterConfiguration](#)
- [InitConfiguration](#)
- [JoinConfiguration](#)

## BootstrapToken

Appears in:

- [InitConfiguration](#)

BootstrapToken describes one bootstrap token, stored as a Secret in the cluster

| Field                                                             | Description                                                                                                                           |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| token [ <b>Required</b> ]<br><a href="#">BootstrapTokenString</a> | token is used for establishing bidirectional trust between nodes and control-planes. Used for joining nodes in the cluster.           |
| description<br>string                                             | description sets a human-friendly message why this token exists and what it's used for, so other administrators can know its purpose. |

| Field                                   | Description                                                                                                                                                     |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ttl<br><a href="#">meta/v1.Duration</a> | ttl defines the time to live for this token. Defaults to 24h. expires and ttl are mutually exclusive.                                                           |
| expires<br><a href="#">meta/v1.Time</a> | expires specifies the timestamp when this token expires. Defaults to being set dynamically at runtime based on the ttl. expires and ttl are mutually exclusive. |
| usages<br>[]string                      | usages describes the ways in which this token can be used. Can by default be used for establishing bidirectional trust, but that can be changed here.           |
| groups<br>[]string                      | groups specifies the extra groups that this token will authenticate as when/if used for authentication                                                          |

## BootstrapTokenString

Appears in:

- [BootstrapToken](#)

BootstrapTokenString is a token of the format abcdef.abcdef0123456789 that is used for both validation of the practicality of the API server from a joining node's point of view and as an authentication method for the node in the bootstrap phase of "kubeadm join". This token is and should be short-lived.

| Field                  | Description              |
|------------------------|--------------------------|
| - [Required]<br>string | No description provided. |
| - [Required]<br>string | No description provided. |

## ClusterConfiguration

ClusterConfiguration contains cluster-wide configuration for a kubeadm cluster.

| Field                                    | Description                            |
|------------------------------------------|----------------------------------------|
| apiVersion<br>string                     | kubeadm.k8s.io/v1beta3                 |
| kind<br>string                           | ClusterConfiguration                   |
| etcd<br><a href="#">Etcd</a>             | etcd holds the configuration for etcd. |
| networking<br><a href="#">Networking</a> |                                        |

| Field                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | networking holds configuration for the networking topology of the cluster.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| kubernetesVersion<br>string                                | kubernetesVersion is the target version of the control plane.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| controlPlaneEndpoint<br>string                             | <p>controlPlaneEndpoint sets a stable IP address or DNS name for the control plane. It can be a valid IP address or a RFC-1123 DNS subdomain, both with optional TCP port. In case the controlPlaneEndpoint is not specified, the advertiseAddress + bindPort are used; in case the controlPlaneEndpoint is specified but without a TCP port, the bindPort is used. Possible usages are:</p> <ul style="list-style-type: none"> <li>• In a cluster with more than one control plane instances, this field should be assigned the address of the external load balancer in front of the control plane instances.</li> <li>• In environments with enforced node recycling, the controlPlaneEndpoint could be used for assigning a stable DNS to the control plane.</li> </ul> |
| apiServer<br><a href="#">APIServer</a>                     | apiServer contains extra settings for the API server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| controllerManager<br><a href="#">ControlPlaneComponent</a> | controllerManager contains extra settings for the controller manager.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| scheduler<br><a href="#">ControlPlaneComponent</a>         | scheduler contains extra settings for the scheduler.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| dns<br><a href="#">DNS</a>                                 | dns defines the options for the DNS add-on installed in the cluster.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| certificatesDir<br>string                                  | certificatesDir specifies where to store or look for all required certificates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| imageRepository<br>string                                  | imageRepository sets the container registry to pull images from. If empty, registry.k8s.io will be used by default. In case of kubernetes version is a CI build (kubernetes version starts with ci/) gcr.io/k8s-staging-ci-images will be used as a default for control plane components and for kube-proxy, while registry.k8s.io will be used for all the other images.                                                                                                                                                                                                                                                                                                                                                                                                   |
| featureGates<br>map[string]bool                            | featureGates contains the feature gates enabled by the user.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| clusterName<br>string                                      | The cluster name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## InitConfiguration

InitConfiguration contains a list of elements that is specific "kubeadm init"-only runtime information. kubeadm init-only information. These fields are solely used the first time kubeadm init runs. After that, the information in the fields IS NOT uploaded to the kubeadm-config ConfigMap that is used by kubeadm upgrade for instance. These fields must be empty.

| Field                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                        | kubeadm.k8s.io/v1beta3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| kind<br>string                                              | InitConfiguration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| bootstrapTokens<br><a href="#">[]BootstrapToken</a>         | bootstrapTokens is respected at kubeadm init time and describes a set of Bootstrap Tokens to create. This information IS NOT uploaded to the kubeadm cluster configmap, partly because of its sensitive nature                                                                                                                                                                                                                                                                                                                                                                                                                     |
| nodeRegistration<br><a href="#">NodeRegistrationOptions</a> | nodeRegistration holds fields that relate to registering the new control-plane node to the cluster.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| localAPIEndpoint<br><a href="#">APIEndpoint</a>             | localAPIEndpoint represents the endpoint of the API server instance that's deployed on this control plane node. In HA setups, this differs from ClusterConfiguration.controlPlaneEndpoint in the sense that controlPlaneEndpoint is the global endpoint for the cluster, which then load-balances the requests to each individual API server. This configuration object lets you customize what IP/DNS name and port the local API server advertises it's accessible on. By default, kubeadm tries to auto-detect the IP of the default interface and use that, but in case that process fails you may set the desired value here. |
| certificateKey<br>string                                    | certificateKey sets the key with which certificates and keys are encrypted prior to being uploaded in a Secret in the cluster during the uploadcerts init phase. The certificate key is a hex encoded string that is an AES key of size 32 bytes.                                                                                                                                                                                                                                                                                                                                                                                  |
| skipPhases<br><a href="#">[]string</a>                      | skipPhases is a list of phases to skip during command execution. The list of phases can be obtained with the kubeadm init --help command. The flag "--skip-phases" takes precedence over this field.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| patches<br><a href="#">Patches</a>                          | patches contains options related to applying patches to components deployed by kubeadm during kubeadm init.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## JoinConfiguration

JoinConfiguration contains elements describing a particular node.



| Field                                                       | Description                                                                                                                                                                                        |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                        | kubeadm.k8s.io/v1beta3                                                                                                                                                                             |
| kind<br>string                                              | JoinConfiguration                                                                                                                                                                                  |
| nodeRegistration<br><a href="#">NodeRegistrationOptions</a> | nodeRegistration holds fields that relate to registering the new control-plane node to the cluster.                                                                                                |
| caCertPath<br>string                                        | caCertPath is the path to the SSL certificate authority used to secure communications between a node and the control-plane. Defaults to "/etc/kubernetes/pki/ca.crt".                              |
| discovery <b>[Required]</b><br><a href="#">Discovery</a>    | discovery specifies the options for the kubelet to use during the TLS bootstrap process.                                                                                                           |
| controlPlane<br><a href="#">JoinControlPlane</a>            | controlPlane defines the additional control plane instance to be deployed on the joining node. If nil, no additional control plane instance will be deployed.                                      |
| skipPhases<br>[]string                                      | skipPhases is a list of phases to skip during command execution. The list of phases can be obtained with the kubeadm join --help command. The flag --skip-phases takes precedence over this field. |
| patches<br><a href="#">Patches</a>                          | patches contains options related to applying patches to components deployed by kubeadm during kubeadm join.                                                                                        |

## APIEndpoint

Appears in:

- [InitConfiguration](#)
- [JoinControlPlane](#)

APIEndpoint struct contains elements of API server instance deployed on a node.

| Field                      | Description                                                                    |
|----------------------------|--------------------------------------------------------------------------------|
| advertiseAddress<br>string | advertiseAddress sets the IP address for the API server to advertise.          |
| bindPort<br>int32          | bindPort sets the secure port for the API Server to bind to. Defaults to 6443. |

## APIServer

Appears in:

- [ClusterConfiguration](#)

APIServer holds settings necessary for API server deployments in the cluster

| Field                                                                                 | Description                                                                                  |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| ControlPlaneComponent<br>[ <b>Required</b> ]<br><a href="#">ControlPlaneComponent</a> | (Members of ControlPlaneComponent are embedded into this type.) No description provided.     |
| certSANs<br>[]string                                                                  | certSANs sets extra Subject Alternative Names (SANs) for the API Server signing certificate. |
| timeoutForControlPlane<br><a href="#">meta/v1.Duration</a>                            | timeoutForControlPlane controls the timeout that we wait for API server to appear.           |

## BootstrapTokenDiscovery

Appears in:

- [Discovery](#)

BootstrapTokenDiscovery is used to set the options for bootstrap token based discovery.

| Field                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| token [ <b>Required</b> ]<br>string | token is a token used to validate cluster information fetched from the control-plane.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| apiServerEndpoint<br>string         | apiServerEndpoint is an IP or domain name to the API server from which information will be fetched.                                                                                                                                                                                                                                                                                                                                                                                                      |
| caCertHashes<br>[]string            | caCertHashes specifies a set of public key pins to verify when token-based discovery is used. The root CA found during discovery must match one of these values. Specifying an empty set disables root CA pinning, which can be unsafe. Each hash is specified as <type>:<value>, where the only currently supported type is "sha256". This is a hex-encoded SHA-256 hash of the Subject Public Key Info (SPKI) object in DER-encoded ASN.1. These hashes can be calculated using, for example, OpenSSL. |

| Field                            | Description                                                                                                                                                                                  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| unsafeSkipCAVerification<br>bool | unsafeSkipCAVerification allows token-based discovery without CA verification via caCertHashes. This can weaken the security of kubeadm since other nodes can impersonate the control-plane. |

## ControlPlaneComponent

Appears in:

- [ClusterConfiguration](#)
- [APIServer](#)

ControlPlaneComponent holds settings common to control plane component of the cluster

| Field                                           | Description                                                                                                                                                                      |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extraArgs<br>map[string]string                  | extraArgs is an extra set of flags to pass to the control plane component. A key in this map is the flag name as it appears on the command line except without leading dash(es). |
| extraVolumes<br><a href="#">[]HostPathMount</a> | extraVolumes is an extra set of host volumes, mounted to the control plane component.                                                                                            |

## DNS

Appears in:

- [ClusterConfiguration](#)

DNS defines the DNS addon that should be used in the cluster

| Field                                                | Description                                                                                                                    |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| ImageMeta<br>[Required]<br><a href="#">ImageMeta</a> | (Members of ImageMeta are embedded into this type.)<br><br>imageMeta allows to customize the image used for the DNS component. |

## Discovery

Appears in:

- [JoinConfiguration](#)

Discovery specifies the options for the kubelet to use during the TLS Bootstrap process.

| Field                                                     | Description                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bootstrapToken<br><a href="#">BootstrapTokenDiscovery</a> | bootstrapToken is used to set the options for bootstrap token based discovery. bootstrapToken and file are mutually exclusive.                                                                                                                                                             |
| file<br><a href="#">FileDiscovery</a>                     | file is used to specify a file or URL to a kubeconfig file from which to load cluster information. bootstrapToken and file are mutually exclusive.                                                                                                                                         |
| tlsBootstrapToken<br>string                               | tlsBootstrapToken is a token used for TLS bootstrapping. If bootstrapToken is set, this field is defaulted to .bootstrapToken.token, but can be overridden. If file is set, this field <b>must be set</b> in case the KubeConfigFile does not contain any other authentication information |
| timeout<br><a href="#">meta/v1.Duration</a>               | timeout modifies the discovery timeout.                                                                                                                                                                                                                                                    |

## Etcd

Appears in:

- [ClusterConfiguration](#)

Etcd contains elements describing Etcd configuration.

| Field                                    | Description                                                                                                            |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| local<br><a href="#">LocalEtcd</a>       | local provides configuration knobs for configuring the local etcd instance. local and external are mutually exclusive. |
| external<br><a href="#">ExternalEtcd</a> | external describes how to connect to an external etcd cluster. local and external are mutually exclusive.              |

## ExternalEtcd

Appears in:

- [Etcd](#)

ExternalEtcd describes an external etcd cluster. Kubeadm has no knowledge of where certificate files live and they must be supplied.

| Field                               | Description                                  |
|-------------------------------------|----------------------------------------------|
| endpoints<br>[Required]<br>[]string | endpoints contains the list of etcd members. |

| Field                            | Description                                                                                                             |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| caFile<br>[Required]<br>string   | caFile is an SSL Certificate Authority (CA) file used to secure etcd communication. Required if using a TLS connection. |
| certFile<br>[Required]<br>string | certFile is an SSL certification file used to secure etcd communication. Required if using a TLS connection.            |
| keyFile<br>[Required]<br>string  | keyFile is an SSL key file used to secure etcd communication. Required if using a TLS connection.                       |

## FileDiscovery

Appears in:

- [Discovery](#)

FileDiscovery is used to specify a file or URL to a kubeconfig file from which to load cluster information.

| Field                                  | Description                                                                                                                  |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| kubeConfigPath<br>[Required]<br>string | kubeConfigPath is used to specify the actual file path or URL to the kubeconfig file from which to load cluster information. |

## HostPathMount

Appears in:

- [ControlPlaneComponent](#)

HostPathMount contains elements describing volumes that are mounted from the host.

| Field                          | Description                                                           |
|--------------------------------|-----------------------------------------------------------------------|
| name [Required]<br>string      | name is the name of the volume inside the Pod template.               |
| hostPath [Required]<br>string  | hostPath is the path in the host that will be mounted inside the Pod. |
| mountPath [Required]<br>string | mountPath is the path inside the Pod where hostPath will be mounted.  |
| readOnly<br>bool               | readOnly controls write access to the volume.                         |

| Field                                            | Description                           |
|--------------------------------------------------|---------------------------------------|
| pathType<br><a href="#">core/v1.HostPathType</a> | pathType is the type of the hostPath. |

## ImageMeta

Appears in:

- [DNS](#)
- [LocalEtcd](#)

ImageMeta allows to customize the image used for components that are not originated from the Kubernetes/Kubernetes release process

| Field                     | Description                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| imageRepository<br>string | imageRepository sets the container registry to pull images from. If not set, the imageRepository defined in ClusterConfiguration will be used instead.                |
| imageTag<br>string        | imageTag allows to specify a tag for the image. In case this value is set, kubeadm does not change automatically the version of the above components during upgrades. |

## JoinControlPlane

Appears in:

- [JoinConfiguration](#)

JoinControlPlane contains elements describing an additional control plane instance to be deployed on the joining node.

| Field                                           | Description                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| localAPIEndpoint<br><a href="#">APIEndpoint</a> | localAPIEndpoint represents the endpoint of the API server instance to be deployed on this node.                                                                                                                                                                                                      |
| certificateKey<br>string                        | certificateKey is the key that is used for decryption of certificates after they are downloaded from the secret upon joining a new control plane node. The corresponding encryption key is in the InitConfiguration. The certificate key is a hex encoded string that is an AES key of size 32 bytes. |

## LocalEtcd

Appears in:

- [Etcd](#)

LocalEtcd describes that kubeadm should run an etcd cluster locally.

| Field                                                | Description                                                                                                                                                                                   |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ImageMeta<br>[Required]<br><a href="#">ImageMeta</a> | (Members of ImageMeta are embedded into this type.)<br><br>ImageMeta allows to customize the container used for etcd.                                                                         |
| dataDir<br>[Required]<br>string                      | dataDir is the directory etcd will place its data. Defaults to "/var/lib/etcd".                                                                                                               |
| extraArgs<br>map[string]string                       | extraArgs are extra arguments provided to the etcd binary when run inside a static Pod. A key in this map is the flag name as it appears on the command line except without leading dash(es). |
| serverCertSANs<br>[]string                           | serverCertSANs sets extra Subject Alternative Names (SANs) for the etcd server signing certificate.                                                                                           |
| peerCertSANs<br>[]string                             | peerCertSANs sets extra Subject Alternative Names (SANs) for the etcd peer signing certificate.                                                                                               |

## Networking

Appears in:

- [ClusterConfiguration](#)

Networking contains elements describing cluster's networking configuration.

| Field                   | Description                                                                           |
|-------------------------|---------------------------------------------------------------------------------------|
| serviceSubnet<br>string | serviceSubnet is the subnet used by Kubernetes Services. Defaults to "10.96.0.0/12".  |
| podSubnet<br>string     | podSubnet is the subnet used by Pods.                                                 |
| dnsDomain<br>string     | dnsDomain is the DNS domain used by Kubernetes Services. Defaults to "cluster.local". |

# NodeRegistrationOptions

Appears in:

- [InitConfiguration](#)
- [JoinConfiguration](#)

NodeRegistrationOptions holds fields that relate to registering a new control-plane or node to the cluster, either via kubeadm init or kubeadm join.

| Field                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name<br>string                                              | name is the .metadata.name field of the Node API object that will be created in this kubeadm init or kubeadm join operation. This field is also used in the CommonName field of the kubelet's client certificate to the API server. Defaults to the hostname of the node if not provided.                                                                                                                                                                                                                            |
| criSocket<br>string                                         | criSocket is used to retrieve container runtime info. This information will be annotated to the Node API object, for later re-use.                                                                                                                                                                                                                                                                                                                                                                                   |
| taints <b>[Required]</b><br><a href="#">[]core/v1.Taint</a> | taints specifies the taints the Node API object should be registered with. If this field is unset, i.e. nil, it will be defaulted with a control-plane taint for control-plane nodes. If you don't want to taint your control-plane node, set this field to an empty list, i.e. taints: [] in the YAML file. This field is solely used for Node registration.                                                                                                                                                        |
| kubeletExtraArgs<br>map[string]string                       | kubeletExtraArgs passes through extra arguments to the kubelet. The arguments here are passed to the kubelet command line via the environment file kubeadm writes at runtime for the kubelet to source. This overrides the generic base-level configuration in the kubelet-config ConfigMap. Flags have higher priority when parsing. These values are local and specific to the node kubeadm is executing on. A key in this map is the flag name as it appears on the command line except without leading dash(es). |
| ignorePreflightErrors<br>[]string                           | ignorePreflightErrors provides a list of pre-flight errors to be ignored when the current node is registered, e.g. IsPrivilegedUser,Swap. Value all ignores errors from all checks.                                                                                                                                                                                                                                                                                                                                  |
| imagePullPolicy<br><a href="#">core/v1.PullPolicy</a>       | imagePullPolicy specifies the policy for image pulling during kubeadm "init" and "join" operations. The value of this field must be one of "Always", "IfNotPresent" or "Never". If this field is not set, kubeadm will default it to "IfNotPresent", or pull the required images if not present on the host.                                                                                                                                                                                                         |



# Patches

Appears in:

- [InitConfiguration](#)
- [JoinConfiguration](#)

Patches contains options related to applying patches to components deployed by kubeadm.

| Field            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| directory string | directory is a path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd". "patchtype" can be one of "strategic" "merge" or "json" and they match the patch formats supported by kubectl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

## kubeadm Configuration (v1beta4)

### Overview

Package v1beta4 defines the v1beta4 version of the kubeadm configuration file format. This version improves on the v1beta3 format by fixing some minor issues and adding a few new fields.

A list of changes since v1beta3:

- TODO <https://github.com/kubernetes/kubeadm/issues/2890>
- Support custom environment variables in control plane components under ClusterConfiguration. Use APIServer.ExtraEnvs, ControllerManager.ExtraEnvs, Scheduler.ExtraEnvs, EtcLocal.ExtraEnvs.
- The ResetConfiguration API type is now supported in v1beta4. Users are able to reset a node by passing a --config file to kubeadm reset.
- dry-run mode is now configureable in InitConfiguration and JoinConfiguration config files.
- Replace the existing string/string extra argument maps with structured extra arguments that support duplicates. The change applies to ClusterConfiguration - APIServer.ExtraArgs, ControllerManager.ExtraArgs, Scheduler.ExtraArgs. Also to NodeRegistrationOptions.KubeletExtraArgs.
- Add ClusterConfiguration.EncryptionAlgorithm that can be used to set the asymmetric encryption algorithm used for this cluster's keys and certificates. Can be one of "RSA-2048" (default), "RSA-3072", "RSA-4096" or "ECDSA-P256".
- Add ClusterConfiguration.DNS.Disabled and ClusterConfiguration.Proxy.Disabled that can be used to disable the CoreDNS and kube-proxy addons during cluster initialization. Skipping the related addons phases, during cluster creation will set the same fields to false.

- Add the `NodeRegistration.ImagePullSerial` field in `'InitConfigurationandJoinConfiguration'`, which can be used to control if kubeadm pulls images serially or in parallel.
- The `UpgradeConfiguration` kubeadm API is now supported in `v1beta4` when passing `--config` to kubeadm upgrade subcommands. Usage of component configuration for kubelet and kube-proxy, `InitConfiguration` and `ClusterConfiguration` is deprecated and will be ignored when passing `--config` to upgrade subcommands.
- Add a `Timeouts` structure to `InitConfiguration`, `JoinConfiguration`, `ResetConfiguration` and `UpgradeConfiguration` that can be used to configure various timeouts.

## Migration from old kubeadm config versions

- kubeadm `v1.15.x` and newer can be used to migrate from `v1beta1` to `v1beta2`.
- kubeadm `v1.22.x` and newer no longer support `v1beta1` and older APIs, but can be used to migrate `v1beta2` to `v1beta3`.
- kubeadm `v1.27.x` and newer no longer support `v1beta2` and older APIs.
- TODO: <https://github.com/kubernetes/kubeadm/issues/2890> add version that can be used to convert to `v1beta4`

## Basics

The preferred way to configure kubeadm is to pass an YAML configuration file with the `'--config'` option. Some of the configuration options defined in the kubeadm config file are also available as command line flags, but only the most common/simple use case are supported with this approach.

A kubeadm config file could contain multiple configuration types separated using three dashes (`---`).

kubeadm supports the following configuration types:

```
apiVersion: kubeadm.k8s.io/v1beta4
kind: InitConfiguration
```

```
apiVersion: kubeadm.k8s.io/v1beta4
kind: ClusterConfiguration
```

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
```

```
apiVersion: kubeproxy.config.k8s.io/v1alpha1
kind: KubeProxyConfiguration
```

```
apiVersion: kubeadm.k8s.io/v1beta4
kind: JoinConfiguration
```

To print the defaults for "init" and "join" actions use the following commands:

```
kubeadm config print init-defaults
kubeadm config print join-defaults
```

The list of configuration types that must be included in a configuration file depends by the action you are performing (init or join) and by the configuration options you are going to use (defaults or advanced customization).

If some configuration types are not provided, or provided only partially, kubeadm will use default values; defaults provided by kubeadm includes also enforcing consistency of values across components when required (e.g. `--cluster-cidr` flag on controller manager and `clusterCIDR` on kube-proxy).

Users are always allowed to override default values, with the only exception of a small subset of setting with relevance for security (e.g. `enforce authorization-mode` Node and RBAC on api server).

If the user provides a configuration types that is not expected for the action you are performing, kubeadm will ignore those types and print a warning.

## Kubeadm init configuration types

When executing kubeadm init with the `--config` option, the following configuration types could be used: `InitConfiguration`, `ClusterConfiguration`, `KubeProxyConfiguration`, `KubeletConfiguration`, but only one between `InitConfiguration` and `ClusterConfiguration` is mandatory.

```
apiVersion: kubeadm.k8s.io/v1beta4
kind: InitConfiguration
bootstrapTokens:
```

```
...
```

```
nodeRegistration:
```

```
...
```

The `InitConfiguration` type should be used to configure runtime settings, that in case of kubeadm init are the configuration of the bootstrap token and all the setting which are specific to the node where kubeadm is executed, including:

- `NodeRegistration`, that holds fields that relate to registering the new node to the cluster; use it to customize the node name, the CRI socket to use or any other settings that should apply to this node only (e.g. the node ip).
- `LocalAPIEndpoint`, that represents the endpoint of the instance of the API server to be deployed on this node; use it e.g. to customize the API server advertise address.

```
apiVersion: kubeadm.k8s.io/v1beta4
kind: ClusterConfiguration
networking:
```

```
...
```

```
etcd:
```

```
...
```

```
apiServer:
```

```
extraArgs:
```

```
...
```

```
extraVolumes:
```

```
...
```

```
...
```

The ClusterConfiguration type should be used to configure cluster-wide settings, including settings for:

- networking that holds configuration for the networking topology of the cluster; use it e.g. to customize Pod subnet or services subnet.
- etcd: use it e.g. to customize the local etcd or to configure the API server for using an external etcd cluster.
- kube-apiserver, kube-scheduler, kube-controller-manager configurations; use it to customize control-plane components by adding customized setting or overriding kubeadm default settings.

```
apiVersion: kubeproxy.config.k8s.io/v1alpha1
kind: KubeProxyConfiguration
```

```
...
```

The KubeProxyConfiguration type should be used to change the configuration passed to kube-proxy instances deployed in the cluster. If this object is not provided or provided only partially, kubeadm applies defaults.

See <https://kubernetes.io/docs/reference/command-line-tools-reference/kube-proxy/> or <https://pkg.go.dev/k8s.io/kube-proxy/config/v1alpha1#KubeProxyConfiguration> for kube-proxy official documentation.

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
```

```
...
```

The KubeletConfiguration type should be used to change the configurations that will be passed to all kubelet instances deployed in the cluster. If this object is not provided or provided only partially, kubeadm applies defaults.

See <https://kubernetes.io/docs/reference/command-line-tools-reference/kubelet/> or <https://pkg.go.dev/k8s.io/kubelet/config/v1beta1#KubeletConfiguration> for kubelet official documentation.

Here is a fully populated example of a single YAML file containing multiple configuration types to be used during a kubeadm init run.

```
apiVersion: kubeadm.k8s.io/v1beta4
kind: InitConfiguration
bootstrapTokens:
- token: "9a08jv.c0izixklcxtmnze7"
 description: "kubeadm bootstrap token"
 ttl: "24h"
- token: "783bde.3f89s0fje9f38fhf"
 description: "another bootstrap token"
 usages:
 - authentication
 - signing
 groups:
 - system:bootstrappers:kubeadm:default-node-token
nodeRegistration:
 name: "ec2-10-100-0-1"
 criSocket: "unix:///var/run/containerd/containerd.sock"
 taints:
 - key: "kubeadmNode"
 value: "someValue"
 effect: "NoSchedule"
 kubeletExtraArgs:
 v: 4
 ignorePreflightErrors:
 - IsPrivilegedUser
 imagePullPolicy: "IfNotPresent"
localAPIEndpoint:
 advertiseAddress: "10.100.0.1"
 bindPort: 6443
certificateKey: "e6a2eb8581237ab72a4f494f30285ec12a9694d750b9785706a83bfcbbbd2204"
skipPhases:
- addon/kube-proxy

apiVersion: kubeadm.k8s.io/v1beta4
kind: ClusterConfiguration
etcd:
```

```
one of local or external
local:
 imageRepository: "registry.k8s.io"
 imageTag: "3.2.24"
 dataDir: "/var/lib/etcd"
 extraArgs:
 listen-client-urls: "http://10.100.0.1:2379"
 serverCertSANs:
 - "ec2-10-100-0-1.compute-1.amazonaws.com"
 peerCertSANs:
 - "10.100.0.1"
external:
endpoints:
- "10.100.0.1:2379"
- "10.100.0.2:2379"
caFile: "/etc/kubernetes/pki/etcd/etcd-ca.crt"
certFile: "/etc/kubernetes/pki/etcd/etcd.crt"
keyFile: "/etc/kubernetes/pki/etcd/etcd.key"
```

networking:

```
serviceSubnet: "10.96.0.0/16"
podSubnet: "10.244.0.0/24"
dnsDomain: "cluster.local"
```

```
kubernetesVersion: "v1.21.0"
controlPlaneEndpoint: "10.100.0.1:6443"
apiServer:
```

```
 extraArgs:
 authorization-mode: "Node,RBAC"
 extraVolumes:
 - name: "some-volume"
 hostPath: "/etc/some-path"
 mountPath: "/etc/some-pod-path"
 readOnly: false
 pathType: File
 certSANs:
 - "10.100.1.1"
 - "ec2-10-100-0-1.compute-1.amazonaws.com"
 timeoutForControlPlane: 4m0s
```

controllerManager:

```
 extraArgs:
 "node-cidr-mask-size": "20"
 extraVolumes:
 - name: "some-volume"
 hostPath: "/etc/some-path"
 mountPath: "/etc/some-pod-path"
 readOnly: false
 pathType: File
```

scheduler:

```
extraArgs:
 address: "10.100.0.1"
extraVolumes:
 - name: "some-volume"
 hostPath: "/etc/some-path"
 mountPath: "/etc/some-pod-path"
 readOnly: false
 pathType: File
```

```
certificatesDir: "/etc/kubernetes/pki"
imageRepository: "registry.k8s.io"
clusterName: "example-cluster"
```

---

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
kubelet specific options here
```

---

```
apiVersion: kubeproxy.config.k8s.io/v1alpha1
kind: KubeProxyConfiguration
kube-proxy specific options here
```

## Kubeadm join configuration types

When executing kubeadm join with the --config option, the JoinConfiguration type should be provided.

```
apiVersion: kubeadm.k8s.io/v1beta4
kind: JoinConfiguration
```

...

The JoinConfiguration type should be used to configure runtime settings, that in case of kubeadm join are the discovery method used for accessing the cluster info and all the setting which are specific to the node where kubeadm is executed, including:

- `nodeRegistration`, that holds fields that relate to registering the new node to the cluster; use it to customize the node name, the CRI socket to use or any other settings that should apply to this node only (e.g. the node ip).
- `apiEndpoint`, that represents the endpoint of the instance of the API server to be eventually deployed on this node.

## Resource Types

- [ClusterConfiguration](#)
- [InitConfiguration](#)
- [JoinConfiguration](#)
- [ResetConfiguration](#)
- [UpgradeConfiguration](#)

# BootstrapToken

Appears in:

- [InitConfiguration](#)
- [InitConfiguration](#)

BootstrapToken describes one bootstrap token, stored as a Secret in the cluster

| Field                                                           | Description                                                                                                                                                     |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| token <b>[Required]</b><br><a href="#">BootstrapTokenString</a> | token is used for establishing bidirectional trust between nodes and control-planes. Used for joining nodes in the cluster.                                     |
| description<br>string                                           | description sets a human-friendly message why this token exists and what it's used for, so other administrators can know its purpose.                           |
| ttl<br><a href="#">meta/v1.Duration</a>                         | ttl defines the time to live for this token. Defaults to 24h. expires and ttl are mutually exclusive.                                                           |
| expires<br><a href="#">meta/v1.Time</a>                         | expires specifies the timestamp when this token expires. Defaults to being set dynamically at runtime based on the ttl. expires and ttl are mutually exclusive. |
| usages<br>[]string                                              | usages describes the ways in which this token can be used. Can by default be used for establishing bidirectional trust, but that can be changed here.           |
| groups<br>[]string                                              | groups specifies the extra groups that this token will authenticate as when/if used for authentication                                                          |

# BootstrapTokenString

Appears in:

- [BootstrapToken](#)

BootstrapTokenString is a token of the format abcdef.abcdef0123456789 that is used for both validation of the practically of the API server from a joining node's point of view and as an authentication method for the node in the bootstrap phase of "kubeadm join". This token is and should be short-lived.

| Field                         | Description              |
|-------------------------------|--------------------------|
| - <b>[Required]</b><br>string | No description provided. |



| Field                  | Description              |
|------------------------|--------------------------|
| - [Required]<br>string | No description provided. |

## ClusterConfiguration

ClusterConfiguration contains cluster-wide configuration for a kubeadm cluster.

| Field                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                       | kubeadm.k8s.io/v1beta4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| kind<br>string                                             | ClusterConfiguration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| etcd<br><a href="#">Etcd</a>                               | etcd holds the configuration for etcd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| networking<br><a href="#">Networking</a>                   | networking holds configuration for the networking topology of the cluster.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| kubernetesVersion<br>string                                | kubernetesVersion is the target version of the control plane.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| controlPlaneEndpoint<br>string                             | controlPlaneEndpoint sets a stable IP address or DNS name for the control plane; It can be a valid IP address or a RFC-1123 DNS subdomain, both with optional TCP port. In case the controlPlaneEndpoint is not specified, the advertiseAddress + bindPort are used; in case the controlPlaneEndpoint is specified but without a TCP port, the bindPort is used. Possible usages are: <ul style="list-style-type: none"> <li>• In a cluster with more than one control plane instances, this field should be assigned the address of the external load balancer in front of the control plane instances.</li> <li>• In environments with enforced node recycling, the controlPlaneEndpoint could be used for assigning a stable DNS to the control plane.</li> </ul> |
| apiServer<br><a href="#">APIServer</a>                     | apiServer contains extra settings for the API server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| controllerManager<br><a href="#">ControlPlaneComponent</a> | controllerManager contains extra settings for the controller manager.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| scheduler<br><a href="#">ControlPlaneComponent</a>         | scheduler contains extra settings for the scheduler.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| dns<br><a href="#">DNS</a>                                 | dns defines the options for the DNS add-on installed in the cluster.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Field                                                          | Description                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proxy <b>[Required]</b><br><a href="#">Proxy</a>               | proxy defines the options for the proxy add-on installed in the cluster.                                                                                                                                                                                                                                                                                                  |
| certificatesDir<br>string                                      | certificatesDir specifies where to store or look for all required certificates.                                                                                                                                                                                                                                                                                           |
| imageRepository<br>string                                      | imageRepository sets the container registry to pull images from. If empty, registry.k8s.io will be used by default. In case of kubernetes version is a CI build (kubernetes version starts with ci/) gcr.io/k8s-staging-ci-images will be used as a default for control plane components and for kube-proxy, while registry.k8s.io will be used for all the other images. |
| featureGates<br>map[string]bool                                | featureGates contains the feature gates enabled by the user.                                                                                                                                                                                                                                                                                                              |
| clusterName<br>string                                          | The cluster name.                                                                                                                                                                                                                                                                                                                                                         |
| encryptionAlgorithm<br><a href="#">EncryptionAlgorithmType</a> | encryptionAlgorithm holds the type of asymmetric encryption algorithm used for keys and certificates. Can be "RSA" (default algorithm, key size is 2048) or "ECDSA" (uses the P-256 elliptic curve).                                                                                                                                                                      |

## InitConfiguration

InitConfiguration contains a list of elements that is specific "kubeadm init"-only runtime information. kubeadm init-only information. These fields are solely used the first time kubeadm init runs. After that, the information in the fields IS NOT uploaded to the kubeadm-config ConfigMap that is used by kubeadm upgrade for instance. These fields must be omitted.

| Field                                               | Description                                                                                                                                                                                                    |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                | kubeadm.k8s.io/v1beta4                                                                                                                                                                                         |
| kind<br>string                                      | InitConfiguration                                                                                                                                                                                              |
| bootstrapTokens<br><a href="#">[]BootstrapToken</a> | bootstrapTokens is respected at kubeadm init time and describes a set of Bootstrap Tokens to create. This information IS NOT uploaded to the kubeadm cluster configmap, partly because of its sensitive nature |
| dryRun <b>[Required]</b><br>bool                    | dryRun tells if the dry run mode is enabled, don't apply any change in dry run mode, just output what would be done.                                                                                           |

| Field                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nodeRegistration<br><a href="#">NodeRegistrationOptions</a> | nodeRegistration holds fields that relate to registering the new control-plane node to the cluster.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| localAPIEndpoint<br><a href="#">APIEndpoint</a>             | localAPIEndpoint represents the endpoint of the API server instance that's deployed on this control plane node. In HA setups, this differs from ClusterConfiguration.controlPlaneEndpoint in the sense that controlPlaneEndpoint is the global endpoint for the cluster, which then loadbalances the requests to each individual API server. This configuration object lets you customize what IP/DNS name and port the local API server advertises it's accessible on. By default, kubeadm tries to auto-detect the IP of the default interface and use that, but in case that process fails you may set the desired value here. |
| certificateKey<br>string                                    | certificateKey sets the key with which certificates and keys are encrypted prior to being uploaded in a Secret in the cluster during the uploadcerts init phase. The certificate key is a hex encoded string that is an AES key of size 32 bytes.                                                                                                                                                                                                                                                                                                                                                                                 |
| skipPhases<br>[]string                                      | skipPhases is a list of phases to skip during command execution. The list of phases can be obtained with the kubeadm init --help command. The flag --skip-phases takes precedence over this field.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| patches<br><a href="#">Patches</a>                          | patches contains options related to applying patches to components deployed by kubeadm during kubeadm init.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## JoinConfiguration

JoinConfiguration contains elements describing a particular node.

| Field                                                       | Description                                                                                                       |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                        | kubeadm.k8s.io/v1beta4                                                                                            |
| kind<br>string                                              | JoinConfiguration                                                                                                 |
| dryRun<br>bool                                              | dryRun tells if the dry run mode is enabled, don't apply any change if it is set, just output what would be done. |
| nodeRegistration<br><a href="#">NodeRegistrationOptions</a> | nodeRegistration holds fields that relate to registering the new control-plane node to the cluster                |
| caCertPath<br>string                                        |                                                                                                                   |

| Field                                                    | Description                                                                                                                                                                                                                  |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | caCertPath is the path to the SSL certificate authority used to secure communications between node and control-plane. Defaults to <code>"/etc/kubernetes/pki/ca.crt"</code> .                                                |
| discovery <b>[Required]</b><br><a href="#">Discovery</a> | discovery specifies the options for the kubelet to use during the TLS bootstrap process.                                                                                                                                     |
| controlPlane<br><a href="#">JoinControlPlane</a>         | controlPlane defines the additional control plane instance to be deployed on the joining node. If nil, no additional control plane instance will be deployed.                                                                |
| skipPhases<br>[]string                                   | skipPhases is a list of phases to skip during command execution. The list of phases can be obtained with the <code>kubeadm join --help</code> command. The flag <code>--skip-phases</code> takes precedence over this field. |
| patches<br><a href="#">Patches</a>                       | patches contains options related to applying patches to components deployed by kubeadm during kubeadm join.                                                                                                                  |

## ResetConfiguration

ResetConfiguration contains a list of fields that are specifically kubeadm reset-only runtime information.

| Field                     | Description                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string      | kubeadm.k8s.io/v1beta4                                                                                                                                                                                                 |
| kind<br>string            | ResetConfiguration                                                                                                                                                                                                     |
| cleanupTmpDir<br>bool     | cleanupTmpDir specifies whether the <code>"/etc/kubernetes/tmp"</code> directory should be cleaned during the reset process.                                                                                           |
| certificatesDir<br>string | certificatesDir specifies the directory where the certificates are stored. If specified, it will be cleaned during the reset process.                                                                                  |
| criSocket<br>string       | criSocket is used to retrieve container runtime information and used for the removal of the containers. If criSocket is not specified by flag or config file, kubeadm will try to detect one valid CRI socket instead. |
| dryRun<br>bool            | dryRun tells if the dry run mode is enabled, don't apply any change if it is set and just output what would be done.                                                                                                   |

| Field                             | Description                                                                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| force<br>bool                     | The force flag instructs kubeadm to reset the node without prompting for confirmation.                                                                                   |
| ignorePreflightErrors<br>[]string | ignorePreflightErrors provides a list of pre-flight errors to be ignored during the reset process, e.g. IsPrivilegedUser,Swap. Value all ignores errors from all checks. |
| skipPhases<br>[]string            | skipPhases is a list of phases to skip during command execution. The list of phases can be obtained with the kubeadm reset phase --help command.                         |

## UpgradeConfiguration

UpgradeConfiguration contains a list of options that are specific to kubeadm upgrade subcommands.

| Field                                              | Description                                                                           |
|----------------------------------------------------|---------------------------------------------------------------------------------------|
| apiVersion<br>string                               | kubeadm.k8s.io/v1beta4                                                                |
| kind<br>string                                     | UpgradeConfiguration                                                                  |
| apply<br><a href="#">UpgradeApplyConfiguration</a> | apply holds a list of options that are specific to the kubeadm upgrade apply command. |
| diff<br><a href="#">UpgradeDiffConfiguration</a>   | diff holds a list of options that are specific to the kubeadm upgrade diff command.   |
| node<br><a href="#">UpgradeNodeConfiguration</a>   | node holds a list of options that are specific to the kubeadm upgrade node command.   |
| plan<br><a href="#">UpgradePlanConfiguration</a>   | plan holds a list of options that are specific to the kubeadm upgrade plan command.   |
| timeouts<br><a href="#">Timeouts</a>               | timeouts holds various timeouts that apply to kubeadm commands.                       |

## APIEndpoint

Appears in:

- [InitConfiguration](#)

- [JoinControlPlane](#)

APIEndpoint struct contains elements of API server instance deployed on a node.

| Field                      | Description                                                                    |
|----------------------------|--------------------------------------------------------------------------------|
| advertiseAddress<br>string | advertiseAddress sets the IP address for the API server to advertise.          |
| bindPort<br>int32          | bindPort sets the secure port for the API Server to bind to. Defaults to 6443. |

## APIServer

Appears in:

- [ClusterConfiguration](#)

APIServer holds settings necessary for API server deployments in the cluster

| Field                                                                                 | Description                                                                                  |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| ControlPlaneComponent<br>[ <b>Required</b> ]<br><a href="#">ControlPlaneComponent</a> | (Members of ControlPlaneComponent are embedded into this type.) No description provided.     |
| certSANs<br>[]string                                                                  | certSANs sets extra Subject Alternative Names (SANs) for the API Server signing certificate. |
| timeoutForControlPlane<br><a href="#">meta/v1.Duration</a>                            | timeoutForControlPlane controls the timeout that we use for API server to appear.            |

## Arg

Appears in:

- [ControlPlaneComponent](#)
- [LocalEtcd](#)
- [NodeRegistrationOptions](#)

Arg represents an argument with a name and a value.

| Field                               | Description                |
|-------------------------------------|----------------------------|
| name [ <b>Required</b> ]<br>string  | The name of the argument.  |
| value [ <b>Required</b> ]<br>string | The value of the argument. |

# BootstrapTokenDiscovery

Appears in:

- [Discovery](#)

BootstrapTokenDiscovery is used to set the options for bootstrap token based discovery.

| Field                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| token <b>[Required]</b><br>string | token is a token used to validate cluster information fetched from the control-plane.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| apiServerEndpoint<br>string       | apiServerEndpoint is an IP or domain name to the API server from which information will be fetched.                                                                                                                                                                                                                                                                                                                                                                                                         |
| caCertHashes<br>[]string          | caCertHashes specifies a set of public key pins to verify when token-based discovery is used. The root CA found during discovery must match one of these values. Specifying an empty set disables root CA pinning, which can be unsafe. Each hash is specified as <type>:<value>, where the only currently supported type is "sha256". This is a hex-encoded SHA-256 hash of the Subject Public Key Info (SPKI) object in DER-encoded ASN.1. These hashes can be // calculated using, for example, OpenSSL. |
| unsafeSkipCAVerification<br>bool  | unsafeSkipCAVerification allows token-based discovery without CA verification via caCertHashes. This can weaken the security of kubeadm since other nodes can impersonate the control-plane.                                                                                                                                                                                                                                                                                                                |

# ControlPlaneComponent

Appears in:

- [ClusterConfiguration](#)
- [APIServer](#)

ControlPlaneComponent holds settings common to control plane component of the cluster

| Field                                           | Description                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extraArgs<br><a href="#">[]Arg</a>              | extraArgs is an extra set of flags to pass to the control plane component. An argument name in this list is the flag name as it appears on the command line except without leading dash(es). Extra arguments will override existing default arguments. Duplicate extra arguments are allowed. |
| extraVolumes<br><a href="#">[]HostPathMount</a> |                                                                                                                                                                                                                                                                                               |

| Field                                 | Description                                                                                                                                                                                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | extraVolumes is an extra set of host volumes, mounted to the control plane component.                                                                                                                                                                    |
| extraEnvs<br><a href="#">[]EnvVar</a> | extraEnvs is an extra set of environment variables to pass to the control plane component. Environment variables passed using extraEnvs will override any existing environment variables, or *_proxy environment variables that kubeadm adds by default. |

## DNS

Appears in:

- [ClusterConfiguration](#)

DNS defines the DNS addon that should be used in the cluster

| Field                                                    | Description                                                                                                            |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| ImageMeta <b>[Required]</b><br><a href="#">ImageMeta</a> | (Members of ImageMeta are embedded into this type.)<br>imageMeta allows to customize the image used for the DNS addon. |
| disabled <b>[Required]</b><br>bool                       | disabled specifies whether to disable this addon in the cluster.                                                       |

## Discovery

Appears in:

- [JoinConfiguration](#)

Discovery specifies the options for the kubelet to use during the TLS Bootstrap process

| Field                                                     | Description                                                                                                                                                                                                          |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bootstrapToken<br><a href="#">BootstrapTokenDiscovery</a> | bootstrapToken is used to set the options for bootstrap token based discovery. bootstrapToken and file are mutually exclusive.                                                                                       |
| file<br><a href="#">FileDiscovery</a>                     | file is used to specify a file or URL to a kubeconfig file from which to load cluster information. bootstrapToken and file are mutually exclusive.                                                                   |
| tlsBootstrapToken<br>string                               | tlsBootstrapToken is a token used for TLS bootstrapping. If bootstrapToken is set, this field is defaulted to bootstrapToken.token, but can be overridden. If file is set, this field <b>must be set</b> in case the |



| Field                                       | Description                                                           |
|---------------------------------------------|-----------------------------------------------------------------------|
|                                             | KubeConfigFile does not contain any other authentication information. |
| timeout<br><a href="#">meta/v1.Duration</a> | timeout modifies the discovery timeout.                               |

## EncryptionAlgorithmType

(Alias of string)

Appears in:

- [ClusterConfiguration](#)

EncryptionAlgorithmType can define an asymmetric encryption algorithm type.

## EnvVar

Appears in:

- [ControlPlaneComponent](#)
- [LocalEtcd](#)

EnvVar represents an environment variable present in a Container.

| Field                                                  | Description                                                               |
|--------------------------------------------------------|---------------------------------------------------------------------------|
| EnvVar<br>[Required]<br><a href="#">core/v1.EnvVar</a> | (Members of EnvVar are embedded into this type.) No description provided. |

## Etcd

Appears in:

- [ClusterConfiguration](#)

Etcd contains elements describing Etcd configuration.

| Field                                    | Description                                                                                                            |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| local<br><a href="#">LocalEtcd</a>       | local provides configuration knobs for configuring the local etcd instance. local and external are mutually exclusive. |
| external<br><a href="#">ExternalEtcd</a> | external describes how to connect to an external etcd cluster. local and external are mutually exclusive.              |

## ExternalEtcd

Appears in:

- [Etcd](#)

ExternalEtcd describes an external etcd cluster. Kubeadm has no knowledge of where certificate files live and they must be supplied.

| Field                               | Description                                                                                                             |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| endpoints<br>[Required]<br>[]string | endpoints contains the list of etcd members.                                                                            |
| caFile<br>[Required]<br>string      | caFile is an SSL Certificate Authority (CA) file used to secure etcd communication. Required if using a TLS connection. |
| certFile<br>[Required]<br>string    | certFile is an SSL certification file used to secure etcd communication. Required if using a TLS connection.            |
| keyFile<br>[Required]<br>string     | keyFile is an SSL key file used to secure etcd communication. Required if using a TLS connection.                       |

## FileDiscovery

Appears in:

- [Discovery](#)

FileDiscovery is used to specify a file or URL to a kubeconfig file from which to load cluster information.

| Field                                  | Description                                                                                                                  |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| kubeConfigPath<br>[Required]<br>string | kubeConfigPath is used to specify the actual file path or URL to the kubeconfig file from which to load cluster information. |

## HostPathMount

Appears in:

- [ControlPlaneComponent](#)

HostPathMount contains elements describing volumes that are mounted from the host.

| Field                                            | Description                                                           |
|--------------------------------------------------|-----------------------------------------------------------------------|
| name <b>[Required]</b><br>string                 | name is the name of the volume inside the Pod template.               |
| hostPath <b>[Required]</b><br>string             | hostPath is the path in the host that will be mounted inside the Pod. |
| mountPath <b>[Required]</b><br>string            | mountPath is the path inside the Pod where hostPath will be mounted.  |
| readOnly<br>bool                                 | readOnly controls write access to the volume.                         |
| pathType<br><a href="#">core/v1.HostPathType</a> | pathType is the type of the hostPath.                                 |

## ImageMeta

Appears in:

- [DNS](#)
- [LocalEtcd](#)

ImageMeta allows to customize the image used for components that are not originated from the Kubernetes/Kubernetes release process

| Field                     | Description                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| imageRepository<br>string | imageRepository sets the container registry to pull images from. if not set, the imageRepository defined in ClusterConfiguration will be used instead.                |
| imageTag<br>string        | imageTag allows to specify a tag for the image. In case this value is set, kubeadm does not change automatically the version of the above components during upgrades. |

## JoinControlPlane

Appears in:

- [JoinConfiguration](#)

JoinControlPlane contains elements describing an additional control plane instance to be deployed on the joining node.

| Field                                           | Description                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| localAPIEndpoint<br><a href="#">APIEndpoint</a> | localAPIEndpoint represents the endpoint of the API server instance to be deployed on this node.                                                                                                                                                                                                      |
| certificateKey<br>string                        | certificateKey is the key that is used for decryption of certificates after they are downloaded from the Secret upon joining a new control plane node. The corresponding encryption key is in the InitConfiguration. The certificate key is a hex encoded string that is an AES key of size 32 bytes. |

## LocalEtcd

Appears in:

- [Etcd](#)

LocalEtcd describes that kubeadm should run an etcd cluster locally.

| Field                                                       | Description                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ImageMeta<br><b>[Required]</b><br><a href="#">ImageMeta</a> | (Members of ImageMeta are embedded into this type.)<br><br>ImageMeta allows to customize the container used for etcd                                                                                                                                                                                       |
| dataDir<br><b>[Required]</b><br>string                      | dataDir is the directory etcd will place its data. Defaults to "/var/lib/etcd".                                                                                                                                                                                                                            |
| extraArgs<br><b>[Required]</b><br><a href="#">[]Arg</a>     | extraArgs are extra arguments provided to the etcd binary when run inside a static Pod. An argument name in this list is the flag name as it appears on the command line except without leading dash(es). Extra arguments will override existing default arguments. Duplicate extra arguments are allowed. |
| extraEnvs<br><a href="#">[]EnvVar</a>                       | extraEnvs is an extra set of environment variables to pass to the control plane component. Environment variables passed using extraEnvs will override any existing environment variables, or *_proxy environment variables that kubeadm adds by default.                                                   |
| serverCertSANs<br><a href="#">[]string</a>                  | serverCertSANs sets extra Subject Alternative Names (SANs) for the etcd server signing certificate.                                                                                                                                                                                                        |
| peerCertSANs<br><a href="#">[]string</a>                    | peerCertSANs sets extra Subject Alternative Names (SANs) for the etcd peer signing certificate.                                                                                                                                                                                                            |

# Networking

Appears in:

- [ClusterConfiguration](#)

Networking contains elements describing cluster's networking configuration.

| Field                   | Description                                                                           |
|-------------------------|---------------------------------------------------------------------------------------|
| serviceSubnet<br>string | serviceSubnet is the subnet used by Kubernetes Services. Defaults to "10.96.0.0/12".  |
| podSubnet<br>string     | podSubnet is the subnet used by Pods.                                                 |
| dnsDomain<br>string     | dnsDomain is the dns domain used by Kubernetes Services. Defaults to "cluster.local". |

# NodeRegistrationOptions

Appears in:

- [InitConfiguration](#)
- [JoinConfiguration](#)

NodeRegistrationOptions holds fields that relate to registering a new control-plane or node to the cluster, either via kubeadm init or kubeadm join.

| Field                                                       | Description                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name<br>string                                              | name is the .Metadata.Name field of the Node API object that will be created in this kubeadm init or kubeadm join operation. This field is also used in the CommonName field of the kubelet's client certificate to the API server. Defaults to the hostname of the node if not provided.                                                                     |
| criSocket<br>string                                         | criSocket is used to retrieve container runtime info. This information will be annotated to the Node API object, for later re-use.                                                                                                                                                                                                                            |
| taints <b>[Required]</b><br><a href="#">[]core/v1.Taint</a> | taints specifies the taints the Node API object should be registered with. If this field is unset, i.e. nil, it will be defaulted with a control-plane taint for control-plane nodes. If you don't want to taint your control-plane node, set this field to an empty list, i.e. taints: [] in the YAML file. This field is solely used for Node registration. |
| kubeletExtraArgs<br><a href="#">[]Arg</a>                   | kubeletExtraArgs passes through extra arguments to the kubelet. The arguments here are passed to the kubelet command line via the                                                                                                                                                                                                                             |

| Field                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | environment file kubeadm writes at runtime for the kubelet to source. This overrides the generic base-level configuration in the kubelet-config ConfigMap. Flags have higher priority when parsing. These values are local and specific to the node kubeadm is executing on. An argument name in this list is the flag name as it appears on the command line except without leading dash(es). Extra arguments will override existing default arguments. Duplicate extra arguments are allowed. |
| ignorePreflightErrors<br>[]string                     | ignorePreflightErrors provides a slice of pre-flight errors to be ignored when the current node is registered, e.g. 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                                                                                                                                                                                                                                                                                                        |
| imagePullPolicy<br><a href="#">core/v1.PullPolicy</a> | imagePullPolicy specifies the policy for image pulling during kubeadm init and join operations. The value of this field must be one of "Always", "IfNotPresent" or "Never". If this field is unset kubeadm will default it to "IfNotPresent", or pull the required images if not present on the host.                                                                                                                                                                                           |

## Patches

Appears in:

- [InitConfiguration](#)
- [JoinConfiguration](#)
- [UpgradeApplyConfiguration](#)
- [UpgradeNodeConfiguration](#)

Patches contains options related to applying patches to components deployed by kubeadm.

| Field               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| directory<br>string | directory is a path to a directory that contains files named "target[suffix][+patchtype].extension". For example, "kube-apiserver0+merge.yaml" or just "etcd.json". "target" can be one of "kube-apiserver", "kube-controller-manager", "kube-scheduler", "etcd", "kubeletconfiguration". "patchtype" can be one of "strategic", "merge" or "json" and they match the patch formats supported by kubectrl. The default "patchtype" is "strategic". "extension" must be either "json" or "yaml". "suffix" is an optional string that can be used to determine which patches are applied first alpha-numerically. |

## Proxy

Appears in:

- [ClusterConfiguration](#)

Proxy defines the proxy addon that should be used in the cluster.

| Field                              | Description                                                      |
|------------------------------------|------------------------------------------------------------------|
| disabled <b>[Required]</b><br>bool | disabled specifies whether to disable this addon in the cluster. |

## Timeouts

Appears in:

- [UpgradeConfiguration](#)

Timeouts holds various timeouts that apply to kubeadm commands.

| Field                                                                  | Description                                                                                                                                                                           |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| controlPlaneComponentHealthCheck<br><a href="#">meta/v1.Duration</a>   | controlPlaneComponentHealthCheck is the amount of time to wait for a control plane component, such as the API server, to be healthy during kubeadm init and kubeadm join. Default: 4m |
| kubeletHealthCheck<br><a href="#">meta/v1.Duration</a>                 | kubeletHealthCheck is the amount of time to wait for the kubelet to be healthy during kubeadm init and kubeadm join. Default: 4m                                                      |
| kubernetesAPICall<br><a href="#">meta/v1.Duration</a>                  | kubernetesAPICall is the amount of time to wait for the kubeadm client to complete a request to the API server. This applies to all types of methods (GET, POST, etc). Default: 1m    |
| etcdAPICall<br><a href="#">meta/v1.Duration</a>                        | etcdAPICall is the amount of time to wait for the kubeadm etcd client to complete a request to the etcd cluster. Default: 2m                                                          |
| tlsBootstrap<br><a href="#">meta/v1.Duration</a>                       | tlsBootstrap is the amount of time to wait for the kubelet to complete TLS bootstrap for a joining node. Default: 5m                                                                  |
| discovery<br><a href="#">meta/v1.Duration</a>                          | discovery is the amount of time to wait for kubeadm to validate the API server identity for a joining node. Default: 5m                                                               |
| upgradeManifests <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | upgradeManifests is the timeout for upgrading static Pod manifests Default: 5m                                                                                                        |

# UpgradeApplyConfiguration

Appears in:

- [UpgradeConfiguration](#)

UpgradeApplyConfiguration contains a list of configurable options which are specific to the "kubeadm upgrade apply" command.

| Field                              | Description                                                                                                                                                                                                 |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kubernetesVersion<br>string        | kubernetesVersion is the target version of the control plane.                                                                                                                                               |
| allowExperimentalUpgrades<br>bool  | allowExperimentalUpgrades instructs kubeadm to show unstable versions of Kubernetes as an upgrade alternative and allows upgrading to an alpha/beta/release candidate version of Kubernetes. Default: false |
| allowRCUpgrades<br>bool            | Enable allowRCUpgrades will show release candidate versions of Kubernetes as an upgrade alternative and allows upgrading to a release candidate version of Kubernetes.                                      |
| certificateRenewal<br>bool         | certificateRenewal instructs kubeadm to execute certificate renewal during upgrades. Defaults to true.                                                                                                      |
| dryRun<br>bool                     | dryRun tells if the dry run mode is enabled, don't apply any change if it is and just output what would be done.                                                                                            |
| etcdUpgrade<br>bool                | etcdUpgrade instructs kubeadm to execute etcd upgrade during upgrades. Defaults to true.                                                                                                                    |
| forceUpgrade<br>bool               | forceUpgrade flag instructs kubeadm to upgrade the cluster without prompting for confirmation.                                                                                                              |
| ignorePreflightErrors<br>[]string  | ignorePreflightErrors provides a slice of pre-flight errors to be ignored during the upgrade process, e.g. 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                             |
| patches<br><a href="#">Patches</a> | patches contains options related to applying patches to components deployed by kubeadm during "kubeadm upgrade".                                                                                            |
| printConfig<br>bool                | printConfig specifies whether the configuration file that will be used in the upgrade should be printed or not.                                                                                             |



| Field                                    | Description                                                                                                                                                                    |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SkipPhases <b>[Required]</b><br>[]string | skipPhases is a list of phases to skip during command execution.<br>NOTE: This field is currently ignored for "kubeadm upgrade apply", but in the future it will be supported. |

## UpgradeDiffConfiguration

Appears in:

- [UpgradeConfiguration](#)

UpgradeDiffConfiguration contains a list of configurable options which are specific to the "kubeadm upgrade diff" command.

| Field                       | Description                                                     |
|-----------------------------|-----------------------------------------------------------------|
| kubernetesVersion<br>string | kubernetesVersion is the target version of the control plane.   |
| contextLines<br>int         | diffContextLines is the number of lines of context in the diff. |

## UpgradeNodeConfiguration

Appears in:

- [UpgradeConfiguration](#)

UpgradeNodeConfiguration contains a list of configurable options which are specific to the "kubeadm upgrade node" command.

| Field                             | Description                                                                                                                                                                     |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| certificateRenewal<br>bool        | certificateRenewal instructs kubeadm to execute certificate renewal during upgrades. Defaults to true.                                                                          |
| dryRun<br>bool                    | dryRun tells if the dry run mode is enabled, don't apply any change if it is and just output what would be done.                                                                |
| etcdUpgrade<br>bool               | etcdUpgrade instructs kubeadm to execute etcd upgrade during upgrades. Defaults to true.                                                                                        |
| ignorePreflightErrors<br>[]string | ignorePreflightErrors provides a slice of pre-flight errors to be ignored during the upgrade process, e.g. 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks. |

| Field                              | Description                                                                                                                                             |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| skipPhases<br>[]string             | skipPhases is a list of phases to skip during command execution. The list of phases can be obtained with the kubeadm upgrade node phase --help command. |
| patches<br><a href="#">Patches</a> | patches contains options related to applying patches to components deployed by kubeadm during kubeadm upgrade.                                          |

## UpgradePlanConfiguration

Appears in:

- [UpgradeConfiguration](#)

UpgradePlanConfiguration contains a list of configurable options which are specific to the "kubeadm upgrade plan" command.

| Field                                            | Description                                                                                                                                                                                                 |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kubernetesVersion<br><b>[Required]</b><br>string | kubernetesVersion is the target version of the control plane.                                                                                                                                               |
| allowExperimentalUpgrades<br>bool                | allowExperimentalUpgrades instructs kubeadm to show unstable versions of Kubernetes as an upgrade alternative and allows upgrading to an alpha/beta/release candidate version of Kubernetes. Default: false |
| allowRCUpgrades<br>bool                          | Enable allowRCUpgrades will show release candidate versions of Kubernetes as an upgrade alternative and allows upgrading to a release candidate version of Kubernetes.                                      |
| dryRun<br>bool                                   | dryRun tells if the dry run mode is enabled, don't apply any change if it is and just output what would be done.                                                                                            |
| ignorePreflightErrors<br>[]string                | ignorePreflightErrors provides a slice of pre-flight errors to be ignored during the upgrade process, e.g. 'IsPrivilegedUser,Swap'. Value 'all' ignores errors from all checks.                             |
| printConfig<br>bool                              | printConfig specifies whether the configuration file that will be used in the upgrade should be printed or not.                                                                                             |

# kubeconfig (v1)

## Resource Types

- [Config](#)

### Config

Config holds the information needed to build connect to remote kubernetes clusters as a given user

| Field                                                             | Description                                                                                                                 |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                              | /v1                                                                                                                         |
| kind<br>string                                                    | Config                                                                                                                      |
| kind<br>string                                                    | Legacy field from pkg/api/types.go TypeMeta. TODO(jlowdermilk): remove this after eliminating downstream dependencies.      |
| apiVersion<br>string                                              | Legacy field from pkg/api/types.go TypeMeta. TODO(jlowdermilk): remove this after eliminating downstream dependencies.      |
| preferences<br>[ <b>Required</b> ]<br><a href="#">Preferences</a> | Preferences holds general information to be use for cli interactions                                                        |
| clusters [ <b>Required</b> ]<br><a href="#">[]NamedCluster</a>    | Clusters is a map of referencable names to cluster configs                                                                  |
| users [ <b>Required</b> ]<br><a href="#">[]NamedAuthInfo</a>      | AuthInfos is a map of referencable names to user configs                                                                    |
| contexts [ <b>Required</b> ]<br><a href="#">[]NamedContext</a>    | Contexts is a map of referencable names to context configs                                                                  |
| current-context<br>[ <b>Required</b> ]<br>string                  | CurrentContext is the name of the context that you would like to use by default                                             |
| extensions<br><a href="#">[]NamedExtension</a>                    | Extensions holds additional information. This is useful for extenders so that reads and writes don't clobber unknown fields |

### AuthInfo

Appears in:

- [NamedAuthInfo](#)

AuthInfo contains information that describes identity information. This is use to tell the kubernetes cluster who you are.

| Field                                               | Description                                                                                                                                          |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| client-certificate-string                           | ClientCertificate is the path to a client cert file for TLS.                                                                                         |
| client-certificate-data<br>[]byte                   | ClientCertificateData contains PEM-encoded data from a client cert file for TLS. Overrides ClientCertificate                                         |
| client-key-string                                   | ClientKey is the path to a client key file for TLS.                                                                                                  |
| client-key-data<br>[]byte                           | ClientKeyData contains PEM-encoded data from a client key file for TLS. Overrides ClientKey                                                          |
| token-string                                        | Token is the bearer token for authentication to the kubernetes cluster.                                                                              |
| tokenFile-string                                    | TokenFile is a pointer to a file that contains a bearer token (as described above). If both Token and TokenFile are present, Token takes precedence. |
| as-string                                           | Impersonate is the username to impersonate. The name matches the flag.                                                                               |
| as-uid-string                                       | ImpersonateUID is the uid to impersonate.                                                                                                            |
| as-groups<br>[]string                               | ImpersonateGroups is the groups to impersonate.                                                                                                      |
| as-user-extra<br>map[string][]string                | ImpersonateUserExtra contains additional information for impersonated user.                                                                          |
| username-string                                     | Username is the username for basic authentication to the kubernetes cluster.                                                                         |
| password-string                                     | Password is the password for basic authentication to the kubernetes cluster.                                                                         |
| auth-provider<br><a href="#">AuthProviderConfig</a> | AuthProvider specifies a custom authentication plugin for the kubernetes cluster.                                                                    |
| exec<br><a href="#">ExecConfig</a>                  | Exec specifies a custom exec-based authentication plugin for the kubernetes cluster.                                                                 |

| Field                                          | Description                                                                                                                 |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| extensions<br><a href="#">[]NamedExtension</a> | Extensions holds additional information. This is useful for extenders so that reads and writes don't clobber unknown fields |

## AuthProviderConfig

Appears in:

- [AuthInfo](#)

AuthProviderConfig holds the configuration for a specified auth provider.

| Field                                         | Description              |
|-----------------------------------------------|--------------------------|
| name <b>[Required]</b><br>string              | No description provided. |
| config <b>[Required]</b><br>map[string]string | No description provided. |

## Cluster

Appears in:

- [NamedCluster](#)

Cluster contains information about how to communicate with a kubernetes cluster

| Field                                | Description                                                                                                                                                                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| server <b>[Required]</b><br>string   | Server is the address of the kubernetes cluster (https://hostname:port).                                                                                                                                                                                                                           |
| tls-server-name<br>string            | TLSServerName is used to check server certificate. If TLSServerName is empty, the hostname used to contact the server is used.                                                                                                                                                                     |
| insecure-skip-tls-verify<br>bool     | InsecureSkipTLSVerify skips the validity check for the server's certificate. This will make your HTTPS connections insecure.                                                                                                                                                                       |
| certificate-authority<br>string      | CertificateAuthority is the path to a cert file for the certificate authority.                                                                                                                                                                                                                     |
| certificate-authority-data<br>[]byte | CertificateAuthorityData contains PEM-encoded certificate authority certificates. Overrides CertificateAuthority                                                                                                                                                                                   |
| proxy-url<br>string                  | ProxyURL is the URL to the proxy to be used for all requests made by this client. URLs with "http", "https", and "socks5" schemes are supported. If this configuration is not provided or the empty string, the client attempts to construct a proxy configuration from http_proxy and https_proxy |

| Field                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | environment variables. If these environment variables are not set, the client does not attempt to proxy requests.<br><br>socks5 proxying does not currently support spdy streaming endpoints (exec, attach, port forward).                                                                                                                                                                                             |
| disable-compression<br>bool                    | DisableCompression allows client to opt-out of response compression for all requests to the server. This is useful to speed up requests (specifically lists) when client-server network bandwidth is ample, by saving time on compression (server-side) and decompression (client-side): <a href="https://github.com/kubernetes/kubernetes/issues/112296">https://github.com/kubernetes/kubernetes/issues/112296</a> . |
| extensions<br><a href="#">[]NamedExtension</a> | Extensions holds additional information. This is useful for extenders so that reads and writes don't clobber unknown fields                                                                                                                                                                                                                                                                                            |

## Context

Appears in:

- [NamedContext](#)

Context is a tuple of references to a cluster (how do I communicate with a kubernetes cluster), a user (how do I identify myself), and a namespace (what subset of resources do I want to work with)

| Field                                          | Description                                                                                                                 |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| cluster<br>[Required]<br>string                | Cluster is the name of the cluster for this context                                                                         |
| user [Required]<br>string                      | AuthInfo is the name of the authInfo for this context                                                                       |
| namespace<br>string                            | Namespace is the default namespace to use on unspecified requests                                                           |
| extensions<br><a href="#">[]NamedExtension</a> | Extensions holds additional information. This is useful for extenders so that reads and writes don't clobber unknown fields |

## ExecConfig

Appears in:

- [AuthInfo](#)

ExecConfig specifies a command to provide client credentials. The command is exec'd and outputs structured stdout holding credentials.

See the `client.authentication.k8s.io` API group for specifications of the exact input and output format

| Field                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>command</code><br><b>[Required]</b><br>string                 | Command to execute.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>args</code><br>[]string                                       | Arguments to pass to the command when executing it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>env</code><br><a href="#">[]ExecEnvVar</a>                    | Env defines additional environment variables to expose to the process. These are unioned with the host's environment, as well as variables <code>client-go</code> uses to pass argument to the plugin.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>apiVersion</code><br><b>[Required]</b><br>string              | Preferred input version of the <code>ExecInfo</code> . The returned <code>ExecCredentials</code> MUST use the same encoding version as the input.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>installHint</code><br><b>[Required]</b><br>string             | This text is shown to the user when the executable doesn't seem to be present. For example, <code>brew install foo-cli</code> might be a good <code>InstallHint</code> for <code>foo-cli</code> on Mac OS systems.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>provideClusterInfo</code><br><b>[Required]</b><br>bool        | <code>ProvideClusterInfo</code> determines whether or not to provide cluster information, which could potentially contain very large CA data, to this exec plugin as a part of the <code>KUBERNETES_EXEC_INFO</code> environment variable. By default, it is set to false. Package <code>k8s.io/client-go/tools/auth/exec</code> provides helper methods for reading this environment variable.                                                                                                                                                                                                                                                 |
| <code>interactiveMode</code><br><a href="#">ExecInteractiveMode</a> | <p><code>InteractiveMode</code> determines this plugin's relationship with standard input. Valid values are "Never" (this exec plugin never uses standard input), "IfAvailable" (this exec plugin wants to use standard input if it is available), or "Always" (this exec plugin requires standard input to function). See <code>ExecInteractiveMode</code> values for more details.</p> <p>If <code>APIVersion</code> is <code>client.authentication.k8s.io/v1alpha1</code> or <code>client.authentication.k8s.io/v1beta1</code>, then this field is optional and defaults to "IfAvailable" when unset. Otherwise, this field is required.</p> |

## ExecEnvVar

Appears in:

- [ExecConfig](#)

`ExecEnvVar` is used for setting environment variables when executing an exec-based credential plugin.

| Field                             | Description              |
|-----------------------------------|--------------------------|
| name <b>[Required]</b><br>string  | No description provided. |
| value <b>[Required]</b><br>string | No description provided. |

## ExecInteractiveMode

(Alias of string)

Appears in:

- [ExecConfig](#)

ExecInteractiveMode is a string that describes an exec plugin's relationship with standard input.

## NamedAuthInfo

Appears in:

- [Config](#)

NamedAuthInfo relates nicknames to auth information

| Field                                              | Description                            |
|----------------------------------------------------|----------------------------------------|
| name <b>[Required]</b><br>string                   | Name is the nickname for this AuthInfo |
| user <b>[Required]</b><br><a href="#">AuthInfo</a> | AuthInfo holds the auth information    |

## NamedCluster

Appears in:

- [Config](#)

NamedCluster relates nicknames to cluster information

| Field                                                | Description                           |
|------------------------------------------------------|---------------------------------------|
| name <b>[Required]</b><br>string                     | Name is the nickname for this Cluster |
| cluster <b>[Required]</b><br><a href="#">Cluster</a> | Cluster holds the cluster information |



## NamedContext

Appears in:

- [Config](#)

NamedContext relates nicknames to context information

| Field                                                  | Description                           |
|--------------------------------------------------------|---------------------------------------|
| name [ <b>Required</b> ]<br>string                     | Name is the nickname for this Context |
| context [ <b>Required</b> ]<br><a href="#">Context</a> | Context holds the context information |

## NamedExtension

Appears in:

- [Config](#)
- [AuthInfo](#)
- [Cluster](#)
- [Context](#)
- [Preferences](#)

NamedExtension relates nicknames to extension information

| Field                                                                                                                                            | Description                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| name [ <b>Required</b> ]<br>string                                                                                                               | Name is the nickname for this Extension   |
| extension [ <b>Required</b> ]<br><a href="https://k8s.io/apimachinery/pkg/runtime.RawExtension">k8s.io/apimachinery/pkg/runtime.RawExtension</a> | Extension holds the extension information |

## Preferences

Appears in:

- [Config](#)

| Field          | Description              |
|----------------|--------------------------|
| colors<br>bool | No description provided. |

| Field                                          | Description                                                                                                                 |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| extensions<br><a href="#">[]NamedExtension</a> | Extensions holds additional information. This is useful for extenders so that reads and writes don't clobber unknown fields |

# Kubelet Configuration (v1)

## Resource Types

- [CredentialProviderConfig](#)

## CredentialProviderConfig

CredentialProviderConfig is the configuration containing information about each exec credential provider. Kubelet reads this configuration from disk and enables each provider as specified by the CredentialProvider type.

| Field                                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                     | kubelet.config.k8s.io/v1                                                                                                                                                                                                                                                                                                                                                                                       |
| kind<br>string                                                           | CredentialProviderConfig                                                                                                                                                                                                                                                                                                                                                                                       |
| providers<br>[ <b>Required</b> ]<br><a href="#">[]CredentialProvider</a> | providers is a list of credential provider plugins that will be enabled by the kubelet. Multiple providers may match against a single image, in which case credentials from all providers will be returned to the kubelet. If multiple providers are called for a single image, the results are combined. If providers return overlapping auth keys, the value from the provider earlier in this list is used. |

## CredentialProvider

Appears in:

- [CredentialProviderConfig](#)

CredentialProvider represents an exec plugin to be invoked by the kubelet. The plugin is only invoked when an image being pulled matches the images handled by the plugin (see matchImages).

| Field                              | Description                                                                                                                                                                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name [ <b>Required</b> ]<br>string | name is the required name of the credential provider. It must match the name of the provider executable as seen by the kubelet. The executable must be in the kubelet's bin directory (set by the --image-credential-provider-bin-dir flag). |

| Field                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| matchImages<br><b>[Required]</b><br>[]string                                  | <p>matchImages is a required list of strings used to match against images in order to determine if this provider should be invoked. If one of the strings matches the requested image from the kubelet, the plugin will be invoked and given a chance to provide credentials. Images are expected to contain the registry domain and URL path.</p> <p>Each entry in matchImages is a pattern which can optionally contain a port and a path. Globs can be used in the domain, but not in the port or the path. Globs are supported as subdomains like '*.k8s.io' or 'k8s.*.io', and top-level-domains such as 'k8s.*'. Matching partial subdomains like 'app*.k8s.io' is also supported. Each glob can only match a single subdomain segment, so '*.io' does not match '*.k8s.io'.</p> <p>A match exists between an image and a matchImage when all of the below are true:</p> <ul style="list-style-type: none"> <li>• Both contain the same number of domain parts and each part matches.</li> <li>• The URL path of an imageMatch must be a prefix of the target image URL path.</li> <li>• If the imageMatch contains a port, then the port must match in the image as well.</li> </ul> <p>Example values of matchImages:</p> <ul style="list-style-type: none"> <li>• 123456789.dkr.ecr.us-east-1.amazonaws.com</li> <li>• *.azurecr.io</li> <li>• gcr.io</li> <li>• *.registry.io</li> <li>• registry.io:8080/path</li> </ul> |
| defaultCacheDuration<br><b>[Required]</b><br><a href="#">meta/v1.Duration</a> | defaultCacheDuration is the default duration the plugin will cache credentials in-memory if a cache duration is not provided in the plugin response. This field is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| apiVersion<br><b>[Required]</b><br>string                                     | <p>Required input version of the exec CredentialProviderRequest. The returned CredentialProviderResponse MUST use the same encoding version as the input. Current supported values are:</p> <ul style="list-style-type: none"> <li>• credentialprovider.kubelet.k8s.io/v1</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| args<br>[]string                                                              | Arguments to pass to the command when executing it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| env<br><a href="#">[]ExecEnvVar</a>                                           | Env defines additional environment variables to expose to the process. These are unioned with the host's environment, as well as variables client-go uses to pass argument to the plugin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## ExecEnvVar

Appears in:

- [CredentialProvider](#)

ExecEnvVar is used for setting environment variables when executing an exec-based credential plugin.

| Field                      | Description              |
|----------------------------|--------------------------|
| name [Required]<br>string  | No description provided. |
| value [Required]<br>string | No description provided. |

## Kubelet Configuration (v1alpha1)

### Resource Types

- [CredentialProviderConfig](#)

## CredentialProviderConfig

CredentialProviderConfig is the configuration containing information about each exec credential provider. Kubelet reads this configuration from disk and enables each provider as specified by the CredentialProvider type.

| Field                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                            | kubelet.config.k8s.io/v1alpha1                                                                                                                                                                                                                                                                                                                                                                                 |
| kind<br>string                                                  | CredentialProviderConfig                                                                                                                                                                                                                                                                                                                                                                                       |
| providers<br>[Required]<br><a href="#">[]CredentialProvider</a> | providers is a list of credential provider plugins that will be enabled by the kubelet. Multiple providers may match against a single image, in which case credentials from all providers will be returned to the kubelet. If multiple providers are called for a single image, the results are combined. If providers return overlapping auth keys, the value from the provider earlier in this list is used. |

## CredentialProvider

Appears in:

- [CredentialProviderConfig](#)

CredentialProvider represents an exec plugin to be invoked by the kubelet. The plugin is only invoked when an image being pulled matches the images handled by the plugin (see `matchImages`).

| Field                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>name</code> <b>[Required]</b><br>string                                           | <code>name</code> is the required name of the credential provider. It must match the name of the provider executable as seen by the kubelet. The executable must be in the kubelet's bin directory (set by the <code>--image-credential-provider-bin-dir</code> flag).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>matchImages</code> <b>[Required]</b><br>[]string                                  | <p><code>matchImages</code> is a required list of strings used to match against images in order to determine if this provider should be invoked. If one of the strings matches the requested image from the kubelet, the plugin will be invoked and given a chance to provide credentials. Images are expected to contain the registry domain and URL path.</p> <p>Each entry in <code>matchImages</code> is a pattern which can optionally contain a port and a path. Globs can be used in the domain, but not in the port or the path. Globs are supported as subdomains like <code>*.k8s.io</code> or <code>k8s*.io</code>, and top-level-domains such as <code>k8s.*</code>. Matching partial subdomains like <code>app*.k8s.io</code> is also supported. Each glob can only match a single subdomain segment, so <code>*.io</code> does not match <code>*.k8s.io</code>.</p> <p>A match exists between an image and a <code>matchImage</code> when all of the below are true:</p> <ul style="list-style-type: none"><li>• Both contain the same number of domain parts and each part matches.</li><li>• The URL path of an <code>imageMatch</code> must be a prefix of the target image URL path.</li><li>• If the <code>imageMatch</code> contains a port, then the port must match in the image as well.</li></ul> <p>Example values of <code>matchImages</code>:</p> <ul style="list-style-type: none"><li>• <code>123456789.dkr.ecr.us-east-1.amazonaws.com</code></li><li>• <code>*.azurecr.io</code></li><li>• <code>gcr.io</code></li><li>• <code>*.*.registry.io</code></li><li>• <code>registry.io:8080/path</code></li></ul> |
| <code>defaultCacheDuration</code> <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | <code>defaultCacheDuration</code> is the default duration the plugin will cache credentials in-memory if a cache duration is not provided in the plugin response. This field is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Field                              | Description                                                                                                                                                                                                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>[Required]<br>string | Required input version of the exec CredentialProviderRequest. The returned CredentialProviderResponse MUST use the same encoding version as the input. Current supported values are: <ul style="list-style-type: none"> <li>credentialprovider.kubelet.k8s.io/v1alpha1</li> </ul> |
| args<br>[]string                   | Arguments to pass to the command when executing it.                                                                                                                                                                                                                               |
| env<br>[]ExecEnvVar                | Env defines additional environment variables to expose to the process. These are unioned with the host's environment, as well as variables client-go uses to pass argument to the plugin.                                                                                         |

## ExecEnvVar

Appears in:

- [CredentialProvider](#)

ExecEnvVar is used for setting environment variables when executing an exec-based credential plugin.

| Field                      | Description              |
|----------------------------|--------------------------|
| name [Required]<br>string  | No description provided. |
| value [Required]<br>string | No description provided. |

# Kubelet Configuration (v1beta1)

## Resource Types

- [CredentialProviderConfig](#)
- [KubeletConfiguration](#)
- [SerializedNodeConfigSource](#)

## FormatOptions

Appears in:

- [LoggingConfiguration](#)

FormatOptions contains options for the different logging formats.

| Field                                                      | Description                                                                                                                   |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| text<br>[ <b>Required</b> ]<br><a href="#">TextOptions</a> | [Alpha] Text contains options for logging format "text". Only available when the LoggingAlphaOptions feature gate is enabled. |
| json<br>[ <b>Required</b> ]<br><a href="#">JSONOptions</a> | [Alpha] JSON contains options for logging format "json". Only available when the LoggingAlphaOptions feature gate is enabled. |

## JSONOptions

Appears in:

- [FormatOptions](#)

JSONOptions contains options for logging format "json".

| Field                                                                               | Description                                                                             |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| OutputRoutingOptions<br>[ <b>Required</b> ]<br><a href="#">OutputRoutingOptions</a> | (Members of OutputRoutingOptions are embedded into this type.) No description provided. |

## LogFormatFactory

LogFormatFactory provides support for a certain additional, non-default log format.

## LoggingConfiguration

Appears in:

- [KubeletConfiguration](#)

LoggingConfiguration contains logging options.

| Field                                                              | Description                                                                                                                                                                                                                         |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| format [Required]<br>string                                        | Format Flag specifies the structure of log messages. default value of format is text                                                                                                                                                |
| flushFrequency<br>[Required]<br><a href="#">TimeOrMetaDuration</a> | Maximum time between log flushes. If a string, parsed as a duration (i.e. "1s") If an int, the maximum number of nanoseconds (i.e. 1s = 1000000000). Ignored if the selected logging backend writes log messages without buffering. |

| Field                                                             | Description                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| verbosity <b>[Required]</b><br><a href="#">VerbosityLevel</a>     | Verbosity is the threshold that determines which log messages are logged. Default is zero which logs only the most important messages. Higher values enable additional messages. Error messages are always logged.                                     |
| vmodule <b>[Required]</b><br><a href="#">VModuleConfiguration</a> | VModule overrides the verbosity threshold for individual files. Only supported for "text" log format.                                                                                                                                                  |
| options <b>[Required]</b><br><a href="#">FormatOptions</a>        | [Alpha] Options holds additional parameters that are specific to the different logging formats. Only the options for the selected format get used, but all of them get validated. Only available when the LoggingAlphaOptions feature gate is enabled. |

## LoggingOptions

LoggingOptions can be used with ValidateAndApplyWithOptions to override certain global defaults.

| Field                                                      | Description                                                |
|------------------------------------------------------------|------------------------------------------------------------|
| ErrorStream <b>[Required]</b><br><a href="#">io.Writer</a> | ErrorStream can be used to override the os.Stderr default. |
| InfoStream <b>[Required]</b><br><a href="#">io.Writer</a>  | InfoStream can be used to override the os.Stdout default.  |

## OutputRoutingOptions

Appears in:

- [JSONOptions](#)
- [TextOptions](#)

OutputRoutingOptions contains options that are supported by both "text" and "json".

| Field                                                                                                  | Description                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| splitStream <b>[Required]</b><br>bool                                                                  | [Alpha] SplitStream redirects error messages to stderr while info messages go to stdout, with buffering. The default is to write both to stdout, without buffering. Only available when the LoggingAlphaOptions feature gate is enabled. |
| infoBufferSize <b>[Required]</b><br><a href="#">k8s.io/apimachinery/pkg/api/resource.QuantityValue</a> | [Alpha] InfoBufferSize sets the size of the info stream when using split streams. The default is zero, which disables buffering.                                                                                                         |



| Field | Description                                                          |
|-------|----------------------------------------------------------------------|
|       | Only available when the LoggingAlphaOptions feature gate is enabled. |

## TextOptions

Appears in:

- [FormatOptions](#)

TextOptions contains options for logging format "text".

| Field                                                                      | Description                                                                             |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| OutputRoutingOptions<br>[Required]<br><a href="#">OutputRoutingOptions</a> | (Members of OutputRoutingOptions are embedded into this type.) No description provided. |

## TimeOrMetaDuration

Appears in:

- [LoggingConfiguration](#)

TimeOrMetaDuration is present only for backwards compatibility for the flushFrequency field, and new fields should use metav1.Duration.

| Field                                                      | Description                                                                          |
|------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Duration<br>[Required]<br><a href="#">meta/v1.Duration</a> | Duration holds the duration                                                          |
| - [Required]<br>bool                                       | SerializeAsString controls whether the value is serialized as a string or an integer |

## TracingConfiguration

Appears in:

- [KubeletConfiguration](#)

TracingConfiguration provides versioned configuration for OpenTelemetry tracing clients.

| Field              | Description                                                                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| endpoint<br>string | Endpoint of the collector this component will report traces to. The connection is insecure, and does not currently support TLS. Recommended is unset, and endpoint is the otel grpc default, localhost:4317. |

| Field                           | Description                                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| samplingRatePerMillion<br>int32 | SamplingRatePerMillion is the number of samples to collect per million spans. Recommended is unset. If unset, sampler respects its parent span's sampling rate, but otherwise never samples. |

## VModuleConfiguration

(Alias of `[]k8s.io/component-base/logs/api/v1.VModuleItem`)

**Appears in:**

- [LoggingConfiguration](#)

VModuleConfiguration is a collection of individual file names or patterns and the corresponding verbosity threshold.

## VerbosityLevel

(Alias of `uint32`)

**Appears in:**

- [LoggingConfiguration](#)

VerbosityLevel represents a klog or logr verbosity threshold.

## CredentialProviderConfig

CredentialProviderConfig is the configuration containing information about each exec credential provider. Kubelet reads this configuration from disk and enables each provider as specified by the CredentialProvider type.

| Field                                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                     | kubelet.config.k8s.io/v1beta1                                                                                                                                                                                                                                                                                                                                                                                  |
| kind<br>string                                                           | CredentialProviderConfig                                                                                                                                                                                                                                                                                                                                                                                       |
| providers<br>[ <b>Required</b> ]<br><a href="#">[]CredentialProvider</a> | providers is a list of credential provider plugins that will be enabled by the kubelet. Multiple providers may match against a single image, in which case credentials from all providers will be returned to the kubelet. If multiple providers are called for a single image, the results are combined. If providers return overlapping auth keys, the value from the provider earlier in this list is used. |

# KubeletConfiguration

KubeletConfiguration contains the configuration for the Kubelet

| Field                                                  | Description                                                                                                                                                                                                                               |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                   | kubelet.config.k8s.io/v1beta1                                                                                                                                                                                                             |
| kind<br>string                                         | KubeletConfiguration                                                                                                                                                                                                                      |
| enableServer <b>[Required]</b><br>bool                 | enableServer enables Kubelet's secured server.<br>Note: Kubelet's insecure port is controlled by the readOnlyPort option. Default: true                                                                                                   |
| staticPodPath<br>string                                | staticPodPath is the path to the directory containing local (static) pods to run, or the path to a single static pod file. Default: ""                                                                                                    |
| podLogsDir<br>string                                   | podLogsDir is a custom root directory path kubelet will use to place pod's log files. Default: "/var/log/pods/" Note: it is not recommended to use the temp folder as a log directory as it may cause unexpected behavior in many places. |
| syncFrequency<br><a href="#">meta/v1.Duration</a>      | syncFrequency is the max period between synchronizing running containers and config. Default: "1m"                                                                                                                                        |
| fileCheckFrequency<br><a href="#">meta/v1.Duration</a> | fileCheckFrequency is the duration between checking config files for new data. Default: "20s"                                                                                                                                             |
| httpCheckFrequency<br><a href="#">meta/v1.Duration</a> | httpCheckFrequency is the duration between checking http for new data. Default: "20s"                                                                                                                                                     |
| staticPodURL<br>string                                 | staticPodURL is the URL for accessing static pods to run. Default: ""                                                                                                                                                                     |
| staticPodURLHeader<br>map[string][]string              | staticPodURLHeader is a map of slices with HTTP headers to use when accessing the podURL. Default: nil                                                                                                                                    |
| address<br>string                                      | address is the IP address for the Kubelet to serve on (set to 0.0.0.0 for all interfaces). Default: "0.0.0.0"                                                                                                                             |

| Field                       | Description                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| port<br>int32               | port is the port for the Kubelet to serve on. The port number must be between 1 and 65535, inclusive. Default: 10250                                                                                                                                                                                                            |
| readOnlyPort<br>int32       | readOnlyPort is the read-only port for the Kubelet to serve on with no authentication/authorization. The port number must be between 1 and 65535, inclusive. Setting this field to 0 disables the read-only service. Default: 0 (disabled)                                                                                      |
| tlsCertFile<br>string       | tlsCertFile is the file containing x509 Certificate for HTTPS. (CA cert, if any, concatenated after server cert). If tlsCertFile and tlsPrivateKeyFile are not provided, a self-signed certificate and key are generated for the public address and saved to the directory passed to the Kubelet's --cert-dir flag. Default: "" |
| tlsPrivateKeyFile<br>string | tlsPrivateKeyFile is the file containing x509 private key matching tlsCertFile. Default: ""                                                                                                                                                                                                                                     |
| tlsCipherSuites<br>[]string | tlsCipherSuites is the list of allowed cipher suites for the server. Note that TLS 1.3 ciphersuites are not configurable. Values are from tls package constants ( <a href="https://golang.org/pkg/crypto/tls/#pkg-constants">https://golang.org/pkg/crypto/tls/#pkg-constants</a> ). Default: nil                               |
| tlsMinVersion<br>string     | tlsMinVersion is the minimum TLS version supported. Values are from tls package constants ( <a href="https://golang.org/pkg/crypto/tls/#pkg-constants">https://golang.org/pkg/crypto/tls/#pkg-constants</a> ). Default: ""                                                                                                      |
| rotateCertificates<br>bool  | rotateCertificates enables client certificate rotation. The Kubelet will request a new certificate from the certificates.k8s.io API. This requires an approver to approve the certificate signing requests. Default: false                                                                                                      |
| serverTLSBootstrap<br>bool  | serverTLSBootstrap enables server certificate bootstrap. Instead of self signing a serving certificate, the Kubelet will request a certificate from the 'certificates.k8s.io' API. This requires an approver to approve the certificate signing requests (CSR). The RotateKubeletServerCertificate                              |

| Field                                                   | Description                                                                                                                                                                                                                                                          |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | feature must be enabled when setting this field.<br>Default: false                                                                                                                                                                                                   |
| authentication<br><a href="#">KubeletAuthentication</a> | authentication specifies how requests to the Kubelet's server are authenticated. Defaults: anonymous: enabled: false webhook: enabled: true cacheTTL: "2m"                                                                                                           |
| authorization<br><a href="#">KubeletAuthorization</a>   | authorization specifies how requests to the Kubelet's server are authorized. Defaults: mode: Webhook webhook: cacheAuthorizedTTL: "5m" cacheUnauthorizedTTL: "30s"                                                                                                   |
| registryPullQPS<br>int32                                | registryPullQPS is the limit of registry pulls per second. The value must not be a negative number. Setting it to 0 means no limit. Default: 5                                                                                                                       |
| registryBurst<br>int32                                  | registryBurst is the maximum size of bursty pulls, temporarily allows pulls to burst to this number, while still not exceeding registryPullQPS. The value must not be a negative number. Only used if registryPullQPS is greater than 0. Default: 10                 |
| eventRecordQPS<br>int32                                 | eventRecordQPS is the maximum event creations per second. If 0, there is no limit enforced. The value cannot be a negative number. Default: 50                                                                                                                       |
| eventBurst<br>int32                                     | eventBurst is the maximum size of a burst of event creations, temporarily allows event creations to burst to this number, while still not exceeding eventRecordQPS. This field cannot be a negative number and it is only used when eventRecordQPS > 0. Default: 100 |
| enableDebuggingHandlers<br>bool                         | enableDebuggingHandlers enables server endpoints for log access and local running of containers and commands, including the exec, attach, logs, and portforward features. Default: true                                                                              |
| enableContentionProfiling<br>bool                       | enableContentionProfiling enables block profiling, if enableDebuggingHandlers is true. Default: false                                                                                                                                                                |

| Field                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| healthzPort<br>int32                                               | healthzPort is the port of the localhost healthz endpoint (set to 0 to disable). A valid number is between 1 and 65535. Default: 10248                                                                                                                                                                                                                                                                                        |
| healthzBindAddress<br>string                                       | healthzBindAddress is the IP address for the healthz server to serve on. Default: "127.0.0.1"                                                                                                                                                                                                                                                                                                                                 |
| oomScoreAdj<br>int32                                               | oomScoreAdj is The oom-score-adj value for kubelet process. Values must be within the range [-1000, 1000]. Default: -999                                                                                                                                                                                                                                                                                                      |
| clusterDomain<br>string                                            | clusterDomain is the DNS domain for this cluster. If set, kubelet will configure all containers to search this domain in addition to the host's search domains. Default: ""                                                                                                                                                                                                                                                   |
| clusterDNS<br>[]string                                             | clusterDNS is a list of IP addresses for the cluster DNS server. If set, kubelet will configure all containers to use this for DNS resolution instead of the host's DNS servers. Default: nil                                                                                                                                                                                                                                 |
| streamingConnectionIdleTimeout<br><a href="#">meta/v1.Duration</a> | streamingConnectionIdleTimeout is the maximum time a streaming connection can be idle before the connection is automatically closed. Default: "4h"                                                                                                                                                                                                                                                                            |
| nodeStatusUpdateFrequency<br><a href="#">meta/v1.Duration</a>      | nodeStatusUpdateFrequency is the frequency that kubelet computes node status. If node lease feature is not enabled, it is also the frequency that kubelet posts node status to master. Note: When node lease feature is not enabled, be cautious when changing the constant, it must work with nodeMonitorGracePeriod in nodecontroller. Default: "10s"                                                                       |
| nodeStatusReportFrequency<br><a href="#">meta/v1.Duration</a>      | nodeStatusReportFrequency is the frequency that kubelet posts node status to master if node status does not change. Kubelet will ignore this frequency and post node status immediately if any change is detected. It is only used when node lease feature is enabled. nodeStatusReportFrequency's default value is 5m. But if nodeStatusUpdateFrequency is set explicitly, nodeStatusReportFrequency's default value will be |

| Field                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | set to nodeStatusUpdateFrequency for backward compatibility. Default: "5m"                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| nodeLeaseDurationSeconds<br>int32                     | nodeLeaseDurationSeconds is the duration the Kubelet will set on its corresponding Lease. NodeLease provides an indicator of node health by having the Kubelet create and periodically renew a lease, named after the node, in the kube-node-lease namespace. If the lease expires, the node can be considered unhealthy. The lease is currently renewed every 10s, per KEP-0009. In the future, the lease renewal interval may be set based on the lease duration. The field value must be greater than 0. Default: 40 |
| imageMinimumGCAge<br><a href="#">meta/v1.Duration</a> | imageMinimumGCAge is the minimum age for an unused image before it is garbage collected. Default: "2m"                                                                                                                                                                                                                                                                                                                                                                                                                  |
| imageMaximumGCAge<br><a href="#">meta/v1.Duration</a> | imageMaximumGCAge is the maximum age an image can be unused before it is garbage collected. The default of this field is "0s", which disables this field--meaning images won't be garbage collected based on being unused for too long. Default: "0s" (disabled)                                                                                                                                                                                                                                                        |
| imageGCHighThresholdPercent<br>int32                  | imageGCHighThresholdPercent is the percent of disk usage after which image garbage collection is always run. The percent is calculated by dividing this field value by 100, so this field must be between 0 and 100, inclusive. When specified, the value must be greater than imageGCLowThresholdPercent. Default: 85                                                                                                                                                                                                  |
| imageGCLowThresholdPercent<br>int32                   | imageGCLowThresholdPercent is the percent of disk usage before which image garbage collection is never run. Lowest disk usage to garbage collect to. The percent is calculated by dividing this field value by 100, so the field value must be between 0 and 100, inclusive. When specified, the value must be less than imageGCHighThresholdPercent. Default: 80                                                                                                                                                       |

| Field                                                    | Description                                                                                                                                                                                                                                                         |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volumeStatsAggPeriod<br><a href="#">meta/v1.Duration</a> | volumeStatsAggPeriod is the frequency for calculating and caching volume disk usage for all pods. Default: "1m"                                                                                                                                                     |
| kubeletCgroups<br>string                                 | kubeletCgroups is the absolute name of cgroups to isolate the kubelet in Default: ""                                                                                                                                                                                |
| systemCgroups<br>string                                  | systemCgroups is absolute name of cgroups in which to place all non-kernel processes that are not already in a container. Empty for no container. Rolling back the flag requires a reboot. The cgroupRoot must be specified if this field is not empty. Default: "" |
| cgroupRoot<br>string                                     | cgroupRoot is the root cgroup to use for pods. This is handled by the container runtime on a best effort basis.                                                                                                                                                     |
| cgroupsPerQOS<br>bool                                    | cgroupsPerQOS enable QoS based CGroup hierarchy: top level CGroups for QoS classes and all Burstable and BestEffort Pods are brought up under their specific top level QoS CGroup. Default: true                                                                    |
| cgroupDriver<br>string                                   | cgroupDriver is the driver kubelet uses to manipulate CGroups on the host (cgroupfs or systemd). Default: "cgroupfs"                                                                                                                                                |
| cpuManagerPolicy<br>string                               | cpuManagerPolicy is the name of the policy to use. Requires the CPUManager feature gate to be enabled. Default: "None"                                                                                                                                              |
| cpuManagerPolicyOptions<br>map[string]string             | cpuManagerPolicyOptions is a set of key=value which allows to set extra options to fine tune the behaviour of the cpu manager policies. Requires both the "CPUManager" and "CPUManagerPolicyOptions" feature gates to be enabled. Default: nil                      |



| Field                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cpuManagerReconcilePeriod<br><a href="#">meta/v1.Duration</a> | cpuManagerReconcilePeriod is the reconciliation period for the CPU Manager. Requires the CPUManager feature gate to be enabled. Default: "10s"                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| memoryManagerPolicy<br>string                                 | memoryManagerPolicy is the name of the policy to use by memory manager. Requires the MemoryManager feature gate to be enabled. Default: "none"                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| topologyManagerPolicy<br>string                               | <p>topologyManagerPolicy is the name of the topology manager policy to use. Valid values include:</p> <ul style="list-style-type: none"> <li>• restricted: kubelet only allows pods with optimal NUMA node alignment for requested resources;</li> <li>• best-effort: kubelet will favor pods with NUMA alignment of CPU and device resources;</li> <li>• none: kubelet has no knowledge of NUMA alignment of a pod's CPU and device resources.</li> <li>• single-numa-node: kubelet only allows pods with a single NUMA alignment of CPU and device resources.</li> </ul> <p>Default: "none"</p> |
| topologyManagerScope<br>string                                | <p>topologyManagerScope represents the scope of topology hint generation that topology manager requests and hint providers generate. Valid values include:</p> <ul style="list-style-type: none"> <li>• container: topology policy is applied on a per-container basis.</li> <li>• pod: topology policy is applied on a per-pod basis.</li> </ul> <p>Default: "container"</p>                                                                                                                                                                                                                     |
| topologyManagerPolicyOptions<br>map[string]string             | TopologyManagerPolicyOptions is a set of key=value which allows to set extra options to fine tune the behaviour of the topology manager policies. Requires both the "TopologyManager" and "TopologyManagerPolicyOptions" feature gates to be enabled. Default: nil                                                                                                                                                                                                                                                                                                                                |

| Field                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| qosReserved<br>map[string]string                          | qosReserved is a set of resource name to percentage pairs that specify the minimum percentage of a resource reserved for exclusive use by the guaranteed QoS tier. Currently supported resources: "memory" Requires the QOSReserved feature gate to be enabled. Default: nil                                                                                                                                                                                                                                                                                                                                                                                                        |
| runtimeRequestTimeout<br><a href="#">meta/v1.Duration</a> | runtimeRequestTimeout is the timeout for all runtime requests except long running requests - pull, logs, exec and attach. Default: "2m"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| hairpinMode<br>string                                     | <p>hairpinMode specifies how the Kubelet should configure the container bridge for hairpin packets. Setting this flag allows endpoints in a Service to loadbalance back to themselves if they should try to access their own Service. Values:</p> <ul style="list-style-type: none"> <li>• "promiscuous-bridge": make the container bridge promiscuous.</li> <li>• "hairpin-veth": set the hairpin flag on container veth interfaces.</li> <li>• "none": do nothing.</li> </ul> <p>Generally, one must set --hairpin-mode=hairpin-veth to achieve hairpin NAT, because promiscuous-bridge assumes the existence of a container bridge named cbr0. Default: "promiscuous-bridge"</p> |
| maxPods<br>int32                                          | maxPods is the maximum number of Pods that can run on this Kubelet. The value must be a non-negative integer. Default: 110                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| podCIDR<br>string                                         | podCIDR is the CIDR to use for pod IP addresses, only used in standalone mode. In cluster mode, this is obtained from the control plane. Default: ""                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| podPidsLimit<br>int64                                     | podPidsLimit is the maximum number of PIDs in any pod. Default: -1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Field                                                 | Description                                                                                                                                                                                                                            |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resolvConf<br>string                                  | resolvConf is the resolver configuration file used as the basis for the container DNS resolution configuration. If set to the empty string, will override the default and effectively disable DNS lookups. Default: "/etc/resolv.conf" |
| runOnce<br>bool                                       | runOnce causes the Kubelet to check the API server once for pods, run those in addition to the pods specified by static pod files, and exit. Default: false                                                                            |
| cpuCFSQuota<br>bool                                   | cpuCFSQuota enables CPU CFS quota enforcement for containers that specify CPU limits. Default: true                                                                                                                                    |
| cpuCFSQuotaPeriod<br><a href="#">meta/v1.Duration</a> | cpuCFSQuotaPeriod is the CPU CFS quota period value, cpu.cfs_period_us. The value must be between 1 ms and 1 second, inclusive. Requires the CustomCPUCFSQuotaPeriod feature gate to be enabled. Default: "100ms"                      |
| nodeStatusMaxImages<br>int32                          | nodeStatusMaxImages caps the number of images reported in Node.status.images. The value must be greater than -2. Note: If -1 is specified, no cap will be applied. If 0 is specified, no image is returned. Default: 50                |
| maxOpenFiles<br>int64                                 | maxOpenFiles is Number of files that can be opened by Kubelet process. The value must be a non-negative number. Default: 1000000                                                                                                       |
| contentType<br>string                                 | contentType is contentType of requests sent to apiserver. Default: "application/vnd.kubernetes.protobuf"                                                                                                                               |
| kubeAPIQPS<br>int32                                   | kubeAPIQPS is the QPS to use while talking with kubernetes apiserver. Default: 50                                                                                                                                                      |
| kubeAPIBurst<br>int32                                 | kubeAPIBurst is the burst to allow while talking with kubernetes API server. This field cannot be a negative number. Default: 100                                                                                                      |

| Field                                                                           | Description                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>serializeImagePulls</p> <p>bool</p>                                          | <p>serializeImagePulls when enabled, tells the Kubelet to pull images one at a time. We recommend <i>not</i> changing the default value on nodes that run docker daemon with version &lt; 1.9 or an Aufs storage backend. Issue #10959 has more details. Default: true</p>                                                             |
| <p>maxParallelImagePulls</p> <p>int32</p>                                       | <p>MaxParallelImagePulls sets the maximum number of image pulls in parallel. This field cannot be set if SerializeImagePulls is true. Setting it to nil means no limit. Default: nil</p>                                                                                                                                               |
| <p>evictionHard</p> <p>map[string]string</p>                                    | <p>evictionHard is a map of signal names to quantities that defines hard eviction thresholds. For example: {"memory.available": "300Mi"}. To explicitly disable, pass a 0% or 100% threshold on an arbitrary resource. Default: memory.available: "100Mi" nodefs.available: "10%" nodefs.inodesFree: "5%" imagefs.available: "15%"</p> |
| <p>evictionSoft</p> <p>map[string]string</p>                                    | <p>evictionSoft is a map of signal names to quantities that defines soft eviction thresholds. For example: {"memory.available": "300Mi"}. Default: nil</p>                                                                                                                                                                             |
| <p>evictionSoftGracePeriod</p> <p>map[string]string</p>                         | <p>evictionSoftGracePeriod is a map of signal names to quantities that defines grace periods for each soft eviction signal. For example: {"memory.available": "30s"}. Default: nil</p>                                                                                                                                                 |
| <p>evictionPressureTransitionPeriod</p> <p><a href="#">meta/v1.Duration</a></p> | <p>evictionPressureTransitionPeriod is the duration for which the kubelet has to wait before transitioning out of an eviction pressure condition. Default: "5m"</p>                                                                                                                                                                    |

| Field                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| evictionMaxPodGracePeriod<br>int32          | evictionMaxPodGracePeriod is the maximum allowed grace period (in seconds) to use when terminating pods in response to a soft eviction threshold being met. This value effectively caps the Pod's terminationGracePeriodSeconds value during soft evictions. Note: Due to issue #64530, the behavior has a bug where this value currently just overrides the grace period during soft eviction, which can increase the grace period from what is set on the Pod. This bug will be fixed in a future release. Default: 0 |
| evictionMinimumReclaim<br>map[string]string | evictionMinimumReclaim is a map of signal names to quantities that defines minimum reclaims, which describe the minimum amount of a given resource the kubelet will reclaim when performing a pod eviction while that resource is under pressure. For example: {"imagefs.available": "2Gi"}. Default: nil                                                                                                                                                                                                               |
| podsPerCore<br>int32                        | podsPerCore is the maximum number of pods per core. Cannot exceed maxPods. The value must be a non-negative integer. If 0, there is no limit on the number of Pods. Default: 0                                                                                                                                                                                                                                                                                                                                          |
| enableControllerAttachDetach<br>bool        | enableControllerAttachDetach enables the Attach/Detach controller to manage attachment/detachment of volumes scheduled to this node, and disables kubelet from executing any attach/detach operations. Note: attaching/detaching CSI volumes is not supported by the kubelet, so this option needs to be true for that use case. Default: true                                                                                                                                                                          |
| protectKernelDefaults<br>bool               | protectKernelDefaults, if true, causes the Kubelet to error if kernel flags are not as it expects. Otherwise the Kubelet will attempt to modify kernel flags to match its expectation. Default: false                                                                                                                                                                                                                                                                                                                   |
| makeIPTablesUtilChains<br>bool              | makeIPTablesUtilChains, if true, causes the Kubelet to create the KUBE-IPTABLES-HINT chain in iptables as a hint to other components about the configuration of iptables on the system. Default: true                                                                                                                                                                                                                                                                                                                   |
| iptablesMasqueradeBit<br>int32              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Field                                                           | Description                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                 | iptablesMasqueradeBit formerly controlled the creation of the KUBE-MARK-MASQ chain. Deprecated: no longer has any effect. Default: 14                                                                                                                                                                        |
| iptablesDropBit<br>int32                                        | iptablesDropBit formerly controlled the creation of the KUBE-MARK-DROP chain. Deprecated: no longer has any effect. Default: 15                                                                                                                                                                              |
| featureGates<br>map[string]bool                                 | featureGates is a map of feature names to bools that enable or disable experimental features. This field modifies piecemeal the built-in default values from "k8s.io/kubernetes/pkg/features/kube_features.go". Default: nil                                                                                 |
| failSwapOn<br>bool                                              | failSwapOn tells the Kubelet to fail to start if swap is enabled on the node. Default: true                                                                                                                                                                                                                  |
| memorySwap<br><a href="#">MemorySwapConfiguration</a>           | memorySwap configures swap memory available to container workloads.                                                                                                                                                                                                                                          |
| containerLogMaxSize<br>string                                   | containerLogMaxSize is a quantity defining the maximum size of the container log file before it is rotated. For example: "5Mi" or "256Ki". Default: "10Mi"                                                                                                                                                   |
| containerLogMaxFiles<br>int32                                   | containerLogMaxFiles specifies the maximum number of container log files that can be present for a container. Default: 5                                                                                                                                                                                     |
| containerLogMaxWorkers<br>int32                                 | ContainerLogMaxWorkers specifies the maximum number of concurrent workers to spawn for performing the log rotate operations. Set this count to 1 for disabling the concurrent log rotation workflows Default: 1                                                                                              |
| containerLogMonitorInterval<br><a href="#">meta/v1.Duration</a> | ContainerLogMonitorInterval specifies the duration at which the container logs are monitored for performing the log rotate operation. This defaults to 10 * time.Seconds. But can be customized to a smaller value based on the log generation rate and the size required to be rotated against Default: 10s |

| Field                                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>configMapAndSecretChangeDetectionStrategy</code><br><a href="#">ResourceChangeDetectionStrategy</a> | <p><code>configMapAndSecretChangeDetectionStrategy</code> is a mode in which ConfigMap and Secret managers are running. Valid values include:</p> <ul style="list-style-type: none"> <li>• Get: kubelet fetches necessary objects directly from the API server;</li> <li>• Cache: kubelet uses TTL cache for object fetched from the API server;</li> <li>• Watch: kubelet uses watches to observe changes to objects that are in its interest.</li> </ul> <p>Default: "Watch"</p>                   |
| <code>systemReserved</code><br><code>map[string]string</code>                                             | <p><code>systemReserved</code> is a set of <code>ResourceName=ResourceQuantity</code> (e.g. <code>cpu=200m,memory=150G</code>) pairs that describe resources reserved for non-kubernetes components. Currently only cpu and memory are supported. See <a href="http://kubernetes.io/docs/user-guide/compute-resources">http://kubernetes.io/docs/user-guide/compute-resources</a> for more detail. Default: nil</p>                                                                                  |
| <code>kubeReserved</code><br><code>map[string]string</code>                                               | <p><code>kubeReserved</code> is a set of <code>ResourceName=ResourceQuantity</code> (e.g. <code>cpu=200m,memory=150G</code>) pairs that describe resources reserved for kubernetes system components. Currently cpu, memory and local storage for root file system are supported. See <a href="https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/">https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/</a> for more details. Default: nil</p> |
| <code>reservedSystemCPUs</code> <b>[Required]</b><br><code>string</code>                                  | <p>The <code>reservedSystemCPUs</code> option specifies the CPU list reserved for the host level system threads and kubernetes related threads. This provide a "static" CPU list rather than the "dynamic" list by <code>systemReserved</code> and <code>kubeReserved</code>. This option does not support <code>systemReservedCgroup</code> or <code>kubeReservedCgroup</code>.</p>                                                                                                                 |
| <code>showHiddenMetricsForVersion</code><br><code>string</code>                                           | <p><code>showHiddenMetricsForVersion</code> is the previous version for which you want to show hidden metrics. Only the previous minor version is meaningful, other values will not be allowed. The format is <code>&lt;major&gt;.&lt;minor&gt;</code>, e.g.: 1.16. The purpose of this format is make sure you have the opportunity to notice if the next release hides</p>                                                                                                                         |

| Field                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | additional metrics, rather than being surprised when they are permanently removed in the release after that. Default: ""                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| systemReservedCgroup<br>string     | systemReservedCgroup helps the kubelet identify absolute name of top level CGroup used to enforce systemReserved compute resource reservation for OS system daemons. Refer to <a href="#">Node Allocatable</a> doc for more information. Default: ""                                                                                                                                                                                                                                                                                                                     |
| kubeReservedCgroup<br>string       | kubeReservedCgroup helps the kubelet identify absolute name of top level CGroup used to enforce KubeReserved compute resource reservation for Kubernetes node system daemons. Refer to <a href="#">Node Allocatable</a> doc for more information. Default: ""                                                                                                                                                                                                                                                                                                            |
| enforceNodeAllocatable<br>[]string | This flag specifies the various Node Allocatable enforcements that Kubelet needs to perform. This flag accepts a list of options. Acceptable options are none, pods, system-reserved and kube-reserved. If none is specified, no other options may be specified. When system-reserved is in the list, systemReservedCgroup must be specified. When kube-reserved is in the list, kubeReservedCgroup must be specified. This field is supported only when cgroupsPerQOS is set to true. Refer to <a href="#">Node Allocatable</a> for more information. Default: ["pods"] |
| allowedUnsafeSysctls<br>[]string   | A comma separated whitelist of unsafe sysctls or sysctl patterns (ending in *). Unsafe sysctl groups are kernel.shm*, kernel.msg*, kernel.sem, fs.mqueue.*, and net.*. For example: "kernel.msg*,net.ipv4.route.min_pmtu" Default: []                                                                                                                                                                                                                                                                                                                                    |
| volumePluginDir<br>string          | volumePluginDir is the full path of the directory in which to search for additional third party volume plugins. Default: "/usr/libexec/kubernetes/kubelet-plugins/volume/exec/"                                                                                                                                                                                                                                                                                                                                                                                          |
| providerID<br>string               | providerID, if set, sets the unique ID of the instance that an external provider (i.e. cloudprovider) can use to identify a specific node. Default: ""                                                                                                                                                                                                                                                                                                                                                                                                                   |
| kernelMemcgNotification<br>bool    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



| Field                                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                        | kernelMemcgNotification, if set, instructs the kubelet to integrate with the kernel memcg notification for determining if memory eviction thresholds are exceeded rather than polling. Default: false                                                                                                                                                                                                                                                                                                                                                 |
| logging <b>[Required]</b><br><a href="#">LoggingConfiguration</a>                      | logging specifies the options of logging. Refer to <a href="#">Logs Options</a> for more information. Default: Format: text                                                                                                                                                                                                                                                                                                                                                                                                                           |
| enableSystemLogHandler<br>bool                                                         | enableSystemLogHandler enables system logs via web interface host:port/logs/ Default: true                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| enableSystemLogQuery<br>bool                                                           | enableSystemLogQuery enables the node log query feature on the /logs endpoint. EnableSystemLogHandler has to be enabled in addition for this feature to work. Default: false                                                                                                                                                                                                                                                                                                                                                                          |
| shutdownGracePeriod<br><a href="#">meta/v1.Duration</a>                                | shutdownGracePeriod specifies the total duration that the node should delay the shutdown and total grace period for pod termination during a node shutdown. Default: "0s"                                                                                                                                                                                                                                                                                                                                                                             |
| shutdownGracePeriodCriticalPods<br><a href="#">meta/v1.Duration</a>                    | shutdownGracePeriodCriticalPods specifies the duration used to terminate critical pods during a node shutdown. This should be less than shutdownGracePeriod. For example, if shutdownGracePeriod=30s, and shutdownGracePeriodCriticalPods=10s, during a node shutdown the first 20 seconds would be reserved for gracefully terminating normal pods, and the last 10 seconds would be reserved for terminating critical pods. Default: "0s"                                                                                                           |
| shutdownGracePeriodByPodPriority<br><a href="#">[]ShutdownGracePeriodByPodPriority</a> | shutdownGracePeriodByPodPriority specifies the shutdown grace period for Pods based on their associated priority class value. When a shutdown request is received, the Kubelet will initiate shutdown on all pods running on the node with a grace period that depends on the priority of the pod, and then wait for all pods to exit. Each entry in the array represents the graceful shutdown time a pod with a priority class value that lies in the range of that value and the next higher entry in the list when the node is shutting down. For |

| Field                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                  | <p>example, to allow critical pods 10s to shutdown, priority&gt;=10000 pods 20s to shutdown, and all remaining pods 30s to shutdown.</p> <p>shutdownGracePeriodByPodPriority:</p> <ul style="list-style-type: none"> <li>• priority: 2000000000<br/>shutdownGracePeriodSeconds: 10</li> <li>• priority: 10000<br/>shutdownGracePeriodSeconds: 20</li> <li>• priority: 0 shutdownGracePeriodSeconds: 30</li> </ul> <p>The time the Kubelet will wait before exiting will at most be the maximum of all shutdownGracePeriodSeconds for each priority class range represented on the node. When all pods have exited or reached their grace periods, the Kubelet will release the shutdown inhibit lock. Requires the GracefulNodeShutdown feature gate to be enabled. This configuration must be empty if either ShutdownGracePeriod or ShutdownGracePeriodCriticalPods is set. Default: nil</p>                                                                                                                                                                                                                                                                                                                  |
| <p>reservedMemory</p> <p><a href="#">[]MemoryReservation</a></p> | <p>reservedMemory specifies a comma-separated list of memory reservations for NUMA nodes. The parameter makes sense only in the context of the memory manager feature. The memory manager will not allocate reserved memory for container workloads. For example, if you have a NUMA0 with 10Gi of memory and the reservedMemory was specified to reserve 1Gi of memory at NUMA0, the memory manager will assume that only 9Gi is available for allocation. You can specify a different amount of NUMA node and memory types. You can omit this parameter at all, but you should be aware that the amount of reserved memory from all NUMA nodes should be equal to the amount of memory specified by the <a href="#">node allocatable</a>. If at least one node allocatable parameter has a non-zero value, you will need to specify at least one NUMA node. Also, avoid specifying:</p> <ol style="list-style-type: none"> <li>1. Duplicates, the same NUMA node, and memory type, but with a different value.</li> <li>2. zero limits for any memory type.</li> <li>3. NUMAs nodes IDs that do not exist under the machine.</li> <li>4. memory types except for memory and hugepages-&lt;size&gt;</li> </ol> |

| Field                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | Default: nil                                                                                                                                                                                                                                                                                                                                                                                                                      |
| enableProfilingHandler<br>bool                        | enableProfilingHandler enables profiling via web interface host:port/debug/pprof/ Default: true                                                                                                                                                                                                                                                                                                                                   |
| enableDebugFlagsHandler<br>bool                       | enableDebugFlagsHandler enables flags endpoint via web interface host:port/debug/flags/v Default: true                                                                                                                                                                                                                                                                                                                            |
| seccompDefault<br>bool                                | SeccompDefault enables the use of RuntimeDefault as the default seccomp profile for all workloads. Default: false                                                                                                                                                                                                                                                                                                                 |
| memoryThrottlingFactor<br>float64                     | MemoryThrottlingFactor specifies the factor multiplied by the memory limit or node allocatable memory when setting the cgroupv2 memory.high value to enforce MemoryQoS. Decreasing this factor will set lower high limit for container cgroups and put heavier reclaim pressure while increasing will put less reclaim pressure. See <a href="https://kep.k8s.io/2570">https://kep.k8s.io/2570</a> for more details. Default: 0.9 |
| registerWithTaints<br><a href="#">[]core/v1.Taint</a> | registerWithTaints are an array of taints to add to a node object when the kubelet registers itself. This only takes effect when registerNode is true and upon the initial registration of the node. Default: nil                                                                                                                                                                                                                 |
| registerNode<br>bool                                  | registerNode enables automatic registration with the apiserver. Default: true                                                                                                                                                                                                                                                                                                                                                     |
| tracing<br><a href="#">TracingConfiguration</a>       | Tracing specifies the versioned configuration for OpenTelemetry tracing clients. See <a href="https://kep.k8s.io/2832">https://kep.k8s.io/2832</a> for more details. Default: nil                                                                                                                                                                                                                                                 |
| localStorageCapacityIsolation<br>bool                 | LocalStorageCapacityIsolation enables local ephemeral storage isolation feature. The default setting is true. This feature allows users to set request/limit for container's ephemeral storage and manage it in a similar way as cpu and memory. It also allows setting sizeLimit for emptyDir volume, which will trigger pod eviction if disk usage from                                                                         |

| Field                                                | Description                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | the volume exceeds the limit. This feature depends on the capability of detecting correct root file system disk usage. For certain systems, such as kind rootless, if this capability cannot be supported, the feature LocalStorageCapacityIsolation should be disabled. Once disabled, user should not set request/limit for container's ephemeral storage, or sizeLimit for emptyDir. Default: true |
| containerRuntimeEndpoint <b>[Required]</b><br>string | ContainerRuntimeEndpoint is the endpoint of container runtime. Unix Domain Sockets are supported on Linux, while npipe and tcp endpoints are supported on Windows. Examples:'unix:///path/to/runtime.sock', 'npipe:////./pipe/runtime'                                                                                                                                                                |
| imageServiceEndpoint<br>string                       | ImageServiceEndpoint is the endpoint of container image service. Unix Domain Socket are supported on Linux, while npipe and tcp endpoints are supported on Windows. Examples:'unix:///path/to/runtime.sock', 'npipe:////./pipe/runtime'. If not specified, the value in containerRuntimeEndpoint is used.                                                                                             |

## SerializedNodeConfigSource

SerializedNodeConfigSource allows us to serialize v1.NodeConfigSource. This type is used internally by the Kubelet for tracking checkpointed dynamic configs. It exists in the kubeletconfig API group because it is classified as a versioned input to the Kubelet.

| Field                                              | Description                                   |
|----------------------------------------------------|-----------------------------------------------|
| apiVersion<br>string                               | kubelet.config.k8s.io/v1beta1                 |
| kind<br>string                                     | SerializedNodeConfigSource                    |
| source<br><a href="#">core/v1.NodeConfigSource</a> | source is the source that we are serializing. |

## CredentialProvider

Appears in:

- [CredentialProviderConfig](#)

CredentialProvider represents an exec plugin to be invoked by the kubelet. The plugin is only invoked when an image being pulled matches the images handled by the plugin (see `matchImages`).

| Field                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>name</code> <b>[Required]</b><br>string                                           | <code>name</code> is the required name of the credential provider. It must match the name of the provider executable as seen by the kubelet. The executable must be in the kubelet's bin directory (set by the <code>--image-credential-provider-bin-dir</code> flag).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>matchImages</code> <b>[Required]</b><br>[]string                                  | <p><code>matchImages</code> is a required list of strings used to match against images in order to determine if this provider should be invoked. If one of the strings matches the requested image from the kubelet, the plugin will be invoked and given a chance to provide credentials. Images are expected to contain the registry domain and URL path.</p> <p>Each entry in <code>matchImages</code> is a pattern which can optionally contain a port and a path. Globs can be used in the domain, but not in the port or the path. Globs are supported as subdomains like <code>*.k8s.io</code> or <code>k8s.*.io</code>, and top-level-domains such as <code>k8s.*</code>. Matching partial subdomains like <code>app*.k8s.io</code> is also supported. Each glob can only match a single subdomain segment, so <code>*.io</code> does not match <code>*.k8s.io</code>.</p> <p>A match exists between an image and a <code>matchImage</code> when all of the below are true:</p> <ul style="list-style-type: none"><li>• Both contain the same number of domain parts and each part matches.</li><li>• The URL path of an <code>imageMatch</code> must be a prefix of the target image URL path.</li><li>• If the <code>imageMatch</code> contains a port, then the port must match in the image as well.</li></ul> <p>Example values of <code>matchImages</code>:</p> <ul style="list-style-type: none"><li>• <code>123456789.dkr.ecr.us-east-1.amazonaws.com</code></li><li>• <code>*.azurecr.io</code></li><li>• <code>gcr.io</code></li><li>• <code>*.*.registry.io</code></li><li>• <code>registry.io:8080/path</code></li></ul> |
| <code>defaultCacheDuration</code> <b>[Required]</b><br><a href="#">meta/v1.Duration</a> | <code>defaultCacheDuration</code> is the default duration the plugin will cache credentials in-memory if a cache duration is not provided in the plugin response. This field is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Field                              | Description                                                                                                                                                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>[Required]<br>string | Required input version of the exec CredentialProviderRequest. The returned CredentialProviderResponse MUST use the same encoding version as the input. Current supported values are: <ul style="list-style-type: none"> <li>credentialprovider.kubelet.k8s.io/v1beta1</li> </ul> |
| args<br>[]string                   | Arguments to pass to the command when executing it.                                                                                                                                                                                                                              |
| env<br>[]ExecEnvVar                | Env defines additional environment variables to expose to the process. These are unioned with the host's environment, as well as variables client-go uses to pass argument to the plugin.                                                                                        |

## ExecEnvVar

Appears in:

- [CredentialProvider](#)

ExecEnvVar is used for setting environment variables when executing an exec-based credential plugin.

| Field                      | Description              |
|----------------------------|--------------------------|
| name [Required]<br>string  | No description provided. |
| value [Required]<br>string | No description provided. |

## KubeletAnonymousAuthentication

Appears in:

- [KubeletAuthentication](#)

| Field           | Description                                                                                                                                                                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| enabled<br>bool | enabled allows anonymous requests to the kubelet server. Requests that are not rejected by another authentication method are treated as anonymous requests. Anonymous requests have a username of system:anonymous, and a group name of system:unauthenticated. |

## KubeletAuthentication

Appears in:

- [KubeletConfiguration](#)

| Field                                                       | Description                                                               |
|-------------------------------------------------------------|---------------------------------------------------------------------------|
| x509<br><a href="#">KubeletX509Authentication</a>           | x509 contains settings related to x509 client certificate authentication. |
| webhook<br><a href="#">KubeletWebhookAuthentication</a>     | webhook contains settings related to webhook bearer token authentication. |
| anonymous<br><a href="#">KubeletAnonymousAuthentication</a> | anonymous contains settings related to anonymous authentication.          |

## KubeletAuthorization

Appears in:

- [KubeletConfiguration](#)

| Field                                                  | Description                                                                                                                                                                                    |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mode<br><a href="#">KubeletAuthorizationMode</a>       | mode is the authorization mode to apply to requests to the kubelet server. Valid values are AlwaysAllow and Webhook. Webhook mode uses the SubjectAccessReview API to determine authorization. |
| webhook<br><a href="#">KubeletWebhookAuthorization</a> | webhook contains settings related to Webhook authorization.                                                                                                                                    |

## KubeletAuthorizationMode

(Alias of string)

Appears in:

- [KubeletAuthorization](#)

## KubeletWebhookAuthentication

Appears in:

- [KubeletAuthentication](#)

| Field           | Description                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------|
| enabled<br>bool | enabled allows bearer token authentication backed by the tokenreviews.authentication.k8s.io API. |

| Field                                        | Description                                        |
|----------------------------------------------|----------------------------------------------------|
| cacheTTL<br><a href="#">meta/v1.Duration</a> | cacheTTL enables caching of authentication results |

## KubeletWebhookAuthorization

Appears in:

- [KubeletAuthorization](#)

| Field                                                    | Description                                                                                         |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| cacheAuthorizedTTL<br><a href="#">meta/v1.Duration</a>   | cacheAuthorizedTTL is the duration to cache 'authorized' responses from the webhook authorizer.     |
| cacheUnauthorizedTTL<br><a href="#">meta/v1.Duration</a> | cacheUnauthorizedTTL is the duration to cache 'unauthorized' responses from the webhook authorizer. |

## KubeletX509Authentication

Appears in:

- [KubeletAuthentication](#)

| Field                  | Description                                                                                                                                                                                                                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clientCAFile<br>string | clientCAFile is the path to a PEM-encoded certificate bundle. If set, any request presenting a client certificate signed by one of the authorities in the bundle is authenticated with a username corresponding to the CommonName, and groups corresponding to the Organization in the client certificate. |

## MemoryReservation

Appears in:

- [KubeletConfiguration](#)

MemoryReservation specifies the memory reservation of different types for each NUMA node

| Field                                                     | Description              |
|-----------------------------------------------------------|--------------------------|
| numaNode [Required]<br>int32                              | No description provided. |
| limits [Required]<br><a href="#">core/v1.ResourceList</a> | No description provided. |



## MemorySwapConfiguration

Appears in:

- [KubeletConfiguration](#)

| Field               | Description                                                                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| swapBehavior string | swapBehavior configures swap memory available to container workloads. May be one of "", "NoSwap": workloads can not use swap, default option. "LimitedSwap": workload swap usage is limited. The swap limit is proportionate to the container's memory request. |

## ResourceChangeDetectionStrategy

(Alias of string)

Appears in:

- [KubeletConfiguration](#)

ResourceChangeDetectionStrategy denotes a mode in which internal managers (secret, configmap) are discovering object changes.

## ShutdownGracePeriodByPodPriority

Appears in:

- [KubeletConfiguration](#)

ShutdownGracePeriodByPodPriority specifies the shutdown grace period for Pods based on their associated priority class value

| Field                                                 | Description                                                              |
|-------------------------------------------------------|--------------------------------------------------------------------------|
| priority <b>[Required]</b><br>int32                   | priority is the priority value associated with the shutdown grace period |
| shutdownGracePeriodSeconds <b>[Required]</b><br>int64 | shutdownGracePeriodSeconds is the shutdown grace period in seconds       |

## Kubelet CredentialProvider (v1)

### Resource Types

- [CredentialProviderRequest](#)
- [CredentialProviderResponse](#)

## CredentialProviderRequest

CredentialProviderRequest includes the image that the kubelet requires authentication for. Kubelet will pass this request object to the plugin via stdin. In general, plugins should prefer responding with the same apiVersion they were sent.

| Field                                | Description                                                                                                                                                                                           |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                 | credentialprovider.kubelet.k8s.io/v1                                                                                                                                                                  |
| kind<br>string                       | CredentialProviderRequest                                                                                                                                                                             |
| image<br><b>[Required]</b><br>string | image is the container image that is being pulled as part of the credential provider plugin request. Plugins may optionally parse the image to extract any information required to fetch credentials. |

## CredentialProviderResponse

CredentialProviderResponse holds credentials that the kubelet should use for the specified image provided in the original request. Kubelet will read the response from the plugin via stdout. This response should be set to the same apiVersion as CredentialProviderRequest.

| Field                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                    | credentialprovider.kubelet.k8s.io/v1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| kind<br>string                                                          | CredentialProviderResponse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| cacheKeyType<br><b>[Required]</b><br><a href="#">PluginCacheKeyType</a> | cacheKeyType indicates the type of caching key to use based on the image provided in the request. There are three valid values for the cache key type: Image, Registry, and Global. If an invalid value is specified, the response will NOT be used by the kubelet.                                                                                                                                                                                                                                                                                                              |
| cacheDuration<br><a href="#">meta/v1.Duration</a>                       | cacheDuration indicates the duration the provided credentials should be cached for. The kubelet will use this field to set the in-memory cache duration for credentials in the AuthConfig. If null, the kubelet will use defaultCacheDuration provided in CredentialProviderConfig. If set to 0, the kubelet will not cache the provided AuthConfig.                                                                                                                                                                                                                             |
| auth<br><a href="#">map[string]AuthConfig</a>                           | <p>auth is a map containing authentication information passed into the kubelet. Each key is a match image string (more on this below). The corresponding authConfig value should be valid for all images that match against this key. A plugin should set this field to null if no valid credentials can be returned for the requested image.</p> <p>Each key in the map is a pattern which can optionally contain a port and a path. Globs can be used in the domain, but not in the port or the path. Globs are supported as subdomains like '*.k8s.io' or 'k8s.*.io', and</p> |

| Field | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <p>top-level-domains such as 'k8s.*'. Matching partial subdomains like 'app*.k8s.io' is also supported. Each glob can only match a single subdomain segment, so '*.io' does not match '*.k8s.io'.</p> <p>The kubelet will match images against the key when all of the below are true:</p> <ul style="list-style-type: none"> <li>• Both contain the same number of domain parts and each part matches.</li> <li>• The URL path of an imageMatch must be a prefix of the target image URL path.</li> <li>• If the imageMatch contains a port, then the port must match in the image as well.</li> </ul> <p>When multiple keys are returned, the kubelet will traverse all keys in reverse order so that:</p> <ul style="list-style-type: none"> <li>• longer keys come before shorter keys with the same prefix</li> <li>• non-wildcard keys come before wildcard keys with the same prefix.</li> </ul> <p>For any given match, the kubelet will attempt an image pull with the provided credentials, stopping after the first successfully authenticated pull.</p> <p>Example keys:</p> <ul style="list-style-type: none"> <li>• 123456789.dkr.ecr.us-east-1.amazonaws.com</li> <li>• *.azurecr.io</li> <li>• gcr.io</li> <li>• *.registry.io</li> <li>• registry.io:8080/path</li> </ul> |

## AuthConfig

Appears in:

- [CredentialProviderResponse](#)

AuthConfig contains authentication information for a container registry. Only username/password based authentication is supported today, but more authentication mechanisms may be added in the future.

| Field                                   | Description                                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| username<br><b>[Required]</b><br>string | username is the username used for authenticating to the container registry<br>An empty username is valid. |

| Field                            | Description                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------|
| password<br>[Required]<br>string | password is the password used for authenticating to the container registry<br>An empty password is valid. |

## PluginCacheKeyType

(Alias of string)

Appears in:

- [CredentialProviderResponse](#)

# WebhookAdmission Configuration (v1)

Package v1 is the v1 version of the API.

## Resource Types

- [WebhookAdmission](#)

## WebhookAdmission

WebhookAdmission provides configuration for the webhook admission controller.

| Field                               | Description                                        |
|-------------------------------------|----------------------------------------------------|
| apiVersion<br>string                | apiserver.config.k8s.io/v1                         |
| kind<br>string                      | WebhookAdmission                                   |
| kubeConfigFile [Required]<br>string | KubeConfigFile is the path to the kubeconfig file. |

## External APIs

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[Kubernetes Custom Metrics \(v1beta2\)](#)

[Kubernetes External Metrics \(v1beta1\)](#)

[Kubernetes Metrics \(v1beta1\)](#)

# Kubernetes Custom Metrics (v1beta2)

Package v1beta2 is the v1beta2 version of the custom\_metrics API.

## Resource Types

- [MetricListOptions](#)
- [MetricValue](#)
- [MetricValueList](#)

## MetricListOptions

MetricListOptions is used to select metrics by their label selectors

| Field                         | Description                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------|
| apiVersion<br>string          | custom.metrics.k8s.io/v1beta2                                                                |
| kind<br>string                | MetricListOptions                                                                            |
| labelSelector<br>string       | A selector to restrict the list of returned objects by their labels. Defaults to everything. |
| metricLabelSelector<br>string | A selector to restrict the list of returned metrics by their labels                          |

## MetricValue

Appears in:

- [MetricValueList](#)

MetricValue is the metric value for some object

| Field                                                                    | Description                                           |
|--------------------------------------------------------------------------|-------------------------------------------------------|
| apiVersion<br>string                                                     | custom.metrics.k8s.io/v1beta2                         |
| kind<br>string                                                           | MetricValue                                           |
| describedObject<br>[Required]<br><a href="#">core/v1.ObjectReference</a> | a reference to the described object                   |
| metric [Required]<br><a href="#">MetricIdentifier</a>                    | No description provided.                              |
| timestamp [Required]<br><a href="#">meta/v1.Time</a>                     | indicates the time at which the metrics were produced |

| Field                                                                                                                                        | Description                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| windowSeconds<br><b>[Required]</b><br>int64                                                                                                  | indicates the window ([Timestamp-Window, Timestamp]) from which these metrics were calculated, when returning rate metrics calculated from cumulative metrics (or zero for non-calculated instantaneous metrics). |
| value <b>[Required]</b><br><a href="https://k8s.io/apimachinery/pkg/api/resource.Quantity">k8s.io/apimachinery/pkg/api/resource.Quantity</a> | the value of the metric for this                                                                                                                                                                                  |

## MetricValueList

MetricValueList is a list of values for a given metric for some set of objects

| Field                                                                                            | Description                                          |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------|
| apiVersion<br>string                                                                             | custom.metrics.k8s.io/v1beta2                        |
| kind<br>string                                                                                   | MetricValueList                                      |
| metadata <b>[Required]</b><br><a href="https://k8s.io/api/meta/v1.ListMeta">meta/v1.ListMeta</a> | No description provided.                             |
| items <b>[Required]</b><br><a href="#">[]MetricValue</a>                                         | the value of the metric across the described objects |

## MetricIdentifier

Appears in:

- [MetricValue](#)

MetricIdentifier identifies a metric by name and, optionally, selector

| Field                                                                                    | Description                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name<br><b>[Required]</b><br>string                                                      | name is the name of the given metric                                                                                                                                                                                                               |
| selector<br><a href="https://k8s.io/api/meta/v1.LabelSelector">meta/v1.LabelSelector</a> | selector represents the label selector that could be used to select this metric, and will generally just be the selector passed in to the query used to fetch this metric. When left blank, only the metric's Name will be used to gather metrics. |

# Kubernetes External Metrics (v1beta1)

Package v1beta1 is the v1beta1 version of the external metrics API.

# Resource Types

- [ExternalMetricValue](#)
- [ExternalMetricValueList](#)

## ExternalMetricValue

Appears in:

- [ExternalMetricValueList](#)

ExternalMetricValue is a metric value for external metric A single metric value is identified by metric name and a set of string labels. For one metric there can be multiple values with different sets of labels.

| Field                                                                                    | Description                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                                                     | external.metrics.k8s.io/v1beta1                                                                                                                                                                                   |
| kind<br>string                                                                           | ExternalMetricValue                                                                                                                                                                                               |
| metricName <b>[Required]</b><br>string                                                   | the name of the metric                                                                                                                                                                                            |
| metricLabels<br><b>[Required]</b><br>map[string]string                                   | a set of labels that identify a single time series for the metric                                                                                                                                                 |
| timestamp <b>[Required]</b><br><a href="#">meta/v1.Time</a>                              | indicates the time at which the metrics were produced                                                                                                                                                             |
| window <b>[Required]</b><br>int64                                                        | indicates the window ([Timestamp-Window, Timestamp]) from which these metrics were calculated, when returning rate metrics calculated from cumulative metrics (or zero for non-calculated instantaneous metrics). |
| value <b>[Required]</b><br><a href="#">k8s.io/apimachinery/pkg/api/resource.Quantity</a> | the value of the metric                                                                                                                                                                                           |

## ExternalMetricValueList

ExternalMetricValueList is a list of values for a given metric for some set labels

| Field                | Description                     |
|----------------------|---------------------------------|
| apiVersion<br>string | external.metrics.k8s.io/v1beta1 |
| kind<br>string       | ExternalMetricValueList         |
|                      | No description provided.        |

| Field                                                              | Description                                        |
|--------------------------------------------------------------------|----------------------------------------------------|
| metadata [ <b>Required</b> ]<br><a href="#">meta/v1.ListMeta</a>   |                                                    |
| items [ <b>Required</b> ]<br><a href="#">[]ExternalMetricValue</a> | value of the metric matching a given set of labels |

# Kubernetes Metrics (v1beta1)

Package v1beta1 is the v1beta1 version of the metrics API.

## Resource Types

- [NodeMetrics](#)
- [NodeMetricsList](#)
- [PodMetrics](#)
- [PodMetricsList](#)

## NodeMetrics

Appears in:

- [NodeMetricsList](#)

NodeMetrics sets resource usage metrics of a node.

| Field                                                             | Description                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                              | metrics.k8s.io/v1beta1                                                                                                                                                                                                                                                                                                                |
| kind<br>string                                                    | NodeMetrics                                                                                                                                                                                                                                                                                                                           |
| metadata<br><a href="#">meta/v1.ObjectMeta</a>                    | Standard object's metadata. More info: <a href="https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata">https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata</a><br><br>Refer to the Kubernetes API documentation for the fields of the metadata field. |
| timestamp<br>[ <b>Required</b> ]<br><a href="#">meta/v1.Time</a>  | The following fields define time interval from which metrics were collected from the interval [Timestamp-Window, Timestamp].                                                                                                                                                                                                          |
| window<br>[ <b>Required</b> ]<br><a href="#">meta/v1.Duration</a> | No description provided.                                                                                                                                                                                                                                                                                                              |
| usage [ <b>Required</b> ]<br><a href="#">core/v1.ResourceList</a> | The memory usage is the memory working set.                                                                                                                                                                                                                                                                                           |



## NodeMetricsList

NodeMetricsList is a list of NodeMetrics.

| Field                                                      | Description                                                                                                                                                                                                                                      |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                       | metrics.k8s.io/v1beta1                                                                                                                                                                                                                           |
| kind<br>string                                             | NodeMetricsList                                                                                                                                                                                                                                  |
| metadata<br>[Required]<br><a href="#">meta/v1.ListMeta</a> | Standard list metadata. More info: <a href="https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds">https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds</a> |
| items<br>[Required]<br><a href="#">[]NodeMetrics</a>       | List of node metrics.                                                                                                                                                                                                                            |

## PodMetrics

Appears in:

- [PodMetricsList](#)

PodMetrics sets resource usage metrics of a pod.

| Field                                                          | Description                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                           | metrics.k8s.io/v1beta1                                                                                                                                                                                                                                                                                                                |
| kind<br>string                                                 | PodMetrics                                                                                                                                                                                                                                                                                                                            |
| metadata<br><a href="#">meta/v1.ObjectMeta</a>                 | Standard object's metadata. More info: <a href="https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata">https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata</a><br><br>Refer to the Kubernetes API documentation for the fields of the metadata field. |
| timestamp<br>[Required]<br><a href="#">meta/v1.Time</a>        | The following fields define time interval from which metrics were collected from the interval [Timestamp-Window, Timestamp].                                                                                                                                                                                                          |
| window [Required]<br><a href="#">meta/v1.Duration</a>          | No description provided.                                                                                                                                                                                                                                                                                                              |
| containers<br>[Required]<br><a href="#">[]ContainerMetrics</a> | Metrics for all containers are collected within the same time window.                                                                                                                                                                                                                                                                 |

## PodMetricsList

PodMetricsList is a list of PodMetrics.

| Field                                                      | Description                                                                                                                                                                                                                                      |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion<br>string                                       | metrics.k8s.io/v1beta1                                                                                                                                                                                                                           |
| kind<br>string                                             | PodMetricsList                                                                                                                                                                                                                                   |
| metadata<br>[Required]<br><a href="#">meta/v1.ListMeta</a> | Standard list metadata. More info: <a href="https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds">https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds</a> |
| items<br>[Required]<br><a href="#">[]PodMetrics</a>        | List of pod metrics.                                                                                                                                                                                                                             |

## ContainerMetrics

Appears in:

- [PodMetrics](#)

ContainerMetrics sets resource usage metrics of a container.

| Field                                                    | Description                                                       |
|----------------------------------------------------------|-------------------------------------------------------------------|
| name [Required]<br>string                                | Container name corresponding to the one from pod.spec.containers. |
| usage [Required]<br><a href="#">core/v1.ResourceList</a> | The memory usage is the memory working set.                       |

## Scheduling

---

[Scheduler Configuration](#)

[Scheduling Policies](#)

## Scheduler Configuration

FEATURE STATE: Kubernetes v1.25 [stable]

You can customize the behavior of the kube-scheduler by writing a configuration file and passing its path as a command line argument.

A scheduling Profile allows you to configure the different stages of scheduling in the [kube-scheduler](#). Each stage is exposed in an extension point. Plugins provide scheduling behaviors by implementing one or more of these extension points.

You can specify scheduling profiles by running kube-scheduler --config <filename>, using the KubeSchedulerConfiguration [v1](#) struct.

A minimal configuration looks as follows:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
clientConnection:
 kubeconfig: /etc/srv/kubernetes/kube-scheduler/kubeconfig
```

**Note:**

KubeSchedulerConfiguration v1beta3 is deprecated in v1.26 and is removed in v1.29. Please migrate KubeSchedulerConfiguration to [v1](#).

## Profiles

A scheduling Profile allows you to configure the different stages of scheduling in the [kube-scheduler](#). Each stage is exposed in an [extension point](#). [Plugins](#) provide scheduling behaviors by implementing one or more of these extension points.

You can configure a single instance of kube-scheduler to run [multiple profiles](#).

## Extension points

Scheduling happens in a series of stages that are exposed through the following extension points:

1. **queueSort**: These plugins provide an ordering function that is used to sort pending Pods in the scheduling queue. Exactly one queue sort plugin may be enabled at a time.
2. **preFilter**: These plugins are used to pre-process or check information about a Pod or the cluster before filtering. They can mark a pod as unschedulable.
3. **filter**: These plugins are the equivalent of Predicates in a scheduling Policy and are used to filter out nodes that can not run the Pod. Filters are called in the configured order. A pod is marked as unschedulable if no nodes pass all the filters.
4. **postFilter**: These plugins are called in their configured order when no feasible nodes were found for the pod. If any postFilter plugin marks the Pod *schedulable*, the remaining plugins are not called.
5. **preScore**: This is an informational extension point that can be used for doing pre-scoring work.
6. **score**: These plugins provide a score to each node that has passed the filtering phase. The scheduler will then select the node with the highest weighted scores sum.
7. **reserve**: This is an informational extension point that notifies plugins when resources have been reserved for a given Pod. Plugins also implement an Unreserve call that gets called in the case of failure during or after Reserve.
8. **permit**: These plugins can prevent or delay the binding of a Pod.
9. **preBind**: These plugins perform any work required before a Pod is bound.
10. **bind**: The plugins bind a Pod to a Node. bind plugins are called in order and once one has done the binding, the remaining plugins are skipped. At least one bind plugin is required.
11. **postBind**: This is an informational extension point that is called after a Pod has been bound.
12. **multiPoint**: This is a config-only field that allows plugins to be enabled or disabled for all of their applicable extension points simultaneously.

For each extension point, you could disable specific [default plugins](#) or enable your own. For example:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- plugins:
 score:
 disabled:
 - name: PodTopologySpread
 enabled:
 - name: MyCustomPluginA
 weight: 2
 - name: MyCustomPluginB
 weight: 1
```

You can use \* as name in the disabled array to disable all default plugins for that extension point. This can also be used to rearrange plugins order, if desired.

## Scheduling plugins

The following plugins, enabled by default, implement one or more of these extension points:

- **ImageLocality**: Favors nodes that already have the container images that the Pod runs. Extension points: score.
- **TaintToleration**: Implements [taints and tolerations](#). Implements extension points: filter, preScore, score.
- **NodeName**: Checks if a Pod spec node name matches the current node. Extension points: filter.
- **NodePorts**: Checks if a node has free ports for the requested Pod ports. Extension points: preFilter, filter.
- **NodeAffinity**: Implements [node selectors](#) and [node affinity](#). Extension points: filter, score.
- **PodTopologySpread**: Implements [Pod topology spread](#). Extension points: preFilter, filter, preScore, score.
- **NodeUnschedulable**: Filters out nodes that have .spec.unschedulable set to true. Extension points: filter.
- **NodeResourcesFit**: Checks if the node has all the resources that the Pod is requesting. The score can use one of three strategies: LeastAllocated (default), MostAllocated and RequestedToCapacityRatio. Extension points: preFilter, filter, score.
- **NodeResourcesBalancedAllocation**: Favors nodes that would obtain a more balanced resource usage if the Pod is scheduled there. Extension points: score.
- **VolumeBinding**: Checks if the node has or if it can bind the requested [volumes](#). Extension points: preFilter, filter, reserve, preBind, score.

### Note:

score extension point is enabled when VolumeCapacityPriority feature is enabled. It prioritizes the smallest PVs that can fit the requested volume size.

- **VolumeRestrictions**: Checks that volumes mounted in the node satisfy restrictions that are specific to the volume provider. Extension points: filter.
- **VolumeZone**: Checks that volumes requested satisfy any zone requirements they might have. Extension points: filter.

- **NodeVolumeLimits**: Checks that CSI volume limits can be satisfied for the node. Extension points: filter.
- **EBSLimits**: Checks that AWS EBS volume limits can be satisfied for the node. Extension points: filter.
- **GCEPDLimits**: Checks that GCP-PD volume limits can be satisfied for the node. Extension points: filter.
- **AzureDiskLimits**: Checks that Azure disk volume limits can be satisfied for the node. Extension points: filter.
- **InterPodAffinity**: Implements [inter-Pod affinity and anti-affinity](#). Extension points: preFilter, filter, preScore, score.
- **PrioritySort**: Provides the default priority based sorting. Extension points: queueSort.
- **DefaultBinder**: Provides the default binding mechanism. Extension points: bind.
- **DefaultPreemption**: Provides the default preemption mechanism. Extension points: postFilter.

You can also enable the following plugins, through the component config APIs, that are not enabled by default:

- **CinderLimits**: Checks that [OpenStack Cinder](#) volume limits can be satisfied for the node. Extension points: filter.

## Multiple profiles

You can configure kube-scheduler to run more than one profile. Each profile has an associated scheduler name and can have a different set of plugins configured in its [extension points](#).

With the following sample configuration, the scheduler will run with two profiles: one with the default plugins and one with all scoring plugins disabled.

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: default-scheduler
- schedulerName: no-scoring-scheduler
plugins:
 preScore:
 disabled:
 - name: '*'
 score:
 disabled:
 - name: '*'
```

Pods that want to be scheduled according to a specific profile can include the corresponding scheduler name in its `.spec.schedulerName`.

By default, one profile with the scheduler name `default-scheduler` is created. This profile includes the default plugins described above. When declaring more than one profile, a unique scheduler name for each of them is required.

If a Pod doesn't specify a scheduler name, kube-apiserver will set it to `default-scheduler`. Therefore, a profile with this scheduler name should exist to get those pods scheduled.

**Note:**

Pod's scheduling events have `.spec.schedulerName` as their `reportingController`. Events for leader election use the scheduler name of the first profile in the list.

For more information, please refer to the `reportingController` section under [Event API Reference](#).

**Note:**

All profiles must use the same plugin in the `queueSort` extension point and have the same configuration parameters (if applicable). This is because the scheduler only has one pending pods queue.

## Plugins that apply to multiple extension points

Starting from `kubescheduler.config.k8s.io/v1beta3`, there is an additional field in the profile config, `multiPoint`, which allows for easily enabling or disabling a plugin across several extension points. The intent of `multiPoint` config is to simplify the configuration needed for users and administrators when using custom profiles.

Consider a plugin, `MyPlugin`, which implements the `preScore`, `score`, `preFilter`, and `filter` extension points. To enable `MyPlugin` for all its available extension points, the profile config looks like:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: multipoint-scheduler
 plugins:
 multiPoint:
 enabled:
 - name: MyPlugin
```

This would equate to manually enabling `MyPlugin` for all of its extension points, like so:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: non-multipoint-scheduler
 plugins:
 preScore:
 enabled:
 - name: MyPlugin
 score:
 enabled:
 - name: MyPlugin
 preFilter:
 enabled:
 - name: MyPlugin
 filter:
 enabled:
 - name: MyPlugin
```

One benefit of using `multiPoint` here is that if `MyPlugin` implements another extension point in the future, the `multiPoint` config will automatically enable it for the new extension.

Specific extension points can be excluded from `MultiPoint` expansion using the `disabled` field for that extension point. This works with disabling default plugins, non-default plugins, or with the wildcard (`*`) to disable all plugins. An example of this, disabling `Score` and `PreScore`, would be:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: non-multipoint-scheduler
 plugins:
 multiPoint:
 enabled:
 - name: 'MyPlugin'
 preScore:
 disabled:
 - name: '*'
 score:
 disabled:
 - name: '*'
```

Starting from `kubescheduler.config.k8s.io/v1beta3`, all [default plugins](#) are enabled internally through `MultiPoint`. However, individual extension points are still available to allow flexible reconfiguration of the default values (such as ordering and `Score` weights). For example, consider two `Score` plugins `DefaultScore1` and `DefaultScore2`, each with a weight of 1. They can be reordered with different weights like so:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: multipoint-scheduler
 plugins:
 score:
 enabled:
 - name: 'DefaultScore2'
 weight: 5
```

In this example, it's unnecessary to specify the plugins in `MultiPoint` explicitly because they are default plugins. And the only plugin specified in `Score` is `DefaultScore2`. This is because plugins set through specific extension points will always take precedence over `MultiPoint` plugins. So, this snippet essentially re-orders the two plugins without needing to specify both of them.

The general hierarchy for precedence when configuring `MultiPoint` plugins is as follows:

1. Specific extension points run first, and their settings override whatever is set elsewhere
2. Plugins manually configured through `MultiPoint` and their settings
3. Default plugins and their default settings

To demonstrate the above hierarchy, the following example is based on these plugins:

| Plugin           | Extension Points |
|------------------|------------------|
| DefaultQueueSort | QueueSort        |

| Plugin          | Extension Points |
|-----------------|------------------|
| CustomQueueSort | QueueSort        |
| DefaultPlugin1  | Score, Filter    |
| DefaultPlugin2  | Score            |
| CustomPlugin1   | Score, Filter    |
| CustomPlugin2   | Score, Filter    |

A valid sample configuration for these plugins would be:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: multipoint-scheduler
 plugins:
 multiPoint:
 enabled:
 - name: 'CustomQueueSort'
 - name: 'CustomPlugin1'
 weight: 3
 disabled:
 - name: 'DefaultQueueSort'
 filter:
 disabled:
 - name: 'DefaultPlugin1'
 score:
 enabled:
 - name: 'DefaultPlugin2'
```

Note that there is no error for re-declaring a MultiPoint plugin in a specific extension point. The re-declaration is ignored (and logged), as specific extension points take precedence.

Besides keeping most of the config in one spot, this sample does a few things:

- Enables the custom queueSort plugin and disables the default one
- Enables CustomPlugin1 and CustomPlugin2, which will run first for all of their extension points
- Disables DefaultPlugin1, but only for filter
- Reorders DefaultPlugin2 to run first in score (even before the custom plugins)

In versions of the config before v1beta3, without multiPoint, the above snippet would equate to this:

```
apiVersion: kubescheduler.config.k8s.io/v1beta2
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: multipoint-scheduler
 plugins:

 # Disable the default QueueSort plugin
 queueSort:
 enabled:
 - name: 'CustomQueueSort'
```



```

disabled:
- name: 'DefaultQueueSort'

Enable custom Filter plugins
filter:
 enabled:
 - name: 'CustomPlugin1'
 - name: 'CustomPlugin2'
 - name: 'DefaultPlugin2'
 disabled:
 - name: 'DefaultPlugin1'

Enable and reorder custom score plugins
score:
 enabled:
 - name: 'DefaultPlugin2'
 weight: 1
 - name: 'DefaultPlugin1'
 weight: 3

```

While this is a complicated example, it demonstrates the flexibility of MultiPoint config as well as its seamless integration with the existing methods for configuring extension points.

## Scheduler configuration migrations

- [v1beta1 → v1beta2](#)
- [v1beta2 → v1beta3](#)
- [v1beta3 → v1](#)
- With the v1beta2 configuration version, you can use a new score extension for the NodeResourcesFit plugin. The new extension combines the functionalities of the NodeResourcesLeastAllocated, NodeResourcesMostAllocated and RequestedToCapacityRatio plugins. For example, if you previously used the NodeResourcesMostAllocated plugin, you would instead use NodeResourcesFit (enabled by default) and add a pluginConfig with a scoreStrategy that is similar to:

```

apiVersion: kubescheduler.config.k8s.io/v1beta2
kind: KubeSchedulerConfiguration
profiles:
- pluginConfig:
 - args:
 scoringStrategy:
 resources:
 - name: cpu
 weight: 1
 type: MostAllocated
 name: NodeResourcesFit

```

- The scheduler plugin NodeLabel is deprecated; instead, use the [NodeAffinity](#) plugin (enabled by default) to achieve similar behavior.
- The scheduler plugin ServiceAffinity is deprecated; instead, use the [InterPodAffinity](#) plugin (enabled by default) to achieve similar behavior.

- The scheduler plugin `NodePreferAvoidPods` is deprecated; instead, use [node taints](#) to achieve similar behavior.
- A plugin enabled in a `v1beta2` configuration file takes precedence over the default configuration for that plugin.
- Invalid host or port configured for scheduler healthz and metrics bind address will cause validation failure.
- Three plugins' weight are increased by default:
  - `InterPodAffinity` from 1 to 2
  - `NodeAffinity` from 1 to 2
  - `TaintToleration` from 1 to 3
- The scheduler plugin `SelectorSpread` is removed, instead, use the `PodTopologySpread` plugin (enabled by default) to achieve similar behavior.

## What's next

- Read the [kube-scheduler reference](#)
- Learn about [scheduling](#)
- Read the [kube-scheduler configuration \(v1\)](#) reference

# Scheduling Policies

In Kubernetes versions before `v1.23`, a scheduling policy can be used to specify the *predicates* and *priorities* process. For example, you can set a scheduling policy by running `kube-scheduler --policy-config-file <filename>` or `kube-scheduler --policy-configmap <ConfigMap>`.

This scheduling policy is not supported since Kubernetes `v1.23`. Associated flags `policy-config-file`, `policy-configmap`, `policy-configmap-namespace` and `use-legacy-policy-config` are also not supported. Instead, use the [Scheduler Configuration](#) to achieve similar behavior.

## What's next

- Learn about [scheduling](#)
- Learn about [kube-scheduler Configuration](#)
- Read the [kube-scheduler configuration reference \(v1\)](#)

# Other Tools

Kubernetes contains several tools to help you work with the Kubernetes system.

## crictl

[crictl](#) is a command-line interface for inspecting and debugging [CRI](#)-compatible container runtimes.

# Dashboard

[Dashboard](#), the web-based user interface of Kubernetes, allows you to deploy containerized applications to a Kubernetes cluster, troubleshoot them, and manage the cluster and its resources itself.

## Helm

This item links to a third party project or product that is not part of Kubernetes itself. [More information](#)

[Helm](#) is a tool for managing packages of pre-configured Kubernetes resources. These packages are known as *Helm charts*.

Use Helm to:

- Find and use popular software packaged as Kubernetes charts
- Share your own applications as Kubernetes charts
- Create reproducible builds of your Kubernetes applications
- Intelligently manage your Kubernetes manifest files
- Manage releases of Helm packages

## Kompose

[Kompose](#) is a tool to help Docker Compose users move to Kubernetes.

Use Kompose to:

- Translate a Docker Compose file into Kubernetes objects
- Go from local Docker development to managing your application via Kubernetes
- Convert v1 or v2 Docker Compose yaml files or [Distributed Application Bundles](#)

## Kui

[Kui](#) is a GUI tool that takes your normal kubectl command line requests and responds with graphics.

Kui takes the normal kubectl command line requests and responds with graphics. Instead of ASCII tables, Kui provides a GUI rendering with tables that you can sort.

Kui lets you:

- Directly click on long, auto-generated resource names instead of copying and pasting
- Type in kubectl commands and see them execute, even sometimes faster than kubectl itself
- Query a [Job](#) and see its execution rendered as a waterfall diagram
- Click through resources in your cluster using a tabbed UI

# Minikube

[minikube](#) is a tool that runs a single-node Kubernetes cluster locally on your workstation for development and testing purposes.

## Mapping from dockercli to crictl

### Note:

This page is being directed to <https://v1-24.docs.kubernetes.io/docs/reference/tools/map-crictl-dockercli/> because of the [removal of dockershim from crictl in v1.24](#). As per our community policy, deprecated documents are not maintained beyond next three versions. The reason for deprecation is explained in [Dockershim-FAQ](#).

This section lists the different ways to set up and run Kubernetes. When you install Kubernetes, choose an installation type based on: ease of maintenance, security, control, available resources, and expertise required to operate and manage a cluster.

You can [download Kubernetes](#) to deploy a Kubernetes cluster on a local machine, into the cloud, or for your own datacenter.

Several [Kubernetes components](#) such as [kube-apiserver](#) or [kube-proxy](#) can also be deployed as [container images](#) within the cluster.

It is **recommended** to run Kubernetes components as container images wherever that is possible, and to have Kubernetes manage those components. Components that run containers - notably, the kubelet - can't be included in this category.

If you don't want to manage a Kubernetes cluster yourself, you could pick a managed service, including [certified platforms](#). There are also other standardized and custom solutions across a wide range of cloud and bare metal environments.

## Learning environment

If you're learning Kubernetes, use the tools supported by the Kubernetes community, or tools in the ecosystem to set up a Kubernetes cluster on a local machine. See [Install tools](#).

## Production environment

When evaluating a solution for a [production environment](#), consider which aspects of operating a Kubernetes cluster (or *abstractions*) you want to manage yourself and which you prefer to hand off to a provider.

For a cluster you're managing yourself, the officially supported tool for deploying Kubernetes is [kubeadm](#).

## What's next

- [Download Kubernetes](#)
- Download and [install tools](#) including kubect1
- Select a [container runtime](#) for your new cluster
- Learn about [best practices](#) for cluster setup

Kubernetes is designed for its [control plane](#) to run on Linux. Within your cluster you can run applications on Linux or other operating systems, including Windows.

- Learn to [set up clusters with Windows nodes](#)

## Install Tools

Set up Kubernetes tools on your computer.

# kubectl

The Kubernetes command-line tool, [kubectl](#), allows you to run commands against Kubernetes clusters. You can use kubectl to deploy applications, inspect and manage cluster resources, and view logs. For more information including a complete list of kubectl operations, see the [kubectl reference documentation](#).

kubectl is installable on a variety of Linux platforms, macOS and Windows. Find your preferred operating system below.

- [Install kubectl on Linux](#)
- [Install kubectl on macOS](#)
- [Install kubectl on Windows](#)

# kind

[kind](#) lets you run Kubernetes on your local computer. This tool requires that you have either [Docker](#) or [Podman](#) installed.

The kind [Quick Start](#) page shows you what you need to do to get up and running with kind.

[View kind Quick Start Guide](#)

# minikube

Like kind, [minikube](#) is a tool that lets you run Kubernetes locally. minikube runs an all-in-one or a multi-node local Kubernetes cluster on your personal computer (including Windows, macOS and Linux PCs) so that you can try out Kubernetes, or for daily development work.

You can follow the official [Get Started!](#) guide if your focus is on getting the tool installed.

[View minikube Get Started! Guide](#)

Once you have minikube working, you can use it to [run a sample application](#).

# kubeadm

You can use the [kubeadm](#) tool to create and manage Kubernetes clusters. It performs the actions necessary to get a minimum viable, secure cluster up and running in a user friendly way.

[Installing kubeadm](#) shows you how to install kubeadm. Once installed, you can use it to [create a cluster](#).

[View kubeadm Install Guide](#)

# Production environment

Create a production-quality Kubernetes cluster

A production-quality Kubernetes cluster requires planning and preparation. If your Kubernetes cluster is to run critical workloads, it must be configured to be resilient. This page explains steps you can take to set up a production-ready cluster, or to promote an existing cluster for production use. If you're already familiar with production setup and want the links, skip to [What's next](#).

## Production considerations

Typically, a production Kubernetes cluster environment has more requirements than a personal learning, development, or test environment Kubernetes. A production environment may require secure access by many users, consistent availability, and the resources to adapt to changing demands.

As you decide where you want your production Kubernetes environment to live (on premises or in a cloud) and the amount of management you want to take on or hand to others, consider how your requirements for a Kubernetes cluster are influenced by the following issues:

- *Availability:* A single-machine Kubernetes [learning environment](#) has a single point of failure. Creating a highly available cluster means considering:
  - Separating the control plane from the worker nodes.
  - Replicating the control plane components on multiple nodes.
  - Load balancing traffic to the cluster's [API server](#).
  - Having enough worker nodes available, or able to quickly become available, as changing workloads warrant it.
- *Scale:* If you expect your production Kubernetes environment to receive a stable amount of demand, you might be able to set up for the capacity you need and be done. However, if you expect demand to grow over time or change dramatically based on things like season or special events, you need to plan how to scale to relieve increased pressure from more requests to the control plane and worker nodes or scale down to reduce unused resources.
- *Security and access management:* You have full admin privileges on your own Kubernetes learning cluster. But shared clusters with important workloads, and more than one or two users, require a more refined approach to who and what can access cluster resources. You can use role-based access control ([RBAC](#)) and other security mechanisms to make sure that users and workloads can get access to the resources they need, while keeping workloads, and the cluster itself, secure. You can set limits on the resources that users and workloads can access by managing [policies](#) and [container resources](#).

Before building a Kubernetes production environment on your own, consider handing off some or all of this job to [Turnkey Cloud Solutions](#) providers or other [Kubernetes Partners](#). Options include:

- *Serverless:* Just run workloads on third-party equipment without managing a cluster at all. You will be charged for things like CPU usage, memory, and disk requests.
- *Managed control plane:* Let the provider manage the scale and availability of the cluster's control plane, as well as handle patches and upgrades.
- *Managed worker nodes:* Configure pools of nodes to meet your needs, then the provider makes sure those nodes are available and ready to implement upgrades when needed.

- *Integration:* There are providers that integrate Kubernetes with other services you may need, such as storage, container registries, authentication methods, and development tools.

Whether you build a production Kubernetes cluster yourself or work with partners, review the following sections to evaluate your needs as they relate to your cluster's *control plane*, *worker nodes*, *user access*, and *workload resources*.

## Production cluster setup

In a production-quality Kubernetes cluster, the control plane manages the cluster from services that can be spread across multiple computers in different ways. Each worker node, however, represents a single entity that is configured to run Kubernetes pods.

### Production control plane

The simplest Kubernetes cluster has the entire control plane and worker node services running on the same machine. You can grow that environment by adding worker nodes, as reflected in the diagram illustrated in [Kubernetes Components](#). If the cluster is meant to be available for a short period of time, or can be discarded if something goes seriously wrong, this might meet your needs.

If you need a more permanent, highly available cluster, however, you should consider ways of extending the control plane. By design, one-machine control plane services running on a single machine are not highly available. If keeping the cluster up and running and ensuring that it can be repaired if something goes wrong is important, consider these steps:

- *Choose deployment tools:* You can deploy a control plane using tools such as kubeadm, kops, and kubespray. See [Installing Kubernetes with deployment tools](#) to learn tips for production-quality deployments using each of those deployment methods. Different [Container Runtimes](#) are available to use with your deployments.
- *Manage certificates:* Secure communications between control plane services are implemented using certificates. Certificates are automatically generated during deployment or you can generate them using your own certificate authority. See [PKI certificates and requirements](#) for details.
- *Configure load balancer for apiserver:* Configure a load balancer to distribute external API requests to the apiserver service instances running on different nodes. See [Create an External Load Balancer](#) for details.
- *Separate and backup etcd service:* The etcd services can either run on the same machines as other control plane services or run on separate machines, for extra security and availability. Because etcd stores cluster configuration data, backing up the etcd database should be done regularly to ensure that you can repair that database if needed. See the [etcd FAQ](#) for details on configuring and using etcd. See [Operating etcd clusters for Kubernetes](#) and [Set up a High Availability etcd cluster with kubeadm](#) for details.
- *Create multiple control plane systems:* For high availability, the control plane should not be limited to a single machine. If the control plane services are run by an init service (such as systemd), each service should run on at least three machines. However, running control plane services as pods in Kubernetes ensures that the replicated number of services that you request will always be available. The scheduler should be fault tolerant, but not highly available. Some deployment tools set up [Raft](#) consensus algorithm to do leader election of Kubernetes services. If the primary goes away, another service elects itself and take over.



- *Span multiple zones:* If keeping your cluster available at all times is critical, consider creating a cluster that runs across multiple data centers, referred to as zones in cloud environments. Groups of zones are referred to as regions. By spreading a cluster across multiple zones in the same region, it can improve the chances that your cluster will continue to function even if one zone becomes unavailable. See [Running in multiple zones](#) for details.
- *Manage on-going features:* If you plan to keep your cluster over time, there are tasks you need to do to maintain its health and security. For example, if you installed with kubeadm, there are instructions to help you with [Certificate Management](#) and [Upgrading kubeadm clusters](#). See [Administer a Cluster](#) for a longer list of Kubernetes administrative tasks.

To learn about available options when you run control plane services, see [kube-apiserver](#), [kube-controller-manager](#), and [kube-scheduler](#) component pages. For highly available control plane examples, see [Options for Highly Available topology](#), [Creating Highly Available clusters with kubeadm](#), and [Operating etcd clusters for Kubernetes](#). See [Backing up an etcd cluster](#) for information on making an etcd backup plan.

## Production worker nodes

Production-quality workloads need to be resilient and anything they rely on needs to be resilient (such as CoreDNS). Whether you manage your own control plane or have a cloud provider do it for you, you still need to consider how you want to manage your worker nodes (also referred to simply as *nodes*).

- *Configure nodes:* Nodes can be physical or virtual machines. If you want to create and manage your own nodes, you can install a supported operating system, then add and run the appropriate [Node services](#). Consider:
  - The demands of your workloads when you set up nodes by having appropriate memory, CPU, and disk speed and storage capacity available.
  - Whether generic computer systems will do or you have workloads that need GPU processors, Windows nodes, or VM isolation.
- *Validate nodes:* See [Valid node setup](#) for information on how to ensure that a node meets the requirements to join a Kubernetes cluster.
- *Add nodes to the cluster:* If you are managing your own cluster you can add nodes by setting up your own machines and either adding them manually or having them register themselves to the cluster's apiserver. See the [Nodes](#) section for information on how to set up Kubernetes to add nodes in these ways.
- *Scale nodes:* Have a plan for expanding the capacity your cluster will eventually need. See [Considerations for large clusters](#) to help determine how many nodes you need, based on the number of pods and containers you need to run. If you are managing nodes yourself, this can mean purchasing and installing your own physical equipment.
- *Autoscale nodes:* Read [Cluster Autoscaling](#) to learn about the tools available to automatically manage your nodes and the capacity they provide.
- *Set up node health checks:* For important workloads, you want to make sure that the nodes and pods running on those nodes are healthy. Using the [Node Problem Detector](#) daemon, you can ensure your nodes are healthy.

## Production user management

In production, you may be moving from a model where you or a small group of people are accessing the cluster to where there may potentially be dozens or hundreds of people. In a

learning environment or platform prototype, you might have a single administrative account for everything you do. In production, you will want more accounts with different levels of access to different namespaces.

Taking on a production-quality cluster means deciding how you want to selectively allow access by other users. In particular, you need to select strategies for validating the identities of those who try to access your cluster (authentication) and deciding if they have permissions to do what they are asking (authorization):

- *Authentication*: The apiserver can authenticate users using client certificates, bearer tokens, an authenticating proxy, or HTTP basic auth. You can choose which authentication methods you want to use. Using plugins, the apiserver can leverage your organization's existing authentication methods, such as LDAP or Kerberos. See [Authentication](#) for a description of these different methods of authenticating Kubernetes users.
- *Authorization*: When you set out to authorize your regular users, you will probably choose between RBAC and ABAC authorization. See [Authorization Overview](#) to review different modes for authorizing user accounts (as well as service account access to your cluster):
  - *Role-based access control (RBAC)*: Lets you assign access to your cluster by allowing specific sets of permissions to authenticated users. Permissions can be assigned for a specific namespace (Role) or across the entire cluster (ClusterRole). Then using RoleBindings and ClusterRoleBindings, those permissions can be attached to particular users.
  - *Attribute-based access control (ABAC)*: Lets you create policies based on resource attributes in the cluster and will allow or deny access based on those attributes. Each line of a policy file identifies versioning properties (apiVersion and kind) and a map of spec properties to match the subject (user or group), resource property, non-resource property (/version or /apis), and readonly. See [Examples](#) for details.

As someone setting up authentication and authorization on your production Kubernetes cluster, here are some things to consider:

- *Set the authorization mode*: When the Kubernetes API server ([kube-apiserver](#)) starts, the supported authentication modes must be set using the `--authorization-mode` flag. For example, that flag in the `kube-adminserver.yaml` file (in `/etc/kubernetes/manifests`) could be set to `Node,RBAC`. This would allow Node and RBAC authorization for authenticated requests.
- *Create user certificates and role bindings (RBAC)*: If you are using RBAC authorization, users can create a CertificateSigningRequest (CSR) that can be signed by the cluster CA. Then you can bind Roles and ClusterRoles to each user. See [Certificate Signing Requests](#) for details.
- *Create policies that combine attributes (ABAC)*: If you are using ABAC authorization, you can assign combinations of attributes to form policies to authorize selected users or groups to access particular resources (such as a pod), namespace, or apiGroup. For more information, see [Examples](#).
- *Consider Admission Controllers*: Additional forms of authorization for requests that can come in through the API server include [Webhook Token Authentication](#). Webhooks and other special authorization types need to be enabled by adding [Admission Controllers](#) to the API server.

# Set limits on workload resources

Demands from production workloads can cause pressure both inside and outside of the Kubernetes control plane. Consider these items when setting up for the needs of your cluster's workloads:

- *Set namespace limits:* Set per-namespace quotas on things like memory and CPU. See [Manage Memory, CPU, and API Resources](#) for details. You can also set [Hierarchical Namespaces](#) for inheriting limits.
- *Prepare for DNS demand:* If you expect workloads to massively scale up, your DNS service must be ready to scale up as well. See [Autoscale the DNS service in a Cluster](#).
- *Create additional service accounts:* User accounts determine what users can do on a cluster, while a service account defines pod access within a particular namespace. By default, a pod takes on the default service account from its namespace. See [Managing Service Accounts](#) for information on creating a new service account. For example, you might want to:
  - Add secrets that a pod could use to pull images from a particular container registry. See [Configure Service Accounts for Pods](#) for an example.
  - Assign RBAC permissions to a service account. See [ServiceAccount permissions](#) for details.

## What's next

- Decide if you want to build your own production Kubernetes or obtain one from available [Turnkey Cloud Solutions](#) or [Kubernetes Partners](#).
- If you choose to build your own cluster, plan how you want to handle [certificates](#) and set up high availability for features such as [etcd](#) and the [API server](#).
- Choose from [kubeadm](#), [kops](#) or [Kubespray](#) deployment methods.
- Configure user management by determining your [Authentication](#) and [Authorization](#) methods.
- Prepare for application workloads by setting up [resource limits](#), [DNS autoscaling](#) and [service accounts](#).

# Container Runtimes

**Note:** Dockershim has been removed from the Kubernetes project as of release 1.24. Read the [Dockershim Removal FAQ](#) for further details.

You need to install a [container runtime](#) into each node in the cluster so that Pods can run there. This page outlines what is involved and describes related tasks for setting up nodes.

Kubernetes 1.30 requires that you use a runtime that conforms with the [Container Runtime Interface](#) (CRI).

See [CRI version support](#) for more information.

This page provides an outline of how to use several common container runtimes with Kubernetes.

- [containerd](#)
- [CRI-O](#)

- [Docker Engine](#)
- [Mirantis Container Runtime](#)

**Note:**

Kubernetes releases before v1.24 included a direct integration with Docker Engine, using a component named *dockershim*. That special direct integration is no longer part of Kubernetes (this removal was [announced](#) as part of the v1.20 release). You can read [Check whether Dockershim removal affects you](#) to understand how this removal might affect you. To learn about migrating from using dockershim, see [Migrating from dockershim](#).

If you are running a version of Kubernetes other than v1.30, check the documentation for that version.

## Install and configure prerequisites

### Network configuration

By default, the Linux kernel does not allow IPv4 packets to be routed between interfaces. Most Kubernetes cluster networking implementations will change this setting (if needed), but some might expect the administrator to do it for them. (Some might also expect other `sysctl` parameters to be set, kernel modules to be loaded, etc; consult the documentation for your specific network implementation.)

### Enable IPv4 packet forwarding

To manually enable IPv4 packet forwarding:

```
sysctl params required by setup, params persist across reboots
cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
net.ipv4.ip_forward = 1
EOF

Apply sysctl params without reboot
sudo sysctl --system
```

Verify that `net.ipv4.ip_forward` is set to 1 with:

```
sysctl net.ipv4.ip_forward
```

## cgroup drivers

On Linux, [control groups](#) are used to constrain resources that are allocated to processes.

Both the [kubelet](#) and the underlying container runtime need to interface with control groups to enforce [resource management for pods and containers](#) and set resources such as cpu/memory requests and limits. To interface with control groups, the kubelet and the container runtime need to use a *cgroup driver*. It's critical that the kubelet and the container runtime use the same cgroup driver and are configured the same.

There are two cgroup drivers available:

- [cgroupfs](#)
- [systemd](#)

## cgroupfs driver

The cgroupfs driver is the [default cgroup driver in the kubelet](#). When the cgroupfs driver is used, the kubelet and the container runtime directly interface with the cgroup filesystem to configure cgroups.

The cgroupfs driver is **not** recommended when [systemd](#) is the init system because systemd expects a single cgroup manager on the system. Additionally, if you use [cgroup v2](#), use the systemd cgroup driver instead of cgroupfs.

## systemd cgroup driver

When [systemd](#) is chosen as the init system for a Linux distribution, the init process generates and consumes a root control group (cgroup) and acts as a cgroup manager.

systemd has a tight integration with cgroups and allocates a cgroup per systemd unit. As a result, if you use systemd as the init system with the cgroupfs driver, the system gets two different cgroup managers.

Two cgroup managers result in two views of the available and in-use resources in the system. In some cases, nodes that are configured to use cgroupfs for the kubelet and container runtime, but use systemd for the rest of the processes become unstable under resource pressure.

The approach to mitigate this instability is to use systemd as the cgroup driver for the kubelet and the container runtime when systemd is the selected init system.

To set systemd as the cgroup driver, edit the [KubeletConfiguration](#) option of cgroupDriver and set it to systemd. For example:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
...
cgroupDriver: systemd
```

### Note:

Starting with v1.22 and later, when creating a cluster with kubeadm, if the user does not set the cgroupDriver field under KubeletConfiguration, kubeadm defaults it to systemd.

In Kubernetes v1.28, with the KubeletCgroupDriverFromCRI [feature gate](#) enabled and a container runtime that supports the RuntimeConfig CRI RPC, the kubelet automatically detects the appropriate cgroup driver from the runtime, and ignores the cgroupDriver setting within the kubelet configuration.

If you configure `systemd` as the cgroup driver for the kubelet, you must also configure `systemd` as the cgroup driver for the container runtime. Refer to the documentation for your container runtime for instructions. For example:

- [containerd](#)
- [CRI-O](#)

### Caution:

Changing the cgroup driver of a Node that has joined a cluster is a sensitive operation. If the kubelet has created Pods using the semantics of one cgroup driver, changing the container runtime to another cgroup driver can cause errors when trying to re-create the Pod sandbox for such existing Pods. Restarting the kubelet may not solve such errors.

If you have automation that makes it feasible, replace the node with another using the updated configuration, or reinstall it using automation.

## Migrating to the `systemd` driver in kubeadm managed clusters

If you wish to migrate to the `systemd` cgroup driver in existing kubeadm managed clusters, follow [configuring a cgroup driver](#).

## CRI version support

Your container runtime must support at least v1alpha2 of the container runtime interface.

Kubernetes [starting v1.26](#) *only works* with v1 of the CRI API. Earlier versions default to v1 version, however if a container runtime does not support the v1 API, the kubelet falls back to using the (deprecated) v1alpha2 API instead.

## Container runtimes

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

### containerd

This section outlines the necessary steps to use containerd as CRI runtime.

To install containerd on your system, follow the instructions on [getting started with containerd](#). Return to this step once you've created a valid `config.toml` configuration file.

- [Linux](#)
- [Windows](#)

You can find this file under the path `/etc/containerd/config.toml`.

You can find this file under the path `C:\Program Files\containerd\config.toml`.



On Linux the default CRI socket for containerd is `/run/containerd/containerd.sock`. On Windows the default CRI endpoint is `npip://./pipe/containerd-containerd`.

## Configuring the systemd cgroup driver

To use the systemd cgroup driver in `/etc/containerd/config.toml` with runc, set

```
[plugins."io.containerd.grpc.v1.cri".containerd.runtimes.runc]
...
[plugins."io.containerd.grpc.v1.cri".containerd.runtimes.runc.options]
 SystemdCgroup = true
```

The systemd cgroup driver is recommended if you use [cgroup v2](#).

### Note:

If you installed containerd from a package (for example, RPM or .deb), you may find that the CRI integration plugin is disabled by default.

You need CRI support enabled to use containerd with Kubernetes. Make sure that `cri` is not included in the `disabled_plugins` list within `/etc/containerd/config.toml`; if you made changes to that file, also restart containerd.

If you experience container crash loops after the initial cluster installation or after installing a CNI, the containerd configuration provided with the package might contain incompatible configuration parameters. Consider resetting the containerd configuration with `containerd config default > /etc/containerd/config.toml` as specified in [getting-started.md](#) and then set the configuration parameters specified above accordingly.

If you apply this change, make sure to restart containerd:

```
sudo systemctl restart containerd
```

When using kubeadm, manually configure the [cgroup driver for kubelet](#).

In Kubernetes v1.28, you can enable automatic detection of the cgroup driver as an alpha feature. See [systemd cgroup driver](#) for more details.

## Overriding the sandbox (pause) image

In your [containerd config](#) you can overwrite the sandbox image by setting the following config:

```
[plugins."io.containerd.grpc.v1.cri"]
 sandbox_image = "registry.k8s.io/pause:3.2"
```

You might need to restart containerd as well once you've updated the config file: `systemctl restart containerd`.

Please note, that it is a best practice for kubelet to declare the matching `pod-infra-container-image`. If not configured, kubelet may attempt to garbage collect the pause image. There is ongoing work in [containerd to pin the pause image](#) and not require this setting on kubelet any longer.

## CRI-O

This section contains the necessary steps to install CRI-O as a container runtime.

To install CRI-O, follow [CRI-O Install Instructions](#).

### cgroup driver

CRI-O uses the systemd cgroup driver per default, which is likely to work fine for you. To switch to the cgroupfs cgroup driver, either edit `/etc/crio/crio.conf` or place a drop-in configuration in `/etc/crio/crio.conf.d/02-cgroup-manager.conf`, for example:

```
[crio.runtime]
common_cgroup = "pod"
cgroup_manager = "cgroupfs"
```

You should also note the changed `common_cgroup`, which has to be set to the value `pod` when using CRI-O with `cgroupfs`. It is generally necessary to keep the cgroup driver configuration of the kubelet (usually done via `kubeadm`) and CRI-O in sync.

In Kubernetes v1.28, you can enable automatic detection of the cgroup driver as an alpha feature. See [systemd cgroup driver](#) for more details.

For CRI-O, the CRI socket is `/var/run/crio/crio.sock` by default.

### Overriding the sandbox (pause) image

In your [CRI-O config](#) you can set the following config value:

```
[crio.image]
pause_image="registry.k8s.io/pause:3.6"
```

This config option supports live configuration reload to apply this change: `systemctl reload crio` or by sending `SIGHUP` to the `crio` process.

## Docker Engine

### Note:

These instructions assume that you are using the [cri-dockerd](#) adapter to integrate Docker Engine with Kubernetes.

1. On each of your nodes, install Docker for your Linux distribution as per [Install Docker Engine](#).
2. Install [cri-dockerd](#), following the directions in the install section of the documentation.

For `cri-dockerd`, the CRI socket is `/run/cri-dockerd.sock` by default.



## Mirantis Container Runtime

[Mirantis Container Runtime](#) (MCR) is a commercially available container runtime that was formerly known as Docker Enterprise Edition.

You can use Mirantis Container Runtime with Kubernetes using the open source [cri-dockerd](#) component, included with MCR.

To learn more about how to install Mirantis Container Runtime, visit [MCR Deployment Guide](#).

Check the systemd unit named `cri-docker.socket` to find out the path to the CRI socket.

### Overriding the sandbox (pause) image

The `cri-dockerd` adapter accepts a command line argument for specifying which container image to use as the Pod infrastructure container (“pause image”). The command line argument to use is `--pod-infra-container-image`.

## What's next

As well as a container runtime, your cluster will need a working [network plugin](#).

# Installing Kubernetes with deployment tools

There are many methods and tools for setting up your own production Kubernetes cluster. For example:

- [kubeadm](#)
- [kops](#): An automated cluster provisioning tool. For tutorials, best practices, configuration options and information on reaching out to the community, please check the [kOps website](#) for details.
- [kubespray](#): A composition of [Ansible](#) playbooks, [inventory](#), provisioning tools, and domain knowledge for generic OS/Kubernetes clusters configuration management tasks. You can reach out to the community on Slack channel [#kubespray](#).

# Bootstrapping clusters with kubeadm

---

[Installing kubeadm](#)

[Troubleshooting kubeadm](#)

[Creating a cluster with kubeadm](#)

[Customizing components with the kubeadm API](#)

[Options for Highly Available Topology](#)

[Creating Highly Available Clusters with kubeadm](#)

[Set up a High Availability etcd Cluster with kubeadm](#)

[Configuring each kubelet in your cluster using kubeadm](#)

[Dual-stack support with kubeadm](#)

# Installing kubeadm

This page shows how to install the kubeadm toolbox. For information on how to create a cluster with kubeadm once you have performed this installation process, see the [Creating a cluster with kubeadm](#) page.

This installation guide is for Kubernetes v1.30. If you want to use a different Kubernetes version, please refer to the following pages instead:

- [Installing kubeadm \(Kubernetes v1.29\)](#)
- [Installing kubeadm \(Kubernetes v1.28\)](#)
- [Installing kubeadm \(Kubernetes v1.27\)](#)
- [Installing kubeadm \(Kubernetes v1.26\)](#)

## Before you begin

- A compatible Linux host. The Kubernetes project provides generic instructions for Linux distributions based on Debian and Red Hat, and those distributions without a package manager.
- 2 GB or more of RAM per machine (any less will leave little room for your apps).
- 2 CPUs or more.
- Full network connectivity between all machines in the cluster (public or private network is fine).
- Unique hostname, MAC address, and product\_uuid for every node. See [here](#) for more details.
- Certain ports are open on your machines. See [here](#) for more details.
- Swap configuration. The default behavior of a kubelet was to fail to start if swap memory was detected on a node. Swap has been supported since v1.22. And since v1.28, Swap is supported for cgroup v2 only; the NodeSwap feature gate of the kubelet is beta but disabled by default.
  - You **MUST** disable swap if the kubelet is not properly configured to use swap. For example, `sudo swapoff -a` will disable swapping temporarily. To make this change persistent across reboots, make sure swap is disabled in config files like `/etc/fstab`, `systemd.swap`, depending how it was configured on your system.

### Note:

The kubeadm installation is done via binaries that use dynamic linking and assumes that your target system provides glibc. This is a reasonable assumption on many Linux distributions (including Debian, Ubuntu, Fedora, CentOS, etc.) but it is not always the case with custom and lightweight distributions which don't include glibc by default, such as Alpine Linux. The

expectation is that the distribution either includes glibc or a [compatibility layer](#) that provides the expected symbols.

## Verify the MAC address and product\_uuid are unique for every node

- You can get the MAC address of the network interfaces using the command `ip link` or `ifconfig -a`
- The `product_uuid` can be checked by using the command `sudo cat /sys/class/dmi/id/product_uuid`

It is very likely that hardware devices will have unique addresses, although some virtual machines may have identical values. Kubernetes uses these values to uniquely identify the nodes in the cluster. If these values are not unique to each node, the installation process may [fail](#).

## Check network adapters

If you have more than one network adapter, and your Kubernetes components are not reachable on the default route, we recommend you add IP route(s) so Kubernetes cluster addresses go via the appropriate adapter.

## Check required ports

These [required ports](#) need to be open in order for Kubernetes components to communicate with each other. You can use tools like [netcat](#) to check if a port is open. For example:

```
nc 127.0.0.1 6443 -v
```

The pod network plugin you use may also require certain ports to be open. Since this differs with each pod network plugin, please see the documentation for the plugins about what port(s) those need.

## Installing a container runtime

To run containers in Pods, Kubernetes uses a [container runtime](#).

By default, Kubernetes uses the [Container Runtime Interface](#) (CRI) to interface with your chosen container runtime.

If you don't specify a runtime, kubeadm automatically tries to detect an installed container runtime by scanning through a list of known endpoints.

If multiple or no container runtimes are detected kubeadm will throw an error and will request that you specify which one you want to use.

See [container runtimes](#) for more information.

## Note:

Docker Engine does not implement the [CRI](#) which is a requirement for a container runtime to work with Kubernetes. For that reason, an additional service [cri-dockerd](#) has to be installed. cri-dockerd is a project based on the legacy built-in Docker Engine support that was [removed](#) from the kubelet in version 1.24.

The tables below include the known endpoints for supported operating systems:

- [Linux](#)
- [Windows](#)

### Linux container runtimes

| Runtime                           | Path to Unix domain socket                 |
|-----------------------------------|--------------------------------------------|
| containerd                        | unix:///var/run/containerd/containerd.sock |
| CRI-O                             | unix:///var/run/crio/crio.sock             |
| Docker Engine (using cri-dockerd) | unix:///var/run/cri-dockerd.sock           |

### Windows container runtimes

| Runtime                           | Path to Windows named pipe             |
|-----------------------------------|----------------------------------------|
| containerd                        | npipes:///./pipe/containerd-containerd |
| Docker Engine (using cri-dockerd) | npipes:///./pipe/cri-dockerd           |

## Installing kubeadm, kubelet and kubectl

You will install these packages on all of your machines:

- kubeadm: the command to bootstrap the cluster.
- kubelet: the component that runs on all of the machines in your cluster and does things like starting pods and containers.
- kubectl: the command line util to talk to your cluster.

kubeadm **will not** install or manage kubelet or kubectl for you, so you will need to ensure they match the version of the Kubernetes control plane you want kubeadm to install for you. If you do not, there is a risk of a version skew occurring that can lead to unexpected, buggy behaviour. However, *one* minor version skew between the kubelet and the control plane is supported, but the kubelet version may never exceed the API server version. For example, the kubelet running 1.7.0 should be fully compatible with a 1.8.0 API server, but not vice versa.

For information about installing kubectl, see [Install and set up kubectl](#).

## Warning:

These instructions exclude all Kubernetes packages from any system upgrades. This is because kubeadm and Kubernetes require [special attention to upgrade](#).

For more information on version skews, see:

- Kubernetes [version and version-skew policy](#)
- Kubeadm-specific [version skew policy](#)

**Note:** The legacy package repositories ([apt.kubernetes.io](https://apt.kubernetes.io) and [yum.kubernetes.io](https://yum.kubernetes.io)) have been [deprecated and frozen starting from September 13, 2023](#). **Using the [new package repositories hosted at pkgs.k8s.io](https://pkgs.k8s.io) is strongly recommended and required in order to install Kubernetes versions released after September 13, 2023.** The deprecated legacy repositories, and their contents, might be removed at any time in the future and without a further notice period. The new package repositories provide downloads for Kubernetes versions starting with v1.24.0.

**Note:**

There's a dedicated package repository for each Kubernetes minor version. If you want to install a minor version other than v1.30, please see the installation guide for your desired minor version.

- [Debian-based distributions](#)
- [Red Hat-based distributions](#)
- [Without a package manager](#)

These instructions are for Kubernetes v1.30.

1. Update the apt package index and install packages needed to use the Kubernetes apt repository:

```
sudo apt-get update
apt-transport-https may be a dummy package; if so, you can skip that package
sudo apt-get install -y apt-transport-https ca-certificates curl gpg
```

2. Download the public signing key for the Kubernetes package repositories. The same signing key is used for all repositories so you can disregard the version in the URL:

```
If the directory `/etc/apt/keyrings` does not exist, it should be created before the curl
command, read the note below.
sudo mkdir -p -m 755 /etc/apt/keyrings
curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.30/deb/Release.key | sudo gpg --dearmor -o /
etc/apt/keyrings/kubernetes-apt-keyring.gpg
```

**Note:**

In releases older than Debian 12 and Ubuntu 22.04, directory `/etc/apt/keyrings` does not exist by default, and it should be created before the curl command.

1. Add the appropriate Kubernetes apt repository. Please note that this repository have packages only for Kubernetes 1.30; for other Kubernetes minor versions, you need to change the Kubernetes minor version in the URL to match your desired minor version (you should also check that you are reading the documentation for the version of Kubernetes that you plan to install).

```
This overwrites any existing configuration in /etc/apt/sources.list.d/kubernetes.list
echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/
core:/stable:/v1.30/deb/ ' | sudo tee /etc/apt/sources.list.d/kubernetes.list
```

2. Update the apt package index, install kubelet, kubeadm and kubectl, and pin their version:

```
sudo apt-get update
sudo apt-get install -y kubelet kubeadm kubectl
sudo apt-mark hold kubelet kubeadm kubectl
```

3. (Optional) Enable the kubelet service before running kubeadm:

```
sudo systemctl enable --now kubelet
```

1. Set SELinux to permissive mode:

These instructions are for Kubernetes 1.30.

```
Set SELinux in permissive mode (effectively disabling it)
sudo setenforce 0
sudo sed -i 's/^SELINUX=enforcing$/SELINUX=permissive/' /etc/selinux/config
```

### Caution:

- Setting SELinux in permissive mode by running `setenforce 0` and `sed ...` effectively disables it. This is required to allow containers to access the host filesystem; for example, some cluster network plugins require that. You have to do this until SELinux support is improved in the kubelet.
  - You can leave SELinux enabled if you know how to configure it but it may require settings that are not supported by kubeadm.
1. Add the Kubernetes yum repository. The `exclude` parameter in the repository definition ensures that the packages related to Kubernetes are not upgraded upon running `yum update` as there's a special procedure that must be followed for upgrading Kubernetes. Please note that this repository has packages only for Kubernetes 1.30; for other Kubernetes minor versions, you need to change the Kubernetes minor version in the URL to match your desired minor version (you should also check that you are reading the documentation for the version of Kubernetes that you plan to install).

```
This overwrites any existing configuration in /etc/yum.repos.d/kubernetes.repo
cat <<EOF | sudo tee /etc/yum.repos.d/kubernetes.repo
[kubernetes]
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key
exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
EOF
```

2. Install kubelet, kubeadm and kubectl:

```
sudo yum install -y kubelet kubeadm kubectl --disableexcludes=kubernetes
```

3. (Optional) Enable the kubelet service before running kubeadm:

```
sudo systemctl enable --now kubelet
```

Install CNI plugins (required for most pod network):

```
CNI_PLUGINS_VERSION="v1.3.0"
ARCH="amd64"
DEST="/opt/cni/bin"
sudo mkdir -p "$DEST"
curl -L "https://github.com/containernetworking/plugins/releases/download/${CNI_PLUGINS_VERSION}/cni-plugins-linux-${ARCH}-${CNI_PLUGINS_VERSION}.tgz" | sudo tar -C "$DEST" -xz
```

Define the directory to download command files:

**Note:**

The DOWNLOAD\_DIR variable must be set to a writable directory. If you are running Flatcar Container Linux, set DOWNLOAD\_DIR="/opt/bin".

```
DOWNLOAD_DIR="/usr/local/bin"
sudo mkdir -p "$DOWNLOAD_DIR"
```

Install crictl (required for kubeadm / Kubelet Container Runtime Interface (CRI)):

```
CRICTL_VERSION="v1.28.0"
ARCH="amd64"
curl -L "https://github.com/kubernetes-sigs/cri-tools/releases/download/${CRICTL_VERSION}/crictl-${CRICTL_VERSION}-linux-${ARCH}.tar.gz" | sudo tar -C $DOWNLOAD_DIR -xz
```

Install kubeadm, kubelet and add a kubelet systemd service:

```
RELEASE="$(curl -sSL https://dl.k8s.io/release/stable.txt)"
ARCH="amd64"
cd $DOWNLOAD_DIR
sudo curl -L --remote-name-all https://dl.k8s.io/release/${RELEASE}/bin/linux/${ARCH}/{kubeadm,kubelet}
sudo chmod +x {kubeadm,kubelet}
```

```
RELEASE_VERSION="v0.16.2"
curl -sSL "https://raw.githubusercontent.com/kubernetes/release/${RELEASE_VERSION}/cmd/krel/templates/latest/kubelet/kubelet.service" | sed "s:/usr/bin:${DOWNLOAD_DIR}:g" | sudo tee /usr/lib/systemd/system/kubelet.service
sudo mkdir -p /usr/lib/systemd/system/kubelet.service.d
curl -sSL "https://raw.githubusercontent.com/kubernetes/release/${RELEASE_VERSION}/cmd/krel/templates/latest/kubeadm/10-kubeadm.conf" | sed "s:/usr/bin:${DOWNLOAD_DIR}:g" | sudo tee /usr/lib/systemd/system/kubelet.service.d/10-kubeadm.conf
```

**Note:**

Please refer to the note in the [Before you begin](#) section for Linux distributions that do not include glibc by default.

Install kubectl by following the instructions on [Install Tools page](#).

Optionally, enable the kubelet service before running kubeadm:

```
sudo systemctl enable --now kubelet
```

**Note:**

The Flatcar Container Linux distribution mounts the `/usr` directory as a read-only filesystem. Before bootstrapping your cluster, you need to take additional steps to configure a writable directory. See the [Kubeadm Troubleshooting guide](#) to learn how to set up a writable directory.

The kubelet is now restarting every few seconds, as it waits in a crashloop for kubeadm to tell it what to do.

## Configuring a cgroup driver

Both the container runtime and the kubelet have a property called "[cgroup driver](#)", which is important for the management of cgroups on Linux machines.

**Warning:**

Matching the container runtime and kubelet cgroup drivers is required or otherwise the kubelet process will fail.

See [Configuring a cgroup driver](#) for more details.

## Troubleshooting

If you are running into difficulties with kubeadm, please consult our [troubleshooting docs](#).

## What's next

- [Using kubeadm to Create a Cluster](#)

## Troubleshooting kubeadm

As with any program, you might run into an error installing or running kubeadm. This page lists some common failure scenarios and have provided steps that can help you understand and fix the problem.

If your problem is not listed below, please follow the following steps:

- If you think your problem is a bug with kubeadm:
  - Go to [github.com/kubernetes/kubeadm](https://github.com/kubernetes/kubeadm) and search for existing issues.
  - If no issue exists, please [open one](#) and follow the issue template.
- If you are unsure about how kubeadm works, you can ask on [Slack](#) in #kubeadm, or open a question on [StackOverflow](#). Please include relevant tags like #kubernetes and #kubeadm so folks can help you.



# Not possible to join a v1.18 Node to a v1.17 cluster due to missing RBAC

In v1.18 kubeadm added prevention for joining a Node in the cluster if a Node with the same name already exists. This required adding RBAC for the bootstrap-token user to be able to GET a Node object.

However this causes an issue where kubeadm join from v1.18 cannot join a cluster created by kubeadm v1.17.

To workaround the issue you have two options:

Execute kubeadm init phase bootstrap-token on a control-plane node using kubeadm v1.18. Note that this enables the rest of the bootstrap-token permissions as well.

or

Apply the following RBAC manually using kubectl apply -f ...:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: kubeadm:get-nodes
rules:
- apiGroups:
 - ""
 resources:
 - nodes
 verbs:
 - get

apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
 name: kubeadm:get-nodes
roleRef:
 apiGroup: rbac.authorization.k8s.io
 kind: ClusterRole
 name: kubeadm:get-nodes
subjects:
- apiGroup: rbac.authorization.k8s.io
 kind: Group
 name: system:bootstrappers:kubeadm:default-node-token
```

## ebtables or some similar executable not found during installation

If you see the following warnings while running kubeadm init

```
[preflight] WARNING: ebtables not found in system path
[preflight] WARNING: ethtool not found in system path
```

Then you may be missing `ebtables`, `ethtool` or a similar executable on your node. You can install them with the following commands:

- For Ubuntu/Debian users, run `apt install ebtables ethtool`.
- For CentOS/Fedora users, run `yum install ebtables ethtool`.

## kubeadm blocks waiting for control plane during installation

If you notice that `kubeadm init` hangs after printing out the following line:

```
[apiclient] Created API client, waiting for the control plane to become ready
```

This may be caused by a number of problems. The most common are:

- network connection problems. Check that your machine has full network connectivity before continuing.
- the cgroup driver of the container runtime differs from that of the kubelet. To understand how to configure it properly, see [Configuring a cgroup driver](#).
- control plane containers are crashlooping or hanging. You can check this by running `docker ps` and investigating each container by running `docker logs`. For other container runtime, see [Debugging Kubernetes nodes with crictl](#).

## kubeadm blocks when removing managed containers

The following could happen if the container runtime halts and does not remove any Kubernetes-managed containers:

```
sudo kubeadm reset
```

```
[preflight] Running pre-flight checks
[reset] Stopping the kubelet service
[reset] Unmounting mounted directories in "/var/lib/kubelet"
[reset] Removing kubernetes-managed containers
(block)
```

A possible solution is to restart the container runtime and then re-run `kubeadm reset`. You can also use `crictl` to debug the state of the container runtime. See [Debugging Kubernetes nodes with crictl](#).

## Pods in RunContainerError, CrashLoopBackOff or Error state

Right after `kubeadm init` there should not be any pods in these states.

- If there are pods in one of these states *right after* `kubeadm init`, please open an issue in the `kubeadm` repo. `coredns` (or `kube-dns`) should be in the Pending state until you have deployed the network add-on.
- If you see Pods in the `RunContainerError`, `CrashLoopBackOff` or `Error` state after deploying the network add-on and nothing happens to `coredns` (or `kube-dns`), it's very likely that the Pod Network add-on that you installed is somehow broken. You might

have to grant it more RBAC privileges or use a newer version. Please file an issue in the Pod Network providers' issue tracker and get the issue triaged there.

## coredns is stuck in the Pending state

This is **expected** and part of the design. kubeadm is network provider-agnostic, so the admin should [install the pod network add-on](#) of choice. You have to install a Pod Network before CoreDNS may be deployed fully. Hence the Pending state before the network is set up.

## HostPort services do not work

The HostPort and HostIP functionality is available depending on your Pod Network provider. Please contact the author of the Pod Network add-on to find out whether HostPort and HostIP functionality are available.

Calico, Canal, and Flannel CNI providers are verified to support HostPort.

For more information, see the [CNI portmap documentation](#).

If your network provider does not support the portmap CNI plugin, you may need to use the [NodePort feature of services](#) or use HostNetwork=true.

## Pods are not accessible via their Service IP

- Many network add-ons do not yet enable [hairpin mode](#) which allows pods to access themselves via their Service IP. This is an issue related to [CNI](#). Please contact the network add-on provider to get the latest status of their support for hairpin mode.
- If you are using VirtualBox (directly or via Vagrant), you will need to ensure that `hostname -i` returns a routable IP address. By default, the first interface is connected to a non-routable host-only network. A work around is to modify `/etc/hosts`, see this [Vagrantfile](#) for an example.

## TLS certificate errors

The following error indicates a possible certificate mismatch.

```
kubectl get pods
```

```
Unable to connect to the server: x509: certificate signed by unknown authority (possibly because of "crypto/rsa: verification error" while trying to verify candidate authority certificate "kubernetes")
```

- Verify that the `$HOME/.kube/config` file contains a valid certificate, and regenerate a certificate if necessary. The certificates in a kubeconfig file are base64 encoded. The `base64 --decode` command can be used to decode the certificate and `openssl x509 -text -noout` can be used for viewing the certificate information.
- Unset the KUBECONFIG environment variable using:

```
unset KUBECONFIG
```

Or set it to the default KUBECONFIG location:

```
export KUBECONFIG=/etc/kubernetes/admin.conf
```

- Another workaround is to overwrite the existing kubeconfig for the "admin" user:

```
mv $HOME/.kube $HOME/.kube.bak
mkdir $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

## Kubelet client certificate rotation fails

By default, kubeadm configures a kubelet with automatic rotation of client certificates by using the `/var/lib/kubelet/pki/kubelet-client-current.pem` symlink specified in `/etc/kubernetes/kubelet.conf`. If this rotation process fails you might see errors such as `x509: certificate has expired or is not yet valid` in kube-apiserver logs. To fix the issue you must follow these steps:

1. Backup and delete `/etc/kubernetes/kubelet.conf` and `/var/lib/kubelet/pki/kubelet-client*` from the failed node.
2. From a working control plane node in the cluster that has `/etc/kubernetes/pki/ca.key` execute `kubeadm kubeconfig user --org system:nodes --client-name system:node:$NODE > kubelet.conf`. `$NODE` must be set to the name of the existing failed node in the cluster. Modify the resulted `kubelet.conf` manually to adjust the cluster name and server endpoint, or pass `kubeconfig user --config` (see [Generating kubeconfig files for additional users](#)). If your cluster does not have the `ca.key` you must sign the embedded certificates in the `kubelet.conf` externally.
3. Copy this resulted `kubelet.conf` to `/etc/kubernetes/kubelet.conf` on the failed node.
4. Restart the kubelet (`systemctl restart kubelet`) on the failed node and wait for `/var/lib/kubelet/pki/kubelet-client-current.pem` to be recreated.
5. Manually edit the `kubelet.conf` to point to the rotated kubelet client certificates, by replacing `client-certificate-data` and `client-key-data` with:

```
client-certificate: /var/lib/kubelet/pki/kubelet-client-current.pem
client-key: /var/lib/kubelet/pki/kubelet-client-current.pem
```

6. Restart the kubelet.
7. Make sure the node becomes Ready.

## Default NIC When using flannel as the pod network in Vagrant

The following error might indicate that something was wrong in the pod network:

Error from server (NotFound): the server could not find the requested resource

- If you're using flannel as the pod network inside Vagrant, then you will have to specify the default interface name for flannel.

Vagrant typically assigns two interfaces to all VMs. The first, for which all hosts are assigned the IP address 10.0.2.15, is for external traffic that gets NATed.

This may lead to problems with flannel, which defaults to the first interface on a host. This leads to all hosts thinking they have the same public IP address. To prevent this, pass the `--iface eth1` flag to flannel so that the second interface is chosen.

## Non-public IP used for containers

In some situations `kubectl logs` and `kubectl run` commands may return with the following errors in an otherwise functional cluster:

```
Error from server: Get https://10.19.0.41:10250/containerLogs/default/mysql-ddc65b868-glc5m/mysql: dial tcp 10.19.0.41:10250: getsockopt: no route to host
```

- This may be due to Kubernetes using an IP that can not communicate with other IPs on the seemingly same subnet, possibly by policy of the machine provider.
- DigitalOcean assigns a public IP to `eth0` as well as a private one to be used internally as anchor for their floating IP feature, yet kubelet will pick the latter as the node's InternalIP instead of the public one.

Use `ip addr show` to check for this scenario instead of `ifconfig` because `ifconfig` will not display the offending alias IP address. Alternatively an API endpoint specific to DigitalOcean allows to query for the anchor IP from the droplet:

```
curl http://169.254.169.254/metadata/v1/interfaces/public/0/anchor_ip_v4/address
```

The workaround is to tell kubelet which IP to use using `--node-ip`. When using DigitalOcean, it can be the public one (assigned to `eth0`) or the private one (assigned to `eth1`) should you want to use the optional private network. The `kubeletExtraArgs` section of the `kubeadm NodeRegistrationOptions` structure can be used for this.

Then restart kubelet:

```
systemctl daemon-reload
systemctl restart kubelet
```

## coredns pods have CrashLoopBackOff or Error state

If you have nodes that are running SELinux with an older version of Docker, you might experience a scenario where the coredns pods are not starting. To solve that, you can try one of the following options:

- Upgrade to a [newer version of Docker](#).
- [Disable SELinux](#).
- Modify the coredns deployment to set `allowPrivilegeEscalation` to true:

```
kubectl -n kube-system get deployment coredns -o yaml | \
sed 's/allowPrivilegeEscalation: false/allowPrivilegeEscalation: true/g' | \
kubectl apply -f -
```

Another cause for CoreDNS to have CrashLoopBackOff is when a CoreDNS Pod deployed in Kubernetes detects a loop. [A number of workarounds](#) are available to avoid Kubernetes trying to restart the CoreDNS Pod every time CoreDNS detects the loop and exits.

### Warning:

Disabling SELinux or setting allowPrivilegeEscalation to true can compromise the security of your cluster.

## etcd pods restart continually

If you encounter the following error:

```
rpc error: code = 2 desc = oci runtime error: exec failed: container_linux.go:247: starting container process caused "process_linux.go:110: decoding init error from pipe caused \"read parent: connection reset by peer\""
```

This issue appears if you run CentOS 7 with Docker 1.13.1.84. This version of Docker can prevent the kubelet from executing into the etcd container.

To work around the issue, choose one of these options:

- Roll back to an earlier version of Docker, such as 1.13.1-75

```
yum downgrade docker-1.13.1-75.git8633870.el7.centos.x86_64 docker-client-1.13.1-75.git8633870.el7.centos.x86_64 docker-common-1.13.1-75.git8633870.el7.centos.x86_64
```

- Install one of the more recent recommended versions, such as 18.06:

```
sudo yum-config-manager --add-repo https://download.docker.com/linux/centos/docker-ce.repo
yum install docker-ce-18.06.1.ce-3.el7.x86_64
```

## Not possible to pass a comma separated list of values to arguments inside a --component-extra-args flag

kubeadm init flags such as --component-extra-args allow you to pass custom arguments to a control-plane component like the kube-apiserver. However, this mechanism is limited due to the underlying type used for parsing the values (mapStringString).

If you decide to pass an argument that supports multiple, comma-separated values such as --apiserver-extra-args "enable-admission-plugins=LimitRanger,NamespaceExists" this flag will fail with flag: malformed pair, expect string=string. This happens because the list of arguments for --apiserver-extra-args expects key=value pairs and in this case NamespaceExists is considered as a key that is missing a value.

Alternatively, you can try separating the key=value pairs like so: --apiserver-extra-args "enable-admission-plugins=LimitRanger,enable-admission-plugins=NamespaceExists" but this will result in the key enable-admission-plugins only having the value of NamespaceExists.

A known workaround is to use the kubeadm [configuration file](#).

## kube-proxy scheduled before node is initialized by cloud-controller-manager

In cloud provider scenarios, kube-proxy can end up being scheduled on new worker nodes before the cloud-controller-manager has initialized the node addresses. This causes kube-proxy to fail to pick up the node's IP address properly and has knock-on effects to the proxy function managing load balancers.

The following error can be seen in kube-proxy Pods:

```
server.go:610] Failed to retrieve node IP: host IP unknown; known addresses: []
proxier.go:340] invalid nodeIP, initializing kube-proxy with 127.0.0.1 as nodeIP
```

A known solution is to patch the kube-proxy DaemonSet to allow scheduling it on control-plane nodes regardless of their conditions, keeping it off of other nodes until their initial guarding conditions abate:

```
kubectl -n kube-system patch ds kube-proxy -p='{
 "spec": {
 "template": {
 "spec": {
 "tolerations": [
 {
 "key": "CriticalAddonsOnly",
 "operator": "Exists"
 },
 {
 "effect": "NoSchedule",
 "key": "node-role.kubernetes.io/control-plane"
 }
]
 }
 }
 }
}'
```

The tracking issue for this problem is [here](#).

## /usr is mounted read-only on nodes

On Linux distributions such as Fedora CoreOS or Flatcar Container Linux, the directory /usr is mounted as a read-only filesystem. For [flex-volume support](#), Kubernetes components like the kubelet and kube-controller-manager use the default path of /usr/libexec/kubernetes/kubelet-plugins/volume/exec/, yet the flex-volume directory *must be writeable* for the feature to work.

### Note:

FlexVolume was deprecated in the Kubernetes v1.23 release.

To workaround this issue, you can configure the flex-volume directory using the [kubeadm configuration file](#).

On the primary control-plane Node (created using `kubeadm init`), pass the following file using `--config`:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: InitConfiguration
nodeRegistration:
 kubeletExtraArgs:
 volume-plugin-dir: "/opt/libexec/kubernetes/kubelet-plugins/volume/exec/"

apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
controllerManager:
 extraArgs:
 flex-volume-plugin-dir: "/opt/libexec/kubernetes/kubelet-plugins/volume/exec/"
```

On joining Nodes:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: JoinConfiguration
nodeRegistration:
 kubeletExtraArgs:
 volume-plugin-dir: "/opt/libexec/kubernetes/kubelet-plugins/volume/exec/"
```

Alternatively, you can modify `/etc/fstab` to make the `/usr` mount writeable, but please be advised that this is modifying a design principle of the Linux distribution.

## kubeadm upgrade plan prints out context deadline exceeded error message

This error message is shown when upgrading a Kubernetes cluster with `kubeadm` in the case of running an external `etcd`. This is not a critical bug and happens because older versions of `kubeadm` perform a version check on the external `etcd` cluster. You can proceed with `kubeadm upgrade apply ...`

This issue is fixed as of version 1.19.

## kubeadm reset unmounts /var/lib/kubelet

If `/var/lib/kubelet` is being mounted, performing a `kubeadm reset` will effectively unmount it.

To workaround the issue, re-mount the `/var/lib/kubelet` directory after performing the `kubeadm reset` operation.

This is a regression introduced in `kubeadm` 1.15. The issue is fixed in 1.20.

## Cannot use the metrics-server securely in a kubeadm cluster

In a `kubeadm` cluster, the [metrics-server](#) can be used insecurely by passing the `--kubelet-insecure-tls` to it. This is not recommended for production clusters.



If you want to use TLS between the metrics-server and the kubelet there is a problem, since kubeadm deploys a self-signed serving certificate for the kubelet. This can cause the following errors on the side of the metrics-server:

```
x509: certificate signed by unknown authority
x509: certificate is valid for IP-foo not IP-bar
```

See [Enabling signed kubelet serving certificates](#) to understand how to configure the kubelets in a kubeadm cluster to have properly signed serving certificates.

Also see [How to run the metrics-server securely](#).

## Upgrade fails due to etcd hash not changing

Only applicable to upgrading a control plane node with a kubeadm binary v1.28.3 or later, where the node is currently managed by kubeadm versions v1.28.0, v1.28.1 or v1.28.2.

Here is the error message you may encounter:

```
[upgrade/etcd] Failed to upgrade etcd: couldn't upgrade control plane. kubeadm has tried to
recover everything into the earlier state. Errors faced: static Pod hash for component etcd on
Node kinder-upgrade-control-plane-1 did not change after 5m0s: timed out waiting for the
condition
[upgrade/etcd] Waiting for previous etcd to become available
I0907 10:10:09.109104 3704 etcd.go:588] [etcd] attempting to see if all cluster endpoints
([https://172.17.0.6:2379/ https://172.17.0.4:2379/ https://172.17.0.3:2379/]) are available 1/10
[upgrade/etcd] Etcd was rolled back and is now available
static Pod hash for component etcd on Node kinder-upgrade-control-plane-1 did not change
after 5m0s: timed out waiting for the condition
couldn't upgrade control plane. kubeadm has tried to recover everything into the earlier state.
Errors faced
k8s.io/kubernetes/cmd/kubeadm/app/phases/upgrade.rollbackOldManifests
cmd/kubeadm/app/phases/upgrade/staticpods.go:525
k8s.io/kubernetes/cmd/kubeadm/app/phases/upgrade.upgradeComponent
cmd/kubeadm/app/phases/upgrade/staticpods.go:254
k8s.io/kubernetes/cmd/kubeadm/app/phases/upgrade.performEtcdStaticPodUpgrade
cmd/kubeadm/app/phases/upgrade/staticpods.go:338
...
```

The reason for this failure is that the affected versions generate an etcd manifest file with unwanted defaults in the PodSpec. This will result in a diff from the manifest comparison, and kubeadm will expect a change in the Pod hash, but the kubelet will never update the hash.

There are two way to workaround this issue if you see it in your cluster:

- The etcd upgrade can be skipped between the affected versions and v1.28.3 (or later) by using:

```
kubeadm upgrade {apply|node} [version] --etcd-upgrade=false
```

This is not recommended in case a new etcd version was introduced by a later v1.28 patch version.

- Before upgrade, patch the manifest for the etcd static pod, to remove the problematic
- defaulted attributes:

```
diff --git a/etc/kubernetes/manifests/etcd_defaults.yaml b/etc/kubernetes/manifests/etcd_origin.yaml
index d807ccbe0aa..46b35f00e15 100644
--- a/etc/kubernetes/manifests/etcd_defaults.yaml
+++ b/etc/kubernetes/manifests/etcd_origin.yaml
@@ -43,7 +43,6 @@ spec:
 scheme: HTTP
 initialDelaySeconds: 10
 periodSeconds: 10
- successThreshold: 1
 timeoutSeconds: 15
 name: etcd
 resources:
@@ -59,26 +58,18 @@ spec:
 scheme: HTTP
 initialDelaySeconds: 10
 periodSeconds: 10
- successThreshold: 1
 timeoutSeconds: 15
- terminationMessagePath: /dev/termination-log
- terminationMessagePolicy: File
 volumeMounts:
- - mountPath: /var/lib/etcd
 name: etcd-data
- - mountPath: /etc/kubernetes/pki/etcd
 name: etcd-certs
- dnsPolicy: ClusterFirst
- enableServiceLinks: true
 hostNetwork: true
 priority: 2000001000
 priorityClassName: system-node-critical
- restartPolicy: Always
- schedulerName: default-scheduler
 securityContext:
 seccompProfile:
 type: RuntimeDefault
- terminationGracePeriodSeconds: 30
 volumes:
- - hostPath:
 path: /etc/kubernetes/pki/etcd
```

More information can be found in the [tracking issue](#) for this bug.

## Creating a cluster with kubeadm

Using kubeadm, you can create a minimum viable Kubernetes cluster that conforms to best practices. In fact, you can use kubeadm to set up a cluster that will pass the [Kubernetes Conformance tests](#). kubeadm also supports other cluster lifecycle functions, such as [bootstrap tokens](#) and cluster upgrades.

The kubeadm tool is good if you need:

- A simple way for you to try out Kubernetes, possibly for the first time.
- A way for existing users to automate setting up a cluster and test their application.
- A building block in other ecosystem and/or installer tools with a larger scope.

You can install and use kubeadm on various machines: your laptop, a set of cloud servers, a Raspberry Pi, and more. Whether you're deploying into the cloud or on-premises, you can integrate kubeadm into provisioning systems such as Ansible or Terraform.

## Before you begin

To follow this guide, you need:

- One or more machines running a deb/rpm-compatible Linux OS; for example: Ubuntu or CentOS.
- 2 GiB or more of RAM per machine--any less leaves little room for your apps.
- At least 2 CPUs on the machine that you use as a control-plane node.
- Full network connectivity among all machines in the cluster. You can use either a public or a private network.

You also need to use a version of kubeadm that can deploy the version of Kubernetes that you want to use in your new cluster.

[Kubernetes' version and version skew support policy](#) applies to kubeadm as well as to Kubernetes overall. Check that policy to learn about what versions of Kubernetes and kubeadm are supported. This page is written for Kubernetes v1.30.

The kubeadm tool's overall feature state is General Availability (GA). Some sub-features are still under active development. The implementation of creating the cluster may change slightly as the tool evolves, but the overall implementation should be pretty stable.

### Note:

Any commands under kubeadm alpha are, by definition, supported on an alpha level.

## Objectives

- Install a single control-plane Kubernetes cluster
- Install a Pod network on the cluster so that your Pods can talk to each other

## Instructions

### Preparing the hosts

#### Component installation

Install a [container runtime](#) and kubeadm on all the hosts. For detailed instructions and other prerequisites, see [Installing kubeadm](#).

**Note:**

If you have already installed kubeadm, see the first two steps of the [Upgrading Linux nodes](#) document for instructions on how to upgrade kubeadm.

When you upgrade, the kubelet restarts every few seconds as it waits in a crashloop for kubeadm to tell it what to do. This crashloop is expected and normal. After you initialize your control-plane, the kubelet runs normally.

**Network setup**

kubeadm similarly to other Kubernetes components tries to find a usable IP on the network interfaces associated with a default gateway on a host. Such an IP is then used for the advertising and/or listening performed by a component.

To find out what this IP is on a Linux host you can use:

```
ip route show # Look for a line starting with "default via"
```

**Note:**

If two or more default gateways are present on the host, a Kubernetes component will try to use the first one it encounters that has a suitable global unicast IP address. While making this choice, the exact ordering of gateways might vary between different operating systems and kernel versions.

Kubernetes components do not accept custom network interface as an option, therefore a custom IP address must be passed as a flag to all components instances that need such a custom configuration.

**Note:**

If the host does not have a default gateway and if a custom IP address is not passed to a Kubernetes component, the component may exit with an error.

To configure the API server advertise address for control plane nodes created with both init and join, the flag `--apiserver-advertise-address` can be used. Preferably, this option can be set in the [kubeadm API](#) as `InitConfiguration.localAPIEndpoint` and `JoinConfiguration.controlPlane.localAPIEndpoint`.

For kubelets on all nodes, the `--node-ip` option can be passed in `.nodeRegistration.kubeletExtraArgs` inside a kubeadm configuration file (`InitConfiguration` or `JoinConfiguration`).

For dual-stack see [Dual-stack support with kubeadm](#).

The IP addresses that you assign to control plane components become part of their X.509 certificates' subject alternative name fields. Changing these IP addresses would require signing new certificates and restarting the affected components, so that the change in certificate files is reflected. See [Manual certificate renewal](#) for more details on this topic.

## Warning:

The Kubernetes project recommends against this approach (configuring all component instances with custom IP addresses). Instead, the Kubernetes maintainers recommend to setup the host network, so that the default gateway IP is the one that Kubernetes components auto-detect and use. On Linux nodes, you can use commands such as `ip route` to configure networking; your operating system might also provide higher level network management tools. If your node's default gateway is a public IP address, you should configure packet filtering or other security measures that protect the nodes and your cluster.

## Preparing the required container images

This step is optional and only applies in case you wish `kubeadm init` and `kubeadm join` to not download the default container images which are hosted at `registry.k8s.io`.

Kubeadm has commands that can help you pre-pull the required images when creating a cluster without an internet connection on its nodes. See [Running kubeadm without an internet connection](#) for more details.

Kubeadm allows you to use a custom image repository for the required images. See [Using custom images](#) for more details.

## Initializing your control-plane node

The control-plane node is the machine where the control plane components run, including [etcd](#) (the cluster database) and the [API Server](#) (which the [kubectx](#) command line tool communicates with).

1. (Recommended) If you have plans to upgrade this single control-plane kubeadm cluster to high availability you should specify the `--control-plane-endpoint` to set the shared endpoint for all control-plane nodes. Such an endpoint can be either a DNS name or an IP address of a load-balancer.
2. Choose a Pod network add-on, and verify whether it requires any arguments to be passed to `kubeadm init`. Depending on which third-party provider you choose, you might need to set the `--pod-network-cidr` to a provider-specific value. See [Installing a Pod network add-on](#).
3. (Optional) `kubeadm` tries to detect the container runtime by using a list of well known endpoints. To use different container runtime or if there are more than one installed on the provisioned node, specify the `--cri-socket` argument to `kubeadm`. See [Installing a runtime](#).

To initialize the control-plane node run:

```
kubeadm init <args>
```

## Considerations about apiserver-advertise-address and ControlPlaneEndpoint

While `--apiserver-advertise-address` can be used to set the advertise address for this particular control-plane node's API server, `--control-plane-endpoint` can be used to set the shared endpoint for all control-plane nodes.

--control-plane-endpoint allows both IP addresses and DNS names that can map to IP addresses. Please contact your network administrator to evaluate possible solutions with respect to such mapping.

Here is an example mapping:

192.168.0.102 cluster-endpoint

Where 192.168.0.102 is the IP address of this node and cluster-endpoint is a custom DNS name that maps to this IP. This will allow you to pass --control-plane-endpoint=cluster-endpoint to kubeadm init and pass the same DNS name to kubeadm join. Later you can modify cluster-endpoint to point to the address of your load-balancer in an high availability scenario.

Turning a single control plane cluster created without --control-plane-endpoint into a highly available cluster is not supported by kubeadm.

## More information

For more information about kubeadm init arguments, see the [kubeadm reference guide](#).

To configure kubeadm init with a configuration file see [Using kubeadm init with a configuration file](#).

To customize control plane components, including optional IPv6 assignment to liveness probe for control plane components and etcd server, provide extra arguments to each component as documented in [custom arguments](#).

To reconfigure a cluster that has already been created see [Reconfiguring a kubeadm cluster](#).

To run kubeadm init again, you must first [tear down the cluster](#).

If you join a node with a different architecture to your cluster, make sure that your deployed DaemonSets have container image support for this architecture.

kubeadm init first runs a series of prechecks to ensure that the machine is ready to run Kubernetes. These prechecks expose warnings and exit on errors. kubeadm init then downloads and installs the cluster control plane components. This may take several minutes. After it finishes you should see:

Your Kubernetes control-plane has initialized successfully!

To start using your cluster, you need to run the following as a regular user:

```
mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

You should now deploy a Pod network to the cluster.

Run "kubectl apply -f [podnetwork].yaml" with one of the options listed at: [/docs/concepts/cluster-administration/addons/](#)

You can now join any number of machines by running the following on each node as root:

```
kubeadm join <control-plane-host>:<control-plane-port> --token <token> --discovery-token-ca-cert-hash sha256:<hash>
```

To make kubectl work for your non-root user, run these commands, which are also part of the kubeadm init output:

```
mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

Alternatively, if you are the root user, you can run:

```
export KUBECONFIG=/etc/kubernetes/admin.conf
```

### Warning:

The kubeconfig file admin.conf that kubeadm init generates contains a certificate with Subject: O = kubeadm:cluster-admins, CN = kubernetes-admin. The group kubeadm:cluster-admins is bound to the built-in cluster-admin ClusterRole. Do not share the admin.conf file with anyone.

kubeadm init generates another kubeconfig file super-admin.conf that contains a certificate with Subject: O = system:masters, CN = kubernetes-super-admin. system:masters is a break-glass, super user group that bypasses the authorization layer (for example RBAC). Do not share the super-admin.conf file with anyone. It is recommended to move the file to a safe location.

See [Generating kubeconfig files for additional users](#) on how to use kubeadm kubeconfig user to generate kubeconfig files for additional users.

Make a record of the kubeadm join command that kubeadm init outputs. You need this command to [join nodes to your cluster](#).

The token is used for mutual authentication between the control-plane node and the joining nodes. The token included here is secret. Keep it safe, because anyone with this token can add authenticated nodes to your cluster. These tokens can be listed, created, and deleted with the kubeadm token command. See the [kubeadm reference guide](#).

## Installing a Pod network add-on

### Caution:

This section contains important information about networking setup and deployment order. Read all of this advice carefully before proceeding.

**You must deploy a [Container Network Interface](#) (CNI) based Pod network add-on so that your Pods can communicate with each other. Cluster DNS (CoreDNS) will not start up before a network is installed.**

- Take care that your Pod network must not overlap with any of the host networks: you are likely to see problems if there is any overlap. (If you find a collision between your network plugin's preferred Pod network and some of your host networks, you should think of a suitable CIDR block to use instead, then use that during kubeadm init with --pod-network-cidr and as a replacement in your network plugin's YAML).



- By default, kubeadm sets up your cluster to use and enforce use of [RBAC](#) (role based access control). Make sure that your Pod network plugin supports RBAC, and so do any manifests that you use to deploy it.
- If you want to use IPv6--either dual-stack, or single-stack IPv6 only networking--for your cluster, make sure that your Pod network plugin supports IPv6. IPv6 support was added to CNI in [v0.6.0](#).

**Note:**

Kubeadm should be CNI agnostic and the validation of CNI providers is out of the scope of our current e2e testing. If you find an issue related to a CNI plugin you should log a ticket in its respective issue tracker instead of the kubeadm or kubernetes issue trackers.

Several external projects provide Kubernetes Pod networks using CNI, some of which also support [Network Policy](#).

See a list of add-ons that implement the [Kubernetes networking model](#).

Please refer to the [Installing Addons](#) page for a non-exhaustive list of networking addons supported by Kubernetes. You can install a Pod network add-on with the following command on the control-plane node or a node that has the kubeconfig credentials:

```
kubectl apply -f <add-on.yaml>
```

You can install only one Pod network per cluster.

Once a Pod network has been installed, you can confirm that it is working by checking that the CoreDNS Pod is Running in the output of `kubectl get pods --all-namespaces`. And once the CoreDNS Pod is up and running, you can continue by joining your nodes.

If your network is not working or CoreDNS is not in the Running state, check out the [troubleshooting guide](#) for kubeadm.

## Managed node labels

By default, kubeadm enables the [NodeRestriction](#) admission controller that restricts what labels can be self-applied by kubelets on node registration. The admission controller documentation covers what labels are permitted to be used with the kubelet `--node-labels` option. The `node-role.kubernetes.io/control-plane` label is such a restricted label and kubeadm manually applies it using a privileged client after a node has been created. To do that manually you can do the same by using `kubectl label` and ensure it is using a privileged kubeconfig such as the `kubeadm managed /etc/kubernetes/admin.conf`.

## Control plane node isolation

By default, your cluster will not schedule Pods on the control plane nodes for security reasons. If you want to be able to schedule Pods on the control plane nodes, for example for a single machine Kubernetes cluster, run:

```
kubectl taint nodes --all node-role.kubernetes.io/control-plane-
```

The output will look something like:



```
node "test-01" untainted
...
```

This will remove the `node-role.kubernetes.io/control-plane:NoSchedule` taint from any nodes that have it, including the control plane nodes, meaning that the scheduler will then be able to schedule Pods everywhere.

Additionally, you can execute the following command to remove the [node.kubernetes.io/exclude-from-external-load-balancers](https://kubernetes.io/docs/concepts/configuration/label/#exclude-from-external-load-balancers) label from the control plane node, which excludes it from the list of backend servers:

```
kubectll label nodes --all node.kubernetes.io/exclude-from-external-load-balancers-
```

## Joining your nodes

The nodes are where your workloads (containers and Pods, etc) run. To add new nodes to your cluster do the following for each machine:

- SSH to the machine
- Become root (e.g. `sudo su -`)
- [Install a runtime](#) if needed
- Run the command that was output by `kubeadm init`. For example:

```
kubeadm join --token <token> <control-plane-host>:<control-plane-port> --discovery-
token-ca-cert-hash sha256:<hash>
```

If you do not have the token, you can get it by running the following command on the control-plane node:

```
kubeadm token list
```

The output is similar to this:

| TOKEN GROUPS            | TTL | EXPIRES              | USAGES                  | DESCRIPTION                        | EXTRA                                                                    |
|-------------------------|-----|----------------------|-------------------------|------------------------------------|--------------------------------------------------------------------------|
| 8ewj1p.9r9hcjoqgajrj4gi | 23h | 2018-06-12T02:51:28Z | authentication, signing | token generated by 'kubeadm init'. | The default bootstrap system: bootstrappers: kubeadm: default-node-token |

By default, tokens expire after 24 hours. If you are joining a node to the cluster after the current token has expired, you can create a new token by running the following command on the control-plane node:

```
kubeadm token create
```

The output is similar to this:

```
5didvk.d09sbcov8ph2amjw
```

If you don't have the value of `--discovery-token-ca-cert-hash`, you can get it by running the following command chain on the control-plane node:

```
openssl x509 -pubkey -in /etc/kubernetes/pki/ca.crt | openssl rsa -pubin -outform der 2>/dev/null | \
 openssl dgst -sha256 -hex | sed 's/^.* //'
```

The output is similar to:

```
8cb2de97839780a412b93877f8507ad6c94f73add17d5d7058e91741c9d5ec78
```

### **Note:**

To specify an IPv6 tuple for `<control-plane-host>:<control-plane-port>`, IPv6 address must be enclosed in square brackets, for example: `[2001:db8::101]:2073`.

The output should look something like:

```
[preflight] Running pre-flight checks
```

```
... (log output of join workflow) ...
```

```
Node join complete:
```

- \* Certificate signing request sent to control-plane and response received.
- \* Kubelet informed of new secure connection details.

```
Run 'kubectl get nodes' on control-plane to see this machine join.
```

A few seconds later, you should notice this node in the output from `kubectl get nodes` when run on the control-plane node.

### **Note:**

As the cluster nodes are usually initialized sequentially, the CoreDNS Pods are likely to all run on the first control-plane node. To provide higher availability, please rebalance the CoreDNS Pods with `kubectl -n kube-system rollout restart deployment coredns` after at least one new node is joined.

## **(Optional) Controlling your cluster from machines other than the control-plane node**

In order to get a `kubectl` on some other computer (e.g. laptop) to talk to your cluster, you need to copy the administrator kubeconfig file from your control-plane node to your workstation like this:

```
scp root@<control-plane-host>:/etc/kubernetes/admin.conf .
kubectl --kubeconfig ./admin.conf get nodes
```

## Note:

The example above assumes SSH access is enabled for root. If that is not the case, you can copy the `admin.conf` file to be accessible by some other user and `scp` using that other user instead.

The `admin.conf` file gives the user *superuser* privileges over the cluster. This file should be used sparingly. For normal users, it's recommended to generate a unique credential to which you grant privileges. You can do this with the `kubeadm kubeconfig user --client-name <CN>` command. That command will print out a KubeConfig file to STDOUT which you should save to a file and distribute to your user. After that, grant privileges by using `kubectl create (cluster)rolebinding`.

## (Optional) Proxying API Server to localhost

If you want to connect to the API Server from outside the cluster you can use `kubectl proxy`:

```
scp root@<control-plane-host>:/etc/kubernetes/admin.conf .
kubectl --kubeconfig ./admin.conf proxy
```

You can now access the API Server locally at `http://localhost:8001/api/v1`

## Clean up

If you used disposable servers for your cluster, for testing, you can switch those off and do no further clean up. You can use `kubectl config delete-cluster` to delete your local references to the cluster.

However, if you want to deprovision your cluster more cleanly, you should first [drain the node](#) and make sure that the node is empty, then deconfigure the node.

## Remove the node

Talking to the control-plane node with the appropriate credentials, run:

```
kubectl drain <node name> --delete-emptydir-data --force --ignore-daemonsets
```

Before removing the node, reset the state installed by `kubeadm`:

```
kubeadm reset
```

The reset process does not reset or clean up iptables rules or IPVS tables. If you wish to reset iptables, you must do so manually:

```
iptables -F && iptables -t nat -F && iptables -t mangle -F && iptables -X
```

If you want to reset the IPVS tables, you must run the following command:

```
ipvsadm -C
```

Now remove the node:

```
kubectl delete node <node name>
```

If you wish to start over, run `kubeadm init` or `kubeadm join` with the appropriate arguments.

## Clean up the control plane

You can use `kubeadm reset` on the control plane host to trigger a best-effort clean up.

See the [kubeadm reset](#) reference documentation for more information about this subcommand and its options.

## Version skew policy

While `kubeadm` allows version skew against some components that it manages, it is recommended that you match the `kubeadm` version with the versions of the control plane components, `kube-proxy` and `kubelet`.

### kubeadm's skew against the Kubernetes version

`kubeadm` can be used with Kubernetes components that are the same version as `kubeadm` or one version older. The Kubernetes version can be specified to `kubeadm` by using the `--kubernetes-version` flag of `kubeadm init` or the [ClusterConfiguration.kubernetesVersion](#) field when using `--config`. This option will control the versions of `kube-apiserver`, `kube-controller-manager`, `kube-scheduler` and `kube-proxy`.

Example:

- `kubeadm` is at 1.30
- `kubernetesVersion` must be at 1.30 or 1.29

### kubeadm's skew against the kubelet

Similarly to the Kubernetes version, `kubeadm` can be used with a `kubelet` version that is the same version as `kubeadm` or three versions older.

Example:

- `kubeadm` is at 1.30
- `kubelet` on the host must be at 1.30, 1.29, 1.28 or 1.27

### kubeadm's skew against kubeadm

There are certain limitations on how `kubeadm` commands can operate on existing nodes or whole clusters managed by `kubeadm`.

If new nodes are joined to the cluster, the `kubeadm` binary used for `kubeadm join` must match the last version of `kubeadm` used to either create the cluster with `kubeadm init` or to upgrade the same node with `kubeadm upgrade`. Similar rules apply to the rest of the `kubeadm` commands with the exception of `kubeadm upgrade`.

Example for `kubeadm join`:

- `kubeadm` version 1.30 was used to create a cluster with `kubeadm init`
- Joining nodes must use a `kubeadm` binary that is at version 1.30

Nodes that are being upgraded must use a version of kubeadm that is the same MINOR version or one MINOR version newer than the version of kubeadm used for managing the node.

Example for kubeadm upgrade:

- kubeadm version 1.29 was used to create or upgrade the node
- The version of kubeadm used for upgrading the node must be at 1.29 or 1.30

To learn more about the version skew between the different Kubernetes component see the [Version Skew Policy](#).

## Limitations

### Cluster resilience

The cluster created here has a single control-plane node, with a single etcd database running on it. This means that if the control-plane node fails, your cluster may lose data and may need to be recreated from scratch.

Workarounds:

- Regularly [back up etcd](#). The etcd data directory configured by kubeadm is at `/var/lib/etcd` on the control-plane node.
- Use multiple control-plane nodes. You can read [Options for Highly Available topology](#) to pick a cluster topology that provides [high-availability](#).

### Platform compatibility

kubeadm deb/rpm packages and binaries are built for amd64, arm (32-bit), arm64, ppc64le, and s390x following the [multi-platform proposal](#).

Multiplatform container images for the control plane and addons are also supported since v1.12.

Only some of the network providers offer solutions for all platforms. Please consult the list of network providers above or the documentation from each provider to figure out whether the provider supports your chosen platform.

## Troubleshooting

If you are running into difficulties with kubeadm, please consult our [troubleshooting docs](#).

## What's next

- Verify that your cluster is running properly with [Sonobuoy](#)
- See [Upgrading kubeadm clusters](#) for details about upgrading your cluster using kubeadm.
- Learn about advanced kubeadm usage in the [kubeadm reference documentation](#)
- Learn more about Kubernetes [concepts](#) and [kubect](#)l.
- See the [Cluster Networking](#) page for a bigger list of Pod network add-ons.
- See the [list of add-ons](#) to explore other add-ons, including tools for logging, monitoring, network policy, visualization & control of your Kubernetes cluster.

- Configure how your cluster handles logs for cluster events and from applications running in Pods. See [Logging Architecture](#) for an overview of what is involved.

## Feedback

- For bugs, visit the [kubeadm GitHub issue tracker](#)
- For support, visit the [#kubeadm](#) Slack channel
- General SIG Cluster Lifecycle development Slack channel: [#sig-cluster-lifecycle](#)
- SIG Cluster Lifecycle [SIG information](#)
- SIG Cluster Lifecycle mailing list: [kubernetes-sig-cluster-lifecycle](#)

# Customizing components with the kubeadm API

This page covers how to customize the components that kubeadm deploys. For control plane components you can use flags in the ClusterConfiguration structure or patches per-node. For the kubelet and kube-proxy you can use KubeletConfiguration and KubeProxyConfiguration, accordingly.

All of these options are possible via the kubeadm configuration API. For more details on each field in the configuration you can navigate to our [API reference pages](#).

### Note:

Customizing the CoreDNS deployment of kubeadm is currently not supported. You must manually patch the kube-system/coredns [ConfigMap](#) and recreate the CoreDNS [Pods](#) after that. Alternatively, you can skip the default CoreDNS deployment and deploy your own variant. For more details on that see [Using init phases with kubeadm](#).

### Note:

To reconfigure a cluster that has already been created see [Reconfiguring a kubeadm cluster](#).

## Customizing the control plane with flags in ClusterConfiguration

The kubeadm ClusterConfiguration object exposes a way for users to override the default flags passed to control plane components such as the APIServer, ControllerManager, Scheduler and Etd. The components are defined using the following structures:

- apiServer
- controllerManager
- scheduler
- etcd

These structures contain a common extraArgs field, that consists of key: value pairs. To override a flag for a control plane component:

1. Add the appropriate extraArgs to your configuration.

2. Add flags to the extraArgs field.
3. Run kubeadm init with --config <YOUR CONFIG YAML>.

**Note:**

You can generate a ClusterConfiguration object with default values by running `kubeadm config print init-defaults` and saving the output to a file of your choice.

**Note:**

The ClusterConfiguration object is currently global in kubeadm clusters. This means that any flags that you add, will apply to all instances of the same component on different nodes. To apply individual configuration per component on different nodes you can use [patches](#).

**Note:**

Duplicate flags (keys), or passing the same flag --foo multiple times, is currently not supported. To workaroud that you must use [patches](#).

## APIServer flags

For details, see the [reference documentation for kube-apiserver](#).

Example usage:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
kubernetesVersion: v1.16.0
apiServer:
 extraArgs:
 anonymous-auth: "false"
 enable-admission-plugins: AlwaysPullImages,DefaultStorageClass
 audit-log-path: /home/johndoe/audit.log
```

## ControllerManager flags

For details, see the [reference documentation for kube-controller-manager](#).

Example usage:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
kubernetesVersion: v1.16.0
controllerManager:
 extraArgs:
 cluster-signing-key-file: /home/johndoe/keys/ca.key
 deployment-controller-sync-period: "50"
```

## Scheduler flags

For details, see the [reference documentation for kube-scheduler](#).

Example usage:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
kubernetesVersion: v1.16.0
scheduler:
 extraArgs:
 config: /etc/kubernetes/scheduler-config.yaml
 extraVolumes:
 - name: schedulerconfig
 hostPath: /home/johndoe/schedconfig.yaml
 mountPath: /etc/kubernetes/scheduler-config.yaml
 readOnly: true
 pathType: "File"
```

## Etcd flags

For details, see the [etcd server documentation](#).

Example usage:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
etcd:
 local:
 extraArgs:
 election-timeout: 1000
```

## Customizing with patches

FEATURE STATE: Kubernetes v1.22 [beta]

Kubeadm allows you to pass a directory with patch files to `InitConfiguration` and `JoinConfiguration` on individual nodes. These patches can be used as the last customization step before component configuration is written to disk.

You can pass this file to `kubeadm init` with `--config <YOUR CONFIG YAML>`:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: InitConfiguration
patches:
 directory: /home/user/somedir
```

### Note:

For `kubeadm init` you can pass a file containing both a `ClusterConfiguration` and `InitConfiguration` separated by `---`.

You can pass this file to `kubeadm join` with `--config <YOUR CONFIG YAML>`:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: JoinConfiguration
```



patches:  
directory: /home/user/somedir

The directory must contain files named `target[suffix][+patchtype].extension`. For example, `kube-apiserver0+merge.yaml` or just `etcd.json`.

- `target` can be one of `kube-apiserver`, `kube-controller-manager`, `kube-scheduler`, `etcd` and `kubeletconfiguration`.
- `suffix` is an optional string that can be used to determine which patches are applied first alpha-numerically.
- `patchtype` can be one of `strategic`, `merge` or `json` and these must match the patching formats [supported by kubectrl](#). The default patchtype is `strategic`.
- `extension` must be either `json` or `yaml`.

#### Note:

If you are using `kubeadm upgrade` to upgrade your `kubeadm` nodes you must again provide the same patches, so that the customization is preserved after upgrade. To do that you can use the `--patches` flag, which must point to the same directory. `kubeadm upgrade` currently does not support a configuration API structure that can be used for the same purpose.

## Customizing the kubelet

To customize the kubelet you can add a [KubeletConfiguration](#) next to the `ClusterConfiguration` or `InitConfiguration` separated by `---` within the same configuration file. This file can then be passed to `kubeadm init` and `kubeadm` will apply the same base `KubeletConfiguration` to all nodes in the cluster.

For applying instance-specific configuration over the base `KubeletConfiguration` you can use the [kubeletconfiguration patch target](#).

Alternatively, you can use kubelet flags as overrides by passing them in the `nodeRegistration.kubeletExtraArgs` field supported by both `InitConfiguration` and `JoinConfiguration`. Some kubelet flags are deprecated, so check their status in the [kubelet reference documentation](#) before using them.

For additional details see [Configuring each kubelet in your cluster using kubeadm](#)

## Customizing kube-proxy

To customize kube-proxy you can pass a `KubeProxyConfiguration` next your `ClusterConfiguration` or `InitConfiguration` to `kubeadm init` separated by `---`.

For more details you can navigate to our [API reference pages](#).

#### Note:

`kubeadm` deploys kube-proxy as a [DaemonSet](#), which means that the `KubeProxyConfiguration` would apply to all instances of kube-proxy in the cluster.

# Options for Highly Available Topology

This page explains the two options for configuring the topology of your highly available (HA) Kubernetes clusters.

You can set up an HA cluster:

- With stacked control plane nodes, where etcd nodes are colocated with control plane nodes
- With external etcd nodes, where etcd runs on separate nodes from the control plane

You should carefully consider the advantages and disadvantages of each topology before setting up an HA cluster.

## Note:

kubeadm bootstraps the etcd cluster statically. Read the etcd [Clustering Guide](#) for more details.

## Stacked etcd topology

A stacked HA cluster is a [topology](#) where the distributed data storage cluster provided by etcd is stacked on top of the cluster formed by the nodes managed by kubeadm that run control plane components.

Each control plane node runs an instance of the kube-apiserver, kube-scheduler, and kube-controller-manager. The kube-apiserver is exposed to worker nodes using a load balancer.

Each control plane node creates a local etcd member and this etcd member communicates only with the kube-apiserver of this node. The same applies to the local kube-controller-manager and kube-scheduler instances.

This topology couples the control planes and etcd members on the same nodes. It is simpler to set up than a cluster with external etcd nodes, and simpler to manage for replication.

However, a stacked cluster runs the risk of failed coupling. If one node goes down, both an etcd member and a control plane instance are lost, and redundancy is compromised. You can mitigate this risk by adding more control plane nodes.

You should therefore run a minimum of three stacked control plane nodes for an HA cluster.

This is the default topology in kubeadm. A local etcd member is created automatically on control plane nodes when using `kubeadm init` and `kubeadm join --control-plane`.

Stacked etcd topology

## External etcd topology

An HA cluster with external etcd is a [topology](#) where the distributed data storage cluster provided by etcd is external to the cluster formed by the nodes that run control plane components.

Like the stacked etcd topology, each control plane node in an external etcd topology runs an instance of the kube-apiserver, kube-scheduler, and kube-controller-manager. And the kube-apiserver is exposed to worker nodes using a load balancer. However, etcd members run on separate hosts, and each etcd host communicates with the kube-apiserver of each control plane node.

This topology decouples the control plane and etcd member. It therefore provides an HA setup where losing a control plane instance or an etcd member has less impact and does not affect the cluster redundancy as much as the stacked HA topology.

However, this topology requires twice the number of hosts as the stacked HA topology. A minimum of three hosts for control plane nodes and three hosts for etcd nodes are required for an HA cluster with this topology.

External etcd topology

## What's next

- [Set up a highly available cluster with kubeadm](#)

# Creating Highly Available Clusters with kubeadm

This page explains two different approaches to setting up a highly available Kubernetes cluster using kubeadm:

- With stacked control plane nodes. This approach requires less infrastructure. The etcd members and control plane nodes are co-located.
- With an external etcd cluster. This approach requires more infrastructure. The control plane nodes and etcd members are separated.

Before proceeding, you should carefully consider which approach best meets the needs of your applications and environment. [Options for Highly Available topology](#) outlines the advantages and disadvantages of each.

If you encounter issues with setting up the HA cluster, please report these in the kubeadm [issue tracker](#).

See also the [upgrade documentation](#).

### Caution:

This page does not address running your cluster on a cloud provider. In a cloud environment, neither approach documented here works with Service objects of type LoadBalancer, or with dynamic PersistentVolumes.

# Before you begin

The prerequisites depend on which topology you have selected for your cluster's control plane:

- [Stacked etcd](#)
- [External etcd](#)

You need:

- Three or more machines that meet [kubeadm's minimum requirements](#) for the control-plane nodes. Having an odd number of control plane nodes can help with leader selection in the case of machine or zone failure.
  - including a [container runtime](#), already set up and working
- Three or more machines that meet [kubeadm's minimum requirements](#) for the workers
  - including a container runtime, already set up and working
- Full network connectivity between all machines in the cluster (public or private network)
- Superuser privileges on all machines using sudo
  - You can use a different tool; this guide uses sudo in the examples.
- SSH access from one device to all nodes in the system
- kubeadm and kubelet already installed on all machines.

See [Stacked etcd topology](#) for context.

You need:

- Three or more machines that meet [kubeadm's minimum requirements](#) for the control-plane nodes. Having an odd number of control plane nodes can help with leader selection in the case of machine or zone failure.
  - including a [container runtime](#), already set up and working
- Three or more machines that meet [kubeadm's minimum requirements](#) for the workers
  - including a container runtime, already set up and working
- Full network connectivity between all machines in the cluster (public or private network)
- Superuser privileges on all machines using sudo
  - You can use a different tool; this guide uses sudo in the examples.
- SSH access from one device to all nodes in the system
- kubeadm and kubelet already installed on all machines.

And you also need:

- Three or more additional machines, that will become etcd cluster members. Having an odd number of members in the etcd cluster is a requirement for achieving optimal voting quorum.
  - These machines again need to have kubeadm and kubelet installed.
  - These machines also require a container runtime, that is already set up and working.

See [External etcd topology](#) for context.

## Container images

Each host should have access read and fetch images from the Kubernetes container image registry, registry.k8s.io. If you want to deploy a highly-available cluster where the hosts do not

have access to pull images, this is possible. You must ensure by some other means that the correct container images are already available on the relevant hosts.

## Command line interface

To manage Kubernetes once your cluster is set up, you should [install kubectl](#) on your PC. It is also useful to install the kubectl tool on each control plane node, as this can be helpful for troubleshooting.

## First steps for both methods

### Create load balancer for kube-apiserver

#### Note:

There are many configurations for load balancers. The following example is only one option. Your cluster requirements may need a different configuration.

1. Create a kube-apiserver load balancer with a name that resolves to DNS.
  - In a cloud environment you should place your control plane nodes behind a TCP forwarding load balancer. This load balancer distributes traffic to all healthy control plane nodes in its target list. The health check for an apiserver is a TCP check on the port the kube-apiserver listens on (default value :6443).
  - It is not recommended to use an IP address directly in a cloud environment.
  - The load balancer must be able to communicate with all control plane nodes on the apiserver port. It must also allow incoming traffic on its listening port.
  - Make sure the address of the load balancer always matches the address of kubeadm's ControlPlaneEndpoint.
  - Read the [Options for Software Load Balancing](#) guide for more details.
2. Add the first control plane node to the load balancer, and test the connection:

```
nc -v <LOAD_BALANCER_IP> <PORT>
```

A connection refused error is expected because the API server is not yet running. A timeout, however, means the load balancer cannot communicate with the control plane node. If a timeout occurs, reconfigure the load balancer to communicate with the control plane node.

3. Add the remaining control plane nodes to the load balancer target group.

## Stacked control plane and etcd nodes

### Steps for the first control plane node

1. Initialize the control plane:

```
sudo kubeadm init --control-plane-endpoint "LOAD_BALANCER_DNS:LOAD_BALANCER_PORT" --upload-certs
```

- You can use the `--kubernetes-version` flag to set the Kubernetes version to use. It is recommended that the versions of kubeadm, kubelet, kubectl and Kubernetes match.
- The `--control-plane-endpoint` flag should be set to the address or DNS and port of the load balancer.
- The `--upload-certs` flag is used to upload the certificates that should be shared across all the control-plane instances to the cluster. If instead, you prefer to copy certs across control-plane nodes manually or using automation tools, please remove this flag and refer to [Manual certificate distribution](#) section below.

**Note:**

The kubeadm init flags `--config` and `--certificate-key` cannot be mixed, therefore if you want to use the [kubeadm configuration](#) you must add the `certificateKey` field in the appropriate config locations (under `InitConfiguration` and `JoinConfiguration: controlPlane`).

**Note:**

Some CNI network plugins require additional configuration, for example specifying the pod IP CIDR, while others do not. See the [CNI network documentation](#). To add a pod CIDR pass the flag `--pod-network-cidr`, or if you are using a kubeadm configuration file set the `podSubnet` field under the `networking` object of `ClusterConfiguration`.

The output looks similar to:

...

You can now join any number of control-plane node by running the following command on each as a root:

```
kubeadm join 192.168.0.200:6443 --token 9vr73a.a8uxyaju799qwdjv --discovery-token-ca-cert-hash sha256:7c2e69131a36ae2a042a339b33381c6d0d43887e2de83720eff5359e26aec866 --control-plane --certificate-key f8902e114ef118304e561c3ecd4d0b543adc226b7a07f675f56564185ffe0c07
```

Please note that the `certificate-key` gives access to cluster sensitive data, keep it secret! As a safeguard, uploaded-certs will be deleted in two hours; If necessary, you can use `kubeadm init phase upload-certs` to reload certs afterward.

Then you can join any number of worker nodes by running the following on each as root:

```
kubeadm join 192.168.0.200:6443 --token 9vr73a.a8uxyaju799qwdjv --discovery-token-ca-cert-hash sha256:7c2e69131a36ae2a042a339b33381c6d0d43887e2de83720eff5359e26aec866
```

- Copy this output to a text file. You will need it later to join control plane and worker nodes to the cluster.

When `--upload-certs` is used with `kubeadm init`, the certificates of the primary control plane are encrypted and uploaded in the `kubeadm-certs` Secret.

- To re-upload the certificates and generate a new decryption key, use the following command on a control plane node that is already joined to the cluster:

```
sudo kubeadm init phase upload-certs --upload-certs
```

- You can also specify a custom `--certificate-key` during `init` that can later be used by `join`. To generate such a key you can use the following command:

```
kubeadm certs certificate-key
```

The certificate key is a hex encoded string that is an AES key of size 32 bytes.

**Note:**

The `kubeadm-certs` Secret and the decryption key expire after two hours.

**Caution:**

As stated in the command output, the certificate key gives access to cluster sensitive data, keep it secret!

2. Apply the CNI plugin of your choice: [Follow these instructions](#) to install the CNI provider. Make sure the configuration corresponds to the Pod CIDR specified in the `kubeadm` configuration file (if applicable).

**Note:**

You must pick a network plugin that suits your use case and deploy it before you move on to next step. If you don't do this, you will not be able to launch your cluster properly.

3. Type the following and watch the pods of the control plane components get started:

```
kubectrl get pod -n kube-system -w
```

## Steps for the rest of the control plane nodes

For each additional control plane node you should:

1. Execute the join command that was previously given to you by the `kubeadm init` output on the first node. It should look something like this:

```
sudo kubeadm join 192.168.0.200:6443 --token 9vr73a.a8uxyaju799qwdjv --discovery-
token-ca-cert-hash
sha256:7c2e69131a36ae2a042a339b33381c6d0d43887e2de83720eff5359e26aec866 --control-
plane --certificate-key
f8902e114ef118304e561c3ecd4d0b543adc226b7a07f675f56564185ffe0c07
```

- The `--control-plane` flag tells `kubeadm join` to create a new control plane.
- The `--certificate-key ...` will cause the control plane certificates to be downloaded from the `kubeadm-certs` Secret in the cluster and be decrypted using the given key.

You can join multiple control-plane nodes in parallel.

## External etcd nodes

Setting up a cluster with external etcd nodes is similar to the procedure used for stacked etcd with the exception that you should setup etcd first, and you should pass the etcd information in the kubeadm config file.

### Set up the etcd cluster

1. Follow these [instructions](#) to set up the etcd cluster.
2. Set up SSH as described [here](#).
3. Copy the following files from any etcd node in the cluster to the first control plane node:

```
export CONTROL_PLANE="ubuntu@10.0.0.7"
scp /etc/kubernetes/pki/etcd/ca.crt "${CONTROL_PLANE}":
scp /etc/kubernetes/pki/apiserver-etcd-client.crt "${CONTROL_PLANE}":
scp /etc/kubernetes/pki/apiserver-etcd-client.key "${CONTROL_PLANE}":
```

- Replace the value of CONTROL\_PLANE with the user@host of the first control-plane node.

### Set up the first control plane node

1. Create a file called kubeadm-config.yaml with the following contents:

```

apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
kubernetesVersion: stable
controlPlaneEndpoint: "LOAD_BALANCER_DNS:LOAD_BALANCER_PORT" # change
this (see below)
etcd:
 external:
 endpoints:
 - https://ETCD_0_IP:2379 # change ETCD_0_IP appropriately
 - https://ETCD_1_IP:2379 # change ETCD_1_IP appropriately
 - https://ETCD_2_IP:2379 # change ETCD_2_IP appropriately
 caFile: /etc/kubernetes/pki/etcd/ca.crt
 certFile: /etc/kubernetes/pki/apiserver-etcd-client.crt
 keyFile: /etc/kubernetes/pki/apiserver-etcd-client.key
```



**Note:**

The difference between stacked etcd and external etcd here is that the external etcd setup requires a configuration file with the etcd endpoints under the external object for etcd. In the case of the stacked etcd topology, this is managed automatically.

- Replace the following variables in the config template with the appropriate values for your cluster:
  - `LOAD_BALANCER_DNS`
  - `LOAD_BALANCER_PORT`
  - `ETCD_0_IP`
  - `ETCD_1_IP`
  - `ETCD_2_IP`

The following steps are similar to the stacked etcd setup:

1. Run `sudo kubeadm init --config kubeadm-config.yaml --upload-certs` on this node.
2. Write the output join commands that are returned to a text file for later use.
3. Apply the CNI plugin of your choice.

**Note:**

You must pick a network plugin that suits your use case and deploy it before you move on to next step. If you don't do this, you will not be able to launch your cluster properly.

## Steps for the rest of the control plane nodes

The steps are the same as for the stacked etcd setup:

- Make sure the first control plane node is fully initialized.
- Join each control plane node with the join command you saved to a text file. It's recommended to join the control plane nodes one at a time.
- Don't forget that the decryption key from `--certificate-key` expires after two hours, by default.

## Common tasks after bootstrapping control plane

### Install workers

Worker nodes can be joined to the cluster with the command you stored previously as the output from the `kubeadm init` command:

```
sudo kubeadm join 192.168.0.200:6443 --token 9vr73a.a8uxyaju799qwdjv --discovery-token-ca-cert-hash sha256:7c2e69131a36ae2a042a339b33381c6d0d43887e2de83720eff5359e26aec866
```

# Manual certificate distribution

If you choose to not use `kubeadm init` with the `--upload-certs` flag this means that you are going to have to manually copy the certificates from the primary control plane node to the joining control plane nodes.

There are many ways to do this. The following example uses `ssh` and `scp`:

SSH is required if you want to control all nodes from a single machine.

1. Enable `ssh-agent` on your main device that has access to all other nodes in the system:

```
eval $(ssh-agent)
```

2. Add your SSH identity to the session:

```
ssh-add ~/.ssh/path_to_private_key
```

3. SSH between nodes to check that the connection is working correctly.

- When you SSH to any node, add the `-A` flag. This flag allows the node that you have logged into via SSH to access the SSH agent on your PC. Consider alternative methods if you do not fully trust the security of your user session on the node.

```
ssh -A 10.0.0.7
```

- When using `sudo` on any node, make sure to preserve the environment so SSH forwarding works:

```
sudo -E -s
```

4. After configuring SSH on all the nodes you should run the following script on the first control plane node after running `kubeadm init`. This script will copy the certificates from the first control plane node to the other control plane nodes:

In the following example, replace `CONTROL_PLANE_IPS` with the IP addresses of the other control plane nodes.

```
USER=ubuntu # customizable
CONTROL_PLANE_IPS="10.0.0.7 10.0.0.8"
for host in ${CONTROL_PLANE_IPS}; do
 scp /etc/kubernetes/pki/ca.crt "${USER}"@$host:
 scp /etc/kubernetes/pki/ca.key "${USER}"@$host:
 scp /etc/kubernetes/pki/sa.key "${USER}"@$host:
 scp /etc/kubernetes/pki/sa.pub "${USER}"@$host:
 scp /etc/kubernetes/pki/front-proxy-ca.crt "${USER}"@$host:
 scp /etc/kubernetes/pki/front-proxy-ca.key "${USER}"@$host:
 scp /etc/kubernetes/pki/etcd/ca.crt "${USER}"@$host:etcd-ca.crt
 # Skip the next line if you are using external etcd
 scp /etc/kubernetes/pki/etcd/ca.key "${USER}"@$host:etcd-ca.key
done
```

### Caution:

Copy only the certificates in the above list. kubeadm will take care of generating the rest of the certificates with the required SANs for the joining control-plane instances. If you copy all the certificates by mistake, the creation of additional nodes could fail due to a lack of required SANs.

5. Then on each joining control plane node you have to run the following script before running kubeadm join. This script will move the previously copied certificates from the home directory to /etc/kubernetes/pki:

```
USER=ubuntu # customizable
mkdir -p /etc/kubernetes/pki/etcd
mv /home/${USER}/ca.crt /etc/kubernetes/pki/
mv /home/${USER}/ca.key /etc/kubernetes/pki/
mv /home/${USER}/sa.pub /etc/kubernetes/pki/
mv /home/${USER}/sa.key /etc/kubernetes/pki/
mv /home/${USER}/front-proxy-ca.crt /etc/kubernetes/pki/
mv /home/${USER}/front-proxy-ca.key /etc/kubernetes/pki/
mv /home/${USER}/etcd-ca.crt /etc/kubernetes/pki/etcd/ca.crt
Skip the next line if you are using external etcd
mv /home/${USER}/etcd-ca.key /etc/kubernetes/pki/etcd/ca.key
```

## Set up a High Availability etcd Cluster with kubeadm

### Note:

While kubeadm is being used as the management tool for external etcd nodes in this guide, please note that kubeadm does not plan to support certificate rotation or upgrades for such nodes. The long-term plan is to empower the tool [etcdadm](#) to manage these aspects.

By default, kubeadm runs a local etcd instance on each control plane node. It is also possible to treat the etcd cluster as external and provision etcd instances on separate hosts. The differences between the two approaches are covered in the [Options for Highly Available topology](#) page.

This task walks through the process of creating a high availability external etcd cluster of three members that can be used by kubeadm during cluster creation.

### Before you begin

- Three hosts that can talk to each other over TCP ports 2379 and 2380. This document assumes these default ports. However, they are configurable through the kubeadm config file.
- Each host must have systemd and a bash compatible shell installed.
- Each host must [have a container runtime, kubelet, and kubeadm installed](#).
- Each host should have access to the Kubernetes container image registry (registry.k8s.io) or list/pull the required etcd image using kubeadm config images list/pull. This guide will set up etcd instances as [static pods](#) managed by a kubelet.

- Some infrastructure to copy files between hosts. For example `ssh` and `scp` can satisfy this requirement.

## Setting up the cluster

The general approach is to generate all certs on one node and only distribute the *necessary* files to the other nodes.

### Note:

kubeadm contains all the necessary cryptographic machinery to generate the certificates described below; no other cryptographic tooling is required for this example.

### Note:

The examples below use IPv4 addresses but you can also configure kubeadm, the kubelet and etcd to use IPv6 addresses. Dual-stack is supported by some Kubernetes options, but not by etcd. For more details on Kubernetes dual-stack support see [Dual-stack support with kubeadm](#).

1. Configure the kubelet to be a service manager for etcd.

### Note:

You must do this on every host where etcd should be running.

Since etcd was created first, you must override the service priority by creating a new unit file that has higher precedence than the kubeadm-provided kubelet unit file.

```
cat << EOF > /etc/systemd/system/kubelet.service.d/kubelet.conf
Replace "systemd" with the cgroup driver of your container runtime. The default value
in the kubelet is "cgroupfs".
Replace the value of "containerRuntimeEndpoint" for a different container runtime if
needed.
#
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
authentication:
 anonymous:
 enabled: false
 webhook:
 enabled: false
authorization:
 mode: AlwaysAllow
cgroupDriver: systemd
address: 127.0.0.1
containerRuntimeEndpoint: unix:///var/run/containerd/containerd.sock
staticPodPath: /etc/kubernetes/manifests
EOF
```

```
cat << EOF > /etc/systemd/system/kubelet.service.d/20-etcd-service-manager.conf
[Service]
ExecStart=
```

```
ExecStart=/usr/bin/kubelet --config=/etc/systemd/system/kubelet.service.d/kubelet.conf
Restart=always
EOF
```

```
systemctl daemon-reload
systemctl restart kubelet
```

Check the kubelet status to ensure it is running.

```
systemctl status kubelet
```

## 2. Create configuration files for kubeadm.

Generate one kubeadm configuration file for each host that will have an etcd member running on it using the following script.

```
Update HOST0, HOST1 and HOST2 with the IPs of your hosts
export HOST0=10.0.0.6
export HOST1=10.0.0.7
export HOST2=10.0.0.8

Update NAME0, NAME1 and NAME2 with the hostnames of your hosts
export NAME0="infra0"
export NAME1="infra1"
export NAME2="infra2"

Create temp directories to store files that will end up on other hosts
mkdir -p /tmp/${HOST0}/ /tmp/${HOST1}/ /tmp/${HOST2}/

HOSTS=(${HOST0} ${HOST1} ${HOST2})
NAMES=(${NAME0} ${NAME1} ${NAME2})

for i in "${!HOSTS[@]}"; do
 HOST=${HOSTS[$i]}
 NAME=${NAMES[$i]}
 cat << EOF > /tmp/${HOST}/kubeadmcfg.yaml

 apiVersion: "kubeadm.k8s.io/v1beta3"
 kind: InitConfiguration
 nodeRegistration:
 name: ${NAME}
 localAPIEndpoint:
 advertiseAddress: ${HOST}

 apiVersion: "kubeadm.k8s.io/v1beta3"
 kind: ClusterConfiguration
 etcd:
 local:
 serverCertSANs:
 - "${HOST}"
 peerCertSANs:
 - "${HOST}"
 extraArgs:
```

```

 initial-cluster: ${NAMES[0]}=https://${HOSTS[0]}:2380,${NAMES[1]}=https://$
 {HOSTS[1]}:2380,${NAMES[2]}=https://${HOSTS[2]}:2380
 initial-cluster-state: new
 name: ${NAME}
 listen-peer-urls: https://${HOST}:2380
 listen-client-urls: https://${HOST}:2379
 advertise-client-urls: https://${HOST}:2379
 initial-advertise-peer-urls: https://${HOST}:2380
EOF
done

```

### 3. Generate the certificate authority.

If you already have a CA then the only action that is copying the CA's crt and key file to /etc/kubernetes/pki/etcd/ca.crt and /etc/kubernetes/pki/etcd/ca.key. After those files have been copied, proceed to the next step, "Create certificates for each member".

If you do not already have a CA then run this command on \$HOST0 (where you generated the configuration files for kubeadm).

```
kubeadm init phase certs etcd-ca
```

This creates two files:

- /etc/kubernetes/pki/etcd/ca.crt
- /etc/kubernetes/pki/etcd/ca.key

### 4. Create certificates for each member.

```

kubeadm init phase certs etcd-server --config=/tmp/${HOST2}/kubeadmcfg.yaml
kubeadm init phase certs etcd-peer --config=/tmp/${HOST2}/kubeadmcfg.yaml
kubeadm init phase certs etcd-healthcheck-client --config=/tmp/${HOST2}/
kubeadmcfg.yaml
kubeadm init phase certs apiserver-etcd-client --config=/tmp/${HOST2}/kubeadmcfg.yaml
cp -R /etc/kubernetes/pki /tmp/${HOST2}/
cleanup non-reusable certificates
find /etc/kubernetes/pki -not -name ca.crt -not -name ca.key -type f -delete

kubeadm init phase certs etcd-server --config=/tmp/${HOST1}/kubeadmcfg.yaml
kubeadm init phase certs etcd-peer --config=/tmp/${HOST1}/kubeadmcfg.yaml
kubeadm init phase certs etcd-healthcheck-client --config=/tmp/${HOST1}/
kubeadmcfg.yaml
kubeadm init phase certs apiserver-etcd-client --config=/tmp/${HOST1}/kubeadmcfg.yaml
cp -R /etc/kubernetes/pki /tmp/${HOST1}/
find /etc/kubernetes/pki -not -name ca.crt -not -name ca.key -type f -delete

kubeadm init phase certs etcd-server --config=/tmp/${HOST0}/kubeadmcfg.yaml
kubeadm init phase certs etcd-peer --config=/tmp/${HOST0}/kubeadmcfg.yaml
kubeadm init phase certs etcd-healthcheck-client --config=/tmp/${HOST0}/
kubeadmcfg.yaml
kubeadm init phase certs apiserver-etcd-client --config=/tmp/${HOST0}/kubeadmcfg.yaml
No need to move the certs because they are for HOST0

clean up certs that should not be copied off this host

```

```
find /tmp/${HOST2} -name ca.key -type f -delete
find /tmp/${HOST1} -name ca.key -type f -delete
```

#### 5. Copy certificates and kubeadm configs.

The certificates have been generated and now they must be moved to their respective hosts.

```
USER=ubuntu
HOST=${HOST1}
scp -r /tmp/${HOST}/* ${USER}@${HOST}:
ssh ${USER}@${HOST}
USER@HOST $ sudo -Es
root@HOST $ chown -R root:root pki
root@HOST $ mv pki /etc/kubernetes/
```

#### 6. Ensure all expected files exist.

The complete list of required files on \$HOST0 is:

```
/tmp/${HOST0}
└─ kubeadmcfg.yaml

/etc/kubernetes/pki
├─ apiserver-etcd-client.crt
├─ apiserver-etcd-client.key
├─ etcd
│ ├─ ca.crt
│ ├─ ca.key
│ ├─ healthcheck-client.crt
│ ├─ healthcheck-client.key
│ ├─ peer.crt
│ ├─ peer.key
│ ├─ server.crt
│ └─ server.key
```

On \$HOST1:

```
$HOME
└─ kubeadmcfg.yaml

/etc/kubernetes/pki
├─ apiserver-etcd-client.crt
├─ apiserver-etcd-client.key
├─ etcd
│ ├─ ca.crt
│ ├─ healthcheck-client.crt
│ ├─ healthcheck-client.key
│ ├─ peer.crt
│ ├─ peer.key
│ ├─ server.crt
│ └─ server.key
```

On \$HOST2:

```

$HOME
└── kubeadmcfg.yaml

/etc/kubernetes/pki
├── apiserver-etcd-client.crt
├── apiserver-etcd-client.key
└── etcd
 ├── ca.crt
 ├── healthcheck-client.crt
 ├── healthcheck-client.key
 ├── peer.crt
 ├── peer.key
 ├── server.crt
 └── server.key

```

## 7. Create the static pod manifests.

Now that the certificates and configs are in place it's time to create the manifests. On each host run the kubeadm command to generate a static manifest for etcd.

```

root@HOST0 $ kubeadm init phase etcd local --config=/tmp/${HOST0}/kubeadmcfg.yaml
root@HOST1 $ kubeadm init phase etcd local --config=$HOME/kubeadmcfg.yaml
root@HOST2 $ kubeadm init phase etcd local --config=$HOME/kubeadmcfg.yaml

```

## 8. Optional: Check the cluster health.

If etcdctl isn't available, you can run this tool inside a container image. You would do that directly with your container runtime using a tool such as crictl run and not through Kubernetes

```

ETCDCTL_API=3 etcdctl \
--cert /etc/kubernetes/pki/etcd/peer.crt \
--key /etc/kubernetes/pki/etcd/peer.key \
--cacert /etc/kubernetes/pki/etcd/ca.crt \
--endpoints https://${HOST0}:2379 endpoint health
...
https://[HOST0 IP]:2379 is healthy: successfully committed proposal: took = 16.283339ms
https://[HOST1 IP]:2379 is healthy: successfully committed proposal: took = 19.44402ms
https://[HOST2 IP]:2379 is healthy: successfully committed proposal: took = 35.926451ms

```

- Set \${HOST0} to the IP address of the host you are testing.

## What's next

Once you have an etcd cluster with 3 working members, you can continue setting up a highly available control plane using the [external etcd method with kubeadm](#).



# Configuring each kubelet in your cluster using kubeadm

**Note:** Dockershim has been removed from the Kubernetes project as of release 1.24. Read the [Dockershim Removal FAQ](#) for further details.

FEATURE STATE: Kubernetes v1.11 [stable]

The lifecycle of the kubeadm CLI tool is decoupled from the [kubelet](#), which is a daemon that runs on each node within the Kubernetes cluster. The kubeadm CLI tool is executed by the user when Kubernetes is initialized or upgraded, whereas the kubelet is always running in the background.

Since the kubelet is a daemon, it needs to be maintained by some kind of an init system or service manager. When the kubelet is installed using DEBs or RPMs, systemd is configured to manage the kubelet. You can use a different service manager instead, but you need to configure it manually.

Some kubelet configuration details need to be the same across all kubelets involved in the cluster, while other configuration aspects need to be set on a per-kubelet basis to accommodate the different characteristics of a given machine (such as OS, storage, and networking). You can manage the configuration of your kubelets manually, but kubeadm now provides a KubeletConfiguration API type for [managing your kubelet configurations centrally](#).

## Kubelet configuration patterns

The following sections describe patterns to kubelet configuration that are simplified by using kubeadm, rather than managing the kubelet configuration for each Node manually.

### Propagating cluster-level configuration to each kubelet

You can provide the kubelet with default values to be used by kubeadm init and kubeadm join commands. Interesting examples include using a different container runtime or setting the default subnet used by services.

If you want your services to use the subnet 10.96.0.0/12 as the default for services, you can pass the --service-cidr parameter to kubeadm:

```
kubeadm init --service-cidr 10.96.0.0/12
```

Virtual IPs for services are now allocated from this subnet. You also need to set the DNS address used by the kubelet, using the --cluster-dns flag. This setting needs to be the same for every kubelet on every manager and Node in the cluster. The kubelet provides a versioned, structured API object that can configure most parameters in the kubelet and push out this configuration to each running kubelet in the cluster. This object is called [KubeletConfiguration](#). The KubeletConfiguration allows the user to specify flags such as the cluster DNS IP addresses expressed as a list of values to a camelCased key, illustrated by the following example:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
clusterDNS:
- 10.96.0.10
```

For more details on the KubeletConfiguration have a look at [this section](#).

## Providing instance-specific configuration details

Some hosts require specific kubelet configurations due to differences in hardware, operating system, networking, or other host-specific parameters. The following list provides a few examples.

- The path to the DNS resolution file, as specified by the `--resolv-conf` kubelet configuration flag, may differ among operating systems, or depending on whether you are using `systemd-resolved`. If this path is wrong, DNS resolution will fail on the Node whose kubelet is configured incorrectly.
- The Node API object `.metadata.name` is set to the machine's hostname by default, unless you are using a cloud provider. You can use the `--hostname-override` flag to override the default behavior if you need to specify a Node name different from the machine's hostname.
- Currently, the kubelet cannot automatically detect the cgroup driver used by the container runtime, but the value of `--cgroup-driver` must match the cgroup driver used by the container runtime to ensure the health of the kubelet.
- To specify the container runtime you must set its endpoint with the `--container-runtime-endpoint=<path>` flag.

The recommended way of applying such instance-specific configuration is by using [KubeletConfiguration patches](#).

## Configure kubelets using kubeadm

It is possible to configure the kubelet that kubeadm will start if a custom [KubeletConfiguration](#) API object is passed with a configuration file like so `kubeadm ... --config some-config-file.yaml`.

By calling `kubeadm config print init-defaults --component-configs KubeletConfiguration` you can see all the default values for this structure.

It is also possible to apply instance-specific patches over the base KubeletConfiguration. Have a look at [Customizing the kubelet](#) for more details.

## Workflow when using kubeadm init

When you call `kubeadm init`, the kubelet configuration is marshalled to disk at `/var/lib/kubelet/config.yaml`, and also uploaded to a `kubelet-config` ConfigMap in the `kube-system` namespace of the cluster. A kubelet configuration file is also written to `/etc/kubernetes/kubelet.conf` with the baseline cluster-wide configuration for all kubelets in the cluster. This configuration file points to the client certificates that allow the kubelet to communicate with the API server. This addresses the need to [propagate cluster-level configuration to each kubelet](#).

To address the second pattern of [providing instance-specific configuration details](#), kubeadm writes an environment file to `/var/lib/kubelet/kubeadm-flags.env`, which contains a list of flags to pass to the kubelet when it starts. The flags are presented in the file like this:

```
KUBELET_KUBEADM_ARGS="--flag1=value1 --flag2=value2 ..."
```

In addition to the flags used when starting the kubelet, the file also contains dynamic parameters such as the cgroup driver and whether to use a different container runtime socket (`--cri-socket`).

After marshalling these two files to disk, kubeadm attempts to run the following two commands, if you are using systemd:

```
systemctl daemon-reload && systemctl restart kubelet
```

If the reload and restart are successful, the normal kubeadm init workflow continues.

## Workflow when using kubeadm join

When you run `kubeadm join`, kubeadm uses the Bootstrap Token credential to perform a TLS bootstrap, which fetches the credential needed to download the kubelet-config ConfigMap and writes it to `/var/lib/kubelet/config.yaml`. The dynamic environment file is generated in exactly the same way as `kubeadm init`.

Next, kubeadm runs the following two commands to load the new configuration into the kubelet:

```
systemctl daemon-reload && systemctl restart kubelet
```

After the kubelet loads the new configuration, kubeadm writes the `/etc/kubernetes/bootstrap-kubelet.conf` KubeConfig file, which contains a CA certificate and Bootstrap Token. These are used by the kubelet to perform the TLS Bootstrap and obtain a unique credential, which is stored in `/etc/kubernetes/kubelet.conf`.

When the `/etc/kubernetes/kubelet.conf` file is written, the kubelet has finished performing the TLS Bootstrap. Kubeadm deletes the `/etc/kubernetes/bootstrap-kubelet.conf` file after completing the TLS Bootstrap.

## The kubelet drop-in file for systemd

kubeadm ships with configuration for how systemd should run the kubelet. Note that the kubeadm CLI command never touches this drop-in file.

This configuration file installed by the kubeadm [package](#) is written to `/usr/lib/systemd/system/kubelet.service.d/10-kubeadm.conf` and is used by systemd. It augments the basic [kubelet.service](#).

If you want to override that further, you can make a directory `/etc/systemd/system/kubelet.service.d/` (not `/usr/lib/systemd/system/kubelet.service.d/`) and put your own customizations into a file there. For example, you might add a new local file `/etc/systemd/system/kubelet.service.d/local-overrides.conf` to override the unit settings configured by kubeadm.

Here is what you are likely to find in `/usr/lib/systemd/system/kubelet.service.d/10-kubeadm.conf`:

## Note:

The contents below are just an example. If you don't want to use a package manager follow the guide outlined in the ([Without a package manager](#)) section.

```
[Service]
Environment="KUBELET_KUBECONFIG_ARGS=--bootstrap-kubeconfig=/etc/kubernetes/
bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf"
Environment="KUBELET_CONFIG_ARGS=--config=/var/lib/kubelet/config.yaml"
This is a file that "kubeadm init" and "kubeadm join" generate at runtime, populating
the KUBELET_KUBEADM_ARGS variable dynamically
EnvironmentFile=-/var/lib/kubelet/kubeadm-flags.env
This is a file that the user can use for overrides of the kubelet args as a last resort. Preferably,
the user should use the .NodeRegistration.KubeletExtraArgs object in the configuration files
instead.
KUBELET_EXTRA_ARGS should be sourced from this file.
EnvironmentFile=-/etc/default/kubelet
ExecStart=
ExecStart=/usr/bin/kubelet $KUBELET_KUBECONFIG_ARGS $KUBELET_CONFIG_ARGS
$KUBELET_KUBEADM_ARGS $KUBELET_EXTRA_ARGS
```

This file specifies the default locations for all of the files managed by kubeadm for the kubelet.

- The KubeConfig file to use for the TLS Bootstrap is `/etc/kubernetes/bootstrap-kubelet.conf`, but it is only used if `/etc/kubernetes/kubelet.conf` does not exist.
- The KubeConfig file with the unique kubelet identity is `/etc/kubernetes/kubelet.conf`.
- The file containing the kubelet's ComponentConfig is `/var/lib/kubelet/config.yaml`.
- The dynamic environment file that contains `KUBELET_KUBEADM_ARGS` is sourced from `/var/lib/kubelet/kubeadm-flags.env`.
- The file that can contain user-specified flag overrides with `KUBELET_EXTRA_ARGS` is sourced from `/etc/default/kubelet` (for DEBs), or `/etc/sysconfig/kubelet` (for RPMs). `KUBELET_EXTRA_ARGS` is last in the flag chain and has the highest priority in the event of conflicting settings.

## Kubernetes binaries and package contents

The DEB and RPM packages shipped with the Kubernetes releases are:

| Package name   | Description                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| kubeadm        | Installs the <code>/usr/bin/kubeadm</code> CLI tool and the <a href="#">kubelet drop-in file</a> for the kubelet. |
| kubelet        | Installs the <code>/usr/bin/kubelet</code> binary.                                                                |
| kubectl        | Installs the <code>/usr/bin/kubectl</code> binary.                                                                |
| cri-tools      | Installs the <code>/usr/bin/crictl</code> binary from the <a href="#">cri-tools git repository</a> .              |
| kubernetes-cni | Installs the <code>/opt/cni/bin</code> binaries from the <a href="#">plugins git repository</a> .                 |

## Dual-stack support with kubeadm

FEATURE STATE: Kubernetes v1.23 [stable]

Your Kubernetes cluster includes [dual-stack](#) networking, which means that cluster networking lets you use either address family. In a cluster, the control plane can assign both an IPv4 address and an IPv6 address to a single [Pod](#) or a [Service](#).

## Before you begin

You need to have installed the [kubeadm](#) tool, following the steps from [Installing kubeadm](#).

For each server that you want to use as a [node](#), make sure it allows IPv6 forwarding. On Linux, you can set this by running `run systemctl -w net.ipv6.conf.all.forwarding=1` as the root user on each server.

You need to have an IPv4 and an IPv6 address range to use. Cluster operators typically use private address ranges for IPv4. For IPv6, a cluster operator typically chooses a global unicast address block from within `2000::/3`, using a range that is assigned to the operator. You don't have to route the cluster's IP address ranges to the public internet.

The size of the IP address allocations should be suitable for the number of Pods and Services that you are planning to run.

### Note:

If you are upgrading an existing cluster with the `kubeadm upgrade` command, `kubeadm` does not support making modifications to the pod IP address range ("cluster CIDR") nor to the cluster's Service address range ("Service CIDR").

## Create a dual-stack cluster

To create a dual-stack cluster with `kubeadm init` you can pass command line arguments similar to the following example:

```
These address ranges are examples
kubeadm init --pod-network-cidr=10.244.0.0/16,2001:db8:42:0::/56 --service-cidr=10.96.0.0/16,2001:db8:42:1::/112
```

To make things clearer, here is an example `kubeadm` [configuration file](#) `kubeadm-config.yaml` for the primary dual-stack control plane node.

```

apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
networking:
 podSubnet: 10.244.0.0/16,2001:db8:42:0::/56
 serviceSubnet: 10.96.0.0/16,2001:db8:42:1::/112

apiVersion: kubeadm.k8s.io/v1beta3
kind: InitConfiguration
localAPIEndpoint:
 advertiseAddress: "10.100.0.1"
 bindPort: 6443
nodeRegistration:
```

```
kubeletExtraArgs:
 node-ip: 10.100.0.2,fd00:1:2:3::2
```

advertiseAddress in InitConfiguration specifies the IP address that the API Server will advertise it is listening on. The value of advertiseAddress equals the `--apiserver-advertise-address` flag of `kubeadm init`.

Run `kubeadm` to initiate the dual-stack control plane node:

```
kubeadm init --config=kubeadm-config.yaml
```

The kube-controller-manager flags `--node-cidr-mask-size-ipv4|--node-cidr-mask-size-ipv6` are set with default values. See [configure IPv4/IPv6 dual stack](#).

### Note:

The `--apiserver-advertise-address` flag does not support dual-stack.

## Join a node to dual-stack cluster

Before joining a node, make sure that the node has IPv6 routable network interface and allows IPv6 forwarding.

Here is an example `kubeadm` [configuration file](#) `kubeadm-config.yaml` for joining a worker node to the cluster.

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: JoinConfiguration
discovery:
 bootstrapToken:
 apiServerEndpoint: 10.100.0.1:6443
 token: "clvldh.vjjwg16ucnhp94qr"
 caCertHashes:
 - "sha256:a4863cde706cfc580a439f842cc65d5ef112b7b2be31628513a9881cf0d9fe0e"
 # change auth info above to match the actual token and CA certificate hash for your cluster
nodeRegistration:
 kubeletExtraArgs:
 node-ip: 10.100.0.3,fd00:1:2:3::3
```

Also, here is an example `kubeadm` [configuration file](#) `kubeadm-config.yaml` for joining another control plane node to the cluster.

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: JoinConfiguration
controlPlane:
 localAPIEndpoint:
 advertiseAddress: "10.100.0.2"
 bindPort: 6443
discovery:
 bootstrapToken:
 apiServerEndpoint: 10.100.0.1:6443
 token: "clvldh.vjjwg16ucnhp94qr"
 caCertHashes:
```

```
- "sha256:a4863cde706cfc580a439f842cc65d5ef112b7b2be31628513a9881cf0d9fe0e"
change auth info above to match the actual token and CA certificate hash for your cluster
nodeRegistration:
 kubeletExtraArgs:
 node-ip: 10.100.0.4,fd00:1:2:3::4
```

advertiseAddress in JoinConfiguration.controlPlane specifies the IP address that the API Server will advertise it is listening on. The value of advertiseAddress equals the --apiserver-advertise-address flag of kubeadm join.

```
kubeadm join --config=kubeadm-config.yaml
```

## Create a single-stack cluster

### Note:

Dual-stack support doesn't mean that you need to use dual-stack addressing. You can deploy a single-stack cluster that has the dual-stack networking feature enabled.

To make things more clear, here is an example kubeadm [configuration file](#) kubeadm-config.yaml for the single-stack control plane node.

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
networking:
 podSubnet: 10.244.0.0/16
 serviceSubnet: 10.96.0.0/16
```

## What's next

- [Validate IPv4/IPv6 dual-stack](#) networking
- Read about [Dual-stack](#) cluster networking
- Learn more about the kubeadm [configuration format](#)

# Turnkey Cloud Solutions

This page provides a list of Kubernetes certified solution providers. From each provider page, you can learn how to install and setup production ready clusters.

## Best practices

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[Considerations for large clusters](#)

[Running in multiple zones](#)

[Validate node setup](#)

[Enforcing Pod Security Standards](#)



# Considerations for large clusters

A cluster is a set of [nodes](#) (physical or virtual machines) running Kubernetes agents, managed by the [control plane](#). Kubernetes v1.30 supports clusters with up to 5,000 nodes. More specifically, Kubernetes is designed to accommodate configurations that meet *all* of the following criteria:

- No more than 110 pods per node
- No more than 5,000 nodes
- No more than 150,000 total pods
- No more than 300,000 total containers

You can scale your cluster by adding or removing nodes. The way you do this depends on how your cluster is deployed.

## Cloud provider resource quotas

To avoid running into cloud provider quota issues, when creating a cluster with many nodes, consider:

- Requesting a quota increase for cloud resources such as:
  - Computer instances
  - CPUs
  - Storage volumes
  - In-use IP addresses
  - Packet filtering rule sets
  - Number of load balancers
  - Network subnets
  - Log streams
- Gating the cluster scaling actions to bring up new nodes in batches, with a pause between batches, because some cloud providers rate limit the creation of new instances.

## Control plane components

For a large cluster, you need a control plane with sufficient compute and other resources.

Typically you would run one or two control plane instances per failure zone, scaling those instances vertically first and then scaling horizontally after reaching the point of falling returns to (vertical) scale.

You should run at least one instance per failure zone to provide fault-tolerance. Kubernetes nodes do not automatically steer traffic towards control-plane endpoints that are in the same failure zone; however, your cloud provider might have its own mechanisms to do this.

For example, using a managed load balancer, you configure the load balancer to send traffic that originates from the kubelet and Pods in failure zone *A*, and direct that traffic only to the control plane hosts that are also in zone *A*. If a single control-plane host or endpoint failure zone *A* goes offline, that means that all the control-plane traffic for nodes in zone *A* is now being sent



between zones. Running multiple control plane hosts in each zone makes that outcome less likely.

## etcd storage

To improve performance of large clusters, you can store Event objects in a separate dedicated etcd instance.

When creating a cluster, you can (using custom tooling):

- start and configure additional etcd instance
- configure the [API server](#) to use it for storing events

See [Operating etcd clusters for Kubernetes](#) and [Set up a High Availability etcd cluster with kubeadm](#) for details on configuring and managing etcd for a large cluster.

## Addon resources

Kubernetes [resource limits](#) help to minimize the impact of memory leaks and other ways that pods and containers can impact on other components. These resource limits apply to [addon](#) resources just as they apply to application workloads.

For example, you can set CPU and memory limits for a logging component:

```
...
containers:
- name: fluentd-cloud-logging
 image: fluent/fluentd-kubernetes-daemonset:v1
 resources:
 limits:
 cpu: 100m
 memory: 200Mi
```

Addons' default limits are typically based on data collected from experience running each addon on small or medium Kubernetes clusters. When running on large clusters, addons often consume more of some resources than their default limits. If a large cluster is deployed without adjusting these values, the addon(s) may continuously get killed because they keep hitting the memory limit. Alternatively, the addon may run but with poor performance due to CPU time slice restrictions.

To avoid running into cluster addon resource issues, when creating a cluster with many nodes, consider the following:

- Some addons scale vertically - there is one replica of the addon for the cluster or serving a whole failure zone. For these addons, increase requests and limits as you scale out your cluster.
- Many addons scale horizontally - you add capacity by running more pods - but with a very large cluster you may also need to raise CPU or memory limits slightly. The VerticalPodAutoscaler can run in *recommender* mode to provide suggested figures for requests and limits.
- Some addons run as one copy per node, controlled by a [DaemonSet](#): for example, a node-level log aggregator. Similar to the case with horizontally-scaled addons, you may also need to raise CPU or memory limits slightly.

## What's next

- VerticalPodAutoscaler is a custom resource that you can deploy into your cluster to help you manage resource requests and limits for pods. Learn more about [Vertical Pod Autoscaler](#) and how you can use it to scale cluster components, including cluster-critical addons.
- Read about [cluster autoscaling](#)
- The [addon resizer](#) helps you in resizing the addons automatically as your cluster's scale changes.

## Running in multiple zones

This page describes running Kubernetes across multiple zones.

### Background

Kubernetes is designed so that a single Kubernetes cluster can run across multiple failure zones, typically where these zones fit within a logical grouping called a *region*. Major cloud providers define a region as a set of failure zones (also called *availability zones*) that provide a consistent set of features: within a region, each zone offers the same APIs and services.

Typical cloud architectures aim to minimize the chance that a failure in one zone also impairs services in another zone.

### Control plane behavior

All [control plane components](#) support running as a pool of interchangeable resources, replicated per component.

When you deploy a cluster control plane, place replicas of control plane components across multiple failure zones. If availability is an important concern, select at least three failure zones and replicate each individual control plane component (API server, scheduler, etcd, cluster controller manager) across at least three failure zones. If you are running a cloud controller manager then you should also replicate this across all the failure zones you selected.

#### Note:

Kubernetes does not provide cross-zone resilience for the API server endpoints. You can use various techniques to improve availability for the cluster API server, including DNS round-robin, SRV records, or a third-party load balancing solution with health checking.

### Node behavior

Kubernetes automatically spreads the Pods for workload resources (such as [Deployment](#) or [StatefulSet](#)) across different nodes in a cluster. This spreading helps reduce the impact of failures.

When nodes start up, the kubelet on each node automatically adds [labels](#) to the Node object that represents that specific kubelet in the Kubernetes API. These labels can include [zone information](#).

If your cluster spans multiple zones or regions, you can use node labels in conjunction with [Pod topology spread constraints](#) to control how Pods are spread across your cluster among fault domains: regions, zones, and even specific nodes. These hints enable the [scheduler](#) to place Pods for better expected availability, reducing the risk that a correlated failure affects your whole workload.

For example, you can set a constraint to make sure that the 3 replicas of a StatefulSet are all running in different zones to each other, whenever that is feasible. You can define this declaratively without explicitly defining which availability zones are in use for each workload.

## Distributing nodes across zones

Kubernetes' core does not create nodes for you; you need to do that yourself, or use a tool such as the [Cluster API](#) to manage nodes on your behalf.

Using tools such as the Cluster API you can define sets of machines to run as worker nodes for your cluster across multiple failure domains, and rules to automatically heal the cluster in case of whole-zone service disruption.

## Manual zone assignment for Pods

You can apply [node selector constraints](#) to Pods that you create, as well as to Pod templates in workload resources such as Deployment, StatefulSet, or Job.

## Storage access for zones

When persistent volumes are created, Kubernetes automatically adds zone labels to any PersistentVolumes that are linked to a specific zone. The [scheduler](#) then ensures, through its NoVolumeZoneConflict predicate, that pods which claim a given PersistentVolume are only placed into the same zone as that volume.

Please note that the method of adding zone labels can depend on your cloud provider and the storage provisioner you're using. Always refer to the specific documentation for your environment to ensure correct configuration.

You can specify a [StorageClass](#) for PersistentVolumeClaims that specifies the failure domains (zones) that the storage in that class may use. To learn about configuring a StorageClass that is aware of failure domains or zones, see [Allowed topologies](#).

## Networking

By itself, Kubernetes does not include zone-aware networking. You can use a [network plugin](#) to configure cluster networking, and that network solution might have zone-specific elements. For example, if your cloud provider supports Services with type=LoadBalancer, the load balancer might only send traffic to Pods running in the same zone as the load balancer element processing a given connection. Check your cloud provider's documentation for details.

For custom or on-premises deployments, similar considerations apply. [Service](#) and [Ingress](#) behavior, including handling of different failure zones, does vary depending on exactly how your cluster is set up.

## Fault recovery

When you set up your cluster, you might also need to consider whether and how your setup can restore service if all the failure zones in a region go off-line at the same time. For example, do you rely on there being at least one node able to run Pods in a zone?

Make sure that any cluster-critical repair work does not rely on there being at least one healthy node in your cluster. For example: if all nodes are unhealthy, you might need to run a repair Job with a special [toleration](#) so that the repair can complete enough to bring at least one node into service.

Kubernetes doesn't come with an answer for this challenge; however, it's something to consider.

## What's next

To learn how the scheduler places Pods in a cluster, honoring the configured constraints, visit [Scheduling and Eviction](#).

# Validate node setup

## Node Conformance Test

*Node conformance test* is a containerized test framework that provides a system verification and functionality test for a node. The test validates whether the node meets the minimum requirements for Kubernetes; a node that passes the test is qualified to join a Kubernetes cluster.

## Node Prerequisite

To run node conformance test, a node must satisfy the same prerequisites as a standard Kubernetes node. At a minimum, the node should have the following daemons installed:

- CRI-compatible container runtimes such as Docker, Containerd and CRI-O
- Kubelet

## Running Node Conformance Test

To run the node conformance test, perform the following steps:

1. Work out the value of the `--kubeconfig` option for the kubelet; for example: `--kubeconfig=/var/lib/kubelet/config.yaml`. Because the test framework starts a local

control plane to test the kubelet, use `http://localhost:8080` as the URL of the API server. There are some other kubelet command line parameters you may want to use:

- `--cloud-provider`: If you are using `--cloud-provider=gce`, you should remove the flag to run the test.

1. Run the node conformance test with command:

```
$CONFIG_DIR is the pod manifest path of your Kubelet.
$LOG_DIR is the test output path.
sudo docker run -it --rm --privileged --net=host \
-v /:/rootfs -v $CONFIG_DIR:$CONFIG_DIR -v $LOG_DIR:/var/result \
registry.k8s.io/node-test:0.2
```

## Running Node Conformance Test for Other Architectures

Kubernetes also provides node conformance test docker images for other architectures:

| Arch  | Image           |
|-------|-----------------|
| amd64 | node-test-amd64 |
| arm   | node-test-arm   |
| arm64 | node-test-arm64 |

## Running Selected Test

To run specific tests, overwrite the environment variable `FOCUS` with the regular expression of tests you want to run.

```
sudo docker run -it --rm --privileged --net=host \
-v /:/rootfs:ro -v $CONFIG_DIR:$CONFIG_DIR -v $LOG_DIR:/var/result \
-e FOCUS=MirrorPod \ # Only run MirrorPod test
registry.k8s.io/node-test:0.2
```

To skip specific tests, overwrite the environment variable `SKIP` with the regular expression of tests you want to skip.

```
sudo docker run -it --rm --privileged --net=host \
-v /:/rootfs:ro -v $CONFIG_DIR:$CONFIG_DIR -v $LOG_DIR:/var/result \
-e SKIP=MirrorPod \ # Run all conformance tests but skip MirrorPod test
registry.k8s.io/node-test:0.2
```

Node conformance test is a containerized version of [node e2e test](#). By default, it runs all conformance tests.

Theoretically, you can run any node e2e test if you configure the container and mount required volumes properly. But **it is strongly recommended to only run conformance test**, because it requires much more complex configuration to run non-conformance test.

## Caveats

- The test leaves some docker images on the node, including the node conformance test image and images of containers used in the functionality test.
- The test leaves dead containers on the node. These containers are created during the functionality test.

# Enforcing Pod Security Standards

This page provides an overview of best practices when it comes to enforcing [Pod Security Standards](#).

## Using the built-in Pod Security Admission Controller

FEATURE STATE: Kubernetes v1.25 [stable]

The [Pod Security Admission Controller](#) intends to replace the deprecated PodSecurityPolicies.

### Configure all cluster namespaces

Namespaces that lack any configuration at all should be considered significant gaps in your cluster security model. We recommend taking the time to analyze the types of workloads occurring in each namespace, and by referencing the Pod Security Standards, decide on an appropriate level for each of them. Unlabeled namespaces should only indicate that they've yet to be evaluated.

In the scenario that all workloads in all namespaces have the same security requirements, we provide an [example](#) that illustrates how the PodSecurity labels can be applied in bulk.

### Embrace the principle of least privilege

In an ideal world, every pod in every namespace would meet the requirements of the restricted policy. However, this is not possible nor practical, as some workloads will require elevated privileges for legitimate reasons.

- Namespaces allowing privileged workloads should establish and enforce appropriate access controls.
- For workloads running in those permissive namespaces, maintain documentation about their unique security requirements. If at all possible, consider how those requirements could be further constrained.

### Adopt a multi-mode strategy

The audit and warn modes of the Pod Security Standards admission controller make it easy to collect important security insights about your pods without breaking existing workloads.

It is good practice to enable these modes for all namespaces, setting them to the *desired* level and version you would eventually like to enforce. The warnings and audit annotations generated in this phase can guide you toward that state. If you expect workload authors to

make changes to fit within the desired level, enable the warn mode. If you expect to use audit logs to monitor/drive changes to fit within the desired level, enable the audit mode.

When you have the enforce mode set to your desired value, these modes can still be useful in a few different ways:

- By setting warn to the same level as enforce, clients will receive warnings when attempting to create Pods (or resources that have Pod templates) that do not pass validation. This will help them update those resources to become compliant.
- In Namespaces that pin enforce to a specific non-latest version, setting the audit and warn modes to the same level as enforce, but to the latest version, gives visibility into settings that were allowed by previous versions but are not allowed per current best practices.

## Third-party alternatives

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information.](#)

Other alternatives for enforcing security profiles are being developed in the Kubernetes ecosystem:

- [Kubewarden](#).
- [Kyverno](#).
- [OPA Gatekeeper](#).

The decision to go with a *built-in* solution (e.g. PodSecurity admission controller) versus a third-party tool is entirely dependent on your own situation. When evaluating any solution, trust of your supply chain is crucial. Ultimately, using *any* of the aforementioned approaches will be better than doing nothing.

## PKI certificates and requirements

Kubernetes requires PKI certificates for authentication over TLS. If you install Kubernetes with [kubeadm](#), the certificates that your cluster requires are automatically generated. You can also generate your own certificates -- for example, to keep your private keys more secure by not storing them on the API server. This page explains the certificates that your cluster requires.

## How certificates are used by your cluster

Kubernetes requires PKI for the following operations:

- Client certificates for the kubelet to authenticate to the API server
- Kubelet [server certificates](#) for the API server to talk to the kubelets
- Server certificate for the API server endpoint
- Client certificates for administrators of the cluster to authenticate to the API server
- Client certificates for the API server to talk to the kubelets
- Client certificate for the API server to talk to etcd
- Client certificate/kubeconfig for the controller manager to talk to the API server

- Client certificate/kubeconfig for the scheduler to talk to the API server.
- Client and server certificates for the [front-proxy](#)

**Note:**

front-proxy certificates are required only if you run kube-proxy to support [an extension API server](#).

etcd also implements mutual TLS to authenticate clients and peers.

## Where certificates are stored

If you install Kubernetes with kubeadm, most certificates are stored in `/etc/kubernetes/pki`. All paths in this documentation are relative to that directory, with the exception of user account certificates which kubeadm places in `/etc/kubernetes`.

## Configure certificates manually

If you don't want kubeadm to generate the required certificates, you can create them using a single root CA or by providing all certificates. See [Certificates](#) for details on creating your own certificate authority. See [Certificate Management with kubeadm](#) for more on managing certificates.

### Single root CA

You can create a single root CA, controlled by an administrator. This root CA can then create multiple intermediate CAs, and delegate all further creation to Kubernetes itself.

Required CAs:

| path                   | Default CN                | description                             |
|------------------------|---------------------------|-----------------------------------------|
| ca.crt,key             | kubernetes-ca             | Kubernetes general CA                   |
| etcd/ca.crt,key        | etcd-ca                   | For all etcd-related functions          |
| front-proxy-ca.crt,key | kubernetes-front-proxy-ca | For the <a href="#">front-end proxy</a> |

On top of the above CAs, it is also necessary to get a public/private key pair for service account management, `sa.key` and `sa.pub`. The following example illustrates the CA key and certificate files shown in the previous table:

```
/etc/kubernetes/pki/ca.crt
/etc/kubernetes/pki/ca.key
/etc/kubernetes/pki/etcd/ca.crt
/etc/kubernetes/pki/etcd/ca.key
/etc/kubernetes/pki/front-proxy-ca.crt
/etc/kubernetes/pki/front-proxy-ca.key
```

### All certificates

If you don't wish to copy the CA private keys to your cluster, you can generate all certificates yourself.



Required certificates:

| Default CN                    | Parent CA                 | O (in Subject) | kind              | hosts (SAN)                                    |
|-------------------------------|---------------------------|----------------|-------------------|------------------------------------------------|
| kube-etcd                     | etcd-ca                   |                | server,<br>client | <hostname>, <Host_IP>, localhost,<br>127.0.0.1 |
| kube-etcd-peer                | etcd-ca                   |                | server,<br>client | <hostname>, <Host_IP>, localhost,<br>127.0.0.1 |
| kube-etcd-healthcheck-client  | etcd-ca                   |                | client            |                                                |
| kube-apiserver-etcd-client    | etcd-ca                   |                | client            |                                                |
| kube-apiserver                | kubernetes-ca             |                | server            | <hostname>, <Host_IP>, <advertise_IP>,<br>[1]  |
| kube-apiserver-kubelet-client | kubernetes-ca             | system:masters | client            |                                                |
| front-proxy-client            | kubernetes-front-proxy-ca |                | client            |                                                |

**Note:**

Instead of using the super-user group `system:masters` for `kube-apiserver-kubelet-client` a less privileged group can be used. `kubeadm` uses the `kubeadm:cluster-admins` group for that purpose.

[1]: any other IP or DNS name you contact your cluster on (as used by [kubeadm](#) the load balancer stable IP and/or DNS name, `kubernetes`, `kubernetes.default`, `kubernetes.default.svc`, `kubernetes.default.svc.cluster`, `kubernetes.default.svc.cluster.local`)

where `kind` maps to one or more of the x509 key usage, which is also documented in the `.spec.usages` of a [CertificateSigningRequest](#) type:

| kind   | Key usage                                        |
|--------|--------------------------------------------------|
| server | digital signature, key encipherment, server auth |
| client | digital signature, key encipherment, client auth |

**Note:**

Hosts/SAN listed above are the recommended ones for getting a working cluster; if required by a specific setup, it is possible to add additional SANs on all the server certificates.

**Note:**

For `kubeadm` users only:

- The scenario where you are copying to your cluster CA certificates without private keys is referred as external CA in the `kubeadm` documentation.

- If you are comparing the above list with a kubeadm generated PKI, please be aware that kube-etcd, kube-etcd-peer and kube-etcd-healthcheck-client certificates are not generated in case of external etcd.

## Certificate paths

Certificates should be placed in a recommended path (as used by [kubeadm](#)). Paths should be specified using the given argument regardless of location.

| Default CN                    | recommended key path         | recommended cert path        | command                 | key argument               | cert argument                                                 |
|-------------------------------|------------------------------|------------------------------|-------------------------|----------------------------|---------------------------------------------------------------|
| etcd-ca                       | etcd/ca.key                  | etcd/ca.crt                  | kube-apiserver          |                            | --etcd-cafile                                                 |
| kube-apiserver-etcd-client    | apiserver-etcd-client.key    | apiserver-etcd-client.crt    | kube-apiserver          | --etcd-keyfile             | --etcd-certfile                                               |
| kubernetes-ca                 | ca.key                       | ca.crt                       | kube-apiserver          |                            | --client-ca-file                                              |
| kubernetes-ca                 | ca.key                       | ca.crt                       | kube-controller-manager | --cluster-signing-key-file | --client-ca-file, --root-ca-file, --cluster-signing-cert-file |
| kube-apiserver                | apiserver.key                | apiserver.crt                | kube-apiserver          | --tls-private-key-file     | --tls-cert-file                                               |
| kube-apiserver-kubelet-client | apiserver-kubelet-client.key | apiserver-kubelet-client.crt | kube-apiserver          | --kubelet-client-key       | --kubelet-client-certificate                                  |
| front-proxy-ca                | front-proxy-ca.key           | front-proxy-ca.crt           | kube-apiserver          |                            | --requestheader-client-ca-file                                |
| front-proxy-ca                | front-proxy-ca.key           | front-proxy-ca.crt           | kube-controller-manager |                            | --requestheader-client-ca-file                                |
| front-proxy-client            | front-proxy-client.key       | front-proxy-client.crt       | kube-apiserver          | --proxy-client-key-file    | --proxy-client-cert-file                                      |
| etcd-ca                       | etcd/ca.key                  | etcd/ca.crt                  | etcd                    |                            | --trusted-ca-file, --peer-trusted-ca-file                     |
| kube-etcd                     | etcd/server.key              | etcd/server.crt              | etcd                    | --key-file                 | --cert-file                                                   |
| kube-etcd-peer                | etcd/peer.key                | etcd/peer.crt                | etcd                    | --peer-key-file            | --peer-cert-file                                              |
| etcd-ca                       |                              | etcd/ca.crt                  | etcdctl                 |                            | --cacert                                                      |
| kube-etcd-healthcheck-client  | etcd/healthcheck-client.key  | etcd/healthcheck-client.crt  | etcdctl                 | --key                      | --cert                                                        |

Same considerations apply for the service account key pair:

| private key path | public key path | command                 | argument                           |
|------------------|-----------------|-------------------------|------------------------------------|
| sa.key           |                 | kube-controller-manager | --service-account-private-key-file |

| private key path | public key path | command        | argument                   |
|------------------|-----------------|----------------|----------------------------|
|                  | sa.pub          | kube-apiserver | --service-account-key-file |

The following example illustrates the file paths [from the previous tables](#) you need to provide if you are generating all of your own keys and certificates:

```

/etc/kubernetes/pki/etcd/ca.key
/etc/kubernetes/pki/etcd/ca.crt
/etc/kubernetes/pki/apiserver-etcd-client.key
/etc/kubernetes/pki/apiserver-etcd-client.crt
/etc/kubernetes/pki/ca.key
/etc/kubernetes/pki/ca.crt
/etc/kubernetes/pki/apiserver.key
/etc/kubernetes/pki/apiserver.crt
/etc/kubernetes/pki/apiserver-kubelet-client.key
/etc/kubernetes/pki/apiserver-kubelet-client.crt
/etc/kubernetes/pki/front-proxy-ca.key
/etc/kubernetes/pki/front-proxy-ca.crt
/etc/kubernetes/pki/front-proxy-client.key
/etc/kubernetes/pki/front-proxy-client.crt
/etc/kubernetes/pki/etcd/server.key
/etc/kubernetes/pki/etcd/server.crt
/etc/kubernetes/pki/etcd/peer.key
/etc/kubernetes/pki/etcd/peer.crt
/etc/kubernetes/pki/etcd/healthcheck-client.key
/etc/kubernetes/pki/etcd/healthcheck-client.crt
/etc/kubernetes/pki/sa.key
/etc/kubernetes/pki/sa.pub

```

## Configure certificates for user accounts

You must manually configure these administrator account and service accounts:

| filename                | credential name            | Default CN                        | O (in Subject) |
|-------------------------|----------------------------|-----------------------------------|----------------|
| admin.conf              | default-admin              | kubernetes-admin                  | <admin-group>  |
| super-admin.conf        | default-super-admin        | kubernetes-super-admin            | system:masters |
| kubelet.conf            | default-auth               | system:node:<nodeName> (see note) | system:nodes   |
| controller-manager.conf | default-controller-manager | system:kube-controller-manager    |                |
| scheduler.conf          | default-scheduler          | system:kube-scheduler             |                |

### Note:

The value of <nodeName> for kubelet.conf **must** match precisely the value of the node name provided by the kubelet as it registers with the apiserver. For further details, read the [Node Authorization](#).

## Note:

In the above example `<admin-group>` is implementation specific. Some tools sign the certificate in the default `admin.conf` to be part of the `system:masters` group. `system:masters` is a break-glass, super user group can bypass the authorization layer of Kubernetes, such as RBAC. Also some tools do not generate a separate `super-admin.conf` with a certificate bound to this super user group.

kubeadm generates two separate administrator certificates in kubeconfig files. One is in `admin.conf` and has Subject: O = kubeadm:cluster-admins, CN = kubernetes-admin. `kubeadm:cluster-admins` is a custom group bound to the cluster-admin ClusterRole. This file is generated on all kubeadm managed control plane machines.

Another is in `super-admin.conf` that has Subject: O = system:masters, CN = kubernetes-super-admin. This file is generated only on the node where kubeadm init was called.

1. For each config, generate an x509 cert/key pair with the given CN and O.
2. Run `kubect`l as follows for each config:

```
KUBECONFIG=<filename> kubectl config set-cluster default-cluster --server=https://<host ip>:6443 --certificate-authority <path-to-kubernetes-ca> --embed-certs
KUBECONFIG=<filename> kubectl config set-credentials <credential-name> --client-key <path-to-key>.pem --client-certificate <path-to-cert>.pem --embed-certs
KUBECONFIG=<filename> kubectl config set-context default-system --cluster default-cluster --user <credential-name>
KUBECONFIG=<filename> kubectl config use-context default-system
```

These files are used as follows:

| filename                | command                 | comment                                                             |
|-------------------------|-------------------------|---------------------------------------------------------------------|
| admin.conf              | kubect                  | Configures administrator user for the cluster                       |
| super-admin.conf        | kubect                  | Configures super administrator user for the cluster                 |
| kubelet.conf            | kubelet                 | One required for each node in the cluster.                          |
| controller-manager.conf | kube-controller-manager | Must be added to manifest in manifests/kube-controller-manager.yaml |
| scheduler.conf          | kube-scheduler          | Must be added to manifest in manifests/kube-scheduler.yaml          |

The following files illustrate full paths to the files listed in the previous table:

```
/etc/kubernetes/admin.conf
/etc/kubernetes/super-admin.conf
/etc/kubernetes/kubelet.conf
/etc/kubernetes/controller-manager.conf
/etc/kubernetes/scheduler.conf
```

This section of the Kubernetes documentation contains pages that show how to do individual tasks. A task page shows how to do a single thing, typically by giving a short sequence of steps.

If you would like to write a task page, see [Creating a Documentation Pull Request](#).

---

### [Install Tools](#)

Set up Kubernetes tools on your computer.

### [Administer a Cluster](#)

Learn common tasks for administering a cluster.

### [Configure Pods and Containers](#)

Perform common configuration tasks for Pods and containers.

### [Monitoring, Logging, and Debugging](#)

Set up monitoring and logging to troubleshoot a cluster, or debug a containerized application.

### [Manage Kubernetes Objects](#)

Declarative and imperative paradigms for interacting with the Kubernetes API.

### [Managing Secrets](#)

Managing confidential settings data using Secrets.

### [Inject Data Into Applications](#)

Specify configuration and other data for the Pods that run your workload.

### [Run Applications](#)

Run and manage both stateless and stateful applications.

### [Run Jobs](#)

Run Jobs using parallel processing.

### [Access Applications in a Cluster](#)

Configure load balancing, port forwarding, or setup firewall or DNS configurations to access applications in a cluster.

### [Extend Kubernetes](#)

Understand advanced ways to adapt your Kubernetes cluster to the needs of your work environment.

### [TLS](#)

Understand how to protect traffic within your cluster using Transport Layer Security (TLS).

### [Manage Cluster Daemons](#)

Perform common tasks for managing a DaemonSet, such as performing a rolling update.

### [Networking](#)

Learn how to configure networking for your cluster.

### [Extend kubectl with plugins](#)

Extend kubectl by creating and installing kubectl plugins.

### [Manage HugePages](#)

Configure and manage huge pages as a schedulable resource in a cluster.

### [Schedule GPUs](#)

Configure and schedule GPUs for use as a resource by nodes in a cluster.

## Install Tools

Set up Kubernetes tools on your computer.

### kubectl

The Kubernetes command-line tool, [kubectl](#), allows you to run commands against Kubernetes clusters. You can use kubectl to deploy applications, inspect and manage cluster resources, and view logs. For more information including a complete list of kubectl operations, see the [kubectl reference documentation](#).

kubectl is installable on a variety of Linux platforms, macOS and Windows. Find your preferred operating system below.

- [Install kubectl on Linux](#)
- [Install kubectl on macOS](#)
- [Install kubectl on Windows](#)

## kind

[kind](#) lets you run Kubernetes on your local computer. This tool requires that you have either [Docker](#) or [Podman](#) installed.

The kind [Quick Start](#) page shows you what you need to do to get up and running with kind.

[View kind Quick Start Guide](#)

## minikube

Like kind, [minikube](#) is a tool that lets you run Kubernetes locally. minikube runs an all-in-one or a multi-node local Kubernetes cluster on your personal computer (including Windows, macOS and Linux PCs) so that you can try out Kubernetes, or for daily development work.

You can follow the official [Get Started!](#) guide if your focus is on getting the tool installed.

[View minikube Get Started! Guide](#)

Once you have minikube working, you can use it to [run a sample application](#).

## kubeadm

You can use the [kubeadm](#) tool to create and manage Kubernetes clusters. It performs the actions necessary to get a minimum viable, secure cluster up and running in a user friendly way.

[Installing kubeadm](#) shows you how to install kubeadm. Once installed, you can use it to [create a cluster](#).

[View kubeadm Install Guide](#)

# Install and Set Up kubectl on Linux

## Before you begin

You must use a kubectl version that is within one minor version difference of your cluster. For example, a v1.30 client can communicate with v1.29, v1.30, and v1.31 control planes. Using the latest compatible version of kubectl helps avoid unforeseen issues.

## Install kubectl on Linux

The following methods exist for installing kubectl on Linux:

- [Install kubectl binary with curl on Linux](#)
- [Install using native package management](#)
- [Install using other package management](#)

## Install kubectl binary with curl on Linux

1. Download the latest release with the command:

- [x86-64](#)
- [ARM64](#)

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/arm64/kubectl"
```

### Note:

To download a specific version, replace the `$(curl -L -s https://dl.k8s.io/release/stable.txt)` portion of the command with the specific version.

For example, to download version 1.30.0 on Linux x86-64, type:

```
curl -LO https://dl.k8s.io/release/v1.30.0/bin/linux/amd64/kubectl
```

And for Linux ARM64, type:

```
curl -LO https://dl.k8s.io/release/v1.30.0/bin/linux/arm64/kubectl
```

2. Validate the binary (optional)

Download the kubectl checksum file:

- [x86-64](#)
- [ARM64](#)

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl.sha256"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/arm64/kubectl.sha256"
```

Validate the kubectl binary against the checksum file:

```
echo "$(cat kubectl.sha256) kubectl" | sha256sum --check
```

If valid, the output is:

```
kubectl: OK
```



If the check fails, sha256 exits with nonzero status and prints output similar to:

```
kubectrl: FAILED
sha256sum: WARNING: 1 computed checksum did NOT match
```

**Note:**

Download the same version of the binary and checksum.

3. Install kubectrl

```
sudo install -o root -g root -m 0755 kubectrl /usr/local/bin/kubectrl
```

**Note:**

If you do not have root access on the target system, you can still install kubectrl to the `~/local/bin` directory:

```
chmod +x kubectrl
mkdir -p ~/.local/bin
mv ./kubectrl ~/.local/bin/kubectrl
and then append (or prepend) ~/.local/bin to $PATH
```

4. Test to ensure the version you installed is up-to-date:

```
kubectrl version --client
```

Or use this for detailed view of version:

```
kubectrl version --client --output=yaml
```

## Install using native package management

- [Debian-based distributions](#)
- [Red Hat-based distributions](#)
- [SUSE-based distributions](#)

1. Update the apt package index and install packages needed to use the Kubernetes apt repository:

```
sudo apt-get update
apt-transport-https may be a dummy package; if so, you can skip that package
sudo apt-get install -y apt-transport-https ca-certificates curl gnupg
```

2. Download the public signing key for the Kubernetes package repositories. The same signing key is used for all repositories so you can disregard the version in the URL:

```
If the folder `/etc/apt/keyrings` does not exist, it should be created before the curl
command, read the note below.
sudo mkdir -p -m 755 /etc/apt/keyrings
curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.30/deb/Release.key | sudo gpg --dearmor -o /
etc/apt/keyrings/kubernetes-apt-keyring.gpg
```

```
sudo chmod 644 /etc/apt/keyrings/kubernetes-apt-keyring.gpg # allow unprivileged APT programs to read this keyring
```

**Note:**

In releases older than Debian 12 and Ubuntu 22.04, folder `/etc/apt/keyrings` does not exist by default, and it should be created before the `curl` command.

1. Add the appropriate Kubernetes apt repository. If you want to use Kubernetes version different than v1.30, replace v1.30 with the desired minor version in the command below:

```
This overwrites any existing configuration in /etc/apt/sources.list.d/kubernetes.list
echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.30/deb/ ' | sudo tee /etc/apt/sources.list.d/kubernetes.list
sudo chmod 644 /etc/apt/sources.list.d/kubernetes.list # helps tools such as command-not-found to work correctly
```

**Note:**

To upgrade `kubectl` to another minor release, you'll need to bump the version in `/etc/apt/sources.list.d/kubernetes.list` before running `apt-get update` and `apt-get upgrade`. This procedure is described in more detail in [Changing The Kubernetes Package Repository](#).

1. Update apt package index, then install `kubectl`:

```
sudo apt-get update
sudo apt-get install -y kubectl
```

1. Add the Kubernetes yum repository. If you want to use Kubernetes version different than v1.30, replace v1.30 with the desired minor version in the command below.

```
This overwrites any existing configuration in /etc/yum.repos.d/kubernetes.repo
cat <<EOF | sudo tee /etc/yum.repos.d/kubernetes.repo
[kubernetes]
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key
EOF
```

**Note:**

To upgrade `kubectl` to another minor release, you'll need to bump the version in `/etc/yum.repos.d/kubernetes.repo` before running `yum update`. This procedure is described in more detail in [Changing The Kubernetes Package Repository](#).

1. Install `kubectl` using `yum`:

```
sudo yum install -y kubectl
```

1. Add the Kubernetes zypper repository. If you want to use Kubernetes version different than v1.30, replace v1.30 with the desired minor version in the command below.

```
This overwrites any existing configuration in /etc/zypp/repos.d/kubernetes.repo
cat <<EOF | sudo tee /etc/zypp/repos.d/kubernetes.repo
[kubernetes]
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key
EOF
```

### Note:

To upgrade kubectl to another minor release, you'll need to bump the version in `/etc/zypp/repos.d/kubernetes.repo` before running zypper update. This procedure is described in more detail in [Changing The Kubernetes Package Repository](#).

1. Install kubectl using zypper:

```
sudo zypper install -y kubectl
```

## Install using other package management

- [Snap](#)
- [Homebrew](#)

If you are on Ubuntu or another Linux distribution that supports the [snap](#) package manager, kubectl is available as a [snap](#) application.

```
snap install kubectl --classic
kubectl version --client
```

If you are on Linux and using [Homebrew](#) package manager, kubectl is available for [installation](#).

```
brew install kubectl
kubectl version --client
```

## Verify kubectl configuration

In order for kubectl to find and access a Kubernetes cluster, it needs a [kubeconfig file](#), which is created automatically when you create a cluster using [kube-up.sh](#) or successfully deploy a Minikube cluster. By default, kubectl configuration is located at `~/.kube/config`.

Check that kubectl is properly configured by getting the cluster state:

```
kubectl cluster-info
```

If you see a URL response, kubectl is correctly configured to access your cluster.

If you see a message similar to the following, kubectl is not configured correctly or is not able to connect to a Kubernetes cluster.

The connection to the server <server-name:port> was refused - did you specify the right host or port?

For example, if you are intending to run a Kubernetes cluster on your laptop (locally), you will need a tool like [Minikube](#) to be installed first and then re-run the commands stated above.

If `kubectl cluster-info` returns the url response but you can't access your cluster, to check whether it is configured properly, use:

```
kubectl cluster-info dump
```

## Troubleshooting the 'No Auth Provider Found' error message

In Kubernetes 1.26, `kubectl` removed the built-in authentication for the following cloud providers' managed Kubernetes offerings. These providers have released `kubectl` plugins to provide the cloud-specific authentication. For instructions, refer to the following provider documentation:

- Azure AKS: [kubelogin plugin](#)
- Google Kubernetes Engine: [gke-gcloud-auth-plugin](#)

(There could also be other reasons to see the same error message, unrelated to that change.)

## Optional kubectl configurations and plugins

### Enable shell autocompletion

`kubectl` provides autocompletion support for Bash, Zsh, Fish, and PowerShell, which can save you a lot of typing.

Below are the procedures to set up autocompletion for Bash, Fish, and Zsh.

- [Bash](#)
- [Fish](#)
- [Zsh](#)

### Introduction

The `kubectl` completion script for Bash can be generated with the command `kubectl completion bash`. Sourcing the completion script in your shell enables `kubectl` autocompletion.

However, the completion script depends on [bash-completion](#), which means that you have to install this software first (you can test if you have `bash-completion` already installed by running `type _init_completion`).

### Install bash-completion

`bash-completion` is provided by many package managers (see [here](#)). You can install it with `apt-get install bash-completion` or `yum install bash-completion`, etc.

The above commands create `/usr/share/bash-completion/bash_completion`, which is the main script of `bash-completion`. Depending on your package manager, you have to manually source this file in your `~/.bashrc` file.

To find out, reload your shell and run `type _init_completion`. If the command succeeds, you're already set, otherwise add the following to your `~/.bashrc` file:

```
source /usr/share/bash-completion/bash_completion
```

Reload your shell and verify that bash-completion is correctly installed by typing `type _init_completion`.

## Enable kubectl autocompletion

### Bash

You now need to ensure that the kubectl completion script gets sourced in all your shell sessions. There are two ways in which you can do this:

- [User](#)
- [System](#)

```
echo 'source <(kubectl completion bash)' >> ~/.bashrc
```

```
kubectl completion bash | sudo tee /etc/bash_completion.d/kubectl > /dev/null
sudo chmod a+r /etc/bash_completion.d/kubectl
```

If you have an alias for kubectl, you can extend shell completion to work with that alias:

```
echo 'alias k=kubectl' >> ~/.bashrc
echo 'complete -o default -F __start_kubectl k' >> ~/.bashrc
```

### Note:

bash-completion sources all completion scripts in `/etc/bash_completion.d`.

Both approaches are equivalent. After reloading your shell, kubectl autocompletion should be working. To enable bash autocompletion in current session of shell, source the `~/.bashrc` file:

```
source ~/.bashrc
```

### Note:

Autocomplete for Fish requires kubectl 1.23 or later.

The kubectl completion script for Fish can be generated with the command `kubectl completion fish`. Sourcing the completion script in your shell enables kubectl autocompletion.

To do so in all your shell sessions, add the following line to your `~/.config/fish/config.fish` file:

```
kubectl completion fish | source
```

After reloading your shell, kubectl autocompletion should be working.

The kubectl completion script for Zsh can be generated with the command `kubectl completion zsh`. Sourcing the completion script in your shell enables kubectl autocompletion.

To do so in all your shell sessions, add the following to your `~/.zshrc` file:

```
source <(kubectl completion zsh)
```

If you have an alias for kubectl, kubectl autocompletion will automatically work with it.

After reloading your shell, kubectl autocompletion should be working.

If you get an error like `2: command not found: compdef`, then add the following to the beginning of your `~/.zshrc` file:

```
autoload -Uz compinit
compinit
```

## Install kubectl convert plugin

A plugin for Kubernetes command-line tool kubectl, which allows you to convert manifests between different API versions. This can be particularly helpful to migrate manifests to a non-deprecated api version with newer Kubernetes release. For more info, visit [migrate to non deprecated apis](#)

1. Download the latest release with the command:

- [x86-64](#)
- [ARM64](#)

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl-convert"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/arm64/kubectl-convert"
```

2. Validate the binary (optional)

Download the kubectl-convert checksum file:

- [x86-64](#)
- [ARM64](#)

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl-convert.sha256"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/
```

```
linux/arm64/kubectl-convert.sha256"
```

Validate the kubectl-convert binary against the checksum file:

```
echo "$(cat kubectl-convert.sha256) kubectl-convert" | sha256sum --check
```

If valid, the output is:

```
kubectl-convert: OK
```

If the check fails, sha256 exits with nonzero status and prints output similar to:

```
kubectl-convert: FAILED
sha256sum: WARNING: 1 computed checksum did NOT match
```

**Note:**

Download the same version of the binary and checksum.

3. Install kubectl-convert

```
sudo install -o root -g root -m 0755 kubectl-convert /usr/local/bin/kubectl-convert
```

4. Verify plugin is successfully installed

```
kubectl convert --help
```

If you do not see an error, it means the plugin is successfully installed.

5. After installing the plugin, clean up the installation files:

```
rm kubectl-convert kubectl-convert.sha256
```

## What's next

- [Install Minikube](#)
- See the [getting started guides](#) for more about creating clusters.
- [Learn how to launch and expose your application.](#)
- If you need access to a cluster you didn't create, see the [Sharing Cluster Access document](#).
- Read the [kubectl reference docs](#)

# Install and Set Up kubectl on macOS

## Before you begin

You must use a kubectl version that is within one minor version difference of your cluster. For example, a v1.30 client can communicate with v1.29, v1.30, and v1.31 control planes. Using the latest compatible version of kubectl helps avoid unforeseen issues.

# Install kubectl on macOS

The following methods exist for installing kubectl on macOS:

- [Install kubectl on macOS](#)
  - [Install kubectl binary with curl on macOS](#)
  - [Install with Homebrew on macOS](#)
  - [Install with Macports on macOS](#)
- [Verify kubectl configuration](#)
- [Optional kubectl configurations and plugins](#)
  - [Enable shell autocompletion](#)
  - [Install kubectl convert plugin](#)

## Install kubectl binary with curl on macOS

1. Download the latest release:

- [Intel](#)
- [Apple Silicon](#)

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/darwin/amd64/kubectl"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/darwin/arm64/kubectl"
```

### Note:

To download a specific version, replace the `$(curl -L -s https://dl.k8s.io/release/stable.txt)` portion of the command with the specific version.

For example, to download version 1.30.0 on Intel macOS, type:

```
curl -LO "https://dl.k8s.io/release/v1.30.0/bin/darwin/amd64/kubectl"
```

And for macOS on Apple Silicon, type:

```
curl -LO "https://dl.k8s.io/release/v1.30.0/bin/darwin/arm64/kubectl"
```

2. Validate the binary (optional)

Download the kubectl checksum file:

- [Intel](#)
- [Apple Silicon](#)

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/
```



```
darwin/amd64/kubect1.sha256"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/darwin/arm64/kubect1.sha256"
```

Validate the kubect1 binary against the checksum file:

```
echo "$(cat kubect1.sha256) kubect1" | shasum -a 256 --check
```

If valid, the output is:

```
kubect1: OK
```

If the check fails, shasum exits with nonzero status and prints output similar to:

```
kubect1: FAILED
shasum: WARNING: 1 computed checksum did NOT match
```

**Note:**

Download the same version of the binary and checksum.

3. Make the kubect1 binary executable.

```
chmod +x ./kubect1
```

4. Move the kubect1 binary to a file location on your system PATH.

```
sudo mv ./kubect1 /usr/local/bin/kubect1
sudo chown root: /usr/local/bin/kubect1
```

**Note:**

Make sure /usr/local/bin is in your PATH environment variable.

5. Test to ensure the version you installed is up-to-date:

```
kubect1 version --client
```

Or use this for detailed view of version:

```
kubect1 version --client --output=yaml
```

6. After installing and validating kubect1, delete the checksum file:

```
rm kubect1.sha256
```

## Install with Homebrew on macOS

If you are on macOS and using [Homebrew](#) package manager, you can install kubectl with Homebrew.

1. Run the installation command:

```
brew install kubectl
```

or

```
brew install kubernetes-cli
```

2. Test to ensure the version you installed is up-to-date:

```
kubectl version --client
```

## Install with Macports on macOS

If you are on macOS and using [Macports](#) package manager, you can install kubectl with Macports.

1. Run the installation command:

```
sudo port selfupdate
sudo port install kubectl
```

2. Test to ensure the version you installed is up-to-date:

```
kubectl version --client
```

## Verify kubectl configuration

In order for kubectl to find and access a Kubernetes cluster, it needs a [kubeconfig file](#), which is created automatically when you create a cluster using [kube-up.sh](#) or successfully deploy a Minikube cluster. By default, kubectl configuration is located at ~/.kube/config.

Check that kubectl is properly configured by getting the cluster state:

```
kubectl cluster-info
```

If you see a URL response, kubectl is correctly configured to access your cluster.

If you see a message similar to the following, kubectl is not configured correctly or is not able to connect to a Kubernetes cluster.

```
The connection to the server <server-name:port> was refused - did you specify the right host or port?
```

For example, if you are intending to run a Kubernetes cluster on your laptop (locally), you will need a tool like [Minikube](#) to be installed first and then re-run the commands stated above.

If kubectl cluster-info returns the url response but you can't access your cluster, to check whether it is configured properly, use:

```
kubectl cluster-info dump
```

## Troubleshooting the 'No Auth Provider Found' error message

In Kubernetes 1.26, kubectl removed the built-in authentication for the following cloud providers' managed Kubernetes offerings. These providers have released kubectl plugins to provide the cloud-specific authentication. For instructions, refer to the following provider documentation:

- Azure AKS: [kubelogin plugin](#)
- Google Kubernetes Engine: [gke-gcloud-auth-plugin](#)

(There could also be other reasons to see the same error message, unrelated to that change.)

## Optional kubectl configurations and plugins

### Enable shell autocompletion

kubectl provides autocompletion support for Bash, Zsh, Fish, and PowerShell which can save you a lot of typing.

Below are the procedures to set up autocompletion for Bash, Fish, and Zsh.

- [Bash](#)
- [Fish](#)
- [Zsh](#)

### Introduction

The kubectl completion script for Bash can be generated with `kubectl completion bash`. Sourcing this script in your shell enables kubectl completion.

However, the kubectl completion script depends on [bash-completion](#) which you thus have to previously install.

#### Warning:

There are two versions of bash-completion, v1 and v2. V1 is for Bash 3.2 (which is the default on macOS), and v2 is for Bash 4.1+. The kubectl completion script **doesn't work** correctly with bash-completion v1 and Bash 3.2. It requires **bash-completion v2** and **Bash 4.1+**. Thus, to be able to correctly use kubectl completion on macOS, you have to install and use Bash 4.1+ ([instructions](#)). The following instructions assume that you use Bash 4.1+ (that is, any Bash version of 4.1 or newer).

### Upgrade Bash

The instructions here assume you use Bash 4.1+. You can check your Bash's version by running:

```
echo $BASH_VERSION
```

If it is too old, you can install/upgrade it using Homebrew:

```
brew install bash
```

Reload your shell and verify that the desired version is being used:

```
echo $BASH_VERSION $SHELL
```

Homebrew usually installs it at `/usr/local/bin/bash`.

## Install bash-completion

### Note:

As mentioned, these instructions assume you use Bash 4.1+, which means you will install bash-completion v2 (in contrast to Bash 3.2 and bash-completion v1, in which case kubectl completion won't work).

You can test if you have bash-completion v2 already installed with `type _init_completion`. If not, you can install it with Homebrew:

```
brew install bash-completion@2
```

As stated in the output of this command, add the following to your `~/.bash_profile` file:

```
brew_etc="$(brew --prefix)/etc" && [[-r "${brew_etc}/profile.d/bash_completion.sh"]] && . "${brew_etc}/profile.d/bash_completion.sh"
```

Reload your shell and verify that bash-completion v2 is correctly installed with `type _init_completion`.

## Enable kubectl autocompletion

You now have to ensure that the kubectl completion script gets sourced in all your shell sessions. There are multiple ways to achieve this:

- Source the completion script in your `~/.bash_profile` file:

```
echo 'source <(kubectl completion bash)' >> ~/.bash_profile
```

- Add the completion script to the `/usr/local/etc/bash_completion.d` directory:

```
kubectl completion bash >/usr/local/etc/bash_completion.d/kubectl
```

- If you have an alias for kubectl, you can extend shell completion to work with that alias:

```
echo 'alias k=kubectl' >> ~/.bash_profile
echo 'complete -o default -F __start_kubectl k' >> ~/.bash_profile
```

- If you installed kubectl with Homebrew (as explained [here](#)), then the kubectl completion script should already be in `/usr/local/etc/bash_completion.d/kubectl`. In that case, you don't need to do anything.

**Note:**

The Homebrew installation of bash-completion v2 sources all the files in the BASH\_COMPLETION\_COMPAT\_DIR directory, that's why the latter two methods work.

In any case, after reloading your shell, kubectl completion should be working.

**Note:**

Autocomplete for Fish requires kubectl 1.23 or later.

The kubectl completion script for Fish can be generated with the command `kubectl completion fish`. Sourcing the completion script in your shell enables kubectl autocompletion.

To do so in all your shell sessions, add the following line to your `~/.config/fish/config.fish` file:

```
kubectl completion fish | source
```

After reloading your shell, kubectl autocompletion should be working.

The kubectl completion script for Zsh can be generated with the command `kubectl completion zsh`. Sourcing the completion script in your shell enables kubectl autocompletion.

To do so in all your shell sessions, add the following to your `~/.zshrc` file:

```
source <(kubectl completion zsh)
```

If you have an alias for kubectl, kubectl autocompletion will automatically work with it.

After reloading your shell, kubectl autocompletion should be working.

If you get an error like `2: command not found: compdef`, then add the following to the beginning of your `~/.zshrc` file:

```
autoload -Uz compinit
compinit
```

## Install kubectl convert plugin

A plugin for Kubernetes command-line tool kubectl, which allows you to convert manifests between different API versions. This can be particularly helpful to migrate manifests to a non-deprecated api version with newer Kubernetes release. For more info, visit [migrate to non deprecated apis](#)

1. Download the latest release with the command:

- [Intel](#)
- [Apple Silicon](#)

```
curl -LO "https://dl.k8s.io/release/${curl -L -s https://dl.k8s.io/release/stable.txt}/bin/darwin/amd64/kubectl-convert"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/darwin/arm64/kubectl-convert"
```

## 2. Validate the binary (optional)

Download the kubectl-convert checksum file:

- [Intel](#)
- [Apple Silicon](#)

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/darwin/amd64/kubectl-convert.sha256"
```

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/darwin/arm64/kubectl-convert.sha256"
```

Validate the kubectl-convert binary against the checksum file:

```
echo "$(cat kubectl-convert.sha256) kubectl-convert" | shasum -a 256 --check
```

If valid, the output is:

```
kubectl-convert: OK
```

If the check fails, shasum exits with nonzero status and prints output similar to:

```
kubectl-convert: FAILED
shasum: WARNING: 1 computed checksum did NOT match
```

### **Note:**

Download the same version of the binary and checksum.

## 3. Make kubectl-convert binary executable

```
chmod +x ./kubectl-convert
```

## 4. Move the kubectl-convert binary to a file location on your system PATH.

```
sudo mv ./kubectl-convert /usr/local/bin/kubectl-convert
sudo chown root: /usr/local/bin/kubectl-convert
```

### **Note:**

Make sure `/usr/local/bin` is in your PATH environment variable.

## 5. Verify plugin is successfully installed

```
kubectrl convert --help
```

If you do not see an error, it means the plugin is successfully installed.

6. After installing the plugin, clean up the installation files:

```
rm kubectrl-convert kubectrl-convert.sha256
```

## Uninstall kubectrl on macOS

Depending on how you installed kubectrl, use one of the following methods.

### Uninstall kubectrl using the command-line

1. Locate the kubectrl binary on your system:

```
which kubectrl
```

2. Remove the kubectrl binary:

```
sudo rm <path>
```

Replace <path> with the path to the kubectrl binary from the previous step. For example, `sudo rm /usr/local/bin/kubectrl`.

### Uninstall kubectrl using homebrew

If you installed kubectrl using Homebrew, run the following command:

```
brew remove kubectrl
```

## What's next

- [Install Minikube](#)
- See the [getting started guides](#) for more about creating clusters.
- [Learn how to launch and expose your application.](#)
- If you need access to a cluster you didn't create, see the [Sharing Cluster Access document](#).
- Read the [kubectrl reference docs](#)

# Install and Set Up kubectrl on Windows

## Before you begin

You must use a kubectrl version that is within one minor version difference of your cluster. For example, a v1.30 client can communicate with v1.29, v1.30, and v1.31 control planes. Using the latest compatible version of kubectrl helps avoid unforeseen issues.

# Install kubectl on Windows

The following methods exist for installing kubectl on Windows:

- [Install kubectl binary with curl on Windows](#)
- [Install on Windows using Chocolatey, Scoop, or winget](#)

## Install kubectl binary with curl on Windows

1. Download the latest 1.30 patch release: [kubectl 1.30.0](#).

Or if you have curl installed, use this command:

```
curl.exe -LO "https://dl.k8s.io/release/v1.30.0/bin/windows/amd64/kubectl.exe"
```

### Note:

To find out the latest stable version (for example, for scripting), take a look at <https://dl.k8s.io/release/stable.txt>.

2. Validate the binary (optional)

Download the kubectl checksum file:

```
curl.exe -LO "https://dl.k8s.io/v1.30.0/bin/windows/amd64/kubectl.exe.sha256"
```

Validate the kubectl binary against the checksum file:

- Using Command Prompt to manually compare CertUtil's output to the checksum file downloaded:

```
CertUtil -hashfile kubectl.exe SHA256
type kubectl.exe.sha256
```

- Using PowerShell to automate the verification using the -eq operator to get a True or False result:

```
$(Get-FileHash -Algorithm SHA256 .\kubectl.exe).Hash -eq $(Get-Content .
\kubectl.exe.sha256)
```

3. Append or prepend the kubectl binary folder to your PATH environment variable.
4. Test to ensure the version of kubectl is the same as downloaded:

```
kubectl version --client
```

Or use this for detailed view of version:

```
kubectl version --client --output=yaml
```



**Note:**

[Docker Desktop for Windows](#) adds its own version of kubectl to PATH. If you have installed Docker Desktop before, you may need to place your PATH entry before the one added by the Docker Desktop installer or remove the Docker Desktop's kubectl.

**Install on Windows using Chocolatey, Scoop, or winget**

1. To install kubectl on Windows you can use either [Chocolatey](#) package manager, [Scoop](#) command-line installer, or [winget](#) package manager.

- [choco](#)
- [scoop](#)
- [winget](#)

```
choco install kubernetes-cli
```

```
scoop install kubectl
```

```
winget install -e --id Kubernetes.kubectl
```

2. Test to ensure the version you installed is up-to-date:

```
kubectl version --client
```

3. Navigate to your home directory:

```
If you're using cmd.exe, run: cd %USERPROFILE%
cd ~
```

4. Create the .kube directory:

```
mkdir .kube
```

5. Change to the .kube directory you just created:

```
cd .kube
```

6. Configure kubectl to use a remote Kubernetes cluster:

```
New-Item config -type file
```

**Note:**

Edit the config file with a text editor of your choice, such as Notepad.

**Verify kubectl configuration**

In order for kubectl to find and access a Kubernetes cluster, it needs a [kubeconfig file](#), which is created automatically when you create a cluster using [kube-up.sh](#) or successfully deploy a Minikube cluster. By default, kubectl configuration is located at ~/.kube/config.

Check that kubectl is properly configured by getting the cluster state:

```
kubectl cluster-info
```

If you see a URL response, kubectl is correctly configured to access your cluster.

If you see a message similar to the following, kubectl is not configured correctly or is not able to connect to a Kubernetes cluster.

The connection to the server <server-name:port> was refused - did you specify the right host or port?

For example, if you are intending to run a Kubernetes cluster on your laptop (locally), you will need a tool like [Minikube](#) to be installed first and then re-run the commands stated above.

If kubectl cluster-info returns the url response but you can't access your cluster, to check whether it is configured properly, use:

```
kubectl cluster-info dump
```

## Troubleshooting the 'No Auth Provider Found' error message

In Kubernetes 1.26, kubectl removed the built-in authentication for the following cloud providers' managed Kubernetes offerings. These providers have released kubectl plugins to provide the cloud-specific authentication. For instructions, refer to the following provider documentation:

- Azure AKS: [kubelogin plugin](#)
- Google Kubernetes Engine: [gke-gcloud-auth-plugin](#)

(There could also be other reasons to see the same error message, unrelated to that change.)

## Optional kubectl configurations and plugins

### Enable shell autocompletion

kubectl provides autocompletion support for Bash, Zsh, Fish, and PowerShell, which can save you a lot of typing.

Below are the procedures to set up autocompletion for PowerShell.

The kubectl completion script for PowerShell can be generated with the command `kubectl completion powershell`.

To do so in all your shell sessions, add the following line to your \$PROFILE file:

```
kubectl completion powershell | Out-String | Invoke-Expression
```

This command will regenerate the auto-completion script on every PowerShell start up. You can also add the generated script directly to your \$PROFILE file.

To add the generated script to your \$PROFILE file, run the following line in your powershell prompt:

```
kubectl completion powershell >> $PROFILE
```

After reloading your shell, kubectl autocompletion should be working.

## Install kubectl convert plugin

A plugin for Kubernetes command-line tool kubectl, which allows you to convert manifests between different API versions. This can be particularly helpful to migrate manifests to a non-deprecated api version with newer Kubernetes release. For more info, visit [migrate to non deprecated apis](#)

1. Download the latest release with the command:

```
curl.exe -LO "https://dl.k8s.io/release/v1.30.0/bin/windows/amd64/kubectl-convert.exe"
```

2. Validate the binary (optional).

Download the kubectl-convert checksum file:

```
curl.exe -LO "https://dl.k8s.io/v1.30.0/bin/windows/amd64/kubectl-convert.exe.sha256"
```

Validate the kubectl-convert binary against the checksum file:

- Using Command Prompt to manually compare CertUtil's output to the checksum file downloaded:

```
CertUtil -hashfile kubectl-convert.exe SHA256
type kubectl-convert.exe.sha256
```

- Using PowerShell to automate the verification using the -eq operator to get a True or False result:

```
$(($($CertUtil -hashfile .\kubectl-convert.exe SHA256)[1] -replace " ", "") -eq $(type .\k
ubectl-convert.exe.sha256)
```

3. Append or prepend the kubectl-convert binary folder to your PATH environment variable.
4. Verify the plugin is successfully installed.

```
kubectl convert --help
```

If you do not see an error, it means the plugin is successfully installed.

5. After installing the plugin, clean up the installation files:

```
del kubectl-convert.exe
del kubectl-convert.exe.sha256
```

## What's next

- [Install Minikube](#)
- See the [getting started guides](#) for more about creating clusters.
- [Learn how to launch and expose your application.](#)
- If you need access to a cluster you didn't create, see the [Sharing Cluster Access document.](#)
- Read the [kubectl reference docs](#)

# Administer a Cluster

Learn common tasks for administering a cluster.

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[Administration with kubeadm](#)

[Migrating from dockershim](#)

[Generate Certificates Manually](#)

[Manage Memory, CPU, and API Resources](#)

[Install a Network Policy Provider](#)

[Access Clusters Using the Kubernetes API](#)

[Advertise Extended Resources for a Node](#)

[Autoscale the DNS Service in a Cluster](#)

[Change the Access Mode of a PersistentVolume to ReadWriteOncePod](#)

[Change the default StorageClass](#)

[Switching from Polling to CRI Event-based Updates to Container Status](#)

[Change the Reclaim Policy of a PersistentVolume](#)

[Cloud Controller Manager Administration](#)

[Configure a kubelet image credential provider](#)

[Configure Quotas for API Objects](#)

[Control CPU Management Policies on the Node](#)

[Control Topology Management Policies on a node](#)

[Customizing DNS Service](#)

[Debugging DNS Resolution](#)

[Declare Network Policy](#)

[Developing Cloud Controller Manager](#)

[Enable Or Disable A Kubernetes API](#)

[Encrypting Confidential Data at Rest](#)

[Decrypt Confidential Data that is Already Encrypted at Rest](#)

[Guaranteed Scheduling For Critical Add-On Pods](#)

[IP Masquerade Agent User Guide](#)

[Limit Storage Consumption](#)

[Migrate Replicated Control Plane To Use Cloud Controller Manager](#)

[Namespaces Walkthrough](#)

[Operating etcd clusters for Kubernetes](#)

[Reserve Compute Resources for System Daemons](#)

[Running Kubernetes Node Components as a Non-root User](#)

[Safely Drain a Node](#)

[Securing a Cluster](#)

[Set Kubelet Parameters Via A Configuration File](#)

[Share a Cluster with Namespaces](#)

[Upgrade A Cluster](#)

[Use Cascading Deletion in a Cluster](#)

[Using a KMS provider for data encryption](#)

[Using CoreDNS for Service Discovery](#)

[Using NodeLocal DNSCache in Kubernetes Clusters](#)

[Using sysctls in a Kubernetes Cluster](#)

[Utilizing the NUMA-aware Memory Manager](#)

[Verify Signed Kubernetes Artifacts](#)

# Administration with kubeadm

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[Certificate Management with kubeadm](#)

[Configuring a cgroup driver](#)

[Reconfiguring a kubeadm cluster](#)

[Upgrading kubeadm clusters](#)

[Upgrading Linux nodes](#)

[Upgrading Windows nodes](#)

[Changing The Kubernetes Package Repository](#)

# Certificate Management with kubeadm

FEATURE STATE: Kubernetes v1.15 [stable]

Client certificates generated by [kubeadm](#) expire after 1 year. This page explains how to manage certificate renewals with kubeadm. It also covers other tasks related to kubeadm certificate management.

## Before you begin

You should be familiar with [PKI certificates and requirements in Kubernetes](#).

## Using custom certificates

By default, kubeadm generates all the certificates needed for a cluster to run. You can override this behavior by providing your own certificates.

To do so, you must place them in whatever directory is specified by the `--cert-dir` flag or the `certificatesDir` field of kubeadm's `ClusterConfiguration`. By default this is `/etc/kubernetes/pki`.

If a given certificate and private key pair exists before running `kubeadm init`, kubeadm does not overwrite them. This means you can, for example, copy an existing CA into `/etc/kubernetes/pki/ca.crt` and `/etc/kubernetes/pki/ca.key`, and kubeadm will use this CA for signing the rest of the certificates.

## External CA mode

It is also possible to provide only the `ca.crt` file and not the `ca.key` file (this is only available for the root CA file, not other cert pairs). If all other certificates and kubeconfig files are in place, kubeadm recognizes this condition and activates the "External CA" mode. kubeadm will proceed without the CA key on disk.

Instead, run the controller-manager standalone with `--controllers=csrsigner` and point to the CA certificate and key.

There are various ways to prepare the component credentials when using external CA mode.

## Manual preparation of component credentials

[PKI certificates and requirements](#) includes information on how to prepare all the required by kubeadm component credentials manually.

## Preparation of credentials by signing CSRs generated by kubeadm

kubeadm can [generate CSR files](#) that you can sign manually with tools like openssl and your external CA. These CSR files will include all the specification for credentials that components deployed by kubeadm require.

## Automated preparation of component credentials by using kubeadm phases

Alternatively, it is possible to use kubeadm phase commands to automate this process.

- Go to a host that you want to prepare as a kubeadm control plane node with external CA.
- Copy the external CA files ca.crt and ca.key that you have into /etc/kubernetes/pki on the node.
- Prepare a temporary [kubeadm configuration file](#) called config.yaml that can be used with kubeadm init. Make sure that this file includes any relevant cluster wide or host-specific information that could be included in certificates, such as, ClusterConfiguration.controlPlaneEndpoint, ClusterConfiguration.certSANs and InitConfiguration.APIEndpoint.
- On the same host execute the commands `kubeadm init phase kubeconfig all --config config.yaml` and `kubeadm init phase certs all --config config.yaml`. This will generate all required kubeconfig files and certificates under /etc/kubernetes/ and its pki sub directory.
- Inspect the generated files. Delete /etc/kubernetes/pki/ca.key, delete or move to a safe location the file /etc/kubernetes/super-admin.conf.
- On nodes where kubeadm join will be called also delete /etc/kubernetes/kubelet.conf. This file is only required on the first node where kubeadm init will be called.
- Note that some files such pki/sa.\*, pki/front-proxy-ca.\* and pki/etc/ca.\* are shared between control plane nodes, You can generate them once and [distribute them manually](#) to nodes where kubeadm join will be called, or you can use the [--upload-certs](#) functionality of kubeadm init and `--certificate-key` of kubeadm join to automate this distribution.

Once the credentials are prepared on all nodes, call kubeadm init and kubeadm join for these nodes to join the cluster. kubeadm will use the existing kubeconfig and certificate files under /etc/kubernetes/ and its pki sub directory.

## Check certificate expiration

You can use the check-expiration subcommand to check when certificates expire:

```
kubeadm certs check-expiration
```

The output is similar to this:

| CERTIFICATE              | EXPIRES                | RESIDUAL TIME | CERTIFICATE AUTHORITY |
|--------------------------|------------------------|---------------|-----------------------|
| EXTERNALLY MANAGED       |                        |               |                       |
| admin.conf               | Dec 30, 2020 23:36 UTC | 364d          | no                    |
| apiserver                | Dec 30, 2020 23:36 UTC | 364d          | ca no                 |
| apiserver-etcd-client    | Dec 30, 2020 23:36 UTC | 364d          | etcd-ca no            |
| apiserver-kubelet-client | Dec 30, 2020 23:36 UTC | 364d          | ca no                 |
| controller-manager.conf  | Dec 30, 2020 23:36 UTC | 364d          | no                    |
| etcd-healthcheck-client  | Dec 30, 2020 23:36 UTC | 364d          | etcd-ca no            |
| etcd-peer                | Dec 30, 2020 23:36 UTC | 364d          | etcd-ca no            |

|                       |                        |         |                |            |
|-----------------------|------------------------|---------|----------------|------------|
| etcd-server           | Dec 30, 2020 23:36 UTC | 364d    | etcd-ca        | no         |
| front-proxy-client    | Dec 30, 2020 23:36 UTC | 364d    | front-proxy-ca | no         |
| scheduler.conf        | Dec 30, 2020 23:36 UTC | 364d    |                | no         |
| CERTIFICATE AUTHORITY |                        | EXPIRES | RESIDUAL TIME  | EXTERNALLY |
| MANAGED               |                        |         |                |            |
| ca                    | Dec 28, 2029 23:36 UTC | 9y      | no             |            |
| etcd-ca               | Dec 28, 2029 23:36 UTC | 9y      | no             |            |
| front-proxy-ca        | Dec 28, 2029 23:36 UTC | 9y      | no             |            |

The command shows expiration/residual time for the client certificates in the `/etc/kubernetes/pki` folder and for the client certificate embedded in the kubeconfig files used by kubeadm (`admin.conf`, `controller-manager.conf` and `scheduler.conf`).

Additionally, kubeadm informs the user if the certificate is externally managed; in this case, the user should take care of managing certificate renewal manually/using other tools.

### Warning:

kubeadm cannot manage certificates signed by an external CA.

### Note:

kubelet.conf is not included in the list above because kubeadm configures kubelet for [automatic certificate renewal](#) with rotatable certificates under `/var/lib/kubelet/pki`. To repair an expired kubelet client certificate see [Kubelet client certificate rotation fails](#).

### Warning:

On nodes created with kubeadm init, prior to kubeadm version 1.17, there is a [bug](#) where you manually have to modify the contents of kubelet.conf. After kubeadm init finishes, you should update kubelet.conf to point to the rotated kubelet client certificates, by replacing `client-certificate-data` and `client-key-data` with:

```
client-certificate: /var/lib/kubelet/pki/kubelet-client-current.pem
client-key: /var/lib/kubelet/pki/kubelet-client-current.pem
```

## Automatic certificate renewal

kubeadm renews all the certificates during control plane [upgrade](#).

This feature is designed for addressing the simplest use cases; if you don't have specific requirements on certificate renewal and perform Kubernetes version upgrades regularly (less than 1 year in between each upgrade), kubeadm will take care of keeping your cluster up to date and reasonably secure.

### Note:

It is a best practice to upgrade your cluster frequently in order to stay secure.



If you have more complex requirements for certificate renewal, you can opt out from the default behavior by passing `--certificate-renewal=false` to `kubeadm upgrade apply` or to `kubeadm upgrade node`.

### Warning:

Prior to `kubeadm` version 1.17 there is a [bug](#) where the default value for `--certificate-renewal` is false for the `kubeadm upgrade node` command. In that case, you should explicitly set `--certificate-renewal=true`.

## Manual certificate renewal

You can renew your certificates manually at any time with the `kubeadm certs renew` command, with the appropriate command line options.

This command performs the renewal using CA (or front-proxy-CA) certificate and key stored in `/etc/kubernetes/pki`.

After running the command you should restart the control plane Pods. This is required since dynamic certificate reload is currently not supported for all components and certificates. [Static Pods](#) are managed by the local kubelet and not by the API Server, thus `kubectl` cannot be used to delete and restart them. To restart a static Pod you can temporarily remove its manifest file from `/etc/kubernetes/manifests/` and wait for 20 seconds (see the `fileCheckFrequency` value in [KubeletConfiguration struct](#)). The kubelet will terminate the Pod if it's no longer in the manifest directory. You can then move the file back and after another `fileCheckFrequency` period, the kubelet will recreate the Pod and the certificate renewal for the component can complete.

### Warning:

If you are running an HA cluster, this command needs to be executed on all the control-plane nodes.

### Note:

`certs renew` uses the existing certificates as the authoritative source for attributes (Common Name, Organization, SAN, etc.) instead of the `kubeadm-config` ConfigMap. It is strongly recommended to keep them both in sync.

`kubeadm certs renew` can renew any specific certificate or, with the subcommand `all`, it can renew all of them, as shown below:

```
kubeadm certs renew all
```

### Note:

Clusters built with `kubeadm` often copy the `admin.conf` certificate into `$HOME/.kube/config`, as instructed in [Creating a cluster with kubeadm](#). On such a system, to update the contents of `$HOME/.kube/config` after renewing the `admin.conf`, you must run the following commands:

```
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

# Renew certificates with the Kubernetes certificates API

This section provides more details about how to execute manual certificate renewal using the Kubernetes certificates API.

## Caution:

These are advanced topics for users who need to integrate their organization's certificate infrastructure into a kubeadm-built cluster. If the default kubeadm configuration satisfies your needs, you should let kubeadm manage certificates instead.

## Set up a signer

The Kubernetes Certificate Authority does not work out of the box. You can configure an external signer such as [cert-manager](#), or you can use the built-in signer.

The built-in signer is part of [kube-controller-manager](#).

To activate the built-in signer, you must pass the `--cluster-signing-cert-file` and `--cluster-signing-key-file` flags.

If you're creating a new cluster, you can use a kubeadm [configuration file](#):

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
controllerManager:
 extraArgs:
 cluster-signing-cert-file: /etc/kubernetes/pki/ca.crt
 cluster-signing-key-file: /etc/kubernetes/pki/ca.key
```

## Create certificate signing requests (CSR)

See [Create CertificateSigningRequest](#) for creating CSRs with the Kubernetes API.

# Renew certificates with external CA

This section provide more details about how to execute manual certificate renewal using an external CA.

To better integrate with external CAs, kubeadm can also produce certificate signing requests (CSRs). A CSR represents a request to a CA for a signed certificate for a client. In kubeadm terms, any certificate that would normally be signed by an on-disk CA can be produced as a CSR instead. A CA, however, cannot be produced as a CSR.

## Renewal by using certificate signing requests (CSR)

Renewal of ceritficates is possible by generating new CSRs and signing them with the external CA. For more details about working with CSRs generated by kubeadm see the section [Signing certificate signing requests \(CSR\) generated by kubeadm](#).

# Certificate authority (CA) rotation

Kubeadm does not support rotation or replacement of CA certificates out of the box.

For more information about manual rotation or replacement of CA, see [manual rotation of CA certificates](#).

## Enabling signed kubelet serving certificates

By default the kubelet serving certificate deployed by kubeadm is self-signed. This means a connection from external services like the [metrics-server](#) to a kubelet cannot be secured with TLS.

To configure the kubelets in a new kubeadm cluster to obtain properly signed serving certificates you must pass the following minimal configuration to kubeadm init:

```
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration

apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
serverTLSBootstrap: true
```

If you have already created the cluster you must adapt it by doing the following:

- Find and edit the kubelet-config-1.30 ConfigMap in the kube-system namespace. In that ConfigMap, the kubelet key has a [KubeletConfiguration](#) document as its value. Edit the KubeletConfiguration document to set serverTLSBootstrap: true.
- On each node, add the serverTLSBootstrap: true field in /var/lib/kubelet/config.yaml and restart the kubelet with systemctl restart kubelet

The field serverTLSBootstrap: true will enable the bootstrap of kubelet serving certificates by requesting them from the certificates.k8s.io API. One known limitation is that the CSRs (Certificate Signing Requests) for these certificates cannot be automatically approved by the default signer in the kube-controller-manager - [kubernetes.io/kubelet-serving](#). This will require action from the user or a third party controller.

These CSRs can be viewed using:

```
kubectl get csr
```

| NAME      | AGE   | SIGNERNAME                    | REQUESTOR                   | CONDITION |
|-----------|-------|-------------------------------|-----------------------------|-----------|
| csr-9wvgt | 112s  | kubernetes.io/kubelet-serving | system:node:worker-1        | Pending   |
| csr-lz97v | 1m58s | kubernetes.io/kubelet-serving | system:node:control-plane-1 | Pending   |

To approve them you can do the following:

```
kubectl certificate approve <CSR-name>
```

By default, these serving certificate will expire after one year. Kubeadm sets the KubeletConfiguration field rotateCertificates to true, which means that close to expiration a new set of CSRs for the serving certificates will be created and must be approved to complete the rotation. To understand more see [Certificate Rotation](#).

If you are looking for a solution for automatic approval of these CSRs it is recommended that you contact your cloud provider and ask if they have a CSR signer that verifies the node identity with an out of band mechanism.

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Third party custom controllers can be used:

- [kubelet-csr-approver](#)

Such a controller is not a secure mechanism unless it not only verifies the CommonName in the CSR but also verifies the requested IPs and domain names. This would prevent a malicious actor that has access to a kubelet client certificate to create CSRs requesting serving certificates for any IP or domain name.

## Generating kubeconfig files for additional users

During cluster creation, kubeadm signs the certificate in the admin.conf to have Subject: O = system:masters, CN = kubernetes-admin. [system:masters](#) is a break-glass, super user group that bypasses the authorization layer (for example, [RBAC](#)). Sharing the admin.conf with additional users is **not recommended**!

Instead, you can use the [kubeadm kubeconfig user](#) command to generate kubeconfig files for additional users. The command accepts a mixture of command line flags and [kubeadm configuration](#) options. The generated kubeconfig will be written to stdout and can be piped to a file using `kubeadm kubeconfig user ... > somefile.conf`.

Example configuration file that can be used with `--config`:

```
example.yaml
apiVersion: kubeadm.k8s.io/v1beta3
kind: ClusterConfiguration
Will be used as the target "cluster" in the kubeconfig
clusterName: "kubernetes"
Will be used as the "server" (IP or DNS name) of this cluster in the kubeconfig
controlPlaneEndpoint: "some-dns-address:6443"
The cluster CA key and certificate will be loaded from this local directory
certificatesDir: "/etc/kubernetes/pki"
```

Make sure that these settings match the desired target cluster settings. To see the settings of an existing cluster use:

```
kubectl get cm kubeadm-config -n kube-system -o=jsonpath="{.data.ClusterConfiguration}"
```

The following example will generate a kubeconfig file with credentials valid for 24 hours for a new user johndoe that is part of the appdevs group:

```
kubeadm kubeconfig user --config example.yaml --org appdevs --client-name johndoe --
validity-period 24h
```

The following example will generate a kubeconfig file with administrator credentials valid for 1 week:

```
kubeadm kubeconfig user --config example.yaml --client-name admin --validity-period 168h
```

## Signing certificate signing requests (CSR) generated by kubeadm

You can create certificate signing requests with `kubeadm certs generate-csr`. Calling this command will generate `.csr` / `.key` file pairs for regular certificates. For certificates embedded in kubeconfig files, the command will generate a `.csr` / `.conf` pair where the key is already embedded in the `.conf` file.

A CSR file contains all relevant information for a CA to sign a certificate. kubeadm uses a [well defined specification](#) for all its certificates and CSRs.

The default certificate directory is `/etc/kubernetes/pki`, while the default directory for kubeconfig files is `/etc/kubernetes`. These defaults can be overridden with the flags `--cert-dir` and `--kubeconfig-dir`, respectively.

To pass custom options to `kubeadm certs generate-csr` use the `--config` flag, which accepts a [kubeadm configuration](#) file, similarly to commands such as `kubeadm init`. Any specification such as extra SANs and custom IP addresses must be stored in the same configuration file and used for all relevant kubeadm commands by passing it as `--config`.

### Note:

This guide will cover the usage of the `openssl` command for signing the CSRs, but you can use your preferred tools.

### Note:

This guide will use the default Kubernetes directory `/etc/kubernetes`, which requires a super user. If you are following this guide with permissive directories (by passing `--cert-dir` and `--kubeconfig-dir`) you can omit the `sudo` command). But note that the resulted files must be copied to the `/etc/kubernetes` tree, so that `kubeadm init` or `kubeadm join` will find them.

## Preparing CA and service account files

On the primary control plane node, where `kubeadm init` will be executed, call the following commands:

```
sudo kubeadm init phase certs ca
sudo kubeadm init phase certs etcd-ca
sudo kubeadm init phase certs front-proxy-ca
sudo kubeadm init phase certs sa
```

This will populate the folders `/etc/kubernetes/pki` and `/etc/kubernetes/pki/etcd` with all self-signed CA files (certificates and keys) and service account (public and private keys) that kubeadm needs for a control plane node.

**Note:**

If you are using an external CA, you must generate the same files out of band and manually copy them to the primary control plane node in `/etc/kubernetes`. Once all CSRs are signed, you can delete the root CA key (`ca.key`) as noted in the [External CA mode](#) section.

For secondary control plane nodes (`kubeadm join --control-plane`) there is no need to call the above commands. Depending on how you setup the [High Availability](#) cluster, you either have to manually copy the same files from the primary control plane node, or use the automated `--upload-certs` functionality of `kubeadm init`.

## Generate CSRs

The `kubeadm certs generate-csr` command generates CSRs for all known certificates managed by `kubeadm`. Once the command is done you must manually delete `.csr`, `.conf` or `.key` files that you don't need.

## Considerations for kubelet.conf

This section applies to both control plane and worker nodes.

If you have deleted the `ca.key` file from control plane nodes ([External CA mode](#)), the active kube-controller-manager in this cluster will not be able to sign kubelet client certificates. If no external method for signing these certificates exists in your setup (such as an [external signer](#), you could manually sign the `kubelet.conf.csr` as explained in this guide.

Note that this also means that the automatic [kubelet client certificate rotation](#) will be disabled. If so, close to certificate expiration, you must generate a new `kubelet.conf.csr`, sign the certificate, embed it in `kubelet.conf` and restart the kubelet.

If this does not apply to your setup, you can skip processing the `kubelet.conf.csr` on secondary control plane and on workers nodes (all nodes that call `kubeadm join ...`). That is because the active kube-controller-manager will be responsible for signing new kubelet client certificates.

**Note:**

Processing the `kubelet.conf.csr` on the primary control plane node (`kubeadm init`) is required, because that is considered the node that bootstraps the cluster and a pre-populated `kubelet.conf` is needed.

## Control plane nodes

Execute the following command on primary (`kubeadm init`) and secondary (`kubeadm join --control-plane`) control plane nodes to generate all CSR files:

```
sudo kubeadm certs generate-csr
```

If external etcd is to be used, follow the [External etcd with kubeadm](#) guide to understand what CSR files are needed on the `kubeadm` and `etcd` nodes. Other `.csr` and `.key` files under `/etc/kubernetes/pki/etcd` can be removed.

Based on the explanation in [Considerations for kubelet.conf](#) keep or delete the kubelet.conf and kubelet.conf.csr files.

## Worker nodes

Based on the explanation in [Considerations for kubelet.conf](#), optionally call:

```
sudo kubeadm certs generate-csr
```

and keep only the kubelet.conf and kubelet.conf.csr files. Alternatively skip the steps for worker nodes entirely.

## Signing CSRs for all certificates

### Note:

If you are using external CA and already have CA serial number files (.srl) for openssl you can copy such files to a kubeadm node where CSRs will be processed. .srl files to copy are /etc/kubernetes/pki/ca.srl, /etc/kubernetes/pki/front-proxy-ca.srl and /etc/kubernetes/pki/etcd/ca.srl. The files can be then moved to a new node where CSR files will be processed.

If a .srl file is missing for a CA on a node, the script below will generate a new SRL file with a random starting serial number.

To read more about .srl files see the [openssl](#) documentation for the --CAserial flag.

Repeat this step for all nodes that have CSR files.

Write the following script in the /etc/kubernetes directory, navigate to the directory and execute the script. The script will generate certificates for all CSR files that are present in the /etc/kubernetes tree.

```
#!/bin/bash

Set certificate expiration time in days
DAYS=365

Process all CSR files except those for front-proxy and etcd
find ./ -name "*.csr" | grep -v "pki/etcd" | grep -v "front-proxy" | while read -r FILE;
do
 echo "* Processing ${FILE} ..."
 FILE=${FILE%.*} # Trim the extension
 if [-f "./pki/ca.srl"]; then
 SERIAL_FLAG="-CAserial ./pki/ca.srl"
 else
 SERIAL_FLAG="-CAcreateserial"
 fi
 openssl x509 -req -days "${DAYS}" -CA ./pki/ca.crt -CAkey ./pki/ca.key ${SERIAL_FLAG} \
 -in "${FILE}.csr" -out "${FILE}.crt"
 sleep 2
done

Process all etcd CSRs
```

```

find ./pki/etcd -name "*.csr" | while read -r FILE;
do
 echo "* Processing ${FILE} ..."
 FILE=${FILE%.*} # Trim the extension
 if [-f "./pki/etcd/ca.srl"]; then
 SERIAL_FLAG=-CAserial ./pki/etcd/ca.srl
 else
 SERIAL_FLAG=-CAcreateserial
 fi
 openssl x509 -req -days "${DAYS}" -CA ./pki/etcd/ca.crt -CAkey ./pki/etcd/ca.key ${SERIAL_F
LAG} \
 -in "${FILE}.csr" -out "${FILE}.crt"
done

Process front-proxy CSRs
echo "* Processing ./pki/front-proxy-client.csr ..."
openssl x509 -req -days "${DAYS}" -CA ./pki/front-proxy-ca.crt -CAkey ./pki/front-proxy-ca.key
-CAcreateserial \
 -in ./pki/front-proxy-client.csr -out ./pki/front-proxy-client.crt

```

## Embedding certificates in kubeconfig files

Repeat this step for all nodes that have CSR files.

Write the following script in the /etc/kubernetes directory, navigate to the directory and execute the script. The script will take the .crt files that were signed for kubeconfig files from CSRs in the previous step and will embed them in the kubeconfig files.

```

#!/bin/bash

CLUSTER=kubernetes
find ./ -name "*.conf" | while read -r FILE;
do
 echo "* Processing ${FILE} ..."
 KUBECONFIG="${FILE}" kubectl config set-cluster "${CLUSTER}" --certificate-authority ./pki/
ca.crt --embed-certs
 USER=$(KUBECONFIG="${FILE}" kubectl config view -o jsonpath='{.users[0].name}')
 KUBECONFIG="${FILE}" kubectl config set-credentials "${USER}" --client-certificate "${FILE}.c
rt" --embed-certs
done

```

## Performing cleanup

Perform this step on all nodes that have CSR files.

Write the following script in the /etc/kubernetes directory, navigate to the directory and execute the script.

```

#!/bin/bash

Cleanup CSR files
rm -f ./*.csr ./pki/*.csr ./pki/etcd/*.csr # Clean all CSR files

```



```
Cleanup CRT files that were already embedded in kubeconfig files
rm -f ./*.crt
```

Optionally, move .srl files to the next node to be processed.

Optionally, if using external CA remove the /etc/kubernetes/pki/ca.key file, as explained in the [External CA node](#) section.

## kubeadm node initialization

Once CSR files have been signed and required certificates are in place on the hosts you want to use as nodes, you can use the commands `kubeadm init` and `kubeadm join` to create a Kubernetes cluster from these nodes. During init and join, kubeadm uses existing certificates, encryption keys and kubeconfig files that it finds in the /etc/kubernetes tree on the host's local filesystem.

# Configuring a cgroup driver

This page explains how to configure the kubelet's cgroup driver to match the container runtime cgroup driver for kubeadm clusters.

## Before you begin

You should be familiar with the Kubernetes [container runtime requirements](#).

## Configuring the container runtime cgroup driver

The [Container runtimes](#) page explains that the systemd driver is recommended for kubeadm based setups instead of the kubelet's [default](#) cgroupfs driver, because kubeadm manages the kubelet as a [systemd service](#).

The page also provides details on how to set up a number of different container runtimes with the systemd driver by default.

## Configuring the kubelet cgroup driver

kubeadm allows you to pass a KubeletConfiguration structure during kubeadm init. This KubeletConfiguration can include the cgroupDriver field which controls the cgroup driver of the kubelet.

### Note:

In v1.22 and later, if the user does not set the cgroupDriver field under KubeletConfiguration, kubeadm defaults it to systemd.

In Kubernetes v1.28, you can enable automatic detection of the cgroup driver as an alpha feature. See [systemd cgroup driver](#) for more details.

A minimal example of configuring the field explicitly:

```
kubeadm-config.yaml
kind: ClusterConfiguration
apiVersion: kubeadm.k8s.io/v1beta3
kubernetesVersion: v1.21.0

kind: KubeletConfiguration
apiVersion: kubelet.config.k8s.io/v1beta1
cgroupDriver: systemd
```

Such a configuration file can then be passed to the kubeadm command:

```
kubeadm init --config kubeadm-config.yaml
```

**Note:**

Kubeadm uses the same KubeletConfiguration for all nodes in the cluster. The KubeletConfiguration is stored in a [ConfigMap](#) object under the kube-system namespace.

Executing the sub commands init, join and upgrade would result in kubeadm writing the KubeletConfiguration as a file under /var/lib/kubelet/config.yaml and passing it to the local node kubelet.

## Using the cgroupfs driver

To use cgroupfs and to prevent kubeadm upgrade from modifying the KubeletConfiguration cgroup driver on existing setups, you must be explicit about its value. This applies to a case where you do not wish future versions of kubeadm to apply the systemd driver by default.

See the below section on "[Modify the kubelet ConfigMap](#)" for details on how to be explicit about the value.

If you wish to configure a container runtime to use the cgroupfs driver, you must refer to the documentation of the container runtime of your choice.

## Migrating to the systemd driver

To change the cgroup driver of an existing kubeadm cluster from cgroupfs to systemd in-place, a similar procedure to a kubelet upgrade is required. This must include both steps outlined below.

**Note:**

Alternatively, it is possible to replace the old nodes in the cluster with new ones that use the systemd driver. This requires executing only the first step below before joining the new nodes and ensuring the workloads can safely move to the new nodes before deleting the old nodes.

### Modify the kubelet ConfigMap

- Call `kubectrl edit cm kubelet-config -n kube-system`.
- Either modify the existing cgroupDriver value or add a new field that looks like this:

```
cgroupDriver: systemd
```

This field must be present under the kubelet: section of the ConfigMap.

## Update the cgroup driver on all nodes

For each node in the cluster:

- [Drain the node](#) using `kubectldrain <node-name> --ignore-daemonsets`
- Stop the kubelet using `systemctl stop kubelet`
- Stop the container runtime
- Modify the container runtime cgroup driver to `systemd`
- Set `cgroupDriver: systemd` in `/var/lib/kubelet/config.yaml`
- Start the container runtime
- Start the kubelet using `systemctl start kubelet`
- [Uncordon the node](#) using `kubectluncordon <node-name>`

Execute these steps on nodes one at a time to ensure workloads have sufficient time to schedule on different nodes.

Once the process is complete ensure that all nodes and workloads are healthy.

# Reconfiguring a kubeadm cluster

kubeadm does not support automated ways of reconfiguring components that were deployed on managed nodes. One way of automating this would be by using a custom [operator](#).

To modify the components configuration you must manually edit associated cluster objects and files on disk.

This guide shows the correct sequence of steps that need to be performed to achieve kubeadm cluster reconfiguration.

## Before you begin

- You need a cluster that was deployed using kubeadm
- Have administrator credentials (`/etc/kubernetes/admin.conf`) and network connectivity to a running kube-apiserver in the cluster from a host that has kubectl installed
- Have a text editor installed on all hosts

## Reconfiguring the cluster

kubeadm writes a set of cluster wide component configuration options in ConfigMaps and other objects. These objects must be manually edited. The command `kubectledit` can be used for that.

The `kubectledit` command will open a text editor where you can edit and save the object directly.

You can use the environment variables `KUBECONFIG` and `KUBE_EDITOR` to specify the location of the kubectl consumed kubeconfig file and preferred text editor.

For example:

```
KUBECONFIG=/etc/kubernetes/admin.conf KUBE_EDITOR=nano kubectl edit <parameters>
```

**Note:**

Upon saving any changes to these cluster objects, components running on nodes may not be automatically updated. The steps below instruct you on how to perform that manually.

**Warning:**

Component configuration in ConfigMaps is stored as unstructured data (YAML string). This means that validation will not be performed upon updating the contents of a ConfigMap. You have to be careful to follow the documented API format for a particular component configuration and avoid introducing typos and YAML indentation mistakes.

## Applying cluster configuration changes

### Updating the ClusterConfiguration

During cluster creation and upgrade, kubeadm writes its [ClusterConfiguration](#) in a ConfigMap called kubeadm-config in the kube-system namespace.

To change a particular option in the ClusterConfiguration you can edit the ConfigMap with this command:

```
kubectl edit cm -n kube-system kubeadm-config
```

The configuration is located under the data.ClusterConfiguration key.

**Note:**

The ClusterConfiguration includes a variety of options that affect the configuration of individual components such as kube-apiserver, kube-scheduler, kube-controller-manager, CoreDNS, etcd and kube-proxy. Changes to the configuration must be reflected on node components manually.

### Reflecting ClusterConfiguration changes on control plane nodes

kubeadm manages the control plane components as static Pod manifests located in the directory /etc/kubernetes/manifests. Any changes to the ClusterConfiguration under the apiServer, controllerManager, scheduler or etcd keys must be reflected in the associated files in the manifests directory on a control plane node.

Such changes may include:

- extraArgs - requires updating the list of flags passed to a component container
- extraMounts - requires updated the volume mounts for a component container
- \*SANs - requires writing new certificates with updated Subject Alternative Names.

Before proceeding with these changes, make sure you have backed up the directory /etc/kubernetes/.

To write new certificates you can use:

```
kubeadm init phase certs <component-name> --config <config-file>
```

To write new manifest files in `/etc/kubernetes/manifests` you can use:

```
For Kubernetes control plane components
kubeadm init phase control-plane <component-name> --config <config-file>
For local etcd
kubeadm init phase etcd local --config <config-file>
```

The `<config-file>` contents must match the updated `ClusterConfiguration`. The `<component-name>` value must be a name of a Kubernetes control plane component (apiserver, controller-manager or scheduler).

**Note:**

Updating a file in `/etc/kubernetes/manifests` will tell the kubelet to restart the static Pod for the corresponding component. Try doing these changes one node at a time to leave the cluster without downtime.

## Applying kubelet configuration changes

### Updating the KubeletConfiguration

During cluster creation and upgrade, kubeadm writes its [KubeletConfiguration](#) in a ConfigMap called `kubelet-config` in the `kube-system` namespace.

You can edit the ConfigMap with this command:

```
kubectrl edit cm -n kube-system kubelet-config
```

The configuration is located under the `data.kubelet` key.

### Reflecting the kubelet changes

To reflect the change on kubeadm nodes you must do the following:

- Log in to a kubeadm node
- Run `kubeadm upgrade node phase kubelet-config` to download the latest kubelet-config ConfigMap contents into the local file `/var/lib/kubelet/config.yaml`
- Edit the file `/var/lib/kubelet/kubeadm-flags.env` to apply additional configuration with flags
- Restart the kubelet service with `systemctl restart kubelet`

**Note:**

Do these changes one node at a time to allow workloads to be rescheduled properly.

**Note:**

During kubeadm upgrade, kubeadm downloads the KubeletConfiguration from the kubelet-config ConfigMap and overwrite the contents of /var/lib/kubelet/config.yaml. This means that node local configuration must be applied either by flags in /var/lib/kubelet/kubeadm-flags.env or by manually updating the contents of /var/lib/kubelet/config.yaml after kubeadm upgrade, and then restarting the kubelet.

## Applying kube-proxy configuration changes

### Updating the KubeProxyConfiguration

During cluster creation and upgrade, kubeadm writes its [KubeProxyConfiguration](#) in a ConfigMap in the kube-system namespace called kube-proxy.

This ConfigMap is used by the kube-proxy DaemonSet in the kube-system namespace.

To change a particular option in the KubeProxyConfiguration, you can edit the ConfigMap with this command:

```
kubectrl edit cm -n kube-system kube-proxy
```

The configuration is located under the data.config.conf key.

### Reflecting the kube-proxy changes

Once the kube-proxy ConfigMap is updated, you can restart all kube-proxy Pods:

Obtain the Pod names:

```
kubectrl get po -n kube-system | grep kube-proxy
```

Delete a Pod with:

```
kubectrl delete po -n kube-system <pod-name>
```

New Pods that use the updated ConfigMap will be created.

**Note:**

Because kubeadm deploys kube-proxy as a DaemonSet, node specific configuration is unsupported.

## Applying CoreDNS configuration changes

### Updating the CoreDNS Deployment and Service

kubeadm deploys CoreDNS as a Deployment called coredns and with a Service kube-dns, both in the kube-system namespace.

To update any of the CoreDNS settings, you can edit the Deployment and Service objects:

```
kubectl edit deployment -n kube-system coredns
kubectl edit service -n kube-system kube-dns
```

## Reflecting the CoreDNS changes

Once the CoreDNS changes are applied you can delete the CoreDNS Pods:

Obtain the Pod names:

```
kubectl get po -n kube-system | grep coredns
```

Delete a Pod with:

```
kubectl delete po -n kube-system <pod-name>
```

New Pods with the updated CoreDNS configuration will be created.

### Note:

kubeadm does not allow CoreDNS configuration during cluster creation and upgrade. This means that if you execute kubeadm upgrade apply, your changes to the CoreDNS objects will be lost and must be reapplied.

## Persisting the reconfiguration

During the execution of kubeadm upgrade on a managed node, kubeadm might overwrite configuration that was applied after the cluster was created (reconfiguration).

### Persisting Node object reconfiguration

kubeadm writes Labels, Taints, CRI socket and other information on the Node object for a particular Kubernetes node. To change any of the contents of this Node object you can use:

```
kubectl edit no <node-name>
```

During kubeadm upgrade the contents of such a Node might get overwritten. If you would like to persist your modifications to the Node object after upgrade, you can prepare a [kubectl patch](#) and apply it to the Node object:

```
kubectl patch no <node-name> --patch-file <patch-file>
```

### Persisting control plane component reconfiguration

The main source of control plane configuration is the ClusterConfiguration object stored in the cluster. To extend the static Pod manifests configuration, [patches](#) can be used.

These patch files must remain as files on the control plane nodes to ensure that they can be used by the kubeadm upgrade ... --patches <directory>.

If reconfiguration is done to the ClusterConfiguration and static Pod manifests on disk, the set of node specific patches must be updated accordingly.

## Persisting kubelet reconfiguration

Any changes to the KubeletConfiguration stored in `/var/lib/kubelet/config.yaml` will be overwritten on kubeadm upgrade by downloading the contents of the cluster wide kubelet-config ConfigMap. To persist kubelet node specific configuration either the file `/var/lib/kubelet/config.yaml` has to be updated manually post-upgrade or the file `/var/lib/kubelet/kubeadm-flags.env` can include flags. The kubelet flags override the associated KubeletConfiguration options, but note that some of the flags are deprecated.

A kubelet restart will be required after changing `/var/lib/kubelet/config.yaml` or `/var/lib/kubelet/kubeadm-flags.env`.

## What's next

- [Upgrading kubeadm clusters](#)
- [Customizing components with the kubeadm API](#)
- [Certificate management with kubeadm](#)
- [Find more about kubeadm set-up](#)

# Upgrading kubeadm clusters

This page explains how to upgrade a Kubernetes cluster created with kubeadm from version 1.29.x to version 1.30.x, and from version 1.30.x to 1.30.y (where  $y > x$ ). Skipping MINOR versions when upgrading is unsupported. For more details, please visit [Version Skew Policy](#).

To see information about upgrading clusters created using older versions of kubeadm, please refer to following pages instead:

- [Upgrading a kubeadm cluster from 1.28 to 1.29](#)
- [Upgrading a kubeadm cluster from 1.27 to 1.28](#)
- [Upgrading a kubeadm cluster from 1.26 to 1.27](#)
- [Upgrading a kubeadm cluster from 1.25 to 1.26](#)

The upgrade workflow at high level is the following:

1. Upgrade a primary control plane node.
2. Upgrade additional control plane nodes.
3. Upgrade worker nodes.

## Before you begin

- Make sure you read the [release notes](#) carefully.
- The cluster should use a static control plane and etcd pods or external etcd.
- Make sure to back up any important components, such as app-level state stored in a database. kubeadm upgrade does not touch your workloads, only components internal to Kubernetes, but backups are always a best practice.
- [Swap must be disabled](#).



## Additional information

- The instructions below outline when to drain each node during the upgrade process. If you are performing a **minor** version upgrade for any kubelet, you **must** first drain the node (or nodes) that you are upgrading. In the case of control plane nodes, they could be running CoreDNS Pods or other critical workloads. For more information see [Draining nodes](#).
- The Kubernetes project recommends that you match your kubelet and kubeadm versions. You can instead use a version of kubelet that is older than kubeadm, provided it is within the range of supported versions. For more details, please visit [kubeadm's skew against the kubelet](#).
- All containers are restarted after upgrade, because the container spec hash value is changed.
- To verify that the kubelet service has successfully restarted after the kubelet has been upgraded, you can execute `systemctl status kubelet` or view the service logs with `journalctl -xeu kubelet`.
- Usage of the `--config` flag of kubeadm upgrade with [kubeadm configuration API types](#) with the purpose of reconfiguring the cluster is not recommended and can have unexpected results. Follow the steps in [Reconfiguring a kubeadm cluster](#) instead.

## Considerations when upgrading etcd

Because the kube-apiserver static pod is running at all times (even if you have drained the node), when you perform a kubeadm upgrade which includes an etcd upgrade, in-flight requests to the server will stall while the new etcd static pod is restarting. As a workaround, it is possible to actively stop the kube-apiserver process a few seconds before starting the kubeadm upgrade apply command. This permits to complete in-flight requests and close existing connections, and minimizes the consequence of the etcd downtime. This can be done as follows on control plane nodes:

```
killall -s SIGTERM kube-apiserver # trigger a graceful kube-apiserver shutdown
sleep 20 # wait a little bit to permit completing in-flight requests
kubeadm upgrade ... # execute a kubeadm upgrade command
```

## Changing the package repository

If you're using the community-owned package repositories (`pkgs.k8s.io`), you need to enable the package repository for the desired Kubernetes minor release. This is explained in [Changing the Kubernetes package repository](#) document.

**Note:** The legacy package repositories (`apt.kubernetes.io` and `yum.kubernetes.io`) have been [deprecated and frozen starting from September 13, 2023](#). **Using the [new package repositories hosted at `pkgs.k8s.io`](#) is strongly recommended and required in order to install Kubernetes versions released after September 13, 2023.** The deprecated legacy repositories, and their contents, might be removed at any time in the future and without a further notice period. The new package repositories provide downloads for Kubernetes versions starting with v1.24.0.

## Determine which version to upgrade to

Find the latest patch release for Kubernetes 1.30 using the OS package manager:

- [Ubuntu, Debian or HyprionOS](#)
- [CentOS, RHEL or Fedora](#)

```
Find the latest 1.30 version in the list.
It should look like 1.30.x-*, where x is the latest patch.
sudo apt update
sudo apt-cache madison kubeadm
```

```
Find the latest 1.30 version in the list.
It should look like 1.30.x-*, where x is the latest patch.
sudo yum list --showduplicates kubeadm --disableexcludes=kubernetes
```

## Upgrading control plane nodes

The upgrade procedure on control plane nodes should be executed one node at a time. Pick a control plane node that you wish to upgrade first. It must have the `/etc/kubernetes/admin.conf` file.

### Call "kubeadm upgrade"

#### For the first control plane node

1. Upgrade kubeadm:

- [Ubuntu, Debian or HyprionOS](#)
- [CentOS, RHEL or Fedora](#)

```
replace x in 1.30.x-* with the latest patch version
sudo apt-mark unhold kubeadm && \
sudo apt-get update && sudo apt-get install -y kubeadm='1.30.x-*' && \
sudo apt-mark hold kubeadm
```

```
replace x in 1.30.x-* with the latest patch version
sudo yum install -y kubeadm-'1.30.x-*' --disableexcludes=kubernetes
```

2. Verify that the download works and has the expected version:

```
kubeadm version
```

3. Verify the upgrade plan:

```
sudo kubeadm upgrade plan
```

This command checks that your cluster can be upgraded, and fetches the versions you can upgrade to. It also shows a table with the component config version states.

**Note:**

kubeadm upgrade also automatically renews the certificates that it manages on this node. To opt-out of certificate renewal the flag `--certificate-renewal=false` can be used. For more information see the [certificate management guide](#).

**Note:**

If kubeadm upgrade plan shows any component configs that require manual upgrade, users must provide a config file with replacement configs to kubeadm upgrade apply via the `--config` command line flag. Failing to do so will cause kubeadm upgrade apply to exit with an error and not perform an upgrade.

4. Choose a version to upgrade to, and run the appropriate command. For example:

```
replace x with the patch version you picked for this upgrade
sudo kubeadm upgrade apply v1.30.x
```

Once the command finishes you should see:

```
[upgrade/successful] SUCCESS! Your cluster was upgraded to "v1.30.x". Enjoy!
```

```
[upgrade/kubelet] Now that your control plane is upgraded, please proceed with
upgrading your kubelets if you haven't already done so.
```

**Note:**

For versions earlier than v1.28, kubeadm defaulted to a mode that upgrades the addons (including CoreDNS and kube-proxy) immediately during kubeadm upgrade apply, regardless of whether there are other control plane instances that have not been upgraded. This may cause compatibility problems. Since v1.28, kubeadm defaults to a mode that checks whether all the control plane instances have been upgraded before starting to upgrade the addons. You must perform control plane instances upgrade sequentially or at least ensure that the last control plane instance upgrade is not started until all the other control plane instances have been upgraded completely, and the addons upgrade will be performed after the last control plane instance is upgraded. If you want to keep the old upgrade behavior, please enable the `UpgradeAddonsBeforeControlPlane` feature gate by `kubeadm upgrade apply --feature-gates=UpgradeAddonsBeforeControlPlane=true`. The Kubernetes project does not in general recommend enabling this feature gate, you should instead change your upgrade process or cluster addons so that you do not need to enable the legacy behavior. The `UpgradeAddonsBeforeControlPlane` feature gate will be removed in a future release.

5. Manually upgrade your CNI provider plugin.

Your Container Network Interface (CNI) provider may have its own upgrade instructions to follow. Check the [addons](#) page to find your CNI provider and see whether additional upgrade steps are required.

This step is not required on additional control plane nodes if the CNI provider runs as a DaemonSet.

**For the other control plane nodes**

Same as the first control plane node but use:

```
sudo kubeadm upgrade node
```

instead of:

```
sudo kubeadm upgrade apply
```

Also calling `kubeadm upgrade plan` and upgrading the CNI provider plugin is no longer needed.

## Drain the node

Prepare the node for maintenance by marking it unschedulable and evicting the workloads:

```
replace <node-to-drain> with the name of your node you are draining
kubectl drain <node-to-drain> --ignore-daemonsets
```

## Upgrade kubelet and kubectl

1. Upgrade the kubelet and kubectl:

- [Ubuntu, Debian or HypriotOS](#)
- [CentOS, RHEL or Fedora](#)

```
replace x in 1.30.x-* with the latest patch version
sudo apt-mark unhold kubelet kubectl && \
sudo apt-get update && sudo apt-get install -y kubelet='1.30.x-*' kubectl='1.30.x-*' && \
sudo apt-mark hold kubelet kubectl
```

```
replace x in 1.30.x-* with the latest patch version
sudo yum install -y kubelet-'1.30.x-*' kubectl-'1.30.x-*' --disableexcludes=kubernetes
```

2. Restart the kubelet:

```
sudo systemctl daemon-reload
sudo systemctl restart kubelet
```

## Uncordon the node

Bring the node back online by marking it schedulable:

```
replace <node-to-uncordon> with the name of your node
kubectl uncordon <node-to-uncordon>
```

## Upgrade worker nodes

The upgrade procedure on worker nodes should be executed one node at a time or few nodes at a time, without compromising the minimum required capacity for running your workloads.

The following pages show how to upgrade Linux and Windows worker nodes:

- [Upgrade Linux nodes](#)
- [Upgrade Windows nodes](#)

## Verify the status of the cluster

After the kubelet is upgraded on all nodes verify that all nodes are available again by running the following command from anywhere kubectl can access the cluster:

```
kubectl get nodes
```

The STATUS column should show Ready for all your nodes, and the version number should be updated.

## Recovering from a failure state

If kubeadm upgrade fails and does not roll back, for example because of an unexpected shutdown during execution, you can run kubeadm upgrade again. This command is idempotent and eventually makes sure that the actual state is the desired state you declare.

To recover from a bad state, you can also run `sudo kubeadm upgrade apply --force` without changing the version that your cluster is running.

During upgrade kubeadm writes the following backup folders under `/etc/kubernetes/tmp`:

- `kubeadm-backup-etcd-<date>--<time>`
- `kubeadm-backup-manifests-<date>--<time>`

`kubeadm-backup-etcd` contains a backup of the local etcd member data for this control plane Node. In case of an etcd upgrade failure and if the automatic rollback does not work, the contents of this folder can be manually restored in `/var/lib/etcd`. In case external etcd is used this backup folder will be empty.

`kubeadm-backup-manifests` contains a backup of the static Pod manifest files for this control plane Node. In case of a upgrade failure and if the automatic rollback does not work, the contents of this folder can be manually restored in `/etc/kubernetes/manifests`. If for some reason there is no difference between a pre-upgrade and post-upgrade manifest file for a certain component, a backup file for it will not be written.

## How it works

`kubeadm upgrade apply` does the following:

- Checks that your cluster is in an upgradeable state:
  - The API server is reachable
  - All nodes are in the Ready state
  - The control plane is healthy
- Enforces the version skew policies.
- Makes sure the control plane images are available or available to pull to the machine.
- Generates replacements and/or uses user supplied overwrites if component configs require version upgrades.
- Upgrades the control plane components or rollbacks if any of them fails to come up.
- Applies the new CoreDNS and kube-proxy manifests and makes sure that all necessary RBAC rules are created.
- Creates new certificate and key files of the API server and backs up old files if they're about to expire in 180 days.

kubeadm upgrade node does the following on additional control plane nodes:

- Fetches the kubeadm ClusterConfiguration from the cluster.
- Optionally backups the kube-apiserver certificate.
- Upgrades the static Pod manifests for the control plane components.
- Upgrades the kubelet configuration for this node.

kubeadm upgrade node does the following on worker nodes:

- Fetches the kubeadm ClusterConfiguration from the cluster.
- Upgrades the kubelet configuration for this node.

## Upgrading Linux nodes

This page explains how to upgrade a Linux Worker Nodes created with kubeadm.

### Before you begin

You need to have shell access to all the nodes, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts.

To check the version, enter kubectl version.

- Familiarize yourself with [the process for upgrading the rest of your kubeadm cluster](#). You will want to upgrade the control plane nodes before upgrading your Linux Worker nodes.

### Changing the package repository

If you're using the community-owned package repositories (pkgs.k8s.io), you need to enable the package repository for the desired Kubernetes minor release. This is explained in [Changing the Kubernetes package repository](#) document.

**Note:** The legacy package repositories (apt.kubernetes.io and yum.kubernetes.io) have been [deprecated and frozen starting from September 13, 2023](#). **Using the [new package repositories hosted at pkgs.k8s.io](#) is strongly recommended and required in order to install Kubernetes versions released after September 13, 2023.** The deprecated legacy repositories, and their contents, might be removed at any time in the future and without a further notice period. The new package repositories provide downloads for Kubernetes versions starting with v1.24.0.

## Upgrading worker nodes

### Upgrade kubeadm

Upgrade kubeadm:

- [Ubuntu, Debian or HypriotOS](#)
- [CentOS, RHEL or Fedora](#)

```
replace x in 1.30.x-* with the latest patch version
sudo apt-mark unhold kubeadm && \
sudo apt-get update && sudo apt-get install -y kubeadm='1.30.x-*' && \
sudo apt-mark hold kubeadm
```

```
replace x in 1.30.x-* with the latest patch version
sudo yum install -y kubeadm-'1.30.x-*' --disableexcludes=kubernetes
```

## Call "kubeadm upgrade"

For worker nodes this upgrades the local kubelet configuration:

```
sudo kubeadm upgrade node
```

## Drain the node

Prepare the node for maintenance by marking it unschedulable and evicting the workloads:

```
execute this command on a control plane node
replace <node-to-drain> with the name of your node you are draining
kubectl drain <node-to-drain> --ignore-daemonsets
```

## Upgrade kubelet and kubectl

1. Upgrade the kubelet and kubectl:

- [Ubuntu, Debian or HyprionOS](#)
- [CentOS, RHEL or Fedora](#)

```
replace x in 1.30.x-* with the latest patch version
sudo apt-mark unhold kubelet kubectl && \
sudo apt-get update && sudo apt-get install -y kubelet='1.30.x-*' kubectl='1.30.x-*' && \
sudo apt-mark hold kubelet kubectl
```

```
replace x in 1.30.x-* with the latest patch version
sudo yum install -y kubelet-'1.30.x-*' kubectl-'1.30.x-*' --disableexcludes=kubernetes
```

2. Restart the kubelet:

```
sudo systemctl daemon-reload
sudo systemctl restart kubelet
```

## Uncordon the node

Bring the node back online by marking it schedulable:

```
execute this command on a control plane node
replace <node-to-uncordon> with the name of your node
kubectl uncordon <node-to-uncordon>
```

## What's next

- See how to [Upgrade Windows nodes](#).

# Upgrading Windows nodes

FEATURE STATE: Kubernetes v1.18 [beta]

This page explains how to upgrade a Windows node created with kubeadm.

## Before you begin

You need to have shell access to all the nodes, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts.

Your Kubernetes server must be at or later than version 1.17. To check the version, enter kubectl version.

- Familiarize yourself with [the process for upgrading the rest of your kubeadm cluster](#). You will want to upgrade the control plane nodes before upgrading your Windows nodes.

## Upgrading worker nodes

### Upgrade kubeadm

1. From the Windows node, upgrade kubeadm:

```
replace 1.30.0 with your desired version
curl.exe -Lo <path-to-kubeadm.exe> "https://dl.k8s.io/v1.30.0/bin/windows/amd64/kubeadm.exe"
```

### Drain the node

1. From a machine with access to the Kubernetes API, prepare the node for maintenance by marking it unschedulable and evicting the workloads:

```
replace <node-to-drain> with the name of your node you are draining
kubectl drain <node-to-drain> --ignore-daemonsets
```

You should see output similar to this:

```
node/ip-172-31-85-18 cordoned
node/ip-172-31-85-18 drained
```

### Upgrade the kubelet configuration

1. From the Windows node, call the following command to sync new kubelet configuration:

```
kubeadm upgrade node
```



## Upgrade kubelet and kube-proxy

1. From the Windows node, upgrade and restart the kubelet:

```
stop-service kubelet
curl.exe -Lo <path-to-kubelet.exe> "https://dl.k8s.io/v1.30.0/bin/windows/amd64/
kubelet.exe"
restart-service kubelet
```

2. From the Windows node, upgrade and restart the kube-proxy.

```
stop-service kube-proxy
curl.exe -Lo <path-to-kube-proxy.exe> "https://dl.k8s.io/v1.30.0/bin/windows/amd64/
kube-proxy.exe"
restart-service kube-proxy
```

### Note:

If you are running kube-proxy in a HostProcess container within a Pod, and not as a Windows Service, you can upgrade kube-proxy by applying a newer version of your kube-proxy manifests.

## Uncordon the node

1. From a machine with access to the Kubernetes API, bring the node back online by marking it schedulable:

```
replace <node-to-drain> with the name of your node
kubectl uncordon <node-to-drain>
```

## What's next

- See how to [Upgrade Linux nodes](#).

# Changing The Kubernetes Package Repository

This page explains how to enable a package repository for the desired Kubernetes minor release upon upgrading a cluster. This is only needed for users of the community-owned package repositories hosted at [pkgs.k8s.io](https://pkgs.k8s.io). Unlike the legacy package repositories, the community-owned package repositories are structured in a way that there's a dedicated package repository for each Kubernetes minor version.

### Note:

This guide only covers a part of the Kubernetes upgrade process. Please see the [upgrade guide](#) for more information about upgrading Kubernetes clusters.

## Note:

This step is only needed upon upgrading a cluster to another **minor** release. If you're upgrading to another patch release within the same minor release (e.g. v1.30.5 to v1.30.7), you don't need to follow this guide. However, if you're still using the legacy package repositories, you'll need to migrate to the new community-owned package repositories before upgrading (see the next section for more details on how to do this).

## Before you begin

This document assumes that you're already using the community-owned package repositories (pkgs.k8s.io). If that's not the case, it's strongly recommended to migrate to the community-owned package repositories as described in the [official announcement](#).

**Note:** The legacy package repositories (apt.kubernetes.io and yum.kubernetes.io) have been [deprecated and frozen starting from September 13, 2023](#). **Using the [new package repositories hosted at pkgs.k8s.io](#) is strongly recommended and required in order to install Kubernetes versions released after September 13, 2023.** The deprecated legacy repositories, and their contents, might be removed at any time in the future and without a further notice period. The new package repositories provide downloads for Kubernetes versions starting with v1.24.0.

## Verifying if the Kubernetes package repositories are used

If you're unsure whether you're using the community-owned package repositories or the legacy package repositories, take the following steps to verify:

- [Ubuntu, Debian or HyprIoTOS](#)
- [CentOS, RHEL or Fedora](#)
- [openSUSE or SLES](#)

Print the contents of the file that defines the Kubernetes apt repository:

```
On your system, this configuration file could have a different name
pager /etc/apt/sources.list.d/kubernetes.list
```

If you see a line similar to:

```
deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.29/deb/ /
```

**You're using the Kubernetes package repositories and this guide applies to you.**

Otherwise, it's strongly recommended to migrate to the Kubernetes package repositories as described in the [official announcement](#).

Print the contents of the file that defines the Kubernetes yum repository:

```
On your system, this configuration file could have a different name
cat /etc/yum.repos.d/kubernetes.repo
```

If you see a baseurl similar to the baseurl in the output below:

```
[kubernetes]
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.29/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.29/rpm/repodata/repomd.xml.key
exclude=kubelet kubeadm kubectl
```

**You're using the Kubernetes package repositories and this guide applies to you.**

Otherwise, it's strongly recommended to migrate to the Kubernetes package repositories as described in the [official announcement](#).

Print the contents of the file that defines the Kubernetes zypper repository:

```
On your system, this configuration file could have a different name
cat /etc/zypp/repos.d/kubernetes.repo
```

If you see a baseurl similar to the baseurl in the output below:

```
[kubernetes]
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.29/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.29/rpm/repodata/repomd.xml.key
exclude=kubelet kubeadm kubectl
```

**You're using the Kubernetes package repositories and this guide applies to you.**

Otherwise, it's strongly recommended to migrate to the Kubernetes package repositories as described in the [official announcement](#).

#### Note:

The URL used for the Kubernetes package repositories is not limited to `pkgs.k8s.io`, it can also be one of:

- `pkgs.k8s.io`
- `pkgs.kubernetes.io`
- `packages.kubernetes.io`

## Switching to another Kubernetes package repository

This step should be done upon upgrading from one to another Kubernetes minor release in order to get access to the packages of the desired Kubernetes minor version.

- [Ubuntu, Debian or HypriotOS](#)
- [CentOS, RHEL or Fedora](#)

1. Open the file that defines the Kubernetes apt repository using a text editor of your choice:

```
nano /etc/apt/sources.list.d/kubernetes.list
```

You should see a single line with the URL that contains your current Kubernetes minor version. For example, if you're using v1.29, you should see this:

```
deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.29/deb/ /
```

2. Change the version in the URL to **the next available minor release**, for example:

```
deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.30/deb/ /
```

3. Save the file and exit your text editor. Continue following the relevant upgrade instructions.

1. Open the file that defines the Kubernetes yum repository using a text editor of your choice:

```
nano /etc/yum.repos.d/kubernetes.repo
```

You should see a file with two URLs that contain your current Kubernetes minor version. For example, if you're using v1.29, you should see this:

```
[kubernetes]
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.29/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.29/rpm/repodata/repomd.xml.key
exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
```

2. Change the version in these URLs to **the next available minor release**, for example:

```
[kubernetes]
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key
exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
```

3. Save the file and exit your text editor. Continue following the relevant upgrade instructions.

## What's next

- See how to [Upgrade Linux nodes](#).
- See how to [Upgrade Windows nodes](#).

## Migrating from dockershim

This section presents information you need to know when migrating from dockershim to other container runtimes.

Since the announcement of [dockershim deprecation](#) in Kubernetes 1.20, there were questions on how this will affect various workloads and Kubernetes installations. Our [Dockershim Removal FAQ](#) is there to help you to understand the problem better.

Dockershim was removed from Kubernetes with the release of v1.24. If you use Docker Engine via dockershim as your container runtime and wish to upgrade to v1.24, it is recommended that you either migrate to another runtime or find an alternative means to obtain Docker Engine support. Check out the [container runtimes](#) section to know your options.

The version of Kubernetes with dockershim (1.23) is out of support and the v1.24 will run out of support [soon](#). Make sure to [report issues](#) you encountered with the migration so the issues can be fixed in a timely manner and your cluster would be ready for dockershim removal. After v1.24 running out of support, you will need to contact your Kubernetes provider for support or upgrade multiple versions at a time if there are critical issues affecting your cluster.

Your cluster might have more than one kind of node, although this is not a common configuration.

These tasks will help you to migrate:

- [Check whether Dockershim removal affects you](#)
- [Migrate Docker Engine nodes from dockershim to cri-dockerd](#)
- [Migrating telemetry and security agents from dockershim](#)

## What's next

- Check out [container runtimes](#) to understand your options for an alternative.
- If you find a defect or other technical concern relating to migrating away from dockershim, you can [report an issue](#) to the Kubernetes project.

# Changing the Container Runtime on a Node from Docker Engine to containerd

This task outlines the steps needed to update your container runtime to containerd from Docker. It is applicable for cluster operators running Kubernetes 1.23 or earlier. This also covers an example scenario for migrating from dockershim to containerd. Alternative container runtimes can be picked from this [page](#).

## Before you begin

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Install containerd. For more information see [containerd's installation documentation](#) and for specific prerequisite follow [the containerd guide](#).

## Drain the node

```
kubectcl drain <node-to-drain> --ignore-daemonsets
```

Replace <node-to-drain> with the name of your node you are draining.

## Stop the Docker daemon

```
systemctl stop kubelet
systemctl disable docker.service --now
```

## Install Containerd

Follow the [guide](#) for detailed steps to install containerd.

- [Linux](#)
- [Windows \(PowerShell\)](#)

1. Install the containerd.io package from the official Docker repositories. Instructions for setting up the Docker repository for your respective Linux distribution and installing the containerd.io package can be found at [Getting started with containerd](#).

2. Configure containerd:

```
sudo mkdir -p /etc/containerd
containerd config default | sudo tee /etc/containerd/config.toml
```

3. Restart containerd:

```
sudo systemctl restart containerd
```

Start a Powershell session, set \$Version to the desired version (ex: \$Version="1.4.3"), and then run the following commands:

1. Download containerd:

```
curl.exe -L https://github.com/containerd/containerd/releases/download/v$Version/
containerd-$Version-windows-amd64.tar.gz -o containerd-windows-amd64.tar.gz
tar.exe xvf .\containerd-windows-amd64.tar.gz
```

2. Extract and configure:

```
Copy-Item -Path ".\bin\" -Destination "$Env:ProgramFiles\containerd" -Recurse -Force
cd $Env:ProgramFiles\containerd\
.\containerd.exe config default | Out-File config.toml -Encoding ascii
```

```
Review the configuration. Depending on setup you may want to adjust:
- the sandbox_image (Kubernetes pause image)
- cni bin_dir and conf_dir locations
Get-Content config.toml
```

```
(Optional - but highly recommended) Exclude containerd from Windows Defender
```

```
Scans
Add-MpPreference -ExclusionProcess "$Env:ProgramFiles\containerd\containerd.exe"
```

3. Start containerd:

```
.\containerd.exe --register-service
Start-Service containerd
```

## Configure the kubelet to use containerd as its container runtime

Edit the file `/var/lib/kubelet/kubeadm-flags.env` and add the containerd runtime to the flags; `--container-runtime-endpoint=unix:///run/containerd/containerd.sock`.

Users using kubeadm should be aware that the kubeadm tool stores the CRI socket for each host as an annotation in the Node object for that host. To change it you can execute the following command on a machine that has the kubeadm `/etc/kubernetes/admin.conf` file.

```
kubectrl edit no <node-name>
```

This will start a text editor where you can edit the Node object. To choose a text editor you can set the `KUBE_EDITOR` environment variable.

- Change the value of `kubeadm.alpha.kubernetes.io/cri-socket` from `/var/run/dockershim.sock` to the CRI socket path of your choice (for example `unix:///run/containerd/containerd.sock`).

Note that new CRI socket paths must be prefixed with `unix://` ideally.

- Save the changes in the text editor, which will update the Node object.

## Restart the kubelet

```
systemctl start kubelet
```

## Verify that the node is healthy

Run `kubectrl get nodes -o wide` and containerd appears as the runtime for the node we just changed.

## Remove Docker Engine

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

If the node appears healthy, remove Docker.

- [CentOS](#)
- [Debian](#)

- [Fedora](#)
- [Ubuntu](#)

```
sudo yum remove docker-ce docker-ce-cli
```

```
sudo apt-get purge docker-ce docker-ce-cli
```

```
sudo dnf remove docker-ce docker-ce-cli
```

```
sudo apt-get purge docker-ce docker-ce-cli
```

The preceding commands don't remove images, containers, volumes, or customized configuration files on your host. To delete them, follow Docker's instructions to [Uninstall Docker Engine](#).

### Caution:

Docker's instructions for uninstalling Docker Engine create a risk of deleting containerd. Be careful when executing commands.

## Uncordon the node

```
kubectrl uncordon <node-to-uncordon>
```

Replace `<node-to-uncordon>` with the name of your node you previously drained.

# Migrate Docker Engine nodes from dockershim to cri-dockerd

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

This page shows you how to migrate your Docker Engine nodes to use cri-dockerd instead of dockershim. You should follow these steps in these scenarios:

- You want to switch away from dockershim and still use Docker Engine to run containers in Kubernetes.
- You want to upgrade to Kubernetes v1.30 and your existing cluster relies on dockershim, in which case you must migrate from dockershim and cri-dockerd is one of your options.

To learn more about the removal of dockershim, read the [FAQ page](#).

## What is cri-dockerd?

In Kubernetes 1.23 and earlier, you could use Docker Engine with Kubernetes, relying on a built-in component of Kubernetes named *dockershim*. The dockershim component was removed in the Kubernetes 1.24 release; however, a third-party replacement, cri-dockerd, is available. The cri-dockerd adapter lets you use Docker Engine through the [Container Runtime Interface](#).



## Note:

If you already use `cri-dockerd`, you aren't affected by the dockershim removal. Before you begin, [Check whether your nodes use the dockershim](#).

If you want to migrate to `cri-dockerd` so that you can continue using Docker Engine as your container runtime, you should do the following for each affected node:

1. Install `cri-dockerd`.
2. Cordon and drain the node.
3. Configure the kubelet to use `cri-dockerd`.
4. Restart the kubelet.
5. Verify that the node is healthy.

Test the migration on non-critical nodes first.

You should perform the following steps for each node that you want to migrate to `cri-dockerd`.

## Before you begin

- [cri-dockerd](#) installed and started on each node.
- A [network plugin](#).

## Cordon and drain the node

1. Cordon the node to stop new Pods scheduling on it:

```
kubectl cordon <NODE_NAME>
```

Replace `<NODE_NAME>` with the name of the node.

2. Drain the node to safely evict running Pods:

```
kubectl drain <NODE_NAME> \
--ignore-daemonsets
```

## Configure the kubelet to use cri-dockerd

The following steps apply to clusters set up using the `kubeadm` tool. If you use a different tool, you should modify the kubelet using the configuration instructions for that tool.

1. Open `/var/lib/kubelet/kubeadm-flags.env` on each affected node.
2. Modify the `--container-runtime-endpoint` flag to `unix:///var/run/cri-dockerd.sock`.
3. Modify the `--container-runtime` flag to `remote` (unavailable in Kubernetes v1.27 and later).

The `kubeadm` tool stores the node's socket as an annotation on the Node object in the control plane. To modify this socket for each affected node:

1. Edit the YAML representation of the Node object:

```
KUBECONFIG=/path/to/admin.conf kubectl edit no <NODE_NAME>
```

Replace the following:

- `/path/to/admin.conf`: the path to the kubectl configuration file, `admin.conf`.
  - `<NODE_NAME>`: the name of the node you want to modify.
2. Change `kubeadm.alpha.kubernetes.io/cri-socket` from `/var/run/dockershim.sock` to `unix:///var/run/cri-dockerd.sock`.
  3. Save the changes. The Node object is updated on save.

## Restart the kubelet

```
systemctl restart kubelet
```

## Verify that the node is healthy

To check whether the node uses the cri-dockerd endpoint, follow the instructions in [Find out which runtime you use](#). The `--container-runtime-endpoint` flag for the kubelet should be `unix:///var/run/cri-dockerd.sock`.

## Uncordon the node

Uncordon the node to let Pods schedule on it:

```
kubectl uncordon <NODE_NAME>
```

## What's next

- Read the [dockershim removal FAQ](#).
- [Learn how to migrate from Docker Engine with dockershim to containerd](#).

# Find Out What Container Runtime is Used on a Node

This page outlines steps to find out what [container runtime](#) the nodes in your cluster use.

Depending on the way you run your cluster, the container runtime for the nodes may have been pre-configured or you need to configure it. If you're using a managed Kubernetes service, there might be vendor-specific ways to check what container runtime is configured for the nodes. The method described on this page should work whenever the execution of kubectl is allowed.

## Before you begin

Install and configure kubectl. See [Install Tools](#) section for details.

## Find out the container runtime used on a Node

Use `kubectl` to fetch and show node information:

```
kubectl get nodes -o wide
```

The output is similar to the following. The column `CONTAINER-RUNTIME` outputs the runtime and its version.

For Docker Engine, the output is similar to this:

| NAME   | STATUS | VERSION  | CONTAINER-RUNTIME |
|--------|--------|----------|-------------------|
| node-1 | Ready  | v1.16.15 | docker://19.3.1   |
| node-2 | Ready  | v1.16.15 | docker://19.3.1   |
| node-3 | Ready  | v1.16.15 | docker://19.3.1   |

If your runtime shows as Docker Engine, you still might not be affected by the removal of dockershim in Kubernetes v1.24. [Check the runtime endpoint](#) to see if you use dockershim. If you don't use dockershim, you aren't affected.

For containerd, the output is similar to this:

| NAME   | STATUS | VERSION | CONTAINER-RUNTIME  |
|--------|--------|---------|--------------------|
| node-1 | Ready  | v1.19.6 | containerd://1.4.1 |
| node-2 | Ready  | v1.19.6 | containerd://1.4.1 |
| node-3 | Ready  | v1.19.6 | containerd://1.4.1 |

Find out more information about container runtimes on [Container Runtimes](#) page.

## Find out what container runtime endpoint you use

The container runtime talks to the kubelet over a Unix socket using the [CRI protocol](#), which is based on the gRPC framework. The kubelet acts as a client, and the runtime acts as the server. In some cases, you might find it useful to know which socket your nodes use. For example, with the removal of dockershim in Kubernetes v1.24 and later, you might want to know whether you use Docker Engine with dockershim.

### Note:

If you currently use Docker Engine in your nodes with `cri-dockerd`, you aren't affected by the dockershim removal.

You can check which socket you use by checking the kubelet configuration on your nodes.

1. Read the starting commands for the kubelet process:

```
tr '\0 ' < /proc/"$(pgrep kubelet)"/cmdline
```

If you don't have `tr` or `pgrep`, check the command line for the kubelet process manually.

In the output, look for the `--container-runtime` flag and the `--container-runtime-endpoint` 2. flag.

- If your nodes use Kubernetes v1.23 and earlier and these flags aren't present or if the `--container-runtime` flag is not remote, you use the dockershim socket with Docker Engine. The `--container-runtime` command line argument is not available in Kubernetes v1.27 and later.
- If the `--container-runtime-endpoint` flag is present, check the socket name to find out which runtime you use. For example, `unix:///run/containerd/containerd.sock` is the containerd endpoint.

If you want to change the Container Runtime on a Node from Docker Engine to containerd, you can find out more information on [migrating from Docker Engine to containerd](#), or, if you want to continue using Docker Engine in Kubernetes v1.24 and later, migrate to a CRI-compatible adapter like [cri-dockerd](#).

## Troubleshooting CNI plugin-related errors

To avoid CNI plugin-related errors, verify that you are using or upgrading to a container runtime that has been tested to work correctly with your version of Kubernetes.

### About the "Incompatible CNI versions" and "Failed to destroy network for sandbox" errors

Service issues exist for pod CNI network setup and tear down in containerd v1.6.0-v1.6.3 when the CNI plugins have not been upgraded and/or the CNI config version is not declared in the CNI config files. The containerd team reports, "these issues are resolved in containerd v1.6.4."

With containerd v1.6.0-v1.6.3, if you do not upgrade the CNI plugins and/or declare the CNI config version, you might encounter the following "Incompatible CNI versions" or "Failed to destroy network for sandbox" error conditions.

#### Incompatible CNI versions error

If the version of your CNI plugin does not correctly match the plugin version in the config because the config version is later than the plugin version, the containerd log will likely show an error message on startup of a pod similar to:

```
incompatible CNI versions; config is \"1.0.0\", plugin supports [\"0.1.0\" \"0.2.0\" \"0.3.0\" \"0.3.1\" \"0.4.0\"]
```

To fix this issue, [update your CNI plugins and CNI config files](#).

#### Failed to destroy network for sandbox error

If the version of the plugin is missing in the CNI plugin config, the pod may run. However, stopping the pod generates an error similar to:

```
ERROR[2022-04-26T00:43:24.518165483Z] StopPodSandbox for "b" failed
error="failed to destroy network for sandbox"
```

```
\\"bbc85f891eaf060c5a879e27bba9b6b06450210161dfdecfb2732959fb6500a\\": invalid version \\"\\":
the version is empty"
```

This error leaves the pod in the not-ready state with a network namespace still attached. To recover from this problem, [edit the CNI config file](#) to add the missing version information. The next attempt to stop the pod should be successful.

## Updating your CNI plugins and CNI config files

If you're using containerd v1.6.0-v1.6.3 and encountered "Incompatible CNI versions" or "Failed to destroy network for sandbox" errors, consider updating your CNI plugins and editing the CNI config files.

Here's an overview of the typical steps for each node:

1. [Safely drain and cordon the node](#).
2. After stopping your container runtime and kubelet services, perform the following upgrade operations:
  - If you're running CNI plugins, upgrade them to the latest version.
  - If you're using non-CNI plugins, replace them with CNI plugins. Use the latest version of the plugins.
  - Update the plugin configuration file to specify or match a version of the CNI specification that the plugin supports, as shown in the following ["An example containerd configuration file"](#) section.
  - For containerd, ensure that you have installed the latest version (v1.0.0 or later) of the CNI loopback plugin.
  - Upgrade node components (for example, the kubelet) to Kubernetes v1.24
  - Upgrade to or install the most current version of the container runtime.
1. Bring the node back into your cluster by restarting your container runtime and kubelet. Uncordon the node (`kubectl uncordon <nodename>`).

## An example containerd configuration file

The following example shows a configuration for containerd runtime v1.6.x, which supports a recent version of the CNI specification (v1.0.0).

Please see the documentation from your plugin and networking provider for further instructions on configuring your system.

On Kubernetes, containerd runtime adds a loopback interface, `lo`, to pods as a default behavior. The containerd runtime configures the loopback interface via a CNI plugin, `loopback`. The `loopback` plugin is distributed as part of the containerd release packages that have the `cni` designation. containerd v1.6.0 and later includes a CNI v1.0.0-compatible loopback plugin as well as other default CNI plugins. The configuration for the loopback plugin is done internally by containerd, and is set to use CNI v1.0.0. This also means that the version of the loopback plugin must be v1.0.0 or later when this newer version containerd is started.

The following bash command generates an example CNI config. Here, the 1.0.0 value for the config version is assigned to the `cniVersion` field for use when containerd invokes the CNI bridge plugin.

```
cat << EOF | tee /etc/cni/net.d/10-containerd-net.conflist
{
 "cniVersion": "1.0.0",
 "name": "containerd-net",
 "plugins": [
 {
 "type": "bridge",
 "bridge": "cni0",
 "isGateway": true,
 "ipMasq": true,
 "promiscMode": true,
 "ipam": {
 "type": "host-local",
 "ranges": [
 [
 {
 "subnet": "10.88.0.0/16"
 }
],
 [
 {
 "subnet": "2001:db8:4860::/64"
 }
]
],
 "routes": [
 { "dst": "0.0.0.0/0" },
 { "dst": "::/0" }
]
 }
 },
 {
 "type": "portmap",
 "capabilities": {"portMappings": true},
 "externalSetMarkChain": "KUBE-MARK-MASQ"
 }
]
}
EOF
```

Update the IP address ranges in the preceding example with ones that are based on your use case and network addressing plan.

## Check whether dockershim removal affects you

The dockershim component of Kubernetes allows the use of Docker as a Kubernetes's [container runtime](#). Kubernetes' built-in dockershim component was removed in release v1.24.

This page explains how your cluster could be using Docker as a container runtime, provides details on the role that dockershim plays when in use, and shows steps you can take to check whether any workloads could be affected by dockershim removal.

# Finding if your app has a dependencies on Docker

If you are using Docker for building your application containers, you can still run these containers on any container runtime. This use of Docker does not count as a dependency on Docker as a container runtime.

When alternative container runtime is used, executing Docker commands may either not work or yield unexpected output. This is how you can find whether you have a dependency on Docker:

1. Make sure no privileged Pods execute Docker commands (like `docker ps`), restart the Docker service (commands such as `systemctl restart docker.service`), or modify Docker-specific files such as `/etc/docker/daemon.json`.
2. Check for any private registries or image mirror settings in the Docker configuration file (like `/etc/docker/daemon.json`). Those typically need to be reconfigured for another container runtime.
3. Check that scripts and apps running on nodes outside of your Kubernetes infrastructure do not execute Docker commands. It might be:
  - SSH to nodes to troubleshoot;
  - Node startup scripts;
  - Monitoring and security agents installed on nodes directly.
4. Third-party tools that perform above mentioned privileged operations. See [Migrating telemetry and security agents from dockershim](#) for more information.
5. Make sure there are no indirect dependencies on dockershim behavior. This is an edge case and unlikely to affect your application. Some tooling may be configured to react to Docker-specific behaviors, for example, raise alert on specific metrics or search for a specific log message as part of troubleshooting instructions. If you have such tooling configured, test the behavior on a test cluster before migration.

## Dependency on Docker explained

A [container runtime](#) is software that can execute the containers that make up a Kubernetes pod. Kubernetes is responsible for orchestration and scheduling of Pods; on each node, the [kubelet](#) uses the container runtime interface as an abstraction so that you can use any compatible container runtime.

In its earliest releases, Kubernetes offered compatibility with one container runtime: Docker. Later in the Kubernetes project's history, cluster operators wanted to adopt additional container runtimes. The CRI was designed to allow this kind of flexibility - and the kubelet began supporting CRI. However, because Docker existed before the CRI specification was invented, the Kubernetes project created an adapter component, `dockershim`. The `dockershim` adapter allows the kubelet to interact with Docker as if Docker were a CRI compatible runtime.

You can read about it in [Kubernetes Containerd integration goes GA](#) blog post.

### Dockershim vs. CRI with Containerd

Switching to Containerd as a container runtime eliminates the middleman. All the same containers can be run by container runtimes like Containerd as before. But now, since containers schedule directly with the container runtime, they are not visible to Docker. So any Docker tooling or fancy UI you might have used before to check on these containers is no longer available.

You cannot get container information using `docker ps` or `docker inspect` commands. As you cannot list containers, you cannot get logs, stop containers, or execute something inside a container using `docker exec`.

#### Note:

If you're running workloads via Kubernetes, the best way to stop a container is through the Kubernetes API rather than directly through the container runtime (this advice applies for all container runtimes, not only Docker).

You can still pull images or build them using `docker build` command. But images built or pulled by Docker would not be visible to container runtime and Kubernetes. They needed to be pushed to some registry to allow them to be used by Kubernetes.

## Known issues

### Some filesystem metrics are missing and the metrics format is different

The Kubelet `/metrics/cadvisor` endpoint provides Prometheus metrics, as documented in [Metrics for Kubernetes system components](#). If you install a metrics collector that depends on that endpoint, you might see the following issues:

- The metrics format on the Docker node is `k8s_<container-name>_<pod-name>_<namespace>_<pod-uid>_<restart-count>` but the format on other runtime is different. For example, on containerd node it is `<container-id>`.
- Some filesystem metrics are missing, as follows:

```
container_fs_inodes_free
container_fs_inodes_total
container_fs_io_current
container_fs_io_time_seconds_total
container_fs_io_time_weighted_seconds_total
container_fs_limit_bytes
container_fs_read_seconds_total
container_fs_reads_merged_total
container_fs_sector_reads_total
container_fs_sector_writes_total
container_fs_usage_bytes
container_fs_write_seconds_total
container_fs_writes_merged_total
```

### Workaround

You can mitigate this issue by using [cAdvisor](#) as a standalone daemonset.

1. Find the latest [cAdvisor release](#) with the name pattern `vX.Y.Z-containerd-cri` (for example, `v0.42.0-containerd-cri`).
2. Follow the steps in [cAdvisor Kubernetes Daemonset](#) to create the daemonset.
3. Point the installed metrics collector to use the cAdvisor `/metrics` endpoint which provides the full set of [Prometheus container metrics](#).



Alternatives:

- Use alternative third party metrics collection solution.
- Collect metrics from the Kubelet summary API that is served at `/stats/summary`.

## What's next

- Read [Migrating from dockershim](#) to understand your next steps
- Read the [dockershim deprecation FAQ](#) article for more information.

# Migrating telemetry and security agents from dockershim

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Kubernetes' support for direct integration with Docker Engine is deprecated and has been removed. Most apps do not have a direct dependency on runtime hosting containers. However, there are still a lot of telemetry and monitoring agents that have a dependency on Docker to collect containers metadata, logs, and metrics. This document aggregates information on how to detect these dependencies as well as links on how to migrate these agents to use generic tools or alternative runtimes.

## Telemetry and security agents

Within a Kubernetes cluster there are a few different ways to run telemetry or security agents. Some agents have a direct dependency on Docker Engine when they run as DaemonSets or directly on nodes.

### Why do some telemetry agents communicate with Docker Engine?

Historically, Kubernetes was written to work specifically with Docker Engine. Kubernetes took care of networking and scheduling, relying on Docker Engine for launching and running containers (within Pods) on a node. Some information that is relevant to telemetry, such as a pod name, is only available from Kubernetes components. Other data, such as container metrics, is not the responsibility of the container runtime. Early telemetry agents needed to query the container runtime *and* Kubernetes to report an accurate picture. Over time, Kubernetes gained the ability to support multiple runtimes, and now supports any runtime that is compatible with the [container runtime interface](#).

Some telemetry agents rely specifically on Docker Engine tooling. For example, an agent might run a command such as [docker ps](#) or [docker top](#) to list containers and processes or [docker logs](#) to receive streamed logs. If nodes in your existing cluster use Docker Engine, and you switch to a different container runtime, these commands will not work any longer.

## Identify DaemonSets that depend on Docker Engine

If a pod wants to make calls to the dockerd running on the node, the pod must either:

- mount the filesystem containing the Docker daemon's privileged socket, as a [volume](#); or
- mount the specific path of the Docker daemon's privileged socket directly, also as a volume.

For example: on COS images, Docker exposes its Unix domain socket at `/var/run/docker.sock`. This means that the pod spec will include a `hostPath` volume mount of `/var/run/docker.sock`.

Here's a sample shell script to find Pods that have a mount directly mapping the Docker socket. This script outputs the namespace and name of the pod. You can remove the `grep '/var/run/docker.sock'` to review other mounts.

```
kubectl get pods --all-namespaces \
-o=jsonpath='{range .items[*]}{"\n"}{{.metadata.namespace}}{"\t"}{{.metadata.name}}{"\t"}\n{{range .spec.volumes[*]}}{{.hostPath.path}}", "{end}}{end}}' \
| sort \
| grep '/var/run/docker.sock'
```

### Note:

There are alternative ways for a pod to access Docker on the host. For instance, the parent directory `/var/run` may be mounted instead of the full path (like in [this example](#)). The script above only detects the most common uses.

## Detecting Docker dependency from node agents

If your cluster nodes are customized and install additional security and telemetry agents on the node, check with the agent vendor to verify whether it has any dependency on Docker.

## Telemetry and security agent vendors

This section is intended to aggregate information about various telemetry and security agents that may have a dependency on container runtimes.

We keep the work in progress version of migration instructions for various telemetry and security agent vendors in [Google doc](#). Please contact the vendor to get up to date instructions for migrating from dockershim.

## Migration from dockershim

### [Aqua](#)

No changes are needed: everything should work seamlessly on the runtime switch.

## Datadog

How to migrate: [Docker deprecation in Kubernetes](#) The pod that accesses Docker Engine may have a name containing any of:

- datadog-agent
- datadog
- dd-agent

## Dynatrace

How to migrate: [Migrating from Docker-only to generic container metrics in Dynatrace](#)

Containerd support announcement: [Get automated full-stack visibility into containerd-based Kubernetes environments](#)

CRI-O support announcement: [Get automated full-stack visibility into your CRI-O Kubernetes containers \(Beta\)](#)

The pod accessing Docker may have name containing:

- dynatrace-oneagent

## Falco

How to migrate:

[Migrate Falco from dockershim](#) Falco supports any CRI-compatible runtime (containerd is used in the default configuration); the documentation explains all details. The pod accessing Docker may have name containing:

- falco

## Prisma Cloud Compute

Check [documentation for Prisma Cloud](#), under the "Install Prisma Cloud on a CRI (non-Docker) cluster" section. The pod accessing Docker may be named like:

- twistlock-defender-ds

## SignalFx (Splunk)

The SignalFx Smart Agent (deprecated) uses several different monitors for Kubernetes including kubernetes-cluster, kubelet-stats/kubelet-metrics, and docker-container-stats. The kubelet-stats monitor was previously deprecated by the vendor, in favor of kubelet-metrics. The docker-container-stats monitor is the one affected by dockershim removal. Do not use the docker-container-stats with container runtimes other than Docker Engine.

How to migrate from dockershim-dependent agent:

1. Remove docker-container-stats from the list of [configured monitors](#). Note, keeping this monitor enabled with non-dockershim runtime will result in incorrect metrics being reported when docker is installed on node and no metrics when docker is not installed.

2. [Enable and configure kubelet-metrics](#) monitor.

**Note:**

The set of collected metrics will change. Review your alerting rules and dashboards.

The Pod accessing Docker may be named something like:

- `signalfx-agent`

## Yahoo Kubectl Flame

Flame does not support container runtimes other than Docker. See <https://github.com/yahoo/kubectl-flame/issues/51>

# Generate Certificates Manually

When using client certificate authentication, you can generate certificates manually through [easysrsa](#), [openssl](#) or [cfssl](#).

## easysrsa

**easysrsa** can manually generate certificates for your cluster.

1. Download, unpack, and initialize the patched version of easysrsa3.

```
curl -LO https://dl.k8s.io/easy-rsa/easy-rsa.tar.gz
tar xzf easy-rsa.tar.gz
cd easy-rsa-master/easysrsa3
./easysrsa init-pki
```

2. Generate a new certificate authority (CA). `--batch` sets automatic mode; `--req-cn` specifies the Common Name (CN) for the CA's new root certificate.

```
./easysrsa --batch "--req-cn=${MASTER_IP}@`date +%s`" build-ca nopass
```

3. Generate server certificate and key.

The argument `--subject-alt-name` sets the possible IPs and DNS names the API server will be accessed with. The `MASTER_CLUSTER_IP` is usually the first IP from the service CIDR that is specified as the `--service-cluster-ip-range` argument for both the API server and the controller manager component. The argument `--days` is used to set the number of days after which the certificate expires. The sample below also assumes that you are using `cluster.local` as the default DNS domain name.

```
./easysrsa --subject-alt-name="IP:${MASTER_IP},\
"IP:${MASTER_CLUSTER_IP},\
"DNS:kubernetes,\
"DNS:kubernetes.default,\
"DNS:kubernetes.default.svc,\
"DNS:kubernetes.default.svc.cluster,\
"DNS:kubernetes.default.svc.cluster.local" \
```

```
--days=10000 \
build-server-full server nopass
```

4. Copy pki/ca.crt, pki/issued/server.crt, and pki/private/server.key to your directory.
5. Fill in and add the following parameters into the API server start parameters:

```
--client-ca-file=/yourdirectory/ca.crt
--tls-cert-file=/yourdirectory/server.crt
--tls-private-key-file=/yourdirectory/server.key
```

## openssl

**openssl** can manually generate certificates for your cluster.

1. Generate a ca.key with 2048bit:

```
openssl genrsa -out ca.key 2048
```

2. According to the ca.key generate a ca.crt (use -days to set the certificate effective time):

```
openssl req -x509 -new -nodes -key ca.key -subj "/CN=${MASTER_IP}" -days 10000 -out
ca.crt
```

3. Generate a server.key with 2048bit:

```
openssl genrsa -out server.key 2048
```

4. Create a config file for generating a Certificate Signing Request (CSR).

Be sure to substitute the values marked with angle brackets (e.g. <MASTER\_IP>) with real values before saving this to a file (e.g. csr.conf). Note that the value for MASTER\_CLUSTER\_IP is the service cluster IP for the API server as described in previous subsection. The sample below also assumes that you are using cluster.local as the default DNS domain name.

```
[req]
default_bits = 2048
prompt = no
default_md = sha256
req_extensions = req_ext
distinguished_name = dn

[dn]
C = <country>
ST = <state>
L = <city>
O = <organization>
OU = <organization unit>
CN = <MASTER_IP>

[req_ext]
subjectAltName = @alt_names
```

```
[alt_names]
DNS.1 = kubernetes
DNS.2 = kubernetes.default
DNS.3 = kubernetes.default.svc
DNS.4 = kubernetes.default.svc.cluster
DNS.5 = kubernetes.default.svc.cluster.local
IP.1 = <MASTER_IP>
IP.2 = <MASTER_CLUSTER_IP>

[v3_ext]
authorityKeyIdentifier=keyid,issuer:always
basicConstraints=CA:FALSE
keyUsage=keyEncipherment,dataEncipherment
extendedKeyUsage=serverAuth,clientAuth
subjectAltName=@alt_names
```

5. Generate the certificate signing request based on the config file:

```
openssl req -new -key server.key -out server.csr -config csr.conf
```

6. Generate the server certificate using the ca.key, ca.crt and server.csr:

```
openssl x509 -req -in server.csr -CA ca.crt -CAkey ca.key \
-CACreateserial -out server.crt -days 10000 \
-extensions v3_ext -extfile csr.conf -sha256
```

7. View the certificate signing request:

```
openssl req -noout -text -in ./server.csr
```

8. View the certificate:

```
openssl x509 -noout -text -in ./server.crt
```

Finally, add the same parameters into the API server start parameters.

## cfssl

**cfssl** is another tool for certificate generation.

1. Download, unpack and prepare the command line tools as shown below.

Note that you may need to adapt the sample commands based on the hardware architecture and cfssl version you are using.

```
curl -L https://github.com/cloudflare/cfssl/releases/download/v1.5.0/
cfssl_1.5.0_linux_amd64 -o cfssl
chmod +x cfssl
curl -L https://github.com/cloudflare/cfssl/releases/download/v1.5.0/
cfssljson_1.5.0_linux_amd64 -o cfssljson
chmod +x cfssljson
curl -L https://github.com/cloudflare/cfssl/releases/download/v1.5.0/cfssl-
certinfo_1.5.0_linux_amd64 -o cfssl-certinfo
chmod +x cfssl-certinfo
```

Create a directory to hold the artifacts and initialize cfssl:

2.

```
mkdir cert
cd cert
../cfssl print-defaults config > config.json
../cfssl print-defaults csr > csr.json
```

3. Create a JSON config file for generating the CA file, for example, `ca-config.json`:

```
{
 "signing": {
 "default": {
 "expiry": "8760h"
 },
 "profiles": {
 "kubernetes": {
 "usages": [
 "signing",
 "key encipherment",
 "server auth",
 "client auth"
],
 "expiry": "8760h"
 }
 }
 }
}
```

4. Create a JSON config file for CA certificate signing request (CSR), for example, `ca-csr.json`. Be sure to replace the values marked with angle brackets with real values you want to use.

```
{
 "CN": "kubernetes",
 "key": {
 "algo": "rsa",
 "size": 2048
 },
 "names": [{
 "C": "<country>",
 "ST": "<state>",
 "L": "<city>",
 "O": "<organization>",
 "OU": "<organization unit>"
 }]
}
```

5. Generate CA key (`ca-key.pem`) and certificate (`ca.pem`):

```
../cfssl gencert -initca ca-csr.json | ../cfssljason -bare ca
```

6. Create a JSON config file for generating keys and certificates for the API server, for example, `server-csr.json`. Be sure to replace the values in angle brackets with real values you want to use. The `<MASTER_CLUSTER_IP>` is the service cluster IP for the API server

as described in previous subsection. The sample below also assumes that you are using cluster.local as the default DNS domain name.

```
{
 "CN": "kubernetes",
 "hosts": [
 "127.0.0.1",
 "<MASTER_IP>",
 "<MASTER_CLUSTER_IP>",
 "kubernetes",
 "kubernetes.default",
 "kubernetes.default.svc",
 "kubernetes.default.svc.cluster",
 "kubernetes.default.svc.cluster.local"
],
 "key": {
 "algo": "rsa",
 "size": 2048
 },
 "names": [{
 "C": "<country>",
 "ST": "<state>",
 "L": "<city>",
 "O": "<organization>",
 "OU": "<organization unit>"
 }]
}
```

7. Generate the key and certificate for the API server, which are by default saved into file server-key.pem and server.pem respectively:

```
../cfssl gencert -ca=ca.pem -ca-key=ca-key.pem \
 --config=ca-config.json -profile=kubernetes \
 server-csr.json | ../cfssljson -bare server
```

## Distributing Self-Signed CA Certificate

A client node may refuse to recognize a self-signed CA certificate as valid. For a non-production deployment, or for a deployment that runs behind a company firewall, you can distribute a self-signed CA certificate to all clients and refresh the local list for valid certificates.

On each client, perform the following operations:

```
sudo cp ca.crt /usr/local/share/ca-certificates/kubernetes.crt
sudo update-ca-certificates
```

```
Updating certificates in /etc/ssl/certs...
1 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d....
done.
```



## Certificates API

You can use the `certificates.k8s.io` API to provision x509 certificates to use for authentication as documented in the [Managing TLS in a cluster](#) task page.

# Manage Memory, CPU, and API Resources

---

### [Configure Default Memory Requests and Limits for a Namespace](#)

Define a default memory resource limit for a namespace, so that every new Pod in that namespace has a memory resource limit configured.

### [Configure Default CPU Requests and Limits for a Namespace](#)

Define a default CPU resource limits for a namespace, so that every new Pod in that namespace has a CPU resource limit configured.

### [Configure Minimum and Maximum Memory Constraints for a Namespace](#)

Define a range of valid memory resource limits for a namespace, so that every new Pod in that namespace falls within the range you configure.

### [Configure Minimum and Maximum CPU Constraints for a Namespace](#)

Define a range of valid CPU resource limits for a namespace, so that every new Pod in that namespace falls within the range you configure.

### [Configure Memory and CPU Quotas for a Namespace](#)

Define overall memory and CPU resource limits for a namespace.

### [Configure a Pod Quota for a Namespace](#)

Restrict how many Pods you can create within a namespace.

## Configure Default Memory Requests and Limits for a Namespace

Define a default memory resource limit for a namespace, so that every new Pod in that namespace has a memory resource limit configured.

This page shows how to configure default memory requests and limits for a [namespace](#).

A Kubernetes cluster can be divided into namespaces. Once you have a namespace that has a default memory [limit](#), and you then try to create a Pod with a container that does not specify its own memory limit, then the [control plane](#) assigns the default memory limit to that container.

Kubernetes assigns a default memory request under certain conditions that are explained later in this topic.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

You must have access to create namespaces in your cluster.

Each node in your cluster must have at least 2 GiB of memory.

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace default-mem-example
```

## Create a LimitRange and a Pod

Here's a manifest for an example [LimitRange](#). The manifest specifies a default memory request and a default memory limit.

[admin/resource/memory-defaults.yaml](#)

```
apiVersion: v1
kind: LimitRange
metadata:
 name: mem-limit-range
spec:
 limits:
 - default:
 memory: 512Mi
 defaultRequest:
 memory: 256Mi
 type: Container
```

Create the `LimitRange` in the `default-mem-example` namespace:

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-defaults.yaml --namespace=default-mem-example
```

Now if you create a Pod in the `default-mem-example` namespace, and any container within that Pod does not specify its own values for memory request and memory limit, then the [control plane](#) applies default values: a memory request of 256MiB and a memory limit of 512MiB.

Here's an example manifest for a Pod that has one container. The container does not specify a memory request and limit.

[admin/resource/memory-defaults-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: default-mem-demo
spec:
 containers:
 - name: default-mem-demo-ctr
 image: nginx
```

Create the Pod.

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-defaults-pod.yaml --
namespace=default-mem-example
```

View detailed information about the Pod:

```
kubectl get pod default-mem-demo --output=yaml --namespace=default-mem-example
```

The output shows that the Pod's container has a memory request of 256 MiB and a memory limit of 512 MiB. These are the default values specified by the LimitRange.

```
containers:
- image: nginx
 imagePullPolicy: Always
 name: default-mem-demo-ctr
 resources:
 limits:
 memory: 512Mi
 requests:
 memory: 256Mi
```

Delete your Pod:

```
kubectl delete pod default-mem-demo --namespace=default-mem-example
```

## What if you specify a container's limit, but not its request?

Here's a manifest for a Pod that has one container. The container specifies a memory limit, but not a request:

[admin/resource/memory-defaults-pod-2.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: default-mem-demo-2
spec:
```

```
containers:
- name: default-mem-demo-2-ctr
 image: nginx
 resources:
 limits:
 memory: "1Gi"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-defaults-pod-2.yaml --
namespace=default-mem-example
```

View detailed information about the Pod:

```
kubectl get pod default-mem-demo-2 --output=yaml --namespace=default-mem-example
```

The output shows that the container's memory request is set to match its memory limit. Notice that the container was not assigned the default memory request value of 256Mi.

```
resources:
 limits:
 memory: 1Gi
 requests:
 memory: 1Gi
```

## What if you specify a container's request, but not its limit?

Here's a manifest for a Pod that has one container. The container specifies a memory request, but not a limit:

[admin/resource/memory-defaults-pod-3.yaml](https://k8s.io/examples/admin/resource/memory-defaults-pod-3.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: default-mem-demo-3
spec:
 containers:
 - name: default-mem-demo-3-ctr
 image: nginx
 resources:
 requests:
 memory: "128Mi"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-defaults-pod-3.yaml --
namespace=default-mem-example
```

View the Pod's specification:

```
kubectl get pod default-mem-demo-3 --output=yaml --namespace=default-mem-example
```

The output shows that the container's memory request is set to the value specified in the container's manifest. The container is limited to use no more than 512MiB of memory, which matches the default memory limit for the namespace.

```
resources:
 limits:
 memory: 512Mi
 requests:
 memory: 128Mi
```

#### Note:

A LimitRange does **not** check the consistency of the default values it applies. This means that a default value for the *limit* that is set by LimitRange may be less than the *request* value specified for the container in the spec that a client submits to the API server. If that happens, the final Pod will not be scheduleable. See [Constraints on resource limits and requests](#) for more details.

## Motivation for default memory limits and requests

If your namespace has a memory [resource quota](#) configured, it is helpful to have a default value in place for memory limit. Here are three of the restrictions that a resource quota imposes on a namespace:

- For every Pod that runs in the namespace, the Pod and each of its containers must have a memory limit. (If you specify a memory limit for every container in a Pod, Kubernetes can infer the Pod-level memory limit by adding up the limits for its containers).
- Memory limits apply a resource reservation on the node where the Pod in question is scheduled. The total amount of memory reserved for all Pods in the namespace must not exceed a specified limit.
- The total amount of memory actually used by all Pods in the namespace must also not exceed a specified limit.

When you add a LimitRange:

If any Pod in that namespace that includes a container does not specify its own memory limit, the control plane applies the default memory limit to that container, and the Pod can be allowed to run in a namespace that is restricted by a memory ResourceQuota.

## Clean up

Delete your namespace:

```
kubectl delete namespace default-mem-example
```

## What's next

### For cluster administrators

- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)

[Configure Minimum and Maximum CPU Constraints for a Namespace](#)

- [Configure Memory and CPU Quotas for a Namespace](#)
- [Configure a Pod Quota for a Namespace](#)
- [Configure Quotas for API Objects](#)

## For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

# Configure Default CPU Requests and Limits for a Namespace

Define a default CPU resource limits for a namespace, so that every new Pod in that namespace has a CPU resource limit configured.

This page shows how to configure default CPU requests and limits for a [namespace](#).

A Kubernetes cluster can be divided into namespaces. If you create a Pod within a namespace that has a default CPU [limit](#), and any container in that Pod does not specify its own CPU limit, then the [control plane](#) assigns the default CPU limit to that container.

Kubernetes assigns a default CPU [request](#), but only under certain conditions that are explained later in this page.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You must have access to create namespaces in your cluster.

If you're not already familiar with what Kubernetes means by 1.0 CPU, read [meaning of CPU](#).

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace default-cpu-example
```

## Create a LimitRange and a Pod

Here's a manifest for an example [LimitRange](#). The manifest specifies a default CPU request and a default CPU limit.

[admin/resource/cpu-defaults.yaml](#)

```
apiVersion: v1
kind: LimitRange
metadata:
 name: cpu-limit-range
spec:
 limits:
 - default:
 cpu: 1
 defaultRequest:
 cpu: 0.5
 type: Container
```

Create the LimitRange in the default-cpu-example namespace:

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-defaults.yaml --namespace=default-cpu-example
```

Now if you create a Pod in the default-cpu-example namespace, and any container in that Pod does not specify its own values for CPU request and CPU limit, then the control plane applies default values: a CPU request of 0.5 and a default CPU limit of 1.

Here's a manifest for a Pod that has one container. The container does not specify a CPU request and limit.

[admin/resource/cpu-defaults-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: default-cpu-demo
spec:
 containers:
 - name: default-cpu-demo-ctr
 image: nginx
```

Create the Pod.

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-defaults-pod.yaml --namespace=default-cpu-example
```

View the Pod's specification:

```
kubectl get pod default-cpu-demo --output=yaml --namespace=default-cpu-example
```

The output shows that the Pod's only container has a CPU request of 500m cpu (which you can read as "500 millicpu"), and a CPU limit of 1 cpu. These are the default values specified by the LimitRange.

```
containers:
- image: nginx
 imagePullPolicy: Always
 name: default-cpu-demo-ctr
 resources:
 limits:
 cpu: "1"
 requests:
 cpu: 500m
```

## What if you specify a container's limit, but not its request?

Here's a manifest for a Pod that has one container. The container specifies a CPU limit, but not a request:

[admin/resource/cpu-defaults-pod-2.yaml](https://k8s.io/examples/admin/resource/cpu-defaults-pod-2.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: default-cpu-demo-2
spec:
 containers:
 - name: default-cpu-demo-2-ctr
 image: nginx
 resources:
 limits:
 cpu: "1"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-defaults-pod-2.yaml --
namespace=default-cpu-example
```

View the [specification](#) of the Pod that you created:

```
kubectl get pod default-cpu-demo-2 --output=yaml --namespace=default-cpu-example
```

The output shows that the container's CPU request is set to match its CPU limit. Notice that the container was not assigned the default CPU request value of 0.5 cpu:

```
resources:
 limits:
 cpu: "1"
 requests:
 cpu: "1"
```



# What if you specify a container's request, but not its limit?

Here's an example manifest for a Pod that has one container. The container specifies a CPU request, but not a limit:

[admin/resource/cpu-defaults-pod-3.yaml](https://k8s.io/examples/admin/resource/cpu-defaults-pod-3.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: default-cpu-demo-3
spec:
 containers:
 - name: default-cpu-demo-3-ctr
 image: nginx
 resources:
 requests:
 cpu: "0.75"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-defaults-pod-3.yaml --
namespace=default-cpu-example
```

View the specification of the Pod that you created:

```
kubectl get pod default-cpu-demo-3 --output=yaml --namespace=default-cpu-example
```

The output shows that the container's CPU request is set to the value you specified at the time you created the Pod (in other words: it matches the manifest). However, the same container's CPU limit is set to 1 cpu, which is the default CPU limit for that namespace.

```
resources:
 limits:
 cpu: "1"
 requests:
 cpu: 750m
```

## Motivation for default CPU limits and requests

If your namespace has a CPU [resource quota](#) configured, it is helpful to have a default value in place for CPU limit. Here are two of the restrictions that a CPU resource quota imposes on a namespace:

- For every Pod that runs in the namespace, each of its containers must have a CPU limit.
- CPU limits apply a resource reservation on the node where the Pod in question is scheduled. The total amount of CPU that is reserved for use by all Pods in the namespace must not exceed a specified limit.

When you add a LimitRange:

If any Pod in that namespace that includes a container does not specify its own CPU limit, the control plane applies the default CPU limit to that container, and the Pod can be allowed to run in a namespace that is restricted by a CPU ResourceQuota.

## Clean up

Delete your namespace:

```
kubectl delete namespace default-cpu-example
```

## What's next

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)
- [Configure a Pod Quota for a Namespace](#)
- [Configure Quotas for API Objects](#)

### For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

# Configure Minimum and Maximum Memory Constraints for a Namespace

Define a range of valid memory resource limits for a namespace, so that every new Pod in that namespace falls within the range you configure.

This page shows how to set minimum and maximum values for memory used by containers running in a [namespace](#). You specify minimum and maximum memory values in a [LimitRange](#) object. If a Pod does not meet the constraints imposed by the LimitRange, it cannot be created in the namespace.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You must have access to create namespaces in your cluster.

Each node in your cluster must have at least 1 GiB of memory available for Pods.

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace constraints-mem-example
```

## Create a LimitRange and a Pod

Here's an example manifest for a LimitRange:

[admin/resource/memory-constraints.yaml](#)

```
apiVersion: v1
kind: LimitRange
metadata:
 name: mem-min-max-demo-lr
spec:
 limits:
 - max:
 memory: 1Gi
 min:
 memory: 500Mi
 type: Container
```

Create the LimitRange:

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-constraints.yaml --
namespace=constraints-mem-example
```

View detailed information about the LimitRange:

```
kubectl get limitrange mem-min-max-demo-lr --namespace=constraints-mem-example --
output=yaml
```

The output shows the minimum and maximum memory constraints as expected. But notice that even though you didn't specify default values in the configuration file for the LimitRange, they were created automatically.

```
limits:
- default:
 memory: 1Gi
 defaultRequest:
```

```
memory: 1Gi
max:
 memory: 1Gi
min:
 memory: 500Mi
type: Container
```

Now whenever you define a Pod within the constraints-mem-example namespace, Kubernetes performs these steps:

- If any container in that Pod does not specify its own memory request and limit, the control plane assigns the default memory request and limit to that container.
- Verify that every container in that Pod requests at least 500 MiB of memory.
- Verify that every container in that Pod requests no more than 1024 MiB (1 GiB) of memory.

Here's a manifest for a Pod that has one container. Within the Pod spec, the sole container specifies a memory request of 600 MiB and a memory limit of 800 MiB. These satisfy the minimum and maximum memory constraints imposed by the LimitRange.

[admin/resource/memory-constraints-pod.yaml](https://k8s.io/examples/admin/resource/memory-constraints-pod.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: constraints-mem-demo
spec:
 containers:
 - name: constraints-mem-demo-ctr
 image: nginx
 resources:
 limits:
 memory: "800Mi"
 requests:
 memory: "600Mi"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-constraints-pod.yaml --
namespace=constraints-mem-example
```

Verify that the Pod is running and that its container is healthy:

```
kubectl get pod constraints-mem-demo --namespace=constraints-mem-example
```

View detailed information about the Pod:

```
kubectl get pod constraints-mem-demo --output=yaml --namespace=constraints-mem-example
```

The output shows that the container within that Pod has a memory request of 600 MiB and a memory limit of 800 MiB. These satisfy the constraints imposed by the LimitRange for this namespace:

```
resources:
 limits:
 memory: 800Mi
 requests:
 memory: 600Mi
```

Delete your Pod:

```
kubectl delete pod constraints-mem-demo --namespace=constraints-mem-example
```

## Attempt to create a Pod that exceeds the maximum memory constraint

Here's a manifest for a Pod that has one container. The container specifies a memory request of 800 MiB and a memory limit of 1.5 GiB.

[admin/resource/memory-constraints-pod-2.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: constraints-mem-demo-2
spec:
 containers:
 - name: constraints-mem-demo-2-ctr
 image: nginx
 resources:
 limits:
 memory: "1.5Gi"
 requests:
 memory: "800Mi"
```

Attempt to create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-constraints-pod-2.yaml --namespace=constraints-mem-example
```

The output shows that the Pod does not get created, because it defines a container that requests more memory than is allowed:

```
Error from server (Forbidden): error when creating "examples/admin/resource/memory-constraints-pod-2.yaml":
pods "constraints-mem-demo-2" is forbidden: maximum memory usage per Container is 1Gi, but limit is 1536Mi.
```

## Attempt to create a Pod that does not meet the minimum memory request

Here's a manifest for a Pod that has one container. That container specifies a memory request of 100 MiB and a memory limit of 800 MiB.

[admin/resource/memory-constraints-pod-3.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: constraints-mem-demo-3
spec:
 containers:
 - name: constraints-mem-demo-3-ctr
 image: nginx
 resources:
 limits:
 memory: "800Mi"
 requests:
 memory: "100Mi"
```

Attempt to create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-constraints-pod-3.yaml --
namespace=constraints-mem-example
```

The output shows that the Pod does not get created, because it defines a container that requests less memory than the enforced minimum:

Error from server (Forbidden): error when creating "examples/admin/resource/memory-constraints-pod-3.yaml":  
pods "constraints-mem-demo-3" is forbidden: minimum memory usage per Container is 500Mi, but request is 100Mi.

## Create a Pod that does not specify any memory request or limit

Here's a manifest for a Pod that has one container. The container does not specify a memory request, and it does not specify a memory limit.

[admin/resource/memory-constraints-pod-4.yaml](https://k8s.io/examples/admin/resource/memory-constraints-pod-4.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: constraints-mem-demo-4
spec:
 containers:
 - name: constraints-mem-demo-4-ctr
 image: nginx
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/memory-constraints-pod-4.yaml --
namespace=constraints-mem-example
```

View detailed information about the Pod:

```
kubectl get pod constraints-mem-demo-4 --namespace=constraints-mem-example --output=yaml
```

The output shows that the Pod's only container has a memory request of 1 GiB and a memory limit of 1 GiB. How did that container get those values?

```
resources:
 limits:
 memory: 1Gi
 requests:
 memory: 1Gi
```

Because your Pod did not define any memory request and limit for that container, the cluster applied a [default memory request and limit](#) from the LimitRange.

This means that the definition of that Pod shows those values. You can check it using `kubectl describe`:

```
Look for the "Requests:" section of the output
kubectl describe pod constraints-mem-demo-4 --namespace=constraints-mem-example
```

At this point, your Pod might be running or it might not be running. Recall that a prerequisite for this task is that your Nodes have at least 1 GiB of memory. If each of your Nodes has only 1 GiB of memory, then there is not enough allocatable memory on any Node to accommodate a memory request of 1 GiB. If you happen to be using Nodes with 2 GiB of memory, then you probably have enough space to accommodate the 1 GiB request.

Delete your Pod:

```
kubectl delete pod constraints-mem-demo-4 --namespace=constraints-mem-example
```

## Enforcement of minimum and maximum memory constraints

The maximum and minimum memory constraints imposed on a namespace by a LimitRange are enforced only when a Pod is created or updated. If you change the LimitRange, it does not affect Pods that were created previously.

## Motivation for minimum and maximum memory constraints

As a cluster administrator, you might want to impose restrictions on the amount of memory that Pods can use. For example:

- Each Node in a cluster has 2 GiB of memory. You do not want to accept any Pod that requests more than 2 GiB of memory, because no Node in the cluster can support the request.
- A cluster is shared by your production and development departments. You want to allow production workloads to consume up to 8 GiB of memory, but you want development workloads to be limited to 512 MiB. You create separate namespaces for production and development, and you apply memory constraints to each namespace.

## Clean up

Delete your namespace:

```
kubectrl delete namespace constraints-mem-example
```

## What's next

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)
- [Configure a Pod Quota for a Namespace](#)
- [Configure Quotas for API Objects](#)

### For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

# Configure Minimum and Maximum CPU Constraints for a Namespace

Define a range of valid CPU resource limits for a namespace, so that every new Pod in that namespace falls within the range you configure.

This page shows how to set minimum and maximum values for the CPU resources used by containers and Pods in a [namespace](#). You specify minimum and maximum CPU values in a [LimitRange](#) object. If a Pod does not meet the constraints imposed by the LimitRange, it cannot be created in the namespace.

## Before you begin

You need to have a Kubernetes cluster, and the kubectrl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)



- [Play with Kubernetes](#)

You must have access to create namespaces in your cluster.

Each node in your cluster must have at least 1.0 CPU available for Pods. See [meaning of CPU](#) to learn what Kubernetes means by “1 CPU”.

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace constraints-cpu-example
```

## Create a LimitRange and a Pod

Here's a manifest for an example [LimitRange](#):

[admin/resource/cpu-constraints.yaml](#)

```
apiVersion: v1
kind: LimitRange
metadata:
 name: cpu-min-max-demo-lr
spec:
 limits:
 - max:
 cpu: "800m"
 min:
 cpu: "200m"
 type: Container
```

Create the LimitRange:

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-constraints.yaml --namespace=constraints-cpu-example
```

View detailed information about the LimitRange:

```
kubectl get limitrange cpu-min-max-demo-lr --output=yaml --namespace=constraints-cpu-example
```

The output shows the minimum and maximum CPU constraints as expected. But notice that even though you didn't specify default values in the configuration file for the LimitRange, they were created automatically.

```
limits:
- default:
 cpu: 800m
 defaultRequest:
 cpu: 800m
 max:
 cpu: 800m
```

```
min:
 cpu: 200m
type: Container
```

Now whenever you create a Pod in the constraints-cpu-example namespace (or some other client of the Kubernetes API creates an equivalent Pod), Kubernetes performs these steps:

- If any container in that Pod does not specify its own CPU request and limit, the control plane assigns the default CPU request and limit to that container.
- Verify that every container in that Pod specifies a CPU request that is greater than or equal to 200 millicpu.
- Verify that every container in that Pod specifies a CPU limit that is less than or equal to 800 millicpu.

**Note:**

When creating a LimitRange object, you can specify limits on huge-pages or GPUs as well. However, when both default and defaultRequest are specified on these resources, the two values must be the same.

Here's a manifest for a Pod that has one container. The container manifest specifies a CPU request of 500 millicpu and a CPU limit of 800 millicpu. These satisfy the minimum and maximum CPU constraints imposed by the LimitRange for this namespace.

[admin/resource/cpu-constraints-pod.yaml](https://k8s.io/examples/admin/resource/cpu-constraints-pod.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: constraints-cpu-demo
spec:
 containers:
 - name: constraints-cpu-demo-ctr
 image: nginx
 resources:
 limits:
 cpu: "800m"
 requests:
 cpu: "500m"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-constraints-pod.yaml --
namespace=constraints-cpu-example
```

Verify that the Pod is running and that its container is healthy:

```
kubectl get pod constraints-cpu-demo --namespace=constraints-cpu-example
```

View detailed information about the Pod:

```
kubectl get pod constraints-cpu-demo --output=yaml --namespace=constraints-cpu-example
```

The output shows that the Pod's only container has a CPU request of 500 millicpu and CPU limit of 800 millicpu. These satisfy the constraints imposed by the LimitRange.

```
resources:
 limits:
 cpu: 800m
 requests:
 cpu: 500m
```

## Delete the Pod

```
kubectl delete pod constraints-cpu-demo --namespace=constraints-cpu-example
```

## Attempt to create a Pod that exceeds the maximum CPU constraint

Here's a manifest for a Pod that has one container. The container specifies a CPU request of 500 millicpu and a cpu limit of 1.5 cpu.

[admin/resource/cpu-constraints-pod-2.yaml](https://k8s.io/examples/admin/resource/cpu-constraints-pod-2.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: constraints-cpu-demo-2
spec:
 containers:
 - name: constraints-cpu-demo-2-ctr
 image: nginx
 resources:
 limits:
 cpu: "1.5"
 requests:
 cpu: "500m"
```

Attempt to create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-constraints-pod-2.yaml --namespace=constraints-cpu-example
```

The output shows that the Pod does not get created, because it defines an unacceptable container. That container is not acceptable because it specifies a CPU limit that is too large:

```
Error from server (Forbidden): error when creating "examples/admin/resource/cpu-constraints-pod-2.yaml":
pods "constraints-cpu-demo-2" is forbidden: maximum cpu usage per Container is 800m, but
limit is 1500m.
```

## Attempt to create a Pod that does not meet the minimum CPU request

Here's a manifest for a Pod that has one container. The container specifies a CPU request of 100 millicpu and a CPU limit of 800 millicpu.

[admin/resource/cpu-constraints-pod-3.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: constraints-cpu-demo-3
spec:
 containers:
 - name: constraints-cpu-demo-3-ctr
 image: nginx
 resources:
 limits:
 cpu: "800m"
 requests:
 cpu: "100m"
```

Attempt to create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-constraints-pod-3.yaml --
namespace=constraints-cpu-example
```

The output shows that the Pod does not get created, because it defines an unacceptable container. That container is not acceptable because it specifies a CPU request that is lower than the enforced minimum:

```
Error from server (Forbidden): error when creating "examples/admin/resource/cpu-constraints-
pod-3.yaml":
pods "constraints-cpu-demo-3" is forbidden: minimum cpu usage per Container is 200m, but
request is 100m.
```

## Create a Pod that does not specify any CPU request or limit

Here's a manifest for a Pod that has one container. The container does not specify a CPU request, nor does it specify a CPU limit.

[admin/resource/cpu-constraints-pod-4.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: constraints-cpu-demo-4
spec:
 containers:
 - name: constraints-cpu-demo-4-ctr
 image: vish/stress
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/cpu-constraints-pod-4.yaml --namespace=constraints-cpu-example
```

View detailed information about the Pod:

```
kubectl get pod constraints-cpu-demo-4 --namespace=constraints-cpu-example --output=yaml
```

The output shows that the Pod's single container has a CPU request of 800 millicpu and a CPU limit of 800 millicpu. How did that container get those values?

```
resources:
 limits:
 cpu: 800m
 requests:
 cpu: 800m
```

Because that container did not specify its own CPU request and limit, the control plane applied the [default CPU request and limit](#) from the LimitRange for this namespace.

At this point, your Pod may or may not be running. Recall that a prerequisite for this task is that your Nodes must have at least 1 CPU available for use. If each of your Nodes has only 1 CPU, then there might not be enough allocatable CPU on any Node to accommodate a request of 800 millicpu. If you happen to be using Nodes with 2 CPU, then you probably have enough CPU to accommodate the 800 millicpu request.

Delete your Pod:

```
kubectl delete pod constraints-cpu-demo-4 --namespace=constraints-cpu-example
```

## Enforcement of minimum and maximum CPU constraints

The maximum and minimum CPU constraints imposed on a namespace by a LimitRange are enforced only when a Pod is created or updated. If you change the LimitRange, it does not affect Pods that were created previously.

## Motivation for minimum and maximum CPU constraints

As a cluster administrator, you might want to impose restrictions on the CPU resources that Pods can use. For example:

- Each Node in a cluster has 2 CPU. You do not want to accept any Pod that requests more than 2 CPU, because no Node in the cluster can support the request.
- A cluster is shared by your production and development departments. You want to allow production workloads to consume up to 3 CPU, but you want development workloads to be limited to 1 CPU. You create separate namespaces for production and development, and you apply CPU constraints to each namespace.

## Clean up

Delete your namespace:

```
kubectrl delete namespace constraints-cpu-example
```

## What's next

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)
- [Configure a Pod Quota for a Namespace](#)
- [Configure Quotas for API Objects](#)

### For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

# Configure Memory and CPU Quotas for a Namespace

Define overall memory and CPU resource limits for a namespace.

This page shows how to set quotas for the total amount memory and CPU that can be used by all Pods running in a [namespace](#). You specify quotas in a [ResourceQuota](#) object.

## Before you begin

You need to have a Kubernetes cluster, and the kubectrl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You must have access to create namespaces in your cluster.

Each node in your cluster must have at least 1 GiB of memory.

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace quota-mem-cpu-example
```

## Create a ResourceQuota

Here is a manifest for an example ResourceQuota:

[admin/resource/quota-mem-cpu.yaml](https://k8s.io/examples/admin/resource/quota-mem-cpu.yaml)

```
apiVersion: v1
kind: ResourceQuota
metadata:
 name: mem-cpu-demo
spec:
 hard:
 requests.cpu: "1"
 requests.memory: 1Gi
 limits.cpu: "2"
 limits.memory: 2Gi
```

Create the ResourceQuota:

```
kubectl apply -f https://k8s.io/examples/admin/resource/quota-mem-cpu.yaml --namespace=quota-mem-cpu-example
```

View detailed information about the ResourceQuota:

```
kubectl get resourcequota mem-cpu-demo --namespace=quota-mem-cpu-example --output=yaml
```

The ResourceQuota places these requirements on the quota-mem-cpu-example namespace:

- For every Pod in the namespace, each container must have a memory request, memory limit, cpu request, and cpu limit.
- The memory request total for all Pods in that namespace must not exceed 1 GiB.
- The memory limit total for all Pods in that namespace must not exceed 2 GiB.
- The CPU request total for all Pods in that namespace must not exceed 1 cpu.
- The CPU limit total for all Pods in that namespace must not exceed 2 cpu.

See [meaning of CPU](#) to learn what Kubernetes means by “1 CPU”.

## Create a Pod

Here is a manifest for an example Pod:

[admin/resource/quota-mem-cpu-pod.yaml](https://k8s.io/examples/admin/resource/quota-mem-cpu-pod.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: quota-mem-cpu-demo
spec:
 containers:
 - name: quota-mem-cpu-demo-ctr
 image: nginx
 resources:
 limits:
 memory: "800Mi"
 cpu: "800m"
 requests:
 memory: "600Mi"
 cpu: "400m"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/quota-mem-cpu-pod.yaml --
namespace=quota-mem-cpu-example
```

Verify that the Pod is running and that its (only) container is healthy:

```
kubectl get pod quota-mem-cpu-demo --namespace=quota-mem-cpu-example
```

Once again, view detailed information about the ResourceQuota:

```
kubectl get resourcequota mem-cpu-demo --namespace=quota-mem-cpu-example --output=ya
ml
```

The output shows the quota along with how much of the quota has been used. You can see that the memory and CPU requests and limits for your Pod do not exceed the quota.

```
status:
 hard:
 limits.cpu: "2"
 limits.memory: 2Gi
 requests.cpu: "1"
 requests.memory: 1Gi
 used:
 limits.cpu: 800m
 limits.memory: 800Mi
 requests.cpu: 400m
 requests.memory: 600Mi
```

If you have the jq tool, you can also query (using [JSONPath](#)) for just the used values, **and** pretty-print that that of the output. For example:

```
kubectl get resourcequota mem-cpu-demo --namespace=quota-mem-cpu-example -o jsonpath='
{.status.used }' | jq .
```



## Attempt to create a second Pod

Here is a manifest for a second Pod:

[admin/resource/quota-mem-cpu-pod-2.yaml](https://k8s.io/examples/admin/resource/quota-mem-cpu-pod-2.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: quota-mem-cpu-demo-2
spec:
 containers:
 - name: quota-mem-cpu-demo-2-ctr
 image: redis
 resources:
 limits:
 memory: "1Gi"
 cpu: "800m"
 requests:
 memory: "700Mi"
 cpu: "400m"
```

In the manifest, you can see that the Pod has a memory request of 700 MiB. Notice that the sum of the used memory request and this new memory request exceeds the memory request quota: 600 MiB + 700 MiB > 1 GiB.

Attempt to create the Pod:

```
kubectl apply -f https://k8s.io/examples/admin/resource/quota-mem-cpu-pod-2.yaml --
namespace=quota-mem-cpu-example
```

The second Pod does not get created. The output shows that creating the second Pod would cause the memory request total to exceed the memory request quota.

```
Error from server (Forbidden): error when creating "examples/admin/resource/quota-mem-cpu-
pod-2.yaml":
pods "quota-mem-cpu-demo-2" is forbidden: exceeded quota: mem-cpu-demo,
requested: requests.memory=700Mi,used: requests.memory=600Mi, limited:
requests.memory=1Gi
```

## Discussion

As you have seen in this exercise, you can use a ResourceQuota to restrict the memory request total for all Pods running in a namespace. You can also restrict the totals for memory limit, cpu request, and cpu limit.

Instead of managing total resource use within a namespace, you might want to restrict individual Pods, or the containers in those Pods. To achieve that kind of limiting, use a [LimitRange](#).

## Clean up

Delete your namespace:

```
kubectrl delete namespace quota-mem-cpu-example
```

## What's next

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure a Pod Quota for a Namespace](#)
- [Configure Quotas for API Objects](#)

### For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

## Configure a Pod Quota for a Namespace

Restrict how many Pods you can create within a namespace.

This page shows how to set a quota for the total number of Pods that can run in a [Namespace](#). You specify quotas in a [ResourceQuota](#) object.

## Before you begin

You need to have a Kubernetes cluster, and the kubectrl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You must have access to create namespaces in your cluster.

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace quota-pod-example
```

## Create a ResourceQuota

Here is an example manifest for a ResourceQuota:

[admin/resource/quota-pod.yaml](#)

```
apiVersion: v1
kind: ResourceQuota
metadata:
 name: pod-demo
spec:
 hard:
 pods: "2"
```

Create the ResourceQuota:

```
kubectl apply -f https://k8s.io/examples/admin/resource/quota-pod.yaml --namespace=quota-pod-example
```

View detailed information about the ResourceQuota:

```
kubectl get resourcequota pod-demo --namespace=quota-pod-example --output=yaml
```

The output shows that the namespace has a quota of two Pods, and that currently there are no Pods; that is, none of the quota is used.

```
spec:
 hard:
 pods: "2"
status:
 hard:
 pods: "2"
 used:
 pods: "0"
```

Here is an example manifest for a [Deployment](#):

[admin/resource/quota-pod-deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: pod-quota-demo
spec:
 selector:
 matchLabels:
```

```
 purpose: quota-demo
replicas: 3
template:
 metadata:
 labels:
 purpose: quota-demo
 spec:
 containers:
 - name: pod-quota-demo
 image: nginx
```

In that manifest, `replicas: 3` tells Kubernetes to attempt to create three new Pods, all running the same application.

Create the Deployment:

```
kubectl apply -f https://k8s.io/examples/admin/resource/quota-pod-deployment.yaml --
namespace=quota-pod-example
```

View detailed information about the Deployment:

```
kubectl get deployment pod-quota-demo --namespace=quota-pod-example --output=yaml
```

The output shows that even though the Deployment specifies three replicas, only two Pods were created because of the quota you defined earlier:

```
spec:
 ...
 replicas: 3
 ...
status:
 availableReplicas: 2
 ...
lastUpdateTime: 2021-04-02T20:57:05Z
 message: 'unable to create pods: pods "pod-quota-demo-1650323038-" is forbidden:
 exceeded quota: pod-demo, requested: pods=1, used: pods=2, limited: pods=2'
```

## Choice of resource

In this task you have defined a `ResourceQuota` that limited the total number of Pods, but you could also limit the total number of other kinds of object. For example, you might decide to limit how many [CronJobs](#) that can live in a single namespace.

## Clean up

Delete your namespace:

```
kubectl delete namespace quota-pod-example
```

## What's next

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)
- [Configure Quotas for API Objects](#)

### For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

## Install a Network Policy Provider

---

[Use Antrea for NetworkPolicy](#)

[Use Calico for NetworkPolicy](#)

[Use Cilium for NetworkPolicy](#)

[Use Kube-router for NetworkPolicy](#)

[Romana for NetworkPolicy](#)

[Weave Net for NetworkPolicy](#)

## Use Antrea for NetworkPolicy

This page shows how to install and use Antrea CNI plugin on Kubernetes. For background on Project Antrea, read the [Introduction to Antrea](#).

### Before you begin

You need to have a Kubernetes cluster. Follow the [kubeadm getting started guide](#) to bootstrap one.

# Deploying Antrea with kubeadm

Follow [Getting Started](#) guide to deploy Antrea for kubeadm.

## What's next

Once your cluster is running, you can follow the [Declare Network Policy](#) to try out Kubernetes NetworkPolicy.

# Use Calico for NetworkPolicy

This page shows a couple of quick ways to create a Calico cluster on Kubernetes.

## Before you begin

Decide whether you want to deploy a [cloud](#) or [local](#) cluster.

## Creating a Calico cluster with Google Kubernetes Engine (GKE)

**Prerequisite:** [gcloud](#).

1. To launch a GKE cluster with Calico, include the `--enable-network-policy` flag.

### Syntax

```
gcloud container clusters create [CLUSTER_NAME] --enable-network-policy
```

### Example

```
gcloud container clusters create my-calico-cluster --enable-network-policy
```

2. To verify the deployment, use the following command.

```
kubectl get pods --namespace=kube-system
```

The Calico pods begin with `calico`. Check to make sure each one has a status of `Running`.

## Creating a local Calico cluster with kubeadm

To get a local single-host Calico cluster in fifteen minutes using kubeadm, refer to the [Calico Quickstart](#).

## What's next

Once your cluster is running, you can follow the [Declare Network Policy](#) to try out Kubernetes NetworkPolicy.

# Use Cilium for NetworkPolicy

This page shows how to use Cilium for NetworkPolicy.

For background on Cilium, read the [Introduction to Cilium](#).

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

## Deploying Cilium on Minikube for Basic Testing

To get familiar with Cilium easily you can follow the [Cilium Kubernetes Getting Started Guide](#) to perform a basic DaemonSet installation of Cilium in minikube.

To start minikube, minimal version required is  $\geq v1.5.2$ , run the with the following arguments:

```
minikube version
```

```
minikube version: v1.5.2
```

```
minikube start --network-plugin=cni
```

For minikube you can install Cilium using its CLI tool. To do so, first download the latest version of the CLI with the following command:

```
curl -LO https://github.com/cilium/cilium-cli/releases/latest/download/cilium-linux-amd64.tar.gz
```

Then extract the downloaded file to your `/usr/local/bin` directory with the following command:

```
sudo tar xzvfC cilium-linux-amd64.tar.gz /usr/local/bin
rm cilium-linux-amd64.tar.gz
```

After running the above commands, you can now install Cilium with the following command:

```
cilium install
```

Cilium will then automatically detect the cluster configuration and create and install the appropriate components for a successful installation. The components are:

- Certificate Authority (CA) in Secret `cilium-ca` and certificates for Hubble (Cilium's observability layer).
- Service accounts.

- Cluster roles.
- ConfigMap.
- Agent DaemonSet and an Operator Deployment.

After the installation, you can view the overall status of the Cilium deployment with the `cilium status` command. See the expected output of the status command [here](#).

The remainder of the Getting Started Guide explains how to enforce both L3/L4 (i.e., IP address + port) security policies, as well as L7 (e.g., HTTP) security policies using an example application.

## Deploying Cilium for Production Use

For detailed instructions around deploying Cilium for production, see: [Cilium Kubernetes Installation Guide](#) This documentation includes detailed requirements, instructions and example production DaemonSet files.

## Understanding Cilium components

Deploying a cluster with Cilium adds Pods to the kube-system namespace. To see this list of Pods run:

```
kubectl get pods --namespace=kube-system -l k8s-app=cilium
```

You'll see a list of Pods similar to this:

| NAME         | READY | STATUS  | RESTARTS | AGE   |
|--------------|-------|---------|----------|-------|
| cilium-kkdhz | 1/1   | Running | 0        | 3m23s |
| ...          |       |         |          |       |

A cilium Pod runs on each node in your cluster and enforces network policy on the traffic to/from Pods on that node using Linux BPF.

## What's next

Once your cluster is running, you can follow the [Declare Network Policy](#) to try out Kubernetes NetworkPolicy with Cilium. Have fun, and if you have questions, contact us using the [Cilium Slack Channel](#).

## Use Kube-router for NetworkPolicy

This page shows how to use [Kube-router](#) for NetworkPolicy.

## Before you begin

You need to have a Kubernetes cluster running. If you do not already have a cluster, you can create one by using any of the cluster installers like Kops, Bootkube, Kubeadm etc.



## Installing Kube-router addon

The Kube-router Addon comes with a Network Policy Controller that watches Kubernetes API server for any NetworkPolicy and pods updated and configures iptables rules and ipsets to allow or block traffic as directed by the policies. Please follow the [trying Kube-router with cluster installers](#) guide to install Kube-router addon.

### What's next

Once you have installed the Kube-router addon, you can follow the [Declare Network Policy](#) to try out Kubernetes NetworkPolicy.

## Romana for NetworkPolicy

This page shows how to use Romana for NetworkPolicy.

### Before you begin

Complete steps 1, 2, and 3 of the [kubeadm getting started guide](#).

### Installing Romana with kubeadm

Follow the [containerized installation guide](#) for kubeadm.

### Applying network policies

To apply network policies use one of the following:

- [Romana network policies](#).
  - [Example of Romana network policy](#).
- The NetworkPolicy API.

### What's next

Once you have installed Romana, you can follow the [Declare Network Policy](#) to try out Kubernetes NetworkPolicy.

## Weave Net for NetworkPolicy

This page shows how to use Weave Net for NetworkPolicy.

### Before you begin

You need to have a Kubernetes cluster. Follow the [kubeadm getting started guide](#) to bootstrap one.

## Install the Weave Net addon

Follow the [Integrating Kubernetes via the Addon](#) guide.

The Weave Net addon for Kubernetes comes with a [Network Policy Controller](#) that automatically monitors Kubernetes for any NetworkPolicy annotations on all namespaces and configures iptables rules to allow or block traffic as directed by the policies.

## Test the installation

Verify that the weave works.

Enter the following command:

```
kubectl get pods -n kube-system -o wide
```

The output is similar to this:

| NAME            | READY | STATUS  | RESTARTS | AGE | IP            | NODE        |
|-----------------|-------|---------|----------|-----|---------------|-------------|
| weave-net-1t1qg | 2/2   | Running | 0        | 9d  | 192.168.2.10  | worknode3   |
| weave-net-231d7 | 2/2   | Running | 1        | 7d  | 10.2.0.17     | worknodegpu |
| weave-net-7nmwt | 2/2   | Running | 3        | 9d  | 192.168.2.131 | masternode  |
| weave-net-pmw8w | 2/2   | Running | 0        | 9d  | 192.168.2.216 | worknode2   |

Each Node has a weave Pod, and all Pods are Running and 2/2 READY. (2/2 means that each Pod has weave and weave-npc.)

## What's next

Once you have installed the Weave Net addon, you can follow the [Declare Network Policy](#) to try out Kubernetes NetworkPolicy. If you have any question, contact us at [#weave-community on Slack or Weave User Group](#).

## Access Clusters Using the Kubernetes API

This page shows how to access clusters using the Kubernetes API.

### Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

# Accessing the Kubernetes API

## Accessing for the first time with kubectl

When accessing the Kubernetes API for the first time, use the Kubernetes command-line tool, kubectl.

To access a cluster, you need to know the location of the cluster and have credentials to access it. Typically, this is automatically set-up when you work through a [Getting started guide](#), or someone else set up the cluster and provided you with credentials and a location.

Check the location and credentials that kubectl knows about with this command:

```
kubectl config view
```

Many of the [examples](#) provide an introduction to using kubectl. Complete documentation is found in the [kubectl manual](#).

## Directly accessing the REST API

kubectl handles locating and authenticating to the API server. If you want to directly access the REST API with an http client like curl or wget, or a browser, there are multiple ways you can locate and authenticate against the API server:

1. Run kubectl in proxy mode (recommended). This method is recommended, since it uses the stored API server location and verifies the identity of the API server using a self-signed certificate. No man-in-the-middle (MITM) attack is possible using this method.
2. Alternatively, you can provide the location and credentials directly to the http client. This works with client code that is confused by proxies. To protect against man in the middle attacks, you'll need to import a root cert into your browser.

Using the Go or Python client libraries provides accessing kubectl in proxy mode.

## Using kubectl proxy

The following command runs kubectl in a mode where it acts as a reverse proxy. It handles locating the API server and authenticating.

Run it like this:

```
kubectl proxy --port=8080 &
```

See [kubectl proxy](#) for more details.

Then you can explore the API with curl, wget, or a browser, like so:

```
curl http://localhost:8080/api/
```

The output is similar to this:

```
{
 "versions": [
 "v1"
```

```

],
"serverAddressByClientCIDRs": [
 {
 "clientCIDR": "0.0.0.0/0",
 "serverAddress": "10.0.1.149:443"
 }
]
}

```

## Without kubectl proxy

It is possible to avoid using kubectl proxy by passing an authentication token directly to the API server, like this:

Using grep/cut approach:

```

Check all possible clusters, as your .KUBECONFIG may have multiple contexts:
kubectl config view -o jsonpath='{ "Cluster name\tServer\n" }{range .clusters[*]}{.name}{ "\t" }{.cluster.server}{ "\n" }{end}'

Select name of cluster you want to interact with from above output:
export CLUSTER_NAME="some_server_name"

Point to the API server referring the cluster name
APISERVER=$(kubectl config view -o jsonpath="{.clusters[?(@.name==\"$CLUSTER_NAME\")].cluster.server}")

Create a secret to hold a token for the default service account
kubectl apply -f - <<EOF
apiVersion: v1
kind: Secret
metadata:
 name: default-token
 annotations:
 kubernetes.io/service-account.name: default
type: kubernetes.io/service-account-token
EOF

Wait for the token controller to populate the secret with a token:
while ! kubectl describe secret default-token | grep -E '^token' >/dev/null; do
 echo "waiting for token..." >&2
 sleep 1
done

Get the token value
TOKEN=$(kubectl get secret default-token -o jsonpath='{.data.token}' | base64 --decode)

Explore the API with TOKEN
curl -X GET $APISERVER/api --header "Authorization: Bearer $TOKEN" --insecure

```

The output is similar to this:

```
{
 "kind": "APIVersions",
 "versions": [
 "v1"
],
 "serverAddressByClientCIDRs": [
 {
 "clientCIDR": "0.0.0.0/0",
 "serverAddress": "10.0.1.149:443"
 }
]
}
```

The above example uses the `--insecure` flag. This leaves it subject to MITM attacks. When `kubectl` accesses the cluster it uses a stored root certificate and client certificates to access the server. (These are installed in the `~/.kube` directory). Since cluster certificates are typically self-signed, it may take special configuration to get your http client to use root certificate.

On some clusters, the API server does not require authentication; it may serve on localhost, or be protected by a firewall. There is not a standard for this. [Controlling Access to the Kubernetes API](#) describes how you can configure this as a cluster administrator.

## Programmatic access to the API

Kubernetes officially supports client libraries for [Go](#), [Python](#), [Java](#), [dotnet](#), [JavaScript](#), and [Haskell](#). There are other client libraries that are provided and maintained by their authors, not the Kubernetes team. See [client libraries](#) for accessing the API from other languages and how they authenticate.

### Go client

- To get the library, run the following command: `go get k8s.io/client-go@kubernetes-<kubernetes-version-number>` See <https://github.com/kubernetes/client-go/releases> to see which versions are supported.
- Write an application atop of the client-go clients.

#### Note:

`client-go` defines its own API objects, so if needed, import API definitions from `client-go` rather than from the main repository. For example, `import "k8s.io/client-go/kubernetes"` is correct.

The Go client can use the same [kubeconfig file](#) as the `kubectl` CLI does to locate and authenticate to the API server. See this [example](#):

```
package main

import (
 "context"
 "fmt"
 "k8s.io/apimachinery/pkg/apis/meta/v1"
 "k8s.io/client-go/kubernetes"
 "k8s.io/client-go/tools/clientcmd"
```

```

)

func main() {
 // uses the current context in kubeconfig
 // path-to-kubeconfig -- for example, /root/.kube/config
 config, _ := clientcmd.BuildConfigFromFlags("", "<path-to-kubeconfig>")
 // creates the clientset
 clientset, _ := kubernetes.NewForConfig(config)
 // access the API to list pods
 pods, _ := clientset.CoreV1().Pods("").List(context.TODO(), v1.ListOptions{})
 fmt.Printf("There are %d pods in the cluster\n", len(pods.Items))
}

```

If the application is deployed as a Pod in the cluster, see [Accessing the API from within a Pod](#).

## Python client

To use [Python client](#), run the following command: `pip install kubernetes`. See [Python Client Library page](#) for more installation options.

The Python client can use the same [kubeconfig file](#) as the kubectl CLI does to locate and authenticate to the API server. See this [example](#):

```

from kubernetes import client, config

config.load_kube_config()

v1=client.CoreV1Api()
print("Listing pods with their IPs:")
ret = v1.list_pod_for_all_namespaces(watch=False)
for i in ret.items:
 print("%s\t%s\t%s" % (i.status.pod_ip, i.metadata.namespace, i.metadata.name))

```

## Java client

To install the [Java Client](#), run:

```

Clone java library
git clone --recursive https://github.com/kubernetes-client/java

Installing project artifacts, POM etc:
cd java
mvn install

```

See <https://github.com/kubernetes-client/java/releases> to see which versions are supported.

The Java client can use the same [kubeconfig file](#) as the kubectl CLI does to locate and authenticate to the API server. See this [example](#):

```

package io.kubernetes.client.examples;

import io.kubernetes.client.ApiClient;
import io.kubernetes.client.ApiException;

```

```

import io.kubernetes.client.Configuration;
import io.kubernetes.client.apis.CoreV1Api;
import io.kubernetes.client.models.V1Pod;
import io.kubernetes.client.models.V1PodList;
import io.kubernetes.client.util.ClientBuilder;
import io.kubernetes.client.util.KubeConfig;
import java.io.FileReader;
import java.io.IOException;

/**
 * A simple example of how to use the Java API from an application outside a kubernetes cluster
 *
 * <p>Easiest way to run this: mvn exec:java
 * -Dexec.mainClass="io.kubernetes.client.examples.KubeConfigFileClientExample"
 *
 */
public class KubeConfigFileClientExample {
 public static void main(String[] args) throws IOException, ApiException {

 // file path to your KubeConfig
 String kubeConfigPath = "~/.kube/config";

 // loading the out-of-cluster config, a kubeconfig from file-system
 ApiClient client =
 ClientBuilder.kubeconfig(KubeConfig.loadKubeConfig(new FileReader(kubeConfigPath))).build();

 // set the global default api-client to the in-cluster one from above
 Configuration.setDefaultApiClient(client);

 // the CoreV1Api loads default api-client from global configuration.
 CoreV1Api api = new CoreV1Api();

 // invokes the CoreV1Api client
 V1PodList list = api.listPodForAllNamespaces(null, null, null, null, null, null, null, null, null);
 System.out.println("Listing all pods: ");
 for (V1Pod item : list.getItems()) {
 System.out.println(item.getMetadata().getName());
 }
 }
}

```

## dotnet client

To use [dotnet client](#), run the following command: `dotnet add package KubernetesClient --version 1.6.1`. See [dotnet Client Library page](#) for more installation options. See <https://github.com/kubernetes-client/csharp/releases> to see which versions are supported.

The dotnet client can use the same [kubeconfig file](#) as the kubectl CLI does to locate and authenticate to the API server. See this [example](#):

```

using System;
using k8s;

```

```
namespace simple
{
 internal class PodList
 {
 private static void Main(string[] args)
 {
 var config = KubernetesClientConfiguration.BuildDefaultConfig();
 IKubernetes client = new Kubernetes(config);
 Console.WriteLine("Starting Request!");

 var list = client.ListNamespacedPod("default");
 foreach (var item in list.Items)
 {
 Console.WriteLine(item.Metadata.Name);
 }
 if (list.Items.Count == 0)
 {
 Console.WriteLine("Empty!");
 }
 }
 }
}
```

## JavaScript client

To install [JavaScript client](#), run the following command: `npm install @kubernetes/client-node`. See <https://github.com/kubernetes-client/javascript/releases> to see which versions are supported.

The JavaScript client can use the same [kubeconfig file](#) as the kubectl CLI does to locate and authenticate to the API server. See this [example](#):

```
const k8s = require('@kubernetes/client-node');

const kc = new k8s.KubeConfig();
kc.loadFromDefault();

const k8sApi = kc.makeApiClient(k8s.CoreV1Api);

k8sApi.listNamespacedPod('default').then((res) => {
 console.log(res.body);
});
```

## Haskell client

See <https://github.com/kubernetes-client/haskell/releases> to see which versions are supported.

The [Haskell client](#) can use the same [kubeconfig file](#) as the kubectl CLI does to locate and authenticate to the API server. See this [example](#):

```
exampleWithKubeConfig :: IO ()
exampleWithKubeConfig = do
```



```
oidcCache <- atomically $ newTVar $ Map.fromList []
(mgr, kcfg) <- mkKubeClientConfig oidcCache $ KubeConfigFile "/path/to/kubeconfig"
dispatchMime
 mgr
 kcfg
 (CoreV1.listPodForAllNamespaces (Accept MIMEJSON))
>>= print
```

## What's next

- [Accessing the Kubernetes API from a Pod](#)

# Advertise Extended Resources for a Node

This page shows how to specify extended resources for a Node. Extended resources allow cluster administrators to advertise node-level resources that would otherwise be unknown to Kubernetes.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

## Get the names of your Nodes

```
kubectl get nodes
```

Choose one of your Nodes to use for this exercise.

## Advertise a new extended resource on one of your Nodes

To advertise a new extended resource on a Node, send an HTTP PATCH request to the Kubernetes API server. For example, suppose one of your Nodes has four dongles attached. Here's an example of a PATCH request that advertises four dongle resources for your Node.

```
PATCH /api/v1/nodes/<your-node-name>/status HTTP/1.1
Accept: application/json
Content-Type: application/json-patch+json
Host: k8s-master:8080
```

```
[
{
```

```
[{"op": "add",
 "path": "/status/capacity/example.com~1dongle",
 "value": "4"}]
```

Note that Kubernetes does not need to know what a dongle is or what a dongle is for. The preceding PATCH request tells Kubernetes that your Node has four things that you call dongles.

Start a proxy, so that you can easily send requests to the Kubernetes API server:

```
kubectrl proxy
```

In another command window, send the HTTP PATCH request. Replace `<your-node-name>` with the name of your Node:

```
curl --header "Content-Type: application/json-patch+json" \
--request PATCH \
--data '[{"op": "add", "path": "/status/capacity/example.com~1dongle", "value": "4"}]' \
http://localhost:8001/api/v1/nodes/<your-node-name>/status
```

#### **Note:**

In the preceding request, `~1` is the encoding for the character `/` in the patch path. The operation path value in JSON-Patch is interpreted as a JSON-Pointer. For more details, see [IETF RFC 6901](https://tools.ietf.org/html/rfc6901), section 3.

The output shows that the Node has a capacity of 4 dongles:

```
"capacity": {
 "cpu": "2",
 "memory": "2049008Ki",
 "example.com/dongle": "4",
```

Describe your Node:

```
kubectrl describe node <your-node-name>
```

Once again, the output shows the dongle resource:

```
Capacity:
cpu: 2
memory: 2049008Ki
example.com/dongle: 4
```

Now, application developers can create Pods that request a certain number of dongles. See [Assign Extended Resources to a Container](#).

## **Discussion**

Extended resources are similar to memory and CPU resources. For example, just as a Node has a certain amount of memory and CPU to be shared by all components running on the Node, it

can have a certain number of dongles to be shared by all components running on the Node. And just as application developers can create Pods that request a certain amount of memory and CPU, they can create Pods that request a certain number of dongles.

Extended resources are opaque to Kubernetes; Kubernetes does not know anything about what they are. Kubernetes knows only that a Node has a certain number of them. Extended resources must be advertised in integer amounts. For example, a Node can advertise four dongles, but not 4.5 dongles.

## Storage example

Suppose a Node has 800 GiB of a special kind of disk storage. You could create a name for the special storage, say `example.com/special-storage`. Then you could advertise it in chunks of a certain size, say 100 GiB. In that case, your Node would advertise that it has eight resources of type `example.com/special-storage`.

Capacity:

...

`example.com/special-storage: 8`

If you want to allow arbitrary requests for special storage, you could advertise special storage in chunks of size 1 byte. In that case, you would advertise 800Gi resources of type `example.com/special-storage`.

Capacity:

...

`example.com/special-storage: 800Gi`

Then a Container could request any number of bytes of special storage, up to 800Gi.

## Clean up

Here is a PATCH request that removes the dongle advertisement from a Node.

```
PATCH /api/v1/nodes/<your-node-name>/status HTTP/1.1
```

```
Accept: application/json
```

```
Content-Type: application/json-patch+json
```

```
Host: k8s-master:8080
```

```
[
 {
 "op": "remove",
 "path": "/status/capacity/example.com~1dongle",
 }
]
```

Start a proxy, so that you can easily send requests to the Kubernetes API server:

```
kubectrl proxy
```

In another command window, send the HTTP PATCH request. Replace `<your-node-name>` with the name of your Node:

```
curl --header "Content-Type: application/json-patch+json" \
--request PATCH \
--data ' [{"op": "remove", "path": "/status/capacity/example.com~1dongle"}]' \
http://localhost:8001/api/v1/nodes/<your-node-name>/status
```

Verify that the dongle advertisement has been removed:

```
kubectl describe node <your-node-name> | grep dongle
```

(you should not see any output)

## What's next

### For application developers

- [Assign Extended Resources to a Container](#)

### For cluster administrators

- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)

# Autoscale the DNS Service in a Cluster

This page shows how to enable and configure autoscaling of the DNS service in your Kubernetes cluster.

## Before you begin

- You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercoda](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

- This guide assumes your nodes use the AMD64 or Intel 64 CPU architecture.
- Make sure [Kubernetes DNS](#) is enabled.

## Determine whether DNS horizontal autoscaling is already enabled

List the [Deployments](#) in your cluster in the kube-system [namespace](#):

```
kubectl get deployment --namespace=kube-system
```

The output is similar to this:

```
NAME READY UP-TO-DATE AVAILABLE AGE
...
kube-dns-autoscaler 1/1 1 1 ...
...
```

If you see "kube-dns-autoscaler" in the output, DNS horizontal autoscaling is already enabled, and you can skip to [Tuning autoscaling parameters](#).

## Get the name of your DNS Deployment

List the DNS deployments in your cluster in the kube-system namespace:

```
kubectl get deployment -l k8s-app=kube-dns --namespace=kube-system
```

The output is similar to this:

```
NAME READY UP-TO-DATE AVAILABLE AGE
...
coredns 2/2 2 2 ...
...
```

If you don't see a Deployment for DNS services, you can also look for it by name:

```
kubectl get deployment --namespace=kube-system
```

and look for a deployment named coredns or kube-dns.

Your scale target is

```
Deployment/<your-deployment-name>
```

where <your-deployment-name> is the name of your DNS Deployment. For example, if the name of your Deployment for DNS is coredns, your scale target is Deployment/coredns.

### Note:

CoreDNS is the default DNS service for Kubernetes. CoreDNS sets the label k8s-app=kube-dns so that it can work in clusters that originally used kube-dns.

## Enable DNS horizontal autoscaling

In this section, you create a new Deployment. The Pods in the Deployment run a container based on the cluster-proportional-autoscaler-amd64 image.

Create a file named dns-horizontal-autoscaler.yaml with this content:

[admin/dns/dns-horizontal-autoscaler.yaml](#)

```
kind: ServiceAccount
apiVersion: v1
metadata:
```

```

name: kube-dns-autoscaler
namespace: kube-system

kind: ClusterRole
apiVersion: rbac.authorization.k8s.io/v1
metadata:
 name: system:kube-dns-autoscaler
rules:
 - apiGroups: [""]
 resources: ["nodes"]
 verbs: ["list", "watch"]
 - apiGroups: [""]
 resources: ["replicationcontrollers/scale"]
 verbs: ["get", "update"]
 - apiGroups: ["apps"]
 resources: ["deployments/scale", "replicasets/scale"]
 verbs: ["get", "update"]
Remove the configmaps rule once below issue is fixed:
kubernetes-incubator/cluster-proportional-autoscaler#16
 - apiGroups: [""]
 resources: ["configmaps"]
 verbs: ["get", "create"]

kind: ClusterRoleBinding
apiVersion: rbac.authorization.k8s.io/v1
metadata:
 name: system:kube-dns-autoscaler
subjects:
 - kind: ServiceAccount
 name: kube-dns-autoscaler
 namespace: kube-system
roleRef:
 kind: ClusterRole
 name: system:kube-dns-autoscaler
 apiGroup: rbac.authorization.k8s.io

apiVersion: apps/v1
kind: Deployment
metadata:
 name: kube-dns-autoscaler
 namespace: kube-system
 labels:
 k8s-app: kube-dns-autoscaler
 kubernetes.io/cluster-service: "true"
spec:
 selector:
 matchLabels:
 k8s-app: kube-dns-autoscaler
 template:
 metadata:
 labels:
 k8s-app: kube-dns-autoscaler

```

```

spec:
 priorityClassName: system-cluster-critical
 securityContext:
 seccompProfile:
 type: RuntimeDefault
 supplementalGroups: [65534]
 fsGroup: 65534
 nodeSelector:
 kubernetes.io/os: linux
 containers:
 - name: autoscaler
 image: registry.k8s.io/cpa/cluster-proportional-autoscaler:1.8.4
 resources:
 requests:
 cpu: "20m"
 memory: "10Mi"
 command:
 - /cluster-proportional-autoscaler
 - --namespace=kube-system
 - --configmap=kube-dns-autoscaler
 # Should keep target in sync with cluster/addons/dns/kube-dns.yaml.base
 - --target=<SCALE_TARGET>
 # When cluster is using large nodes(with more cores), "coresPerReplica" should dominate.
 # If using small nodes, "nodesPerReplica" should dominate.
 - --default-params={"linear":{"coresPerReplica":256,"nodesPerReplica":
16,"preventSinglePointFailure":true,"includeUnschedulableNodes":true}}
 - --logtostderr=true
 - --v=2
 tolerations:
 - key: "CriticalAddonsOnly"
 operator: "Exists"
 serviceAccountName: kube-dns-autoscaler

```

In the file, replace `<SCALE_TARGET>` with your scale target.

Go to the directory that contains your configuration file, and enter this command to create the Deployment:

```
kubectl apply -f dns-horizontal-autoscaler.yaml
```

The output of a successful command is:

```
deployment.apps/kube-dns-autoscaler created
```

DNS horizontal autoscaling is now enabled.

## Tune DNS autoscaling parameters

Verify that the kube-dns-autoscaler [ConfigMap](#) exists:

```
kubectl get configmap --namespace=kube-system
```

The output is similar to this:

| NAME                | DATA | AGE |
|---------------------|------|-----|
| ...                 |      |     |
| kube-dns-autoscaler | 1    | ... |
| ...                 |      |     |

Modify the data in the ConfigMap:

```
kubectrl edit configmap kube-dns-autoscaler --namespace=kube-system
```

Look for this line:

```
linear: '{"coresPerReplica":256,"min":1,"nodesPerReplica":16}'
```

Modify the fields according to your needs. The "min" field indicates the minimal number of DNS backends. The actual number of backends is calculated using this equation:

$$\text{replicas} = \max(\text{ceil}(\text{cores} \times 1/\text{coresPerReplica}), \text{ceil}(\text{nodes} \times 1/\text{nodesPerReplica}))$$

Note that the values of both `coresPerReplica` and `nodesPerReplica` are floats.

The idea is that when a cluster is using nodes that have many cores, `coresPerReplica` dominates. When a cluster is using nodes that have fewer cores, `nodesPerReplica` dominates.

There are other supported scaling patterns. For details, see [cluster-proportional-autoscaler](#).

## Disable DNS horizontal autoscaling

There are a few options for tuning DNS horizontal autoscaling. Which option to use depends on different conditions.

### Option 1: Scale down the kube-dns-autoscaler deployment to 0 replicas

This option works for all situations. Enter this command:

```
kubectrl scale deployment --replicas=0 kube-dns-autoscaler --namespace=kube-system
```

The output is:

```
deployment.apps/kube-dns-autoscaler scaled
```

Verify that the replica count is zero:

```
kubectrl get rs --namespace=kube-system
```

The output displays 0 in the DESIRED and CURRENT columns:

| NAME                           | DESIRED | CURRENT | READY | AGE |
|--------------------------------|---------|---------|-------|-----|
| ...                            |         |         |       |     |
| kube-dns-autoscaler-6b59789fc8 | 0       | 0       | 0     | ... |
| ...                            |         |         |       |     |



## Option 2: Delete the kube-dns-autoscaler deployment

This option works if kube-dns-autoscaler is under your own control, which means no one will re-create it:

```
kubectl delete deployment kube-dns-autoscaler --namespace=kube-system
```

The output is:

```
deployment.apps "kube-dns-autoscaler" deleted
```

## Option 3: Delete the kube-dns-autoscaler manifest file from the master node

This option works if kube-dns-autoscaler is under control of the (deprecated) [Addon Manager](#), and you have write access to the master node.

Sign in to the master node and delete the corresponding manifest file. The common path for this kube-dns-autoscaler is:

```
/etc/kubernetes/addons/dns-horizontal-autoscaler/dns-horizontal-autoscaler.yaml
```

After the manifest file is deleted, the Addon Manager will delete the kube-dns-autoscaler Deployment.

## Understanding how DNS horizontal autoscaling works

- The cluster-proportional-autoscaler application is deployed separately from the DNS service.
- An autoscaler Pod runs a client that polls the Kubernetes API server for the number of nodes and cores in the cluster.
- A desired replica count is calculated and applied to the DNS backends based on the current schedulable nodes and cores and the given scaling parameters.
- The scaling parameters and data points are provided via a ConfigMap to the autoscaler, and it refreshes its parameters table every poll interval to be up to date with the latest desired scaling parameters.
- Changes to the scaling parameters are allowed without rebuilding or restarting the autoscaler Pod.
- The autoscaler provides a controller interface to support two control patterns: *linear* and *ladder*.

## What's next

- Read about [Guaranteed Scheduling For Critical Add-On Pods](#).
- Learn more about the [implementation of cluster-proportional-autoscaler](#).

# Change the Access Mode of a PersistentVolume to ReadWriteOncePod

This page shows how to change the access mode on an existing PersistentVolume to use ReadWriteOncePod.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.22. To check the version, enter kubectl version.

### Note:

The ReadWriteOncePod access mode graduated to stable in the Kubernetes v1.29 release. If you are running a version of Kubernetes older than v1.29, you might need to enable a feature gate. Check the documentation for your version of Kubernetes.

### Note:

The ReadWriteOncePod access mode is only supported for [CSI](#) volumes. To use this volume access mode you will need to update the following [CSI sidecars](#) to these versions or greater:

- [csi-provisioner:v3.0.0+](#)
- [csi-attacher:v3.3.0+](#)
- [csi-resizer:v1.3.0+](#)

## Why should I use ReadWriteOncePod?

Prior to Kubernetes v1.22, the ReadWriteOnce access mode was commonly used to restrict PersistentVolume access for workloads that required single-writer access to storage. However, this access mode had a limitation: it restricted volume access to a single *node*, allowing multiple pods on the same node to read from and write to the same volume simultaneously. This could pose a risk for applications that demand strict single-writer access for data safety.

If ensuring single-writer access is critical for your workloads, consider migrating your volumes to ReadWriteOncePod.

# Migrating existing PersistentVolumes

If you have existing PersistentVolumes, they can be migrated to use ReadWriteOncePod. Only migrations from ReadWriteOnce to ReadWriteOncePod are supported.

In this example, there is already a ReadWriteOnce "cat-pictures-pvc" PersistentVolumeClaim that is bound to a "cat-pictures-pv" PersistentVolume, and a "cat-pictures-writer" Deployment that uses this PersistentVolumeClaim.

## Note:

If your storage plugin supports [Dynamic provisioning](#), the "cat-pictures-pv" will be created for you, but its name may differ. To get your PersistentVolume's name run:

```
kubectl get pvc cat-pictures-pvc -o jsonpath='{.spec.volumeName}'
```

And you can view the PVC before you make changes. Either view the manifest locally, or run `kubectl get pvc <name-of-pvc> -o yaml`. The output is similar to:

```
cat-pictures-pvc.yaml
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
 name: cat-pictures-pvc
spec:
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 1Gi
```

Here's an example Deployment that relies on that PersistentVolumeClaim:

```
cat-pictures-writer-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: cat-pictures-writer
spec:
 replicas: 3
 selector:
 matchLabels:
 app: cat-pictures-writer
 template:
 metadata:
 labels:
 app: cat-pictures-writer
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
```

```
volumeMounts:
- name: cat-pictures
 mountPath: /mnt
volumes:
- name: cat-pictures
 persistentVolumeClaim:
 claimName: cat-pictures-pvc
 readOnly: false
```

As a first step, you need to edit your PersistentVolume's `spec.persistentVolumeReclaimPolicy` and set it to `Retain`. This ensures your PersistentVolume will not be deleted when you delete the corresponding PersistentVolumeClaim:

```
kubectl patch pv cat-pictures-pv -p '{"spec":{"persistentVolumeReclaimPolicy":"Retain"}}'
```

Next you need to stop any workloads that are using the PersistentVolumeClaim bound to the PersistentVolume you want to migrate, and then delete the PersistentVolumeClaim. Avoid making any other changes to the PersistentVolumeClaim, such as volume resizes, until after the migration is complete.

Once that is done, you need to clear your PersistentVolume's `spec.claimRef.uid` to ensure PersistentVolumeClaims can bind to it upon recreation:

```
kubectl scale --replicas=0 deployment cat-pictures-writer
kubectl delete pvc cat-pictures-pvc
kubectl patch pv cat-pictures-pv -p '{"spec":{"claimRef":{"uid":""}}}'
```

After that, replace the PersistentVolume's list of valid access modes to be (only) `ReadWriteOncePod`:

```
kubectl patch pv cat-pictures-pv -p '{"spec":{"accessModes":["ReadWriteOncePod"]}}'
```

### **Note:**

The `ReadWriteOncePod` access mode cannot be combined with other access modes. Make sure `ReadWriteOncePod` is the only access mode on the PersistentVolume when updating, otherwise the request will fail.

Next you need to modify your PersistentVolumeClaim to set `ReadWriteOncePod` as the only access mode. You should also set the PersistentVolumeClaim's `spec.volumeName` to the name of your PersistentVolume to ensure it binds to this specific PersistentVolume.

Once this is done, you can recreate your PersistentVolumeClaim and start up your workloads:

```
IMPORTANT: Make sure to edit your PVC in cat-pictures-pvc.yaml before applying. You need
to:
- Set ReadWriteOncePod as the only access mode
- Set spec.volumeName to "cat-pictures-pv"

kubectl apply -f cat-pictures-pvc.yaml
kubectl apply -f cat-pictures-writer-deployment.yaml
```

Lastly you may edit your PersistentVolume's `spec.persistentVolumeReclaimPolicy` and set it back to `Delete` if you previously changed it.

```
kubectl patch pv cat-pictures-pv -p '{"spec":{"persistentVolumeReclaimPolicy":"Delete"}}'
```

## What's next

- Learn more about [PersistentVolumes](#).
- Learn more about [PersistentVolumeClaims](#).
- Learn more about [Configuring a Pod to Use a PersistentVolume for Storage](#)

# Change the default StorageClass

This page shows how to change the default Storage Class that is used to provision volumes for PersistentVolumeClaims that have no special requirements.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Why change the default storage class?

Depending on the installation method, your Kubernetes cluster may be deployed with an existing StorageClass that is marked as default. This default StorageClass is then used to dynamically provision storage for PersistentVolumeClaims that do not require any specific storage class. See [PersistentVolumeClaim documentation](#) for details.

The pre-installed default StorageClass may not fit well with your expected workload; for example, it might provision storage that is too expensive. If this is the case, you can either change the default StorageClass or disable it completely to avoid dynamic provisioning of storage.

Deleting the default StorageClass may not work, as it may be re-created automatically by the addon manager running in your cluster. Please consult the docs for your installation for details about addon manager and how to disable individual addons.

## Changing the default StorageClass

1. List the StorageClasses in your cluster:

```
kubectl get storageclass
```

The output is similar to this:

| NAME               | PROVISIONER          | AGE |
|--------------------|----------------------|-----|
| standard (default) | kubernetes.io/gce-pd | 1d  |
| gold               | kubernetes.io/gce-pd | 1d  |

The default StorageClass is marked by (default).

## 2. Mark the default StorageClass as non-default:

The default StorageClass has an annotation `storageclass.kubernetes.io/is-default-class` set to `true`. Any other value or absence of the annotation is interpreted as `false`.

To mark a StorageClass as non-default, you need to change its value to `false`:

```
kubectrl patch storageclass standard -p '{"metadata": {"annotations": {"storageclass.kubernetes.io/is-default-class": "false"}}}'
```

where `standard` is the name of your chosen StorageClass.

## 3. Mark a StorageClass as default:

Similar to the previous step, you need to add/set the annotation `storageclass.kubernetes.io/is-default-class=true`.

```
kubectrl patch storageclass gold -p '{"metadata": {"annotations": {"storageclass.kubernetes.io/is-default-class": "true"}}}'
```

Please note that at most one StorageClass can be marked as default. If two or more of them are marked as default, a `PersistentVolumeClaim` without `storageClassName` explicitly specified cannot be created.

## 4. Verify that your chosen StorageClass is default:

```
kubectrl get storageclass
```

The output is similar to this:

| NAME           | PROVISIONER          | AGE |
|----------------|----------------------|-----|
| standard       | kubernetes.io/gce-pd | 1d  |
| gold (default) | kubernetes.io/gce-pd | 1d  |

## What's next

- Learn more about [PersistentVolumes](#).

# Switching from Polling to CRI Event-based Updates to Container Status

FEATURE STATE: Kubernetes v1.27 [beta]

This page shows how to migrate nodes to use event based updates for container status. The event-based implementation reduces node resource consumption by the kubelet, compared to the legacy approach that relies on polling. You may know this feature as *evented Pod lifecycle*

*event generator (PLEG)*. That's the name used internally within the Kubernetes project for a key implementation detail.

The polling based approach is referred to as *generic PLEG*.

## Before you begin

- You need to run a version of Kubernetes that provides this feature. Kubernetes v1.27 includes beta support for event-based container status updates. The feature is beta but is *disabled* by default because it requires support from the container runtime.
- Your Kubernetes server must be at or later than version 1.26. To check the version, enter `kubectl version`. If you are running a different version of Kubernetes, check the documentation for that release.
- The container runtime in use must support container lifecycle events. The kubelet automatically switches back to the legacy generic PLEG mechanism if the container runtime does not announce support for container lifecycle events, even if you have this feature gate enabled.

## Why switch to Evented PLEG?

- The *Generic PLEG* incurs non-negligible overhead due to frequent polling of container statuses.
- This overhead is exacerbated by Kubelet's parallelized polling of container states, thus limiting its scalability and causing poor performance and reliability problems.
- The goal of *Evented PLEG* is to reduce unnecessary work during inactivity by replacing periodic polling.

## Switching to Evented PLEG

1. Start the Kubelet with the [feature gate](#) `EventedPLEG` enabled. You can manage the kubelet feature gates editing the kubelet [config file](#) and restarting the kubelet service. You need to do this on each node where you are using this feature.
2. Make sure the node is [drained](#) before proceeding.
3. Start the container runtime with the container event generation enabled.
  - [Containerd](#)
  - [CRI-O](#)

Version 1.7+

Version 1.26+

Check if the CRI-O is already configured to emit CRI events by verifying the configuration,

```
crio config | grep enable_pod_events
```

If it is enabled, the output should be similar to the following:

```
enable_pod_events = true
```

To enable it, start the CRI-O daemon with the flag `--enable-pod-events=true` or use a dropin config with the following lines:

```
[crio.runtime]
enable_pod_events: true
```

Your Kubernetes server must be at or later than version 1.26. To check the version, enter `kubectl version`.

4. Verify that the kubelet is using event-based container stage change monitoring. To check, look for the term `EventedPLEG` in the kubelet logs.

The output should be similar to this:

```
I0314 11:10:13.909915 1105457 feature_gate.go:249] feature gates:
&{map[EventedPLEG:true]}
```

If you have set `--v` to 4 and above, you might see more entries that indicate that the kubelet is using event-based container state monitoring.

```
I0314 11:12:42.009542 1110177 evented.go:238] "Evented PLEG: Generated pod status from
the received event" podUID=3b2c6172-b112-447a-ba96-94e7022912dc
I0314 11:12:44.623326 1110177 evented.go:238] "Evented PLEG: Generated pod status from
the received event" podUID=b3fba5ea-a8c5-4b76-8f43-481e17e8ec40
I0314 11:12:44.714564 1110177 evented.go:238] "Evented PLEG: Generated pod status from
the received event" podUID=b3fba5ea-a8c5-4b76-8f43-481e17e8ec40
```

## What's next

- Learn more about the design in the Kubernetes Enhancement Proposal (KEP): [Kubelet Evented PLEG for Better Performance](#).

# Change the Reclaim Policy of a PersistentVolume

This page shows how to change the reclaim policy of a Kubernetes PersistentVolume.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercoda](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.



# Why change reclaim policy of a PersistentVolume

PersistentVolumes can have various reclaim policies, including "Retain", "Recycle", and "Delete". For dynamically provisioned PersistentVolumes, the default reclaim policy is "Delete". This means that a dynamically provisioned volume is automatically deleted when a user deletes the corresponding PersistentVolumeClaim. This automatic behavior might be inappropriate if the volume contains precious data. In that case, it is more appropriate to use the "Retain" policy. With the "Retain" policy, if a user deletes a PersistentVolumeClaim, the corresponding PersistentVolume will not be deleted. Instead, it is moved to the Released phase, where all of its data can be manually recovered.

## Changing the reclaim policy of a PersistentVolume

1. List the PersistentVolumes in your cluster:

```
kubectl get pv
```

The output is similar to this:

| NAME                                     |        | CAPACITY     | ACCESSMODES | RECLAIMPOLICY |       |
|------------------------------------------|--------|--------------|-------------|---------------|-------|
| STATUS                                   | CLAIM  | STORAGECLASS | REASON      | AGE           |       |
| pvc-b6efd8da-b7b5-11e6-9d58-0ed433a7dd94 |        | 4Gi          | RWO         | Delete        | Bound |
| default/claim1                           | manual | 10s          |             |               |       |
| pvc-b95650f8-b7b5-11e6-9d58-0ed433a7dd94 |        | 4Gi          | RWO         | Delete        | Bound |
| default/claim2                           | manual | 6s           |             |               |       |
| pvc-bb3ca71d-b7b5-11e6-9d58-0ed433a7dd94 |        | 4Gi          | RWO         | Delete        | Bound |
| default/claim3                           | manual | 3s           |             |               |       |

This list also includes the name of the claims that are bound to each volume for easier identification of dynamically provisioned volumes.

2. Choose one of your PersistentVolumes and change its reclaim policy:

```
kubectl patch pv <your-pv-name> -p '{"spec":{"persistentVolumeReclaimPolicy":"Retain"}}'
```

where <your-pv-name> is the name of your chosen PersistentVolume.

### Note:

On Windows, you must *double* quote any JSONPath template that contains spaces (not single quote as shown above for bash). This in turn means that you must use a single quote or escaped double quote around any literals in the template. For example:

```
kubectl patch pv <your-pv-name> -p '{"spec":{"persistentVolumeReclaimPolicy":"Retain\\\""}}'
```

3. Verify that your chosen PersistentVolume has the right policy:

```
kubectl get pv
```

The output is similar to this:

| NAME                                     |        | CAPACITY     | ACCESSMODES | RECLAIMPOLICY |       |
|------------------------------------------|--------|--------------|-------------|---------------|-------|
| STATUS                                   | CLAIM  | STORAGECLASS | REASON      | AGE           |       |
| pvc-b6efd8da-b7b5-11e6-9d58-0ed433a7dd94 |        | 4Gi          | RWO         | Delete        | Bound |
| default/claim1                           | manual | 40s          |             |               |       |
| pvc-b95650f8-b7b5-11e6-9d58-0ed433a7dd94 |        | 4Gi          | RWO         | Delete        | Bound |
| default/claim2                           | manual | 36s          |             |               |       |
| pvc-bb3ca71d-b7b5-11e6-9d58-0ed433a7dd94 |        | 4Gi          | RWO         | Retain        | Bound |
| default/claim3                           | manual | 33s          |             |               |       |

In the preceding output, you can see that the volume bound to claim default/claim3 has reclaim policy Retain. It will not be automatically deleted when a user deletes claim default/claim3.

## What's next

- Learn more about [PersistentVolumes](#).
- Learn more about [PersistentVolumeClaims](#).

## References

- [PersistentVolume](#)
  - Pay attention to the .spec.persistentVolumeReclaimPolicy [field](#) of PersistentVolume.
- [PersistentVolumeClaim](#)

# Cloud Controller Manager Administration

FEATURE STATE: Kubernetes v1.11 [beta]

Since cloud providers develop and release at a different pace compared to the Kubernetes project, abstracting the provider-specific code to the [cloud-controller-manager](#) binary allows cloud vendors to evolve independently from the core Kubernetes code.

The cloud-controller-manager can be linked to any cloud provider that satisfies [cloudprovider.Interface](#). For backwards compatibility, the [cloud-controller-manager](#) provided in the core Kubernetes project uses the same cloud libraries as kube-controller-manager. Cloud providers already supported in Kubernetes core are expected to use the in-tree cloud-controller-manager to transition out of Kubernetes core.

## Administration

### Requirements

Every cloud has their own set of requirements for running their own cloud provider integration, it should not be too different from the requirements when running kube-controller-manager. As a general rule of thumb you'll need:

- cloud authentication/authorization: your cloud may require a token or IAM rules to allow access to their APIs
- kubernetes authentication/authorization: cloud-controller-manager may need RBAC rules set to speak to the kubernetes apiserver

- high availability: like kube-controller-manager, you may want a high available setup for cloud controller manager using leader election (on by default).

## Running cloud-controller-manager

Successfully running cloud-controller-manager requires some changes to your cluster configuration.

- kubelet, kube-apiserver, and kube-controller-manager must be set according to the user's usage of external CCM. If the user has an external CCM (not the internal cloud controller loops in the Kubernetes Controller Manager), then `--cloud-provider=external` must be specified. Otherwise, it should not be specified.

Keep in mind that setting up your cluster to use cloud controller manager will change your cluster behaviour in a few ways:

- Components that specify `--cloud-provider=external` will add a taint `node.cloudprovider.kubernetes.io/uninitialized` with an effect `NoSchedule` during initialization. This marks the node as needing a second initialization from an external controller before it can be scheduled work. Note that in the event that cloud controller manager is not available, new nodes in the cluster will be left unschedulable. The taint is important since the scheduler may require cloud specific information about nodes such as their region or type (high cpu, gpu, high memory, spot instance, etc).
- cloud information about nodes in the cluster will no longer be retrieved using local metadata, but instead all API calls to retrieve node information will go through cloud controller manager. This may mean you can restrict access to your cloud API on the kubelets for better security. For larger clusters you may want to consider if cloud controller manager will hit rate limits since it is now responsible for almost all API calls to your cloud from within the cluster.

The cloud controller manager can implement:

- Node controller - responsible for updating kubernetes nodes using cloud APIs and deleting kubernetes nodes that were deleted on your cloud.
- Service controller - responsible for loadbalancers on your cloud against services of type `LoadBalancer`.
- Route controller - responsible for setting up network routes on your cloud
- any other features you would like to implement if you are running an out-of-tree provider.

## Examples

If you are using a cloud that is currently supported in Kubernetes core and would like to adopt cloud controller manager, see the [cloud controller manager in kubernetes core](#).

For cloud controller managers not in Kubernetes core, you can find the respective projects in repositories maintained by cloud vendors or by SIGs.

For providers already in Kubernetes core, you can run the in-tree cloud controller manager as a DaemonSet in your cluster, use the following as a guideline:

[admin/cloud/ccm-example.yaml](#)

```
This is an example of how to set up cloud-controller-manager as a Daemonset in your cluster.
It assumes that your masters can run pods and has the role node-role.kubernetes.io/master
Note that this Daemonset will not work straight out of the box for your cloud, this is
meant to be a guideline.
```

```

```

```
apiVersion: v1
kind: ServiceAccount
metadata:
 name: cloud-controller-manager
 namespace: kube-system
```

```

```

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
 name: system:cloud-controller-manager
roleRef:
 apiGroup: rbac.authorization.k8s.io
 kind: ClusterRole
 name: cluster-admin
```

```
subjects:
- kind: ServiceAccount
 name: cloud-controller-manager
 namespace: kube-system
```

```

```

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
 labels:
 k8s-app: cloud-controller-manager
 name: cloud-controller-manager
 namespace: kube-system
```

```
spec:
 selector:
 matchLabels:
 k8s-app: cloud-controller-manager
 template:
 metadata:
 labels:
 k8s-app: cloud-controller-manager
 spec:
 serviceAccountName: cloud-controller-manager
 containers:
 - name: cloud-controller-manager
 # for in-tree providers we use registry.k8s.io/cloud-controller-manager
 # this can be replaced with any other image for out-of-tree providers
 image: registry.k8s.io/cloud-controller-manager:v1.8.0
 command:
 - /usr/local/bin/cloud-controller-manager
 - --cloud-provider=[YOUR_CLOUD_PROVIDER] # Add your own cloud provider here!
 - --leader-elect=true
 - --use-service-account-credentials
 # these flags will vary for every cloud provider
```

```
- --allocate-node-cidrs=true
- --configure-cloud-routes=true
- --cluster-cidr=172.17.0.0/16
tolerations:
this is required so CCM can bootstrap itself
- key: node.cloudprovider.kubernetes.io/uninitialized
 value: "true"
 effect: NoSchedule
these tolerations are to have the daemonset runnable on control plane nodes
remove them if your control plane nodes should not run pods
- key: node-role.kubernetes.io/control-plane
 operator: Exists
 effect: NoSchedule
- key: node-role.kubernetes.io/master
 operator: Exists
 effect: NoSchedule
this is to restrict CCM to only run on master nodes
the node selector may vary depending on your cluster setup
nodeSelector:
 node-role.kubernetes.io/master: ""
```

## Limitations

Running cloud controller manager comes with a few possible limitations. Although these limitations are being addressed in upcoming releases, it's important that you are aware of these limitations for production workloads.

## Support for Volumes

Cloud controller manager does not implement any of the volume controllers found in kube-controller-manager as the volume integrations also require coordination with kubelets. As we evolve CSI (container storage interface) and add stronger support for flex volume plugins, necessary support will be added to cloud controller manager so that clouds can fully integrate with volumes. Learn more about out-of-tree CSI volume plugins [here](#).

## Scalability

The cloud-controller-manager queries your cloud provider's APIs to retrieve information for all nodes. For very large clusters, consider possible bottlenecks such as resource requirements and API rate limiting.

## Chicken and Egg

The goal of the cloud controller manager project is to decouple development of cloud features from the core Kubernetes project. Unfortunately, many aspects of the Kubernetes project has assumptions that cloud provider features are tightly integrated into the project. As a result, adopting this new architecture can create several situations where a request is being made for information from a cloud provider, but the cloud controller manager may not be able to return that information without the original request being complete.

A good example of this is the TLS bootstrapping feature in the Kubelet. TLS bootstrapping assumes that the Kubelet has the ability to ask the cloud provider (or a local metadata service)

for all its address types (private, public, etc) but cloud controller manager cannot set a node's address types without being initialized in the first place which requires that the kubelet has TLS certificates to communicate with the apiserver.

As this initiative evolves, changes will be made to address these issues in upcoming releases.

## What's next

To build and develop your own cloud controller manager, read [Developing Cloud Controller Manager](#).

# Configure a kubelet image credential provider

FEATURE STATE: Kubernetes v1.26 [stable]

Starting from Kubernetes v1.20, the kubelet can dynamically retrieve credentials for a container image registry using exec plugins. The kubelet and the exec plugin communicate through stdio (stdin, stdout, and stderr) using Kubernetes versioned APIs. These plugins allow the kubelet to request credentials for a container registry dynamically as opposed to storing static credentials on disk. For example, the plugin may talk to a local metadata server to retrieve short-lived credentials for an image that is being pulled by the kubelet.

You may be interested in using this capability if any of the below are true:

- API calls to a cloud provider service are required to retrieve authentication information for a registry.
- Credentials have short expiration times and requesting new credentials frequently is required.
- Storing registry credentials on disk or in `imagePullSecrets` is not acceptable.

This guide demonstrates how to configure the kubelet's image credential provider plugin mechanism.

## Before you begin

- You need a Kubernetes cluster with nodes that support kubelet credential provider plugins. This support is available in Kubernetes 1.30; Kubernetes v1.24 and v1.25 included this as a beta feature, enabled by default.
- A working implementation of a credential provider exec plugin. You can build your own plugin or use one provided by cloud providers.

Your Kubernetes server must be at or later than version v1.26. To check the version, enter `kubectl version`.

# Installing Plugins on Nodes

A credential provider plugin is an executable binary that will be run by the kubelet. Ensure that the plugin binary exists on every node in your cluster and stored in a known directory. The directory will be required later when configuring kubelet flags.

## Configuring the Kubelet

In order to use this feature, the kubelet expects two flags to be set:

- `--image-credential-provider-config` - the path to the credential provider plugin config file.
- `--image-credential-provider-bin-dir` - the path to the directory where credential provider plugin binaries are located.

### Configure a kubelet credential provider

The configuration file passed into `--image-credential-provider-config` is read by the kubelet to determine which exec plugins should be invoked for which container images. Here's an example configuration file you may end up using if you are using the [ECR-based plugin](#):

```
apiVersion: kubelet.config.k8s.io/v1
kind: CredentialProviderConfig
providers is a list of credential provider helper plugins that will be enabled by the kubelet.
Multiple providers may match against a single image, in which case credentials
from all providers will be returned to the kubelet. If multiple providers are called
for a single image, the results are combined. If providers return overlapping
auth keys, the value from the provider earlier in this list is used.
providers:
 # name is the required name of the credential provider. It must match the name of the
 # provider executable as seen by the kubelet. The executable must be in the kubelet's
 # bin directory (set by the --image-credential-provider-bin-dir flag).
 - name: ecr-credential-provider
 # matchImages is a required list of strings used to match against images in order to
 # determine if this provider should be invoked. If one of the strings matches the
 # requested image from the kubelet, the plugin will be invoked and given a chance
 # to provide credentials. Images are expected to contain the registry domain
 # and URL path.
 #
 # Each entry in matchImages is a pattern which can optionally contain a port and a path.
 # Globs can be used in the domain, but not in the port or the path. Globs are supported
 # as subdomains like '*.k8s.io' or 'k8s*.io', and top-level-domains such as 'k8s.*'.
 # Matching partial subdomains like 'app*.k8s.io' is also supported. Each glob can only match
 # a single subdomain segment, so '*.io' does not match '*.k8s.io'.
 #
 # A match exists between an image and a matchImage when all of the below are true:
 # - Both contain the same number of domain parts and each part matches.
 # - The URL path of an matchImages must be a prefix of the target image URL path.
 # - If the matchImages contains a port, then the port must match in the image as well.
 #
 # Example values of matchImages:
 # - 123456789.dkr.ecr.us-east-1.amazonaws.com
 # - *.azurecr.io
```

```

- gcr.io
- *.registry.io
- registry.io:8080/path
matchImages:
 - "*.dkr.ecr.*.amazonaws.com"
 - "*.dkr.ecr.*.amazonaws.com.cn"
 - "*.dkr.ecr-fips.*.amazonaws.com"
 - "*.dkr.ecr.us-iso-east-1.c2s.ic.gov"
 - "*.dkr.ecr.us-isob-east-1.sc2s.sgov.gov"
defaultCacheDuration is the default duration the plugin will cache credentials in-memory
if a cache duration is not provided in the plugin response. This field is required.
defaultCacheDuration: "12h"
Required input version of the exec CredentialProviderRequest. The returned
CredentialProviderResponse
MUST use the same encoding version as the input. Current supported values are:
- credentialprovider.kubelet.k8s.io/v1
apiVersion: credentialprovider.kubelet.k8s.io/v1
Arguments to pass to the command when executing it.
+optional
args:
- --example-argument
Env defines additional environment variables to expose to the process. These
are unioned with the host's environment, as well as variables client-go uses
to pass argument to the plugin.
+optional
env:
 - name: AWS_PROFILE
 value: example_profile

```

The providers field is a list of enabled plugins used by the kubelet. Each entry has a few required fields:

- **name**: the name of the plugin which MUST match the name of the executable binary that exists in the directory passed into `--image-credential-provider-bin-dir`.
- **matchImages**: a list of strings used to match against images in order to determine if this provider should be invoked. More on this below.
- **defaultCacheDuration**: the default duration the kubelet will cache credentials in-memory if a cache duration was not specified by the plugin.
- **apiVersion**: the API version that the kubelet and the exec plugin will use when communicating.

Each credential provider can also be given optional args and environment variables as well. Consult the plugin implementors to determine what set of arguments and environment variables are required for a given plugin.

## Configure image matching

The `matchImages` field for each credential provider is used by the kubelet to determine whether a plugin should be invoked for a given image that a Pod is using. Each entry in `matchImages` is an image pattern which can optionally contain a port and a path. Globs can be used in the domain, but not in the port or the path. Globs are supported as subdomains like `*.k8s.io` or `k8s*.io`, and top-level domains such as `k8s.*`. Matching partial subdomains like `app*.k8s.io` is also



supported. Each glob can only match a single subdomain segment, so \*.io does NOT match \*.k8s.io.

A match exists between an image name and a matchImage entry when all of the below are true:

- Both contain the same number of domain parts and each part matches.
- The URL path of match image must be a prefix of the target image URL path.
- If the matchImages contains a port, then the port must match in the image as well.

Some example values of matchImages patterns are:

- 123456789.dkr.ecr.us-east-1.amazonaws.com
- \*.azurecr.io
- gcr.io
- \*.registry.io
- foo.registry.io:8080/path

## What's next

- Read the details about CredentialProviderConfig in the [kubelet configuration API \(v1\) reference](#).
- Read the [kubelet credential provider API reference \(v1\)](#).

# Configure Quotas for API Objects

This page shows how to configure quotas for API objects, including PersistentVolumeClaims and Services. A quota restricts the number of objects, of a particular type, that can be created in a namespace. You specify quotas in a [ResourceQuota](#) object.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace quota-object-example
```

## Create a ResourceQuota

Here is the configuration file for a ResourceQuota object:

[admin/resource/quota-objects.yaml](https://k8s.io/examples/admin/resource/quota-objects.yaml)

```
apiVersion: v1
kind: ResourceQuota
metadata:
 name: object-quota-demo
spec:
 hard:
 persistentvolumeclaims: "1"
 services.loadbalancers: "2"
 services.nodeports: "0"
```

Create the ResourceQuota:

```
kubectl apply -f https://k8s.io/examples/admin/resource/quota-objects.yaml --namespace=quota-object-example
```

View detailed information about the ResourceQuota:

```
kubectl get resourcequota object-quota-demo --namespace=quota-object-example --output=yaml
```

The output shows that in the quota-object-example namespace, there can be at most one PersistentVolumeClaim, at most two Services of type LoadBalancer, and no Services of type NodePort.

```
status:
 hard:
 persistentvolumeclaims: "1"
 services.loadbalancers: "2"
 services.nodeports: "0"
 used:
 persistentvolumeclaims: "0"
 services.loadbalancers: "0"
 services.nodeports: "0"
```

## Create a PersistentVolumeClaim

Here is the configuration file for a PersistentVolumeClaim object:

[admin/resource/quota-objects-pvc.yaml](https://k8s.io/examples/admin/resource/quota-objects-pvc.yaml)

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: pvc-quota-demo
spec:
 storageClassName: manual
 accessModes:
```

```
- ReadWriteOnce
resources:
 requests:
 storage: 3Gi
```

Create the PersistentVolumeClaim:

```
kubectl apply -f https://k8s.io/examples/admin/resource/quota-objects-pvc.yaml --namespace=quota-object-example
```

Verify that the PersistentVolumeClaim was created:

```
kubectl get persistentvolumeclaims --namespace=quota-object-example
```

The output shows that the PersistentVolumeClaim exists and has status Pending:

| NAME           | STATUS  |
|----------------|---------|
| pvc-quota-demo | Pending |

## Attempt to create a second PersistentVolumeClaim

Here is the configuration file for a second PersistentVolumeClaim:

[admin/resource/quota-objects-pvc-2.yaml](https://k8s.io/examples/admin/resource/quota-objects-pvc-2.yaml)

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: pvc-quota-demo-2
spec:
 storageClassName: manual
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 4Gi
```

Attempt to create the second PersistentVolumeClaim:

```
kubectl apply -f https://k8s.io/examples/admin/resource/quota-objects-pvc-2.yaml --namespace=quota-object-example
```

The output shows that the second PersistentVolumeClaim was not created, because it would have exceeded the quota for the namespace.

```
persistentvolumeclaims "pvc-quota-demo-2" is forbidden:
exceeded quota: object-quota-demo, requested: persistentvolumeclaims=1,
used: persistentvolumeclaims=1, limited: persistentvolumeclaims=1
```

## Notes

These are the strings used to identify API resources that can be constrained by quotas:

| String                   | API Object                   |
|--------------------------|------------------------------|
| "pods"                   | Pod                          |
| "services"               | Service                      |
| "replicationcontrollers" | ReplicationController        |
| "resourcequotas"         | ResourceQuota                |
| "secrets"                | Secret                       |
| "configmaps"             | ConfigMap                    |
| "persistentvolumeclaims" | PersistentVolumeClaim        |
| "services.nodeports"     | Service of type NodePort     |
| "services.loadbalancers" | Service of type LoadBalancer |

## Clean up

Delete your namespace:

```
kubectl delete namespace quota-object-example
```

## What's next

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)
- [Configure a Pod Quota for a Namespace](#)

### For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

# Control CPU Management Policies on the Node

FEATURE STATE: Kubernetes v1.26 [stable]

Kubernetes keeps many aspects of how pods execute on nodes abstracted from the user. This is by design. However, some workloads require stronger guarantees in terms of latency and/or

performance in order to operate acceptably. The kubelet provides methods to enable more complex workload placement policies while keeping the abstraction free from explicit placement directives.

For detailed information on resource management, please refer to the [Resource Management for Pods and Containers](#) documentation.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.26. To check the version, enter kubectl version.

If you are running an older version of Kubernetes, please look at the documentation for the version you are actually running.

## CPU Management Policies

By default, the kubelet uses [CFS quota](#) to enforce pod CPU limits. When the node runs many CPU-bound pods, the workload can move to different CPU cores depending on whether the pod is throttled and which CPU cores are available at scheduling time. Many workloads are not sensitive to this migration and thus work fine without any intervention.

However, in workloads where CPU cache affinity and scheduling latency significantly affect workload performance, the kubelet allows alternative CPU management policies to determine some placement preferences on the node.

## Configuration

The CPU Manager policy is set with the `--cpu-manager-policy` kubelet flag or the `cpuManagerPolicy` field in [KubeletConfiguration](#). There are two supported policies:

- [none](#): the default policy.
- [static](#): allows pods with certain resource characteristics to be granted increased CPU affinity and exclusivity on the node.

The CPU manager periodically writes resource updates through the CRI in order to reconcile in-memory CPU assignments with cgroups. The reconcile frequency is set through a new Kubelet configuration value `--cpu-manager-reconcile-period`. If not specified, it defaults to the same duration as `--node-status-update-frequency`.

The behavior of the static policy can be fine-tuned using the `--cpu-manager-policy-options` flag. The flag takes a comma-separated list of key=value policy options. If you disable the `CPUManagerPolicyOptions` [feature gate](#) then you cannot fine-tune CPU manager policies. In that case, the CPU manager operates only using its default settings.

In addition to the top-level `CPUManagerPolicyOptions` feature gate, the policy options are split into two groups: alpha quality (hidden by default) and beta quality (visible by default). The groups are guarded respectively by the `CPUManagerPolicyAlphaOptions` and `CPUManagerPolicyBetaOptions` feature gates. Diverging from the Kubernetes standard, these feature gates guard groups of options, because it would have been too cumbersome to add a feature gate for each individual option.

## Changing the CPU Manager Policy

Since the CPU manager policy can only be applied when kubelet spawns new pods, simply changing from "none" to "static" won't apply to existing pods. So in order to properly change the CPU manager policy on a node, perform the following steps:

1. [Drain](#) the node.
2. Stop kubelet.
3. Remove the old CPU manager state file. The path to this file is `/var/lib/kubelet/cpu_manager_state` by default. This clears the state maintained by the `CPUManager` so that the `cpu-sets` set up by the new policy won't conflict with it.
4. Edit the kubelet configuration to change the CPU manager policy to the desired value.
5. Start kubelet.

Repeat this process for every node that needs its CPU manager policy changed. Skipping this process will result in kubelet crashlooping with the following error:

```
could not restore state from checkpoint: configured policy "static" differs from state checkpoint policy "none", please drain this node and delete the CPU manager checkpoint file "/var/lib/kubelet/cpu_manager_state" before restarting Kubelet
```

## None policy

The none policy explicitly enables the existing default CPU affinity scheme, providing no affinity beyond what the OS scheduler does automatically. Limits on CPU usage for [Guaranteed pods](#) and [Burstable pods](#) are enforced using CFS quota.

## Static policy

The static policy allows containers in `Guaranteed` pods with integer CPU requests access to exclusive CPUs on the node. This exclusivity is enforced using the [cpuset cgroup controller](#).

### Note:

System services such as the container runtime and the kubelet itself can continue to run on these exclusive CPUs. The exclusivity only extends to other pods.

### Note:

CPU Manager doesn't support offlining and onlining of CPUs at runtime. Also, if the set of online CPUs changes on the node, the node must be drained and CPU manager manually reset by deleting the state file `cpu_manager_state` in the kubelet root directory.

This policy manages a shared pool of CPUs that initially contains all CPUs in the node. The amount of exclusively allocatable CPUs is equal to the total number of CPUs in the node minus

any CPU reservations by the kubelet `--kube-reserved` or `--system-reserved` options. From 1.17, the CPU reservation list can be specified explicitly by kubelet `--reserved-cpus` option. The explicit CPU list specified by `--reserved-cpus` takes precedence over the CPU reservation specified by `--kube-reserved` and `--system-reserved`. CPUs reserved by these options are taken, in integer quantity, from the initial shared pool in ascending order by physical core ID. This shared pool is the set of CPUs on which any containers in BestEffort and Burstable pods run. Containers in Guaranteed pods with fractional CPU requests also run on CPUs in the shared pool. Only containers that are both part of a Guaranteed pod and have integer CPU requests are assigned exclusive CPUs.

**Note:**

The kubelet requires a CPU reservation greater than zero be made using either `--kube-reserved` and/or `--system-reserved` or `--reserved-cpus` when the static policy is enabled. This is because zero CPU reservation would allow the shared pool to become empty.

As Guaranteed pods whose containers fit the requirements for being statically assigned are scheduled to the node, CPUs are removed from the shared pool and placed in the cpuset for the container. CFS quota is not used to bound the CPU usage of these containers as their usage is bound by the scheduling domain itself. In other words, the number of CPUs in the container cpuset is equal to the integer CPU limit specified in the pod spec. This static assignment increases CPU affinity and decreases context switches due to throttling for the CPU-bound workload.

Consider the containers in the following pod specs:

```
spec:
 containers:
 - name: nginx
 image: nginx
```

This pod runs in the BestEffort QoS class because no resource requests or limits are specified. It runs in the shared pool.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 requests:
 memory: "100Mi"
```

This pod runs in the Burstable QoS class because resource requests do not equal limits and the cpu quantity is not specified. It runs in the shared pool.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
```

```
memory: "200Mi"
cpu: "2"
requests:
 memory: "100Mi"
 cpu: "1"
```

This pod runs in the **Burstable QoS** class because resource requests do not equal limits. It runs in the shared pool.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 cpu: "2"
 requests:
 memory: "200Mi"
 cpu: "2"
```

This pod runs in the **Guaranteed QoS** class because requests are equal to limits. And the container's resource limit for the CPU resource is an integer greater than or equal to one. The nginx container is granted 2 exclusive CPUs.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 cpu: "1.5"
 requests:
 memory: "200Mi"
 cpu: "1.5"
```

This pod runs in the **Guaranteed QoS** class because requests are equal to limits. But the container's resource limit for the CPU resource is a fraction. It runs in the shared pool.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 cpu: "2"
```

This pod runs in the **Guaranteed QoS** class because only limits are specified and requests are set equal to limits when not explicitly specified. And the container's resource limit for the CPU resource is an integer greater than or equal to one. The nginx container is granted 2 exclusive CPUs.



## Static policy options

You can toggle groups of options on and off based upon their maturity level using the following feature gates:

- `CPUManagerPolicyBetaOptions` default enabled. Disable to hide beta-level options.
- `CPUManagerPolicyAlphaOptions` default disabled. Enable to show alpha-level options. You will still have to enable each option using the `CPUManagerPolicyOptions` kubelet option.

The following policy options exist for the static `CPUManager` policy:

- `full-pcpus-only` (beta, visible by default) (1.22 or higher)
- `distribute-cpus-across-numa` (alpha, hidden by default) (1.23 or higher)
- `align-by-socket` (alpha, hidden by default) (1.25 or higher)

If the `full-pcpus-only` policy option is specified, the static policy will always allocate full physical cores. By default, without this option, the static policy allocates CPUs using a topology-aware best-fit allocation. On SMT enabled systems, the policy can allocate individual virtual cores, which correspond to hardware threads. This can lead to different containers sharing the same physical cores; this behaviour in turn contributes to the [noisy neighbours problem](#). With the option enabled, the pod will be admitted by the kubelet only if the CPU request of all its containers can be fulfilled by allocating full physical cores. If the pod does not pass the admission, it will be put in Failed state with the message `SMTAlignmentError`.

If the `distribute-cpus-across-numa` policy option is specified, the static policy will evenly distribute CPUs across NUMA nodes in cases where more than one NUMA node is required to satisfy the allocation. By default, the `CPUManager` will pack CPUs onto one NUMA node until it is filled, with any remaining CPUs simply spilling over to the next NUMA node. This can cause undesired bottlenecks in parallel code relying on barriers (and similar synchronization primitives), as this type of code tends to run only as fast as its slowest worker (which is slowed down by the fact that fewer CPUs are available on at least one NUMA node). By distributing CPUs evenly across NUMA nodes, application developers can more easily ensure that no single worker suffers from NUMA effects more than any other, improving the overall performance of these types of applications.

If the `align-by-socket` policy option is specified, CPUs will be considered aligned at the socket boundary when deciding how to allocate CPUs to a container. By default, the `CPUManager` aligns CPU allocations at the NUMA boundary, which could result in performance degradation if CPUs need to be pulled from more than one NUMA node to satisfy the allocation. Although it tries to ensure that all CPUs are allocated from the *minimum* number of NUMA nodes, there is no guarantee that those NUMA nodes will be on the same socket. By directing the `CPUManager` to explicitly align CPUs at the socket boundary rather than the NUMA boundary, we are able to avoid such issues. Note, this policy option is not compatible with `TopologyManager` `single-numa-node` policy and does not apply to hardware where the number of sockets is greater than number of NUMA nodes.

The `full-pcpus-only` option can be enabled by adding `full-pcpus-only=true` to the `CPUManager` policy options. Likewise, the `distribute-cpus-across-numa` option can be enabled by adding `distribute-cpus-across-numa=true` to the `CPUManager` policy options. When both are set, they are "additive" in the sense that CPUs will be distributed across NUMA nodes in chunks of full-pcpus rather than individual cores. The `align-by-socket` policy option can be enabled by adding `align-by-socket=true` to the `CPUManager` policy options. It is also additive to the `full-pcpus-only` and `distribute-cpus-across-numa` policy options.

# Control Topology Management Policies on a node

FEATURE STATE: Kubernetes v1.27 [stable]

An increasing number of systems leverage a combination of CPUs and hardware accelerators to support latency-critical execution and high-throughput parallel computation. These include workloads in fields such as telecommunications, scientific computing, machine learning, financial services and data analytics. Such hybrid systems comprise a high performance environment.

In order to extract the best performance, optimizations related to CPU isolation, memory and device locality are required. However, in Kubernetes, these optimizations are handled by a disjoint set of components.

*Topology Manager* is a Kubelet component that aims to coordinate the set of components that are responsible for these optimizations.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercoda](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.18. To check the version, enter `kubectl version`.

## How Topology Manager Works

Prior to the introduction of Topology Manager, the CPU and Device Manager in Kubernetes make resource allocation decisions independently of each other. This can result in undesirable allocations on multiple-socketed systems, performance/latency sensitive applications will suffer due to these undesirable allocations. Undesirable in this case meaning for example, CPUs and devices being allocated from different NUMA Nodes thus, incurring additional latency.

The Topology Manager is a Kubelet component, which acts as a source of truth so that other Kubelet components can make topology aligned resource allocation choices.

The Topology Manager provides an interface for components, called *Hint Providers*, to send and receive topology information. Topology Manager has a set of node level policies which are explained below.

The Topology manager receives Topology information from the *Hint Providers* as a bitmask denoting NUMA Nodes available and a preferred allocation indication. The Topology Manager policies perform a set of operations on the hints provided and converge on the hint determined by the policy to give the optimal result, if an undesirable hint is stored the preferred field for

the hint will be set to false. In the current policies preferred is the narrowest preferred mask. The selected hint is stored as part of the Topology Manager. Depending on the policy configured the pod can be accepted or rejected from the node based on the selected hint. The hint is then stored in the Topology Manager for use by the *Hint Providers* when making the resource allocation decisions.

## Topology Manager Scopes and Policies

The Topology Manager currently:

- Aligns Pods of all QoS classes.
- Aligns the requested resources that Hint Provider provides topology hints for.

If these conditions are met, the Topology Manager will align the requested resources.

In order to customise how this alignment is carried out, the Topology Manager provides two distinct knobs: scope and policy.

The scope defines the granularity at which you would like resource alignment to be performed (e.g. at the pod or container level). And the policy defines the actual strategy used to carry out the alignment (e.g. best-effort, restricted, single-numa-node, etc.). Details on the various scopes and policies available today can be found below.

### Note:

To align CPU resources with other requested resources in a Pod Spec, the CPU Manager should be enabled and proper CPU Manager policy should be configured on a Node. See [control CPU Management Policies](#).

### Note:

To align memory (and hugepages) resources with other requested resources in a Pod Spec, the Memory Manager should be enabled and proper Memory Manager policy should be configured on a Node. Examine [Memory Manager](#) documentation.

## Topology Manager Scopes

The Topology Manager can deal with the alignment of resources in a couple of distinct scopes:

- container (default)
- pod

Either option can be selected at a time of the kubelet startup, with `--topology-manager-scope` flag.

### container scope

The container scope is used by default.

Within this scope, the Topology Manager performs a number of sequential resource alignments, i.e., for each container (in a pod) a separate alignment is computed. In other words, there is no notion of grouping the containers to a specific set of NUMA nodes, for this particular scope. In

effect, the Topology Manager performs an arbitrary alignment of individual containers to NUMA nodes.

The notion of grouping the containers was endorsed and implemented on purpose in the following scope, for example the pod scope.

## pod scope

To select the pod scope, start the kubelet with the command line option `--topology-manager-scope=pod`.

This scope allows for grouping all containers in a pod to a common set of NUMA nodes. That is, the Topology Manager treats a pod as a whole and attempts to allocate the entire pod (all containers) to either a single NUMA node or a common set of NUMA nodes. The following examples illustrate the alignments produced by the Topology Manager on different occasions:

- all containers can be and are allocated to a single NUMA node;
- all containers can be and are allocated to a shared set of NUMA nodes.

The total amount of particular resource demanded for the entire pod is calculated according to [effective requests/limits](#) formula, and thus, this total value is equal to the maximum of:

- the sum of all app container requests,
- the maximum of init container requests,

for a resource.

Using the pod scope in tandem with single-numa-node Topology Manager policy is specifically valuable for workloads that are latency sensitive or for high-throughput applications that perform IPC. By combining both options, you are able to place all containers in a pod onto a single NUMA node; hence, the inter-NUMA communication overhead can be eliminated for that pod.

In the case of single-numa-node policy, a pod is accepted only if a suitable set of NUMA nodes is present among possible allocations. Reconsider the example above:

- a set containing only a single NUMA node - it leads to pod being admitted,
- whereas a set containing more NUMA nodes - it results in pod rejection (because instead of one NUMA node, two or more NUMA nodes are required to satisfy the allocation).

To recap, Topology Manager first computes a set of NUMA nodes and then tests it against Topology Manager policy, which either leads to the rejection or admission of the pod.

## Topology Manager Policies

Topology Manager supports four allocation policies. You can set a policy via a Kubelet flag, `--topology-manager-policy`. There are four supported policies:

- none (default)
- best-effort
- restricted
- single-numa-node

**Note:**

If Topology Manager is configured with the **pod** scope, the container, which is considered by the policy, is reflecting requirements of the entire pod, and thus each container from the pod will result with **the same** topology alignment decision.

**none policy**

This is the default policy and does not perform any topology alignment.

**best-effort policy**

For each container in a Pod, the kubelet, with best-effort topology management policy, calls each Hint Provider to discover their resource availability. Using this information, the Topology Manager stores the preferred NUMA Node affinity for that container. If the affinity is not preferred, Topology Manager will store this and admit the pod to the node anyway.

The *Hint Providers* can then use this information when making the resource allocation decision.

**restricted policy**

For each container in a Pod, the kubelet, with restricted topology management policy, calls each Hint Provider to discover their resource availability. Using this information, the Topology Manager stores the preferred NUMA Node affinity for that container. If the affinity is not preferred, Topology Manager will reject this pod from the node. This will result in a pod in a Terminated state with a pod admission failure.

Once the pod is in a Terminated state, the Kubernetes scheduler will **not** attempt to reschedule the pod. It is recommended to use a ReplicaSet or Deployment to trigger a redeploy of the pod. An external control loop could be also implemented to trigger a redeployment of pods that have the Topology Affinity error.

If the pod is admitted, the *Hint Providers* can then use this information when making the resource allocation decision.

**single-numa-node policy**

For each container in a Pod, the kubelet, with single-numa-node topology management policy, calls each Hint Provider to discover their resource availability. Using this information, the Topology Manager determines if a single NUMA Node affinity is possible. If it is, Topology Manager will store this and the *Hint Providers* can then use this information when making the resource allocation decision. If, however, this is not possible then the Topology Manager will reject the pod from the node. This will result in a pod in a Terminated state with a pod admission failure.

Once the pod is in a Terminated state, the Kubernetes scheduler will **not** attempt to reschedule the pod. It is recommended to use a Deployment with replicas to trigger a redeploy of the Pod. An external control loop could be also implemented to trigger a redeployment of pods that have the Topology Affinity error.

## Topology manager policy options

Support for the Topology Manager policy options requires `TopologyManagerPolicyOptions` [feature gate](#) to be enabled (it is enabled by default).

You can toggle groups of options on and off based upon their maturity level using the following feature gates:

- `TopologyManagerPolicyBetaOptions` default enabled. Enable to show beta-level options.
- `TopologyManagerPolicyAlphaOptions` default disabled. Enable to show alpha-level options.

You will still have to enable each option using the `TopologyManagerPolicyOptions` kubelet option.

The following policy options exists:

- `prefer-closest-numa-nodes` (beta, visible by default; `TopologyManagerPolicyOptions` and `TopologyManagerPolicyBetaOptions` feature gates have to be enabled). The `prefer-closest-numa-nodes` policy option is beta in Kubernetes 1.30.

If the `prefer-closest-numa-nodes` policy option is specified, the best-effort and restricted policies will favor sets of NUMA nodes with shorter distance between them when making admission decisions. You can enable this option by adding `prefer-closest-numa-nodes=true` to the Topology Manager policy options. By default, without this option, Topology Manager aligns resources on either a single NUMA node or the minimum number of NUMA nodes (in cases where more than one NUMA node is required). However, the TopologyManager is not aware of NUMA distances and does not take them into account when making admission decisions. This limitation surfaces in multi-socket, as well as single-socket multi NUMA systems, and can cause significant performance degradation in latency-critical execution and high-throughput applications if the Topology Manager decides to align resources on non-adjacent NUMA nodes.

## Pod Interactions with Topology Manager Policies

Consider the containers in the following pod specs:

```
spec:
 containers:
 - name: nginx
 image: nginx
```

This pod runs in the BestEffort QoS class because no resource requests or limits are specified.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 requests:
 memory: "100Mi"
```

This pod runs in the Burstable QoS class because requests are less than limits.

If the selected policy is anything other than none, Topology Manager would consider these Pod specifications. The Topology Manager would consult the Hint Providers to get topology hints. In the case of the static, the CPU Manager policy would return default topology hint, because these Pods do not have explicitly request CPU resources.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 cpu: "2"
 example.com/device: "1"
 requests:
 memory: "200Mi"
 cpu: "2"
 example.com/device: "1"
```

This pod with integer CPU request runs in the Guaranteed QoS class because requests are equal to limits.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 cpu: "300m"
 example.com/device: "1"
 requests:
 memory: "200Mi"
 cpu: "300m"
 example.com/device: "1"
```

This pod with sharing CPU request runs in the Guaranteed QoS class because requests are equal to limits.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 example.com/deviceA: "1"
 example.com/deviceB: "1"
 requests:
 example.com/deviceA: "1"
 example.com/deviceB: "1"
```

This pod runs in the BestEffort QoS class because there are no CPU and memory requests.

The Topology Manager would consider the above pods. The Topology Manager would consult the Hint Providers, which are CPU and Device Manager to get topology hints for the pods.

In the case of the Guaranteed pod with integer CPU request, the static CPU Manager policy would return topology hints relating to the exclusive CPU and the Device Manager would send back hints for the requested device.

In the case of the Guaranteed pod with sharing CPU request, the static CPU Manager policy would return default topology hint as there is no exclusive CPU request and the Device Manager would send back hints for the requested device.

In the above two cases of the Guaranteed pod, the none CPU Manager policy would return default topology hint.

In the case of the BestEffort pod, the static CPU Manager policy would send back the default topology hint as there is no CPU request and the Device Manager would send back the hints for each of the requested devices.

Using this information the Topology Manager calculates the optimal hint for the pod and stores this information, which will be used by the Hint Providers when they are making their resource assignments.

## Known Limitations

1. The maximum number of NUMA nodes that Topology Manager allows is 8. With more than 8 NUMA nodes there will be a state explosion when trying to enumerate the possible NUMA affinities and generating their hints.
2. The scheduler is not topology-aware, so it is possible to be scheduled on a node and then fail on the node due to the Topology Manager.

# Customizing DNS Service

This page explains how to configure your DNS [Pod\(s\)](#) and customize the DNS resolution process in your cluster.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your cluster must be running the CoreDNS add-on.

Your Kubernetes server must be at or later than version v1.12. To check the version, enter kubectl version.



# Introduction

DNS is a built-in Kubernetes service launched automatically using the *addon manager* [cluster add-on](#).

## Note:

The CoreDNS Service is named kube-dns in the metadata.name field.

The intent is to ensure greater interoperability with workloads that relied on the legacy kube-dns Service name to resolve addresses internal to the cluster. Using a Service named kube-dns abstracts away the implementation detail of which DNS provider is running behind that common name.

If you are running CoreDNS as a Deployment, it will typically be exposed as a Kubernetes Service with a static IP address. The kubelet passes DNS resolver information to each container with the `--cluster-dns=<dns-service-ip>` flag.

DNS names also need domains. You configure the local domain in the kubelet with the flag `--cluster-domain=<default-local-domain>`.

The DNS server supports forward lookups (A and AAAA records), port lookups (SRV records), reverse IP address lookups (PTR records), and more. For more information, see [DNS for Services and Pods](#).

If a Pod's dnsPolicy is set to default, it inherits the name resolution configuration from the node that the Pod runs on. The Pod's DNS resolution should behave the same as the node. But see [Known issues](#).

If you don't want this, or if you want a different DNS config for pods, you can use the kubelet's `--resolv-conf` flag. Set this flag to `""` to prevent Pods from inheriting DNS. Set it to a valid file path to specify a file other than `/etc/resolv.conf` for DNS inheritance.

## CoreDNS

CoreDNS is a general-purpose authoritative DNS server that can serve as cluster DNS, complying with the [DNS specifications](#).

### CoreDNS ConfigMap options

CoreDNS is a DNS server that is modular and pluggable, with plugins adding new functionalities. The CoreDNS server can be configured by maintaining a [Corefile](#), which is the CoreDNS configuration file. As a cluster administrator, you can modify the [ConfigMap](#) for the CoreDNS Corefile to change how DNS service discovery behaves for that cluster.

In Kubernetes, CoreDNS is installed with the following default Corefile configuration:

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: coredns
 namespace: kube-system
data:
```

```
Corefile: |
.:53 {
 errors
 health {
 lameduck 5s
 }
 ready
 kubernetes cluster.local in-addr.arpa ip6.arpa {
 pods insecure
 fallthrough in-addr.arpa ip6.arpa
 ttl 30
 }
 prometheus :9153
 forward . /etc/resolv.conf
 cache 30
 loop
 reload
 loadbalance
}
```

The Corefile configuration includes the following [plugins](#) of CoreDNS:

- [errors](#): Errors are logged to stdout.
- [health](#): Health of CoreDNS is reported to <http://localhost:8080/health>. In this extended syntax lameduck will make the process unhealthy then wait for 5 seconds before the process is shut down.
- [ready](#): An HTTP endpoint on port 8181 will return 200 OK, when all plugins that are able to signal readiness have done so.
- [kubernetes](#): CoreDNS will reply to DNS queries based on IP of the Services and Pods. You can find [more details](#) about this plugin on the CoreDNS website.
  - ttl allows you to set a custom TTL for responses. The default is 5 seconds. The minimum TTL allowed is 0 seconds, and the maximum is capped at 3600 seconds. Setting TTL to 0 will prevent records from being cached.
  - The pods insecure option is provided for backward compatibility with kube-dns.
  - You can use the pods verified option, which returns an A record only if there exists a pod in the same namespace with a matching IP.
  - The pods disabled option can be used if you don't use pod records.
- [prometheus](#): Metrics of CoreDNS are available at <http://localhost:9153/metrics> in the [Prometheus](#) format (also known as OpenMetrics).
- [forward](#): Any queries that are not within the Kubernetes cluster domain are forwarded to predefined resolvers (/etc/resolv.conf).
- [cache](#): This enables a frontend cache.
- [loop](#): Detects simple forwarding loops and halts the CoreDNS process if a loop is found.
- [reload](#): Allows automatic reload of a changed Corefile. After you edit the ConfigMap configuration, allow two minutes for your changes to take effect.
- [loadbalance](#): This is a round-robin DNS loadbalancer that randomizes the order of A, AAAA, and MX records in the answer.

You can modify the default CoreDNS behavior by modifying the ConfigMap.

## Configuration of Stub-domain and upstream nameserver using CoreDNS

CoreDNS has the ability to configure stub-domains and upstream nameservers using the [forward plugin](#).

### Example

If a cluster operator has a [Consul](#) domain server located at "10.150.0.1", and all Consul names have the suffix ".consul.local". To configure it in CoreDNS, the cluster administrator creates the following stanza in the CoreDNS ConfigMap.

```
consul.local:53 {
 errors
 cache 30
 forward . 10.150.0.1
}
```

To explicitly force all non-cluster DNS lookups to go through a specific nameserver at 172.16.0.1, point the forward to the nameserver instead of /etc/resolv.conf

```
forward . 172.16.0.1
```

The final ConfigMap along with the default Corefile configuration looks like:

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: coredns
 namespace: kube-system
data:
 Corefile: |
 .:53 {
 errors
 health
 kubernetes cluster.local in-addr.arpa ip6.arpa {
 pods insecure
 fallthrough in-addr.arpa ip6.arpa
 }
 prometheus :9153
 forward . 172.16.0.1
 cache 30
 loop
 reload
 loadbalance
 }
 consul.local:53 {
 errors
 cache 30
 forward . 10.150.0.1
 }
```

## Note:

CoreDNS does not support FQDNs for stub-domains and nameservers (eg: "ns.foo.com"). During translation, all FQDN nameservers will be omitted from the CoreDNS config.

## What's next

- Read [Debugging DNS Resolution](#)

# Debugging DNS Resolution

This page provides hints on diagnosing DNS problems.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your cluster must be configured to use the CoreDNS [addon](#) or its precursor, kube-dns.

Your Kubernetes server must be at or later than version v1.6. To check the version, enter kubectl version.

## Create a simple Pod to use as a test environment

[admin/dns/dnsutils.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: dnsutils
 namespace: default
spec:
 containers:
 - name: dnsutils
 image: registry.k8s.io/e2e-test-images/jessie-dnsutils:1.3
 command:
 - sleep
 - "infinity"
 imagePullPolicy: IfNotPresent
 restartPolicy: Always
```

**Note:**

This example creates a pod in the default namespace. DNS name resolution for services depends on the namespace of the pod. For more information, review [DNS for Services and Pods](#).

Use that manifest to create a Pod:

```
kubectl apply -f https://k8s.io/examples/admin/dns/dnsutils.yaml
```

```
pod/dnsutils created
```

...and verify its status:

```
kubectl get pods dnsutils
```

| NAME     | READY | STATUS  | RESTARTS | AGE         |
|----------|-------|---------|----------|-------------|
| dnsutils | 1/1   | Running | 0        | <some-time> |

Once that Pod is running, you can exec nslookup in that environment. If you see something like the following, DNS is working correctly.

```
kubectl exec -i -t dnsutils -- nslookup kubernetes.default
```

```
Server: 10.0.0.10
Address 1: 10.0.0.10
```

```
Name: kubernetes.default
Address 1: 10.0.0.1
```

If the nslookup command fails, check the following:

## Check the local DNS configuration first

Take a look inside the resolv.conf file. (See [Customizing DNS Service](#) and [Known issues](#) below for more information)

```
kubectl exec -ti dnsutils -- cat /etc/resolv.conf
```

Verify that the search path and name server are set up like the following (note that search path may vary for different cloud providers):

```
search default.svc.cluster.local svc.cluster.local cluster.local google.internal
c.gce_project_id.internal
nameserver 10.0.0.10
options ndots:5
```

Errors such as the following indicate a problem with the CoreDNS (or kube-dns) add-on or with associated Services:

```
kubectl exec -i -t dnsutils -- nslookup kubernetes.default
```

```
Server: 10.0.0.10
Address 1: 10.0.0.10
```

```
nslookup: can't resolve 'kubernetes.default'
```

or

```
kubectl exec -i -t dnsutils -- nslookup kubernetes.default
```

```
Server: 10.0.0.10
```

```
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local
```

```
nslookup: can't resolve 'kubernetes.default'
```

## Check if the DNS pod is running

Use the `kubectl get pods` command to verify that the DNS pod is running.

```
kubectl get pods --namespace=kube-system -l k8s-app=kube-dns
```

| NAME                     | READY | STATUS  | RESTARTS | AGE |
|--------------------------|-------|---------|----------|-----|
| ...                      |       |         |          |     |
| coredns-7b96bf9f76-5hsxb | 1/1   | Running | 0        | 1h  |
| coredns-7b96bf9f76-mvmmt | 1/1   | Running | 0        | 1h  |
| ...                      |       |         |          |     |

### Note:

The value for label `k8s-app` is `kube-dns` for both CoreDNS and kube-dns deployments.

If you see that no CoreDNS Pod is running or that the Pod has failed/completed, the DNS add-on may not be deployed by default in your current environment and you will have to deploy it manually.

## Check for errors in the DNS pod

Use the `kubectl logs` command to see logs for the DNS containers.

For CoreDNS:

```
kubectl logs --namespace=kube-system -l k8s-app=kube-dns
```

Here is an example of a healthy CoreDNS log:

```
..:53
2018/08/15 14:37:17 [INFO] CoreDNS-1.2.2
2018/08/15 14:37:17 [INFO] linux/amd64, go1.10.3, 2e322f6
CoreDNS-1.2.2
linux/amd64, go1.10.3, 2e322f6
2018/08/15 14:37:17 [INFO] plugin/reload: Running configuration MD5 =
24e6c59e83ce706f07bcc82c31b1ea1c
```

See if there are any suspicious or unexpected messages in the logs.

## Is DNS service up?

Verify that the DNS service is up by using the `kubectl get service` command.

```
kubectl get svc --namespace=kube-system
```

| NAME     | TYPE      | CLUSTER-IP | EXTERNAL-IP | PORT(S)       | AGE |
|----------|-----------|------------|-------------|---------------|-----|
| ...      |           |            |             |               |     |
| kube-dns | ClusterIP | 10.0.0.10  | <none>      | 53/UDP,53/TCP | 1h  |
| ...      |           |            |             |               |     |

### Note:

The service name is `kube-dns` for both CoreDNS and `kube-dns` deployments.

If you have created the Service or in the case it should be created by default but it does not appear, see [debugging Services](#) for more information.

## Are DNS endpoints exposed?

You can verify that DNS endpoints are exposed by using the `kubectl get endpoints` command.

```
kubectl get endpoints kube-dns --namespace=kube-system
```

| NAME     | ENDPOINTS                     | AGE |
|----------|-------------------------------|-----|
| kube-dns | 10.180.3.17:53,10.180.3.17:53 | 1h  |

If you do not see the endpoints, see the endpoints section in the [debugging Services](#) documentation.

For additional Kubernetes DNS examples, see the [cluster-dns examples](#) in the Kubernetes GitHub repository.

## Are DNS queries being received/processed?

You can verify if queries are being received by CoreDNS by adding the `log` plugin to the CoreDNS configuration (aka Corefile). The CoreDNS Corefile is held in a [ConfigMap](#) named `coredns`. To edit it, use the command:

```
kubectl -n kube-system edit configmap coredns
```

Then add `log` in the Corefile section per the example below:

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: coredns
 namespace: kube-system
data:
 Corefile: |
 .:53 {
 log
 errors
```

```

health
kubernetes cluster.local in-addr.arpa ip6.arpa {
 pods insecure
 upstream
 fallthrough in-addr.arpa ip6.arpa
}
prometheus :9153
forward . /etc/resolv.conf
cache 30
loop
reload
loadbalance
}

```

After saving the changes, it may take up to minute or two for Kubernetes to propagate these changes to the CoreDNS pods.

Next, make some queries and view the logs per the sections above in this document. If CoreDNS pods are receiving the queries, you should see them in the logs.

Here is an example of a query in the log:

```

.:53
2018/08/15 14:37:15 [INFO] CoreDNS-1.2.0
2018/08/15 14:37:15 [INFO] linux/amd64, go1.10.3, 2e322f6
CoreDNS-1.2.0
linux/amd64, go1.10.3, 2e322f6
2018/09/07 15:29:04 [INFO] plugin/reload: Running configuration MD5 =
162475cdf272d8aa601e6fe67a6ad42f
2018/09/07 15:29:04 [INFO] Reloading complete
172.17.0.18:41675 - [07/Sep/2018:15:29:11 +0000] 59925 "A IN kubernetes.default.svc.cluster.local.
udp 54 false 512" NOERROR qr,aa,rd,ra 106 0.000066649s

```

## Does CoreDNS have sufficient permissions?

CoreDNS must be able to list [service](#) and [endpoint](#) related resources to properly resolve service names.

Sample error message:

```

2022-03-18T07:12:15.699431183Z [INFO] 10.96.144.227:52299 - 3686 "A IN
serverproxy.contoso.net.cluster.local. udp 52 false 512" SERVFAIL qr,aa,rd 145 0.000091221s

```

First, get the current ClusterRole of system:coredns:

```
kubectl describe clusterrole system:coredns -n kube-system
```

Expected output:

| PolicyRule: |                   |                |              |
|-------------|-------------------|----------------|--------------|
| Resources   | Non-Resource URLs | Resource Names | Verbs        |
| -----       | -----             | -----          | ----         |
| endpoints   | []                | []             | [list watch] |
| namespaces  | []                | []             | [list watch] |



```

pods [] [] [list watch]
services [] [] [list watch]
endpointslices.discovery.k8s.io [] [] [list watch]
```

If any permissions are missing, edit the ClusterRole to add them:

```
kubectcl edit clusterrole system:coredns -n kube-system
```

Example insertion of EndpointSlices permissions:

```
...
- apiGroups:
 - discovery.k8s.io
resources:
- endpointslices
verbs:
- list
- watch
...
```

## Are you in the right namespace for the service?

DNS queries that don't specify a namespace are limited to the pod's namespace.

If the namespace of the pod and service differ, the DNS query must include the namespace of the service.

This query is limited to the pod's namespace:

```
kubectcl exec -i -t dnsutils -- nslookup <service-name>
```

This query specifies the namespace:

```
kubectcl exec -i -t dnsutils -- nslookup <service-name>.<namespace>
```

To learn more about name resolution, see [DNS for Services and Pods](#).

## Known issues

Some Linux distributions (e.g. Ubuntu) use a local DNS resolver by default (systemd-resolved). Systemd-resolved moves and replaces /etc/resolv.conf with a stub file that can cause a fatal forwarding loop when resolving names in upstream servers. This can be fixed manually by using kubelet's --resolv-conf flag to point to the correct resolv.conf (With systemd-resolved, this is /run/systemd/resolve/resolv.conf). kubeadm automatically detects systemd-resolved, and adjusts the kubelet flags accordingly.

Kubernetes installs do not configure the nodes' resolv.conf files to use the cluster DNS by default, because that process is inherently distribution-specific. This should probably be implemented eventually.

Linux's libc (a.k.a. glibc) has a limit for the DNS nameserver records to 3 by default and Kubernetes needs to consume 1 nameserver record. This means that if a local installation already uses 3 nameservers, some of those entries will be lost. To work around this limit, the

node can run `dnsmasq`, which will provide more `nameserver` entries. You can also use kubelet's `--resolv-conf` flag.

If you are using Alpine version 3.17 or earlier as your base image, DNS may not work properly due to a design issue with Alpine. Until `musl` version 1.24 didn't include TCP fallback to the DNS stub resolver meaning any DNS call above 512 bytes would fail. Please upgrade your images to Alpine version 3.18 or above.

## What's next

- See [Autoscaling the DNS Service in a Cluster](#).
- Read [DNS for Services and Pods](#)

# Declare Network Policy

This document helps you get started using the Kubernetes [NetworkPolicy API](#) to declare network policies that govern how pods communicate with each other.

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.8. To check the version, enter `kubectl version`.

Make sure you've configured a network provider with network policy support. There are a number of network providers that support NetworkPolicy, including:

- [Antrea](#)
- [Calico](#)
- [Cilium](#)
- [Kube-router](#)
- [Romana](#)
- [Weave Net](#)

## Create an `nginx` deployment and expose it via a service

To see how Kubernetes network policy works, start off by creating an `nginx` Deployment.

```
kubectl create deployment nginx --image=nginx
```

```
deployment.apps/nginx created
```

Expose the Deployment through a Service called nginx.

```
kubectl expose deployment nginx --port=80
```

```
service/nginx exposed
```

The above commands create a Deployment with an nginx Pod and expose the Deployment through a Service named nginx. The nginx Pod and Deployment are found in the default namespace.

```
kubectl get svc,pod
```

| NAME               | CLUSTER-IP  | EXTERNAL-IP | PORT(S) | AGE |
|--------------------|-------------|-------------|---------|-----|
| service/kubernetes | 10.100.0.1  | <none>      | 443/TCP | 46m |
| service/nginx      | 10.100.0.16 | <none>      | 80/TCP  | 33s |

| NAME                      | READY | STATUS  | RESTARTS | AGE |
|---------------------------|-------|---------|----------|-----|
| pod/nginx-701339712-e0qfq | 1/1   | Running | 0        | 35s |

## Test the service by accessing it from another Pod

You should be able to access the new nginx service from other Pods. To access the nginx Service from another Pod in the default namespace, start a busybox container:

```
kubectl run busybox --rm -ti --image=busybox:1.28 -- /bin/sh
```

In your shell, run the following command:

```
wget --spider --timeout=1 nginx
```

```
Connecting to nginx (10.100.0.16:80)
remote file exists
```

## Limit access to the nginx service

To limit the access to the nginx service so that only Pods with the label `access: true` can query it, create a NetworkPolicy object as follows:

[service/networking/nginx-policy.yaml](#)

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: access-nginx
spec:
 podSelector:
 matchLabels:
 app: nginx
 ingress:
```

```
- from:
- podSelector:
 matchLabels:
 access: "true"
```

The name of a NetworkPolicy object must be a valid [DNS subdomain name](#).

### Note:

NetworkPolicy includes a `podSelector` which selects the grouping of Pods to which the policy applies. You can see this policy selects Pods with the label `app=nginx`. The label was automatically added to the Pod in the nginx Deployment. An empty `podSelector` selects all pods in the namespace.

## Assign the policy to the service

Use `kubectl` to create a NetworkPolicy from the above `nginx-policy.yaml` file:

```
kubectl apply -f https://k8s.io/examples/service/networking/nginx-policy.yaml
```

```
networkpolicy.networking.k8s.io/access-nginx created
```

## Test access to the service when access label is not defined

When you attempt to access the nginx Service from a Pod without the correct labels, the request times out:

```
kubectl run busybox --rm -ti --image=busybox:1.28 -- /bin/sh
```

In your shell, run the command:

```
wget --spider --timeout=1 nginx
```

```
Connecting to nginx (10.100.0.16:80)
wget: download timed out
```

## Define access label and test again

You can create a Pod with the correct labels to see that the request is allowed:

```
kubectl run busybox --rm -ti --labels="access=true" --image=busybox:1.28 -- /bin/sh
```

In your shell, run the command:

```
wget --spider --timeout=1 nginx
```

```
Connecting to nginx (10.100.0.16:80)
remote file exists
```

# Developing Cloud Controller Manager

FEATURE STATE: Kubernetes v1.11 [beta]

The cloud-controller-manager is a Kubernetes [control plane](#) component that embeds cloud-specific control logic. The [cloud controller manager](#) lets you link your cluster into your cloud provider's API, and separates out the components that interact with that cloud platform from components that only interact with your cluster.

By decoupling the interoperability logic between Kubernetes and the underlying cloud infrastructure, the cloud-controller-manager component enables cloud providers to release features at a different pace compared to the main Kubernetes project.

## Background

Since cloud providers develop and release at a different pace compared to the Kubernetes project, abstracting the provider-specific code to the cloud-controller-manager binary allows cloud vendors to evolve independently from the core Kubernetes code.

The Kubernetes project provides skeleton cloud-controller-manager code with Go interfaces to allow you (or your cloud provider) to plug in your own implementations. This means that a cloud provider can implement a cloud-controller-manager by importing packages from Kubernetes core; each cloudprovider will register their own code by calling `cloudprovider.RegisterCloudProvider` to update a global variable of available cloud providers.

## Developing

### Out of tree

To build an out-of-tree cloud-controller-manager for your cloud:

1. Create a go package with an implementation that satisfies [cloudprovider.Interface](#).
2. Use [main.go in cloud-controller-manager](#) from Kubernetes core as a template for your `main.go`. As mentioned above, the only difference should be the cloud package that will be imported.
3. Import your cloud package in `main.go`, ensure your package has an `init` block to run [cloudprovider.RegisterCloudProvider](#).

Many cloud providers publish their controller manager code as open source. If you are creating a new cloud-controller-manager from scratch, you could take an existing out-of-tree cloud controller manager as your starting point.

### In tree

For in-tree cloud providers, you can run the in-tree cloud controller manager as a [DaemonSet](#) in your cluster. See [Cloud Controller Manager Administration](#) for more details.

# Enable Or Disable A Kubernetes API

This page shows how to enable or disable an API version from your cluster's [control plane](#).

Specific API versions can be turned on or off by passing `--runtime-config=api/<version>` as a command line argument to the API server. The values for this argument are a comma-separated list of API versions. Later values override earlier values.

The `runtime-config` command line argument also supports 2 special keys:

- `api/all`, representing all known APIs
- `api/legacy`, representing only legacy APIs. Legacy APIs are any APIs that have been explicitly [deprecated](#).

For example, to turn off all API versions except v1, pass `--runtime-config=api/all=false,api/v1=true` to the kube-apiserver.

## What's next

Read the [full documentation](#) for the kube-apiserver component.

# Encrypting Confidential Data at Rest

All of the APIs in Kubernetes that let you write persistent API resource data support at-rest encryption. For example, you can enable at-rest encryption for [Secrets](#). This at-rest encryption is additional to any system-level encryption for the etcd cluster or for the filesystem(s) on hosts where you are running the kube-apiserver.

This page shows how to enable and configure encryption of API data at rest.

### Note:

This task covers encryption for resource data stored using the [Kubernetes API](#). For example, you can encrypt Secret objects, including the key-value data they contain.

If you want to encrypt data in filesystems that are mounted into containers, you instead need to either:

- use a storage integration that provides encrypted [volumes](#)
- encrypt the data within your own application

## Before you begin

- You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:
  - [Killercodea](#)

- [Play with Kubernetes](#)

- This task assumes that you are running the Kubernetes API server as a [static pod](#) on each control plane node.
- Your cluster's control plane **must** use etcd v3.x (major version 3, any minor version).
- To encrypt a custom resource, your cluster must be running Kubernetes v1.26 or newer.
- To use a wildcard to match resources, your cluster must be running Kubernetes v1.27 or newer.

To check the version, enter `kubectl version`.

## Determine whether encryption at rest is already enabled

By default, the API server stores plain-text representations of resources into etcd, with no at-rest encryption.

The `kube-apiserver` process accepts an argument `--encryption-provider-config` that specifies a path to a configuration file. The contents of that file, if you specify one, control how Kubernetes API data is encrypted in etcd. If you are running the `kube-apiserver` without the `--encryption-provider-config` command line argument, you do not have encryption at rest enabled. If you are running the `kube-apiserver` with the `--encryption-provider-config` command line argument, and the file that it references specifies the identity provider as the first encryption provider in the list, then you do not have at-rest encryption enabled (**the default identity provider does not provide any confidentiality protection.**)

If you are running the `kube-apiserver` with the `--encryption-provider-config` command line argument, and the file that it references specifies a provider other than identity as the first encryption provider in the list, then you already have at-rest encryption enabled. However, that check does not tell you whether a previous migration to encrypted storage has succeeded. If you are not sure, see [ensure all relevant data are encrypted](#).

## Understanding the encryption at rest configuration

```

#
CAUTION: this is an example configuration.
Do not use this for your own cluster!
#
apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
- resources:
 - secrets
 - configmaps
 - pandas.awesome.bears.example # a custom resource API
providers:
This configuration does not provide data confidentiality. The first
configured provider is specifying the "identity" mechanism, which
stores resources as plain text.
```

```
#
- identity: {} # plain text, in other words NO encryption
- aesgcm:
 keys:
 - name: key1
 secret: c2VjcmV0IGlzIHNIY3VyZQ==
 - name: key2
 secret: dGhpcyBpcyBwYXNzd29yZA==
- aescbc:
 keys:
 - name: key1
 secret: c2VjcmV0IGlzIHNIY3VyZQ==
 - name: key2
 secret: dGhpcyBpcyBwYXNzd29yZA==
- secretbox:
 keys:
 - name: key1
 secret: YWJjZGVmZ2hpamtsbW5vcHFyc3R1dnd4eXoxMjM0NTY=
- resources:
 - events
providers:
- identity: {} # do not encrypt Events even though '*' is specified below
- resources:
 - '*.apps' # wildcard match requires Kubernetes 1.27 or later
providers:
- aescbc:
 keys:
 - name: key2
 secret: c2VjcmV0IGlzIHNIY3VyZSwgb3IgaXMgaXQ/Cg==
- resources:
 - '*.*' # wildcard match requires Kubernetes 1.27 or later
providers:
- aescbc:
 keys:
 - name: key3
 secret: c2VjcmV0IGlzIHNIY3VyZSwgSSB0aGluaw==
```

Each resources array item is a separate config and contains a complete configuration. The `resources.resources` field is an array of Kubernetes resource names (resource or resource.group) that should be encrypted like Secrets, ConfigMaps, or other resources.

If custom resources are added to `EncryptionConfiguration` and the cluster version is 1.26 or newer, any newly created custom resources mentioned in the `EncryptionConfiguration` will be encrypted. Any custom resources that existed in etcd prior to that version and configuration will be unencrypted until they are next written to storage. This is the same behavior as built-in resources. See the [Ensure all secrets are encrypted](#) section.

The `providers` array is an ordered list of the possible encryption providers to use for the APIs that you listed. Each provider supports multiple keys - the keys are tried in order for decryption, and if the provider is the first provider, the first key is used for encryption.

Only one provider type may be specified per entry (identity or aescbc may be provided, but not both in the same item). The first provider in the list is used to encrypt resources written into the



storage. When reading resources from storage, each provider that matches the stored data attempts in order to decrypt the data. If no provider can read the stored data due to a mismatch in format or secret key, an error is returned which prevents clients from accessing that resource.

`EncryptionConfiguration` supports the use of wildcards to specify the resources that should be encrypted. Use `'*.<group>'` to encrypt all resources within a group (for eg `'*.apps'` in above example) or `'**'` to encrypt all resources. `'*.'` can be used to encrypt all resource in the core group. `'**'` will encrypt all resources, even custom resources that are added after API server start.

#### Note:

Use of wildcards that overlap within the same resource list or across multiple entries are not allowed since part of the configuration would be ineffective. The resources list's processing order and precedence are determined by the order it's listed in the configuration.

If you have a wildcard covering resources and want to opt out of at-rest encryption for a particular kind of resource, you achieve that by adding a separate resources array item with the name of the resource that you want to exempt, followed by a providers array item where you specify the identity provider. You add this item to the list so that it appears earlier than the configuration where you do specify encryption (a provider that is not identity).

For example, if `'*.'` is enabled and you want to opt out of encryption for Events and ConfigMaps, add a new **earlier** item to the resources, followed by the providers array item with identity as the provider. The more specific entry must come before the wildcard entry.

The new item would look similar to:

```
...
- resources:
 - configmaps. # specifically from the core API group,
 # because of trailing "."
 - events
providers:
 - identity: {}
and then other entries in resources
```

Ensure that the exemption is listed *before* the wildcard `'**'` item in the resources array to give it precedence.

For more detailed information about the `EncryptionConfiguration` struct, please refer to the [encryption configuration API](#).

#### Caution:

If any resource is not readable via the encryption configuration (because keys were changed), and you cannot restore a working configuration, your only recourse is to delete that entry from the underlying etcd directly.

Any calls to the Kubernetes API that attempt to read that resource will fail until it is deleted or a valid decryption key is provided.

## Available providers

Before you configure encryption-at-rest for data in your cluster's Kubernetes API, you need to select which provider(s) you will use.

The following table describes each available provider.

Providers for Kubernetes encryption at rest

| Name                                                   | Encryption                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Strength                             | Speed                                     | Key length         |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------|--------------------|
| identity                                               | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | N/A                                  | N/A                                       | N/A                |
|                                                        | Resources written as-is without encryption. When set as the first provider, the resource will be decrypted as new values are written. Existing encrypted resources are <b>not</b> automatically overwritten with the plaintext data. The identity provider is the default if you do not specify otherwise.                                                                                                                                                                 |                                      |                                           |                    |
| aescbc                                                 | AES-CBC with <a href="#">PKCS#7</a> padding                                                                                                                                                                                                                                                                                                                                                                                                                                | Weak                                 | Fast                                      | 32-byte            |
|                                                        | Not recommended due to CBC's vulnerability to padding oracle attacks. Key material accessible from control plane host.                                                                                                                                                                                                                                                                                                                                                     |                                      |                                           |                    |
| aesgcm                                                 | AES-GCM with random nonce                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Must be rotated every 200,000 writes | Fastest                                   | 16, 24, or 32-byte |
|                                                        | Not recommended for use except when an automated key rotation scheme is implemented. Key material accessible from control plane host.                                                                                                                                                                                                                                                                                                                                      |                                      |                                           |                    |
| kms v1<br>( <i>deprecated since Kubernetes v1.28</i> ) | Uses envelope encryption scheme with DEK per resource.                                                                                                                                                                                                                                                                                                                                                                                                                     | Strongest                            | Slow ( <i>compared to kms version 2</i> ) | 32-bytes           |
|                                                        | Data is encrypted by data encryption keys (DEKs) using AES-GCM; DEKs are encrypted by key encryption keys (KEKs) according to configuration in Key Management Service (KMS). Simple key rotation, with a new DEK generated for each encryption, and KEK rotation controlled by the user. Read how to <a href="#">configure the KMS V1 provider</a> .                                                                                                                       |                                      |                                           |                    |
| kms v2                                                 | Uses envelope encryption scheme with DEK per API server.                                                                                                                                                                                                                                                                                                                                                                                                                   | Strongest                            | Fast                                      | 32-bytes           |
|                                                        | Data is encrypted by data encryption keys (DEKs) using AES-GCM; DEKs are encrypted by key encryption keys (KEKs) according to configuration in Key Management Service (KMS). Kubernetes generates a new DEK per encryption from a secret seed. The seed is rotated whenever the KEK is rotated.<br>A good choice if using a third party tool for key management. Available as stable from Kubernetes v1.29.<br>Read how to <a href="#">configure the KMS V2 provider</a> . |                                      |                                           |                    |
| secretbox                                              | XSalsa20 and Poly1305                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Strong                               | Faster                                    | 32-byte            |
|                                                        | Uses relatively new encryption technologies that may not be considered acceptable in environments that require high levels of review. Key material accessible from control plane host.                                                                                                                                                                                                                                                                                     |                                      |                                           |                    |

The identity provider is the default if you do not specify otherwise. **The identity provider does not encrypt stored data and provides no additional confidentiality protection.**

## Key storage

### Local key storage

Encrypting secret data with a locally managed key protects against an etcd compromise, but it fails to protect against a host compromise. Since the encryption keys are stored on the host in the EncryptionConfiguration YAML file, a skilled attacker can access that file and extract the encryption keys.

### Managed (KMS) key storage

The KMS provider uses *envelope encryption*: Kubernetes encrypts resources using a data key, and then encrypts that data key using the managed encryption service. Kubernetes generates a unique data key for each resource. The API server stores an encrypted version of the data key in etcd alongside the ciphertext; when reading the resource, the API server calls the managed encryption service and provides both the ciphertext and the (encrypted) data key. Within the managed encryption service, the provider uses a *key encryption key* to decipher the data key, deciphers the data key, and finally recovers the plain text. Communication between the control plane and the KMS requires in-transit protection, such as TLS.

Using envelope encryption creates dependence on the key encryption key, which is not stored in Kubernetes. In the KMS case, an attacker who intends to get unauthorised access to the plaintext values would need to compromise etcd **and** the third-party KMS provider.

## Protection for encryption keys

You should take appropriate measures to protect the confidential information that allows decryption, whether that is a local encryption key, or an authentication token that allows the API server to call KMS.

Even when you rely on a provider to manage the use and lifecycle of the main encryption key (or keys), you are still responsible for making sure that access controls and other security measures for the managed encryption service are appropriate for your security needs.

## Encrypt your data

### Generate the encryption key

The following steps assume that you are not using KMS, and therefore the steps also assume that you need to generate an encryption key. If you already have an encryption key, skip to [Write an encryption configuration file](#).

#### Caution:

Storing the raw encryption key in the EncryptionConfig only moderately improves your security posture, compared to no encryption.

For additional secrecy, consider using the kms provider as this relies on keys held outside your Kubernetes cluster. Implementations of kms can work with hardware security modules or with encryption services managed by your cloud provider.

To learn about setting up encryption at rest using KMS, see [Using a KMS provider for data encryption](#). The KMS provider plugin that you use may also come with additional specific documentation.

Start by generating a new encryption key, and then encode it using base64:

- [Linux](#)
- [macOS](#)
- [Windows](#)

Generate a 32-byte random key and base64 encode it. You can use this command:

```
head -c 32 /dev/urandom | base64
```

You can use `/dev/hwrng` instead of `/dev/urandom` if you want to use your PC's built-in hardware entropy source. Not all Linux devices provide a hardware random generator.

Generate a 32-byte random key and base64 encode it. You can use this command:

```
head -c 32 /dev/urandom | base64
```

Generate a 32-byte random key and base64 encode it. You can use this command:

```
Do not run this in a session where you have set a random number
generator seed.
[Convert]::ToBase64String((1..32|%{[byte](Get-Random -Max 256)}))
```

#### **Note:**

Keep the encryption key confidential, including while you generate it and ideally even after you are no longer actively using it.

## **Replicate the encryption key**

Using a secure mechanism for file transfer, make a copy of that encryption key available to every other control plane host.

At a minimum, use encryption in transit - for example, secure shell (SSH). For more security, use asymmetric encryption between hosts, or change the approach you are using so that you're relying on KMS encryption.

## **Write an encryption configuration file**

#### **Caution:**

The encryption configuration file may contain keys that can decrypt content in etcd. If the configuration file contains any key material, you must properly restrict permissions on all your control plane hosts so only the user who runs the kube-apiserver can read this configuration.

Create a new encryption configuration file. The contents should be similar to:

```

apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
 - resources:
 - secrets
 - configmaps
 - pandas.awesome.bears.example
 providers:
 - aescbc:
 keys:
 - name: key1
 # See the following text for more details about the secret value
 secret: <BASE 64 ENCODED SECRET>
 - identity: {} # this fallback allows reading unencrypted secrets;
 # for example, during initial migration
```

To create a new encryption key (that does not use KMS), see [Generate the encryption key](#).

## Use the new encryption configuration file

You will need to mount the new encryption config file to the kube-apiserver static pod. Here is an example on how to do that:

1. Save the new encryption config file to `/etc/kubernetes/enc/enc.yaml` on the control-plane node.
2. Edit the manifest for the kube-apiserver static pod: `/etc/kubernetes/manifests/kube-apiserver.yaml` so that it is similar to:

```

#
This is a fragment of a manifest for a static Pod.
Check whether this is correct for your cluster and for your API server.
#
apiVersion: v1
kind: Pod
metadata:
 annotations:
 kubeadm.kubernetes.io/kube-apiserver.advertise-address.endpoint: 10.20.30.40:443
 creationTimestamp: null
 labels:
 app.kubernetes.io/component: kube-apiserver
 tier: control-plane
 name: kube-apiserver
 namespace: kube-system
spec:
 containers:
 - command:
 - kube-apiserver
 ...
```

```

- --encryption-provider-config=/etc/kubernetes/enc/enc.yaml # add this line
volumeMounts:
...
- name: enc # add this line
 mountPath: /etc/kubernetes/enc # add this line
 readOnly: true # add this line
...
volumes:
...
- name: enc # add this line
 hostPath: # add this line
 path: /etc/kubernetes/enc # add this line
 type: DirectoryOrCreate # add this line
...

```

3. Restart your API server.

### Caution:

Your config file contains keys that can decrypt the contents in etcd, so you must properly restrict permissions on your control-plane nodes so only the user who runs the kube-apiserver can read it.

You now have encryption in place for **one** control plane host. A typical Kubernetes cluster has multiple control plane hosts, so there is more to do.

## Reconfigure other control plane hosts

If you have multiple API servers in your cluster, you should deploy the changes in turn to each API server.

### Caution:

For cluster configurations with two or more control plane nodes, the encryption configuration should be identical across each control plane node.

If there is a difference in the encryption provider configuration between control plane nodes, this difference may mean that the kube-apiserver can't decrypt data.

When you are planning to update the encryption configuration of your cluster, plan this so that the API servers in your control plane can always decrypt the stored data (even part way through rolling out the change).

Make sure that you use the **same** encryption configuration on each control plane host.

## Verify that newly written data is encrypted

Data is encrypted when written to etcd. After restarting your kube-apiserver, any newly created or updated Secret (or other resource kinds configured in EncryptionConfiguration) should be encrypted when stored.

To check this, you can use the `etcdctl` command line program to retrieve the contents of your secret data.

This example shows how to check this for encrypting the Secret API.

1. Create a new Secret called `secret1` in the default namespace:

```
kubectl create secret generic secret1 -n default --from-literal=mykey=mydata
```

2. Using the `etcdctl` command line tool, read that Secret out of etcd:

```
ETCDCTL_API=3 etcdctl get /registry/secrets/default/secret1 [...] | hexdump -C
```

where [...] must be the additional arguments for connecting to the etcd server.

For example:

```
ETCDCTL_API=3 etcdctl \
 --cacert=/etc/kubernetes/pki/etcd/ca.crt \
 --cert=/etc/kubernetes/pki/etcd/server.crt \
 --key=/etc/kubernetes/pki/etcd/server.key \
 get /registry/secrets/default/secret1 | hexdump -C
```

The output is similar to this (abbreviated):

```
00000000 2f 72 65 67 69 73 74 72 79 2f 73 65 63 72 65 74 |/registry/secret|
00000010 73 2f 64 65 66 61 75 6c 74 2f 73 65 63 72 65 74 |s/default/secret|
00000020 31 0a 6b 38 73 3a 65 6e 63 3a 61 65 73 63 62 63 |1.k8s:enc:aescbc|
00000030 3a 76 31 3a 6b 65 79 31 3a c7 6c e7 d3 09 bc 06 |:v1:key1:l.....|
00000040 25 51 91 e4 e0 6c e5 b1 4d 7a 8b 3d b9 c2 7c 6e |%Q...l..Mz.=..|n|
00000050 b4 79 df 05 28 ae 0d 8e 5f 35 13 2c c0 18 99 3e |.y..(..._5,...>|
[...]
00000110 23 3a 0d fc 28 ca 48 2d 6b 2d 46 cc 72 0b 70 4c |#..(.H-k-F.r.pL|
00000120 a5 fc 35 43 12 4e 60 ef bf 6f fe cf df 0b ad 1f |..5C.N`..o.....|
00000130 82 c4 88 53 02 da 3e 66 ff 0a |...S..>f..|
0000013a
```

3. Verify the stored Secret is prefixed with `k8s:enc:aescbc:v1:` which indicates the `aescbc` provider has encrypted the resulting data. Confirm that the key name shown in `etcd` matches the key name specified in the `EncryptionConfiguration` mentioned above. In this example, you can see that the encryption key named `key1` is used in `etcd` and in `EncryptionConfiguration`.
4. Verify the Secret is correctly decrypted when retrieved via the API:

```
kubectl get secret secret1 -n default -o yaml
```

The output should contain `mykey: bXlkYXRh`, with contents of `mydata` encoded using base64; read [decoding a Secret](#) to learn how to completely decode the Secret.

## Ensure all relevant data are encrypted

It's often not enough to make sure that new objects get encrypted: you also want that encryption to apply to the objects that are already stored.

For this example, you have configured your cluster so that Secrets are encrypted on write. Performing a replace operation for each Secret will encrypt that content at rest, where the objects are unchanged.

You can make this change across all Secrets in your cluster:

```
Run this as an administrator that can read and write all Secrets
kubectl get secrets --all-namespaces -o json | kubectl replace -f -
```

The command above reads all Secrets and then updates them with the same data, in order to apply server side encryption.

**Note:**

If an error occurs due to a conflicting write, retry the command. It is safe to run that command more than once.

For larger clusters, you may wish to subdivide the Secrets by namespace, or script an update.

## Prevent plain text retrieval

If you want to make sure that the only access to a particular API kind is done using encryption, you can remove the API server's ability to read that API's backing data as plaintext.

**Warning:**

Making this change prevents the API server from retrieving resources that are marked as encrypted at rest, but are actually stored in the clear.

When you have configured encryption at rest for an API (for example: the API kind Secret, representing secrets resources in the core API group), you **must** ensure that all those resources in this cluster really are encrypted at rest. Check this before you carry on with the next steps.

Once all Secrets in your cluster are encrypted, you can remove the identity part of the encryption configuration. For example:

```

apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
- resources:
 - secrets
providers:
- aescbc:
 keys:
 - name: key1
 secret: <BASE 64 ENCODED SECRET>
- identity: {} # REMOVE THIS LINE
```

...and then restart each API server in turn. This change prevents the API server from accessing a plain-text Secret, even by accident.



## Rotate a decryption key

Changing an encryption key for Kubernetes without incurring downtime requires a multi-step operation, especially in the presence of a highly-available deployment where multiple kube-apiserver processes are running.

1. Generate a new key and add it as the second key entry for the current provider on all control plane nodes.
2. Restart **all** kube-apiserver processes, to ensure each server can decrypt any data that are encrypted with the new key.
3. Make a secure backup of the new encryption key. If you lose all copies of this key you would need to delete all the resources were encrypted under the lost key, and workloads may not operate as expected during the time that at-rest encryption is broken.
4. Make the new key the first entry in the keys array so that it is used for encryption-at-rest for new writes
5. Restart all kube-apiserver processes to ensure each control plane host now encrypts using the new key
6. As a privileged user, run `kubectl get secrets --all-namespaces -o json | kubectl replace -f -` to encrypt all existing Secrets with the new key
7. After you have updated all existing Secrets to use the new key and have made a secure backup of the new key, remove the old decryption key from the configuration.

## Decrypt all data

This example shows how to stop encrypting the Secret API at rest. If you are encrypting other API kinds, adjust the steps to match.

To disable encryption at rest, place the identity provider as the first entry in your encryption configuration file:

```

apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
 - resources:
 - secrets
 # list any other resources here that you previously were
 # encrypting at rest
 providers:
 - identity: {} # add this line
 - aescbc:
 keys:
 - name: key1
 secret: <BASE 64 ENCODED SECRET> # keep this in place
 # make sure it comes after "identity"
```

Then run the following command to force decryption of all Secrets:

```
kubectl get secrets --all-namespaces -o json | kubectl replace -f -
```

Once you have replaced all existing encrypted resources with backing data that don't use encryption, you can remove the encryption settings from the kube-apiserver.

## Configure automatic reloading

You can configure automatic reloading of encryption provider configuration. That setting determines whether the [API server](#) should load the file you specify for `--encryption-provider-config` only once at startup, or automatically whenever you change that file. Enabling this option allows you to change the keys for encryption at rest without restarting the API server.

To allow automatic reloading, configure the API server to run with: `--encryption-provider-config-automatic-reload=true`. When enabled, file changes are polled every minute to observe the modifications. The `apiserver_encryption_config_controller_automatic_reload_last_timestamp_seconds` metric identifies when the new config becomes effective. This allows encryption keys to be rotated without restarting the API server.

## What's next

- Read about [decrypting data that are already stored at rest](#)
- Learn more about the [EncryptionConfiguration configuration API \(v1\)](#).

## Decrypt Confidential Data that is Already Encrypted at Rest

All of the APIs in Kubernetes that let you write persistent API resource data support at-rest encryption. For example, you can enable at-rest encryption for [Secrets](#). This at-rest encryption is additional to any system-level encryption for the etcd cluster or for the filesystem(s) on hosts where you are running the kube-apiserver.

This page shows how to switch from encryption of API data at rest, so that API data are stored unencrypted. You might want to do this to improve performance; usually, though, if it was a good idea to encrypt some data, it's also a good idea to leave them encrypted.

### Note:

This task covers encryption for resource data stored using the [Kubernetes API](#). For example, you can encrypt Secret objects, including the key-value data they contain.

If you wanted to manage encryption for data in filesystems that are mounted into containers, you instead need to either:

- use a storage integration that provides encrypted [volumes](#)
- encrypt the data within your own application

## Before you begin

- You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not

already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

- This task assumes that you are running the Kubernetes API server as a [static pod](#) on each control plane node.
- Your cluster's control plane **must** use etcd v3.x (major version 3, any minor version).
- To encrypt a custom resource, your cluster must be running Kubernetes v1.26 or newer.
- You should have some API data that are already encrypted.

To check the version, enter `kubectl version`.

## Determine whether encryption at rest is already enabled

By default, the API server uses an identity provider that stores plain-text representations of resources. **The default identity provider does not provide any confidentiality protection.**

The kube-apiserver process accepts an argument `--encryption-provider-config` that specifies a path to a configuration file. The contents of that file, if you specify one, control how Kubernetes API data is encrypted in etcd. If it is not specified, you do not have encryption at rest enabled.

The format of that configuration file is YAML, representing a configuration API kind named [EncryptionConfiguration](#). You can see an example configuration in [Encryption at rest configuration](#).

If `--encryption-provider-config` is set, check which resources (such as secrets) are configured for encryption, and what provider is used. Make sure that the preferred provider for that resource type is **not** identity; you only set identity (*no encryption*) as default when you want to disable encryption at rest. Verify that the first-listed provider for a resource is something **other** than identity, which means that any new information written to resources of that type will be encrypted as configured. If you do see identity as the first-listed provider for any resource, this means that those resources are being written out to etcd without encryption.

## Decrypt all data

This example shows how to stop encrypting the Secret API at rest. If you are encrypting other API kinds, adjust the steps to match.

### Locate the encryption configuration file

First, find the API server configuration files. On each control plane node, static Pod manifest for the kube-apiserver specifies a command line argument, `--encryption-provider-config`. You are likely to find that this file is mounted into the static Pod using a [hostPath](#) volume mount. Once you locate the volume you can find the file on the node filesystem and inspect it.

## Configure the API server to decrypt objects

To disable encryption at rest, place the identity provider as the first entry in your encryption configuration file.

For example, if your existing EncryptionConfiguration file reads:

```

apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
- resources:
 - secrets
providers:
- aescbc:
 keys:
 # Do not use this (invalid) example key for encryption
 - name: example
 secret: 2KfZgdiq2K0g2YrYpyDYs9mF2LPZhQ==
```

then change it to:

```

apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
- resources:
 - secrets
providers:
- identity: {} # add this line
- aescbc:
 keys:
 - name: example
 secret: 2KfZgdiq2K0g2YrYpyDYs9mF2LPZhQ==
```

and restart the kube-apiserver Pod on this node.

## Reconfigure other control plane hosts

If you have multiple API servers in your cluster, you should deploy the changes in turn to each API server.

Make sure that you use the same encryption configuration on each control plane host.

## Force decryption

Then run the following command to force decryption of all Secrets:

```
If you are decrypting a different kind of object, change "secrets" to match.
kubectl get secrets --all-namespaces -o json | kubectl replace -f -
```

Once you have replaced **all** existing encrypted resources with backing data that don't use encryption, you can remove the encryption settings from the kube-apiserver.

The command line options to remove are:

- `--encryption-provider-config`
- `--encryption-provider-config-automatic-reload`

Restart the kube-apiserver Pod again to apply the new configuration.

## Reconfigure other control plane hosts

If you have multiple API servers in your cluster, you should again deploy the changes in turn to each API server.

Make sure that you use the same encryption configuration on each control plane host.

## What's next

- Learn more about the [EncryptionConfiguration configuration API \(v1\)](#).

# Guaranteed Scheduling For Critical Add-On Pods

Kubernetes core components such as the API server, scheduler, and controller-manager run on a control plane node. However, add-ons must run on a regular cluster node. Some of these add-ons are critical to a fully functional cluster, such as metrics-server, DNS, and UI. A cluster may stop working properly if a critical add-on is evicted (either manually or as a side effect of another operation like upgrade) and becomes pending (for example when the cluster is highly utilized and either there are other pending pods that schedule into the space vacated by the evicted critical add-on pod or the amount of resources available on the node changed for some other reason).

Note that marking a pod as critical is not meant to prevent evictions entirely; it only prevents the pod from becoming permanently unavailable. A static pod marked as critical can't be evicted. However, non-static pods marked as critical are always rescheduled.

## Marking pod as critical

To mark a Pod as critical, set `priorityClassName` for that Pod to `system-cluster-critical` or `system-node-critical`. `system-node-critical` is the highest available priority, even higher than `system-cluster-critical`.

# IP Masquerade Agent User Guide

This page shows how to configure and enable the `ip-masq-agent`.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## IP Masquerade Agent User Guide

The ip-masq-agent configures iptables rules to hide a pod's IP address behind the cluster node's IP address. This is typically done when sending traffic to destinations outside the cluster's pod [CIDR](#) range.

### Key Terms

- **NAT (Network Address Translation):** Is a method of remapping one IP address to another by modifying either the source and/or destination address information in the IP header. Typically performed by a device doing IP routing.
- **Masquerading:** A form of NAT that is typically used to perform a many to one address translation, where multiple source IP addresses are masked behind a single address, which is typically the device doing the IP routing. In Kubernetes this is the Node's IP address.
- **CIDR (Classless Inter-Domain Routing):** Based on the variable-length subnet masking, allows specifying arbitrary-length prefixes. CIDR introduced a new method of representation for IP addresses, now commonly known as **CIDR notation**, in which an address or routing prefix is written with a suffix indicating the number of bits of the prefix, such as 192.168.2.0/24.
- **Link Local:** A link-local address is a network address that is valid only for communications within the network segment or the broadcast domain that the host is connected to. Link-local addresses for IPv4 are defined in the address block 169.254.0.0/16 in CIDR notation.

The ip-masq-agent configures iptables rules to handle masquerading node/pod IP addresses when sending traffic to destinations outside the cluster node's IP and the Cluster IP range. This essentially hides pod IP addresses behind the cluster node's IP address. In some environments, traffic to "external" addresses must come from a known machine address. For example, in Google Cloud, any traffic to the internet must come from a VM's IP. When containers are used, as in Google Kubernetes Engine, the Pod IP will be rejected for egress. To avoid this, we must hide the Pod IP behind the VM's own IP address - generally known as "masquerade". By default, the agent is configured to treat the three private IP ranges specified by [RFC 1918](#) as non-masquerade [CIDR](#). These ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16. The agent will also treat link-local (169.254.0.0/16) as a non-masquerade CIDR by default. The agent is configured to reload its configuration from the location `/etc/config/ip-masq-agent` every 60 seconds, which is also configurable.

### masq/non-masq example

The agent configuration file must be written in YAML or JSON syntax, and may contain three optional keys:

- `nonMasqueradeCIDRs`: A list of strings in [CIDR](#) notation that specify the non-masquerade ranges.

- `masqLinkLocal`: A Boolean (true/false) which indicates whether to masquerade traffic to the link local prefix 169.254.0.0/16. False by default.
- `resyncInterval`: A time interval at which the agent attempts to reload config from disk. For example: '30s', where 's' means seconds, 'ms' means milliseconds.

Traffic to 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16 ranges will NOT be masqueraded. Any other traffic (assumed to be internet) will be masqueraded. An example of a local destination from a pod could be its Node's IP address as well as another node's address or one of the IP addresses in Cluster's IP range. Any other traffic will be masqueraded by default. The below entries show the default set of rules that are applied by the ip-masq-agent:

```
iptables -t nat -L IP-MASQ-AGENT
```

```
target prot opt source destination
RETURN all -- anywhere 169.254.0.0/16 /* ip-masq-agent: cluster-local traffic
should not be subject to MASQUERADE */ ADDRTYPE match dst-type !LOCAL
RETURN all -- anywhere 10.0.0.0/8 /* ip-masq-agent: cluster-local traffic
should not be subject to MASQUERADE */ ADDRTYPE match dst-type !LOCAL
RETURN all -- anywhere 172.16.0.0/12 /* ip-masq-agent: cluster-local traffic
should not be subject to MASQUERADE */ ADDRTYPE match dst-type !LOCAL
RETURN all -- anywhere 192.168.0.0/16 /* ip-masq-agent: cluster-local traffic
should not be subject to MASQUERADE */ ADDRTYPE match dst-type !LOCAL
MASQUERADE all -- anywhere anywhere /* ip-masq-agent: outbound traffic
should be subject to MASQUERADE (this match must come after cluster-local CIDR matches) */
ADDRTYPE match dst-type !LOCAL
```

By default, in GCE/Google Kubernetes Engine, if network policy is enabled or you are using a cluster CIDR not in the 10.0.0.0/8 range, the ip-masq-agent will run in your cluster. If you are running in another environment, you can add the ip-masq-agent [DaemonSet](#) to your cluster.

## Create an ip-masq-agent

To create an ip-masq-agent, run the following kubectl command:

```
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/ip-masq-agent/master/ip-masq-agent.yaml
```

You must also apply the appropriate node label to any nodes in your cluster that you want the agent to run on.

```
kubectl label nodes my-node node.kubernetes.io/masq-agent-ds-ready=true
```

More information can be found in the ip-masq-agent documentation [here](#).

In most cases, the default set of rules should be sufficient; however, if this is not the case for your cluster, you can create and apply a [ConfigMap](#) to customize the IP ranges that are affected. For example, to allow only 10.0.0.0/8 to be considered by the ip-masq-agent, you can create the following [ConfigMap](#) in a file called "config".

### Note:

It is important that the file is called config since, by default, that will be used as the key for lookup by the ip-masq-agent:

```
nonMasqueradeCIDRs:
- 10.0.0.0/8
resyncInterval: 60s
```

Run the following command to add the configmap to your cluster:

```
kubectl create configmap ip-masq-agent --from-file=config --namespace=kube-system
```

This will update a file located at `/etc/config/ip-masq-agent` which is periodically checked every `resyncInterval` and applied to the cluster node. After the `resync` interval has expired, you should see the `iptables` rules reflect your changes:

```
iptables -t nat -L IP-MASQ-AGENT
```

Chain IP-MASQ-AGENT (1 references)

```
target prot opt source destination
RETURN all -- anywhere 169.254.0.0/16 /* ip-masq-agent: cluster-local traffic
should not be subject to MASQUERADE */ ADDRTYPE match dst-type !LOCAL
RETURN all -- anywhere 10.0.0.0/8 /* ip-masq-agent: cluster-local
MASQUERADE all -- anywhere anywhere /* ip-masq-agent: outbound traffic
should be subject to MASQUERADE (this match must come after cluster-local CIDR matches) */
ADDRTYPE match dst-type !LOCAL
```

By default, the link local range (169.254.0.0/16) is also handled by the `ip-masq` agent, which sets up the appropriate `iptables` rules. To have the `ip-masq-agent` ignore link local, you can set `masqLinkLocal` to `true` in the `ConfigMap`.

```
nonMasqueradeCIDRs:
- 10.0.0.0/8
resyncInterval: 60s
masqLinkLocal: true
```

## Limit Storage Consumption

This example demonstrates how to limit the amount of storage consumed in a namespace.

The following resources are used in the demonstration: [ResourceQuota](#), [LimitRange](#), and [PersistentVolumeClaim](#).

### Before you begin

- You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.



## Scenario: Limiting Storage Consumption

The cluster-admin is operating a cluster on behalf of a user population and the admin wants to control how much storage a single namespace can consume in order to control cost.

The admin would like to limit:

1. The number of persistent volume claims in a namespace
2. The amount of storage each claim can request
3. The amount of cumulative storage the namespace can have

## LimitRange to limit requests for storage

Adding a `LimitRange` to a namespace enforces storage request sizes to a minimum and maximum. Storage is requested via `PersistentVolumeClaim`. The admission controller that enforces limit ranges will reject any PVC that is above or below the values set by the admin.

In this example, a PVC requesting 10Gi of storage would be rejected because it exceeds the 2Gi max.

```
apiVersion: v1
kind: LimitRange
metadata:
 name: storagelimits
spec:
 limits:
 - type: PersistentVolumeClaim
 max:
 storage: 2Gi
 min:
 storage: 1Gi
```

Minimum storage requests are used when the underlying storage provider requires certain minimums. For example, AWS EBS volumes have a 1Gi minimum requirement.

## StorageQuota to limit PVC count and cumulative storage capacity

Admins can limit the number of PVCs in a namespace as well as the cumulative capacity of those PVCs. New PVCs that exceed either maximum value will be rejected.

In this example, a 6th PVC in the namespace would be rejected because it exceeds the maximum count of 5. Alternatively, a 5Gi maximum quota when combined with the 2Gi max limit above, cannot have 3 PVCs where each has 2Gi. That would be 6Gi requested for a namespace capped at 5Gi.

```
apiVersion: v1
kind: ResourceQuota
metadata:
 name: storagequota
spec:
```

```
hard:
 persistentvolumeclaims: "5"
 requests.storage: "5Gi"
```

## Summary

A limit range can put a ceiling on how much storage is requested while a resource quota can effectively cap the storage consumed by a namespace through claim counts and cumulative storage capacity. This allows a cluster-admin to plan their cluster's storage budget without risk of any one project going over their allotment.

# Migrate Replicated Control Plane To Use Cloud Controller Manager

The cloud-controller-manager is a Kubernetes [control plane](#) component that embeds cloud-specific control logic. The [cloud controller manager](#) lets you link your cluster into your cloud provider's API, and separates out the components that interact with that cloud platform from components that only interact with your cluster.

By decoupling the interoperability logic between Kubernetes and the underlying cloud infrastructure, the cloud-controller-manager component enables cloud providers to release features at a different pace compared to the main Kubernetes project.

## Background

As part of the [cloud provider extraction effort](#), all cloud specific controllers must be moved out of the kube-controller-manager. All existing clusters that run cloud controllers in the kube-controller-manager must migrate to instead run the controllers in a cloud provider specific cloud-controller-manager.

Leader Migration provides a mechanism in which HA clusters can safely migrate "cloud specific" controllers between the kube-controller-manager and the cloud-controller-manager via a shared resource lock between the two components while upgrading the replicated control plane. For a single-node control plane, or if unavailability of controller managers can be tolerated during the upgrade, Leader Migration is not needed and this guide can be ignored.

Leader Migration can be enabled by setting `--enable-leader-migration` on kube-controller-manager or cloud-controller-manager. Leader Migration only applies during the upgrade and can be safely disabled or left enabled after the upgrade is complete.

This guide walks you through the manual process of upgrading the control plane from kube-controller-manager with built-in cloud provider to running both kube-controller-manager and cloud-controller-manager. If you use a tool to deploy and manage the cluster, please refer to the documentation of the tool and the cloud provider for specific instructions of the migration.

## Before you begin

It is assumed that the control plane is running Kubernetes version N and to be upgraded to version N + 1. Although it is possible to migrate within the same version, ideally the migration

should be performed as part of an upgrade so that changes of configuration can be aligned to each release. The exact versions of N and N + 1 depend on each cloud provider. For example, if a cloud provider builds a cloud-controller-manager to work with Kubernetes 1.24, then N can be 1.23 and N + 1 can be 1.24.

The control plane nodes should run kube-controller-manager with Leader Election enabled, which is the default. As of version N, an in-tree cloud provider must be set with --cloud-provider flag and cloud-controller-manager should not yet be deployed.

The out-of-tree cloud provider must have built a cloud-controller-manager with Leader Migration implementation. If the cloud provider imports k8s.io/cloud-provider and k8s.io/controller-manager of version v0.21.0 or later, Leader Migration will be available. However, for version before v0.22.0, Leader Migration is alpha and requires feature gate ControllerManagerLeaderMigration to be enabled in cloud-controller-manager.

This guide assumes that kubelet of each control plane node starts kube-controller-manager and cloud-controller-manager as static pods defined by their manifests. If the components run in a different setting, please adjust the steps accordingly.

For authorization, this guide assumes that the cluster uses RBAC. If another authorization mode grants permissions to kube-controller-manager and cloud-controller-manager components, please grant the needed access in a way that matches the mode.

## Grant access to Migration Lease

The default permissions of the controller manager allow only accesses to their main Lease. In order for the migration to work, accesses to another Lease are required.

You can grant kube-controller-manager full access to the leases API by modifying the system::leader-locking-kube-controller-manager role. This task guide assumes that the name of the migration lease is cloud-provider-extraction-migration.

```
kubectl patch -n kube-system role 'system::leader-locking-kube-controller-manager' -p '{"rules": [{"apiGroups": ["coordination.k8s.io"], "resources": ["leases"], "resourceNames": ["cloud-provider-extraction-migration"], "verbs": ["create", "list", "get", "update"] }]}' --type=merge`
```

Do the same to the system::leader-locking-cloud-controller-manager role.

```
kubectl patch -n kube-system role 'system::leader-locking-cloud-controller-manager' -p '{"rules": [{"apiGroups": ["coordination.k8s.io"], "resources": ["leases"], "resourceNames": ["cloud-provider-extraction-migration"], "verbs": ["create", "list", "get", "update"] }]}' --type=merge`
```

## Initial Leader Migration configuration

Leader Migration optionally takes a configuration file representing the state of controller-to-manager assignment. At this moment, with in-tree cloud provider, kube-controller-manager runs route, service, and cloud-node-lifecycle. The following example configuration shows the assignment.

Leader Migration can be enabled without a configuration. Please see [Default Configuration](#) for details.

```
kind: LeaderMigrationConfiguration
apiVersion: controllermanager.config.k8s.io/v1
leaderName: cloud-provider-extraction-migration
controllerLeaders:
 - name: route
 component: kube-controller-manager
 - name: service
 component: kube-controller-manager
 - name: cloud-node-lifecycle
 component: kube-controller-manager
```

Alternatively, because the controllers can run under either controller managers, setting component to `*` for both sides makes the configuration file consistent between both parties of the migration.

```
wildcard version
kind: LeaderMigrationConfiguration
apiVersion: controllermanager.config.k8s.io/v1
leaderName: cloud-provider-extraction-migration
controllerLeaders:
 - name: route
 component: *
 - name: service
 component: *
 - name: cloud-node-lifecycle
 component: *
```

On each control plane node, save the content to `/etc/leadermigration.conf`, and update the manifest of kube-controller-manager so that the file is mounted inside the container at the same location. Also, update the same manifest to add the following arguments:

- `--enable-leader-migration` to enable Leader Migration on the controller manager
- `--leader-migration-config=/etc/leadermigration.conf` to set configuration file

Restart kube-controller-manager on each node. At this moment, kube-controller-manager has leader migration enabled and is ready for the migration.

## Deploy Cloud Controller Manager

In version  $N + 1$ , the desired state of controller-to-manager assignment can be represented by a new configuration file, shown as follows. Please note component field of each controllerLeaders changing from kube-controller-manager to cloud-controller-manager. Alternatively, use the wildcard version mentioned above, which has the same effect.

```
kind: LeaderMigrationConfiguration
apiVersion: controllermanager.config.k8s.io/v1
leaderName: cloud-provider-extraction-migration
controllerLeaders:
 - name: route
 component: cloud-controller-manager
 - name: service
 component: cloud-controller-manager
```

```
- name: cloud-node-lifecycle
 component: cloud-controller-manager
```

When creating control plane nodes of version  $N + 1$ , the content should be deployed to `/etc/leadermigration.conf`. The manifest of `cloud-controller-manager` should be updated to mount the configuration file in the same manner as `kube-controller-manager` of version  $N$ . Similarly, add `--enable-leader-migration` and `--leader-migration-config=/etc/leadermigration.conf` to the arguments of `cloud-controller-manager`.

Create a new control plane node of version  $N + 1$  with the updated `cloud-controller-manager` manifest, and with the `--cloud-provider` flag set to `external` for `kube-controller-manager`. `kube-controller-manager` of version  $N + 1$  **MUST NOT** have Leader Migration enabled because, with an external cloud provider, it does not run the migrated controllers anymore, and thus it is not involved in the migration.

Please refer to [Cloud Controller Manager Administration](#) for more detail on how to deploy `cloud-controller-manager`.

## Upgrade Control Plane

The control plane now contains nodes of both version  $N$  and  $N + 1$ . The nodes of version  $N$  run `kube-controller-manager` only, and these of version  $N + 1$  run both `kube-controller-manager` and `cloud-controller-manager`. The migrated controllers, as specified in the configuration, are running under either `kube-controller-manager` of version  $N$  or `cloud-controller-manager` of version  $N + 1$  depending on which controller manager holds the migration lease. No controller will ever be running under both controller managers at any time.

In a rolling manner, create a new control plane node of version  $N + 1$  and bring down one of version  $N$  until the control plane contains only nodes of version  $N + 1$ . If a rollback from version  $N + 1$  to  $N$  is required, add nodes of version  $N$  with Leader Migration enabled for `kube-controller-manager` back to the control plane, replacing one of version  $N + 1$  each time until there are only nodes of version  $N$ .

## (Optional) Disable Leader Migration

Now that the control plane has been upgraded to run both `kube-controller-manager` and `cloud-controller-manager` of version  $N + 1$ , Leader Migration has finished its job and can be safely disabled to save one Lease resource. It is safe to re-enable Leader Migration for the rollback in the future.

In a rolling manner, update manifest of `cloud-controller-manager` to unset both `--enable-leader-migration` and `--leader-migration-config=` flag, also remove the mount of `/etc/leadermigration.conf`, and finally remove `/etc/leadermigration.conf`. To re-enable Leader Migration, recreate the configuration file and add its mount and the flags that enable Leader Migration back to `cloud-controller-manager`.

## Default Configuration

Starting Kubernetes 1.22, Leader Migration provides a default configuration suitable for the default controller-to-manager assignment. The default configuration can be enabled by setting `--enable-leader-migration` but without `--leader-migration-config=`.

For kube-controller-manager and cloud-controller-manager, if there are no flags that enable any in-tree cloud provider or change ownership of controllers, the default configuration can be used to avoid manual creation of the configuration file.

## Special case: migrating the Node IPAM controller

If your cloud provider provides an implementation of Node IPAM controller, you should switch to the implementation in cloud-controller-manager. Disable Node IPAM controller in kube-controller-manager of version N + 1 by adding `--controllers=*,-nodeipam` to its flags. Then add nodeipam to the list of migrated controllers.

```
wildcard version, with nodeipam
kind: LeaderMigrationConfiguration
apiVersion: controllermanager.config.k8s.io/v1
leaderName: cloud-provider-extraction-migration
controllerLeaders:
- name: route
 component: *
- name: service
 component: *
- name: cloud-node-lifecycle
 component: *
- name: nodeipam
- component: *
```

## What's next

- Read the [Controller Manager Leader Migration](#) enhancement proposal.

# Namespaces Walkthrough

Kubernetes [namespaces](#) help different projects, teams, or customers to share a Kubernetes cluster.

It does this by providing the following:

1. A scope for [Names](#).
2. A mechanism to attach authorization and policy to a subsection of the cluster.

Use of multiple namespaces is optional.

This example demonstrates how to use Kubernetes namespaces to subdivide your cluster.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)

- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Prerequisites

This example assumes the following:

1. You have an [existing Kubernetes cluster](#).
2. You have a basic understanding of Kubernetes [Pods](#), [Services](#), and [Deployments](#).

## Understand the default namespace

By default, a Kubernetes cluster will instantiate a default namespace when provisioning the cluster to hold the default set of Pods, Services, and Deployments used by the cluster.

Assuming you have a fresh cluster, you can inspect the available namespaces by doing the following:

```
kubectl get namespaces
```

| NAME    | STATUS | AGE |
|---------|--------|-----|
| default | Active | 13m |

## Create new namespaces

For this exercise, we will create two additional Kubernetes namespaces to hold our content.

Let's imagine a scenario where an organization is using a shared Kubernetes cluster for development and production use cases.

The development team would like to maintain a space in the cluster where they can get a view on the list of Pods, Services, and Deployments they use to build and run their application. In this space, Kubernetes resources come and go, and the restrictions on who can or cannot modify resources are relaxed to enable agile development.

The operations team would like to maintain a space in the cluster where they can enforce strict procedures on who can or cannot manipulate the set of Pods, Services, and Deployments that run the production site.

One pattern this organization could follow is to partition the Kubernetes cluster into two namespaces: development and production.

Let's create two new namespaces to hold our work.

Use the file [namespace-dev.yaml](#) which describes a development namespace:

[admin/namespace-dev.yaml](#)

```
apiVersion: v1
kind: Namespace
metadata:
```

```
name: development
labels:
 name: development
```

Create the development namespace using kubectl.

```
kubectl create -f https://k8s.io/examples/admin/namespace-dev.yaml
```

Save the following contents into file [namespace-prod.yaml](#) which describes a production namespace:

[admin/namespace-prod.yaml](#)

```
apiVersion: v1
kind: Namespace
metadata:
 name: production
 labels:
 name: production
```

And then let's create the production namespace using kubectl.

```
kubectl create -f https://k8s.io/examples/admin/namespace-prod.yaml
```

To be sure things are right, let's list all of the namespaces in our cluster.

```
kubectl get namespaces --show-labels
```

| NAME        | STATUS | AGE | LABELS           |
|-------------|--------|-----|------------------|
| default     | Active | 32m | <none>           |
| development | Active | 29s | name=development |
| production  | Active | 23s | name=production  |

## Create pods in each namespace

A Kubernetes namespace provides the scope for Pods, Services, and Deployments in the cluster.

Users interacting with one namespace do not see the content in another namespace.

To demonstrate this, let's spin up a simple Deployment and Pods in the development namespace.

We first check what is the current context:

```
kubectl config view
```

```
apiVersion: v1
clusters:
- cluster:
 certificate-authority-data: REDACTED
 server: https://130.211.122.180
 name: lithe-cocoa-92103_kubernetes
contexts:
- context:
```



```
cluster: lithe-cocoa-92103_kubernetes
user: lithe-cocoa-92103_kubernetes
name: lithe-cocoa-92103_kubernetes
current-context: lithe-cocoa-92103_kubernetes
kind: Config
preferences: {}
users:
- name: lithe-cocoa-92103_kubernetes
 user:
 client-certificate-data: REDACTED
 client-key-data: REDACTED
 token: 65rZW78y8HbwXXtSXuUw9DbP4FLjHi4b
- name: lithe-cocoa-92103_kubernetes-basic-auth
 user:
 password: h5M0FtUUIfBSdI7
 username: admin
```

```
kubectl config current-context
```

```
lithe-cocoa-92103_kubernetes
```

The next step is to define a context for the kubectl client to work in each namespace. The value of "cluster" and "user" fields are copied from the current context.

```
kubectl config set-context dev --namespace=development \
--cluster=lithe-cocoa-92103_kubernetes \
--user=lithe-cocoa-92103_kubernetes

kubectl config set-context prod --namespace=production \
--cluster=lithe-cocoa-92103_kubernetes \
--user=lithe-cocoa-92103_kubernetes
```

By default, the above commands add two contexts that are saved into file `.kube/config`. You can now view the contexts and alternate against the two new request contexts depending on which namespace you wish to work against.

To view the new contexts:

```
kubectl config view
```

```
apiVersion: v1
clusters:
- cluster:
 certificate-authority-data: REDACTED
 server: https://130.211.122.180
 name: lithe-cocoa-92103_kubernetes
contexts:
- context:
 cluster: lithe-cocoa-92103_kubernetes
 user: lithe-cocoa-92103_kubernetes
 name: lithe-cocoa-92103_kubernetes
- context:
 cluster: lithe-cocoa-92103_kubernetes
 namespace: development
```

```
 user: lithe-cocoa-92103_kubernetes
name: dev
- context:
 cluster: lithe-cocoa-92103_kubernetes
 namespace: production
 user: lithe-cocoa-92103_kubernetes
 name: prod
current-context: lithe-cocoa-92103_kubernetes
kind: Config
preferences: {}
users:
- name: lithe-cocoa-92103_kubernetes
 user:
 client-certificate-data: REDACTED
 client-key-data: REDACTED
 token: 65rZW78y8HbwXXtSXuUw9DbP4FLjHi4b
- name: lithe-cocoa-92103_kubernetes-basic-auth
 user:
 password: h5M0FtUUIfBSdI7
 username: admin
```

Let's switch to operate in the development namespace.

```
kubectl config use-context dev
```

You can verify your current context by doing the following:

```
kubectl config current-context
```

```
dev
```

At this point, all requests we make to the Kubernetes cluster from the command line are scoped to the development namespace.

Let's create some contents.

[admin/snowflake-deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 labels:
 app: snowflake
 name: snowflake
spec:
 replicas: 2
 selector:
 matchLabels:
 app: snowflake
 template:
 metadata:
 labels:
 app: snowflake
 spec:
```

```
containers:
- image: registry.k8s.io/serve_hostname
 imagePullPolicy: Always
 name: snowflake
```

Apply the manifest to create a Deployment

```
kubectl apply -f https://k8s.io/examples/admin/snowflake-deployment.yaml
```

We have created a deployment whose replica size is 2 that is running the pod called snowflake with a basic container that serves the hostname.

```
kubectl get deployment
```

| NAME      | READY | UP-TO-DATE | AVAILABLE | AGE |
|-----------|-------|------------|-----------|-----|
| snowflake | 2/2   | 2          | 2         | 2m  |

```
kubectl get pods -l app=snowflake
```

| NAME                       | READY | STATUS  | RESTARTS | AGE |
|----------------------------|-------|---------|----------|-----|
| snowflake-3968820950-9dgr8 | 1/1   | Running | 0        | 2m  |
| snowflake-3968820950-vgc4n | 1/1   | Running | 0        | 2m  |

And this is great, developers are able to do what they want, and they do not have to worry about affecting content in the production namespace.

Let's switch to the production namespace and show how resources in one namespace are hidden from the other.

```
kubectl config use-context prod
```

The production namespace should be empty, and the following commands should return nothing.

```
kubectl get deployment
kubectl get pods
```

Production likes to run cattle, so let's create some cattle pods.

```
kubectl create deployment cattle --image=registry.k8s.io/serve_hostname --replicas=5
```

```
kubectl get deployment
```

| NAME   | READY | UP-TO-DATE | AVAILABLE | AGE |
|--------|-------|------------|-----------|-----|
| cattle | 5/5   | 5          | 5         | 10s |

```
kubectl get pods -l app=cattle
```

| NAME                    | READY | STATUS  | RESTARTS | AGE |
|-------------------------|-------|---------|----------|-----|
| cattle-2263376956-41xy6 | 1/1   | Running | 0        | 34s |
| cattle-2263376956-kw466 | 1/1   | Running | 0        | 34s |
| cattle-2263376956-n4v97 | 1/1   | Running | 0        | 34s |
| cattle-2263376956-p5p3i | 1/1   | Running | 0        | 34s |
| cattle-2263376956-sxpth | 1/1   | Running | 0        | 34s |

At this point, it should be clear that the resources users create in one namespace are hidden from the other namespace.

As the policy support in Kubernetes evolves, we will extend this scenario to show how you can provide different authorization rules for each namespace.

## Operating etcd clusters for Kubernetes

etcd is a consistent and highly-available key value store used as Kubernetes' backing store for all cluster data.

If your Kubernetes cluster uses etcd as its backing store, make sure you have a [back up](#) plan for the data.

You can find in-depth information about etcd in the official [documentation](#).

### Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to follow this guide on a cluster with at least two nodes that are not acting as control plane nodes. If you do not already have a cluster, you can create one by using [minikube](#).

### Prerequisites

- Run etcd as a cluster of odd members.
- etcd is a leader-based distributed system. Ensure that the leader periodically send heartbeats on time to all followers to keep the cluster stable.
- Ensure that no resource starvation occurs.

Performance and stability of the cluster is sensitive to network and disk I/O. Any resource starvation can lead to heartbeat timeout, causing instability of the cluster. An unstable etcd indicates that no leader is elected. Under such circumstances, a cluster cannot make any changes to its current state, which implies no new pods can be scheduled.

- Keeping etcd clusters stable is critical to the stability of Kubernetes clusters. Therefore, run etcd clusters on dedicated machines or isolated environments for [guaranteed resource requirements](#).
- The minimum recommended etcd versions to run in production are 3.4.22+ and 3.5.6+.

### Resource requirements

Operating etcd with limited resources is suitable only for testing purposes. For deploying in production, advanced hardware configuration is required. Before deploying etcd in production, see [resource requirement reference](#).

# Starting etcd clusters

This section covers starting a single-node and multi-node etcd cluster.

## Single-node etcd cluster

Use a single-node etcd cluster only for testing purposes.

1. Run the following:

```
etcd --listen-client-urls=http://$PRIVATE_IP:2379 \
--advertise-client-urls=http://$PRIVATE_IP:2379
```

2. Start the Kubernetes API server with the flag `--etcd-servers=$PRIVATE_IP:2379`.

Make sure `PRIVATE_IP` is set to your etcd client IP.

## Multi-node etcd cluster

For durability and high availability, run etcd as a multi-node cluster in production and back it up periodically. A five-member cluster is recommended in production. For more information, see [FAQ documentation](#).

Configure an etcd cluster either by static member information or by dynamic discovery. For more information on clustering, see [etcd clustering documentation](#).

For an example, consider a five-member etcd cluster running with the following client URLs: `http://$IP1:2379`, `http://$IP2:2379`, `http://$IP3:2379`, `http://$IP4:2379`, and `http://$IP5:2379`. To start a Kubernetes API server:

1. Run the following:

```
etcd --listen-client-urls=http://$IP1:2379,http://$IP2:2379,http://$IP3:2379,http://$IP4:2379,http://$IP5:2379 --advertise-client-urls=http://$IP1:2379,http://$IP2:2379,http://$IP3:2379,http://$IP4:2379,http://$IP5:2379
```

2. Start the Kubernetes API servers with the flag `--etcd-servers=$IP1:2379,$IP2:2379,$IP3:2379,$IP4:2379,$IP5:2379`.

Make sure the `IP<n>` variables are set to your client IP addresses.

## Multi-node etcd cluster with load balancer

To run a load balancing etcd cluster:

1. Set up an etcd cluster.
2. Configure a load balancer in front of the etcd cluster. For example, let the address of the load balancer be `$LB`.
3. Start Kubernetes API Servers with the flag `--etcd-servers=$LB:2379`.

## Securing etcd clusters

Access to etcd is equivalent to root permission in the cluster so ideally only the API server should have access to it. Considering the sensitivity of the data, it is recommended to grant permission to only those nodes that require access to etcd clusters.

To secure etcd, either set up firewall rules or use the security features provided by etcd. etcd security features depend on x509 Public Key Infrastructure (PKI). To begin, establish secure communication channels by generating a key and certificate pair. For example, use key pairs `peer.key` and `peer.cert` for securing communication between etcd members, and `client.key` and `client.cert` for securing communication between etcd and its clients. See the [example scripts](#) provided by the etcd project to generate key pairs and CA files for client authentication.

### Securing communication

To configure etcd with secure peer communication, specify flags `--peer-key-file=peer.key` and `--peer-cert-file=peer.cert`, and use HTTPS as the URL schema.

Similarly, to configure etcd with secure client communication, specify flags `--key-file=k8sclient.key` and `--cert-file=k8sclient.cert`, and use HTTPS as the URL schema. Here is an example on a client command that uses secure communication:

```
ETCDCTL_API=3 etcdctl --endpoints 10.2.0.9:2379 \
--cert=/etc/kubernetes/pki/etcd/server.crt \
--key=/etc/kubernetes/pki/etcd/server.key \
--cacert=/etc/kubernetes/pki/etcd/ca.crt \
member list
```

### Limiting access of etcd clusters

After configuring secure communication, restrict the access of the etcd cluster to only the Kubernetes API servers using TLS authentication.

For example, consider key pairs `k8sclient.key` and `k8sclient.cert` that are trusted by the CA `etcd.ca`. When etcd is configured with `--client-cert-auth` along with TLS, it verifies the certificates from clients by using system CAs or the CA passed in by `--trusted-ca-file` flag. Specifying flags `--client-cert-auth=true` and `--trusted-ca-file=etcd.ca` will restrict the access to clients with the certificate `k8sclient.cert`.

Once etcd is configured correctly, only clients with valid certificates can access it. To give Kubernetes API servers the access, configure them with the flags `--etcd-certfile=k8sclient.cert`, `--etcd-keyfile=k8sclient.key` and `--etcd-cafile=ca.cert`.

#### Note:

etcd authentication is not planned for Kubernetes.

## Replacing a failed etcd member

etcd cluster achieves high availability by tolerating minor member failures. However, to improve the overall health of the cluster, replace failed members immediately. When multiple

members fail, replace them one by one. Replacing a failed member involves two steps: removing the failed member and adding a new member.

Though etcd keeps unique member IDs internally, it is recommended to use a unique name for each member to avoid human errors. For example, consider a three-member etcd cluster. Let the URLs be, member1=http://10.0.0.1, member2=http://10.0.0.2, and member3=http://10.0.0.3. When member1 fails, replace it with member4=http://10.0.0.4.

1. Get the member ID of the failed member1:

```
etcdctl --endpoints=http://10.0.0.2,http://10.0.0.3 member list
```

The following message is displayed:

```
8211f1d0f64f3269, started, member1, http://10.0.0.1:2380, http://10.0.0.1:2379
91bc3c398fb3c146, started, member2, http://10.0.0.2:2380, http://10.0.0.2:2379
fd422379fda50e48, started, member3, http://10.0.0.3:2380, http://10.0.0.3:2379
```

2. Do either of the following:

1. If each Kubernetes API server is configured to communicate with all etcd members, remove the failed member from the --etcd-servers flag, then restart each Kubernetes API server.
2. If each Kubernetes API server communicates with a single etcd member, then stop the Kubernetes API server that communicates with the failed etcd.

3. Stop the etcd server on the broken node. It is possible that other clients besides the Kubernetes API server are causing traffic to etcd and it is desirable to stop all traffic to prevent writes to the data directory.

4. Remove the failed member:

```
etcdctl member remove 8211f1d0f64f3269
```

The following message is displayed:

```
Removed member 8211f1d0f64f3269 from cluster
```

5. Add the new member:

```
etcdctl member add member4 --peer-urls=http://10.0.0.4:2380
```

The following message is displayed:

```
Member 2be1eb8f84b7f63e added to cluster ef37ad9dc622a7c4
```

6. Start the newly added member on a machine with the IP 10.0.0.4:

```
export ETCD_NAME="member4"
export ETCD_INITIAL_CLUSTER="member2=http://10.0.0.2:2380,member3=http://
10.0.0.3:2380,member4=http://10.0.0.4:2380"
export ETCD_INITIAL_CLUSTER_STATE=existing
etcd [flags]
```

Do either of the following:

7.
  1. If each Kubernetes API server is configured to communicate with all etcd members, add the newly added member to the `--etcd-servers` flag, then restart each Kubernetes API server.
  2. If each Kubernetes API server communicates with a single etcd member, start the Kubernetes API server that was stopped in step 2. Then configure Kubernetes API server clients to again route requests to the Kubernetes API server that was stopped. This can often be done by configuring a load balancer.

For more information on cluster reconfiguration, see [etcd reconfiguration documentation](#).

## Backing up an etcd cluster

All Kubernetes objects are stored in etcd. Periodically backing up the etcd cluster data is important to recover Kubernetes clusters under disaster scenarios, such as losing all control plane nodes. The snapshot file contains all the Kubernetes state and critical information. In order to keep the sensitive Kubernetes data safe, encrypt the snapshot files.

Backing up an etcd cluster can be accomplished in two ways: etcd built-in snapshot and volume snapshot.

### Built-in snapshot

etcd supports built-in snapshot. A snapshot may either be created from a live member with the `etcdctl snapshot save` command or by copying the `member/snap/db` file from an etcd [data directory](#) that is not currently used by an etcd process. Creating the snapshot will not affect the performance of the member.

Below is an example for creating a snapshot of the keyspace served by \$ENDPOINT to the file `snapshot.db`:

```
ETCDCTL_API=3 etcdctl --endpoints $ENDPOINT snapshot save snapshot.db
```

Verify the snapshot:

```
ETCDCTL_API=3 etcdctl --write-out=table snapshot status snapshot.db
```

```
+-----+-----+-----+-----+
| HASH | REVISION | TOTAL KEYS | TOTAL SIZE |
+-----+-----+-----+-----+
| fe01cf57 | 10 | 7 | 2.1 MB |
+-----+-----+-----+-----+
```

### Volume snapshot

If etcd is running on a storage volume that supports backup, such as Amazon Elastic Block Store, back up etcd data by creating a snapshot of the storage volume.

### Snapshot using etcdctl options

We can also create the snapshot using various options given by `etcdctl`. For example:



```
ETCDCTL_API=3 etcdctl -h
```

will list various options available from etcdctl. For example, you can create a snapshot by specifying the endpoint, certificates and key as shown below:

```
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379 \
--cacert=<trusted-ca-file> --cert=<cert-file> --key=<key-file> \
snapshot save <backup-file-location>
```

where trusted-ca-file, cert-file and key-file can be obtained from the description of the etcd Pod.

## Scaling out etcd clusters

Scaling out etcd clusters increases availability by trading off performance. Scaling does not increase cluster performance nor capability. A general rule is not to scale out or in etcd clusters. Do not configure any auto scaling groups for etcd clusters. It is strongly recommended to always run a static five-member etcd cluster for production Kubernetes clusters at any officially supported scale.

A reasonable scaling is to upgrade a three-member cluster to a five-member one, when more reliability is desired. See [etcd reconfiguration documentation](#) for information on how to add members into an existing cluster.

## Restoring an etcd cluster

etcd supports restoring from snapshots that are taken from an etcd process of the [major.minor](#) version. Restoring a version from a different patch version of etcd is also supported. A restore operation is employed to recover the data of a failed cluster.

Before starting the restore operation, a snapshot file must be present. It can either be a snapshot file from a previous backup operation, or from a remaining [data directory](#).

When restoring the cluster, use the --data-dir option to specify to which folder the cluster should be restored:

```
ETCDCTL_API=3 etcdctl --data-dir <data-dir-location> snapshot restore snapshot.db
```

where <data-dir-location> is a directory that will be created during the restore process.

Yet another example would be to first export the ETCDCTL\_API environment variable:

```
export ETCDCTL_API=3
etcdctl --data-dir <data-dir-location> snapshot restore snapshot.db
```

If <data-dir-location> is the same folder as before, delete it and stop the etcd process before restoring the cluster. Otherwise, change etcd configuration and restart the etcd process after restoration to have it use the new data directory.

For more information and examples on restoring a cluster from a snapshot file, see [etcd disaster recovery documentation](#).

If the access URLs of the restored cluster are changed from the previous cluster, the Kubernetes API server must be reconfigured accordingly. In this case, restart Kubernetes API servers with

the flag `--etcd-servers=$NEW_ETCD_CLUSTER` instead of the flag `--etcd-servers=$OLD_ETCD_CLUSTER`. Replace `$NEW_ETCD_CLUSTER` and `$OLD_ETCD_CLUSTER` with the respective IP addresses. If a load balancer is used in front of an etcd cluster, you might need to update the load balancer instead.

If the majority of etcd members have permanently failed, the etcd cluster is considered failed. In this scenario, Kubernetes cannot make any changes to its current state. Although the scheduled pods might continue to run, no new pods can be scheduled. In such cases, recover the etcd cluster and potentially reconfigure Kubernetes API servers to fix the issue.

**Note:**

If any API servers are running in your cluster, you should not attempt to restore instances of etcd. Instead, follow these steps to restore etcd:

- stop *all* API server instances
- restore state in all etcd instances
- restart all API server instances

We also recommend restarting any components (e.g. kube-scheduler, kube-controller-manager, kubelet) to ensure that they don't rely on some stale data. Note that in practice, the restore takes a bit of time. During the restoration, critical components will lose leader lock and restart themselves.

## Upgrading etcd clusters

For more details on etcd upgrade, please refer to the [etcd upgrades](#) documentation.

**Note:**

Before you start an upgrade, please back up your etcd cluster first.

## Maintaining etcd clusters

For more details on etcd maintenance, please refer to the [etcd maintenance](#) documentation.

This item links to a third party project or product that is not part of Kubernetes itself. [More information](#)

**Note:**

Defragmentation is an expensive operation, so it should be executed as infrequently as possible. On the other hand, it's also necessary to make sure any etcd member will not exceed the storage quota. The Kubernetes project recommends that when you perform defragmentation, you use a tool such as [etcd-defrag](#).

You can also run the defragmentation tool as a Kubernetes CronJob, to make sure that defragmentation happens regularly. See [etcd-defrag-cronjob.yaml](#) for details.

# Reserve Compute Resources for System Daemons

Kubernetes nodes can be scheduled to Capacity. Pods can consume all the available capacity on a node by default. This is an issue because nodes typically run quite a few system daemons that power the OS and Kubernetes itself. Unless resources are set aside for these system daemons, pods and system daemons compete for resources and lead to resource starvation issues on the node.

The kubelet exposes a feature named 'Node Allocatable' that helps to reserve compute resources for system daemons. Kubernetes recommends cluster administrators to configure 'Node Allocatable' based on their workload density on each node.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version 1.8. To check the version, enter kubectl version. Your Kubernetes server must be at or later than version 1.17 to use the kubelet command line option --reserved-cpus to set an [explicitly reserved CPU list](#).

## Node Allocatable

node capacity

'Allocatable' on a Kubernetes node is defined as the amount of compute resources that are available for pods. The scheduler does not over-subscribe 'Allocatable'. 'CPU', 'memory' and 'ephemeral-storage' are supported as of now.

Node Allocatable is exposed as part of v1.Node object in the API and as part of kubectl describe node in the CLI.

Resources can be reserved for two categories of system daemons in the kubelet.

## Enabling QoS and Pod level cgroups

To properly enforce node allocatable constraints on the node, you must enable the new cgroup hierarchy via the --cgroups-per-qos flag. This flag is enabled by default. When enabled, the kubelet will parent all end-user pods under a cgroup hierarchy managed by the kubelet.

## Configuring a cgroup driver

The kubelet supports manipulation of the cgroup hierarchy on the host using a cgroup driver. The driver is configured via the --cgroup-driver flag.

The supported values are the following:

- `cgroupfs` is the default driver that performs direct manipulation of the cgroup filesystem on the host in order to manage cgroup sandboxes.
- `systemd` is an alternative driver that manages cgroup sandboxes using transient slices for resources that are supported by that init system.

Depending on the configuration of the associated container runtime, operators may have to choose a particular cgroup driver to ensure proper system behavior. For example, if operators use the `systemd` cgroup driver provided by the `containerd` runtime, the `kubelet` must be configured to use the `systemd` cgroup driver.

## Kube Reserved

- **Kubelet Flag:** `--kube-reserved=[cpu=100m][,][memory=100Mi][,][ephemeral-storage=1Gi][,][pid=1000]`
- **Kubelet Flag:** `--kube-reserved-cgroup=`

`kube-reserved` is meant to capture resource reservation for kubernetes system daemons like the `kubelet`, container runtime, node problem detector, etc. It is not meant to reserve resources for system daemons that are run as pods. `kube-reserved` is typically a function of pod density on the nodes.

In addition to `cpu`, `memory`, and `ephemeral-storage`, `pid` may be specified to reserve the specified number of process IDs for kubernetes system daemons.

To optionally enforce `kube-reserved` on kubernetes system daemons, specify the parent control group for kube daemons as the value for `--kube-reserved-cgroup` kubelet flag.

It is recommended that the kubernetes system daemons are placed under a top level control group (`runtime.slice` on `systemd` machines for example). Each system daemon should ideally run within its own child control group. Refer to [the design proposal](#) for more details on recommended control group hierarchy.

Note that Kubelet **does not** create `--kube-reserved-cgroup` if it doesn't exist. The kubelet will fail to start if an invalid cgroup is specified. With `systemd` cgroup driver, you should follow a specific pattern for the name of the cgroup you define: the name should be the value you set for `--kube-reserved-cgroup`, with `.slice` appended.

## System Reserved

- **Kubelet Flag:** `--system-reserved=[cpu=100m][,][memory=100Mi][,][ephemeral-storage=1Gi][,][pid=1000]`
- **Kubelet Flag:** `--system-reserved-cgroup=`

`system-reserved` is meant to capture resource reservation for OS system daemons like `sshd`, `udev`, etc. `system-reserved` should reserve memory for the kernel too since kernel memory is not accounted to pods in Kubernetes at this time. Reserving resources for user login sessions is also recommended (`user.slice` in `systemd` world).

In addition to `cpu`, `memory`, and `ephemeral-storage`, `pid` may be specified to reserve the specified number of process IDs for OS system daemons.

To optionally enforce system-reserved on system daemons, specify the parent control group for OS system daemons as the value for --system-reserved-cgroup kubelet flag.

It is recommended that the OS system daemons are placed under a top level control group (system.slice on systemd machines for example).

Note that kubelet **does not** create --system-reserved-cgroup if it doesn't exist. kubelet will fail if an invalid cgroup is specified. With systemd cgroup driver, you should follow a specific pattern for the name of the cgroup you define: the name should be the value you set for --system-reserved-cgroup, with .slice appended.

## Explicitly Reserved CPU List

FEATURE STATE: Kubernetes v1.17 [stable]

**Kubelet Flag:** --reserved-cpus=0-3 **KubeletConfiguration Flag:** reservedSystemCPUs: 0-3

reserved-cpus is meant to define an explicit CPU set for OS system daemons and kubernetes system daemons. reserved-cpus is for systems that do not intend to define separate top level cgroups for OS system daemons and kubernetes system daemons with regard to cpuset resource. If the Kubelet **does not** have --system-reserved-cgroup and --kube-reserved-cgroup, the explicit cpuset provided by reserved-cpus will take precedence over the CPUs defined by --kube-reserved and --system-reserved options.

This option is specifically designed for Telco/NFV use cases where uncontrolled interrupts/timers may impact the workload performance. you can use this option to define the explicit cpuset for the system/kubernetes daemons as well as the interrupts/timers, so the rest CPUs on the system can be used exclusively for workloads, with less impact from uncontrolled interrupts/timers. To move the system daemon, kubernetes daemons and interrupts/timers to the explicit cpuset defined by this option, other mechanism outside Kubernetes should be used. For example: in Centos, you can do this using the tuned toolset.

## Eviction Thresholds

**Kubelet Flag:** --eviction-hard=[memory.available<500Mi]

Memory pressure at the node level leads to System OOMs which affects the entire node and all pods running on it. Nodes can go offline temporarily until memory has been reclaimed. To avoid (or reduce the probability of) system OOMs kubelet provides [out of resource](#) management. Evictions are supported for memory and ephemeral-storage only. By reserving some memory via --eviction-hard flag, the kubelet attempts to evict pods whenever memory availability on the node drops below the reserved value. Hypothetically, if system daemons did not exist on a node, pods cannot use more than capacity - eviction-hard. For this reason, resources reserved for evictions are not available for pods.

## Enforcing Node Allocatable

**Kubelet Flag:** --enforce-node-allocatable=pods[,][system-reserved][,][kube-reserved]

The scheduler treats 'Allocatable' as the available capacity for pods.

kubelet enforce 'Allocatable' across pods by default. Enforcement is performed by evicting pods whenever the overall usage across all pods exceeds 'Allocatable'. More details on eviction policy

can be found on the [node pressure eviction](#) page. This enforcement is controlled by specifying pods value to the kubelet flag `--enforce-node-allocatable`.

Optionally, kubelet can be made to enforce kube-reserved and system-reserved by specifying kube-reserved & system-reserved values in the same flag. Note that to enforce kube-reserved or system-reserved, `--kube-reserved-cgroup` or `--system-reserved-cgroup` needs to be specified respectively.

## General Guidelines

System daemons are expected to be treated similar to [Guaranteed pods](#). System daemons can burst within their bounding control groups and this behavior needs to be managed as part of kubernetes deployments. For example, kubelet should have its own control group and share kube-reserved resources with the container runtime. However, Kubelet cannot burst and use up all available Node resources if kube-reserved is enforced.

Be extra careful while enforcing system-reserved reservation since it can lead to critical system services being CPU starved, OOM killed, or unable to fork on the node. The recommendation is to enforce system-reserved only if a user has profiled their nodes exhaustively to come up with precise estimates and is confident in their ability to recover if any process in that group is oom-killed.

- To begin with enforce 'Allocatable' on pods.
- Once adequate monitoring and alerting is in place to track kube system daemons, attempt to enforce kube-reserved based on usage heuristics.
- If absolutely necessary, enforce system-reserved over time.

The resource requirements of kube system daemons may grow over time as more and more features are added. Over time, kubernetes project will attempt to bring down utilization of node system daemons, but that is not a priority as of now. So expect a drop in Allocatable capacity in future releases.

## Example Scenario

Here is an example to illustrate Node Allocatable computation:

- Node has 32Gi of memory, 16 CPUs and 100Gi of Storage
- `--kube-reserved` is set to `cpu=1,memory=2Gi,ephemeral-storage=1Gi`
- `--system-reserved` is set to `cpu=500m,memory=1Gi,ephemeral-storage=1Gi`
- `--eviction-hard` is set to `memory.available<500Mi,nodefs.available<10%`

Under this scenario, 'Allocatable' will be 14.5 CPUs, 28.5Gi of memory and 88Gi of local storage. Scheduler ensures that the total memory requests across all pods on this node does not exceed 28.5Gi and storage doesn't exceed 88Gi. Kubelet evicts pods whenever the overall memory usage across pods exceeds 28.5Gi, or if overall disk usage exceeds 88Gi. If all processes on the node consume as much CPU as they can, pods together cannot consume more than 14.5 CPUs.

If kube-reserved and/or system-reserved is not enforced and system daemons exceed their reservation, kubelet evicts pods whenever the overall node memory usage is higher than 31.5Gi or storage is greater than 90Gi.

# Running Kubernetes Node Components as a Non-root User

FEATURE STATE: Kubernetes v1.22 [alpha]

This document describes how to run Kubernetes Node components such as kubelet, CRI, OCI, and CNI without root privileges, by using a [user namespace](#).

This technique is also known as *rootless mode*.

## Note:

This document describes how to run Kubernetes Node components (and hence pods) as a non-root user.

If you are just looking for how to run a pod as a non-root user, see [SecurityContext](#).

## Before you begin

Your Kubernetes server must be at or later than version 1.22. To check the version, enter `kubectl version`.

- [Enable Cgroup v2](#)
- [Enable systemd with user session](#)
- [Configure several sysctl values, depending on host Linux distribution](#)
- [Ensure that your unprivileged user is listed in /etc/subuid and /etc/subgid](#)
- Enable the KubeletInUserNamespace [feature gate](#)

## Running Kubernetes inside Rootless Docker/Podman

### kind

[kind](#) supports running Kubernetes inside Rootless Docker or Rootless Podman.

See [Running kind with Rootless Docker](#).

### minikube

[minikube](#) also supports running Kubernetes inside Rootless Docker or Rootless Podman.

See the Minikube documentation:

- [Rootless Docker](#)
- [Rootless Podman](#)

## Running Kubernetes inside Unprivileged Containers

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are



listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information.](#)

## **sysbox**

[Sysbox](#) is an open-source container runtime (similar to "runc") that supports running system-level workloads such as Docker and Kubernetes inside unprivileged containers isolated with the Linux user namespace.

See [Sysbox Quick Start Guide: Kubernetes-in-Docker](#) for more info.

Sysbox supports running Kubernetes inside unprivileged containers without requiring Cgroup v2 and without the KubeletInUserNamespace feature gate. It does this by exposing specially crafted /proc and /sys filesystems inside the container plus several other advanced OS virtualization techniques.

## **Running Rootless Kubernetes directly on a host**

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information.](#)

### **K3s**

[K3s](#) experimentally supports rootless mode.

See [Running K3s with Rootless mode](#) for the usage.

### **Usernetes**

[Usernetes](#) is a reference distribution of Kubernetes that can be installed under \$HOME directory without the root privilege.

Usernetes supports both containerd and CRI-O as CRI runtimes. Usernetes supports multi-node clusters using Flannel (VXLAN).

See [the Usernetes repo](#) for the usage.

## **Manually deploy a node that runs the kubelet in a user namespace**

This section provides hints for running Kubernetes in a user namespace manually.

### **Note:**

This section is intended to be read by developers of Kubernetes distributions, not by end users.



## Creating a user namespace

The first step is to create a [user namespace](#).

If you are trying to run Kubernetes in a user-namespaced container such as Rootless Docker/Podman or LXC/LXD, you are all set, and you can go to the next subsection.

Otherwise you have to create a user namespace by yourself, by calling `unshare(2)` with `CLONE_NEWUSER`.

A user namespace can be also unshared by using command line tools such as:

- [unshare\(1\)](#)
- [RootlessKit](#)
- [become-root](#)

After unsharing the user namespace, you will also have to unshare other namespaces such as mount namespace.

You do *not* need to call `chroot()` nor `pivot_root()` after unsharing the mount namespace, however, you have to mount writable filesystems on several directories *in* the namespace.

At least, the following directories need to be writable *in* the namespace (not *outside* the namespace):

- `/etc`
- `/run`
- `/var/logs`
- `/var/lib/kubelet`
- `/var/lib/cni`
- `/var/lib/containerd` (for containerd)
- `/var/lib/containers` (for CRI-O)

## Creating a delegated cgroup tree

In addition to the user namespace, you also need to have a writable cgroup tree with cgroup v2.

### Note:

Kubernetes support for running Node components in user namespaces requires cgroup v2. Cgroup v1 is not supported.

If you are trying to run Kubernetes in Rootless Docker/Podman or LXC/LXD on a systemd-based host, you are all set.

Otherwise you have to create a systemd unit with `Delegate=yes` property to delegate a cgroup tree with writable permission.

On your node, systemd must already be configured to allow delegation; for more details, see [cgroup v2](#) in the Rootless Containers documentation.

## Configuring network

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

The network namespace of the Node components has to have a non-loopback interface, which can be for example configured with [slirp4netns](#), [VPNKit](#), or [lxc-user-nic\(1\)](#).

The network namespaces of the Pods can be configured with regular CNI plugins. For multi-node networking, Flannel (VXLAN, 8472/UDP) is known to work.

Ports such as the kubelet port (10250/TCP) and NodePort service ports have to be exposed from the Node network namespace to the host with an external port forwarder, such as RootlessKit, slirp4netns, or [socat\(1\)](#).

You can use the port forwarder from K3s. See [Running K3s in Rootless Mode](#) for more details. The implementation can be found in [the pkg/rootlessports package](#) of k3s.

## Configuring CRI

The kubelet relies on a container runtime. You should deploy a container runtime such as containerd or CRI-O and ensure that it is running within the user namespace before the kubelet starts.

- [containerd](#)
- [CRI-O](#)

Running CRI plugin of containerd in a user namespace is supported since containerd 1.4.

Running containerd within a user namespace requires the following configurations.

```
version = 2

[plugins."io.containerd.grpc.v1.cri"]
Disable AppArmor
 disable_apparmor = true
Ignore an error during setting oom_score_adj
 restrict_oom_score_adj = true
Disable hugetlb cgroup v2 controller (because systemd does not support delegating hugetlb
controller)
 disable_hugetlb_controller = true

[plugins."io.containerd.grpc.v1.cri".containerd]
Using non-fuse overlayfs is also possible for kernel >= 5.11, but requires SELinux to be
disabled
 snapshotter = "fuse-overlayfs"

[plugins."io.containerd.grpc.v1.cri".containerd.runtimes.runc.options]
We use cgroupfs that is delegated by systemd, so we do not use SystemdCgroup driver
(unless you run another systemd in the namespace)
 SystemdCgroup = false
```

The default path of the configuration file is `/etc/containerd/config.toml`. The path can be specified with `containerd -c /path/to/containerd/config.toml`.

Running CRI-O in a user namespace is supported since CRI-O 1.22.

CRI-O requires an environment variable `_CRIO_ROOTLESS=1` to be set.

The following configurations are also recommended:

```
[crio]
storage_driver = "overlay"
Using non-fuse overlayfs is also possible for kernel >= 5.11, but requires SELinux to be
disabled
storage_option = ["overlay.mount_program=/usr/local/bin/fuse-overlayfs"]

[crio.runtime]
We use cgroupfs that is delegated by systemd, so we do not use "systemd" driver
(unless you run another systemd in the namespace)
cgroup_manager = "cgroupfs"
```

The default path of the configuration file is `/etc/crio/crio.conf`. The path can be specified with `crio --config /path/to/crio/crio.conf`.

## Configuring kubelet

Running kubelet in a user namespace requires the following configuration:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
featureGates:
 KubeletInUserNamespace: true
We use cgroupfs that is delegated by systemd, so we do not use "systemd" driver
(unless you run another systemd in the namespace)
cgroupDriver: "cgroupfs"
```

When the `KubeletInUserNamespace` feature gate is enabled, the kubelet ignores errors that may happen during setting the following `sysctl` values on the node.

- `vm.overcommit_memory`
- `vm.panic_on_oom`
- `kernel.panic`
- `kernel.panic_on_oops`
- `kernel.keys.root_maxkeys`
- `kernel.keys.root_maxbytes`.

Within a user namespace, the kubelet also ignores any error raised from trying to open `/dev/kmsg`. This feature gate also allows kube-proxy to ignore an error during setting `RLIMIT_NOFILE`.

The `KubeletInUserNamespace` feature gate was introduced in Kubernetes v1.22 with "alpha" status.

Running kubelet in a user namespace without using this feature gate is also possible by mounting a specially crafted proc filesystem (as done by [Sysbox](#)), but not officially supported.

## Configuring kube-proxy

Running kube-proxy in a user namespace requires the following configuration:

```
apiVersion: kubeproxy.config.k8s.io/v1alpha1
kind: KubeProxyConfiguration
mode: "iptables" # or "userspace"
conntrack:
Skip setting sysctl value "net.netfilter.nf_conntrack_max"
 maxPerCore: 0
Skip setting "net.netfilter.nf_conntrack_tcp_timeout_established"
 tcpEstablishedTimeout: 0s
Skip setting "net.netfilter.nf_conntrack_tcp_timeout_close"
 tcpCloseWaitTimeout: 0s
```

## Caveats

- Most of "non-local" volume drivers such as nfs and iscsi do not work. Local volumes like local, hostPath, emptyDir, configMap, secret, and downwardAPI are known to work.
- Some CNI plugins may not work. Flannel (VXLAN) is known to work.

For more on this, see the [Caveats and Future work](#) page on the rootlesscontainers website.

## See Also

- [rootlesscontainers.rs](#)
- [Rootless Containers 2020 \(KubeCon NA 2020\)](#)
- [Running kind with Rootless Docker](#)
- [Usernetes](#)
- [Running K3s with rootless mode](#)
- [KEP-2033: Kubelet-in-UserNS \(aka Rootless mode\)](#)

## Safely Drain a Node

This page shows how to safely drain a [node](#), optionally respecting the PodDisruptionBudget you have defined.

## Before you begin

This task assumes that you have met the following prerequisites:

1. You do not require your applications to be highly available during the node drain, or
2. You have read about the [PodDisruptionBudget](#) concept, and have [configured PodDisruptionBudgets](#) for applications that need them.

## (Optional) Configure a disruption budget

To ensure that your workloads remain available during maintenance, you can configure a [PodDisruptionBudget](#).

If availability is important for any applications that run or could run on the node(s) that you are draining, [configure a PodDisruptionBudgets](#) first and then continue following this guide.

It is recommended to set AlwaysAllow [Unhealthy Pod Eviction Policy](#) to your PodDisruptionBudgets to support eviction of misbehaving applications during a node drain. The default behavior is to wait for the application pods to become [healthy](#) before the drain can proceed.

## Use kubectl drain to remove a node from service

You can use kubectl drain to safely evict all of your pods from a node before you perform maintenance on the node (e.g. kernel upgrade, hardware maintenance, etc.). Safe evictions allow the pod's containers to [gracefully terminate](#) and will respect the PodDisruptionBudgets you have specified.

### Note:

By default kubectl drain ignores certain system pods on the node that cannot be killed; see the [kubectl drain](#) documentation for more details.

When kubectl drain returns successfully, that indicates that all of the pods (except the ones excluded as described in the previous paragraph) have been safely evicted (respecting the desired graceful termination period, and respecting the PodDisruptionBudget you have defined). It is then safe to bring down the node by powering down its physical machine or, if running on a cloud platform, deleting its virtual machine.

### Note:

If any new Pods tolerate the node.kubernetes.io/unschedulable taint, then those Pods might be scheduled to the node you have drained. Avoid tolerating that taint other than for DaemonSets.

If you or another API user directly set the [nodeName](#) field for a Pod (bypassing the scheduler), then the Pod is bound to the specified node and will run there, even though you have drained that node and marked it unschedulable.

First, identify the name of the node you wish to drain. You can list all of the nodes in your cluster with

```
kubectl get nodes
```

Next, tell Kubernetes to drain the node:

```
kubectl drain --ignore-daemonsets <node name>
```

If there are pods managed by a DaemonSet, you will need to specify `--ignore-daemonsets` with kubectl to successfully drain the node. The kubectl drain subcommand on its own does not actually drain a node of its DaemonSet pods: the DaemonSet controller (part of the control

plane) immediately replaces missing Pods with new equivalent Pods. The DaemonSet controller also creates Pods that ignore unschedulable taints, which allows the new Pods to launch onto a node that you are draining.

Once it returns (without giving an error), you can power down the node (or equivalently, if on a cloud platform, delete the virtual machine backing the node). If you leave the node in the cluster during the maintenance operation, you need to run

```
kubectl uncordon <node name>
```

afterwards to tell Kubernetes that it can resume scheduling new pods onto the node.

## Draining multiple nodes in parallel

The `kubectl drain` command should only be issued to a single node at a time. However, you can run multiple `kubectl drain` commands for different nodes in parallel, in different terminals or in the background. Multiple drain commands running concurrently will still respect the `PodDisruptionBudget` you specify.

For example, if you have a `StatefulSet` with three replicas and have set a `PodDisruptionBudget` for that set specifying `minAvailable: 2`, `kubectl drain` only evicts a pod from the `StatefulSet` if all three replicas pods are [healthy](#); if then you issue multiple drain commands in parallel, Kubernetes respects the `PodDisruptionBudget` and ensures that only 1 (calculated as `replicas - minAvailable`) Pod is unavailable at any given time. Any drains that would cause the number of [healthy](#) replicas to fall below the specified budget are blocked.

## The Eviction API

If you prefer not to use [kubectl drain](#) (such as to avoid calling to an external command, or to get finer control over the pod eviction process), you can also programmatically cause evictions using the eviction API.

For more information, see [API-initiated eviction](#).

## What's next

- Follow steps to protect your application by [configuring a Pod Disruption Budget](#).

# Securing a Cluster

This document covers topics related to protecting a cluster from accidental or malicious access and provides recommendations on overall security.

## Before you begin

- You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not

already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Controlling access to the Kubernetes API

As Kubernetes is entirely API-driven, controlling and limiting who can access the cluster and what actions they are allowed to perform is the first line of defense.

### Use Transport Layer Security (TLS) for all API traffic

Kubernetes expects that all API communication in the cluster is encrypted by default with TLS, and the majority of installation methods will allow the necessary certificates to be created and distributed to the cluster components. Note that some components and installation methods may enable local ports over HTTP and administrators should familiarize themselves with the settings of each component to identify potentially unsecured traffic.

### API Authentication

Choose an authentication mechanism for the API servers to use that matches the common access patterns when you install a cluster. For instance, small, single-user clusters may wish to use a simple certificate or static Bearer token approach. Larger clusters may wish to integrate an existing OIDC or LDAP server that allow users to be subdivided into groups.

All API clients must be authenticated, even those that are part of the infrastructure like nodes, proxies, the scheduler, and volume plugins. These clients are typically [service accounts](#) or use x509 client certificates, and they are created automatically at cluster startup or are setup as part of the cluster installation.

Consult the [authentication reference document](#) for more information.

### API Authorization

Once authenticated, every API call is also expected to pass an authorization check. Kubernetes ships an integrated [Role-Based Access Control \(RBAC\)](#) component that matches an incoming user or group to a set of permissions bundled into roles. These permissions combine verbs (get, create, delete) with resources (pods, services, nodes) and can be namespace-scoped or cluster-scoped. A set of out-of-the-box roles are provided that offer reasonable default separation of responsibility depending on what actions a client might want to perform. It is recommended that you use the [Node](#) and [RBAC](#) authorizers together, in combination with the [NodeRestriction](#) admission plugin.

As with authentication, simple and broad roles may be appropriate for smaller clusters, but as more users interact with the cluster, it may become necessary to separate teams into separate [namespaces](#) with more limited roles.

With authorization, it is important to understand how updates on one object may cause actions in other places. For instance, a user may not be able to create pods directly, but allowing them to create a deployment, which creates pods on their behalf, will let them create those pods

indirectly. Likewise, deleting a node from the API will result in the pods scheduled to that node being terminated and recreated on other nodes. The out-of-the box roles represent a balance between flexibility and common use cases, but more limited roles should be carefully reviewed to prevent accidental escalation. You can make roles specific to your use case if the out-of-box ones don't meet your needs.

Consult the [authorization reference section](#) for more information.

## Controlling access to the Kubelet

Kubelets expose HTTPS endpoints which grant powerful control over the node and containers. By default Kubelets allow unauthenticated access to this API.

Production clusters should enable Kubelet authentication and authorization.

Consult the [Kubelet authentication/authorization reference](#) for more information.

## Controlling the capabilities of a workload or user at runtime

Authorization in Kubernetes is intentionally high level, focused on coarse actions on resources. More powerful controls exist as **policies** to limit by use case how those objects act on the cluster, themselves, and other resources.

### Limiting resource usage on a cluster

[Resource quota](#) limits the number or capacity of resources granted to a namespace. This is most often used to limit the amount of CPU, memory, or persistent disk a namespace can allocate, but can also control how many pods, services, or volumes exist in each namespace.

[Limit ranges](#) restrict the maximum or minimum size of some of the resources above, to prevent users from requesting unreasonably high or low values for commonly reserved resources like memory, or to provide default limits when none are specified.

### Controlling what privileges containers run with

A pod definition contains a [security context](#) that allows it to request access to run as a specific Linux user on a node (like root), access to run privileged or access the host network, and other controls that would otherwise allow it to run unfettered on a hosting node.

You can configure [Pod security admission](#) to enforce use of a particular [Pod Security Standard](#) in a [namespace](#), or to detect breaches.

Generally, most application workloads need limited access to host resources so they can successfully run as a root process (uid 0) without access to host information. However, considering the privileges associated with the root user, you should write application containers to run as a non-root user. Similarly, administrators who wish to prevent client applications from escaping their containers should apply the **Baseline** or **Restricted** Pod Security Standard.



## Preventing containers from loading unwanted kernel modules

The Linux kernel automatically loads kernel modules from disk if needed in certain circumstances, such as when a piece of hardware is attached or a filesystem is mounted. Of particular relevance to Kubernetes, even unprivileged processes can cause certain network-protocol-related kernel modules to be loaded, just by creating a socket of the appropriate type. This may allow an attacker to exploit a security hole in a kernel module that the administrator assumed was not in use.

To prevent specific modules from being automatically loaded, you can uninstall them from the node, or add rules to block them. On most Linux distributions, you can do that by creating a file such as `/etc/modprobe.d/kubernetes-blacklist.conf` with contents like:

```
DCCP is unlikely to be needed, has had multiple serious
vulnerabilities, and is not well-maintained.
blacklist dccp

SCTP is not used in most Kubernetes clusters, and has also had
vulnerabilities in the past.
blacklist sctp
```

To block module loading more generically, you can use a Linux Security Module (such as SELinux) to completely deny the `module_request` permission to containers, preventing the kernel from loading modules for containers under any circumstances. (Pods would still be able to use modules that had been loaded manually, or modules that were loaded by the kernel on behalf of some more-privileged process.)

## Restricting network access

The [network policies](#) for a namespace allows application authors to restrict which pods in other namespaces may access pods and ports within their namespaces. Many of the supported [Kubernetes networking providers](#) now respect network policy.

Quota and limit ranges can also be used to control whether users may request node ports or load-balanced services, which on many clusters can control whether those users applications are visible outside of the cluster.

Additional protections may be available that control network rules on a per-plugin or per-environment basis, such as per-node firewalls, physically separating cluster nodes to prevent cross talk, or advanced networking policy.

## Restricting cloud metadata API access

Cloud platforms (AWS, Azure, GCE, etc.) often expose metadata services locally to instances. By default these APIs are accessible by pods running on an instance and can contain cloud credentials for that node, or provisioning data such as kubelet credentials. These credentials can be used to escalate within the cluster or to other cloud services under the same account.

When running Kubernetes on a cloud platform, limit permissions given to instance credentials, use [network policies](#) to restrict pod access to the metadata API, and avoid using provisioning data to deliver secrets.

## Controlling which nodes pods may access

By default, there are no restrictions on which nodes may run a pod. Kubernetes offers a [rich set of policies for controlling placement of pods onto nodes](#) and the [taint-based pod placement and eviction](#) that are available to end users. For many clusters use of these policies to separate workloads can be a convention that authors adopt or enforce via tooling.

As an administrator, a beta admission plugin PodNodeSelector can be used to force pods within a namespace to default or require a specific node selector, and if end users cannot alter namespaces, this can strongly limit the placement of all of the pods in a specific workload.

## Protecting cluster components from compromise

This section describes some common patterns for protecting clusters from compromise.

### Restrict access to etcd

Write access to the etcd backend for the API is equivalent to gaining root on the entire cluster, and read access can be used to escalate fairly quickly. Administrators should always use strong credentials from the API servers to their etcd server, such as mutual auth via TLS client certificates, and it is often recommended to isolate the etcd servers behind a firewall that only the API servers may access.

#### Caution:

Allowing other components within the cluster to access the master etcd instance with read or write access to the full keypace is equivalent to granting cluster-admin access. Using separate etcd instances for non-master components or using etcd ACLs to restrict read and write access to a subset of the keypace is strongly recommended.

### Enable audit logging

The [audit logger](#) is a beta feature that records actions taken by the API for later analysis in the event of a compromise. It is recommended to enable audit logging and archive the audit file on a secure server.

### Restrict access to alpha or beta features

Alpha and beta Kubernetes features are in active development and may have limitations or bugs that result in security vulnerabilities. Always assess the value an alpha or beta feature may provide against the possible risk to your security posture. When in doubt, disable features you do not use.

### Rotate infrastructure credentials frequently

The shorter the lifetime of a secret or credential the harder it is for an attacker to make use of that credential. Set short lifetimes on certificates and automate their rotation. Use an authentication provider that can control how long issued tokens are available and use short lifetimes where possible. If you use service-account tokens in external integrations, plan to rotate those tokens frequently. For example, once the bootstrap phase is complete, a bootstrap token used for setting up nodes should be revoked or its authorization removed.

## Review third party integrations before enabling them

Many third party integrations to Kubernetes may alter the security profile of your cluster. When enabling an integration, always review the permissions that an extension requests before granting it access. For example, many security integrations may request access to view all secrets on your cluster which is effectively making that component a cluster admin. When in doubt, restrict the integration to functioning in a single namespace if possible.

Components that create pods may also be unexpectedly powerful if they can do so inside namespaces like the kube-system namespace, because those pods can gain access to service account secrets or run with elevated permissions if those service accounts are granted access to permissive [PodSecurityPolicies](#).

If you use [Pod Security admission](#) and allow any component to create Pods within a namespace that permits privileged Pods, those Pods may be able to escape their containers and use this widened access to elevate their privileges.

You should not allow untrusted components to create Pods in any system namespace (those with names that start with kube-) nor in any namespace where that access grant allows the possibility of privilege escalation.

## Encrypt secrets at rest

In general, the etcd database will contain any information accessible via the Kubernetes API and may grant an attacker significant visibility into the state of your cluster. Always encrypt your backups using a well reviewed backup and encryption solution, and consider using full disk encryption where possible.

Kubernetes supports optional [encryption at rest](#) for information in the Kubernetes API. This lets you ensure that when Kubernetes stores data for objects (for example, Secret or ConfigMap objects), the API server writes an encrypted representation of the object. That encryption means that even someone who has access to etcd backup data is unable to view the content of those objects. In Kubernetes 1.30 you can also encrypt custom resources; encryption-at-rest for extension APIs defined in CustomResourceDefinitions was added to Kubernetes as part of the v1.26 release.

## Receiving alerts for security updates and reporting vulnerabilities

Join the [kubernetes-announce](#) group for emails about security announcements. See the [security reporting](#) page for more on how to report vulnerabilities.

## What's next

- [Security Checklist](#) for additional information on Kubernetes security guidance.

# Set Kubelet Parameters Via A Configuration File

## Before you begin

Some steps in this page use the jq tool. If you don't have jq, you can install it via your operating system's software sources, or fetch it from <https://jqlang.github.io/jq/>.

Some steps also involve installing curl, which can be installed via your operating system's software sources.

A subset of the kubelet's configuration parameters may be set via an on-disk config file, as a substitute for command-line flags.

Providing parameters via a config file is the recommended approach because it simplifies node deployment and configuration management.

## Create the config file

The subset of the kubelet's configuration that can be configured via a file is defined by the [KubeletConfiguration](#) struct.

The configuration file must be a JSON or YAML representation of the parameters in this struct. Make sure the kubelet has read permissions on the file.

Here is an example of what this file might look like:

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
address: "192.168.0.8"
port: 20250
serializeImagePulls: false
evictionHard:
 memory.available: "100Mi"
 nodefs.available: "10%"
 nodefs.inodesFree: "5%"
 imagefs.available: "15%"
```

In this example, the kubelet is configured with the following settings:

1. address: The kubelet will serve on IP address 192.168.0.8.
2. port: The kubelet will serve on port 20250.
3. serializeImagePulls: Image pulls will be done in parallel.
4. evictionHard: The kubelet will evict Pods under one of the following conditions:
  - When the node's available memory drops below 100MiB.
  - When the node's main filesystem's available space is less than 10%.
  - When the image filesystem's available space is less than 15%.

- When more than 95% of the node's main filesystem's inodes are in use.

**Note:**

In the example, by changing the default value of only one parameter for `evictionHard`, the default values of other parameters will not be inherited and will be set to zero. In order to provide custom values, you should provide all the threshold values respectively.

The `imagefs` is an optional filesystem that container runtimes use to store container images and container writable layers.

## Start a kubelet process configured via the config file

**Note:**

If you use `kubeadm` to initialize your cluster, use the `kubelet-config` while creating your cluster with `kubeadm init`. See [configuring kubelet using kubeadm](#) for details.

Start the kubelet with the `--config` flag set to the path of the kubelet's config file. The kubelet will then load its config from this file.

Note that command line flags which target the same value as a config file will override that value. This helps ensure backwards compatibility with the command-line API.

Note that relative file paths in the kubelet config file are resolved relative to the location of the kubelet config file, whereas relative paths in command line flags are resolved relative to the kubelet's current working directory.

Note that some default values differ between command-line flags and the kubelet config file. If `--config` is provided and the values are not specified via the command line, the defaults for the `KubeletConfiguration` version apply. In the above example, this version is `kubelet.config.k8s.io/v1beta1`.

## Drop-in directory for kubelet configuration files

FEATURE STATE: Kubernetes v1.30 [beta]

You can specify a drop-in configuration directory for the kubelet. By default, the kubelet does not look for drop-in configuration files anywhere - you must specify a path. For example: `--config-dir=/etc/kubernetes/kubelet.conf.d`

For Kubernetes v1.28 to v1.29, you can only specify `--config-dir` if you also set the environment variable `KUBELET_CONFIG_DROPIN_DIR_ALPHA` for the kubelet process (the value of that variable does not matter).

**Note:**

The suffix of a valid kubelet drop-in configuration file **must** be `.conf`. For instance: `99-kubelet-address.conf`

The kubelet processes files in its config drop-in directory by sorting the **entire file name** alphanumerically. For instance, 00-kubelet.conf is processed first, and then overridden with a file named 01-kubelet.conf.

These files may contain partial configurations but should not be invalid and must include type metadata, specifically `apiVersion` and `kind`. Validation is only performed on the final resulting configuration structure stored internally in the kubelet. This offers flexibility in managing and merging kubelet configurations from different sources while preventing undesirable configurations. However, it is important to note that behavior varies based on the data type of the configuration fields.

Different data types in the kubelet configuration structure merge differently. See the [reference document](#) for more information.

## Kubelet configuration merging order

On startup, the kubelet merges configuration from:

- Feature gates specified over the command line (lowest precedence).
- The kubelet configuration.
- Drop-in configuration files, according to sort order.
- Command line arguments excluding feature gates (highest precedence).

### Note:

The config drop-in dir mechanism for the kubelet is similar but different from how the kubeadm tool allows you to patch configuration. The kubeadm tool uses a specific [patching strategy](#) for its configuration, whereas the only patch strategy for kubelet configuration drop-in files is replace. The kubelet determines the order of merges based on sorting the **suffixes** alphanumerically, and replaces every field present in a higher priority file.

## Viewing the kubelet configuration

Since the configuration could now be spread over multiple files with this feature, if someone wants to inspect the final actuated configuration, they can follow these steps to inspect the kubelet configuration:

1. Start a proxy server using [kubectrl proxy](#) in your terminal.

```
kubectrl proxy
```

Which gives output like:

```
Starting to serve on 127.0.0.1:8001
```

2. Open another terminal window and use `curl` to fetch the kubelet configuration. Replace `<node-name>` with the actual name of your node:

```
curl -X GET http://127.0.0.1:8001/api/v1/nodes/<node-name>/proxy/configz | jq .
```

```
{
 "kubeletconfig": {
 "enableServer": true,
```

```
"staticPodPath": "/var/run/kubernetes/static-pods",
"syncFrequency": "1m0s",
"fileCheckFrequency": "20s",
"httpCheckFrequency": "20s",
"address": "192.168.1.16",
"port": 10250,
"readOnlyPort": 10255,
"tlsCertFile": "/var/lib/kubelet/pki/kubelet.crt",
"tlsPrivateKeyFile": "/var/lib/kubelet/pki/kubelet.key",
"rotateCertificates": true,
"authentication": {
 "x509": {
 "clientCAFile": "/var/run/kubernetes/client-ca.crt"
 },
 "webhook": {
 "enabled": true,
 "cacheTTL": "2m0s"
 },
 "anonymous": {
 "enabled": true
 }
},
"authorization": {
 "mode": "AlwaysAllow",
 "webhook": {
 "cacheAuthorizedTTL": "5m0s",
 "cacheUnauthorizedTTL": "30s"
 }
},
"registryPullQPS": 5,
"registryBurst": 10,
"eventRecordQPS": 50,
"eventBurst": 100,
"enableDebuggingHandlers": true,
"healthzPort": 10248,
"healthzBindAddress": "127.0.0.1",
"oomScoreAdj": -999,
"clusterDomain": "cluster.local",
"clusterDNS": [
 "10.0.0.10"
],
"streamingConnectionIdleTimeout": "4h0m0s",
"nodeStatusUpdateFrequency": "10s",
"nodeStatusReportFrequency": "5m0s",
"nodeLeaseDurationSeconds": 40,
"imageMinimumGCAge": "2m0s",
"imageMaximumGCAge": "0s",
"imageGCHighThresholdPercent": 85,
"imageGCLowThresholdPercent": 80,
"volumeStatsAggPeriod": "1m0s",
"cgroupsPerQOS": true,
"cgroupDriver": "systemd",
"cpuManagerPolicy": "none",
```

```
"cpuManagerReconcilePeriod": "10s",
"memoryManagerPolicy": "None",
"topologyManagerPolicy": "none",
"topologyManagerScope": "container",
"runtimeRequestTimeout": "2m0s",
"hairpinMode": "promiscuous-bridge",
"maxPods": 110,
"podPidsLimit": -1,
"resolvConf": "/run/systemd/resolve/resolv.conf",
"cpuCFSQuota": true,
"cpuCFSQuotaPeriod": "100ms",
"nodeStatusMaxImages": 50,
"maxOpenFiles": 1000000,
"contentType": "application/vnd.kubernetes.protobuf",
"kubeAPIQPS": 50,
"kubeAPIBurst": 100,
"serializeImagePulls": true,
"evictionHard": {
 "imagefs.available": "15%",
 "memory.available": "100Mi",
 "nodefs.available": "10%",
 "nodefs.inodesFree": "5%"
},
"evictionPressureTransitionPeriod": "1m0s",
"enableControllerAttachDetach": true,
"makeIPTablesUtilChains": true,
"iptablesMasqueradeBit": 14,
"iptablesDropBit": 15,
"featureGates": {
 "AllAlpha": false
},
"failSwapOn": false,
"memorySwap": {},
"containerLogMaxSize": "10Mi",
"containerLogMaxFiles": 5,
"configMapAndSecretChangeDetectionStrategy": "Watch",
"enforceNodeAllocatable": [
 "pods"
],
"volumePluginDir": "/usr/libexec/kubernetes/kubelet-plugins/volume/exec/",
"logging": {
 "format": "text",
 "flushFrequency": "5s",
 "verbosity": 3,
 "options": {
 "json": {
 "infoBufferSize": "0"
 }
 }
},
"enableSystemLogHandler": true,
"enableSystemLogQuery": false,
"shutdownGracePeriod": "0s",
```



```
"shutdownGracePeriodCriticalPods": "0s",
"enableProfilingHandler": true,
"enableDebugFlagsHandler": true,
"seccompDefault": false,
"memoryThrottlingFactor": 0.9,
"registerNode": true,
"localStorageCapacityIsolation": true,
"containerRuntimeEndpoint": "unix:///var/run/crio/crio.sock"
}
}
```

## What's next

- Learn more about kubelet configuration by checking the [KubeletConfiguration](#) reference.
- Learn more about kubelet configuration merging in the [reference document](#).

# Share a Cluster with Namespaces

This page shows how to view, work in, and delete [namespaces](#). The page also shows how to use Kubernetes namespaces to subdivide your cluster.

## Before you begin

- Have an [existing Kubernetes cluster](#).
- You have a basic understanding of Kubernetes [Pods](#), [Services](#), and [Deployments](#).

## Viewing namespaces

List the current namespaces in a cluster using:

```
kubectl get namespaces
```

| NAME            | STATUS | AGE |
|-----------------|--------|-----|
| default         | Active | 11d |
| kube-node-lease | Active | 11d |
| kube-public     | Active | 11d |
| kube-system     | Active | 11d |

Kubernetes starts with four initial namespaces:

- default The default namespace for objects with no other namespace
- kube-node-lease This namespace holds [Lease](#) objects associated with each node. Node leases allow the kubelet to send [heartbeats](#) so that the control plane can detect node failure.
- kube-public This namespace is created automatically and is readable by all users (including those not authenticated). This namespace is mostly reserved for cluster usage, in case that some resources should be visible and readable publicly throughout the whole cluster. The public aspect of this namespace is only a convention, not a requirement.
- kube-system The namespace for objects created by the Kubernetes system

You can also get the summary of a specific namespace using:

```
kubectl get namespaces <name>
```

Or you can get detailed information with:

```
kubectl describe namespaces <name>
```

```
Name: default
Labels: <none>
Annotations: <none>
Status: Active
```

No resource quota.

#### Resource Limits

| Type      | Resource | Min | Max | Default |
|-----------|----------|-----|-----|---------|
| Container | cpu      | -   | -   | 100m    |

Note that these details show both resource quota (if present) as well as resource limit ranges.

Resource quota tracks aggregate usage of resources in the Namespace and allows cluster operators to define *Hard* resource usage limits that a Namespace may consume.

A limit range defines min/max constraints on the amount of resources a single entity can consume in a Namespace.

See [Admission control: Limit Range](#)

A namespace can be in one of two phases:

- Active the namespace is in use
- Terminating the namespace is being deleted, and can not be used for new objects

For more details, see [Namespace](#) in the API reference.

## Creating a new namespace

### Note:

Avoid creating namespace with prefix kube-, since it is reserved for Kubernetes system namespaces.

Create a new YAML file called my-namespace.yaml with the contents:

```
apiVersion: v1
kind: Namespace
metadata:
 name: <insert-namespace-name-here>
```

Then run:

```
kubectl create -f ./my-namespace.yaml
```

Alternatively, you can create namespace using below command:

```
kubectl create namespace <insert-namespace-name-here>
```

The name of your namespace must be a valid [DNS label](#).

There's an optional field `finalizers`, which allows observables to purge resources whenever the namespace is deleted. Keep in mind that if you specify a nonexistent finalizer, the namespace will be created but will get stuck in the Terminating state if the user tries to delete it.

More information on finalizers can be found in the namespace [design doc](#).

## Deleting a namespace

Delete a namespace with

```
kubectl delete namespaces <insert-some-namespace-name>
```

### Warning:

This deletes *everything* under the namespace!

This delete is asynchronous, so for a time you will see the namespace in the Terminating state.

## Subdividing your cluster using Kubernetes namespaces

By default, a Kubernetes cluster will instantiate a default namespace when provisioning the cluster to hold the default set of Pods, Services, and Deployments used by the cluster.

Assuming you have a fresh cluster, you can introspect the available namespaces by doing the following:

```
kubectl get namespaces
```

| NAME    | STATUS | AGE |
|---------|--------|-----|
| default | Active | 13m |

### Create new namespaces

For this exercise, we will create two additional Kubernetes namespaces to hold our content.

In a scenario where an organization is using a shared Kubernetes cluster for development and production use cases:

- The development team would like to maintain a space in the cluster where they can get a view on the list of Pods, Services, and Deployments they use to build and run their application. In this space, Kubernetes resources come and go, and the restrictions on who can or cannot modify resources are relaxed to enable agile development.
- The operations team would like to maintain a space in the cluster where they can enforce strict procedures on who can or cannot manipulate the set of Pods, Services, and Deployments that run the production site.

One pattern this organization could follow is to partition the Kubernetes cluster into two namespaces: development and production. Let's create two new namespaces to hold our work.

Create the development namespace using kubectl:

```
kubectl create -f https://k8s.io/examples/admin/namespace-dev.json
```

And then let's create the production namespace using kubectl:

```
kubectl create -f https://k8s.io/examples/admin/namespace-prod.json
```

To be sure things are right, list all of the namespaces in our cluster.

```
kubectl get namespaces --show-labels
```

| NAME        | STATUS | AGE | LABELS           |
|-------------|--------|-----|------------------|
| default     | Active | 32m | <none>           |
| development | Active | 29s | name=development |
| production  | Active | 23s | name=production  |

## Create pods in each namespace

A Kubernetes namespace provides the scope for Pods, Services, and Deployments in the cluster. Users interacting with one namespace do not see the content in another namespace. To demonstrate this, let's spin up a simple Deployment and Pods in the development namespace.

```
kubectl create deployment snowflake \
--image=registry.k8s.io/serve_hostname \
-n=development --replicas=2
```

We have created a deployment whose replica size is 2 that is running the pod called snowflake with a basic container that serves the hostname.

```
kubectl get deployment -n=development
```

| NAME      | READY | UP-TO-DATE | AVAILABLE | AGE |
|-----------|-------|------------|-----------|-----|
| snowflake | 2/2   | 2          | 2         | 2m  |

```
kubectl get pods -l app=snowflake -n=development
```

| NAME                       | READY | STATUS  | RESTARTS | AGE |
|----------------------------|-------|---------|----------|-----|
| snowflake-3968820950-9dgr8 | 1/1   | Running | 0        | 2m  |
| snowflake-3968820950-vgc4n | 1/1   | Running | 0        | 2m  |

And this is great, developers are able to do what they want, and they do not have to worry about affecting content in the production namespace.

Let's switch to the production namespace and show how resources in one namespace are hidden from the other. The production namespace should be empty, and the following commands should return nothing.

```
kubectl get deployment -n=production
kubectl get pods -n=production
```

Production likes to run cattle, so let's create some cattle pods.

```
kubectl create deployment cattle --image=registry.k8s.io/serve_hostname -n=production
kubectl scale deployment cattle --replicas=5 -n=production

kubectl get deployment -n=production
```

| NAME   | READY | UP-TO-DATE | AVAILABLE | AGE |
|--------|-------|------------|-----------|-----|
| cattle | 5/5   | 5          | 5         | 10s |

```
kubectl get pods -l app=cattle -n=production
```

| NAME                    | READY | STATUS  | RESTARTS | AGE |
|-------------------------|-------|---------|----------|-----|
| cattle-2263376956-41xy6 | 1/1   | Running | 0        | 34s |
| cattle-2263376956-kw466 | 1/1   | Running | 0        | 34s |
| cattle-2263376956-n4v97 | 1/1   | Running | 0        | 34s |
| cattle-2263376956-p5p3i | 1/1   | Running | 0        | 34s |
| cattle-2263376956-sxpth | 1/1   | Running | 0        | 34s |

At this point, it should be clear that the resources users create in one namespace are hidden from the other namespace.

As the policy support in Kubernetes evolves, we will extend this scenario to show how you can provide different authorization rules for each namespace.

## Understanding the motivation for using namespaces

A single cluster should be able to satisfy the needs of multiple users or groups of users (henceforth in this document a *user community*).

Kubernetes *namespaces* help different projects, teams, or customers to share a Kubernetes cluster.

It does this by providing the following:

1. A scope for [names](#).
2. A mechanism to attach authorization and policy to a subsection of the cluster.

Use of multiple namespaces is optional.

Each user community wants to be able to work in isolation from other communities. Each user community has its own:

1. resources (pods, services, replication controllers, etc.)
2. policies (who can or cannot perform actions in their community)
3. constraints (this community is allowed this much quota, etc.)

A cluster operator may create a Namespace for each unique user community.

The Namespace provides a unique scope for:

1. named resources (to avoid basic naming collisions)
2. delegated management authority to trusted users
3. ability to limit community resource consumption

Use cases include:

1. As a cluster operator, I want to support multiple user communities on a single cluster.
2. As a cluster operator, I want to delegate authority to partitions of the cluster to trusted users in those communities.
3. As a cluster operator, I want to limit the amount of resources each community can consume in order to limit the impact to other communities using the cluster.
4. As a cluster user, I want to interact with resources that are pertinent to my user community in isolation of what other user communities are doing on the cluster.

## Understanding namespaces and DNS

When you create a [Service](#), it creates a corresponding [DNS entry](#). This entry is of the form `<service-name>.<namespace-name>.svc.cluster.local`, which means that if a container uses `<service-name>` it will resolve to the service which is local to a namespace. This is useful for using the same configuration across multiple namespaces such as Development, Staging and Production. If you want to reach across namespaces, you need to use the fully qualified domain name (FQDN).

## What's next

- Learn more about [setting the namespace preference](#).
- Learn more about [setting the namespace for a request](#)
- See [namespaces design](#).

## Upgrade A Cluster

This page provides an overview of the steps you should follow to upgrade a Kubernetes cluster.

The way that you upgrade a cluster depends on how you initially deployed it and on any subsequent changes.

At a high level, the steps you perform are:

- Upgrade the [control plane](#)
- Upgrade the nodes in your cluster
- Upgrade clients such as [kubect](#)
- Adjust manifests and other resources based on the API changes that accompany the new Kubernetes version

## Before you begin

You must have an existing cluster. This page is about upgrading from Kubernetes 1.29 to Kubernetes 1.30. If your cluster is not currently running Kubernetes 1.29 then please check the documentation for the version of Kubernetes that you plan to upgrade to.

# Upgrade approaches

## kubeadm

If your cluster was deployed using the kubeadm tool, refer to [Upgrading kubeadm clusters](#) for detailed information on how to upgrade the cluster.

Once you have upgraded the cluster, remember to [install the latest version of kubectl](#).

## Manual deployments

### Caution:

These steps do not account for third-party extensions such as network and storage plugins.

You should manually update the control plane following this sequence:

- etcd (all instances)
- kube-apiserver (all control plane hosts)
- kube-controller-manager
- kube-scheduler
- cloud controller manager, if you use one

At this point you should [install the latest version of kubectl](#).

For each node in your cluster, [drain](#) that node and then either replace it with a new node that uses the 1.30 kubelet, or upgrade the kubelet on that node and bring the node back into service.

### Caution:

Draining nodes before upgrading kubelet ensures that pods are re-admitted and containers are re-created, which may be necessary to resolve some security issues or other important bugs.

## Other deployments

Refer to the documentation for your cluster deployment tool to learn the recommended set up steps for maintenance.

# Post-upgrade tasks

## Switch your cluster's storage API version

The objects that are serialized into etcd for a cluster's internal representation of the Kubernetes resources active in the cluster are written using a particular version of the API.

When the supported API changes, these objects may need to be rewritten in the newer API. Failure to do this will eventually result in resources that are no longer decodable or usable by the Kubernetes API server.

For each affected object, fetch it using the latest supported API and then write it back also using the latest supported API.

## Update manifests

Upgrading to a new Kubernetes version can provide new APIs.

You can use `kubectl convert` command to convert manifests between different API versions. For example:

```
kubectl convert -f pod.yaml --output-version v1
```

The `kubectl` tool replaces the contents of `pod.yaml` with a manifest that sets `kind` to `Pod` (unchanged), but with a revised `apiVersion`.

## Device Plugins

If your cluster is running device plugins and the node needs to be upgraded to a Kubernetes release with a newer device plugin API version, device plugins must be upgraded to support both version before the node is upgraded in order to guarantee that device allocations continue to complete successfully during the upgrade.

Refer to [API compatibility](#) and [Kubelet Device Manager API Versions](#) for more details.

# Use Cascading Deletion in a Cluster

This page shows you how to specify the type of [cascading deletion](#) to use in your cluster during [garbage collection](#).

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You also need to [create a sample Deployment](#) to experiment with the different types of cascading deletion. You will need to recreate the Deployment for each type.

## Check owner references on your pods

Check that the `ownerReferences` field is present on your pods:

```
kubectl get pods -l app=nginx --output=yaml
```

The output has an `ownerReferences` field similar to this:

```
apiVersion: v1
...
ownerReferences:
- apiVersion: apps/v1
```



```
blockOwnerDeletion: true
controller: true
kind: ReplicaSet
name: nginx-deployment-6b474476c4
uid: 4fdcd81c-bd5d-41f7-97af-3a3b759af9a7
...
```

## Use foreground cascading deletion

By default, Kubernetes uses [background cascading deletion](#) to delete dependents of an object. You can switch to foreground cascading deletion using either `kubectl` or the Kubernetes API, depending on the Kubernetes version your cluster runs. To check the version, enter `kubectl version`.

You can delete objects using foreground cascading deletion using `kubectl` or the Kubernetes API.

### Using `kubectl`

Run the following command:

```
kubectl delete deployment nginx-deployment --cascade=foreground
```

### Using the Kubernetes API

1. Start a local proxy session:

```
kubectl proxy --port=8080
```

2. Use `curl` to trigger deletion:

```
curl -X DELETE localhost:8080/apis/apps/v1/namespaces/default/deployments/nginx-deployment \
-d '{"kind":"DeleteOptions","apiVersion":"v1","propagationPolicy":"Foreground"}' \
-H "Content-Type: application/json"
```

The output contains a foregroundDeletion [finalizer](#) like this:

```
"kind": "Deployment",
"apiVersion": "apps/v1",
"metadata": {
 "name": "nginx-deployment",
 "namespace": "default",
 "uid": "d1ce1b02-cae8-4288-8a53-30e84d8fa505",
 "resourceVersion": "1363097",
 "creationTimestamp": "2021-07-08T20:24:37Z",
 "deletionTimestamp": "2021-07-08T20:27:39Z",
 "finalizers": [
 "foregroundDeletion"
]
}
...
```

# Use background cascading deletion

1. [Create a sample Deployment](#).
2. Use either kubectl or the Kubernetes API to delete the Deployment, depending on the Kubernetes version your cluster runs. To check the version, enter kubectl version.

You can delete objects using background cascading deletion using kubectl or the Kubernetes API.

Kubernetes uses background cascading deletion by default, and does so even if you run the following commands without the `--cascade` flag or the `propagationPolicy` argument.

## Using kubectl

Run the following command:

```
kubectl delete deployment nginx-deployment --cascade=background
```

## Using the Kubernetes API

1. Start a local proxy session:

```
kubectl proxy --port=8080
```

2. Use curl to trigger deletion:

```
curl -X DELETE localhost:8080/apis/apps/v1/namespaces/default/deployments/nginx-deployment \
-d '{"kind":"DeleteOptions","apiVersion":"v1","propagationPolicy":"Background"}' \
-H "Content-Type: application/json"
```

The output is similar to this:

```
"kind": "Status",
"apiVersion": "v1",
...
"status": "Success",
"details": {
 "name": "nginx-deployment",
 "group": "apps",
 "kind": "deployments",
 "uid": "cc9ee9b9-2d49-4445-b1c1-d261c9396456"
}
```

# Delete owner objects and orphan dependents

By default, when you tell Kubernetes to delete an object, the [controller](#) also deletes dependent objects. You can make Kubernetes *orphan* these dependents using kubectl or the Kubernetes API, depending on the Kubernetes version your cluster runs. To check the version, enter kubectl version.

## Using kubectl

Run the following command:

```
kubectl delete deployment nginx-deployment --cascade=orphan
```

## Using the Kubernetes API

1. Start a local proxy session:

```
kubectl proxy --port=8080
```

2. Use curl to trigger deletion:

```
curl -X DELETE localhost:8080/apis/apps/v1/namespaces/default/deployments/nginx-
deployment \
-d '{"kind":"DeleteOptions","apiVersion":"v1","propagationPolicy":"Orphan"}' \
-H "Content-Type: application/json"
```

The output contains orphan in the finalizers field, similar to this:

```
"kind": "Deployment",
"apiVersion": "apps/v1",
"namespace": "default",
"uid": "6f577034-42a0-479d-be21-78018c466f1f",
"creationTimestamp": "2021-07-09T16:46:37Z",
"deletionTimestamp": "2021-07-09T16:47:08Z",
"deletionGracePeriodSeconds": 0,
"finalizers": [
 "orphan"
],
...
```

You can check that the Pods managed by the Deployment are still running:

```
kubectl get pods -l app=nginx
```

## What's next

- Learn about [owners and dependents](#) in Kubernetes.
- Learn about Kubernetes [finalizers](#).
- Learn about [garbage collection](#).

# Using a KMS provider for data encryption

This page shows how to configure a Key Management Service (KMS) provider and plugin to enable secret data encryption. In Kubernetes 1.30 there are two versions of KMS at-rest encryption. You should use KMS v2 if feasible because KMS v1 is deprecated (since Kubernetes v1.28) and disabled by default (since Kubernetes v1.29). KMS v2 offers significantly better performance characteristics than KMS v1.

## Caution:

This documentation is for the generally available implementation of KMS v2 (and for the deprecated version 1 implementation). If you are using any control plane components older than Kubernetes v1.29, please check the equivalent page in the documentation for the version of Kubernetes that your cluster is running. Earlier releases of Kubernetes had different behavior that may be relevant for information security.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

The version of Kubernetes that you need depends on which KMS API version you have selected. Kubernetes recommends using KMS v2.

- If you selected KMS API v2, you should use Kubernetes v1.30 (if you are running a different version of Kubernetes that also supports the v2 KMS API, switch to the documentation for that version of Kubernetes).
- If you selected KMS API v1 to support clusters prior to version v1.27 or if you have a legacy KMS plugin that only supports KMS v1, any supported Kubernetes version will work. This API is deprecated as of Kubernetes v1.28. Kubernetes does not recommend the use of this API.

To check the version, enter `kubectl version`.

### KMS v1

FEATURE STATE: Kubernetes v1.28 [deprecated]

- Kubernetes version 1.10.0 or later is required
- For version 1.29 and later, the v1 implementation of KMS is disabled by default. To enable the feature, set `--feature-gates=KMSv1=true` to configure a KMS v1 provider.
- Your cluster must use etcd v3 or later

### KMS v2

FEATURE STATE: Kubernetes v1.29 [stable]

- Your cluster must use etcd v3 or later

# KMS encryption and per-object encryption keys

The KMS encryption provider uses an envelope encryption scheme to encrypt data in etcd. The data is encrypted using a data encryption key (DEK). The DEKs are encrypted with a key encryption key (KEK) that is stored and managed in a remote KMS.

If you use the (deprecated) v1 implementation of KMS, a new DEK is generated for each encryption.

With KMS v2, a new DEK is generated **per encryption**: the API server uses a *key derivation function* to generate single use data encryption keys from a secret seed combined with some random data. The seed is rotated whenever the KEK is rotated (see the *Understanding key\_id and Key Rotation* section below for more details).

The KMS provider uses gRPC to communicate with a specific KMS plugin over a UNIX domain socket. The KMS plugin, which is implemented as a gRPC server and deployed on the same host(s) as the Kubernetes control plane, is responsible for all communication with the remote KMS.

## Configuring the KMS provider

To configure a KMS provider on the API server, include a provider of type `kms` in the `providers` array in the encryption configuration file and set the following properties:

### KMS v1

- `apiVersion`: API Version for KMS provider. Leave this value empty or set it to `v1`.
- `name`: Display name of the KMS plugin. Cannot be changed once set.
- `endpoint`: Listen address of the gRPC server (KMS plugin). The endpoint is a UNIX domain socket.
- `cacheSize`: Number of data encryption keys (DEKs) to be cached in the clear. When cached, DEKs can be used without another call to the KMS; whereas DEKs that are not cached require a call to the KMS to unwrap.
- `timeout`: How long should kube-apiserver wait for kms-plugin to respond before returning an error (default is 3 seconds).

### KMS v2

- `apiVersion`: API Version for KMS provider. Set this to `v2`.
- `name`: Display name of the KMS plugin. Cannot be changed once set.
- `endpoint`: Listen address of the gRPC server (KMS plugin). The endpoint is a UNIX domain socket.
- `timeout`: How long should kube-apiserver wait for kms-plugin to respond before returning an error (default is 3 seconds).

KMS v2 does not support the `cacheSize` property. All data encryption keys (DEKs) will be cached in the clear once the server has unwrapped them via a call to the KMS. Once cached, DEKs can be used to perform decryption indefinitely without making a call to the KMS.

See [Understanding the encryption at rest configuration](#).

# Implementing a KMS plugin

To implement a KMS plugin, you can develop a new plugin gRPC server or enable a KMS plugin already provided by your cloud provider. You then integrate the plugin with the remote KMS and deploy it on the Kubernetes control plane.

## Enabling the KMS supported by your cloud provider

Refer to your cloud provider for instructions on enabling the cloud provider-specific KMS plugin.

## Developing a KMS plugin gRPC server

You can develop a KMS plugin gRPC server using a stub file available for Go. For other languages, you use a proto file to create a stub file that you can use to develop the gRPC server code.

### KMS v1

- Using Go: Use the functions and data structures in the stub file: [api.pb.go](#) to develop the gRPC server code
- Using languages other than Go: Use the protoc compiler with the proto file: [api.proto](#) to generate a stub file for the specific language

### KMS v2

- Using Go: A high level [library](#) is provided to make the process easier. Low level implementations can use the functions and data structures in the stub file: [api.pb.go](#) to develop the gRPC server code
- Using languages other than Go: Use the protoc compiler with the proto file: [api.proto](#) to generate a stub file for the specific language

Then use the functions and data structures in the stub file to develop the server code.

## Notes

### KMS v1

- kms plugin version: v1beta1

In response to procedure call Version, a compatible KMS plugin should return v1beta1 as VersionResponse.version.

- message version: v1beta1

All messages from KMS provider have the version field set to v1beta1.

- protocol: UNIX domain socket (unix)

The plugin is implemented as a gRPC server that listens at UNIX domain socket. The plugin deployment should create a file on the file system to run the gRPC unix domain socket connection. The API server (gRPC client) is configured with the KMS provider (gRPC server) unix domain socket endpoint in order to communicate with it. An abstract Linux socket may be used by starting the endpoint with `/@`, i.e. `unix:///@foo`. Care must be taken when using this type of socket as they do not have concept of ACL (unlike traditional file based sockets). However, they are subject to Linux networking namespace, so will only be accessible to containers within the same pod unless host networking is used.

## KMS v2

- KMS plugin version: v2

In response to the `Status` remote procedure call, a compatible KMS plugin should return its KMS compatibility version as `StatusResponse.version`. That status response should also include "ok" as `StatusResponse.healthz` and a `key_id` (remote KMS KEK ID) as `StatusResponse.key_id`. The Kubernetes project recommends you make your plugin compatible with the stable v2 KMS API. Kubernetes 1.30 also supports the v2beta1 API for KMS; future Kubernetes releases are likely to continue supporting that beta version.

The API server polls the `Status` procedure call approximately every minute when everything is healthy, and every 10 seconds when the plugin is not healthy. Plugins must take care to optimize this call as it will be under constant load.

- Encryption

The `EncryptRequest` procedure call provides the plaintext and a UID for logging purposes. The response must include the ciphertext, the `key_id` for the KEK used, and, optionally, any metadata that the KMS plugin needs to aid in future `DecryptRequest` calls (via the `annotations` field). The plugin must guarantee that any distinct plaintext results in a distinct response (ciphertext, `key_id`, `annotations`).

If the plugin returns a non-empty `annotations` map, all map keys must be fully qualified domain names such as `example.com`. An example use case of annotation is `{"kms.example.io/remote-kms-auditid": "<audit ID used by the remote KMS>"}`

The API server does not perform the `EncryptRequest` procedure call at a high rate. Plugin implementations should still aim to keep each request's latency at under 100 milliseconds.

- Decryption

The `DecryptRequest` procedure call provides the (ciphertext, `key_id`, `annotations`) from `EncryptRequest` and a UID for logging purposes. As expected, it is the inverse of the `EncryptRequest` call. Plugins must verify that the `key_id` is one that they understand - they must not attempt to decrypt data unless they are sure that it was encrypted by them at an earlier time.

The API server may perform thousands of `DecryptRequest` procedure calls on startup to fill its watch cache. Thus plugin implementations must perform these calls as quickly as possible, and should aim to keep each request's latency at under 10 milliseconds.

- Understanding `key_id` and Key Rotation

The `key_id` is the public, non-secret name of the remote KMS KEK that is currently in use. It may be logged during regular operation of the API server, and thus must not contain any private data. Plugin implementations are encouraged to use a hash to avoid leaking any data. The KMS v2 metrics take care to hash this value before exposing it via the `/metrics` endpoint.

The API server considers the `key_id` returned from the `Status` procedure call to be authoritative. Thus, a change to this value signals to the API server that the remote KEK has changed, and data encrypted with the old KEK should be marked stale when a no-op write is performed (as described below). If an `EncryptRequest` procedure call returns a `key_id` that is different from `Status`, the response is thrown away and the plugin is considered unhealthy. Thus implementations must guarantee that the `key_id` returned from `Status` will be the same as the one returned by `EncryptRequest`. Furthermore, plugins must ensure that the `key_id` is stable and does not flip-flop between values (i.e. during a remote KEK rotation).

Plugins must not re-use `key_ids`, even in situations where a previously used remote KEK has been reinstated. For example, if a plugin was using `key_id=A`, switched to `key_id=B`, and then went back to `key_id=A` - instead of reporting `key_id=A` the plugin should report some derivative value such as `key_id=A_001` or use a new value such as `key_id=C`.

Since the API server polls `Status` about every minute, `key_id` rotation is not immediate. Furthermore, the API server will coast on the last valid state for about three minutes. Thus if a user wants to take a passive approach to storage migration (i.e. by waiting), they must schedule a migration to occur at  $3 + N + M$  minutes after the remote KEK has been rotated ( $N$  is how long it takes the plugin to observe the `key_id` change and  $M$  is the desired buffer to allow config changes to be processed - a minimum  $M$  of five minutes is recommend). Note that no API server restart is required to perform KEK rotation.

### **Caution:**

Because you don't control the number of writes performed with the DEK, the Kubernetes project recommends rotating the KEK at least every 90 days.

- protocol: UNIX domain socket (unix)

The plugin is implemented as a gRPC server that listens at UNIX domain socket. The plugin deployment should create a file on the file system to run the gRPC unix domain socket connection. The API server (gRPC client) is configured with the KMS provider (gRPC server) unix domain socket endpoint in order to communicate with it. An abstract Linux socket may be used by starting the endpoint with `/@`, i.e. `unix:///@foo`. Care must be taken when using this type of socket as they do not have concept of ACL (unlike traditional file based sockets). However, they are subject to Linux networking namespace, so will only be accessible to containers within the same pod unless host networking is used.

## **Integrating a KMS plugin with the remote KMS**

The KMS plugin can communicate with the remote KMS using any protocol supported by the KMS. All configuration data, including authentication credentials the KMS plugin uses to communicate with the remote KMS, are stored and managed by the KMS plugin independently. The KMS plugin can encode the ciphertext with additional metadata that may be required



before sending it to the KMS for decryption (KMS v2 makes this process easier by providing a dedicated annotations field).

## Deploying the KMS plugin

Ensure that the KMS plugin runs on the same host(s) as the Kubernetes API server(s).

## Encrypting your data with the KMS provider

To encrypt the data:

1. Create a new EncryptionConfiguration file using the appropriate properties for the kms provider to encrypt resources like Secrets and ConfigMaps. If you want to encrypt an extension API that is defined in a CustomResourceDefinition, your cluster must be running Kubernetes v1.26 or newer.
2. Set the `--encryption-provider-config` flag on the kube-apiserver to point to the location of the configuration file.
3. `--encryption-provider-config-automatic-reload` boolean argument determines if the file set by `--encryption-provider-config` should be [automatically reloaded](#) if the disk contents change.
4. Restart your API server.

### KMS v1

```
apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
- resources:
 - secrets
 - configmaps
 - pandas.awesome.bears.example
providers:
- kms:
 name: myKmsPluginFoo
 endpoint: unix:///tmp/socketfile-foo.sock
 cachesize: 100
 timeout: 3s
- kms:
 name: myKmsPluginBar
 endpoint: unix:///tmp/socketfile-bar.sock
 cachesize: 100
 timeout: 3s
```

### KMS v2

```
apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
- resources:
```

```
- secrets
- configmaps
- pandas.awesome.bears.example
providers:
- kms:
 apiVersion: v2
 name: myKmsPluginFoo
 endpoint: unix:///tmp/socketfile-foo.sock
 timeout: 3s
- kms:
 apiVersion: v2
 name: myKmsPluginBar
 endpoint: unix:///tmp/socketfile-bar.sock
 timeout: 3s
```

Setting `--encryption-provider-config-automatic-reload` to `true` collapses all health checks to a single health check endpoint. Individual health checks are only available when KMS v1 providers are in use and the encryption config is not auto-reloaded.

The following table summarizes the health check endpoints for each KMS version:

| KMS configurations | Without Automatic Reload | With Automatic Reload |
|--------------------|--------------------------|-----------------------|
| KMS v1 only        | Individual Healthchecks  | Single Healthcheck    |
| KMS v2 only        | Single Healthcheck       | Single Healthcheck    |
| Both KMS v1 and v2 | Individual Healthchecks  | Single Healthcheck    |
| No KMS             | None                     | Single Healthcheck    |

Single Healthcheck means that the only health check endpoint is `/healthz/kms-providers`.

Individual Healthchecks means that each KMS plugin has an associated health check endpoint based on its location in the encryption config: `/healthz/kms-provider-0`, `/healthz/kms-provider-1` etc.

These healthcheck endpoint paths are hard coded and generated/controlled by the server. The indices for individual healthchecks corresponds to the order in which the KMS encryption config is processed.

Until the steps defined in [Ensuring all secrets are encrypted](#) are performed, the providers list should end with the identity: `{}` provider to allow unencrypted data to be read. Once all resources are encrypted, the identity provider should be removed to prevent the API server from honoring unencrypted data.

For details about the EncryptionConfiguration format, please check the [API server encryption API reference](#).

## Verifying that the data is encrypted

When encryption at rest is correctly configured, resources are encrypted on write. After restarting your kube-apiserver, any newly created or updated Secret or other resource types

configured in EncryptionConfiguration should be encrypted when stored. To verify, you can use the etcdctl command line program to retrieve the contents of your secret data.

1. Create a new secret called secret1 in the default namespace:

```
kubectl create secret generic secret1 -n default --from-literal=mykey=mydata
```

2. Using the etcdctl command line, read that secret out of etcd:

```
ETCDCTL_API=3 etcdctl get /kubernetes.io/secrets/default/secret1 [...] | hexdump -C
```

where [...] contains the additional arguments for connecting to the etcd server.

3. Verify the stored secret is prefixed with k8s:enc:kms:v1: for KMS v1 or prefixed with k8s:enc:kms:v2: for KMS v2, which indicates that the kms provider has encrypted the resulting data.
4. Verify that the secret is correctly decrypted when retrieved via the API:

```
kubectl describe secret secret1 -n default
```

The Secret should contain mykey: mydata

## Ensuring all secrets are encrypted

When encryption at rest is correctly configured, resources are encrypted on write. Thus we can perform an in-place no-op update to ensure that data is encrypted.

The following command reads all secrets and then updates them to apply server side encryption. If an error occurs due to a conflicting write, retry the command. For larger clusters, you may wish to subdivide the secrets by namespace or script an update.

```
kubectl get secrets --all-namespaces -o json | kubectl replace -f -
```

## Switching from a local encryption provider to the KMS provider

To switch from a local encryption provider to the kms provider and re-encrypt all of the secrets:

1. Add the kms provider as the first entry in the configuration file as shown in the following example.

```
apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
- resources:
 - secrets
providers:
- kms:
 apiVersion: v2
 name : myKmsPlugin
 endpoint: unix:///tmp/socketfile.sock
- aescbc:
```

```
keys:
- name: key1
 secret: <BASE 64 ENCODED SECRET>
```

2. Restart all kube-apiserver processes.
3. Run the following command to force all secrets to be re-encrypted using the kms provider.

```
kubectl get secrets --all-namespaces -o json | kubectl replace -f -
```

## What's next

If you no longer want to use encryption for data persisted in the Kubernetes API, read [decrypt data that are already stored at rest](#).

# Using CoreDNS for Service Discovery

This page describes the CoreDNS upgrade process and how to install CoreDNS instead of kube-dns.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.9. To check the version, enter `kubectl version`.

## About CoreDNS

[CoreDNS](#) is a flexible, extensible DNS server that can serve as the Kubernetes cluster DNS. Like Kubernetes, the CoreDNS project is hosted by the [CNCF](#).

You can use CoreDNS instead of kube-dns in your cluster by replacing kube-dns in an existing deployment, or by using tools like kubeadm that will deploy and upgrade the cluster for you.

## Installing CoreDNS

For manual deployment or replacement of kube-dns, see the documentation at the [CoreDNS website](#).

# Migrating to CoreDNS

## Upgrading an existing cluster with kubeadm

In Kubernetes version 1.21, kubeadm removed its support for kube-dns as a DNS application. For kubeadm v1.30, the only supported cluster DNS application is CoreDNS.

You can move to CoreDNS when you use kubeadm to upgrade a cluster that is using kube-dns. In this case, kubeadm generates the CoreDNS configuration ("Corefile") based upon the kube-dns ConfigMap, preserving configurations for stub domains, and upstream name server.

## Upgrading CoreDNS

You can check the version of CoreDNS that kubeadm installs for each version of Kubernetes in the page [CoreDNS version in Kubernetes](#).

CoreDNS can be upgraded manually in case you want to only upgrade CoreDNS or use your own custom image. There is a helpful [guideline and walkthrough](#) available to ensure a smooth upgrade. Make sure the existing CoreDNS configuration ("Corefile") is retained when upgrading your cluster.

If you are upgrading your cluster using the kubeadm tool, kubeadm can take care of retaining the existing CoreDNS configuration automatically.

## Tuning CoreDNS

When resource utilisation is a concern, it may be useful to tune the configuration of CoreDNS. For more details, check out the [documentation on scaling CoreDNS](#).

## What's next

You can configure [CoreDNS](#) to support many more use cases than kube-dns does by modifying the CoreDNS configuration ("Corefile"). For more information, see the [documentation](#) for the Kubernetes CoreDNS plugin, or read the [Custom DNS Entries for Kubernetes](#) in the CoreDNS blog.

# Using NodeLocal DNSCache in Kubernetes Clusters

FEATURE STATE: Kubernetes v1.18 [stable]

This page provides an overview of NodeLocal DNSCache feature in Kubernetes.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Introduction

NodeLocal DNSCache improves Cluster DNS performance by running a DNS caching agent on cluster nodes as a DaemonSet. In today's architecture, Pods in 'ClusterFirst' DNS mode reach out to a kube-dns serviceIP for DNS queries. This is translated to a kube-dns/CoreDNS endpoint via iptables rules added by kube-proxy. With this new architecture, Pods will reach out to the DNS caching agent running on the same node, thereby avoiding iptables DNAT rules and connection tracking. The local caching agent will query kube-dns service for cache misses of cluster hostnames ("cluster.local" suffix by default).

## Motivation

- With the current DNS architecture, it is possible that Pods with the highest DNS QPS have to reach out to a different node, if there is no local kube-dns/CoreDNS instance. Having a local cache will help improve the latency in such scenarios.
- Skipping iptables DNAT and connection tracking will help reduce [conntrack races](#) and avoid UDP DNS entries filling up conntrack table.
- Connections from the local caching agent to kube-dns service can be upgraded to TCP. TCP conntrack entries will be removed on connection close in contrast with UDP entries that have to timeout ([default](#) `nf_conntrack_udp_timeout` is 30 seconds)
- Upgrading DNS queries from UDP to TCP would reduce tail latency attributed to dropped UDP packets and DNS timeouts usually up to 30s (3 retries + 10s timeout). Since the nodelocal cache listens for UDP DNS queries, applications don't need to be changed.
- Metrics & visibility into DNS requests at a node level.
- Negative caching can be re-enabled, thereby reducing the number of queries for the kube-dns service.

## Architecture Diagram

This is the path followed by DNS Queries after NodeLocal DNSCache is enabled:

NodeLocal DNSCache flow

**Nodelocal DNSCache flow**

This image shows how NodeLocal DNSCache handles DNS queries.

# Configuration

## Note:

The local listen IP address for NodeLocal DNSCache can be any address that can be guaranteed to not collide with any existing IP in your cluster. It's recommended to use an address with a local scope, for example, from the 'link-local' range '169.254.0.0/16' for IPv4 or from the 'Unique Local Address' range in IPv6 'fd00::/8'.

This feature can be enabled using the following steps:

- Prepare a manifest similar to the sample [nodelocaldns.yaml](#) and save it as `nodelocaldns.yaml`.
- If using IPv6, the CoreDNS configuration file needs to enclose all the IPv6 addresses into square brackets if used in 'IP:Port' format. If you are using the sample manifest from the previous point, this will require you to modify [the configuration line L70](#) like this:  
"health [\_\_PILLAR\_\_LOCAL\_\_DNS\_\_]:8080"
- Substitute the variables in the manifest with the right values:

```
kubedns=`kubectl get svc kube-dns -n kube-system -o jsonpath={.spec.clusterIP}`
domain=<cluster-domain>
localdns=<node-local-address>
```

<cluster-domain> is "cluster.local" by default. <node-local-address> is the local listen IP address chosen for NodeLocal DNSCache.

- If kube-proxy is running in IPTABLES mode:

```
sed -i "s/__PILLAR__LOCAL__DNS__/$localdns/g; s/
__PILLAR__DNS__DOMAIN__/$domain/g; s/__PILLAR__DNS__SERVER__/$kubedns/g" nodelocaldns.yaml
```

\_\_PILLAR\_\_CLUSTER\_\_DNS\_\_ and \_\_PILLAR\_\_UPSTREAM\_\_SERVERS\_\_ will be populated by the node-local-dns pods. In this mode, the node-local-dns pods listen on both the kube-dns service IP as well as <node-local-address>, so pods can look up DNS records using either IP address.

- If kube-proxy is running in IPVS mode:

```
sed -i "s/__PILLAR__LOCAL__DNS__/$localdns/g; s/
__PILLAR__DNS__DOMAIN__/$domain/g; s/,__PILLAR__DNS__SERVER__//g; s/
__PILLAR__CLUSTER__DNS__/$kubedns/g" nodelocaldns.yaml
```

In this mode, the node-local-dns pods listen only on <node-local-address>. The node-local-dns interface cannot bind the kube-dns cluster IP since the interface used for IPVS loadbalancing already uses this address.

\_\_PILLAR\_\_UPSTREAM\_\_SERVERS\_\_ will be populated by the node-local-dns pods.

- Run `kubectl create -f nodelocaldns.yaml`

- If using kube-proxy in IPVS mode, `--cluster-dns` flag to kubelet needs to be modified to use `<node-local-address>` that NodeLocal DNSCache is listening on. Otherwise, there is no need to modify the value of the `--cluster-dns` flag, since NodeLocal DNSCache listens on both the kube-dns service IP as well as `<node-local-address>`.

Once enabled, the `node-local-dns` Pods will run in the `kube-system` namespace on each of the cluster nodes. This Pod runs [CoreDNS](#) in cache mode, so all CoreDNS metrics exposed by the different plugins will be available on a per-node basis.

You can disable this feature by removing the DaemonSet, using `kubectl delete -f <manifest>`. You should also revert any changes you made to the kubelet configuration.

## StubDomains and Upstream server Configuration

StubDomains and upstream servers specified in the `kube-dns` ConfigMap in the `kube-system` namespace are automatically picked up by `node-local-dns` pods. The ConfigMap contents need to follow the format shown in [the example](#). The `node-local-dns` ConfigMap can also be modified directly with the stubDomain configuration in the Corefile format. Some cloud providers might not allow modifying `node-local-dns` ConfigMap directly. In those cases, the `kube-dns` ConfigMap can be updated.

## Setting memory limits

The `node-local-dns` Pods use memory for storing cache entries and processing queries. Since they do not watch Kubernetes objects, the cluster size or the number of Services / EndpointSlices do not directly affect memory usage. Memory usage is influenced by the DNS query pattern. From [CoreDNS docs](#),

The default cache size is 10000 entries, which uses about 30 MB when completely filled.

This would be the memory usage for each server block (if the cache gets completely filled). Memory usage can be reduced by specifying smaller cache sizes.

The number of concurrent queries is linked to the memory demand, because each extra goroutine used for handling a query requires an amount of memory. You can set an upper limit using the `max_concurrent` option in the forward plugin.

If a `node-local-dns` Pod attempts to use more memory than is available (because of total system resources, or because of a configured [resource limit](#)), the operating system may shut down that pod's container. If this happens, the container that is terminated ("OOMKilled") does not clean up the custom packet filtering rules that it previously added during startup. The `node-local-dns` container should get restarted (since managed as part of a DaemonSet), but this will lead to a brief DNS downtime each time that the container fails: the packet filtering rules direct DNS queries to a local Pod that is unhealthy.

You can determine a suitable memory limit by running `node-local-dns` pods without a limit and measuring the peak usage. You can also set up and use a [VerticalPodAutoscaler](#) in *recommender mode*, and then check its recommendations.



# Using sysctls in a Kubernetes Cluster

FEATURE STATE: Kubernetes v1.21 [stable]

This document describes how to configure and use kernel parameters within a Kubernetes cluster using the [sysctl](#) interface.

## Note:

Starting from Kubernetes version 1.23, the kubelet supports the use of either / or . as separators for sysctl names. Starting from Kubernetes version 1.25, setting Sysctls for a Pod supports setting sysctls with slashes. For example, you can represent the same sysctl name as kernel.shm\_rmid\_forced using a period as the separator, or as kernel/shm\_rmid\_forced using a slash as a separator. For more sysctl parameter conversion method details, please refer to the page [sysctl.d\(5\)](#) from the Linux man-pages project.

## Before you begin

### Note:

sysctl is a Linux-specific command-line tool used to configure various kernel parameters and it is not available on non-Linux operating systems.

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

For some steps, you also need to be able to reconfigure the command line options for the kubelets running on your cluster.

## Listing all Sysctl Parameters

In Linux, the sysctl interface allows an administrator to modify kernel parameters at runtime. Parameters are available via the /proc/sys/ virtual process file system. The parameters cover various subsystems such as:

- kernel (common prefix: kernel.)
- networking (common prefix: net.)
- virtual memory (common prefix: vm.)
- MDADM (common prefix: dev.)
- More subsystems are described in [Kernel docs](#).

To get a list of all parameters, you can run

```
sudo sysctl -a
```

# Safe and Unsafe Sysctls

Kubernetes classes sysctls as either *safe* or *unsafe*. In addition to proper namespacing, a *safe* sysctl must be properly *isolated* between pods on the same node. This means that setting a *safe* sysctl for one pod

- must not have any influence on any other pod on the node
- must not allow to harm the node's health
- must not allow to gain CPU or memory resources outside of the resource limits of a pod.

By far, most of the *namespaced* sysctls are not necessarily considered *safe*. The following sysctls are supported in the *safe* set:

- `kernel.shm_rmid_forced`;
- `net.ipv4.ip_local_port_range`;
- `net.ipv4.tcp_syncookies`;
- `net.ipv4.ping_group_range` (since Kubernetes 1.18);
- `net.ipv4.ip_unprivileged_port_start` (since Kubernetes 1.22);
- `net.ipv4.ip_local_reserved_ports` (since Kubernetes 1.27, needs kernel 3.16+);
- `net.ipv4.tcp_keepalive_time` (since Kubernetes 1.29, needs kernel 4.5+);
- `net.ipv4.tcp_fin_timeout` (since Kubernetes 1.29, needs kernel 4.6+);
- `net.ipv4.tcp_keepalive_intvl` (since Kubernetes 1.29, needs kernel 4.5+);
- `net.ipv4.tcp_keepalive_probes` (since Kubernetes 1.29, needs kernel 4.5+).

## Note:

There are some exceptions to the set of safe sysctls:

- The `net.*` sysctls are not allowed with host networking enabled.
- The `net.ipv4.tcp_syncookies` sysctl is not namespaced on Linux kernel version 4.4 or lower.

This list will be extended in future Kubernetes versions when the kubelet supports better isolation mechanisms.

## Enabling Unsafe Sysctls

All *safe* sysctls are enabled by default.

All *unsafe* sysctls are disabled by default and must be allowed manually by the cluster admin on a per-node basis. Pods with disabled unsafe sysctls will be scheduled, but will fail to launch.

With the warning above in mind, the cluster admin can allow certain *unsafe* sysctls for very special situations such as high-performance or real-time application tuning. *Unsafe* sysctls are enabled on a node-by-node basis with a flag of the kubelet; for example:

```
kubelet --allowed-unsafe-sysctls \
'kernel.msg*,net.core.somaxconn' ...
```

For [Minikube](#), this can be done via the extra-config flag:

```
minikube start --extra-config="kubelet.allowed-unsafe-
sysctls=kernel.msg*,net.core.somaxconn"...
```

Only *namespaced* sysctls can be enabled this way.

## Setting Sysctls for a Pod

A number of sysctls are *namespaced* in today's Linux kernels. This means that they can be set independently for each pod on a node. Only namespaced sysctls are configurable via the pod securityContext within Kubernetes.

The following sysctls are known to be namespaced. This list could change in future versions of the Linux kernel.

- kernel.shm\*,
- kernel.msg\*,
- kernel.sem,
- fs.mqueue.\*,
- Those net.\* that can be set in container networking namespace. However, there are exceptions (e.g., net.netfilter.nf\_conntrack\_max and net.netfilter.nf\_conntrack\_expect\_max can be set in container networking namespace but are unnamespaced before Linux 5.12.2).

Sysctls with no namespace are called *node-level* sysctls. If you need to set them, you must manually configure them on each node's operating system, or by using a DaemonSet with privileged containers.

Use the pod securityContext to configure namespaced sysctls. The securityContext applies to all containers in the same pod.

This example uses the pod securityContext to set a safe sysctl kernel.shm\_rmid\_forced and two unsafe sysctls net.core.somaxconn and kernel.msgmax. There is no distinction between *safe* and *unsafe* sysctls in the specification.

### Warning:

Only modify sysctl parameters after you understand their effects, to avoid destabilizing your operating system.

```
apiVersion: v1
kind: Pod
metadata:
 name: sysctl-example
spec:
 securityContext:
 sysctls:
 - name: kernel.shm_rmid_forced
 value: "0"
 - name: net.core.somaxconn
 value: "1024"
 - name: kernel.msgmax
 value: "65536"
 ...
```

## Warning:

Due to their nature of being *unsafe*, the use of *unsafe* sysctls is at-your-own-risk and can lead to severe problems like wrong behavior of containers, resource shortage or complete breakage of a node.

It is good practice to consider nodes with special sysctl settings as *tainted* within a cluster, and only schedule pods onto them which need those sysctl settings. It is suggested to use the Kubernetes [taints and toleration feature](#) to implement this.

A pod with the *unsafe* sysctls will fail to launch on any node which has not enabled those two *unsafe* sysctls explicitly. As with *node-level* sysctls it is recommended to use [taints and toleration feature](#) or [taints on nodes](#) to schedule those pods onto the right nodes.

# Utilizing the NUMA-aware Memory Manager

FEATURE STATE: Kubernetes v1.22 [beta]

The Kubernetes *Memory Manager* enables the feature of guaranteed memory (and hugepages) allocation for pods in the Guaranteed [QoS class](#).

The Memory Manager employs hint generation protocol to yield the most suitable NUMA affinity for a pod. The Memory Manager feeds the central manager (*Topology Manager*) with these affinity hints. Based on both the hints and Topology Manager policy, the pod is rejected or admitted to the node.

Moreover, the Memory Manager ensures that the memory which a pod requests is allocated from a minimum number of NUMA nodes.

The Memory Manager is only pertinent to Linux based hosts.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.21. To check the version, enter kubectl version.

To align memory resources with other requested resources in a Pod spec:

- the CPU Manager should be enabled and proper CPU Manager policy should be configured on a Node. See [control CPU Management Policies](#);
- the Topology Manager should be enabled and proper Topology Manager policy should be configured on a Node. See [control Topology Management Policies](#).

Starting from v1.22, the Memory Manager is enabled by default through `MemoryManager` [feature gate](#).

Preceding v1.22, the kubelet must be started with the following flag:

```
--feature-gates=MemoryManager=true
```

in order to enable the Memory Manager feature.

## How Memory Manager Operates?

The Memory Manager currently offers the guaranteed memory (and hugepages) allocation for Pods in Guaranteed QoS class. To immediately put the Memory Manager into operation follow the guidelines in the section [Memory Manager configuration](#), and subsequently, prepare and deploy a Guaranteed pod as illustrated in the section [Placing a Pod in the Guaranteed QoS class](#).

The Memory Manager is a Hint Provider, and it provides topology hints for the Topology Manager which then aligns the requested resources according to these topology hints. It also enforces cgroups (i.e. `cpuset.mems`) for pods. The complete flow diagram concerning pod admission and deployment process is illustrated in [Memory Manager KEP: Design Overview](#) and below:

Memory Manager in the pod admission and deployment process

During this process, the Memory Manager updates its internal counters stored in [Node Map and Memory Maps](#) to manage guaranteed memory allocation.

The Memory Manager updates the Node Map during the startup and runtime as follows.

### Startup

This occurs once a node administrator employs `--reserved-memory` (section [Reserved memory flag](#)). In this case, the Node Map becomes updated to reflect this reservation as illustrated in [Memory Manager KEP: Memory Maps at start-up \(with examples\)](#).

The administrator must provide `--reserved-memory` flag when Static policy is configured.

### Runtime

Reference [Memory Manager KEP: Memory Maps at runtime \(with examples\)](#) illustrates how a successful pod deployment affects the Node Map, and it also relates to how potential Out-of-Memory (OOM) situations are handled further by Kubernetes or operating system.

Important topic in the context of Memory Manager operation is the management of NUMA groups. Each time pod's memory request is in excess of single NUMA node capacity, the Memory Manager attempts to create a group that comprises several NUMA nodes and features extend memory capacity. The problem has been solved as elaborated in [Memory Manager KEP: How to enable the guaranteed memory allocation over many NUMA nodes?](#). Also, reference [Memory Manager KEP: Simulation - how the Memory Manager works? \(by examples\)](#) illustrates how the management of groups occurs.

# Memory Manager configuration

Other Managers should be first pre-configured. Next, the Memory Manager feature should be enabled and be run with Static policy (section [Static policy](#)). Optionally, some amount of memory can be reserved for system or kubelet processes to increase node stability (section [Reserved memory flag](#)).

## Policies

Memory Manager supports two policies. You can select a policy via a kubelet flag `--memory-manager-policy`:

- None (default)
- Static

### None policy

This is the default policy and does not affect the memory allocation in any way. It acts the same as if the Memory Manager is not present at all.

The None policy returns default topology hint. This special hint denotes that Hint Provider (Memory Manager in this case) has no preference for NUMA affinity with any resource.

### Static policy

In the case of the Guaranteed pod, the Static Memory Manager policy returns topology hints relating to the set of NUMA nodes where the memory can be guaranteed, and reserves the memory through updating the internal [NodeMap](#) object.

In the case of the BestEffort or Burstable pod, the Static Memory Manager policy sends back the default topology hint as there is no request for the guaranteed memory, and does not reserve the memory in the internal [NodeMap](#) object.

## Reserved memory flag

The [Node Allocatable](#) mechanism is commonly used by node administrators to reserve K8S node system resources for the kubelet or operating system processes in order to enhance the node stability. A dedicated set of flags can be used for this purpose to set the total amount of reserved memory for a node. This pre-configured value is subsequently utilized to calculate the real amount of node's "allocatable" memory available to pods.

The Kubernetes scheduler incorporates "allocatable" to optimise pod scheduling process. The foregoing flags include `--kube-reserved`, `--system-reserved` and `--eviction-threshold`. The sum of their values will account for the total amount of reserved memory.

A new `--reserved-memory` flag was added to Memory Manager to allow for this total reserved memory to be split (by a node administrator) and accordingly reserved across many NUMA nodes.

The flag specifies a comma-separated list of memory reservations of different memory types per NUMA node. Memory reservations across multiple NUMA nodes can be specified using semicolon as separator. This parameter is only useful in the context of the Memory Manager

feature. The Memory Manager will not use this reserved memory for the allocation of container workloads.

For example, if you have a NUMA node "NUMA0" with 10Gi of memory available, and the `--reserved-memory` was specified to reserve 1Gi of memory at "NUMA0", the Memory Manager assumes that only 9Gi is available for containers.

You can omit this parameter, however, you should be aware that the quantity of reserved memory from all NUMA nodes should be equal to the quantity of memory specified by the [Node Allocatable feature](#). If at least one node allocatable parameter is non-zero, you will need to specify `--reserved-memory` for at least one NUMA node. In fact, eviction-hard threshold value is equal to 100Mi by default, so if Static policy is used, `--reserved-memory` is obligatory.

Also, avoid the following configurations:

1. duplicates, i.e. the same NUMA node or memory type, but with a different value;
2. setting zero limit for any of memory types;
3. NUMA node IDs that do not exist in the machine hardware;
4. memory type names different than memory or hugepages-`<size>` (hugepages of particular `<size>` should also exist).

Syntax:

```
--reserved-memory N:memory-type1=value1,memory-type2=value2,...
```

- N (integer) - NUMA node index, e.g. 0
- memory-type (string) - represents memory type:
  - memory - conventional memory
  - hugepages-2Mi or hugepages-1Gi - hugepages
- value (string) - the quantity of reserved memory, e.g. 1Gi

Example usage:

```
--reserved-memory 0:memory=1Gi,hugepages-1Gi=2Gi
```

or

```
--reserved-memory 0:memory=1Gi --reserved-memory 1:memory=2Gi
```

or

```
--reserved-memory '0:memory=1Gi;1:memory=2Gi'
```

When you specify values for `--reserved-memory` flag, you must comply with the setting that you prior provided via Node Allocatable Feature flags. That is, the following rule must be obeyed for each memory type:

$$\text{sum}(\text{reserved-memory}(i)) = \text{kube-reserved} + \text{system-reserved} + \text{eviction-threshold},$$

where *i* is an index of a NUMA node.

If you do not follow the formula above, the Memory Manager will show an error on startup.

In other words, the example above illustrates that for the conventional memory (type=memory), we reserve 3Gi in total, i.e.:

$\text{sum}(\text{reserved-memory}(i)) = \text{reserved-memory}(0) + \text{reserved-memory}(1) = 1\text{Gi} + 2\text{Gi} = 3\text{Gi}$

An example of kubelet command-line arguments relevant to the node Allocatable configuration:

- `--kube-reserved=cpu=500m,memory=50Mi`
- `--system-reserved=cpu=123m,memory=333Mi`
- `--eviction-hard=memory.available<500Mi`

#### Note:

The default hard eviction threshold is 100MiB, and **not** zero. Remember to increase the quantity of memory that you reserve by setting `--reserved-memory` by that hard eviction threshold. Otherwise, the kubelet will not start Memory Manager and display an error.

Here is an example of a correct configuration:

```
--feature-gates=MemoryManager=true
--kube-reserved=cpu=4,memory=4Gi
--system-reserved=cpu=1,memory=1Gi
--memory-manager-policy=Static
--reserved-memory '0:memory=3Gi;1:memory=2148Mi'
```

Let us validate the configuration above:

1. `kube-reserved + system-reserved + eviction-hard(default) = reserved-memory(0) + reserved-memory(1)`
2. `4GiB + 1GiB + 100MiB = 3GiB + 2148MiB`
3. `5120MiB + 100MiB = 3072MiB + 2148MiB`
4. `5220MiB = 5220MiB` (which is correct)

## Placing a Pod in the Guaranteed QoS class

If the selected policy is anything other than None, the Memory Manager identifies pods that are in the Guaranteed QoS class. The Memory Manager provides specific topology hints to the Topology Manager for each Guaranteed pod. For pods in a QoS class other than Guaranteed, the Memory Manager provides default topology hints to the Topology Manager.

The following excerpts from pod manifests assign a pod to the Guaranteed QoS class.

Pod with integer CPU(s) runs in the Guaranteed QoS class, when requests are equal to limits:

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 cpu: "2"
 example.com/device: "1"
 requests:
 memory: "200Mi"
```



```
cpu: "2"
example.com/device: "1"
```

Also, a pod sharing CPU(s) runs in the Guaranteed QoS class, when requests are equal to limits.

```
spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 cpu: "300m"
 example.com/device: "1"
 requests:
 memory: "200Mi"
 cpu: "300m"
 example.com/device: "1"
```

Notice that both CPU and memory requests must be specified for a Pod to lend it to Guaranteed QoS class.

## Troubleshooting

The following means can be used to troubleshoot the reason why a pod could not be deployed or became rejected at a node:

- pod status - indicates topology affinity errors
- system logs - include valuable information for debugging, e.g., about generated hints
- state file - the dump of internal state of the Memory Manager (includes [Node Map and Memory Maps](#))
- starting from v1.22, the [device plugin resource API](#) can be used to retrieve information about the memory reserved for containers

### Pod status (TopologyAffinityError)

This error typically occurs in the following situations:

- a node has not enough resources available to satisfy the pod's request
- the pod's request is rejected due to particular Topology Manager policy constraints

The error appears in the status of a pod:

```
kubectl get pods
```

| NAME       | READY | STATUS                | RESTARTS | AGE  |
|------------|-------|-----------------------|----------|------|
| guaranteed | 0/1   | TopologyAffinityError | 0        | 113s |

Use `kubectl describe pod <id>` or `kubectl get events` to obtain detailed error message:

```
Warning TopologyAffinityError 10m kubelet, dell8 Resources cannot be allocated with
Topology locality
```

## System logs

Search system logs with respect to a particular pod.

The set of hints that Memory Manager generated for the pod can be found in the logs. Also, the set of hints generated by CPU Manager should be present in the logs.

Topology Manager merges these hints to calculate a single best hint. The best hint should be also present in the logs.

The best hint indicates where to allocate all the resources. Topology Manager tests this hint against its current policy, and based on the verdict, it either admits the pod to the node or rejects it.

Also, search the logs for occurrences associated with the Memory Manager, e.g. to find out information about cgroups and cpuset.mems updates.

## Examine the memory manager state on a node

Let us first deploy a sample Guaranteed pod whose specification is as follows:

```
apiVersion: v1
kind: Pod
metadata:
 name: guaranteed
spec:
 containers:
 - name: guaranteed
 image: consumer
 imagePullPolicy: Never
 resources:
 limits:
 cpu: "2"
 memory: 150Gi
 requests:
 cpu: "2"
 memory: 150Gi
 command: ["sleep", "infinity"]
```

Next, let us log into the node where it was deployed and examine the state file in `/var/lib/kubelet/memory_manager_state`:

```
{
 "policyName": "Static",
 "machineState": {
 "0": {
 "numberOfAssignments": 1,
 "memoryMap": {
 "hugepages-1Gi": {
 "total": 0,
 "systemReserved": 0,
 "allocatable": 0,
 "reserved": 0,
```

```
 "free":0
 },
 "memory":{
 "total":134987354112,
 "systemReserved":3221225472,
 "allocatable":131766128640,
 "reserved":131766128640,
 "free":0
 }
},
"nodes":[
 0,
 1
]
},
"1":{
 "numberOfAssignments":1,
 "memoryMap":{
 "hugepages-1Gi":{
 "total":0,
 "systemReserved":0,
 "allocatable":0,
 "reserved":0,
 "free":0
 },
 "memory":{
 "total":135286722560,
 "systemReserved":2252341248,
 "allocatable":133034381312,
 "reserved":29295144960,
 "free":103739236352
 }
 },
 "nodes":[
 0,
 1
]
}
},
"entries":{
 "fa9bdd38-6df9-4cf9-aa67-8c4814da37a8":{
 "guaranteed":[
 {
 "numaAffinity":[
 0,
 1
],
 "type":"memory",
 "size":161061273600
 }
]
 }
}
},
```

```
"checksum":4142013182
}
```

It can be deduced from the state file that the pod was pinned to both NUMA nodes, i.e.:

```
"numaAffinity":[
 0,
 1
],
```

Pinned term means that pod's memory consumption is constrained (through cgroups configuration) to these NUMA nodes.

This automatically implies that Memory Manager instantiated a new group that comprises these two NUMA nodes, i.e. 0 and 1 indexed NUMA nodes.

Notice that the management of groups is handled in a relatively complex manner, and further elaboration is provided in Memory Manager KEP in [this](#) and [this](#) sections.

In order to analyse memory resources available in a group, the corresponding entries from NUMA nodes belonging to the group must be added up.

For example, the total amount of free "conventional" memory in the group can be computed by adding up the free memory available at every NUMA node in the group, i.e., in the "memory" section of NUMA node 0 ("free":0) and NUMA node 1 ("free":103739236352). So, the total amount of free "conventional" memory in this group is equal to 0 + 103739236352 bytes.

The line "systemReserved":3221225472 indicates that the administrator of this node reserved 3221225472 bytes (i.e. 3Gi) to serve kubelet and system processes at NUMA node 0, by using --reserved-memory flag.

## Device plugin resource API

The kubelet provides a PodResourceLister gRPC service to enable discovery of resources and associated metadata. By using its [List gRPC endpoint](#), information about reserved memory for each container can be retrieved, which is contained in protobuf ContainerMemory message. This information can be retrieved solely for pods in Guaranteed QoS class.

## What's next

- [Memory Manager KEP: Design Overview](#)
- [Memory Manager KEP: Memory Maps at start-up \(with examples\)](#)
- [Memory Manager KEP: Memory Maps at runtime \(with examples\)](#)
- [Memory Manager KEP: Simulation - how the Memory Manager works? \(by examples\)](#)
- [Memory Manager KEP: The Concept of Node Map and Memory Maps](#)
- [Memory Manager KEP: How to enable the guaranteed memory allocation over many NUMA nodes?](#)

# Verify Signed Kubernetes Artifacts

FEATURE STATE: Kubernetes v1.26 [beta]

## Before you begin

You will need to have the following tools installed:

- cosign ([install guide](#))
- curl (often provided by your operating system)
- jq ([download jq](#))

## Verifying binary signatures

The Kubernetes release process signs all binary artifacts (tarballs, SPDX files, standalone binaries) by using cosign's keyless signing. To verify a particular binary, retrieve it together with its signature and certificate:

```
URL=https://dl.k8s.io/release/v1.30.0/bin/linux/amd64
BINARY=kubectl

FILES=(
 "$BINARY"
 "$BINARY.sig"
 "$BINARY.cert"
)

for FILE in "${FILES[@]"; do
 curl -sSfL --retry 3 --retry-delay 3 "$URL/$FILE" -o "$FILE"
done
```

Then verify the blob by using `cosign verify-blob`:

```
cosign verify-blob "$BINARY" \
 --signature "$BINARY".sig \
 --certificate "$BINARY".cert \
 --certificate-identity krel-staging@k8s-releng-prod.iam.gserviceaccount.com \
 --certificate-oidc-issuer https://accounts.google.com
```

### Note:

Cosign 2.0 requires the `--certificate-identity` and `--certificate-oidc-issuer` options.

To learn more about keyless signing, please refer to [Keyless Signatures](#).

Previous versions of Cosign required that you set `COSIGN_EXPERIMENTAL=1`.

For additional information, please refer to the [sigstore Blog](#)

## Verifying image signatures

For a complete list of images that are signed please refer to [Releases](#).

Pick one image from this list and verify its signature using the `cosign verify` command:

```
cosign verify registry.k8s.io/kube-apiserver-amd64:v1.30.0 \
--certificate-identity krel-trust@k8s-releng-prod.iam.gserviceaccount.com \
--certificate-oidc-issuer https://accounts.google.com \
| jq .
```

## Verifying images for all control plane components

To verify all signed control plane images for the latest stable version (v1.30.0), please run the following commands:

```
curl -Ls "https://sbom.k8s.io/$(curl -Ls https://dl.k8s.io/release/stable.txt)/release" \
| grep "SPDXID: SPDXRef-Package-registry.k8s.io" \
| grep -v sha256 | cut -d- -f3- | sed 's/-/\//' | sed 's/-v1/:v1/' \
| sort > images.txt
input=images.txt
while IFS= read -r image
do
 cosign verify "$image" \
 --certificate-identity krel-trust@k8s-releng-prod.iam.gserviceaccount.com \
 --certificate-oidc-issuer https://accounts.google.com \
 | jq .
done < "$input"
```

Once you have verified an image, you can specify the image by its digest in your Pod manifests as per this example:

```
registry-url/image-
name@sha256:45b23dee08af5e43a7fea6c4cf9c25ccf269ee113168c19722f87876677c5cb2
```

For more information, please refer to the [Image Pull Policy](#) section.

## Verifying Image Signatures with Admission Controller

For non-control plane images (for example [conformance image](#)), signatures can also be verified at deploy time using [sigstore policy-controller](#) admission controller.

Here are some helpful resources to get started with policy-controller:

- [Installation](#)
- [Configuration Options](#)

## Verify the Software Bill Of Materials

You can verify the Kubernetes Software Bill of Materials (SBOM) by using the sigstore certificate and signature, or the corresponding SHA files:

```
Retrieve the latest available Kubernetes release version
VERSION=$(curl -Ls https://dl.k8s.io/release/stable.txt)

Verify the SHA512 sum
curl -Ls "https://sbom.k8s.io/$VERSION/release" -o "$VERSION.spdx"
echo "$(curl -Ls "https://sbom.k8s.io/$VERSION/release.sha512") $VERSION.spdx" | sha512sum
```

```
--check

Verify the SHA256 sum
echo "$(curl -Ls "https://sbom.k8s.io/$VERSION/release.sha256") $VERSION.spdx" | sha256sum
--check

Retrieve sigstore signature and certificate
curl -Ls "https://sbom.k8s.io/$VERSION/release.sig" -o "$VERSION.spdx.sig"
curl -Ls "https://sbom.k8s.io/$VERSION/release.cert" -o "$VERSION.spdx.cert"

Verify the sigstore signature
cosign verify-blob \
 --certificate "$VERSION.spdx.cert" \
 --signature "$VERSION.spdx.sig" \
 --certificate-identity krel-staging@k8s-releng-prod.iam.gserviceaccount.com \
 --certificate-oidc-issuer https://accounts.google.com \
 "$VERSION.spdx"
```

# Configure Pods and Containers

Perform common configuration tasks for Pods and containers.

---

[Assign Memory Resources to Containers and Pods](#)

[Assign CPU Resources to Containers and Pods](#)

[Configure GMSA for Windows Pods and containers](#)

[Resize CPU and Memory Resources assigned to Containers](#)

[Configure RunAsUserName for Windows pods and containers](#)

[Create a Windows HostProcess Pod](#)

[Configure Quality of Service for Pods](#)

[Assign Extended Resources to a Container](#)

[Configure a Pod to Use a Volume for Storage](#)

[Configure a Pod to Use a PersistentVolume for Storage](#)

[Configure a Pod to Use a Projected Volume for Storage](#)

[Configure a Security Context for a Pod or Container](#)

[Configure Service Accounts for Pods](#)

[Pull an Image from a Private Registry](#)

[Configure Liveness, Readiness and Startup Probes](#)

[Assign Pods to Nodes](#)

[Assign Pods to Nodes using Node Affinity](#)

[Configure Pod Initialization](#)

[Attach Handlers to Container Lifecycle Events](#)

[Configure a Pod to Use a ConfigMap](#)

[Share Process Namespace between Containers in a Pod](#)

[Use a User Namespace With a Pod](#)

[Create static Pods](#)

[Translate a Docker Compose File to Kubernetes Resources](#)

[Enforce Pod Security Standards by Configuring the Built-in Admission Controller](#)

[Enforce Pod Security Standards with Namespace Labels](#)

[Migrate from PodSecurityPolicy to the Built-In PodSecurity Admission Controller](#)

# Assign Memory Resources to Containers and Pods

This page shows how to assign a memory *request* and a memory *limit* to a Container. A Container is guaranteed to have as much memory as it requests, but is not allowed to use more memory than its limit.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

Each node in your cluster must have at least 300 MiB of memory.

A few of the steps on this page require you to run the [metrics-server](#) service in your cluster. If you have the metrics-server running, you can skip those steps.



If you are running Minikube, run the following command to enable the metrics-server:

```
minikube addons enable metrics-server
```

To see whether the metrics-server is running, or another provider of the resource metrics API (metrics.k8s.io), run the following command:

```
kubectl get apiservices
```

If the resource metrics API is available, the output includes a reference to metrics.k8s.io.

NAME

v1beta1.metrics.k8s.io

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace mem-example
```

## Specify a memory request and a memory limit

To specify a memory request for a Container, include the resources:requests field in the Container's resource manifest. To specify a memory limit, include resources:limits.

In this exercise, you create a Pod that has one Container. The Container has a memory request of 100 MiB and a memory limit of 200 MiB. Here's the configuration file for the Pod:

[pods/resource/memory-request-limit.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: memory-demo
 namespace: mem-example
spec:
 containers:
 - name: memory-demo-ctr
 image: polinux/stress
 resources:
 requests:
 memory: "100Mi"
 limits:
 memory: "200Mi"
 command: ["stress"]
 args: ["--vm", "1", "--vm-bytes", "150M", "--vm-hang", "1"]
```

The args section in the configuration file provides arguments for the Container when it starts. The "--vm-bytes", "150M" arguments tell the Container to attempt to allocate 150 MiB of memory.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/resource/memory-request-limit.yaml --namespace=mem-example
```

Verify that the Pod Container is running:

```
kubectl get pod memory-demo --namespace=mem-example
```

View detailed information about the Pod:

```
kubectl get pod memory-demo --output=yaml --namespace=mem-example
```

The output shows that the one Container in the Pod has a memory request of 100 MiB and a memory limit of 200 MiB.

```
...
resources:
 requests:
 memory: 100Mi
 limits:
 memory: 200Mi
...
```

Run `kubectl top` to fetch the metrics for the pod:

```
kubectl top pod memory-demo --namespace=mem-example
```

The output shows that the Pod is using about 162,900,000 bytes of memory, which is about 150 MiB. This is greater than the Pod's 100 MiB request, but within the Pod's 200 MiB limit.

| NAME        | CPU(cores)  | MEMORY(bytes) |
|-------------|-------------|---------------|
| memory-demo | <something> | 162856960     |

Delete your Pod:

```
kubectl delete pod memory-demo --namespace=mem-example
```

## Exceed a Container's memory limit

A Container can exceed its memory request if the Node has memory available. But a Container is not allowed to use more than its memory limit. If a Container allocates more memory than its limit, the Container becomes a candidate for termination. If the Container continues to consume memory beyond its limit, the Container is terminated. If a terminated Container can be restarted, the kubelet restarts it, as with any other type of runtime failure.

In this exercise, you create a Pod that attempts to allocate more memory than its limit. Here is the configuration file for a Pod that has one Container with a memory request of 50 MiB and a memory limit of 100 MiB:

[pods/resource/memory-request-limit-2.yaml](https://k8s.io/examples/pods/resource/memory-request-limit-2.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: memory-demo-2
```

```

namespace: mem-example
spec:
 containers:
 - name: memory-demo-2-ctr
 image: polinux/stress
 resources:
 requests:
 memory: "50Mi"
 limits:
 memory: "100Mi"
 command: ["stress"]
 args: ["--vm", "1", "--vm-bytes", "250M", "--vm-hang", "1"]

```

In the args section of the configuration file, you can see that the Container will attempt to allocate 250 MiB of memory, which is well above the 100 MiB limit.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/resource/memory-request-limit-2.yaml --namespace=mem-example
```

View detailed information about the Pod:

```
kubectl get pod memory-demo-2 --namespace=mem-example
```

At this point, the Container might be running or killed. Repeat the preceding command until the Container is killed:

```

NAME READY STATUS RESTARTS AGE
memory-demo-2 0/1 OOMKilled 1 24s

```

Get a more detailed view of the Container status:

```
kubectl get pod memory-demo-2 --output=yaml --namespace=mem-example
```

The output shows that the Container was killed because it is out of memory (OOM):

```

lastState:
 terminated:
 containerID: 65183c1877aaec2e8427bc95609cc52677a454b56fcb24340dbd22917c23b10f
 exitCode: 137
 finishedAt: 2017-06-20T20:52:19Z
 reason: OOMKilled
 startedAt: null

```

The Container in this exercise can be restarted, so the kubelet restarts it. Repeat this command several times to see that the Container is repeatedly killed and restarted:

```
kubectl get pod memory-demo-2 --namespace=mem-example
```

The output shows that the Container is killed, restarted, killed again, restarted again, and so on:

```

kubectl get pod memory-demo-2 --namespace=mem-example
NAME READY STATUS RESTARTS AGE
memory-demo-2 0/1 OOMKilled 1 37s

```

```
kubectrl get pod memory-demo-2 --namespace=mem-example
NAME READY STATUS RESTARTS AGE
memory-demo-2 1/1 Running 2 40s
```

View detailed information about the Pod history:

```
kubectrl describe pod memory-demo-2 --namespace=mem-example
```

The output shows that the Container starts and fails repeatedly:

```
... Normal Created Created container with id
66a3a20aa7980e61be4922780bf9d24d1a1d8b7395c09861225b0eba1b1f8511
... Warning BackOff Back-off restarting failed container
```

View detailed information about your cluster's Nodes:

```
kubectrl describe nodes
```

The output includes a record of the Container being killed because of an out-of-memory condition:

```
Warning OOMKilling Memory cgroup out of memory: Kill process 4481 (stress) score 1994 or
sacrifice child
```

Delete your Pod:

```
kubectrl delete pod memory-demo-2 --namespace=mem-example
```

## Specify a memory request that is too big for your Nodes

Memory requests and limits are associated with Containers, but it is useful to think of a Pod as having a memory request and limit. The memory request for the Pod is the sum of the memory requests for all the Containers in the Pod. Likewise, the memory limit for the Pod is the sum of the limits of all the Containers in the Pod.

Pod scheduling is based on requests. A Pod is scheduled to run on a Node only if the Node has enough available memory to satisfy the Pod's memory request.

In this exercise, you create a Pod that has a memory request so big that it exceeds the capacity of any Node in your cluster. Here is the configuration file for a Pod that has one Container with a request for 1000 GiB of memory, which likely exceeds the capacity of any Node in your cluster.

[pods/resource/memory-request-limit-3.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: memory-demo-3
 namespace: mem-example
spec:
 containers:
 - name: memory-demo-3-ctr
```

```
image: polinux/stress
resources:
 requests:
 memory: "1000Gi"
 limits:
 memory: "1000Gi"
command: ["stress"]
args: ["--vm", "1", "--vm-bytes", "150M", "--vm-hang", "1"]
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/resource/memory-request-limit-3.yaml --namespace=mem-example
```

View the Pod status:

```
kubectl get pod memory-demo-3 --namespace=mem-example
```

The output shows that the Pod status is PENDING. That is, the Pod is not scheduled to run on any Node, and it will remain in the PENDING state indefinitely:

```
kubectl get pod memory-demo-3 --namespace=mem-example
NAME READY STATUS RESTARTS AGE
memory-demo-3 0/1 Pending 0 25s
```

View detailed information about the Pod, including events:

```
kubectl describe pod memory-demo-3 --namespace=mem-example
```

The output shows that the Container cannot be scheduled because of insufficient memory on the Nodes:

```
Events:
... Reason Message

... FailedScheduling No nodes are available that match all of the following predicates::
Insufficient memory (3).
```

## Memory units

The memory resource is measured in bytes. You can express memory as a plain integer or a fixed-point integer with one of these suffixes: E, P, T, G, M, K, Ei, Pi, Ti, Gi, Mi, Ki. For example, the following represent approximately the same value:

```
128974848, 129e6, 129M, 123Mi
```

Delete your Pod:

```
kubectl delete pod memory-demo-3 --namespace=mem-example
```

## If you do not specify a memory limit

If you do not specify a memory limit for a Container, one of the following situations applies:

- The Container has no upper bound on the amount of memory it uses. The Container could use all of the memory available on the Node where it is running which in turn could invoke the OOM Killer. Further, in case of an OOM Kill, a container with no resource limits will have a greater chance of being killed.
- The Container is running in a namespace that has a default memory limit, and the Container is automatically assigned the default limit. Cluster administrators can use a [LimitRange](#) to specify a default value for the memory limit.

## Motivation for memory requests and limits

By configuring memory requests and limits for the Containers that run in your cluster, you can make efficient use of the memory resources available on your cluster's Nodes. By keeping a Pod's memory request low, you give the Pod a good chance of being scheduled. By having a memory limit that is greater than the memory request, you accomplish two things:

- The Pod can have bursts of activity where it makes use of memory that happens to be available.
- The amount of memory a Pod can use during a burst is limited to some reasonable amount.

## Clean up

Delete your namespace. This deletes all the Pods that you created for this task:

```
kubectrl delete namespace mem-example
```

## What's next

### For app developers

- [Assign CPU Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)

[Configure a Pod Quota for a Namespace](#)

- [Configure Quotas for API Objects](#)
- [Resize CPU and Memory Resources assigned to Containers](#)

# Assign CPU Resources to Containers and Pods

This page shows how to assign a CPU *request* and a CPU *limit* to a container. Containers cannot use more CPU than the configured limit. Provided the system has CPU time free, a container is guaranteed to be allocated as much CPU as it requests.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

Your cluster must have at least 1 CPU available for use to run the task examples.

A few of the steps on this page require you to run the [metrics-server](#) service in your cluster. If you have the metrics-server running, you can skip those steps.

If you are running [Minikube](#), run the following command to enable metrics-server:

```
minikube addons enable metrics-server
```

To see whether metrics-server (or another provider of the resource metrics API, `metrics.k8s.io`) is running, type the following command:

```
kubectl get apiservices
```

If the resource metrics API is available, the output will include a reference to `metrics.k8s.io`.

```
NAME
v1beta1.metrics.k8s.io
```

## Create a namespace

Create a [Namespace](#) so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace cpu-example
```

# Specify a CPU request and a CPU limit

To specify a CPU request for a container, include the `resources:requests` field in the Container resource manifest. To specify a CPU limit, include `resources:limits`.

In this exercise, you create a Pod that has one container. The container has a request of 0.5 CPU and a limit of 1 CPU. Here is the configuration file for the Pod:

[pods/resource/cpu-request-limit.yaml](https://k8s.io/examples/pods/resource/cpu-request-limit.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: cpu-demo
 namespace: cpu-example
spec:
 containers:
 - name: cpu-demo-ctr
 image: vish/stress
 resources:
 limits:
 cpu: "1"
 requests:
 cpu: "0.5"
 args:
 - -cpus
 - "2"
```

The `args` section of the configuration file provides arguments for the container when it starts. The `-cpus "2"` argument tells the Container to attempt to use 2 CPUs.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/resource/cpu-request-limit.yaml --namespace=cpu-example
```

Verify that the Pod is running:

```
kubectl get pod cpu-demo --namespace=cpu-example
```

View detailed information about the Pod:

```
kubectl get pod cpu-demo --output=yaml --namespace=cpu-example
```

The output shows that the one container in the Pod has a CPU request of 500 milliCPU and a CPU limit of 1 CPU.

```
resources:
 limits:
 cpu: "1"
 requests:
 cpu: 500m
```

Use `kubectl top` to fetch the metrics for the Pod:



```
kubecttl top pod cpu-demo --namespace=cpu-example
```

This example output shows that the Pod is using 974 milliCPU, which is slightly less than the limit of 1 CPU specified in the Pod configuration.

| NAME     | CPU(cores) | MEMORY(bytes) |
|----------|------------|---------------|
| cpu-demo | 974m       | <something>   |

Recall that by setting `-cpu "2"`, you configured the Container to attempt to use 2 CPUs, but the Container is only being allowed to use about 1 CPU. The container's CPU use is being throttled, because the container is attempting to use more CPU resources than its limit.

#### Note:

Another possible explanation for the CPU use being below 1.0 is that the Node might not have enough CPU resources available. Recall that the prerequisites for this exercise require your cluster to have at least 1 CPU available for use. If your Container runs on a Node that has only 1 CPU, the Container cannot use more than 1 CPU regardless of the CPU limit specified for the Container.

## CPU units

The CPU resource is measured in *CPU* units. One CPU, in Kubernetes, is equivalent to:

- 1 AWS vCPU
- 1 GCP Core
- 1 Azure vCore
- 1 Hyperthread on a bare-metal Intel processor with Hyperthreading

Fractional values are allowed. A Container that requests 0.5 CPU is guaranteed half as much CPU as a Container that requests 1 CPU. You can use the suffix `m` to mean milli. For example 100m CPU, 100 milliCPU, and 0.1 CPU are all the same. Precision finer than 1m is not allowed.

CPU is always requested as an absolute quantity, never as a relative quantity; 0.1 is the same amount of CPU on a single-core, dual-core, or 48-core machine.

Delete your Pod:

```
kubecttl delete pod cpu-demo --namespace=cpu-example
```

## Specify a CPU request that is too big for your Nodes

CPU requests and limits are associated with Containers, but it is useful to think of a Pod as having a CPU request and limit. The CPU request for a Pod is the sum of the CPU requests for all the Containers in the Pod. Likewise, the CPU limit for a Pod is the sum of the CPU limits for all the Containers in the Pod.

Pod scheduling is based on requests. A Pod is scheduled to run on a Node only if the Node has enough CPU resources available to satisfy the Pod CPU request.

In this exercise, you create a Pod that has a CPU request so big that it exceeds the capacity of any Node in your cluster. Here is the configuration file for a Pod that has one Container. The Container requests 100 CPU, which is likely to exceed the capacity of any Node in your cluster.

[pods/resource/cpu-request-limit-2.yaml](https://k8s.io/examples/pods/resource/cpu-request-limit-2.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: cpu-demo-2
 namespace: cpu-example
spec:
 containers:
 - name: cpu-demo-ctr-2
 image: vish/stress
 resources:
 limits:
 cpu: "100"
 requests:
 cpu: "100"
 args:
 - -cpus
 - "2"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/resource/cpu-request-limit-2.yaml --namespace=cpu-example
```

View the Pod status:

```
kubectl get pod cpu-demo-2 --namespace=cpu-example
```

The output shows that the Pod status is Pending. That is, the Pod has not been scheduled to run on any Node, and it will remain in the Pending state indefinitely:

| NAME       | READY | STATUS  | RESTARTS | AGE |
|------------|-------|---------|----------|-----|
| cpu-demo-2 | 0/1   | Pending | 0        | 7m  |

View detailed information about the Pod, including events:

```
kubectl describe pod cpu-demo-2 --namespace=cpu-example
```

The output shows that the Container cannot be scheduled because of insufficient CPU resources on the Nodes:

Events:

| Reason           | Message                                                                                   |
|------------------|-------------------------------------------------------------------------------------------|
| FailedScheduling | No nodes are available that match all of the following predicates:: Insufficient cpu (3). |

Delete your Pod:

```
kubectl delete pod cpu-demo-2 --namespace=cpu-example
```

## If you do not specify a CPU limit

If you do not specify a CPU limit for a Container, then one of these situations applies:

- The Container has no upper bound on the CPU resources it can use. The Container could use all of the CPU resources available on the Node where it is running.
- The Container is running in a namespace that has a default CPU limit, and the Container is automatically assigned the default limit. Cluster administrators can use a [LimitRange](#) to specify a default value for the CPU limit.

## If you specify a CPU limit but do not specify a CPU request

If you specify a CPU limit for a Container but do not specify a CPU request, Kubernetes automatically assigns a CPU request that matches the limit. Similarly, if a Container specifies its own memory limit, but does not specify a memory request, Kubernetes automatically assigns a memory request that matches the limit.

## Motivation for CPU requests and limits

By configuring the CPU requests and limits of the Containers that run in your cluster, you can make efficient use of the CPU resources available on your cluster Nodes. By keeping a Pod CPU request low, you give the Pod a good chance of being scheduled. By having a CPU limit that is greater than the CPU request, you accomplish two things:

- The Pod can have bursts of activity where it makes use of CPU resources that happen to be available.
- The amount of CPU resources a Pod can use during a burst is limited to some reasonable amount.

## Clean up

Delete your namespace:

```
kubectl delete namespace cpu-example
```

## What's next

### For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Configure Quality of Service for Pods](#)

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)

- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)
- [Configure a Pod Quota for a Namespace](#)
- [Configure Quotas for API Objects](#)
- [Resize CPU and Memory Resources assigned to Containers](#)

# Configure GMSA for Windows Pods and containers

FEATURE STATE: Kubernetes v1.18 [stable]

This page shows how to configure [Group Managed Service Accounts](#) (GMSA) for Pods and containers that will run on Windows nodes. Group Managed Service Accounts are a specific type of Active Directory account that provides automatic password management, simplified service principal name (SPN) management, and the ability to delegate the management to other administrators across multiple servers.

In Kubernetes, GMSA credential specs are configured at a Kubernetes cluster-wide scope as Custom Resources. Windows Pods, as well as individual containers within a Pod, can be configured to use a GMSA for domain based functions (e.g. Kerberos authentication) when interacting with other Windows services.

## Before you begin

You need to have a Kubernetes cluster and the `kubectl` command-line tool must be configured to communicate with your cluster. The cluster is expected to have Windows worker nodes. This section covers a set of initial steps required once for each cluster:

### Install the `GMSACredentialSpec` CRD

A [CustomResourceDefinition](#) (CRD) for GMSA credential spec resources needs to be configured on the cluster to define the custom resource type `GMSACredentialSpec`. Download the GMSA CRD [YAML](#) and save it as `gmsa-crd.yaml`. Next, install the CRD with `kubectl apply -f gmsa-crd.yaml`

### Install webhooks to validate GMSA users

Two webhooks need to be configured on the Kubernetes cluster to populate and validate GMSA credential spec references at the Pod or container level:

1. A mutating webhook that expands references to GMSAs (by name from a Pod specification) into the full credential spec in JSON form within the Pod spec.

- A validating webhook ensures all references to GSAs are authorized to be used by the
2. Pod service account.

Installing the above webhooks and associated objects require the steps below:

1. Create a certificate key pair (that will be used to allow the webhook container to communicate to the cluster)
2. Install a secret with the certificate from above.
3. Create a deployment for the core webhook logic.
4. Create the validating and mutating webhook configurations referring to the deployment.

A [script](#) can be used to deploy and configure the GSA webhooks and associated objects mentioned above. The script can be run with a `--dry-run=server` option to allow you to review the changes that would be made to your cluster.

The [YAML template](#) used by the script may also be used to deploy the webhooks and associated objects manually (with appropriate substitutions for the parameters)

## Configure GSAs and Windows nodes in Active Directory

Before Pods in Kubernetes can be configured to use GSAs, the desired GSAs need to be provisioned in Active Directory as described in the [Windows GSA documentation](#). Windows worker nodes (that are part of the Kubernetes cluster) need to be configured in Active Directory to access the secret credentials associated with the desired GSA as described in the [Windows GSA documentation](#).

## Create GSA credential spec resources

With the GSACredentialSpec CRD installed (as described earlier), custom resources containing GSA credential specs can be configured. The GSA credential spec does not contain secret or sensitive data. It is information that a container runtime can use to describe the desired GSA of a container to Windows. GSA credential specs can be generated in YAML format with a utility [PowerShell script](#).

Following are the steps for generating a GSA credential spec YAML manually in JSON format and then converting it:

1. Import the CredentialSpec [module](#): `ipmo CredentialSpec.psm1`
2. Create a credential spec in JSON format using `New-CredentialSpec`. To create a GSA credential spec named `WebApp1`, invoke `New-CredentialSpec -Name WebApp1 -AccountName WebApp1 -Domain $(Get-AddDomain -Current LocalComputer)`
3. Use `Get-CredentialSpec` to show the path of the JSON file.
4. Convert the credspec file from JSON to YAML format and apply the necessary header fields `apiVersion`, `kind`, `metadata` and `credspec` to make it a `GSACredentialSpec` custom resource that can be configured in Kubernetes.

The following YAML configuration describes a GMSA credential spec named gmsa-WebApp1:

```
apiVersion: windows.k8s.io/v1
kind: GMSACredentialSpec
metadata:
 name: gmsa-WebApp1 # This is an arbitrary name but it will be used as a reference
credspec:
 ActiveDirectoryConfig:
 GroupManagedServiceAccounts:
 - Name: WebApp1 # Username of the GMSA account
 Scope: CONTOSO # NETBIOS Domain Name
 - Name: WebApp1 # Username of the GMSA account
 Scope: contoso.com # DNS Domain Name
 CmsPlugins:
 - ActiveDirectory
 DomainJoinConfig:
 DnsName: contoso.com # DNS Domain Name
 DnsTreeName: contoso.com # DNS Domain Name Root
 Guid: 244818ae-87ac-4fcd-92ec-e79e5252348a # GUID
 MachineAccountName: WebApp1 # Username of the GMSA account
 NetBiosName: CONTOSO # NETBIOS Domain Name
 Sid: S-1-5-21-2126449477-2524075714-3094792973 # SID of GMSA
```

The above credential spec resource may be saved as gmsa-Webapp1-credspec.yaml and applied to the cluster using: `kubectl apply -f gmsa-Webapp1-credspec.yml`

## Configure cluster role to enable RBAC on specific GMSA credential specs

A cluster role needs to be defined for each GMSA credential spec resource. This authorizes the use verb on a specific GMSA resource by a subject which is typically a service account. The following example shows a cluster role that authorizes usage of the gmsa-WebApp1 credential spec from above. Save the file as gmsa-webapp1-role.yaml and apply using `kubectl apply -f gmsa-webapp1-role.yaml`

```
Create the Role to read the credspec
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: webapp1-role
rules:
- apiGroups: ["windows.k8s.io"]
 resources: ["gmsacredentialspecs"]
 verbs: ["use"]
 resourceNames: ["gmsa-WebApp1"]
```

## Assign role to service accounts to use specific GMSA credspecs

A service account (that Pods will be configured with) needs to be bound to the cluster role create above. This authorizes the service account to use the desired GMSA credential spec

resource. The following shows the default service account being bound to a cluster role webapp1-role to use gmsa-WebApp1 credential spec resource created above.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
 name: allow-default-svc-account-read-on-gmsa-WebApp1
 namespace: default
subjects:
- kind: ServiceAccount
 name: default
 namespace: default
roleRef:
 kind: ClusterRole
 name: webapp1-role
 apiGroup: rbac.authorization.k8s.io
```

## Configure GMSA credential spec reference in Pod spec

The Pod spec field `securityContext.windowsOptions.gmsaCredentialSpecName` is used to specify references to desired GMSA credential spec custom resources in Pod specs. This configures all containers in the Pod spec to use the specified GMSA. A sample Pod spec with the annotation populated to refer to gmsa-WebApp1:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 labels:
 run: with-creds
 name: with-creds
 namespace: default
spec:
 replicas: 1
 selector:
 matchLabels:
 run: with-creds
 template:
 metadata:
 labels:
 run: with-creds
 spec:
 securityContext:
 windowsOptions:
 gmsaCredentialSpecName: gmsa-webapp1
 containers:
 - image: mcr.microsoft.com/windows/servercore/iis:windowsservercore-ltsc2019
 imagePullPolicy: Always
 name: iis
 nodeSelector:
 kubernetes.io/os: windows
```

Individual containers in a Pod spec can also specify the desired GMSA credspec using a per-container securityContext.windowsOptions.gmsaCredentialSpecName field. For example:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 labels:
 run: with-creds
 name: with-creds
 namespace: default
spec:
 replicas: 1
 selector:
 matchLabels:
 run: with-creds
 template:
 metadata:
 labels:
 run: with-creds
 spec:
 containers:
 - image: mcr.microsoft.com/windows/servercore/iis:windowsservercore-ltsc2019
 imagePullPolicy: Always
 name: iis
 securityContext:
 windowsOptions:
 gmsaCredentialSpecName: gmsa-Webapp1
 nodeSelector:
 kubernetes.io/os: windows
```

As Pod specs with GMSA fields populated (as described above) are applied in a cluster, the following sequence of events take place:

1. The mutating webhook resolves and expands all references to GMSA credential spec resources to the contents of the GMSA credential spec.
2. The validating webhook ensures the service account associated with the Pod is authorized for the use verb on the specified GMSA credential spec.
3. The container runtime configures each Windows container with the specified GMSA credential spec so that the container can assume the identity of the GMSA in Active Directory and access services in the domain using that identity.

## Authenticating to network shares using hostname or FQDN

If you are experiencing issues connecting to SMB shares from Pods using hostname or FQDN, but are able to access the shares via their IPv4 address then make sure the following registry key is set on the Windows nodes.

```
reg add "HKLM\SYSTEM\CurrentControlSet\Services\hns\State" /v
EnableCompartmentNamespace /t REG_DWORD /d 1
```



Running Pods will then need to be recreated to pick up the behavior changes. More information on how this registry key is used can be found [here](#)

## Troubleshooting

If you are having difficulties getting GMSA to work in your environment, there are a few troubleshooting steps you can take.

First, make sure the credspec has been passed to the Pod. To do this you will need to exec into one of your Pods and check the output of the `nltest.exe /parentdomain` command.

In the example below the Pod did not get the credspec correctly:

```
kubect exec -it iis-auth-7776966999-n5nzs powershell.exe
```

`nltest.exe /parentdomain` results in the following error:

```
Getting parent domain failed: Status = 1722 0x6ba RPC_S_SERVER_UNAVAILABLE
```

If your Pod did get the credspec correctly, then next check communication with the domain. First, from inside of your Pod, quickly do an `nslookup` to find the root of your domain.

This will tell us 3 things:

1. The Pod can reach the DC
2. The DC can reach the Pod
3. DNS is working correctly.

If the DNS and communication test passes, next you will need to check if the Pod has established secure channel communication with the domain. To do this, again, exec into your Pod and run the `nltest.exe /query` command.

```
nltest.exe /query
```

Results in the following output:

```
I_NetLogonControl failed: Status = 1722 0x6ba RPC_S_SERVER_UNAVAILABLE
```

This tells us that for some reason, the Pod was unable to logon to the domain using the account specified in the credspec. You can try to repair the secure channel by running the following:

```
nltest /sc_reset:domain.example
```

If the command is successful you will see an output similar to this:

```
Flags: 30 HAS_IP HAS_TIMESERV
Trusted DC Name \\dc10.domain.example
Trusted DC Connection Status Status = 0 0x0 NERR_Success
The command completed successfully
```

If the above corrects the error, you can automate the step by adding the following lifecycle hook to your Pod spec. If it did not correct the error, you will need to examine your credspec again and confirm that it is correct and complete.

```
image: registry.domain.example/iis-auth:1809v1
lifecycle:
 postStart:
 exec:
 command: ["powershell.exe","-command","do { Restart-Service -Name netlogon } while
($($Result = (nltest.exe /query)); if ($Result -like '*0x0 NERR_Success*') {return $true} else
{return $false}) -eq $false)"]
 imagePullPolicy: IfNotPresent
```

If you add the lifecycle section shown above to your Pod spec, the Pod will execute the commands listed to restart the netlogon service until the `nltest.exe /query` command exits without error.

## Resize CPU and Memory Resources assigned to Containers

FEATURE STATE: Kubernetes v1.27 [alpha]

This page assumes that you are familiar with [Quality of Service](#) for Kubernetes Pods.

This page shows how to resize CPU and memory resources assigned to containers of a running pod without restarting the pod or its containers. A Kubernetes node allocates resources for a pod based on its requests, and restricts the pod's resource usage based on the limits specified in the pod's containers.

Changing the resource allocation for a running Pod requires the `InPlacePodVerticalScaling` [feature gate](#) to be enabled. The alternative is to delete the Pod and let the [workload controller](#) make a replacement Pod that has a different resource requirement.

For in-place resize of pod resources:

- Container's resource requests and limits are *mutable* for CPU and memory resources.
- `allocatedResources` field in `containerStatuses` of the Pod's status reflects the resources allocated to the pod's containers.
- `resources` field in `containerStatuses` of the Pod's status reflects the actual resource requests and limits that are configured on the running containers as reported by the container runtime.
- `resize` field in the Pod's status shows the status of the last requested pending resize. It can have the following values:
  - `Proposed`: This value indicates an acknowledgement of the requested resize and that the request was validated and recorded.
  - `InProgress`: This value indicates that the node has accepted the resize request and is in the process of applying it to the pod's containers.
  - `Deferred`: This value means that the requested resize cannot be granted at this time, and the node will keep retrying. The resize may be granted when other pods leave and free up node resources.
  - `Infeasible`: is a signal that the node cannot accommodate the requested resize. This can happen if the requested resize exceeds the maximum resources the node can ever allocate for a pod.

### Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version 1.27. To check the version, enter kubectl version.

The InPlacePodVerticalScaling [feature gate](#) must be enabled for your control plane and for all nodes in your cluster.

## Container Resize Policies

Resize policies allow for a more fine-grained control over how pod's containers are resized for CPU and memory resources. For example, the container's application may be able to handle CPU resources resized without being restarted, but resizing memory may require that the application hence the containers be restarted.

To enable this, the Container specification allows users to specify a `resizePolicy`. The following restart policies can be specified for resizing CPU and memory:

- `NotRequired`: Resize the container's resources while it is running.
- `RestartContainer`: Restart the container and apply new resources upon restart.

If `resizePolicy[*].restartPolicy` is not specified, it defaults to `NotRequired`.

### Note:

If the Pod's `restartPolicy` is `Never`, container's resize restart policy must be set to `NotRequired` for all Containers in the Pod.

Below example shows a Pod whose Container's CPU can be resized without restart, but resizing memory requires the container to be restarted.

```
apiVersion: v1
kind: Pod
metadata:
 name: qos-demo-5
 namespace: qos-example
spec:
 containers:
 - name: qos-demo-ctr-5
 image: nginx
 resizePolicy:
 - resourceName: cpu
 restartPolicy: NotRequired
 - resourceName: memory
 restartPolicy: RestartContainer
 resources:
 limits:
```

```
memory: "200Mi"
cpu: "700m"
requests:
 memory: "200Mi"
 cpu: "700m"
```

**Note:**

In the above example, if desired requests or limits for both CPU *and* memory have changed, the container will be restarted in order to resize its memory.

## Create a pod with resource requests and limits

You can create a Guaranteed or Burstable [Quality of Service](#) class pod by specifying requests and/or limits for a pod's containers.

Consider the following manifest for a Pod that has one Container.

[pods/qos/qos-pod-5.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: qos-demo-5
 namespace: qos-example
spec:
 containers:
 - name: qos-demo-ctr-5
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 cpu: "700m"
 requests:
 memory: "200Mi"
 cpu: "700m"
```

Create the pod in the qos-example namespace:

```
kubectl create namespace qos-example
kubectl create -f https://k8s.io/examples/pods/qos/qos-pod-5.yaml
```

This pod is classified as a Guaranteed QoS class requesting 700m CPU and 200Mi memory.

View detailed information about the pod:

```
kubectl get pod qos-demo-5 --output=yaml --namespace=qos-example
```

Also notice that the values of `resizePolicy[*].restartPolicy` defaulted to `NotRequired`, indicating that CPU and memory can be resized while container is running.

```
spec:
 containers:
```

```

...
resizePolicy:
- resourceName: cpu
 restartPolicy: NotRequired
- resourceName: memory
 restartPolicy: NotRequired
resources:
 limits:
 cpu: 700m
 memory: 200Mi
 requests:
 cpu: 700m
 memory: 200Mi
...
containerStatuses:
...
 name: qos-demo-ctr-5
 ready: true
...
 allocatedResources:
 cpu: 700m
 memory: 200Mi
 resources:
 limits:
 cpu: 700m
 memory: 200Mi
 requests:
 cpu: 700m
 memory: 200Mi
 restartCount: 0
 started: true
...
qosClass: Guaranteed

```

## Updating the pod's resources

Let's say the CPU requirements have increased, and 0.8 CPU is now desired. This may be specified manually, or determined and programmatically applied by an entity such as [VerticalPodAutoscaler](#) (VPA).

### Note:

While you can change a Pod's requests and limits to express new desired resources, you cannot change the QoS class in which the Pod was created.

Now, patch the Pod's Container with CPU requests & limits both set to 800m:

```
kubectrl -n qos-example patch pod qos-demo-5 --patch '{"spec":{"containers":[{"name":"qos-demo-ctr-5", "resources":{"requests":{"cpu":"800m"}, "limits":{"cpu":"800m"}}}]}'
```

Query the Pod's detailed information after the Pod has been patched.

```
kubectl get pod qos-demo-5 --output=yaml --namespace=qos-example
```

The Pod's spec below reflects the updated CPU requests and limits.

```
spec:
 containers:
 ...
 resources:
 limits:
 cpu: 800m
 memory: 200Mi
 requests:
 cpu: 800m
 memory: 200Mi
 ...
 containerStatuses:
 ...
 allocatedResources:
 cpu: 800m
 memory: 200Mi
 resources:
 limits:
 cpu: 800m
 memory: 200Mi
 requests:
 cpu: 800m
 memory: 200Mi
 restartCount: 0
 started: true
```

Observe that the `allocatedResources` values have been updated to reflect the new desired CPU requests. This indicates that node was able to accommodate the increased CPU resource needs.

In the Container's status, updated CPU resource values shows that new CPU resources have been applied. The Container's `restartCount` remains unchanged, indicating that container's CPU resources were resized without restarting the container.

## Clean up

Delete your namespace:

```
kubectl delete namespace qos-example
```

## What's next

### For application developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)

## For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)

# Configure RunAsUserName for Windows pods and containers

FEATURE STATE: Kubernetes v1.18 [stable]

This page shows how to use the `runAsUserName` setting for Pods and containers that will run on Windows nodes. This is roughly equivalent of the Linux-specific `runAsUser` setting, allowing you to run applications in a container as a different username than the default.

## Before you begin

You need to have a Kubernetes cluster and the `kubectl` command-line tool must be configured to communicate with your cluster. The cluster is expected to have Windows worker nodes where pods with containers running Windows workloads will get scheduled.

## Set the Username for a Pod

To specify the username with which to execute the Pod's container processes, include the `securityContext` field ([PodSecurityContext](#)) in the Pod specification, and within it, the `windowsOptions` ([WindowsSecurityContextOptions](#)) field containing the `runAsUserName` field.

The Windows security context options that you specify for a Pod apply to all Containers and init Containers in the Pod.

Here is a configuration file for a Windows Pod that has the `runAsUserName` field set:

[windows/run-as-username-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: run-as-username-pod-demo
spec:
 securityContext:
 windowsOptions:
 runAsUserName: "ContainerUser"
 containers:
 - name: run-as-username-demo
```

```
image: mcr.microsoft.com/windows/servercore:ltsc2019
command: ["ping", "-t", "localhost"]
nodeSelector:
 kubernetes.io/os: windows
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/windows/run-as-username-pod.yaml
```

Verify that the Pod's Container is running:

```
kubectl get pod run-as-username-pod-demo
```

Get a shell to the running Container:

```
kubectl exec -it run-as-username-pod-demo -- powershell
```

Check that the shell is running user the correct username:

```
echo $env:USERNAME
```

The output should be:

```
ContainerUser
```

## Set the Username for a Container

To specify the username with which to execute a Container's processes, include the `securityContext` field ([SecurityContext](#)) in the Container manifest, and within it, the `windowsOptions` ([WindowsSecurityContextOptions](#)) field containing the `runAsUserName` field.

The Windows security context options that you specify for a Container apply only to that individual Container, and they override the settings made at the Pod level.

Here is the configuration file for a Pod that has one Container, and the `runAsUserName` field is set at the Pod level and the Container level:

[windows/run-as-username-container.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: run-as-username-container-demo
spec:
 securityContext:
 windowsOptions:
 runAsUserName: "ContainerUser"
 containers:
 - name: run-as-username-demo
 image: mcr.microsoft.com/windows/servercore:ltsc2019
 command: ["ping", "-t", "localhost"]
 securityContext:
 windowsOptions:
 runAsUserName: "ContainerAdministrator"
```



```
nodeSelector:
 kubernetes.io/os: windows
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/windows/run-as-username-container.yaml
```

Verify that the Pod's Container is running:

```
kubectl get pod run-as-username-container-demo
```

Get a shell to the running Container:

```
kubectl exec -it run-as-username-container-demo -- powershell
```

Check that the shell is running user the correct username (the one set at the Container level):

```
echo $env:USERNAME
```

The output should be:

```
ContainerAdministrator
```

## Windows Username limitations

In order to use this feature, the value set in the `runAsUserName` field must be a valid username. It must have the following format: `DOMAIN\USER`, where `DOMAIN\` is optional. Windows user names are case insensitive. Additionally, there are some restrictions regarding the `DOMAIN` and `USER`:

- The `runAsUserName` field cannot be empty, and it cannot contain control characters (ASCII values: 0x00-0x1F, 0x7F)
- The `DOMAIN` must be either a NetBios name, or a DNS name, each with their own restrictions:
  - NetBios names: maximum 15 characters, cannot start with `.` (dot), and cannot contain the following characters: `\ / : * ? " < > |`
  - DNS names: maximum 255 characters, contains only alphanumeric characters, dots, and dashes, and it cannot start or end with a `.` (dot) or `-` (dash).
- The `USER` must have at most 20 characters, it cannot contain *only* dots or spaces, and it cannot contain the following characters: `" / \ [ ] : ; | = , + * ? < > @`.

Examples of acceptable values for the `runAsUserName` field: `ContainerAdministrator`, `ContainerUser`, `NT AUTHORITY\NETWORK SERVICE`, `NT AUTHORITY\LOCAL SERVICE`.

For more information about these limitations, check [here](#) and [here](#).

## What's next

- [Guide for scheduling Windows containers in Kubernetes](#)
- [Managing Workload Identity with Group Managed Service Accounts \(GMSA\)](#)
- [Configure GMSA for Windows pods and containers](#)

# Create a Windows HostProcess Pod

FEATURE STATE: Kubernetes v1.26 [stable]

Windows HostProcess containers enable you to run containerized workloads on a Windows host. These containers operate as normal processes but have access to the host network namespace, storage, and devices when given the appropriate user privileges. HostProcess containers can be used to deploy network plugins, storage configurations, device plugins, kube-proxy, and other components to Windows nodes without the need for dedicated proxies or the direct installation of host services.

Administrative tasks such as installation of security patches, event log collection, and more can be performed without requiring cluster operators to log onto each Windows node. HostProcess containers can run as any user that is available on the host or is in the domain of the host machine, allowing administrators to restrict resource access through user permissions. While neither filesystem or process isolation are supported, a new volume is created on the host upon starting the container to give it a clean and consolidated workspace. HostProcess containers can also be built on top of existing Windows base images and do not inherit the same [compatibility requirements](#) as Windows server containers, meaning that the version of the base images does not need to match that of the host. It is, however, recommended that you use the same base image version as your Windows Server container workloads to ensure you do not have any unused images taking up space on the node. HostProcess containers also support [volume mounts](#) within the container volume.

## When should I use a Windows HostProcess container?

- When you need to perform tasks which require the networking namespace of the host. HostProcess containers have access to the host's network interfaces and IP addresses.
- You need access to resources on the host such as the filesystem, event logs, etc.
- Installation of specific device drivers or Windows services.
- Consolidation of administrative tasks and security policies. This reduces the degree of privileges needed by Windows nodes.

## Before you begin

This task guide is specific to Kubernetes v1.30. If you are not running Kubernetes v1.30, check the documentation for that version of Kubernetes.

In Kubernetes 1.30, the HostProcess container feature is enabled by default. The kubelet will communicate with containerd directly by passing the hostprocess flag via CRI. You can use the latest version of containerd (v1.6+) to run HostProcess containers. [How to install containerd.](#)

## Limitations

These limitations are relevant for Kubernetes v1.30:

- HostProcess containers require containerd 1.6 or higher [container runtime](#) and containerd 1.7 is recommended.
- HostProcess pods can only contain HostProcess containers. This is a current limitation of the Windows OS; non-privileged Windows containers cannot share a vNIC with the host IP namespace.

- HostProcess containers run as a process on the host and do not have any degree of isolation other than resource constraints imposed on the HostProcess user account. Neither filesystem or Hyper-V isolation are supported for HostProcess containers.
- Volume mounts are supported and are mounted under the container volume. See [Volume Mounts](#)
- A limited set of host user accounts are available for HostProcess containers by default. See [Choosing a User Account](#).
- Resource limits (disk, memory, cpu count) are supported in the same fashion as processes on the host.
- Both Named pipe mounts and Unix domain sockets are **not** supported and should instead be accessed via their path on the host (e.g. \\.\pipe\\*)

## HostProcess Pod configuration requirements

Enabling a Windows HostProcess pod requires setting the right configurations in the pod security configuration. Of the policies defined in the [Pod Security Standards](#) HostProcess pods are disallowed by the baseline and restricted policies. It is therefore recommended that HostProcess pods run in alignment with the privileged profile.

When running under the privileged policy, here are the configurations which need to be set to enable the creation of a HostProcess pod:

Privileged policy specification

| Control                                                      | Policy                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">securityContext.windowsOptions.hostProcess</a>   | <p>Windows pods offer the ability to run <a href="#">HostProcess containers</a> which enables privileged access to the Windows node.</p> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• true</li> </ul>                                                                                                          |
| <a href="#">hostNetwork</a>                                  | <p>Pods container HostProcess containers must use the host's network namespace.</p> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• true</li> </ul>                                                                                                                                                               |
| <a href="#">securityContext.windowsOptions.runAsUserName</a> | <p>Specification of which user the HostProcess container should run as is required for the pod spec.</p> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• NT AUTHORITY\SYSTEM</li> <li>• NT AUTHORITY\Local service</li> <li>• NT AUTHORITY\NetworkService</li> <li>• Local usergroup names (see below)</li> </ul> |

| Control                      | Policy                                                                                                                                                                                                                                  |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">runAsNonRoot</a> | <p>Because HostProcess containers have privileged access to the host, the runAsNonRoot field cannot be set to true.</p> <p><b>Allowed Values</b></p> <ul style="list-style-type: none"> <li>• Undefined/Nil</li> <li>• false</li> </ul> |

## Example manifest (excerpt)

```
spec:
 securityContext:
 windowsOptions:
 hostProcess: true
 runAsUserName: "NT AUTHORITY\\Local service"
 hostNetwork: true
 containers:
 - name: test
 image: image1:latest
 command:
 - ping
 - -t
 - 127.0.0.1
 nodeSelector:
 "kubernetes.io/os": windows
```

## Volume mounts

HostProcess containers support the ability to mount volumes within the container volume space. Volume mount behavior differs depending on the version of containerd runtime used by on the node.

### Containerd v1.6

Applications running inside the container can access volume mounts directly via relative or absolute paths. An environment variable `$CONTAINER_SANDBOX_MOUNT_POINT` is set upon container creation and provides the absolute host path to the container volume. Relative paths are based upon the `.spec.containers.volumeMounts.mountPath` configuration.

To access service account tokens (for example) the following path structures are supported within the container:

- `.\var\run\secrets\kubernetes.io\serviceaccount\`
- `$CONTAINER_SANDBOX_MOUNT_POINT\var\run\secrets\kubernetes.io\serviceaccount\`

## Containerd v1.7 (and greater)

Applications running inside the container can access volume mounts directly via the volumeMount's specified mountPath (just like Linux and non-HostProcess Windows containers).

For backwards compatibility volumes can also be accessed via using the same relative paths configured by containerd v1.6.

As an example, to access service account tokens within the container you would use one of the following paths:

- c:\var\run\secrets\kubernetes.io\serviceaccount
- /var/run/secrets/kubernetes.io/serviceaccount/
- \$CONTAINER\_SANDBOX\_MOUNT\_POINT\var\run\secrets\kubernetes.io\serviceaccount\

## Resource limits

Resource limits (disk, memory, cpu count) are applied to the job and are job wide. For example, with a limit of 10MB set, the memory allocated for any HostProcess job object will be capped at 10MB. This is the same behavior as other Windows container types. These limits would be specified the same way they are currently for whatever orchestrator or runtime is being used. The only difference is in the disk resource usage calculation used for resource tracking due to the difference in how HostProcess containers are bootstrapped.

## Choosing a user account

### System accounts

By default, HostProcess containers support the ability to run as one of three supported Windows service accounts:

- [LocalSystem](#)
- [LocalService](#)
- [NetworkService](#)

You should select an appropriate Windows service account for each HostProcess container, aiming to limit the degree of privileges so as to avoid accidental (or even malicious) damage to the host. The LocalSystem service account has the highest level of privilege of the three and should be used only if absolutely necessary. Where possible, use the LocalService service account as it is the least privileged of the three options.

### Local accounts

If configured, HostProcess containers can also run as local user accounts which allows for node operators to give fine-grained access to workloads.

To run HostProcess containers as a local user; A local usergroup must first be created on the node and the name of that local usergroup must be specified in the runAsUserName field in the deployment. Prior to initializing the HostProcess container, a new **ephemeral** local user account to be created and joined to the specified usergroup, from which the container is run.

This provides a number a benefits including eliminating the need to manage passwords for local user accounts. An initial HostProcess container running as a service account can be used to prepare the user groups for later HostProcess containers.

**Note:**

Running HostProcess containers as local user accounts requires containerd v1.7+

Example:

1. Create a local user group on the node (this can be done in another HostProcess container).

```
net localgroup hpc-localgroup /add
```

2. Grant access to desired resources on the node to the local usergroup. This can be done with tools like [icacis](#).
3. Set runAsUserName to the name of the local usergroup for the pod or individual containers.

```
securityContext:
 windowsOptions:
 hostProcess: true
 runAsUserName: hpc-localgroup
```

4. Schedule the pod!

## Base Image for HostProcess Containers

HostProcess containers can be built from any of the existing [Windows Container base images](#).

Additionally a new base mage has been created just for HostProcess containers! For more information please check out the [windows-host-process-containers-base-image github project](#).

## Troubleshooting HostProcess containers

- HostProcess containers fail to start with failed to create user process token: failed to logon user: Access is denied.: unknown

Ensure containerd is running as LocalSystem or LocalService service accounts. User accounts (even Administrator accounts) do not have permissions to create logon tokens for any of the supported [user accounts](#).

## Configure Quality of Service for Pods

This page shows how to configure Pods so that they will be assigned particular [Quality of Service \(QoS\) classes](#). Kubernetes uses QoS classes to make decisions about evicting Pods when Node resources are exceeded.

When Kubernetes creates a Pod it assigns one of these QoS classes to the Pod:

- [Guaranteed](#)
- [Burstable](#)
- [BestEffort](#)

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You also need to be able to create and delete namespaces.

## Create a namespace

Create a namespace so that the resources you create in this exercise are isolated from the rest of your cluster.

```
kubectl create namespace qos-example
```

## Create a Pod that gets assigned a QoS class of Guaranteed

For a Pod to be given a QoS class of Guaranteed:

- Every Container in the Pod must have a memory limit and a memory request.
- For every Container in the Pod, the memory limit must equal the memory request.
- Every Container in the Pod must have a CPU limit and a CPU request.
- For every Container in the Pod, the CPU limit must equal the CPU request.

These restrictions apply to init containers and app containers equally. [Ephemeral containers](#) cannot define resources so these restrictions do not apply.

Here is a manifest for a Pod that has one Container. The Container has a memory limit and a memory request, both equal to 200 MiB. The Container has a CPU limit and a CPU request, both equal to 700 milliCPU:

[pods/qos/qos-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: qos-demo
 namespace: qos-example
spec:
 containers:
 - name: qos-demo-ctr
 image: nginx
```

```
resources:
 limits:
 memory: "200Mi"
 cpu: "700m"
 requests:
 memory: "200Mi"
 cpu: "700m"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/qos/qos-pod.yaml --namespace=qos-example
```

View detailed information about the Pod:

```
kubectl get pod qos-demo --namespace=qos-example --output=yaml
```

The output shows that Kubernetes gave the Pod a QoS class of Guaranteed. The output also verifies that the Pod Container has a memory request that matches its memory limit, and it has a CPU request that matches its CPU limit.

```
spec:
 containers:
 ...
 resources:
 limits:
 cpu: 700m
 memory: 200Mi
 requests:
 cpu: 700m
 memory: 200Mi
 ...
status:
 qosClass: Guaranteed
```

### Note:

If a Container specifies its own memory limit, but does not specify a memory request, Kubernetes automatically assigns a memory request that matches the limit. Similarly, if a Container specifies its own CPU limit, but does not specify a CPU request, Kubernetes automatically assigns a CPU request that matches the limit.

### Clean up

Delete your Pod:

```
kubectl delete pod qos-demo --namespace=qos-example
```

## Create a Pod that gets assigned a QoS class of Burstable

A Pod is given a QoS class of Burstable if:

- The Pod does not meet the criteria for QoS class Guaranteed.



- At least one Container in the Pod has a memory or CPU request or limit.

Here is a manifest for a Pod that has one Container. The Container has a memory limit of 200 MiB and a memory request of 100 MiB.

[pods/qos/qos-pod-2.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: qos-demo-2
 namespace: qos-example
spec:
 containers:
 - name: qos-demo-2-ctr
 image: nginx
 resources:
 limits:
 memory: "200Mi"
 requests:
 memory: "100Mi"
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/qos/qos-pod-2.yaml --namespace=qos-example
```

View detailed information about the Pod:

```
kubectl get pod qos-demo-2 --namespace=qos-example --output=yaml
```

The output shows that Kubernetes gave the Pod a QoS class of Burstable:

```
spec:
 containers:
 - image: nginx
 imagePullPolicy: Always
 name: qos-demo-2-ctr
 resources:
 limits:
 memory: 200Mi
 requests:
 memory: 100Mi
 ...
status:
 qosClass: Burstable
```

## Clean up

Delete your Pod:

```
kubectl delete pod qos-demo-2 --namespace=qos-example
```

## Create a Pod that gets assigned a QoS class of BestEffort

For a Pod to be given a QoS class of BestEffort, the Containers in the Pod must not have any memory or CPU limits or requests.

Here is a manifest for a Pod that has one Container. The Container has no memory or CPU limits or requests:

[pods/qos/qos-pod-3.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: qos-demo-3
 namespace: qos-example
spec:
 containers:
 - name: qos-demo-3-ctr
 image: nginx
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/qos/qos-pod-3.yaml --namespace=qos-example
```

View detailed information about the Pod:

```
kubectl get pod qos-demo-3 --namespace=qos-example --output=yaml
```

The output shows that Kubernetes gave the Pod a QoS class of BestEffort:

```
spec:
 containers:
 ...
 resources: {}
 ...
status:
 qosClass: BestEffort
```

### Clean up

Delete your Pod:

```
kubectl delete pod qos-demo-3 --namespace=qos-example
```

## Create a Pod that has two Containers

Here is a manifest for a Pod that has two Containers. One container specifies a memory request of 200 MiB. The other Container does not specify any requests or limits.

[pods/qos/qos-pod-4.yaml](#)

```
apiVersion: v1
kind: Pod
```

```
metadata:
 name: qos-demo-4
 namespace: qos-example
spec:
 containers:

 - name: qos-demo-4-ctr-1
 image: nginx
 resources:
 requests:
 memory: "200Mi"

 - name: qos-demo-4-ctr-2
 image: redis
```

Notice that this Pod meets the criteria for QoS class Burstable. That is, it does not meet the criteria for QoS class Guaranteed, and one of its Containers has a memory request.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/qos/qos-pod-4.yaml --namespace=qos-example
```

View detailed information about the Pod:

```
kubectl get pod qos-demo-4 --namespace=qos-example --output=yaml
```

The output shows that Kubernetes gave the Pod a QoS class of Burstable:

```
spec:
 containers:
 ...
 name: qos-demo-4-ctr-1
 resources:
 requests:
 memory: 200Mi
 ...
 name: qos-demo-4-ctr-2
 resources: {}
 ...
status:
 qosClass: Burstable
```

## Retrieve the QoS class for a Pod

Rather than see all the fields, you can view just the field you need:

```
kubectl --namespace=qos-example get pod qos-demo-4 -o jsonpath='{.status.qosClass}'
```

Burstable

## Clean up

Delete your namespace:

```
kubectl delete namespace qos-example
```

## What's next

### For app developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)

### For cluster administrators

- [Configure Default Memory Requests and Limits for a Namespace](#)
- [Configure Default CPU Requests and Limits for a Namespace](#)
- [Configure Minimum and Maximum Memory Constraints for a Namespace](#)
- [Configure Minimum and Maximum CPU Constraints for a Namespace](#)
- [Configure Memory and CPU Quotas for a Namespace](#)
- [Configure a Pod Quota for a Namespace](#)
- [Configure Quotas for API Objects](#)
- [Control Topology Management policies on a node](#)

# Assign Extended Resources to a Container

FEATURE STATE: Kubernetes v1.30 [stable]

This page shows how to assign extended resources to a Container.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

Before you do this exercise, do the exercise in [Advertise Extended Resources for a Node](#). That will configure one of your Nodes to advertise a dongle resource.

## Assign an extended resource to a Pod

To request an extended resource, include the `resources:requests` field in your Container manifest. Extended resources are fully qualified with any domain outside of `*.kubernetes.io/`. Valid extended resource names have the form `example.com/foo` where `example.com` is replaced with your organization's domain and `foo` is a descriptive resource name.

Here is the configuration file for a Pod that has one Container:

[pods/resource/extended-resource-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: extended-resource-demo
spec:
 containers:
 - name: extended-resource-demo-ctr
 image: nginx
 resources:
 requests:
 example.com/dongle: 3
 limits:
 example.com/dongle: 3
```

In the configuration file, you can see that the Container requests 3 dongles.

Create a Pod:

```
kubectl apply -f https://k8s.io/examples/pods/resource/extended-resource-pod.yaml
```

Verify that the Pod is running:

```
kubectl get pod extended-resource-demo
```

Describe the Pod:

```
kubectl describe pod extended-resource-demo
```

The output shows dongle requests:

```
Limits:
 example.com/dongle: 3
Requests:
 example.com/dongle: 3
```

## Attempt to create a second Pod

Here is the configuration file for a Pod that has one Container. The Container requests two dongles.

[pods/resource/extended-resource-pod-2.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: extended-resource-demo-2
spec:
 containers:
 - name: extended-resource-demo-2-ctr
 image: nginx
 resources:
 requests:
 example.com/dongle: 2
 limits:
 example.com/dongle: 2
```

Kubernetes will not be able to satisfy the request for two dongles, because the first Pod used three of the four available dongles.

Attempt to create a Pod:

```
kubectl apply -f https://k8s.io/examples/pods/resource/extended-resource-pod-2.yaml
```

Describe the Pod

```
kubectl describe pod extended-resource-demo-2
```

The output shows that the Pod cannot be scheduled, because there is no Node that has 2 dongles available:

```
Conditions:
 Type Status
 PodScheduled False
...
Events:
 ...
 ... Warning FailedScheduling pod (extended-resource-demo-2) failed to fit in any node
fit failure summary on nodes : Insufficient example.com/dongle (1)
```

View the Pod status:

```
kubectl get pod extended-resource-demo-2
```

The output shows that the Pod was created, but not scheduled to run on a Node. It has a status of Pending:

| NAME                     | READY | STATUS  | RESTARTS | AGE |
|--------------------------|-------|---------|----------|-----|
| extended-resource-demo-2 | 0/1   | Pending | 0        | 6m  |

## Clean up

Delete the Pods that you created for this exercise:

```
kubectl delete pod extended-resource-demo
kubectl delete pod extended-resource-demo-2
```

## What's next

### For application developers

- [Assign Memory Resources to Containers and Pods](#)
- [Assign CPU Resources to Containers and Pods](#)

### For cluster administrators

- [Advertise Extended Resources for a Node](#)

# Configure a Pod to Use a Volume for Storage

This page shows how to configure a Pod to use a Volume for storage.

A Container's file system lives only as long as the Container does. So when a Container terminates and restarts, filesystem changes are lost. For more consistent storage that is independent of the Container, you can use a [Volume](#). This is especially important for stateful applications, such as key-value stores (such as Redis) and databases.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Configure a volume for a Pod

In this exercise, you create a Pod that runs one Container. This Pod has a Volume of type [emptyDir](#) that lasts for the life of the Pod, even if the Container terminates and restarts. Here is the configuration file for the Pod:

[pods/storage/redis.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: redis
spec:
 containers:
 - name: redis
 image: redis
```

```
volumeMounts:
- name: redis-storage
 mountPath: /data/redis
volumes:
- name: redis-storage
 emptyDir: {}
```

1. Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/storage/redis.yaml
```

2. Verify that the Pod's Container is running, and then watch for changes to the Pod:

```
kubectl get pod redis --watch
```

The output looks like this:

| NAME  | READY | STATUS  | RESTARTS | AGE |
|-------|-------|---------|----------|-----|
| redis | 1/1   | Running | 0        | 13s |

3. In another terminal, get a shell to the running Container:

```
kubectl exec -it redis -- /bin/bash
```

4. In your shell, go to /data/redis, and then create a file:

```
root@redis:/data# cd /data/redis/
root@redis:/data/redis# echo Hello > test-file
```

5. In your shell, list the running processes:

```
root@redis:/data/redis# apt-get update
root@redis:/data/redis# apt-get install procps
root@redis:/data/redis# ps aux
```

The output is similar to this:

| USER  | PID | %CPU | %MEM | VSZ   | RSS  | TTY | STAT | START | TIME | COMMAND             |
|-------|-----|------|------|-------|------|-----|------|-------|------|---------------------|
| redis | 1   | 0.1  | 0.1  | 33308 | 3828 | ?   | Ssl  | 00:46 | 0:00 | redis-server *:6379 |
| root  | 12  | 0.0  | 0.0  | 20228 | 3020 | ?   | Ss   | 00:47 | 0:00 | /bin/bash           |
| root  | 15  | 0.0  | 0.0  | 17500 | 2072 | ?   | R+   | 00:48 | 0:00 | ps aux              |

6. In your shell, kill the Redis process:

```
root@redis:/data/redis# kill <pid>
```

where <pid> is the Redis process ID (PID).

7. In your original terminal, watch for changes to the Redis Pod. Eventually, you will see something like this:

| NAME  | READY | STATUS    | RESTARTS | AGE |
|-------|-------|-----------|----------|-----|
| redis | 1/1   | Running   | 0        | 13s |
| redis | 0/1   | Completed | 0        | 6m  |
| redis | 1/1   | Running   | 1        | 6m  |



At this point, the Container has terminated and restarted. This is because the Redis Pod has a [restartPolicy](#) of Always.

1. Get a shell into the restarted Container:

```
kubectl exec -it redis -- /bin/bash
```

2. In your shell, go to /data/redis, and verify that test-file is still there.

```
root@redis:/data/redis# cd /data/redis/
root@redis:/data/redis# ls
test-file
```

3. Delete the Pod that you created for this exercise:

```
kubectl delete pod redis
```

## What's next

- See [Volume](#).
- See [Pod](#).
- In addition to the local disk storage provided by emptyDir, Kubernetes supports many different network-attached storage solutions, including PD on GCE and EBS on EC2, which are preferred for critical data and will handle details such as mounting and unmounting the devices on the nodes. See [Volumes](#) for more details.

# Configure a Pod to Use a PersistentVolume for Storage

This page shows you how to configure a Pod to use a [PersistentVolumeClaim](#) for storage. Here is a summary of the process:

1. You, as cluster administrator, create a PersistentVolume backed by physical storage. You do not associate the volume with any Pod.
2. You, now taking the role of a developer / cluster user, create a PersistentVolumeClaim that is automatically bound to a suitable PersistentVolume.
3. You create a Pod that uses the above PersistentVolumeClaim for storage.

## Before you begin

- You need to have a Kubernetes cluster that has only one Node, and the [kubectl](#) command-line tool must be configured to communicate with your cluster. If you do not already have a single-node cluster, you can create one by using [Minikube](#).
- Familiarize yourself with the material in [Persistent Volumes](#).

## Create an index.html file on your Node

Open a shell to the single Node in your cluster. How you open a shell depends on how you set up your cluster. For example, if you are using Minikube, you can open a shell to your Node by entering `minikube ssh`.

In your shell on that Node, create a `/mnt/data` directory:

```
This assumes that your Node uses "sudo" to run commands
as the superuser
sudo mkdir /mnt/data
```

In the `/mnt/data` directory, create an `index.html` file:

```
This again assumes that your Node uses "sudo" to run commands
as the superuser
sudo sh -c "echo 'Hello from Kubernetes storage' > /mnt/data/index.html"
```

### Note:

If your Node uses a tool for superuser access other than `sudo`, you can usually make this work if you replace `sudo` with the name of the other tool.

Test that the `index.html` file exists:

```
cat /mnt/data/index.html
```

The output should be:

```
Hello from Kubernetes storage
```

You can now close the shell to your Node.

## Create a PersistentVolume

In this exercise, you create a *hostPath* PersistentVolume. Kubernetes supports *hostPath* for development and testing on a single-node cluster. A *hostPath* PersistentVolume uses a file or directory on the Node to emulate network-attached storage.

In a production cluster, you would not use *hostPath*. Instead a cluster administrator would provision a network resource like a Google Compute Engine persistent disk, an NFS share, or an Amazon Elastic Block Store volume. Cluster administrators can also use [StorageClasses](#) to set up [dynamic provisioning](#).

Here is the configuration file for the *hostPath* PersistentVolume:

[pods/storage/pv-volume.yaml](#)

```
apiVersion: v1
kind: PersistentVolume
metadata:
 name: task-pv-volume
 labels:
```

```
 type: local
spec:
 storageClassName: manual
 capacity:
 storage: 10Gi
 accessModes:
 - ReadWriteOnce
 hostPath:
 path: "/mnt/data"
```

The configuration file specifies that the volume is at `/mnt/data` on the cluster's Node. The configuration also specifies a size of 10 gibibytes and an access mode of `ReadWriteOnce`, which means the volume can be mounted as read-write by a single Node. It defines the [StorageClass name](#) `manual` for the `PersistentVolume`, which will be used to bind `PersistentVolumeClaim` requests to this `PersistentVolume`.

### Note:

This example uses the `ReadWriteOnce` access mode, for simplicity. For production use, the Kubernetes project recommends using the `ReadWriteOncePod` access mode instead.

Create the `PersistentVolume`:

```
kubectl apply -f https://k8s.io/examples/pods/storage/pv-volume.yaml
```

View information about the `PersistentVolume`:

```
kubectl get pv task-pv-volume
```

The output shows that the `PersistentVolume` has a `STATUS` of `Available`. This means it has not yet been bound to a `PersistentVolumeClaim`.

| NAME           | CAPACITY | ACCESSMODES | RECLAIMPOLICY | STATUS    | CLAIM  |
|----------------|----------|-------------|---------------|-----------|--------|
| task-pv-volume | 10Gi     | RWO         | Retain        | Available | manual |

## Create a PersistentVolumeClaim

The next step is to create a `PersistentVolumeClaim`. Pods use `PersistentVolumeClaims` to request physical storage. In this exercise, you create a `PersistentVolumeClaim` that requests a volume of at least three gibibytes that can provide read-write access for at most one Node at a time.

Here is the configuration file for the `PersistentVolumeClaim`:

[pods/storage/pv-claim.yaml](#)

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: task-pv-claim
spec:
 storageClassName: manual
 accessModes:
```

```
- ReadWriteOnce
resources:
 requests:
 storage: 3Gi
```

Create the PersistentVolumeClaim:

```
kubectcl apply -f https://k8s.io/examples/pods/storage/pv-claim.yaml
```

After you create the PersistentVolumeClaim, the Kubernetes control plane looks for a PersistentVolume that satisfies the claim's requirements. If the control plane finds a suitable PersistentVolume with the same StorageClass, it binds the claim to the volume.

Look again at the PersistentVolume:

```
kubectcl get pv task-pv-volume
```

Now the output shows a STATUS of Bound.

| NAME           | CAPACITY | ACCESSMODES | RECLAIMPOLICY | STATUS | CLAIM                 |
|----------------|----------|-------------|---------------|--------|-----------------------|
| task-pv-volume | 10Gi     | RWO         | Retain        | Bound  | default/task-pv-claim |
| manual         | 2m       |             |               |        |                       |

Look at the PersistentVolumeClaim:

```
kubectcl get pvc task-pv-claim
```

The output shows that the PersistentVolumeClaim is bound to your PersistentVolume, task-pv-volume.

| NAME          | STATUS | VOLUME         | CAPACITY | ACCESSMODES | STORAGECLASS |
|---------------|--------|----------------|----------|-------------|--------------|
| task-pv-claim | Bound  | task-pv-volume | 10Gi     | RWO         | manual       |
|               |        |                |          |             | 30s          |

## Create a Pod

The next step is to create a Pod that uses your PersistentVolumeClaim as a volume.

Here is the configuration file for the Pod:

[pods/storage/pv-pod.yaml](https://k8s.io/examples/pods/storage/pv-pod.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: task-pv-pod
spec:
 volumes:
 - name: task-pv-storage
 persistentVolumeClaim:
 claimName: task-pv-claim
 containers:
 - name: task-pv-container
```

```
image: nginx
ports:
 - containerPort: 80
 name: "http-server"
volumeMounts:
 - mountPath: "/usr/share/nginx/html"
 name: task-pv-storage
```

Notice that the Pod's configuration file specifies a `PersistentVolumeClaim`, but it does not specify a `PersistentVolume`. From the Pod's point of view, the claim is a volume.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/storage/pv-pod.yaml
```

Verify that the container in the Pod is running:

```
kubectl get pod task-pv-pod
```

Get a shell to the container running in your Pod:

```
kubectl exec -it task-pv-pod -- /bin/bash
```

In your shell, verify that nginx is serving the `index.html` file from the `hostPath` volume:

```
Be sure to run these 3 commands inside the root shell that comes from
running "kubectl exec" in the previous step
apt update
apt install curl
curl http://localhost/
```

The output shows the text that you wrote to the `index.html` file on the `hostPath` volume:

```
Hello from Kubernetes storage
```

If you see that message, you have successfully configured a Pod to use storage from a `PersistentVolumeClaim`.

## Clean up

Delete the Pod, the `PersistentVolumeClaim` and the `PersistentVolume`:

```
kubectl delete pod task-pv-pod
kubectl delete pvc task-pv-claim
kubectl delete pv task-pv-volume
```

If you don't already have a shell open to the Node in your cluster, open a new shell the same way that you did earlier.

In the shell on your Node, remove the file and directory that you created:

```
This assumes that your Node uses "sudo" to run commands
as the superuser
sudo rm /mnt/data/index.html
sudo rmdir /mnt/data
```

You can now close the shell to your Node.

## Mounting the same persistentVolume in two places

[pods/storage/pv-duplicate.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: test
spec:
 containers:
 - name: test
 image: nginx
 volumeMounts:
 # a mount for site-data
 - name: config
 mountPath: /usr/share/nginx/html
 subPath: html
 # another mount for nginx config
 - name: config
 mountPath: /etc/nginx/nginx.conf
 subPath: nginx.conf
 volumes:
 - name: config
 persistentVolumeClaim:
 claimName: test-nfs-claim
```

You can perform 2 volume mounts on your nginx container:

- /usr/share/nginx/html for the static website
- /etc/nginx/nginx.conf for the default config

## Access control

Storage configured with a group ID (GID) allows writing only by Pods using the same GID. Mismatched or missing GIDs cause permission denied errors. To reduce the need for coordination with users, an administrator can annotate a PersistentVolume with a GID. Then the GID is automatically added to any Pod that uses the PersistentVolume.

Use the `pv.beta.kubernetes.io/gid` annotation as follows:

```
apiVersion: v1
kind: PersistentVolume
metadata:
 name: pv1
```

```
annotations:
 pv.beta.kubernetes.io/gid: "1234"
```

When a Pod consumes a PersistentVolume that has a GID annotation, the annotated GID is applied to all containers in the Pod in the same way that GIDs specified in the Pod's security context are. Every GID, whether it originates from a PersistentVolume annotation or the Pod's specification, is applied to the first process run in each container.

**Note:**

When a Pod consumes a PersistentVolume, the GIDs associated with the PersistentVolume are not present on the Pod resource itself.

## What's next

- Learn more about [PersistentVolumes](#).
- Read the [Persistent Storage design document](#).

## Reference

- [PersistentVolume](#)
- [PersistentVolumeSpec](#)
- [PersistentVolumeClaim](#)
- [PersistentVolumeClaimSpec](#)

# Configure a Pod to Use a Projected Volume for Storage

This page shows how to use a [projected](#) Volume to mount several existing volume sources into the same directory. Currently, secret, configMap, downwardAPI, and serviceAccountToken volumes can be projected.

**Note:**

serviceAccountToken is not a volume type.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

# Configure a projected volume for a pod

In this exercise, you create username and password [Secrets](#) from local files. You then create a Pod that runs one container, using a [projected](#) Volume to mount the Secrets into the same shared directory.

Here is the configuration file for the Pod:

[pods/storage/projected.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: test-projected-volume
spec:
 containers:
 - name: test-projected-volume
 image: busybox:1.28
 args:
 - sleep
 - "86400"
 volumeMounts:
 - name: all-in-one
 mountPath: "/projected-volume"
 readOnly: true
 volumes:
 - name: all-in-one
 projected:
 sources:
 - secret:
 name: user
 - secret:
 name: pass
```

## 1. Create the Secrets:

```
Create files containing the username and password:
echo -n "admin" > ./username.txt
echo -n "1f2d1e2e67df" > ./password.txt

Package these files into secrets:
kubectl create secret generic user --from-file=./username.txt
kubectl create secret generic pass --from-file=./password.txt
```

## 2. Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/storage/projected.yaml
```

## 3. Verify that the Pod's container is running, and then watch for changes to the Pod:

```
kubectl get --watch pod test-projected-volume
```

The output looks like this:



| NAME                  | READY | STATUS  | RESTARTS | AGE |
|-----------------------|-------|---------|----------|-----|
| test-projected-volume | 1/1   | Running | 0        | 14s |

4. In another terminal, get a shell to the running container:

```
kubectl exec -it test-projected-volume -- /bin/sh
```

5. In your shell, verify that the projected-volume directory contains your projected sources:

```
ls /projected-volume/
```

## Clean up

Delete the Pod and the Secrets:

```
kubectl delete pod test-projected-volume
kubectl delete secret user pass
```

## What's next

- Learn more about [projected](#) volumes.
- Read the [all-in-one volume](#) design document.

# Configure a Security Context for a Pod or Container

A security context defines privilege and access control settings for a Pod or Container. Security context settings include, but are not limited to:

- Discretionary Access Control: Permission to access an object, like a file, is based on [user ID \(UID\) and group ID \(GID\)](#).
- [Security Enhanced Linux \(SELinux\)](#): Objects are assigned security labels.
- Running as privileged or unprivileged.
- [Linux Capabilities](#): Give a process some privileges, but not all the privileges of the root user.
- [AppArmor](#): Use program profiles to restrict the capabilities of individual programs.
- [Seccomp](#): Filter a process's system calls.
- `allowPrivilegeEscalation`: Controls whether a process can gain more privileges than its parent process. This bool directly controls whether the `no_new_privs` flag gets set on the container process. `allowPrivilegeEscalation` is always true when the container:
  - is run as privileged, or
  - has `CAP_SYS_ADMIN`
- `readOnlyRootFilesystem`: Mounts the container's root filesystem as read-only.

The above bullets are not a complete set of security context settings -- please see [SecurityContext](#) for a comprehensive list.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

## Set the security context for a Pod

To specify security settings for a Pod, include the securityContext field in the Pod specification. The securityContext field is a [PodSecurityContext](#) object. The security settings that you specify for a Pod apply to all Containers in the Pod. Here is a configuration file for a Pod that has a securityContext and an emptyDir volume:

[pods/security/security-context.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: security-context-demo
spec:
 securityContext:
 runAsUser: 1000
 runAsGroup: 3000
 fsGroup: 2000
 volumes:
 - name: sec-ctx-vol
 emptyDir: {}
 containers:
 - name: sec-ctx-demo
 image: busybox:1.28
 command: ["sh", "-c", "sleep 1h"]
 volumeMounts:
 - name: sec-ctx-vol
 mountPath: /data/demo
 securityContext:
 allowPrivilegeEscalation: false
```

In the configuration file, the runAsUser field specifies that for any Containers in the Pod, all processes run with user ID 1000. The runAsGroup field specifies the primary group ID of 3000 for all processes within any containers of the Pod. If this field is omitted, the primary group ID of the containers will be root(0). Any files created will also be owned by user 1000 and group 3000 when runAsGroup is specified. Since fsGroup field is specified, all processes of the

container are also part of the supplementary group ID 2000. The owner for volume /data/demo and any files created in that volume will be Group ID 2000.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/security/security-context.yaml
```

Verify that the Pod's Container is running:

```
kubectl get pod security-context-demo
```

Get a shell to the running Container:

```
kubectl exec -it security-context-demo -- sh
```

In your shell, list the running processes:

```
ps
```

The output shows that the processes are running as user 1000, which is the value of runAsUser:

```
PID USER TIME COMMAND
 1 1000 0:00 sleep 1h
 6 1000 0:00 sh
...
```

In your shell, navigate to /data, and list the one directory:

```
cd /data
ls -l
```

The output shows that the /data/demo directory has group ID 2000, which is the value of fsGroup.

```
drwxrwsrwx 2 root 2000 4096 Jun 6 20:08 demo
```

In your shell, navigate to /data/demo, and create a file:

```
cd demo
echo hello > testfile
```

List the file in the /data/demo directory:

```
ls -l
```

The output shows that testfile has group ID 2000, which is the value of fsGroup.

```
-rw-r--r-- 1 1000 2000 6 Jun 6 20:08 testfile
```

Run the following command:

```
id
```

The output is similar to this:

```
uid=1000 gid=3000 groups=2000
```

From the output, you can see that `gid` is 3000 which is same as the `runAsGroup` field. If the `runAsGroup` was omitted, the `gid` would remain as 0 (root) and the process will be able to interact with files that are owned by the root(0) group and groups that have the required group permissions for the root (0) group.

Exit your shell:

```
exit
```

## Configure volume permission and ownership change policy for Pods

FEATURE STATE: Kubernetes v1.23 [stable]

By default, Kubernetes recursively changes ownership and permissions for the contents of each volume to match the `fsGroup` specified in a Pod's `securityContext` when that volume is mounted. For large volumes, checking and changing ownership and permissions can take a lot of time, slowing Pod startup. You can use the `fsGroupChangePolicy` field inside a `securityContext` to control the way that Kubernetes checks and manages ownership and permissions for a volume.

**fsGroupChangePolicy** - `fsGroupChangePolicy` defines behavior for changing ownership and permission of the volume before being exposed inside a Pod. This field only applies to volume types that support `fsGroup` controlled ownership and permissions. This field has two possible values:

- *OnRootMismatch*: Only change permissions and ownership if the permission and the ownership of root directory does not match with expected permissions of the volume. This could help shorten the time it takes to change ownership and permission of a volume.
- *Always*: Always change permission and ownership of the volume when volume is mounted.

For example:

```
securityContext:
 runAsUser: 1000
 runAsGroup: 3000
 fsGroup: 2000
 fsGroupChangePolicy: "OnRootMismatch"
```

### Note:

This field has no effect on ephemeral volume types such as [secret](#), [configMap](#), and [emptydir](#).

## Delegating volume permission and ownership change to CSI driver

FEATURE STATE: Kubernetes v1.26 [stable]

If you deploy a [Container Storage Interface \(CSI\)](#) driver which supports the `VOLUME_MOUNT_GROUP` `NodeServiceCapability`, the process of setting file ownership and permissions based on the `fsGroup` specified in the `securityContext` will be performed by the CSI driver instead of Kubernetes. In this case, since Kubernetes doesn't perform any ownership and permission change, `fsGroupChangePolicy` does not take effect, and as specified by CSI, the driver is expected to mount the volume with the provided `fsGroup`, resulting in a volume that is readable/writable by the `fsGroup`.

## Set the security context for a Container

To specify security settings for a Container, include the `securityContext` field in the Container manifest. The `securityContext` field is a [SecurityContext](#) object. Security settings that you specify for a Container apply only to the individual Container, and they override settings made at the Pod level when there is overlap. Container settings do not affect the Pod's Volumes.

Here is the configuration file for a Pod that has one Container. Both the Pod and the Container have a `securityContext` field:

[pods/security/security-context-2.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: security-context-demo-2
spec:
 securityContext:
 runAsUser: 1000
 containers:
 - name: sec-ctx-demo-2
 image: gcr.io/google-samples/node-hello:1.0
 securityContext:
 runAsUser: 2000
 allowPrivilegeEscalation: false
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/security/security-context-2.yaml
```

Verify that the Pod's Container is running:

```
kubectl get pod security-context-demo-2
```

Get a shell into the running Container:

```
kubectl exec -it security-context-demo-2 -- sh
```

In your shell, list the running processes:

```
ps aux
```

The output shows that the processes are running as user 2000. This is the value of `runAsUser` specified for the Container. It overrides the value 1000 that is specified for the Pod.

| USER | PID | %CPU | %MEM | VSZ    | RSS   | TTY | STAT | START | TIME | COMMAND                   |
|------|-----|------|------|--------|-------|-----|------|-------|------|---------------------------|
| 2000 | 1   | 0.0  | 0.0  | 4336   | 764   | ?   | Ss   | 20:36 | 0:00 | /bin/sh -c node server.js |
| 2000 | 8   | 0.1  | 0.5  | 772124 | 22604 | ?   | Sl   | 20:36 | 0:00 | node server.js            |
| ...  |     |      |      |        |       |     |      |       |      |                           |

Exit your shell:

```
exit
```

## Set capabilities for a Container

With [Linux capabilities](#), you can grant certain privileges to a process without granting all the privileges of the root user. To add or remove Linux capabilities for a Container, include the capabilities field in the securityContext section of the Container manifest.

First, see what happens when you don't include a capabilities field. Here is configuration file that does not add or remove any Container capabilities:

[pods/security/security-context-3.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: security-context-demo-3
spec:
 containers:
 - name: sec-ctx-3
 image: gcr.io/google-samples/node-hello:1.0
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/security/security-context-3.yaml
```

Verify that the Pod's Container is running:

```
kubectl get pod security-context-demo-3
```

Get a shell into the running Container:

```
kubectl exec -it security-context-demo-3 -- sh
```

In your shell, list the running processes:

```
ps aux
```

The output shows the process IDs (PIDs) for the Container:

| USER | PID | %CPU | %MEM | VSZ    | RSS   | TTY | STAT | START | TIME | COMMAND                   |
|------|-----|------|------|--------|-------|-----|------|-------|------|---------------------------|
| root | 1   | 0.0  | 0.0  | 4336   | 796   | ?   | Ss   | 18:17 | 0:00 | /bin/sh -c node server.js |
| root | 5   | 0.1  | 0.5  | 772124 | 22700 | ?   | Sl   | 18:17 | 0:00 | node server.js            |

In your shell, view the status for process 1:

```
cd /proc/1
cat status
```

The output shows the capabilities bitmap for the process:

```
...
CapPrm: 00000000a80425fb
CapEff: 00000000a80425fb
...
```

Make a note of the capabilities bitmap, and then exit your shell:

```
exit
```

Next, run a Container that is the same as the preceding container, except that it has additional capabilities set.

Here is the configuration file for a Pod that runs one Container. The configuration adds the CAP\_NET\_ADMIN and CAP\_SYS\_TIME capabilities:

[pods/security/security-context-4.yaml](https://k8s.io/examples/pods/security/security-context-4.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: security-context-demo-4
spec:
 containers:
 - name: sec-ctx-4
 image: gcr.io/google-samples/node-hello:1.0
 securityContext:
 capabilities:
 add: ["NET_ADMIN", "SYS_TIME"]
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/security/security-context-4.yaml
```

Get a shell into the running Container:

```
kubectl exec -it security-context-demo-4 -- sh
```

In your shell, view the capabilities for process 1:

```
cd /proc/1
cat status
```

The output shows capabilities bitmap for the process:

```
...
CapPrm: 00000000aa0435fb
CapEff: 00000000aa0435fb
...
```

Compare the capabilities of the two Containers:

```
00000000a80425fb
00000000aa0435fb
```

In the capability bitmap of the first container, bits 12 and 25 are clear. In the second container, bits 12 and 25 are set. Bit 12 is CAP\_NET\_ADMIN, and bit 25 is CAP\_SYS\_TIME. See [capability.h](#) for definitions of the capability constants.

#### Note:

Linux capability constants have the form CAP\_XXX. But when you list capabilities in your container manifest, you must omit the CAP\_ portion of the constant. For example, to add CAP\_SYS\_TIME, include SYS\_TIME in your list of capabilities.

## Set the Seccomp Profile for a Container

To set the Seccomp profile for a Container, include the `seccompProfile` field in the `securityContext` section of your Pod or Container manifest. The `seccompProfile` field is a [SeccompProfile](#) object consisting of `type` and `localhostProfile`. Valid options for `type` include `RuntimeDefault`, `Unconfined`, and `Localhost`. `localhostProfile` must only be set if `type: Localhost`. It indicates the path of the pre-configured profile on the node, relative to the kubelet's configured Seccomp profile location (configured with the `--root-dir` flag).

Here is an example that sets the Seccomp profile to the node's container runtime default profile:

```
...
securityContext:
 seccompProfile:
 type: RuntimeDefault
```

Here is an example that sets the Seccomp profile to a pre-configured file at `<kubelet-root-dir>/seccomp/my-profiles/profile-allow.json`:

```
...
securityContext:
 seccompProfile:
 type: Localhost
 localhostProfile: my-profiles/profile-allow.json
```

## Assign SELinux labels to a Container

To assign SELinux labels to a Container, include the `seLinuxOptions` field in the `securityContext` section of your Pod or Container manifest. The `seLinuxOptions` field is an [SELinuxOptions](#) object. Here's an example that applies an SELinux level:

```
...
securityContext:
 seLinuxOptions:
 level: "s0:c123,c456"
```



## Note:

To assign SELinux labels, the SELinux security module must be loaded on the host operating system.

## Efficient SELinux volume relabeling

FEATURE STATE: Kubernetes v1.28 [beta]

## Note:

Kubernetes v1.27 introduced an early limited form of this behavior that was only applicable to volumes (and PersistentVolumeClaims) using the ReadWriteOncePod access mode.

As an alpha feature, you can enable the SELinuxMount [feature gate](#) to widen that performance improvement to other kinds of PersistentVolumeClaims, as explained in detail below.

By default, the container runtime recursively assigns SELinux label to all files on all Pod volumes. To speed up this process, Kubernetes can change the SELinux label of a volume instantly by using a mount option `-o context=<label>`.

To benefit from this speedup, all these conditions must be met:

- The [feature gates](#) `ReadWriteOncePod` and `SELinuxMountReadWriteOncePod` must be enabled.
- Pod must use PersistentVolumeClaim with applicable accessModes and [feature gates](#):
  - Either the volume has accessModes: `["ReadWriteOncePod"]`, and feature gate `SELinuxMountReadWriteOncePod` is enabled.
  - Or the volume can use any other access modes and both feature gates `SELinuxMountReadWriteOncePod` and `SELinuxMount` must be enabled.
- Pod (or all its Containers that use the PersistentVolumeClaim) must have `seLinuxOptions` set.
- The corresponding PersistentVolume must be either:
  - A volume that uses the legacy in-tree `iscsi`, `rbd` or `fc` volume type.
  - Or a volume that uses a [CSI](#) driver. The CSI driver must announce that it supports mounting with `-o context` by setting `spec.seLinuxMount: true` in its `CSIDriver` instance.

For any other volume types, SELinux relabelling happens another way: the container runtime recursively changes the SELinux label for all inodes (files and directories) in the volume. The more files and directories in the volume, the longer that relabelling takes.

## Managing access to the /proc filesystem

FEATURE STATE: Kubernetes v1.12 [alpha]

For runtimes that follow the OCI runtime specification, containers default to running in a mode where there are multiple paths that are both masked and read-only. The result of this is the container has these paths present inside the container's mount namespace, and they can

function similarly to if the container was an isolated host, but the container process cannot write to them. The list of masked and read-only paths are as follows:

- Masked Paths:

- /proc/asound
- /proc/acpi
- /proc/kcore
- /proc/keys
- /proc/latency\_stats
- /proc/timer\_list
- /proc/timer\_stats
- /proc/sched\_debug
- /proc/scsi
- /sys/firmware

- Read-Only Paths:

- /proc/bus
- /proc/fs
- /proc/irq
- /proc/sys
- /proc/sysrq-trigger

For some Pods, you might want to bypass that default masking of paths. The most common context for wanting this is if you are trying to run containers within a Kubernetes container (within a pod).

The securityContext field procMount allows a user to request a container's /proc be Unmasked, or be mounted as read-write by the container process. This also applies to /sys/firmware which is not in /proc.

```
...
securityContext:
 procMount: Unmasked
```

**Note:**

Setting procMount to Unmasked requires the spec.hostUsers value in the pod spec to be false. In other words: a container that wishes to have an Unmasked /proc or unmasked /sys must also be in a [user namespace](#). Kubernetes v1.12 to v1.29 did not enforce that requirement.

## Discussion

The security context for a Pod applies to the Pod's Containers and also to the Pod's Volumes when applicable. Specifically fsGroup and seLinuxOptions are applied to Volumes as follows:

- fsGroup: Volumes that support ownership management are modified to be owned and writable by the GID specified in fsGroup. See the [Ownership Management design document](#) for more details.
- seLinuxOptions: Volumes that support SELinux labeling are relabeled to be accessible by the label specified under seLinuxOptions. Usually you only need to set the level section.

This sets the [Multi-Category Security \(MCS\)](#) label given to all Containers in the Pod as well as the Volumes.

### Warning:

After you specify an MCS label for a Pod, all Pods with the same label can access the Volume. If you need inter-Pod protection, you must assign a unique MCS label to each Pod.

## Clean up

Delete the Pod:

```
kubectl delete pod security-context-demo
kubectl delete pod security-context-demo-2
kubectl delete pod security-context-demo-3
kubectl delete pod security-context-demo-4
```

## What's next

- [PodSecurityContext](#)
- [SecurityContext](#)
- [CRI Plugin Config Guide](#)
- [Security Contexts design document](#)
- [Ownership Management design document](#)
- [PodSecurity Admission](#)
- [AllowPrivilegeEscalation design document](#)
- For more information about security mechanisms in Linux, see [Overview of Linux Kernel Security Features](#) (Note: Some information is out of date)
- Read about [User Namespaces](#) for Linux pods.
- [Masked Paths in the OCI Runtime Specification](#)

# Configure Service Accounts for Pods

Kubernetes offers two distinct ways for clients that run within your cluster, or that otherwise have a relationship to your cluster's [control plane](#) to authenticate to the [API server](#).

A *service account* provides an identity for processes that run in a Pod, and maps to a ServiceAccount object. When you authenticate to the API server, you identify yourself as a particular *user*. Kubernetes recognises the concept of a user, however, Kubernetes itself does **not** have a User API.

This task guide is about ServiceAccounts, which do exist in the Kubernetes API. The guide shows you some ways to configure ServiceAccounts for Pods.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

## Use the default service account to access the API server

When Pods contact the API server, Pods authenticate as a particular ServiceAccount (for example, default). There is always at least one ServiceAccount in each [namespace](#).

Every Kubernetes namespace contains at least one ServiceAccount: the default ServiceAccount for that namespace, named default. If you do not specify a ServiceAccount when you create a Pod, Kubernetes automatically assigns the ServiceAccount named default in that namespace.

You can fetch the details for a Pod you have created. For example:

```
kubectl get pods/<podname> -o yaml
```

In the output, you see a field `spec.serviceAccountName`. Kubernetes automatically sets that value if you don't specify it when you create a Pod.

An application running inside a Pod can access the Kubernetes API using automatically mounted service account credentials. See [accessing the Cluster](#) to learn more.

When a Pod authenticates as a ServiceAccount, its level of access depends on the [authorization plugin and policy](#) in use.

## Opt out of API credential automounting

If you don't want the [kubelet](#) to automatically mount a ServiceAccount's API credentials, you can opt out of the default behavior. You can opt out of automounting API credentials on `/var/run/secrets/kubernetes.io/serviceaccount/token` for a service account by setting `automountServiceAccountToken: false` on the ServiceAccount:

For example:

```
apiVersion: v1
kind: ServiceAccount
metadata:
 name: build-robot
automountServiceAccountToken: false
...
```

You can also opt out of automounting API credentials for a particular Pod:

```
apiVersion: v1
kind: Pod
metadata:
 name: my-pod
spec:
 serviceAccountName: build-robot
```

```
automountServiceAccountToken: false
...
```

If both the ServiceAccount and the Pod's .spec specify a value for automountServiceAccountToken, the Pod spec takes precedence.

## Use more than one ServiceAccount

Every namespace has at least one ServiceAccount: the default ServiceAccount resource, called default. You can list all ServiceAccount resources in your [current namespace](#) with:

```
kubectl get serviceaccounts
```

The output is similar to this:

| NAME    | SECRETS | AGE |
|---------|---------|-----|
| default | 1       | 1d  |

You can create additional ServiceAccount objects like this:

```
kubectl apply -f - <<EOF
apiVersion: v1
kind: ServiceAccount
metadata:
 name: build-robot
EOF
```

The name of a ServiceAccount object must be a valid [DNS subdomain name](#).

If you get a complete dump of the service account object, like this:

```
kubectl get serviceaccounts/build-robot -o yaml
```

The output is similar to this:

```
apiVersion: v1
kind: ServiceAccount
metadata:
 creationTimestamp: 2019-06-16T00:12:34Z
 name: build-robot
 namespace: default
 resourceVersion: "272500"
 uid: 721ab723-13bc-11e5-aec2-42010af0021e
```

You can use authorization plugins to [set permissions on service accounts](#).

To use a non-default service account, set the spec.serviceAccountName field of a Pod to the name of the ServiceAccount you wish to use.

You can only set the serviceAccountName field when creating a Pod, or in a template for a new Pod. You cannot update the .spec.serviceAccountName field of a Pod that already exists.

**Note:**

The `.spec.serviceAccount` field is a deprecated alias for `.spec.serviceAccountName`. If you want to remove the fields from a workload resource, set both fields to empty explicitly on the [pod template](#).

## Cleanup

If you tried creating build-robot ServiceAccount from the example above, you can clean it up by running:

```
kubectl delete serviceaccount/build-robot
```

## Manually create an API token for a ServiceAccount

Suppose you have an existing service account named "build-robot" as mentioned earlier.

You can get a time-limited API token for that ServiceAccount using kubectl:

```
kubectl create token build-robot
```

The output from that command is a token that you can use to authenticate as that ServiceAccount. You can request a specific token duration using the `--duration` command line argument to `kubectl create token` (the actual duration of the issued token might be shorter, or could even be longer).

When the `ServiceAccountTokenNodeBinding` and `ServiceAccountTokenNodeBindingValidation` features are enabled and the `KUBECTL_NODE_BOUND_TOKENS` environment variable is set to true, it is possible to create a service account token that is directly bound to a Node:

```
KUBECTL_NODE_BOUND_TOKENS=true kubectl create token build-robot --bound-object-kind Node --bound-object-name node-001 --bound-object-uid 123...456
```

The token will be valid until it expires or either the associated Node or service account are deleted.

**Note:**

Versions of Kubernetes before v1.22 automatically created long term credentials for accessing the Kubernetes API. This older mechanism was based on creating token Secrets that could then be mounted into running Pods. In more recent versions, including Kubernetes v1.30, API credentials are obtained directly by using the [TokenRequest](#) API, and are mounted into Pods using a [projected volume](#). The tokens obtained using this method have bounded lifetimes, and are automatically invalidated when the Pod they are mounted into is deleted.

You can still manually create a service account token Secret; for example, if you need a token that never expires. However, using the [TokenRequest](#) subresource to obtain a token to access the API is recommended instead.

## Manually create a long-lived API token for a ServiceAccount

If you want to obtain an API token for a ServiceAccount, you create a new Secret with a special annotation, `kubernetes.io/service-account.name`.

```
kubectl apply -f - <<EOF
apiVersion: v1
kind: Secret
metadata:
 name: build-robot-secret
 annotations:
 kubernetes.io/service-account.name: build-robot
type: kubernetes.io/service-account-token
EOF
```

If you view the Secret using:

```
kubectl get secret/build-robot-secret -o yaml
```

you can see that the Secret now contains an API token for the "build-robot" ServiceAccount.

Because of the annotation you set, the control plane automatically generates a token for that ServiceAccounts, and stores them into the associated Secret. The control plane also cleans up tokens for deleted ServiceAccounts.

```
kubectl describe secrets/build-robot-secret
```

The output is similar to this:

```
Name: build-robot-secret
Namespace: default
Labels: <none>
Annotations: kubernetes.io/service-account.name: build-robot
 kubernetes.io/service-account.uid: da68f9c6-9d26-11e7-b84e-002dc52800da

Type: kubernetes.io/service-account-token

Data
====
ca.crt: 1338 bytes
namespace: 7 bytes
token: ...
```

### Note:

The content of token is elided here.

Take care not to display the contents of a `kubernetes.io/service-account-token` Secret somewhere that your terminal / computer screen could be seen by an onlooker.

When you delete a ServiceAccount that has an associated Secret, the Kubernetes control plane automatically cleans up the long-lived token from that Secret.

## Note:

If you view the ServiceAccount using:

```
kubectl get serviceaccount build-robot -o yaml
```

You can't see the build-robot-secret Secret in the ServiceAccount API objects [.secrets](#) field because that field is only populated with auto-generated Secrets.

## Add ImagePullSecrets to a service account

First, [create an imagePullSecret](#). Next, verify it has been created. For example:

- Create an imagePullSecret, as described in [Specifying ImagePullSecrets on a Pod](#).

```
kubectl create secret docker-registry myregistrykey --docker-server=<registry name> \
 --docker-username=DUMMY_USERNAME --docker-password=DUMMY_DOCKER_
PASSWORD \
 --docker-email=DUMMY_DOCKER_EMAIL
```

- Verify it has been created.

```
kubectl get secrets myregistrykey
```

The output is similar to this:

| NAME          | TYPE                           | DATA | AGE |
|---------------|--------------------------------|------|-----|
| myregistrykey | kubernetes.io/dockerconfigjson | 1    | 1d  |

## Add image pull secret to service account

Next, modify the default service account for the namespace to use this Secret as an imagePullSecret.

```
kubectl patch serviceaccount default -p '{"imagePullSecrets": [{"name": "myregistrykey"}]}'
```

You can achieve the same outcome by editing the object manually:

```
kubectl edit serviceaccount/default
```

The output of the sa.yaml file is similar to this:

Your selected text editor will open with a configuration looking something like this:

```
apiVersion: v1
kind: ServiceAccount
metadata:
 creationTimestamp: 2021-07-07T22:02:39Z
 name: default
 namespace: default
 resourceVersion: "243024"
 uid: 052fb0f4-3d50-11e5-b066-42010af0d7b6
```



Using your editor, delete the line with key `resourceVersion`, add lines for `imagePullSecrets`: and save it. Leave the `uid` value set the same as you found it.

After you made those changes, the edited `ServiceAccount` looks something like this:

```
apiVersion: v1
kind: ServiceAccount
metadata:
 creationTimestamp: 2021-07-07T22:02:39Z
 name: default
 namespace: default
 uid: 052fb0f4-3d50-11e5-b066-42010af0d7b6
imagePullSecrets:
 - name: myregistrykey
```

## Verify that `imagePullSecrets` are set for new Pods

Now, when a new Pod is created in the current namespace and using the default `ServiceAccount`, the new Pod has its `spec.imagePullSecrets` field set automatically:

```
kubectl run nginx --image=<registry name>/nginx --restart=Never
kubectl get pod nginx -o=jsonpath='{.spec.imagePullSecrets[0].name}'
```

The output is:

```
myregistrykey
```

## ServiceAccount token volume projection

FEATURE STATE: Kubernetes v1.20 [stable]

### Note:

To enable and use token request projection, you must specify each of the following command line arguments to `kube-apiserver`:

#### `--service-account-issuer`

defines the Identifier of the service account token issuer. You can specify the `--service-account-issuer` argument multiple times, this can be useful to enable a non-disruptive change of the issuer. When this flag is specified multiple times, the first is used to generate tokens and all are used to determine which issuers are accepted. You must be running Kubernetes v1.22 or later to be able to specify `--service-account-issuer` multiple times.

#### `--service-account-key-file`

specifies the path to a file containing PEM-encoded X.509 private or public keys (RSA or ECDSA), used to verify `ServiceAccount` tokens. The specified file can contain multiple keys, and the flag can be specified multiple times with different files. If specified multiple times, tokens signed by any of the specified keys are considered valid by the Kubernetes API server.

#### `--service-account-signing-key-file`

specifies the path to a file that contains the current private key of the service account token issuer. The issuer signs issued ID tokens with this private key.

--api-audiences (can be omitted)

defines audiences for ServiceAccount tokens. The service account token authenticator validates that tokens used against the API are bound to at least one of these audiences. If api-audiences is specified multiple times, tokens for any of the specified audiences are considered valid by the Kubernetes API server. If you specify the --service-account-issuer command line argument but you don't set --api-audiences, the control plane defaults to a single element audience list that contains only the issuer URL.

The kubelet can also project a ServiceAccount token into a Pod. You can specify desired properties of the token, such as the audience and the validity duration. These properties are *not* configurable on the default ServiceAccount token. The token will also become invalid against the API when either the Pod or the ServiceAccount is deleted.

You can configure this behavior for the spec of a Pod using a [projected volume](#) type called ServiceAccountToken.

The token from this projected volume is a [JSON Web Token](#) (JWT). The JSON payload of this token follows a well defined schema - an example payload for a pod bound token:

```
{
 "aud": [# matches the requested audiences, or the API server's default audiences when none
 "https://kubernetes.default.svc"
],
 "exp": 1731613413,
 "iat": 1700077413,
 "iss": "https://kubernetes.default.svc", # matches the first value passed to the --service-
 account-issuer flag
 "jti": "ea28ed49-2e11-4280-9ec5-bc3d1d84661a", # ServiceAccountTokenJTI feature must be
 enabled for the claim to be present
 "kubernetes.io": {
 "namespace": "kube-system",
 "node": { # ServiceAccountTokenPodNodeInfo feature must be enabled for the API server to
 add this node reference claim
 "name": "127.0.0.1",
 "uid": "58456cb0-dd00-45ed-b797-5578fdceaced"
 },
 "pod": {
 "name": "coredns-69cbfb9798-jv9gn",
 "uid": "778a530c-b3f4-47c0-9cd5-ab018fb64f33"
 },
 "serviceaccount": {
 "name": "coredns",
 "uid": "a087d5a0-e1dd-43ec-93ac-f13d89cd13af"
 },
 "warnafter": 1700081020
 },
 "nbf": 1700077413,
 "sub": "system:serviceaccount:kube-system:coredns"
}
```

## Launch a Pod using service account token projection

To provide a Pod with a token with an audience of `vault` and a validity duration of two hours, you could define a Pod manifest that is similar to:

[pods/pod-projected-svc-token.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: nginx
spec:
 containers:
 - image: nginx
 name: nginx
 volumeMounts:
 - mountPath: /var/run/secrets/tokens
 name: vault-token
 serviceAccountName: build-robot
 volumes:
 - name: vault-token
 projected:
 sources:
 - serviceAccountToken:
 path: vault-token
 expirationSeconds: 7200
 audience: vault
```

Create the Pod:

```
kubectl create -f https://k8s.io/examples/pods/pod-projected-svc-token.yaml
```

The kubelet will: request and store the token on behalf of the Pod; make the token available to the Pod at a configurable file path; and refresh the token as it approaches expiration. The kubelet proactively requests rotation for the token if it is older than 80% of its total time-to-live (TTL), or if the token is older than 24 hours.

The application is responsible for reloading the token when it rotates. It's often good enough for the application to load the token on a schedule (for example: once every 5 minutes), without tracking the actual expiry time.

## Service account issuer discovery

FEATURE STATE: Kubernetes v1.21 [stable]

If you have enabled [token projection](#) for ServiceAccounts in your cluster, then you can also make use of the discovery feature. Kubernetes provides a way for clients to federate as an *identity provider*, so that one or more external systems can act as a *relying party*.

## Note:

The issuer URL must comply with the [OIDC Discovery Spec](#). In practice, this means it must use the https scheme, and should serve an OpenID provider configuration at `{service-account-issuer}/.well-known/openid-configuration`.

If the URL does not comply, ServiceAccount issuer discovery endpoints are not registered or accessible.

When enabled, the Kubernetes API server publishes an OpenID Provider Configuration document via HTTP. The configuration document is published at `/.well-known/openid-configuration`. The OpenID Provider Configuration is sometimes referred to as the *discovery document*. The Kubernetes API server publishes the related JSON Web Key Set (JWKS), also via HTTP, at `/openid/v1/jwks`.

## Note:

The responses served at `/.well-known/openid-configuration` and `/openid/v1/jwks` are designed to be OIDC compatible, but not strictly OIDC compliant. Those documents contain only the parameters necessary to perform validation of Kubernetes service account tokens.

Clusters that use [RBAC](#) include a default ClusterRole called `system:service-account-issuer-discovery`. A default ClusterRoleBinding assigns this role to the `system:serviceaccounts` group, which all ServiceAccounts implicitly belong to. This allows pods running on the cluster to access the service account discovery document via their mounted service account token. Administrators may, additionally, choose to bind the role to `system:authenticated` or `system:unauthenticated` depending on their security requirements and which external systems they intend to federate with.

The JWKS response contains public keys that a relying party can use to validate the Kubernetes service account tokens. Relying parties first query for the OpenID Provider Configuration, and use the `jwks_uri` field in the response to find the JWKS.

In many cases, Kubernetes API servers are not available on the public internet, but public endpoints that serve cached responses from the API server can be made available by users or by service providers. In these cases, it is possible to override the `jwks_uri` in the OpenID Provider Configuration so that it points to the public endpoint, rather than the API server's address, by passing the `--service-account-jwks-uri` flag to the API server. Like the issuer URL, the JWKS URI is required to use the https scheme.

## What's next

See also:

- Read the [Cluster Admin Guide to Service Accounts](#)
- Read about [Authorization in Kubernetes](#)
- Read about [Secrets](#)
  - or learn to [distribute credentials securely using Secrets](#)
  - but also bear in mind that using Secrets for authenticating as a ServiceAccount is deprecated. The recommended alternative is [ServiceAccount token volume projection](#).
- Read about [projected volumes](#).

- For background on OIDC discovery, read the [ServiceAccount signing key retrieval](#) Kubernetes Enhancement Proposal
- Read the [OIDC Discovery Spec](#)

# Pull an Image from a Private Registry

This page shows how to create a Pod that uses a [Secret](#) to pull an image from a private container image registry or repository. There are many private registries in use. This task uses [Docker Hub](#) as an example registry.

This item links to a third party project or product that is not part of Kubernetes itself. [More information](#)

## Before you begin

- You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:
  - [Killercoda](#)
  - [Play with Kubernetes](#)
- To do this exercise, you need the docker command line tool, and a [Docker ID](#) for which you know the password.
- If you are using a different private container registry, you need the command line tool for that registry and any login information for the registry.

## Log in to Docker Hub

On your laptop, you must authenticate with a registry in order to pull a private image.

Use the docker tool to log in to Docker Hub. See the *log in* section of [Docker ID accounts](#) for more information.

```
docker login
```

When prompted, enter your Docker ID, and then the credential you want to use (access token, or the password for your Docker ID).

The login process creates or updates a config.json file that holds an authorization token. Review [how Kubernetes interprets this file](#).

View the config.json file:

```
cat ~/.docker/config.json
```

The output contains a section similar to this:

```
{
 "auths": {
 "https://index.docker.io/v1/": {
 "auth": "c3R...zE2"
 }
 }
}
```

**Note:**

If you use a Docker credentials store, you won't see that auth entry but a credsStore entry with the name of the store as value. In that case, you can create a secret directly. See [Create a Secret by providing credentials on the command line](#).

## Create a Secret based on existing credentials

A Kubernetes cluster uses the Secret of `kubernetes.io/dockerconfigjson` type to authenticate with a container registry to pull a private image.

If you already ran `docker login`, you can copy that credential into Kubernetes:

```
kubectl create secret generic regcred \
 --from-file=.dockerconfigjson=<path/to/.docker/config.json> \
 --type=kubernetes.io/dockerconfigjson
```

If you need more control (for example, to set a namespace or a label on the new secret) then you can customise the Secret before storing it. Be sure to:

- set the name of the data item to `.dockerconfigjson`
- base64 encode the Docker configuration file and then paste that string, unbroken as the value for field `data[".dockerconfigjson"]`
- set type to `kubernetes.io/dockerconfigjson`

Example:

[illegible]

If you get the error message `error: no objects passed to create`, it may mean the base64 encoded string is invalid. If you get an error message like `Secret "myregistrykey" is invalid: data[.dockerconfigjson]: invalid value ...`, it means the base64 encoded string in the data was successfully decoded, but could not be parsed as a `.docker/config.json` file.

# Create a Secret by providing credentials on the command line

Create this Secret, naming it regcred:

```
kubectl create secret docker-registry regcred --docker-server=<your-registry-server> --docker-username=<your-name> --docker-password=<your-pword> --docker-email=<your-email>
```

where:

- <your-registry-server> is your Private Docker Registry FQDN. Use <https://index.docker.io/v1/> for DockerHub.
- <your-name> is your Docker username.
- <your-pword> is your Docker password.
- <your-email> is your Docker email.

You have successfully set your Docker credentials in the cluster as a Secret called regcred.

## Note:

Typing secrets on the command line may store them in your shell history unprotected, and those secrets might also be visible to other users on your PC during the time that kubectl is running.

## Inspecting the Secret regcred

To understand the contents of the regcred Secret you created, start by viewing the Secret in YAML format:

```
kubectl get secret regcred --output=yaml
```

The output is similar to this:

```
apiVersion: v1
kind: Secret
metadata:
 ...
 name: regcred
 ...
data:
 .dockerconfigjson: eyJodHRwczovL2luZGV4L ... J0QUl6RTlifX0=
type: kubernetes.io/dockerconfigjson
```

The value of the .dockerconfigjson field is a base64 representation of your Docker credentials.

To understand what is in the .dockerconfigjson field, convert the secret data to a readable format:

```
kubectl get secret regcred --output="jsonpath={.data.\.dockerconfigjson}" | base64 --decode
```

The output is similar to this:

```
{"auths":{"your.private.registry.example.com":{"username":"janedoe","password":"xxxxxxxxxxxx","email":"jdoe@example.com","auth":"c3R...zE2"}}}
```

To understand what is in the auth field, convert the base64-encoded data to a readable format:

```
echo "c3R...zE2" | base64 --decode
```

The output, username and password concatenated with a :, is similar to this:

```
janedoe:xxxxxxxxxxxx
```

Notice that the Secret data contains the authorization token similar to your local `~/.docker/config.json` file.

You have successfully set your Docker credentials as a Secret called `regcred` in the cluster.

## Create a Pod that uses your Secret

Here is a manifest for an example Pod that needs access to your Docker credentials in `regcred`:

[pods/private-reg-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: private-reg
spec:
 containers:
 - name: private-reg-container
 image: <your-private-image>
 imagePullSecrets:
 - name: regcred
```

Download the above file onto your computer:

```
curl -L -o my-private-reg-pod.yaml https://k8s.io/examples/pods/private-reg-pod.yaml
```

In file `my-private-reg-pod.yaml`, replace `<your-private-image>` with the path to an image in a private registry such as:

```
your.private.registry.example.com/janedoe/jdoe-private:v1
```

To pull the image from the private registry, Kubernetes needs credentials. The `imagePullSecrets` field in the configuration file specifies that Kubernetes should get the credentials from a Secret named `regcred`.

Create a Pod that uses your Secret, and verify that the Pod is running:

```
kubectl apply -f my-private-reg-pod.yaml
kubectl get pod private-reg
```



## Note:

To use image pull secrets for a Pod (or a Deployment, or other object that has a pod template that you are using), you need to make sure that the appropriate Secret does exist in the right namespace. The namespace to use is the same namespace where you defined the Pod.

Also, in case the Pod fails to start with the status `ImagePullBackOff`, view the Pod events:

```
kubectl describe pod private-reg
```

If you then see an event with the reason set to `FailedToRetrieveImagePullSecret`, Kubernetes can't find a Secret with name (`regcred`, in this example). If you specify that a Pod needs image pull credentials, the kubelet checks that it can access that Secret before attempting to pull the image.

Make sure that the Secret you have specified exists, and that its name is spelled properly.

Events:

| ... Reason                          | ... Message                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------|
| -----                               | -----                                                                                                     |
| ... FailedToRetrieveImagePullSecret | ... Unable to retrieve some image pull secrets (<regcred>); attempting to pull the image may not succeed. |

## What's next

- Learn more about [Secrets](#)
  - or read the API reference for [Secret](#)
- Learn more about [using a private registry](#).
- Learn more about [adding image pull secrets to a service account](#).
- See [kubectl create secret docker-registry](#).
- See the `imagePullSecrets` field within the [container definitions](#) of a Pod

# Configure Liveness, Readiness and Startup Probes

This page shows how to configure liveness, readiness and startup probes for containers.

The [kubelet](#) uses liveness probes to know when to restart a container. For example, liveness probes could catch a deadlock, where an application is running, but unable to make progress. Restarting a container in such a state can help to make the application more available despite bugs.

A common pattern for liveness probes is to use the same low-cost HTTP endpoint as for readiness probes, but with a higher `failureThreshold`. This ensures that the pod is observed as not-ready for some period of time before it is hard killed.

The kubelet uses readiness probes to know when a container is ready to start accepting traffic. A Pod is considered ready when all of its containers are ready. One use of this signal is to control which Pods are used as backends for Services. When a Pod is not ready, it is removed from Service load balancers.

The kubelet uses startup probes to know when a container application has started. If such a probe is configured, liveness and readiness probes do not start until it succeeds, making sure those probes don't interfere with the application startup. This can be used to adopt liveness checks on slow starting containers, avoiding them getting killed by the kubelet before they are up and running.

### Caution:

Liveness probes can be a powerful way to recover from application failures, but they should be used with caution. Liveness probes must be configured carefully to ensure that they truly indicate unrecoverable application failure, for example a deadlock.

### Note:

Incorrect implementation of liveness probes can lead to cascading failures. This results in restarting of container under high load; failed client requests as your application became less scalable; and increased workload on remaining pods due to some failed pods. Understand the difference between readiness and liveness probes and when to apply them for your app.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

## Define a liveness command

Many applications running for long periods of time eventually transition to broken states, and cannot recover except by being restarted. Kubernetes provides liveness probes to detect and remedy such situations.

In this exercise, you create a Pod that runs a container based on the `registry.k8s.io/busybox` image. Here is the configuration file for the Pod:

[pods/probe/exec-liveness.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 labels:
 test: liveness
 name: liveness-exec
spec:
 containers:
 - name: liveness
 image: registry.k8s.io/busybox
 args:
```

```
- /bin/sh
- -c
- touch /tmp/healthy; sleep 30; rm -f /tmp/healthy; sleep 600
livenessProbe:
 exec:
 command:
 - cat
 - /tmp/healthy
 initialDelaySeconds: 5
 periodSeconds: 5
```

In the configuration file, you can see that the Pod has a single Container. The `periodSeconds` field specifies that the kubelet should perform a liveness probe every 5 seconds. The `initialDelaySeconds` field tells the kubelet that it should wait 5 seconds before performing the first probe. To perform a probe, the kubelet executes the command `cat /tmp/healthy` in the target container. If the command succeeds, it returns 0, and the kubelet considers the container to be alive and healthy. If the command returns a non-zero value, the kubelet kills the container and restarts it.

When the container starts, it executes this command:

```
/bin/sh -c "touch /tmp/healthy; sleep 30; rm -f /tmp/healthy; sleep 600"
```

For the first 30 seconds of the container's life, there is a `/tmp/healthy` file. So during the first 30 seconds, the command `cat /tmp/healthy` returns a success code. After 30 seconds, `cat /tmp/healthy` returns a failure code.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/probe/exec-liveness.yaml
```

Within 30 seconds, view the Pod events:

```
kubectl describe pod liveness-exec
```

The output indicates that no liveness probes have failed yet:

| Type   | Reason    | Age | From              | Message                                               |
|--------|-----------|-----|-------------------|-------------------------------------------------------|
| Normal | Scheduled | 11s | default-scheduler | Successfully assigned default/liveness-exec to node01 |
| Normal | Pulling   | 9s  | kubelet, node01   | Pulling image "registry.k8s.io/busybox"               |
| Normal | Pulled    | 7s  | kubelet, node01   | Successfully pulled image "registry.k8s.io/busybox"   |
| Normal | Created   | 7s  | kubelet, node01   | Created container liveness                            |
| Normal | Started   | 7s  | kubelet, node01   | Started container liveness                            |

After 35 seconds, view the Pod events again:

```
kubectl describe pod liveness-exec
```

At the bottom of the output, there are messages indicating that the liveness probes have failed, and the failed containers have been killed and recreated.

| Type | Reason | Age | From | Message |
|------|--------|-----|------|---------|
|------|--------|-----|------|---------|

|         |           |                   |                   |                                                                                  |
|---------|-----------|-------------------|-------------------|----------------------------------------------------------------------------------|
| Normal  | Scheduled | 57s               | default-scheduler | Successfully assigned default/liveness-exec to node01                            |
| Normal  | Pulling   | 55s               | kubelet, node01   | Pulling image "registry.k8s.io/busybox"                                          |
| Normal  | Pulled    | 53s               | kubelet, node01   | Successfully pulled image "registry.k8s.io/busybox"                              |
| Normal  | Created   | 53s               | kubelet, node01   | Created container liveness                                                       |
| Normal  | Started   | 53s               | kubelet, node01   | Started container liveness                                                       |
| Warning | Unhealthy | 10s (x3 over 20s) | kubelet, node01   | Liveness probe failed: cat: can't open '/tmp/healthy': No such file or directory |
| Normal  | Killing   | 10s               | kubelet, node01   | Container liveness failed liveness probe, will be restarted                      |

Wait another 30 seconds, and verify that the container has been restarted:

```
kubectl get pod liveness-exec
```

The output shows that RESTARTS has been incremented. Note that the RESTARTS counter increments as soon as a failed container comes back to the running state:

| NAME          | READY | STATUS  | RESTARTS | AGE |
|---------------|-------|---------|----------|-----|
| liveness-exec | 1/1   | Running | 1        | 1m  |

## Define a liveness HTTP request

Another kind of liveness probe uses an HTTP GET request. Here is the configuration file for a Pod that runs a container based on the registry.k8s.io/e2e-test-images/agnhost image.

[pods/probe/http-liveness.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 labels:
 test: liveness
 name: liveness-http
spec:
 containers:
 - name: liveness
 image: registry.k8s.io/e2e-test-images/agnhost:2.40
 args:
 - liveness
 livenessProbe:
 httpGet:
 path: /healthz
 port: 8080
 httpHeaders:
 - name: Custom-Header
 value: Awesome
 initialDelaySeconds: 3
 periodSeconds: 3
```

In the configuration file, you can see that the Pod has a single container. The periodSeconds field specifies that the kubelet should perform a liveness probe every 3 seconds. The

`initialDelaySeconds` field tells the kubelet that it should wait 3 seconds before performing the first probe. To perform a probe, the kubelet sends an HTTP GET request to the server that is running in the container and listening on port 8080. If the handler for the server's `/healthz` path returns a success code, the kubelet considers the container to be alive and healthy. If the handler returns a failure code, the kubelet kills the container and restarts it.

Any code greater than or equal to 200 and less than 400 indicates success. Any other code indicates failure.

You can see the source code for the server in [server.go](#).

For the first 10 seconds that the container is alive, the `/healthz` handler returns a status of 200. After that, the handler returns a status of 500.

```
http.HandleFunc("/healthz", func(w http.ResponseWriter, r *http.Request) {
 duration := time.Now().Sub(started)
 if duration.Seconds() > 10 {
 w.WriteHeader(500)
 w.Write([]byte(fmt.Sprintf("error: %v", duration.Seconds())))
 } else {
 w.WriteHeader(200)
 w.Write([]byte("ok"))
 }
})
```

The kubelet starts performing health checks 3 seconds after the container starts. So the first couple of health checks will succeed. But after 10 seconds, the health checks will fail, and the kubelet will kill and restart the container.

To try the HTTP liveness check, create a Pod:

```
kubectl apply -f https://k8s.io/examples/pods/probe/http-liveness.yaml
```

After 10 seconds, view Pod events to verify that liveness probes have failed and the container has been restarted:

```
kubectl describe pod liveness-http
```

In releases after v1.13, local HTTP proxy environment variable settings do not affect the HTTP liveness probe.

## Define a TCP liveness probe

A third type of liveness probe uses a TCP socket. With this configuration, the kubelet will attempt to open a socket to your container on the specified port. If it can establish a connection, the container is considered healthy, if it can't it is considered a failure.

[pods/probe/tcp-liveness-readiness.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: goproxy
 labels:
```

```
 app: goproxy
spec:
 containers:
 - name: goproxy
 image: registry.k8s.io/goproxy:0.1
 ports:
 - containerPort: 8080
 readinessProbe:
 tcpSocket:
 port: 8080
 initialDelaySeconds: 15
 periodSeconds: 10
 livenessProbe:
 tcpSocket:
 port: 8080
 initialDelaySeconds: 15
 periodSeconds: 10
```

As you can see, configuration for a TCP check is quite similar to an HTTP check. This example uses both readiness and liveness probes. The kubelet will send the first readiness probe 15 seconds after the container starts. This will attempt to connect to the goproxy container on port 8080. If the probe succeeds, the Pod will be marked as ready. The kubelet will continue to run this check every 10 seconds.

In addition to the readiness probe, this configuration includes a liveness probe. The kubelet will run the first liveness probe 15 seconds after the container starts. Similar to the readiness probe, this will attempt to connect to the goproxy container on port 8080. If the liveness probe fails, the container will be restarted.

To try the TCP liveness check, create a Pod:

```
kubectl apply -f https://k8s.io/examples/pods/probe/tcp-liveness-readiness.yaml
```

After 15 seconds, view Pod events to verify that liveness probes:

```
kubectl describe pod goproxy
```

## Define a gRPC liveness probe

FEATURE STATE: Kubernetes v1.27 [stable]

If your application implements the [gRPC Health Checking Protocol](#), this example shows how to configure Kubernetes to use it for application liveness checks. Similarly you can configure readiness and startup probes.

Here is an example manifest:

[pods/probe/grpc-liveness.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: etcd-with-grpc
```

```
spec:
 containers:
 - name: etcd
 image: registry.k8s.io/etcd:3.5.1-0
 command: ["/usr/local/bin/etcd", "--data-dir", "/var/lib/etcd", "--listen-client-urls", "http://0.0.0.0:2379", "--advertise-client-urls", "http://127.0.0.1:2379", "--log-level", "debug"]
 ports:
 - containerPort: 2379
 livenessProbe:
 grpc:
 port: 2379
 initialDelaySeconds: 10
```

To use a gRPC probe, port must be configured. If you want to distinguish probes of different types and probes for different features you can use the service field. You can set service to the value liveness and make your gRPC Health Checking endpoint respond to this request differently than when you set service set to readiness. This lets you use the same endpoint for different kinds of container health check rather than listening on two different ports. If you want to specify your own custom service name and also specify a probe type, the Kubernetes project recommends that you use a name that concatenates those. For example: myservice-liveness (using - as a separator).

#### Note:

Unlike HTTP or TCP probes, you cannot specify the health check port by name, and you cannot configure a custom hostname.

Configuration problems (for example: incorrect port or service, unimplemented health checking protocol) are considered a probe failure, similar to HTTP and TCP probes.

To try the gRPC liveness check, create a Pod using the command below. In the example below, the etcd pod is configured to use gRPC liveness probe.

```
kubectl apply -f https://k8s.io/examples/pods/probe/grpc-liveness.yaml
```

After 15 seconds, view Pod events to verify that the liveness check has not failed:

```
kubectl describe pod etcd-with-grpc
```

When using a gRPC probe, there are some technical details to be aware of:

- The probes run against the pod IP address or its hostname. Be sure to configure your gRPC endpoint to listen on the Pod's IP address.
- The probes do not support any authentication parameters (like -tls).
- There are no error codes for built-in probes. All errors are considered as probe failures.
- If ExecProbeTimeout feature gate is set to false, grpc-health-probe does **not** respect the timeoutSeconds setting (which defaults to 1s), while built-in probe would fail on timeout.

## Use a named port

You can use a named [port](#) for HTTP and TCP probes. gRPC probes do not support named ports.

For example:

```
ports:
- name: liveness-port
 containerPort: 8080
```

```
livenessProbe:
 httpGet:
 path: /healthz
 port: liveness-port
```

## Protect slow starting containers with startup probes

Sometimes, you have to deal with legacy applications that might require an additional startup time on their first initialization. In such cases, it can be tricky to set up liveness probe parameters without compromising the fast response to deadlocks that motivated such a probe. The trick is to set up a startup probe with the same command, HTTP or TCP check, with a `failureThreshold` \* `periodSeconds` long enough to cover the worst case startup time.

So, the previous example would become:

```
ports:
- name: liveness-port
 containerPort: 8080
```

```
livenessProbe:
 httpGet:
 path: /healthz
 port: liveness-port
 failureThreshold: 1
 periodSeconds: 10
```

```
startupProbe:
 httpGet:
 path: /healthz
 port: liveness-port
 failureThreshold: 30
 periodSeconds: 10
```

Thanks to the startup probe, the application will have a maximum of 5 minutes ( $30 * 10 = 300s$ ) to finish its startup. Once the startup probe has succeeded once, the liveness probe takes over to provide a fast response to container deadlocks. If the startup probe never succeeds, the container is killed after 300s and subject to the pod's `restartPolicy`.

## Define readiness probes

Sometimes, applications are temporarily unable to serve traffic. For example, an application might need to load large data or configuration files during startup, or depend on external services after startup. In such cases, you don't want to kill the application, but you don't want to send it requests either. Kubernetes provides readiness probes to detect and mitigate these situations. A pod with containers reporting that they are not ready does not receive traffic through Kubernetes Services.



## Note:

Readiness probes runs on the container during its whole lifecycle.

## Caution:

The readiness and liveness probes do not depend on each other to succeed. If you want to wait before executing a readiness probe, you should use `initialDelaySeconds` or a `startupProbe`.

Readiness probes are configured similarly to liveness probes. The only difference is that you use the `readinessProbe` field instead of the `livenessProbe` field.

```
readinessProbe:
 exec:
 command:
 - cat
 - /tmp/healthy
 initialDelaySeconds: 5
 periodSeconds: 5
```

Configuration for HTTP and TCP readiness probes also remains identical to liveness probes.

Readiness and liveness probes can be used in parallel for the same container. Using both can ensure that traffic does not reach a container that is not ready for it, and that containers are restarted when they fail.

## Configure Probes

[Probes](#) have a number of fields that you can use to more precisely control the behavior of startup, liveness and readiness checks:

- `initialDelaySeconds`: Number of seconds after the container has started before startup, liveness or readiness probes are initiated. If a startup probe is defined, liveness and readiness probe delays do not begin until the startup probe has succeeded. If the value of `periodSeconds` is greater than `initialDelaySeconds` then the `initialDelaySeconds` would be ignored. Defaults to 0 seconds. Minimum value is 0.
- `periodSeconds`: How often (in seconds) to perform the probe. Default to 10 seconds. The minimum value is 1.
- `timeoutSeconds`: Number of seconds after which the probe times out. Defaults to 1 second. Minimum value is 1.
- `successThreshold`: Minimum consecutive successes for the probe to be considered successful after having failed. Defaults to 1. Must be 1 for liveness and startup Probes. Minimum value is 1.
- `failureThreshold`: After a probe fails `failureThreshold` times in a row, Kubernetes considers that the overall check has failed: the container is *not* ready/healthy/live. For the case of a startup or liveness probe, if at least `failureThreshold` probes have failed, Kubernetes treats the container as unhealthy and triggers a restart for that specific container. The kubelet honors the setting of `terminationGracePeriodSeconds` for that container. For a failed readiness probe, the kubelet continues running the container that failed checks, and also continues to run more probes; because the check failed, the kubelet sets the Ready [condition](#) on the Pod to false.

- `terminationGracePeriodSeconds`: configure a grace period for the kubelet to wait between triggering a shut down of the failed container, and then forcing the container runtime to stop that container. The default is to inherit the Pod-level value for `terminationGracePeriodSeconds` (30 seconds if not specified), and the minimum value is 1. See [probe-level terminationGracePeriodSeconds](#) for more detail.

### Caution:

Incorrect implementation of readiness probes may result in an ever growing number of processes in the container, and resource starvation if this is left unchecked.

## HTTP probes

[HTTP probes](#) have additional fields that can be set on `httpGet`:

- `host`: Host name to connect to, defaults to the pod IP. You probably want to set "Host" in `httpHeaders` instead.
- `scheme`: Scheme to use for connecting to the host (HTTP or HTTPS). Defaults to "HTTP".
- `path`: Path to access on the HTTP server. Defaults to "/".
- `httpHeaders`: Custom headers to set in the request. HTTP allows repeated headers.
- `port`: Name or number of the port to access on the container. Number must be in the range 1 to 65535.

For an HTTP probe, the kubelet sends an HTTP request to the specified port and path to perform the check. The kubelet sends the probe to the Pod's IP address, unless the address is overridden by the optional `host` field in `httpGet`. If `scheme` field is set to HTTPS, the kubelet sends an HTTPS request skipping the certificate verification. In most scenarios, you do not want to set the `host` field. Here's one scenario where you would set it. Suppose the container listens on 127.0.0.1 and the Pod's `hostNetwork` field is true. Then `host`, under `httpGet`, should be set to 127.0.0.1. If your pod relies on virtual hosts, which is probably the more common case, you should not use `host`, but rather set the `Host` header in `httpHeaders`.

For an HTTP probe, the kubelet sends two request headers in addition to the mandatory `Host` header:

- `User-Agent`: The default value is `kube-probe/1.30`, where 1.30 is the version of the kubelet.
- `Accept`: The default value is `*/*`.

You can override the default headers by defining `httpHeaders` for the probe. For example:

```
livenessProbe:
 httpGet:
 httpHeaders:
 - name: Accept
 value: application/json

startupProbe:
 httpGet:
 httpHeaders:
 - name: User-Agent
 value: MyUserAgent
```

You can also remove these two headers by defining them with an empty value.

```

livenessProbe:
 httpGet:
 httpHeaders:
 - name: Accept
 value: ""

startupProbe:
 httpGet:
 httpHeaders:
 - name: User-Agent
 value: ""

```

## Note:

When the kubelet probes a Pod using HTTP, it only follows redirects if the redirect is to the same host. If the kubelet receives 11 or more redirects during probing, the probe is considered successful and a related Event is created:

```

Events:
 Type Reason Age From Message
 ---- -
Normal Scheduled 29m default-scheduler Successfully assigned default/
httpbin-7b8bc9cb85-bjzwn to daocloud
Normal Pulling 29m kubelet Pulling image "docker.io/kennethreitz/
httpbin"
Normal Pulled 24m kubelet Successfully pulled image "docker.io/
kennethreitz/httpbin" in 5m12.402735213s
Normal Created 24m kubelet Created container httpbin
Normal Started 24m kubelet Started container httpbin
Warning ProbeWarning 4m11s (x1197 over 24m) kubelet Readiness probe warning:
Probe terminated redirects

```

If the kubelet receives a redirect where the hostname is different from the request, the outcome of the probe is treated as successful and kubelet creates an event to report the redirect failure.

## TCP probes

For a TCP probe, the kubelet makes the probe connection at the node, not in the Pod, which means that you can not use a service name in the host parameter since the kubelet is unable to resolve it.

## Probe-level terminationGracePeriodSeconds

FEATURE STATE: Kubernetes v1.28 [stable]

In 1.25 and above, users can specify a probe-level `terminationGracePeriodSeconds` as part of the probe specification. When both a pod- and probe-level `terminationGracePeriodSeconds` are set, the kubelet will use the probe-level value.

When setting the `terminationGracePeriodSeconds`, please note the following:

- The kubelet always honors the probe-level `terminationGracePeriodSeconds` field if it is present on a Pod.

- If you have existing Pods where the `terminationGracePeriodSeconds` field is set and you no longer wish to use per-probe termination grace periods, you must delete those existing Pods.

For example:

```
spec:
 terminationGracePeriodSeconds: 3600 # pod-level
 containers:
 - name: test
 image: ...

 ports:
 - name: liveness-port
 containerPort: 8080

 livenessProbe:
 httpGet:
 path: /healthz
 port: liveness-port
 failureThreshold: 1
 periodSeconds: 60
 # Override pod-level terminationGracePeriodSeconds #
 terminationGracePeriodSeconds: 60
```

Probe-level `terminationGracePeriodSeconds` cannot be set for readiness probes. It will be rejected by the API server.

## What's next

- Learn more about [Container Probes](#).

You can also read the API references for:

- [Pod](#), and specifically:
  - [container\(s\)](#)
  - [probe\(s\)](#)

## Assign Pods to Nodes

This page shows how to assign a Kubernetes Pod to a particular node in a Kubernetes cluster.

### Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Add a label to a node

1. List the [nodes](#) in your cluster, along with their labels:

```
kubectl get nodes --show-labels
```

The output is similar to this:

| NAME    | STATUS | ROLES  | AGE | VERSION | LABELS                             |
|---------|--------|--------|-----|---------|------------------------------------|
| worker0 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker0 |
| worker1 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker1 |
| worker2 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker2 |

2. Choose one of your nodes, and add a label to it:

```
kubectl label nodes <your-node-name> disktype=ssd
```

where `<your-node-name>` is the name of your chosen node.

3. Verify that your chosen node has a `disktype=ssd` label:

```
kubectl get nodes --show-labels
```

The output is similar to this:

| NAME    | STATUS | ROLES  | AGE | VERSION | LABELS                                          |
|---------|--------|--------|-----|---------|-------------------------------------------------|
| worker0 | Ready  | <none> | 1d  | v1.13.0 | ...,disktype=ssd,kubernetes.io/hostname=worker0 |
| worker1 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker1              |
| worker2 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker2              |

In the preceding output, you can see that the worker0 node has a `disktype=ssd` label.

## Create a pod that gets scheduled to your chosen node

This pod configuration file describes a pod that has a node selector, `disktype: ssd`. This means that the pod will get scheduled on a node that has a `disktype=ssd` label.

[pods/pod-nginx.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: nginx
 labels:
 env: test
spec:
 containers:
 - name: nginx
 image: nginx
 imagePullPolicy: IfNotPresent
```

```
nodeSelector:
 disktype: ssd
```

1. Use the configuration file to create a pod that will get scheduled on your chosen node:

```
kubectl apply -f https://k8s.io/examples/pods/pod-nginx.yaml
```

2. Verify that the pod is running on your chosen node:

```
kubectl get pods --output=wide
```

The output is similar to this:

| NAME  | READY | STATUS  | RESTARTS | AGE | IP         | NODE    |
|-------|-------|---------|----------|-----|------------|---------|
| nginx | 1/1   | Running | 0        | 13s | 10.200.0.4 | worker0 |

## Create a pod that gets scheduled to specific node

You can also schedule a pod to one specific node via setting nodeName.

[pods/pod-nginx-specific-node.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: nginx
spec:
 nodeName: foo-node # schedule pod to specific node
 containers:
 - name: nginx
 image: nginx
 imagePullPolicy: IfNotPresent
```

Use the configuration file to create a pod that will get scheduled on foo-node only.

## What's next

- Learn more about [labels and selectors](#).
- Learn more about [nodes](#).

## Assign Pods to Nodes using Node Affinity

This page shows how to assign a Kubernetes Pod to a particular node using Node Affinity in a Kubernetes cluster.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.10. To check the version, enter `kubectl version`.

## Add a label to a node

1. List the nodes in your cluster, along with their labels:

```
kubectl get nodes --show-labels
```

The output is similar to this:

| NAME    | STATUS | ROLES  | AGE | VERSION | LABELS                             |
|---------|--------|--------|-----|---------|------------------------------------|
| worker0 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker0 |
| worker1 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker1 |
| worker2 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker2 |

2. Choose one of your nodes, and add a label to it:

```
kubectl label nodes <your-node-name> disktype=ssd
```

where <your-node-name> is the name of your chosen node.

3. Verify that your chosen node has a disktype=ssd label:

```
kubectl get nodes --show-labels
```

The output is similar to this:

| NAME    | STATUS | ROLES  | AGE | VERSION | LABELS                                          |
|---------|--------|--------|-----|---------|-------------------------------------------------|
| worker0 | Ready  | <none> | 1d  | v1.13.0 | ...,disktype=ssd,kubernetes.io/hostname=worker0 |
| worker1 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker1              |
| worker2 | Ready  | <none> | 1d  | v1.13.0 | ...,kubernetes.io/hostname=worker2              |

In the preceding output, you can see that the worker0 node has a disktype=ssd label.

## Schedule a Pod using required node affinity

This manifest describes a Pod that has a requiredDuringSchedulingIgnoredDuringExecution node affinity, disktype: ssd. This means that the pod will get scheduled only on a node that has a disktype=ssd label.

[pods/pod-nginx-required-affinity.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
```

```

name: nginx
spec:
 affinity:
 nodeAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 nodeSelectorTerms:
 - matchExpressions:
 - key: disktype
 operator: In
 values:
 - ssd
 containers:
 - name: nginx
 image: nginx
 imagePullPolicy: IfNotPresent

```

1. Apply the manifest to create a Pod that is scheduled onto your chosen node:

```
kubectl apply -f https://k8s.io/examples/pods/pod-nginx-required-affinity.yaml
```

2. Verify that the pod is running on your chosen node:

```
kubectl get pods --output=wide
```

The output is similar to this:

| NAME  | READY | STATUS  | RESTARTS | AGE | IP         | NODE    |
|-------|-------|---------|----------|-----|------------|---------|
| nginx | 1/1   | Running | 0        | 13s | 10.200.0.4 | worker0 |

## Schedule a Pod using preferred node affinity

This manifest describes a Pod that has a `preferredDuringSchedulingIgnoredDuringExecution` node affinity, `disktype: ssd`. This means that the pod will prefer a node that has a `disktype=ssd` label.

[pods/pod-nginx-preferred-affinity.yaml](#)

```

apiVersion: v1
kind: Pod
metadata:
 name: nginx
spec:
 affinity:
 nodeAffinity:
 preferredDuringSchedulingIgnoredDuringExecution:
 - weight: 1
 preference:
 matchExpressions:
 - key: disktype
 operator: In
 values:
 - ssd
 containers:

```



```
- name: nginx
 image: nginx
 imagePullPolicy: IfNotPresent
```

1. Apply the manifest to create a Pod that is scheduled onto your chosen node:

```
kubectl apply -f https://k8s.io/examples/pods/pod-nginx-preferred-affinity.yaml
```

2. Verify that the pod is running on your chosen node:

```
kubectl get pods --output=wide
```

The output is similar to this:

| NAME  | READY | STATUS  | RESTARTS | AGE | IP         | NODE    |
|-------|-------|---------|----------|-----|------------|---------|
| nginx | 1/1   | Running | 0        | 13s | 10.200.0.4 | worker0 |

## What's next

Learn more about [Node Affinity](#).

# Configure Pod Initialization

This page shows how to use an Init Container to initialize a Pod before an application Container runs.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Create a Pod that has an Init Container

In this exercise you create a Pod that has one application Container and one Init Container. The init container runs to completion before the application container starts.

Here is the configuration file for the Pod:

[pods/init-containers.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: init-demo
```

```
spec:
 containers:
 - name: nginx
 image: nginx
 ports:
 - containerPort: 80
 volumeMounts:
 - name: workdir
 mountPath: /usr/share/nginx/html
 # These containers are run during pod initialization
 initContainers:
 - name: install
 image: busybox:1.28
 command:
 - wget
 - "-O"
 - "/work-dir/index.html"
 - "http://info.cern.ch"
 volumeMounts:
 - name: workdir
 mountPath: "/work-dir"
 dnsPolicy: Default
 volumes:
 - name: workdir
 emptyDir: {}
```

In the configuration file, you can see that the Pod has a Volume that the init container and the application container share.

The init container mounts the shared Volume at /work-dir, and the application container mounts the shared Volume at /usr/share/nginx/html. The init container runs the following command and then terminates:

```
wget -O /work-dir/index.html http://info.cern.ch
```

Notice that the init container writes the index.html file in the root directory of the nginx server.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/init-containers.yaml
```

Verify that the nginx container is running:

```
kubectl get pod init-demo
```

The output shows that the nginx container is running:

| NAME      | READY | STATUS  | RESTARTS | AGE |
|-----------|-------|---------|----------|-----|
| init-demo | 1/1   | Running | 0        | 1m  |

Get a shell into the nginx container running in the init-demo Pod:

```
kubectl exec -it init-demo -- /bin/bash
```

In your shell, send a GET request to the nginx server:

```
root@nginx:~# apt-get update
root@nginx:~# apt-get install curl
root@nginx:~# curl localhost
```

The output shows that nginx is serving the web page that was written by the init container:

```
<html><head></head><body><header>
<title>http://info.cern.ch</title>
</header>

<h1>http://info.cern.ch - home of the first website</h1>
...
Browse the first website</
a>
...
```

## What's next

- Learn more about [communicating between Containers running in the same Pod](#).
- Learn more about [Init Containers](#).
- Learn more about [Volumes](#).
- Learn more about [Debugging Init Containers](#)

# Attach Handlers to Container Lifecycle Events

This page shows how to attach handlers to Container lifecycle events. Kubernetes supports the postStart and preStop events. Kubernetes sends the postStart event immediately after a Container is started, and it sends the preStop event immediately before the Container is terminated. A Container may specify one handler per event.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Define postStart and preStop handlers

In this exercise, you create a Pod that has one Container. The Container has handlers for the postStart and preStop events.

Here is the configuration file for the Pod:

[pods/lifecycle-events.yaml](https://k8s.io/examples/pods/lifecycle-events.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: lifecycle-demo
spec:
 containers:
 - name: lifecycle-demo-container
 image: nginx
 lifecycle:
 postStart:
 exec:
 command: ["/bin/sh", "-c", "echo Hello from the postStart handler > /usr/share/message"]
 preStop:
 exec:
 command: ["/bin/sh", "-c", "nginx -s quit; while killall -0 nginx; do sleep 1; done"]
```

In the configuration file, you can see that the postStart command writes a message file to the Container's /usr/share directory. The preStop command shuts down nginx gracefully. This is helpful if the Container is being terminated because of a failure.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/lifecycle-events.yaml
```

Verify that the Container in the Pod is running:

```
kubectl get pod lifecycle-demo
```

Get a shell into the Container running in your Pod:

```
kubectl exec -it lifecycle-demo -- /bin/bash
```

In your shell, verify that the postStart handler created the message file:

```
root@lifecycle-demo:/# cat /usr/share/message
```

The output shows the text written by the postStart handler:

```
Hello from the postStart handler
```

## Discussion

Kubernetes sends the postStart event immediately after the Container is created. There is no guarantee, however, that the postStart handler is called before the Container's entrypoint is called. The postStart handler runs asynchronously relative to the Container's code, but Kubernetes' management of the container blocks until the postStart handler completes. The Container's status is not set to RUNNING until the postStart handler completes.

Kubernetes sends the `preStop` event immediately before the Container is terminated. Kubernetes' management of the Container blocks until the `preStop` handler completes, unless the Pod's grace period expires. For more details, see [Pod Lifecycle](#).

#### Note:

Kubernetes only sends the `preStop` event when a Pod or a container in the Pod is *terminated*. This means that the `preStop` hook is not invoked when the Pod is *completed*. About this limitation, please see [Container hooks](#) for the detail.

## What's next

- Learn more about [Container lifecycle hooks](#).
- Learn more about the [lifecycle of a Pod](#).

## Reference

- [Lifecycle](#)
- [Container](#)
- See `terminationGracePeriodSeconds` in [PodSpec](#)

# Configure a Pod to Use a ConfigMap

Many applications rely on configuration which is used during either application initialization or runtime. Most times, there is a requirement to adjust values assigned to configuration parameters. ConfigMaps are a Kubernetes mechanism that let you inject configuration data into application [pods](#).

The ConfigMap concept allow you to decouple configuration artifacts from image content to keep containerized applications portable. For example, you can download and run the same [container image](#) to spin up containers for the purposes of local development, system test, or running a live end-user workload.

This page provides a series of usage examples demonstrating how to create ConfigMaps and configure Pods using data stored in ConfigMaps.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You need to have the `wget` tool installed. If you have a different tool such as `curl`, and you do not have `wget`, you will need to adapt the step that downloads example data.

# Create a ConfigMap

You can use either `kubectl create configmap` or a ConfigMap generator in `kustomization.yaml` to create a ConfigMap.

## Create a ConfigMap using `kubectl create configmap`

Use the `kubectl create configmap` command to create ConfigMaps from [directories](#), [files](#), or [literal values](#):

```
kubectl create configmap <map-name> <data-source>
```

where `<map-name>` is the name you want to assign to the ConfigMap and `<data-source>` is the directory, file, or literal value to draw the data from. The name of a ConfigMap object must be a valid [DNS subdomain name](#).

When you are creating a ConfigMap based on a file, the key in the `<data-source>` defaults to the basename of the file, and the value defaults to the file content.

You can use [kubectl describe](#) or [kubectl get](#) to retrieve information about a ConfigMap.

## Create a ConfigMap from a directory

You can use `kubectl create configmap` to create a ConfigMap from multiple files in the same directory. When you are creating a ConfigMap based on a directory, `kubectl` identifies files whose filename is a valid key in the directory and packages each of those files into the new ConfigMap. Any directory entries except regular files are ignored (for example: subdirectories, symlinks, devices, pipes, and more).

### Note:

Each filename being used for ConfigMap creation must consist of only acceptable characters, which are: letters (A to Z and a to z), digits (0 to 9), '-', '\_', or '.'. If you use `kubectl create configmap` with a directory where any of the file names contains an unacceptable character, the `kubectl` command may fail.

The `kubectl` command does not print an error when it encounters an invalid filename.

Create the local directory:

```
mkdir -p configure-pod-container/configmap/
```

Now, download the sample configuration and create the ConfigMap:

```
Download the sample files into `configure-pod-container/configmap/` directory
wget https://kubernetes.io/examples/configmap/game.properties -O configure-pod-container/
configmap/game.properties
wget https://kubernetes.io/examples/configmap/ui.properties -O configure-pod-container/
configmap/ui.properties
```

```
Create the ConfigMap
```

```
kubectl create configmap game-config --from-file=configure-pod-container/configmap/
```

The above command packages each file, in this case, `game.properties` and `ui.properties` in the `configure-pod-container/configmap/` directory into the `game-config` ConfigMap. You can display details of the ConfigMap using the following command:

```
kubectl describe configmaps game-config
```

The output is similar to this:

```
Name: game-config
Namespace: default
Labels: <none>
Annotations: <none>

Data
====
game.properties:

enemies=aliens
lives=3
enemies.cheat=true
enemies.cheat.level=noGoodRotten
secret.code.passphrase=UDDLRBABAS
secret.code.allowed=true
secret.code.lives=30
ui.properties:

color.good=purple
color.bad=yellow
allow.textmode=true
how.nice.to.look=fairlyNice
```

The `game.properties` and `ui.properties` files in the `configure-pod-container/configmap/` directory are represented in the `data` section of the ConfigMap.

```
kubectl get configmaps game-config -o yaml
```

The output is similar to this:

```
apiVersion: v1
kind: ConfigMap
metadata:
 creationTimestamp: 2022-02-18T18:52:05Z
 name: game-config
 namespace: default
 resourceVersion: "516"
 uid: b4952dc3-d670-11e5-8cd0-68f728db1985
data:
 game.properties: |
 enemies=aliens
 lives=3
 enemies.cheat=true
 enemies.cheat.level=noGoodRotten
 secret.code.passphrase=UDDLRBABAS
 secret.code.allowed=true
```

```
secret.code.lives=30
ui.properties: |
 color.good=purple
 color.bad=yellow
 allow.textmode=true
 how.nice.to.look=fairlyNice
```

## Create ConfigMaps from files

You can use `kubect`l create configmap to create a ConfigMap from an individual file, or from multiple files.

For example,

```
kubect
```

l create configmap game-config-2 --from-file=configure-pod-container/configmap/
game.properties

would produce the following ConfigMap:

```
kubect
```

l describe configmaps game-config-2

where the output is similar to this:

```
Name: game-config-2
Namespace: default
Labels: <none>
Annotations: <none>

Data
====
game.properties:

enemies=aliens
lives=3
enemies.cheat=true
enemies.cheat.level=noGoodRotten
secret.code.passphrase=UDDLRRLRBABAS
secret.code.allowed=true
secret.code.lives=30
```

You can pass in the `--from-file` argument multiple times to create a ConfigMap from multiple data sources.

```
kubect
```

l create configmap game-config-2 --from-file=configure-pod-container/configmap/
game.properties --from-file=configure-pod-container/configmap/ui.properties

You can display details of the game-config-2 ConfigMap using the following command:

```
kubect
```

l describe configmaps game-config-2

The output is similar to this:

```
Name: game-config-2
Namespace: default
```



```
Labels: <none>
Annotations: <none>
```

Data

====

game.properties:

----

enemies=aliens

lives=3

enemies.cheat=true

enemies.cheat.level=noGoodRotten

secret.code.passphrase=UDDLRBABAS

secret.code.allowed=true

secret.code.lives=30

ui.properties:

----

color.good=purple

color.bad=yellow

allow.textmode=true

how.nice.to.look=fairlyNice

Use the option `--from-env-file` to create a ConfigMap from an env-file, for example:

```
Env-files contain a list of environment variables.
These syntax rules apply:
Each line in an env file has to be in VAR=VAL format.
Lines beginning with # (i.e. comments) are ignored.
Blank lines are ignored.
There is no special handling of quotation marks (i.e. they will be part of the ConfigMap
value)).

Download the sample files into `configure-pod-container/configmap/` directory
wget https://kubernetes.io/examples/configmap/game-env-file.properties -O configure-pod-
container/configmap/game-env-file.properties
wget https://kubernetes.io/examples/configmap/ui-env-file.properties -O configure-pod-
container/configmap/ui-env-file.properties

The env-file `game-env-file.properties` looks like below
cat configure-pod-container/configmap/game-env-file.properties
enemies=aliens
lives=3
allowed="true"

This comment and the empty line above it are ignored
```

```
kubectl create configmap game-config-env-file \
 --from-env-file=configure-pod-container/configmap/game-env-file.properties
```

would produce a ConfigMap. View the ConfigMap:

```
kubectl get configmap game-config-env-file -o yaml
```

the output is similar to:

```
apiVersion: v1
kind: ConfigMap
metadata:
 creationTimestamp: 2019-12-27T18:36:28Z
 name: game-config-env-file
 namespace: default
 resourceVersion: "809965"
 uid: d9d1ca5b-eb34-11e7-887b-42010a8002b8
data:
 allowed: "true"
 enemies: aliens
 lives: "3"
```

Starting with Kubernetes v1.23, `kubectl` supports the `--from-env-file` argument to be specified multiple times to create a ConfigMap from multiple data sources.

```
kubectl create configmap config-multi-env-files \
 --from-env-file=configure-pod-container/configmap/game-env-file.properties \
 --from-env-file=configure-pod-container/configmap/ui-env-file.properties
```

would produce the following ConfigMap:

```
kubectl get configmap config-multi-env-files -o yaml
```

where the output is similar to this:

```
apiVersion: v1
kind: ConfigMap
metadata:
 creationTimestamp: 2019-12-27T18:38:34Z
 name: config-multi-env-files
 namespace: default
 resourceVersion: "810136"
 uid: 252c4572-eb35-11e7-887b-42010a8002b8
data:
 allowed: "true"
 color: purple
 enemies: aliens
 how: fairlyNice
 lives: "3"
 textmode: "true"
```

## Define the key to use when creating a ConfigMap from a file

You can define a key other than the file name to use in the data section of your ConfigMap when using the `--from-file` argument:

```
kubectl create configmap game-config-3 --from-file=<my-key-name>=<path-to-file>
```

where `<my-key-name>` is the key you want to use in the ConfigMap and `<path-to-file>` is the location of the data source file you want the key to represent.

For example:

```
kubectl create configmap game-config-3 --from-file=game-special-key=configure-pod-container/configmap/game.properties
```

would produce the following ConfigMap:

```
kubectl get configmaps game-config-3 -o yaml
```

where the output is similar to this:

```
apiVersion: v1
kind: ConfigMap
metadata:
 creationTimestamp: 2022-02-18T18:54:22Z
 name: game-config-3
 namespace: default
 resourceVersion: "530"
 uid: 05f8da22-d671-11e5-8cd0-68f728db1985
data:
 game-special-key: |
 enemies=aliens
 lives=3
 enemies.cheat=true
 enemies.cheat.level=noGoodRotten
 secret.code.passphrase=UDDLRBABAS
 secret.code.allowed=true
 secret.code.lives=30
```

## Create ConfigMaps from literal values

You can use `kubectl create configmap` with the `--from-literal` argument to define a literal value from the command line:

```
kubectl create configmap special-config --from-literal=special.how=very --from-literal=special.type=charm
```

You can pass in multiple key-value pairs. Each pair provided on the command line is represented as a separate entry in the data section of the ConfigMap.

```
kubectl get configmaps special-config -o yaml
```

The output is similar to this:

```
apiVersion: v1
kind: ConfigMap
metadata:
 creationTimestamp: 2022-02-18T19:14:38Z
 name: special-config
 namespace: default
 resourceVersion: "651"
 uid: dadce046-d673-11e5-8cd0-68f728db1985
data:
 special.how: very
 special.type: charm
```

## Create a ConfigMap from generator

You can also create a ConfigMap from generators and then apply it to create the object in the cluster's API server. You should specify the generators in a kustomization.yaml file within a directory.

### Generate ConfigMaps from files

For example, to generate a ConfigMap from files `configure-pod-container/configmap/game.properties`

```
Create a kustomization.yaml file with ConfigMapGenerator
cat <<EOF >./kustomization.yaml
configMapGenerator:
- name: game-config-4
 options:
 labels:
 game-config: config-4
 files:
 - configure-pod-container/configmap/game.properties
EOF
```

Apply the kustomization directory to create the ConfigMap object:

```
kubectl apply -k .
```

```
configmap/game-config-4-m9dm2f92bt created
```

You can check that the ConfigMap was created like this:

```
kubectl get configmap
```

NAME	DATA	AGE
game-config-4-m9dm2f92bt	1	37s

and also:

```
kubectl describe configmaps/game-config-4-m9dm2f92bt
```

```
Name: game-config-4-m9dm2f92bt
Namespace: default
Labels: game-config=config-4
Annotations: kubectl.kubernetes.io/last-applied-configuration:
 {"apiVersion":"v1","data":
{"game.properties":{"enemies=aliens\nlives=3\nenemies.cheat=true\nenemies.cheat.level=noGoo
dRotten\nsecret.code.p...

Data
====
game.properties:

enemies=aliens
lives=3
enemies.cheat=true
```

```
enemies.cheat.level=noGoodRotten
secret.code.passphrase=UUDDLRLRBABAS
secret.code.allowed=true
secret.code.lives=30
Events: <none>
```

Notice that the generated ConfigMap name has a suffix appended by hashing the contents. This ensures that a new ConfigMap is generated each time the content is modified.

### Define the key to use when generating a ConfigMap from a file

You can define a key other than the file name to use in the ConfigMap generator. For example, to generate a ConfigMap from files `configure-pod-container/configmap/game.properties` with the key `game-special-key`

```
Create a kustomization.yaml file with ConfigMapGenerator
cat <<EOF >./kustomization.yaml
configMapGenerator:
- name: game-config-5
 options:
 labels:
 game-config: config-5
 files:
 - game-special-key=configure-pod-container/configmap/game.properties
EOF
```

Apply the kustomization directory to create the ConfigMap object.

```
kubectl apply -k .
```

```
configmap/game-config-5-m67dt67794 created
```

### Generate ConfigMaps from literals

This example shows you how to create a ConfigMap from two literal key/value pairs: `special.type=charm` and `special.how=very`, using Kustomize and kubectl. To achieve this, you can specify the ConfigMap generator. Create (or replace) `kustomization.yaml` so that it has the following contents:

```

kustomization.yaml contents for creating a ConfigMap from literals
configMapGenerator:
- name: special-config-2
 literals:
 - special.how=very
 - special.type=charm
```

Apply the kustomization directory to create the ConfigMap object:

```
kubectl apply -k .
```

```
configmap/special-config-2-c92b5mmcf2 created
```

## Interim cleanup

Before proceeding, clean up some of the ConfigMaps you made:

```
kubectrl delete configmap special-config
kubectrl delete configmap env-config
kubectrl delete configmap -l 'game-config in (config-4,config-5)'
```

Now that you have learned to define ConfigMaps, you can move on to the next section, and learn how to use these objects with Pods.

---

## Define container environment variables using ConfigMap data

### Define a container environment variable with data from a single ConfigMap

1. Define an environment variable as a key-value pair in a ConfigMap:

```
kubectrl create configmap special-config --from-literal=special.how=very
```

2. Assign the special.how value defined in the ConfigMap to the SPECIAL\_LEVEL\_KEY environment variable in the Pod specification.

[pods/pod-single-configmap-env-variable.yaml](https://kubernetes.io/examples/pods/pod-single-configmap-env-variable.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: dapi-test-pod
spec:
 containers:
 - name: test-container
 image: registry.k8s.io/busybox
 command: ["/bin/sh", "-c", "env"]
 env:
 # Define the environment variable
 - name: SPECIAL_LEVEL_KEY
 valueFrom:
 configMapKeyRef:
 # The ConfigMap containing the value you want to assign to
 SPECIAL_LEVEL_KEY
 name: special-config
 # Specify the key associated with the value
 key: special.how
 restartPolicy: Never
```

Create the Pod:

```
kubectrl create -f https://kubernetes.io/examples/pods/pod-single-configmap-env-
variable.yaml
```

Now, the Pod's output includes environment variable `SPECIAL_LEVEL_KEY=very`.

## Define container environment variables with data from multiple ConfigMaps

As with the previous example, create the ConfigMaps first. Here is the manifest you will use:

[configmap/configmaps.yaml](#)

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: special-config
 namespace: default
data:
 special.how: very

apiVersion: v1
kind: ConfigMap
metadata:
 name: env-config
 namespace: default
data:
 log_level: INFO
```

- Create the ConfigMap:

```
kubectl create -f https://kubernetes.io/examples/configmap/configmaps.yaml
```

- Define the environment variables in the Pod specification.

[pods/pod-multiple-configmap-env-variable.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: dapi-test-pod
spec:
 containers:
 - name: test-container
 image: registry.k8s.io/busybox
 command: ["/bin/sh", "-c", "env"]
 env:
 - name: SPECIAL_LEVEL_KEY
 valueFrom:
 configMapKeyRef:
 name: special-config
 key: special.how
 - name: LOG_LEVEL
 valueFrom:
 configMapKeyRef:
 name: env-config
```

```
 key: log_level
restartPolicy: Never
```

Create the Pod:

```
kubectl create -f https://kubernetes.io/examples/pods/pod-multiple-configmap-env-variable.yaml
```

Now, the Pod's output includes environment variables SPECIAL\_LEVEL\_KEY=very and LOG\_LEVEL=INFO.

Once you're happy to move on, delete that Pod:

```
kubectl delete pod dapi-test-pod --now
```

## Configure all key-value pairs in a ConfigMap as container environment variables

- Create a ConfigMap containing multiple key-value pairs.

[configmap/configmap-multikeys.yaml](https://kubernetes.io/examples/configmap/configmap-multikeys.yaml)

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: special-config
 namespace: default
data:
 SPECIAL_LEVEL: very
 SPECIAL_TYPE: charm
```

Create the ConfigMap:

```
kubectl create -f https://kubernetes.io/examples/configmap/configmap-multikeys.yaml
```

- Use envFrom to define all of the ConfigMap's data as container environment variables. The key from the ConfigMap becomes the environment variable name in the Pod.

[pods/pod-configmap-envFrom.yaml](https://kubernetes.io/examples/pods/pod-configmap-envFrom.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: dapi-test-pod
spec:
 containers:
 - name: test-container
 image: registry.k8s.io/busybox
 command: ["/bin/sh", "-c", "env"]
 envFrom:
 - configMapRef:
 name: special-config
 restartPolicy: Never
```



Create the Pod:

```
kubectl create -f https://kubernetes.io/examples/pods/pod-configmap-envFrom.yaml
```

Now, the Pod's output includes environment variables `SPECIAL_LEVEL=very` and `SPECIAL_TYPE=charm`.

Once you're happy to move on, delete that Pod:

```
kubectl delete pod dapi-test-pod --now
```

## Use ConfigMap-defined environment variables in Pod commands

You can use ConfigMap-defined environment variables in the command and args of a container using the `$(VAR_NAME)` Kubernetes substitution syntax.

For example, the following Pod manifest:

[pods/pod-configmap-env-var-valueFrom.yaml](https://kubernetes.io/examples/pods/pod-configmap-env-var-valueFrom.yaml)

```
apiVersion: v1
kind: Pod
metadata:
 name: dapi-test-pod
spec:
 containers:
 - name: test-container
 image: registry.k8s.io/busybox
 command: ["/bin/echo", "$(SPECIAL_LEVEL_KEY) $(SPECIAL_TYPE_KEY)"]
 env:
 - name: SPECIAL_LEVEL_KEY
 valueFrom:
 configMapKeyRef:
 name: special-config
 key: SPECIAL_LEVEL
 - name: SPECIAL_TYPE_KEY
 valueFrom:
 configMapKeyRef:
 name: special-config
 key: SPECIAL_TYPE
 restartPolicy: Never
```

Create that Pod, by running:

```
kubectl create -f https://kubernetes.io/examples/pods/pod-configmap-env-var-valueFrom.yaml
```

That pod produces the following output from the test-container container:

```
kubectl logs dapi-test-pod
```

```
very charm
```

Once you're happy to move on, delete that Pod:

```
kubectl delete pod dapi-test-pod --now
```

## Add ConfigMap data to a Volume

As explained in [Create ConfigMaps from files](#), when you create a ConfigMap using `--from-file`, the filename becomes a key stored in the data section of the ConfigMap. The file contents become the key's value.

The examples in this section refer to a ConfigMap named `special-config`:

[configmap/configmap-multikeys.yaml](#)

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: special-config
 namespace: default
data:
 SPECIAL_LEVEL: very
 SPECIAL_TYPE: charm
```

Create the ConfigMap:

```
kubectl create -f https://kubernetes.io/examples/configmap/configmap-multikeys.yaml
```

## Populate a Volume with data stored in a ConfigMap

Add the ConfigMap name under the `volumes` section of the Pod specification. This adds the ConfigMap data to the directory specified as `volumeMounts.mountPath` (in this case, `/etc/config`). The `command` section lists directory files with names that match the keys in ConfigMap.

[pods/pod-configmap-volume.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: dapi-test-pod
spec:
 containers:
 - name: test-container
 image: registry.k8s.io/busybox
 command: ["/bin/sh", "-c", "ls /etc/config/"]
 volumeMounts:
 - name: config-volume
 mountPath: /etc/config
 volumes:
 - name: config-volume
 configMap:
 # Provide the name of the ConfigMap containing the files you want
 # to add to the container
```

```
name: special-config
restartPolicy: Never
```

Create the Pod:

```
kubectl create -f https://kubernetes.io/examples/pods/pod-configmap-volume.yaml
```

When the pod runs, the command `ls /etc/config/` produces the output below:

```
SPECIAL_LEVEL
SPECIAL_TYPE
```

Text data is exposed as files using the UTF-8 character encoding. To use some other character encoding, use `binaryData` (see [ConfigMap object](#) for more details).

### Note:

If there are any files in the `/etc/config` directory of that container image, the volume mount will make those files from the image inaccessible.

Once you're happy to move on, delete that Pod:

```
kubectl delete pod dapi-test-pod --now
```

## Add ConfigMap data to a specific path in the Volume

Use the `path` field to specify the desired file path for specific ConfigMap items. In this case, the `SPECIAL_LEVEL` item will be mounted in the `config-volume` volume at `/etc/config/keys`.

[pods/pod-configmap-volume-specific-key.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: dapi-test-pod
spec:
 containers:
 - name: test-container
 image: registry.k8s.io/busybox
 command: ["/bin/sh", "-c", "cat /etc/config/keys"]
 volumeMounts:
 - name: config-volume
 mountPath: /etc/config
 volumes:
 - name: config-volume
 configMap:
 name: special-config
 items:
 - key: SPECIAL_LEVEL
 path: keys
 restartPolicy: Never
```

Create the Pod:

```
kubectl create -f https://kubernetes.io/examples/pods/pod-configmap-volume-specific-key.yaml
```

When the pod runs, the command `cat /etc/config/keys` produces the output below:

```
very
```

### Caution:

Like before, all previous files in the `/etc/config/` directory will be deleted.

Delete that Pod:

```
kubectl delete pod dapi-test-pod --now
```

## Project keys to specific paths and file permissions

You can project keys to specific paths. Refer to the corresponding section in the [Secrets](#) guide for the syntax.

You can set POSIX permissions for keys. Refer to the corresponding section in the [Secrets](#) guide for the syntax.

## Optional references

A ConfigMap reference may be marked *optional*. If the ConfigMap is non-existent, the mounted volume will be empty. If the ConfigMap exists, but the referenced key is non-existent, the path will be absent beneath the mount point. See [Optional ConfigMaps](#) for more details.

## Mounted ConfigMaps are updated automatically

When a mounted ConfigMap is updated, the projected content is eventually updated too. This applies in the case where an optionally referenced ConfigMap comes into existence after a pod has started.

Kubelet checks whether the mounted ConfigMap is fresh on every periodic sync. However, it uses its local TTL-based cache for getting the current value of the ConfigMap. As a result, the total delay from the moment when the ConfigMap is updated to the moment when new keys are projected to the pod can be as long as kubelet sync period (1 minute by default) + TTL of ConfigMaps cache (1 minute by default) in kubelet. You can trigger an immediate refresh by updating one of the pod's annotations.

### Note:

A container using a ConfigMap as a [subPath](#) volume will not receive ConfigMap updates.

## Understanding ConfigMaps and Pods

The ConfigMap API resource stores configuration data as key-value pairs. The data can be consumed in pods or provide the configurations for system components such as controllers. ConfigMap is similar to [Secrets](#), but provides a means of working with strings that don't contain sensitive information. Users and system components alike can store configuration data in ConfigMap.

## Note:

ConfigMaps should reference properties files, not replace them. Think of the ConfigMap as representing something similar to the Linux `/etc` directory and its contents. For example, if you create a [Kubernetes Volume](#) from a ConfigMap, each data item in the ConfigMap is represented by an individual file in the volume.

The ConfigMap's `data` field contains the configuration data. As shown in the example below, this can be simple (like individual properties defined using `--from-literal`) or complex (like configuration files or JSON blobs defined using `--from-file`).

```
apiVersion: v1
kind: ConfigMap
metadata:
 creationTimestamp: 2016-02-18T19:14:38Z
 name: example-config
 namespace: default
data:
 # example of a simple property defined using --from-literal
 example.property.1: hello
 example.property.2: world
 # example of a complex property defined using --from-file
 example.property.file: |-
 property.1=value-1
 property.2=value-2
 property.3=value-3
```

When `kubectl` creates a ConfigMap from inputs that are not ASCII or UTF-8, the tool puts these into the `binaryData` field of the ConfigMap, and not in `data`. Both text and binary data sources can be combined in one ConfigMap.

If you want to view the `binaryData` keys (and their values) in a ConfigMap, you can run `kubectl get configmap -o jsonpath='{.binaryData}' <name>`.

Pods can load data from a ConfigMap that uses either `data` or `binaryData`.

## Optional ConfigMaps

You can mark a reference to a ConfigMap as *optional* in a Pod specification. If the ConfigMap doesn't exist, the configuration for which it provides data in the Pod (for example: environment variable, mounted volume) will be empty. If the ConfigMap exists, but the referenced key is non-existent the data is also empty.

For example, the following Pod specification marks an environment variable from a ConfigMap as optional:

```
apiVersion: v1
kind: Pod
metadata:
 name: dapi-test-pod
spec:
 containers:
 - name: test-container
```

```
image: gcr.io/google_containers/busybox
command: ["/bin/sh", "-c", "env"]
env:
- name: SPECIAL_LEVEL_KEY
 valueFrom:
 configMapKeyRef:
 name: a-config
 key: akey
 optional: true # mark the variable as optional
restartPolicy: Never
```

If you run this pod, and there is no ConfigMap named a-config, the output is empty. If you run this pod, and there is a ConfigMap named a-config but that ConfigMap doesn't have a key named akey, the output is also empty. If you do set a value for akey in the a-config ConfigMap, this pod prints that value and then terminates.

You can also mark the volumes and files provided by a ConfigMap as optional. Kubernetes always creates the mount paths for the volume, even if the referenced ConfigMap or key doesn't exist. For example, the following Pod specification marks a volume that references a ConfigMap as optional:

```
apiVersion: v1
kind: Pod
metadata:
 name: dapi-test-pod
spec:
 containers:
 - name: test-container
 image: gcr.io/google_containers/busybox
 command: ["/bin/sh", "-c", "ls /etc/config"]
 volumeMounts:
 - name: config-volume
 mountPath: /etc/config
 volumes:
 - name: config-volume
 configMap:
 name: no-config
 optional: true # mark the source ConfigMap as optional
 restartPolicy: Never
```

## Restrictions

- You must create the ConfigMap object before you reference it in a Pod specification. Alternatively, mark the ConfigMap reference as optional in the Pod spec (see [Optional ConfigMaps](#)). If you reference a ConfigMap that doesn't exist and you don't mark the reference as optional, the Pod won't start. Similarly, references to keys that don't exist in the ConfigMap will also prevent the Pod from starting, unless you mark the key references as optional.
- If you use envFrom to define environment variables from ConfigMaps, keys that are considered invalid will be skipped. The pod will be allowed to start, but the invalid names will be recorded in the event log (InvalidVariableNames). The log message lists each skipped key. For example:

```
kubectl get events
```

The output is similar to this:

LASTSEEN	FIRSTSEEN	COUNT	NAME	KIND	SUBOBJECT	TYPE
REASON			SOURCE			MESSAGE
0s	0s	1	dapi-test-pod	Pod		Warning
InvalidEnvironmentVariableNames {kubelet, 127.0.0.1} Keys [1badkey, 2alsobad] from the EnvFrom configMap default/myconfig were skipped since they are considered invalid environment variable names.						

- ConfigMaps reside in a specific [Namespace](#). Pods can only refer to ConfigMaps that are in the same namespace as the Pod.
- You can't use ConfigMaps for [static pods](#), because the kubelet does not support this.

## Cleaning up

Delete the ConfigMaps and Pods that you made:

```
kubectl delete configmaps/game-config configmaps/game-config-2 configmaps/game-config-3 \
 configmaps/game-config-env-file
kubectl delete pod dapi-test-pod --now
```

```
You might already have removed the next set
kubectl delete configmaps/special-config configmaps/env-config
kubectl delete configmap -l 'game-config in (config-4,config-5)'
```

If you created a directory `configure-pod-container` and no longer need it, you should remove that too, or move it into the trash can / deleted files location.

## What's next

- Follow a real world example of [Configuring Redis using a ConfigMap](#).
- Follow an example of [Updating configuration via a ConfigMap](#).

# Share Process Namespace between Containers in a Pod

This page shows how to configure process namespace sharing for a pod. When process namespace sharing is enabled, processes in a container are visible to all other containers in the same pod.

You can use this feature to configure cooperating containers, such as a log handler sidecar container, or to troubleshoot container images that don't include debugging utilities like a shell.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

## Configure a Pod

Process namespace sharing is enabled using the `shareProcessNamespace` field of `.spec` for a Pod. For example:

[pods/share-process-namespace.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: nginx
spec:
 shareProcessNamespace: true
 containers:
 - name: nginx
 image: nginx
 - name: shell
 image: busybox:1.28
 command: ["sleep", "3600"]
 securityContext:
 capabilities:
 add:
 - SYS_PTRACE
 stdin: true
 tty: true
```

1. Create the pod nginx on your cluster:

```
kubectl apply -f https://k8s.io/examples/pods/share-process-namespace.yaml
```

2. Attach to the shell container and run ps:

```
kubectl attach -it nginx -c shell
```

If you don't see a command prompt, try pressing enter. In the container shell:

```
run this inside the "shell" container
ps ax
```

The output is similar to this:

```
PID USER TIME COMMAND
1 root 0:00 /pause
```



```
8 root 0:00 nginx: master process nginx -g daemon off;
14 101 0:00 nginx: worker process
15 root 0:00 sh
21 root 0:00 ps ax
```

You can signal processes in other containers. For example, send `SIGHUP` to `nginx` to restart the worker process. This requires the `SYS_PTRACE` capability.

```
run this inside the "shell" container
kill -HUP 8 # change "8" to match the PID of the nginx leader process, if necessary
ps ax
```

The output is similar to this:

```
PID USER TIME COMMAND
 1 root 0:00 /pause
 8 root 0:00 nginx: master process nginx -g daemon off;
15 root 0:00 sh
22 101 0:00 nginx: worker process
23 root 0:00 ps ax
```

It's even possible to access the file system of another container using the `/proc/$pid/root` link.

```
run this inside the "shell" container
change "8" to the PID of the Nginx process, if necessary
head /proc/8/root/etc/nginx/nginx.conf
```

The output is similar to this:

```
user nginx;
worker_processes 1;

error_log /var/log/nginx/error.log warn;
pid /var/run/nginx.pid;

events {
 worker_connections 1024;
```

## Understanding process namespace sharing

Pods share many resources so it makes sense they would also share a process namespace. Some containers may expect to be isolated from others, though, so it's important to understand the differences:

1. **The container process no longer has PID 1.** Some containers refuse to start without PID 1 (for example, containers using `systemd`) or run commands like `kill -HUP 1` to signal the container process. In pods with a shared process namespace, `kill -HUP 1` will signal the pod sandbox (`/pause` in the above example).
2. **Processes are visible to other containers in the pod.** This includes all information visible in `/proc`, such as passwords that were passed as arguments or environment variables. These are protected only by regular Unix permissions.

- Container filesystems are visible to other containers in the pod through the /proc/\$pid/root link.** This makes debugging easier, but it also means that filesystem secrets are protected only by filesystem permissions.

# Use a User Namespace With a Pod

FEATURE STATE: Kubernetes v1.30 [beta]

This page shows how to configure a user namespace for pods. This allows you to isolate the user running inside the container from the one in the host.

A process running as root in a container can run as a different (non-root) user in the host; in other words, the process has full privileges for operations inside the user namespace, but is unprivileged for operations outside the namespace.

You can use this feature to reduce the damage a compromised container can do to the host or other pods in the same node. There are [several security vulnerabilities](#) rated either **HIGH** or **CRITICAL** that were not exploitable when user namespaces is active. It is expected user namespace will mitigate some future vulnerabilities too.

Without using a user namespace a container running as root, in the case of a container breakout, has root privileges on the node. And if some capability were granted to the container, the capabilities are valid on the host too. None of this is true when user namespaces are used.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercoda](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.25. To check the version, enter kubectl version.

This item links to a third party project or product that is not part of Kubernetes itself. [More information](#)

- The node OS needs to be Linux
- You need to exec commands in the host
- You need to be able to exec into pods
- You need to enable the UserNamespacesSupport [feature gate](#)

### Note:

The feature gate to enable user namespaces was previously named UserNamespacesStatelessPodsSupport, when only stateless pods were supported. Only Kubernetes v1.25 through to v1.27 recognise UserNamespacesStatelessPodsSupport.

The cluster that you're using **must** include at least one node that meets the [requirements](#) for using user namespaces with Pods.

If you have a mixture of nodes and only some of the nodes provide user namespace support for Pods, you also need to ensure that the user namespace Pods are [scheduled](#) to suitable nodes.

## Run a Pod that uses a user namespace

A user namespace for a pod is enabled setting the `hostUsers` field of `.spec` to `false`. For example:

[pods/user-namespaces-stateless.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: userns
spec:
 hostUsers: false
 containers:
 - name: shell
 command: ["sleep", "infinity"]
 image: debian
```

1. Create the pod on your cluster:

```
kubectl apply -f https://k8s.io/examples/pods/user-namespaces-stateless.yaml
```

2. Attach to the container and run `readlink /proc/self/ns/user`:

```
kubectl attach -it userns bash
```

Run this command:

```
readlink /proc/self/ns/user
```

The output is similar to:

```
user:[4026531837]
```

Also run:

```
cat /proc/self/uid_map
```

The output is similar to:

```
0 833617920 65536
```

Then, open a shell in the host and run the same commands.

The `readlink` command shows the user namespace the process is running in. It should be different when it is run on the host and inside the container.

The last number of the `uid_map` file inside the container must be 65536, on the host it must be a bigger number.

If you are running the kubelet inside a user namespace, you need to compare the output from running the command in the pod to the output of running in the host:

```
readlink /proc/$pid/ns/user
```

replacing \$pid with the kubelet PID.

## Create static Pods

*Static Pods* are managed directly by the kubelet daemon on a specific node, without the [API server](#) observing them. Unlike Pods that are managed by the control plane (for example, a [Deployment](#)); instead, the kubelet watches each static Pod (and restarts it if it fails).

Static Pods are always bound to one [Kubelet](#) on a specific node.

The kubelet automatically tries to create a [mirror Pod](#) on the Kubernetes API server for each static Pod. This means that the Pods running on a node are visible on the API server, but cannot be controlled from there. The Pod names will be suffixed with the node hostname with a leading hyphen.

### Note:

If you are running clustered Kubernetes and are using static Pods to run a Pod on every node, you should probably be using a [DaemonSet](#) instead.

### Note:

The spec of a static Pod cannot refer to other API objects (e.g., [ServiceAccount](#), [ConfigMap](#), [Secret](#), etc).

### Note:

Static pods do not support [ephemeral containers](#).

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

This page assumes you're using [CRI-O](#) to run Pods, and that your nodes are running the Fedora operating system. Instructions for other distributions or Kubernetes installations may vary.

# Create a static pod

You can configure a static Pod with either a [file system hosted configuration file](#) or a [web hosted configuration file](#).

## Filesystem-hosted static Pod manifest

Manifests are standard Pod definitions in JSON or YAML format in a specific directory. Use the `staticPodPath: <the directory>` field in the [kubelet configuration file](#), which periodically scans the directory and creates/deletes static Pods as YAML/JSON files appear/disappear there. Note that the kubelet will ignore files starting with dots when scanning the specified directory.

For example, this is how to start a simple web server as a static Pod:

1. Choose a node where you want to run the static Pod. In this example, it's `my-node1`.

```
ssh my-node1
```

2. Choose a directory, say `/etc/kubernetes/manifests` and place a web server Pod definition there, for example `/etc/kubernetes/manifests/static-web.yaml`:

```
Run this command on the node where kubelet is running
mkdir -p /etc/kubernetes/manifests/
cat <<EOF >/etc/kubernetes/manifests/static-web.yaml
apiVersion: v1
kind: Pod
metadata:
 name: static-web
 labels:
 role: myrole
spec:
 containers:
 - name: web
 image: nginx
 ports:
 - name: web
 containerPort: 80
 protocol: TCP
EOF
```

3. Configure the kubelet on that node to set a `staticPodPath` value in the [kubelet configuration file](#).  
See [Set Kubelet Parameters Via A Configuration File](#) for more information.

An alternative and deprecated method is to configure the kubelet on that node to look for static Pod manifests locally, using a command line argument. To use the deprecated approach, start the kubelet with the `--pod-manifest-path=/etc/kubernetes/manifests/` argument.

4. Restart the kubelet. On Fedora, you would run:

```
Run this command on the node where the kubelet is running
systemctl restart kubelet
```

## Web-hosted static pod manifest

Kubelet periodically downloads a file specified by `--manifest-url=<URL>` argument and interprets it as a JSON/YAML file that contains Pod definitions. Similar to how [filesystem-hosted manifests](#) work, the kubelet refetches the manifest on a schedule. If there are changes to the list of static Pods, the kubelet applies them.

To use this approach:

1. Create a YAML file and store it on a web server so that you can pass the URL of that file to the kubelet.

```
apiVersion: v1
kind: Pod
metadata:
 name: static-web
 labels:
 role: myrole
spec:
 containers:
 - name: web
 image: nginx
 ports:
 - name: web
 containerPort: 80
 protocol: TCP
```

2. Configure the kubelet on your selected node to use this web manifest by running it with `--manifest-url=<manifest-url>`. On Fedora, edit `/etc/kubernetes/kubelet` to include this line:

```
KUBELET_ARGS="--cluster-dns=10.254.0.10 --cluster-domain=kube.local --manifest-
url=<manifest-url>"
```

3. Restart the kubelet. On Fedora, you would run:

```
Run this command on the node where the kubelet is running
systemctl restart kubelet
```

## Observe static pod behavior

When the kubelet starts, it automatically starts all defined static Pods. As you have defined a static Pod and restarted the kubelet, the new static Pod should already be running.

You can view running containers (including static Pods) by running (on the node):

```
Run this command on the node where the kubelet is running
crictl ps
```

The output might be something like:

CONTAINER POD ID	IMAGE	CREATED	STATE	NAME	ATTEMPT
---------------------	-------	---------	-------	------	---------

```
129fd7d382018 docker.io/library/nginx@sha256:... 11 minutes ago Running web 0
34533c6729106
```

**Note:**

crictl outputs the image URI and SHA-256 checksum. NAME will look more like: docker.io/library/nginx@sha256:0d17b565c37bcbd895e9d92315a05c1c3c9a29f762b011a10c54a66cd53c9b31.

You can see the mirror Pod on the API server:

```
kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
static-web-my-node1	1/1	Running	0	2m

**Note:**

Make sure the kubelet has permission to create the mirror Pod in the API server. If not, the creation request is rejected by the API server.

[Labels](#) from the static Pod are propagated into the mirror Pod. You can use those labels as normal via [selectors](#), etc.

If you try to use kubectl to delete the mirror Pod from the API server, the kubelet *doesn't* remove the static Pod:

```
kubectl delete pod static-web-my-node1
```

```
pod "static-web-my-node1" deleted
```

You can see that the Pod is still running:

```
kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
static-web-my-node1	1/1	Running	0	4s

Back on your node where the kubelet is running, you can try to stop the container manually. You'll see that, after a time, the kubelet will notice and will restart the Pod automatically:

```
Run these commands on the node where the kubelet is running
crictl stop 129fd7d382018 # replace with the ID of your container
sleep 20
crictl ps
```

CONTAINER POD ID	IMAGE	CREATED	STATE	NAME	ATTEMPT
89db4553e1eeb 34533c6729106	docker.io/library/nginx@sha256:...	19 seconds ago	Running	web	1

Once you identify the right container, you can get the logs for that container with crictl:

```
Run these commands on the node where the container is running
crictl logs <container_id>
```

```
10.240.0.48 - - [16/Nov/2022:12:45:49 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.47.0" "-"
10.240.0.48 - - [16/Nov/2022:12:45:50 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.47.0" "-"
10.240.0.48 - - [16/Nov/2022:12:45:51 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.47.0" "-"
```

To find more about how to debug using `crictl`, please visit [Debugging Kubernetes nodes with crictl](#).

## Dynamic addition and removal of static pods

The running kubelet periodically scans the configured directory (`/etc/kubernetes/manifests` in our example) for changes and adds/removes Pods as files appear/disappear in this directory.

```
This assumes you are using filesystem-hosted static Pod configuration
Run these commands on the node where the container is running
#
mv /etc/kubernetes/manifests/static-web.yaml /tmp
sleep 20
crictl ps
You see that no nginx container is running
mv /tmp/static-web.yaml /etc/kubernetes/manifests/
sleep 20
crictl ps
```

CONTAINER POD ID	IMAGE	CREATED	STATE	NAME	ATTEMPT
f427638871c35 34533c6729106	docker.io/library/nginx@sha256:...	19 seconds ago	Running	web	1

## What's next

- [Generate static Pod manifests for control plane components](#)
- [Generate static Pod manifest for local etcd](#)
- [Debugging Kubernetes nodes with crictl](#)
- [Learn more about crictl](#).
- [Map docker CLI commands to crictl](#).
- [Set up etcd instances as static pods managed by a kubelet](#)

## Translate a Docker Compose File to Kubernetes Resources

What's Kompose? It's a conversion tool for all things compose (namely Docker Compose) to container orchestrators (Kubernetes or OpenShift).

More information can be found on the Kompose website at <http://kompose.io>.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at



least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Install Kompose

We have multiple ways to install Kompose. Our preferred method is downloading the binary from the latest GitHub release.

- [GitHub download](#)
- [Build from source](#)
- [CentOS package](#)
- [Fedora package](#)
- [Homebrew \(macOS\)](#)

Kompose is released via GitHub on a three-week cycle, you can see all current releases on the [GitHub release page](#).

```
Linux
curl -L https://github.com/kubernetes/kompose/releases/download/v1.26.0/kompose-linux-amd64 -o kompose

macOS
curl -L https://github.com/kubernetes/kompose/releases/download/v1.26.0/kompose-darwin-amd64 -o kompose

Windows
curl -L https://github.com/kubernetes/kompose/releases/download/v1.26.0/kompose-windows-amd64.exe -o kompose.exe

chmod +x kompose
sudo mv ./kompose /usr/local/bin/kompose
```

Alternatively, you can download the [tarball](#).

Installing using go get pulls from the master branch with the latest development changes.

```
go get -u github.com/kubernetes/kompose
```

Kompose is in [EPEL](#) CentOS repository. If you don't have [EPEL](#) repository already installed and enabled you can do it by running `sudo yum install epel-release`.

If you have [EPEL](#) enabled in your system, you can install Kompose like any other package.

```
sudo yum -y install kompose
```

Kompose is in Fedora 24, 25 and 26 repositories. You can install it like any other package.

```
sudo dnf -y install kompose
```

On macOS you can install the latest release via [Homebrew](#):

```
brew install kompose
```

## Use Kompose

In a few steps, we'll take you from Docker Compose to Kubernetes. All you need is an existing docker-compose.yml file.

1. Go to the directory containing your docker-compose.yml file. If you don't have one, test using this one.

```
version: "2"

services:

 redis-master:
 image: registry.k8s.io/redis:e2e
 ports:
 - "6379"

 redis-slave:
 image: gcr.io/google_samples/gb-redisslave:v3
 ports:
 - "6379"
 environment:
 - GET_HOSTS_FROM=dns

 frontend:
 image: gcr.io/google-samples/gb-frontend:v4
 ports:
 - "80:80"
 environment:
 - GET_HOSTS_FROM=dns
 labels:
 kompose.service.type: LoadBalancer
```

2. To convert the docker-compose.yml file to files that you can use with kubectl, run kompose convert and then kubectl apply -f <output file>.

```
kompose convert
```

The output is similar to:

```
INFO Kubernetes file "frontend-tcp-service.yaml" created
INFO Kubernetes file "redis-master-service.yaml" created
INFO Kubernetes file "redis-slave-service.yaml" created
INFO Kubernetes file "frontend-deployment.yaml" created
INFO Kubernetes file "redis-master-deployment.yaml" created
INFO Kubernetes file "redis-slave-deployment.yaml" created
```

```
kubectl apply -f frontend-tcp-service.yaml,redis-master-service.yaml,redis-slave-
service.yaml,frontend-deployment.yaml,redis-master-deployment.yaml,redis-slave-
deployment.yaml
```

The output is similar to:

```
service/frontend-tcp created
service/redis-master created
service/redis-slave created
deployment.apps/frontend created
deployment.apps/redis-master created
deployment.apps/redis-slave created
```

Your deployments are running in Kubernetes.

### 3. Access your application.

If you're already using minikube for your development process:

```
minikube service frontend
```

Otherwise, let's look up what IP your service is using!

```
kubectl describe svc frontend
```

```
Name: frontend-tcp
Namespace: default
Labels: io.kompose.service=frontend-tcp
Annotations: kompose.cmd: kompose convert
 kompose.service.type: LoadBalancer
 kompose.version: 1.26.0 (40646f47)
Selector: io.kompose.service=frontend
Type: LoadBalancer
IP Family Policy: SingleStack
IP Families: IPv4
IP: 10.43.67.174
IPs: 10.43.67.174
Port: 80 80/TCP
TargetPort: 80/TCP
NodePort: 80 31254/TCP
Endpoints: 10.42.0.25:80
Session Affinity: None
External Traffic Policy: Cluster
Events:
 Type Reason Age From Message
 ---- -
Normal EnsuringLoadBalancer 62s service-controller Ensuring load balancer
Normal AppliedDaemonSet 62s service-controller Applied LoadBalancer
DaemonSet kube-system/svclb-frontend-tcp-9362d276
```

If you're using a cloud provider, your IP will be listed next to LoadBalancer Ingress.

```
curl http://192.0.2.89
```

Clean-up.

4.

After you are finished testing out the example application deployment, simply run the following command in your shell to delete the resources used.

```
kubectl delete -f frontend-tcp-service.yaml,redis-master-service.yaml,redis-slave-
service.yaml,frontend-deployment.yaml,redis-master-deployment.yaml,redis-slave-
deployment.yaml
```

## User Guide

- CLI
  - [kompose convert](#)
- Documentation
  - [Alternative Conversions](#)
  - [Labels](#)
  - [Restart](#)
  - [Docker Compose Versions](#)

Kompose has support for two providers: OpenShift and Kubernetes. You can choose a targeted provider using global option `--provider`. If no provider is specified, Kubernetes is set by default.

## kompose convert

Kompose supports conversion of V1, V2, and V3 Docker Compose files into Kubernetes and OpenShift objects.

### Kubernetes kompose convert example

```
kompose --file docker-voting.yml convert
```

```
WARN Unsupported key networks - ignoring
WARN Unsupported key build - ignoring
INFO Kubernetes file "worker-svc.yaml" created
INFO Kubernetes file "db-svc.yaml" created
INFO Kubernetes file "redis-svc.yaml" created
INFO Kubernetes file "result-svc.yaml" created
INFO Kubernetes file "vote-svc.yaml" created
INFO Kubernetes file "redis-deployment.yaml" created
INFO Kubernetes file "result-deployment.yaml" created
INFO Kubernetes file "vote-deployment.yaml" created
INFO Kubernetes file "worker-deployment.yaml" created
INFO Kubernetes file "db-deployment.yaml" created
```

```
ls
```

```
db-deployment.yaml docker-compose.yml docker-gitlab.yml redis-deployment.yaml
result-deployment.yaml vote-deployment.yaml worker-deployment.yaml
db-svc.yaml docker-voting.yml redis-svc.yaml result-svc.yaml vote-
svc.yaml worker-svc.yaml
```

You can also provide multiple docker-compose files at the same time:

```
kompose -f docker-compose.yml -f docker-guestbook.yml convert
```

```
INFO Kubernetes file "frontend-service.yaml" created
INFO Kubernetes file "mlbparks-service.yaml" created
INFO Kubernetes file "mongodb-service.yaml" created
INFO Kubernetes file "redis-master-service.yaml" created
INFO Kubernetes file "redis-slave-service.yaml" created
INFO Kubernetes file "frontend-deployment.yaml" created
INFO Kubernetes file "mlbparks-deployment.yaml" created
INFO Kubernetes file "mongodb-deployment.yaml" created
INFO Kubernetes file "mongodb-claim0-persistentvolumeclaim.yaml" created
INFO Kubernetes file "redis-master-deployment.yaml" created
INFO Kubernetes file "redis-slave-deployment.yaml" created
```

```
ls
```

```
mlbparks-deployment.yaml mongodb-service.yaml redis-slave-
service.jsonmlbparks-service.yaml
frontend-deployment.yaml mongodb-claim0-persistentvolumeclaim.yaml redis-master-
service.yaml
frontend-service.yaml mongodb-deployment.yaml redis-slave-deployment.yaml
redis-master-deployment.yaml
```

When multiple docker-compose files are provided the configuration is merged. Any configuration that is common will be overridden by subsequent file.

## OpenShift kompose convert example

```
kompose --provider openshift --file docker-voting.yml convert
```

```
WARN [worker] Service cannot be created because of missing port.
INFO OpenShift file "vote-service.yaml" created
INFO OpenShift file "db-service.yaml" created
INFO OpenShift file "redis-service.yaml" created
INFO OpenShift file "result-service.yaml" created
INFO OpenShift file "vote-deploymentconfig.yaml" created
INFO OpenShift file "vote-imagestream.yaml" created
INFO OpenShift file "worker-deploymentconfig.yaml" created
INFO OpenShift file "worker-imagestream.yaml" created
INFO OpenShift file "db-deploymentconfig.yaml" created
INFO OpenShift file "db-imagestream.yaml" created
INFO OpenShift file "redis-deploymentconfig.yaml" created
INFO OpenShift file "redis-imagestream.yaml" created
INFO OpenShift file "result-deploymentconfig.yaml" created
INFO OpenShift file "result-imagestream.yaml" created
```

It also supports creating buildconfig for build directive in a service. By default, it uses the remote repo for the current git branch as the source repo, and the current branch as the source branch for the build. You can specify a different source repo and branch using `--build-repo` and `--build-branch` options respectively.

```
kompose --provider openshift --file buildconfig/docker-compose.yml convert
```

```
WARN [foo] Service cannot be created because of missing port.
INFO OpenShift Buildconfig using git@github.com:rtnpro/kompose.git::master as source.
INFO OpenShift file "foo-deploymentconfig.yaml" created
INFO OpenShift file "foo-imagestream.yaml" created
INFO OpenShift file "foo-buildconfig.yaml" created
```

#### Note:

If you are manually pushing the OpenShift artifacts using `oc create -f`, you need to ensure that you push the imagestream artifact before the buildconfig artifact, to workaround this OpenShift issue: <https://github.com/openshift/origin/issues/4518>.

## Alternative Conversions

The default kompose transformation will generate Kubernetes [Deployments](#) and [Services](#), in yaml format. You have alternative option to generate json with `-j`. Also, you can alternatively generate [Replication Controllers](#) objects, [Daemon Sets](#), or [Helm](#) charts.

```
kompose convert -j
INFO Kubernetes file "redis-svc.json" created
INFO Kubernetes file "web-svc.json" created
INFO Kubernetes file "redis-deployment.json" created
INFO Kubernetes file "web-deployment.json" created
```

The `*-deployment.json` files contain the Deployment objects.

```
kompose convert --replication-controller
INFO Kubernetes file "redis-svc.yaml" created
INFO Kubernetes file "web-svc.yaml" created
INFO Kubernetes file "redis-replicationcontroller.yaml" created
INFO Kubernetes file "web-replicationcontroller.yaml" created
```

The `*-replicationcontroller.yaml` files contain the Replication Controller objects. If you want to specify replicas (default is 1), use `--replicas` flag: `kompose convert --replication-controller --replicas 3`.

```
kompose convert --daemon-set
INFO Kubernetes file "redis-svc.yaml" created
INFO Kubernetes file "web-svc.yaml" created
INFO Kubernetes file "redis-daemonset.yaml" created
INFO Kubernetes file "web-daemonset.yaml" created
```

The `*-daemonset.yaml` files contain the DaemonSet objects.

If you want to generate a Chart to be used with [Helm](#) run:

```
kompose convert -c

INFO Kubernetes file "web-svc.yaml" created
INFO Kubernetes file "redis-svc.yaml" created
INFO Kubernetes file "web-deployment.yaml" created
INFO Kubernetes file "redis-deployment.yaml" created
chart created in "./docker-compose/"
```

```
tree docker-compose/
```

```
docker-compose
├── Chart.yaml
├── README.md
└── templates
 ├── redis-deployment.yaml
 ├── redis-svc.yaml
 ├── web-deployment.yaml
 └── web-svc.yaml
```

The chart structure is aimed at providing a skeleton for building your Helm charts.

## Labels

kompose supports Kompose-specific labels within the `docker-compose.yml` file in order to explicitly define a service's behavior upon conversion.

- `kompose.service.type` defines the type of service to be created.

For example:

```
version: "2"
services:
 nginx:
 image: nginx
 dockerfile: foobar
 build: ./foobar
 cap_add:
 - ALL
 container_name: foobar
 labels:
 kompose.service.type: nodeport
```

- `kompose.service.expose` defines if the service needs to be made accessible from outside the cluster or not. If the value is set to `"true"`, the provider sets the endpoint automatically, and for any other value, the value is set as the hostname. If multiple ports are defined in a service, the first one is chosen to be the exposed.
  - For the Kubernetes provider, an ingress resource is created and it is assumed that an ingress controller has already been configured.
  - For the OpenShift provider, a route is created.

For example:

```
version: "2"
services:
 web:
 image: tuna/docker-counter23
 ports:
 - "5000:5000"
 links:
 - redis
```

```
labels:
 kompose.service.expose: "counter.example.com"
redis:
 image: redis:3.0
ports:
 - "6379"
```

The currently supported options are:

Key	Value
kompose.service.type	nodeport / clusterip / loadbalancer
kompose.service.expose	true / hostname

**Note:**

The kompose.service.type label should be defined with ports only, otherwise kompose will fail.

## Restart

If you want to create normal pods without controllers you can use restart construct of docker-compose to define that. Follow table below to see what happens on the restart value.

docker-compose restart	object created	Pod restartPolicy
""	controller object	Always
always	controller object	Always
on-failure	Pod	OnFailure
no	Pod	Never

**Note:**

The controller object could be deployment or replicationcontroller.

For example, the pival service will become pod down here. This container calculated value of pi.

```
version: '2'

services:
 pival:
 image: perl
 command: ["perl", "-Mbignum=bpi", "-wle", "print bpi(2000)"]
 restart: "on-failure"
```

## Warning about Deployment Configurations

If the Docker Compose file has a volume specified for a service, the Deployment (Kubernetes) or DeploymentConfig (OpenShift) strategy is changed to "Recreate" instead of "RollingUpdate" (default). This is done to avoid multiple instances of a service from accessing a volume at the same time.



If the Docker Compose file has service name with `_` in it (for example, `web_service`), then it will be replaced by `-` and the service name will be renamed accordingly (for example, `web-service`). Kompose does this because "Kubernetes" doesn't allow `_` in object name.

Please note that changing service name might break some docker-compose files.

## Docker Compose Versions

Kompose supports Docker Compose versions: 1, 2 and 3. We have limited support on versions 2.1 and 3.2 due to their experimental nature.

A full list on compatibility between all three versions is listed in our [conversion document](#) including a list of all incompatible Docker Compose keys.

# Enforce Pod Security Standards by Configuring the Built-in Admission Controller

Kubernetes provides a built-in [admission controller](#) to enforce the [Pod Security Standards](#). You can configure this admission controller to set cluster-wide defaults and [exemptions](#).

## Before you begin

Following an alpha release in Kubernetes v1.22, Pod Security Admission became available by default in Kubernetes v1.23, as a beta. From version 1.25 onwards, Pod Security Admission is generally available.

To check the version, enter `kubectl version`.

If you are not running Kubernetes 1.30, you can switch to viewing this page in the documentation for the Kubernetes version that you are running.

## Configure the Admission Controller

### Note:

`pod-security.admission.config.k8s.io/v1` configuration requires v1.25+. For v1.23 and v1.24, use [v1beta1](#). For v1.22, use [v1alpha1](#).

```
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
- name: PodSecurity
 configuration:
 apiVersion: pod-security.admission.config.k8s.io/v1 # see compatibility note
 kind: PodSecurityConfiguration
 # Defaults applied when a mode label is not set.
 #
```

```
Level label values must be one of:
- "privileged" (default)
- "baseline"
- "restricted"
#
Version label values must be one of:
- "latest" (default)
- specific version like "v1.30"
defaults:
 enforce: "privileged"
 enforce-version: "latest"
 audit: "privileged"
 audit-version: "latest"
 warn: "privileged"
 warn-version: "latest"
exemptions:
 # Array of authenticated usernames to exempt.
 usernames: []
 # Array of runtime class names to exempt.
 runtimeClasses: []
 # Array of namespaces to exempt.
 namespaces: []
```

#### Note:

The above manifest needs to be specified via the `--admission-control-config-file` to kube-apiserver.

## Enforce Pod Security Standards with Namespace Labels

Namespaces can be labeled to enforce the [Pod Security Standards](#). The three policies [privileged](#), [baseline](#) and [restricted](#) broadly cover the security spectrum and are implemented by the [Pod Security admission controller](#).

### Before you begin

Pod Security Admission was available by default in Kubernetes v1.23, as a beta. From version 1.25 onwards, Pod Security Admission is generally available.

To check the version, enter `kubectl version`.

### Requiring the baseline Pod Security Standard with namespace labels

This manifest defines a Namespace `my-baseline-namespace` that:

- *Blocks* any pods that don't satisfy the baseline policy requirements.

- Generates a user-facing warning and adds an audit annotation to any created pod that does not meet the restricted policy requirements.
- Pins the versions of the baseline and restricted policies to v1.30.

```
apiVersion: v1
kind: Namespace
metadata:
 name: my-baseline-namespace
 labels:
 pod-security.kubernetes.io/enforce: baseline
 pod-security.kubernetes.io/enforce-version: v1.30

 # We are setting these to our _desired_ `enforce` level.
 pod-security.kubernetes.io/audit: restricted
 pod-security.kubernetes.io/audit-version: v1.30
 pod-security.kubernetes.io/warn: restricted
 pod-security.kubernetes.io/warn-version: v1.30
```

## Add labels to existing namespaces with kubectl label

### Note:

When an enforce policy (or version) label is added or changed, the admission plugin will test each pod in the namespace against the new policy. Violations are returned to the user as warnings.

It is helpful to apply the `--dry-run` flag when initially evaluating security profile changes for namespaces. The Pod Security Standard checks will still be run in *dry run* mode, giving you information about how the new policy would treat existing pods, without actually updating a policy.

```
kubectl label --dry-run=server --overwrite ns --all \
 pod-security.kubernetes.io/enforce=baseline
```

### Applying to all namespaces

If you're just getting started with the Pod Security Standards, a suitable first step would be to configure all namespaces with audit annotations for a stricter level such as baseline:

```
kubectl label --overwrite ns --all \
 pod-security.kubernetes.io/audit=baseline \
 pod-security.kubernetes.io/warn=baseline
```

Note that this is not setting an enforce level, so that namespaces that haven't been explicitly evaluated can be distinguished. You can list namespaces without an explicitly set enforce level using this command:

```
kubectl get namespaces --selector='!pod-security.kubernetes.io/enforce'
```

## Applying to a single namespace

You can update a specific namespace as well. This command adds the `enforce=restricted` policy to my-existing-namespace, pinning the restricted policy version to v1.30.

```
kubectrl label --overwrite ns my-existing-namespace \
 pod-security.kubernetes.io/enforce=restricted \
 pod-security.kubernetes.io/enforce-version=v1.30
```

# Migrate from PodSecurityPolicy to the Built-In PodSecurity Admission Controller

This page describes the process of migrating from PodSecurityPolicies to the built-in PodSecurity admission controller. This can be done effectively using a combination of dry-run and audit and warn modes, although this becomes harder if mutating PSPs are used.

## Before you begin

Your Kubernetes server must be at or later than version v1.22. To check the version, enter `kubectrl version`.

If you are currently running a version of Kubernetes other than 1.30, you may want to switch to viewing this page in the documentation for the version of Kubernetes that you are actually running.

This page assumes you are already familiar with the basic [Pod Security Admission](#) concepts.

## Overall approach

There are multiple strategies you can take for migrating from PodSecurityPolicy to Pod Security Admission. The following steps are one possible migration path, with a goal of minimizing both the risks of a production outage and of a security gap.

1. Decide whether Pod Security Admission is the right fit for your use case.
2. Review namespace permissions
3. Simplify & standardize PodSecurityPolicies
4. Update namespaces
  1. Identify an appropriate Pod Security level
  2. Verify the Pod Security level
  3. Enforce the Pod Security level
  4. Bypass PodSecurityPolicy
5. Review namespace creation processes
6. Disable PodSecurityPolicy

## 0. Decide whether Pod Security Admission is right for you

Pod Security Admission was designed to meet the most common security needs out of the box, and to provide a standard set of security levels across clusters. However, it is less flexible than PodSecurityPolicy. Notably, the following features are supported by PodSecurityPolicy but not Pod Security Admission:

- **Setting default security constraints** - Pod Security Admission is a non-mutating admission controller, meaning it won't modify pods before validating them. If you were relying on this aspect of PSP, you will need to either modify your workloads to meet the Pod Security constraints, or use a [Mutating Admission Webhook](#) to make those changes. See [Simplify & Standardize PodSecurityPolicies](#) below for more detail.
- **Fine-grained control over policy definition** - Pod Security Admission only supports [3 standard levels](#). If you require more control over specific constraints, then you will need to use a [Validating Admission Webhook](#) to enforce those policies.
- **Sub-namespace policy granularity** - PodSecurityPolicy lets you bind different policies to different Service Accounts or users, even within a single namespace. This approach has many pitfalls and is not recommended, but if you require this feature anyway you will need to use a 3rd party webhook instead. The exception to this is if you only need to completely exempt specific users or [RuntimeClasses](#), in which case Pod Security Admission does expose some [static configuration for exemptions](#).

Even if Pod Security Admission does not meet all of your needs it was designed to be *complementary* to other policy enforcement mechanisms, and can provide a useful fallback running alongside other admission webhooks.

## 1. Review namespace permissions

Pod Security Admission is controlled by [labels on namespaces](#). This means that anyone who can update (or patch or create) a namespace can also modify the Pod Security level for that namespace, which could be used to bypass a more restrictive policy. Before proceeding, ensure that only trusted, privileged users have these namespace permissions. It is not recommended to grant these powerful permissions to users that shouldn't have elevated permissions, but if you must you will need to use an [admission webhook](#) to place additional restrictions on setting Pod Security labels on Namespace objects.

## 2. Simplify & standardize PodSecurityPolicies

In this section, you will reduce mutating PodSecurityPolicies and remove options that are outside the scope of the Pod Security Standards. You should make the changes recommended here to an offline copy of the original PodSecurityPolicy being modified. The cloned PSP should have a different name that is alphabetically before the original (for example, prepend a 0 to it). Do not create the new policies in Kubernetes yet - that will be covered in the [Rollout the updated policies](#) section below.

### 2.a. Eliminate purely mutating fields

If a PodSecurityPolicy is mutating pods, then you could end up with pods that don't meet the Pod Security level requirements when you finally turn PodSecurityPolicy off. In order to avoid

this, you should eliminate all PSP mutation prior to switching over. Unfortunately PSP does not cleanly separate mutating & validating fields, so this is not a straightforward migration.

You can start by eliminating the fields that are purely mutating, and don't have any bearing on the validating policy. These fields (also listed in the [Mapping PodSecurityPolicies to Pod Security Standards](#) reference) are:

- `.spec.defaultAllowPrivilegeEscalation`
- `.spec.runtimeClass.defaultRuntimeClassName`
- `.metadata.annotations['seccomp.security.alpha.kubernetes.io/defaultProfileName']`
- `.metadata.annotations['apparmor.security.beta.kubernetes.io/defaultProfileName']`
- `.spec.defaultAddCapabilities` - Although technically a mutating & validating field, these should be merged into `.spec.allowedCapabilities` which performs the same validation without mutation.

### **Caution:**

Removing these could result in workloads missing required configuration, and cause problems. See [Rollout the updated policies](#) below for advice on how to roll these changes out safely.

## **2.b. Eliminate options not covered by the Pod Security Standards**

There are several fields in PodSecurityPolicy that are not covered by the Pod Security Standards. If you must enforce these options, you will need to supplement Pod Security Admission with an [admission webhook](#), which is outside the scope of this guide.

First, you can remove the purely validating fields that the Pod Security Standards do not cover. These fields (also listed in the [Mapping PodSecurityPolicies to Pod Security Standards](#) reference with "no opinion") are:

- `.spec.allowedHostPaths`
- `.spec.allowedFlexVolumes`
- `.spec.allowedCSIDrivers`
- `.spec.forbiddenSysctls`
- `.spec.runtimeClass`

You can also remove the following fields, that are related to POSIX / UNIX group controls.

### **Caution:**

If any of these use the MustRunAs strategy they may be mutating! Removing these could result in workloads not setting the required groups, and cause problems. See [Rollout the updated policies](#) below for advice on how to roll these changes out safely.

- `.spec.runAsGroup`
- `.spec.supplementalGroups`
- `.spec.fsGroup`

The remaining mutating fields are required to properly support the Pod Security Standards, and will need to be handled on a case-by-case basis later:

- `.spec.requiredDropCapabilities` - Required to drop ALL for the Restricted profile.

- `.spec.seLinux` - (Only mutating with the `MustRunAs` rule) required to enforce the SELinux requirements of the Baseline & Restricted profiles.
- `.spec.runAsUser` - (Non-mutating with the `RunAsAny` rule) required to enforce `RunAsNonRoot` for the Restricted profile.
- `.spec.allowPrivilegeEscalation` - (Only mutating if set to false) required for the Restricted profile.

## 2.c. Rollout the updated PSPs

Next, you can rollout the updated policies to your cluster. You should proceed with caution, as removing the mutating options may result in workloads missing required configuration.

For each updated PodSecurityPolicy:

1. Identify pods running under the original PSP. This can be done using the `kubernetes.io/psp` annotation. For example, using `kubectl`:

```
PSP_NAME="original" # Set the name of the PSP you're checking for
kubectl get pods --all-namespaces -o jsonpath="{range .items[?
(@.metadata.annotations.kubernetes.io\psp=='$PSP_NAME')]}{.metadata.namespace}
{.metadata.name}}{'\n'}{end}"
```

2. Compare these running pods against the original pod spec to determine whether PodSecurityPolicy has modified the pod. For pods created by a [workload resource](#) you can compare the pod with the PodTemplate in the controller resource. If any changes are identified, the original Pod or PodTemplate should be updated with the desired configuration. The fields to review are:
  - `.metadata.annotations['container.apparmor.security.beta.kubernetes.io/*']` (replace \* with each container name)
  - `.spec.runtimeClassName`
  - `.spec.securityContext.fsGroup`
  - `.spec.securityContext.seccompProfile`
  - `.spec.securityContext.seLinuxOptions`
  - `.spec.securityContext.supplementalGroups`
  - On containers, under `.spec.containers[*]` and `.spec.initContainers[*]`:
    - `.securityContext.allowPrivilegeEscalation`
    - `.securityContext.capabilities.add`
    - `.securityContext.capabilities.drop`
    - `.securityContext.readOnlyRootFilesystem`
    - `.securityContext.runAsGroup`
    - `.securityContext.runAsNonRoot`
    - `.securityContext.runAsUser`
    - `.securityContext.seccompProfile`
    - `.securityContext.seLinuxOptions`
3. Create the new PodSecurityPolicies. If any Roles or ClusterRoles are granting use on all PSPs this could cause the new PSPs to be used instead of their mutating counter-parts.
4. Update your authorization to grant access to the new PSPs. In RBAC this means updating any Roles or ClusterRoles that grant the use permission on the original PSP to also grant it to the updated PSP.
5. Verify: after some soak time, rerun the command from step 1 to see if any pods are still using the original PSPs. Note that pods need to be recreated after the new policies have been rolled out before they can be fully verified.

6. (optional) Once you have verified that the original PSPs are no longer in use, you can delete them.

## 3. Update Namespaces

The following steps will need to be performed on every namespace in the cluster. Commands referenced in these steps use the `$NAMESPACE` variable to refer to the namespace being updated.

### 3.a. Identify an appropriate Pod Security level

Start reviewing the [Pod Security Standards](#) and familiarizing yourself with the 3 different levels.

There are several ways to choose a Pod Security level for your namespace:

1. **By security requirements for the namespace** - If you are familiar with the expected access level for the namespace, you can choose an appropriate level based on those requirements, similar to how one might approach this on a new cluster.
2. **By existing PodSecurityPolicies** - Using the [Mapping PodSecurityPolicies to Pod Security Standards](#) reference you can map each PSP to a Pod Security Standard level. If your PSPs aren't based on the Pod Security Standards, you may need to decide between choosing a level that is at least as permissive as the PSP, and a level that is at least as restrictive. You can see which PSPs are in use for pods in a given namespace with this command:

```
kubectl get pods -n $NAMESPACE -o jsonpath="{.items[*].metadata.annotations.kubernetes.io\psp}" | tr " " "\n" | sort -u
```

3. **By existing pods** - Using the strategies under [Verify the Pod Security level](#), you can test out both the Baseline and Restricted levels to see whether they are sufficiently permissive for existing workloads, and chose the least-privileged valid level.

#### Caution:

Options 2 & 3 above are based on *existing* pods, and may miss workloads that aren't currently running, such as CronJobs, scale-to-zero workloads, or other workloads that haven't rolled out.

### 3.b. Verify the Pod Security level

Once you have selected a Pod Security level for the namespace (or if you're trying several), it's a good idea to test it out first (you can skip this step if using the Privileged level). Pod Security includes several tools to help test and safely roll out profiles.

First, you can dry-run the policy, which will evaluate pods currently running in the namespace against the applied policy, without making the new policy take effect:

```
$LEVEL is the level to dry-run, either "baseline" or "restricted".
kubectl label --dry-run=server --overwrite ns $NAMESPACE pod-security.kubernetes.io/
enforce=$LEVEL
```

This command will return a warning for any *existing* pods that are not valid under the proposed level.



The second option is better for catching workloads that are not currently running: audit mode. When running under audit-mode (as opposed to enforcing), pods that violate the policy level are recorded in the audit logs, which can be reviewed later after some soak time, but are not forbidden. Warning mode works similarly, but returns the warning to the user immediately. You can set the audit level on a namespace with this command:

```
kubectl label --overwrite ns $NAMESPACE pod-security.kubernetes.io/audit=$LEVEL
```

If either of these approaches yield unexpected violations, you will need to either update the violating workloads to meet the policy requirements, or relax the namespace Pod Security level.

### 3.c. Enforce the Pod Security level

When you are satisfied that the chosen level can safely be enforced on the namespace, you can update the namespace to enforce the desired level:

```
kubectl label --overwrite ns $NAMESPACE pod-security.kubernetes.io/enforce=$LEVEL
```

### 3.d. Bypass PodSecurityPolicy

Finally, you can effectively bypass PodSecurityPolicy at the namespace level by binding the fully [privileged PSP](#) to all service accounts in the namespace.

```
The following cluster-scoped commands are only needed once.
kubectl apply -f privileged-psp.yaml
kubectl create clusterrole privileged-psp --verb use --resource podsecuritypolicies.policy --
resource-name privileged

Per-namespace disable
kubectl create -n $NAMESPACE rolebinding disable-psp --clusterrole privileged-psp --group
system:serviceaccounts:$NAMESPACE
```

Since the privileged PSP is non-mutating, and the PSP admission controller always prefers non-mutating PSPs, this will ensure that pods in this namespace are no longer being modified or restricted by PodSecurityPolicy.

The advantage to disabling PodSecurityPolicy on a per-namespace basis like this is if a problem arises you can easily roll the change back by deleting the RoleBinding. Just make sure the pre-existing PodSecurityPolicies are still in place!

```
Undo PodSecurityPolicy disablement.
kubectl delete -n $NAMESPACE rolebinding disable-psp
```

## 4. Review namespace creation processes

Now that existing namespaces have been updated to enforce Pod Security Admission, you should ensure that your processes and/or policies for creating new namespaces are updated to ensure that an appropriate Pod Security profile is applied to new namespaces.

You can also statically configure the Pod Security admission controller to set a default enforce, audit, and/or warn level for unlabeled namespaces. See [Configure the Admission Controller](#) for more information.

## 5. Disable PodSecurityPolicy

Finally, you're ready to disable PodSecurityPolicy. To do so, you will need to modify the admission configuration of the API server: [How do I turn off an admission controller?](#).

To verify that the PodSecurityPolicy admission controller is no longer enabled, you can manually run a test by impersonating a user without access to any PodSecurityPolicies (see the [PodSecurityPolicy example](#)), or by verifying in the API server logs. At startup, the API server outputs log lines listing the loaded admission controller plugins:

```
I0218 00:59:44.903329 13 plugins.go:158] Loaded 16 mutating admission controller(s)
successfully in the following order:
NamespaceLifecycle,LimitRanger,ServiceAccount,NodeRestriction,TaintNodesByCondition,Priority,DefaultTolerationSeconds,ExtendedResourceToleration,PersistentVolumeLabel,DefaultStorageClass,StorageObjectInUseProtection,RuntimeClass,DefaultIngressClass,MutatingAdmissionWebhook.
I0218 00:59:44.903350 13 plugins.go:161] Loaded 14 validating admission controller(s)
successfully in the following order:
LimitRanger,ServiceAccount,PodSecurity,Priority,PersistentVolumeClaimResize,RuntimeClass,CertificateApproval,CertificateSigning,CertificateSubjectRestriction,DenyServiceExternalIPs,ValidatingAdmissionWebhook,ResourceQuota.
```

You should see PodSecurity (in the validating admission controllers), and neither list should contain PodSecurityPolicy.

Once you are certain the PSP admission controller is disabled (and after sufficient soak time to be confident you won't need to roll back), you are free to delete your PodSecurityPolicies and any associated Roles, ClusterRoles, RoleBindings and ClusterRoleBindings (just make sure they don't grant any other unrelated permissions).

## Monitoring, Logging, and Debugging

Set up monitoring and logging to troubleshoot a cluster, or debug a containerized application.

Sometimes things go wrong. This guide is aimed at making them right. It has two sections:

- [Debugging your application](#) - Useful for users who are deploying code into Kubernetes and wondering why it is not working.
- [Debugging your cluster](#) - Useful for cluster administrators and people whose Kubernetes cluster is unhappy.

You should also check the known issues for the [release](#) you're using.

## Getting help

If your problem isn't answered by any of the guides above, there are variety of ways for you to get help from the Kubernetes community.

## Questions

The documentation on this site has been structured to provide answers to a wide range of questions. [Concepts](#) explain the Kubernetes architecture and how each component works, while [Setup](#) provides practical instructions for getting started. [Tasks](#) show how to accomplish commonly used tasks, and [Tutorials](#) are more comprehensive walkthroughs of real-world, industry-specific, or end-to-end development scenarios. The [Reference](#) section provides detailed documentation on the [Kubernetes API](#) and command-line interfaces (CLIs), such as [kubectl](#).

## Help! My question isn't covered! I need help now!

### Stack Exchange, Stack Overflow, or Server Fault

If you have questions related to *software development* for your containerized app, you can ask those on [Stack Overflow](#).

If you have Kubernetes questions related to *cluster management* or *configuration*, you can ask those on [Server Fault](#).

There are also several more specific Stack Exchange network sites which might be the right place to ask Kubernetes questions in areas such as [DevOps](#), [Software Engineering](#), or [InfoSec](#).

Someone else from the community may have already asked a similar question or may be able to help with your problem.

The Kubernetes team will also monitor [posts tagged Kubernetes](#). If there aren't any existing questions that help, **please ensure that your question is [on-topic on Stack Overflow](#), [Server Fault](#), or the Stack Exchange Network site you're asking on**, and read through the guidance on [how to ask a new question](#), before asking a new one!

### Slack

Many people from the Kubernetes community hang out on Kubernetes Slack in the [#kubernetes-users](#) channel. Slack requires registration; you can [request an invitation](#), and registration is open to everyone). Feel free to come and ask any and all questions. Once registered, access the [Kubernetes organisation in Slack](#) via your web browser or via Slack's own dedicated app.

Once you are registered, browse the growing list of channels for various subjects of interest. For example, people new to Kubernetes may also want to join the [#kubernetes-novice](#) channel. As another example, developers should join the [#kubernetes-contributors](#) channel.

There are also many country specific / local language channels. Feel free to join these channels for localized support and info:

Country / language specific Slack channels

Country	Channels
China	<a href="#">#cn-users</a> , <a href="#">#cn-events</a>
Finland	<a href="#">#fi-users</a>
France	<a href="#">#fr-users</a> , <a href="#">#fr-events</a>
Germany	<a href="#">#de-users</a> , <a href="#">#de-events</a>

Country	Channels
India	<a href="#">#in-users</a> , <a href="#">#in-events</a>
Italy	<a href="#">#it-users</a> , <a href="#">#it-events</a>
Japan	<a href="#">#jp-users</a> , <a href="#">#jp-events</a>
Korea	<a href="#">#kr-users</a>
Netherlands	<a href="#">#nl-users</a>
Norway	<a href="#">#norw-users</a>
Poland	<a href="#">#pl-users</a>
Russia	<a href="#">#ru-users</a>
Spain	<a href="#">#es-users</a>
Sweden	<a href="#">#se-users</a>
Turkey	<a href="#">#tr-users</a> , <a href="#">#tr-events</a>

## Forum

You're welcome to join the official Kubernetes Forum: [discuss.kubernetes.io](https://discuss.kubernetes.io).

## Bugs and feature requests

If you have what looks like a bug, or you would like to make a feature request, please use the [GitHub issue tracking system](#).

Before you file an issue, please search existing issues to see if your issue is already covered.

If filing a bug, please include detailed information about how to reproduce the problem, such as:

- Kubernetes version: kubectl version
- Cloud provider, OS distro, network configuration, and container runtime version
- Steps to reproduce the problem

# Troubleshooting Applications

Debugging common containerized application issues.

This doc contains a set of resources for fixing issues with containerized applications. It covers things like common issues with Kubernetes resources (like Pods, Services, or StatefulSets), advice on making sense of container termination messages, and ways to debug running containers.

---

[Debug Pods](#)

[Debug Services](#)

[Debug a StatefulSet](#)

[Determine the Reason for Pod Failure](#)

[Debug Init Containers](#)

[Debug Running Pods](#)

[Get a Shell to a Running Container](#)

# Debug Pods

This guide is to help users debug applications that are deployed into Kubernetes and not behaving correctly. This is *not* a guide for people who want to debug their cluster. For that you should check out [this guide](#).

## Diagnosing the problem

The first step in troubleshooting is triage. What is the problem? Is it your Pods, your Replication Controller or your Service?

- [Debugging Pods](#)
- [Debugging Replication Controllers](#)
- [Debugging Services](#)

## Debugging Pods

The first step in debugging a Pod is taking a look at it. Check the current state of the Pod and recent events with the following command:

```
kubectl describe pods ${POD_NAME}
```

Look at the state of the containers in the pod. Are they all Running? Have there been recent restarts?

Continue debugging depending on the state of the pods.

### My pod stays pending

If a Pod is stuck in Pending it means that it can not be scheduled onto a node. Generally this is because there are insufficient resources of one type or another that prevent scheduling. Look at the output of the `kubectl describe ...` command above. There should be messages from the scheduler about why it can not schedule your pod. Reasons include:

- **You don't have enough resources:** You may have exhausted the supply of CPU or Memory in your cluster, in this case you need to delete Pods, adjust resource requests, or add new nodes to your cluster. See [Compute Resources document](#) for more information.
- **You are using hostPort:** When you bind a Pod to a hostPort there are a limited number of places that pod can be scheduled. In most cases, hostPort is unnecessary, try using a Service object to expose your Pod. If you do require hostPort then you can only schedule as many Pods as there are nodes in your Kubernetes cluster.

### My pod stays waiting

If a Pod is stuck in the Waiting state, then it has been scheduled to a worker node, but it can't run on that machine. Again, the information from `kubectl describe ...` should be informative.

The most common cause of `Waiting` pods is a failure to pull the image. There are three things to check:

- Make sure that you have the name of the image correct.
- Have you pushed the image to the registry?
- Try to manually pull the image to see if the image can be pulled. For example, if you use Docker on your PC, run `docker pull <image>`.

### **My pod stays terminating**

If a Pod is stuck in the `Terminating` state, it means that a deletion has been issued for the Pod, but the control plane is unable to delete the Pod object.

This typically happens if the Pod has a [finalizer](#) and there is an [admission webhook](#) installed in the cluster that prevents the control plane from removing the finalizer.

To identify this scenario, check if your cluster has any `ValidatingWebhookConfiguration` or `MutatingWebhookConfiguration` that target `UPDATE` operations for pods resources.

If the webhook is provided by a third-party:

- Make sure you are using the latest version.
- Disable the webhook for `UPDATE` operations.
- Report an issue with the corresponding provider.

If you are the author of the webhook:

- For a mutating webhook, make sure it never changes immutable fields on `UPDATE` operations. For example, changes to containers are usually not allowed.
- For a validating webhook, make sure that your validation policies only apply to new changes. In other words, you should allow Pods with existing violations to pass validation. This allows Pods that were created before the validating webhook was installed to continue running.

### **My pod is crashing or otherwise unhealthy**

Once your pod has been scheduled, the methods described in [Debug Running Pods](#) are available for debugging.

### **My pod is running but not doing what I told it to do**

If your pod is not behaving as you expected, it may be that there was an error in your pod description (e.g. `mypod.yaml` file on your local machine), and that the error was silently ignored when you created the pod. Often a section of the pod description is nested incorrectly, or a key name is typed incorrectly, and so the key is ignored. For example, if you misspelled `command` as `commnd` then the pod will be created but will not use the command line you intended it to use.

The first thing to do is to delete your pod and try creating it again with the `--validate` option. For example, run `kubectl apply --validate -f mypod.yaml`. If you misspelled `command` as `commnd` then will give an error like this:

```
I0805 10:43:25.129850 46757 schema.go:126] unknown field: commnd
I0805 10:43:25.129973 46757 schema.go:129] this may be a false alarm, see https://github.com/
kubernetes/kubernetes/issues/6842
pods/mypod
```

The next thing to check is whether the pod on the apiserver matches the pod you meant to create (e.g. in a yaml file on your local machine). For example, run `kubectl get pods/mypod -o yaml > mypod-on-apiserver.yaml` and then manually compare the original pod description, `mypod.yaml` with the one you got back from apiserver, `mypod-on-apiserver.yaml`. There will typically be some lines on the "apiserver" version that are not on the original version. This is expected. However, if there are lines on the original that are not on the apiserver version, then this may indicate a problem with your pod spec.

## Debugging Replication Controllers

Replication controllers are fairly straightforward. They can either create Pods or they can't. If they can't create pods, then please refer to the [instructions above](#) to debug your pods.

You can also use `kubectl describe rc ${CONTROLLER_NAME}` to introspect events related to the replication controller.

## Debugging Services

Services provide load balancing across a set of pods. There are several common problems that can make Services not work properly. The following instructions should help debug Service problems.

First, verify that there are endpoints for the service. For every Service object, the apiserver makes an endpoints resource available.

You can view this resource with:

```
kubectl get endpoints ${SERVICE_NAME}
```

Make sure that the endpoints match up with the number of pods that you expect to be members of your service. For example, if your Service is for an nginx container with 3 replicas, you would expect to see three different IP addresses in the Service's endpoints.

### My service is missing endpoints

If you are missing endpoints, try listing pods using the labels that Service uses. Imagine that you have a Service where the labels are:

```
...
spec:
 - selector:
 name: nginx
 type: frontend
```

You can use:

```
kubectl get pods --selector=name=nginx,type=frontend
```



to list pods that match this selector. Verify that the list matches the Pods that you expect to provide your Service. Verify that the pod's containerPort matches up with the Service's targetPort

### Network traffic is not forwarded

Please see [debugging service](#) for more information.

## What's next

If none of the above solves your problem, follow the instructions in [Debugging Service document](#) to make sure that your Service is running, has Endpoints, and your Pods are actually serving; you have DNS working, iptables rules installed, and kube-proxy does not seem to be misbehaving.

You may also visit [troubleshooting document](#) for more information.

# Debug Services

An issue that comes up rather frequently for new installations of Kubernetes is that a Service is not working properly. You've run your Pods through a Deployment (or other workload controller) and created a Service, but you get no response when you try to access it. This document will hopefully help you to figure out what's going wrong.

## Running commands in a Pod

For many steps here you will want to see what a Pod running in the cluster sees. The simplest way to do this is to run an interactive busybox Pod:

```
kubectrl run -it --rm --restart=Never busybox --image=gcr.io/google-containers/busybox sh
```

### Note:

If you don't see a command prompt, try pressing enter.

If you already have a running Pod that you prefer to use, you can run a command in it using:

```
kubectrl exec <POD-NAME> -c <CONTAINER-NAME> -- <COMMAND>
```

## Setup

For the purposes of this walk-through, let's run some Pods. Since you're probably debugging your own Service you can substitute your own details, or you can follow along and get a second data point.

```
kubectrl create deployment hostnames --image=registry.k8s.io/serve_hostname
```

```
deployment.apps/hostnames created
```



kubectl commands will print the type and name of the resource created or mutated, which can then be used in subsequent commands.

Let's scale the deployment to 3 replicas.

```
kubectl scale deployment hostnames --replicas=3
```

```
deployment.apps/hostnames scaled
```

Note that this is the same as if you had started the Deployment with the following YAML:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 labels:
 app: hostnames
 name: hostnames
spec:
 selector:
 matchLabels:
 app: hostnames
 replicas: 3
 template:
 metadata:
 labels:
 app: hostnames
 spec:
 containers:
 - name: hostnames
 image: registry.k8s.io/serve_hostname
```

The label "app" is automatically set by kubectl create deployment to the name of the Deployment.

You can confirm your Pods are running:

```
kubectl get pods -l app=hostnames
```

NAME	READY	STATUS	RESTARTS	AGE
hostnames-632524106-bbpiw	1/1	Running	0	2m
hostnames-632524106-ly40y	1/1	Running	0	2m
hostnames-632524106-tlaok	1/1	Running	0	2m

You can also confirm that your Pods are serving. You can get the list of Pod IP addresses and test them directly.

```
kubectl get pods -l app=hostnames \
-o go-template='{{range .items}}{{.status.podIP}}{"\n"}}{{end}}'
```

```
10.244.0.5
10.244.0.6
10.244.0.7
```

The example container used for this walk-through serves its own hostname via HTTP on port 9376, but if you are debugging your own app, you'll want to use whatever port number your Pods are listening on.

From within a pod:

```
for ep in 10.244.0.5:9376 10.244.0.6:9376 10.244.0.7:9376; do
 wget -qO- $ep
done
```

This should produce something like:

```
hostnames-632524106-bbpiw
hostnames-632524106-ly40y
hostnames-632524106-tlaok
```

If you are not getting the responses you expect at this point, your Pods might not be healthy or might not be listening on the port you think they are. You might find `kubect` logs to be useful for seeing what is happening, or perhaps you need to `kubect` exec directly into your Pods and debug from there.

Assuming everything has gone to plan so far, you can start to investigate why your Service doesn't work.

## Does the Service exist?

The astute reader will have noticed that you did not actually create a Service yet - that is intentional. This is a step that sometimes gets forgotten, and is the first thing to check.

What would happen if you tried to access a non-existent Service? If you have another Pod that consumes this Service by name you would get something like:

```
wget -O- hostnames
```

```
Resolving hostnames (hostnames)... failed: Name or service not known.
wget: unable to resolve host address 'hostnames'
```

The first thing to check is whether that Service actually exists:

```
kubect
```

 get svc hostnames

```
No resources found.
Error from server (NotFound): services "hostnames" not found
```

Let's create the Service. As before, this is for the walk-through - you can use your own Service's details here.

```
kubect
```

 expose deployment hostnames --port=80 --target-port=9376

```
service/hostnames exposed
```

And read it back:

```
kubect
```

 get svc hostnames

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
hostnames	ClusterIP	10.0.1.175	<none>	80/TCP	5s

Now you know that the Service exists.

As before, this is the same as if you had started the Service with YAML:

```
apiVersion: v1
kind: Service
metadata:
 labels:
 app: hostnames
 name: hostnames
spec:
 selector:
 app: hostnames
 ports:
 - name: default
 protocol: TCP
 port: 80
 targetPort: 9376
```

In order to highlight the full range of configuration, the Service you created here uses a different port number than the Pods. For many real-world Services, these values might be the same.

## Any Network Policy Ingress rules affecting the target Pods?

If you have deployed any Network Policy Ingress rules which may affect incoming traffic to `hostnames-*` Pods, these need to be reviewed.

Please refer to [Network Policies](#) for more details.

## Does the Service work by DNS name?

One of the most common ways that clients consume a Service is through a DNS name.

From a Pod in the same Namespace:

```
nslookup hostnames
```

```
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local
```

```
Name: hostnames
```

```
Address 1: 10.0.1.175 hostnames.default.svc.cluster.local
```

If this fails, perhaps your Pod and Service are in different Namespaces, try a namespace-qualified name (again, from within a Pod):

```
nslookup hostnames.default
```

```
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local
```

```
Name: hostnames.default
```

```
Address 1: 10.0.1.175 hostnames.default.svc.cluster.local
```

If this works, you'll need to adjust your app to use a cross-namespace name, or run your app and Service in the same Namespace. If this still fails, try a fully-qualified name:

```
nslookup hostnames.default.svc.cluster.local
```

```
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local
```

```
Name: hostnames.default.svc.cluster.local
```

```
Address 1: 10.0.1.175 hostnames.default.svc.cluster.local
```

Note the suffix here: "default.svc.cluster.local". The "default" is the Namespace you're operating in. The "svc" denotes that this is a Service. The "cluster.local" is your cluster domain, which COULD be different in your own cluster.

You can also try this from a Node in the cluster:

#### **Note:**

10.0.0.10 is the cluster's DNS Service IP, yours might be different.

```
nslookup hostnames.default.svc.cluster.local 10.0.0.10
```

```
Server: 10.0.0.10
```

```
Address: 10.0.0.10#53
```

```
Name: hostnames.default.svc.cluster.local
```

```
Address: 10.0.1.175
```

If you are able to do a fully-qualified name lookup but not a relative one, you need to check that your /etc/resolv.conf file in your Pod is correct. From within a Pod:

```
cat /etc/resolv.conf
```

You should see something like:

```
nameserver 10.0.0.10
```

```
search default.svc.cluster.local svc.cluster.local cluster.local example.com
```

```
options ndots:5
```

The nameserver line must indicate your cluster's DNS Service. This is passed into kubelet with the --cluster-dns flag.

The search line must include an appropriate suffix for you to find the Service name. In this case it is looking for Services in the local Namespace ("default.svc.cluster.local"), Services in all Namespaces ("svc.cluster.local"), and lastly for names in the cluster ("cluster.local"). Depending on your own install you might have additional records after that (up to 6 total). The cluster suffix is passed into kubelet with the --cluster-domain flag. Throughout this document, the cluster suffix is assumed to be "cluster.local". Your own clusters might be configured differently, in which case you should change that in all of the previous commands.

The options line must set ndots high enough that your DNS client library considers search paths at all. Kubernetes sets this to 5 by default, which is high enough to cover all of the DNS names it generates.

## Does any Service work by DNS name?

If the above still fails, DNS lookups are not working for your Service. You can take a step back and see what else is not working. The Kubernetes master Service should always work. From within a Pod:

```
nslookup kubernetes.default
```

```
Server: 10.0.0.10
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local

Name: kubernetes.default
Address 1: 10.0.0.1 kubernetes.default.svc.cluster.local
```

If this fails, please see the [kube-proxy](#) section of this document, or even go back to the top of this document and start over, but instead of debugging your own Service, debug the DNS Service.

## Does the Service work by IP?

Assuming you have confirmed that DNS works, the next thing to test is whether your Service works by its IP address. From a Pod in your cluster, access the Service's IP (from `kubectl get` above).

```
for i in $(seq 1 3); do
 wget -qO- 10.0.1.175:80
done
```

This should produce something like:

```
hostnames-632524106-bbpiw
hostnames-632524106-ly40y
hostnames-632524106-tlaok
```

If your Service is working, you should get correct responses. If not, there are a number of things that could be going wrong. Read on.

## Is the Service defined correctly?

It might sound silly, but you should really double and triple check that your Service is correct and matches your Pod's port. Read back your Service and verify it:

```
kubectl get service hostnames -o json
```

```
{
 "kind": "Service",
 "apiVersion": "v1",
 "metadata": {
```

```

 "name": "hostnames",
 "namespace": "default",
 "uid": "428c8b6c-24bc-11e5-936d-42010af0a9bc",
 "resourceVersion": "347189",
 "creationTimestamp": "2015-07-07T15:24:29Z",
 "labels": {
 "app": "hostnames"
 }
 },
 "spec": {
 "ports": [
 {
 "name": "default",
 "protocol": "TCP",
 "port": 80,
 "targetPort": 9376,
 "nodePort": 0
 }
],
 "selector": {
 "app": "hostnames"
 },
 "clusterIP": "10.0.1.175",
 "type": "ClusterIP",
 "sessionAffinity": "None"
 },
 "status": {
 "loadBalancer": {}
 }
}

```

- Is the Service port you are trying to access listed in `spec.ports[]`?
- Is the `targetPort` correct for your Pods (some Pods use a different port than the Service)?
- If you meant to use a numeric port, is it a number (9376) or a string "9376"?
- If you meant to use a named port, do your Pods expose a port with the same name?
- Is the port's protocol correct for your Pods?

## Does the Service have any Endpoints?

If you got this far, you have confirmed that your Service is correctly defined and is resolved by DNS. Now let's check that the Pods you ran are actually being selected by the Service.

Earlier you saw that the Pods were running. You can re-check that:

```
kubectl get pods -l app=hostnames
```

NAME	READY	STATUS	RESTARTS	AGE
hostnames-632524106-bbpiw	1/1	Running	0	1h
hostnames-632524106-ly40y	1/1	Running	0	1h
hostnames-632524106-tlaok	1/1	Running	0	1h

The `-l app=hostnames` argument is a label selector configured on the Service.

The "AGE" column says that these Pods are about an hour old, which implies that they are running fine and not crashing.

The "RESTARTS" column says that these pods are not crashing frequently or being restarted. Frequent restarts could lead to intermittent connectivity issues. If the restart count is high, read more about how to [debug pods](#).

Inside the Kubernetes system is a control loop which evaluates the selector of every Service and saves the results into a corresponding Endpoints object.

```
kubectl get endpoints hostnames
```

NAME	ENDPOINTS
hostnames	10.244.0.5:9376,10.244.0.6:9376,10.244.0.7:9376

This confirms that the endpoints controller has found the correct Pods for your Service. If the ENDPOINTS column is <none>, you should check that the spec.selector field of your Service actually selects for metadata.labels values on your Pods. A common mistake is to have a typo or other error, such as the Service selecting for app=hostnames, but the Deployment specifying run=hostnames, as in versions previous to 1.18, where the kubectl run command could have been also used to create a Deployment.

## Are the Pods working?

At this point, you know that your Service exists and has selected your Pods. At the beginning of this walk-through, you verified the Pods themselves. Let's check again that the Pods are actually working - you can bypass the Service mechanism and go straight to the Pods, as listed by the Endpoints above.

### Note:

These commands use the Pod port (9376), rather than the Service port (80).

From within a Pod:

```
for ep in 10.244.0.5:9376 10.244.0.6:9376 10.244.0.7:9376; do
 wget -qO- $ep
done
```

This should produce something like:

```
hostnames-632524106-bbpiw
hostnames-632524106-ly40y
hostnames-632524106-tlaok
```

You expect each Pod in the Endpoints list to return its own hostname. If this is not what happens (or whatever the correct behavior is for your own Pods), you should investigate what's happening there.

# Is the kube-proxy working?

If you get here, your Service is running, has Endpoints, and your Pods are actually serving. At this point, the whole Service proxy mechanism is suspect. Let's confirm it, piece by piece.

The default implementation of Services, and the one used on most clusters, is kube-proxy. This is a program that runs on every node and configures one of a small set of mechanisms for providing the Service abstraction. If your cluster does not use kube-proxy, the following sections will not apply, and you will have to investigate whatever implementation of Services you are using.

## Is kube-proxy running?

Confirm that kube-proxy is running on your Nodes. Running directly on a Node, you should get something like the below:

```
ps auxw | grep kube-proxy
```

```
root 4194 0.4 0.1 101864 17696 ? Sl Jul04 25:43 /usr/local/bin/kube-proxy --master=https://kubernetes-master --kubeconfig=/var/lib/kube-proxy/kubeconfig --v=2
```

Next, confirm that it is not failing something obvious, like contacting the master. To do this, you'll have to look at the logs. Accessing the logs depends on your Node OS. On some OSes it is a file, such as /var/log/kube-proxy.log, while other OSes use journalctl to access logs. You should see something like:

```
I1027 22:14:53.995134 5063 server.go:200] Running in resource-only container "/kube-proxy"
I1027 22:14:53.998163 5063 server.go:247] Using iptables Proxier.
I1027 22:14:54.038140 5063 proxier.go:352] Setting endpoints for "kube-system/kube-dns:dns-tcp" to [10.244.1.3:53]
I1027 22:14:54.038164 5063 proxier.go:352] Setting endpoints for "kube-system/kube-dns:dns" to [10.244.1.3:53]
I1027 22:14:54.038209 5063 proxier.go:352] Setting endpoints for "default/kubernetes:https" to [10.240.0.2:443]
I1027 22:14:54.038238 5063 proxier.go:429] Not syncing iptables until Services and Endpoints have been received from master
I1027 22:14:54.040048 5063 proxier.go:294] Adding new service "default/kubernetes:https" at 10.0.0.1:443/TCP
I1027 22:14:54.040154 5063 proxier.go:294] Adding new service "kube-system/kube-dns:dns" at 10.0.0.10:53/UDP
I1027 22:14:54.040223 5063 proxier.go:294] Adding new service "kube-system/kube-dns:dns-tcp" at 10.0.0.10:53/TCP
```

If you see error messages about not being able to contact the master, you should double-check your Node configuration and installation steps.

One of the possible reasons that kube-proxy cannot run correctly is that the required conntrack binary cannot be found. This may happen on some Linux systems, depending on how you are installing the cluster, for example, you are installing Kubernetes from scratch. If this is the case, you need to manually install the conntrack package (e.g. `sudo apt install conntrack` on Ubuntu) and then retry.



Kube-proxy can run in one of a few modes. In the log listed above, the line Using iptables Proxier indicates that kube-proxy is running in "iptables" mode. The most common other mode is "ipvs".

## Iptables mode

In "iptables" mode, you should see something like the following on a Node:

```
iptables-save | grep hostnames
```

```
-A KUBE-SEP-57KPRZ3JQVENLNBR -s 10.244.3.6/32 -m comment --comment "default/
hostnames:" -j MARK --set-xmark 0x00004000/0x00004000
-A KUBE-SEP-57KPRZ3JQVENLNBR -p tcp -m comment --comment "default/hostnames:" -m
tcp -j DNAT --to-destination 10.244.3.6:9376
-A KUBE-SEP-WNBA2IHDGP2BOBGZ -s 10.244.1.7/32 -m comment --comment "default/
hostnames:" -j MARK --set-xmark 0x00004000/0x00004000
-A KUBE-SEP-WNBA2IHDGP2BOBGZ -p tcp -m comment --comment "default/hostnames:" -m
tcp -j DNAT --to-destination 10.244.1.7:9376
-A KUBE-SEP-X3P2623AGDH6CDF3 -s 10.244.2.3/32 -m comment --comment "default/
hostnames:" -j MARK --set-xmark 0x00004000/0x00004000
-A KUBE-SEP-X3P2623AGDH6CDF3 -p tcp -m comment --comment "default/hostnames:" -m
tcp -j DNAT --to-destination 10.244.2.3:9376
-A KUBE-SERVICES -d 10.0.1.175/32 -p tcp -m comment --comment "default/hostnames: cluster
IP" -m tcp --dport 80 -j KUBE-SVC-NWV5X2332I4OT4T3
-A KUBE-SVC-NWV5X2332I4OT4T3 -m comment --comment "default/hostnames:" -m statistic
--mode random --probability 0.33332999982 -j KUBE-SEP-WNBA2IHDGP2BOBGZ
-A KUBE-SVC-NWV5X2332I4OT4T3 -m comment --comment "default/hostnames:" -m statistic
--mode random --probability 0.50000000000 -j KUBE-SEP-X3P2623AGDH6CDF3
-A KUBE-SVC-NWV5X2332I4OT4T3 -m comment --comment "default/hostnames:" -j KUBE-
SEP-57KPRZ3JQVENLNBR
```

For each port of each Service, there should be 1 rule in KUBE-SERVICES and one KUBE-SVC-`<hash>` chain. For each Pod endpoint, there should be a small number of rules in that KUBE-SVC-`<hash>` and one KUBE-SEP-`<hash>` chain with a small number of rules in it. The exact rules will vary based on your exact config (including node-ports and load-balancers).

## IPVS mode

In "ipvs" mode, you should see something like the following on a Node:

```
ipvsadm -ln
```

```
Prot LocalAddress:Port Scheduler Flags
-> RemoteAddress:Port Forward Weight ActiveConn InActConn
...
TCP 10.0.1.175:80 rr
-> 10.244.0.5:9376 Masq 1 0 0
-> 10.244.0.6:9376 Masq 1 0 0
-> 10.244.0.7:9376 Masq 1 0 0
...
```

For each port of each Service, plus any NodePorts, external IPs, and load-balancer IPs, kube-proxy will create a virtual server. For each Pod endpoint, it will create corresponding real

servers. In this example, service hostnames(10.0.1.175:80) has 3 endpoints(10.244.0.5:9376, 10.244.0.6:9376, 10.244.0.7:9376).

## Is kube-proxy proxying?

Assuming you do see one the above cases, try again to access your Service by IP from one of your Nodes:

```
curl 10.0.1.175:80
```

```
hostnames-632524106-bbpiw
```

If this still fails, look at the kube-proxy logs for specific lines like:

```
Setting endpoints for default/hostnames:default to [10.244.0.5:9376 10.244.0.6:9376 10.244.0.7:9376]
```

If you don't see those, try restarting kube-proxy with the `-v` flag set to 4, and then look at the logs again.

## Edge case: A Pod fails to reach itself via the Service IP

This might sound unlikely, but it does happen and it is supposed to work.

This can happen when the network is not properly configured for "hairpin" traffic, usually when kube-proxy is running in iptables mode and Pods are connected with bridge network. The Kubelet exposes a hairpin-mode [flag](#) that allows endpoints of a Service to loadbalance back to themselves if they try to access their own Service VIP. The hairpin-mode flag must either be set to hairpin-veth or promiscuous-bridge.

The common steps to trouble shoot this are as follows:

- Confirm hairpin-mode is set to hairpin-veth or promiscuous-bridge. You should see something like the below. hairpin-mode is set to promiscuous-bridge in the following example.

```
ps auxw | grep kubelet
```

```
root 3392 1.1 0.8 186804 65208 ? Sl 00:51 11:11 /usr/local/bin/kubelet --enable-debugging-handlers=true --config=/etc/kubernetes/manifests --allow-privileged=True --v=4 --cluster-dns=10.0.0.10 --cluster-domain=cluster.local --configure-cbr0=true --cgroup-root=/ --system-cgroups=/system --hairpin-mode=promiscuous-bridge --runtime-cgroups=/docker-daemon --kubelet-cgroups=/kubelet --babysit-daemons=true --max-pods=110 --serialize-image-pulls=false --outofdisk-transition-frequency=0
```

- Confirm the effective hairpin-mode. To do this, you'll have to look at kubelet log. Accessing the logs depends on your Node OS. On some OSes it is a file, such as `/var/log/kubelet.log`, while other OSes use `journalctl` to access logs. Please be noted that the effective hairpin mode may not match `--hairpin-mode` flag due to compatibility. Check if there is any log lines with key word hairpin in kubelet.log. There should be log lines indicating the effective hairpin mode, like something below.

```
I0629 00:51:43.648698 3252 kubelet.go:380] Hairpin mode set to "promiscuous-bridge"
```

- If the effective hairpin mode is `hairpin-veth`, ensure the Kubelet has the permission to operate in `/sys` on node. If everything works properly, you should see something like:

```
for intf in /sys/devices/virtual/net/cbr0/brif/*; do cat $intf/hairpin_mode; done
```

```
1
1
1
1
```

- If the effective hairpin mode is `promiscuous-bridge`, ensure Kubelet has the permission to manipulate linux bridge on node. If `cbr0` bridge is used and configured properly, you should see:

```
ifconfig cbr0 |grep PROMISC
```

```
UP BROADCAST RUNNING PROMISC MULTICAST MTU:1460 Metric:1
```

- Seek help if none of above works out.

## Seek help

If you get this far, something very strange is happening. Your Service is running, has Endpoints, and your Pods are actually serving. You have DNS working, and kube-proxy does not seem to be misbehaving. And yet your Service is not working. Please let us know what is going on, so we can help investigate!

Contact us on [Slack](#) or [Forum](#) or [GitHub](#).

## What's next

Visit the [troubleshooting overview document](#) for more information.

# Debug a StatefulSet

This task shows you how to debug a StatefulSet.

## Before you begin

- You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster.
- You should have a StatefulSet running that you want to investigate.

## Debugging a StatefulSet

In order to list all the pods which belong to a StatefulSet, which have a label `app.kubernetes.io/name=MyApp` set on them, you can use the following:

```
kubectl get pods -l app.kubernetes.io/name=MyApp
```

If you find that any Pods listed are in Unknown or Terminating state for an extended period of time, refer to the [Deleting StatefulSet Pods](#) task for instructions on how to deal with them. You can debug individual Pods in a StatefulSet using the [Debugging Pods](#) guide.

## What's next

Learn more about [debugging an init-container](#).

# Determine the Reason for Pod Failure

This page shows how to write and read a Container termination message.

Termination messages provide a way for containers to write information about fatal events to a location where it can be easily retrieved and surfaced by tools like dashboards and monitoring software. In most cases, information that you put in a termination message should also be written to the general [Kubernetes logs](#).

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

## Writing and reading a termination message

In this exercise, you create a Pod that runs one container. The manifest for that Pod specifies a command that runs when the container starts:

[debug/termination.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: termination-demo
spec:
 containers:
 - name: termination-demo-container
 image: debian
 command: ["/bin/sh"]
 args: ["-c", "sleep 10 && echo Sleep expired > /dev/termination-log"]
```

1. Create a Pod based on the YAML configuration file:

```
kubectl apply -f https://k8s.io/examples/debug/termination.yaml
```

In the YAML file, in the `command` and `args` fields, you can see that the container sleeps for 10 seconds and then writes "Sleep expired" to the `/dev/termination-log` file. After the container writes the "Sleep expired" message, it terminates.

2. Display information about the Pod:

```
kubectl get pod termination-demo
```

Repeat the preceding command until the Pod is no longer running.

3. Display detailed information about the Pod:

```
kubectl get pod termination-demo --output=yaml
```

The output includes the "Sleep expired" message:

```
apiVersion: v1
kind: Pod
...
 lastState:
 terminated:
 containerID: ...
 exitCode: 0
 finishedAt: ...
 message: |
 Sleep expired
 ...
```

4. Use a Go template to filter the output so that it includes only the termination message:

```
kubectl get pod termination-demo -o go-template="{{range .status.containerStatuses}}
{{.lastState.terminated.message}}{{end}}"
```

If you are running a multi-container Pod, you can use a Go template to include the container's name. By doing so, you can discover which of the containers is failing:

```
kubectl get pod multi-container-pod -o go-template='{{range .status.containerStatuses}}{{printf
"%s:\n%s\n\n" .name .lastState.terminated.message}}{{end}}'
```

## Customizing the termination message

Kubernetes retrieves termination messages from the termination message file specified in the `terminationMessagePath` field of a Container, which has a default value of `/dev/termination-log`. By customizing this field, you can tell Kubernetes to use a different file. Kubernetes use the contents from the specified file to populate the Container's status message on both success and failure.

The termination message is intended to be brief final status, such as an assertion failure message. The kubelet truncates messages that are longer than 4096 bytes.

The total message length across all containers is limited to 12KiB, divided equally among each container. For example, if there are 12 containers (`initContainers` or `containers`), each has 1024 bytes of available termination message space.

The default termination message path is `/dev/termination-log`. You cannot set the termination message path after a Pod is launched.

In the following example, the container writes termination messages to `/tmp/my-log` for Kubernetes to retrieve:

```
apiVersion: v1
kind: Pod
metadata:
 name: msg-path-demo
spec:
 containers:
 - name: msg-path-demo-container
 image: debian
 terminationMessagePath: "/tmp/my-log"
```

Moreover, users can set the `terminationMessagePolicy` field of a Container for further customization. This field defaults to "File" which means the termination messages are retrieved only from the termination message file. By setting the `terminationMessagePolicy` to "FallbackToLogsOnError", you can tell Kubernetes to use the last chunk of container log output if the termination message file is empty and the container exited with an error. The log output is limited to 2048 bytes or 80 lines, whichever is smaller.

## What's next

- See the `terminationMessagePath` field in [Container](#).
- See [ImagePullBackOff](#) in [Images](#).
- Learn about [retrieving logs](#).
- Learn about [Go templates](#).
- Learn about [Pod status](#) and [Pod phase](#).
- Learn about [container states](#).

## Debug Init Containers

This page shows how to investigate problems related to the execution of Init Containers. The example command lines below refer to the Pod as `<pod-name>` and the Init Containers as `<init-container-1>` and `<init-container-2>`.

### Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

- You should be familiar with the basics of [Init Containers](#).

- You should have [Configured an Init Container](#).

## Checking the status of Init Containers

Display the status of your pod:

```
kubectl get pod <pod-name>
```

For example, a status of Init:1/2 indicates that one of two Init Containers has completed successfully:

NAME	READY	STATUS	RESTARTS	AGE
<pod-name>	0/1	Init:1/2	0	7s

See [Understanding Pod status](#) for more examples of status values and their meanings.

## Getting details about Init Containers

View more detailed information about Init Container execution:

```
kubectl describe pod <pod-name>
```

For example, a Pod with two Init Containers might show the following:

Init Containers:

<init-container-1>:

Container ID: ...

...

State: Terminated

Reason: Completed

Exit Code: 0

Started: ...

Finished: ...

Ready: True

Restart Count: 0

...

<init-container-2>:

Container ID: ...

...

State: Waiting

Reason: CrashLoopBackOff

Last State: Terminated

Reason: Error

Exit Code: 1

Started: ...

Finished: ...

Ready: False

Restart Count: 3

...

You can also access the Init Container statuses programmatically by reading the `status.initContainerStatuses` field on the Pod Spec:

```
kubectl get pod nginx --template '{{.status.initContainerStatuses}}'
```

This command will return the same information as above in raw JSON.

## Accessing logs from Init Containers

Pass the Init Container name along with the Pod name to access its logs.

```
kubectl logs <pod-name> -c <init-container-2>
```

Init Containers that run a shell script print commands as they're executed. For example, you can do this in Bash by running `set -x` at the beginning of the script.

## Understanding Pod status

A Pod status beginning with `Init:` summarizes the status of Init Container execution. The table below describes some example status values that you might see while debugging Init Containers.

Status	Meaning
Init:N/M	The Pod has M Init Containers, and N have completed so far.
Init:Error	An Init Container has failed to execute.
Init:CrashLoopBackOff	An Init Container has failed repeatedly.
Pending	The Pod has not yet begun executing Init Containers.
PodInitializing or Running	The Pod has already finished executing Init Containers.

## Debug Running Pods

This page explains how to debug Pods running (or crashing) on a Node.

### Before you begin

- Your [Pod](#) should already be scheduled and running. If your Pod is not yet running, start with [Debugging Pods](#).
- For some of the advanced debugging steps you need to know on which Node the Pod is running and have shell access to run commands on that Node. You don't need that access to run the standard debug steps that use `kubectl`.

## Using `kubectl describe pod` to fetch details about pods

For this example we'll use a Deployment to create two pods, similar to the earlier example.

[application/nginx-with-request.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
```



```
spec:
 selector:
 matchLabels:
 app: nginx
 replicas: 2
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx
 resources:
 limits:
 memory: "128Mi"
 cpu: "500m"
 ports:
 - containerPort: 80
```

Create deployment by running following command:

```
kubectl apply -f https://k8s.io/examples/application/nginx-with-request.yaml
```

```
deployment.apps/nginx-deployment created
```

Check pod status by following command:

```
kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-67d4bdd6f5-cx2nz	1/1	Running	0	13s
nginx-deployment-67d4bdd6f5-w6kd7	1/1	Running	0	13s

We can retrieve a lot more information about each of these pods using `kubectl describe pod`. For example:

```
kubectl describe pod nginx-deployment-67d4bdd6f5-w6kd7
```

```
Name: nginx-deployment-67d4bdd6f5-w6kd7
Namespace: default
Priority: 0
Node: kube-worker-1/192.168.0.113
Start Time: Thu, 17 Feb 2022 16:51:01 -0500
Labels: app=nginx
 pod-template-hash=67d4bdd6f5
Annotations: <none>
Status: Running
IP: 10.88.0.3
IPs:
 IP: 10.88.0.3
 IP: 2001:db8::1
Controlled By: ReplicaSet/nginx-deployment-67d4bdd6f5
Containers:
```

```

nginx:
 Container ID: containerd://
5403af59a2b46ee5a23fb0ae4b1e077f7ca5c5fb7af16e1ab21c00e0e616462a
 Image: nginx
 Image ID: docker.io/library/
nginx@sha256:2834dc507516af02784808c5f48b7cbe38b8ed5d0f4837f16e78d00deb7e7767
 Port: 80/TCP
 Host Port: 0/TCP
 State: Running
 Started: Thu, 17 Feb 2022 16:51:05 -0500
 Ready: True
 Restart Count: 0
 Limits:
 cpu: 500m
 memory: 128Mi
 Requests:
 cpu: 500m
 memory: 128Mi
 Environment: <none>
 Mounts:
 /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-bgsgp (ro)
Conditions:
 Type Status
 Initialized True
 Ready True
 ContainersReady True
 PodScheduled True
Volumes:
 kube-api-access-bgsgp:
 Type: Projected (a volume that contains injected data from multiple sources)
 TokenExpirationSeconds: 3607
 ConfigMapName: kube-root-ca.crt
 ConfigMapOptional: <nil>
 DownwardAPI: true
QoS Class: Guaranteed
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
 node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
 Type Reason Age From Message
 ---- -
Normal Scheduled 34s default-scheduler Successfully assigned default/nginx-
deployment-67d4bdd6f5-w6kd7 to kube-worker-1
Normal Pulling 31s kubelet Pulling image "nginx"
Normal Pulled 30s kubelet Successfully pulled image "nginx" in 1.146417389s
Normal Created 30s kubelet Created container nginx
Normal Started 30s kubelet Started container nginx

```

Here you can see configuration information about the container(s) and Pod (labels, resource requirements, etc.), as well as status information about the container(s) and Pod (state, readiness, restart count, events, etc.).

The container state is one of Waiting, Running, or Terminated. Depending on the state, additional information will be provided -- here you can see that for a container in Running state, the system tells you when the container started.

Ready tells you whether the container passed its last readiness probe. (In this case, the container does not have a readiness probe configured; the container is assumed to be ready if no readiness probe is configured.)

Restart Count tells you how many times the container has been restarted; this information can be useful for detecting crash loops in containers that are configured with a restart policy of 'always.'

Currently the only Condition associated with a Pod is the binary Ready condition, which indicates that the pod is able to service requests and should be added to the load balancing pools of all matching services.

Lastly, you see a log of recent events related to your Pod. "From" indicates the component that is logging the event. "Reason" and "Message" tell you what happened.

## Example: debugging Pending Pods

A common scenario that you can detect using events is when you've created a Pod that won't fit on any node. For example, the Pod might request more resources than are free on any node, or it might specify a label selector that doesn't match any nodes. Let's say we created the previous Deployment with 5 replicas (instead of 2) and requesting 600 millicores instead of 500, on a four-node cluster where each (virtual) machine has 1 CPU. In that case one of the Pods will not be able to schedule. (Note that because of the cluster add-on pods such as fluentd, skydns, etc., that run on each node, if we requested 1000 millicores then none of the Pods would be able to schedule.)

```
kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-1006230814-6winp	1/1	Running	0	7m
nginx-deployment-1006230814-fmgu3	1/1	Running	0	7m
nginx-deployment-1370807587-6ekbw	1/1	Running	0	1m
nginx-deployment-1370807587-fg172	0/1	Pending	0	1m
nginx-deployment-1370807587-fz9sd	0/1	Pending	0	1m

To find out why the nginx-deployment-1370807587-fz9sd pod is not running, we can use kubectl describe pod on the pending Pod and look at its events:

```
kubectl describe pod nginx-deployment-1370807587-fz9sd
```

```
Name: nginx-deployment-1370807587-fz9sd
Namespace: default
Node: /
Labels: app=nginx,pod-template-hash=1370807587
Status: Pending
IP:
Controllers: ReplicaSet/nginx-deployment-1370807587
Containers:
 nginx:
```

```

Image: nginx
Port: 80/TCP
QoS Tier:
 memory: Guaranteed
 cpu: Guaranteed
Limits:
 cpu: 1
 memory: 128Mi
Requests:
 cpu: 1
 memory: 128Mi
Environment Variables:
Volumes:
 default-token-4bcbi:
 Type: Secret (a volume populated by a Secret)
 SecretName: default-token-4bcbi
Events:
 FirstSeen LastSeen Count From SubobjectPath Type
Reason Message

1m 48s 7 {default-scheduler } Warning
FailedScheduling pod (nginx-deployment-1370807587-fz9sd) failed to fit in any node
fit failure on node (kubernetes-node-6ta5): Node didn't have enough resource: CPU, requested:
1000, used: 1420, capacity: 2000
fit failure on node (kubernetes-node-wul5): Node didn't have enough resource: CPU,
requested: 1000, used: 1100, capacity: 2000

```

Here you can see the event generated by the scheduler saying that the Pod failed to schedule for reason FailedScheduling (and possibly others). The message tells us that there were not enough resources for the Pod on any of the nodes.

To correct this situation, you can use `kubectl scale` to update your Deployment to specify four or fewer replicas. (Or you could leave the one Pod pending, which is harmless.)

Events such as the ones you saw at the end of `kubectl describe pod` are persisted in etcd and provide high-level information on what is happening in the cluster. To list all events you can use

```
kubectl get events
```

but you have to remember that events are namespaced. This means that if you're interested in events for some namespaced object (e.g. what happened with Pods in namespace my-namespace) you need to explicitly provide a namespace to the command:

```
kubectl get events --namespace=my-namespace
```

To see events from all namespaces, you can use the `--all-namespaces` argument.

In addition to `kubectl describe pod`, another way to get extra information about a pod (beyond what is provided by `kubectl get pod`) is to pass the `-o yaml` output format flag to `kubectl get pod`. This will give you, in YAML format, even more information than `kubectl describe pod`--essentially all of the information the system has about the Pod. Here you will see things like

annotations (which are key-value metadata without the label restrictions, that is used internally by Kubernetes system components), restart policy, ports, and volumes.

```
kubectl get pod nginx-deployment-1006230814-6winp -o yaml
```

```
apiVersion: v1
kind: Pod
metadata:
 creationTimestamp: "2022-02-17T21:51:01Z"
 generateName: nginx-deployment-67d4bdd6f5-
 labels:
 app: nginx
 pod-template-hash: 67d4bdd6f5
 name: nginx-deployment-67d4bdd6f5-w6kd7
 namespace: default
 ownerReferences:
 - apiVersion: apps/v1
 blockOwnerDeletion: true
 controller: true
 kind: ReplicaSet
 name: nginx-deployment-67d4bdd6f5
 uid: 7d41dfd4-84c0-4be4-88ab-cedbe626ad82
 resourceVersion: "1364"
 uid: a6501da1-0447-4262-98eb-c03d4002222e
spec:
 containers:
 - image: nginx
 imagePullPolicy: Always
 name: nginx
 ports:
 - containerPort: 80
 protocol: TCP
 resources:
 limits:
 cpu: 500m
 memory: 128Mi
 requests:
 cpu: 500m
 memory: 128Mi
 terminationMessagePath: /dev/termination-log
 terminationMessagePolicy: File
 volumeMounts:
 - mountPath: /var/run/secrets/kubernetes.io/serviceaccount
 name: kube-api-access-bgsgp
 readOnly: true
 dnsPolicy: ClusterFirst
 enableServiceLinks: true
 nodeName: kube-worker-1
 preemptionPolicy: PreemptLowerPriority
 priority: 0
 restartPolicy: Always
 schedulerName: default-scheduler
 securityContext: {}
```

```
serviceAccount: default
serviceAccountName: default
terminationGracePeriodSeconds: 30
tolerations:
- effect: NoExecute
 key: node.kubernetes.io/not-ready
 operator: Exists
 tolerationSeconds: 300
- effect: NoExecute
 key: node.kubernetes.io/unreachable
 operator: Exists
 tolerationSeconds: 300
volumes:
- name: kube-api-access-bgsgp
 projected:
 defaultMode: 420
 sources:
 - serviceAccountToken:
 expirationSeconds: 3607
 path: token
 - configMap:
 items:
 - key: ca.crt
 path: ca.crt
 name: kube-root-ca.crt
 - downwardAPI:
 items:
 - fieldRef:
 apiVersion: v1
 fieldPath: metadata.namespace
 path: namespace
status:
 conditions:
 - lastProbeTime: null
 lastTransitionTime: "2022-02-17T21:51:01Z"
 status: "True"
 type: Initialized
 - lastProbeTime: null
 lastTransitionTime: "2022-02-17T21:51:06Z"
 status: "True"
 type: Ready
 - lastProbeTime: null
 lastTransitionTime: "2022-02-17T21:51:06Z"
 status: "True"
 type: ContainersReady
 - lastProbeTime: null
 lastTransitionTime: "2022-02-17T21:51:01Z"
 status: "True"
 type: PodScheduled
 containerStatuses:
 - containerID: containerd://
5403af59a2b46ee5a23fb0ae4b1e077f7ca5c5fb7af16e1ab21c00e0e616462a
 image: docker.io/library/nginx:latest
```

```
imageID: docker.io/library/
nginx@sha256:2834dc507516af02784808c5f48b7cbe38b8ed5d0f4837f16e78d00deb7e7767
lastState: {}
name: nginx
ready: true
restartCount: 0
started: true
state:
 running:
 startedAt: "2022-02-17T21:51:05Z"
hostIP: 192.168.0.113
phase: Running
podIP: 10.88.0.3
podIPs:
- ip: 10.88.0.3
- ip: 2001:db8::1
qosClass: Guaranteed
startTime: "2022-02-17T21:51:01Z"
```

## Examining pod logs

First, look at the logs of the affected container:

```
kubectl logs ${POD_NAME} ${CONTAINER_NAME}
```

If your container has previously crashed, you can access the previous container's crash log with:

```
kubectl logs --previous ${POD_NAME} ${CONTAINER_NAME}
```

## Debugging with container exec

If the [container image](#) includes debugging utilities, as is the case with images built from Linux and Windows OS base images, you can run commands inside a specific container with `kubectl exec`:

```
kubectl exec ${POD_NAME} -c ${CONTAINER_NAME} -- ${CMD} ${ARG1} ${ARG2} ... ${ARGN}
```

### Note:

`-c ${CONTAINER_NAME}` is optional. You can omit it for Pods that only contain a single container.

As an example, to look at the logs from a running Cassandra pod, you might run

```
kubectl exec cassandra -- cat /var/log/cassandra/system.log
```

You can run a shell that's connected to your terminal using the `-i` and `-t` arguments to `kubectl exec`, for example:

```
kubectl exec -it cassandra -- sh
```

For more details, see [Get a Shell to a Running Container](#).

## Debugging with an ephemeral debug container

FEATURE STATE: Kubernetes v1.25 [stable]

[Ephemeral containers](#) are useful for interactive troubleshooting when `kubectl exec` is insufficient because a container has crashed or a container image doesn't include debugging utilities, such as with [distroless images](#).

### Example debugging using ephemeral containers

You can use the `kubectl debug` command to add ephemeral containers to a running Pod. First, create a pod for the example:

```
kubectl run ephemeral-demo --image=registry.k8s.io/pause:3.1 --restart=Never
```

The examples in this section use the `pause` container image because it does not contain debugging utilities, but this method works with all container images.

If you attempt to use `kubectl exec` to create a shell you will see an error because there is no shell in this container image.

```
kubectl exec -it ephemeral-demo -- sh
```

```
OCI runtime exec failed: exec failed: container_linux.go:346: starting container process caused "exec: \"sh\": executable file not found in $PATH": unknown
```

You can instead add a debugging container using `kubectl debug`. If you specify the `-i/--interactive` argument, `kubectl` will automatically attach to the console of the Ephemeral Container.

```
kubectl debug -it ephemeral-demo --image=busybox:1.28 --target=ephemeral-demo
```

```
Defaulting debug container name to debugger-8xzrl.
If you don't see a command prompt, try pressing enter.
/ #
```

This command adds a new `busybox` container and attaches to it. The `--target` parameter targets the process namespace of another container. It's necessary here because `kubectl run` does not enable [process namespace sharing](#) in the pod it creates.

#### Note:

The `--target` parameter must be supported by the [Container Runtime](#). When not supported, the Ephemeral Container may not be started, or it may be started with an isolated process namespace so that `ps` does not reveal processes in other containers.

You can view the state of the newly created ephemeral container using `kubectl describe`:

```
kubectl describe pod ephemeral-demo
```



```
...
Ephemeral Containers:
 debugger-8xzrl:
 Container ID: docker://
b888f9adfd15bd5739fefa39e1df4dd3c617b9902082b1cfdc29c4028ffb2eb
 Image: busybox
 Image ID: docker-pullable://
busybox@sha256:1828edd60c5efd34b2bf5dd3282ec0cc04d47b2ff9caa0b6d4f07a21d1c08084
 Port: <none>
 Host Port: <none>
 State: Running
 Started: Wed, 12 Feb 2020 14:25:42 +0100
 Ready: False
 Restart Count: 0
 Environment: <none>
 Mounts: <none>
...
```

Use `kubectl delete` to remove the Pod when you're finished:

```
kubectl delete pod ephemeral-demo
```

## Debugging using a copy of the Pod

Sometimes Pod configuration options make it difficult to troubleshoot in certain situations. For example, you can't run `kubectl exec` to troubleshoot your container if your container image does not include a shell or if your application crashes on startup. In these situations you can use `kubectl debug` to create a copy of the Pod with configuration values changed to aid debugging.

### Copying a Pod while adding a new container

Adding a new container can be useful when your application is running but not behaving as you expect and you'd like to add additional troubleshooting utilities to the Pod.

For example, maybe your application's container images are built on busybox but you need debugging utilities not included in busybox. You can simulate this scenario using `kubectl run`:

```
kubectl run myapp --image=busybox:1.28 --restart=Never -- sleep 1d
```

Run this command to create a copy of `myapp` named `myapp-debug` that adds a new Ubuntu container for debugging:

```
kubectl debug myapp -it --image=ubuntu --share-processes --copy-to=myapp-debug
```

Defaulting debug container name to `debugger-w7xmf`.  
If you don't see a command prompt, try pressing enter.  
`root@myapp-debug:/#`

#### Note:

- `kubectl debug` automatically generates a container name if you don't choose one using the `--container` flag.

- The `-i` flag causes `kubectl debug` to attach to the new container by default. You can prevent this by specifying `--attach=false`. If your session becomes disconnected you can reattach using `kubectl attach`.
- The `--share-processes` allows the containers in this Pod to see processes from the other containers in the Pod. For more information about how this works, see [Share Process Namespace between Containers in a Pod](#).

Don't forget to clean up the debugging Pod when you're finished with it:

```
kubectl delete pod myapp myapp-debug
```

## Copying a Pod while changing its command

Sometimes it's useful to change the command for a container, for example to add a debugging flag or because the application is crashing.

To simulate a crashing application, use `kubectl run` to create a container that immediately exits:

```
kubectl run --image=busybox:1.28 myapp -- false
```

You can see using `kubectl describe pod myapp` that this container is crashing:

```
Containers:
 myapp:
 Image: busybox
 ...
 Args:
 false
 State: Waiting
 Reason: CrashLoopBackOff
 Last State: Terminated
 Reason: Error
 Exit Code: 1
```

You can use `kubectl debug` to create a copy of this Pod with the command changed to an interactive shell:

```
kubectl debug myapp -it --copy-to=myapp-debug --container=myapp -- sh
```

If you don't see a command prompt, try pressing enter.  
/ #

Now you have an interactive shell that you can use to perform tasks like checking filesystem paths or running the container command manually.

### Note:

- To change the command of a specific container you must specify its name using `--container` or `kubectl debug` will instead create a new container to run the command you specified.
- The `-i` flag causes `kubectl debug` to attach to the container by default. You can prevent this by specifying `--attach=false`. If your session becomes disconnected you can reattach using `kubectl attach`.

Don't forget to clean up the debugging Pod when you're finished with it:

```
kubectl delete pod myapp myapp-debug
```

## Copying a Pod while changing container images

In some situations you may want to change a misbehaving Pod from its normal production container images to an image containing a debugging build or additional utilities.

As an example, create a Pod using `kubectl run`:

```
kubectl run myapp --image=busybox:1.28 --restart=Never -- sleep 1d
```

Now use `kubectl debug` to make a copy and change its container image to `ubuntu`:

```
kubectl debug myapp --copy-to=myapp-debug --set-image *=ubuntu
```

The syntax of `--set-image` uses the same `container_name=image` syntax as `kubectl set image`. `*=ubuntu` means change the image of all containers to `ubuntu`.

Don't forget to clean up the debugging Pod when you're finished with it:

```
kubectl delete pod myapp myapp-debug
```

## Debugging via a shell on the node

If none of these approaches work, you can find the Node on which the Pod is running and create a Pod running on the Node. To create an interactive shell on a Node using `kubectl debug`, run:

```
kubectl debug node/mynode -it --image=ubuntu
```

Creating debugging pod `node-debugger-mynode-pdx84` with container debugger on node `mynode`.

If you don't see a command prompt, try pressing enter.

```
root@ek8s:/#
```

When creating a debugging session on a node, keep in mind that:

- `kubectl debug` automatically generates the name of the new Pod based on the name of the Node.
- The root filesystem of the Node will be mounted at `/host`.
- The container runs in the host IPC, Network, and PID namespaces, although the pod isn't privileged, so reading some process information may fail, and `chroot /host` may fail.
- If you need a privileged pod, create it manually.

Don't forget to clean up the debugging Pod when you're finished with it:

```
kubectl delete pod node-debugger-mynode-pdx84
```

# Get a Shell to a Running Container

This page shows how to use `kubectl exec` to get a shell to a running container.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

## Getting a shell to a container

In this exercise, you create a Pod that has one container. The container runs the `nginx` image. Here is the configuration file for the Pod:

[application/shell-demo.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
 name: shell-demo
spec:
 volumes:
 - name: shared-data
 emptyDir: {}
 containers:
 - name: nginx
 image: nginx
 volumeMounts:
 - name: shared-data
 mountPath: /usr/share/nginx/html
 hostNetwork: true
 dnsPolicy: Default
```

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/application/shell-demo.yaml
```

Verify that the container is running:

```
kubectl get pod shell-demo
```

Get a shell to the running container:

```
kubectl exec --stdin --tty shell-demo -- /bin/bash
```

**Note:**

The double dash (--) separates the arguments you want to pass to the command from the kubectl arguments.

In your shell, list the root directory:

```
Run this inside the container
ls /
```

In your shell, experiment with other commands. Here are some examples:

```
You can run these example commands inside the container
ls /
cat /proc/mounts
cat /proc/1/maps
apt-get update
apt-get install -y tcpdump
tcpdump
apt-get install -y lsof
lsof
apt-get install -y procps
ps aux
ps aux | grep nginx
```

## Writing the root page for nginx

Look again at the configuration file for your Pod. The Pod has an `emptyDir` volume, and the container mounts the volume at `/usr/share/nginx/html`.

In your shell, create an `index.html` file in the `/usr/share/nginx/html` directory:

```
Run this inside the container
echo 'Hello shell demo' > /usr/share/nginx/html/index.html
```

In your shell, send a GET request to the nginx server:

```
Run this in the shell inside your container
apt-get update
apt-get install curl
curl http://localhost/
```

The output shows the text that you wrote to the `index.html` file:

```
Hello shell demo
```

When you are finished with your shell, enter `exit`.

```
exit # To quit the shell in the container
```

## Running individual commands in a container

In an ordinary command window, not your shell, list the environment variables in the running container:

```
kubectl exec shell-demo -- env
```

Experiment with running other commands. Here are some examples:

```
kubectl exec shell-demo -- ps aux
kubectl exec shell-demo -- ls /
kubectl exec shell-demo -- cat /proc/1/mounts
```

## Opening a shell when a Pod has more than one container

If a Pod has more than one container, use `--container` or `-c` to specify a container in the `kubectl exec` command. For example, suppose you have a Pod named `my-pod`, and the Pod has two containers named *main-app* and *helper-app*. The following command would open a shell to the *main-app* container.

```
kubectl exec -i -t my-pod --container main-app -- /bin/bash
```

**Note:**

The short options `-i` and `-t` are the same as the long options `--stdin` and `--tty`

## What's next

- Read about [kubectl exec](#)

## Troubleshooting Clusters

Debugging common cluster issues.

This doc is about cluster troubleshooting; we assume you have already ruled out your application as the root cause of the problem you are experiencing. See the [application troubleshooting guide](#) for tips on application debugging. You may also visit the [troubleshooting overview document](#) for more information.

For troubleshooting [kubectl](#), refer to [Troubleshooting kubectl](#).

## Listing your cluster

The first thing to debug in your cluster is if your nodes are all registered correctly.

Run the following command:

```
kubectl get nodes
```

And verify that all of the nodes you expect to see are present and that they are all in the Ready state.

To get detailed information about the overall health of your cluster, you can run:

```
kubectl cluster-info dump
```

## Example: debugging a down/unreachable node

Sometimes when debugging it can be useful to look at the status of a node -- for example, because you've noticed strange behavior of a Pod that's running on the node, or to find out why a Pod won't schedule onto the node. As with Pods, you can use `kubectl describe node` and `kubectl get node -o yaml` to retrieve detailed information about nodes. For example, here's what you'll see if a node is down (disconnected from the network, or kubelet dies and won't restart, etc.). Notice the events that show the node is NotReady, and also notice that the pods are no longer running (they are evicted after five minutes of NotReady status).

```
kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
kube-worker-1	NotReady	<none>	1h	v1.23.3
kubernetes-node-bols	Ready	<none>	1h	v1.23.3
kubernetes-node-st6x	Ready	<none>	1h	v1.23.3
kubernetes-node-unaj	Ready	<none>	1h	v1.23.3

```
kubectl describe node kube-worker-1
```

```
Name: kube-worker-1
Roles: <none>
Labels: beta.kubernetes.io/arch=amd64
 beta.kubernetes.io/os=linux
 kubernetes.io/arch=amd64
 kubernetes.io/hostname=kube-worker-1
 kubernetes.io/os=linux
Annotations: kubeadm.alpha.kubernetes.io/cri-socket: /run/containerd/containerd.sock
 node.alpha.kubernetes.io/ttl: 0
 volumes.kubernetes.io/controller-managed-attach-detach: true
CreationTimestamp: Thu, 17 Feb 2022 16:46:30 -0500
Taints: node.kubernetes.io/unreachable:NoExecute
 node.kubernetes.io/unreachable:NoSchedule
Unschedulable: false
Lease:
 HolderIdentity: kube-worker-1
 AcquireTime: <unset>
 RenewTime: Thu, 17 Feb 2022 17:13:09 -0500
Conditions:
 Type Status LastHeartbeatTime LastTransitionTime
Reason Message

NetworkUnavailable False Thu, 17 Feb 2022 17:09:13 -0500 Thu, 17 Feb 2022 17:09:13
-0500 WeaveIsUp Weave pod has set this
MemoryPressure Unknown Thu, 17 Feb 2022 17:12:40 -0500 Thu, 17 Feb 2022 17:13:52
-0500 NodeStatusUnknown Kubelet stopped posting node status.
```

DiskPressure Unknown Thu, 17 Feb 2022 17:12:40 -0500 Thu, 17 Feb 2022 17:13:52 -0500 NodeStatusUnknown Kubelet stopped posting node status.  
PIDPressure Unknown Thu, 17 Feb 2022 17:12:40 -0500 Thu, 17 Feb 2022 17:13:52 -0500 NodeStatusUnknown Kubelet stopped posting node status.  
Ready Unknown Thu, 17 Feb 2022 17:12:40 -0500 Thu, 17 Feb 2022 17:13:52 -0500 NodeStatusUnknown Kubelet stopped posting node status.

Addresses:

InternalIP: 192.168.0.113  
Hostname: kube-worker-1

Capacity:

cpu: 2  
ephemeral-storage: 15372232Ki  
hugepages-2Mi: 0  
memory: 2025188Ki  
pods: 110

Allocatable:

cpu: 2  
ephemeral-storage: 14167048988  
hugepages-2Mi: 0  
memory: 1922788Ki  
pods: 110

System Info:

Machine ID: 9384e2927f544209b5d7b67474bbf92b  
System UUID: aa829ca9-73d7-064d-9019-df07404ad448  
Boot ID: 5a295a03-aaca-4340-af20-1327fa5dab5c  
Kernel Version: 5.13.0-28-generic  
OS Image: Ubuntu 21.10  
Operating System: linux  
Architecture: amd64  
Container Runtime Version: containerd://1.5.9  
Kubelet Version: v1.23.3  
Kube-Proxy Version: v1.23.3

Non-terminated Pods: (4 in total)

Namespace	Name	CPU Requests	CPU Limits	Memory Requests	Memory Limits	Age
default	nginx-deployment-67d4bdd6f5-cx2nz	500m (25%)	500m (25%)	128Mi (6%)	128Mi (6%)	23m
default	nginx-deployment-67d4bdd6f5-w6kd7	500m (25%)	500m (25%)	128Mi (6%)	128Mi (6%)	23m
kube-system	kube-proxy-dnxbz	0 (0%)	0 (0%)	28m (0%)	28m (0%)	
kube-system	weave-net-gjxxp	100m (5%)	0 (0%)	200Mi (10%)	200Mi (10%)	0

Allocated resources:

(Total limits may be over 100 percent, i.e., overcommitted.)

Resource	Requests	Limits
cpu	1100m (55%)	1 (50%)
memory	456Mi (24%)	256Mi (13%)
ephemeral-storage	0 (0%)	0 (0%)
hugepages-2Mi	0 (0%)	0 (0%)



Events:

...

kubectl get node kube-worker-1 -o yaml

```
apiVersion: v1
kind: Node
metadata:
 annotations:
 kubeadm.alpha.kubernetes.io/cri-socket: /run/containerd/containerd.sock
 node.alpha.kubernetes.io/ttl: "0"
 volumes.kubernetes.io/controller-managed-attach-detach: "true"
 creationTimestamp: "2022-02-17T21:46:30Z"
 labels:
 beta.kubernetes.io/arch: amd64
 beta.kubernetes.io/os: linux
 kubernetes.io/arch: amd64
 kubernetes.io/hostname: kube-worker-1
 kubernetes.io/os: linux
 name: kube-worker-1
 resourceVersion: "4026"
 uid: 98efe7cb-2978-4a0b-842a-1a7bf12c05f8
spec: {}
status:
 addresses:
 - address: 192.168.0.113
 type: InternalIP
 - address: kube-worker-1
 type: Hostname
 allocatable:
 cpu: "2"
 ephemeral-storage: "14167048988"
 hugepages-2Mi: "0"
 memory: 1922788Ki
 pods: "110"
 capacity:
 cpu: "2"
 ephemeral-storage: 15372232Ki
 hugepages-2Mi: "0"
 memory: 2025188Ki
 pods: "110"
 conditions:
 - lastHeartbeatTime: "2022-02-17T22:20:32Z"
 lastTransitionTime: "2022-02-17T22:20:32Z"
 message: Weave pod has set this
 reason: WeaveIsUp
 status: "False"
 type: NetworkUnavailable
 - lastHeartbeatTime: "2022-02-17T22:20:15Z"
 lastTransitionTime: "2022-02-17T22:13:25Z"
 message: kubelet has sufficient memory available
 reason: KubeletHasSufficientMemory
 status: "False"
```

```
type: MemoryPressure
- lastHeartbeatTime: "2022-02-17T22:20:15Z"
 lastTransitionTime: "2022-02-17T22:13:25Z"
 message: kubelet has no disk pressure
 reason: KubeletHasNoDiskPressure
 status: "False"
type: DiskPressure
- lastHeartbeatTime: "2022-02-17T22:20:15Z"
 lastTransitionTime: "2022-02-17T22:13:25Z"
 message: kubelet has sufficient PID available
 reason: KubeletHasSufficientPID
 status: "False"
type: PIDPressure
- lastHeartbeatTime: "2022-02-17T22:20:15Z"
 lastTransitionTime: "2022-02-17T22:15:15Z"
 message: kubelet is posting ready status
 reason: KubeletReady
 status: "True"
type: Ready
daemonEndpoints:
 kubeletEndpoint:
 Port: 10250
nodeInfo:
 architecture: amd64
 bootID: 22333234-7a6b-44d4-9ce1-67e31dc7e369
 containerRuntimeVersion: containerd://1.5.9
 kernelVersion: 5.13.0-28-generic
 kubeProxyVersion: v1.23.3
 kubeletVersion: v1.23.3
 machineID: 9384e2927f544209b5d7b67474bbf92b
 operatingSystem: linux
 osImage: Ubuntu 21.10
 systemUUID: aa829ca9-73d7-064d-9019-df07404ad448
```

## Looking at logs

For now, digging deeper into the cluster requires logging into the relevant machines. Here are the locations of the relevant log files. On systemd-based systems, you may need to use `journalctl` instead of examining log files.

### Control Plane nodes

- `/var/log/kube-apiserver.log` - API Server, responsible for serving the API
- `/var/log/kube-scheduler.log` - Scheduler, responsible for making scheduling decisions
- `/var/log/kube-controller-manager.log` - a component that runs most Kubernetes built-in [controllers](#), with the notable exception of scheduling (the kube-scheduler handles scheduling).

### Worker Nodes

- `/var/log/kubelet.log` - logs from the kubelet, responsible for running containers on the node

- `/var/log/kube-proxy.log` - logs from kube-proxy, which is responsible for directing traffic to Service endpoints

## Cluster failure modes

This is an incomplete list of things that could go wrong, and how to adjust your cluster setup to mitigate the problems.

### Contributing causes

- VM(s) shutdown
- Network partition within cluster, or between cluster and users
- Crashes in Kubernetes software
- Data loss or unavailability of persistent storage (e.g. GCE PD or AWS EBS volume)
- Operator error, for example, misconfigured Kubernetes software or application software

### Specific scenarios

- API server VM shutdown or apiserver crashing
  - Results
    - unable to stop, update, or start new pods, services, replication controller
    - existing pods and services should continue to work normally unless they depend on the Kubernetes API
- API server backing storage lost
  - Results
    - the kube-apiserver component fails to start successfully and become healthy
    - kubelets will not be able to reach it but will continue to run the same pods and provide the same service proxying
    - manual recovery or recreation of apiserver state necessary before apiserver is restarted
- Supporting services (node controller, replication controller manager, scheduler, etc) VM shutdown or crashes
  - currently those are colocated with the apiserver, and their unavailability has similar consequences as apiserver
  - in future, these will be replicated as well and may not be co-located
  - they do not have their own persistent state
- Individual node (VM or physical machine) shuts down
  - Results
    - pods on that Node stop running
- Network partition
  - Results
    - partition A thinks the nodes in partition B are down; partition B thinks the apiserver is down. (Assuming the master VM ends up in partition A.)
- Kubelet software fault
  - Results
    - crashing kubelet cannot start new pods on the node
    - kubelet might delete the pods or not
    - node marked unhealthy
    - replication controllers start new pods elsewhere
- Cluster operator error
  - Results
    - loss of pods, services, etc

- lost of apiserver backing store
- users unable to read API
- etc.

## Mitigations

- Action: Use the IaaS provider's automatic VM restarting feature for IaaS VMs
  - Mitigates: Apiserver VM shutdown or apiserver crashing
  - Mitigates: Supporting services VM shutdown or crashes
- Action: Use IaaS providers reliable storage (e.g. GCE PD or AWS EBS volume) for VMs with apiserver+etcd
  - Mitigates: Apiserver backing storage lost
- Action: Use [high-availability](#) configuration
  - Mitigates: Control plane node shutdown or control plane components (scheduler, API server, controller-manager) crashing
    - Will tolerate one or more simultaneous node or component failures
  - Mitigates: API server backing storage (i.e., etcd's data directory) lost
    - Assumes HA (highly-available) etcd configuration
- Action: Snapshot apiserver PDs/EBS-volumes periodically
  - Mitigates: Apiserver backing storage lost
  - Mitigates: Some cases of operator error
  - Mitigates: Some cases of Kubernetes software fault
- Action: use replication controller and services in front of pods
  - Mitigates: Node shutdown
  - Mitigates: Kubelet software fault
- Action: applications (containers) designed to tolerate unexpected restarts
  - Mitigates: Node shutdown
  - Mitigates: Kubelet software fault

## What's next

- Learn about the metrics available in the [Resource Metrics Pipeline](#)
- Discover additional tools for [monitoring resource usage](#)
- Use Node Problem Detector to [monitor node health](#)
- Use kubectl debug node to [debug Kubernetes nodes](#)
- Use crictl to [debug Kubernetes nodes](#)
- Get more information about [Kubernetes auditing](#)
- Use telepresence to [develop and debug services locally](#)

# Troubleshooting kubectl

This documentation is about investigating and diagnosing [kubectl](#) related issues. If you encounter issues accessing kubectl or connecting to your cluster, this document outlines various common scenarios and potential solutions to help identify and address the likely cause.

## Before you begin

- You need to have a Kubernetes cluster.
- You also need to have kubectl installed - see [install tools](#)

## Verify kubectl setup

Make sure you have installed and configured kubectl correctly on your local machine. Check the kubectl version to ensure it is up-to-date and compatible with your cluster.

Check kubectl version:

```
kubectl version
```

You'll see a similar output:

```
Client Version: version.Info{Major:"1", Minor:"27",
GitVersion:"v1.27.4",GitCommit:"fa3d7990104d7c1f16943a67f11b154b71f6a132",
GitTreeState:"clean",BuildDate:"2023-07-19T12:20:54Z", GoVersion:"go1.20.6", Compiler:"gc",
Platform:"linux/amd64"}
Kustomize Version: v5.0.1
Server Version: version.Info{Major:"1", Minor:"27",
GitVersion:"v1.27.3",GitCommit:"25b4e43193bcda6c7328a6d147b1fb73a33f1598",
GitTreeState:"clean",BuildDate:"2023-06-14T09:47:40Z", GoVersion:"go1.20.5", Compiler:"gc",
Platform:"linux/amd64"}
```

If you see Unable to connect to the server: dial tcp <server-ip>:8443: i/o timeout, instead of Server Version, you need to troubleshoot kubectl connectivity with your cluster.

Make sure you have installed the kubectl by following the [official documentation for installing kubectl](#), and you have properly configured the \$PATH environment variable.

## Check kubeconfig

The kubectl requires a kubeconfig file to connect to a Kubernetes cluster. The kubeconfig file is usually located under the ~/.kube/config directory. Make sure that you have a valid kubeconfig file. If you don't have a kubeconfig file, you can obtain it from your Kubernetes administrator, or you can copy it from your Kubernetes control plane's /etc/kubernetes/admin.conf directory. If you have deployed your Kubernetes cluster on a cloud platform and lost your kubeconfig file, you can re-generate it using your cloud provider's tools. Refer the cloud provider's documentation for re-generating a kubeconfig file.

Check if the \$KUBECONFIG environment variable is configured correctly. You can set \$KUBECONFIGenvironment variable or use the --kubeconfig parameter with the kubectl to specify the directory of a kubeconfig file.

## Check VPN connectivity

If you are using a Virtual Private Network (VPN) to access your Kubernetes cluster, make sure that your VPN connection is active and stable. Sometimes, VPN disconnections can lead to connection issues with the cluster. Reconnect to the VPN and try accessing the cluster again.

## Authentication and authorization

If you are using the token based authentication and the kubectl is returning an error regarding the authentication token or authentication server address, validate the Kubernetes authentication token and the authentication server address are configured properly.

If kubectl is returning an error regarding the authorization, make sure that you are using the valid user credentials. And you have the permission to access the resource that you have requested.

## Verify contexts

Kubernetes supports [multiple clusters and contexts](#). Ensure that you are using the correct context to interact with your cluster.

List available contexts:

```
kubectl config get-contexts
```

Switch to the appropriate context:

```
kubectl config use-context <context-name>
```

## API server and load balancer

The [kube-apiserver](#) server is the central component of a Kubernetes cluster. If the API server or the load balancer that runs in front of your API servers is not reachable or not responding, you won't be able to interact with the cluster.

Check the if the API server's host is reachable by using ping command. Check cluster's network connectivity and firewall. If your are using a cloud provider for deploying the cluster, check your cloud provider's health check status for the cluster's API server.

Verify the status of the load balancer (if used) to ensure it is healthy and forwarding traffic to the API server.

## TLS problems

The Kubernetes API server only serves HTTPS requests by default. In that case TLS problems may occur due to various reasons, such as certificate expiry or chain of trust validity.

You can find the TLS certificate in the kubeconfig file, located in the `~/.kube/config` directory. The `certificate-authority` attribute contains the CA certificate and the `client-certificate` attribute contains the client certificate.

Verify the expiry of these certificates:

```
openssl x509 -noout -dates -in $(kubectl config view --minify --output 'jsonpath={.clusters[0].cluster.certificate-authority}')
```

output:

```
notBefore=Sep 2 08:34:12 2023 GMT
notAfter=Aug 31 08:34:12 2033 GMT
```

```
openssl x509 -noout -dates -in $(kubectl config view --minify --output 'jsonpath={.users[0].user.client-certificate}')
```

output:

```
notBefore=Sep 2 08:34:12 2023 GMT
notAfter=Sep 2 08:34:12 2026 GMT
```

## Verify kubectl helpers

Some kubectl authentication helpers provide easy access to Kubernetes clusters. If you have used such helpers and are facing connectivity issues, ensure that the necessary configurations are still present.

Check kubectl configuration for authentication details:

```
kubectl config view
```

If you previously used a helper tool (for example, `kubectl-oidc-login`), ensure that it is still installed and configured correctly.

## Resource metrics pipeline

For Kubernetes, the *Metrics API* offers a basic set of metrics to support automatic scaling and similar use cases. This API makes information available about resource usage for node and pod, including metrics for CPU and memory. If you deploy the Metrics API into your cluster, clients of the Kubernetes API can then query for this information, and you can use Kubernetes' access control mechanisms to manage permissions to do so.

The [HorizontalPodAutoscaler](#) (HPA) and [VerticalPodAutoscaler](#) (VPA) use data from the metrics API to adjust workload replicas and resources to meet customer demand.

You can also view the resource metrics using the [kubectl top](#) command.

### Note:

The Metrics API, and the metrics pipeline that it enables, only offers the minimum CPU and memory metrics to enable automatic scaling using HPA and / or VPA. If you would like to

provide a more complete set of metrics, you can complement the simpler Metrics API by deploying a second [metrics pipeline](#) that uses the *Custom Metrics API*.

Figure 1 illustrates the architecture of the resource metrics pipeline.

```

flowchart RL
 subgraph cluster [Cluster]
 direction RL
 S[cAdvisor] --> C[kubelet]
 C --> E[Container runtime]
 E --> D1[Container runtime]
 D1 --> D2[Pod data]
 D2 --> C
 end
 L[API server] --> W[HPA]
 C -.-> L
 L --> A[metrics resource metrics]
 A --> L
 L --> W
 W --> K[kubectl]
 top
 classDef box fill:#fff,stroke:#000,stroke-width:1px,color:#000;
 class W,B,P,K,cluster,D,E,E1 box;
 classDef spacewhite fill:#ffffff,stroke:#fff,stroke-width:0px,color:#000;
 class S spacewhite;
 classDef k8s fill:#326ce5,stroke:#fff,stroke-width:1px,color:#fff;
 class A,L,C k8s;

```

JavaScript must be [enabled](#) to view this content

Figure 1. Resource Metrics Pipeline

The architecture components, from right to left in the figure, consist of the following:

- [cAdvisor](#): Daemon for collecting, aggregating and exposing container metrics included in Kubelet.
- [kubelet](#): Node agent for managing container resources. Resource metrics are accessible using the `/metrics/resource` and `/stats kubelet` API endpoints.
- [node level resource metrics](#): API provided by the kubelet for discovering and retrieving per-node summarized stats available through the `/metrics/resource` endpoint.
- [metrics-server](#): Cluster addon component that collects and aggregates resource metrics pulled from each kubelet. The API server serves Metrics API for use by HPA, VPA, and by the `kubectl top` command. Metrics Server is a reference implementation of the Metrics API.
- [Metrics API](#): Kubernetes API supporting access to CPU and memory used for workload autoscaling. To make this work in your cluster, you need an API extension server that provides the Metrics API.

#### Note:

cAdvisor supports reading metrics from cgroups, which works with typical container runtimes on Linux. If you use a container runtime that uses another resource isolation mechanism, for example virtualization, then that container runtime must support [CRI Container Metrics](#) in order for metrics to be available to the kubelet.

## Metrics API

FEATURE STATE: Kubernetes 1.8 [beta]



The metrics-server implements the Metrics API. This API allows you to access CPU and memory usage for the nodes and pods in your cluster. Its primary role is to feed resource usage metrics to K8s autoscaler components.

Here is an example of the Metrics API request for a minikube node piped through jq for easier reading:

```
kubectl get --raw "/apis/metrics.k8s.io/v1beta1/nodes/minikube" | jq "
```

Here is the same API call using curl:

```
curl http://localhost:8080/apis/metrics.k8s.io/v1beta1/nodes/minikube
```

Sample response:

```
{
 "kind": "NodeMetrics",
 "apiVersion": "metrics.k8s.io/v1beta1",
 "metadata": {
 "name": "minikube",
 "selfLink": "/apis/metrics.k8s.io/v1beta1/nodes/minikube",
 "creationTimestamp": "2022-01-27T18:48:43Z"
 },
 "timestamp": "2022-01-27T18:48:33Z",
 "window": "30s",
 "usage": {
 "cpu": "487558164n",
 "memory": "732212Ki"
 }
}
```

Here is an example of the Metrics API request for a kube-scheduler-minikube pod contained in the kube-system namespace and piped through jq for easier reading:

```
kubectl get --raw "/apis/metrics.k8s.io/v1beta1/namespaces/kube-system/pods/kube-scheduler-minikube" | jq "
```

Here is the same API call using curl:

```
curl http://localhost:8080/apis/metrics.k8s.io/v1beta1/namespaces/kube-system/pods/kube-scheduler-minikube
```

Sample response:

```
{
 "kind": "PodMetrics",
 "apiVersion": "metrics.k8s.io/v1beta1",
 "metadata": {
 "name": "kube-scheduler-minikube",
 "namespace": "kube-system",
 "selfLink": "/apis/metrics.k8s.io/v1beta1/namespaces/kube-system/pods/kube-scheduler-minikube",
 "creationTimestamp": "2022-01-27T19:25:00Z"
 },
}
```

```
"timestamp": "2022-01-27T19:24:31Z",
"window": "30s",
"containers": [
 {
 "name": "kube-scheduler",
 "usage": {
 "cpu": "9559630n",
 "memory": "22244Ki"
 }
 }
]
```

The Metrics API is defined in the [k8s.io/metrics](https://github.com/kubernetes/k8s.io/tree/master/metrics) repository. You must enable the [API aggregation layer](#) and register an [APIService](#) for the metrics.k8s.io API.

To learn more about the Metrics API, see [resource metrics API design](#), the [metrics-server repository](#) and the [resource metrics API](#).

#### **Note:**

You must deploy the metrics-server or alternative adapter that serves the Metrics API to be able to access it.

## **Measuring resource usage**

### **CPU**

CPU is reported as the average core usage measured in cpu units. One cpu, in Kubernetes, is equivalent to 1 vCPU/Core for cloud providers, and 1 hyper-thread on bare-metal Intel processors.

This value is derived by taking a rate over a cumulative CPU counter provided by the kernel (in both Linux and Windows kernels). The time window used to calculate CPU is shown under window field in Metrics API.

To learn more about how Kubernetes allocates and measures CPU resources, see [meaning of CPU](#).

### **Memory**

Memory is reported as the working set, measured in bytes, at the instant the metric was collected.

In an ideal world, the "working set" is the amount of memory in-use that cannot be freed under memory pressure. However, calculation of the working set varies by host OS, and generally makes heavy use of heuristics to produce an estimate.

The Kubernetes model for a container's working set expects that the container runtime counts anonymous memory associated with the container in question. The working set metric typically also includes some cached (file-backed) memory, because the host OS cannot always reclaim pages.

To learn more about how Kubernetes allocates and measures memory resources, see [meaning of memory](#).

## Metrics Server

The metrics-server fetches resource metrics from the kubelets and exposes them in the Kubernetes API server through the Metrics API for use by the HPA and VPA. You can also view these metrics using the `kubectl top` command.

The metrics-server uses the Kubernetes API to track nodes and pods in your cluster. The metrics-server queries each node over HTTP to fetch metrics. The metrics-server also builds an internal view of pod metadata, and keeps a cache of pod health. That cached pod health information is available via the extension API that the metrics-server makes available.

For example with an HPA query, the metrics-server needs to identify which pods fulfill the label selectors in the deployment.

The metrics-server calls the [kubelet](#) API to collect metrics from each node. Depending on the metrics-server version it uses:

- Metrics resource endpoint `/metrics/resource` in version v0.6.0+ or
- Summary API endpoint `/stats/summary` in older versions

## What's next

To learn more about the metrics-server, see the [metrics-server repository](#).

You can also check out the following:

- [metrics-server design](#)
- [metrics-server FAQ](#)
- [metrics-server known issues](#)
- [metrics-server releases](#)
- [Horizontal Pod Autoscaling](#)

To learn about how the kubelet serves node metrics, and how you can access those via the Kubernetes API, read [Node Metrics Data](#).

## Tools for Monitoring Resources

To scale an application and provide a reliable service, you need to understand how the application behaves when it is deployed. You can examine application performance in a Kubernetes cluster by examining the containers, [pods](#), [services](#), and the characteristics of the overall cluster. Kubernetes provides detailed information about an application's resource usage at each of these levels. This information allows you to evaluate your application's performance and where bottlenecks can be removed to improve overall performance.

In Kubernetes, application monitoring does not depend on a single monitoring solution. On new clusters, you can use [resource metrics](#) or [full metrics](#) pipelines to collect monitoring statistics.

# Resource metrics pipeline

The resource metrics pipeline provides a limited set of metrics related to cluster components such as the [Horizontal Pod Autoscaler](#) controller, as well as the `kubectl top` utility. These metrics are collected by the lightweight, short-term, in-memory [metrics-server](#) and are exposed via the `metrics.k8s.io` API.

`metrics-server` discovers all nodes on the cluster and queries each node's [kubelet](#) for CPU and memory usage. The kubelet acts as a bridge between the Kubernetes master and the nodes, managing the pods and containers running on a machine. The kubelet translates each pod into its constituent containers and fetches individual container usage statistics from the container runtime through the container runtime interface. If you use a container runtime that uses Linux cgroups and namespaces to implement containers, and the container runtime does not publish usage statistics, then the kubelet can look up those statistics directly (using code from [cAdvisor](#)). No matter how those statistics arrive, the kubelet then exposes the aggregated pod resource usage statistics through the `metrics-server` Resource Metrics API. This API is served at `/metrics/resource/v1beta1` on the kubelet's authenticated and read-only ports.

## Full metrics pipeline

A full metrics pipeline gives you access to richer metrics. Kubernetes can respond to these metrics by automatically scaling or adapting the cluster based on its current state, using mechanisms such as the Horizontal Pod Autoscaler. The monitoring pipeline fetches metrics from the kubelet and then exposes them to Kubernetes via an adapter by implementing either the `custom.metrics.k8s.io` or `external.metrics.k8s.io` API.

Kubernetes is designed to work with [OpenMetrics](#), which is one of the [CNCF Observability and Analysis - Monitoring Projects](#), built upon and carefully extending [Prometheus exposition format](#) in almost 100% backwards-compatible ways.

If you glance over at the [CNCF Landscape](#), you can see a number of monitoring projects that can work with Kubernetes by *scraping* metric data and using that to help you observe your cluster. It is up to you to select the tool or tools that suit your needs. The CNCF landscape for observability and analytics includes a mix of open-source software, paid-for software-as-a-service, and other commercial products.

When you design and implement a full metrics pipeline you can make that monitoring data available back to Kubernetes. For example, a `HorizontalPodAutoscaler` can use the processed metrics to work out how many Pods to run for a component of your workload.

Integration of a full metrics pipeline into your Kubernetes implementation is outside the scope of Kubernetes documentation because of the very wide scope of possible solutions.

The choice of monitoring platform depends heavily on your needs, budget, and technical resources. Kubernetes does not recommend any specific metrics pipeline; [many options](#) are available. Your monitoring system should be capable of handling the [OpenMetrics](#) metrics transmission standard and needs to be chosen to best fit into your overall design and deployment of your infrastructure platform.

## What's next

Learn about additional debugging tools, including:

- [Logging](#)
- [Monitoring](#)
- [Getting into containers via exec](#)
- [Connecting to containers via proxies](#)
- [Connecting to containers via port forwarding](#)
- [Inspect Kubernetes node with crictl](#)

## Monitor Node Health

*Node Problem Detector* is a daemon for monitoring and reporting about a node's health. You can run Node Problem Detector as a DaemonSet or as a standalone daemon. Node Problem Detector collects information about node problems from various daemons and reports these conditions to the API server as Node [Conditions](#) or as [Events](#).

To learn how to install and use Node Problem Detector, see [Node Problem Detector project documentation](#).

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

## Limitations

- Node Problem Detector uses the kernel log format for reporting kernel issues. To learn how to extend the kernel log format, see [Add support for another log format](#).

## Enabling Node Problem Detector

Some cloud providers enable Node Problem Detector as an [Addon](#). You can also enable Node Problem Detector with kubectl or by creating an Addon DaemonSet.

### Using kubectl to enable Node Problem Detector

kubectl provides the most flexible management of Node Problem Detector. You can overwrite the default configuration to fit it into your environment or to detect customized node problems. For example:

1. Create a Node Problem Detector configuration similar to `node-problem-detector.yaml`:

[debug/node-problem-detector.yaml](#)

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
 name: node-problem-detector-v0.1
 namespace: kube-system
 labels:
 k8s-app: node-problem-detector
 version: v0.1
 kubernetes.io/cluster-service: "true"
spec:
 selector:
 matchLabels:
 k8s-app: node-problem-detector
 version: v0.1
 kubernetes.io/cluster-service: "true"
 template:
 metadata:
 labels:
 k8s-app: node-problem-detector
 version: v0.1
 kubernetes.io/cluster-service: "true"
 spec:
 hostNetwork: true
 containers:
 - name: node-problem-detector
 image: registry.k8s.io/node-problem-detector:v0.1
 securityContext:
 privileged: true
 resources:
 limits:
 cpu: "200m"
 memory: "100Mi"
 requests:
 cpu: "20m"
 memory: "20Mi"
 volumeMounts:
 - name: log
 mountPath: /log
 readOnly: true
 volumes:
 - name: log
 hostPath:
 path: /var/log/
```

**Note:**

You should verify that the system log directory is right for your operating system distribution.

2. Start node problem detector with kubectl:

```
kubectly apply -f https://k8s.io/examples/debug/node-problem-detector.yaml
```

## Using an Addon pod to enable Node Problem Detector

If you are using a custom cluster bootstrap solution and don't need to overwrite the default configuration, you can leverage the Addon pod to further automate the deployment.

Create `node-problem-detector.yaml`, and save the configuration in the Addon pod's directory /`etc/kubernetes/addons/node-problem-detector` on a control plane node.

## Overwrite the configuration

The [default configuration](#) is embedded when building the Docker image of Node Problem Detector.

However, you can use a [ConfigMap](#) to overwrite the configuration:

1. Change the configuration files in `config/`
2. Create the ConfigMap `node-problem-detector-config`:

```
kubectly create configmap node-problem-detector-config --from-file=config/
```

3. Change the `node-problem-detector.yaml` to use the ConfigMap:

[debug/node-problem-detector-configmap.yaml](#)

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
 name: node-problem-detector-v0.1
 namespace: kube-system
 labels:
 k8s-app: node-problem-detector
 version: v0.1
 kubernetes.io/cluster-service: "true"
spec:
 selector:
 matchLabels:
 k8s-app: node-problem-detector
 version: v0.1
 kubernetes.io/cluster-service: "true"
 template:
 metadata:
 labels:
 k8s-app: node-problem-detector
 version: v0.1
 kubernetes.io/cluster-service: "true"
 spec:
 hostNetwork: true
 containers:
 - name: node-problem-detector
 image: registry.k8s.io/node-problem-detector:v0.1
```

```

securityContext:
 privileged: true
resources:
 limits:
 cpu: "200m"
 memory: "100Mi"
 requests:
 cpu: "20m"
 memory: "20Mi"
volumeMounts:
- name: log
 mountPath: /log
 readOnly: true
- name: config # Overwrite the config/ directory with ConfigMap volume
 mountPath: /config
 readOnly: true
volumes:
- name: log
 hostPath:
 path: /var/log/
- name: config # Define ConfigMap volume
 configMap:
 name: node-problem-detector-config

```

4. Recreate the Node Problem Detector with the new configuration file:

```

If you have a node-problem-detector running, delete before recreating
kubectl delete -f https://k8s.io/examples/debug/node-problem-detector.yaml
kubectl apply -f https://k8s.io/examples/debug/node-problem-detector-configmap.yaml

```

### Note:

This approach only applies to a Node Problem Detector started with kubectl.

Overwriting a configuration is not supported if a Node Problem Detector runs as a cluster Addon. The Addon manager does not support ConfigMap.

## Problem Daemons

A problem daemon is a sub-daemon of the Node Problem Detector. It monitors specific kinds of node problems and reports them to the Node Problem Detector. There are several types of supported problem daemons.

- A `SystemLogMonitor` type of daemon monitors the system logs and reports problems and metrics according to predefined rules. You can customize the configurations for different log sources such as [filelog](#), [kmsg](#), [kernel](#), [abrt](#), and [systemd](#).
- A `SystemStatsMonitor` type of daemon collects various health-related system stats as metrics. You can customize its behavior by updating its [configuration file](#).
- A `CustomPluginMonitor` type of daemon invokes and checks various node problems by running user-defined scripts. You can use different custom plugin monitors to monitor different problems and customize the daemon behavior by updating the [configuration file](#).



- A HealthChecker type of daemon checks the health of the kubelet and container runtime on a node.

## Adding support for other log format

The system log monitor currently supports file-based logs, journald, and kmsg. Additional sources can be added by implementing a new [log watcher](#).

## Adding custom plugin monitors

You can extend the Node Problem Detector to execute any monitor scripts written in any language by developing a custom plugin. The monitor scripts must conform to the plugin protocol in exit code and standard output. For more information, please refer to the [plugin interface proposal](#).

## Exporter

An exporter reports the node problems and/or metrics to certain backends. The following exporters are supported:

- **Kubernetes exporter:** this exporter reports node problems to the Kubernetes API server. Temporary problems are reported as Events and permanent problems are reported as Node Conditions.
- **Prometheus exporter:** this exporter reports node problems and metrics locally as Prometheus (or OpenMetrics) metrics. You can specify the IP address and port for the exporter using command line arguments.
- **Stackdriver exporter:** this exporter reports node problems and metrics to the Stackdriver Monitoring API. The exporting behavior can be customized using a [configuration file](#).

## Recommendations and restrictions

It is recommended to run the Node Problem Detector in your cluster to monitor node health. When running the Node Problem Detector, you can expect extra resource overhead on each node. Usually this is fine, because:

- The kernel log grows relatively slowly.
- A resource limit is set for the Node Problem Detector.
- Even under high load, the resource usage is acceptable. For more information, see the Node Problem Detector [benchmark result](#).

## Debugging Kubernetes nodes with crictl

FEATURE STATE: Kubernetes v1.11 [stable]

crictl is a command-line interface for CRI-compatible container runtimes. You can use it to inspect and debug container runtimes and applications on a Kubernetes node. crictl and its source are hosted in the [cri-tools](#) repository.

# Before you begin

crictl requires a Linux operating system with a CRI runtime.

## Installing crictl

You can download a compressed archive crictl from the cri-tools [release page](#), for several different architectures. Download the version that corresponds to your version of Kubernetes. Extract it and move it to a location on your system path, such as `/usr/local/bin/`.

## General usage

The crictl command has several subcommands and runtime flags. Use `crictl help` or `crictl <subcommand> help` for more details.

You can set the endpoint for crictl by doing one of the following:

- Set the `--runtime-endpoint` and `--image-endpoint` flags.
- Set the `CONTAINER_RUNTIME_ENDPOINT` and `IMAGE_SERVICE_ENDPOINT` environment variables.
- Set the endpoint in the configuration file `/etc/crictl.yaml`. To specify a different file, use the `--config=PATH_TO_FILE` flag when you run crictl.

### Note:

If you don't set an endpoint, crictl attempts to connect to a list of known endpoints, which might result in an impact to performance.

You can also specify timeout values when connecting to the server and enable or disable debugging, by specifying timeout or debug values in the configuration file or using the `--timeout` and `--debug` command-line flags.

To view or edit the current configuration, view or edit the contents of `/etc/crictl.yaml`. For example, the configuration when using the containerd container runtime would be similar to this:

```
runtime-endpoint: unix:///var/run/containerd/containerd.sock
image-endpoint: unix:///var/run/containerd/containerd.sock
timeout: 10
debug: true
```

To learn more about crictl, refer to the [crictl documentation](#).

## Example crictl commands

The following examples show some crictl commands and example output.

## Warning:

If you use `crictl` to create pod sandboxes or containers on a running Kubernetes cluster, the Kubelet will eventually delete them. `crictl` is not a general purpose workflow tool, but a tool that is useful for debugging.

## List pods

List all pods:

```
crictl pods
```

The output is similar to this:

POD ID	CREATED	STATE	NAME	NAMESPACE
ATTEMPT				
926f1b5a1d33a	About a minute ago	Ready	sh-84d7dcf559-4r2gq	
default 0				
4dccb216c4adb	About a minute ago	Ready	nginx-65899c769f-wv2gp	
default 0				
a86316e96fa89	17 hours ago	Ready	kube-proxy-gblk4	kube-system
0				
919630b8f81f1	17 hours ago	Ready	nvidia-device-plugin-zgbbv	kube-
system 0				

List pods by name:

```
crictl pods --name nginx-65899c769f-wv2gp
```

The output is similar to this:

POD ID	CREATED	STATE	NAME	NAMESPACE
ATTEMPT				
4dccb216c4adb	2 minutes ago	Ready	nginx-65899c769f-wv2gp	default 0

List pods by label:

```
crictl pods --label run=nginx
```

The output is similar to this:

POD ID	CREATED	STATE	NAME	NAMESPACE
ATTEMPT				
4dccb216c4adb	2 minutes ago	Ready	nginx-65899c769f-wv2gp	default 0

## List images

List all images:

```
crictl images
```

The output is similar to this:

IMAGE	TAG	IMAGE ID	SIZE
busybox	latest	8c811b4aec35f	1.15MB
k8s-gcrio.azureedge.net/hyperkube-amd64	v1.10.3	e179bbfe5d238	665MB
k8s-gcrio.azureedge.net/pause-amd64	3.1	da86e6ba6ca19	742kB
nginx	latest	cd5239a0906a6	109MB

List images by repository:

```
crictl images nginx
```

The output is similar to this:

IMAGE	TAG	IMAGE ID	SIZE
nginx	latest	cd5239a0906a6	109MB

Only list image IDs:

```
crictl images -q
```

The output is similar to this:

```
sha256:8c811b4aec35f259572d0f79207bc0678df4c736eeec50bc9fec37ed936a472a
sha256:e179bbfe5d238de6069f3b03fccbecc3fb4f2019af741bfff1233c4d7b2970c5
sha256:da86e6ba6ca197bf6bc5e9d900febd906b133eaa4750e6bed647b0f50ed43e
sha256:cd5239a0906a6ccf0562354852fae04bc5b52d72a2aff9a871ddb6bd57553569
```

## List containers

List all containers:

```
crictl ps -a
```

The output is similar to this:

CONTAINER ID	IMAGE	STATE	NAME	ATTEMPT	
1f73f2d81bf98					
busybox@sha256:141c253bc4c3fd0a201d32dc1f493bcf3fff003b6df416dea4f41046e0f37d47		7 minutes ago	Running	sh	1
9c5951df22c78					
busybox@sha256:141c253bc4c3fd0a201d32dc1f493bcf3fff003b6df416dea4f41046e0f37d47		8 minutes ago	Exited	sh	0
87d3992f84f74					
nginx@sha256:d0a8828cccb73397acb0073bf34f4d7d8aa315263f1e7806bf8c55d8ac139d5f		8 minutes ago	Running	nginx	0
1941fb4da154f	k8s-gcrio.azureedge.net/hyperkube-				
amd64@sha256:00d814b1f7763f4ab5be80c58e98140dfc69df107f253d7fdd714b30a714260a		18 hours ago	Running	kube-proxy	0

List running containers:

```
crictl ps
```

The output is similar to this:

CONTAINER ID	IMAGE	NAME	STATE	ATTEMPT
1f73f2d81bf98	busybox@sha256:141c253bc4c3fd0a201d32dc1f493bcf3fff003b6df416dea4f41046e0f37d47		Running	1
87d3992f84f74	nginx@sha256:d0a8828cccb73397acb0073bf34f4d7d8aa315263f1e7806bf8c55d8ac139d5f		Running	0
1941fb4da154f	k8s-gcrio.azureedge.net/hyperkube-amd64@sha256:00d814b1f7763f4ab5be80c58e98140dfc69df107f253d7fdd714b30a714260a	kube-proxy	Running	0

## Execute a command in a running container

```
crictl exec -i -t 1f73f2d81bf98 ls
```

The output is similar to this:

```
bin dev etc home proc root sys tmp usr var
```

## Get a container's logs

Get all container logs:

```
crictl logs 87d3992f84f74
```

The output is similar to this:

```
10.240.0.96 - - [06/Jun/2018:02:45:49 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.47.0" "-"
10.240.0.96 - - [06/Jun/2018:02:45:50 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.47.0" "-"
10.240.0.96 - - [06/Jun/2018:02:45:51 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.47.0" "-"
```

Get only the latest N lines of logs:

```
crictl logs --tail=1 87d3992f84f74
```

The output is similar to this:

```
10.240.0.96 - - [06/Jun/2018:02:45:51 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.47.0" "-"
```

## Run a pod sandbox

Using crictl to run a pod sandbox is useful for debugging container runtimes. On a running Kubernetes cluster, the sandbox will eventually be stopped and deleted by the Kubelet.

1. Create a JSON file like the following:

```
{
 "metadata": {
 "name": "nginx-sandbox",
 "namespace": "default",
 "attempt": 1,
 "uid": "hdishd83djaidwnduwk28bcsb"
```

```
},
"log_directory": "/tmp",
"linux": {
}
}
```

2. Use the `crictl runp` command to apply the JSON and run the sandbox.

```
crictl runp pod-config.json
```

The ID of the sandbox is returned.

## Create a container

Using `crictl` to create a container is useful for debugging container runtimes. On a running Kubernetes cluster, the container will eventually be stopped and deleted by the Kubelet.

1. Pull a busybox image

```
crictl pull busybox
```

```
Image is up to date for
busybox@sha256:141c253bc4c3fd0a201d32dc1f493bcf3fff003b6df416dea4f41046e0f37d47
```

2. Create configs for the pod and the container:

### Pod config:

```
{
 "metadata": {
 "name": "busybox-sandbox",
 "namespace": "default",
 "attempt": 1,
 "uid": "aewi4aeThua7ooShohbo1phoj"
 },
 "log_directory": "/tmp",
 "linux": {
 }
}
```

### Container config:

```
{
 "metadata": {
 "name": "busybox"
 },
 "image": {
 "image": "busybox"
 },
 "command": [
 "top"
],
 "log_path": "busybox.log",
 "linux": {

```

```
}
}
```

3. Create the container, passing the ID of the previously-created pod, the container config file, and the pod config file. The ID of the container is returned.

```
crictl create f84dd361f8dc51518ed291fbadd6db537b0496536c1d2d6c05ff943ce8c9a54f
container-config.json pod-config.json
```

4. List all containers and verify that the newly-created container has its state set to Created.

```
crictl ps -a
```

The output is similar to this:

CONTAINER ID	IMAGE	CREATED	STATE	NAME	ATTEMPT
3e025dd50a72d	busybox	32 seconds ago	Created	busybox	0

## Start a container

To start a container, pass its ID to `crictl start`:

```
crictl start 3e025dd50a72d956c4f14881fbb5b1080c9275674e95fb67f965f6478a957d60
```

The output is similar to this:

```
3e025dd50a72d956c4f14881fbb5b1080c9275674e95fb67f965f6478a957d60
```

Check the container has its state set to Running.

```
crictl ps
```

The output is similar to this:

CONTAINER ID	IMAGE	CREATED	STATE	NAME	ATTEMPT
3e025dd50a72d	busybox	About a minute ago	Running	busybox	0

## What's next

- [Learn more about crictl.](#)
- [Map docker CLI commands to crictl.](#)

# Auditing

Kubernetes *auditing* provides a security-relevant, chronological set of records documenting the sequence of actions in a cluster. The cluster audits the activities generated by users, by applications that use the Kubernetes API, and by the control plane itself.

Auditing allows cluster administrators to answer the following questions:

- what happened?

- when did it happen?
- who initiated it?
- on what did it happen?
- where was it observed?
- from where was it initiated?
- to where was it going?

Audit records begin their lifecycle inside the [kube-apiserver](#) component. Each request on each stage of its execution generates an audit event, which is then pre-processed according to a certain policy and written to a backend. The policy determines what's recorded and the backends persist the records. The current backend implementations include logs files and webhooks.

Each request can be recorded with an associated *stage*. The defined stages are:

- RequestReceived - The stage for events generated as soon as the audit handler receives the request, and before it is delegated down the handler chain.
- ResponseStarted - Once the response headers are sent, but before the response body is sent. This stage is only generated for long-running requests (e.g. watch).
- ResponseComplete - The response body has been completed and no more bytes will be sent.
- Panic - Events generated when a panic occurred.

#### Note:

The configuration of an [Audit Event configuration](#) is different from the [Event](#) API object.

The audit logging feature increases the memory consumption of the API server because some context required for auditing is stored for each request. Memory consumption depends on the audit logging configuration.

## Audit policy

Audit policy defines rules about what events should be recorded and what data they should include. The audit policy object structure is defined in the [audit.k8s.io API group](#). When an event is processed, it's compared against the list of rules in order. The first matching rule sets the *audit level* of the event. The defined audit levels are:

- None - don't log events that match this rule.
- Metadata - log request metadata (requesting user, timestamp, resource, verb, etc.) but not request or response body.
- Request - log event metadata and request body but not response body. This does not apply for non-resource requests.
- RequestResponse - log event metadata, request and response bodies. This does not apply for non-resource requests.

You can pass a file with the policy to kube-apiserver using the `--audit-policy-file` flag. If the flag is omitted, no events are logged. Note that the rules field **must** be provided in the audit policy file. A policy with no (0) rules is treated as illegal.

Below is an example audit policy file:

[audit/audit-policy.yaml](#)



```

apiVersion: audit.k8s.io/v1 # This is required.
kind: Policy
Don't generate audit events for all requests in RequestReceived stage.
omitStages:
 - "RequestReceived"
rules:
 # Log pod changes at RequestResponse level
 - level: RequestResponse
 resources:
 - group: ""
 # Resource "pods" doesn't match requests to any subresource of pods,
 # which is consistent with the RBAC policy.
 resources: ["pods"]
 # Log "pods/log", "pods/status" at Metadata level
 - level: Metadata
 resources:
 - group: ""
 resources: ["pods/log", "pods/status"]

 # Don't log requests to a configmap called "controller-leader"
 - level: None
 resources:
 - group: ""
 resources: ["configmaps"]
 resourceNames: ["controller-leader"]

 # Don't log watch requests by the "system:kube-proxy" on endpoints or services
 - level: None
 users: ["system:kube-proxy"]
 verbs: ["watch"]
 resources:
 - group: "" # core API group
 resources: ["endpoints", "services"]

 # Don't log authenticated requests to certain non-resource URL paths.
 - level: None
 userGroups: ["system:authenticated"]
 nonResourceURLs:
 - "/api*" # Wildcard matching.
 - "/version"

 # Log the request body of configmap changes in kube-system.
 - level: Request
 resources:
 - group: "" # core API group
 resources: ["configmaps"]
 # This rule only applies to resources in the "kube-system" namespace.
 # The empty string "" can be used to select non-namespaced resources.
 namespaces: ["kube-system"]

 # Log configmap and secret changes in all other namespaces at the Metadata level.
 - level: Metadata
 resources:

```

```

- group: "" # core API group
 resources: ["secrets", "configmaps"]

Log all other resources in core and extensions at the Request level.
- level: Request
 resources:
 - group: "" # core API group
 - group: "extensions" # Version of group should NOT be included.

A catch-all rule to log all other requests at the Metadata level.
- level: Metadata
 # Long-running requests like watches that fall under this rule will not
 # generate an audit event in RequestReceived.
 omitStages:
 - "RequestReceived"

```

You can use a minimal audit policy file to log all requests at the Metadata level:

```

Log all requests at the Metadata level.
apiVersion: audit.k8s.io/v1
kind: Policy
rules:
- level: Metadata

```

If you're crafting your own audit profile, you can use the audit profile for Google Container-Optimized OS as a starting point. You can check the [configure-helper.sh](#) script, which generates an audit policy file. You can see most of the audit policy file by looking directly at the script.

You can also refer to the [Policy configuration reference](#) for details about the fields defined.

## Audit backends

Audit backends persist audit events to an external storage. Out of the box, the kube-apiserver provides two backends:

- Log backend, which writes events into the filesystem
- Webhook backend, which sends events to an external HTTP API

In all cases, audit events follow a structure defined by the Kubernetes API in the [audit.k8s.io API group](#).

### Note:

In case of patches, request body is a JSON array with patch operations, not a JSON object with an appropriate Kubernetes API object. For example, the following request body is a valid patch request to `/apis/batch/v1/namespaces/some-namespace/jobs/some-job-name`:

```

[
 {
 "op": "replace",
 "path": "/spec/parallelism",
 "value": 0
 },

```

```
{
 "op": "remove",
 "path": "/spec/template/spec/containers/0/terminationMessagePolicy"
}
]
```

## Log backend

The log backend writes audit events to a file in [JSONlines](#) format. You can configure the log audit backend using the following kube-apiserver flags:

- `--audit-log-path` specifies the log file path that log backend uses to write audit events. Not specifying this flag disables log backend. `-` means standard out
- `--audit-log-maxage` defines the maximum number of days to retain old audit log files
- `--audit-log-maxbackup` defines the maximum number of audit log files to retain
- `--audit-log-maxsize` defines the maximum size in megabytes of the audit log file before it gets rotated

If your cluster's control plane runs the kube-apiserver as a Pod, remember to mount the `hostPath` to the location of the policy file and log file, so that audit records are persisted. For example:

```
- --audit-policy-file=/etc/kubernetes/audit-policy.yaml
- --audit-log-path=/var/log/kubernetes/audit/audit.log
```

then mount the volumes:

```
...
volumeMounts:
- mountPath: /etc/kubernetes/audit-policy.yaml
 name: audit
 readOnly: true
- mountPath: /var/log/kubernetes/audit/
 name: audit-log
 readOnly: false
```

and finally configure the `hostPath`:

```
...
volumes:
- name: audit
 hostPath:
 path: /etc/kubernetes/audit-policy.yaml
 type: File
- name: audit-log
 hostPath:
 path: /var/log/kubernetes/audit/
 type: DirectoryOrCreate
```

## Webhook backend

The webhook audit backend sends audit events to a remote web API, which is assumed to be a form of the Kubernetes API, including means of authentication. You can configure a webhook audit backend using the following kube-apiserver flags:

- `--audit-webhook-config-file` specifies the path to a file with a webhook configuration. The webhook configuration is effectively a specialized [kubeconfig](#).
- `--audit-webhook-initial-backoff` specifies the amount of time to wait after the first failed request before retrying. Subsequent requests are retried with exponential backoff.

The webhook config file uses the kubeconfig format to specify the remote address of the service and credentials used to connect to it.

## Event batching

Both log and webhook backends support batching. Using webhook as an example, here's the list of available flags. To get the same flag for log backend, replace `webhook` with `log` in the flag name. By default, batching is enabled in webhook and disabled in log. Similarly, by default throttling is enabled in webhook and disabled in log.

- `--audit-webhook-mode` defines the buffering strategy. One of the following:
  - `batch` - buffer events and asynchronously process them in batches. This is the default.
  - `blocking` - block API server responses on processing each individual event.
  - `blocking-strict` - Same as `blocking`, but when there is a failure during audit logging at the `RequestReceived` stage, the whole request to the kube-apiserver fails.

The following flags are used only in the batch mode:

- `--audit-webhook-batch-buffer-size` defines the number of events to buffer before batching. If the rate of incoming events overflows the buffer, events are dropped.
- `--audit-webhook-batch-max-size` defines the maximum number of events in one batch.
- `--audit-webhook-batch-max-wait` defines the maximum amount of time to wait before unconditionally batching events in the queue.
- `--audit-webhook-batch-throttle-qps` defines the maximum average number of batches generated per second.
- `--audit-webhook-batch-throttle-burst` defines the maximum number of batches generated at the same moment if the allowed QPS was underutilized previously.

## Parameter tuning

Parameters should be set to accommodate the load on the API server.

For example, if kube-apiserver receives 100 requests each second, and each request is audited only on `ResponseStarted` and `ResponseComplete` stages, you should account for  $\approx 200$  audit events being generated each second. Assuming that there are up to 100 events in a batch, you should set throttling level at least 2 queries per second. Assuming that the backend can take up to 5 seconds to write events, you should set the buffer size to hold up to 5 seconds of events; that is: 10 batches, or 1000 events.

In most cases however, the default parameters should be sufficient and you don't have to worry about setting them manually. You can look at the following Prometheus metrics exposed by kube-apiserver and in the logs to monitor the state of the auditing subsystem.

- `apiserver_audit_event_total` metric contains the total number of audit events exported.
- `apiserver_audit_error_total` metric contains the total number of events dropped due to an error during exporting.

## Log entry truncation

Both log and webhook backends support limiting the size of events that are logged. As an example, the following is the list of flags available for the log backend:

- `audit-log-truncate-enabled` whether event and batch truncating is enabled.
- `audit-log-truncate-max-batch-size` maximum size in bytes of the batch sent to the underlying backend.
- `audit-log-truncate-max-event-size` maximum size in bytes of the audit event sent to the underlying backend.

By default truncate is disabled in both webhook and log, a cluster administrator should set `audit-log-truncate-enabled` or `audit-webhook-truncate-enabled` to enable the feature.

## What's next

- Learn about [Mutating webhook auditing annotations](#).
- Learn more about [Event](#) and the [Policy](#) resource types by reading the Audit configuration reference.

# Debugging Kubernetes Nodes With Kubectl

This page shows how to debug a [node](#) running on the Kubernetes cluster using `kubectl debug` command.

## Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version 1.2. To check the version, enter `kubectl version`.

You need to have permission to create Pods and to assign those new Pods to arbitrary nodes. You also need to be authorized to create Pods that access filesystems from the host.

# Debugging a Node using kubectl debug node

Use the `kubectl debug node` command to deploy a Pod to a Node that you want to troubleshoot. This command is helpful in scenarios where you can't access your Node by using an SSH connection. When the Pod is created, the Pod opens an interactive shell on the Node. To create an interactive shell on a Node named “mynode”, run:

```
kubectl debug node/mynode -it --image=ubuntu
```

Creating debugging pod node-debugger-mynode-pdx84 with container debugger on node mynode.

If you don't see a command prompt, try pressing enter.

```
root@mynode:/#
```

The debug command helps to gather information and troubleshoot issues. Commands that you might use include `ip`, `ifconfig`, `nc`, `ping`, and `ps` and so on. You can also install other tools, such as `mtr`, `tcpdump`, and `curl`, from the respective package manager.

## Note:

The debug commands may differ based on the image the debugging pod is using and these commands might need to be installed.

The debugging Pod can access the root filesystem of the Node, mounted at `/host` in the Pod. If you run your kubelet in a filesystem namespace, the debugging Pod sees the root for that namespace, not for the entire node. For a typical Linux node, you can look at the following paths to find relevant logs:

```
/host/var/log/kubelet.log
```

Logs from the kubelet, responsible for running containers on the node.

```
/host/var/log/kube-proxy.log
```

Logs from kube-proxy, which is responsible for directing traffic to Service endpoints.

```
/host/var/log/containerd.log
```

Logs from the containerd process running on the node.

```
/host/var/log/syslog
```

Shows general messages and information regarding the system.

```
/host/var/log/kern.log
```

Shows kernel logs.

When creating a debugging session on a Node, keep in mind that:

- `kubectl debug` automatically generates the name of the new pod, based on the name of the node.
- The root filesystem of the Node will be mounted at `/host`.
- Although the container runs in the host IPC, Network, and PID namespaces, the pod isn't privileged. This means that reading some process information might fail because access to that information is restricted to superusers. For example, `chroot /host` will fail. If you need a privileged pod, create it manually.

## Cleaning up

When you finish using the debugging Pod, delete it:

```
kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
node-debugger-mynode-pdx84	0/1	Completed	0	8m1s

```
Change the pod name accordingly
kubectl delete pod node-debugger-mynode-pdx84 --now
```

```
pod "node-debugger-mynode-pdx84" deleted
```

#### Note:

The `kubectl debug node` command won't work if the Node is down (disconnected from the network, or kubelet dies and won't restart, etc.). Check [debugging a down/unreachable node](#) in that case.

## Developing and debugging services locally using telepresence

**Note:** This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

Kubernetes applications usually consist of multiple, separate services, each running in its own container. Developing and debugging these services on a remote Kubernetes cluster can be cumbersome, requiring you to [get a shell on a running container](#) in order to run debugging tools.

telepresence is a tool to ease the process of developing and debugging services locally while proxying the service to a remote Kubernetes cluster. Using telepresence allows you to use custom tools, such as a debugger and IDE, for a local service and provides the service full access to ConfigMap, secrets, and the services running on the remote cluster.

This document describes using telepresence to develop and debug services running on a remote cluster locally.

### Before you begin

- Kubernetes cluster is installed
- `kubectl` is configured to communicate with the cluster
- [Telepresence](#) is installed

## Connecting your local machine to a remote Kubernetes cluster

After installing telepresence, run `telepresence connect` to launch its Daemon and connect your local workstation to the cluster.

```
$ telepresence connect
```

```
Launching Telepresence Daemon
```

```
...
```

```
Connected to context default (https://<cluster public IP>)
```

You can curl services using the Kubernetes syntax e.g. `curl -ik https://kubernetes.default`

## Developing or debugging an existing service

When developing an application on Kubernetes, you typically program or debug a single service. The service might require access to other services for testing and debugging. One option is to use the continuous deployment pipeline, but even the fastest deployment pipeline introduces a delay in the program or debug cycle.

Use the `telepresence intercept $SERVICE_NAME --port $LOCAL_PORT:$REMOTE_PORT` command to create an "intercept" for rerouting remote service traffic.

Where:

- `$SERVICE_NAME` is the name of your local service
- `$LOCAL_PORT` is the port that your service is running on your local workstation
- And `$REMOTE_PORT` is the port your service listens to in the cluster

Running this command tells Telepresence to send remote traffic to your local service instead of the service in the remote Kubernetes cluster. Make edits to your service source code locally, save, and see the corresponding changes when accessing your remote application take effect immediately. You can also run your local service using a debugger or any other local development tool.

## How does Telepresence work?

Telepresence installs a traffic-agent sidecar next to your existing application's container running in the remote cluster. It then captures all traffic requests going into the Pod, and instead of forwarding this to the application in the remote cluster, it routes all traffic (when you create a [global intercept](#) or a subset of the traffic (when you create a [personal intercept](#)) to your local development environment.

## What's next

If you're interested in a hands-on tutorial, check out [this tutorial](#) that walks through locally developing the Guestbook application on Google Kubernetes Engine.

For further reading, visit the [Telepresence website](#).

## Windows debugging tips

### Node-level troubleshooting

1. My Pods are stuck at "Container Creating" or restarting over and over



Ensure that your pause image is compatible with your Windows OS version. See [Pause container](#) to see the latest / recommended pause image and/or get more information.

**Note:**

If using containerd as your container runtime the pause image is specified in the `plugins.plugins.cri.sandbox_image` field of the `config.toml` configuration file.

2. My pods show status as `ErrImagePull` or `ImagePullBackOff`

Ensure that your Pod is getting scheduled to a [compatible](#) Windows Node.

More information on how to specify a compatible node for your Pod can be found in [this guide](#).

## Network troubleshooting

1. My Windows Pods do not have network connectivity

If you are using virtual machines, ensure that MAC spoofing is **enabled** on all the VM network adapter(s).

2. My Windows Pods cannot ping external resources

Windows Pods do not have outbound rules programmed for the ICMP protocol. However, TCP/UDP is supported. When trying to demonstrate connectivity to resources outside of the cluster, substitute `ping <IP>` with corresponding `curl <IP>` commands.

If you are still facing problems, most likely your network configuration in [cni.conf](#) deserves some extra attention. You can always edit this static file. The configuration update will apply to any new Kubernetes resources.

One of the Kubernetes networking requirements (see [Kubernetes model](#)) is for cluster communication to occur without NAT internally. To honor this requirement, there is an [ExceptionList](#) for all the communication where you do not want outbound NAT to occur. However, this also means that you need to exclude the external IP you are trying to query from the `ExceptionList`. Only then will the traffic originating from your Windows pods be SNAT'ed correctly to receive a response from the outside world. In this regard, your `ExceptionList` in `cni.conf` should look as follows:

```
"ExceptionList": [
 "10.244.0.0/16", # Cluster subnet
 "10.96.0.0/12", # Service subnet
 "10.127.130.0/24" # Management (host) subnet
]
```

3. My Windows node cannot access NodePort type Services

Local NodePort access from the node itself fails. This is a known limitation. NodePort access works from other nodes or external clients.

4. vNICs and HNS endpoints of containers are being deleted

This issue can be caused when the `hostname-override` parameter is not passed to [kube-proxy](#). To resolve it, users need to pass the hostname to kube-proxy as follows:

```
C:\k\kube-proxy.exe --hostname-override=$(hostname)
```

#### 5. My Windows node cannot access my services using the service IP

This is a known limitation of the networking stack on Windows. However, Windows Pods can access the Service IP.

#### 6. No network adapter is found when starting the kubelet

The Windows networking stack needs a virtual adapter for Kubernetes networking to work. If the following commands return no results (in an admin shell), virtual network creation — a necessary prerequisite for the kubelet to work — has failed:

```
Get-HnsNetwork | ? Name -ieq "cbr0"
Get-NetAdapter | ? Name -Like "vEthernet (Ethernet*)"
```

Often it is worthwhile to modify the [InterfaceName](#) parameter of the `start.ps1` script, in cases where the host's network adapter isn't "Ethernet". Otherwise, consult the output of the `start-kubelet.ps1` script to see if there are errors during virtual network creation.

#### 7. DNS resolution is not properly working

Check the DNS limitations for Windows in this [section](#).

#### 8. `kubectl port-forward` fails with "unable to do port forwarding: wincat not found"

This was implemented in Kubernetes 1.15 by including `wincat.exe` in the pause infrastructure container `mcr.microsoft.com/oss/kubernetes/pause:3.6`. Be sure to use a supported version of Kubernetes. If you would like to build your own pause infrastructure container be sure to include [wincat](#).

#### 9. My Kubernetes installation is failing because my Windows Server node is behind a proxy

If you are behind a proxy, the following PowerShell environment variables must be defined:

```
[Environment]::SetEnvironmentVariable("HTTP_PROXY", "http://proxy.example.com:80/", [EnvironmentVariableTarget]::Machine)
[Environment]::SetEnvironmentVariable("HTTPS_PROXY", "http://proxy.example.com:443/", [EnvironmentVariableTarget]::Machine)
```

## Flannel troubleshooting

#### 1. With Flannel, my nodes are having issues after rejoining a cluster

Whenever a previously deleted node is being re-joined to the cluster, flannelD tries to assign a new pod subnet to the node. Users should remove the old pod subnet configuration files in the following paths:

```
Remove-Item C:\k\SourceVip.json
Remove-Item C:\k\SourceVipRequest.json
```

Flanneld is stuck in "Waiting for the Network to be created"

2.

There are numerous reports of this [issue](#); most likely it is a timing issue for when the management IP of the flannel network is set. A workaround is to relaunch start.ps1 or relaunch it manually as follows:

```
[Environment]::SetEnvironmentVariable("NODE_NAME", "<Windows_Worker_Hostname>")
C:\flannel\flanneld.exe --kubeconfig-file=c:\k\config --
iface=<Windows_Worker_Node_IP> --ip-masq=1 --kube-subnet-mgr=1
```

3. My Windows Pods cannot launch because of missing /run/flannel/subnet.env

This indicates that Flannel didn't launch correctly. You can either try to restart flanneld.exe or you can copy the files over manually from /run/flannel/subnet.env on the Kubernetes master to C:\run\flannel\subnet.env on the Windows worker node and modify the FLANNEL\_SUBNET row to a different number. For example, if node subnet 10.244.4.1/24 is desired:

```
FLANNEL_NETWORK=10.244.0.0/16
FLANNEL_SUBNET=10.244.4.1/24
FLANNEL_MTU=1500
FLANNEL_IPMASQ=true
```

## Further investigation

If these steps don't resolve your problem, you can get help running Windows containers on Windows nodes in Kubernetes through:

- StackOverflow [Windows Server Container](#) topic
- Kubernetes Official Forum [discuss.kubernetes.io](#)
- Kubernetes Slack [#SIG-Windows Channel](#)

# Manage Kubernetes Objects

Declarative and imperative paradigms for interacting with the Kubernetes API.

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[Declarative Management of Kubernetes Objects Using Configuration Files](#)

[Declarative Management of Kubernetes Objects Using Kustomize](#)

[Managing Kubernetes Objects Using Imperative Commands](#)

[Imperative Management of Kubernetes Objects Using Configuration Files](#)

[Update API Objects in Place Using kubectl patch](#)

Use kubectl patch to update Kubernetes API objects in place. Do a strategic merge patch or a JSON merge patch.

[Migrate Kubernetes Objects Using Storage Version Migration](#)

# Declarative Management of Kubernetes Objects Using Configuration Files

Kubernetes objects can be created, updated, and deleted by storing multiple object configuration files in a directory and using `kubectl` apply to recursively create and update those objects as needed. This method retains writes made to live objects without merging the changes back into the object configuration files. `kubectl diff` also gives you a preview of what changes apply will make.

## Before you begin

Install [kubectl](#).

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Trade-offs

The `kubectl` tool supports three kinds of object management:

- Imperative commands
- Imperative object configuration
- Declarative object configuration

See [Kubernetes Object Management](#) for a discussion of the advantages and disadvantage of each kind of object management.

## Overview

Declarative object configuration requires a firm understanding of the Kubernetes object definitions and configuration. Read and complete the following documents if you have not already:

- [Managing Kubernetes Objects Using Imperative Commands](#)
- [Imperative Management of Kubernetes Objects Using Configuration Files](#)

Following are definitions for terms used in this document:

- *object configuration file / configuration file*: A file that defines the configuration for a Kubernetes object. This topic shows how to pass configuration files to `kubectl` apply. Configuration files are typically stored in source control, such as Git.

- *live object configuration / live configuration*: The live configuration values of an object, as observed by the Kubernetes cluster. These are kept in the Kubernetes cluster storage, typically etcd.
- *declarative configuration writer / declarative writer*: A person or software component that makes updates to a live object. The live writers referred to in this topic make changes to object configuration files and run `kubectl apply` to write the changes.

## How to create objects

Use `kubectl apply` to create all objects, except those that already exist, defined by configuration files in a specified directory:

```
kubectl apply -f <directory>
```

This sets the `kubectl.kubernetes.io/last-applied-configuration: '{...}'` annotation on each object. The annotation contains the contents of the object configuration file that was used to create the object.

### Note:

Add the `-R` flag to recursively process directories.

Here's an example of an object configuration file:

[application/simple\\_deployment.yaml](https://k8s.io/examples/application/simple_deployment.yaml)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 selector:
 matchLabels:
 app: nginx
 minReadySeconds: 5
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
```

Run `kubectl diff` to print the object that will be created:

```
kubectl diff -f https://k8s.io/examples/application/simple_deployment.yaml
```

**Note:**

diff uses [server-side dry-run](#), which needs to be enabled on kube-apiserver.

Since diff performs a server-side apply request in dry-run mode, it requires granting PATCH, CREATE, and UPDATE permissions. See [Dry-Run Authorization](#) for details.

Create the object using kubectl apply:

```
kubectl apply -f https://k8s.io/examples/application/simple_deployment.yaml
```

Print the live configuration using kubectl get:

```
kubectl get -f https://k8s.io/examples/application/simple_deployment.yaml -o yaml
```

The output shows that the kubectl.kubernetes.io/last-applied-configuration annotation was written to the live configuration, and it matches the configuration file:

```
kind: Deployment
metadata:
 annotations:
 # ...
 # This is the json representation of simple_deployment.yaml
 # It was written by kubectl apply when the object was created
 kubectl.kubernetes.io/last-applied-configuration: |
 {"apiVersion":"apps/v1","kind":"Deployment",
 "metadata":{"annotations":{},"name":"nginx-deployment","namespace":"default"},
 "spec":{"minReadySeconds":5,"selector":{"matchLabels":{"app":"nginx"}},
 "template":{"metadata":
{"labels":{"app":"nginx"}},
 "spec":{"containers":[{"image":"nginx:1.14.2","name":"nginx",
 "ports":[{"containerPort":80}]}}}}}}
 # ...
spec:
 # ...
 minReadySeconds: 5
 selector:
 matchLabels:
 # ...
 app: nginx
 template:
 metadata:
 # ...
 labels:
 app: nginx
 spec:
 containers:
 - image: nginx:1.14.2
 # ...
 name: nginx
 ports:
 - containerPort: 80
 # ...
 # ...
```

```
...
...
```

## How to update objects

You can also use `kubect apply` to update all objects defined in a directory, even if those objects already exist. This approach accomplishes the following:

1. Sets fields that appear in the configuration file in the live configuration.
2. Clears fields removed from the configuration file in the live configuration.

```
kubect diff -f <directory>
kubect apply -f <directory>
```

### Note:

Add the `-R` flag to recursively process directories.

Here's an example configuration file:

[application/simple\\_deployment.yaml](https://k8s.io/examples/application/simple_deployment.yaml)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 selector:
 matchLabels:
 app: nginx
 minReadySeconds: 5
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
```

Create the object using `kubect apply`:

```
kubect apply -f https://k8s.io/examples/application/simple_deployment.yaml
```

### Note:

For purposes of illustration, the preceding command refers to a single configuration file instead of a directory.

Print the live configuration using `kubect get`:

```
kubectl get -f https://k8s.io/examples/application/simple_deployment.yaml -o yaml
```

The output shows that the `kubectl.kubernetes.io/last-applied-configuration` annotation was written to the live configuration, and it matches the configuration file:

```
kind: Deployment
metadata:
 annotations:
 # ...
 # This is the json representation of simple_deployment.yaml
 # It was written by kubectl apply when the object was created
 kubectl.kubernetes.io/last-applied-configuration: |
 {"apiVersion":"apps/v1","kind":"Deployment",
 "metadata":{"annotations":{"name":"nginx-deployment","namespace":"default"},
 "spec":{"minReadySeconds":5,"selector":{"matchLabels":{"app":"nginx"},"template":{"metadata":
{"labels":{"app":"nginx"},
 "spec":{"containers":[{"image":"nginx:1.14.2","name":"nginx",
 "ports":[{"containerPort":80}]}}}}}}}}
 # ...
spec:
 # ...
 minReadySeconds: 5
 selector:
 matchLabels:
 # ...
 app: nginx
 template:
 metadata:
 # ...
 labels:
 app: nginx
 spec:
 containers:
 - image: nginx:1.14.2
 # ...
 name: nginx
 ports:
 - containerPort: 80
 # ...
 # ...
 # ...
 # ...
```

Directly update the `replicas` field in the live configuration by using `kubectl scale`. This does not use `kubectl apply`:

```
kubectl scale deployment/nginx-deployment --replicas=2
```

Print the live configuration using `kubectl get`:

```
kubectl get deployment nginx-deployment -o yaml
```

The output shows that the `replicas` field has been set to 2, and the `last-applied-configuration` annotation does not contain a `replicas` field:



```

apiVersion: apps/v1
kind: Deployment
metadata:
 annotations:
 # ...
 # note that the annotation does not contain replicas
 # because it was not updated through apply
 kubectl.kubernetes.io/last-applied-configuration: |
 {"apiVersion":"apps/v1","kind":"Deployment",
 "metadata":{"annotations":{},"name":"nginx-deployment","namespace":"default"},
 "spec":{"minReadySeconds":5,"selector":{"matchLabels":{"app":"nginx"},"template":{"metadata":
{"labels":{"app":"nginx"}},
 "spec":{"containers":[{"image":"nginx:1.14.2","name":"nginx",
 "ports":[{"containerPort":80}]}]}}}}
 # ...
spec:
 replicas: 2 # written by scale
 # ...
 minReadySeconds: 5
 selector:
 matchLabels:
 # ...
 app: nginx
 template:
 metadata:
 # ...
 labels:
 app: nginx
 spec:
 containers:
 - image: nginx:1.14.2
 # ...
 name: nginx
 ports:
 - containerPort: 80
 # ...

```

Update the `simple_deployment.yaml` configuration file to change the image from `nginx:1.14.2` to `nginx:1.16.1`, and delete the `minReadySeconds` field:

[application/update\\_deployment.yaml](#)

```

apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:

```

```
 app: nginx
spec:
 containers:
 - name: nginx
 image: nginx:1.16.1 # update the image
 ports:
 - containerPort: 80
```

Apply the changes made to the configuration file:

```
kubectl diff -f https://k8s.io/examples/application/update_deployment.yaml
kubectl apply -f https://k8s.io/examples/application/update_deployment.yaml
```

Print the live configuration using kubectl get:

```
kubectl get -f https://k8s.io/examples/application/update_deployment.yaml -o yaml
```

The output shows the following changes to the live configuration:

- The replicas field retains the value of 2 set by kubectl scale. This is possible because it is omitted from the configuration file.
- The image field has been updated to nginx:1.16.1 from nginx:1.14.2.
- The last-applied-configuration annotation has been updated with the new image.
- The minReadySeconds field has been cleared.
- The last-applied-configuration annotation no longer contains the minReadySeconds field.

```
apiVersion: apps/v1
kind: Deployment
metadata:
 annotations:
 # ...
 # The annotation contains the updated image to nginx 1.16.1,
 # but does not contain the updated replicas to 2
 kubectl.kubernetes.io/last-applied-configuration: |
 {"apiVersion":"apps/v1","kind":"Deployment",
 "metadata":{"annotations":{},"name":"nginx-deployment","namespace":"default"},
 "spec":{"selector":{"matchLabels":{"app":"nginx"},"template":{"metadata":{"labels":
{"app":"nginx"},
 "spec":{"containers":[{"image":"nginx:1.16.1","name":"nginx",
 "ports":[{"containerPort":80}]}}]]}}}}
 # ...
spec:
 replicas: 2 # Set by `kubectl scale`. Ignored by `kubectl apply`.
 # minReadySeconds cleared by `kubectl apply`
 # ...
 selector:
 matchLabels:
 # ...
 app: nginx
 template:
 metadata:
 # ...
 labels:
 app: nginx
```

```
spec:
 containers:
 - image: nginx:1.16.1 # Set by `kubectl apply`
 # ...
 name: nginx
 ports:
 - containerPort: 80
 # ...
 # ...
 # ...
...
```

### Warning:

Mixing `kubectl apply` with the imperative object configuration commands `create` and `replace` is not supported. This is because `create` and `replace` do not retain the `kubectl.kubernetes.io/last-applied-configuration` that `kubectl apply` uses to compute updates.

## How to delete objects

There are two approaches to delete objects managed by `kubectl apply`.

### Recommended: `kubectl delete -f <filename>`

Manually deleting objects using the imperative command is the recommended approach, as it is more explicit about what is being deleted, and less likely to result in the user deleting something unintentionally:

```
kubectl delete -f <filename>
```

### Alternative: `kubectl apply -f <directory> --prune`

As an alternative to `kubectl delete`, you can use `kubectl apply` to identify objects to be deleted after their manifests have been removed from a directory in the local filesystem.

In Kubernetes 1.30, there are two pruning modes available in `kubectl apply`:

- Allowlist-based pruning: This mode has existed since `kubectl` v1.5 but is still in alpha due to usability, correctness and performance issues with its design. The `ApplySet`-based mode is designed to replace it.
  - `ApplySet`-based pruning: An *apply set* is a server-side object (by default, a `Secret`) that `kubectl` can use to accurately and efficiently track set membership across **apply** operations. This mode was introduced in alpha in `kubectl` v1.27 as a replacement for allowlist-based pruning.
- [Allow list](#)
  - [Apply set](#)

FEATURE STATE: Kubernetes v1.5 [alpha]

## Warning:

Take care when using `--prune` with `kubectl apply` in allow list mode. Which objects are pruned depends on the values of the `--prune-allowlist`, `--selector` and `--namespace` flags, and relies on dynamic discovery of the objects in scope. Especially if flag values are changed between invocations, this can lead to objects being unexpectedly deleted or retained.

To use allowlist-based pruning, add the following flags to your `kubectl apply` invocation:

- `--prune`: Delete previously applied objects that are not in the set passed to the current invocation.
- `--prune-allowlist`: A list of group-version-kinds (GVKs) to consider for pruning. This flag is optional but strongly encouraged, as its default value is a partial list of both namespaced and cluster-scoped types, which can lead to surprising results.
- `--selector/-l`: Use a label selector to constrain the set of objects selected for pruning. This flag is optional but strongly encouraged.
- `--all`: use instead of `--selector/-l` to explicitly select all previously applied objects of the allowlisted types.

Allowlist-based pruning queries the API server for all objects of the allowlisted GVKs that match the given labels (if any), and attempts to match the returned live object configurations against the object manifest files. If an object matches the query, and it does not have a manifest in the directory, and it has a `kubectl.kubernetes.io/last-applied-configuration` annotation, it is deleted.

```
kubectl apply -f <directory> --prune -l <labels> --prune-allowlist=<gvk-list>
```

## Warning:

Apply with prune should only be run against the root directory containing the object manifests. Running against sub-directories can cause objects to be unintentionally deleted if they were previously applied, have the labels given (if any), and do not appear in the subdirectory.

FEATURE STATE: Kubernetes v1.27 [alpha]

## Caution:

`kubectl apply --prune --applyset` is in alpha, and backwards incompatible changes might be introduced in subsequent releases.

To use ApplySet-based pruning, set the `KUBECTL_APPLYSET=true` environment variable, and add the following flags to your `kubectl apply` invocation:

- `--prune`: Delete previously applied objects that are not in the set passed to the current invocation.
- `--applyset`: The name of an object that `kubectl` can use to accurately and efficiently track set membership across apply operations.

```
KUBECTL_APPLYSET=true kubectl apply -f <directory> --prune --applyset=<name>
```

By default, the type of the ApplySet parent object used is a Secret. However, ConfigMaps can also be used in the format: `--applyset=configmaps/<name>`. When using a Secret or ConfigMap, `kubectl` will create the object if it does not already exist.

It is also possible to use custom resources as ApplySet parent objects. To enable this, label the Custom Resource Definition (CRD) that defines the resource you want to use with the following: `applyset.kubernetes.io/is-parent-type: true`. Then, create the object you want to use as an ApplySet parent (kubectl does not do this automatically for custom resources). Finally, refer to that object in the applyset flag as follows: `--applyset=<resource>.<group>/<name>` (for example, `widgets.custom.example.com/widget-name`).

With ApplySet-based pruning, kubectl adds the `applyset.kubernetes.io/part-of=<parentID>` label to each object in the set before they are sent to the server. For performance reasons, it also collects the list of resource types and namespaces that the set contains and adds these in annotations on the live parent object. Finally, at the end of the apply operation, it queries the API server for objects of those types in those namespaces (or in the cluster scope, as applicable) that belong to the set, as defined by the `applyset.kubernetes.io/part-of=<parentID>` label.

Caveats and restrictions:

- Each object may be a member of at most one set.
- The `--namespace` flag is required when using any namespaced parent, including the default Secret. This means that ApplySets spanning multiple namespaces must use a cluster-scoped custom resource as the parent object.
- To safely use ApplySet-based pruning with multiple directories, use a unique ApplySet name for each.

## How to view an object

You can use `kubectl get` with `-o yaml` to view the configuration of a live object:

```
kubectl get -f <filename|url> -o yaml
```

## How apply calculates differences and merges changes

**Caution:**

A *patch* is an update operation that is scoped to specific fields of an object instead of the entire object. This enables updating only a specific set of fields on an object without reading the object first.

When `kubectl apply` updates the live configuration for an object, it does so by sending a patch request to the API server. The patch defines updates scoped to specific fields of the live object configuration. The `kubectl apply` command calculates this patch request using the configuration file, the live configuration, and the last-applied-configuration annotation stored in the live configuration.

### Merge patch calculation

The `kubectl apply` command writes the contents of the configuration file to the `kubectl.kubernetes.io/last-applied-configuration` annotation. This is used to identify fields that have been removed from the configuration file and need to be cleared from the live configuration. Here are the steps used to calculate which fields should be deleted or set:

1. Calculate the fields to delete. These are the fields present in last-applied-configuration and missing from the configuration file.

2. Calculate the fields to add or set. These are the fields present in the configuration file whose values don't match the live configuration.

Here's an example. Suppose this is the configuration file for a Deployment object:

[application/update\\_deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.16.1 # update the image
 ports:
 - containerPort: 80
```

Also, suppose this is the live configuration for the same Deployment object:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 annotations:
 # ...
 # note that the annotation does not contain replicas
 # because it was not updated through apply
 kubectl.kubernetes.io/last-applied-configuration: |
 {"apiVersion":"apps/v1","kind":"Deployment",
 "metadata":{"annotations":{},"name":"nginx-deployment","namespace":"default"},
 "spec":{"minReadySeconds":5,"selector":{"matchLabels":{"app":"nginx"},"template":{"metadata":
{"labels":{"app":"nginx"},
"spec":{"containers":[{"image":"nginx:1.14.2","name":"nginx",
"ports":[{"containerPort":80}]}}]}}}}
 # ...
spec:
 replicas: 2 # written by scale
 # ...
 minReadySeconds: 5
 selector:
 matchLabels:
 # ...
 app: nginx
 template:
 metadata:
```

```
...
labels:
 app: nginx
spec:
 containers:
 - image: nginx:1.14.2
 # ...
 name: nginx
 ports:
 - containerPort: 80
 # ...
```

Here are the merge calculations that would be performed by `kubect apply`:

1. Calculate the fields to delete by reading values from last-applied-configuration and comparing them to values in the configuration file. Clear fields explicitly set to null in the local object configuration file regardless of whether they appear in the last-applied-configuration. In this example, `minReadySeconds` appears in the last-applied-configuration annotation, but does not appear in the configuration file. **Action:** Clear `minReadySeconds` from the live configuration.
2. Calculate the fields to set by reading values from the configuration file and comparing them to values in the live configuration. In this example, the value of `image` in the configuration file does not match the value in the live configuration. **Action:** Set the value of `image` in the live configuration.
3. Set the last-applied-configuration annotation to match the value of the configuration file.
4. Merge the results from 1, 2, 3 into a single patch request to the API server.

Here is the live configuration that is the result of the merge:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 annotations:
 # ...
 # The annotation contains the updated image to nginx 1.16.1,
 # but does not contain the updated replicas to 2
 kubectl.kubernetes.io/last-applied-configuration: |
 {"apiVersion":"apps/v1","kind":"Deployment",
 "metadata":{"annotations":{},"name":"nginx-deployment","namespace":"default"},
 "spec":{"selector":{"matchLabels":{"app":"nginx"},"template":{"metadata":{"labels":
{"app":"nginx"},"spec":{"containers":[{"image":"nginx:1.16.1","name":"nginx",
"ports":[{"containerPort":80}]}}}}}}}
 # ...
 spec:
 selector:
 matchLabels:
 # ...
 app: nginx
 replicas: 2 # Set by `kubectl scale`. Ignored by `kubectl apply`.
 # minReadySeconds cleared by `kubectl apply`
 # ...
 template:
```

```

metadata:
 # ...
 labels:
 app: nginx
spec:
 containers:
 - image: nginx:1.16.1 # Set by `kubectl apply`
 # ...
 name: nginx
 ports:
 - containerPort: 80
 # ...
 # ...
 # ...
 # ...

```

## How different types of fields are merged

How a particular field in a configuration file is merged with the live configuration depends on the type of the field. There are several types of fields:

- *primitive*: A field of type string, integer, or boolean. For example, `image` and `replicas` are primitive fields. **Action**: Replace.
- *map*, also called *object*: A field of type map or a complex type that contains subfields. For example, `labels`, `annotations`, `spec` and `metadata` are all maps. **Action**: Merge elements or subfields.
- *list*: A field containing a list of items that can be either primitive types or maps. For example, `containers`, `ports`, and `args` are lists. **Action**: Varies.

When `kubectl apply` updates a map or list field, it typically does not replace the entire field, but instead updates the individual subelements. For instance, when merging the `spec` on a `Deployment`, the entire `spec` is not replaced. Instead the subfields of `spec`, such as `replicas`, are compared and merged.

## Merging changes to primitive fields

Primitive fields are replaced or cleared.

**Note:**

- is used for "not applicable" because the value is not used.

Field in object configuration file	Field in live object configuration	Field in last-applied-configuration	Action
Yes	Yes	-	Set live to configuration file value.
Yes	No	-	Set live to local configuration.
No	-	Yes	Clear from live configuration.



Field in object configuration file	Field in live object configuration	Field in last-applied-configuration	Action
No	-	No	Do nothing. Keep live value.

## Merging changes to map fields

Fields that represent maps are merged by comparing each of the subfields or elements of the map:

### Note:

- is used for "not applicable" because the value is not used.

Key in object configuration file	Key in live object configuration	Field in last-applied-configuration	Action
Yes	Yes	-	Compare sub fields values.
Yes	No	-	Set live to local configuration.
No	-	Yes	Delete from live configuration.
No	-	No	Do nothing. Keep live value.

## Merging changes for fields of type list

Merging changes to a list uses one of three strategies:

- Replace the list if all its elements are primitives.
- Merge individual elements in a list of complex elements.
- Merge a list of primitive elements.

The choice of strategy is made on a per-field basis.

### Replace the list if all its elements are primitives

Treat the list the same as a primitive field. Replace or delete the entire list. This preserves ordering.

**Example:** Use `kubectl apply` to update the `args` field of a `Container` in a `Pod`. This sets the value of `args` in the live configuration to the value in the configuration file. Any `args` elements that had previously been added to the live configuration are lost. The order of the `args` elements defined in the configuration file is retained in the live configuration.

```
last-applied-configuration value
args: ["a", "b"]
```

```
configuration file value
args: ["a", "c"]
```

```
live configuration
```

```
args: ["a", "b", "d"]
```

```
result after merge
args: ["a", "c"]
```

**Explanation:** The merge used the configuration file value as the new list value.

### Merge individual elements of a list of complex elements:

Treat the list as a map, and treat a specific field of each element as a key. Add, delete, or update individual elements. This does not preserve ordering.

This merge strategy uses a special tag on each field called a `patchMergeKey`. The `patchMergeKey` is defined for each field in the Kubernetes source code: [types.go](https://types.go) When merging a list of maps, the field specified as the `patchMergeKey` for a given element is used like a map key for that element.

**Example:** Use `kubectl apply` to update the `containers` field of a `PodSpec`. This merges the list as though it was a map where each element is keyed by `name`.

```
last-applied-configuration value
containers:
- name: nginx
 image: nginx:1.16
- name: nginx-helper-a # key: nginx-helper-a; will be deleted in result
 image: helper:1.3
- name: nginx-helper-b # key: nginx-helper-b; will be retained
 image: helper:1.3

configuration file value
containers:
- name: nginx
 image: nginx:1.16
- name: nginx-helper-b
 image: helper:1.3
- name: nginx-helper-c # key: nginx-helper-c; will be added in result
 image: helper:1.3

live configuration
containers:
- name: nginx
 image: nginx:1.16
- name: nginx-helper-a
 image: helper:1.3
- name: nginx-helper-b
 image: helper:1.3
 args: ["run"] # Field will be retained
- name: nginx-helper-d # key: nginx-helper-d; will be retained
 image: helper:1.3

result after merge
containers:
- name: nginx
```

```
image: nginx:1.16
Element nginx-helper-a was deleted
- name: nginx-helper-b
 image: helper:1.3
 args: ["run"] # Field was retained
- name: nginx-helper-c # Element was added
 image: helper:1.3
- name: nginx-helper-d # Element was ignored
 image: helper:1.3
```

### Explanation:

- The container named "nginx-helper-a" was deleted because no container named "nginx-helper-a" appeared in the configuration file.
- The container named "nginx-helper-b" retained the changes to args in the live configuration. kubectl apply was able to identify that "nginx-helper-b" in the live configuration was the same "nginx-helper-b" as in the configuration file, even though their fields had different values (no args in the configuration file). This is because the patchMergeKey field value (name) was identical in both.
- The container named "nginx-helper-c" was added because no container with that name appeared in the live configuration, but one with that name appeared in the configuration file.
- The container named "nginx-helper-d" was retained because no element with that name appeared in the last-applied-configuration.

### Merge a list of primitive elements

As of Kubernetes 1.5, merging lists of primitive elements is not supported.

### Note:

Which of the above strategies is chosen for a given field is controlled by the patchStrategy tag in [types.go](https://kubernetes.io/docs/reference/generated/kubernetes-api/v1.10/#listmeta-1.10). If no patchStrategy is specified for a field of type list, then the list is replaced.

## Default field values

The API server sets certain fields to default values in the live configuration if they are not specified when the object is created.

Here's a configuration file for a Deployment. The file does not specify strategy:

[application/simple\\_deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 selector:
 matchLabels:
 app: nginx
 minReadySeconds: 5
```

```
template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
```

Create the object using kubectl apply:

```
kubectl apply -f https://k8s.io/examples/application/simple_deployment.yaml
```

Print the live configuration using kubectl get:

```
kubectl get -f https://k8s.io/examples/application/simple_deployment.yaml -o yaml
```

The output shows that the API server set several fields to default values in the live configuration. These fields were not specified in the configuration file.

```
apiVersion: apps/v1
kind: Deployment
...
spec:
 selector:
 matchLabels:
 app: nginx
 minReadySeconds: 5
 replicas: 1 # defaulted by apiserver
 strategy:
 rollingUpdate: # defaulted by apiserver - derived from strategy.type
 maxSurge: 1
 maxUnavailable: 1
 type: RollingUpdate # defaulted by apiserver
 template:
 metadata:
 creationTimestamp: null
 labels:
 app: nginx
 spec:
 containers:
 - image: nginx:1.14.2
 imagePullPolicy: IfNotPresent # defaulted by apiserver
 name: nginx
 ports:
 - containerPort: 80
 protocol: TCP # defaulted by apiserver
 resources: {} # defaulted by apiserver
 terminationMessagePath: /dev/termination-log # defaulted by apiserver
 dnsPolicy: ClusterFirst # defaulted by apiserver
 restartPolicy: Always # defaulted by apiserver
 securityContext: {} # defaulted by apiserver
```

```
terminationGracePeriodSeconds: 30 # defaulted by apiserver
...
```

In a patch request, defaulted fields are not re-defaulted unless they are explicitly cleared as part of a patch request. This can cause unexpected behavior for fields that are defaulted based on the values of other fields. When the other fields are later changed, the values defaulted from them will not be updated unless they are explicitly cleared.

For this reason, it is recommended that certain fields defaulted by the server are explicitly defined in the configuration file, even if the desired values match the server defaults. This makes it easier to recognize conflicting values that will not be re-defaulted by the server.

### Example:

```
last-applied-configuration
spec:
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80

configuration file
spec:
 strategy:
 type: Recreate # updated value
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80

live configuration
spec:
 strategy:
 type: RollingUpdate # defaulted value
 rollingUpdate: # defaulted value derived from type
 maxSurge : 1
 maxUnavailable: 1
 template:
 metadata:
 labels:
 app: nginx
```

```
spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
```

# result after merge - ERROR!

```
spec:
 strategy:
 type: Recreate # updated value: incompatible with rollingUpdate
 rollingUpdate: # defaulted value: incompatible with "type: Recreate"
 maxSurge : 1
 maxUnavailable: 1
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
```

### Explanation:

1. The user creates a Deployment without defining `strategy.type`.
2. The server defaults `strategy.type` to `RollingUpdate` and defaults the `strategy.rollingUpdate` values.
3. The user changes `strategy.type` to `Recreate`. The `strategy.rollingUpdate` values remain at their defaulted values, though the server expects them to be cleared. If the `strategy.rollingUpdate` values had been defined initially in the configuration file, it would have been more clear that they needed to be deleted.
4. Apply fails because `strategy.rollingUpdate` is not cleared. The `strategy.rollingupdate` field cannot be defined with a `strategy.type` of `Recreate`.

Recommendation: These fields should be explicitly defined in the object configuration file:

- Selectors and PodTemplate labels on workloads, such as Deployment, StatefulSet, Job, DaemonSet, ReplicaSet, and ReplicationController
- Deployment rollout strategy

### How to clear server-defaulted fields or fields set by other writers

Fields that do not appear in the configuration file can be cleared by setting their values to null and then applying the configuration file. For fields defaulted by the server, this triggers re-defaulting the values.

# How to change ownership of a field between the configuration file and direct imperative writers

These are the only methods you should use to change an individual object field:

- Use `kubectl apply`.
- Write directly to the live configuration without modifying the configuration file: for example, use `kubectl scale`.

## Changing the owner from a direct imperative writer to a configuration file

Add the field to the configuration file. For the field, discontinue direct updates to the live configuration that do not go through `kubectl apply`.

## Changing the owner from a configuration file to a direct imperative writer

As of Kubernetes 1.5, changing ownership of a field from a configuration file to an imperative writer requires manual steps:

- Remove the field from the configuration file.
- Remove the field from the `kubectl.kubernetes.io/last-applied-configuration` annotation on the live object.

## Changing management methods

Kubernetes objects should be managed using only one method at a time. Switching from one method to another is possible, but is a manual process.

### Note:

It is OK to use imperative deletion with declarative management.

## Migrating from imperative command management to declarative object configuration

Migrating from imperative command management to declarative object configuration involves several manual steps:

1. Export the live object to a local configuration file:

```
kubectl get <kind>/<name> -o yaml > <kind>_<name>.yaml
```

2. Manually remove the `status` field from the configuration file.

### Note:

This step is optional, as `kubectl apply` does not update the `status` field even if it is present in the configuration file.

- Set the `kubectl.kubernetes.io/last-applied-configuration` annotation on the object:
3. 

```
kubectl replace --save-config -f <kind>_<name>.yaml
```
  4. Change processes to use `kubectl apply` for managing the object exclusively.

## Migrating from imperative object configuration to declarative object configuration

1. Set the `kubectl.kubernetes.io/last-applied-configuration` annotation on the object:  

```
kubectl replace --save-config -f <kind>_<name>.yaml
```
2. Change processes to use `kubectl apply` for managing the object exclusively.

## Defining controller selectors and PodTemplate labels

### Warning:

Updating selectors on controllers is strongly discouraged.

The recommended approach is to define a single, immutable PodTemplate label used only by the controller selector with no other semantic meaning.

### Example:

```
selector:
 matchLabels:
 controller-selector: "apps/v1/deployment/nginx"
template:
 metadata:
 labels:
 controller-selector: "apps/v1/deployment/nginx"
```

## What's next

- [Managing Kubernetes Objects Using Imperative Commands](#)
- [Imperative Management of Kubernetes Objects Using Configuration Files](#)
- [Kubectl Command Reference](#)
- [Kubernetes API Reference](#)

## Declarative Management of Kubernetes Objects Using Kustomize

[Kustomize](#) is a standalone tool to customize Kubernetes objects through a [kustomization file](#).

Since 1.14, Kubectl also supports the management of Kubernetes objects using a kustomization file. To view Resources found in a directory containing a kustomization file, run the following command:



```
kubectl kustomize <kustomization_directory>
```

To apply those Resources, run kubectl apply with --kustomize or -k flag:

```
kubectl apply -k <kustomization_directory>
```

## Before you begin

Install [kubectl](#).

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

## Overview of Kustomize

Kustomize is a tool for customizing Kubernetes configurations. It has the following features to manage application configuration files:

- generating resources from other sources
- setting cross-cutting fields for resources
- composing and customizing collections of resources

## Generating Resources

ConfigMaps and Secrets hold configuration or sensitive data that are used by other Kubernetes objects, such as Pods. The source of truth of ConfigMaps or Secrets are usually external to a cluster, such as a .properties file or an SSH keyfile. Kustomize has secretGenerator and configMapGenerator, which generate Secret and ConfigMap from files or literals.

### configMapGenerator

To generate a ConfigMap from a file, add an entry to the files list in configMapGenerator. Here is an example of generating a ConfigMap with a data item from a .properties file:

```
Create a application.properties file
cat <<EOF >application.properties
FOO=Bar
EOF

cat <<EOF >./kustomization.yaml
configMapGenerator:
- name: example-configmap-1
 files:
 - application.properties
EOF
```

The generated ConfigMap can be examined with the following command:

```
kubectl kustomize ./
```

The generated ConfigMap is:

```
apiVersion: v1
data:
 application.properties: |
 FOO=Bar
kind: ConfigMap
metadata:
 name: example-configmap-1-8mbdf7882g
```

To generate a ConfigMap from an env file, add an entry to the envs list in configMapGenerator. Here is an example of generating a ConfigMap with a data item from a .env file:

```
Create a .env file
cat <<EOF >.env
FOO=Bar
EOF

cat <<EOF >./kustomization.yaml
configMapGenerator:
- name: example-configmap-1
 envs:
 - .env
EOF
```

The generated ConfigMap can be examined with the following command:

```
kubectl kustomize ./
```

The generated ConfigMap is:

```
apiVersion: v1
data:
 FOO: Bar
kind: ConfigMap
metadata:
 name: example-configmap-1-42cbbf598f
```

**Note:**

Each variable in the .env file becomes a separate key in the ConfigMap that you generate. This is different from the previous example which embeds a file named application.properties (and all its entries) as the value for a single key.

ConfigMaps can also be generated from literal key-value pairs. To generate a ConfigMap from a literal key-value pair, add an entry to the literals list in configMapGenerator. Here is an example of generating a ConfigMap with a data item from a key-value pair:

```
cat <<EOF >./kustomization.yaml
configMapGenerator:
```

```
- name: example-configmap-2
 literals:
 - FOO=Bar
EOF
```

The generated ConfigMap can be checked by the following command:

```
kubectl kustomize ./
```

The generated ConfigMap is:

```
apiVersion: v1
data:
 FOO: Bar
kind: ConfigMap
metadata:
 name: example-configmap-2-g2hdhfc6tk
```

To use a generated ConfigMap in a Deployment, reference it by the name of the configMapGenerator. Kustomize will automatically replace this name with the generated name.

This is an example deployment that uses a generated ConfigMap:

```
Create a application.properties file
cat <<EOF >application.properties
FOO=Bar
EOF
```

```
cat <<EOF >deployment.yaml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-app
 labels:
 app: my-app
spec:
 selector:
 matchLabels:
 app: my-app
 template:
 metadata:
 labels:
 app: my-app
 spec:
 containers:
 - name: app
 image: my-app
 volumeMounts:
 - name: config
 mountPath: /config
 volumes:
 - name: config
 configMap:
 name: example-configmap-1
```

EOF

```
cat <<EOF >./kustomization.yaml
resources:
- deployment.yaml
configMapGenerator:
- name: example-configmap-1
 files:
 - application.properties
EOF
```

Generate the ConfigMap and Deployment:

```
kubectl kustomize ./
```

The generated Deployment will refer to the generated ConfigMap by name:

```
apiVersion: v1
data:
 application.properties: |
 FOO=Bar
kind: ConfigMap
metadata:
 name: example-configmap-1-g4hk9g2ff8

apiVersion: apps/v1
kind: Deployment
metadata:
 labels:
 app: my-app
 name: my-app
spec:
 selector:
 matchLabels:
 app: my-app
 template:
 metadata:
 labels:
 app: my-app
 spec:
 containers:
 - image: my-app
 name: app
 volumeMounts:
 - mountPath: /config
 name: config
 volumes:
 - configMap:
 name: example-configmap-1-g4hk9g2ff8
 name: config
```

## secretGenerator

You can generate Secrets from files or literal key-value pairs. To generate a Secret from a file, add an entry to the files list in secretGenerator. Here is an example of generating a Secret with a data item from a file:

```
Create a password.txt file
cat <<EOF >./password.txt
username=admin
password=secret
EOF

cat <<EOF >./kustomization.yaml
secretGenerator:
- name: example-secret-1
 files:
 - password.txt
EOF
```

The generated Secret is as follows:

```
apiVersion: v1
data:
 password.txt: dXNlcm5hbWU9YWRTaW4KcGFzc3dvcmQ9c2VjcmV0Cg==
kind: Secret
metadata:
 name: example-secret-1-t2kt65hgtb
type: Opaque
```

To generate a Secret from a literal key-value pair, add an entry to literals list in secretGenerator. Here is an example of generating a Secret with a data item from a key-value pair:

```
cat <<EOF >./kustomization.yaml
secretGenerator:
- name: example-secret-2
 literals:
 - username=admin
 - password=secret
EOF
```

The generated Secret is as follows:

```
apiVersion: v1
data:
 password: c2VjcmV0
 username: YWRtaW4=
kind: Secret
metadata:
 name: example-secret-2-t52t6g96d8
type: Opaque
```

Like ConfigMaps, generated Secrets can be used in Deployments by referring to the name of the secretGenerator:

```
Create a password.txt file
cat <<EOF >./password.txt
username=admin
password=secret
EOF

cat <<EOF >deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-app
 labels:
 app: my-app
spec:
 selector:
 matchLabels:
 app: my-app
 template:
 metadata:
 labels:
 app: my-app
 spec:
 containers:
 - name: app
 image: my-app
 volumeMounts:
 - name: password
 mountPath: /secrets
 volumes:
 - name: password
 secret:
 secretName: example-secret-1
EOF

cat <<EOF >./kustomization.yaml
resources:
- deployment.yaml
secretGenerator:
- name: example-secret-1
 files:
 - password.txt
EOF
```

## generatorOptions

The generated ConfigMaps and Secrets have a content hash suffix appended. This ensures that a new ConfigMap or Secret is generated when the contents are changed. To disable the behavior of appending a suffix, one can use generatorOptions. Besides that, it is also possible to specify cross-cutting options for generated ConfigMaps and Secrets.

```
cat <<EOF >./kustomization.yaml
configMapGenerator:
- name: example-configmap-3
```

```
literals:
- FOO=Bar
generatorOptions:
 disableNameSuffixHash: true
labels:
 type: generated
annotations:
 note: generated
EOF
```

Runkubectl kustomize ./ to view the generated ConfigMap:

```
apiVersion: v1
data:
 FOO: Bar
kind: ConfigMap
metadata:
 annotations:
 note: generated
 labels:
 type: generated
 name: example-configmap-3
```

## Setting cross-cutting fields

It is quite common to set cross-cutting fields for all Kubernetes resources in a project. Some use cases for setting cross-cutting fields:

- setting the same namespace for all Resources
- adding the same name prefix or suffix
- adding the same set of labels
- adding the same set of annotations

Here is an example:

```
Create a deployment.yaml
cat <<EOF >./deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
 labels:
 app: nginx
spec:
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
```

```
- name: nginx
 image: nginx
EOF

cat <<EOF >./kustomization.yaml
namespace: my-namespace
namePrefix: dev-
nameSuffix: "-001"
commonLabels:
 app: bingo
commonAnnotations:
 oncallPager: 800-555-1212
resources:
- deployment.yaml
EOF
```

Run `kubectl kustomize ./` to view those fields are all set in the Deployment Resource:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 annotations:
 oncallPager: 800-555-1212
 labels:
 app: bingo
 name: dev-nginx-deployment-001
 namespace: my-namespace
spec:
 selector:
 matchLabels:
 app: bingo
 template:
 metadata:
 annotations:
 oncallPager: 800-555-1212
 labels:
 app: bingo
 spec:
 containers:
 - image: nginx
 name: nginx
```

## Composing and Customizing Resources

It is common to compose a set of Resources in a project and manage them inside the same file or directory. Kustomize offers composing Resources from different files and applying patches or other customization to them.

### Composing

Kustomize supports composition of different resources. The `resources` field, in the `kustomization.yaml` file, defines the list of resources to include in a configuration. Set the path



to a resource's configuration file in the resources list. Here is an example of an NGINX application comprised of a Deployment and a Service:

```
Create a deployment.yaml file
```

```
cat <<EOF > deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
 name: my-nginx
```

```
spec:
```

```
 selector:
```

```
 matchLabels:
```

```
 run: my-nginx
```

```
 replicas: 2
```

```
 template:
```

```
 metadata:
```

```
 labels:
```

```
 run: my-nginx
```

```
 spec:
```

```
 containers:
```

```
 - name: my-nginx
```

```
 image: nginx
```

```
 ports:
```

```
 - containerPort: 80
```

```
EOF
```

```
Create a service.yaml file
```

```
cat <<EOF > service.yaml
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
 name: my-nginx
```

```
 labels:
```

```
 run: my-nginx
```

```
spec:
```

```
 ports:
```

```
 - port: 80
```

```
 protocol: TCP
```

```
 selector:
```

```
 run: my-nginx
```

```
EOF
```

```
Create a kustomization.yaml composing them
```

```
cat <<EOF > ./kustomization.yaml
```

```
resources:
```

```
- deployment.yaml
```

```
- service.yaml
```

```
EOF
```

The Resources from `kubectl kustomize ./` contain both the Deployment and the Service objects.

## Customizing

Patches can be used to apply different customizations to Resources. Kustomize supports different patching mechanisms through `patchesStrategicMerge` and `patchesJson6902`. `patchesStrategicMerge` is a list of file paths. Each file should be resolved to a [strategic merge patch](#). The names inside the patches must match Resource names that are already loaded. Small patches that do one thing are recommended. For example, create one patch for increasing the deployment replica number and another patch for setting the memory limit.

```
Create a deployment.yaml file
cat <<EOF > deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
spec:
 selector:
 matchLabels:
 run: my-nginx
 replicas: 2
 template:
 metadata:
 labels:
 run: my-nginx
 spec:
 containers:
 - name: my-nginx
 image: nginx
 ports:
 - containerPort: 80
EOF

Create a patch increase_replicas.yaml
cat <<EOF > increase_replicas.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
spec:
 replicas: 3
EOF

Create another patch set_memory.yaml
cat <<EOF > set_memory.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
spec:
 template:
 spec:
 containers:
 - name: my-nginx
```

```
resources:
 limits:
 memory: 512Mi
EOF
```

```
cat <<EOF >./kustomization.yaml
resources:
- deployment.yaml
patchesStrategicMerge:
- increase_replicas.yaml
- set_memory.yaml
EOF
```

Run `kubectl kustomize ./` to view the Deployment:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
spec:
 replicas: 3
 selector:
 matchLabels:
 run: my-nginx
 template:
 metadata:
 labels:
 run: my-nginx
 spec:
 containers:
 - image: nginx
 name: my-nginx
 ports:
 - containerPort: 80
 resources:
 limits:
 memory: 512Mi
```

Not all Resources or fields support strategic merge patches. To support modifying arbitrary fields in arbitrary Resources, Kustomize offers applying [JSON patch](#) through `patchesJson6902`. To find the correct Resource for a Json patch, the group, version, kind and name of that Resource need to be specified in `kustomization.yaml`. For example, increasing the replica number of a Deployment object can also be done through `patchesJson6902`.

```
Create a deployment.yaml file
cat <<EOF > deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
spec:
 selector:
 matchLabels:
```

```
 run: my-nginx
replicas: 2
template:
 metadata:
 labels:
 run: my-nginx
 spec:
 containers:
 - name: my-nginx
 image: nginx
 ports:
 - containerPort: 80
EOF
```

```
Create a json patch
cat <<EOF > patch.yaml
- op: replace
 path: /spec/replicas
 value: 3
EOF
```

```
Create a kustomization.yaml
cat <<EOF > ./kustomization.yaml
resources:
- deployment.yaml

patchesJson6902:
- target:
 group: apps
 version: v1
 kind: Deployment
 name: my-nginx
 path: patch.yaml
EOF
```

Run `kubectl kustomize ./` to see the `replicas` field is updated:

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
spec:
 replicas: 3
 selector:
 matchLabels:
 run: my-nginx
 template:
 metadata:
 labels:
 run: my-nginx
 spec:
 containers:
 - image: nginx
```

```
name: my-nginx
ports:
- containerPort: 80
```

In addition to patches, Kustomize also offers customizing container images or injecting field values from other objects into containers without creating patches. For example, you can change the image used inside containers by specifying the new image in images field in kustomization.yaml.

```
cat <<EOF > deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
 name: my-nginx
```

```
spec:
```

```
 selector:
```

```
 matchLabels:
```

```
 run: my-nginx
```

```
 replicas: 2
```

```
 template:
```

```
 metadata:
```

```
 labels:
```

```
 run: my-nginx
```

```
 spec:
```

```
 containers:
```

```
 - name: my-nginx
```

```
 image: nginx
```

```
 ports:
```

```
 - containerPort: 80
```

```
EOF
```

```
cat <<EOF > ./kustomization.yaml
```

```
resources:
```

```
- deployment.yaml
```

```
images:
```

```
- name: nginx
```

```
 newName: my.image.registry/nginx
```

```
 newTag: 1.4.0
```

```
EOF
```

Run `kubectl kustomize ./` to see that the image being used is updated:

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
 name: my-nginx
```

```
spec:
```

```
 replicas: 2
```

```
 selector:
```

```
 matchLabels:
```

```
 run: my-nginx
```

```
 template:
```

```
 metadata:
```

```
labels:
 run: my-nginx
spec:
 containers:
 - image: my.image.registry/nginx:1.4.0
 name: my-nginx
 ports:
 - containerPort: 80
```

Sometimes, the application running in a Pod may need to use configuration values from other objects. For example, a Pod from a Deployment object need to read the corresponding Service name from Env or as a command argument. Since the Service name may change as namePrefix or nameSuffix is added in the kustomization.yaml file. It is not recommended to hard code the Service name in the command argument. For this usage, Kustomize can inject the Service name into containers through vars.

```
Create a deployment.yaml file (quoting the here doc delimiter)
```

```
cat <<'EOF' > deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
 name: my-nginx
```

```
spec:
```

```
 selector:
```

```
 matchLabels:
```

```
 run: my-nginx
```

```
 replicas: 2
```

```
 template:
```

```
 metadata:
```

```
 labels:
```

```
 run: my-nginx
```

```
 spec:
```

```
 containers:
```

```
 - name: my-nginx
```

```
 image: nginx
```

```
 command: ["start", "--host", "$(MY_SERVICE_NAME)"]
```

```
EOF
```

```
Create a service.yaml file
```

```
cat <<EOF > service.yaml
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
 name: my-nginx
```

```
 labels:
```

```
 run: my-nginx
```

```
spec:
```

```
 ports:
```

```
 - port: 80
```

```
 protocol: TCP
```

```
 selector:
```

```
 run: my-nginx
```

```
EOF
```

```

cat <<EOF >./kustomization.yaml
namePrefix: dev-
nameSuffix: "-001"

resources:
- deployment.yaml
- service.yaml

vars:
- name: MY_SERVICE_NAME
 objref:
 kind: Service
 name: my-nginx
 apiVersion: v1
EOF

```

Run `kubectl kustomize ./` to see that the Service name injected into containers is `dev-my-nginx-001`:

```

apiVersion: apps/v1
kind: Deployment
metadata:
 name: dev-my-nginx-001
spec:
 replicas: 2
 selector:
 matchLabels:
 run: my-nginx
 template:
 metadata:
 labels:
 run: my-nginx
 spec:
 containers:
 - command:
 - start
 - --host
 - dev-my-nginx-001
 image: nginx
 name: my-nginx

```

## Bases and Overlays

Kustomize has the concepts of **bases** and **overlays**. A **base** is a directory with a `kustomization.yaml`, which contains a set of resources and associated customization. A base could be either a local directory or a directory from a remote repo, as long as a `kustomization.yaml` is present inside. An **overlay** is a directory with a `kustomization.yaml` that refers to other kustomization directories as its bases. A **base** has no knowledge of an overlay and can be used in multiple overlays. An overlay may have multiple bases and it composes all resources from bases and may also have customization on top of them.

Here is an example of a base:

```
Create a directory to hold the base
mkdir base
Create a base/deployment.yaml
cat <<EOF > base/deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
spec:
 selector:
 matchLabels:
 run: my-nginx
 replicas: 2
 template:
 metadata:
 labels:
 run: my-nginx
 spec:
 containers:
 - name: my-nginx
 image: nginx
EOF

Create a base/service.yaml file
cat <<EOF > base/service.yaml
apiVersion: v1
kind: Service
metadata:
 name: my-nginx
 labels:
 run: my-nginx
spec:
 ports:
 - port: 80
 protocol: TCP
 selector:
 run: my-nginx
EOF

Create a base/kustomization.yaml
cat <<EOF > base/kustomization.yaml
resources:
- deployment.yaml
- service.yaml
EOF
```

This base can be used in multiple overlays. You can add different `namePrefix` or other cross-cutting fields in different overlays. Here are two overlays using the same base.

```
mkdir dev
cat <<EOF > dev/kustomization.yaml
resources:
```



```
- ../base
namePrefix: dev-
EOF

mkdir prod
cat <<EOF > prod/kustomization.yaml
resources:
- ../base
namePrefix: prod-
EOF
```

## How to apply/view/delete objects using Kustomize

Use `--kustomize` or `-k` in `kubectl` commands to recognize Resources managed by `kustomization.yaml`. Note that `-k` should point to a `kustomization` directory, such as

```
kubectl apply -k <kustomization directory>/
```

Given the following `kustomization.yaml`,

```
Create a deployment.yaml file
cat <<EOF > deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: my-nginx
spec:
 selector:
 matchLabels:
 run: my-nginx
 replicas: 2
 template:
 metadata:
 labels:
 run: my-nginx
 spec:
 containers:
 - name: my-nginx
 image: nginx
 ports:
 - containerPort: 80
EOF

Create a kustomization.yaml
cat <<EOF > ./kustomization.yaml
namePrefix: dev-
commonLabels:
 app: my-nginx
resources:
- deployment.yaml
EOF
```

Run the following command to apply the Deployment object `dev-my-nginx`:

```
> kubectl apply -k ./
deployment.apps/dev-my-nginx created
```

Run one of the following commands to view the Deployment object dev-my-nginx:

```
kubectl get -k ./
```

```
kubectl describe -k ./
```

Run the following command to compare the Deployment object dev-my-nginx against the state that the cluster would be in if the manifest was applied:

```
kubectl diff -k ./
```

Run the following command to delete the Deployment object dev-my-nginx:

```
> kubectl delete -k ./
deployment.apps "dev-my-nginx" deleted
```

## Kustomize Feature List

Field	Type	Explanation
namespace	string	add namespace to all resources
namePrefix	string	value of this field is prepended to the names of all resources
nameSuffix	string	value of this field is appended to the names of all resources
commonLabels	map[string]string	labels to add to all resources and selectors
commonAnnotations	map[string]string	annotations to add to all resources
resources	[]string	each entry in this list must resolve to an existing resource configuration file
configMapGenerator	[] <a href="#">ConfigMapArgs</a>	Each entry in this list generates a ConfigMap
secretGenerator	[] <a href="#">SecretArgs</a>	Each entry in this list generates a Secret
generatorOptions	<a href="#">GeneratorOptions</a>	Modify behaviors of all ConfigMap and Secret generator
bases	[]string	Each entry in this list should resolve to a directory containing a kustomization.yaml file
patchesStrategicMerge	[]string	Each entry in this list should resolve a strategic merge patch of a Kubernetes object
patchesJson6902	[] <a href="#">Patch</a>	Each entry in this list should resolve to a Kubernetes object and a Json Patch
vars	[] <a href="#">Var</a>	Each entry is to capture text from one resource's field
images	[] <a href="#">Image</a>	Each entry is to modify the name, tags and/or digest for one image without creating patches
configurations	[]string	Each entry in this list should resolve to a file containing <a href="#">Kustomize transformer configurations</a>
crds	[]string	Each entry in this list should resolve to an OpenAPI definition file for Kubernetes types

## What's next

- [Kustomize](#)
- [Kubectl Book](#)
- [Kubectl Command Reference](#)
- [Kubernetes API Reference](#)

# Managing Kubernetes Objects Using Imperative Commands

Kubernetes objects can quickly be created, updated, and deleted directly using imperative commands built into the kubectl command-line tool. This document explains how those commands are organized and how to use them to manage live objects.

## Before you begin

Install [kubectl](#).

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

## Trade-offs

The kubectl tool supports three kinds of object management:

- Imperative commands
- Imperative object configuration
- Declarative object configuration

See [Kubernetes Object Management](#) for a discussion of the advantages and disadvantage of each kind of object management.

## How to create objects

The kubectl tool supports verb-driven commands for creating some of the most common object types. The commands are named to be recognizable to users unfamiliar with the Kubernetes object types.

- `run`: Create a new Pod to run a Container.
- `expose`: Create a new Service object to load balance traffic across Pods.
- `autoscale`: Create a new Autoscaler object to automatically horizontally scale a controller, such as a Deployment.

The `kubectl` tool also supports creation commands driven by object type. These commands support more object types and are more explicit about their intent, but require users to know the type of objects they intend to create.

- `create <objecttype> [<subtype>] <instancename>`

Some objects types have subtypes that you can specify in the `create` command. For example, the `Service` object has several subtypes including `ClusterIP`, `LoadBalancer`, and `NodePort`. Here's an example that creates a `Service` with subtype `NodePort`:

```
kubectl create service nodeport <myservicename>
```

In the preceding example, the `create service nodeport` command is called a subcommand of the `create service` command.

You can use the `-h` flag to find the arguments and flags supported by a subcommand:

```
kubectl create service nodeport -h
```

## How to update objects

The `kubectl` command supports verb-driven commands for some common update operations. These commands are named to enable users unfamiliar with Kubernetes objects to perform updates without knowing the specific fields that must be set:

- `scale`: Horizontally scale a controller to add or remove Pods by updating the replica count of the controller.
- `annotate`: Add or remove an annotation from an object.
- `label`: Add or remove a label from an object.

The `kubectl` command also supports update commands driven by an aspect of the object. Setting this aspect may set different fields for different object types:

- `set <field>`: Set an aspect of an object.

### Note:

In Kubernetes version 1.5, not every verb-driven command has an associated aspect-driven command.

The `kubectl` tool supports these additional ways to update a live object directly, however they require a better understanding of the Kubernetes object schema.

- `edit`: Directly edit the raw configuration of a live object by opening its configuration in an editor.
- `patch`: Directly modify specific fields of a live object by using a patch string. For more details on patch strings, see the patch section in [API Conventions](#).

## How to delete objects

You can use the `delete` command to delete an object from a cluster:

- `delete <type>/<name>`

## Note:

You can use `kubectl delete` for both imperative commands and imperative object configuration. The difference is in the arguments passed to the command. To use `kubectl delete` as an imperative command, pass the object to be deleted as an argument. Here's an example that passes a Deployment object named `nginx`:

```
kubectl delete deployment/nginx
```

## How to view an object

There are several commands for printing information about an object:

- `get`: Prints basic information about matching objects. Use `get -h` to see a list of options.
- `describe`: Prints aggregated detailed information about matching objects.
- `logs`: Prints the stdout and stderr for a container running in a Pod.

## Using set commands to modify objects before creation

There are some object fields that don't have a flag you can use in a `create` command. In some of those cases, you can use a combination of `set` and `create` to specify a value for the field before object creation. This is done by piping the output of the `create` command to the `set` command, and then back to the `create` command. Here's an example:

```
kubectl create service clusterip my-svc --clusterip="None" -o yaml --dry-run=client | kubectl set selector --local -f - 'environment=qa' -o yaml | kubectl create -f -
```

1. The `kubectl create service -o yaml --dry-run=client` command creates the configuration for the Service, but prints it to stdout as YAML instead of sending it to the Kubernetes API server.
2. The `kubectl set selector --local -f - -o yaml` command reads the configuration from stdin, and writes the updated configuration to stdout as YAML.
3. The `kubectl create -f -` command creates the object using the configuration provided via stdin.

## Using --edit to modify objects before creation

You can use `kubectl create --edit` to make arbitrary changes to an object before it is created. Here's an example:

```
kubectl create service clusterip my-svc --clusterip="None" -o yaml --dry-run=client > /tmp/srv.yaml
kubectl create --edit -f /tmp/srv.yaml
```

1. The `kubectl create service` command creates the configuration for the Service and saves it to `/tmp/srv.yaml`.
2. The `kubectl create --edit` command opens the configuration file for editing before it creates the object.

## What's next

- [Imperative Management of Kubernetes Objects Using Configuration Files](#)
- [Declarative Management of Kubernetes Objects Using Configuration Files](#)
- [Kubectl Command Reference](#)
- [Kubernetes API Reference](#)

# Imperative Management of Kubernetes Objects Using Configuration Files

Kubernetes objects can be created, updated, and deleted by using the `kubectl` command-line tool along with an object configuration file written in YAML or JSON. This document explains how to define and manage objects using configuration files.

## Before you begin

Install [kubectl](#).

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

## Trade-offs

The `kubectl` tool supports three kinds of object management:

- Imperative commands
- Imperative object configuration
- Declarative object configuration

See [Kubernetes Object Management](#) for a discussion of the advantages and disadvantage of each kind of object management.

## How to create objects

You can use `kubectl create -f` to create an object from a configuration file. Refer to the [kubernetes API reference](#) for details.

- `kubectl create -f <filename|url>`

# How to update objects

## Warning:

Updating objects with the `replace` command drops all parts of the spec not specified in the configuration file. This should not be used with objects whose specs are partially managed by the cluster, such as Services of type LoadBalancer, where the `externalIPs` field is managed independently from the configuration file. Independently managed fields must be copied to the configuration file to prevent `replace` from dropping them.

You can use `kubectl replace -f` to update a live object according to a configuration file.

- `kubectl replace -f <filename|url>`

# How to delete objects

You can use `kubectl delete -f` to delete an object that is described in a configuration file.

- `kubectl delete -f <filename|url>`

## Note:

If configuration file has specified the `generateName` field in the metadata section instead of the `name` field, you cannot delete the object using `kubectl delete -f <filename|url>`. You will have to use other flags for deleting the object. For example:

```
kubectl delete <type> <name>
kubectl delete <type> -l <label>
```

# How to view an object

You can use `kubectl get -f` to view information about an object that is described in a configuration file.

- `kubectl get -f <filename|url> -o yaml`

The `-o yaml` flag specifies that the full object configuration is printed. Use `kubectl get -h` to see a list of options.

# Limitations

The `create`, `replace`, and `delete` commands work well when each object's configuration is fully defined and recorded in its configuration file. However when a live object is updated, and the updates are not merged into its configuration file, the updates will be lost the next time a `replace` is executed. This can happen if a controller, such as a HorizontalPodAutoscaler, makes updates directly to a live object. Here's an example:

1. You create an object from a configuration file.
2. Another source updates the object by changing some field.

3. You replace the object from the configuration file. Changes made by the other source in step 2 are lost.

If you need to support multiple writers to the same object, you can use `kubectl apply` to manage the object.

## Creating and editing an object from a URL without saving the configuration

Suppose you have the URL of an object configuration file. You can use `kubectl create --edit` to make changes to the configuration before the object is created. This is particularly useful for tutorials and tasks that point to a configuration file that could be modified by the reader.

```
kubectl create -f <url> --edit
```

## Migrating from imperative commands to imperative object configuration

Migrating from imperative commands to imperative object configuration involves several manual steps.

1. Export the live object to a local object configuration file:

```
kubectl get <kind>/<name> -o yaml > <kind>_<name>.yaml
```

2. Manually remove the status field from the object configuration file.
3. For subsequent object management, use `replace` exclusively.

```
kubectl replace -f <kind>_<name>.yaml
```

## Defining controller selectors and PodTemplate labels

### Warning:

Updating selectors on controllers is strongly discouraged.

The recommended approach is to define a single, immutable PodTemplate label used only by the controller selector with no other semantic meaning.

Example label:

```
selector:
 matchLabels:
 controller-selector: "apps/v1/deployment/nginx"
template:
 metadata:
 labels:
 controller-selector: "apps/v1/deployment/nginx"
```



## What's next

- [Managing Kubernetes Objects Using Imperative Commands](#)
- [Declarative Management of Kubernetes Objects Using Configuration Files](#)
- [Kubectl Command Reference](#)
- [Kubernetes API Reference](#)

# Update API Objects in Place Using kubectl patch

Use kubectl patch to update Kubernetes API objects in place. Do a strategic merge patch or a JSON merge patch.

This task shows how to use kubectl patch to update an API object in place. The exercises in this task demonstrate a strategic merge patch and a JSON merge patch.

## Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

## Use a strategic merge patch to update a Deployment

Here's the configuration file for a Deployment that has two replicas. Each replica is a Pod that has one container:

[application/deployment-patch.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: patch-demo
spec:
 replicas: 2
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
 spec:
```

```
containers:
- name: patch-demo-ctr
 image: nginx
tolerations:
- effect: NoSchedule
 key: dedicated
 value: test-team
```

Create the Deployment:

```
kubectl apply -f https://k8s.io/examples/application/deployment-patch.yaml
```

View the Pods associated with your Deployment:

```
kubectl get pods
```

The output shows that the Deployment has two Pods. The 1/1 indicates that each Pod has one container:

NAME	READY	STATUS	RESTARTS	AGE
patch-demo-28633765-670qr	1/1	Running	0	23s
patch-demo-28633765-j5qs3	1/1	Running	0	23s

Make a note of the names of the running Pods. Later, you will see that these Pods get terminated and replaced by new ones.

At this point, each Pod has one Container that runs the nginx image. Now suppose you want each Pod to have two containers: one that runs nginx and one that runs redis.

Create a file named patch-file.yaml that has this content:

```
spec:
 template:
 spec:
 containers:
 - name: patch-demo-ctr-2
 image: redis
```

Patch your Deployment:

```
kubectl patch deployment patch-demo --patch-file patch-file.yaml
```

View the patched Deployment:

```
kubectl get deployment patch-demo --output yaml
```

The output shows that the PodSpec in the Deployment has two Containers:

```
containers:
- image: redis
 imagePullPolicy: Always
 name: patch-demo-ctr-2
...
- image: nginx
 imagePullPolicy: Always
```

```
name: patch-demo-ctr
...
```

View the Pods associated with your patched Deployment:

```
kubectl get pods
```

The output shows that the running Pods have different names from the Pods that were running previously. The Deployment terminated the old Pods and created two new Pods that comply with the updated Deployment spec. The 2/2 indicates that each Pod has two Containers:

NAME	READY	STATUS	RESTARTS	AGE
patch-demo-1081991389-2wrn5	2/2	Running	0	1m
patch-demo-1081991389-jmg7b	2/2	Running	0	1m

Take a closer look at one of the patch-demo Pods:

```
kubectl get pod <your-pod-name> --output yaml
```

The output shows that the Pod has two Containers: one running nginx and one running redis:

```
containers:
- image: redis
...
- image: nginx
...
```

## Notes on the strategic merge patch

The patch you did in the preceding exercise is called a *strategic merge patch*. Notice that the patch did not replace the containers list. Instead it added a new Container to the list. In other words, the list in the patch was merged with the existing list. This is not always what happens when you use a strategic merge patch on a list. In some cases, the list is replaced, not merged.

With a strategic merge patch, a list is either replaced or merged depending on its patch strategy. The patch strategy is specified by the value of the patchStrategy key in a field tag in the Kubernetes source code. For example, the Containers field of PodSpec struct has a patchStrategy of merge:

```
type PodSpec struct {
...
Containers []Container `json:"containers" patchStrategy:"merge" patchMergeKey:"name" ...`
...
}
```

You can also see the patch strategy in the [OpenApi spec](#):

```
"io.k8s.api.core.v1.PodSpec": {
 ...,
 "containers": {
 "description": "List of containers belonging to the pod.",
 },
 "x-kubernetes-patch-merge-key": "name",
}
```

```
"x-kubernetes-patch-strategy": "merge"
}
```

And you can see the patch strategy in the [Kubernetes API documentation](#).

Create a file named patch-file-tolerations.yaml that has this content:

```
spec:
 template:
 spec:
 tolerations:
 - effect: NoSchedule
 key: disktype
 value: ssd
```

Patch your Deployment:

```
kubectl patch deployment patch-demo --patch-file patch-file-tolerations.yaml
```

View the patched Deployment:

```
kubectl get deployment patch-demo --output yaml
```

The output shows that the PodSpec in the Deployment has only one Tolerations:

```
tolerations:
- effect: NoSchedule
 key: disktype
 value: ssd
```

Notice that the tolerations list in the PodSpec was replaced, not merged. This is because the Tolerations field of PodSpec does not have a patchStrategy key in its field tag. So the strategic merge patch uses the default patch strategy, which is replace.

```
type PodSpec struct {
 ...
 Tolerations []Toleration `json:"tolerations,omitempty" protobuf:"bytes,
22,opt,name=tolerations"`
 ...
}
```

## Use a JSON merge patch to update a Deployment

A strategic merge patch is different from a [JSON merge patch](#). With a JSON merge patch, if you want to update a list, you have to specify the entire new list. And the new list completely replaces the existing list.

The kubectl patch command has a type parameter that you can set to one of these values:

Parameter value	Merge type
json	<a href="#">JSON Patch, RFC 6902</a>
merge	<a href="#">JSON Merge Patch, RFC 7386</a>
strategic	Strategic merge patch

For a comparison of JSON patch and JSON merge patch, see [JSON Patch and JSON Merge Patch](#).

The default value for the type parameter is strategic. So in the preceding exercise, you did a strategic merge patch.

Next, do a JSON merge patch on your same Deployment. Create a file named patch-file-2.yaml that has this content:

```
spec:
 template:
 spec:
 containers:
 - name: patch-demo-ctr-3
 image: gcr.io/google-samples/node-hello:1.0
```

In your patch command, set type to merge:

```
kubectl patch deployment patch-demo --type merge --patch-file patch-file-2.yaml
```

View the patched Deployment:

```
kubectl get deployment patch-demo --output yaml
```

The containers list that you specified in the patch has only one Container. The output shows that your list of one Container replaced the existing containers list.

```
spec:
 containers:
 - image: gcr.io/google-samples/node-hello:1.0
 ...
 name: patch-demo-ctr-3
```

List the running Pods:

```
kubectl get pods
```

In the output, you can see that the existing Pods were terminated, and new Pods were created. The 1/1 indicates that each new Pod is running only one Container.

NAME	READY	STATUS	RESTARTS	AGE
patch-demo-1307768864-69308	1/1	Running	0	1m
patch-demo-1307768864-c86dc	1/1	Running	0	1m

## Use strategic merge patch to update a Deployment using the retainKeys strategy

Here's the configuration file for a Deployment that uses the RollingUpdate strategy:

[application/deployment-retainkeys.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
```

```
name: retainkeys-demo
spec:
 selector:
 matchLabels:
 app: nginx
 strategy:
 rollingUpdate:
 maxSurge: 30%
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: retainkeys-demo-ctr
 image: nginx
```

Create the deployment:

```
kubectl apply -f https://k8s.io/examples/application/deployment-retainkeys.yaml
```

At this point, the deployment is created and is using the RollingUpdate strategy.

Create a file named patch-file-no-retainkeys.yaml that has this content:

```
spec:
 strategy:
 type: Recreate
```

Patch your Deployment:

```
kubectl patch deployment retainkeys-demo --type strategic --patch-file patch-file-no-retainkeys.yaml
```

In the output, you can see that it is not possible to set type as Recreate when a value is defined for spec.strategy.rollingUpdate:

The Deployment "retainkeys-demo" is invalid: spec.strategy.rollingUpdate: Forbidden: may not be specified when strategy `type` is 'Recreate'

The way to remove the value for spec.strategy.rollingUpdate when updating the value for type is to use the retainKeys strategy for the strategic merge.

Create another file named patch-file-retainkeys.yaml that has this content:

```
spec:
 strategy:
 $retainKeys:
 - type
 type: Recreate
```

With this patch, we indicate that we want to retain only the type key of the strategy object. Thus, the rollingUpdate will be removed during the patch operation.

Patch your Deployment again with this new patch:

```
kubectl patch deployment retainkeys-demo --type strategic --patch-file patch-file-retainkeys.yaml
```

Examine the content of the Deployment:

```
kubectl get deployment retainkeys-demo --output yaml
```

The output shows that the strategy object in the Deployment does not contain the rollingUpdate key anymore:

```
spec:
 strategy:
 type: Recreate
 template:
```

## Notes on the strategic merge patch using the retainKeys strategy

The patch you did in the preceding exercise is called a *strategic merge patch with retainKeys strategy*. This method introduces a new directive \$retainKeys that has the following strategies:

- It contains a list of strings.
- All fields needing to be preserved must be present in the \$retainKeys list.
- The fields that are present will be merged with live object.
- All of the missing fields will be cleared when patching.
- All fields in the \$retainKeys list must be a superset or the same as the fields present in the patch.

The retainKeys strategy does not work for all objects. It only works when the value of the patchStrategy key in a field tag in the Kubernetes source code contains retainKeys. For example, the Strategy field of the DeploymentSpec struct has a patchStrategy of retainKeys:

```
type DeploymentSpec struct {
 ...
 // +patchStrategy=retainKeys
 Strategy DeploymentStrategy `json:"strategy,omitempty" patchStrategy:"retainKeys" ...`
 ...
}
```

You can also see the retainKeys strategy in the [OpenApi spec](#):

```
"io.k8s.api.apps.v1.DeploymentSpec": {
 ...,
 "strategy": {
 "$ref": "#/definitions/io.k8s.api.apps.v1.DeploymentStrategy",
 "description": "The deployment strategy to use to replace existing pods with new ones.",
 "x-kubernetes-patch-strategy": "retainKeys"
 },

}
```

And you can see the retainKeys strategy in the [Kubernetes API documentation](#).

## Alternate forms of the kubectl patch command

The `kubectl patch` command takes YAML or JSON. It can take the patch as a file or directly on the command line.

Create a file named `patch-file.json` that has this content:

```
{
 "spec": {
 "template": {
 "spec": {
 "containers": [
 {
 "name": "patch-demo-ctr-2",
 "image": "redis"
 }
]
 }
 }
 }
}
```

The following commands are equivalent:

```
kubectl patch deployment patch-demo --patch-file patch-file.yaml
kubectl patch deployment patch-demo --patch 'spec:\n template:\n spec:\n containers:\n -\n name: patch-demo-ctr-2\n image: redis'

kubectl patch deployment patch-demo --patch-file patch-file.json
kubectl patch deployment patch-demo --patch '{"spec": {"template": {"spec": {"containers": [{"name": "patch-demo-ctr-2", "image": "redis"}]}}}'
```

## Update an object's replica count using kubectl patch with --subresource

FEATURE STATE: Kubernetes v1.24 [alpha]

The flag `--subresource=[subresource-name]` is used with `kubectl` commands like `get`, `patch`, `edit` and `replace` to fetch and update status and scale subresources of the resources (applicable for `kubectl` version v1.24 or more). This flag is used with all the API resources (built-in and CRs) that have status or scale subresource. Deployment is one of the examples which supports these subresources.

Here's a manifest for a Deployment that has two replicas:

[application/deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 selector:
 matchLabels:
```



```
 app: nginx
replicas: 2 # tells deployment to run 2 pods matching the template
template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80
```

Create the Deployment:

```
kubectl apply -f https://k8s.io/examples/application/deployment.yaml
```

View the Pods associated with your Deployment:

```
kubectl get pods -l app=nginx
```

In the output, you can see that Deployment has two Pods. For example:

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-7fb96c846b-22567	1/1	Running	0	47s
nginx-deployment-7fb96c846b-mlgns	1/1	Running	0	47s

Now, patch that Deployment with `--subresource=[subresource-name]` flag:

```
kubectl patch deployment nginx-deployment --subresource='scale' --type='merge' -p '{"spec": {"replicas":3}}'
```

The output is:

```
scale.autoscaling/nginx-deployment patched
```

View the Pods associated with your patched Deployment:

```
kubectl get pods -l app=nginx
```

In the output, you can see one new pod is created, so now you have 3 running pods.

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-7fb96c846b-22567	1/1	Running	0	107s
nginx-deployment-7fb96c846b-lxfr2	1/1	Running	0	14s
nginx-deployment-7fb96c846b-mlgns	1/1	Running	0	107s

View the patched Deployment:

```
kubectl get deployment nginx-deployment -o yaml
```

```
...
spec:
 replicas: 3
...
```

```
status:
...
availableReplicas: 3
readyReplicas: 3
replicas: 3
```

**Note:**

If you run `kubectl patch` and specify `--subresource` flag for resource that doesn't support that particular subresource, the API server returns a 404 Not Found error.

## Summary

In this exercise, you used `kubectl patch` to change the live configuration of a Deployment object. You did not change the configuration file that you originally used to create the Deployment object. Other commands for updating API objects include [kubectl annotate](#), [kubectl edit](#), [kubectl replace](#), [kubectl scale](#), and [kubectl apply](#).

**Note:**

Strategic merge patch is not supported for custom resources.

## What's next

- [Kubernetes Object Management](#)
- [Managing Kubernetes Objects Using Imperative Commands](#)
- [Imperative Management of Kubernetes Objects Using Configuration Files](#)
- [Declarative Management of Kubernetes Objects Using Configuration Files](#)

# Migrate Kubernetes Objects Using Storage Version Migration

FEATURE STATE: Kubernetes v1.30 [alpha]

Kubernetes relies on API data being actively re-written, to support some maintenance activities related to at rest storage. Two prominent examples are the versioned schema of stored resources (that is, the preferred storage schema changing from v1 to v2 for a given resource) and encryption at rest (that is, rewriting stale data based on a change in how the data should be encrypted).

## Before you begin

Install [kubectl](#).

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be version v1.30. To check the version, enter `kubectl version`.

## Re-encrypt Kubernetes secrets using storage version migration

- To begin with, [configure KMS provider](#) to encrypt data at rest in etcd using following encryption configuration.

```
kind: EncryptionConfiguration
apiVersion: apiserver.config.k8s.io/v1
resources:
- resources:
 - secrets
providers:
- aescbc:
 keys:
 - name: key1
 secret: c2VjcmV0IGlzIHNIY3VyZQ==
```

Make sure to enable automatic reload of encryption configuration file by setting `--encryption-provider-config-automatic-reload` to true.

- Create a Secret using `kubectl`.

```
kubectl create secret generic my-secret --from-literal=key1=supersecret
```

- [Verify](#) the serialized data for that Secret object is prefixed with `k8s:enc:aescbc:v1:key1`.
- Update the encryption configuration file as follows to rotate the encryption key.

```
kind: EncryptionConfiguration
apiVersion: apiserver.config.k8s.io/v1
resources:
- resources:
 - secrets
providers:
- aescbc:
 keys:
 - name: key2
 secret: c2VjcmV0IGlzIHNIY3VyZSwgaXMgaXQ/
 - aescbc:
 keys:
 - name: key1
 secret: c2VjcmV0IGlzIHNIY3VyZQ==
```

- To ensure that previously created secret `my-secret` is re-encrypted with new key `key2`, you will use *Storage Version Migration*.

Create a StorageVersionMigration manifest named `migrate-secret.yaml` as follows:

- ```
kind: StorageVersionMigration
apiVersion: storagemigration.k8s.io/v1alpha1
metadata:
  name: secrets-migration
spec:
  resource:
    group: ""
    version: v1
    resource: secrets
```

Create the object using *kubectl* as follows:

```
kubectl apply -f migrate-secret.yaml
```

- Monitor migration of Secrets by checking the `.status` of the StorageVersionMigration. A successful migration should have its Succeeded condition set to true. Get the StorageVersionMigration object as follows:

```
kubectl get storageversionmigration.storagemigration.k8s.io/secrets-migration -o yaml
```

The output is similar to:

```
kind: StorageVersionMigration
apiVersion: storagemigration.k8s.io/v1alpha1
metadata:
  name: secrets-migration
  uid: 628f6922-a9cb-4514-b076-12d3c178967c
  resourceVersion: "90"
  creationTimestamp: "2024-03-12T20:29:45Z"
spec:
  resource:
    group: ""
    version: v1
    resource: secrets
status:
  conditions:
  - type: Running
    status: "False"
    lastUpdateTime: "2024-03-12T20:29:46Z"
    reason: StorageVersionMigrationInProgress
  - type: Succeeded
    status: "True"
    lastUpdateTime: "2024-03-12T20:29:46Z"
    reason: StorageVersionMigrationSucceeded
  resourceVersion: "84"
```

- [Verify](#) the stored secret is now prefixed with `k8s:enc:aescbc:v1:key2`.

Update the preferred storage schema of a CRD

Consider a scenario where a [CustomResourceDefinition](#) (CRD) is created to serve custom resources (CRs) and is set as the preferred storage schema. When it's time to introduce v2 of the CRD, it can be added for serving only with a conversion webhook. This enables a smoother transition where users can create CRs using either the v1 or v2 schema, with the webhook in place to perform the necessary schema conversion between them. Before setting v2 as the preferred storage schema version, it's important to ensure that all existing CRs stored as v1 are migrated to v2. This migration can be achieved through *Storage Version Migration* to migrate all CRs from v1 to v2.

- Create a manifest for the CRD, named test-crd.yaml, as follows:

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: selfierequests.stable.example.com
spec:
  group: stable.example.com
  names:
    plural: SelfieRequests
    singular: SelfieRequest
    kind: SelfieRequest
    listKind: SelfieRequestList
  scope: Namespaced
  versions:
  - name: v1
    served: true
    storage: true
    schema:
      openAPIV3Schema:
        type: object
        properties:
          hostPort:
            type: string
  conversion:
    strategy: Webhook
    webhook:
      clientConfig:
        url: "https://127.0.0.1:9443/crdconvert"
        caBundle: <CABundle info>
    conversionReviewVersions:
    - v1
    - v2
```

Create CRD using kubectl:

```
kubectl apply -f test-crd.yaml
```

- Create a manifest for an example testcrd. Name the manifest cr1.yaml and use these contents:

```
apiVersion: stable.example.com/v1
kind: SelfieRequest
metadata:
  name: cr1
  namespace: default
```

Create CR using kubectl:

```
kubectl apply -f cr1.yaml
```

- Verify that CR is written and stored as v1 by getting the object from etcd.

```
ETCDCTL_API=3 etcdctl get /kubernetes.io/stable.example.com/testcrds/default/cr1 [...] |
hexdump -C
```

where [...] contains the additional arguments for connecting to the etcd server.

- Update the CRD test-crd.yaml to include v2 version for serving and storage and v1 as serving only, as follows:

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: selfierequests.stable.example.com
spec:
  group: stable.example.com
  names:
    plural: SelfieRequests
    singular: SelfieRequest
    kind: SelfieRequest
    listKind: SelfieRequestList
    scope: Namespaced
  versions:
    - name: v2
      served: true
      storage: true
      schema:
        openAPIV3Schema:
          type: object
          properties:
            host:
              type: string
            port:
              type: string
    - name: v1
      served: true
      storage: false
      schema:
        openAPIV3Schema:
          type: object
          properties:
            hostPort:
              type: string
  conversion:
```

```
strategy: Webhook
webhook:
  clientConfig:
    url: "https://127.0.0.1:9443/crdconvert"
    caBundle: <CABundle info>
  conversionReviewVersions:
    - v1
    - v2
```

Update CRD using kubectl:

```
kubectl apply -f test-crd.yaml
```

- Create CR resource file with name cr2.yaml as follows:

```
apiVersion: stable.example.com/v2
kind: SelfieRequest
metadata:
  name: cr2
  namespace: default
```

- Create CR using kubectl:

```
kubectl apply -f cr2.yaml
```

- Verify that CR is written and stored as v2 by getting the object from etcd.

```
ETCDCTL_API=3 etcdctl get /kubernetes.io/stable.example.com/testcrds/default/cr2 [...] |
hexdump -C
```

where [...] contains the additional arguments for connecting to the etcd server.

- Create a StorageVersionMigration manifest named migrate-crd.yaml, with the contents as follows:

```
kind: StorageVersionMigration
apiVersion: storagemigration.k8s.io/v1alpha1
metadata:
  name: crdsvm
spec:
  resource:
    group: stable.example.com
    version: v1
  resource: SelfieRequest
```

Create the object using *kubectl* as follows:

```
kubectl apply -f migrate-crd.yaml
```

- Monitor migration of secrets using status. Successful migration should have Succeeded condition set to "True" in the status field. Get the migration resource as follows:

```
kubectl get storageversionmigration.storagemigration.k8s.io/crdsvm -o yaml
```

The output is similar to:

```
kind: StorageVersionMigration
apiVersion: storagemigration.k8s.io/v1alpha1
metadata:
  name: crdsvm
  uid: 13062fe4-32d7-47cc-9528-5067fa0c6ac8
  resourceVersion: "111"
  creationTimestamp: "2024-03-12T22:40:01Z"
spec:
  resource:
    group: stable.example.com
    version: v1
    resource: testcrds
status:
  conditions:
    - type: Running
      status: "False"
      lastUpdateTime: "2024-03-12T22:40:03Z"
      reason: StorageVersionMigrationInProgress
    - type: Succeeded
      status: "True"
      lastUpdateTime: "2024-03-12T22:40:03Z"
      reason: StorageVersionMigrationSucceeded
  resourceVersion: "106"
```

- Verify that previously created cr1 is now written and stored as v2 by getting the object from etcd.

```
ETCDCTL_API=3 etcdctl get /kubernetes.io/stable.example.com/testcrds/default/cr1 [...] | hexdump -C
```

where [...] contains the additional arguments for connecting to the etcd server.

Managing Secrets

Managing confidential settings data using Secrets.

[Managing Secrets using kubectl](#)

Creating Secret objects using kubectl command line.

[Managing Secrets using Configuration File](#)

Creating Secret objects using resource configuration file.

[Managing Secrets using Kustomize](#)

Creating Secret objects using kustomization.yaml file.

Managing Secrets using kubectl

Creating Secret objects using kubectl command line.

This page shows you how to create, edit, manage, and delete Kubernetes [Secrets](#) using the kubectl command-line tool.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Create a Secret

A Secret object stores sensitive data such as credentials used by Pods to access services. For example, you might need a Secret to store the username and password needed to access a database.

You can create the Secret by passing the raw data in the command, or by storing the credentials in files that you pass in the command. The following commands create a Secret that stores the username admin and the password S!B*d\$zDsb=.

Use raw data

Run the following command:

```
kubectl create secret generic db-user-pass \
  --from-literal=username=admin \
  --from-literal=password='S!B\*d$zDsb='
```

You must use single quotes " to escape special characters such as \$, \, *, =, and ! in your strings. If you don't, your shell will interpret these characters.

Note:

The stringData field for a Secret does not work well with server-side apply.

Use source files

1. Store the credentials in files:

```
echo -n 'admin' > ./username.txt
echo -n 'S!B\*d$zDsb=' > ./password.txt
```

The `-n` flag ensures that the generated files do not have an extra newline character at the end of the text. This is important because when `kubectl` reads a file and encodes the content into a base64 string, the extra newline character gets encoded too. You do not need to escape special characters in strings that you include in a file.

2. Pass the file paths in the `kubectl` command:

```
kubectl create secret generic db-user-pass \  
  --from-file=./username.txt \  
  --from-file=./password.txt
```

The default key name is the file name. You can optionally set the key name using `--from-file=[key=]source`. For example:

```
kubectl create secret generic db-user-pass \  
  --from-file=username=./username.txt \  
  --from-file=password=./password.txt
```

With either method, the output is similar to:

```
secret/db-user-pass created
```

Verify the Secret

Check that the Secret was created:

```
kubectl get secrets
```

The output is similar to:

| NAME | TYPE | DATA | AGE |
|--------------|--------|------|-----|
| db-user-pass | Opaque | 2 | 51s |

View the details of the Secret:

```
kubectl describe secret db-user-pass
```

The output is similar to:

```
Name:      db-user-pass  
Namespace: default  
Labels:    <none>  
Annotations: <none>  
  
Type:      Opaque  
  
Data  
====  
password:  12 bytes  
username:  5 bytes
```

The commands `kubectl get` and `kubectl describe` avoid showing the contents of a Secret by default. This is to protect the Secret from being exposed accidentally, or from being stored in a terminal log.

Decode the Secret

1. View the contents of the Secret you created:

```
kubectl get secret db-user-pass -o jsonpath='{.data}'
```

The output is similar to:

```
{ "password": "UyFCXCpkJHpEc2I9", "username": "YWRtaW4=" }
```

2. Decode the password data:

```
echo 'UyFCXCpkJHpEc2I9' | base64 --decode
```

The output is similar to:

```
S!B\*d$zDsb=
```

Caution:

This is an example for documentation purposes. In practice, this method could cause the command with the encoded data to be stored in your shell history. Anyone with access to your computer could find the command and decode the secret. A better approach is to combine the view and decode commands.

```
kubectl get secret db-user-pass -o jsonpath='{.data.password}' | base64 --decode
```

Edit a Secret

You can edit an existing Secret object unless it is [immutable](#). To edit a Secret, run the following command:

```
kubectl edit secrets <secret-name>
```

This opens your default editor and allows you to update the base64 encoded Secret values in the data field, such as in the following example:

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file, it will be
# reopened with the relevant failures.
#
apiVersion: v1
data:
  password: UyFCXCpkJHpEc2I9
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2022-06-28T17:44:13Z"
  name: db-user-pass
  namespace: default
  resourceVersion: "12708504"
  uid: 91becd59-78fa-4c85-823f-6d44436242ac
type: Opaque
```

Clean up

To delete a Secret, run the following command:

```
kubectl delete secret db-user-pass
```

What's next

- Read more about the [Secret concept](#)
- Learn how to [manage Secrets using config file](#)
- Learn how to [manage Secrets using kustomize](#)

Managing Secrets using Configuration File

Creating Secret objects using resource configuration file.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Create the Secret

You can define the Secret object in a manifest first, in JSON or YAML format, and then create that object. The [Secret](#) resource contains two maps: data and stringData. The data field is used to store arbitrary data, encoded using base64. The stringData field is provided for convenience, and it allows you to provide the same data as unencoded strings. The keys of data and stringData must consist of alphanumeric characters, -, _ or ..

The following example stores two strings in a Secret using the data field.

1. Convert the strings to base64:

```
echo -n 'admin' | base64  
echo -n '1f2d1e2e67df' | base64
```

Note:

The serialized JSON and YAML values of Secret data are encoded as base64 strings. Newlines are not valid within these strings and must be omitted. When using the base64 utility on Darwin/macOS, users should avoid using the -b option to split long lines. Conversely, Linux users *should* add the option -w 0 to base64 commands or the pipeline `base64 | tr -d '\n'` if the -w option is not available.

The output is similar to:

```
YWRtaW4=  
MWYyZDFlMmU2N2Rm
```

2. Create the manifest:

```
apiVersion: v1  
kind: Secret  
metadata:  
  name: mysecret  
type: Opaque  
data:  
  username: YWRtaW4=  
  password: MWYyZDFlMmU2N2Rm
```

Note that the name of a Secret object must be a valid [DNS subdomain name](#).

3. Create the Secret using [kubectl apply](#):

```
kubectl apply -f ./secret.yaml
```

The output is similar to:

```
secret/mysecret created
```

To verify that the Secret was created and to decode the Secret data, refer to [Managing Secrets using kubectl](#).

Specify unencoded data when creating a Secret

For certain scenarios, you may wish to use the `stringData` field instead. This field allows you to put a non-base64 encoded string directly into the Secret, and the string will be encoded for you when the Secret is created or updated.

A practical example of this might be where you are deploying an application that uses a Secret to store a configuration file, and you want to populate parts of that configuration file during your deployment process.

For example, if your application uses the following configuration file:

```
apiUrl: "https://my.api.com/api/v1"  
username: "<user>"  
password: "<password>"
```

You could store this in a Secret using the following definition:

```
apiVersion: v1  
kind: Secret  
metadata:  
  name: mysecret  
type: Opaque  
stringData:  
  config.yaml: |
```

```
apiUrl: "https://my.api.com/api/v1"
username: <user>
password: <password>
```

Note:

The `stringData` field for a Secret does not work well with server-side apply.

When you retrieve the Secret data, the command returns the encoded values, and not the plaintext values you provided in `stringData`.

For example, if you run the following command:

```
kubectl get secret mysecret -o yaml
```

The output is similar to:

```
apiVersion: v1
data:
  config.yaml: YXBpVXJsOiAiaHR0cHM6Ly9teS5hcGkuY29tL2FwaS92MSIKdXNlcm5hbWU6IHt7dXNlcm5hbWV9fQpwYXNzd29yZDoge3twYXNzd29yZHU19
kind: Secret
metadata:
  creationTimestamp: 2018-11-15T20:40:59Z
  name: mysecret
  namespace: default
  resourceVersion: "7225"
  uid: c280ad2e-e916-11e8-98f2-025000000001
type: Opaque
```

Specify both data and stringData

If you specify a field in both `data` and `stringData`, the value from `stringData` is used.

For example, if you define the following Secret:

```
apiVersion: v1
kind: Secret
metadata:
  name: mysecret
type: Opaque
data:
  username: YWRtaW4=
stringData:
  username: administrator
```

Note:

The `stringData` field for a Secret does not work well with server-side apply.

The Secret object is created as follows:

```
apiVersion: v1
data:
  username: YWRtaW5pc3RyYXRvcg==
kind: Secret
metadata:
  creationTimestamp: 2018-11-15T20:46:46Z
  name: mysecret
  namespace: default
  resourceVersion: "7579"
  uid: 91460ecb-e917-11e8-98f2-025000000001
type: Opaque
```

YWRtaW5pc3RyYXRvcg== decodes to administrator.

Edit a Secret

To edit the data in the Secret you created using a manifest, modify the data or stringData field in your manifest and apply the file to your cluster. You can edit an existing Secret object unless it is [immutable](#).

For example, if you want to change the password from the previous example to birdsarentreal, do the following:

1. Encode the new password string:

```
echo -n 'birdsarentreal' | base64
```

The output is similar to:

```
YmlyZHNhcmVudHJlYWw=
```

2. Update the data field with your new password string:

```
apiVersion: v1
kind: Secret
metadata:
  name: mysecret
type: Opaque
data:
  username: YWRtaW4=
  password: YmlyZHNhcmVudHJlYWw=
```

3. Apply the manifest to your cluster:

```
kubectl apply -f ./secret.yaml
```

The output is similar to:

```
secret/mysecret configured
```

Kubernetes updates the existing Secret object. In detail, the kubectl tool notices that there is an existing Secret object with the same name. kubectl fetches the existing object, plans changes to it, and submits the changed Secret object to your cluster control plane.

If you specified `kubectl apply --server-side` instead, `kubectl` uses [Server Side Apply](#) instead.

Clean up

To delete the Secret you have created:

```
kubectl delete secret mysecret
```

What's next

- Read more about the [Secret concept](#)
- Learn how to [manage Secrets using kubectl](#)
- Learn how to [manage Secrets using kustomize](#)

Managing Secrets using Kustomize

Creating Secret objects using `kustomization.yaml` file.

`kubectl` supports using the [Kustomize object management tool](#) to manage Secrets and ConfigMaps. You create a *resource generator* using Kustomize, which generates a Secret that you can apply to the API server using `kubectl`.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Create a Secret

You can generate a Secret by defining a `secretGenerator` in a `kustomization.yaml` file that references other existing files, `.env` files, or literal values. For example, the following instructions create a Kustomization file for the username `admin` and the password `1f2d1e2e67df`.

Note:

The `stringData` field for a Secret does not work well with server-side apply.

Create the Kustomization file

- [Literals](#)
- [Files](#)
- [.env files](#)


```
secretGenerator:
- name: database-creds
  literals:
  - username=admin
  - password=1f2d1e2e67df
```

1. Store the credentials in files. The filenames are the keys of the secret:

```
echo -n 'admin' > ./username.txt
echo -n '1f2d1e2e67df' > ./password.txt
```

The `-n` flag ensures that there's no newline character at the end of your files.

2. Create the `kustomization.yaml` file:

```
secretGenerator:
- name: database-creds
  files:
  - username.txt
  - password.txt
```

You can also define the `secretGenerator` in the `kustomization.yaml` file by providing `.env` files. For example, the following `kustomization.yaml` file pulls in data from an `.env.secret` file:

```
secretGenerator:
- name: db-user-pass
  envs:
  - .env.secret
```

In all cases, you don't need to base64 encode the values. The name of the YAML file **must** be `kustomization.yaml` or `kustomization.yml`.

Apply the kustomization file

To create the Secret, apply the directory that contains the kustomization file:

```
kubectl apply -k <directory-path>
```

The output is similar to:

```
secret/database-creds-5hdk7hhgfk created
```

When a Secret is generated, the Secret name is created by hashing the Secret data and appending the hash value to the name. This ensures that a new Secret is generated each time the data is modified.

To verify that the Secret was created and to decode the Secret data,

```
kubectl get -k <directory-path> -o jsonpath='{.data}'
```

The output is similar to:

```
{ "password": "UyFCXCpkJHpEc2I9", "username": "YWRtaW4=" }
```

```
echo 'UyFCXCpkJHpEc2I9' | base64 --decode
```

The output is similar to:

```
S!B\*d$zDsb=
```

For more information, refer to [Managing Secrets using kubectl](#) and [Declarative Management of Kubernetes Objects Using Kustomize](#).

Edit a Secret

1. In your kustomization.yaml file, modify the data, such as the password.
2. Apply the directory that contains the kustomization file:

```
kubectl apply -k <directory-path>
```

The output is similar to:

```
secret/db-user-pass-6f24b56cc8 created
```

The edited Secret is created as a new Secret object, instead of updating the existing Secret object. You might need to update references to the Secret in your Pods.

Clean up

To delete a Secret, use kubectl:

```
kubectl delete secret db-user-pass
```

What's next

- Read more about the [Secret concept](#)
- Learn how to [manage Secrets using kubectl](#)
- Learn how to [manage Secrets using config file](#)

Inject Data Into Applications

Specify configuration and other data for the Pods that run your workload.

[Define a Command and Arguments for a Container](#)

[Define Dependent Environment Variables](#)

[Define Environment Variables for a Container](#)

[Expose Pod Information to Containers Through Environment Variables](#)

[Expose Pod Information to Containers Through Files](#)

Define a Command and Arguments for a Container

This page shows how to define commands and arguments when you run a container in a [Pod](#).

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

Define a command and arguments when you create a Pod

When you create a Pod, you can define a command and arguments for the containers that run in the Pod. To define a command, include the command field in the configuration file. To define arguments for the command, include the args field in the configuration file. The command and arguments that you define cannot be changed after the Pod is created.

The command and arguments that you define in the configuration file override the default command and arguments provided by the container image. If you define args, but do not define a command, the default command is used with your new arguments.

Note:

The command field corresponds to ENTRYPOINT, and the args field corresponds to CMD in some container runtimes.

In this exercise, you create a Pod that runs one container. The configuration file for the Pod defines a command and two arguments:

[pods/commands.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: command-demo
  labels:
    purpose: demonstrate-command
spec:
  containers:
  - name: command-demo-container
    image: debian
```

```
command: ["printenv"]
args: ["HOSTNAME", "KUBERNETES_PORT"]
restartPolicy: OnFailure
```

1. Create a Pod based on the YAML configuration file:

```
kubectl apply -f https://k8s.io/examples/pods/commands.yaml
```

2. List the running Pods:

```
kubectl get pods
```

The output shows that the container that ran in the command-demo Pod has completed.

3. To see the output of the command that ran in the container, view the logs from the Pod:

```
kubectl logs command-demo
```

The output shows the values of the HOSTNAME and KUBERNETES_PORT environment variables:

```
command-demo
tcp://10.3.240.1:443
```

Use environment variables to define arguments

In the preceding example, you defined the arguments directly by providing strings. As an alternative to providing strings directly, you can define arguments by using environment variables:

```
env:
- name: MESSAGE
  value: "hello world"
command: ["/bin/echo"]
args: ["$(MESSAGE)"]
```

This means you can define an argument for a Pod using any of the techniques available for defining environment variables, including [ConfigMaps](#) and [Secrets](#).

Note:

The environment variable appears in parentheses, "\$(VAR)". This is required for the variable to be expanded in the command or args field.

Run a command in a shell

In some cases, you need your command to run in a shell. For example, your command might consist of several commands piped together, or it might be a shell script. To run your command in a shell, wrap it like this:

```
command: ["/bin/sh"]
args: ["-c", "while true; do echo hello; sleep 10;done"]
```

What's next

- Learn more about [configuring pods and containers](#).
- Learn more about [running commands in a container](#).
- See [Container](#).

Define Dependent Environment Variables

This page shows how to define dependent environment variables for a container in a Kubernetes Pod.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Define an environment dependent variable for a container

When you create a Pod, you can set dependent environment variables for the containers that run in the Pod. To set dependent environment variables, you can use `$(VAR_NAME)` in the value of `env` in the configuration file.

In this exercise, you create a Pod that runs one container. The configuration file for the Pod defines a dependent environment variable with common usage defined. Here is the configuration manifest for the Pod:

[pods/inject/dependent-envvars.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: dependent-envvars-demo
spec:
  containers:
    - name: dependent-envvars-demo
      args:
        - while true; do echo -en '\n'; printf
UNCHANGED_REFERENCE=$UNCHANGED_REFERENCE'\n'; printf
SERVICE_ADDRESS=$SERVICE_ADDRESS'\n';printf
ESCAPED_REFERENCE=$ESCAPED_REFERENCE'\n'; sleep 30; done;
      command:
        - sh
        - -c
```

```
image: busybox:1.28
env:
- name: SERVICE_PORT
  value: "80"
- name: SERVICE_IP
  value: "172.17.0.1"
- name: UNCHANGED_REFERENCE
  value: "${(PROTOCOL)}://${(SERVICE_IP)}:${(SERVICE_PORT)}"
- name: PROTOCOL
  value: "https"
- name: SERVICE_ADDRESS
  value: "${(PROTOCOL)}://${(SERVICE_IP)}:${(SERVICE_PORT)}"
- name: ESCAPED_REFERENCE
  value: "${$(PROTOCOL)}://${(SERVICE_IP)}:${(SERVICE_PORT)}"
```

1. Create a Pod based on that manifest:

```
kubectl apply -f https://k8s.io/examples/pods/inject/dependent-envvars.yaml
```

```
pod/dependent-envvars-demo created
```

2. List the running Pods:

```
kubectl get pods dependent-envvars-demo
```

| NAME | READY | STATUS | RESTARTS | AGE |
|------------------------|-------|---------|----------|-----|
| dependent-envvars-demo | 1/1 | Running | 0 | 9s |

3. Check the logs for the container running in your Pod:

```
kubectl logs pod/dependent-envvars-demo
```

```
UNCHANGED_REFERENCE=${(PROTOCOL)}://172.17.0.1:80
SERVICE_ADDRESS=https://172.17.0.1:80
ESCAPED_REFERENCE=${$(PROTOCOL)}://172.17.0.1:80
```

As shown above, you have defined the correct dependency reference of `SERVICE_ADDRESS`, bad dependency reference of `UNCHANGED_REFERENCE` and skip dependent references of `ESCAPED_REFERENCE`.

When an environment variable is already defined when being referenced, the reference can be correctly resolved, such as in the `SERVICE_ADDRESS` case.

Note that order matters in the `env` list. An environment variable is not considered "defined" if it is specified further down the list. That is why `UNCHANGED_REFERENCE` fails to resolve `$(PROTOCOL)` in the example above.

When the environment variable is undefined or only includes some variables, the undefined environment variable is treated as a normal string, such as `UNCHANGED_REFERENCE`. Note that incorrectly parsed environment variables, in general, will not block the container from starting.

The `$(VAR_NAME)` syntax can be escaped with a double `$`, ie: `$$ (VAR_NAME)`. Escaped references are never expanded, regardless of whether the referenced variable is defined or not. This can be seen from the `ESCAPED_REFERENCE` case above.

What's next

- Learn more about [environment variables](#).
- See [EnvVarSource](#).

Define Environment Variables for a Container

This page shows how to define environment variables for a container in a Kubernetes Pod.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Define an environment variable for a container

When you create a Pod, you can set environment variables for the containers that run in the Pod. To set environment variables, include the `env` or `envFrom` field in the configuration file.

The `env` and `envFrom` fields have different effects.

`env`

allows you to set environment variables for a container, specifying a value directly for each variable that you name.

`envFrom`

allows you to set environment variables for a container by referencing either a `ConfigMap` or a `Secret`. When you use `envFrom`, all the key-value pairs in the referenced `ConfigMap` or `Secret` are set as environment variables for the container. You can also specify a common prefix string.

You can read more about [ConfigMap](#) and [Secret](#).

This page explains how to use `env`.

In this exercise, you create a Pod that runs one container. The configuration file for the Pod defines an environment variable with name `DEMO_GREETING` and value "Hello from the environment". Here is the configuration manifest for the Pod:

[pods/inject/envvars.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: envar-demo
  labels:
    purpose: demonstrate-envvars
spec:
  containers:
  - name: envar-demo-container
    image: gcr.io/google-samples/node-hello:1.0
    env:
    - name: DEMO_GREETING
      value: "Hello from the environment"
    - name: DEMO_FAREWELL
      value: "Such a sweet sorrow"
```

1. Create a Pod based on that manifest:

```
kubectl apply -f https://k8s.io/examples/pods/inject/envvars.yaml
```

2. List the running Pods:

```
kubectl get pods -l purpose=demonstrate-envvars
```

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|------------|-------|---------|----------|-----|
| envar-demo | 1/1 | Running | 0 | 9s |

3. List the Pod's container environment variables:

```
kubectl exec envar-demo -- printenv
```

The output is similar to this:

```
NODE_VERSION=4.4.2
EXAMPLE_SERVICE_PORT_8080_TCP_ADDR=10.3.245.237
HOSTNAME=envar-demo
...
DEMO_GREETING=Hello from the environment
DEMO_FAREWELL=Such a sweet sorrow
```

Note:

The environment variables set using the `env` or `envFrom` field override any environment variables specified in the container image.

Note:

Environment variables may reference each other, however ordering is important. Variables making use of others defined in the same context must come later in the list. Similarly, avoid circular references.

Using environment variables inside of your config

Environment variables that you define in a Pod's configuration under `.spec.containers[*].env[*]` can be used elsewhere in the configuration, for example in commands and arguments that you set for the Pod's containers. In the example configuration below, the `GREETING`, `HONORIFIC`, and `NAME` environment variables are set to Warm greetings to, The Most Honorable, and Kubernetes, respectively. The environment variable `MESSAGE` combines the set of all these environment variables and then uses it as a CLI argument passed to the `env-print-demo` container.

Environment variable names consist of letters, numbers, underscores, dots, or hyphens, but the first character cannot be a digit. If the `RelaxedEnvironmentVariableValidation` [feature gate](#) is enabled, all [printable ASCII characters](#) except `"=`" may be used for environment variable names.

```
apiVersion: v1
kind: Pod
metadata:
  name: print-greeting
spec:
  containers:
  - name: env-print-demo
    image: bash
    env:
    - name: GREETING
      value: "Warm greetings to"
    - name: HONORIFIC
      value: "The Most Honorable"
    - name: NAME
      value: "Kubernetes"
    - name: MESSAGE
      value: "${GREETING} ${HONORIFIC} ${NAME}"
    command: ["echo"]
    args: ["${MESSAGE}"]
```

Upon creation, the command `echo Warm greetings to The Most Honorable Kubernetes` is run on the container.

What's next

- Learn more about [environment variables](#).
- Learn about [using secrets as environment variables](#).
- See [EnvVarSource](#).

Expose Pod Information to Containers Through Environment Variables

This page shows how a Pod can use environment variables to expose information about itself to containers running in the Pod, using the *downward API*. You can use environment variables to expose Pod fields, container fields, or both.

In Kubernetes, there are two ways to expose Pod and container fields to a running container:

- *Environment variables*, as explained in this task
- [Volume files](#)

Together, these two ways of exposing Pod and container fields are called the downward API.

As Services are the primary mode of communication between containerized applications managed by Kubernetes, it is helpful to be able to discover them at runtime.

Read more about accessing Services [here](#).

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Use Pod fields as values for environment variables

In this part of exercise, you create a Pod that has one container, and you project Pod-level fields into the running container as environment variables.

[pods/inject/dapi-envvars-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: dapi-envvars-fieldref
spec:
  containers:
    - name: test-container
      image: registry.k8s.io/busybox
      command: [ "sh", "-c" ]
      args:
        - while true; do
          echo -en '\n';
          printenv MY_NODE_NAME MY_POD_NAME MY_POD_NAMESPACE;
          printenv MY_POD_IP MY_POD_SERVICE_ACCOUNT;
          sleep 10;
        done;
      env:
        - name: MY_NODE_NAME
          valueFrom:
            fieldRef:
              fieldPath: spec.nodeName
        - name: MY_POD_NAME
          valueFrom:
```

```
    fieldRef:
      fieldPath: metadata.name
- name: MY_POD_NAMESPACE
valueFrom:
  fieldRef:
    fieldPath: metadata.namespace
- name: MY_POD_IP
valueFrom:
  fieldRef:
    fieldPath: status.podIP
- name: MY_POD_SERVICE_ACCOUNT
valueFrom:
  fieldRef:
    fieldPath: spec.serviceAccountName
restartPolicy: Never
```

In that manifest, you can see five environment variables. The `env` field is an array of environment variable definitions. The first element in the array specifies that the `MY_NODE_NAME` environment variable gets its value from the Pod's `spec.nodeName` field. Similarly, the other environment variables get their names from Pod fields.

Note:

The fields in this example are Pod fields. They are not fields of the container in the Pod.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/inject/dapi-envvars-pod.yaml
```

Verify that the container in the Pod is running:

```
# If the new Pod isn't yet healthy, rerun this command a few times.
kubectl get pods
```

View the container's logs:

```
kubectl logs dapi-envvars-fieldref
```

The output shows the values of selected environment variables:

```
minikube
dapi-envvars-fieldref
default
172.17.0.4
default
```

To see why these values are in the log, look at the `command` and `args` fields in the configuration file. When the container starts, it writes the values of five environment variables to stdout. It repeats this every ten seconds.

Next, get a shell into the container that is running in your Pod:

```
kubectl exec -it dapi-envvars-fieldref -- sh
```

In your shell, view the environment variables:

```
# Run this in a shell inside the container
printenv
```

The output shows that certain environment variables have been assigned the values of Pod fields:

```
MY_POD_SERVICE_ACCOUNT=default
...
MY_POD_NAMESPACE=default
MY_POD_IP=172.17.0.4
...
MY_NODE_NAME=minikube
...
MY_POD_NAME=dapi-envvars-fieldref
```

Use container fields as values for environment variables

In the preceding exercise, you used information from Pod-level fields as the values for environment variables. In this next exercise, you are going to pass fields that are part of the Pod definition, but taken from the specific [container](#) rather than from the Pod overall.

Here is a manifest for another Pod that again has just one container:

[pods/inject/dapi-envvars-container.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: dapi-envvars-resourcefieldref
spec:
  containers:
    - name: test-container
      image: registry.k8s.io/busybox:1.24
      command: [ "sh", "-c" ]
      args:
        - while true; do
          echo -en '\n';
          printenv MY_CPU_REQUEST MY_CPU_LIMIT;
          printenv MY_MEM_REQUEST MY_MEM_LIMIT;
          sleep 10;
        done;
      resources:
        requests:
          memory: "32Mi"
          cpu: "125m"
        limits:
          memory: "64Mi"
          cpu: "250m"
      env:
        - name: MY_CPU_REQUEST
          valueFrom:
```

```

    resourceFieldRef:
      containerName: test-container
      resource: requests.cpu
- name: MY_CPU_LIMIT
valueFrom:
  resourceFieldRef:
    containerName: test-container
    resource: limits.cpu
- name: MY_MEM_REQUEST
valueFrom:
  resourceFieldRef:
    containerName: test-container
    resource: requests.memory
- name: MY_MEM_LIMIT
valueFrom:
  resourceFieldRef:
    containerName: test-container
    resource: limits.memory
restartPolicy: Never

```

In this manifest, you can see four environment variables. The `env` field is an array of environment variable definitions. The first element in the array specifies that the `MY_CPU_REQUEST` environment variable gets its value from the `requests.cpu` field of a container named `test-container`. Similarly, the other environment variables get their values from fields that are specific to this container.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/inject/dapi-envvars-container.yaml
```

Verify that the container in the Pod is running:

```
# If the new Pod isn't yet healthy, rerun this command a few times.
kubectl get pods
```

View the container's logs:

```
kubectl logs dapi-envvars-resourcefieldref
```

The output shows the values of selected environment variables:

```

1
1
33554432
67108864

```

What's next

- Read [Defining Environment Variables for a Container](#)
- Read the [spec](#) API definition for Pod. This includes the definition of Container (part of Pod).
- Read the list of [available fields](#) that you can expose using the downward API.

Read about Pods, containers and environment variables in the legacy API reference:

- [PodSpec](#)
- [Container](#)
- [EnvVar](#)
- [EnvVarSource](#)
- [ObjectFieldSelector](#)
- [ResourceFieldSelector](#)

Expose Pod Information to Containers Through Files

This page shows how a Pod can use a [downwardAPI volume](#), to expose information about itself to containers running in the Pod. A downwardAPI volume can expose Pod fields and container fields.

In Kubernetes, there are two ways to expose Pod and container fields to a running container:

- [Environment variables](#)
- Volume files, as explained in this task

Together, these two ways of exposing Pod and container fields are called the *downward API*.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Store Pod fields

In this part of exercise, you create a Pod that has one container, and you project Pod-level fields into the running container as files. Here is the manifest for the Pod:

[pods/inject/dapi-volume.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: kubernetes-downwardapi-volume-example
  labels:
    zone: us-est-coast
    cluster: test-cluster1
    rack: rack-22
  annotations:
    build: two
```

```

  builder: john-doe
spec:
  containers:
  - name: client-container
    image: registry.k8s.io/busybox
    command: ["sh", "-c"]
    args:
    - while true; do
      if [[ -e /etc/podinfo/labels ]]; then
        echo -en '\n\n'; cat /etc/podinfo/labels; fi;
      if [[ -e /etc/podinfo/annotations ]]; then
        echo -en '\n\n'; cat /etc/podinfo/annotations; fi;
      sleep 5;
    done;
  volumeMounts:
  - name: podinfo
    mountPath: /etc/podinfo
  volumes:
  - name: podinfo
    downwardAPI:
      items:
      - path: "labels"
        fieldRef:
          fieldPath: metadata.labels
      - path: "annotations"
        fieldRef:
          fieldPath: metadata.annotations

```

In the manifest, you can see that the Pod has a downwardAPI Volume, and the container mounts the volume at /etc/podinfo.

Look at the items array under downwardAPI. Each element of the array defines a downwardAPI volume. The first element specifies that the value of the Pod's metadata.labels field should be stored in a file named labels. The second element specifies that the value of the Pod's annotations field should be stored in a file named annotations.

Note:

The fields in this example are Pod fields. They are not fields of the container in the Pod.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/inject/dapi-volume.yaml
```

Verify that the container in the Pod is running:

```
kubectl get pods
```

View the container's logs:

```
kubectl logs kubernetes-downwardapi-volume-example
```

The output shows the contents of the labels file and the annotations file:

```
cluster="test-cluster1"
rack="rack-22"
zone="us-est-coast"

build="two"
builder="john-doe"
```

Get a shell into the container that is running in your Pod:

```
kubectl exec -it kubernetes-downwardapi-volume-example -- sh
```

In your shell, view the labels file:

```
/# cat /etc/podinfo/labels
```

The output shows that all of the Pod's labels have been written to the labels file:

```
cluster="test-cluster1"
rack="rack-22"
zone="us-est-coast"
```

Similarly, view the annotations file:

```
/# cat /etc/podinfo/annotations
```

View the files in the /etc/podinfo directory:

```
/# ls -laR /etc/podinfo
```

In the output, you can see that the labels and annotations files are in a temporary subdirectory: in this example, `..2982_06_02_21_47_53.299460680`. In the `/etc/podinfo` directory, `..data` is a symbolic link to the temporary subdirectory. Also in the `/etc/podinfo` directory, `labels` and `annotations` are symbolic links.

```
drwxr-xr-x ... Feb 6 21:47 ..2982_06_02_21_47_53.299460680
lrwxrwxrwx ... Feb 6 21:47 ..data -> ..2982_06_02_21_47_53.299460680
lrwxrwxrwx ... Feb 6 21:47 annotations -> ..data/annotations
lrwxrwxrwx ... Feb 6 21:47 labels -> ..data/labels
```

```
/etc/..2982_06_02_21_47_53.299460680:
total 8
-rw-r--r-- ... Feb 6 21:47 annotations
-rw-r--r-- ... Feb 6 21:47 labels
```

Using symbolic links enables dynamic atomic refresh of the metadata; updates are written to a new temporary directory, and the `..data` symlink is updated atomically using [rename\(2\)](#).

Note:

A container using Downward API as a [subPath](#) volume mount will not receive Downward API updates.

Exit the shell:


```
/# exit
```

Store container fields

The preceding exercise, you made Pod-level fields accessible using the downward API. In this next exercise, you are going to pass fields that are part of the Pod definition, but taken from the specific [container](#) rather than from the Pod overall. Here is a manifest for a Pod that again has just one container:

[pods/inject/dapi-volume-resources.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: kubernetes-downwardapi-volume-example-2
spec:
  containers:
    - name: client-container
      image: registry.k8s.io/busybox:1.24
      command: ["sh", "-c"]
      args:
        - while true; do
          echo -en '\n';
          if [[ -e /etc/podinfo/cpu_limit ]]; then
            echo -en '\n'; cat /etc/podinfo/cpu_limit; fi;
          if [[ -e /etc/podinfo/cpu_request ]]; then
            echo -en '\n'; cat /etc/podinfo/cpu_request; fi;
          if [[ -e /etc/podinfo/mem_limit ]]; then
            echo -en '\n'; cat /etc/podinfo/mem_limit; fi;
          if [[ -e /etc/podinfo/mem_request ]]; then
            echo -en '\n'; cat /etc/podinfo/mem_request; fi;
          sleep 5;
        done;
      resources:
        requests:
          memory: "32Mi"
          cpu: "125m"
        limits:
          memory: "64Mi"
          cpu: "250m"
      volumeMounts:
        - name: podinfo
          mountPath: /etc/podinfo
  volumes:
    - name: podinfo
      downwardAPI:
        items:
          - path: "cpu_limit"
            resourceFieldRef:
              containerName: client-container
              resource: limits.cpu
              divisor: 1m
```

```
- path: "cpu_request"
  resourceFieldRef:
    containerName: client-container
    resource: requests.cpu
    divisor: 1m
- path: "mem_limit"
  resourceFieldRef:
    containerName: client-container
    resource: limits.memory
    divisor: 1Mi
- path: "mem_request"
  resourceFieldRef:
    containerName: client-container
    resource: requests.memory
    divisor: 1Mi
```

In the manifest, you can see that the Pod has a [downwardAPI volume](#), and that the single container in that Pod mounts the volume at `/etc/podinfo`.

Look at the `items` array under `downwardAPI`. Each element of the array defines a file in the downward API volume.

The first element specifies that in the container named `client-container`, the value of the `limits.cpu` field in the format specified by `1m` should be published as a file named `cpu_limit`. The `divisor` field is optional and has the default value of 1. A divisor of 1 means cores for cpu resources, or bytes for memory resources.

Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/inject/dapi-volume-resources.yaml
```

Get a shell into the container that is running in your Pod:

```
kubectl exec -it kubernetes-downwardapi-volume-example-2 -- sh
```

In your shell, view the `cpu_limit` file:

```
# Run this in a shell inside the container
cat /etc/podinfo/cpu_limit
```

You can use similar commands to view the `cpu_request`, `mem_limit` and `mem_request` files.

Project keys to specific paths and file permissions

You can project keys to specific paths and specific permissions on a per-file basis. For more information, see [Secrets](#).

What's next

- Read the [spec](#) API definition for Pod. This includes the definition of Container (part of Pod).
- Read the list of [available fields](#) that you can expose using the downward API.

Read about volumes in the legacy API reference:

- Check the [Volume](#) API definition which defines a generic volume in a Pod for containers to access.
- Check the [DownwardAPIVolumeSource](#) API definition which defines a volume that contains Downward API information.
- Check the [DownwardAPIVolumeFile](#) API definition which contains references to object or resource fields for populating a file in the Downward API volume.
- Check the [ResourceFieldSelector](#) API definition which specifies the container resources and their output format.

Distribute Credentials Securely Using Secrets

This page shows how to securely inject sensitive data, such as passwords and encryption keys, into Pods.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Convert your secret data to a base-64 representation

Suppose you want to have two pieces of secret data: a username `my-app` and a password `39528$vdg7Jb`. First, use a base64 encoding tool to convert your username and password to a base64 representation. Here's an example using the commonly available base64 program:

```
echo -n 'my-app' | base64
echo -n '39528$vdg7Jb' | base64
```

The output shows that the base-64 representation of your username is `bXktYXBw`, and the base-64 representation of your password is `Mzk1MjgkdmRnN0pi`.

Caution:

Use a local tool trusted by your OS to decrease the security risks of external tools.

Create a Secret

Here is a configuration file you can use to create a Secret that holds your username and password:

[pods/inject/secret.yaml](#)

```
apiVersion: v1
kind: Secret
metadata:
  name: test-secret
data:
  username: bXktYXBw
  password: Mzk1MjgkdmRnN0pi
```

1. Create the Secret

```
kubectl apply -f https://k8s.io/examples/pods/inject/secret.yaml
```

2. View information about the Secret:

```
kubectl get secret test-secret
```

Output:

| NAME | TYPE | DATA | AGE |
|-------------|--------|------|-----|
| test-secret | Opaque | 2 | 1m |

3. View more detailed information about the Secret:

```
kubectl describe secret test-secret
```

Output:

```
Name:      test-secret
Namespace: default
Labels:    <none>
Annotations: <none>
```

```
Type: Opaque
```

```
Data
```

```
====
```

```
password: 13 bytes
```

```
username: 7 bytes
```

Create a Secret directly with kubectl

If you want to skip the Base64 encoding step, you can create the same Secret using the `kubectl create secret` command. For example:

```
kubectl create secret generic test-secret --from-literal=username=my-app --from-literal=password=39528$vdg7Jb'
```

This is more convenient. The detailed approach shown earlier runs through each step explicitly to demonstrate what is happening.

Create a Pod that has access to the secret data through a Volume

Here is a configuration file you can use to create a Pod:

[pods/inject/secret-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: secret-test-pod
spec:
  containers:
    - name: test-container
      image: nginx
      volumeMounts:
        # name must match the volume name below
        - name: secret-volume
          mountPath: /etc/secret-volume
          readOnly: true
  # The secret data is exposed to Containers in the Pod through a Volume.
  volumes:
    - name: secret-volume
      secret:
        secretName: test-secret
```

1. Create the Pod:

```
kubectl apply -f https://k8s.io/examples/pods/inject/secret-pod.yaml
```

2. Verify that your Pod is running:

```
kubectl get pod secret-test-pod
```

Output:

| NAME | READY | STATUS | RESTARTS | AGE |
|-----------------|-------|---------|----------|-----|
| secret-test-pod | 1/1 | Running | 0 | 42m |

3. Get a shell into the Container that is running in your Pod:

```
kubectl exec -i -t secret-test-pod -- /bin/bash
```

4. The secret data is exposed to the Container through a Volume mounted under `/etc/secret-volume`.

In your shell, list the files in the `/etc/secret-volume` directory:

```
# Run this in the shell inside the container
ls /etc/secret-volume
```

The output shows two files, one for each piece of secret data:

```
password username
```

In your shell, display the contents of the `username` and `password` files:

5.

```
# Run this in the shell inside the container
echo "$( cat /etc/secret-volume/username )"
echo "$( cat /etc/secret-volume/password )"
```

The output is your username and password:

```
my-app
39528$vdg7Jb
```

Modify your image or command line so that the program looks for files in the `mountPath` directory. Each key in the Secret data map becomes a file name in this directory.

Project Secret keys to specific file paths

You can also control the paths within the volume where Secret keys are projected. Use the `.spec.volumes[].secret.items` field to change the target path of each key:

```
apiVersion: v1
kind: Pod
metadata:
  name: mypod
spec:
  containers:
  - name: mypod
    image: redis
    volumeMounts:
    - name: foo
      mountPath: "/etc/foo"
      readOnly: true
  volumes:
  - name: foo
    secret:
      secretName: mysecret
      items:
      - key: username
        path: my-group/my-username
```

When you deploy this Pod, the following happens:

- The `username` key from `mysecret` is available to the container at the path `/etc/foo/my-group/my-username` instead of at `/etc/foo/username`.
- The `password` key from that Secret object is not projected.

If you list keys explicitly using `.spec.volumes[].secret.items`, consider the following:

- Only keys specified in `items` are projected.
- To consume all keys from the Secret, all of them must be listed in the `items` field.
- All listed keys must exist in the corresponding Secret. Otherwise, the volume is not created.

Set POSIX permissions for Secret keys

You can set the POSIX file access permission bits for a single Secret key. If you don't specify any permissions, 0644 is used by default. You can also set a default POSIX file mode for the entire Secret volume, and you can override per key if needed.

For example, you can specify a default mode like this:

```
apiVersion: v1
kind: Pod
metadata:
  name: mypod
spec:
  containers:
  - name: mypod
    image: redis
    volumeMounts:
    - name: foo
      mountPath: "/etc/foo"
  volumes:
  - name: foo
    secret:
      secretName: mysecret
      defaultMode: 0400
```

The Secret is mounted on /etc/foo; all the files created by the secret volume mount have permission 0400.

Note:

If you're defining a Pod or a Pod template using JSON, beware that the JSON specification doesn't support octal literals for numbers because JSON considers 0400 to be the *decimal* value 400. In JSON, use decimal values for the defaultMode instead. If you're writing YAML, you can write the defaultMode in octal.

Define container environment variables using Secret data

You can consume the data in Secrets as environment variables in your containers.

If a container already consumes a Secret in an environment variable, a Secret update will not be seen by the container unless it is restarted. There are third party solutions for triggering restarts when secrets change.

Define a container environment variable with data from a single Secret

- Define an environment variable as a key-value pair in a Secret:

```
kubectl create secret generic backend-user --from-literal=backend-username='backend-admin'
```

- Assign the backend-username value defined in the Secret to the SECRET_USERNAME environment variable in the Pod specification.

[pods/inject/pod-single-secret-env-variable.yaml](https://k8s.io/examples/pods/inject/pod-single-secret-env-variable.yaml)

```
apiVersion: v1
kind: Pod
metadata:
  name: env-single-secret
spec:
  containers:
  - name: envvars-test-container
    image: nginx
    env:
    - name: SECRET_USERNAME
      valueFrom:
        secretKeyRef:
          name: backend-user
          key: backend-username
```

- Create the Pod:

```
kubectl create -f https://k8s.io/examples/pods/inject/pod-single-secret-env-variable.yaml
```

- In your shell, display the content of SECRET_USERNAME container environment variable.

```
kubectl exec -i -t env-single-secret -- /bin/sh -c 'echo $SECRET_USERNAME'
```

The output is similar to:

```
backend-admin
```

Define container environment variables with data from multiple Secrets

- As with the previous example, create the Secrets first.

```
kubectl create secret generic backend-user --from-literal=backend-username='backend-admin'
kubectl create secret generic db-user --from-literal=db-username='db-admin'
```

- Define the environment variables in the Pod specification.

[pods/inject/pod-multiple-secret-env-variable.yaml](https://k8s.io/examples/pods/inject/pod-multiple-secret-env-variable.yaml)

```
apiVersion: v1
kind: Pod
metadata:
  name: envvars-multiple-secrets
spec:
  containers:
  - name: envvars-test-container
    image: nginx
    env:
    - name: BACKEND_USERNAME
      valueFrom:
        secretKeyRef:
```



```
  name: backend-user
  key: backend-username
- name: DB_USERNAME
  valueFrom:
    secretKeyRef:
      name: db-user
      key: db-username
```

- Create the Pod:

```
kubectl create -f https://k8s.io/examples/pods/inject/pod-multiple-secret-env-variable.yaml
```

- In your shell, display the container environment variables.

```
kubectl exec -i -t envvars-multiple-secrets -- /bin/sh -c 'env | grep _USERNAME'
```

The output is similar to:

```
DB_USERNAME=db-admin
BACKEND_USERNAME=backend-admin
```

Configure all key-value pairs in a Secret as container environment variables

Note:

This functionality is available in Kubernetes v1.6 and later.

- Create a Secret containing multiple key-value pairs

```
kubectl create secret generic test-secret --from-literal=username='my-app' --from-literal=password='39528$vdg7Jb'
```

- Use envFrom to define all of the Secret's data as container environment variables. The key from the Secret becomes the environment variable name in the Pod.

[pods/inject/pod-secret-envFrom.yaml](https://k8s.io/examples/pods/inject/pod-secret-envFrom.yaml)

```
apiVersion: v1
kind: Pod
metadata:
  name: envfrom-secret
spec:
  containers:
  - name: envvars-test-container
    image: nginx
    envFrom:
    - secretRef:
        name: test-secret
```

- Create the Pod:

```
kubectl create -f https://k8s.io/examples/pods/inject/pod-secret-envFrom.yaml
```

- In your shell, display username and password container environment variables.

```
kubectl exec -i -t envfrom-secret -- /bin/sh -c 'echo "username: $username\npassword: $password\n"'
```

The output is similar to:

```
username: my-app
password: 39528$vdg7Jb
```

Example: Provide prod/test credentials to Pods using Secrets

This example illustrates a Pod which consumes a secret containing production credentials and another Pod which consumes a secret with test environment credentials.

1. Create a secret for prod environment credentials:

```
kubectl create secret generic prod-db-secret --from-literal=username=produser --from-literal=password=Y4nys7f11
```

The output is similar to:

```
secret "prod-db-secret" created
```

2. Create a secret for test environment credentials.

```
kubectl create secret generic test-db-secret --from-literal=username=testuser --from-literal=password=iluvtests
```

The output is similar to:

```
secret "test-db-secret" created
```

Note:

Special characters such as \$, \, *, =, and ! will be interpreted by your [shell](#) and require escaping.

In most shells, the easiest way to escape the password is to surround it with single quotes ('). For example, if your actual password is S!B*d\$zDsb=, you should execute the command as follows:

```
kubectl create secret generic dev-db-secret --from-literal=username=devuser --from-literal=password='S!B\*d$zDsb='
```

You do not need to escape special characters in passwords from files (--from-file).

3. Create the Pod manifests:

```

cat <<EOF > pod.yaml
apiVersion: v1
kind: List
items:
- kind: Pod
  apiVersion: v1
  metadata:
    name: prod-db-client-pod
    labels:
      name: prod-db-client
  spec:
    volumes:
    - name: secret-volume
      secret:
        secretName: prod-db-secret
    containers:
    - name: db-client-container
      image: myClientImage
      volumeMounts:
      - name: secret-volume
        readOnly: true
        mountPath: "/etc/secret-volume"
- kind: Pod
  apiVersion: v1
  metadata:
    name: test-db-client-pod
    labels:
      name: test-db-client
  spec:
    volumes:
    - name: secret-volume
      secret:
        secretName: test-db-secret
    containers:
    - name: db-client-container
      image: myClientImage
      volumeMounts:
      - name: secret-volume
        readOnly: true
        mountPath: "/etc/secret-volume"
EOF

```

Note:

How the specs for the two Pods differ only in one field; this facilitates creating Pods with different capabilities from a common Pod template.

4. Apply all those objects on the API server by running:

```
kubectl create -f pod.yaml
```

Both containers will have the following files present on their filesystems with the values for each container's environment:

```
/etc/secret-volume/username
/etc/secret-volume/password
```

You could further simplify the base Pod specification by using two service accounts:

1. `prod-user` with the `prod-db-secret`
2. `test-user` with the `test-db-secret`

The Pod specification is shortened to:

```
apiVersion: v1
kind: Pod
metadata:
  name: prod-db-client-pod
  labels:
    name: prod-db-client
spec:
  serviceAccount: prod-db-client
  containers:
  - name: db-client-container
    image: myClientImage
```

References

- [Secret](#)
- [Volume](#)
- [Pod](#)

What's next

- Learn more about [Secrets](#).
- Learn about [Volumes](#).

Run Applications

Run and manage both stateless and stateful applications.

[Run a Stateless Application Using a Deployment](#)

[Run a Single-Instance Stateful Application](#)

[Run a Replicated Stateful Application](#)

[Scale a StatefulSet](#)

[Delete a StatefulSet](#)

[Force Delete StatefulSet Pods](#)

[Horizontal Pod Autoscaling](#)

[HorizontalPodAutoscaler Walkthrough](#)

[Specifying a Disruption Budget for your Application](#)

[Accessing the Kubernetes API from a Pod](#)

Run a Stateless Application Using a Deployment

This page shows how to run an application using a Kubernetes Deployment object.

Objectives

- Create an nginx deployment.
- Use kubectl to list information about the deployment.
- Update the deployment.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.9. To check the version, enter kubectl version.

Creating and exploring an nginx deployment

You can run an application by creating a Kubernetes Deployment object, and you can describe a Deployment in a YAML file. For example, this YAML file describes a Deployment that runs the nginx:1.14.2 Docker image:

[application/deployment.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  selector:
    matchLabels:
      app: nginx
  replicas: 2 # tells deployment to run 2 pods matching the template
  template:
    metadata:
```

```
labels:
  app: nginx
spec:
  containers:
  - name: nginx
    image: nginx:1.14.2
    ports:
    - containerPort: 80
```

1. Create a Deployment based on the YAML file:

```
kubectl apply -f https://k8s.io/examples/application/deployment.yaml
```

2. Display information about the Deployment:

```
kubectl describe deployment nginx-deployment
```

The output is similar to this:

```
Name:      nginx-deployment
Namespace: default
CreationTimestamp: Tue, 30 Aug 2016 18:11:37 -0700
Labels:    app=nginx
Annotations: deployment.kubernetes.io/revision=1
Selector:  app=nginx
Replicas:  2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=nginx
  Containers:
    nginx:
      Image:      nginx:1.14.2
      Port:       80/TCP
      Environment: <none>
      Mounts:      <none>
      Volumes:     <none>
  Conditions:
    Type      Status Reason
    ----      -
    Available  True   MinimumReplicasAvailable
    Progressing True   NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet:  nginx-deployment-1771418926 (2/2 replicas created)
No events.
```

3. List the Pods created by the deployment:

```
kubectl get pods -l app=nginx
```

The output is similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|-----------------------------------|-------|---------|----------|-----|
| nginx-deployment-1771418926-7o5ns | 1/1 | Running | 0 | 16h |
| nginx-deployment-1771418926-r18az | 1/1 | Running | 0 | 16h |

4. Display information about a Pod:

```
kubectl describe pod <pod-name>
```

where <pod-name> is the name of one of your Pods.

Updating the deployment

You can update the deployment by applying a new YAML file. This YAML file specifies that the deployment should be updated to use nginx 1.16.1.

[application/deployment-update.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  selector:
    matchLabels:
      app: nginx
  replicas: 2
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:1.16.1 # Update the version of nginx from 1.14.2 to 1.16.1
          ports:
            - containerPort: 80
```

1. Apply the new YAML file:

```
kubectl apply -f https://k8s.io/examples/application/deployment-update.yaml
```

2. Watch the deployment create pods with new names and delete the old pods:

```
kubectl get pods -l app=nginx
```

Scaling the application by increasing the replica count

You can increase the number of Pods in your Deployment by applying a new YAML file. This YAML file sets replicas to 4, which specifies that the Deployment should have four Pods:

[application/deployment-scale.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  selector:
    matchLabels:
      app: nginx
  replicas: 4 # Update the replicas from 2 to 4
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.16.1
        ports:
        - containerPort: 80
```

1. Apply the new YAML file:

```
kubectl apply -f https://k8s.io/examples/application/deployment-scale.yaml
```

2. Verify that the Deployment has four Pods:

```
kubectl get pods -l app=nginx
```

The output is similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|----------------------------------|-------|---------|----------|-----|
| nginx-deployment-148880595-4zdqq | 1/1 | Running | 0 | 25s |
| nginx-deployment-148880595-6zgi1 | 1/1 | Running | 0 | 25s |
| nginx-deployment-148880595-fxcez | 1/1 | Running | 0 | 2m |
| nginx-deployment-148880595-rwovn | 1/1 | Running | 0 | 2m |

Deleting a deployment

Delete the deployment by name:

```
kubectl delete deployment nginx-deployment
```

ReplicationControllers -- the Old Way

The preferred way to create a replicated application is to use a Deployment, which in turn uses a ReplicaSet. Before the Deployment and ReplicaSet were added to Kubernetes, replicated applications were configured using a [ReplicationController](#).

What's next

- Learn more about [Deployment objects](#).

Run a Single-Instance Stateful Application

This page shows you how to run a single-instance stateful application in Kubernetes using a PersistentVolume and a Deployment. The application is MySQL.

Objectives

- Create a PersistentVolume referencing a disk in your environment.
- Create a MySQL Deployment.
- Expose MySQL to other pods in the cluster at a known DNS name.

Before you begin

- You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

- You need to either have a [dynamic PersistentVolume provisioner](#) with a default [StorageClass](#), or [statically provision PersistentVolumes](#) yourself to satisfy the [PersistentVolumeClaims](#) used here.

Deploy MySQL

You can run a stateful application by creating a Kubernetes Deployment and connecting it to an existing PersistentVolume using a PersistentVolumeClaim. For example, this YAML file describes a Deployment that runs MySQL and references the PersistentVolumeClaim. The file defines a volume mount for /var/lib/mysql, and then creates a PersistentVolumeClaim that looks for a 20G volume. This claim is satisfied by any existing volume that meets the requirements, or by a dynamic provisioner.

Note: The password is defined in the config yaml, and this is insecure. See [Kubernetes Secrets](#) for a secure solution.

[application/mysql/mysql-deployment.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: mysql
spec:
  ports:
    - port: 3306
  selector:
    app: mysql
```

```

clusterIP: None
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mysql
spec:
  selector:
    matchLabels:
      app: mysql
  strategy:
    type: Recreate
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
        - image: mysql:5.6
          name: mysql
          env:
            # Use secret in real usage
            - name: MYSQL_ROOT_PASSWORD
              value: password
          ports:
            - containerPort: 3306
              name: mysql
          volumeMounts:
            - name: mysql-persistent-storage
              mountPath: /var/lib/mysql
          volumes:
            - name: mysql-persistent-storage
              persistentVolumeClaim:
                claimName: mysql-pv-claim

```

[application/mysql/mysql-pv.yaml](#)

```

apiVersion: v1
kind: PersistentVolume
metadata:
  name: mysql-pv-volume
  labels:
    type: local
spec:
  storageClassName: manual
  capacity:
    storage: 20Gi
  accessModes:
    - ReadWriteOnce
  hostPath:
    path: "/mnt/data"
---
apiVersion: v1

```

```

kind: PersistentVolumeClaim
metadata:
  name: mysql-pv-claim
spec:
  storageClassName: manual
  accessModes:
    - ReadWriteOnce
resources:
  requests:
    storage: 20Gi

```

1. Deploy the PV and PVC of the YAML file:

```
kubectl apply -f https://k8s.io/examples/application/mysql/mysql-pv.yaml
```

2. Deploy the contents of the YAML file:

```
kubectl apply -f https://k8s.io/examples/application/mysql/mysql-deployment.yaml
```

3. Display information about the Deployment:

```
kubectl describe deployment mysql
```

The output is similar to this:

```

Name:          mysql
Namespace:     default
CreationTimestamp:  Tue, 01 Nov 2016 11:18:45 -0700
Labels:        app=mysql
Annotations:    deployment.kubernetes.io/revision=1
Selector:      app=mysql
Replicas:      1 desired | 1 updated | 1 total | 0 available | 1 unavailable
StrategyType:  Recreate
MinReadySeconds:  0
Pod Template:
  Labels:  app=mysql
  Containers:
    mysql:
      Image:  mysql:5.6
      Port:   3306/TCP
      Environment:
        MYSQL_ROOT_PASSWORD:  password
      Mounts:
        /var/lib/mysql from mysql-persistent-storage (rw)
  Volumes:
    mysql-persistent-storage:
      Type:  PersistentVolumeClaim (a reference to a PersistentVolumeClaim in the same
namespace)
      ClaimName: mysql-pv-claim
      ReadOnly:  false
  Conditions:
    Type      Status  Reason
    ----      -
    Available  False   MinimumReplicasUnavailable

```

```

Progressing True ReplicaSetUpdated
OldReplicaSets: <none>
NewReplicaSet:  mysql-63082529 (1/1 replicas created)
Events:
  FirstSeen    LastSeen    Count   From              SubobjectPath  Type      Reason
  Message
  -----
  33s          33s         1       {deployment-controller }      Normal    ScalingReplicaSet
  Scaled up replica set mysql-63082529 to 1

```

4. List the pods created by the Deployment:

```
kubectl get pods -l app=mysql
```

The output is similar to this:

```

NAME                READY   STATUS    RESTARTS   AGE
mysql-63082529-2z3ki 1/1     Running   0          3m

```

5. Inspect the PersistentVolumeClaim:

```
kubectl describe pvc mysql-pv-claim
```

The output is similar to this:

```

Name:          mysql-pv-claim
Namespace:     default
StorageClass:
Status:        Bound
Volume:        mysql-pv-volume
Labels:        <none>
Annotations:   pv.kubernetes.io/bind-completed=yes
               pv.kubernetes.io/bound-by-controller=yes
Capacity:      20Gi
Access Modes:  RWO
Events:        <none>

```

Accessing the MySQL instance

The preceding YAML file creates a service that allows other Pods in the cluster to access the database. The Service option `clusterIP: None` lets the Service DNS name resolve directly to the Pod's IP address. This is optimal when you have only one Pod behind a Service and you don't intend to increase the number of Pods.

Run a MySQL client to connect to the server:

```
kubectl run -it --rm --image=mysql:5.6 --restart=Never mysql-client -- mysql -h mysql -ppassword
```

This command creates a new Pod in the cluster running a MySQL client and connects it to the server through the Service. If it connects, you know your stateful MySQL database is up and running.

Waiting for pod default/mysql-client-274442439-zyp6i to be running, status is Pending, pod ready: false
If you don't see a command prompt, try pressing enter.

```
mysql>
```

Updating

The image or any other part of the Deployment can be updated as usual with the `kubectl apply` command. Here are some precautions that are specific to stateful apps:

- Don't scale the app. This setup is for single-instance apps only. The underlying `PersistentVolume` can only be mounted to one Pod. For clustered stateful apps, see the [StatefulSet documentation](#).
- Use `strategy: type: Recreate` in the Deployment configuration YAML file. This instructs Kubernetes to *not* use rolling updates. Rolling updates will not work, as you cannot have more than one Pod running at a time. The Recreate strategy will stop the first pod before creating a new one with the updated configuration.

Deleting a deployment

Delete the deployed objects by name:

```
kubectl delete deployment,svc mysql
kubectl delete pvc mysql-pv-claim
kubectl delete pv mysql-pv-volume
```

If you manually provisioned a `PersistentVolume`, you also need to manually delete it, as well as release the underlying resource. If you used a dynamic provisioner, it automatically deletes the `PersistentVolume` when it sees that you deleted the `PersistentVolumeClaim`. Some dynamic provisioners (such as those for EBS and PD) also release the underlying resource upon deleting the `PersistentVolume`.

What's next

- Learn more about [Deployment objects](#).
- Learn more about [Deploying applications](#)
- [kubectl run documentation](#)
- [Volumes](#) and [Persistent Volumes](#)

Run a Replicated Stateful Application

This page shows how to run a replicated stateful application using a [StatefulSet](#). This application is a replicated MySQL database. The example topology has a single primary server and multiple replicas, using asynchronous row-based replication.

Note:

This is not a production configuration. MySQL settings remain on insecure defaults to keep the focus on general patterns for running stateful applications in Kubernetes.

Before you begin

- You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:
 - [Killercodea](#)
 - [Play with Kubernetes](#)
- You need to either have a [dynamic PersistentVolume provisioner](#) with a default [StorageClass](#), or [statically provision PersistentVolumes](#) yourself to satisfy the [PersistentVolumeClaims](#) used here.
- This tutorial assumes you are familiar with [PersistentVolumes](#) and [StatefulSets](#), as well as other core concepts like [Pods](#), [Services](#), and [ConfigMaps](#).
- Some familiarity with MySQL helps, but this tutorial aims to present general patterns that should be useful for other systems.
- You are using the default namespace or another namespace that does not contain any conflicting objects.

Objectives

- Deploy a replicated MySQL topology with a StatefulSet.
- Send MySQL client traffic.
- Observe resistance to downtime.
- Scale the StatefulSet up and down.

Deploy MySQL

The example MySQL deployment consists of a ConfigMap, two Services, and a StatefulSet.

Create a ConfigMap

Create the ConfigMap from the following YAML configuration file:

[application/mysql/mysql-configmap.yaml](#)

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: mysql
  labels:
    app: mysql
    app.kubernetes.io/name: mysql
```

```
data:
  primary.cnf: |
    # Apply this config only on the primary.
    [mysqld]
    log-bin
  replica.cnf: |
    # Apply this config only on replicas.
    [mysqld]
    super-read-only
```

```
kubectl apply -f https://k8s.io/examples/application/mysql/mysql-configmap.yaml
```

This ConfigMap provides my.cnf overrides that let you independently control configuration on the primary MySQL server and its replicas. In this case, you want the primary server to be able to serve replication logs to replicas and you want replicas to reject any writes that don't come via replication.

There's nothing special about the ConfigMap itself that causes different portions to apply to different Pods. Each Pod decides which portion to look at as it's initializing, based on information provided by the StatefulSet controller.

Create Services

Create the Services from the following YAML configuration file:

[application/mysql/mysql-services.yaml](https://k8s.io/examples/application/mysql/mysql-services.yaml)

```
# Headless service for stable DNS entries of StatefulSet members.
apiVersion: v1
kind: Service
metadata:
  name: mysql
  labels:
    app: mysql
    app.kubernetes.io/name: mysql
spec:
  ports:
  - name: mysql
    port: 3306
  clusterIP: None
  selector:
    app: mysql
---
# Client service for connecting to any MySQL instance for reads.
# For writes, you must instead connect to the primary: mysql-0.mysql.
apiVersion: v1
kind: Service
metadata:
  name: mysql-read
  labels:
    app: mysql
    app.kubernetes.io/name: mysql
```

```
  readonly: "true"
spec:
  ports:
  - name: mysql
    port: 3306
  selector:
    app: mysql
```

```
kubectl apply -f https://k8s.io/examples/application/mysql/mysql-services.yaml
```

The headless Service provides a home for the DNS entries that the StatefulSet [controllers](#) creates for each Pod that's part of the set. Because the headless Service is named mysql, the Pods are accessible by resolving <pod-name>.mysql from within any other Pod in the same Kubernetes cluster and namespace.

The client Service, called mysql-read, is a normal Service with its own cluster IP that distributes connections across all MySQL Pods that report being Ready. The set of potential endpoints includes the primary MySQL server and all replicas.

Note that only read queries can use the load-balanced client Service. Because there is only one primary MySQL server, clients should connect directly to the primary MySQL Pod (through its DNS entry within the headless Service) to execute writes.

Create the StatefulSet

Finally, create the StatefulSet from the following YAML configuration file:

[application/mysql/mysql-statefulset.yaml](#)

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: mysql
spec:
  selector:
    matchLabels:
      app: mysql
      app.kubernetes.io/name: mysql
  serviceName: mysql
  replicas: 3
  template:
    metadata:
      labels:
        app: mysql
        app.kubernetes.io/name: mysql
    spec:
      initContainers:
      - name: init-mysql
        image: mysql:5.7
        command:
        - bash
        - "-c"
        - |
```



```

set -ex
# Generate mysql server-id from pod ordinal index.
[[ $HOSTNAME =~ -([0-9]+)$ ]] || exit 1
ordinal=${BASH_REMATCH[1]}
echo [mysqld] > /mnt/conf.d/server-id.cnf
# Add an offset to avoid reserved server-id=0 value.
echo server-id=$((100 + $ordinal)) >> /mnt/conf.d/server-id.cnf
# Copy appropriate conf.d files from config-map to emptyDir.
if [[ $ordinal -eq 0 ]]; then
    cp /mnt/config-map/primary.cnf /mnt/conf.d/
else
    cp /mnt/config-map/replica.cnf /mnt/conf.d/
fi
volumeMounts:
- name: conf
  mountPath: /mnt/conf.d
- name: config-map
  mountPath: /mnt/config-map
- name: clone-mysql
  image: gcr.io/google-samples/xtrabackup:1.0
  command:
  - bash
  - "-c"
  - |
    set -ex
    # Skip the clone if data already exists.
    [[ -d /var/lib/mysql/mysql ]] && exit 0
    # Skip the clone on primary (ordinal index 0).
    [[ `hostname` =~ -([0-9]+)$ ]] || exit 1
    ordinal=${BASH_REMATCH[1]}
    [[ $ordinal -eq 0 ]] && exit 0
    # Clone data from previous peer.
    ncat --recv-only mysql-$$($ordinal-1).mysql 3307 | xbstream -x -C /var/lib/mysql
    # Prepare the backup.
    xtrabackup --prepare --target-dir=/var/lib/mysql
  volumeMounts:
  - name: data
    mountPath: /var/lib/mysql
    subPath: mysql
  - name: conf
    mountPath: /etc/mysql/conf.d
containers:
- name: mysql
  image: mysql:5.7
  env:
  - name: MYSQL_ALLOW_EMPTY_PASSWORD
    value: "1"
  ports:
  - name: mysql
    containerPort: 3306
  volumeMounts:
  - name: data
    mountPath: /var/lib/mysql

```

```

    subPath: mysql
- name: conf
  mountPath: /etc/mysql/conf.d
resources:
  requests:
    cpu: 500m
    memory: 1Gi
livenessProbe:
  exec:
    command: ["mysqladmin", "ping"]
  initialDelaySeconds: 30
  periodSeconds: 10
  timeoutSeconds: 5
readinessProbe:
  exec:
    # Check we can execute queries over TCP (skip-networking is off).
    command: ["mysql", "-h", "127.0.0.1", "-e", "SELECT 1"]
  initialDelaySeconds: 5
  periodSeconds: 2
  timeoutSeconds: 1
- name: xtrabackup
  image: gcr.io/google-samples/xtrabackup:1.0
  ports:
    - name: xtrabackup
      containerPort: 3307
  command:
    - bash
    - "-c"
    - |
      set -ex
      cd /var/lib/mysql

      # Determine binlog position of cloned data, if any.
      if [[ -f xtrabackup_slave_info && "x$(<xtrabackup_slave_info)" != "x" ]]; then
        # XtraBackup already generated a partial "CHANGE MASTER TO" query
        # because we're cloning from an existing replica. (Need to remove the trailing semicolon!)
        cat xtrabackup_slave_info | sed -E 's/;$//g' > change_master_to.sql.in
        # Ignore xtrabackup_binlog_info in this case (it's useless).
        rm -f xtrabackup_slave_info xtrabackup_binlog_info
      elif [[ -f xtrabackup_binlog_info ]]; then
        # We're cloning directly from primary. Parse binlog position.
        [[ `cat xtrabackup_binlog_info` =~ ^(.?)[[:space:]]+(.?)$ ]] || exit 1
        rm -f xtrabackup_binlog_info xtrabackup_slave_info
        echo "CHANGE MASTER TO MASTER_LOG_FILE='${BASH_REMATCH[1]}',\
          MASTER_LOG_POS=${BASH_REMATCH[2]}" > change_master_to.sql.in
      fi

      # Check if we need to complete a clone by starting replication.
      if [[ -f change_master_to.sql.in ]]; then
        echo "Waiting for mysqld to be ready (accepting connections)"
        until mysql -h 127.0.0.1 -e "SELECT 1"; do sleep 1; done

        echo "Initializing replication from clone position"

```

```

mysql -h 127.0.0.1 \
-e "$(<change_master_to.sql.in), \
    MASTER_HOST='mysql-0.mysql', \
    MASTER_USER='root', \
    MASTER_PASSWORD='', \
    MASTER_CONNECT_RETRY=10; \
    START SLAVE;" || exit 1
# In case of container restart, attempt this at-most-once.
mv change_master_to.sql.in change_master_to.sql.orig
fi

# Start a server to send backups when requested by peers.
exec ncat --listen --keep-open --send-only --max-conns=1 3307 -c \
    "xtrabackup --backup --slave-info --stream=xbstream --host=127.0.0.1 --user=root"
volumeMounts:
- name: data
  mountPath: /var/lib/mysql
  subPath: mysql
- name: conf
  mountPath: /etc/mysql/conf.d
resources:
  requests:
    cpu: 100m
    memory: 100Mi
volumes:
- name: conf
  emptyDir: {}
- name: config-map
  configMap:
    name: mysql
volumeClaimTemplates:
- metadata:
    name: data
  spec:
    accessModes: ["ReadWriteOnce"]
    resources:
      requests:
        storage: 10Gi

```

```
kubectl apply -f https://k8s.io/examples/application/mysql/mysql-statefulset.yaml
```

You can watch the startup progress by running:

```
kubectl get pods -l app=mysql --watch
```

After a while, you should see all 3 Pods become Running:

| NAME | READY | STATUS | RESTARTS | AGE |
|---------|-------|---------|----------|-----|
| mysql-0 | 2/2 | Running | 0 | 2m |
| mysql-1 | 2/2 | Running | 0 | 1m |
| mysql-2 | 2/2 | Running | 0 | 1m |

Press **Ctrl+C** to cancel the watch.

Note:

If you don't see any progress, make sure you have a dynamic PersistentVolume provisioner enabled, as mentioned in the [prerequisites](#).

This manifest uses a variety of techniques for managing stateful Pods as part of a StatefulSet. The next section highlights some of these techniques to explain what happens as the StatefulSet creates Pods.

Understanding stateful Pod initialization

The StatefulSet controller starts Pods one at a time, in order by their ordinal index. It waits until each Pod reports being Ready before starting the next one.

In addition, the controller assigns each Pod a unique, stable name of the form `<statefulset-name>--<ordinal-index>`, which results in Pods named `mysql-0`, `mysql-1`, and `mysql-2`.

The Pod template in the above StatefulSet manifest takes advantage of these properties to perform orderly startup of MySQL replication.

Generating configuration

Before starting any of the containers in the Pod spec, the Pod first runs any [init containers](#) in the order defined.

The first init container, named `init-mysql`, generates special MySQL config files based on the ordinal index.

The script determines its own ordinal index by extracting it from the end of the Pod name, which is returned by the `hostname` command. Then it saves the ordinal (with a numeric offset to avoid reserved values) into a file called `server-id.cnf` in the MySQL `conf.d` directory. This translates the unique, stable identity provided by the StatefulSet into the domain of MySQL server IDs, which require the same properties.

The script in the `init-mysql` container also applies either `primary.cnf` or `replica.cnf` from the ConfigMap by copying the contents into `conf.d`. Because the example topology consists of a single primary MySQL server and any number of replicas, the script assigns ordinal 0 to be the primary server, and everyone else to be replicas. Combined with the StatefulSet controller's [deployment order guarantee](#), this ensures the primary MySQL server is Ready before creating replicas, so they can begin replicating.

Cloning existing data

In general, when a new Pod joins the set as a replica, it must assume the primary MySQL server might already have data on it. It also must assume that the replication logs might not go all the way back to the beginning of time. These conservative assumptions are the key to allow a running StatefulSet to scale up and down over time, rather than being fixed at its initial size.

The second init container, named `clone-mysql`, performs a clone operation on a replica Pod the first time it starts up on an empty PersistentVolume. That means it copies all existing data from another running Pod, so its local state is consistent enough to begin replicating from the primary server.

MySQL itself does not provide a mechanism to do this, so the example uses a popular open-source tool called Percona XtraBackup. During the clone, the source MySQL server might suffer reduced performance. To minimize impact on the primary MySQL server, the script instructs each Pod to clone from the Pod whose ordinal index is one lower. This works because the StatefulSet controller always ensures Pod N is Ready before starting Pod N+1.

Starting replication

After the init containers complete successfully, the regular containers run. The MySQL Pods consist of a mysql container that runs the actual mysqld server, and an xtrabackup container that acts as a [sidecar](#).

The xtrabackup sidecar looks at the cloned data files and determines if it's necessary to initialize MySQL replication on the replica. If so, it waits for mysqld to be ready and then executes the CHANGE MASTER TO and START SLAVE commands with replication parameters extracted from the XtraBackup clone files.

Once a replica begins replication, it remembers its primary MySQL server and reconnects automatically if the server restarts or the connection dies. Also, because replicas look for the primary server at its stable DNS name (mysql-0.mysql), they automatically find the primary server even if it gets a new Pod IP due to being rescheduled.

Lastly, after starting replication, the xtrabackup container listens for connections from other Pods requesting a data clone. This server remains up indefinitely in case the StatefulSet scales up, or in case the next Pod loses its PersistentVolumeClaim and needs to redo the clone.

Sending client traffic

You can send test queries to the primary MySQL server (hostname mysql-0.mysql) by running a temporary container with the mysql:5.7 image and running the mysql client binary.

```
kubectrl run mysql-client --image=mysql:5.7 -i --rm --restart=Never --\
mysql -h mysql-0.mysql <<EOF
CREATE DATABASE test;
CREATE TABLE test.messages (message VARCHAR(250));
INSERT INTO test.messages VALUES ('hello');
EOF
```

Use the hostname mysql-read to send test queries to any server that reports being Ready:

```
kubectrl run mysql-client --image=mysql:5.7 -i -t --rm --restart=Never --\
mysql -h mysql-read -e "SELECT * FROM test.messages"
```

You should get output like this:

```
Waiting for pod default/mysql-client to be running, status is Pending, pod ready: false
+-----+
| message |
+-----+
| hello   |
+-----+
pod "mysql-client" deleted
```

To demonstrate that the mysql-read Service distributes connections across servers, you can run `SELECT @@server_id` in a loop:

```
kubectrl run mysql-client-loop --image=mysql:5.7 -i -t --rm --restart=Never --\
  bash -ic "while sleep 1; do mysql -h mysql-read -e 'SELECT @@server_id,NOW()'; done"
```

You should see the reported `@@server_id` change randomly, because a different endpoint might be selected upon each connection attempt:

```
+-----+-----+
| @@server_id | NOW() |
+-----+-----+
|      100 | 2006-01-02 15:04:05 |
+-----+-----+
| @@server_id | NOW() |
+-----+-----+
|      102 | 2006-01-02 15:04:06 |
+-----+-----+
| @@server_id | NOW() |
+-----+-----+
|      101 | 2006-01-02 15:04:07 |
+-----+-----+
```

You can press **Ctrl+C** when you want to stop the loop, but it's useful to keep it running in another window so you can see the effects of the following steps.

Simulate Pod and Node failure

To demonstrate the increased availability of reading from the pool of replicas instead of a single server, keep the `SELECT @@server_id` loop from above running while you force a Pod out of the Ready state.

Break the Readiness probe

The [readiness probe](#) for the mysql container runs the command `mysql -h 127.0.0.1 -e 'SELECT 1'` to make sure the server is up and able to execute queries.

One way to force this readiness probe to fail is to break that command:

```
kubectrl exec mysql-2 -c mysql -- mv /usr/bin/mysql /usr/bin/mysql.off
```

This reaches into the actual container's filesystem for Pod `mysql-2` and renames the `mysql` command so the readiness probe can't find it. After a few seconds, the Pod should report one of its containers as not Ready, which you can check by running:

```
kubectrl get pod mysql-2
```

Look for 1/2 in the READY column:

| NAME | READY | STATUS | RESTARTS | AGE |
|---------|-------|---------|----------|-----|
| mysql-2 | 1/2 | Running | 0 | 3m |

At this point, you should see your `SELECT @@server_id` loop continue to run, although it never reports 102 anymore. Recall that the `init-mysql` script defined `server-id` as `100 + $ordinal`, so server ID 102 corresponds to Pod `mysql-2`.

Now repair the Pod and it should reappear in the loop output after a few seconds:

```
kubectl exec mysql-2 -c mysql -- mv /usr/bin/mysql.off /usr/bin/mysql
```

Delete Pods

The `StatefulSet` also recreates Pods if they're deleted, similar to what a `ReplicaSet` does for stateless Pods.

```
kubectl delete pod mysql-2
```

The `StatefulSet` controller notices that no `mysql-2` Pod exists anymore, and creates a new one with the same name and linked to the same `PersistentVolumeClaim`. You should see server ID 102 disappear from the loop output for a while and then return on its own.

Drain a Node

If your Kubernetes cluster has multiple Nodes, you can simulate Node downtime (such as when Nodes are upgraded) by issuing a [drain](#).

First determine which Node one of the MySQL Pods is on:

```
kubectl get pod mysql-2 -o wide
```

The Node name should show up in the last column:

| NAME | READY | STATUS | RESTARTS | AGE | IP | NODE |
|---------|-------|---------|----------|-----|-------------|----------------------|
| mysql-2 | 2/2 | Running | 0 | 15m | 10.244.5.27 | kubernetes-node-9l2t |

Then, drain the Node by running the following command, which cordons it so no new Pods may schedule there, and then evicts any existing Pods. Replace `<node-name>` with the name of the Node you found in the last step.

Caution:

Draining a Node can impact other workloads and applications running on the same node. Only perform the following step in a test cluster.

```
# See above advice about impact on other workloads
kubectl drain <node-name> --force --delete-emptydir-data --ignore-daemonsets
```

Now you can watch as the Pod reschedules on a different Node:

```
kubectl get pod mysql-2 -o wide --watch
```

It should look something like this:

| NAME | READY | STATUS | RESTARTS | AGE | IP | NODE |
|---------|-------|-------------|----------|-----|-------------|----------------------|
| mysql-2 | 2/2 | Terminating | 0 | 15m | 10.244.1.56 | kubernetes-node-9l2t |
| [...] | | | | | | |

| | | | | | | |
|---------|-----|-----------------|---|-----|-------------|----------------------|
| mysql-2 | 0/2 | Pending | 0 | 0s | <none> | kubernetes-node-fjlm |
| mysql-2 | 0/2 | Init:0/2 | 0 | 0s | <none> | kubernetes-node-fjlm |
| mysql-2 | 0/2 | Init:1/2 | 0 | 20s | 10.244.5.32 | kubernetes-node-fjlm |
| mysql-2 | 0/2 | PodInitializing | 0 | 21s | 10.244.5.32 | kubernetes-node-fjlm |
| mysql-2 | 1/2 | Running | 0 | 22s | 10.244.5.32 | kubernetes-node-fjlm |
| mysql-2 | 2/2 | Running | 0 | 30s | 10.244.5.32 | kubernetes-node-fjlm |

And again, you should see server ID 102 disappear from the SELECT @@server_id loop output for a while and then return.

Now uncordon the Node to return it to a normal state:

```
kubectl uncordon <node-name>
```

Scaling the number of replicas

When you use MySQL replication, you can scale your read query capacity by adding replicas. For a StatefulSet, you can achieve this with a single command:

```
kubectl scale statefulset mysql --replicas=5
```

Watch the new Pods come up by running:

```
kubectl get pods -l app=mysql --watch
```

Once they're up, you should see server IDs 103 and 104 start appearing in the SELECT @@server_id loop output.

You can also verify that these new servers have the data you added before they existed:

```
kubectl run mysql-client --image=mysql:5.7 -i -t --rm --restart=Never --\
mysql -h mysql-3.mysql -e "SELECT * FROM test.messages"
```

Waiting for pod default/mysql-client to be running, status is Pending, pod ready: false

```
+-----+
| message |
+-----+
| hello  |
+-----+
```

pod "mysql-client" deleted

Scaling back down is also seamless:

```
kubectl scale statefulset mysql --replicas=3
```

Note:

Although scaling up creates new PersistentVolumeClaims automatically, scaling down does not automatically delete these PVCs.

This gives you the choice to keep those initialized PVCs around to make scaling back up quicker, or to extract data before deleting them.

You can see this by running:


```
kubectrl get pvc -l app=mysql
```

Which shows that all 5 PVCs still exist, despite having scaled the StatefulSet down to 3:

| NAME | STATUS | VOLUME | CAPACITY | ACCESSMODES | AGE |
|--------------|--------|--|----------|-------------|-----|
| data-mysql-0 | Bound | pvc-8acbf5dc-b103-11e6-93fa-42010a800002 | 10Gi | RWO | 20m |
| data-mysql-1 | Bound | pvc-8ad39820-b103-11e6-93fa-42010a800002 | 10Gi | RWO | 20m |
| data-mysql-2 | Bound | pvc-8ad69a6d-b103-11e6-93fa-42010a800002 | 10Gi | RWO | 20m |
| data-mysql-3 | Bound | pvc-50043c45-b1c5-11e6-93fa-42010a800002 | 10Gi | RWO | 2m |
| data-mysql-4 | Bound | pvc-500a9957-b1c5-11e6-93fa-42010a800002 | 10Gi | RWO | 2m |

If you don't intend to reuse the extra PVCs, you can delete them:

```
kubectrl delete pvc data-mysql-3  
kubectrl delete pvc data-mysql-4
```

Cleaning up

1. Cancel the SELECT @@server_id loop by pressing **Ctrl+C** in its terminal, or running the following from another terminal:

```
kubectrl delete pod mysql-client-loop --now
```

2. Delete the StatefulSet. This also begins terminating the Pods.

```
kubectrl delete statefulset mysql
```

3. Verify that the Pods disappear. They might take some time to finish terminating.

```
kubectrl get pods -l app=mysql
```

You'll know the Pods have terminated when the above returns:

```
No resources found.
```

4. Delete the ConfigMap, Services, and PersistentVolumeClaims.

```
kubectrl delete configmap,service,pvc -l app=mysql
```

5. If you manually provisioned PersistentVolumes, you also need to manually delete them, as well as release the underlying resources. If you used a dynamic provisioner, it automatically deletes the PersistentVolumes when it sees that you deleted the PersistentVolumeClaims. Some dynamic provisioners (such as those for EBS and PD) also release the underlying resources upon deleting the PersistentVolumes.

What's next

- Learn more about [scaling a StatefulSet](#).
- Learn more about [debugging a StatefulSet](#).
- Learn more about [deleting a StatefulSet](#).
- Learn more about [force deleting StatefulSet Pods](#).
- Look in the [Helm Charts repository](#) for other stateful application examples.

Scale a StatefulSet

This task shows how to scale a StatefulSet. Scaling a StatefulSet refers to increasing or decreasing the number of replicas.

Before you begin

- StatefulSets are only available in Kubernetes version 1.5 or later. To check your version of Kubernetes, run `kubectl version`.
- Not all stateful applications scale nicely. If you are unsure about whether to scale your StatefulSets, see [StatefulSet concepts](#) or [StatefulSet tutorial](#) for further information.
- You should perform scaling only when you are confident that your stateful application cluster is completely healthy.

Scaling StatefulSets

Use kubectl to scale StatefulSets

First, find the StatefulSet you want to scale.

```
kubectl get statefulsets <stateful-set-name>
```

Change the number of replicas of your StatefulSet:

```
kubectl scale statefulsets <stateful-set-name> --replicas=<new-replicas>
```

Make in-place updates on your StatefulSets

Alternatively, you can do [in-place updates](#) on your StatefulSets.

If your StatefulSet was initially created with `kubectl apply`, update `.spec.replicas` of the StatefulSet manifests, and then do a `kubectl apply`:

```
kubectl apply -f <stateful-set-file-updated>
```

Otherwise, edit that field with `kubectl edit`:

```
kubectl edit statefulsets <stateful-set-name>
```

Or use `kubectl patch`:

```
kubectl patch statefulsets <stateful-set-name> -p '{"spec":{"replicas":<new-replicas>}}'
```

Troubleshooting

Scaling down does not work right

You cannot scale down a StatefulSet when any of the stateful Pods it manages is unhealthy. Scaling down only takes place after those stateful Pods become running and ready.

If `spec.replicas > 1`, Kubernetes cannot determine the reason for an unhealthy Pod. It might be the result of a permanent fault or of a transient fault. A transient fault can be caused by a restart required by upgrading or maintenance.

If the Pod is unhealthy due to a permanent fault, scaling without correcting the fault may lead to a state where the StatefulSet membership drops below a certain minimum number of replicas that are needed to function correctly. This may cause your StatefulSet to become unavailable.

If the Pod is unhealthy due to a transient fault and the Pod might become available again, the transient error may interfere with your scale-up or scale-down operation. Some distributed databases have issues when nodes join and leave at the same time. It is better to reason about scaling operations at the application level in these cases, and perform scaling only when you are sure that your stateful application cluster is completely healthy.

What's next

- Learn more about [deleting a StatefulSet](#).

Delete a StatefulSet

This task shows you how to delete a [StatefulSet](#).

Before you begin

- This task assumes you have an application running on your cluster represented by a StatefulSet.

Deleting a StatefulSet

You can delete a StatefulSet in the same way you delete other resources in Kubernetes: use the `kubectl delete` command, and specify the StatefulSet either by file or by name.

```
kubectl delete -f <file.yaml>
```

```
kubectl delete statefulsets <statefulset-name>
```

You may need to delete the associated headless service separately after the StatefulSet itself is deleted.

```
kubectl delete service <service-name>
```

When deleting a StatefulSet through `kubectl`, the StatefulSet scales down to 0. All Pods that are part of this workload are also deleted. If you want to delete only the StatefulSet and not the Pods, use `--cascade=orphan`. For example:

```
kubectl delete -f <file.yaml> --cascade=orphan
```

By passing `--cascade=orphan` to `kubectl delete`, the Pods managed by the StatefulSet are left behind even after the StatefulSet object itself is deleted. If the pods have a label `app.kubernetes.io/name=MyApp`, you can then delete them as follows:

```
kubectl delete pods -l app.kubernetes.io/name=MyApp
```

Persistent Volumes

Deleting the Pods in a StatefulSet will not delete the associated volumes. This is to ensure that you have the chance to copy data off the volume before deleting it. Deleting the PVC after the pods have terminated might trigger deletion of the backing Persistent Volumes depending on the storage class and reclaim policy. You should never assume ability to access a volume after claim deletion.

Note:

Use caution when deleting a PVC, as it may lead to data loss.

Complete deletion of a StatefulSet

To delete everything in a StatefulSet, including the associated pods, you can run a series of commands similar to the following:

```
grace=$(kubectl get pods <stateful-set-pod> --template '{{.spec.terminationGracePeriodSeconds}}')
kubectl delete statefulset -l app.kubernetes.io/name=MyApp
sleep $grace
kubectl delete pvc -l app.kubernetes.io/name=MyApp
```

In the example above, the Pods have the label `app.kubernetes.io/name=MyApp`; substitute your own label as appropriate.

Force deletion of StatefulSet pods

If you find that some pods in your StatefulSet are stuck in the 'Terminating' or 'Unknown' states for an extended period of time, you may need to manually intervene to forcefully delete the pods from the apiserver. This is a potentially dangerous task. Refer to [Force Delete StatefulSet Pods](#) for details.

What's next

Learn more about [force deleting StatefulSet Pods](#).

Force Delete StatefulSet Pods

This page shows how to delete Pods which are part of a [stateful set](#), and explains the considerations to keep in mind when doing so.

Before you begin

- This is a fairly advanced task and has the potential to violate some of the properties inherent to StatefulSet.
- Before proceeding, make yourself familiar with the considerations enumerated below.

StatefulSet considerations

In normal operation of a StatefulSet, there is **never** a need to force delete a StatefulSet Pod. The [StatefulSet controller](#) is responsible for creating, scaling and deleting members of the StatefulSet. It tries to ensure that the specified number of Pods from ordinal 0 through N-1 are alive and ready. StatefulSet ensures that, at any time, there is at most one Pod with a given identity running in a cluster. This is referred to as *at most one* semantics provided by a StatefulSet.

Manual force deletion should be undertaken with caution, as it has the potential to violate the at most one semantics inherent to StatefulSet. StatefulSets may be used to run distributed and clustered applications which have a need for a stable network identity and stable storage. These applications often have configuration which relies on an ensemble of a fixed number of members with fixed identities. Having multiple members with the same identity can be disastrous and may lead to data loss (e.g. split brain scenario in quorum-based systems).

Delete Pods

You can perform a graceful pod deletion with the following command:

```
kubectl delete pods <pod>
```

For the above to lead to graceful termination, the Pod **must not** specify a `pod.Spec.TerminationGracePeriodSeconds` of 0. The practice of setting a `pod.Spec.TerminationGracePeriodSeconds` of 0 seconds is unsafe and strongly discouraged for StatefulSet Pods. Graceful deletion is safe and will ensure that the Pod [shuts down gracefully](#) before the kubelet deletes the name from the apiserver.

A Pod is not deleted automatically when a node is unreachable. The Pods running on an unreachable Node enter the 'Terminating' or 'Unknown' state after a [timeout](#). Pods may also enter these states when the user attempts graceful deletion of a Pod on an unreachable Node. The only ways in which a Pod in such a state can be removed from the apiserver are as follows:

- The Node object is deleted (either by you, or by the [Node Controller](#)).
- The kubelet on the unresponsive Node starts responding, kills the Pod and removes the entry from the apiserver.
- Force deletion of the Pod by the user.

The recommended best practice is to use the first or second approach. If a Node is confirmed to be dead (e.g. permanently disconnected from the network, powered down, etc), then delete the Node object. If the Node is suffering from a network partition, then try to resolve this or wait for it to resolve. When the partition heals, the kubelet will complete the deletion of the Pod and free up its name in the apiserver.

Normally, the system completes the deletion once the Pod is no longer running on a Node, or the Node is deleted by an administrator. You may override this by force deleting the Pod.

Force Deletion

Force deletions **do not** wait for confirmation from the kubelet that the Pod has been terminated. Irrespective of whether a force deletion is successful in killing a Pod, it will immediately free up the name from the apiserver. This would let the StatefulSet controller create a replacement Pod with that same identity; this can lead to the duplication of a still-running Pod, and if said Pod can still communicate with the other members of the StatefulSet, will violate the at most one semantics that StatefulSet is designed to guarantee.

When you force delete a StatefulSet pod, you are asserting that the Pod in question will never again make contact with other Pods in the StatefulSet and its name can be safely freed up for a replacement to be created.

If you want to delete a Pod forcibly using kubectl version ≥ 1.5 , do the following:

```
kubectl delete pods <pod> --grace-period=0 --force
```

If you're using any version of kubectl ≤ 1.4 , you should omit the `--force` option and use:

```
kubectl delete pods <pod> --grace-period=0
```

If even after these commands the pod is stuck on Unknown state, use the following command to remove the pod from the cluster:

```
kubectl patch pod <pod> -p '{"metadata":{"finalizers":null}}'
```

Always perform force deletion of StatefulSet Pods carefully and with complete knowledge of the risks involved.

What's next

Learn more about [debugging a StatefulSet](#).

Horizontal Pod Autoscaling

In Kubernetes, a *HorizontalPodAutoscaler* automatically updates a workload resource (such as a [Deployment](#) or [StatefulSet](#)), with the aim of automatically scaling the workload to match demand.

Horizontal scaling means that the response to increased load is to deploy more [Pods](#). This is different from *vertical* scaling, which for Kubernetes would mean assigning more resources (for example: memory or CPU) to the Pods that are already running for the workload.

If the load decreases, and the number of Pods is above the configured minimum, the HorizontalPodAutoscaler instructs the workload resource (the Deployment, StatefulSet, or other similar resource) to scale back down.

Horizontal pod autoscaling does not apply to objects that can't be scaled (for example: a [DaemonSet](#).)

The HorizontalPodAutoscaler is implemented as a Kubernetes API resource and a [controller](#). The resource determines the behavior of the controller. The horizontal pod autoscaling controller, running within the Kubernetes [control plane](#), periodically adjusts the desired scale of its target (for example, a Deployment) to match observed metrics such as average CPU utilization, average memory utilization, or any other custom metric you specify.

There is [walkthrough example](#) of using horizontal pod autoscaling.

How does a HorizontalPodAutoscaler work?

```
graph BT
    hpa[Horizontal Pod Autoscaler] --> scale[Scale]
    subgraph rc [RC / Deployment]
        scale --> pod1[Pod 1]
        scale --> pod2[Pod 2]
        scale --> pod3[Pod N]
    end
    classDef hpa fill:#D5A6BD,stroke:#1E1E1D,stroke-width:1px,color:#1E1E1D;
    classDef rc fill:#F9CB9C,stroke:#1E1E1D,stroke-width:1px,color:#1E1E1D;
    classDef scale fill:#B6D7A8,stroke:#1E1E1D,stroke-width:1px,color:#1E1E1D;
    classDef pod fill:#9FC5E8,stroke:#1E1E1D,stroke-width:1px,color:#1E1E1D;
    class hpa hpa;
    class rc rc;
    class scale scale;
    class pod1,pod2,pod3 pod;
```

JavaScript must be [enabled](#) to view this content

Figure 1. HorizontalPodAutoscaler controls the scale of a Deployment and its ReplicaSet

Kubernetes implements horizontal pod autoscaling as a control loop that runs intermittently (it is not a continuous process). The interval is set by the `--horizontal-pod-autoscaler-sync-period` parameter to the [kube-controller-manager](#) (and the default interval is 15 seconds).

Once during each period, the controller manager queries the resource utilization against the metrics specified in each HorizontalPodAutoscaler definition. The controller manager finds the target resource defined by the `scaleTargetRef`, then selects the pods based on the target resource's `.spec.selector` labels, and obtains the metrics from either the resource metrics API (for per-pod resource metrics), or the custom metrics API (for all other metrics).

- For per-pod resource metrics (like CPU), the controller fetches the metrics from the resource metrics API for each Pod targeted by the HorizontalPodAutoscaler. Then, if a target utilization value is set, the controller calculates the utilization value as a percentage of the equivalent [resource request](#) on the containers in each Pod. If a target raw value is set, the raw metric values are used directly. The controller then takes the mean of the utilization or the raw value (depending on the type of target specified) across all targeted Pods, and produces a ratio used to scale the number of desired replicas.

Please note that if some of the Pod's containers do not have the relevant resource request set, CPU utilization for the Pod will not be defined and the autoscaler will not take any action for that metric. See the [algorithm details](#) section below for more information about how the autoscaling algorithm works.

- For per-pod custom metrics, the controller functions similarly to per-pod resource metrics, except that it works with raw values, not utilization values.
- For object metrics and external metrics, a single metric is fetched, which describes the object in question. This metric is compared to the target value, to produce a ratio as above. In the autoscaling/v2 API version, this value can optionally be divided by the number of Pods before the comparison is made.

The common use for HorizontalPodAutoscaler is to configure it to fetch metrics from [aggregated APIs](#) (metrics.k8s.io, custom.metrics.k8s.io, or external.metrics.k8s.io). The metrics.k8s.io API is usually provided by an add-on named Metrics Server, which needs to be launched separately. For more information about resource metrics, see [Metrics Server](#).

[Support for metrics APIs](#) explains the stability guarantees and support status for these different APIs.

The HorizontalPodAutoscaler controller accesses corresponding workload resources that support scaling (such as Deployments and StatefulSet). These resources each have a subresource named scale, an interface that allows you to dynamically set the number of replicas and examine each of their current states. For general information about subresources in the Kubernetes API, see [Kubernetes API Concepts](#).

Algorithm details

From the most basic perspective, the HorizontalPodAutoscaler controller operates on the ratio between desired metric value and current metric value:

```
desiredReplicas = ceil[currentReplicas * ( currentMetricValue / desiredMetricValue )]
```

For example, if the current metric value is 200m, and the desired value is 100m, the number of replicas will be doubled, since $200.0 / 100.0 == 2.0$. If the current value is instead 50m, you'll halve the number of replicas, since $50.0 / 100.0 == 0.5$. The control plane skips any scaling action if the ratio is sufficiently close to 1.0 (within a globally-configurable tolerance, 0.1 by default).

When a targetAverageValue or targetAverageUtilization is specified, the currentMetricValue is computed by taking the average of the given metric across all Pods in the HorizontalPodAutoscaler's scale target.

Before checking the tolerance and deciding on the final values, the control plane also considers whether any metrics are missing, and how many Pods are [Ready](#). All Pods with a deletion timestamp set (objects with a deletion timestamp are in the process of being shut down / removed) are ignored, and all failed Pods are discarded.

If a particular Pod is missing metrics, it is set aside for later; Pods with missing metrics will be used to adjust the final scaling amount.

When scaling on CPU, if any pod has yet to become ready (it's still initializing, or possibly is unhealthy) or the most recent metric point for the pod was before it became ready, that pod is set aside as well.

Due to technical constraints, the HorizontalPodAutoscaler controller cannot exactly determine the first time a pod becomes ready when determining whether to set aside certain CPU metrics. Instead, it considers a Pod "not yet ready" if it's unready and transitioned to ready within a

short, configurable window of time since it started. This value is configured with the `--horizontal-pod-autoscaler-initial-readiness-delay` flag, and its default is 30 seconds. Once a pod has become ready, it considers any transition to ready to be the first if it occurred within a longer, configurable time since it started. This value is configured with the `--horizontal-pod-autoscaler-cpu-initialization-period` flag, and its default is 5 minutes.

The $\text{currentMetricValue} / \text{desiredMetricValue}$ base scale ratio is then calculated using the remaining pods not set aside or discarded from above.

If there were any missing metrics, the control plane recomputes the average more conservatively, assuming those pods were consuming 100% of the desired value in case of a scale down, and 0% in case of a scale up. This dampens the magnitude of any potential scale.

Furthermore, if any not-yet-ready pods were present, and the workload would have scaled up without factoring in missing metrics or not-yet-ready pods, the controller conservatively assumes that the not-yet-ready pods are consuming 0% of the desired metric, further dampening the magnitude of a scale up.

After factoring in the not-yet-ready pods and missing metrics, the controller recalculates the usage ratio. If the new ratio reverses the scale direction, or is within the tolerance, the controller doesn't take any scaling action. In other cases, the new ratio is used to decide any change to the number of Pods.

Note that the *original* value for the average utilization is reported back via the HorizontalPodAutoscaler status, without factoring in the not-yet-ready pods or missing metrics, even when the new usage ratio is used.

If multiple metrics are specified in a HorizontalPodAutoscaler, this calculation is done for each metric, and then the largest of the desired replica counts is chosen. If any of these metrics cannot be converted into a desired replica count (e.g. due to an error fetching the metrics from the metrics APIs) and a scale down is suggested by the metrics which can be fetched, scaling is skipped. This means that the HPA is still capable of scaling up if one or more metrics give a `desiredReplicas` greater than the current value.

Finally, right before HPA scales the target, the scale recommendation is recorded. The controller considers all recommendations within a configurable window choosing the highest recommendation from within that window. This value can be configured using the `--horizontal-pod-autoscaler-downscale-stabilization` flag, which defaults to 5 minutes. This means that scaledowns will occur gradually, smoothing out the impact of rapidly fluctuating metric values.

API Object

The Horizontal Pod Autoscaler is an API resource in the Kubernetes autoscaling API group. The current stable version can be found in the `autoscaling/v2` API version which includes support for scaling on memory and custom metrics. The new fields introduced in `autoscaling/v2` are preserved as annotations when working with `autoscaling/v1`.

When you create a HorizontalPodAutoscaler API object, make sure the name specified is a valid [DNS subdomain name](#). More details about the API object can be found at [HorizontalPodAutoscaler Object](#).

Stability of workload scale

When managing the scale of a group of replicas using the HorizontalPodAutoscaler, it is possible that the number of replicas keeps fluctuating frequently due to the dynamic nature of the metrics evaluated. This is sometimes referred to as *thrashing*, or *flapping*. It's similar to the concept of *hysteresis* in cybernetics.

Autoscaling during rolling update

Kubernetes lets you perform a rolling update on a Deployment. In that case, the Deployment manages the underlying ReplicaSets for you. When you configure autoscaling for a Deployment, you bind a HorizontalPodAutoscaler to a single Deployment. The HorizontalPodAutoscaler manages the replicas field of the Deployment. The deployment controller is responsible for setting the replicas of the underlying ReplicaSets so that they add up to a suitable number during the rollout and also afterwards.

If you perform a rolling update of a StatefulSet that has an autoscaled number of replicas, the StatefulSet directly manages its set of Pods (there is no intermediate resource similar to ReplicaSet).

Support for resource metrics

Any HPA target can be scaled based on the resource usage of the pods in the scaling target. When defining the pod specification the resource requests like cpu and memory should be specified. This is used to determine the resource utilization and used by the HPA controller to scale the target up or down. To use resource utilization based scaling specify a metric source like this:

```
type: Resource
resource:
  name: cpu
  target:
    type: Utilization
    averageUtilization: 60
```

With this metric the HPA controller will keep the average utilization of the pods in the scaling target at 60%. Utilization is the ratio between the current usage of resource to the requested resources of the pod. See [Algorithm](#) for more details about how the utilization is calculated and averaged.

Note:

Since the resource usages of all the containers are summed up the total pod utilization may not accurately represent the individual container resource usage. This could lead to situations where a single container might be running with high usage and the HPA will not scale out because the overall pod usage is still within acceptable limits.

Container resource metrics

FEATURE STATE: Kubernetes v1.30 [stable]

The HorizontalPodAutoscaler API also supports a container metric source where the HPA can track the resource usage of individual containers across a set of Pods, in order to scale the target resource. This lets you configure scaling thresholds for the containers that matter most in a particular Pod. For example, if you have a web application and a sidecar container that provides logging, you can scale based on the resource use of the web application, ignoring the sidecar container and its resource use.

If you revise the target resource to have a new Pod specification with a different set of containers, you should revise the HPA spec if that newly added container should also be used for scaling. If the specified container in the metric source is not present or only present in a subset of the pods then those pods are ignored and the recommendation is recalculated. See [Algorithm](#) for more details about the calculation. To use container resources for autoscaling define a metric source as follows:

```
type: ContainerResource
containerResource:
  name: cpu
  container: application
  target:
    type: Utilization
    averageUtilization: 60
```

In the above example the HPA controller scales the target such that the average utilization of the cpu in the application container of all the pods is 60%.

Note:

If you change the name of a container that a HorizontalPodAutoscaler is tracking, you can make that change in a specific order to ensure scaling remains available and effective whilst the change is being applied. Before you update the resource that defines the container (such as a Deployment), you should update the associated HPA to track both the new and old container names. This way, the HPA is able to calculate a scaling recommendation throughout the update process.

Once you have rolled out the container name change to the workload resource, tidy up by removing the old container name from the HPA specification.

Scaling on custom metrics

FEATURE STATE: Kubernetes v1.23 [stable]

(the autoscaling/v2beta2 API version previously provided this ability as a beta feature)

Provided that you use the autoscaling/v2 API version, you can configure a HorizontalPodAutoscaler to scale based on a custom metric (that is not built in to Kubernetes or any Kubernetes component). The HorizontalPodAutoscaler controller then queries for these custom metrics from the Kubernetes API.

See [Support for metrics APIs](#) for the requirements.

Scaling on multiple metrics

FEATURE STATE: Kubernetes v1.23 [stable]

(the autoscaling/v2beta2 API version previously provided this ability as a beta feature)

Provided that you use the autoscaling/v2 API version, you can specify multiple metrics for a HorizontalPodAutoscaler to scale on. Then, the HorizontalPodAutoscaler controller evaluates each metric, and proposes a new scale based on that metric. The HorizontalPodAutoscaler takes the maximum scale recommended for each metric and sets the workload to that size (provided that this isn't larger than the overall maximum that you configured).

Support for metrics APIs

By default, the HorizontalPodAutoscaler controller retrieves metrics from a series of APIs. In order for it to access these APIs, cluster administrators must ensure that:

- The [API aggregation layer](#) is enabled.
- The corresponding APIs are registered:
 - For resource metrics, this is the metrics.k8s.io [API](#), generally provided by [metrics-server](#). It can be launched as a cluster add-on.
 - For custom metrics, this is the custom.metrics.k8s.io [API](#). It's provided by "adapter" API servers provided by metrics solution vendors. Check with your metrics pipeline to see if there is a Kubernetes metrics adapter available.
 - For external metrics, this is the external.metrics.k8s.io [API](#). It may be provided by the custom metrics adapters provided above.

For more information on these different metrics paths and how they differ please see the relevant design proposals for [the HPA V2](#), [custom.metrics.k8s.io](#) and [external.metrics.k8s.io](#).

For examples of how to use them see [the walkthrough for using custom metrics](#) and [the walkthrough for using external metrics](#).

Configurable scaling behavior

FEATURE STATE: Kubernetes v1.23 [stable]

(the autoscaling/v2beta2 API version previously provided this ability as a beta feature)

If you use the v2 HorizontalPodAutoscaler API, you can use the behavior field (see the [API reference](#)) to configure separate scale-up and scale-down behaviors. You specify these behaviours by setting scaleUp and / or scaleDown under the behavior field.

You can specify a *stabilization window* that prevents [flapping](#) the replica count for a scaling target. Scaling policies also let you control the rate of change of replicas while scaling.

Scaling policies

One or more scaling policies can be specified in the `behavior` section of the spec. When multiple policies are specified the policy which allows the highest amount of change is the policy which is selected by default. The following example shows this behavior while scaling down:

```
behavior:
  scaleDown:
    policies:
      - type: Pods
        value: 4
        periodSeconds: 60
      - type: Percent
        value: 10
        periodSeconds: 60
```

`periodSeconds` indicates the length of time in the past for which the policy must hold true. The maximum value that you can set for `periodSeconds` is 1800 (half an hour). The first policy (*Pods*) allows at most 4 replicas to be scaled down in one minute. The second policy (*Percent*) allows at most 10% of the current replicas to be scaled down in one minute.

Since by default the policy which allows the highest amount of change is selected, the second policy will only be used when the number of pod replicas is more than 40. With 40 or less replicas, the first policy will be applied. For instance if there are 80 replicas and the target has to be scaled down to 10 replicas then during the first step 8 replicas will be reduced. In the next iteration when the number of replicas is 72, 10% of the pods is 7.2 but the number is rounded up to 8. On each loop of the autoscaler controller the number of pods to be change is re-calculated based on the number of current replicas. When the number of replicas falls below 40 the first policy (*Pods*) is applied and 4 replicas will be reduced at a time.

The policy selection can be changed by specifying the `selectPolicy` field for a scaling direction. By setting the value to `Min` which would select the policy which allows the smallest change in the replica count. Setting the value to `Disabled` completely disables scaling in that direction.

Stabilization window

The stabilization window is used to restrict the [flapping](#) of replica count when the metrics used for scaling keep fluctuating. The autoscaling algorithm uses this window to infer a previous desired state and avoid unwanted changes to workload scale.

For example, in the following example snippet, a stabilization window is specified for `scaleDown`.

```
behavior:
  scaleDown:
    stabilizationWindowSeconds: 300
```

When the metrics indicate that the target should be scaled down the algorithm looks into previously computed desired states, and uses the highest value from the specified interval. In the above example, all desired states from the past 5 minutes will be considered.

This approximates a rolling maximum, and avoids having the scaling algorithm frequently remove Pods only to trigger recreating an equivalent Pod just moments later.

Default Behavior

To use the custom scaling not all fields have to be specified. Only values which need to be customized can be specified. These custom values are merged with default values. The default values match the existing behavior in the HPA algorithm.

```
behavior:
  scaleDown:
    stabilizationWindowSeconds: 300
  policies:
    - type: Percent
      value: 100
      periodSeconds: 15
  scaleUp:
    stabilizationWindowSeconds: 0
  policies:
    - type: Percent
      value: 100
      periodSeconds: 15
    - type: Pods
      value: 4
      periodSeconds: 15
  selectPolicy: Max
```

For scaling down the stabilization window is 300 seconds (or the value of the `--horizontal-pod-autoscaler-downscale-stabilization` flag if provided). There is only a single policy for scaling down which allows a 100% of the currently running replicas to be removed which means the scaling target can be scaled down to the minimum allowed replicas. For scaling up there is no stabilization window. When the metrics indicate that the target should be scaled up the target is scaled up immediately. There are 2 policies where 4 pods or a 100% of the currently running replicas may at most be added every 15 seconds till the HPA reaches its steady state.

Example: change downscale stabilization window

To provide a custom downscale stabilization window of 1 minute, the following behavior would be added to the HPA:

```
behavior:
  scaleDown:
    stabilizationWindowSeconds: 60
```

Example: limit scale down rate

To limit the rate at which pods are removed by the HPA to 10% per minute, the following behavior would be added to the HPA:

```
behavior:
  scaleDown:
    policies:
      - type: Percent
        value: 10
        periodSeconds: 60
```

To ensure that no more than 5 Pods are removed per minute, you can add a second scale-down policy with a fixed size of 5, and set `selectPolicy` to `minimum`. Setting `selectPolicy` to `Min` means that the autoscaler chooses the policy that affects the smallest number of Pods:

```
behavior:
  scaleDown:
    policies:
      - type: Percent
        value: 10
        periodSeconds: 60
      - type: Pods
        value: 5
        periodSeconds: 60
    selectPolicy: Min
```

Example: disable scale down

The `selectPolicy` value of `Disabled` turns off scaling the given direction. So to prevent downscaling the following policy would be used:

```
behavior:
  scaleDown:
    selectPolicy: Disabled
```

Support for HorizontalPodAutoscaler in kubectl

`HorizontalPodAutoscaler`, like every API resource, is supported in a standard way by `kubectl`. You can create a new autoscaler using `kubectl create` command. You can list autoscalers by `kubectl get hpa` or get detailed description by `kubectl describe hpa`. Finally, you can delete an autoscaler using `kubectl delete hpa`.

In addition, there is a special `kubectl autoscale` command for creating a `HorizontalPodAutoscaler` object. For instance, executing `kubectl autoscale rs foo --min=2 --max=5 --cpu-percent=80` will create an autoscaler for `ReplicaSet foo`, with target CPU utilization set to 80% and the number of replicas between 2 and 5.

Implicit maintenance-mode deactivation

You can implicitly deactivate the HPA for a target without the need to change the HPA configuration itself. If the target's desired replica count is set to 0, and the HPA's minimum replica count is greater than 0, the HPA stops adjusting the target (and sets the `ScalingActiveCondition` on itself to `false`) until you reactivate it by manually adjusting the target's desired replica count or HPA's minimum replica count.

Migrating Deployments and StatefulSets to horizontal autoscaling

When an HPA is enabled, it is recommended that the value of `spec.replicas` of the `Deployment` and / or `StatefulSet` be removed from their [manifest\(s\)](#). If this isn't done, any time a change to that object is applied, for example via `kubectl apply -f deployment.yaml`, this will instruct Kubernetes to scale the current number of Pods to the value of the `spec.replicas` key. This may not be desired and could be troublesome when an HPA is active.

Keep in mind that the removal of `spec.replicas` may incur a one-time degradation of Pod counts as the default value of this key is 1 (reference [Deployment Replicas](#)). Upon the update, all Pods except 1 will begin their termination procedures. Any deployment application afterwards will behave as normal and respect a rolling update configuration as desired. You can avoid this degradation by choosing one of the following two methods based on how you are modifying your deployments:

- [Client Side Apply \(this is the default\)](#)
- [Server Side Apply](#)

1. `kubectl apply edit-last-applied deployment/<deployment_name>`
2. In the editor, remove `spec.replicas`. When you save and exit the editor, `kubectl` applies the update. No changes to Pod counts happen at this step.
3. You can now remove `spec.replicas` from the manifest. If you use source code management, also commit your changes or take whatever other steps for revising the source code are appropriate for how you track updates.
4. From here on out you can run `kubectl apply -f deployment.yaml`

When using the [Server-Side Apply](#) you can follow the [transferring ownership](#) guidelines, which cover this exact use case.

What's next

If you configure autoscaling in your cluster, you may also want to consider using [cluster autoscaling](#) to ensure you are running the right number of nodes.

For more information on HorizontalPodAutoscaler:

- Read a [walkthrough example](#) for horizontal pod autoscaling.
- Read documentation for [kubectl autoscale](#).
- If you would like to write your own custom metrics adapter, check out the [boilerplate](#) to get started.
- Read the [API reference](#) for HorizontalPodAutoscaler.

HorizontalPodAutoscaler Walkthrough

A [HorizontalPodAutoscaler](#) (HPA for short) automatically updates a workload resource (such as a [Deployment](#) or [StatefulSet](#)), with the aim of automatically scaling the workload to match demand.

Horizontal scaling means that the response to increased load is to deploy more [Pods](#). This is different from *vertical* scaling, which for Kubernetes would mean assigning more resources (for example: memory or CPU) to the Pods that are already running for the workload.

If the load decreases, and the number of Pods is above the configured minimum, the HorizontalPodAutoscaler instructs the workload resource (the Deployment, StatefulSet, or other similar resource) to scale back down.

This document walks you through an example of enabling HorizontalPodAutoscaler to automatically manage scale for an example web app. This example workload is Apache httpd running some PHP code.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version 1.23. To check the version, enter kubectl version. If you're running an older release of Kubernetes, refer to the version of the documentation for that release (see [available documentation versions](#)).

To follow this walkthrough, you also need to use a cluster that has a [Metrics Server](#) deployed and configured. The Kubernetes Metrics Server collects resource metrics from the [kubelets](#) in your cluster, and exposes those metrics through the [Kubernetes API](#), using an [APIService](#) to add new kinds of resource that represent metric readings.

To learn how to deploy the Metrics Server, see the [metrics-server documentation](#).

If you are running [Minikube](#), run the following command to enable metrics-server:

```
minikube addons enable metrics-server
```

Run and expose php-apache server

To demonstrate a HorizontalPodAutoscaler, you will first start a Deployment that runs a container using the hpa-example image, and expose it as a [Service](#) using the following manifest:

[application/php-apache.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: php-apache
spec:
  selector:
    matchLabels:
      run: php-apache
  template:
    metadata:
      labels:
        run: php-apache
    spec:
      containers:
        - name: php-apache
          image: registry.k8s.io/hpa-example
          ports:
            - containerPort: 80
          resources:
            limits:
              cpu: 500m
```

```
    requests:
      cpu: 200m
---
apiVersion: v1
kind: Service
metadata:
  name: php-apache
  labels:
    run: php-apache
spec:
  ports:
    - port: 80
  selector:
    run: php-apache
```

To do so, run the following command:

```
kubectl apply -f https://k8s.io/examples/application/php-apache.yaml
```

```
deployment.apps/php-apache created
service/php-apache created
```

Create the HorizontalPodAutoscaler

Now that the server is running, create the autoscaler using kubectl. There is [kubectl autoscale](#) subcommand, part of kubectl, that helps you do this.

You will shortly run a command that creates a HorizontalPodAutoscaler that maintains between 1 and 10 replicas of the Pods controlled by the php-apache Deployment that you created in the first step of these instructions.

Roughly speaking, the HPA [controller](#) will increase and decrease the number of replicas (by updating the Deployment) to maintain an average CPU utilization across all Pods of 50%. The Deployment then updates the ReplicaSet - this is part of how all Deployments work in Kubernetes - and then the ReplicaSet either adds or removes Pods based on the change to its .spec.

Since each pod requests 200 milli-cores by kubectl run, this means an average CPU usage of 100 milli-cores. See [Algorithm details](#) for more details on the algorithm.

Create the HorizontalPodAutoscaler:

```
kubectl autoscale deployment php-apache --cpu-percent=50 --min=1 --max=10
```

```
horizontalpodautoscaler.autoscaling/php-apache autoscaled
```

You can check the current status of the newly-made HorizontalPodAutoscaler, by running:

```
# You can use "hpa" or "horizontalpodautoscaler"; either name works OK.
kubectl get hpa
```

The output is similar to:

| NAME | REFERENCE | TARGET | MINPODS | MAXPODS | REPLICAS | AGE |
|------------|-----------------------------|----------|---------|---------|----------|-----|
| php-apache | Deployment/php-apache/scale | 0% / 50% | 1 | 10 | 1 | 18s |

(if you see other HorizontalPodAutoscalers with different names, that means they already existed, and isn't usually a problem).

Please note that the current CPU consumption is 0% as there are no clients sending requests to the server (the TARGET column shows the average across all the Pods controlled by the corresponding deployment).

Increase the load

Next, see how the autoscaler reacts to increased load. To do this, you'll start a different Pod to act as a client. The container within the client Pod runs in an infinite loop, sending queries to the php-apache service.

```
# Run this in a separate terminal
# so that the load generation continues and you can carry on with the rest of the steps
kubectl run -i --tty load-generator --rm --image=busybox:1.28 --restart=Never -- /bin/sh -c "while sleep 0.01; do wget -q -O- http://php-apache; done"
```

Now run:

```
# type Ctrl+C to end the watch when you're ready
kubectl get hpa php-apache --watch
```

Within a minute or so, you should see the higher CPU load; for example:

| NAME | REFERENCE | TARGET | MINPODS | MAXPODS | REPLICAS | AGE |
|------------|-----------------------------|------------|---------|---------|----------|-----|
| php-apache | Deployment/php-apache/scale | 305% / 50% | 1 | 10 | 1 | 3m |

and then, more replicas. For example:

| NAME | REFERENCE | TARGET | MINPODS | MAXPODS | REPLICAS | AGE |
|------------|-----------------------------|------------|---------|---------|----------|-----|
| php-apache | Deployment/php-apache/scale | 305% / 50% | 1 | 10 | 7 | 3m |

Here, CPU consumption has increased to 305% of the request. As a result, the Deployment was resized to 7 replicas:

```
kubectl get deployment php-apache
```

You should see the replica count matching the figure from the HorizontalPodAutoscaler

| NAME | READY | UP-TO-DATE | AVAILABLE | AGE |
|------------|-------|------------|-----------|-----|
| php-apache | 7/7 | 7 | 7 | 19m |

Note:

It may take a few minutes to stabilize the number of replicas. Since the amount of load is not controlled in any way it may happen that the final number of replicas will differ from this example.

Stop generating load

To finish the example, stop sending the load.

In the terminal where you created the Pod that runs a busybox image, terminate the load generation by typing `<Ctrl> + C`.

Then verify the result state (after a minute or so):

```
# type Ctrl+C to end the watch when you're ready
kubectl get hpa php-apache --watch
```

The output is similar to:

| NAME | REFERENCE | TARGET | MINPODS | MAXPODS | REPLICAS | AGE |
|------------|-----------------------------|----------|---------|---------|----------|-----|
| php-apache | Deployment/php-apache/scale | 0% / 50% | 1 | 10 | 1 | 11m |

and the Deployment also shows that it has scaled down:

```
kubectl get deployment php-apache
```

| NAME | READY | UP-TO-DATE | AVAILABLE | AGE |
|------------|-------|------------|-----------|-----|
| php-apache | 1/1 | 1 | 1 | 27m |

Once CPU utilization dropped to 0, the HPA automatically scaled the number of replicas back down to 1.

Autoscaling the replicas may take a few minutes.

Autoscaling on multiple metrics and custom metrics

You can introduce additional metrics to use when autoscaling the php-apache Deployment by making use of the autoscaling/v2 API version.

First, get the YAML of your HorizontalPodAutoscaler in the autoscaling/v2 form:

```
kubectl get hpa php-apache -o yaml > /tmp/hpa-v2.yaml
```

Open the `/tmp/hpa-v2.yaml` file in an editor, and you should see YAML which looks like this:

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: php-apache
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: php-apache
  minReplicas: 1
  maxReplicas: 10
  metrics:
  - type: Resource
```

```
resource:
  name: cpu
  target:
    type: Utilization
    averageUtilization: 50
status:
  observedGeneration: 1
  lastScaleTime: <some-time>
  currentReplicas: 1
  desiredReplicas: 1
  currentMetrics:
  - type: Resource
    resource:
      name: cpu
      current:
        averageUtilization: 0
        averageValue: 0
```

Notice that the `targetCPUUtilizationPercentage` field has been replaced with an array called `metrics`. The CPU utilization metric is a *resource metric*, since it is represented as a percentage of a resource specified on pod containers. Notice that you can specify other resource metrics besides CPU. By default, the only other supported resource metric is memory. These resources do not change names from cluster to cluster, and should always be available, as long as the `metrics.k8s.io` API is available.

You can also specify resource metrics in terms of direct values, instead of as percentages of the requested value, by using a `target.type` of `AverageValue` instead of `Utilization`, and setting the corresponding `target.averageValue` field instead of the `target.averageUtilization`.

There are two other types of metrics, both of which are considered *custom metrics*: pod metrics and object metrics. These metrics may have names which are cluster specific, and require a more advanced cluster monitoring setup.

The first of these alternative metric types is *pod metrics*. These metrics describe Pods, and are averaged together across Pods and compared with a target value to determine the replica count. They work much like resource metrics, except that they *only* support a target type of `AverageValue`.

Pod metrics are specified using a metric block like this:

```
type: Pods
pods:
  metric:
    name: packets-per-second
  target:
    type: AverageValue
    averageValue: 1k
```

The second alternative metric type is *object metrics*. These metrics describe a different object in the same namespace, instead of describing Pods. The metrics are not necessarily fetched from the object; they only describe it. Object metrics support target types of both `Value` and `AverageValue`. With `Value`, the target is compared directly to the returned metric from the API. With `AverageValue`, the value returned from the custom metrics API is divided by the number

of Pods before being compared to the target. The following example is the YAML representation of the requests-per-second metric.

```
type: Object
object:
  metric:
    name: requests-per-second
  describedObject:
    apiVersion: networking.k8s.io/v1
    kind: Ingress
    name: main-route
target:
  type: Value
  value: 2k
```

If you provide multiple such metric blocks, the HorizontalPodAutoscaler will consider each metric in turn. The HorizontalPodAutoscaler will calculate proposed replica counts for each metric, and then choose the one with the highest replica count.

For example, if you had your monitoring system collecting metrics about network traffic, you could update the definition above using `kubectl edit` to look like this:

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: php-apache
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: php-apache
  minReplicas: 1
  maxReplicas: 10
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 50
  - type: Pods
    pods:
      metric:
        name: packets-per-second
      target:
        type: AverageValue
        averageValue: 1k
  - type: Object
    object:
      metric:
        name: requests-per-second
      describedObject:
        apiVersion: networking.k8s.io/v1
```

```

    kind: Ingress
    name: main-route
  target:
    type: Value
    value: 10k
status:
  observedGeneration: 1
  lastScaleTime: <some-time>
  currentReplicas: 1
  desiredReplicas: 1
  currentMetrics:
  - type: Resource
    resource:
      name: cpu
    current:
      averageUtilization: 0
      averageValue: 0
  - type: Object
    object:
      metric:
        name: requests-per-second
      describedObject:
        apiVersion: networking.k8s.io/v1
        kind: Ingress
        name: main-route
      current:
        value: 10k

```

Then, your HorizontalPodAutoscaler would attempt to ensure that each pod was consuming roughly 50% of its requested CPU, serving 1000 packets per second, and that all pods behind the main-route Ingress were serving a total of 10000 requests per second.

Autoscaling on more specific metrics

Many metrics pipelines allow you to describe metrics either by name or by a set of additional descriptors called *labels*. For all non-resource metric types (pod, object, and external, described below), you can specify an additional label selector which is passed to your metric pipeline. For instance, if you collect a metric `http_requests` with the verb label, you can specify the following metric block to scale only on GET requests:

```

type: Object
object:
  metric:
    name: http_requests
    selector: {matchLabels: {verb: GET}}

```

This selector uses the same syntax as the full Kubernetes label selectors. The monitoring pipeline determines how to collapse multiple series into a single value, if the name and selector match multiple series. The selector is additive, and cannot select metrics that describe objects that are **not** the target object (the target pods in the case of the Pods type, and the described object in the case of the Object type).

Autoscaling on metrics not related to Kubernetes objects

Applications running on Kubernetes may need to autoscale based on metrics that don't have an obvious relationship to any object in the Kubernetes cluster, such as metrics describing a hosted service with no direct correlation to Kubernetes namespaces. In Kubernetes 1.10 and later, you can address this use case with *external metrics*.

Using external metrics requires knowledge of your monitoring system; the setup is similar to that required when using custom metrics. External metrics allow you to autoscale your cluster based on any metric available in your monitoring system. Provide a metric block with a name and selector, as above, and use the External metric type instead of Object. If multiple time series are matched by the metricSelector, the sum of their values is used by the HorizontalPodAutoscaler. External metrics support both the Value and AverageValue target types, which function exactly the same as when you use the Object type.

For example if your application processes tasks from a hosted queue service, you could add the following section to your HorizontalPodAutoscaler manifest to specify that you need one worker per 30 outstanding tasks.

```
- type: External
  external:
    metric:
      name: queue_messages_ready
      selector:
        matchLabels:
          queue: "worker_tasks"
    target:
      type: AverageValue
      averageValue: 30
```

When possible, it's preferable to use the custom metric target types instead of external metrics, since it's easier for cluster administrators to secure the custom metrics API. The external metrics API potentially allows access to any metric, so cluster administrators should take care when exposing it.

Appendix: Horizontal Pod Autoscaler Status Conditions

When using the autoscaling/v2 form of the HorizontalPodAutoscaler, you will be able to see *status conditions* set by Kubernetes on the HorizontalPodAutoscaler. These status conditions indicate whether or not the HorizontalPodAutoscaler is able to scale, and whether or not it is currently restricted in any way.

The conditions appear in the status.conditions field. To see the conditions affecting a HorizontalPodAutoscaler, we can use `kubectl describe hpa`:

```
kubectl describe hpa cm-test
```

```
Name:          cm-test
Namespace:     prom
Labels:        <none>
Annotations:   <none>
CreationTimestamp:  Fri, 16 Jun 2017 18:09:22 +0000
Reference:     ReplicationController/cm-test
```



```

Metrics:                ( current / target )
"http_requests" on pods: 66m / 500m
Min replicas:           1
Max replicas:           4
ReplicationController pods: 1 current / 1 desired
Conditions:
  Type                Status Reason                Message
  ----                -
  AbleToScale         True   ReadyForNewScale    the last scale time was sufficiently old as to
warrant a new scale
  ScalingActive        True   ValidMetricFound    the HPA was able to successfully calculate a
replica count from pods metric http_requests
  ScalingLimited       False  DesiredWithinRange  the desired replica count is within the
acceptable range
Events:

```

For this HorizontalPodAutoscaler, you can see several conditions in a healthy state. The first, `AbleToScale`, indicates whether or not the HPA is able to fetch and update scales, as well as whether or not any backoff-related conditions would prevent scaling. The second, `ScalingActive`, indicates whether or not the HPA is enabled (i.e. the replica count of the target is not zero) and is able to calculate desired scales. When it is `False`, it generally indicates problems with fetching metrics. Finally, the last condition, `ScalingLimited`, indicates that the desired scale was capped by the maximum or minimum of the HorizontalPodAutoscaler. This is an indication that you may wish to raise or lower the minimum or maximum replica count constraints on your HorizontalPodAutoscaler.

Quantities

All metrics in the HorizontalPodAutoscaler and metrics APIs are specified using a special whole-number notation known in Kubernetes as a [quantity](#). For example, the quantity 10500m would be written as 10.5 in decimal notation. The metrics APIs will return whole numbers without a suffix when possible, and will generally return quantities in milli-units otherwise. This means you might see your metric value fluctuate between 1 and 1500m, or 1 and 1.5 when written in decimal notation.

Other possible scenarios

Creating the autoscaler declaratively

Instead of using `kubectl autoscale` command to create a HorizontalPodAutoscaler imperatively we can use the following manifest to create it declaratively:

[application/hpa/php-apache.yaml](#)

```

apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: php-apache
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment

```

```
name: php-apache
minReplicas: 1
maxReplicas: 10
metrics:
- type: Resource
  resource:
    name: cpu
    target:
      type: Utilization
      averageUtilization: 50
```

Then, create the autoscaler by executing the following command:

```
kubectl create -f https://k8s.io/examples/application/hpa/php-apache.yaml
```

```
horizontalpodautoscaler.autoscaling/php-apache created
```

Specifying a Disruption Budget for your Application

FEATURE STATE: Kubernetes v1.21 [stable]

This page shows how to limit the number of concurrent disruptions that your application experiences, allowing for higher availability while permitting the cluster administrator to manage the clusters nodes.

Before you begin

Your Kubernetes server must be at or later than version v1.21. To check the version, enter `kubectl version`.

- You are the owner of an application running on a Kubernetes cluster that requires high availability.
- You should know how to deploy [Replicated Stateless Applications](#) and/or [Replicated Stateful Applications](#).
- You should have read about [Pod Disruptions](#).
- You should confirm with your cluster owner or service provider that they respect Pod Disruption Budgets.

Protecting an Application with a PodDisruptionBudget

1. Identify what application you want to protect with a PodDisruptionBudget (PDB).
2. Think about how your application reacts to disruptions.
3. Create a PDB definition as a YAML file.
4. Create the PDB object from the YAML file.

Identify an Application to Protect

The most common use case when you want to protect an application specified by one of the built-in Kubernetes controllers:

- Deployment
- ReplicationController
- ReplicaSet
- StatefulSet

In this case, make a note of the controller's `.spec.selector`; the same selector goes into the PDBs `.spec.selector`.

From version 1.15 PDBs support custom controllers where the [scale subresource](#) is enabled.

You can also use PDBs with pods which are not controlled by one of the above controllers, or arbitrary groups of pods, but there are some restrictions, described in [Arbitrary workloads and arbitrary selectors](#).

Think about how your application reacts to disruptions

Decide how many instances can be down at the same time for a short period due to a voluntary disruption.

- Stateless frontends:
 - Concern: don't reduce serving capacity by more than 10%.
 - Solution: use PDB with `minAvailable 90%` for example.
- Single-instance Stateful Application:
 - Concern: do not terminate this application without talking to me.
 - Possible Solution 1: Do not use a PDB and tolerate occasional downtime.
 - Possible Solution 2: Set PDB with `maxUnavailable=0`. Have an understanding (outside of Kubernetes) that the cluster operator needs to consult you before termination. When the cluster operator contacts you, prepare for downtime, and then delete the PDB to indicate readiness for disruption. Recreate afterwards.
- Multiple-instance Stateful application such as Consul, ZooKeeper, or etcd:
 - Concern: Do not reduce number of instances below quorum, otherwise writes fail.
 - Possible Solution 1: set `maxUnavailable` to 1 (works with varying scale of application).
 - Possible Solution 2: set `minAvailable` to quorum-size (e.g. 3 when scale is 5). (Allows more disruptions at once).
- Restartable Batch Job:
 - Concern: Job needs to complete in case of voluntary disruption.
 - Possible solution: Do not create a PDB. The Job controller will create a replacement pod.

Rounding logic when specifying percentages

Values for `minAvailable` or `maxUnavailable` can be expressed as integers or as a percentage.

- When you specify an integer, it represents a number of Pods. For instance, if you set `minAvailable` to 10, then 10 Pods must always be available, even during a disruption.

- When you specify a percentage by setting the value to a string representation of a percentage (eg. "50%"), it represents a percentage of total Pods. For instance, if you set `minAvailable` to "50%", then at least 50% of the Pods remain available during a disruption.

When you specify the value as a percentage, it may not map to an exact number of Pods. For example, if you have 7 Pods and you set `minAvailable` to "50%", it's not immediately obvious whether that means 3 Pods or 4 Pods must be available. Kubernetes rounds up to the nearest integer, so in this case, 4 Pods must be available. When you specify the value `maxUnavailable` as a percentage, Kubernetes rounds up the number of Pods that may be disrupted. Thereby a disruption can exceed your defined `maxUnavailable` percentage. You can examine the [code](#) that controls this behavior.

Specifying a PodDisruptionBudget

A `PodDisruptionBudget` has three fields:

- A label selector `.spec.selector` to specify the set of pods to which it applies. This field is required.
- `.spec.minAvailable` which is a description of the number of pods from that set that must still be available after the eviction, even in the absence of the evicted pod. `minAvailable` can be either an absolute number or a percentage.
- `.spec.maxUnavailable` (available in Kubernetes 1.7 and higher) which is a description of the number of pods from that set that can be unavailable after the eviction. It can be either an absolute number or a percentage.

Note:

The behavior for an empty selector differs between the `policy/v1beta1` and `policy/v1` APIs for `PodDisruptionBudgets`. For `policy/v1beta1` an empty selector matches zero pods, while for `policy/v1` an empty selector matches every pod in the namespace.

You can specify only one of `maxUnavailable` and `minAvailable` in a single `PodDisruptionBudget`. `maxUnavailable` can only be used to control the eviction of pods that have an associated controller managing them. In the examples below, "desired replicas" is the scale of the controller managing the pods being selected by the `PodDisruptionBudget`.

Example 1: With a `minAvailable` of 5, evictions are allowed as long as they leave behind 5 or more [healthy](#) pods among those selected by the `PodDisruptionBudget`'s selector.

Example 2: With a `minAvailable` of 30%, evictions are allowed as long as at least 30% of the number of desired replicas are healthy.

Example 3: With a `maxUnavailable` of 5, evictions are allowed as long as there are at most 5 unhealthy replicas among the total number of desired replicas.

Example 4: With a `maxUnavailable` of 30%, evictions are allowed as long as the number of unhealthy replicas does not exceed 30% of the total number of desired replica rounded up to the nearest integer. If the total number of desired replicas is just one, that single replica is still allowed for disruption, leading to an effective unavailability of 100%.

In typical usage, a single budget would be used for a collection of pods managed by a controller—for example, the pods in a single `ReplicaSet` or `StatefulSet`.

Note:

A disruption budget does not truly guarantee that the specified number/percentage of pods will always be up. For example, a node that hosts a pod from the collection may fail when the collection is at the minimum size specified in the budget, thus bringing the number of available pods from the collection below the specified size. The budget can only protect against voluntary evictions, not all causes of unavailability.

If you set `maxUnavailable` to 0% or 0, or you set `minAvailable` to 100% or the number of replicas, you are requiring zero voluntary evictions. When you set zero voluntary evictions for a workload object such as `ReplicaSet`, then you cannot successfully drain a Node running one of those Pods. If you try to drain a Node where an unevictable Pod is running, the drain never completes. This is permitted as per the semantics of `PodDisruptionBudget`.

You can find examples of pod disruption budgets defined below. They match pods with the label `app: zookeeper`.

Example PDB Using `minAvailable`:

[policy/zookeeper-pod-disruption-budget-minavailable.yaml](#)

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: zk-pdb
spec:
  minAvailable: 2
  selector:
    matchLabels:
      app: zookeeper
```

Example PDB Using `maxUnavailable`:

[policy/zookeeper-pod-disruption-budget-maxunavailable.yaml](#)

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: zk-pdb
spec:
  maxUnavailable: 1
  selector:
    matchLabels:
      app: zookeeper
```

For example, if the above `zk-pdb` object selects the pods of a `StatefulSet` of size 3, both specifications have the exact same meaning. The use of `maxUnavailable` is recommended as it automatically responds to changes in the number of replicas of the corresponding controller.

Create the PDB object

You can create or update the PDB object using `kubectl`.

```
kubectl apply -f mypdb.yaml
```

Check the status of the PDB

Use `kubectl` to check that your PDB is created.

Assuming you don't actually have pods matching `app: zookeeper` in your namespace, then you'll see something like this:

```
kubectl get poddisruptionbudgets
```

| NAME | MIN AVAILABLE | MAX UNAVAILABLE | ALLOWED DISRUPTIONS | AGE |
|--------|---------------|-----------------|---------------------|-----|
| zk-pdb | 2 | N/A | 0 | 7s |

If there are matching pods (say, 3), then you would see something like this:

```
kubectl get poddisruptionbudgets
```

| NAME | MIN AVAILABLE | MAX UNAVAILABLE | ALLOWED DISRUPTIONS | AGE |
|--------|---------------|-----------------|---------------------|-----|
| zk-pdb | 2 | N/A | 1 | 7s |

The non-zero value for `ALLOWED DISRUPTIONS` means that the disruption controller has seen the pods, counted the matching pods, and updated the status of the PDB.

You can get more information about the status of a PDB with this command:

```
kubectl get poddisruptionbudgets zk-pdb -o yaml
```

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  annotations:
  ...
  creationTimestamp: "2020-03-04T04:22:56Z"
  generation: 1
  name: zk-pdb
  ...
status:
  currentHealthy: 3
  desiredHealthy: 2
  disruptionsAllowed: 1
  expectedPods: 3
  observedGeneration: 1
```

Healthiness of a Pod

The current implementation considers healthy pods, as pods that have `.status.conditions` item with `type="Ready"` and `status="True"`. These pods are tracked via `.status.currentHealthy` field in the PDB status.

Unhealthy Pod Eviction Policy

FEATURE STATE: Kubernetes v1.27 [beta]

Note:

This feature is enabled by default. You can disable it by disabling the `PDBUnhealthyPodEvictionPolicy` [feature gate](#) on the [API server](#).

`PodDisruptionBudget` guarding an application ensures that `.status.currentHealthy` number of pods does not fall below the number specified in `.status.desiredHealthy` by disallowing eviction of healthy pods. By using `.spec.unhealthyPodEvictionPolicy`, you can also define the criteria when unhealthy pods should be considered for eviction. The default behavior when no policy is specified corresponds to the `IfHealthyBudget` policy.

Policies:

IfHealthyBudget

Running pods (`.status.phase="Running"`), but not yet healthy can be evicted only if the guarded application is not disrupted (`.status.currentHealthy` is at least equal to `.status.desiredHealthy`).

This policy ensures that running pods of an already disrupted application have the best chance to become healthy. This has negative implications for draining nodes, which can be blocked by misbehaving applications that are guarded by a PDB. More specifically applications with pods in `CrashLoopBackOff` state (due to a bug or misconfiguration), or pods that are just failing to report the Ready condition.

AlwaysAllow

Running pods (`.status.phase="Running"`), but not yet healthy are considered disrupted and can be evicted regardless of whether the criteria in a PDB is met.

This means prospective running pods of a disrupted application might not get a chance to become healthy. By using this policy, cluster managers can easily evict misbehaving applications that are guarded by a PDB. More specifically applications with pods in `CrashLoopBackOff` state (due to a bug or misconfiguration), or pods that are just failing to report the Ready condition.

Note:

Pods in `Pending`, `Succeeded` or `Failed` phase are always considered for eviction.

Arbitrary workloads and arbitrary selectors

You can skip this section if you only use PDBs with the built-in workload resources (`Deployment`, `ReplicaSet`, `StatefulSet` and `ReplicationController`) or with [custom resources](#) that implement a scale [subresource](#), and where the PDB selector exactly matches the selector of the Pod's owning resource.

You can use a PDB with pods controlled by another resource, by an "operator", or bare pods, but with these restrictions:

- only `.spec.minAvailable` can be used, not `.spec.maxUnavailable`.
- only an integer value can be used with `.spec.minAvailable`, not a percentage.

It is not possible to use other availability configurations, because Kubernetes cannot derive a total number of pods without a supported owning resource.

You can use a selector which selects a subset or superset of the pods belonging to a workload resource. The eviction API will disallow eviction of any pod covered by multiple PDBs, so most users will want to avoid overlapping selectors. One reasonable use of overlapping PDBs is when pods are being transitioned from one PDB to another.

Accessing the Kubernetes API from a Pod

This guide demonstrates how to access the Kubernetes API from within a pod.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Accessing the API from within a Pod

When accessing the API from within a Pod, locating and authenticating to the API server are slightly different to the external client case.

The easiest way to use the Kubernetes API from a Pod is to use one of the official [client libraries](#). These libraries can automatically discover the API server and authenticate.

Using Official Client Libraries

From within a Pod, the recommended ways to connect to the Kubernetes API are:

- For a Go client, use the official [Go client library](#). The `rest.InClusterConfig()` function handles API host discovery and authentication automatically. See [an example here](#).
- For a Python client, use the official [Python client library](#). The `config.load_incluster_config()` function handles API host discovery and authentication automatically. See [an example here](#).
- There are a number of other libraries available, please refer to the [Client Libraries](#) page.

In each case, the service account credentials of the Pod are used to communicate securely with the API server.

Directly accessing the REST API

While running in a Pod, your container can create an HTTPS URL for the Kubernetes API server by fetching the `KUBERNETES_SERVICE_HOST` and `KUBERNETES_SERVICE_PORT_HTTPS` environment variables. The API server's in-cluster address is also published to a Service named `kubernetes` in the default namespace so that pods may reference `kubernetes.default.svc` as a DNS name for the local API server.

Note:

Kubernetes does not guarantee that the API server has a valid certificate for the hostname `kubernetes.default.svc`; however, the control plane **is** expected to present a valid certificate for the hostname or IP address that `$KUBERNETES_SERVICE_HOST` represents.

The recommended way to authenticate to the API server is with a [service account](#) credential. By default, a Pod is associated with a service account, and a credential (token) for that service account is placed into the filesystem tree of each container in that Pod, at `/var/run/secrets/kubernetes.io/serviceaccount/token`.

If available, a certificate bundle is placed into the filesystem tree of each container at `/var/run/secrets/kubernetes.io/serviceaccount/ca.crt`, and should be used to verify the serving certificate of the API server.

Finally, the default namespace to be used for namespaced API operations is placed in a file at `/var/run/secrets/kubernetes.io/serviceaccount/namespace` in each container.

Using kubectl proxy

If you would like to query the API without an official client library, you can run `kubectl proxy` as the [command](#) of a new sidecar container in the Pod. This way, `kubectl proxy` will authenticate to the API and expose it on the localhost interface of the Pod, so that other containers in the Pod can use it directly.

Without using a proxy

It is possible to avoid using the `kubectl proxy` by passing the authentication token directly to the API server. The internal certificate secures the connection.

```
# Point to the internal API server hostname
APISERVER=https://kubernetes.default.svc

# Path to ServiceAccount token
SERVICEACCOUNT=/var/run/secrets/kubernetes.io/serviceaccount

# Read this Pod's namespace
NAMESPACE=$(cat ${SERVICEACCOUNT}/namespace)

# Read the ServiceAccount bearer token
TOKEN=$(cat ${SERVICEACCOUNT}/token)

# Reference the internal certificate authority (CA)
CACERT=${SERVICEACCOUNT}/ca.crt
```

```
# Explore the API with TOKEN
```

```
curl --cacert ${CACERT} --header "Authorization: Bearer ${TOKEN}" -X GET ${APISERVER}/api
```

The output will be similar to this:

```
{
  "kind": "APIVersions",
  "versions": ["v1"],
  "serverAddressByClientCIDRs": [
    {
      "clientCIDR": "0.0.0.0/0",
      "serverAddress": "10.0.1.149:443"
    }
  ]
}
```

Run Jobs

Run Jobs using parallel processing.

[Running Automated Tasks with a CronJob](#)

[Coarse Parallel Processing Using a Work Queue](#)

[Fine Parallel Processing Using a Work Queue](#)

[Indexed Job for Parallel Processing with Static Work Assignment](#)

[Job with Pod-to-Pod Communication](#)

[Parallel Processing using Expansions](#)

[Handling retrieable and non-retrieable pod failures with Pod failure policy](#)

Running Automated Tasks with a CronJob

This page shows how to run automated tasks using Kubernetes [CronJob](#) object.

Before you begin

- You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:
 - [Killercodea](#)
 - [Play with Kubernetes](#)

Creating a CronJob

Cron jobs require a config file. Here is a manifest for a CronJob that runs a simple demonstration task every minute:

[application/job/cronjob.yaml](https://k8s.io/examples/application/job/cronjob.yaml)

```
apiVersion: batch/v1
kind: CronJob
metadata:
  name: hello
spec:
  schedule: "* * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: hello
              image: busybox:1.28
              imagePullPolicy: IfNotPresent
              command:
                - /bin/sh
                - -c
                - date; echo Hello from the Kubernetes cluster
          restartPolicy: OnFailure
```

Run the example CronJob by using this command:

```
kubectl create -f https://k8s.io/examples/application/job/cronjob.yaml
```

The output is similar to this:

```
cronjob.batch/hello created
```

After creating the cron job, get its status using this command:

```
kubectl get cronjob hello
```

The output is similar to this:

| NAME | SCHEDULE | SUSPEND | ACTIVE | LAST SCHEDULE | AGE |
|-------|-------------|---------|--------|---------------|-----|
| hello | */* * * * * | False | 0 | <none> | 10s |

As you can see from the results of the command, the cron job has not scheduled or run any jobs yet. Watch for the job to be created in around one minute:

```
kubectl get jobs --watch
```

The output is similar to this:

| NAME | COMPLETIONS | DURATION | AGE |
|------------------|-------------|----------|-----|
| hello-4111706356 | 0/1 | 0s | |

```
hello-4111706356 0/1      0s      0s
hello-4111706356 1/1      5s      5s
```

Now you've seen one running job scheduled by the "hello" cron job. You can stop watching the job and view the cron job again to see that it scheduled the job:

```
kubectrl get cronjob hello
```

The output is similar to this:

| NAME | SCHEDULE | SUSPEND | ACTIVE | LAST SCHEDULE | AGE |
|-------|--------------|---------|--------|---------------|-----|
| hello | * /1 * * * * | False | 0 | 50s | 75s |

You should see that the cron job hello successfully scheduled a job at the time specified in LAST SCHEDULE. There are currently 0 active jobs, meaning that the job has completed or failed.

Now, find the pods that the last scheduled job created and view the standard output of one of the pods.

Note:

The job name is different from the pod name.

```
# Replace "hello-4111706356" with the job name in your system
pods=$(kubectrl get pods --selector=job-name=hello-4111706356 --output=jsonpath={.items[*].metadata.name})
```

Show the pod log:

```
kubectrl logs $pods
```

The output is similar to this:

```
Fri Feb 22 11:02:09 UTC 2019
Hello from the Kubernetes cluster
```

Deleting a CronJob

When you don't need a cron job any more, delete it with `kubectrl delete cronjob <cronjob name>`:

```
kubectrl delete cronjob hello
```

Deleting the cron job removes all the jobs and pods it created and stops it from creating additional jobs. You can read more about removing jobs in [garbage collection](#).

Coarse Parallel Processing Using a Work Queue

In this example, you will run a Kubernetes Job with multiple parallel worker processes.

In this example, as each pod is created, it picks up one unit of work from a task queue, completes it, deletes it from the queue, and exits.

Here is an overview of the steps in this example:

1. **Start a message queue service.** In this example, you use RabbitMQ, but you could use another one. In practice you would set up a message queue service once and reuse it for many jobs.
2. **Create a queue, and fill it with messages.** Each message represents one task to be done. In this example, a message is an integer that we will do a lengthy computation on.
3. **Start a Job that works on tasks from the queue.** The Job starts several pods. Each pod takes one task from the message queue, processes it, and exits.

Before you begin

You should already be familiar with the basic, non-parallel, use of [Job](#).

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

You will need a container image registry where you can upload images to run in your cluster.

This task example also assumes that you have Docker installed locally.

Starting a message queue service

This example uses RabbitMQ, however, you can adapt the example to use another AMQP-type message service.

In practice you could set up a message queue service once in a cluster and reuse it for many jobs, as well as for long-running services.

Start RabbitMQ as follows:

```
# make a Service for the StatefulSet to use
kubectl create -f https://kubernetes.io/examples/application/job/rabbitmq/rabbitmq-
service.yaml
```

```
service "rabbitmq-service" created
```

```
kubectl create -f https://kubernetes.io/examples/application/job/rabbitmq/rabbitmq-
statefulset.yaml
```

```
statefulset "rabbitmq" created
```

Testing the message queue service

Now, we can experiment with accessing the message queue. We will create a temporary interactive pod, install some tools on it, and experiment with queues.

First create a temporary interactive Pod.

```
# Create a temporary interactive container
kubectl run -i --tty temp --image ubuntu:22.04
```

```
Waiting for pod default/temp-loe07 to be running, status is Pending, pod ready: false
... [ previous line repeats several times .. hit return when it stops ] ...
```

Note that your pod name and command prompt will be different.

Next install the `amqp-tools` so you can work with message queues. The next commands show what you need to run inside the interactive shell in that Pod:

```
apt-get update && apt-get install -y curl ca-certificates amqp-tools python3 dnsutils
```

Later, you will make a container image that includes these packages.

Next, you will check that you can discover the Service for RabbitMQ:

```
# Run these commands inside the Pod
# Note the rabbitmq-service has a DNS name, provided by Kubernetes:
nslookup rabbitmq-service
```

```
Server:      10.0.0.10
Address:     10.0.0.10#53
```

```
Name:  rabbitmq-service.default.svc.cluster.local
Address: 10.0.147.152
```

(the IP addresses will vary)

If the kube-dns addon is not set up correctly, the previous step may not work for you. You can also find the IP address for that Service in an environment variable:

```
# run this check inside the Pod
env | grep RABBITMQ_SERVICE | grep HOST
```

```
RABBITMQ_SERVICE_SERVICE_HOST=10.0.147.152
```

(the IP address will vary)

Next you will verify that you can create a queue, and publish and consume messages.

```
# Run these commands inside the Pod
# In the next line, rabbitmq-service is the hostname where the rabbitmq-service
# can be reached. 5672 is the standard port for rabbitmq.
export BROKER_URL=amqp://guest:guest@rabbitmq-service:5672
# If you could not resolve "rabbitmq-service" in the previous step,
# then use this command instead:
```

```
BROKER_URL=amqp://guest:guest@$RABBITMQ_SERVICE_SERVICE_HOST:5672
```

Now create a queue:

```
/usr/bin/amqp-declare-queue --url=$BROKER_URL -q foo -d
```

```
foo
```

Publish one message to the queue:

```
/usr/bin/amqp-publish --url=$BROKER_URL -r foo -p -b Hello
```

And get it back.

```
/usr/bin/amqp-consume --url=$BROKER_URL -q foo -c 1 cat && echo 1>&2
```

```
Hello
```

In the last command, the `amqp-consume` tool took one message (`-c 1`) from the queue, and passes that message to the standard input of an arbitrary command. In this case, the program `cat` prints out the characters read from standard input, and the `echo` adds a carriage return so the example is readable.

Fill the queue with tasks

Now, fill the queue with some simulated tasks. In this example, the tasks are strings to be printed.

In a practice, the content of the messages might be:

- names of files to that need to be processed
- extra flags to the program
- ranges of keys in a database table
- configuration parameters to a simulation
- frame numbers of a scene to be rendered

If there is large data that is needed in a read-only mode by all pods of the Job, you typically put that in a shared file system like NFS and mount that readonly on all the pods, or write the program in the pod so that it can natively read data from a cluster file system (for example: HDFS).

For this example, you will create the queue and fill it using the AMQP command line tools. In practice, you might write a program to fill the queue using an AMQP client library.

Run this on your computer, not in the Pod

```
/usr/bin/amqp-declare-queue --url=$BROKER_URL -q job1 -d
```

```
job1
```

Add items to the queue:

```
for f in apple banana cherry date fig grape lemon melon  
do
```

```
/usr/bin/amqp-publish --url=$BROKER_URL -r job1 -p -b $f
done
```

You added 8 messages to the queue.

Create a container image

Now you are ready to create an image that you will run as a Job.

The job will use the `amqp-consume` utility to read the message from the queue and run the actual work. Here is a very simple example program:

[application/job/rabbitmq/worker.py](#)

```
#!/usr/bin/env python

# Just prints standard out and sleeps for 10 seconds.
import sys
import time
print("Processing " + sys.stdin.readlines()[0])
time.sleep(10)
```

Give the script execution permission:

```
chmod +x worker.py
```

Now, build an image. Make a temporary directory, change to it, download the [Dockerfile](#), and [worker.py](#). In either case, build the image with this command:

```
docker build -t job-wq-1 .
```

For the [Docker Hub](#), tag your app image with your username and push to the Hub with the below commands. Replace `<username>` with your Hub username.

```
docker tag job-wq-1 <username>/job-wq-1
docker push <username>/job-wq-1
```

If you are using an alternative container image registry, tag the image and push it there instead.

Defining a Job

Here is a manifest for a Job. You'll need to make a copy of the Job manifest (call it `./job.yaml`), and edit the name of the container image to match the name you used.

[application/job/rabbitmq/job.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
  name: job-wq-1
spec:
  completions: 8
  parallelism: 2
```



```

template:
  metadata:
    name: job-wq-1
  spec:
    containers:
      - name: c
        image: gcr.io/<project>/job-wq-1
        env:
          - name: BROKER_URL
            value: amqp://guest:guest@rabbitmq-service:5672
          - name: QUEUE
            value: job1
        restartPolicy: OnFailure

```

In this example, each pod works on one item from the queue and then exits. So, the completion count of the Job corresponds to the number of work items done. That is why the example manifest has `.spec.completions` set to 8.

Running the Job

Now, run the Job:

```
# this assumes you downloaded and then edited the manifest already
kubectl apply -f ./job.yaml
```

You can wait for the Job to succeed, with a timeout:

```
# The check for condition name is case insensitive
kubectl wait --for=condition=complete --timeout=300s job/job-wq-1
```

Next, check on the Job:

```
kubectl describe jobs/job-wq-1
```

```

Name:          job-wq-1
Namespace:     default
Selector:      controller-uid=41d75705-92df-11e7-b85e-fa163ee3c11f
Labels:        controller-uid=41d75705-92df-11e7-b85e-fa163ee3c11f
                job-name=job-wq-1
Annotations:   <none>
Parallelism:   2
Completions:   8
Start Time:    Wed, 06 Sep 2022 16:42:02 +0000
Pods Statuses: 0 Running / 8 Succeeded / 0 Failed
Pod Template:
  Labels:  controller-uid=41d75705-92df-11e7-b85e-fa163ee3c11f
           job-name=job-wq-1
  Containers:
  c:
    Image:  container-registry.example/causal-jigsaw-637/job-wq-1
    Port:
    Environment:
      BROKER_URL:  amqp://guest:guest@rabbitmq-service:5672

```

| | | | | | | | | |
|-----------|----------|-------|--------|---------------|------------------|------------------------|---------|--|
| QUEUE: | job1 | | | | | | | |
| Mounts: | <none> | | | | | | | |
| Volumes: | <none> | | | | | | | |
| Events: | | | | | | | | |
| FirstSeen | LastSeen | Count | From | SubobjectPath | Type | Reason | Message | |
| | | | | | | | | |
| 27s | 27s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |
| hcobb | 27s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |
| weytj | 27s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |
| qaam5 | 27s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |
| b67sr | 27s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |
| xe5hj | 26s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |
| w2zqe | 15s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |
| d6ppa | 14s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |
| p17e0 | 14s | 1 | {job } | Normal | SuccessfulCreate | Created pod: job-wq-1- | | |

All the pods for that Job succeeded! You're done.

Alternatives

This approach has the advantage that you do not need to modify your "worker" program to be aware that there is a work queue. You can include the worker program unmodified in your container image.

Using this approach does require that you run a message queue service. If running a queue service is inconvenient, you may want to consider one of the other [job patterns](#).

This approach creates a pod for every work item. If your work items only take a few seconds, though, creating a Pod for every work item may add a lot of overhead. Consider another design, such as in the [fine parallel work queue example](#), that executes multiple work items per Pod.

In this example, you used the `amqp-consume` utility to read the message from the queue and run the actual program. This has the advantage that you do not need to modify your program to be aware of the queue. The [fine parallel work queue example](#) shows how to communicate with the work queue using a client library.

Caveats

If the number of completions is set to less than the number of items in the queue, then not all items will be processed.

If the number of completions is set to more than the number of items in the queue, then the Job will not appear to be completed, even though all items in the queue have been processed. It will start additional pods which will block waiting for a message. You would need to make your

own mechanism to spot when there is work to do and measure the size of the queue, setting the number of completions to match.

There is an unlikely race with this pattern. If the container is killed in between the time that the message is acknowledged by the `amqp-consume` command and the time that the container exits with success, or if the node crashes before the kubelet is able to post the success of the pod back to the API server, then the Job will not appear to be complete, even though all items in the queue have been processed.

Fine Parallel Processing Using a Work Queue

In this example, you will run a Kubernetes Job that runs multiple parallel tasks as worker processes, each running as a separate Pod.

In this example, as each pod is created, it picks up one unit of work from a task queue, processes it, and repeats until the end of the queue is reached.

Here is an overview of the steps in this example:

1. **Start a storage service to hold the work queue.** In this example, you will use Redis to store work items. In the [previous example](#), you used RabbitMQ. In this example, you will use Redis and a custom work-queue client library; this is because AMQP does not provide a good way for clients to detect when a finite-length work queue is empty. In practice you would set up a store such as Redis once and reuse it for the work queues of many jobs, and other things.
2. **Create a queue, and fill it with messages.** Each message represents one task to be done. In this example, a message is an integer that we will do a lengthy computation on.
3. **Start a Job that works on tasks from the queue.** The Job starts several pods. Each pod takes one task from the message queue, processes it, and repeats until the end of the queue is reached.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You will need a container image registry where you can upload images to run in your cluster. The example uses [Docker Hub](#), but you could adapt it to a different container image registry.

This task example also assumes that you have Docker installed locally. You use Docker to build container images.

Be familiar with the basic, non-parallel, use of [Job](#).

Starting Redis

For this example, for simplicity, you will start a single instance of Redis. See the [Redis Example](#) for an example of deploying Redis scalably and redundantly.

You could also download the following files directly:

- [redis-pod.yaml](#)
- [redis-service.yaml](#)
- [Dockerfile](#)
- [job.yaml](#)
- [rediswq.py](#)
- [worker.py](#)

To start a single instance of Redis, you need to create the redis pod and redis service:

```
kubectl apply -f https://k8s.io/examples/application/job/redis/redis-pod.yaml
kubectl apply -f https://k8s.io/examples/application/job/redis/redis-service.yaml
```

Filling the queue with tasks

Now let's fill the queue with some "tasks". In this example, the tasks are strings to be printed.

Start a temporary interactive pod for running the Redis CLI.

```
kubectl run -i --tty temp --image redis --command "/bin/sh"
```

```
Waiting for pod default/redis2-c7h78 to be running, status is Pending, pod ready: false
Hit enter for command prompt
```

Now hit enter, start the Redis CLI, and create a list with some work items in it.

```
redis-cli -h redis
```

```
redis:6379> rpush job2 "apple"
(integer) 1
redis:6379> rpush job2 "banana"
(integer) 2
redis:6379> rpush job2 "cherry"
(integer) 3
redis:6379> rpush job2 "date"
(integer) 4
redis:6379> rpush job2 "fig"
(integer) 5
redis:6379> rpush job2 "grape"
(integer) 6
redis:6379> rpush job2 "lemon"
(integer) 7
redis:6379> rpush job2 "melon"
(integer) 8
redis:6379> rpush job2 "orange"
(integer) 9
redis:6379> lrange job2 0 -1
```

- 1) "apple"
- 2) "banana"
- 3) "cherry"
- 4) "date"
- 5) "fig"
- 6) "grape"
- 7) "lemon"
- 8) "melon"
- 9) "orange"

So, the list with key job2 will be the work queue.

Note: if you do not have Kube DNS setup correctly, you may need to change the first step of the above block to `redis-cli -h $REDIS_SERVICE_HOST`.

Create a container image

Now you are ready to create an image that will process the work in that queue.

You're going to use a Python worker program with a Redis client to read the messages from the message queue.

A simple Redis work queue client library is provided, called `rediswq.py` ([Download](#)).

The "worker" program in each Pod of the Job uses the work queue client library to get work. Here it is:

[application/job/redis/worker.py](#)

```
#!/usr/bin/env python

import time
import rediswq

host="redis"
# Uncomment next two lines if you do not have Kube-DNS working.
# import os
# host = os.getenv("REDIS_SERVICE_HOST")

q = rediswq.RedisWQ(name="job2", host=host)
print("Worker with sessionID: " + q.sessionID())
print("Initial queue state: empty=" + str(q.empty()))
while not q.empty():
    item = q.lease(lease_secs=10, block=True, timeout=2)
    if item is not None:
        itemstr = item.decode("utf-8")
        print("Working on " + itemstr)
        time.sleep(10) # Put your actual work here instead of sleep.
        q.complete(item)
    else:
        print("Waiting for work")
print("Queue empty, exiting")
```

You could also download [worker.py](#), [rediswq.py](#), and [Dockerfile](#) files, then build the container image. Here's an example using Docker to do the image build:

```
docker build -t job-wq-2 .
```

Push the image

For the [Docker Hub](#), tag your app image with your username and push to the Hub with the below commands. Replace <username> with your Hub username.

```
docker tag job-wq-2 <username>/job-wq-2
docker push <username>/job-wq-2
```

You need to push to a public repository or [configure your cluster to be able to access your private repository](#).

Defining a Job

Here is a manifest for the Job you will create:

[application/job/redis/job.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
  name: job-wq-2
spec:
  parallelism: 2
  template:
    metadata:
      name: job-wq-2
    spec:
      containers:
        - name: c
          image: gcr.io/myproject/job-wq-2
      restartPolicy: OnFailure
```

Note:

Be sure to edit the manifest to change gcr.io/myproject to your own path.

In this example, each pod works on several items from the queue and then exits when there are no more items. Since the workers themselves detect when the workqueue is empty, and the Job controller does not know about the workqueue, it relies on the workers to signal when they are done working. The workers signal that the queue is empty by exiting with success. So, as soon as **any** worker exits with success, the controller knows the work is done, and that the Pods will exit soon. So, you need to leave the completion count of the Job unset. The job controller will wait for the other pods to complete too.

Running the Job

So, now run the Job:

```
# this assumes you downloaded and then edited the manifest already
kubectl apply -f ./job.yaml
```

Now wait a bit, then check on the Job:

```
kubectl describe jobs/job-wq-2
```

```
Name:          job-wq-2
Namespace:     default
Selector:      controller-uid=b1c7e4e3-92e1-11e7-b85e-fa163ee3c11f
Labels:        controller-uid=b1c7e4e3-92e1-11e7-b85e-fa163ee3c11f
                job-name=job-wq-2
Annotations:   <none>
Parallelism:   2
Completions:   <unset>
Start Time:    Mon, 11 Jan 2022 17:07:59 +0000
Pods Statuses: 1 Running / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  controller-uid=b1c7e4e3-92e1-11e7-b85e-fa163ee3c11f
           job-name=job-wq-2
  Containers:
  c:
    Image:          container-registry.example/exampleproject/job-wq-2
    Port:
    Environment:    <none>
    Mounts:          <none>
    Volumes:         <none>
Events:
  FirstSeen    LastSeen    Count   From          SubobjectPath  Type      Reason          Message
  -----
  33s          33s         1       {job-controller }      Normal        SuccessfulCreate Created pod:
job-wq-2-lglf8
```

You can wait for the Job to succeed, with a timeout:

```
# The check for condition name is case insensitive
kubectl wait --for=condition=complete --timeout=300s job/job-wq-2
```

```
kubectl logs pods/job-wq-2-7r7b2
```

```
Worker with sessionID: bbd72d0a-9e5c-4dd6-abf6-416cc267991f
Initial queue state: empty=False
Working on banana
Working on date
Working on lemon
```

As you can see, one of the pods for this Job worked on several work units.

Alternatives

If running a queue service or modifying your containers to use a work queue is inconvenient, you may want to consider one of the other [job patterns](#).

If you have a continuous stream of background processing work to run, then consider running your background workers with a ReplicaSet instead, and consider running a background processing library such as <https://github.com/resque/resque>.

Indexed Job for Parallel Processing with Static Work Assignment

FEATURE STATE: Kubernetes v1.24 [stable]

In this example, you will run a Kubernetes Job that uses multiple parallel worker processes. Each worker is a different container running in its own Pod. The Pods have an *index number* that the control plane sets automatically, which allows each Pod to identify which part of the overall task to work on.

The pod index is available in the [annotation](#) `batch.kubernetes.io/job-completion-index` as a string representing its decimal value. In order for the containerized task process to obtain this index, you can publish the value of the annotation using the [downward API](#) mechanism. For convenience, the control plane automatically sets the downward API to expose the index in the `JOB_COMPLETION_INDEX` environment variable.

Here is an overview of the steps in this example:

1. **Define a Job manifest using indexed completion.** The downward API allows you to pass the pod index annotation as an environment variable or file to the container.
2. **Start an Indexed Job based on that manifest.**

Before you begin

You should already be familiar with the basic, non-parallel, use of [Job](#).

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.21. To check the version, enter `kubectl version`.

Choose an approach

To access the work item from the worker program, you have a few options:

1. Read the `JOB_COMPLETION_INDEX` environment variable. The Job [controller](#) automatically links this variable to the annotation containing the completion index.
2. Read a file that contains the completion index.

3. Assuming that you can't modify the program, you can wrap it with a script that reads the index using any of the methods above and converts it into something that the program can use as input.

For this example, imagine that you chose option 3 and you want to run the [rev](#) utility. This program accepts a file as an argument and prints its content reversed.

```
rev data.txt
```

You'll use the rev tool from the [busybox](#) container image.

As this is only an example, each Pod only does a tiny piece of work (reversing a short string). In a real workload you might, for example, create a Job that represents the task of producing 60 seconds of video based on scene data. Each work item in the video rendering Job would be to render a particular frame of that video clip. Indexed completion would mean that each Pod in the Job knows which frame to render and publish, by counting frames from the start of the clip.

Define an Indexed Job

Here is a sample Job manifest that uses Indexed completion mode:

[application/job/indexed-job.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
  name: 'indexed-job'
spec:
  completions: 5
  parallelism: 3
  completionMode: Indexed
  template:
    spec:
      restartPolicy: Never
      initContainers:
        - name: 'input'
          image: 'docker.io/library/bash'
          command:
            - "bash"
            - "-c"
            - |
              items=(foo bar baz qux xyz)
              echo ${items[$JOB_COMPLETION_INDEX]} > /input/data.txt
      volumeMounts:
        - mountPath: /input
          name: input
      containers:
        - name: 'worker'
          image: 'docker.io/library/busybox'
          command:
            - "rev"
            - "/input/data.txt"
          volumeMounts:
```

```
- mountPath: /input
  name: input
volumes:
- name: input
  emptyDir: {}
```

In the example above, you use the builtin `JOB_COMPLETION_INDEX` environment variable set by the Job controller for all containers. An [init container](#) maps the index to a static value and writes it to a file that is shared with the container running the worker through an [emptyDir volume](#). Optionally, you can [define your own environment variable through the downward API](#) to publish the index to containers. You can also choose to load a list of values from a [ConfigMap as an environment variable or file](#).

Alternatively, you can directly [use the downward API to pass the annotation value as a volume file](#), like shown in the following example:

[application/job/indexed-job-vol.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
  name: 'indexed-job'
spec:
  completions: 5
  parallelism: 3
  completionMode: Indexed
  template:
    spec:
      restartPolicy: Never
      containers:
      - name: 'worker'
        image: 'docker.io/library/busybox'
        command:
        - "rev"
        - "/input/data.txt"
        volumeMounts:
        - mountPath: /input
          name: input
      volumes:
      - name: input
        downwardAPI:
          items:
          - path: "data.txt"
            fieldRef:
              fieldPath: metadata.annotations['batch.kubernetes.io/job-completion-index']
```

Running the Job

Now run the Job:

```
# This uses the first approach (relying on $JOB_COMPLETION_INDEX)
kubectl apply -f https://kubernetes.io/examples/application/job/indexed-job.yaml
```

When you create this Job, the control plane creates a series of Pods, one for each index you specified. The value of `.spec.parallelism` determines how many can run at once whereas `.spec.completions` determines how many Pods the Job creates in total.

Because `.spec.parallelism` is less than `.spec.completions`, the control plane waits for some of the first Pods to complete before starting more of them.

You can wait for the Job to succeed, with a timeout:

```
# The check for condition name is case insensitive
kubectl wait --for=condition=complete --timeout=300s job/indexed-job
```

Now, describe the Job and check that it was successful.

```
kubectl describe jobs/indexed-job
```

The output is similar to:

```
Name:          indexed-job
Namespace:     default
Selector:      controller-uid=bf865e04-0b67-483b-9a90-74cfc4c3e756
Labels:        controller-uid=bf865e04-0b67-483b-9a90-74cfc4c3e756
                job-name=indexed-job
Annotations:    <none>
Parallelism:    3
Completions:    5
Start Time:     Thu, 11 Mar 2021 15:47:34 +0000
Pods Statuses:  2 Running / 3 Succeeded / 0 Failed
Completed Indexes: 0-2
Pod Template:
  Labels:  controller-uid=bf865e04-0b67-483b-9a90-74cfc4c3e756
          job-name=indexed-job
  Init Containers:
  input:
    Image:   docker.io/library/bash
    Port:    <none>
    Host Port: <none>
    Command:
    bash
    -c
    items=(foo bar baz qux xyz)
    echo ${items[$JOB_COMPLETION_INDEX]} > /input/data.txt

  Environment: <none>
  Mounts:
    /input from input (rw)
  Containers:
  worker:
    Image:   docker.io/library/busybox
    Port:    <none>
    Host Port: <none>
    Command:
    rev
    /input/data.txt
```

```

Environment: <none>
Mounts:
  /input from input (rw)
Volumes:
input:
  Type:      EmptyDir (a temporary directory that shares a pod's lifetime)
  Medium:
  SizeLimit: <unset>
Events:
Type      Reason      Age   From          Message
----      -
Normal    SuccessfulCreate 4s    job-controller Created pod: indexed-job-njkjj
Normal    SuccessfulCreate 4s    job-controller Created pod: indexed-job-9kd4h
Normal    SuccessfulCreate 4s    job-controller Created pod: indexed-job-qjwsz
Normal    SuccessfulCreate 1s    job-controller Created pod: indexed-job-fdhq5
Normal    SuccessfulCreate 1s    job-controller Created pod: indexed-job-ncslj

```

In this example, you run the Job with custom values for each index. You can inspect the output of one of the pods:

```
kubectl logs indexed-job-fdhq5 # Change this to match the name of a Pod from that Job
```

The output is similar to:

```
xuq
```

Job with Pod-to-Pod Communication

In this example, you will run a Job in [Indexed completion mode](#) configured such that the pods created by the Job can communicate with each other using pod hostnames rather than pod IP addresses.

Pods within a Job might need to communicate among themselves. The user workload running in each pod could query the Kubernetes API server to learn the IPs of the other Pods, but it's much simpler to rely on Kubernetes' built-in DNS resolution.

Jobs in Indexed completion mode automatically set the pods' hostname to be in the format of `{jobName}-{completionIndex}`. You can use this format to deterministically build pod hostnames and enable pod communication *without* needing to create a client connection to the Kubernetes control plane to obtain pod hostnames/IPs via API requests.

This configuration is useful for use cases where pod networking is required but you don't want to depend on a network connection with the Kubernetes API server.

Before you begin

You should already be familiar with the basic use of [Job](#).

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.21. To check the version, enter `kubectl version`.

Note:

If you are using MiniKube or a similar tool, you may need to take [extra steps](#) to ensure you have DNS.

Starting a Job with Pod-to-Pod Communication

To enable pod-to-pod communication using pod hostnames in a Job, you must do the following:

1. Set up a [headless Service](#) with a valid label selector for the pods created by your Job. The headless service must be in the same namespace as the Job. One easy way to do this is to use the job-name: `<your-job-name>` selector, since the job-name label will be automatically added by Kubernetes. This configuration will trigger the DNS system to create records of the hostnames of the pods running your Job.
2. Configure the headless service as subdomain service for the Job pods by including the following value in your Job template spec:

```
subdomain: <headless-svc-name>
```

Example

Below is a working example of a Job with pod-to-pod communication via pod hostnames enabled. The Job is completed only after all pods successfully ping each other using hostnames.

Note:

In the Bash script executed on each pod in the example below, the pod hostnames can be prefixed by the namespace as well if the pod needs to be reached from outside the namespace.

```
apiVersion: v1
kind: Service
metadata:
  name: headless-svc
spec:
  clusterIP: None # clusterIP must be None to create a headless service
  selector:
    job-name: example-job # must match Job name
---
apiVersion: batch/v1
kind: Job
metadata:
```

```

name: example-job
spec:
  completions: 3
  parallelism: 3
  completionMode: Indexed
  template:
    spec:
      subdomain: headless-svc # has to match Service name
      restartPolicy: Never
      containers:
      - name: example-workload
        image: bash:latest
        command:
        - bash
        - -c
        - |
          for i in 0 1 2
          do
            gotStatus="-1"
            wantStatus="0"
            while [ $gotStatus -ne $wantStatus ]
            do
              ping -c 1 example-job-{i}.headless-svc > /dev/null 2>&1
              gotStatus=$?
              if [ $gotStatus -ne $wantStatus ]; then
                echo "Failed to ping pod example-job-{i}.headless-svc, retrying in 1 second..."
                sleep 1
              fi
            done
            echo "Successfully pinged pod: example-job-{i}.headless-svc"
          done

```

After applying the example above, reach each other over the network using: `<pod-hostname>.<headless-service-name>`. You should see output similar to the following:

```
kubectl logs example-job-0-qws42
```

```

Failed to ping pod example-job-0.headless-svc, retrying in 1 second...
Successfully pinged pod: example-job-0.headless-svc
Successfully pinged pod: example-job-1.headless-svc
Successfully pinged pod: example-job-2.headless-svc

```

Note:

Keep in mind that the `<pod-hostname>.<headless-service-name>` name format used in this example would not work with DNS policy set to None or Default. You can learn more about pod DNS policies [here](#).

Parallel Processing using Expansions

This task demonstrates running multiple [Jobs](#) based on a common template. You can use this approach to process batches of work in parallel.

For this example there are only three items: *apple*, *banana*, and *cherry*. The sample Jobs process each item by printing a string then pausing.

See [using Jobs in real workloads](#) to learn about how this pattern fits more realistic use cases.

Before you begin

You should be familiar with the basic, non-parallel, use of [Job](#).

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercoda](#)
- [Play with Kubernetes](#)

For basic templating you need the command-line utility sed.

To follow the advanced templating example, you need a working installation of [Python](#), and the Jinja2 template library for Python.

Once you have Python set up, you can install Jinja2 by running:

```
pip install --user jinja2
```

Create Jobs based on a template

First, download the following template of a Job to a file called job-tmpl.yaml. Here's what you'll download:

[application/job/job-tmpl.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
  name: process-item-$ITEM
  labels:
    jobgroup: jobexample
spec:
  template:
    metadata:
      name: jobexample
      labels:
        jobgroup: jobexample
    spec:
```

```
containers:
- name: c
  image: busybox:1.28
  command: ["sh", "-c", "echo Processing item $ITEM && sleep 5"]
restartPolicy: Never
```

```
# Use curl to download job-tmpl.yaml
curl -L -s -O https://k8s.io/examples/application/job/job-tmpl.yaml
```

The file you downloaded is not yet a valid Kubernetes [manifest](#). Instead that template is a YAML representation of a Job object with some placeholders that need to be filled in before it can be used. The \$ITEM syntax is not meaningful to Kubernetes.

Create manifests from the template

The following shell snippet uses sed to replace the string \$ITEM with the loop variable, writing into a temporary directory named jobs. Run this now:

```
# Expand the template into multiple files, one for each item to be processed.
mkdir ./jobs
for i in apple banana cherry
do
  cat job-tmpl.yaml | sed "s/\$ITEM/\$i/" > ./jobs/job-$.yaml
done
```

Check if it worked:

```
ls jobs/
```

The output is similar to this:

```
job-apple.yaml
job-banana.yaml
job-cherry.yaml
```

You could use any type of template language (for example: Jinja2; ERB), or write a program to generate the Job manifests.

Create Jobs from the manifests

Next, create all the Jobs with one kubectl command:

```
kubectl create -f ./jobs
```

The output is similar to this:

```
job.batch/process-item-apple created
job.batch/process-item-banana created
job.batch/process-item-cherry created
```

Now, check on the jobs:

```
kubectl get jobs -l jobgroup=jobexample
```


The output is similar to this:

| NAME | COMPLETIONS | DURATION | AGE |
|---------------------|-------------|----------|-----|
| process-item-apple | 1/1 | 14s | 22s |
| process-item-banana | 1/1 | 12s | 21s |
| process-item-cherry | 1/1 | 12s | 20s |

Using the `-l` option to `kubectl` selects only the Jobs that are part of this group of jobs (there might be other unrelated jobs in the system).

You can check on the Pods as well using the same [label selector](#):

```
kubectl get pods -l jobgroup=jobexample
```

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|---------------------------|-------|-----------|----------|-----|
| process-item-apple-kixwv | 0/1 | Completed | 0 | 4m |
| process-item-banana-wrsf7 | 0/1 | Completed | 0 | 4m |
| process-item-cherry-dnfu9 | 0/1 | Completed | 0 | 4m |

We can use this single command to check on the output of all jobs at once:

```
kubectl logs -f -l jobgroup=jobexample
```

The output should be:

```
Processing item apple
Processing item banana
Processing item cherry
```

Clean up

```
# Remove the Jobs you created
# Your cluster automatically cleans up their Pods
kubectl delete job -l jobgroup=jobexample
```

Use advanced template parameters

In the [first example](#), each instance of the template had one parameter, and that parameter was also used in the Job's name. However, [names](#) are restricted to contain only certain characters.

This slightly more complex example uses the [Jinja template language](#) to generate manifests and then objects from those manifests, with a multiple parameters for each Job.

For this part of the task, you are going to use a one-line Python script to convert the template to a set of manifests.

First, copy and paste the following template of a Job object, into a file called `job.yaml.jinja2`:

```
{% set params = [{ "name": "apple", "url": "http://dbpedia.org/resource/Apple", },
                  { "name": "banana", "url": "http://dbpedia.org/resource/Banana", },
                  { "name": "cherry", "url": "http://dbpedia.org/resource/Cherry" }]
%}
```

```
{% for p in params %}
{% set name = p["name"] %}
{% set url = p["url"] %}
---
apiVersion: batch/v1
kind: Job
metadata:
  name: jobexample-{{ name }}
  labels:
    jobgroup: jobexample
spec:
  template:
    metadata:
      name: jobexample
      labels:
        jobgroup: jobexample
    spec:
      containers:
      - name: c
        image: busybox:1.28
        command: ["sh", "-c", "echo Processing URL {{ url }} && sleep 5"]
        restartPolicy: Never
{% endfor %}
```

The above template defines two parameters for each Job object using a list of python dicts (lines 1-4). A for loop emits one Job manifest for each set of parameters (remaining lines).

This example relies on a feature of YAML. One YAML file can contain multiple documents (Kubernetes manifests, in this case), separated by `---` on a line by itself. You can pipe the output directly to `kubectl` to create the Jobs.

Next, use this one-line Python program to expand the template:

```
alias render_template='python -c "from jinja2 import Template; import sys;
print(Template(sys.stdin.read()).render());"'
```

Use `render_template` to convert the parameters and template into a single YAML file containing Kubernetes manifests:

```
# This requires the alias you defined earlier
cat job.yaml.jinja2 | render_template > jobs.yaml
```

You can view `jobs.yaml` to verify that the `render_template` script worked correctly.

Once you are happy that `render_template` is working how you intend, you can pipe its output into `kubectl`:

```
cat job.yaml.jinja2 | render_template | kubectl apply -f -
```

Kubernetes accepts and runs the Jobs you created.

Clean up

```
# Remove the Jobs you created
# Your cluster automatically cleans up their Pods
kubectl delete job -l jobgroup=jobexample
```

Using Jobs in real workloads

In a real use case, each Job performs some substantial computation, such as rendering a frame of a movie, or processing a range of rows in a database. If you were rendering a movie you would set \$ITEM to the frame number. If you were processing rows from a database table, you would set \$ITEM to represent the range of database rows to process.

In the task, you ran a command to collect the output from Pods by fetching their logs. In a real use case, each Pod for a Job writes its output to durable storage before completing. You can use a PersistentVolume for each Job, or an external storage service. For example, if you are rendering frames for a movie, use HTTP to PUT the rendered frame data to a URL, using a different URL for each frame.

Labels on Jobs and Pods

After you create a Job, Kubernetes automatically adds additional [labels](#) that distinguish one Job's pods from another Job's pods.

In this example, each Job and its Pod template have a label: jobgroup=jobexample.

Kubernetes itself pays no attention to labels named jobgroup. Setting a label for all the Jobs you create from a template makes it convenient to operate on all those Jobs at once. In the [first example](#) you used a template to create several Jobs. The template ensures that each Pod also gets the same label, so you can check on all Pods for these templated Jobs with a single command.

Note:

The label key jobgroup is not special or reserved. You can pick your own labelling scheme. There are [recommended labels](#) that you can use if you wish.

Alternatives

If you plan to create a large number of Job objects, you may find that:

- Even using labels, managing so many Jobs is cumbersome.
- If you create many Jobs in a batch, you might place high load on the Kubernetes control plane. Alternatively, the Kubernetes API server could rate limit you, temporarily rejecting your requests with a 429 status.
- You are limited by a [resource quota](#) on Jobs: the API server permanently rejects some of your requests when you create a great deal of work in one batch.

There are other [job patterns](#) that you can use to process large amounts of work without creating very many Job objects.

You could also consider writing your own [controller](#) to manage Job objects automatically.

Handling retrieable and non-retrieable pod failures with Pod failure policy

FEATURE STATE: Kubernetes v1.26 [beta]

This document shows you how to use the [Pod failure policy](#), in combination with the default [Pod backoff failure policy](#), to improve the control over the handling of container- or Pod-level failure within a [Job](#).

The definition of Pod failure policy may help you to:

- better utilize the computational resources by avoiding unnecessary Pod retries.
- avoid Job failures due to Pod disruptions (such [preemption](#), [API-initiated eviction](#) or [taint](#)-based eviction).

Before you begin

You should already be familiar with the basic use of [Job](#).

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.25. To check the version, enter kubectl version.

Ensure that the [feature gates](#) PodDisruptionConditions and JobPodFailurePolicy are both enabled in your cluster.

Using Pod failure policy to avoid unnecessary Pod retries

With the following example, you can learn how to use Pod failure policy to avoid unnecessary Pod restarts when a Pod failure indicates a non-retrieable software bug.

First, create a Job based on the config:

[/controllers/job-pod-failure-policy-failjob.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
  name: job-pod-failure-policy-failjob
spec:
  completions: 8
```

```
parallelism: 2
template:
  spec:
    restartPolicy: Never
    containers:
      - name: main
        image: docker.io/library/bash:5
        command: ["bash"]
        args:
          - -c
          - echo "Hello world! I'm going to exit with 42 to simulate a software bug." && sleep 30 &&
            exit 42
    backoffLimit: 6
    podFailurePolicy:
      rules:
        - action: FailJob
      onExitCodes:
        containerName: main
        operator: In
        values: [42]
```

by running:

```
kubectl create -f job-pod-failure-policy-failjob.yaml
```

After around 30s the entire Job should be terminated. Inspect the status of the Job by running:

```
kubectl get jobs -l job-name=job-pod-failure-policy-failjob -o yaml
```

In the Job status, see a job Failed condition with the field reason equal PodFailurePolicy. Additionally, the message field contains a more detailed information about the Job termination, such as: Container main for pod default/job-pod-failure-policy-failjob-8ckj8 failed with exit code 42 matching FailJob rule at index 0.

For comparison, if the Pod failure policy was disabled it would take 6 retries of the Pod, taking at least 2 minutes.

Clean up

Delete the Job you created:

```
kubectl delete jobs/job-pod-failure-policy-failjob
```

The cluster automatically cleans up the Pods.

Using Pod failure policy to ignore Pod disruptions

With the following example, you can learn how to use Pod failure policy to ignore Pod disruptions from incrementing the Pod retry counter towards the `.spec.backoffLimit` limit.

Caution:

Timing is important for this example, so you may want to read the steps before execution. In order to trigger a Pod disruption it is important to drain the node while the Pod is running on it (within 90s since the Pod is scheduled).

1. Create a Job based on the config:

[/controllers/job-pod-failure-policy-ignore.yaml](#)

```
apiVersion: batch/v1
kind: Job
metadata:
  name: job-pod-failure-policy-ignore
spec:
  completions: 4
  parallelism: 2
  template:
    spec:
      restartPolicy: Never
      containers:
      - name: main
        image: docker.io/library/bash:5
        command: ["bash"]
        args:
        - -c
        - echo "Hello world! I'm going to exit with 0 (success)." && sleep 90 && exit 0
      backoffLimit: 0
      podFailurePolicy:
        rules:
        - action: Ignore
          onPodConditions:
          - type: DisruptionTarget
```

by running:

```
kubectl create -f job-pod-failure-policy-ignore.yaml
```

2. Run this command to check the nodeName the Pod is scheduled to:

```
nodeName=$(kubectl get pods -l job-name=job-pod-failure-policy-ignore -o jsonpath='{.items[0].spec.nodeName}')
```

3. Drain the node to evict the Pod before it completes (within 90s):

```
kubectl drain nodes/$nodeName --ignore-daemonsets --grace-period=0
```

4. Inspect the .status.failed to check the counter for the Job is not incremented:

```
kubectl get jobs -l job-name=job-pod-failure-policy-ignore -o yaml
```

5. Uncordon the node:

```
kubectl uncordon nodes/$nodeName
```

The Job resumes and succeeds.

For comparison, if the Pod failure policy was disabled the Pod disruption would result in terminating the entire Job (as the `.spec.backoffLimit` is set to 0).

Cleaning up

Delete the Job you created:

```
kubectl delete jobs/job-pod-failure-policy-ignore
```

The cluster automatically cleans up the Pods.

Using Pod failure policy to avoid unnecessary Pod retries based on custom Pod Conditions

With the following example, you can learn how to use Pod failure policy to avoid unnecessary Pod restarts based on custom Pod Conditions.

Note:

The example below works since version 1.27 as it relies on transitioning of deleted pods, in the Pending phase, to a terminal phase (see: [Pod Phase](#)).

1. First, create a Job based on the config:

</controllers/job-pod-failure-policy-config-issue.yaml>

```
apiVersion: batch/v1
kind: Job
metadata:
  name: job-pod-failure-policy-config-issue
spec:
  completions: 8
  parallelism: 2
  template:
    spec:
      restartPolicy: Never
      containers:
        - name: main
          image: "non-existing-repo/non-existing-image:example"
      backoffLimit: 6
      podFailurePolicy:
        rules:
          - action: FailJob
            onPodConditions:
              - type: ConfigIssue
```

by running:

```
kubectl create -f job-pod-failure-policy-config-issue.yaml
```

Note that, the image is misconfigured, as it does not exist.

2. Inspect the status of the job's Pods by running:

```
kubectl get pods -l job-name=job-pod-failure-policy-config-issue -o yaml
```

You will see output similar to this:

```
containerStatuses:
- image: non-existing-repo/non-existing-image:example
  ...
  state:
    waiting:
      message: Back-off pulling image "non-existing-repo/non-existing-image:example"
      reason: ImagePullBackOff
    ...
phase: Pending
```

Note that the pod remains in the Pending phase as it fails to pull the misconfigured image. This, in principle, could be a transient issue and the image could get pulled. However, in this case, the image does not exist so we indicate this fact by a custom condition.

3. Add the custom condition. First prepare the patch by running:

```
cat <<EOF > patch.yaml
status:
  conditions:
  - type: ConfigIssue
    status: "True"
    reason: "NonExistingImage"
    lastTransitionTime: "$(date -u +"%Y-%m-%dT%H:%M:%SZ")"
EOF
```

Second, select one of the pods created by the job by running:

```
podName=$(kubectl get pods -l job-name=job-pod-failure-policy-config-issue -o
jsonpath='{.items[0].metadata.name}')
```

Then, apply the patch on one of the pods by running the following command:

```
kubectl patch pod $podName --subresource=status --patch-file=patch.yaml
```

If applied successfully, you will get a notification like this:

```
pod/job-pod-failure-policy-config-issue-k6pvp patched
```

4. Delete the pod to transition it to Failed phase, by running the command:

```
kubectl delete pods/$podName
```

5. Inspect the status of the Job by running:

```
kubectl get jobs -l job-name=job-pod-failure-policy-config-issue -o yaml
```


In the Job status, see a job Failed condition with the field reason equal PodFailurePolicy. Additionally, the message field contains a more detailed information about the Job termination, such as:
Pod default/job-pod-failure-policy-config-issue-k6pvp has condition ConfigIssue matching FailJob rule at index 0.

Note:

In a production environment, the steps 3 and 4 should be automated by a user-provided controller.

Cleaning up

Delete the Job you created:

```
kubectrl delete jobs/job-pod-failure-policy-config-issue
```

The cluster automatically cleans up the Pods.

Alternatives

You could rely solely on the [Pod backoff failure policy](#), by specifying the Job's .spec.backoffLimit field. However, in many situations it is problematic to find a balance between setting a low value for .spec.backoffLimit to avoid unnecessary Pod retries, yet high enough to make sure the Job would not be terminated by Pod disruptions.

Access Applications in a Cluster

Configure load balancing, port forwarding, or setup firewall or DNS configurations to access applications in a cluster.

[Deploy and Access the Kubernetes Dashboard](#)

Deploy the web UI (Kubernetes Dashboard) and access it.

[Accessing Clusters](#)

[Configure Access to Multiple Clusters](#)

[Use Port Forwarding to Access Applications in a Cluster](#)

[Use a Service to Access an Application in a Cluster](#)

[Connect a Frontend to a Backend Using Services](#)

[Create an External Load Balancer](#)

[List All Container Images Running in a Cluster](#)

[Set up Ingress on Minikube with the NGINX Ingress Controller](#)

[Communicate Between Containers in the Same Pod Using a Shared Volume](#)

[Configure DNS for a Cluster](#)

[Access Services Running on Clusters](#)

Deploy and Access the Kubernetes Dashboard

Deploy the web UI (Kubernetes Dashboard) and access it.

Dashboard is a web-based Kubernetes user interface. You can use Dashboard to deploy containerized applications to a Kubernetes cluster, troubleshoot your containerized application, and manage the cluster resources. You can use Dashboard to get an overview of applications running on your cluster, as well as for creating or modifying individual Kubernetes resources (such as Deployments, Jobs, DaemonSets, etc). For example, you can scale a Deployment, initiate a rolling update, restart a pod or deploy new applications using a deploy wizard.

Dashboard also provides information on the state of Kubernetes resources in your cluster and on any errors that may have occurred.

Kubernetes Dashboard UI

Deploying the Dashboard UI

Note:

Kubernetes Dashboard supports only Helm-based installation currently as it is faster and gives us better control over all dependencies required by Dashboard to run.

The Dashboard UI is not deployed by default. To deploy it, run the following command:

```
# Add kubernetes-dashboard repository
helm repo add kubernetes-dashboard https://kubernetes.github.io/dashboard/
# Deploy a Helm Release named "kubernetes-dashboard" using the kubernetes-dashboard chart
helm upgrade --install kubernetes-dashboard kubernetes-dashboard/kubernetes-dashboard --
create-namespace --namespace kubernetes-dashboard
```

Accessing the Dashboard UI

To protect your cluster data, Dashboard deploys with a minimal RBAC configuration by default. Currently, Dashboard only supports logging in with a Bearer Token. To create a token for this demo, you can follow our guide on [creating a sample user](#).

Warning:

The sample user created in the tutorial will have administrative privileges and is for educational purposes only.

Command line proxy

You can enable access to the Dashboard using the `kubectl` command-line tool, by running the following command:

```
kubectl proxy
```

Kubectl will make Dashboard available at <http://localhost:8001/api/v1/namespaces/kubernetes-dashboard/services/https:kubernetes-dashboard:/proxy/>.

The UI can *only* be accessed from the machine where the command is executed. See `kubectl proxy --help` for more options.

Note:

The kubeconfig authentication method does **not** support external identity providers or X.509 certificate-based authentication.

Welcome view

When you access Dashboard on an empty cluster, you'll see the welcome page. This page contains a link to this document as well as a button to deploy your first application. In addition, you can view which system applications are running by default in the kube-system [namespace](#) of your cluster, for example the Dashboard itself.

Kubernetes Dashboard welcome page

Deploying containerized applications

Dashboard lets you create and deploy a containerized application as a Deployment and optional Service with a simple wizard. You can either manually specify application details, or upload a YAML or JSON *manifest* file containing application configuration.

Click the **CREATE** button in the upper right corner of any page to begin.

Specifying application details

The deploy wizard expects that you provide the following information:

- **App name** (mandatory): Name for your application. A [label](#) with the name will be added to the Deployment and Service, if any, that will be deployed.

The application name must be unique within the selected Kubernetes [namespace](#). It must start with a lowercase character, and end with a lowercase character or a number, and contain only lowercase letters, numbers and dashes (-). It is limited to 24 characters. Leading and trailing spaces are ignored.

- **Container image** (mandatory): The URL of a public Docker [container image](#) on any registry, or a private image (commonly hosted on the Google Container Registry or Docker Hub). The container image specification must end with a colon.
- **Number of pods** (mandatory): The target number of Pods you want your application to be deployed in. The value must be a positive integer.

A [Deployment](#) will be created to maintain the desired number of Pods across your cluster.

- **Service** (optional): For some parts of your application (e.g. frontends) you may want to expose a [Service](#) onto an external, maybe public IP address outside of your cluster (external Service).

Note:

For external Services, you may need to open up one or more ports to do so.

Other Services that are only visible from inside the cluster are called internal Services.

Irrespective of the Service type, if you choose to create a Service and your container listens on a port (incoming), you need to specify two ports. The Service will be created mapping the port (incoming) to the target port seen by the container. This Service will route to your deployed Pods. Supported protocols are TCP and UDP. The internal DNS name for this Service will be the value you specified as application name above.

If needed, you can expand the **Advanced options** section where you can specify more settings:

- **Description:** The text you enter here will be added as an [annotation](#) to the Deployment and displayed in the application's details.
- **Labels:** Default [labels](#) to be used for your application are application name and version. You can specify additional labels to be applied to the Deployment, Service (if any), and Pods, such as release, environment, tier, partition, and release track.

Example:

```
release=1.0
tier=frontend
environment=pod
track=stable
```

- **Namespace:** Kubernetes supports multiple virtual clusters backed by the same physical cluster. These virtual clusters are called [namespaces](#). They let you partition resources into logically named groups.

Dashboard offers all available namespaces in a dropdown list, and allows you to create a new namespace. The namespace name may contain a maximum of 63 alphanumeric characters and dashes (-) but can not contain capital letters. Namespace names should not consist of only numbers. If the name is set as a number, such as 10, the pod will be put in the default namespace.

In case the creation of the namespace is successful, it is selected by default. If the creation fails, the first namespace is selected.

- **Image Pull Secret:** In case the specified Docker container image is private, it may require [pull secret](#) credentials.

Dashboard offers all available secrets in a dropdown list, and allows you to create a new secret. The secret name must follow the DNS domain name syntax, for example `new.image-pull.secret`. The content of a secret must be base64-encoded and specified in a [.dockercfg](#) file. The secret name may consist of a maximum of 253 characters.

In case the creation of the image pull secret is successful, it is selected by default. If the creation fails, no secret is applied.

- **CPU requirement (cores) and Memory requirement (MiB):** You can specify the minimum [resource limits](#) for the container. By default, Pods run with unbounded CPU and memory limits.
- **Run command and Run command arguments:** By default, your containers run the specified Docker image's default [entrypoint command](#). You can use the command options and arguments to override the default.
- **Run as privileged:** This setting determines whether processes in [privileged containers](#) are equivalent to processes running as root on the host. Privileged containers can make use of capabilities like manipulating the network stack and accessing devices.
- **Environment variables:** Kubernetes exposes Services through [environment variables](#). You can compose environment variable or pass arguments to your commands using the values of environment variables. They can be used in applications to find a Service. Values can reference other variables using the `$(VAR_NAME)` syntax.

Uploading a YAML or JSON file

Kubernetes supports declarative configuration. In this style, all configuration is stored in manifests (YAML or JSON configuration files). The manifests use Kubernetes [API](#) resource schemas.

As an alternative to specifying application details in the deploy wizard, you can define your application in one or more manifests, and upload the files using Dashboard.

Using Dashboard

Following sections describe views of the Kubernetes Dashboard UI; what they provide and how can they be used.

Navigation

When there are Kubernetes objects defined in the cluster, Dashboard shows them in the initial view. By default only objects from the *default* namespace are shown and this can be changed using the namespace selector located in the navigation menu.

Dashboard shows most Kubernetes object kinds and groups them in a few menu categories.

Admin overview

For cluster and namespace administrators, Dashboard lists Nodes, Namespaces and PersistentVolumes and has detail views for them. Node list view contains CPU and memory usage metrics aggregated across all Nodes. The details view shows the metrics for a Node, its specification, status, allocated resources, events and pods running on the node.

Workloads

Shows all applications running in the selected namespace. The view lists applications by workload kind (for example: Deployments, ReplicaSets, StatefulSets). Each workload kind can be viewed separately. The lists summarize actionable information about the workloads, such as the number of ready pods for a ReplicaSet or current memory usage for a Pod.

Detail views for workloads show status and specification information and surface relationships between objects. For example, Pods that ReplicaSet is controlling or new ReplicaSets and HorizontalPodAutoscalers for Deployments.

Services

Shows Kubernetes resources that allow for exposing services to external world and discovering them within a cluster. For that reason, Service and Ingress views show Pods targeted by them, internal endpoints for cluster connections and external endpoints for external users.

Storage

Storage view shows PersistentVolumeClaim resources which are used by applications for storing data.

ConfigMaps and Secrets

Shows all Kubernetes resources that are used for live configuration of applications running in clusters. The view allows for editing and managing config objects and displays secrets hidden by default.

Logs viewer

Pod lists and detail pages link to a logs viewer that is built into Dashboard. The viewer allows for drilling down logs from containers belonging to a single Pod.

Logs viewer

What's next

For more information, see the [Kubernetes Dashboard project page](#).

Accessing Clusters

This topic discusses multiple ways to interact with clusters.

Accessing for the first time with kubectl

When accessing the Kubernetes API for the first time, we suggest using the Kubernetes CLI, kubectl.

To access a cluster, you need to know the location of the cluster and have credentials to access it. Typically, this is automatically set-up when you work through a [Getting started guide](#), or someone else set up the cluster and provided you with credentials and a location.

Check the location and credentials that kubectl knows about with this command:

```
kubectl config view
```

Many of the [examples](#) provide an introduction to using kubectl, and complete documentation is found in the [kubectl reference](#).

Directly accessing the REST API

Kubectl handles locating and authenticating to the apiserver. If you want to directly access the REST API with an http client like curl or wget, or a browser, there are several ways to locate and authenticate:

- Run kubectl in proxy mode.
 - Recommended approach.
 - Uses stored apiserver location.
 - Verifies identity of apiserver using self-signed cert. No MITM possible.
 - Authenticates to apiserver.
 - In future, may do intelligent client-side load-balancing and failover.
- Provide the location and credentials directly to the http client.
 - Alternate approach.
 - Works with some types of client code that are confused by using a proxy.
 - Need to import a root cert into your browser to protect against MITM.

Using kubectl proxy

The following command runs kubectl in a mode where it acts as a reverse proxy. It handles locating the apiserver and authenticating. Run it like this:

```
kubectl proxy --port=8080
```

See [kubectl proxy](#) for more details.

Then you can explore the API with curl, wget, or a browser, replacing localhost with [::1] for IPv6, like so:

```
curl http://localhost:8080/api/
```

The output is similar to this:

```
{
  "kind": "APIVersions",
  "versions": [
    "v1"
```

```

],
"serverAddressByClientCIDRs": [
  {
    "clientCIDR": "0.0.0.0/0",
    "serverAddress": "10.0.1.149:443"
  }
]
}

```

Without kubectl proxy

Use `kubectl apply` and `kubectl describe secret...` to create a token for the default service account with `grep/cut`:

First, create the Secret, requesting a token for the default ServiceAccount:

```

kubectl apply -f - <<EOF
apiVersion: v1
kind: Secret
metadata:
  name: default-token
  annotations:
    kubernetes.io/service-account.name: default
type: kubernetes.io/service-account-token
EOF

```

Next, wait for the token controller to populate the Secret with a token:

```

while ! kubectl describe secret default-token | grep -E '^token' >/dev/null; do
  echo "waiting for token..." >&2
  sleep 1
done

```

Capture and use the generated token:

```

APISERVER=$(kubectl config view --minify | grep server | cut -f 2- -d ":" | tr -d " ")
TOKEN=$(kubectl describe secret default-token | grep -E '^token' | cut -f2 -d ':' | tr -d " ")

curl $APISERVER/api --header "Authorization: Bearer $TOKEN" --insecure

```

The output is similar to this:

```

{
  "kind": "APIVersions",
  "versions": [
    "v1"
  ],
  "serverAddressByClientCIDRs": [
    {
      "clientCIDR": "0.0.0.0/0",
      "serverAddress": "10.0.1.149:443"
    }
  ]
}

```


Using jsonpath:

```
APISERVER=$(kubectl config view --minify -o jsonpath='{.clusters[0].cluster.server}')
TOKEN=$(kubectl get secret default-token -o jsonpath='{.data.token}' | base64 --decode)

curl $APISERVER/api --header "Authorization: Bearer $TOKEN" --insecure
```

The output is similar to this:

```
{
  "kind": "APIVersions",
  "versions": [
    "v1"
  ],
  "serverAddressByClientCIDRs": [
    {
      "clientCIDR": "0.0.0.0/0",
      "serverAddress": "10.0.1.149:443"
    }
  ]
}
```

The above examples use the `--insecure` flag. This leaves it subject to MITM attacks. When `kubectl` accesses the cluster it uses a stored root certificate and client certificates to access the server. (These are installed in the `~/.kube` directory). Since cluster certificates are typically self-signed, it may take special configuration to get your http client to use root certificate.

On some clusters, the apiserver does not require authentication; it may serve on localhost, or be protected by a firewall. There is not a standard for this. [Controlling Access to the API](#) describes how a cluster admin can configure this.

Programmatic access to the API

Kubernetes officially supports [Go](#) and [Python](#) client libraries.

Go client

- To get the library, run the following command: `go get k8s.io/client-go@kubernetes-<kubernetes-version-number>`, see [INSTALL.md](#) for detailed installation instructions. See <https://github.com/kubernetes/client-go> to see which versions are supported.
- Write an application atop of the client-go clients. Note that client-go defines its own API objects, so if needed, please import API definitions from client-go rather than from the main repository, e.g., `import "k8s.io/client-go/kubernetes"` is correct.

The Go client can use the same [kubeconfig file](#) as the `kubectl` CLI does to locate and authenticate to the apiserver. See this [example](#).

If the application is deployed as a Pod in the cluster, please refer to the [next section](#).

Python client

To use [Python client](#), run the following command: `pip install kubernetes`. See [Python Client Library page](#) for more installation options.

The Python client can use the same [kubeconfig file](#) as the kubectl CLI does to locate and authenticate to the apiserver. See this [example](#).

Other languages

There are [client libraries](#) for accessing the API from other languages. See documentation for other libraries for how they authenticate.

Accessing the API from a Pod

When accessing the API from a pod, locating and authenticating to the API server are somewhat different.

Please check [Accessing the API from within a Pod](#) for more details.

Accessing services running on the cluster

The previous section describes how to connect to the Kubernetes API server. For information about connecting to other services running on a Kubernetes cluster, see [Access Cluster Services](#).

Requesting redirects

The redirect capabilities have been deprecated and removed. Please use a proxy (see below) instead.

So many proxies

There are several different proxies you may encounter when using Kubernetes:

1. The [kubectl proxy](#):

- runs on a user's desktop or in a pod
- proxies from a localhost address to the Kubernetes apiserver
- client to proxy uses HTTP
- proxy to apiserver uses HTTPS
- locates apiserver
- adds authentication headers

2. The [apiserver proxy](#):

- is a bastion built into the apiserver
- connects a user outside of the cluster to cluster IPs which otherwise might not be reachable
- runs in the apiserver processes
- client to proxy uses HTTPS (or http if apiserver so configured)
- proxy to target may use HTTP or HTTPS as chosen by proxy using available information
- can be used to reach a Node, Pod, or Service
- does load balancing when used to reach a Service

The [kube proxy](#):

3.

- runs on each node
- proxies UDP and TCP
- does not understand HTTP
- provides load balancing
- is only used to reach services

4. A Proxy/Load-balancer in front of apiserver(s):

- existence and implementation varies from cluster to cluster (e.g. nginx)
- sits between all clients and one or more apiservers
- acts as load balancer if there are several apiservers.

5. Cloud Load Balancers on external services:

- are provided by some cloud providers (e.g. AWS ELB, Google Cloud Load Balancer)
- are created automatically when the Kubernetes service has type LoadBalancer
- use UDP/TCP only
- implementation varies by cloud provider.

Kubernetes users will typically not need to worry about anything other than the first two types. The cluster admin will typically ensure that the latter types are set up correctly.

Configure Access to Multiple Clusters

This page shows how to configure access to multiple clusters by using configuration files. After your clusters, users, and contexts are defined in one or more configuration files, you can quickly switch between clusters by using the `kubectl config use-context` command.

Note:

A file that is used to configure access to a cluster is sometimes called a *kubeconfig file*. This is a generic way of referring to configuration files. It does not mean that there is a file named `kubeconfig`.

Warning:

Only use kubeconfig files from trusted sources. Using a specially-crafted kubeconfig file could result in malicious code execution or file exposure. If you must use an untrusted kubeconfig file, inspect it carefully first, much as you would a shell script.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercoda](#)
- [Play with Kubernetes](#)

To check that [kubect](#)l is installed, run `kubect`l version --client. The kubectl version should be [within one minor version](#) of your cluster's API server.

Define clusters, users, and contexts

Suppose you have two clusters, one for development work and one for test work. In the development cluster, your frontend developers work in a namespace called frontend, and your storage developers work in a namespace called storage. In your test cluster, developers work in the default namespace, or they create auxiliary namespaces as they see fit. Access to the development cluster requires authentication by certificate. Access to the test cluster requires authentication by username and password.

Create a directory named config-exercise. In your config-exercise directory, create a file named config-demo with this content:

```
apiVersion: v1
kind: Config
preferences: {}

clusters:
- cluster:
  name: development
- cluster:
  name: test

users:
- name: developer
- name: experimenter

contexts:
- context:
  name: dev-frontend
- context:
  name: dev-storage
- context:
  name: exp-test
```

A configuration file describes clusters, users, and contexts. Your config-demo file has the framework to describe two clusters, two users, and three contexts.

Go to your config-exercise directory. Enter these commands to add cluster details to your configuration file:

```
kubectl config --kubeconfig=config-demo set-cluster development --server=https://1.2.3.4 --
certificate-authority=fake-ca-file
kubectl config --kubeconfig=config-demo set-cluster test --server=https://5.6.7.8 --insecure-
skip-tls-verify
```

Add user details to your configuration file:

Caution:

Storing passwords in Kubernetes client config is risky. A better alternative would be to use a credential plugin and store them separately. See: [client-go credential plugins](#)

```
kubectcl config --kubeconfig=config-demo set-credentials developer --client-certificate=fake-  
cert-file --client-key=fake-key-seefile  
kubectcl config --kubeconfig=config-demo set-credentials experimenter --username=exp --  
password=some-password
```

Note:

- To delete a user you can run `kubectcl --kubeconfig=config-demo config unset users.<name>`
- To remove a cluster, you can run `kubectcl --kubeconfig=config-demo config unset clusters.<name>`
- To remove a context, you can run `kubectcl --kubeconfig=config-demo config unset contexts.<name>`

Add context details to your configuration file:

```
kubectcl config --kubeconfig=config-demo set-context dev-frontend --cluster=development --  
namespace=frontend --user=developer  
kubectcl config --kubeconfig=config-demo set-context dev-storage --cluster=development --  
namespace=storage --user=developer  
kubectcl config --kubeconfig=config-demo set-context exp-test --cluster=test --namespace=defau  
lt --user=experimenter
```

Open your config-demo file to see the added details. As an alternative to opening the config-demo file, you can use the config view command.

```
kubectcl config --kubeconfig=config-demo view
```

The output shows the two clusters, two users, and three contexts:

```
apiVersion: v1  
clusters:  
- cluster:  
  certificate-authority: fake-ca-file  
  server: https://1.2.3.4  
  name: development  
- cluster:  
  insecure-skip-tls-verify: true  
  server: https://5.6.7.8  
  name: test  
contexts:  
- context:  
  cluster: development  
  namespace: frontend  
  user: developer  
  name: dev-frontend  
- context:  
  cluster: development
```

```

  namespace: storage
  user: developer
name: dev-storage
- context:
  cluster: test
  namespace: default
  user: experimenter
  name: exp-test
current-context: ""
kind: Config
preferences: {}
users:
- name: developer
  user:
    client-certificate: fake-cert-file
    client-key: fake-key-file
- name: experimenter
  user:
    # Documentation note (this comment is NOT part of the command output).
    # Storing passwords in Kubernetes client config is risky.
    # A better alternative would be to use a credential plugin
    # and store the credentials separately.
    # See https://kubernetes.io/docs/reference/access-authn-authz/authentication/#client-go-credential-plugins
    password: some-password
    username: exp

```

The fake-ca-file, fake-cert-file and fake-key-file above are the placeholders for the pathnames of the certificate files. You need to change these to the actual pathnames of certificate files in your environment.

Sometimes you may want to use Base64-encoded data embedded here instead of separate certificate files; in that case you need to add the suffix -data to the keys, for example, certificate-authority-data, client-certificate-data, client-key-data.

Each context is a triple (cluster, user, namespace). For example, the dev-frontend context says, "Use the credentials of the developer user to access the frontend namespace of the development cluster".

Set the current context:

```
kubectrl config --kubeconfig=config-demo use-context dev-frontend
```

Now whenever you enter a kubectrl command, the action will apply to the cluster, and namespace listed in the dev-frontend context. And the command will use the credentials of the user listed in the dev-frontend context.

To see only the configuration information associated with the current context, use the --minify flag.

```
kubectrl config --kubeconfig=config-demo view --minify
```

The output shows configuration information associated with the dev-frontend context:

```
apiVersion: v1
clusters:
- cluster:
  certificate-authority: fake-ca-file
  server: https://1.2.3.4
  name: development
contexts:
- context:
  cluster: development
  namespace: frontend
  user: developer
  name: dev-frontend
current-context: dev-frontend
kind: Config
preferences: {}
users:
- name: developer
  user:
    client-certificate: fake-cert-file
    client-key: fake-key-file
```

Now suppose you want to work for a while in the test cluster.

Change the current context to exp-test:

```
kubectl config --kubeconfig=config-demo use-context exp-test
```

Now any kubectl command you give will apply to the default namespace of the test cluster. And the command will use the credentials of the user listed in the exp-test context.

View configuration associated with the new current context, exp-test.

```
kubectl config --kubeconfig=config-demo view --minify
```

Finally, suppose you want to work for a while in the storage namespace of the development cluster.

Change the current context to dev-storage:

```
kubectl config --kubeconfig=config-demo use-context dev-storage
```

View configuration associated with the new current context, dev-storage.

```
kubectl config --kubeconfig=config-demo view --minify
```

Create a second configuration file

In your config-exercise directory, create a file named config-demo-2 with this content:

```
apiVersion: v1
kind: Config
preferences: {}

contexts:
```

```
- context:
  cluster: development
  namespace: ramp
  user: developer
  name: dev-ramp-up
```

The preceding configuration file defines a new context named `dev-ramp-up`.

Set the KUBECONFIG environment variable

See whether you have an environment variable named `KUBECONFIG`. If so, save the current value of your `KUBECONFIG` environment variable, so you can restore it later. For example:

Linux

```
export KUBECONFIG_SAVED="$KUBECONFIG"
```

Windows PowerShell

```
$Env:KUBECONFIG_SAVED=$ENV:KUBECONFIG
```

The `KUBECONFIG` environment variable is a list of paths to configuration files. The list is colon-delimited for Linux and Mac, and semicolon-delimited for Windows. If you have a `KUBECONFIG` environment variable, familiarize yourself with the configuration files in the list.

Temporarily append two paths to your `KUBECONFIG` environment variable. For example:

Linux

```
export KUBECONFIG="${KUBECONFIG}:config-demo:config-demo-2"
```

Windows PowerShell

```
$Env:KUBECONFIG=("config-demo;config-demo-2")
```

In your `config-exercise` directory, enter this command:

```
kubectl config view
```

The output shows merged information from all the files listed in your `KUBECONFIG` environment variable. In particular, notice that the merged information has the `dev-ramp-up` context from the `config-demo-2` file and the three contexts from the `config-demo` file:

```
contexts:
- context:
  cluster: development
  namespace: frontend
  user: developer
  name: dev-frontend
- context:
  cluster: development
  namespace: ramp
```



```
  user: developer
  name: dev-ramp-up
- context:
  cluster: development
  namespace: storage
  user: developer
  name: dev-storage
- context:
  cluster: test
  namespace: default
  user: experimenter
  name: exp-test
```

For more information about how kubeconfig files are merged, see [Organizing Cluster Access Using kubeconfig Files](#)

Explore the \$HOME/.kube directory

If you already have a cluster, and you can use `kubectl` to interact with the cluster, then you probably have a file named `config` in the `$HOME/.kube` directory.

Go to `$HOME/.kube`, and see what files are there. Typically, there is a file named `config`. There might also be other configuration files in this directory. Briefly familiarize yourself with the contents of these files.

Append \$HOME/.kube/config to your KUBECONFIG environment variable

If you have a `$HOME/.kube/config` file, and it's not already listed in your `KUBECONFIG` environment variable, append it to your `KUBECONFIG` environment variable now. For example:

Linux

```
export KUBECONFIG="${KUBECONFIG};${HOME}/.kube/config"
```

Windows Powershell

```
$Env:KUBECONFIG="$Env:KUBECONFIG;$HOME\.kube\config"
```

View configuration information merged from all the files that are now listed in your `KUBECONFIG` environment variable. In your `config-exercise` directory, enter:

```
kubectl config view
```

Clean up

Return your `KUBECONFIG` environment variable to its original value. For example:

Linux

```
export KUBECONFIG="$KUBECONFIG_SAVED"
```

Windows PowerShell

```
$Env:KUBECONFIG=$ENV:KUBECONFIG_SAVED
```

Check the subject represented by the kubeconfig

It is not always obvious what attributes (username, groups) you will get after authenticating to the cluster. It can be even more challenging if you are managing more than one cluster at the same time.

There is a `kubectl` subcommand to check subject attributes, such as username, for your selected Kubernetes client context: `kubectl auth whoami`.

Read [API access to authentication information for a client](#) to learn about this in more detail.

What's next

- [Organizing Cluster Access Using kubeconfig Files](#)
- [kubectl config](#)

Use Port Forwarding to Access Applications in a Cluster

This page shows how to use `kubectl port-forward` to connect to a MongoDB server running in a Kubernetes cluster. This type of connection can be useful for database debugging.

Before you begin

- You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.10. To check the version, enter `kubectl version`.

- Install [MongoDB Shell](#).

Creating MongoDB deployment and service

1. Create a Deployment that runs MongoDB:

```
kubectl apply -f https://k8s.io/examples/application/mongodb/mongo-deployment.yaml
```

The output of a successful command verifies that the deployment was created:

```
deployment.apps/mongo created
```

View the pod status to check that it is ready:

```
kubectl get pods
```

The output displays the pod created:

| NAME | READY | STATUS | RESTARTS | AGE |
|------------------------|-------|---------|----------|------|
| mongo-75f59d57f4-4nd6q | 1/1 | Running | 0 | 2m4s |

View the Deployment's status:

```
kubectl get deployment
```

The output displays that the Deployment was created:

| NAME | READY | UP-TO-DATE | AVAILABLE | AGE |
|-------|-------|------------|-----------|-------|
| mongo | 1/1 | 1 | 1 | 2m21s |

The Deployment automatically manages a ReplicaSet. View the ReplicaSet status using:

```
kubectl get replicaset
```

The output displays that the ReplicaSet was created:

| NAME | DESIRED | CURRENT | READY | AGE |
|------------------|---------|---------|-------|-------|
| mongo-75f59d57f4 | 1 | 1 | 1 | 3m12s |

2. Create a Service to expose MongoDB on the network:

```
kubectl apply -f https://k8s.io/examples/application/mongodb/mongo-service.yaml
```

The output of a successful command verifies that the Service was created:

```
service/mongo created
```

Check the Service created:

```
kubectl get service mongo
```

The output displays the service created:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|-------|-----------|--------------|-------------|-----------|-----|
| mongo | ClusterIP | 10.96.41.183 | <none> | 27017/TCP | 11s |

3. Verify that the MongoDB server is running in the Pod, and listening on port 27017:

```
# Change mongo-75f59d57f4-4nd6q to the name of the Pod
kubectl get pod mongo-75f59d57f4-4nd6q --template='{{(index (index .spec.containers 0).ports 0).containerPort}}{{"\n"}}'
```

The output displays the port for MongoDB in that Pod:

```
27017
```

27017 is the official TCP port for MongoDB.

Forward a local port to a port on the Pod

1. `kubectl port-forward` allows using resource name, such as a pod name, to select a matching pod to port forward to.

```
# Change mongo-75f59d57f4-4nd6q to the name of the Pod
kubectl port-forward mongo-75f59d57f4-4nd6q 28015:27017
```

which is the same as

```
kubectl port-forward pods/mongo-75f59d57f4-4nd6q 28015:27017
```

or

```
kubectl port-forward deployment/mongo 28015:27017
```

or

```
kubectl port-forward replicaset/mongo-75f59d57f4 28015:27017
```

or

```
kubectl port-forward service/mongo 28015:27017
```

Any of the above commands works. The output is similar to this:

```
Forwarding from 127.0.0.1:28015 -> 27017
Forwarding from [::1]:28015 -> 27017
```

Note:

`kubectl port-forward` does not return. To continue with the exercises, you will need to open another terminal.

2. Start the MongoDB command line interface:

```
mongosh --port 28015
```

3. At the MongoDB command line prompt, enter the `ping` command:

```
db.runCommand( { ping: 1 } )
```

A successful ping request returns:

```
{ ok: 1 }
```

Optionally let *kubect* choose the local port

If you don't need a specific local port, you can let *kubect* choose and allocate the local port and thus relieve you from having to manage local port conflicts, with the slightly simpler syntax:

```
kubect port-forward deployment/mongo :27017
```

The *kubect* tool finds a local port number that is not in use (avoiding low ports numbers, because these might be used by other applications). The output is similar to:

```
Forwarding from 127.0.0.1:63753 -> 27017  
Forwarding from [::1]:63753 -> 27017
```

Discussion

Connections made to local port 28015 are forwarded to port 27017 of the Pod that is running the MongoDB server. With this connection in place, you can use your local workstation to debug the database that is running in the Pod.

Note:

kubect port-forward is implemented for TCP ports only. The support for UDP protocol is tracked in [issue 47862](#).

What's next

Learn more about [kubect port-forward](#).

Use a Service to Access an Application in a Cluster

This page shows how to create a Kubernetes Service object that external clients can use to access an application running in a cluster. The Service provides load balancing for an application that has two running instances.

Before you begin

You need to have a Kubernetes cluster, and the *kubect* command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Objectives

- Run two instances of a Hello World application.

- Create a Service object that exposes a node port.
- Use the Service object to access the running application.

Creating a service for an application running in two pods

Here is the configuration file for the application Deployment:

[service/access/hello-application.yaml](https://k8s.io/examples/service/access/hello-application.yaml)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: hello-world
spec:
  selector:
    matchLabels:
      run: load-balancer-example
  replicas: 2
  template:
    metadata:
      labels:
        run: load-balancer-example
    spec:
      containers:
        - name: hello-world
          image: gcr.io/google-samples/node-hello:1.0
          ports:
            - containerPort: 8080
              protocol: TCP
```

1. Run a Hello World application in your cluster: Create the application Deployment using the file above:

```
kubectl apply -f https://k8s.io/examples/service/access/hello-application.yaml
```

The preceding command creates a [Deployment](#) and an associated [ReplicaSet](#). The ReplicaSet has two [Pods](#) each of which runs the Hello World application.

2. Display information about the Deployment:

```
kubectl get deployments hello-world
kubectl describe deployments hello-world
```

3. Display information about your ReplicaSet objects:

```
kubectl get replicaset
kubectl describe replicaset
```

4. Create a Service object that exposes the deployment:

```
kubectl expose deployment hello-world --type=NodePort --name=example-service
```

5. Display information about the Service:

```
kubectl describe services example-service
```

The output is similar to this:

```
Name:          example-service
Namespace:     default
Labels:        run=load-balancer-example
Annotations:   <none>
Selector:      run=load-balancer-example
Type:          NodePort
IP:            10.32.0.16
Port:          <unset> 8080/TCP
TargetPort:    8080/TCP
NodePort:      <unset> 31496/TCP
Endpoints:     10.200.1.4:8080,10.200.2.5:8080
Session Affinity: None
Events:        <none>
```

Make a note of the NodePort value for the Service. For example, in the preceding output, the NodePort value is 31496.

6. List the pods that are running the Hello World application:

```
kubectl get pods --selector="run=load-balancer-example" --output=wide
```

The output is similar to this:

| NAME | READY | STATUS | ... | IP | NODE |
|------------------------------|-------|---------|-----|------------|---------|
| hello-world-2895499144-bsbk5 | 1/1 | Running | ... | 10.200.1.4 | worker1 |
| hello-world-2895499144-m1pwt | 1/1 | Running | ... | 10.200.2.5 | worker2 |

7. Get the public IP address of one of your nodes that is running a Hello World pod. How you get this address depends on how you set up your cluster. For example, if you are using Minikube, you can see the node address by running `kubectl cluster-info`. If you are using Google Compute Engine instances, you can use the `gcloud compute instances list` command to see the public addresses of your nodes.
8. On your chosen node, create a firewall rule that allows TCP traffic on your node port. For example, if your Service has a NodePort value of 31568, create a firewall rule that allows TCP traffic on port 31568. Different cloud providers offer different ways of configuring firewall rules.
9. Use the node address and node port to access the Hello World application:

```
curl http://<public-node-ip>:<node-port>
```

where `<public-node-ip>` is the public IP address of your node, and `<node-port>` is the NodePort value for your service. The response to a successful request is a hello message:

```
Hello Kubernetes!
```

Using a service configuration file

As an alternative to using `kubectl expose`, you can use a [service configuration file](#) to create a Service.

Cleaning up

To delete the Service, enter this command:

```
kubectl delete services example-service
```

To delete the Deployment, the ReplicaSet, and the Pods that are running the Hello World application, enter this command:

```
kubectl delete deployment hello-world
```

What's next

Follow the [Connecting Applications with Services](#) tutorial.

Connect a Frontend to a Backend Using Services

This task shows how to create a *frontend* and a *backend* microservice. The backend microservice is a hello greeter. The frontend exposes the backend using nginx and a Kubernetes [Service](#) object.

Objectives

- Create and run a sample hello backend microservice using a [Deployment](#) object.
- Use a Service object to send traffic to the backend microservice's multiple replicas.
- Create and run a nginx frontend microservice, also using a Deployment object.
- Configure the frontend microservice to send traffic to the backend microservice.
- Use a Service object of type=LoadBalancer to expose the frontend microservice outside the cluster.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

This task uses [Services with external load balancers](#), which require a supported environment. If your environment does not support this, you can use a Service of type [NodePort](#) instead.

Creating the backend using a Deployment

The backend is a simple hello greeter microservice. Here is the configuration file for the backend Deployment:

[service/access/backend-deployment.yaml](#)

```
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend
spec:
  selector:
    matchLabels:
      app: hello
      tier: backend
      track: stable
  replicas: 3
  template:
    metadata:
      labels:
        app: hello
        tier: backend
        track: stable
    spec:
      containers:
        - name: hello
          image: "gcr.io/google-samples/hello-go-gke:1.0"
          ports:
            - name: http
              containerPort: 80
...

```

Create the backend Deployment:

```
kubectl apply -f https://k8s.io/examples/service/access/backend-deployment.yaml
```

View information about the backend Deployment:

```
kubectl describe deployment backend
```

The output is similar to this:

```
Name:                backend
Namespace:           default
CreationTimestamp:    Mon, 24 Oct 2016 14:21:02 -0700
Labels:               app=hello
                     tier=backend
                     track=stable

```

```

Annotations:      deployment.kubernetes.io/revision=1
Selector:         app=hello,tier=backend,track=stable
Replicas:         3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:     RollingUpdate
MinReadySeconds:  0
RollingUpdateStrategy: 1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=hello
           tier=backend
           track=stable
Containers:
  hello:
    Image:      "gcr.io/google-samples/hello-go-gke:1.0"
    Port:       80/TCP
    Environment: <none>
    Mounts:      <none>
    Volumes:     <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   hello-3621623197 (3/3 replicas created)
Events:
...

```

Creating the hello Service object

The key to sending requests from a frontend to a backend is the backend Service. A Service creates a persistent IP address and DNS name entry so that the backend microservice can always be reached. A Service uses [selectors](#) to find the Pods that it routes traffic to.

First, explore the Service configuration file:

[service/access/backend-service.yaml](#)

```

---
apiVersion: v1
kind: Service
metadata:
  name: hello
spec:
  selector:
    app: hello
    tier: backend
  ports:
  - protocol: TCP
    port: 80
    targetPort: http
...

```

In the configuration file, you can see that the Service, named `hello` routes traffic to Pods that have the labels `app: hello` and `tier: backend`.

Create the backend Service:

```
kubectl apply -f https://k8s.io/examples/service/access/backend-service.yaml
```

At this point, you have a backend Deployment running three replicas of your `hello` application, and you have a Service that can route traffic to them. However, this service is neither available nor resolvable outside the cluster.

Creating the frontend

Now that you have your backend running, you can create a frontend that is accessible outside the cluster, and connects to the backend by proxying requests to it.

The frontend sends requests to the backend worker Pods by using the DNS name given to the backend Service. The DNS name is `hello`, which is the value of the `name` field in the `examples/service/access/backend-service.yaml` configuration file.

The Pods in the frontend Deployment run a `nginx` image that is configured to proxy requests to the `hello` backend Service. Here is the `nginx` configuration file:

[service/access/frontend-nginx.conf](https://k8s.io/examples/service/access/frontend-nginx.conf)

```
# The identifier Backend is internal to nginx, and used to name this specific upstream
upstream Backend {
    # hello is the internal DNS name used by the backend Service inside Kubernetes
    server hello;
}

server {
    listen 80;

    location / {
        # The following statement will proxy traffic to the upstream named Backend
        proxy_pass http://Backend;
    }
}
```

Similar to the backend, the frontend has a Deployment and a Service. An important difference to notice between the backend and frontend services, is that the configuration for the frontend Service has `type: LoadBalancer`, which means that the Service uses a load balancer provisioned by your cloud provider and will be accessible from outside the cluster.

[service/access/frontend-service.yaml](https://k8s.io/examples/service/access/frontend-service.yaml)

```
---
apiVersion: v1
```

```
kind: Service
metadata:
  name: frontend
spec:
  selector:
    app: hello
    tier: frontend
  ports:
  - protocol: "TCP"
    port: 80
    targetPort: 80
  type: LoadBalancer
...
```

[service/access/frontend-deployment.yaml](https://k8s.io/examples/service/access/frontend-deployment.yaml)

```
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend
spec:
  selector:
    matchLabels:
      app: hello
      tier: frontend
      track: stable
  replicas: 1
  template:
    metadata:
      labels:
        app: hello
        tier: frontend
        track: stable
    spec:
      containers:
      - name: nginx
        image: "gcr.io/google-samples/hello-frontend:1.0"
        lifecycle:
          preStop:
            exec:
              command: ["/usr/sbin/nginx","-s","quit"]
...
```

Create the frontend Deployment and Service:

```
kubectl apply -f https://k8s.io/examples/service/access/frontend-deployment.yaml
kubectl apply -f https://k8s.io/examples/service/access/frontend-service.yaml
```

The output verifies that both resources were created:

```
deployment.apps/frontend created
service/frontend created
```

Note:

The nginx configuration is baked into the [container image](#). A better way to do this would be to use a [ConfigMap](#), so that you can change the configuration more easily.

Interact with the frontend Service

Once you've created a Service of type LoadBalancer, you can use this command to find the external IP:

```
kubectl get service frontend --watch
```

This displays the configuration for the frontend Service and watches for changes. Initially, the external IP is listed as <pending>:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|----------|--------------|---------------|-------------|---------|-----|
| frontend | LoadBalancer | 10.51.252.116 | <pending> | 80/TCP | 10s |

As soon as an external IP is provisioned, however, the configuration updates to include the new IP under the EXTERNAL-IP heading:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|----------|--------------|---------------|-----------------|---------|-----|
| frontend | LoadBalancer | 10.51.252.116 | XXX.XXX.XXX.XXX | 80/TCP | 1m |

That IP can now be used to interact with the frontend service from outside the cluster.

Send traffic through the frontend

The frontend and backend are now connected. You can hit the endpoint by using the curl command on the external IP of your frontend Service.

```
curl http://${EXTERNAL_IP} # replace this with the EXTERNAL-IP you saw earlier
```

The output shows the message generated by the backend:

```
{"message":"Hello"}
```

Cleaning up

To delete the Services, enter this command:

```
kubectl delete services frontend backend
```

To delete the Deployments, the ReplicaSets and the Pods that are running the backend and frontend applications, enter this command:

```
kubectl delete deployment frontend backend
```

What's next

- Learn more about [Services](#)

- Learn more about [ConfigMaps](#)
- Learn more about [DNS for Service and Pods](#)

Create an External Load Balancer

This page shows how to create an external load balancer.

When creating a [Service](#), you have the option of automatically creating a cloud load balancer. This provides an externally-accessible IP address that sends traffic to the correct port on your cluster nodes, *provided your cluster runs in a supported environment and is configured with the correct cloud load balancer provider package.*

You can also use an [Ingress](#) in place of Service. For more information, check the [Ingress](#) documentation.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your cluster must be running in a cloud or other environment that already has support for configuring external load balancers.

Create a Service

Create a Service from a manifest

To create an external load balancer, add the following line to your Service manifest:

```
type: LoadBalancer
```

Your manifest might then look like:

```
apiVersion: v1
kind: Service
metadata:
  name: example-service
spec:
  selector:
    app: example
  ports:
    - port: 8765
      targetPort: 9376
  type: LoadBalancer
```

Create a Service using kubectl

You can alternatively create the service with the `kubectl expose` command and its `--type=LoadBalancer` flag:

```
kubectl expose deployment example --port=8765 --target-port=9376 \
  --name=example-service --type=LoadBalancer
```

This command creates a new Service using the same selectors as the referenced resource (in the case of the example above, a [Deployment](#) named example).

For more information, including optional flags, refer to the [kubectl expose reference](#).

Finding your IP address

You can find the IP address created for your service by getting the service information through `kubectl`:

```
kubectl describe services example-service
```

which should produce output similar to:

```
Name:          example-service
Namespace:     default
Labels:        app=example
Annotations:   <none>
Selector:      app=example
Type:          LoadBalancer
IP Families:   <none>
IP:            10.3.22.96
IPs:           10.3.22.96
LoadBalancer Ingress: 192.0.2.89
Port:          <unset> 8765/TCP
TargetPort:    9376/TCP
NodePort:      <unset> 30593/TCP
Endpoints:     172.17.0.3:9376
Session Affinity: None
External Traffic Policy: Cluster
Events:        <none>
```

The load balancer's IP address is listed next to `LoadBalancer Ingress`.

Note:

If you are running your service on Minikube, you can find the assigned IP address and port with:

```
minikube service example-service --url
```

Preserving the client source IP

By default, the source IP seen in the target container is *not the original source IP* of the client. To enable preservation of the client IP, the following fields can be configured in the `.spec` of the Service:

- `.spec.externalTrafficPolicy` - denotes if this Service desires to route external traffic to node-local or cluster-wide endpoints. There are two available options: Cluster (default) and Local. Cluster obscures the client source IP and may cause a second hop to another node, but should have good overall load-spreading. Local preserves the client source IP and avoids a second hop for LoadBalancer and NodePort type Services, but risks potentially imbalanced traffic spreading.
- `.spec.healthCheckNodePort` - specifies the health check node port (numeric port number) for the service. If you don't specify `healthCheckNodePort`, the service controller allocates a port from your cluster's NodePort range.
You can configure that range by setting an API server command line option, `--service-node-port-range`. The Service will use the user-specified `healthCheckNodePort` value if you specify it, provided that the Service type is set to LoadBalancer and `externalTrafficPolicy` is set to Local.

Setting `externalTrafficPolicy` to Local in the Service manifest activates this feature. For example:

```
apiVersion: v1
kind: Service
metadata:
  name: example-service
spec:
  selector:
    app: example
  ports:
    - port: 8765
      targetPort: 9376
  externalTrafficPolicy: Local
  type: LoadBalancer
```

Caveats and limitations when preserving source IPs

Load balancing services from some cloud providers do not let you configure different weights for each target.

With each target weighted equally in terms of sending traffic to Nodes, external traffic is not equally load balanced across different Pods. The external load balancer is unaware of the number of Pods on each node that are used as a target.

Where $\text{NumServicePods} \ll \text{NumNodes}$ or $\text{NumServicePods} \gg \text{NumNodes}$, a fairly close-to-equal distribution will be seen, even without weights.

Internal pod to pod traffic should behave similar to ClusterIP services, with equal probability across all pods.

Garbage collecting load balancers

FEATURE STATE: Kubernetes v1.17 [stable]

In usual case, the correlating load balancer resources in cloud provider should be cleaned up soon after a LoadBalancer type Service is deleted. But it is known that there are various corner cases where cloud resources are orphaned after the associated Service is deleted. Finalizer Protection for Service LoadBalancers was introduced to prevent this from happening. By using finalizers, a Service resource will never be deleted until the correlating load balancer resources are also deleted.

Specifically, if a Service has type LoadBalancer, the service controller will attach a finalizer named `service.kubernetes.io/load-balancer-cleanup`. The finalizer will only be removed after the load balancer resource is cleaned up. This prevents dangling load balancer resources even in corner cases such as the service controller crashing.

External load balancer providers

It is important to note that the datapath for this functionality is provided by a load balancer external to the Kubernetes cluster.

When the Service type is set to LoadBalancer, Kubernetes provides functionality equivalent to type equals ClusterIP to pods within the cluster and extends it by programming the (external to Kubernetes) load balancer with entries for the nodes hosting the relevant Kubernetes pods. The Kubernetes control plane automates the creation of the external load balancer, health checks (if needed), and packet filtering rules (if needed). Once the cloud provider allocates an IP address for the load balancer, the control plane looks up that external IP address and populates it into the Service object.

What's next

- Follow the [Connecting Applications with Services](#) tutorial
- Read about [Service](#)
- Read about [Ingress](#)

List All Container Images Running in a Cluster

This page shows how to use `kubectl` to list all of the Container images for Pods running in a cluster.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

In this exercise you will use `kubectl` to fetch all of the Pods running in a cluster, and format the output to pull out the list of Containers for each.

List all Container images in all namespaces

- Fetch all Pods in all namespaces using `kubectl get pods --all-namespaces`
- Format the output to include only the list of Container image names using `-o jsonpath={.items[*].spec['initContainers', 'containers'][*].image}`. This will recursively parse out the image field from the returned json.
 - See the [jsonpath reference](#) for further information on how to use jsonpath.
- Format the output using standard tools: `tr`, `sort`, `uniq`
 - Use `tr` to replace spaces with newlines
 - Use `sort` to sort the results
 - Use `uniq` to aggregate image counts

```
kubectl get pods --all-namespaces -o jsonpath="{.items[*].spec['initContainers', 'containers']
[*].image}" | \
tr -s '[:space:]' '\n' | \
sort | \
uniq -c
```

The jsonpath is interpreted as follows:

- `.items[*]`: for each returned value
- `.spec`: get the spec
- `['initContainers', 'containers'][*]`: for each container
- `.image`: get the image

Note:

When fetching a single Pod by name, for example `kubectl get pod nginx`, the `.items[*]` portion of the path should be omitted because a single Pod is returned instead of a list of items.

List Container images by Pod

The formatting can be controlled further by using the `range` operation to iterate over elements individually.

```
kubectl get pods --all-namespaces -o jsonpath='{range .items[*]}{"\n"}{{.metadata.name}}{":\t"}
{range .spec.containers[*]}{.image}}{"", ""}{end}{end}' | \
sort
```

List Container images filtering by Pod label

To target only Pods matching a specific label, use the `-l` flag. The following matches only Pods with labels matching `app=nginx`.

```
kubectl get pods --all-namespaces -o jsonpath="{.items[*].spec.containers[*].image}" -l app=nginx
```

List Container images filtering by Pod namespace

To target only pods in a specific namespace, use the `namespace` flag. The following matches only Pods in the `kube-system` namespace.

```
kubectl get pods --namespace kube-system -o jsonpath="{.items[*].spec.containers[*].image}"
```

List Container images using a go-template instead of jsonpath

As an alternative to `jsonpath`, Kubectl supports using [go-templates](#) for formatting the output:

```
kubectl get pods --all-namespaces -o go-template --template="{{range .items}}\n{{range .spec.containers}}{{.image}} {{end}}{{end}}\n"
```

What's next

Reference

- [Jsonpath](#) reference guide
- [Go template](#) reference guide

Set up Ingress on Minikube with the NGINX Ingress Controller

An [Ingress](#) is an API object that defines rules which allow external access to services in a cluster. An [Ingress controller](#) fulfills the rules set in the Ingress.

This page shows you how to set up a simple Ingress which routes requests to Service 'web' or 'web2' depending on the HTTP URI.

Before you begin

This tutorial assumes that you are using `minikube` to run a local Kubernetes cluster. Visit [Install tools](#) to learn how to install `minikube`.

Note:

This tutorial uses a container that requires the AMD64 architecture. If you are using minikube on a computer with a different CPU architecture, you could try using minikube with a driver that can emulate AMD64. For example, the Docker Desktop driver can do this.

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version 1.19. To check the version, enter kubectl version. If you are using an older Kubernetes version, switch to the documentation for that version.

Create a minikube cluster

If you haven't already set up a cluster locally, run minikube start to create a cluster.

Enable the Ingress controller

1. To enable the NGINX Ingress controller, run the following command:

```
minikube addons enable ingress
```

2. Verify that the NGINX Ingress controller is running

```
kubectl get pods -n ingress-nginx
```

Note:

It can take up to a minute before you see these pods running OK.

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|---|-------|-----------|----------|-----|
| ingress-nginx-admission-create-g9g49 | 0/1 | Completed | 0 | 11m |
| ingress-nginx-admission-patch-rqp78 | 0/1 | Completed | 1 | 11m |
| ingress-nginx-controller-59b45fb494-26npt | 1/1 | Running | 0 | 11m |

Deploy a hello, world app

1. Create a Deployment using the following command:

```
kubectl create deployment web --image=gcr.io/google-samples/hello-app:1.0
```

The output should be:

```
deployment.apps/web created
```

Verify that the Deployment is in a Ready state:

```
kubectl get deployment web
```

The output should be similar to:

| NAME | READY | UP-TO-DATE | AVAILABLE | AGE |
|------|-------|------------|-----------|-----|
| web | 1/1 | 1 | 1 | 53s |

2. Expose the Deployment:

```
kubectl expose deployment web --type=NodePort --port=8080
```

The output should be:

```
service/web exposed
```

3. Verify the Service is created and is available on a node port:

```
kubectl get service web
```

The output is similar to:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|------|----------|----------------|-------------|----------------|-----|
| web | NodePort | 10.104.133.249 | <none> | 8080:31637/TCP | 12m |

4. Visit the Service via NodePort, using the [minikube service](#) command. Follow the instructions for your platform:

- [Linux](#)
- [MacOS](#)

```
minikube service web --url
```

The output is similar to:

```
http://172.17.0.15:31637
```

Invoke the URL obtained in the output of the previous step:

```
curl http://172.17.0.15:31637
```

```
# The command must be run in a separate terminal.  
minikube service web --url
```

The output is similar to:

```
http://127.0.0.1:62445
```

! Because you are using a Docker driver on darwin, the terminal needs to be open to run it.

From a different terminal, invoke the URL obtained in the output of the previous step:

```
curl http://127.0.0.1:62445
```

The output is similar to:

```
Hello, world!  
Version: 1.0.0  
Hostname: web-55b8c6998d-8k564
```

You can now access the sample application via the Minikube IP address and NodePort. The next step lets you access the application using the Ingress resource.

Create an Ingress

The following manifest defines an Ingress that sends traffic to your Service via `hello-world.example`.

1. Create `example-ingress.yaml` from the following file:

[service/networking/example-ingress.yaml](https://k8s.io/examples/service/networking/example-ingress.yaml)

```
apiVersion: networking.k8s.io/v1  
kind: Ingress  
metadata:  
  name: example-ingress  
  annotations:  
    nginx.ingress.kubernetes.io/rewrite-target: /$1  
spec:  
  rules:  
    - host: hello-world.example  
      http:  
        paths:  
          - path: /  
            pathType: Prefix  
            backend:  
              service:  
                name: web  
                port:  
                  number: 8080
```

2. Create the Ingress object by running the following command:

```
kubectl apply -f https://k8s.io/examples/service/networking/example-ingress.yaml
```

The output should be:

```
ingress.networking.k8s.io/example-ingress created
```

3. Verify the IP address is set:

```
kubectl get ingress
```

Note:

This can take a couple of minutes.

You should see an IPv4 address in the ADDRESS column; for example:

| NAME | CLASS | HOSTS | ADDRESS | PORTS | AGE |
|-----------------|-------|---------------------|-------------|-------|-----|
| example-ingress | nginx | hello-world.example | 172.17.0.15 | 80 | 38s |

4. Verify that the Ingress controller is directing traffic, by following the instructions for your platform:

Note:

The network is limited if using the Docker driver on MacOS (Darwin) and the Node IP is not reachable directly. To get ingress to work you'll need to open a new terminal and run minikube tunnel.

sudo permission is required for it, so provide the password when prompted.

- [Linux](#)
- [MacOS](#)

```
curl --resolve "hello-world.example:80:$( minikube ip )" -i http://hello-world.example
```

```
minikube tunnel
```

The output is similar to:

```
Tunnel successfully started
```

```
NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...
```

```
The service/ingress example-ingress requires privileged ports to be exposed: [80 443]
sudo permission will be asked for it.
Starting tunnel for service example-ingress.
```

From within a new terminal, invoke the following command:

```
curl --resolve "hello-world.example:80:127.0.0.1" -i http://hello-world.example
```

You should see:

```
Hello, world!
Version: 1.0.0
Hostname: web-55b8c6998d-8k564
```

5. Optionally, you can also visit hello-world.example from your browser.

Add a line to the bottom of the /etc/hosts file on your computer (you will need administrator access):

- [Linux](#)
- [MacOS](#)

Look up the external IP address as reported by minikube

```
minikube ip
```

```
172.17.0.15 hello-world.example
```

Note:

Change the IP address to match the output from minikube ip.

```
127.0.0.1 hello-world.example
```

After you make this change, your web browser sends requests for hello-world.example URLs to Minikube.

Create a second Deployment

1. Create another Deployment using the following command:

```
kubectl create deployment web2 --image=gcr.io/google-samples/hello-app:2.0
```

The output should be:

```
deployment.apps/web2 created
```

Verify that the Deployment is in a Ready state:

```
kubectl get deployment web2
```

The output should be similar to:

| NAME | READY | UP-TO-DATE | AVAILABLE | AGE |
|------|-------|------------|-----------|-----|
| web2 | 1/1 | 1 | 1 | 16s |

2. Expose the second Deployment:

```
kubectl expose deployment web2 --port=8080 --type=NodePort
```

The output should be:

```
service/web2 exposed
```

Edit the existing Ingress

1. Edit the existing example-ingress.yaml manifest, and add the following lines at the end:

```
- path: /v2
  pathType: Prefix
  backend:
    service:
      name: web2
```



```
port:
  number: 8080
```

2. Apply the changes:

```
kubectl apply -f example-ingress.yaml
```

You should see:

```
ingress.networking/example-ingress configured
```

Test your Ingress

1. Access the 1st version of the Hello World app.

- [Linux](#)
- [MacOS](#)

```
curl --resolve "hello-world.example:80:$( minikube ip )" -i http://hello-world.example
```

```
minikube tunnel
```

The output is similar to:

```
Tunnel successfully started
```

```
NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...
```

```
The service/ingress example-ingress requires privileged ports to be exposed: [80 443]
sudo permission will be asked for it.
Starting tunnel for service example-ingress.
```

From within a new terminal, invoke the following command:

```
curl --resolve "hello-world.example:80:127.0.0.1" -i http://hello-world.example
```

The output is similar to:

```
Hello, world!
Version: 1.0.0
Hostname: web-55b8c6998d-8k564
```

2. Access the 2nd version of the Hello World app.

- [Linux](#)
- [MacOS](#)

```
curl --resolve "hello-world.example:80:$( minikube ip )" -i http://hello-world.example/v2
```

```
minikube tunnel
```

The output is similar to:

```
Tunnel successfully started
```

NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...

The service/ingress example-ingress requires privileged ports to be exposed: [80 443] sudo permission will be asked for it.
Starting tunnel for service example-ingress.

From within a new terminal, invoke the following command:

```
curl --resolve "hello-world.example:80:127.0.0.1" -i http://hello-world.example/v2
```

The output is similar to:

```
Hello, world!  
Version: 2.0.0  
Hostname: web2-75cd47646f-t8cjk
```

Note:

If you did the optional step to update `/etc/hosts`, you can also visit `hello-world.example` and `hello-world.example/v2` from your browser.

What's next

- Read more about [Ingress](#)
- Read more about [Ingress Controllers](#)
- Read more about [Services](#)

Communicate Between Containers in the Same Pod Using a Shared Volume

This page shows how to use a Volume to communicate between two Containers running in the same Pod. See also how to allow processes to communicate by [sharing process namespace](#) between containers.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

Creating a Pod that runs two Containers

In this exercise, you create a Pod that runs two Containers. The two containers share a Volume that they can use to communicate. Here is the configuration file for the Pod:

[pods/two-container-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: two-containers
spec:

  restartPolicy: Never

  volumes:
  - name: shared-data
    emptyDir: {}

  containers:

  - name: nginx-container
    image: nginx
    volumeMounts:
    - name: shared-data
      mountPath: /usr/share/nginx/html

  - name: debian-container
    image: debian
    volumeMounts:
    - name: shared-data
      mountPath: /pod-data
    command: ["/bin/sh"]
    args: ["-c", "echo Hello from the debian container > /pod-data/index.html"]
```

In the configuration file, you can see that the Pod has a Volume named shared-data.

The first container listed in the configuration file runs an nginx server. The mount path for the shared Volume is /usr/share/nginx/html. The second container is based on the debian image, and has a mount path of /pod-data. The second container runs the following command and then terminates.

```
echo Hello from the debian container > /pod-data/index.html
```

Notice that the second container writes the index.html file in the root directory of the nginx server.

Create the Pod and the two Containers:

```
kubectl apply -f https://k8s.io/examples/pods/two-container-pod.yaml
```

View information about the Pod and the Containers:

```
kubectl get pod two-containers --output=yaml
```

Here is a portion of the output:

```
apiVersion: v1
kind: Pod
metadata:
  ...
  name: two-containers
  namespace: default
  ...
spec:
  ...
  containerStatuses:

  - containerID: docker://c1d8abd1 ...
    image: debian
    ...
    lastState:
      terminated:
        ...
    name: debian-container
    ...

  - containerID: docker://96c1ff2c5bb ...
    image: nginx
    ...
    name: nginx-container
    ...
    state:
      running:
        ...
```

You can see that the debian Container has terminated, and the nginx Container is still running.

Get a shell to nginx Container:

```
kubectl exec -it two-containers -c nginx-container -- /bin/bash
```

In your shell, verify that nginx is running:

```
root@two-containers:/# apt-get update
root@two-containers:/# apt-get install curl procps
root@two-containers:/# ps aux
```

The output is similar to this:

| USER | PID | ... | STAT | START | TIME | COMMAND |
|------|-----|-----|------|-------|------|--|
| root | 1 | ... | Ss | 21:12 | 0:00 | nginx: master process nginx -g daemon off; |

Recall that the debian Container created the index.html file in the nginx root directory. Use curl to send a GET request to the nginx server:

```
root@two-containers:/# curl localhost
```

The output shows that nginx serves a web page written by the debian container:

```
Hello from the debian container
```

Discussion

The primary reason that Pods can have multiple containers is to support helper applications that assist a primary application. Typical examples of helper applications are data pullers, data pushers, and proxies. Helper and primary applications often need to communicate with each other. Typically this is done through a shared filesystem, as shown in this exercise, or through the loopback network interface, localhost. An example of this pattern is a web server along with a helper program that polls a Git repository for new updates.

The Volume in this exercise provides a way for Containers to communicate during the life of the Pod. If the Pod is deleted and recreated, any data stored in the shared Volume is lost.

What's next

- Learn more about [patterns for composite containers](#).
- Learn about [composite containers for modular architecture](#).
- See [Configuring a Pod to Use a Volume for Storage](#).
- See [Configure a Pod to share process namespace between containers in a Pod](#)
- See [Volume](#).
- See [Pod](#).

Configure DNS for a Cluster

Kubernetes offers a DNS cluster addon, which most of the supported environments enable by default. In Kubernetes version 1.11 and later, CoreDNS is recommended and is installed by default with kubeadm.

For more information on how to configure CoreDNS for a Kubernetes cluster, see the [Customizing DNS Service](#). An example demonstrating how to use Kubernetes DNS with kube-dns, see the [Kubernetes DNS sample plugin](#).

Access Services Running on Clusters

This page shows how to connect to services running on the Kubernetes cluster.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at

least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

Accessing services running on the cluster

In Kubernetes, [nodes](#), [pods](#) and [services](#) all have their own IPs. In many cases, the node IPs, pod IPs, and some service IPs on a cluster will not be routable, so they will not be reachable from a machine outside the cluster, such as your desktop machine.

Ways to connect

You have several options for connecting to nodes, pods and services from outside the cluster:

- Access services through public IPs.
 - Use a service with type NodePort or LoadBalancer to make the service reachable outside the cluster. See the [services](#) and [kubectl expose](#) documentation.
 - Depending on your cluster environment, this may only expose the service to your corporate network, or it may expose it to the internet. Think about whether the service being exposed is secure. Does it do its own authentication?
 - Place pods behind services. To access one specific pod from a set of replicas, such as for debugging, place a unique label on the pod and create a new service which selects this label.
 - In most cases, it should not be necessary for application developer to directly access nodes via their nodeIPs.
- Access services, nodes, or pods using the Proxy Verb.
 - Does apiserver authentication and authorization prior to accessing the remote service. Use this if the services are not secure enough to expose to the internet, or to gain access to ports on the node IP, or for debugging.
 - Proxies may cause problems for some web applications.
 - Only works for HTTP/HTTPS.
 - Described [here](#).
- Access from a node or pod in the cluster.
 - Run a pod, and then connect to a shell in it using [kubectl exec](#). Connect to other nodes, pods, and services from that shell.
 - Some clusters may allow you to ssh to a node in the cluster. From there you may be able to access cluster services. This is a non-standard method, and will work on some clusters but not others. Browsers and other tools may or may not be installed. Cluster DNS may not work.

Discovering builtin services

Typically, there are several services which are started on a cluster by kube-system. Get a list of these with the `kubectl cluster-info` command:

```
kubectl cluster-info
```

The output is similar to this:

Kubernetes master is running at `https://192.0.2.1`
elasticsearch-logging is running at `https://192.0.2.1/api/v1/namespaces/kube-system/services/elasticsearch-logging/proxy`
kibana-logging is running at `https://192.0.2.1/api/v1/namespaces/kube-system/services/kibana-logging/proxy`
kube-dns is running at `https://192.0.2.1/api/v1/namespaces/kube-system/services/kube-dns/proxy`
grafana is running at `https://192.0.2.1/api/v1/namespaces/kube-system/services/monitoring-grafana/proxy`
heapster is running at `https://192.0.2.1/api/v1/namespaces/kube-system/services/monitoring-heapster/proxy`

This shows the proxy-verb URL for accessing each service. For example, this cluster has cluster-level logging enabled (using Elasticsearch), which can be reached at `https://192.0.2.1/api/v1/namespaces/kube-system/services/elasticsearch-logging/proxy/` if suitable credentials are passed, or through a kubectl proxy at, for example: `http://localhost:8080/api/v1/namespaces/kube-system/services/elasticsearch-logging/proxy/`.

Note:

See [Access Clusters Using the Kubernetes API](#) for how to pass credentials or use kubectl proxy.

Manually constructing apiserver proxy URLs

As mentioned above, you use the `kubectl cluster-info` command to retrieve the service's proxy URL. To create proxy URLs that include service endpoints, suffixes, and parameters, you append to the service's proxy URL: `http://<kubernetes_master_address>/api/v1/namespaces/<namespace_name>/services/[<https>]<service_name>[:<port_name>]/proxy`

If you haven't specified a name for your port, you don't have to specify `port_name` in the URL. You can also use the port number in place of the `port_name` for both named and unnamed ports.

By default, the API server proxies to your service using HTTP. To use HTTPS, prefix the service name with `https::`: `http://<kubernetes_master_address>/api/v1/namespaces/<namespace_name>/services/<service_name>/proxy`

The supported formats for the `<service_name>` segment of the URL are:

- `<service_name>` - proxies to the default or unnamed port using http
- `<service_name>:<port_name>` - proxies to the specified port name or port number using http
- `https:<service_name>:` - proxies to the default or unnamed port using https (note the trailing colon)
- `https:<service_name>:<port_name>` - proxies to the specified port name or port number using https

Examples

- To access the Elasticsearch service endpoint `_search?q=user:kimchy`, you would use:

```
http://192.0.2.1/api/v1/namespaces/kube-system/services/elasticsearch-logging/proxy/_search?q=user:kimchy
```

- To access the Elasticsearch cluster health information `_cluster/health?pretty=true`, you
- would use:

```
https://192.0.2.1/api/v1/namespaces/kube-system/services/elasticsearch-logging/proxy/_cluster/health?pretty=true
```

The health information is similar to this:

```
{
  "cluster_name" : "kubernetes_logging",
  "status" : "yellow",
  "timed_out" : false,
  "number_of_nodes" : 1,
  "number_of_data_nodes" : 1,
  "active_primary_shards" : 5,
  "active_shards" : 5,
  "relocating_shards" : 0,
  "initializing_shards" : 0,
  "unassigned_shards" : 5
}
```

- To access the *https* Elasticsearch service health information `_cluster/health?pretty=true`, you would use:

```
https://192.0.2.1/api/v1/namespaces/kube-system/services/https:elasticsearch-logging:/proxy/_cluster/health?pretty=true
```

Using web browsers to access services running on the cluster

You may be able to put an apiserver proxy URL into the address bar of a browser. However:

- Web browsers cannot usually pass tokens, so you may need to use basic (password) auth. Apiserver can be configured to accept basic auth, but your cluster may not be configured to accept basic auth.
- Some web apps may not work, particularly those with client side javascript that construct URLs in a way that is unaware of the proxy path prefix.

Extend Kubernetes

Understand advanced ways to adapt your Kubernetes cluster to the needs of your work environment.

[Configure the Aggregation Layer](#)

[Use Custom Resources](#)

[Set up an Extension API Server](#)

[Configure Multiple Schedulers](#)

[Use an HTTP Proxy to Access the Kubernetes API](#)

Configure the Aggregation Layer

Configuring the [aggregation layer](#) allows the Kubernetes apiserver to be extended with additional APIs, which are not part of the core Kubernetes APIs.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

Note:

There are a few setup requirements for getting the aggregation layer working in your environment to support mutual TLS auth between the proxy and extension apiservers. Kubernetes and the kube-apiserver have multiple CAs, so make sure that the proxy is signed by the aggregation layer CA and not by something else, like the Kubernetes general CA.

Caution:

Reusing the same CA for different client types can negatively impact the cluster's ability to function. For more information, see [CA Reusage and Conflicts](#).

Authentication Flow

Unlike Custom Resource Definitions (CRDs), the Aggregation API involves another server - your Extension apiserver - in addition to the standard Kubernetes apiserver. The Kubernetes apiserver will need to communicate with your extension apiserver, and your extension apiserver will need to communicate with the Kubernetes apiserver. In order for this communication to be secured, the Kubernetes apiserver uses x509 certificates to authenticate itself to the extension apiserver.

This section describes how the authentication and authorization flows work, and how to configure them.

The high-level flow is as follows:

1. Kubernetes apiserver: authenticate the requesting user and authorize their rights to the requested API path.
2. Kubernetes apiserver: proxy the request to the extension apiserver

3. Extension apiserver: authenticate the request from the Kubernetes apiserver
4. Extension apiserver: authorize the request from the original user
5. Extension apiserver: execute

The rest of this section describes these steps in detail.

The flow can be seen in the following diagram.

aggregation auth flows

The source for the above swimlanes can be found in the source of this document.

Kubernetes Apiserver Authentication and Authorization

A request to an API path that is served by an extension apiserver begins the same way as all API requests: communication to the Kubernetes apiserver. This path already has been registered with the Kubernetes apiserver by the extension apiserver.

The user communicates with the Kubernetes apiserver, requesting access to the path. The Kubernetes apiserver uses standard authentication and authorization configured with the Kubernetes apiserver to authenticate the user and authorize access to the specific path.

For an overview of authenticating to a Kubernetes cluster, see ["Authenticating to a Cluster"](#). For an overview of authorization of access to Kubernetes cluster resources, see ["Authorization Overview"](#).

Everything to this point has been standard Kubernetes API requests, authentication and authorization.

The Kubernetes apiserver now is prepared to send the request to the extension apiserver.

Kubernetes Apiserver Proxies the Request

The Kubernetes apiserver now will send, or proxy, the request to the extension apiserver that registered to handle the request. In order to do so, it needs to know several things:

1. How should the Kubernetes apiserver authenticate to the extension apiserver, informing the extension apiserver that the request, which comes over the network, is coming from a valid Kubernetes apiserver?
2. How should the Kubernetes apiserver inform the extension apiserver of the username and group for which the original request was authenticated?

In order to provide for these two, you must configure the Kubernetes apiserver using several flags.

Kubernetes Apiserver Client Authentication

The Kubernetes apiserver connects to the extension apiserver over TLS, authenticating itself using a client certificate. You must provide the following to the Kubernetes apiserver upon startup, using the provided flags:

- private key file via `--proxy-client-key-file`
- signed client certificate file via `--proxy-client-cert-file`

- certificate of the CA that signed the client certificate file via `--requestheader-client-ca-file`
- valid Common Name values (CNs) in the signed client certificate via `--requestheader-allowed-names`

The Kubernetes apiserver will use the files indicated by `--proxy-client-*file` to authenticate to the extension apiserver. In order for the request to be considered valid by a compliant extension apiserver, the following conditions must be met:

1. The connection must be made using a client certificate that is signed by the CA whose certificate is in `--requestheader-client-ca-file`.
2. The connection must be made using a client certificate whose CN is one of those listed in `--requestheader-allowed-names`.

Note:

You can set this option to blank as `--requestheader-allowed-names=""`. This will indicate to an extension apiserver that *any* CN is acceptable.

When started with these options, the Kubernetes apiserver will:

1. Use them to authenticate to the extension apiserver.
2. Create a configmap in the kube-system namespace called `extension-apiserver-authentication`, in which it will place the CA certificate and the allowed CNs. These in turn can be retrieved by extension apiservers to validate requests.

Note that the same client certificate is used by the Kubernetes apiserver to authenticate against *all* extension apiservers. It does not create a client certificate per extension apiserver, but rather a single one to authenticate as the Kubernetes apiserver. This same one is reused for all extension apiserver requests.

Original Request Username and Group

When the Kubernetes apiserver proxies the request to the extension apiserver, it informs the extension apiserver of the username and group with which the original request successfully authenticated. It provides these in http headers of its proxied request. You must inform the Kubernetes apiserver of the names of the headers to be used.

- the header in which to store the username via `--requestheader-username-headers`
- the header in which to store the group via `--requestheader-group-headers`
- the prefix to append to all extra headers via `--requestheader-extra-headers-prefix`

These header names are also placed in the `extension-apiserver-authentication` configmap, so they can be retrieved and used by extension apiservers.

Extension Apiserver Authenticates the Request

The extension apiserver, upon receiving a proxied request from the Kubernetes apiserver, must validate that the request actually did come from a valid authenticating proxy, which role the Kubernetes apiserver is fulfilling. The extension apiserver validates it via:

1. Retrieve the following from the configmap in kube-system, as described above:
 - Client CA certificate
 - List of allowed names (CNs)

- Header names for username, group and extra info
- 2. Check that the TLS connection was authenticated using a client certificate which:
 - Was signed by the CA whose certificate matches the retrieved CA certificate.
 - Has a CN in the list of allowed CNs, unless the list is blank, in which case all CNs are allowed.
 - Extract the username and group from the appropriate headers

If the above passes, then the request is a valid proxied request from a legitimate authenticating proxy, in this case the Kubernetes apiserver.

Note that it is the responsibility of the extension apiserver implementation to provide the above. Many do it by default, leveraging the `k8s.io/apiserver/` package. Others may provide options to override it using command-line options.

In order to have permission to retrieve the configmap, an extension apiserver requires the appropriate role. There is a default role named `extension-apiserver-authentication-reader` in the `kube-system` namespace which can be assigned.

Extension Apiserver Authorizes the Request

The extension apiserver now can validate that the user/group retrieved from the headers are authorized to execute the given request. It does so by sending a standard [SubjectAccessReview](#) request to the Kubernetes apiserver.

In order for the extension apiserver to be authorized itself to submit the `SubjectAccessReview` request to the Kubernetes apiserver, it needs the correct permissions. Kubernetes includes a default `ClusterRole` named `system:auth-delegator` that has the appropriate permissions. It can be granted to the extension apiserver's service account.

Extension Apiserver Executes

If the `SubjectAccessReview` passes, the extension apiserver executes the request.

Enable Kubernetes Apiserver flags

Enable the aggregation layer via the following kube-apiserver flags. They may have already been taken care of by your provider.

```
--requestheader-client-ca-file=<path to aggregator CA cert>
--requestheader-allowed-names=front-proxy-client
--requestheader-extra-headers-prefix=X-Remote-Extra-
--requestheader-group-headers=X-Remote-Group
--requestheader-username-headers=X-Remote-User
--proxy-client-cert-file=<path to aggregator proxy cert>
--proxy-client-key-file=<path to aggregator proxy key>
```

CA Reusage and Conflicts

The Kubernetes apiserver has two client CA options:

- `--client-ca-file`
- `--requestheader-client-ca-file`

Each of these functions independently and can conflict with each other, if not used correctly.

- `--client-ca-file`: When a request arrives to the Kubernetes apiserver, if this option is enabled, the Kubernetes apiserver checks the certificate of the request. If it is signed by one of the CA certificates in the file referenced by `--client-ca-file`, then the request is treated as a legitimate request, and the user is the value of the common name `CN=`, while the group is the organization `O=`. See the [documentation on TLS authentication](#).
- `--requestheader-client-ca-file`: When a request arrives to the Kubernetes apiserver, if this option is enabled, the Kubernetes apiserver checks the certificate of the request. If it is signed by one of the CA certificates in the file reference by `--requestheader-client-ca-file`, then the request is treated as a potentially legitimate request. The Kubernetes apiserver then checks if the common name `CN=` is one of the names in the list provided by `--requestheader-allowed-names`. If the name is allowed, the request is approved; if it is not, the request is not.

If *both* `--client-ca-file` and `--requestheader-client-ca-file` are provided, then the request first checks the `--requestheader-client-ca-file` CA and then the `--client-ca-file`. Normally, different CAs, either root CAs or intermediate CAs, are used for each of these options; regular client requests match against `--client-ca-file`, while aggregation requests match against `--requestheader-client-ca-file`. However, if both use the *same* CA, then client requests that normally would pass via `--client-ca-file` will fail, because the CA will match the CA in `--requestheader-client-ca-file`, but the common name `CN=` will **not** match one of the acceptable common names in `--requestheader-allowed-names`. This can cause your kubelets and other control plane components, as well as end-users, to be unable to authenticate to the Kubernetes apiserver.

For this reason, use different CA certs for the `--client-ca-file` option - to authorize control plane components and end-users - and the `--requestheader-client-ca-file` option - to authorize aggregation apiserver requests.

Warning:

Do **not** reuse a CA that is used in a different context unless you understand the risks and the mechanisms to protect the CA's usage.

If you are not running kube-proxy on a host running the API server, then you must make sure that the system is enabled with the following kube-apiserver flag:

```
--enable-aggregator-routing=true
```

Register APIService objects

You can dynamically configure what client requests are proxied to extension apiserver. The following is an example registration:

```
apiVersion: apiregistration.k8s.io/v1
kind: APIService
metadata:
  name: <name of the registration object>
spec:
  group: <API group name this extension apiserver hosts>
  version: <API version this extension apiserver hosts>
```

```
groupPriorityMinimum: <priority this APIService for this group, see API documentation>
versionPriority: <prioritizes ordering of this version within a group, see API documentation>
service:
  namespace: <namespace of the extension apiserver service>
  name: <name of the extension apiserver service>
caBundle: <pem encoded ca cert that signs the server cert used by the webhook>
```

The name of an APIService object must be a valid [path segment name](#).

Contacting the extension apiserver

Once the Kubernetes apiserver has determined a request should be sent to an extension apiserver, it needs to know how to contact it.

The service stanza is a reference to the service for an extension apiserver. The service namespace and name are required. The port is optional and defaults to 443.

Here is an example of an extension apiserver that is configured to be called on port "1234", and to verify the TLS connection against the ServerName my-service-name.my-service-namespace.svc using a custom CA bundle.

```
apiVersion: apiregistration.k8s.io/v1
kind: APIService
...
spec:
  ...
  service:
    namespace: my-service-namespace
    name: my-service-name
    port: 1234
  caBundle: "Ci0tLS0tQk...<base64-encoded PEM bundle>...tLS0K"
...
```

What's next

- [Set up an extension api-server](#) to work with the aggregation layer.
- For a high level overview, see [Extending the Kubernetes API with the aggregation layer](#).
- Learn how to [Extend the Kubernetes API Using Custom Resource Definitions](#).

Use Custom Resources

[Extend the Kubernetes API with CustomResourceDefinitions](#)

[Versions in CustomResourceDefinitions](#)

Extend the Kubernetes API with CustomResourceDefinitions

This page shows how to install a [custom resource](#) into the Kubernetes API by creating a [CustomResourceDefinition](#).

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version 1.16. To check the version, enter `kubectl version`. If you are using an older version of Kubernetes that is still supported, switch to the documentation for that version to see advice that is relevant for your cluster.

Create a CustomResourceDefinition

When you create a new CustomResourceDefinition (CRD), the Kubernetes API Server creates a new RESTful resource path for each version you specify. The custom resource created from a CRD object can be either namespaced or cluster-scoped, as specified in the CRD's `spec.scope` field. As with existing built-in objects, deleting a namespace deletes all custom objects in that namespace. CustomResourceDefinitions themselves are non-namespaced and are available to all namespaces.

For example, if you save the following CustomResourceDefinition to `resourcedefinition.yaml`:

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  # name must match the spec fields below, and be in the form: <plural>.<group>
  name: crontabs.stable.example.com
spec:
  # group name to use for REST API: /apis/<group>/<version>
  group: stable.example.com
  # list of versions supported by this CustomResourceDefinition
  versions:
    - name: v1
      # Each version can be enabled/disabled by Served flag.
      served: true
      # One and only one version must be marked as the storage version.
      storage: true
    schema:
      openAPIV3Schema:
        type: object
        properties:
```

```

spec:
  type: object
  properties:
    cronSpec:
      type: string
    image:
      type: string
    replicas:
      type: integer
# either Namespaced or Cluster
scope: Namespaced
names:
  # plural name to be used in the URL: /apis/<group>/<version>/<plural>
  plural: crontabs
  # singular name to be used as an alias on the CLI and for display
  singular: crontab
  # kind is normally the CamelCased singular type. Your resource manifests use this.
  kind: CronTab
  # shortNames allow shorter string to match your resource on the CLI
  shortNames:
    - ct

```

and create it:

```
kubectl apply -f resourcedefinition.yaml
```

Then a new namespaced RESTful API endpoint is created at:

```
/apis/stable.example.com/v1/namespaces/*/crontabs/...
```

This endpoint URL can then be used to create and manage custom objects. The kind of these objects will be CronTab from the spec of the CustomResourceDefinition object you created above.

It might take a few seconds for the endpoint to be created. You can watch the Established condition of your CustomResourceDefinition to be true or watch the discovery information of the API server for your resource to show up.

Create custom objects

After the CustomResourceDefinition object has been created, you can create custom objects. Custom objects can contain custom fields. These fields can contain arbitrary JSON. In the following example, the cronSpec and image custom fields are set in a custom object of kind CronTab. The kind CronTab comes from the spec of the CustomResourceDefinition object you created above.

If you save the following YAML to my-crontab.yaml:

```

apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  name: my-new-cron-object
spec:

```



```
cronSpec: "* * * * */5"
image: my-awesome-cron-image
```

and create it:

```
kubectl apply -f my-crontab.yaml
```

You can then manage your CronTab objects using kubectl. For example:

```
kubectl get crontab
```

Should print a list like this:

| NAME | AGE |
|--------------------|-----|
| my-new-cron-object | 6s |

Resource names are not case-sensitive when using kubectl, and you can use either the singular or plural forms defined in the CRD, as well as any short names.

You can also view the raw YAML data:

```
kubectl get ct -o yaml
```

You should see that it contains the custom cronSpec and image fields from the YAML you used to create it:

```
apiVersion: v1
items:
- apiVersion: stable.example.com/v1
  kind: CronTab
  metadata:
    annotations:
      kubectl.kubernetes.io/last-applied-configuration: |
        {"apiVersion":"stable.example.com/v1","kind":"CronTab","metadata":{"annotations":
{"name":"my-new-cron-object","namespace":"default"},"spec":{"cronSpec":"* * * * */
5","image":"my-awesome-cron-image"}}
    creationTimestamp: "2021-06-20T07:35:27Z"
    generation: 1
    name: my-new-cron-object
    namespace: default
    resourceVersion: "1326"
    uid: 9aab1d66-628e-41bb-a422-57b8b3b1f5a9
  spec:
    cronSpec: '* * * * */5'
    image: my-awesome-cron-image
kind: List
metadata:
  resourceVersion: ""
  selfLink: ""
```

Delete a CustomResourceDefinition

When you delete a CustomResourceDefinition, the server will uninstall the RESTful API endpoint and delete all custom objects stored in it.

```
kubectl delete -f resourcedefinition.yaml
kubectl get crontabs
```

Error from server (NotFound): Unable to list {"stable.example.com" "v1" "crontabs"}: the server could not find the requested resource (get crontabs.stable.example.com)

If you later recreate the same CustomResourceDefinition, it will start out empty.

Specifying a structural schema

CustomResources store structured data in custom fields (alongside the built-in fields `apiVersion`, `kind` and `metadata`, which the API server validates implicitly). With [OpenAPI v3.0 validation](#) a schema can be specified, which is validated during creation and updates, compare below for details and limits of such a schema.

With `apiextensions.k8s.io/v1` the definition of a structural schema is mandatory for CustomResourceDefinitions. In the beta version of CustomResourceDefinition, the structural schema was optional.

A structural schema is an [OpenAPI v3.0 validation schema](#) which:

1. specifies a non-empty type (via `type` in OpenAPI) for the root, for each specified field of an object node (via `properties` or `additionalProperties` in OpenAPI) and for each item in an array node (via `items` in OpenAPI), with the exception of:
 - a node with `x-kubernetes-int-or-string: true`
 - a node with `x-kubernetes-preserve-unknown-fields: true`
2. for each field in an object and each item in an array which is specified within any of `allOf`, `anyOf`, `oneOf` or `not`, the schema also specifies the field/item outside of those logical junctors (compare example 1 and 2).
3. does not set `description`, `type`, `default`, `additionalProperties`, `nullable` within an `allOf`, `anyOf`, `oneOf` or `not`, with the exception of the two pattern for `x-kubernetes-int-or-string: true` (see below).
4. if `metadata` is specified, then only restrictions on `metadata.name` and `metadata.generateName` are allowed.

Non-structural example 1:

```
allOf:
- properties:
  foo:
  ...
```

conflicts with rule 2. The following would be correct:

```
properties:
  foo:
  ...
```

```
allOf:  
- properties:  
  foo:  
  ...
```

Non-structural example 2:

```
allOf:  
- items:  
  properties:  
  foo:  
  ...
```

conflicts with rule 2. The following would be correct:

```
items:  
  properties:  
  foo:  
  ...  
allOf:  
- items:  
  properties:  
  foo:  
  ...
```

Non-structural example 3:

```
properties:  
  foo:  
    pattern: "abc"  
  metadata:  
    type: object  
    properties:  
      name:  
        type: string  
        pattern: "^a"  
      finalizers:  
        type: array  
        items:  
          type: string  
          pattern: "my-finalizer"  
anyOf:  
- properties:  
  bar:  
    type: integer  
    minimum: 42  
required: ["bar"]  
description: "foo bar object"
```

is not a structural schema because of the following violations:

- the type at the root is missing (rule 1).
- the type of foo is missing (rule 1).
- bar inside of anyOf is not specified outside (rule 2).

- bar's type is within anyOf (rule 3).
- the description is set within anyOf (rule 3).
- metadata.finalizers might not be restricted (rule 4).

In contrast, the following, corresponding schema is structural:

```

type: object
description: "foo bar object"
properties:
  foo:
    type: string
    pattern: "abc"
  bar:
    type: integer
  metadata:
    type: object
    properties:
      name:
        type: string
        pattern: "^a"
anyOf:
- properties:
  bar:
    minimum: 42
    required: ["bar"]

```

Violations of the structural schema rules are reported in the NonStructural condition in the CustomResourceDefinition.

Field pruning

CustomResourceDefinitions store validated resource data in the cluster's persistence store, [etcd](#). As with native Kubernetes resources such as [ConfigMap](#), if you specify a field that the API server does not recognize, the unknown field is *pruned* (removed) before being persisted.

CRDs converted from `apiextensions.k8s.io/v1beta1` to `apiextensions.k8s.io/v1` might lack structural schemas, and `spec.preserveUnknownFields` might be true.

For legacy CustomResourceDefinition objects created as `apiextensions.k8s.io/v1beta1` with `spec.preserveUnknownFields` set to true, the following is also true:

- Pruning is not enabled.
- You can store arbitrary data.

For compatibility with `apiextensions.k8s.io/v1`, update your custom resource definitions to:

1. Use a structural OpenAPI schema.
2. Set `spec.preserveUnknownFields` to false.

If you save the following YAML to `my-crontab.yaml`:

```

apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:

```

```
name: my-new-cron-object
spec:
  cronSpec: "* * * * */5"
  image: my-awesome-cron-image
  someRandomField: 42
```

and create it:

```
kubectl create --validate=false -f my-crontab.yaml -o yaml
```

Your output is similar to:

```
apiVersion: stable.example.com/v1
kind: CronTab
metadata:
  creationTimestamp: 2017-05-31T12:56:35Z
  generation: 1
  name: my-new-cron-object
  namespace: default
  resourceVersion: "285"
  uid: 9423255b-4600-11e7-af6a-28d2447dc82b
spec:
  cronSpec: '* * * * */5'
  image: my-awesome-cron-image
```

Notice that the field `someRandomField` was pruned.

This example turned off client-side validation to demonstrate the API server's behavior, by adding the `--validate=false` command line option. Because the [OpenAPI validation schemas are also published](#) to clients, `kubectl` also checks for unknown fields and rejects those objects well before they would be sent to the API server.

Controlling pruning

By default, all unspecified fields for a custom resource, across all versions, are pruned. It is possible though to opt-out of that for specific sub-trees of fields by adding `x-kubernetes-preserve-unknown-fields: true` in the [structural OpenAPI v3 validation schema](#).

For example:

```
type: object
properties:
  json:
    x-kubernetes-preserve-unknown-fields: true
```

The field `json` can store any JSON value, without anything being pruned.

You can also partially specify the permitted JSON; for example:

```
type: object
properties:
  json:
    x-kubernetes-preserve-unknown-fields: true
```

```
type: object
description: this is arbitrary JSON
```

With this, only object type values are allowed.

Pruning is enabled again for each specified property (or `additionalProperties`):

```
type: object
properties:
  json:
    x-kubernetes-preserve-unknown-fields: true
    type: object
    properties:
      spec:
        type: object
        properties:
          foo:
            type: string
          bar:
            type: string
```

With this, the value:

```
json:
  spec:
    foo: abc
    bar: def
    something: x
  status:
    something: x
```

is pruned to:

```
json:
  spec:
    foo: abc
    bar: def
  status:
    something: x
```

This means that the `something` field in the specified `spec` object is pruned, but everything outside is not.

IntOrString

Nodes in a schema with `x-kubernetes-int-or-string: true` are excluded from rule 1, such that the following is structural:

```
type: object
properties:
  foo:
    x-kubernetes-int-or-string: true
```

Also those nodes are partially excluded from rule 3 in the sense that the following two patterns are allowed (exactly those, without variations in order to additional fields):

```
x-kubernetes-int-or-string: true
anyOf:
  - type: integer
  - type: string
...
```

and

```
x-kubernetes-int-or-string: true
allOf:
  - anyOf:
    - type: integer
    - type: string
  - ... # zero or more
...
```

With one of those specification, both an integer and a string validate.

In [Validation Schema Publishing](#), `x-kubernetes-int-or-string: true` is unfolded to one of the two patterns shown above.

RawExtension

RawExtensions (as in [runtime.RawExtension](#)) holds complete Kubernetes objects, i.e. with `apiVersion` and `kind` fields.

It is possible to specify those embedded objects (both completely without constraints or partially specified) by setting `x-kubernetes-embedded-resource: true`. For example:

```
type: object
properties:
  foo:
    x-kubernetes-embedded-resource: true
    x-kubernetes-preserve-unknown-fields: true
```

Here, the field `foo` holds a complete object, e.g.:

```
foo:
  apiVersion: v1
  kind: Pod
  spec:
  ...
```

Because `x-kubernetes-preserve-unknown-fields: true` is specified alongside, nothing is pruned. The use of `x-kubernetes-preserve-unknown-fields: true` is optional though.

With `x-kubernetes-embedded-resource: true`, the `apiVersion`, `kind` and `metadata` are implicitly specified and validated.

Serving multiple versions of a CRD

See [Custom resource definition versioning](#) for more information about serving multiple versions of your CustomResourceDefinition and migrating your objects from one version to another.

Advanced topics

Finalizers

Finalizers allow controllers to implement asynchronous pre-delete hooks. Custom objects support finalizers similar to built-in objects.

You can add a finalizer to a custom object like this:

```
apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  finalizers:
    - stable.example.com/finalizer
```

Identifiers of custom finalizers consist of a domain name, a forward slash and the name of the finalizer. Any controller can add a finalizer to any object's list of finalizers.

The first delete request on an object with finalizers sets a value for the `metadata.deletionTimestamp` field but does not delete it. Once this value is set, entries in the finalizers list can only be removed. While any finalizers remain it is also impossible to force the deletion of an object.

When the `metadata.deletionTimestamp` field is set, controllers watching the object execute any finalizers they handle and remove the finalizer from the list after they are done. It is the responsibility of each controller to remove its finalizer from the list.

The value of `metadata.deletionGracePeriodSeconds` controls the interval between polling updates.

Once the list of finalizers is empty, meaning all finalizers have been executed, the resource is deleted by Kubernetes.

Validation

Custom resources are validated via [OpenAPI v3 schemas](#), by x-kubernetes-validations when the [Validation Rules feature](#) is enabled, and you can add additional validation using [admission webhooks](#).

Additionally, the following restrictions are applied to the schema:

- These fields cannot be set:
 - definitions,
 - dependencies,
 - deprecated,

- discriminator,
 - id,
 - patternProperties,
 - readOnly,
 - writeOnly,
 - xml,
 - \$ref.
- The field `uniqueItems` cannot be set to `true`.
 - The field `additionalProperties` cannot be set to `false`.
 - The field `additionalProperties` is mutually exclusive with `properties`.

The `x-kubernetes-validations` extension can be used to validate custom resources using [Common Expression Language \(CEL\)](#) expressions when the [Validation rules](#) feature is enabled and the `CustomResourceDefinition` schema is a [structural schema](#).

Refer to the [structural schemas](#) section for other restrictions and `CustomResourceDefinition` features.

The schema is defined in the `CustomResourceDefinition`. In the following example, the `CustomResourceDefinition` applies the following validations on the custom object:

- `spec.cronSpec` must be a string and must be of the form described by the regular expression.
- `spec.replicas` must be an integer and must have a minimum value of 1 and a maximum value of 10.

Save the `CustomResourceDefinition` to `resourcedefinition.yaml`:

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: crontabs.stable.example.com
spec:
  group: stable.example.com
  versions:
    - name: v1
      served: true
      storage: true
      schema:
        # openAPIV3Schema is the schema for validating custom objects.
        openAPIV3Schema:
          type: object
          properties:
            spec:
              type: object
              properties:
                cronSpec:
                  type: string
                  pattern: '^(\\d+|\\*)(\\/\\d+)?(\\s+(\\d+|\\*)(\\/\\d+)?){4}$'
            image:
              type: string
```

```
    replicas:
      type: integer
      minimum: 1
      maximum: 10
  scope: Namespaced
  names:
    plural: crontabs
    singular: crontab
    kind: CronTab
    shortNames:
    - ct
```

and create it:

```
kubectl apply -f resourcedefinition.yaml
```

A request to create a custom object of kind CronTab is rejected if there are invalid values in its fields. In the following example, the custom object contains fields with invalid values:

- spec.cronSpec does not match the regular expression.
- spec.replicas is greater than 10.

If you save the following YAML to my-crontab.yaml:

```
apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  name: my-new-cron-object
spec:
  cronSpec: "* * * *"
  image: my-awesome-cron-image
  replicas: 15
```

and attempt to create it:

```
kubectl apply -f my-crontab.yaml
```

then you get an error:

```
The CronTab "my-new-cron-object" is invalid: [: Invalid value: map[string]interface {}
{"apiVersion":"stable.example.com/v1", "kind":"CronTab", "metadata":map[string]interface {}
{"name":"my-new-cron-object", "namespace":"default", "deletionTimestamp":interface {}(nil),
"deletionGracePeriodSeconds":(*int64)(nil), "creationTimestamp":"2017-09-05T05:20:07Z",
"uid":"e14d79e7-91f9-11e7-a598-f0761cb232d1", "clusterName":""}, "spec":map[string]interface {}
{"cronSpec":"* * * *", "image":"my-awesome-cron-image", "replicas":15}}:
validation failure list:
spec.cronSpec in body should match '^(\\d+|\\*)(\\d+)?(\\s+(\\d+|\\*)(\\d+)?){4}$'
spec.replicas in body should be less than or equal to 10
```

If the fields contain valid values, the object creation request is accepted.

Save the following YAML to my-crontab.yaml:

```
apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  name: my-new-cron-object
spec:
  cronSpec: "* * * * */5"
  image: my-awesome-cron-image
  replicas: 5
```

And create it:

```
kubectl apply -f my-crontab.yaml
crontab "my-new-cron-object" created
```

Validation ratcheting

FEATURE STATE: Kubernetes v1.30 [beta]

If you are using a version of Kubernetes older than v1.30, you need to explicitly enable the `CRDValidationRatcheting` [feature gate](#) to use this behavior, which then applies to all `CustomResourceDefinitions` in your cluster.

Provided you enabled the feature gate, Kubernetes implements *validation ratcheting* for `CustomResourceDefinitions`. The API server is willing to accept updates to resources that are not valid after the update, provided that each part of the resource that failed to validate was not changed by the update operation. In other words, any invalid part of the resource that remains invalid must have already been wrong. You cannot use this mechanism to update a valid resource so that it becomes invalid.

This feature allows authors of CRDs to confidently add new validations to the OpenAPIV3 schema under certain conditions. Users can update to the new schema safely without bumping the version of the object or breaking workflows.

While most validations placed in the OpenAPIV3 schema of a CRD support ratcheting, there are a few exceptions. The following OpenAPIV3 schema validations are not supported by ratcheting under the implementation in Kubernetes 1.30 and if violated will continue to throw an error as normally:

- Quantors
 - `allOf`
 - `oneOf`
 - `anyOf`
 - `not`
 - any validations in a descendent of one of these fields
- `x-kubernetes-validations` For Kubernetes 1.28, CRD validation rules][`#validation-rules`) are ignored by ratcheting. Starting with Alpha 2 in Kubernetes 1.29, `x-kubernetes-validations` are ratcheted only if they do not refer to `oldSelf`.

Transition Rules are never ratcheted: only errors raised by rules that do not use `oldSelf` will be automatically ratcheted if their values are unchanged.

To write custom ratcheting logic for CEL expressions, check out [optionalOldSelf](#).

- **x-kubernetes-list-type** Errors arising from changing the list type of a subschema will not be ratcheted. For example adding set onto a list with duplicates will always result in an error.
- **x-kubernetes-map-keys** Errors arising from changing the map keys of a list schema will not be ratcheted.
- **required** Errors arising from changing the list of required fields will not be ratcheted.
- **properties** Adding/removing/modifying the names of properties is not ratcheted, but changes to validations in each properties' schemas and subschemas may be ratcheted if the name of the property stays the same.
- **additionalProperties** To remove a previously specified additionalProperties validation will not be ratcheted.
- **metadata** Errors that come from Kubernetes' built-in validation of an object's metadata are not ratcheted (such as object name, or characters in a label value). If you specify your own additional rules for the metadata of a custom resource, that additional validation will be ratcheted.

Validation rules

FEATURE STATE: Kubernetes v1.29 [stable]

Validation rules use the [Common Expression Language \(CEL\)](#) to validate custom resource values. Validation rules are included in CustomResourceDefinition schemas using the x-kubernetes-validations extension.

The Rule is scoped to the location of the x-kubernetes-validations extension in the schema. And self variable in the CEL expression is bound to the scoped value.

All validation rules are scoped to the current object: no cross-object or stateful validation rules are supported.

For example:

```
...
openAPIV3Schema:
  type: object
  properties:
    spec:
      type: object
      x-kubernetes-validations:
        - rule: "self.minReplicas <= self.replicas"
          message: "replicas should be greater than or equal to minReplicas."
        - rule: "self.replicas <= self.maxReplicas"
          message: "replicas should be smaller than or equal to maxReplicas."
      properties:
        ...
        minReplicas:
          type: integer
        replicas:
          type: integer
```

```
maxReplicas:
  type: integer
required:
- minReplicas
- replicas
- maxReplicas
```

will reject a request to create this custom resource:

```
apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  name: my-new-cron-object
spec:
  minReplicas: 0
  replicas: 20
  maxReplicas: 10
```

with the response:

The CronTab "my-new-cron-object" is invalid:

* spec: Invalid value: map[string]interface {}{"maxReplicas":10, "minReplicas":0, "replicas":20}: replicas should be smaller than or equal to maxReplicas.

x-kubernetes-validations could have multiple rules. The rule under x-kubernetes-validations represents the expression which will be evaluated by CEL. The message represents the message displayed when validation fails. If message is unset, the above response would be:

The CronTab "my-new-cron-object" is invalid:

* spec: Invalid value: map[string]interface {}{"maxReplicas":10, "minReplicas":0, "replicas":20}: failed rule: self.replicas <= self.maxReplicas

Note:

You can quickly test CEL expressions in [CEL Playground](#).

Validation rules are compiled when CRDs are created/updated. The request of CRDs create/update will fail if compilation of validation rules fail. Compilation process includes type checking as well.

The compilation failure:

- `no_matching_overload`: this function has no overload for the types of the arguments.

For example, a rule like `self == true` against a field of integer type will get error:

```
Invalid value: apiextensions.ValidationRule{Rule:"self == true", Message:"": compilation
failed: ERROR: \<input>:1:6: found no matching overload for '_==' applied to '(int, bool)'
```

- `no_such_field`: does not contain the desired field.

For example, a rule like `self.nonExistingField > 0` against a non-existing field will return the following error:

```
Invalid value: apiextensions.ValidationRule{Rule:"self.nonExistingField > 0", Message:""}:
compilation failed: ERROR: \<input>:1:5: undefined field 'nonExistingField'
```

- invalid argument: invalid argument to macros.

For example, a rule like `has(self)` will return error:

```
Invalid value: apiextensions.ValidationRule{Rule:"has(self)", Message:""}: compilation
failed: ERROR: <input>:1:4: invalid argument to has() macro
```

Validation Rules Examples:

| Rule | Purpose |
|---|--|
| <code>self.minReplicas <= self.replicas && self.replicas <= self.maxReplicas</code> | Validate that the three fields defining replicas are ordered appropriately |
| <code>'Available' in self.stateCounts</code> | Validate that an entry with the 'Available' key exists in a map |
| <code>(size(self.list1) == 0) != (size(self.list2) == 0)</code> | Validate that one of two lists is non-empty, but not both |
| <code>!('MY_KEY' in self.map1) self['MY_KEY'].matches('^ [a-zA-Z]*\$')</code> | Validate the value of a map for a specific key, if it is in the map |
| <code>self.envvars.filter(e, e.name == 'MY_ENV').all(e, e.value.matches('^ [a-zA-Z]*\$'))</code> | Validate the 'value' field of a listMap entry where key field 'name' is 'MY_ENV' |
| <code>has(self.expired) && self.created + self.ttl < self.expired</code> | Validate that 'expired' date is after a 'create' date plus a 'ttl' duration |
| <code>self.health.startsWith('ok')</code> | Validate a 'health' string field has the prefix 'ok' |
| <code>self.widgets.exists(w, w.key == 'x' && w.foo < 10)</code> | Validate that the 'foo' property of a listMap item with a key 'x' is less than 10 |
| <code>type(self) == string ? self == '100%' : self == 1000</code> | Validate an int-or-string field for both the int and string cases |
| <code>self.metadata.name.startsWith(self.prefix)</code> | Validate that an object's name has the prefix of another field value |
| <code>self.set1.all(e, !(e in self.set2))</code> | Validate that two listSets are disjoint |
| <code>size(self.names) == size(self.details) && self.names.all(n, n in self.details)</code> | Validate the 'details' map is keyed by the items in the 'names' listSet |
| <code>size(self.clusters.filter(c, c.name == self.primary)) == 1</code> | Validate that the 'primary' property has one and only one occurrence in the 'clusters' listMap |

Xref: [Supported evaluation on CEL](#)

- If the Rule is scoped to the root of a resource, it may make field selection into any fields declared in the OpenAPIv3 schema of the CRD as well as `apiVersion`, `kind`, `metadata.name` and `metadata.generateName`. This includes selection of fields in both the spec and status in the same expression:

```
...
openAPIV3Schema:
  type: object
```

```

x-kubernetes-validations:
  - rule: "self.status.availableReplicas >= self.spec.minReplicas"
properties:
  spec:
    type: object
    properties:
      minReplicas:
        type: integer
    ...
  status:
    type: object
    properties:
      availableReplicas:
        type: integer

```

- If the Rule is scoped to an object with properties, the accessible properties of the object are field selectable via `self.field` and field presence can be checked via `has(self.field)`. Null valued fields are treated as absent fields in CEL expressions.

```

...
openAPIV3Schema:
  type: object
  properties:
    spec:
      type: object
      x-kubernetes-validations:
        - rule: "has(self.foo)"
      properties:
        ...
        foo:
          type: integer

```

- If the Rule is scoped to an object with `additionalProperties` (i.e. a map) the value of the map are accessible via `self[mapKey]`, map containment can be checked via `mapKey` in `self` and all entries of the map are accessible via CEL macros and functions such as `self.all(...)`.

```

...
openAPIV3Schema:
  type: object
  properties:
    spec:
      type: object
      x-kubernetes-validations:
        - rule: "self['xyz'].foo > 0"
      additionalProperties:
        ...
        type: object
        properties:
          foo:
            type: integer

```

- If the Rule is scoped to an array, the elements of the array are accessible via `self[i]` and also by macros and functions.

```

...
openAPIV3Schema:
  type: object
  properties:
    ...
    foo:
      type: array
      x-kubernetes-validations:
        - rule: "size(self) == 1"
      items:
        type: string

```

- If the Rule is scoped to a scalar, `self` is bound to the scalar value.

```

...
openAPIV3Schema:
  type: object
  properties:
    spec:
      type: object
      properties:
        ...
        foo:
          type: integer
          x-kubernetes-validations:
            - rule: "self > 0"

```

Examples:

| type of the field rule scoped to | Rule example |
|----------------------------------|--|
| root object | <code>self.status.actual <= self.spec.maxDesired</code> |
| map of objects | <code>self.components['Widget'].priority < 10</code> |
| list of integers | <code>self.values.all(value, value >= 0 && value < 100)</code> |
| string | <code>self.startsWith('kube')</code> |

The `apiVersion`, `kind`, `metadata.name` and `metadata.generateName` are always accessible from the root of the object and from any x-kubernetes-embedded-resource annotated objects. No other metadata properties are accessible.

Unknown data preserved in custom resources via `x-kubernetes-preserve-unknown-fields` is not accessible in CEL expressions. This includes:

- Unknown field values that are preserved by object schemas with `x-kubernetes-preserve-unknown-fields`.
- Object properties where the property schema is of an "unknown type". An "unknown type" is recursively defined as:
 - A schema with no type and `x-kubernetes-preserve-unknown-fields` set to true
 - An array where the items schema is of an "unknown type"
 - An object where the `additionalProperties` schema is of an "unknown type"

Only property names of the form `[a-zA-Z_.-][a-zA-Z0-9_.-]*` are accessible. Accessible property names are escaped according to the following rules when accessed in the expression:

| escape sequence | property name equivalent |
|------------------------------|--------------------------------------|
| <code>__underscores__</code> | <code>__</code> |
| <code>__dot__</code> | <code>.</code> |
| <code>__dash__</code> | <code>-</code> |
| <code>__slash__</code> | <code>/</code> |
| <code>__keyword__</code> | CEL RESERVED keyword |

Note: CEL RESERVED keyword needs to match the exact property name to be escaped (e.g. `int` in the word `sprint` would not be escaped).

Examples on escaping:

| property name | rule with escaped property name |
|------------------------|---|
| <code>namespace</code> | <code>self.__namespace__ > 0</code> |
| <code>x-prop</code> | <code>self.x__dash__prop > 0</code> |
| <code>redact__d</code> | <code>self.redact__underscores__d > 0</code> |
| <code>string</code> | <code>self.startsWith('kube')</code> |

Equality on arrays with `x-kubernetes-list-type` of `set` or `map` ignores element order, i.e., `[1, 2] == [2, 1]`. Concatenation on arrays with `x-kubernetes-list-type` use the semantics of the list type:

- `set`: `X + Y` performs a union where the array positions of all elements in `X` are preserved and non-intersecting elements in `Y` are appended, retaining their partial order.
- `map`: `X + Y` performs a merge where the array positions of all keys in `X` are preserved but the values are overwritten by values in `Y` when the key sets of `X` and `Y` intersect. Elements in `Y` with non-intersecting keys are appended, retaining their partial order.

Here is the declarations type mapping between OpenAPIv3 and CEL type:

| OpenAPIv3 type | CEL type |
|--|--|
| 'object' with Properties | object / "message type" |
| 'object' with AdditionalProperties | map |
| 'object' with x-kubernetes-embedded-type | object / "message type", 'apiVersion', 'kind', 'metadata.name' and 'metadata.generateName' are implicitly included in schema |
| 'object' with x-kubernetes-preserve-unknown-fields | object / "message type", unknown fields are NOT accessible in CEL expression |
| x-kubernetes-int-or-string | dynamic object that is either an int or a string, <code>type(value)</code> can be used to check the type |
| 'array' | list |
| 'array' with x-kubernetes-list-type=map | list with map based Equality & unique key guarantees |
| 'array' with x-kubernetes-list-type=set | list with set based Equality & unique entry guarantees |

| OpenAPIv3 type | CEL type |
|--|---------------------------------------|
| 'boolean' | boolean |
| 'number' (all formats) | double |
| 'integer' (all formats) | int (64) |
| 'null' | null_type |
| 'string' | string |
| 'string' with format=byte (base64 encoded) | bytes |
| 'string' with format=date | timestamp (google.protobuf.Timestamp) |
| 'string' with format=datetime | timestamp (google.protobuf.Timestamp) |
| 'string' with format=duration | duration (google.protobuf.Duration) |

xref: [CEL types](#), [OpenAPI types](#), [Kubernetes Structural Schemas](#).

The messageExpression field

Similar to the message field, which defines the string reported for a validation rule failure, messageExpression allows you to use a CEL expression to construct the message string. This allows you to insert more descriptive information into the validation failure message. messageExpression must evaluate a string and may use the same variables that are available to the rule field. For example:

```
x-kubernetes-validations:
- rule: "self.x <= self.maxLimit"
  messageExpression: "'x exceeded max limit of ' + string(self.maxLimit)'
```

Keep in mind that CEL string concatenation (+ operator) does not auto-cast to string. If you have a non-string scalar, use the string(<value>) function to cast the scalar to a string like shown in the above example.

messageExpression must evaluate to a string, and this is checked while the CRD is being written. Note that it is possible to set message and messageExpression on the same rule, and if both are present, messageExpression will be used. However, if messageExpression evaluates to an error, the string defined in message will be used instead, and the messageExpression error will be logged. This fallback will also occur if the CEL expression defined in messageExpression generates an empty string, or a string containing line breaks.

If one of the above conditions are met and no message has been set, then the default validation failure message will be used instead.

messageExpression is a CEL expression, so the restrictions listed in [Resource use by validation functions](#) apply. If evaluation halts due to resource constraints during messageExpression execution, then no further validation rules will be executed.

Setting messageExpression is optional.

The message field

If you want to set a static message, you can supply message rather than messageExpression. The value of message is used as an opaque error string if validation fails.

Setting message is optional.

The reason field

You can add a machine-readable validation failure reason within a validation, to be returned whenever a request fails this validation rule.

For example:

```
x-kubernetes-validations:  
- rule: "self.x <= self.maxLimit"  
  reason: "FieldValueInvalid"
```

The HTTP status code returned to the caller will match the reason of the first failed validation rule. The currently supported reasons are: "FieldValueInvalid", "FieldValueForbidden", "FieldValueRequired", "FieldValueDuplicate". If not set or unknown reasons, default to use "FieldValueInvalid".

Setting reason is optional.

The fieldPath field

You can specify the field path returned when the validation fails.

For example:

```
x-kubernetes-validations:  
- rule: "self.foo.test.x <= self.maxLimit"  
  fieldPath: ".foo.test.x"
```

In the example above, the validation checks the value of field x should be less than the value of maxLimit. If no fieldPath specified, when validation fails, the fieldPath would be default to wherever self scoped. With fieldPath specified, the returned error will have fieldPath properly refer to the location of field x.

The fieldPath value must be a relative JSON path that is scoped to the location of this x-kubernetes-validations extension in the schema. Additionally, it should refer to an existing field within the schema. For example when validation checks if a specific attribute foo under a map testMap, you could set fieldPath to ".testMap.foo" or ".testMap['foo']". If the validation requires checking for unique attributes in two lists, the fieldPath can be set to either of the lists. For example, it can be set to .testList1 or .testList2. It supports child operation to refer to an existing field currently. Refer to [JSONPath support in Kubernetes](#) for more info. The fieldPath field does not support indexing arrays numerically.

Setting fieldPath is optional.

The optionalOldSelf field

FEATURE STATE: Kubernetes v1.30 [beta]

If your cluster does not have [CRD validation ratcheting](#) enabled, the CustomResourceDefinition API doesn't include this field, and trying to set it may result in an error.

The optionalOldSelf field is a boolean field that alters the behavior of [Transition Rules](#) described below. Normally, a transition rule will not evaluate if oldSelf cannot be determined: during object creation or when a new value is introduced in an update.

If optionalOldSelf is set to true, then transition rules will always be evaluated and the type of oldSelf be changed to a CEL [Optional](#) type.

optionalOldSelf is useful in cases where schema authors would like a more control tool [than provided by the default equality based behavior of](#) to introduce newer, usually stricter constraints on new values, while still allowing old values to be "grandfathered" or ratcheted using the older validation.

Example Usage:

| CEL | Description |
|---|-------------|
| <code>`self.foo == "foo"</code> | |
| <code>[oldSelf.orValue(""), self].all(x, ["OldCase1", "OldCase2"].exists(case, x == case))</code> | |
| <code>oldSelf.optMap(o, o.size()).orValue(0) < 4</code> | |

Validation functions

Functions available include:

- CEL standard functions, defined in the [list of standard definitions](#)
- CEL standard [macros](#)
- CEL [extended string function library](#)
- Kubernetes [CEL extension library](#)

Transition rules

A rule that contains an expression referencing the identifier oldSelf is implicitly considered a *transition rule*. Transition rules allow schema authors to prevent certain transitions between two otherwise valid states. For example:

```
type: string
enum: ["low", "medium", "high"]
x-kubernetes-validations:
- rule: "!(self == 'high' && oldSelf == 'low') && !(self == 'low' && oldSelf == 'high')"
  message: cannot transition directly between 'low' and 'high'
```

Unlike other rules, transition rules apply only to operations meeting the following criteria:

- The operation updates an existing object. Transition rules never apply to create operations.
- Both an old and a new value exist. It remains possible to check if a value has been added or removed by placing a transition rule on the parent node. Transition rules are never applied to custom resource creation. When placed on an optional field, a transition rule will not apply to update operations that set or unset the field.
- The path to the schema node being validated by a transition rule must resolve to a node that is comparable between the old object and the new object. For example, list items and

their descendants (spec.foo[10].bar) can't necessarily be correlated between an existing object and a later update to the same object.

Errors will be generated on CRD writes if a schema node contains a transition rule that can never be applied, e.g. "*path*: update rule *rule* cannot be set on schema because the schema or its parent schema is not mergeable".

Transition rules are only allowed on *correlatable portions* of a schema. A portion of the schema is correlatable if all array parent schemas are of type x-kubernetes-list-type=map; any set or atomicarray parent schemas make it impossible to unambiguously correlate a self with oldSelf.

Here are some examples for transition rules:

Transition rules examples

| Use Case | Rule |
|---|---|
| Immutability | self.foo == oldSelf.foo |
| Prevent modification/removal once assigned | oldSelf != 'bar' self == 'bar' or !has(oldSelf.field) has(self.field) |
| Append-only set | self.all(element, element in oldSelf) |
| If previous value was X, new value can only be A or B, not Y or Z | oldSelf != 'X' self in ['A', 'B'] |
| Monotonic (non-decreasing) counters | self >= oldSelf |

Resource use by validation functions

When you create or update a CustomResourceDefinition that uses validation rules, the API server checks the likely impact of running those validation rules. If a rule is estimated to be prohibitively expensive to execute, the API server rejects the create or update operation, and returns an error message. A similar system is used at runtime that observes the actions the interpreter takes. If the interpreter executes too many instructions, execution of the rule will be halted, and an error will result. Each CustomResourceDefinition is also allowed a certain amount of resources to finish executing all of its validation rules. If the sum total of its rules are estimated at creation time to go over that limit, then a validation error will also occur.

You are unlikely to encounter issues with the resource budget for validation if you only specify rules that always take the same amount of time regardless of how large their input is. For example, a rule that asserts that self.foo == 1 does not by itself have any risk of rejection on validation resource budget groups. But if foo is a string and you define a validation rule self.foo.contains("someString"), that rule takes longer to execute depending on how long foo is. Another example would be if foo were an array, and you specified a validation rule self.foo.all(x, x > 5). The cost system always assumes the worst-case scenario if a limit on the length of foo is not given, and this will happen for anything that can be iterated over (lists, maps, etc.).

Because of this, it is considered best practice to put a limit via maxItems, maxProperties, and maxLength for anything that will be processed in a validation rule in order to prevent validation errors during cost estimation. For example, given this schema with one rule:

```
openAPIV3Schema:
  type: object
  properties:
    foo:
```

```
type: array
items:
  type: string
x-kubernetes-validations:
  - rule: "self.all(x, x.contains('a string'))"
```

then the API server rejects this rule on validation budget grounds with error:

```
spec.validation.openAPIV3Schema.properties[spec].properties[foo].x-kubernetes-
validations[0].rule: Forbidden:
CEL rule exceeded budget by more than 100x (try simplifying the rule, or adding maxItems,
maxProperties, and
maxLength where arrays, maps, and strings are used)
```

The rejection happens because `self.all` implies calling `contains()` on every string in `foo`, which in turn will check the given string to see if it contains 'a string'. Without limits, this is a very expensive rule.

If you do not specify any validation limit, the estimated cost of this rule will exceed the per-rule cost limit. But if you add limits in the appropriate places, the rule will be allowed:

```
openAPIV3Schema:
  type: object
  properties:
    foo:
      type: array
      maxItems: 25
      items:
        type: string
        maxLength: 10
      x-kubernetes-validations:
        - rule: "self.all(x, x.contains('a string'))"
```

The cost estimation system takes into account how many times the rule will be executed in addition to the estimated cost of the rule itself. For instance, the following rule will have the same estimated cost as the previous example (despite the rule now being defined on the individual array items):

```
openAPIV3Schema:
  type: object
  properties:
    foo:
      type: array
      maxItems: 25
      items:
        type: string
        x-kubernetes-validations:
          - rule: "self.contains('a string')"
```

If a list inside of a list has a validation rule that uses `self.all`, that is significantly more expensive than a non-nested list with the same rule. A rule that would have been allowed on a non-nested list might need lower limits set on both nested lists in order to be allowed. For example, even without having limits set, the following rule is allowed:

```
openAPIV3Schema:
  type: object
  properties:
    foo:
      type: array
      items:
        type: integer
      x-kubernetes-validations:
        - rule: "self.all(x, x == 5)"
```

But the same rule on the following schema (with a nested array added) produces a validation error:

```
openAPIV3Schema:
  type: object
  properties:
    foo:
      type: array
      items:
        type: array
        items:
          type: integer
      x-kubernetes-validations:
        - rule: "self.all(x, x == 5)"
```

This is because each item of `foo` is itself an array, and each subarray in turn calls `self.all`. Avoid nested lists and maps if possible where validation rules are used.

Defaulting

Note:

To use defaulting, your CustomResourceDefinition must use API version `apiextensions.k8s.io/v1`.

Defaulting allows to specify default values in the [OpenAPI v3 validation schema](#):

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: crontabs.stable.example.com
spec:
  group: stable.example.com
  versions:
    - name: v1
      served: true
      storage: true
      schema:
        # openAPIV3Schema is the schema for validating custom objects.
        openAPIV3Schema:
          type: object
          properties:
            spec:
```

```
type: object
properties:
  cronSpec:
    type: string
    pattern: '^(\\d+|\\*)(\\/\\d+)?(\\s+(\\d+|\\*)(\\/\\d+)?){4}$'
    default: "5 0 * * *"
  image:
    type: string
  replicas:
    type: integer
    minimum: 1
    maximum: 10
    default: 1
scope: Namespaced
names:
  plural: crontabs
  singular: crontab
  kind: CronTab
  shortNames:
  - ct
```

With this both cronSpec and replicas are defaulted:

```
apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  name: my-new-cron-object
spec:
  image: my-awesome-cron-image
```

leads to

```
apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  name: my-new-cron-object
spec:
  cronSpec: "5 0 * * *"
  image: my-awesome-cron-image
  replicas: 1
```

Defaulting happens on the object

- in the request to the API server using the request version defaults,
- when reading from etcd using the storage version defaults,
- after mutating admission plugins with non-empty patches using the admission webhook object version defaults.

Defaults applied when reading data from etcd are not automatically written back to etcd. An update request via the API is required to persist those defaults back into etcd.

Default values must be pruned (with the exception of defaults for metadata fields) and must validate against a provided schema.

Default values for metadata fields of x-kubernetes-embedded-resources: true nodes (or parts of a default value covering metadata) are not pruned during CustomResourceDefinition creation, but through the pruning step during handling of requests.

Defaulting and Nullable

Null values for fields that either don't specify the nullable flag, or give it a false value, will be pruned before defaulting happens. If a default is present, it will be applied. When nullable is true, null values will be conserved and won't be defaulted.

For example, given the OpenAPI schema below:

```
type: object
properties:
  spec:
    type: object
    properties:
      foo:
        type: string
        nullable: false
        default: "default"
      bar:
        type: string
        nullable: true
      baz:
        type: string
```

creating an object with null values for foo and bar and baz

```
spec:
  foo: null
  bar: null
  baz: null
```

leads to

```
spec:
  foo: "default"
  bar: null
```

with foo pruned and defaulted because the field is non-nullable, bar maintaining the null value due to nullable: true, and baz pruned because the field is non-nullable and has no default.

Publish Validation Schema in OpenAPI

CustomResourceDefinition [OpenAPI v3 validation schemas](#) which are [structural](#) and [enable pruning](#) are published as [OpenAPI v3](#) and OpenAPI v2 from Kubernetes API server. It is recommended to use the OpenAPI v3 document as it is a lossless representation of the CustomResourceDefinition OpenAPI v3 validation schema while OpenAPI v2 represents a lossy conversion.

The [kubectrl](#) command-line tool consumes the published schema to perform client-side validation (kubectrl create and kubectrl apply), schema explanation (kubectrl explain) on custom

resources. The published schema can be consumed for other purposes as well, like client generation or documentation.

Compatibility with OpenAPI V2

For compatibility with OpenAPI V2, the OpenAPI v3 validation schema performs a lossy conversion to the OpenAPI v2 schema. The schema show up in definitions and paths fields in the [OpenAPI v2 spec](#).

The following modifications are applied during the conversion to keep backwards compatibility with kubectl in previous 1.13 version. These modifications prevent kubectl from being over-strict and rejecting valid OpenAPI schemas that it doesn't understand. The conversion won't modify the validation schema defined in CRD, and therefore won't affect [validation](#) in the API server.

1. The following fields are removed as they aren't supported by OpenAPI v2.
 - The fields allOf, anyOf, oneOf and not are removed
2. If nullable: true is set, we drop type, nullable, items and properties because OpenAPI v2 is not able to express nullable. To avoid kubectl to reject good objects, this is necessary.

Additional printer columns

The kubectl tool relies on server-side output formatting. Your cluster's API server decides which columns are shown by the kubectl get command. You can customize these columns for a CustomResourceDefinition. The following example adds the Spec, Replicas, and Age columns.

Save the CustomResourceDefinition to resourcedefinition.yaml:

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: crontabs.stable.example.com
spec:
  group: stable.example.com
  scope: Namespaced
  names:
    plural: crontabs
    singular: crontab
    kind: CronTab
    shortNames:
      - ct
  versions:
    - name: v1
      served: true
      storage: true
      schema:
        openAPIV3Schema:
          type: object
          properties:
            spec:
              type: object
```

```

    properties:
      cronSpec:
        type: string
      image:
        type: string
      replicas:
        type: integer
    additionalPrinterColumns:
    - name: Spec
      type: string
      description: The cron spec defining the interval a CronJob is run
      jsonPath: .spec.cronSpec
    - name: Replicas
      type: integer
      description: The number of jobs launched by the CronJob
      jsonPath: .spec.replicas
    - name: Age
      type: date
      jsonPath: .metadata.creationTimestamp

```

Create the CustomResourceDefinition:

```
kubectl apply -f resourcedefinition.yaml
```

Create an instance using the my-crontab.yaml from the previous section.

Invoke the server-side printing:

```
kubectl get crontab my-new-cron-object
```

Notice the NAME, SPEC, REPLICAS, and AGE columns in the output:

```

NAME          SPEC      REPLICAS  AGE
my-new-cron-object  * * * * * 1      7s

```

Note:

The NAME column is implicit and does not need to be defined in the CustomResourceDefinition.

Field selectors

[Field Selectors](#) let clients select custom resources based on the value of one or more resource fields.

All custom resources support the `metadata.name` and `metadata.namespace` field selectors.

Fields declared in a [CustomResourceDefinition](#) may also be used with field selectors when included in the `spec.versions[*].selectableFields` field of the [CustomResourceDefinition](#).

Selectable fields for custom resources

FEATURE STATE: Kubernetes v1.30 [alpha]

FEATURE STATE: Kubernetes v1.30 [alpha]

You need to enable the CustomResourceFieldSelectors [feature gate](#) to use this behavior, which then applies to all CustomResourceDefinitions in your cluster.

The spec.versions[*].selectableFields field of a [CustomResourceDefinition](#) may be used to declare which other fields in a custom resource may be used in field selectors. The following example adds the .spec.color and .spec.size fields as selectable fields.

Save the CustomResourceDefinition to shirt-resource-definition.yaml:

[customresourcedefinition/shirt-resource-definition.yaml](#)

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: shirts.stable.example.com
spec:
  group: stable.example.com
  scope: Namespaced
  names:
    plural: shirts
    singular: shirt
    kind: Shirt
  versions:
    - name: v1
      served: true
      storage: true
      schema:
        openAPIV3Schema:
          type: object
          properties:
            spec:
              type: object
              properties:
                color:
                  type: string
                size:
                  type: string
            selectableFields:
              - jsonPath: .spec.color
              - jsonPath: .spec.size
          additionalPrinterColumns:
            - jsonPath: .spec.color
              name: Color
              type: string
            - jsonPath: .spec.size
              name: Size
              type: string
```

Create the CustomResourceDefinition:

```
kubectl apply -f https://k8s.io/examples/customresourcedefinition/shirt-resource-
definition.yaml
```

Define some Shirts by editing `shirt-resources.yaml`; for example:

[customresourcedefinition/shirt-resources.yaml](#)

```
---
apiVersion: stable.example.com/v1
kind: Shirt
metadata:
  name: example1
spec:
  color: blue
  size: S
---
apiVersion: stable.example.com/v1
kind: Shirt
metadata:
  name: example2
spec:
  color: blue
  size: M
---
apiVersion: stable.example.com/v1
kind: Shirt
metadata:
  name: example3
spec:
  color: green
  size: M
```

Create the custom resources:

```
kubectl apply -f https://k8s.io/examples/customresourcedefinition/shirt-resources.yaml
```

Get all the resources:

```
kubectl get shirts.stable.example.com
```

The output is:

| NAME | COLOR | SIZE |
|----------|-------|------|
| example1 | blue | S |
| example2 | blue | M |
| example3 | green | M |

Fetch blue shirts (retrieve Shirts with a color of blue):

```
kubectl get shirts.stable.example.com --field-selector spec.color=blue
```

Should output:

| NAME | COLOR | SIZE |
|----------|-------|------|
| example1 | blue | S |
| example2 | blue | M |

Get only resources with a color of green and a size of M:

```
kubectl get shirts.stable.example.com --field-selector spec.color=green,spec.size=M
```

Should output:

| NAME | COLOR | SIZE |
|----------|-------|------|
| example2 | blue | M |

Priority

Each column includes a priority field. Currently, the priority differentiates between columns shown in standard view or wide view (using the `-o wide` flag).

- Columns with priority 0 are shown in standard view.
- Columns with priority greater than 0 are shown only in wide view.

Type

A column's type field can be any of the following (compare [OpenAPI v3 data types](#)):

- integer – non-floating-point numbers
- number – floating point numbers
- string – strings
- boolean – true or false
- date – rendered differentially as time since this timestamp.

If the value inside a CustomResource does not match the type specified for the column, the value is omitted. Use CustomResource validation to ensure that the value types are correct.

Format

A column's format field can be any of the following:

- int32
- int64
- float
- double
- byte
- date
- date-time
- password

The column's format controls the style used when kubectl prints the value.

Subresources

Custom resources support `/status` and `/scale` subresources.

The status and scale subresources can be optionally enabled by defining them in the CustomResourceDefinition.

Status subresource

When the status subresource is enabled, the `/status` subresource for the custom resource is exposed.

- The status and the spec stanzas are represented by the `.status` and `.spec` JSONPaths respectively inside of a custom resource.
- PUT requests to the `/status` subresource take a custom resource object and ignore changes to anything except the status stanza.
- PUT requests to the `/status` subresource only validate the status stanza of the custom resource.
- PUT/POST/PATCH requests to the custom resource ignore changes to the status stanza.
- The `.metadata.generation` value is incremented for all changes, except for changes to `.metadata` or `.status`.
- Only the following constructs are allowed at the root of the CRD OpenAPI validation schema:
 - `description`
 - `example`
 - `exclusiveMaximum`
 - `exclusiveMinimum`
 - `externalDocs`
 - `format`
 - `items`
 - `maximum`
 - `maxItems`
 - `maxLength`
 - `minimum`
 - `minItems`
 - `minLength`
 - `multipleOf`
 - `pattern`
 - `properties`
 - `required`
 - `title`
 - `type`
 - `uniqueItems`

Scale subresource

When the scale subresource is enabled, the `/scale` subresource for the custom resource is exposed. The `autoscaling/v1.Scale` object is sent as the payload for `/scale`.

To enable the scale subresource, the following fields are defined in the CustomResourceDefinition.

- `specReplicasPath` defines the JSONPath inside of a custom resource that corresponds to `scale.spec.replicas`.
 - It is a required value.
 - Only JSONPaths under `.spec` and with the dot notation are allowed.
 - If there is no value under the `specReplicasPath` in the custom resource, the `/scale` subresource will return an error on GET.
- `statusReplicasPath` defines the JSONPath inside of a custom resource that corresponds to `scale.status.replicas`.
 - It is a required value.
 - Only JSONPaths under `.status` and with the dot notation are allowed.
 - If there is no value under the `statusReplicasPath` in the custom resource, the status replica value in the `/scale` subresource will default to 0.
- `labelSelectorPath` defines the JSONPath inside of a custom resource that corresponds to `Scale.Status.Selector`.
 - It is an optional value.
 - It must be set to work with HPA and VPA.
 - Only JSONPaths under `.status` or `.spec` and with the dot notation are allowed.
 - If there is no value under the `labelSelectorPath` in the custom resource, the status selector value in the `/scale` subresource will default to the empty string.
 - The field pointed by this JSON path must be a string field (not a complex selector struct) which contains a serialized label selector in string form.

In the following example, both status and scale subresources are enabled.

Save the CustomResourceDefinition to `resourcedefinition.yaml`:

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: crontabs.stable.example.com
spec:
  group: stable.example.com
  versions:
    - name: v1
      served: true
      storage: true
      schema:
        openAPIV3Schema:
          type: object
          properties:
            spec:
              type: object
              properties:
                cronSpec:
                  type: string
                image:
```



```

    type: string
  replicas:
    type: integer
  status:
    type: object
  properties:
    replicas:
      type: integer
    labelSelector:
      type: string
# subresources describes the subresources for custom resources.
subresources:
# status enables the status subresource.
  status: {}
# scale enables the scale subresource.
  scale:
# specReplicasPath defines the JSONPath inside of a custom resource that corresponds to
Scale.Spec.Replicas.
    specReplicasPath: .spec.replicas
# statusReplicasPath defines the JSONPath inside of a custom resource that corresponds
to Scale.Status.Replicas.
    statusReplicasPath: .status.replicas
# labelSelectorPath defines the JSONPath inside of a custom resource that corresponds to
Scale.Status.Selector.
    labelSelectorPath: .status.labelSelector
scope: Namespaced
names:
  plural: crontabs
  singular: crontab
  kind: CronTab
  shortNames:
  - ct

```

And create it:

```
kubectl apply -f resourcedefinition.yaml
```

After the CustomResourceDefinition object has been created, you can create custom objects.

If you save the following YAML to my-crontab.yaml:

```

apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  name: my-new-cron-object
spec:
  cronSpec: "* * * * */5"
  image: my-awesome-cron-image
  replicas: 3

```

and create it:

```
kubectl apply -f my-crontab.yaml
```

Then new namespaced RESTful API endpoints are created at:

```
/apis/stable.example.com/v1/namespaces/*/crontabs/status
```

and

```
/apis/stable.example.com/v1/namespaces/*/crontabs/scale
```

A custom resource can be scaled using the `kubectl scale` command. For example, the following command sets `.spec.replicas` of the custom resource created above to 5:

```
kubectl scale --replicas=5 crontabs/my-new-cron-object
crontabs "my-new-cron-object" scaled

kubectl get crontabs my-new-cron-object -o jsonpath='{.spec.replicas}'
5
```

You can use a [PodDisruptionBudget](#) to protect custom resources that have the scale subresource enabled.

Categories

Categories is a list of grouped resources the custom resource belongs to (eg. all). You can use `kubectl get <category-name>` to list the resources belonging to the category.

The following example adds all in the list of categories in the CustomResourceDefinition and illustrates how to output the custom resource using `kubectl get all`.

Save the following CustomResourceDefinition to `resourcedefinition.yaml`:

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: crontabs.stable.example.com
spec:
  group: stable.example.com
  versions:
    - name: v1
      served: true
      storage: true
      schema:
        openAPIV3Schema:
          type: object
          properties:
            spec:
              type: object
              properties:
                cronSpec:
                  type: string
                image:
                  type: string
                replicas:
                  type: integer
  scope: Namespaced
```

```
names:
  plural: crontabs
  singular: crontab
  kind: CronTab
  shortNames:
  - ct
  # categories is a list of grouped resources the custom resource belongs to.
  categories:
  - all
```

and create it:

```
kubectl apply -f resourcedefinition.yaml
```

After the CustomResourceDefinition object has been created, you can create custom objects.

Save the following YAML to `my-crontab.yaml`:

```
apiVersion: "stable.example.com/v1"
kind: CronTab
metadata:
  name: my-new-cron-object
spec:
  cronSpec: "** * * * */5"
  image: my-awesome-cron-image
```

and create it:

```
kubectl apply -f my-crontab.yaml
```

You can specify the category when using `kubectl get`:

```
kubectl get all
```

and it will include the custom resources of kind CronTab:

| NAME | AGE |
|-----------------------------|-----|
| crontabs/my-new-cron-object | 3s |

What's next

- Read about [custom resources](#).
- See [CustomResourceDefinition](#).
- Serve [multiple versions](#) of a CustomResourceDefinition.

Versions in CustomResourceDefinitions

This page explains how to add versioning information to [CustomResourceDefinitions](#), to indicate the stability level of your CustomResourceDefinitions or advance your API to a new

version with conversion between API representations. It also describes how to upgrade an object from one version to another.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You should have an initial understanding of [custom resources](#).

Your Kubernetes server must be at or later than version v1.16. To check the version, enter kubectl version.

Overview

The CustomResourceDefinition API provides a workflow for introducing and upgrading to new versions of a CustomResourceDefinition.

When a CustomResourceDefinition is created, the first version is set in the CustomResourceDefinition spec.versions list to an appropriate stability level and a version number. For example v1beta1 would indicate that the first version is not yet stable. All custom resource objects will initially be stored at this version.

Once the CustomResourceDefinition is created, clients may begin using the v1beta1 API.

Later it might be necessary to add new version such as v1.

Adding a new version:

1. Pick a conversion strategy. Since custom resource objects need the ability to be served at both versions, that means they will sometimes be served in a different version than the one stored. To make this possible, the custom resource objects must sometimes be converted between the version they are stored at and the version they are served at. If the conversion involves schema changes and requires custom logic, a conversion webhook should be used. If there are no schema changes, the default None conversion strategy may be used and only the apiVersion field will be modified when serving different versions.
2. If using conversion webhooks, create and deploy the conversion webhook. See the [Webhook conversion](#) for more details.
3. Update the CustomResourceDefinition to include the new version in the spec.versions list with served:true. Also, set spec.conversion field to the selected conversion strategy. If using a conversion webhook, configure spec.conversion.webhookClientConfig field to call the webhook.

Once the new version is added, clients may incrementally migrate to the new version. It is perfectly safe for some clients to use the old version while others use the new version.

Migrate stored objects to the new version:

1. See the [upgrade existing objects to a new stored version](#) section.

It is safe for clients to use both the old and new version before, during and after upgrading the objects to a new stored version.

Removing an old version:

1. Ensure all clients are fully migrated to the new version. The kube-apiserver logs can be reviewed to help identify any clients that are still accessing via the old version.
2. Set served to false for the old version in the spec.versions list. If any clients are still unexpectedly using the old version they may begin reporting errors attempting to access the custom resource objects at the old version. If this occurs, switch back to using served:true on the old version, migrate the remaining clients to the new version and repeat this step.
3. Ensure the [upgrade of existing objects to the new stored version](#) step has been completed.
 1. Verify that the storage is set to true for the new version in the spec.versions list in the CustomResourceDefinition.
 2. Verify that the old version is no longer listed in the CustomResourceDefinition status.storedVersions.
4. Remove the old version from the CustomResourceDefinition spec.versions list.
5. Drop conversion support for the old version in conversion webhooks.

Specify multiple versions

The CustomResourceDefinition API versions field can be used to support multiple versions of custom resources that you have developed. Versions can have different schemas, and conversion webhooks can convert custom resources between versions. Webhook conversions should follow the [Kubernetes API conventions](#) wherever applicable. Specifically, See the [API change documentation](#) for a set of useful gotchas and suggestions.

Note:

In apiextensions.k8s.io/v1beta1, there was a version field instead of versions. The version field is deprecated and optional, but if it is not empty, it must match the first item in the versions field.

This example shows a CustomResourceDefinition with two versions. For the first example, the assumption is all versions share the same schema with no conversion between them. The comments in the YAML provide more context.

- [apiextensions.k8s.io/v1](#)
- [apiextensions.k8s.io/v1beta1](#)

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  # name must match the spec fields below, and be in the form: <plural>.<group>
  name: crontabs.example.com
spec:
  # group name to use for REST API: /apis/<group>/<version>
  group: example.com
  # list of versions supported by this CustomResourceDefinition
```

```

versions:
- name: v1beta1
  # Each version can be enabled/disabled by Served flag.
  served: true
  # One and only one version must be marked as the storage version.
  storage: true
  # A schema is required
  schema:
    openAPIV3Schema:
      type: object
      properties:
        host:
          type: string
        port:
          type: string
- name: v1
  served: true
  storage: false
  schema:
    openAPIV3Schema:
      type: object
      properties:
        host:
          type: string
        port:
          type: string
# The conversion section is introduced in Kubernetes 1.13+ with a default value of
# None conversion (strategy sub-field set to None).
conversion:
  # None conversion assumes the same schema for all versions and only sets the apiVersion
  # field of custom resources to the proper value
  strategy: None
# either Namespaced or Cluster
scope: Namespaced
names:
  # plural name to be used in the URL: /apis/<group>/<version>/<plural>
  plural: crontabs
  # singular name to be used as an alias on the CLI and for display
  singular: crontab
  # kind is normally the CamelCased singular type. Your resource manifests use this.
  kind: CronTab
  # shortNames allow shorter string to match your resource on the CLI
  shortNames:
  - ct

```

```

# Deprecated in v1.16 in favor of apiextensions.k8s.io/v1
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
metadata:
  # name must match the spec fields below, and be in the form: <plural>.<group>
  name: crontabs.example.com
spec:
  # group name to use for REST API: /apis/<group>/<version>

```

```

group: example.com
# list of versions supported by this CustomResourceDefinition
versions:
- name: v1beta1
  # Each version can be enabled/disabled by Served flag.
  served: true
  # One and only one version must be marked as the storage version.
  storage: true
- name: v1
  served: true
  storage: false
validation:
  openAPIV3Schema:
    type: object
    properties:
      host:
        type: string
      port:
        type: string
# The conversion section is introduced in Kubernetes 1.13+ with a default value of
# None conversion (strategy sub-field set to None).
conversion:
  # None conversion assumes the same schema for all versions and only sets the apiVersion
  # field of custom resources to the proper value
  strategy: None
# either Namespaced or Cluster
scope: Namespaced
names:
  # plural name to be used in the URL: /apis/<group>/<version>/<plural>
  plural: crontabs
  # singular name to be used as an alias on the CLI and for display
  singular: crontab
  # kind is normally the PascalCased singular type. Your resource manifests use this.
  kind: CronTab
  # shortNames allow shorter string to match your resource on the CLI
  shortNames:
  - ct

```

You can save the CustomResourceDefinition in a YAML file, then use `kubectl apply` to create it.

```
kubectl apply -f my-versioned-crontab.yaml
```

After creation, the API server starts to serve each enabled version at an HTTP REST endpoint. In the above example, the API versions are available at `/apis/example.com/v1beta1` and `/apis/example.com/v1`.

Version priority

Regardless of the order in which versions are defined in a CustomResourceDefinition, the version with the highest priority is used by `kubectl` as the default version to access objects. The priority is determined by parsing the *name* field to determine the version number, the stability (GA, Beta, or Alpha), and the sequence within that stability level.

The algorithm used for sorting the versions is designed to sort versions in the same way that the Kubernetes project sorts Kubernetes versions. Versions start with a v followed by a number, an optional beta or alpha designation, and optional additional numeric versioning information. Broadly, a version string might look like v2 or v2beta1. Versions are sorted using the following algorithm:

- Entries that follow Kubernetes version patterns are sorted before those that do not.
- For entries that follow Kubernetes version patterns, the numeric portions of the version string is sorted largest to smallest.
- If the strings beta or alpha follow the first numeric portion, they sorted in that order, after the equivalent string without the beta or alpha suffix (which is presumed to be the GA version).
- If another number follows the beta, or alpha, those numbers are also sorted from largest to smallest.
- Strings that don't fit the above format are sorted alphabetically and the numeric portions are not treated specially. Notice that in the example below, foo1 is sorted above foo10. This is different from the sorting of the numeric portion of entries that do follow the Kubernetes version patterns.

This might make sense if you look at the following sorted version list:

```
- v10
- v2
- v1
- v11beta2
- v10beta3
- v3beta1
- v12alpha1
- v11alpha2
- foo1
- foo10
```

For the example in [Specify multiple versions](#), the version sort order is v1, followed by v1beta1. This causes the kubectl command to use v1 as the default version unless the provided object specifies the version.

Version deprecation

FEATURE STATE: Kubernetes v1.19 [stable]

Starting in v1.19, a CustomResourceDefinition can indicate a particular version of the resource it defines is deprecated. When API requests to a deprecated version of that resource are made, a warning message is returned in the API response as a header. The warning message for each deprecated version of the resource can be customized if desired.

A customized warning message should indicate the deprecated API group, version, and kind, and should indicate what API group, version, and kind should be used instead, if applicable.

- apiextensions.k8s.io/v1
- apiextensions.k8s.io/v1beta1

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
name: crontabs.example.com
```



```

spec:
  group: example.com
  names:
    plural: crontabs
    singular: crontab
    kind: CronTab
  scope: Namespaced
  versions:
  - name: v1alpha1
    served: true
    storage: false
    # This indicates the v1alpha1 version of the custom resource is deprecated.
    # API requests to this version receive a warning header in the server response.
    deprecated: true
    # This overrides the default warning returned to API clients making v1alpha1 API requests.
    deprecationWarning: "example.com/v1alpha1 CronTab is deprecated; see http://example.com/
v1alpha1-v1 for instructions to migrate to example.com/v1 CronTab"

    schema: ...
  - name: v1beta1
    served: true
    # This indicates the v1beta1 version of the custom resource is deprecated.
    # API requests to this version receive a warning header in the server response.
    # A default warning message is returned for this version.
    deprecated: true
    schema: ...
  - name: v1
    served: true
    storage: true
    schema: ...

```

```

# Deprecated in v1.16 in favor of apiextensions.k8s.io/v1
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
metadata:
  name: crontabs.example.com
spec:
  group: example.com
  names:
    plural: crontabs
    singular: crontab
    kind: CronTab
  scope: Namespaced
  validation: ...
  versions:
  - name: v1alpha1
    served: true
    storage: false
    # This indicates the v1alpha1 version of the custom resource is deprecated.
    # API requests to this version receive a warning header in the server response.
    deprecated: true
    # This overrides the default warning returned to API clients making v1alpha1 API requests.
    deprecationWarning: "example.com/v1alpha1 CronTab is deprecated; see http://example.com/

```

```
v1alpha1-v1 for instructions to migrate to example.com/v1 CronTab"
- name: v1beta1
  served: true
  # This indicates the v1beta1 version of the custom resource is deprecated.
  # API requests to this version receive a warning header in the server response.
  # A default warning message is returned for this version.
  deprecated: true
- name: v1
  served: true
  storage: true
```

Version removal

An older API version cannot be dropped from a CustomResourceDefinition manifest until existing stored data has been migrated to the newer API version for all clusters that served the older version of the custom resource, and the old version is removed from the `status.storedVersions` of the CustomResourceDefinition.

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
  name: crontabs.example.com
spec:
  group: example.com
  names:
    plural: crontabs
    singular: crontab
    kind: CronTab
  scope: Namespaced
  versions:
    - name: v1beta1
      # This indicates the v1beta1 version of the custom resource is no longer served.
      # API requests to this version receive a not found error in the server response.
      served: false
      schema: ...
    - name: v1
      served: true
      # The new served version should be set as the storage version
      storage: true
      schema: ...
```

Webhook conversion

FEATURE STATE: Kubernetes v1.16 [stable]

Note:

Webhook conversion is available as beta since 1.15, and as alpha since Kubernetes 1.13. The CustomResourceWebhookConversion feature must be enabled, which is the case automatically for many clusters for beta features. Please refer to the [feature gate](#) documentation for more information.

The above example has a None conversion between versions which only sets the `apiVersion` field on conversion and does not change the rest of the object. The API server also supports webhook conversions that call an external service in case a conversion is required. For example when:

- custom resource is requested in a different version than stored version.
- Watch is created in one version but the changed object is stored in another version.
- custom resource PUT request is in a different version than storage version.

To cover all of these cases and to optimize conversion by the API server, the conversion requests may contain multiple objects in order to minimize the external calls. The webhook should perform these conversions independently.

Write a conversion webhook server

Please refer to the implementation of the [custom resource conversion webhook server](#) that is validated in a Kubernetes e2e test. The webhook handles the `ConversionReview` requests sent by the API servers, and sends back conversion results wrapped in `ConversionResponse`. Note that the request contains a list of custom resources that need to be converted independently without changing the order of objects. The example server is organized in a way to be reused for other conversions. Most of the common code are located in the [framework file](#) that leaves only [one function](#) to be implemented for different conversions.

Note:

The example conversion webhook server leaves the `ClientAuth` field [empty](#), which defaults to `NoClientCert`. This means that the webhook server does not authenticate the identity of the clients, supposedly API servers. If you need mutual TLS or other ways to authenticate the clients, see how to [authenticate API servers](#).

Permissible mutations

A conversion webhook must not mutate anything inside of metadata of the converted object other than labels and annotations. Attempted changes to name, UID and namespace are rejected and fail the request which caused the conversion. All other changes are ignored.

Deploy the conversion webhook service

Documentation for deploying the conversion webhook is the same as for the [admission webhook example service](#). The assumption for next sections is that the conversion webhook server is deployed to a service named `example-conversion-webhook-server` in default namespace and serving traffic on path `/crdconvert`.

Note:

When the webhook server is deployed into the Kubernetes cluster as a service, it has to be exposed via a service on port 443 (The server itself can have an arbitrary port but the service object should map it to port 443). The communication between the API server and the webhook service may fail if a different port is used for the service.

Configure CustomResourceDefinition to use conversion webhooks

The None conversion example can be extended to use the conversion webhook by modifying conversion section of the spec:

- apiextensions.k8s.io/v1
- apiextensions.k8s.io/v1beta1

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  # name must match the spec fields below, and be in the form: <plural>.<group>
  name: crontabs.example.com
spec:
  # group name to use for REST API: /apis/<group>/<version>
  group: example.com
  # list of versions supported by this CustomResourceDefinition
  versions:
    - name: v1beta1
      # Each version can be enabled/disabled by Served flag.
      served: true
      # One and only one version must be marked as the storage version.
      storage: true
      # Each version can define its own schema when there is no top-level
      # schema is defined.
      schema:
        openAPIV3Schema:
          type: object
          properties:
            hostPort:
              type: string
    - name: v1
      served: true
      storage: false
      schema:
        openAPIV3Schema:
          type: object
          properties:
            host:
              type: string
            port:
              type: string
  conversion:
    # the Webhook strategy instructs the API server to call an external webhook for any
    # conversion between custom resources.
    strategy: Webhook
    # webhook is required when strategy is `Webhook` and it configures the webhook endpoint
    # to be called by API server.
    webhook:
      # conversionReviewVersions indicates what ConversionReview versions are understood/
      # preferred by the webhook.
      # The first version in the list understood by the API server is sent to the webhook.
      # The webhook must respond with a ConversionReview object in the same version it
```

received.

```
conversionReviewVersions: ["v1","v1beta1"]
clientConfig:
  service:
    namespace: default
    name: example-conversion-webhook-server
    path: /crdconvert
    caBundle: "Ci0tLS0tQk...<base64-encoded PEM bundle>...tLS0K"
```

either Namespaced or Cluster

scope: Namespaced

names:

plural name to be used in the URL: /apis/<group>/<version>/<plural>

plural: crontabs

singular name to be used as an alias on the CLI and for display

singular: crontab

kind is normally the CamelCased singular type. Your resource manifests use this.

kind: CronTab

shortNames allow shorter string to match your resource on the CLI

shortNames:

- ct

Deprecated in v1.16 in favor of apiextensions.k8s.io/v1

apiVersion: apiextensions.k8s.io/v1beta1

kind: CustomResourceDefinition

metadata:

name must match the spec fields below, and be in the form: <plural>.<group>

name: crontabs.example.com

spec:

group name to use for REST API: /apis/<group>/<version>

group: example.com

prunes object fields that are not specified in OpenAPI schemas below.

preserveUnknownFields: false

list of versions supported by this CustomResourceDefinition

versions:

- name: v1beta1

Each version can be enabled/disabled by Served flag.

served: true

One and only one version must be marked as the storage version.

storage: true

Each version can define its own schema when there is no top-level

schema is defined.

schema:

openAPIV3Schema:

type: object

properties:

hostPort:

type: string

- name: v1

served: true

storage: false

schema:

openAPIV3Schema:

type: object

```

properties:
  host:
    type: string
  port:
    type: string
conversion:
  # the Webhook strategy instructs the API server to call an external webhook for any
  # conversion between custom resources.
  strategy: Webhook
  # webhookClientConfig is required when strategy is `Webhook` and it configures the
  # webhook endpoint to be called by API server.
  webhookClientConfig:
    service:
      namespace: default
      name: example-conversion-webhook-server
      path: /crdconvert
      caBundle: "Ci0tLS0tQk...<base64-encoded PEM bundle>...tLS0K"
  # either Namespaced or Cluster
  scope: Namespaced
  names:
    # plural name to be used in the URL: /apis/<group>/<version>/<plural>
    plural: crontabs
    # singular name to be used as an alias on the CLI and for display
    singular: crontab
    # kind is normally the CamelCased singular type. Your resource manifests use this.
    kind: CronTab
    # shortNames allow shorter string to match your resource on the CLI
  shortNames:
    - ct

```

You can save the CustomResourceDefinition in a YAML file, then use `kubectl apply` to apply it.

```
kubectl apply -f my-versioned-crontab-with-conversion.yaml
```

Make sure the conversion service is up and running before applying new changes.

Contacting the webhook

Once the API server has determined a request should be sent to a conversion webhook, it needs to know how to contact the webhook. This is specified in the `webhookClientConfig` stanza of the webhook configuration.

Conversion webhooks can either be called via a URL or a service reference, and can optionally include a custom CA bundle to use to verify the TLS connection.

URL

`url` gives the location of the webhook, in standard URL form (`scheme://host:port/path`).

The host should not refer to a service running in the cluster; use a service reference by specifying the `service` field instead. The host might be resolved via external DNS in some apiservers (i.e., kube-apiserver cannot resolve in-cluster DNS as that would be a layering violation). host may also be an IP address.

Please note that using localhost or 127.0.0.1 as a host is risky unless you take great care to run this webhook on all hosts which run an apiserver which might need to make calls to this webhook. Such installations are likely to be non-portable or not readily run in a new cluster.

The scheme must be "https"; the URL must begin with "https://".

Attempting to use a user or basic auth (for example "user:password@") is not allowed. Fragments ("#...") and query parameters ("?...") are also not allowed.

Here is an example of a conversion webhook configured to call a URL (and expects the TLS certificate to be verified using system trust roots, so does not specify a caBundle):

- apiextensions.k8s.io/v1
- apiextensions.k8s.io/v1beta1

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
...
spec:
  ...
  conversion:
    strategy: Webhook
    webhook:
      clientConfig:
        url: "https://my-webhook.example.com:9443/my-webhook-path"
  ...
```

```
# Deprecated in v1.16 in favor of apiextensions.k8s.io/v1
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
...
spec:
  ...
  conversion:
    strategy: Webhook
    webhookClientConfig:
      url: "https://my-webhook.example.com:9443/my-webhook-path"
  ...
```

Service Reference

The service stanza inside `webhookClientConfig` is a reference to the service for a conversion webhook. If the webhook is running within the cluster, then you should use service instead of url. The service namespace and name are required. The port is optional and defaults to 443. The path is optional and defaults to "/".

Here is an example of a webhook that is configured to call a service on port "1234" at the subpath "/my-path", and to verify the TLS connection against the ServerName my-service-name.my-service-namespace.svc using a custom CA bundle.

- apiextensions.k8s.io/v1
- apiextensions.k8s.io/v1beta1

```

apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
...
spec:
  ...
  conversion:
    strategy: Webhook
    webhook:
      clientConfig:
        service:
          namespace: my-service-namespace
          name: my-service-name
          path: /my-path
          port: 1234
      caBundle: "Ci0tLS0tQk...<base64-encoded PEM bundle>...tLS0K"
  ...

```

```

# Deprecated in v1.16 in favor of apiextensions.k8s.io/v1
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
...
spec:
  ...
  conversion:
    strategy: Webhook
    webhookClientConfig:
      service:
        namespace: my-service-namespace
        name: my-service-name
        path: /my-path
        port: 1234
      caBundle: "Ci0tLS0tQk...<base64-encoded PEM bundle>...tLS0K"
  ...

```

Webhook request and response

Request

Webhooks are sent a POST request, with Content-Type: application/json, with a ConversionReview API object in the apiextensions.k8s.io API group serialized to JSON as the body.

Webhooks can specify what versions of ConversionReview objects they accept with the conversionReviewVersions field in their CustomResourceDefinition:

- apiextensions.k8s.io/v1
- apiextensions.k8s.io/v1beta1

```

apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
...
spec:

```



```
...
conversion:
  strategy: Webhook
  webhook:
    conversionReviewVersions: ["v1", "v1beta1"]
...
```

conversionReviewVersions is a required field when creating apiextensions.k8s.io/v1 custom resource definitions. Webhooks are required to support at least one ConversionReview version understood by the current and previous API server.

```
# Deprecated in v1.16 in favor of apiextensions.k8s.io/v1
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
...
spec:
  ...
  conversion:
    strategy: Webhook
    conversionReviewVersions: ["v1", "v1beta1"]
  ...
```

If no conversionReviewVersions are specified, the default when creating apiextensions.k8s.io/v1beta1 custom resource definitions is v1beta1.

API servers send the first ConversionReview version in the conversionReviewVersions list they support. If none of the versions in the list are supported by the API server, the custom resource definition will not be allowed to be created. If an API server encounters a conversion webhook configuration that was previously created and does not support any of the ConversionReview versions the API server knows how to send, attempts to call to the webhook will fail.

This example shows the data contained in an ConversionReview object for a request to convert CronTab objects to example.com/v1:

- [apiextensions.k8s.io/v1](#)
- [apiextensions.k8s.io/v1beta1](#)

```
{
  "apiVersion": "apiextensions.k8s.io/v1",
  "kind": "ConversionReview",
  "request": {
    # Random uid uniquely identifying this conversion call
    "uid": "705ab4f5-6393-11e8-b7cc-42010a800002",

    # The API group and version the objects should be converted to
    "desiredAPIVersion": "example.com/v1",

    # The list of objects to convert.
    # May contain one or more objects, in one or more versions.
    "objects": [
      {
        "kind": "CronTab",
        "apiVersion": "example.com/v1beta1",
        "metadata": {
```

```

    "creationTimestamp": "2019-09-04T14:03:02Z",
    "name": "local-crontab",
    "namespace": "default",
    "resourceVersion": "143",
    "uid": "3415a7fc-162b-4300-b5da-fd6083580d66"
  },
  "hostPort": "localhost:1234"
},
{
  "kind": "CronTab",
  "apiVersion": "example.com/v1beta1",
  "metadata": {
    "creationTimestamp": "2019-09-03T13:02:01Z",
    "name": "remote-crontab",
    "resourceVersion": "12893",
    "uid": "359a83ec-b575-460d-b553-d859cedde8a0"
  },
  "hostPort": "example.com:2345"
}
]
}
}

```

```

{
  # Deprecated in v1.16 in favor of apiextensions.k8s.io/v1
  "apiVersion": "apiextensions.k8s.io/v1beta1",
  "kind": "ConversionReview",
  "request": {
    # Random uid uniquely identifying this conversion call
    "uid": "705ab4f5-6393-11e8-b7cc-42010a800002",

    # The API group and version the objects should be converted to
    "desiredAPIVersion": "example.com/v1",

    # The list of objects to convert.
    # May contain one or more objects, in one or more versions.
    "objects": [
      {
        "kind": "CronTab",
        "apiVersion": "example.com/v1beta1",
        "metadata": {
          "creationTimestamp": "2019-09-04T14:03:02Z",
          "name": "local-crontab",
          "namespace": "default",
          "resourceVersion": "143",
          "uid": "3415a7fc-162b-4300-b5da-fd6083580d66"
        },
        "hostPort": "localhost:1234"
      },
      {
        "kind": "CronTab",
        "apiVersion": "example.com/v1beta1",
        "metadata": {

```

```

    "creationTimestamp": "2019-09-03T13:02:01Z",
    "name": "remote-crontab",
    "resourceVersion": "12893",
    "uid": "359a83ec-b575-460d-b553-d859cedde8a0"
  },
  "hostPort": "example.com:2345"
}
]
}
}

```

Response

Webhooks respond with a 200 HTTP status code, Content-Type: application/json, and a body containing a ConversionReview object (in the same version they were sent), with the response stanza populated, serialized to JSON.

If conversion succeeds, a webhook should return a response stanza containing the following fields:

- uid, copied from the request.uid sent to the webhook
- result, set to {"status": "Success"}
- convertedObjects, containing all of the objects from request.objects, converted to request.desiredVersion

Example of a minimal successful response from a webhook:

- apiextensions.k8s.io/v1
- apiextensions.k8s.io/v1beta1

```

{
  "apiVersion": "apiextensions.k8s.io/v1",
  "kind": "ConversionReview",
  "response": {
    # must match <request.uid>
    "uid": "705ab4f5-6393-11e8-b7cc-42010a800002",
    "result": {
      "status": "Success"
    },
    # Objects must match the order of request.objects, and have apiVersion set to
    <request.desiredAPIVersion>.
  }
}

```

kind, metadata.uid, metadata.name, and metadata.namespace fields must not be changed by the webhook.

metadata.labels and metadata.annotations fields may be changed by the webhook.

All other changes to metadata fields by the webhook are ignored.

```

"convertedObjects": [
  {
    "kind": "CronTab",
    "apiVersion": "example.com/v1",
    "metadata": {
      "creationTimestamp": "2019-09-04T14:03:02Z",
      "name": "local-crontab",

```

```

    "namespace": "default",
    "resourceVersion": "143",
    "uid": "3415a7fc-162b-4300-b5da-fd6083580d66"
  },
  "host": "localhost",
  "port": "1234"
},
{
  "kind": "CronTab",
  "apiVersion": "example.com/v1",
  "metadata": {
    "creationTimestamp": "2019-09-03T13:02:01Z",
    "name": "remote-crontab",
    "resourceVersion": "12893",
    "uid": "359a83ec-b575-460d-b553-d859cedde8a0"
  },
  "host": "example.com",
  "port": "2345"
}
]
}
}

```

```

{
  # Deprecated in v1.16 in favor of apiextensions.k8s.io/v1
  "apiVersion": "apiextensions.k8s.io/v1beta1",
  "kind": "ConversionReview",
  "response": {
    # must match <request.uid>
    "uid": "705ab4f5-6393-11e8-b7cc-42010a800002",
    "result": {
      "status": "Failed"
    }
  },
  # Objects must match the order of request.objects, and have apiVersion set to
  <request.desiredAPIVersion>.

```

kind, metadata.uid, metadata.name, and metadata.namespace fields must not be changed by the webhook.

metadata.labels and metadata.annotations fields may be changed by the webhook.

All other changes to metadata fields by the webhook are ignored.

```

"convertedObjects": [
  {
    "kind": "CronTab",
    "apiVersion": "example.com/v1",
    "metadata": {
      "creationTimestamp": "2019-09-04T14:03:02Z",
      "name": "local-crontab",
      "namespace": "default",
      "resourceVersion": "143",
      "uid": "3415a7fc-162b-4300-b5da-fd6083580d66"
    },
    "host": "localhost",
    "port": "1234"
  }
]

```

```

    },
    {
      "kind": "CronTab",
      "apiVersion": "example.com/v1",
      "metadata": {
        "creationTimestamp": "2019-09-03T13:02:01Z",
        "name": "remote-crontab",
        "resourceVersion": "12893",
        "uid": "359a83ec-b575-460d-b553-d859cedde8a0"
      },
      "host": "example.com",
      "port": "2345"
    }
  ]
}

```

If conversion fails, a webhook should return a response stanza containing the following fields:

- uid, copied from the request.uid sent to the webhook
- result, set to {"status":"Failed"}

Warning:

Failing conversion can disrupt read and write access to the custom resources, including the ability to update or delete the resources. Conversion failures should be avoided whenever possible, and should not be used to enforce validation constraints (use validation schemas or webhook admission instead).

Example of a response from a webhook indicating a conversion request failed, with an optional message:

- apiextensions.k8s.io/v1
- apiextensions.k8s.io/v1beta1

```

{
  "apiVersion": "apiextensions.k8s.io/v1",
  "kind": "ConversionReview",
  "response": {
    "uid": "<value from request.uid>",
    "result": {
      "status": "Failed",
      "message": "hostPort could not be parsed into a separate host and port"
    }
  }
}

```

```

{
  # Deprecated in v1.16 in favor of apiextensions.k8s.io/v1
  "apiVersion": "apiextensions.k8s.io/v1beta1",
  "kind": "ConversionReview",
  "response": {
    "uid": "<value from request.uid>",

```

```
"result": {
  "status": "Failed",
  "message": "hostPort could not be parsed into a separate host and port"
}
}
```

Writing, reading, and updating versioned CustomResourceDefinition objects

When an object is written, it is stored at the version designated as the storage version at the time of the write. If the storage version changes, existing objects are never converted automatically. However, newly-created or updated objects are written at the new storage version. It is possible for an object to have been written at a version that is no longer served.

When you read an object, you specify the version as part of the path. You can request an object at any version that is currently served. If you specify a version that is different from the object's stored version, Kubernetes returns the object to you at the version you requested, but the stored object is not changed on disk.

What happens to the object that is being returned while serving the read request depends on what is specified in the CRD's `spec.conversion`:

- if the default strategy value `None` is specified, the only modifications to the object are changing the `apiVersion` string and perhaps [pruning unknown fields](#) (depending on the configuration). Note that this is unlikely to lead to good results if the schemas differ between the storage and requested version. In particular, you should not use this strategy if the same data is represented in different fields between versions.
- if [webhook conversion](#) is specified, then this mechanism controls the conversion.

If you update an existing object, it is rewritten at the version that is currently the storage version. This is the only way that objects can change from one version to another.

To illustrate this, consider the following hypothetical series of events:

1. The storage version is `v1beta1`. You create an object. It is stored at version `v1beta1`
2. You add version `v1` to your CustomResourceDefinition and designate it as the storage version. Here the schemas for `v1` and `v1beta1` are identical, which is typically the case when promoting an API to stable in the Kubernetes ecosystem.
3. You read your object at version `v1beta1`, then you read the object again at version `v1`. Both returned objects are identical except for the `apiVersion` field.
4. You create a new object. It is stored at version `v1`. You now have two objects, one of which is at `v1beta1`, and the other of which is at `v1`.
5. You update the first object. It is now stored at version `v1` since that is the current storage version.

Previous storage versions

The API server records each version which has ever been marked as the storage version in the status field `storedVersions`. Objects may have been stored at any version that has ever been designated as a storage version. No objects can exist in storage at a version that has never been a storage version.

Upgrade existing objects to a new stored version

When deprecating versions and dropping support, select a storage upgrade procedure.

Option 1: Use the Storage Version Migrator

1. Run the [storage Version migrator](#)
2. Remove the old version from the CustomResourceDefinition status.storedVersions field.

Option 2: Manually upgrade the existing objects to a new stored version

The following is an example procedure to upgrade from v1beta1 to v1.

1. Set v1 as the storage in the CustomResourceDefinition file and apply it using kubectl. The storedVersions is now v1beta1, v1.
2. Write an upgrade procedure to list all existing objects and write them with the same content. This forces the backend to write objects in the current storage version, which is v1.
3. Remove v1beta1 from the CustomResourceDefinition status.storedVersions field.

Note:

The flag `--subresource` is used with the kubectl get, patch, edit, and replace commands to fetch and update the subresources, status and scale, for all the API resources that support them. This flag is available starting from kubectl version v1.24. Previously, reading subresources (like status) via kubectl involved using `kubectl --raw`, and updating subresources using kubectl was not possible at all. Starting from v1.24, the kubectl tool can be used to edit or patch the status subresource on a CRD object. See [How to patch a Deployment using the subresource flag](#).

This page is part of the documentation for Kubernetes v1.30. If you are running a different version of Kubernetes, consult the documentation for that release.

Here is an example of how to patch the status subresource for a CRD object using kubectl:

```
kubectl patch customresourcedefinitions <CRD_Name> --subresource='status' --type='merge' -p '{"status":{"storedVersions":["v1"]}}'
```

Set up an Extension API Server

Setting up an extension API server to work with the aggregation layer allows the Kubernetes apiserver to be extended with additional APIs, which are not part of the core Kubernetes APIs.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

- You must [configure the aggregation layer](#) and enable the apiserver flags.

Set up an extension api-server to work with the aggregation layer

The following steps describe how to set up an extension-apiserver *at a high level*. These steps apply regardless if you're using YAML configs or using APIs. An attempt is made to specifically identify any differences between the two. For a concrete example of how they can be implemented using YAML configs, you can look at the [sample-apiserver](#) in the Kubernetes repo.

Alternatively, you can use an existing 3rd party solution, such as [apiserver-builder](#), which should generate a skeleton and automate all of the following steps for you.

1. Make sure the APIService API is enabled (check `--runtime-config`). It should be on by default, unless it's been deliberately turned off in your cluster.
2. You may need to make an RBAC rule allowing you to add APIService objects, or get your cluster administrator to make one. (Since API extensions affect the entire cluster, it is not recommended to do testing/development/debug of an API extension in a live cluster.)
3. Create the Kubernetes namespace you want to run your extension api-service in.
4. Create/get a CA cert to be used to sign the server cert the extension api-server uses for HTTPS.
5. Create a server cert/key for the api-server to use for HTTPS. This cert should be signed by the above CA. It should also have a CN of the Kube DNS name. This is derived from the Kubernetes service and be of the form `<service name>.<service name namespace>.svc`
6. Create a Kubernetes secret with the server cert/key in your namespace.
7. Create a Kubernetes deployment for the extension api-server and make sure you are loading the secret as a volume. It should contain a reference to a working image of your extension api-server. The deployment should also be in your namespace.
8. Make sure that your extension-apiserver loads those certs from that volume and that they are used in the HTTPS handshake.
9. Create a Kubernetes service account in your namespace.
10. Create a Kubernetes cluster role for the operations you want to allow on your resources.
11. Create a Kubernetes cluster role binding from the service account in your namespace to the cluster role you created.
12. Create a Kubernetes cluster role binding from the service account in your namespace to the `system:auth-delegator` cluster role to delegate auth decisions to the Kubernetes core API server.
13. Create a Kubernetes role binding from the service account in your namespace to the `extension-apiserver-authentication-reader` role. This allows your extension api-server to access the `extension-apiserver-authentication` configmap.
14. Create a Kubernetes apiservice. The CA cert above should be base64 encoded, stripped of new lines and used as the `spec.caBundle` in the apiservice. This should not be namespaced. If using the [kube-aggregator API](#), only pass in the PEM encoded CA bundle because the base 64 encoding is done for you.
15. Use `kubectl` to get your resource. When run, `kubectl` should return "No resources found.". This message indicates that everything worked but you currently have no objects of that resource type created.

What's next

- Walk through the steps to [configure the API aggregation layer](#) and enable the apiserver flags.
- For a high level overview, see [Extending the Kubernetes API with the aggregation layer](#).
- Learn how to [Extend the Kubernetes API using Custom Resource Definitions](#).

Configure Multiple Schedulers

Kubernetes ships with a default scheduler that is described [here](#). If the default scheduler does not suit your needs you can implement your own scheduler. Moreover, you can even run multiple schedulers simultaneously alongside the default scheduler and instruct Kubernetes what scheduler to use for each of your pods. Let's learn how to run multiple schedulers in Kubernetes with an example.

A detailed description of how to implement a scheduler is outside the scope of this document. Please refer to the kube-scheduler implementation in [pkg/scheduler](#) in the Kubernetes source directory for a canonical example.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

Package the scheduler

Package your scheduler binary into a container image. For the purposes of this example, you can use the default scheduler (kube-scheduler) as your second scheduler. Clone the [Kubernetes source code from GitHub](#) and build the source.

```
git clone https://github.com/kubernetes/kubernetes.git
cd kubernetes
make
```

Create a container image containing the kube-scheduler binary. Here is the Dockerfile to build the image:

```
FROM busybox
ADD ./_output/local/bin/linux/amd64/kube-scheduler /usr/local/bin/kube-scheduler
```

Save the file as Dockerfile, build the image and push it to a registry. This example pushes the image to [Google Container Registry \(GCR\)](#). For more details, please read the GCR

[documentation](#). Alternatively you can also use the [docker hub](#). For more details refer to the docker hub [documentation](#).

```
docker build -t gcr.io/my-gcp-project/my-kube-scheduler:1.0 .    # The image name and the repository
gcloud docker -- push gcr.io/my-gcp-project/my-kube-scheduler:1.0 # used in here is just an example
```

Define a Kubernetes Deployment for the scheduler

Now that you have your scheduler in a container image, create a pod configuration for it and run it in your Kubernetes cluster. But instead of creating a pod directly in the cluster, you can use a [Deployment](#) for this example. A [Deployment](#) manages a [Replica Set](#) which in turn manages the pods, thereby making the scheduler resilient to failures. Here is the deployment config. Save it as my-scheduler.yaml:

[admin/sched/my-scheduler.yaml](#)

```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: my-scheduler
  namespace: kube-system
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: my-scheduler-as-kube-scheduler
subjects:
- kind: ServiceAccount
  name: my-scheduler
  namespace: kube-system
roleRef:
  kind: ClusterRole
  name: system:kube-scheduler
  apiGroup: rbac.authorization.k8s.io
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: my-scheduler-as-volume-scheduler
subjects:
- kind: ServiceAccount
  name: my-scheduler
  namespace: kube-system
roleRef:
  kind: ClusterRole
  name: system:volume-scheduler
  apiGroup: rbac.authorization.k8s.io
---
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
```

```

metadata:
  name: my-scheduler-extension-apiserver-authentication-reader
  namespace: kube-system
roleRef:
  kind: Role
  name: extension-apiserver-authentication-reader
  apiGroup: rbac.authorization.k8s.io
subjects:
- kind: ServiceAccount
  name: my-scheduler
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: my-scheduler-config
  namespace: kube-system
data:
  my-scheduler-config.yaml: |
    apiVersion: kubescheduler.config.k8s.io/v1beta2
    kind: KubeSchedulerConfiguration
    profiles:
      - schedulerName: my-scheduler
    leaderElection:
      leaderElect: false
---
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    component: scheduler
    tier: control-plane
  name: my-scheduler
  namespace: kube-system
spec:
  selector:
    matchLabels:
      component: scheduler
      tier: control-plane
  replicas: 1
  template:
    metadata:
      labels:
        component: scheduler
        tier: control-plane
        version: second
    spec:
      serviceAccountName: my-scheduler
      containers:
      - command:
        - /usr/local/bin/kube-scheduler
        - --config=/etc/kubernetes/my-scheduler/my-scheduler-config.yaml
        image: gcr.io/my-gcp-project/my-kube-scheduler:1.0

```

```
livenessProbe:
  httpGet:
    path: /healthz
    port: 10259
    scheme: HTTPS
  initialDelaySeconds: 15
name: kube-second-scheduler
readinessProbe:
  httpGet:
    path: /healthz
    port: 10259
    scheme: HTTPS
resources:
  requests:
    cpu: '0.1'
securityContext:
  privileged: false
volumeMounts:
  - name: config-volume
    mountPath: /etc/kubernetes/my-scheduler
hostNetwork: false
hostPID: false
volumes:
  - name: config-volume
    configMap:
      name: my-scheduler-config
```

In the above manifest, you use a [KubeSchedulerConfiguration](#) to customize the behavior of your scheduler implementation. This configuration has been passed to the kube-scheduler during initialization with the `--config` option. The `my-scheduler-config` ConfigMap stores the configuration file. The Pod of the `my-scheduler` Deployment mounts the `my-scheduler-config` ConfigMap as a volume.

In the aforementioned Scheduler Configuration, your scheduler implementation is represented via a [KubeSchedulerProfile](#).

Note:

To determine if a scheduler is responsible for scheduling a specific Pod, the `spec.schedulerName` field in a PodTemplate or Pod manifest must match the `schedulerName` field of the `KubeSchedulerProfile`. All schedulers running in the cluster must have unique names.

Also, note that you create a dedicated service account `my-scheduler` and bind the `ClusterRole` `system:kube-scheduler` to it so that it can acquire the same privileges as `kube-scheduler`.

Please see the [kube-scheduler documentation](#) for detailed description of other command line arguments and [Scheduler Configuration reference](#) for detailed description of other customizable kube-scheduler configurations.

Run the second scheduler in the cluster

In order to run your scheduler in a Kubernetes cluster, create the deployment specified in the config above in a Kubernetes cluster:

```
kubectl create -f my-scheduler.yaml
```

Verify that the scheduler pod is running:

```
kubectl get pods --namespace=kube-system
```

| NAME | READY | STATUS | RESTARTS | AGE |
|--------------------------|-------|---------|----------|-----|
| | | | | |
| my-scheduler-lnf4s-4744f | 1/1 | Running | 0 | 2m |
| ... | | | | |

You should see a "Running" my-scheduler pod, in addition to the default kube-scheduler pod in this list.

Enable leader election

To run multiple-scheduler with leader election enabled, you must do the following:

Update the following fields for the KubeSchedulerConfiguration in the my-scheduler-config ConfigMap in your YAML file:

- leaderElection.leaderElect to true
- leaderElection.resourceNamespace to <lock-object-namespace>
- leaderElection.resourceName to <lock-object-name>

Note:

The control plane creates the lock objects for you, but the namespace must already exist. You can use the kube-system namespace.

If RBAC is enabled on your cluster, you must update the system:kube-scheduler cluster role. Add your scheduler name to the resourceNames of the rule applied for endpoints and leases resources, as in the following example:

```
kubectl edit clusterrole system:kube-scheduler
```

[admin/sched/clusterrole.yaml](#)

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:kube-scheduler
rules:
- apiGroups:
```

```

- coordination.k8s.io
resources:
- leases
verbs:
- create
- apiGroups:
- coordination.k8s.io
resourceNames:
- kube-scheduler
- my-scheduler
resources:
- leases
verbs:
- get
- update
- apiGroups:
- ""
resourceNames:
- kube-scheduler
- my-scheduler
resources:
- endpoints
verbs:
- delete
- get
- patch
- update

```

Specify schedulers for pods

Now that your second scheduler is running, create some pods, and direct them to be scheduled by either the default scheduler or the one you deployed. In order to schedule a given pod using a specific scheduler, specify the name of the scheduler in that pod spec. Let's look at three examples.

- Pod spec without any scheduler name

[admin/sched/pod1.yaml](#)

```

apiVersion: v1
kind: Pod
metadata:
  name: no-annotation
  labels:
    name: multischeduler-example
spec:
  containers:
  - name: pod-with-no-annotation-container
    image: registry.k8s.io/pause:2.0

```

When no scheduler name is supplied, the pod is automatically scheduled using the default-scheduler.

Save this file as `pod1.yaml` and submit it to the Kubernetes cluster.

```
kubectrl create -f pod1.yaml
```

- Pod spec with default-scheduler

[admin/sched/pod2.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: annotation-default-scheduler
  labels:
    name: multischeduler-example
spec:
  schedulerName: default-scheduler
  containers:
  - name: pod-with-default-annotation-container
    image: registry.k8s.io/pause:2.0
```

A scheduler is specified by supplying the scheduler name as a value to `spec.schedulerName`. In this case, we supply the name of the default scheduler which is `default-scheduler`.

Save this file as `pod2.yaml` and submit it to the Kubernetes cluster.

```
kubectrl create -f pod2.yaml
```

- Pod spec with my-scheduler

[admin/sched/pod3.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: annotation-second-scheduler
  labels:
    name: multischeduler-example
spec:
  schedulerName: my-scheduler
  containers:
  - name: pod-with-second-annotation-container
    image: registry.k8s.io/pause:2.0
```

In this case, we specify that this pod should be scheduled using the scheduler that we deployed - `my-scheduler`. Note that the value of `spec.schedulerName` should match the name supplied for the scheduler in the `schedulerName` field of the mapping `KubeSchedulerProfile`.

Save this file as `pod3.yaml` and submit it to the Kubernetes cluster.

```
kubectrl create -f pod3.yaml
```

Verify that all three pods are running.

```
kubectrl get pods
```

Verifying that the pods were scheduled using the desired schedulers

In order to make it easier to work through these examples, we did not verify that the pods were actually scheduled using the desired schedulers. We can verify that by changing the order of pod and deployment config submissions above. If we submit all the pod configs to a Kubernetes cluster before submitting the scheduler deployment config, we see that the pod annotation-second-scheduler remains in "Pending" state forever while the other two pods get scheduled. Once we submit the scheduler deployment config and our new scheduler starts running, the annotation-second-scheduler pod gets scheduled as well.

Alternatively, you can look at the "Scheduled" entries in the event logs to verify that the pods were scheduled by the desired schedulers.

```
kubectrl get events
```

You can also use a [custom scheduler configuration](#) or a custom container image for the cluster's main scheduler by modifying its static pod manifest on the relevant control plane nodes.

Use an HTTP Proxy to Access the Kubernetes API

This page shows how to use an HTTP proxy to access the Kubernetes API.

Before you begin

You need to have a Kubernetes cluster, and the kubectrl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectrl version.

If you do not already have an application running in your cluster, start a Hello world application by entering this command:

```
kubectrl create deployment node-hello --image=gcr.io/google-samples/node-hello:1.0 --port=8080
```

Using kubectrl to start a proxy server

This command starts a proxy to the Kubernetes API server:

```
kubectrl proxy --port=8080
```


Exploring the Kubernetes API

When the proxy server is running, you can explore the API using `curl`, `wget`, or a browser.

Get the API versions:

```
curl http://localhost:8080/api/
```

The output should look similar to this:

```
{
  "kind": "APIVersions",
  "versions": [
    "v1"
  ],
  "serverAddressByClientCIDRs": [
    {
      "clientCIDR": "0.0.0.0/0",
      "serverAddress": "10.0.2.15:8443"
    }
  ]
}
```

Get a list of pods:

```
curl http://localhost:8080/api/v1/namespaces/default/pods
```

The output should look similar to this:

```
{
  "kind": "PodList",
  "apiVersion": "v1",
  "metadata": {
    "resourceVersion": "33074"
  },
  "items": [
    {
      "metadata": {
        "name": "kubernetes-bootcamp-2321272333-ix8pt",
        "generateName": "kubernetes-bootcamp-2321272333-",
        "namespace": "default",
        "uid": "ba21457c-6b1d-11e6-85f7-1ef9f1dab92b",
        "resourceVersion": "33003",
        "creationTimestamp": "2016-08-25T23:43:30Z",
        "labels": {
          "pod-template-hash": "2321272333",
          "run": "kubernetes-bootcamp"
        }
      },
      ...
    }
  ]
}
```

What's next

Learn more about [kubect proxy](#).

Use a SOCKS5 Proxy to Access the Kubernetes API

FEATURE STATE: Kubernetes v1.24 [stable]

This page shows how to use a SOCKS5 proxy to access the API of a remote Kubernetes cluster. This is useful when the cluster you want to access does not expose its API directly on the public internet.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.24. To check the version, enter `kubectl version`.

You need SSH client software (the `ssh` tool), and an SSH service running on the remote server. You must be able to log in to the SSH service on the remote server.

Task context

Note:

This example tunnels traffic using SSH, with the SSH client and server acting as a SOCKS proxy. You can instead use any other kind of [SOCKS5](#) proxies.

Figure 1 represents what you're going to achieve in this task.

- You have a client computer, referred to as local in the steps ahead, from where you're going to create requests to talk to the Kubernetes API.
- The Kubernetes server/API is hosted on a remote server.
- You will use SSH client and server software to create a secure SOCKS5 tunnel between the local and the remote server. The HTTPS traffic between the client and the Kubernetes API will flow over the SOCKS5 tunnel, which is itself tunnelled over SSH.

```
graph LR; subgraph local[Local client machine] client([client])-. local traffic .-> local_ssh[Local SSH SOCKS5 proxy]; end local_ssh[SSH SOCKS5
```

```

proxy]-- SSH Tunnel -->sshd subgraph remote[Remote server] sshd[SSH
server]-- local traffic -->service1; end client([client])-. proxied HTTPs traffic
going through the proxy .->service1[Kubernetes API]; classDef plain
fill:#ddd,stroke:#fff,stroke-width:4px,color:#000; classDef k8s
fill:#326ce5,stroke:#fff,stroke-width:4px,color:#fff; classDef cluster
fill:#fff,stroke:#bbb,stroke-width:2px,color:#326ce5; class
ingress,service1,service2,pod1,pod2,pod3,pod4 k8s; class client plain; class cluster
cluster;

```

JavaScript must be [enabled](#) to view this content

Figure 1. SOCKS5 tutorial components

Using ssh to create a SOCKS5 proxy

The following command starts a SOCKS5 proxy between your client machine and the remote SOCKS server:

```
# The SSH tunnel continues running in the foreground after you run this
ssh -D 1080 -q -N username@kubernetes-remote-server.example
```

The SOCKS5 proxy lets you connect to your cluster's API server based on the following configuration:

- `-D 1080`: opens a SOCKS proxy on local port `:1080`.
- `-q`: quiet mode. Causes most warning and diagnostic messages to be suppressed.
- `-N`: Do not execute a remote command. Useful for just forwarding ports.
- `username@kubernetes-remote-server.example`: the remote SSH server behind which the Kubernetes cluster is running (eg: a bastion host).

Client configuration

To access the Kubernetes API server through the proxy you must instruct `kubectl` to send queries through the SOCKS proxy we created earlier. Do this by either setting the appropriate environment variable, or via the `proxy-url` attribute in the `kubeconfig` file. Using an environment variable:

```
export HTTPS_PROXY=socks5://localhost:1080
```

To always use this setting on a specific `kubectl` context, specify the `proxy-url` attribute in the relevant cluster entry within the `~/.kube/config` file. For example:

```

apiVersion: v1
clusters:
- cluster:
  certificate-authority-data: LRMEMMW2 # shortened for readability
  server: https://<API_SERVER_IP_ADDRESS>:6443 # the "Kubernetes API" server, in other
words the IP address of kubernetes-remote-server.example
  proxy-url: socks5://localhost:1080 # the "SSH SOCKS5 proxy" in the diagram above
  name: default
contexts:
- context:
  cluster: default

```

```
user: default
name: default
current-context: default
kind: Config
preferences: {}
users:
- name: default
  user:
    client-certificate-data: LS0tLS1CR== # shortened for readability
    client-key-data: LS0tLS1CRUdJT= # shortened for readability
```

Once you have created the tunnel via the ssh command mentioned earlier, and defined either the environment variable or the proxy-url attribute, you can interact with your cluster through that proxy. For example:

```
kubectl get pods
```

| NAMESPACE | NAME | READY | STATUS | RESTARTS | AGE |
|-------------|-------------------------|-------|---------|----------|-------|
| kube-system | coredns-85cb69466-klwq8 | 1/1 | Running | 0 | 5m46s |

Note:

- Before kubectl 1.24, most kubectl commands worked when using a socks proxy, except kubectl exec.
- kubectl supports both HTTPS_PROXY and https_proxy environment variables. These are used by other programs that support SOCKS, such as curl. Therefore in some cases it will be better to define the environment variable on the command line:

```
HTTPS_PROXY=socks5://localhost:1080 kubectl get pods
```

- When using proxy-url, the proxy is used only for the relevant kubectl context, whereas the environment variable will affect all contexts.
- The k8s API server hostname can be further protected from DNS leakage by using the socks5h protocol name instead of the more commonly known socks5 protocol shown above. In this case, kubectl will ask the proxy server (such as an ssh bastion) to resolve the k8s API server domain name, instead of resolving it on the system running kubectl. Note also that with socks5h, a k8s API server URL like https://localhost:6443/api does not refer to your local client computer. Instead, it refers to localhost as known on the proxy server (eg the ssh bastion).

Clean up

Stop the ssh port-forwarding process by pressing CTRL+C on the terminal where it is running.

Type unset https_proxy in a terminal to stop forwarding http traffic through the proxy.

Further reading

- [OpenSSH remote login client](#)

Set up Konnectivity service

The Konnectivity service provides a TCP level proxy for the control plane to cluster communication.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#).

Configure the Konnectivity service

The following steps require an egress configuration, for example:

[admin/konnectivity/egress-selector-configuration.yaml](#)

```
apiVersion: apiserver.k8s.io/v1beta1
kind: EgressSelectorConfiguration
egressSelections:
# Since we want to control the egress traffic to the cluster, we use the
# "cluster" as the name. Other supported values are "etcd", and "controlplane".
- name: cluster
  connection:
    # This controls the protocol between the API Server and the Konnectivity
    # server. Supported values are "GRPC" and "HTTPConnect". There is no
    # end user visible difference between the two modes. You need to set the
    # Konnectivity server to work in the same mode.
    proxyProtocol: GRPC
  transport:
    # This controls what transport the API Server uses to communicate with the
    # Konnectivity server. UDS is recommended if the Konnectivity server
    # locates on the same machine as the API Server. You need to configure the
    # Konnectivity server to listen on the same UDS socket.
    # The other supported transport is "tcp". You will need to set up TLS
    # config to secure the TCP transport.
    uds:
      udsName: /etc/kubernetes/konnectivity-server/konnectivity-server.socket
```

You need to configure the API Server to use the Konnectivity service and direct the network traffic to the cluster nodes:

1. Make sure that [Service Account Token Volume Projection](#) feature enabled in your cluster. It is enabled by default since Kubernetes v1.20.
2. Create an egress configuration file such as `admin/konnectivity/egress-selector-configuration.yaml`.
3. Set the `--egress-selector-config-file` flag of the API Server to the path of your API Server egress configuration file.
4. If you use UDS connection, add volumes config to the kube-apiserver:

```
spec:
  containers:
    volumeMounts:
      - name: konnectivity-uds
        mountPath: /etc/kubernetes/konnectivity-server
        readOnly: false
  volumes:
    - name: konnectivity-uds
      hostPath:
        path: /etc/kubernetes/konnectivity-server
        type: DirectoryOrCreate
```

Generate or obtain a certificate and kubeconfig for konnectivity-server. For example, you can use the OpenSSL command line tool to issue a X.509 certificate, using the cluster CA certificate /etc/kubernetes/pki/ca.crt from a control-plane host.

```
openssl req -subj "/CN=system:konnectivity-server" -new -newkey rsa:2048 -nodes -out
konnectivity.csr -keyout konnectivity.key
openssl x509 -req -in konnectivity.csr -CA /etc/kubernetes/pki/ca.crt -CAkey /etc/kubernetes/
pki/ca.key -CAcreateserial -out konnectivity.crt -days 375 -sha256
SERVER=$(kubectl config view -o jsonpath='{.clusters..server}')
kubectl --kubeconfig /etc/kubernetes/konnectivity-server.conf config set-credentials
system:konnectivity-server --client-certificate konnectivity.crt --client-key konnectivity.key --
embed-certs=true
kubectl --kubeconfig /etc/kubernetes/konnectivity-server.conf config set-cluster kubernet
es --server "$SERVER" --certificate-authority /etc/kubernetes/pki/ca.crt --embed-certs=true
kubectl --kubeconfig /etc/kubernetes/konnectivity-server.conf config set-context
system:konnectivity-server@kubernetes --cluster kubernet
es --user system:konnectivity-server
kubectl --kubeconfig /etc/kubernetes/konnectivity-server.conf config use-context
system:konnectivity-server@kubernetes
rm -f konnectivity.crt konnectivity.key konnectivity.csr
```

Next, you need to deploy the Konnectivity server and agents. [kubernetes-sigs/apiserver-network-proxy](#) is a reference implementation.

Deploy the Konnectivity server on your control plane node. The provided konnectivity-server.yaml manifest assumes that the Kubernetes components are deployed as a [static Pod](#) in your cluster. If not, you can deploy the Konnectivity server as a DaemonSet.

[admin/konnectivity/konnectivity-server.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: konnectivity-server
  namespace: kube-system
spec:
  priorityClassName: system-cluster-critical
  hostNetwork: true
  containers:
    - name: konnectivity-server-container
      image: registry.k8s.io/kas-network-proxy/proxy-server:v0.0.37
      command: ["/proxy-server"]
      args: [
```

```

"--logtostderr=true",
# This needs to be consistent with the value set in egressSelectorConfiguration.
"--uds-name=/etc/kubernetes/konnectivity-server/konnectivity-server.socket",
"--delete-existing-uds-file",
# The following two lines assume the Konnectivity server is
# deployed on the same machine as the apiserver, and the certs and
# key of the API Server are at the specified location.
"--cluster-cert=/etc/kubernetes/pki/apiserver.crt",
"--cluster-key=/etc/kubernetes/pki/apiserver.key",
# This needs to be consistent with the value set in egressSelectorConfiguration.
"--mode=grpc",
"--server-port=0",
"--agent-port=8132",
"--admin-port=8133",
"--health-port=8134",
"--agent-namespace=kube-system",
"--agent-service-account=konnectivity-agent",
"--kubeconfig=/etc/kubernetes/konnectivity-server.conf",
"--authentication-audience=system:konnectivity-server"
]

```

livenessProbe:

httpGet:

scheme: HTTP

host: 127.0.0.1

port: 8134

path: /healthz

initialDelaySeconds: 30

timeoutSeconds: 60

ports:

- name: agentport

containerPort: 8132

hostPort: 8132

- name: adminport

containerPort: 8133

hostPort: 8133

- name: healthport

containerPort: 8134

hostPort: 8134

volumeMounts:

- name: k8s-certs

mountPath: /etc/kubernetes/pki

readOnly: true

- name: kubeconfig

mountPath: /etc/kubernetes/konnectivity-server.conf

readOnly: true

- name: konnectivity-uds

mountPath: /etc/kubernetes/konnectivity-server

readOnly: false

volumes:

- name: k8s-certs

hostPath:

path: /etc/kubernetes/pki

- name: kubeconfig

```
hostPath:
  path: /etc/kubernetes/konnectivity-server.conf
  type: FileOrCreate
- name: konnectivity-uds
  hostPath:
    path: /etc/kubernetes/konnectivity-server
    type: DirectoryOrCreate
```

Then deploy the Konnectivity agents in your cluster:

[admin/konnectivity/konnectivity-agent.yaml](#)

```
apiVersion: apps/v1
# Alternatively, you can deploy the agents as Deployments. It is not necessary
# to have an agent on each node.
kind: DaemonSet
metadata:
  labels:
    addonmanager.kubernetes.io/mode: Reconcile
    k8s-app: konnectivity-agent
  namespace: kube-system
  name: konnectivity-agent
spec:
  selector:
    matchLabels:
      k8s-app: konnectivity-agent
  template:
    metadata:
      labels:
        k8s-app: konnectivity-agent
    spec:
      priorityClassName: system-cluster-critical
      tolerations:
        - key: "CriticalAddonsOnly"
          operator: "Exists"
      containers:
        - image: us.gcr.io/k8s-artifacts-prod/kas-network-proxy/proxy-agent:v0.0.37
          name: konnectivity-agent
          command: ["/proxy-agent"]
          args: [
            "--logtostderr=true",
            "--ca-cert=/var/run/secrets/kubernetes.io/serviceaccount/ca.crt",
            # Since the konnectivity server runs with hostNetwork=true,
            # this is the IP address of the master machine.
            "--proxy-server-host=35.225.206.7",
            "--proxy-server-port=8132",
            "--admin-server-port=8133",
            "--health-server-port=8134",
            "--service-account-token-path=/var/run/secrets/tokens/konnectivity-agent-token"
          ]
      volumeMounts:
        - mountPath: /var/run/secrets/tokens
          name: konnectivity-agent-token
```



```
livenessProbe:
  httpGet:
    port: 8134
    path: /healthz
  initialDelaySeconds: 15
  timeoutSeconds: 15
serviceAccountName: konnectivity-agent
volumes:
- name: konnectivity-agent-token
  projected:
    sources:
    - serviceAccountToken:
        path: konnectivity-agent-token
        audience: system:konnectivity-server
```

Last, if RBAC is enabled in your cluster, create the relevant RBAC rules:

[admin/konnectivity/konnectivity-rbac.yaml](#)

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: system:konnectivity-server
  labels:
    kubernetes.io/cluster-service: "true"
    addonmanager.kubernetes.io/mode: Reconcile
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:auth-delegator
subjects:
- apiGroup: rbac.authorization.k8s.io
  kind: User
  name: system:konnectivity-server
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: konnectivity-agent
  namespace: kube-system
  labels:
    kubernetes.io/cluster-service: "true"
    addonmanager.kubernetes.io/mode: Reconcile
```

TLS

Understand how to protect traffic within your cluster using Transport Layer Security (TLS).

[Configure Certificate Rotation for the Kubelet](#)

[Manage TLS Certificates in a Cluster](#)

Configure Certificate Rotation for the Kubelet

This page shows how to enable and configure certificate rotation for the kubelet.

FEATURE STATE: [Kubernetes v1.19](#) [stable]

Before you begin

- Kubernetes version 1.8.0 or later is required

Overview

The kubelet uses certificates for authenticating to the Kubernetes API. By default, these certificates are issued with one year expiration so that they do not need to be renewed too frequently.

Kubernetes contains [kubelet certificate rotation](#), that will automatically generate a new key and request a new certificate from the Kubernetes API as the current certificate approaches expiration. Once the new certificate is available, it will be used for authenticating connections to the Kubernetes API.

Enabling client certificate rotation

The kubelet process accepts an argument `--rotate-certificates` that controls if the kubelet will automatically request a new certificate as the expiration of the certificate currently in use approaches.

The kube-controller-manager process accepts an argument `--cluster-signing-duration` (`--experimental-cluster-signing-duration` prior to 1.19) that controls how long certificates will be issued for.

Understanding the certificate rotation configuration

When a kubelet starts up, if it is configured to bootstrap (using the `--bootstrap-kubeconfig` flag), it will use its initial certificate to connect to the Kubernetes API and issue a certificate signing request. You can view the status of certificate signing requests using:

```
kubectl get csr
```

Initially a certificate signing request from the kubelet on a node will have a status of `Pending`. If the certificate signing requests meets specific criteria, it will be auto approved by the controller manager, then it will have a status of `Approved`. Next, the controller manager will sign a certificate, issued for the duration specified by the `--cluster-signing-duration` parameter, and the signed certificate will be attached to the certificate signing request.

The kubelet will retrieve the signed certificate from the Kubernetes API and write that to disk, in the location specified by `--cert-dir`. Then the kubelet will use the new certificate to connect to the Kubernetes API.

As the expiration of the signed certificate approaches, the kubelet will automatically issue a new certificate signing request, using the Kubernetes API. This can happen at any point between 30% and 10% of the time remaining on the certificate. Again, the controller manager will automatically approve the certificate request and attach a signed certificate to the certificate signing request. The kubelet will retrieve the new signed certificate from the Kubernetes API and write that to disk. Then it will update the connections it has to the Kubernetes API to reconnect using the new certificate.

Manage TLS Certificates in a Cluster

Kubernetes provides a `certificates.k8s.io` API, which lets you provision TLS certificates signed by a Certificate Authority (CA) that you control. These CA and certificates can be used by your workloads to establish trust.

`certificates.k8s.io` API uses a protocol that is similar to the [ACME draft](#).

Note:

Certificates created using the `certificates.k8s.io` API are signed by a [dedicated CA](#). It is possible to configure your cluster to use the cluster root CA for this purpose, but you should never rely on this. Do not assume that these certificates will validate against the cluster root CA.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercode](#)
- [Play with Kubernetes](#)

You need the `cfssl` tool. You can download `cfssl` from <https://github.com/cloudflare/cfssl/releases>.

Some steps in this page use the `jq` tool. If you don't have `jq`, you can install it via your operating system's software sources, or fetch it from <https://jqlang.github.io/jq/>.

Trusting TLS in a cluster

Trusting the [custom CA](#) from an application running as a pod usually requires some extra application configuration. You will need to add the CA certificate bundle to the list of CA certificates that the TLS client or server trusts. For example, you would do this with a `golang` TLS config by parsing the certificate chain and adding the parsed certificates to the `RootCAs` field in the [tls.Config](#) struct.

Note:

Even though the custom CA certificate may be included in the filesystem (in the ConfigMap `kube-root-ca.crt`), you should not use that certificate authority for any purpose other than to verify internal Kubernetes endpoints. An example of an internal Kubernetes endpoint is the Service named `kubernetes` in the default namespace.

If you want to use a custom certificate authority for your workloads, you should generate that CA separately, and distribute its CA certificate using a [ConfigMap](#) that your pods have access to read.

Requesting a certificate

The following section demonstrates how to create a TLS certificate for a Kubernetes service accessed through DNS.

Note:

This tutorial uses CFSSL: Cloudflare's PKI and TLS toolkit [click here](#) to know more.

Create a certificate signing request

Generate a private key and certificate signing request (or CSR) by running the following command:

```
cat <<EOF | cfssl genkey - | cfssljson -bare server
{
  "hosts": [
    "my-svc.my-namespace.svc.cluster.local",
    "my-pod.my-namespace.pod.cluster.local",
    "192.0.2.24",
    "10.0.34.2"
  ],
  "CN": "my-pod.my-namespace.pod.cluster.local",
  "key": {
    "algo": "ecdsa",
    "size": 256
  }
}
EOF
```

Where `192.0.2.24` is the service's cluster IP, `my-svc.my-namespace.svc.cluster.local` is the service's DNS name, `10.0.34.2` is the pod's IP and `my-pod.my-namespace.pod.cluster.local` is the pod's DNS name. You should see the output similar to:

```
2022/02/01 11:45:32 [INFO] generate received request
2022/02/01 11:45:32 [INFO] received CSR
2022/02/01 11:45:32 [INFO] generating key: ecdsa-256
2022/02/01 11:45:32 [INFO] encoded CSR
```

This command generates two files; it generates `server.csr` containing the PEM encoded [PKCS#10](#) certification request, and `server-key.pem` containing the PEM encoded key to the certificate that is still to be created.

Create a CertificateSigningRequest object to send to the Kubernetes API

Generate a CSR manifest (in YAML), and send it to the API server. You can do that by running the following command:

```
cat <<EOF | kubectl apply -f -
apiVersion: certificates.k8s.io/v1
kind: CertificateSigningRequest
metadata:
  name: my-svc.my-namespace
spec:
  request: $(cat server.csr | base64 | tr -d '\n')
  signerName: example.com/serving
  usages:
    - digital signature
    - key encipherment
    - server auth
EOF
```

Notice that the `server.csr` file created in step 1 is base64 encoded and stashed in the `.spec.request` field. You are also requesting a certificate with the "digital signature", "key encipherment", and "server auth" key usages, signed by an example `example.com/serving` signer. A specific `signerName` must be requested. View documentation for [supported signer names](#) for more information.

The CSR should now be visible from the API in a Pending state. You can see it by running:

```
kubectl describe csr my-svc.my-namespace
```

```
Name:          my-svc.my-namespace
Labels:        <none>
Annotations:   <none>
CreationTimestamp:  Tue, 01 Feb 2022 11:49:15 -0500
Requesting User:  yourname@example.com
Signer:         example.com/serving
Status:        Pending
Subject:
  Common Name:  my-pod.my-namespace.pod.cluster.local
  Serial Number:
Subject Alternative Names:
  DNS Names:   my-pod.my-namespace.pod.cluster.local
               my-svc.my-namespace.svc.cluster.local
  IP Addresses: 192.0.2.24
               10.0.34.2
Events: <none>
```

Get the CertificateSigningRequest approved

Approving the [certificate signing request](#) is either done by an automated approval process or on a one off basis by a cluster administrator. If you're authorized to approve a certificate request, you can do that manually using kubectl; for example:

```
kubectl certificate approve my-svc.my-namespace
```

```
certificatesigningrequest.certificates.k8s.io/my-svc.my-namespace approved
```

You should now see the following:

```
kubectl get csr
```

| NAME | AGE | SIGNERNAME | REQUESTOR | REQUESTEDDURATION | CONDITION |
|---------------------|----------|---------------------|----------------------|-------------------|-----------|
| my-svc.my-namespace | 10m | example.com/serving | yourname@example.com | | |
| <none> | Approved | | | | |

This means the certificate request has been approved and is waiting for the requested signer to sign it.

Sign the CertificateSigningRequest

Next, you'll play the part of a certificate signer, issue the certificate, and upload it to the API.

A signer would typically watch the CertificateSigningRequest API for objects with its signerName, check that they have been approved, sign certificates for those requests, and update the API object status with the issued certificate.

Create a Certificate Authority

You need an authority to provide the digital signature on the new certificate.

First, create a signing certificate by running the following:

```
cat <<EOF | cfssl gencert -initca - | cfssljson -bare ca
{
  "CN": "My Example Signer",
  "key": {
    "algo": "rsa",
    "size": 2048
  }
}
EOF
```

You should see output similar to:

```
2022/02/01 11:50:39 [INFO] generating a new CA key and certificate from CSR
2022/02/01 11:50:39 [INFO] generate received request
2022/02/01 11:50:39 [INFO] received CSR
2022/02/01 11:50:39 [INFO] generating key: rsa-2048
2022/02/01 11:50:39 [INFO] encoded CSR
```

```
2022/02/01 11:50:39 [INFO] signed certificate with serial number
263983151013686720899716354349605500797834580472
```

This produces a certificate authority key file (ca-key.pem) and certificate (ca.pem).

Issue a certificate

[tls/server-signing-config.json](#)

```
{
  "signing": {
    "default": {
      "usages": [
        "digital signature",
        "key encipherment",
        "server auth"
      ],
      "expiry": "876000h",
      "ca_constraint": {
        "is_ca": false
      }
    }
  }
}
```

Use a server-signing-config.json signing configuration and the certificate authority key file and certificate to sign the certificate request:

```
kubectl get csr my-svc.my-namespace -o jsonpath='{.spec.request}' | \
base64 --decode | \
cfssl sign -ca ca.pem -ca-key ca-key.pem -config server-signing-config.json - | \
cfssljson -bare ca-signed-server
```

You should see the output similar to:

```
2022/02/01 11:52:26 [INFO] signed certificate with serial number
576048928624926584381415936700914530534472870337
```

This produces a signed serving certificate file, ca-signed-server.pem.

Upload the signed certificate

Finally, populate the signed certificate in the API object's status:

```
kubectl get csr my-svc.my-namespace -o json | \
jq '.status.certificate = "$(base64 ca-signed-server.pem | tr -d '\n')"' | \
kubectl replace --raw /apis/certificates.k8s.io/v1/certificatesigningrequests/my-svc.my-namespace/status -f -
```

Note:

This uses the command line tool [jq](#) to populate the base64-encoded content in the `.status.certificate` field. If you do not have `jq`, you can also save the JSON output to a file, populate this field manually, and upload the resulting file.

Once the CSR is approved and the signed certificate is uploaded, run:

```
kubectl get csr
```

The output is similar to:

| NAME | AGE | SIGNERNAME | REQUESTOR | REQUESTEDDURATION | CONDITION |
|---------------------|-----|---------------------|----------------------|-------------------|-----------|
| my-svc.my-namespace | 20m | example.com/serving | yourname@example.com | | |
| <none> | | Approved,Issued | | | |

Download the certificate and use it

Now, as the requesting user, you can download the issued certificate and save it to a `server.crt` file by running the following:

```
kubectl get csr my-svc.my-namespace -o jsonpath='{.status.certificate}' \
| base64 --decode > server.crt
```

Now you can populate `server.crt` and `server-key.pem` in a [Secret](#) that you could later mount into a Pod (for example, to use with a webserver that serves HTTPS).

```
kubectl create secret tls server --cert server.crt --key server-key.pem
```

```
secret/server created
```

Finally, you can populate `ca.pem` into a [ConfigMap](#) and use it as the trust root to verify the serving certificate:

```
kubectl create configmap example-serving-ca --from-file ca.crt=ca.pem
```

```
configmap/example-serving-ca created
```

Approving CertificateSigningRequests

A Kubernetes administrator (with appropriate permissions) can manually approve (or deny) `CertificateSigningRequests` by using the `kubectl certificate approve` and `kubectl certificate deny` commands. However if you intend to make heavy usage of this API, you might consider writing an automated certificates controller.

Caution:

The ability to approve CSRs decides who trusts whom within your environment. The ability to approve CSRs should not be granted broadly or lightly.

You should make sure that you confidently understand both the verification requirements that fall on the approver **and** the repercussions of issuing a specific certificate before you grant the approve permission.

Whether a machine or a human using `kubectl` as above, the role of the *approver* is to verify that the CSR satisfies two requirements:

1. The subject of the CSR controls the private key used to sign the CSR. This addresses the threat of a third party masquerading as an authorized subject. In the above example, this step would be to verify that the pod controls the private key used to generate the CSR.
2. The subject of the CSR is authorized to act in the requested context. This addresses the threat of an undesired subject joining the cluster. In the above example, this step would be to verify that the pod is allowed to participate in the requested service.

If and only if these two requirements are met, the approver should approve the CSR and otherwise should deny the CSR.

For more information on certificate approval and access control, read the [Certificate Signing Requests](#) reference page.

Configuring your cluster to provide signing

This page assumes that a signer is set up to serve the certificates API. The Kubernetes controller manager provides a default implementation of a signer. To enable it, pass the `--cluster-signing-cert-file` and `--cluster-signing-key-file` parameters to the controller manager with paths to your Certificate Authority's keypair.

Manual Rotation of CA Certificates

This page shows how to manually rotate the certificate authority (CA) certificates.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)
- For more information about authentication in Kubernetes, see [Authenticating](#).
- For more information about best practices for CA certificates, see [Single root CA](#).

Rotate the CA certificates manually

Caution:

Make sure to back up your certificate directory along with configuration files and any other necessary files.

This approach assumes operation of the Kubernetes control plane in a HA configuration with multiple API servers. Graceful termination of the API server is also assumed so clients can cleanly disconnect from one API server and reconnect to another.

Configurations with a single API server will experience unavailability while the API server is being restarted.

1. Distribute the new CA certificates and private keys (for example: `ca.crt`, `ca.key`, `front-proxy-ca.crt`, and `front-proxy-ca.key`) to all your control plane nodes in the Kubernetes certificates directory.
2. Update the `--root-ca-file` flag for the [kube-controller-manager](#) to include both old and new CA, then restart the kube-controller-manager.

Any [ServiceAccount](#) created after this point will get Secrets that include both old and new CAs.

Note:

The files specified by the kube-controller-manager flags `--client-ca-file` and `--cluster-signing-cert-file` cannot be CA bundles. If these flags and `--root-ca-file` point to the same `ca.crt` file which is now a bundle (includes both old and new CA) you will face an error. To workaround this problem you can copy the new CA to a separate file and make the flags `--client-ca-file` and `--cluster-signing-cert-file` point to the copy. Once `ca.crt` is no longer a bundle you can restore the problem flags to point to `ca.crt` and delete the copy.

[Issue 1350](#) for kubeadm tracks a bug with the kube-controller-manager being unable to accept a CA bundle.

3. Wait for the controller manager to update `ca.crt` in the service account Secrets to include both old and new CA certificates.

If any Pods are started before new CA is used by API servers, the new Pods get this update and will trust both old and new CAs.

4. Restart all pods using in-cluster configurations (for example: kube-proxy, CoreDNS, etc) so they can use the updated certificate authority data from Secrets that link to ServiceAccounts.

- Make sure CoreDNS, kube-proxy and other Pods using in-cluster configurations are working as expected.

5. Append the both old and new CA to the file against `--client-ca-file` and `--kubelet-certificate-authority` flag in the kube-apiserver configuration.
6. Append the both old and new CA to the file against `--client-ca-file` flag in the kube-scheduler configuration.
7. Update certificates for user accounts by replacing the content of `client-certificate-data` and `client-key-data` respectively.

For information about creating certificates for individual user accounts, see [Configure certificates for user accounts](#).

Additionally, update the `certificate-authority-data` section in the kubeconfig files, respectively with Base64-encoded old and new certificate authority data

8. Update the `--root-ca-file` flag for the [Cloud Controller Manager](#) to include both old and new CA, then restart the cloud-controller-manager.

Note:

If your cluster does not have a cloud-controller-manager, you can skip this step.

9. Follow the steps below in a rolling fashion.

1. Restart any other [aggregated API servers](#) or webhook handlers to trust the new CA certificates.
2. Restart the kubelet by update the file against `clientCAFile` in kubelet configuration and `certificate-authority-data` in `kubelet.conf` to use both the old and new CA on all nodes.

If your kubelet is not using client certificate rotation, update `client-certificate-data` and `client-key-data` in `kubelet.conf` on all nodes along with the kubelet client certificate file usually found in `/var/lib/kubelet/pki`.

3. Restart API servers with the certificates (`apiserver.crt`, `apiserver-kubelet-client.crt` and `front-proxy-client.crt`) signed by new CA. You can use the existing private keys or new private keys. If you changed the private keys then update these in the Kubernetes certificates directory as well.

Since the Pods in your cluster trust both old and new CAs, there will be a momentarily disconnection after which pods' Kubernetes clients reconnect to the new API server. The new API server uses a certificate signed by the new CA.

- Restart the [kube-scheduler](#) to use and trust the new CAs.
- Make sure control plane components logs no TLS errors.

Note:

To generate certificates and private keys for your cluster using the `openssl` command line tool, see [\[Certificates \('openssl'\)\]\(/docs/tasks/administer-cluster/certificates/#openssl\)](#). You can also use [\['cfssl'\]\(/docs/tasks/administer-cluster/certificates/#cfssl\)](#).

4. Annotate any DaemonSets and Deployments to trigger pod replacement in a safer rolling fashion.

```
for namespace in $(kubectl get namespace -o jsonpath='{.items[*].metadata.name}'); do
  for name in $(kubectl get deployments -n $namespace -o jsonpath='{.items[*].metadata.name}'); do
    kubectl patch deployment -n ${namespace} ${name} -p '{"spec":{"template":{"metadata":{"annotations":{"ca-rotation": "1"}}}}}'
  done
  for name in $(kubectl get daemonset -n $namespace -o jsonpath='{.items[*].metadata.name}'); do
```

```
kubectl patch daemonset -n ${namespace} ${name} -p '{"spec":{"template":{"metadata":{"annotations":{"ca-rotation": "1"}}}}}'  
done  
done
```

Note:

To limit the number of concurrent disruptions that your application experiences, see [\[configure pod disruption budget\]\(/docs/tasks/run-application/configure-pdb/\)](#).

Depending on how you use StatefulSets you may also need to perform similar rolling replacement.

10. If your cluster is using bootstrap tokens to join nodes, update the ConfigMap `cluster-info` in the `kube-public` namespace with new CA.

```
base64_encoded_ca="$(base64 -w0 /etc/kubernetes/pki/ca.crt)"  
  
kubectl get cm/cluster-info --namespace kube-public -o yaml | \  
/bin/sed "s/\\(certificate-authority-data:\\).*/\\1 ${base64_encoded_ca}/" | \  
kubectl apply -f -
```

11. Verify the cluster functionality.

1. Check the logs from control plane components, along with the kubelet and the kube-proxy. Ensure those components are not reporting any TLS errors; see [looking at the logs](#) for more details.
2. Validate logs from any aggregated api servers and pods using in-cluster config.

12. Once the cluster functionality is successfully verified:

1. Update all service account tokens to include new CA certificate only.
 - All pods using an in-cluster kubeconfig will eventually need to be restarted to pick up the new Secret, so that no Pods are relying on the old cluster CA.
2. Restart the control plane components by removing the old CA from the kubeconfig files and the files against `--client-ca-file`, `--root-ca-file` flags resp.
3. On each node, restart the kubelet by removing the old CA from file against the `clientCAFile` flag and from the kubelet kubeconfig file. You should carry this out as a rolling update.

If your cluster lets you make this change, you can also roll it out by replacing nodes rather than reconfiguring them.

Manage Cluster Daemons

Perform common tasks for managing a DaemonSet, such as performing a rolling update.

[Perform a Rolling Update on a DaemonSet](#)

[Perform a Rollback on a DaemonSet](#)

[Running Pods on Only Some Nodes](#)

Perform a Rolling Update on a DaemonSet

This page shows how to perform a rolling update on a DaemonSet.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

DaemonSet Update Strategy

DaemonSet has two update strategy types:

- OnDelete: With OnDelete update strategy, after you update a DaemonSet template, new DaemonSet pods will *only* be created when you manually delete old DaemonSet pods. This is the same behavior of DaemonSet in Kubernetes version 1.5 or before.
- RollingUpdate: This is the default update strategy.
With RollingUpdate update strategy, after you update a DaemonSet template, old DaemonSet pods will be killed, and new DaemonSet pods will be created automatically, in a controlled fashion. At most one pod of the DaemonSet will be running on each node during the whole update process.

Performing a Rolling Update

To enable the rolling update feature of a DaemonSet, you must set its `.spec.updateStrategy.type` to RollingUpdate.

You may want to set `.spec.updateStrategy.rollingUpdate.maxUnavailable` (default to 1), `.spec.minReadySeconds` (default to 0) and `.spec.updateStrategy.rollingUpdate.maxSurge` (defaults to 0) as well.

Creating a DaemonSet with RollingUpdate update strategy

This YAML file specifies a DaemonSet with an update strategy as 'RollingUpdate'

[controllers/fluentd-daemonset.yaml](#)

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: fluentd-elasticsearch
```

```

namespace: kube-system
labels:
  k8s-app: fluentd-logging
spec:
  selector:
    matchLabels:
      name: fluentd-elasticsearch
  updateStrategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
  template:
    metadata:
      labels:
        name: fluentd-elasticsearch
    spec:
      tolerations:
        # these tolerations are to have the daemonset runnable on control plane nodes
        # remove them if your control plane nodes should not run pods
        - key: node-role.kubernetes.io/control-plane
          operator: Exists
          effect: NoSchedule
        - key: node-role.kubernetes.io/master
          operator: Exists
          effect: NoSchedule
      containers:
        - name: fluentd-elasticsearch
          image: quay.io/fluentd_elasticsearch/fluentd:v2.5.2
          volumeMounts:
            - name: varlog
              mountPath: /var/log
            - name: varlibdockercontainers
              mountPath: /var/lib/docker/containers
              readOnly: true
      terminationGracePeriodSeconds: 30
      volumes:
        - name: varlog
          hostPath:
            path: /var/log
        - name: varlibdockercontainers
          hostPath:
            path: /var/lib/docker/containers

```

After verifying the update strategy of the DaemonSet manifest, create the DaemonSet:

```
kubectl create -f https://k8s.io/examples/controllers/fluentd-daemonset.yaml
```

Alternatively, use `kubectl apply` to create the same DaemonSet if you plan to update the DaemonSet with `kubectl apply`.

```
kubectl apply -f https://k8s.io/examples/controllers/fluentd-daemonset.yaml
```

Checking DaemonSet RollingUpdate update strategy

Check the update strategy of your DaemonSet, and make sure it's set to RollingUpdate:

```
kubectl get ds/fluentd-elasticsearch -o go-template='{{.spec.updateStrategy.type}}{{"\n"}}' -n kube-system
```

If you haven't created the DaemonSet in the system, check your DaemonSet manifest with the following command instead:

```
kubectl apply -f https://k8s.io/examples/controllers/fluentd-daemonset.yaml --dry-run=client -o go-template='{{.spec.updateStrategy.type}}{{"\n"}}'
```

The output from both commands should be:

RollingUpdate

If the output isn't RollingUpdate, go back and modify the DaemonSet object or manifest accordingly.

Updating a DaemonSet template

Any updates to a RollingUpdate DaemonSet .spec.template will trigger a rolling update. Let's update the DaemonSet by applying a new YAML file. This can be done with several different kubectl commands.

[controllers/fluentd-daemonset-update.yaml](https://k8s.io/examples/controllers/fluentd-daemonset-update.yaml)

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: fluentd-elasticsearch
  namespace: kube-system
  labels:
    k8s-app: fluentd-logging
spec:
  selector:
    matchLabels:
      name: fluentd-elasticsearch
  updateStrategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
  template:
    metadata:
      labels:
        name: fluentd-elasticsearch
    spec:
      tolerations:
        # these tolerations are to have the daemonset runnable on control plane nodes
        # remove them if your control plane nodes should not run pods
        - key: node-role.kubernetes.io/control-plane
          operator: Exists
```

```

    effect: NoSchedule
  - key: node-role.kubernetes.io/master
    operator: Exists
    effect: NoSchedule
containers:
- name: fluentd-elasticsearch
  image: quay.io/fluentd_elasticsearch/fluentd:v2.5.2
resources:
  limits:
    memory: 200Mi
  requests:
    cpu: 100m
    memory: 200Mi
volumeMounts:
- name: varlog
  mountPath: /var/log
- name: varlibdockercontainers
  mountPath: /var/lib/docker/containers
  readOnly: true
terminationGracePeriodSeconds: 30
volumes:
- name: varlog
  hostPath:
    path: /var/log
- name: varlibdockercontainers
  hostPath:
    path: /var/lib/docker/containers

```

Declarative commands

If you update DaemonSets using [configuration files](#), use kubectl apply:

```
kubectl apply -f https://k8s.io/examples/controllers/fluentd-daemonset-update.yaml
```

Imperative commands

If you update DaemonSets using [imperative commands](#), use kubectl edit :

```
kubectl edit ds/fluentd-elasticsearch -n kube-system
```

Updating only the container image

If you only need to update the container image in the DaemonSet template, i.e. `.spec.template.spec.containers[*].image`, use kubectl set image:

```
kubectl set image ds/fluentd-elasticsearch fluentd-elasticsearch=quay.io/fluentd_elasticsearch/fluentd:v2.6.0 -n kube-system
```

Watching the rolling update status

Finally, watch the rollout status of the latest DaemonSet rolling update:


```
kubectl rollout status ds/fluentd-elasticsearch -n kube-system
```

When the rollout is complete, the output is similar to this:

```
daemonset "fluentd-elasticsearch" successfully rolled out
```

Troubleshooting

DaemonSet rolling update is stuck

Sometimes, a DaemonSet rolling update may be stuck. Here are some possible causes:

Some nodes run out of resources

The rollout is stuck because new DaemonSet pods can't be scheduled on at least one node. This is possible when the node is [running out of resources](#).

When this happens, find the nodes that don't have the DaemonSet pods scheduled on by comparing the output of `kubectl get nodes` and the output of:

```
kubectl get pods -l name=fluentd-elasticsearch -o wide -n kube-system
```

Once you've found those nodes, delete some non-DaemonSet pods from the node to make room for new DaemonSet pods.

Note:

This will cause service disruption when deleted pods are not controlled by any controllers or pods are not replicated. This does not respect [PodDisruptionBudget](#) either.

Broken rollout

If the recent DaemonSet template update is broken, for example, the container is crash looping, or the container image doesn't exist (often due to a typo), DaemonSet rollout won't progress.

To fix this, update the DaemonSet template again. New rollout won't be blocked by previous unhealthy rollouts.

Clock skew

If `.spec.minReadySeconds` is specified in the DaemonSet, clock skew between master and nodes will make DaemonSet unable to detect the right rollout progress.

Clean up

Delete DaemonSet from a namespace :

```
kubectl delete ds fluentd-elasticsearch -n kube-system
```

What's next

- See [Performing a rollback on a DaemonSet](#)
- See [Creating a DaemonSet to adopt existing DaemonSet pods](#)

Perform a Rollback on a DaemonSet

This page shows how to perform a rollback on a [DaemonSet](#).

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version 1.7. To check the version, enter kubectl version.

You should already know how to [perform a rolling update on a DaemonSet](#).

Performing a rollback on a DaemonSet

Step 1: Find the DaemonSet revision you want to roll back to

You can skip this step if you only want to roll back to the last revision.

List all revisions of a DaemonSet:

```
kubectl rollout history daemonset <daemonset-name>
```

This returns a list of DaemonSet revisions:

```
daemonsets "<daemonset-name>"
REVISION    CHANGE-CAUSE
1           ...
2           ...
...
```

- Change cause is copied from DaemonSet annotation `kubernetes.io/change-cause` to its revisions upon creation. You may specify `--record=true` in kubectl to record the command executed in the change cause annotation.

To see the details of a specific revision:

```
kubectl rollout history daemonset <daemonset-name> --revision=1
```

This returns the details of that revision:

```
daemonsets "<daemonset-name>" with revision #1
Pod Template:
Labels:    foo=bar
Containers:
app:
  Image:    ...
  Port:     ...
  Environment: ...
  Mounts:   ...
  Volumes:  ...
```

Step 2: Roll back to a specific revision

```
# Specify the revision number you get from Step 1 in --to-revision
kubectl rollout undo daemonset <daemonset-name> --to-revision=<revision>
```

If it succeeds, the command returns:

```
daemonset "<daemonset-name>" rolled back
```

Note:

If `--to-revision` flag is not specified, kubectl picks the most recent revision.

Step 3: Watch the progress of the DaemonSet rollback

kubectl rollout undo daemonset tells the server to start rolling back the DaemonSet. The real rollback is done asynchronously inside the cluster [control plane](#).

To watch the progress of the rollback:

```
kubectl rollout status ds/<daemonset-name>
```

When the rollback is complete, the output is similar to:

```
daemonset "<daemonset-name>" successfully rolled out
```

Understanding DaemonSet revisions

In the previous kubectl rollout history step, you got a list of DaemonSet revisions. Each revision is stored in a resource named ControllerRevision.

To see what is stored in each revision, find the DaemonSet revision raw resources:

```
kubectl get controllerrevision -l <daemonset-selector-key>=<daemonset-selector-value>
```

This returns a list of ControllerRevisions:

| NAME | CONTROLLER | REVISION | AGE |
|----------------------------------|----------------------------|----------|-----|
| <daemonset-name>-<revision-hash> | DaemonSet/<daemonset-name> | 1 | 1h |
| <daemonset-name>-<revision-hash> | DaemonSet/<daemonset-name> | 2 | 1h |

Each ControllerRevision stores the annotations and template of a DaemonSet revision.

kubectl rollout undo takes a specific ControllerRevision and replaces DaemonSet template with the template stored in the ControllerRevision. kubectl rollout undo is equivalent to updating DaemonSet template to a previous revision through other commands, such as kubectl edit or kubectl apply.

Note:

DaemonSet revisions only roll forward. That is to say, after a rollback completes, the revision number (.revision field) of the ControllerRevision being rolled back to will advance. For example, if you have revision 1 and 2 in the system, and roll back from revision 2 to revision 1, the ControllerRevision with .revision: 1 will become .revision: 3.

Troubleshooting

- See [troubleshooting DaemonSet rolling update](#).

Running Pods on Only Some Nodes

This page demonstrates how can you run [Pods](#) on only some [Nodes](#) as part of a [DaemonSet](#)

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Running Pods on only some Nodes

Imagine that you want to run a [DaemonSet](#), but you only need to run those daemon pods on nodes that have local solid state (SSD) storage. For example, the Pod might provide cache service to the node, and the cache is only useful when low-latency local storage is available.

Step 1: Add labels to your nodes

Add the label `ssd=true` to the nodes which have SSDs.

```
kubectl label nodes example-node-1 example-node-2 ssd=true
```

Step 2: Create the manifest

Let's create a [DaemonSet](#) which will provision the daemon pods on the SSD labeled [nodes](#) only.

Next, use a nodeSelector to ensure that the DaemonSet only runs Pods on nodes with the `ssd` label set to "true".

[controllers/daemonset-label-selector.yaml](#)

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: ssd-driver
  labels:
    app: nginx
spec:
  selector:
    matchLabels:
      app: ssd-driver-pod
  template:
    metadata:
      labels:
        app: ssd-driver-pod
    spec:
      nodeSelector:
        ssd: "true"
      containers:
        - name: example-container
          image: example-image
```

Step 3: Create the DaemonSet

Create the DaemonSet from the manifest by using `kubectl create` or `kubectl apply`

Let's label another node as `ssd=true`.

```
kubectl label nodes example-node-3 ssd=true
```

Labelling the node automatically triggers the control plane (specifically, the DaemonSet controller) to run a new daemon pod on that node.

```
kubectl get pods -o wide
```

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE | IP | NODE |
|--------------------------------|-------|---------|----------|-----|-------|----------------|
| <daemonset-name><some-hash-01> | 1/1 | Running | 0 | 13s | | example-node-1 |
| <daemonset-name><some-hash-02> | 1/1 | Running | 0 | 13s | | example-node-2 |
| <daemonset-name><some-hash-03> | 1/1 | Running | 0 | 5s | | example-node-3 |

Networking

Learn how to configure networking for your cluster.

[Adding entries to Pod /etc/hosts with HostAliases](#)

[Extend Service IP Ranges](#)

Adding entries to Pod /etc/hosts with HostAliases

Adding entries to a Pod's /etc/hosts file provides Pod-level override of hostname resolution when DNS and other options are not applicable. You can add these custom entries with the HostAliases field in PodSpec.

Modification not using HostAliases is not suggested because the file is managed by the kubelet and can be overwritten on during Pod creation/restart.

Default hosts file content

Start an Nginx Pod which is assigned a Pod IP:

```
kubectl run nginx --image nginx
```

```
pod/nginx created
```

Examine a Pod IP:

```
kubectl get pods --output=wide
```

| NAME | READY | STATUS | RESTARTS | AGE | IP | NODE |
|-------|-------|---------|----------|-----|------------|---------|
| nginx | 1/1 | Running | 0 | 13s | 10.200.0.4 | worker0 |

The hosts file content would look like this:

```
kubectl exec nginx -- cat /etc/hosts
```

```
# Kubernetes-managed hosts file.
127.0.0.1 localhost
::1 localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
fe00::0 ip6-mcastprefix
fe00::1 ip6-allnodes
fe00::2 ip6-allrouters
10.200.0.4nginx
```

By default, the hosts file only includes IPv4 and IPv6 boilerplates like localhost and its own hostname.

Adding additional entries with hostAliases

In addition to the default boilerplate, you can add additional entries to the hosts file. For example: to resolve foo.local, bar.local to 127.0.0.1 and foo.remote, bar.remote to 10.1.2.3, you can configure HostAliases for a Pod under .spec.hostAliases:

[service/networking/hostaliases-pod.yaml](#)

```

apiVersion: v1
kind: Pod
metadata:
  name: hostaliases-pod
spec:
  restartPolicy: Never
  hostAliases:
  - ip: "127.0.0.1"
    hostnames:
    - "foo.local"
    - "bar.local"
  - ip: "10.1.2.3"
    hostnames:
    - "foo.remote"
    - "bar.remote"
  containers:
  - name: cat-hosts
    image: busybox:1.28
    command:
    - cat
    args:
    - "/etc/hosts"

```

You can start a Pod with that configuration by running:

```
kubectl apply -f https://k8s.io/examples/service/networking/hostaliases-pod.yaml
```

```
pod/hostaliases-pod created
```

Examine a Pod's details to see its IPv4 address and its status:

```
kubectl get pod --output=wide
```

| NAME | READY | STATUS | RESTARTS | AGE | IP | NODE |
|-----------------|-------|-----------|----------|-----|------------|---------|
| hostaliases-pod | 0/1 | Completed | 0 | 6s | 10.200.0.5 | worker0 |

The hosts file content looks like this:

```
kubectl logs hostaliases-pod
```

```

# Kubernetes-managed hosts file.
127.0.0.1 localhost
::1 localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
fe00::0 ip6-mcastprefix
fe00::1 ip6-allnodes
fe00::2 ip6-allrouters
10.200.0.5 hostaliases-pod

# Entries added by HostAliases.
127.0.0.1 foo.local bar.local
10.1.2.3 foo.remote bar.remote

```

with the additional entries specified at the bottom.

Why does the kubelet manage the hosts file?

The kubelet manages the hosts file for each container of the Pod to prevent the container runtime from modifying the file after the containers have already been started. Historically, Kubernetes always used Docker Engine as its container runtime, and Docker Engine would then modify the `/etc/hosts` file after each container had started.

Current Kubernetes can use a variety of container runtimes; even so, the kubelet manages the hosts file within each container so that the outcome is as intended regardless of which container runtime you use.

Caution:

Avoid making manual changes to the hosts file inside a container.

If you make manual changes to the hosts file, those changes are lost when the container exits.

Extend Service IP Ranges

FEATURE STATE: Kubernetes v1.27 [alpha]

This document shares how to extend the existing Service IP range assigned to a cluster.

Before you begin

You need to have a Kubernetes cluster, and the `kubectl` command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.29. To check the version, enter `kubectl version`.

API

Kubernetes clusters with kube-apiservers that have enabled the `MultiCIDRServiceAllocator` [feature gate](#) and the `networking.k8s.io/v1alpha1` API, will create a new `ServiceCIDR` object that takes the well-known name `kubernetes`, and that uses an IP address range based on the value of the `--service-cluster-ip-range` command line argument to kube-apiserver.

```
kubectl get servicecidr
```

| NAME | CIDRS | AGE |
|------------|--------------|-----|
| kubernetes | 10.96.0.0/28 | 17d |

The well-known kubernetes Service, that exposes the kube-apiserver endpoint to the Pods, calculates the first IP address from the default ServiceCIDR range and uses that IP address as its cluster IP address.

```
kubectl get service kubernetes
```

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|------------|-----------|------------|-------------|---------|-----|
| kubernetes | ClusterIP | 10.96.0.1 | <none> | 443/TCP | 17d |

The default Service, in this case, uses the ClusterIP 10.96.0.1, that has the corresponding IPAddress object.

```
kubectl get ipaddress 10.96.0.1
```

| NAME | PARENTREF |
|-----------|-----------------------------|
| 10.96.0.1 | services/default/kubernetes |

The ServiceCIDRs are protected with [finalizers](#), to avoid leaving Service ClusterIPs orphans; the finalizer is only removed if there is another subnet that contains the existing IPAddresses or there are no IPAddresses belonging to the subnet.

Extend the number of available IPs for Services

There are cases that users will need to increase the number addresses available to Services, previously, increasing the Service range was a disruptive operation that could also cause data loss. With this new feature users only need to add a new ServiceCIDR to increase the number of available addresses.

Adding a new ServiceCIDR

On a cluster with a 10.96.0.0/28 range for Services, there is only $2^{(32-28)} - 2 = 14$ IP addresses available. The kubernetes.default Service is always created; for this example, that leaves you with only 13 possible Services.

```
for i in $(seq 1 13); do kubectl create service clusterip "test-$i" --tcp 80 -o json | jq -r .spec.clusterIP; done
```

```
10.96.0.11
10.96.0.5
10.96.0.12
10.96.0.13
10.96.0.14
10.96.0.2
10.96.0.3
10.96.0.4
10.96.0.6
10.96.0.7
10.96.0.8
10.96.0.9
```

```
error: failed to create ClusterIP service: Internal error occurred: failed to allocate a serviceIP: range is full
```

You can increase the number of IP addresses available for Services, by creating a new ServiceCIDR that extends or adds new IP address ranges.

```
cat <EOF | kubectl apply -f -
apiVersion: networking.k8s.io/v1alpha1
kind: ServiceCIDR
metadata:
  name: newcidr1
spec:
  cidrs:
  - 10.96.0.0/24
EOF
```

```
servicecidr.networking.k8s.io/newcidr1 created
```

and this will allow you to create new Services with ClusterIPs that will be picked from this new range.

```
for i in $(seq 13 16); do kubectl create service clusterip "test-$i" --tcp 80 -o json | jq -
r .spec.clusterIP; done
```

```
10.96.0.48
10.96.0.200
10.96.0.121
10.96.0.144
```

Deleting a ServiceCIDR

You cannot delete a ServiceCIDR if there are IPAddresses that depend on the ServiceCIDR.

```
kubectl delete servicecidr newcidr1
```

```
servicecidr.networking.k8s.io "newcidr1" deleted
```

Kubernetes uses a finalizer on the ServiceCIDR to track this dependent relationship.

```
kubectl get servicecidr newcidr1 -o yaml
```

```
apiVersion: networking.k8s.io/v1alpha1
kind: ServiceCIDR
metadata:
  creationTimestamp: "2023-10-12T15:11:07Z"
  deletionGracePeriodSeconds: 0
  deletionTimestamp: "2023-10-12T15:12:45Z"
  finalizers:
  - networking.k8s.io/service-cidr-finalizer
  name: newcidr1
  resourceVersion: "1133"
  uid: 5ffd8afe-c78f-4e60-ae76-cec448a8af40
spec:
  cidrs:
  - 10.96.0.0/24
status:
  conditions:
```

```
- lastTransitionTime: "2023-10-12T15:12:45Z"
  message: There are still IPAddresses referencing the ServiceCIDR, please remove
    them or create a new ServiceCIDR
  reason: OrphanIPAddress
  status: "False"
  type: Ready
```

By removing the Services containing the IP addresses that are blocking the deletion of the ServiceCIDR

```
for i in $(seq 13 16); do kubectl delete service "test-$i" ; done
```

```
service "test-13" deleted
service "test-14" deleted
service "test-15" deleted
service "test-16" deleted
```

the control plane notices the removal. The control plane then removes its finalizer, so that the ServiceCIDR that was pending deletion will actually be removed.

```
kubectl get servicecidr newcidr1
```

```
Error from server (NotFound): servicecidrs.networking.k8s.io "newcidr1" not found
```

Validate IPv4/IPv6 dual-stack

This document shares how to validate IPv4/IPv6 dual-stack enabled Kubernetes clusters.

Before you begin

- Provider support for dual-stack networking (Cloud provider or otherwise must be able to provide Kubernetes nodes with routable IPv4/IPv6 network interfaces)
- A [network plugin](#) that supports dual-stack networking.
- [Dual-stack enabled](#) cluster

Your Kubernetes server must be at or later than version v1.23. To check the version, enter `kubectl version`.

Note:

While you can validate with an earlier version, the feature is only GA and officially supported since v1.23.

Validate addressing

Validate node addressing

Each dual-stack Node should have a single IPv4 block and a single IPv6 block allocated. Validate that IPv4/IPv6 Pod address ranges are configured by running the following command. Replace

the sample node name with a valid dual-stack Node from your cluster. In this example, the Node's name is k8s-linuxpool1-34450317-0:

```
kubectl get nodes k8s-linuxpool1-34450317-0 -o go-template --template='{{range .spec.podCIDRs}}{{printf "%s\n" .}}{{end}}'
```

```
10.244.1.0/24
2001:db8::/64
```

There should be one IPv4 block and one IPv6 block allocated.

Validate that the node has an IPv4 and IPv6 interface detected. Replace node name with a valid node from the cluster. In this example the node name is k8s-linuxpool1-34450317-0:

```
kubectl get nodes k8s-linuxpool1-34450317-0 -o go-template --template='{{range .status.addresses}}{{printf "%s: %s\n" .type .address}}{{end}}'
```

```
Hostname: k8s-linuxpool1-34450317-0
InternalIP: 10.0.0.5
InternalIP: 2001:db8:10::5
```

Validate Pod addressing

Validate that a Pod has an IPv4 and IPv6 address assigned. Replace the Pod name with a valid Pod in your cluster. In this example the Pod name is pod01:

```
kubectl get pods pod01 -o go-template --template='{{range .status.podIPs}}{{printf "%s\n" .ip}}{{end}}'
```

```
10.244.1.4
2001:db8::4
```

You can also validate Pod IPs using the Downward API via the status.podIPs fieldPath. The following snippet demonstrates how you can expose the Pod IPs via an environment variable called MY_POD_IPS within a container.

```
env:
- name: MY_POD_IPS
  valueFrom:
    fieldRef:
      fieldPath: status.podIPs
```

The following command prints the value of the MY_POD_IPS environment variable from within a container. The value is a comma separated list that corresponds to the Pod's IPv4 and IPv6 addresses.

```
kubectl exec -it pod01 -- set | grep MY_POD_IPS
```

```
MY_POD_IPS=10.244.1.4,2001:db8::4
```

The Pod's IP addresses will also be written to /etc/hosts within a container. The following command executes a cat on /etc/hosts on a dual stack Pod. From the output you can verify both the IPv4 and IPv6 IP address for the Pod.

```
kubect exec -it pod01 -- cat /etc/hosts
```

```
# Kubernetes-managed hosts file.
127.0.0.1 localhost
::1 localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
fe00::0 ip6-mcastprefix
fe00::1 ip6-allnodes
fe00::2 ip6-allrouters
10.244.1.4 pod01
2001:db8::4 pod01
```

Validate Services

Create the following Service that does not explicitly define `.spec.ipFamilyPolicy`. Kubernetes will assign a cluster IP for the Service from the first configured service-cluster-ip-range and set the `.spec.ipFamilyPolicy` to SingleStack.

[service/networking/dual-stack-default-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  labels:
    app.kubernetes.io/name: MyApp
spec:
  selector:
    app.kubernetes.io/name: MyApp
  ports:
    - protocol: TCP
      port: 80
```

Use `kubectl` to view the YAML for the Service.

```
kubectl get svc my-service -o yaml
```

The Service has `.spec.ipFamilyPolicy` set to SingleStack and `.spec.clusterIP` set to an IPv4 address from the first configured range set via `--service-cluster-ip-range` flag on kube-controller-manager.

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  namespace: default
spec:
  clusterIP: 10.0.217.164
  clusterIPs:
    - 10.0.217.164
  ipFamilies:
    - IPv4
  ipFamilyPolicy: SingleStack
```

```
ports:
- port: 80
  protocol: TCP
  targetPort: 9376
selector:
  app.kubernetes.io/name: MyApp
sessionAffinity: None
type: ClusterIP
status:
  loadBalancer: {}
```

Create the following Service that explicitly defines IPv6 as the first array element in `.spec.ipFamilies`. Kubernetes will assign a cluster IP for the Service from the IPv6 range configured `service-cluster-ip-range` and set the `.spec.ipFamilyPolicy` to `SingleStack`.

[service/networking/dual-stack-ipfamilies-ipv6.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  labels:
    app.kubernetes.io/name: MyApp
spec:
  ipFamilies:
  - IPv6
  selector:
    app.kubernetes.io/name: MyApp
  ports:
  - protocol: TCP
    port: 80
```

Use `kubectl` to view the YAML for the Service.

```
kubectl get svc my-service -o yaml
```

The Service has `.spec.ipFamilyPolicy` set to `SingleStack` and `.spec.clusterIP` set to an IPv6 address from the IPv6 range set via `--service-cluster-ip-range` flag on `kube-controller-manager`.

```
apiVersion: v1
kind: Service
metadata:
  labels:
    app.kubernetes.io/name: MyApp
  name: my-service
spec:
  clusterIP: 2001:db8:fd00::5118
  clusterIPs:
  - 2001:db8:fd00::5118
  ipFamilies:
  - IPv6
  ipFamilyPolicy: SingleStack
  ports:
  - port: 80
```

```
protocol: TCP
targetPort: 80
selector:
  app.kubernetes.io/name: MyApp
sessionAffinity: None
type: ClusterIP
status:
  loadBalancer: {}
```

Create the following Service that explicitly defines `PreferDualStack` in `.spec.ipFamilyPolicy`. Kubernetes will assign both IPv4 and IPv6 addresses (as this cluster has dual-stack enabled) and select the `.spec.ClusterIP` from the list of `.spec.ClusterIPs` based on the address family of the first element in the `.spec.ipFamilies` array.

[service/networking/dual-stack-preferred-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  labels:
    app.kubernetes.io/name: MyApp
spec:
  ipFamilyPolicy: PreferDualStack
  selector:
    app.kubernetes.io/name: MyApp
  ports:
    - protocol: TCP
      port: 80
```

Note:

The `kubectl get svc` command will only show the primary IP in the `CLUSTER-IP` field.

```
kubectl get svc -l app.kubernetes.io/name=MyApp
```

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|------------|-----------|--------------|-------------|---------|-----|
| my-service | ClusterIP | 10.0.216.242 | <none> | 80/TCP | 5s |

Validate that the Service gets cluster IPs from the IPv4 and IPv6 address blocks using `kubectl describe`. You may then validate access to the service via the IPs and ports.

```
kubectl describe svc -l app.kubernetes.io/name=MyApp
```

```
Name:          my-service
Namespace:     default
Labels:        app.kubernetes.io/name=MyApp
Annotations:   <none>
Selector:      app.kubernetes.io/name=MyApp
Type:          ClusterIP
IP Family Policy: PreferDualStack
IP Families:   IPv4,IPv6
IP:            10.0.216.242
```

```
IPs:      10.0.216.242,2001:db8:fd00::af55
Port:      <unset> 80/TCP
TargetPort: 9376/TCP
Endpoints: <none>
Session Affinity: None
Events:    <none>
```

Create a dual-stack load balanced Service

If the cloud provider supports the provisioning of IPv6 enabled external load balancers, create the following Service with `PreferDualStack` in `.spec.ipFamilyPolicy`, IPv6 as the first element of the `.spec.ipFamilies` array and the type field set to `LoadBalancer`.

[service/networking/dual-stack-prefer-ipv6-lb-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  labels:
    app.kubernetes.io/name: MyApp
spec:
  ipFamilyPolicy: PreferDualStack
  ipFamilies:
    - IPv6
  type: LoadBalancer
  selector:
    app.kubernetes.io/name: MyApp
  ports:
    - protocol: TCP
      port: 80
```

Check the Service:

```
kubectl get svc -l app.kubernetes.io/name=MyApp
```

Validate that the Service receives a CLUSTER-IP address from the IPv6 address block along with an EXTERNAL-IP. You may then validate access to the service via the IP and port.

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|------------|--------------|---------------------|------------------|--------------|-----|
| my-service | LoadBalancer | 2001:db8:fd00::7ebc | 2603:1030:805::5 | 80:30790/TCP | 35s |

Extend kubectl with plugins

Extend kubectl by creating and installing kubectl plugins.

This guide demonstrates how to install and write extensions for [kubectl](#). By thinking of core kubectl commands as essential building blocks for interacting with a Kubernetes cluster, a cluster administrator can think of plugins as a means of utilizing these building blocks to create more complex behavior. Plugins extend kubectl with new sub-commands, allowing for new and custom features not included in the main distribution of kubectl.

Before you begin

You need to have a working `kubectl` binary installed.

Installing `kubectl` plugins

A plugin is a standalone executable file, whose name begins with `kubectl-`. To install a plugin, move its executable file to anywhere on your `PATH`.

You can also discover and install `kubectl` plugins available in the open source using [Krew](#). Krew is a plugin manager maintained by the Kubernetes SIG CLI community.

Caution:

`Kubectl` plugins available via the Krew [plugin index](#) are not audited for security. You should install and run third-party plugins at your own risk, since they are arbitrary programs running on your machine.

Discovering plugins

`kubectl` provides a command `kubectl plugin list` that searches your `PATH` for valid plugin executables. Executing this command causes a traversal of all files in your `PATH`. Any files that are executable, and begin with `kubectl-` will show up *in the order in which they are present in your PATH* in this command's output. A warning will be included for any files beginning with `kubectl-` that are *not* executable. A warning will also be included for any valid plugin files that overlap each other's name.

You can use [Krew](#) to discover and install `kubectl` plugins from a community-curated [plugin index](#).

Limitations

It is currently not possible to create plugins that overwrite existing `kubectl` commands. For example, creating a plugin `kubectl-version` will cause that plugin to never be executed, as the existing `kubectl version` command will always take precedence over it. Due to this limitation, it is also *not* possible to use plugins to add new subcommands to existing `kubectl` commands. For example, adding a subcommand `kubectl create foo` by naming your plugin `kubectl-create-foo` will cause that plugin to be ignored.

`kubectl plugin list` shows warnings for any valid plugins that attempt to do this.

Writing `kubectl` plugins

You can write a plugin in any programming language or script that allows you to write command-line commands.

There is no plugin installation or pre-loading required. Plugin executables receive the inherited environment from the `kubectl` binary. A plugin determines which command path it wishes to implement based on its name. For example, a plugin named `kubectl-foo` provides a command `kubectl foo`. You must install the plugin executable somewhere in your `PATH`.

Example plugin

```
#!/bin/bash

# optional argument handling
if [[ "$1" == "version" ]]
then
    echo "1.0.0"
    exit 0
fi

# optional argument handling
if [[ "$1" == "config" ]]
then
    echo "$KUBECONFIG"
    exit 0
fi

echo "I am a plugin named kubectl-foo"
```

Using a plugin

To use a plugin, make the plugin executable:

```
sudo chmod +x ./kubectl-foo
```

and place it anywhere in your PATH:

```
sudo mv ./kubectl-foo /usr/local/bin
```

You may now invoke your plugin as a kubectl command:

```
kubectl foo
```

```
I am a plugin named kubectl-foo
```

All args and flags are passed as-is to the executable:

```
kubectl foo version
```

```
1.0.0
```

All environment variables are also passed as-is to the executable:

```
export KUBECONFIG=~/.kube/config
kubectl foo config
```

```
/home/<user>/.kube/config
```

```
KUBECONFIG=/etc/kube/config kubectl foo config
```

```
/etc/kube/config
```

Additionally, the first argument that is passed to a plugin will always be the full path to the location where it was invoked (\$0 would equal /usr/local/bin/kubectl-foo in the example above).

Naming a plugin

As seen in the example above, a plugin determines the command path that it will implement based on its filename. Every sub-command in the command path that a plugin targets, is separated by a dash (-). For example, a plugin that wishes to be invoked whenever the command `kubectl foo bar baz` is invoked by the user, would have the filename of `kubectl-foo-bar-baz`.

Flags and argument handling

Note:

The plugin mechanism does *not* create any custom, plugin-specific values or environment variables for a plugin process.

An older kubectl plugin mechanism provided environment variables such as `KUBECTL_PLUGINS_CURRENT_NAMESPACE`; that no longer happens.

kubectl plugins must parse and validate all of the arguments passed to them. See [using the command line runtime package](#) for details of a Go library aimed at plugin authors.

Here are some additional cases where users invoke your plugin while providing additional flags and arguments. This builds upon the `kubectl-foo-bar-baz` plugin from the scenario above.

If you run `kubectl foo bar baz arg1 --flag=value arg2`, kubectl's plugin mechanism will first try to find the plugin with the longest possible name, which in this case would be `kubectl-foo-bar-baz-arg1`. Upon not finding that plugin, kubectl then treats the last dash-separated value as an argument (`arg1` in this case), and attempts to find the next longest possible name, `kubectl-foo-bar-baz`. Upon having found a plugin with this name, kubectl then invokes that plugin, passing all args and flags after the plugin's name as arguments to the plugin process.

Example:

```
# create a plugin
echo -e '#!/bin/bash\n\nnecho "My first command-line argument was $1"' > kubectl-foo-bar-baz
sudo chmod +x ./kubectl-foo-bar-baz

# "install" your plugin by moving it to a directory in your $PATH
sudo mv ./kubectl-foo-bar-baz /usr/local/bin

# check that kubectl recognizes your plugin
kubectl plugin list
```

The following kubectl-compatible plugins are available:

```
/usr/local/bin/kubectl-foo-bar-baz
```

```
# test that calling your plugin via a "kubectl" command works
# even when additional arguments and flags are passed to your
```

```
# plugin executable by the user.
kubectl foo bar baz arg1 --meaningless-flag=true
```

My first command-line argument was arg1

As you can see, your plugin was found based on the kubectl command specified by a user, and all extra arguments and flags were passed as-is to the plugin executable once it was found.

Names with dashes and underscores

Although the kubectl plugin mechanism uses the dash (-) in plugin filenames to separate the sequence of sub-commands processed by the plugin, it is still possible to create a plugin command containing dashes in its commandline invocation by using underscores (_) in its filename.

Example:

```
# create a plugin containing an underscore in its filename
echo -e '#!/bin/bash\n\necho "I am a plugin with a dash in my name"' > ./kubectl-foo_bar
sudo chmod +x ./kubectl-foo_bar

# move the plugin into your $PATH
sudo mv ./kubectl-foo_bar /usr/local/bin

# You can now invoke your plugin via kubectl:
kubectl foo-bar
```

I am a plugin with a dash in my name

Note that the introduction of underscores to a plugin filename does not prevent you from having commands such as kubectl foo_bar. The command from the above example, can be invoked using either a dash (-) or an underscore (_):

```
# You can invoke your custom command with a dash
kubectl foo-bar
```

I am a plugin with a dash in my name

```
# You can also invoke your custom command with an underscore
kubectl foo_bar
```

I am a plugin with a dash in my name

Name conflicts and overshadowing

It is possible to have multiple plugins with the same filename in different locations throughout your PATH. For example, given a PATH with the following value: PATH=/usr/local/bin/plugins:/usr/local/bin/moreplugins, a copy of plugin kubectl-foo could exist in /usr/local/bin/plugins and /usr/local/bin/moreplugins, such that the output of the kubectl plugin list command is:

```
PATH=/usr/local/bin/plugins:/usr/local/bin/moreplugins kubectl plugin list
```

The following kubectl-compatible plugins are available:

```
/usr/local/bin/plugins/kubectl-foo
/usr/local/bin/moreplugins/kubectl-foo
- warning: /usr/local/bin/moreplugins/kubectl-foo is overshadowed by a similarly named
plugin: /usr/local/bin/plugins/kubectl-foo

error: one plugin warning was found
```

In the above scenario, the warning under `/usr/local/bin/moreplugins/kubectl-foo` tells you that this plugin will never be executed. Instead, the executable that appears first in your PATH, `/usr/local/bin/plugins/kubectl-foo`, will always be found and executed first by the kubectl plugin mechanism.

A way to resolve this issue is to ensure that the location of the plugin that you wish to use with kubectl always comes first in your PATH. For example, if you want to always use `/usr/local/bin/moreplugins/kubectl-foo` anytime that the kubectl command `kubectl foo` was invoked, change the value of your PATH to be `/usr/local/bin/moreplugins:/usr/local/bin/plugins`.

Invocation of the longest executable filename

There is another kind of overshadowing that can occur with plugin filenames. Given two plugins present in a user's PATH: `kubectl-foo-bar` and `kubectl-foo-bar-baz`, the kubectl plugin mechanism will always choose the longest possible plugin name for a given user command. Some examples below, clarify this further:

```
# for a given kubectl command, the plugin with the longest possible filename will always be
preferred
kubectl foo bar baz
```

Plugin `kubectl-foo-bar-baz` is executed

```
kubectl foo bar
```

Plugin `kubectl-foo-bar` is executed

```
kubectl foo bar baz buz
```

Plugin `kubectl-foo-bar-baz` is executed, with "buz" as its first argument

```
kubectl foo bar buz
```

Plugin `kubectl-foo-bar` is executed, with "buz" as its first argument

This design choice ensures that plugin sub-commands can be implemented across multiple files, if needed, and that these sub-commands can be nested under a "parent" plugin command:

```
ls ./plugin_command_tree
```

```
kubectl-parent
kubectl-parent-subcommand
kubectl-parent-subcommand-subsubcommand
```

Checking for plugin warnings

You can use the aforementioned `kubectl plugin list` command to ensure that your plugin is visible by `kubectl`, and verify that there are no warnings preventing it from being called as a `kubectl` command.

```
kubectl plugin list
```

The following `kubectl`-compatible plugins are available:

```
test/fixtures/pkg/kubectl/plugins/kubectl-foo
/usr/local/bin/kubectl-foo
- warning: /usr/local/bin/kubectl-foo is overshadowed by a similarly named plugin: test/
fixtures/pkg/kubectl/plugins/kubectl-foo
plugins/kubectl-invalid
- warning: plugins/kubectl-invalid identified as a kubectl plugin, but it is not executable

error: 2 plugin warnings were found
```

Using the command line runtime package

If you're writing a plugin for `kubectl` and you're using Go, you can make use of the [cli-runtime](#) utility libraries.

These libraries provide helpers for parsing or updating a user's [kubeconfig](#) file, for making REST-style requests to the API server, or to bind flags associated with configuration and printing.

See the [Sample CLI Plugin](#) for an example usage of the tools provided in the CLI Runtime repo.

Distributing kubectl plugins

If you have developed a plugin for others to use, you should consider how you package it, distribute it and deliver updates to your users.

Krew

[Krew](#) offers a cross-platform way to package and distribute your plugins. This way, you use a single packaging format for all target platforms (Linux, Windows, macOS etc) and deliver updates to your users. Krew also maintains a [plugin index](#) so that other people can discover your plugin and install it.

Native / platform specific package management

Alternatively, you can use traditional package managers such as, `apt` or `yum` on Linux, Chocolatey on Windows, and Homebrew on macOS. Any package manager will be suitable if it can place new executables placed somewhere in the user's `PATH`. As a plugin author, if you pick this option then you also have the burden of updating your `kubectl` plugin's distribution package across multiple platforms for each release.

Source code

You can publish the source code; for example, as a Git repository. If you choose this option, someone who wants to use that plugin must fetch the code, set up a build environment (if it needs compiling), and deploy the plugin. If you also make compiled packages available, or use Krew, that will make installs easier.

What's next

- Check the Sample CLI Plugin repository for a [detailed example](#) of a plugin written in Go. In case of any questions, feel free to reach out to the [SIG CLI team](#).
- Read about [Krew](#), a package manager for kubectl plugins.

Manage HugePages

Configure and manage huge pages as a schedulable resource in a cluster.

FEATURE STATE: Kubernetes v1.30 [stable]

Kubernetes supports the allocation and consumption of pre-allocated huge pages by applications in a Pod. This page describes how users can consume huge pages.

Before you begin

Kubernetes nodes must [pre-allocate huge pages](#) in order for the node to report its huge page capacity.

A node can pre-allocate huge pages for multiple sizes, for instance, the following line in `/etc/default/grub` allocates 2*1GiB of 1 GiB and 512*2 MiB of 2 MiB pages:

```
GRUB_CMDLINE_LINUX="hugepagesz=1G hugepages=2 hugepagesz=2M hugepages=512"
```

The nodes will automatically discover and report all huge page resources as schedulable resources.

When you describe the Node, you should see something similar to the following in the following in the Capacity and Allocatable sections:

Capacity:

```
cpu:      ...
ephemeral-storage: ...
hugepages-1Gi:    2Gi
hugepages-2Mi:    1Gi
memory:      ...
pods:        ...
```

Allocatable:

```
cpu:      ...
ephemeral-storage: ...
hugepages-1Gi:    2Gi
hugepages-2Mi:    1Gi
memory:      ...
pods:        ...
```

Note:

For dynamically allocated pages (after boot), the Kubelet needs to be restarted for the new allocations to be refrelected.

API

Huge pages can be consumed via container level resource requirements using the resource name `hugepages-<size>`, where `<size>` is the most compact binary notation using integer values supported on a particular node. For example, if a node supports 2048KiB and 1048576KiB page sizes, it will expose a schedulable resources `hugepages-2Mi` and `hugepages-1Gi`. Unlike CPU or memory, huge pages do not support overcommit. Note that when requesting hugepage resources, either memory or CPU resources must be requested as well.

A pod may consume multiple huge page sizes in a single pod spec. In this case it must use `medium: HugePages-<hugepagesize>` notation for all volume mounts.

```
apiVersion: v1
kind: Pod
metadata:
  name: huge-pages-example
spec:
  containers:
    - name: example
      image: fedora:latest
      command:
        - sleep
        - inf
      volumeMounts:
        - mountPath: /hugepages-2Mi
          name: hugepage-2mi
        - mountPath: /hugepages-1Gi
          name: hugepage-1gi
      resources:
        limits:
          hugepages-2Mi: 100Mi
          hugepages-1Gi: 2Gi
          memory: 100Mi
        requests:
          memory: 100Mi
  volumes:
    - name: hugepage-2mi
      emptyDir:
        medium: HugePages-2Mi
    - name: hugepage-1gi
      emptyDir:
        medium: HugePages-1Gi
```

A pod may use `medium: HugePages` only if it requests huge pages of one size.

```
apiVersion: v1
kind: Pod
```



```
metadata:
  name: huge-pages-example
spec:
  containers:
  - name: example
    image: fedora:latest
    command:
    - sleep
    - inf
    volumeMounts:
    - mountPath: /hugepages
      name: hugepage
  resources:
    limits:
      hugepages-2Mi: 100Mi
      memory: 100Mi
    requests:
      memory: 100Mi
  volumes:
  - name: hugepage
    emptyDir:
      medium: HugePages
```

- Huge page requests must equal the limits. This is the default if limits are specified, but requests are not.
- Huge pages are isolated at a container scope, so each container has own limit on their cgroup sandbox as requested in a container spec.
- EmptyDir volumes backed by huge pages may not consume more huge page memory than the pod request.
- Applications that consume huge pages via `shmget()` with `SHM_HUGETLB` must run with a supplemental group that matches `proc/sys/vm/hugetlb_shm_group`.
- Huge page usage in a namespace is controllable via `ResourceQuota` similar to other compute resources like `cpu` or `memory` using the `hugepages-<size>` token.

Schedule GPUs

Configure and schedule GPUs for use as a resource by nodes in a cluster.

FEATURE STATE: Kubernetes v1.26 [stable]

Kubernetes includes **stable** support for managing AMD and NVIDIA GPUs (graphical processing units) across different nodes in your cluster, using [device plugins](#).

This page describes how users can consume GPUs, and outlines some of the limitations in the implementation.

Using device plugins

Kubernetes implements device plugins to let Pods access specialized hardware features such as GPUs.

Note: This section links to third party projects that provide functionality required by Kubernetes. The Kubernetes project authors aren't responsible for these projects, which are listed alphabetically. To add a project to this list, read the [content guide](#) before submitting a change. [More information](#).

As an administrator, you have to install GPU drivers from the corresponding hardware vendor on the nodes and run the corresponding device plugin from the GPU vendor. Here are some links to vendors' instructions:

- [AMD](#)
- [Intel](#)
- [NVIDIA](#)

Once you have installed the plugin, your cluster exposes a custom schedulable resource such as `amd.com/gpu` or `nvidia.com/gpu`.

You can consume these GPUs from your containers by requesting the custom GPU resource, the same way you request `cpu` or `memory`. However, there are some limitations in how you specify the resource requirements for custom devices.

GPUs are only supposed to be specified in the `limits` section, which means:

- You can specify GPU limits without specifying requests, because Kubernetes will use the limit as the request value by default.
- You can specify GPU in both `limits` and `requests` but these two values must be equal.
- You cannot specify GPU requests without specifying limits.

Here's an example manifest for a Pod that requests a GPU:

```
apiVersion: v1
kind: Pod
metadata:
  name: example-vector-add
spec:
  restartPolicy: OnFailure
  containers:
  - name: example-vector-add
    image: "registry.example/example-vector-add:v42"
    resources:
      limits:
        gpu-vendor.example/example-gpu: 1 # requesting 1 GPU
```

Manage clusters with different types of GPUs

If different nodes in your cluster have different types of GPUs, then you can use [Node Labels and Node Selectors](#) to schedule pods to appropriate nodes.

For example:

```
# Label your nodes with the accelerator type they have.
kubectl label nodes node1 accelerator=example-gpu-x100
kubectl label nodes node2 accelerator=other-gpu-k915
```

That label key `accelerator` is just an example; you can use a different label key if you prefer.

Automatic node labelling

As an administrator, you can automatically discover and label all your GPU enabled nodes by deploying Kubernetes [Node Feature Discovery](#) (NFD). NFD detects the hardware features that are available on each node in a Kubernetes cluster. Typically, NFD is configured to advertise those features as node labels, but NFD can also add extended resources, annotations, and node taints. NFD is compatible with all [supported versions](#) of Kubernetes. By default NFD create the [feature labels](#) for the detected features. Administrators can leverage NFD to also taint nodes with specific features, so that only pods that request those features can be scheduled on those nodes.

You also need a plugin for NFD that adds appropriate labels to your nodes; these might be generic labels or they could be vendor specific. Your GPU vendor may provide a third party plugin for NFD; check their documentation for more details.

```
apiVersion: v1
kind: Pod
metadata:
  name: example-vector-add
spec:
  restartPolicy: OnFailure
  # You can use Kubernetes node affinity to schedule this Pod onto a node
  # that provides the kind of GPU that its container needs in order to work
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: "gpu.gpu-vendor.example/installed-memory"
                operator: Gt # (greater than)
                values: ["40535"]
              - key: "feature.node.kubernetes.io/pci-10.present" # NFD Feature label
                values: ["true"] # (optional) only schedule on nodes with PCI device 10
  containers:
    - name: example-vector-add
      image: "registry.example/example-vector-add:v42"
      resources:
        limits:
          gpu-vendor.example/example-gpu: 1 # requesting 1 GPU
```

GPU vendor implementations

- [Intel](#)
- [NVIDIA](#)

This section of the Kubernetes documentation contains tutorials. A tutorial shows how to accomplish a goal that is larger than a single [task](#). Typically a tutorial has several sections, each of which has a sequence of steps. Before walking through each tutorial, you may want to bookmark the [Standardized Glossary](#) page for later references.

Basics

- [Kubernetes Basics](#) is an in-depth interactive tutorial that helps you understand the Kubernetes system and try out some basic Kubernetes features.
- [Introduction to Kubernetes \(edX\)](#)
- [Hello Minikube](#)

Configuration

- [Example: Configuring a Java Microservice](#)
- [Configuring Redis Using a ConfigMap](#)

Stateless Applications

- [Exposing an External IP Address to Access an Application in a Cluster](#)
- [Example: Deploying PHP Guestbook application with Redis](#)

Stateful Applications

- [StatefulSet Basics](#)
- [Example: WordPress and MySQL with Persistent Volumes](#)
- [Example: Deploying Cassandra with Stateful Sets](#)
- [Running ZooKeeper, A CP Distributed System](#)

Services

- [Connecting Applications with Services](#)
- [Using Source IP](#)

Security

- [Apply Pod Security Standards at Cluster level](#)
- [Apply Pod Security Standards at Namespace level](#)
- [AppArmor](#)
- [Seccomp](#)

What's next

If you would like to write a tutorial, see [Content Page Types](#) for information about the tutorial page type.

Hello Minikube

This tutorial shows you how to run a sample app on Kubernetes using minikube. The tutorial provides a container image that uses NGINX to echo back all the requests.

Objectives

- Deploy a sample application to minikube.
- Run the app.
- View application logs.

Before you begin

This tutorial assumes that you have already set up minikube. See **Step 1** in [minikube start](#) for installation instructions.

Note:

Only execute the instructions in **Step 1, Installation**. The rest is covered on this page.

You also need to install kubectl. See [Install tools](#) for installation instructions.

Create a minikube cluster

```
minikube start
```

Open the Dashboard

Open the Kubernetes dashboard. You can do this two different ways:

- [Launch a browser](#)
- [URL copy and paste](#)

Open a **new** terminal, and run:

```
# Start a new terminal, and leave this running.  
minikube dashboard
```

Now, switch back to the terminal where you ran minikube start.

Note:

The dashboard command enables the dashboard add-on and opens the proxy in the default web browser. You can create Kubernetes resources on the dashboard such as Deployment and Service.

To find out how to avoid directly invoking the browser from the terminal and get a URL for the web dashboard, see the "URL copy and paste" tab.

By default, the dashboard is only accessible from within the internal Kubernetes virtual network. The dashboard command creates a temporary proxy to make the dashboard accessible from outside the Kubernetes virtual network.

To stop the proxy, run Ctrl+C to exit the process. After the command exits, the dashboard remains running in the Kubernetes cluster. You can run the dashboard command again to create another proxy to access the dashboard.

If you don't want minikube to open a web browser for you, run the dashboard subcommand with the --url flag. minikube outputs a URL that you can open in the browser you prefer.

Open a **new** terminal, and run:

```
# Start a new terminal, and leave this running.
minikube dashboard --url
```

Now, you can use this URL and switch back to the terminal where you ran minikube start.

Create a Deployment

A Kubernetes [Pod](#) is a group of one or more Containers, tied together for the purposes of administration and networking. The Pod in this tutorial has only one Container. A Kubernetes [Deployment](#) checks on the health of your Pod and restarts the Pod's Container if it terminates. Deployments are the recommended way to manage the creation and scaling of Pods.

1. Use the kubectl create command to create a Deployment that manages a Pod. The Pod runs a Container based on the provided Docker image.

```
# Run a test container image that includes a webserver
kubectl create deployment hello-node --image=registry.k8s.io/e2e-test-images/agnhost:
2.39 -- /agnhost netexec --http-port=8080
```

2. View the Deployment:

```
kubectl get deployments
```

The output is similar to:

| NAME | READY | UP-TO-DATE | AVAILABLE | AGE |
|------------|-------|------------|-----------|-----|
| hello-node | 1/1 | 1 | 1 | 1m |

(It may take some time for the pod to become available. If you see "0/1", try again in a few seconds.)

3. View the Pod:

```
kubectl get pods
```

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|-----------------------------|-------|---------|----------|-----|
| hello-node-5f76cf6ccf-br9b5 | 1/1 | Running | 0 | 1m |

4. View cluster events:

```
kubectctl get events
```

5. View the kubectctl configuration:

```
kubectctl config view
```

6. View application logs for a container in a pod.

```
kubectctl logs hello-node-5f76cf6ccf-br9b5
```

The output is similar to:

```
I0911 09:19:26.677397    1 log.go:195] Started HTTP server on port 8080
I0911 09:19:26.677586    1 log.go:195] Started UDP server on port 8081
```

Note:

For more information about kubectctl commands, see the [kubectctl overview](#).

Create a Service

By default, the Pod is only accessible by its internal IP address within the Kubernetes cluster. To make the hello-node Container accessible from outside the Kubernetes virtual network, you have to expose the Pod as a Kubernetes [Service](#).

1. Expose the Pod to the public internet using the kubectctl expose command:

```
kubectctl expose deployment hello-node --type=LoadBalancer --port=8080
```

The `--type=LoadBalancer` flag indicates that you want to expose your Service outside of the cluster.

The application code inside the test image only listens on TCP port 8080. If you used `kubectctl expose` to expose a different port, clients could not connect to that other port.

2. View the Service you created:

```
kubectctl get services
```

The output is similar to:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|------------|--------------|---------------|-------------|----------------|-----|
| hello-node | LoadBalancer | 10.108.144.78 | <pending> | 8080:30369/TCP | 21s |
| kubernetes | ClusterIP | 10.96.0.1 | <none> | 443/TCP | 23m |

On cloud providers that support load balancers, an external IP address would be provisioned to access the Service. On minikube, the LoadBalancer type makes the Service accessible through the `minikube service` command.

3. Run the following command:

```
minikube service hello-node
```

This opens up a browser window that serves your app and shows the app's response.

Enable addons

The minikube tool includes a set of built-in [addons](#) that can be enabled, disabled and opened in the local Kubernetes environment.

1. List the currently supported addons:

```
minikube addons list
```

The output is similar to:

```
addon-manager: enabled
dashboard: enabled
default-storageclass: enabled
efk: disabled
freshpod: disabled
gvisor: disabled
helm-tiller: disabled
ingress: disabled
ingress-dns: disabled
logviewer: disabled
metrics-server: disabled
nvidia-driver-installer: disabled
nvidia-gpu-device-plugin: disabled
registry: disabled
registry-creds: disabled
storage-provisioner: enabled
storage-provisioner-gluster: disabled
```

2. Enable an addon, for example, metrics-server:

```
minikube addons enable metrics-server
```

The output is similar to:

```
The 'metrics-server' addon is enabled
```

3. View the Pod and Service you created by installing that addon:

```
kubectl get pod,svc -n kube-system
```

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|--------------------------------------|-------|---------|----------|-----|
| pod/coredns-5644d7b6d9-mh9ll | 1/1 | Running | 0 | 34m |
| pod/coredns-5644d7b6d9-pqd2t | 1/1 | Running | 0 | 34m |
| pod/metrics-server-67fb648c5 | 1/1 | Running | 0 | 26s |
| pod/etcd-minikube | 1/1 | Running | 0 | 34m |
| pod/influxdb-grafana-b29w8 | 2/2 | Running | 0 | 26s |
| pod/kube-addon-manager-minikube | 1/1 | Running | 0 | 34m |
| pod/kube-apiserver-minikube | 1/1 | Running | 0 | 34m |
| pod/kube-controller-manager-minikube | 1/1 | Running | 0 | 34m |
| pod/kube-proxy-rnlps | 1/1 | Running | 0 | 34m |
| pod/kube-scheduler-minikube | 1/1 | Running | 0 | 34m |

| pod/storage-provisioner | | 1/1 | Running | 0 | 34m | |
|-----------------------------|-----------|---------------|-------------|-------------------|-----|--|
| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE | |
| service/metrics-server | ClusterIP | 10.96.241.45 | <none> | 80/TCP | 26s | |
| service/kube-dns | ClusterIP | 10.96.0.10 | <none> | 53/UDP,53/TCP | 34m | |
| service/monitoring-grafana | NodePort | 10.99.24.54 | <none> | 80:30002/TCP | 26s | |
| service/monitoring-influxdb | ClusterIP | 10.111.169.94 | <none> | 8083/TCP,8086/TCP | 26s | |

4. Check the output from metrics-server:

```
kubectl top pods
```

The output is similar to:

| NAME | CPU(cores) | MEMORY(bytes) |
|----------------------------|------------|---------------|
| hello-node-ccf4b9788-4jn97 | 1m | 6Mi |

If you see the following message, wait, and try again:

```
error: Metrics API not available
```

5. Disable metrics-server:

```
minikube addons disable metrics-server
```

The output is similar to:

```
metrics-server was successfully disabled
```

Clean up

Now you can clean up the resources you created in your cluster:

```
kubectl delete service hello-node
kubectl delete deployment hello-node
```

Stop the Minikube cluster

```
minikube stop
```

Optionally, delete the Minikube VM:

```
# Optional
minikube delete
```

If you want to use minikube again to learn more about Kubernetes, you don't need to delete it.

Conclusion

This page covered the basic aspects to get a minikube cluster up and running. You are now ready to deploy applications.

What's next

- Tutorial to [deploy your first app on Kubernetes with kubectl](#).
- Learn more about [Deployment objects](#).
- Learn more about [Deploying applications](#).
- Learn more about [Service objects](#).

Learn Kubernetes Basics

html

Kubernetes Basics

This tutorial provides a walkthrough of the basics of the Kubernetes cluster orchestration system. Each module contains some background information on major Kubernetes features and concepts, and a tutorial for you to follow along.

Using the tutorials, you can learn to:

- Deploy a containerized application on a cluster.
- Scale the deployment.
- Update the containerized application with a new software version.
- Debug the containerized application.

What can Kubernetes do for you?

With modern web services, users expect applications to be available 24/7, and developers expect to deploy new versions of those applications several times a day. Containerization helps package software to serve these goals, enabling applications to be released and updated without downtime. Kubernetes helps you make sure those containerized applications run where and when you want, and helps them find the resources and tools they need to work. Kubernetes is a production-ready, open source platform designed with Google's accumulated experience in container orchestration, combined with best-of-breed ideas from the community.

Kubernetes Basics Modules

[1. Create a Kubernetes cluster](#)

[2. Deploy an app](#)

[3. Explore your app](#)

[4. Expose your app publicly](#)

[5. Scale up your app](#)

[6. Update your app](#)

Create a Cluster

Learn about Kubernetes [cluster](#) and create a simple cluster using Minikube.

[Using Minikube to Create a Cluster](#)

Learn what a Kubernetes cluster is. Learn what Minikube is. Start a Kubernetes cluster.

Using Minikube to Create a Cluster

Learn what a Kubernetes cluster is. Learn what Minikube is. Start a Kubernetes cluster.
html

Objectives

- Learn what a Kubernetes cluster is.
- Learn what Minikube is.
- Start a Kubernetes cluster on your computer.

Kubernetes Clusters

Kubernetes coordinates a highly available cluster of computers that are connected to work as a single unit. The abstractions in Kubernetes allow you to deploy containerized applications to a cluster without tying them specifically to individual machines. To make use of this new model of deployment, applications need to be packaged in a way that decouples them from individual hosts: they need to be containerized. Containerized applications are more flexible and available than in past deployment models, where applications were installed directly onto specific machines as packages deeply integrated into the host. **Kubernetes automates the distribution and scheduling of application containers across a cluster in a more efficient way.** Kubernetes is an open-source platform and is production-ready.

A Kubernetes cluster consists of two types of resources:

- The **Control Plane** coordinates the cluster
- **Nodes** are the workers that run applications

Summary:

- Kubernetes cluster

- Minikube

Kubernetes is a production-grade, open-source platform that orchestrates the placement (scheduling) and execution of application containers within and across computer clusters.

Cluster Diagram

The Control Plane is responsible for managing the cluster. The Control Plane coordinates all activities in your cluster, such as scheduling applications, maintaining applications' desired state, scaling applications, and rolling out new updates.

A node is a VM or a physical computer that serves as a worker machine in a Kubernetes cluster. Each node has a Kubelet, which is an agent for managing the node and communicating with the Kubernetes control plane. The node should also have tools for handling container operations, such as [containerd](#) or [CRI-O](#). A Kubernetes cluster that handles production traffic should have a minimum of three nodes because if one node goes down, both an [etcd](#) member and a control plane instance are lost, and redundancy is compromised. You can mitigate this risk by adding more control plane nodes.

Control Planes manage the cluster and the nodes that are used to host the running applications.

When you deploy applications on Kubernetes, you tell the control plane to start the application containers. The control plane schedules the containers to run on the cluster's nodes. **Node-level components, such as the kubelet, communicate with the control plane using the [Kubernetes API](#)**, which the control plane exposes. End users can also use the Kubernetes API directly to interact with the cluster.

A Kubernetes cluster can be deployed on either physical or virtual machines. To get started with Kubernetes development, you can use Minikube. Minikube is a lightweight Kubernetes implementation that creates a VM on your local machine and deploys a simple cluster containing only one node. Minikube is available for Linux, macOS, and Windows systems. The Minikube CLI provides basic bootstrapping operations for working with your cluster, including start, stop, status, and delete.

Now that you know more about what Kubernetes is, visit [Hello Minikube](#) to try this out on your computer.

Deploy an App

[Using kubectl to Create a Deployment](#)

Learn about application Deployments. Deploy your first app on Kubernetes with kubectl.

Using kubectl to Create a Deployment

Learn about application Deployments. Deploy your first app on Kubernetes with kubectl.
html

Objectives

- Learn about application Deployments.
- Deploy your first app on Kubernetes with kubectl.

Kubernetes Deployments

Note:

This tutorial uses a container that requires the AMD64 architecture. If you are using minikube on a computer with a different CPU architecture, you could try using minikube with a driver that can emulate AMD64. For example, the Docker Desktop driver can do this.

Once you have a [running Kubernetes cluster](#), you can deploy your containerized applications on top of it. To do so, you create a Kubernetes **Deployment**. The Deployment instructs Kubernetes how to create and update instances of your application. Once you've created a Deployment, the Kubernetes control plane schedules the application instances included in that Deployment to run on individual Nodes in the cluster.

Once the application instances are created, a Kubernetes Deployment controller continuously monitors those instances. If the Node hosting an instance goes down or is deleted, the Deployment controller replaces the instance with an instance on another Node in the cluster. **This provides a self-healing mechanism to address machine failure or maintenance.**

In a pre-orchestration world, installation scripts would often be used to start applications, but they did not allow recovery from machine failure. By both creating your application instances and keeping them running across Nodes, Kubernetes Deployments provide a fundamentally different approach to application management.

Summary:

- Deployments
- Kubectl

A Deployment is responsible for creating and updating instances of your application

Deploying your first app on Kubernetes

You can create and manage a Deployment by using the Kubernetes command line interface, **kubectl**. Kubectl uses the Kubernetes API to interact with the cluster. In this module, you'll

learn the most common kubectl commands needed to create Deployments that run your applications on a Kubernetes cluster.

When you create a Deployment, you'll need to specify the container image for your application and the number of replicas that you want to run. You can change that information later by updating your Deployment; Modules [5](#) and [6](#) of the bootcamp discuss how you can scale and update your Deployments.

Applications need to be packaged into one of the supported container formats in order to be deployed on Kubernetes

For your first Deployment, you'll use a hello-node application packaged in a Docker container that uses NGINX to echo back all the requests. (If you didn't already try creating a hello-node application and deploying it using a container, you can do that first by following the instructions from the [Hello Minikube tutorial](#)).

You will need to have installed kubectl as well. If you need to install it, visit [install tools](#).

Now that you know what Deployments are, let's deploy our first app!

kubectl basics

The common format of a kubectl command is: `kubectl action resource`

This performs the specified *action* (like create, describe or delete) on the specified *resource* (like node or deployment). You can use `--help` after the subcommand to get additional info about possible parameters (for example: `kubectl get nodes --help`).

Check that kubectl is configured to talk to your cluster, by running the **kubectl version** command.

Check that kubectl is installed and you can see both the client and the server versions.

To view the nodes in the cluster, run the **kubectl get nodes** command.

You see the available nodes. Later, Kubernetes will choose where to deploy our application based on Node available resources.

Deploy an app

Let's deploy our first app on Kubernetes with the `kubectl create deployment` command. We need to provide the deployment name and app image location (include the full repository url for images hosted outside Docker Hub).

```
kubectl create deployment kubernetes-bootcamp --image=gcr.io/google-samples/  
kubernetes-bootcamp:v1
```

Great! You just deployed your first application by creating a deployment. This performed a few things for you:

- searched for a suitable node where an instance of the application could be run (we have only 1 available node)
- scheduled the application to run on that Node
- configured the cluster to reschedule the instance on a new Node when needed

To list your deployments use the `kubectl get deployments` command:

kubectl get deployments

We see that there is 1 deployment running a single instance of your app. The instance is running inside a container on your node.

View the app

[Pods](#) that are running inside Kubernetes are running on a private, isolated network. By default they are visible from other pods and services within the same Kubernetes cluster, but not outside that network. When we use `kubectl`, we're interacting through an API endpoint to communicate with our application.

We will cover other options on how to expose your application outside the Kubernetes cluster later, in [Module 4](#). Also as a basic tutorial, we're not explaining what Pods are in any detail here, it will be covered in later topics.

The `kubectl proxy` command can create a proxy that will forward communications into the cluster-wide, private network. The proxy can be terminated by pressing control-C and won't show any output while it's running.

You need to open a second terminal window to run the proxy.

kubectl proxy

We now have a connection between our host (the terminal) and the Kubernetes cluster. The proxy enables direct access to the API from these terminals.

You can see all those APIs hosted through the proxy endpoint. For example, we can query the version directly through the API using the `curl` command:

`curl http://localhost:8001/version`

Note: If port 8001 is not accessible, ensure that the `kubectl proxy` that you started above is running in the second terminal.

The API server will automatically create an endpoint for each pod, based on the pod name, that is also accessible through the proxy.

First we need to get the Pod name, and we'll store it in the environment variable `POD_NAME`:

```
export POD_NAME=$(kubectl get pods -o go-template --template '{{range .items}}  
{{.metadata.name}}{{"\n"}}{{end}}')  
echo Name of the Pod: $POD_NAME
```

You can access the Pod through the proxied API, by running:

```
curl http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME:8080/proxy/
```

In order for the new Deployment to be accessible without using the proxy, a Service is required which will be explained in [Module 4](#).

Once you're ready, move on to [Viewing Pods and Nodes](#).

Explore Your App

[Viewing Pods and Nodes](#)

Learn how to troubleshoot Kubernetes applications using `kubectl get`, `kubectl describe`, `kubectl logs` and `kubectl exec`.

Viewing Pods and Nodes

Learn how to troubleshoot Kubernetes applications using `kubectl get`, `kubectl describe`, `kubectl logs` and `kubectl exec`.
html

Objectives

- Learn about Kubernetes Pods.
- Learn about Kubernetes Nodes.
- Troubleshoot deployed applications.

Kubernetes Pods

When you created a Deployment in [Module 2](#), Kubernetes created a **Pod** to host your application instance. A Pod is a Kubernetes abstraction that represents a group of one or more application containers (such as Docker), and some shared resources for those containers. Those resources include:

- Shared storage, as Volumes
- Networking, as a unique cluster IP address
- Information about how to run each container, such as the container image version or specific ports to use

A Pod models an application-specific "logical host" and can contain different application containers which are relatively tightly coupled. For example, a Pod might include both the container with your Node.js app as well as a different container that feeds the data to be published by the Node.js webserver. The containers in a Pod share an IP Address and port space, are always co-located and co-scheduled, and run in a shared context on the same Node.

Pods are the atomic unit on the Kubernetes platform. When we create a Deployment on Kubernetes, that Deployment creates Pods with containers inside them (as opposed to creating containers directly). Each Pod is tied to the Node where it is scheduled, and remains there until

termination (according to restart policy) or deletion. In case of a Node failure, identical Pods are scheduled on other available Nodes in the cluster.

Summary:

- Pods
- Nodes
- Kubectl main commands

A Pod is a group of one or more application containers (such as Docker) and includes shared storage (volumes), IP address and information about how to run them.

Pods overview

Nodes

A Pod always runs on a **Node**. A Node is a worker machine in Kubernetes and may be either a virtual or a physical machine, depending on the cluster. Each Node is managed by the control plane. A Node can have multiple pods, and the Kubernetes control plane automatically handles scheduling the pods across the Nodes in the cluster. The control plane's automatic scheduling takes into account the available resources on each Node.

Every Kubernetes Node runs at least:

- Kubelet, a process responsible for communication between the Kubernetes control plane and the Node; it manages the Pods and the containers running on a machine.
- A container runtime (like Docker) responsible for pulling the container image from a registry, unpacking the container, and running the application.

Containers should only be scheduled together in a single Pod if they are tightly coupled and need to share resources such as disk.

Node overview

Troubleshooting with kubectl

In Module [2](#), you used the kubectl command-line interface. You'll continue to use it in Module 3 to get information about deployed applications and their environments. The most common operations can be done with the following kubectl subcommands:

- **kubectl get** - list resources

- **kubectl describe** - show detailed information about a resource
- **kubectl logs** - print the logs from a container in a pod
- **kubectl exec** - execute a command on a container in a pod

You can use these commands to see when applications were deployed, what their current statuses are, where they are running and what their configurations are.

Now that we know more about our cluster components and the command line, let's explore our application.

A node is a worker machine in Kubernetes and may be a VM or physical machine, depending on the cluster. Multiple Pods can run on one Node.

Check application configuration

Let's verify that the application we deployed in the previous scenario is running. We'll use the kubectl get command and look for existing Pods:

kubectl get pods

If no pods are running, please wait a couple of seconds and list the Pods again. You can continue once you see one Pod running.

Next, to view what containers are inside that Pod and what images are used to build those containers we run the kubectl describe pods command:

kubectl describe pods

We see here details about the Pod's container: IP address, the ports used and a list of events related to the lifecycle of the Pod.

The output of the describe subcommand is extensive and covers some concepts that we didn't explain yet, but don't worry, they will become familiar by the end of this bootcamp.

***Note:** the describe subcommand can be used to get detailed information about most of the Kubernetes primitives, including Nodes, Pods, and Deployments. The describe output is designed to be human readable, not to be scripted against.*

Show the app in the terminal

Recall that Pods are running in an isolated, private network - so we need to proxy access to them so we can debug and interact with them. To do this, we'll use the kubectl proxy command to run a proxy in a **second terminal**. Open a new terminal window, and in that new terminal, run:

kubectl proxy

Now again, we'll get the Pod name and query that pod directly through the proxy. To get the Pod name and store it in the POD_NAME environment variable:

```
export POD_NAME="$(kubectl get pods -o go-template --template '{{range .items}}
{{.metadata.name}}{{"\n"}}{{end}}')"
```

echo Name of the Pod: \$POD_NAME

To see the output of our application, run a `curl` request:

```
curl http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME:8080/proxy/
```

The URL is the route to the API of the Pod.

View the container logs

Anything that the application would normally send to standard output becomes logs for the container within the Pod. We can retrieve these logs using the `kubectl logs` command:

```
kubectl logs "$POD_NAME"
```

***Note:** We don't need to specify the container name, because we only have one container inside the pod.*

Executing command on the container

We can execute commands directly on the container once the Pod is up and running. For this, we use the `exec` subcommand and use the name of the Pod as a parameter. Let's list the environment variables:

```
kubectl exec "$POD_NAME" -- env
```

Again, it's worth mentioning that the name of the container itself can be omitted since we only have a single container in the Pod.

Next let's start a bash session in the Pod's container:

```
kubectl exec -ti $POD_NAME -- bash
```

We have now an open console on the container where we run our NodeJS application. The source code of the app is in the `server.js` file:

```
cat server.js
```

You can check that the application is up by running a `curl` command:

```
curl http://localhost:8080
```

***Note:** here we used localhost because we executed the command inside the NodeJS Pod. If you cannot connect to localhost:8080, check to make sure you have run the `kubectl exec` command and are launching the command from within the Pod*

To close your container connection, type `exit`.

Once you're ready, move on to [Using A Service To Expose Your App](#).

Expose Your App Publicly

[Using a Service to Expose Your App](#)

Learn about a Service in Kubernetes. Understand how labels and selectors relate to a Service. Expose an application outside a Kubernetes cluster.

Using a Service to Expose Your App

Learn about a Service in Kubernetes. Understand how labels and selectors relate to a Service. Expose an application outside a Kubernetes cluster.
html

Objectives

- Learn about a Service in Kubernetes
- Understand how labels and selectors relate to a Service
- Expose an application outside a Kubernetes cluster using a Service

Overview of Kubernetes Services

Kubernetes [Pods](#) are mortal. Pods have a [lifecycle](#). When a worker node dies, the Pods running on the Node are also lost. A [ReplicaSet](#) might then dynamically drive the cluster back to the desired state via the creation of new Pods to keep your application running. As another example, consider an image-processing backend with 3 replicas. Those replicas are exchangeable; the front-end system should not care about backend replicas or even if a Pod is lost and recreated. That said, each Pod in a Kubernetes cluster has a unique IP address, even Pods on the same Node, so there needs to be a way of automatically reconciling changes among Pods so that your applications continue to function.

A Service in Kubernetes is an abstraction which defines a logical set of Pods and a policy by which to access them. Services enable a loose coupling between dependent Pods. A Service is defined using YAML or JSON, like all Kubernetes object manifests. The set of Pods targeted by a Service is usually determined by a *label selector* (see below for why you might want a Service without including a selector in the spec).

Although each Pod has a unique IP address, those IPs are not exposed outside the cluster without a Service. Services allow your applications to receive traffic. Services can be exposed in different ways by specifying a type in the spec of the Service:

- *ClusterIP* (default) - Exposes the Service on an internal IP in the cluster. This type makes the Service only reachable from within the cluster.
- *NodePort* - Exposes the Service on the same port of each selected Node in the cluster using NAT. Makes a Service accessible from outside the cluster using `<NodeIP>:<NodePort>`. Superset of ClusterIP.
- *LoadBalancer* - Creates an external load balancer in the current cloud (if supported) and assigns a fixed, external IP to the Service. Superset of NodePort.
- *ExternalName* - Maps the Service to the contents of the externalName field (e.g. `foo.bar.example.com`), by returning a CNAME record with its value. No proxying of any kind is set up. This type requires v1.7 or higher of kube-dns, or CoreDNS version 0.0.8 or higher.

More information about the different types of Services can be found in the [Using Source IP](#) tutorial. Also see [Connecting Applications with Services](#).

Additionally, note that there are some use cases with Services that involve not defining a selector in the spec. A Service created without selector will also not create the corresponding Endpoints object. This allows users to manually map a Service to specific endpoints. Another possibility why there may be no selector is you are strictly using type: ExternalName.

Summary

- Exposing Pods to external traffic
- Load balancing traffic across multiple Pods
- Using labels

A Kubernetes Service is an abstraction layer which defines a logical set of Pods and enables external traffic exposure, load balancing and service discovery for those Pods.

Services and Labels

A Service routes traffic across a set of Pods. Services are the abstraction that allows pods to die and replicate in Kubernetes without impacting your application. Discovery and routing among dependent Pods (such as the frontend and backend components in an application) are handled by Kubernetes Services.

Services match a set of Pods using [labels and selectors](#), a grouping primitive that allows logical operation on objects in Kubernetes. Labels are key/value pairs attached to objects and can be used in any number of ways:

- Designate objects for development, test, and production
- Embed version tags
- Classify an object using tags

Labels can be attached to objects at creation time or later on. They can be modified at any time. Let's expose our application now using a Service and apply some labels.

Step 1: Creating a new Service

Let's verify that our application is running. We'll use the `kubectl get` command and look for existing Pods:

kubectl get pods

If no Pods are running then it means the objects from the previous tutorials were cleaned up. In this case, go back and recreate the deployment from the [Using kubectl to create a Deployment](#) tutorial. Please wait a couple of seconds and list the Pods again. You can continue once you see the one Pod running.

Next, let's list the current Services from our cluster:

kubectl get services

We have a Service called kubernetes that is created by default when minikube starts the cluster. To create a new service and expose it to external traffic we'll use the expose command with NodePort as parameter.

kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080

Let's run again the get services subcommand:

kubectl get services

We have now a running Service called kubernetes-bootcamp. Here we see that the Service received a unique cluster-IP, an internal port and an external-IP (the IP of the Node).

To find out what port was opened externally (for the type: NodePort Service) we'll run the describe service subcommand:

kubectl describe services/kubernetes-bootcamp

Create an environment variable called NODE_PORT that has the value of the Node port assigned:

```
export NODE_PORT="$(kubectl get services/kubernetes-bootcamp -o go-template='{{(index .spec.ports 0).nodePort}}')"  
echo "NODE_PORT=$NODE_PORT"
```

Now we can test that the app is exposed outside of the cluster using curl, the IP address of the Node and the externally exposed port:

```
curl http://$(minikube ip):$NODE_PORT"
```

Note:

If you're running minikube with Docker Desktop as the container driver, a minikube tunnel is needed. This is because containers inside Docker Desktop are isolated from your host computer.

In a separate terminal window, execute:

```
minikube service kubernetes-bootcamp --url
```

The output looks like this:

```
http://127.0.0.1:51082
```

```
! Because you are using a Docker driver on darwin, the terminal needs to be open to run it.
```

Then use the given URL to access the app:

```
curl 127.0.0.1:51082
```

And we get a response from the server. The Service is exposed.

Step 2: Using labels

The Deployment created automatically a label for our Pod. With the `describe deployment` subcommand you can see the name (the *key*) of that label:

`kubectl describe deployment`

Let's use this label to query our list of Pods. We'll use the `kubectl get pods` command with `-l` as a parameter, followed by the label values:

`kubectl get pods -l app=kubernetes-bootcamp`

You can do the same to list the existing Services:

`kubectl get services -l app=kubernetes-bootcamp`

Get the name of the Pod and store it in the `POD_NAME` environment variable:

```
export POD_NAME="$(kubectl get pods -o go-template --template '{{range .items}}
{{.metadata.name}}{{"\n"}}{{end}}')"
```

`echo "Name of the Pod: $POD_NAME"`

To apply a new label we use the `label` subcommand followed by the object type, object name and the new label:

`kubectl label pods "$POD_NAME" version=v1`

This will apply a new label to our Pod (we pinned the application version to the Pod), and we can check it with the `describe pod` command:

`kubectl describe pods "$POD_NAME"`

We see here that the label is attached now to our Pod. And we can query now the list of pods using the new label:

`kubectl get pods -l version=v1`

And we see the Pod.

Step 3: Deleting a service

To delete Services you can use the `delete service` subcommand. Labels can be used also here:

`kubectl delete service -l app=kubernetes-bootcamp`

Confirm that the Service is gone:

`kubectl get services`

This confirms that our Service was removed. To confirm that route is not exposed anymore you can `curl` the previously exposed IP and port:

```
curl http://"$(minikube ip):$NODE_PORT"
```

This proves that the application is not reachable anymore from outside of the cluster. You can confirm that the app is still running with a curl from inside the pod:

```
kubectl exec -ti $POD_NAME -- curl http://localhost:8080
```

We see here that the application is up. This is because the Deployment is managing the application. To shut down the application, you would need to delete the Deployment as well.

Once you're ready, move on to [Running Multiple Instances of Your App](#).

Scale Your App

[Running Multiple Instances of Your App](#)

Scale an existing app manually using kubectl.

Running Multiple Instances of Your App

Scale an existing app manually using kubectl.
html

Objectives

- Scale an app using kubectl.

Scaling an application

Previously we created a [Deployment](#), and then exposed it publicly via a [Service](#). The Deployment created only one Pod for running our application. When traffic increases, we will need to scale the application to keep up with user demand.

If you haven't worked through the earlier sections, start from [Using minikube to create a cluster](#).

Scaling is accomplished by changing the number of replicas in a Deployment

Summary:

- Scaling a Deployment

You can create from the start a Deployment with multiple instances using the --replicas parameter for the kubectl create deployment command

Note:

If you are trying this after [the previous section](#), then you may have deleted the service you created, or have created a Service of type: NodePort. In this section, it is assumed that a service with type: LoadBalancer is created for the kubernetes-bootcamp Deployment.

If you have *not* deleted the Service created in [the previous section](#), first delete that Service and then run the following command to create a new Service with its type set to LoadBalancer:

```
kubectl expose deployment/kubernetes-bootcamp --type="LoadBalancer" --port 8080
```

Scaling overview

- 1.
- 2.

[Previous](#) [Next](#)

Scaling out a Deployment will ensure new Pods are created and scheduled to Nodes with available resources. Scaling will increase the number of Pods to the new desired state. Kubernetes also supports [autoscaling](#) of Pods, but it is outside of the scope of this tutorial. Scaling to zero is also possible, and it will terminate all Pods of the specified Deployment.

Running multiple instances of an application will require a way to distribute the traffic to all of them. Services have an integrated load-balancer that will distribute network traffic to all Pods of an exposed Deployment. Services will monitor continuously the running Pods using endpoints, to ensure the traffic is sent only to available Pods.

Scaling is accomplished by changing the number of replicas in a Deployment.

Once you have multiple instances of an application running, you would be able to do Rolling updates without downtime. We'll cover that in the next section of the tutorial. Now, let's go to the terminal and scale our application.

Scaling a Deployment

To list your Deployments, use the `get deployments` subcommand:

```
kubectl get deployments
```

The output should be similar to:

| NAME | READY | UP-TO-DATE | AVAILABLE | AGE |
|---------------------|-------|------------|-----------|-----|
| kubernetes-bootcamp | 1/1 | 1 | 1 | 11m |

We should have 1 Pod. If not, run the command again. This shows:

- *NAME* lists the names of the Deployments in the cluster.
- *READY* shows the ratio of CURRENT/DESIRED replicas
- *UP-TO-DATE* displays the number of replicas that have been updated to achieve the desired state.
- *AVAILABLE* displays how many replicas of the application are available to your users.
- *AGE* displays the amount of time that the application has been running.

To see the ReplicaSet created by the Deployment, run:

kubectl get rs

Notice that the name of the ReplicaSet is always formatted as [DEPLOYMENT-NAME]-[RANDOM-STRING]. The random string is randomly generated and uses the *pod-template-hash* as a seed.

Two important columns of this output are:

- *DESIRED* displays the desired number of replicas of the application, which you define when you create the Deployment. This is the desired state.
- *CURRENT* displays how many replicas are currently running.

Next, let's scale the Deployment to 4 replicas. We'll use the `kubectl scale` command, followed by the Deployment type, name and desired number of instances:

```
kubectl scale deployments/kubernetes-bootcamp --replicas=4
```

To list your Deployments once again, use `get deployments`:

kubectl get deployments

The change was applied, and we have 4 instances of the application available. Next, let's check if the number of Pods changed:

kubectl get pods -o wide

There are 4 Pods now, with different IP addresses. The change was registered in the Deployment events log. To check that, use the `describe` subcommand:

kubectl describe deployments/kubernetes-bootcamp

You can also view in the output of this command that there are 4 replicas now.

Load Balancing

Let's check that the Service is load-balancing the traffic. To find out the exposed IP and Port we can use the `describe service` as we learned in the previous part of the tutorial:

kubectl describe services/kubernetes-bootcamp

Create an environment variable called `NODE_PORT` that has a value as the Node port:

```
export NODE_PORT="$(kubectl get services/kubernetes-bootcamp -o go-template='{{(index .spec.ports 0).nodePort}}')"
```

```
echo NODE_PORT=$NODE_PORT
```

Next, we'll do a `curl` to the exposed IP address and port. Execute the command multiple times:

```
curl http://"$(minikube ip):$NODE_PORT"
```

We hit a different Pod with every request. This demonstrates that the load-balancing is working.

Note:

If you're running minikube with Docker Desktop as the container driver, a minikube tunnel is needed. This is because containers inside Docker Desktop are isolated from your host computer.

In a separate terminal window, execute:

```
minikube service kubernetes-bootcamp --url
```

The output looks like this:

```
http://127.0.0.1:51082
```

```
! Because you are using a Docker driver on darwin, the terminal needs to be open to run it.
```

Then use the given URL to access the app:

```
curl 127.0.0.1:51082
```

Scale Down

To scale down the Deployment to 2 replicas, run again the `scale` subcommand:

```
kubectl scale deployments/kubernetes-bootcamp --replicas=2
```

List the Deployments to check if the change was applied with the `get deployments` subcommand:

```
kubectl get deployments
```

The number of replicas decreased to 2. List the number of Pods, with `get pods`:

```
kubectl get pods -o wide
```

This confirms that 2 Pods were terminated.

Once you're ready, move on to [Performing a Rolling Update](#).

Update Your App

[Performing a Rolling Update](#)

Perform a rolling update using `kubectl`.

Performing a Rolling Update

Perform a rolling update using `kubectl`.

```
html
```

Objectives

- Perform a rolling update using kubectl.

Updating an application

Users expect applications to be available all the time, and developers are expected to deploy new versions of them several times a day. In Kubernetes this is done with rolling updates. A **rolling update** allows a Deployment update to take place with zero downtime. It does this by incrementally replacing the current Pods with new ones. The new Pods are scheduled on Nodes with available resources, and Kubernetes waits for those new Pods to start before removing the old Pods.

In the previous module we scaled our application to run multiple instances. This is a requirement for performing updates without affecting application availability. By default, the maximum number of Pods that can be unavailable during the update and the maximum number of new Pods that can be created, is one. Both options can be configured to either numbers or percentages (of Pods). In Kubernetes, updates are versioned and any Deployment update can be reverted to a previous (stable) version.

Summary:

- Updating an app

Rolling updates allow Deployments' update to take place with zero downtime by incrementally updating Pods instances with new ones.

Rolling updates overview

- 1.
- 2.
- 3.
- 4.

[Previous](#) [Next](#)

Similar to application Scaling, if a Deployment is exposed publicly, the Service will load-balance the traffic only to available Pods during the update. An available Pod is an instance that is available to the users of the application.

Rolling updates allow the following actions:

- Promote an application from one environment to another (via container image updates)
- Rollback to previous versions
- Continuous Integration and Continuous Delivery of applications with zero downtime

If a Deployment is exposed publicly, the Service will load-balance the traffic only to available Pods during the update.

In the following interactive tutorial, we'll update our application to a new version, and also perform a rollback.

Update the version of the app

To list your Deployments, run the get deployments subcommand: **kubectl get deployments**

To list the running Pods, run the get pods subcommand:

kubectl get pods

To view the current image version of the app, run the describe pods subcommand and look for the Image field:

kubectl describe pods

To update the image of the application to version 2, use the set image subcommand, followed by the deployment name and the new image version:

```
kubectl set image deployments/kubernetes-bootcamp kubernetes-  
bootcamp=docker.io/jocatalin/kubernetes-bootcamp:v2
```

The command notified the Deployment to use a different image for your app and initiated a rolling update. Check the status of the new Pods, and view the old one terminating with the get pods subcommand:

kubectl get pods

Verify an update

First, check that the service is running, as you might have deleted it in previous tutorial step, run describe services/kubernetes-bootcamp. If it's missing, you can create it again with:

```
kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
```

Create an environment variable called NODE_PORT that has the value of the Node port assigned:

```
export NODE_PORT="$(kubectl get services/kubernetes-bootcamp -o go-  
template='{{(index .spec.ports 0).nodePort}}')"  
echo "NODE_PORT=$NODE_PORT"
```

Next, do a curl to the exposed IP and port:

```
curl http://$(minikube ip):$NODE_PORT
```

Every time you run the curl command, you will hit a different Pod. Notice that all Pods are now running the latest version (v2).

You can also confirm the update by running the rollout status subcommand:

kubectl rollout status deployments/kubernetes-bootcamp

To view the current image version of the app, run the `describe pods` subcommand:

kubectl describe pods

In the `Image` field of the output, verify that you are running the latest image version (v2).

Roll back an update

Let's perform another update, and try to deploy an image tagged with v10:

```
kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=gcr.io/google-samples/kubernetes-bootcamp:v10
```

Use `get deployments` to see the status of the deployment:

kubectl get deployments

Notice that the output doesn't list the desired number of available Pods. Run the `get pods` subcommand to list all Pods:

kubectl get pods

Notice that some of the Pods have a status of `ImagePullBackOff`.

To get more insight into the problem, run the `describe pods` subcommand:

kubectl describe pods

In the `Events` section of the output for the affected Pods, notice that the v10 image version did not exist in the repository.

To roll back the deployment to your last working version, use the `rollout undo` subcommand:

kubectl rollout undo deployments/kubernetes-bootcamp

The `rollout undo` command reverts the deployment to the previous known state (v2 of the image). Updates are versioned and you can revert to any previously known state of a Deployment.

Use the `get pods` subcommand to list the Pods again:

kubectl get pods

Four Pods are running. To check the image deployed on these Pods, use the `describe pods` subcommand:

kubectl describe pods

The Deployment is once again using a stable version of the app (v2). The rollback was successful.

Remember to clean up your local cluster

kubectl delete deployments/kubernetes-bootcamp services/kubernetes-bootcamp

Configuration

[Example: Configuring a Java Microservice](#)

[Updating Configuration via a ConfigMap](#)

[Configuring Redis using a ConfigMap](#)

Example: Configuring a Java Microservice

[Externalizing config using MicroProfile, ConfigMaps and Secrets](#)

Externalizing config using MicroProfile, ConfigMaps and Secrets

In this tutorial you will learn how and why to externalize your microservice's configuration. Specifically, you will learn how to use Kubernetes ConfigMaps and Secrets to set environment variables and then consume them using MicroProfile Config.

Before you begin

Creating Kubernetes ConfigMaps & Secrets

There are several ways to set environment variables for a Docker container in Kubernetes, including: Dockerfile, kubernetes.yml, Kubernetes ConfigMaps, and Kubernetes Secrets. In the tutorial, you will learn how to use the latter two for setting your environment variables whose values will be injected into your microservices. One of the benefits for using ConfigMaps and Secrets is that they can be re-used across multiple containers, including being assigned to different environment variables for the different containers.

ConfigMaps are API Objects that store non-confidential key-value pairs. In the Interactive Tutorial you will learn how to use a ConfigMap to store the application's name. For more information regarding ConfigMaps, you can find the documentation [here](#).

Although Secrets are also used to store key-value pairs, they differ from ConfigMaps in that they're intended for confidential/sensitive information and are stored using Base64 encoding. This makes secrets the appropriate choice for storing such things as credentials, keys, and tokens, the former of which you'll do in the Interactive Tutorial. For more information on Secrets, you can find the documentation [here](#).

Externalizing Config from Code

Externalized application configuration is useful because configuration usually changes depending on your environment. In order to accomplish this, we'll use Java's Contexts and Dependency Injection (CDI) and MicroProfile Config. MicroProfile Config is a feature of MicroProfile, a set of open Java technologies for developing and deploying cloud-native microservices.

CDI provides a standard dependency injection capability enabling an application to be assembled from collaborating, loosely-coupled beans. MicroProfile Config provides apps and microservices a standard way to obtain config properties from various sources, including the application, runtime, and environment. Based on the source's defined priority, the properties are automatically combined into a single set of properties that the application can access via an API. Together, CDI & MicroProfile will be used in the Interactive Tutorial to retrieve the externally provided properties from the Kubernetes ConfigMaps and Secrets and get injected into your application code.

Many open source frameworks and runtimes implement and support MicroProfile Config. Throughout the interactive tutorial, you'll be using Open Liberty, a flexible open-source Java runtime for building and running cloud-native apps and microservices. However, any MicroProfile compatible runtime could be used instead.

Objectives

- Create a Kubernetes ConfigMap and Secret
- Inject microservice configuration using MicroProfile Config

Example: Externalizing config using MicroProfile, ConfigMaps and Secrets

[Start Interactive Tutorial](#)

Updating Configuration via a ConfigMap

This page provides a step-by-step example of updating configuration within a Pod via a ConfigMap and builds upon the [Configure a Pod to Use a ConfigMap](#) task.

At the end of this tutorial, you will understand how to change the configuration for a running application.

This tutorial uses the alpine and nginx images as examples.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You need to have the [curl](#) command-line tool for making HTTP requests from the terminal or command prompt. If you do not have curl available, you can install it. Check the documentation for your local operating system.

Objectives

- Update configuration via a ConfigMap mounted as a Volume
- Update environment variables of a Pod via a ConfigMap
- Update configuration via a ConfigMap in a multi-container Pod
- Update configuration via a ConfigMap in a Pod possessing a Sidecar Container

Update configuration via a ConfigMap mounted as a Volume

Use the `kubectl create configmap sport --from-literal=sport=football` command to create a ConfigMap from [literal values](#):

```
kubectl create configmap sport --from-literal=sport=football
```

Below is an example of a Deployment manifest with the ConfigMap `sport` mounted as a [volume](#) into the Pod's only container.

[deployments/deployment-with-configmap-as-volume.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: configmap-volume
  labels:
    app.kubernetes.io/name: configmap-volume
spec:
  replicas: 3
  selector:
    matchLabels:
      app.kubernetes.io/name: configmap-volume
  template:
    metadata:
      labels:
        app.kubernetes.io/name: configmap-volume
    spec:
      containers:
        - name: alpine
          image: alpine:3
          command:
            - /bin/sh
            - -c
            - while true; do echo "$(date) My preferred sport is $(cat /etc/config/sport)";
              sleep 10; done;
          ports:
            - containerPort: 80
          volumeMounts:
            - name: config-volume
```

```
    mountPath: /etc/config
volumes:
- name: config-volume
  configMap:
    name: sport
```

Create the Deployment:

```
kubectl apply -f https://k8s.io/examples/deployments/deployment-with-configmap-as-volume.yaml
```

Check the pods for this Deployment to ensure they are ready (matching by [selector](#)):

```
kubectl get pods --selector=app.kubernetes.io/name=configmap-volume
```

You should see an output similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|-----------------------------------|-------|---------|----------|-----|
| configmap-volume-6b976dfdcf-qxvbm | 1/1 | Running | 0 | 72s |
| configmap-volume-6b976dfdcf-skpvm | 1/1 | Running | 0 | 72s |
| configmap-volume-6b976dfdcf-tbc6r | 1/1 | Running | 0 | 72s |

On each node where one of these Pods is running, the kubelet fetches the data for that ConfigMap and translates it to files in a local volume. The kubelet then mounts that volume into the container, as specified in the Pod template. The code running in that container loads the information from the file and uses it to print a report to stdout. You can check this report by viewing the logs for one of the Pods in that Deployment:

```
# Pick one Pod that belongs to the Deployment, and view its logs
kubectl logs deployments/configmap-volume
```

You should see an output similar to:

```
Found 3 pods, using pod/configmap-volume-76d9c5678f-x5rgj
Thu Jan 4 14:06:46 UTC 2024 My preferred sport is football
Thu Jan 4 14:06:56 UTC 2024 My preferred sport is football
Thu Jan 4 14:07:06 UTC 2024 My preferred sport is football
Thu Jan 4 14:07:16 UTC 2024 My preferred sport is football
Thu Jan 4 14:07:26 UTC 2024 My preferred sport is football
```

Edit the ConfigMap:

```
kubectl edit configmap sport
```

In the editor that appears, change the value of key sport from football to cricket. Save your changes. The kubectl tool updates the ConfigMap accordingly (if you see an error, try again).

Here's an example of how that manifest could look after you edit it:

```
apiVersion: v1
data:
  sport: cricket
kind: ConfigMap
# You can leave the existing metadata as they are.
# The values you'll see won't exactly match these.
```

```
metadata:
  creationTimestamp: "2024-01-04T14:05:06Z"
  name: sport
  namespace: default
  resourceVersion: "1743935"
  uid: 024ee001-fe72-487e-872e-34d6464a8a23
```

You should see the following output:

```
configmap/sport edited
```

Tail (follow the latest entries in) the logs of one of the pods that belongs to this Deployment:

```
kubectl logs deployments/configmap-volume --follow
```

After few seconds, you should see the log output change as follows:

```
Thu Jan 4 14:11:36 UTC 2024 My preferred sport is football
Thu Jan 4 14:11:46 UTC 2024 My preferred sport is football
Thu Jan 4 14:11:56 UTC 2024 My preferred sport is football
Thu Jan 4 14:12:06 UTC 2024 My preferred sport is cricket
Thu Jan 4 14:12:16 UTC 2024 My preferred sport is cricket
```

When you have a ConfigMap that is mapped into a running Pod using either a configMap volume or a projected volume, and you update that ConfigMap, the running Pod sees the update almost immediately.

However, your application only sees the change if it is written to either poll for changes, or watch for file updates.

An application that loads its configuration once at startup will not notice a change.

Note:

The total delay from the moment when the ConfigMap is updated to the moment when new keys are projected to the Pod can be as long as kubelet sync period.

Also check [Mounted ConfigMaps are updated automatically](#).

Update environment variables of a Pod via a ConfigMap

Use the `kubectl create configmap` command to create a ConfigMap from [literal values](#):

```
kubectl create configmap fruits --from-literal=fruits=apples
```

Below is an example of a Deployment manifest with an environment variable configured via the ConfigMap `fruits`.

[deployments/deployment-with-configmap-as-envvar.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: configmap-env-var
  labels:
    app.kubernetes.io/name: configmap-env-var
```

```
spec:
  replicas: 3
  selector:
    matchLabels:
      app.kubernetes.io/name: configmap-env-var
  template:
    metadata:
      labels:
        app.kubernetes.io/name: configmap-env-var
    spec:
      containers:
        - name: alpine
          image: alpine:3
          env:
            - name: FRUITS
              valueFrom:
                configMapKeyRef:
                  key: fruits
                  name: fruits
          command:
            - /bin/sh
            - -c
            - while true; do echo "$(date) The basket is full of $FRUITS";
              sleep 10; done;
          ports:
            - containerPort: 80
```

Create the Deployment:

```
kubectl apply -f https://k8s.io/examples/deployments/deployment-with-configmap-as-envvar.yaml
```

Check the pods for this Deployment to ensure they are ready (matching by [selector](#)):

```
kubectl get pods --selector=app.kubernetes.io/name=configmap-env-var
```

You should see an output similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|------------------------------------|-------|---------|----------|-----|
| configmap-env-var-59cfc64f7d-74d7z | 1/1 | Running | 0 | 46s |
| configmap-env-var-59cfc64f7d-c4wmj | 1/1 | Running | 0 | 46s |
| configmap-env-var-59cfc64f7d-dpr98 | 1/1 | Running | 0 | 46s |

The key-value pair in the ConfigMap is configured as an environment variable in the container of the Pod. Check this by viewing the logs of one Pod that belongs to the Deployment.

```
kubectl logs deployment/configmap-env-var
```

You should see an output similar to:

```
Found 3 pods, using pod/configmap-env-var-7c994f7769-l74nq
Thu Jan 4 16:07:06 UTC 2024 The basket is full of apples
Thu Jan 4 16:07:16 UTC 2024 The basket is full of apples
Thu Jan 4 16:07:26 UTC 2024 The basket is full of apples
```

Edit the ConfigMap:

```
kubectl edit configmap fruits
```

In the editor that appears, change the value of key fruits from apples to mangoes. Save your changes. The kubectl tool updates the ConfigMap accordingly (if you see an error, try again).

Here's an example of how that manifest could look after you edit it:

```
apiVersion: v1
data:
  fruits: mangoes
kind: ConfigMap
# You can leave the existing metadata as they are.
# The values you'll see won't exactly match these.
metadata:
  creationTimestamp: "2024-01-04T16:04:19Z"
  name: fruits
  namespace: default
  resourceVersion: "1749472"
```

You should see the following output:

```
configmap/fruits edited
```

Tail the logs of the Deployment and observe the output for few seconds:

```
# As the text explains, the output does NOT change
kubectl logs deployments/configmap-env-var --follow
```

Notice that the output remains **unchanged**, even though you edited the ConfigMap:

```
Thu Jan 4 16:12:56 UTC 2024 The basket is full of apples
Thu Jan 4 16:13:06 UTC 2024 The basket is full of apples
Thu Jan 4 16:13:16 UTC 2024 The basket is full of apples
Thu Jan 4 16:13:26 UTC 2024 The basket is full of apples
```

Note:

Although the value of the key inside the ConfigMap has changed, the environment variable in the Pod still shows the earlier value. This is because environment variables for a process running inside a Pod are **not** updated when the source data changes; if you wanted to force an update, you would need to have Kubernetes replace your existing Pods. The new Pods would then run with the updated information.

You can trigger that replacement. Perform a rollout for the Deployment, using [kubectl rollout](#):

```
# Trigger the rollout
kubectl rollout restart deployment configmap-env-var

# Wait for the rollout to complete
kubectl rollout status deployment configmap-env-var --watch=true
```

Next, check the Deployment:

```
kubectl get deployment configmap-env-var
```

You should see an output similar to:

| NAME | READY | UP-TO-DATE | AVAILABLE | AGE |
|-------------------|-------|------------|-----------|-----|
| configmap-env-var | 3/3 | 3 | 3 | 12m |

Check the Pods:

```
kubectl get pods --selector=app.kubernetes.io/name=configmap-env-var
```

The rollout causes Kubernetes to make a new [ReplicaSet](#) for the Deployment; that means the existing Pods eventually terminate, and new ones are created. After few seconds, you should see an output similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|------------------------------------|-------|---------|----------|-----|
| configmap-env-var-6d94d89bf5-2ph2l | 1/1 | Running | 0 | 13s |
| configmap-env-var-6d94d89bf5-74twx | 1/1 | Running | 0 | 8s |
| configmap-env-var-6d94d89bf5-d5vx8 | 1/1 | Running | 0 | 11s |

Note:

Please wait for the older Pods to fully terminate before proceeding with the next steps.

View the logs for a Pod in this Deployment:

```
# Pick one Pod that belongs to the Deployment, and view its logs
kubectl logs deployment/configmap-env-var
```

You should see an output similar to the below:

```
Found 3 pods, using pod/configmap-env-var-6d9ff89fb6-bzcf6
Thu Jan 4 16:30:35 UTC 2024 The basket is full of mangoes
Thu Jan 4 16:30:45 UTC 2024 The basket is full of mangoes
Thu Jan 4 16:30:55 UTC 2024 The basket is full of mangoes
```

This demonstrates the scenario of updating environment variables in a Pod that are derived from a ConfigMap. Changes to the ConfigMap values are applied to the Pod during the subsequent rollout. If Pods get created for another reason, such as scaling up the Deployment, then the new Pods also use the latest configuration values; if you don't trigger a rollout, then you might find that your app is running with a mix of old and new environment variable values.

Update configuration via a ConfigMap in a multi-container Pod

Use the `kubectl create configmap` command to create a ConfigMap from [literal values](#):

```
kubectl create configmap color --from-literal=color=red
```

Below is an example manifest for a Deployment that manages a set of Pods, each with two containers. The two containers share an emptyDir volume that they use to communicate. The first container runs a web server (nginx). The mount path for the shared volume in the web

server container is `/usr/share/nginx/html`. The second helper container is based on `alpine`, and for this container the `emptyDir` volume is mounted at `/pod-data`. The helper container writes a file in HTML that has its content based on a `ConfigMap`. The web server container serves the HTML via HTTP.

[deployments/deployment-with-configmap-two-containers.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: configmap-two-containers
  labels:
    app.kubernetes.io/name: configmap-two-containers
spec:
  replicas: 3
  selector:
    matchLabels:
      app.kubernetes.io/name: configmap-two-containers
  template:
    metadata:
      labels:
        app.kubernetes.io/name: configmap-two-containers
    spec:
      volumes:
        - name: shared-data
          emptyDir: {}
        - name: config-volume
          configMap:
            name: color
      containers:
        - name: nginx
          image: nginx
          volumeMounts:
            - name: shared-data
              mountPath: /usr/share/nginx/html
        - name: alpine
          image: alpine:3
          volumeMounts:
            - name: shared-data
              mountPath: /pod-data
            - name: config-volume
              mountPath: /etc/config
          command:
            - /bin/sh
            - -c
            - while true; do echo "$(date) My preferred color is $(cat /etc/config/color)" > /pod-data/
              index.html;
              sleep 10; done;
```

Create the Deployment:

```
kubectl apply -f https://k8s.io/examples/deployments/deployment-with-configmap-two-containers.yaml
```

Check the pods for this Deployment to ensure they are ready (matching by [selector](#)):

```
kubectl get pods --selector=app.kubernetes.io/name=configmap-two-containers
```

You should see an output similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|---|-------|---------|----------|-----|
| configmap-two-containers-565fb6d4f4-2xhxf | 2/2 | Running | 0 | 20s |
| configmap-two-containers-565fb6d4f4-g5v4j | 2/2 | Running | 0 | 20s |
| configmap-two-containers-565fb6d4f4-mzsmf | 2/2 | Running | 0 | 20s |

Expose the Deployment (the kubectl tool creates a [Service](#) for you):

```
kubectl expose deployment configmap-two-containers --name=configmap-service --port=8080 --target-port=80
```

Use kubectl to forward the port:

```
kubectl port-forward service/configmap-service 8080:8080 & # this stays running in the background
```

Access the service.

```
curl http://localhost:8080
```

You should see an output similar to:

```
Fri Jan 5 08:08:22 UTC 2024 My preferred color is red
```

Edit the ConfigMap:

```
kubectl edit configmap color
```

In the editor that appears, change the value of key color from red to blue. Save your changes. The kubectl tool updates the ConfigMap accordingly (if you see an error, try again).

Here's an example of how that manifest could look after you edit it:

```
apiVersion: v1
data:
  color: blue
kind: ConfigMap
# You can leave the existing metadata as they are.
# The values you'll see won't exactly match these.
metadata:
  creationTimestamp: "2024-01-05T08:12:05Z"
  name: color
  namespace: configmap
  resourceVersion: "1801272"
  uid: 80d33e4a-cbb4-4bc9-ba8c-544c68e425d6
```

Loop over the service URL for few seconds.

```
# Cancel this when you're happy with it (Ctrl-C)
while true; do curl --connect-timeout 7.5 http://localhost:8080; sleep 10; done
```


You should see the output change as follows:

```
Fri Jan 5 08:14:00 UTC 2024 My preferred color is red
Fri Jan 5 08:14:02 UTC 2024 My preferred color is red
Fri Jan 5 08:14:20 UTC 2024 My preferred color is red
Fri Jan 5 08:14:22 UTC 2024 My preferred color is red
Fri Jan 5 08:14:32 UTC 2024 My preferred color is blue
Fri Jan 5 08:14:43 UTC 2024 My preferred color is blue
Fri Jan 5 08:15:00 UTC 2024 My preferred color is blue
```

Update configuration via a ConfigMap in a Pod possessing a sidecar container

The above scenario can be replicated by using a [Sidecar Container](#) as a helper container to write the HTML file.

As a Sidecar Container is conceptually an Init Container, it is guaranteed to start before the main web server container.

This ensures that the HTML file is always available when the web server is ready to serve it. Please see [Enabling sidecar containers](#) to utilize this feature.

If you are continuing from the previous scenario, you can reuse the ConfigMap named color for this scenario.

If you are executing this scenario independently, use the `kubectl create configmap` command to create a ConfigMap from [literal values](#):

```
kubectl create configmap color --from-literal=color=blue
```

Below is an example manifest for a Deployment that manages a set of Pods, each with a main container and a sidecar container. The two containers share an emptyDir volume that they use to communicate. The main container runs a web server (NGINX). The mount path for the shared volume in the web server container is `/usr/share/nginx/html`. The second container is a Sidecar Container based on Alpine Linux which acts as a helper container. For this container the emptyDir volume is mounted at `/pod-data`. The Sidecar Container writes a file in HTML that has its content based on a ConfigMap. The web server container serves the HTML via HTTP.

[deployments/deployment-with-configmap-and-sidecar-container.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: configmap-sidecar-container
  labels:
    app.kubernetes.io/name: configmap-sidecar-container
spec:
  replicas: 3
  selector:
    matchLabels:
      app.kubernetes.io/name: configmap-sidecar-container
  template:
    metadata:
      labels:
```

```

    app.kubernetes.io/name: configmap-sidecar-container
spec:
  volumes:
    - name: shared-data
      emptyDir: {}
    - name: config-volume
      configMap:
        name: color
  containers:
    - name: nginx
      image: nginx
      volumeMounts:
        - name: shared-data
          mountPath: /usr/share/nginx/html
  initContainers:
    - name: alpine
      image: alpine:3
      restartPolicy: Always
      volumeMounts:
        - name: shared-data
          mountPath: /pod-data
        - name: config-volume
          mountPath: /etc/config
      command:
        - /bin/sh
        - -c
        - while true; do echo "$(date) My preferred color is $(cat /etc/config/color)" > /pod-data/
index.html;
        sleep 10; done;

```

Create the Deployment:

```
kubectl apply -f https://k8s.io/examples/deployments/deployment-with-configmap-and-sidecar-container.yaml
```

Check the pods for this Deployment to ensure they are ready (matching by [selector](#)):

```
kubectl get pods --selector=app.kubernetes.io/name=configmap-sidecar-container
```

You should see an output similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|--|-------|---------|----------|-----|
| configmap-sidecar-container-5fb59f558b-87rp7 | 2/2 | Running | 0 | 94s |
| configmap-sidecar-container-5fb59f558b-ccs7s | 2/2 | Running | 0 | 94s |
| configmap-sidecar-container-5fb59f558b-wnmgk | 2/2 | Running | 0 | 94s |

Expose the Deployment (the kubectl tool creates a [Service](#) for you):

```
kubectl expose deployment configmap-sidecar-container --name=configmap-sidecar-service --port=8081 --target-port=80
```

Use kubectl to forward the port:

```
kubectrl port-forward service/configmap-sidecar-service 8081:8081 & # this stays running in the background
```

Access the service.

```
curl http://localhost:8081
```

You should see an output similar to:

```
Sat Feb 17 13:09:05 UTC 2024 My preferred color is blue
```

Edit the ConfigMap:

```
kubectrl edit configmap color
```

In the editor that appears, change the value of key `color` from `blue` to `green`. Save your changes. The `kubectrl` tool updates the ConfigMap accordingly (if you see an error, try again).

Here's an example of how that manifest could look after you edit it:

```
apiVersion: v1
data:
  color: green
kind: ConfigMap
# You can leave the existing metadata as they are.
# The values you'll see won't exactly match these.
metadata:
  creationTimestamp: "2024-02-17T12:20:30Z"
  name: color
  namespace: default
  resourceVersion: "1054"
  uid: e40bb34c-58df-4280-8bea-6ed16edccfaa
```

Loop over the service URL for few seconds.

```
# Cancel this when you're happy with it (Ctrl-C)
while true; do curl --connect-timeout 7.5 http://localhost:8081; sleep 10; done
```

You should see the output change as follows:

```
Sat Feb 17 13:12:35 UTC 2024 My preferred color is blue
Sat Feb 17 13:12:45 UTC 2024 My preferred color is blue
Sat Feb 17 13:12:55 UTC 2024 My preferred color is blue
Sat Feb 17 13:13:05 UTC 2024 My preferred color is blue
Sat Feb 17 13:13:15 UTC 2024 My preferred color is green
Sat Feb 17 13:13:25 UTC 2024 My preferred color is green
Sat Feb 17 13:13:35 UTC 2024 My preferred color is green
```

Update configuration via an immutable ConfigMap that is mounted as a volume

Note:

Immutable ConfigMaps are especially used for configuration that is constant and is **not** expected to change over time. Marking a ConfigMap as immutable allows a performance improvement where the kubelet does not watch for changes.

If you do need to make a change, you should plan to either:

- change the name of the ConfigMap, and switch to running Pods that reference the new name
- replace all the nodes in your cluster that have previously run a Pod that used the old value
- restart the kubelet on any node where the kubelet previously loaded the old ConfigMap

An example manifest for an [Immutable ConfigMap](#) is shown below.

[configmap/immutable-configmap.yaml](#)

```
apiVersion: v1
data:
  company_name: "ACME, Inc." # existing fictional company name
kind: ConfigMap
immutable: true
metadata:
  name: company-name-20150801
```

Create the Immutable ConfigMap:

```
kubectl apply -f https://k8s.io/examples/configmap/immutable-configmap.yaml
```

Below is an example of a Deployment manifest with the Immutable ConfigMap company-name-20150801 mounted as a [volume](#) into the Pod's only container.

[deployments/deployment-with-immutable-configmap-as-volume.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: immutable-configmap-volume
  labels:
    app.kubernetes.io/name: immutable-configmap-volume
spec:
  replicas: 3
  selector:
    matchLabels:
      app.kubernetes.io/name: immutable-configmap-volume
  template:
    metadata:
      labels:
        app.kubernetes.io/name: immutable-configmap-volume
```

```
spec:
  containers:
  - name: alpine
    image: alpine:3
    command:
    - /bin/sh
    - -c
    - while true; do echo "$(date) The name of the company is $(cat /etc/config/
company_name)";
      sleep 10; done;
    ports:
    - containerPort: 80
    volumeMounts:
    - name: config-volume
      mountPath: /etc/config
  volumes:
  - name: config-volume
    configMap:
      name: company-name-20150801
```

Create the Deployment:

```
kubectl apply -f https://k8s.io/examples/deployments/deployment-with-immutable-configmap-
as-volume.yaml
```

Check the pods for this Deployment to ensure they are ready (matching by [selector](#)):

```
kubectl get pods --selector=app.kubernetes.io/name=immutable-configmap-volume
```

You should see an output similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|---|-------|---------|----------|-----|
| immutable-configmap-volume-78b6fbff95-5gsfh | 1/1 | Running | 0 | 62s |
| immutable-configmap-volume-78b6fbff95-7vcj4 | 1/1 | Running | 0 | 62s |
| immutable-configmap-volume-78b6fbff95-vdslm | 1/1 | Running | 0 | 62s |

The Pod's container refers to the data defined in the ConfigMap and uses it to print a report to stdout. You can check this report by viewing the logs for one of the Pods in that Deployment:

```
# Pick one Pod that belongs to the Deployment, and view its logs
kubectl logs deployments/immutable-configmap-volume
```

You should see an output similar to:

```
Found 3 pods, using pod/immutable-configmap-volume-78b6fbff95-5gsfh
Wed Mar 20 03:52:34 UTC 2024 The name of the company is ACME, Inc.
Wed Mar 20 03:52:44 UTC 2024 The name of the company is ACME, Inc.
Wed Mar 20 03:52:54 UTC 2024 The name of the company is ACME, Inc.
```

Note:

Once a ConfigMap is marked as immutable, it is not possible to revert this change nor to mutate the contents of the data or the binaryData field.

In order to modify the behavior of the Pods that use this configuration, you will create a new

immutable ConfigMap and edit the Deployment to define a slightly different pod template, referencing the new ConfigMap.

Create a new immutable ConfigMap by using the manifest shown below:

[configmap/new-immutable-configmap.yaml](https://k8s.io/examples/configmap/new-immutable-configmap.yaml)

```
apiVersion: v1
data:
  company_name: "Fiktivesunternehmen GmbH" # new fictional company name
kind: ConfigMap
immutable: true
metadata:
  name: company-name-20240312
```

```
kubectl apply -f https://k8s.io/examples/configmap/new-immutable-configmap.yaml
```

You should see an output similar to:

```
configmap/company-name-20240312 created
```

Check the newly created ConfigMap:

```
kubectl get configmap
```

You should see an output displaying both the old and new ConfigMaps:

| NAME | DATA | AGE |
|-----------------------|------|-----|
| company-name-20150801 | 1 | 22m |
| company-name-20240312 | 1 | 24s |

Modify the Deployment to reference the new ConfigMap.

Edit the Deployment:

```
kubectl edit deployment immutable-configmap-volume
```

In the editor that appears, update the existing volume definition to use the new ConfigMap.

```
volumes:
- configMap:
  defaultMode: 420
  name: company-name-20240312 # Update this field
  name: config-volume
```

You should see the following output:

```
deployment.apps/immutable-configmap-volume edited
```

This will trigger a rollout. Wait for all the previous Pods to terminate and the new Pods to be in a ready state.

Monitor the status of the Pods:

```
kubectl get pods --selector=app.kubernetes.io/name=immutable-configmap-volume
```

| NAME | READY | STATUS | RESTARTS | AGE |
|---|-------|-------------|----------|-----|
| immutable-configmap-volume-5fdb88fcc8-29v8n | 1/1 | Running | 0 | 13s |
| immutable-configmap-volume-5fdb88fcc8-52ddd | 1/1 | Running | 0 | 14s |
| immutable-configmap-volume-5fdb88fcc8-n5jx4 | 1/1 | Running | 0 | 15s |
| immutable-configmap-volume-78b6fbff95-5gsfh | 1/1 | Terminating | 0 | 32m |
| immutable-configmap-volume-78b6fbff95-7vcj4 | 1/1 | Terminating | 0 | 32m |
| immutable-configmap-volume-78b6fbff95-vdslm | 1/1 | Terminating | 0 | 32m |

You should eventually see an output similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|---|-------|---------|----------|-----|
| immutable-configmap-volume-5fdb88fcc8-29v8n | 1/1 | Running | 0 | 43s |
| immutable-configmap-volume-5fdb88fcc8-52ddd | 1/1 | Running | 0 | 44s |
| immutable-configmap-volume-5fdb88fcc8-n5jx4 | 1/1 | Running | 0 | 45s |

View the logs for a Pod in this Deployment:

```
# Pick one Pod that belongs to the Deployment, and view its logs
kubectl logs deployment/immutable-configmap-volume
```

You should see an output similar to the below:

```
Found 3 pods, using pod/immutable-configmap-volume-5fdb88fcc8-n5jx4
Wed Mar 20 04:24:17 UTC 2024 The name of the company is Fiktivesunternehmen GmbH
Wed Mar 20 04:24:27 UTC 2024 The name of the company is Fiktivesunternehmen GmbH
Wed Mar 20 04:24:37 UTC 2024 The name of the company is Fiktivesunternehmen GmbH
```

Once all the deployments have migrated to use the new immutable ConfigMap, it is advised to delete the old one.

```
kubectl delete configmap company-name-20150801
```

Summary

Changes to a ConfigMap mounted as a Volume on a Pod are available seamlessly after the subsequent kubelet sync.

Changes to a ConfigMap that configures environment variables for a Pod are available after the subsequent rollout for the Pod.

Once a ConfigMap is marked as immutable, it is not possible to revert this change (you cannot make an immutable ConfigMap mutable), and you also cannot make any change to the contents of the data or the binaryData field. You can delete and recreate the ConfigMap, or you can make a new different ConfigMap. When you delete a ConfigMap, running containers and their Pods maintain a mount point to any volume that referenced that existing ConfigMap.

Cleaning up

Terminate the kubectl port-forward commands in case they are running.

Delete the resources created during the tutorial:

```
kubectl delete deployment configmap-volume configmap-env-var configmap-two-containers  
configmap-sidecar-container immutable-configmap-volume  
kubectl delete service configmap-service configmap-sidecar-service  
kubectl delete configmap sport fruits color company-name-20240312
```

```
kubectl delete configmap company-name-20150801 # In case it was not handled during the task  
execution
```

Configuring Redis using a ConfigMap

This page provides a real world example of how to configure Redis using a ConfigMap and builds upon the [Configure a Pod to Use a ConfigMap](#) task.

Objectives

- Create a ConfigMap with Redis configuration values
- Create a Redis Pod that mounts and uses the created ConfigMap
- Verify that the configuration was correctly applied.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter kubectl version.

- The example shown on this page works with kubectl 1.14 and above.
- Understand [Configure a Pod to Use a ConfigMap](#).

Real World Example: Configuring Redis using a ConfigMap

Follow the steps below to configure a Redis cache using data stored in a ConfigMap.

First create a ConfigMap with an empty configuration block:

```
cat <<EOF >./example-redis-config.yaml  
apiVersion: v1  
kind: ConfigMap  
metadata:  
  name: example-redis-config  
data:  
  redis-config: ""  
EOF
```


Apply the ConfigMap created above, along with a Redis pod manifest:

```
kubectl apply -f example-redis-config.yaml
kubectl apply -f https://raw.githubusercontent.com/kubernetes/website/main/content/en/
examples/pods/config/redis-pod.yaml
```

Examine the contents of the Redis pod manifest and note the following:

- A volume named config is created by spec.volumes[1]
- The key and path under spec.volumes[1].configMap.items[0] exposes the redis-config key from the example-redis-config ConfigMap as a file named redis.conf on the config volume.
- The config volume is then mounted at /redis-master by spec.containers[0].volumeMounts[1].

This has the net effect of exposing the data in data.redis-config from the example-redis-config ConfigMap above as /redis-master/redis.conf inside the Pod.

[pods/config/redis-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: redis
spec:
  containers:
  - name: redis
    image: redis:5.0.4
    command:
      - redis-server
      - "/redis-master/redis.conf"
    env:
      - name: MASTER
        value: "true"
    ports:
      - containerPort: 6379
    resources:
      limits:
        cpu: "0.1"
    volumeMounts:
      - mountPath: /redis-master-data
        name: data
      - mountPath: /redis-master
        name: config
  volumes:
  - name: data
    emptyDir: {}
  - name: config
    configMap:
      name: example-redis-config
      items:
      - key: redis-config
        path: redis.conf
```

Examine the created objects:

```
kubectl get pod/redis configmap/example-redis-config
```

You should see the following output:

| NAME | READY | STATUS | RESTARTS | AGE |
|-----------|-------|---------|----------|-----|
| pod/redis | 1/1 | Running | 0 | 8s |

| NAME | DATA | AGE |
|--------------------------------|------|-----|
| configmap/example-redis-config | 1 | 14s |

Recall that we left redis-config key in the example-redis-config ConfigMap blank:

```
kubectl describe configmap/example-redis-config
```

You should see an empty redis-config key:

```
Name:      example-redis-config
Namespace:  default
Labels:     <none>
Annotations: <none>
```

```
Data
====
redis-config:
```

Use kubectl exec to enter the pod and run the redis-cli tool to check the current configuration:

```
kubectl exec -it redis -- redis-cli
```

Check maxmemory:

```
127.0.0.1:6379> CONFIG GET maxmemory
```

It should show the default value of 0:

- 1) "maxmemory"
- 2) "0"

Similarly, check maxmemory-policy:

```
127.0.0.1:6379> CONFIG GET maxmemory-policy
```

Which should also yield its default value of noeviction:

- 1) "maxmemory-policy"
- 2) "noeviction"

Now let's add some configuration values to the example-redis-config ConfigMap:

[pods/config/example-redis-config.yaml](#)

```
apiVersion: v1
kind: ConfigMap
```

```
metadata:
  name: example-redis-config
data:
  redis-config: |
    maxmemory 2mb
    maxmemory-policy allkeys-lru
```

Apply the updated ConfigMap:

```
kubectl apply -f example-redis-config.yaml
```

Confirm that the ConfigMap was updated:

```
kubectl describe configmap/example-redis-config
```

You should see the configuration values we just added:

```
Name:      example-redis-config
Namespace:  default
Labels:     <none>
Annotations: <none>
```

Data

====

```
redis-config:
```

```
maxmemory 2mb
maxmemory-policy allkeys-lru
```

Check the Redis Pod again using `redis-cli` via `kubectl exec` to see if the configuration was applied:

```
kubectl exec -it redis -- redis-cli
```

Check `maxmemory`:

```
127.0.0.1:6379> CONFIG GET maxmemory
```

It remains at the default value of 0:

- 1) "maxmemory"
- 2) "0"

Similarly, `maxmemory-policy` remains at the `noeviction` default setting:

```
127.0.0.1:6379> CONFIG GET maxmemory-policy
```

Returns:

- 1) "maxmemory-policy"
- 2) "noeviction"

The configuration values have not changed because the Pod needs to be restarted to grab updated values from associated ConfigMaps. Let's delete and recreate the Pod:

```
kubectrl delete pod redis
kubectrl apply -f https://raw.githubusercontent.com/kubernetes/website/main/content/en/examples/pods/config/redis-pod.yaml
```

Now re-check the configuration values one last time:

```
kubectrl exec -it redis -- redis-cli
```

Check maxmemory:

```
127.0.0.1:6379> CONFIG GET maxmemory
```

It should now return the updated value of 2097152:

```
1) "maxmemory"
2) "2097152"
```

Similarly, maxmemory-policy has also been updated:

```
127.0.0.1:6379> CONFIG GET maxmemory-policy
```

It now reflects the desired value of allkeys-lru:

```
1) "maxmemory-policy"
2) "allkeys-lru"
```

Clean up your work by deleting the created resources:

```
kubectrl delete pod/redis configmap/example-redis-config
```

What's next

- Learn more about [ConfigMaps](#).
- Follow an example of [Updating configuration via a ConfigMap](#).

Security

Security is an important concern for most organizations and people who run Kubernetes clusters. You can find a basic [security checklist](#) elsewhere in the Kubernetes documentation.

To learn how to deploy and manage security aspects of Kubernetes, you can follow the tutorials in this section.

[Apply Pod Security Standards at the Cluster Level](#)

[Apply Pod Security Standards at the Namespace Level](#)

[Restrict a Container's Access to Resources with AppArmor](#)

[Restrict a Container's Syscalls with seccomp](#)

Apply Pod Security Standards at the Cluster Level

Note

This tutorial applies only for new clusters.

Pod Security is an admission controller that carries out checks against the Kubernetes [Pod Security Standards](#) when new pods are created. It is a feature GA'ed in v1.25. This tutorial shows you how to enforce the baseline Pod Security Standard at the cluster level which applies a standard configuration to all namespaces in a cluster.

To apply Pod Security Standards to specific namespaces, refer to [Apply Pod Security Standards at the namespace level](#).

If you are running a version of Kubernetes other than v1.30, check the documentation for that version.

Before you begin

Install the following on your workstation:

- [kind](#)
- [kubectrl](#)

This tutorial demonstrates what you can configure for a Kubernetes cluster that you fully control. If you are learning how to configure Pod Security Admission for a managed cluster where you are not able to configure the control plane, read [Apply Pod Security Standards at the namespace level](#).

Choose the right Pod Security Standard to apply

[Pod Security Admission](#) lets you apply built-in [Pod Security Standards](#) with the following modes: enforce, audit, and warn.

To gather information that helps you to choose the Pod Security Standards that are most appropriate for your configuration, do the following:

1. Create a cluster with no Pod Security Standards applied:

```
kind create cluster --name psa-wo-cluster-pss
```

The output is similar to:

```
Creating cluster "psa-wo-cluster-pss" ...
Ensuring node image (kindest/node:v1.30.0)
Preparing nodes
Writing configuration
Starting control-plane
Installing CNI
```

```
Installing StorageClass
Set kubectl context to "kind-psa-wo-cluster-pss"
You can now use your cluster with:
```

```
kubectl cluster-info --context kind-psa-wo-cluster-pss
```

```
Thanks for using kind!
```

2. Set the kubectl context to the new cluster:

```
kubectl cluster-info --context kind-psa-wo-cluster-pss
```

The output is similar to this:

```
Kubernetes control plane is running at https://127.0.0.1:61350
```

```
CoreDNS is running at https://127.0.0.1:61350/api/v1/namespaces/kube-system/services/
kube-dns:dns/proxy
```

```
To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

3. Get a list of namespaces in the cluster:

```
kubectl get ns
```

The output is similar to this:

| NAME | STATUS | AGE |
|--------------------|--------|-------|
| default | Active | 9m30s |
| kube-node-lease | Active | 9m32s |
| kube-public | Active | 9m32s |
| kube-system | Active | 9m32s |
| local-path-storage | Active | 9m26s |

4. Use --dry-run=server to understand what happens when different Pod Security Standards are applied:

1. Privileged

```
kubectl label --dry-run=server --overwrite ns --all \
pod-security.kubernetes.io/enforce=privileged
```

The output is similar to:

```
namespace/default labeled
namespace/kube-node-lease labeled
namespace/kube-public labeled
namespace/kube-system labeled
namespace/local-path-storage labeled
```

2. Baseline

```
kubectl label --dry-run=server --overwrite ns --all \
pod-security.kubernetes.io/enforce=baseline
```

The output is similar to:

```
namespace/default labeled
namespace/kube-node-lease labeled
namespace/kube-public labeled
Warning: existing pods in namespace "kube-system" violate the new PodSecurity
enforce level "baseline:latest"
Warning: etcd-psa-wo-cluster-pss-control-plane (and 3 other pods): host
namespaces, hostPath volumes
Warning: kindnet-vzj42: non-default capabilities, host namespaces, hostPath
volumes
Warning: kube-proxy-m6hwf: host namespaces, hostPath volumes, privileged
namespace/kube-system labeled
namespace/local-path-storage labeled
```

3. Restricted

```
kubectl label --dry-run=server --overwrite ns --all \
pod-security.kubernetes.io/enforce=restricted
```

The output is similar to:

```
namespace/default labeled
namespace/kube-node-lease labeled
namespace/kube-public labeled
Warning: existing pods in namespace "kube-system" violate the new PodSecurity
enforce level "restricted:latest"
Warning: coredns-7bb9c7b568-hsptc (and 1 other pod): unrestricted capabilities,
runAsNonRoot != true, seccompProfile
Warning: etcd-psa-wo-cluster-pss-control-plane (and 3 other pods): host
namespaces, hostPath volumes, allowPrivilegeEscalation != false, unrestricted
capabilities, restricted volume types, runAsNonRoot != true
Warning: kindnet-vzj42: non-default capabilities, host namespaces, hostPath
volumes, allowPrivilegeEscalation != false, unrestricted capabilities, restricted
volume types, runAsNonRoot != true, seccompProfile
Warning: kube-proxy-m6hwf: host namespaces, hostPath volumes, privileged,
allowPrivilegeEscalation != false, unrestricted capabilities, restricted volume types,
runAsNonRoot != true, seccompProfile
namespace/kube-system labeled
Warning: existing pods in namespace "local-path-storage" violate the new
PodSecurity enforce level "restricted:latest"
Warning: local-path-provisioner-d6d9f7ffc-lw9lh: allowPrivilegeEscalation != false,
unrestricted capabilities, runAsNonRoot != true, seccompProfile
namespace/local-path-storage labeled
```

From the previous output, you'll notice that applying the privileged Pod Security Standard shows no warnings for any namespaces. However, baseline and restricted standards both have warnings, specifically in the kube-system namespace.

Set modes, versions and standards

In this section, you apply the following Pod Security Standards to the latest version:

- baseline standard in enforce mode.
- restricted standard in warn and audit mode.

The baseline Pod Security Standard provides a convenient middle ground that allows keeping the exemption list short and prevents known privilege escalations.

Additionally, to prevent pods from failing in kube-system, you'll exempt the namespace from having Pod Security Standards applied.

When you implement Pod Security Admission in your own environment, consider the following:

1. Based on the risk posture applied to a cluster, a stricter Pod Security Standard like restricted might be a better choice.
2. Exempting the kube-system namespace allows pods to run as privileged in this namespace. For real world use, the Kubernetes project strongly recommends that you apply strict RBAC policies that limit access to kube-system, following the principle of least privilege. To implement the preceding standards, do the following:
3. Create a configuration file that can be consumed by the Pod Security Admission Controller to implement these Pod Security Standards:

```
mkdir -p /tmp/pss
cat <<EOF > /tmp/pss/cluster-level-pss.yaml
apiVersion: apiserver.config.k8s.io/v1
kind: AdmissionConfiguration
plugins:
- name: PodSecurity
  configuration:
    apiVersion: pod-security.admission.config.k8s.io/v1
    kind: PodSecurityConfiguration
    defaults:
      enforce: "baseline"
      enforce-version: "latest"
      audit: "restricted"
      audit-version: "latest"
      warn: "restricted"
      warn-version: "latest"
    exemptions:
      usernames: []
      runtimeClasses: []
      namespaces: [kube-system]
EOF
```

Note:

pod-security.admission.config.k8s.io/v1 configuration requires v1.25+. For v1.23 and v1.24, use [v1beta1](#). For v1.22, use [v1alpha1](#).

Configure the API server to consume this file during cluster creation:

4.

```
cat <<EOF > /tmp/pss/cluster-config.yaml
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
- role: control-plane
  kubeadmConfigPatches:
  - |
    kind: ClusterConfiguration
    apiServer:
      extraArgs:
        admission-control-config-file: /etc/config/cluster-level-pss.yaml
      extraVolumes:
        - name: accf
          hostPath: /etc/config
          mountPath: /etc/config
          readOnly: false
          pathType: "DirectoryOrCreate"
    extraMounts:
      - hostPath: /tmp/pss
        containerPath: /etc/config
        # optional: if set, the mount is read-only.
        # default false
        readOnly: false
        # optional: if set, the mount needs SELinux relabeling.
        # default false
        selinuxRelabel: false
        # optional: set propagation mode (None, HostToContainer or Bidirectional)
        # see https://kubernetes.io/docs/concepts/storage/volumes/#mount-propagation
        # default None
        propagation: None
EOF
```

Note:

If you use Docker Desktop with *kind* on macOS, you can add /tmp as a Shared Directory under the menu item **Preferences > Resources > File Sharing**.

5. Create a cluster that uses Pod Security Admission to apply these Pod Security Standards:

```
kind create cluster --name psa-with-cluster-pss --config /tmp/pss/cluster-config.yaml
```

The output is similar to this:

```
Creating cluster "psa-with-cluster-pss" ...
  Ensuring node image (kindest/node:v1.30.0)
  Preparing nodes
  Writing configuration
  Starting control-plane
  Installing CNI
  Installing StorageClass
Set kubectl context to "kind-psa-with-cluster-pss"
```

You can now use your cluster with:

```
kubectl cluster-info --context kind-psa-with-cluster-pss
```

Have a question, bug, or feature request? Let us know! <https://kind.sigs.k8s.io/#community>

6. Point kubectl to the cluster:

```
kubectl cluster-info --context kind-psa-with-cluster-pss
```

The output is similar to this:

```
Kubernetes control plane is running at https://127.0.0.1:63855
CoreDNS is running at https://127.0.0.1:63855/api/v1/namespaces/kube-system/services/
kube-dns:dns/proxy
```

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

7. Create a Pod in the default namespace:

[security/example-baseline-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
  - image: nginx
    name: nginx
    ports:
    - containerPort: 80
```

```
kubectl apply -f https://k8s.io/examples/security/example-baseline-pod.yaml
```

The pod is started normally, but the output includes a warning:

```
Warning: would violate PodSecurity "restricted:latest": allowPrivilegeEscalation != false
(container "nginx" must set securityContext.allowPrivilegeEscalation=false), unrestricted
capabilities (container "nginx" must set securityContext.capabilities.drop=["ALL"]),
runAsNonRoot != true (pod or container "nginx" must set
securityContext.runAsNonRoot=true), seccompProfile (pod or container "nginx" must set
securityContext.seccompProfile.type to "RuntimeDefault" or "Localhost")
pod/nginx created
```

Clean up

Now delete the clusters which you created above by running the following command:

```
kind delete cluster --name psa-with-cluster-pss
```

```
kind delete cluster --name psa-wo-cluster-pss
```

What's next

- Run a [shell script](#) to perform all the preceding steps at once:
 1. Create a Pod Security Standards based cluster level Configuration
 2. Create a file to let API server consume this configuration
 3. Create a cluster that creates an API server with this configuration
 4. Set kubectl context to this new cluster
 5. Create a minimal pod yaml file
 6. Apply this file to create a Pod in the new cluster
- [Pod Security Admission](#)
- [Pod Security Standards](#)
- [Apply Pod Security Standards at the namespace level](#)

Apply Pod Security Standards at the Namespace Level

Note

This tutorial applies only for new clusters.

Pod Security Admission is an admission controller that applies [Pod Security Standards](#) when pods are created. It is a feature GA'ed in v1.25. In this tutorial, you will enforce the baseline Pod Security Standard, one namespace at a time.

You can also apply Pod Security Standards to multiple namespaces at once at the cluster level. For instructions, refer to [Apply Pod Security Standards at the cluster level](#).

Before you begin

Install the following on your workstation:

- [kind](#)
- [kubectl](#)

Create cluster

1. Create a kind cluster as follows:

```
kind create cluster --name psa-ns-level
```

The output is similar to this:

```
Creating cluster "psa-ns-level" ...
  Ensuring node image (kindest/node:v1.30.0)
  Preparing nodes
  Writing configuration
  Starting control-plane
  Installing CNI
  Installing StorageClass
```

Set kubectl context to "kind-psa-ns-level"

You can now use your cluster with:

```
kubectl cluster-info --context kind-psa-ns-level
```

Not sure what to do next? Check out <https://kind.sigs.k8s.io/docs/user/quick-start/>

2. Set the kubectl context to the new cluster:

```
kubectl cluster-info --context kind-psa-ns-level
```

The output is similar to this:

Kubernetes control plane is running at <https://127.0.0.1:50996>

CoreDNS is running at <https://127.0.0.1:50996/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy>

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

Create a namespace

Create a new namespace called example:

```
kubectl create ns example
```

The output is similar to this:

```
namespace/example created
```

Enable Pod Security Standards checking for that namespace

1. Enable Pod Security Standards on this namespace using labels supported by built-in Pod Security Admission. In this step you will configure a check to warn on Pods that don't meet the latest version of the *baseline* pod security standard.

```
kubectl label --overwrite ns example \
  pod-security.kubernetes.io/warn=baseline \
  pod-security.kubernetes.io/warn-version=latest
```

2. You can configure multiple pod security standard checks on any namespace, using labels. The following command will enforce the baseline Pod Security Standard, but warn and audit for restricted Pod Security Standards as per the latest version (default value)

```
kubectl label --overwrite ns example \
  pod-security.kubernetes.io/enforce=baseline \
  pod-security.kubernetes.io/enforce-version=latest \
  pod-security.kubernetes.io/warn=restricted \
  pod-security.kubernetes.io/warn-version=latest \
  pod-security.kubernetes.io/audit=restricted \
  pod-security.kubernetes.io/audit-version=latest
```

Verify the Pod Security Standard enforcement

1. Create a baseline Pod in the example namespace:

```
kubectl apply -n example -f https://k8s.io/examples/security/example-baseline-pod.yaml
```

The Pod does start OK; the output includes a warning. For example:

```
Warning: would violate PodSecurity "restricted:latest": allowPrivilegeEscalation != false (container "nginx" must set securityContext.allowPrivilegeEscalation=false), unrestricted capabilities (container "nginx" must set securityContext.capabilities.drop=["ALL"]), runAsNonRoot != true (pod or container "nginx" must set securityContext.runAsNonRoot=true), seccompProfile (pod or container "nginx" must set securityContext.seccompProfile.type to "RuntimeDefault" or "Localhost")
pod/nginx created
```

2. Create a baseline Pod in the default namespace:

```
kubectl apply -n default -f https://k8s.io/examples/security/example-baseline-pod.yaml
```

Output is similar to this:

```
pod/nginx created
```

The Pod Security Standards enforcement and warning settings were applied only to the example namespace. You could create the same Pod in the default namespace with no warnings.

Clean up

Now delete the cluster which you created above by running the following command:

```
kind delete cluster --name psa-ns-level
```

What's next

- Run a [shell script](#) to perform all the preceding steps all at once.
 1. Create kind cluster
 2. Create new namespace
 3. Apply baseline Pod Security Standard in enforce mode while applying restricted Pod Security Standard also in warn and audit mode.
 4. Create a new pod with the following pod security standards applied
- [Pod Security Admission](#)
- [Pod Security Standards](#)
- [Apply Pod Security Standards at the cluster level](#)

Restrict a Container's Access to Resources with AppArmor

FEATURE STATE: Kubernetes v1.4 [beta]

This page shows you how to load AppArmor profiles on your nodes and enforce those profiles in Pods. To learn more about how Kubernetes can confine Pods using AppArmor, see [Linux kernel security constraints for Pods and containers](#).

Objectives

- See an example of how to load a profile on a Node
- Learn how to enforce the profile on a Pod
- Learn how to check that the profile is loaded
- See what happens when a profile is violated
- See what happens when a profile cannot be loaded

Before you begin

AppArmor is an optional kernel module and Kubernetes feature, so verify it is supported on your Nodes before proceeding:

1. AppArmor kernel module is enabled -- For the Linux kernel to enforce an AppArmor profile, the AppArmor kernel module must be installed and enabled. Several distributions enable the module by default, such as Ubuntu and SUSE, and many others provide optional support. To check whether the module is enabled, check the `/sys/module/apparmor/parameters/enabled` file:

```
cat /sys/module/apparmor/parameters/enabled
Y
```

The kubelet verifies that AppArmor is enabled on the host before admitting a pod with AppArmor explicitly configured.

2. Container runtime supports AppArmor -- All common Kubernetes-supported container runtimes should support AppArmor, including [containerd](#) and [CRI-O](#). Please refer to the corresponding runtime documentation and verify that the cluster fulfills the requirements to use AppArmor.
3. Profile is loaded -- AppArmor is applied to a Pod by specifying an AppArmor profile that each container should be run with. If any of the specified profiles are not loaded in the kernel, the kubelet will reject the Pod. You can view which profiles are loaded on a node by checking the `/sys/kernel/security/apparmor/profiles` file. For example:

```
ssh gke-test-default-pool-239f5d02-gyn2 "sudo cat /sys/kernel/security/apparmor/profiles | sort"
```

```
apparmor-test-deny-write (enforce)
apparmor-test-audit-write (enforce)
```

```
docker-default (enforce)
k8s-nginx (enforce)
```

For more details on loading profiles on nodes, see [Setting up nodes with profiles](#).

Securing a Pod

Note:

Prior to Kubernetes v1.30, AppArmor was specified through annotations. Use the documentation version selector to view the documentation with this deprecated API.

AppArmor profiles can be specified at the pod level or container level. The container AppArmor profile takes precedence over the pod profile.

```
securityContext:
  appArmorProfile:
    type: <profile_type>
```

Where <profile_type> is one of:

- RuntimeDefault to use the runtime's default profile
- Localhost to use a profile loaded on the host (see below)
- Unconfined to run without AppArmor

See the [API Reference](#) for the full details on the AppArmor profile API.

To verify that the profile was applied, you can check that the container's root process is running with the correct profile by examining its proc attr:

```
kubectl exec <pod_name> -- cat /proc/1/attr/current
```

The output should look something like this:

```
cri-containerd.apparmor.d (enforce)
```

Example

This example assumes you have already set up a cluster with AppArmor support.

First, load the profile you want to use onto your Nodes. This profile blocks all file write operations:

```
#include <tunables/global>

profile k8s-apparmor-example-deny-write flags=(attach_disconnected) {
  #include <abstractions/base>

  file,

  # Deny all file writes.
  deny /** w,
}
```

The profile needs to be loaded onto all nodes, since you don't know where the pod will be scheduled. For this example you can use SSH to install the profiles, but other approaches are discussed in [Setting up nodes with profiles](#).

```
# This example assumes that node names match host names, and are reachable via SSH.
NODES=$(kubectl get nodes -o name)

for NODE in ${NODES[*]}; do ssh $NODE 'sudo apparmor_parser -q <EOF
#include <tunables/global>

profile k8s-apparmor-example-deny-write flags=(attach_disconnected) {
#include <abstractions/base>

file,

# Deny all file writes.
deny /** w,
}
EOF'
done
```

Next, run a simple "Hello AppArmor" Pod with the deny-write profile:

[pods/security/hello-apparmor.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: hello-apparmor
spec:
  securityContext:
    appArmorProfile:
      type: Localhost
      localhostProfile: k8s-apparmor-example-deny-write
  containers:
  - name: hello
    image: busybox:1.28
    command: [ "sh", "-c", "echo 'Hello AppArmor!' && sleep 1h" ]
```

```
kubectl create -f hello-apparmor.yaml
```

You can verify that the container is actually running with that profile by checking `/proc/1/attr/current`:

```
kubectl exec hello-apparmor -- cat /proc/1/attr/current
```

The output should be:

```
k8s-apparmor-example-deny-write (enforce)
```

Finally, you can see what happens if you violate the profile by writing to a file:

```
kubectl exec hello-apparmor -- touch /tmp/test
```



```
touch: /tmp/test: Permission denied
error: error executing remote command: command terminated with non-zero exit code: Error
executing in Docker Container: 1
```

To wrap up, see what happens if you try to specify a profile that hasn't been loaded:

```
kubectl create -f /dev/stdin <<EOF
apiVersion: v1
kind: Pod
metadata:
  name: hello-apparmor-2
spec:
  securityContext:
    appArmorProfile:
      type: Localhost
      localhostProfile: k8s-apparmor-example-allow-write
  containers:
  - name: hello
    image: busybox:1.28
    command: [ "sh", "-c", "echo 'Hello AppArmor!' && sleep 1h" ]
EOF
```

```
pod/hello-apparmor-2 created
```

Although the Pod was created successfully, further examination will show that it is stuck in pending:

```
kubectl describe pod hello-apparmor-2
```

```
Name:      hello-apparmor-2
Namespace: default
Node:      gke-test-default-pool-239f5d02-x1kf/10.128.0.27
Start Time: Tue, 30 Aug 2016 17:58:56 -0700
Labels:    <none>
Annotations: container.apparmor.security.beta.kubernetes.io/hello=localhost/k8s-apparmor-
example-allow-write
Status:    Pending
...
Events:
  Type     Reason      Age           From          Message
  ----     -
  Normal   Scheduled   10s           default-scheduler Successfully assigned default/hello-
apparmor to gke-test-default-pool-239f5d02-x1kf
  Normal   Pulled      8s            kubelet       Successfully pulled image "busybox:1.28" in
370.157088ms (370.172701ms including waiting)
  Normal   Pulling     7s (x2 over 9s) kubelet       Pulling image "busybox:1.28"
  Warning   Failed      7s (x2 over 8s) kubelet       Error: failed to get container spec opts: failed
to generate apparmor spec opts: apparmor profile not found k8s-apparmor-example-allow-write
  Normal   Pulled      7s            kubelet       Successfully pulled image "busybox:1.28" in
90.980331ms (91.005869ms including waiting)
```

An Event provides the error message with the reason, the specific wording is runtime-dependent:

```
Warning Failed 7s (x2 over 8s) kubelet Error: failed to get container spec opts: failed to generate apparmor spec opts: apparmor profile not found
```

Administration

Setting up Nodes with profiles

Kubernetes 1.30 does not provide any built-in mechanisms for loading AppArmor profiles onto Nodes. Profiles can be loaded through custom infrastructure or tools like the [Kubernetes Security Profiles Operator](#).

The scheduler is not aware of which profiles are loaded onto which Node, so the full set of profiles must be loaded onto every Node. An alternative approach is to add a Node label for each profile (or class of profiles) on the Node, and use a [node selector](#) to ensure the Pod is run on a Node with the required profile.

Authoring Profiles

Getting AppArmor profiles specified correctly can be a tricky business. Fortunately there are some tools to help with that:

- aa-genprof and aa-logprof generate profile rules by monitoring an application's activity and logs, and admitting the actions it takes. Further instructions are provided by the [AppArmor documentation](#).
- [bane](#) is an AppArmor profile generator for Docker that uses a simplified profile language.

To debug problems with AppArmor, you can check the system logs to see what, specifically, was denied. AppArmor logs verbose messages to dmesg, and errors can usually be found in the system logs or through journalctl. More information is provided in [AppArmor failures](#).

Specifying AppArmor confinement

Caution:

Prior to Kubernetes v1.30, AppArmor was specified through annotations. Use the documentation version selector to view the documentation with this deprecated API.

AppArmor profile within security context

You can specify the appArmorProfile on either a container's securityContext or on a Pod's securityContext. If the profile is set at the pod level, it will be used as the default profile for all containers in the pod (including init, sidecar, and ephemeral containers). If both a pod & container AppArmor profile are set, the container's profile will be used.

An AppArmor profile has 2 fields:

type (*required*) - indicates which kind of AppArmor profile will be applied. Valid options are:

Localhost

a profile pre-loaded on the node (specified by localhostProfile).

RuntimeDefault

the container runtime's default profile.
Unconfined
no AppArmor enforcement.

localhostProfile - The name of a profile loaded on the node that should be used. The profile must be preconfigured on the node to work. This option must be provided if and only if the type is Localhost.

What's next

Additional resources:

- [Quick guide to the AppArmor profile language](#)
- [AppArmor core policy reference](#)

Restrict a Container's Syscalls with seccomp

FEATURE STATE: Kubernetes v1.19 [stable]

Seccomp stands for secure computing mode and has been a feature of the Linux kernel since version 2.6.12. It can be used to sandbox the privileges of a process, restricting the calls it is able to make from userspace into the kernel. Kubernetes lets you automatically apply seccomp profiles loaded onto a [node](#) to your Pods and containers.

Identifying the privileges required for your workloads can be difficult. In this tutorial, you will go through how to load seccomp profiles into a local Kubernetes cluster, how to apply them to a Pod, and how you can begin to craft profiles that give only the necessary privileges to your container processes.

Objectives

- Learn how to load seccomp profiles on a node
- Learn how to apply a seccomp profile to a container
- Observe auditing of syscalls made by a container process
- Observe behavior when a missing profile is specified
- Observe a violation of a seccomp profile
- Learn how to create fine-grained seccomp profiles
- Learn how to apply a container runtime default seccomp profile

Before you begin

In order to complete all steps in this tutorial, you must install [kind](#) and [kubectl](#).

The commands used in the tutorial assume that you are using [Docker](#) as your container runtime. (The cluster that kind creates may use a different container runtime internally). You could also use [Podman](#) but in that case, you would have to follow specific [instructions](#) in order to complete the tasks successfully.

This tutorial shows some examples that are still beta (since v1.25) and others that use only generally available seccomp functionality. You should make sure that your cluster is [configured correctly](#) for the version you are using.

The tutorial also uses the `curl` tool for downloading examples to your computer. You can adapt the steps to use a different tool if you prefer.

Note:

It is not possible to apply a seccomp profile to a container running with `privileged: true` set in the container's `securityContext`. Privileged containers always run as Unconfined.

Download example seccomp profiles

The contents of these profiles will be explored later on, but for now go ahead and download them into a directory named `profiles/` so that they can be loaded into the cluster.

- [audit.json](#)
- [violation.json](#)
- [fine-grained.json](#)

[pods/security/seccomp/profiles/audit.json](#)

```
{
  "defaultAction": "SCMP_ACT_LOG"
}
```

[pods/security/seccomp/profiles/violation.json](#)

```
{
  "defaultAction": "SCMP_ACT_ERRNO"
}
```

[pods/security/seccomp/profiles/fine-grained.json](#)

```
{
  "defaultAction": "SCMP_ACT_ERRNO",
  "architectures": [
    "SCMP_ARCH_X86_64",
    "SCMP_ARCH_X86",
    "SCMP_ARCH_X32"
  ],
  "syscalls": [
    {
      "names": [
        "accept4",
        "epoll_wait",
        "pselect6",
        "futex",
        "madvise",
        "epoll_ctl",
        "getsockname",
        "setsockopt",

```

```

        "vfork",
        "mmap",
        "read",
        "write",
        "close",
        "arch_prctl",
        "sched_getaffinity",
        "munmap",
        "brk",
        "rt_sigaction",
        "rt_sigprocmask",
        "sigaltstack",
        "gettid",
        "clone",
        "bind",
        "socket",
        "openat",
        "readlinkat",
        "exit_group",
        "epoll_create1",
        "listen",
        "rt_sigreturn",
        "sched_yield",
        "clock_gettime",
        "connect",
        "dup2",
        "epoll_pwait",
        "execve",
        "exit",
        "fcntl",
        "getpid",
        "getuid",
        "ioctl",
        "mprotect",
        "nanosleep",
        "open",
        "poll",
        "recvfrom",
        "sendto",
        "set_tid_address",
        "setitimer",
        "writev",
        "fstatfs",
        "getdents64",
        "pipe2",
        "getrlimit"
    ],
    "action": "SCMP_ACT_ALLOW"
}
]
}

```

Run these commands:

```
mkdir ./profiles
curl -L -o profiles/audit.json https://k8s.io/examples/pods/security/seccomp/profiles/audit.json
curl -L -o profiles/violation.json https://k8s.io/examples/pods/security/seccomp/profiles/
violation.json
curl -L -o profiles/fine-grained.json https://k8s.io/examples/pods/security/seccomp/profiles/
fine-grained.json
ls profiles
```

You should see three profiles listed at the end of the final step:

```
audit.json fine-grained.json violation.json
```

Create a local Kubernetes cluster with kind

For simplicity, [kind](#) can be used to create a single node cluster with the seccomp profiles loaded. Kind runs Kubernetes in Docker, so each node of the cluster is a container. This allows for files to be mounted in the filesystem of each container similar to loading files onto a node.

[pods/security/seccomp/kind.yaml](#)

```
apiVersion: kind.x-k8s.io/v1alpha4
kind: Cluster
nodes:
- role: control-plane
  extraMounts:
  - hostPath: "./profiles"
    containerPath: "/var/lib/kubelet/seccomp/profiles"
```

Download that example kind configuration, and save it to a file named kind.yaml:

```
curl -L -O https://k8s.io/examples/pods/security/seccomp/kind.yaml
```

You can set a specific Kubernetes version by setting the node's container image. See [Nodes](#) within the kind documentation about configuration for more details on this. This tutorial assumes you are using Kubernetes v1.30.

As a beta feature, you can configure Kubernetes to use the profile that the [container runtime](#) prefers by default, rather than falling back to Unconfined. If you want to try that, see [enable the use of RuntimeDefault as the default seccomp profile for all workloads](#) before you continue.

Once you have a kind configuration in place, create the kind cluster with that configuration:

```
kind create cluster --config=kind.yaml
```

After the new Kubernetes cluster is ready, identify the Docker container running as the single node cluster:

```
docker ps
```

You should see output indicating that a container is running with name kind-control-plane. The output is similar to:

| CONTAINER ID | IMAGE | COMMAND | CREATED |
|--------------|-------|---------|---------|
| STATUS | PORTS | NAMES | |

```
6a96207fed4b    kindest/node:v1.18.2  "/usr/local/bin/entr..." 27 seconds ago    Up 24
seconds        127.0.0.1:42223->6443/tcp  kind-control-plane
```

If observing the filesystem of that container, you should see that the profiles/ directory has been successfully loaded into the default seccomp path of the kubelet. Use `docker exec` to run a command in the Pod:

```
# Change 6a96207fed4b to the container ID you saw from "docker ps"
docker exec -it 6a96207fed4b ls /var/lib/kubelet/seccomp/profiles
```

```
audit.json  fine-grained.json  violation.json
```

You have verified that these seccomp profiles are available to the kubelet running within kind.

Create a Pod that uses the container runtime default seccomp profile

Most container runtimes provide a sane set of default syscalls that are allowed or not. You can adopt these defaults for your workload by setting the seccomp type in the security context of a pod or container to `RuntimeDefault`.

Note:

If you have the seccompDefault [configuration](#) enabled, then Pods use the `RuntimeDefault` seccomp profile whenever no other seccomp profile is specified. Otherwise, the default is `Unconfined`.

Here's a manifest for a Pod that requests the `RuntimeDefault` seccomp profile for all its containers:

[pods/security/seccomp/ga/default-pod.yaml](#)

```
apiVersion: v1
kind: Pod
metadata:
  name: default-pod
  labels:
    app: default-pod
spec:
  securityContext:
    seccompProfile:
      type: RuntimeDefault
  containers:
    - name: test-container
      image: hashicorp/http-echo:1.0
      args:
        - "-text=just made some more syscalls!"
      securityContext:
        allowPrivilegeEscalation: false
```

Create that Pod:

```
kubectl apply -f https://k8s.io/examples/pods/security/seccomp/ga/default-pod.yaml
```

```
kubectl get pod default-pod
```

The Pod should be showing as having started successfully:

| NAME | READY | STATUS | RESTARTS | AGE |
|-------------|-------|---------|----------|-----|
| default-pod | 1/1 | Running | 0 | 20s |

Delete the Pod before moving to the next section:

```
kubectl delete pod default-pod --wait --now
```

Create a Pod with a seccomp profile for syscall auditing

To start off, apply the `audit.json` profile, which will log all syscalls of the process, to a new Pod.

Here's a manifest for that Pod:

[pods/security/seccomp/ga/audit-pod.yaml](https://k8s.io/examples/pods/security/seccomp/ga/audit-pod.yaml)

```
apiVersion: v1
kind: Pod
metadata:
  name: audit-pod
  labels:
    app: audit-pod
spec:
  securityContext:
    seccompProfile:
      type: Localhost
      localhostProfile: profiles/audit.json
  containers:
  - name: test-container
    image: hashicorp/http-echo:1.0
    args:
    - "-text=just made some syscalls!"
    securityContext:
      allowPrivilegeEscalation: false
```

Note:

Older versions of Kubernetes allowed you to configure seccomp behavior using [annotations](#). Kubernetes 1.30 only supports using fields within `.spec.securityContext` to configure seccomp, and this tutorial explains that approach.

Create the Pod in the cluster:

```
kubectl apply -f https://k8s.io/examples/pods/security/seccomp/ga/audit-pod.yaml
```

This profile does not restrict any syscalls, so the Pod should start successfully.

```
kubectl get pod audit-pod
```


| NAME | READY | STATUS | RESTARTS | AGE |
|-----------|-------|---------|----------|-----|
| audit-pod | 1/1 | Running | 0 | 30s |

In order to be able to interact with this endpoint exposed by this container, create a [NodePort Service](#) that allows access to the endpoint from inside the kind control plane container.

```
kubectrl expose pod audit-pod --type NodePort --port 5678
```

Check what port the Service has been assigned on the node.

```
kubectrl get service audit-pod
```

The output is similar to:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|-----------|----------|---------------|-------------|----------------|-----|
| audit-pod | NodePort | 10.111.36.142 | <none> | 5678:32373/TCP | 72s |

Now you can use curl to access that endpoint from inside the kind control plane container, at the port exposed by this Service. Use docker exec to run the curl command within the container belonging to that control plane container:

```
# Change 6a96207fed4b to the control plane container ID and 32373 to the port number you saw from "docker ps"
docker exec -it 6a96207fed4b curl localhost:32373
```

just made some syscalls!

You can see that the process is running, but what syscalls did it actually make? Because this Pod is running in a local cluster, you should be able to see those in /var/log/syslog on your local system. Open up a new terminal window and tail the output for calls from http-echo:

```
# The log path on your computer might be different from "/var/log/syslog"
tail -f /var/log/syslog | grep 'http-echo'
```

You should already see some logs of syscalls made by http-echo, and if you run curl again inside the control plane container you will see more output written to the log.

For example:

```
Jul 6 15:37:40 my-machine kernel: [369128.669452] audit: type=1326
audit(1594067860.484:14536): auid=4294967295 uid=0 gid=0 ses=4294967295 pid=29064
comm="http-echo" exe="/http-echo" sig=0 arch=c000003e syscall=51 compat=0 ip=0x46fe1f
code=0x7ffc0000
Jul 6 15:37:40 my-machine kernel: [369128.669453] audit: type=1326
audit(1594067860.484:14537): auid=4294967295 uid=0 gid=0 ses=4294967295 pid=29064
comm="http-echo" exe="/http-echo" sig=0 arch=c000003e syscall=54 compat=0 ip=0x46fdbb
code=0x7ffc0000
Jul 6 15:37:40 my-machine kernel: [369128.669455] audit: type=1326
audit(1594067860.484:14538): auid=4294967295 uid=0 gid=0 ses=4294967295 pid=29064
comm="http-echo" exe="/http-echo" sig=0 arch=c000003e syscall=202 compat=0 ip=0x455e53
code=0x7ffc0000
Jul 6 15:37:40 my-machine kernel: [369128.669456] audit: type=1326
audit(1594067860.484:14539): auid=4294967295 uid=0 gid=0 ses=4294967295 pid=29064
comm="http-echo" exe="/http-echo" sig=0 arch=c000003e syscall=288 compat=0 ip=0x46fdbb
code=0x7ffc0000
```

```

Jul 6 15:37:40 my-machine kernel: [369128.669517] audit: type=1326
audit(1594067860.484:14540): auid=4294967295 uid=0 gid=0 ses=4294967295 pid=29064
comm="http-echo" exe="/http-echo" sig=0 arch=c000003e syscall=0 compat=0 ip=0x46fd44
code=0x7ffc0000
Jul 6 15:37:40 my-machine kernel: [369128.669519] audit: type=1326
audit(1594067860.484:14541): auid=4294967295 uid=0 gid=0 ses=4294967295 pid=29064
comm="http-echo" exe="/http-echo" sig=0 arch=c000003e syscall=270 compat=0 ip=0x4559b1
code=0x7ffc0000
Jul 6 15:38:40 my-machine kernel: [369188.671648] audit: type=1326
audit(1594067920.488:14559): auid=4294967295 uid=0 gid=0 ses=4294967295 pid=29064
comm="http-echo" exe="/http-echo" sig=0 arch=c000003e syscall=270 compat=0 ip=0x4559b1
code=0x7ffc0000
Jul 6 15:38:40 my-machine kernel: [369188.671726] audit: type=1326
audit(1594067920.488:14560): auid=4294967295 uid=0 gid=0 ses=4294967295 pid=29064
comm="http-echo" exe="/http-echo" sig=0 arch=c000003e syscall=202 compat=0 ip=0x455e53
code=0x7ffc0000

```

You can begin to understand the syscalls required by the http-echo process by looking at the syscall= entry on each line. While these are unlikely to encompass all syscalls it uses, it can serve as a basis for a seccomp profile for this container.

Delete the Service and the Pod before moving to the next section:

```

kubectl delete service audit-pod --wait
kubectl delete pod audit-pod --wait --now

```

Create a Pod with a seccomp profile that causes violation

For demonstration, apply a profile to the Pod that does not allow for any syscalls.

The manifest for this demonstration is:

<pods/security/seccomp/ga/violation-pod.yaml>

```

apiVersion: v1
kind: Pod
metadata:
  name: violation-pod
  labels:
    app: violation-pod
spec:
  securityContext:
    seccompProfile:
      type: Localhost
      localhostProfile: profiles/violation.json
  containers:
    - name: test-container
      image: hashicorp/http-echo:1.0
      args:
        - "-text=just made some syscalls!"
      securityContext:
        allowPrivilegeEscalation: false

```

Attempt to create the Pod in the cluster:

```
kubectl apply -f https://k8s.io/examples/pods/security/seccomp/ga/violation-pod.yaml
```

The Pod creates, but there is an issue. If you check the status of the Pod, you should see that it failed to start.

```
kubectl get pod violation-pod
```

| NAME | READY | STATUS | RESTARTS | AGE |
|---------------|-------|------------------|----------|-----|
| violation-pod | 0/1 | CrashLoopBackOff | 1 | 6s |

As seen in the previous example, the http-echo process requires quite a few syscalls. Here seccomp has been instructed to error on any syscall by setting "defaultAction": "SCMP_ACT_ERRNO". This is extremely secure, but removes the ability to do anything meaningful. What you really want is to give workloads only the privileges they need.

Delete the Pod before moving to the next section:

```
kubectl delete pod violation-pod --wait --now
```

Create a Pod with a seccomp profile that only allows necessary syscalls

If you take a look at the fine-grained.json profile, you will notice some of the syscalls seen in syslog of the first example where the profile set "defaultAction": "SCMP_ACT_LOG". Now the profile is setting "defaultAction": "SCMP_ACT_ERRNO", but explicitly allowing a set of syscalls in the "action": "SCMP_ACT_ALLOW" block. Ideally, the container will run successfully and you will see no messages sent to syslog.

The manifest for this example is:

[pods/security/seccomp/ga/fine-pod.yaml](https://k8s.io/examples/pods/security/seccomp/ga/fine-pod.yaml)

```
apiVersion: v1
kind: Pod
metadata:
  name: fine-pod
  labels:
    app: fine-pod
spec:
  securityContext:
    seccompProfile:
      type: Localhost
      localhostProfile: profiles/fine-grained.json
  containers:
  - name: test-container
    image: hashicorp/http-echo:1.0
    args:
    - "-text=just made some syscalls!"
    securityContext:
      allowPrivilegeEscalation: false
```

Create the Pod in your cluster:

```
kubectl apply -f https://k8s.io/examples/pods/security/seccomp/ga/fine-pod.yaml
```

```
kubectl get pod fine-pod
```

The Pod should be showing as having started successfully:

| NAME | READY | STATUS | RESTARTS | AGE |
|----------|-------|---------|----------|-----|
| fine-pod | 1/1 | Running | 0 | 30s |

Open up a new terminal window and use tail to monitor for log entries that mention calls from http-echo:

```
# The log path on your computer might be different from "/var/log/syslog"  
tail -f /var/log/syslog | grep 'http-echo'
```

Next, expose the Pod with a NodePort Service:

```
kubectl expose pod fine-pod --type NodePort --port 5678
```

Check what port the Service has been assigned on the node:

```
kubectl get service fine-pod
```

The output is similar to:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|----------|----------|---------------|-------------|----------------|-----|
| fine-pod | NodePort | 10.111.36.142 | <none> | 5678:32373/TCP | 72s |

Use curl to access that endpoint from inside the kind control plane container:

```
# Change 6a96207fed4b to the control plane container ID and 32373 to the port number you saw  
from "docker ps"  
docker exec -it 6a96207fed4b curl localhost:32373
```

```
just made some syscalls!
```

You should see no output in the syslog. This is because the profile allowed all necessary syscalls and specified that an error should occur if one outside of the list is invoked. This is an ideal situation from a security perspective, but required some effort in analyzing the program. It would be nice if there was a simple way to get closer to this security without requiring as much effort.

Delete the Service and the Pod before moving to the next section:

```
kubectl delete service fine-pod --wait  
kubectl delete pod fine-pod --wait --now
```

Enable the use of RuntimeDefault as the default seccomp profile for all workloads

FEATURE STATE: Kubernetes v1.27 [stable]

To use seccomp profile defaulting, you must run the kubelet with the `--seccomp-default` [command line flag](#) enabled for each node where you want to use it.

If enabled, the kubelet will use the RuntimeDefault seccomp profile by default, which is defined by the container runtime, instead of using the Unconfined (seccomp disabled) mode. The default profiles aim to provide a strong set of security defaults while preserving the functionality of the workload. It is possible that the default profiles differ between container runtimes and their release versions, for example when comparing those from CRI-O and containerd.

Note:

Enabling the feature will neither change the Kubernetes `securityContext.seccompProfile` API field nor add the deprecated annotations of the workload. This provides users the possibility to rollback anytime without actually changing the workload configuration. Tools like [crictrl inspect](#) can be used to verify which seccomp profile is being used by a container.

Some workloads may require a lower amount of syscall restrictions than others. This means that they can fail during runtime even with the RuntimeDefault profile. To mitigate such a failure, you can:

- Run the workload explicitly as Unconfined.
- Disable the SeccompDefault feature for the nodes. Also making sure that workloads get scheduled on nodes where the feature is disabled.
- Create a custom seccomp profile for the workload.

If you were introducing this feature into production-like cluster, the Kubernetes project recommends that you enable this feature gate on a subset of your nodes and then test workload execution before rolling the change out cluster-wide.

You can find more detailed information about a possible upgrade and downgrade strategy in the related Kubernetes Enhancement Proposal (KEP): [Enable seccomp by default](#).

Kubernetes 1.30 lets you configure the seccomp profile that applies when the spec for a Pod doesn't define a specific seccomp profile. However, you still need to enable this defaulting for each node where you would like to use it.

If you are running a Kubernetes 1.30 cluster and want to enable the feature, either run the kubelet with the `--seccomp-default` command line flag, or enable it through the [kubelet configuration file](#). To enable the feature gate in [kind](#), ensure that kind provides the minimum required Kubernetes version and enables the SeccompDefault feature [in the kind configuration](#):

```
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
- role: control-plane
  image: kindest/
node:v1.28.0@sha256:9f3ff58f19dcf1a0611d11e8ac989fdb30a28f40f236f59f0bea31fb956ccf5c
  kubeadmConfigPatches:
  - |
    kind: JoinConfiguration
    nodeRegistration:
      kubeletExtraArgs:
        seccomp-default: "true"
```

```
- role: worker
  image: kindest/
node:v1.28.0@sha256:9f3ff58f19dcf1a0611d11e8ac989fdb30a28f40f236f59f0bea31fb956ccf5c
  kubeadmConfigPatches:
    - |
      kind: JoinConfiguration
      nodeRegistration:
        kubeletExtraArgs:
          seccomp-default: "true"
```

If the cluster is ready, then running a pod:

```
kubectrl run --rm -it --restart=Never --image=alpine alpine -- sh
```

Should now have the default seccomp profile attached. This can be verified by using docker exec to run crictl inspect for the container on the kind worker:

```
docker exec -it kind-worker bash -c \
'crictl inspect $(crictl ps --name=alpine -q) | jq .info.runtimeSpec.linux.seccomp'
```

```
{
  "defaultAction": "SCMP_ACT_ERRNO",
  "architectures": ["SCMP_ARCH_X86_64", "SCMP_ARCH_X86", "SCMP_ARCH_X32"],
  "syscalls": [
    {
      "names": ["..."]
    }
  ]
}
```

What's next

You can learn more about Linux seccomp:

- [A seccomp Overview](#)
- [Seccomp Security Profiles for Docker](#)

Stateless Applications

[Exposing an External IP Address to Access an Application in a Cluster](#)

[Example: Deploying PHP Guestbook application with Redis](#)

Exposing an External IP Address to Access an Application in a Cluster

This page shows how to create a Kubernetes Service object that exposes an external IP address.

Before you begin

- Install [kubectl](#).
- Use a cloud provider like Google Kubernetes Engine or Amazon Web Services to create a Kubernetes cluster. This tutorial creates an [external load balancer](#), which requires a cloud provider.
- Configure kubectl to communicate with your Kubernetes API server. For instructions, see the documentation for your cloud provider.

Objectives

- Run five instances of a Hello World application.
- Create a Service object that exposes an external IP address.
- Use the Service object to access the running application.

Creating a service for an application running in five pods

1. Run a Hello World application in your cluster:

[service/load-balancer-example.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app.kubernetes.io/name: load-balancer-example
  name: hello-world
spec:
  replicas: 5
  selector:
    matchLabels:
      app.kubernetes.io/name: load-balancer-example
  template:
    metadata:
      labels:
        app.kubernetes.io/name: load-balancer-example
    spec:
      containers:
        - image: gcr.io/google-samples/node-hello:1.0
          name: hello-world
          ports:
            - containerPort: 8080
```

```
kubectl apply -f https://k8s.io/examples/service/load-balancer-example.yaml
```

The preceding command creates a [Deployment](#) and an associated [ReplicaSet](#). The ReplicaSet has five [Pods](#) each of which runs the Hello World application.

2. Display information about the Deployment:

```
kubectl get deployments hello-world
kubectl describe deployments hello-world
```

Display information about your ReplicaSet objects:

3.

```
kubectl get replicaset  
kubectl describe replicaset
```

4. Create a Service object that exposes the deployment:

```
kubectl expose deployment hello-world --type=LoadBalancer --name=my-service
```

5. Display information about the Service:

```
kubectl get services my-service
```

The output is similar to:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|------------|--------------|--------------|----------------|----------|-----|
| my-service | LoadBalancer | 10.3.245.137 | 104.198.205.71 | 8080/TCP | 54s |

Note:

The type=LoadBalancer service is backed by external cloud providers, which is not covered in this example, please refer to [this page](#) for the details.

Note:

If the external IP address is shown as <pending>, wait for a minute and enter the same command again.

6. Display detailed information about the Service:

```
kubectl describe services my-service
```

The output is similar to:

```
Name:      my-service  
Namespace: default  
Labels:    app.kubernetes.io/name=load-balancer-example  
Annotations: <none>  
Selector:  app.kubernetes.io/name=load-balancer-example  
Type:      LoadBalancer  
IP:        10.3.245.137  
LoadBalancer Ingress: 104.198.205.71  
Port:      <unset> 8080/TCP  
NodePort:  <unset> 32377/TCP  
Endpoints: 10.0.0.6:8080,10.0.1.6:8080,10.0.1.7:8080 + 2 more...  
Session Affinity: None  
Events:    <none>
```

Make a note of the external IP address (LoadBalancer Ingress) exposed by your service. In this example, the external IP address is 104.198.205.71. Also note the value of Port and NodePort. In this example, the Port is 8080 and the NodePort is 32377.

7. In the preceding output, you can see that the service has several endpoints:

10.0.0.6:8080,10.0.1.6:8080,10.0.1.7:8080 + 2 more. These are internal addresses of the pods

that are running the Hello World application. To verify these are pod addresses, enter this command:

```
kubectl get pods --output=wide
```

The output is similar to:

| NAME | ... | IP | NODE |
|------------------------------|-----|----------|--|
| hello-world-2895499144-1jaz9 | ... | 10.0.1.6 | gke-cluster-1-default-pool-e0b8d269-1afc |
| hello-world-2895499144-2e5uh | ... | 10.0.1.8 | gke-cluster-1-default-pool-e0b8d269-1afc |
| hello-world-2895499144-9m4h1 | ... | 10.0.0.6 | gke-cluster-1-default-pool-e0b8d269-5v7a |
| hello-world-2895499144-o4z13 | ... | 10.0.1.7 | gke-cluster-1-default-pool-e0b8d269-1afc |
| hello-world-2895499144-segjf | ... | 10.0.2.5 | gke-cluster-1-default-pool-e0b8d269-cpuc |

8. Use the external IP address (LoadBalancer Ingress) to access the Hello World application:

```
curl http://<external-ip>:<port>
```

where <external-ip> is the external IP address (LoadBalancer Ingress) of your Service, and <port> is the value of Port in your Service description. If you are using minikube, typing minikube service my-service will automatically open the Hello World application in a browser.

The response to a successful request is a hello message:

```
Hello Kubernetes!
```

Cleaning up

To delete the Service, enter this command:

```
kubectl delete services my-service
```

To delete the Deployment, the ReplicaSet, and the Pods that are running the Hello World application, enter this command:

```
kubectl delete deployment hello-world
```

What's next

Learn more about [connecting applications with services](#).

Example: Deploying PHP Guestbook application with Redis

This tutorial shows you how to build and deploy a simple (*not production ready*), multi-tier web application using Kubernetes and [Docker](#). This example consists of the following components:

- A single-instance [Redis](#) to store guestbook entries
- Multiple web frontend instances

Objectives

- Start up a Redis leader.
- Start up two Redis followers.
- Start up the guestbook frontend.
- Expose and view the Frontend Service.
- Clean up.

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

Your Kubernetes server must be at or later than version v1.14. To check the version, enter kubectl version.

Start up the Redis Database

The guestbook application uses Redis to store its data.

Creating the Redis Deployment

The manifest file, included below, specifies a Deployment controller that runs a single replica Redis Pod.

[application/guestbook/redis-leader-deployment.yaml](#)

```
# SOURCE: https://cloud.google.com/kubernetes-engine/docs/tutorials/guestbook
apiVersion: apps/v1
kind: Deployment
metadata:
  name: redis-leader
  labels:
    app: redis
    role: leader
    tier: backend
spec:
  replicas: 1
  selector:
    matchLabels:
      app: redis
  template:
    metadata:
      labels:
        app: redis
        role: leader
```

```
    tier: backend
spec:
  containers:
  - name: leader
    image: "docker.io/redis:6.0.5"
    resources:
      requests:
        cpu: 100m
        memory: 100Mi
    ports:
    - containerPort: 6379
```

1. Launch a terminal window in the directory you downloaded the manifest files.
2. Apply the Redis Deployment from the redis-leader-deployment.yaml file:

```
kubectl apply -f https://k8s.io/examples/application/guestbook/redis-leader-
deployment.yaml
```

3. Query the list of Pods to verify that the Redis Pod is running:

```
kubectl get pods
```

The response should be similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|------------------------------|-------|---------|----------|-----|
| redis-leader-fb76b4755-xjr2n | 1/1 | Running | 0 | 13s |

4. Run the following command to view the logs from the Redis leader Pod:

```
kubectl logs -f deployment/redis-leader
```

Creating the Redis leader Service

The guestbook application needs to communicate to the Redis to write its data. You need to apply a [Service](#) to proxy the traffic to the Redis Pod. A Service defines a policy to access the Pods.

[application/guestbook/redis-leader-service.yaml](#)

```
# SOURCE: https://cloud.google.com/kubernetes-engine/docs/tutorials/guestbook
apiVersion: v1
kind: Service
metadata:
  name: redis-leader
  labels:
    app: redis
    role: leader
    tier: backend
spec:
  ports:
  - port: 6379
    targetPort: 6379
  selector:
```

```
app: redis
role: leader
tier: backend
```

1. Apply the Redis Service from the following `redis-leader-service.yaml` file:

```
kubectl apply -f https://k8s.io/examples/application/guestbook/redis-leader-service.yaml
```

2. Query the list of Services to verify that the Redis Service is running:

```
kubectl get service
```

The response should be similar to this:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|--------------|-----------|--------------|-------------|----------|-----|
| kubernetes | ClusterIP | 10.0.0.1 | <none> | 443/TCP | 1m |
| redis-leader | ClusterIP | 10.103.78.24 | <none> | 6379/TCP | 16s |

Note:

This manifest file creates a Service named `redis-leader` with a set of labels that match the labels previously defined, so the Service routes network traffic to the Redis Pod.

Set up Redis followers

Although the Redis leader is a single Pod, you can make it highly available and meet traffic demands by adding a few Redis followers, or replicas.

[application/guestbook/redis-follower-deployment.yaml](https://k8s.io/examples/application/guestbook/redis-follower-deployment.yaml)

```
# SOURCE: https://cloud.google.com/kubernetes-engine/docs/tutorials/guestbook
apiVersion: apps/v1
kind: Deployment
metadata:
  name: redis-follower
  labels:
    app: redis
    role: follower
    tier: backend
spec:
  replicas: 2
  selector:
    matchLabels:
      app: redis
  template:
    metadata:
      labels:
        app: redis
        role: follower
        tier: backend
    spec:
      containers:
        - name: follower
```

```
image: us-docker.pkg.dev/google-samples/containers/gke/gb-redis-follower:v2
resources:
  requests:
    cpu: 100m
    memory: 100Mi
ports:
- containerPort: 6379
```

1. Apply the Redis Deployment from the following redis-follower-deployment.yaml file:

```
kubectl apply -f https://k8s.io/examples/application/guestbook/redis-follower-deployment.yaml
```

2. Verify that the two Redis follower replicas are running by querying the list of Pods:

```
kubectl get pods
```

The response should be similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|--------------------------------|-------|---------|----------|-----|
| redis-follower-dddfbdcc9-82sfr | 1/1 | Running | 0 | 37s |
| redis-follower-dddfbdcc9-qrt5k | 1/1 | Running | 0 | 38s |
| redis-leader-fb76b4755-xjr2n | 1/1 | Running | 0 | 11m |

Creating the Redis follower service

The guestbook application needs to communicate with the Redis followers to read data. To make the Redis followers discoverable, you must set up another [Service](#).

[application/guestbook/redis-follower-service.yaml](https://k8s.io/examples/application/guestbook/redis-follower-service.yaml)

```
# SOURCE: https://cloud.google.com/kubernetes-engine/docs/tutorials/guestbook
apiVersion: v1
kind: Service
metadata:
  name: redis-follower
labels:
  app: redis
  role: follower
  tier: backend
spec:
  ports:
    # the port that this service should serve on
    - port: 6379
  selector:
    app: redis
    role: follower
    tier: backend
```

1. Apply the Redis Service from the following redis-follower-service.yaml file:

```
kubectl apply -f https://k8s.io/examples/application/guestbook/redis-follower-service.yaml
```

- Query the list of Services to verify that the Redis Service is running:
2. `kubectl get service`

The response should be similar to this:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|----------------|-----------|---------------|-------------|----------|-------|
| kubernetes | ClusterIP | 10.96.0.1 | <none> | 443/TCP | 3d19h |
| redis-follower | ClusterIP | 10.110.162.42 | <none> | 6379/TCP | 9s |
| redis-leader | ClusterIP | 10.103.78.24 | <none> | 6379/TCP | 6m10s |

Note:

This manifest file creates a Service named `redis-follower` with a set of labels that match the labels previously defined, so the Service routes network traffic to the Redis Pod.

Set up and Expose the Guestbook Frontend

Now that you have the Redis storage of your guestbook up and running, start the guestbook web servers. Like the Redis followers, the frontend is deployed using a Kubernetes Deployment.

The guestbook app uses a PHP frontend. It is configured to communicate with either the Redis follower or leader Services, depending on whether the request is a read or a write. The frontend exposes a JSON interface, and serves a jQuery-Ajax-based UX.

Creating the Guestbook Frontend Deployment

[application/guestbook/frontend-deployment.yaml](#)

```
# SOURCE: https://cloud.google.com/kubernetes-engine/docs/tutorials/guestbook
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend
spec:
  replicas: 3
  selector:
    matchLabels:
      app: guestbook
      tier: frontend
  template:
    metadata:
      labels:
        app: guestbook
        tier: frontend
    spec:
      containers:
        - name: php-redis
          image: us-docker.pkg.dev/google-samples/containers/gke/gb-frontend:v5
          env:
            - name: GET_HOSTS_FROM
              value: "dns"
```

```
resources:
  requests:
    cpu: 100m
    memory: 100Mi
ports:
- containerPort: 80
```

1. Apply the frontend Deployment from the frontend-deployment.yaml file:

```
kubectl apply -f https://k8s.io/examples/application/guestbook/frontend-deployment.yaml
```

2. Query the list of Pods to verify that the three frontend replicas are running:

```
kubectl get pods -l app=guestbook -l tier=frontend
```

The response should be similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|---------------------------|-------|---------|----------|-----|
| frontend-85595f5bf9-5tqhb | 1/1 | Running | 0 | 47s |
| frontend-85595f5bf9-qbzwm | 1/1 | Running | 0 | 47s |
| frontend-85595f5bf9-zchwc | 1/1 | Running | 0 | 47s |

Creating the Frontend Service

The Redis Services you applied is only accessible within the Kubernetes cluster because the default type for a Service is [ClusterIP](#). ClusterIP provides a single IP address for the set of Pods the Service is pointing to. This IP address is accessible only within the cluster.

If you want guests to be able to access your guestbook, you must configure the frontend Service to be externally visible, so a client can request the Service from outside the Kubernetes cluster. However a Kubernetes user can use `kubectl port-forward` to access the service even though it uses a ClusterIP.

Note:

Some cloud providers, like Google Compute Engine or Google Kubernetes Engine, support external load balancers. If your cloud provider supports load balancers and you want to use it, uncomment type: LoadBalancer.

[application/guestbook/frontend-service.yaml](#)

```
# SOURCE: https://cloud.google.com/kubernetes-engine/docs/tutorials/guestbook
apiVersion: v1
kind: Service
metadata:
  name: frontend
  labels:
    app: guestbook
    tier: frontend
spec:
  # if your cluster supports it, uncomment the following to automatically create
  # an external load-balanced IP for the frontend service.
  # type: LoadBalancer
```

```
#type: LoadBalancer
ports:
  # the port that this service should serve on
  - port: 80
selector:
  app: guestbook
  tier: frontend
```

1. Apply the frontend Service from the frontend-service.yaml file:

```
kubectl apply -f https://k8s.io/examples/application/guestbook/frontend-service.yaml
```

2. Query the list of Services to verify that the frontend Service is running:

```
kubectl get services
```

The response should be similar to this:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|----------------|-----------|---------------|-------------|----------|-------|
| frontend | ClusterIP | 10.97.28.230 | <none> | 80/TCP | 19s |
| kubernetes | ClusterIP | 10.96.0.1 | <none> | 443/TCP | 3d19h |
| redis-follower | ClusterIP | 10.110.162.42 | <none> | 6379/TCP | 5m48s |
| redis-leader | ClusterIP | 10.103.78.24 | <none> | 6379/TCP | 11m |

Viewing the Frontend Service via kubectl port-forward

1. Run the following command to forward port 8080 on your local machine to port 80 on the service.

```
kubectl port-forward svc/frontend 8080:80
```

The response should be similar to this:

```
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
```

2. load the page <http://localhost:8080> in your browser to view your guestbook.

Viewing the Frontend Service via LoadBalancer

If you deployed the frontend-service.yaml manifest with type: LoadBalancer you need to find the IP address to view your Guestbook.

1. Run the following command to get the IP address for the frontend Service.

```
kubectl get service frontend
```

The response should be similar to this:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|----------|--------------|---------------|----------------|--------------|-----|
| frontend | LoadBalancer | 10.51.242.136 | 109.197.92.229 | 80:32372/TCP | 1m |

2. Copy the external IP address, and load the page in your browser to view your guestbook.

Note:

Try adding some guestbook entries by typing in a message, and clicking Submit. The message you typed appears in the frontend. This message indicates that data is successfully added to Redis through the Services you created earlier.

Scale the Web Frontend

You can scale up or down as needed because your servers are defined as a Service that uses a Deployment controller.

1. Run the following command to scale up the number of frontend Pods:

```
kubectl scale deployment frontend --replicas=5
```

2. Query the list of Pods to verify the number of frontend Pods running:

```
kubectl get pods
```

The response should look similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|--------------------------------|-------|---------|----------|------|
| frontend-85595f5bf9-5df5m | 1/1 | Running | 0 | 83s |
| frontend-85595f5bf9-7zmg5 | 1/1 | Running | 0 | 83s |
| frontend-85595f5bf9-cpskg | 1/1 | Running | 0 | 15m |
| frontend-85595f5bf9-l2l54 | 1/1 | Running | 0 | 14m |
| frontend-85595f5bf9-l9c8z | 1/1 | Running | 0 | 14m |
| redis-follower-dddfbdcc9-82sfr | 1/1 | Running | 0 | 97m |
| redis-follower-dddfbdcc9-qrt5k | 1/1 | Running | 0 | 97m |
| redis-leader-fb76b4755-xjr2n | 1/1 | Running | 0 | 108m |

3. Run the following command to scale down the number of frontend Pods:

```
kubectl scale deployment frontend --replicas=2
```

4. Query the list of Pods to verify the number of frontend Pods running:

```
kubectl get pods
```

The response should look similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|--------------------------------|-------|---------|----------|------|
| frontend-85595f5bf9-cpskg | 1/1 | Running | 0 | 16m |
| frontend-85595f5bf9-l9c8z | 1/1 | Running | 0 | 15m |
| redis-follower-dddfbdcc9-82sfr | 1/1 | Running | 0 | 98m |
| redis-follower-dddfbdcc9-qrt5k | 1/1 | Running | 0 | 98m |
| redis-leader-fb76b4755-xjr2n | 1/1 | Running | 0 | 109m |

Cleaning up

Deleting the Deployments and Services also deletes any running Pods. Use labels to delete multiple resources with one command.

1. Run the following commands to delete all Pods, Deployments, and Services.

```
kubectl delete deployment -l app=redis
kubectl delete service -l app=redis
kubectl delete deployment frontend
kubectl delete service frontend
```

The response should look similar to this:

```
deployment.apps "redis-follower" deleted
deployment.apps "redis-leader" deleted
deployment.apps "frontend" deleted
service "frontend" deleted
```

2. Query the list of Pods to verify that no Pods are running:

```
kubectl get pods
```

The response should look similar to this:

```
No resources found in default namespace.
```

What's next

- Complete the [Kubernetes Basics](#) Interactive Tutorials
- Use Kubernetes to create a blog using [Persistent Volumes for MySQL and Wordpress](#)
- Read more about [connecting applications with services](#)
- Read more about [using labels effectively](#)

Stateful Applications

[StatefulSet Basics](#)

[Example: Deploying WordPress and MySQL with Persistent Volumes](#)

[Example: Deploying Cassandra with a StatefulSet](#)

[Running ZooKeeper, A Distributed System Coordinator](#)

StatefulSet Basics

This tutorial provides an introduction to managing applications with [StatefulSets](#). It demonstrates how to create, delete, scale, and update the Pods of StatefulSets.

Before you begin

Before you begin this tutorial, you should familiarize yourself with the following Kubernetes concepts:

- [Pods](#)
- [Cluster DNS](#)
- [Headless Services](#)
- [PersistentVolumes](#)
- [PersistentVolume Provisioning](#)
- The [kubectl](#) command line tool

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

You should configure kubectl to use a context that uses the default namespace. If you are using an existing cluster, make sure that it's OK to use that cluster's default namespace to practice. Ideally, practice in a cluster that doesn't run any real workloads.

It's also useful to read the concept page about [StatefulSets](#).

Note:

This tutorial assumes that your cluster is configured to dynamically provision PersistentVolumes. You'll also need to have a [default StorageClass](#). If your cluster is not configured to provision storage dynamically, you will have to manually provision two 1 GiB volumes prior to starting this tutorial and set up your cluster so that those PersistentVolumes map to the PersistentVolumeClaim templates that the StatefulSet defines.

Objectives

StatefulSets are intended to be used with stateful applications and distributed systems. However, the administration of stateful applications and distributed systems on Kubernetes is a broad, complex topic. In order to demonstrate the basic features of a StatefulSet, and not to conflate the former topic with the latter, you will deploy a simple web application using a StatefulSet.

After this tutorial, you will be familiar with the following.

- How to create a StatefulSet
- How a StatefulSet manages its Pods
- How to delete a StatefulSet
- How to scale a StatefulSet
- How to update a StatefulSet's Pods

Creating a StatefulSet

Begin by creating a StatefulSet (and the Service that it relies upon) using the example below. It is similar to the example presented in the [StatefulSets](#) concept. It creates a [headless Service](#), nginx, to publish the IP addresses of Pods in the StatefulSet, web.

[application/web/web.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: nginx
  labels:
    app: nginx
spec:
  ports:
    - port: 80
      name: web
  clusterIP: None
  selector:
    app: nginx
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: web
spec:
  serviceName: "nginx"
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: registry.k8s.io/nginx-slim:0.8
          ports:
            - containerPort: 80
              name: web
          volumeMounts:
            - name: www
              mountPath: /usr/share/nginx/html
      volumeClaimTemplates:
        - metadata:
            name: www
          spec:
            accessModes: [ "ReadWriteOnce" ]
            resources:
```

```
requests:
  storage: 1Gi
```

You will need to use at least two terminal windows. In the first terminal, use [kubectl get](#) to watch the creation of the StatefulSet's Pods.

```
# use this terminal to run commands that specify --watch
# end this watch when you are asked to start a new watch
kubectl get pods --watch -l app=nginx
```

In the second terminal, use [kubectl apply](#) to create the headless Service and StatefulSet:

```
kubectl apply -f https://k8s.io/examples/application/web/web.yaml
```

```
service/nginx created
statefulset.apps/web created
```

The command above creates two Pods, each running an [NGINX](#) webserver. Get the nginx Service...

```
kubectl get service nginx
```

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|-------|-----------|------------|-------------|---------|-----|
| nginx | ClusterIP | None | <none> | 80/TCP | 12s |

...then get the web StatefulSet, to verify that both were created successfully:

```
kubectl get statefulset web
```

| NAME | DESIRED | CURRENT | AGE |
|------|---------|---------|-----|
| web | 2 | 1 | 20s |

Ordered Pod creation

A StatefulSet defaults to creating its Pods in a strict order.

For a StatefulSet with n replicas, when Pods are being deployed, they are created sequentially, ordered from $\{0..n-1\}$. Examine the output of the `kubectl get` command in the first terminal. Eventually, the output will look like the example below.

```
# Do not start a new watch;
# this should already be running
kubectl get pods --watch -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 0/1 | Pending | 0 | 0s |
| web-0 | 0/1 | Pending | 0 | 0s |
| web-0 | 0/1 | ContainerCreating | 0 | 0s |
| web-0 | 1/1 | Running | 0 | 19s |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | ContainerCreating | 0 | 0s |
| web-1 | 1/1 | Running | 0 | 18s |

Notice that the web-1 Pod is not launched until the web-0 Pod is *Running* (see [Pod Phase](#)) and *Ready* (see type in [Pod Conditions](#)).

Later in this tutorial you will practice [parallel startup](#).

Note:

To configure the integer ordinal assigned to each Pod in a StatefulSet, see [Start ordinal](#).

Pods in a StatefulSet

Pods in a StatefulSet have a unique ordinal index and a stable network identity.

Examining the Pod's ordinal index

Get the StatefulSet's Pods:

```
kubectl get pods -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|---------|----------|-----|
| web-0 | 1/1 | Running | 0 | 1m |
| web-1 | 1/1 | Running | 0 | 1m |

As mentioned in the [StatefulSets](#) concept, the Pods in a StatefulSet have a sticky, unique identity. This identity is based on a unique ordinal index that is assigned to each Pod by the StatefulSet [controller](#).

The Pods' names take the form <statefulset name>--<ordinal index>. Since the web StatefulSet has two replicas, it creates two Pods, web-0 and web-1.

Using stable network identities

Each Pod has a stable hostname based on its ordinal index. Use [kubectl exec](#) to execute the `hostname` command in each Pod:

```
for i in 0 1; do kubectl exec "web-$i" -- sh -c 'hostname'; done
```

```
web-0
web-1
```

Use [kubectl run](#) to execute a container that provides the `nslookup` command from the `dnsutils` package. Using `nslookup` on the Pods' hostnames, you can examine their in-cluster DNS addresses:

```
kubectl run -i --tty --image busybox:1.28 dns-test --restart=Never --rm
```

which starts a new shell. In that new shell, run:

```
# Run this in the dns-test container shell
nslookup web-0.nginx
```

The output is similar to:

```
Server: 10.0.0.10
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local

Name: web-0.nginx
Address 1: 10.244.1.6

nslookup web-1.nginx
Server: 10.0.0.10
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local

Name: web-1.nginx
Address 1: 10.244.2.6
```

(and now exit the container shell: exit)

The CNAME of the headless service points to SRV records (one for each Pod that is Running and Ready). The SRV records point to A record entries that contain the Pods' IP addresses.

In one terminal, watch the StatefulSet's Pods:

```
# Start a new watch
# End this watch when you've seen that the delete is finished
kubectrl get pod --watch -l app=nginx
```

In a second terminal, use [kubectrl delete](#) to delete all the Pods in the StatefulSet:

```
kubectrl delete pod -l app=nginx
```

```
pod "web-0" deleted
pod "web-1" deleted
```

Wait for the StatefulSet to restart them, and for both Pods to transition to Running and Ready:

```
# This should already be running
kubectrl get pod --watch -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 0/1 | ContainerCreating | 0 | 0s |

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 1/1 | Running | 0 | 2s |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | ContainerCreating | 0 | 0s |
| web-1 | 1/1 | Running | 0 | 34s |

Use `kubectrl exec` and `kubectrl run` to view the Pods' hostnames and in-cluster DNS entries. First, view the Pods' hostnames:

```
for i in 0 1; do kubectrl exec web-$i -- sh -c 'hostname'; done
```

```
web-0
web-1
```

then, run:

```
kubectrl run -i --tty --image busybox:1.28 dns-test --restart=Never --rm
```

which starts a new shell.

In that new shell, run:

```
# Run this in the dns-test container shell
nslookup web-0.nginx
```

The output is similar to:

```
Server: 10.0.0.10
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local

Name: web-0.nginx
Address 1: 10.244.1.7

nslookup web-1.nginx
Server: 10.0.0.10
Address 1: 10.0.0.10 kube-dns.kube-system.svc.cluster.local

Name: web-1.nginx
Address 1: 10.244.2.8
```

(and now exit the container shell: exit)

The Pods' ordinals, hostnames, SRV records, and A record names have not changed, but the IP addresses associated with the Pods may have changed. In the cluster used for this tutorial, they have. This is why it is important not to configure other applications to connect to Pods in a StatefulSet by the IP address of a particular Pod (it is OK to connect to Pods by resolving their hostname).

Discovery for specific Pods in a StatefulSet

If you need to find and connect to the active members of a StatefulSet, you should query the CNAME of the headless Service (nginx.default.svc.cluster.local). The SRV records associated with the CNAME will contain only the Pods in the StatefulSet that are Running and Ready.

If your application already implements connection logic that tests for liveness and readiness, you can use the SRV records of the Pods (web-0.nginx.default.svc.cluster.local, web-1.nginx.default.svc.cluster.local), as they are stable, and your application will be able to discover the Pods' addresses when they transition to Running and Ready.

If your application wants to find any healthy Pod in a StatefulSet, and therefore does not need to track each specific Pod, you could also connect to the IP address of a type: ClusterIP Service, backed by the Pods in that StatefulSet. You can use the same Service that tracks the StatefulSet (specified in the serviceName of the StatefulSet) or a separate Service that selects the right set of Pods.

Writing to stable storage

Get the PersistentVolumeClaims for web-0 and web-1:

```
kubectrl get pvc -l app=nginx
```


The output is similar to:

| NAME | STATUS | VOLUME | CAPACITY | ACCESSMODES | AGE |
|-----------|--------|--|----------|-------------|-----|
| www-web-0 | Bound | pvc-15c268c7-b507-11e6-932f-42010a800002 | 1Gi | RWO | 48s |
| www-web-1 | Bound | pvc-15c79307-b507-11e6-932f-42010a800002 | 1Gi | RWO | 48s |

The StatefulSet controller created two [PersistentVolumeClaims](#) that are bound to two [PersistentVolumes](#).

As the cluster used in this tutorial is configured to dynamically provision PersistentVolumes, the PersistentVolumes were created and bound automatically.

The NGINX webserver, by default, serves an index file from `/usr/share/nginx/html/index.html`. The `volumeMounts` field in the StatefulSet's spec ensures that the `/usr/share/nginx/html` directory is backed by a PersistentVolume.

Write the Pods' hostnames to their `index.html` files and verify that the NGINX webserver serves the hostnames:

```
for i in 0 1; do kubectl exec "web-$i" -- sh -c 'echo "${hostname}" > /usr/share/nginx/html/index.html'; done
```

```
for i in 0 1; do kubectl exec -i -t "web-$i" -- curl http://localhost/; done
```

```
web-0
web-1
```

Note:

If you instead see **403 Forbidden** responses for the above `curl` command, you will need to fix the permissions of the directory mounted by the `volumeMounts` (due to a [bug when using hostPath volumes](#)), by running:

```
for i in 0 1; do kubectl exec web-$i -- chmod 755 /usr/share/nginx/html; done
```

before retrying the `curl` command above.

In one terminal, watch the StatefulSet's Pods:

```
# End this watch when you've reached the end of the section.
# At the start of "Scaling a StatefulSet" you'll start a new watch.
kubectl get pod --watch -l app=nginx
```

In a second terminal, delete all of the StatefulSet's Pods:

```
kubectl delete pod -l app=nginx
```

```
pod "web-0" deleted
pod "web-1" deleted
```

Examine the output of the `kubectl get` command in the first terminal, and wait for all of the Pods to transition to Running and Ready.

```
# This should already be running
kubectl get pod --watch -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 0/1 | ContainerCreating | 0 | 0s |

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 1/1 | Running | 0 | 2s |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | ContainerCreating | 0 | 0s |
| web-1 | 1/1 | Running | 0 | 34s |

Verify the web servers continue to serve their hostnames:

```
for i in 0 1; do kubectl exec -i -t "web-$i" -- curl http://localhost/; done
```

```
web-0
web-1
```

Even though web-0 and web-1 were rescheduled, they continue to serve their hostnames because the PersistentVolumes associated with their PersistentVolumeClaims are remounted to their volumeMounts. No matter what node web-0 and web-1 are scheduled on, their PersistentVolumes will be mounted to the appropriate mount points.

Scaling a StatefulSet

Scaling a StatefulSet refers to increasing or decreasing the number of replicas (horizontal scaling). This is accomplished by updating the replicas field. You can use either [kubectl scale](#) or [kubectl patch](#) to scale a StatefulSet.

Scaling up

Scaling up means adding more replicas. Provided that your app is able to distribute work across the StatefulSet, the new larger set of Pods can perform more of that work.

In one terminal window, watch the Pods in the StatefulSet:

```
# If you already have a watch running, you can continue using that.
# Otherwise, start one.
# End this watch when there are 5 healthy Pods for the StatefulSet
kubectl get pods --watch -l app=nginx
```

In another terminal window, use `kubectl scale` to scale the number of replicas to 5:

```
kubectl scale sts web --replicas=5
```

```
statefulset.apps/web scaled
```

Examine the output of the `kubectl get` command in the first terminal, and wait for the three additional Pods to transition to Running and Ready.

```
# This should already be running
kubectl get pod --watch -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|---------|----------|-----|
| web-0 | 1/1 | Running | 0 | 2h |
| web-1 | 1/1 | Running | 0 | 2h |

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-2 | 0/1 | Pending | 0 | 0s |
| web-2 | 0/1 | Pending | 0 | 0s |
| web-2 | 0/1 | ContainerCreating | 0 | 0s |
| web-2 | 1/1 | Running | 0 | 19s |
| web-3 | 0/1 | Pending | 0 | 0s |
| web-3 | 0/1 | Pending | 0 | 0s |
| web-3 | 0/1 | ContainerCreating | 0 | 0s |
| web-3 | 1/1 | Running | 0 | 18s |
| web-4 | 0/1 | Pending | 0 | 0s |
| web-4 | 0/1 | Pending | 0 | 0s |
| web-4 | 0/1 | ContainerCreating | 0 | 0s |
| web-4 | 1/1 | Running | 0 | 19s |

The StatefulSet controller scaled the number of replicas. As with [StatefulSet creation](#), the StatefulSet controller created each Pod sequentially with respect to its ordinal index, and it waited for each Pod's predecessor to be Running and Ready before launching the subsequent Pod.

Scaling down

Scaling down means reducing the number of replicas. For example, you might do this because the level of traffic to a service has decreased, and at the current scale there are idle resources.

In one terminal, watch the StatefulSet's Pods:

```
# End this watch when there are only 3 Pods for the StatefulSet
kubectl get pod --watch -l app=nginx
```

In another terminal, use kubectl patch to scale the StatefulSet back down to three replicas:

```
kubectl patch sts web -p '{"spec":{"replicas":3}}'
```

```
statefulset.apps/web patched
```

Wait for web-4 and web-3 to transition to Terminating.

```
# This should already be running
kubectl get pods --watch -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 1/1 | Running | 0 | 3h |
| web-1 | 1/1 | Running | 0 | 3h |
| web-2 | 1/1 | Running | 0 | 55s |
| web-3 | 1/1 | Running | 0 | 36s |
| web-4 | 0/1 | ContainerCreating | 0 | 18s |

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------|----------|-----|
| web-4 | 1/1 | Running | 0 | 19s |
| web-4 | 1/1 | Terminating | 0 | 24s |
| web-4 | 1/1 | Terminating | 0 | 24s |

```
web-3    1/1    Terminating    0    42s
web-3    1/1    Terminating    0    42s
```

Ordered Pod termination

The control plane deleted one Pod at a time, in reverse order with respect to its ordinal index, and it waited for each Pod to be completely shut down before deleting the next one.

Get the StatefulSet's PersistentVolumeClaims:

```
kubectl get pvc -l app=nginx
```

| NAME | STATUS | VOLUME | CAPACITY | ACCESSMODES | AGE |
|-----------|--------|--|----------|-------------|-----|
| www-web-0 | Bound | pvc-15c268c7-b507-11e6-932f-42010a800002 | 1Gi | RWO | 13h |
| www-web-1 | Bound | pvc-15c79307-b507-11e6-932f-42010a800002 | 1Gi | RWO | 13h |
| www-web-2 | Bound | pvc-e1125b27-b508-11e6-932f-42010a800002 | 1Gi | RWO | 13h |
| www-web-3 | Bound | pvc-e1176df6-b508-11e6-932f-42010a800002 | 1Gi | RWO | 13h |
| www-web-4 | Bound | pvc-e11bb5f8-b508-11e6-932f-42010a800002 | 1Gi | RWO | 13h |

There are still five PersistentVolumeClaims and five PersistentVolumes. When exploring a Pod's [stable storage](#), you saw that the PersistentVolumes mounted to the Pods of a StatefulSet are not deleted when the StatefulSet's Pods are deleted. This is still true when Pod deletion is caused by scaling the StatefulSet down.

Updating StatefulSets

The StatefulSet controller supports automated updates. The strategy used is determined by the `spec.updateStrategy` field of the StatefulSet API object. This feature can be used to upgrade the container images, resource requests and/or limits, labels, and annotations of the Pods in a StatefulSet.

There are two valid update strategies, `RollingUpdate` (the default) and `OnDelete`.

RollingUpdate

The `RollingUpdate` update strategy will update all Pods in a StatefulSet, in reverse ordinal order, while respecting the StatefulSet guarantees.

You can split updates to a StatefulSet that uses the `RollingUpdate` strategy into *partitions*, by specifying `.spec.updateStrategy.rollingUpdate.partition`. You'll practice that later in this tutorial.

First, try a simple rolling update.

In one terminal window, patch the web StatefulSet to change the container image again:

```
kubectl patch statefulset web --type='json' -p='[{"op": "replace", "path": "/spec/template/spec/containers/0/image", "value": "gcr.io/google_containers/nginx-slim:0.8"}]'
```

```
statefulset.apps/web patched
```

In another terminal, watch the Pods in the StatefulSet:

```
# End this watch when the rollout is complete
#
# If you're not sure, leave it running one more minute
kubectl get pod -l app=nginx --watch
```

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 1/1 | Running | 0 | 7m |
| web-1 | 1/1 | Running | 0 | 7m |
| web-2 | 1/1 | Running | 0 | 8m |
| web-2 | 1/1 | Terminating | 0 | 8m |
| web-2 | 1/1 | Terminating | 0 | 8m |
| web-2 | 0/1 | Terminating | 0 | 8m |
| web-2 | 0/1 | Terminating | 0 | 8m |
| web-2 | 0/1 | Terminating | 0 | 8m |
| web-2 | 0/1 | Terminating | 0 | 8m |
| web-2 | 0/1 | Pending | 0 | 0s |
| web-2 | 0/1 | Pending | 0 | 0s |
| web-2 | 0/1 | ContainerCreating | 0 | 0s |
| web-2 | 1/1 | Running | 0 | 19s |
| web-1 | 1/1 | Terminating | 0 | 8m |
| web-1 | 0/1 | Terminating | 0 | 8m |
| web-1 | 0/1 | Terminating | 0 | 8m |
| web-1 | 0/1 | Terminating | 0 | 8m |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | ContainerCreating | 0 | 0s |
| web-1 | 1/1 | Running | 0 | 6s |
| web-0 | 1/1 | Terminating | 0 | 7m |
| web-0 | 1/1 | Terminating | 0 | 7m |
| web-0 | 0/1 | Terminating | 0 | 7m |
| web-0 | 0/1 | Terminating | 0 | 7m |
| web-0 | 0/1 | Terminating | 0 | 7m |
| web-0 | 0/1 | Terminating | 0 | 7m |
| web-0 | 0/1 | Pending | 0 | 0s |
| web-0 | 0/1 | Pending | 0 | 0s |
| web-0 | 0/1 | ContainerCreating | 0 | 0s |
| web-0 | 1/1 | Running | 0 | 10s |

The Pods in the StatefulSet are updated in reverse ordinal order. The StatefulSet controller terminates each Pod, and waits for it to transition to Running and Ready prior to updating the next Pod. Note that, even though the StatefulSet controller will not proceed to update the next Pod until its ordinal successor is Running and Ready, it will restore any Pod that fails during the update to that Pod's existing version.

Pods that have already received the update will be restored to the updated version, and Pods that have not yet received the update will be restored to the previous version. In this way, the controller attempts to continue to keep the application healthy and the update consistent in the presence of intermittent failures.

Get the Pods to view their container images:

```
for p in 0 1 2; do kubectl get pod "web-$p" --template '{{range $i, $c := .spec.containers}}{{$c.image}}{{end}}'; echo; done
```

```
registry.k8s.io/nginx-slim:0.8
registry.k8s.io/nginx-slim:0.8
registry.k8s.io/nginx-slim:0.8
```

All the Pods in the StatefulSet are now running the previous container image.

Note:

You can also use `kubectl rollout status sts/<name>` to view the status of a rolling update to a StatefulSet

Staging an update

You can split updates to a StatefulSet that uses the RollingUpdate strategy into *partitions*, by specifying `.spec.updateStrategy.rollingUpdate.partition`.

For more context, you can read [Partitioned rolling updates](#) in the StatefulSet concept page.

You can stage an update to a StatefulSet by using the `partition` field within `.spec.updateStrategy.rollingUpdate`. For this update, you will keep the existing Pods in the StatefulSet unchanged whilst you change the pod template for the StatefulSet. Then you - or, outside of a tutorial, some external automation - can trigger that prepared update.

First, patch the web StatefulSet to add a partition to the `updateStrategy` field:

```
# The value of "partition" determines which ordinals a change applies to
# Make sure to use a number bigger than the last ordinal for the
# StatefulSet
kubectl patch statefulset web -p '{"spec":{"updateStrategy":
{"type":"RollingUpdate","rollingUpdate":{"partition":3}}}}'
```

```
statefulset.apps/web patched
```

Patch the StatefulSet again to change the container image that this StatefulSet uses:

```
kubectl patch statefulset web --type='json' -p='[{"op": "replace", "path": "/spec/template/spec/containers/0/image", "value": "registry.k8s.io/nginx-slim:0.7"}]'
```

```
statefulset.apps/web patched
```

Delete a Pod in the StatefulSet:

```
kubectl delete pod web-2
```

```
pod "web-2" deleted
```

Wait for the replacement web-2 Pod to be Running and Ready:

```
# End the watch when you see that web-2 is healthy
kubectl get pod -l app=nginx --watch
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 1/1 | Running | 0 | 4m |
| web-1 | 1/1 | Running | 0 | 4m |
| web-2 | 0/1 | ContainerCreating | 0 | 11s |
| web-2 | 1/1 | Running | 0 | 18s |

Get the Pod's container image:

```
kubectl get pod web-2 --template '{{range $i, $c := .spec.containers}}{{ $c.image}}{{end}}'
```

```
registry.k8s.io/nginx-slim:0.8
```

Notice that, even though the update strategy is RollingUpdate the StatefulSet restored the Pod with the original container image. This is because the ordinal of the Pod is less than the partition specified by the updateStrategy.

Rolling out a canary

You're now going to try a [canary rollout](#) of that staged change.

You can roll out a canary (to test the modified template) by decrementing the partition you specified [above](#).

Patch the StatefulSet to decrement the partition:

```
# The value of "partition" should match the highest existing ordinal for
# the StatefulSet
kubectl patch statefulset web -p '{"spec":{"updateStrategy":
{"type":"RollingUpdate","rollingUpdate":{"partition":2}}}]'
```

```
statefulset.apps/web patched
```

The control plane triggers replacement for web-2 (implemented by a graceful **delete** followed by creating a new Pod once the deletion is complete). Wait for the new web-2 Pod to be Running and Ready.

```
# This should already be running
kubectl get pod -l app=nginx --watch
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 1/1 | Running | 0 | 4m |
| web-1 | 1/1 | Running | 0 | 4m |
| web-2 | 0/1 | ContainerCreating | 0 | 11s |
| web-2 | 1/1 | Running | 0 | 18s |

Get the Pod's container:

```
kubectl get pod web-2 --template '{{range $i, $c := .spec.containers}}{{ $c.image}}{{end}}'
```

```
registry.k8s.io/nginx-slim:0.7
```

When you changed the partition, the StatefulSet controller automatically updated the web-2 Pod because the Pod's ordinal was greater than or equal to the partition.

Delete the web-1 Pod:

```
kubectl delete pod web-1
```

```
pod "web-1" deleted
```

Wait for the web-1 Pod to be Running and Ready.

```
# This should already be running
kubectl get pod -l app=nginx --watch
```

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 1/1 | Running | 0 | 6m |
| web-1 | 0/1 | Terminating | 0 | 6m |
| web-2 | 1/1 | Running | 0 | 2m |
| web-1 | 0/1 | Terminating | 0 | 6m |
| web-1 | 0/1 | Terminating | 0 | 6m |
| web-1 | 0/1 | Terminating | 0 | 6m |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | Pending | 0 | 0s |
| web-1 | 0/1 | ContainerCreating | 0 | 0s |
| web-1 | 1/1 | Running | 0 | 18s |

Get the web-1 Pod's container image:

```
kubectl get pod web-1 --template '{{range $i, $c := .spec.containers}}{{ $c.image}}{{end}}'
```

```
registry.k8s.io/nginx-slim:0.8
```

web-1 was restored to its original configuration because the Pod's ordinal was less than the partition. When a partition is specified, all Pods with an ordinal that is greater than or equal to the partition will be updated when the StatefulSet's `.spec.template` is updated. If a Pod that has an ordinal less than the partition is deleted or otherwise terminated, it will be restored to its original configuration.

Phased roll outs

You can perform a phased roll out (e.g. a linear, geometric, or exponential roll out) using a partitioned rolling update in a similar manner to how you rolled out a [canary](#). To perform a phased roll out, set the partition to the ordinal at which you want the controller to pause the update.

The partition is currently set to 2. Set the partition to 0:

```
kubectl patch statefulset web -p '{"spec":{"updateStrategy":  
{"type":"RollingUpdate","rollingUpdate":{"partition":0}}}}'
```

```
statefulset.apps/web patched
```

Wait for all of the Pods in the StatefulSet to become Running and Ready.


```
# This should already be running
kubectl get pod -l app=nginx --watch
```

The output is similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 1/1 | Running | 0 | 3m |
| web-1 | 0/1 | ContainerCreating | 0 | 11s |
| web-2 | 1/1 | Running | 0 | 2m |
| web-1 | 1/1 | Running | 0 | 18s |
| web-0 | 1/1 | Terminating | 0 | 3m |
| web-0 | 1/1 | Terminating | 0 | 3m |
| web-0 | 0/1 | Terminating | 0 | 3m |
| web-0 | 0/1 | Terminating | 0 | 3m |
| web-0 | 0/1 | Terminating | 0 | 3m |
| web-0 | 0/1 | Terminating | 0 | 3m |
| web-0 | 0/1 | Pending | 0 | 0s |
| web-0 | 0/1 | Pending | 0 | 0s |
| web-0 | 0/1 | ContainerCreating | 0 | 0s |
| web-0 | 1/1 | Running | 0 | 3s |

Get the container image details for the Pods in the StatefulSet:

```
for p in 0 1 2; do kubectl get pod "web-$p" --template '{{range $i, $c := .spec.containers}}
{{$c.image}}{{end}}'; echo; done
```

```
registry.k8s.io/nginx-slim:0.7
registry.k8s.io/nginx-slim:0.7
registry.k8s.io/nginx-slim:0.7
```

By moving the partition to 0, you allowed the StatefulSet to continue the update process.

OnDelete

You select this update strategy for a StatefulSet by setting the `.spec.template.updateStrategy.type` to `OnDelete`.

Patch the web StatefulSet to use the `OnDelete` update strategy:

```
kubectl patch statefulset web -p '{"spec":{"updateStrategy":{"type":"OnDelete"}}}'
```

```
statefulset.apps/web patched
```

When you select this update strategy, the StatefulSet controller does not automatically update Pods when a modification is made to the StatefulSet's `.spec.template` field. You need to manage the rollout yourself - either manually, or using separate automation.

Deleting StatefulSets

StatefulSet supports both *non-cascading* and *cascading* deletion. In a non-cascading **delete**, the StatefulSet's Pods are not deleted when the StatefulSet is deleted. In a cascading **delete**, both the StatefulSet and its Pods are deleted.

Read [Use Cascading Deletion in a Cluster](#) to learn about cascading deletion generally.

Non-cascading delete

In one terminal window, watch the Pods in the StatefulSet.

```
# End this watch when there are no Pods for the StatefulSet
kubectl get pods --watch -l app=nginx
```

Use [kubectl delete](#) to delete the StatefulSet. Make sure to supply the `--cascade=orphan` parameter to the command. This parameter tells Kubernetes to only delete the StatefulSet, and to **not** delete any of its Pods.

```
kubectl delete statefulset web --cascade=orphan
```

```
statefulset.apps "web" deleted
```

Get the Pods, to examine their status:

```
kubectl get pods -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|---------|----------|-----|
| web-0 | 1/1 | Running | 0 | 6m |
| web-1 | 1/1 | Running | 0 | 7m |
| web-2 | 1/1 | Running | 0 | 5m |

Even though web has been deleted, all of the Pods are still Running and Ready. Delete web-0:

```
kubectl delete pod web-0
```

```
pod "web-0" deleted
```

Get the StatefulSet's Pods:

```
kubectl get pods -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|---------|----------|-----|
| web-1 | 1/1 | Running | 0 | 10m |
| web-2 | 1/1 | Running | 0 | 7m |

As the web StatefulSet has been deleted, web-0 has not been relaunched.

In one terminal, watch the StatefulSet's Pods.

```
# Leave this watch running until the next time you start a watch
kubectl get pods --watch -l app=nginx
```

In a second terminal, recreate the StatefulSet. Note that, unless you deleted the nginx Service (which you should not have), you will see an error indicating that the Service already exists.

```
kubectl apply -f https://k8s.io/examples/application/web/web.yaml
```

```
statefulset.apps/web created
service/nginx unchanged
```

Ignore the error. It only indicates that an attempt was made to create the *nginx* headless Service even though that Service already exists.

Examine the output of the `kubectl get` command running in the first terminal.

```
# This should already be running
kubectl get pods --watch -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|---------|----------|-----|
| web-1 | 1/1 | Running | 0 | 16m |
| web-2 | 1/1 | Running | 0 | 2m |

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------------|----------|-----|
| web-0 | 0/1 | Pending | 0 | 0s |
| web-0 | 0/1 | Pending | 0 | 0s |
| web-0 | 0/1 | ContainerCreating | 0 | 0s |
| web-0 | 1/1 | Running | 0 | 18s |
| web-2 | 1/1 | Terminating | 0 | 3m |
| web-2 | 0/1 | Terminating | 0 | 3m |
| web-2 | 0/1 | Terminating | 0 | 3m |
| web-2 | 0/1 | Terminating | 0 | 3m |

When the web StatefulSet was recreated, it first relaunched web-0. Since web-1 was already Running and Ready, when web-0 transitioned to Running and Ready, it adopted this Pod. Since you recreated the StatefulSet with replicas equal to 2, once web-0 had been recreated, and once web-1 had been determined to already be Running and Ready, web-2 was terminated.

Now take another look at the contents of the `index.html` file served by the Pods' web servers:

```
for i in 0 1; do kubectl exec -i -t "web-$i" -- curl http://localhost/; done
```

```
web-0
web-1
```

Even though you deleted both the StatefulSet and the web-0 Pod, it still serves the hostname originally entered into its `index.html` file. This is because the StatefulSet never deletes the PersistentVolumes associated with a Pod. When you recreated the StatefulSet and it relaunched web-0, its original PersistentVolume was remounted.

Cascading delete

In one terminal window, watch the Pods in the StatefulSet.

```
# Leave this running until the next page section
kubectl get pods --watch -l app=nginx
```

In another terminal, delete the StatefulSet again. This time, omit the `--cascade=orphan` parameter.

```
kubectl delete statefulset web
```

```
statefulset.apps "web" deleted
```

Examine the output of the `kubectl get` command running in the first terminal, and wait for all of the Pods to transition to Terminating.

```
# This should already be running
kubectl get pods --watch -l app=nginx
```

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|---------|----------|-----|
| web-0 | 1/1 | Running | 0 | 11m |
| web-1 | 1/1 | Running | 0 | 27m |

| NAME | READY | STATUS | RESTARTS | AGE |
|-------|-------|-------------|----------|-----|
| web-0 | 1/1 | Terminating | 0 | 12m |
| web-1 | 1/1 | Terminating | 0 | 29m |
| web-0 | 0/1 | Terminating | 0 | 12m |
| web-0 | 0/1 | Terminating | 0 | 12m |
| web-0 | 0/1 | Terminating | 0 | 12m |
| web-1 | 0/1 | Terminating | 0 | 29m |
| web-1 | 0/1 | Terminating | 0 | 29m |
| web-1 | 0/1 | Terminating | 0 | 29m |

As you saw in the [Scaling Down](#) section, the Pods are terminated one at a time, with respect to the reverse order of their ordinal indices. Before terminating a Pod, the StatefulSet controller waits for the Pod's successor to be completely terminated.

Note:

Although a cascading delete removes a StatefulSet together with its Pods, the cascade does **not** delete the headless Service associated with the StatefulSet. You must delete the nginx Service manually.

```
kubectl delete service nginx
```

```
service "nginx" deleted
```

Recreate the StatefulSet and headless Service one more time:

```
kubectl apply -f https://k8s.io/examples/application/web/web.yaml
```

```
service/nginx created
statefulset.apps/web created
```

When all of the StatefulSet's Pods transition to Running and Ready, retrieve the contents of their index.html files:

```
for i in 0 1; do kubectl exec -i -t "web-$i" -- curl http://localhost/; done
```

```
web-0
web-1
```

Even though you completely deleted the StatefulSet, and all of its Pods, the Pods are recreated with their PersistentVolumes mounted, and web-0 and web-1 continue to serve their hostnames.

Finally, delete the nginx Service...

```
kubectl delete service nginx
```

```
service "nginx" deleted
```

...and the web StatefulSet:

```
kubectrl delete statefulset web
```

```
statefulset "web" deleted
```

Pod management policy

For some distributed systems, the StatefulSet ordering guarantees are unnecessary and/or undesirable. These systems require only uniqueness and identity.

You can specify a [Pod management policy](#) to avoid this strict ordering; either OrderedReady (the default), or Parallel.

OrderedReady Pod management

OrderedReady pod management is the default for StatefulSets. It tells the StatefulSet controller to respect the ordering guarantees demonstrated above.

Use this when your application requires or expects that changes, such as rolling out a new version of your application, happen in the strict order of the ordinal (pod number) that the StatefulSet provides. In other words, if you have Pods app-0, app-1 and app-2, Kubernetes will update app-0 first and check it. Once the checks are good, Kubernetes updates app-1 and finally app-2.

If you added two more Pods, Kubernetes would set up app-3 and wait for that to become healthy before deploying app-4.

Because this is the default setting, you've already practised using it.

Parallel Pod management

The alternative, Parallel pod management, tells the StatefulSet controller to launch or terminate all Pods in parallel, and not to wait for Pods to become Running and Ready or completely terminated prior to launching or terminating another Pod.

The Parallel pod management option only affects the behavior for scaling operations. Updates are not affected; Kubernetes still rolls out changes in order. For this tutorial, the application is very simple: a webserver that tells you its hostname (because this is a StatefulSet, the hostname for each Pod is different and predictable).

[application/web/web-parallel.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: nginx
  labels:
    app: nginx
spec:
  ports:
    - port: 80
    name: web
```

```

clusterIP: None
selector:
  app: nginx
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: web
spec:
  serviceName: "nginx"
  podManagementPolicy: "Parallel"
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: registry.k8s.io/nginx-slim:0.8
          ports:
            - containerPort: 80
              name: web
          volumeMounts:
            - name: www
              mountPath: /usr/share/nginx/html
      volumeClaimTemplates:
        - metadata:
            name: www
          spec:
            accessModes: [ "ReadWriteOnce" ]
            resources:
              requests:
                storage: 1Gi

```

This manifest is identical to the one you downloaded above except that the `.spec.podManagementPolicy` of the web StatefulSet is set to Parallel.

In one terminal, watch the Pods in the StatefulSet.

```
# Leave this watch running until the end of the section
kubectl get pod -l app=nginx --watch
```

In another terminal, reconfigure the StatefulSet for Parallel Pod management:

```
kubectl apply -f https://k8s.io/examples/application/web/web-parallel.yaml
```

```
service/nginx updated
statefulset.apps/web updated
```

Keep the terminal open where you're running the watch. In another terminal window, scale the StatefulSet:

```
kubectl scale statefulset/web --replicas=5
```

```
statefulset.apps/web scaled
```

Examine the output of the terminal where the `kubectl get` command is running. It may look something like

```
web-3    0/1    Pending 0    0s
web-3    0/1    Pending 0    0s
web-3    0/1    Pending 0    7s
web-3    0/1    ContainerCreating 0    7s
web-2    0/1    Pending 0    0s
web-4    0/1    Pending 0    0s
web-2    1/1    Running 0    8s
web-4    0/1    ContainerCreating 0    4s
web-3    1/1    Running 0    26s
web-4    1/1    Running 0    2s
```

The StatefulSet launched three new Pods, and it did not wait for the first to become Running and Ready prior to launching the second and third Pods.

This approach is useful if your workload has a stateful element, or needs Pods to be able to identify each other with predictable naming, and especially if you sometimes need to provide a lot more capacity quickly. If this simple web service for the tutorial suddenly got an extra 1,000,000 requests per minute then you would want to run some more Pods - but you also would not want to wait for each new Pod to launch. Starting the extra Pods in parallel cuts the time between requesting the extra capacity and having it available for use.

Cleaning up

You should have two terminals open, ready for you to run `kubectl` commands as part of cleanup.

```
kubectl delete sts web
# sts is an abbreviation for statefulset
```

You can watch `kubectl get` to see those Pods being deleted.

```
# end the watch when you've seen what you need to
kubectl get pod -l app=nginx --watch
```

```
web-3    1/1    Terminating 0    9m
web-2    1/1    Terminating 0    9m
web-3    1/1    Terminating 0    9m
web-2    1/1    Terminating 0    9m
web-1    1/1    Terminating 0    44m
web-0    1/1    Terminating 0    44m
web-0    0/1    Terminating 0    44m
web-3    0/1    Terminating 0    9m
web-2    0/1    Terminating 0    9m
```

| | | | | |
|-------|-----|-------------|---|-----|
| web-1 | 0/1 | Terminating | 0 | 44m |
| web-0 | 0/1 | Terminating | 0 | 44m |
| web-2 | 0/1 | Terminating | 0 | 9m |
| web-2 | 0/1 | Terminating | 0 | 9m |
| web-2 | 0/1 | Terminating | 0 | 9m |
| web-1 | 0/1 | Terminating | 0 | 44m |
| web-1 | 0/1 | Terminating | 0 | 44m |
| web-1 | 0/1 | Terminating | 0 | 44m |
| web-0 | 0/1 | Terminating | 0 | 44m |
| web-0 | 0/1 | Terminating | 0 | 44m |
| web-0 | 0/1 | Terminating | 0 | 44m |
| web-3 | 0/1 | Terminating | 0 | 9m |
| web-3 | 0/1 | Terminating | 0 | 9m |
| web-3 | 0/1 | Terminating | 0 | 9m |

During deletion, a StatefulSet removes all Pods concurrently; it does not wait for a Pod's ordinal successor to terminate prior to deleting that Pod.

Close the terminal where the `kubectl get` command is running and delete the `nginx` Service:

```
kubectl delete svc nginx
```

Delete the persistent storage media for the PersistentVolumes used in this tutorial.

```
kubectl get pvc
```

| NAME | STATUS | VOLUME | CAPACITY | ACCESS MODES |
|--------------|--------|--|----------|--------------|
| STORAGECLASS | AGE | | | |
| www-web-0 | Bound | pvc-2bf00408-d366-4a12-bad0-1869c65d0bee | 1Gi | RWO |
| standard | 25m | | | |
| www-web-1 | Bound | pvc-ba3bfe9c-413e-4b95-a2c0-3ea8a54dbab4 | 1Gi | RWO |
| standard | 24m | | | |
| www-web-2 | Bound | pvc-cba6cfa6-3a47-486b-a138-db5930207eaf | 1Gi | RWO |
| standard | 15m | | | |
| www-web-3 | Bound | pvc-0c04d7f0-787a-4977-8da3-d9d3a6d8d752 | 1Gi | RWO |
| standard | 15m | | | |
| www-web-4 | Bound | pvc-b2c73489-e70b-4a4e-9ec1-9eab439aa43e | 1Gi | RWO |
| standard | 14m | | | |

```
kubectl get pv
```

| NAME | CAPACITY | ACCESS MODES | RECLAIM POLICY | STATUS |
|--|--------------|--------------|----------------|--------|
| CLAIM | STORAGECLASS | REASON | AGE | |
| pvc-0c04d7f0-787a-4977-8da3-d9d3a6d8d752 | 1Gi | RWO | Delete | Bound |
| default/www-web-3 | standard | 15m | | |
| pvc-2bf00408-d366-4a12-bad0-1869c65d0bee | 1Gi | RWO | Delete | Bound |
| default/www-web-0 | standard | 25m | | |
| pvc-b2c73489-e70b-4a4e-9ec1-9eab439aa43e | 1Gi | RWO | Delete | Bound |
| default/www-web-4 | standard | 14m | | |
| pvc-ba3bfe9c-413e-4b95-a2c0-3ea8a54dbab4 | 1Gi | RWO | Delete | Bound |
| default/www-web-1 | standard | 24m | | |
| pvc-cba6cfa6-3a47-486b-a138-db5930207eaf | 1Gi | RWO | Delete | Bound |
| default/www-web-2 | standard | 15m | | |


```
kubectl delete pvc www-web-0 www-web-1 www-web-2 www-web-3 www-web-4
```

```
persistentvolumeclaim "www-web-0" deleted  
persistentvolumeclaim "www-web-1" deleted  
persistentvolumeclaim "www-web-2" deleted  
persistentvolumeclaim "www-web-3" deleted  
persistentvolumeclaim "www-web-4" deleted
```

```
kubectl get pvc
```

No resources found in default namespace.

Note:

You also need to delete the persistent storage media for the PersistentVolumes used in this tutorial. Follow the necessary steps, based on your environment, storage configuration, and provisioning method, to ensure that all storage is reclaimed.

Example: Deploying WordPress and MySQL with Persistent Volumes

This tutorial shows you how to deploy a WordPress site and a MySQL database using Minikube. Both applications use PersistentVolumes and PersistentVolumeClaims to store data.

A [PersistentVolume](#) (PV) is a piece of storage in the cluster that has been manually provisioned by an administrator, or dynamically provisioned by Kubernetes using a [StorageClass](#). A [PersistentVolumeClaim](#) (PVC) is a request for storage by a user that can be fulfilled by a PV. PersistentVolumes and PersistentVolumeClaims are independent from Pod lifecycles and preserve data through restarting, rescheduling, and even deleting Pods.

Warning:

This deployment is not suitable for production use cases, as it uses single instance WordPress and MySQL Pods. Consider using [WordPress Helm Chart](#) to deploy WordPress in production.

Note:

The files provided in this tutorial are using GA Deployment APIs and are specific to Kubernetes version 1.9 and later. If you wish to use this tutorial with an earlier version of Kubernetes, please update the API version appropriately, or reference earlier versions of this tutorial.

Objectives

- Create PersistentVolumeClaims and PersistentVolumes
- Create a kustomization.yaml with
 - a Secret generator
 - MySQL resource configs
 - WordPress resource configs
- Apply the kustomization directory by `kubectl apply -k ./`

- Clean up

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To check the version, enter `kubectl version`.

The example shown on this page works with kubectl 1.27 and above.

Download the following configuration files:

1. [mysql-deployment.yaml](#)
2. [wordpress-deployment.yaml](#)

Create PersistentVolumeClaims and PersistentVolumes

MySQL and Wordpress each require a PersistentVolume to store data. Their PersistentVolumeClaims will be created at the deployment step.

Many cluster environments have a default StorageClass installed. When a StorageClass is not specified in the PersistentVolumeClaim, the cluster's default StorageClass is used instead.

When a PersistentVolumeClaim is created, a PersistentVolume is dynamically provisioned based on the StorageClass configuration.

Warning:

In local clusters, the default StorageClass uses the `hostPath` provisioner. `hostPath` volumes are only suitable for development and testing. With `hostPath` volumes, your data lives in `/tmp` on the node the Pod is scheduled onto and does not move between nodes. If a Pod dies and gets scheduled to another node in the cluster, or the node is rebooted, the data is lost.

Note:

If you are bringing up a cluster that needs to use the `hostPath` provisioner, the `--enable-hostpath-provisioner` flag must be set in the controller-manager component.

Note:

If you have a Kubernetes cluster running on Google Kubernetes Engine, please follow [this guide](#).

Create a kustomization.yaml

Add a Secret generator

A [Secret](#) is an object that stores a piece of sensitive data like a password or key. Since 1.14, kubectl supports the management of Kubernetes objects using a kustomization file. You can create a Secret by generators in kustomization.yaml.

Add a Secret generator in kustomization.yaml from the following command. You will need to replace YOUR_PASSWORD with the password you want to use.

```
cat <<EOF >./kustomization.yaml
secretGenerator:
- name: mysql-pass
  literals:
  - password=YOUR_PASSWORD
EOF
```

Add resource configs for MySQL and WordPress

The following manifest describes a single-instance MySQL Deployment. The MySQL container mounts the PersistentVolume at /var/lib/mysql. The MYSQL_ROOT_PASSWORD environment variable sets the database password from the Secret.

[application/wordpress/mysql-deployment.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: wordpress-mysql
  labels:
    app: wordpress
spec:
  ports:
    - port: 3306
  selector:
    app: wordpress
    tier: mysql
  clusterIP: None
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: mysql-pv-claim
  labels:
    app: wordpress
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 20Gi
```

```

---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wordpress-mysql
  labels:
    app: wordpress
spec:
  selector:
    matchLabels:
      app: wordpress
      tier: mysql
  strategy:
    type: Recreate
  template:
    metadata:
      labels:
        app: wordpress
        tier: mysql
    spec:
      containers:
        - image: mysql:8.0
          name: mysql
          env:
            - name: MYSQL_ROOT_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: mysql-pass
                  key: password
            - name: MYSQL_DATABASE
              value: wordpress
            - name: MYSQL_USER
              value: wordpress
            - name: MYSQL_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: mysql-pass
                  key: password
          ports:
            - containerPort: 3306
              name: mysql
          volumeMounts:
            - name: mysql-persistent-storage
              mountPath: /var/lib/mysql
      volumes:
        - name: mysql-persistent-storage
          persistentVolumeClaim:
            claimName: mysql-pv-claim

```

The following manifest describes a single-instance WordPress Deployment. The WordPress container mounts the PersistentVolume at `/var/www/html` for website data files. The `WORDPRESS_DB_HOST` environment variable sets the name of the MySQL Service defined

above, and WordPress will access the database by Service. The WORDPRESS_DB_PASSWORD environment variable sets the database password from the Secret kustomize generated.

[application/wordpress/wordpress-deployment.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: wordpress
  labels:
    app: wordpress
spec:
  ports:
    - port: 80
  selector:
    app: wordpress
    tier: frontend
  type: LoadBalancer
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: wp-pv-claim
  labels:
    app: wordpress
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 20Gi
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wordpress
  labels:
    app: wordpress
spec:
  selector:
    matchLabels:
      app: wordpress
      tier: frontend
  strategy:
    type: Recreate
  template:
    metadata:
      labels:
        app: wordpress
        tier: frontend
    spec:
      containers:
        - image: wordpress:6.2.1-apache
```

```

name: wordpress
env:
- name: WORDPRESS_DB_HOST
  value: wordpress-mysql
- name: WORDPRESS_DB_PASSWORD
  valueFrom:
    secretKeyRef:
      name: mysql-pass
      key: password
- name: WORDPRESS_DB_USER
  value: wordpress
ports:
- containerPort: 80
  name: wordpress
volumeMounts:
- name: wordpress-persistent-storage
  mountPath: /var/www/html
volumes:
- name: wordpress-persistent-storage
  persistentVolumeClaim:
    claimName: wp-pv-claim

```

1. Download the MySQL deployment configuration file.

```
curl -LO https://k8s.io/examples/application/wordpress/mysql-deployment.yaml
```

2. Download the WordPress configuration file.

```
curl -LO https://k8s.io/examples/application/wordpress/wordpress-deployment.yaml
```

3. Add them to kustomization.yaml file.

```

cat <<EOF >>./kustomization.yaml
resources:
- mysql-deployment.yaml
- wordpress-deployment.yaml
EOF

```

Apply and Verify

The kustomization.yaml contains all the resources for deploying a WordPress site and a MySQL database. You can apply the directory by

```
kubectl apply -k ./
```

Now you can verify that all objects exist.

1. Verify that the Secret exists by running the following command:

```
kubectl get secrets
```

The response should be like this:

| NAME | TYPE | DATA | AGE |
|-----------------------|--------|------|-----|
| mysql-pass-c57bb4t7mf | Opaque | 1 | 9s |

2. Verify that a PersistentVolume got dynamically provisioned.

```
kubectl get pvc
```

Note:

It can take up to a few minutes for the PVs to be provisioned and bound.

The response should be like this:

| NAME | STATUS | VOLUME | CAPACITY | ACCESS MODES |
|----------------|----------|--|----------|--------------|
| STORAGECLASS | AGE | | | |
| mysql-pv-claim | Bound | pvc-8cbd7b2e-4044-11e9-b2bb-42010a800002 | 20Gi | |
| RWO | standard | 77s | | |
| wp-pv-claim | Bound | pvc-8cd0df54-4044-11e9-b2bb-42010a800002 | 20Gi | |
| RWO | standard | 77s | | |

3. Verify that the Pod is running by running the following command:

```
kubectl get pods
```

Note:

It can take up to a few minutes for the Pod's Status to be RUNNING.

The response should be like this:

| NAME | READY | STATUS | RESTARTS | AGE |
|----------------------------------|-------|---------|----------|-----|
| wordpress-mysql-1894417608-x5dzt | 1/1 | Running | 0 | 40s |

4. Verify that the Service is running by running the following command:

```
kubectl get services wordpress
```

The response should be like this:

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|-----------|--------------|------------|-------------|--------------|-----|
| wordpress | LoadBalancer | 10.0.0.89 | <pending> | 80:32406/TCP | 4m |

Note:

Minikube can only expose Services through NodePort. The EXTERNAL-IP is always pending.

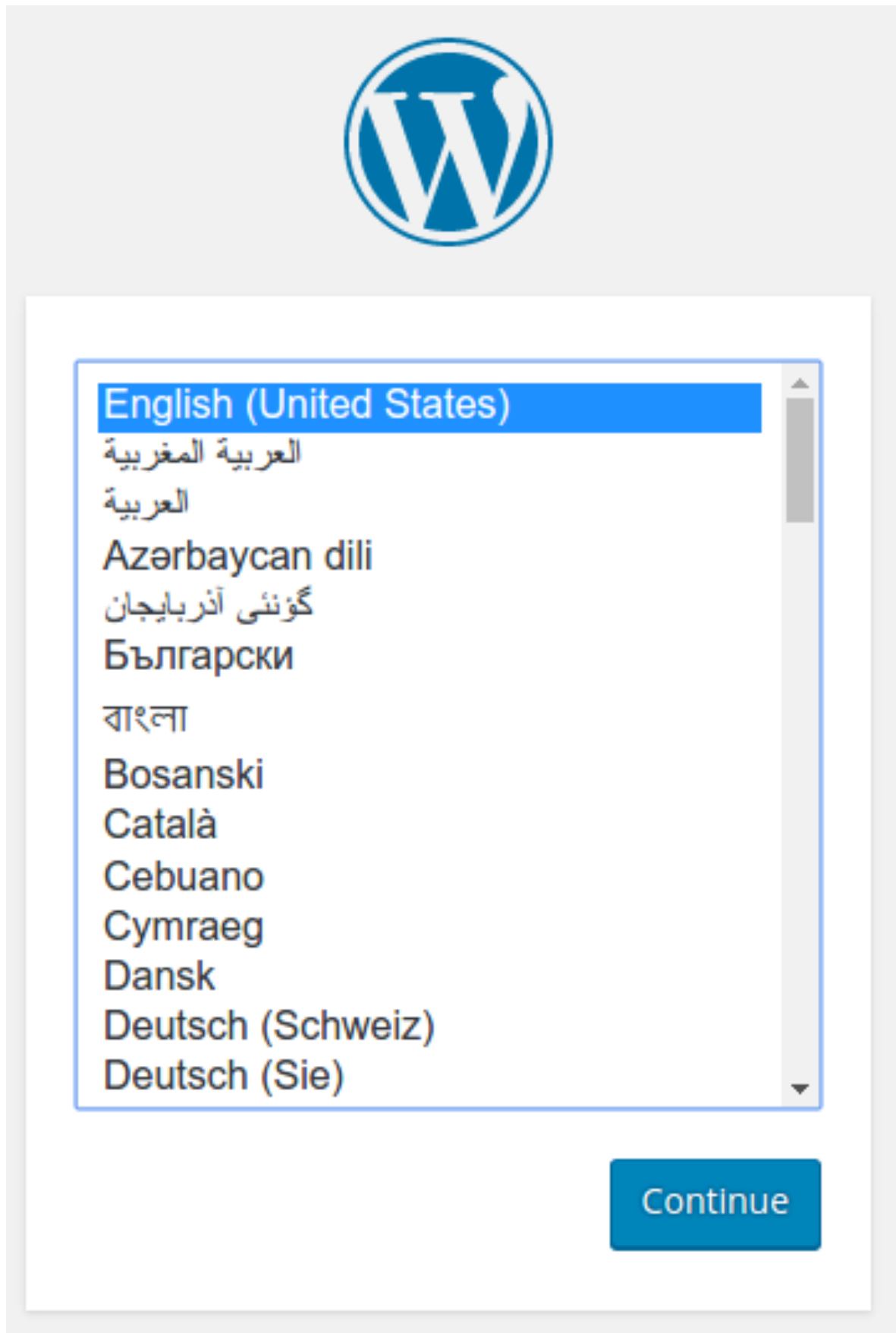
5. Run the following command to get the IP Address for the WordPress Service:

```
minikube service wordpress --url
```

The response should be like this:

```
http://1.2.3.4:32406
```

- Copy the IP address, and load the page in your browser to view your site.
6. You should see the WordPress set up page similar to the following screenshot.



Warning:

Do not leave your WordPress installation on this page. If another user finds it, they can set up a website on your instance and use it to serve malicious content.

Either install WordPress by creating a username and password or delete your instance.

Cleaning up

1. Run the following command to delete your Secret, Deployments, Services and PersistentVolumeClaims:

```
kubectrl delete -k ./
```

What's next

- Learn more about [Introspection and Debugging](#)
- Learn more about [Jobs](#)
- Learn more about [Port Forwarding](#)
- Learn how to [Get a Shell to a Container](#)

Example: Deploying Cassandra with a StatefulSet

This tutorial shows you how to run [Apache Cassandra](#) on Kubernetes. Cassandra, a database, needs persistent storage to provide data durability (application *state*). In this example, a custom Cassandra seed provider lets the database discover new Cassandra instances as they join the Cassandra cluster.

StatefulSets make it easier to deploy stateful applications into your Kubernetes cluster. For more information on the features used in this tutorial, see [StatefulSet](#).

Note:

Cassandra and Kubernetes both use the term *node* to mean a member of a cluster. In this tutorial, the Pods that belong to the StatefulSet are Cassandra nodes and are members of the Cassandra cluster (called a *ring*). When those Pods run in your Kubernetes cluster, the Kubernetes control plane schedules those Pods onto Kubernetes [Nodes](#).

When a Cassandra node starts, it uses a *seed list* to bootstrap discovery of other nodes in the ring. This tutorial deploys a custom Cassandra seed provider that lets the database discover new Cassandra Pods as they appear inside your Kubernetes cluster.

Objectives

- Create and validate a Cassandra headless [Service](#).
- Use a [StatefulSet](#) to create a Cassandra ring.
- Validate the StatefulSet.

- Modify the StatefulSet.
- Delete the StatefulSet and its [Pods](#).

Before you begin

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

To complete this tutorial, you should already have a basic familiarity with [Pods](#), [Services](#), and [StatefulSets](#).

Additional Minikube setup instructions

Caution:

[Minikube](#) defaults to 2048MB of memory and 2 CPU. Running Minikube with the default resource configuration results in insufficient resource errors during this tutorial. To avoid these errors, start Minikube with the following settings:

```
minikube start --memory 5120 --cpus=4
```

Creating a headless Service for Cassandra

In Kubernetes, a [Service](#) describes a set of [Pods](#) that perform the same task.

The following Service is used for DNS lookups between Cassandra Pods and clients within your cluster:

[application/cassandra/cassandra-service.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  labels:
    app: cassandra
    name: cassandra
spec:
  clusterIP: None
  ports:
    - port: 9042
  selector:
    app: cassandra
```

Create a Service to track all Cassandra StatefulSet members from the cassandra-service.yaml file:

```
kubectl apply -f https://k8s.io/examples/application/cassandra/cassandra-service.yaml
```

Validating (optional)

Get the Cassandra Service.

```
kubectl get svc cassandra
```

The response is

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|-----------|-----------|------------|-------------|----------|-----|
| cassandra | ClusterIP | None | <none> | 9042/TCP | 45s |

If you don't see a Service named cassandra, that means creation failed. Read [Debug Services](#) for help troubleshooting common issues.

Using a StatefulSet to create a Cassandra ring

The StatefulSet manifest, included below, creates a Cassandra ring that consists of three Pods.

Note:

This example uses the default provisioner for Minikube. Please update the following StatefulSet for the cloud you are working with.

[application/cassandra/cassandra-statefulset.yaml](#)

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: cassandra
  labels:
    app: cassandra
spec:
  serviceName: cassandra
  replicas: 3
  selector:
    matchLabels:
      app: cassandra
  template:
    metadata:
      labels:
        app: cassandra
    spec:
      terminationGracePeriodSeconds: 1800
      containers:
        - name: cassandra
          image: gcr.io/google-samples/cassandra:v13
          imagePullPolicy: Always
          ports:
            - containerPort: 7000
              name: intra-node
            - containerPort: 7001
              name: tls-intra-node
            - containerPort: 7199
```

```

    name: jmx
  - containerPort: 9042
    name: cql
resources:
  limits:
    cpu: "500m"
    memory: 1Gi
  requests:
    cpu: "500m"
    memory: 1Gi
securityContext:
  capabilities:
    add:
      - IPC_LOCK
lifecycle:
  preStop:
    exec:
      command:
        - /bin/sh
        - -c
        - nodetool drain
env:
  - name: MAX_HEAP_SIZE
    value: 512M
  - name: HEAP_NEWSIZE
    value: 100M
  - name: CASSANDRA_SEEDS
    value: "cassandra-0.cassandra.default.svc.cluster.local"
  - name: CASSANDRA_CLUSTER_NAME
    value: "K8Demo"
  - name: CASSANDRA_DC
    value: "DC1-K8Demo"
  - name: CASSANDRA_RACK
    value: "Rack1-K8Demo"
  - name: POD_IP
    valueFrom:
      fieldRef:
        fieldPath: status.podIP
readinessProbe:
  exec:
    command:
      - /bin/bash
      - -c
      - /ready-probe.sh
  initialDelaySeconds: 15
  timeoutSeconds: 5
# These volume mounts are persistent. They are like inline claims,
# but not exactly because the names need to match exactly one of
# the stateful pod volumes.
volumeMounts:
  - name: cassandra-data
    mountPath: /cassandra_data
# These are converted to volume claims by the controller

```

```
# and mounted at the paths mentioned above.
# do not use these in production until ssd GCEPersistentDisk or other ssd pd
volumeClaimTemplates:
- metadata:
  name: cassandra-data
spec:
  accessModes: [ "ReadWriteOnce" ]
  storageClassName: fast
  resources:
    requests:
      storage: 1Gi
---
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: fast
provisioner: k8s.io/minikube-hostpath
parameters:
  type: pd-ssd
```

Create the Cassandra StatefulSet from the `cassandra-statefulset.yaml` file:

```
# Use this if you are able to apply cassandra-statefulset.yaml unmodified
kubectl apply -f https://k8s.io/examples/application/cassandra/cassandra-statefulset.yaml
```

If you need to modify `cassandra-statefulset.yaml` to suit your cluster, download <https://k8s.io/examples/application/cassandra/cassandra-statefulset.yaml> and then apply that manifest, from the folder you saved the modified version into:

```
# Use this if you needed to modify cassandra-statefulset.yaml locally
kubectl apply -f cassandra-statefulset.yaml
```

Validating the Cassandra StatefulSet

1. Get the Cassandra StatefulSet:

```
kubectl get statefulset cassandra
```

The response should be similar to:

| NAME | DESIRED | CURRENT | AGE |
|-----------|---------|---------|-----|
| cassandra | 3 | 0 | 13s |

The StatefulSet resource deploys Pods sequentially.

2. Get the Pods to see the ordered creation status:

```
kubectl get pods -l="app=cassandra"
```

The response should be similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|-------------|-------|-------------------|----------|-----|
| cassandra-0 | 1/1 | Running | 0 | 1m |
| cassandra-1 | 0/1 | ContainerCreating | 0 | 8s |

It can take several minutes for all three Pods to deploy. Once they are deployed, the same command returns output similar to:

| NAME | READY | STATUS | RESTARTS | AGE |
|-------------|-------|---------|----------|-----|
| cassandra-0 | 1/1 | Running | 0 | 10m |
| cassandra-1 | 1/1 | Running | 0 | 9m |
| cassandra-2 | 1/1 | Running | 0 | 8m |

3. Run the Cassandra [nodetool](#) inside the first Pod, to display the status of the ring.

```
kubectrl exec -it cassandra-0 -- nodetool status
```

The response should look something like:

```
Datacenter: DC1-K8Demo
=====
Status=Up/Down
|/ State=Normal/Leaving/Joining/Moving
-- Address    Load      Tokens     Owns (effective)  Host ID                               Rack
UN 172.17.0.5  83.57 KiB  32         74.0%             e2dd09e6-d9d3-477e-96c5-45094c08db0f  Rack1-K8Demo
UN 172.17.0.4  101.04 KiB 32         58.8%             f89d6835-3a42-4419-92b3-0e62cae1479c  Rack1-K8Demo
UN 172.17.0.6  84.74 KiB  32         67.1%             a6a1e8c2-3dc5-4417-b1a0-26507af2aaad  Rack1-K8Demo
```

Modifying the Cassandra StatefulSet

Use `kubectrl edit` to modify the size of a Cassandra StatefulSet.

1. Run the following command:

```
kubectrl edit statefulset cassandra
```

This command opens an editor in your terminal. The line you need to change is the `replicas` field. The following sample is an excerpt of the StatefulSet file:

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: apps/v1
kind: StatefulSet
metadata:
  creationTimestamp: 2016-08-13T18:40:58Z
  generation: 1
  labels:
    app: cassandra
    name: cassandra
    namespace: default
  resourceVersion: "323"
  uid: 7a219483-6185-11e6-a910-42010a8a0fc0
```

```
spec:
  replicas: 3
```

2. Change the number of replicas to 4, and then save the manifest.

The StatefulSet now scales to run with 4 Pods.

3. Get the Cassandra StatefulSet to verify your change:

```
kubectl get statefulset cassandra
```

The response should be similar to:

| NAME | DESIRED | CURRENT | AGE |
|-----------|---------|---------|-----|
| cassandra | 4 | 4 | 36m |

Cleaning up

Deleting or scaling a StatefulSet down does not delete the volumes associated with the StatefulSet. This setting is for your safety because your data is more valuable than automatically purging all related StatefulSet resources.

Warning:

Depending on the storage class and reclaim policy, deleting the *PersistentVolumeClaims* may cause the associated volumes to also be deleted. Never assume you'll be able to access data if its volume claims are deleted.

1. Run the following commands (chained together into a single command) to delete everything in the Cassandra StatefulSet:

```
grace=$(kubectl get pod cassandra-0 -o=jsonpath='{.spec.terminationGracePeriodSeconds}') \
&& kubectl delete statefulset -l app=cassandra \
&& echo "Sleeping ${grace} seconds" 1>&2 \
&& sleep $grace \
&& kubectl delete persistentvolumeclaim -l app=cassandra
```

2. Run the following command to delete the Service you set up for Cassandra:

```
kubectl delete service -l app=cassandra
```

Cassandra container environment variables

The Pods in this tutorial use the gcr.io/google-samples/cassandra:v13 image from Google's [container registry](#). The Docker image above is based on [debian-base](#) and includes OpenJDK 8.

This image includes a standard Cassandra installation from the Apache Debian repo. By using environment variables you can change values that are inserted into `cassandra.yaml`.

| Environment variable | Default value |
|------------------------|----------------|
| CASSANDRA_CLUSTER_NAME | 'Test Cluster' |

| Environment variable | Default value |
|-----------------------|---------------|
| CASSANDRA_NUM_TOKENS | 32 |
| CASSANDRA_RPC_ADDRESS | 0.0.0.0 |

What's next

- Learn how to [Scale a StatefulSet](#).
- Learn more about the [KubernetesSeedProvider](#)
- See more custom [Seed Provider Configurations](#)

Running ZooKeeper, A Distributed System Coordinator

This tutorial demonstrates running [Apache Zookeeper](#) on Kubernetes using [StatefulSets](#), [PodDisruptionBudgets](#), and [PodAntiAffinity](#).

Before you begin

Before starting this tutorial, you should be familiar with the following Kubernetes concepts:

- [Pods](#)
- [Cluster DNS](#)
- [Headless Services](#)
- [PersistentVolumes](#)
- [PersistentVolume Provisioning](#)
- [StatefulSets](#)
- [PodDisruptionBudgets](#)
- [PodAntiAffinity](#)
- [kubectl CLI](#)

You must have a cluster with at least four nodes, and each node requires at least 2 CPUs and 4 GiB of memory. In this tutorial you will cordon and drain the cluster's nodes. **This means that the cluster will terminate and evict all Pods on its nodes, and the nodes will temporarily become unschedulable.** You should use a dedicated cluster for this tutorial, or you should ensure that the disruption you cause will not interfere with other tenants.

This tutorial assumes that you have configured your cluster to dynamically provision PersistentVolumes. If your cluster is not configured to do so, you will have to manually provision three 20 GiB volumes before starting this tutorial.

Objectives

After this tutorial, you will know the following.

- How to deploy a ZooKeeper ensemble using StatefulSet.
- How to consistently configure the ensemble.
- How to spread the deployment of ZooKeeper servers in the ensemble.

- How to use `PodDisruptionBudgets` to ensure service availability during planned maintenance.

ZooKeeper

[Apache ZooKeeper](#) is a distributed, open-source coordination service for distributed applications. ZooKeeper allows you to read, write, and observe updates to data. Data are organized in a file system like hierarchy and replicated to all ZooKeeper servers in the ensemble (a set of ZooKeeper servers). All operations on data are atomic and sequentially consistent. ZooKeeper ensures this by using the [Zab](#) consensus protocol to replicate a state machine across all servers in the ensemble.

The ensemble uses the Zab protocol to elect a leader, and the ensemble cannot write data until that election is complete. Once complete, the ensemble uses Zab to ensure that it replicates all writes to a quorum before it acknowledges and makes them visible to clients. Without respect to weighted quorums, a quorum is a majority component of the ensemble containing the current leader. For instance, if the ensemble has three servers, a component that contains the leader and one other server constitutes a quorum. If the ensemble can not achieve a quorum, the ensemble cannot write data.

ZooKeeper servers keep their entire state machine in memory, and write every mutation to a durable WAL (Write Ahead Log) on storage media. When a server crashes, it can recover its previous state by replaying the WAL. To prevent the WAL from growing without bound, ZooKeeper servers will periodically snapshot them in memory state to storage media. These snapshots can be loaded directly into memory, and all WAL entries that preceded the snapshot may be discarded.

Creating a ZooKeeper ensemble

The manifest below contains a [Headless Service](#), a [Service](#), a [PodDisruptionBudget](#), and a [StatefulSet](#).

[application/zookeeper/zookeeper.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: zk-hs
  labels:
    app: zk
spec:
  ports:
    - port: 2888
      name: server
    - port: 3888
      name: leader-election
  clusterIP: None
  selector:
    app: zk
---
apiVersion: v1
kind: Service
```

```
metadata:
  name: zk-cs
  labels:
    app: zk
spec:
  ports:
    - port: 2181
      name: client
  selector:
    app: zk
---
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: zk-pdb
spec:
  selector:
    matchLabels:
      app: zk
  maxUnavailable: 1
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: zk
spec:
  selector:
    matchLabels:
      app: zk
  serviceName: zk-hs
  replicas: 3
  updateStrategy:
    type: RollingUpdate
  podManagementPolicy: OrderedReady
  template:
    metadata:
      labels:
        app: zk
    spec:
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            - labelSelector:
                matchExpressions:
                  - key: "app"
                    operator: In
                    values:
                      - zk
              topologyKey: "kubernetes.io/hostname"
      containers:
        - name: kubernetes-zookeeper
          imagePullPolicy: Always
          image: "registry.k8s.io/kubernetes-zookeeper:1.0-3.4.10"
```

```
resources:
  requests:
    memory: "1Gi"
    cpu: "0.5"
ports:
- containerPort: 2181
  name: client
- containerPort: 2888
  name: server
- containerPort: 3888
  name: leader-election
command:
- sh
- -c
- "start-zookeeper \
  --servers=3 \
  --data_dir=/var/lib/zookeeper/data \
  --data_log_dir=/var/lib/zookeeper/data/log \
  --conf_dir=/opt/zookeeper/conf \
  --client_port=2181 \
  --election_port=3888 \
  --server_port=2888 \
  --tick_time=2000 \
  --init_limit=10 \
  --sync_limit=5 \
  --heap=512M \
  --max_client_cnxns=60 \
  --snap_retain_count=3 \
  --purge_interval=12 \
  --max_session_timeout=40000 \
  --min_session_timeout=4000 \
  --log_level=INFO"
readinessProbe:
  exec:
    command:
      - sh
      - -c
      - "zookeeper-ready 2181"
  initialDelaySeconds: 10
  timeoutSeconds: 5
livenessProbe:
  exec:
    command:
      - sh
      - -c
      - "zookeeper-ready 2181"
  initialDelaySeconds: 10
  timeoutSeconds: 5
volumeMounts:
- name: datadir
  mountPath: /var/lib/zookeeper
securityContext:
  runAsUser: 1000
```

```

    fsGroup: 1000
  volumeClaimTemplates:
  - metadata:
    name: datadir
    spec:
    accessModes: [ "ReadWriteOnce" ]
    resources:
    requests:
    storage: 10Gi

```

Open a terminal, and use the [kubectl apply](#) command to create the manifest.

```
kubectl apply -f https://k8s.io/examples/application/zookeeper/zookeeper.yaml
```

This creates the zk-hs Headless Service, the zk-cs Service, the zk-pdb PodDisruptionBudget, and the zk StatefulSet.

```

service/zk-hs created
service/zk-cs created
poddisruptionbudget.policy/zk-pdb created
statefulset.apps/zk created

```

Use [kubectl get](#) to watch the StatefulSet controller create the StatefulSet's Pods.

```
kubectl get pods -w -l app=zk
```

Once the zk-2 Pod is Running and Ready, use CTRL-C to terminate kubectl.

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|-------------------|----------|-----|
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | ContainerCreating | 0 | 0s |
| zk-0 | 0/1 | Running | 0 | 19s |
| zk-0 | 1/1 | Running | 0 | 40s |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | ContainerCreating | 0 | 0s |
| zk-1 | 0/1 | Running | 0 | 18s |
| zk-1 | 1/1 | Running | 0 | 40s |
| zk-2 | 0/1 | Pending | 0 | 0s |
| zk-2 | 0/1 | Pending | 0 | 0s |
| zk-2 | 0/1 | ContainerCreating | 0 | 0s |
| zk-2 | 0/1 | Running | 0 | 19s |
| zk-2 | 1/1 | Running | 0 | 40s |

The StatefulSet controller creates three Pods, and each Pod has a container with a [ZooKeeper](#) server.

Facilitating leader election

Because there is no terminating algorithm for electing a leader in an anonymous network, Zab requires explicit membership configuration to perform leader election. Each server in the ensemble needs to have a unique identifier, all servers need to know the global set of identifiers, and each identifier needs to be associated with a network address.

Use [kubectl exec](#) to get the hostnames of the Pods in the zk StatefulSet.

```
for i in 0 1 2; do kubectl exec zk-$i -- hostname; done
```

The StatefulSet controller provides each Pod with a unique hostname based on its ordinal index. The hostnames take the form of <statefulset name>-<ordinal index>. Because the replicas field of the zk StatefulSet is set to 3, the Set's controller creates three Pods with their hostnames set to zk-0, zk-1, and zk-2.

```
zk-0
zk-1
zk-2
```

The servers in a ZooKeeper ensemble use natural numbers as unique identifiers, and store each server's identifier in a file called myid in the server's data directory.

To examine the contents of the myid file for each server use the following command.

```
for i in 0 1 2; do echo "myid zk-$i"; kubectl exec zk-$i -- cat /var/lib/zookeeper/data/myid; done
```

Because the identifiers are natural numbers and the ordinal indices are non-negative integers, you can generate an identifier by adding 1 to the ordinal.

```
myid zk-0
1
myid zk-1
2
myid zk-2
3
```

To get the Fully Qualified Domain Name (FQDN) of each Pod in the zk StatefulSet use the following command.

```
for i in 0 1 2; do kubectl exec zk-$i -- hostname -f; done
```

The zk-hs Service creates a domain for all of the Pods, zk-hs.default.svc.cluster.local.

```
zk-0.zk-hs.default.svc.cluster.local
zk-1.zk-hs.default.svc.cluster.local
zk-2.zk-hs.default.svc.cluster.local
```

The A records in [Kubernetes DNS](#) resolve the FQDNs to the Pods' IP addresses. If Kubernetes reschedules the Pods, it will update the A records with the Pods' new IP addresses, but the A records names will not change.

ZooKeeper stores its application configuration in a file named zoo.cfg. Use kubectl exec to view the contents of the zoo.cfg file in the zk-0 Pod.

```
kubectl exec zk-0 -- cat /opt/zookeeper/conf/zoo.cfg
```

In the server.1, server.2, and server.3 properties at the bottom of the file, the 1, 2, and 3 correspond to the identifiers in the ZooKeeper servers' myid files. They are set to the FQDNs for the Pods in the zk StatefulSet.

```
clientPort=2181
dataDir=/var/lib/zookeeper/data
dataLogDir=/var/lib/zookeeper/log
tickTime=2000
initLimit=10
syncLimit=2000
maxClientCnxns=60
minSessionTimeout= 4000
maxSessionTimeout= 40000
autopurge.snapRetainCount=3
autopurge.purgeInterval=0
server.1=zk-0.zk-hs.default.svc.cluster.local:2888:3888
server.2=zk-1.zk-hs.default.svc.cluster.local:2888:3888
server.3=zk-2.zk-hs.default.svc.cluster.local:2888:3888
```

Achieving consensus

Consensus protocols require that the identifiers of each participant be unique. No two participants in the Zab protocol should claim the same unique identifier. This is necessary to allow the processes in the system to agree on which processes have committed which data. If two Pods are launched with the same ordinal, two ZooKeeper servers would both identify themselves as the same server.

```
kubectll get pods -w -l app=zk
```

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|-------------------|----------|-----|
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | ContainerCreating | 0 | 0s |
| zk-0 | 0/1 | Running | 0 | 19s |
| zk-0 | 1/1 | Running | 0 | 40s |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | ContainerCreating | 0 | 0s |
| zk-1 | 0/1 | Running | 0 | 18s |
| zk-1 | 1/1 | Running | 0 | 40s |
| zk-2 | 0/1 | Pending | 0 | 0s |
| zk-2 | 0/1 | Pending | 0 | 0s |
| zk-2 | 0/1 | ContainerCreating | 0 | 0s |
| zk-2 | 0/1 | Running | 0 | 19s |
| zk-2 | 1/1 | Running | 0 | 40s |

The A records for each Pod are entered when the Pod becomes Ready. Therefore, the FQDNs of the ZooKeeper servers will resolve to a single endpoint, and that endpoint will be the unique ZooKeeper server claiming the identity configured in its myid file.

```
zk-0.zk-hs.default.svc.cluster.local
zk-1.zk-hs.default.svc.cluster.local
zk-2.zk-hs.default.svc.cluster.local
```

This ensures that the servers properties in the ZooKeepers' zoo.cfg files represents a correctly configured ensemble.

```
server.1=zk-0.zk-hs.default.svc.cluster.local:2888:3888
server.2=zk-1.zk-hs.default.svc.cluster.local:2888:3888
server.3=zk-2.zk-hs.default.svc.cluster.local:2888:3888
```

When the servers use the Zab protocol to attempt to commit a value, they will either achieve consensus and commit the value (if leader election has succeeded and at least two of the Pods are Running and Ready), or they will fail to do so (if either of the conditions are not met). No state will arise where one server acknowledges a write on behalf of another.

Sanity testing the ensemble

The most basic sanity test is to write data to one ZooKeeper server and to read the data from another.

The command below executes the `zkCli.sh` script to write `world` to the path `/hello` on the `zk-0` Pod in the ensemble.

```
kubectrl exec zk-0 -- zkCli.sh create /hello world
```

WATCHER::

```
WatchedEvent state:SyncConnected type:None path:null
Created /hello
```

To get the data from the `zk-1` Pod use the following command.

```
kubectrl exec zk-1 -- zkCli.sh get /hello
```

The data that you created on `zk-0` is available on all the servers in the ensemble.

WATCHER::

```
WatchedEvent state:SyncConnected type:None path:null
world
cZxid = 0x100000002
ctime = Thu Dec 08 15:13:30 UTC 2016
mZxid = 0x100000002
mtime = Thu Dec 08 15:13:30 UTC 2016
pZxid = 0x100000002
cversion = 0
dataVersion = 0
aclVersion = 0
ephemeralOwner = 0x0
dataLength = 5
numChildren = 0
```

Providing durable storage

As mentioned in the [ZooKeeper Basics](#) section, ZooKeeper commits all entries to a durable WAL, and periodically writes snapshots in memory state, to storage media. Using WALs to provide durability is a common technique for applications that use consensus protocols to achieve a replicated state machine.

Use the [kubectl delete](#) command to delete the zk StatefulSet.

```
kubectl delete statefulset zk
```

```
statefulset.apps "zk" deleted
```

Watch the termination of the Pods in the StatefulSet.

```
kubectl get pods -w -l app=zk
```

When zk-0 is fully terminated, use CTRL-C to terminate kubectl.

```
zk-2  1/1  Terminating  0    9m
zk-0  1/1  Terminating  0   11m
zk-1  1/1  Terminating  0   10m
zk-2  0/1  Terminating  0    9m
zk-2  0/1  Terminating  0    9m
zk-2  0/1  Terminating  0    9m
zk-1  0/1  Terminating  0   10m
zk-1  0/1  Terminating  0   10m
zk-1  0/1  Terminating  0   10m
zk-0  0/1  Terminating  0   11m
zk-0  0/1  Terminating  0   11m
zk-0  0/1  Terminating  0   11m
```

Reapply the manifest in zookeeper.yaml.

```
kubectl apply -f https://k8s.io/examples/application/zookeeper/zookeeper.yaml
```

This creates the zk StatefulSet object, but the other API objects in the manifest are not modified because they already exist.

Watch the StatefulSet controller recreate the StatefulSet's Pods.

```
kubectl get pods -w -l app=zk
```

Once the zk-2 Pod is Running and Ready, use CTRL-C to terminate kubectl.

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|-------------------|----------|-----|
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | ContainerCreating | 0 | 0s |
| zk-0 | 0/1 | Running | 0 | 19s |
| zk-0 | 1/1 | Running | 0 | 40s |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | ContainerCreating | 0 | 0s |
| zk-1 | 0/1 | Running | 0 | 18s |
| zk-1 | 1/1 | Running | 0 | 40s |
| zk-2 | 0/1 | Pending | 0 | 0s |
| zk-2 | 0/1 | Pending | 0 | 0s |
| zk-2 | 0/1 | ContainerCreating | 0 | 0s |
| zk-2 | 0/1 | Running | 0 | 19s |
| zk-2 | 1/1 | Running | 0 | 40s |

Use the command below to get the value you entered during the [sanity test](#), from the zk-2 Pod.

```
kubectl exec zk-2 zkCli.sh get /hello
```

Even though you terminated and recreated all of the Pods in the zk StatefulSet, the ensemble still serves the original value.

WATCHER::

```
WatchedEvent state:SyncConnected type:None path:null
world
cZxid = 0x1000000002
ctime = Thu Dec 08 15:13:30 UTC 2016
mZxid = 0x1000000002
mtime = Thu Dec 08 15:13:30 UTC 2016
pZxid = 0x1000000002
cversion = 0
dataVersion = 0
aclVersion = 0
ephemeralOwner = 0x0
dataLength = 5
numChildren = 0
```

The volumeClaimTemplates field of the zk StatefulSet's spec specifies a PersistentVolume provisioned for each Pod.

```
volumeClaimTemplates:
- metadata:
  name: datadir
  annotations:
    volume.alpha.kubernetes.io/storage-class: anything
  spec:
    accessModes: [ "ReadWriteOnce" ]
    resources:
      requests:
        storage: 20Gi
```

The StatefulSet controller generates a PersistentVolumeClaim for each Pod in the StatefulSet.

Use the following command to get the StatefulSet's PersistentVolumeClaims.

```
kubectl get pvc -l app=zk
```

When the StatefulSet recreated its Pods, it remounts the Pods' PersistentVolumes.

| NAME | STATUS | VOLUME | CAPACITY | ACCESSMODES | AGE |
|--------------|--------|--|----------|-------------|-----|
| datadir-zk-0 | Bound | pvc-bed742cd-bcb1-11e6-994f-42010a800002 | 20Gi | RWO | 1h |
| datadir-zk-1 | Bound | pvc-bedd27d2-bcb1-11e6-994f-42010a800002 | 20Gi | RWO | 1h |
| datadir-zk-2 | Bound | pvc-bee0817e-bcb1-11e6-994f-42010a800002 | 20Gi | RWO | 1h |

The volumeMounts section of the StatefulSet's container template mounts the PersistentVolumes in the ZooKeeper servers' data directories.

```
volumeMounts:
- name: datadir
  mountPath: /var/lib/zookeeper
```

When a Pod in the zk StatefulSet is (re)scheduled, it will always have the same PersistentVolume mounted to the ZooKeeper server's data directory. Even when the Pods are rescheduled, all the writes made to the ZooKeeper servers' WALs, and all their snapshots, remain durable.

Ensuring consistent configuration

As noted in the [Facilitating Leader Election](#) and [Achieving Consensus](#) sections, the servers in a ZooKeeper ensemble require consistent configuration to elect a leader and form a quorum. They also require consistent configuration of the Zab protocol in order for the protocol to work correctly over a network. In our example we achieve consistent configuration by embedding the configuration directly into the manifest.

Get the zk StatefulSet.

```
kubectrl get sts zk -o yaml
```

```
...
command:
  - sh
  - -c
  - "start-zookeeper \
    --servers=3 \
    --data_dir=/var/lib/zookeeper/data \
    --data_log_dir=/var/lib/zookeeper/data/log \
    --conf_dir=/opt/zookeeper/conf \
    --client_port=2181 \
    --election_port=3888 \
    --server_port=2888 \
    --tick_time=2000 \
    --init_limit=10 \
    --sync_limit=5 \
    --heap=512M \
    --max_client_cnxns=60 \
    --snap_retain_count=3 \
    --purge_interval=12 \
    --max_session_timeout=40000 \
    --min_session_timeout=4000 \
    --log_level=INFO"
...
```

The command used to start the ZooKeeper servers passed the configuration as command line parameter. You can also use environment variables to pass configuration to the ensemble.

Configuring logging

One of the files generated by the zkGenConfig.sh script controls ZooKeeper's logging. ZooKeeper uses [Log4j](#), and, by default, it uses a time and size based rolling file appender for its logging configuration.

Use the command below to get the logging configuration from one of Pods in the zk StatefulSet.

```
kubectl exec zk-0 cat /usr/etc/zookeeper/log4j.properties
```

The logging configuration below will cause the ZooKeeper process to write all of its logs to the standard output file stream.

```
zookeeper.root.logger=CONSOLE
zookeeper.console.threshold=INFO
log4j.rootLogger=${zookeeper.root.logger}
log4j.appender.CONSOLE=org.apache.log4j.ConsoleAppender
log4j.appender.CONSOLE.Threshold=${zookeeper.console.threshold}
log4j.appender.CONSOLE.layout=org.apache.log4j.PatternLayout
log4j.appender.CONSOLE.layout.ConversionPattern=%d{ISO8601} [myid:%X{myid}] - %-5p [%t:
%C{1}@%L] - %m%n
```

This is the simplest possible way to safely log inside the container. Because the applications write logs to standard out, Kubernetes will handle log rotation for you. Kubernetes also implements a sane retention policy that ensures application logs written to standard out and standard error do not exhaust local storage media.

Use [kubectl logs](#) to retrieve the last 20 log lines from one of the Pods.

```
kubectl logs zk-0 --tail 20
```

You can view application logs written to standard out or standard error using `kubectl logs` and from the Kubernetes Dashboard.

```
2016-12-06 19:34:16,236 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxn@827] - Processing ruok command from /127.0.0.1:52740
2016-12-06 19:34:16,237 [myid:1] - INFO [Thread-1136:NIOServerCnxn@1008] - Closed socket
connection for client /127.0.0.1:52740 (no session established for client)
2016-12-06 19:34:26,155 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@192] - Accepted socket connection from /
127.0.0.1:52749
2016-12-06 19:34:26,155 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxn@827] - Processing ruok command from /127.0.0.1:52749
2016-12-06 19:34:26,156 [myid:1] - INFO [Thread-1137:NIOServerCnxn@1008] - Closed socket
connection for client /127.0.0.1:52749 (no session established for client)
2016-12-06 19:34:26,222 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@192] - Accepted socket connection from /
127.0.0.1:52750
2016-12-06 19:34:26,222 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxn@827] - Processing ruok command from /127.0.0.1:52750
2016-12-06 19:34:26,226 [myid:1] - INFO [Thread-1138:NIOServerCnxn@1008] - Closed socket
connection for client /127.0.0.1:52750 (no session established for client)
2016-12-06 19:34:36,151 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@192] - Accepted socket connection from /
127.0.0.1:52760
2016-12-06 19:34:36,152 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxn@827] - Processing ruok command from /127.0.0.1:52760
2016-12-06 19:34:36,152 [myid:1] - INFO [Thread-1139:NIOServerCnxn@1008] - Closed socket
connection for client /127.0.0.1:52760 (no session established for client)
2016-12-06 19:34:36,230 [myid:1] - INFO [NIOServerCxn.Factory:
```

```

0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@192] - Accepted socket connection from /
127.0.0.1:52761
2016-12-06 19:34:36,231 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxn@827] - Processing ruok command from /127.0.0.1:52761
2016-12-06 19:34:36,231 [myid:1] - INFO [Thread-1140:NIOServerCnxn@1008] - Closed socket
connection for client /127.0.0.1:52761 (no session established for client)
2016-12-06 19:34:46,149 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@192] - Accepted socket connection from /
127.0.0.1:52767
2016-12-06 19:34:46,149 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxn@827] - Processing ruok command from /127.0.0.1:52767
2016-12-06 19:34:46,149 [myid:1] - INFO [Thread-1141:NIOServerCnxn@1008] - Closed socket
connection for client /127.0.0.1:52767 (no session established for client)
2016-12-06 19:34:46,230 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@192] - Accepted socket connection from /
127.0.0.1:52768
2016-12-06 19:34:46,230 [myid:1] - INFO [NIOServerCxn.Factory:
0.0.0.0/0.0.0.0:2181:NIOServerCnxn@827] - Processing ruok command from /127.0.0.1:52768
2016-12-06 19:34:46,230 [myid:1] - INFO [Thread-1142:NIOServerCnxn@1008] - Closed socket
connection for client /127.0.0.1:52768 (no session established for client)

```

Kubernetes integrates with many logging solutions. You can choose a logging solution that best fits your cluster and applications. For cluster-level logging and aggregation, consider deploying a [sidecar container](#) to rotate and ship your logs.

Configuring a non-privileged user

The best practices to allow an application to run as a privileged user inside of a container are a matter of debate. If your organization requires that applications run as a non-privileged user you can use a [SecurityContext](#) to control the user that the entry point runs as.

The zk StatefulSet's Pod template contains a SecurityContext.

```

securityContext:
  runAsUser: 1000
  fsGroup: 1000

```

In the Pods' containers, UID 1000 corresponds to the zookeeper user and GID 1000 corresponds to the zookeeper group.

Get the ZooKeeper process information from the zk-0 Pod.

```
kubectl exec zk-0 -- ps -elf
```

As the runAsUser field of the securityContext object is set to 1000, instead of running as root, the ZooKeeper process runs as the zookeeper user.

```

F S UID      PID PPID C PRI NI ADDR SZ WCHAN  STIME TTY      TIME CMD
4 S zookeep+  1   0  0  80  0 - 1127 -   20:46 ?      00:00:00 sh -c zkGenConfig.sh &&
zkServer.sh start-foreground
0 S zookeep+ 27   1  0  80  0 - 1155556 - 20:46 ?      00:00:19 /usr/lib/jvm/java-8-openjdk-
amd64/bin/java -Dzookeeper.log.dir=/var/log/zookeeper -
Dzookeeper.root.logger=INFO,CONSOLE -cp /usr/bin/../build/classes:/usr/bin/../build/lib/*.jar:/
usr/bin/../share/zookeeper/zookeeper-3.4.9.jar:/usr/bin/../share/zookeeper/slf4j-

```

```
log4j12-1.6.1.jar:/usr/bin/../../share/zookeeper/slf4j-api-1.6.1.jar:/usr/bin/../../share/zookeeper/netty-3.10.5.Final.jar:/usr/bin/../../share/zookeeper/log4j-1.2.16.jar:/usr/bin/../../share/zookeeper/jline-0.9.94.jar:/usr/bin/../../src/java/lib/*.jar:/usr/bin/../../etc/zookeeper: -Xmx2G -Xms2G -Dcom.sun.management.jmxremote -Dcom.sun.management.jmxremote.local.only=false org.apache.zookeeper.server.quorum.QuorumPeerMain /usr/bin/../../etc/zookeeper/zoo.cfg
```

By default, when the Pod's PersistentVolumes is mounted to the ZooKeeper server's data directory, it is only accessible by the root user. This configuration prevents the ZooKeeper process from writing to its WAL and storing its snapshots.

Use the command below to get the file permissions of the ZooKeeper data directory on the zk-0 Pod.

```
kubectl exec -ti zk-0 -- ls -ld /var/lib/zookeeper/data
```

Because the fsGroup field of the securityContext object is set to 1000, the ownership of the Pods' PersistentVolumes is set to the zookeeper group, and the ZooKeeper process is able to read and write its data.

```
drwxr-sr-x 3 zookeeper zookeeper 4096 Dec  5 20:45 /var/lib/zookeeper/data
```

Managing the ZooKeeper process

The [ZooKeeper documentation](#) mentions that "You will want to have a supervisory process that manages each of your ZooKeeper server processes (JVM)." Utilizing a watchdog (supervisory process) to restart failed processes in a distributed system is a common pattern. When deploying an application in Kubernetes, rather than using an external utility as a supervisory process, you should use Kubernetes as the watchdog for your application.

Updating the ensemble

The zk StatefulSet is configured to use the RollingUpdate update strategy.

You can use kubectl patch to update the number of cpus allocated to the servers.

```
kubectl patch sts zk --type='json' -p='[{"op": "replace", "path": "/spec/template/spec/containers/0/resources/requests/cpu", "value": "0.3"}]'
```

```
statefulset.apps/zk patched
```

Use kubectl rollout status to watch the status of the update.

```
kubectl rollout status sts/zk
```

```
waiting for statefulset rolling update to complete 0 pods at revision zk-5db4499664...
Waiting for 1 pods to be ready...
Waiting for 1 pods to be ready...
waiting for statefulset rolling update to complete 1 pods at revision zk-5db4499664...
Waiting for 1 pods to be ready...
Waiting for 1 pods to be ready...
waiting for statefulset rolling update to complete 2 pods at revision zk-5db4499664...
Waiting for 1 pods to be ready...
```

```
Waiting for 1 pods to be ready...
statefulset rolling update complete 3 pods at revision zk-5db4499664...
```

This terminates the Pods, one at a time, in reverse ordinal order, and recreates them with the new configuration. This ensures that quorum is maintained during a rolling update.

Use the `kubectl rollout history` command to view a history of previous configurations.

```
kubectl rollout history sts/zk
```

The output is similar to this:

```
statefulsets "zk"
REVISION
1
2
```

Use the `kubectl rollout undo` command to roll back the modification.

```
kubectl rollout undo sts/zk
```

The output is similar to this:

```
statefulset.apps/zk rolled back
```

Handling process failure

[Restart Policies](#) control how Kubernetes handles process failures for the entry point of the container in a Pod. For Pods in a StatefulSet, the only appropriate RestartPolicy is Always, and this is the default value. For stateful applications you should **never** override the default policy.

Use the following command to examine the process tree for the ZooKeeper server running in the zk-0 Pod.

```
kubectl exec zk-0 -- ps -ef
```

The command used as the container's entry point has PID 1, and the ZooKeeper process, a child of the entry point, has PID 27.

```
UID      PID  PPID  C  STIME TTY      TIME CMD
zookeeper+  1    0  0 15:03 ?        00:00:00 sh -c zkGenConfig.sh && zkServer.sh start-foreground
zookeeper+  27   1  0 15:03 ?        00:00:03 /usr/lib/jvm/java-8-openjdk-amd64/bin/java -
Dzookeeper.log.dir=/var/log/zookeeper -Dzookeeper.root.logger=INFO,CONSOLE -cp /usr/
bin/./build/classes:/usr/bin/./build/lib/*:./share/zookeeper/zookeeper-3.4.9.jar:/usr/
bin/./share/zookeeper/slf4j-log4j12-1.6.1.jar:/usr/bin/./share/zookeeper/slf4j-api-1.6.1.jar:/usr/
bin/./share/zookeeper/netty-3.10.5.Final.jar:/usr/bin/./share/zookeeper/log4j-1.2.16.jar:/usr/
bin/./share/zookeeper/jline-0.9.94.jar:/usr/bin/./src/java/lib/*:./etc/zookeeper: -
Xmx2G -Xms2G -Dcom.sun.management.jmxremote -
Dcom.sun.management.jmxremote.local.only=false
org.apache.zookeeper.server.quorum.QuorumPeerMain /usr/bin/./etc/zookeeper/zoo.cfg
```

In another terminal watch the Pods in the zk StatefulSet with the following command.

```
kubectl get pod -w -l app=zk
```

In another terminal, terminate the ZooKeeper process in Pod zk-0 with the following command.

```
kubectrl exec zk-0 -- pkill java
```

The termination of the ZooKeeper process caused its parent process to terminate. Because the RestartPolicy of the container is Always, it restarted the parent process.

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|---------|----------|-----|
| zk-0 | 1/1 | Running | 0 | 21m |
| zk-1 | 1/1 | Running | 0 | 20m |
| zk-2 | 1/1 | Running | 0 | 19m |

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|---------|----------|-----|
| zk-0 | 0/1 | Error | 0 | 29m |
| zk-0 | 0/1 | Running | 1 | 29m |
| zk-0 | 1/1 | Running | 1 | 29m |

If your application uses a script (such as zkServer.sh) to launch the process that implements the application's business logic, the script must terminate with the child process. This ensures that Kubernetes will restart the application's container when the process implementing the application's business logic fails.

Testing for liveness

Configuring your application to restart failed processes is not enough to keep a distributed system healthy. There are scenarios where a system's processes can be both alive and unresponsive, or otherwise unhealthy. You should use liveness probes to notify Kubernetes that your application's processes are unhealthy and it should restart them.

The Pod template for the zk StatefulSet specifies a liveness probe.

```
livenessProbe:
  exec:
    command:
      - sh
      - -c
      - "zookeeper-ready 2181"
  initialDelaySeconds: 15
  timeoutSeconds: 5
```

The probe calls a bash script that uses the ZooKeeper ruok four letter word to test the server's health.

```
OK=$(echo ruok | nc 127.0.0.1 $1)
if [ "$OK" == "imok" ]; then
  exit 0
else
  exit 1
fi
```

In one terminal window, use the following command to watch the Pods in the zk StatefulSet.

```
kubectrl get pod -w -l app=zk
```

In another window, using the following command to delete the `zookeeper-ready` script from the file system of Pod `zk-0`.

```
kubectrl exec zk-0 -- rm /opt/zookeeper/bin/zookeeper-ready
```

When the liveness probe for the ZooKeeper process fails, Kubernetes will automatically restart the process for you, ensuring that unhealthy processes in the ensemble are restarted.

```
kubectrl get pod -w -l app=zk
```

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|---------|----------|-----|
| zk-0 | 1/1 | Running | 0 | 1h |
| zk-1 | 1/1 | Running | 0 | 1h |
| zk-2 | 1/1 | Running | 0 | 1h |

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|---------|----------|-----|
| zk-0 | 0/1 | Running | 0 | 1h |
| zk-0 | 0/1 | Running | 1 | 1h |
| zk-0 | 1/1 | Running | 1 | 1h |

Testing for readiness

Readiness is not the same as liveness. If a process is alive, it is scheduled and healthy. If a process is ready, it is able to process input. Liveness is a necessary, but not sufficient, condition for readiness. There are cases, particularly during initialization and termination, when a process can be alive but not ready.

If you specify a readiness probe, Kubernetes will ensure that your application's processes will not receive network traffic until their readiness checks pass.

For a ZooKeeper server, liveness implies readiness. Therefore, the readiness probe from the `zookeeper.yaml` manifest is identical to the liveness probe.

```
readinessProbe:
  exec:
    command:
      - sh
      - -c
      - "zookeeper-ready 2181"
  initialDelaySeconds: 15
  timeoutSeconds: 5
```

Even though the liveness and readiness probes are identical, it is important to specify both. This ensures that only healthy servers in the ZooKeeper ensemble receive network traffic.

Tolerating Node failure

ZooKeeper needs a quorum of servers to successfully commit mutations to data. For a three server ensemble, two servers must be healthy for writes to succeed. In quorum based systems, members are deployed across failure domains to ensure availability. To avoid an outage, due to the loss of an individual machine, best practices preclude co-locating multiple instances of the application on the same machine.

By default, Kubernetes may co-locate Pods in a StatefulSet on the same node. For the three server ensemble you created, if two servers are on the same node, and that node fails, the clients of your ZooKeeper service will experience an outage until at least one of the Pods can be rescheduled.

You should always provision additional capacity to allow the processes of critical systems to be rescheduled in the event of node failures. If you do so, then the outage will only last until the Kubernetes scheduler reschedules one of the ZooKeeper servers. However, if you want your service to tolerate node failures with no downtime, you should set `podAntiAffinity`.

Use the command below to get the nodes for Pods in the `zk` StatefulSet.

```
for i in 0 1 2; do kubectl get pod zk-$i --template {{.spec.nodeName}}; echo ""; done
```

All of the Pods in the `zk` StatefulSet are deployed on different nodes.

```
kubernetes-node-cxpk
kubernetes-node-a5aq
kubernetes-node-2g2d
```

This is because the Pods in the `zk` StatefulSet have a `PodAntiAffinity` specified.

```
affinity:
  podAntiAffinity:
    requiredDuringSchedulingIgnoredDuringExecution:
      - labelSelector:
          matchExpressions:
            - key: "app"
              operator: In
              values:
                - zk
        topologyKey: "kubernetes.io/hostname"
```

The `requiredDuringSchedulingIgnoredDuringExecution` field tells the Kubernetes Scheduler that it should never co-locate two Pods which have `app` label as `zk` in the domain defined by the `topologyKey`. The `topologyKey` `kubernetes.io/hostname` indicates that the domain is an individual node. Using different rules, labels, and selectors, you can extend this technique to spread your ensemble across physical, network, and power failure domains.

Surviving maintenance

In this section you will cordon and drain nodes. If you are using this tutorial on a shared cluster, be sure that this will not adversely affect other tenants.

The previous section showed you how to spread your Pods across nodes to survive unplanned node failures, but you also need to plan for temporary node failures that occur due to planned maintenance.

Use this command to get the nodes in your cluster.

```
kubectl get nodes
```

This tutorial assumes a cluster with at least four nodes. If the cluster has more than four, use [kubectl cordon](#) to cordon all but four nodes. Constraining to four nodes will ensure Kubernetes

encounters affinity and PodDisruptionBudget constraints when scheduling zookeeper Pods in the following maintenance simulation.

```
kubectcl cordon <node-name>
```

Use this command to get the zk-pdb PodDisruptionBudget.

```
kubectcl get pdb zk-pdb
```

The max-unavailable field indicates to Kubernetes that at most one Pod from zk StatefulSet can be unavailable at any time.

| NAME | MIN-AVAILABLE | MAX-UNAVAILABLE | ALLOWED-DISRUPTIONS | AGE |
|--------|---------------|-----------------|---------------------|-----|
| zk-pdb | N/A | 1 | 1 | |

In one terminal, use this command to watch the Pods in the zk StatefulSet.

```
kubectcl get pods -w -l app=zk
```

In another terminal, use this command to get the nodes that the Pods are currently scheduled on.

```
for i in 0 1 2; do kubectcl get pod zk-$i --template {{.spec.nodeName}}; echo ""; done
```

The output is similar to this:

```
kubernetes-node-pb41
kubernetes-node-ixsl
kubernetes-node-i4c4
```

Use [kubectcl drain](#) to cordon and drain the node on which the zk-0 Pod is scheduled.

```
kubectcl drain $(kubectcl get pod zk-0 --template {{.spec.nodeName}}) --ignore-daemonsets --force
--delete-emptydir-data
```

The output is similar to this:

```
node "kubernetes-node-pb41" cordoned

WARNING: Deleting pods not managed by ReplicationController, ReplicaSet, Job, or
DaemonSet: fluentd-cloud-logging-kubernetes-node-pb41, kube-proxy-kubernetes-node-pb41;
Ignoring DaemonSet-managed pods: node-problem-detector-v0.1-o5elz
pod "zk-0" deleted
node "kubernetes-node-pb41" drained
```

As there are four nodes in your cluster, kubectcl drain, succeeds and the zk-0 is rescheduled to another node.

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|---------|----------|-----|
| zk-0 | 1/1 | Running | 2 | 1h |
| zk-1 | 1/1 | Running | 0 | 1h |
| zk-2 | 1/1 | Running | 0 | 1h |

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|-------------|----------|-----|
| zk-0 | 1/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Terminating | 2 | 2h |

| | | | | |
|------|-----|-------------------|---|-----|
| zk-0 | 0/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | ContainerCreating | 0 | 0s |
| zk-0 | 0/1 | Running | 0 | 51s |
| zk-0 | 1/1 | Running | 0 | 1m |

Keep watching the StatefulSet's Pods in the first terminal and drain the node on which zk-1 is scheduled.

```
kubectldrain $(kubectldget pod zk-1 --template {{.spec.nodeName}}) --ignore-daemonsets --force
--delete-emptydir-data
```

The output is similar to this:

```
"kubernetes-node-ixsl" cordoned
WARNING: Deleting pods not managed by ReplicationController, ReplicaSet, Job, or
DaemonSet: fluentd-cloud-logging-kubernetes-node-ixsl, kube-proxy-kubernetes-node-ixsl;
Ignoring DaemonSet-managed pods: node-problem-detector-v0.1-voc74
pod "zk-1" deleted
node "kubernetes-node-ixsl" drained
```

The zk-1 Pod cannot be scheduled because the zk StatefulSet contains a PodAntiAffinity rule preventing co-location of the Pods, and as only two nodes are schedulable, the Pod will remain in a Pending state.

```
kubectldget pods -w -l app=zk
```

The output is similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|---------|----------|-----|
| zk-0 | 1/1 | Running | 2 | 1h |
| zk-1 | 1/1 | Running | 0 | 1h |
| zk-2 | 1/1 | Running | 0 | 1h |

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|-------------------|----------|-----|
| zk-0 | 1/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | ContainerCreating | 0 | 0s |
| zk-0 | 0/1 | Running | 0 | 51s |
| zk-0 | 1/1 | Running | 0 | 1m |
| zk-1 | 1/1 | Terminating | 0 | 2h |
| zk-1 | 0/1 | Terminating | 0 | 2h |
| zk-1 | 0/1 | Terminating | 0 | 2h |
| zk-1 | 0/1 | Terminating | 0 | 2h |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | Pending | 0 | 0s |

Continue to watch the Pods of the StatefulSet, and drain the node on which zk-2 is scheduled.

```
kubectldrain $(kubectldget pod zk-2 --template {{.spec.nodeName}}) --ignore-daemonsets --force
--delete-emptydir-data
```

The output is similar to this:

```
node "kubernetes-node-i4c4" cordoned
```

```
WARNING: Deleting pods not managed by ReplicationController, ReplicaSet, Job, or
DaemonSet: fluentd-cloud-logging-kubernetes-node-i4c4, kube-proxy-kubernetes-node-i4c4;
Ignoring DaemonSet-managed pods: node-problem-detector-v0.1-dyrog
WARNING: Ignoring DaemonSet-managed pods: node-problem-detector-v0.1-dyrog; Deleting
pods not managed by ReplicationController, ReplicaSet, Job, or DaemonSet: fluentd-cloud-
logging-kubernetes-node-i4c4, kube-proxy-kubernetes-node-i4c4
There are pending pods when an error occurred: Cannot evict pod as it would violate the pod's
disruption budget.
pod/zk-2
```

Use CTRL-C to terminate kubectld.

You cannot drain the third node because evicting zk-2 would violate zk-budget. However, the node will remain cordoned.

Use zkCli.sh to retrieve the value you entered during the sanity test from zk-0.

```
kubectldexec zk-0 zkCli.sh get /hello
```

The service is still available because its PodDisruptionBudget is respected.

```
WatchedEvent state:SyncConnected type:None path:null
world
cZxid = 0x2000000002
ctime = Wed Dec 07 00:08:59 UTC 2016
mZxid = 0x2000000002
mtime = Wed Dec 07 00:08:59 UTC 2016
pZxid = 0x2000000002
cversion = 0
dataVersion = 0
aclVersion = 0
ephemeralOwner = 0x0
dataLength = 5
numChildren = 0
```

Use [kubectlduncordon](#) to uncordon the first node.

```
kubectlduncordon kubernetes-node-pb41
```

The output is similar to this:

```
node "kubernetes-node-pb41" uncordoned
```

zk-1 is rescheduled on this node. Wait until zk-1 is Running and Ready.

```
kubectldget pods -w -l app=zk
```

The output is similar to this:

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|---------|----------|-----|
| zk-0 | 1/1 | Running | 2 | 1h |
| zk-1 | 1/1 | Running | 0 | 1h |
| zk-2 | 1/1 | Running | 0 | 1h |

| NAME | READY | STATUS | RESTARTS | AGE |
|------|-------|-------------------|----------|-----|
| zk-0 | 1/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Terminating | 2 | 2h |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | Pending | 0 | 0s |
| zk-0 | 0/1 | ContainerCreating | 0 | 0s |
| zk-0 | 0/1 | Running | 0 | 51s |
| zk-0 | 1/1 | Running | 0 | 1m |
| zk-1 | 1/1 | Terminating | 0 | 2h |
| zk-1 | 0/1 | Terminating | 0 | 2h |
| zk-1 | 0/1 | Terminating | 0 | 2h |
| zk-1 | 0/1 | Terminating | 0 | 2h |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | Pending | 0 | 0s |
| zk-1 | 0/1 | Pending | 0 | 12m |
| zk-1 | 0/1 | ContainerCreating | 0 | 12m |
| zk-1 | 0/1 | Running | 0 | 13m |
| zk-1 | 1/1 | Running | 0 | 13m |

Attempt to drain the node on which zk-2 is scheduled.

```
kubectl drain $(kubectl get pod zk-2 --template {{.spec.nodeName}}) --ignore-daemonsets --force --delete-emptydir-data
```

The output is similar to this:

```
node "kubernetes-node-i4c4" already cordoned
WARNING: Deleting pods not managed by ReplicationController, ReplicaSet, Job, or
DaemonSet: fluentd-cloud-logging-kubernetes-node-i4c4, kube-proxy-kubernetes-node-i4c4;
Ignoring DaemonSet-managed pods: node-problem-detector-v0.1-dyrog
pod "heapster-v1.2.0-2604621511-wht1r" deleted
pod "zk-2" deleted
node "kubernetes-node-i4c4" drained
```

This time kubectl drain succeeds.

Uncordon the second node to allow zk-2 to be rescheduled.

```
kubectl uncordon kubernetes-node-ixsl
```

The output is similar to this:

```
node "kubernetes-node-ixsl" uncordoned
```

You can use kubectl drain in conjunction with PodDisruptionBudgets to ensure that your services remain available during maintenance. If drain is used to cordon nodes and evict pods prior to taking the node offline for maintenance, services that express a disruption budget will

have that budget respected. You should always allocate additional capacity for critical services so that their Pods can be immediately rescheduled.

Cleaning up

- Use `kubectl uncordon` to uncordon all the nodes in your cluster.
- You must delete the persistent storage media for the PersistentVolumes used in this tutorial. Follow the necessary steps, based on your environment, storage configuration, and provisioning method, to ensure that all storage is reclaimed.

Services

[Connecting Applications with Services](#)

[Using Source IP](#)

[Explore Termination Behavior for Pods And Their Endpoints](#)

Connecting Applications with Services

The Kubernetes model for connecting containers

Now that you have a continuously running, replicated application you can expose it on a network.

Kubernetes assumes that pods can communicate with other pods, regardless of which host they land on. Kubernetes gives every pod its own cluster-private IP address, so you do not need to explicitly create links between pods or map container ports to host ports. This means that containers within a Pod can all reach each other's ports on localhost, and all pods in a cluster can see each other without NAT. The rest of this document elaborates on how you can run reliable services on such a networking model.

This tutorial uses a simple nginx web server to demonstrate the concept.

Exposing pods to the cluster

We did this in a previous example, but let's do it once again and focus on the networking perspective. Create an nginx Pod, and note that it has a container port specification:

[service/networking/run-my-nginx.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-nginx
spec:
  selector:
```

```
matchLabels:
  run: my-nginx
replicas: 2
template:
  metadata:
    labels:
      run: my-nginx
  spec:
    containers:
      - name: my-nginx
        image: nginx
        ports:
          - containerPort: 80
```

This makes it accessible from any node in your cluster. Check the nodes the Pod is running on:

```
kubectl apply -f ./run-my-nginx.yaml
kubectl get pods -l run=my-nginx -o wide
```

| NAME | READY | STATUS | RESTARTS | AGE | IP | NODE |
|---------------------------|-------|---------|----------|-----|------------|------------------------|
| my-nginx-3800858182-jr4a2 | 1/1 | Running | 0 | 13s | 10.244.3.4 | kubernetes-minion-905m |
| my-nginx-3800858182-kna2y | 1/1 | Running | 0 | 13s | 10.244.2.5 | kubernetes-minion-ljyd |

Check your pods' IPs:

```
kubectl get pods -l run=my-nginx -o custom-columns=POD_IP:.status.podIPs
POD_IP
[map[ip:10.244.3.4]]
[map[ip:10.244.2.5]]
```

You should be able to ssh into any node in your cluster and use a tool such as curl to make queries against both IPs. Note that the containers are *not* using port 80 on the node, nor are there any special NAT rules to route traffic to the pod. This means you can run multiple nginx pods on the same node all using the same containerPort, and access them from any other pod or node in your cluster using the assigned IP address for the pod. If you want to arrange for a specific port on the host Node to be forwarded to backing Pods, you can - but the networking model should mean that you do not need to do so.

You can read more about the [Kubernetes Networking Model](#) if you're curious.

Creating a Service

So we have pods running nginx in a flat, cluster wide, address space. In theory, you could talk to these pods directly, but what happens when a node dies? The pods die with it, and the ReplicaSet inside the Deployment will create new ones, with different IPs. This is the problem a Service solves.

A Kubernetes Service is an abstraction which defines a logical set of Pods running somewhere in your cluster, that all provide the same functionality. When created, each Service is assigned a unique IP address (also called clusterIP). This address is tied to the lifespan of the Service, and

will not change while the Service is alive. Pods can be configured to talk to the Service, and know that communication to the Service will be automatically load-balanced out to some pod that is a member of the Service.

You can create a Service for your 2 nginx replicas with `kubectl expose`:

```
kubectl expose deployment/my-nginx
```

```
service/my-nginx exposed
```

This is equivalent to `kubectl apply -f` the following yaml:

[service/networking/nginx-svc.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: my-nginx
  labels:
    run: my-nginx
spec:
  ports:
    - port: 80
      protocol: TCP
  selector:
    run: my-nginx
```

This specification will create a Service which targets TCP port 80 on any Pod with the `run: my-nginx` label, and expose it on an abstracted Service port (`targetPort`: is the port the container accepts traffic on, `port`: is the abstracted Service port, which can be any port other pods use to access the Service). View [Service](#) API object to see the list of supported fields in service definition. Check your Service:

```
kubectl get svc my-nginx
```

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|----------|-----------|--------------|-------------|---------|-----|
| my-nginx | ClusterIP | 10.0.162.149 | <none> | 80/TCP | 21s |

As mentioned previously, a Service is backed by a group of Pods. These Pods are exposed through [EndpointSlices](#). The Service's selector will be evaluated continuously and the results will be POSTed to an EndpointSlice that is connected to the Service using [labels](#). When a Pod dies, it is automatically removed from the EndpointSlices that contain it as an endpoint. New Pods that match the Service's selector will automatically get added to an EndpointSlice for that Service. Check the endpoints, and note that the IPs are the same as the Pods created in the first step:

```
kubectl describe svc my-nginx
```

```
Name:          my-nginx
Namespace:     default
Labels:        run=my-nginx
Annotations:   <none>
Selector:      run=my-nginx
Type:          ClusterIP
```



```
IP Family Policy: SingleStack
IP Families:      IPv4
IP:              10.0.162.149
IPs:             10.0.162.149
Port:            <unset> 80/TCP
TargetPort:      80/TCP
Endpoints:       10.244.2.5:80,10.244.3.4:80
Session Affinity: None
Events:          <none>
```

```
kubectl get endpointslices -l kubernetes.io/service-name=my-nginx
```

| NAME | ADDRESSTYPE | PORTS | ENDPOINTS | AGE |
|----------------|-------------|-------|-----------------------|-----|
| my-nginx-7vzhx | IPv4 | 80 | 10.244.2.5,10.244.3.4 | 21s |

You should now be able to curl the nginx Service on `<CLUSTER-IP>:<PORT>` from any node in your cluster. Note that the Service IP is completely virtual, it never hits the wire. If you're curious about how this works you can read more about the [service proxy](#).

Accessing the Service

Kubernetes supports 2 primary modes of finding a Service - environment variables and DNS. The former works out of the box while the latter requires the [CoreDNS cluster addon](#).

Note:

If the service environment variables are not desired (because possible clashing with expected program ones, too many variables to process, only using DNS, etc) you can disable this mode by setting the `enableServiceLinks` flag to false on the [pod spec](#).

Environment Variables

When a Pod runs on a Node, the kubelet adds a set of environment variables for each active Service. This introduces an ordering problem. To see why, inspect the environment of your running nginx Pods (your Pod name will be different):

```
kubectl exec my-nginx-3800858182-jr4a2 -- printenv | grep SERVICE
```

```
KUBERNETES_SERVICE_HOST=10.0.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
```

Note there's no mention of your Service. This is because you created the replicas before the Service. Another disadvantage of doing this is that the scheduler might put both Pods on the same machine, which will take your entire Service down if it dies. We can do this the right way by killing the 2 Pods and waiting for the Deployment to recreate them. This time the Service exists *before* the replicas. This will give you scheduler-level Service spreading of your Pods (provided all your nodes have equal capacity), as well as the right environment variables:

```
kubectl scale deployment my-nginx --replicas=0; kubectl scale deployment my-nginx --replicas=2;
```

```
kubectrl get pods -l run=my-nginx -o wide
```

| NAME | READY | STATUS | RESTARTS | AGE | IP | NODE |
|---------------------------------------|-------|---------|----------|-----|------------|--------------------|
| my-nginx-3800858182-e9ihh-ljyd | 1/1 | Running | 0 | 5s | 10.244.2.7 | kubernetes-minion- |
| my-nginx-3800858182-j4rm4-minion-905m | 1/1 | Running | 0 | 5s | 10.244.3.8 | kubernetes- |

You may notice that the pods have different names, since they are killed and recreated.

```
kubectrl exec my-nginx-3800858182-e9ihh -- printenv | grep SERVICE
```

```
KUBERNETES_SERVICE_PORT=443
MY_NGINX_SERVICE_HOST=10.0.162.149
KUBERNETES_SERVICE_HOST=10.0.0.1
MY_NGINX_SERVICE_PORT=80
KUBERNETES_SERVICE_PORT_HTTPS=443
```

DNS

Kubernetes offers a DNS cluster addon Service that automatically assigns dns names to other Services. You can check if it's running on your cluster:

```
kubectrl get services kube-dns --namespace=kube-system
```

| NAME | TYPE | CLUSTER-IP | EXTERNAL-IP | PORT(S) | AGE |
|----------|-----------|------------|-------------|---------------|-----|
| kube-dns | ClusterIP | 10.0.0.10 | <none> | 53/UDP,53/TCP | 8m |

The rest of this section will assume you have a Service with a long lived IP (my-nginx), and a DNS server that has assigned a name to that IP. Here we use the CoreDNS cluster addon (application name kube-dns), so you can talk to the Service from any pod in your cluster using standard methods (e.g. `gethostbyname()`). If CoreDNS isn't running, you can enable it referring to the [CoreDNS README](#) or [Installing CoreDNS](#). Let's run another curl application to test this:

```
kubectrl run curl --image=radial/busyboxplus:curl -i --tty --rm
```

Waiting for pod default/curl-131556218-9fnch to be running, status is Pending, pod ready: false
Hit enter for command prompt

Then, hit enter and run `nslookup my-nginx`:

```
[ root@curl-131556218-9fnch:/ ]$ nslookup my-nginx
Server: 10.0.0.10
Address 1: 10.0.0.10

Name: my-nginx
Address 1: 10.0.162.149
```

Securing the Service

Till now we have only accessed the nginx server from within the cluster. Before exposing the Service to the internet, you want to make sure the communication channel is secure. For this, you will need:

- Self signed certificates for https (unless you already have an identity certificate)
- An nginx server configured to use the certificates
- A [secret](#) that makes the certificates accessible to pods

You can acquire all these from the [nginx https example](#). This requires having go and make tools installed. If you don't want to install those, then follow the manual steps later. In short:

```
make keys KEY=/tmp/nginx.key CERT=/tmp/nginx.crt
kubectl create secret tls nginxsecret --key /tmp/nginx.key --cert /tmp/nginx.crt
```

```
secret/nginxsecret created
```

```
kubectl get secrets
```

| NAME | TYPE | DATA | AGE |
|-------------|-------------------|------|-----|
| nginxsecret | kubernetes.io/tls | 2 | 1m |

And also the configmap:

```
kubectl create configmap nginxconfigmap --from-file=default.conf
```

You can find an example for default.conf in [the Kubernetes examples project repo](#).

```
configmap/nginxconfigmap created
```

```
kubectl get configmaps
```

| NAME | DATA | AGE |
|----------------|------|------|
| nginxconfigmap | 1 | 114s |

You can view the details of the nginxconfigmap ConfigMap using the following command:

```
kubectl describe configmap nginxconfigmap
```

The output is similar to:

```
Name:      nginxconfigmap
Namespace: default
Labels:    <none>
Annotations: <none>

Data
====
default.conf:
----
server {
    listen 80 default_server;
    listen [::]:80 default_server ipv6only=on;
```

```

listen 443 ssl;

root /usr/share/nginx/html;
index index.html;

server_name localhost;
ssl_certificate /etc/nginx/ssl/tls.crt;
ssl_certificate_key /etc/nginx/ssl/tls.key;

location / {
    try_files $uri $uri/ =404;
}
}

```

BinaryData

====

Events: <none>

Following are the manual steps to follow in case you run into problems running make (on windows for example):

```

# Create a public private key pair
openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /d/tmp/nginx.key -out /d/tmp/nginx.crt -subj "/CN=my-nginx/O=my-nginx"
# Convert the keys to base64 encoding
cat /d/tmp/nginx.crt | base64
cat /d/tmp/nginx.key | base64

```

Use the output from the previous commands to create a yaml file as follows. The base64 encoded value should all be on a single line.

```

apiVersion: "v1"
kind: "Secret"
metadata:
  name: "nginxsecret"
  namespace: "default"
type: kubernetes.io/tls
data:
  tls.crt: "LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSURlekNDQWdlZ0F3SUJBZ0lKQUp5M3lQK0pzMlplJTUEwR0NTcUdTSWlZRFFFQkJRUVFNQ1l4RVRBUEJnTIYKQkFNVENHNW5hVzU0YzNaak1SRXdEd1lEVlFRS0V3aHVVaMmx1ZUhOMll6QWVGdzB4TnpFd01qWXdOekEzTVRKYQpGdzB4T0RFd01qWXdOekEzTVRKYU1DWXhFVEFQQmdOVk1BTVRDRzVuYVc1NGMzWm1pNUkV3RHdZRFZRUUtFd2h1CloybHVlSE4yWXPdQ0FTSXdeUUVlKS29aSWWh2Y05BUUVCQlFBRGdnRVB BRENDQVFvQ2dnRUJBSjFhSU1SOVdWM0IKMlZlQlRMRmtobDRONXljMEJxYUhlQktMSnJMcY8vdzZhU3hRS29GbHlJSU94NGUrMlN5ajBFcndCLzlyTnBwbQppeW1CL3JkRldkOXg5UWWhBQUxCZkVaTmNiV3NsTVFVcnhBZW50VWt1dk1vLzgvMHRpbGhjc3paenJEYVJ4NEo5Ci82UVRtVVI3a0ZTWUpOWTVQZkR3cGc3dlVvaDZmZ1Voam92VG42eHNVR0M2QURVODBpNXFlZWWhNeVI1N2lmU2YKNHJpaXdlY3hnL3lZR1JBRS9mRTRqakxCdmdONjc2SU90S01rZXV3R0ljNDFhd05tNnNTS2RqYUNGeGpYSnZaZQp2by9kTlEyYHhHWCTKT2l3SEhXbXNhdGp4WTRaNVk3R1ZoK0QrWnYvcW1mMFgvdVY0Rmo1NzV3ajFMWVBocWtsCmdhSXZYRyt4U1FVQ0F3RUFBYU5RTUU0d0hRWURWUjBPQkJZRUZPNG9OWWk3YXc1OUlsYkROMzhIYkduYnhFVjcKT"

```

```
UI4R0ExVWRJd1FZTUJhQUZPNG9OWkI3YXc1OUlsYkROMzhIYkduYnhFVjdNQXdHQTfVZE
V3UUNZQU1CQWY4dwpEUUVKS29aSWH2Y05BUUVGQlFBRGdnRUJBRVhTMW9FU0lFaXdyM
DhWcVA0K2NwTHI3TW5FMTducDBvMm14aIFvCjRGb0RvRjdRZnZqeE04Tzd2TjB0clcb2pGS
W0vWDE4ZnZaL3k4ZzVaWG40Vm8zc3hKVmRBcStNZC9jTStzUGEKNmJjTkNUekZqeFpUV0U
rKzE5NS9zb2dmOUZ3VDVdK3U2Q3B5N0M3MTZvUXRUakViV05VdEt4cXI0Nk1OZWNCMAp
wRFhWZmdWQTRadkR4NFo3S2RiZDY5eXM3OVFHYYmg5ZW1PZ05NZFlsSUswSGt0ejF5WU4v
bVpmK3FqTkJqbWZjCkNnMnlwbGQ0Wi8rUUNQZjl3SkoybFlrY2FnT0R4elBWcGxNSEcybzgvT
HFDdnh6elZPUDUxeXdLZEtxaUMwSVEKQ0I5T2wwWW5scE9UNEh1b2hSUzBPOSTlMm9KdF
ZsNUlYczRpbDlhZ3RTVXFxUIU9Ci0tLS0tRU5EIENFUlRJRklDQVRFLS0tLS0K"
```

```
tls.key: "LS0tLS1CRUdJTiBQUklWQVRFIEtFWS0tLS0tCk1JSUV2UUlCQURBTkJna3Foa2lHOXc
wQkFRRUZBQVNDQktjd2dnU2pBZ0VBQW9JQkFRQ2RhaURFZlZsZHDkbFIkd1V5eFpJWmVE
ZWNuTkFhbWh4d1NpeWF5N1AvOE9ta3NVQ3FCWmNpQ0RzZUh2dGtzbzlCSzhBZi9WemFh
Wm9zcApnZjYzUlZuZmNmVUIRQUN3WHhHVfHhMXJKVEVGSzhRSHA3VkpMcnplUC9QO
UxZcFIYTE0yYzZ3MmtjZUNmZitrCkU1bEVlNUJvbnUNUV09UM3c4S1PNzFLSWVuNEZJWTZ
MMDUrc2JGQmd1Z0ExUE5JdWFubm9UTWtZTRuMG4rTDQKb3NCM01ZUDhtQmtRQlAzeE9
JNHl3YjREZXUraURyU2pKSHJzQmlIT05Xc0RadXJfFaXVJMmdoY1kxeWIyWHI2UAozVFVOCG
NSbC9pVG9zQngxcHJHcl4V09HZVdPeGxZZmcvbWlVnNBUOUYvNWxlQlkrZStjSTITMkQ0YX
BKWUdpCkwxeHZzVWtGQWdNqkFBRUNnZ0VBZFhCK0xkbk8ySElOTGo5bWRsb25IUGlHW
WVzZ294RGQwci9hQ1Zkank4dlEKTjIwL3FQWkUxek1yall6Ry9kVGhTMmMwc0QxaTBXSjdw
R1lGb0xtXlWTjltY0FXUTM5SjM0VHZaU2FFSWZWNGo5TE1jUHhNTmFsNjRLMFRVbUFQZy
tGam9QSFlhUUxLOERLOUtnNXNRSE5pOWNzMIY5ckd6VWlVZWtBL0RBUIBTCII3L2ZjUFBac
DRuRWVBZmI3WTk1R1llb1p5V21SU3VKdlNyblBESGtUdW1vVlVWdkxMRHRzaG9reUxiTWV
tN3oKMmJzVmpwSW1GTHJqbGtmQXlpNHg0WjJrV3YyMFRrdWtsZU1jaVlMbjk4QWxiRi9DS
mRLM3QraTRoMTVIR2ZQegpoTnh3bk9QdlVTaDR2Q0o3c2Q5TmtEUGJvS2JneVVHOXBYamZ
hRGR2UVFLQmdRRFFLM01nUkhkQ1pKNVFqZWFKClFGdXF4cHdnNzhZTjQyL1NwenlUYmtG
cVfoQWtyczJxWGx1MDZBRzhrZzIzQkswaHkzaE9zSGgxcXRVK3NHZVAKOWRERHBsUWV0
ODZsY2FIR3hoc0V0L1R6cEdtNGFKSm5oNzVVaTVGZk9QTDhPTm1FZ3MxMVRhUldhNzZxelR
yMgphRlpjQ2pWV1g0YnRSTHVwSkgrMjZnY0FhUUtCZ1FEQmxVSUUzTnNVOFBZEYvL25sQ
VB5VWs1T3lDdWc3dmVyClUycXlrdXFzYnBkSi9hODViT1JhM05IVmpVM25uRGpHVHbWae9J
eXg5TEFrc2RwZEFjVmxvcG9HODhXYk9lMTAKMUdqbnkySmdDK3JVWUZiRGtpUGx1K09IYn
RnOXFYcGJMSHBzUVpsMGhucDBYSFNyVm9CMUliQndnMGEyOFVadApCbFBtWmc2d1BRS0
JnRHVIUVV2SDZHYTNDVUsxNFdmOFhlcFFnMU16M2VvWTBPM5iSDRvZUZKZmcraEppS
XlnCm9RN3hqWldVR3Blc3AyblRtcHErQWlSNzdyRVhsdlhtOEIVU2FsbkNiRGlKY01Pc29RdFBZ
NS9NczJMRm5LQTQKaENmL0pWb2FtZm1nZEN0ZGtFMXNINE9MR2lJVHdEbTRpb0dWZGIw
MllnbzFyb2htNUplMUI3MkpBb0dBUW01UQpHNDhXOTVhL0w1eSt5dCsyZ3YvUHM2VnBvMj
ZlTzRNQ3ljazJVem9ZWE9IYnNkODJkaC8xT2sybGdHZlI2K3VuCnc1YytZUXRSTHlhQmd3MUt
pbGhFZDBKTWU3cGpUSVpnQWJ0LzVPbnlDak9OVXN2aDJsS2lrQ1Z2dTZsZlBjNkQKckliT2ZI
aHhXV0RZK2Q1TGN1YSt2NzJ0RkxhenJsSlBsRzI0ZHhrQ2dZRUF5ellzT3UyMDNRVVV6bUlCR
kwzZAp4Wm5XZ0JLSEo3TnNxcGFwb2RjL0d5aGVycjFDZzE2MmJaSjJdV2RsZkI0VEdtUjZZdm
xTZEFOOFRwUWhFbUtKCnFBLzVzdHdxNWd0WGVLOVJmMWxXK29xNThRNTBxMmk1NVd
UTThoSDZhTjlaMTltZ0FGdE5VdGNqQUx2dFYxdEYKWSs4WFJkSHJaRnBIWlI2NWkwVW1Vb
Gc9Ci0tLS0tRU5EIFBSSVZBVEUgS0VZLS0tLS0K"
```

Now create the secrets using the file:

```
kubectl apply -f nginxsecrets.yaml
kubectl get secrets
```

| NAME | TYPE | DATA | AGE |
|-------------|-------------------|------|-----|
| nginxsecret | kubernetes.io/tls | 2 | 1m |

Now modify your nginx replicas to start an https server using the certificate in the secret, and the Service, to expose both ports (80 and 443):

[service/networking/nginx-secure-app.yaml](#)

```
apiVersion: v1
kind: Service
metadata:
  name: my-nginx
  labels:
    run: my-nginx
spec:
  type: NodePort
  ports:
    - port: 8080
      targetPort: 80
      protocol: TCP
      name: http
    - port: 443
      protocol: TCP
      name: https
  selector:
    run: my-nginx
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-nginx
spec:
  selector:
    matchLabels:
      run: my-nginx
  replicas: 1
  template:
    metadata:
      labels:
        run: my-nginx
    spec:
      volumes:
        - name: secret-volume
          secret:
            secretName: nginxsecret
        - name: configmap-volume
          configMap:
            name: nginxconfigmap
      containers:
        - name: nginxhttps
          image: bprashanth/nginxhttps:1.0
          ports:
            - containerPort: 443
            - containerPort: 80
          volumeMounts:
            - mountPath: /etc/nginx/ssl
              name: secret-volume
            - mountPath: /etc/nginx/conf.d
              name: configmap-volume
```

Noteworthy points about the nginx-secure-app manifest:

- It contains both Deployment and Service specification in the same file.
- The [nginx server](#) serves HTTP traffic on port 80 and HTTPS traffic on 443, and nginx Service exposes both ports.
- Each container has access to the keys through a volume mounted at /etc/nginx/ssl. This is set up *before* the nginx server is started.

```
kubectl delete deployments,svc my-nginx; kubectl create -f ./nginx-secure-app.yaml
```

At this point you can reach the nginx server from any node.

```
kubectl get pods -l run=my-nginx -o custom-columns=POD_IP:.status.podIPs
POD_IP
[map[ip:10.244.3.5]]
```

```
node $ curl -k https://10.244.3.5
...
<h1>Welcome to nginx!</h1>
```

Note how we supplied the `-k` parameter to curl in the last step, this is because we don't know anything about the pods running nginx at certificate generation time, so we have to tell curl to ignore the CName mismatch. By creating a Service we linked the CName used in the certificate with the actual DNS name used by pods during Service lookup. Let's test this from a pod (the same secret is being reused for simplicity, the pod only needs nginx.crt to access the Service):

[service/networking/curlpod.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: curl-deployment
spec:
  selector:
    matchLabels:
      app: curlpod
  replicas: 1
  template:
    metadata:
      labels:
        app: curlpod
    spec:
      volumes:
        - name: secret-volume
          secret:
            secretName: nginxsecret
      containers:
        - name: curlpod
          command:
            - sh
            - -c
            - while true; do sleep 1; done
          image: radial/busyboxplus:curl
          volumeMounts:
```

```
- mountPath: /etc/nginx/ssl
  name: secret-volume
```

```
kubectl apply -f ./curlpod.yaml
kubectl get pods -l app=curlpod
```

NAME	READY	STATUS	RESTARTS	AGE
curl-deployment-1515033274-1410r	1/1	Running	0	1m

```
kubectl exec curl-deployment-1515033274-1410r -- curl https://my-nginx --cacert /etc/nginx/ssl/
tls.crt
```

```
...
<title>Welcome to nginx!</title>
...
```

Exposing the Service

For some parts of your applications you may want to expose a Service onto an external IP address. Kubernetes supports two ways of doing this: NodePorts and LoadBalancers. The Service created in the last section already used NodePort, so your nginx HTTPS replica is ready to serve traffic on the internet if your node has a public IP.

```
kubectl get svc my-nginx -o yaml | grep nodePort -C 5
uid: 07191fb3-f61a-11e5-8ae5-42010af00002
```

```
spec:
  clusterIP: 10.0.162.149
  ports:
  - name: http
    nodePort: 31704
    port: 8080
    protocol: TCP
    targetPort: 80
  - name: https
    nodePort: 32453
    port: 443
    protocol: TCP
    targetPort: 443
  selector:
    run: my-nginx
```

```
kubectl get nodes -o yaml | grep ExternalIP -C 1
```

```
- address: 104.197.41.11
  type: ExternalIP
  allocatable:
```

```
--
```

```
- address: 23.251.152.56
  type: ExternalIP
  allocatable:
```

```
...
```

```
$ curl https://<EXTERNAL-IP>:<NODE-PORT> -k
```

```
...
<h1>Welcome to nginx!</h1>
```


Let's now recreate the Service to use a cloud load balancer. Change the Type of my-nginx Service from NodePort to LoadBalancer:

```
kubectl edit svc my-nginx
kubectl get svc my-nginx
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
my-nginx	LoadBalancer	10.0.162.149	xx.xxx.xxx.xxx	8080:30163/TCP	21s

```
curl https://<EXTERNAL-IP> -k
...
<title>Welcome to nginx!</title>
```

The IP address in the EXTERNAL-IP column is the one that is available on the public internet. The CLUSTER-IP is only available inside your cluster/private cloud network.

Note that on AWS, type LoadBalancer creates an ELB, which uses a (long) hostname, not an IP. It's too long to fit in the standard kubectl get svc output, in fact, so you'll need to do kubectl describe service my-nginx to see it. You'll see something like this:

```
kubectl describe service my-nginx
...
LoadBalancer Ingress: a320587ffd19711e5a37606cf4a74574-1142138393.us-
east-1.elb.amazonaws.com
...
```

What's next

- Learn more about [Using a Service to Access an Application in a Cluster](#)
- Learn more about [Connecting a Front End to a Back End Using a Service](#)
- Learn more about [Creating an External Load Balancer](#)

Using Source IP

Applications running in a Kubernetes cluster find and communicate with each other, and the outside world, through the Service abstraction. This document explains what happens to the source IP of packets sent to different types of Services, and how you can toggle this behavior according to your needs.

Before you begin

Terminology

This document makes use of the following terms:

[NAT](#)

Network address translation

[Source NAT](#)

Replacing the source IP on a packet; in this page, that usually means replacing with the IP address of a node.

[Destination NAT](#)

Replacing the destination IP on a packet; in this page, that usually means replacing with the IP address of a [Pod](#)

VIP

A virtual IP address, such as the one assigned to every [Service](#) in Kubernetes

kube-proxy

A network daemon that orchestrates Service VIP management on every node

Prerequisites

You need to have a Kubernetes cluster, and the kubectl command-line tool must be configured to communicate with your cluster. It is recommended to run this tutorial on a cluster with at least two nodes that are not acting as control plane hosts. If you do not already have a cluster, you can create one by using [minikube](#) or you can use one of these Kubernetes playgrounds:

- [Killercodea](#)
- [Play with Kubernetes](#)

The examples use a small nginx webserver that echoes back the source IP of requests it receives through an HTTP header. You can create it as follows:

Note:

The image in the following command only runs on AMD64 architectures.

```
kubectl create deployment source-ip-app --image=registry.k8s.io/echoserver:1.4
```

The output is:

```
deployment.apps/source-ip-app created
```

Objectives

- Expose a simple application through various types of Services
- Understand how each Service type handles source IP NAT
- Understand the tradeoffs involved in preserving source IP

Source IP for Services with Type=ClusterIP

Packets sent to ClusterIP from within the cluster are never source NAT'd if you're running kube-proxy in [iptables mode](#), (the default). You can query the kube-proxy mode by fetching `http://localhost:10249/proxyMode` on the node where kube-proxy is running.

```
kubectl get nodes
```

The output is similar to this:

NAME	STATUS	ROLES	AGE	VERSION
kubernetes-node-6jst	Ready	<none>	2h	v1.13.0
kubernetes-node-cx31	Ready	<none>	2h	v1.13.0
kubernetes-node-jj1t	Ready	<none>	2h	v1.13.0

Get the proxy mode on one of the nodes (kube-proxy listens on port 10249):

```
# Run this in a shell on the node you want to query.  
curl http://localhost:10249/proxyMode
```

The output is:

```
iptables
```

You can test source IP preservation by creating a Service over the source IP app:

```
kubectcl expose deployment source-ip-app --name=clusterip --port=80 --target-port=8080
```

The output is:

```
service/clusterip exposed
```

```
kubectcl get svc clusterip
```

The output is similar to:

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
clusterip	ClusterIP	10.0.170.92	<none>	80/TCP	51s

And hitting the ClusterIP from a pod in the same cluster:

```
kubectcl run busybox -it --image=busybox:1.28 --restart=Never --rm
```

The output is similar to this:

Waiting for pod default/busybox to be running, status is Pending, pod ready: false
If you don't see a command prompt, try pressing enter.

You can then run a command inside that Pod:

```
# Run this inside the terminal from "kubectcl run"  
ip addr
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
3: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue  
    link/ether 0a:58:0a:f4:03:08 brd ff:ff:ff:ff:ff:ff  
    inet 10.244.3.8/24 scope global eth0  
        valid_lft forever preferred_lft forever  
    inet6 fe80::188a:84ff:feb0:26a5/64 scope link  
        valid_lft forever preferred_lft forever
```

...then use wget to query the local webserver

```
# Replace "10.0.170.92" with the IPv4 address of the Service named "clusterip"  
wget -qO - 10.0.170.92
```

```
CLIENT VALUES:
client_address=10.244.3.8
command=GET
...
```

The `client_address` is always the client pod's IP address, whether the client pod and server pod are in the same node or in different nodes.

Source IP for Services with `Type=NodePort`

Packets sent to Services with [Type=NodePort](#) are source NAT'd by default. You can test this by creating a NodePort Service:

```
kubectl expose deployment source-ip-app --name=nodeport --port=80 --target-port=8080 --type=NodePort
```

The output is:

```
service/nodeport exposed
```

```
NODEPORT=$(kubectl get -o jsonpath="{.spec.ports[0].nodePort}" services nodeport)
NODES=$(kubectl get nodes -o jsonpath='{ $.items[*].status.addresses[? (@.type=="InternalIP")].address }')
```

If you're running on a cloud provider, you may need to open up a firewall-rule for the `nodes:nodeport` reported above. Now you can try reaching the Service from outside the cluster through the node port allocated above.

```
for node in $NODES; do curl -s $node:$NODEPORT | grep -i client_address; done
```

The output is similar to:

```
client_address=10.180.1.1
client_address=10.240.0.5
client_address=10.240.0.3
```

Note that these are not the correct client IPs, they're cluster internal IPs. This is what happens:

- Client sends packet to `node2:nodePort`
- `node2` replaces the source IP address (SNAT) in the packet with its own IP address
- `node2` replaces the destination IP on the packet with the pod IP
- packet is routed to node 1, and then to the endpoint
- the pod's reply is routed back to `node2`
- the pod's reply is sent back to the client

Visually:

[source IP nodeport figure 01](#)

Figure. Source IP `Type=NodePort` using SNAT

To avoid this, Kubernetes has a feature to [preserve the client source IP](#). If you set `service.spec.externalTrafficPolicy` to the value `Local`, kube-proxy only proxies proxy requests to local endpoints, and does not forward traffic to other nodes. This approach preserves the

original source IP address. If there are no local endpoints, packets sent to the node are dropped, so you can rely on the correct source-ip in any packet processing rules you might apply a packet that make it through to the endpoint.

Set the `service.spec.externalTrafficPolicy` field as follows:

```
kubectl patch svc nodeport -p '{"spec":{"externalTrafficPolicy":"Local"}}'
```

The output is:

```
service/nodeport patched
```

Now, re-run the test:

```
for node in $NODES; do curl --connect-timeout 1 -s $node:$NODEPORT | grep -i  
client_address; done
```

The output is similar to:

```
client_address=198.51.100.79
```

Note that you only got one reply, with the *right* client IP, from the one node on which the endpoint pod is running.

This is what happens:

- client sends packet to `node2:nodePort`, which doesn't have any endpoints
- packet is dropped
- client sends packet to `node1:nodePort`, which *does* have endpoints
- node1 routes packet to endpoint with the correct source IP

Visually:

source IP nodeport figure 02

Figure. Source IP `Type=NodePort` preserves client source IP address

Source IP for Services with `Type=LoadBalancer`

Packets sent to Services with `Type=LoadBalancer` are source NAT'd by default, because all schedulable Kubernetes nodes in the Ready state are eligible for load-balanced traffic. So if packets arrive at a node without an endpoint, the system proxies it to a node *with* an endpoint, replacing the source IP on the packet with the IP of the node (as described in the previous section).

You can test this by exposing the `source-ip-app` through a load balancer:

```
kubectl expose deployment source-ip-app --name=loadbalancer --port=80 --target-port=8080 --  
type=LoadBalancer
```

The output is:

```
service/loadbalancer exposed
```

Print out the IP addresses of the Service:

```
kubectl get svc loadbalancer
```

The output is similar to this:

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
loadbalancer	LoadBalancer	10.0.65.118	203.0.113.140	80/TCP	5m

Next, send a request to this Service's external-ip:

```
curl 203.0.113.140
```

The output is similar to this:

```
CLIENT VALUES:
client_address=10.240.0.5
...
```

However, if you're running on Google Kubernetes Engine/GCE, setting the same `service.spec.externalTrafficPolicy` field to `Local` forces nodes *without* Service endpoints to remove themselves from the list of nodes eligible for loadbalanced traffic by deliberately failing health checks.

Visually:

Source IP with `externalTrafficPolicy`

You can test this by setting the annotation:

```
kubectl patch svc loadbalancer -p '{"spec":{"externalTrafficPolicy":"Local"}}'
```

You should immediately see the `service.spec.healthCheckNodePort` field allocated by Kubernetes:

```
kubectl get svc loadbalancer -o yaml | grep -i healthCheckNodePort
```

The output is similar to this:

```
healthCheckNodePort: 32122
```

The `service.spec.healthCheckNodePort` field points to a port on every node serving the health check at `/healthz`. You can test this:

```
kubectl get pod -o wide -l app=source-ip-app
```

The output is similar to this:

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE
source-ip-app-826191075-qehz4	1/1	Running	0	20h	10.180.1.136	kubernetes-node-6jst

Use `curl` to fetch the `/healthz` endpoint on various nodes:

```
# Run this locally on a node you choose
curl localhost:32122/healthz
```

1 Service Endpoints found

On a different node you might get a different result:

```
# Run this locally on a node you choose
curl localhost:32122/healthz
```

No Service Endpoints Found

A controller running on the [control plane](#) is responsible for allocating the cloud load balancer. The same controller also allocates HTTP health checks pointing to this port/path on each node. Wait about 10 seconds for the 2 nodes without endpoints to fail health checks, then use curl to query the IPv4 address of the load balancer:

```
curl 203.0.113.140
```

The output is similar to this:

```
CLIENT VALUES:
client_address=198.51.100.79
...
```

Cross-platform support

Only some cloud providers offer support for source IP preservation through Services with Type=LoadBalancer. The cloud provider you're running on might fulfill the request for a loadbalancer in a few different ways:

1. With a proxy that terminates the client connection and opens a new connection to your nodes/endpoints. In such cases the source IP will always be that of the cloud LB, not that of the client.
2. With a packet forwarder, such that requests from the client sent to the loadbalancer VIP end up at the node with the source IP of the client, not an intermediate proxy.

Load balancers in the first category must use an agreed upon protocol between the loadbalancer and backend to communicate the true client IP such as the HTTP [Forwarded](#) or [X-FORWARDED-FOR](#) headers, or the [proxy protocol](#). Load balancers in the second category can leverage the feature described above by creating an HTTP health check pointing at the port stored in the service.spec.healthCheckNodePort field on the Service.

Cleaning up

Delete the Services:

```
kubectl delete svc -l app=source-ip-app
```

Delete the Deployment, ReplicaSet and Pod:

```
kubectl delete deployment source-ip-app
```

What's next

- Learn more about [connecting applications via services](#)
- Read how to [Create an External Load Balancer](#)

Explore Termination Behavior for Pods And Their Endpoints

Once you connected your Application with Service following steps like those outlined in [Connecting Applications with Services](#), you have a continuously running, replicated application, that is exposed on a network. This tutorial helps you look at the termination flow for Pods and to explore ways to implement graceful connection draining.

Termination process for Pods and their endpoints

There are often cases when you need to terminate a Pod - be it for upgrade or scale down. In order to improve application availability, it may be important to implement a proper active connections draining.

This tutorial explains the flow of Pod termination in connection with the corresponding endpoint state and removal by using a simple nginx web server to demonstrate the concept.

Example flow with endpoint termination

The following is the example of the flow described in the [Termination of Pods](#) document.

Let's say you have a Deployment containing of a single nginx replica (just for demonstration purposes) and a Service:

[service/pod-with-graceful-termination.yaml](#)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      terminationGracePeriodSeconds: 120 # extra long grace period
      containers:
```



```

- name: nginx
  image: nginx:latest
  ports:
  - containerPort: 80
  lifecycle:
    preStop:
      exec:
        # Real life termination may take any time up to terminationGracePeriodSeconds.
        # In this example - just hang around for at least the duration of
        terminationGracePeriodSeconds,
        # at 120 seconds container will be forcibly terminated.
        # Note, all this time nginx will keep processing requests.
        command: [
          "/bin/sh", "-c", "sleep 180"
        ]

```

[service/explore-graceful-termination-nginx.yaml](#)

```

apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80

```

Now create the Deployment Pod and Service using the above files:

```

kubectl apply -f pod-with-graceful-termination.yaml
kubectl apply -f explore-graceful-termination-nginx.yaml

```

Once the Pod and Service are running, you can get the name of any associated EndpointSlices:

```

kubectl get endpointslice

```

The output is similar to this:

NAME	ADDRESSTYPE	PORTS	ENDPOINTS	AGE
nginx-service-6tjbr	IPv4	80	10.12.1.199,10.12.1.201	22m

You can see its status, and validate that there is one endpoint registered:

```

kubectl get endpointslices -o json -l kubernetes.io/service-name=nginx-service

```

The output is similar to this:

```

{
  "addressType": "IPv4",
  "apiVersion": "discovery.k8s.io/v1",
  "endpoints": [

```

```
{
  "addresses": [
    "10.12.1.201"
  ],
  "conditions": {
    "ready": true,
    "serving": true,
    "terminating": false
  }
}
```

Now let's terminate the Pod and validate that the Pod is being terminated respecting the graceful termination period configuration:

```
kubectl delete pod nginx-deployment-7768647bf9-b4b9s
```

All pods:

```
kubectl get pods
```

The output is similar to this:

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-7768647bf9-b4b9s	1/1	Terminating	0	4m1s
nginx-deployment-7768647bf9-rkxlw	1/1	Running	0	8s

You can see that the new pod got scheduled.

While the new endpoint is being created for the new Pod, the old endpoint is still around in the terminating state:

```
kubectl get endpointslice -o json nginx-service-6tjbr
```

The output is similar to this:

```
{
  "addressType": "IPv4",
  "apiVersion": "discovery.k8s.io/v1",
  "endpoints": [
    {
      "addresses": [
        "10.12.1.201"
      ],
      "conditions": {
        "ready": false,
        "serving": true,
        "terminating": true
      },
      "nodeName": "gke-main-default-pool-dca1511c-d17b",
      "targetRef": {
        "kind": "Pod",
        "name": "nginx-deployment-7768647bf9-b4b9s",
        "namespace": "default",
        "uid": "66fa831c-7eb2-407f-bd2c-f96dfe841478"
      },
      "zone": "us-central1-c"
    }
  ]
}
```

```
},
{
  "addresses": [
    "10.12.1.202"
  ],
  "conditions": {
    "ready": true,
    "serving": true,
    "terminating": false
  },
  "nodeName": "gke-main-default-pool-dca1511c-d17b",
  "targetRef": {
    "kind": "Pod",
    "name": "nginx-deployment-7768647bf9-rkxlw",
    "namespace": "default",
    "uid": "722b1cbe-dcd7-4ed4-8928-4a4d0e2bbe35"
  },
  "zone": "us-central1-c"
}
```

This allows applications to communicate their state during termination and clients (such as load balancers) to implement a connections draining functionality. These clients may detect terminating endpoints and implement a special logic for them.

In Kubernetes, endpoints that are terminating always have their ready status set as false. This needs to happen for backward compatibility, so existing load balancers will not use it for regular traffic. If traffic draining on terminating pod is needed, the actual readiness can be checked as a condition serving.

When Pod is deleted, the old endpoint will also be deleted.

What's next

- Learn how to [Connect Applications with Services](#)
- Learn more about [Using a Service to Access an Application in a Cluster](#)
- Learn more about [Connecting a Front End to a Back End Using a Service](#)
- Learn more about [Creating an External Load Balancer](#)